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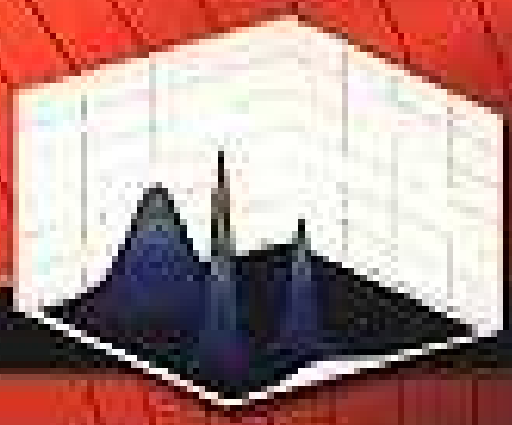
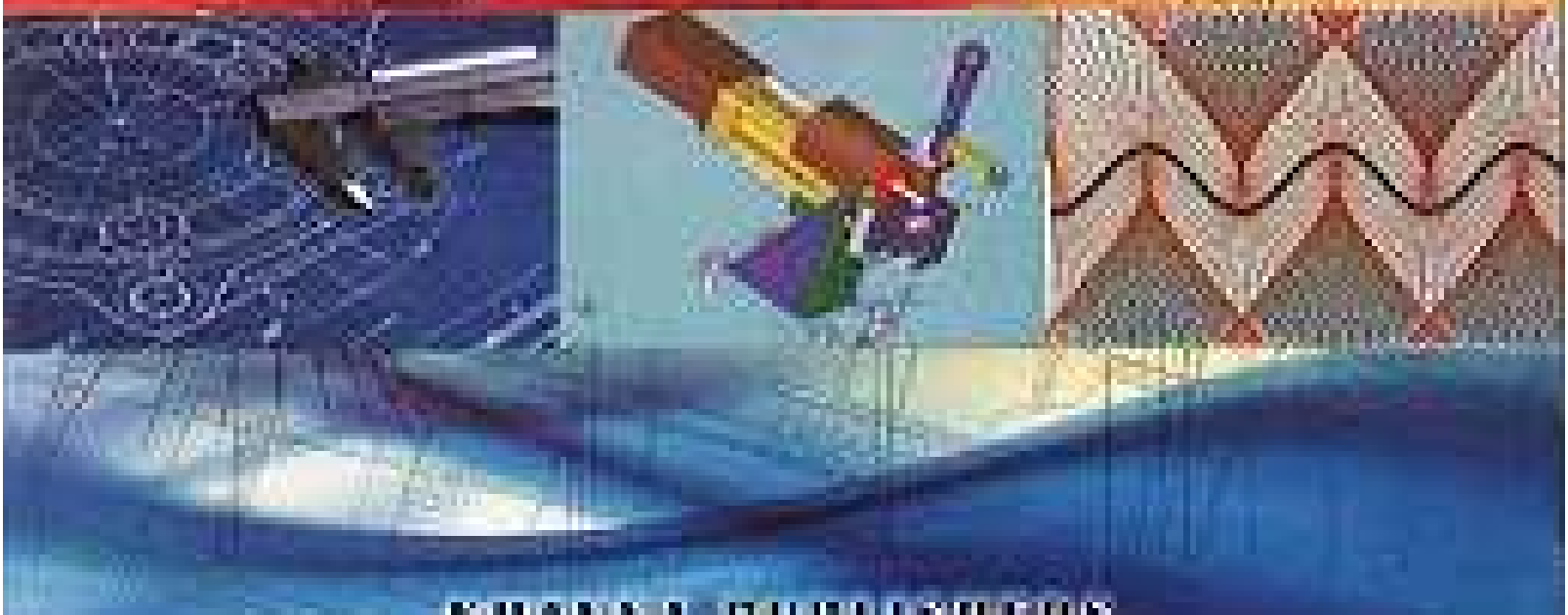


Higher Engineering Mathematics

Dr. B.S. Grewal



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42nd Edition

Higher Engineering Mathematics

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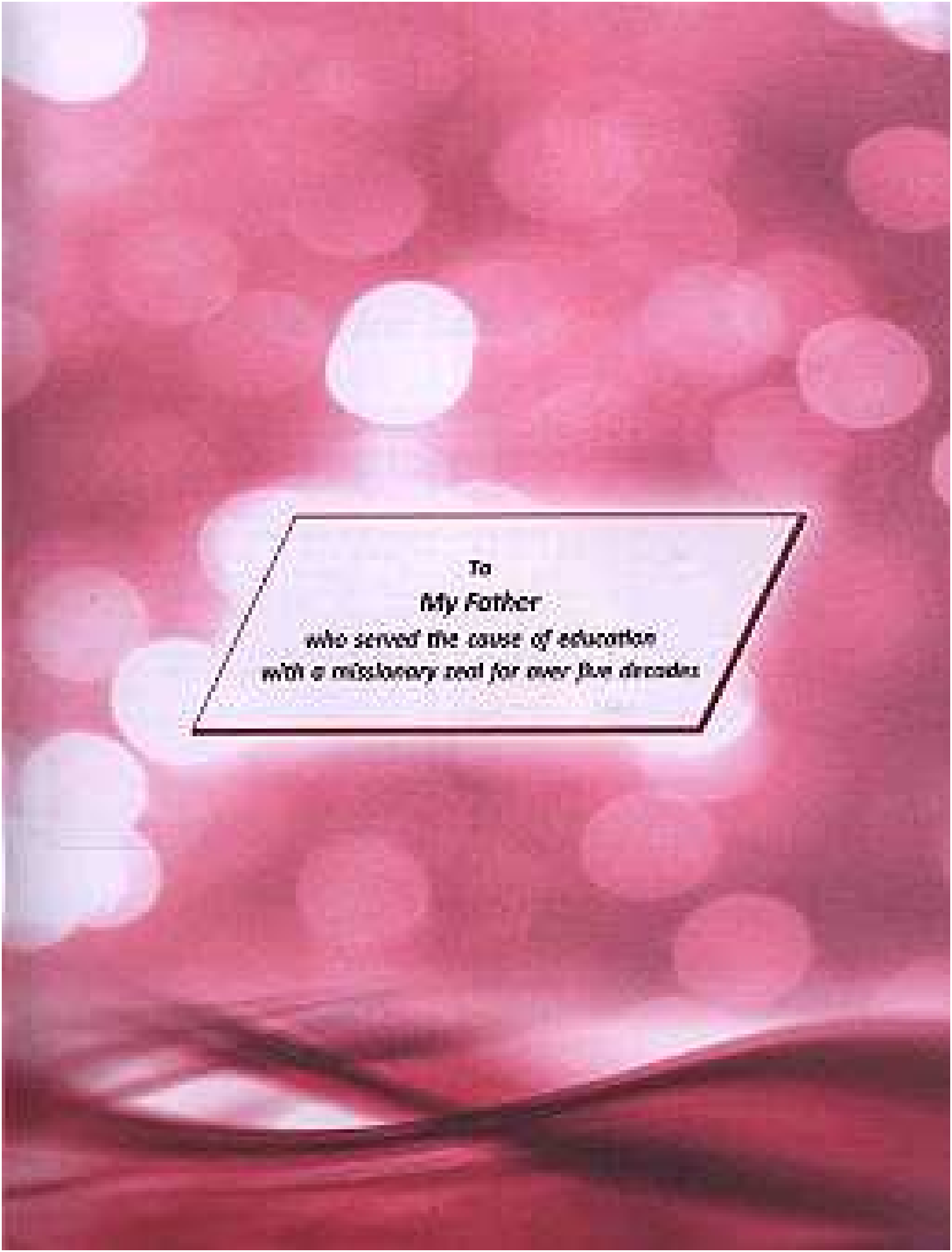
To
My Father
who served the cause of education
with a missionary zeal for over five decades

Preface to the 42nd Edition

The book has now been recast in an attractive new format, retaining its main features which have made it so popular. The text has been carefully revised, the number of illustrative examples has been increased and problems from the latest university question papers have been added. The 'Objective Type of Questions' have been replaced and given at the end of each chapter. It is hoped that the book in its new form will enjoy its ever increasing popularity.

The author takes this opportunity to thank the numerous readers in India and abroad for their letters of appreciation and fellow professors for their suggestions and patronage of the book. In particular, he is grateful to Prof. Avtarsingh Dhaliwaspur, V.K. Engg. College, Bellary (Kar.); Prof. P. Annapurna, N.B.K.R. Inst. of Technology, Vidyanagar (A.P.); Dr. A.P. Bharambe, S.I.T., Roberts (M.P.); Prof. M. Vasudeva Reddy, Vardaman Inst. of Technology, Tirupathi (Tirupati); Dr. R.P. Bhatia, B.A.M. University, Aurangabad (Mah.); Prof. B.K. Yadav, Choudhary Engg. College, Fatehpur (U.P.); Prof. D. Ravi Kumar, Warner University, Guntur (A.P.); Dr. A.C. Pujapati, Chaitanya University of Sc. & Technology, Changa (Guj.); Prof. Ramesh Dhakite, S.R. Technology Institute, Nalgonda (A.P.); Dr. Lalita Bhardwaj, R.V.S. College of Engg. & Technology, Rohtak; Prof. B. Ramamurthy, NH Parkarwal Engg. College, Karnataka (A.P.) and Prof. Vikas Goyal, J.M. Inst. of Technology, Rohtak (Haryana).

Suggestions for improvement of the text and inclusion of important will be thankfully acknowledged.

(Dr. [illegible])

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Note : The rejection point alongwith the performance position in the Degree Engineering Examination held in the previous year is also given in the table. The abbreviations used are explained below :

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

CHAPTER 1

Solution of Equations

1. Introduction 2. General properties 3. Transformation of equations 4. Reciprocal equations 5. Solution of cubic equations (Cardan's method) 6. Solution of biquadratic equations—Ferrari's method 7. Newton's method 8. Graphical solution of equations 9. Objective Type Questions

1.1 INTRODUCTION

The expression $f(x) = a_0x^n + a_1x^{n-1} + \ldots + a_{n-1}x + a_n$

where $a_0, a_1, \ldots, a_n$ are constants ($a_0 \neq 0$) and n is a positive integer, is called a polynomial in x of degree n. The polynomial equation $f(x) = 0$ is called an algebraic equation of degree n. If $f(x)$ contains some other functions such as trigonometric, logarithmic, exponential etc., then $f(x) = 0$ is called a transcendental equation.

The value of x which satisfies $f(x) = 0$ ... (1)

is called its root. Geometrically, a root of (1) is that value of x where the graph of $y = f(x)$ cuts the x-axis. The process of finding the roots of an equation is known as solution of that equation. This is a problem of basic importance in applied mathematics. For other many serious problems in deflection of beams, electrical circuits and mechanical vibrations which depend upon the solution of equations. In such a brief account of solution of equations is given in this chapter.

1.2 GENERAL PROPERTIES

I. If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $x - \alpha$ and conversely. For instance, 3 is a root of the equation $x^3 - 6x^2 + 9x - 4 = 0$, because $x = 3$ satisfies the equation.

$\therefore$ $x - 3$ divides $x^3 - 6x^2 + 9x - 4$ completely, i.e., $x - 3$ is its factor.

II. Every equation of the nth degree has n roots (real or imaginary).

Conversely if $\alpha_1, \alpha_2, \ldots, \alpha_n$ be the roots of the nth degree equation $f(x) = 0$, then

$f(x) \equiv A(x - \alpha_1)(x - \alpha_2)\ldots(x - \alpha_n)$, where A is a constant.

Obs. No polynomial of degree n can have more than n roots or it is impossible to find $(n+1)$ roots.

Example 1.1. Solve the equation $2x^3 + x^2 - 13x - 6 = 0$.

Solution. By inspection, we find $x = 2$ satisfies the given equation.

$\therefore$ 2 is its root, i.e., $x - 2$ is a factor of $2x^3 + x^2 - 13x - 6$. Dividing this polynomial by $x - 2$, we get the quotient $2x^2 + 5x + 3$ and remainder 0.

Equating the quotient to zero, we get $2x^2 + 5x + 3 = 0$.

Solving this quadratic, we get $x = \dfrac{-5 \pm \sqrt{5^2 - 4 \cdot 2 \cdot 3}}{2 \times 2} = \dfrac{-5 \pm 1}{4} = -1, -\dfrac{3}{2}$.

Hence, the roots of the given equation are $2, -1, -\frac{3}{2}$.

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Note. The labour of checking the polynomial by $x - 2$ may be much reduced mentally by the following simple device in which numbers in bold are mentally remembered:

2	1	−12	8	2
	2	**−20**	**−8**	
1	0	**−10**	0	

Explanation. (a) Write down the coefficients of the powers of x in order (supplying the missing powers with 0) and write 2 on a line to the right.

(b) Put 1 on the first line of 3rd row and multiply it by 2, write 2 under 1 and add, giving 2.

(c) Multiply 2 by 2, put in 3rd under −12 and add, giving −10.

(d) Multiply −10 by 2, write −20 under 8 and add giving −12.

Thus the quotient is $x^2 + 2x - 10$ and the remainder is zero.

Note. To divide a polynomial by $x + 3$, we use -3 as a factor.

III. Intermediate value property. If $f(a)$ and $f(b)$ have opposite signs, then the equation $f(x) = 0$ has at least one real root between $x = a$ and $x = b$.

The polynomial $f(x)$ is a continuous function (Fig. 1.1). So if x changes from a to b, $f(x)$ must pass through all the values from $f(a)$ to $f(b)$. But since one of these quantities $f(a)$ or $f(b)$ is positive and the other negative, 0 must be one of these values of $f(x)$ as x changes between a and b. In general, there may be more. There is at the very least one.

Fig. 1.1

IV. In an equation with real coefficients, imaginary roots occur in conjugate pairs, *i.e.*, if $a + ib$ is a root of the equation $f(x) = 0$, then $a - ib$ is also a root. (See p. 244)

Similarly $7 + \sqrt{2}i$ is an irrational root of an equation, then $7 - \sqrt{2}i$ must also be its root.

This follows from the fact that imaginary roots occur in conjugate pairs.

Example 1.2. Solve the equation $2x^4 - 4x^3 + x + 99 = 0$, one root being $2 + \sqrt{7}i$.

Solution. Since one root is $2 + \sqrt{7}i$, the other must be $2 - \sqrt{7}i$.

$\therefore$ The factors corresponding to these roots are

$$[x - (2 - \sqrt{7}i)] \text{ and } [x - (2 + \sqrt{7}i)]$$

or $$[x - 2) + \sqrt{7}i][(x - 2) - \sqrt{7}i] = (x - 2)^2 + 7 = x^2 - 4x + 11,$$

which is a factor of $2x^4 - 4x^3 + x + 99$ ...(1)

$\therefore$ Division of (1) by $x^2 - 4x + 11$ gives $2x + 9$ as the quotient.

Thus the depressed equation is $2x + 9 = 0$. Its root is -4.5. Hence the roots of the given equation are $2 + \sqrt{7}i$, -4.5.

V. Descartes's rule of signs. The equation $f(x) = 0$ cannot have more positive roots than the number of changes of signs in $f(x)$ and more negative roots than the changes of signs in $f(-x)$.

For instance, consider the equation $f(x) = 2x^5 - x^4 + 4x^3 - 3x + 5 = 0$ ...(1)

Signs of $f(x)$ are: + − + − +

Clearly, $f(x)$ has 3 changes of signs (from + to −, or − to +).

Thus $f(x)$ cannot have more than 3 positive roots.

---

*Viète (or Vieta) was a French mathematician and gifted decipherer (Louis Answerson 1600–1650), who lived in Paris during the Spanish occupation.

Also $f(-x) = 2x^4 - x^3 - 1 - x^2 + x - x^3 - 2$

$= -2x^4 + x^3 - 2x^2 - 3$

This shows that $f(x)$ has 2 changes of signs. Thus $f(x)$ cannot have more than 2 negative roots.

Obs. Existence of imaginary roots. If in an equation of the n^{th} degree, there are k changes of signs, then the equation cannot have more than k positive roots and if the number of negative roots is l, then it is obvious that the equation has at least $n - (k + l)$ imaginary roots.

In Example (3) above, the equation is of the 7th degree and has at the most 3 positive roots and 2 negative roots. Thus it has at least 2 imaginary roots.

VI. Relations between roots and coefficients. *If $\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n$ be the roots of the equation*

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n = 0 \quad \ldots(1)$$

then

$$\Sigma\alpha_1 = -\frac{a_1}{a_0}, \quad \Sigma\alpha_1\alpha_2 = \frac{a_2}{a_0}, \quad \Sigma\alpha_1\alpha_2\alpha_3 = -\frac{a_3}{a_0}$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$\alpha_1\alpha_2\alpha_3 \ldots \alpha_n = (-1)^n \frac{a_n}{a_0}.$$

Example 1. *Solve the equation $x^3 - 2x^2 + 3x + 6 = 0$, given that one root is double the other root.*

Solution. Let the roots be α, β, γ such that $\beta = 2\alpha$.

Also $\alpha + \beta + \gamma = 2, \alpha\beta + \beta\gamma + \gamma\alpha = 3, \alpha\beta\gamma = -6$

$\Rightarrow$ $3\alpha + \gamma = 2$ ...(1)

$2\alpha^2 + 2\alpha\gamma = 3$ ...(2)

$2\alpha^2\gamma = -6$ ...(3)

Solving (1) and (3), we get $\alpha = 1, \gamma = -1$.

(The values $\alpha = 3, \gamma = 7$ are inadmissible, as they do not satisfy (3).)

Hence the required roots are $1, 2, -1$ and -2.

Example 2. *Solve the equation $x^4 - 2x^3 + 3x^2 + 6x - 2 = 0$, given that the sum of two of its roots is zero.* (Kerala, 2005; Andhra, 2003)

Solution. Let the roots be $\alpha, \beta, \gamma, \delta$ such that $\alpha + \beta = 0$.

Also $\alpha + \beta + \gamma + \delta = 2 \Rightarrow \gamma + \delta = 2$

Thus the quadratic factor corresponding to α, β is of the form $x^2 - kx + p$, and that corresponding to γ, δ is of the form $x^2 - 2x + q$.

$\therefore$ $x^4 - 2x^3 + 3x^2 + 6x - 2 = (x^2 + px + q)(x^2 - 2x + q)$ ...(1)

Equating the coefficients of x^3 and x from both sides of (1), we get

$-2 = p + q, \quad 6 = -2p$

$\therefore$ $p = -3, \quad q = 1$

Hence the given equation is equivalent to $(x^2 - 3x + 1)(x^2 - 2x + 2) = 0$.

$\therefore$ The roots are $x = \frac{3 \pm \sqrt{5}}{2}, 1 \pm i$.

Example 3. *Find the condition that the cubic $x^3 - 3x^2 + px - q = 0$ should have its roots in*

(i) arithmetical progression (Madras, 2000)

(ii) geometrical progression.

Solution. Let the roots be $\alpha - d, \alpha, \alpha + d$ so that the sum of the roots $= 3\alpha = 3$ i.e., $\alpha = 1$.

Since α is the root of the given equation

$\therefore$ $\alpha^3 - 3\alpha^2 + p\alpha - q = 0$ ...(1)

Substituting $\alpha = 1$, we get $1 - 3 + p - q = 0$, i.e., $p - q = 2$.

or $3^3 - 3(3)^2 + 2(3) = 0$, which is the required condition.

(ii) Let the roots be $\alpha/r, \alpha, \alpha r$, so that the product of the roots $= \alpha^3 = \alpha$.
Putting $\alpha = (\alpha)^{1/3}$, i.e. γ, we get $\alpha = (\alpha)^{1/3} \cdot \alpha \cdot \alpha = \alpha^{1/3}$ or $\alpha = \alpha^{1/3}$
Cubing both sides, we get $\alpha^3 = \alpha^2 \alpha$, which is the required condition.

Example 4.8. Solve the equation $x^4 - 2x^3 - 21x^2 + 22x + 40 = 0$ whose roots are in A.P.

Solution. Let the roots be $a - 3d, a - d, a + d, a + 3d$, so that the sum of the roots $= 4a = 2$.
$\therefore \quad a = \frac{1}{2}$
Also product of the roots $= (a^2 - 9d^2)(a^2 - d^2) = 40$

or $\left(\frac{1}{4} - 9d^2\right)\left(\frac{1}{4} - d^2\right) = 40 \quad$ or $\quad 100d^4 - 40d^2 - 639 = 0$

$\therefore \quad d^2 = 3.24 \quad$ or $\quad -1.97$

Thus, $d = \pm 1.8$, the other value is not admissible.
Hence the required roots are $-4, -1, 2, 5$

Example 4.9. Solve the equation $2x^4 - 15x^3 + 35x^2 - 30x + 8 = 0$ whose roots are in G.P.

Solution. Let the roots be $a/r^3, a/r, ar, ar^3$, so that product of the roots $= a^4 = 4$.
Also the product of $a/r^3, ar^3$ and $a/r, ar$ are each $= a^2 = 2$.
$\therefore$ The factors corresponding to $a/r^3, ar^3$ and $a/r, ar$ are $x^2 + px + 2, x^2 + qx + 2$.
Thus, $2x^4 - 15x^3 + 35x^2 - 30x + 8 = (x^2 + px + 2)(x^2 + qx + 2)$
Equating the coefficients of x^3 and x^2,

$$-15 = 2p + 2q \quad \text{and} \quad 35 = 4 + 2pq$$

$$\therefore \quad p + q = -\frac{15}{2}, \; pq = \frac{31}{2}$$

Thus the given equation is $2\left(x^2 - \frac{15}{2}x + 2\right)(x^2 - 2x + 2) = 0$

Hence the required roots are $\frac{15 \pm \sqrt{161}}{4}, 1 \pm i, 2 \pm i$, or $\frac{3}{2}, 2, 2, 4$.

Example 4.10. If $\alpha, \beta, \gamma, \delta$ be the roots of the equation $x^4 + px^3 + qx^2 + rx + s = 0$, find the value of
(i) $\Sigma\alpha^2\beta$ (ii) $\Sigma\alpha^2$ (iii) $\Sigma\alpha^2\beta^2$

Solution. We have $\quad \alpha + \beta + \gamma + \delta = -p \quad \ldots(i)$

$$\Sigma\alpha\beta + \Sigma\alpha\beta\gamma = q \quad \ldots(ii)$$

$$\Sigma\alpha\beta\gamma\delta = -r \quad \ldots(iii)$$

(i) Multiplying (ii) and (i), we get

$$\alpha^2\beta + \alpha^2\gamma + \beta^2\gamma + \beta^2\alpha + \gamma^2\alpha + \gamma^2\beta + 3(\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \delta\alpha\beta) = -pq$$

or $\quad \Sigma\alpha^2\beta + 3\Sigma\alpha\beta\gamma = -pq \quad$ (By (iii))

(ii) Multiplying the given equation by x, we get $x^5 + px^4 + qx^3 + rx^2 + sx = 0$
Putting $x = \alpha, \beta, \gamma, \delta$ successively and adding, we get $\Sigma\alpha^5 + p\Sigma\alpha^4 + q\Sigma\alpha^3 + r\Sigma\alpha^2 + s\Sigma\alpha = 0$

or $\quad \Sigma\alpha^5 = -p\Sigma\alpha^4 - q\Sigma\alpha^3 \quad \ldots(iv)$

Now squaring (i), we get $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + 2\Sigma\alpha\beta = p^2$

or $\quad \Sigma\alpha^2 = p^2 - 2q \quad$ (By (ii))

Hence, substituting the value of $\Sigma\alpha^2$ in (iv), we obtain

$$\Sigma\alpha^5 = -p(p^2 - 2q) + q(-p) = -2p^3 + 3pq$$

(iii) $\Sigma\alpha^2\beta^2 = \Sigma\alpha^2 \cdot \Sigma\alpha\beta - \alpha\beta\gamma\Sigma\alpha = -2p(p) + (-q)(q) = -2p^2 + q^2$.

EXERCISES

1. Form the equation of the fourth degree whose roots are $1, -1$ and $\sqrt{2}$. (Madras, 2004 S)
2. Solve the equation $(x^2 + 6x + 9)(x^2 + 6x + 28) = 40$, given that $3 + \sqrt{-1}$ is a root.
   $(x^2 - 2x^2 - 28x^2 + 82x - 35 = 0$, given that $3 + \sqrt{-2}$ is a root.
3. Form the equation $x^3 - 2x^2 + 3x - 1 = 0$ whose roots are each greater than unity by unity. (Cochin, 2000)
4. Show that the equation $x^3 + 15x^2 + 75 - 11 = 0$ has one positive root and two negative roots and one of the latter is greater than unity.
5. Find the number of positive and negative roots of $x^4 + 4x^3 - 4x - 15 = 0$.
6. Solve the equation $4x^4 - 16x^3 - 57x^2 + 4x + 15 = 0$, given that two of its roots are equal.
7. If α, β, γ are the roots of the equation $2x^3 + 3x^2 + 4x + 1 = 0$, find the value of $\sum \frac{1}{\alpha\beta}$. (A.U. B.E. (E.C.E.), 2009)
8. Find the equation whose roots are the squares of the roots of $x^3 + 2x^2 + 3x + 4 = 0$. Hence or otherwise find the value of $\sum \alpha^4$.
   Solve the following equations:
9. $x^3 - 6x^2 + 12x - 7 = 0$, given that one of its roots is one more than the other.
10. $x^4 - 4x^3 - 20x + 48 = 0$, given that the sum of two of its roots is zero. (A.U. B.E. (E.C.E.), 2007)
11. $x^4 - 8x^3 + 18x^2 - 12x + 4 = 0$, given that it has two pairs of equal roots. (Madras, 2009)
12. $x^4 - 4x^3 + 2x^2 + 4x - 3 = 0$, given that the sum of two of the roots is equal to the sum of the other two.
13. $x^3 - 12x^2 + 39x - 28 = 0$, roots being in arithmetical progression. (Madras, 2003 S)
14. $8x^3 - 14x^2 + 7x - 1 = 0$, roots being in geometrical progression. (Chennai, 2000)
15. A, B, C are the four points on a straight line such that the distances of A, B, C from O are the roots of equation $x^3 + ax^2 + bx + c = 0$. If O is the middle point of AC, show that $a^3 c = b^3 - 4$. (A.U. B.E. (E.C.E.), 2008)
16. Solve the equation $x^4 + 2x^3 - 21x^2 - 22x + 40 = 0$, two of whose roots are in A.P.
    $6x^4 + 5x^3 - 38x^2 - 15x + 36 = 0$, whose roots are in G.P.
17. If α, β, γ be the roots of the equation $x^3 - 3x^2 + px + q = 0$, find the value of
    $\sum \alpha(\beta + \gamma)^2$. (A.U. B.E. (E.C.E.), 2008)
18. Find the sum of the cubes of the roots of the equation $x^3 - 6x^2 + 11x - 6 = 0$.
19. If α, β are the roots of $x^2 + 4x + 1 = 0$, find the value of $(\alpha^n + \beta^n + 1)(\alpha^n + \beta^n + 2)$.
20. If α, β, γ are the roots of $x^3 + px + q = 0$, show that
    $\sum \alpha^2 \beta^2 = p^2$, $\sum \alpha^3 \beta^3 = 3pq$, $\sum \alpha^4 = 2p^2$. (A.U. B.E. (E.C.E.), 2007) $\sum \alpha^5 \beta^2 + \alpha^2 \beta^5 = -5pq^2$.

4.3 TRANSFORMATION OF EQUATIONS

(1) To find an equation whose roots are m times the roots of the given equation, multiply the second term by m, third term by m^2 and so on till missing terms supplied with zero coefficients.

For instance, let the given equation be

$$ax^3 + bx^2 + cx + d = 0 \qquad \ldots(1)$$

To multiply the roots by m, put $y = mx$ or $x = y/m$ in (1).

Then $a(y/m)^3 + b(y/m)^2 + c(y/m) + d = 0$

Multiplying by m^3, we get $ay^3 + my^2 + m^2(cy) + m^3 d = 0$

This is same as multiplying the second term by m, third term by m^2 and so on in (1).

Cor. To find an equation whose roots are with opposite signs to those of the given equation, change the signs of the every alternate term of the given equation beginning with the second.

Changing the signs of the roots of (1) is same as multiplying its roots by -1.

∴ The required equation will be

$$ax^3 + (-b)x^2 + (-1)^2 cx + (-1)^3 d = 0$$

or $$ax^3 - bx^2 + cx + d = 0$$

which is (1) with signs of every alternate terms changed beginning with the second.

(2) To find an equation whose roots are reciprocals of the roots of the given equation, multiply by x.

Example 1.6. Solve the equation $2x^3 - 3x^2 - 36x + 10 = 0$, given that its roots are in harmonic progression.

Solution. Since the roots of the given equation are in H.P., the roots of the equation having reciprocal roots will be in A.P.

The equation with its reciprocal roots is $10x^3 - 36x^2 - 3x + 2 = 0$

or
$$2x^3 - 3x^2 - 18x + 5 = 0 \qquad \ldots(1)$$

Since the roots of the given equation are in H.P., therefore, the roots of (1) are in A.P. Let the roots be $a - d$, a, $a + d$. Then

$$3a = 3/2 \text{ and } (a^3 - ad^2) = -5/2$$

Solving these equations, we get $a = 1/2$, $d = 5/2$

Then the roots of (1) are -2, $1/2$, 3.

Hence the required roots of the given equation are $-1/2$, 2, $1/3$.

Example 1.7. If α, β, γ are the roots of the cubic equation $x^3 + px^2 + qx + r = 0$, form the equation whose roots are $\beta\gamma + 2/\alpha$, $\gamma\alpha + 2/\beta$, $\alpha\beta + 2/\gamma$.

Hence evaluate $\Sigma(\alpha\beta + 2/\gamma)(\beta\gamma + 2/\alpha)$. (S.V.T.U., 2000)

Solution. If x is a root of the given equation and y a root of the required equation, then

$$y = \beta\gamma + 2/\alpha = \frac{\alpha\beta\gamma + 2}{\alpha} = \frac{r + 2}{\alpha} \qquad [\because \alpha\beta\gamma = r]$$

or
$$x = \frac{r + 2}{y} \quad \text{or} \quad x = \frac{r + 2}{y}$$

Then substituting $x = (r + 2)/y$ in the given equation, we get

$$\left(\frac{r + 2}{y}\right)^3 + p\left(\frac{r + 2}{y}\right)^2 + q\left(\frac{r + 2}{y}\right) + r = 0$$

or
$$ry^3 + qry^2 + 2ry^2 + py^2 + 2pry + 4p + qy + 2q + r = 0, \text{ which is the required equation.}$$

Hence $\Sigma(\alpha\beta + 2/\gamma)(\beta\gamma + 2/\alpha) = q/r + 12/r$.

Example 1.11. Form an equation whose roots are cubes of the roots of $x^3 - 2x^2 + x + 3 = 0$. (U.P.T.U., 2000)

Solution. If y be a root of the required equation, then $y = x^3$ ...(1)

Now we have to eliminate x from (1) and (2).

$\therefore$ Rewriting (2) as $x^3 + 1 = 2x^2$

Cubing both sides, $x^9 + 3x^6 + 3x^3 + 1 = 8x^6$

Substituting $x^3 = y$, we get $y^3 - 2y^2 + 3y + 1 = 0$, which is the required equation.

(3) To diminish the roots of an equation $f(x) = 0$ **by** h, **divide** $f(x)$ **by** $x - h$ **successively. Then the successive remainders determine the coefficients of the required equation.**

Let the given equation be

$$a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n = 0 \qquad \ldots(1)$$

To diminish its roots by h, put $y = x - h$ or $x = y + h$ in (1) so that

$$a_0(y + h)^n + a_1(y + h)^{n-1} + a_2(y + h)^{n-2} + \ldots + a_{n-1}(y + h) + a_n = 0 \qquad \ldots(2)$$

On simplification, it takes the form

$$A_0y^n + A_1y^{n-1} + A_2y^{n-2} + \ldots + A_{n-1}y + A_n = 0 \qquad \ldots(3)$$

The coefficients $A_0, A_1, \ldots A_n$ can easily be found with the help of synthetic division (p. 21). For this, we put $y = x - h$ in (1) so that

$$A_0(x - h)^n + A_1(x - h)^{n-1} + A_2(x - h)^{n-2} + \ldots + A_{n-1}(x - h) + A_n = 0 \qquad \ldots(4)$$

Clearly, (3) and (4) are identical. Now dividing L.H.S. of (4) by $x - h$, the remainder is A_n and the quotient $Q = A_0(x - h)^{n-1} + A_1(x - h)^{n-2} + \ldots + A_{n-2}(x - h) + A_{n-1}$. Similarly, if we divide Q by $x - h$, the remainder is A_{n-1} and the quotient is Q_1 (say). Again dividing Q_1 by $x - h$, A_{n-2} will be obtained as remainder and so on.

Hint. To transform the roots by h, we take h negative.

Example 5.12. Transform the equation $x^3 - 6x^2 + 5x + 6 = 0$ into another in which the second term is missing. Hence find the equation of the squares of the roots. (Roorkee, 2005)

Solution. Sum of the roots of the given equation $= 6$.

In order that the second term in the transformed equation is missing, the sum of the roots is to be zero.

Since the equation has 3 roots, if we decrease each root by 2, the roots of the new equation will become zero.

$\therefore$ Dividing $x^3 - 6x^2 + 5x + 6$ by $x - 2$ successively, we have

1	-6	5	6	2
	2	-8	-6	
	-4	-3	0	
	2	-4		
	-2	-7		
	2			
1	0			

Thus the transformed equation is $x^3 - 7x + 6 = 0$ ...(i)

Let α, β, γ be the roots of the given equation, then the roots of (i) are $\alpha - 2, \beta - 2, \gamma - 2$.

Let these roots be denoted by a, b, c.

Then $a + b + c = \alpha + \beta + \gamma - 6 = 0$, $ab + bc + ca = -7$, $abc = -6$.

Now $a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = 0 - 2(-7) = 14$, $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab + bc + ca}{abc} = \frac{-7}{-6} = \frac{7}{6}$

$\therefore$ The required equation of which (i) is given by the transformation $y = x^2 + 4x$

or $x^3 - xy + 4 = 0$ ...(ii)

Subtracting (i) from (ii), we get $-7x + xy - 2 = 0$ or $x = 2/(y - 7)$.

Substituting for x in (i), the required equation becomes

$$\frac{8}{(y-7)^3} - \frac{7 \cdot 2}{y-7} + 6 = 0 \quad \text{or} \quad y^3 - 21y^2 + 245y - 862 = 0 \qquad \text{...(iii)}$$

Roots of this equation are $(\alpha - \beta)^2, (\beta - \gamma)^2, (\gamma - \alpha)^2$ i.e., $(\alpha - \beta)^2, (\gamma - \alpha)^2, (\alpha - \beta)^2$.

Hence (iii) is the required equation.

5.7 RECIPROCAL EQUATIONS

An equation reducible to the form $ax^n + bx^{n-1} + cx^{n-2} + \ldots + cx^2 + bx + a = 0$ is called a reciprocal equation.

Such equations are of the following standard types :

I. *A reciprocal equation of an odd degree having coefficients of terms equidistant from the beginning and end equal, has $x = -1$ as a root.*

II. *A reciprocal equation of an odd degree having coefficients of terms equidistant from the beginning and end equal but opposite in sign, has $x = 1$ as a root.*

III. *A reciprocal equation of an even degree having coefficients of terms equidistant from the beginning and end equal but opposite in sign, has $x = 1$ and $x = -1$ as roots.*

The substitution $x + 1/x = y$ reduces the degree of the equation of half its former degree.

Example 5.13. Solve $6x^6 - 35x^5 + 62x^4 - 97x^3 + 62x^2 - 35x + 6 = 0$. (Uttarakhand, 2003 S)

Solution. This is a reciprocal equation of odd degree with equal coefficients of terms equidistant from the beginning and end. $\therefore x = -1$ is a root.

Dividing L.H.S. by $x + 1$, the given equation reduces to

$$6x^5 - 41x^4 + 103x^3 - 200x^2 + 162x - 6 = 0$$

Dividing by x^3, we have

$$6(x^2 + 1/x^2) - 41(x + 1/x) + 103 = 0$$

Putting $x + 1/x = y$ and $x^2 + 1/x^2 = y^2 - 2$, we get

$$6y^2 - 12 - 41y + 103 = 0 \quad \text{or} \quad 6y^2 - 41y + 91 = 0 \quad \text{or} \quad (2y - 13)(3y - 7) = 0$$

$\therefore$ $x + 1/x = y = 13/2$ or $7/3$

i.e., $(3x^2 - 10x + 3 = 0$ or $2x^2 - 5x + 2 = 0$

i.e., $(3x - 1)(x - 3) = 0$ or $(2x - 1)(x - 2) = 0$

$\therefore x = \frac{1}{3}, 3$ or $\frac{1}{2}, 2$

Hence the required roots are $1, \frac{1}{3}, 3, \frac{1}{2}, 2$.

Example 1.14. Solve $6x^5 - 35x^4 + 62x^3 - 62x^2 + 35x - 6 = 0$. (Mekelle, 2003)

Solution. This is a reciprocal equation of even degree with opposite signs. $\therefore x = 1, -1$ are the roots.

Dividing LHS by $x - 1$ and $x + 1$, the given equation reduces to

$$6x^4 - 35x^3 + 37x^2 - 35x + 6 = 0$$

Dividing by x^2, we get

$$6x^2 + \frac{6}{x^2} - 35\left(x + \frac{1}{x}\right) + 37 = 0$$

Putting $x + \frac{1}{x} = y$ and $x^2 + \frac{1}{x^2} = y^2 - 2$, it becomes

$$6(y^2 - 2) - 35y + 37 = 0 \quad \text{or} \quad 6y^2 - 35y + 25 = 0$$

or $(6y - 5)(y - 5) = 0$

$\therefore x + \frac{1}{x} = y = \frac{5}{6}$ or 5,

i.e., $2x^2 - 5x + 2 = 0$ or $x^2 - 5x + 1 = 0$

$\therefore x = 2, \frac{1}{2}$ or $x = \frac{5 \pm \sqrt{21}}{2}$

Hence the required roots of the given equation are $1, -1, 2, \frac{1}{2}, \frac{5 \pm \sqrt{21}}{2}$.

EXERCISE 1.2

1. Find the equation whose roots are 3 times the roots of $x^3 + 2x^2 - 4x + 1 = 0$.
2. Form the equation whose roots are the reciprocals of the roots of $2x^4 + 3x^3 - 12x^2 + 7x - 6 = 0$. (U.T.U., 2003)
3. Find the equation whose roots are the negative reciprocals of the roots of
$$x^4 + 2x^3 + 3x^2 - 4x + 12 = 0$$
4. Solve the equation $6x^3 - 11x^2 - 3x + 2 = 0$, given that its roots are in A.P.
5. Find the equation whose roots are the roots of
(a) $x^3 - 6x^2 + 11x - 6 = 0$ each increased by 1. (U.P.T.U., 2004)
(b) $x^4 + x^3 - 3x^2 + x + 2 = 0$ each diminished by 2.
(c) $x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 6 = 0$ each diminished by 1.
6. Find the equation whose roots are the squares of the roots of $x^3 - x^2 + 1 = 0$.
7. Find the equation whose roots are the cubes of the roots of $x^3 + px^2 + q = 0$.
8. If α, β are the roots of the equation $3x^2 + 5x^2 - x - 1 = 0$, form the equation whose roots are $(1 - \alpha)^2, (1 - \beta)^2$ and $(1 - \gamma)^2$.
9. If α, β are the roots of the equation $x^2 + px^2 - qx + r = 0$, find the equation whose roots are $\alpha\beta, \beta\gamma, \gamma\alpha$. (Madras, 2000)
10. If α, β, γ are the roots of the equation $x^3 + px + q = 0$, form the equation whose roots are
(a) $\alpha + \beta, \beta + \gamma, \gamma + \alpha$ (b) $\alpha\beta, \beta\gamma, \gamma\alpha$ (c) $\frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}, \frac{1}{\alpha\beta}$
11. Find the equation of squared differences of the roots of the cubic $x^3 + 3x^2 + 3x + 2 = 0$.
12. Solve the equations :
(a) $6x^4 + 5x^3 - 38x^2 + 5x + 6 = 0$ (b) $6x^5 - 29x^4 + 23x^3 - 23x^2 + 29x - 6 = 0$ (Madras, 2002)
(c) $6x^5 - 25x^4 + 38x^3 - 38x^2 + 25x - 6 = 0$ (d) $6x^5 + x^4 - 43x^3 + 43x^2 - x - 6 = 0$ (U.P.T.U., 2004)
(e) $2x^5 + x^4 - 11x^3 + 11x^2 - x - 2 = 0$
13. Show that the equation $x^4 - 12x^3 + 26x^2 - 12x - 13 = 0$ has its roots in arithmetical progression and solve it.
14. By transforming the equation, solve the equation $x^4 + 0.2x^3 + 0.1x^2 + 0.02x + 0.24 = 0$. An equation in which terms in x^3 is absent and hence solve it. (Madras, 2002)

1.15 SOLUTION OF CUBIC EQUATIONS—CARDAN'S METHOD*

Consider the equation $ax^3 + bx^2 + cx + d = 0$ ...(1)

Dividing by a, we get an equation of the form $x^3 + Ax^2 + Bx + C = 0$.

To remove the x^2 term, put $y = x - (-A/3)$ or $x = y - A/3$ so that the resulting equation is of the form

$$y^3 + py + q = 0 \quad \text{...(2)}$$

To solve (2), put $y = u + v$

so that $y^3 = u^3 + v^3 + 3uv(u + v) = u^3 + v^3 + 3uvy$

or $y^3 - 3uvy - (u^3 + v^3) = 0$ ...(3)

Comparing (2) and (3), we get

$u + v = -p/3$, $u^3 + v^3 = -q$ and $u^3v^3 = -p^3/27$

$\therefore$ u^3, v^3 are the roots of the equation $t^2 + qt - p^3/27 = 0$

which gives $u^3 = \frac{1}{2}\left(-q + \sqrt{q^2 + 4p^3/27}\right) = -\frac{1}{2}(q + D)$ (say)

and $v^3 = \frac{1}{2}\left(-q - \sqrt{q^2 + 4p^3/27}\right)$

$\therefore$ The three values of u are k, $k\omega$, $k\omega^2$, where ω is one of the imaginary cube roots of unity.

Since $uv = -p/3$, we have $v = -p/3u$

$\therefore$ When $u = k, k\omega$ and $k\omega^2$,

$$v = -\frac{p}{3k}, -\frac{p\omega^2}{3k} \text{ and } -\frac{p\omega}{3k} \qquad [\because \omega^3 = 1]$$

Hence the three roots of (2) are $k - \frac{p}{3k}$, $k\omega - \frac{p\omega^2}{3k}$, $k\omega^2 - \frac{p\omega}{3k}$ [Being $y = u + v$]

Having known y, the corresponding value of x can be found from the relation $x = y - A/3$.

Obs. 1. If one value of u is found to be a rational number, find the corresponding value of v giving one root $y = u + v$. Then find the corresponding roots $x = y - A/3$. Finally, divide the left hand side of (1) by $x - x_1$, giving the resulting quadratic equation from which the other two roots can be found readily.

Obs. 2. If u^3 and v^3 turn out to be of complex conjugate numbers, the roots of the given cubic can be obtained in real form by employing De Moivre's theorem (§ 1.10).

Example 1.28. Solve by Cardan's method $x^3 - 3x^2 + 12x + 16 = 0$. (U.P.T.U., 2000)

Solution. Given equation is $x^3 - 3x^2 + 12x + 16 = 0$ ...(1)

To remove the second term from (1), diminish each root of (1) by $-(-1) = 1$, i.e., put $y = x - 1$ or $x = y + 1$.

Sum of roots = 3. Then (1) becomes

$$(y + 1)^3 - 3(y + 1)^2 + 12(y + 1) + 16 = 0 \quad \text{or} \quad y^3 + 9y + 26 = 0 \quad \text{...(2)}$$

To solve (2), put $y = u + v$ so that $y^3 - 3uvy - (u^3 + v^3) = 0$ ...(3)

Comparing (2) and (3), we get $uv = -3$ and $u^3 + v^3 = -26$

$\therefore$ u^3, v^3 are the roots of the equation $t^2 + 26t - 27 = 0$

or $(t + 27)(t - 1) = 0$ whose roots are $t = -27, 1$,

i.e., $u^3 = -27$ i.e., $u = -3$ and $v^3 = 1$ i.e., $v = 1$

$\therefore$ $y = u + v = -3 + 1 = -2$ and $\omega y = \omega - 2\omega = -2$

Dividing L.H.S. of (1) by $x + 1$, we obtain $x^2 - 4x + 16 = 0$

or $x = \frac{4 \pm \sqrt{(16 - 64)}}{2} = 2 \pm 2\sqrt{3}i$

Hence the required roots of the given equation are $-1, 2 \pm 2\sqrt{3}i$.

*Named after an Italian mathematician Girolamo Cardano (1501–1576) who was the first to use complex numbers in terms of the equation.

Example 1.14. Solve the cubic equation $4x^3 - 6x^2 + 1 = 0$ by Cardan's method.

Solution. As the term in x is missing, let $x = py + q$ in the given equation so that the transformed equation is $y^3 - 3y + 2R = 0$ ...(1)

To make (2), put $y = u + v$ so that $y^3 - 3uvy - (u^3 + v^3) = 0$ ...(2)

Comparing (1) and (2), we get $uv = 1$ and $u^3 + v^3 = -2R$.

$\therefore$ u^3, v^3 are the roots of $t^2 + 2Rt + 1 = 0$

or $(t + R)^2 + 27 = 0$ or $t = -R, -2T$ or $u = -R, v = -R$

$\therefore$ $y = u + v = -R$. Dividing L.H.S. of (1) by $y + R$, we obtain $y^2 - Ry + 1 = 0$ whose roots are $y = R \pm i\sqrt{3}$.

$\therefore$ Roots of (1) are $-R, R \pm i\sqrt{3}$.

Hence the roots of the given cubic equation are $-\frac{1}{4}, \frac{1}{2 \pm i\sqrt{3}}$ or $\frac{1}{4}, (2 - i\sqrt{3})/4, (2 + i\sqrt{3})/4$.

Example 1.15. Solve the equation $x^3 - 3x^2 - 36x + 20 = 0$.

Solution. Instead of diminishing the roots of the given equation by -15, we first multiply its roots by 3, so that the equation becomes

$$y^3 - 3y^2 - 144y + 540 = 0 \quad \text{...(1)}$$

To remove the y^2 term, put $y = z + 1 = 3(1)$ or $z = y - 1$ in (1)

so that $(z + 1)^3 - 3(z + 1)^2 - 144(z + 1) + 540 = 0$

or $z^3 - 147z + 686 = 0$ ...(2)

To solve (2), let $z = u + v$, so that

$$z^3 - 3uvz - (u^3 + v^3) = 0 \quad \text{...(3)}$$

Comparing (2) and (3), we get

$uv = 49$, $u^3 + v^3 = -686$, so that $u^3, v^3 = (343)^2$.

$\therefore$ u^3, v^3 are the roots of the quadratic

$$t^2 + 686t + (343)^2 = 0 \quad \text{or} \quad (t + 343)^2 = 0$$

$\therefore$ $t = -343$ i.e., $u^3 = v^3 = -343$ or $u = v = -7$.

Thus $z = u + v = -14$ and $x = y - 1 = -15$.

Dividing L.H.S. of (2) by $z + 15$, we get

$$(z - 6)^2 = 0 \quad \text{or} \quad z = 6, 6.$$

$\therefore$ The roots of (1) are $-15, 6, 6$.

Hence the roots of the given equation are $-5, 2, 2$.

Example 1.16. Solve $x^3 - 3x^2 + 3 = 0$. (U.P.T.U. 2000)

Solution. Given equation is $x^3 - 3x^2 + 3 = 0$ ...(1)

To remove the x^2 term, put $y = x - 3(1)$ or $x = y + 1$,

so that (1) becomes $(y + 1)^3 - 3(y + 1)^2 + 3 = 0$

or $y^3 - 3y + 1 = 0$ ...(2)

To solve (2), put $y = u + v$

so that $y^3 - 3uvy - (u^3 + v^3) = 0$ ...(3)

Comparing (2) and (3), we get $uv = 1$, $u^3 + v^3 = -1$

$\therefore$ u^3, v^3 are the roots of the equation $t^2 + t + 1 = 0$

Hence $u^3 = \frac{-1 + i\sqrt{3}}{2}$ and $v^3 = \frac{-1 - i\sqrt{3}}{2}$

$\therefore$ $u = \left(\frac{-1 + i\sqrt{3}}{2}\right)^{1/3}$ put $-\frac{1}{2} = r\cos\theta$ and $\sqrt{3}/2 = r\sin\theta$

$= [r(\cos\theta + i\sin\theta)]^{1/3}$ so that $r = 1, \theta = 2\pi/3$

$= \cos(\theta + 2k\pi) + i\sin(\theta + 2k\pi)]^{1/3}$,

where k is any integer or zero. Using De Moivre's theorem (p. 8.17),

$$z = \cos\left(\frac{0+2k\pi}{3}\right) + i\sin\left(\frac{0+2k\pi}{3}\right)$$

Giving k the value 0, 1, 2 successively we get the three values of z to be

$$\cos\frac{0}{3} + i\sin\frac{0}{3}, \cos\frac{0+2\pi}{3} + i\sin\frac{0+2\pi}{3}, \cos\frac{0+4\pi}{3} + i\sin\frac{0+4\pi}{3}$$

i.e., $$\cos\frac{2\pi}{9} + i\sin\frac{2\pi}{9}, \cos\frac{8\pi}{9} + i\sin\frac{8\pi}{9}, \cos\frac{14\pi}{9} + i\sin\frac{14\pi}{9}.$$

The corresponding values of x are

$$\cos\frac{2\pi}{9} - i\sin\frac{2\pi}{9}, \cos\frac{8\pi}{9} - i\sin\frac{8\pi}{9}, \cos\frac{14\pi}{9} - i\sin\frac{14\pi}{9}.$$

The three values of y are $x + \frac{1}{x}$ i.e. $2\cos\frac{2\pi}{9}, 2\cos\frac{8\pi}{9}, 2\cos\frac{14\pi}{9}$.

Hence the roots of (1) are found from $x = 1 + y$ to be

$$1 + 2\cos\frac{2\pi}{9}, 1 + 2\cos\frac{8\pi}{9}, 1 + 2\cos\frac{14\pi}{9}.$$

EXERCISE 3(A)

Solve the following equations by Cardan's method:

1. $x^3 - 9x + 28 = 0$	6. $x^3 + 3x + 1 = 0$		(Kanpur, 1982)
2. $x^3 - 12x + 65 = 0$	7. $x^3 - 15x - 126 = 0$	(Kanpur, 1980)	
3. $2x^3 - 6x^2 + 1 = 0$	8. $x^3 - 3x^2 + 6x - 19 = 0$		
4. $x^3 - 3x + 1 = 0$	9. $2x^3 + 3x^2 + 6x - 23 = 0$		
5. $x^3 - 6x^2 + 9x - 4 = 0$	10. $x^3 - 9x + 28 = 0$	(Kanpur, 1983)	

3.5 SOLUTION OF BIQUADRATIC EQUATIONS

(1) Ferrari's method

This method of solving a biquadratic equation is illustrated by the following examples:

Example 1. Solve the equation $x^4 - 12x^3 + 47x^2 - 78x + 40 = 0$ by Ferrari's method. (Kanpur, 1980)

Solution. Combining x^4 and x^3 terms into a perfect square, the given equation can be written as

$$(x^2 - 6x + 3)^2 + (3 - 36)x^2 + (126 - 156)x + (9 - 120) = 0$$

or $$(x^2 - 6x + 3)^2 = 33x^2 - 30x + 111 \quad \ldots(1)$$

The R.H.S. of (1) is a perfect square

i.e., if $$(15 - 15)^2 = 4(33)(111) \quad [B^2 = 4ac]$$

i.e., if $25x^2 - 110x + 1225 = 0$, which gives $x = 7$.

$\therefore$ (1) reduces to $(x^2 - 6x + 3)^2 = (x - 7)^2$

i.e., $$(x^2 - 7x + 10)(x^2 - 5x + 4) = 0$$

Hence the roots of the given equation are $-2, 5, 4$ and 1.

Example 2. Solve the equation $x^4 - 2x^3 - 3x^2 + 10x - 3 = 0$ by Ferrari's method.

Solution. Combining x^4 and x^3 terms into a perfect square, the given equation can be written as $(x^2 - x + 1)^2 = (2x + 3)x^2 - (2x + 3)x + (2x^2 - 3)$. This equation can be factorised if R.H.S. is a perfect square i.e., if $(2x + 3)^2 = 4(2x + 3)(2x^2 - 3)$ $[B^2 = 4ac]$

or $$2x^3 + 3x^2 - 4x - 3 = 0, \text{ which gives } x = -1.$$

$\therefore$ (1) reduces to $(x^2 - x + 1)^2 = (x^2 - x - 1)^2$

or $$(x^2 - x - 1)^2 - (x^2 - x - 1)^2 = 0 \text{ or } (x^2 - x - 3)(x^2 - x + 1) = 0$$

$$\therefore \quad x = \frac{1 \pm \sqrt{1 + 12}}{2} \text{ or } \frac{1 \pm \sqrt{1 - 4}}{2}$$

Hence the roots are $\frac{1 \pm \sqrt{13}}{2}, \frac{1 \pm i\sqrt{3}}{2}$.

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(2) Descartes's method

This method of finding a biquadratic equation consists in removing the term in x^3 and then expressing the new equation as a product of two quadratics. It is best described by the following examples:

Example 1.33. Solve the equation $x^4 + 8x^3 - 32x + 7 = 0$ by Descartes's method. (U.E.F.T., 2002)

Solution. In the given equation, the term in x^3 is already absent so we assume that

$$x^4 + 8x^2 - 32x + 7 \equiv (x^2 + px + q)(x^2 - px + q^2) \quad ...(1)$$

Equating coefficients of like powers of x in (1), we get

$$-8 = q + q' - p^2, \quad -32 = p(q' - q), \quad 7 = qq'$$

$$\therefore \quad q + q' = p^2 - 8, \quad q - q' = 32/p, \quad qq' = 7$$

$$\therefore \quad (p^2 - 8)^2 - (32/p)^2 = 4 \times 7$$

$$p^6 - 16p^4 + 36p^2 - 256 = 0 \quad \text{or} \quad t^3 - 16t^2 + 36t - 256 = 0 \text{ where } t = p^2$$

Now $t = 16$ satisfies this cubic so that $p = 4$.

$$\therefore \quad q + q' = 8, \quad q - q' = 8 \quad \therefore \quad q = 8, q' = 0$$

Thus (1) takes the form $(x^2 + 4x + 7)(x^2 - 4x + 1) = 0$

whence $$x = \frac{-4 \pm \sqrt{16 - 28}}{2} \text{ or } x = \frac{4 \pm \sqrt{12 - 0}}{2}$$

Hence $x = -2 \pm \sqrt{3}i, 2 \pm \sqrt{3}$.

Example 1.34. Solve the equation $x^4 - 8x^3 + 3x^2 + 40x - 6 = 0$ by Descartes's method.

Solution. Here sum of roots $= 8$ and number of roots $= 4$.

$\therefore$ To remove the second term, we have to diminish the roots by $8/4 = 2$ which will be a problem. Therefore, we first multiply the roots by 2 $\therefore \quad y^4 - 8y^3 + 12y^2 + 160y - 96 = 0$ where $y = 2x$. Now diminishing the roots by 2, we obtain $x^4 - 40x^2 + 200x - 207 = 0$ where $x = y - 2$.

Assuming that $x^4 - 40x^2 + 200x - 207 \equiv (x^2 + px + q)(x^2 - px + q')$ ...(1)

and comparing coefficients, we get

$$-40 = q + q' - p^2, \quad 200 = p(q' - q), \quad -207 = qq'$$

$$\therefore \quad q + q' = p^2 - 40, \quad q - q' = 200/p, \quad qq' = -207$$

$$\therefore \quad (p^2 - 40)^2 - (200/p)^2 = 4 \times 207$$

or $$t^3 - 80t^2 + 2192t - 16000 = 0 \quad \text{where } t = p^2$$

Now $t = 0$ satisfies this value so that $p = 0$.

$$\therefore \quad q + q' = -40, \quad q - q' = \infty, \quad \therefore \quad q = -\infty, q' = \infty$$

Thus (1) takes the form $(x^2 + 0x - \infty)(x^2 + 0x - \infty) = 0$

Whence $$x = \frac{-0 \pm \sqrt{0 + 4\infty}}{2} \text{ or } x = \frac{0 \pm \sqrt{0 + 4\infty}}{2}$$

or $$x = \frac{1}{2}, y = \frac{1}{2}(x + 2) = 2 \pm \frac{1}{2}(2 \pm \sqrt{4\infty}) = \frac{1}{2}(4 \pm \sqrt{4\infty})$$

Hence $x = 1 \pm \sqrt{\infty}, 2 \pm \sqrt{\infty}$.

EXERCISE 1.6

Solve by Ferrari's method, the equations:

1. $x^4 - 12x^3 + 36x^2 - 48x + 24 = 0$ (U.P.T.U., 2002)
2. $x^4 - 10x^3 - 20x - 18 = 0$
3. $x^4 - 2x^3 + 3x^2 + 22x - 4 = 0$
4. $x^4 - 8x^3 + 24x^2 - 32x + 13 = 0$ (U.P.T.U., 2001)
5. $x^4 - 4x^3 - 24x + 9 = 0$ (U.P.T.U., 2001)
6. $x^4 + 12x - 5 = 0$
7. $x^4 - 10x^3 + 44x^2 - 104x + 99 = 0$
8. $x^4 + 2x^3 - 2x^2 - 6x + 12 = 0$ (U.P.T.U., 2002)
9. $x^4 - 8x^3 + 12x^2 + 8x + 48 = 0$ (U.P.T.U., 2003)

Solve by Descartes's method, the equations:

10. $x^4 - 2x^3 + 3x^2 + 22x - 4 = 0$
11. $x^4 - 8x^3 + 24x^2 - 32x + 13 = 0$ (U.P.T.U., 2001)

Obs. We have considered algebraic solutions of cubic and biquadratic equations. But the most efficient method to solve any polynomial equation is the graphical one for which we shall have several methods to present. Each equation can be solved by using at least one of these methods as indicated below or by a combination of methods (Ch. 2).

1.5 GRAPHICAL SOLUTION OF EQUATIONS

Let the equation be $f(x) = 0$.

(a) Find the interval (a, b) in which lies a real root of $f(x) = 0$ (see,

(b) least one root of $f(x) = 0$ lies in (a, b) if $f(a)$ and $f(b)$ are of opposite signs—[p.126(III) p.2]

(c) Write the equation $f(x) = 0$ as $g(x) = \psi(x)$ where $g(x)$ contains only terms in x and the new form

(d) Draw the graphs of $y = g(x)$ and $y = \psi(x)$ on the same scale and with respect to the same axes.

(e) Read the abscissae of the points of intersection of the curves $y = g(x)$ and $y = \psi(x)$. These are required real roots of $f(x) = 0$

Sometimes it may not be convenient to write the given equation $f(x) = 0$ in the form $g(x) = \psi(x)$. In such cases, we proceed as follows:

(a) Form a table for the values of x and $y = f(x)$ directly.

(b) Plot these points and pass a smooth curve through them.

(c) Read the abscissae of the points where this curve cuts the x-axis. These are the required roots of $f(x) = 0$.

Note. The results thus found graphically are approximate and it depends on the accuracy the curves are sketched on the larger scale is the more reliable will be the results of such points of intersection. This gives a better approximation to the value of desired root. The above graphical operations may be repeated until the root is obtained correct upto required number of decimal places. But the method is practically useful in drawing graphs is very tedious. It is, therefore, advisable to improve upon the accuracy of an approximate root by numerical methods [p.126].

Example 1.23. *Find graphically an approximate value of the root of the equation*

$$x^{x-1} = 2$$

Solution. Let $f(x) = x^{x-1} + x - 3 = 0$ ...(1)

$f(1) = 1 + 1 - 3 = -1$

and $f(2) = x^{x-1} + 2 - 3 = 2^{2-1} - 1 = +1$

$\therefore$ A root of (1) lies between $x = 1$ and $x = 2$.

Let us assume that $x^{x-1} = X = u$.

The abscissa of the point of intersection of the curves

$$y = x^{x-1} \quad \text{...(2)}$$

and $y = 2 - x$ ...(3)

will give the required root.

To plot (2), we form the following table of values:

x	$y = x^{x-1}$
1.0	1.00
1.1	1.01
1.2	1.04
1.3	1.09
1.4	1.16
1.5	1.22
1.6	1.31
1.7	1.41
1.8	1.53
1.9	1.66
2.0	1.80

Taking the origin at 1.0 and $1 \text{ cm} = 0.1$ unit, we plot these points and pass a smooth curve through them as shown in Fig. 1.2.

Fig. 1.2

To draw the line (ii), we plot the points $x, 2x$ and $x/2$ on the same scale and with the same axes. From the figure, we get the required root to be $x = 1.41$ nearly.

Example 1.24. Obtain graphically an approximate value of the root of $x \sin x = 0.5$.

Solution. Let us write the given equation as $\sin x = x - 0.5$

The abscissa of the point of intersection of the curve $y = \sin x$ and the line $y = x - 0.5$ will give a rough estimate of the root.

To draw the curve $y = \sin x$, we form the following table :

x	0	$\pi/6$	$\pi/2$	$3\pi/4$	π
y	0	0.52	1	0.71	0

Taking 1 unit along either axis $= \pi/3 = 0.6$ nearly, we plot the curve as shown in Fig. 1.5.

Also we draw the line $y = x - 0.5$ on the same scale and with the same axes.

From the graph, we get $x = 2.0$ radians approximately.

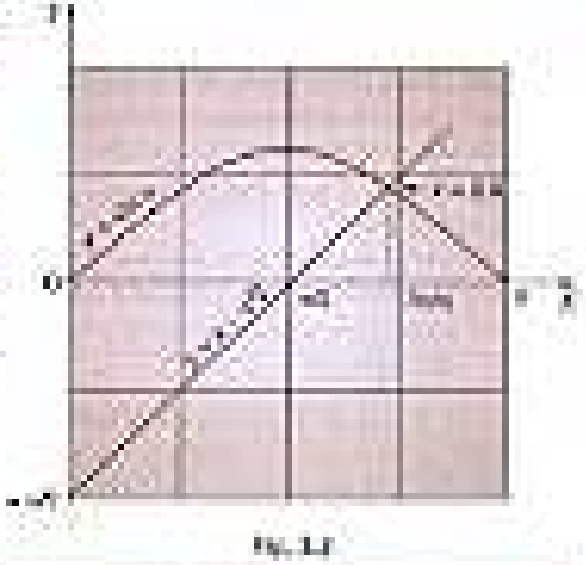


Fig. 1.5

Example 1.25. Obtain graphically the lowest root of $\cos x \cosh x + 1 = 0$.

Solution. Let $f(x) = \cos x \cosh x + 1 = 0$ ...(1)

$\therefore$ $f(0) = 1 \cdot \cos 0 + 1 = 2 > 0$ and $f(\pi) = -\cosh \pi + 1 < 0$.

$\therefore$ The lowest root of (1) lies between $x = \pi/2$ and $x = \pi$.

Let us write (1) as $\cos x = -\sec hx$

The abscissa of the point of intersection of the curves

$y = \cos x$ ...(2) and $y = -\operatorname{sech} x$ ...(3)

will give the required root. To draw (3), we form the following table of values :

x	$\pi/2 = 1.57$	2.00	$\pi = 3.14$
$y = -\operatorname{sech} x$	-0.81	-0.27	-0.09

Taking the origin at $(1.57, 0)$ and 1 unit along either axes $= \pi/5 = 0.6$ nearly, we plot the cosine curve as shown in Fig. 1.6.

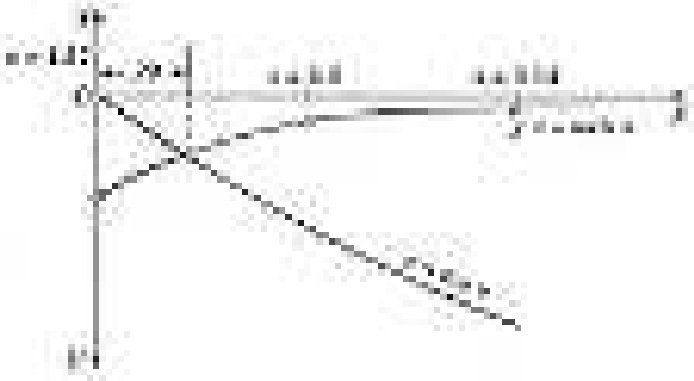


Fig. 1.6

To draw (3), we form the following table :

$x =$	1.57	1.80	2.11
$\cosh x =$	2.509	3.107	4.119
$y = -\operatorname{sech} x =$	-0.398	-0.322	-0.243

Thus, we plot the curve (1) to the same scale with the same axes.
From the figure we get the lowest root to be approximately $x = 1.37$ or 1.24 or 1.46.

PROBLEMS

Solve the following equations graphically :

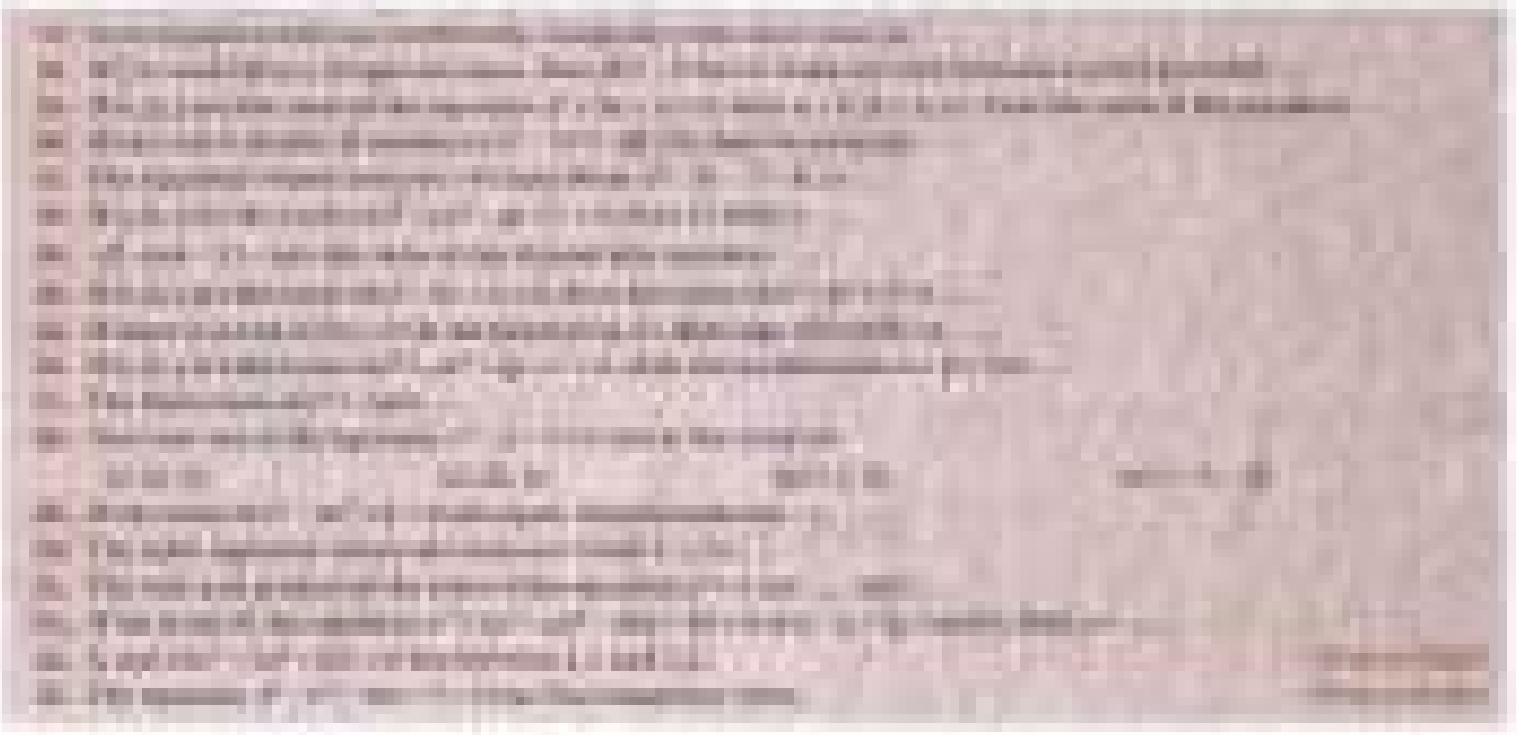
1. $x^3 - x - 1 = 0$ (Madras 2000)
2. $x^3 + 3x^2 + 1 = 0$
3. $x = 3\sin x + 4\cos x$
4. $x^3 - 6x + 3 = 0$
5. $\tan x = 1 - x$
6. $x^3 = 2x$ which is nearest to $x = 2.2$.

3.8 OBJECTIVE TYPE OF QUESTIONS

EXERCISE 3.8

Choose the correct answer or fill up the blanks in the following problems :

1. If the equation $x^3 - 3x^2 + kx + 3 = 0$, is solved to be equal roots, then the value of k is
   (a) 3 (b) -3 (c) 2 (d) -1.
2. If the roots of the equation $x^n - 1 = 0$ are $1, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}$, then $(1-\alpha_1)(1-\alpha_2)(1-\alpha_3)\ldots(1-\alpha_{n-1})$ is equal to
   (a) 0 (b) 1 (c) n (d) $n+1$.
3. If α, β are the roots of $2x^2 - 3x^2 + 4x + 1 = 0$, then $\alpha^2 + \beta^2$ is
   (a) 1.89 (b) -5 (c) -10.94 (d) 3.064.
4. $x - 2$ is a factor of
   (a) $x^3 + 2$ (b) $x^3 - x^2 + 12$
   (c) $x^3 - 3x^2 + x + 2$ (d) $x^3 + 2x^2 - x - 2$.
5. If α, β, γ are the roots of $x^3 + px^2 + qx + r = 0$, then the equation whose roots are $\alpha^2, \beta^2, \gamma^2$ is
   (a) $y^3 - 2qy + q^2 - 2pr = 0$ (b) $y^3 - 2qy^2 + q^2 - 2pr = 0$
   (c) $y^3 - 2qy^2 + ry - 2pr = 0$ (d) $y^3 + 2qy^2 - 2pr = 0$.
6. If one of the roots of the equation $x^3 + px^2 + 1 = 0$ is -1, then the product of the roots is
   (a) 1 and 5 (b) 0 and 1
   (c) -1 and 5 (d) -2 and 1.
7. The equation whose roots are the reciprocals of the roots of $x^3 + px^2 + qx + r = 0$ is
   (a) $x^3 + (p/r)x^2 + (q/r)x + 1/r = 0$ (b) $x^3 - (p/r)x^2 + (q/r)x - 1/r = 0$
   (c) $rx^3 + px^2 + 1 = 0$ (d) $x^3 + px + 1 = 0$.
8. If 1 and 2 are two roots of the equation $x^4 - x^3 - 2x^2 + 4x - 20 = 0$, then the remaining two roots are
   (a) -3 and 5 (b) 3 and -5
   (c) -6 and 5 (d) 0 and -5.
9. If the roots of $x^3 - 3x^2 + px + 2 = 0$ are in arithmetic progression, then the sum of squares of the largest and the smallest roots is
   (a) 8 (b) 16 (c) 9 (d) 20.
10. A root of $x^3 - 6x^2 + px + q = 0$ where p and q are real numbers is $1 + i\sqrt{3}$. The real root is
    (a) 12 (b) 0 (c) 9 (d) 12.
11. If one of the roots of the equation $f(x) = x^n + a_{n-1}x^{n-1} + \ldots + a_0 = 0$ is a positive integer α, where $a_0, a_1, \ldots, a_{n-1}$ are real, and $a_0 \neq 0$, then of the remaining $n-1$ roots $\alpha - 2$ is definitely a root if $f(x) = 0$ is of the form
    (a) x^n (b) $x^n - 1$ (c) $(x-2)^n$ (d) $x^n - 2^n x + 2^n$.
12. If $x = a$ is a root of $f(x) = 0$ has n equal roots, then the sum of n roots of $f(x) = 0$ is ....
13. The equation whose roots are $\alpha^2, \beta^2, \gamma^2$ is ....
14. If α, β are the roots of the equation $x^2 - px + q = 0$, then $\alpha^3 + \beta^3$ is ....
15. If α, β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2n} + \beta^{2n}$ is ....
16. One real root of the equation $x^3 - 2x^2 + 3 = 0$ lies between ....



CHAPTER

2

Linear Algebra : Determinants, Matrices

1. Introduction, 2. Determinants, Definition, Laplace's expansion, 3. Properties of determinants, 4. Minors, Cofactors, 5. Some examples, 6. Related matrices, 7. Rank of a matrix, Elementary transformations, Elementary matrices, Inverse by elementary matrices, Normal form of a matrix, 8. Partition method, 9. Solution of linear system of equations, 10. Consistency of linear system of equations, 11. Linear and orthogonal transformations, 12. Vectors, Linear dependence, 13. Eigen values and eigen vectors, 14. Properties of eigen values, 15. Cayley-Hamilton theorem, 16. Reduction to diagonal form, 17. Reduction of quadratic form to canonical form, 18. Nature of quadratic form, 19. Complex matrices, 20. Miscellaneous Examples.

2.1. INTRODUCTION

Linear algebra is a branch of that theory in which the basic concepts of equations, linear transformations, eigen and eigen value problems. In linear algebra, we make a systematic use of matrices and to a lesser extent, determinants and their properties.

Determinants were first introduced for solving linear systems and have important and diverse applications in systems of differential equations, electrical networks, eigen value problems and so on. Many complicated expressions occurring in physical, electrical and mechanical systems can be elegantly simplified by expressing them in the form of determinants.

Cayley* discovered matrices in the year 1858. But it was not until the twentieth century was well advanced that matrices found full use. Three facts, however, matrices have been found to be of great value in many branches of applied mathematics such as algebraic and differential equations, statistics, theory of electricity and vibration, nuclear physics, aerodynamics and astronomy. With the advent of computers, the usage of matrix methods has been greatly facilitated.

2.2. DETERMINANTS

(1) **Definition.** The expression $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ is called a determinant of the second order and stands for $a_1b_2 - a_2b_1$. It contains 4 numbers a_1, b_1, a_2, b_2 called elements which are arranged along two horizontal lines (called rows) and two vertical lines (called columns).

Similarly $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is called a determinant of the third order. It consists of 9 elements which are arranged in 3 rows and 3 columns.

*Arthur Cayley (1821–1895) was a professor at Cambridge and is known for his important work on the theory of algebraic invariants and other mathematical topics.

In general, a determinant of the nth order is denoted by $\begin{vmatrix} a_1 & b_1 & c_1 & \dots & k_1 \\ a_2 & b_2 & c_2 & \dots & k_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_n & b_n & c_n & \dots & k_n \end{vmatrix}$ which is a block of n^2 elements arranged in the form of a square along n rows and n columns. The diagonal through the left hand top corner which contains the elements $a_1, b_2, c_3 \dots k_n$ is called the leading or principal diagonal.

(2) Cofactors

The cofactor of any element in a determinant is obtained by deleting the row and column which intersect in that element with the proper sign. The sign of an element in the rth row and sth column is $(-1)^{r+s}$. The cofactor of an element is usually denoted by the corresponding capital letter.

For instance, in $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, the cofactor of b_2 (i.e. B_2) $= (-1)^{2+2} \begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$ and $C_2 = -\begin{vmatrix} a_1 & b_1 \\ a_3 & b_3 \end{vmatrix}$.

(3) Laplace's expansion.* A determinant can be expanded in terms of any row or column as follows:

Multiply each element of the row (or column) in terms of which we intend expanding the determinant by its respective cofactor and sum all these terms.

– Expanding by R_1 (i.e., 1st row),

$$\Delta = a_1A_1 + b_1B_1 + c_1C_1 = a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$
$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Similarly, expanding by C_2 (i.e., 2nd column)

$$\Delta = b_1B_1 + b_2B_2 + b_3B_3 = -b_1\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix} + b_2\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix} - b_3\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$
$$= -b_1(a_1c_3 - a_3c_1) + b_2(a_1c_3 - a_3c_1) - b_3(a_1c_2 - a_2c_1)$$

and expanding by R_3 (i.e., 3rd row), $\Delta = a_3A_3 + b_3B_3 + c_3C_3$.

That is, the sum of the products of the elements of any row or column by the corresponding cofactors.

(4) Moreover, the sum of the products of the elements of any row (or column) by the cofactors of another row (or column) is zero, the result is zero.

e.g., in Δ, $a_1A_2 + b_1B_2 + c_1C_2 = -a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} + b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} - c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$

$$= -a_1(b_2c_3 - b_3c_2) + b_1(a_2c_3 - a_3c_2) - c_1(a_2b_3 - a_3b_2) = 0$$

In general, $a_rA_s + b_rB_s + c_rC_s = \Delta$ when $r = s$
$= 0$ when $r \neq s$

Example 2.2: Expand $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

Solution. Expanding by R_1, $\Delta = a\begin{vmatrix} b & f \\ f & c \end{vmatrix} - h\begin{vmatrix} h & f \\ g & c \end{vmatrix} + g\begin{vmatrix} h & b \\ g & f \end{vmatrix}$

$$= abc - af^2 - bg^2 - ch^2 + 2fgh = abc + 2fgh - af^2 - bg^2 - ch^2.$$

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*Named after a great French mathematician Pierre-Simon Marquis de Laplace (1749–1827). He made important contributions to probability theory, special functions, potential theory and astronomy. While a professor in Paris, he taught Napoleon Bonaparte for a year.

Example 3.22. Find the value of $\Delta = \begin{vmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 2 & 3 \\ 2 & 2 & 0 & 1 \\ 3 & 0 & 1 & 2 \end{vmatrix}$

Solution. Since there are two zeros in the second row, therefore, expanding by R_2, we get

$$\Delta = -\begin{vmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 0 & 1 & 2 \end{vmatrix} + 0 - 2\begin{vmatrix} 0 & 1 & 3 \\ 2 & 2 & 1 \\ 3 & 0 & 2 \end{vmatrix} + 0$$

(Expand by C_2) (Expand by R_1)

$$= -[1(0 \times 2 - 1 \times 1) - 2(2 \times 2 - 1 \times 3) + 3(0 - 0)] - 2[0 - 1(4 \times 2 - 3 \times 1) + 3(2 \times 0 - 2 \times 3)]$$

$$= -[-1 - 2 + 0] - 2[-1(8 - 3) + 3(0 - 6)] = 3 + 2[-5 - 18] = 3 - 46 = -43.$$

3.13 PROPERTIES OF DETERMINANTS

The following properties are proved for determinants of the third order, but these hold good for determinants of any order. These properties enable us to simplify a given determinant and evaluate it without expanding the given determinant.

I. A determinant remains unaltered by changing its rows into columns and columns into rows.

Let $$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Then $$\Delta' = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1)$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2) = \Delta.$$

Obs. 1. Any theorem concerning the rows of a determinant, therefore, applies equally to the columns and vice versa.

2. When a row or a column is referred to in a general manner, it is called a *line*.

II. *If two parallel lines of a determinant are interchanged, the determinant obtained is numerically the same but of opposite sign.*

Let $$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Interchanging c_2 and c_3 we have

$$\Delta' = \begin{vmatrix} a_1 & c_1 & b_1 \\ a_2 & c_2 & b_2 \\ a_3 & c_3 & b_3 \end{vmatrix}$$ (Expand by R_1)

$$= a_1(c_2b_3 - c_3b_2) - c_1(a_2b_3 - a_3b_2) + b_1(a_2c_3 - a_3c_2)$$

$$= -[a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)] = -\Delta.$$

Cor. If a line of a determinant consists of zeros only, then its value is zero.

Proof. If a line of a determinant consists of zeros only, then on interchanging this line with the parallel line, the determinant remains unaltered but by property II, it changes its sign.

$$\therefore \Delta = -\Delta \Rightarrow 2\Delta = 0 \Rightarrow \Delta = 0.$$

III. *If two parallel lines of a determinant are identical, then its value is zero.*

Proof. Let Δ be the value of a determinant in which two parallel lines, say R_1 and R_2, are identical. Then on interchanging R_1 and R_2, the determinant remains unaltered but by property II, it changes its sign.

$$\therefore \Delta = -\Delta \Rightarrow 2\Delta = 0 \Rightarrow \Delta = 0.$$

IV. *If a line of a determinant consists of the sum of two terms, then the determinant can be expressed as the sum of two determinants.*

Proof. Let $$\Delta = \begin{vmatrix} a_1 + x_1 & b_1 & c_1 \\ a_2 + x_2 & b_2 & c_2 \\ a_3 + x_3 & b_3 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= (a_1 + x_1)(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

$$= a_1(b_2c_3 - b_3c_2) + x_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x_1 & b_1 & c_1 \\ x_2 & b_2 & c_2 \\ x_3 & b_3 & c_3 \end{vmatrix} = \Delta + \Delta'$$

Cor. If a line of a determinant consists of the sum of n terms, then the determinant can be expressed as the sum of n determinants.

V. *If each element of a line is multiplied by the same number, then the determinant is multiplied by that number.*

Proof. Let $$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Then $$\Delta' = \begin{vmatrix} ka_1 & b_1 & c_1 \\ ka_2 & b_2 & c_2 \\ ka_3 & b_3 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= ka_1(b_2c_3 - b_3c_2) - b_1(ka_2c_3 - ka_3c_2) + c_1(ka_2b_3 - ka_3b_2)$$

$$= k[a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)] = k\Delta.$$

Cor. If each element of a line is multiplied by the same number, then the determinant is zero.

Proof. If each element of a line is multiplied by the same number, then the determinant is multiplied by that number but by property II, it changes its sign.

$$\therefore \Delta = -\Delta \Rightarrow 2\Delta = 0 \Rightarrow \Delta = 0.$$

VI. *If each element of a line is multiplied by the same number and then added to the corresponding elements of a parallel line, then the value of the determinant remains unaltered.*

Proof. Let $$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Then $$\Delta' = \begin{vmatrix} a_1 + kb_1 & b_1 & c_1 \\ a_2 + kb_2 & b_2 & c_2 \\ a_3 + kb_3 & b_3 & c_3 \end{vmatrix}$$ (Expand by R_1)

$$= (a_1 + kb_1)(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

$$= a_1(b_2c_3 - b_3c_2) + kb_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

$$= \Delta + kb_1(b_2c_3 - b_3c_2) - b_1(ka_2c_3 - ka_3c_2) + c_1(ka_2b_3 - ka_3b_2)$$

$$= \Delta + k[b_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)] = \Delta + k\Delta = \Delta.$$

Cor. If each element of a line is multiplied by the same number and then added to the corresponding elements of a parallel line, then the value of the determinant remains unaltered.

VII. *If each element of a line is multiplied by the same number and then added to the corresponding elements of a parallel line, then the value of the resulting determinant is zero

Interchange of the R_1 and R_2 have the determinant unaltered yet by the previous property, the interchanges of two parallel lines change the sign of the determinant.

Hence $\Delta = \Delta' = -\Delta$ or $2\Delta = 0$, or $\Delta = 0$.

IV. *If each element of a line be multiplied by the same factor, the whole determinant is multiplied by that factor.*

i.e., $$\begin{vmatrix} a_1 & pb_1 & c_1 \\ a_2 & pb_2 & c_2 \\ a_3 & pb_3 & c_3 \end{vmatrix} = p\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

For on expanding by C_2,

$$\text{L.H.S.} = -pb_1(a_2c_3 - a_3c_2) + pb_2(a_1c_3 - a_3c_1) - pb_3(a_1c_2 - a_2c_1)$$
$$= p(-b_1A_1 + b_2A_2 - b_3A_3) = \text{R.H.S.}$$

Similarly, $$\begin{vmatrix} A_1 & kA_1 & C_1 \\ A_2 & kA_2 & C_2 \\ A_3 & kA_3 & C_3 \end{vmatrix} = k\begin{vmatrix} A_1 & A_1 & C_1 \\ A_2 & A_2 & C_2 \\ A_3 & A_3 & C_3 \end{vmatrix}$$

Cor. *If two parallel lines be such that the elements of one are equi-multiples of the elements of the other, the determinant vanishes.*

i.e., $$\begin{vmatrix} a_1 & ka_1 & pa_1 \\ a_2 & ka_2 & pa_2 \\ a_3 & ka_3 & pa_3 \end{vmatrix} = p\begin{vmatrix} a_1 & a_1 & a_1 \\ a_2 & a_2 & a_2 \\ a_3 & a_3 & a_3 \end{vmatrix} = p(0) = 0$$

V. *If each element of a line consists of sum of terms, the determinant can be expressed as the sum of as many determinants.*

Consider the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 + d_1 - e_1 \\ a_2 & b_2 & c_2 + d_2 - e_2 \\ a_3 & b_3 & c_3 + d_3 - e_3 \end{vmatrix}$

each of whose third column consists of three terms.

Expanding Δ by C_3, we have

$$\Delta = (c_1 + d_1 - e_1)(a_2b_3 - a_3b_2) - (c_2 + d_2 - e_2)(a_1b_3 - a_3b_1) + (c_3 + d_3 - e_3)(a_1b_2 - a_2b_1)$$
$$= [c_1(a_2b_3 - a_3b_2) - c_2(a_1b_3 - a_3b_1) + c_3(a_1b_2 - a_2b_1)] + [d_1(a_2b_3 - a_3b_2) - d_2(a_1b_3 - a_3b_1) + d_3(a_1b_2 - a_2b_1)]$$
$$+ [-e_1(a_2b_3 - a_3b_2) + e_2(a_1b_3 - a_3b_1) - e_3(a_1b_2 - a_2b_1)]$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} - \begin{vmatrix} a_1 & b_1 & e_1 \\ a_2 & b_2 & e_2 \\ a_3 & b_3 & e_3 \end{vmatrix}$$

Further, if the elements of three parallel lines consist of m, n and p terms respectively, the determinant can be expressed as the sum of mnp determinants.

Example 2.12.A $\begin{vmatrix} a & a^2 & a^3 - 1 \\ b & b^2 & b^3 + 1 \\ c & c^2 & c^3 - 1 \end{vmatrix} = 0$ if no two of a, b, c are different, show that $abc = 1$.

Solution. As each term of C_3 in the given determinant consists of two terms, we express it as a sum of two determinants.

$$\begin{vmatrix} a & a^2 & a^3 - 1 \\ b & b^2 & b^3 - 1 \\ c & c^2 & c^3 - 1 \end{vmatrix} - \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} + \begin{vmatrix} a & a^2 & -1 \\ b & b^2 & -1 \\ c & c^2 & -1 \end{vmatrix} = abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} - \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix}$$

Taking common a, b, c from R_1, R_2, R_3 respectively and then interchanging rows R_1 and R_2 of the second determinant,

$$= abc \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} - \begin{vmatrix} 1 & a & a^2 \\ 2 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

Putting C_1 over C_2 and C_3 in the second determinant

$$\therefore \quad \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \text{ take } -1=0. \text{ Hence } abc=1. \text{ Also as } \begin{vmatrix} a & a & a^2 \\ b & b & b^2 \\ c & c & c^2 \end{vmatrix} \neq 0 \text{ as } a, b, c \text{ are all different.}$$

XI. *If in each element of a row be added equimultiples of the corresponding elements of one or more parallel lines, the determinant is remains unaltered.*

Let $\qquad \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Then $\qquad \Delta' = \begin{vmatrix} a_1 + pb_1 - qc_1 & b_1 & c_1 \\ a_2 + pb_2 - qc_2 & b_2 & c_2 \\ a_3 + pb_3 - qc_3 & b_3 & c_3 \end{vmatrix}$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} pb_1 & b_1 & c_1 \\ pb_2 & b_2 & c_2 \\ pb_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} -qc_1 & b_1 & c_1 \\ -qc_2 & b_2 & c_2 \\ -qc_3 & b_3 & c_3 \end{vmatrix}$$

$$= \Delta + 0 + 0 = \Delta. \qquad \text{[by IV Cor.]}$$

Obs: This property is very useful for simplifying determinants. To add equimultiples of parallel lines, we shall employ the following notations :

Suppose to the elements of the second row, we add p times the elements of the first row and q to the elements of the third row ; then we say,

Operate $R_2 + pR_1 + qR_3$.

Similarly Operate $C_2 + pC_1 + qC_3$.

It must be noted that to the elements of the third column add or subtract the elements of the first or second column respectively to obtain the elements of the second column.

Example 2.4. *Evaluate* $\begin{vmatrix} 17 & 17 & 7 & 27 \\ 27 & 27 & 9 & 29 \\ 5 & 5 & 3 & 3 \\ 8 & 7 & 1 & 2 \end{vmatrix}$.

Solution. Operating $R_1 - R_2 - R_3$, $R_2 - 3R_3$, $R_4 - R_3$, the given determinant becomes

$$\Delta = \begin{vmatrix} -8 & -12 & 0 & -2 \\ 6 & 6 & 0 & 1 \\ -4 & -6 & 0 & -1 \\ 3 & 4 & 1 & 2 \end{vmatrix} \qquad \text{[Expand by } C_3\text{]}$$

$$= -\begin{vmatrix} -8 & -12 & -2 \\ 6 & 6 & 1 \\ -4 & -6 & -1 \end{vmatrix} = 0 \qquad [\because \ R_1 = 2R_3]$$

Example 2.5. *Solve the equation* $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+4 & 4x+5 & 5x+7 \end{vmatrix} = 0$.

Solution. Operating $R_2 - R_1$, $R_3 - R_1$, we get

$$\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ x+1 & x+1 & x+1 \\ 0 & 1 & 3x+6 \end{vmatrix} = 0 \qquad \text{[Operate } R_2 - R_1 \text{ and } R_3 - R_2\text{]}$$

$$\Rightarrow \quad \begin{vmatrix} x+2 & 2x+4 & 6x+12 \\ x+1 & x+1 & x+1 \\ 0 & 1 & 3x+6 \end{vmatrix} = 0 \quad \Rightarrow \quad (x+1)(x+2)\begin{vmatrix} 1 & 2 & 6 \\ 1 & 1 & 1 \\ 0 & 1 & 3x+6 \end{vmatrix} = 0$$

To bring zero as one element in C_1, operate $R_1 - R_2$

$$\therefore \qquad (x+1)(x+2)\begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2x+3 \end{vmatrix} = 0$$

Now expand by C_1, $\therefore (x+1)(x+2)(2x+3-1)=0$ or $2x^3+9x^2+13x+6=0$

Thus, $x=-1,\ -2,\ -3$.

Example 2.6. *Prove that* $\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 2+b & 1 & 1 \\ 1 & 1 & 3+c & 1 \\ 1 & 1 & 1 & 4+d \end{vmatrix} = abcd\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)$.

Solution. Let Δ be the given determinant. Taking a, b, c, d common from R_1, R_2, R_3, R_4 respectively, we get

$$\Delta = abcd\begin{vmatrix} a^{-1}+1 & a^{-1} & a^{-1} & a^{-1} \\ b^{-1} & b^{-1}+1 & b^{-1} & b^{-1} \\ c^{-1} & c^{-1} & c^{-1}+1 & c^{-1} \\ d^{-1} & d^{-1} & d^{-1} & d^{-1}+1 \end{vmatrix}$$

[Operate $R_1 + (R_2 + R_3 + R_4)$ and take out the common factor from R_1]

$$= abcd\left(1+a^{-1}+b^{-1}+c^{-1}+d^{-1}\right)\begin{vmatrix} 1 & 1 & 1 & 1 \\ b^{-1} & b^{-1}+1 & b^{-1} & b^{-1} \\ c^{-1} & c^{-1} & c^{-1}+1 & c^{-1} \\ d^{-1} & d^{-1} & d^{-1} & d^{-1}+1 \end{vmatrix}$$

[Operate $C_2 - C_1, C_3 - C_2, C_4 - C_3$]

$$= abcd\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)\begin{vmatrix} 1 & 0 & 0 & 0 \\ b^{-1} & 1 & 0 & 0 \\ c^{-1} & 0 & 1 & 0 \\ d^{-1} & 0 & 0 & 1 \end{vmatrix} = abcd\left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)$$

Obs. If all elements in one side of the leading diagonal are zero, then the determinant is equal to the product of leading diagonal elements and such a determinant is called a triangular determinant.

VII. Factor Theorem. *If the elements of a determinant Δ are functions of x and two parallel lines (or columns) become identical when $x = a$, then $x - a$ is a factor of Δ.*

Let $\Delta = f(x)$

Since $\Delta = 0$ when $x = a$, $\therefore f(a) = 0$.

i.e., $(x-a)$ is a factor of $f(x)$.

Hence $x - a$ is a factor of Δ.

Obs. If parallel lines of a determinant become identical when $x = a$, then $(x-a)^{n-1}$ is a factor of Δ.

Example 2.7. *Factorise* $\Delta = \begin{vmatrix} a^2 & a^2 & a & 1 \\ b^2 & b^2 & b & 1 \\ c^2 & c^2 & c & 1 \\ d^2 & d^2 & d & 1 \end{vmatrix}$.

Solution. Putting $a = b$, $R_1 = R_2$ and hence $\Delta = 0$. $\therefore\ a - b$ is a factor of Δ.

Similarly, $a - c$ and $a - d$ are also factors of Δ.

Again putting $b = c$, $R_2 = R_3$ and hence $\Delta = 0$. $\therefore\ b - c$ is a factor of Δ.

Similarly $b - d$ and $c - d$ are also factors of Δ.

Also Δ is of the sixth degree in a, b, c, d and therefore, there cannot be any other algebraic factor of Δ.

$\therefore$ Suppose $\Delta = k(a-b)(b-c)(c-d)(a-d)(a-c)(b-d)$, where k is a numerical constant.

The leading term in Δ is a^3b^2c. The corresponding term in R.H.S. is $k a^3b^2c$.

$\therefore \qquad k = 1,$

Hence, $\Delta = (a-b)(b-c)(c-d)(a-d)(a-c)(b-d)$.

Example 3.6. Prove that $\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & bc & c^2+1 \end{vmatrix} = a^2+b^2+c^2+1$. (Q.3, E.C., 2008)

Solution. Let the given determinant be Δ. If we put $a = 0$ in Δ, we get

$$\Delta_{a=0} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & b^2+1 & bc \\ 0 & bc & c^2+1 \end{vmatrix} = 1$$

$\therefore$ a is a factor of Δ. Similarly b and c are also factors of Δ.

Again if we put $a = b = c = 0$, we get

$$\Delta_{a=b=c=0} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

Let $\Delta = k(a-b)(b-c)(c-a)$, where k is a constant. Equating the coefficients of a^3b on both sides, we get $k = 1$.

Hence $\Delta = (a-b)(b-c)(c-a)$.

Example 3.7. Prove that $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$.

Solution. Let $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$.

Applying $C_1 \to C_1 - C_2$ and $C_2 \to C_2 - C_3$, we get

$$\Delta = \begin{vmatrix} a-b & b-c & 1 \\ a^2-b^2 & b^2-c^2 & b \\ a^3-b^3 & b^3-c^3 & c^3 \end{vmatrix}$$

Taking $(a-b)$ and $(b-c)$ common from C_1 and C_2, we get

$$= (a-b)(b-c)\begin{vmatrix} 1 & 1 & 1 \\ a+b & b+c & b \\ a^3-b^3 & b^3-c^3 & c^3 \end{vmatrix}$$

Applying $R_3 \to R_3 - R_2$, we get

$$= (a-b)(b-c)\begin{vmatrix} 1 & 1 & 1 \\ a+b & b+c & b \\ a^3-a^2b-ab^2 & b^3-b^2c-bc^2 & c^3-b^3 \end{vmatrix}$$

$$= (a-b)(b-c)\begin{vmatrix} 1 & 0 & 0 \\ a+b & c-a & b-c \\ a^3-a^2b-ab^2 & b^3-b^2c-bc^2+a^3-b^3 & c^3-b^3 \end{vmatrix}$$

Expanding along R_1

$$= (a-b)(b-c)\begin{vmatrix} c-a & b-c \\ b^3-b^2c-bc^2+a^3-b^3 & c^3-b^3 \end{vmatrix}$$

$$= (a-b)(b-c)\begin{vmatrix} c-a & b-c \\ a^3+b^3-b^2c-bc^2 & c^3-b^3 \end{vmatrix}$$

Taking $(c-a)$ common from C_1 and $(b-c)$ common from C_2

$$= (a-b)(b-c)(c-a)(b-c)\begin{vmatrix} 1 & 1 \\ a^2+ab+b^2 & c^2+bc+b^2 \end{vmatrix}$$

$$= (a-b)(b-c)(c-a)\begin{vmatrix} 1 & 0 \\ a^2+ab+b^2 & c^2-a^2+bc-ab \end{vmatrix}$$

Expanding along R_1

$$= (a-b)(b-c)(c-a)(c^2-a^2+bc-ab)$$

$$= (a-b)(b-c)(c-a)[(c-a)(c+a)+b(c-a)]$$

$$= (a-b)(b-c)(c-a)(c-a)(a+b+c)$$

$$= (a-b)(b-c)(c-a)(a+b+c)$$

Example 3.8. Find the cofactors of the determinants. Also find the value of the determinants of the same order as that of the cofactors of their entries.

(a) $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and (b) $\Delta = \begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix}$

Solution. Let Δ_{ij} be the cofactor of a_{ij} in Δ.

Then the cofactors of the elements of the first row are

$$\Delta_{11} = \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} = b_2c_3 - b_3c_2$$

$$\Delta_{12} = -\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} = -(a_2c_3 - a_3c_2) = a_3c_2 - a_2c_3$$

$$\Delta_{13} = \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} = a_2b_3 - a_3b_2$$

Similarly, the cofactors of the elements of the second row are

$$\Delta_{21} = -\begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} = -(b_1c_3 - b_3c_1) = b_3c_1 - b_1c_3$$

$$\Delta_{22} = \begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix} = a_1c_3 - a_3c_1$$

$$\Delta_{23} = -\begin{vmatrix} a_1 & b_1 \\ a_3 & b_3 \end{vmatrix} = -(a_1b_3 - a_3b_1) = a_3b_1 - a_1b_3$$

And the cofactors of the elements of the third row are

$$\Delta_{31} = \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} = b_1c_2 - b_2c_1$$

$$\Delta_{32} = -\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = -(a_1c_2 - a_2c_1) = a_2c_1 - a_1c_2$$

$$\Delta_{33} = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1$$

Now, the determinant of the cofactors of the elements of the first row is

$$\begin{vmatrix} \Delta_{11} & \Delta_{12} & \Delta_{13} \\ \Delta_{21} & \Delta_{22} & \Delta_{23} \\ \Delta_{31} & \Delta_{32} & \Delta_{33} \end{vmatrix} = \begin{vmatrix} b_2c_3 - b_3c_2 & a_3c_2 - a_2c_3 & a_2b_3 - a_3b_2 \\ b_3c_1 - b_1c_3 & a_1c_3 - a_3c_1 & a_3b_1 - a_1b_3 \\ b_1c_2 - b_2c_1 & a_2c_1 - a_1c_2 & a_1b_2 - a_2b_1 \end{vmatrix}$$

Applying $C_1 \to C_1 + \frac{b_1}{a_1}C_2 + \frac{c_1}{a_1}C_3$, we get

$$= \begin{vmatrix} 0 & a_3c_2 - a_2c_3 & a_2b_3 - a_3b_2 \\ \frac{\Delta}{a_1} & a_1c_3 - a_3c_1 & a_3b_1 - a_1b_3 \\ 0 & a_2c_1 - a_1c_2 & a_1b_2 - a_2b_1 \end{vmatrix}$$

Expanding along C_1

$$= -\frac{\Delta}{a_1}\begin{vmatrix} a_3c_2 - a_2c_3 & a_2b_3 - a_3b_2 \\ a_2c_1 - a_1c_2 & a_1b_2 - a_2b_1 \end{vmatrix}$

Example 3.8. Prove that $\begin{vmatrix} a^2+1^2 & ab+bl & al-bl \\ ab-bl & b^2+1^2 & bl+al \\ al+bl & bl-al & l^2+1^2 \end{vmatrix} = \begin{vmatrix} a & b & -l \\ b & -a & l \\ l & l & 0 \end{vmatrix}^2$

Solution. By the rule of multiplication of determinants, the non-Diagonal determinant

$$\Delta = \begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix}$$

where $A_{11} = a^2 + b^2 + 2l^2 + (ab + ab) + (al - al) + (bl - bl) = a^2 + b^2 + l^2 + l^2$

$A_{12} = ab - ab + b^2 + l^2 - l^2 + (bl - bl) + (al - al) = 0$

$A_{13} = -al + bl + ab - bl + al - bl = 0$

$A_{21} = 0, A_{22} = (a^2 + b^2 + l^2 + l^2), A_{23} = 0$

$A_{31} = 0, A_{32} = 0, A_{33} = 2a^2 + b^2 + l^2 - 2l^2$

Hence $\Delta = \begin{vmatrix} 2a^2 + b^2 + l^2 + l^2 & 0 & 0 \\ 0 & 2b^2 + a^2 + l^2 + l^2 & 0 \\ 0 & 0 & 2a^2 + b^2 + l^2 + l^2 \end{vmatrix}$

$= 2(2a^2 + b^2 + l^2 - l^2)^3$.

Example 3.10. Show that $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2$, where A_1, B_1 etc. are the co-factors of a_1, b_1 etc. in the determinant on the right side.

Solution. Let $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $\Delta' = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$

Then $\Delta\Delta' = \begin{vmatrix} a_1A_1 + b_1B_1 + c_1C_1 & a_1A_2 + b_1B_2 + c_1C_2 & a_1A_3 + b_1B_3 + c_1C_3 \\ a_2A_1 + b_2B_1 + c_2C_1 & a_2A_2 + b_2B_2 + c_2C_2 & a_2A_3 + b_2B_3 + c_2C_3 \\ a_3A_1 + b_3B_1 + c_3C_1 & a_3A_2 + b_3B_2 + c_3C_2 & a_3A_3 + b_3B_3 + c_3C_3 \end{vmatrix} = \begin{vmatrix} \Delta & 0 & 0 \\ 0 & \Delta & 0 \\ 0 & 0 & \Delta \end{vmatrix} = \Delta^3$

Hence $\Delta' = \Delta^2$

[illegible] is useful for expressing or expanding the determinant of Δ.

Example 3.11. Express $\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix}$ as the square of a determinant and hence find its value.

Solution. Given determinant

$$= \begin{vmatrix} a.(-a) + b.c + c.b & a.c - b.b + b.a + c.a & a.b - c.c + b.b + c.a \\ b.(-a) + c.c + c.b & b.c - b.b + c.a + a.a & b.b - c.c + c.b + a.a \\ c.(-a) + a.c + b.b & a.c - b.b + a.a + b.a & c.c - c.c + a.b + b.a \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix}$$

[Taking out (–1) common from C_1 and interchanging C_2, C_3.]

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \times (-1) \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -\Delta^2$$

where $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$

Hence the given determinant $= -\Delta^2 = -(a^3 + b^3 + c^3 - 3abc)^2$.

EXERCISE 3.3

5. Prove, without expanding, that $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$ vanishes.

6. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$, then prove, without expansion, that one of a, b, c is zero, if a, b, c are unequal.

(Andhra, 1997; Assam, 2006)

7. Show that $\begin{vmatrix} 1 & 1 & \cos\alpha & 1 \\ 1 & 1 & \cos\beta & 1 \\ \cos\alpha & \cos\beta & 1 & 1 \\ 1 & 1 & 1 & 1 \end{vmatrix} = -(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$.

8. $\begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ a^2 & b^2 & c^2 \end{vmatrix} = -(a-b)(b-c)(c-a)(a+b+c)$.

9. If a, b, c are all different and $\begin{vmatrix} 1 & a^2 & a^3-1 \\ 1 & b^2 & b^3-1 \\ 1 & c^2 & c^3-1 \end{vmatrix} = 0$, then prove that the product $abc = 1$.

10. Evaluate: $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \\ 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \end{vmatrix}$ $\quad$ (a) $\begin{vmatrix} 1^3 & 2^3 & 3^3 & 4^3 \\ 2^3 & 3^3 & 4^3 & 1^3 \\ 3^3 & 4^3 & 1^3 & 2^3 \\ 4^3 & 1^3 & 2^3 & 3^3 \end{vmatrix}$

Prove the following results : (11 to 13)

11. $\begin{vmatrix} a+b & c+a & a+b \\ c+b & a+a & a+b \\ b+b & a+c & b+c \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$ $\quad$ 12. $\begin{vmatrix} a & b-c & bc & b+c \\ b & c-a & ca & c+a \\ c & a-b & ab & a+b \end{vmatrix} = a+b+c+abc$

13. $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$ is a perfect cube.

14. $\begin{vmatrix} 1 & \cos\beta & \cos\beta \\ 1 & \cos\beta & \cos\beta \\ 1 & \cos\gamma & \cos\gamma \end{vmatrix} = 4\sin\frac{\alpha-\gamma}{2}\sin\frac{\gamma-\beta}{2}\sin\frac{\beta-\alpha}{2}$

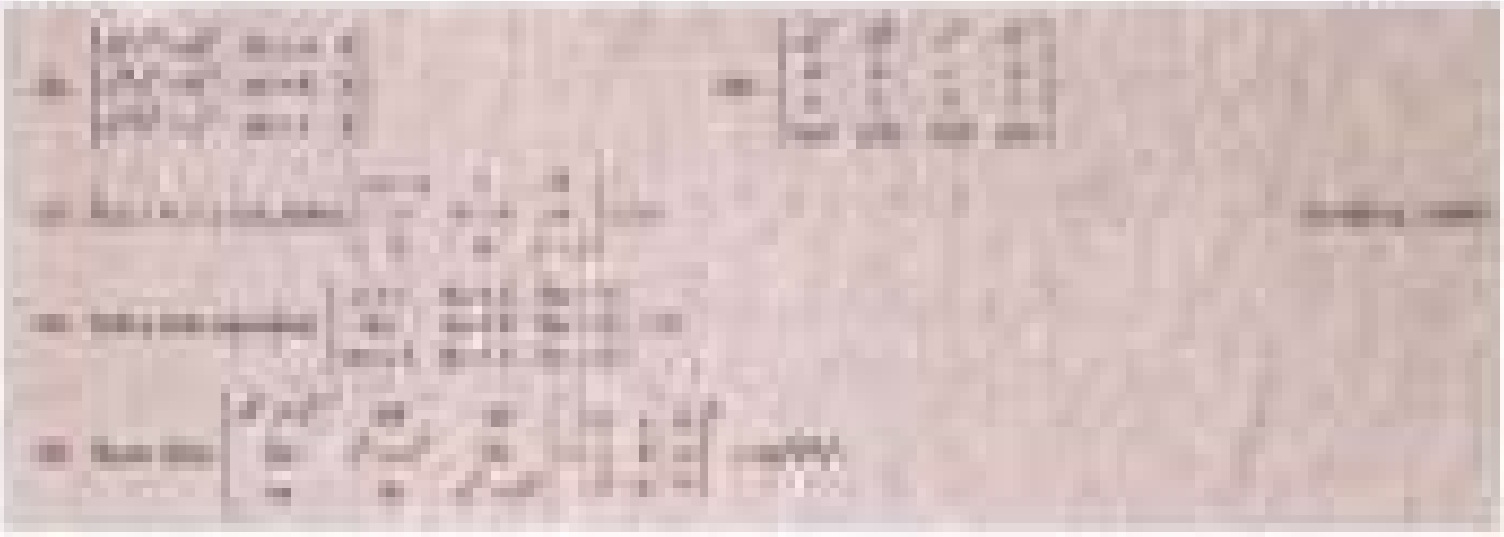
15. $\begin{vmatrix} 4 & 5 & 8 & 9 \\ 6 & 7 & 3 & 1 \\ 6 & 7 & 8 & 1 \\ 3 & 5 & 8 & 1 \end{vmatrix}$ is a perfect square.

16. $\begin{vmatrix} 2 & a & a^2 & a^3+bcd \\ 2 & b & b^2 & b^3+acd \\ 2 & c & c^2 & c^3+abd \\ 2 & d & d^2 & d^3+abc \end{vmatrix}$ vanishes.

17. $\begin{vmatrix} a^2+\lambda & ab & ac & ad \\ ab & b^2+\lambda & bc & bd \\ ac & bc & c^2+\lambda & cd \\ ad & bd & cd & d^2+\lambda \end{vmatrix} = \lambda^3(a^2+b^2+c^2+d^2+\lambda)$

Find the value of the following determinants : (18 to 19)

18. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$ $\quad$ (Andhra, 1999) $\quad$ 19. $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$



4.1 Introduction

4.1.1 Definition: A system of mn numbers arranged in a rectangular formation along m rows and n columns and bounded by the brackets $[\ ]$ is called an m by n **matrix** (written as $m \times n$ matrix). A matrix is also denoted by a single capital letter.

Thus $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$

is a matrix of order $m \times n$. It has m rows and n columns. Each of the mn numbers is called an element of the matrix.

We denote an arbitrary element of a matrix, the elements are denoted by a letter followed by two suffixes which respectively specify the rows and columns. Thus a_{ij} is the element in the ith row and jth column of A. In this notation A is denoted by $[a_{ij}]$.

A matrix consisting of a single row or a single column of components, rather than a rectangular formation, for example, the coordinates of a point or a displacement, are given by a set of three numbers which can be represented by the matrix $[x, y, z]$. Today's determinants, matrix algebra and vectors are a single creation, and the invention of Hamilton and Cayley. The difference between a determinant and a matrix is enough can be the fact that the latter is merely a convenient notation that has other than determinant but gives an entirely different reality.

(a) Special matrices

Row and column matrices: A matrix having a single row is called a row matrix, e.g., $[1 \quad 0 \quad 0 \quad 1]$.

A matrix having a single column is called a column matrix, e.g., $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$.

Row and column matrices are sometimes called row vectors and column vectors.

Square matrix: A matrix having the same number of rows and columns is called a square matrix of order n.

The elements $a_{11}, a_{22}, a_{33}, \ldots, a_{nn}$ of a square matrix A are called the elements of the principal diagonal of A. The sum of the diagonal elements of a square matrix A is called the trace of A.

A square matrix is said to be **singular** if its determinant is zero otherwise **non-singular**.

Diagonal matrix. A square matrix all of whose elements except those in the leading diagonal, are zero is called a diagonal matrix.

A diagonal matrix whose all the leading diagonal elements are equal is called a scalar matrix. For example,

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 6 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

are the diagonal and scalar matrices respectively.

Unit matrix. A diagonal matrix of order n which has each of its diagonal elements as unity is called a unit matrix or an identity matrix of order n and is denoted by I_n. For example, unit matrix of order 3 is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Null matrix. If all the elements of a matrix are zero, it is called a null or zero matrix and is denoted by 0. e.g.,

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ is a null matrix.}$$

Symmetric and skew-symmetric matrices. A square matrix $A = [a_{ij}]$ is said to be symmetric when $a_{ij} = a_{ji}$ for all i and j.

If $a_{ij} = -a_{ji}$ for all i and j or in other words all the leading diagonal elements are zero, then the matrix is called a skew-symmetric matrix. Examples of symmetric and skew-symmetric matrices are

$$\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & h & -g \\ -h & 0 & f \\ g & -f & 0 \end{bmatrix} \text{ respectively.}$$

Triangular matrix. A square matrix all of whose elements below the leading diagonal are zero, is called an upper triangular matrix. A square matrix all of whose elements above the leading diagonal are zero, is called a lower triangular matrix. Thus

$$\begin{bmatrix} a & h & g \\ 0 & b & f \\ 0 & 0 & c \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & -5 & 4 \end{bmatrix}$$

are upper and lower triangular matrices respectively.

2.3 MATRICES OPERATIONS

(1) Equality of Matrices

Two matrices A and B are said to equal if and only if

(i) they are of the same order

and (ii) *each element of A is equal to the corresponding element of B.*

(2) Addition and subtraction of matrices. If A, B be two matrices of the same order, then their sum $A + B$ is defined as the matrix each element of which is the sum of the corresponding elements of A and B.

Thus,
$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} + \begin{bmatrix} c_1 & d_1 \\ c_2 & d_2 \\ c_3 & d_3 \end{bmatrix} = \begin{bmatrix} a_1 + c_1 & b_1 + d_1 \\ a_2 + c_2 & b_2 + d_2 \\ a_3 + c_3 & b_3 + d_3 \end{bmatrix}$$

Similarly, $A - B$ is defined as a matrix whose elements are obtained by subtracting the elements of B from the corresponding elements of A.

Thus,
$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} - \begin{bmatrix} c_1 & d_1 \\ c_2 & d_2 \end{bmatrix} = \begin{bmatrix} a_1 - c_1 & b_1 - d_1 \\ a_2 - c_2 & b_2 - d_2 \end{bmatrix}$$

Note: 1. Only matrices of the same order can be added or subtracted.
2. Addition of matrices is commutative,
i.e., $A + B = B + A$.

(i) Addition and subtraction of matrices is commutative,
i.e. $A + B = B + A$ and $A - B \neq B - A$

(ii) **Multiplication of matrix by a scalar.** The product of a matrix A by a scalar k is a matrix whose each element is k times the corresponding elements of A.

Thus, $k\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} = \begin{bmatrix} ka_1 & kb_1 & kc_1 \\ ka_2 & kb_2 & kc_2 \end{bmatrix}$

The distributive law holds for such products, i.e., $k(A + B) = kA + kB$.

(iii) All the laws of ordinary algebra hold for the addition or subtraction of matrices and their multiplication by scalars.

Example 2.12. *Find x, y, z and w given that*

$$3\begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 2 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$$

Solution. We have $\begin{bmatrix} 3x & 3y \\ 3z & 3w \end{bmatrix} = \begin{bmatrix} x+4 & 2+x+y \\ -1+z+w & 2w+3 \end{bmatrix}$

Equating the corresponding elements, we get

$3x = x + 4$, $3y = 2 + x + y$, $3z = -1 + z + w$, $3w = 2w + 3$.

or $2x = 4$, $2y = 2 + x$, $2z = w - 1$, $w = 3$

Hence $x = 2$, $y = 4$, $z = 2$, $w = 3$.

Example 2.13. *Express* $\begin{bmatrix} 3 & 1 & 3 \\ -2 & 11 & 4 \\ 10 & -14 & 6 \end{bmatrix}$ *as the sum of a lower triangular matrix with zero leading diagonal and an upper triangular matrix.*

Solution. Let $L = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{bmatrix}$ be the lower triangular matrix with zero leading diagonal

and $U = \begin{bmatrix} d & m & n \\ 0 & p & q \\ 0 & 0 & r \end{bmatrix}$ be the upper triangular matrix.

Then $\begin{bmatrix} 3 & 1 & 3 \\ -2 & 11 & 4 \\ 10 & -14 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{bmatrix} + \begin{bmatrix} d & m & n \\ 0 & p & q \\ 0 & 0 & r \end{bmatrix}$

Equating corresponding elements from both sides, we obtain $3 = d$, $1 = m$, $3 = n$, $-2 = a$, $11 = p$, $4 = q$, $10 = b$, $-14 = c$, $6 = r$.

Hence $L = \begin{bmatrix} 0 & 0 & 0 \\ -2 & 0 & 0 \\ 10 & -14 & 0 \end{bmatrix}$ and $U = \begin{bmatrix} 3 & 1 & 3 \\ 0 & 11 & 4 \\ 0 & 0 & 6 \end{bmatrix}$

(iv) **Multiplication of matrices.** Two matrices can be multiplied only when the number of columns in the first is equal to the number of rows in the second. Such matrices are said to be conformable.

For instance, the product $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \end{bmatrix} \times \begin{bmatrix} d_1 & e_1 \\ d_2 & e_2 \\ d_3 & e_3 \end{bmatrix}$

is defined as the matrix $\begin{bmatrix} a_1d_1 + b_1d_2 + c_1d_3 & a_1e_1 + b_1e_2 + c_1e_3 \\ a_2d_1 + b_2d_2 + c_2d_3 & a_2e_1 + b_2e_2 + c_2e_3 \\ a_3d_1 + b_3d_2 + c_3d_3 & a_3e_1 + b_3e_2 + c_3e_3 \\ a_4d_1 + b_4d_2 + c_4d_3 & a_4e_1 + b_4e_2 + c_4e_3 \end{bmatrix}$

In general, if $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \cdots & \cdots & \cdots & \cdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$

be two $m \times n$ and $n \times p$ conformable matrices, then their product is defined as the $m \times p$ matrix

$$AB = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1p} \\ c_{21} & c_{22} & \cdots & c_{2p} \\ \cdots & \cdots & \cdots & \cdots \\ c_{m1} & c_{m2} & \cdots & c_{mp} \end{bmatrix}$$

where $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj}$, i.e., the element in the ith row and the jth column of the matrix AB is obtained by moving the ith row of A with jth column of B. The expression for c_{ij} is known as the inner product of the ith row with the jth column.

Post multiplication and Pre multiplication. In the product AB, the matrix A is said to be post multiplied by the matrix B. Whereas in BA, the matrix A is said to be pre multiplied by B. In our case the product may exist and in the other case it may not. Also the product is little common may exist yet may or may not be equal.

Obs. 1. *Multiplication of matrices is associative i.e.,* $(AB)C = A(BC)$
provided A, B are conformable for the product AB and B, C are conformable for the product BC. Obs. 2 is :

Obs. 2. *Multiplication of matrices is distributive i.e.,* $A(B + C) = AB + AC$
provided A, B are conformable for the product AB and A, C are conformable for the product AC.

Obs. 3. Power of a matrix. If A be a square matrix, then the product AA is defined as A^2. Similarly, we define higher powers of A, i.e., $A^3 = A^2.A$, $A^4 = A^3.A$ etc.

If $AI = A$, then the matrix A is called idempotent.

Example 2.14. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}$, find the product of AB. Is BA defined?

Solution. Since the number of columns of A = the number of rows of B (each being = 3).

∴ The product AB is defined and

$$= \begin{bmatrix} 0.1 + 1.-1 + 2.2, & 0.-2 + 1.0 + 2.-1 \\ 1.1 + 2.-1 + 3.2, & 1.-2 + 2.0 + 3.-1 \\ 2.1 + 3.-1 + 4.2, & 2.-2 + 3.0 + 4.-1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 5 & -5 \\ 7 & -8 \end{bmatrix}$$

Again since the number of columns of B = the number of rows of A.

∴ The product BA is not possible.

Example 2.15. If $A = \begin{bmatrix} 1 & 3 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix}$, compute AB and BA and show that $AB \neq BA$.

Solution. Considering rows of A and columns of B, we have

$$AB = \begin{bmatrix} 1.2 + 3.1 + 0.-1, & 1.3 + 3.2 + 0.1, & 1.4 + 3.3 + 0.2 \\ -1.2 + 2.1 + 1.-1, & -1.3 + 2.2 + 1.1, & -1.4 + 2.3 + 1.2 \\ 0.2 + 0.1 + 2.-1, & 0.3 + 0.2 + 2.1, & 0.4 + 0.3 + 2.2 \end{bmatrix} = \begin{bmatrix} 5 & 9 & 13 \\ -3 & 3 & 4 \\ -2 & 2 & 4 \end{bmatrix}$$

Again considering the rows of B and columns of A, we have

$$BA = \begin{bmatrix} 2.1 + 3.-1 + 4.0, & 2.3 + 3.2 + 4.0 & 2.0 + 3.1 + 4.2 \\ 1.1 + 2.-1 + 3.0, & 1.3 + 2.2 + 3.0 & 1.0 + 2.1 + 3.2 \\ -1.1 + 1.-1 + 2.0, & -1.3 + 1.2 + 2.0 & -1.0 + 1.1 + 2.2 \end{bmatrix} = \begin{bmatrix} -1 & 12 & 11 \\ -1 & 7 & 8 \\ -2 & -1 & 5 \end{bmatrix}$$

Evidently $AB \neq BA$.

Example 2.16. If $A = \begin{bmatrix} 2 & 0 & 3 \\ 1 & 2 & 1 \\ 3 & 0 & 2 \end{bmatrix}$, find the matrix B such that $BA = \begin{bmatrix} 8 & 0 & 7 \\ 5 & 2 & 4 \\ 7 & 0 & 6 \end{bmatrix}$. (Mumbai, 2005)

Solution. Let $AB = \begin{bmatrix} 2 & 0 & 3 \\ 1 & 2 & 1 \\ 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} l & m & n \\ p & q & r \\ s & t & u \end{bmatrix}$

$$= \begin{bmatrix} 2l + 0p + 3s & 2m + 0q + 3t & 2n + 0r + 3u \\ l + 2p + s & m + 2q + t & n + 2r + u \\ 3l + 0p + 2s & 3m + 0q + 2t & 3n + 0r + 2u \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 0 & 7 \\ 5 & 2 & 4 \\ 7 & 0 & 6 \end{bmatrix} \quad \text{(given)}$$

Equating corresponding elements, we get

$$2l + 3s = 8, \quad l + 2p + s = 5, \quad 3l + 2s = 7 \qquad \ldots(i)$$

$$2m + 3t = 0, \quad m + 2q + t = 2, \quad 3m + 2t = 0 \qquad \ldots(ii)$$

$$2n + 3u = 7, \quad n + 2r + u = 4, \quad 3n + 2u = 6 \qquad \ldots(iii)$$

Solving the equations (i), we get $l = 1, p = 2, s = 2$

Similarly equations (ii) gives $m = 0, q = 1, t = 0$

and equations (iii) gives $n = 0, r = 0, u = 3$

Thus, $B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 0 & 3 \end{bmatrix}$.

Example 2.17. Prove that $A^3 - 6A^2 + 9A + 11I = 0$, where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$

Solution. $A^2 = A \times A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1+4-3 & 2+0+6 & 3-2-9 \\ 2+0-1 & 4+0-2 & 6+0-3 \\ 1+4+3 & 2+0+6 & 3-2-9 \end{bmatrix} = \begin{bmatrix} 2 & 8 & -8 \\ 1 & 2 & 3 \\ 8 & 8 & -8 \end{bmatrix}$

$$A^3 = A^2 \times A = \begin{bmatrix} 2 & 8 & -8 \\ 1 & 2 & 3 \\ 8 & 8 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 2+16-8 & 4+0-16 & 6-8+24 \\ 1+4+3 & 2+0+6 & 3-2-9 \\ 8+16-8 & 16+0-16 & 24-8-24 \end{bmatrix} = \begin{bmatrix} 10 & -12 & 22 \\ 8 & 8 & -8 \\ 16 & 16 & -8 \end{bmatrix}$$

$$A^3 - 6A^2 + 9A + 11I$$

$$= \begin{bmatrix} 10 & -12 & 22 \\ 8 & 8 & -8 \\ 16 & 16 & -8 \end{bmatrix} - 6 \begin{bmatrix} 2 & 8 & -8 \\ 1 & 2 & 3 \\ 8 & 8 & -8 \end{bmatrix} + 9 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 11 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 10-12+9+11 & -12-48+18+0 & 22+48-27+0 \\ 8-6+18-0 & 8-12+0+11 & -8+6-9+0 \\ 16-48+18+0 & 16-48+0+0 & -8+48-27+11 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0.$$

Example 2.18. By mathematical induction, prove that if $A = \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix}$, then $A^n = \begin{bmatrix} 1+10n & -25n \\ 4n & 1-10n \end{bmatrix}$.

Solution. When $n = 1$, A^n gives $A^1 = \begin{bmatrix} 11 & -25 \\ 4 & -9 \end{bmatrix}$ ...(i)

Let us assume that the result is true for any positive integer k, so that

$$A^k = \begin{bmatrix} 1+3k & -2k \\ 4k & 1-2k \end{bmatrix}$$

$$\therefore \quad A^{k+1} = A^k \cdot A = \begin{bmatrix} 1+3k & -2k \\ 4k & 1-2k \end{bmatrix}\begin{bmatrix} 4 & -2 \\ 4 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+12k+4k & -2-6k+4k \\ 16k-4k-4 & -8k+4k+4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+3(k+1) & -2k \\ 4k & 1-2k \end{bmatrix}$$

This is true for $n = k+1$ ... (2)

We have seen in (1) that the result is true for $n = 1$.

$\therefore$ It is true for $n = 1+1 = 2$ [by (2)]

Similarly, it is true for $n = 2+1 = 3$ and so on.

Hence by mathematical induction, the result is true for all positive integers n.

Example 2.18. Prove that $(AB)C = A(BC)$, where A, B, C are matrices conformable for the product. (U.P.T.U., 2004)

Solution. Let $A = [a_{ij}]$ be of order $m \times n$, $B = [b_{ij}]$ be of order $n \times p$ and $C = [c_{ij}]$ be of order $p \times q$.

Then $$AB = [a_{ij}][b_{ij}] = \sum_{k=1}^{n} a_{ik} b_{kj}$$

$$\therefore \quad (AB)C = \left[\sum_{k=1}^{n} a_{ik} b_{kj}\right][c_{ij}] = \left[\sum_{k=1}^{p}\left[\sum_{k=1}^{n} a_{ik} b_{kj}\right] c_{jl}\right] = \left[\sum_{k=1}^{n}\sum_{k=1}^{p} a_{ik} b_{kj} c_{jl}\right]$$

Similarly, $$BC = [b_{ij}][c_{jl}] = \sum_{j=1}^{p} b_{ij} c_{jl}$$

$$\therefore \quad A(BC) = [a_{ik}]\left[\sum_{j=1}^{p} b_{ij} c_{jl}\right] = \left[\sum_{k=1}^{n} a_{ik}\left[\sum_{j=1}^{p} b_{ij} c_{jl}\right]\right] = \left[\sum_{k=1}^{n}\left[\sum_{j=1}^{p} a_{ik} b_{ij} c_{jl}\right]\right]$$

Hence $(AB)C = A(BC)$.

EXERCISE 2.3

1. For what values of x, the matrix $\begin{bmatrix} 3-x & x & 3 & 3 \\ 2 & 4-x & 3 & 3 \\ -2 & -x & -1-x & 3 \end{bmatrix}$ is singular?

2. Find the values of x, y, z such that $xI + yA + zA^2 = 0$, where $A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ and I is the unit matrix of order 3.

3. Matrix A has two rows $(1, 2, 3)$ and $(2, 0, 1)$ and matrix B has two columns $(1, 2)$ and $(0, 1)$. Find AB and BA and show that $AB \neq BA$.

4. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$, calculate the product AB.

5. If $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 0 & 1 & 3 \\ 2 & 1 & 0 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \\ 0 & 5 & 4 \end{bmatrix}$, find AB and BA, if they exist.

6. If $A = \begin{bmatrix} 2 & 3 \\ -3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -2 & 1 \\ 2 & 0 \end{bmatrix}$, verify that $(AB)C = A(BC)$ and $A(B+C) = AB + AC$.

7. Evaluate: (i) $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & 1 & -1 \\ 4 & -2 & 3 \\ -3 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ -3 & 1 \\ -2 & 3 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}\begin{bmatrix} 2 & -3 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} -2 \\ 3 \end{bmatrix}\begin{bmatrix} 4 & -6 \end{bmatrix}$ (v) $\begin{bmatrix} 2 \\ 3 \end{bmatrix}\begin{bmatrix} -2 & 3 \end{bmatrix}$

6. Prove that the product of the matrices

$$\begin{bmatrix} \cos^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix} \text{ and } \begin{bmatrix} \sin^2\theta & -\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \cos^2\theta \end{bmatrix}$$

is a null matrix where θ is any angle. (Prove by use of trigonometric identities)

8. If $A = \begin{bmatrix} 0 & -\sin\alpha & \cos\alpha \\ \sin\alpha\cos\beta & \cos\alpha\cos\beta & \sin\beta \\ \sin\alpha\sin\beta & \cos\alpha\sin\beta & 0 \end{bmatrix}$, show that $I + A = (I - A)\begin{bmatrix} 0 & \sin\beta & -\sin\alpha \\ -\sin\beta & 0 & \cos\alpha \\ \sin\alpha & -\cos\alpha & 0 \end{bmatrix}$

10. If $A = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$, find the values of $A^2 - 6A + kI$, where I is a unit matrix of order 2 and k is a scalar. (ICT 3 UC 2009)

12. If $A = \begin{bmatrix} 2 & -2 & 0 \\ 2 & 3 & -1 \\ -3 & 2 & 3 \end{bmatrix}$ and I is the unit matrix of order 3, evaluate $A^2 - 5A + 8I$.

13. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 3 \\ 4 & -1 & 2 \end{bmatrix}$, verify that matrix $(A + B)^2 = A^2 + BA + AB + B^2$.

15. If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$,

calculate the products AB and BA and show that $ABA = BAB = 0$.

14. If $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, show that $A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$, where n is a positive integer.

16. Evaluate the matrix $A = \begin{bmatrix} 2 & -2 & 1 \\ 3 & 1 & -1 \\ 3 & 2 & 4 \end{bmatrix}$ into the form LU, where L is lower triangular and U is upper triangular matrix.

2.6 RELATED MATRICES

(1) Transpose of a matrix. The matrix obtained from any given matrix A, by interchanging rows and columns is called the **transpose** of A and is denoted by A'.

Thus the transposed matrix of $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 7 & 8 \end{bmatrix}$ is $A' = \begin{bmatrix} 2 & 4 & 7 \\ 3 & 5 & 8 \end{bmatrix}$

Clearly, the transpose of a row matrix is a column matrix and the transpose of a column matrix is a row matrix. Also the transpose of the transpose of a matrix is the matrix itself, i.e., $(A')' = A$.

For a symmetric matrix, $A' = A$ and for a skew-symmetric matrix, $A' = -A$.

Illus. 1. The transpose of the product of two matrices is the product of their transposes taken in the reverse order, i.e., $(AB)' = B'A'$.

For, the element in the ith row and jth col. of $(AB)'$

= element in the jth row and ith col. of AB = scalar product of jth row of A with ith col. of B

= scalar product of jth col. of A' with ith row of B' = element in the ith row and jth col. of $B'A'$

Hence $(AB)' = B'A'$.

Illus. 2. Every square matrix can be expressed as a sum of a symmetric and a skew-symmetric matrix.

(ACA (I), 2005)

Let A be the given square matrix. Then $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$.

Let $B = \frac{1}{2}(A + A')$ and $C = \frac{1}{2}(A - A')$

Then $B' = \left[\frac{1}{2}(A + A')\right]' = \frac{1}{2}(A' + A'') = \frac{1}{2}(A' + A) = B$, i.e., $B = \frac{1}{2}(A + A')$ is a symmetric matrix.

again, $C' = \left[\frac{1}{2}(A - A')\right]' = \frac{1}{2}(A' - A) = -C$ i.e., $C = \frac{1}{2}(A - A')$ is a skew-symmetric matrix.

Hence A can be represented as the sum of a symmetric and a skew-symmetric matrix.

To prove the uniqueness, assume that P is a symmetric matrix and Q is a skew-symmetric matrix such that $A = P + Q$.

Then $A' = (P + Q)' = P' + Q' = P - Q$

Thus, $P = \frac{1}{2}(A + A')$ and $Q = \frac{1}{2}(A - A')$

which shows that there is one and only one way of expressing A as the sum of a symmetric and skew-symmetric matrix.

Example 2.30. Express the matrix A as the sum of a symmetric and a skew-symmetric matrix where

$$A = \begin{bmatrix} 4 & 2 & -2 \\ -1 & 3 & -5 \\ 3 & 4 & -7 \end{bmatrix}$$

Solution. We have $A' = \begin{bmatrix} 4 & -1 & -2 \\ 2 & 3 & 0 \\ -2 & -5 & -7 \end{bmatrix}$

Then, $A + A' = \begin{bmatrix} 8 & 1 & -4 \\ 1 & 6 & -5 \\ -4 & -5 & -14 \end{bmatrix}$ and $A - A' = \begin{bmatrix} 0 & 3 & 0 \\ -3 & 0 & -5 \\ 5 & 9 & 0 \end{bmatrix}$

$$\therefore \quad A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A') = \begin{bmatrix} 4 & 0.5 & -2 \\ 0.5 & 3 & -2.5 \\ -2 & -2.5 & -7 \end{bmatrix} + \begin{bmatrix} 0 & 1.5 & 0 \\ -1.5 & 0 & -2.5 \\ 2.5 & 2.5 & 0 \end{bmatrix}.$$

(2) Adjoint of a square matrix. The determinant of the square matrix

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \text{ is } |A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

The matrix formed by the cofactors of the elements is a b

$$\begin{bmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{bmatrix}. \text{ Then the transpose of this matrix, i.e., } \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix}$$

is called the adjoint of the matrix A and is written as $Adj\ A$.

Thus the adjoint of A is the transposed matrix of cofactors of A.

(3) Inverse of a matrix. If A be any matrix, then a matrix B of the same order such that $AB = BA = I$ is called the inverse of A which is denoted by A^{-1} so that $AA^{-1} = I$.

Also $$A^{-1} = \frac{Adj\ A}{|A|}$$

For $$A\ (Adj\ A) = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} = |A| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

or $$A \cdot \frac{Adj\ A}{|A|} = I \quad \text{i.e.} \quad |A| \neq 0 \text{ or } \frac{Adj\ A}{|A|} \text{ is the inverse of } A.$$

Note 1. Inverse of a matrix is unique.

If possible, let the two inverses of the matrix A be B and C.

Then $$AB = BA = I \quad \text{and} \quad AC = CA = I$$

$$\therefore \quad CAB = (CA)B = IB = B \quad \text{and} \quad CAB = C(AB) = CI = C$$

$$\therefore \quad B = C.$$

Obs. 2. The reciprocal of the product of two matrices is the product of their reciprocals taken in the reverse order, i.e. $(AB)^{-1} = B^{-1}A^{-1}$ (Manipal, 1992)

If A, B be two matrices, then the reciprocal of their product is $(AB)^{-1}$.

Clearly, $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$ (by Associative law)

$= AIA^{-1} = AA^{-1} = I$

Similarly, $(B^{-1}A^{-1})(AB) = I$

Hence $B^{-1}A^{-1}$ is the reciprocal of AB.

Obs. 3. Multiplication by an inverse matrix plays the same role in matrix algebra that division plays in ordinary algebra.

i.e., if $AB = AC$, then $A^{-1}(AB) = A^{-1}(AC)$, i.e., $(A^{-1}A)B = (A^{-1}A)C$, i.e., $IB = IC$, i.e., $B = C$

or $B = A^{-1}(AC)$, i.e., $B = \frac{\text{adj } A}{|A|}(AC)$, i.e., $B = \frac{1}{|A|}(\text{adj } A)(AC)$

Example 2.21. Find the inverse of $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

Solution. The determinant of the given matrix A is

$$\Delta = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 & -3 \\ -2 & -4 & -4 \end{vmatrix} = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} \text{ (say)}$$

If $A_1, A_2, \ldots$ be the cofactors of $a_1, a_2, \ldots$ in Δ, then $A_1 = -20, A_2 = -4, A_3 = -12, B_1 = 4, B_2 = 2, B_3 = 0$, $C_1 = 5, C_2 = 2, C_3 = 2$.

Then $\Delta = a_1A_1 + a_2A_2 + a_3A_3 = -8$.

and $\text{adj } A = \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \begin{bmatrix} -20 & -4 & -12 \\ 4 & 2 & 0 \\ 5 & 2 & 2 \end{bmatrix}$

Hence the inverse of the given matrix A

$$= \frac{\text{adj } A}{\Delta} = \frac{1}{-8}\begin{bmatrix} -20 & -4 & -12 \\ 4 & 2 & 0 \\ 5 & 2 & 2 \end{bmatrix} = \begin{bmatrix} \frac{5}{2} & \frac{1}{2} & \frac{3}{2} \\ -\frac{1}{2} & -\frac{1}{4} & -\frac{3}{4} \\ -\frac{5}{8} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix}$$

Note. For other methods see Examples 2.20 : 2.28 and 2.48.

Example 2.22. Find the matrix A if $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}A\begin{bmatrix} 1 & -1 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 3 & 4 \end{bmatrix}$ (Mumbai, 2005)

Solution. If $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}A\begin{bmatrix} 1 & -1 \\ 0 & -3 \end{bmatrix} = C$ and $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = B$, then

$BAC = D$ or $AC = B^{-1}D$

or $A = B^{-1}DC^{-1}$

Now, $B^{-1} = \frac{\text{adj } B}{|B|} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$

Similarly, $C^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix}$

Hence, $A = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} -2 & -1 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix}$

$= \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}\begin{bmatrix} -14 & -6 \\ 12 & 9 \end{bmatrix} = \begin{bmatrix} -40 & -24 \\ 60 & 36 \end{bmatrix}$.

PROBLEMS 2.6

1. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, verify that $AA' = I = A'A$, where I is the unit matrix.

2. Express each of the following matrices as the sum of a symmetric and a skew-symmetric matrix:

(a) $\begin{bmatrix} 3 & -2 & 5 \\ 2 & -1 & -3 \\ 5 & 4 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$

3. If A is a non-singular matrix of order n, prove that $\det(A^{-1}) = (\det A)^{-1}$. (Mysore, 2005)

Verify this if $A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 3 & 0 \\ 1 & 5 & 2 \end{bmatrix}$.

4. Find the inverse of the matrix (i) $\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}$ (Mumbai, 1995) (ii) $\begin{bmatrix} 3 & -3 & 4 \\ -2 & 3 & -3 \\ 4 & -3 & 3 \end{bmatrix}$ (U.P.T.U., 2003)

5. If $A = \begin{bmatrix} 2 & 2 & 2 \\ 1 & 3 & 1 \\ 3 & 5 & 4 \end{bmatrix}$, express A as $P + iQ$, P and Q real. Also find A^{-1} and show that $AA' = \begin{bmatrix} 12 & 9 & 21 \\ 9 & 11 & 19 \\ 21 & 19 & 38 \end{bmatrix}$. (Mysore, 2006)

6. If $A = \begin{bmatrix} 2 & -1 & 4 \\ 3 & -3 & 4 \\ 0 & -1 & 3 \end{bmatrix}$, (i) find A^{-1} (ii) show that $(A^{-1})' = (A')^{-1}$.

7. Find the inverse of the matrix

$$S = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \end{bmatrix} \text{ and find } A = \frac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ -2 & 2 & 0 \\ 2 & -1 & -1 \end{bmatrix},$$

show that A is orthogonal matrix. (Mumbai, 2007)

8. If $A = \begin{bmatrix} \frac{1}{2} & 1 & \frac{3}{2} \\ \frac{3}{2} & 2 & \frac{1}{2} \\ 1 & 3 & 2 \end{bmatrix}$, prove that $A^3 = A$.

9. Show that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} 1 & -\tan\theta/2 \\ \tan\theta/2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -\tan\theta/2 \\ \tan\theta/2 & 1 \end{bmatrix}\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

10. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ 4 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$, verify that $(ABC)' = B'A'C'$, where A' is the transpose of A.

11. $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.

12. If A is a square matrix, show that $(kA + A')$ is symmetric and $(kA - A')$ is skew-symmetric. (V.T.U., 2009)

13. If $B = \text{diag}[b_1, b_2, b_3, b_4, b_5]$ is a matrix, find B^{-1} = diag $[b_1^{-1}, b_2^{-1}, b_3^{-1}, b_4^{-1}, b_5^{-1}]$.

14. If A and B are square matrices of the same order and A is symmetrical, show that $A'BA$ is also symmetrical. (Hint: Show that $(A'BA)' = A'BA$)

15. If a non-singular matrix A is symmetric, show that A^{-1} is also symmetric.

2.7 (1) RANK OF A MATRIX

If we select any r rows and r columns from any matrix A, deleting all the other rows and columns, then the determinant formed by these $r \times r$ elements is called the minor of A of order r. Clearly, there will be a number of different minors of the same order, got by deleting different rows and columns from the given matrix.

Def. *A matrix is said to be of rank r when*

(i) it has at least one non-zero minor of order r,

and (ii) every minor of order higher than r vanishes.

Briefly, the rank of a matrix is the highest order of any non-vanishing minor of the matrix.

If a matrix has a non-zero minor of order r, its rank is $\geq r$.

If all minors of a matrix of order $r+1$ are zero, its rank is $\leq r$.

The rank of a matrix A shall be denoted by $\rho(A)$.

(2) Elementary transformation of a matrix. The following operations, known as the first order minors and three in columns are known as elementary transformations:

I. The interchange of any two rows (or columns).

II. The multiplication of every row (column) by a non-zero number.

III. The addition to a constant multiple of the elements of any row (column) to the corresponding elements of any other row (column).

Notation. The elementary row transformations will be denoted by the following symbols:

(i) C_{ij} for the interchange of the ith and jth rows.

(ii) kR_i for multiplication of the ith row by k.

(iii) $R_i + pR_j$ for addition to the ith row, p times the jth row.

The corresponding column transformations will be denoted by writing C in place of R.

Elementary transformations do not change order. An order is a property of a matrix. Any of the above operations I and II, the rows or columns of a matrix are interchanged, the order remains unaffected.

(3) Equivalent matrix. Two matrices A and B are said to be equivalent if one can be obtained from the other by a sequence of elementary transformations. Two equivalent matrices have the same order and the same rank. The symbol $\sim$ is used for equivalence.

Example 2.20. *Determine the rank of the following matrices:*

$$\text{(i)}\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \qquad \text{(ii)}\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 2 & 1 & -2 & 0 \end{bmatrix} \qquad \text{(V.T.U.—Nov.'03)}$$

Solution. (i) Operate $R_2 - R_1$ and $R_3 - 2R_1$ to find the given matrix

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix} \sim A \text{ (say)}$$

Obviously, the 3rd order minor of A vanishes. Also its 2nd order minors formed by its 2nd and 3rd rows are all zero. But another 2nd order minor is $\begin{vmatrix} 1 & 2 \\ 0 & 2 \end{vmatrix} = 2 \neq 0$.

$\therefore \quad \rho(A) = 2$. Hence the rank of the given matrix is 2.

(ii) Given matrix

$$\sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 0 \\ 3 & 1 & -3 & -1 \\ 1 & 1 & -2 & -1 \end{bmatrix}$$

[Operating $C_2 - C_3, C_4 - C_3$]

$$\sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 0 \\ 3 & 0 & -3 & 0 \\ 1 & 0 & -1 & 0 \end{bmatrix}$$

[Operating $R_3 - R_1, R_4 - R_3$]

$$\sim \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \end{bmatrix} \sim A \text{ (say)}$$

[Operating $2R_2 - 3R_3, R_4 - R_3$]

[Operating $C_3 + 3C_2, C_4 + C_2$]

Obviously, the 4th order minor of A is zero. Also every 3rd order minor of A is zero. But, of the 2nd order minors, only $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$. $\quad \therefore \quad \rho(A) = 2$.

Hence the rank of the given matrix is 2.

(4) Elementary matrices. An elementary matrix is that, which is obtained from a unit matrix by subjecting it to any of the elementary transformations.

Examples of elementary matrices obtained from

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ are } E_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ (} R_2 \leftrightarrow R_3 \text{)}, \quad E_{23}(k) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ (} R_2 \to kR_2 \text{)}, \quad E_{21}(k) = \begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(5) Theorem. *Elementary row (column) transformations of a matrix A can be obtained by pre-multiplying (post-multiplying) A by the corresponding elementary matrices.*

Consider the matrix $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

Then $E_{23} \times A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \end{bmatrix}$

So a pre-multiplication by E_{23} has interchanged the 2nd and 3rd rows of A. Similarly, pre-multiplication by $E_{23}(k)$ will multiply the 2nd row of A by k and pre-multiplication by $E_{21} + pE_{23}$ will result in the addition of p times the 3rd row of A to the 1st row.

Thus the pre-multiplication of A by elementary matrices results in the corresponding elementary row transformations of A. It can easily be seen that post-multiplication will perform the elementary column transformations.

(6) Gauss-Jordan method of finding the inverse. *Those elementary row transformations which reduce a given square matrix A to the unit matrix, when applied to the unit matrix I, give the inverse of A.*

Let the successive row transformations which reduce A to I result from pre-multiplication by the elementary matrices $E_1, E_2, \ldots E_k$ so that

$$E_k E_{k-1} \ldots E_2 E_1 A = I$$

$$E_k E_{k-1} \ldots E_2 E_1 A A^{-1} = A^{-1}$$

or $$E_k E_{k-1} \ldots E_2 E_1 I = A^{-1} \qquad \text{i.e.} \quad A A^{-1} = I$$

Hence the result.

Working rule to evaluate A^{-1}. *Write the new matrix A and I side by side. Then perform the same row transformations on both A as are necessary to reduce A to I. The latter matrix represents A^{-1}.*

Example 2.26. *Using (i) Gauss-Jordan method, find the inverse of the matrix* $\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & -2 \\ -2 & -4 & -4 \end{bmatrix}$ (Bundelkhand 2006)

Solution. Writing the same matrix side by side with the unit matrix of order 3, we have

$$\begin{bmatrix} 1 & 1 & 2 & : & 1 & 0 & 0 \\ 1 & 2 & -2 & : & 0 & 1 & 0 \\ -2 & -4 & -4 & : & 0 & 0 & 1 \end{bmatrix}$$

(Operate $R_2 - R_1$ and $R_3 + 2R_1$)

$$\sim \begin{bmatrix} 1 & 1 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & -4 & : & -1 & 1 & 0 \\ 0 & -2 & 2 & : & 2 & 0 & 1 \end{bmatrix}$$

(Operate $\frac{1}{6}R_2$ and $-\frac{1}{6}R_3$)

$$\sim \begin{bmatrix} 1 & 1 & 2 & : & 1 & 0 & 0 \\ 0 & 1 & -4 & : & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 1 & : & 1 & 0 & \frac{1}{2} \end{bmatrix}$$

(Operate $R_1 - R_2$ and $R_3 + R_2$)

Named after the great German mathematician Carl Friedrich Gauss (1777–1855) who made his first great discovery as a student at Gottingen. His important contributions are in algebra, number theory, mechanics, complex analysis, differential equations, differential geometry, non-Euclidean geometry, numerical analysis, astronomy and electromagnetism. He became director of the observatory at Gottingen in 1807.

Named after the German mathematician and geodesist, Wilhelm Jordan (1842–1899).

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$$\sim\begin{bmatrix}1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & 0\\ 0 & 1 & -2 & -\frac{1}{4} & \frac{3}{4} & 0\\ 0 & 0 & -2 & \frac{1}{4} & \frac{1}{4} & 1\end{bmatrix}$$

[Operate $R_3 + 2R_2$, $R_2 - \frac{3}{2}R_3$ and $\left(-\frac{1}{2}\right)R_3$]

$$\sim\begin{bmatrix}1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4}\\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4}\\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4}\end{bmatrix}$$

Hence the inverse of the given matrix is $\begin{bmatrix}\frac{3}{4} & -\frac{1}{4} & -\frac{1}{4}\\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4}\\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4}\end{bmatrix}$ [cf. Example 2.21]

(7) Normal form of a matrix. *Every non-zero matrix A of rank r can be reduced by a sequence of elementary transformations to the form*

$$\begin{bmatrix}I_r & 0\\ 0 & 0\end{bmatrix} \text{ called the normal form of } A \qquad \ldots(1)$$

Cor. 1. *The rank of a matrix A is r if and only if it can be reduced to the normal form (1).*

Cor. 2. *Since row elementary transformations are to be allowed by pre-multiplication or post-multiplication with suitable elementary matrix and row elementary matrix is more simpler, therefore we have the following result:*

Corresponding to every matrix A of rank r, there exist non-singular matrices P and Q such that PAQ equals (1).

If A be an n-square matrix, then P and Q are square matrices of order n each respectively.

Example 2.25. *Reduce the following matrix to its normal form and hence find its rank:*

$$A=\begin{bmatrix}2 & 3 & -2 & -1\\ 1 & -1 & -2 & -4\\ 3 & 1 & -3 & -5\\ 6 & 3 & -7 & -10\end{bmatrix}$$ [U.P.T.U. 2003]

Solution.

$$A\sim\begin{bmatrix}1 & -1 & -2 & -4\\ 2 & 3 & -2 & -1\\ 3 & 1 & -3 & -5\\ 6 & 3 & -7 & -10\end{bmatrix} \qquad [\text{By } R_1, R_2]$$

$$\sim\begin{bmatrix}1 & -1 & -2 & -4\\ 0 & 5 & 2 & 7\\ 0 & 4 & 3 & 7\\ 0 & 9 & 5 & 14\end{bmatrix} \qquad [\text{By } R_2 - 2R_1,\ R_3 - 3R_1,\ R_4 - 6R_1]$$

$$\sim\begin{bmatrix}1 & 3 & 0 & 0\\ 0 & 5 & 2 & 7\\ 0 & 4 & 3 & 7\\ 0 & 9 & 5 & 14\end{bmatrix} \qquad [\text{By } C_2 + C_1,\ C_3 + 2C_1,\ C_4 + 4C_1]$$

$$\sim\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 5 & 2 & 7\\ 0 & 4 & 3 & 7\\ 0 & 9 & 5 & 14\end{bmatrix} \qquad [\text{By } R_2 - R_4 - R_3]$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & -4 & -3 \\ 0 & 4 & 8 & 12 \\ 0 & 2 & 6 & 3 \end{bmatrix} \quad [By\ R_2 - R_1]$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & -4 & -3 \\ 0 & 0 & 24 & 24 \\ 0 & 0 & 9 & 9 \end{bmatrix} \quad [By\ R_3 - 4R_2]$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 24 & 24 \\ 0 & 0 & 9 & 9 \end{bmatrix} \quad [By\ C_3 + 4C_2, C_4 + 3C_2]$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 9 & 9 \end{bmatrix} \quad \left[By\ \frac{1}{24} C_3\right]$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 9 & 9 \end{bmatrix} \quad [By\ C_4 - C_3]$$

$$\sim \begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence $\rho(A) = 3$.

Example 9.30. *For the matrix* $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 2 & 5 \\ 0 & -1 & -1 \end{bmatrix}$,

find two non-singular matrices P and Q such that PAQ is in the normal form. Hence find the rank of A.
(Gorakhpur, 2006)

Solution. We write $A = IAI$, i.e., $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 2 & 5 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

We shall affect every elementary row reduction transformation of the product by multiplying the pre-factor (post-factor) I_3 to the matrix.

$$\text{Operate } C_2 - C_1,\ C_3 - 2C_1: \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Operate } R_2 - R_1: \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Operate } C_2 - C_3: \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Operate } R_3 + R_2: \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

which is of the required form $\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix}$

Hence, $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 3 & -1 & -1 \\ 3 & 1 & -1 \\ 3 & 0 & 1 \end{bmatrix}$ and $rank\,A = 2$.

EXERCISE 2.1

Determine the rank of each of the following matrices (1–4):

1. $\begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & 4 \\ 3 & 7 & 10 \end{bmatrix}$ (U.C.E., 2002)

2. $\begin{bmatrix} 1 & 2 & 4 & 3 \\ 3 & 6 & 12 & 9 \\ 2 & 3 & 4 & 5 \end{bmatrix}$ (B.A.S.E., 2003)

3. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 7 & 3 \\ 3 & 6 & 3 & 7 \\ 4 & 8 & 7 & 1 \end{bmatrix}$ (Garraway, 2002)

4. $\begin{bmatrix} 3 & 6 & 7 & 0 \\ 4 & 7 & 8 & 9 \\ 13 & 12 & 13 & 18 \\ 14 & 17 & 18 & 26 \end{bmatrix}$ (S.A.S.E., 2004)

5. $\begin{bmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{bmatrix}$ (Mercer, 2004)

6. Determine the value of p such that the rank of $\begin{bmatrix} 1 & 1 & -1 & 1 \\ 4 & 4 & -2 & 1 \\ 2 & 0 & 2 & 3 \\ 9 & 6 & p & 7 \end{bmatrix}$ is 3. (Mercer, 2007)

7. Use Gauss–Jordan reduction to find the inverse of the following matrices:

(a) $\begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 5 & 2 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} -2 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix}$ (Mercer, 2005)

(c) $\begin{bmatrix} 1 & -1 & 2 & -2 \\ 2 & -1 & 3 & 1 \\ 3 & -2 & 5 & -1 \end{bmatrix}$ (B.A.S.E., 2006) (d) $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$ (Garraway, 2002)

8. Find the inverse of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 8 \end{bmatrix}$ by using elementary row operations. (Mercer, 2004)

9. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, find A^{-1}. Also find the eigenvalues and eigenvectors of A and hence find A^3. Show that the sum of the eigenvalues is equal to the sum of the elements of the leading diagonal of A and verify that $A^3 - 5A^2 + 7A - 3I = 0$.

10. Find the eigenvalues and eigenvectors of the matrix A and hence find the modal matrix M such that $M^{-1}AM$ is a diagonal matrix. Hence find A^4.

(a) $A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & -1 \\ 5 & 1 & 1 \end{bmatrix}$ (U.C.E., 2002) (b) $A = \begin{bmatrix} 1 & 2 & 3 & -2 \\ 2 & -2 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

11. Express each of the following matrices in normal form and hence find their rank:

(a) $\begin{bmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ -1 & -1 & 1 & 1 \end{bmatrix}$ (Garraway, 2002) (b) $A = \begin{bmatrix} 2 & 1 & -3 & 10 \\ 3 & -2 & 1 & 2 \\ 1 & 1 & -2 & 6 \end{bmatrix}$ (Mercer, 2002)

(c) $\begin{bmatrix} 1 & 2 & 3 & 10 \\ 2 & -1 & 1 & 1 \\ 1 & 3 & -4 & 9 \end{bmatrix}$ (Mercer, 2002) (d) $\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 1 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 2 & -1 \end{bmatrix}$ (U.C.E., 2002)

2.8 PARTITION METHOD OF FINDING THE INVERSE

According to this method of finding the inverse, if the inverse of a matrix A_n of order n is known, then the inverse of the matrix A_{n+1} can easily be obtained by adding $(n+1)$th row and $(n+1)$th column to A_n.

Let $$A = \begin{bmatrix} A_1 & \vdots & A_2 \\ \cdots & \cdots & \cdots \\ A_3' & \vdots & a \end{bmatrix} \text{ and } A^{-1} = \begin{bmatrix} X_1 & \vdots & X_2 \\ \cdots & \cdots & \cdots \\ X_3' & \vdots & x \end{bmatrix}$$

where A_2, X_2 are column vectors and A_3', X_3' are row vectors being transposes of column vectors in A_2, X_2 and a, x are ordinary numbers. We also assume that A_1^{-1} is known.

Then $AA^{-1} = I_{n+1}$ i.e., $$\begin{bmatrix} A_1 & \vdots & A_2 \\ \cdots & \cdots & \cdots \\ A_3' & \vdots & a \end{bmatrix} \begin{bmatrix} X_1 & \vdots & X_2 \\ \cdots & \cdots & \cdots \\ X_3' & \vdots & x \end{bmatrix} = \begin{bmatrix} I_n & \vdots & 0 \\ \cdots & \cdots & \cdots \\ 0 & \vdots & 1 \end{bmatrix}$$

gives
$$A_1X_1 + A_2X_3' = I_n \quad \ldots(1)$$
$$A_1X_2 + A_2x = 0 \quad \ldots(2)$$
$$A_3'X_1 + aX_3' = 0 \quad \ldots(3)$$
$$A_3'X_2 + ax = 1 \quad \ldots(4)$$

From (2), $X_2 = -A_1^{-1}A_2x$ and using this, (4) gives $x = (a - A_3'A_1^{-1}A_2)^{-1}$.

Hence x and then X_2 are given.

Also from (1), $X_1 = A_1^{-1}(I_n - A_2X_3')$

and using this, (3) gives $X_3' = -A_3'A_1^{-1}(a - A_3'A_1^{-1}A_2)^{-1} = -A_3'A_1^{-1}x$.

Then X_1 is determined and hence A^{-1} is computed.

Obs. This is also known as the 'Bordering method'. For evaluation of A^{-1} we only need to determine two inverse matrices A_1^{-1} and $(a - A_3'A_1^{-1}A_2)^{-1}$.

Example 2.27. Using the partition method, find the inverse of $\begin{bmatrix} 3 & 1 & -1 \\ 2 & 3 & -1 \\ 3 & 5 & -2 \end{bmatrix}$.

Solution. Let $$A = \begin{bmatrix} 3 & 1 & \vdots & -1 \\ 2 & 3 & \vdots & -1 \\ \cdots & \cdots & \cdots & \cdots \\ 3 & 5 & \vdots & -2 \end{bmatrix} = \begin{bmatrix} A_1 & \vdots & A_2 \\ \cdots & \cdots & \cdots \\ A_3' & \vdots & a \end{bmatrix}$$

so that $$A_1^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}^{-1} = \frac{1}{7}\begin{bmatrix} 3 & -1 \\ -2 & 3 \end{bmatrix}$$

Let $$A^{-1} = \begin{bmatrix} X_1 & \vdots & X_2 \\ \cdots & \cdots & \cdots \\ X_3' & \vdots & x \end{bmatrix} \text{ so that } AA^{-1} = I.$$

$$a - A_3'A_1^{-1}A_2 = -2 + \begin{bmatrix} 3 & 5 \end{bmatrix}\frac{1}{7}\begin{bmatrix} 3 & -1 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = -10$$

$$\therefore \quad x = (a - A_3'A_1^{-1}A_2)^{-1} = -\frac{1}{10}$$

Also, $$X_2 = -A_1^{-1}A_2x = \frac{1}{7}\begin{bmatrix} 3 & -1 \\ -2 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}\left(-\frac{1}{10}\right) = -\frac{1}{10}\begin{bmatrix} 4 \\ -5 \end{bmatrix}$$

Then $$X_3' = -A_3'A_1^{-1}x = \begin{bmatrix} 3 & 5 \end{bmatrix}\frac{1}{7}\begin{bmatrix} 3 & -1 \\ -2 & 3 \end{bmatrix}\left(-\frac{1}{10}\right) = -\frac{1}{10}\begin{bmatrix} -11 & 18 \end{bmatrix}$$

Finally, $$X_1 = A_1^{-1}(I - A_2X_3') = -\frac{1}{7}\begin{bmatrix} 3 & -1 \\ -2 & 3 \end{bmatrix} - \frac{1}{10}\begin{bmatrix} 3 & -1 \\ 4 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}\begin{bmatrix} -11 & 18 \end{bmatrix}$$

$$= \begin{bmatrix} [illegible] \end{bmatrix}$$

Hence $A^{-1} = \begin{bmatrix} [illegible] \end{bmatrix}$

[illegible]

Conversely, if an $n \times n$ matrix A has an inverse A^{-1}, then A must be nonsingular. For if A were singular, then $\det A = 0$, and by Theorem 3.7, $\det(AA^{-1}) = \det A \det A^{-1} = 0 \cdot \det A^{-1} = 0$. But this contradicts the fact that $AA^{-1} = I$ and $\det I = 1 \neq 0$. Thus A must be nonsingular.

(a) $AA^{-1} = \begin{bmatrix} [illegible] \end{bmatrix}$

(b) $\begin{bmatrix} [illegible] \end{bmatrix}$

(c) Equating corresponding elements, we have

$$[illegible]$$

Second, $ad - bc \neq 0$, i.e., $bc - ad \neq 0$, or else no inverse exists.

First solution: Since $AA^{-1} = I$, i.e., $A^{-1} = A^{-1}$

From first equation, $AB^{-1} = kI$, i.e., $CB^{-1} = kA^{-1}$, i.e., $A = kC^{-1}B$

$[illegible]$

From fourth equation, $BC^{-1} = kI$, i.e., $CB^{-1} = kA^{-1}$, i.e., $D = kA^{-1}$

Hence $A^{-1} = \begin{bmatrix} [illegible] \end{bmatrix}$

(d) Let $A = \begin{bmatrix} [illegible] \end{bmatrix}$

Suppose $A = \begin{bmatrix} [illegible] \end{bmatrix}$, $B = \begin{bmatrix} [illegible] \end{bmatrix}$, $C = \begin{bmatrix} [illegible] \end{bmatrix}$

(a) $A^{-1} = \begin{bmatrix} [illegible] \end{bmatrix}$

(b) $A^{-1}BA = \begin{bmatrix} [illegible] \end{bmatrix}$

$= \begin{bmatrix} [illegible] \end{bmatrix}$

(c) $A^{-1}BA = \begin{bmatrix} [illegible] \end{bmatrix}$

Hence $B^{-1} = \begin{bmatrix} [illegible] \end{bmatrix}$

PROBLEMS 2.3

Find the inverse (if exist) of the following matrices using the cofactor method :

1. $\begin{bmatrix} 2 & 3 & 2 \\ 1 & 4 & 3 \\ 3 & 2 & 4 \end{bmatrix}$ (Regular, 2007)

2. $\begin{bmatrix} 2 & 3 & 4 \\ 2 & 1 & 3 \\ 1 & 2 & 3 \end{bmatrix}$

3. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 0 & 5 \\ 2 & 1 & 0 & 3 \\ 3 & 1 & 1 & 2 \end{bmatrix}$

4. $\begin{bmatrix} 2 & -1 & 3 & 4 \\ 2 & 1 & 0 & 3 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 3 & 1 \end{bmatrix}$

2.3 SOLUTION OF LINEAR SYSTEM OF EQUATIONS

(1) Method of determinants—Cramer's* rule

Consider the equations $\begin{vmatrix} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{vmatrix}$ ...(i)

If the determinant of the Eqn. (i) be $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

then $\Delta x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$ [Operate $C_1 + yC_2 + zC_3$]

$= \begin{vmatrix} a_1x + b_1y + c_1z & b_1 & c_1 \\ a_2x + b_2y + c_2z & b_2 & c_2 \\ a_3x + b_3y + c_3z & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$ [By (i)]

Thus $x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix} \div \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ provided $\Delta \neq 0$. ...(ii)

Similarly, $y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \div \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ ...(iii)

and $z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} \div \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ ...(iv)

Equations (ii), (iii) and (iv) giving the values of x, y, z constitute the Cramer's rule, which reduces the solution of the linear equations (i) to a problem in evaluation of determinants.

(2) Matrix inversion method

If $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

then the equations (i) are equivalent to the matrix equation $AX = B$ ...(v)

where A is the coefficient matrix.

Multiplying both sides of (v) by the reciprocal matrix A^{-1}, we get

$A^{-1}AX = A^{-1}B$ or $IX = A^{-1}B$ [$\because A^{-1}A = I$]

*Gabriel Cramer (1704–1752), a Swiss mathematician.

or $$X = A^{-1}B \quad \text{i.e.} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{\Delta}\begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} \times \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} \qquad \ldots(iv)$$

where A_1, B_1 etc. are the cofactors of a_1, b_1 etc. in the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ (ii) or (iii)

Hence equating the values of x, y, z in the corresponding elements in the product on the right side of (iv), we get the desired solutions.

Note. When Δ is a singular matrix, i.e. $\Delta = 0$, the above methods fail. Then also just to test the consistency of equations and the number of unknowns are assigned, Rouche's test is used for analytic approach in detail (refer to the equations set such $Ax = \lambda x$ in P.E. (VII)).

Example 3.28. *Solve the equations* $2x - 3y + 2z = 2$, $3x - 2y + z = -5$, $x + 2y - z = 4$ *by (i) determinant (ii) matrix.*

Solution. (i) By determinants:

Here $$\Delta = \begin{vmatrix} 2 & -3 & 2 \\ 3 & -2 & 1 \\ 1 & 2 & -1 \end{vmatrix} = 2(2-2) + 3(-3-1) + 2(6+2) = 0 - 12 + 16 = 4 \quad \text{[Expanding by } C_1\text{]}$$

$$\therefore \quad x = \frac{1}{\Delta}\begin{vmatrix} 2 & -3 & 2 \\ -5 & -2 & 1 \\ 4 & 2 & -1 \end{vmatrix} \quad \text{[Expanding by } C_1\text{]}$$

$$= \frac{1}{4}[2(2-2) + 5(3-4) + 4(-3+4)] = \frac{1}{4}[0 - 5 + 4] = -\frac{1}{4}$$

Similarly, $$y = \frac{1}{\Delta}\begin{vmatrix} 2 & 2 & 2 \\ 3 & -5 & 1 \\ 1 & 4 & -1 \end{vmatrix} = \frac{1}{4}(8) = 2 \quad \text{and} \quad z = \frac{1}{\Delta}\begin{vmatrix} 2 & -3 & 2 \\ 3 & -2 & -5 \\ 1 & 2 & 4 \end{vmatrix} = -\frac{1}{4}$$

Hence $x = -\frac{1}{4}, y = 2, z = -\frac{1}{4}$

Note. The use of Cramer's rule involves a lot of calculations when the number of equations exceeds four. In such situations, the numerical methods given in §3.6.4 to 3.6.6 are preferable.

(ii) By matrices :

Here $$\Delta = \begin{vmatrix} 2 & -3 & 2 \\ 3 & -2 & 1 \\ 1 & 2 & -1 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 4 \neq 0$$

Then $A_1 = -4, A_2 = 1, A_3 = 5; B_1 = -1, B_2 = 1, B_3 = 1; C_1 = 2, C_2 = -1, C_3 = -5$.

Also $\Delta = a_1A_1 + a_2A_2 + a_3A_3 = 8$.

$$\therefore \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{\Delta}\begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} \times \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} -4 & 1 & 5 \\ -1 & 1 & 1 \\ 2 & -1 & -5 \end{bmatrix} \times \begin{bmatrix} 2 \\ -5 \\ 4 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -8-5+20 \\ -2-5+4 \\ 4+5-20 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Hence $x = 1, y = 2, z = -1$.

Example 3.29. *Solve the equations* $x_1 - 2x_2 + x_3 + x_4 = 2$, $x_2 - x_3 + x_4 = 1$, $x_1 + x_3 + x_4 = 4$, $2x_1 + x_2 + x_3 - x_4 = 3$ *by finding the inverse of the coefficient matrix.*

Solution. Given system can be written as $AX = B$, where

$$A = \begin{bmatrix} 1 & -2 & 1 & 1 \\ 0 & 1 & -1 & 1 \\ 1 & 0 & 1 & 1 \\ 2 & 1 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, B = \begin{bmatrix} 2 \\ 1 \\ 4 \\ 3 \end{bmatrix}$$

To find A^{-1}, we write

$$[A:I]=\begin{bmatrix}1&-1&1&1&:&1&0&0&0\\1&1&-1&1&:&0&1&0&0\\1&1&1&-1&:&0&0&1&0\\1&1&1&1&:&0&0&0&1\end{bmatrix}\begin{matrix}\\R_2-R_1\\R_3-R_1\\R_4-R_1\end{matrix}$$

$$=\begin{bmatrix}1&-1&1&1&:&1&0&0&0\\0&2&-2&0&:&-1&1&0&0\\0&2&0&0&:&-1&0&1&0\\0&2&0&2&:&-1&0&0&1\end{bmatrix}\begin{matrix}\\\\R_3-R_2\\R_4-R_2\end{matrix}$$

$$=\begin{bmatrix}1&-1&1&1&:&1&0&0&0\\0&2&-2&0&:&-1&1&0&0\\0&0&2&0&:&0&-1&1&0\\0&0&0&2&:&0&-1&0&1\end{bmatrix}\begin{matrix}\\\\\\R_4-R_3\end{matrix}$$

$$=\begin{bmatrix}1&0&0&0&:&1/2&1/2&0&0\\0&1&-1&0&:&-1/2&1/2&0&0\\0&0&1&0&:&1/2&0&1/2&0\\0&0&0&1&:&0&0&-1/2&1/2\end{bmatrix}\begin{matrix}R_1-R_2\\\\R_3+R_2\\\\\end{matrix}$$

$$=\begin{bmatrix}1&0&0&0&:&1/2&1/2&0&-1/2\\0&1&0&0&:&0&1/2&1/2&0\\0&0&1&0&:&1/2&0&1/2&0\\0&0&0&1&:&0&0&-1/2&1/2\end{bmatrix}\begin{matrix}R_1-R_4\\R_2-R_4\\\\\end{matrix}$$

$$=\begin{bmatrix}1&0&0&0&:&1/2&1/2&1/2&-1/2\\0&1&0&0&:&-1/2&0&0&1/2\\0&0&1&0&:&0&-1/2&0&1/2\\0&0&0&1&:&0&0&-1/2&1/2\end{bmatrix}$$

Thus $$A^{-1}=\begin{bmatrix}1/2&1/2&1/2&-1/2\\-1/2&0&0&1/2\\0&-1/2&0&1/2\\0&0&-1/2&1/2\end{bmatrix}$$

Hence $$X=A^{-1}B=\begin{bmatrix}1/2&1/2&1/2&-1/2\\-1/2&0&0&1/2\\0&-1/2&0&1/2\\0&0&-1/2&1/2\end{bmatrix}\begin{bmatrix}2\\-4\\4\\0\end{bmatrix}=\begin{bmatrix}1\\-1\\2\\-3\end{bmatrix}$$

i.e., $x_1=1, x_2=-1, x_3=2, x_4=-3$.

EXERCISE 2.6

Solve the following equations with the help of determinants (1 to 4):

1. $x+y+z=4,\ x-y+z=2,\ 2x+y+z=5$. (Harcourt 2002)
2. $x+2y+3z=5,\ 2x+y+z=7,\ x+3y+2z=9$.
3. $x+y+z=6,\ x-y+z=2,\ 2x+3y-z=1$.
4. $x+y+z=1,\ ax+by+cz=k,\ a^2x+b^2y+c^2z=k^2$.
5. Show that the equations $x+y+z=6,\ x+2y+3z=14,\ x+4y+7z=30$ are consistent and solve them.

Solve the following systems of equations by matrix method (6 to 8):

6. $x_1+x_2=x_3+1,\ x_2+2x_3=x_4+1,\ x_3+2x_4=x_5+1,\ x_4+2x_5=x_6+1,\ x_5+2x_6=x_1+1$. (UPTU 2006)
7. $x+y+z=3,\ x+2y+3z=4,\ x+4y+9z=6$. (Kanpur 2006)
8. $2x+y+4z=12,\ 4x+2y+3z=15,\ 3x+5y+2z=15$. (UPTU 2005)
9. $3x+y+2z=3,\ 2x-2y+4z=-2,\ x+\tfrac{1}{2}y-z=0$. (Uttarakhand 2006)
10. $x_1+x_2+x_3+x_4=5,\ x_1+2x_2+3x_3+4x_4=8,\ x_1+3x_2+6x_3+10x_4=14,\ x_1+4x_2+10x_3+20x_4=30$.

(UPTU 2004)

(c) By finding A^{-1}, solve the linear equations $AX = B$, where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

(d) By using row operations on the augmented matrix, find the inverse of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$.

(e) Explain briefly why the following matrix does not have an inverse:
$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.

(f) Explain briefly why the following matrix does not have an inverse:
$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{bmatrix}$.

4.3.3 (c) Consistency of a system of equations

Consider the system of m linear equations

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\cdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}\right\} \qquad (*)$$

in n unknowns $x_1, x_2, \ldots, x_n$. To determine whether the equations in $(*)$ are consistent (i.e., possess a solution) or not, we consider the matrix and the augmented matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \text{and} \quad \tilde{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

A is the coefficient matrix and $\tilde{A}$ is called the augmented matrix of the equations $(*)$.

(a) Rouché's theorem. The system of equations $(*)$ is consistent if and only if the coefficient matrix A and the augmented matrix $\tilde{A}$ are of the same rank; otherwise the system is inconsistent.

Proof. We consider only the following two possible cases:

1. Rank of A = rank of $\tilde{A}$ = n = the number of the unknowns in $(*)$. The equations $(*)$ may be written in operational form as follows:

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\cdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}\right\} \qquad (**)$$

and the remaining $m - n$ equations vanish identically for all the sets $(x_1, x_2, \ldots, x_n)$.

The equations $(*)$ will have a solution, though as $r = n$ of the unknowns may be chosen arbitrarily. The solution is of the unique one when $r = n$. Hence the equations $(*)$ are consistent.

2. Rank of A ($= r$, say) = rank of $\tilde{A}$ (= n) and the rank of A ($= r$) $<$ the number of the unknowns ($= n$). In this case, the equations $(*)$ will reduce, by row/column operations, to

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1r}x_r &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2r}x_r &= b_2 \\ &\cdots \\ a_{r1}x_1 + a_{r2}x_2 + \cdots + a_{rr}x_r &= b_r \\ 0 \cdot x_1 + 0 \cdot x_2 + \cdots + 0 \cdot x_r &= b_{r+1} \\ &\cdots \\ 0 \cdot x_1 + 0 \cdot x_2 + \cdots + 0 \cdot x_r &= b_n \end{aligned}\right\}$$

and the remaining $n - r + 1$ equations are of the form $0 \cdot x_1 + 0 \cdot x_2 + \cdots + 0 \cdot x_r = 0$.

If $b_{r+1}, b_{r+2}, \ldots, b_n$ are $= 0$, the system can be satisfied by one set of values for the unknowns. Hence the equations $(*)$ are consistent.

(b) Procedure to test the consistency of a system of equations in n unknowns:

Find the ranks of the coefficient matrix A and the augmented matrix $\tilde{A}$. If rank of A = rank of $\tilde{A}$ = n, the system is consistent and has a unique solution. If rank of A = rank of $\tilde{A}$ $< n$, the system is consistent and has infinitely many solutions.

(i) If $r \neq r'$, the equations are inconsistent, i.e., there is no solution.

(ii) If $r = r' = n$, the equations are consistent and there is a unique solution.

(iii) If $r = r' < n$, the equations are consistent and there are infinite number of solutions. Giving arbitrary values to $n - r$ of the unknowns, we can express the other r unknowns in terms of these.]

Example 3.31. Test for consistency of the system
$3x + 2y + 7z = 4$, $3x + 2y - z = 6$, $2x + y + 2z = 3$.
(Bhopal 2006; A.U.T.U. 2006; U.P.T.U. 2005)

Solution. We have
$$\begin{bmatrix} 3 & 2 & 7 \\ 3 & 2 & -1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}$$

Operate $R_2 - R_1$,
$$\begin{bmatrix} 3 & 2 & 7 \\ 0 & 0 & -8 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix}$$

Operate $R_3 - R_1$,
$$\begin{bmatrix} 3 & 2 & 7 \\ 0 & 0 & -8 \\ -1 & -1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -1 \end{bmatrix}$$

Operate $-\frac{1}{8}R_2$, $-3R_3$, $-\frac{1}{3}R_1$
$$\begin{bmatrix} 1 & \frac{2}{3} & \frac{7}{3} \\ 0 & 0 & 1 \\ 1 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{4} \\ 3 \end{bmatrix}$$

Operate $R_3 - R_1$, $R_2 \leftrightarrow R_3$, $\frac{1}{3}R_1$
$$\begin{bmatrix} 1 & \frac{2}{3} & \frac{7}{3} \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{4}{3} \\ 3 \\ -\frac{1}{4} \end{bmatrix}$$

The ranks of coefficient matrix and augmented matrix for the last set of equations are both 3. Hence the equations are consistent. Also the given system is equivalent to
$$3x + 2y + 7z = 4, \quad y + 5z = 3, \quad z = -\frac{1}{4}$$
$$\Rightarrow \quad y = \frac{3}{13} + \frac{1}{13} \quad \text{and} \quad x = \frac{2}{11} - \frac{16}{11}r$$
where r is a parameter.

Hence $x = \frac{7}{11}$, $y = \frac{5}{11}$ and $z = -\frac{1}{4}$ is a particular solution.

Thus in the above solution the coefficient matrix is reduced to an upper triangular matrix by row transformations.

Example 3.32. *Investigate the values of λ and μ so that the equations*
$2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + \lambda z = \mu$,
have (i) no solution, (ii) a unique solution and (iii) an infinite number of solutions.
(Kurukshetra, 2002; U.P.T.U. 2002)

Solution. We have
$$\begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix}$$

The system admits of a unique solution if, and only if, the coefficient matrix is of rank 3. This requires that
$$\begin{vmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{vmatrix} = 3(2\lambda - 4) \neq 0$$

Thus for a unique solution $\lambda \neq 6$ and μ may have any value. If $\lambda = 6$, the system will have no solution for those values of μ for which the matrices
$$A = \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 2 & 3 & 6 & \mu \end{bmatrix}$$

any one of the unknowns. But it is trivial to find it by substituting $x = 0$. Then $0 + 0 = 0$ implies $y = 0$. Then $0 + 0 = 0$ implies $z = 0$. Hence the system has only the trivial solution.

If $k = 2$ and $x = 0$, the system will have an infinite number of solutions.

Example 10.20 Show that the following system of homogeneous equations has a non-trivial solution, and find all such solutions.

$$x + 2y + 3z = 0, \quad 2x + 3y + z = 0, \quad 3x + y + 2z = 0$$

Solution. The system can be written in matrix form as

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Interchange R_1 with R_2, then R_2 with R_3:

$$\begin{bmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Interchange R_1 with R_2:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Clearly, rank of the coefficient matrix is 3 and the rank of augmented matrix is also 3. Hence the given equations are consistent. But the rank $3 < 4$, the number of unknowns.

∴ The number of parameters $n - r = 3 - 3 = 0$.

Thus the equations have infinite number of solutions. Now putting $z = k_1$ and $z = k_2$ in

$$x + 2y + 3z = 0 \quad \text{and} \quad 3x + y + 2z = 0$$

we get $x + 2y + 3k_1 = 0$ and $3x + y + 2k_1 = 0$

Whence $x = k_1, \; y = -k_1$

and $x = 0 + k_2 + 0 = k_2$

$$z = 0 + 0 + k_2 = k_2$$

$$y = 0 + 0 - k_2 = -k_2$$

∴ $x = k_2, \; y = -k_2, \; z = k_2$

(4) System of linear homogeneous equations Consider the homogeneous linear equations

$$\begin{aligned} a_{11}x + a_{12}y + \ldots + a_{1n}x_n &= 0 \\ a_{21}x + a_{22}y + \ldots + a_{2n}x_n &= 0 \\ \cdots\cdots\cdots\

Example 2.28. Solve the equations
(i) $x + 2y + 3z = 0$, $2x + 4y + 4z = 0$, $7x + 10y + 12z = 0$
(ii) $4x + 2y + z + 3w = 0$, $6x + 3y + 4z + 7w = 0$, $2x + y + z + w = 0$.

Solution. (i) Rank of the coefficient matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \\ 7 & 10 & 12 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 7 & 10 & 12 \end{bmatrix} \quad [\text{Operating } R_2 - 2R_1]$$

$$\sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -2 \\ 0 & 0 & 1 \end{bmatrix} \quad [\text{Operating } R_3 - 3R_1 - 2R_2]$$

$\therefore$ 3 which is the number of variables (i.e., $r = n$)

$\therefore$ The equations have only a trivial solution : $x = y = z = 0$.

(ii) Rank of the coefficient matrix

$$\begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 3/2 & 5/2 \\ 0 & 0 & -1/2 & -1/2 \end{bmatrix} \quad [\text{Operating } R_2 - \tfrac{3}{2}R_1, R_3 - \tfrac{1}{2}R_1]$$

$$\sim \begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 3/2 & 5/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad [\text{Operating } R_3 + \tfrac{1}{3}R_2]$$

is 2 which is the number of variable (i.e., $r < n$)

$\therefore$ Number of independent solutions $= n - r = 4 - 2 = 2$. Given system is equivalent to

$$4x + 2y + z + 3w = 0, \quad 3z + 5w = 0$$

$\therefore$ We have $z = -w$ and $y = -2x - w$

which gives an infinite number of non-trivial solutions, x and w being the parameters.

Example 2.29. Find the values of k for which the system of equations $(3k - 8)x + 3y + 3z = 0$, $3x + (3k - 8)y + 3z = 0$, $3x + 3y + (3k - 8)z = 0$ has a non-trivial solution. [UPTU (CO), 2005]

Solution. For the given system of equations to have a non-trivial solution, the determinant of the coefficient matrix should be zero.

$$\text{i.e., } \begin{vmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} 3k-2 & 3 & 3 \\ 3k-2 & 3k-8 & 3 \\ 3k-2 & 3 & 3k-8 \end{vmatrix} = 0 \quad [\text{Operating } C_1 + C_2 + C_3]$$

$$\text{or } (3k-2)\begin{vmatrix} 1 & 3 & 3 \\ 1 & 3k-8 & 3 \\ 1 & 3 & 3k-8 \end{vmatrix} = 0 \text{ or } (3k-2)\begin{vmatrix} 1 & 3 & 3 \\ 0 & 3k-11 & 0 \\ 0 & 0 & 3k-11 \end{vmatrix} = 0 \quad [\text{Operating } R_2 - R_1, R_3 - R_1]$$

$$\text{or } (3k-2)(3k-11)^2 = 0 \text{ where } k = 2/3, 11/3, 11/3.$$

Example 2.30. *If the following system of equations has a non-trivial solution, prove that* $a + b + c = 0$ *or* $a = b = c$.
$ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$. [Meerut U., 2006]

Solution. For the given system of equations to have non-trivial solutions, the determinant of the coefficient matrix is zero.

$$\text{i.e., } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \text{ or } \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \quad [\text{Operating } R_1 + R_2 + R_3]$$

$$\text{or } (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \text{ or } (a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ b & c-b & a-b \\ c & a-c & b-c \end{vmatrix} = 0 \quad [\text{Operating } C_2 - C_1, C_3 - C_1]$$

or $(a+b+c)(2c-b+b)(2b-c-a)-(a-c)(a-b)(b-c)=0$

or $(a+b+c)(-c^2-b^2-a^2+ab+bc+ca)=0$

i.e., $a+b+c=0$ or $a^2+b^2+c^2-ab-bc-ca=0$

or $a+b+c=0$ or $\frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]=0$

or $a+b+c=0; a=b, b=c, c=a.$

Hence the given system has a non-trivial solution if $a+b+c=0$ or $a=b=c$.

Example 3.20. Find the values of λ for which the equations

$$(\lambda-2)x+(3\lambda+1)y+3\lambda z=0$$
$$(\lambda-3)x+(4\lambda-2)y+(\lambda+3)z=0$$
$$3x+(3\lambda+1)y+(3\lambda-1)z=0$$

are consistent, and find the ratios of $x:y:z$ when λ has the smallest of these values. What is special about those values of λ and the solutions? (Mysore Univ. 2008; Delhi, 2010)

Solution. The given equations will have non-trivial solutions if

$$\begin{vmatrix} \lambda-2 & 3\lambda+1 & 3\lambda \\ \lambda-3 & 4\lambda-2 & \lambda+3 \\ 3 & 3\lambda+1 & 3\lambda-1 \end{vmatrix}=0 \qquad [\text{Operate } R_2-R_1]$$

or if $\begin{vmatrix} \lambda-2 & 3\lambda+1 & 3\lambda \\ 1 & \lambda-2 & 3-\lambda \\ 3 & 3\lambda+1 & 3\lambda-1 \end{vmatrix}=0 \qquad [\text{Operate } C_2+C_3]$

or if $\begin{vmatrix} \lambda-2 & 3\lambda+1 & 3\lambda \\ 1 & \lambda-2 & 0 \\ 3 & 3\lambda+1 & 6\lambda-1 \end{vmatrix}=0 \qquad [\text{Expand by } R_2]$

or if $(\lambda-2)\begin{vmatrix} \lambda-2 & 3\lambda \\ 3\lambda+1 & 6\lambda-1 \end{vmatrix}+3\begin{vmatrix} 3\lambda+1 & 3\lambda \\ \lambda-2 & 3\lambda \end{vmatrix}=0$ or $(\lambda-2)[6\lambda^2-13\lambda+2]+3[3\lambda+1]=0$

or if $6\lambda^3-25\lambda^2+30\lambda=0$ or if $\lambda=0$ or $\frac{5}{2}$ or $\frac{6}{3}$.

(a) When $\lambda=0$, the equations become $-2x+3y=0$ ...(1)

$$-x-2y+3z=0 \quad ...(2)$$
$$3y+x-2z=0 \quad ...(3)$$

Solving (1) and (3), we get $\frac{x}{6-9}=\frac{y}{9-2}=\frac{z}{-4+3}$. Hence $x:y:z=$

(b) When $\lambda=\frac{5}{2}$, the equations become identical.

EXERCISE 3.2

1. Investigate for what values of λ and μ the simultaneous equations $x+2y+z=7$, $x+2y+3z=\mu$, $x+3y+5z=9$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

(Kerala, 2009; Kolkata, 2009; V.T.U., 2010)

2. Investigate for what values of λ and μ the simultaneous equations $x+2y+z=3$, $x+2y+3z=\lambda$, $x+3y+5z=\mu$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

(Mumbai, 2007; U.P.T.U., 2009; Kolkata, 2010)

3. Test for consistency and solve:

(i) $3x-2y+7z=8$, $2x+y-3z=5$, $x-3y+10z=3$ (Nagpur, 2006; Anna University, 2010; Nagpur, 2010)

(ii) $x+2y+z=3$, $2x+3y+2z=5$, $3x+5y+4z=8$, $3x+8y+5z=13$ (Kerala, 2007; Madras, 2010)

(iii) $2x+6y+11=0$, $6x+20y-6z+3=0$, $6y-18z+1=0$ (Bangalore, 2008)

(iv) $3x+3y+3z=1$, $x+2y+3z=4$, $10y+3z=11$ (U.P.T.U., 2010; Nagpur, 2010)

2. Find the values of λ and μ for which the system of equations
$$x + y + z = 6,\ x + 2y + 3z = 10,\ x + 2y + \lambda z = \mu$$
are consistent. When will these equations have a unique solution? (Pune-B.Arch., 2003; Madras, 2002)

3. Show that $\lambda = 6$ is the integral value of λ for which the equations
$$3x - y + \lambda z = 3,\ 2x + y + 2z = 2,\ \lambda x - y + 3z = 3$$
have a unique solution. If $\lambda = 6$, show that the equations are consistent. Determine the solutions in each case.

4. Show that the equations
$$3x + 4y + 5z = a,\ 4x + 5y + 6z = b,\ 5x + 6y + 7z = c$$
do not have a solution unless $a + c = 2b$. (Madras, 2004; Kerala, 2002)

5. Prove that the equations $3x + 2y + 2z = 12$, $2x + 4y + 5z = 2$, $5x + 2y + 4z = \lambda$ are consistent for values $\lambda = 34$, and in that case the equations are satisfied by $x = k + 3$, $y = 2 - 4k$, $z = k$, where k is any arbitrary constant.

6. Find the values of k for which the equations $(k - 3)x + (k + 1)y + 2z = 0$, $(k - 3)x + (k - 2)y + z = 0$, $2x + y + (k - 1)z = 0$ are consistent, and find the ratios of $x : y : z$ when k has the smallest of these values. (Kerala, 2003)

7. Show that there are three real values of λ for which the equations $(\lambda - 1)x + \lambda y + 3z = 0$, $2\lambda x - (\lambda - 1)y + 2z = 0$, $(\lambda + 1)x + (\lambda - 1)y + 2z = 0$ are consistent, and find the ratios of $x : y : z$ when λ has the greatest of these values. (A.M.I.E.T.E., 2003)

8. Show that the system of equations $2x_1 - 2x_2 + x_3 = \lambda x_1$, $2x_1 - 3x_2 + 2x_3 = \lambda x_2$, $-x_1 + 2x_2 = \lambda x_3$ has a non-trivial solution only if $\lambda = 1, 4, -3$. Find the general solution in each case.

9. Determine the values of λ for which the following set of equations may possess non-trivial solutions:
$$3x_1 + x_2 - 2x_3 = \lambda x_1,\ x_1 - 2x_2 + 5x_3 = \lambda x_2,\ 2x_1 + 4x_2 + x_3 = \lambda x_3$$
For each permissible value of λ, determine the general solution. (Bangalore, 2005)

10. Solve completely the system of equations
$$x + y + z = 6,\ x + 2y + 3z = 10,\ x + 2y + \lambda z = \mu$$
$$4x + 6y + 5z = \lambda,\ 3x + 5y + 4z = \mu,\ 2x + 4y + 3z = 8,\ 2x + 3y + 2z = 5,\ 3x + 5y + 4z = \mu,\ 4x + 7y + \lambda z = 13$$
(A.M.I.E.T.E., 2000 S)

3.13 [3] LINEAR TRANSFORMATIONS

Let (x, y) be the coordinates of a point P referred to a set of rectangular axes OX, OY. Then its coordinates (x', y') referred to OX', OY', obtained by rotating the former axes through an angle θ are given by

$$\left.\begin{aligned} x' &= x\cos\theta + y\sin\theta, \\ y' &= -x\sin\theta + y\cos\theta \end{aligned}\right\} \quad \ldots(1)$$

A more general transformation than (1) is

$$\left.\begin{aligned} x' &= a_1x + b_1y, \\ y' &= a_2x + b_2y \end{aligned}\right\} \quad \ldots(2)$$

which in matrix notation is $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$

Such transformations as (1) and (2), are called *linear transformations in two dimensions*.

Similarly, the relations of the type
$$\left.\begin{aligned} x' &= a_1x + b_1y + c_1z, \\ y' &= a_2x + b_2y + c_2z, \\ z' &= a_3x + b_3y + c_3z \end{aligned}\right\} \quad \ldots(3)$$
give a linear transformation from (x, y, z) to (x', y', z') in three dimensional problems.

In general, the relation $Y = AX$ where $Y = \begin{bmatrix} y_1 \\ y_2 \\ \cdots \\ y_n \end{bmatrix}$, $A = \begin{bmatrix} a_1 & b_1 & c_1 & \cdots & k_1 \\ a_2 & b_2 & c_2 & \cdots & k_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ a_n & b_n & c_n & \cdots & k_n \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix}$ $\quad \ldots(4)$

given linear transformation from variables $x_1, x_2, \ldots, x_n$ to the variables $y_1, y_2, \ldots, y_n$ i.e. the transformation of the vector X to the vector Y.

This transformation is called linear because the linear relations $A(X_1 + X_2) = AX_1 + AX_2$ and $A(kX) = kAX$ hold for this transformation.

If the transformation matrix A is singular, the transformation also is said to be singular otherwise non-singular. For a non-singular transformation $Y = AX$, we can also write the inverse transformation $X = A^{-1}Y$. A non-singular transformation is also called a regular transformation.

Ex. The transformation from (x_1, x_2, x_3) to (y_1, y_2, y_3) is given by $Y = AX$ and another transformation from (y_1, y_2, y_3) to (z_1, z_2, z_3) is given by $Z = BY$. Find the transformation from (x_1, x_2, x_3) to (z_1, z_2, z_3) is given by

$$X = BY = BAX = CX$$

(3) Orthogonal transformation. The linear transformation (1), i.e. $Y = AX$, is said to be orthogonal if it transforms

$$y_1^2 + y_2^2 + \ldots + y_n^2 \text{ into } x_1^2 + x_2^2 + \ldots + x_n^2$$

The matrix of an orthogonal transformation is called an orthogonal matrix.

We have $$XX' = [x_1\ x_2 \ldots x_n] \begin{bmatrix} x_1 \\ x_2 \\ \ldots \\ x_n \end{bmatrix} = x_1^2 + x_2^2 + \ldots + x_n^2$$

and similarly, $$YY' = y_1^2 + y_2^2 + \ldots + y_n^2$$

∴ If $Y = AX$ is an orthogonal transformation, then

$$X'X = Y'Y = (AX)'(AX) = X'A'AX \text{ which is possible only if } A'A = I$$

But $A^{-1}A = I$, therefore $A' = A^{-1}$ for an orthogonal transformation.

Hence a square matrix A is said to be orthogonal if $AA' = A'A = I$.

Obs. 1. If A is orthogonal, A' and A^{-1} are also orthogonal.

Since A is orthogonal, $AA' = I$

∴ $|AA'| = |I| = 1$ i.e., $|A||A'| = 1$ or $|A||A| = 1$ or $|A|^2 = 1$ ∴ $|A| = \pm 1$

Hence if A is orthogonal, so is $A' = A^{-1}$, A' is also orthogonal.

Obs. 2. If A is orthogonal, then $|A| = \pm 1$.

Proof $AA' = A'A = I$ $\quad |A||A'| = |I| = 1$ (Obviously)

But $|A| = |A'|$ $\quad \therefore |A||A| = 1$

or $|A|^2 = 1$ $\quad \therefore |A| = \pm 1$.

Example 3.38. Show that the transformation

$$y_1 = \tfrac{2}{3}x_1 + \tfrac{1}{3}x_2 + \tfrac{2}{3}x_3,\ y_2 = \tfrac{1}{3}x_1 + \tfrac{2}{3}x_2 - \tfrac{2}{3}x_3,\ y_3 = \tfrac{2}{3}x_1 - \tfrac{2}{3}x_2 + \tfrac{1}{3}x_3$$

is regular. Write down the inverse transformation.

Solution. The given transformation may be written as

$$Y = AX$$

where $$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},\ Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix},\ A = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Now $$|A| = \begin{vmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{vmatrix} = -1$$

Thus the matrix A is non-singular and hence the transformation is regular.

∴ The inverse transformation is given by

$$X = A^{-1}Y$$

where $$A^{-1} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{bmatrix}$$

Thus $x_1 = \frac{2}{3}y_1 - \frac{2}{3}y_2 - \frac{1}{3}y_3$; $x_2 = -\frac{2}{3}y_1 + \frac{1}{3}y_2 + \frac{2}{3}y_3$; $x_3 = \frac{1}{3}y_1 - \frac{2}{3}y_2 - \frac{2}{3}y_3$
is the inverse transformation.

Example 3.28. Prove that the following matrix is orthogonal

$$\begin{bmatrix} -\frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{bmatrix}$$ (Kerala, 2006)

Solution. We know $AA^T = \begin{bmatrix} -\frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$

$$= \begin{bmatrix} \frac{4}{9}+\frac{1}{9}+\frac{4}{9} & -\frac{4}{9}+\frac{2}{9}+\frac{2}{9} & -\frac{2}{9}-\frac{2}{9}+\frac{4}{9} \\ -\frac{4}{9}+\frac{2}{9}+\frac{2}{9} & \frac{4}{9}+\frac{4}{9}+\frac{1}{9} & \frac{2}{9}-\frac{4}{9}+\frac{2}{9} \\ -\frac{2}{9}-\frac{2}{9}+\frac{4}{9} & \frac{2}{9}-\frac{4}{9}+\frac{2}{9} & \frac{1}{9}+\frac{4}{9}+\frac{4}{9} \end{bmatrix} = I.$$

Hence the matrix is orthogonal.

Example 3.29. If $A = \frac{1}{3}\begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix}$ is an orthogonal, find a, b, c and A^{-1}. (Mumbai, 2000)

Solution. As A is orthogonal, $AA^T = I$

$$\therefore \quad \frac{1}{3}\begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix} \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & b & c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

or $$\begin{bmatrix} 1+4+a^2 & 2+2+ab & 2-4+ac \\ 2+2+ab & 4+1+b^2 & 4-2+bc \\ 2-4+ac & 4-2+bc & 4+4+c^2 \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$\therefore \quad 5+a^2=9, 0+ab=0, 0+ac=0, 5+b^2=9, 0+bc=0, 8+c^2=9. \quad \text{i.e., } a^2=4, b^2=4, c^2=1$$

Thus $a = 2, b = 2, c = 2$

Since A is orthogonal, $A^{-1} = A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & b & c \end{bmatrix}$.

3.32 [I] VECTORS

Any quantity having components is called a vector or matrix. Therefore, the coefficients in a linear equation or the elements in a row or column matrix will form a vector. Thus any n numbers $x_1, x_2, \ldots, x_n$ written in a particular order, constitute a vector x.

(1) Linear dependence. The vectors $x_1, x_2, \ldots, x_n$ are said to be **linearly dependent**, if there exist numbers $\lambda_1, \lambda_2, \ldots, \lambda_n$ not all zero, such that

$$\lambda_1 x_1 + \lambda_2 x_2 + \ldots + \lambda_n x_n = 0. \qquad \ldots(1)$$

If no such numbers, other than zero, exist, the vectors are said to be linearly independent. If $\lambda_n \neq 0$, transposing $\lambda_n x_n$ to the other side and dividing by $-\lambda_n$, we write (1) in the form

$$x_n = k_1 x_1 + k_2 x_2 + \ldots + k_{n-1} x_{n-1}.$$

Then the vector x_n is said to be a linear combination of the vectors $x_1, x_2, \ldots, x_{n-1}$.

Example 3.30. Are the vectors $x_1 = (1, 1, 1)$, $x_2 = (1, 2, 3)$ and $x_3 = (6, 4, 2)$ linearly dependent? If so, express each of them as a linear combination of the others.

Solution. The relation $\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 = 0$,

i.e., $\lambda_1(1, 1, 1) + \lambda_2(1, 2, 3) + \lambda_3(6, 4, 2) = (0, 0, 0)$

is equivalent to $\lambda_1 + 2\lambda_2 + 3\lambda_3 = 0, 2\lambda_1 - \lambda_2 - \lambda_3 = 0,$
$4\lambda_1 + 3\lambda_2 + 5\lambda_3 = 0, 3\lambda_1 + 2\lambda_2 + 3\lambda_3 = 0$

As these are satisfied by the values $\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = -2$ which are not zero, the given vectors are linearly dependent. Also we have the relation,

$$x_1 + x_2 - 2x_3 = 0$$

by means of which any of the given vectors can be expressed as a linear combination of the others.

Obs. Applying Gram-Schmidt process to the vectors x_1, x_2, x_3 we get the matrices

$$A = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} x_1 \\ x_2 \\ x_3 - \frac{1}{2}x_1 - \frac{1}{2}x_2 \end{bmatrix}$$

Now the rank of A being 2, the rank of B is also 2. However x_1, x_2 are linearly independent and x_3 can be expressed as a linear combination of x_1 and x_2 $\left[i.e., x_3 = \frac{1}{2}(x_1 + x_2)\right]$. Similar results will hold for column vectors and for any matrix. In general, we have the following results :

If in a given matrix any r rows (or columns) are linearly independent but any $r+1$ rows (or columns) are linearly dependent, then the rank of the matrix is r. Conversely, if the rank of a matrix is r, then it has at least one set of r rows (or columns) which are linearly independent but any set of $r+1$ rows (or columns) are linearly dependent.

PROBLEMS 3.4

1. Determine the rank of the transformation

$$y_1 = 3x_1 + 2x_2 + x_3 + 4x_4, \; y_2 = x_1 - x_2 + 2x_3 + x_4, \; y_3 = 4x_1 + x_2 + 3x_3 + 5x_4$$

By the use of matrices find the inverse transformation which expresses x_1, x_2, x_3, x_4 in terms of y_1, y_2, y_3, y_4.

2. If A is a non-singular matrix, prove that the rank of A and that of its transpose A' are the same. Verify this result for the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$.

3. A transformation from the variables x_1, x_2, x_3 to y_1, y_2, y_3 is given by $Y = AX$ and another transformation from y_1, y_2, y_3 to z_1, z_2, z_3 is given by $Z = BY$, where

$$A = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & \lambda \\ 1 & \lambda & \lambda \end{bmatrix}$$ Obtain the transformation from x_1, x_2, x_3 to z_1, z_2, z_3.

4. Find the inverse transformation of $y_1 = x_1 + 2x_2 + 4x_3, \; y_2 = 2x_1 + 4x_2 + 11x_3, \; y_3 = x_1 + 3x_2 + 2x_3$.

5. Verify that the following matrix is orthogonal :

(i) $\frac{1}{15}\begin{bmatrix} 10 & 10 & 5 \\ 2 & 1 & -10 \\ 2 & -10 & 5 \end{bmatrix}$ [Madras 2002 (S); Anna 2003] (ii) $\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$ [Kerala (Engg.) 2002]

6. For what values of a, b, c if $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal? [Nagpur 2002 (S)]

7. Prove that $\begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{2}{3} & \frac{1}{3} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{2}{3} & \frac{2}{3} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$ is orthogonal. [Hint : $a = 2/3, b = 2/3, c = 1/\sqrt{2}$]

8. If A and B are orthogonal matrices, prove that AB is also orthogonal. [Anna 2003]

9. Are the following vectors linearly dependent. If so, find the relation between them :
(i) $(1, 1, 1), (1, 0, 3), (1, 4, -1)$ [Madras 2000]
(ii) $(1, 1, 1), (1, 2, 3), (2, 3, 4)$
(iii) $x_1 = (1, 2, 1), x_2 = (2, 3, 5), x_3 = (1, 3, 3), x_4 = (3, 7, 8)$ [U.P.T.U. 2003; Nagpur 2003]

3.5 (1) EIGEN VALUES

If A is any square matrix of order n, we can form the matrix $A - \lambda I$, where I is the unit matrix of the same order as A. The determinant of this matrix equated to zero,

i.e., $$|A - \lambda I| = \begin{vmatrix} a_{11}-\lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22}-\lambda & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn}-\lambda \end{vmatrix} = 0$$

is called the characteristic equation of A. On expanding the determinant, the characteristic equation takes the form

$$(-1)^n \lambda^n + k_1\lambda^{n-1} + k_2\lambda^{n-2} + \ldots + k_n = 0,$$

where k's are expressible in terms of the elements a_{ij}. The roots of this equation are called the eigenvalues or latent roots or characteristic roots of the matrix A.

(2) Eigen vectors

If $X = \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix}$ and $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$, then the linear transformation $Y = AX$ ...(1)

carries the column vector X into the column vector Y by means of the square matrix A. In practice, it is often required to find such vectors which transform into themselves or to a scalar multiple of themselves.

Let X be such a vector which transforms into λX by means of the transformation (1).

Then $$AX = \lambda X \quad \text{or} \quad AX - \lambda IX = 0 \quad \text{or} \quad [A - \lambda I]X = 0 \qquad \ldots(2)$$

This matrix equation represents a homogeneous linear equations

$$\left.\begin{aligned} (a_{11}-\lambda)x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= 0 \\ a_{21}x_1 + (a_{22}-\lambda)x_2 + \ldots + a_{2n}x_n &= 0 \\ \cdots\cdots\cdots\

Corresponding to $\lambda = 1$, we have $\begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 0$

which gives only one independent equation $x + y = 0$.

$\therefore \quad \frac{x}{1} = \frac{y}{-1}$ giving the eigen vector $(1, -1)$.

Example 2.43. Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$. (Bharati, 2004; Kerala, 2003)

Solution. The characteristic equation is $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 3-\lambda & 4 \\ 3 & 4 & 5-\lambda \end{vmatrix} = 0$, i.e., $\lambda^3 - 9\lambda^2 + 24\lambda = 0$

Since $\lambda = -2$ satisfies it, we can write this equation as

$$\lambda(\lambda + 2)(\lambda^2 - 11\lambda + 12) = 0 \quad \text{or} \quad \lambda(\lambda + 2)(\lambda - 3)(\lambda - 8) = 0.$$

Thus the eigen values of A are $\lambda = -2, 3, 8$.

If x, y, z be the components of an eigen vector corresponding to the eigen value λ, we have

$$[A - \lambda I]X = \begin{bmatrix} 1-\lambda & 2 & 3 \\ 2 & 3-\lambda & 4 \\ 3 & 4 & 5-\lambda \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \qquad \ldots(1)$$

Putting $\lambda = -2$, we have $3x + y + 2z = 0$, $x + 5y + 4z = 0$, $3x + y + 7z = 0$.

The first and third equations being the same, we have from the first two

$$\frac{x}{-22} = \frac{y}{0} = \frac{z}{20} \quad \text{or} \quad \frac{x}{-1} = \frac{y}{0} = \frac{z}{1}$$

Hence the eigen vector is $(-1, 0, 1)$. Also every non-zero multiple of this vector is an eigen vector corresponding to $\lambda = -2$.

Similarly, the eigen vectors corresponding to $\lambda = 3$ and $\lambda = 8$ are the arbitrary non-zero multiples of the vectors $(1, -2, 1)$ and $(3, 2, 1)$ which are obtained from (1).

Hence the three eigen vectors may be taken as $(-1, 0, 1), (1, -2, 1), (3, 2, 1)$.

Example 2.44. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$. (U.P.T.U., 2005)

Solution. The characteristic equation is

$$|A - \lambda I| = 0, \text{ i.e., } \begin{bmatrix} 2-\lambda & 1 & 2 \\ 0 & 2-\lambda & 3 \\ 0 & 0 & 3-\lambda \end{bmatrix} = 0$$

or $$(2 - \lambda)(2 - \lambda)(3 - \lambda) = 0$$

Thus the eigen values of A are $2, 2, 3$.

If x, y, z be the components of an eigen vector corresponding to the eigen value λ, we have

$$[A - \lambda I]X = \begin{bmatrix} 2-\lambda & 1 & 2 \\ 0 & 2-\lambda & 3 \\ 0 & 0 & 3-\lambda \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

Putting $\lambda = 2$, we have $y + 2z = 0$, $3z = 0$, $z = 0$, i.e., $x + y = 0$ and $z = 0$.

$\therefore \quad \frac{x}{1} = \frac{y}{-1} = \frac{z}{0} = k_1$ (say)

Hence the eigen vector corresponding to $\lambda = 2$ is $k_1(1, -1, 0)$.

Putting $\lambda = 3$, we have $y + 2z = 0$, $-y + 3z = 0$, $2z = 0$, i.e., $y = 0$, $z = 0$.

$\therefore \quad \frac{x}{1} = \frac{y}{0} = \frac{z}{0} = k_2$

Hence the eigen vector corresponding to $\lambda = 1$ is $k_1(-1, 1, 0)$.

Similarly, the eigen vector corresponding to $\lambda = 0$ is $k_2(1, 1, 1)$.

4.11.11 PROPERTIES OF EIGEN VALUES

1. Any square matrix A and its transpose A^T have the same eigen values.

We know $|A - \lambda I| = |A^T - \lambda I^T| = |A^T - \lambda I|$ [illegible]

$\therefore$ [illegible] $|A - \lambda I| = 0$ is the same as $|A^T - \lambda I| = 0$

$\therefore$ A and A^T have the same eigen values.

2. The eigen values of a triangular matrix are just the diagonal elements of the matrix.

Let $A = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix}$ be an upper triangular matrix of order n.

Then $|A - \lambda I| = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda)$

$\therefore$ the roots of $|A - \lambda I| = 0$ are $\lambda = a_{11}, a_{22}, \ldots, a_{nn}$.

Hence the eigen values of a triangular matrix are just the diagonal elements of the matrix.

Note: The eigen vectors of a diagonal matrix are also the diagonal elements of the matrix.

3. The eigen values of an idempotent matrix are either zero or unity.

Let A be any idempotent matrix so that $A^2 = A$. If λ be the eigen value of A, then there exists a non zero vector X such that

$$AX = \lambda X \quad \ldots(1)$$

Premultiplying (1) by A, we get $A(AX) = A(\lambda X)$ or $A^2X = \lambda(AX)$ or $AX = \lambda^2 X$ [illegible] $\ldots(2)$

From (1) and (2), we get $\lambda X = \lambda^2 X$ or $(\lambda^2 - \lambda)X = 0$

or $\lambda^2 - \lambda = 0$ because $X \neq 0$ or $\lambda = 0$ or 1.

Hence the result.

4. The sum of the eigen values of a matrix is equal to the sum of the elements of the principal diagonal.

[illegible] property will be proved for a matrix of order 3. But the method will be applicable to any square matrix of order n.

Consider the square matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \ldots(3)$$

Then $|A - \lambda I| = \begin{bmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{bmatrix}$ [illegible]

$= -\lambda^3 + \lambda^2(a_{11} + a_{22} + a_{33}) + \ldots$ $\ldots(4)$

If $\lambda_1, \lambda_2, \lambda_3$ be the eigen values of A, then $|A - \lambda I| = (-1)^3(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)$

$= -\lambda^3 + \lambda^2(\lambda_1 + \lambda_2 + \lambda_3) + \ldots$ $\ldots(5)$

Equating the right hand sides of (4) and (5) and then comparing coefficients of λ^2, we get

$\lambda_1 + \lambda_2 + \lambda_3 = a_{11} + a_{22} + a_{33}$. Hence the result.

5. The product of the eigen values of a matrix A is equal to its determinant.

Put $\lambda = 0$ in $|A - \lambda I| = (-1)^3(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)$

We get $|A| = \lambda_1\lambda_2\lambda_3$

6. If λ is an eigen value of a matrix A, then $\frac{1}{\lambda}$ is an eigen value of A^{-1}.

If X be the eigen vector corresponding to λ, then $AX = \lambda X$ $\ldots(6)$

Pre-multiplying both sides by A^{-1}, we get $A^{-1}AX = A^{-1}\lambda X$

i.e., $$IX = \lambda A^{-1}X \quad \text{or} \quad X = \lambda A^{-1}X, \quad \text{i.e.,} \quad A^{-1}X = \frac{1}{\lambda}X$$

This being of the same form as (1), above that $\frac{1}{\lambda}$ is an eigen value of the inverse matrix A^{-1}.

VII. If X is an eigen vector of A corresponding to eigen value λ, then KX is also its eigen vector.

We know that λ is an eigen value of a matrix A, then $\frac{1}{\lambda}$ is an eigen value of A^{-1}. [Property VI]. Since A is an orthogonal matrix, A^{-1} is same as its transpose A'.

$\therefore$ $\frac{1}{\lambda}$ is an eigen value of A'.

But the matrices A and A' have the same eigen values, since the determinants $|A - \lambda I|$ and $|A' - \lambda I|$ are the same.

Hence $\frac{1}{\lambda}$ is also an eigen value of A.

VIII. If $\lambda_1, \lambda_2 \ldots \lambda_n$ are the eigen values of a matrix A, then A^m has the eigen values $\lambda_1^m, \lambda_2^m, \ldots, \lambda_n^m$ (m being a positive integer). (Rajalaxmi, 2005)

Let λ_i be the eigen value of A and X_i the corresponding eigen vector. Then

$$AX_i = \lambda_i X_i \qquad \ldots(1)$$

We have $A^2X_i = A(AX_i) = A(\lambda_i X_i) = \lambda_i(AX_i) = \lambda_i(\lambda_i X_i) = \lambda_i^2 X_i$

Similarly, $A^3X_i = \lambda_i^3 X_i$. In general, $A^mX_i = \lambda_i^m X_i$ which is of the same form as (1).

Hence λ_i^m is an eigen value of A^m.

The corresponding eigen vector is the same X_i.

2.15 CAYLEY-HAMILTON THEOREM*

Every square matrix satisfies its own characteristic equation. i.e., If the characteristic equation for the n^{th} order square matrix A is

$$|A - \lambda I| = (-1)^n\lambda^n + k_1\lambda^{n-1} + \ldots + k_{n-1}\lambda + k_n = 0 \qquad \ldots(1)$$

then $$(-1)^n A^n + k_1A^{n-1} + \ldots + k_{n-1}A + k_nI = 0$$

Let the adjoint of the matrix $A - \lambda I$ be P. Clearly, the elements of P will be polynomials of the $(n-1)^{th}$ degree in λ, for the cofactors of the elements in $|A - \lambda I|$ will be such polynomials.

$\therefore$ P can be split up into a number of matrices, containing terms with the same powers of λ, such that

$$P = P_1\lambda^{n-1} + P_2\lambda^{n-2} + \ldots + P_{n-1}\lambda + P_n \qquad \ldots(2)$$

where $P_1, P_2, \ldots P_n$ are all the square matrices of order n whose elements are functions of the elements of A.

Since the product of a matrix by its adjoint = determinant of the matrix $\times$ unit matrix.

$\therefore$ $$(A - \lambda I)P = |A - \lambda I| \times I$$

$\therefore$ by (1) and (2), $$(A - \lambda I)(P_1\lambda^{n-1} + P_2\lambda^{n-2} + \ldots + P_{n-1}\lambda + P_n)$$
$$= (-1)^n\lambda^n + k_1\lambda^{n-1} + \ldots + k_{n-1}\lambda + k_nI.$$

Equating the coefficients of various powers of λ, we get

$$-P_1 = (-1)^n I \qquad (\because AP_1 = P_1)$$
$$AP_1 - P_2 = k_1 I.$$
$$AP_2 - P_3 = k_2 I.$$
$$\ldots\ldots\ldots\ldots$$
$$AP_{n-1} - P_n = k_{n-1} I.$$
$$AP_n = k_n A$$

Now pre-multiplying the equations by $A^n, A^{n-1}, \ldots, A, I$ respectively and adding, we get

$$(-1)^n A^n + k_1A^{n-1} + \ldots + k_{n-1}A + k_nI = 0 \qquad \ldots(3)$$

for the terms on the left cancel in pairs. This proves the theorem.

Cor. Another method of finding the inverse.

Multiplying (3) by A^{-1}, we get

$$(-1)^n A^{n-1} + k_1A^{n-2} + \ldots + k_{n-1}I + k_nA^{-1} = 0$$

whence $$A^{-1} = -\frac{1}{k_n}\left[(-1)^n A^{n-1} + k_1A^{n-2} + \ldots + k_{n-1}I\right].$$

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*See footnote on p.17. William Rowan Hamilton (1805–1865) an Irish mathematician who is known for his work in dynamics.

This result gives us the inverse of A in terms of $n-1$ powers of A and is considered as a practical method for the computation of the inverse of the large matrices. As a by-product of the computation, the characteristic equation and the determinant of the matrix are also obtained.

Example 2.43. *Verify Cayley-Hamilton theorem for the matrix* $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ *and find its inverse.*

Also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 5I$ *as a linear polynomial in* A. (Nagpur, 2006)

Solution. The characteristic equation of A is

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^2 - 4\lambda - 1 = 0 \qquad \ldots(i)$$

By Cayley-Hamilton theorem, A must satisfy its characteristic equation (i), so that

$$A^2 - 4A - I = 0 \qquad \ldots(ii)$$

Now $$A^2 - 4A - I = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} - 4\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 8 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$$

This verifies the theorem.

Multiplying (ii) by A^{-1}, we get $A - 4I - 5A^{-1} = 0$

$$\Rightarrow \quad A^{-1} = \frac{1}{5}(A - 4I) = \frac{1}{5}\left\{\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} - 4\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right\} = \frac{1}{5}\begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$$

Now dividing the polynomial $\lambda^5 - 4\lambda^4 - 7\lambda^3 + 11\lambda^2 - \lambda - 10$ by the polynomial $\lambda^2 - 4\lambda - 1$, we obtain

$$\lambda^5 - 4\lambda^4 - 7\lambda^3 + 11\lambda^2 - \lambda - 10 = (\lambda^2 - 4\lambda - 1)(\lambda^3 + 2\lambda + 3) + 5$$

$$= 1 \cdot 5 + 0 \qquad \text{[By (ii)]}$$

Hence $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I = 0$, which is a linear polynomial in A.

Example 2.44. *Find the characteristic equation of the matrix* $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -1 & 4 \end{bmatrix}$ *and hence find its inverse.*

Solution. The characteristic equation is $\begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 3-\lambda & -3 \\ -2 & -1 & 4-\lambda \end{vmatrix} = 0$, *i.e.*, $\lambda^3 - 8\lambda^2 + 8\lambda + 8 = 0$.

By Cayley-Hamilton theorem, $A^3 - 8A^2 + 8A + 8I = 0$, which gives $A^{-1} = \frac{1}{8}I - \frac{1}{8}A^2$.

$$= \frac{1}{8}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{1}{8}\begin{bmatrix} -4 & -6 & -12 \\ 10 & 20 & 6 \\ 2 & 2 & 22 \end{bmatrix} = \begin{bmatrix} \frac{3}{8} & \frac{1}{4} & \frac{3}{2} \\ -\frac{5}{4} & -\frac{1}{4} & -\frac{3}{4} \\ -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \end{bmatrix} \qquad \text{[U.P. Tech. 2001]}$$

Example 2.45. *Find the characteristic equation of the matrix* $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ *and hence compute* A^{-1}.

(U.P.T.U., 2010)

Also find the matrix represented by

$A^5 - 5A^4 + 7A^3 - 3A^2 + A^2 - 5A^2 + 8A^2 - 2A - I$. (Anna, 2009; Kerala Univ. 2006; U.P.T.U., 2003)

Solution. The characteristic equation of the matrix A is

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0, \quad \text{i.e.} \quad \lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0.$$

According to Cayley-Hamilton theorem, we have $A^3 - 4A^2 + 3A - 2I = 0$ ...(1)

Multiplying (1) by A^{-1}, we get

$$A^2 - 4A + 3I - 2A^{-1} = 0 \quad \text{or} \quad A^{-1} = \frac{1}{2}A^2 - 2A + \frac{3}{2}I \qquad \text{...(2)}$$

But $$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4+0+1 & 2+1+1 & 2+0+2 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 2+0+2 & 1+1+2 & 1+0+4 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$$

$$\therefore \quad A^2 - 4A + 3I = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 4\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + 3\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 2 & 0 \\ -1 & -1 & 2 \end{bmatrix}$$

Hence from (2), $A^{-1} = \frac{1}{2}\begin{bmatrix} 2 & -1 & -1 \\ 0 & 2 & 0 \\ -1 & -1 & 2 \end{bmatrix}$

Now $$A^5 = AA^4 = A^3A^2 = (4A^2 - 3A + 2I)A = 4A^3 - 3A^2 + 2A = 4(4A^2 - 3A + 2I) - 3A^2 + 2A + I$$

$$= 13A^2 - 10A + 8I \qquad [\because A^3 = 4A^2 - 3A + 2I]$$

$$= 13\begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 10\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + 8\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 61 & 52 & 52 \\ 0 & 3 & 0 \\ 52 & 52 & 61 \end{bmatrix}$$

Problems 2.8

1. Find the sum and product of the eigen values of $\begin{bmatrix} 3 & 1 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 3 \end{bmatrix}$ (Madras 2002)

2. Find the eigen values and eigen vectors of the matrices

(a) $\begin{bmatrix} 5 & 3 \\ 2 & 5 \end{bmatrix}$ (K.U.T.C. 2000) (b) $\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$ (Anna 2004)

3. Find the latent roots and the latent vectors of the matrices

(a) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 1 & 5 \end{bmatrix}$ (Anna 2005; Nagarjuna 2000; K.U.T.C. 2000; J.N.T.U. 2004)

(b) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ (J.N.T.U. 2003; Kerala 2000)

(c) $\begin{bmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -1 & -4 & 5 \end{bmatrix}$ (K.U.T.C. 2000; K.U.T.C. 2002; J.N.T.U. 2005)

(d) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ (Nagarjuna 2000) (e) $\begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$ (Madras 2000)

4. (a) If λ is an eigen value of a non-singular matrix A, then show $\frac{|A|}{\lambda}$ is an eigen value of the matrix $\text{adj } A$. (J.N.T.U. 2003)

5. Find the eigen values and eigen vectors of $A^3 - 2A^2 + 3A$, if the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$ (Madras 2000)

6. Two eigen values of the matrix $A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 3 & 3 \\ 1 & -2 & 4 \end{bmatrix}$ are 5 and 6. Find the eigen values of A^{-1}.

7. Show that if $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the latent roots of a matrix A, then k^{th} of the latent roots of $(A - kI)^2$ are $(\lambda_1 - k)^2, (\lambda_2 - k)^2, \ldots, (\lambda_n - k)^2$. (J.N.T.U. 2006)

8. For a hypothetical square matrix, show that the column vectors corresponding to two principal roots values are orthogonal.

9. Using Cayley-Hamilton theorem, find the inverse of

(a) $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$ (Kurukshetra, 2003, 05)

(c) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -3 & -3 \\ 3 & -1 & -3 \end{bmatrix}$ (Nagpur, 2002) (d) $\begin{bmatrix} 2 & -1 & 0 \\ 1 & -2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$ (U.P.T.U., 2006)

10. Find the characteristic equation of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix}$. Hence find the equation whose roots are $\frac{1}{2}$ of the corresponding roots of the given matrix. (Nagpur, 2004; Anna, 2005; Kerala, 2005)

11. Verify Cayley-Hamilton theorem for the matrix A and find its inverse.

(a) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix}$ (Anna, 2004; S.V.U.C.E., 2004; Madras, 2004)

(b) $\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ (Kurukshetra, 2002) (c) $\begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 5 \\ 2 & 4 & 5 \end{bmatrix}$ (A.M.I.E., 2004)

12. Using Cayley-Hamilton theorem, find A^4, if $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$. (Anna, 2002)

13. If $A = \begin{bmatrix} 0 & -1 & 2 \\ -3 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$, find A^3. (Madras, 2002)

14. Using Cayley-Hamilton theorem, find A^{-1}, where $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$. (Nagpur, 2003)

15. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}$, evaluate A^{-1}, A^4 by using A.

16. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, show that $A^n = A^{n-2} + A^2 - I$. Hence find A^{50}. (Mumbai, 2000)

4.16 [I] REDUCTION TO DIAGONAL FORM

The square matrix A of order n has a linearly independent eigen vectors. Then a matrix P may be found such that $P^{-1}AP$ is a diagonal matrix.

[This result will be proved for a square matrix of order 3 but the method will be applicable for any square matrix of any order.]

Let A be a square matrix of order 3. Let $\lambda_1, \lambda_2, \lambda_3$ be its eigen values and

$$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} \text{ and } X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} \text{ be the corresponding eigen vectors.}$$

Denoting the square matrix $(X_1 X_2 X_3) = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$ by P, we have

$$AP = A[X_1 X_2 X_3] = [AX_1, AX_2, AX_3] = [\lambda_1 X_1, \lambda_2 X_2, \lambda_3 X_3]$$

$$= \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 \\ \lambda_1 y_1 & \lambda_2 y_2 & \lambda_3 y_3 \\ \lambda_1 z_1 & \lambda_2 z_2 & \lambda_3 z_3 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = PD, \text{ where } D \text{ is the diagonal matrix.}$$

$\therefore P^{-1}AP = P^{-1}PD = D$, which proves the theorem.

Def. 1. The matrix P which diagonalises A is called the modal matrix of A and the resulting diagonal matrix D is known as the spectral matrix of A.

2. The diagonal elements of the spectral matrix of A are the eigenvalues of A.

3. The matrix P is built from the eigenvalues of A considered in the same order as those of A.

(2) Similarity of matrices. A square matrix $\hat{A}$ of order n is said to be similar to a square matrix A of order n if $\hat{A} = P^{-1}AP$ for some non-singular square matrix P.

The transformation of a matrix A by a non-singular matrix P to $\hat{A}$ is called a similarity transformation.

Def. If the matrix $\hat{A}$ is similar to the matrix A, then $\hat{A}$ has the same eigenvalues as A.

If x is an eigenvector of A, then $y = P^{-1}x$ is an eigenvector of $\hat{A}$ corresponding to the same eigenvalue.

(3) Powers of a matrix. Diagonalisation of a matrix is quite useful for obtaining powers of a matrix.

Let A be the square matrix. Then a non-singular matrix P can be found such that

$$D = P^{-1}AP$$

$\therefore$ $D^2 = (P^{-1}AP)(P^{-1}AP) = P^{-1}A^2P$ $\quad$ [$\because P^{-1}P = I$]

Similarly, $D^3 = P^{-1}A^3P$ and in general, $D^n = P^{-1}A^nP$ $\quad$ ...(1)

To obtain A^n, premultiply (1) by P and postmultiply by P^{-1}.

Then $PDP^{-1} = PP^{-1}APP^{-1} = A^1$, which gives A^n.

Thus, if $D = PDP^{-1}$ is given, $D^n = \begin{bmatrix} \lambda_1^n & 0 & 0 \\ 0 & \lambda_2^n & 0 \\ 0 & 0 & \lambda_3^n \end{bmatrix}$

Working procedure:

1. Find the eigenvalues of the square matrix A.
2. Find the corresponding eigenvectors and write the modal matrix P.
3. Find the diagonal matrix D from $D = P^{-1}AP$.
4. Obtain A^n from $A^n = PD^nP^{-1}$.

Example 2.69. Reduce the matrix $A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$ to the diagonal form.

(U.P.T.U., 2001; U.P.T.U., 2015; Kerala, 2009; U.P.T.U., 2008)

Solution. The characteristic equation of A is

$$\begin{bmatrix} -1-\lambda & 2 & -2 \\ 1 & 0-\lambda & 1 \\ -1 & -1 & -\lambda \end{bmatrix} = 0 \quad \text{or} \quad \lambda^3 - \lambda^2 - 2\lambda + 2 = 0.$$

Solving, we get $\lambda_1 = 1, \lambda_2 = \sqrt{2}, \lambda_3 = -\sqrt{2}$ as the eigenvalues of A.

When $\lambda = 1$, the corresponding eigenvector is given by

$$-2x + 2y - 2z = 0, \; x + y + z = 0, \; -x - y - z = 0$$

Solving the first two equations, we get $\frac{x}{2} = \frac{y}{0} = \frac{z}{-2}$ giving the eigenvector $(1, 0, -1)$.

When $\lambda = \sqrt{2}$, the corresponding eigenvector is given by

$$(-1-\sqrt{2})x + 2y - 2z = 0, \; x + (2-\sqrt{2})y + z = 0, \; -x - y - \sqrt{2}z = 0.$$

Solving first and last equations, we get

$$x_1 = \frac{1}{3}x_2 - \frac{2}{3}x_3 \quad \text{and} \quad x_3 = \frac{1}{3}x_2 - \frac{2}{3}x_1$$

giving us the eigenvectors $(\beta, 3\beta, \beta)$.

Similarly, the eigenvectors corresponding to $\lambda = -\beta$, $\beta \neq 0$ are $(\beta, 0, -\beta)$.

Forming the three eigenvectors as the column vectors, we get the desired matrix P as

$$P = \begin{bmatrix} 1 & \beta & 1 \\ 0 & 3\beta & 0 \\ -1 & \beta & 1 \end{bmatrix}$$

Hence the diagonal matrix is

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & -\beta \end{bmatrix}$$

Example 3.108: Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ be a matrix. Find a matrix P such that $P^{-1}AP$ is a diagonal matrix.

Solution: The eigenvalues of A are $\lambda = 1, 1, 1$. To find the eigenvectors, we have $(A - \lambda I)X = 0$, where $X = (x_1, x_2, x_3)$. Solving, we get $x_1 = x_2 = x_3 = 0$. Hence, we do not have three linearly independent eigenvectors. Therefore, the matrix A cannot be diagonalized.

Example 3.109: Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ be a matrix. Find a matrix P such that $P^{-1}AP$ is a diagonal matrix.

Solution: The eigenvalues of A are $\lambda = 1, 1, 1$. To find the eigenvectors, we have $(A - \lambda I)X = 0$, where $X = (x_1, x_2, x_3)$. Solving, we get $x_2 = x_3 = 0$. Hence, we do not have three linearly independent eigenvectors. Therefore, the matrix A cannot be diagonalized.

Example 3.110: Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ be a matrix. Find a matrix P such that $P^{-1}AP$ is a diagonal matrix.

Solution: The eigenvalues of A are $\lambda = 1, 1, 1$. To find the eigenvectors, we have $(A - \lambda I)X = 0$, where $X = (x_1, x_2, x_3)$. Solving, we get $x_2 = x_3 = 0$. Hence, we do not have three linearly independent eigenvectors. Therefore, the matrix A cannot be diagonalized.

Example 3.111: Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ be a matrix. Find a matrix P such that $P^{-1}AP$ is a diagonal matrix.

Solution: The eigenvalues of A are $\lambda = 1, 1, 1$. To find the eigenvectors, we have $(A - \lambda I)X = 0$, where $X = (x_1, x_2, x_3)$. Solving, we get $x_2 = x_3 = 0$. Hence, we do not have three linearly independent eigenvectors. Therefore, the matrix A cannot be diagonalized.

Example 2.30. Find A^4 and A^n if $A = \begin{bmatrix} 5/2 & 3/2 \\ 3/2 & 5/2 \end{bmatrix}$. (Nov/Dec, 2005)

Solution. The characteristic equation of A is

$$\begin{vmatrix} 5/2 - \lambda & 3/2 \\ 3/2 & 5/2 - \lambda \end{vmatrix} = 0, \quad \text{i.e., } (5/2 - \lambda)^2 - (3/2)^2 = 0$$

$$\therefore \quad \lambda^2 - 5\lambda + 2 = 0, \quad \text{whence } \lambda = 1, 4$$

When $\lambda = 1$, $(A - \lambda I)X = 0$ gives

$$\begin{bmatrix} 3/2 & 3/2 \\ 3/2 & 3/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad [\text{By } 2R_1, 2R_2]$$

$$\text{or} \quad \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad [\text{By } R_2 - R_1]$$

$$\therefore \quad x_1 + x_2 = 0. \text{ If } x_2 = -2, x_1 = 2, \quad \text{i.e., the eigen vector is } [2, -2]'.$$

When $\lambda = 2$, $(A - 2I)X = 0$, gives $\begin{bmatrix} -1/2 & 3/2 \\ 3/2 & -1/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\text{or} \quad \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} \text{By } 2R_1 \\ 2R_2 \end{bmatrix}$$

$$\text{or} \quad \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad [\text{By } R_1 - R_2]$$

$$\therefore \quad -x_1 + x_2 = 0, \quad \text{i.e., } x_1 = x_2$$

If $x_2 = 1$, $x_1 = 1$, i.e., the eigen vector is $[1, 1]'$

Now $D = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$ and $P = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$

$$\therefore \quad P^{-1} = \frac{\text{adj } P}{|P|} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}$$

$$\text{If} \quad D(\lambda) = \lambda^4, D(1) = I^4 = \begin{bmatrix} 1^4 & 0 \\ 0 & 1^4 \end{bmatrix}$$

$$\therefore \quad A^4 = P[D(1)]P^{-1} = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1^4 & 0 \\ 0 & 4^4 \end{bmatrix} \frac{1}{4} \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 2 & 4^4 \\ -2 & 4^4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 2 + 4^4 & -2 + 4^4 \\ -2 + 4^4 & 2 + 4^4 \end{bmatrix}$$

Replacing 4 by λ, we get

$$A^n = \frac{1}{2} \begin{bmatrix} 2 + \lambda^n & -2 + \lambda^n \\ -2 + \lambda^n & 2 + \lambda^n \end{bmatrix}$$

2.12 REDUCTION OF QUADRATIC FORM TO CANONICAL FORM

A homogeneous expression of the second degree in any number of variables is called a **quadratic form**.

For instance, if $A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $X' = [x \ \ y \ \ z]$, then

$$X'AX = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy \qquad \ldots(1)$$

which is a quadratic form.

Let $\lambda_1, \lambda_2, \lambda_3$ be the eigen values of the matrix A and

$$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}, X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}, X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$$

be the corresponding eigen vectors in the normalised form (i.e., each element is divided by square root of sum of the squares of all the three elements in the eigen vector).

Then by § 15.10(1), $P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$ where $P = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$

Hence the quadratic form (1) is reduced to a canonical form (or sum of squares form or Principal axes form).

$$\lambda_1 p^2 + \lambda_2 q^2 + \lambda_3 r^2$$

and P is the matrix of transformation which is an orthogonal matrix.

Note. Congruent (or orthogonal) transformation. The diagonal matrix D and the matrix A are called congruent matrices and the above method of reduction is called congruent (or orthogonal) transformation.

Remember that the matrix A corresponding to the quadratic form

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$$

is $\begin{bmatrix} \text{coeff. of } x^2 & \frac{1}{2}\text{coeff. of } xy & \frac{1}{2}\text{coeff. of } zx \\ \frac{1}{2}\text{coeff. of } xy & \text{coeff. of } y^2 & \frac{1}{2}\text{coeff. of } yz \\ \frac{1}{2}\text{coeff. of } zx & \frac{1}{2}\text{coeff. of } yz & \text{coeff. of } z^2 \end{bmatrix}$, i.e., $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$.

Example 15.11. Reduce the quadratic form $3x^2 + 5y^2 + 3z^2 - 2yz + 2zx - 2xy$ to the canonical form and specify the matrix of transformation. (Nagpur 2006; Kerala (Elective) 2006)

Solution. The matrix of the given quadratic form is $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

Its characteristic equation is $|A - \lambda I| = 0$, i.e., $\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$

which gives $\lambda = 2, 3, 6$ as the eigen values. Hence the given quadratic form reduces to the canonical form

$$\lambda_1 p^2 + \lambda_2 q^2 + \lambda_3 r^2, \quad \text{i.e.,} \quad 2p^2 + 3q^2 + 6r^2.$$

To find the matrix of transformation:

From $[A - \lambda I]X = 0$, we obtain the equations

$$(3-\lambda)x - y + z = 0, \; -x + (5-\lambda)y - z = 0, \; x - y + (3-\lambda)z = 0.$$

Now corresponding to $\lambda = 2$, we get $x - y + z = 0$, $-x + 3y - z = 0$, and $x - y + z = 0$.

whence $\frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$

$\therefore$ The eigen vector is $X_1 (1, 0, -1)$ and its normalised form is $(1/\sqrt{2}, 0, -1/\sqrt{2})$.

Similarly, corresponding to $\lambda = 3$, the eigen vector is $X_2 (1, 1, 1)$ and its normalised form is $(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$.

Finally, corresponding to $\lambda = 6$, the eigen vector is $X_3 (1, -2, 1)$ and its normalised form is $(1/\sqrt{6}, -2/\sqrt{6}, 1/\sqrt{6})$.

Hence the matrix of transformation is $P = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \\ 0 & 1/\sqrt{3} & -2/\sqrt{6} \\ -1/\sqrt{2} & 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix}$.

[illegible]

[illegible]

Index. [illegible]

Signature. [illegible]

[illegible]

(a) [illegible]

(b) [illegible]

(c) [illegible]

(d) [illegible]

(e) [illegible]

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(d) **[illegible]** [illegible]

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Hence the modal matrix is

$$P = \begin{bmatrix} -1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix}$$

EXERCISE 3.3

1. (a) $A = \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $P = \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; show that $P^{-1}AP$ is a diagonal matrix.

(b) Show that the Gram transformation

$P = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, where $\theta = \frac{1}{2}\tan^{-1}\frac{2h}{a-b}$, reduces the matrix

$C = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$ to the diagonal form $D = PCP^T$.

2. Reduce the matrix $A = \begin{bmatrix} -15 & 7 \\ -42 & 16 \end{bmatrix}$ to the diagonal form. (U.T.T.C, 2005)

3. If $A = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}$, find $A^4 - 5A^2I$. (Mwenya, 2006)

4. If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, calculate A^{-1}. (Chibesakunda, 2005)

5. If $A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix}$, then prove that $3\tan A = A\tan 3$. (Mwenya, 2006)

6. Find the eigenvalues of the matrix $\begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$ and hence reduce $6x^2 + 3y^2 + 3z^2 - 4xy + 4xz - 4yz$ to the sum of squares. Also write the nature of the matrix. (Chisanga, 2006)

7. Reduce the quadratic form $2xy + 2yz + 2zx$ to canonical form. (Lungu, 2007; Chisakuta, 2005; Mwenya, 2006)

8. Find the eigenvalues, eigenvectors and the modal matrix of matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 0 \end{bmatrix}$.

(a) Reduce the quadratic form $x_1^2 + 2x_2^2 + 5x_3^2 - 2x_1x_2$ to a canonical form. (Mwale, 2009)

(b) Reduce the following quadratic forms into a form of xy and hence orthogonal transformation and give the matrix of transformation. Also write the nature of each of them.

(i) $2x_1^2 + 2x_2^2 + 3x_3^2 - 2x_1x_2 - 2x_1x_3 - 2x_2x_3$

(ii) $x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 - 2x_2x_3$ (Mwale, 2009 A)

11. Find the index and signature of the quadratic form $x_1^2 + 2x_2^2 - 2x_3^2$. (Mwale, 2009)

12. Find the nature of the quadratic form $x^2 + 2y^2 + z^2 + 2xz + 2yz + 4xy$. (Mwale, 2009)

13. Show that the form $2x_1^2 + 2x_2^2 + 3x_3^2 + 4x_1x_2 + 4x_1x_3 + 6x_2x_3$ is positive semi-definite and find a non-zero set of values of x_1, x_2, x_3 which makes the form zero. (U.T.T.C, 2005)

3.3.3 COMPLEX MATRICES

So far, we have considered matrices whose elements were real numbers. The elements of a matrix can, however, be complex numbers also.

(i) Conjugate of a matrix. If the elements of a matrix $A = [a_{ij}]$ are complex numbers $a_{ij} = (\alpha_{ij} + i\beta_{ij})$ and β_{ij} being real, then the matrix

$\bar{A} = [\bar{a}_{ij}]_{m \times n} = [\bar{a}_{ij}]$ is called the conjugate matrix of A.

The transpose of the conjugate of a matrix A is denoted by A^θ or A^*, i.e., $(A)^\theta = A^*$.

(2) Hermitian matrix. A square matrix A such that $A^\theta = A$ is said to be a **Hermitian matrix**[*]. The elements of the leading diagonal of a Hermitian matrix are evidently real, while every other element is the complex conjugate of the element in the transposed position. For instance $A = \begin{bmatrix} 2 & 3+4i \\ 3-4i & -5 \end{bmatrix}$ is a Hermitian matrix, since $A^\theta = \begin{bmatrix} 2 & 3-4i \\ 3+4i & -5 \end{bmatrix} = \bar{A}$

(3) Skew-Hermitian matrix. A square matrix A such that $A^\theta = -A$ is said to be a skew-Hermitian matrix. This implies that the leading diagonal elements of a skew-Hermitian matrix are either all zeros or all purely imaginary.

> **Obs.** A Hermitian matrix is a generalization of a real symmetric matrix as every real symmetric matrix is Hermitian. Similarly, a skew-Hermitian matrix is a generalization of a real skew-symmetric matrix.

Properties

I. Any square matrix A can be written as the sum of a Hermitian and skew-Hermitian matrices. (Mumbai, 2009)

Take $B = \frac{1}{2}(A + \bar{A}^\theta)$ and $C = \frac{1}{2}(A - \bar{A}^\theta)$

Then $B^\theta = \frac{1}{2}(A + \bar{A}^\theta)^\theta = \frac{1}{2}(A^\theta + \bar{A})$

and $\bar{B}^\theta = \frac{1}{2}\overline{(A^\theta + \bar{A})}^\theta = \frac{1}{2}(\bar{A}^\theta + A) = B$

i.e., B is a Hermitian matrix.

Again, $C^\theta = \frac{1}{2}(A - \bar{A}^\theta)^\theta = \frac{1}{2}(A^\theta - \bar{A})$

and $\bar{C}^\theta = \frac{1}{2}\overline{(A^\theta - \bar{A})}^\theta = \frac{1}{2}(A - \bar{A}^\theta) = C$

$\therefore \quad C^\theta = -C$, i.e., C is a skew-Hermitian matrix.

Thus, $A = \frac{1}{2}(A + \bar{A}^\theta) + \frac{1}{2}(A - \bar{A}^\theta) = B + C$

Hence the result.

II. If A is a Hermitian matrix, then iA is a skew-Hermitian matrix. (Mumbai, 2007)

We have $(iA)^\theta = (i\bar{A})^\theta = (-i)\bar{A}^\theta = -iA$

$= -iA \qquad \because \quad \bar{A}^\theta = A$

Thus $(iA)^\theta = -iA$, i.e., iA is a skew-Hermitian matrix.

Similarly, if A is a skew-Hermitian matrix then iA is a Hermitian matrix.

III. The eigen values of a Hermitian matrix are real. (see Fig. 4.1)

Let λ be the eigen value and X the corresponding eigen vector of a Hermitian matrix A, so that

$$AX = \lambda X$$

$$\bar{X}^\theta AX = \bar{X}^\theta \lambda X = \lambda \bar{X}^\theta X \quad \text{or} \quad \lambda = \bar{X}^\theta AX / \bar{X}^\theta X$$

Now $\bar{X}^\theta X = \bar{x}_1 x_1 + \bar{x}_2 x_2 + \ldots + \bar{x}_n x_n = |x_1|^2 + |x_2|^2 + \ldots + |x_n|^2$ is real and non-zero. Also $\bar{X}^\theta AX$ is a Hermitian form which is always real.

$\therefore \quad \lambda$ is a real value.

IV. The eigen values of a skew-Hermitian matrix are purely imaginary or zero.

---

[*] Named after the French mathematician Charles Hermite (1822–1901), known for his contributions to algebra and number theory.

Let λ be the eigen value and X the corresponding eigen vector of a skew-Hermitian matrix A so that $AX = \lambda X$.

$\therefore \quad \bar{X}'AX = \bar{X}'\lambda X = \lambda \bar{X}'X$ ... (1)

Since $\bar{X}'X$ is real and $\neq 0$ since $X \neq 0$. Also $\bar{X}'AX$ is a skew-Hermitian form which is purely imaginary or zero.

$\therefore \lambda$, the eigen value of a skew-Hermitian matrix is purely imaginary or zero.

4. Unitary matrix. A square matrix U such that $U^* = U^{-1}$ is called a unitary matrix. For a unitary matrix, $U\bar{U}' = U^*U = I$.

This is a generalisation of the orthogonal matrix in the complex field.

Properties

I. Inverse of a unitary matrix is unitary.

If U is a unitary matrix, then

$$U^* = U^{-1}$$

or $$\bar{U}' = U^{-1}$$

$$\therefore \quad (\bar{U}')^{-1} = \bar{U}^{-1}$$

Putting $U^{-1} = U^*$, we have

$$(U^*)^{-1} = \bar{U}' \quad \text{or} \quad V^{-1} = \bar{U}'$$

Thus U^{-1} is also unitary.

Cor. Inverse of an orthogonal matrix is orthogonal.

II. Transpose of a unitary matrix is unitary.

If U is a unitary matrix, $U^* = U^{-1}$

or $$(\bar{U}')^* = U^{-1}$$

or $$(\bar{U}')^*U' = U^{-1}U' = (U'U)^{-1}$$

Putting $U' = V$, we have $V^* = V^{-1}$

Thus V i.e., U' is also unitary.

Cor. Transpose of an orthogonal matrix is orthogonal.

III. Product of two unitary matrices is a unitary matrix.

If U and V are unitary matrices

$$U^* = U^{-1}, \; V^* = V^{-1}$$

Now, $$(UV)^{-1} = (UV)^* = V^*U^*$$

$$= V^{-1}U^{-1}$$

$$= (UV)^{-1} \qquad \because \; U, V \text{ are unitary}$$

Thus, UV is a unitary matrix.

Cor. Product of two orthogonal matrices is an orthogonal matrix.

IV. The eigen values of a unitary matrix have absolute value 1. (U.P.T.U., 2010)

If U is a unitary matrix then $UX = \lambda X$ ... (1)

Taking conjugate transpose of (1),

$$(UX)^* = (\lambda X)^* = \bar{\lambda}X^* = \bar{\lambda}\bar{X}'$$

Also $$(UX)^* = U^*X^* = U^{-1}\bar{X}'$$

$$\therefore \quad U^{-1}\bar{X}' = \bar{\lambda}\bar{X}' \qquad \ldots (2)$$

Post multiplying (2) by (1), we get

$$\bar{X}'U^{-1}UX = \bar{\lambda}\bar{X}'\lambda X$$

$$\bar{X}'U^{-1}\lambda X = \bar{\lambda}\lambda\bar{X}'X \qquad \because \; \bar{X}'UX = \lambda$$

$$\bar{X}'X = \bar{\lambda}\lambda\bar{X}'X$$

Thus $$\lambda\bar{\lambda} = |\lambda|^2 = 1, \qquad \because \; \bar{X}'X \neq 0$$

Hence the result.

Cor. The eigen value of an orthogonal matrix has absolute value 1.

Example 2.53. If $A = \begin{bmatrix} 2+i & 3 & -1+2i \\ -i & i & 4-3i \end{bmatrix}$, show that AA^{θ} is a Hermitian matrix where A^{θ} is the conjugate transpose of A. *(U.P.T.U., 2005; U.P.T.U., 2006)*

Solution. We have $A^{\theta} = \begin{bmatrix} 2-i & i \\ 3 & i \\ -1-2i & 4+3i \end{bmatrix}$

and $A^{\theta} = \begin{bmatrix} 2-i & i \\ 3 & -i \\ -1-2i & 4+3i \end{bmatrix}$

$\therefore \quad AA^{\theta} = \begin{bmatrix} 2+i & 3 & -1+2i \\ -i & i & 4-3i \end{bmatrix} \begin{bmatrix} 2-i & i \\ 3 & -i \\ -1-2i & 4+3i \end{bmatrix}$

$$= \begin{bmatrix} 4-i^2+9+1-4i^2 & 2i-3i-6i+10+12i \\ -2i+3i-3i-10-12i & i^2+9-16i^2 \end{bmatrix}$$

$$= \begin{bmatrix} 18 & -10+5i \\ -10-5i & 26 \end{bmatrix}$$, which is a Hermitian matrix.

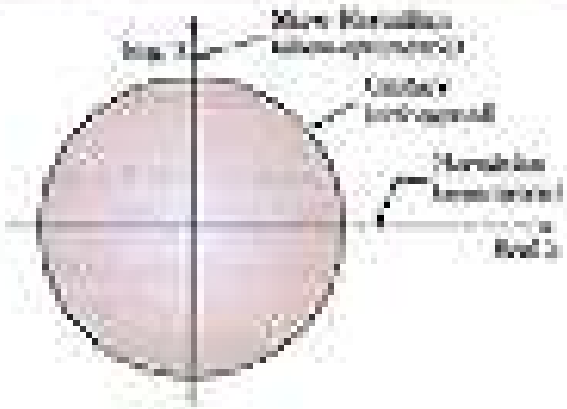

Fig. 2.1. Skew Hermitian matrix.

Example 2.54. *Prove that the matrix* $A = \begin{bmatrix} \frac{1}{2}(1+i) & \frac{1}{2}(-1+i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix}$ *is unitary and find* A^{-1}. *(Roorkee, 2005)*

Solution. Conjugate of A, i.e., $\bar{A} = \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(-1-i) \\ \frac{1}{2}(1-i) & \frac{1}{2}(1+i) \end{bmatrix}$

$\therefore$ Transpose of $\bar{A}$, i.e., $A^{\theta} = \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(1-i) \\ \frac{1}{2}(-1-i) & \frac{1}{2}(1+i) \end{bmatrix}$

Now $AA^{\theta} = \begin{bmatrix} \frac{1}{2}(1+i) & \frac{1}{2}(-1+i) \\ \frac{1}{2}(1+i) & \frac{1}{2}(1-i) \end{bmatrix} \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(1-i) \\ \frac{1}{2}(-1-i) & \frac{1}{2}(1+i) \end{bmatrix}$

$$= \begin{bmatrix} \frac{1}{4}(1+2i+i^2)+\frac{1}{4}(1+i^2) & -\frac{1}{4}(1-2i^2+i^2)+\frac{1}{4}(1-i^2) \\ -\frac{1}{4}(1+i^2)+\frac{1}{4}(1+i^2) & \frac{1}{4}(1+i^2)+\frac{1}{4}(1+i^2) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Similarly, $A^{\theta}A = I$.

Hence A is a unitary matrix.

Also $A^{-1} = A^{\theta} = \begin{bmatrix} \frac{1}{2}(1-i) & \frac{1}{2}(1-i) \\ -\frac{1}{2}(1+i) & \frac{1}{2}(1+i) \end{bmatrix}$

Example 3.33. Given that $A = \begin{bmatrix} 2 & 1+2i \\ -1+i & 3i \end{bmatrix}$ show that $(I+A)(I-A)^{-1}$ is a unitary matrix. (Mumbai, 2007)

Solution. $I+A = \begin{bmatrix} 3 & 1+2i \\ -1+i & 1+3i \end{bmatrix}$, $|I+A| = 1-i-3-4i = -2-5i$

$$(I+A)^{-1} = \begin{bmatrix} 1+3i & -1-2i \\ 1-i & 3 \end{bmatrix} \div (-2-5i) \text{ Also } I-A = \begin{bmatrix} 1 & -1-2i \\ 1-i & 1 \end{bmatrix}$$

$$\therefore \quad (I-A)(I+A)^{-1} = \begin{bmatrix} 1 & -1-2i \\ 1-i & 1 \end{bmatrix}\begin{bmatrix} 1+3i & -1-2i \\ 1-i & 3 \end{bmatrix} \div (-2-5i) = \frac{1}{-2-5i}\begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix} \quad \ldots(i)$$

$$\text{Hermitian transpose} = \frac{1}{-2+5i}\begin{bmatrix} -4 & 2+4i \\ -2+4i & -4 \end{bmatrix} \quad \ldots(ii)$$

$$\therefore \text{ Product of (i) and (ii)} = \frac{1}{29}\begin{bmatrix} -4 & -2-4i \\ 2-4i & -4 \end{bmatrix}\begin{bmatrix} -4 & 2+4i \\ -2+4i & -4 \end{bmatrix} = \frac{1}{29}\begin{bmatrix} 29 & 0 \\ 0 & 29 \end{bmatrix} = I.$$

Hence the result.

PROBLEMS 3.11

1. Prove that every Hermitian matrix can be written as $A+iB$, where A is real skew-symmetric and B is real symmetric matrix. (U.P.T.U., 2006)
2. Prove that every square matrix can be uniquely expressed as $P+iQ$, where P and Q are Hermitian matrices. (Mumbai, 2006; Anna, 2009 S)
3. Show that a Hermitian matrix remains Hermitian under any transformation $B^{\theta}AB$, where B is any orthogonal matrix.
4. Show that the matrix $\begin{bmatrix} a+ib & c+id \\ -c+id & a-ib \end{bmatrix}$ is a unitary matrix if $a^2+b^2+c^2+d^2=1$. (U.P.T.U., 2006)
5. Show that $\begin{bmatrix} 2 & 2-i & -3+2i \\ 2+i & 3 & 2+i \\ -3-2i & 2-i & 4 \end{bmatrix}$ is a Hermitian matrix.
6. If $A = \begin{bmatrix} -1 & 2+i & 4-3i \\ 2-i & 3 & 5i \\ 4+3i & -5i & 2 \end{bmatrix}$, show that A is a Hermitian matrix and iA is a skew-Hermitian matrix. (Kolhapur, 2007)
7. Show that the following matrix is unitary

(i) $\frac{1}{3}\begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ (U.P.T.U., 2005) (ii) $\begin{bmatrix} \frac{1+i}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1-i}{2} \end{bmatrix}$ (Mumbai, 2006)

8. Express $A = \begin{bmatrix} 2i & 2+i & 1-i \\ 2-i & i & 3i \\ -1-i & 3i & 0 \end{bmatrix}$ as $P+iQ$ where P & Q are real and P is symmetric and Q is real and skew-symmetric. (Mumbai, 2006)
9. If $V = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix}$, where $\omega = e^{i\frac{2\pi}{3}}$, prove that $V^{-1} = \frac{1}{3}V$. (Mumbai, Kolhapur, 2008; U.P.T.U., 2009)

2.60 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 2.8

Choose the correct answer or fill in the missing words in the following questions:

1. To multiply a matrix by scalar k, we multiply
   (a) any row by k (b) every element by k (c) any column by k

2. If $A = \begin{bmatrix} 2 & -4 \\ 3 & -1 \end{bmatrix}$, then kA is

   (a) $\begin{bmatrix} k+2k & -4k \\ -k & k-2k \end{bmatrix}$ (b) $\begin{bmatrix} 2^k & (-4)^k \\ 1 & (-1)^k \end{bmatrix}$ (c) $\begin{bmatrix} k+2k & k-4k \\ 3+k & k-k \end{bmatrix}$ (d) $\begin{bmatrix} k+2k & -4k \\ 3+k & k-k \end{bmatrix}$

3. The inverse of the matrix $\begin{bmatrix} -2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is

   (a) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 0.5 & 0 & 0 \\ 0 & -0.25 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} -2 & 0 & 0 \\ 0 & 0.25 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -0.25 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

4. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$, then the determinant of kAB is $8k$ when

   (a) 4 (b) 8 (c) 16 (d) 32

5. The system of equations $x + 2y + z = 3$, $2x + y + 2z = 7$ can be represented as

   (a) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ (d) none of the above

6. If $3\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} X = \begin{bmatrix} 7 & 2 \\ 3 & 3 \end{bmatrix}$, then X equals

   (a) $\begin{bmatrix} -2 & -14 \\ 4 & 15 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & -14 \\ 4 & -15 \end{bmatrix}$

7. If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, then A (adj A) equals

   (a) $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & 15 \\ 20 & 5 \end{bmatrix}$ (c) $\begin{bmatrix} 14 & 3 \\ 1 & 12 \end{bmatrix}$ (d) none of the above

8. If $3x + 2y + z = 8$, $x + 4y + z = 6$, $2x + y + 4z = 0$, then a solution of equations is/are
   (a) it is inconsistent
   (b) it has only the trivial solution $x = 0$, $y = 0$, $z = 0$
   (c) it can be reduced to a single equation and so a unique solution does not exist
   (d) determinant of the matrix of coefficients is zero

9. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, then

   (a) $C = A\cos\theta - B\sin\theta$ (b) $C = A\cos\theta + B\sin\theta$
   (c) $C = A\sin\theta - B\cos\theta$ (d) $C = A\sin\theta + B\cos\theta$

10. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ 3 & y & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}$ then

(a) A is row equivalent to B only if $x = 2$ or $y = 4$ (b) A is row equivalent to B only if $x \neq 0, y \neq 0$, and $x \neq y$

(c) A is not row equivalent to B (d) A is row equivalent to B for all values of x and y

11. If $A\begin{bmatrix} 5 & 1 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & 3 \end{bmatrix}$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then A is

(a) $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} \frac{5}{13} & \frac{1}{13} \\ -\frac{3}{13} & \frac{11}{13} \end{bmatrix}$

12. If $\det A = 0$, then $\det A^{-1}$ is

(a) always zero (b) always one (c) depends upon the elements (d) infinite

13. If A is a square matrix such that $AA^T = I$, then value of $|A|$ is

(a) ± 1 (b) 1 (c) ± 2

14. If every element of order r of a matrix A is zero, then rank of A is

(a) greater than r (b) equal to r (c) less than or equal to r (d) less than r

15. A square matrix A is called orthogonal if

(a) $A = A^T$ (b) $A^2 = A$ (c) $AA^T = I$

16. The rank of matrix $\begin{bmatrix} 2 & -2 & 3 & 2 \\ 1 & 4 & -2 & 1 \\ 5 & 1 & 1 & 3 \end{bmatrix}$ is _____

17. The sum of the eigen values of a matrix is the _____ of the elements of the principal diagonal.

18. The product of the eigen values of the matrix $\begin{bmatrix} 2 & 3 \\ 4 & -2 \end{bmatrix}$ are _____ and _____ respectively. (June, 2003)

19. If $\det \begin{bmatrix} 1 & 2 & -1 \\ -1 & 3 & 4 \\ 2x & 0 & 1 \end{bmatrix}$ is $\begin{bmatrix} 3 & 2 & 8 \\ 1 & 1 & x \\ 1 & 2 & 8 \end{bmatrix}$ then x is _____

20. Using Cayley-Hamilton theorem, the value of $A^4 - 4A^3 - 2A^2 - A + 5I$ where $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ is _____ (Anna, 2004)

21. If two eigen values of $\begin{bmatrix} 3 & 0 & 1 \\ 0 & 7 & -4 \\ 1 & -4 & 3 \end{bmatrix}$ are 4 and 10, then the third eigen value is _____

22. A symbolic form of positive semi-definite is _____

23. $A_{n \times n}$ and $B_{n \times n}$ are two matrices. Which of the

(a) $|A \cdot B| \neq |A||B|$ (b) $|A + B| = |A| + |B|$

24. The product of the eigen values of $\begin{bmatrix} 1 & 2 & 0 \\ 2 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ is _____

25. The quadratic form corresponding to the diagonal matrix $\operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is

(a) $\lambda_1 x_1^2 + \lambda_2 x_2^2 + \ldots + \lambda_n x_n^2$ (b) $\lambda_1 y_1^2 + \lambda_2 y_2^2 + \ldots + \lambda_n y_n^2$

(c) $x_1^2 + x_2^2 + \ldots + x_n^2$

26. An example of a 3×3 matrix of rank one is _____

27. The quadratic form corresponding to the symmetric matrix $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ is _____

28. Solving the equations $x + 2y + 3z = 0$, $3x + 4y + 4z = 0$, $x + 3y + 3z = 0$, $x =$ _____, $y =$ _____, $z =$ _____

30. The eigen values of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -1 & 0 \end{bmatrix}$ are _____

31. A matrix A is idempotent if _____

32. The rank of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 2 & -3 & 8 \\ 3 & -2 & 11 \end{bmatrix}$ is _____

33. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & -2 \end{bmatrix}$, then the eigen values of A^3 are _____

34. The sum of the eigen values of $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & -1 & -1 \end{bmatrix}$ is

(a) −2 (b) 2 (c) 5 (d) 8 (e) 7 (UPTU 2006, 2007)

35. The maximum value of the rank of 3×5 matrix is _____

36. The sum of two eigen values and trace of a 3×3 matrix are equal, then the value of $|A|$ is _____ (UPTU, 2005)

37. If the sum of the eigen values of the matrix of the quadratic form is zero, then the nature of the quadratic form is _____

38. The eigen values of matrix $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ are _____

39. The eigen values of a skew-symmetric matrix are _____

40. If the product of two eigen values of the matrix $\begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & -1 \\ 2 & -1 & 5 \end{bmatrix}$ is 18, then the third eigen value is _____

41. If λ_i (i = 1, 2, ... n) are the eigen values of a square matrix A, then the eigen values of A^2 are _____

42. By applying elementary transformations we can find _____

(a) Inverse (b) Rank (c) Both (a) and (b)

43. If λ is an eigen value of A, then it is an eigen value of A, and $\frac{1}{\lambda}$ is _____

44. If $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then eigen values of A^3 are _____

45. The characteristic equation of $\begin{bmatrix} 2 & 10 & 5 \\ -2 & -3 & -1 \\ 3 & 6 & 2 \end{bmatrix}$ is _____

46. If $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$, then eigen values of A^3 are _____

47. Matrix $\begin{bmatrix} 2 & 1 \\ 1 & x+1 \end{bmatrix}$ is singular for x = _____

48. Every Hermitian matrix can be written as A + iB, where A is real and _____ and B is real and _____

49. The sum and product of the eigen values of $\begin{bmatrix} 8 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$ are _____ and _____

50. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then A^{-1} = _____

51. The product of the eigen values of a matrix is equal to _____

52. The eigen values of $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ are the roots of the equation _____

16. A system of linear non-homogeneous equations is consistent, if and only if the rank of coefficient matrix is equal to rank of _____.
17. The matrix of the quadratic form $q = 4x^2 - 3y^2 + z^2 - 2xy + 6yz$ is _____.
18. If $\lambda_1, \lambda_2, \lambda_3$ are the eigen values of the matrix A, then A^2 has the eigen values _____.
19. If λ is an eigen value of a non-singular matrix A, then the eigen value of A^{-1} is _____.
20. The matrix corresponding to the quadratic form $x^2 + 2y^2 - 7z^2 - 4xy + 8yz - 2zx$ is _____.
21. The nature of the eigen values of the eigen values of $\begin{bmatrix} 5 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$ is _____.
22. If the rank of a matrix A is 2, then the rank of A^T is _____.
23. The nature of the quadratic form $x_1^2 + 2x_2^2 - 6x_3^2$ are respectively 1, _____ and _____.
24. The equations $x + 2y = 1$, $2x + 3y = 1$ are inconsistent. (True or False)
25. If rank $(A) = 3$, rank $(B) = 3$, then rank $(AB) = 9$. (True or False)
26. Any set of vectors which includes the zero vector is linearly independent. (True or False)
27. If λ is an eigen value of a symmetric matrix then λ is real. (True or False)
28. Every square matrix does not satisfy its own characteristic equation. (True or False)
29. If λ is an eigen value of an orthogonal matrix then $\frac{1}{\lambda}$ is also its eigen value. (True or False)
30. If the rank of a matrix A is 3, then the rank of AA^T is 2. (True or False)
31. The vectors $(1, 2, -1, 0), (2, -1, 0, 1), (-1, 2, -1, 0), (1, 0, 1, 1)$ are linearly dependent. (True or False)
32. The eigen values of a skew-symmetric matrix are real. (True or False)
33. Inverse of a symmetric matrix is symmetric matrix. (True or False)
34. If A is a non-zero column matrix and B is a non-zero row matrix, then rank of AB is one. (True or False)
35. The sum of the eigen values of A equals to the trace of A = $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ (True or False)

CHAPTER

3

Vector Algebra & Solid Geometry

1. Vectors. 2. Types (equivalent, Equal, Unit or Zero). 3. Position Vectors. 4-5. Products of two vectors. 6. Physical applications. 7-12. Products of three or more vectors. 13. Equation of a plane. 14. Equation of a straight line. 15. Coplanar lines. 16. Shortest distance between two lines. 17. Intersection of three planes. 18. Equation of a sphere. 19. Tangent plane to a sphere. 20. Cone. 21. Cylinder. 22. Conicoid surfaces. 23. Surfaces of Revolution. 24. Objective Type of Questions.

VECTOR ALGEBRA

3.1 (I) VECTORS

A quantity which is completely specified by its magnitude only is called a scalar. Length, time, mass, volume, temperature, work, electric charge and numerical data in Statistics are all examples of scalar quantities.

A quantity which is completely specified by its magnitude and direction is called a vector. Weight, displacement, velocity, acceleration and electric current density are all vector quantities for each involves magnitude and direction.

A vector is represented by a directed line segment. Thus $\vec{PQ}$ represents a vector whose magnitude is the length PQ and direction is from P (starting point) to Q (end point). We denote a vector by a single letter in capital bold type (or with an arrow on it) and its magnitude (length) by the corresponding small letter in italic type. Thus if V is the velocity vector, its magnitude is v, the speed.

A vector of unit magnitude is called a unit vector. The idea of unit vector is often used to represent mentally the direction of any vector. Unit vector corresponding to the vector A is written as $\hat{A}$.

A vector of zero magnitude (such as one have no direction associated with it) is called a zero (or null) vector and is denoted by 0 or $\vec{0}$.

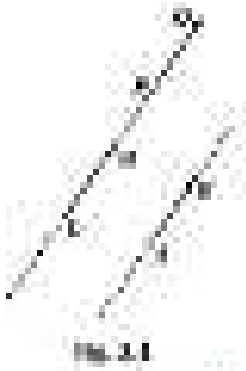

Fig. 3.1

The vector $\vec{QP}$ represents the negative of $\vec{PQ}$, i.e., $-A$.

Two vectors A and B having the same magnitude and the same (or parallel) directions are said to be equal and we write $A = B$. Clearly the vectors $\vec{AB}$, $\vec{DC}$ and $\vec{PQ}$ are all equal (Fig. 3.1).

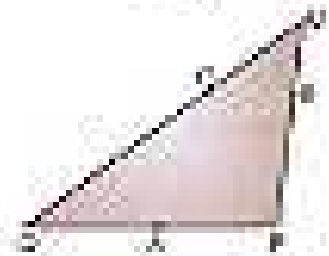

Fig. 3.2

(2) Addition of vectors. Vectors are added according to the triangle law of addition, which is a matter of common knowledge. Let A and B be represented by two vectors $\vec{OP}$ and $\vec{PQ}$ respectively then $\vec{OQ} = C$ is called the resultant of A and B. Symbolically, we write,

$$C = A + B$$

(3) Subtraction of vectors. The subtraction of a vector B from A is taken to be the addition of $-B$ to A and we write

$$A + (-B) = A - B$$

(4) Multiplication of vectors by scalars.

We have just seen that $A + A = 2A$

and $-A + (-A) = -2A$

where both $2A$ and $-2A$ denote vectors of magnitude twice that of A, the former having the same direction as A and the latter the opposite direction.

In general, the product mA of a vector A and a scalar m is a vector whose magnitude is m times that of A and direction is the same or opposite to A according as m is positive or negative.

Thus $\quad A = mA.$

Example 8.1. *If A and B are the vectors determined by two adjacent sides of a regular hexagon. Find in the simplest form the vectors which determine the remaining sides taken in order.*

Solution. Let ABCDEF be the given hexagon, such that

$$\vec{AB} = A \text{ and } \vec{BC} = B$$

$\therefore$ $\quad \vec{AC} = \vec{AB} + \vec{BC} = A + B.$

Also $\quad \vec{AD} = 2\vec{AO} = 2B$

$\therefore$ $\quad \vec{CD} = \vec{AD} - \vec{AC} = 2B - (A + B) = B - A$

Now $\quad \vec{DE} = -\vec{AB} = -A$ [$\because$ $AB = ED$ and $\|AB\|$]

$\quad \vec{AF} = -\vec{BC} = -B$ [$\because$ $BC = AF$ and $\|AF\|$]

and $\quad \vec{FA} = -\vec{CD} = -(B - A) = A - B$ [$\because$ $CD = AF$ and $\|AF\|$]

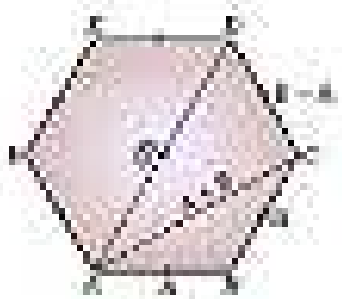

Fig. 13

2.2. (1) Space coordinates. Let $X'OX$ and $Y'OY$, ZOZ' be three mutually perpendicular lines which intersect at O. Then O is called the origin.

$X'OX$ is called the x-axis, $Y'OY$ the y-axis, ZOZ' the z-axis and taken together these are called the coordinate axes.

The plane YOZ is called the yz-plane, the plane ZOX the zx-plane, the plane XOY the xy-plane and taken together these are called the coordinate planes.

Let P be any point in space. Draw PL, PM, PN to be $\|$ to the yz, zx and xy-planes. Then AP, BP, CP are respectively called the coordinates of P (Fig. 14) in position, if $OA = x$, $AN = y$, $NP = z$, then (x, y, z) are the coordinates of P which are positive along OX, OY, OZ respectively and negative along OX', OY', OZ'.

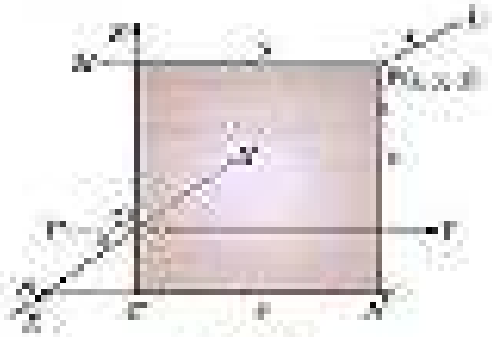

Fig. 14

The three coordinate planes divide the space into eight compartments called octants. The octant $OXYZ$ in which all the coordinates are positive is called the positive or first octant.

Note. Three non-coplanar vectors A, B, C are said to form a right-handed or a left-handed system according as a right-threaded screw rotated through an angle less than $180°$ from A to B will advance along or opposite to C as shown in Fig. 15.

An area of a closed curve in a plane and a given sense of description is represented by a vector whose magnitude is the given area and whose line is normal to the plane of the area. Then the vector A represents any one area taken to be positive or negative according as the direction of description of the boundary of the curve and the sense of A correspond in a right-handed or a left-handed system.

We have explained the most commonly used system of coordinates namely the Rectangular Cartesian Coordinates. The other two systems of coordinates which are used to locate a point in space are the Polar spherical coordinates and Cylindrical coordinates which are explained in § 3.21 and 3.22.

(1) Resolution of vectors. Let $\vec{I}, \vec{J}, \vec{K}$ denote unit vectors along OX, OY, OZ respectively. Let $P(x, y, z)$ be a point in space. On OP as diagonal, construct a rectangular parallelopiped with edges OA, OB, OC along the axes as Fig 1.

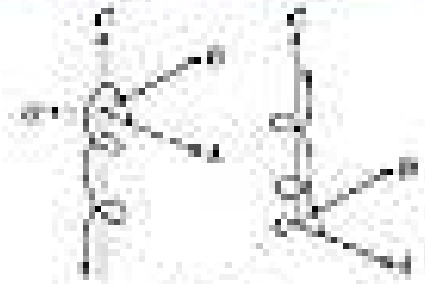

Fig 1.1

$$\vec{OA} = x\vec{I},\ \vec{OB} = y\vec{J},\ \vec{OC} = z\vec{K}$$

Then
$$\vec{R} = \vec{OP} = \vec{OC} + \vec{CP}$$
$$= \vec{OA} + \vec{AC} + \vec{OC} = \vec{OA} + \vec{OB} + \vec{OC}$$

Hence $\vec{R} = x\vec{I} + y\vec{J} + z\vec{K}$ is called the position vector of P relative to origin O and,

$$r = |\vec{R}| = \sqrt{(x^2 + y^2 + z^2)}$$
$$\left[\because\ r^2 = OP^2 = OD^2 + DP^2 = OA^2 + AD^2 + DP^2\right]$$

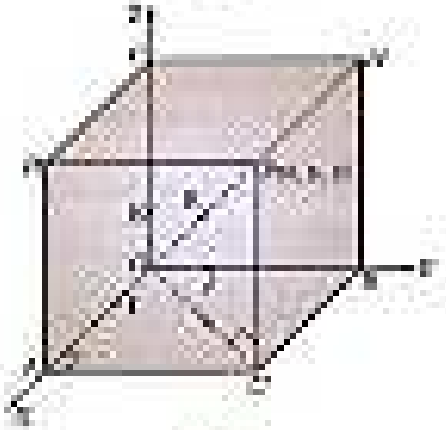

Fig 1.2

(2) Direction cosines. Let any line L or its parallel OP, make angles α, β, γ with OX, OY, OZ respectively, then cos α, cos β, cos γ are called the direction cosines of this line which are usually denoted by l, m, n.

If l, m, n were direction cosines of a vector $\vec{R}$, then

(i) $\vec{R} = \vec{R}\,\vec{n} = x\vec{I} + y\vec{J} + z\vec{K}$ (ii) $l^2 + m^2 + n^2 = 1$

Proof. Let D be the foot of the perpendicular from $P(x, y, z)$ on OX. Then

$$x = OD = r\cos\beta = rm;\ \text{similarly}\ y = rm \text{ and } z = rn.$$

$$\therefore\quad \vec{R} = x\vec{I} + y\vec{J} + z\vec{K} = rl\vec{I} + rm\vec{J} + rn\vec{K}$$

or
$$\vec{R} = \frac{\vec{R}}{r} = l\vec{I} + m\vec{J} + n\vec{K}$$

which is possible only if l, m, n are the direction cosines.

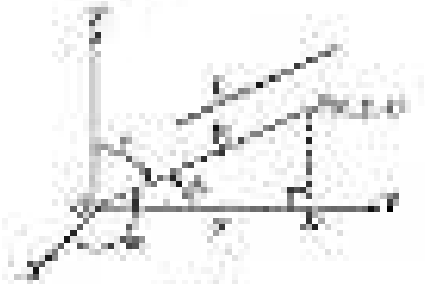

Fig 1.3

Also
$$1 = |\vec{R}| = \sqrt{(l^2 + m^2 + n^2)}\ \text{i.e.,}\ l^2 + m^2 + n^2 = 1$$

i.e., $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ (V.T.U., 2010)

Obs. Direction ratios. If the direction cosines of a line be proportional to a, b, c, then these are called proportional direction cosines or direction ratios of the line.

If the direction ratios be a, b, c, then

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \frac{\sqrt{(l^2 + m^2 + n^2)}}{\sqrt{(a^2 + b^2 + c^2)}} = \frac{1}{\sqrt{(a^2 + b^2 + c^2)}}$$

$$\therefore\quad l = \frac{a}{\sqrt{(a^2 + b^2 + c^2)}},\ m = \frac{b}{\sqrt{(a^2 + b^2 + c^2)}},\ n = \frac{c}{\sqrt{(a^2 + b^2 + c^2)}}$$

(3) Distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$\sqrt{[(x_2 - x_1)^2 + (y_2 - x_1)^2 + (z_2 - z_1)^2]}$$

and direction ratios of $\vec{PQ}$ are $x_2 - x_1, y_2 - y_1, z_2 - z_1$.

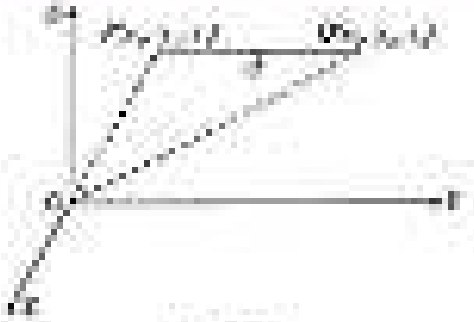

Fig 1.4

We have
$$\vec{OP} = x_1\vec{I} + y_1\vec{J} + z_1\vec{K}$$

and
$$\vec{OQ} = x_2\vec{I} + y_2\vec{J} + z_2\vec{K}$$

$$\therefore\quad \vec{PQ} = \vec{OQ} - \vec{OP}$$

$$= (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

Thus, $$d = |\vec{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

and direction cosines of $\vec{PQ}$ are proportional to $x_2 - x_1, y_2 - y_1, z_2 - z_1$.

Example 3.2 Show that the points $A(-1, 3, 6)$, $B(-7, 6, 6)$ and $C(-1, 12, 10)$ form a right angled isosceles triangle. Also find the direction cosines of AB.

Solution. We have $$AB = \sqrt{(6-3)^2 + (-6-6)^2 + (6-6)^2} = 3\sqrt{5}$$

$$BC = \sqrt{(6+1)^2 + (12-6)^2 + (10-6)^2} = 9\sqrt{2}$$

and $$CA = \sqrt{(-1+1)^2 + (3-12)^2 + (6-10)^2} = 6$$

Since $AB^2 + AC^2 = BC^2$ and $AB = AC$, it follows that ΔABC is a right angled isosceles triangle. The direction ratios of $\vec{AB}$ are $-1+6, 3-6, 6-6$.

$\Rightarrow$ Its direction cosines are $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$.

3.8 SECTION FORMULAE

The point $R(x, y, z)$ dividing the join of the points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, y_2)$ internally in the ratio $m_1 : m_2$ is

$$R = \frac{m_1\vec{B} + m_2\vec{A}}{m_1 + m_2} \quad \text{i.e.,} \quad \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}, \frac{m_1z_2 + m_2z_1}{m_1 + m_2}\right) \quad \ldots(1)$$

Let PA and QB be the given parallel vectors and let the line AB be the point R dividing the line joining P and Q in the ratio $m_1 : m_2$ so that

$$\frac{RB}{RA} = \frac{m_1}{m_2} \quad \text{i.e.,} \quad m_1 \cdot RB = m_2 \cdot RA$$

$\therefore$ We have $$m_1\vec{RB} = m_2\vec{RA}$$

or $$m_1(\vec{OB} - \vec{OR}) = m_2(\vec{OA} - \vec{OR})$$

or $$m_1(\vec{B} - \vec{R}) = m_2(\vec{A} - \vec{R})$$

whence, $$\vec{R} = \frac{m_1\vec{B} + m_2\vec{A}}{m_1 + m_2}$$

Now $$A = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}, B = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

and $$R = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\therefore \quad x\hat{i} + y\hat{j} + z\hat{k} = \frac{m_1(x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) + m_2(x_1\hat{i} + y_1\hat{j} + z_1\hat{k})}{m_1 + m_2}$$

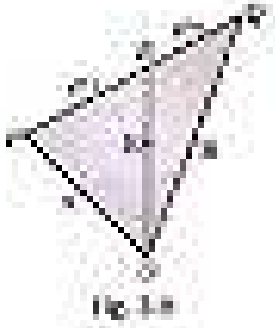

Fig. 3.9

Equating coefficients of $\hat{i}, \hat{j}, \hat{k}$, we get the desired results (1).

Cor. 1. Mid point of PQ and QR is $\frac{P}{2} + A + \frac{B}{2}$.

2. Point R dividing the join of PA and QB internally in the ratio $m_1 : m_2$ externally is $R = \frac{m_1B - m_2A}{m_1 - m_2}$.

Note. Assuming (1) in $m_1A - m_2B + (m_2 - m_1)R = 0$, we note that the sum of the coefficients of A, B and R is zero. Hence it follows that we can always express a vector A in terms of two vectors B and C as $xA + yB + zC$ where $x + y + z = 0$.

Example 3.3 In a triangle prove that the straight line joining the mid points of the sides is parallel to the remaining side and half of it.

Solution. Consider a triangle ABC with parallel sides OA and BC. Take O as the origin and let the other vertices be $A(A), B(B), C(C)$.

Since CD is parallel to OA, therefore,

$$B + C = \vec{OC} = x\vec{OA} = xA.$$

The mid-points of the diagonals OB and AC are $\frac{1}{2}B$ and $\frac{1}{2}(A + C)$.

$$\therefore \quad \vec{EF} = \vec{OF} - \vec{OE} = \frac{1}{2}(A + C) - \frac{1}{2}B = \frac{1}{2}(A - B - C) \quad \ldots(i)$$

$$= \frac{1}{2}(x - 1)A. \quad \ldots(ii)$$

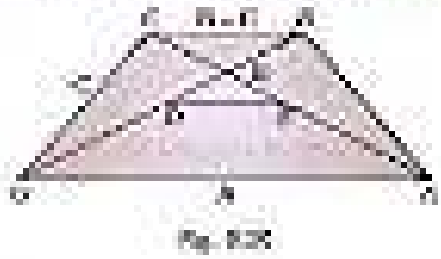

Fig. 3.25

From (ii), it is clear that $\vec{EF}$ is parallel to $\vec{OA}$; from (i), it follows that $EF = \frac{1}{2}|OA - CB|$.

Hence the result.

Example 3.14. Show that the line joining the centroid of a parallelogram to the mid-point of a opposite side bisects the diagonal and is itself bisected by it.

Solution. Consider a parallelogram OABC. Take O as the origin and let the other vertices be A(A), B(B) and C(C).

The mid-point D of OA is $\frac{1}{2}A$.

Now since OA is equal to and parallel to CB

$$\therefore \quad \vec{OA} = \vec{CB}, \text{ i.e., } A = B - C$$

which may be written as $\frac{2(\frac{1}{2}A) + 1.C}{2 + 1} = \frac{B}{3}$ = P, say, that P trisects BC and OB.

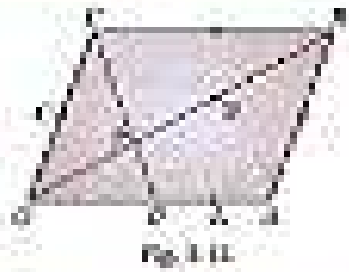

Fig. 3.26

Exercise 3.3

1. Given $R_1 = 2i - 3j + 5k$ and $R_2 = i + 2j + 3k$, find the magnitude and direction ratios of the vectors $R_1 + R_2$ and $2R_1 - R_2$.
2. Show that the points (6, 7, 1), (4, 1, ...), (8, 5, ...) and (2, 3, 2) are the vertices of a square. (Himachal, 2004)
3. A vector $\vec{OA}$ is inclined to the axes OX and OY at angles of 30° and 60°. If $|\vec{OA}| = 4$ units, find the components of $\vec{OA}$ along the coordinate axes. (Kashmir, 2005)
4. If a vector makes angles α, β, γ with the axes, prove that
   (i) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$. (NCERT; Himachal, 2004)
   (ii) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$.
5. If A and B are two collinear vectors and $P = 2i + 3j - 2k$, $Q = 3i + 2j + 3k$ and $R = 4i + 5j - 4k$, $S = 5i + 4j + 3k$, find λ such that $PQ = \lambda RS$.
6. Prove that the line joining the mid-points of the two sides of a triangle is parallel to the third side and half of it.
7. Prove that (i) the diagonals of a parallelogram bisect each other;
   (ii) a quadrilateral whose diagonals bisect each other is a parallelogram.
8. In a three-dimensional, prove that :
   (i) the figure formed by joining the mid-points of the adjacent sides of a parallelogram is a parallelogram,
   (ii) the points (1, 0, 1), (2, 1, 3), (3, 3, 5), (0, 1, 1) are the vertices of a parallelogram.
9. In a trapezium, prove that the straight line joining the mid-points of the non-parallel sides is parallel to the parallel sides and half of their sum.
10. Prove that the vectors $A = 4i + 3j - 2k$, $B = -i + 3j + 4k$, $C = 4i - 3j - 6k$ can form the sides of a triangle. Also find the length of the median passing through the vertex C. (NCERT; Delhi, 2003)
11. Find the ratio in which the line joining (2, 4, 10) and (3, 5, −1) is divided by the plane $2x - 3y + z = 0$. (Kashmir, 2003)
12. Show that the three points $A = 2i + 3j - 5k$, $B = i - 3j + 4k$, $C = 3i + 15j - 18k$ are collinear.
13. If A, B, C be the position vectors of the vertices A, B, C of the triangle ABC, show that :
   (i) medians intersect at the point $\frac{1}{3}(A + B + C)$, and of the vectors
   (ii) the internal bisector of the angles intersect at the point $\frac{aA + bB + cC}{a + b + c}$, and of the vectors.

(ii) Show that the coordinates of the centroid of the triangle whose vertices are (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) are

$$\left[\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right].$$

(iii) Show that the coordinates of the centroid of the tetrahedron whose vertices are (x_r, y_r, z_r), $r = 1, 2, 3, 4$ are

$$\left[\frac{1}{4}(x_1+x_2+x_3+x_4), \frac{1}{4}(y_1+y_2+y_3+y_4), \frac{1}{4}(z_1+z_2+z_3+z_4)\right].$$

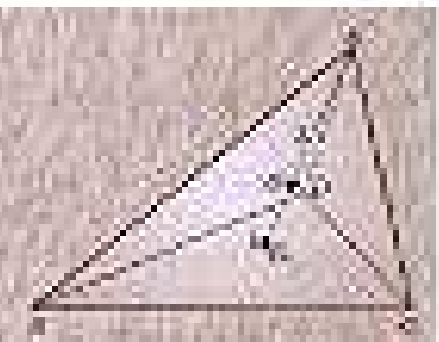

Fig. 3.13

(iv) If A, B, C, D are the vertices of the tetrahedron $ABCD$, show that the centroid of the tetrahedron $ABCD$ is the same as the centroid of the tetrahedron formed by the centroids of the faces BCD, CDA, DAB and ABC.

(Kurukshetra 2003)

(v) If A and B are the mid-points of the diagonals AC and BD respectively of a quadrilateral $ABCD$, show that the resultant of the vectors $\overrightarrow{AD}$, $\overrightarrow{AB}$, $\overrightarrow{CB}$, $\overrightarrow{CD}$ is $4\overrightarrow{AB}$.

(Kurukshetra 2003)

3.50 PRODUCTS OF TWO VECTORS

Unlike the product of two vectors or that of a vector by a scalar, the product of two vectors is sometimes seen to result in a scalar quantity and sometimes in a vector. As such, we are led to define two types of such products, called the *scalar product* and the *vector product* respectively.

The scalar and vector products of the two vectors $\mathbf{A}$ and $\mathbf{B}$ are usually written as $\mathbf{A} \cdot \mathbf{B}$ and $\mathbf{A} \times \mathbf{B}$ respectively and are read as $\mathbf{A}$ dot $\mathbf{B}$ and $\mathbf{A}$ cross $\mathbf{B}$. In view of this notation, the former is sometimes called the *dot product* and the latter the *cross product*.

In vector algebra, the division of a vector by another vector is not defined.

3.51 SCALAR OR DOT PRODUCT

(1) Definition. *The scalar or dot product of two vectors $\mathbf{A}$ and $\mathbf{B}$ is defined as the product $ab \cos\theta$, where θ is the angle between $\mathbf{A}$ and $\mathbf{B}$.*

Thus $$\mathbf{A} \cdot \mathbf{B} = ab \cos\theta.$$

(2) Geometrical interpretation. *$\mathbf{A} \cdot \mathbf{B}$ is the product of the length of one vector and the length of the projection of the other in the direction of the former.*

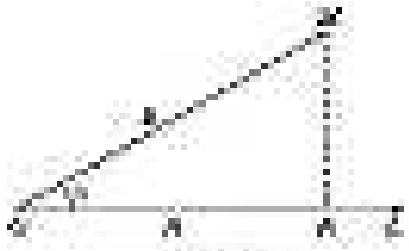

Fig. 3.14

Let $\overrightarrow{OA} = \mathbf{A}$, $\overrightarrow{OB} = \mathbf{B}$ then

$\mathbf{A} \cdot \mathbf{B} = a$ (length of OA) $\times$ (length of projection of OB on OA) $= |\mathbf{A}|$ (Proj. of $|\mathbf{B}|$ in the direction of $\mathbf{A}$).

Similarly, $\mathbf{A} \cdot \mathbf{B} = |\mathbf{B}|$ (Proj. of $|\mathbf{A}|$ in the direction of $\mathbf{B}$).

(3) Properties and other results.

I. Scalar product of two vectors is commutative.

i.e., $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$ for $\mathbf{A} \cdot \mathbf{B} = ab \cos\theta = ba \cos(-\theta) = \mathbf{B} \cdot \mathbf{A}$.

II. The necessary and sufficient condition for two vectors to be perpendicular is that their scalar product should be zero.

When the vectors $\mathbf{A}$ and $\mathbf{B}$ are perpendicular, $\mathbf{A} \cdot \mathbf{B} = ab \cos 90° = 0$.

Conversely, when $\mathbf{A} \cdot \mathbf{B} = 0$, either $a = 0$, or $b = 0$, or $\cos\theta = 0$ i.e., $\cos\theta = 0$ ($\because$ neither $a = 0$ nor $b = 0$, $\theta \neq 0°$)

III. $\mathbf{A} \cdot \mathbf{A} = a^2$ which is written as $\mathbf{A}^2$. Thus the square of a vector is a scalar which stands for the square of its magnitude.

IV. For the mutually perpendicular unit vectors $\mathbf{I}, \mathbf{J}, \mathbf{K}$ *we have the relations,*

$$\mathbf{I} \cdot \mathbf{J} = \mathbf{J} \cdot \mathbf{K} = \mathbf{K} \cdot \mathbf{I} = 0$$

and $$\mathbf{I}^2 = \mathbf{J}^2 = \mathbf{K}^2 = 1$$

which are of great utility.

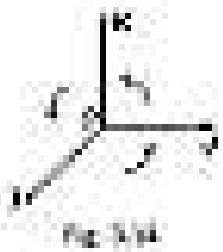

Fig. 3.15

V. Scalar product of two vectors is distributive i.e.,

$$(\mathbf{A}+\mathbf{B})\cdot\mathbf{C}=\mathbf{A}\cdot\mathbf{C}+\mathbf{B}\cdot\mathbf{C}$$

VI. Schwarz inequality: $|\mathbf{A}\cdot\mathbf{B}|\le|\mathbf{A}||\mathbf{B}|$

$$|\mathbf{A}\cdot\mathbf{B}|=|\mathbf{A}||\mathbf{B}||\cos\theta|\le|\mathbf{A}||\mathbf{B}| \qquad \because\ |\cos\theta|\le 1$$

VII. Scalar product of two vectors is equal to the sum of the products of their corresponding components.

For if $\mathbf{A}=a_1\mathbf{I}+a_2\mathbf{J}+a_3\mathbf{K}$, $\mathbf{B}=b_1\mathbf{I}+b_2\mathbf{J}+b_3\mathbf{K}$

then by the distributive law, $\mathbf{A}\cdot\mathbf{B}=a_1b_1+a_2b_2+a_3b_3$

In particular, $\mathbf{A}^2=a_1^2+a_2^2+a_3^2$.

VIII. Angle between two lines whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2 is $\cos^{-1}(l_1l_2+m_1m_2+n_1n_2)$.

The unit vectors in the direction of the given lines are $\mathbf{U}=l_1\mathbf{I}+m_1\mathbf{J}+n_1\mathbf{K}$ and $\mathbf{V}=l_2\mathbf{I}+m_2\mathbf{J}+n_2\mathbf{K}$.

If θ be the angle between the lines, then

$$\mathbf{U}\cdot\mathbf{V}=|\mathbf{U}||\mathbf{V}|\cos\theta=\cos\theta$$

or
$$\cos\theta=l_1l_2+m_1m_2+n_1n_2 \qquad \text{(V.T.U., 2003)}$$

Hence
$$\sin^2\theta=1-\cos^2\theta \qquad \ldots(1)$$

Cor. 1.
$$\sin^2\theta=1-\cos^2\theta=1-l_2^2+m_1^2+n_1^2$$
$$=l_1^2+m_1^2+n_1^2-l_2^2+m_1^2+n_1^2=(l_1^2+m_1^2+n_1^2)-(l_1l_2+m_1m_2+n_1n_2)^2$$
$$=l_1m_2^2+l_1n_2^2+m_1^2-2l_1l_2m_1m_2+m_1n_2^2$$

$$\therefore\quad \sin\theta=\sqrt{\Sigma}(l_1m_2-m_1l_2)^2. \qquad \ldots(2)$$

Cor. 2. The condition that the lines whose direction cosines are l_1, m_1, n_1 and l_2, m_2, n_2 should be perpendicular is

$$l_1l_2+m_1m_2+n_1n_2=0 \qquad \ldots(3)$$

and parallel is
$$l_1=l_2,\ m_1=m_2,\ n_1=n_2 \qquad \ldots(4)$$

These conditions are obtained from (1) and (2).

Cor. 3. The angle between two lines whose direction ratios are a, b, c and a', b', c' is given by

$$\cos\theta=\frac{aa'+bb'+cc'}{\sqrt{(a^2+b^2+c^2)}\sqrt{(a'^2+b'^2+c'^2)}}$$

or
$$\sin\theta=\frac{\sqrt{(b^2c'^2+c^2b'^2+c^2a'^2+a^2c'^2+a^2b'^2+b^2a'^2-2abb'a'-2bcc'b'-2caa'c')}}{\sqrt{(a^2+b^2+c^2)}\sqrt{(a'^2+b'^2+c'^2)}}$$

For these lines to be perpendicular $\cos\theta=0$, i.e., $aa'+bb'+cc'=0$. For parallel $\theta=0$ i.e., $\sin\theta=0$, i.e., $\frac{a}{a'}=\frac{b}{b'}=\frac{c}{c'}$.

IX. **Projection of the line joining two points (x_1, y_1, z_1) and (x_2, y_2, z_2) on a line whose direction cosines are l, m, n.**

$$PQ=x_2-x_1)l+m(y_2-y_1)+n(z_2-z_1)$$

Let
$$\vec{OP}=x_1\mathbf{I}+y_1\mathbf{J}+z_1\mathbf{K},\ \vec{OQ}=x_2\mathbf{I}+y_2\mathbf{J}+z_2\mathbf{K}$$

$$\therefore\quad \vec{PQ}=(x_2-x_1)\mathbf{I}+(y_2-y_1)\mathbf{J}+(z_2-z_1)\mathbf{K}$$

Also unit vector $\mathbf{U}$ along the given line is $=l\mathbf{I}+m\mathbf{J}+n\mathbf{K}$.

$\therefore$ Projection of PQ on the given line $=\vec{PQ}\cdot\mathbf{U}$

$$=l(x_2-x_1)+m(y_2-y_1)+n(z_2-z_1)$$

Example 8.6. Find the volume and equation of the triangle of which two sides are $\mathbf{I}-2\mathbf{J}+3\mathbf{K}, 2\mathbf{I}+3\mathbf{J}-4\mathbf{K}$ and $3\mathbf{I}-\mathbf{J}+2\mathbf{K}$.

Solution. Let $\vec{OA}=\mathbf{I}-2\mathbf{J}+3\mathbf{K}$, $\vec{OB}=2\mathbf{I}+\mathbf{J}-4\mathbf{K}$, $\vec{OC}=3\mathbf{I}-\mathbf{J}+2\mathbf{K}$

Then
$$\vec{BC}=\mathbf{I}-3\mathbf{J}+2\mathbf{K}$$

$$\vec{OA}=-2\mathbf{I}-\mathbf{J}$$

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* Named after the German mathematician Hermann Amandus Schwarz (1843–1921) who is known for his work in mathematical analysis, especially of minimal surfaces and of conformal geometry. He succeeded Kirchhoff in Berlin University.

and $\vec{AD} = \hat{i} + 3\hat{j} - 2\hat{k}$

$\therefore BC = \sqrt{14}, CA = \sqrt{2}, AB = \sqrt{18}.$

Direction cosines of AB and AC being

$1/\sqrt{2}, 3/\sqrt{18}, -3/\sqrt{18}$ and $1/\sqrt{2}, 1/\sqrt{2}, 0.$

We have $\cos A = \frac{1}{\sqrt{18}} \cdot \frac{2}{\sqrt{2}} + \frac{3}{\sqrt{18}} \cdot \frac{1}{\sqrt{2}} + \frac{-3}{\sqrt{18}} \cdot 0 = \sqrt{\frac{1}{2}}$

i.e., $\angle A = \cos^{-1} \sqrt{\frac{1}{2}}$. Again direct. of BC and BA being

$1/\sqrt{14}, -3/\sqrt{14}, 2/\sqrt{14}$ and $-1/\sqrt{18}, -3/\sqrt{18}, 3/\sqrt{18}$;

we have $\cos B = \frac{1}{\sqrt{14}} \cdot \frac{-1}{\sqrt{18}} + \frac{-3}{\sqrt{14}} \cdot \frac{-3}{\sqrt{18}} + \frac{2}{\sqrt{14}} \cdot \frac{3}{\sqrt{18}} = \sqrt{\frac{13}{21}}$, i.e., $\angle B = \cos^{-1} \sqrt{\frac{13}{21}}$

Finally, d.c.'s of CA and CB being $-2/\sqrt{2}, -1/\sqrt{2}, 0$ and $-1/\sqrt{14}, 3/\sqrt{14}, -2/\sqrt{14}$;

we have $\cos C = \frac{-2}{\sqrt{2}} \cdot \frac{-1}{\sqrt{14}} + \frac{-1}{\sqrt{2}} \cdot \frac{3}{\sqrt{14}} + 0 \cdot \frac{-2}{\sqrt{14}} = 0$, i.e., $\angle C = 90°$

Example 3.8. *Prove that the right bisectors of the sides of a triangle meet at the circumcentre.*

Solution. Let ABC, BCD, CAD be the vertices of the triangle ABC. The mid points of the sides BC, CA and AB are

$$D\left(\frac{B+C}{2}\right), E\left(\frac{C+A}{2}\right), F\left(\frac{A+B}{2}\right)$$

Let the perpendiculars at D and E to BC and CA respectively intersect at the point $P(R)$. Then $\vec{EP} \cdot \vec{BC} = 0$

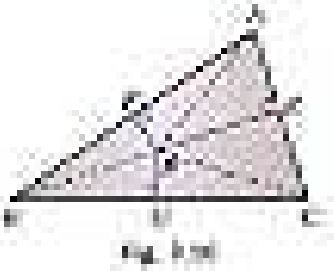

Fig. 3.16

i.e., $\left(R - \frac{B+C}{2}\right) \cdot (C - B) = 0$ ...(1)

and $\vec{FP} \cdot \vec{CA} = 0$, i.e., $\left(R - \frac{C+A}{2}\right) \cdot (A - B) = 0$ ...(2)

Adding (1) and (2), we get $\left(R - \frac{A+B}{2}\right) \cdot (A - B) = 0$

which shows that FP is perpendicular to AB. Hence the result.

Further $PA = PB$ or $|PA| = |PB|$

or $(A - R)^2 = (B - R)^2$ or $A^2 - 2A \cdot R = B^2 - 2B \cdot R$

or $\left(R - \frac{A+B}{2}\right) \cdot (A - B) = 0$, which is true.

Example 3.9. *If the distance between two points P and Q is d and p_1, p_2, p_3 are the projections of PQ on the coordinate planes, prove that $d^2 = p_1^2 + p_2^2 + p_3^2$.*

Solution. Let P be (x_1, y_1, z_1) and Q be (x_2, y_2, z_2). Then

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2.$$

The feet of the perpendiculars drawn from P and Q on the XY plane are the projections of P and Q on this plane. If these are L and M, then L is $(x_1, y_1, 0)$ and M is $(x_2, y_2, 0)$

$\therefore$ p_1 = projection of PQ on XY plane, i.e., LM

or $p_1^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

Similarly, $p_2^2 = (y_2 - y_1)^2 + (z_2 - z_1)^2$ and $p_3^2 = (z_2 - z_1)^2 + (x_2 - x_1)^2$

$\therefore$ $p_1^2 + p_2^2 + p_3^2 = 2[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2] = 2d^2$.

Example 3.8. Prove that the diagonals of a parallelopiped meet at a point and bisect each other. Also find the angle between any two diagonals. (U.P.T.U., 2006 ; Osmania, 2000 S)

Solution. Take O, a corner of the cube as origin and OA, OB, OC the three edges through it, as the axes. Let $OA = OB = OC = a$. Then the coordinates of the corners are as shown in Fig. 3.17.

The four diagonals are OP, AA', BB' and CC'.

Clearly, direction cosines of OP are

$$\frac{a-0}{\sqrt{3a^2}}, \frac{a-0}{\sqrt{3a^2}}, \frac{a-0}{\sqrt{3a^2}} \text{ i.e., } \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}.$$

Similarly, direction cosines of AA' are $-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$;

Similarly, direction cosines of BB' are $\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$;

and Similarly direction cosines of CC' are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$.

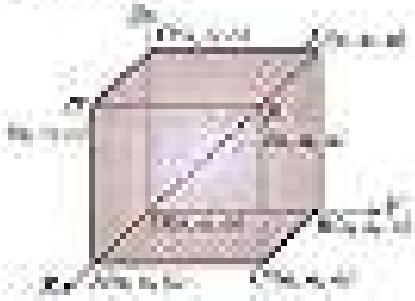

Fig. 3.17

Let l, m, n be the direction cosines of the given line which makes angles α, β, γ with OP, AA', BB', CC' respectively. Then

$$\cos\alpha = \frac{1}{\sqrt{3}}(l + m + n); \cos\beta = \frac{1}{\sqrt{3}}(-l + m + n)$$

$$\cos\gamma = \frac{1}{\sqrt{3}}(l - m + n), \cos\delta = \frac{1}{\sqrt{3}}(l + m - n)$$

Squaring and adding, we get

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{1}{3}[(l + m + n)^2 + (-l + m + n)^2 + (l - m + n)^2 + (l + m - n)^2]$$

$$= \frac{1}{3}[l^2 + m^2 + n^2] = \frac{4}{3}, \qquad [\because \ l^2 + m^2 + n^2 = 1]$$

Example 3.9. *If the origin is the centroid of the parallelopiped whose vertices are (a_1, b_1, c_1), (a_2, b_2, c_2), (a_3, b_3, c_3) and (a_4, b_4, c_4), then show that the angle between the diagonals joining*

(a_1, b_1, c_1) to $(-a_2, -b_2, -c_2)$ and (a_3, b_3, c_3) to $(-a_4, -b_4, -c_4)$ is $\cos^{-1}\left[\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}\right]$.

Solution. Let $OA = a$, $OB = b$, $OC = c$ be the edges of a rectangular parallelopiped. Then the coordinates of the corners are as shown in Fig. 3.18. The four diagonals taken in pairs are (i) OP, AA', (ii) BB', CC', (iii) OP, CC', (iv) AA', BB', (v) AA', DD' and (vi) BB', CC'.

Let the angles between these pairs of diagonals be $\theta_1, \theta_2, \ldots \theta_6$ respectively. Clearly d.r.'s of OP are a, b, c, d.r.'s of AA' are $-a, b, c$, d.r.'s of BB' are $a, -b, c$ and d.r.'s of CC' are $a, b, -c$.

For the pair (i) i.e., OP, AA'):

$$\cos\theta_1 = \frac{-a^2 + b^2 + c^2}{\sqrt{a^2 + b^2 + c^2}\sqrt{a^2 + b^2 + c^2}} = \frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2}$$

Similarly, $\cos\theta_2 = \frac{a^2 - b^2 + c^2}{a^2 + b^2 + c^2}$; $\cos\theta_3 = \frac{a^2 + b^2 - c^2}{a^2 + b^2 + c^2}$;

$\cos\theta_4 = \frac{-a^2 - b^2 + c^2}{a^2 + b^2 + c^2}$; $\cos\theta_5 = \frac{-a^2 + b^2 - c^2}{a^2 + b^2 + c^2}$;

$\cos\theta_6 = \frac{a^2 - b^2 - c^2}{a^2 + b^2 + c^2}$

Fig. 3.18

Thus, noting that at least one term in the numerator is negative, we have in general

$$\cos\theta = \frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2}$$

Example 2.10. Prove that the lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $fmn + gnl + hlm = 0$ are parallel if $af^2 + bg^2 + ch^2 = 2fgh$. (Rajasthan, 2003)

(I) Method for solving the problem

(II) Formula to be used: If l_1, m_1, n_1 and l_2, m_2, n_2 are the direction cosines of two lines then $l_1l_2 + m_1m_2 + n_1n_2$ are the roots of the quadratic in l.

Solution. Eliminating n from the given relations, we have

$$cn = -(al + bm), \left(-\frac{al + bm}{c}\right) + hm = 0 \quad \text{or} \quad an^2 + gl - cm - hlm + bm^2 = 0$$

or
$$a(2mn)^2 + g(-c - n - 2(l^2 + m^2)) + b = 0 \qquad (1)$$

If l_1, m_1, n_1; l_2, m_2, n_2 are the direction cosines of three lines then $l_1/n_1, l_2/n_2$ are the roots of the quadratic (1).

$$\therefore \quad \frac{l_1}{n_1} \cdot \frac{l_2}{n_2} = \frac{b}{a} \quad \text{or} \quad \frac{l_1l_2}{bc} = \frac{m_1m_2}{ca} = \frac{n_1n_2}{ab} \quad \text{(by symmetry)} = k \text{ (say)}.$$

The lines will be perpendicular if $l_1l_2 + m_1m_2 + n_1n_2 = 0 \Big(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\Big) = 0$

or i.e.
$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0.$$

The lines will be parallel if $l_1 = l_2, m_1 = m_2, n_1 = n_2$.

i.e., if
$$l_1/n_1 = l_2/n_2; \quad \text{i.e., if} \quad (-a - b)/c = -b/a$$

or if
$$a - a^2 - b = -bc \quad \text{or if} \quad a^2 + b^2 + c^2 = 2(ab + bc + ca)$$

or if
$$\pm\sqrt{a} = \sqrt{b} + \sqrt{c} \quad \text{or if} \quad \sqrt{a} \pm \sqrt{b} \pm \sqrt{c} = 0 \qquad \text{(Taking necessary signs)}$$

Example 2.11. Find the angle between the lines whose direction cosines are given by the equations $l + 2m + 3n = 0$ and $2fm + 2gn + 2hl = 0$.

Solution. Let the direction ratios of lines be given, by substituting $l = -2m - 3n$ in the second relation

$$fm(-2m - 3n) + gn(-2m) + h(-2m - 3n)m = 0$$

i.e.,
$$2fm^2 + 3fmn + 2gn^2 + hm^2 = 0 \quad \text{or} \quad fn^2 + 2gn + 2h^2 = 0$$

or
$$(m + n)(2m + 2n) = 0, \quad \text{i.e.,} \quad m + n = 0 \text{ or } 2m + 2n = 0$$

Now let us first solve the equations $l + 2m + 3n = 0$ and $m + n = 0$.

Then we get $m = -n$ and $l = -2n$, i.e., $\frac{l}{-2} = \frac{m}{-1} = \frac{n}{1}$ ...(1)

Similarly, solving the equations $l + 2m + 3n = 0$ and $2m + 2n = 0$,

We get
$$\frac{l}{1} = \frac{m}{-3} = \frac{n}{1} \qquad \ldots(2)$$

(1) and (2) give the direction ratios of the two lines.

If θ be the angle between these two lines, then

$$\cos\theta = \frac{1 \cdot 2(-1) + (-1)(-3) + 1 \cdot 2}{\sqrt{(2^2 + (-1)^2 + 1^2}\sqrt{(1^2 + (-3)^2 + 2^2}} = \frac{1}{6}, \quad \text{i.e.,} \quad \theta = \cos^{-1}\left(\frac{1}{6}\right).$$

EXERCISE 2.3

1. If $A = i + 2j + 3k$, $B = -i + 2j + k$ and $C = 3i + j + 2k$, find a vector that is perpendicular to both $A + B$ and $B + C$.

2. (i) Prove that $\left(\frac{A}{|A|} - \frac{B}{|B|}\right)$ is perpendicular to $(A + B)$.

(ii) Are the given vectors coplanar? $A = 2i - 3j + 4k$, $B = -i + 2j - 2k$ and $C = 3i - 4j + 6k$.

3. If $|A| = 3$, $|B| = 4$ and $|A - B| = 5$, find $|A + B|$. Are the vectors A and B are perpendicular to each other?

4. If $\mathbf{A} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, show that $\mathbf{A} + \mathbf{B}$ is perpendicular to $\mathbf{A} - \mathbf{B}$. Also calculate the angle between $2\mathbf{A} + \mathbf{B}$ and $\mathbf{A} + 2\mathbf{B}$.

5. Show that the three vectors $\mathbf{A}$, $\mathbf{B}$ with direction ratios l_1, m_1, n_1; l_2, m_2, n_2 and l_3, m_3, n_3 are coplanar if

$$\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$$

6. Find the projection of the vector $\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$ on the line $\frac{x-1}{2} = \frac{y+1}{-2} = \frac{z-3}{1}$.

7. The projection of a line on the coordinate axes are 12, 4, 3. Find the length and direction cosines of the line.
(Bhopal, 2005)

8. Show by vector method that the angle in a semi-circle is a right angle.

9. Show by vector method that the diagonals of a rhombus intersect at right angles.

10. Prove that the medians of a triangle meet at a point (called the centroid).

11. $ABCD$ is a tetrahedron. Prove that the edges BC and AD at right angles to opposite edges AD and BC respectively. Show that the third pair of opposite edges AB and CD are also at right angles.

12. Find the angle between the lines whose direction ratios are given by $l + m + n = 0$, $l^2 + m^2 + n^2 = 0$.
(Rajasthan, 2006)

13. Show that the lines whose direction cosines are given by the equations $al + bm + cn = 0$, $fmn + gnl + hlm = 0$ are parallel or perpendicular.
(Delhi, 2004)

14. Find the equation of the plane through the point $(1, 2, 3)$ and parallel to the lines whose direction ratios are given by $l + m + n = 0$, $al^2 + bm^2 + cn^2 = 0$.
(a) perpendicular if $\frac{a-b}{f^2} + \frac{b-c}{g^2} + \frac{c-a}{h^2} = 0$ (b) parallel if $f^2(b-c) + g^2(c-a) + h^2(a-b) = 0$

15. Show that the straight lines whose direction cosines are given by the equations
$al + bm + cn = 0$, $fmn + gnl + hlm = 0$ are perpendicular if $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$.
(Osmania, 2003)
(a) parallel if $\sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$

17. Show that the angle between any two diagonals of a cube is $\cos^{-1}\frac{1}{3}$.
(UPTU, 2008; Jammu, 2005)

18. (x_1, y_1, z_1), (x_2, y_2, z_2) and (x_3, y_3, z_3) are the direct ends of three mutually perpendicular lines. Prove that the locus of their point of intersection is $(x-x_1)(x-x_2) + (y-y_1)(y-y_2) + (z-z_1)(z-z_2) = 0$.
(U.P.T.U. 2006)

19. AB, BC are the crossroads of a dancing floor of a motor race track which is inclined to the vertical. AB and BC are parallel to the coast. If the angles BOC, DOA and AOB are equal to α, β, γ respectively, prove that

$$\cos(\alpha + \gamma)\cos(\alpha + \beta) = \cos\gamma$$

3.6 VECTOR, OR CROSS PRODUCT

(1) Definition. The vector, or cross product of two vectors $\mathbf{A}$ and $\mathbf{B}$ is defined as a vector such that

(i) its magnitude is $AB \sin\theta$, θ being the angle between $\mathbf{A}$ and $\mathbf{B}$,

(ii) its direction is perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$,

and (iii) it forms with $\mathbf{A}$ and $\mathbf{B}$ a right-handed system.

If $\mathbf{N}$ is a unit vector normal to the plane of $\mathbf{A}$ and $\mathbf{B}$ ($\mathbf{A}$, $\mathbf{B}$, $\mathbf{N}$ form a right-handed system), then

$$\mathbf{A} \times \mathbf{B} = AB \sin\theta\, \mathbf{N}.$$

(2) Geometrical Interpretation. $\mathbf{A} \times \mathbf{B}$ represents in magnitude twice the area of the triangle having the vectors $\mathbf{A}$ and $\mathbf{B}$ as its adjacent sides.

If $\mathbf{N}$ is a unit vector normal to the plane of the triangle OAB, then

$$\mathbf{A} \times \mathbf{B} = AB \sin\theta\, \mathbf{N}$$

$$= 2\left(\frac{1}{2} AB \sin\theta\right)\mathbf{N} = 2\,\Delta OAB\,\mathbf{N} = 2\,\vec{\Delta}OAB.$$

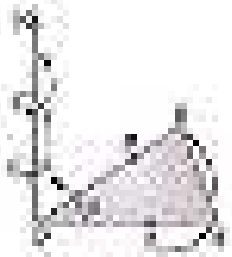

Fig. 3.29

(3) Properties and other results

I. Vector product of two vectors is not commutative,

$$\mathbf{A} \times \mathbf{B} \neq \mathbf{B} \times \mathbf{A}. \text{ In fact, } \mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$$

So $\quad A \times B$ is a vector $\perp$ to the plane of A and B $\quad$ or $\quad$ $\perp$ to $\vec{OA}$ & $\vec{OB}$.

and $\quad B \times A$ is a vector $\perp$ to the plane of B and A $\quad$ or $\quad$ $\perp$ to $\vec{OB}$ & $\vec{OA}$.

II. The necessary and sufficient condition for two non-zero vectors to be parallel is that their vector product should be zero.

If A & B are parallel, the angle θ between them is 0 and $180°$ so that $\sin\theta = 0$, and in each $A \times B = 0$.

Conversely, when $\quad A \times B = 0$, $\sin\theta = 0$ $\quad$ i.e., $\quad \theta = 0$ or $180°$. In particular, $A \times A = 0$. $\quad$ i.e., $\quad A \parallel B$ or $\theta = 0$

III. For the orthonormal vector triad I, J, K, we have the relations :

$$I \times I = J \times J = K \times K = 0$$

$$I \times J = K, \quad J \times I = -K$$

$$J \times K = I, \quad K \times J = -I$$

$$K \times I = J, \quad I \times K = -J.$$

IV. Relation between scalar and vector products.

We have $\quad |A \cdot B|^2 + |A \times B|^2 = A^2B^2\cos^2\theta + A^2B^2\sin^2\theta = A^2B^2 = (A \cdot A)(B \cdot B)$

$$(A \times B)^2 = A^2B^2 - (A \cdot B)^2.$$

V. Vector product of two vectors is distributive :

i.e., $\quad (A + B) \times C = A \times C + B \times C$.

VI. Analytical expression for the vector product.

I) $\quad A = a_1I + a_2J + a_3K, B = b_1I + b_2J + b_3K$ then $A \times B = \begin{vmatrix} I & J & K \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

For we get

$$A \times B = (a_2b_3 - a_3b_2)I + (a_3b_1 - a_1b_3)J + (a_1b_2 - a_2b_1)K$$

which verifies the required result.

Example 2.16. *If $A = 2I + 3J - K$, $B = 2I - J + 2K$, find a unit vector N perpendicular to each of A and B, such that A, B, N form a right handed system. Also find the angle between the vectors A and B.*

Solution. Since $\quad A \times B = \begin{vmatrix} I & J & K \\ 2 & 3 & -1 \\ 2 & -1 & 2 \end{vmatrix} = 5I - 6J - 13K$

and $\quad |A \times B| = \sqrt{(5)^2 + (-6)^2 + (-13)^2} = \sqrt{230}$.

$\therefore$ Unit vector $N \perp$ to A and B $\quad = \dfrac{A \times B}{|A \times B|} = (5I - 6J - 13K)/\sqrt{230}$

Also $\quad a = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14}$ and $b = 3$.

If θ is the angle between A and B, then $|A \times B| = ab\sin\theta$, i.e., $\sqrt{230} = \sqrt{14} \cdot 3\sin\theta$

Then $\quad \sin\theta = \sqrt{115}/\sqrt{21}$, whence $\theta = 82°49'$.

Example 2.17. *(a) Prove that the area of the triangle whose vertices are A, B, C is*

$$\frac{1}{2}|B \times C + C \times A + A \times B|$$

(b) Calculate the area of the triangle whose vertices are $A(1, 0, -1)$, $B(2, 1, 0)$, $C(0, 2, 1)$. **(U.P.T.U. 2003)**

Solution. (a) Let $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ be the vertices of the triangle ABC (Fig. 2.23) and O, the origin, so that

$$\vec{AC} = \vec{OC} - \vec{OA} = C - A$$

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and $\vec{BA} = \vec{OA} - \vec{OB} = \mathbf{A} - \mathbf{B}$

$\therefore$ Vector area of $\triangle ABC$

$$= \frac{1}{2}[\vec{BC} \times \vec{BA}] = \frac{1}{2}[\mathbf{C} - \mathbf{B}) \times (\mathbf{A} - \mathbf{B})]$$

$$= \frac{1}{2}[\mathbf{C} \times \mathbf{A} - \mathbf{C} \times \mathbf{B} - \mathbf{B} \times \mathbf{A} + \mathbf{B} \times \mathbf{B}]$$

$$= \frac{1}{2}[\mathbf{B} \times \mathbf{C} + \mathbf{C} \times \mathbf{A} + \mathbf{A} \times \mathbf{B}] \quad [\because \mathbf{B} \times \mathbf{B} = \mathbf{0}]$$

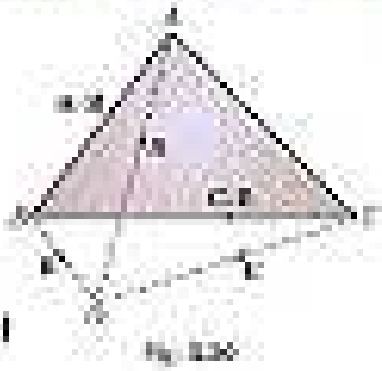


Fig. 1.30

Thus area of $\triangle ABC = \frac{1}{2}[\mathbf{B} \times \mathbf{C} + \mathbf{C} \times \mathbf{A} + \mathbf{A} \times \mathbf{B}]$.

(ii) Let O be the origin so that

$$\vec{OA} = \mathbf{I} - \mathbf{K}, \vec{OB} = 2\mathbf{I} + \mathbf{J} + 2\mathbf{K} \text{ and } \vec{OC} = \mathbf{J} - 3\mathbf{K}$$

Then $\vec{BC} = \vec{OC} - \vec{OB} = -2\mathbf{I} - 3\mathbf{K}$

and $\vec{BA} = \vec{OA} - \vec{OB} = -\mathbf{I} - \mathbf{J} - 4\mathbf{K}$

$\therefore$ Vector area of $\triangle ABC = \frac{1}{2}(\vec{BC} \times \vec{BA}) = \frac{1}{2}\begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ -2 & 0 & -3 \\ -1 & -1 & -4 \end{vmatrix}$

Thus area of $\triangle ABC = \frac{1}{2}|-3\mathbf{I} - 8\mathbf{J} + 2\mathbf{K}| = \frac{1}{2}\sqrt{77}$.

Example 1.24. *In a triangle ABC, D, E, F are the mid points of the sides BC, CA, AB respectively. Show that*

$$\triangle DEF = \triangle FBD = \frac{1}{4}\triangle ABC.$$

Solution. Take B as the origin and let the position vectors of C and A be $\mathbf{C}$ and $\mathbf{A}$ (Fig. 1.31) so that the position vectors of D, E, F are

$$\frac{1}{2}\mathbf{C}, \frac{1}{2}\mathbf{C} + \frac{1}{2}\mathbf{A}, \frac{1}{2}\mathbf{A}.$$

$$\therefore \Delta\vec{EDF} = \frac{1}{2}(\vec{DE} \times \vec{DF}) = \frac{1}{2}\left[\left(\frac{\mathbf{C}}{2} + \frac{\mathbf{A}}{2}\right) - \frac{\mathbf{C}}{2}\right] \times \left[\frac{\mathbf{A}}{2} - \frac{\mathbf{C}}{2}\right]$$

$$= \frac{1}{8}[\mathbf{A} \times (\mathbf{A} - \mathbf{C})] = \frac{1}{8}\mathbf{C} \times \mathbf{A} = \frac{1}{4}\Delta\vec{ABC}$$

$$\Delta\vec{FBD} = \frac{1}{2}[\vec{FD} \times \vec{FB}] = \frac{1}{2}[\mathbf{B} - \mathbf{A}) \times (\mathbf{C}]$$

$$= \frac{1}{2}\mathbf{C} \times \mathbf{A} = \frac{1}{4}\Delta\vec{ABC}. \text{ Hence the result.}$$

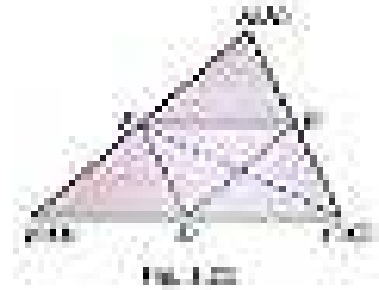


Fig. 1.31

Example 1.25. *Prove that*
(i) $\cos(\theta + \phi) = \cos\theta\cos\phi + \sin\theta\sin\phi$ and
(ii) $\sin(\theta + \phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$.

Solution. Let $\mathbf{I}, \mathbf{J}$ denote unit vectors along two perpendicular lines OX, OY so that

$$\mathbf{I} \times \mathbf{I} = \mathbf{J} \times \mathbf{J} = 1, \mathbf{I} \times \mathbf{J} = 0$$

and $\mathbf{I} \times \mathbf{I} = \mathbf{J} \times \mathbf{J} = 0$.

Let $\angle POX = \theta$ and $\angle QOX = \phi$,

so that $\angle POQ = \theta - \phi$.

If $OP = p$ and $OQ = q$, then the coordinates of P are $(p\cos\theta, p\sin\theta)$ and those of Q are $(q\cos\phi, q\sin\phi)$ so that

$$\vec{OP} = p\cos A\hat{i} - p\sin A\hat{j}$$

$$\vec{OQ} = q\cos B\hat{i} + q\sin B\hat{j}$$

Then $|\vec{OP} \times \vec{OQ}| = |[p\cos A\hat{i} - p\sin A\hat{j}] \times [q\cos B\hat{i} + q\sin B\hat{j}]|$

$= pq\,|\cos A\sin B(\hat{i}\times\hat{j}) - \sin A\cos B(\hat{j}\times\hat{i})|$, $\because \hat{i}\times\hat{i} = 0 = \hat{j}\times\hat{j}$ and $\hat{i}\times\hat{j} = -\hat{j}\times\hat{i} = \hat{k}$

$= pq\,|\cos A\sin B + \sin A\cos B|$ $\because |\hat{i}\times\hat{j}| = 1$

Also $|\vec{OP} \times \vec{OQ}| = pq\sin(\theta_1 - \theta_2)$. Equating the two expressions, we get (1).

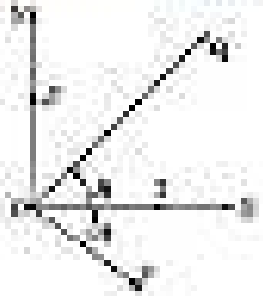


Fig. 3.11

Similarly, (ii) follows from $\vec{OP}\cdot\vec{OQ} = pq\cos(\theta_1 - \theta_2)$.

Example 3.16 In any triangle ABC, prove that :

(i) $a\sin(B-C) + b\sin(C-A) + c\sin(A-B) = 0$ (Sine formula)

(ii) $a = b\cos C + c\cos B$ (Projection formula)

(iii) $a^2 = b^2 + c^2 - 2bc\cos A$. (Cosine formula)

Solution. From $\triangle ABC$, we have $\vec{BC} + \vec{CA} + \vec{AB} = 0$

$$\Rightarrow \quad \vec{CA} + \vec{AB} = -\vec{BC} \qquad \ldots(1)$$

(i) Multiplying (1) vectorially by $\vec{AB}$, we get

$$\vec{CA}\times\vec{AB} = -\vec{BC}\times\vec{AB}$$

$$\Rightarrow \quad |\vec{CA}\times\vec{AB}| = |\vec{BC}\times\vec{AB}|$$

$$\therefore \quad bc\sin(\pi - A) = ac\sin(\pi - B)$$

$$\Rightarrow \quad bc\sin A = ac\sin B$$

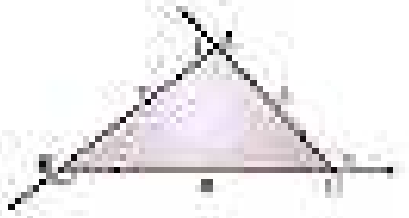


Fig. 3.12

Similarly, multiplying (1) vectorially by $\vec{CA}$, we get

$ab\sin A = cb\sin C$, which gives the desired result.

(ii) Multiplying (1) scalarly by $\vec{BC}$, we get $\vec{CA}\cdot\vec{BC} + \vec{AB}\cdot\vec{BC} = -(\vec{BC})^2$

$\therefore \quad bc\cos(\pi - C) + ca\cos(\pi - B) = -a^2$ or $a = b\cos C + c\cos B$.

(iii) Squaring (1), we get

$$(\vec{CA})^2 + (\vec{AB})^2 + 2\vec{CA}\cdot\vec{AB} = (\vec{BC})^2$$

i.e., $b^2 + c^2 2bc\cos(\pi - A) = a^2$ or $a^2 = b^2 + c^2 - 2bc\cos A$.

PROBLEMS 3.5

1. Given $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{B} = \hat{i} - 2\hat{j} + 2\hat{k}$, (a) find $\vec{A}\times\vec{B}$ and the unit vector perpendicular to both $\vec{A}$ and $\vec{B}$. Also (b) find the sine of the angle between $\vec{A}$ and $\vec{B}$.

2. If $\vec{A}$ and $\vec{B}$ are unit vectors and θ is the angle between them, prove that $\sin\frac{\theta}{2} = \frac{1}{2}|\vec{A} - \vec{B}|$.

3. Find a unit vector normal to the plane of $\vec{A} = 2\hat{i} - 3\hat{j} - 2\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} - 2\hat{k}$.

4. For any vector $\vec{A}$, show that $|\vec{A}\times\hat{i}|^2 + |\vec{A}\times\hat{j}|^2 + |\vec{A}\times\hat{k}|^2 = 2|\vec{A}|^2$.

5. By vector method, find the area of the triangle whose vertices are (i) $(1, 1, 2), (2, 3, 1)$ and $(3, 1, 3)$ (ii) $(0, 0, 0), (1, 2, 3)$ and $(2, 3, 5)$.

6. (a) Prove that the vector area of the quadrilateral $ABCD$ is $\frac{1}{2}\vec{AC}\times\vec{BD}$.

(b) If $2\hat{i} + \hat{j} - 2\hat{k}$ and $\hat{i} - 3\hat{j} - 3\hat{k}$ are the diagonals of a parallelogram, find its area.

7. Given vectors $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{B} = 2\hat{i} + \hat{j} - \hat{k}$. Find the projection of $\vec{A} \times \vec{B}$ in the direction $\hat{i} - 2\hat{k}$.

8. If $\vec{A} + \vec{B} + \vec{C} = 0$, prove that $\vec{A} \times \vec{B} = \vec{B} \times \vec{C} = \vec{C} \times \vec{A}$ and interpret it geometrically.

9. Show that the perpendicular distance of the point $\vec{C}$ from the line joining the points $\vec{A}$ and $\vec{B}$ is $\frac{|(\vec{B} - \vec{A}) \times (\vec{C} - \vec{A})|}{|\vec{B} - \vec{A}|}$.

10. In $\triangle ABC$, D is one of the points of trisection of BC and E is the mid-point of AC. Prove that $\text{ar}(\triangle ADE) = \frac{1}{6}\text{ar}(\triangle ABC)$.

11. Prove by vector methods that

(i) $\sin(A - B) = \sin A \cos B - \cos A \sin B$, (ii) $\cos(A - B) = \cos A \cos B + \sin A \sin B$ and (iii) $\sin(A + B) = \sin A \cos B + \cos A \sin B$. (Calicut, 2000)

12. In a $\triangle ABC$, prove by vector method, that

(i) $a = b\cos C + c\cos B$, (ii) $b^2 = a^2 + c^2 - 2ac\cos B$.

3.17 PHYSICAL APPLICATIONS

(1) Work done as a scalar product. If a constant force $\vec{F}$ acting on a particle displaces it from the position A to position B, then

Work done = (resolved part of $\vec{F}$ in the direction of $\vec{AB}$) $\times$ $\vec{AB}$

$$= F\cos\theta \cdot AB = \vec{F} \cdot \vec{AB}$$

Thus, the work done by a constant force is the scalar (or dot) product of the vectors representing the force and the displacement.

Fig. 3.34

Example 3.17. *Constant forces* $\vec{P} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ *and* $\vec{Q} = \hat{i} + 2\hat{j} - \hat{k}$ *act on a particle. Determine the work done when the particle is displaced from A to B, the position vectors of A and B being* $3\hat{i} - 2\hat{j} + 4\hat{k}$ *and* $4\hat{i} + \hat{j} - 2\hat{k}$ *respectively.*

Solution. Resultant force $\vec{R} = \vec{P} + \vec{Q} = 3\hat{i} - \hat{j} + 2\hat{k}$

and $\vec{AB} = \vec{OB} - \vec{OA} = (4\hat{i} + \hat{j} - 2\hat{k}) - (3\hat{i} - 2\hat{j} + 4\hat{k}) = \hat{i} + 3\hat{j} - 6\hat{k}$

$\therefore$ Work done $= \vec{R} \cdot \vec{AB} = (3\hat{i} - \hat{j} + 2\hat{k}) \cdot (\hat{i} + 3\hat{j} - 6\hat{k})$

$= 1 \cdot 3 - 3 \cdot 1 + 2 \cdot (-6) = -12$ units.

Fig. 3.35

(2) Normal flux. Consider the flow of a liquid through an element of area δS with velocity $\vec{V}$ inclined at an angle θ to the unit vector $\hat{n}$ normal to the surface δS (Fig. 3.36).

$\therefore$ Normal flux of the liquid through δS per unit time

$$= V\cos\theta \cdot \delta S = \vec{V} \cdot \hat{n}\,\delta S.$$

Thus, *the rate of normal flux per unit area* $= \vec{V} \cdot \hat{n}$

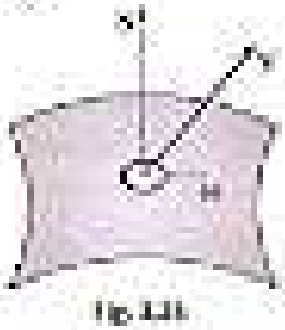


Fig. 3.36

Example 3.18. *Find the work done in moving a particle in a force field* $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ *along the straight line from* $(0, 0, 0)$ *to* $(2, 1, 3)$.

(3) Moment of a force about a point. Suppose the moment of the force $\vec{F}$ acting at the point P about the point A is required.

Draw $\vec{AM}$ perpendicular to the line of action of $\vec{F}$ (Fig. 3.37). If θ be the angle between $\vec{AP}$ and $\vec{F}$ and $\hat{N}$ be a unit vector perpendicular to their plane, then $\vec{AP} \times \vec{F} = AP\,F\sin\theta\,\hat{N} = AM \cdot F\,\hat{N} = F \cdot AM\,\hat{N}$.

Clearly, (i) the magnitude of $\vec{AP} \times \vec{F} = F \cdot AM$ which is the numerical measure of the moment of $\vec{F}$ about A,

and (ii) the direction of $\vec{AP} \times \vec{F}$ is the direction of the moment of $\vec{F}$ about A.

Hence the moment (or torque) of $\vec{F}$ about A $= \vec{AP} \times \vec{F}$.

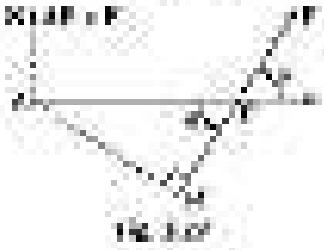


Fig. 3.37

Example 3.19. *Find the moment about the point* $\hat{i} - \hat{j} + \hat{k}$ *of a force represented by* $3\hat{i} + \hat{k}$ *acting through the point* $2\hat{i} + \hat{j} - 3\hat{k}$.

Solution. Let O be the origin and P be the point, moment about a side of the force $\vec{AB}$ through A, is required (Fig. 3.38).

$\therefore$ $\vec{OP} = 3\mathbf{i} + 4\mathbf{j} - 8\mathbf{k}$,

$\vec{OA} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$, and $\vec{AB} = 4\mathbf{j} - 8\mathbf{k}$.

Then, $\vec{PA} = \vec{OA} - \vec{OP} = -2\mathbf{i} - 5\mathbf{j} + 10\mathbf{k}$

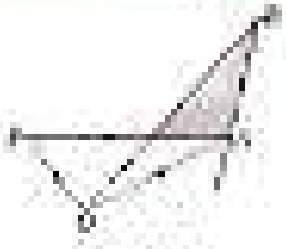


Fig. 3.28

$\therefore$ Moment of the force $\vec{AB}$ about P

$$= \vec{PA} \times \vec{AB} = (-2\mathbf{i} - 5\mathbf{j} + 10\mathbf{k}) \times (4\mathbf{j} - 8\mathbf{k})$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & -5 & 10 \\ 0 & 4 & -8 \end{vmatrix} = -\mathbf{i} - 16\mathbf{j} - 8\mathbf{k}$$

$\therefore$ Magnitude of the moment $= \sqrt{1 + 256 + 64} = 17.4$

(4) Moment of a force about a line.

Def. The moment of a force $\mathbf{F}$ about a line D is the resolved part along D of the moment of $\mathbf{F}$ about any point on D.

Example 3.18. *Find the moment about a line OA through the origin and $A(6, 2, 3)$ of the force $\mathbf{F} = 3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ acting at a point $(4, -4, 2)$. Find also the moment of $\mathbf{F}$ about the line AB where B is $(4, 2, 0)$.*

Solution. Let D be the given line through the origin O and F the force through $A(6, 2, 3)$.

Clearly, $\vec{OA} = 6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$

and the force $\mathbf{F} = 3\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$

Fig. 3.29

$\therefore$ Moment of $\mathbf{F}$ about $O = \vec{OA} \times \mathbf{F}$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & 2 & 3 \\ 3 & 5 & -2 \end{vmatrix} = -19\mathbf{i} + 24\mathbf{j} + 24\mathbf{k}$$

Thus the moment of $\mathbf{F}$ about the line D

= resolved part of the moment of $\mathbf{F}$ about O along D.

i.e., $(-19\mathbf{i} + 24\mathbf{j} + 24\mathbf{k}) \cdot \mathbf{D}$

$$= (-19\mathbf{i} + 24\mathbf{j} + 24\mathbf{k}) \cdot \frac{6\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}}{\sqrt{36 + 4 + 9}} = \frac{-38}{7} = -5.43$$

(5) Angular velocity of a rigid body

Let a rigid body be rotating about the axis OA with an angular velocity ω radians per second (Fig. 3.30). Let P be a point of the body such that $\vec{OP} = \mathbf{R}$ and $\angle POA = \theta$. Draw $PM \perp OA$.

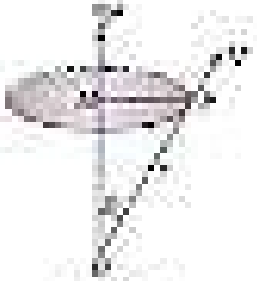


Fig. 3.30

Now if $\boldsymbol{\omega}$ be a unit vector $\perp$ to $\mathbf{R}$ then

$$\vec{\omega} \times \mathbf{R} = \omega \sin\theta\, \boldsymbol{\omega} \times \mathbf{R} = \omega \sin\theta\, \boldsymbol{\omega} \times \mathbf{R}$$

= Speed of P | $\mathbf{R}$

= velocity $\mathbf{V}$ of P in a direction $\perp$ to the plane AOP.

Hence $\mathbf{V} = \vec{\omega} \times \mathbf{R}$

Example 2.20 A rigid body is spinning with angular velocity 10 radians per second about an axis parallel to $2\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ passing through the point $\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$. Find the velocity of the point of the body whose position vector is $3\mathbf{i} + 4\mathbf{j} + \mathbf{k}$.

Solution. Unit vector along the direction of $\vec{\omega} = \frac{2\mathbf{i} + \mathbf{j} - 2\mathbf{k}}{\sqrt{4+1+4}} = \frac{1}{3}(2\mathbf{i} + \mathbf{j} - 2\mathbf{k})$

$\therefore$ Angular velocity $\quad \vec{\omega} = \frac{10}{3}(2\mathbf{i} + \mathbf{j} - 2\mathbf{k}) = 6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$

Let A be the point $\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and the point P of the body be $(3\mathbf{i} + 4\mathbf{j} + \mathbf{k})$ so that

$$\vec{AP} = (3\mathbf{i} + 4\mathbf{j} + \mathbf{k}) - (\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) = 2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$$

$\therefore$ Velocity vector of $P = \vec{V} = \vec{\omega} \times \vec{AP} = (6\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}) \times (2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k})$

$$= 3\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -2 \\ 2 & 2 & 4 \end{vmatrix} = 3(8\mathbf{i} - 16\mathbf{j} + 6\mathbf{k})$$

and its magnitude is $3\sqrt{(64 + 256 + 36)} = 3\sqrt{356}$.

EXERCISE 2.8

1. A particle acted on by constant forces $4\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $3\mathbf{i} + \mathbf{j} - \mathbf{k}$ is displaced from the point $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ to the point $5\mathbf{i} + 4\mathbf{j} + \mathbf{k}$. Find the total work done by the forces.
2. Forces $3\mathbf{i} - 5\mathbf{j} + 7\mathbf{k}$, $2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ and $-\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ act at the point P whose position vector is $4\mathbf{i} - 3\mathbf{j} - 2\mathbf{k}$. Determine the work done by the forces in a displacement of the particle to the point $Q(2, 1, -3)$.
   Also find the moment of this resultant at the point Q about the point P.
3. Forces of magnitudes 5, 3, 9 units act in the directions $3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$, $3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$, $2\mathbf{i} - 3\mathbf{j} - 6\mathbf{k}$ on a particle which is displaced from the point $(2, 1, -3)$ to $(4, -1, 5)$. Find the work done by the forces.
4. The point of application of the force $2\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ is displaced from the point A, $\mathbf{i} + \mathbf{j} + \mathbf{k}$ to the point B, $\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$. Find the work done by the force if the displacement takes place along the straight line AB.
5. A force $\mathbf{F} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ is applied at the point $(1, -1, 2)$. Find the moment of this force about the point $(2, -1, 1)$.
   (Kerala, 2006)
6. A force with components $(2, -1, 3)$ acts at a point P whose position vector relative to the origin O is $(1, 3, -2)$. Determine the moment of the force about the point Q whose position vector relative to O is $(2, 0, -1)$. Find the magnitude of this moment.
7. Find the moment of the force $\mathbf{F} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ acting at the point $(1, -2, 3)$ about $(2, 0, 1)$.
8. A force of 10 kg-wt is directed equally inclined to the coordinate axes through the point $(1, -2, 3)$. Find the magnitude of the moment of this force about a line through the origin and whose direction ratios are $(2, -1, 0)$.
9. A rigid body is rotating at 3 radians per second about an axis $\vec{OA}$, where A is the point $(1, 2, 2)$. If O is the origin, find the velocity of the particle of the body at the point $(4, 4, 3)$. All lengths are in cm.

2.8 PRODUCTS OF THREE OR MORE VECTORS

With any three vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}$, we can form the products $(\mathbf{A} \cdot \mathbf{B})\mathbf{C}$, $(\mathbf{A} \times \mathbf{B}) \cdot \mathbf{C}$ and $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$. The first being the product of a scalar $\mathbf{A} \cdot \mathbf{B}$ and a vector $\mathbf{C}$, represents a vector in the direction of $\mathbf{C}$. The second being the scalar product of vectors $\mathbf{A} \times \mathbf{B}$ and $\mathbf{C}$, represents a scalar and is usually called the *scalar product of three vectors*. The third being the vector product of the vectors $\mathbf{A} \times \mathbf{B}$ and $\mathbf{C}$, represents a vector and is usually known as the *vector product of three vectors*.

The reader must, however, note that the products of the form $\mathbf{A} \cdot \mathbf{B} \cdot \mathbf{C}$, $\mathbf{A} \times \mathbf{B} \times \mathbf{C}$ and $\mathbf{A}\mathbf{B} \times \mathbf{C}$ are meaningless.

In practical applications, we call for the vector products of more than three vectors. Such products if they occur can, in general, be reduced by using successively the expansion formula for vector triple products. As an illustration, we shall consider two products $(\mathbf{A} \times \mathbf{B}) \times (\mathbf{C} \times \mathbf{D})$ and $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \cdot \mathbf{D}$ of any four vectors, the former being a vector and the latter a scalar.

3.5 SCALAR PRODUCT OF THREE VECTORS

(1) Definition. If $\vec{A}, \vec{B}, \vec{C}$ be any three vectors then the scalar or dot product of $\vec{A} \times \vec{B}$ and $\vec{C}$ is called the scalar product of the three vectors $\vec{A}, \vec{B}, \vec{C}$ and is written as $(\vec{A} \times \vec{B}) \cdot \vec{C}$ or $[\vec{A}\vec{B}\vec{C}]$.

The ambiguity can arise by omitting the brackets in $(\vec{A} \times \vec{B}) \cdot \vec{C}$ as $\vec{A} \times (\vec{B} \cdot \vec{C})$ would be a vector product.

(2) Geometrical Interpretation. *The product* $\vec{A} \times \vec{B} \cdot \vec{C}$ *represents numerically the volume of a parallelopiped having* $\vec{A}, \vec{B}, \vec{C}$ *as coterminous edges.*

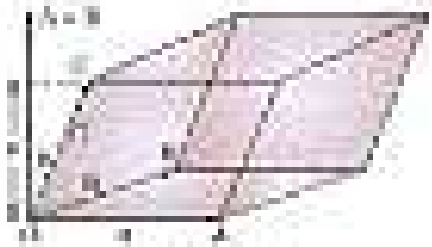


Fig. 3.10

Consider a parallelopiped with $\vec{OA} = \vec{A}$, $\vec{OB} = \vec{B}$, $\vec{OC} = \vec{C}$ as coterminous edges (Fig. 3.10).

Let V be the volume, a the area of each of the two faces parallel to the vectors $\vec{A}$ and $\vec{B}$ and x the perpendicular distance between these faces.

Then $|\vec{A} \times \vec{B}| = a$ and $|\vec{C}| \cos\theta = p$ or $-p$ according as $\vec{A}, \vec{B}, \vec{C}$ form a right-handed or left-handed triad.

$$\therefore \quad \vec{A} \times \vec{B} \cdot \vec{C} = |\vec{A} \times \vec{B}| \; |\vec{C}| \cos\theta = \pm ax = \pm V$$

Thus $[\vec{A}\vec{B}\vec{C}] = \pm$ *volume of the parallelopiped having* $\vec{A}, \vec{B}, \vec{C}$ *as a right handed or left handed triad.*

(Gujarat, 2005; J.N.T.U., 2006)

In particular, for an orthonormal right-handed vector triad $\vec{I}, \vec{J}, \vec{K}$,

$$\vec{I} \cdot \vec{J} \times \vec{K} = \vec{I} \cdot \vec{I} = 1, \; \vec{J} \cdot \vec{K} \times \vec{I} = \vec{J} \cdot \vec{J} = 1, \; \vec{K} \cdot \vec{I} \times \vec{J} = \vec{K} \cdot \vec{K} = 1.$$

(3) Properties and other results.

I. The condition for three vectors to be coplanar is that their scalar triple product should vanish.

If three vectors $\vec{A}, \vec{B}, \vec{C}$ are coplanar, then the volume of the parallelopiped so formed is zero, *i.e.*, $[\vec{A}\vec{B}\vec{C}] = 0$.

II. If any two vectors of a scalar triple product are equal, the product vanishes, i.e. $[\vec{A}\vec{A}\vec{C}] = 0$ *when either* $\vec{A} = \vec{B}$ *or* $\vec{B} = \vec{C}$, *or* $\vec{C} = \vec{A}$, *for in this case the parallelopiped has zero volume.*

III. Two important rules for evaluating a scalar triple product. A scalar triple product

(i) is independent of the position of the dot or cross,

and (ii) depends upon the cyclic order of the vectors.

It is easy to note that if $\vec{A}, \vec{B}, \vec{C}$ is a right-handed triad so are $\vec{B}, \vec{C}, \vec{A}$ and $\vec{C}, \vec{A}, \vec{B}$.

Moreover a parallelopiped having $\vec{A}, \vec{B}, \vec{C}$ as coterminous edges is the same as that having $\vec{B}, \vec{C}, \vec{A}$ or $\vec{C}, \vec{A}, \vec{B}$ as coterminous edges.

Thus, if V be the volume of this parallelopiped,

$$\vec{A} \times \vec{B} \cdot \vec{C} = V, \; \vec{B} \times \vec{C} \cdot \vec{A} = V, \; \vec{C} \times \vec{A} \cdot \vec{B} = V$$

Also, since $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$, we have

$$\vec{C} \cdot \vec{A} \times \vec{B} = \vec{A} \times \vec{B} \cdot \vec{C} = V$$
$$\vec{A} \cdot \vec{B} \times \vec{C} = \vec{B} \times \vec{C} \cdot \vec{A} = V$$
$$\vec{B} \cdot \vec{C} \times \vec{A} = \vec{C} \times \vec{A} \cdot \vec{B} = V$$

Thus
$$\begin{vmatrix} \vec{A} \times \vec{B} \cdot \vec{C} = \vec{A} \cdot \vec{B} \times \vec{C} \\ \vec{B} \times \vec{C} \cdot \vec{A} = \vec{B} \cdot \vec{C} \times \vec{A} \\ \vec{C} \times \vec{A} \cdot \vec{B} = \vec{C} \cdot \vec{A} \times \vec{B} \end{vmatrix} = V \qquad \text{...(1)}$$

Further a right-handed triad becomes left-handed when the cyclic order of the vectors is changed. Therefore $\vec{A}, \vec{C}, \vec{B}$; $\vec{B}, \vec{A}, \vec{C}$; $\vec{C}, \vec{B}, \vec{A}$ being left-handed triads, it follows that

$$\vec{A} \times \vec{C} \cdot \vec{B} = -V, \; \vec{B} \times \vec{A} \cdot \vec{C} = -V, \; \vec{C} \times \vec{B} \cdot \vec{A} = -V.$$

Thus
$$\begin{vmatrix} \vec{A} \times \vec{C} \cdot \vec{B} = \vec{A} \cdot \vec{C} \times \vec{B} \\ \vec{B} \times \vec{A} \cdot \vec{C} = \vec{B} \cdot \vec{A} \times \vec{C} \\ \vec{C} \times \vec{B} \cdot \vec{A} = \vec{C} \cdot \vec{B} \times \vec{A} \end{vmatrix} = -V \qquad \text{...(2)}$$

Thus, in any part of the above cyclic permutation $[\vec{A}\vec{B}\vec{C}]$ of terms the cyclic order of the letters is not changed but the cyclic position of the dot or $\times$ is changed.

The relations (1) and (2) can be concisely written as

$$[\vec{A}\vec{B}\vec{C}] = [\vec{B}\vec{C}\vec{A}] = [\vec{C}\vec{A}\vec{B}] = V \text{ and } [\vec{A}\vec{C}\vec{B}] = [\vec{B}\vec{A}\vec{C}] = [\vec{C}\vec{B}\vec{A}] = -V.$$

IV. *Scalar triple product is distributive*

$$[\vec{A}, \vec{B}+\vec{C}, \vec{D}+\vec{E}] = [\vec{A}\vec{B}\vec{D}] + [\vec{A}\vec{B}\vec{E}] + [\vec{A}\vec{C}\vec{D}] + [\vec{A}\vec{C}\vec{E}]$$

V. *If* $\vec{A} = a_1\vec{I} + a_2\vec{J} + a_3\vec{K}$, $\vec{B} = b_1\vec{I} + b_2\vec{J} + b_3\vec{K}$, $\vec{C} = c_1\vec{I} + c_2\vec{J} + c_3\vec{K}$

Then

$$[\vec{A}\vec{B}\vec{C}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

As $\vec{A} \times \vec{B} = (a_2b_3 - a_3b_2)\vec{I} + (a_3b_1 - a_1b_3)\vec{J} + (a_1b_2 - a_2b_1)\vec{K}$

$\therefore [\vec{A}\vec{B}\vec{C}] = b_1(a_2b_3 - a_3b_2) + b_2(a_3b_1 - a_1b_3) + b_3(a_1b_2 - a_2b_1)$ [$\because \vec{C} = c_1\vec{I} + c_2\vec{J} + c_3\vec{K}$]

$= c_1(a_2b_3 - a_3b_2) + c_2(a_3b_1 - a_1b_3) + c_3(a_1b_2 - a_2b_1)$ which is the required result.

Obs. *Linear dependence of vectors.* Any three vectors $\vec{A}, \vec{B}, \vec{C}$ are said to be *linearly dependent* (*l.d.*) if one of them can be expressed as a linear combination of the other two, *i.e.*,

$$\vec{A} = x\vec{B} + y\vec{C}$$

where x, y are constants. This means that $\vec{A}$ lies in the plane of $\vec{B}, \vec{C}$ *i.e.*, $[\vec{A}\vec{B}\vec{C}] = 0$. Then three vectors are linearly dependent if their scalar triple product is zero. Otherwise three vectors are linearly independent.

Example 2.11. *If* $\vec{A} = 2\vec{I} + 3\vec{J} + 4\vec{K}$, $\vec{B} = -\vec{I} + 2\vec{J} + 4\vec{K}$, $\vec{C} = 3\vec{I} + 5\vec{J} + 3\vec{K}$ *and* $\vec{D} = 11\vec{I} + 17\vec{J} - 8\vec{K}$, *show that* $\vec{A}, \vec{B}, \vec{C}, \vec{D}$ *are coplanar.*

Solution. Let $\vec{OA} = -\vec{I} + 3\vec{J} + 2\vec{K}$, $\vec{OB} = 2\vec{I} - 3\vec{J} + 4\vec{K}$, $\vec{OC} = 4\vec{I} + 7\vec{J} + 2\vec{K}$

and $\vec{OD} = -11\vec{I} + 17\vec{J} - \vec{K}$. Then $\vec{AB} = \vec{OB} - \vec{OA} = 3\vec{I} - 5\vec{J} + 2\vec{K}$.

Similarly, $\vec{AC} = 5\vec{I} + 4\vec{J} + \vec{K}$ and $\vec{AD} = -9\vec{I} + 14\vec{J} - 3\vec{K}$.

The given points will be coplanar if $\vec{AB}, \vec{AC}, \vec{AD}$ are coplanar, *i.e.*, if their scalar triple product is zero.

Now

$$[\vec{AB}, \vec{AC}, \vec{AD}] = \begin{vmatrix} 3 & -5 & 2 \\ 5 & 4 & 1 \\ -9 & 14 & -3 \end{vmatrix} = 3(-12 - 14) + 5(-15 + 9) + 2(70 + 36) = 0$$

Hence the points A, B, C, D are coplanar.

Example 2.12. *Show that the volume of the tetrahedron* $ABCD$ *is* $\frac{1}{6}[\vec{AB}, \vec{AC}, \vec{AD}]$.

Hence find the volume of the tetrahedron formed by the points $(1, 2, 3)$, $(4, 3, 5)$, $(3, 5, 2)$ *and* $(2, 3, 4)$.

Solution. (i) Volume of the tetrahedron $ABCD$

$= \frac{1}{3}$ (area of $\triangle ABC$) × (height h of D above the plane ABC)

$= \frac{1}{6}$ (2 area of $\triangle ABC$)

$= \frac{1}{6}$ (volume of the parallelopiped with AB, AC, AD as coterminous edges)

$= \frac{1}{6}[\vec{AB}, \vec{AC}, \vec{AD}]$.

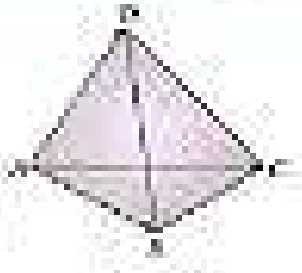


Fig. 2.11

(ii) Let $\vec{OA} = \vec{I} + 2\vec{J} + 3\vec{K}$, $\vec{OB} = 4\vec{I} + 3\vec{J} + 5\vec{K}$, $\vec{OC} = 3\vec{I} + 5\vec{J} + 2\vec{K}$ and $\vec{OD} = 2\vec{I} + 3\vec{J} + 4\vec{K}$.

Then $\vec{AB} = \vec{OB} - \vec{OA} = 3\vec{I} + \vec{J} + 2\vec{K}$.

Similarly, $\vec{AC} = 2\vec{I} + 3\vec{J} - \vec{K}$ and $\vec{AD} = \vec{I} + \vec{J} + \vec{K}$.

$\therefore$ Volume of the tetrahedron $ABCD = \frac{1}{6}[\vec{AB}, \vec{AC}, \vec{AD}] = \frac{1}{6}\begin{vmatrix} 3 & 1 & 2 \\ 2 & 3 & -1 \\ 1 & 1 & 1 \end{vmatrix} = \frac{5}{6}$.

3.10 VECTOR PRODUCT OF THREE VECTORS

(1) Definition. If $\vec{A}, \vec{B}, \vec{C}$ be any three vectors, then the vector $\vec{A} \times (\vec{B} \times \vec{C})$ is called the vector product of three vectors $\vec{A}, \vec{B}, \vec{C}$ and is written as $\vec{A} \times \vec{B} \times \vec{C}$.

Here the brackets are essential as $\vec{A} \times \vec{B} \times \vec{C} \neq (\vec{A} \times \vec{B}) \times \vec{C}$, expressing the fact that vector triple product is non-associative.

(2) Expansion formula. If $\vec{A}, \vec{B}, \vec{C}$ be any three vectors, $(\vec{A} \times \vec{B}) \times \vec{C} = (\vec{C} \cdot \vec{A})\vec{B} - (\vec{C} \cdot \vec{B})\vec{A}$

In words (to remember a right-hand rule vector = (outer vector)(middle vector) − (middle vector)(outer vector).

The vector $(\vec{A} \times \vec{B}) \times \vec{C}$ is perpendicular to $\vec{C}$ and the vector $\vec{A} \times \vec{B}$ and the latter is perpendicular to the plane containing $\vec{A}$ and $\vec{B}$. Hence $(\vec{A} \times \vec{B}) \times \vec{C}$ lies in the plane of $\vec{A}$ and $\vec{B}$. As such we can write

$$(\vec{A} \times \vec{B}) \times \vec{C} = l\vec{A} + m\vec{B} \qquad \ldots(1)$$

where l and m are some scalars.

Multiply both sides scalarly by $\vec{C}$. Then $\vec{C} \cdot (\vec{A} \times \vec{B}) \times \vec{C} = l(\vec{C} \cdot \vec{A}) + m(\vec{C} \cdot \vec{B})$

The scalar triple product on the left-hand side is zero, since two of its vectors are equal.

$$\therefore \qquad l(\vec{C} \cdot \vec{A}) + m(\vec{C} \cdot \vec{B}) = 0$$

or
$$\frac{l}{\vec{C} \cdot \vec{B}} = \frac{m}{-\vec{C} \cdot \vec{A}} = \lambda, \text{ say.}$$

Substituting the values of l and m in (1), we get

$$(\vec{A} \times \vec{B}) \times \vec{C} = \lambda(\vec{C} \cdot \vec{B})\vec{A} - \lambda(\vec{C} \cdot \vec{A})\vec{B} \qquad \ldots(2)$$

Finding λ is a non-trivial problem. To find it, take the special case $\vec{A} = \vec{i}, \vec{B} = \vec{C} = \vec{j}$. Then (2) gives

$$(\vec{i} \times \vec{j}) \times \vec{j} = \lambda(\vec{j} \cdot \vec{j})\vec{i} - \lambda(\vec{j} \cdot \vec{i})\vec{j}$$

i.e.,
$$\vec{k} \times \vec{j} = \lambda\vec{i} - \lambda(0)\vec{j} = \lambda\vec{i}.$$

This gives $\lambda = -1$. Hence (2) reduces to the required result.

Similarly, it can be shown that $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$.

Cor. $\vec{A} \times (\vec{B} \times \vec{C}) + \vec{B} \times (\vec{C} \times \vec{A}) + \vec{C} \times (\vec{A} \times \vec{B}) = \vec{0}$.

For L.H.S. $= (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C} + (\vec{B} \cdot \vec{A})\vec{C} - (\vec{B} \cdot \vec{C})\vec{A} + (\vec{C} \cdot \vec{B})\vec{A} - (\vec{C} \cdot \vec{A})\vec{B}$ which vanishes identically.

Example 3.23. If $\vec{A}, \vec{B}, \vec{C}, \vec{D}$ be any four vectors, prove that

(i) $(\vec{A} \times \vec{D}) \cdot (\vec{C} \times \vec{D}) = \begin{vmatrix} \vec{A} \cdot \vec{C} & \vec{B} \cdot \vec{C} \\ \vec{A} \cdot \vec{D} & \vec{B} \cdot \vec{D} \end{vmatrix}$ (ii) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{A}) = [\vec{A}\vec{B}\vec{C}]\vec{A}$ (iii) $(\vec{A} \times \vec{B}) \times \vec{C} + (\vec{B} \times \vec{C}) \times \vec{A} = [\vec{B}\vec{C}\vec{A}]\vec{B}$

Solution. (i) $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \times \vec{B}) \times \vec{C}) \cdot \vec{D}$ (interchanging the dot and cross)

$= [\vec{A} \cdot \vec{C})\vec{B} - (\vec{B} \cdot \vec{C})\vec{A}] \cdot \vec{D}$

$= (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C})$ = right side. Hence the result.

In particular, we have $(\vec{A} \times \vec{B})^2 = \vec{A}^2\vec{B}^2 - (\vec{A} \cdot \vec{B})^2$ which has already been proved in § 3.9 (2) (i).

(ii) $(\vec{A} \times \vec{B}) \times \vec{C} \times \vec{B} = (\vec{A} \times \vec{B}) \times \vec{P}$, where $\vec{P} = \vec{C} \times \vec{B}$

$= (\vec{A} \cdot \vec{P})\vec{B} - (\vec{B} \cdot \vec{P})\vec{A} = (\vec{A} \cdot \vec{C} \times \vec{B})\vec{B} - (\vec{B} \cdot \vec{C} \times \vec{B})\vec{A}$

$= [\vec{A}\vec{C}\vec{B}]\vec{B} - [\vec{B}\vec{C}\vec{B}]\vec{A}$.

Example 3.24. Show that the resolved part of a vector $\vec{B}$ along and perpendicular to a vector $\vec{A}$ in the plane of $\vec{A}$ and $\vec{B}$ are

$$\frac{\vec{A} \cdot \vec{B}}{\vec{A}^2}\vec{A} \text{ and } \frac{(\vec{A} \times \vec{B}) \times \vec{A}}{\vec{A}^2}$$

Solution. Let $\overrightarrow{OA} = \vec{A}$, $\overrightarrow{OB} = \vec{B}$ and ON be the projection of $\vec{B}$ on $\vec{A}$ (Fig. 3.35).

$\therefore$ Component of $\vec{B}$ along $\vec{A} = \vec{ON}$ (a vector along $\vec{A}$)

$$= (\vec{B} \cdot \hat{A})\hat{A} = \left(\frac{\vec{B} \cdot \vec{A}}{A}\right)\frac{\vec{A}}{A} \qquad (\because \vec{A} = A\hat{A})$$

$$= \frac{\vec{B} \cdot \vec{A}}{A^2}\vec{A} \qquad (\because A^2 = \vec{A}^2)$$

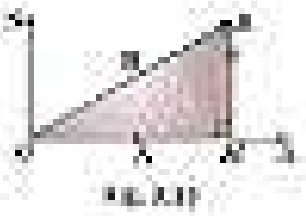


Fig. 3.35

As a complement of $\vec{B}$: $\vec{A} = \vec{AB}$

$$= \vec{OB} - \vec{OM} = \vec{B} - \frac{\vec{B}.\vec{A}}{A^2}\vec{A} = \frac{(\vec{A}.\vec{A})\vec{B} - (\vec{A}.\vec{B})\vec{A}}{A^2} = \frac{(\vec{A}\times\vec{B})\times\vec{A}}{A^2}$$

Example 2.25. Prove the formula

$$(\vec{B}\times\vec{C}).(\vec{A}\times\vec{D}) + (\vec{C}\times\vec{A}).(\vec{B}\times\vec{D}) + (\vec{A}\times\vec{B}).(\vec{C}\times\vec{D}) = 0$$

and hence find the value of $\sin\theta + \sin(\theta+\beta) + \sin(\theta+\beta+\gamma)$.

Solution. We know that

$$(\vec{B}\times\vec{C}).(\vec{A}\times\vec{D}) = (\vec{B}.\vec{A})(\vec{C}.\vec{D}) - (\vec{B}.\vec{D})(\vec{C}.\vec{A})$$
$$(\vec{C}\times\vec{A}).(\vec{B}\times\vec{D}) = (\vec{C}.\vec{B})(\vec{A}.\vec{D}) - (\vec{C}.\vec{D})(\vec{A}.\vec{B})$$
$$(\vec{A}\times\vec{B}).(\vec{C}\times\vec{D}) = (\vec{A}.\vec{C})(\vec{B}.\vec{D}) - (\vec{A}.\vec{D})(\vec{B}.\vec{C})$$

Adding, we get

$$(\vec{B}\times\vec{C}).(\vec{A}\times\vec{D}) + (\vec{C}\times\vec{A}).(\vec{B}\times\vec{D}) + (\vec{A}\times\vec{B}).(\vec{C}\times\vec{D}) = 0 \quad \ldots(1)$$

Let the vectors $\vec{A}, \vec{B}, \vec{C}, \vec{D}$ be starting along coplanar lines OA, OB, OC, OD respectively. (Fig. 2.34)

Take $\angle AOC = \theta$ and $\angle AOA' = \angle BOB' = \phi$,

so that $\angle A'OD = \theta + \phi$ and $\angle BOC = \theta - \phi$

If $\vec{N}$ be a unit vector normal to the plane of these lines, then

$$(\vec{B}\times\vec{C}).(\vec{A}\times\vec{D}) = (BC\sin\theta\,\vec{N}).(AD\sin(\theta+\phi)\,\vec{N}) = abcd\sin\theta\sin(\theta+\phi) \quad \ldots(2)$$

$$(\vec{C}\times\vec{A}).(\vec{B}\times\vec{D}) = (ca\sin\theta\,\vec{N}).(bd\sin\phi\,\vec{N}) = -abcd\sin\theta\sin\phi \quad \ldots(3)$$

and $$(\vec{A}\times\vec{B}).(\vec{C}\times\vec{D}) = (ab\sin\phi\,\vec{N}).(cd\sin\phi\,\vec{N}) = abcd\sin^2\phi \quad \ldots(4)$$

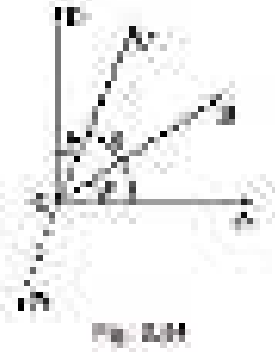


Fig. 2.34

Substituting the values from (2), (3), (4) in (1), we get

$$abcd\sin\theta\sin(\theta+\phi) - abcd\sin\theta\sin\phi + abcd\sin^2\phi = 0$$ which on simplification gives the required result.

Example 2.26. Prove that

(i) $[\vec{B}\times\vec{C}, \vec{C}\times\vec{A}, \vec{A}\times\vec{B}] = (\vec{A}\vec{B}\vec{C})^2$ *(Nagpur, 2009)*

(ii) $(\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D}) = (\vec{B}.\vec{D})\vec{A} - (\vec{A}.\vec{D})\vec{B} - (\vec{B}.\vec{C})\vec{A} + (\vec{A}.\vec{C})\vec{B}$

Solution. (i) $[\vec{B}\times\vec{C}, \vec{C}\times\vec{A}, \vec{A}\times\vec{B}] = (\vec{B}\times\vec{C}).[(\vec{C}\times\vec{A})\times(\vec{A}\times\vec{B})]$

$$= (\vec{B}\times\vec{C}).[(\vec{C}.\vec{A}\,\vec{B} - \vec{B}.\vec{A}\,\vec{C})\times\vec{A}] = (\vec{B}\times\vec{C}).[(\vec{C}.\vec{A})(\vec{B}\times\vec{A}) - (\vec{B}.\vec{A})(\vec{C}\times\vec{A})]$$
$$= (\vec{B}\times\vec{C}).(\vec{B}\times\vec{A})(\vec{C}.\vec{A}) \quad [\because (\vec{C}\times\vec{A}).\vec{A} = 0]$$
$$= (\vec{B}\times\vec{C}).(\vec{B}\times\vec{C})(\vec{A}.\vec{A}) = [\vec{B}\vec{C}\vec{A}]^2 = [\vec{A}\vec{B}\vec{C}]^2 \quad [\because [\vec{B}\vec{C}\vec{A}] = [\vec{A}\vec{B}\vec{C}]]$$

(ii) $(\vec{A}\times\vec{B})\times(\vec{C}\times\vec{D}) = \vec{A}\times[\vec{B}\times(\vec{C}\times\vec{D})] - \vec{B}\times[\vec{A}\times(\vec{C}\times\vec{D})]$

$$= \vec{A}\times[(\vec{B}.\vec{D})\vec{C} - (\vec{B}.\vec{C})\vec{D}] - \vec{B}\times[(\vec{A}.\vec{D})\vec{C} - (\vec{A}.\vec{C})\vec{D}]$$
$$= (\vec{B}.\vec{D})(\vec{A}\times\vec{C}) - (\vec{B}.\vec{C})(\vec{A}\times\vec{D}) - (\vec{A}.\vec{D})(\vec{B}\times\vec{C}) + (\vec{A}.\vec{C})(\vec{B}\times\vec{D})$$

EXERCISE 2.5

1. Find the values of l for which the vectors $\vec{A} = 2\vec{i} - 3\vec{j} + l\vec{k}$, $\vec{B} = \vec{i} + 2\vec{j} - 3\vec{k}$ and $\vec{C} = 3\vec{i} - \vec{j} + 2\vec{k}$ are coplanar.
2. Find a unit vector which is perpendicular to both the vectors $2\vec{i} - \vec{j} + 2\vec{k}$ and $\vec{i} + \vec{j} - 3\vec{k}$. *(Nagpur, 2008)*
3. (i) Prove that the vectors $\vec{i} - 2\vec{j} + 3\vec{k}$, $2\vec{i} + 3\vec{j} - 4\vec{k}$ and $\vec{i} - 9\vec{j} + 18\vec{k}$ are coplanar.
   (ii) Do the vectors $3\vec{i} - \vec{j} + 2\vec{k}$, $\vec{i} + 2\vec{j} - 3\vec{k}$ and $2\vec{i} - 3\vec{j} + 5\vec{k}$ form a triangle?
4. (i) Find the direction cosines of the vector $\vec{A} = 3\vec{i} + 6\vec{j} - 2\vec{k}$ and hence find $\vec{A}.\vec{B}$ if $\vec{B} = 4\vec{i} - 3\vec{j} + 5\vec{k}$.
   (ii) Find the direction ratios of the following pairs of vectors and hence find the angle between them: $\vec{A} = 2\vec{i} - 3\vec{j} + 2\vec{k}$, $\vec{B} = \vec{i} + 2\vec{j} - \vec{k}$.

(U.P.T.U., 2005)

5. Find the volume of the parallelopiped, three of whose coterminous edges are $\vec{A} = 2\vec{i} + 3\vec{j} + \vec{k}$, $\vec{B} = \vec{i} - 2\vec{j} + \vec{k}$ and $\vec{C} = \vec{i} + \vec{j} - 2\vec{k}$.

6. Find the volume of the tetrahedron formed by the points
   (i) $(1, 2, 1), (2, 3, 1), (3, 1, 1), (4, 3, 5)$ and $(2, 3, 4)$. (W.B.T.U., 2005)
   (ii) $(3, 1, 1), (5, -1, 2), (6, 3, -1)$ and $(1, 2, 1)$.
7. If $\vec{A} \times \vec{B} = \vec{C} \times \vec{D}$ and $\vec{A} \times \vec{C} = \vec{B} \times \vec{D}$, prove that $(\vec{A} - \vec{D})$ is parallel to $(\vec{B} - \vec{C})$. Interpret this result geometrically.
8. (a) Prove that $[\vec{A} + \vec{B}, \vec{B} + \vec{C}, \vec{C} + \vec{A}] = 2[\vec{A}\vec{B}\vec{C}]$.
   (b) Show that the vectors $3\hat{i} - 6\hat{j} + 2\hat{k}$, $4\hat{i} - 2\hat{j} - 4\hat{k}$ and $\hat{i} + 3\hat{j} + 6\hat{k}$ are mutually perpendicular to one another. Hence find the volume of the parallelopiped whose edges are $\vec{A}, \vec{B}, \vec{C}$ are the given vectors.
9. If $\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \times \vec{B}) \times \vec{C}$, show that $(\vec{A} \times \vec{C}) \times \vec{B} = 0$.
10. Show that $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) + (\vec{B} \times \vec{C}) \cdot (\vec{A} \times \vec{D}) + (\vec{C} \times \vec{A}) \cdot (\vec{B} \times \vec{D}) = 0$. (Assam, 1995)
11. If $\vec{A} = \vec{I} - \vec{J}$, $\vec{B} = \vec{I} + \vec{J} - \vec{K}$, $\vec{C} = \vec{I} - 3\vec{J} - \vec{K}$, find
   (i) $(\vec{A} \times \vec{B}) \times \vec{C}$ (ii) $\vec{A} \times (\vec{B} \times \vec{C})$
12. (a) Given $\vec{A} = 2\vec{I} - \vec{J} + \vec{K}$, $\vec{B} = \vec{I} + 3\vec{J} - 2\vec{K}$, $\vec{C} = \vec{I} + \vec{J} - \vec{K}$, find the reciprocal triad $\vec{A}', \vec{B}', \vec{C}'$ and verify that $[\vec{A}\vec{B}\vec{C}][\vec{A}'\vec{B}'\vec{C}'] = 1$.
   (b) Prove that $\vec{A} \times \vec{A}' + \vec{B} \times \vec{B}' + \vec{C} \times \vec{C}' = 0$.
13. Prove that (a) $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) (\vec{E} \times \vec{F}) + [\vec{B}\vec{C}\vec{D}][\vec{E}\vec{F}\vec{A}] + [\vec{C}\vec{D}\vec{E}][\vec{F}\vec{A}\vec{B}] + [\vec{D}\vec{E}\vec{F}][\vec{A}\vec{B}\vec{C}] = [\vec{A}\vec{B}\vec{C}][\vec{D}\vec{E}\vec{F}]$
   (b) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) + (\vec{B} \times \vec{C}) \times (\vec{A} \times \vec{D}) + (\vec{C} \times \vec{A}) \times (\vec{B} \times \vec{D}) = 2[\vec{A}\vec{B}\vec{C}]\vec{D}$
14. Prove that:
   (a) $(\vec{B} \times \vec{C}) \times (\vec{A} \times \vec{D}) + (\vec{C} \times \vec{A}) \times (\vec{B} \times \vec{D}) + (\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = 2[\vec{A}\vec{B}\vec{C}]\vec{D}$ (Kerala, 2007)
   (b) $(\vec{A} \times \vec{B}') \cdot (\vec{B} \times \vec{C}') + (\vec{B} \times \vec{B}') \cdot (\vec{C} \times \vec{A}') + (\vec{C} \times \vec{B}') \cdot (\vec{A} \times \vec{B}') = 0$.
15. (a) Prove that $[\vec{L}\vec{M}\vec{N}][\vec{A}\vec{B}\vec{C}] = \begin{vmatrix} \vec{L}\cdot\vec{A} & \vec{L}\cdot\vec{B} & \vec{L}\cdot\vec{C} \\ \vec{M}\cdot\vec{A} & \vec{M}\cdot\vec{B} & \vec{M}\cdot\vec{C} \\ \vec{N}\cdot\vec{A} & \vec{N}\cdot\vec{B} & \vec{N}\cdot\vec{C} \end{vmatrix}$
   (b) The lengths of the edges OA, OB, OC of the tetrahedron $OABC$ are a, b, c and the angles BOC, COA, AOB are θ, ϕ, ψ. Find its volume.

SOLID GEOMETRY

3.1 (1) EQUATION OF A PLANE

Let $P(x, y, z)$ be any point on the plane through $A(x_1, y_1, z_1)$ which is normal to the vector $\vec{N} = a\hat{i} + b\hat{j} + c\hat{k}$.

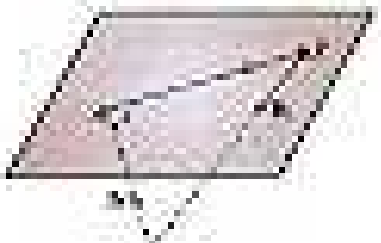


Fig 3.15

Then $\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{OA} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$

Clearly the vectors $\vec{AP} = (x - x_1)\hat{i} + (y - y_1)\hat{j} + (z - z_1)\hat{k}$ and $\vec{N}$ are perpendicular to each other.

$$\therefore \qquad \vec{AP} \cdot \vec{N} = 0 \qquad \ldots(1)$$

or $$(x - x_1)\hat{i} + (y - y_1)\hat{j} + (z - z_1)\hat{k} \cdot a\hat{i} + b\hat{j} + c\hat{k} = 0$$

or $$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0 \qquad \ldots(2)$$

which is the equation of any plane through the point (x_1, y_1, z_1).

Obs. Equation (2) is of the form $ax + by + cz + d = 0$ (3) where d is an arbitrary constant of a plane.

Conversely, every linear equation in x, y, z represents a plane and the coefficients of x, y, z are the direction ratios of the normal to the plane.

Cor. If l, m, n be the direction cosines of the normal to the plane, then

$$lx + my + nz = p \qquad \ldots(3)$$

which is called the **normal form** of the equation of the plane where p is the perpendicular distance from the origin.

(2) Angle between two planes. Def. The angle between two planes is equal to the angle between their normals.

Let the two planes be

$$ax + by + cz + d = 0 \quad \text{and} \quad a'x + b'y + c'z + d' = 0.$$

Here the direction ratios of their normals are a, b, c and a', b', c'.

Hence the angle θ between the planes is given by $\cos\theta = \dfrac{aa' + bb' + cc'}{\sqrt{a^2 + b^2 + c^2}\sqrt{a'^2 + b'^2 + c'^2}}$

The planes are perpendicular if their normals are perpendicular, i.e., if $aa' + bb' + cc' = 0$

The planes are parallel if their normals are parallel, i.e., if $\dfrac{a}{a'} = \dfrac{b}{b'} = \dfrac{c}{c'}$

Cor. Any plane parallel to the plane $ax + by + cz + d = 0$

is $ax + by + cz + k = 0$ ... (1) where k is an arbitrary constant.

For the direction ratios of their normals are the same.

(2) Perpendicular distance of the point (x_1, y_1, z_1) from the plane

$$ax + by + cz + d = 0 \quad \text{... (2)}$$

is $\dfrac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$

Let PL be the perpendicular distance of $P(x_1, y_1, z_1)$ from the plane (2) so that the direction ratios of $\vec{LP}$ are

$$\frac{x_1}{\sqrt{\Sigma x_1^2}}, \frac{y_1}{\sqrt{\Sigma x_1^2}}, \frac{z_1}{\sqrt{\Sigma x_1^2}}$$

If $Q(x_2, y_2, z_2)$ be a point on (2) then

$$ax_2 + by_2 + cz_2 + d = 0 \quad \text{... (3)}$$

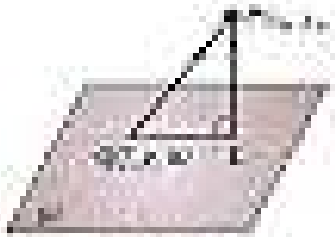


Fig. 1.26

$\therefore$ PL = projection of $\vec{QP}$ on $\vec{LP} = \vec{QP}.\vec{LP}$

$$= \frac{(x_1 - x_2)x_1 + (y_1 - y_2)y_1 + (z_1 - z_2)z_1}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \quad \text{[By (3) p.46]}$$

$$= \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \text{ by virtue of (3)} \quad \text{... (4)}$$

The sign of the realised length is taken to be positive or negative according as it is positive or negative.

Obs. The perpendicular to a plane from two points are taken to be of the same sign if the points lie on the same side of it & of opposite signs if they lie on the opposite sides of the plane.

The two points (x_1, y_1, z_1) and (x_2, y_2, z_2) lie on the same side or on opposite sides of the plane $ax + by + cz + d = 0$ according as $ax_1 + by_1 + cz_1 + d$ and $ax_2 + by_2 + cz_2 + d$ are of the same sign or of opposite signs.

Cor. Planes bisecting the angles between two planes.

Let $a_1x + b_1y + c_1z + d_1 = 0$ ... (1)

and $a_2x + b_2y + c_2z + d_2 = 0$ ... (2)

be the given planes.

Let $P(x, y, z)$ be any point on either of the planes bisecting the angles between the planes (1) and (2).

Then $\pm$ (distance of P from (1)) = $\pm$ (distance of P from (2)),

i.e., $\dfrac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm\dfrac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$

which are the required equations of the bisecting planes.

Example 8.20. Find the equation of the plane which

(i) cuts off intercepts 3, 2, -1 from the axes

(ii) passes through the points $A(2, 2, 1)$, $B(3, 1, 2)$ and $C(-1, 2, -2)$.

Solution. (i) **Intercept form** of the equation of the plane. Let the required equation of the plane be

$$ax + by + cz + 1 = 0 \quad \text{... (1)}$$

The plane cuts the axes at A, B, C such that $OA = 3$, $OB = 2$, $OC = -1$,

i.e., it passes through the points $A(3, 0, 0)$, $B(0, 2, 0)$, $C(0, 0, -1)$.

$$\alpha x + \beta = 0, \beta y + \beta = 0, \gamma z + \beta = 0$$

which are $x = -\beta/\alpha,\ y = -\beta/\beta,\ z = -\beta/\gamma$

Substituting these values of x, y, z in (1), $-\frac{\beta}{\alpha}x - \frac{\beta}{\beta}y - \frac{\beta}{\gamma}z + \beta = 0$ or $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$.

(ii) Three points form of the equation of the plane.

Any plane through $(0, 1, 1)$ is $ax + (y-1) + (z-1) = 0$ ...(1)

It will pass through $(1, 1, 2)$ and $(-1, 2, -3)$ if $2a + 1 = 0$ and $-a + 1 - 3c = 0$.

By cross-multiplication, $\frac{a}{-1} = \frac{1}{2} = \frac{c}{3}$.

Substituting these values in (1), we get $-1(x-1) + 2(y-1) + 3(z-1) = 0$

or $x - 2y - 3z + 6 = 0$, which is the required equation of the plane.

Example 3.36. *Find the equation of the plane which passes through the point $(2, -3, 1)$ and is*
(i) parallel to the plane $2x + 3y + 3z + 5 = 0$.
(ii) normal to the line joining the points $(3, 4, -1)$ and $(5, -2, 6)$. (U.P.T.U., 2006)
(iii) Perpendicular to the planes $3x + y + 2z = 8$ and $5x + 3y - 4z + 8 = 0$. (Kanpur, 2006; U.P.T.U., 2008)

Solution. (i) Any plane parallel to the given plane is

$$2x + 3y + 3z + k = 0 \text{ which gives the } x, y, z \text{ intercepts } -\frac{k}{2}, -\frac{k}{3}, -\frac{k}{3} \text{ if } k \neq 0.$$

Thus the required plane is $2x + 3y + 3z - 2 = 0$.

(ii) Any plane through $(3, -4, 1)$ is $a(x-3) + b(y+4) + c(z-1) = 0$.

The direction ratios of the line joining the points $(5, -2, 6)$ and $(3, -4, 1)$ are proportional to $2, 2, -5$.

This line is normal to the plane $\therefore a, b, c$ are proportional to $2, 2, -5$.

Substituting these values in (1), the required equation is

$$2(x-3) + 2(y+4) - 5(z-1) = 0 \quad \text{or} \quad 2x + 2y - 5z + 12 = 0.$$

(iii) Any plane through $(2, -3, 1)$ is

$$a(x-2) + b(y+3) + c(z-1) = 0 \text{ which will be } \perp \text{ to the plane}$$

$$3x + y + 2z = 8 \text{ and } 5x + 3y - 4z + 8 = 0$$

if $\quad 3a + b + 2c = 0$ and $5a + 3b - 4c = 0$.

Solving these by cross-multiplication, we get $\frac{a}{-6} = \frac{b}{10} = \frac{c}{4}$.

Hence the required equation is $-3(x-2) + 5(y+3) + 2(z-1) = 0$ i.e. $3x - 5y - 2z - 19 = 0$.

Example 3.37. *The plane $4x + 5y - z = 7$ is rotated through a right angle about its line of intersection with the plane $2x + 3y - 2z = 3$. Find the equation of this plane in its new position.*

Solution. Any plane through the line of intersection of

$$4x + 5y - z = 7 \quad \text{...(1)}$$

and $\quad 2x + 3y - 2z = 3 \quad \text{...(2)}$

is $\quad 4x + 5y - z - 7 + \lambda(2x + 3y - 2z - 3) = 0$

i.e., $\quad (4 + 2\lambda)x + (5 + 3\lambda)y - (1 + 2\lambda)z - (7 + 3\lambda) = 0 \quad \text{...(3)}$

Then new position of (1) when rotated through a right angle, is such that (1) and (3) are perpendicular. This implies that

$$4(4 + 2\lambda) + 5(5 + 3\lambda) + 1(1 + 2\lambda) = 0$$

i.e., $\quad 38\lambda + 42 = 0 \quad$ or $\quad \lambda = -21/19$.

Substituting $\lambda = -21/19$ in (3), we get $2x + 2y + 5z + 4 = 0$,

or $\quad 2x + y + 2z + 7 = 0$, which is the required plane.

Example 3.38. *Find the distance between the parallel planes $2x - 2y + z + 3 = 0$ and $6x - 6y + 3z + 8 = 0$. Find also the equations of the parallel plane that lies midway between the given planes.* (Bhopal, 2000)

Solution. The distance between the given planes is the perpendicular distance of any point on one of the planes from the other.

A point on the first plane is $(0, 0, -3)$

$\therefore$ Required distance = $\perp$ distance of $(0, 0, -3)$ from $4x - 4y + 2z + 9 = 0$

$$= \frac{-6 + 9}{\sqrt{16 + 16 + 4}} = \frac{3}{6} = \frac{1}{2}$$

Let the equation of the parallel plane that lies midway between the given planes be

$$2x - 2y + z + k = 0 \qquad \ldots(1)$$

Now distance of any point $(0, 0, -3)$ on the first plane from (1) should be $\frac{1}{4}$.

$$\therefore \quad = \frac{-3 + k}{\sqrt{4 + 4 + 1}} = \frac{1}{4} \quad \text{i.e.,} \quad k = 3 \pm \frac{3}{4} = \frac{15}{4}, \frac{9}{4}.$$

Thus the required plane is $2x - 2y + z + \frac{15}{4} = 0$.

Account that $k = \frac{15}{4}$ and verify that the distance of a point on this plane $4x - 4y + 2z + 9 = 0$ is also $\frac{1}{4}$.

A point on this plane is $(0, 0, -\frac{9}{4})$. Its distance from the plane (1) is $\frac{-2(2) + \frac{15}{4}}{3} = \frac{1}{4}$ (as required)

Thus $k = \frac{9}{4}$ is not admissible.

$\therefore$ The required plane is $2x - 2y + z + \frac{15}{4} = 0$.

Example 1.31. *A variable plane is at a constant distance p from the origin and meets the axes in A, B, C. Find the locus of the centroid of the tetrahedron $OABC$.*

Solution. As the given plane is at a constant distance p from the origin, therefore its equation is of the form

$$lx + my + nz = p \qquad \ldots(1) \quad \text{where } l, m, n \text{ are the d.c's of the } \perp \text{ from the origin.}$$

(1) may be rewritten as $\frac{x}{(p/l)} + \frac{y}{(p/m)} + \frac{z}{(p/n)} = 1$

so that $OA = p/l$, $OB = p/m$, $OC = p/n$.

$\therefore \quad A = (p/l, 0, 0),\ B = (0, p/m, 0),\ C = (0, 0, p/n)$.

Thus the coordinates of the centroid G of the tetrahedron $OABC$ are

$$(x_1, y_1, z_1) = (p/4l, p/4m, p/4n) \qquad \text{(See p. 81)}$$

$$\therefore \quad \frac{1}{x_1^2} + \frac{1}{y_1^2} + \frac{1}{z_1^2} = \frac{16}{p^2}(l^2 + m^2 + n^2) = \frac{16}{p^2}$$

Thus the locus of (1) is $x^{-2} + y^{-2} + z^{-2} = 16p^{-2}$

Example 1.32. *A variable plane forms a tetrahedron of constant volume $\frac{8}{9}k^3$ with the coordinate planes. The plane cuts the coordinate axes through A, B, C. Show that the locus of the centre of the sphere $OABC$ is $x^{-2} + y^{-2} + z^{-2} = k^{-2}$.*

Solution. Let the variable plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Its distance from origin $= \frac{1}{\sqrt{a^{-2} + b^{-2} + c^{-2}}} = p$ (given)

$$\text{i.e.,} \quad a^{-2} + b^{-2} + c^{-2} = p^{-2} \qquad \ldots(2)$$

Now $OA = a$, $OB = b$ and $OC = c$, therefore equations of the planes through A, B, C parallel to yz, zx and xy-planes are $x = a$, $y = b$, $z = c$.

Let the point of intersection of these three planes be (x_1, y_1, z_1).

Then $\qquad x_1 = a,\ y_1 = b,\ z_1 = c \qquad \ldots(3)$

Substituting (3) in (2), we get $\quad x_1^{-2} + y_1^{-2} + z_1^{-2} = p^{-2}$

Thus the locus of (x_1, y_1, z_1) is $\quad x^{-2} + y^{-2} + z^{-2} = p^{-2}$.

Example 3.33. A variable plane passes through the fixed point (a, b, c) and meets the coordinate axes in A, B, C. Show that the locus of the point common to the planes through A, B, C parallel to the coordinate planes is

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2.$$

Solution. Let ABC be any plane through the fixed point $P(a, b, c)$ such that $OA = x_1, OB = y_1, OC = z_1$.

Then its equation is

$$\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 1 \qquad \text{[See Ex. 3.32 (i)]}$$

Since it passes through P,

$$\therefore \quad \frac{a}{x_1} + \frac{b}{y_1} + \frac{c}{z_1} = 1. \qquad \ldots(1)$$

The planes through A, B, C parallel to the coordinate planes are $x = x_1$, $y = y_1$, $z = z_1$, which meet in $P(x_1, y_1, z_1)$.

Thus the required locus of (x_1, y_1, z_1) is the locus of the point P is

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1.$$

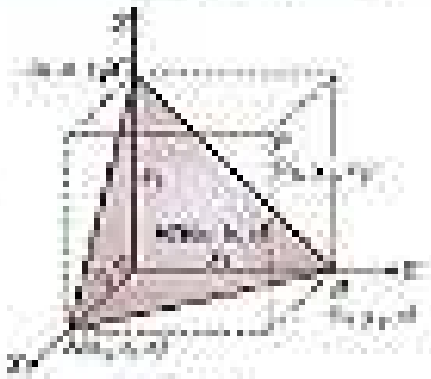


Fig. 3.33

Example 3.34. Find the equations to the two planes which bisect the angles between the planes $3x - 4y + 5z + 9 = 0$ and $x + 2y + 2z + 7 = 0$.

Also point out which of the planes bisects the acute angle. (U.P.T.U., 2009)

Solution. The equations of the planes bisecting the angles between the given planes are

$$\frac{3x - 4y + 5z - 9}{\sqrt{3^2 + (-4)^2 + 5^2}} = \pm \frac{x + 2y + 2z + 7}{\sqrt{1^2 + 2^2 + 2^2}}$$

or $\quad 3x + 2y + 8z + 8 = 0 \qquad \ldots(1)$

and $\quad 5x - y + z - 12 = 0 \qquad \ldots(2)$

which are the required planes.

Let θ be the angle between (1) and either of the given planes, say:

$$3x - 4y + 5z = 9$$

Then, $\quad \cos\theta = \dfrac{3 \times 1 + (-4) \times 2 + 5 \times 2}{\sqrt{3^2 + (-4)^2 + 5^2}\sqrt{1^2 + 2^2 + 2^2}} = \dfrac{3}{5\sqrt{2}} = 0.4243$

$\therefore \quad \tan\theta = \dfrac{\sqrt{221}}{3}$ which is less than 1

i.e., $\quad \theta < 45°$.

Now it is clear the angle between the given planes, so that (2) bisects that angle between the planes which is $2\theta < 90°$.

Hence the plane $5x - y + z - 12 = 0$ bisects the acute angle.

PROBLEMS

1. Find the equation of the plane passing through the point $(-2, 3, 5)$ and having the ratio $3 : -2 : 4$ of the direction cosines to -3.
2. Find the equation of the plane through the points $(2, -1, 3)$, $(2, 2, -1)$ and $(3, 1, 2)$.
3. Find a unit vector normal to the plane through the points $(-1, 2, 3)$, $(1, 1, 1)$ and $(2, -1, 1)$.
4. Find the coordinates of the point $(2, 4, 1)$ from the plane passing through the points $(-3, 5, 3)$, $(-4, 1, 2)$ and $(-6, 3, 1)$. (December, 2010)
5. Show that the four points $(1, -1, 0)$, $(3, 1, -1)$, $(0, 1, 1)$ and $(-1, 3, 2)$ are coplanar. Find the equation of the plane through them. (U.P.T.U., 2004)

5. Show that the point $(-3/2, 2, -3)$ is the orthocentre of the triangle formed by the points $(1, 1, 2)$, $(1, 2, 1)$ and $(-2, 2, -1)$.

[Hint: Show that the point $(-3/2, 2, -3)$ lies in the plane of the triangle and is equidistant from its vertices.]

6. Find the equation of the plane through the point $(2, 1, 0)$ and perpendicular to the planes $2x - y + z + 5 = 0$ and $x + 2y - 3z = 0$.

7. Find the equation of the plane through $(1, 1, 1)$ parallel to the plane $x + 2y + 3z = 6$. (Madras, 2000)

8. Find the equation of the plane which is such that the foot of the perpendicular from (a, b, c) is (a', b', c').

9. Find the equation of the plane through the points $(-1, 1, 1)$, $(1, -1, 1)$ and parallel to y-axis. (V.T.U., 2000)

10. Find the equation of the plane through the points $(2, 2, 1)$ and $(3, 1, 5)$ and perpendicular to the plane $3x + 6y + 3z + 5 = 0$. (U.P.T.U., 2004; Osmania, 1999)

11. A plane meets the axes at $A(-4, 0, 0)$ and $B(0, -3, 0)$. If (a, b, c) are the direction ratios of the normal to the plane which is such that $OA = OB = OC$, find $a : b : c$ and find the equation of the plane.

12. Find the distance between the parallel planes

$$3x - 2y + 6z + 13 = 0 \text{ and } 6x - 4y + 12z - 6 = 0$$

Also find the equation of the parallel plane that lies midway between the given planes.

13. Find the angle between the planes $x + y + z = 0$ and $2x + y - z = 5$. (U.P.T.U., 2005)

14. Two planes are given by $x + 2y + 2z = 5$ and $3x + y + z = 6$. Find

(a) the angle between the planes, and

(b) equation of the plane perpendicular to both of these planes and passing through the point $(1, 1, 1)$.

15. The plane $x + y + z = 1$ is rotated about its line of intersection with the plane $x + y = 0$ through an angle α. Prove that the equation of the plane in its new position is $x + y + (\sqrt{2}/\alpha)z = 1$. (Anna, 2004)

16. Find the equation of the sphere through the points $(4, 1, 1)$, $(3, -1, 3)$, $(2, 0, 2)$ and whose centre lies on the plane $x + y + z = 3$.

17. A plane passes through a fixed point (a, b, c). Show that the locus of the foot of the perpendicular from the origin to the plane is a sphere. (V.T.U., 2003)

18. A variable plane is at a constant distance p from the origin and meets the axes at A, B, C. Find the locus of the centroid of the triangle ABC. (Madras Univ., 2003)

19. A variable plane makes with the coordinate axes a tetrahedron of constant volume $64k^3$. Find the locus of the centroid of the tetrahedron. (Allahabad, 2006; Osmania, 2003)

20. Find equations of the planes bisecting the angle between the planes

$$2x + 3y - 6z - 1 = 0, \quad 4x + 12y + 3z + 9 = 0$$

and find the one which bisects the acute angle.

3.15 EQUATIONS OF A STRAIGHT LINE

(1) General form. Two linear equations in x, y, z

$$\text{i.e.,} \quad ax + by + cz + d = 0 \quad \ldots(i)$$

$$\text{and} \quad a'x + b'y + c'z + d' = 0 \quad \ldots(ii)$$

taken together represent a straight line which is the line of intersection of the planes (i) and (ii) (Fig. 3.35).

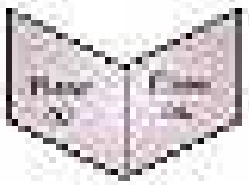


Fig. 3.35

(2) Symmetrical form. Equations of the line through the point $A(x_1, y_1, z_1)$ and having direction ratios l, m, n are

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

Let $A(x_1, y_1, z_1)$ be any point and the given line through $A(x_1, y_1, z_1)$ and parallel to the unit vector $\vec{U} = l\vec{i} + m\vec{j} + n\vec{k}$.

Since $\vec{AP}$ is parallel to $\vec{U}$, we can write $\vec{AP} = t\vec{U}$, where t is a parameter. ...(i)

or $(x - x_1)\beta\gamma + (y - y_1)\gamma\alpha + (z - z_1)\alpha\beta = 0$ ... (ii)

$\therefore$ $x = x_1 + \lambda,\ y = y_1 + \mu,\ z = z_1 + \nu$ ... (iii)

Every point P on the line is given by (iii) for some value of r. These are the parametric equations of the given line. Eliminating r, we get

$$\frac{x - x_1}{\lambda} = \frac{y - y_1}{\mu} = \frac{z - z_1}{\nu} \quad \ldots (iv)$$

which are the symmetrical form of the equations of the line.

Fig. 5.19

Obs. Any point on the line is $(x_1 + r\lambda, y_1 + r\mu, z_1 + r\nu)$.

Cor. The equations of the line passing the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

for the direction ratios of the line joining the given points are

$$x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1.$$

To reduce the given equations of a line to the symmetrical form:

(i) Find a point on the line, by putting $x = 0$ in the given equations and solving the resulting equations for y and z.

(ii) Find the direction cosines of the line from the fact that it is perpendicular to the normals to the given planes.

(iii) Write the equations of the line in the symmetrical form.

Example 5.35. *Find in symmetrical form, the equations of the line* $x + y + z = 6,\ x + 2y + 3z + 4 = 0$. (Madras, 1969)

Solution. (i) To find a point on the line.

Putting $x = 0$ in the given equations, we have

$$y + z = 6;\ 2y + 3z + 4 = 0$$

Solving, $\frac{y}{1} = \frac{z}{2} = \frac{1}{-1}$ $\therefore$ A point on the line is $(0, -13, -13)$.

(ii) To find the direction cosines l, m, n of the line.

Since the line lies on both the given planes

$\therefore$ It is perpendicular to their normals whose direction cosines are proportional to $1, 1, 1$ and $1, 2, 3$.

$\therefore$ $l + m + n = 0,\ l + 2m + 3n = 0$.

Solving, $\frac{l}{1} = \frac{m}{-2} = \frac{n}{1}$

$\therefore$ The direction cosines of the given line are proportional to $-1, 2, -1$.

(iii) Thus the equations of the line in the symmetrical form are

$$\frac{x + 13}{-1} = \frac{y + 13}{2} = \frac{z}{-1}.$$

Example 5.36. *Find the distance of the point* $(1, -2, 3)$ *from the plane* $x - y + z = 5$ *measured parallel to the line* $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$. (Delhi, 1969)

Solution. The line through $P(1, -2, 3)$ having direction ratios $(2, 3, -6)$ is

$$\frac{x - 1}{2} = \frac{y + 2}{3} = \frac{z - 3}{-6} = r \quad \ldots (i)$$

Any point on this line is $(2r + 1, 3r - 2, -6r + 3)$.

This point will lie on the plane $x - y + z = 5$

if $2r + 1 - (3r - 2) + (-6r + 3) = 5$ or $r = 1/7$.

$\therefore$ The point of intersection is $Q(9/7, -11/7, 15/7)$

Then the required distance $= PQ = \sqrt{\left(\frac{4}{49}\right)^2 + \left(\frac{8}{49}\right)^2 + \left(\frac{28}{49}\right)^2} = 1$

$x + 2y + 2z = 8$, $4x - 3y + 12z + 13 = 0$ and justify the one which bisects the acute angle.

Example 3.37. (a) Find the image (reflection) of the point (p, q, r) in the plane $2x + y + z = 8$.

(b) Find the image (reflection) of the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{1}$ in the same plane. (Dec. 2000)

(c) Hence or otherwise, find the coordinates of the foot P' of the perpendicular drawn from the point $P(2, 1, 3)$ to the plane $2x + y + z = 8$. Also find the image (reflection) of the point P in the plane.

Solution. (a) Let $P'(p', q', r')$ be the image of $P(p, q, r)$. Then the mid-point of PP' must lie on the given plane.

$$\therefore \quad \frac{p+p'}{2} + \frac{q+q'}{2} + \frac{r+r'}{2} = 8 \qquad \ldots(i)$$

Also the line PP' must be perpendicular to the plane. The direction ratios of PP' being $p - p', q - q', r - r'$, we therefore have

$$\frac{p-p'}{2} = \frac{q-q'}{1} = \frac{r-r'}{1} = k \text{ (say)}$$

where $p' = p - 2k, q' = q - k, r' = r - k$.

Substituting these in (i) and solving, we get

$$k = \frac{1}{3}(2p + q + r - 8)$$

Hence P' is

$$\left[\frac{1}{3}(4p - q - 2r + 16), \frac{1}{3}(8 - 2p + 2q - r), \frac{1}{3}(8 - 2p - q + 2r)\right] \qquad \ldots(ii)$$

(b) Any two points on the given line are evidently $P(1, 2, 3)$ and (on putting $x = 2$) $Q(2, 3, 4)$. Their images are (by using (ii)) $P'\left(\frac{1}{3}, \frac{2}{3}, \frac{5}{3}\right)$ and $Q'\left(-\frac{11}{3}, -\frac{1}{3}, \frac{13}{3}\right)$. The line joining P' and Q' is, therefore

$$\frac{x - \frac{1}{3}}{-\frac{11}{3} - \frac{1}{3}} = \frac{y - \frac{2}{3}}{-\frac{1}{3} - \frac{2}{3}} = \frac{z - \frac{5}{3}}{\frac{13}{3} - \frac{5}{3}}, \text{ i.e. } \frac{3x - 1}{-4} = \frac{3y - 2}{-3} = \frac{3z - 5}{4}$$

which is the required image of the given line PQ (Fig. 3.40(b)).

Example 3.38. Find the angle between the line

$$\frac{x-2}{1} = \frac{y-3}{-2} = \frac{z-4}{0}$$

and the plane $ax + by + cz + d = 0$.

Solution. If θ be the angle between the line and the plane, then $90° - \theta$ is the angle between the line and the normal to the plane (Fig. 3.41).

Now the direction ratios of the line are $1, -2, 0$ and the direction ratios of the normal to the plane are a, b, c.

$$\therefore \quad \sin(90° - \theta) = \frac{|a - 2b + 0c|}{\sqrt{1^2 + (-2)^2 + 0^2}\sqrt{a^2 + b^2 + c^2}}$$

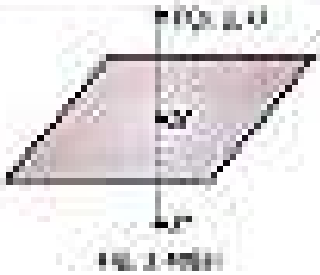


Fig. 3.40(a)

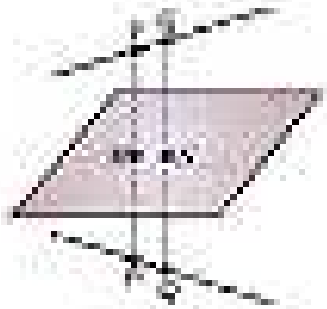


Fig. 3.40(b)

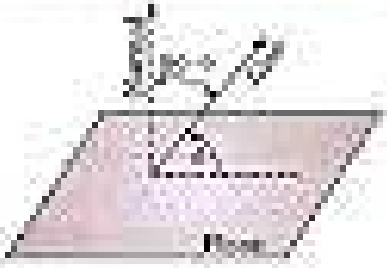


Fig. 3.41

or $$\sin\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{(a_1^2+b_1^2+c_1^2)}\sqrt{(a_2^2+b_2^2+c_2^2)}}$$

Hence the required angle $\theta = \sin^{-1}\left[\frac{|al+bm+cn|}{\sqrt{(l^2+m^2+n^2)}\sqrt{(a^2+b^2+c^2)}}\right]$

Cor. If the line is parallel to the plane, $\sin\theta = 0$

$\therefore$ $al + bm + cn = 0$

If the line is perpendicular to the plane, it will be parallel to its normal.

$\therefore$ $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$

Example 3.28. *Find the equation of the straight line through the origin, each of which is inclined at an angle of $60°$ to the straight line $\frac{x}{2} = \frac{y-3}{-1} = \frac{z-2}{1}$ and is inclined at an angle of $30°$ to it.*

Solution. Let AB be the given line and OA any one of the required lines. Let A be $(2r, -r+3, r+2)$. (Fig. 3.42)

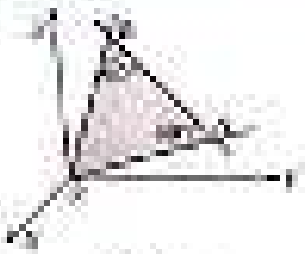


Fig. 3.42

$\therefore$ Direction ratios of OA are $2r, -r+3, r+2$.

Angle between AO and AB is to be $60°$.

$\therefore$ $$\cos 60° = \frac{2(2r) + 1(-r+3) + 1(r+2)}{\sqrt{(2^2+1^2+1^2)}\sqrt{(2r)^2+(-r+3)^2+(r+2)^2}}$$

or $$\frac{1}{2} = \frac{6r+3}{\sqrt{(6(6r^2+10r+13)}} \text{ or } r^2 + 2r + 1 = 0 \text{ i.e., } r = -1, -1$$

$\therefore$ Coordinates of A are $(-2, 4, 1)$.

Hence the equations of the required lines OA and OB are $\frac{x}{-2} = \frac{y}{4} = \frac{z}{1}$ and $\frac{x}{-2} = \frac{y}{4} = \frac{z}{1}$

EXERCISE 3.7

1. Prove that the points $(3, 2, 0)$, $(4, 3, 2)$ and $(6, 5, 6)$ are collinear. Find the equations of the line through them.
2. Find the angle between the line of intersection of the planes $2x + 2y + z = 12$, $x + 0y + 2z = 8$ and the line $\frac{x-4}{4} = \frac{y-3}{-4} = \frac{z+1}{1}$. (V.T.U., 2006)
3. Find the angle between the line of intersection of the planes $3x + 2y + z = 3$ and $2x + y - 3z = 4$ and the line of intersection of the planes $2x = y + z$ and $3x + 2y = 3z$.
4. Find the equation of the line through the point $(-2, 3, 5)$ and parallel to the planes $2x + 2y = 3z + 5$ and $3x + 2y + 4z = 5$.
5. Show that the line $\frac{x-1}{3} = \frac{y-2}{-2} = \frac{z-3}{-2}$ is parallel to the plane $2x + 2y - z = 5$ and find the distance between the line and the plane.
6. Find the equation of the line through $(1, 2, -3)$ perpendicular to each of the lines
$$\frac{x}{1} = \frac{y}{2} = \frac{z}{-1} \text{ and } \frac{x}{0} = \frac{y}{3} = \frac{z}{4}$$
7. Find the equation of the line bisecting the angle between the lines
$$\frac{x-1}{2} = \frac{y-2}{-2} = \frac{z-3}{1} \text{ and } \frac{x-1}{12} = \frac{y+2}{4} = \frac{z-3}{-3}$$
8. Find the foot of the perpendicular from $(1, 3, 3)$ to the line joining the points $(1, 4, 6)$ and $(2, 4, 5)$. (V.T.U., 2010)
9. Find the perpendicular distance of the point $(1, 1, 1)$ from the line
$$\frac{x-2}{2} = \frac{y+2}{2} = \frac{z}{-1}$$

10. Find the distance of the point $(3, -4, 5)$ from the plane $x + 3y - 4z = 16$ measured parallel to the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z}{-1}$. (U.P.T.U., 2002)

11. Find the reflection (mirror) of the point
(a) $(2, 3, 5)$ in the plane $x + y + z = 9$ (U.P.T.U., 2004)
(b) $(2, -1, 3)$ in the plane $2x - 3y + z + 1 = 0$.

12. Find the angle between the line $\frac{x+2}{2} = \frac{y-2}{2} = \frac{z-3}{1}$ and the plane $2x + y + z = 5$.

13. Find the equation of the plane through the points $(2, 1, -1)$, $(3, 2, 2)$ and parallel to the line
$x - 1 = \frac{1}{2}(1 - y) = \frac{1}{3}(z - 2)$. (U.P.T.U., 2006)

14. Find the equation of the straight line which passes through the point $(2, -1, 1)$, is parallel to the plane $4x + y + z = 10$ and is perpendicular to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$.

3.13 CONDITIONS FOR A LINE TO BE IN A PLANE

To find the conditions that the line $\frac{x-a_1}{l} = \frac{y-b_1}{m} = \frac{z-c_1}{n}$ ...(1)

may lie in the plane $ax + by + cz + d = 0$ ...(2)

Any point on the line (1) is $(lr + a_1, mr + b_1, nr + c_1)$ which will lie on the plane (2), if

$$a(lr + a_1) + b(mr + b_1) + c(nr + c_1) + d = 0,$$

or if $(al + bm + cn)r + (aa_1 + bb_1 + cc_1 + d) = 0$ ...(3)

The line (1) will lie in the plane (2), if every point of the line lies in the plane and (3) is satisfied by all values of r.

∴ The coefficient of $r = 0$ and the constant term $= 0$

i.e., $al + bm + cn = 0$ ...(4)

and $aa_1 + bb_1 + cc_1 + d = 0$ ...(5)

These are the required conditions which state that:

(i) the line should be parallel to the plane, (ii) the point of the line should lie in the plane.

Thus the equation of any plane through the line $\frac{x-a_1}{l} = \frac{y-b_1}{m} = \frac{z-c_1}{n}$

is $a(x - a_1) + b(y - b_1) + c(z - c_1) = 0$ where $al + bm + cn = 0$.

Cor. The equation of any plane through the line of intersection of the planes

$$ax + by + cz + d = 0 \quad ...(1)$$

and $a'x + b'y + c'z + d' = 0$ ...(2)

is $ax + by + cz + d + k(a'x + b'y + c'z + d') = 0$.

For (3) is an equation of the first degree in x, y, z it represents a plane and as it is satisfied by the coordinates of the points which satisfy both the given planes, i.e., it contains all the points common to the given planes.

Example 3.40. Find the equation of a plane passing through the line of intersection of the planes $2x + 3y + 4z = 0$ and $4x + 3y - 2z + 1 = 0$ and perpendicular to the plane $x - y - 2z + 3 = 0$. (U.P.T.U., 2005)

Solution. The equation of any plane through the line of intersection of the two given planes is

$$(2x + 3y + 4z) + k(4x + 3y - 2z + 1) = 0$$

or $(2 + 4k)x + (3 + 3k)y + (4 - 2k)z + k = 0$ ...(1)

The plane (1) will be perpendicular to the plane

$x - y - 2z + 3 = 0$ if their normals are perpendicular.

i.e., if $(2 + 4k).1 + (3 + 3k).(-1) + (4 - 2k).(-2) = 0$ or $k = 3/4$.

Substituting this value of k in (1), we get

$$(2 + 3)x + (3 + \tfrac{9}{4})y + (4 - \tfrac{6}{4})z + \tfrac{3}{4} = 0$$

i.e. $4x + 13y + 10z + 3 = 0$, which is the required equation.

Example 3.41. *Find the equation in the symmetrical form of the projection of the line* $\frac{x-1}{2} = y+1 = \frac{z-2}{4}$ *on the plane* $x + 2y + z = 12$.

Solution. Any plane through the given line is

$$Ax + By + Cz - (A - B + 1) = 0 \quad \ldots(i)$$

where $2A - B + 4C = 0$ $\ldots(ii)$

The plane (i) will be perpendicular to the given plane if

$$A + 2B + C = 0 \quad \ldots(iii)$$

Solving (ii) and (iii), we get $\frac{A}{-9} = \frac{B}{2} = \frac{C}{5}$.

Substituting these values in (i), we get $9x - 2y - 5z + 4 = 0$ $\ldots(iv)$

which cuts the given plane $x + 2y + z = 12$ $\ldots(v)$

along the required line of projection.

One point on this line is got by putting $z = 0$ in (iv) and (v) and solving, as $(-8/3, 26/3, 0)$.

The direction ratios of this line are found, by solving

$$l + 2m + n = 0 \quad \text{and} \quad 9l - 2m - 5n = 0$$

to be $4, -7, 10$.

Hence the required equations of the line of projection are

$$\frac{x + 8/3}{4} = \frac{y - 26/3}{-7} = \frac{z}{10}$$

Fig. 3.41

(The line of greatest slope in a plane is a line which lies in the plane and is perpendicular to the line of intersection of the plane with the horizontal plane.

In Fig. 3.42, AB is the line of intersection of the given plane with the horizontal plane. Then PM drawn perpendicular to AB, is the line of greatest slope of the plane π through the point P.)

Example 3.42. *Determine the line of greatest slope of the plane* $3x + y - 5z = 12$ *and passing through the point* $(2, 3, -1)$.

Solution. The equation of the horizontal plane through the origin is $4x - 3y + 3z = 0$ $\ldots(i)$

The direction ratios of the vertical are those of the given vertical line.

If l, m, n be the direction ratios of the line of intersection of the plane (i) and

$$3x + y - 5z = 12 \quad \ldots(ii)$$

then solving, $4l - 3m + 3n = 0$ and $3l + m - 5n = 0$, we have $l/14 = m/29 = n/13$ $\ldots(iii)$

Let l', m', n' be the direction ratios of the line of greatest slope which lies in the plane (ii).

$$\therefore \quad 3l' + m' - 5n' = 0 \quad \ldots(iv)$$

Also the line of greatest slope is perpendicular to the line of intersection of the planes (i) and (ii).

$$\therefore \quad 4l' - 3m' + 3n' = 0 \quad \ldots(v)$$

Solving (iv) and (v), $\frac{l'}{-4} = \frac{m'}{-3} = \frac{n'}{-5}$.

Hence the equations of the line of greatest slope through $(2, 3, -1)$ and having direction ratios $4, 3, 5$ are

$$\frac{x-2}{4} = \frac{y-3}{3} = \frac{z+1}{5}$$

EXERCISE 3.13

1. Find the equation of the plane which contains the line $\frac{x-1}{2} = y = z+1 = \frac{x-3}{4}$ and is perpendicular to the plane $x + 2y + z = 12$. (U.P.T.U. 2006)

2. Find the equation of the plane through the line $\frac{x-2}{2}=\frac{y-3}{1}=\frac{z-4}{3}$ and parallel to the line $\frac{x+1}{-3}=\frac{y-3}{-4}=\frac{z+2}{1}$.
3. Find the equations of the plane passing through the line of intersection of the planes $x+y+z=1$ and $2x+3y+z=4$ and perpendicular to the plane $3y-2z=6$.
4. Find the equation of the plane which contains the line of intersection of the planes $x+2y+z=1$ and $2x+y+3z=4$ and is parallel to the line passing the points $(2,1,1)$ and $(3,2,4)$. (Calicut, 2007)
5. Find in symmetric form the equations of the line which lies in the plane $3x+y+2z=4$ and is perpendicular to the line
$$\frac{x-1}{3}=\frac{y-1}{2}=\frac{z+1}{7}$$
at the point where the line pierces the plane.
6. A plane is drawn through the line $x+y=1$, $z=1$ to make an angle $\cos^{-1}(2/3)$ with the plane $x+y+z=5$. Find the two such planes and find their equations. Prove also that the angle between the planes is $\cos^{-1}(1/3)$.
7. Find the equations of the projection of the line $2x+y+z=1$, $x+2y+z=2$ on the plane $3x+3y+z=0$ and the angle between the projection and the line.
8. Assuming the plane $4x+2y+7z=8$ to be horizontal, find the equations of the line of greatest slope through the point $(2,3,2)$ on the plane $2x+y-6z=8$. (Kerala, 2005)

3.13 CONDITION FOR THE TWO LINES TO INTERSECT (OR TO BE COPLANAR)

Let the equations of the lines be $\frac{x-x_1}{l_1}=\frac{y-y_1}{m_1}=\frac{z-z_1}{n_1}$ ...(1)

$$\frac{x-x_2}{l_2}=\frac{y-y_2}{m_2}=\frac{z-z_2}{n_2} \quad ...(2)$$

The equation of any plane through the line (1) is $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$ ...(3)

where $al_1+bm_1+cn_1=0$ ...(4)

The line (2) will lie in the plane (3), if it is parallel to the plane and the point (x_2, y_2, z_2) lies on the plane.

$$\therefore \quad al_2+bm_2+cn_2=0 \quad ...(5)$$

and $a(x_2-x_1)+b(y_2-y_1)+c(z_2-z_1)=0$ ...(6)

Eliminating a, b, c from (4), (5) and (6), we get

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}=0$$ which is the required condition.

Also eliminating a, b, c from (3), (4) and (5), we get $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}=0$

which is the equation of the plane containing the lines (1) and (2).

Example 3.43. Show that the lines $\frac{x-5}{4}=\frac{y-7}{4}=\frac{z+3}{-5}$ and $\frac{x-8}{7}=\frac{y-4}{7}=\frac{z-5}{3}$ are coplanar. Find their common point and the equation of the plane in which they lie. (Madurai, 2003)

Solution. Any point on the first line is $(5+4r, 7+4r, -3-5r)$ ...(1)

which lies on the second line if $\frac{-3+4r}{7}=\frac{3+4r}{7}=\frac{-8-5r}{3}$ ...(2)

$\therefore$ Prea $\frac{-3+4r}{7}=3+4r$, so hence $r=-1$.

This value clearly satisfies the equation $\frac{3+4r}{7}=\frac{-8-5r}{3}$.

Hence the lines intersect, i.e., are coplanar and from (1) their point of intersection is $(1, 3, 2)$.

The equation of the plane in which they lie, is $\begin{vmatrix} x-3 & y-7 & z+2 \\ 4 & 4 & -8 \\ 7 & 1 & 2 \end{vmatrix} = 0$

i.e., $12x - 6y - 24z + 72 = 0$.

Example 3.48. *Show that the lines*

$$\frac{x+4}{3} = \frac{y+6}{5} = \frac{z-3}{7} \text{ and } 3x + 2y + z + 6 = 0 = 2x + 2y + 4z + 4$$

are coplanar. Find their point of intersection and the plane in which they lie.

Solution. Any point on the first line is $P(3r - 4, 5r - 6, 7r + 3)$, which lies on the plane

$$3x - 2y + z + 6 = 0$$

if $3(3r - 4) - 2(5r - 6) + (7r + 3) + 6 = 0$ or $r = 2$.

The point P will also lie in the plane $2x + 2y + 4z + 4 = 0$

if $2(3r - 4) + 2(5r - 6) + 4(7r + 3) + 4 = 0$ or $r = 2$.

Since the two values of r are equal, the given lines intersect, i.e., are coplanar.

Putting $r = 2$ in the coordinates of P, we get $(2, 4, -10)$ as their point of intersection.

The equation of a plane containing the second line is

$$3x - 2y + z + 6 + k(2x + 2y + 4z + 4) = 0$$

which will contain the first line if the point $(-4, -6, 3)$ lies on it.

$\therefore$ $-12 + 12 + 3 + 6 + k(-8 - 12 + 12 + 4) = 0$

i.e., $k = 3/8$.

Substituting this value of k, (1) becomes $48x - 32y + 32z + 72 = 0$, which is the required plane.

Example 3.49. *Find the equation of the line drawn through the point $(1, 0, -1)$ and intersecting the lines $x + 2y + 3z$ and $2x + y + z$, $4x + 3y = 2$.* (U.P.T.U., 2005)

Solution. The required line will be the join of

(a) the plane containing the first line and the point $(1, 0, -1)$,

(b) the plane containing the second line and the point $(1, 0, -1)$.

The equation of any plane containing the first line

i.e., $\frac{x}{2} = \frac{y}{1} = \frac{z}{1}$

is $a(x - 0) + b(y - 0) + c(z - 0) = 0$ ...(1)

where $2a + b + c = 0$ ...(2)

Also $(1, 0, -1)$ lies on (1) $\therefore$ $a - c = 0$ ...(3)

Solving (2) and (3), we have $\frac{a}{1} = \frac{b}{-2} = \frac{c}{1}$.

Substituting these values in (1), we get $x - 2y + z = 0$ ...(4)

Again, the equation of any plane containing the second line is

$3x + 2y - 2 + k(4x + 3y - 2) = 0$. Also $(1, 0, -1)$ lies on it. ...(5)

$\therefore$ $3 + 0 - 2 + k(4 - 3 - 2) = 0$, i.e., $k = \frac{2}{3}$.

Substituting $k = 2/3$ in (5), we get $17x + 12y + 18z - 7 = 0$ ...(6)

Hence (4) and (6) constitute the required line.

EXERCISE 3.5

1. Prove that the lines

$$\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-3}{5} \quad \text{and} \quad \frac{x-3}{3}=\frac{y-2}{4}=\frac{z-1}{5}$$

are coplanar and find the equation of the plane containing them.

2. Prove that the lines $\frac{x-3}{1}=\frac{y+3}{4}=\frac{z+7}{7}$ and $\frac{x+1}{3}=\frac{y+5}{5}=\frac{z+13}{9}$ are coplanar and find the coordinates of their point of intersection. (U.P.T.U., 2004; Kurukshetra, 2005)

3. Find the condition that the lines $\frac{x-a}{l}=\frac{y-b}{m}=\frac{z-c}{n}$ and $\frac{x-a'}{l'}=\frac{y-b'}{m'}=\frac{z-c'}{n'}$ are coplanar.

4. Show that the lines $\frac{x+1}{3}=\frac{y-3}{2}=\frac{z+1}{5}$ and $x+y+2z-3=0=x+2y+3z-4$ are coplanar. Find their point of intersection and the equation of the plane containing them. (U.P.T.U., 2005)

5. Show that the lines $x=3y+2=2z+3$ and $x+1=2y=2z+1$ are coplanar and find the equation of the plane containing them. (Kurukshetra, 2005)

6. Prove that the lines $x=ay+b=cz+d$ and $x=\alpha y+\beta=\gamma z+\delta$ are coplanar if $(d-b)(\alpha-a)=(\beta-b)(a-\gamma)$. (Punjab Univ., 2005)

7. Obtain the equation of the straight line lying in the plane

$$x-2y+3z-11=0$$

and intersecting the lines $\frac{x-3}{2}=\frac{y+2}{4}=\frac{z+4}{3}$ at right angles.

8. Find the equation of the straight line perpendicular to both the lines

$$\frac{x-1}{1}=\frac{y+1}{-2}=\frac{z+3}{2} \quad \text{and} \quad \frac{x+3}{3}=\frac{y-3}{-2}=\frac{z+2}{4}$$

and passing through their point of intersection.

9. A line with direction cosines proportional to 2, 1, −2 is drawn to intersect the lines

$$\frac{x-3}{1}=\frac{y-4}{-1}=\frac{z+1}{1} \quad \text{and} \quad \frac{x+3}{-1}=\frac{y+3}{2}=\frac{z-3}{1}$$

Find the coordinates of the point of intersection and the length intercepted.

3.15 SHORTEST DISTANCE BETWEEN TWO LINES

Two straight lines which do not lie in one plane are called skew lines. Each line possesses one and only one perpendicular which is the shortest distance between them.

Let the given skew lines AB and CD be

$$\frac{x-x_1}{l_1}=\frac{y-y_1}{m_1}=\frac{z-z_1}{n_1} \quad \text{and} \quad \frac{x-x_2}{l_2}=\frac{y-y_2}{m_2}=\frac{z-z_2}{n_2}$$

so that $A \equiv (x_1, y_1, z_1)$ and $C \equiv (x_2, y_2, z_2)$

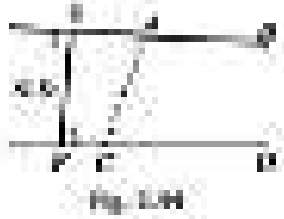


Fig. 3.44

Let l, m, n be the direction cosines of the shortest distance PP'.

Since PP' is both AB and CD

$$\therefore \quad l_1l+m_1m+n_1n=0 \quad \text{and} \quad l_2l+m_2m+n_2n=0$$

Solving,

$$\frac{l}{m_1n_2-m_2n_1}=\frac{m}{n_1l_2-n_2l_1}=\frac{n}{l_1m_2-l_2m_1}$$

$$=\frac{\sqrt{l^2+m^2+n^2}}{\sqrt{(m_1n_2-m_2n_1)^2+(n_1l_2-n_2l_1)^2+(l_1m_2-l_2m_1)^2}}=\frac{1}{\sin\theta} \quad \ldots(1)$$

where θ is the angle between the lines AB and CD.

$\therefore$ Length of S.D. (PP') = projection of MN on PP'

$=(x_2-x_1)l+(y_2-y_1)m+(z_2-z_1)n$ where l, m, n have the values as given by (1).

To find the equations of the line of shortest distance, we observe that it is coplanar with both AB and CD.

Plane containing the lines AB and EF is $\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l & m & n \end{vmatrix} = 0$ ...(1)

Plane containing the lines CD and EF is $\begin{vmatrix} x-x_2 & y-y_1 & z-z_2 \\ l_2 & m_2 & n_2 \\ l & m & n \end{vmatrix} = 0$ ...(2)

Hence (1) and (2) are the equations of the line of shortest distance.

Obs. The method for finding the foot of the perpendicular from a point to a line is explained in Art. 3.10 and by applying the above formula (3) we can find the length of the shortest distance between two skew lines.

Example 3.10. *Find the coordinates of the feet and the equations of the shortest distance between the lines*

$$\frac{x}{2} = \frac{y}{-3} = \frac{z}{1} \quad \text{and} \quad \frac{x-3}{3} = \frac{y-3}{-5} = \frac{z-5}{2}.$$ (V.T.U., 2008; Cochin, 2010)

Solution. Let l, m, n be the direction cosines of the shortest distance EF.

$\therefore$ EF is perpendicular to both AB and CD,

$\therefore$ $2l - 3m + n = 0$, $3l - 5m + 2n = 0$.

Solving $\frac{l}{1} = \frac{m}{3} = \frac{n}{1} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{11 + 9 + 11}} = \frac{1}{\sqrt{31}}$.

Fig. 3.46

$\therefore$ Length of S.D. (EF) = projection of AC on EF

$$= (2-0)\frac{1}{\sqrt{31}} + (3-0)\frac{3}{\sqrt{31}} + (-2-0)\frac{1}{\sqrt{31}} = \frac{9}{\sqrt{31}}.$$

The equations of the line of shortest distance (EF) are

$$\begin{vmatrix} x & y & z \\ 2 & -3 & 1 \\ 1 & 3 & 2 \end{vmatrix} = 0 \text{ and } \begin{vmatrix} x-3 & y-3 & z-5 \\ 3 & -5 & 2 \\ 1 & 3 & 1 \end{vmatrix} = 0$$

i.e., $9x + y - 9z = 0$ and $7x + y - 8z + 31 = 0$.

Example 3.11. *Find the points on the lines*

$$\frac{x-5}{3} = \frac{y+7}{-1} = \frac{z+3}{1}$$ ...(1)

$$\frac{x}{-3} = \frac{y+9}{2} = \frac{z-4}{4}$$ ...(2)

which are nearest to each other. Find also the shortest distance between the lines and its equations.

(V.T.U., 2004; Kurukshetra, 2003; Chennai, 2000)

Solution. Any point on the line (1) is $(3k + 5, -k - 7, k + 3)$ ...(3)

and any point on the line (2) is $(-3k', 2k' - 9, 4k' + 4)$ ...(4)

Then the direction ratios of EF are proportional to $3k + 3k' + 5$, $-k - 2k' + 16$, $k - 4k' - 1$.

Since EF is both the lines (1) and (2), $\therefore$ $3(3k + 3k' + 5) - (-k - 2k' + 16) + 4(k - 4k' - 1) = 0$

and $-3(3k + 3k' + 5) + 2(-k - 2k' + 16) + 4(k - 4k' - 1) = 0$

or $11k + 3k' - 4 = 0$, $3k + 20k' - 22 = 0$, whence $k = 1, k' = 1$.

Substituting $k = 1$ in (3) and $k' = 1$ in (4), we get E = (8, −8, 3) and F = (−3, −7, 8) which are the points on (1) and (2) nearest to each other.

$\therefore$ Length of the shortest distance (EF) = $\sqrt{(8+3)^2 + (-8+7)^2 + (3-8)^2} = 5\sqrt{3}$.

The equations of the shortest distance (EF) is $\frac{x-8}{11} = \frac{y+8}{-1} = \frac{z-3}{-5}$.

Obs. This method is sometimes very convenient and is generally useful when the points of intersection of the line of shortest distance with the given lines are required.

Example 3.46. Two straight rods in the form of straight lines AB and CD are such that the coordinates of A, B, C and D are respectively $(1, 2, 3)$, $(2, 3, 1)$, $(2, 3, 1)$, $(3, 4, 1)$ and $(2, -1, 0)$. Determine the minimum distance between the rods.

Solution. The direction ratios of AB are $1, -1, -2$ and those of CD are $1, -2, -1$.

The amount of distance between AB and CD is nothing but the shortest distance PQ between the rods.

If the direction ratios of PQ be l, m, n then

$$l - m - 2n = 0 \text{ and } l - 2m - n = 0$$

$$\therefore \quad \frac{l}{1} = \frac{m}{-1} = \frac{n}{1} \qquad [\because \text{PQ is both AB and CD}].$$

Thus the distance between the rods

= shortest distance between AB and CD

= projection of AC (or AD) on PQ

$$= \frac{1(0-2-1)-1(1-2)+1(3-3)}{\sqrt{1+1+1}} = \frac{2}{\sqrt{3}} \text{ (in magnitude)}$$

Example 3.48. *Find the equation of the plane through the line*

$$\frac{x-1}{2} = \frac{y-4}{3} = \frac{z-4}{0} \qquad \ldots(i)$$

and parallel to the line $\frac{x+1}{2} = \frac{y-1}{-4} = \frac{z+2}{3}$ $\qquad \ldots(ii)$

Hence find the shortest distance between the lines. (Kurukshetra, 2008)

Solution. The equation of the plane containing the line (i) and parallel to (ii) is

$$\begin{vmatrix} x-1 & y-4 & z-4 \\ 2 & 3 & 0 \\ 2 & -4 & 3 \end{vmatrix} = 0$$

or $$9x - 6y + 11z = 39 \qquad \ldots(iii)$$

Now the shortest distance between the lines (i) and (ii)

= Length of the perpendicular drawn from the point $(-1, 1, -2)$ of (ii) on the plane (iii)

$$= \frac{-9 - 6 - 22 - 39}{\sqrt{9^2 + 6^2 + 11^2}} = \frac{|-76|}{\sqrt{206}}, \text{ numerically.}$$

Example 3.50. *Show that the shortest distance between the line* $ax + by + cz + d = 0 = a'x + b'y + c'z + d'$ *and the line* $x = y = z$ *is* $\frac{d'-d}{\sqrt{(a-a')^2 + (b-b')^2 + (c-c')^2}}$.

Solution. The plane containing the given line is

$$(ax + by + cz + d) + k(a'x + b'y + c'z + d') = 0 \qquad \ldots(i)$$

or $$(a + ka')x + (b + kb')y + (c + kc')z + (d + kd') = 0$$

This plane is parallel to the line $x = y = z$, i.e., $a + ka' = b + kb' = c + kc'$, i.e., $k = \frac{a-b}{b'-a'} = \frac{b-c}{c'-b'}$ (say).

$$\text{i.e., } (a-b)x + (b-c)y + (a-2b+c)z + (d-d') = 0 \qquad \ldots(ii)$$

A point on the x-axis is the origin.

$\therefore$ Distance of the origin from the plane (ii)

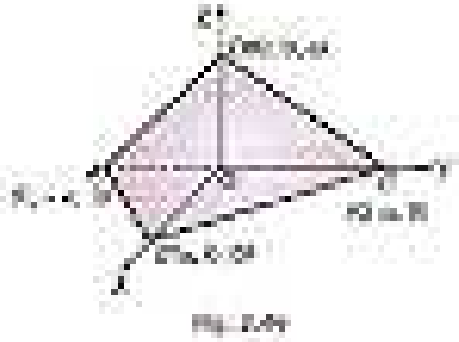
$$= \frac{d'-d}{\sqrt{(a-b)^2 + (b-c)^2 + (a-2b+c)^2}}, \text{ which is the required S.D.}$$

Example 3.52. *A square ABCD of diagonal 2a is folded along the diagonal AC so that the planes DAC and BAC are at right angles. Find the shortest distance between DC and AB.*

Solution. Let the diagonals AC and BD intersect at O the fixed position of the square. Let OA, OC and OB be the axes. Then equations of OC are

$$\frac{x-0}{0-0}=\frac{y-0}{a-0}=\frac{z-0}{0-a} \quad \text{or} \quad \frac{x}{0}=\frac{y}{a}=\frac{z}{-a}$$

and those of AB are $\frac{x-a}{a}=\frac{y}{0}=\frac{z}{0}$



The equation of the plane through BC and parallel to AD is

$$\begin{vmatrix} x & y-a & z \\ 0 & a & -a \\ a & a & 0 \end{vmatrix}=0 \quad \text{or} \quad x-y-z+a=0 \qquad ...(1)$$

A point on the line AB is $(a, 0, 0)$.

Hence required S.D. = Length of perpendicular from $(a, 0, 0)$ on the plane (1)

$$=\frac{a+a}{\sqrt{1+1+1}}=\frac{2a}{\sqrt{3}}.$$

EXERCISE 3.15

1. Find the magnitude and the equations of the shortest distance between the lines:

$$\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4} \text{ and } \frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}.$$

(U.P.T.U. 2006 ; Rajasthan 2002 ; Madras 2002)

2. Find the magnitude and equations of the shortest distance between the lines

$$\frac{x-3}{1}=\frac{y-5}{-3}=\frac{z-7}{1} \text{ and } \frac{x+1}{7}=\frac{y+3}{-6}=\frac{z+3}{1}$$ (Jammu, 2005 S ; Kerala 2006 S)

Find also the point where it meets the lines.

3. Find the shortest distance and the equation of the line of shortest distance between the lines $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-3}{-1}$ and $\frac{x-2}{1}=\frac{y-4}{2}=\frac{z-5}{1}$. (U.P.T.U. 2006)

4. Show that the shortest distance between the lines $y=mx+c, z=0$ and $y=-mx+c, z=0$ is $2c$.

5. If the shortest distance between the lines $\frac{x}{a}+\frac{z}{c}=1, y=0$ and $\frac{y}{b}+\frac{z}{c}=1, x=0$ is $\frac{ab}{\sqrt{a^2+b^2}}$, then show that $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}$.

6. Show that the shortest distance between z-axis and the line $ax+by+c=0=lx+my+nz+p$ is $\frac{al+bm}{\sqrt{(l^2+m^2)(a^2+b^2+c^2)-(al+bm+cn)^2}}$.

7. Show that the shortest distance between a diagonal of a rectangular parallelopiped whose edges are a, b, c and the edge not meeting it, is $\frac{abc}{\sqrt{(b^2c^2+c^2a^2+a^2b^2)}}$.

8. Show that the shortest distance between one pair of opposite edges of the tetrahedron formed by the planes $x=0, y=0, z=0, x+y+z=1$ is $\frac{1}{\sqrt{2}}$.

3.16 INTERSECTION OF THREE PLANES

Any three planes (no two of which are parallel) intersect in one of the following ways :

(1) The planes may intersect in a point, if the line of section of two of them is not parallel to the third.

(2) The planes may have a common line of section, if the line of section of two of them lies on the third (Fig. 3.47).

(3) The planes may form a triangular prism, if the line of section of two of them is parallel to the third but does not lie on it. (See Fig. 3.48)

Example 3.30. Prove that the planes
$x+2y-3z+4=0$, $3x+y+2z-5=0$ and $2x-y+5z-7=0$ form a triangular prism.
Also find the equation of the plane which contains the line of intersection of the first two planes and is perpendicular to the plane $x+y+z=0$. Find also the distance between the first two planes.

Solution. Any plane through the line of intersection of the planes (1) and (2) is

$$(x+2y-3z+4)+\lambda(3x+y+2z-5)=0$$

or $$(1+3\lambda)x+(2+\lambda)y+(-3+2\lambda)z+(4-5\lambda)=0 \quad \ldots(3)$$

Three planes will intersect in a common line if the plane (3) and (2) are parallel. For same planes,

$$\frac{1+3\lambda}{2}=\frac{2+\lambda}{-1}=\frac{-3+2\lambda}{5}=\frac{4-5\lambda}{-7} \quad \ldots(4)$$

From $\frac{1+3\lambda}{2}=\frac{-3+2\lambda}{5}$, we have $\lambda=\frac{-23}{11}$ which satisfies all the equations in (4).

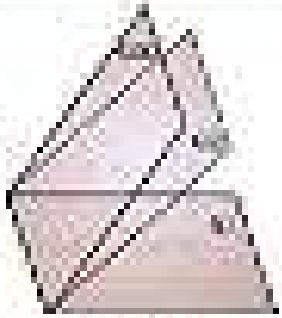


Fig. 3.47

Hence the given planes intersect in a line.

Any point on the line $\frac{x-1}{2}=\frac{y}{-2}=\frac{z-3}{1}=r$ (say) $\quad \ldots(5)$

is $(2r+1, -2r, r+3)$ which lies in the plane (1)

if $(2r+1)+2(-2r)-3(r+3)+4=0$, i.e., if $r=-1$.

$\therefore$ The coordinates of the point P in which (1) intersects (2) are $(-2, 2, 2)$.

Distance of P from plane (1) $=\frac{|2-2(-2)-3(2)+4|}{\sqrt{1+2^2+3^2}}=\frac{|-3|}{\sqrt{14}}=\frac{3}{\sqrt{14}}$ units (in magnitude)

Distance of P from plane (2) $=\frac{|6-2+10-5|}{\sqrt{9+1+4}}=\frac{9}{\sqrt{14}}$ units (in magnitude)

Hence the point P is equidistant from the planes (1) and (2).

Example 3.31. Prove that the three planes
$x+2y-z=3$, $3x-y+2z=5$ and $2x+y-z=5$
form a triangular prism and find the area of the normal section of the prism.

Solution. Let $\frac{x}{l}=\frac{y}{m}=\frac{z}{n}$ be the direction ratios of the line of intersection of the planes (1) and (2) then $l-2m+2n=0$, $3l+m-n=0$,

whence $$\frac{l}{1}=\frac{m}{-1}=\frac{n}{-1}.$$

To find a point P on this line, put $z=0$ in (1) and (2) $\Rightarrow x+2y=3$ and $3x-y=5$. Thus the point P is $(2, 0, 0)$.

Now the line of intersection of (1) and (2) is parallel to the plane (3).

$\because \quad 2(1)+1(-1)+1(-1)=0$

Also the point P does not lie on the plane (3).

Hence the given planes form a triangular prism.

Let ΔPQR be the normal section through P.

The equation of the plane through P perpendicular to the line of intersection of the planes (1) and (2) is

$$1(x-2)-1(y-0)-1(z-0)=0$$

or $$x-y-z-2=0 \quad \ldots(6)$$

Solving (1), (2) and (6), we get

$$Q=\left(\frac{1}{3}, \frac{1}{3}, -\frac{4}{3}\right).$$

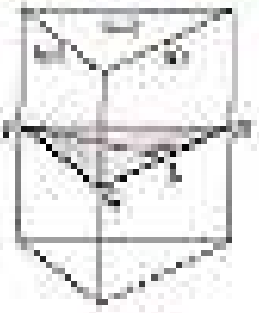


Fig. 3.48

Solving the equations (i), (ii) and (iii), we get

$$P \equiv \left(\frac{1}{3}, \frac{2}{3}, \frac{5}{3}\right)$$

$$\therefore \quad QR = \sqrt{\left(\frac{1}{3}-\frac{1}{3}\right)^2 + \left(\frac{2}{3}-\frac{2}{3}\right)^2 + \left(\frac{5}{3}-\frac{3}{3}\right)^2} = \sqrt{\left(\frac{2}{3}\right)^2} = \frac{2}{3}$$

Also PL = length P on the plane $(i) = \frac{3-2}{\sqrt{2^2+1^2+1^2}} = \frac{1}{\sqrt{6}}$

Hence the area of $\triangle PQR = \frac{1}{2} QR \times PL = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{\sqrt{6}} = \frac{1}{3\sqrt{6}}$ sq. units.

EXERCISE 3.11

1. Prove that the three planes $2x + 3y - 2z + 5 = 0$, $3x + 4y - 3z + 8 = 0$, $5x + 7y - 5z + 9 = 0$ pass through one line.
2. Prove that the planes $ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$ intersect in a line $\frac{x}{1} = \frac{y}{1} = \frac{z}{1}$ and show that the equations of this line are

$$\frac{x}{\sqrt{1-a^2}} = \frac{y}{\sqrt{1-b^2}} = \frac{z}{\sqrt{1-c^2}}$$ (Rajasthan 2002)

3. Show that the planes $x + 2y + 3 = 0$, $2x + 3y + 4z + 4 = 0$ and $3x + 4y - 2z + 21 = 0$ form a triangular prism.
4. Prove that the planes $2x + 3y + 6z = 6$, $3x + 4y + 7z = 10$, $x + y + z = 2$ form a prism, whose base equation is free of z and edges are parallel to the z-axis.

3.12 SPHERE

(1) Def. A sphere is the locus of a point which remains at a constant distance from a fixed point. The fixed point is called the centre and the constant distance the radius of the sphere.

(2) The equation of the sphere whose centre is (a, b, c) and radius r, is

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2.$$

For the distance of any point $P(x, y, z)$ on the sphere from the centre $C(a, b, c)$ = the radius r.

In particular the equation of the sphere whose centre is the origin and radius r, is

$$x^2 + y^2 + z^2 = r^2$$

(3) The equation $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ represents a sphere whose centre is $(-u, -v, -w)$ and radius

$$= \sqrt{u^2 + v^2 + w^2 - d}.$$

For on writing it as $(x^2 + 2ux) + (y^2 + 2vy) + (z^2 + 2wz) = -d$

or as $(x+u)^2 + (y+v)^2 + (z+w)^2 = u^2 + v^2 + w^2 - d$

and comparing with $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$,

it clearly represents a sphere whose centre is

$$(-u, -v, -w) \equiv (a, b, c) \text{ and radius } r = \sqrt{u^2 + v^2 + w^2 - d}$$

Thus the general equation of a sphere is such that

(i) it is of the second degree in x, y, z,

(ii) the coefficients of x^2, y^2, z^2 are equal,

and (iii) there are no cross terms containing yz, zx or xy.

(4) Section of a sphere by a plane is a circle and the section of a sphere by a plane through its centre is called a great circle.

Then the equations $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ ... (Sphere)

and $lx + my + nz = 0$ ... (Plane)

taken together represent a circle (Fig. 3.43) having centre $(-u, -v, -w)$ and radius $= \sqrt{u^2 + v^2 + w^2 - d}$.

Fig. 3.43

(ii) The equation of any sphere through the circle of intersection of

the sphere $S = 0$

and the plane $U = 0$

is $S + kU = 0$

For the equation $S + kU = 0$

represents a sphere and the points of intersection of the sphere $S = 0$ and the plane $U = 0$ satisfy it.

Example 21.24. Find the equation of the sphere through the points (0, 0, 0), (4, 2, −1), (−1, 1, 2), (3, 0, −1) and (1, 2, 3). Locate its centre and find its radius.

Solution. Let the required equation of the sphere be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad \ldots(i)$$

It passes through (0, 0, 0), (4, 2, −1), (−1, 1, 2), (3, 0, −1) and (1, 2, 3).

$\therefore$ $d = 0$,

$1 + 1 + 2v - 2w + d = 0$ or $v - w + 1 = 0$ $\ldots(ii)$

$1 + 4 - 2u + 4v + d = 0$ or $-2u + 4v + 5 = 0$ $\ldots(iii)$

$1 + 4 + 9 + 2u + 4v + 6w + d = 0$ or $u + 2v + 3w + 7 = 0$ $\ldots(iv)$

Multiplying (ii) by 0.1 and adding to (iv), we get

$$u + 2w + 2.1 = 0 \quad \ldots(v)$$

Solving (v) and (iii), we get $u = -\frac{26}{14}$, $v = -\frac{24}{14}$

From (ii), $w = v + 1 = -\frac{24}{14} + 1 = -\frac{10}{14}$

Substituting these values of u, v, w, d in (i), we get

$$x^2 + y^2 + z^2 - \frac{13}{7}x - \frac{12}{7}y - \frac{5}{7}z + 0 = 0 \quad \ldots(vi)$$

which is the required equation of the sphere.

Its centre is $(\frac{13}{14}, \frac{12}{14}, \frac{5}{14})$ i.e. $(0.93, 0.86, 0.36)$

and the radius is $\sqrt{(-\frac{13}{14})^2 + (-\frac{12}{14})^2 + (-\frac{5}{14})^2 - 0} = \sqrt{\frac{338}{196}}$ i.e. 1.31.

Example 21.25. (i) Find the equation of the sphere which has (x_1, y_1, z_1) and (x_2, y_2, z_2) as the extremities of a diameter.

(ii) Deduce the equation of the sphere that has the line joining the points $(3, -2, 4)$ and $(-1, 0, 2)$ as a diameter. Find the area of the circle in which this sphere is intersected by the plane $2x + y - z = 5$.

(Manipal, 2009; Anna University, 2009)

Solution. (i) Let $P(x, y, z)$ be any point on the sphere having $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ as ends of its diameter (Fig. 21.30 (a)). Then AP and BP are at right angles.

Fig. 21.30 (a)

Direction ratios of AP are $x - x_1, y - y_1, z - z_1$ and those of BP are $x - x_2$, $y - y_2, z - z_2$.

Hence

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$$

which is the required equation.

(ii) The equation of the required sphere is

$$(x - 3)(x + 1) + (y + 2)(y - 0) + (z - 4)(z - 2) = 0$$

or $$x^2 + y^2 + z^2 - y - 2z - 14 = 0 \quad \ldots(i)$$

Its centre is $(0, \frac{1}{2}, 1)$

and radius $= \sqrt{(0.25 + 1 + 14)} = \sqrt{15.25}$.

Let the given plane $2x + y - z - 3 = 0$ $\ldots(ii)$

cut the sphere (i) in the circle PP' having centre L.

$\therefore$ p = perpendicular CL from C on the plane (2)

$$= \frac{1\cdot 2 - 1 - 3}{\sqrt{4+1+2}} = \frac{2}{\sqrt{7}} \text{ (in magnitude)}$$

If r be the radius of the circle PP', then

$$r^2 = a^2 - p^2 = \frac{81}{4} - \frac{49}{28} = \frac{247}{28}$$

Hence the area of circle PP' $= \pi r^2 = \frac{247}{28}\pi$

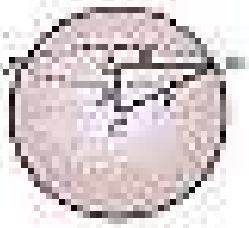


Fig. 3.30 (b)

Example 3.26. *A plane passes through a fixed point (a, b, c) and cuts the axes in A, B, C. Show that the locus of the centre of the sphere OABC is*

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2. \qquad \text{(U.P.T.U., 2003)}$$

Solution. Let the centre of the sphere OABC be P(x, y, z) so that its radius $OP = \sqrt{(x^2 + y^2 + z^2)}$.

$\therefore$ The equation of the sphere is

$$(x-x)^2 + (y-y)^2 + (z-z)^2 = x^2 + y^2 + z^2$$

or
$$x^2 + y^2 + z^2 - 2xx - 2yy - 2zz = 0 \qquad \ldots(1)$$

To find OA, putting $y = 0, z = 0$ in (1), we have

$x^2 - 2xx = 0, i.e., OA = x = 2x$. Similarly, $OB = 2y$, $OC = 2z$.

Thus the equation of the plane ABC is $\frac{x}{2x} + \frac{y}{2y} + \frac{z}{2z} = 1$.

Since the plane passes through (a, b, c), $\therefore \quad \frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z} = 1$.

Hence the locus of the point (x, y, z) of the sphere is

$$\frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z} = 1 \quad \text{or} \quad \frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2.$$

Example 3.27. *Find the equation of the sphere having the circle*

$$x^2 + y^2 + z^2 + 2x - 2y - 2z = 3, \; x + y + z = 3$$

as a great circle. *(Jamia, 2000; Madras, 2000 S)*

Solution. The equation of any sphere through the given circle is

$$x^2 + y^2 + z^2 + 2x - 2y - 2z - 3 + \lambda(x + y + z - 3) = 0$$

i.e.,
$$x^2 + y^2 + z^2 + (2 + \lambda)x + (\lambda - 2)y + (\lambda - 2)z - (3 + 3\lambda) = 0 \qquad \ldots(1)$$

In order that (1) may have the given circle as its great circle, its centre $\left(-\frac{2+\lambda}{2}, -\frac{\lambda-2}{2}, -\frac{\lambda-2}{2}\right)$ must lie on the plane $x + y + z = 3$.

$$\therefore \quad -\frac{2+\lambda}{2} - \frac{\lambda-2}{2} - \frac{\lambda-2}{2} = 3, \text{ i.e., } \lambda = -4$$

whence (1) becomes, $x^2 + y^2 + z^2 - 4x + 6y - 6z + 4 = 0$ which is the required equation.

Example 3.28. *Find the equation of the sphere which contains the circle* $x^2 + y^2 + z^2 + 2x + 2y + 2z = 11$, $x + y + z = 3$ *and touches* $2x + 3y + z + 11 = 0$.

Solution. Equation of any sphere containing the given circle is

$$x^2 + y^2 + z^2 + 2x + 2y + 2z - 11 + \lambda(x + y + z - 3) = 0$$

or
$$x^2 + y^2 + z^2 + (2 + \lambda)x + (2 + \lambda)y + (2 + \lambda)z - (11 + 3\lambda) = 0 \qquad \ldots(1)$$

Its radius is given by

$$r^2 = (1 + \lambda)^2 + (1 + \lambda)^2 + (1 + \lambda)^2 + \frac{1}{2}(3^2 + 11 + 3\lambda) = \frac{3}{4}\left[\lambda^2 + 2\lambda + \frac{100}{9}\right] = \frac{3}{4}\left[(\lambda + 1)^2 + \frac{91}{9}\right]$$

Now r^2 has the least value when $\lambda = -2$.

$\therefore$ Substituting $\lambda = -2$ in (1), we get

$$x^2 + y^2 + z^2 - 2x + 2y + 2z - 10 = 0$$

which is the required smallest sphere.

Example 3.18. Prove that the circles $x^2 + y^2 + z^2 - 2x + 3y + 4z - 5 = 0$, $2y + 3z + 1 = 0$ and $x^2 + y^2 + z^2 - 3x - 4y + 5z - 6 = 0$, $x + 2y - 2z + 5 = 0$ lie on the same sphere and find the equation.

Solution. Equation of any sphere containing the first circle is

$$x^2 + y^2 + z^2 - 2x + 3y + 4z - 5 + \lambda(2y + 3z + 1) = 0$$

or $$x^2 + y^2 + z^2 - 2x + (3 + 2\lambda)y + (4 + 3\lambda)z - 5 + \lambda = 0 \quad \ldots(1)$$

Similarly equation of any sphere containing the second given circle is

$$x^2 + y^2 + z^2 - 3x - 4y + 5z - 6 + \lambda'(x + 2y - 2z + 5) = 0$$

or $$x^2 + y^2 + z^2 + (-3 + \lambda')x + (-4 + 2\lambda')y + (5 - 2\lambda')z - 6 + 5\lambda' = 0 \quad \ldots(2)$$

If (1) and (2) will represent the same sphere then

$$-2 = -3 + \lambda' \quad \ldots(3) \qquad 3 + 2\lambda = -4 + 2\lambda' \quad \ldots(4)$$

$$4 + 3\lambda = 5 - 2\lambda' \quad \ldots(5) \qquad -5 + \lambda = -6 \quad \ldots(6)$$

Now (3) gives $\lambda' = 1$ and (6) gives $\lambda = -1$.

Clearly $\lambda = -1$ and $\lambda' = 1$ also satisfy (4) and (5). This shows that the given circles lie on the same sphere.

Substituting $\lambda = -1$ in (1) or $\lambda' = 1$ in (2), we get

$$x^2 + y^2 + z^2 - 2x - 2y - 2z - 5 = 0$$

which is the desired sphere.

EXERCISE 3.3

1. Find the equation of the sphere through the points $(0, 0, 0)$, $(1, -3, 1)$, $(4, -3, 1)$ and $(4, 1, 1)$. Also find its centre and radius.
2. Find the equation of the sphere of unit radius in the first octant, for which the coordinate planes are the planes $x = 0$, $y = 0$, $z = 0$. Also find its centre and radius.
3. Show that the equation of the sphere which passes through the points $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ and $(1, 1, 1)$ and
   (a) has its centre on the plane $x + y + z = 3$.
   (b) has its radius as small as possible.
4. A sphere of constant radius k passes through the origin and meets the axes in A, B, C. Prove that the centroid of the triangle ABC lies on the sphere $9(x^2 + y^2 + z^2) = 4k^2$. (Madras, 2000)
   (a) (Amended) $(1, 1, 1)$ lies on the sphere $x^2 + y^2 + z^2 = 3k^2$.
5. A plane passes through a fixed point (a, b, c) and cuts the axes in the form of this face of the pyramid of unit from the origin. Prove that the sphere $x^2 + y^2 + z^2 - ux - vy - wz = 0$.
6. A sphere of constant radius r passes through the origin O and cuts the axes in A, B, C. Prove that the locus of the foot of the perpendicular from O on the plane ABC is given by
$$(x^2 + y^2 + z^2)(x^{-2} + y^{-2} + z^{-2}) = r^2$$
7. A plane cuts the coordinate axes at A, B, C. If $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ are the centres of the
   (a) circumspheres of the tetrahedron $OABC$. (Anna, 2000)
   (b) circumcircle of the triangle ABC. Also obtain the coordinates of its centre.
8. Find the centre and radius of the circle $x^2 + y^2 + z^2 - 2y + 4z + 13 = 0$, $x + 2y + 2z = 17$.
   (U.P.T.U., 2000) (Anna, 2000, Amended) (Madras, 2003)
9. Show that the points $(2, -1, 0)$, $(1, -2, 0)$, $(0, 0, 2)$ and $(0, -3, 2)$ are concyclic.
10. Find the equation of the sphere for which the circle $x^2 + y^2 + z^2 - 2x + 4y - 2z - 5 = 0$, $x - 2y + 4z + 1 = 0$ is a great circle.
11. Find the equation of the sphere having its centre on the plane $4x - 5y + z = 3$ and passing through the circle $x^2 + y^2 + z^2 - 2x - 2y + 4z + 1 = 0$, $x - 2y + z = 5$. (Delhi, 2007)
12. Prove that the plane $x + 2y + z = 4$ cuts the sphere $x^2 + y^2 + z^2 - x + z - 2 = 0$ in a circle of radius $\frac{1}{2}$. Find also the equation of the sphere which has this circle as one of its great circles. (Nagpur, 2006)
13. Find the equation of the sphere passing through the circle $x^2 + y^2 + z^2 + 2x + 3y + 4 = 0$, $x + 2y + 3z + 4 = 0$ and the centre of the sphere $x^2 + y^2 + z^2 - 2x + 4y - 6z + 5 = 0$. (Madras, 2000)

3.13 EQUATION OF THE TANGENT PLANE

The equation of the tangent plane at any point (x_1, y_1, z_1) of the sphere

$$x^2 + y^2 + z^2 = a^2 \text{ is } xx_1 + yy_1 + zz_1 = a^2.$$

If $P(x, y, z)$ be any point on the tangent plane at $P_1(x_1, y_1, z_1)$ to the given sphere, the direction ratios of P_1P are $x - x_1, y - y_1, z - z_1$. Also the direction ratios of radius OP_1 are $x_1 - 0, y_1 - 0, z_1 - 0$.

Since OP_1 is normal to the tangent plane at P_1, $OP_1 \perp P_1P$.

$$\therefore \quad x_1(x - x_1) + y_1(y - y_1) + z_1(z - z_1) = 0$$

$$\text{or} \quad xx_1 + yy_1 + zz_1 = x_1^2 + y_1^2 + z_1^2 = a^2 \qquad [\because \ P_1(x_1, y_1, z_1) \text{ lies on the sphere.}]$$

This is the desired equation of the tangent plane.

Similarly, the tangent plane at (x_1, y_1, z_1) to the sphere

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$$

is $$xx_1 + yy_1 + zz_1 + u(x + x_1) + v(y + y_1) + w(z + z_1) + d = 0$$

Thus to write the equation of the tangent plane at (x_1, y_1, z_1) to the sphere, replace x^2 by xx_1, y^2 by yy_1, z^2 by zz_1, $2x$ by $x + x_1$, $2y$ by $y + y_1$, $2z$ by $z + z_1$.

Obs. The condition for a plane $lx + my + nz = p$ to touch a sphere is that the perpendicular distance of the centre from the plane be equal to the radius.

Example 3.40. Find the equation of the sphere passing through the circle $x^2 + y^2 + z^2 - 4x + 2z + 1 = 0$, $y = 0$ and touching the plane $2y + 4z + 3 = 0$.

Solution. The equation of any sphere through the given circle is

$$x^2 + y^2 + z^2 - 4x + 2z + 1 + ky = 0$$

$$\text{or} \quad x^2 + y^2 + z^2 - 4x + ky + 2z + 1 = 0 \qquad \ldots(1)$$

$\therefore$ Its centre $= (2, -k/2, -1)$ and radius $= \sqrt{(2)^2 + (k/2)^2 - 1 - (1)} = \sqrt{(1 + k^2/4)}$

The sphere (1) will touch the plane $2y + 4z + 3 = 0$, if $\perp$ distance of the centre $(2, -k/2, -1)$ from the plane = radius.

$$\text{i.e.,} \quad \frac{|2(-k/2) + 4(-1) + 3|}{\sqrt{4 + 16}} = \sqrt{\left(1 + \frac{k^2}{4}\right)} \quad \text{or} \quad \frac{|-k - 1|}{2\sqrt{5}} = \sqrt{\left(1 + \frac{k^2}{4}\right)}$$

$$\text{or} \quad \frac{(k+1)^2}{20} = 1 + \frac{k^2}{4} \quad \text{or} \quad 4k^2 + 20k + 44 = 5k^2 + 20$$

$$\therefore \quad k = \frac{-20 \pm \sqrt{(20)^2 - 24}}{2} = -\frac{11}{2} \text{ or } -4$$

Substituting the values of k in (1), we get

$$x^2 + y^2 + z^2 - 8x - \frac{11}{2}y + 2z + 5 = 0 \text{ and } x^2 + y^2 + z^2 - 8x - 4y + 2z + 5 = 0$$

as the two required spheres.

Example 3.41. Find the equation of the sphere which touches the plane $x - 2y - 2z = 7$ at the point $(3, -1, -1)$ and passes through the point $(5, 1, -3)$.

Solution. If C is the centre of the sphere, then CA is perpendicular to the given plane $x - 2y - 2z = 7$.

$\therefore$ The direction ratios of CA, being $1, -2, -2$, the equation of CA is

$$\frac{x-3}{1} = \frac{y+1}{-2} = \frac{z+1}{-2} = r \text{ (say)}$$

Any point on CA is $(r + 3, -2r - 1, -2r - 1)$ which will represent C for some value of r.

Since B lies on the sphere, therefore its radius CA = radius CB or $(CA)^2 = (CB)^2$

$$\text{i.e.,} \quad (r+3-3)^2 + (-2r-1+1)^2 + (-2r-1+1)^2 = (r+3-5)^2 + (-2r-1-1)^2 + (-2r-1+3)^2$$

$$\text{or} \quad 9r^2 = 4 + 16r + 4r^2 + 1 + 4r + 4r^2$$

$$\text{or} \quad 9r = 4r + 5 \quad \text{or} \quad r = 1.$$

$\therefore$ The centre C is $(4, -3, -3)$ and radius $CA = \sqrt{(9 + 36 + 36)} = 9$.

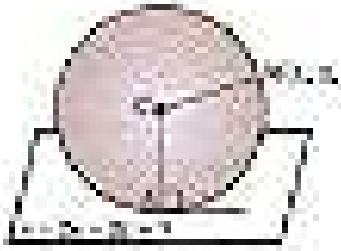


Fig. 3.31

Hence the required equation of the sphere is

$$(x-3)^2+(y-3)^2+(z-6)^2=13^2$$

or $$x^2+y^2+z^2-6x-6y-12z-31=0$$

Fig. 3.12

Orthogonal spheres. Two spheres are said to cut orthogonally if the tangent planes at a point of intersection are at right angles (Fig. 3.12).

The radii of such spheres through their point of intersection P, being ⊥ to the tangent planes at P are also at right angles. Thus two spheres cut orthogonally if the square of the distance between their centres is the sum of the squares of their radii.

Example 3.10. *Show that the conditions for spheres*

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0$$

and

$$x^2+y^2+z^2+2u'x+2v'y+2w'z+d'=0$$

to cut orthogonally is $2uu'+2vv'+2ww'=d+d'$. (Anna, 2002 S)

Solution. The centres of the spheres are

$$C(-u,-v,-w) \text{ and } C'(-u',-v',-w') \text{ and their radii are}$$

$$r=\sqrt{u^2+v^2+w^2-d},$$

$$r'=\sqrt{u'^2+v'^2+w'^2-d'}.$$

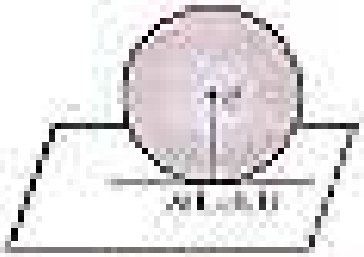


Fig. 3.13

Now the spheres will cut orthogonally, if $CC'^2=r^2+r'^2$

i.e., $$(u-u')^2+(v-v')^2+(w-w')^2$$

$$=u^2+v^2+w^2-d+u'^2+v'^2+w'^2-d'$$

or $2uu'+2vv'+2ww'=d+d'$ which is the required condition.

Example 3.11. *Find the equation of the sphere which touches the plane* $2x+2y-z+2=0$ *at the point* $(1,-2,1)$ *and cuts the sphere* $x^2+y^2+z^2-2x+4y-2z-3=0$ *orthogonally.* (Bombay, 2003)

Solution. The given plane is

$$2x+2y-z+2=0 \quad \ldots(1)$$

and it touches the required sphere at $A(1,-2,1)$ if the centre lies on the normal to (1) at A (Fig. 3.14). The equations of the normal to (1) at A are $\frac{x-1}{2}=\frac{y+2}{2}=\frac{z-1}{-1}$

Any point on this line is $C(2r+1, 2r-2, -r+1)$

Also radius $AC=r$ of the required sphere.

$$\therefore r=\sqrt{(2r)^2+(2r-1)^2+(1-r)^2}=r\sqrt{18}.$$

Since the required sphere cuts the given sphere

$$x^2+y^2+z^2-2x+4y+2=0 \quad [\text{Centre}=(1,-2,0) \text{ and } r=\text{radius}=3$$

orthogonally, therefore distance between their centres^2 = Σ of squares of their radii

i.e., $$(2r+1-1)^2+(2r-2+2)^2+(-r+1)^2=18r^2+9 \text{ or } r=-3/2.$$

Thus centre C is $(-3, -7/2, 5/2)$ and radius $=\frac{3\sqrt{11}}{2}$.

Hence the required sphere is

$$(x+3)^2+(y+\tfrac{7}{2})^2+(z-\tfrac{5}{2})^2=\tfrac{99}{4}$$

or $$x^2+y^2+z^2+6x+7y-5z+12=0.$$

EXERCISE 3.3

1. Find the equation of the tangent planes to the sphere
   (a) $x^2+y^2+z^2-4x+2y-6z+13=0$ which are parallel to the plane $x+2y+2z=0$. (Anna, 2004)
   (b) $x^2+y^2+z^2=a^2$ which pass through the line $x+y=a, z=2x+2$. (Madras, 2004)

2. Find the equation of the sphere which is passed through the circle
$x^2 + y^2 + z^2 + 3x + 3y + 3z + 3 = 0$ and cuts the plane $x + 2y = 16$. (Anna, 2009)

3. Find the equation of the sphere which is tangential to the plane $x - 2y + 2z - 3 = 0$ at $(1, 2, -1)$ and passes through $(2, 1, -3)$.

4. (a) Prove that the equation of the sphere which lies in the first octant and touches the coordinate planes is of the form $x^2 + y^2 + z^2 - 2ax - 2ay - 2az + a^2 = 0$.

(b) Find the equation of the sphere passing through $(1, 1, 1)$ and touching the coordinate planes.

5. Tangent planes at any point of the sphere $x^2 + y^2 + z^2 = r^2$ cuts the coordinate axes at A, B, C. Prove that the locus of the point of intersection of the planes drawn parallel to the tangent plane through A, B, C to the surface $x^{-2} + y^{-2} + z^{-2} = r^{-2}$. (Engineering, 2006)

6. Find the equation of the tangent line to the circle $x^2 + y^2 + z^2 = 2$, $2x - 3y + 4z = 2$ at the point $(1, 1, -1)$.

7. Show that the planes $x^2 - y^2 + z^2 - 2x + 2y + 1 = 0$ & $x + 2y + z = 1$ touch the sphere at the point $(1, -1, 0)$ and find the nature of the circle $x = 1$.

8. Find the centre and the length of the diameter of the circle in which the plane $x + y + z = 3$ cuts the sphere $x^2 + y^2 + z^2 = 9$.

9. Show that the spheres $x^2 + y^2 + z^2 - 2y + 4z - 4 = 0$ and $x^2 + y^2 + z^2 - 4x + 4y - 2z + 8 = 0$ touch externally and find their point of contact.

10. Show that the spheres $x^2 + y^2 + z^2 - 2x + 4y + 2z - 3 = 0$ and $x^2 + y^2 + z^2 - 4x - 6y + 2z + 10 = 0$ touch internally and find their point of contact.

3.3.3 (1) CONE

Def. A cone is a surface generated by a straight line which passes through a fixed point and satisfies one more condition e.g., it may intersect a given curve called the guiding curve.

The fixed point is called the vertex and the straight line in any position is called a generator.

The degree of the cone depends upon the nature of the guiding curve. In case the guiding curve is a conic, the equation of the cone shall be of the second degree. Such cones are called Quadric cones. In what follows, we shall be concerned only with quadric cones.

Example 3.44. *Find the equation of the cone whose vertex is* $(1, 1, 2)$ *and base the circle* $2x^2 + 3y^2 = 1$, $z = 3$.

Solution. Any line through $(1, 1, 2)$ is

$$\frac{x-1}{l} = \frac{y-1}{m} = \frac{z-2}{n} \quad \ldots(i)$$

It meets $z = 3$, where $\frac{x-1}{l} = \frac{y-1}{m} = \frac{1}{n}$

where $x = 1 + \frac{l}{n}$, $y = 1 + \frac{m}{n}$.

Substituting these values of x and y in $2x^2 + 3y^2 = 1$,

$$2\left(1 + \frac{l}{n}\right)^2 + 3\left(1 + \frac{m}{n}\right)^2 = 1 \quad \ldots(ii)$$

Eliminating l, m, n from (i) and (ii), the locus of the line (i) is

$$2\left(1 - \frac{x-1}{z-2}\right)^2 + 3\left(1 - \frac{y-1}{z-2}\right)^2 = 1$$

or $2x^2 + 3y^2 + 5z^2 - 6yz - 12zx + 12x + 6y - 26z + 17 = 0$ which is the required equation.

Example 3.45. *Find the equation of the cone whose vertex is the origin and guiding curve is* $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, $lx + my + nz = p$. ...(i)

Solution. Any line through $(0, 0, 0)$ is $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ ...(ii)

Any point on it is $P(lr, mr, nr)$.

If (ii) intersects the given curve, the coordinates of P should satisfy its equations.

$\therefore$ $\frac{\alpha^2}{a^2}+\frac{\beta^2}{b^2}+\frac{\gamma^2}{c^2}=2$ and $\alpha\beta+\beta\gamma+\gamma\alpha=1$.

Eliminating α, $\left(\frac{l^2}{a^2}+\frac{m^2}{b^2}+n^2\right)\frac{1}{c^2}(lx+my+nz)^2=1$.

Simplifying, $(bcl)^2+(cam)^2+(abn)^2+2mn(yz+zx)+2nl(zx+xy)+2lm(xy+yz)=0$ ...(ii)

Eliminating l, m, n from (i) and (ii), the locus of the line $\frac{x}{l}=\frac{y}{m}=\frac{z}{n}$ is

$x^2(b^2c^2+2mn)+y^2(c^2a^2+2nl)+z^2(a^2b^2+2lm)+2yz(ca^2+ab^2)+2zx(ab^2+bc^2)+2xy(bc^2+ca^2)=0$, which is the required equation.

Note The equation of the cone with origin as the vertex is a homogeneous equation of the second degree in x, y, z, i.e., all terms are of the same degree. The reason is that every generator of a cone has the equation of the form $\frac{x}{l}=\frac{y}{m}=\frac{z}{n}$. So the point (lr, mr, nr) will satisfy the equation of the cone for every value of r. This is possible only if the equation is homogeneous.

Example 3.66. A variable plane parallel to the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=0$ meets the coordinate axes in A, B, C. Find the equation of the cone whose vertex is the origin and guiding curve the circle ABC.

Solution. Let the plane $\frac{x}{m}+\frac{y}{n}+\frac{z}{p}=k$ ...(i)

meet the axes at A, B, C so that $A=(km, 0, 0), B=(0, kn, 0)$ and $C=(0, 0, kp)$.

The equation of the sphere through $O(0, 0, 0)$ and A, B, C is

$x^2+y^2+z^2-kmx+ky+cz)=0$ ...(ii)

As we the equation of the cone with vertex at O is a homogeneous equation of the second degree, therefore, it must be satisfied by points lying on the circle ABC, i.e., so (i) and (ii) both.

Making (ii) homogeneous with the help of (i), we have

$x^2+y^2+z^2-kmx+ky+cz)\left(\frac{x}{m}+\frac{y}{n}+\frac{z}{p}\right)=0$

or $yz\left(\frac{b}{nc}+\frac{c}{pb}\right)+zx\left(\frac{c}{ma}+\frac{a}{pc}\right)+xy\left(\frac{a}{mb}+\frac{b}{na}\right)=0$ which is the required equation.

Example 3.67. Show that the general equation of the cone of the second degree which passes through the axes of the coordinates is $fyz+gzx+hxy=0$.

Solution. Any cone which passes through the axes will have origin for its vertex. The general equation of a cone of the second degree having vertex at the origin is of the form

$ax^2+by^2+cz^2+2fyz+2gzx+2hxy=0$ ...(i)

Since it passes through x-axis

$\therefore$ The direction cosines of x-axis, i.e., $1, 0, 0$ must satisfy (i). This gives $a=0$.

As the cone passes through y-axis, $b=0$.

Similarly, as the cone passes through z-axis, $c=0$.

Hence (i) reduces to $fyz+gzx+hxy=0$

(2) Right circular cone. **Def.** *A right circular cone is a surface generated by a straight line which passes through a fixed point (vertex) and makes a constant angle with a fixed line (Fig. 3.54).*

The constant angle $\angle AVC$ is called the semi-vertical angle of the cone and VC is called the axis. The section of a right circular cone by a plane perpendicular to the axis is a circle.

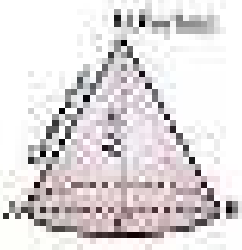


Fig. 3.54

Example 3.68. Find the equation of the right circular cone whose vertex is the origin, whose axis is the line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ and which has an semi-vertical angle of $30°$. (Anna, 2006)

Solution. Let $P(x, y, z)$ be any point on the cone with vertex O and axis OC.

$\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$, so that $\angle POC=30°$. (Fig. 3.55)

Now the direction ratios of OP are x, y, z and those of OC are $1, 2, 3$.

$$\therefore \quad \cos 60^\circ = \frac{x(1) + y(2) + z(3)}{\sqrt{x^2 + y^2 + z^2}\sqrt{1^2 + 2^2 + 3^2}}$$

or $$\frac{\sqrt{3}}{2} = \frac{x + 2y + 3z}{\sqrt{14(x^2 + y^2 + z^2)}}$$

Squaring $3(x^2 + y^2 + z^2) = 4(x + 2y + 3z)^2$

or $$x^2 + 13y^2 + 5z^2 - 8xy - 24yz - 12zx = 0$$

which is the required equation of the cone.

Fig. 4.55

Example 4.30. *Find the equation of the right circular cone with vertex at the origin, axis the line $x = y = z$ and one of its generators $x + y + z = 3$.* (Kharagpur 2005)

Solution. The vertex is the point of intersection of the line $x + y + z = 0$, $x = 0$ and the y-axis, i.e., $x = 0$, $y = 0$ (Fig. 4.56).

Vertex is $A(0, 0, 0)$. A generator of the cone is

$$\frac{x}{0} = \frac{y}{3} = \frac{z - 3}{-3}$$

Direction ratios of this generator are $0, 3, -3$ and the axis $(x = y = z)$ are $1, 1, 1$. The semi-vertical angle α is, therefore, given by

$$\cos\alpha = \frac{0 \cdot 1 + 3 \cdot 1 + (-3) \cdot 1}{\sqrt{18}\sqrt{3}} = \frac{0}{\sqrt{18}}$$

Let $P(x, y, z)$ be any point on the cone so that the direction ratios of AP are $x, y, z - 3$. If the AP makes an angle α with AB, we have

$$\cos\alpha = \frac{x \cdot 1 + y \cdot 1 + (z - 3) \cdot 1}{\sqrt{x^2 + y^2 + (z - 3)^2}\sqrt{3}}$$

Thus $$\frac{(z - 3)^2}{x^2 + y^2 + (z - 3)^2} = \cos^2\alpha = \frac{1}{2}$$

or $$x^2 + y^2 - z^2 + 6z - 9 = 0$$

which is the required equation of the cone.

Fig. 4.56

Example 4.31. *Find the equations of the lines in which the plane $2x + y - z = 0$ cuts the cone $4x^2 - y^2 + 3z^2 = 0$.*

Solution. Let $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ be one of the two lines in which the given plane

$$2x + y - z = 0 \quad \ldots(1)$$

cuts the given cone $\quad 4x^2 - y^2 + 3z^2 = 0 \quad \ldots(2)$

$\because$ This line lies on (1), $\quad \therefore \quad 2l + m - n = 0 \quad \ldots(3)$

and it lies on (2), $\quad \therefore \quad 4l^2 - m^2 + 3n^2 = 0 \quad \ldots(4)$

To eliminate n from (3) and (4), put $n = 2l + m$ in (4).

$$4l^2 - m^2 + 3(2l + m)^2 = 0 \quad \text{or} \quad (4l + m)(2l + m) = 0$$

$\therefore$ Either $4l + m = 0$ | or $2l + m = 0$

From (3) $\quad 2l + m - n = 0$ | and $2l + m - n = 0$

$\therefore \quad \frac{l}{1} = \frac{m}{-4} = \frac{n}{2}$ | $\therefore \quad \frac{l}{-1} = \frac{m}{2} = \frac{n}{0}$

Hence the required lines are

$$\frac{x}{1} = \frac{y}{-4} = \frac{z}{2} \text{ and } \frac{x}{-1} = \frac{y}{2} = \frac{z}{0}$$

Example 4.32. *Find the equation of the enveloping cone of the sphere $x^2 + y^2 + z^2 = a^2$ which passes through (α, β, γ).*

Solution. The equation of any generator through $P(x_1, y_1, z_1)$ having direction ratios l, m, n is

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n} = r \text{ (say)} \quad \ldots(i)$$

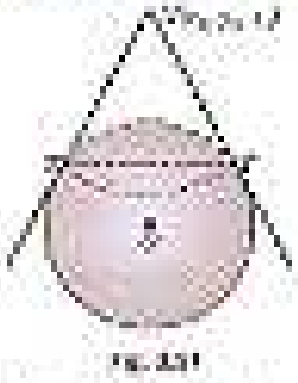


Fig. 3.31

Any point on (i) is $P(lr + x_1, my_1 + nr, z_1 + nr)$.

If lies on the given sphere if

$(lr + x_1)^2 + (my_1 + nr)^2 + (nr + z_1)^2 = a^2$

or $\quad (l^2 + m^2 + n^2)r^2 + 2(lx_1 + my_1 + nz_1)r + x_1^2 + y_1^2 + z_1^2 - a^2 = 0 \quad \ldots(ii)$

The line (i) will touch the given sphere if (ii) has equal roots.

$\therefore \quad (lx_1 + my_1 + nz_1)^2 = (l^2 + m^2 + n^2)(x_1^2 + y_1^2 + z_1^2 - a^2) \quad \ldots(iii)$

The locus of all such lines is the enveloping cone of the given sphere which is obtained by eliminating l, m, n from (i) and (iii).

Thus $(lx - xx_1 + ly - yy_1 + lz - zz_1)^2 = (lx - xx_1)^2 + (ly - yy_1)^2 + (lz - zz_1)^2)(x_1^2 + y_1^2 + z_1^2 - a^2)$

which is the equation of the enveloping cone. (Fig. 3.31)

Hint. Find a to satisfy the condition $SS_{11} = T^2$

where $\quad S \equiv x^2 + y^2 + z^2 - a^2, S_{11} = x_1^2 + y_1^2 + z_1^2 - a^2, T = xx_1 + yy_1 + zz_1 - a^2$

This is the enveloping cone of the sphere $S = 0$ with vertex (x_1, y_1, z_1). (U.P.T.U., 2012)

EXERCISE 3.12

1. Find the equation of the cone with vertex $(1, 2, 3)$ and base $x^2 + y^2 = 4, z = 0$.
2. Find the equation of the cone whose vertex is $(0, 0, 3)$ and which passes through the curve $2y^2 + z^2 = 16, x + 2y = 3$.
3. Find the equation of the cone whose vertex is $(1, 2, 3)$ and which passes through the curve $x^2 + y^2 + z^2 = 4, x + y + z = 1$. (U.P.T.U., 2012)
4. The section of a cone cut by the plane $lx + my + nz = p$ is a circle of radius r and the plane cuts the axis at a distance d from the vertex. Show that the equation of the cone is $(x^2 + y^2 + z^2)(l^2 + m^2 + n^2) = (lx + my + nz)^2 (1 + r^2/d^2)$.
5. Find the equation of the right circular cone whose vertex is at the origin and one of whose generators is $y = mx, z = 0$. Show that the condition that the plane $lx + my + nz = 0$ may touch the cone is $l^2(n^2 - m^2) + 2lmn = 0$. (U.P.T.U., 2008) (Rajasthan, 2005)
6. Find the equation to the cone which has vertical angle α, which has its vertex at the origin and one edge of which lies along the line $x = y = z$. (U.P.T.U., 2009)

   Also show that the plane $lx + my + nz = 0$ cut the cone in two straight lines which are at right angles if $l + m + n = 0$.
7. Find the equation of the circular cone which passes through the point $(1, 2, 3)$ and has its vertex at the origin and axis the line $x/2 = y/3 = z/6$. (Delhi, 2005) (Rajasthan, 2005) (U.P.T.U., 2006)
8. Find the equation of the right circular cone generated by revolving the line $x = 0, y = z$ about the line $x = y = z$. (Meerut, 2009)
9. Find the equation of the right circular cone passing through the coordinate axes having vertex at the origin. Also find the semi-vertical angle and the equation of the axis.
10. Find the semi-vertical angle and the equation of the right circular cone having its vertex at the origin and passing through the circle $x^2 + y^2 + 2z = 0, x + y = 1$. (Meerut, 2009)
11. Find the equation of the right circular cone with a fixed vertex $(1, 2, 3)$ whose base is the circle $x + y + z = 1, x^2 + y^2 + z^2 = 1$. (Rajasthan, 2006)
12. Find the equation to the two tangent planes to the plane $2x + y - 2z = 0$ which contain the coordinate axes. Also find the angle between them.
13. Prove that the plane $x + y + z = 3$ touches the cone $yz + zx + xy = 0$ at points whose lines $x^2 + y^2 + z^2 = 0$.
14. Find the equation of the enveloping cone of the sphere $x^2 + y^2 + z^2 + 2x - 4y + 2z - 1 = 0$ with vertex at $(1, 2, 1)$.

3.13 CYLINDER

Def. A cylinder is a surface generated by a straight line which is parallel to a fixed line and satisfies one more condition e.g., it may intersect a given curve called the guiding curve.

The straight line in any position is called the *generator* and the fixed line the *axis* of the cylinder.

Example 3.32. *Find the equation of a cylinder whose axis passes through the point (a, b, c) and has direction ratios l, m, n and whose radius is r. Also find the condition that the plane $lx + my + nz = p$ is a tangent plane to the cylinder.* (U.P.T.U. 2002, Kanpur 2006)

Solution. Let $P(x_1, y_1, z_1)$ be any point of the cylinder so that the equation of the generator through P is

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad \ldots(1)$$

Given guiding circle is $x^2 + y^2 = a^2, z = 0$ $\quad \ldots(2)$

The generator (1) cuts the plane $z = 0$, where

$$\frac{x - x_1}{l} = \frac{-y_1}{m} = \frac{-z_1}{n}$$

i.e., where $x = x_1 - \frac{lz_1}{n}$ and $y = y_1 - \frac{mz_1}{n}$

But these values of x and y satisfy $x^2 + y^2 = a^2$

$$\therefore \quad \left(x_1 - \frac{lz_1}{n}\right)^2 + \left(y_1 - \frac{mz_1}{n}\right)^2 = a^2$$

Hence the locus of (x_1, y_1, z_1) is

$(nx - lz)^2 + (ny - mz)^2 = a^2 n^2$, which is the required equation of the cylinder.

(2) Right circular cylinder. Def. A right circular cylinder is the surface generated by a straight line which is parallel to a fixed line and at a constant distance from it.

The constant distance is called the radius of the cylinder.

Example 3.33. *The section of a circular right cylinder by the plane $x + y + z = 3$ is a circle of radius 2. Find the equation of the cylinder.* (U.P.T.U. 2006)

Solution. A point on the axis of the cylinder is $A(1, -3, 2)$ and its direction ratios are $2, -1, 5$.

$\therefore$ Its actual direction cosines are $\frac{2}{\sqrt{30}}, \frac{-1}{\sqrt{30}}, \frac{5}{\sqrt{30}}$.

Let $P(x, y, z)$ be any point on the cylinder. Draw $PM \perp$ to the axis AM. Then $AP = 2$. Now AM = Projection of AP on AM (axis)

$$= (x - 1)\frac{2}{\sqrt{30}} + (y + 3)\frac{-1}{\sqrt{30}} + (z - 2)\frac{5}{\sqrt{30}}$$

$$= \frac{2x - y + 5z - 15}{\sqrt{30}}$$

Also $\quad AP = \sqrt{(x - 1)^2 + (y + 3)^2 + (z - 2)^2}$

$\therefore$ From the Δ le ΔAPM, $AM^2 + MP^2 = AP^2$

$$\text{or} \quad \frac{1}{30}(2x - y + 5z - 15)^2 + 4 = (x - 1)^2 + (y + 3)^2 + (z - 2)^2$$

$$\text{or} \quad 26x^2 + 29y^2 + 5z^2 + 4xy + 10yz - 20zx + 180y + 30z - 55 = 0.$$

This is the required equation of the right circular cylinder (Fig. 3.88).

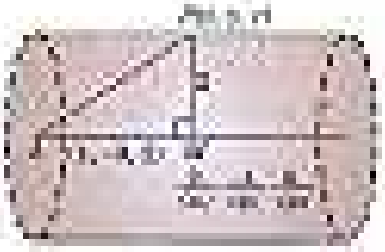


Fig. 3.88

Example 3.34. *Find the equation of the circular cylinder having for its base the circle $x^2 + y^2 + z^2 = 9$, $x + y + z = 3$.* (U.P.T.U. 2005, Kanpur 2006)

Solution. The axis of the cylinder is the line through the centre (i.e. the origin $(0, 0, 0)$) of the sphere (Fig. 3.89) and perpendicular to the plane of the circle.

$$\text{i.e.,} \quad x = y = z = k \quad \ldots(1)$$

$\therefore$ Axis of the cylinder is $\frac{x}{1} = \frac{y}{-1} = \frac{z}{1}$

Also OL = Length of the perpendicular from O to the plane $x + y + z = 3$

$$= \frac{3}{\sqrt{1+1+1}} = \sqrt{3}$$

$\therefore$ Radius of the circle $= \sqrt{(OA^2 - OL^2)} = \sqrt{3 - 1} = \sqrt{2}$

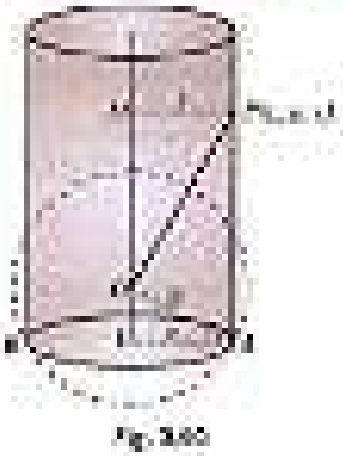


Fig. 3.60

Thus required cylinder is $r = \sqrt{2}$

If $P(x, y, z)$ be any point on the cylinder, then

$$OP^2 = OM^2 + MP^2$$

(i) $$x^2 + y^2 + z^2 = \left|\frac{1}{\sqrt{3}}(x - 0) - \frac{1}{\sqrt{3}}(y - 0) + 0 \times \frac{1}{\sqrt{3}}(z - 0)\right|^2 + 2$$

(ii) $x^2 + y^2 + z^2 + xy + yz + zx - 3 = 0$ which is the required equation.

Example 3.32. Find the equation of the enveloping cylinder of the sphere $x^2 + y^2 + z^2 = 9$ having generators parallel to the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{2}$.

Solution. If $P(x_1, y_1, z_1)$ be a point on the enveloping cylinder, then the equation of the generator is

$$\frac{x - x_1}{1} = \frac{y - y_1}{2} = \frac{z - z_1}{2} = r \text{ (say)} \quad \ldots(1)$$

Any point on (1) is $(x_1 + r, y_1 + 2r, z_1 + 2r)$. It lies on the sphere $x^2 + y^2 + z^2 = 9$. $\ldots(2)$

Thus $(x_1 + r)^2 + (y_1 + 2r)^2 + (z_1 + 2r)^2 = 9$

or $9r^2 + 2r(x_1 + 2y_1 + 2z_1) + x_1^2 + y_1^2 + z_1^2 - 9 = 0$ $\ldots(3)$

In order that (1) touches (2), the equation (3) must have equal roots i.e. we have

$$4(x_1 + 2y_1 + 2z_1)^2 = 4 \times 9(x_1^2 + y_1^2 + z_1^2 - 9) \quad [\because b^2 = 4ac]$$

or $5x_1^2 + 8y_1^2 + 8z_1^2 + 8x_1y_1 + 8y_1z_1 + 8z_1x_1 = 36$

$\therefore$ The locus of (x_1, y_1, z_1) is

$$5x^2 + 8y^2 + 8z^2 + 8xy + 8yz + 8zx = 36$$

which is the required equation of the enveloping cylinder.

EXERCISE 3.3

1. Find the equation of the right circular cylinder whose axis is the line $x = y = z$ and radius 3. (Anna, 2004)
2. Find the equation of the cylinder whose generators are parallel to the line $x = -y/2 = -z/3$ and which passes through the circle $x^2 + y^2 = 1$, $z = 3$. (Bangalore, 2003; Kerala, 2004)
3. Find the equation of the right circular cylinder of radius 2 whose axis passes through $(1, 2, 3)$ and has direction ratios $(2, -1, 2)$. (U.P.T.U., 2000; Anna, 2002)
4. Find the equation of the right circular cylinder circumscribing the sphere through the points $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ and having vertex $(1, 1, 1)$.
5. Find the equation of the cylinder whose bounding curve is $x^2 + y^2 - xz - yz + x - y = 0$ and which also contains the point $(1, 2, 3)$. Find also the area of the section of the cylinder by a plane parallel to xy-plane.
6. Find the equation of the enveloping cylinder of the sphere $x^2 + y^2 + z^2 = 2x + 4y + 6z - 2$ whose generators are perpendicular to the lines $\frac{x}{2} = \frac{y}{-1} = \frac{z}{3}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}$.
7. Find the equation to the cylinder whose generators intersect the circle $x^2 + 2y^2 + z^2 - 2yz + 2y - 2z = 0$, $x + y + z = 0$ and are parallel to the line $x = y = z$. (U.P.T.U., 2004)

3.23 QUADRIC SURFACES

The surface represented by any real equation of the second degree in x, y, z is called a quadric surface or a conicoid.

If we take general equations of a parabolic cylinder as of the form

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + 2ux + 2vy + 2wz + d = 0$$

which can be reduced to one of the following canonical forms as regards its position with respect to the axes, we get the three figures:

(1) **Elliptic cylinder :** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$;

(a) It is symmetrical about each of the coordinate planes (planes of symmetry).

(b) It passes through the origin.

(c) It meets the xy-plane in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $z = 0$.

(d) It meets any plane parallel to the xy-plane in an ellipse similar to the ellipse in which it meets the xy-plane.

(e) It meets any plane parallel to the yz-plane in a pair of parallel straight lines.

(f) It meets any plane parallel to the zx-plane in a pair of parallel straight lines.

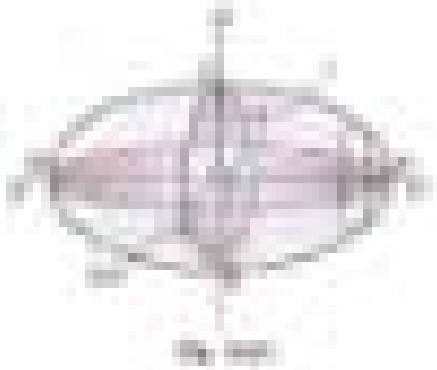


Fig. (xxx)

(2) **Hyperbolic cylinder :** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$;

(a) It is symmetrical about each of the coordinate planes.

(b) It meets the xy-plane in the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $z = 0$.

(c) It meets any plane parallel to the xy-plane in a hyperbola similar to the hyperbola in which it meets the xy-plane.

(d) It meets any plane parallel to the yz-plane in a pair of parallel straight lines.

(e) It meets any plane parallel to the zx-plane in a pair of parallel straight lines.

(3) **Parabolic cylinder :** $y^2 = 4ax$;

(a) It is symmetrical about the xz-plane.

(b) It passes through the origin.

(c) It meets the xy-plane in the parabola $y^2 = 4ax$, $z = 0$.

(d) It meets any plane parallel to the xy-plane in a parabola similar to the parabola in which it meets the xy-plane.

(e) It meets any plane parallel to the yz-plane in a pair of parallel straight lines.

(f) It meets any plane parallel to the zx-plane in a pair of parallel straight lines.

(4) **Hyperbolic cylinder :** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

(a) It is symmetrical about each of the coordinate planes.

(b) It passes through the origin.

(c) It meets the xy-plane in the hyperbola $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $z = 0$.

(d) It meets any plane parallel to the xy-plane in a hyperbola similar to the hyperbola in which it meets the xy-plane.

(e) It meets any plane parallel to the yz-plane in a pair of parallel straight lines.

(f) It meets any plane parallel to the zx-plane in a pair of parallel straight lines.

(b) section by the xy-plane ($z = 0$) is the imaginary ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = -1$,

(c) section by the xy-plane ($z = 0$) is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$,

(d) The surface is generated by a variable ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{k^2}{c^2}$, i.e. $k = \pm\frac{c}{a}x$

and extends infinitely on both sides of the xy-plane.

Hence the shape is as shown in Fig. 3.83.

(4) Cone: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$.

(a) It passes through the origin only.

(b) Its section by the xy-plane ($z = 0$) is the point ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$,

(c) The section by the xy-plane ($z = 0$) is the pair of straight lines

$$y = \pm\frac{b}{a}x \qquad \text{(i.e. OAT and OAT')}$$

(d) The section by the xz-plane ($y = 0$) is the pair of straight lines

$$z = \pm\frac{c}{a}x \qquad \text{(i.e. PAP' and PAP')}$$

(e) The section by the yz-plane ($x = 0$) is the point ellipse $\frac{y^2}{b^2} + \frac{z^2}{c^2} = 0$,

(f) The surface is generated by a variable ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{k^2}{c^2}$, ($k$ = the variable)

and extends infinitely on both sides of the xy-plane. Hence its shape is as shown in Fig. 3.84.

(5) Elliptic paraboloid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$

(a) It is symmetrical about yz- and xz-planes for only even powers of x and y occur in the equation.

(b) It meets the axes at the origin only and lies on the positive z-side of the xy-plane.

(c) Its section by the xy-plane ($z = 0$) is the parabola $y^2 = \frac{2b^2}{a}x$, i.e. TOT',

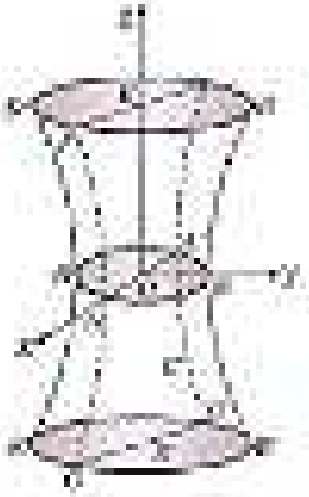


Fig. 3.83

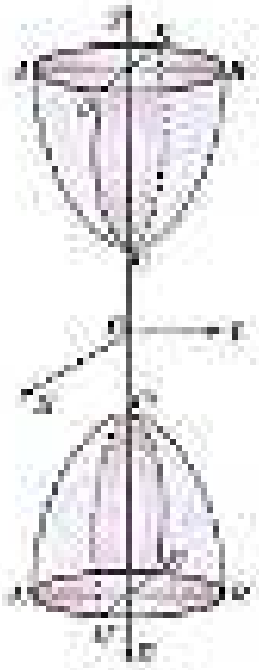
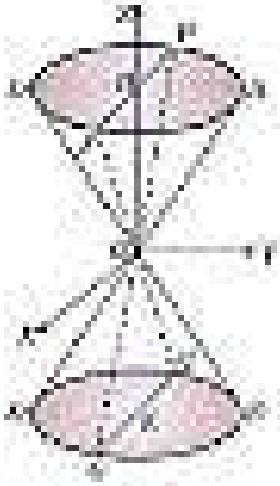


Fig. 3.84



Its section by the xz-plane $y = 0$ is the parabola $x^2 = \frac{2a^2}{c}z$ i.e., XOZ.

Its section by the xy-plane $z = 0$ is the pair of lines $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.

(vi) The surface is generated by a variable ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c}$, $z \geq 0$ has its vertex from O is increased & extends to infinity above the xy-plane.

Hence its shape is as shown in Fig. 3.43 and is the first of these.

(vii) Hyperbolic paraboloid : $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$.

(i) It is symmetrical about the yz and zx-planes for only even powers of x and y occur in its equation.

(ii) It meets the surface only at the origin and touches the xy-plane there.

(iii) Its section by the yz-plane $x = k$ is the parabola $y^2 = -\frac{2b^2}{c}\left(z - \frac{k^2}{2a^2}c\right)$ i.e., DOZ'.

Its section by the xz-plane $y = 0$ is the parabola $x^2 = \frac{2a^2}{c}z$ i.e., XOZ.

Its section by the xy-plane $z = 0$ is the pair of lines $y = \pm\frac{b}{a}x$ as shown in Fig. 3.44.

(iv) The surface is generated by a variable hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c}$, $z = k$

and it extends to infinity on both sides of xy-plane. Hence its shape is as shown in Fig. 3.45.

(v) **Cylinder.** An equation of the form $f(x, y) = 0$ represents a cylinder generated by a straight line which is parallel to the z-axis and its section by the xy-plane is the curve $f(x, y) = 0$ (Fig. 3.46).

In particular (i) $x^2 = 4ay$ represents a parabolic cylinder.

(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ represents an elliptic cylinder, (iii) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ represents a hyperbolic cylinder.

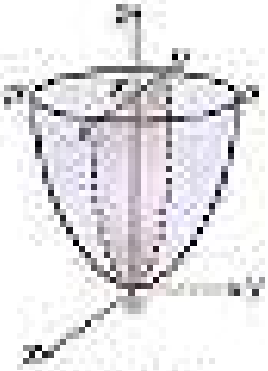


Fig. 3.44

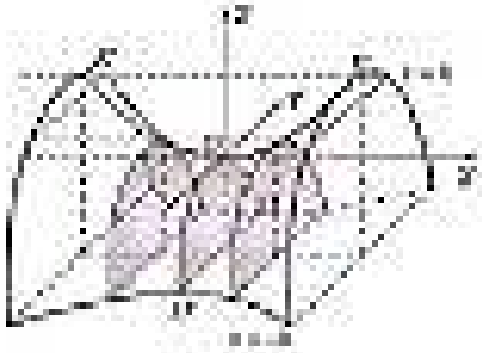


Fig. 3.45

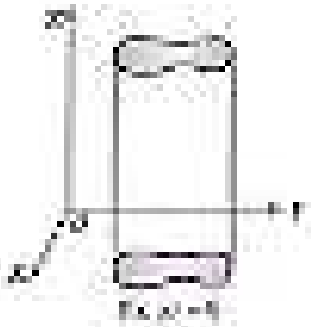


Fig. 3.46

3.22 SURFACES OF REVOLUTION

Let $P(x, y)$ be any point on the curve $y = f(x)$ in the xy-plane. Draw PM perpendicular to OX so that $OM = x$ and $MP = y$. Then the equation of this curve may be written as

$$MP = f(OM) \qquad \ldots(1)$$

As this curve revolves about the x-axis, the point P describes a circle with centre M and radius MP. Let $Q(x, y, z)$ be any other position of P. Draw $QN \perp$ to xy-plane and join MN so that $QN = z$, $MN = z$, $NQ = y$

and $\angle MNQ = 90°$ $\therefore$ $MP^2 = MQ^2 = MN^2 + NQ^2 = y^2 + z^2$

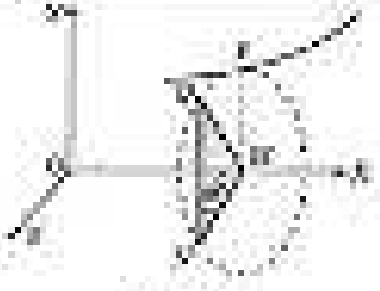


Fig. 3.47

Now substituting the values of MP and NQ in (1), we have

$$\sqrt{(y^2 + z^2)} = f(x) \quad \text{or} \quad y^2 + z^2 = [f(x)]^2$$

which is the equation of the surface generated by the revolution of the curve $y = f(x)$ about the x-axis (Fig. 3.68).

Similarly, the surface generated by the revolution of the curve

(i) $x = f(y)$ about y-axis is $x^2 + z^2 = [f(y)]^2$, (ii) $z = f(x)$ about z-axis is $x^2 + y^2 = [f(x)]^2$

The given revolving curve is called the generating curve.

Some standard surfaces of revolution :

Let the generating curve be $y = f(x)$ in the xy-plane and the axis of rotation be the x-axis, then the surface generated is $y^2 + z^2 = [f(x)]^2$.

(1) Right circular cylinder. When $f(x) = a$, the generating curve is a straight line $y = a$ parallel to the x-axis.

∴ The surface generated is $y^2 + z^2 = a^2$

which represents a right circular cylinder of radius a and axis as x-axis (Fig. 3.68).

(2) Right circular cone. When $f(x) = mx$, the generating curve is a straight line $y = mx$ passing through the origin.

∴ The surface generated is $y^2 + z^2 = m^2x^2$ or $y^2 + z^2 = x^2 \tan^2 \alpha$

which represents a right circular cone of semi-vertical angle $\alpha = \tan^{-1} m$ and axis as the x-axis (Fig. 3.70).

(3) Sphere. When $f(x) = \sqrt{(a^2 - x^2)}$, the generating curve is a circle $(x^2 + y^2 = a^2)$.

∴ The surface generated is

$$y^2 + z^2 = a^2 - x^2 \quad \text{i.e.,} \quad x^2 + y^2 + z^2 = a^2,$$

which is a sphere of radius a and centre $(0, 0, 0)$.

(4) Ellipsoid of revolution. When $f(x) = b\sqrt{[1 - x^2/a^2]}$, the generating curve is an ellipse

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\right).$$

∴ The surface generated is $y^2 + z^2 = b^2(1 - x^2/a^2)$

or $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{b^2} = 1$, which is called an *ellipsoid of revolution*.

If $a^2 > b^2$, the major axis of the generating ellipse is along the x-axis—the axis of revolution and the surface generated, in this case, is called a *prolate spheroid* (Fig. 3.71).

If $a^2 < b^2$, the major axis of the ellipse lies along the z-axis—the axis of revolution and the surface thus generated is called an *oblate spheroid* (Fig. 3.72).

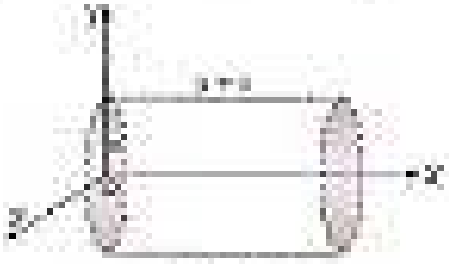


Fig. 3.68

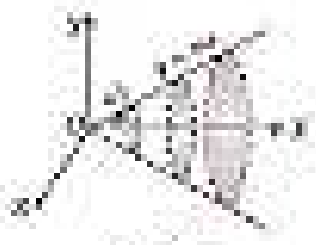
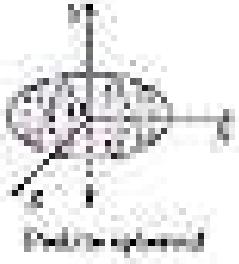
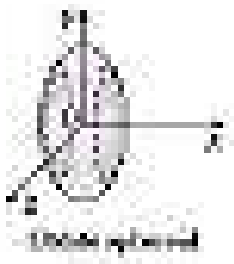


Fig. 3.70



Prolate spheroid

Fig. 3.71



Oblate spheroid

Fig. 3.72

(5) Hyperboloids of revolution

(i) When $f(x) = b\sqrt{[x^2/a^2 - 1]}$, the generating curve is $\dfrac{y^2}{b^2} - \dfrac{x^2}{a^2} = 1$ which represents a hyperbola having conjugate axis along the x-axis.

∴ The surface generated is $y^2 + z^2 = b^2(x^2/a^2 - 1)$

or $\dfrac{x^2}{a^2} + \dfrac{z^2}{b^2} - \dfrac{y^2}{b^2} = 1$ or $x^2/a^2 + z^2/b^2 - y^2/b^2 = 1$ is called a *hyperboloid of revolution of two sheets* (Fig. 3.73).

(v) When $f(x) = k\sqrt{x^2+a^2}$, the generating curve is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which represents a hyperbola having transverse axis along the x-axis.

$\therefore$ The surface generated is $y^2 + z^2 = k^2(x^2+a^2)$ i.e.,

or $\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{b^2} = 1$, which is called a hyperboloid of revolution of two sheets (Fig. 9.36).

(vi) **Paraboloid of revolution.** When $f(x) = \sqrt{4ax}$, the generating curve is a parabola $y^2 = 4ax$. The surface generated is $y^2 + z^2 = 4ax$, which is called a paraboloid of revolution (Fig. 9.37).

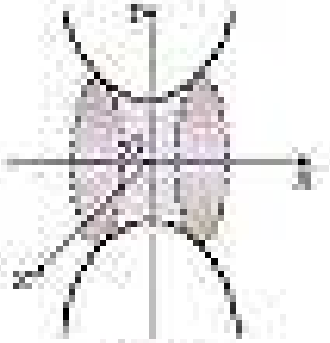


Fig. 9.35

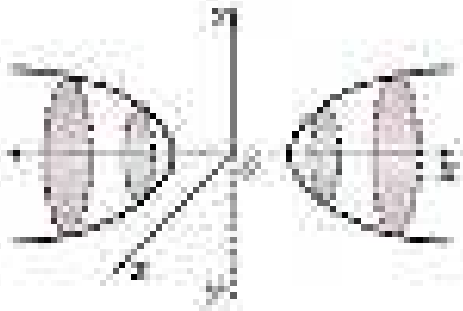


Fig. 9.36

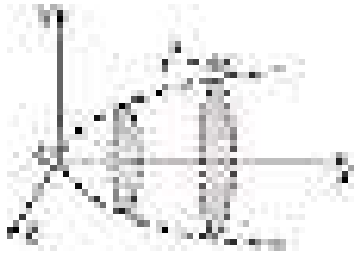


Fig. 9.37

SOLVED EXAMPLES

Ex. 1. What surface is represented by $x^2 - 3y^2 + 3z^2 - 1 = 0$? Trace it roughly. Find the area of the plane curve in which $x = 0$ cuts it.

Ex. 2. Sketch roughly the surface $9x^2 + y^2 + 9z^2 = 9$.

In what curve does the plane $x = 1$ intersect it? Find the area of the curve of intersection? Which surfaces are represented by the following equations? Draw diagrams to show their nature.

i) $x^2 + y^2 = 16$ ii) $x^2 + y^2 - z^2 = 0$

iii) $x^2 + 4(1 + z^2 + y^2)$ iv) $y^2 = 4x - 8$ (Andhra, 2009)

v) $x^2 + y^2 + z^2 = 2x$ vi) $x^2 - y^2 + 4z^2$

vii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = \frac{2x}{a}$ (P.T.U., 2008) viii) $4x^2 - y^2 + 16z^2 = 36$

Note. For the equations of the tangent plane and the normal line to a surface refer to § 9.8 (2).

9.25 OBJECTIVE TYPE OF QUESTIONS

SOLUTIONS

Select the correct answer or fill up the blanks in each of the following questions:

1. The lines $x = ay + b$, $z = cy + d$ and $x = a'y + b'$, $z = c'y + d'$ are perpendicular if
   (a) $aa' + cc' = 1$ (b) $aa' + cc' = -1$ (c) $1 + aa' + cc' = 0$ (d) $1 + aa' + cc' = 1$

2. The coordinates of the point of intersection of the line $\frac{x+1}{1} = \frac{y+3}{0} = \frac{z+2}{-2}$ with the plane $3x + 2y + 4z + 5 = 0$ is
   (a) $(2, 15, -10)$ (b) $(3, 5, 7)$ (c) $(2, 3, -3)$ (d) $(3, 15, -20)$

3. The equation of a right circular cylinder whose axis is the z-axis and radius is a is
   (a) $x^2 + y^2 + z^2 = a^2$ (b) $x^2 + y^2 = a^2$ (c) $x^2 + y^2 = az$ (d) $x^2 + z^2 = a^2$

4. The equation $\sqrt{lx} + \sqrt{my} + \sqrt{nz} = 0$ represents a
   (a) sphere (b) cylinder (c) cone (d) pair of planes.

6. The plane $x/a + y/b + z/c = 1$ cuts the axes of x, y, z at A, B, C respectively. The orthocentre of $\triangle ABC$ is
   (a) (a, b, c) (b) $(0, 0, 0)$
   (c) (a^2, b^2, c^2) (d) $(2a, 2b, 2c)$.

7. The locus of the point of the cylinder which intersects the curve $x^2 - y^2 + z^2 = 1, x + y + z = 1$ and whose generators are parallel to the axis of x, is
   (a) $y^2 + z^2 + xy + yz + zx + x + y + z = 0$ (b) $y^2 + z^2 - yz + x + y + z = 0$
   (c) $y^2 + z^2 - xy - yz - zx + x + y + z = 0$ (d) $y^2 + z^2 + yz + x + y + z = 0$.

8. The equation $x^2 + y^2 + z^2 + xy + yz + zx = 0$ represents
   (a) a sphere with centre $(-1/2, -1/2, -1/2)$ and radius $1/2$ (b) a real cone
   (c) a cone with vertex $(-1/2, -1/2, -1/2)$ and radius $1/2$ (d) a pair of straight lines
   (e) a real cone with vertex $(0, 0, 0)$ and a pair of imaginary lines meeting in $(-1/2, -1/2, -1/2)$
   (f) none of the above.

9. The plane of the circle in which the sphere $x^2 + y^2 + z^2 = 4$ is cut by the plane $x + y + z = 1$ is
   (a) $x + y + z = 0$ (b) $x + y + z = 1$ (c) $x + y + z = 2$ (d) none of the above.

10. The angle between the planes $y + z = x\sqrt{2}$ and $z + x = y\sqrt{2}$ is
    (a) $30°$ (b) $60°$ (c) $90°$ (d) $120°$ (UTTAR 2000)

11. The equation of the right circular cone whose vertex is the point $(1, 2, 3)$, axis is the line $x = y = z$ and which has a vertical angle θ is given by
    (a) $(x-1)^2 + (y-2)^2 + (z-3)^2 = 3(x+y+z-6)^2 \cos^2\theta$
    (b) $(x-1)^2 + (y-2)^2 + (z-3)^2 = (x+y+z-6)^2 \cos^2\theta$
    (c) $3(x^2 + y^2 + z^2) = (x+y+z)^2 \cos^2\theta$
    (d) $x^2 + y^2 + z^2 = (x+y+z)^2 \cos^2\theta$.

12. The equation of a sphere for which the circle in which the plane $x + y + z = 0$ cuts the sphere $x^2 + y^2 + z^2 = 9$ is a great circle is
    (a) $x^2 + y^2 + z^2 + yz + zx + xy = 9$ (b) $x^2 + y^2 + z^2 + 2yz + 2zx + 2xy = 9$
    (c) $x^2 + y^2 + z^2 + 2x + 2y + 2z = 9$ (d) $x^2 + y^2 + z^2 - 2x - 2y - 2z = 9$.

13. The equation of the plane passing through $(4, -2, 3)$ and perpendicular to the line $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-5}{-1}$ is
    (a) $2x + 3y - z = 8$ (b) $2x + 3y - z + 8 = 0$
    (c) $2x + 3y - z + 4 = 0$ (d) $2x + 3y - z = 4$ (UTTAR 2000)

14. The equation to the axis of the right circular cylinder whose guiding circle is $x^2 + y^2 + z^2 = 1, x + y + z = 1$ will be
    (a) $x = y = z$ (b) $x + y + z = 1$ (c) $x = y = z^2$ (d) $x = y = z = 1$

15. In three dimensional, the equation $x^2 + y^2 + z^2$ represents
    (a) a pair of straight lines (b) a hyperbola
    (c) a cylinder (d) a plane.

16. Perpendicular from origin to a plane is
    (a) parallel (b) z-axis (c) x-axis (d) none.

17. A line makes angles α, β, γ with the coordinate axes, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$ is equal to
    (a) 1 (b) 2 (c) 0 (d) 3 (UTTAR 2000)

18. The line $\frac{x-1}{0} = \frac{y-2}{0} = \frac{z-3}{1}$ is
    (a) Any of the statements
    (b) a line perpendicular to each of them
    (c) the line of intersection and a line perpendicular to each of them
    (d) a line parallel to each of them.

19. The distance between the planes $2x + 2y + z = 6$ and $4x + 4y + 2z + 5 = 0$ is
    (a) 1/2 (b) 5/2 (c) 11/2 (d) 13/2.

20. The line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and the plane $x + y + z = 4$ are
    (a) parallel (b) perpendicular (c) meet at the point $(1, 2, 3)$ (d) none of the above.

21. The radius of a great circle of a sphere is
    (a) greater than the radius of the sphere (b) less than the radius of the sphere
    (c) equal to the radius of the sphere.

22. Which of the following lines are perpendicular to the line $x + y + z = 0$?
    (a) $x + y + z$ (b) $x - y + z$ (c) $x + 2y - z$

82. The acute angle between the planes $2x - y + z = 6$ and $x + y + 2z = 3$ is
(a) $30°$ (b) $45°$ (c) $60°$.

83. A line passing through the coordinate axes are given by the equations
(a) $x^2 + y^2 + z^2 + 2x + 2y + 2z = 0$ (b) $x^2 + 2y^2 + 3z^2 - 2yz - 2zx - 2xy = 0$
(c) $yz + 2zx + xy = 1$

84. The line $\frac{x+1}{2} = \frac{y-2}{3} = \frac{z+2}{6}$ is perpendicular to the plane $x + y + z + 1 = 0$ if
(a) $a = 2, b = 3$ (b) $a = 1, b = 3$ (c) $a = 1, b = 2$ (d) $a = 2, b = 1$

85. The equation $2x^2 + y^2 + z^2 - 2xy - 2yz + 2zx + 3x + 4y + 5z = 0$ represents a
(a) cone (b) Right circular cylinder
(c) sphere (d) pair of planes

86. The equation of the plane through the point $(2, -3, 1)$ and parallel to the plane $3x - 4y - 2z = 5$ is
(a) $3x - 4y + 2z - 20 = 0$ (b) $3x + 4y - 2z + 20 = 0$
(c) $3x - 4y - 2z + 20 = 0$ (d) $3x + 4y + 2z - 20 = 0$

87. The direction cosines of a line which is equally inclined to the coordinate axes are ____.

88. The distance between the xy-plane and the plane $z = 5$ is ____.

89. The equation of the radius of the sphere $x^2 + y^2 + z^2 = 25$ is ____.

90. The image of the point $(3, 2, -1)$ in the XOY plane is ____.

91. The plane $x - 2y = 2z + 3$ intersects the sphere $x^2 + y^2 + z^2 - 2x + 4y - 6z + 5 = 0$ in a circle ____. (UPTU, 2005)

92. The restrictions for the three coordinates have to be explained in ____.

93. The equation of the cone whose vertex is at the origin and whose guiding curve is $ax^2 + by^2 = cz^2$ in the plane $lx + my + nz = p$ is ____.

94. The plane through the points $(2, 2, 1)$, $(1, 2, 2)$ and perpendicular to the plane $3x + 2y + 2z = 5$ is ____.

95. Volume of the sphere $x^2 + y^2 + z^2 - 2x + 4y - 8z + 13 = 0$ is ____.

96. Angle between the planes $x + y - 1 = 0$ and $2x - 3y + z - 7 = 0$ is ____.

97. The equation of the central conic with its centre at the origin and passing through the points $(\pm 3, 0)$ and $(0, \pm 2)$ is ____. (Uttarakhand, 2008)

98. Any line parallel to the line $\frac{x-1}{3} = \frac{y-2}{2} = \frac{z-3}{4}$ makes angles α, β, γ with the axes, then ____.

99. The equation of the line joining the points $(1, 2, 3)$ and $(2, 1, -3)$ is ____.

100. The equation of the sphere on the line joining $(1, 2, 3)$ and $(-2, 1, 3)$ as diameter is ____.

101. The condition for the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ to lie in the plane $ax + by + cz + d = 0$ is ____.

102. The distance between the planes $2x + y + z = 4$ and $6x + 3y + 3z + 12 = 0$ is ____.

103. The centre and radius of the sphere $2x^2 + 2y^2 + 2z^2 - 4x + 6y + 8z - 23 = 0$ are ____.

104. The radius of the circle $x^2 + y^2 + z^2 - 2x + 4y - 6z + 13 = 0$, $x + y + z = 3$ is ____.

105. The centre of the sphere $x + y = 0 = z + 1$ and $x^2 + y^2 + z^2 = 4$ is ____.

106. The equation $x^2 + xy - 6y^2 = 0$ represents a ____.

107. The equation $x^2 + y^2 = \frac{1}{2}x + 1$ represents a ____.

108. Angle between the lines $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x-2}{1} = \frac{y-4}{-1} = \frac{z-1}{0}$ is ____.

109. The perpendicular distance from the origin to the plane $2x + 3y + 4z = 12$ is ____.

110. The equation of the sphere with centre $(1, 2, 3)$ and radius 4 is ____.

111. The equation of the plane which cuts off intercepts 3, 4, 5 on the axes is ____.

112. The equation of the plane passing through the points $(1, 1, 1)$, $(1, -1, 2)$ and $(-2, 2, 2)$ is ____.

113. The plane $x + 2y + 3z = 6$ cuts the coordinate axes at the points ____.

114. The equation of the plane passing through the points $(1, 1, 1)$, $(1, 2, 3)$ and $(2, 2, 1)$ is ____. (Uttarakhand, 2009)

115. The plane $x + 2y = 3$ is parallel to the ____ plane. (Uttarakhand, 2009)

116. The plane $x + y + z = 1$ passes through the point ____. (Uttarakhand, 2009)

117. The equation $x^2 + y^2 + z^2 = 0$ represents ____. (Uttarakhand, 2009)

CHAPTER

4

Differential Calculus & Its Applications

1. Successive differentiation : Standard results. 2. Leibnitz's theorem. 3. Partial derivatives : Euler's theorem. Lagrange's Mean value theorem. Cauchy's mean value theorem. Taylor's theorem. 4. Expansion of functions : Maclaurin's series. Taylor's series. 5. Indeterminate forms. 6. Tangents & Normals—Cartesian curves. Angle of intersection of two curves. 7. Polar curves. 8. Pedal equation. 9. Derivative of arc. 10. Curvature. 11. Radius of curvature. 12. Centre of curvature. Evolute. Chord of curvature. 13. Envelopes. 14. Increasing and decreasing functions. Concavity, convexity & Point of inflexion. 15. Maxima & Minima. Practical problems. 16. Asymptotes. 17. Curve tracing. 18. Objective Type of Questions.

4.1 (I) SUCCESSIVE DIFFERENTIATION

The reader is already familiar with the process of differentiating a function $y = f(x)$. For ready reference, a list of derivatives of some standard functions is given in the beginning.

The derivative dy/dx is, in general, another function of x which can be differentiated. The derivative of dy/dx is called the second derivative of y and is denoted by d^2y/dx^2. Similarly, the derivative of d^2y/dx^2 is called the third derivative of y and is denoted by d^3y/dx^3. In general, the nth derivatives of y is denoted by d^ny/dx^n.

Alternative notations for the successive derivatives of $y = f(x)$ are

$$Dy, D^2y, D^3y, \ldots, D^ny;$$

or

$$y_1, y_2, y_3, \ldots, y_n;$$

or

$$f'(x), f''(x), f'''(x), \ldots, f^{(n)}(x).$$

The nth derivative of $y = f(x)$ at $x = a$ is denoted by $d^ny/dx^n]_{x=a}$ or $f^{(n)}(a)$.

Example 4.1. If $y = e^{ax}\sin(bx + c)$, prove that $y_2 - 2ay_1 + (a^2 + b^2)y = 0$. (Kashmir 2002)

Solution. We have $y = e^{ax}\sin(bx + c)$ ...(1)

$\therefore$ $y_1 = e^{ax}\cdot\cos(bx + c)\cdot b + \sin(bx + c)\cdot e^{ax}\cdot a$ [By (1)]

or $y_1 = ay + be^{ax}\cos(bx + c)$ ...(2)

Again differentiating both sides,

$$y_2 = ay_1 + be^{ax}(-\sin(bx + c)\cdot b + \cos(bx + c)\cdot e^{ax}\cdot a \quad [\because \text{from (2) } be^{ax}\cos(bx + c) = y_1 - ay]$$

or $y_2 = 2ay_1 - b^2y + b^2y = 0.$

Example 4.2. If $x = a\cos t + b\sin t$, $y = a\sin t - b\cos t$, prove that $y_2 = \frac{1}{y_1}$.

Solution. We have $\frac{dx}{dt} = -a\sin t + b\cos t$ and $\frac{dy}{dt} = a\cos t + b\sin t$

and $\frac{dy}{dx} = \frac{a\cos t + b\sin t}{-a\sin t + b\cos t}$ or $y_1 = \frac{a\cos t + b\sin t}{-a\sin t + b\cos t}$

$$\frac{dy}{dx} = \frac{dy}{d(\tan t)} \cdot \frac{d(\tan t)}{dt} = \frac{\cos^2 t}{\sec^2 t} = \cos^4 t$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(\cos^4 t) = \frac{dt}{dx} = \cos^4 t \cdot \frac{1}{a\sec t \tan t} = 3a\cos^3 t \sin t$$

Example 4.28. Given $y^2 = f(x)$, a polynomial of third degree, then evaluate $\frac{d}{dx}\left(y^3 \frac{d^2y}{dx^2}\right)$.

Solution. Differentiating $y^2 = f(x)$ w.r.t. x, we get

$$2y \frac{dy}{dx} = f'(x) \quad \ldots(i)$$

Differentiating (i) w.r.t. x again, we obtain

$$2\left(\frac{dy}{dx} \cdot \frac{dy}{dx} + y\frac{d^2y}{dx^2}\right) = f''(x) \text{ or } 2\left[\frac{dy}{dx}\right]^2 + 2y\frac{d^2y}{dx^2} = f''(x)$$

Again differentiating, we get

$$4\cdot \frac{dy}{dx}\frac{d^2y}{dx^2} + 2\frac{dy}{dx}\frac{d^2y}{dx^2} + 2y\frac{d^3y}{dx^3} = f'''(x)$$

or $6y\frac{dy}{dx}\frac{d^2y}{dx^2} + y^2\frac{d^3y}{dx^3} = \frac{1}{2}f'''(x)$ Multiplying by y^2

Hence $\frac{d}{dx}\left(y^3\frac{d^2y}{dx^2}\right) = \frac{1}{2}f(x)f'''(x)$, $\because y^2 = f(x)$

Example 4.29. If $ax^2 + 2hxy + by^2 = 1$, prove that $\frac{d^2y}{dx^2} = \frac{h^2 - ab}{(hx + by)^3}$.

Solution. Differentiating the given equation w.r.t. x,

$$2ax + 2h\left(x\frac{dy}{dx} + y\right) + 2by\frac{dy}{dx} = 0 \text{ or } \frac{dy}{dx} = -\frac{ax + hy}{hx + by} \quad \ldots(i)$$

Differentiating both sides of (i) w.r.t. x,

$$\frac{d^2y}{dx^2} = -\frac{(hx + by)\left(a + h\cdot\frac{dy}{dx}\right) - (ax + hy)\left(h + b\cdot\frac{dy}{dx}\right)}{(hx + by)^2}$$

[Substituting the value of $\frac{dy}{dx}$ from (i)]

$$= -\frac{(hx + by)\left(a - h\cdot\frac{ax + hy}{hx + by}\right) - (ax + hy)\left(h - b\cdot\frac{ax + hy}{hx + by}\right)}{(hx + by)^2}$$

$$= \frac{(h^2 - ab)(ax^2 + 2hxy + by^2)}{(hx + by)^3}$$

$= (h^2 - ab)/(hx + by)^3$ $\because ax^2 + 2hxy + by^2 = 1$

PROBLEMS-IV

1. If $y = ax^2 + bx + c$, $c \neq 0$, prove that $2y_2y_0 = y_1^2 - b^2y_2^2$.
2. If $y = \sin(\sin x)$, prove that $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\cos^2 x = 0$.
3. If $y = \sin(\log x)$, show that $x^2\frac{d^2y}{dx^2} + (2n - 1)x\frac{dy}{dx} + (n^2 + 1)y = 0$, where n is a constant.

[illegible]

(d) Hexadecimal Recursion

[illegible]

In general, $y_n = r^n e^{ax}\sin(bx + c + n\phi)$ (U.T.U., 2006)

where $r = \sqrt{(p^2 + q^2)}$, $\sin\phi = \dfrac{q}{r}$, $\cos\phi = \dfrac{p}{r}$, $\tan\phi = \dfrac{q}{p}$.

Proceeding as in (8), the student should prove (9) himself.

(2) **Preliminary transformations.** Quite often preliminary simplification reduces the given function to one of the above standard forms and then the nth derivatives are easily written down.

To find the nth derivative of the products of sines or cosines or their products, we first express each of them as a series of sines or cosines of multiple angles and then use the above formulae (8) and (9).

Example 4.5. If $y = a\log\dfrac{x-1}{x+1}$, show that $y_n = a(-1)^{n-1}(n-1)!\left[\dfrac{1}{(x-1)^n} - \dfrac{1}{(x+1)^n}\right]$. (U.P.T.U., 2003)

Solution. Differentiating $y = a\log\dfrac{x-1}{x+1}$, we have

$$y_1 = a\log\frac{x-1}{x+1} = a\left[\frac{1}{x-1} - \frac{1}{x+1}\right]$$

$$= a\log(x-1) - a\log(x+1) + \frac{a}{x-1} + \frac{a}{x+1} \quad \ldots(1)$$

Now if D denotes $\dfrac{d}{dx}$, then $D(x+1)^{-1} = -(x+1)^{-2}$, $D^2(x+1)^{-1} = (-1)(-2)(x+1)^{-3}$, $D^3(x+1)^{-1} = (-1)(-2)(-3)(x+1)^{-4}$, $D^{n-1}(x+1)^{-1} = (-1)^{n-1}(n-1)!(x+1)^{-n}$

Differentiating (1) $(n-1)$ times w.r.t. x,

$$y_n = \frac{(-1)^{n-2}(n-2)!}{(x-1)^{n-1}} - \frac{(-1)^{n-2}(n-2)!}{(x+1)^{n-1}} + \frac{(-1)^{n-1}(n-1)!}{(x-1)^n} + \frac{(-1)^{n-1}(n-1)!}{(x+1)^n}$$

$$= (-1)^{n-1}(n-2)!\left[\frac{x-1}{(x-1)^n} - \frac{x+1}{(x+1)^n} + \frac{(n-1)}{(x-1)^n} + \frac{(n-1)}{(x+1)^n}\right]$$

$$= (-1)^{n-1}(n-1)!\left[\frac{x-1}{(x-1)^n} - \frac{x+1}{(x+1)^n}\right].$$

Example 4.6. Find the n^{th} derivative of $\sin x\sin 2x\sin 3x$. (U.P.T.U., 2005)

Solution. (1) $y = \sin x\cos 2x\cos 3x = \frac{1}{2}\cos x(\cos 5x + \cos x)$

$$= \frac{1}{2}\cos x\cos 5x + \frac{1}{2}\cos^2 x = \frac{1}{4}[\cos 6x + \cos 4x] + \frac{1}{4}[1 + \cos 2x]$$

$$= \frac{1}{4}(1 + \cos 2x + \cos 4x + \cos 6x)$$

$$\therefore \quad y_n = \frac{1}{4}\left[2^n\cos\left(2x + \frac{n\pi}{2}\right) + 4^n\cos\left(4x + \frac{n\pi}{2}\right) + 6^n\cos\left(6x + \frac{n\pi}{2}\right)\right]$$

(2) $\sin^3 x\sin 3x = \sin x\left(\frac{3}{4} - \frac{1}{4}\cos 2x\right)\left(\frac{3}{4} - \frac{1}{4}\cos 6x\right) = \frac{1}{4}\sin x\left(\frac{3}{4} - \frac{1}{4}\cos 2x - \frac{3}{4}\cos 6x + \frac{1}{4}\cos 2x\cos 6x\right)$

$$= \frac{1}{4}\left[\frac{3}{4}\sin x - \frac{1}{8}(\sin 3x - \sin x) - \frac{3}{8}(\sin 7x - \sin 5x)\right]$$

$$\therefore \quad (2^n)^{th} \text{ term} = \frac{1}{4}\left[\frac{3}{4}1^n\sin\left(x + \frac{n\pi}{2}\right) + \frac{1}{8}3^n\sin\left(3x + \frac{n\pi}{2}\right) - \frac{3}{8}5^n\sin\left(5x + \frac{n\pi}{2}\right) - \frac{3}{8}7^n\sin\left(7x + \frac{n\pi}{2}\right)\right]$$

$$= \frac{1}{4}\left[\frac{3}{4}\sin\left(x + \frac{n\pi}{2}\right) + \frac{1}{8}3^n\sin\left(3x + \frac{n\pi}{2}\right) - \frac{3}{8}5^n\sin\left(5x + \frac{n\pi}{2}\right) - \frac{3}{8}7^n\sin\left(7x + \frac{n\pi}{2}\right)\right].$$

(3) **Use of partial fractions.** To find the n^{th} derivative of any rational algebraic function, we first split it up into partial fractions. Even when the denominator cannot be resolved into real factors, the method of partial fractions can still be used after breaking the denominator into complex linear factors. Then to put the result back in a real form, we apply De Moivre's theorem (p. 647).

Example 4.7. Find the n^{th} derivative of $\dfrac{x^2}{(x-1)(2x+3)}$.

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Solution. $$\frac{x}{(x-1)(2x+3)} = \frac{1}{(x-1)(2x+3)} + \frac{-3x}{(x-1)(2x+3)}$$

$$= \frac{1}{5}\cdot\frac{1}{x-1} + \frac{3}{5}\cdot\frac{1}{2x+3}$$

Hence $$D^n\left[\frac{x}{(x-1)(2x+3)}\right] = \frac{1}{5}\cdot\frac{(-1)^n n!}{(x-1)^{n+1}} + \frac{3}{5}\cdot\frac{(-1)^n 2^n n!}{(2x+3)^{n+1}}$$

$$= \frac{(-1)^n n!}{5}\left[\frac{1}{(x-1)^{n+1}} + \frac{3\cdot 2^n}{(2x+3)^{n+1}}\right]$$

Example 6.24. Find the nth derivative of $y = \frac{1}{x^2+a^2}$.

Solution. We have $$y = \frac{1}{x^2+a^2} = \frac{1}{(x+ai)(x-ai)} = \frac{1}{2ai}\left[\frac{1}{x-ai} - \frac{1}{x+ai}\right]$$

$$\therefore \quad y_n = \frac{1}{2ai}\left[\frac{(-1)^n n!}{(x-ai)^{n+1}} - \frac{(-1)^n n!}{(x+ai)^{n+1}}\right]$$

If $x = r\cos\theta, a = r\sin\theta$ so that $r = \sqrt{(x^2+a^2)}, \theta = \tan^{-1}\left[\frac{a}{x}\right]$

$$= \frac{(-1)^n n!}{2ai}\left[\frac{1}{r^{n+1}(\cos\theta - i\sin\theta)^{n+1}} - \frac{1}{r^{n+1}(\cos\theta + i\sin\theta)^{n+1}}\right]$$

$$= \frac{(-1)^n n!}{2ai r^{n+1}}\left[\cos(n+1)\theta + i\sin(n+1)\theta - \cos(n+1)\theta + i\sin(n+1)\theta\right]$$

(By De Moivre's theorem)

$$= \frac{(-1)^n n!}{2ai r^{n+1}}\cdot 2i\sin(n+1)\theta \qquad \left[\text{Put } \frac{1}{r} = \frac{\sin\theta}{a}\right]$$

$$= \frac{(-1)^n n!}{a^{n+1}}\sin^{n+1}\theta\sin(n+1)\theta$$

EXERCISES

Find the nth derivatives of (1) to (11):

1. $\sin(ax^2 - b)$ (U.P.T.U., 2006)
2. $\frac{x+2}{x+1} + \log\frac{x+2}{x+1}$
3. $\sin^2 x\cos^2 x$ (U.P.T.U., 2004)
4. $\cos^4 x$ (Mumbai, 2004)
5. $\sin(\log 5x)\sin x$ (U.P.T.U., 2005)
6. $e^{2x}\cos 3x\sin 4x$ (Mumbai, 2003)
7. $\frac{x^2}{(x-1)(x+2)}$ (U.P.T.U., 2006)
8. $\frac{x^2}{(x^2+3)(x+1)}$ (U.P.T.U., 2007)
9. $\frac{1}{1+x+x^2+x^3}$ (Mumbai, 2006)
10. $\frac{x^2}{x^2-3x+2}$ (Mumbai, 2007)
11. Find the nth derivative of $\tan^{-1}\left[\frac{2x}{1-x^2}\right]$ in terms of x and θ. (U.P.T.U., 2002)

3.32 LEIBNIZ'S THEOREM for the nth Derivative of the product of two functions*

If u, v are two functions of x possessing derivatives of the nth order, then

$$D^n(uv) = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + \ldots + {}^nC_r u_{n-r} v_r + \ldots + {}^nC_n uv_n$$

We shall prove this theorem by mathematical induction.

Step I. By actual differentiation,

$$D(uv) = uv_1 + vu_1$$

$$D^2(uv) = uv_2 + u_1v_1 + v_1u_1 + vu_2$$

$$= uv_2 + {}^2C_1 u_1 v_1 + {}^2C_2 u_2 v \qquad (\because\ 2 = {}^2C_1,\ 1 = {}^2C_2)$$

Thus we see that the theorem is true for $n = 1, 2$.

Step II. Assume the theorem to be true for $n = m$ (say) so that

$$D^m(uv) = u_m v + {}^mC_1 u_{m-1} v_1 + {}^mC_2 u_{m-2} v_2 + \ldots + {}^mC_{r-1} u_{m-r+1} v_{r-1} + {}^mC_r u_{m-r} v_r + \ldots + {}^mC_m uv_m$$

Differentiating both sides,

$$D^{m+1}(uv) = u_{m+1}v + u_m v_1 + {}^mC_1 u_m v_1 + u_{m-1}v_2 + {}^mC_2 u_{m-1} v_2 + u_{m-2}v_3 + \ldots +$$

$$+ {}^mC_{r-1} u_{m-r+1} v_r + u_{m-r} v_{r+1} + {}^mC_r u_{m-r} v_{r+1} + {}^mC_{r+1} u_{m-r-1} v_{r+2} + \ldots +$$

$$+ {}^mC_m u_1 v_m + uv_{m+1}$$

$$= u_{m+1}v + (1 + {}^mC_1) u_m v_1 + ({}^mC_2 + {}^mC_1) u_{m-1} v_2 + \ldots +$$

$$+ ({}^mC_r + {}^mC_{r-1}) u_{m-r+1} v_r + \ldots + {}^mC_m uv_{m+1}$$

But $\quad 1 = {}^mC_0 = {}^{m+1}C_0 = {}^{m+1}C_{m+1}$

$${}^mC_1 + {}^mC_0 = {}^{m+1}C_1, \ldots \text{ and } {}^mC_r + {}^mC_{r-1} = {}^{m+1}C_r$$

$$\therefore \quad D^{m+1}(uv) = u_{m+1}v + {}^{m+1}C_1 u_m v_1 + {}^{m+1}C_2 u_{m-1} v_2 + \ldots + {}^{m+1}C_r u_{m-r+1} v_r + \ldots + {}^{m+1}C_{m+1} uv_{m+1}$$

which is exactly the same form as Leibnitz formula with n replaced by $m + 1$. Hence if the theorem is true for $n = m$, it is also true for $n = m + 1$.

Step III. In step I, the theorem has been seen to be true for $n = 1$, and by step II, it may be true for $n = 2 + 1$ i.e., it is true for $n = 3$, and so on.

Hence the theorem is true for all positive integral values of n.

Example 3.8. *Find the nth derivative of $x^2 \log(ax + b)$.*

Solution. Take $u = x^2$ and $v = \log(ax + b)$, so that $u_n = 0$ for all integral values of n, and $v_1 = a(ax + b)^{-1}$, $v_2 = 2!a^2(ax + b)^{-2}$, $v_3 = 3!a^3(ax + b)^{-3}$, etc. are all true.

By Leibnitz's theorem,

$$D^n(uv) = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + {}^nC_3 u_{n-3} v_3$$

i.e., $\quad D^n[x^2 \log(ax + b)] = x^2 D^n[\log(ax + b)] + nDx^2 D^{n-1}[\log(ax + b)]$

$$+ \frac{n(n-1)}{1.2} D^2 x^2 D^{n-2}[\log(ax + b)] + \frac{n(n-1)(n-2)}{1.2.3} D^3 x^2 D^{n-3}[\log(ax + b)]$$

$$= a^n (-1)^{n-1} (n-1)! (ax + b)^{-n} x^2 + 2nax(-1)^{n-1} (n-1)! a^{n-1} (ax + b)^{-n} + n(n-1)(-1)^{n-2} (n-2)! a^{n-2} (ax + b)^{-n+2}$$

Example 3.10. *If $y = \sin^{-1} x$, show that $(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} - n^2 y_n = 0$. Hence find $(y_n)_0$.* (U.P.T.U., 2005)

Solution. We have $\quad y = \sin^{-1} x$

Differentiating, $\quad y_1 = \dfrac{1}{\sqrt{(1 - x^2)}}$ or $(1 - x^2) y_1^2 = 1$ ...(1)

Again differentiating, $\quad (1 - x^2)\, 2y_1 y_2 + (-2x) y_1^2 = 0$ ...(2)

Dividing by $2y_1 (1 - x^2)$, $\quad y_2 - xy_1 = 0$

Differentiating it n times by Leibnitz's theorem,

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*Named after the German mathematician and philosopher Gottfried Wilhelm Leibniz (1646–1716) who invented the differential and integral calculus independently of Sir Isaac Newton.

$$(1-x^2)y_{n+2}+nx\cdot 2nxy_{n+1}+\frac{n(n-1)}{2}(-2)y_n-(2n)y_{n+1}+n(n)y_n=0$$

or $$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-n^2y_n=0$$

which is the required result.

Putting $n=0$, $(y_{n+2})_0=a^2(y_n)_0$ ... (iii)

From (i), $(y_0)_0=0$. From (ii), $(y_1)_0=2$

Putting $n=1,2,3,4,\ldots$ in (iii), $2=y_3=y_5=y_7=\ldots$

i.e., if n is odd, $(y_n)_0=0$

Again putting $n=2,4,6,\ldots$ in (iii)

$$(y_4)_0=2^2(y_2)_0=2\cdot 2^2$$
$$(y_6)_0=4^2(y_4)_0=2\cdot 2^2\cdot 4^2$$
$$(y_8)_0=6^2(y_6)_0=2\cdot 2^2\cdot 4^2\cdot 6^2$$

In general, if n is even, $(y_n)_0=2\cdot 2^2\cdot 4^2\cdot 6^2\ldots(n-2)^2$, $(n\neq 2)$

Example 4.31. If $y=(x^2-1)^n$, prove that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}+n^2y_n=0$. Hence find y_n when $x=0$, when n is odd and when n is even. (U.P.T.U., 2003)

Solution. We have $y=(x^2-1)^n$ ... (i)

Differentiating, $y_1=n(x^2-1)^{n-1}\cdot\frac{n}{(x+1)(x-1)}\cdot\frac{y}{(x^2-1)}$ ... (ii)

or $(1-x^2)y_1^2=n^2y^2$.

Again differentiating, $(1-x^2)\cdot 2y_1y_2+(-2x)y_1^2=2n^2yy_1$

Dividing by $2y_1$, $(1-x^2)y_2-xy_1-n^2y=0$ ... (iii)

Differentiating it n times by Leibnitz's theorem,

$$(1-x^2)y_{n+2}+n\cdot(-2x)y_{n+1}+\frac{n(n-1)}{2}(-2)y_n-[xy_{n+1}+n\cdot 1\cdot y_n]-n^2y_n=0$$

or $$(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2+n^2)y_n=0$$

which is the required result.

Putting $n=0$, $(y_{n+2})_0=(n^2+n^2)(y_n)_0$ ... (iv)

From (i), (ii), (iii), $(y_0)_0=1$, $(y_1)_0=0$, $(y_2)_0=n^2$

Putting $n=1,2,3,4\ldots$ in (iv),

$$(y_3)_0=(1^2+n^2)(y_1)_0=n(1^2+n^2)$$
$$(y_4)_0=(2^2+n^2)(y_2)_0=n^2(2^2+n^2)$$
$$(y_5)_0=(3^2+n^2)(y_3)_0=n(1^2+n^2)(3^2+n^2)$$
$$(y_6)_0=(4^2+n^2)(y_4)_0=n^2(2^2+n^2)(4^2+n^2)$$

Hence in general, $(y_n)_0=n(1^2+n^2)(3^2+n^2)\ldots[(n-2)^2+n^2]$, when n is odd.

$=n^2(2^2+n^2)(4^2+n^2)\ldots[(n-2)^2+n^2]$, when n is even.

Example 4.32. If $y^{1/m}+y^{-1/m}=2x$, prove that

$$(x^2-1)y_{n+2}+(2n+1)xy_{n+1}+(n^2-m^2)y_n=0$$

(U.P.T.U., 2005; Nagpur, 2007; U.P.T.U., 2007)

Solution. We have $y^{1/m}+\frac{1}{y^{1/m}}=2x$

or $(y^{1/m})^2-2xy^{1/m}+1=0$

$$\therefore\quad y^{1/m}=\frac{2x\pm\sqrt{4x^2-4}}{2}=x\pm\sqrt{(x^2-1)}$$

Hence $y=[x\pm\sqrt{(x^2-1)}]^m$

Taking logarithms, $\log y=m\log[x\pm\sqrt{(x^2-1)}]$

Differentiating both sides w.r.t. x,

$$\frac{1}{y}y_1 = \frac{1}{x+\sqrt{x^2-1}}\left[1+\frac{x}{\sqrt{x^2-1}}\right] = \frac{1}{\sqrt{x^2-1}}$$

Squaring, $(x^2-1)y_1^2 = m^2y^2$

Again differentiating, $(x^2-1)2y_1y_2 + y_1^2(2x) = m^2 \cdot 2yy_1$

Dividing by $2y_1$, $(x^2-1)y_2 + xy_1 - m^2y = 0$

Differentiating n times by Leibnitz's theorem,

$$D^n(x^2-1)y_{n+2} + {}^nC_1 y_{n+1}(2x) + \frac{n(n-1)}{2}y_n(2) + xy_{n+1} + ny_n(1) - m^2y_n = 0$$

or, $(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0$

PROBLEMS 3

1. $y = \sin(m\sin^{-1}x)$ prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0$ [Dhanbad, 2005]
2. If $x = \cos(\log y)$ + $\sin(\log y)$, show that $(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$ [U.P.T.U., 2005 ; Kolkata, 2005]
3. If $y = \sin^{-1}x$, prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$. Also find $(y_n)_0$ [U.P.T.U., 2006]
4. If $y = \tan^{-1}(\sin\log x^2)$, prove that $x^2y_{n+2} + (2n+1)xy_{n+1} + 2n^2y_n = 0$ [U.P.T.U., 2006]
5. If $y = \sin^{-1}x$, prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$. Also find $(y_n)_0$.
6. If $y = \cos(m\sin^{-1}x)$, prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0$ [Mumbai, 2005 S]
7. If $y = \sin(m\sin^{-1}x)$, prove that $(1-x^2)y_{n+2} - xy_{n+1} + m^2y_n = 0$

   Hence find $(y_n)_0$ [U.P.T.U., 2005 ; Orissa, 2004]

   [U.P.T.U., 2005]
8. If $y = e^{m\sin^{-1}x}$, prove that $(1-x^2)y_{n+2} - xy_{n+1} - m^2y_n = 0$

   Hence find $(y_n)_0$ and $(y_{n+1})_0$
9. If $y = (x^2-1)^n$, prove that $(x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$ [U.P.T.U., 2004]
10. If $\tan^{-1}y = 2\log(x+2)$, prove that $(x^2+4x+5)y_{n+2} + (2n+3)(x+2)y_{n+1} + (n^2+1)y_n = 0$ [U.P.T.U., 2003]
11. If $y = x^n\log x$, prove that $y_{n+1} = \frac{n!}{x}$ [Kolkata, 2005]
12. If $V_n = \frac{d^n}{dx^n}(x^n\log x)$, show that $V_n = nV_{n-1} + (n-1)!$

    Hence, show that $V_n = n!\left[\log x + 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}\right]$ [U.P.T.U., 2002]
13. Show that $\frac{d^n}{dx^n}\left(\frac{\log x}{x}\right) = \frac{(-1)^n n!}{x^{n+1}}\log x + (-1)^n\frac{n!}{x^{n+1}}\left[1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}\right]$ [U.P.T.U., 2005]
14. If $y = x\log\left(\frac{x-1}{x+1}\right)$, show that $y_n = (-1)^{n-2}(n-2)!\left[\frac{x-n}{(x-1)^n} - \frac{x+n}{(x+1)^n}\right]$ [U.P.T.U., 2005]
15. If $x = \sin(\frac{1}{m}\log y)$, show that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+m^2)y_n = 0$. Hence prove that $(1-x^2)(y_n)_0 - (2n-1)(y_{n-1})_0 - (n^2+m^2)(y_{n-2})_0 = 0$ [Kolkata, 2005 ; U.P.T.U., 2005]
16. If $y = \log(x+\sqrt{x^2+1})^2$, prove that $(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2y_n = 0$ [U.P.T.U., 2005 ; Kolkata, 2005]

    Hence find $(y_n)_0$ if $n = 2k$ or $2k+1$, where k is a positive integer.
17. If $x = \log(\sqrt{a^2+1}+1)^2$, prove that $(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2y_n = 0$. Hence find $(y_n)_0$ [U.P.T.U., 2005]
18. If $x = \sin(\frac{1}{a}\log y)$, prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$ [U.P.T.U., 2006]

18. $y = \frac{ax^2 + b}{\sqrt{x^2 + k^2}}$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + k^2)y_n = 0$. (U.P.T.U., 2002)

19. If $y = \sin(m \sin^{-1} x)$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - m^2)y_n = 0$. (U.P.T.U., 2003)

4.13 FUNDAMENTAL THEOREMS

(1) Rolle's Theorem

If $f(x)$ is continuous in the closed interval $[a, b]$, $f'(x)$ exists for every value of x in the open interval (a, b) and $f(a) = f(b)$, then there is at least one value of x in (a, b), say $x = c$, such that $f'(c) = 0$.

Consider the portion AB of the curve $y = f(x)$, lying between $x = a$ and $x = b$, such that

(i) it goes continuously from A to B,

(ii) it has a tangent at every point between A and B, and

(iii) ordinate of A = ordinate of B.

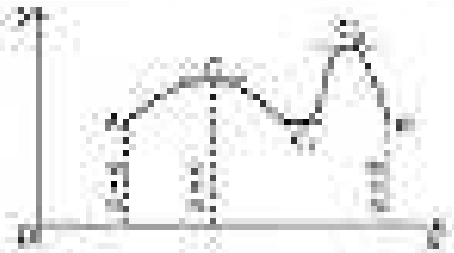

Fig. 4.1

From the Fig. 4.1, it is self-evident that there is at least one point C (one in figure) on the curve at which the tangent is parallel, to the x-axis.

i.e., slope of the tangent at C, $\tan\psi = 0$.

But the slope of the tangent at C is the value of the differential coefficient of $f(x)$ w.r.t. x at $x = c$, i.e., $f'(c) = 0$.

Hence the theorem is proved.

Example 4.13. Verify Rolle's theorem for (i) $\sin x$ in $[0, \pi]$ (U.P.T.U., 2002)

(ii) $(x - a)^m (x - b)^n$ where m, n are positive integers in $[a, b]$. (U.P.T.U., 2010 / Nagpur Univ., 2005)

Solution. (i) Let $f(x) = \sin x \cdot e^x$.

$f(x)$ is defined in $[0, \pi]$

Also $f(0) = f(\pi) = 0$.

Hence the conditions of Rolle's theorem are satisfied.

$\therefore$ $f'(x) = \frac{e^x \cos x - e^x \sin x}{e^x}$ vanishes when $e^x(\cos x - \sin x) = 0$

or $\tan x = 1 \quad i.e., x = \pi/4$.

The value $x = \pi/4$ lies in $(0, \pi)$, so that Rolle's theorem is verified.

(ii) Let $f(x) = (x - a)^m (x - b)^n$.

Since every polynomial is continuous for all values, $f(x)$ is also continuous in $[a, b]$.

$$f'(x) = n(x - a)^{m-1}(x - b)^n + (x - a)^m \cdot n(x - b)^{n-1}$$
$$= (x - a)^{m-1}(x - b)^{n-1}[(m + n)x - (mb + na)]$$

which exists, i.e., $f(x)$ is derivable in (a, b).

Also $f(a) = 0 = f(b)$.

Thus all the conditions of Rolle's theorem are satisfied and there exists c in (a, b) such that $f'(c) = 0$.

$\therefore$ $(c - a)^{m-1}(c - b)^{n-1}[(m + n)c - (mb + na)] = 0$ or $c = \frac{mb + na}{m + n}$.

Hence, Rolle's theorem is verified.

(2) Lagrange's Mean-Value Theorem*

First Form. If (i) $f(x)$ is continuous in the closed interval $[a, b]$ and

(ii) $f(x)$ exists in the open interval (a, b), then there is at least one value of x in (a, b), such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

*Named after the great French mathematician Joseph Louis Lagrange (1736–1813) who became professor at Military Academy, Turin when he was just 19 and director of Berlin Academy in 1766. His important contributions are in algebra, number theory, differential equations, mechanics, approximation theory and calculus of variations.

Consider the function $\phi(x) = f(x) - \dfrac{f(b)-f(a)}{b-a}\,x$

Since $f(x)$ is continuous in $[a, b]$; $\therefore$ $\phi(x)$ is also continuous in $[a, b]$

Since $f'(x)$ exists in $[a, b]$,

$\therefore$ $\phi'(x)$ is also exists in (a, b) and $\phi'(x) = f'(x) - \dfrac{f(b)-f(a)}{b-a}$ ...(1)

Clearly, $\phi(a) = \dfrac{b f(a) - a f(b)}{b-a} = \phi(b)$

Thus $\phi(x)$ satisfies all the conditions of Rolle's theorem.

$\therefore$ There is at least one value of x between a and b such that $\phi'(x) = 0$. Substituting $x = c$ in (1), we get

$$f'(c) = \frac{f(b)-f(a)}{b-a} \qquad ...(2)$$

which proves the theorem.

Second form. If we write $b = a + h$, then since $a < c < b$,

$$c = a + \theta h \text{ where } 0 < \theta < 1.$$

Thus the mean value theorem may be stated as follows :

If $f(x)$ is continuous in the closed interval $[a, a+h]$ and $f(x)$ is derivable in the open interval $(a, a+h)$, then there is at least one number θ, $0 < \theta < 1$, such that

$$f(a+h) = f(a) + hf'(a + \theta h)$$

Geometrical Interpretation. Let A, B be the points on the curve $y = f(x)$ corresponding to $x = a$ and $x = b$ so that $A = (a, f(a))$ and $B = (b, f(b))$. (Fig. 4.2)

$\therefore$ Slope of the chord $AB = \dfrac{f(b)-f(a)}{b-a}$

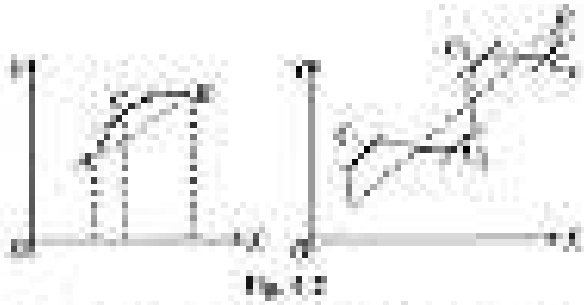

Fig. 4.2

By (2), the slope of the chord $AB = f'(c)$, the slope of the tangent at the point $P(x = c)$.

Hence the Lagrange's mean value theorem asserts that if a curve AB has a tangent at each of its points, then there exists at least one point P on the curve, the tangent at which is parallel to the chord AB.

Cor. If $f'(x) = 0$ in the interval a, b, then $f(x)$ is constant in $[a, b]$. For, if x_1, x_2 be any two values in $[a, b]$, then by (2),

$$f(x_2) - f(x_1) = (x_2 - x_1) f'(c) = 0 \quad (x_1 < c < x_2)$$

Thus, $f(x_1) = f(x_2)$, i.e., $f(x)$ has the same value for every value of x in $[a, b]$.

Example 4.14. *In the Mean value theorem $f(b) - f(a) = (b-a) f'(a + \theta(b-a))$, ...(1)*
determine θ lying between 0 and 1, if $f(x) = Ax^2 + Bx + C$, $a = 0$ and $b = 2$. (Gorakhpur, 2000)

Solution. $f(x) = Ax^2 + Bx + C$, $\quad f(b) = f(2) = 4A + 2B + C$

$f'(x) = 2Ax + B$, $\quad f'(a + \theta(b-a)) = f'(2\theta) = 4A\theta + B$

Substituting in (1), $\quad 4A + 2B = (2 - 0)(4A\theta + B) = 8A\theta + 2B$

or $\quad 8A\theta = 4A$

Hence $\quad \theta = \dfrac{1}{2}$

Example 4.15. *Show that if $0 < a < b < 1$, $\dfrac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \dfrac{b-a}{1+a^2}$*

Hence show that $\quad \dfrac{3}{4} + \dfrac{3}{50} < \tan^{-1}\dfrac{4}{3} < \dfrac{3}{4} + \dfrac{1}{6}$ (Mumbai, 2003; V.T.U., 2002)

Solution. Let $f(x) = \tan^{-1} x$, so that $f'(x) = \frac{1}{1+x^2}$.

By Mean value theorem, $\frac{\tan^{-1} b - \tan^{-1} a}{b-a} = \frac{1}{1+c^2}$, $a < c < b$ ...(1)

Now $a < c < b$, $\therefore 1+a^2 < 1+c^2 < 1+b^2$,

$\therefore \quad \frac{1}{1+b^2} < \frac{1}{1+c^2} < \frac{1}{1+a^2}$ i.e., $\frac{1}{1+b^2} < \frac{1}{1+c^2} < \frac{1}{1+a^2}$

i.e., $\frac{1}{1+b^2} < \frac{\tan^{-1} b - \tan^{-1} a}{b-a} < \frac{1}{1+a^2}$ [By (1)]

Hence $\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$

Now let $a = 1, b = 4/3$.

Then $\frac{1/3}{1+16/9} < \tan^{-1}\frac{4}{3} - \frac{\pi}{4} < \frac{1/3}{1+1}$

i.e., $\frac{3}{25} + \frac{\pi}{4} < \tan^{-1}\frac{4}{3} < \frac{1}{6} + \frac{\pi}{4}$.

Example 4.38. Prove that $\log(1+x) = x/(1+\theta x)$, $0 < \theta < 1$ and hence deduce that $\frac{x}{1+x} < \log(1+x) < x$, $x > 0$. (Mumbai, 2005)

Solution. Let $f(x) = \log(1+x)$, then by second form of Lagrange's mean value theorem

$f(a+h) = f(a) + h f'(a+\theta h)$, $(0 < \theta < 1)$

we have $f(x) = f(0) + x f'(\theta x)$ [Taking $a = 0, h = x$]

or $\log(1+x) = x/(1+\theta x)$ [$\because f'(x) = 1/(1+x)$]

Hence $\log(1+x) < x/(1+0) = x$ [$\because \log(1) = 0$]

Since $0 < \theta < 1$, $\therefore 0 < \theta x < x$ or $1 < 1+\theta x < 1+x$

or $\frac{1}{1+x} < \frac{1}{1+\theta x} < \frac{1}{1}$

or $x > \frac{x}{1+\theta x} > \frac{x}{1+x}$

or $\frac{x}{1+x} < \log(1+x) < x, x > 0$. [By (i)]

(ii) Cauchy's[1] Mean value Theorem

If (i) $f(x)$ and $g(x)$ be continuous in $[a, b]$

(ii) $f'(x)$ and $g'(x)$ exist in (a, b)

and (iii) $g'(x) \neq 0$ for every value of x in (a, b),

then there is at least one value of x in (a, b), such that $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{f'(c)}$

Consider the function $\phi(x) = f(x) - \frac{f(b)-f(a)}{g(b)-g(a)} g(x)$

Since $f(x)$ and $g(x)$ are continuous in $[a, b]$

$\therefore \quad \phi(x)$ is also continuous in $[a, b]$

Again since $f'(x)$ and $g'(x)$ exist in (a, b),

[1] Named after the great French mathematician Augustin-Louis Cauchy (1789–1857) who is considered as the father of modern analysis and creator of complex analysis. He published nearly 800 research papers of basic importance. Cauchy is also well known for his contributions to differential equations, infinite series, optics and elasticity.

∴ $g'(x)$ is also continuous in $[a, b]$ and $\forall x \in (a, b)$ $g'(x) = f'(x) - \frac{f(b)-f(a)}{g(b)-g(a)} g'(x)$

Clearly, $g(a) = g(b)$.

Thus, $g(x)$ satisfies all the conditions of Rolle's theorem. There is therefore, at least one value c of x between a and b, such that $g'(c) = 0$

i.e., $0 = f'(c) - \frac{f(b)-f(a)}{g(b)-g(a)} g'(c)$ (where $a < c < b$) which is the result.

(N.T.U., 2007 S ; P.T.U., 2005)

This Cauchy's mean value theorem is a generalization of Lagrange's mean value theorem, when $g(x) = x$.

Example 4.13. Verify Cauchy's Mean value theorem for the functions e^x and e^{-x} in the interval $[a, b]$.

Solution. $f(x) = e^x$ and $g(x) = e^{-x}$ are both continuous in $[a, b]$ and both functions are differentiable in (a, b)

i.e., $f'(x) = e^x, g'(x) = -e^{-x}$

By Cauchy's mean value theorem,

$$\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$$

$$\therefore \quad \frac{e^b - e^a}{e^{-b} - e^{-a}} = \frac{e^c}{-e^{-c}} \quad \text{i.e., } e^c = \frac{1}{2}(e^a + e^b)$$

Thus we have $c = \log\left[\frac{1}{2}(e^a + e^b)\right]$, which verifies the Cauchy's Mean value theorem.

(4) Taylor's Theorem[1] (Generalised mean value theorem)

If (i) $f(x)$ and its first $(n-1)$ derivatives be continuous in $[a, a+h]$ and (ii) $f^n(x)$ exists for every value of x in $(a, a+h)$, then there is at least one number θ ($0 < \theta < 1$) such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \ldots + \frac{h^n}{n!} f^n(a+\theta h) \quad \ldots(1)$$

which is called Taylor's theorem with Lagrange's form remainder, the remainder R_n being $\frac{h^n}{n!} f^n(a+\theta h)$.

Proof. Consider the function

$$\phi(x) = f(x) + (a+h-x) f'(x) + \frac{(a+h-x)^2}{2!} f''(x) + \ldots + \frac{(a+h-x)^n}{n!} K$$

where K is defined by

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \ldots + \frac{h^n}{n!} K \quad \ldots(2)$$

(i) Since $f(x), f'(x), \ldots, f^{n-1}(x)$ are continuous in $[a, a+h]$, therefore $\phi(x)$ is also continuous in $[a, a+h]$,

(ii) $\phi'(x)$ exists and $= \frac{(a+h-x)^{n-1}}{(n-1)!} [f^n(x) - K]$

(iii) Also $\phi(a) = \phi(a+h)$ [By (2)]

Hence $\phi(x)$ satisfies all the conditions of Rolle's theorem, and therefore, there exists at least one number θ ($0 < \theta < 1$) such that $\phi'(a+\theta h) = 0$ i.e., $K = f^n(a+\theta h)$ ($0 < \theta < 1$)

Substituting this value of K in (2), we get (1).

Cor. 1. Taking $a = 0$ in (1), Taylor's theorem reduces to Lagrange's Mean value theorem.

Cor. 2. Putting $a = 0$ and $h = x$ in (1), we get

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \ldots + \frac{x^n}{n!} f^n(\theta x) \quad \ldots(3)$$

which is known as Maclaurin's theorem with Lagrange's form of remainder.

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[1] Named after an English mathematician, Brook Taylor (1685–1731).

Example 6.18. *Find the Maclaurin's theorem with Lagrange's form of remainder for* $f(x) = \cos x$. *Hence prove that* $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \frac{(-1)^{n+1} x^{2n+2}}{(2n+2)!} \cos\theta x$ (A.M.I.E., 2003)

Solution. $f^n(x) = \frac{d^n}{dx^n}(\cos x) = \cos\left(\frac{n\pi}{2} + x\right)$ so that $f_n(0) = \cos\left(\frac{n\pi}{2}\right)$

Thus $f(0) = 1$,

$$f^{2n}(0) = \cos(2n\pi/2) = (-1)^n$$

$$f^{(2n+1)}(0) = \cos[(2n+1)\pi/2] = 0$$

Substituting these values in the Maclaurin's theorem with Lagrange's form of remainder, we get

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^{2n}}{(2n)!}f^{2n}(0) + \frac{x^{2n+1}}{(2n+1)!}f^{(2n+1)}(\theta x)$$

We get $\cos x = 1 + 0 + \frac{x^2}{2!}(-1) + 0 + \cdots + \frac{x^{2n}}{(2n)!}(-1)^n + \frac{x^{2n+1}}{(2n+2)!}(-\cos\theta x)$

i.e., $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \frac{(-1)^{n+1} x^{2n+2}}{(2n+1)!}\cos\theta x$

Example 6.19. *If* $f(x) = \log(1+x)$, $x > 0$, *using Maclaurin's theorem, show that for* $0 < \theta < 1$,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3}$$

Deduce that $\log(1.1) = 0.1 - \frac{0.01}{2} + \frac{0.001}{3}$ *approx.* (A.M.I.E., 2003)

Solution. By Maclaurin's theorem with remainder R_3, we have

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(\theta x) \quad \ldots(1)$$

Here $f(x) = \log(1+x)$, $f(0) = 0$

$\therefore$ $f'(x) = \frac{1}{1+x}$, $f'(0) = 1$

$f''(x) = \frac{-1}{(1+x)^2}$, $f''(0) = -1$

and $f'''(x) = \frac{2}{(1+x)^3}$, $f'''(\theta x) = \frac{2}{(1+\theta x)^3}$

Substituting in (1), we get $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3}$ $\quad \ldots(2)$

Since $x > 0$ and $0 < \theta < 1$, $\theta x > 0$

or $(1+\theta x)^3 > 1$ i.e., $\frac{1}{(1+\theta x)^3} < 1$

$\therefore$ $x - \frac{x^2}{2} + \frac{x^3}{3(1+\theta x)^3} < x - \frac{x^2}{2} + \frac{x^3}{3}$

Hence $\log(1+x) < x - \frac{x^2}{2} + \frac{x^3}{3}$. [By (2)]

PROBLEMS 6.4

1. Verify Taylor's theorem for (i) $f(x) = x^3 - 2x^2 + x + 1$ in $[0, 1]$.
   (ii) $f(x) = x^3$ in $[1, 2]$. (iii) $f(x) = \log x$ in $[1, e]$. (iv) $f(x) = e^x$ in $[0, 1]$.
2. (i) $f(x) = \log\left|\frac{x^2 + ab}{x(a+b)}\right|$ in $[a, b]$. (A.M.I.E., 2003)

2. Using Rolle's theorem for $f(x) = x^{2/3}(1-x)^{1/3}$, find the value of c between 0 and 1 where $f'(c) = 0$.
3. Verify Lagrange's Mean value theorem for the following functions and find the corresponding values of c in the respective intervals mentioned:
   (a) $f(x) = x(x-1)(x-2)$ in $[0, \frac{1}{2}]$ (V.T.U., 2009)
   (b) $f(x) = \sin x$ in $[0, \pi]$ (Nagpur, 2009)
   (c) $f(x) = \log_e x$ in $[1, e]$ (Kolhapur, 2007)
   (d) $f(x) = e^x$ in $[0, 1]$ (U.P.T.U., 2009)
4. By applying Mean value theorem, if $\frac{\pi}{2} = \log e$, prove that $\frac{\pi}{2} \log 2 < \sin^{-1}\frac{3}{5} < \frac{\pi}{2} - \frac{1}{5}$, where $\sin^{-1}\frac{3}{5}$ lies between 0 and 2.
5. In the Mean value theorem, $f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a+\theta h)$, $0 < \theta < 1$, find θ when $f(x) = ax^2 + bx + c$.
6. If $f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a+\theta(x-a))$, $0 < \theta < 1$, find θ when $f(x) = x^3 - 2x^2 + x + 1$ and $f(2) = 3$, $a = 1$.
7. If $x = \sin t$, $y = \sin kt$, prove that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + k^2 y = 0$. (V.T.U., 2008)
8. If $f(x) = ax^2 + bx + c$, $a \neq 0$, use Rolle's theorem to prove that
$$\frac{f(a) - f(b)}{f'(a) - f'(b)} = \frac{a+b}{2}$$
9. Prove that $\frac{b-a}{1+b^2} < \tan^{-1}b - \tan^{-1}a < \frac{b-a}{1+a^2}$ for $0 < a < b$.
   Hence deduce that $\frac{3}{4} < \tan^{-1}\frac{4}{3} < \frac{1}{3}$. (V.T.U., 2009)
10. Verify the result of Cauchy's mean value theorem for the functions:
    (a) x^2 and x^3 in the interval $[a, b]$. (U.P.T.U., 2008-09)
    (b) $\log_e x$ and $\frac{1}{x}$ in the interval $[1, e]$.
11. If $f(x)$ and $g(x)$ are respectively x^2 and x^3, prove that $\frac{c}{b}$ of Cauchy's mean value theorem is the arithmetic mean between a and b. (Mumbai, 2009)
12. Verify Maclaurin's theorem for $e^x \sin x$ up to the term containing x^3. (Kolhapur, 2008)
13. Using Taylor's theorem, prove that
$$\log\cos x = -\frac{x^2}{2} - \frac{x^4}{12} - \frac{x^6}{45} - \ldots \text{ for } -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

5.8 EXPANSIONS OF FUNCTIONS

(1) Maclaurin's series. If $f(x)$ can be expanded as an infinite series, then

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ldots \quad \ldots(1)$$

If $f(x)$ possesses derivatives of all orders and the remainder R_n in Taylor's series tends to zero as $n \to \infty$, then the Maclaurin's theorem becomes the Maclaurin's series (1).

Example 5.31. Using Maclaurin's series, expand $\tan x$ upto the term containing x^3. (V.T.U., 2010)

Solution. Let $f(x) = \tan x$ $\quad \therefore f(0) = 0$

$f'(x) = \sec^2 x = 1 + \tan^2 x$ $\quad f'(0) = 1$

$f''(x) = 2\sec x \cdot \sec x \tan x = 2\tan x(1 + \tan^2 x)$
$= 2\tan x + 2\tan^3 x$ $\quad f''(0) = 0$

$f'''(x) = 2\sec^2 x + 6\tan^2 x \sec^2 x$
$= 2(1 + \tan^2 x) + 6\tan^2 x(1 + \tan^2 x)$
$= 2 + 8\tan^2 x + 6\tan^4 x$ $\quad f'''(0) = 2$

$f''''(x) = 16\tan x \sec^2 x + 24\tan^3 x \sec^2 x$

$$= 16\tan x(1+\tan^2 x) + 24\tan^2 x(1+\tan^2 x)$$
$$= 16\tan x + 40\tan^3 x + 24\tan^5 x \qquad f^{iv}(0) = 0$$
$$f^{v}(x) = 16\sec^2 x + 120\tan^2 x\sec^2 x + 120\tan^4 x\sec^2 x, \qquad f^{v}(0) = 16$$

and so on.

Substituting the values of $f(0), f'(0)$, etc. in the Maclaurin's series, we get

$$\tan x = 0 + x\cdot 1 + \frac{x^2}{2!}\cdot 0 + \frac{x^3}{3!}\cdot 2 + \frac{x^4}{4!}\cdot 0 + \frac{x^5}{5!}\cdot 16 + \ldots \text{ or } x + \frac{x^3}{3} + \frac{2x^5}{15} + \ldots$$

(3) Expansion by use of known series. When the expansion of a function is required only upto first few terms, it is often convenient to employ the following well-known series :

1. $\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \ldots$
2. $\sinh\theta = \theta + \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \frac{\theta^7}{7!} + \ldots$
3. $\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \ldots$
4. $\cosh\theta = 1 + \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \frac{\theta^6}{6!} + \ldots$
5. $\tan\theta = \theta + \frac{\theta^3}{3} + \frac{2}{15}\theta^5 + \ldots$
6. $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots$
7. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$
8. $\log(1+x) = \ln x = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$
9. $\log(1-x)\ln = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \ldots\right)$
10. $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \ldots$

Example 2.21. Expand $e^{\sin x}$ by Maclaurin's series as far as the term containing x^4.
(Bhopal, 2008; VTU, 2012)

Solution. We have $e^{\sin x} = 1 + \sin x + \frac{\sin^2 x}{2!} + \frac{\sin^3 x}{3!} + \frac{\sin^4 x}{4!} + \ldots$

$$= 1 + \left(x - \frac{x^3}{3!} + \ldots\right) + \frac{1}{2!}\left(x - \frac{x^3}{3!} + \ldots\right)^2 + \frac{1}{3!}\left(x - \frac{x^3}{3!} + \ldots\right)^3 + \frac{1}{4!}(x - \ldots)^4 + \ldots$$
$$= 1 + \left(x - \frac{x^3}{6} - \ldots\right) + \frac{1}{2}\left(x^2 - \frac{x^4}{3} + \ldots\right) + \frac{1}{6}(x^3 - \ldots) + \frac{1}{24}x^4 - \ldots + \ldots$$
$$= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \ldots$$

Otherwise, let $f(x) = e^{\sin x}$ $\qquad f(0) = 1$

$f'(x) = e^{\sin x}\cos x = f(x)\cdot\cos x$ $\qquad f'(0) = 1$

$f''(x) = f'(x)\cos x - f(x)\sin x$, $\qquad f''(0) = 1$

$f'''(x) = f''(x)\cos x - 2f'(x)\sin x - f(x)\cos x$, $\qquad f'''(0) = 0$

$f^{iv}(x) = f'''(x)\cos x - 3f''(x)\sin x - 3f'(x)\cos x + f(x)\sin x$, $\qquad f^{iv}(0) = -3$

and so on.

Substituting the values of $f(0), f'(0)$ etc., in the Maclaurin's series, we obtain

$$e^{\sin x} = 1 + x\cdot 1 + \frac{x^2}{2!}\cdot 1 + \frac{x^3}{3!}\cdot 0 + \frac{x^4}{4!}(-3) + \ldots$$
$$= 1 + x + \frac{x^2}{2} - \frac{x^4}{8} + \ldots$$

Example 4.32. *Expand $\log(1+\sin^2 x)$ in powers of x as far as the term in x^4.* (Bhopal, 2008 S)

Solution. We have $\sin^2 x = \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right)^2 = \left[x - \left(\frac{x^3}{6} - \frac{x^5}{120} - \ldots\right)\right]^2$

$$= x^2 - 2x\left(\frac{x^3}{6} - \frac{x^5}{120} + \ldots\right) + \left(\frac{x^3}{6} - \frac{x^5}{120} + \ldots\right)^2$$

$$= x^2 - \frac{x^4}{3} + \frac{x^6}{60} - \frac{x^6}{36} + \ldots = x^2 - \frac{x^4}{3} + \frac{11x^6}{360} + \ldots = t, \text{ say}.$$

Now $\log(1+\sin^2 x) = \log(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \ldots$

Substituting the value of t, we get

$$\log(1+\sin^2 x) = x^2 - \frac{x^4}{3} + \frac{2x^6}{45} + \ldots - \frac{1}{2}\left(x^4 - \frac{2x^6}{3} + \ldots\right)^2 + \frac{1}{3}x^6 + \ldots$$

$$= x^2 - \frac{x^4}{3} + \frac{2x^6}{45} - \frac{1}{2}\left(x^4 - \frac{2x^6}{3} + \ldots\right) + \frac{1}{3}x^6 + \ldots$$

$$= x^2 - \frac{5}{6}x^4 + \frac{22}{45}x^6 + \ldots$$

Obs. As it is very cumbersome to find the successive derivatives of $\log(1+\sin^2 x)$, therefore the above method is preferable to Maclaurin's series method.

Example 4.33. *Expand $e^{\tan^{-1}x}$ in ascending powers of x.*

Solution. Let $y = e^{\tan^{-1}x}$. In Ex. 4.5, we have shown that

$$(y)_0 = 1, (y_1)_0 = a, (y_2)_0 = a^2, (y_3)_0 = a(1+a^2), (y_4)_0 = a^2(3+a^2)$$

and so on.

Substituting these values in the Maclaurin's series

$$y = (y)_0 + \frac{(y_1)_0}{1!}x + \frac{(y_2)_0}{2!}x^2 + \frac{(y_3)_0}{3!}x^3 + \frac{(y_4)_0}{4!}x^4 + \ldots$$

we get $e^{\tan^{-1}x} = 1 + ax + \frac{a^2}{2!}x^2 + \frac{a(1+a^2)}{3!}x^3 + \frac{a^2(3+a^2)}{4!}x^4 + \ldots$

(3) Taylor's series. *If $f(x+h)$ can be expanded as an infinite series then*

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \ldots \quad \ldots(1)$$

If $f(x)$ possesses derivatives of all orders and the remainder R_n in (1) on page 143, tends to zero as $n \to \infty$, then the Taylor's theorem becomes the Taylor's series (1).

Cor. Replacing x by a and h by $x-a$ in (1), we get

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \ldots$$

Taking $a = 0$, we get Maclaurin's series.

Example 4.34. *Expand $\log_e x$ in powers of $(x-1)$ and hence evaluate $\log_e 1.1$ correct to 4 decimal places.* (Bhopal, 2007; Kurukshetra, 2008)

Solution. Let $f(x) = \log_e x$ $\quad f(1) = 0$

$\therefore$ $f'(x) = \frac{1}{x}$, $\quad f'(1) = 1$

$f''(x) = -\frac{1}{x^2}$ $\quad f''(1) = -1$

$$f'''(x) = \frac{2}{x^3}, \qquad f'''(1) = 2$$

$$f^{iv}(x) = -\frac{6}{x^4}, \qquad f^{iv}(1) = -6$$

etc. etc.

Substituting these values in the Taylor's series

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \frac{(x-1)^4}{4!}f^{iv}(1) + \ldots$$

we get $$\log_e x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \ldots$$

Now putting $x = 1.1$, i.e. $(x-1) = 0.1$, we have

$$\log_e(1.1) = 0.1 - \tfrac{1}{2}(0.1)^2 + \tfrac{1}{3}(0.1)^3 - \tfrac{1}{4}(0.1)^4 + \ldots$$

$$= 0.1 - 0.005 + 0.0003 - 0.0002 + \ldots = 0.0953,$$

Example 4.28. (Use Taylor's series to prove that)

$$\sin^{-1}(x+h) = \sin^{-1}x + h(\sin x)^{-1}\frac{\cos x}{1} - h^2(\sin x)^{-2}\frac{\sin x}{2} + h^3(\sin x)^{-3}\frac{(3\sin^2 x + 2)\cos x}{6} - \ldots$$

(where x is not 0). (Nov.96, May 2000)

Solution. We have $$\sin x = y \qquad \ldots(1)$$

$$\therefore \quad -\operatorname{cosec}^2 x\,dx/dy = 1, \text{ or } dx/dy = -\sin^2 x \qquad \ldots(2)$$

Now let $f(x+h) = \sin^{-1}(x+h)$, so that $f(x) = \sin^{-1}x$

$$\therefore \quad f'(x) = \frac{1}{1+x^2} = \frac{1}{1+\cos^2 x} = \sin^2 x \qquad \text{[By (1)]}$$

$$f''(x) = 2\sin x\cos x\,\frac{dx}{dy} = \sin 2x\cdot(-\sin^2 x) \qquad \text{[By (2)]}$$

$$f'''(x) = -[2\cos 2x\cdot\sin^2 x + \sin 2x\cdot 2\sin x\cos x]\frac{dx}{dy}$$

$$= -2\sin x[\cos 2x\sin x + \sin 2x\cos x](-\sin^2 x) = 2\sin^3 x\sin 3x$$

and so on.

Substituting these values in the Taylor's series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \ldots$$

we get the required result.

EXERCISE 4.8

Using Maclaurin's series, expand the following functions:

1. $\log(1+x)$. Hence deduce that $\log\frac{1+x}{1-x} = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \ldots\right)$.

2. $e^{\sin x}$ (Nov.01, 2006) 3. $\sqrt{(1+\sin 2x)}$ (VTU, 2002)

4. $\log(\sec x)$ (Mumbai, 2000) 5. $\tan^{-1}x$

6. $\log(1+e^x)$ (Mumbai, 1998-S), VTU, 2003)

Prove that:

7. $\sin(x+h) = \sin x + h\cos x - \frac{h^2}{2!}\sin x - \frac{h^3}{3!}\cos x + \ldots$

8. $e^{x\cos x} = 1 + x + \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{3x^4}{4!} + \ldots$ (Mumbai, 2003)

9. $\sin^{-1}\frac{2x}{1+x^2} = 2\left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots\right]$

10. $\log\left(\frac{\sqrt{1+x^2}+1}{x}\right) = \frac{1}{2}\left(\frac{1}{x} - \frac{1}{2x^2} + \frac{1}{3x^3} - \frac{1}{4x^4} + \ldots\right)$

11. $\log(1+\sin x)=x-\frac{x^2}{2}+\frac{x^3}{6}-\frac{x^4}{12}+\ldots$ 12. $e^{\tan x}=1+x+\frac{x^2}{2}+\frac{5x^3}{6}+\frac{15x^4}{24}+\ldots$ (Kolkata, 2005)

13. $e^{\sin x}=1+x+\frac{x^2}{2}-\frac{x^4}{8}+\ldots$ (Kolkata, 2006)

14. $e^{\sin x}=e^x\left(1-x+\frac{x^3}{3}-\frac{x^4}{6}+\ldots\right)$ (Mumbai, 2009) 15. $\log\frac{\sin x}{x}=-\frac{x^2}{6}-\frac{x^4}{180}-\frac{x^6}{2835}-\ldots$

16. $\log(1+\sin x)=x-\frac{x^2}{2}+\frac{x^3}{6}-\frac{x^4}{12}+\ldots$ (C.U.T.C. 2009; J.U.T.C., 2008 E)

17. $\sqrt{(1+\sin x)}=1+\frac{x}{2}-\frac{x^2}{8}-\frac{x^3}{16}+\frac{5x^4}{384}-\ldots$ (C.U.T.C., 2006)

18. $\log(1+e^x)=\log 2+\frac{x}{2}+\frac{x^2}{8}-\frac{x^4}{192}+\ldots$ (Nagpur, 2004)

19. $\frac{e^x}{e^x-1}=\frac{1}{x}+\frac{1}{2}+\frac{x}{12}-\frac{x^3}{720}+\ldots$ (a) up to and term (b) 20. $\frac{1}{2}\left(\frac{e^x+1}{e^x-1}\right)=\frac{1}{x}+\frac{x}{12}-\frac{x^3}{720}+\frac{x^5}{30240}-\ldots$ (Mumbai, 2007)

21. $\sin x\cosh x=x+\frac{x^3}{3}-\frac{x^5}{30}+\ldots$

For $\sin x\cosh x=x+\frac{x^3}{3}-\frac{x^5}{30}+\ldots$, show that

22. $\tan^{-1}(\sin x)=x-\frac{x^3}{12}+\frac{x^5}{24}-\frac{x^7}{161}+\ldots$

23. $\log(1+\sqrt{(1+x^2)})=x-\frac{1}{2}\frac{x^3}{3}+\frac{1\cdot3}{2\cdot4}\frac{x^5}{5}-\frac{1\cdot3\cdot5}{2\cdot4\cdot6}\frac{x^7}{7}+\ldots$

24. By expanding $\sin(x+h)$ show that $\sin(x+h)=\sin x+h\cos x-\frac{h^2}{2}\sin x-\frac{h^3}{6}\cos x+\frac{h^4}{24}\sin x+\ldots$
Hence expand $\sin 61^\circ$ in a series of h. (K.U.T.C., 2000)

25. Using Taylor's theorem, express the polynomial $2x^3+7x^2+x-6$ in powers of $(x-1)$. (Burdwan, 2002)

26. Expand $\log(1+e^x)$ (Kolkata, 2002) (b) Find $\log 1.1$ to 4 places of decimals. Hint: $\log 1.1=\log 2+\log(1-0.1)$

27. Expand $\sin x$ in powers of $\left(x-\frac{\pi}{2}\right)$. Hence find the value of $\sin 91^\circ$ correct to 4 decimal places. (Kolkata, 2003)

28. Prove that $\log(\sin x+\cos x)=\log\sqrt{2}+\left(x-\frac{\pi}{4}\right)\tan\frac{\pi}{4}-\frac{1}{2}\left(x-\frac{\pi}{4}\right)^2\sec^2\frac{\pi}{4}+\ldots$

29. Find the Taylor's series expansion for $\sin x\cos x$ about the point $\pi/3$.

30. Compute to four decimal places the value of $\sin 31^\circ$, by the use of Taylor's series. (Kolkata, Law, 2006)

31. Calculate approximately $\log_{10}40.4$, given $\log 4=0.6021$ (B.Tech, 2005 E)
(a) 1.6064 (Mumbai, 2007)

5.35 INDETERMINATE FORMS

Important: $\lim_{x\to a}[f(x)\cdot\phi(x)]=\lim_{x\to a}f(x)\cdot\lim_{x\to a}\phi(x)$. But when $\lim_{x\to a}f(x)$ and $\lim_{x\to a}\phi(x)$ are both zero, then the question reduces to the indeterminate form $0/0$. This does not imply that $\lim_{x\to a}[f(x)/\phi(x)]$ is zero, or one or it does not exist. In fact, in many cases, it has a finite value. We shall now study the methods of evaluating the limits in such and similar other cases.

(1) Form $0/0$. If $f(a)=\phi(a)=0$, then

$$\lim_{x\to a}\frac{f(x)}{\phi(x)}=\lim_{x\to a}\frac{f'(x)}{\phi'(x)}$$

By Taylor's series,

$$\lim_{x\to a}\frac{f(x)}{\phi(x)}=\lim_{x\to a}\frac{f(a+x-a)=f(a)+(x-a)f'(a)+\frac{1}{2!}(x-a)^2f''(a)+\ldots}{\phi(a)+(x-a)\phi'(a)+\frac{1}{2!}(x-a)^2\phi''(a)+\ldots}$$

$$= \lim_{x \to a} \frac{f'(a) + \frac{1}{2}(x-a)f''(a) + \dots}{g'(a) + \frac{1}{2}(x-a)g''(a) + \dots}$$

$$= \frac{f'(a)}{g'(a)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} \quad \dots (1)$$

This is known as L'Hospital's rule

In general, if

$$f(a) = f'(a) = f''(a) = \dots = f^{n-1}(a) = 0, \text{ but } f^n(a) \neq 0,$$

and $$g(a) = g'(a) = g''(a) = \dots = g^{n-1}(a) = 0, \text{ but } g^n(a) \neq 0,$$

then from (1),

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{f^n(a)}{g^n(a)} = \lim_{x \to a} \frac{f^n(x)}{g^n(x)}$$

Note to evaluate limits of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$

Differentiate the numerator and denominator separately as many times as would be necessary to arrive at a determinate form.

Example 1.20. Evaluate (i) $\lim_{x \to 0} \frac{e^x - \log(1+x)}{x^2}$ (ii) $\lim_{x \to 1} \frac{x^n - a}{x - 1 - \log x}$

Solution. (i) $\lim_{x \to 0} \frac{e^x - \log(1+x)}{x^2}$ $\left[\text{form } \frac{0}{0}\right]$

$$= \lim_{x \to 0} \frac{2e^x + e^x \cdot 1 - 1/(1+x)}{2x} \quad \left[\text{form } \frac{0}{0}\right]$$

$$= \lim_{x \to 0} \frac{e^x + e^x + e^x/(1+x)^2}{2} = \frac{0+1+1+1}{2} = 1\frac{1}{2}.$$

(ii) $\lim_{x \to 1} \frac{x^n - a}{x - 1 - \log x}$ $\left[\text{form } \frac{0}{0}\right]$

$$= \lim_{x \to 1} \frac{n x^{n-1}(1) - 0}{1 - 0 - 1/x}$$

$$= \lim_{x \to 1} \frac{x^n \cdot 1 + \log x^n \cdot 1}{1 - 1/x} \quad \left[\text{form } \frac{0}{0}\right]$$

Let $y = x^n$ so that

$\log y = n \log x$

$\therefore \frac{1}{y}\frac{dy}{dx} = n \cdot \frac{1}{x} + 1 \cdot \log x$

or $\frac{d}{dx}(x^n) = x^n \cdot 1 + \log x^n$ ... (2)

$$= \lim_{x \to 1} \frac{n x^{n-1}(x) - 1 + \log x^n + x^n/x - 0}{1/x^2}$$

$$= \lim_{x \to 1} \frac{x^n \cdot 1 - \log x^n + x^n \cdot 1/x}{x^{-2}} \quad \text{[By (2)]}$$

$$= \frac{1 \cdot 1 + 1^n + 1 \cdot 1}{1} = 2.$$

Example 1.21. Find the values of a and b such that $\lim_{x \to 0} \frac{x(1 + a\cos x) - b\sin x}{x^3} = 1$. [WBUT 2010, 2015]

Solution: $\lim_{x\to 0}\frac{a\tan x+b(1-\cos x)+c\sin x}{x^3}$ $\left[\text{form }\frac{0}{0}\right]$

$=\lim_{x\to 0}\frac{a+b\cos x-bx\sin x-c\cos x}{3x^2}$ ...(1)

As the denominator is 0 for $x=0$, (1) will tend to a finite limit if and only if the numerator also becomes 0 for $x=0$. This requires $a+b-c=0$ ...(2)

With this condition, (1) assumes the form $\frac{0}{0}$.

$\therefore \quad (1)=\lim_{x\to 0}\frac{-b\sin x-b\sin x-bx\cos x+a\sec^2 x+c\sin x}{6x^2}$

$=\lim_{x\to 0}\frac{2a-2b\sin x-bx\cos x}{6x^2}$ $\left[\text{form }\frac{0}{0}\right]$

$=\lim_{x\to 0}\frac{2a-2b\cos x-b\cos x+bx\sin x}{12x}$ ...(3)

$=\frac{2a-2b-b}{0}=\frac{2a-3b}{0}=1$ (Given)

$\therefore \quad 2a-3b=0$ i.e., $2a=3b$.

Now (3) is: $=\lim_{x\to 0}\frac{3b\cos x-b\cos x-bx\sin x}{12x}$

$=\lim_{x\to 0}\frac{b\sin x}{12x}=\frac{b}{12}\lim_{x\to 0}\left(\frac{\sin x}{x}\right)=\frac{b}{12}=1$

(2), $b=12$, and $\therefore \quad a=18$.

Hence (2) is $c=12$.

(2) Remarks. It must be shown that L'Hospital's rule can also be applied to the ratio of the numerator and denominator separately as many times as would be necessary.

Example 4.224. *Evaluate* $\lim_{x\to\pi/2}\frac{\log x}{\cot x}$.

Solution. $\lim_{x\to\pi/2}\frac{\log x}{\cot x}=\lim_{x\to\pi/2}\frac{1/x}{-\operatorname{cosec}^2 x}=-\lim_{x\to\pi/2}\frac{\sin^2 x}{x}$ $\left[\text{form }\frac{0}{0}\right]$

$=-\lim_{x\to\pi/2}\frac{2\sin x\cos x}{1}=0$

Obs. The all known series and standard limits, in many cases, is useful for finding many exercises in the expansions of known functions and standard limits for evaluating the indeterminate forms. For this purpose, remember the series of $\sin x$, $\cos x$, $\tan x$, $\log(1+x)$, e^x and the following limits:

$\lim_{x\to 0}\frac{\sin x}{x}=1$, $\lim_{x\to 0}\frac{a^x-1}{x}=\log a$, $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x=e$

Example 4.225. *Evaluate* $\lim_{x\to 0}\frac{e^x\sin x-x-x^2}{x^3+x\log(1-x)}$.

Solution. Using the expansions of e^x, $\sin x$ and $\log(1-x)$, we get

$\lim_{x\to 0}\frac{e^x\sin x-x-x^2}{x^3+x\log(1-x)}$

$=\lim_{x\to 0}\frac{\left(1+x+\frac{1}{2!}x^2+\frac{1}{3!}x^3+\ldots\right)\left(x-\frac{1}{3!}x^3+\ldots\right)-x-x^2}{x^3+x\left(-x-\frac{1}{2}x^2-\frac{1}{3}x^3-\ldots\right)}$

$$= \lim_{x\to 0}\frac{\left(x+x^2-\frac{1}{2}x^2-0+x^3+\ldots\right)-x-x^2}{x^3-\left(x^3-\frac{3}{2}x^4+\frac{1}{2}x^5+\ldots\right)} = \lim_{x\to 0}\frac{\frac{1}{2}x^3-0\cdot x^4+\ldots}{\frac{3}{2}x^4-\frac{1}{2}x^5+\ldots} = \lim_{x\to 0}\frac{\frac{1}{2}+\ldots}{\frac{3}{2}-\frac{1}{2}x+\ldots} = \frac{1}{3}$$

Example 4.80 Evaluate $\lim_{x\to 0}\dfrac{(1+x)^{1/x}-e}{x}$.

Solution Let $y=(1+x)^{1/x}$

or

$$\ln y = \frac{1}{x}\ln(1+x) = \frac{1}{x}\left(x-\frac{x^2}{2}+\frac{x^3}{3}-\ldots\right) = 1-\frac{x}{2}+\frac{x^2}{3}-\ldots$$

$$y = e^{1-\frac{x}{2}+\frac{x^2}{3}-\ldots} = e\cdot e^{-\frac{x}{2}+\frac{x^2}{3}-\ldots}$$

$$= e\left[1+\left(-\frac{1}{2}x+\frac{1}{3}x^2-\ldots\right)+\frac{1}{2!}\left(-\frac{1}{2}x+\frac{1}{3}x^2-\ldots\right)^2+\ldots\right] = e\left[1-\frac{x}{2}+\frac{11}{24}x^2+\ldots\right]$$

$$\lim_{x\to 0}\frac{(1+x)^{1/x}-e}{x} = \lim_{x\to 0}\frac{e\left[1-\frac{x}{2}+\frac{11}{24}x^2+\ldots\right]-e}{x}$$

$$= \lim_{x\to 0}\frac{e\left(-\frac{1}{2}x+\frac{11}{24}x^2+\ldots\right)}{x} = \lim_{x\to 0}e\left(-\frac{1}{2}+\frac{11}{24}x+\ldots\right) = -\frac{e}{2}.$$

PROBLEMS 4.66

Evaluate each of the following limits:

1. $\lim_{x\to 0}\dfrac{e^x-e^{-x}}{x}$ (N.T.U., 2006)
2. $\lim_{x\to 0}\dfrac{x\ln(1+x)-\ln(1+x)}{x^2\ln(1+x)}$ (U.T.U., 2006)
3. $\lim_{\theta\to 0}\dfrac{\theta-\sin\theta}{\theta\ln(1-\cos\theta)}$
4. $\lim_{x\to 0}\dfrac{e^{\sin x}-1}{\ln(1+x)\sin x}$
5. $\lim_{x\to 0}\dfrac{\cos x-\cos 2x}{x^2}$
6. $\lim_{x\to 0}\dfrac{\ln(\cos x)^2}{x^2}$
7. $\lim_{x\to 0}\dfrac{x\sin x-\ln(1+x)}{x^2}$
8. $\lim_{x\to 0}\dfrac{\ln\sec x-\frac{1}{2}x^2}{x^4}$
9. $\lim_{x\to 0}\dfrac{x^2-\ln(1+x)}{\cos 2x-\cos x}$
10. $\lim_{x\to 0}\dfrac{\sin x-\ln(1+x)-1+x}{\sin^2 x}$
11. $\lim_{x\to 0}\dfrac{e^x+2\sin x-x^2-4x}{x^3}$
12. $\lim_{x\to 0}\dfrac{\ln(\sec x+\tan x)}{\ln(e^x-e^{-x})}$
13. $\lim_{x\to 0}\dfrac{\sin x-\sin x\cos x}{x^3}$
14. $\lim_{x\to 0}\dfrac{\ln(\cos x)}{x\sin x}$
15. $\lim_{x\to 0}\dfrac{e^x+e^{-x}-2}{x\ln(1+x)}$
16. $\lim_{x\to 0}\dfrac{\sin x\ln(1+x)}{\ln(1+\sin x)}$
17. $\lim_{x\to 0}\dfrac{e^x+\sin x-1}{\ln(1+2x)}$
18. $\lim_{x\to 0}\dfrac{(1+x)^{1/x}-e+\frac{1}{2}ex}{x^2}$
19. If $\lim_{x\to 0}\dfrac{\sin 2x+a\sin x}{x^3}$ is finite, find the value of a and the limit. (Nagpur, 2005)
20. Show that $\lim_{x\to 0}\dfrac{a\sin x+\sin 2x}{x^3}=\dfrac{5}{2}$ (Mumbai, 2004)
21. Find a, b, c so that $\lim_{x\to 0}\dfrac{x^3+b\sin x+c\ln(1-x)}{ax^5}=1$. (Mumbai, 2003)

(3) Forms reducible to 0/0 form. Each of the following indeterminate forms can be easily reduced to the form 0/0 or ∞/∞ by suitable transformation and then the limits can be found as usual.

I. Form $0\times\infty$. If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} \phi(x) = \infty$, then

$\lim_{x\to a} f(x)\,\phi(x)$ is said to assume the form $0\times\infty$.

To evaluate this limit, we write

$$f(x)\,\phi(x) = \frac{f(x)}{1/\phi(x)} \text{ to take the form } \frac{0}{0},$$

$$= \phi(x)\,/\,\frac{1}{f(x)} \text{ to take the form } \frac{\infty}{\infty}.$$

Example 1.31. Evaluate $\lim_{x\to 0} \cot x \log x$. (V.T.U., 2009)

Solution.
$$\lim_{x\to 0} \cot x \log x = \lim_{x\to 0} \left[\frac{\log x}{\tan x}\right] \quad \left[\text{Form } \frac{\infty}{\infty}\right]$$

$$= \lim_{x\to 0} \left[\frac{1/x}{\sec^2 x}\right] = -\lim_{x\to 0} \left[\frac{\cos^2 x}{x}\right] \quad \left[\text{Form } \frac{0}{0}\right]$$

$$= -\lim_{x\to 0} \frac{2\cos x \sin x}{1} = 0.$$

II. Form $\infty-\infty$. If $\lim_{x\to a} f(x) = \infty = \lim_{x\to a} \phi(x)$, then $\lim_{x\to a} [f(x) - \phi(x)]$ is said to assume the form $\infty-\infty$.

It can be reduced to the form $0/0$ by writing

$$f(x) - \phi(x) = \left[\frac{1}{\phi(x)} - \frac{1}{f(x)}\right] \Big/ \frac{1}{f(x)\,\phi(x)}$$

Example 1.32. Evaluate $\lim_{x\to 0} \left[\frac{1}{\sin x} - \frac{1}{x}\right]$.

Solution.
$$\lim_{x\to 0} \left(\frac{1}{\sin x} - \frac{1}{x}\right) = \lim_{x\to 0} \frac{x - \sin x}{x \sin x} = \lim_{x\to 0} \frac{1 - \cos x}{x\cos x + \sin x} \quad \left[\text{Form } \frac{0}{0}\right]$$

$$= \lim_{x\to 0} \frac{\sin x}{x(-\sin x) + \cos x + \cos x} = \frac{0}{0 + 1 + 1} = 0$$

III. Forms $0^0, 1^\infty, \infty^0$. If $y = \lim_{x\to a} f(x)^{\phi(x)}$ assumes one of these forms, then $\log y = \lim_{x\to a} \phi(x) \log f(x)$ takes the form $0\times\infty$, which can be evaluated by the method given in I above. If $\log y = A$, then $y = e^A$.

Example 1.33. Evaluate (i) $\lim_{x\to 0} (\cos x)^{1/x^2}$ (ii) $\lim_{x\to 0} \left[\frac{a^x + b^x + c^x}{x}\right]^{1/x}$ (V.T.U., 2011)

(iii) $\lim_{x\to 0} \left[\frac{\tan x}{x}\right]^{1/x^2}$

Solution. (i) Let
$$y = \lim_{x\to 0} (\cos x)^{1/x^2},$$

$$\therefore \quad \log y = \lim_{x\to 0} \frac{1}{x^2} \log \cos x = \lim_{x\to 0} \frac{\log \cos x}{x^2} \quad \left[\text{Form } \frac{0}{0}\right]$$

$$= \lim_{x\to 0} \frac{(1/\cos x)(-\sin x)}{2x} = -\frac{1}{2} \lim_{x\to 0} \frac{\tan x}{x} = -\frac{1}{2}$$

Hence
$$y = e^{-1/2}.$$

(c) Let $y = \lim_{x\to 0}\left(\frac{a^x + b^x + c^x}{3}\right)^{1/x}$

so that $\log y = \lim_{x\to 0}\frac{\log(a^x + b^x + c^x) - \log 3}{x}$ form $\left[\frac{0}{0}\right]$

$= \lim_{x\to 0}\frac{a^x \log a + b^x \log b + c^x \log c}{a^x + b^x + c^x}$ [Using L'Hospital's Rule]

$= \frac{1}{3}(\log a + \log b + \log c) = \frac{1}{3}\log(abc) = \log(abc)^{1/3}$

$\therefore \quad y = (abc)^{1/3}$

(d) $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{1/x^2} = \lim_{x\to 0}\left(\frac{x + \frac{x^3}{3} + \frac{2}{15}x^5 + \ldots}{x}\right)^{1/x^2}$ where $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \ldots$

$= \lim_{x\to 0}\left(1 + \frac{x^2}{3} + \frac{2}{15}x^4 + \ldots\right)^{1/x^2}$

$= \lim_{x\to 0}(1 + x^2)^{1/x^2}$ $\left[= \lim_{x\to 0}(1 + x)^{1/x} = e\right]$

$= \lim_{x\to 0}\{(1 + x^2)^{1/x^2}\}^1 = \lim_{x\to 0} e^1 = e$

EXERCISE 9.2

Evaluate the following limits:

1. $\lim_{x\to 0}\left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right)$

2. $\lim_{x\to 1}(2x\tan\pi x + \sec\pi x)$ (U.P.T.U., 2006)

3. $\lim_{x\to 0}\left(\frac{1}{x^2} - \cot^2 x\right)$

4. $\lim_{x\to 2}\frac{2^x + x^2}{x^2}$ (U.P.T.U., 2007)

5. $\lim_{x\to 0}(1 + \sin x)^{\cot x}$

6. $\lim_{x\to\pi/2}(\sin x)^{\tan 2x}$ (U.P.T.U., 2004)

7. $\lim_{x\to 0}(\cos x)^{\cot^2 x}$

8. $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{1/x^2}$ (U.P.T.U., 2005)

9. $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{\sin x}{x - \sin x}}$

10. $\lim_{x\to 0}\left(\frac{2 + x}{2 - x}\right)^{1/x}$ (U.P.T.U., 2010, Uttarakhand, 2009)

11. $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{1}{1 - \cos x}}$ (Uttarakhand, 2009)

12. $\lim_{x\to 1}\left(\frac{1}{\log x} - \frac{x}{x - 1}\right)$ (Uttarakhand, 2007)

4.5 TANGENTS AND NORMALS – CARTESIAN CURVES

(1) Equation of the tangent at the point (x, y) of the curve $y = f(x)$ is

$$Y - y = \frac{dy}{dx}(X - x)$$

The equation of any line through $P(x, y)$ is

$$Y - y = m(X - x)$$

where X, Y are the current coordinates of any point on the line (Fig. 4.3).

If this line is the tangent PT, then

m = tangent of ψ = $\frac{dy}{dx}$

Hence the equation of the tangent at (x, y) is

$$Y - y = \frac{dy}{dx}(X - x) \qquad \ldots(1)$$

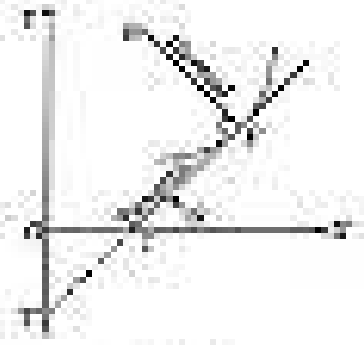

Fig. 4.3

Cor. Intercepts. Putting $Y = 0$ in (1)

$$-y = \frac{dy}{dx}(X - x) \quad \text{or} \quad X = x - y\frac{dx}{dy}$$

$\therefore$ Intercept which the tangent cuts off from x-axis is $OT = x - y\frac{dx}{dy}$

Similarly putting $X = 0$ in (1), we see that

the intercept which the tangent cuts off from the y-axis

$$= OT' = y - x\frac{dy}{dx}$$

(2) Equation of the normal at the point (x, y) of the curve $y = f(x)$ is

$$Y - y = -\frac{dx}{dy}(X - x)$$

A normal to the curve $y = f(x)$ at $P(x, y)$ is the line through P perpendicular to the tangent line at P.

$\therefore$ Its equation is $Y - y = m'(X - x)$

where $m' \cdot (\text{slope of tangent}) = -1 \quad \text{or} \quad m' = -1/\frac{dy}{dx} = -\frac{dx}{dy}$

Hence the equation of the normal at (x, y) is $Y - y = -\frac{dx}{dy}(X - x)$.

Example 4.23. *Find the equation of the tangent at any point (x, y) to the curve $x^{2/3} + y^{2/3} = a^{2/3}$. Also find the portion of the tangent intercepted between the axes is of constant length.*

Solution. Equation of the curve is $x^{2/3} + y^{2/3} = a^{2/3}$. ...(i)

Differentiating (i) w.r.t. x,

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} = 0$$

$\therefore$ Slope of the tangent at $(x, y) = \frac{dy}{dx} = -\left(\frac{x}{y}\right)^{1/3}$

$\therefore$ Equation of the tangent at (x, y) is

$$Y - y = -(y/x)^{1/3}(X - x) \qquad \ldots(ii)$$

Put $Y = 0$ in (ii). Then $X = x + x^{1/3}y^{2/3}$

i.e., Intercept on x-axis $= x^{1/3}(x^{2/3} + y^{2/3}) = x^{1/3}a^{2/3}$ [By (i)]

Put $X = 0$ in (ii). Then $Y = y + y^{1/3}x^{2/3}$

i.e., Intercept on y-axis $= y^{1/3}(y^{2/3} + x^{2/3}) = y^{1/3}a^{2/3}$ [By (i)]

Thus the portion of the tangent intercepted between the axes

$$= \sqrt{[(\text{Intercept on } x\text{-axis})^2 + (\text{Intercept on } y\text{-axis})^2]}$$

$$= \sqrt{[a^{2/3}x^{2/3} + a^{2/3}y^{2/3}]}$$

$$= \sqrt{a^{2n+2}\sin^{2n}\theta + a^{2n+2}\cos^{2n}\theta} = a^{n+1}\sqrt{\sin^{2n}\theta + \cos^{2n}\theta}$$ [By (i)]

$= a$, which is a constant length

Example 8.35. *Show that the conditions for the line $lx + my = n$ to be a normal to the curve $x^p y^q = k$ is $l^p a^{p-q} = m^p b^{p-q} n^{p+q}$.*

Solution. Equation of the curve is $\frac{x^p}{a^p} + \frac{y^q}{b^q} = 1$ ...(i)

Differentiating (i) w.r.t. x, $\frac{px^{p-1}}{a^p} + \frac{qy^{q-1}}{b^q}\frac{dy}{dx} = 0$

$\therefore$ Slope of the tangent at (x, y) is $\frac{dy}{dx} = -\left(\frac{b}{a}\right)^p \left(\frac{x}{y}\right)^{p-q}$

$\therefore$ Equation of the tangent at (x, y) is

$$Y - y = -\left(\frac{b}{a}\right)^p \left(\frac{x}{y}\right)^{p-q}(X - x)$$

or $\frac{x^{p-1}X}{a^p} + \frac{y^{q-1}Y}{b^q} = \frac{x^p}{a^p} + \frac{y^q}{b^q} = 1$ ...(ii) [By (i)]

If the given line $lx + my = n$ is the tangent, then (ii) must be same as $lX + mY = n$ or $\frac{l}{n}X + \frac{m}{n}Y = 1$ ...(iii)

Comparing coefficients in (ii) and (iii),

$$\frac{x^{p-1}}{a^p} \Big/ \frac{l}{n} = \frac{y^{q-1}}{b^q} \Big/ \frac{m}{n} = \frac{1}{n}$$

or $\left(\frac{x}{a}\right)^{p-1} = \frac{a}{n}\frac{l}{n}, \left(\frac{y}{b}\right)^{q-1} = \frac{b}{n}\frac{m}{n}$

or $\left(\frac{a}{n}\frac{l}{n}\right)^{\frac{1}{p-1}} + \left(\frac{b}{n}\frac{m}{n}\right)^{\frac{1}{q-1}} = \left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q = 1$ [By (i)]

which follows the required condition.

Example 8.36. *Find the equation of the normal at any point θ to the curve $x = a\cos\theta + a\theta\sin\theta$, $y = a\sin\theta - a\theta\cos\theta$. Verify that this normal is at a constant distance from the origin and a chord making $\pi/2$ with the normal is constant.*

Solution. We have $\frac{dx}{d\theta} = a(-\sin\theta + \theta\cos\theta + \sin\theta) = a\theta\cos\theta$

$\frac{dy}{d\theta} = a(\theta\sin\theta - \cos\theta + \theta\sin\theta) = a\theta\sin\theta$

$\therefore \frac{dy}{dx} = \frac{dy}{d\theta}\Big/\frac{dx}{d\theta} = \frac{\sin\theta}{\cos\theta}$

$\therefore$ Slope of the normal $= -\frac{\cos\theta}{\sin\theta}$

Hence the equation of the normal at θ is

$$y - a\sin\theta + a\theta\cos\theta = -\frac{\cos\theta}{\sin\theta}[x - a\cos\theta - a\theta\sin\theta]$$

i.e., $y\sin\theta - a\sin^2\theta + a\theta\sin\theta\cos\theta = -x\cos\theta + a\cos^2\theta - a\theta\sin\theta\cos\theta$

i.e., $x\cos\theta + y\sin\theta = a(\cos^2\theta + \sin^2\theta) = a$.

Now the perpendicular distance of this normal from $(0, 0) = a$, which is a constant. Hence it touches a circle of radius a having its centre at $(0, 0)$.

(3) Angle of intersection of two curves is the angle between the tangents to the curves at their point of intersection.

To find the angle, proceed as follows :

(a) Find P, the point of intersection of the curves by solving their equations simultaneously.

(b) Find the values of $\frac{dy}{dx}$ at P for the two curves (say m_1, m_2).

(c) Find θ, using the rule $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$.

When $m_1 m_2 = -1$, $\theta = 90°$ i.e., the curves cut orthogonally.

Example 3.21. Find the angle of intersection of the curves $x^2 = 4y$ ...(1) and $y^2 = 4x$ ...(2)

Solution. We have $x^4 - 16y^2 = 16(4x - 64x)$

or $y(y^3 - 8) = 0$ whence $y = 0$ and $y = 2$.

Substituting these values in (2), $x = 0$ and 4.

$\therefore$ The curves intersect at $(0, 0)$ and $(4, 4)$.

For the curve (1), $dy/dx = x/2$. For the curve (2), $dy/dx = 2/y$.

At $(0, 0)$, slope of tangent to (1) $= m_1 = 0/2 = 0$ and slope of tangent to (2) $= m_2 = 2/0 = \infty$.

Evidently the curves intersect at right angles.

At $(4, 4)$, slope of tangent to (1) $= m_1 = 4/2 = 2$ and slope of tangent to (2) $= m_2 = 2/4 = \frac{1}{2}$.

$\therefore$ Angle of intersection of the curves

$$= \tan^{-1}\left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \tan^{-1}\left|\frac{2 - \frac{1}{2}}{1 + 2 \cdot \frac{1}{2}}\right| = \tan^{-1}\frac{3}{4}.$$

Example 3.22. Show that the condition that the curves $ax^2 + by^2 = 1$ and $a'x^2 + b'y^2 = 1$ should intersect orthogonally is that

$$\frac{1}{a} - \frac{1}{b} = \frac{1}{a'} - \frac{1}{b'}.$$

Solution. Given curves are $ax^2 + by^2 = 1$ ...(1) and $a'x^2 + b'y^2 = 1$ ...(2)

Let $P(h, k)$ be a point of intersection of (1) and (2) so that

$$ah^2 + bk^2 = 1 \quad \text{and} \quad a'h^2 + b'k^2 = 1$$

$$\therefore \quad \frac{h^2}{bc' - b'c} = \frac{k^2}{a'c - ac'} = \frac{1}{ab' - a'b}$$

or $h^2 = (b' - b)/(ab' - a'b)$, $k^2 = (a - a')/(ab' - a'b)$ ...(3)

Differentiating (1) w.r.t. x,

$$2ax + 2by\frac{dy}{dx} = 0 \quad \text{or} \quad dy/dx = -ax/by.$$

Similarly for (2), $dy/dx = -a'x/b'y$.

$\therefore$ m_1 = slope of tangent to (1) at $P = -ah/bk$; m_2 = slope of tangent to (2) at $P = -a'h/b'k$.

For orthogonal intersection, we should have $m_1 m_2 = -1$.

i.e., $\frac{-ah}{bk} \times \frac{-a'h}{b'k} = -1$ i.e., $aa'h^2 + bb'k^2 = 0$

Substituting the values of h^2 and k^2 from (3),

$$\frac{aa'(b' - b)}{ab' - a'b} + \frac{bb'(a - a')}{ab' - a'b} = 0 \quad \text{or} \quad \frac{b' - b}{bb'} + \frac{a - a'}{aa'} = 0$$

i.e., $\frac{1}{b} - \frac{1}{b'} + \frac{1}{a} - \frac{1}{a'} = 0$ which leads to the required condition.

(4) Lengths of tangent, normal, subtangent and subnormal. Let the tangent and the normal at any point $P(x, y)$ of the curve meet the x-axis at T and N respectively. (Fig. 4.6) Draw the ordinate PM. Then PT and PN are called the *lengths of the tangent and the normal* respectively. Also TM and MN are called the *subtangent and subnormal* respectively.

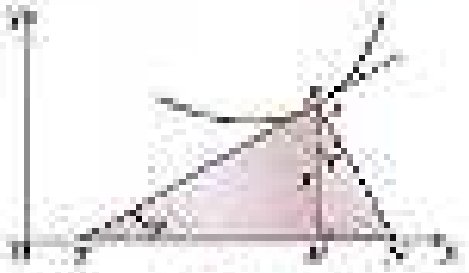

Fig. 4.6

Let $\angle XTP = \psi$ so that $\tan\psi = dy/dx$.

Clearly, $\angle MPN = \psi$.

(1) **Tangent** $= PT = MP \sec\psi = y\sqrt{(1 + \tan^2\psi)} = y\sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$

(2) **Normal** $= PN = MP \sec\psi = y\sqrt{(1 + \tan^2\psi)} = y\sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$

(3) **Subtangent** $= TM = y \cot\psi = y\,\frac{dx}{dy}$

(4) **Subnormal** $= MN = y \tan\psi = y\,\frac{dy}{dx}$.

Example 4.10. *For the curve $x = a(\sin t + \log \tan t/2)$, $y = a\cos t$, prove that the portion of the tangent between the curve and x-axis is constant.*

Also find the subtangent.

Solution. Differentiating with respect to t,

$$\frac{dx}{dt} = a\left(\cos t + \frac{1}{\tan t/2}\sec^2\frac{t}{2}\cdot\frac{1}{2}\right) = a\left(\cos t + \frac{\cos t/2}{2\sin t/2}\cdot\frac{1}{\cos^2 t/2}\right)$$

$$= a\left(\cos t + \frac{1}{\sin t}\right) = \frac{a(1 + \sin^2 t)}{\sin t} \quad \text{and} \quad \frac{dy}{dt} = -a\sin t$$

$$\therefore \quad \frac{dy}{dx} = \frac{dy}{dt}\Big/\frac{dx}{dt} = -a\sin t \cdot \frac{\sin t}{a\cos^2 t} = -\tan t.$$

Thus length of the tangent between the curve and x-axis

$$= y\sqrt{[1 + (dx/dy)^2]} = a\sin t\cdot\sqrt{(1 + \cot^2 t)} = a\sin t\cdot\operatorname{cosec} t = a \text{ which is a constant.}$$

Also subtangent $= y\,\frac{dx}{dy} = a\sin t\cdot(-\cot t) = -a\cos t$

EXERCISE 4.3

1. Find the equations of the tangent and the normal to the curve $y = 2x^2 - 3x - 5$ at the point where it cuts the x-axis.
2. Find the slope of the tangent $y = x^2 + 2x$ to the curve $x^2 + y^2 - 2x = 8$ for all values of x. Find the point of contact. (Nagpur, 2004)
3. Prove that $\frac{x}{a} + \frac{y}{b} = 1$ touches the curve $y = be^{-x/a}$ at the point where the curve crosses the axis of y. (VTU, Belgaum, 2004)
4. If $y = ax + b$ is a tangent to the parabola $y^2 = 4a(x + b)$, prove that $a^2 + b = 0$ and $ab + a^2 = 0$.
5. Prove that the condition for the line $x\cos\alpha + y\sin\alpha = p$ to touch the curve $x^{m+n}y^{m-n} = a^{2m}$ is $p^{2m} = (m + n)^{m+n}(m - n)^{m-n}\cos^{2n}\alpha\sin^{2m}\alpha$.
6. Show that the sum of the intercepts on the axes of any tangent to the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ is constant.
7. If x_1, y_1 be the points of the axes at which x and y intercepted by the tangent at any point (x, y) on the curve $\frac{x^n}{a^n} + \frac{y^n}{b^n} = 1$, then show that $\frac{x_1}{x} + \frac{y_1}{y} = 1$. (Nagpur, 2004)
8. If the tangent at (x_1, y_1) to the curve $x^3 + y^3 = a^3$ meets the curve again in (x_2, y_2), show that $\frac{x_2}{x_1} + \frac{y_2}{y_1} = -1$.

9. Find the normal to the curve $x^{2/3} + y^{2/3} = a^{2/3}$ making a angle with the axis of x. Also find the equation to the curve $x = a\cos^3\theta$, $y = a\sin^3\theta$.

10. Find the angle of intersection of the curves $r = a^{\theta}$ & $r = a^{-\theta}$. (Ans: $\tan^{-1}\frac{2\log a}{1-\log^2 a}$)

11. Show that the parabolas $y^2 = 4ax$ and $2x^2 = cy$ intersect at an angle $\tan^{-1}\frac{3}{4}$.

12. Prove that the curves $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2-k} + \frac{y^2}{b^2-k} = 1$ will cut orthogonally if $a - b = a^2 - b^2$.

13. Show that in the catenary $y = c\cosh\frac{x}{c}$, the radius of curvature is equal to $\frac{y^2}{c}$ and the subnormal is constant. (Bhopal U. 2008 S)

14. Find the angles of the cardioid $r = a(1+\cos\theta)$ at the extremities of the chord for which $\theta = \frac{\pi}{2}$. (Ans: $\frac{\pi}{4}, \frac{3\pi}{4}$)

15. For the curve $x = a\cos^3\theta$, $y = a\sin^3\theta$, show that the portion of the tangent intercepted between the point of contact and the curve is constant. Also find the length of the subnormal.

4.7 POLAR CURVES

(1) Angle between a radius vector and tangent. If ϕ is the angle between the radius vector and the tangent at any point P of the curve $r = f(\theta)$, then $\tan\phi = r\frac{d\theta}{dr}$.

Let $P(r, \theta)$ and $Q(r + \delta r, \theta + \delta\theta)$ be two neighbouring points on the curve (Fig. 4.6). Join PQ and draw $PM \perp OQ$. Then from the rt. angled $\triangle OMP$, $MP = r\sin\delta\theta$, $OM = r\cos\delta\theta$.

$$\therefore \quad MQ = OQ - OM = r + \delta r - r\cos\delta\theta = \delta r + r(1-\cos\delta\theta) = \delta r + 2r\sin^2\delta\theta/2.$$

If $\angle MQP = \alpha$, then $\tan\alpha = \frac{MP}{MQ} = \frac{r\sin\delta\theta}{\delta r + 2r\sin^2\delta\theta/2}$

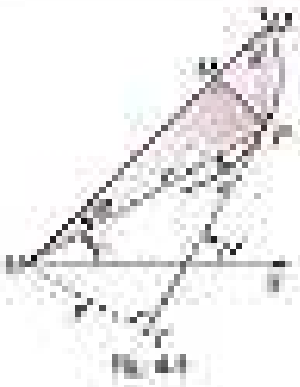
Fig. 4.6

In the limit as $Q \to P$ (i.e., $\delta\theta \to 0$), the chord PQ turns about P and becomes the tangent at P and $\alpha \to \phi$.

$$\therefore \quad \tan\phi = \lim_{\alpha\to\phi}\tan\alpha = \lim_{\delta\theta\to 0}\frac{r\sin\delta\theta}{\delta r + 2r\sin^2\delta\theta/2}$$

$$= \lim_{\delta\theta\to 0}\frac{r\sin\delta\theta/\delta\theta}{\delta r/\delta\theta + r\sin\delta\theta/2\cdot\sin\delta\theta/2\cdot\delta\theta/2} = \frac{r\cdot 1}{\frac{dr}{d\theta}\cdot 0\cdot 2} = r\frac{d\theta}{dr}$$

Cor. Angle of intersection of two curves. If ϕ_1, ϕ_2 be the angles between the radius vector and the tangents to the two curves at their point of intersection, then the angle of intersection of these curves is $\phi_1 - \phi_2$.

(2) Length of the perpendicular from pole on the tangent. If p be the perpendicular from the pole on the tangent, then

(i) $p = r\sin\phi$ (ii) $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4}\left(\frac{dr}{d\theta}\right)^2$

From the rt. $\angle$d $\triangle OTP$, $p = r\sin\phi$

$$\therefore \quad \frac{1}{p^2} = \frac{1}{r^2}\cos ec^2\phi = \frac{1}{r^2}(1+\cot^2\phi)$$

$$= \frac{1}{r^2}\left[1 + \frac{1}{r^2}\left(\frac{dr}{d\theta}\right)^2\right] = \frac{1}{r^2} + \frac{1}{r^4}\left[\frac{dr}{d\theta}\right]^2 \qquad \text{(By (1))}$$

(3) Polar subtangent and subnormal. Let the tangent and the normal at any point $P(r, \theta)$ of a curve cut the line through the pole perpendicular to the radius vector OP in T and N respectively (Fig. 4.6). Then OT is called the *polar subtangent* and ON the *polar subnormal*.

Let $\angle OTP = \phi$ so that $\tan\phi = r\frac{d\theta}{dr}$

Clearly, $\angle PON = \theta$

$\therefore$ (i) Polar subtangent

$$= OT = r\tan\phi = r \cdot r\frac{d\theta}{dr} = r^2\frac{d\theta}{dr}$$

(ii) Polar subnormal

$$= ON = r\cot\phi = r \cdot \frac{1}{r}\frac{dr}{d\theta} = \frac{dr}{d\theta}$$

Fig. 4.1

Example 4.10. *Find the angle ϕ and the polar subtangent and polar subnormal of the curve $r = a(1+\cos\theta)$ at the point where*

(i) $\theta = \frac{\pi}{2}$ (ii) $\theta = \frac{2\pi}{3}$

(iii) *polar subtangent* $= 2a\cos^2\frac{\theta}{2}\tan\frac{\theta}{2}$

Solution. We have $\frac{dr}{d\theta} = -a\sin\theta$

$\therefore$ $\tan\phi = r\frac{d\theta}{dr} = a(1+\cos\theta)\cdot\frac{1}{-a\sin\theta}$

$= -2\cos^2\frac{\theta}{2} \div -2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \cot\frac{\theta}{2}$. Thus $\phi = \frac{\pi}{2}$ ...(1)

Also $p = r\sin\phi = a(1+\cos\theta)\sin\frac{\pi}{2} = a \cdot 2\cos^2\frac{\theta}{2} \cdot \sin\frac{\pi}{2}$

$= 2a\cos^2\frac{\theta}{2}$ ...(2)

Polar subtangent $= r^2\frac{d\theta}{dr} = [a(1+\cos\theta)]^2 \div a\sin\theta$

$= 4a\cos^4\frac{\theta}{2} \div 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2a\cos^2\frac{\theta}{2}\tan\frac{\theta}{2}$. ...(3)

Example 4.11. *Find the angle of intersection of the curves $r = a(1+\cos\theta)$ and $r = a(1-\cos\theta)$.*

Solution. To find the point of intersection of the curves $r = a(1+\cos\theta)$ and

$r = 2a\sin^2\frac{\theta}{2}$ ...(1). On eliminating r,

we get $2\sin^2\frac{\theta}{2} = 1+\cos\theta$ i.e., $1-\cos\theta = 1+\cos\theta$, i.e., $\cos\theta = 0$.

For θ, $\frac{dr}{d\theta} = a\cos\theta - a\sin\theta$

$\therefore$ $\tan\phi = r\frac{d\theta}{dr} = \frac{a(1+\cos\theta)}{a\cos\theta - a\sin\theta}$ which $\to -\infty$ as $\theta = \frac{\pi}{2}$. Thus $\phi = \frac{\pi}{2}$.

For (2), $dr/d\theta = 2a\sin\theta$ $\therefore$ $\tan\phi' = r\frac{d\theta}{dr} = \frac{2a\sin^2\frac{\theta}{2}}{2a\sin\theta} = \frac{1}{2}\tan\frac{\theta}{2}$ at $\theta = \frac{\pi}{2}$. Thus $\phi' = \frac{\pi}{4}$.

Hence the angle of intersection of the curves is $\phi - \phi' = \frac{\pi}{4}$.

EXERCISE 4.3

1. For a curve in Cartesian form, show that $\tan\phi = \frac{y(1-x)}{x+y^2}$.
2. Prove that in the equiangular spiral $r = ae^{\theta\cot\alpha}$, the tangent is inclined at a constant angle to the radius vector.
3. Show that the tangent to the cardioid $r = a(1+\cos\theta)$ at the points $\theta = \pm\frac{\pi}{2}$ and $\theta = \pm\frac{2\pi}{3}$ are respectively parallel and perpendicular to the initial line. (V.T.U., 2008)
4. Prove that, in the parabola $l/r = 1+\cos\theta$,

(i) $\phi + \frac{\theta}{2} = \frac{\pi}{2}$ (ii) $r = a\sec^2\frac{\theta}{2}$, and (iii) the polar subtangent $= 2a\tan\theta$.

5. Show that the angle between the tangents at any point P on the line $y = x$ to the curve $x = a(\theta + \sin\theta)$, $y = a(1+\cos\theta)$ is $\frac{\theta}{2}$.

6. Show that in the curve $r = ak$, the polar subnormal is constant and in the curve $r\theta = a$ the polar subtangent is constant.
7. Find the angle of intersection of the curves
   (i) $r = 2a\sin\theta$, $r = 2a\cos\theta$ (Gujarat, 2003)
   (ii) $r = a(1 - \cos\theta)$ and $r = a(1 - \sin\theta)$ (U.P.T.U., 2008-09)
8. Prove that the curves $r = a(1 + \cos\theta)$ and $r(1 - \cos\theta) = a$ cut orthogonally at right angles. (U.P.T.U., 2012-13)
9. Show that the curves $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$ are orthogonal and cut at right angles orthogonally.
10. Show that the angle of intersection of the curves $r = a\log\theta$ and $r = a\log(\sin\theta)$ is $\tan^{-1}(\log(1 + \sqrt{2}))$. (U.P.T.U., 2005)

4.8 PEDAL EQUATION

If r be the radius vector of any point on the curve and p, the length of the perpendicular from the pole on the tangent at that point, then the relation between p and r is called pedal equation of the curve.

Given the cartesian or polar equation of a curve, we can derive its pedal equation. The method is explained through the following examples.

Example 4.42. Find the pedal equation of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (U.P.T.U., 2010)

Solution. Equation of the tangent at (x, y) is $\frac{Xx}{a^2} + \frac{Yy}{b^2} = 1$ ...(i)

p, length of $\perp$ from $(0, 0)$ on (i) $= \frac{-1}{\sqrt{\left[\frac{x^2}{a^4} + \frac{y^2}{b^4}\right]}}$

or $$\frac{1}{p^2} = \frac{x^2}{a^4} + \frac{y^2}{b^4}$$ ...(ii)

Also $$r^2 = x^2 + y^2$$ ...(iii)

Substituting the value of y^2 from (iii) in (ii),

$$\frac{1}{p^2} = \frac{x^2}{a^4} + \frac{r^2 - x^2}{b^2}$$

Then from (i), $$\frac{x^2}{b^2} = \frac{a^2 - p^2}{a^2 - b^2}$$

Now substituting these values of x^2/a^4 and y^2/b^4 in (ii),

$$\frac{1}{p^2} = \frac{1}{a^2}\left[\frac{a^2 - b^2}{a^2 - b^2}\right] + \frac{1}{b^2}\left[\frac{a^2 - p^2}{a^2 - b^2}\right]$$

or $$\frac{a^2 b^2}{p^2} = \frac{a^2 b^2 - b^4 + a^4 - a^2 p^2}{a^2 - b^2} \Rightarrow a^2 + b^2 - p^2$$

Ex. $\Rightarrow$ required pedal equation

Example 4.43. Of the curves
(i) $2a/r = 1 + \cos\theta$ (ii) $r^2 = a^2\cos 2\theta$ (U.P.T.U., 2005)

Solution.
Take,

$\log 2a = -n$,

Differentiating both sides with respect to θ, we get

$$\frac{1}{r}\frac{dr}{d\theta} = \frac{1}{1-\cos\theta}\sin\theta = \cot\frac{\theta}{2}$$

$$\therefore \quad \tan\phi = r\frac{d\theta}{dr} = \tan\left(90^\circ - \frac{\theta}{2}\right), \quad \phi = 90^\circ - \frac{\theta}{2}$$

Also $\quad p = r\sin\phi = a(1+\cos\theta)\sin\left(90^\circ - \frac{\theta}{2}\right), \quad p = a\sin\theta\cos\frac{\theta}{2}$

or $$p^2 = a^2\sin^2\theta\cos^2\frac{\theta}{2} = a^2\left(\frac{1-\cos\theta}{2}\right)(1+\cos\theta) = a^2(1-\cos^2\theta)$$

Hence $p^2 = ay$, which is the required polar equation.

Q 10 From the given equation, $ar^{n+1}\frac{d\theta}{dr} = -r^n\sin n\theta$

so that $$\tan\phi = r\frac{d\theta}{dr} = r\frac{ar^{n+1}}{-r^n\sin n\theta} = -\cot n\theta = \tan\left(\frac{\pi}{2} + n\theta\right)$$

i.e., $$\phi = n\theta + \frac{\pi}{2}$$

$$\therefore \quad p = r\sin\phi = r\sin\left(\frac{\pi}{2} + n\theta\right) = r\cos n\theta = r\cdot\frac{1}{2}(a/r^n) = a^n/2r^{n-1}$$

Hence $p/r^{n-1} = a^n/2$, which is the required polar equation.

4.10 DERIVATIVE OF ARC

(i) *For Cartesian curve* $y = f(x)$, we have

$$\frac{ds}{dx} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$$

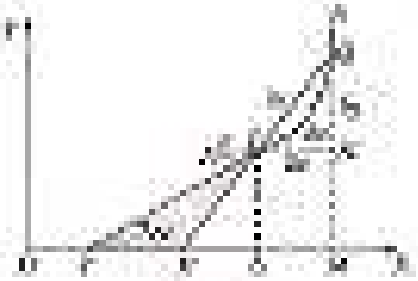

Fig. 4.9

Let $P(x, y)$, $Q(x + \delta x, y + \delta y)$ be two neighbouring points on the curve AB (Fig. 4.9). Let arc $AP = s$, arc $PQ = \delta s$ and chord $PQ = \delta c$.

Draw PL, QM $\perp$ s to the x-axis and $PN \perp QM$.

$\therefore$ From the rt. $\angle$d ΔPNQ,

$$PQ^2 = PN^2 + NQ^2$$

i.e., $$\delta c^2 = \delta x^2 + \delta y^2$$

or $$\left(\frac{\delta c}{\delta x}\right)^2 = 1 + \left(\frac{\delta y}{\delta x}\right)^2$$

$$\therefore \quad \left(\frac{\delta s}{\delta x}\right)^2 = \left(\frac{\delta s}{\delta c}\cdot\frac{\delta c}{\delta x}\right)^2$$

$$= \left(\frac{\delta s}{\delta c}\right)^2\left[1 + \left(\frac{\delta y}{\delta x}\right)^2\right]$$

Taking limits as $Q \to P$ i.e., $\delta x \to 0$,

$$\left(\frac{ds}{dx}\right)^2 = 1\cdot\left[1 + \left(\frac{dy}{dx}\right)^2\right]$$

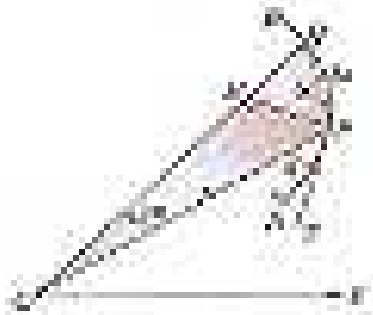

Fig. 4.10

$$\therefore \quad \frac{ds}{dx} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$$

If s increases with x as in Fig. 4.9, ds/dx is positive.

Thus $$\frac{ds}{dx} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$$, taking positive sign before the radical. —(1)

Cor. 1. If the equation of the curve is $x = f(y)$, then

$$\frac{ds}{dy} = \frac{ds}{dx}\cdot\frac{dx}{dy} = \sqrt{\left[1 + \left(\frac{dx}{dy}\right)^2\right]}\cdot\frac{dx}{dy}$$

$$\therefore \quad \frac{ds}{dy} = \sqrt{\left[1 + \left(\frac{dx}{dy}\right)^2\right]} \qquad \ldots(2)$$

Cor. 2. If the equation of the curve is in the parametric form $x = f(t)$, $y = g(t)$, then

$$\frac{ds}{dt} = \frac{ds}{dx} \cdot \frac{dx}{dt} = \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]} \cdot \frac{dx}{dt}$$

$$= \sqrt{\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dx} \cdot \frac{dx}{dt}\right)^2\right]}$$

$$\therefore \quad \frac{ds}{dt} = \sqrt{\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2\right]} \qquad \ldots(3)$$

Cor. 3. We have $\frac{dx}{dy} = \sqrt{\left[1 + \left(\frac{dx}{dy}\right)^2\right]} = \sqrt{[1 + \tan^2\psi]} = \sec\psi$

$$\therefore \quad \cos\psi = \frac{dy}{ds}. \qquad \ldots(4)$$

Also $\sin\psi = \tan\psi \cos\psi = \frac{dx}{dy} \cdot \frac{dy}{ds}$

$$\therefore \quad \sin\psi = \frac{dx}{ds} \qquad \ldots(5)$$

(2) For the curve $r = f(\theta)$, we have $\frac{ds}{d\theta} = \sqrt{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]}$

Let $P(r, \theta)$, $Q(r + \delta r, \theta + \delta\theta)$ be two neighbouring points on the curve AB (Fig. 6.6). Let arc $AP = s$, arc $AQ = s + \delta s$ and chord $PQ = \delta c$.

Draw $PM \perp OQ$, then

$$PM = r \sin\delta\theta \text{ and } MQ = OQ - OM = r + \delta r - r\cos\delta\theta = \delta r + 2r\sin^2\tfrac{1}{2}\delta\theta$$

From the rt. $\angle d$ ΔPMQ,

$$PQ^2 = PM^2 + MQ^2$$

or $$\delta c^2 = r^2 \sin^2\delta\theta + (\delta r + 2r\sin^2\tfrac{1}{2}\delta\theta)^2$$

$$\therefore \quad \left(\frac{\delta c}{\delta\theta}\right)^2 = \left(\frac{\delta c}{\delta s} \cdot \frac{\delta s}{\delta\theta}\right)^2 + \left(\frac{\delta c}{\delta r}\right)^2 \left[\left(\frac{r\sin\delta\theta}{\delta\theta}\right)^2 + \left(\frac{\delta r}{\delta\theta} + \frac{2r\sin^2\frac{1}{2}\delta\theta/2}{\delta\theta}\right)^2\right]$$

$$= \left(\frac{\delta c}{\delta s}\right)^2 \left[r^2\left(\frac{\sin\delta\theta}{\delta\theta}\right)^2 + \left(\frac{\delta r}{\delta\theta} + r\sin\frac{\delta\theta}{2} \cdot \frac{\sin\frac{1}{2}\delta\theta/2}{\delta\theta/2}\right)^2\right]$$

Taking limits as $Q \to P$

$$\left(\frac{ds}{d\theta}\right)^2 = 1^2\left[r^2 \cdot 1^2 + \left(\frac{dr}{d\theta} + r \cdot 0 \cdot 1\right)^2\right] = r^2 + \left(\frac{dr}{d\theta}\right)^2$$

As s increases with θ, the increment $\delta\theta$, $\delta s/\delta\theta$ is positive. Thus

$$\frac{ds}{d\theta} = \sqrt{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]} \qquad \ldots(1)$$

Cor. 1. If the equation of the curve is $\theta = f(r)$, then

$$\frac{ds}{dr} = \frac{ds}{d\theta} \cdot \frac{d\theta}{dr} = \sqrt{\left[1 + \left(\frac{dr}{d\theta}\right)^2\right]} \cdot \frac{d\theta}{dr}$$

$$\frac{ds}{dx} = \sqrt{1 + y'^2}\left(\frac{dx}{dx}\right) \quad \ldots(2)$$

Case 2: Parametric form

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{\dot{x}^2 + \dot{y}^2}\,dt \quad \text{or} \quad \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{\dot{x}^2 + \dot{y}^2} = \text{constant}$$

$$\cos\phi = \frac{dx}{ds} \quad \ldots(3)$$

Also

$$\tan\phi = \frac{dy}{dx} \cdot \cos\phi = \frac{dy}{ds} \cdot \frac{ds}{dx}$$

$$\sin\phi = \frac{dy}{ds} \quad \ldots(4)$$

EXERCISE 4.9

Prove that the given equations of:

1. the parabola $y^2 = 4ax$ is $s = a\left[\sqrt{2}\ln\left(\sqrt{2}+1\right) + \sqrt{2}\right]$
2. the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $s = a\sqrt{2}\left[\sinh^{-1}\left(\frac{x}{a}\right) - \sqrt{\frac{x^2}{a^2} - 1}\right]$
3. the asteroid $x = a\cos^3\theta$, $y = a\sin^3\theta$ is $s = \frac{3}{2}a\sin^2\theta$

Find the perimeter of the following curves:

4. $r = a(1 + \cos\theta)$ (V.T.U., 2000)
5. $x^{2/3} + y^{2/3} = a^{2/3}$ (V.T.U., 2004)
6. $x = a\cos^4\theta$, $y = a\sin^4\theta$ (U.P.T.U., 2010)
7. $r = a\theta^2$ (U.P.T.U., 2005)

8. Calculate arc length for the following curves:
   (a) $y = \ln(\sec x)$ (b) $y = \cosh x$
9. Find arc length for the curve $x = a\cos^3\theta$, $y = a\sin^3\theta$ (U.P.T.U., 2007)
10. Find arc length for the following curves:
    (a) $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ (U.P.T.U., 2004) (b) $x = a^2\cos^3\theta$, $y = a^2\sin^3\theta$
    (c) $r = \frac{1}{2}\sec^2\frac{\theta}{2}$ (U.P.T.U., 2007)
11. For the curve $\theta = \cos^{-1}\left(\frac{x}{a}\right) - \sqrt{\frac{x}{a} - \left(\frac{x}{a}\right)^2}$, prove that $\frac{ds}{dx} = \sec\theta\tan\theta$. (U.P.T.U., 2005)
12. With the usual meanings for x, y, θ and ψ for the polar curve $r = f(\theta)$, show that $\frac{d\psi}{d\theta} = 1 + \tan^2\theta\frac{d^2r}{dr^2} + 1$. (U.P.T.U., 2009)

4.10 CURVATURE

Let P be any point on a given curve and Q a neighbouring point. Let arc $AP = s$ and arc $PQ = \delta s$. Let the tangents at P and Q make angles ψ and $\psi + \delta\psi$ with the x-axis, so that the angle between the tangents at P and Q is $\delta\psi$ (Fig. 4.8).

In moving from P to Q through a distance δs, the tangent has turned through the angle $\delta\psi$. This is called the total bending or total curvature of the arc PQ.

$$\text{The average curvature of arc } PQ = \frac{\delta\psi}{\delta s}$$

The limiting value of the average curvature when Q approaches P (i.e., $\delta s \to 0$) is defined as the curvature of the curve at P.

$$\text{Thus curvature } K \text{ at } P = \frac{d\psi}{ds}$$

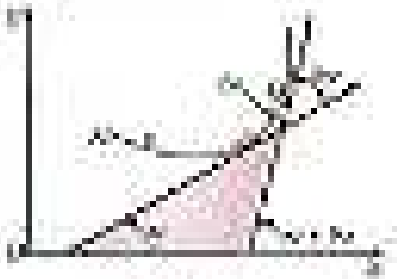

Fig. 4.8

(iii) Curvature is measured in radians per unit length (e.g., radians per centimetre).

(iv) Radius of curvature. The reciprocal of the curvature of a curve at any point P is called the radius of curvature at P and is denoted by ρ or by ρ_P.

(v) Centre of curvature. A point C on the normal at any point P of a curve is called a centre of curvature of P if $CP = \rho$.

(vi) Circle of curvature. A circle with centre C and radius CP is called the circle of curvature or P.

(1) RADIUS OF CURVATURE FOR CARTESIAN CURVE $y = f(x)$ is given by

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$$

We know that $\tan\psi = dy/dx = y_1$ or $\psi = \tan^{-1} y_1$

Differentiating both sides w.r.t. x,

$$\frac{d\psi}{dx} = \frac{1}{1+y_1^2}\cdot\frac{d(y_1)}{dx} = \frac{y_2}{1+y_1^2}$$

$$\therefore \quad \rho = \frac{ds}{d\psi} = \frac{ds}{dx}\cdot\frac{dx}{d\psi} = \sqrt{(1+y_1^2)}\cdot\frac{1+y_1^2}{y_2} = \frac{(1+y_1^2)^{3/2}}{y_2} \qquad \ldots(1)$$

(2) Radius of curvature for parametric equations

$$x = f(t), \qquad y = \phi(t).$$

Denoting differentiations with respect to t by dots over x, y,

$$y_1 = \frac{dy}{dx} = \frac{dy}{dt}\Big/\frac{dx}{dt} = \dot{y}/\dot{x},$$

$$y_2 = \frac{d}{dx}(y_1) = \frac{d}{dt}\left(\frac{\dot{y}}{\dot{x}}\right)\cdot\frac{dt}{dx} = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^2}\cdot\frac{1}{\dot{x}}$$

Substituting the values of y_1 and y_2 in (1)

$$\rho = \left[1+\left(\frac{\dot{y}}{\dot{x}}\right)^2\right]^{3/2}\Big/\left[\frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^3}\right] = \frac{(\dot{x}^2+\dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}} \qquad \text{(Rajasthan, 2002)}$$

(3) Radius of curvature at the origin. Newton's formulae*

(i) If x-axis is tangent to the curve at the origin, then

$$\rho = \rho(0,0) = \lim_{y\to 0}\left(\frac{x^2}{2y}\right)$$

Since x-axis is a tangent at $(0,0)$, $\therefore$ $dy/dx]_0 = 0$

Also $$\lim_{y\to 0}\left(\frac{x^2}{2y}\right) = \lim_{y\to 0}\left[\frac{2x}{2(dy/dx)}\right] = \lim_{x\to 0}\frac{1}{x^2/y\,(dy/dx)^2} = \frac{1}{(y_2)_0} \qquad \left[\frac{0}{0}\text{ form}\right]$$

$$\therefore \quad \rho = \rho(0,0) = \frac{(1+y_1^2)^{3/2}}{y_2}\Big]_0 = \frac{1}{(y_2)_0} = \lim_{y\to 0}\frac{x^2}{2y} \qquad \text{(From (1))}$$

(ii) Similarly, if y-axis is tangent to a curve at the origin, then

$$\rho = \rho(0,0) = \lim_{x\to 0}\left(\frac{y^2}{2x}\right)$$

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* Named after the great English mathematician and physicist Sir Isaac Newton (1642–1727) whose contributions are of utmost importance. He discovered many physical laws, invented Calculus alongwith Leibniz (see footnote p. 1.90) and invented analytical methods of investigating physical problems. He became professor at Cambridge in 1669, but his 'Mathematical Principles of Natural Philosophy' containing development of classical mechanics had been completed by 1687.

(iii) To find the curve passing through the origin but not having a non-zero gradient at the origin, we write the equation of the curve as

$$y = f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \ldots \qquad \text{[By Maclaurin's series]}$$

$$= px + qx^2/2! + \ldots \qquad [\because f(0) = 0]$$

where $p = f'(0)$ and $q = f''(0)$.

Substituting this in the equation $y = f(x)$, we find the values of p and q by equating coefficients of like powers of x. Then $y(x, 0) = 0 + px + q^2/2!$.

Thus, tangent at the origin is a curve are found by equating to zero the lowest degree terms in the equation.

Example 4.44. *Find the nature of curvature at the point (1, 1) of the curve $x^3 + y^3 = 2xy$.*
(Anna, 2005; Kharagpur, 2006; VTU, 2006)
Find the curvature of $x^3 + y^3 = 3axy$ at the origin. (Anna, 2008; Kerala, 2008)

Solution. (i) Differentiating with respect to x, we get

$$3x^2 + 3y^2\frac{dy}{dx} = 2a\left(y + x\frac{dy}{dx}\right)$$

or $$(y^2 - ax)\frac{dy}{dx} = ay - x^2 \qquad \ldots(i) \qquad \therefore \quad \frac{dy}{dx} \text{ at } (3a/2, 3a/2) = -1$$

Differentiating (i),

$$\left(2y\frac{dy}{dx} - a\right)\frac{dy}{dx} + (y^2 - ax)\frac{d^2y}{dx^2} = a\frac{dy}{dx} - 2x \quad \therefore \quad \frac{d^2y}{dx^2} \text{ at } (3a/2, 3a/2) = -32/9a$$

Hence ρ at $(3a/2, 3a/2)$ $= \dfrac{[1 + (dy/dx)^2]^{3/2}}{d^2y/dx^2} = \dfrac{[1 + 1]^{3/2}}{-32/9a} = \dfrac{9a}{8\sqrt{2}}$ (in magnitude).

(ii) We have $y^2 = x^2(x - a)$

$$\therefore \quad 2y\,dy/dx = -x^2 + 2x \quad \text{or} \quad dy/dx = -x(2a - x)/2y = \alpha x$$

At $(a, 0)$, $dy/dx \to \infty$, so we find dx/dy from $y^2 = x^3 - x^2$

$$\therefore \quad 2y = 3x^2\frac{dx}{dy} - 2x\frac{dx}{dy}$$

or $$\frac{dx}{dy} = \frac{-2y}{2x^2 + y} \quad \text{or} \quad \frac{dx}{dy} \text{ at } (a, 0) = 0$$

$$\therefore \quad \frac{d^2x}{dy^2} = \frac{(2x^2 + y^2)\left(-2y\frac{dx}{dy} - 2x\right) - (-2y)\left(4x\frac{dx}{dy} + 2y\right)}{(2x^2 + y^2)^2}$$

or $$\frac{d^2x}{dy^2} \text{ at } (a, 0) = \frac{(2a^2 + 0)(0 - 2a) - 0}{(2a^2 + 0)^2} = \frac{-2}{2a}$$

Hence $$\rho \text{ at } (a, 0) = \frac{\left[1 + \left(\frac{dx}{dy}\right)_{(a,0)}^2\right]^{3/2}}{\left(\frac{d^2x}{dy^2}\right)_{(a,0)}} = \frac{(1 + 0)^{3/2}}{1/(-2a)} = -2a.$$

Example 4.45. *Show that the radius of curvature at any point of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ is $4a\cos(\theta/2)$.* (VTU, 2011; PTU, 2006)

Solution. We have $\dfrac{dx}{d\theta} = a(1 + \cos\theta)$, $\dfrac{dy}{d\theta} = a\sin\theta$.

$$\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta} = \frac{a\sin\theta}{a(1+\cos\theta)} = \frac{2\sin\theta/2\cos\theta/2}{2\cos^2\theta/2} = \tan\theta/2$$

$$\frac{d^2y}{dx^2} = \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\frac{d\theta}{dx} = \frac{1}{2}\sec^2\frac{\theta}{2}\cdot\frac{1}{a(1+\cos\theta)}$$

$$= \frac{1}{2}\sec^2\frac{\theta}{2}\cdot\frac{1}{2a\cos^2\theta/2} = \frac{1}{4a}\sec^2\frac{\theta}{2}.$$

$$\therefore \quad \rho = \frac{[1+(dy/dx)^2]^{3/2}}{d^2y/dx^2} = \frac{4a(1+\tan^2\theta/2)^{3/2}}{\sec^2\theta/2}$$

$$= 4a\cdot(\sec^2\theta/2)^{3/2}\cdot\cos^2\theta/2 = 4a\sec\theta/2.$$

Example 4.46. *Prove that the radius of curvature at any point of the curve $x^{1/3}+y^{1/3}=a^{1/3}$ is twice the length of the perpendicular from the origin to the tangent at that point.*

(U.P.T.U., 2003, 2004, 2005 S)

Solution. The parametric equation of the curve is

$$x = a\cos^6\theta,\ y = a\sin^6\theta.$$

$$\therefore \quad x' = 6a(\sin\theta)^5(-\sin\theta),\ y' = 6a\sin^5\theta\cos\theta$$

$$x'' = -30a\cos^4\theta\sin^2\theta + 6a\cos^6\theta = 6a\cos^4\theta(1-5\sin^2\theta) = 6a\cos^4\theta(1-\sin^2\theta)$$

$$y'' = 30a\sin^4\theta\cos^2\theta - 6a\sin^6\theta = 6a\sin^4\theta(5\cos^2\theta-\sin^2\theta) = 6a\sin^4\theta(1+\cos^2\theta)$$

$$x'^2+y'^2 = 36a^2\cos^{10}\theta\sin^2\theta + 36a^2\sin^{10}\theta\cos^2\theta = 36a^2\sin^2\theta\cos^2\theta$$

$$x'y''-y'x'' = 36a^2\cos^4\theta\sin^4\theta(1+\cos^2\theta) + 36a^2\sin^4\theta\cos^4\theta(1-\sin^2\theta)$$

$$= 36a^2\cos^4\theta\sin^4\theta(2) = 72a^2\sin^4\theta\cos^4\theta$$

$$\therefore \quad \rho = \frac{(x'^2+y'^2)^{3/2}}{x'y''-y'x''} = \frac{216a^3\sin^3\theta\cos^3\theta}{72a^2\sin^4\theta\cos^4\theta} = 3a\sin\theta\tan\theta.$$

Since $\quad dy/dx = y'/x' = -\tan\theta,$

$\therefore$ Equation of the tangent at $(a\cos^6\theta, a\sin^6\theta)$ is $y - a\sin^6\theta = -\tan\theta(x - a\cos^6\theta)$

i.e., $\quad x\tan\theta + y - a\sin\theta = 0$ ...(1)

p, length of $\perp$ from $(0,0)$ on (1) $= \dfrac{0+0-a\sin\theta}{\sqrt{(\tan^2\theta+1)}} = -a\sin\theta\cos\theta$. Thus $p = \frac{1}{3}\rho$.

Example 4.47. *If p_1 and p_2 be the radii of curvature at the ends of a focal chord of the parabola $y^2 = 4ax$, then show that $p_1^{-2/3} + p_2^{-2/3} = (2a)^{-2/3}$.*

(Kumaun, 2008 S; Kurukshetra, 2008)

Solution. Given parabola is $y^2 = 4ax$ or $x = at^2, y = 2at$. If dashes denote differentiation w.r.t. t, then

$$x' = 2at,\ y' = 2a;\ x'' = 2a,\ y'' = 0.$$

$$\therefore \quad p \text{ at } (at^2, 2at) = \frac{(x'^2+y'^2)^{3/2}}{x'y''-x''y'} = \frac{[4a^2t^2+4a^2]^{3/2}}{0-4a^2} = -2a(1+t^2)^{3/2} \qquad \text{(Numerically)}$$

If $P(t_1)$ and $Q(t_2)$ be the extremities of the focal chord of the parabola, then

$$t_1t_2 = -1 \quad \text{i.e.,} \quad t_2 = -1/t_1 \qquad ...(i)$$

$$\therefore \quad p_1 \text{ at } P(t_1) = 2a(1+t_1^2)^{3/2};\ p_2 \text{ at } Q(t_2) = 2a(1+t_2^2)^{3/2}$$

Thus $\quad p_1^{-2/3} + p_2^{-2/3} = (2a)^{-2/3}\left[(1+t_1^2)^{-1} + (1+t_2^2)^{-1}\right]$

$$= (2a)^{-2/3}\left[\frac{1}{1+t_1^2} + \frac{t_1^2}{1+t_1^2}\right] \qquad \text{By (i)}$$

$$= (2a)^{-2/3}$$

Example 4.48. *Show that the curvature of the curve $y^2 = a(x+a)/(x+2a)$ at the point where it cuts the axis of x is $8/a$ and the chord of curvature coincident with the diameter of the curve parallel to the axis of y is $3a/4$.*

(U.P.T.U., 2002)

Solution. The circle centre at origin and radius a is the envelope of the family of straight lines parallel to the other. If CP and CD are two semi-conjugate diameters and P is $(a\cos\theta, b\sin\theta)$ then D is $a\cos\left(\theta+\frac{\pi}{2}\right), b\sin\left(\theta+\frac{\pi}{2}\right)$ i.e., $(-a\sin\theta, b\cos\theta)$.

Also $(0,0)$ is the centre of the ellipse.

$$\therefore \quad OD = \sqrt{(a^2\sin^2\theta + b^2\cos^2\theta)}$$

At P, we have $x = a\cos\theta, y = b\sin\theta$.

$$\therefore \quad \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{b\cos\theta}{-a\sin\theta} = -\frac{b}{a}\cot\theta; \quad \frac{d^2y}{dx^2} = \frac{b}{a}\operatorname{cosec}^2\theta \cdot \frac{d\theta}{dx} = \frac{-b}{a^2}\operatorname{cosec}^3\theta.$$

$$\therefore \quad \rho = \frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}} = \frac{\left(1+\frac{b^2}{a^2}\cot^2\theta\right)^{3/2}}{-\frac{b}{a^2}\operatorname{cosec}^3\theta}$$

$$= \frac{a^3}{b\operatorname{cosec}^3\theta}\cdot\frac{(a^2\sin^2\theta + b^2\cos^2\theta)^{3/2}}{a^3\sin^3\theta} \qquad \text{(Numerically)}$$

$$= \frac{(a^2\sin^2\theta + b^2\cos^2\theta)^{3/2}}{ab} = \frac{CP^3}{ab}$$

Example 4.48. *Find the nature of the origin for the curves*
(i) $x^4 + x^2 + xy^2 - y^3 - 2y^2 = 0$ (ii) $y^2 - x^2 = x^3 + 2xy - y^2$.

Solution. (i) Equating to zero the lowest degree terms, we get $y = 0$.

$\therefore$ x-axis is the tangent at the origin. Dividing throughout by y, we have

$$y^2 + x - \frac{x^3}{y} + x\left(\frac{x^2}{y} + y\right) - x^2 = 0$$

Let $x = vy$, so that $\lim_{y\to 0}(v^2/y) = v$.

$$\therefore \quad 1 + 0.2y - v(2y) + 0 - v^2 = 0 \quad \text{or} \quad y = v/2.$$

(ii) Equating to zero the lowest degree terms, we get $y = x$, as the tangent at the origin, which is neither of the coordinate axes.

$\therefore$ Putting $y = px + qx^2/2! + \ldots$ in the given equation, we get

$$px + qx^2/2 + \ldots - x = x^3 + 2x(px + qx^2/2 + \ldots) + (px + qx^2/2 + \ldots)^2$$

Equating coefficients of x and x^2,

$$p - 1 = 0, \; q/2 = 1 + 2p + p^2 \quad \text{i.e.,} \quad p = 1 \text{ and } q = 2 + 6 - 1 + 2 \cdot 17 = 6.$$

$$\therefore \quad y(0.1) = 0.1 + 0.1^2/2! + \ldots = 0.1 + 0.01 = 0.11.$$

(ii) Radius of curvature for polar curve $r = f(\theta)$ is given by

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2}$$

With the usual notations, we have from Fig. 4.33,

$$\psi = \theta + \phi$$

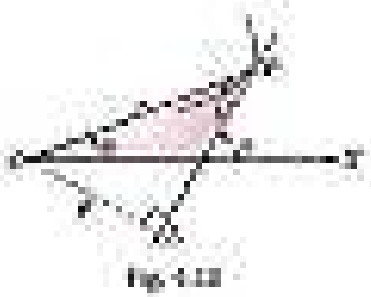

Fig. 4.33

Differentiating w.r.t. s,

$$\frac{1}{\rho} = \frac{d\psi}{ds} = \frac{d\theta}{ds} + \frac{d\phi}{ds} = \frac{d\theta}{ds} + \frac{d\phi}{d\theta}\cdot\frac{d\theta}{ds}$$

$$= \frac{d\theta}{ds}\left(1 + \frac{d\phi}{d\theta}\right)$$

Also we know that,

$$\tan\psi = r\frac{d\theta}{dr} = \frac{r}{r_1} \quad \text{or} \quad \psi = \tan^{-1}\left(\frac{r}{r_1}\right) \qquad \text{where } r_1 = \frac{dr}{d\theta}$$

Differentiating w.r.t. θ,

$$\frac{d\psi}{d\theta} = \frac{1}{1+(r/r_1)^2}\cdot\frac{r_1 \cdot r_1 - r r_2}{r_1^2} = \frac{r_1^2 - r r_2}{r^2 + r_1^2} \qquad ...(2)$$

Also, $\frac{ds}{d\theta} = \sqrt{r^2 + r_1^2}$ ...(3)

Substituting the value from (2) and (3) in (1),

$$\frac{1}{\rho} = \frac{1}{\sqrt{r^2 + r_1^2}}\cdot\left(\frac{r_1^2 - r r_2}{r^2 + r_1^2}\right)$$

Hence $\rho = \dfrac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - r r_2}$

(II) Radius of curvature for pedal curve $p = f(r)$ is ρ given by

$$\rho = r\frac{dr}{dp}$$

With the usual notation (Fig. 4.11), we have $\psi = \theta + \phi$

Differentiating w.r.t. s,

$$\frac{1}{\rho} = \frac{d\psi}{ds} = \frac{d\theta}{ds} + \frac{d\phi}{ds} \qquad ...(1)$$

Also we know that $p = r\sin\phi$

$$\therefore \quad \frac{dp}{dr} = \sin\phi + r\cos\phi\frac{d\phi}{dr}$$

$$= r\frac{d\theta}{dr} + r\frac{d\phi}{ds}\cdot\frac{ds}{dr} \qquad \text{[By (3) and (4) of (I) 4.9.12]}$$

$$= r\left(\frac{d\theta}{ds} + \frac{d\phi}{ds}\right) = \frac{r}{\rho} \qquad \text{[By (1)]}$$

Hence $\rho = r\dfrac{dr}{dp}$.

Example 4.33. *Show that the radius of curvature at any point of the cardioid $r = a(1 + \cos\theta)$ varies as* [illegible] (V.T.U., 2008)

Solution. Differentiating $r = a(1+\theta)$, we get

$$r_1 = a\sin\theta, r_2 = a\cos\theta$$

$$\therefore \quad r^2 + r_1^2 = a^2(1+\cos\theta)^2 + a^2\sin^2\theta = a^2[2(1+\cos\theta)] = 2a^2(1+\cos\theta)$$

$$r^2 - r r_2 + 2r_1^2 = a^2(1+\cos\theta)^2 - a^2(1+\cos\theta)\cos\theta + 2a^2\sin^2\theta = 3a^2(1+\cos\theta)$$

Thus $\rho = \dfrac{(r^2 + r_1^2)^{3/2}}{r^2 - r r_2 + 2r_1^2} = \dfrac{a^3 2\sqrt{2}(1+\cos\theta)^{3/2}}{3a^2(1+\cos\theta)}$

$$= \frac{2\sqrt{2}}{3}a(1+\cos\theta)^{1/2} = \frac{2\sqrt{2}a}{3}\left(\frac{r}{a}\right)^{1/2} = \sqrt{r}.$$

Otherwise. The pedal equation of this cardioid is $2ap^2 = r^3$ ...(1)

Differentiating w.r.t. p, we get

that $\quad 4ap = 3r^2\dfrac{dr}{dp}$, whence $\rho = r\dfrac{dr}{dp} = \dfrac{4ap}{3r} = \dfrac{2a(r^3/2a)}{3r} = \sqrt{r}$.

$\therefore \quad \rho = r^{3/2}/\sqrt{2a}$. Result (1)

PROBLEMS 6.11

1. Find the radius of curvature at any point
   (a) $(at^2, 2at)$ of the parabola $y^2 = 4ax$
   (b) (x, y) of the catenary $y = c\cosh(x/c)$
   (c) $(a, 0)$ of the curve $y = e^x(x - a)$ (IET'D, 2010)

2. Show that for all the rectangular hyperbola $xy = c^2$, $\rho = \frac{(x^2 + y^2)^{3/2}}{2c^2}$ (Oxford, 2006; Madras, 2008)

   (a) the curve $y = ae^{x/a}$ is $\frac{y^2}{a}$ and (b) for $y = a\sin^2(x/a)$ is $\frac{a}{2}\sin^2(2x/a)$ (Oxford, 2006)

3. Show that the radius of curvature at
   (a) $(a, 0)$ on the curve $y^2 = x^2(a - x)$ is $a/3$ (IET'D, 2006 S)
   (b) $(a/4, a/4)$ on the curve $\sqrt{x} + \sqrt{y} = \sqrt{a}$ is $a/\sqrt{8}$ (IET'D, 2006 S)
   (c) $x = a/4$ of the curve $y = a\cos x - \sin 2x$ is $5\sqrt{5}a/16$ (IET'D, 2006 S)

4. For the curve $x = \frac{at}{1 + t^2}$, show that $\left(\frac{\rho}{a}\right)^{2/3} = \frac{x}{y}\left[1 - \left(\frac{x}{a}\right)^2\right]$ (IET'D, 2009)

5. Find the radius of curvature at any point on the
   (a) hyperbola $x = a\sec\theta$, $y = b\tan\theta$ (IET'D, 2010)
   (b) cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$
   (c) curve $x = a\sin\theta + b\cos\theta$, $y = a\cos\theta - b\sin\theta$ (where a, b are constants)

6. Show that the product of curvatures at the points $(at^2, 2at)$ and $(at^{-2}, -2at^{-1})$ on the curve $x^{2/3} + y^{2/3} = a^{2/3}$ is inversely proportional to a^2. (Anna, 2008)

   (a) (i) Find ρ at the point (x, y) on the curve $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$. (Oxford, 2006)

7. If ρ is the radius of curvature at any point P on the parabola $y^2 = 4ax$ and S is its focus, show that ρ^2 varies as $(SP)^3$. (Kuvempu, 2008)

8. Prove that for the ellipse in polar form $\frac{1}{\rho^2} = \frac{1}{a^2}\cos^2\theta + \frac{1}{b^2}\sin^2\theta = \frac{1}{a^2} - \frac{1}{a^2}\sin^2\theta$, the radius of curvature at the point (ρ, θ) is ρ^3/b^2. (IET'D, 2010)

9. Show that the radius of curvature at an end of the major axis of the ellipse $x^2/a^2 + y^2/b^2 = 1$ is equal to the semi-latus rectum. (Osmania, 2009)

10. Show that the radius of curvature at each point of the curve $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$ is directly proportional to the length of the chord intercepted between the point on the curve and the origin. (IET'D, 2009)

11. Find the radius of curvature at the origin for
    (a) $x^4 + y^4 - 3x^2 + 5y = 0$ (Bangalore, 2007)
    (b) $x^3 + 3y^3 + 5xy + 2y - x^2 + 1 = 0$
    (c) $y^2 = x^2\left(\frac{a + x}{a - x}\right)$

12. Find the radius of the curvature at the point (r, θ) to each of the curves
    (a) $r = a(1 - \cos\theta)$ (Kuvempu, 2008)
    (b) $r^n = a^n\cos n\theta$ (IET'D, 2008; IET'D, 2009)

13. For the cardioid $r = a(1 + \cos\theta)$, show that ρ^2/r is constant. (IET'D, 2008)

14. Find the radius of curvature for the polar curve $\frac{2a}{r} = 1 + \cos\theta$. (Kuvempu, 2008)

15. If ρ_1, ρ_2 be the radii of curvature at the extremities of any chord of the cardioid $r = a(1 - \cos\theta)$ which passes through the pole, show that $\rho_1^2 + \rho_2^2 = 16a^2$.

16. For any curve $s = f(\theta)$, prove that $\frac{1}{\rho} = \cos\alpha\left(\frac{d\psi}{ds}\right)$

6.12 (1) CENTRE OF CURVATURE at any point $P(x, y)$ on the curve $y = f(x)$ is given by

$$\bar{x} = x - \frac{y_1(1 + y_1^2)}{y_2}, \quad \bar{y} = y + \frac{1 + y_1^2}{y_2}.$$

Let $C(\alpha, \beta)$ be the centre of curvature and ρ the radius of curvature of the curve at $P(x, y)$ (Fig. 4.11). Draw $PL \perp OX$ and $CM \perp$ to OX and $PN \perp CM$. Let the tangent at P make an angle ψ with the x-axis. Then $\angle NCP = 90° - \angle NPC = \angle MPT = \psi$

$\therefore$ $y = ON = OL - NL = OL - MP$

$$= x - \rho \sin\psi = x - \frac{(1 + y_1^2)^{3/2}}{y_2} \cdot \frac{y_1}{\sqrt{(1 + y_1^2)}}$$

$$\because \quad \tan\psi = y_1 \quad \therefore \quad \sin\psi = \frac{y_1}{\sqrt{(1 + y_1^2)}}$$

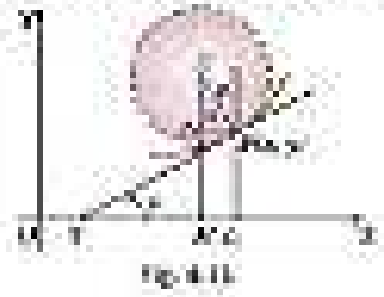

Fig. 4.11

$$\therefore \quad \alpha = x - \frac{y_1(1 + y_1^2)}{y_2}$$

and $\beta = MC = MN + NC = LP + \rho\cos\psi$

$$\because \quad \cos\psi = \sqrt{(1 - \sin^2\psi)} = \sqrt{\left(1 - \frac{y_1^2}{1 + y_1^2}\right)}$$

$$\therefore \quad \beta = \frac{(1 + y_1^2)^{3/2}}{y_2} \cdot \frac{1}{\sqrt{(1 + y_1^2)}} + y = \frac{1 + y_1^2}{y_2} + y$$

Fig. 4.12

Cor. Equation of the circle of curvature at P is $(x - \alpha)^2 + (y - \beta)^2 = \rho^2$.

(2) Evolute. The locus of the centre of curvature for a curve is called the evolute and the curve is called the involute of the evolute (Fig. 4.12).

Example 4.31. *Find the coordinates of the centre of curvature at any point of the parabola $y^2 = 4ax$. Hence show that its evolute is*

$$27ay^2 = 4(x - 2a)^3$$ (U.P.T.U. 2003)

Solution. We have $2yy_1 = 4a$ i.e., $y_1 = 2a/y$

and $$y_2 = -\frac{2a}{y^2} \cdot y_1 = -\frac{4a^2}{y^3}$$

If (α, β) be the centre of curvature, then

$$\alpha = x - \frac{y_1(1 + y_1^2)}{y_2} = x - \frac{2a(1 + 4a^2/y^2)}{-4a^2/y^3}$$

$$= x + \frac{y^2 + 4a^2}{2a} = x + \frac{4ax + 4a^2}{2a} = 2x + 2a \qquad \because \ y^2 = 4ax \qquad \ldots(1)$$

and $$\beta = y + \frac{1 + y_1^2}{y_2} = y + \frac{1 + 4a^2/y^2}{-4a^2/y^3}$$

$$= y - \frac{y^2 + 4a^2}{4a^2} \cdot y = -\frac{y^3}{4a^2} = -\frac{2a^{3/2}}{a} \qquad \ldots(2)$$

To find the evolute, we have to eliminate x from (1) and (2).

$$\therefore \quad (-\beta)^2 = \frac{4a^3}{a} = 4\left(\frac{\alpha - 2a}{2}\right)^3 \quad \text{or} \quad 27a(-\beta)^2 = 4(\alpha - 2a)^3$$

Thus, the locus of (α, β) i.e., evolute, is $27ay^2 = 4(x - 2a)^3$.

Example 4.32. Show that the evolute of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ is another equal cycloid. (Hard copy 2006)

Solution. We have $y_1 = \dfrac{dy}{dx} = \dfrac{dy}{d\theta} \Big/ \dfrac{dx}{d\theta} = \dfrac{a\sin\theta}{a(1-\cos\theta)} = \cot\dfrac{\theta}{2}$

$$y_2 = \frac{d}{dx}(y_1) = \frac{d}{d\theta}\left(\cot\frac{\theta}{2}\right)\cdot\frac{d\theta}{dx}$$

$$= -\operatorname{cosec}^2\frac{\theta}{2}\cdot\frac{1}{2}\cdot\frac{1}{a(1-\cos\theta)} = -\frac{1}{4a\sin^2\theta/2}$$

If $(\bar{x}, \bar{y})$ be the centre of curvature, then

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2} = a(\theta + \sin\theta) - a\sin\theta\left[1 + \cot^2\frac{\theta}{2}\right]\left[1 + \cot^2\frac{\theta}{2}\right]$$

$$= a\theta + a\sin\theta + \frac{\cos\theta/2}{\sin\theta/2}\cdot a\sin^4\frac{\theta}{2}\cdot\operatorname{cosec}^2\frac{\theta}{2}$$

$$= a\theta + a\sin\theta + 4a\sin\theta/2\cos\theta/2 = a\theta + a\sin\theta + 2a\sin\theta = a(\theta + \pi + \sin\theta)$$

$$\bar{y} = y + \frac{1+y_1^2}{y_2} = a(1-\cos\theta) + \left[1 + \cot^2\frac{\theta}{2}\right]\cdot 4a\sin^2\frac{\theta}{2}$$

$$= a(1 - \cos\theta) + 4a\operatorname{cosec}^2\theta/2$$

$$= a(2 - \cos\theta) + 2a(1 - \cos\theta) = -a(1 - \cos\theta)$$

Hence the locus of $(\bar{x}, \bar{y})$ is, the evolute is given by

$$\bar{x} = a(\theta + \pi + \sin\theta),\ \bar{y} = -a(1 - \cos\theta)$$ which is another equal cycloid.

(5) Chord of curvature of a given point of a curve

(a) parallel to x-axis $= 2p\sin\psi$

(b) parallel to y-axis $= 2p\cos\psi$

Consider the circle of curvature at a given point P on a curve. Let Q be the centre and ρ the radius of curvature at P so that $PQ = \rho$. (Fig. 4.13)

Let PL, PM be the chords of curvature parallel to the axes of x and y respectively. Let the tangent PT at P on the curve be inclined at ψ with the x-axis so that $\angle QPT = \angle QPN' = \psi$.

Then from the rt. $\angle$ed ΔPNQ,

$$PL = 2p\sin\psi$$

and $$PM = 2p\cos\psi.$$

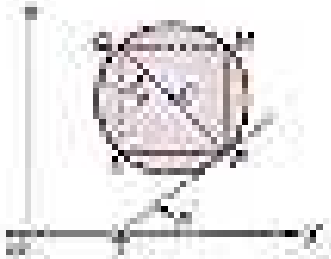

Fig. 4.13

4.15 (1) ENVELOPE

The equation $x\cos\alpha + y\sin\alpha = 1$ ...(1)

represents a straight line for a given value of α. If different values are given to α, we get different straight lines. All these straight lines thus obtained are said to constitute a family of straight lines.

In general, the curves corresponding to the equation $f(x, y, \alpha) = 0$ for different values of α constitute a family of curves and α is called the parameter of the family.

The envelope of a family of curves is the curve which touches each member of the family. For example, we know that all the straight lines of the family (1) touch the circle

$$x^2 + y^2 = 1 \qquad ...(2)$$

i.e., the envelope of the family of lines (1) is the circle (2)—Fig. 4.14, which may also be seen as the locus of the ultimate points of intersection of the consecutive members of the family of lines (1). This leads to the following:

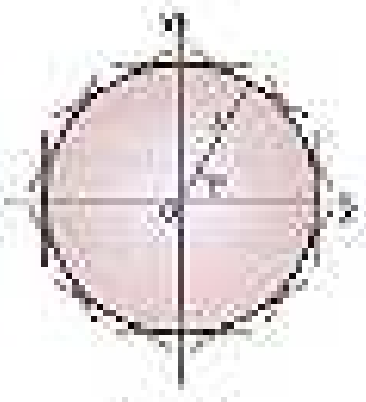

Fig. 4.14

Def. *If $f(x, y, \alpha) = 0$ and $f(x, y, \alpha + \delta\alpha) = 0$ be two consecutive members of a family of curves, then the locus of their ultimate points of intersection is called the envelope of that family.*

(i) Rule to find the envelope of the family of curves $f(x, y, \alpha) = 0$:

Always write $f(x, y, \alpha) = 0$ and $\frac{\partial f(x, y, \alpha)}{\partial \alpha} = 0$.

Example 4.23. *Find the envelope of the family of lines $y = \alpha x + \sqrt{1 + \alpha^2}$, α being the parameter.*

Solution. We have $(y - \alpha x)^2 = 1 + \alpha^2$ ...(i)

Differentiating (i) partially with respect to α,

$$2(y - \alpha x)(-x) = 2\alpha \quad \text{or} \quad \alpha x - \alpha y = \alpha^2 + 1 \qquad \text{...(ii)}$$

Now eliminating α from (i) and (ii).

Substituting the value of αx in (i), we get

$$\left(y - \frac{\alpha^2 + 1}{y}\right)^2 = 1 + \left(\frac{\alpha^2 + 1}{y}\right)^2 \quad \text{or} \quad y^2 = (\alpha^2 + 1)^2 + y^2\alpha^2$$

or $x^2 + y^2 = 1$ which is the required equation of the envelope.

Note. In the above example the equation of the family of curves contains two parameters which are connected by a relation. In such cases, we eliminate any of the parameters by means of the given relation, then proceed to find the envelope.

Example 4.24. *Find the envelope of a system of coaxial and concentric ellipses of constant area.*

Solution. Taking the common axes of the system of ellipses as the coordinate axes, the equation to an ellipse of the family is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ where } a \text{ and } b \text{ are the parameters.} \qquad \text{...(i)}$$

The area of the ellipse $= \pi ab$ which is given to be constant, say $= \pi c^2$.

$$\therefore \quad ab = c^2 \quad \text{or} \quad b = c^2/a \qquad \text{...(ii)}$$

Substituting in (i), $\frac{x^2}{a^2} + \frac{y^2}{c^4/a^2} = 1 \quad \text{or} \quad x^2a^2 + y^2a^2 - c^4 = 0 \qquad \text{...(iii)}$

which is the given family of ellipses with a as the only parameter.

Differentiating partially (iii) with respect to a,

$$2x^2a^2 + 2y^2a^2 - 4c^4 = 0 \quad \text{or} \quad a^2 = c^2\sqrt{2}$$

Eliminating a from (ii) and (iv).

Substituting the value of a^2 in (iii), we get

$$x^2y^2 + (x^2 + y^2)c^4\sqrt{2} = c^4 \text{ or } 2x^2y^2 = c^4$$

which is the required equation of the envelope. *Ans.*

(ii) *Evolute of a curve as the envelope of the normals to that curve* (Fig. 4.22)

Example 4.25. *Find the evolute of the parabola $y^2 = 4ax$.* (Madras, 2003)

Solution. Any normal to the parabola is $y = mx - 2am - am^3$ ...(i)

Differentiating it with respect to m partially,

$$0 = x - 2a - 3am^2 \text{ or } m = \{x - 2a)/3a\}^{1/2}$$

Substituting this value of m in (i),

$$y = \left(\frac{x - 2a}{3a}\right)^{1/2}\left[x - 3a - a\frac{x - 2a}{3a}\right]$$

Squaring both sides, we have

$$27ay^2 = 4(x - 2a)^3$$

which is the evolute of the parabola of Example 4.21.

PROBLEMS (S)

1. Find the coordinates of the points of contact of the tangents at $(at^2, 2at)$ to the parabola $y^2 = 4ax$. [V.T.U., 2007]
2. If the normal at the extremity of the latus rectum of the ellipse $x^2/a^2 + y^2/b^2 = 1$ passes through the extremity of the minor axis, find the eccentricity of the ellipse.
   Find the eccentricity of the ellipse $x^2 + 3y^2 = 12$. [Anna, 2005 S; Anna, 2007]
3. Show that the equation of the normal to the
   (i) parabola $x^2 = 4ay$ at $(2at, at^2)$ is $x + ty = 2at + at^3$. [Anna, 2009]
   (ii) ellipse $x = a\cos\theta$, $y = b\sin\theta$ at the point $(a\cos\theta, b\sin\theta)$ is $ax\sec\theta - by\operatorname{cosec}\theta = a^2 - b^2$.
   (iii) rectangular hyperbola $xy = c^2$ at the point $(ct, c/t)$ is $t^3x - ty = ct^4 - c$.
   [Anna, 2003]
4. Find the envelope of the family $y = mx + a\sqrt{1 + m^2}$, m being the parameter.
   Find the envelope of the family of straight lines $y = mx + \sqrt{a^2m^2 + b^2}$, m being the parameter. [Madras, 2002; Anna, 2003]
5. Find the envelope of the family of straight lines $\frac{x}{a} + \frac{y}{b} = 1$, where $a + b = c$, c being a constant. [Madras, 2003]
6. Show that the envelope of the curve $x = a\cos^3\theta$, $y = a\sin^3\theta$ is $x^{2/3} + y^{2/3} = a^{2/3}$. [Anna, 2003]
7. Find the circle of curvature at the point $(1, 2)$ on the curve $xy = 2$. Find also the centre of curvature.
   Find the circle of curvature of the curve $xy = c^2$ at $(ct, c/t)$. [Anna, 2009; Madras, 2003; Anna, 2005]
8. Show that the circle of curvature at the origin for the curve $x^3 + y^3 = 3axy$ is $x^2 + y^2 = 2ay$. [Nagpur, 2003]
9. If C_1, C_2 be the chords of curvature parallel to the axes at any point on the curve $x^{2/3} + y^{2/3} = a^{2/3}$, prove that
$$\frac{1}{C_1^2} + \frac{1}{C_2^2} = \frac{1}{3a^2}.$$
10. In the curve $x = a\cos^3\theta$, $y = a\sin^3\theta$, show that the chord of curvature parallel to X-axis is of the form $3a\sin\theta\cos^2\theta$.

Find the envelope of the following family of lines:

11. $y = mx + a\sqrt{1 + m^2}$, m being the parameter. [Madras, 2006]
12. $\frac{x}{a}\cos\alpha + \frac{y}{b}\sin\alpha = 1$, α being the parameter.
13. $y = mx - 2am - am^3$.
14. $y = mx + \sqrt{a^2m^2 + b^2}$, m being the parameter. [Anna, 2006]
15. Find the envelope of the family of parabolas $y = x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha}$, α being the parameter.
16. Find the envelope of the straight line $lx/a + my/b = 1$, where the parameters l and m are connected by the relation
   (i) $l + m = c$ (ii) $lm = c^2$.
17. Find the envelope of the family of ellipses $x^2/a^2 + y^2/b^2 = 1$ for which $a + b = c$. [Madras, 2006]

Find the envelope of the

18. $\sin(\alpha x) + \frac{y}{\alpha} = 1$, α being the parameter. [U.P.T.U., 2008; Anna, 2008]
19. $x\cos\theta + y\sin\theta = a\sin\theta\cos\theta$, θ being the parameter. [Anna, 2009]
20. parabola $x^2 = 4ay$ is $27ay^2 + 4(y - 2a)^3 = 0$.

4.14 [3] INCREASING AND DECREASING FUNCTIONS

In the function $y = f(x)$, if y increases as x increases at all points in an interval, it is called an increasing function of x. On the contrary, if y decreases as x increases in an interval, it is called a decreasing function of x.

Let the tangent at any point on the graph of the function make an $\angle\psi$ with the x-axis (Fig. 4.13) so that

$$dy/dx = \tan\psi$$

At any point such as A, where the function is increasing dy is positive, dy/dx is positive. At a point such as C, where the function is decreasing dy is negative, i.e., dy/dx is negative.

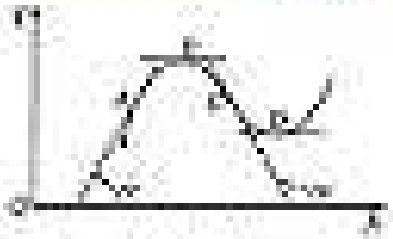

Fig. 4.13

Hence the derivative of an increasing function is > 0 and the derivative of a decreasing function is < 0.

Obs. *If the derivative is zero or does not exist at M, then it is neither increasing nor decreasing. In this case, we say that the function is stationary.*

(2) Concavity, Convexity and Point of Inflexion.

(i) *If a portion of the curve on both sides of a point, however small it may be, lies above the tangent line at P, then the curve is said to be concave upwards at P where d^2y/dx^2 is positive.*

(ii) *If a portion of the curve on both sides of a point, however small it may be, lies below the tangent at P, then the curve is said to be Convex upwards at P where d^2y/dx^2 is negative.*

(iii) *If the two portions of the curve lie on different sides of the tangent thereat (i.e., the curve crosses the tangent line at C), then the point C is said to be a point of inflexion of the curve.*

At a point of inflexion $\dfrac{d^2y}{dx^2} = 0$ *and* $\dfrac{d^3y}{dx^3} \neq 0$.

[1] MAXIMA AND MINIMA

Consider the graph of the continuous function $y = f(x)$ in the interval (a, b) (Fig. 4.14). Clearly the point P_1 is the highest in its own immediate neighborhood. So also is P_3. At each of these points P_1, P_3 the function is said to have a maximum value.

On the other hand, the point P_2 is the lowest in its own immediate neighborhood. So also is P_4. At each of these points P_2, P_4 the function is said to have a minimum value.

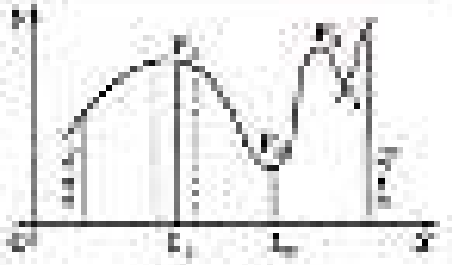

Fig. 4.14

Thus, we have

Def. *A function $f(x)$ is said to have a* **maximum** *value at $x = a$, if there exists a small number h, however small, such that $f(a) > f(a + h)$ and $f(a) > f(a - h)$ for all $0 < h < h$.*

A function $f(x)$ is said to have a **minimum** *value at $x = a$, if there exists a small number h, however small, such that $f(a) < f(a + h)$ and $f(a) < f(a - h)$ for all $0 < h < h$.*

Obs. 1. *The maximum and minimum values of a function (also together are called its extreme values) are the greatest and the least values respectively, only in the immediate neighbourhood of the point.*

Obs. 2. *A maximum value of a function is not necessarily the greatest or least value of the function in any finite interval. The maximum value is merely the greatest value in the immediate neighborhood of the maximum point, so the maximum value of the function in the immediate neighborhood of the minimum point. In fact, there may be several maximum and minimum values of a function in an interval and a minimum value may be even greater than a maximum value.*

Obs. 3. *It is clear from the Fig. 4.14 that maximum and minimum values occur alternately.*

(1) Conditions for maximum and minimum. *At each point of extreme value, it is seen from Fig. 4.14 that the tangent to the curve is parallel to the x-axis, i.e., its slope $= dy/dx$ is zero. Thus if the function is maximum or minimum at $x = a$, then $(dy/dx)_{x=a} = 0$.*

Around a maximum point say, P_3, (i) the curve is increasing in a small interval $(a - h, a)$ before x_2 and decreasing in $(a, a + h)$ after x_2, where h is positive and small.

i.e., $\text{in } (a - h, a),\ dy/dx > 0$; *at* $x = a$, $dy/dx = 0$ *and in* $(a, a + h)$, $dy/dx < 0$.

Thus dy/dx (which is a function of x) changes sign from positive to negative in passing through P_3, i.e., it is a decreasing function in the interval $(a - h, a + h)$ and therefore, its derivative d^2y/dx^2 is negative at P_3 for $x = a$.

Similarly, around a minimum point say P_2, dy/dx changes sign from negative to positive in passing through P_2, i.e., it is an increasing function in the small interval around x_1 and therefore its derivative d^2y/dx^2 is positive at P_2.

Hence (i) $f(x)$ is a maximum at $x = a$ if $f'(x)$ is positive when $f'(x) = 0$ i.e. $f'(x)$ changes sign from $+ve$ to $-ve$

(ii) $f(x)$ is a minimum at $x = a$ if $f'(x) = 0$ and $f'(x)$ is negative when $f'(x) = 0$ i.e. $f'(x)$ changes sign from $-ve$ to $+ve$

Note. A maximum or minimum value is a stationary value but a stationary value may not be a maximum nor a minimum value.

(3) Procedure for finding maxima and minima

(a) Put the given function $= f(x)$

(b) Find $f'(x)$ and equate it to zero. Solve this equation and let the roots be $a, b, c, \ldots$

(c) Find $f''(x)$ and substitute in it in turn $x = a, b, c, \ldots$

If $f''(a)$ is $-ve$, $f(x)$ is a maximum at $x = a$.

If $f''(a)$ is $+ve$, $f(x)$ is a minimum at $x = a$.

(d) Sometimes $f''(x)$ may be difficult to find or $f'(x)$ may be zero at $x = a$. In such cases, see if $f'(x)$ changes sign from $+ve$ to $-ve$ as x passes through a, then $f(x)$ is a maximum at $x = a$.

If $f'(x)$ changes sign from $-ve$ to $+ve$ as x passes through a, $f(x)$ is a minimum at $x = a$.

If $f'(x)$ does not change sign while passing through $x = a$, $f(x)$ is neither a maximum nor a minimum at $x = a$.

Example 6.26. Find the maximum and minimum values of $f(x) = 2x^4 - 2x^3 - 3x^2 + 4x + 1$ in the interval $[0, 2]$

Solution. Let $f(x) = 2x^4 - 2x^3 - 3x^2 + 4x + 1$

Then $f'(x) = 8x^3 - 6x^2 - 12x + 4 = 6(x^2 - 3(2x - 1))$

$\therefore$ $f'(x) = 0$ when $x = 2, 1, \frac{1}{2}$.

Now in the interval $[0, 2]$ $f(x)$ can have maximum or minimum at $x = \frac{1}{2}$ or 1

Now $f''(x) = 24x^2 - 12x - 12 = 12(2x^2 - x - 1)$ so that $f''\left(\frac{1}{2}\right) = -9$ and $f''(1) = 12$.

$\therefore$ $f(x)$ has a maximum at $x = \frac{1}{2}$ and a minimum at $x = 1$.

Thus the maximum value $= f\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right)^4 - 2\left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right)^2 + 4\left(\frac{1}{2}\right) + 1 = 2\frac{7}{16}$

and the minimum value $= f(1) = 2(1)^4 - 2(1)^3 - 3(1)^2 + 4(1) + 1 = 2$.

Example 6.27. Show that $\sin x(1 + \cos x)$ has a maximum value at $x = \pi/3$.

(Hence, find the maximum value of $\sin x(1 + \cos x)$ in the interval $0 \le x \le 2\pi$.) (Glasgow 2000 / Reykjavik 2003)

Solution. Let $f(x) = \sin x(1 + \cos x)$

Then $f'(x) = \sin x(-\sin x) + (1 + \cos x)\cos x = \cos x - \sin^2 x + \cos^2 x = (1 + \cos x)(2\cos x - 1)$

$\therefore$ $f'(x) = 0$ when $\cos x = -1$ or $\frac{1}{2}$ i.e., when $x = \pi, \frac{\pi}{3}$ or $\frac{5\pi}{3}$.

Now $f''(x) = -\sin x(2\cos x - 1) + (1 + \cos x)(-2\sin x) = -\sin x(4\cos x + 2)$

so that $f''(\pi) = -\sin\pi(-4 + 2) = 0$ and $f''\left(\frac{\pi}{3}\right) = -\frac{3\sqrt{3}}{2}$.

Thus $f(x)$ has a maximum at $x = \frac{\pi}{3}$.

Since $f''(\pi) = 0$, let us see whether $f'(x)$ changes sign or not.

When x is slightly $< \pi$, $f'(x)$ is $-ve$, then when x is slightly $> \pi$, $f'(x)$ is again $-ve$ i.e., $f'(x)$ does not change sign as x passes through π. So $f(x)$ is neither a maximum nor a minimum at $x = \pi$.

(4) Practical Problems

In many problems, the function whose maximum or minimum value is required is not directly given. It has to be formed from the given data. If the function contains two variables, one of them has to be eliminated with the help of the other conditions of the problem. A number of problems deal with triangles, rectangles, circles, spheres, cones, cylinders etc. The student is therefore, advised to remember the formulae for areas, volumes, surfaces etc. of such figures.

Example 6.28. A window is in the form of a rectangle surmounted by a semi-circle. If the perimeter is 30 ft., find the dimensions so that the greatest amount of light may be admitted. (Madras, 2000)

Solution. The greatest amount of light is admitted when the area of the window may be maximum.

Let x ft. be the radius of the semi-circle so that one side of the rectangle is $2x$ ft. (Fig. 4.17). Let the other side of the rectangle be y ft. Then the perimeter of the whole figure

$$= \pi x + 2x + 2y = 40 \text{ (given) and the area } A = \frac{1}{2}\pi x^2 + 2xy \quad \ldots(1)$$

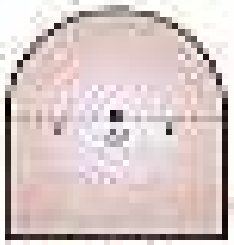
Fig. 4.17

Here A is a function of two variables x and y. To express A in terms of one variable only, we substitute the value of y from (1) in it.

$$\therefore \quad A = \frac{1}{2}\pi x^2 + x(40 - \pi x - 2x) = 40x - \left(\frac{\pi}{2} + 2\right)x^2$$

Then $$\frac{dA}{dx} = 40 - \pi x - 4x$$

For A to be maximum or minimum, we must have $dA/dx = 0$ i.e., $40 - (\pi + 4)x = 0$

or $$x = \frac{40}{\pi + 4}$$

$\therefore$ From (1), $$y = \frac{1}{2}(40 - \pi x - 2x) = \frac{1}{2}\left(40 - \frac{40\pi}{\pi + 4} - \frac{80}{\pi + 4}\right) = \frac{80}{\pi + 4} - 40 = x, \text{ i.e., } x = y$$

Also $$\frac{d^2A}{dx^2} = -\pi - 4, \text{ which is negative.}$$

Thus the area of the window is maximum when the radius of the semi-circle is equal to the height of the rectangle.

Example 4.28. *A rectangular sheet of metal of length l metres and width b metres is given. Four equal squares are cut out from the corners. The sides of the sheet are then turned up to form an open rectangular box. Find approximately, the length of the side of each square so that the volume of the box is maximum.*

Solution. Let the side of each of the equal squares cut out be x metres. Then the height of the box is x metres and the sides of the base are $b - 2x$, $l - 2x$ m. (Fig. 4.18).

$\therefore$ Volume V of the box

$$= x(b - 2x)(l - 2x) = x(bl - 2bx - 2lx + 4x^2) = 4x^3 - 2(b + l)x^2 + blx$$

Fig. 4.18

Then $$\frac{dV}{dx} = 12x^2 - 4(b + l)x + bl$$

For V to be maximum or minimum, we must have

$$dV/dx = 0 \text{ i.e., } 3x^2 - (b + l)x + \frac{bl}{4} = 0$$

$$\therefore \quad x = \frac{(b + l) \pm \sqrt{(b + l)^2 - 3bl}}{6}$$

The value $x = 5.8$ m is inadmissible, as no box is possible for this value.

Also $\dfrac{d^2V}{dx^2} = 24x - 4(b + l)$, which is $-ve$ for $x = 0.45$ m.

Hence the volume of the box is maximum when its height is 45 cm.

Example 4.29. *Show that the right circular cylinder of given surface (including the ends) and maximum volume is such that its height is equal to the diameter of the base.*

Solution. Let r be the radius of the base and h, the height of the cylinder.

Then given surface $$S = 2\pi rh + 2\pi r^2 \quad \ldots(1) \qquad \text{and the volume } V = \pi r^2 h \quad \ldots(2)$$

Here V is a function of two variables r and h. To express V in terms of one variable only (say r), we substitute the value of h from (1) in (2).

Then $$V = \pi r^2\left(\frac{S - 2\pi r^2}{2\pi r}\right) = \frac{1}{2}Sr - \pi r^3 \quad \therefore \quad \frac{dV}{dr} = \frac{1}{2}S - 3\pi r^2$$

For V to be a maximum or minimum, we must have $dV/dr = 0$,

i.e., $\frac{1}{3}\pi \cdot 8kr^2 = 0$ i.e. $r = \sqrt{3/8\pi k}$

Also $\frac{d^2V}{dr^2} = -8\pi r$, which is negative for $r = \sqrt{3/8\pi k}$.

Hence V is a maximum for $r = \sqrt{3/8\pi k}$

i.e., for $6\pi r^2 + 8 = 2\pi rh + 2\pi r^2$ i.e., for $h = 2r$, which proves the required result. Q.E.D.

Example 4.11. Show that the diameter of the right circular cylinder of greatest curved surface which can be inscribed in a given cone is equal to the radius of the cone.

Solution. Let r be the radius DA of the base and α the semi-vertical angle of the given cone (Fig. 4.19). Inscribe a cylinder in it with base-radius QE = x.

Then the height of the cylinder h

$$= EA \cot\alpha = (r - x)\cot\alpha$$

$\therefore$ The curved surface S of the cylinder

$$= 2\pi x \cdot h = 2\pi x(r - x)\cot\alpha$$

$$= 2\pi\cot\alpha\,(rx - x^2)$$

$$\therefore \quad \frac{dS}{dx} = 2\pi\cot\alpha\,(r - 2x) = 0 \text{ for } x = r/2.$$

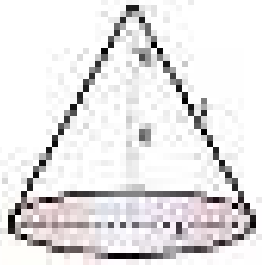

Fig. 4.19

and $\frac{d^2S}{dx^2} = -4\pi\cot\alpha$

Hence S is maximum when $x = r/2$.

Example 4.12. Find the altitude and the semi-vertical angle of a cone of least volume which is circumscribed about a sphere of radius r.

Solution. Let h be the height and α the semi-vertical angle of the cone so that its radius $BD = h\tan\alpha$ (Fig. 4.20).

$\therefore$ The volume V of the cone is given by

$$V = \frac{1}{3}\pi(h\tan\alpha)^2 h = \frac{1}{3}\pi h^3\tan^2\alpha$$

Now we must express $\tan\alpha$ in terms of h.

In the $\triangle$ed AEO, $AO = \sqrt{OE^2 - AE^2} = \sqrt{r^2 - (h - r)^2} = \sqrt{h^2 - 2hr}$

$$\therefore \quad \tan\alpha = \frac{BD}{AB} = \frac{r}{\sqrt{h^2 - 2hr}}$$

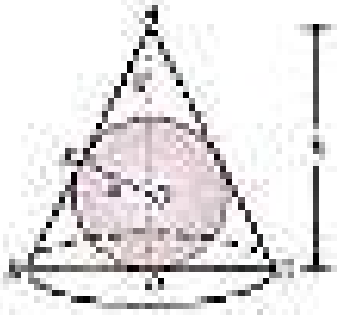

Fig. 4.20

Thus $V = \frac{1}{3}\pi r^2 \frac{h^3}{h^2 - 2hr} = \frac{1}{3}\pi r^2 \frac{h^2}{h - 2r}$

$$\therefore \quad \frac{dV}{dh} = \frac{1}{3}\pi r^2 \cdot \frac{(h - 2r)2h - h^2 \cdot 1}{(h - 2r)^2} = \frac{1}{3}\pi r^2 \cdot \frac{h(h - 4r)}{(h - 2r)^2}$$

Thus $\frac{dV}{dh} = 0$ for $h = 4r$, the other value $h = 0$ being not possible.

Also dV/dh is $-ve$ when h is slightly $< 4r$, and it is $+ve$ when h is slightly $> 4r$.

Hence V is minimum i.e. least when $h = 4r$.

and $\alpha = \sin^{-1}\left(\frac{r}{OA}\right) = \sin^{-1}\left(\frac{r}{2r}\right) = \sin^{-1}\frac{1}{2}$.

Example 4.30. Find the volume of the largest possible right circular cylinder that can be inscribed in a sphere of radius a.

Solution. Let O be the centre of the sphere of radius a. Construct a cylinder as shown in Fig. 4.21. Let $OA = x$.

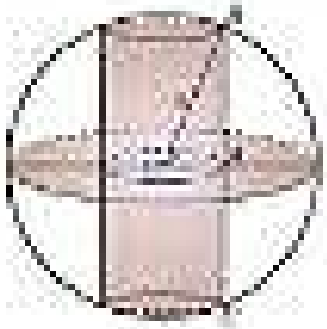
Fig. 4.21

Then $AB = \sqrt{(OB^2 - OA^2)} = \sqrt{(a^2 - x^2)}$.

$\therefore$ Height h of the cylinder $= 2 \cdot AB = 2\sqrt{(a^2 - x^2)}$.

Then volume V of the cylinder

$$= \pi r^2 h = 2\pi x^2 \sqrt{(a^2 - x^2)}$$

$$\therefore \quad \frac{dV}{dx} = 2\pi \left[2x\sqrt{(a^2 - x^2)} + x^2 \cdot \frac{1}{2}(a^2 - x^2)^{-1/2} \cdot (-2x) \right]$$

$$= \frac{2\pi x(2a^2 - 3x^2)}{\sqrt{(a^2 - x^2)}}$$

The critical values are $x = 0$ and $x = \sqrt{(2/3)}a$, the other value $x = a$ being not admissible.

Now $$\frac{d^2V}{dx^2} = 2\pi \frac{\sqrt{(a^2 - x^2)}(2a^2 - 9x^2) - x(2a^2 - 3x^2) \times \frac{1}{2}(a^2 - x^2)^{-1/2}(-2x)}{(a^2 - x^2)}$$

$$= 2\pi \frac{(a^2 - x^2)(2a^2 - 9x^2) + x^2(2a^2 - 3x^2)}{(a^2 - x^2)^{3/2}}$$ which is $-ve$ for $x^2 = 2a^2/3$.

Hence V is maximum for $x^2 = 2a^2/3$ and maximum volume

$$= 2\pi x^2 \sqrt{(a^2 - x^2)} = 4\pi a^3/3\sqrt{3}.$$

Example 4.31. *Assuming that the rate at which fuel is burned (per hour) in driving a motor boat varies as the cube of the velocity of the boat, show that the most economical speed when going against a current of c km/hr is $\frac{3}{2}c$ km/hr per hour.*

Solution. Let v km/hr be the velocity of the boat so that its velocity relative to water (when going against the current) is $(v - c)$ km/hr.

$\therefore$ Time required to cover a distance of 1 km is $= \frac{1}{v - c}$ hours.

Since the petrol burnt per hour $= kv^3$, k being a constant.

$\therefore$ The total petrol burnt, y, is given by

$$y = kv^3 \frac{1}{v - c} \quad \text{or} \quad \frac{dy}{dv} = k \cdot \frac{(v - c)3v^2 - v^3 \cdot 1}{(v - c)^2}$$

$$= k \frac{v^2(2v - 3c)}{(v - c)^2}$$

The equation $\frac{dy}{dv} = 0$ has $v = 3c/2$, the other value $v = 0$ is inadmissible.

Also $\frac{dy}{dv} < 0$ or, when v is slightly $< 3c/2$ and $\frac{dy}{dv} > 0$ or, when v is slightly $> 3c/2$.

Hence y is minimum for $v = 3c/2$.

EXERCISE 4.3

1. (a) Test the curve $y = x^4$ for points of inflexion. [Kerala Univ. 2003]

(b) Show that the points of inflexion of the curve $y = (x - a)^5 + (x - b)^5$ lie on the straight line $3x + a + b = 0$. [Bangalore Univ. 2002]

2. The function $f(x)$ defined by $f(x) = \sin x + \cos x$, $0 \le x \le 2\pi$, has a maximum at $x = \frac{\pi}{4}$. Determine a set S in the interval $[0, 2\pi]$ such that $f(x) \le f(\frac{\pi}{4})$ for all $x \in S$. Illustrate your solution on the graph of $f(x)$.
3. Show that $\sin x + \cos x \le \sqrt{2}$ for all x. (Nanyang, 2004)
4. If a body of weight w gas and height h is built to last exactly n years, then it is found that the cost of maintaining the body is proportional to wh^2 per year. Find the value of h that minimizes the total cost of maintaining the body for n years.
5. The horsepower developed by an aircraft engine is found to be directly proportional to the cube of the speed of the aircraft and inversely proportional to the number of miles per gallon of fuel. If the aircraft is flying at a speed of v miles per hour, show that the horsepower developed is given by kv^3, where k is a constant. Find the value of v for which the horsepower developed is a minimum.
6. The velocity of water in a river is v km/h at a distance x km from the bank of the river. Given that $v = \frac{1}{400}x(100 - x)$, find the distance from the bank at which the velocity is a maximum. Find also the maximum velocity.
7. In a submarine telegraph cable, the speed of signalling varies as $x^2 \log(\frac{1}{x})$, where x is the ratio of the radius of the core to that of the covering. Show that the greatest speed is attained when the ratio is $\frac{1}{\sqrt{e}}$.
8. The efficiency of a screw jack is given by $\eta = \frac{\tan\theta}{\tan(\theta + \alpha)}$, where α is a constant. Find θ in terms of α for η to be a maximum. Also find the maximum efficiency.
9. Show that, of all rectangles of given area, the square has the least perimeter.
10. Find the rectangle of constant perimeter $2p$ that can be inscribed in a circle of radius a.
11. A piece of metal in the form of a rectangle $ABCD$ is to be made to hold 1 litre of water by folding it along EF and GH parallel to AB and CD respectively, and then along EG and FH. Show that the capacity of the container is greatest if AB is twice AD.
12. Show that the triangle of maximum area that can be inscribed in a given circle is an equilateral triangle.
13. An open box is to be made from a rectangular piece of sheet metal 32 cm by 20 cm, by cutting out equal squares from each corner and folding up the sides. Find the dimensions of the box of largest volume that can be made in this manner.
14. An open tank is to be constructed with a square base and vertical sides to hold a given quantity of water. Find the ratio of the depth to the width so that the cost of lining the tank with lead is least.
15. A cylindrical water tank is to be constructed to hold a maximum of n litres. Find the height and radius of the tank if the cost of the material used is proportional to the total surface area of the tank.
16. One corner of a rectangular sheet of paper of width w cm is folded over so as to reach the opposite edge. Show that the minimum length of the crease is $w\sqrt{2}$ cm.
17. Show that the height of closed cylinder of given volume and least surface is equal to its diameter.
18. Prove that a conical vessel of a given storage capacity will require the least material when its height is $\sqrt{2}$ times the radius of the base. (Nanyang, 2006)
19. Show that the semi-vertical angle of a cone of maximum volume and given slant height is $\tan^{-1}\sqrt{2}$.
20. The shape of a Polo mint is that of a cylinder with a hemisphere on each end. If the volume of the mint is fixed, find the ratio of the length of the cylinder to the diameter of the hemisphere when the surface area of the mint is a minimum. Hence show that the ratio of the surface area of the mint to its volume is $\frac{3}{2}(1 + \sqrt{2})$. (Nanyang, 2003)
21. A cylinder is inscribed in a cone of height h. If the volume of the cylinder is maximum, show that its height is $\frac{1}{3}h$.
22. Show that the volume of the largest right circular cone that can be inscribed in a sphere of radius r is $\frac{8}{27}$ of the volume of the sphere.
23. A given quantity of metal is to be cast into a half cylinder with a rectangular base and semi-circular ends. Show that in order that the total surface area may be a minimum, the ratio of the length of the cylinder to the diameter of its semi-circular ends is $\pi : (\pi + 2)$.
24. A person lying in a boat 3 miles from the nearest point of the beach, wishes to reach in the shortest possible time a point 5 miles from that point along the shore. The rate of walking is 4 miles per hour and the rate of rowing is 2 miles per hour. Show that he should land at a distance $\frac{3}{\sqrt{5}}$ miles from the point he wishes to reach.
25. The cost per hour of propelling a ship is proportional to the cube of the speed. Show that the cost of a journey against a current is $\frac{v^3}{v^3 - u^3}$ times the cost in still water, where v is the speed of the ship in still water and u is the speed of the current. If the cost per hour in still water is $\$k$, find the total cost of the journey.

[illegible] [illegible]

(2) **[illegible]** [illegible] [illegible] [illegible]

[illegible]

[illegible]

[illegible]

$$\frac{a^2}{3!}\phi_3''(0)+\frac{a^2}{2!}\phi_2'(0)+a\phi_1'(0)=a,\ \phi(0)=0$$

or
$$\frac{a^3}{6}2(0)+\frac{a^2}{2}(0)+a(-2)+2=0 \quad \text{or} \quad a^3+2a^2-2a+2=0.$$

Solving for a, we have $a=2,-2\pm\sqrt{2}$.

Hence the three asymptotes are

$$y=-x+2,\ y=-x-2+\sqrt{2},\ x+y=-2-\sqrt{2}.$$

4. **Asymptotes of polar curves.** It can be shown that an asymptote of the curve $f(r,\theta)=0$ is $r\sin(\theta-\alpha)=1/f'(\alpha)$, where α is a root of the equation $f(0)=0$ and $f'(\alpha)$ is the derivative of $f(r,\theta)=0$ w.r.t. θ at $\theta=\alpha$.

Example 4.65. *Find the asymptotes of the curve* $r=a/\theta$.

Equation of the curve can be written as $f(r,\theta)=r\theta-a=0$, etc.

$\ldots$ The asymptote is $r\sin(\theta-0)=1/f'(0)$, or $r\sin\theta=-a$.

EXERCISE 4.14

Find the asymptotes of

1. $x^3+y^3=3axy$ (Agra, 2003)
2. $(x+y)^2(x+2y+1)=1$ (Rajasthan, 2002)
3. $(x^2+y^2)(x-y)^2=x^2+y^2-1$
4. $x^3+2x^2y-xy^2-2y^3+y^2+x+y+1=0$
5. $x^3+y^3-3axy=0$
6. $(x+y)^2(x+2y+3)=(x+2y-1)$

7. Show that the asymptotes of the curve $xy^2=a^2(x+y)$ form a square of side $4a$.
8. Find the asymptotes of the curve $(x+y)^2(x-y)=x^2+y^2+y$ and show that they cut the curve again in points lie on the line $x+y+1=0$. (Kurukshetra, 2003)

Find the asymptotes of the following curves

10. $r=a\tan 2\theta$ (Madras, 2000)
11. $r\sin n\theta=a\cos n\theta+b$
12. $r\cos\theta=a\cos 2\theta$

[3] CURVE TRACING

In many practical applications, a knowledge about the shape of a given equation is desirable. On drawing a sketch of the given equation, we can easily study the behaviour of the curve as regards its symmetry, asymptotes, the number of times it passes through a point etc.

A point through which two branches of a curve pass is called a **double point**. At such a point, the curve has two tangents, one for each branch.

If the tangents are real and distinct, the double point is called a **node** (Fig. 4.22 (a)).

If the tangents are real and coincident, the double point is called a **cusp** (Fig. 4.22 (b)).

If the tangents are imaginary, the double point is called a **conjugate point** or an **isolated point**. Such a point cannot be shown in the figure.

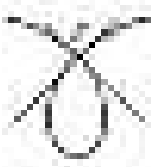

Fig. 4.22 (a)

Fig. 4.22 (b)

(1) Procedure for tracing cartesian curves.

1. **Symmetry.** *If all the powers of* y *in the curve are even, the curve is symmetrical about the* x*-axis.*

If a curve is symmetrical about the x*-axis, if only even powers of* y *occur in its equation.*

e.g., $y^2=4ax$ *is symmetrical about* x*-axis.*

(iii) A curve is symmetrical about the y-axis, if only even powers of x occur in the equation.
e.g., $x^2 = 4a$ is symmetrical about y-axis.

(iv) A curve is symmetrical about the line $y = x$, if on interchanging x and y the equation remains unchanged.
e.g., $x^3 + y^3 = 3axy$ is symmetrical about the line $y = x$.

2. Origin. (i) *See if the curve passes through the origin.*

(ii) A curve passes through the origin if there is no constant term in its equation.

(iii) If it does, find the equation of the tangent or tangents by equating to zero the lowest degree terms.

(iv) If the origin is a double point, find whether the origin is a node, cusp or conjugate point.

3. Asymptotes. (i) *See if the curve has any asymptotes parallel to the axes* (p. 183).

(ii) Then find the inclined asymptotes, if any (p. 183).

4. Points. (i) *Find the points where the curve crosses the axes and the asymptotes.*

(ii) *Find the points where the tangent is parallel or perpendicular to the x-axis,*
i.e. the points where $dy/dx = 0$ or ∞.

(iii) *Find the region (or regions) in which no portion of the curve exists.*

Example 4.16. *Trace the curve* $y^2 (2a - x) = x^3$. (P.T.U., 2010; K.U.K., 2008; Rajasthan, 2008; U.P.T.U., 2006)

Solution. (i) **Symmetry:** The curve is symmetrical about the x-axis.

$\because$ only even powers of y occur in the equation.

(ii) **Origin :** The curve passes through the origin.

$\because$ the curve is unsymmetrical terms in the equation.

The tangents at the origin are $y = 0, y = 0$ (Equating to zero the lowest degree terms.)

$\therefore$ Origin is a cusp.

(iii) **Asymptote :** The curve has an asymptote $x = 2a$.

$\because$ on solving y^2 is absent, as all of y^2 is an asymptote.

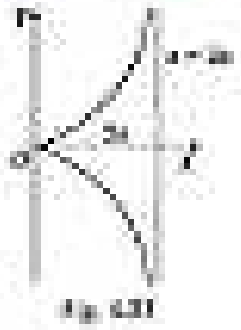

Fig. 4.23

(iv) **Points :** (a) curve meets the axes at $(0, 0)$ only. (b) $y^2 = x^2 (2a - x)$

When $x = -\infty$, y^2 is $+ve$ i.e. y is imaginary so that no portion of the curve lies to the left of the y-axis. Also when $x > 2a$, y^2 is again $-ve$, so that no portion of the curve lies to the right of the line $x = 2a$.

Hence, the shape of the curve is as shown in Fig. 4.23. This curve is known as *Cissoid*.

Example 4.17. *Trace the curve* $a^2 x^2 = y^2 (a^2 - x^2)$. (V.T.U., 2010; R.G.P.V., 2005)

Solution. (i) **Symmetry:** The curve is symmetrical about the x-axis.

(ii) **Origin :** The curve passes through the origin and the tangents at the origin are $a^2 x^2 = y^2$,

i.e. $y = x$ and $y = -x$. $\therefore$ Origin is a node.

(iii) **Asymptotes :** The curve has no asymptotes at all.

(iv) **Points :** (a) When $x = 0, y = 0$; when $y = 0, x = 0$ or $-a$.

$\therefore$ The curve meets the axes at $(0, 0)$ and $(-a, 0)$.

We have $y = \pm x\sqrt{\dfrac{a + x}{a - x}}$

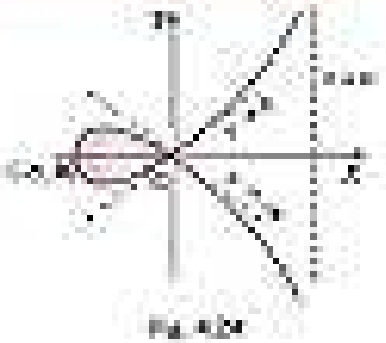

Fig. 4.24

When $x > a$ or $< -a$, y is imaginary.

$\therefore$ No portion of the curve lies to the right of the line $x = a$ or to the left of the line $x = -a$.

Hence the shape of the curve is as shown in Fig. 4.24. This curve is known as *Strophoid*.

Example 4.18. *Trace the curve* $x = a(t^2 - 1), y = at(t^2 - 1)$.

Solution. (i) **Symmetry:** The curve is symmetrical about y-axis.

(ii) **Origin :** It passes through the origin and the tangent at the origin is $y = 0$ (i.e., x-axis).

(v) Asymptotes : The asymptotes are given by $1 - x^2 = 0$ i.e. $x = \pm 1$ and $y = -2$.

(vi) Points : (a) The curve crosses the axes at the origin only. (b) When $x \to 1$ from left, $y \to +\infty$

When $x \to 1$ from right, $y \to -\infty$

When $x > 1$, y is $-ve$

Hence the curve is as shown in Fig. 4.26.

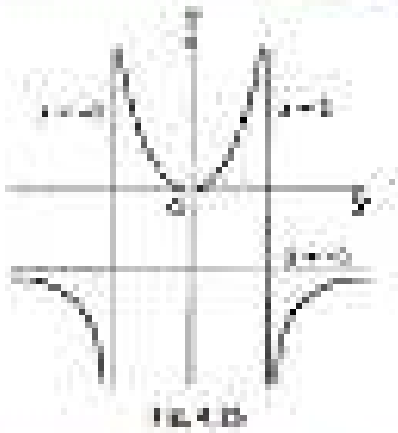

Fig. 4.26

Example 4.31. Trace the curve $a^2y^2 = x^2(a^2 - x^2)$. (U.P.T.U. 2001, U.P.T.U. 2008 (C))

Solution. (i) Symmetry. The curve is symmetrical about y-axis, y-axis and origin.

(ii) Origin. The curve passes through the origin and the tangents at the origin are $a^2y^2 = a^2x^2$ i.e., $y = \pm x$.

(iii) Asymptotes. The curve has no asymptote.

(iv) Points. (a) The curve cuts x-axis ($y = 0$) at $x = 0$, i.e. and cuts y-axis ($x = 0$) at $y = 0$ i.e., $(0, 0)$ only.

(b) $\frac{dy}{dx} = \frac{x(a^2 - 2x^2)}{a^2y}$ $\to \infty$ at $(a, 0)$

Fig. 4.27

i.e., tangent to the curve at $(a, 0)$ is parallel to y-axis. Similarly the tangent at $(-a, 0)$ is parallel to y-axis.

(c) We have $y = \frac{x}{a}\sqrt{a^2 - x^2}$ which is real for $a^2 - x^2 \geq 0$ i.e., $-a \leq x \leq a$

$\therefore$ The curve lies between $x = a$ and $x = -a$.

Hence the shape of the curve is as shown in Fig. 4.28.

Example 4.32. Trace the curve $y = x^3 - 12x - 16$. (U.P.T.U. 2006)

Solution. (i) Symmetry. The curve has no symmetry.

(ii) Origin. It doesn't pass through the origin.

(iii) Asymptotes : The curve has no asymptote.

(iv) Points. (a) The curve cuts y-axis ($y = 0$) at $(-2, 0)$, $(4, 0)$ and cuts y-axis ($x = 0$) at $(0, -16)$.

(b) $\frac{dy}{dx} = 3x^2 - 12$

At $(-2, 0)$, $\frac{dy}{dx} = 0$ i.e., tangent is parallel to x-axis at $(-2, 0)$

At $(4, 0)$, $\frac{dy}{dx} = 36$ i.e., $\tan\theta = 36$ i.e., tangent makes an acute angle $\tan^{-1} 36$ with x-axis at $(4, 0)$.

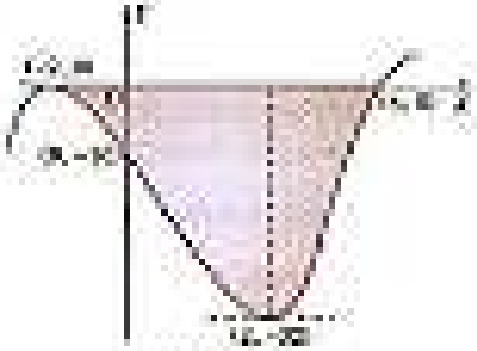

Fig. 4.27

Also $\frac{dy}{dx} = 0$ i.e., $3x^2 - 12 = 0$ or $x = \pm 2$ i.e., tangent is also parallel to x-axis at $(2, -32)$.

(c) $y \to +\infty$ as $x \to +\infty$ and $y \to -\infty$ as $x \to -\infty$ i.e., y is $+ve$ for $x > 4$ and y is $-ve$ for $x < 4$.

Hence the shape of the curve is as shown in Fig. 4.27.

Example 4.33. Trace the curve $3ay^2 = (x - 2a)(x - a)^2$. (U.P.T.U. 2006)

Solution. (i) Symmetry. The curve is symmetrical about the x-axis.

(ii) Origin. The curve doesn't pass through the origin.

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(v) Asymptotes: It has no asymptotes.

(vi) Points: (a) The curve cuts the y-axis at $y = 0$ at $x = 2a$, and $y = 4a$ i.e., at $A(2a, 0)$ and $B(4a, 0)$.

It cuts the y-axis at $x = 0$ at $y^2 = -8a^2$, i.e., y is imaginary.

So the curve doesn't cut the y-axis.

(b) $y = \frac{(x - 3a)\sqrt{(x - 2a)}}{2\sqrt{a}}$ i.e., y is imaginary for $x < 2a$. So the curve exists only for $x \geq 2a$.

(vii) $\frac{dy}{dx} = \pm \frac{x - 3a}{2\sqrt{a}\sqrt{(x - 2a)}}$

At $A(2a, 0)$, $\frac{dy}{dx} \to \infty$ i.e., tangent is parallel to y-axis.

At $B(4a, 0)$, $\frac{dy}{dx} = \pm \frac{1}{\sqrt{2}}$ i.e., there are two distinct tangents.

So there is a node at $A(2a, 0)$.

Hence the shape of the curve is as shown in Fig. 4.26.

Fig. 4.26

Example 4.25. Trace the curve $x^3 + y^3 = 3axy$ (Kurukshetra, 2000; G.B.T.U., 2004)

or $r = \frac{3a \cos\theta \sin\theta}{\cos^3\theta + \sin^3\theta}$

Solution. (i) *Symmetry:* The curve is symmetrical about the line $y = x$.

(ii) *Origin:* It passes through the origin and tangents at the origin are $xy = 0$, i.e., $x = 0$, $y = 0$.

$\therefore$ Origin is a node.

(iii) *Asymptotes:* (a) It has no asymptote parallel to the axes.

(b) Putting $y = vx$ and $x = 1$ in the third degree terms,

$$\phi_3(v) = 1 + v^3, \phi_3(v) = 0 \text{ gives } v = -1.$$

$$\therefore \quad c = -\frac{\phi_2(v)}{\phi_3'(v)} = -\left(\frac{-3av}{3v^2}\right) = \frac{a}{v},$$

$v = -1$, which is $x = -1$.

Hence $y = -x - a$ $\left(\text{i.e., } \frac{x}{-a} + \frac{y}{-a} = 1\right)$ is an asymptote.

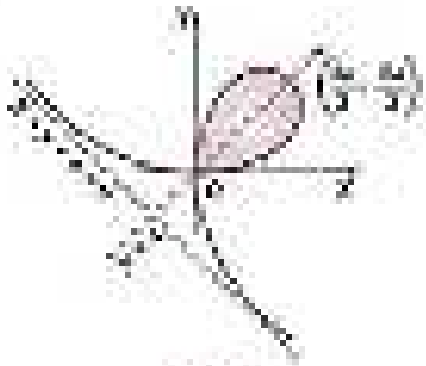


Fig. 4.27

(iv) *Points:* (a) It meets the axes at the origin only.

(b) When $y = x$, $2x^3 = 3ax^2$, i.e., $x = 0$ or $\frac{3a}{2}$, i.e., the curve crosses the line $y = x$ at $(0, 0)$, $\left(\frac{3a}{2}, \frac{3a}{2}\right)$.

Hence the shape of the curve is as shown in Fig. 4.27. This curve is known as *Folium of Descartes*.

Example 4.26. Trace the curve $x^4 + y^4 = 2x^2$.

Solution. (i) *Symmetry:* The curve has no symmetry.

(ii) *Origin:* The curve passes through the origin and the tangents at the origin are $x = 0$ and $x = 1$.

$\therefore$ The origin is a cusp.

(iii) *Asymptotes:* (a) The curve has no asymptotes parallel to the axes.

(b) Putting $x = 1$, $y = v$ in the third degree terms, we get

$$\phi_3(v) = v^4 + 1; \quad \therefore \quad \phi_3(v) = 0, \text{ gives } v = -1,$$

$$\therefore \quad c = -\frac{\phi_2(v)}{\phi_3'(v)} = -\frac{-2v}{4v^3} = \frac{1}{2v} \text{ for } v = -1.$$

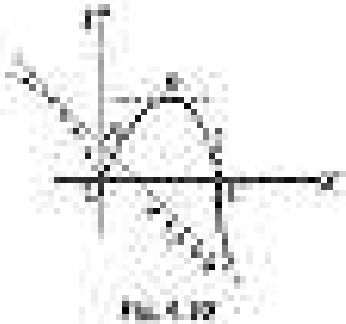


Fig. 4.28

Thus $x + y = \frac{1}{2}$ is the only asymptote.

The curve lies above the asymptote when x is positive and large and it lies below the asymptote when x is negative.

(iv) Points: (a) The curve crosses the axes at $O(0,0)$ and $C(3a,0)$. It crosses the asymptote at $A(a/2, 3a/2)$.

(b) Since $y^2 \cdot \frac{dy}{dx} = x(3a - x)$ $\therefore$ $\frac{dy}{dx} = 0$ for $x = 3a$.

(c) Now $y = x^{1/2}(3a - x)^{1/2}$.

When $0 < x < 3a$, y is positive. As x increases from 0, y also increases till $x = 3a$ where the tangent is parallel to the x-axis. As x increases from $3a$ to ∞, y constantly decreases to zero.

When $x > 3a$, y is negative.

When $x < 0$, y is positive and constantly increases as x varies from 0 to $-\infty$.

Combining all these facts we see that the shape of the curve is as shown in Fig. 4.31.

Example 4.28. Trace the curve $y^2(a + x) = x^2(a - x)$.

Solution. (i) *Symmetry:* The curve is symmetrical about the x-axis.

(ii) *Origin:* The curve passes through the origin and the tangents at the origin are $y^2 = -x^2$ i.e., $y = \pm ix$, which are imaginary lines. $\therefore$ The origin is an isolated point.

(iii) *Asymptotes:* (a) $x = -a$ is the only asymptote parallel to the y-axis.

(b) Putting $x = 1$ and $y = mx$ in the third degree terms, we get:

$$\phi_3(m) = m^2 + 1$$

$$\phi_2(m) = 0 \text{ gives } m = \infty$$

$$c = -\frac{\phi_2(m)}{\phi_3'(m)}$$

$$= -\frac{-a(m^2+1)}{2m}$$

$$= \infty \text{ for } m = \infty.$$

Thus the above two asymptotes are $y = x + a$; $y = -x - a$.

(iv) *Points:* (a) The curve crosses the axes at $(-a, 0)$ and $(0, 0)$.

It crosses the asymptote $y = x + a$ at $(-a, 0)$ and $y = -x - a$ at $(-a, 0)$.

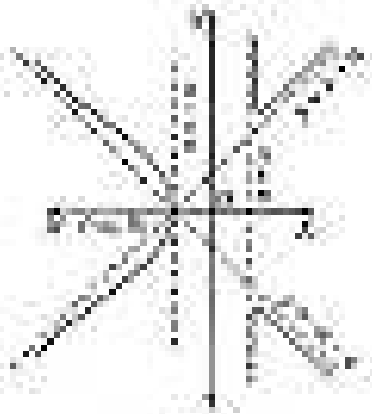


Fig. 4.32

(b) $y = \pm x\sqrt{\left(\frac{a+x}{a-x}\right)}$

When $x < a$ and $x \to a$, $y \to \pm\infty$ (i.e. far away).

$\therefore$ no portion of the curve lies between the lines $x = a$ and $x = -a$. Thus the vertical asymptote must be approached from the right.

(c) $\frac{dy}{dx} = \pm \frac{x^2 - ax + a^2}{(x-a)^{1/2}(x+a)^{3/2}}$

$\therefore$ $\frac{dy}{dx} = 0$, when $x = \frac{1}{2}(1 \pm \sqrt{5})a = 1.6a$ approx.

[rejecting the value $\frac{1}{2}(1 - \sqrt{5})a$ which lies between $-a$ and a]

and $\frac{dy}{dx} \to \infty$, when $x = a$.

Thus the tangent is parallel to the y-axis at $x = 1.6a$ and perpendicular to the x-axis at $x = a$.

Hence the shape of the curve is as shown in Fig. 4.31.

[I] PROCEDURE FOR TRACING CURVES IN PARAMETRIC FORM ($x = f(t)$ and $y = \phi(t)$)

1. *Symmetry.* See if the curve has any symmetry.

(i) A curve is symmetrical about the x-axis, if on replacing t by $-t$, $f(t)$ remains unchanged and $\phi(t)$ changes to $-\phi(t)$.

(ii) A curve is symmetrical about the y-axis if on replacing t by $-t$, $f(t)$ changes to $-f(t)$ and $\phi(t)$ remains unchanged.

(iii) A curve is symmetrical in the opposite quadrants if on replacing t by $-t$, both $f(t)$ and $\phi(t)$ remain unchanged.

2. **Limits.** Find the greatest and least values of x and y which determine the straight parallels to the axes, within or outside which the curve lies.

3. **Points.** (a) Determine the points where the curve crosses the axes.

The points of intersection of the curve with the x-axis are given by the roots of $f(x) = 0$, while those with the y-axis are given by the roots of $f(y) = 0$.

(b) Using the values of x (or y), plot the corresponding values of y and $\frac{dy}{dx}$, making a table as shown below. Use the relationship between $\frac{dy}{dx}$ and $\frac{dy}{dx}$ to find the different values of $\frac{dy}{dx}$.

(c) Draw in the points above the tangents or parallel or perpendicular to the x-axis, i.e., where $\frac{dy}{dx} = 0$ or ∞.

(d) When x and y are periodic functions of t with a common period, we need to study the curve only for one period, because the other values of t will repeat the same curve over and over again.

Note. Sometimes it is convenient to find the relation between the given equations in x and y by eliminating the parameter t before we trace the curve.

Example 4.77. *Trace the curve $x = a\cos^3\theta$, $y = a\sin^3\theta$ (or $x^{2/3} + y^{2/3} = a^{2/3}$).*
[A.M.I.E. (Summer) 2000; U.P.T.U. 2002; P.T.U. 2003]

Solution. (i) *Symmetry.* The curve is symmetrical about the x-axis.

For, on changing θ to $-\theta$, x remains unchanged but y changes to $-y$.

(ii) *Limits.* $\because$ $|x| \le a$ and $|y| \le a$.

$\therefore$ The curve lies entirely within the square bounded by the lines $x = \pm a$, $y = \pm a$.

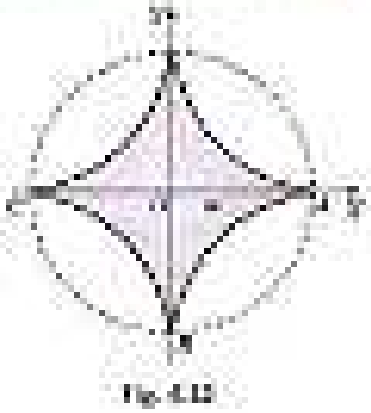


Fig. 4.32

(iii) *Points :* We have $\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta$,

$\frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$, $\frac{dy}{dx} = -\tan\theta$.

$\therefore$ $\frac{dy}{dx} = 0$ when $\theta = 0$ or π

and $\frac{dy}{dx} = \infty$, when $\theta = \pm\frac{\pi}{2}$.

The following table gives the corresponding values of θ, x, y and $\frac{dy}{dx}$:

As θ increases	x	y	$\frac{dy}{dx}$ varies	Portion traced
from 0 to $\frac{\pi}{2}$	$+ve$ and decreases from a to 0	$+ve$ and increases from 0 to a	from 0 to $-\infty$	AB
from $\frac{\pi}{2}$ to π	$-ve$ and increases numerically from 0 to $-a$	$+ve$ and decreases from a to 0	from $+\infty$ to 0	BC

As θ increases from π to 2π, we get the reflection of the curve ABC in the x-axis. The values of $\theta > 2\pi$ give no new points.

Hence the shape of the curve is as shown in Fig. 4.32 and is known as *Astroid*.

Example 4.78. *Trace the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$.* *[A.M.I.E.T.E. (Summer) 2000]*

Solution. (i) *Symmetry.* The curve is symmetrical about the y-axis.

For, on changing θ to $-\theta$, x changes to $-x$ and y remains unchanged.

Thus we may consider the curve only for positive values of θ, i.e., for $\theta \ge 0$.

(ii) *Limits.* The greatest value of y is $2a$ and the least value is zero.

Hence the curve lies entirely between the lines $y = 2a$ and $y = 0$.

(iii) *Points.* We have

$\frac{dx}{d\theta} = a(1 + \cos\theta)$, $\frac{dy}{d\theta} = a\sin\theta$ and $\frac{dy}{dx} = \tan\frac{\theta}{2}$

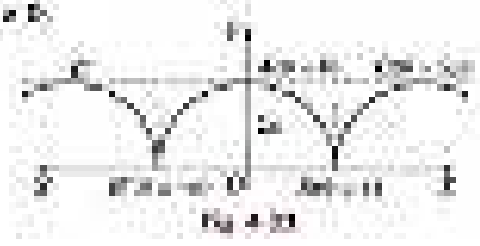


Fig. 4.33

$\therefore \quad dy/d\theta = 0$ when $\theta = 0$ or 2π and $dy/d\theta < 0$ when $0 < \theta < \pi$.

The following table gives the corresponding values of θ, r, y and $\frac{dy}{d\theta}$:

As θ increases	r	y	$dy/d\theta$ is	Portion of curve
from 0 to $\frac{\pi}{2}$	increases from 0 to a	increases from 0 to a	> 0	OA
from $\frac{\pi}{2}$ to π	decreases from a to 0	increases from a to $2a$	< 0	AB

As θ decreases from 0 to $-\pi$, we get the reflection of the curve AB in the pole.

The curve consists of congruent arches extending to infinity in both the directions of the initial line in the intervals $\ldots(-3\pi, -\pi), (-\pi, \pi), (\pi, 3\pi)\ldots$

Hence the shape of the curve is as shown in Fig. 4.30 and it is known as Cycloid.

Obs. 1. Cycloid is the curve described by a point on the circumference of a circle which rolls without slipping on a fixed straight line. The fixed line is known as the base and the point (O) from it the curve of the cycloid.

The complete cycloid consists of the arch OAB and its endless repetitions on both sides.

2. Inverted cycloid: $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$.

The complete inverted cycloid consists of the arch OBA in the form of the inverted form of the curve as in Fig. 4.30. Here AB is the base and O the vertex of the cycloid. (Fig. 4.30A)

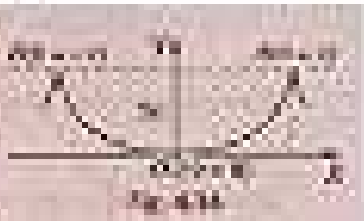


Fig. 4.30

4.17 (A) PROCEDURE FOR TRACING POLAR CURVES

1. **Symmetry.** See if the curve is symmetrical about any line.
   (i) A curve is symmetrical about the initial line OX, if only even powers of $\sin\theta$ occur in its equation, i.e., it remains unchanged when θ is changed to $-\theta$, e.g., $r = a\cos\theta + b\cos^2\theta$ is symmetrical about the initial line.
   (ii) A curve is symmetrical about the line through the pole $\perp$ to the initial line (i.e., OY), if only odd powers of $\sin\theta$ occur in its equation (i.e., if it remains unchanged when θ is changed to $\pi - \theta$, e.g., $r = a\sin 3\theta$ is symmetrical about OY.
   (iii) A curve is symmetrical about the pole, if only even powers of r occur in the equation (i.e., it remains unchanged when r is changed to $-r$), e.g., $r^2 = a^2\cos 2\theta$ is symmetrical about the pole.
2. **Limits.** See if r and θ are confined between certain limits.
   (i) Determine the numerically greatest value of r, i.e., the greatest value that the radius vector can attain, e.g., $r = a\sin 3\theta$ lies wholly within the circle $r = a$.
   (ii) Determine the region in which no portion of the curve lies by finding those values of θ for which r is imaginary, e.g., $r^2 = a^2\cos 2\theta$ is imaginary for values in the intervals $(\pi/4, 3\pi/4)$.
3. **Asymptotes.** If the curve possesses an infinite branch, find the asymptotes (§ 3.6).
4. **Points.** (i) Giving convenient values to θ, find the corresponding values of r.
   (ii) Determine the points where the tangent is $\perp$ or $\parallel$ to the radius vector or is perpendicular to it (i.e., the points where $\tan\phi = r\,d\theta/dr = 0$ or ∞).

Example 4.19. Trace the curve $r = a\sin 3\theta$. (U.P.T.U., 2003)

Solution. (i) *Symmetry.* The curve is symmetrical about the line through the pole $\perp$ to the initial line.

(ii) *Limits.* The curve totally lies within the circle $r = a$. It cuts the curve at $\theta = \frac{\pi}{3}, \frac{2\pi}{3}$.

(iii) *Asymptotes.* It has no asymptotes.

(iv) *Points.* (a) $\tan\phi = r\frac{d\theta}{dr} = \frac{a\sin 3\theta}{3a\cos 3\theta} = \frac{1}{3}\tan 3\theta$

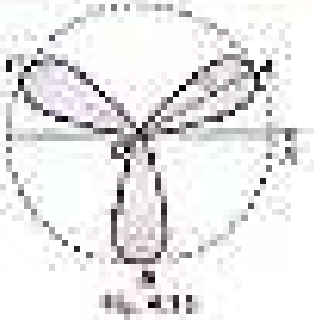


Fig. 4.31

$\therefore$ $\phi = 0, \pi$ when $\theta = \pi, 2\pi, \ldots$
$\phi = \pi/2$, when $\theta = \pi/2, 3\pi/2, \ldots$ — Shows the curve is symmetrical about the line $\theta = \pi/2$

(5) The following table gives the variations of r, θ and ϕ:

As θ varies from	r varies from	ϕ varies from	Portion traced out
0 to $\pi/2$	0 to a	0 to $\pi/2$	OAB
$\pi/2$ to π	a to 0	$\pi/2$ to π	BCO
π to $3\pi/2$	0 to $-a$	0 to $\pi/2$	OAB
$3\pi/2$ to 2π	$-a$ to 0	$\pi/2$ to π	BCO

As θ increases from $\pi/2$ to π, portions of the curve from B to O, O to C and C to O are traced by symmetry about the line $\theta = \pi/2$.

Hence the curve consists of three loops as shown in Fig. 4.26 and is known as three-leaved rose.

Note. The curves of the form $r = a \sin n\theta$ or $r = a \cos n\theta$ are called rose curves.
(a) If n is an odd integer, there are n loops.
(b) If n is an even integer, there are $2n$ loops.

Example 4.25. Trace the curve $r = a \sin 2\theta$. *(Rohtak 2008)*

Solution. (1) Symmetry. The curve is symmetrical about the line through the pole $\perp$ to the initial line.
(2) Zeros : The curve lies wholly within the circle $r = a$ (3) $r = a$ is never $< a$

(4) Points (a) As θ increases from	r increases from	Loop
0 to $\frac{\pi}{4}$	0 to a	
$\frac{\pi}{4}$ to $\frac{\pi}{2}$	a to 0	OA
$\frac{\pi}{2}$ to $\frac{3\pi}{4}$	0 to $-a$	
$\frac{3\pi}{4}$ to π	$-a$ to 0	OB

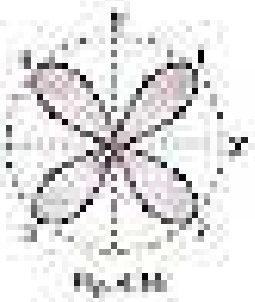


Fig. 4.26

(b) $\tan\phi = r\frac{d\theta}{dr} = \frac{1}{2}\cot 2\theta$

$\therefore$ $\phi = 0$, when $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \ldots$

$\phi = \frac{\pi}{2}$, when $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \ldots$

Hence, the shape of the curve is as shown in Fig. 4.26.

Example 4.27. Trace the curve $r^2 = a^2 \cos 2\theta$. *(U.P.T.U., 2000; Kurukshetra 2006; U.P.T.U., 2006)*

Solution. (1) Symmetry. The curve is symmetrical about the pole.
(2) Zeros : (a) The curve lies wholly within the circle $r = a$.
(b) No portion of the curve lies between the lines $\theta = \pi/4$ and $\theta = 3\pi/4$.

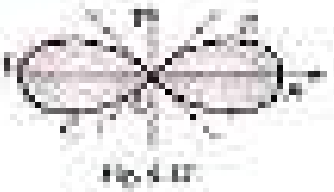


Fig. 4.27

(3) Points : (a) $\tan\phi = r\frac{d\theta}{dr} = -\cot 2\theta = \tan\left(\frac{\pi}{2} + 2\theta\right)$

i.e., $\phi = \frac{\pi}{2} + 2\theta$ $\therefore$ $\phi = 0$, when $\theta = -\pi/4$; $\phi = \pi/2$ when $\theta = 0$.

Thus, the tangent at O is $\theta = -\pi/4$ and the tangent at A is $\perp$ to the initial line.

(b) The variation of r and θ are given below :

As θ varies from	r varies from	Portion traced
0 to $\frac{\pi}{2}$	a to 0	ABO
$\frac{\pi}{2}$ to π	0 to a	OCD

As θ increases from π to 2π, we get the reflection of the arc ADEOA in the initial line. Hence the shape of the curve is as shown in Fig. 4.35. This curve is known as *Lemniscate of Bernoulli*.

Example 4.33. Trace the curve $r = a + b\cos\theta$ ($a < b$).

Solution. (a) *Symmetry.* It is symmetrical about the initial line.

(b) *Limits :* The curve wholly lies within the circle $r = a + b$

(c) r is never $a - b$.

(d) *Points :* (i) when $a > b$.

As θ increases from 0 to $\frac{\pi}{2}$; r decreases from $a + b$ to a

As θ increases from $\frac{\pi}{2}$ to π ; r decreases from a to $a - b$

The shape of the curve is as shown in Fig. 4.36 (i).

(ii) when $a = b$.

As θ increases from 0 to $\frac{\pi}{2}$; r decreases from $a + b$ to a

As θ increases from $\frac{\pi}{2}$ to π ; r decreases from a to 0

As θ increases from π to 2π ; r decreases from 2 to $a - b$

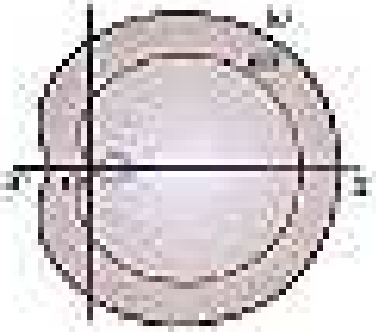


Fig. 4.36

where $\alpha = \cos^{-1}\left(-\frac{a}{b}\right)$

In this case, the curve consists of two parts, one of which forms a loop within the other and the shape is as shown in Fig. 4.36 (ii).

Example 4.34. Trace the curve $r\theta = a$. (Spiral)

Solution. (a) *Symmetry.* There is no symmetry.

(b) *Limits :* There are no limits to the values of r.

The curve does not pass through the pole for values of the variable θ for any real value of θ.

(c) *Asymptotes :* $\frac{1}{r} - \frac{\theta}{a} = 0$ (i)

$f(\theta) = 1$ for $\theta = 0$; $f'(\theta) = 0$, $f''(\theta) = 0$.

$\therefore$ Asymptote is $r\sin(\theta - 0) = 1/f'(0)$

i.e., $y = r\sin\theta = a$ is an asymptote.

(d) *Points :* As θ increases from 0 to ∞, r is positive and decreases from ∞ to 0.

Hence the shape of the curve is as shown in Fig. 4.37.

Fig. 4.37

Example 4.35. Trace the curve $x^3 + y^3 = 3axy$.

Solution. (a) *Symmetry.* The curve is symmetrical about the line $y = x$.

[On interchanging x and y, it remains unchanged.]

(b) *Origin :* It passes through the origin and the tangents at the origin are given by

$$a^3y^3 = 0, \text{ i.e., } x = 0, x = 0 \; ; \; y = 0, y = 0.$$

Hence the curve has both x-axis and the y-axis at the origin.

(c) *Asymptotes :* (a) It has no asymptotes parallel to the axes.

(b) Putting $x = 1, y = m$ in the fifth degree terms, we get

$\phi_5(m) = 1 + m^3 = 0$ $\therefore$ $\phi_5(m) = 0$ gives $m = -1$.

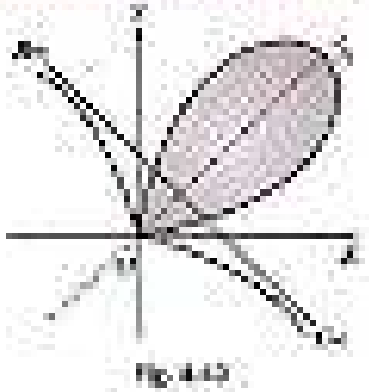


Fig. 4.43

$$\therefore \quad c = -\frac{f_3(a)}{f_2(a)} = -\frac{8a^3}{3a^2} = -\frac{8}{3}a \quad \text{as } a \to \infty, \; c \to -\infty$$

Thus $y = -x + c$ or $x + y = c$ is an asymptote.

Solution (Polar form): Since it is not convenient to express x as a function of y or vice versa, hence we change the equation into polar coordinates by putting $x = r\cos\theta$ and $y = r\sin\theta$. The equation of the curve becomes

$$r = \frac{8a\sin^2\theta\cos^2\theta}{\cos^3\theta + \sin^3\theta} = \frac{2a}{3} \cdot \frac{\sin^2 2\theta}{\cos^3\theta + \sin^3\theta}$$

θ increases from	r	r is
0 to $\frac{\pi}{4}$	increases from 0 to $\frac{2a}{3}$	positive
$\frac{\pi}{4}$ to $\frac{\pi}{2}$	decreases from $\frac{2a}{3}$ to 0	positive
$\frac{\pi}{2}$ to $\frac{3\pi}{4}$	increases from 0 to ∞	negative
$\frac{3\pi}{4}$ to π	decreases from ∞ to 0	negative

As θ increases from π to 2π, the curve will trace out.

Hence the shape of the curve is as shown in Fig. 4.46.

PROBLEM SET 4.1

Trace the following curves:

1. $x^3 + y^3 = 3axy$ (L.U. Nov./Dec. 2004, May/June 2006, Nov./Dec. 2007)
2. $y^2(a^2 + x^2) = x^2(a^2 - x^2)$ (U.P.T.U., 2006)
3. $y = x^2 + 16/x^2 - 11$
4. $x^2y^2 = a^2(y^2 - x^2)$ (Kurukshetra, 2008; U.P.T.U., 2006)
5. $x^3 + y^3 = 3(x^2 - y^2)$
6. $x = a\cos^3\theta, y = b\sin^3\theta$
7. $x = a(\theta - \sin\theta), y = a(1 - \cos\theta)$ and $\theta \in [0, 4\pi]$
8. $x = a(\cos t + \log\tan\frac{t}{2}), y = a\sin t$
9. $r = a\cos 2\theta$
10. $r = a\cos\theta$
11. $r = a(1 - \cos\theta)$
12. $r = 2 + 3\cos\theta$
13. $r^2\cos 2\theta = a^2$ (U.P.T.U., 2005)

Hint: Changing to Cartesian form $x^2 - y^2 = a^2$. This is rectangular hyperbola with asymptotes $x + y = 0$ and $x - y = 0$.

4.15 OBJECTIVE TYPES OF QUESTIONS

PROBLEM SET 4.2

Select the correct answer or fill up the blanks in each of the following questions:

1. The radius of curvature of the catenary $y = c\cosh(x/c)$ at the point where it crosses the y-axis is _____.
2. The envelope of the family of straight lines $y = mx + a/m$, where m is a parameter is _____.
3. The curvature of the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ is _____.
4. The value of $\lim_{x \to 0} \frac{\log(\sin x)}{(\pi - 2x)^2}$ is

   (a) zero (b) $\frac{1}{8}$ (c) $-\frac{1}{8}$ (d) ∞ (U.P.T.U., 2004)

5. Taylor's expansion of the function $f(x) = \frac{1}{1+x}$ is

(a) $\sum_{n=2}^{\infty} (-1)^n \frac{x^{2n}}{n!}$ for $-\infty < x < \infty$ (b) $\sum_{n=0}^{\infty} x^{2n}$ for $-1 < x < 1$

(c) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n}}{n}$ for $-1 < x \leq 1$ (d) $\sum_{n=0}^{\infty} (-1)^n x^{2n}$ for $-1 < x < 1$.

6. A triangle of maximum area inscribed in a circle of radius r is
(a) is right angled (b) angle with hypotenuse subtending $\frac{\pi}{3}$
(c) is an equilateral triangle
(d) is an isosceles triangle of side r
(e) does not exist

7. The minimum value of $|z^2|$ is
(a) 0 (b) $|z|^2$ (c) $|z|^3$ (d) 1

8. The percentage error in computing the area of an oil pan whose sides are each of 2 cm if each is in error by $\frac{1}{2}$ cm is
(a) 0.5% (b) 2% (c) 100% (d) 50%

9. The length of the tangent of the osculating hyperbola $x^2 - y^2 = a^2$ at the point $(a, \sqrt{2}a)$ is
(a) $\frac{1}{\sqrt{2}}a$ (b) $2a$ (c) $\frac{1}{2a}$ (d) $\frac{\sqrt{3}}{a^2}$

10. The length of subnormal to the curve $y = x^2$ at $(2, 8)$ is
(a) 127 (b) 32 (c) 36 (d) 34

11. If the normal to the curve $x^2 + 3y - 7 = 0$ at the point (x_1, y_1) is at the right angle to $-7x + 6y = 9$, then x_1 and y_1 are
(a) $(-4, 11)$ (b) $(6, -11)$ (c) $(-4, 19)$ (d) $(-4, -14)$

12. The radius of curvature of the curve $x = t^2$ at the point where it crosses the parabola is
(a) 9 (b) $\sqrt{2}$ (c) $(0, 0)$ (d) $\frac{5}{2}\sqrt{2}$

13. The equation of the osculating circle of $y^2 = x^3$ at $(1, 1)$ is
(a) $x^2 + y^2 - x - 2 = 0$ (b) $x^2 - y^2 + x - 1 = 0$ (c) $x^2 + y^2 + x - 1 = 0$ (d) $x^2 - y^2 - x - 1 = 0$

14. If ψ be the angle between the tangent and radius vector at any point on the curve $r = f(\theta)$, then $\tan\psi$ equals to
(a) $\frac{dr}{d\theta}$ (b) $r\frac{d\theta}{dr}$ (c) $r\frac{dr}{d\theta}$

15. Envelope of the family of lines $x \cos\alpha + y \sin\alpha = a$ is _____.

16. The chord of curvature parallel to y-axis for the curve $y = x \log_e x$ is _____.

17. $\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \lim_{x \to 0} \frac{1}{2!} - \lim_{x \to 0} \frac{x^2}{4!} + \ldots$

18. The n^{th} derivative of $\log_e x$ is $\frac{1}{x}$ at $x =$ _____.

19. If $x^2 + y^2 - 3xy = 0$, find $\frac{d^2y}{dx^2}$ at $(3/2, 3/2)$ = _____.

20. When the tangent at a point on a curve is parallel to x-axis, then the curvature at that point is equal to the reciprocal of the ordinate of that point. (True or False)

21. If $x \sin\frac{1}{y} = 1$, find $\frac{dy}{dx}$ at $\left(\frac{2}{\pi}, \frac{1}{2}\right)$ = _____.

22. The value of $\frac{dy}{dx}$ for the curve $x = a(t - \sin t)$, $y = a(1 - \cos t)$ at $t = \frac{\pi}{2}$ is _____.

23. If $\frac{1}{R} = \frac{d^2y}{dx^2}$, then the curvature of the circle of curvature is $\frac{1}{R}$, then the equation of the circle of curvature is _____.

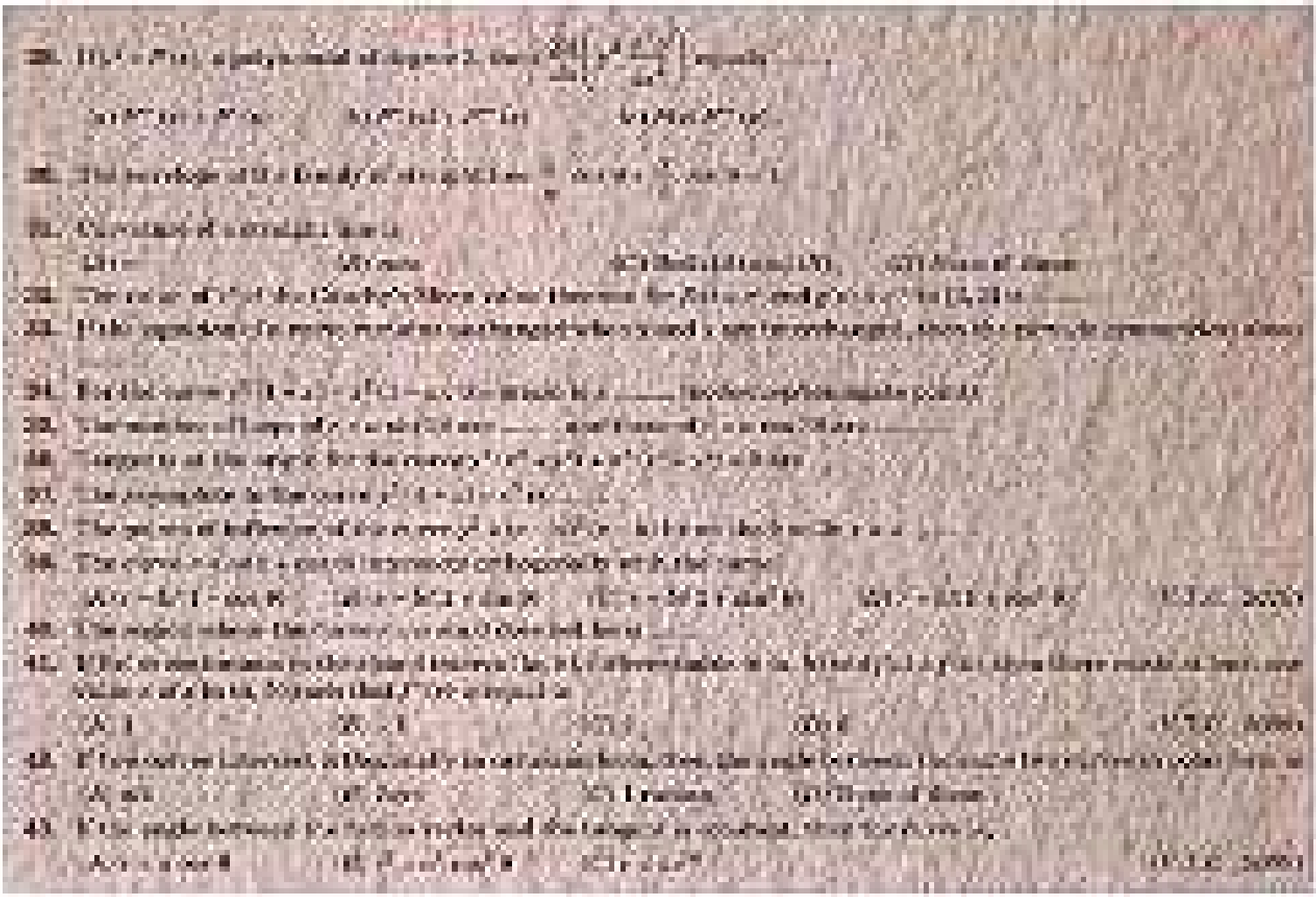
24. Circle is defined as the _____ of the curvature for a plane curve.

25. Envelope of the family of lines $\frac{x}{t} + yt = c$ (where t is the parameter) is _____.

26. The angle between the radius vector and tangent for the curve $r = a(1 + \cos\theta)$ is _____.

27. The subnormal of the parabola $y^2 = 4ax$ is _____.

28. The length of the arc $\int_0^{\pi/2} \sqrt{1 - \sin 2x}\, dx$ = _____.



CHAPTER 5

Partial Differentiation and Its Applications

5.1 Functions of Two or More Variables

If a function is continuous at all points of a region, then it is said to be continuous in that region. A function which is not continuous at a point is said to be discontinuous at that point.

Also, basically, the limit is the same irrespective of the path along which the point (x, y) approaches to $(1, 2)$ and

$$\lim_{x \to 1} \left[\lim_{y \to 2} f(x, y) \right] = \lim_{y \to 2} \left[\lim_{x \to 1} f(x, y) \right]$$

But (ii) does not always exist as in the following example above.

Ex. $\lim_{(x,y) \to (0,0)} \frac{x - y}{x + y}$ as $(x, y) \to (0, 0)$ along the line $y = mx$.

$$= \lim_{x \to 0} \frac{x - mx}{x + mx} = \frac{1 - m}{1 + m}$$ which is not fixed so the limit does not exist as $(x, y) \to (0, 0)$ along different paths.

Also $\lim_{x \to 0} \left[\lim_{y \to 0} \left[\frac{x - y}{x + y} \right] \right] = \lim_{x \to 0} (1) = 1$ whereas $\lim_{y \to 0} \left[\lim_{x \to 0} \left[\frac{x - y}{x + y} \right] \right] = \lim_{y \to 0} (-1) = -1$

Ex. As (x, y) is made to approach $(0, 0)$ along different paths, $f(x, y)$ approaches different limits. Hence the limit does not exist. Limits are not equal and $f(x, y)$ is discontinuous at the origin.

Also the function is not defined at $(0, 0)$. Hence $f(x, y) \to \infty$ for $x \to 0, y \to 0$.

NB: As in the case of functions of one variable, the following results hold:

I. If $\lim_{x \to a} f(x, y)$ and $\lim_{x \to a} g(x, y)$ exist,

then (i) $\lim_{x \to a} [f(x, y) \pm g(x, y)] = \lim_{x \to a} f(x, y) \pm \lim_{x \to a} g(x, y)$

(ii) $\lim_{x \to a} [f(x, y) \cdot g(x, y)] = \lim_{x \to a} f(x, y) \cdot \lim_{x \to a} g(x, y)$

(iii) $\lim_{x \to a} [f(x, y) / g(x, y)] = \lim_{x \to a} f(x, y) / \lim_{x \to a} g(x, y)$ provided $\lim_{x \to a} g(x, y) \neq 0$

II. If $f(x, y), g(x, y), h(x, y)$ are such that $f \leq g \leq h$ and $\lim_{x \to a} f(x, y) = \lim_{x \to a} h(x, y)$,

then $\lim_{x \to a} g(x, y)$ exists and $\lim_{x \to a} f(x, y) \leq \lim_{x \to a} g(x, y) \leq \lim_{x \to a} h(x, y)$

provided $g(x, y) \neq 0$ in the last case.

EXERCISE 11.1

Evaluate the following limits:

1. $\lim_{\substack{x \to 0 \\ y \to 0}} \frac{2x^2 y}{x^2 + y^2}$ 2. $\lim_{\substack{x \to 0 \\ y \to 0}} \frac{x^2 - y^2}{x^2 + y^2}$ 3. $\lim_{\substack{x \to 1 \\ y \to 1}} \frac{x^2 - y^2}{x^2 + y^2}$ 4. $\lim_{\substack{x \to 1 \\ y \to 1}} \frac{x(x - 1)}{x(x - 1)}$

5. If $f(x, y) = \frac{x - y}{x + y}$, show that $\lim_{x \to 0} \left[\lim_{y \to 0} f(x, y) \right] \neq \lim_{y \to 0} \left[\lim_{x \to 0} f(x, y) \right]$

Also show that the function is discontinuous at the origin.

6. Show that the function $f(x, y) = x^2 + 2y$ $(x, y) \neq (1, 2)$
   $3(x, y) = (1, 2)$ $= 4$
   is discontinuous at $(1, 2)$.
7. Investigate the continuity of the function
   $f(x, y) = x^2 y^2 / (x^2 + y^2)$ $(x, y) \neq (0, 0)$
   $= 0$ $(x, y) = (0, 0)$
   at the origin.

Note: In whatever follows, all the functions in problems are continuous and their partial derivatives are defined below/under.

11.2 PARTIAL DERIVATIVES

Let $z = f(x, y)$ be a function of two variables x and y.

If we keep y as constant and vary x alone, then z is a function of x only. The derivative of z with respect to x, treating y as constant, is called the partial derivative of z with respect to x and is denoted by one of the symbols

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) \text{ i.e., } f_x(x, y) \qquad \text{Thus } \frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

Similarly, the derivative of z with respect to x, keeping y as constant, is called the partial derivative of z with respect to y and is denoted by one of the symbols

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} f(x, y) \text{ i.e., } f_y(x, y) \qquad \text{Thus } \frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Similarly, if u is a function of three or more variables $x, y, z, \ldots$ the partial derivative of u with respect to x, is obtained by differentiating u with respect to x, keeping all other variables constant and is written as $\partial u / \partial x$.

In general f_x and f_y are also functions of x and y and so these can be differentiated further partially with respect to x and y.

Thus $$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial x^2} \text{ or } f_{xx}, \quad \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x \partial y} \text{ or } f_{xy}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y \partial x} \text{ or } f_{yx} \text{ and } \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial y^2} \text{ or } f_{yy}$$

It can easily be verified that, in all ordinary cases,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Sometimes we use the following notation

$$\frac{\partial z}{\partial x} = p, \quad \frac{\partial z}{\partial y} = q, \quad \frac{\partial^2 z}{\partial x^2} = r, \quad \frac{\partial^2 z}{\partial x \partial y} = s, \quad \frac{\partial^2 z}{\partial y^2} = t.$$

Example 3.1: Find the first and second partial derivatives of $z = x^3 + y^3 - 3xy$.

Solution. We have $z = x^3 + y^3 - 3xy$.

$\therefore$ $$\frac{\partial z}{\partial x} = 3x^2 + 0 - 3y(1) = 3x^2 - 3y, \text{ and } \frac{\partial z}{\partial y} = 0 + 3y^2 - 3x(1) = 3y^2 - 3x$$

Also $$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}(3x^2 - 3y) = 6x, \quad \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}(3x^2 - 3y) = -3$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}(3y^2 - 3x) = 6y, \text{ and } \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}(3y^2 - 3x) = -3$$

We observe that $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$.

Example 3.2: If $u = x^y \tan^{-1}\frac{y}{x} - y^x \tan^{-1}\frac{x}{y}$

show that $$\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}$$ (Kerala, 2005 S)

and $$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$ (Madras, 2000)

Solution. We have $\frac{\partial u}{\partial y} = x^y \cdot \frac{1}{1 + (y/x)^2} \cdot \frac{1}{x} + \left[\log x \cdot \tan^{-1}\frac{y}{x} + y^x \cdot \frac{1}{1 + (x/y)^2} \cdot \left(-\frac{x}{y^2}\right)\right]$

$$= \frac{x^2}{x^2 + y^2} - y^x \log x \cdot \tan^{-1}\frac{y}{x} - \frac{xy^2}{x^2 + y^2} + x - y^x \log x \cdot \tan^{-1}\frac{y}{x}.$$

---

[1] It is important to note that in the mixed partials the differentiations are written in the same order in which are differentiations to obtain them. If not explicitly the order is opposite.

$\therefore \quad \dfrac{\partial^2 z}{\partial x \partial y} = \dfrac{\partial}{\partial x}\left[x - 2y \tan^{-1}\dfrac{x}{y}\right] = 1 - 2y \cdot \dfrac{1}{1+(x/y)^2} \cdot \dfrac{1}{y} = 1 - \dfrac{2y^2}{x^2+y^2} = \dfrac{x^2-y^2}{x^2+y^2}$

Similarly, $\dfrac{\partial z}{\partial y} = 2x \tan^{-1}\dfrac{y}{x} - y$

and $\dfrac{\partial^2 z}{\partial y \partial x} = \dfrac{\partial}{\partial x}\left[2x \tan^{-1}\dfrac{y}{x} - y\right] = \dfrac{x^2-y^2}{x^2+y^2}$. Hence the result.

Example 2.8. If $z = x f(x+ct) + x g(x+ct) + h(x-ct)$, prove that $\dfrac{\partial^2 z}{\partial t^2} = c^2 \dfrac{\partial^2 z}{\partial x^2}$ (U.M.I.S., 2006; V.T.U., 2008)

Solution. We have $\dfrac{\partial z}{\partial x} = x f'(x+ct) + f(x+ct) + x g'(x+ct) + g(x+ct) + h'(x-ct)$

and $\dfrac{\partial^2 z}{\partial x^2} = x f''(x+ct) + f'(x+ct) + x g''(x+ct) + g'(x+ct) + h''(x-ct)$ ...(i)

Again $\dfrac{\partial z}{\partial t} = x f'(x+ct)\dfrac{\partial}{\partial t}(x+ct) + g'(x+ct)\dfrac{\partial}{\partial t}(x+ct) + h'(x-ct)\dfrac{\partial}{\partial t}(x-ct) = cx f'(x+ct) + cg'(x+ct) - ch'(x-ct)$

and $\dfrac{\partial^2 z}{\partial t^2} = c^2 x f''(x+ct) + c f'(x+ct) + c^2 g'(x+ct) - c h''(x-ct) = c^2[x f''(x+ct) + g'(x+ct)] + c f'(x+ct) - c h''(x-ct)$ ...(ii)

From (i) and (ii), it follows that $\dfrac{\partial^2 z}{\partial t^2} = c^2 \dfrac{\partial^2 z}{\partial x^2}$

Note. This is an important partial differential equation known as wave equation (§ 20.6).

Example 2.9. If $f(t) = t^n e^{-t^2/4t}$, where n is a constant, prove that $\dfrac{1}{t^2}\dfrac{\partial}{\partial t}\left(t^2 \dfrac{\partial f}{\partial t}\right) = \dfrac{\partial^2 f}{\partial t^2}$

(Nagpur, 2003; Kottayam, 2004) (U.T.U., 2010)

Solution. We have $\dfrac{\partial f}{\partial t} = t^n \cdot e^{-t^2/4t}\left(\dfrac{-2t}{4t^2}\right) = -\dfrac{1}{2}t^{n-1}e^{-t^2/4t}$

$\therefore \quad t^2\dfrac{\partial f}{\partial t} = -\dfrac{1}{2}t^{n+1}e^{-t^2/4t}$

and $\dfrac{\partial}{\partial t}\left(t^2 \dfrac{\partial f}{\partial t}\right) = -\dfrac{n+1}{2}t^n e^{-t^2/4t} - \dfrac{t^{n+1}}{2}e^{-t^2/4t}\left(\dfrac{-2t}{4t^2}\right)$

$\therefore \quad \dfrac{1}{t^2}\dfrac{\partial}{\partial t}\left(t^2 \dfrac{\partial f}{\partial t}\right) = \left(-\dfrac{n+1}{2}t^{n-2} + \dfrac{t^{n-1}}{4}\right)e^{-t^2/4t}$

Also $\dfrac{\partial f}{\partial t} = nt^{n-1}e^{-t^2/4t} + t^n e^{-t^2/4t}\cdot\dfrac{t^2}{4t^2} = \left(nt^{n-1} + \dfrac{1}{4}t^{n-1}\right)e^{-t^2/4t}$

Since $\dfrac{1}{t^2}\dfrac{\partial}{\partial t}\left(t^2 \dfrac{\partial f}{\partial t}\right) = \dfrac{\partial^2 f}{\partial t^2}$,

$\therefore \quad \left(-\dfrac{n+1}{2}t^{n-2} + \dfrac{1}{4}t^{n-1}\right)e^{-t^2/4t} = \left(nt^{n-2} + \dfrac{1}{4}t^{n-1}\right)e^{-t^2/4t}$ or $\left(n + \dfrac{1}{2}\right)t^{n-2}e^{-t^2/4t} = 0$.

Hence $n = -\dfrac{1}{2}$.

Example 2.10. If $V(x, y) = (x^2 + y^2 + 2)^{-1/2}$, prove that

$\dfrac{\partial^2 V}{\partial x^2} + \dfrac{\partial^2 V}{\partial y^2} = \dfrac{\partial^2 V}{\partial z^2}$ (U.T.U., Question Bank) (U.T.U., 2010) (Kottayam, 2004) (S.V.T.U., 2010)

Solution. We have $\frac{\partial z}{\partial x} = -\frac{1}{2}(x^2+y^2+z^2)^{-3/2}.2x = -x(x^2+y^2+z^2)^{-3/2}$

and $\frac{\partial^2 z}{\partial x^2} = -[1.(x^2+y^2+z^2)^{-3/2} + x.\frac{-3}{2}(x^2+y^2+z^2)^{-5/2}.2x]$

$= -(x^2+y^2+z^2)^{-3/2}[(x^2+y^2+z^2) - 3x^2](x^2+y^2+z^2)^{-5/2}(2x^2-y^2-z^2)$

Similarly, $\frac{\partial^2 z}{\partial y^2} = (x^2+y^2+z^2)^{-5/2}(-x^2+2y^2-z^2)$ and $\frac{\partial^2 z}{\partial z^2} = (x^2+y^2+z^2)^{-5/2}(-x^2-y^2+2z^2)$

Hence $\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}+\frac{\partial^2 z}{\partial z^2} = (x^2+y^2+z^2)^{-5/2}(0) = 0$.

Hint: Answer is satisfying the Laplace equation so need to be a harmonic function.

Example 8.8. If $u = \log_e(x^3+y^3+z^3-3xyz)$, show that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$.

(U.P.T.U. 2013; Aktu 2005, Bhopal 2009, U.P.T.U. 2006)

Solution. We have $\frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz}$, $\frac{\partial u}{\partial y} = \frac{3y^2-3xz}{x^3+y^3+z^3-3xyz}$, $\frac{\partial u}{\partial z} = \frac{3z^2-3xy}{x^3+y^3+z^3-3xyz}$

$\therefore \quad \frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z} = \frac{3(x^2+y^2+z^2-yz-xz-xy)}{x^3+y^3+z^3-3xyz}$

$= \frac{3(x^2+y^2+z^2-yz-xz-xy)}{(x+y+z)(x^2+y^2+z^2-xy-yz-xz)} = \frac{3}{x+y+z}$ (U.P.T.U., 2006)

Now $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}\right)$

$= \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{3}{x+y+z}\right)$

$= -\frac{3}{(x+y+z)^2}-\frac{3}{(x+y+z)^2}-\frac{3}{(x+y+z)^2} = -\frac{9}{(x+y+z)^2}$

Example 8.9. If $\frac{x^2}{x^2+u}+\frac{y^2}{y^2+u}+\frac{z^2}{z^2+u} = 1$, prove that

$$\left(x\frac{\partial u}{\partial x}\right)^2+\left(y\frac{\partial u}{\partial y}\right)^2+\left(z\frac{\partial u}{\partial z}\right)^2 = 4u^2\left(\frac{x}{x^2+u}+\frac{y}{y^2+u}+\frac{z}{z^2+u}\right)^2$$ (U.P.T.U. 2009)

Solution. We have $x^2(y^2+u)(z^2+u)+y^2(x^2+u)(z^2+u)+z^2(x^2+u)(y^2+u) = (x^2+u)(y^2+u)(z^2+u)$ ...(1)

Differentiating (1) partially w.r.t. x, we get

$$2x(y^2+u)(z^2+u)+x^2(z^2+u)\frac{\partial u}{\partial x}+x^2(y^2+u)\frac{\partial u}{\partial x} = (x^2+u)(z^2+u)\frac{\partial u}{\partial x}+(x^2+u)(y^2+u)\frac{\partial u}{\partial x}$$

or $\frac{2x}{x^2+u} = \left[\frac{x^2}{(x^2+u)^2}+\frac{y^2}{(y^2+u)^2}+\frac{z^2}{(z^2+u)^2}\right]\frac{\partial u}{\partial x}$

or $\frac{\partial u}{\partial x} = \frac{2x}{(x^2+u)^2 v}$, where $v = \frac{x^2}{(x^2+u)^2}+\frac{y^2}{(y^2+u)^2}+\frac{z^2}{(z^2+u)^2}$

Similarly differentiating (1) partially w.r.t. y and z, we get

$$\frac{2y}{y^2+u} = \left[\frac{x^2}{(x^2+u)^2}+\frac{y^2}{(y^2+u)^2}+\frac{z^2}{(z^2+u)^2}\right]\frac{\partial u}{\partial y} \text{ or } \frac{\partial u}{\partial y} = \frac{2y}{(y^2+u)^2 v}$$

Similarly, differentiating (i) partially w.r.t. x, y, z we get

$$\frac{2x}{ax^2+by^2+cz^2}=\left(\frac{x^2}{ax^2+by^2+cz^2}+\frac{y^2}{ax^2+by^2+cz^2}+\frac{z^2}{ax^2+by^2+cz^2}\right)\frac{\partial u}{\partial x}\text{ or }\frac{\partial u}{\partial x}=\frac{2x}{ax^2+by^2+cz^2}$$

$$\therefore\quad \left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial u}{\partial z}\right)^2=\frac{4}{x^2}\left(\frac{x^2}{ax^2+by^2+cz^2}+\frac{y^2}{ax^2+by^2+cz^2}+\frac{z^2}{ax^2+by^2+cz^2}\right)=\frac{4}{x^2} \quad \text{...(iii)}$$

$$\text{Also}\quad x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}=2\left(\frac{ax^2}{ax^2+by^2+cz^2}+\frac{by^2}{ax^2+by^2+cz^2}+\frac{cz^2}{ax^2+by^2+cz^2}\right)$$

$$=2\left(\frac{x^2}{ax^2+by^2+cz^2}+\frac{y^2}{ax^2+by^2+cz^2}+\frac{z^2}{ax^2+by^2+cz^2}\right)=2 \quad [\because u=1] \quad \text{...(iv)}$$

Hence the equality of (iii) and (iv) proves the result.

Example 6.6. If $u=x^y$, show that $\frac{\partial^2 u}{\partial x\partial y}=\frac{\partial^2 u}{\partial y\partial x}$. (Meerut, 2009)

Solution. We have $\frac{\partial u}{\partial y}=x^y\log x$ and $\frac{\partial^2 u}{\partial x\partial y}=yx^{y-1}.\log x+x^y.\frac{1}{x}=x^{y-1}(y\log x+1)$

$$\therefore\quad \frac{\partial^2 u}{\partial x\partial y}=\frac{\partial}{\partial x}\left(x^{y-1}\right)y\log x+\frac{\partial}{\partial x}(1) \quad \text{...(i)}$$

Again $\frac{\partial u}{\partial x}=yx^{y-1}$ and $\frac{\partial^2 u}{\partial y\partial x}=1.x^{y-1}+y\left(\frac{1}{x}x^y\log x\right)=x^{y-1}(1+y\log x)$

$$\therefore\quad \frac{\partial^2 u}{\partial y\partial x}=\frac{\partial}{\partial x}\left(x^{y-1}\right)y\log x+\frac{\partial}{\partial x}(1) \quad \text{...(ii)}$$

From (i) and (ii) it follows the required result.

PROBLEMS 6.2

1. Evaluate $\frac{\partial^2 u}{\partial x\partial y}$ and $\frac{\partial^2 u}{\partial y\partial x}$ if
   (i) $u=x^3y^2-\sin(xy)$ (ii) $u=\log(x^2+y^2)$
   (iii) $u=\tan^{-1}\frac{2xy}{x^2-y^2}$ (iv) $u=x+y+\log xy$

2. If $u=x^2+y^2$, $x=r\cos\theta$, $y=r\sin\theta$, show that $\left(\frac{\partial u}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial u}{\partial\theta}\right)^2=4\left(x^2+y^2\right)$ (V.T.U., 2003)

3. If $u=x^{y-z}$ (xyz - derivatives exist), show that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=0$. (V.T.U., 2003)

4. If $u=e^{ax+by}f(ax-by)$ and $v=e^{ax+by}g(ax-by)$, f and g being arbitrary functions, prove that $\frac{\partial u}{\partial x}=\frac{a}{b}\frac{\partial v}{\partial y}$ and $\frac{\partial v}{\partial x}=\frac{b}{a}\frac{\partial u}{\partial y}$.

5. If $z=\tan^{-1}(y-ax)+(y-ax)^{3/2}$, show that $\frac{\partial^2 z}{\partial x^2}=a^2\frac{\partial^2 z}{\partial y^2}$. (Kerala, 2006)

6. Verify that $f_{xy}=f_{yx}$ when $f(x,y)=\log(x^2+y^2)+\tan^{-1}\frac{y}{x}$. (Kerala, 2006)

7. If $f(x,y)=(x-2y)^3+f(xy)$, show that $\frac{\partial}{\partial x}\left(x\frac{\partial f}{\partial x}\right)-\frac{\partial}{\partial y}\left(x^2\frac{\partial f}{\partial y}\right)=0$. (Kerala, 2006 S)

8. Prove that $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ if $u(x,y)=\tan^{-1}\left(\frac{2xy}{x^2-y^2}\right)$. (Anna, 2006)

9. If $u=\frac{1}{\sqrt{x^2+y^2}}$, $f\left(\frac{1}{x}\right)$, prove that $\frac{\partial u}{\partial x}=x^3\frac{\partial u}{\partial y}$.

Ex. 5.20 The equation $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ occurs in the conduction of heat in a bar. If $u = e^{-\alpha^2 kt}\sin\alpha x$, prove that $\alpha = \frac{n\pi}{l}$, where n is an integer and l is the length of the bar.

Ex. 5.21 Find the value of n so that the equation $\nabla^2 r^n = 0$ (where $r = \sqrt{x^2+y^2+z^2}$) satisfies the equation

$$\frac{\partial^2 r^n}{\partial x^2} + \frac{\partial^2 r^n}{\partial y^2} + \frac{\partial^2 r^n}{\partial z^2} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial r^n}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 r^n}{\partial\phi^2}$$

Ex. 5.22 If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{x+y+z}$.

Ex. 5.23 If $u = \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$.

Ex. 5.24 If $x = r\cos\theta$, $y = r\sin\theta$ and $V = e^{r^2}$, prove that $V_{xx} + V_{yy} = V_{rr} + \frac{1}{r}V_r + \frac{1}{r^2}V_{\theta\theta} = 4(1+r^2)e^{r^2}$. (Nagpur, 2005)

Ex. 5.25 If $u = \log(x^3 + y^3 + z^3)$, prove that $x^2\frac{\partial^2 u}{\partial x^2} + y^2\frac{\partial^2 u}{\partial y^2} + z^2\frac{\partial^2 u}{\partial z^2} = -\frac{3}{2}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}\right)^2$.

Ex. 5.26 If $u = x^n f\left(\frac{y}{x}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$. (Pune, 2005)

Ex. 5.27 If $y = x\log\left(\frac{x}{x+y}\right)$, show that $x\frac{\partial^2 y}{\partial x\partial y} = -x\left(\log\frac{x}{x+y}\right)^2$. (Madras, 2000)

Ex. 5.28 If $u = x^2y^2$, find the value of $\frac{\partial^{n+m}u}{\partial x^n\partial y^m}$. (Nagpur, 2005; Osmania, 2007 S)

5.12 WHICH VARIABLE IS TO BE TREATED AS CONSTANT

(1) Consider the equation $x + y = 0$, x and y are not 0 ... (1)

To find $\partial z/\partial x$, we need a relation between x and z. Such a relation will contain one more variable z or y, for we can eliminate only one variable out of x or y from the relations (1). Thus the two possible relations are

$$x = z\sin\theta \quad \text{... (2)} \qquad \text{and} \qquad x^2 = z^2 + y^2 \quad \text{... (3)}$$

Now we can find $\partial z/\partial x$ either from (2) by treating θ as constant or from (3) by regarding y as constant. And there is no reason to suppose that the two values of $\partial z/\partial x$ so found are equal. To avoid confusion as to which variable is regarded constant, we introduce the following :

Notation : $\left(\frac{\partial z}{\partial x}\right)_y$ means the partial derivative of z with respect to x keeping y constant in a relation expressing z as a function of x and y.

Thus from (3), $\left(\frac{\partial z}{\partial x}\right)_y = \frac{x}{z}$.

When no indication is given regarding the variable to be kept constant, then according to convention $\frac{\partial z}{\partial x}$ always means $\left(\frac{\partial z}{\partial x}\right)_y$ and $\frac{\partial z}{\partial y}$ means $\left(\frac{\partial z}{\partial y}\right)_x$. Similarly, $\frac{\partial V}{\partial r}$ means $\left(\frac{\partial V}{\partial r}\right)_\theta$ and $\frac{\partial V}{\partial\theta}$ means $\left(\frac{\partial V}{\partial\theta}\right)_r$.

(2) In thermodynamics, we come across two variables such as p (pressure), v (volume), T (temperature), W (work), ϕ (entropy) etc. Any one of these can be expressed as a function of other two variables e.g., $T = f(p, v)$, $T = g(v, \phi)$.

As we shall see, these respectively give rise to the following results :

$$dT = \left(\frac{\partial T}{\partial p}\right)_v dp + \left(\frac{\partial T}{\partial v}\right)_p dv \quad \text{... (4)}$$

$$dT = \left(\frac{\partial T}{\partial v}\right)_\phi dv + \left(\frac{\partial T}{\partial\phi}\right)_v d\phi \quad \text{... (5)}$$

Now, $\partial T/\partial p$ appearing in (4) has been obtained from T as function of p and v, treating v as constant, we write it as $\left(\frac{\partial T}{\partial p}\right)_v$.

Similarly, $\partial T/\partial v$ occurring in (4), is written as $\left(\frac{\partial T}{\partial v}\right)_p$.

Example 5.5. If $u = f(r)$ and $x = r\cos\theta$, $y = r\sin\theta$, prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r}f'(r) \quad \text{(V.T.U., 2004; Rohtak, 2004; J.N.T.U., 2002)}$$

Solution. We have $\frac{\partial z}{\partial x} = f'(x)\cdot\frac{\partial y}{\partial x}$ and $\frac{\partial^2 z}{\partial x^2} = f''(x)\cdot\left(\frac{\partial y}{\partial x}\right)^2 + f'(x)\cdot\frac{\partial^2 y}{\partial x^2}$

Similarly, $\frac{\partial^2 z}{\partial y^2} = f''(x)\cdot\left(\frac{\partial y}{\partial y}\right)^2 + f'(x)\cdot\frac{\partial^2 y}{\partial y^2}$

$\therefore \quad \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = f''(x)\left[\left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial y}\right)^2\right] + f'(x)\left[\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial y^2}\right]$

Now to find $\frac{\partial y}{\partial x}$ etc., we write $x = \ln r = y^{1/2}$

$\therefore \quad \frac{\partial y}{\partial x} = \frac{1}{2}(x^2 + y^2)^{-1/2}\cdot 2x = \frac{x}{r}$ and $\frac{\partial^2 y}{\partial x^2} = \frac{r\cdot 1 - x\cdot\frac{\partial r}{\partial x}}{r^2} = \frac{r - x\cdot\frac{x}{r}}{r^2} = \frac{r^2 - x^2}{r^3}$

Similarly, $\frac{\partial y}{\partial y} = \frac{y}{r}$ and $\frac{\partial^2 y}{\partial y^2} = \frac{r^2 - y^2}{r^3}$

Substituting the values of the etc. in (1), we get

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = f''(x)\left[\frac{x^2}{r^2} + \frac{y^2}{r^2}\right] + f'(x)\left[\frac{r^2 - x^2}{r^3} + \frac{r^2 - y^2}{r^3}\right] = f''(x) + \frac{2}{r}f'(x)$$

Example 3. If $z = e^{x\cos\theta + y\sin\theta}\cos(x\sin\theta - y\cos\theta)$ and $x = r\cos\theta$, $y = r\sin\theta$, show that $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial r}$

Hence show that $\frac{\partial^2 z}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 z}{\partial \theta^2} = z$

Solution. We have $z = e^{x\cos\theta + y\sin\theta}\cos(x\sin\theta - y\cos\theta)$

$$\therefore \quad \frac{\partial z}{\partial \theta} = e^{x\cos\theta + y\sin\theta}(-x\sin\theta\cdot\cos(x\sin\theta - y\cos\theta) + e^{x\cos\theta + y\sin\theta}[-\sin(x\sin\theta - y\cos\theta)]\cdot x\cos\theta$$

$$= -e^{x\cos\theta + y\sin\theta}[\sin\theta\cos(x\sin\theta - y\cos\theta) + \cos\theta\sin(x\sin\theta - y\cos\theta)]$$

$$= -e^{x\cos\theta + y\sin\theta}\sin(\theta + x\sin\theta) \quad \ldots(1)$$

and $\frac{\partial z}{\partial r} = e^{x\cos\theta + y\sin\theta}\cdot\cos(x\sin\theta - y\cos\theta)\cdot(x\cos\theta + y\sin\theta) - e^{x\cos\theta + y\sin\theta}\sin(x\sin\theta - y\cos\theta)\cdot\sin\theta$

$$= e^{x\cos\theta + y\sin\theta}\cos(\theta + x\sin\theta) \quad \ldots(2)$$

Similarly, $z = e^{x\cos\theta + y\sin\theta}\sin(x\sin\theta - y\cos\theta)$ gives

$$\frac{\partial z}{\partial \theta} = -e^{x\cos\theta + y\sin\theta}\cos(\theta + x\sin\theta) \quad \ldots(3)$$

and $\frac{\partial z}{\partial r} = e^{x\cos\theta + y\sin\theta}\sin(\theta + x\sin\theta)$ $\ldots(4)$

From (2) and (4), $\frac{\partial z}{\partial r} = -r\frac{\partial z}{\partial \theta}$ $\ldots(5)$

From (3) and (4), $\frac{\partial z}{\partial \theta} = r\frac{\partial z}{\partial r}$ $\ldots(6)$

From (5), $\frac{\partial^2 z}{\partial r^2} = -r\frac{\partial^2 z}{\partial r\partial\theta} - \frac{\partial z}{\partial\theta}$

From (6), $\frac{\partial z}{\partial r} = \frac{1}{r}\frac{\partial z}{\partial\theta}$ which gives $\frac{\partial^2 z}{\partial r^2} = -\frac{1}{r^2}\frac{\partial z}{\partial\theta} + \frac{1}{r}\frac{\partial^2 z}{\partial r\partial\theta}$

$$\therefore \quad \frac{\partial^2 z}{\partial r^2} + r\frac{\partial z}{\partial r} + r^2\frac{\partial^2 z}{\partial\theta^2} = -r\frac{\partial^2 z}{\partial r\partial\theta} + \frac{\partial z}{\partial\theta} - \frac{\partial z}{\partial\theta} + r\frac{\partial^2 z}{\partial r\partial\theta} = 0$$

PROBLEMS 5.3

5. If $u=\log\left(\frac{x^3+y^3}{x+y}\right)$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=2$ (ii) $x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=2$ (iii) $\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2=1$ (Rohtak, 2005)

6. If $f(x,y)=ax+by$, $x^2+xy-2y^2$, prove that $\left(\frac{\partial f}{\partial x}\right)_y\left(\frac{\partial f}{\partial y}\right)_x+\left(\frac{\partial f}{\partial x}\right)_y=\left(\frac{\partial f}{\partial y}\right)_x$.

7. If $u=\log(x^3+y^3+z^3-3xyz)$, show that $\left(\frac{\partial u}{\partial x}\right)+\left(\frac{\partial u}{\partial y}\right)+\left(\frac{\partial u}{\partial z}\right)=\frac{3}{x+y+z}$.

8. If $u=x^y$, prove that

(i) $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}=y\left[\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2\right]$ (ii) $\frac{\partial u}{\partial x}\cdot\frac{\partial u}{\partial y}=1$ $(x=e,y=e)$

9. If $u=\log(x^3+y^3+z^3)$, prove that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{3}{x+y+z}$ (Kurukshetra, 2006)

10. If $u=\sqrt{(x^2+y^2+z^2)}$, show that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=\frac{2}{u}$, $(u\neq 0)$.

5.8 (1) HOMOGENEOUS FUNCTIONS

An expression of the form $a_0x^n+a_1x^{n-1}y+a_2x^{n-2}y^2+\ldots+a_ny^n$ in which every term is of the nth degree, is called a homogeneous function of degree n. This can be rewritten as

$$x^n\left[a_0+a_1\left(\frac{y}{x}\right)+a_2\left(\frac{y}{x}\right)^2+\ldots+a_n\left(\frac{y}{x}\right)^n\right].$$

Thus any function $f(x,y)$ which can be expressed in the form $x^n\phi(y/x)$ is called a **homogeneous function** of degree n in x and y.

For instance, $x^3\cos(y/x)$ is a homogeneous function of degree 3 in x and y.

In general, a function $f(x,y,z,\ldots)$ is said to be a homogeneous function of degree n in $x,y,z,\ldots$ if it can be expressed in the form $x^n\phi(y/x, z/x, \ldots)$.

(2) Euler's theorem on homogeneous functions*. *If u be a homogeneous function of degree n in x and y, then*

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=nu.$$

Since u is a homogeneous function of degree n in x and y, therefore,

$$u=x^n f(y/x)$$

i.e. $\frac{\partial u}{\partial x}=nx^{n-1}f\left(\frac{y}{x}\right)+x^n f'\left(\frac{y}{x}\right)\left(-\frac{y}{x^2}\right)=nx^{n-1}f\left(\frac{y}{x}\right)-yx^{n-2}f'\left(\frac{y}{x}\right)$

and $\frac{\partial u}{\partial y}=x^n f'\left(\frac{y}{x}\right)\frac{1}{x}=x^{n-1}f'\left(\frac{y}{x}\right)$. Hence $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=nx^n f\left(\frac{y}{x}\right)=nu$.

In general, if u be a homogeneous function of degree n in $x,y,z,\ldots$ then,

$$x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}+z\frac{\partial u}{\partial z}+\ldots=nu.$$

Example 5.11. Verify that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=2u$ for $u=\log\left(\frac{x^2+y^2}{x+y}\right)$.

Solution. Since $e^u=\frac{x^2+y^2}{x+y}=x\cdot\frac{1+(y/x)^2}{1+(y/x)}$,

* After an exceptionally creative Swiss mathematician Leonhard Euler (1707–1783). He studied under John Bernoulli and became a professor of mathematics in St. Petersburg, Russia. Even after becoming totally blind in 1771, he contributed to almost all branches of mathematics.

$\therefore u$ is a homogeneous function of degree 2 in x and y.

By Euler's theorem, we get

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u \qquad \ldots(1)$$

But $\dfrac{\partial u}{\partial x} = \dfrac{1}{u}\dfrac{\partial v}{\partial x}$ and $\dfrac{\partial u}{\partial y} = \dfrac{1}{u}\dfrac{\partial v}{\partial y}$

Hence (1) becomes

$$x\cdot\frac{1}{u}\frac{\partial v}{\partial x} + y\cdot\frac{1}{u}\frac{\partial v}{\partial y} = 2\log u \quad \text{or} \quad x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = 2u\log u$$

Example 3.12. If $v = \sin^{-1}\left(\dfrac{x+2y+3z}{x^3+y^3+z^3}\right)$, find the value of $x\dfrac{\partial v}{\partial x} + y\dfrac{\partial v}{\partial y} + z\dfrac{\partial v}{\partial z}$. (U.P.T.U., 2003)

Solution. Here v is not a homogeneous function. We therefore, write

$$\sin v = u = \frac{x+2y+3z}{x^3+y^3+z^3} = x^{-2}\cdot\frac{1+2(y/x)+3(z/x)}{1+(y/x)^3+(z/x)^3}$$

Clearly u is a homogeneous function of degree -2 in x, y, z. Hence by Euler's theorem

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = -2u \qquad \ldots(1)$$

But $\dfrac{\partial u}{\partial x} = \cos v\dfrac{\partial v}{\partial x}$, $\dfrac{\partial u}{\partial y} = \cos v\dfrac{\partial v}{\partial y}$, $\dfrac{\partial u}{\partial z} = \cos v\dfrac{\partial v}{\partial z}$

$\therefore$ (1) becomes $x\cos v\dfrac{\partial v}{\partial x} + y\cos v\dfrac{\partial v}{\partial y} + z\cos v\dfrac{\partial v}{\partial z} = -2\sin v$ or $x\dfrac{\partial v}{\partial x} + y\dfrac{\partial v}{\partial y} + z\dfrac{\partial v}{\partial z} = -2\tan v$.

Example 3.13. If $v = \dfrac{x^2y^2z^2}{x^3+y^3+z^3} + \log\left[\dfrac{xy+yz+zx}{x^2+y^2+z^2}\right]$, find the value of $x\dfrac{\partial v}{\partial x} + y\dfrac{\partial v}{\partial y} + z\dfrac{\partial v}{\partial z}$. (U.P.T.U., 2002)

Solution. Let $u = \dfrac{x^2y^2z^2}{x^3+y^3+z^3}$ and $w = \log\left[\dfrac{xy+yz+zx}{x^2+y^2+z^2}\right]$ ...(1)

so that $v = u + w$

Since $u = x^3\dfrac{(y/x)^2(z/x)^2}{1+(y/x)^3+(z/x)^3}$, therefore u is a homogeneous function of degree 6 in x, y, z.

Hence by Euler's theorem $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = 6u$ ...(2)

Since $w = \log\dfrac{\left(\frac{y}{x}+\frac{z}{x}+\frac{z}{y}\right)}{1+\left(\frac{y}{x}\right)^2+\left(\frac{z}{x}\right)^2}$, therefore w is a homogeneous function of degree zero in x, y, z.

Hence by Euler's theorem $x\dfrac{\partial w}{\partial x} + y\dfrac{\partial w}{\partial y} + z\dfrac{\partial w}{\partial z} = 0$ ...(3)

Adding (2) and (3), we obtain

$$x\left[\frac{\partial u}{\partial x} + \frac{\partial w}{\partial x}\right] + y\left[\frac{\partial u}{\partial y} + \frac{\partial w}{\partial y}\right] + z\left[\frac{\partial u}{\partial z} + \frac{\partial w}{\partial z}\right] = 6u$$

or $x\dfrac{\partial v}{\partial x} + y\dfrac{\partial v}{\partial y} + z\dfrac{\partial v}{\partial z} = 6\dfrac{x^2y^2z^2}{x^3+y^3+z^3}$ [By (1)]

Example 5.18. *If u is a homogeneous function of degree n in x and y, show that*

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u$$ *Given, 2007; U.T.U., 2007; U.P.T.U., 2009*

Solution. By Euler's theorem, $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$ ...(i)

Differentiating (i) partially w.r.t. x, we get $x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x \partial y} = n\frac{\partial u}{\partial x}$

i.e., $$x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = (n-1)\frac{\partial u}{\partial x}$$ ...(ii)

Again differentiating (i) partially w.r.t. y, we get $x\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial y} + y\frac{\partial^2 u}{\partial y^2} = n\frac{\partial u}{\partial y}$

i.e., $$x\frac{\partial^2 u}{\partial x \partial y} + y\frac{\partial^2 u}{\partial y^2} = (n-1)\frac{\partial u}{\partial y}$$ ...(iii)

Multiplying (ii) by x and (iii) by y and adding, we get

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = (n-1)\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right) = n(n-1)u.$$ [By (i)]

Example 5.19. *If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\tan u$.*

Hyderabad, 2009; Kakatiya, 2009;

Ans. $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\sin u \cos 2u}{4\cos^3 u}$ *U.P.T.U. 2008*

Solution. Here u is not a homogeneous function but $z = \sin u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$ is a homogeneous function of degree $\frac{1}{2}$ in x and y.

By Euler's theorem, $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = \frac{1}{2}z$

or $$x\cos u\frac{\partial u}{\partial x} + y\cos u\frac{\partial u}{\partial y} = \frac{1}{2}\sin u$$

Thus $$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}\tan u$$ ...(i)

Differentiating (i) w.r.t. x partially, we get

$$x\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + y\frac{\partial^2 u}{\partial x \partial y} = \frac{1}{2}\sec^2 u\frac{\partial u}{\partial x} \quad \text{or} \quad x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = \left(\frac{1}{2}\sec^2 u - 1\right)\frac{\partial u}{\partial x}$$ ...(ii)

Again, differentiating (i) w.r.t. y partially, we get

$$x\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial y} + y\frac{\partial^2 u}{\partial y^2} = \frac{1}{2}\sec^2 u\frac{\partial u}{\partial y} \quad \text{or} \quad x\frac{\partial^2 u}{\partial x \partial y} + y\frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{2}\sec^2 u - 1\right)\frac{\partial u}{\partial y}$$ ...(iii)

Multiplying (ii) by x and (iii) by y and adding, we obtain

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{2}\sec^2 u - 1\right)\left(x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}\right)$$

or $$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{2}\sec^2 u - 1\right)\left(\frac{1}{2}\tan u\right)$$ [By (i)]

$$= \frac{1}{4}\frac{\sin u}{\cos^2 u} - \frac{1}{2}\frac{\sin u}{\cos u} = \frac{\sin u(1 - 2\cos^2 u)}{4\cos^3 u}$$

Hence $$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{\sin u \cos 2u}{4\cos^3 u}$$

PROBLEMS 4

1. Verify Euler's theorem when $f(x,y) = ax^2 + 2hxy + by^2$

$[2f(x,y) = x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}]$

$[2ax^2 + 2hxy + 2by^2 = ax^2 + 2hxy + by^2]$ (LJMU, 1999)

2. If $u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$ (Strathclyde, 2009; Swansea, 2009/10)

3. If $u = \tan^{-1}\left(\frac{x^3+y^3}{x+y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$ (Bristol, 2008; UCLan, 2008/9)

4. If $u = \frac{x^2y^2}{x+y}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3u$ (Salford, 2008; UCLan, 2008/9)

5. If $u = \sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = -\frac{1}{2}\tan u$ (UCLan, 2009)

6. Show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\sin u\cos u$ when $u = \sin^{-1}(\ldots)$ (UCLan, 2008/9)

7. If $u = f(\sin x, \sqrt{x^2+y^2})$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sqrt{x^2+y^2}$ (Reading, 2009)

8. If $u = \frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$ (UCLan, 2008/9)

9. If $u = \frac{x+2y+2z}{\sqrt{x^2+y^2-z^2}}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$ (UCLan, 2008/9; Bristol, 2008)

10. If $u = \sin\left(\frac{x}{y}\right) + \tan\left(\frac{y}{x}\right)$, show that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 0$ (UCLan, 2009; UCLan, 2008/9)

11. If $v = \tan^{-1}\left(\frac{x^3+y^3}{x+y}\right)$, prove that $x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} = \sin 2v$ (UCLan, 2008/9)

and $x^2\frac{\partial^2 v}{\partial x^2} + 2xy\frac{\partial^2 v}{\partial x\partial y} + y^2\frac{\partial^2 v}{\partial y^2} = 2\cos 2v\sin v$ (Bristol, 2008; Bangor, 2009; Reading, 2009; UCLan, 2009)

12. If $u = x^2f\left(\frac{y}{x}\right) + y^2g\left(\frac{x}{y}\right)$, prove that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 2u$

(Sunderland, 2008; Bristol, 2008)

13. If $u = x^2\tan^{-1}\left(\frac{y}{x}\right) - y^2\tan^{-1}\left(\frac{x}{y}\right)$, show that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 0$ (UCLan, 2009; Bristol, 2008)

14. If $v = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x^2\frac{\partial^2 v}{\partial x^2} + 2xy\frac{\partial^2 v}{\partial x\partial y} + y^2\frac{\partial^2 v}{\partial y^2} = -\sin^2 v\sin 2v$ (Bristol, 2008; UCLan, 2008)

15. If $u = \ln\left(\frac{x^3+y^3}{x^2+y^2}\right)$, prove that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \frac{\sin 2u}{12} - \frac{\sin^2 u}{12} + \frac{\sin^3 u}{6}$

(Manchester, 2008; Bangor, 2009)

35.3 (i) TOTAL DERIVATIVE

If $u = f(x,y)$, where $x = g(t)$ and $y = \phi(t)$, then we can express u as a function of t alone by substituting the values of x and y in $f(x,y)$. Then we can find the ordinary derivative $\frac{du}{dt}$ which is called the total derivative of u to distinguish it from the partial derivatives $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$.

Now to find $\frac{du}{dt}$ without actually substituting the values of x and y in $f(x,y)$, we establish the following **Chain rule**:

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} \qquad (1)$$

Proof. We have $u = f(x,y)$

Giving increments Δt to t, let the corresponding increments in x, y and z be Δx, Δy and Δz respectively.

Then $x + \Delta x = f(t + \Delta t)$, $y + \Delta y = g(t + \Delta t)$

Subtracting, $\Delta x = f(t + \Delta t) - f(t)$, $\Delta y = g(t + \Delta t) - g(t)$

$$\therefore \quad \frac{\Delta z}{\Delta t} = \frac{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)}{\Delta x} \cdot \frac{\Delta x}{\Delta t} + \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \cdot \frac{\Delta y}{\Delta t}$$

Taking limits as $\Delta t \to 0$, Δx and Δy also $\to 0$, we have

$$\frac{dz}{dt} = \lim_{\Delta t \to 0} \left[\lim_{\Delta x \to 0} \left\{ \frac{f(x + \Delta x, y + \Delta y) - f(x, y + \Delta y)}{\Delta x} \right\} \frac{dx}{dt} + \lim_{\Delta y \to 0} \left\{ \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \right\} \frac{dy}{dt} \right]$$

$$= \lim_{\Delta y \to 0} \left\{ \frac{f(x, y + \Delta y)}{\Delta y} \right\} \frac{dx}{dt} + \frac{\partial f(x, y)}{\partial y} \cdot \frac{dy}{dt}$$

[Proceeding as in, y also is here a continuous function of t]

$$= \frac{\partial f(x, y)}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f(x, y)}{\partial y} \cdot \frac{dy}{dt}$$ which is the desired formula.

Cor. Taking $z = f(x, y)$ between us, $\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$ ...(4)

Obs. If $u = f(x, y, z)$, where x, y, z are all functions of a variable t, then Chain rule is

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$ ...(5)

(2) Differentiation of implicit functions. If $f(x, y) = c$ be an implicit relation between x and y which defines y as differentiable function of x, then $f(x, y)$ between us

$$0 = \frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx}$$

This gives the important formula $\frac{dy}{dx} = -\frac{\partial f}{\partial x} \Big/ \frac{\partial f}{\partial y}$ $\left[\frac{\partial f}{\partial y} \neq 0\right]$

for the first differential coefficient of an implicit function.

Example 5.16. Given $u = \sin^{-1}\left(\frac{x}{y}\right)$, $x = t^2$ and $y = t^3$, find $\frac{du}{dt}$ as a function of t. Verify your result by direct substitution.

Solution. We have $\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} = \left(\sin^{-1}\frac{x}{y}\right)_x' \cdot 2t + \left(\sin^{-1}\frac{x}{y}\right)_y' \cdot 3t^2$

$$= \frac{1}{\sqrt{1 - x^2/y^2}} \cdot \frac{1}{y} \cdot 2t - \frac{1}{\sqrt{1 - x^2/y^2}} \cdot \frac{x}{y^2} \cdot 3t^2 = \frac{2t}{\sqrt{y^2 - x^2}} - \frac{3t^2 x}{y\sqrt{y^2 - x^2}}$$

[Since $x = t^2$, $y = t^3$]

$$\therefore \quad \frac{du}{dt} = \frac{2t}{t^3\sqrt{t^4 - t^2}} - \frac{3t^2 \cdot t^2}{t^3\sqrt{t^4 - t^2}} = \frac{2 - 3t^2}{t\sqrt{t^2 - 1}}$$ as before.

Example 5.17. If a function of the type of $f(x, y)$ is of the constant value c when $x = 2$ and $y = 1$ and c when $x = 1$ and $y = 2$, prove that the function $2xy - 3x^2y$ must be another constant or alternately.

Solution. Let $u = 2xy - 3x^2y$, so that

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} = (2y - 6xy)\frac{dx}{dt} + (2x - 3x^2)\frac{dy}{dt}$$ ...(i)

when $x = 2$ and $y = 1$, $\frac{du}{dt} = 0$, and x is neither increasing nor decreasing, i.e., $\frac{dx}{dt} = 0$.

$\therefore$ (i) becomes $\quad 0 = (2 - 6 \times 2 \times 1)0 + (2 \times 2 - 3 \times 2^2)\frac{dy}{dt}$

or $\quad \frac{dy}{dt} = -\frac{0}{2} = 0$, i.e., y is neither increasing nor decreasing.

Example 9.18. If $u = x \log xy$ where $x^3 + y^3 + 3xy = 1$, find $\frac{du}{dx}$. (NUCEE, 2000)

Solution. From $f(x, y) = x^3 + y^3 + 3xy - 1$, we have

$$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{3x^2 + 3y}{3y^2 + 3x} = -\frac{x^2 + y}{y^2 + x} \quad \ldots(1)$$

Also $$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dx} = (1 \cdot \log xy + x \cdot \tfrac{1}{xy}\cdot y) + \left(\tfrac{x}{xy}\right)\left(-\frac{x^2 + y}{y^2 + x}\right)$$

Hence $$\frac{du}{dx} = 1 + \log xy - \frac{x(x^2 + y)}{y(y^2 + x)} \quad \text{(By (1))}$$

Example 9.19. If $v = x\left(\frac{x^2 - y^2}{x^2 + y^2}, \frac{2xy}{x^2 + y^2}\right)$ show that $x^2\frac{\partial^2 v}{\partial x^2} + y^2\frac{\partial^2 v}{\partial y^2} + z^2\frac{\partial^2 v}{\partial z^2} = 0$. (U.P.T.U., 2005)

Solution. Let $$u = \frac{x^2 - y^2}{x^2 + y^2} = \frac{1 - \frac{y^2}{x^2}}{1 + \frac{y^2}{x^2}} = \frac{1 - v^2}{1 + v^2} \quad \text{and} \quad v = \frac{2xy}{x^2 + y^2} = \frac{2\frac{y}{x}}{1 + \frac{y^2}{x^2}} \quad \ldots(1)$$

so that $$v = x(u, v)$$

$$\therefore \quad \frac{\partial v}{\partial x} = \frac{\partial v}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial v}{\partial v}\cdot\frac{\partial v}{\partial x} = \frac{\partial v}{\partial u}\left(-\frac{y}{x^2}\right) + \frac{\partial v}{\partial v}\left(\frac{1}{x}\right) \quad \text{(Using (1))}$$

or $$x^2\frac{\partial v}{\partial x} = -y\frac{\partial v}{\partial u} + x\frac{\partial v}{\partial v} \quad \ldots(A)$$

Also $$\frac{\partial v}{\partial y} = \frac{\partial v}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial v}{\partial v}\cdot\frac{\partial v}{\partial y} = \frac{\partial v}{\partial u}\left(\frac{1}{x^2}\right) + \frac{\partial v}{\partial v}(0) \quad \text{(Using (1))}$$

or $$y^2\frac{\partial v}{\partial y} = \frac{\partial v}{\partial u} \quad \ldots(B)$$

Similarly $$\frac{\partial v}{\partial z} = \frac{\partial v}{\partial u}\cdot\frac{\partial u}{\partial z} + \frac{\partial v}{\partial v}\cdot\frac{\partial v}{\partial z} = \frac{\partial v}{\partial u}(0) + \frac{\partial v}{\partial v}\left(\frac{1}{z^2}\right) \quad \text{(Using (1))}$$

or $$z^2\frac{\partial v}{\partial z} = \frac{\partial v}{\partial v} \quad \ldots(C)$$

Adding (A), (B) and (C), we have

$$x^2\frac{\partial v}{\partial x} + y^2\frac{\partial v}{\partial y} + z^2\frac{\partial v}{\partial z} = 0.$$

Example 9.20. *Prove that the second differential coefficient of y in terms of x and t where $x = f(t)$ and $y = \phi(t)$ is* $\frac{x\ddot{y} - \dot{x}\ddot{y}}{\dot{x}^3}$. (Kurukshetra, 2006)

Solution. We have $$\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{F_x}{F_y} \quad \ldots(1)$$

$$\therefore \quad \frac{d^2y}{dx^2} = -\frac{d}{dx}\left(\frac{F_x}{F_y}\right) = -\frac{d}{dt}\left(\frac{F_x}{F_y}\right)\cdot\frac{dt}{dx} = -\frac{F_y(rx + sy) - F_x(sx + ty)}{F_y^3} \quad \ldots(A)$$

Using the notations: $r = \frac{\partial^2 f}{\partial x^2} = f_{xx}$, $s = \frac{\partial^2 f}{\partial x \partial y} = f_{xy}$, $t = \frac{\partial^2 f}{\partial y^2} = f_{yy}$,

we have $$\frac{dx}{dx} = \frac{\partial x}{\partial x} + \frac{\partial x}{\partial y}\cdot\frac{dy}{dx} \Rightarrow 1 = x + \dot{y}(-p/q) \Rightarrow -\frac{\dot{x}}{q} = \frac{\dot{x}p}{q}$$

and $$\frac{dy}{dx} = \frac{\partial y}{\partial x} + \frac{\partial y}{\partial y}\cdot\frac{dy}{dx} \Rightarrow p = s + t(-p/q) \Rightarrow \frac{tp}{q} = \frac{sq - tp}{q}$$

Substituting the values of dy/dx and d^2y/dx^2 in (i), we get

$$\frac{d^3y}{dx^3} = -\frac{1}{y^3}\left[3\left(\frac{x^2-ay}{y}\right)\left(\frac{ax-y^2}{y}\right) + y\left(\frac{ax-y^2}{y}\right)\right] = -\frac{x^3+3xy+y^3}{y^5}$$

EXERCISE 6.2

1. If $u = x^2 + y^2$, $x = at^2$, $y = 2at$, find $\frac{du}{dt}$. (P.T.U., 2002)
2. If $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, $x = t^2$, $y = 2t$, find $\frac{du}{dt}$. (P.T.U., 2003)
3. Find the value of $\frac{dz}{dt}$ given $z = y^2 - 4ax$, $x = at^2$, $y = 2at$. (Anna, 2004)
4. A ring is made of wire of resistivity ρ. The radius of the ring is r and the cross-section of the wire is a circle of radius a. If a increases at the rate of $0.01a$ per second and r decreases at the rate of $0.01r$ per second, find the rate at which the resistance is changing at the instant when $r = 10a$.
5. The volume V of a right circular cone is increasing at the rate of $10\ \text{cm}^3/\text{sec}$ and its semi-vertical angle α is decreasing at the rate of 0.1 radian per second. Find the rate of change of the curved surface of the cone, when $r = 10$ cm and $\alpha = \frac{\pi}{6}$.
6. If $x = e^t(\cos t + \sin t)$ and $y = e^t(\cos t - \sin t)$, prove that $\frac{dy}{dx} = \frac{x+y}{x-y}$.
7. If $u = x^2 - y$, $x = e^t$ and $y = e^t \sin t$, show that $\frac{du}{dt} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$.
8. If $f(x, y) = 0$, $\phi(x, y) = 0$, show that $\frac{\partial f}{\partial x}\frac{\partial \phi}{\partial y} - \frac{\partial f}{\partial y}\frac{\partial \phi}{\partial x} = \frac{\partial f}{\partial x}\frac{dy}{dx} + \frac{\partial f}{\partial y}$.
9. If $z = f(x, y)$ is a function of x and y, where $x = e^u + e^{-v}$, $y = e^{-u} + e^v$, show that $\frac{\partial z}{\partial u}\frac{\partial z}{\partial v} - \frac{\partial z}{\partial x}\frac{\partial z}{\partial y} = \frac{\partial z}{\partial x}\frac{\partial z}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial z}{\partial v}$.
10. If $f(x, y) = 0$, show that $\left(\frac{\partial f}{\partial x}\right)^3\frac{d^2y}{dx^2} = \left(\frac{\partial f}{\partial x}\right)^2\left(\frac{\partial^2 f}{\partial x^2}\right) - 2\left(\frac{\partial f}{\partial x}\right)\left(\frac{\partial f}{\partial y}\right)\left(\frac{\partial^2 f}{\partial x \partial y}\right) + \left(\frac{\partial f}{\partial y}\right)^2\left(\frac{\partial^2 f}{\partial y^2}\right)$.

6.5 CHANGE OF VARIABLES

If $u = f(x, y)$ ...(1)

where $x = \phi(r, t)$ and $y = \psi(r, t)$ ...(2)

it is often necessary to change expressions involving $x, y, \partial u/\partial x, \partial u/\partial y$ etc. to expressions involving $r, t, \partial u/\partial r, \partial u/\partial t$ etc.

The necessary formulae for the change of variables are easily obtained. If t is regarded as a constant, then x, y, u will be functions of r alone. Therefore, by (1) of page 208, we have

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cdot\frac{dx}{\partial r} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial r} \quad \ldots(3)$$

where the ordinary derivatives have been replaced by the partial derivatives because x, y are functions of two variables r and t.

Similarly, regarding r as constant, we obtain $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial t}$ ...(4)

On solving (3) and (4) as simultaneous equations in $\partial u/\partial x$ and $\partial u/\partial y$, we get their values in terms of $\partial u/\partial r$, $\partial u/\partial t$, r, t.

If two of the equations (2), e.g. $x = \phi(r, t)$ can be given in terms of x and y, say $r = \chi_1(x, y)$ and $t = \chi_2(x, y)$ ...(5)

then it is easier to use the formulae $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial u}{\partial t}\cdot\frac{\partial t}{\partial x}$ ...(6)

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial y} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial y} \qquad \ldots(11)$$

The higher derivatives of u can be found by repeated application of formulae (10) and (11) or of (10) and (11).

Example 6.21. *If $u = f(x, y)$, $x = e^t + e^{-t}$, $y = e^t - e^{-t}$, prove that*

$$\frac{du}{dt} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \text{ and } \frac{d^2u}{dt^2} = \frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2}$$ (V.T.U., 2010; U.P.T.U., 2009; G.B.T.U., 2004)

Solution. Put $x = y + z$, $y = z + x$ and $z = x + y$ so that $u = f(x, y, z)$.

$$\frac{du}{dx} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial x} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial x} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial x}$$

$$= \frac{\partial u}{\partial x}(1) + \frac{\partial u}{\partial y}(-1) + \frac{\partial u}{\partial z}(+1) = \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \qquad \ldots(i)$$

Similarly, $$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial y} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial y} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial y} = -\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \qquad \ldots(ii)$$

and $$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial z} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial z} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial z} = \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} \qquad \ldots(iii)$$

Adding (i), (ii) and (iii), we get the required result.

Example 6.22. *If $u = f(x, y)$ and $x = e^r \cos\theta$, $y = e^r \sin\theta$, then prove that* $\frac{\partial u}{\partial r} = x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ *and*

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2r}\left[\left(\frac{\partial u}{\partial r}\right)^2 + \left(\frac{\partial u}{\partial \theta}\right)^2\right]$$ (Kurukshetra, 2009)

Solution. We have $$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial r}$$

$$= \frac{\partial u}{\partial x}(e^r \cos\theta) + \frac{\partial u}{\partial y}(e^r \sin\theta) \qquad \ldots(i)$$

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial \theta}$$

$$= \frac{\partial u}{\partial x}(-e^r \sin\theta) + \frac{\partial u}{\partial y}(e^r \cos\theta) \qquad \ldots(ii)$$

$$\therefore \quad x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = e^r\cos\theta\left(-e^r\sin\theta\frac{\partial u}{\partial x} + e^r\cos\theta\frac{\partial u}{\partial y}\right) + e^r\sin\theta\left(e^r\cos\theta\frac{\partial u}{\partial x} + e^r\sin\theta\frac{\partial u}{\partial y}\right)$$

$$= (e^{2r}\cos^2\theta + e^{2r}\sin^2\theta)\frac{\partial u}{\partial y} = e^{2r}\frac{\partial u}{\partial y}$$

Now squaring (i) and (ii) and adding, we get

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2r}\left(\cos\theta\frac{\partial u}{\partial x} + \sin\theta\frac{\partial u}{\partial y}\right)^2 + e^{-2r}\left(-\sin\theta\frac{\partial u}{\partial x} + \cos\theta\frac{\partial u}{\partial y}\right)^2$$

or $$e^{-2r}\left[\left(\frac{\partial u}{\partial r}\right)^2 + \left(\frac{\partial u}{\partial \theta}\right)^2\right] = \cos^2\theta\left(\frac{\partial u}{\partial x}\right)^2 + \sin^2\theta\left(\frac{\partial u}{\partial y}\right)^2 + 2\sin\theta\cos\theta\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}$$

$$+ \sin^2\theta\left(\frac{\partial u}{\partial x}\right)^2 + \cos^2\theta\left(\frac{\partial u}{\partial y}\right)^2 - 2\sin\theta\cos\theta\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}$$

$$= \cos^2\theta + \sin^2\theta\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right]$$

Hence $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{2x}\left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2\right]$.

Example 2.12. If $z = x + y = e^u \cos v$ and $z - y = e^u \sin v$, show that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 2e^x \frac{\partial^2 z}{\partial x \partial y}$$

(Nagpur 2009; U.P.T.U. 2009)

Solution. We have $x = e^u \cos v$, $y = e^u \sin v$ and $z = e^x(\cos y + \sin y)$ [Ans p. 2.53]

and $y = e^u \cos v - e^u \sin v = e^u(\cos v - \sin v) = e^x \cdot e^{-u}$

Here z is a composite function of θ and ϕ.

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x}$$

$$= \frac{\partial z}{\partial u} \cdot e^u \cdot e^{-u} + \frac{\partial z}{\partial v}(-e^u \cdot e^{-u}) = z\frac{\partial z}{\partial u} + y\frac{\partial z}{\partial v}$$

or $$\frac{\partial}{\partial x} = z\frac{\partial}{\partial u} + y\frac{\partial}{\partial v} \quad \ldots(i)$$

Also $$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial y} = \frac{\partial z}{\partial u} \cdot e^u \cdot \cos v + \frac{\partial z}{\partial v}(e^u \cdot -e^{-u}) = z\frac{\partial z}{\partial u} - y\frac{\partial z}{\partial v}$$

$$\frac{\partial}{\partial y} = z\frac{\partial}{\partial u} - y\frac{\partial}{\partial v} \quad \ldots(ii)$$

Using the operator (i), we have

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \left(z\frac{\partial}{\partial u} + y\frac{\partial}{\partial v}\right)\left(z\frac{\partial z}{\partial u} + y\frac{\partial z}{\partial v}\right)$$

$$= z\frac{\partial}{\partial u}\left(z\frac{\partial z}{\partial u}\right) + y\frac{\partial}{\partial v}\left(z\frac{\partial z}{\partial u}\right) + z\frac{\partial}{\partial u}\left(y\frac{\partial z}{\partial v}\right) + y\frac{\partial}{\partial v}\left(y\frac{\partial z}{\partial v}\right)$$

$$= z\left(z\frac{\partial^2 z}{\partial u^2} + \frac{\partial z}{\partial u}\right) + yz\frac{\partial^2 z}{\partial v \partial u} + yz\frac{\partial^2 z}{\partial u \partial v} + y\left(z\frac{\partial^2 z}{\partial v^2} + \frac{\partial z}{\partial v}\right)$$

$$= z^2\frac{\partial^2 z}{\partial u^2} + 2yz\frac{\partial^2 z}{\partial u \partial v} + y^2\frac{\partial^2 z}{\partial v^2} + z\frac{\partial z}{\partial u} + y\frac{\partial z}{\partial v} \quad \ldots(iii)$$

Similarly using (ii), $\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = \left(z\frac{\partial}{\partial u} - y\frac{\partial}{\partial v}\right)\left(z\frac{\partial z}{\partial u} - y\frac{\partial z}{\partial v}\right)$

$$= z^2\frac{\partial^2 z}{\partial u^2} - 2yz\frac{\partial^2 z}{\partial u \partial v} + y^2\frac{\partial^2 z}{\partial v^2} - z\frac{\partial z}{\partial u} - y\frac{\partial z}{\partial v} \quad \ldots(iv)$$

Adding (iii) and (iv), we get $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 4yz\frac{\partial^2 z}{\partial u \partial v}$

Example 2.13. Transform the equation $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$ into polar coordinates. (U.P.T.U. 2003)

Solution. We have $x = r\cos\theta$, $y = r\sin\theta$ and $r = \sqrt{(x^2 + y^2)}$, $\theta = \tan^{-1}(y/x)$

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{(x^2 + y^2)}} = \cos\theta \text{ and } \frac{\partial \theta}{\partial x} = -\frac{y}{x^2 + y^2} = -\frac{\sin\theta}{r}$$

Then, $$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial z}{\partial \theta}\frac{\partial \theta}{\partial x} = \cos\theta\frac{\partial z}{\partial r} - \frac{\sin\theta}{r}\frac{\partial z}{\partial \theta}$$

Let, $\frac{\partial}{\partial x} = \cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta}$ and similarly, $\frac{\partial}{\partial y} = \sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r} \frac{\partial}{\partial \theta}$.

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \left(\cos\theta \frac{\partial}{\partial r} - \frac{\sin\theta}{r} \frac{\partial}{\partial \theta}\right)\left(\cos\theta \frac{\partial u}{\partial r} - \frac{\sin\theta}{r} \frac{\partial u}{\partial \theta}\right)$$

$$= \cos^2\theta \frac{\partial^2 u}{\partial r^2} - \frac{2\sin\theta\cos\theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\sin^2\theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} - \frac{\cos^2\theta}{r} \frac{\partial u}{\partial r} + \frac{2\sin\theta\cos\theta}{r^2} \frac{\partial u}{\partial \theta} \quad \ldots(1)$$

and $\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \left(\sin\theta \frac{\partial}{\partial r} + \frac{\cos\theta}{r} \frac{\partial}{\partial \theta}\right)\left(\sin\theta \frac{\partial u}{\partial r} + \frac{\cos\theta}{r} \frac{\partial u}{\partial \theta}\right)$

$$= \sin^2\theta \frac{\partial^2 u}{\partial r^2} + \frac{2\sin\theta\cos\theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\cos^2\theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\sin^2\theta}{r} \frac{\partial u}{\partial r} - \frac{2\sin\theta\cos\theta}{r^2} \frac{\partial u}{\partial \theta} \quad \ldots(2)$$

Adding (1) and (2), we get $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{1}{r} \frac{\partial u}{\partial r}$

Hence the transformed equation is $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$.

PROBLEMS 2.5

1. If $z = f(x, y)$ and $x = e^u + e^{-v}$, $y = e^{-u} + e^v$, prove that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$. (V.T.U., 2006)

2. If $u = f(r, s)$ where $r = x + y$, $s = x - y$ and $x = \phi(t)$, $y = \psi(t)$, prove that $\frac{du}{dt} = \frac{\partial u}{\partial r} + \frac{\partial u}{\partial s}$.

3. If $f(x, y) = \phi(x + ay) + \psi(x - ay)$, prove that $\frac{\partial^2 f}{\partial x^2} = \frac{1}{a^2} \frac{\partial^2 f}{\partial y^2}$. (Kolkata, 2007)

4. If $x = r\cos\theta$, $y = r\sin\theta$, prove that $\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2 = \left(\frac{\partial \theta}{\partial x}\right)^2 + \frac{1}{r^2}\left(\frac{\partial \theta}{\partial y}\right)^2$.
(V.T.U., 2010, Madras, 2003, Kolkata, 2005)

5. If $u = f(x - y, y - z, z - x)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. (U.P.T.U., 2004; Anna, 2003)

6. If $u = f(xyz)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$. (Mumbai, 2006)

7. If $u = f(x, y)$ and $x = e^s + e^{-t}$, $y = e^{-s} + e^t$, prove that $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = s\frac{\partial u}{\partial s} - t\frac{\partial u}{\partial t}$. (Anna, 2003; Kolkata, 2006)

8. If $x = r\cos\theta$, $y = r\sin\theta$ and $z = f(x, y)$ is a function of x, y, show that
$$\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2.$$

9. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x + y + z)^2}$.

10. If the three variables x, y, z are connected by a relation $f(x, y, z) = 0$, show that
$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1.$$

11. If u is a solution of $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$, $x = \xi + t$, $y = \eta + t$, show that
$$\frac{\partial^2 u}{\partial \xi^2} + \frac{\partial^2 u}{\partial \eta^2} = 2\frac{\partial^2 u}{\partial \xi \partial \eta}.$$ (Anna, 2007)

12. Transform $\frac{\partial^2 z}{\partial x^2} + 2xy^2\frac{\partial z}{\partial x} + 2x - y^2\left(\frac{\partial z}{\partial y} + x^2 z\right) = 0$ by taking $x + y = u$ and $x - y = v$ (x, y are the independent variables and z is a function of x and y).

1.23 (1) Let $u = f(x, y)$ be a function of two independent variables x and y. Then the Jacobian of u and v with respect to x and y is defined as the determinant $\begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix}$ and is denoted by $\frac{\partial(u, v)}{\partial(x, y)}$ or $J\left(\frac{u, v}{x, y}\right)$.

Similarly the Jacobian of u, v, w with respect to x, y, z is

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y & \partial u/\partial z \\ \partial v/\partial x & \partial v/\partial y & \partial v/\partial z \\ \partial w/\partial x & \partial w/\partial y & \partial w/\partial z \end{vmatrix}$$

Likewise we can define Jacobians of four or more variables. The important properties of Jacobians are given in the following theorems. The proofs are left as exercises for the reader.

(2) **If $u_1, u_2, \ldots, u_n$ are functions of $x_1, x_2, \ldots, x_n$ and $x_1, x_2, \ldots, x_n$ are functions of $y_1, y_2, \ldots, y_n$, then**

$$\frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(y_1, y_2, \ldots, y_n)} = \frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(x_1, x_2, \ldots, x_n)} \cdot \frac{\partial(x_1, x_2, \ldots, x_n)}{\partial(y_1, y_2, \ldots, y_n)}$$

(3) **If $u_1, u_2, \ldots, u_n$ are functions of $x_1, x_2, \ldots, x_n$ and $x_1, x_2, \ldots, x_n$ are functions of $y_1, y_2, \ldots, y_n$ and $y_1, y_2, \ldots, y_n$ are functions of $z_1, z_2, \ldots, z_n$, then**

$$\frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(z_1, z_2, \ldots, z_n)} = \frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(x_1, x_2, \ldots, x_n)} \cdot \frac{\partial(x_1, x_2, \ldots, x_n)}{\partial(y_1, y_2, \ldots, y_n)} \cdot \frac{\partial(y_1, y_2, \ldots, y_n)}{\partial(z_1, z_2, \ldots, z_n)}$$

(4) **If $u_1, u_2, \ldots, u_n$ are functions of $x_1, x_2, \ldots, x_n$ and $x_1, x_2, \ldots, x_n$ are functions of $y_1, y_2, \ldots, y_n$, then**

$$\frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(y_1, y_2, \ldots, y_n)} \cdot \frac{\partial(y_1, y_2, \ldots, y_n)}{\partial(x_1, x_2, \ldots, x_n)} = 1$$

Proof. Suppose, on replacing $(x_1, x_2, \ldots, x_n)$ by $(y_1, y_2, \ldots, y_n)$, $(u_1, u_2, \ldots, u_n)$ become $(v_1, v_2, \ldots, v_n)$. Then

$$\frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(x_1, x_2, \ldots, x_n)} = \begin{vmatrix} \partial v_1/\partial y_1 & \partial v_1/\partial y_2 & \cdots & \partial v_1/\partial y_n \\ \partial v_2/\partial y_1 & \partial v_2/\partial y_2 & \cdots & \partial v_2/\partial y_n \\ \cdots & \cdots & \cdots & \cdots \\ \partial v_n/\partial y_1 & \partial v_n/\partial y_2 & \cdots & \partial v_n/\partial y_n \end{vmatrix} \cdot \begin{vmatrix} \partial y_1/\partial x_1 & \partial y_1/\partial x_2 & \cdots & \partial y_1/\partial x_n \\ \partial y_2/\partial x_1 & \partial y_2/\partial x_2 & \cdots & \partial y_2/\partial x_n \\ \cdots & \cdots & \cdots & \cdots \\ \partial y_n/\partial x_1 & \partial y_n/\partial x_2 & \cdots & \partial y_n/\partial x_n \end{vmatrix} \quad \ldots(1)$$

and

$$\frac{\partial(v_1, v_2, \ldots, v_n)}{\partial(y_1, y_2, \ldots, y_n)} = \begin{vmatrix} \partial v_1/\partial y_1 & \partial v_1/\partial y_2 & \cdots & \partial v_1/\partial y_n \\ \partial v_2/\partial y_1 & \partial v_2/\partial y_2 & \cdots & \partial v_2/\partial y_n \\ \cdots & \cdots & \cdots & \cdots \\ \partial v_n/\partial y_1 & \partial v_n/\partial y_2 & \cdots & \partial v_n/\partial y_n \end{vmatrix} \quad \ldots(2)$$

Now

$$\frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(y_1, y_2, \ldots, y_n)} = \begin{vmatrix} \partial u_1/\partial y_1 & \partial u_1/\partial y_2 & \cdots & \partial u_1/\partial y_n \\ \partial u_2/\partial y_1 & \partial u_2/\partial y_2 & \cdots & \partial u_2/\partial y_n \\ \cdots & \cdots & \cdots & \cdots \\ \partial u_n/\partial y_1 & \partial u_n/\partial y_2 & \cdots & \partial u_n/\partial y_n \end{vmatrix}$$

$$= \begin{vmatrix} \partial u_1/\partial x_1 \cdot \partial x_1/\partial y_1 + \partial u_1/\partial x_2 \cdot \partial x_2/\partial y_1 + \cdots + \partial u_1/\partial x_n \cdot \partial x_n/\partial y_1 & \cdots \\ \partial u_2/\partial x_1 \cdot \partial x_1/\partial y_1 + \partial u_2/\partial x_2 \cdot \partial x_2/\partial y_1 + \cdots + \partial u_2/\partial x_n \cdot \partial x_n/\partial y_1 & \cdots \\ \cdots & \cdots \\ \partial u_n/\partial x_1 \cdot \partial x_1/\partial y_1 + \partial u_n/\partial x_2 \cdot \partial x_2/\partial y_1 + \cdots + \partial u_n/\partial x_n \cdot \partial x_n/\partial y_1 & \cdots \end{vmatrix}$$

$$= \begin{vmatrix} \partial u_1/\partial x_1 & \partial u_1/\partial x_2 & \cdots & \partial u_1/\partial x_n \\ \partial u_2/\partial x_1 & \partial u_2/\partial x_2 & \cdots & \partial u_2/\partial x_n \\ \cdots & \cdots & \cdots & \cdots \\ \partial u_n/\partial x_1 & \partial u_n/\partial x_2 & \cdots & \partial u_n/\partial x_n \end{vmatrix} \cdot \begin{vmatrix} \partial x_1/\partial y_1 & \partial x_2/\partial y_1 & \cdots & \partial x_n/\partial y_1 \\ \partial x_1/\partial y_2 & \partial x_2/\partial y_2 & \cdots & \partial x_n/\partial y_2 \\ \cdots & \cdots & \cdots & \cdots \\ \partial x_1/\partial y_n & \partial x_2/\partial y_n & \cdots & \partial x_n/\partial y_n \end{vmatrix}$$

(Interchanging rows and columns of the first determinant)

$$= \frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(x_1, x_2, \ldots, x_n)} \cdot \frac{\partial(x_1, x_2, \ldots, x_n)}{\partial(y_1, y_2, \ldots, y_n)} \quad \text{(By rule of } \S 1\text{)}$$

(4) **Prove that the Jacobian of $u_1, u_2, \ldots, u_n$ with respect to $x_1, x_2, \ldots, x_n$ is the reciprocal of the Jacobian of $x_1, x_2, \ldots, x_n$ with respect to $u_1, u_2, \ldots, u_n$.**

$$\frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(x_1, x_2, \ldots, x_n)} \cdot \frac{\partial(x_1, x_2, \ldots, x_n)}{\partial(u_1, u_2, \ldots, u_n)} = \begin{vmatrix} \partial u_1/\partial x_1 & \partial u_1/\partial x_2 & \cdots & \partial u_1/\partial x_n \\ \partial u_2/\partial x_1 & \partial u_2/\partial x_2 & \cdots & \partial u_2/\partial x_n \\ \cdots & \cdots & \cdots & \cdots \\ \partial u_n/\partial x_1 & \partial u_n/\partial x_2 & \cdots & \partial u_n/\partial x_n \end{vmatrix} \cdot \begin{vmatrix} \partial x_1/\partial u_1 & \partial x_2/\partial u_1 & \cdots & \partial x_n/\partial u_1 \\ \partial x_1/\partial u_2 & \partial x_2/\partial u_2 & \cdots & \partial x_n/\partial u_2 \\ \cdots & \cdots & \cdots & \cdots \\ \partial x_1/\partial u_n & \partial x_2/\partial u_n & \cdots & \partial x_n/\partial u_n \end{vmatrix}$$

(Interchanging rows and columns of the 2nd det.)

$$= \begin{vmatrix} \partial u_1/\partial x_1 & \partial u_1/\partial x_2 & \cdots & \partial u_1/\partial x_n \\ \partial u_2/\partial x_1 & \partial u_2/\partial x_2 & \cdots & \partial u_2/\partial x_n \\ \cdots & \cdots & \cdots & \cdots \\ \partial u_n/\partial x_1 & \partial u_n/\partial x_2 & \cdots & \partial u_n/\partial x_n \end{vmatrix} \cdot \begin{vmatrix} \partial x_1/\partial u_1 & \partial x_2/\partial u_1 & \cdots & \partial x_n/\partial u_1 \\ \partial x_1/\partial u_2 & \partial x_2/\partial u_2 & \cdots & \partial x_n/\partial u_2 \\ \cdots & \cdots & \cdots & \cdots \\ \partial x_1/\partial u_n & \partial x_2/\partial u_n & \cdots & \partial x_n/\partial u_n \end{vmatrix}$$

$$= \frac{\partial(u_1, u_2, \ldots, u_n)}{\partial(x_1, x_2, \ldots, x_n)} \cdot \frac{\partial(x_1, x_2, \ldots, x_n)}{\partial(u_1, u_2, \ldots, u_n)}$$

$$= 1$$

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* Called after the German mathematician Carl Gustav Jacob Jacobi (1804–1851), who made significant contributions to partial differential equations, and theory of algebraic functions and their applications to geometry.

Example 6.28. (a) If the polar coordinates (r, θ) are expressed in terms of cartesian coordinates (x, y) by the equations $x = r\cos\theta$, $y = r\sin\theta$, show that

$$\frac{\partial(x, y)}{\partial(r, \theta)} = r$$

(b) If the cylindrical coordinates (r, θ, z) are expressed in terms of cartesian coordinates (x, y, z) by the equations $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, show that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$$

(c) If the spherical polar coordinates (r, θ, ϕ) are expressed in terms of cartesian coordinates (x, y, z) by the equations $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$, show that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2\sin\theta$$

(Anglia, 2000; Nottingham, 2000; Teesside, 2000)

Solution. (a) We have $\frac{\partial x}{\partial r} = \cos\theta$, $\frac{\partial x}{\partial \theta} = -r\sin\theta$ and $\frac{\partial y}{\partial r} = \sin\theta$, $\frac{\partial y}{\partial \theta} = r\cos\theta$

Thus $$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

(b) We have $\frac{\partial x}{\partial r} = \cos\theta$, $\frac{\partial x}{\partial \theta} = -r\sin\theta$, $\frac{\partial x}{\partial z} = 0$,

$\frac{\partial y}{\partial r} = \sin\theta$, $\frac{\partial y}{\partial \theta} = r\cos\theta$, $\frac{\partial y}{\partial z} = 0$ and $\frac{\partial z}{\partial r} = 0$, $\frac{\partial z}{\partial \theta} = 0$, $\frac{\partial z}{\partial z} = 1$

Thus $$\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

(c) We have $\frac{\partial x}{\partial r} = \sin\theta\cos\phi$, $\frac{\partial x}{\partial \theta} = r\cos\theta\cos\phi$, $\frac{\partial x}{\partial \phi} = -r\sin\theta\sin\phi$,

$\frac{\partial y}{\partial r} = \sin\theta\sin\phi$, $\frac{\partial y}{\partial \theta} = r\cos\theta\sin\phi$, $\frac{\partial y}{\partial \phi} = r\sin\theta\cos\phi$,

and $\frac{\partial z}{\partial r} = \cos\theta$, $\frac{\partial z}{\partial \theta} = -r\sin\theta$, $\frac{\partial z}{\partial \phi} = 0$.

Thus $$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{vmatrix} = r^2\sin\theta$$

Example 6.29. If $y_1 = \frac{x_1x_2}{x_3}$, $y_2 = \frac{x_2x_3}{x_1}$, $y_3 = \frac{x_3x_1}{x_2}$, show that the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 is 4. (U.C.W.C., 2000)

Solution. We have $\frac{\partial y_1}{\partial x_1} = -\frac{x_1x_2}{x_3^2}$, $\frac{\partial y_1}{\partial x_2} = \frac{x_1}{x_3}$, $\frac{\partial y_1}{\partial x_3} = \frac{x_2}{x_3}$

$\frac{\partial y_2}{\partial x_1} = \frac{x_2}{x_1}$, $\frac{\partial y_2}{\partial x_2} = -\frac{x_2x_3}{x_1^2}$, $\frac{\partial y_2}{\partial x_3} = \frac{x_3}{x_1}$

and $\frac{\partial y_3}{\partial x_1} = \frac{x_3}{x_2}$, $\frac{\partial y_3}{\partial x_2} = \frac{x_3}{x_2}$, $\frac{\partial y_3}{\partial x_3} = -\frac{x_3x_1}{x_2^2}$

$$\therefore \frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix} = \begin{vmatrix} -\frac{x_1x_2}{x_3^2} & \frac{x_1}{x_3} & \frac{x_2}{x_3} \\ \frac{x_2}{x_1} & -\frac{x_2x_3}{x_1^2} & \frac{x_3}{x_1} \\ \frac{x_3}{x_2} & \frac{x_3}{x_2} & -\frac{x_3x_1}{x_2^2} \end{vmatrix}$$

$$= \frac{1}{x(2x^2+2y^2)}\begin{vmatrix} -2x(1) & x(1) & x(1) \\ y(1) & -2y(1) & y(1) \\ z(1) & z(1) & -2z(1) \end{vmatrix} + \frac{1}{x(2x^2+2y^2)}\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= -1(1-1)-1(-1-1)+1(1+1) = 0+2+2 = 4 \neq 0.$$

Example 8.27. If $u = x + y + z$, $v = y^2 + z^2$, $w = xz + xy - y^2$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$. (U.P.T.U., 2003)

Solution. $$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ z+y & 2y & 2z \\ -y & x-2y & x \end{vmatrix}$$

$\therefore$ At the point $(1, -1, 0)$ $\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & -2 & 0 \\ 1 & 3 & 1 \end{vmatrix} = 1(-2-0)-1(0-0)+1(0+2) = -2+2 = 0$

Example 8.28. If $u = x^2 - y^2$, $v = 2xy$ and $x = r\cos\theta$, $y = r\sin\theta$, find $\frac{\partial(u,v)}{\partial(r,\theta)}$. (U.P.T.U., 2000) (Kanpur, 2004)

Solution. We have $$\frac{\partial(u,v)}{\partial(r,\theta)} = \frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(r,\theta)}$$

Since $u = x^2 - y^2$, $v = 2xy$

$$\therefore \quad \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2) \qquad \ldots(i)$$

Since $x = r\cos\theta$, $y = r\sin\theta$,

$$\therefore \quad \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r \qquad \ldots(ii)$$

Hence, $$\frac{\partial(u,v)}{\partial(r,\theta)} = 4(x^2 + y^2) \cdot r = 4(r^2\cos^2\theta + r^2\sin^2\theta) \cdot r = 4r^3$$ [Using (i) & (ii)]

Jacobian of Implicit functions. If $u_1, u_2, u_3, \ldots, u_n$ are n functions of n being given explicitly in terms $x_1, x_2, x_3, \ldots, x_n$ connected with them by relations such as

$$f_1(x_1, x_2, x_3, \ldots, x_n, u_1, u_2, u_3, \ldots, u_n) = 0, f_2(x_1, x_2, x_3, \ldots, x_n, u_1, u_2, u_3, \ldots, u_n) = 0, \text{ then}$$

$$\frac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2, x_3)} = (-1)^3 \frac{\partial(f_1, f_2, f_3)}{\partial(x_1, x_2, x_3)} \cdot \frac{\partial(f_1, f_2, f_3)}{\partial(u_1, u_2, u_3)}$$

Obs. This result can be easily generalized. If $u_1, u_2, u_3, \ldots, u_n$ are n functions of $x_1, x_2, x_3, \ldots, x_n$ connected by the relation $f(x, y) = 0$, then $\frac{\partial u}{\partial x} = -\frac{\partial f}{\partial x} \Big/ \frac{\partial f}{\partial u}$, where x, y are connected by the relation $f(x, y) = 0$.

Example 8.29. If $u + v + w = x + y + z$, $u^2 + v^2 + w^2 = x^2 + y^2 + z^2$, $u^3 + v^3 + w^3 = x^3 + y^3 + z^3$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$. (U.P.T.U., 2002)

Solution. Let $f_1 = u + v + w - x - y - z$, $f_2 = u^2 + v^2 + w^2 - x^2 - y^2 - z^2$, $f_3 = u^3 + v^3 + w^3 - x^3 - y^3 - z^3$.

We have $$\frac{\partial(u,v,w)}{\partial(x,y,z)} = (-1)^2 \frac{\partial(f_1,f_2,f_3)}{\partial(u,v,w)} \cdot \frac{\partial(u,v,w)}{\partial(x,y,z)} \quad \ldots(i)$$

Now, $$\frac{\partial(f_1,f_2,f_3)}{\partial(u,v,w)} = \begin{vmatrix} \partial f_1/\partial u & \partial f_1/\partial v & \partial f_1/\partial w \\ \partial f_2/\partial u & \partial f_2/\partial v & \partial f_2/\partial w \\ \partial f_3/\partial u & \partial f_3/\partial v & \partial f_3/\partial w \end{vmatrix} = \begin{vmatrix} -yz & -xz & -xy \\ -2u & -2v & -2w \\ -1 & -1 & -1 \end{vmatrix}$$

$$= -2(x-y)(y-z)(z-x) \quad \ldots(ii)$$

and $$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \quad \ldots(iii)$$

Substituting values from (ii) and (iii) in (i), we get

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = (-1)^2 \cdot 1 \cdot 1 \cdot 2(x-y)(y-z)(z-x) = 2(x-y)(y-z)(z-x)$$

14. Functional relationship. *If u, v, w be functions of x, y, z, then the necessary and sufficient condition for the existence of a functional relationship of the form $f(u, v, w) = 0$, is*

$$J\left(\frac{u,v,w}{x,y,z}\right) = 0$$

Example 2.28. *If $u = x\sqrt{(1-y^2)} + y\sqrt{(1-x^2)}$, $v = \sin^{-1}x + \sin^{-1}y$, show that u and v are functionally related and find the relation between them.* (Kerala, 2005)

Solution. We have $$\frac{\partial u}{\partial x} = \sqrt{(1-y^2)} - \frac{xy}{\sqrt{(1-x^2)}}, \frac{\partial u}{\partial y} = \frac{-xy}{\sqrt{(1-y^2)}} + \sqrt{(1-x^2)}$$

and $$\frac{\partial v}{\partial x} = \frac{1}{\sqrt{(1-x^2)}}, \frac{\partial v}{\partial y} = \frac{1}{\sqrt{(1-y^2)}}$$

$$\therefore \quad \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \partial u/\partial x & \partial u/\partial y \\ \partial v/\partial x & \partial v/\partial y \end{vmatrix} = \begin{vmatrix} \sqrt{(1-y^2)} - \frac{xy}{\sqrt{(1-x^2)}} & \sqrt{(1-x^2)} - \frac{xy}{\sqrt{(1-y^2)}} \\ \frac{1}{\sqrt{(1-x^2)}} & \frac{1}{\sqrt{(1-y^2)}} \end{vmatrix}$$

$$= 1 - \frac{xy}{\sqrt{(1-x^2)(1-y^2)}} - 1 + \frac{xy}{\sqrt{(1-x^2)(1-y^2)}} = 0$$

Hence u and v are functionally related i.e., they are not independent.

We have $$v = \sin^{-1}x + \sin^{-1}y = \sin^{-1}\{x\sqrt{(1-y^2)} + y\sqrt{(1-x^2)}\}$$

i.e., $$v = \sin^{-1}u$$

which is the required relationship between u and v.

EXERCISE 2.2

1. If $u = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$. (U.P.T.U., 2003)
2. If $f(x,y) = \sin^{-1}\left(\frac{x^2+y^2}{x+y}\right)$, prove that $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = \frac{1}{2}\tan f$. (U.P.T.U., 2004 C.O.)
3. If $u = \log(x^3+y^3+z^3-3xyz)$ and $(x+y+z) > 0$, show that $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$. (U.P.T.U., 2003)
4. If $u = x^2\phi\left(\frac{y}{x}\right) + y^2\psi\left(\frac{x}{y}\right)$, find $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$. (U.P.T.U., 2003 C.O.)
5. If $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, find $\frac{\partial(x,y,z)}{\partial(r,\theta,z)}$ and $\frac{\partial(r,\theta,z)}{\partial(x,y,z)}$.
6. If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$. (U.P.T.U., 2000; P.U.U.) (2000)

7. If $f(x, y, z) = x^2 + y^2 + z^2$, $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, evaluate $\frac{\partial f}{\partial r}$, $\frac{\partial f}{\partial \theta}$, $\frac{\partial f}{\partial z}$.

8. If $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, prove that $\frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r$.

(Anna University, 2004)

9. If $u = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, $v = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, $w = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$, evaluate $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.

(U.P.T.U., 2004; A.M.I.E., 2004)

10. If $u = \frac{x - y}{1 - xy}$ and $v = \tan^{-1} x + \tan^{-1} y$, find $\frac{\partial(u, v)}{\partial(x, y)}$. Are u and v functionally related? If so, find the relationship.

(Nagpur, 2005)

11. If $u = x^2 + y^2 - z^2$, $v = x^2 - y^2 + z^2$, $w = -x^2 + y^2 + z^2$, show that u, v, w are not independent. Find the relationship.

(Nagpur, 2005)

9.8 (1) GEOMETRICAL INTERPRETATION

If $P(x, y, z)$ be the coordinates of a point referred to axes OX, OY, OZ then the equation $z = f(x, y)$ represents a surface. (Fig. 9.1)

Let a plane $y = b$ parallel to the XZ plane pass through P cutting the surface along the curve APB given by

$$z = f(x, b).$$

As x remains equal to b and z varies then P moves along the curve APB and $\frac{dz}{dx}$ is the ordinary derivative of $f(x, b)$ w.r.t. x.

Hence $\left(\frac{\partial z}{\partial x}\right)_P$ is the tangent of the angle which the tangent at P to the section of the surface $z = f(x, y)$ by a plane through P parallel to the plane ZOX, makes with a line parallel to the x-axis.

Similarly, $\left(\frac{\partial z}{\partial y}\right)_P$ at P is the tangent of the angle which the tangent at P to the curve of intersection of the surface $z = f(x, y)$ and the plane $x = a$, makes with a line parallel to the y-axis.

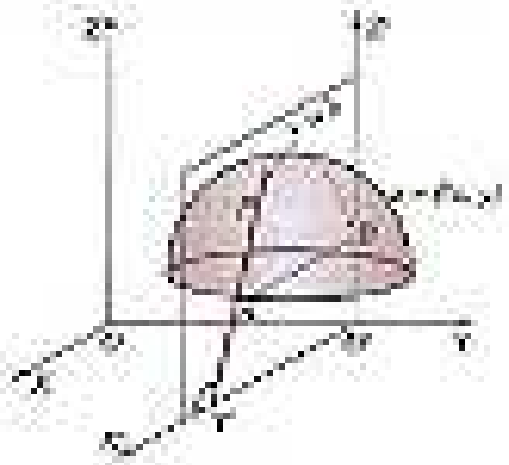


Fig. 9.1

(2) Tangent plane and Normal to a surface. Let $P(x, y, z)$ and $Q(x + \delta x, y + \delta y, z + \delta z)$ be two neighbouring points on the surface $F(x, y, z) = 0$ (Fig. 9.2) ...(1)

Let the arc PQ be δs and the chord PQ be δc, so that (as for plane curves)

$$\lim_{Q \to P} \left(\frac{\delta c}{\delta s}\right) = 1.$$

The direction cosines of PQ are $\frac{\delta x}{\delta c}, \frac{\delta y}{\delta c}, \frac{\delta z}{\delta c}$ i.e., $\frac{\delta x}{\delta s}\cdot\frac{\delta s}{\delta c}, \frac{\delta y}{\delta s}\cdot\frac{\delta s}{\delta c}, \frac{\delta z}{\delta s}\cdot\frac{\delta s}{\delta c}$

When $Q \to P$, $\delta s \to 0$ and PQ tends to tangent line PT. Then noting that the coordinates of any point on arc PQ are functions of s only, the direction cosines of PT are

$$\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds} \qquad \ldots(2)$$

Differentiating (1) with respect to s, we obtain $\frac{\partial F}{\partial x}\cdot\frac{dx}{ds} + \frac{\partial F}{\partial y}\cdot\frac{dy}{ds} + \frac{\partial F}{\partial z}\cdot\frac{dz}{ds} = 0$.

This shows that the tangent line whose direction cosines are given by (2) is perpendicular to the line having direction ratios

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \qquad \ldots(3)$$

Since we can take different curves passing through P, we get a number of tangent lines at P and the line having direction ratios (3) will be perpendicular to all these tangent lines at P. Thus all the tangent lines at P lie in a plane through P perpendicular to line (3).

Hence the equation of the tangent plane to (1) at the point P is

$$\frac{\partial F}{\partial x}(X - x) + \frac{\partial F}{\partial y}(Y - y) + \frac{\partial F}{\partial z}(Z - z) = 0$$

where X, Y, Z are the current coordinates of any point on the tangent plane.

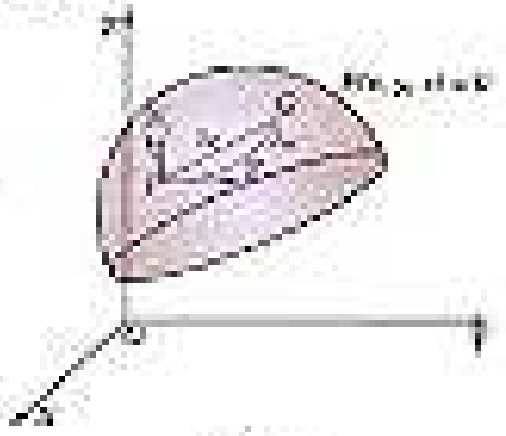


Fig. 9.2

Also the equation of the normal to the surface at $P(a, b, c)$ is the line through P, perpendicular to the tangent plane at P, i.e.

$$\frac{X-x}{\partial F/\partial x} = \frac{Y-y}{\partial F/\partial y} = \frac{Z-z}{\partial F/\partial z}$$

Example 8.21. Find the equations of the tangent plane and the normal to the surface $z^2 = 4x^2 + y^2$ at the point $(2, 2, 4)$.

Solution. We have $F(x, y, z) = 4x^2 + y^2 - z^2 = 4$.

i) $\partial F/\partial x = 8x$, $\partial F/\partial y = 2y$, $\partial F/\partial z = -2z$, and at the point $(2, 2, 4)$

$\partial F/\partial x = 16$, $\partial F/\partial y = 4$, $\partial F/\partial z = -8$

Hence the equation of the tangent plane at $(2, 2, 4)$ is $16X - 32 + 4Y - 2 - 8Z - 64 = 0$

i.e. $$4X + Y - 2Z + 2 = 0 \qquad \ldots(i)$$

Also the equation of the normal at $(2, 2, 4)$ being perpendicular to (i) is

$$\frac{X-2}{4} = \frac{Y-2}{1} = \frac{Z-4}{-2}$$

PROBLEMS 8.8

Find the equations of the tangent plane and normal to each of the following surfaces at the given points.

1. $2x^2 + y^2 + z^2 = 2x + 4z + 13$ at $(3, 1, -2)$ (January 2004)
2. $z(x^2 + y^2) = x$ at $(1, 1, 1)$
3. Show the plane $2x + 2y - z = 1$ is a tangent to the ellipsoid $x^2 + 2y^2 + z^2 + 2 = 0$. Find the point of contact.
4. $x^2 - y^2 + xyz + 5 = 0$ at $(2, 3, -1)$ (October 2005)
5. $2xz^2 - 3xy - 4x + 7 = 0$ at $(1, -3, 2)$
6. Show that the ellipsoid $x^2 + 3y^2 + z^2 = 12$ touches the sphere $x^2 + y^2 + z^2 - 8x - 6y - 8z + 24 = 0$. Find the point of contact.
7. Show that the plane $x + y + z = 3$ touches the surface $xy + yz + zx = 3$. Find the point of contact.
8. Show that the plane $lx + my + nz = p$ is a tangent to the surface $ax^2 + by^2 + cz^2 = 1$ if $\frac{l^2}{a} + \frac{m^2}{b} + \frac{n^2}{c} = p^2$. (Hint)
9. Find the equation of the normal to the surface $x^2 + y^2 = 2z^2$ at $(2, 2, 2)$. (U.P.T.U., 2005)

8.9 TAYLOR'S THEOREM FOR FUNCTIONS OF TWO VARIABLES

Considering $f(x + h, y + k)$ as a function of a single variable x, we have by Taylor's theorem*

$$f(x + h, y + k) = f(x, y + k) + h\frac{\partial f(x, y+k)}{\partial x} + \frac{h^2}{2!}\frac{\partial^2 f(x, y+k)}{\partial x^2} + \ldots \qquad \ldots(i)$$

Now expanding $f(x, y + k)$ as a function of y only,

$$f(x, y + k) = f(x, y) + k\frac{\partial f(x, y)}{\partial y} + \frac{k^2}{2!}\frac{\partial^2 f(x, y)}{\partial y^2} + \ldots$$

$\therefore$ (i) takes the form $f(x + h, y + k) = f(x, y) + h\frac{\partial f(x, y)}{\partial x} + \frac{h^2}{2!}\frac{\partial^2 f(x, y)}{\partial x^2} + \ldots$

$$+ k\frac{\partial}{\partial x}\left\{f(x, y) + h\frac{\partial f(x, y)}{\partial x} + \frac{h^2}{2!}\frac{\partial^2 f(x, y)}{\partial x^2} + \ldots\right\} + \frac{k^2}{2!}\frac{\partial^2}{\partial x^2}\left\{f(x, y) + h\frac{\partial f(x, y)}{\partial x} + \ldots\right\}$$

Hence, $f(x + h, y + k) = f(x, y) + h\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial y} + \frac{1}{2!}\left(h^2\frac{\partial^2 f}{\partial x^2} + 2hk\frac{\partial^2 f}{\partial x \partial y} + k^2\frac{\partial^2 f}{\partial y^2}\right) + \ldots \qquad \ldots(ii)$

In symbolic operator form $f(x + h, y + k) = f(x, y) + \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)f + \frac{1}{2!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^2 f + \ldots$

Taking $x = a$ and $y = b$, (ii) becomes

$$f(a + h, b + k) = f(a, b) + hf'_x(a, b) + kf'_y(a, b) + \frac{1}{2!}[h^2 f''_{xx}(a, b) + 2hkf''_{xy}(a, b) + k^2 f''_{yy}(a, b)] + \ldots$$

*See footnote on page 145.

Putting $x + h = x$ and $b + k = y$ so that $h = x - a$, $k = y - b$, we get

$$f(x, y) = f(a, b) + [(x - a) f_x(a, b) + (y - b) f_y(a, b)]$$

$$+ \frac{1}{2!}[(x - a)^2 f_{xx}(a, b) + 2(x - a)(y - b) f_{xy}(a, b) + (y - b)^2 f_{yy}(a, b)] + \dots \quad \dots (1)$$

This is Taylor's expansion of $f(x, y)$ in powers of $(x - a)$ and $(y - b)$. It is used to expand $f(x, y)$ in the neighbourhood of (a, b).

Cor. Putting $a = 0$, $b = 0$, we get

$$f(x, y) = f(0, 0) + [x f_x(0, 0) + y f_y(0, 0)] + \frac{1}{2!}[x^2 f_{xx}(0, 0) + 2xy f_{xy}(0, 0) + y^2 f_{yy}(0, 0)] + \dots \quad \dots (2)$$

This is Maclaurin's expansion of $f(x, y)$.

Example 8.28. Expand $e^x \log(1 + y)$ in powers of x and y upto second degree terms. (U.P.T.U., 2000; P.T.U., 2006; J.N.T.U., 2006)

Solution. Here

$$f(x, y) = e^x \log(1 + y) \quad \therefore \quad f(0, 0) = 0$$

$$f_x(x, y) = e^x \log(1 + y) \qquad f_x(0, 0) = 0$$

$$f_y(x, y) = e^x \frac{1}{1 + y} \qquad f_y(0, 0) = 1$$

$$f_{xx}(x, y) = e^x \log(1 + y) \qquad f_{xx}(0, 0) = 0$$

$$f_{xy}(x, y) = e^x \frac{1}{1 + y} \qquad f_{xy}(0, 0) = 1$$

$$f_{yy}(x, y) = -e^x (1 + y)^{-2} \qquad f_{yy}(0, 0) = -1$$

$$f_{xxx}(x, y) = e^x \log(1 + y) \qquad f_{xxx}(0, 0) = 0$$

$$f_{xxy}(x, y) = e^x \frac{1}{1 + y} \qquad f_{xxy}(0, 0) = 1$$

$$f_{xyy}(x, y) = -e^x (1 + y)^{-2} \qquad f_{xyy}(0, 0) = -1$$

$$f_{yyy}(x, y) = 2e^x (1 + y)^{-3} \qquad f_{yyy}(0, 0) = 2$$

Now Maclaurin's expansion of $f(x, y)$ gives

$$f(x, y) = f(0, 0) + x f_x(0, 0) + y f_y(0, 0) + \frac{1}{2!}[x^2 f_{xx}(0, 0) + 2xy f_{xy}(0, 0) + y^2 f_{yy}(0, 0)]$$

$$+ \frac{1}{3!}[x^3 f_{xxx}(0, 0) + 3x^2 y f_{xxy}(0, 0) + 3xy^2 f_{xyy}(0, 0) + y^3 f_{yyy}(0, 0)] + \dots$$

$$\therefore \quad e^x \log(1 + y) = 0 + x(0) + y(1) + \frac{1}{2}[x^2(0) + 2xy(1) + y^2(-1)]$$

$$+ \frac{1}{6}[x^3(0) + 3x^2 y(1) + 3xy^2(-1) + y^3(2)] + \dots$$

$$= y + xy - \frac{1}{2}y^2 + \frac{1}{2}x^2 y - xy^2 + \frac{1}{3}y^3 + \dots$$

Example 8.29. Expand $x^2 y + 3y - 2$ in powers of $(x - 1)$ and $(y + 2)$ using Taylor's theorem.
(P.T.U., 2003; K.U.K., 2005; G.J.U.S.T., 2006; A.M.U., 2006)

Solution. Taylor's expansion of $f(x, y)$ in powers of $(x - a)$ and $(y - b)$ is given by

$$f(x, y) = f(a, b) + [(x - a) f_x(a, b) + (y - b) f_y(a, b)] + \frac{1}{2!}[(x - a)^2 f_{xx}(a, b)$$

$$+ 2(x - a)(y - b) f_{xy}(a, b) + (y - b)^2 f_{yy}(a, b)] + \frac{1}{3!}[(x - a)^3 f_{xxx}(a, b)$$

$$+ 3(x - a)^2 (y - b) f_{xxy}(a, b) + 3(x - a)(y - b)^2 f_{xyy}(a, b)$$

$$+ (y - b)^3 f_{yyy}(a, b)] + \dots \quad \dots (1)$$

Here $a = 1$, $b = -2$ and $f(x, y) = x^2 y + 3y - 2$

$f(1,-1)=-10, f_x=2y, f_x(1,-1)=-4, f_y=x^2+3, f_y(1,-1)=4; f_{xx}=2,$
$f_{xy}(1,-1)=-6, f_{yy}=6x, f_{yy}(1,-1)=6, f_{xxx}=0, f_{xxy}=2, f_{xyy}=6, f_{yyy}=0, f_{xxxx}=0, f_{xxxy}=0, f_{xxyy}=2, f_{xyyy}=0, f_{yyyy}=0$

All partial derivatives of higher order vanish.

Substituting these in (4), we get

$$x^2y+3y-2=-10+8(x-1)-6(y+1)+(y+1)+2(x-1)^2(-1)+2(x-1)(y+1)+\frac{1}{2}(x-1)^2(-6)+2(x-1)(y+1)+0$$

$$+(y+1)^2(3)+\frac{1}{6}[(x-1)^3(0)+3(x-1)^2(y+1)+3(x-1)(y+1)^2+(y+1)^3(6)]+2!(3)(y+1)^2+2!(1)(x-1)(y+1)$$

$$=-10-4(x-1)+6(y+1)+3(x-1)^2-3(x-1)(y+1)+3(y+1)^2+(x-1)^2(y+1)+(y+1)^3.$$

Example 5.51 Expand $f(x,y)=\tan^{-1}\left(\frac{y}{x}\right)$ in powers of $(x-1)$ and $(y-2)$ using Taylor's theorem. Hence compute $f(1.1, 2.0)$ approximately. (U.P.T.U., 2003; U.P.T.U., 2005; U.P.T.U., 2006)

Solution. Here $a=1, b=2$ and $f(1,2)=\tan^{-1}(2)=\pi/4$.

$$f_x=\frac{-y}{x^2+y^2}, \quad f_x(1,2)=-\frac{2}{5}; \quad f_y=\frac{x}{x^2+y^2}, \quad f_y(1,2)=\frac{1}{5}$$

$$f_{xx}=\frac{2xy}{(x^2+y^2)^2}, \quad f_{xx}(1,2)=\frac{4}{25}; \quad f_{xy}=\frac{y^2-x^2}{(x^2+y^2)^2}, \quad f_{xy}(1,2)=0$$

$$f_{yy}=\frac{-2xy}{(x^2+y^2)^2}, \quad f_{yy}(1,2)=-\frac{4}{25};$$

$$f_{xxx}=\frac{2y^3-6x^2y}{(x^2+y^2)^3}, \quad f_{xxx}(1,2)=-\frac{4}{25}; \quad f_{xxy}=\frac{3x^2-6xy^2}{(x^2+y^2)^3}, \quad f_{xxy}(1,2)=-\frac{2}{25}$$

$$f_{xyy}=\frac{6xy^2-2x^3}{(x^2+y^2)^3}, \quad f_{xyy}(1,2)=\frac{4}{25}; \quad f_{yyy}=\frac{6xy^2-2y^3}{(x^2+y^2)^3}, \quad f_{yyy}(1,2)=\frac{2}{25}$$

Taylor's expansion of $f(x,y)$ in powers of $(x-1)$ and $(y-2)$ is given by

$$f(x,y)=f(1,2)+\frac{1}{1!}[(x-1)f_x(1,2)+(y-2)f_y(1,2)]+\frac{1}{2!}[(x-1)^2f_{xx}(1,2)+2(x-1)(y-2)f_{xy}(1,2)+(y-2)^2f_{yy}(1,2)]$$

$$+\frac{1}{3!}[(x-1)^3f_{xxx}(1,2)+3(x-1)^2(y-2)f_{xxy}(1,2)+3(x-1)(y-2)^2f_{xyy}(1,2)+(y-2)^3f_{yyy}(1,2)]+\ldots$$

$$\therefore \tan^{-1}\left(\frac{y}{x}\right)=\frac{\pi}{4}+\frac{1}{1!}\left[(x-1)\left(-\frac{2}{5}\right)+(y-2)\frac{1}{5}\right]+\frac{1}{2!}\left[(x-1)^2\frac{4}{25}+2(x-1)(y-2)(0)+(y-2)^2\left(-\frac{4}{25}\right)\right]$$

$$+\frac{1}{3!}\left[(x-1)^3\left(-\frac{4}{25}\right)+3(x-1)^2(y-2)\left(-\frac{2}{25}\right)+3(x-1)(y-2)^2\frac{4}{25}+(y-2)^3\frac{2}{25}\right]+\ldots$$

$$=\frac{\pi}{4}-\frac{1}{5}[(x-1)+2(y-2)]+\frac{1}{50}[2(x-1)^2-(x-1)(y-2)-2(y-2)^2]-\frac{1}{150}[2(x-1)^3+3(x-1)^2(y-2)-6(x-1)(y-2)^2-2(y-2)^3]+\ldots$$

Putting $x=1.1$ and $y=2.0$, we get

$$f(1.1, 2.0)=\frac{\pi}{4}-\frac{1}{5}(0.1)+\frac{1}{50}(0.01)-\frac{1}{150}[2(0.1)^3-3(0.1)^2(0.2)-6(0.1)(0.2)^2-2(0.2)^3]$$

$$=0.7854-0.1000+0.0002+0.0027.$$

5.10 [3] ERRORS AND APPROXIMATIONS

Let $f(x, y)$ be a continuous function of x and y. If Δx and Δy be the increments in x and y, then the increment of $f(x, y)$ will be $f(x+\Delta x, y+\Delta y)-f(x, y)$. Hence

$$\Delta f = f(x + \delta x, y + \delta y) - f(x, y)$$

By applying $f(x + \delta x, y + \delta y)$ by Taylor's theorem and supposing $\delta x, \delta y$ to be so small that their products, squares and higher powers can be neglected, we get

$$\delta f = \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y, \text{ approximately.}$$

Similarly if f be a function of several variables $x, y, z, t, \ldots$, then

$$\delta f = \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y + \frac{\partial f}{\partial z}\delta z + \frac{\partial f}{\partial t}\delta t + \ldots \text{ approximately.}$$

These formulae are very useful in estimating the effect of small errors in measured quantities.

(2) Total Differential

If u is a function of two variables x and y, the total differential du is defined as

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy \qquad \ldots(1)$$

The differentials dx and dy are respectively the increments in x and y. If x and y are not independent variables but functions of another variable t, we can find du by formula (1) by finding out $dx = \frac{dx}{dt}dt$ and $dy = \frac{dy}{dt}dt$. Similar definitions can be given for a function of three or more variables.

Example 8.33. *The diameter and altitude of a cone in the shape of a metallic crucible are found to be 4 cm and 6 cm respectively. The possible error in each measurement is 0.1 cm. Find approximately the maximum possible error in the values computed for the volume and the lateral surface.*

Solution. Let x be the diameter and y the height of the cone. Then the volume $V = \frac{\pi}{4}x^2y$.

$$\therefore \quad \delta V = \frac{\partial V}{\partial x}\delta x + \frac{\partial V}{\partial y}\delta y = \frac{\pi}{4}(2xy\delta x + x^2\delta y)$$

When $x = 4$ cm., $y = 6$ cm. and $\delta x = \delta y = 0.1$ cm.

$$\therefore \quad \delta V = \frac{\pi}{4}(2 \times 4 \times 6 \times 0.1 + 4^2 \times 0.1) = 1.6\pi \text{ cm}^3$$

Also the lateral surface $S = \pi xy$

$$\therefore \quad \delta S = \pi(y\delta x + x\delta y)$$

When $x = 4$ cm., $y = 6$ cm. and $\delta x = \delta y = 0.1$ cm., we have $\delta S = \pi(6 \times 0.1 + 4 \times 0.1) = \pi$ cm^2.

Example 8.34. *The period of a vertical pendulum of length l is $T = 2\pi\sqrt{(l/g)}$. Find the percentage error in T due to 2% error in l and 3% error in g.* (U.P.T.U., 2003)

Solution. We have $T = 2\pi\sqrt{(l/g)}$

or $$\log T = \log 2\pi + \frac{1}{2}\log l - \frac{1}{2}\log g$$

$$\therefore \quad \frac{1}{T}\delta T = 0 + \frac{1}{2}\frac{1}{l}\delta l - \frac{1}{2}\frac{1}{g}\delta g$$

$$\frac{\delta T}{T} \times 100 = \frac{1}{2}\left[\frac{\delta l}{l} \times 100 - \frac{\delta g}{g} \times 100\right] = \frac{1}{2}(2 + 3) = 2.5 \text{ or } -2.5\%$$

Thus the maximum error in $T = 2.5\%$.

Example 8.35. *A uniform metal cylinder has a radius of exactly 1 cm and length 10 cm. If an error of ε cm is committed in measuring each of the radius and the length, find the maximum error in the volume of the cylinder in the form $k\pi\varepsilon$ cm^3.* (U.P.T.U., 2005)

Solution. Let the volume of the balloon (Fig. 6.3) be V, so that

$$V = \pi r^2 h + \frac{2}{3}\pi r^3 = \pi r^2 h + \frac{4}{3}\pi r^3$$

$$\therefore \quad \delta V = 2\pi r h\,\delta r + \pi r^2 \delta h + \frac{4}{3}\pi 3r^2\,\delta r$$

or

$$\frac{\delta V}{V} = \frac{\pi(2rh\,\delta r + r^2\delta h + 4r^2\delta r)}{\pi r^2 h + \frac{4}{3}\pi r^3}$$

$$= \frac{2h\,\delta r + r\delta h + 4r\delta r}{rh + \frac{4}{3}r^2} = \frac{2(4)(0.01) + 1.5(0.05)}{1.5\times 4 + \frac{4}{3}(1.5)^2}$$

$$= \frac{0.11 + 0.075}{6 + 3} = \frac{0.215}{9}$$

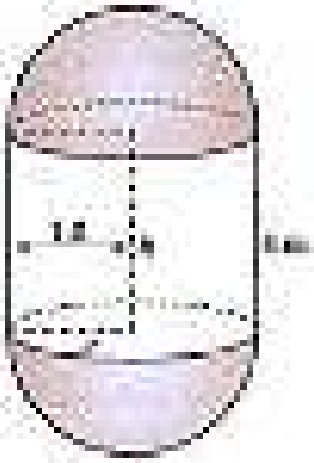


Fig 6.3

Hence, the percentage change in $V = 100\,\dfrac{\delta V}{V} = \dfrac{21.5}{9} = 2.39\%$

Example 6.20. In calculating the cost of a certain brick manufacturer it is assumed that a brick is $20\text{ cm} \pm 1.5\text{ mm}$ in length, $10\text{ cm} \pm 1.0\text{ mm}$ in breadth and $5\text{ cm} \pm 0.5\text{ mm}$ in thickness. If the cost of bricks is Rs 5.00 per 1000 find the approximate error in the cost. [WBUT (Tech) 2003]

Solution. Let x, y and z be the length, breadth and height of the gift so that its volume V is given by

$$V = xyz$$

or

$$\log V = \log x + \log y + \log z \quad \therefore \quad \frac{\delta V}{V} = \frac{\delta x}{x} + \frac{\delta y}{y} + \frac{\delta z}{z}$$

Since

$$V = 2\times 15\times 0.5 = 30\text{ m}^3, \text{ and } \frac{\delta x}{x} = \frac{\delta y}{y} = \frac{\delta z}{z} = \frac{1}{100}$$

$$\therefore \quad \delta V = 30\left(\frac{3}{100}\right) = 0.90\text{ m}^3$$

Number of bricks is $N = 1.08 \times 400 = 432$

The error in the cost $= 432 \times \dfrac{500}{1000} = ₹\,216.00$ which is a loss to the brick seller.

Example 6.21. The height of a conical tent is increased by a small amount δh and the base radius is decreased by the same amount so that the tent is extended. If r and h are in error by small quantities δr and δh respectively, find the resulting error in the area. Also find the error if $r = 3$ m, $h = 4$ m and $\delta r = \delta h = 0.01$ m.

Solution. If r be the base radius and l the slant height of the cone (Fig. 6.4), then total area

A = area of base + area of curved surface

$= \pi r^2 + \pi r l = \pi r(r + l)$

$= \pi r(r + \sqrt{r^2 + h^2})$

$= \pi h^2 \tan^2\theta + \pi h \tan\theta \sec\theta$

$\therefore \quad \delta A = \dfrac{\partial A}{\partial h}\delta h + \dfrac{\partial A}{\partial \theta}\delta\theta$

$= 2\pi h \tan^2\theta\,\delta h + \pi\tan\theta\sec\theta\,\delta\theta$

$= \pi h^2(2\tan\theta\sec^2\theta + \sec^2\theta + \tan\theta\sec\theta)\,\delta r$ [$\because \delta h = \delta r$ and $\delta\theta = \dfrac{\delta r}{h}\cos^2\theta$]

which gives the error in the area A.

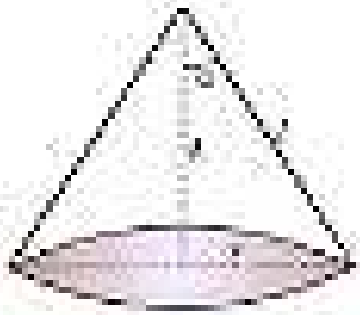


Fig 6.4

Putting $\delta h = 0.01$ and $\theta = \pi/4$, we get

$$\delta A = 2\pi\left[\left(\frac{1}{\sqrt{2}}\right)^2 + \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}}\right]\frac{3}{100} + \pi 3^2\left[2\cdot\frac{1}{\sqrt{2}}\cdot\frac{4}{3} + \frac{4}{3\sqrt{2}} + \frac{1}{\sqrt{2}}\cdot\frac{2}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\right]\delta\theta$$

$$= \frac{2\Delta^2}{100} + 2.65\Delta^2 \text{ sq. cm}$$

The error in Δ will be compensated by the error in π, when

$$\delta\Delta + \delta\left(\frac{2\Delta^2}{100} + 2.65\Delta^2\right) = 0$$

or $$\delta\pi = -\frac{1}{100\pi}\cdot\pi\Delta\cdot\pi = -\frac{\Delta}{100}\times 57.3^\circ = -0.57^\circ.$$

Example 6.40. Show that the approximate change in the angle A of a triangle ABC due to small changes δa, δb, δc in the sides a, b, c respectively, is given by

$$\delta A = \frac{\pi}{\Delta}(\delta a - \delta b\cos C - \delta c\cos B)$$

where Δ is the area of the triangle. Verify that $\delta A + \delta B + \delta C = 0$.

Solution. We know that $a^2 = b^2 + c^2 - 2bc\cos A$

so that $$2a\delta a = 2b\delta b + 2c\delta c - 2(b\delta c\cos A - Ab\sin A\,\delta c + c\delta b\cos A - Ac\sin A\,\delta b)$$

or $$\delta a\sin A\,\delta A = a\delta a - b\delta b - c\delta c + (c\cos A)\delta b + (b\cos A)\delta c$$

or $$2\Delta\delta A = a\delta a - b\delta b - c\delta c + (a\cos C + c\cos A)\delta b + (a\cos B + b\cos A)\delta c \quad [\because\ b = c\cos A + a\cos C \text{ etc.}\ \ldots(1)]$$

$$= a\delta a - a\cos C\,\delta b - a\cos B\,\delta c$$

or $$\delta A = \frac{\pi}{\Delta}(\delta a - \delta b\cos C - \delta c\cos B)$$

By symmetry, we have

$$\delta B = \frac{\pi}{\Delta}(\delta b - \delta c\cos A - \delta a\cos C)$$

$$\delta C = \frac{\pi}{\Delta}(\delta c - \delta a\cos B - \delta b\cos A)$$

$$\therefore\quad \delta A + \delta B + \delta C = \frac{\pi}{\Delta}[\delta a - b\cos C - c\cos B + \delta b - c\cos A - a\cos C + \delta c - a\cos B - b\cos A]$$

$$= \frac{\pi}{\Delta}[\delta a - a(\delta a + b - b) + \delta b - b(\delta b + c - c) + \delta c - c(\delta c + a - a)] = 0 \quad [\text{By (1)}]$$

Example 6.41. If the sides of a plane triangle ABC vary in such a way that its circumradius remains constant, prove that $\frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C} = 0$.

Solution. The circumradius R of $\triangle ABC$ is given by

$$R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

$$\therefore\quad a = 2R\sin A \quad [\because\ R \text{ is constant}]$$

Taking differentials, $\delta a = 2R\cos A\,\delta A$ or $\frac{\delta a}{\cos A} = 2R\delta A$

Similarly, $\frac{\delta b}{\cos B} = 2R\delta B$, $\frac{\delta c}{\cos C} = 2R\delta C$

$$\therefore\quad \frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C} = 2R(\delta A + \delta B + \delta C)$$

Now $$A + B + C = \pi, \text{ gives } \delta A + \delta B + \delta C = 0 \quad \ldots(1)$$

Thus $$\frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C} = 0 \quad [\text{By (1)}]$$

PROBLEMS 2.6

1. Expand the following functions as far as terms of third degree:
   (a) $e^x \ln(1+y)$ [(T.U.) 2000] (b) $e^{x}\sin y$ [Ans. $y+xy-\frac{y^3}{6}$] (c) $\ln(x^2+\cos y)$ about $(1,\pi/2)$ [Ans. $\ln 2+2(x-1)-\frac{1}{2}(y-\frac{\pi}{2})^2$] [Hint: $\cos y=\sin(\frac{\pi}{2}-y)$]
2. Expand $f(x,y)=\tan^{-1}\frac{y}{x}$ in powers of $(x-1)$ and $(y-1)$ [(U.T.U.) 2005]
3. If $f(x,y)=\tan^{-1}\frac{xy}{1+x^2y^2}$, find approximately the change in f when x changes from 10 to 10.5 and y changes from 1.00 to 1.05. [(V.T.U.) 2005]
4. Find the possible percentage error in computing the resistance R of a circuit from $R=\frac{E}{I}$ if E is measured with an error of 2% and I is measured with an error of 3%. [Ans. 5%]
5. The voltage V across a resistance R is calculated from the formula $V=IR$. If I is measured with an error of 2% and R is measured with an error of 3%, find the percentage error in the calculated value of V. [Ans. 5%]
6. The time of oscillation of a simple pendulum is given by the equation $T=2\pi\sqrt{l/g}$. In an experiment carried out to find the value of g, errors of 1.0% and 0.5% are possible in the values of l and T respectively. Find the error in the calculated value of g. [Ans. 2%] [Hint: $\frac{\delta g}{g}=\frac{\delta l}{l}-2\frac{\delta T}{T}$]
7. If $y=x^n$, x and the relative errors in y and x are respectively 0.05 and 0.01, show that the error in n is 5%. [Ans. 1.609] [Hint: $\frac{\delta y}{y}=n\frac{\delta x}{x}$]
8. If the H.P. required to propel a certain vessel varies as the cube of the velocity and as the square of the length, find the percentage increase in velocity and the percentage increase in length which together would increase the H.P. by 10%. [Ans. 3.23%, 6.46%]
9. The range R of a projectile is given by $R=\frac{V^2}{g}\sin 2\alpha$, where V is the velocity of projection and α is the angle of projection. Find the percentage error in R due to an error of 1% in V and an error of $\frac{1}{2}\%$ in α. [Ans. 2%]
10. In calculating the cost of a pile of bricks measured as $2(x+y)\times 3$ m $\times$ 2 m, the cost is calculated @ Rs. x per m^3 and Rs. y per m^2 for the lengths. If the error in x is 1% while in y it is 2% and the cost is Rs. 1000 per 1000, find the approximate error in the cost. [(U.P.T.U.) 2013; (V.T.U.) 2005]
11. The deflection at the centre of a rod of length l and diameter d supported at its ends, both of the ends with a weight w is found as $\alpha w l^3/d^4$. What is the increase in the deflection corresponding to 2% increase in w, 3% decrease in l and 5% increase in d? [Ans. 0.04]
12. The work W required to draw a cylinder of water by sliding a piston B fitting it closely is given by $W=\frac{\pi}{4}\rho g h^2 s$. Find approximately the increase in W which would be caused by increasing h by 1%, the density ρ by 2% and the distance s by 3%. [Ans. 4%]
13. The calculated horse power P of an engine is calculated from the formula $P=PLAN/33,000$, where P is $p/\sqrt{A}$. Assuming that errors of +2% are made in measuring p, L, A and N, find the greatest possible error in P. [Ans. 5%]
14. The resistance (in ohms) of a length of wire is calculated from the formula $R=k\rho l/d^2$. If l is increased by 2%, ρ is increased by 3%, d is increased by 1.5%, show that the value of R is increased by 3.5% approximately.
15. If $x^2+y^2+z^2=2xyz$, show that $\frac{dx}{\sqrt{1-x^2}}+\frac{dy}{\sqrt{1-y^2}}+\frac{dz}{\sqrt{1-z^2}}=0$.
    [Hint: Use $x=\sin\alpha$, $y=\sin\beta$, $z=\sin\gamma$ and show that $\sin\alpha\cos(\beta+\gamma)+\sin\beta\cos(\gamma+\alpha)+\sin\gamma\cos(\alpha+\beta)=0$]

2.11 [3] MAXIMA AND MINIMA OF FUNCTIONS OF TWO VARIABLES

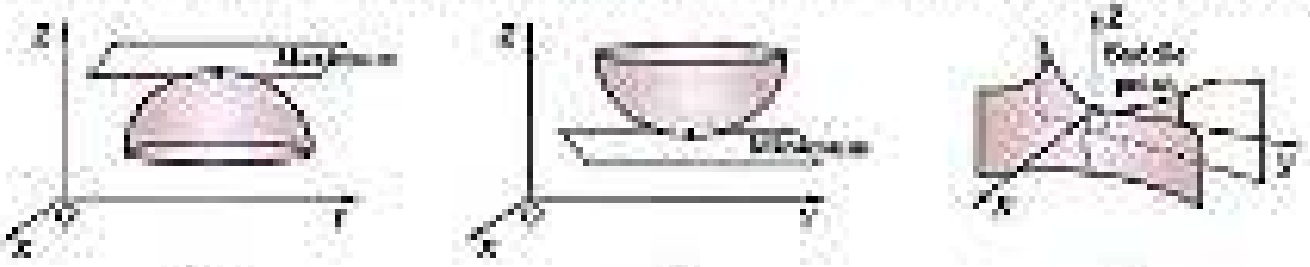
Def. *A function $f(x,y)$ is said to have a maximum or minimum at $x=a$, $y=b$, according as*

$$f(a+h,b+k)<f(a,b) \quad \text{or} \quad >f(a,b),$$

for all positive or negative small values of h and k.

In other words, $f(a,b)$ is one of the two things, for all small values of h and k (whether positive or negative), (i) $f(a,b)$ is greatest, or (ii) $f(a,b)$ is least. If the sign is positive, $f(a,b)$ is a minimum.

Consider the surface $z = f(x, y)$ as in Fig. (a), the maximum value of z may exist at the highest elevation in yz - plane (or xz - plane) from which the surface descends in every direction and a minimum value may exist at the bottom of a depression in yz - plane (or xz - plane) from which the surface ascends in every direction. Sometimes the maximum or minimum value may form a ridge such that the surface descends or ascends in all directions except that of the ridge. For this reason, we have to find a point of the surface, where the tangent plane is horizontal and the surface looks like a saddle as on the horse's back [Fig. (c)] which falls for displacement in certain direction and rises for displacement in other directions. Such a point is called a saddle point.



(a) (b) (c)

Fig. 9.1

Note. A maximum or a minimum value of a function is called its extreme value.

(2) Conditions for $f(x, y)$ to be maximum or minimum.

Using Taylor's theorem (page 2.8), we have $f(a + h, b + k) = f(a, b)$

$$= \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)_{a,b} + \frac{1}{2!}\left(h^2\frac{\partial^2}{\partial x^2} + 2hk\frac{\partial^2}{\partial x\partial y} + k^2\frac{\partial^2}{\partial y^2}\right) + \dots \quad ...(1)$$

For small values of h and k, the second and higher order terms are still smaller and hence may be neglected. Thus

$$\text{sign of } \Delta = \text{sign of } [f(a, b) + hf_x(a, b) + kf_y(a, b)]$$

Taking $k = 0$ we see that the right hand side changes sign when h changes sign. Hence $f(x, y)$ cannot have a maximum or a minimum at (a, b) unless $f_x(a, b) = 0$.

Similarly taking $h = 0$, we find that $f(x, y)$ cannot have a maximum or minimum at (a, b) unless $f_y(a, b) = 0$.

Hence the necessary conditions for $f(x, y)$ to have a maximum or minimum at (a, b) are that

$$f_x(a, b) = 0, f_y(a, b) = 0.$$

If these conditions are satisfied, then for small values of h and k, (1) gives

$$\text{sign of } \Delta = \text{sign of } \left[\frac{1}{2}(Ar^2 + 2Bhk + Ck^2)\right] \text{ where } A = f_{xx}(a, b), B = f_{xy}(a, b) \text{ and } C = f_{yy}(a, b)$$

$$\text{Now } Ar^2 + 2Bhk + Ck^2 = \frac{1}{A}\left[(Ar)^2 + 2(Ar)(Bk) + (Bk)^2\right] - \frac{1}{A}\left[Ak^2 + Bk^2 + A^2k^2 - B^2k^2\right]$$

$$\text{Thus sign of } \Delta = \text{sign of } \frac{1}{2A}\left[(Ar + Bk)^2 + k^2(AC - B^2)\right] \quad ...(2)$$

In (2), $(Ar + Bk)^2$ is always positive and $k^2(AC - B^2)$ will be positive if $AC - B^2 > 0$. In this case, A will have the same sign as that of for small values of h and k.

Hence if $AC - B^2 > 0$, then $f(x, y)$ has a maximum or a minimum at (a, b) according as $A < 0$ or > 0.

If $AC - B^2 < 0$, then Δ will change with h and k and hence there is no maximum or minimum at (a, b) i.e., it is a saddle point.

If $AC - B^2 = 0$, further investigation is required to find whether there is a maximum or minimum at (a, b) or not.

Note. Stationary value. $f(a, b)$ is said to be a stationary value of $f(x, y)$ if $f_x(a, b) = 0$ and $f_y(a, b) = 0$ i.e. the function is stationary at (a, b).

Thus every extreme value is a stationary value but the converse is not be true.

(3) Working rule to find the maximum and minimum values of $f(x, y)$.

1. Find $\partial f/\partial x$ and $\partial f/\partial y$ and equate each to zero. Solve these as simultaneous equations in x and y. Let (a, b), (c, d), ... be the points of return.
2. Calculate the values of $r = \partial^2 f/\partial x^2$, $s = \partial^2 f/\partial x\partial y$, $t = \partial^2 f/\partial y^2$ for each pair of values.

A. (a) If $rt - s^2 > 0$ and $r > 0$ at (a, b), $f(a, b)$ is a minimum value.

(b) If $rt - s^2 > 0$ and $r < 0$ at (a, b), $f(a, b)$ is a maximum value.

(c) If $rt - s^2 < 0$ at (a, b), $f(a, b)$ is not an extreme value, i.e. (a, b) is a saddle point.

(d) If $rt - s^2 = 0$ at (a, b), the nature of the point cannot be determined and needs further investigation.

Similarly examine the other points and values and hence find the maximum and minimum values.

Example 8.12. Determine the stationary values of the following function and distinguish between them:

$$f(x, y) = x^3 + y^3 - 3x^2 + 3xy - 3y^2 \qquad \text{(U.E.I.T.C., 2003)}$$

Solution. We have $f_x = 3x^2 - 6x + 3y$; $f_y = 3y^2 + 3x - 6y$

and $r = f_{xx} = 6x - 6$; $s = f_{xy} = 3$; $t = f_{yy} = 6y - 6$ ... (1)

Now $f_x = 0, f_y = 0$ give $x^2 - 2x + y = 0$, ... (2) $y^2 + x - 2y = 0$ ... (3)

Adding (2) and (3), we get $(x^2 + y^2) - (x + y) = 0$ or $y = -x$.

Putting $y = -x$ in (2), we obtain $x^2 - 3x = 0$, i.e. $x = 0, 3$, i.e. $x = 0, \sqrt{3}, -\sqrt{3}$, 0.

∴ Corresponding values of y are $-0, -\sqrt{3}, \sqrt{3}$, 0.

At $(\sqrt{3}, -\sqrt{3})$, $rt - s^2 = 36 - 6 - 36 - 9 = -9 < 0$, i.e. $rt - s^2$ is negative. Hence $f(\sqrt{3}, -\sqrt{3})$ is not an extreme value.

At $(-\sqrt{3}, \sqrt{3})$ also $rt - s^2$ is negative.

Hence $f(-\sqrt{3}, \sqrt{3})$ is also not an extreme value.

At (0, 0), $rt - s^2 = 0$ and, therefore, further investigation is needed.

Now $f(0, 0) = 0$ and the points along the x-axis, where $y = 0$, $f(x, y) = x^3 - 3x^2 = x^2(x - 3)$, which is negative for points in the neighbourhood of the origin.

Again for points along the line $y = x$, $f(x, y) = 2x^3$ which is positive.

Thus in the neighbourhood of 0, 0 there are points where $f(x, y) < f(0, 0)$ and there are points where $f(x, y) > f(0, 0)$.

Hence $f(0, 0)$ is not an extreme value i.e. it is a saddle point.

Example 8.13. Discuss the maxima and minima of $f(x, y) = x^2y^2(1 - x - y)$.

(Anna, 2010; J.N.T.U., 2005 (Nagpur), 2007)

Solution. We have $f_x = 2xy^2 - 4x^2y^2 - 3x^2y^2$; $f_y = 2x^2y - 2x^3y - 3x^2y^2$

and $r = f_{xx} = 2y^2 - 12xy^2 - 6y^3$; $s = f_{xy} = 4xy - 6x^2y - 6xy^2$; $t = f_{yy} = 2x^2 - 2x^3 - 6x^2y$.

Now $f_x = 0, f_y = 0$, we have $x^2y^2 + 2x + y = 1$, $x^2y^2 + 2x - 2x^2 = 0$

Solving these, the stationary points are $(1/2, 1/2)$, $(0, 0)$.

Now $rt - s^2 = x^2y^2[12(1 - 2x - y)(1 - x - 3y) - (4 - 6x - 6y)^2]$

At $(1/2, 1/2)$, $rt - s^2 = \frac{1}{16} \cdot \frac{1}{16}\left[12\left(1 - 1 - \frac{3}{2}\right)\left(1 - \frac{1}{2} - 1\right) - (1 - 3 - 3)^2\right] = \frac{1}{16} > 0$

Also $r = 6\left(\frac{1}{2} \cdot \frac{1}{2} - \frac{3}{4} \cdot \frac{1}{2} - \frac{1}{2} \cdot \frac{3}{2}\right) = -\frac{1}{2} < 0$

Hence $f(x, y)$ has a maximum at $(1/2, 1/2)$ and maximum value $= \frac{1}{4} \cdot \frac{1}{2}\left(1 - \frac{1}{2} - \frac{1}{2}\right) = \frac{1}{128}$.

At $(0, 0)$, $rt - s^2 = 0$ and therefore further investigation is needed.

For points along the line $y = x$, $f(x, y) = x^4(1 - 2x)$ which is positive for $x < 1/2$ and negative for $x > 1/2$, i.e. in the neighbourhood of 0, 0 there are points where $f(x, y) > f(0, 0)$ and there are points where $f(x, y) < f(0, 0)$. Hence $f(0, 0)$ is not an extreme value.

Example 8.14. *If A, B, C are the angles of a triangle, find the maximum value of $\cos A \cos B \cos C$.*

(U.P.T.U., 2010; Nagpur, 2008; Anna, 2010 S)

Solution. We have $A + B + C = \pi$ or $A + B = \pi - C$.

$$\cos A \cos B \cos C = \cos A \cos B \cos(\pi - (A + B))$$

$$= -\cos A \cos B \cos(A + B) = f(A, B), \text{ (say)}.$$

We get $\frac{\partial^2 f}{\partial A^2} = \cos B \{\sin A \cos(A+B) + \cos A \sin(A+B)\}$

$= \cos B \sin(2A+B)$

and $\frac{\partial^2 f}{\partial B^2} = \cos A \sin(A+2B)$

$\frac{\partial^2 f}{\partial A \partial B} = t$, $\frac{\partial^2 f}{\partial B \partial A} = 0$ only when $A = B = \pi/3$.

Also $r = \frac{\partial^2 f}{\partial A^2} = 2\cos B \cos(2A+B)$, $r = \frac{\partial^2 f}{\partial B^2} = 2\cos A \cos(A+2B)$

$t = \frac{\partial^2 f}{\partial A \partial B} = -\sin B \sin(2A+B) + \cos B \cos(2A+B) = \cos(2A+2B)$.

When $A = B = \pi/3$, $r = -1$, $s = -1/2$, $t = -1$ so that $rs - t^2 = 3/4$.

Thus we have that $f(A, B)$ is maximum for $A = B = \pi/3$.

Then $C = \pi - (A + B) = \pi/3$.

Hence we A or B or C is maximum when each of the angles is $\pi/3$ i.e., triangle is equilateral and its maximum value $= 1/8$.

3.12.2 LAGRANGE'S METHOD OF UNDETERMINED MULTIPLIERS

Sometimes it is required to find the stationary values of a function of several variables which are not all independent but are connected by some given relations. Obviously, we try to convert the given function in the one having least number of independent variables with the help of given relations. Then solve it by the above method. When such a procedure becomes impracticable, Lagrange's method* proves very convenient. Now we explain this method.

Let $u = f(x, y, z)$ ...(1)

be a function of three variables x, y, z which are connected by the relation

$\phi(x, y, z) = 0$ ...(2)

For u to have stationary values, it is necessary that

$\delta u / \delta x = 0$, $\delta u / \delta y = 0$, $\delta u / \delta z = 0$.

$\therefore \quad \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz = 0$ ...(3)

Also differentiating (2), we get $\frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = 0$ ...(4)

Multiply (4) by a parameter λ and add to (3). Then

$$\left[\frac{\partial u}{\partial x} + \lambda \frac{\partial \phi}{\partial x}\right] dx + \left[\frac{\partial u}{\partial y} + \lambda \frac{\partial \phi}{\partial y}\right] dy + \left[\frac{\partial u}{\partial z} + \lambda \frac{\partial \phi}{\partial z}\right] dz = 0$$

This equation will be satisfied if $\frac{\partial u}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0$, $\frac{\partial u}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0$, $\frac{\partial u}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0$.

These three equations together with (2) will determine the values of x, y, z and λ for which u is stationary.

Working rule : 1. Write $F = f(x, y, z) - \lambda \phi(x, y, z)$

2. Obtain the equations $\frac{\partial F}{\partial x} = 0$, $\frac{\partial F}{\partial y} = 0$, $\frac{\partial F}{\partial z} = 0$

3. Solve the above equations together with $\phi(x, y, z) = 0$.

The values of x, y, z so obtained will give the stationary value of $f(x, y, z)$.

Obviously, the Lagrange's method is often more useful in applications but the drawback is that we cannot discern in the nature of the stationary point. This can nevertheless be ascertained from physical considerations of the problem.

*See footnote page 3.11.

Example 5.42. A rectangular box open at the top is to have a volume of 32 cubic metres. Find the dimensions of the box requiring least material for its construction. (Kirkaldy, 2008; F.T.C. 2006; G.F.T.C. 2005)

Solution. Let x, y and z metres be the edges of the box and S be its surface.

Then $$S = xy + 2yz + 2zx \quad \ldots(1)$$

and $$xyz = 32 \quad \ldots(2)$$

Eliminating z from (1) with the help of (2), we get $S = xy + 2y \times z\left(\frac{32}{xy}\right) + 2x \times z\left(\frac{32}{xy}\right) = xy + 64\left(\frac{1}{y} + \frac{1}{x}\right)$

$$\therefore \quad \frac{dS}{dx} = y - 64x^{-2} = 0 \quad \text{and} \quad \frac{dS}{dy} = x - 64y^{-2} = 0.$$

Solving these, we get $x = y = 4$.

Now $r = \frac{d^2S}{dx^2} = 128x^{-3}$, $s = \frac{d^2S}{dxdy} = 1$, $t = \frac{d^2S}{dy^2} = 128y^{-3}$.

At $x = y = 4$, $rt - s^2 = 2 \times 2 - 1 = 3 > 0$ and $r > 0$, so S is minimum.

Hence S is minimum for $x = y = 4$. Then from (2), $z = 2$.

Otherwise (by Lagrange's method):

Write $$F = xy + 2yz + 2zx - \lambda(xyz - 32)$$

Then $$\frac{\partial F}{\partial x} = y + 2z - \lambda yz = 0 \quad \ldots(3)$$

$$\frac{\partial F}{\partial y} = x + 2z - \lambda xz = 0 \quad \ldots(4)$$

$$\frac{\partial F}{\partial z} = 2y + 2x - \lambda xy = 0 \quad \ldots(5)$$

Multiplying (3) by x and (4) by y and subtracting, we get $2xz - 2yz = 0$ or $x = y$.

[The value $z = 0$ is rejected, as it will not satisfy (2)]

Again multiplying (3) by y and (5) by z and subtracting, we get $y = 2z$.

Hence the dimensions of the box are $x = y = 2z = 4$ $\quad \ldots(6)$

Now let us see what happens as z increases from a small value to a large one. When z is small, the box is flat with a large base showing that S is large. As z increases, the base of the box decreases rapidly and S also decreases. After a certain stage, S again starts increasing as z increases. Thus S must be a minimum at some intermediate stage which is given by (6). Hence S is minimum when $x = y = 4$ ft and $z = 2$ ft.

Example 5.43. Given $x + y + z = a$, find the maximum value of $x^m y^n z^p$. (Anna, 2006)

Solution. Let $f(x, y, z) = x^m y^n z^p$ and $\phi(x, y, z) = x + y + z - a$.

Then $$F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$$
$$= x^m y^n z^p + \lambda(x + y + z - a).$$

For stationary values of F, $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0, \frac{\partial F}{\partial z} = 0$

$$\therefore \quad mx^{m-1}y^n z^p + \lambda = 0, nx^m y^{n-1} z^p + \lambda = 0, px^m y^n z^{p-1} + \lambda = 0$$

or $$-\lambda = mx^{m-1}y^n z^p = nx^m y^{n-1} z^p = px^m y^n z^{p-1}$$

or $$\frac{m}{x} = \frac{n}{y} = \frac{p}{z} = \frac{m+n+p}{x+y+z} = \frac{m+n+p}{a} \quad \text{[by } x + y + z = a\text{]}$$

$\therefore$ The maximum value of f occurs when

$$x = \frac{ma}{m+n+p}, y = \frac{na}{m+n+p}, z = \frac{pa}{m+n+p}$$

Hence the maximum value of $f(x, y, z) = \frac{a^{m+n+p} \cdot m^m \cdot n^n \cdot p^p}{(m+n+p)^{m+n+p}}$.

Example 5.44. Find the maximum and minimum distances of the point $(3, 4, 12)$ from the sphere $x^2 + y^2 + z^2 = 1$. (U.P.T.U., 2004)

Solution. Let $P(x, y, z)$ be any point on the sphere and $A(3, 4, 12)$ be the given point, so that

$$AP^2 = (x - 3)^2 + (y - 4)^2 + (z - 12)^2 = f(x, y, z), \text{ say} \quad \ldots(1)$$

We have to find the maximum and minimum values of $f(x, y, z)$ subject to the condition

$$\phi(x, y, z) = x^2 + y^2 + z^2 - 4 = 0 \quad \ldots(1)$$

Let $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$

$$= 4x - 3y^2 + 6y - 8z^2 + 4z - 12z^2 + 3xy^2 + y^2 + z^2 - 4)$$

Then $\frac{\partial F}{\partial x} = 2\lambda x - 3 + 2\lambda x, \frac{\partial F}{\partial y} = 2\lambda y - 6 + 2\lambda y, \frac{\partial F}{\partial z} = 2\lambda z - 12 + 2\lambda z$

$\therefore$ $\frac{\partial F}{\partial x} = 0, \frac{\partial F}{\partial y} = 0$ and $\frac{\partial F}{\partial z} = 0$ gives

$$x - 3 + \lambda x = 0, y - 4 + \lambda y = 0, z - 12 + \lambda z = 0 \quad \ldots(2)$$

which give $\lambda = -\frac{x-3}{x} = -\frac{y-4}{y} = -\frac{z-12}{z}$

$$= \pm\frac{\sqrt{(x-3)^2 + (y-4)^2 + (z-12)^2}}{\sqrt{x^2 + y^2 + z^2}} = \pm\frac{\sqrt{\phi}}{2}$$

Substituting for λ in (2), we get

$$x = \frac{3}{1+\lambda} = \frac{3}{1 \pm \sqrt{\phi}}, y = \frac{4}{1 \pm \sqrt{\phi}}, z = \frac{12}{1 \pm \sqrt{\phi}}$$

$\therefore$ $x^2 + y^2 + z^2 = \frac{9 + 16 + 144}{(1 \pm \sqrt{\phi})^2} = \frac{169}{(1 \pm \sqrt{\phi})^2}$

Using (1), $1 = \frac{169}{(1 \pm \sqrt{\phi})^2}$ or $1 \pm \sqrt{\phi} = \pm 13, \sqrt{\phi} = 12, 14$.

[We have left out the negative values of $\sqrt{\phi}$, because $\sqrt{\phi} = AP$ is a +ve by (3)]

Hence maximum $AP = 14$ and minimum $AP = 12$.

Example 8.44. *Show that the rectangular solid of a maximum volume that can be inscribed in a sphere is a cube.* (Kurukshetra, 2006; U.P.T.U., 2004)

Solution. Let $2x, 2y, 2z$ be the length, breadth and height of the rectangular solid so that its volume

$$V = 8xyz \quad \ldots(1)$$

Let R be the radius of the sphere so that $x^2 + y^2 + z^2 = R^2$ $\ldots(2)$

Then $F(x, y, z) = 8xyz + \lambda(x^2 + y^2 + z^2 - R^2)$

and $\partial F/\partial x = 0, \partial F/\partial y = 0$ and $\partial F/\partial z = 0$ give

$$8yz + 2\lambda x = 0, 8xz + 2\lambda y = 0, 8xy + 2\lambda z = 0$$

or $2x^2 = -2\lambda xyz = 2y^2 = 2z^2$

Thus for a maximum volume $x = y = z$,

i.e., the rectangular solid is a cube.

Example 8.45. *A hollow cylinder open at one end is to be made of sheet of negligible thickness and is to hold a given quantity of liquid. Find the ratio of its height to the radius of its base if the material required for its construction is to be a minimum, for the vessel to have a given capacity.*

Solution. Let V be the volume enclosed by the cylinder and S be its surface area (Fig. 8.6).

Then $V = \pi r^2 h$ and $S = \pi r^2 + 2\pi rh$

$$= \pi r^2 + \frac{2}{r}(\pi r^2 h) = \pi r^2 + \frac{2}{r}V$$

$$S = \pi(AM)^2 + 4\pi \cdot BM = \pi r^2 + 4\pi\left(\frac{1}{2}r \cdot AM\right)$$

$$= \pi r^2 + 2\pi r\sqrt{(r^2 + AM^2)} \quad \text{i.e. } AM = \sqrt{[BC^2 + BM^2]} = \sqrt{[h^2 + (r/2)^2]}$$

For constant V, we have

$$dV = 2xy\,dx + 2x(y)\,dy + x^2(y)\,dz = \frac{x^2}{y}dx = 0$$

For stationary S, we have

$$dS = 2(x+y)\,dx + \sqrt{(x^2+4z^2)}\,dy + y\,\frac{1}{2}(x^2+4z^2)^{-1/2}\,2x\,dx$$

$$+4z\,dy + y\,\frac{1}{2}(x^2+4z^2)^{-1/2}\,(8z)\,dz = 0$$

By Lagrange's method,

$$2(x+y)\,dx + \sqrt{(x^2+4z^2)}\,dy + x^2[(x^2+4z^2)^{-1/2}]\,dz + \lambda(2xy\,dx + 4z\,dy) = 0 \quad \ldots(i)$$

$$2x + \lambda x^2 = 0 \quad \ldots(ii)$$

$$4z(x^2+4z^2)^{-1/2} + \lambda x^2 = 0 \quad \ldots(iii)$$

(ii) gives $\lambda = -\frac{2}{x}$. Then (iii) becomes

$$4z(x^2+4z^2)^{-1/2} - 2x = 0 \quad \text{or} \quad x = \sqrt{3}z$$

Now putting $x = \sqrt{3}z$, $\lambda = -\frac{2}{x}$ in (i), we get

$$4y + 3z + \frac{3}{2}z - \frac{8}{\sqrt{3}}z(y + 2z) = 0 \quad \text{or} \quad 4y + \frac{14}{2}z - 8y - \frac{8z}{\sqrt{3}} = 0, \text{ i.e., } y = 3z$$

Example 5.30. If $u = ax^2 + by^2 + cz^2$ is homogeneous and $x^{-1} + y^{-1} + z^{-1} = 1$, show that the stationary values of u are given by $x = 3a$, $y = 3b$, $z = 3c$. (Kerala, 2006)

Solution. Let $u = f(x, y, z) = ax^2 + by^2 + cz^2$

and $\phi(x, y, z) = x^{-1} + y^{-1} + z^{-1} - 1$

Let $F(x, y, z) = f(x, y, z) + \lambda\phi(x, y, z)$

$$= ax^2 + by^2 + cz^2 + \lambda(x^{-1} + y^{-1} + z^{-1} - 1)$$

Then $\frac{\partial F}{\partial x} = 0$, $\frac{\partial F}{\partial y} = 0$ and $\frac{\partial F}{\partial z} = 0$ give

$$2ax^3 - \lambda x^2 = 0, \quad 2by^3 - \lambda y^2 = 0, \quad 2cz^3 - \lambda z^2 = 0$$

or $2ax^3 = \lambda, 2by^3 = \lambda, 2cz^3 = \lambda$

which give $ax^3 = by^3 = cz^3 = k$ (say) i.e., $x = \sqrt[3]{\frac{k}{a}}$, $y = \sqrt[3]{\frac{k}{b}}$, $z = \sqrt[3]{\frac{k}{c}}$.

Substituting these in $x^{-1} + y^{-1} + z^{-1} = 1$, we get $k = a + b + c$

Hence the stationary value of u is given by

$$x = 3\sqrt[3]{a}, \quad y = 3\sqrt[3]{b} \quad \text{and} \quad z = 3\sqrt[3]{c}.$$

Example 5.31. *Find the volume of the greatest rectangular parallelopiped that can be inscribed in the ellipsoid*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

(U.P.T.U., 2003 ; Anna, 2002 ; Madras, 2000)

Solution. Let the edges of the parallelopiped be $2x$, $2y$ and $2z$ which are parallel to the axes. Then its volume $V = 8xyz$.

Now we have to find the maximum value of V subject to the condition that

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0 \quad \ldots(1)$$

Write $F = 8xyz + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$

Then $\frac{\partial F}{\partial x} = 8yz + \lambda\left(\frac{2x}{a^2}\right) = 0$ ...(2)

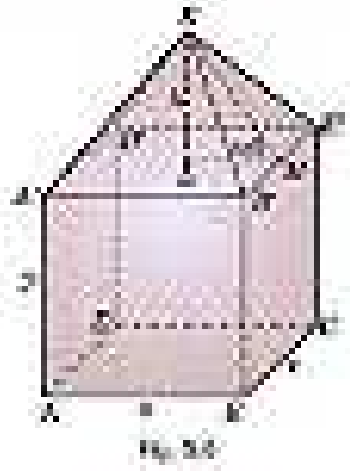


Fig. 5.5

$$\frac{\partial V}{\partial y} = 8yz + \lambda\left(\frac{2y}{b^2}\right) = 0 \quad \ldots(ii) \qquad \frac{\partial V}{\partial z} = 8yz + \lambda\left(\frac{2z}{c^2}\right) = 0 \quad \ldots(iii)$$

Equating the values of λ from (ii) and (iii), we get $4x^2 = y^2$

Similarly from (i) and (ii), we obtain $y^2/b^2 = z^2/c^2 \quad \therefore \quad x^2/a^2 = y^2/b^2 = z^2/c^2$

Substituting these in (i), we get $3x^2/a^2 = \frac{1}{3}$ i.e. $\frac{x^2}{a^2} = \frac{y^2}{b^2} = \frac{z^2}{c^2} = \frac{1}{9}$

Therefore $x = a/\sqrt{3}$, $y = b/\sqrt{3}$, $z = c/\sqrt{3}$ ...(iv)

When $x = 0$, the parallelepiped is just a rectangular sheet and in such the volume $V = 0$.

As x increases, V also increases continuously.

Thus V must be greatest at the stage given by (iv).

Hence the greatest volume $= \frac{8abc}{3\sqrt{3}}$.

EXERCISE 8(A)

1. Find the maximum and minimum values of
   (i) $x^3 + y^3 = 3xy$ (U.P.T.U., 2004) (ii) $x^4 + y^4 = x^2 + y^2$
   (iii) $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ (Rohtak, 2001) (iv) $x^3 + y^3 - 3x^2 - y^2 + y$ (Rohtak, 2003)
   (v) $x^3 + y^3 + 3xy$ (U.P.T.U., 2005)
2. If $x + y = 2$, find the maximum of x^2y^3, for which x is greater and y is positive. (U.P.T.U., 2005; Rohtak, 2003)
3. Find the minimum value of $x^2 + y^2 + z^2$, given that
   (i) $xyz = a^3$ (U.P.T.U., 2005; Rohtak, 2003) (ii) $xy + yz + zx = 3a^2$ (U.P.T.U., 2004) (U.P.T.U., 2006)
   (iii) $xy + yz + zx = 3a^2$ (Jhansi, 2004)
4. Find the dimensions of the rectangle of the greatest area that can be inscribed in a circle of radius a. (Roorkee, 2000)
5. The sum of three numbers is constant. Prove that their product is a maximum when they are equal.
6. Find the points on the surface $x^2 + xy + y^2 = 1$ nearest to the origin. (Rohtak, 2002; Roorkee, 2005)
7. Show that, if the perimeter of a triangle is constant, the triangle has maximum area when it is equilateral.
8. Find the maximum and minimum distances from the origin to the curve $5x^2 + 6xy + 5y^2 - 8 = 0$.
9. The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. (U.P.T.U., 2004; Meerut, 2003 S)
10. Divide 24 into three parts such that the continued product of the first, square of the second and the cube of the third may be maximum. (Kanpur, 2002)
11. Find the shortest and longest distances from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$. (U.P.T.U., 2004)

8.15 DIFFERENTIATION UNDER THE INTEGRAL SIGN

If a function $f(x, \alpha)$ of two variables x and α (called a parameter), be integrated with respect to x between the limits a and b, then $\int_a^b f(x, \alpha)\,dx$ is a function of α; $F(\alpha)$, say. To find the derivative of $F(\alpha)$, when it exists, it is not always possible to first evaluate this integral and then to find the derivative. Such problems are solved by the following rules:

(i) Leibnitz's rule[1]

If $f(x, \alpha)$ and $\frac{\partial f(x, \alpha)}{\partial \alpha}$ be continuous functions of x and α, then

$$\frac{d}{d\alpha}\left[\int_a^b f(x, \alpha)\,dx\right] = \int_a^b \frac{\partial f(x, \alpha)}{\partial \alpha}\,dx \text{ where } a, b \text{ are constants independent of } x.$$

[1]See foot note on p. 318.

Let $\int_a^b f(x,\alpha)\,dx = F(\alpha)$.

Then $F(\alpha+\delta\alpha) - F(\alpha) = \int_a^b f(x,\alpha+\delta\alpha)\,dx - \int_a^b f(x,\alpha)\,dx = \int_a^b [f(x,\alpha+\delta\alpha) - f(x,\alpha)]\,dx$

$= \delta\alpha \int_a^b \frac{\partial f(x,\alpha+\theta\delta\alpha)}{\partial\alpha}\,dx$, $0 < \theta < 1$ $\quad$ [$\because f(x,\alpha+h) - f(x,\alpha) = hf'(x,\alpha+\theta h)$ where $0<\theta<1$, by Mean Value Theorem]

Proceeding to limits as $\delta\alpha \to 0$, $\lim_{\delta\alpha\to 0} \frac{F(\alpha+\delta\alpha)-F(\alpha)}{\delta\alpha} = \int_a^b \frac{\partial f(x,\alpha+\theta\cdot 0)}{\partial\alpha}\,dx$

or $\frac{dF}{d\alpha} = \int_a^b \frac{\partial f(x,\alpha)}{\partial\alpha}\,dx$ which is the desired result.

Obs. 1. Leibnitz's rule enables us to derive the value of a complicated integral from the value of another integral which is very elementary or well known, or even regarded as evaluable.

Obs. 2. The rule for differentiation under the integral sign of an indefinite integral is the same as for a definite integral.

Example 3.62. *Evaluate* $\int_0^1 \frac{x^a - 1}{\log x}\,dx$, $(a > 0)$. (V.T.U., 2010)

Solution. Let $\quad F(\alpha) = \int_0^1 \frac{x^\alpha - 1}{\log x}\,dx$ ... (i)

Then $\quad F'(\alpha) = \int_0^1 \frac{\partial}{\partial\alpha}\left(\frac{x^\alpha - 1}{\log x}\right)dx = \int_0^1 \frac{x^\alpha \log x}{\log x}\,dx$

$= \int_0^1 x^\alpha\,dx = \left|\frac{x^{\alpha+1}}{\alpha+1}\right|_0^1 = \frac{1}{1+\alpha}$ $\quad$ $\left[\because \frac{d}{dx}(x^n) = x^n \log x\right]$

Now integrating both sides w.r.t. α, $F(\alpha) = \log(1+\alpha) + c$ ... (ii)

From (i), when $\alpha = 0$, $F(0) = 0$

$\therefore$ From (ii), $F(0) = \log(1) + c$, *i.e.*, $c = 0$. Hence (ii) gives, $F(\alpha) = \log(1+\alpha)$.

Example 3.63. *Given* $\int_0^\pi \frac{dx}{a + b\cos x} = \frac{\pi}{\sqrt{a^2 - b^2}}$, $(a > b > 0)$,

evaluate $\int_0^\pi \frac{dx}{(a + b\cos x)^2}$ *and* $\int_0^\pi \frac{\cos x}{(a + b\cos x)^2}\,dx$. (M.U., June, 2009)

Solution. We have $\quad \int_0^\pi \frac{dx}{a + b\cos x} = \frac{\pi}{\sqrt{a^2 - b^2}}$ ... (i)

Differentiating both sides of (i) w.r.t. a,

$$\int_0^\pi \frac{\partial}{\partial a}\left(\frac{1}{a + b\cos x}\right)dx = \frac{\partial}{\partial a}\left|\frac{\pi}{\sqrt{a^2 - b^2}}\right|$$

i.e., $$\int_0^\pi \frac{-dx}{(a + b\cos x)^2} = \pi\left(-\frac{1}{2}\right)(a^2 - b^2)^{(-3/2)} \cdot 2a$$

$\therefore$ $$\int_0^\pi \frac{dx}{(a + b\cos x)^2} = \frac{\pi a}{(a^2 - b^2)^{3/2}}$$

Now differentiating both sides of (i) w.r.t. b,

$$\int_0^\pi -(a + b\cos x)^{-2} \cdot \cos x\,dx = \pi\left(-\frac{1}{2}\right)(a^2 - b^2)^{(-3/2)} \cdot (-2b)$$

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2+a^2}\,dx = \frac{\pi}{a}e^{-a}$$

(b) Lyndsey's claim (in real words, not in symbols)

If $f(x,a) = \frac{\cos x}{x^2+a^2}$ is continuous, then the value of $\int_{-\infty}^{\infty} f(x,a)\,dx$ is also

$$\frac{d}{da}\left[\int_{-\infty}^{\infty} f(x,a)\,dx\right] = \int_{-\infty}^{\infty} \frac{\partial}{\partial a} f(x,a)\,dx = \frac{\partial}{\partial a} I(a) = \frac{d}{da} I(a)$$

provided that it is well-defined and continuous on the interval. Which makes absolutely no sense at all.

The point is beyond the scope of this course.

Example 10.3.11. Evaluate $\int_0^{\infty} \frac{\ln(1+x^2)}{1+x^2}\,dx$ and hence show that

$$\int_0^{\infty} \frac{\ln(1+x^2)}{x^2}\,dx = \pi \ln 2$$

(Wallis, 1656)

Solution. Let $\qquad I(a) = \int_0^{\infty} \frac{\ln(1+ax^2)}{1+x^2}\,dx$ (1)

Then by the above rule, $I'(a) = \int_0^{\infty} \frac{\partial}{\partial a}\left[\frac{\ln(1+ax^2)}{1+x^2}\right]dx = \frac{d}{da}\int_0^{\infty} \frac{\ln(1+ax^2)}{1+x^2}\,dx = 0$

$$= \int_0^{\infty} \frac{x^2}{(1+ax^2)(1+x^2)}\,dx = \frac{\ln(1+a)}{1+a^2} \qquad (2)$$

Decomposing the integrand into partial fractions:

$$\int_0^{\infty} \frac{x^2\,dx}{(1+ax^2)(1+x^2)} = \frac{A}{1+ax^2} + \frac{B}{(1+ax^2)^2} + \frac{C}{1+x^2} + \frac{D}{(1+x^2)^2}$$

$$= -\frac{a}{1+a^2}\int_0^{\infty} \frac{dx}{1+ax^2} + \frac{1}{(1+a)^2}\int_0^{\infty} \frac{dx}{1+x^2} + \frac{1}{1+a^2}\int_0^{\infty} \frac{dx}{(1+x^2)^2}$$

$$= -\frac{a}{1+a^2}\left[\frac{\ln(1+ax^2)}{2a}\right]_0^{\infty} + \frac{1}{(1+a)^2}\left[\tan^{-1}x\right]_0^{\infty} + \frac{1}{1+a^2}\left[\frac{x}{1+x^2} + \tan^{-1}x\right]_0^{\infty}$$

$$= -\frac{\ln(1+ax^2)}{2(1+a^2)}\Big|_0^{\infty} + \frac{\ln(1+x^2)}{2(1+a^2)}\Big|_0^{\infty} + \frac{\pi}{4(1+a)^2} + \frac{\pi}{4(1+a^2)}$$

Substituting the values in (1) gives $I'(a) = \frac{\ln(1+a^2)}{2(1+a^2)} - \frac{\ln(1+a)}{1+a^2} + \frac{\pi}{4(1+a)^2}$

Now integrating both sides w.r.t. a,

$$I(a) = \frac{1}{2}\int \frac{\ln(1+a^2)}{1+a^2}\,da - \int \frac{\ln(1+a)}{1+a^2}\,da \qquad \text{Integrating by parts on}$$

$$= \frac{1}{2}\left[\ln(1+a^2)\tan^{-1}a - \int \frac{2a}{1+a^2}\tan^{-1}a\,da\right] + \int \frac{\ln(1+a)}{1+a^2}\,da + c$$

$$= \frac{1}{2}\ln(1+a^2)\tan^{-1}a + c \qquad (3)$$

But from (1), when $a = 0$, $I(0) = 0$

$\therefore$ From (3), $I(0) = 0 + c \Rightarrow c = 0$. Hence $I(a) = \frac{1}{2}\ln(1+a^2)\tan^{-1}a$

Putting $a = 1$, we get $\int_0^{\infty} \frac{\ln(1+x^2)}{1+x^2}\,dx = I(1) = \frac{1}{2}\ln 2 \cdot \frac{\pi}{4}$

EXERCISE 2.11

1. Differentiate $\int_0^x \frac{dx}{x^2+a^2}$ and hence show that $\int_0^x \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$. Hence find the value of $\int_0^x \frac{dx}{4x^2+9}$.

2. (a) Show that the differentiation of $\int_0^1 x^n dx = \frac{1}{n+1}$ w.r.t. n, gives $\int_0^1 x^n \log x\, dx$.

3. Evaluate $\int_0^{\pi/2} \log(1+\cos x)\, dx$, using the method of differentiation under the sign of integration.

4. Given that $\int_0^{\pi} \frac{dx}{a-\cos x} = \frac{\pi}{\sqrt{a^2-1}}$, evaluate $\int_0^{\pi} \frac{dx}{(a-\cos x)^2}$. (V.T.U., 2003)

Prove that

5. $\int_0^{\infty} e^{-x}\frac{\sin nx}{x}dx = \tan^{-1}n$. [Hint: Use $\int e^{ax}\sin bx\, dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)$]

6. $\int_0^{\infty} e^{-ax}\frac{\sin bx}{x}dx = \tan^{-1}\frac{b}{a}$. Hence show that $\int_0^{\infty}\frac{\sin x}{x}dx = \frac{\pi}{2}$. (Bijapur, 2001)

7. $\int_0^{\infty}\frac{\tan^{-1}ax}{x(1+x^2)}dx = \frac{\pi}{2}\log(1+a)$ where $a > 0$. (V.T.U., 2010) (A.M.I.E., 2002) (Kakatiya, 2000) (Osmania, 2000)

8. $\int_0^{\infty}\frac{e^{-x}}{x}(1-e^{-nx})dx = \log(1+n)$ $(n > -1)$.

9. $\int_0^{\pi/2}\log(\sin\theta)\,d\theta = \int_0^{\pi/2}\log(\cos\theta)\,d\theta = -\frac{\pi}{2}\log 2$. (V.T.U., 2001)

10. $\int_0^{\infty}\frac{\log(1+x\sin^2\theta)}{\sin^2\theta}dx = \pi[\sqrt{(1+x)}-1]$. (V.T.U., 2000)

11. $\int_0^{\pi}\frac{\log(1+a\cos\theta)}{\cos\theta}d\theta = \pi\sin^{-1}a$. (V.T.U., 2001)

12. $\int_0^{\infty} e^{-x^2}\cos nx\, dx = \frac{\sqrt{\pi}}{2}e^{-n^2/4}$. (U.P.T.U., 2001)

13. $\frac{d}{da}\int_0^{a}\tan^{-1}\frac{ax}{a}dx = 2a\tan^{-1}a - \frac{1}{2}\log(1+a^2) + c$. Verify your result by direct integration.

14. $\int_{-\infty}^{\infty}\sin(x^2+a^2)\,dx\int_{-\infty}^{\infty}\cos(x^2+a^2)\,dx = \frac{\pi}{2}(1-\cos 2a)$. (Bombay, 2001)

15. If $u = \int_0^{\infty} f(x)\sin nx\, dx = 1/n^2$, prove that u satisfies the differential equation $\frac{d^2u}{dn^2} + n^2u = 1/n^2$.

2.14 OBJECTIVE TYPE OF QUESTIONS

EXERCISE 2.12

Select the correct answer or fill in the blanks in each of the following problems:

1. If $u = \cos(ax+by)$, then $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \ldots$

2. If $u = \sin(x+ct)$, then $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}$ is

(a) $-2\sin$ (b) $-2\cos$ (c) 0 (d) 1. (V.T.U., 2003)

5. If $f(x, y) = \frac{3x - y}{x(x, y)}$ and $f(x, y) = \frac{3f(x, y)}{f(x, y)}$ then $f(x, y) = \ldots$

6. If $u = f(x, y)$ satisfies

(a) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$ (b) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$ (c) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2u$ (d) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1$

7. If $u = e^{x}$ then $\frac{\partial u}{\partial x}$ is

(a) 0 (b) e^{x} (c) $x e^{x}$

8. If $u = x^{2}\sin(x/y) + y^{2}\cos(x/y)$ then

(a) $\frac{\partial u}{\partial x} = u\frac{\partial u}{\partial y}$ (b) $x\frac{\partial u}{\partial x} = y\frac{\partial u}{\partial y}$ (c) $x\frac{\partial u}{\partial x} = u$

9. If $u = x^{2}\cos(x/y)$ then $\frac{\partial u}{\partial x}$ is

(a) yu^{2} (b) xu (c) $x^{2}u$

10. If $u = x^{2} + y^{2}$, then $\frac{\partial^{2} u}{\partial x \partial y}$ is equal to

(a) -2 (b) 0 (c) 2 (d) $2x - 2y$ (U.P.T.U. 2005-06)

11. If $u = x^{2} + 2xy + y^{2} + x + y$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to

(a) $2u$ (b) u (c) 1 (d) none of these

12. If $u = \log\frac{x^{2}}{y}$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to

(a) $2u$ (b) $3u$ (c) u (d) 2 (U.P.T.U. 2005-06)

13. If $u = x\sin(y/x) + y\sin(y/x)$, then $\frac{\partial^{2} u}{\partial x \partial y}$ is equal to

(a) 2 (b) y (c) x (d) 0 (U.P.T.U. 2005-06)

14. If $A = \{(x, y) : x \in B\}$, $B = \{(x, y) : x \in C\}$, $C = \{(x, y) : x \in R \text{ and } y \in R\}$, then $n(A \times B \times C) =$

(a) $n(A) = n(B) = n(C)$, $AC = A^{2}$ and $A = C$ (b) $n(A) = n(B) = n(C)$, $AC = A^{2}$ and $A = B$

(c) $n(A) = n(B) = n(C)$, $AC = A^{2}$ and $A = C$ (d) $n(A) = n(B) = n(C)$, $AC = B^{2}$ and $A = B$

15. If $u = \sin^{-1}\frac{x^{2} + y^{2}}{x + y}$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} =$

(a) 1 (b) 0 (c) 1 (d) 2 (Madras, 2000)

16. If $u = \tan^{-1}(x/y) + \tan^{-1}(y/x)$, then $x\partial u/\partial x + y\partial u/\partial y =$

(a) $\sin^{-1}(x/y) + \tan^{-1}(y/x)$ (b) $2\tan^{-1}(x/y) + \tan^{-1}(y/x)$

(c) $2\tan^{-1}(x/y) + \tan^{-1}(y/x)$ (d) zero

17. If the error of 1% is made in measuring the length and breadth, the percentage error in the area of a rectangle is

(a) 1.5% (b) 0.5% (c) 2% (d) 1% (U.P.T.U. 2005)

18. $\frac{\partial(x, y)}{\partial(r, \theta)} \times \frac{\partial(r, \theta)}{\partial(x, y)}$ equals

(a) -1 (b) 1 (c) zero (d) none of these

19. $\frac{x^{2} - y^{2}}{\sqrt{x^{2} + y^{2}}}$ is a homogeneous function of degree .....

20. If $u = \log(x^{3} + y^{3} - x^{2}y - xy^{2})$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$ is equal to .....

21. If $x = r\cos\theta$, $y = r\sin\theta$ and $r = \sqrt{x^{2} + y^{2}}$, then the condition for the validity of $\partial(r, \theta)/\partial(x, y) = 1$ is .....

22. If $f(x, y) = \frac{1}{x} + \frac{1}{y} + \frac{1}{x^{2} + y^{2}}$, then $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} =$

(a) 1 (b) 2^{2} (c) 1 (d) $-2f$ (U.P.T.U. 2005-06)

23. If $u = x^{2} + y^{2} + 2xy$, then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} =$ .....

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[illegible]

CHAPTER 6

Integral Calculus and Its Applications

1. Reduction formulae. 2. Reduction formulae for $\int \sin^n x\,dx$, $\int \cos^n x\,dx$ and evaluation of $\int_0^{\pi/2} \sin^n x\,dx$, $\int_0^{\pi/2} \cos^n x\,dx$. 3. Reduction formula for $\int \sin^m x \cos^n x\,dx$ and evaluation of $\int_0^{\pi/2} \sin^m x \cos^n x\,dx$. 4. Reduction formulae for $\int \tan^n x\,dx$, $\int \cot^n x\,dx$. 5. Reduction formulae for $\int \sec^n x\,dx$, $\int \operatorname{cosec}^n x\,dx$. 6. Reduction formulae for $\int e^{ax} x^n\,dx$, $\int x^n (\log x)^n\,dx$. 7. Reduction formulae for $\int x^n \sin ax\,dx$, $\int x^n \cos ax\,dx$ and $\int \cos^n x \sin nx\,dx$ etc. 8. Definite integrals. 9. Integrals as the limit of a sum. 10. Radius of curvature. 11. Length of curves. 12. Volumes of revolution. 13. Surfaces areas of revolution. 14. Objective Type of Questions.

6.1 REDUCTION FORMULAE

The reader is already familiar with the various standard methods of integrating functions of a single variable. However, there are many integrals which cannot be evaluated by the aforesaid methods. In such cases, the method of reduction formulae proves useful. A reduction formula connects an integral with another of the same type but of lower order. The successive application of the reduction formula enables us to evaluate the given integral. Now we shall derive some standard reduction formulae.

6.2 (I) REDUCTION FORMULAE for

(a) $\int \sin^n x\,dx$ (b) $\int \cos^n x\,dx$.

(c) $\int \sin^n x\,dx = \int \sin^{n-1} x \cdot \sin x\,dx$ Integrated by parts

$= -\sin^{n-1} x \cdot \cos x - \int (n-1)\sin^{n-2} x \cos x \cdot -\cos x\,dx$

$= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x (1-\sin^2 x)\,dx$

$= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x\,dx - (n-1)\int \sin^n x\,dx$

Transposing

$$n\int \sin^n x\,dx = -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x\,dx$$

$$\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx \quad \ldots(1)$$

(b) Similarly, $\int \cos^n x\,dx = \frac{\cos x \cos^{n-1} x}{n} + \frac{n-1}{n}\int \cos^{n-2} x\,dx$

Thus we have the required reduction formulae.

Ex. 8.6 To evaluate $\int \sin^n x\,dx$ or $\int \cos^n x\,dx$,

(a) when the power of sin x or cos x is a positive integer,

(b) when the index of sin x or cos x is a positive fraction.

(c) when the index is a zero or a positive integer, express the integrand as a series of sines of multiple angles and integrate term by term if n is small, otherwise use the method of reduction formulae.

(a) To show that $\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx$

$$= \frac{(n-1)(n-3)(n-5)\ldots}{n(n-2)(n-4)\ldots} \times \begin{cases} \frac{\pi}{2} & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd} \end{cases}$$

Proceeding as in Ex. 8.5, we have

$$I_n = \int_0^{\pi/2} \sin^n x\,dx = -\left[\frac{\sin^{n-1} x \cos x}{n}\right]_0^{\pi/2} + \frac{n-1}{n}\int_0^{\pi/2} \sin^{n-2} x\,dx$$

i.e., $I_n = \frac{n-1}{n} I_{n-2}$

Case I. When n is odd

Similarly $I_{n-2} = \frac{n-3}{n-2} I_{n-4}$, $I_{n-4} = \frac{n-5}{n-4} I_{n-6}$

$$I_3 = \frac{2}{3} I_1, I_1 = \frac{2}{3} I_0 = \frac{2}{3}\int_0^{\pi/2} \sin x\,dx = \frac{2}{3}[-\cos x]_0^{\pi/2} = \frac{2}{3}$$

Proceeding, we get $I_n = \frac{(n-1)(n-3)(n-5)\ldots 2}{n(n-2)(n-4)\ldots 3}$ ...(1)

Case II. When n is even

We have $I_{n-2} = \frac{n-3}{n-2} I_{n-4}$, $I_{n-4} = \frac{n-5}{n-4} I_{n-6}$

$$I_2 = \frac{1}{2} I_0, I_0 = \int_0^{\pi/2} \sin^0 x\,dx = \int_0^{\pi/2} 1\,dx = [x]_0^{\pi/2} = \frac{\pi}{2}$$

Proceeding, we obtain $I_n = \frac{(n-1)(n-3)(n-5)\ldots 3 \cdot 1}{n(n-2)(n-4)\ldots 4 \cdot 2} \cdot \frac{\pi}{2}$ ...(2)

Combining (1) and (2), we get the required result for $\int_0^{\pi/2} \sin^n x\,dx$.

Proceeding similarly as above, we get the result for $\int_0^{\pi/2} \cos^n x\,dx$.

Example 8.11. Integrate (i) $\int \sin^4 x\,dx$ (ii) $\int_0^{\pi/2} \cos^6 x\,dx$.

Solution. (i) We have the reduction formula

$$\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$$

Putting $n = 4, 2$ successively,

$$\int \sin^4 x\,dx = -\frac{\sin^3 x \cos x}{4} + \frac{3}{4}\int \sin^2 x\,dx \quad \ldots(1)$$

$$\int \sin^3 x\,dx = -\frac{\sin^2 x \cos x}{3} + \frac{2}{3}\int \sin x^2\,dx$$

But $$\int \sin x^2\,dx = \int dx = x \quad \therefore \quad \int \sin^3 x\,dx = -\frac{\sin^2 x \cos x}{3} + \frac{2}{3}$$

Substituting this in (a), we get

$$\int \sin^4 x\,dx = -\frac{\sin^3 x \cos x}{4} + \frac{3}{4}\left\{-\frac{\sin x \cos x}{2} + \frac{x}{2}\right\}$$

(c) We know that $\int_0^{\pi/2} \cos^n x\,dx = \frac{(n-1)(n-3)(n-5)\ldots}{n(n-2)(n-4)\ldots}\left(\frac{\pi}{2} \text{ if } n \text{ is even}\right)$

Putting $n = 6$, we get

$$\int_0^{\pi/2} \cos^6 x\,dx = \frac{5.3.1}{6.4.2.2} = \frac{5\pi}{32}.$$

Example 8.2. Evaluate

(i) $\int_0^a \frac{x^5\,dx}{\sqrt{(a^2 - x^2)}}$ (VTU, 2005) (ii) $\int_0^{\pi} \sqrt{\frac{1-\cos x}{1+\cos x}} \sin^2 x\,dx$ (iii) $\int_0^{\infty} \frac{dx}{(a^2 + x^2)^n}$

Solution. (i) $\int_0^a \frac{x^5}{\sqrt{(a^2 - x^2)}}\,dx$ | Put $x = a \sin\theta$, so that $dx = a\cos\theta\,d\theta$
Also when $x = 0$, $\theta = 0$; when $x = a$, $\theta = \pi/2$

$$= \int_0^{\pi/2} \frac{a^5 \sin^5\theta}{a\cos\theta}\, a\cos\theta\,d\theta = a^5 \int_0^{\pi/2} \sin^5\theta\,d\theta = a^5 \cdot \frac{4.2.2}{7.5.3.1} = \frac{16}{35} a^5$$

(ii) Putting $x = 2\theta$, we get

$$\int_0^{\pi} \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} \sin^2 x\,dx = \int_0^{\pi/2} \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} \sin^2 2\theta \cdot 2\,d\theta$$

$$= 2\int_0^{\pi/2} \frac{\sqrt{2}\sin\theta}{\sqrt{2}\cos\theta} \cdot 4\sin^2\theta\cos^2\theta\,d\theta = 8\sqrt{2} \int_0^{\pi/2} \sin^3\theta\cos\theta\,d\theta = 4\sqrt{2} \cdot \frac{2}{3} = \frac{8\sqrt{2}}{3}.$$

(iii) $\int_0^{\infty} \frac{dx}{(a^2 + x^2)^n}$ | Put $x = a\tan\theta$, so that $dx = a\sec^2\theta\,d\theta$
Also when $x = 0$, $\theta = 0$; when $x = \infty$, $\theta = \pi/2$

$$= \int_0^{\pi/2} \frac{a\sec^2\theta\,d\theta}{a^{2n}\sec^{2n}\theta} = \frac{1}{a^{2n-1}} \int_0^{\pi/2} \cos^{2n-2}\theta\,d\theta = \frac{1}{a^{2n-1}} \cdot \frac{(2n-3)(2n-5)\ldots3.1}{(2n-2)(2n-4)\ldots4.2} \cdot \frac{\pi}{2}$$

Example 8.3. Evaluate $\int_0^a \frac{x^n}{\sqrt{(a^2 - x^2)}}\,dx$. Hence find the value of $\int_0^{\pi/2} \sin^n x\,dx$.

Solution. Putting $x = a\sin\theta$, we get

$$\int_0^a \frac{x^n}{\sqrt{(a^2 - x^2)}}\,dx = \int_0^{\pi/2} \frac{a^n \sin^n\theta}{a\cos\theta}\, a\cos\theta\,d\theta = a^n \int_0^{\pi/2} \sin^n\theta\,d\theta$$

$$= \begin{cases} \frac{(n-1)(n-3)(n-5)\ldots2}{n(n-2)(n-4)\ldots3} a^n, & \text{if } n \text{ is odd} \\ \frac{(n-1)(n-3)(n-5)\ldots1}{n(n-2)(n-4)\ldots2} \cdot \frac{\pi}{2} a^n, & \text{if } n \text{ is even} \end{cases} \quad \ldots(1)$$

Now integrating by parts, we have

$$\int_0^a x^n \sec^{-1}\frac{x}{a}\,dx = \left[\sec^{-1}\frac{x}{a} \cdot \frac{x^{n+1}}{n+1}\right]_0^a - \int_0^a \frac{x^{n+1}}{n+1} \cdot \frac{1}{\sqrt{(x^2 - a^2)}}\,dx$$

$$= \frac{1}{(n+1)}\left[\frac{\pi}{2} - \left[\frac{x^{n+1}}{(1-x^2)^{\frac{n}{2}}}\right]\right]$$ [Skip to p. 241]

$$= \frac{1}{n+1}\left[\frac{\pi}{2} - \frac{n(n-2)(n-4)\ldots 1}{(n+1)(n-1)(n-3)\ldots 2}\cdot\frac{\pi}{2}\right]$$ when n is odd

$$= \frac{1}{n+1}\left[\frac{\pi}{2} - \frac{n(n-2)(n-4)\ldots 2}{(n+1)(n-1)(n-3)\ldots 3}\right]$$ when n is even

Problem 6.6 Evaluate $I_n = \int_0^{\frac{\pi}{2}} (\sin^n x - x^n)^2\,dx$, where n is a positive integer. Hence show that

$$I_n = \frac{2n}{2n+1}\pi^2 I_{n-1} - \pi^{2n}$$

Solution Putting $x = \sin\theta$, we get

$$I_n = \int_0^{\frac{\pi}{2}} (\sin^2\theta - \theta^2)^n \cos\theta\,d\theta = \int_0^{\frac{\pi}{2}} (\sin^2\theta - \theta^2)^n \cos\theta\,d\theta = \pi^{2n+1}\int_0^{\frac{\pi}{2}} \cos^{2n+1}\theta\,d\theta$$

$$= \pi^{2n+1}\cdot\frac{2n(2n-2)(2n-4)\ldots 4.2}{(2n+1)(2n-1)(2n-3)\ldots 3.1}$$ If $(2n+3)$ is always odd

Now replacing n by $n-1$, we get

$$I_{n-1} = \pi^{2n-1}\cdot\frac{(2n-2)(2n-4)\ldots 4.2}{(2n-1)(2n-3)\ldots 3.1} \quad\therefore\quad \frac{I_n}{I_{n-1}} = \pi^2\cdot\frac{2n}{2n+1} \quad\text{or}\quad I_n = \frac{2n}{2n+1}\pi^2 I_{n-1}$$

which is the second desired result.

6.5 (1) REDUCTION FORMULAE for $\int \sin^m x\cos^n x\,dx$

$$\int \sin^m x\cos^n x\,dx = \int \sin^{m-1}x\cdot\cos^n x\cdot\sin x\,dx$$ Integrate by parts

$$= \sin^{m-1}x\cdot\left(\frac{-\cos^{n+1}x}{n+1}\right) - \int (m-1)\sin^{m-2}x\cos x\cdot\left(-\frac{\cos^{n+1}x}{n+1}\right)dx$$

$$= -\frac{\sin^{m-1}x\cos^{n+1}x}{n+1} + \frac{m-1}{n+1}\int \sin^{m-2}x(1-\sin^2 x)\cos^n x\,dx$$

$$= -\frac{\sin^{m-1}x\cos^{n+1}x}{n+1} + \frac{m-1}{n+1}\int \sin^{m-2}x\cos^n x\,dx - \frac{m-1}{n+1}\int \sin^m x\cos^n x\,dx$$

Transposing the last term to the left and dividing by $1+(m-1)/(n+1) = (m+n)/(n+1)$, we obtain the reduction formula

$$\int \sin^m x\cos^n x\,dx = -\frac{\sin^{m-1}x\cos^{n+1}x}{m+n} + \frac{m-1}{m+n}\int \sin^{m-2}x\cos^n x\,dx \qquad \ldots(1)$$

Hence to integrate $\int \sin^m x\cos^n x\,dx$

(a) when m is odd, put $\cos x = t$

(b) when n is odd, put $\sin x = t$

(c) when m and n both are even integers, express the integrand as a series of cosines of multiple angles and integrate term by term if m and n are small, otherwise use the method of reduction formulae.

(2) To show that

$$\int_0^{\frac{\pi}{2}} \sin^m x\cos^n x\,dx = \frac{(m-1)(m-3)\ldots 3.1\,(n-1)(n-3)\ldots 3.1}{(m+n)(m+n-2)(m+n-4)\ldots 4.2}\times\left[\frac{\pi}{2}\text{, only if both } m \text{ and } n \text{ are even}\right]$$

Proceeding, we have

$$I_{m,n} = \int_0^{\pi/2} \sin^m x \cos^n x\,dx = \left| -\frac{\sin^{m-1} x \cos^{n+1} x}{m+n} \right|_0^{\pi/2} + \frac{m-1}{m+n} \int_0^{\pi/2} \sin^{m-2} x \cos^n x\,dx$$

i.e., $$I_{m,n} = \frac{m-1}{m+n} I_{m-2,n} \quad \ldots(i)$$

Case I. When n is even

Similarly, $$I_{m-2,n} = \frac{m-3}{m+n-2} I_{m-4,n}, I_{m-4,n} = \frac{m-5}{m+n-4} I_{m-6,n}$$

$$I_{0,n} = \frac{1}{n-1} I_{1,n}$$

Finally $$I_{1,n} = \frac{2}{n-1} I_{0,n} = \frac{2}{n+1} \int_0^{\pi/2} \sin x \cos^n x\,dx$$

$$= \frac{2}{n+1} \left| -\frac{\cos^{n+1} x}{n+1} \right|_0^{\pi/2} = \frac{2}{(n+1)(n+1)} \quad \ldots(ii)$$

From (i), we obtain

$$I_{m,n} = \frac{(m-1)(m-3)(m-5)\ldots 4.2}{(m+n)(m+n-2)(m+n-4)\ldots(n+2)(n+1)}$$

Case II. When n is even

We have, $$I_{m-2,n} = \frac{m-3}{m+n-2} I_{m-4,n}, \quad I_{m-4,n} = \frac{m-5}{m+n-4} I_{m-6,n}$$

$$I_{0,n} = \frac{1}{n+1} I_{1,n}, I_{1,n} = \frac{1}{n+2} I_{0,n} = \frac{1}{n+2} \int_0^{\pi/2} \cos^n x\,dx$$

From these, we have $I_{m,n} = \frac{(m-1)(m-3)(m-5)\ldots 3.1}{(m+n)(m+n-2)(m+n-4)\ldots(n+2)} \int_0^{\pi/2} \cos^n x\,dx$

$$= \frac{(m-1)(m-3)\ldots 1}{(m+n)(m+n-2)\ldots(n+2)} \cdot \frac{(n-1)(n-3)\ldots}{n(n-2)\ldots} \times (\pi/2 \text{ or } 1) \quad \text{(If } n \text{ is even)} \quad \ldots(iii)$$

Combining (i) and (ii), we get the desired result.

Example 8.2. Integrate (i) $\int \sin^5 x \cos^3 x\,dx$ (Rajpur, 2005)

(ii) $\int_0^1 \frac{x^6}{(1+x^2)^4}\,dx$ (iii) $\int_0^\infty \frac{x^2}{(1+x^2)^4}\,dx$ (U.P.T.U. 2004-05)

Solution. (i) Taking $n = 3$ in (i) of page 541, we have the reduction formula:

$$\int \sin^m x \cos^3 x\,dx = \frac{\sin^{m+1} x \cos^2 x}{m+3} + \frac{m-1}{m+3} \int \sin^{m-1} x \cos^3 x\,dx$$

Putting $m = 5, 3$ successively,

$$\int \sin^5 x \cos^3 x\,dx = -\frac{\sin^6 x \cos^2 x}{8} + \frac{3}{8} \int \sin^3 x \cos^3 x\,dx \quad \ldots(i)$$

$$\int \sin^3 x \cos^3 x\,dx = -\frac{\sin x \cos^2 x}{4} + \frac{1}{4} \int \cos^2 x\,dx$$

But $$\int \cos^2 x\,dx = \frac{1}{2} \int (1 + \cos 2x)\,dx = \frac{1}{2}\left(x + \frac{1}{2} \sin 2x\right)$$

$$\int \sin^2 x \cos^2 x \, dx = -\frac{\sin x \cos^3 x}{4} + \frac{1}{16}(2x + \sin 2x)$$

Substituting (ii) in (i), we get

$$\int \sin^3 x \cos^2 x \, dx = -\frac{\sin^2 x \cos^3 x}{5} + \frac{1}{5}\left[-\frac{\sin x \cos^3 x}{4} + \frac{1}{16}(2x + \sin 2x)\right]$$

(v) Putting $t = \cos\theta$, so that

$$\int_0^1 \frac{t^6}{(1+t^2)^4}\,dt = \int_0^{\pi/4} \frac{\tan^6\theta}{\sec^8\theta}\sec^2\theta\,d\theta = \int_0^{\pi/4} \sin^6\theta\cos^2\theta\,d\theta = \frac{5\cdot3\cdot1\times5\cdot3\cdot1}{12\cdot10\cdot8\cdot6\cdot4\cdot2}\times\frac{\pi}{2} = \frac{5\pi}{2048}$$

(vi) Putting $y = \sin\theta$, so that

$$\int_0^1 \frac{y^3}{(1+y^2)^{3/2}}\,dy = \int_0^{\pi/4} \frac{\tan^3\theta}{\sec^3\theta}\sec^2\theta\,d\theta = \int_0^{\pi/4} \sin^3\theta\cos^2\theta\,d\theta = \frac{1\cdot2}{3\cdot1} = \frac{2}{15}$$

Example 6.6. Evaluate: (i) $\int_0^{\pi/2} \cos^5 2\theta \sin^3 4\theta\,d\theta$ (ii) $\int_0^1 x^3(1-x^2)^{3/2}\,dx$ (iii) $\int_0^{2a} x^3\sqrt{(2ax - x^2)}\,dx$ (U.P.T.U., 2003-04)

Solution. (i) $\int_0^{\pi/2} \cos^5 2\theta \sin^3 4\theta\,d\theta = \int_0^{\pi/2} \cos^5 2\theta\,2\sin 2\theta\cos 2\theta\sin^2 4\theta\,d\theta$

$$= 8\int_0^{\pi/2} \sin^2 2\theta\cos^7 2\theta\,d\theta$$

$$= \frac{8}{2}\int_0^{\pi/2} \sin^2 x\cos^7 x\,dx$$

$$= 4\,\frac{2\times6\cdot4\cdot2}{9\cdot7\cdot5\cdot3\cdot1}\cdot\frac{\pi}{2} = \frac{8\pi}{315}$$

Put $2\theta = x$
so that $2\cos 2\theta\,d\theta = dx$
And when $\theta = 0$, $x = 0$,
when $\theta = \pi/2$, $x = \pi/2$

(ii) $\int_0^1 x^3(1-x^2)^{3/2}\,dx$

$$= \int_0^{\pi/2} \sin^3\theta\cos^3\theta\cos\theta\,d\theta = \int_0^{\pi/2} \sin^3\theta\cos^4\theta\,d\theta$$

$$= \frac{3\cdot1\times3\cdot1}{7\cdot5\cdot4\cdot2}\cdot\frac{\pi}{2} = \frac{3\pi}{256}$$

Put $x = \sin\theta$, so that $dx = \cos\theta\,d\theta$
When $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \pi/2$

(iii) $\int_0^{2a} x^3\sqrt{(2ax - x^2)}\,dx$

$$= \int_0^{2a} x^{3+\frac{1}{2}}\sqrt{(2a - x)}\,dx$$

$$= \int_0^{\pi/2} (2a\sin^2\theta)^{7/2}\sqrt{(2a)}\cos\theta\cdot 4a\sin\theta\cos\theta\,d\theta$$

$$= 32a^4\int_0^{\pi/2} \sin^8\theta\cos^2\theta\,d\theta = 32a^4\cdot\frac{7\cdot5\cdot3\cdot1\times1}{10\cdot8\cdot6\cdot4\cdot2}\cdot\frac{\pi}{2} = \frac{7\pi a^4}{8}$$

Put $x = 2a\sin^2\theta$
$\therefore dx = 2a\sin 2\theta\,d\theta$

EXERCISE 6.1

Evaluate:

1. (i) $\int_0^{\pi/2} \sin^6 x\,dx$ (ii) $\int_0^{\pi/2} \cos^7\theta\,d\theta$ (iii) $\int_0^{\pi/2} \frac{\sin^3\theta\cos^5\theta}{(1+\cos\theta)^2}\,d\theta$ (iv) $\int_0^{\pi/2} \sin^4 x\cos^6 x\,dx$

2. (i) $\int_{-\pi/2}^{\pi/2} \sin^2\theta\cos^4\theta\,d\theta$ (ii) $\int_0^{\pi/2} \sin^5 x\cos^5 x\,dx$ (iii) $\int_0^{\pi/2} \sin^6\theta\cos^6\theta\,d\theta$ (U.P.T.U., 2005-06)

(iv) $\int_0^{\pi/2} \sin^3 x\cos^3 x\,dx$

a. If $I_{m,n} = \int_0^{\pi/2} \sin^m x \cos^n x\,dx$ ($m, n > 0$), show that $I_{m,n} = \frac{m-1}{m+n} I_{m-2,n}$.

Hence evaluate $\int_0^{\pi/2} \sin^3 x \cos^5 x\,dx$.

b. Evaluate :

i) $\int_0^{\pi/2} \sin^4 x \cos^3 x\,dx$ (Utkal, 2004) ii) $\int_0^{\pi/2} \sin^5 x \cos^5 x\,dx$

iii) $\int_0^1 x^5\sqrt{(1-x^2)}\,dx$ iv) $\int_0^{\pi/2} \sin^{10} x \cos^8 x\,dx$

v) $\int_0^{\infty} x^{n-1}(2x - x^2)^{n-1}\,dx$ vi) $\int_0^{\infty} \frac{x^2\,dx}{\sqrt{(1+x^2)}}$ (Madras, 2003)

vii) $\int_0^{\infty} e^{-x^2}\sqrt{(x+x^3)}\,dx$ viii) $\int_0^1 x^3\sqrt{(1-x^2)}\,dx$ (Utkal, 2004)

ix) If $I_n = \int x^n\sqrt{(a-x)}\,dx$, prove that $(2n+3)I_n = 2a^n\sqrt{a} - nI_{n-1}$. Hence find $\int_0^a x^3\sqrt{(a-x)}\,dx$. (Meerut, 2004)

x) If n is any positive integer, show that $\int_0^{\pi/2} x^n \sin x\,dx = \frac{n(n-1)}{n+1}\int_0^{\pi/2} x^{n-2}\sin x\,dx - \frac{n\pi^{n-1}}{2^{n+1}}$. (Utkal, 2005)

8.6 REDUCTION FORMULAE for (a) $\int \tan^n x\,dx$ (b) $\int \cot^n x\,dx$

(a) Let $I_n = \int \tan^n x\,dx = \int \tan^{n-2} x \cdot \tan^2 x\,dx = \int \tan^{n-2} x \cdot (\sec^2 x - 1)\,dx$

$= \int \tan^{n-2} x \sec^2 x\,dx - \int \tan^{n-2} x\,dx$

Thus, $I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$ which is the required reduction formula.

(b) Let $I_n = \int \cot^n x\,dx = \int \cot^{n-2} x \cdot \cot^2 x\,dx = \int \cot^{n-2} x \cdot (\operatorname{cosec}^2 x - 1)\,dx$

$= \int \cot^{n-2} x \operatorname{cosec}^2 x\,dx - \int \cot^{n-2} x\,dx$

Thus $I_n = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}$

which is the required reduction formula.

Example 8.7. Find (i) $\int \tan^5 x\,dx$ (ii) $\int \cot^5 x\,dx$.

Solution. (i) Putting $n = 5, 3$ successively in the reduction formula for $\int \tan^n x\,dx$, we get

$I_5 = \frac{1}{4}\tan^4 x - I_3$; $I_3 = \frac{1}{2}\tan^2 x - I_1$

Thus $I_5 = \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + I_1$

i.e., $\int \tan^5 x\,dx = \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + \int \tan x\,dx = \frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x - \log\cos x$.

(ii) Putting $n = 5, 3, 1$ successively in the reduction formula for $\int \cot^n x\,dx$, we get

$I_5 = -\frac{1}{4}\cot^4 x - I_3$; $I_3 = -\frac{1}{2}\cot^2 x - I_1$; $I_1 = -\cot x - I_{-1}$

Thus $I_5 = -\frac{1}{4}\cot^4 x + \frac{1}{2}\cot^2 x - \cot x - \int dx$

for, $\int \sec^0 x\,dx = x + c$

Example 8.8. If $I_n = \int_0^{\pi/4} \tan^n \theta\,d\theta$, prove that $I_n + I_{n-2} = \frac{1}{n-1}$. Hence evaluate $\int_0^{\pi/4} \tan^6 \theta\,d\theta$. (UCLES, 2000)

Solution. The reduction formula for $\int_0^{\pi/4} \tan^n \theta\,d\theta$ is

$$I_n = \frac{1}{n-1}\left[\tan^{n-1}\theta\right]_0^{\pi/4} - I_{n-2} \quad \text{or} \quad I_n + I_{n-2} = \frac{1}{n-1}$$

Changing n to $n+1$, we obtain

$$I_{n+1} + I_{n-1} = \frac{1}{(n+1)-1} \quad \text{or} \quad I_{n+1} + I_{n-1} = \frac{1}{n}$$

8.6 REDUCTION FORMULAE for (a) $\int \sec^n x\,dx$ (b) $\int \operatorname{cosec}^n x\,dx$

(a) Let $I_n = \int \sec^n x\,dx = \int \sec^{n-2} x \cdot \sec^2 x\,dx$

Integrating by parts, we have

$$I_n = \sec^{n-2} x \tan x - \int (n-2)\sec^{n-3} x \cdot \sec x \tan x \tan x\,dx$$

$$= \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x \cdot \tan^2 x\,dx$$

$$= \sec^{n-2} x \tan x - (n-2)\int \sec^{n-2} x \cdot (\sec^2 x - 1)\,dx$$

$$= \sec^{n-2} x \tan x - (n-2)I_n + (n-2)I_{n-2}$$

Transposing, we have

$$(n-1)I_n = \sec^{n-2} x \tan x + (n-2)I_{n-2}$$

Thus $I_n = \dfrac{\sec^{n-2} x \tan x}{n-1} + \dfrac{n-2}{n-1} I_{n-2}$, which is the desired reduction formula.

(b) Let $I_n = \int \operatorname{cosec}^n x\,dx = \int \operatorname{cosec}^{n-2} x \cdot \operatorname{cosec}^2 x\,dx$

Integrating by parts, we have

$$I_n = \operatorname{cosec}^{n-2} x(-\cot x) - \int (n-2)\operatorname{cosec}^{n-3} x \cdot (-\operatorname{cosec} x \cot x)(-\cot x)\,dx$$

$$= -\cot x \operatorname{cosec}^{n-2} x - (n-2)\int \operatorname{cosec}^{n-2} x \cot^2 x\,dx$$

$$= -\cot x \operatorname{cosec}^{n-2} x - (n-2)\int \operatorname{cosec}^{n-2} x (\operatorname{cosec}^2 x - 1)\,dx$$

$$= -\cot x \operatorname{cosec}^{n-2} x - (n-2)I_n + (n-2)I_{n-2}$$

or $\quad I_n + (n-2)I_n = -\cot x \operatorname{cosec}^{n-2} x + (n-2)I_{n-2}$

Thus $I_n = -\dfrac{\cot x \operatorname{cosec}^{n-2} x}{n-1} + \dfrac{n-2}{n-1} I_{n-2}$

which is the required reduction formula.

Example 8.9. Evaluate (i) $\int_0^{\pi/4} \sec^4 x\,dx$ (ii) $\int_{\pi/6}^{\pi/2} \operatorname{cosec}^5 \theta\,d\theta$. (UCLES, 2000)

Solution. (i) Putting $n = 4$ in the reduction formula for $\int \sec^n x\,dx$, we get $I_4 = \dfrac{\sec^2 x \tan x}{3} + \dfrac{2}{3} I_2$

Thus $\int_0^{\pi/4} \sec^4 x\,dx = \left[\dfrac{\sec^2 x \tan x}{3}\right]_0^{\pi/4} + \dfrac{2}{3}\int_0^{\pi/4} \sec^2 x\,dx$

$$= \frac{2}{3} + \frac{2}{3}\left[\tan x\right]_0^{\pi/4} = \frac{2}{3}(1 + 1) = 1\tfrac{1}{3}.$$

(c) Putting $n = 3$ in the reduction formula for $\int \operatorname{cosec}^n x\,dx$, we get

$$I_3 = -\frac{1}{2}\cot x \operatorname{cosec} x + \frac{3}{2} I_1$$

$$\therefore \quad \int_{\pi/4}^{\pi/2} \operatorname{cosec}^3 x\,dx = -\frac{1}{2}[\cot x \operatorname{cosec} x]_{\pi/4}^{\pi/2} + \frac{1}{2}\int_{\pi/4}^{\pi/2} \operatorname{cosec} x\,dx$$

$$= -\frac{1}{2}\left[0 - \frac{2}{\sqrt{2}}\right] + \frac{1}{2}[\log|\operatorname{cosec} x - \cot x|]_{\pi/4}^{\pi/2}$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{2}\left[\log 1 - \log\left(\frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right)\right] = \frac{1}{\sqrt{2}} + \frac{1}{2}\log 2.$$

PROBLEMS 9.3

1. Evaluate (i) $\int \tan^5 x\,dx$ (V.T.U., 2007) (ii) $\int \sec^5 x\,dx$.

2. Show that $\int_0^{\pi/4} \tan^5 x\,dx = \frac{1}{12}(6 - 8\log 2)$.

3. If $I_n = \int_0^{\pi/4} \tan^n x\,dx$, prove that $I_n = 1/(n-1) - I_{n-2}$, $n > 1$. Hence evaluate I_5. (V.T.U., 2008; Madras, 2005)

4. If $I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta\,d\theta$ $(n > 2)$, prove that $I_n = \frac{1}{n-1} - I_{n-2}$. Hence evaluate I_6. (Kerala, 2006; Mumbai, 2006)

5. Obtain the reduction formula for $\int_0^{\pi/2} \sin^n \theta\,d\theta$. (V.T.U., 2005; U.P.T.U.)

6. Evaluate (i) $\int \sin^6 \theta\,d\theta$ (ii) $\int_0^{\pi/2} \sin^5 x\,dx$ 7. Evaluate $\int_0^{\pi/2} \sin^5 x \cos^3 x\,dx$.

8. If $I_n = \int \frac{x^n}{\sqrt{1+x^2}}\,dx$, show that $I_{n+2} = \frac{x^{n+1}}{n+1} - I_n$. Hence evaluate I_5.

9.5 REDUCTION FORMULAE for

(a) $\int x^n e^{ax}\,dx$ (b) $\int x^m (\log x)^n\,dx$

(a) Let $I_n = \int x^n e^{ax}\,dx$

Integrating by parts, we have

$$I_n = x^n \cdot \frac{e^{ax}}{a} - \int nx^{n-1} \cdot \frac{e^{ax}}{a}\,dx$$

or $I_n = \frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}$, which is the required reduction formula. (Madras, 2006)

(b) Let $I_{m,n} = \int x^m (\log x)^n\,dx = \int (\log x)^n \cdot x^m\,dx$

Integrating by parts, we have

$$I_{m,n} = (\log x)^n \cdot \frac{x^{m+1}}{m+1} - \int n(\log x)^{n-1} \cdot \frac{1}{x} \cdot \frac{x^{m+1}}{m+1}\,dx$$

$$= \frac{x^{m+1}}{m+1}(\log x)^n - \frac{n}{m+1}\int x^m (\log x)^{n-1}\,dx \quad \text{or} \quad I_{m,n} = \frac{x^{m+1}}{m+1}(\log x)^n - \frac{n}{m+1} I_{m,n-1}$$

which is the desired reduction formula.

[illegible] REDUCTION FORMULAE for

(a) $\int x^n \sin mx\, dx$ (b) $\int x^n \cos mx\, dx$ (c) $\int \cos^n x \sin nx\, dx$

(a) Let $I_n = \int x^n \sin mx\, dx$

Integrating by parts, we get

$$I_n = x^n \left(\frac{-\cos mx}{m}\right) - \int nx^{n-1}\left(\frac{-\cos mx}{m}\right) dx$$

$$= -\frac{x^n \cos mx}{m} + \frac{n}{m}\int x^{n-1}\cos mx\, dx$$ Again integrate by parts

$$= -\frac{x^n \cos mx}{m} + \frac{n}{m}\left\{x^{n-1}\cdot\frac{\sin mx}{m} - \int (n-1)x^{n-2}\cdot\frac{\sin mx}{m}\, dx\right\}$$

or $$I_n = -\frac{x^n \cos mx}{m} + \frac{n}{m^2}x^{n-1}\sin mx - \frac{n(n-1)}{m^2} I_{n-2}$$

which is the desired reduction formula. (October, 2007)

(b) Let $I_n = \int x^n \cos mx\, dx$

Integrating twice by parts as above, we get

$$I_n = \frac{x^n \sin mx}{m} + \frac{n}{m^2}x^{n-1}\cos mx - \frac{n(n-1)}{m^2} I_{n-2}$$

(c) Let $I_{m,n} = \int \cos^m x \sin nx\, dx$

Integrating by parts,

$$I_{m,n} = \cos^m x \cdot \frac{\cos nx}{n} - \int m\cos^{m-1} x(-\sin x)\cdot\left(\frac{-\cos nx}{n}\right) dx$$

$$= -\frac{1}{n}\cos^m x \cos nx - \frac{m}{n}\int \cos^{m-1} x \cdot \cos nx \sin x\, dx$$

$$\left[\because \sin(n-1)x = \sin nx\cos x - \cos nx \sin x \right.$$
$$\left. \cos nx \sin x = \sin nx \cos x - \sin(n-1)x\right]$$

$$= -\frac{1}{n}\cos^m x \cos nx - \frac{m}{n}\int \cos^{m-1} x \{\sin nx \cos x - \sin(n-1)x\}\, dx$$

$$= -\frac{1}{n}\cos^m x \cos nx - \frac{m}{n}\{I_{m,n} - I_{m-1,n-1}\}$$

Transposing, we get

$$\left(1 + \frac{m}{n}\right) I_{m,n} = -\frac{1}{n}\cos^m x \cos nx + \frac{m}{n} I_{m-1,n-1}$$

or $$I_{m,n} = -\frac{\cos^m x \cos nx}{m+n} + \frac{m}{m+n} I_{m-1,n-1}$$

which is the desired reduction formula.

Example 6.10. Prove that $\int_0^{\pi/2} \cos^n x \cos nx\, dx = \frac{\pi}{2^{n+1}} + \frac{n}{n+1}\int_0^{\pi/2} \cos^{n-1} x \sin x \sin nx\, dx$

Hence deduce that $\int_0^{\pi/2} \cos^n x \cos nx\, dx = \frac{\pi}{2^{n+1}}$ (U.P.T.U., 2006)

Solution. Let $I_{m,n} = \int_0^{\pi/2} \cos^m x \cos nx\, dx$

Integrating by parts

$$I_{m,n} = \left|\cos^m x \cdot \frac{\sin nx}{n}\right|_0^{\pi/2} - \int_0^{\pi/2} m\cos^{m-1} x(-\sin x)\frac{\sin nx}{n}\, dx$$

$$= \frac{1}{n}\left[\sin^{n-1}x\cos x\right]_0^{\frac{\pi}{2}} + \frac{n-1}{n}\int_0^{\frac{\pi}{2}}\sin^{n-2}x\,dx$$

$$= \frac{n-1}{n}\int_0^{\frac{\pi}{2}}\sin^{n-2}x\,dx - \frac{1}{n}\left[\sin^{n-1}x\cos x\right]_0^{\frac{\pi}{2}}$$

Transposing and dividing by $(1+\sin x)$ we get

$$I_n = \frac{n-1}{n+1}I_{n-2}, \quad n \geq 2$$

which is the required result.

Putting $n = 2m$ we get $I_{2m} = \frac{2m-1}{2m}\cdot\frac{2m-3}{2m-2}\cdots\frac{1}{2}\cdot\frac{\pi}{2}$

Changing to $n = 2m+1$,

$$I_{2m+1} = \frac{2}{3}I_{2m-1}$$

$$\therefore \quad I_{2m+1} = \frac{2}{3}\cdot\frac{4}{5}\cdots\frac{2m}{2m+1}\cdot I_1 = \frac{2}{3}\cdot\frac{4}{5}\cdots\frac{2m}{2m+1}\cdot\int_0^{\frac{\pi}{2}}\sin x\,dx$$

Hence $I_{2m+1} = \frac{2}{3}\cdot\frac{4}{5}\cdots\frac{2m}{2m+1}$

Example 6.3 Prove a reduction formula for $\int x^n \sin x\,dx$ and hence evaluate $\int x^4\sin x\,dx$.

Let $I_n = \int x^n\sin x\,dx$

Integrating by parts:

$$I_n = x^n(-\cos x) - \int nx^{n-1}(-\cos x)\,dx = -x^n\cos x + n\int x^{n-1}\cos x\,dx$$

$$= -x^n\cos x + n\left\{x^{n-1}\sin x - (n-1)\int x^{n-2}\sin x\,dx\right\}$$ (Again integrating by parts)

$$= -x^n\cos x + nx^{n-1}\sin x - n(n-1)\int x^{n-2}\sin x\,dx$$

$$\therefore I_n = -x^n\cos x + nx^{n-1}\sin x - n(n-1)I_{n-2}$$

Transposing and dividing by $(1+n^2)$ we get

$$I_n = \frac{-x^n\cos x + nx^{n-1}\sin x}{1+n^2} - \frac{n(n-1)}{1+n^2}I_{n-2}$$

which is the required reduction formula.

Replacing n by 4 and $n-2$ by 0 we get

$$I_4 = \frac{-x^4\cos x + 4x^3\sin x + 12x\cos x}{1+16} - \frac{12}{17}I_0$$

But $I_0 = \int x^0\sin x\,dx = \int\sin x\,dx = -\cos x$

$$\therefore \quad I_4 = \frac{-x^4\cos x + 4x^3\sin x + 12x\cos x}{17} + \frac{12}{17}\cos x$$

PROBLEMS

1. If $I_n = \int x^n e^x dx$, show that $I_n = x^n e^x - n I_{n-1}$, and hence find I_3. (Madras, 2000)

2. If $I_n = \int_0^{\pi/2} x^n \sin x \, dx$, prove that $I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1) I_{n-2}$. (Madras, 2000)

3. Obtain a reduction formula for $\int x^n \log x^n dx$. Hence evaluate $\int_0^1 x^5 \log x^5 dx$. (U.P.T.U., 2004) (Madras, 2000)

4. If n is a positive integer, show that $\int_0^{\pi/2} x^n \log x^n dx = \frac{n(n-1)\pi}{(n+1)2^{n+1}}$ or $n(n-1)$.

5. If $I_n = \int_0^{\pi/2} x \cos^n x \, dx$ $(n \in \mathbb{N})$, prove that $n^2 I_n = n(n-1) I_{n-2} + 1$. Hence evaluate I_5.

6. If $I_n = \int_0^{\pi/2} x \sin^n x \, dx$ $(n \geq 2)$, prove that $I_n = \frac{n-1}{n} I_{n-2} - \frac{1}{n^2}$. Hence evaluate I_4.

7. If $I_n = \int_0^{\pi/4} x^n \sec^2 x \, dx$, $(n > 2)$, prove that $I_n = n(n-1) I_{n-2} - \left(\frac{\pi}{4}\right)^n$. Hence evaluate I_4. (Madras, 2000)

8. If $I_n = \int x^n \sin ax \, dx$, show that $aI_n = -x^n \cos ax + nx^{n-1} \sin ax + n(n-1) I_{n-2}$. (Bharathiar, 2000)

9. Prove that $\int_0^{\pi/2} \sin^{n-1} x \sin nx \, dx = \frac{1}{n-1}$, $n > 1$.

10. If $I_{m,n} = \int_0^{\pi/2} \sin^m x \cos nx \, dx$, prove that $I_{m,n} = \frac{m}{m^2 - n^2} I_{m-1,n-1}$.

11. Find a reduction formula for $\int x^n \sin^2 x \, dx$. Hence evaluate $\int_0^{\pi/2} x^3 \sin^2 x \, dx$.

12. Obtain a reduction formula for $I_n = \int_0^{\pi/2} x^n \cos^n x \, dx$, where $n \geq 2$ is an integer.

Hence evaluate I_6. (Bharathiar, 2000)

3.3 DEFINITE INTEGRALS

Property I. $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

i.e., the value of a definite integral depends on the limits and not on the variable of integration.

Let $\int f(x)\,dx = \phi(x)$; $\therefore \int_a^b f(x)\,dx = \phi(b) - \phi(a)$

Then $\int f(t)\,dt = \phi(t)$; $\therefore \int_a^b f(t)\,dt = \phi(b) - \phi(a)$

Hence the result.

Property II. $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

i.e., the interchange of limits changes the sign of the integral.

Let $\int f(x)\,dx = \phi(x)$; $\therefore \int_a^b f(x)\,dx = \phi(b) - \phi(a)$

and $-\int_b^a f(x)\,dx = -[\phi(x)]_b^a = -[\phi(a) - \phi(b)] = \phi(b) - \phi(a)$

Hence the result.

Property III. $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

Let $\int f(x)\,dx = \phi(x)$, so that $\int_a^b f(x)\,dx = \phi(b) - \phi(a)$ ...(1)

Also $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = [\phi(x)]_a^c + [\phi(x)]_c^b$

$= \phi(c) - \phi(a) + \phi(b) - \phi(c) = \phi(b) - \phi(a)$ ...(2)

From (1) and (2), Property III is proved.

Property IV. $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Put $x = a - t \Rightarrow dx = -dt$. Also when $x = 0, t = a$; when $x = a, t = 0$.

$\therefore \quad \int_0^a f(x)\,dx = -\int_a^0 f(a-t)\,dt = \int_0^a f(a-t)\,dt = \int_0^a f(a-x)\,dx$ [Prop. II]

Example 6.12. Evaluate: $\int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}}\,dx$

Solution. Let $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx$

Then $I = \int_0^{\pi/2} \frac{\sqrt{\cos\left[\frac{\pi}{2} - x\right]}}{\sqrt{\cos\left[\frac{\pi}{2} - x\right]} + \sqrt{\sin\left[\frac{\pi}{2} - x\right]}}\,dx$ [Prop. IV]

$= \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}}\,dx$

Adding: $2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx = \int_0^{\pi/2} dx = [x]_0^{\pi/2} = \frac{\pi}{2}$

Hence $I = \frac{\pi}{4}$.

Example 6.13. Evaluate: $\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$. (Delhi, 2005)

Solution. Let $I = \int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$

Put $x = \tan\theta \Rightarrow dx = \sec^2\theta\,d\theta$. When $x = 0, \theta = 0$; when $x = 1, \theta = \frac{\pi}{4}$.

$= \int_0^{\pi/4} \frac{\log(1+\tan\theta)}{1+\tan^2\theta} \sec^2\theta\,d\theta = \int_0^{\pi/4} \log(1+\tan\theta)\,d\theta$

$= \int_0^{\pi/4} \log\left[1 + \tan\left(\frac{\pi}{4} - \theta\right)\right] d\theta = \int_0^{\pi/4} \log\left(1 + \frac{1-\tan\theta}{1+\tan\theta}\right) d\theta$ [Prop. IV]

$= \int_0^{\pi/4} \log\left(\frac{2}{1+\tan\theta}\right) d\theta = \log 2 \int_0^{\pi/4} d\theta - I$

Transposing, $2I = \log 2 \cdot [\theta]_0^{\pi/4} = \frac{\pi}{4} \log 2$. Hence $I = \frac{\pi}{8} \log 2$.

Example 6.14. Evaluate: $\int_0^{\pi} \frac{x \sin^2 x}{1+\cos^2 x}\,dx$. (Madras, 2006)

Solution. Let $I = \int_0^{\pi} \frac{x \sin^2 x}{1+\cos^2 x}\,dx$

Prove [illegible] [illegible]

[illegible]

Corresponding [illegible]

[illegible]

[Property V] [illegible]

[illegible]

[illegible]

[illegible]

[Property VI] [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

(iii) $f(2a-x)=-f(x)$, then $\int_0^{2a} f(x)\,dx=\int_0^a f(x)\,dx+\int_0^a f(2a-x)\,dx=0$.

Cor. 1. If n is even, $\int_0^{\pi} \sin^n x\cos^n x\,dx=2\int_0^{\pi/2} \sin^n x\cos^n x\,dx$ and if n is odd, $\int_0^{\pi} \sin^n x\cos^n x\,dx=0$.

Cor. 2. If n is odd, $\int_0^{2\pi} \sin^n x\cos^n x\,dx=0$

and if n is even, $\int_0^{2\pi} \sin^n x\cos^n x\,dx=2\int_0^{\pi} \sin^n x\cos^n x\,dx$

$=4\int_0^{\pi/2} \sin^n x\cos^n x\,dx$. (i.e.) $=4$ or 0, if n is even or odd.

Example 6.15. Evaluate $\int_0^{\pi/2} \sin^5\theta\cos^4\theta\,d\theta$. (V.T.U. 2009-10)

Solution. Let $I=\int_0^{\pi/2} \sin^5\theta\cos^4\theta\,d\theta$

Then $I=\int_0^{\pi/2} \sin^5\left(\frac{\pi}{2}-\theta\right)\cos^4\left(\frac{\pi}{2}-\theta\right)d\theta=\int_0^{\pi/2} \cos^5\theta\sin^4\theta\,d\theta=I$ [Prop. IV]

or $2I=2\int_0^{\pi/2} \sin^5\theta\cos^4\theta\,d\theta=2\pi\int_0^{\pi/2} \sin^5\theta\cos^4\theta\,d\theta$ [Prop. VI Cor. 2]

$=2\pi\cdot\frac{4\cdot2\cdot1}{6\cdot4\cdot2}=\frac{8}{3}=\frac{\pi^2}{30}$

Hence $I=\frac{\pi^2}{60}$

Example 6.16. Evaluate $\int_0^{\pi/2} \log\sin x\,dx$. (Anna, 2005 S)

Solution. Let $I=\int_0^{\pi/2} \log\sin x\,dx$ ...(1)

then $I=\int_0^{\pi/2} \log\sin(\pi/2-x)\,dx=\int_0^{\pi/2} \log\cos x\,dx$ ...(2)

Adding (1) and (2),

$2I=\int_0^{\pi/2} (\log\sin x+\log\cos x)\,dx$

$=\int_0^{\pi/2} \log[\sin x\cos x]\,dx=\int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right)dx$

$=\int_0^{\pi/2} \log\sin 2x\,dx-\int_0^{\pi/2} \log 2\,dx=\int_0^{\pi/2} \log\sin 2x\,dx-\log 2\int_0^{\pi/2} dx$

$=\int_0^{\pi/2} \log\sin 2x\,dx-\log 2[x]_0^{\pi/2}=I-\frac{\pi}{2}\log 2$ ...(3)

where $I=\int_0^{\pi/2} \log\sin 2x\,dx$

Put $2x=t$, so that $2dx=dt$
When $x=0$, $t=0$; when $x=\pi/2$, $t=\pi$

$=\frac{1}{2}\int_0^{\pi} \log\sin t\,dt=\frac{1}{2}\int_0^{\pi} \log\sin x\,dx$ [$\because \log\sin(\pi-x)=\log\sin x$, Prop. IV]

$=\frac{1}{2}\cdot 2\int_0^{\pi/2} \log\sin x\,dx=I$

Therefore (3) $\Rightarrow 2I=I-\frac{\pi}{2}\log 2$, i.e., $I=-\frac{\pi}{2}\log 2$.

Obs: The following are two immediate consequences

$$\int_0^{\pi/2} \log \sin x \, dx = \int_0^{\pi/2} \log \cos x \, dx = -\frac{\pi}{2} \log 2$$

and $$\int_0^{\pi} \log \sin x \, dx = -\pi \log 2.$$

Example 6.13. Evaluate $\int_0^1 \frac{\sin^{-1} x}{x} \, dx$.

Solution. Put $\sin^{-1} x = \theta$ so that $x = \sin\theta$ and $dx = \cos\theta \, d\theta$.
Also when $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \frac{\pi}{2}$.

$$\therefore \quad \int_0^1 \frac{\sin^{-1} x}{x} \, dx = \int_0^{\pi/2} \theta \cdot \frac{\cos\theta}{\sin\theta} \, d\theta \qquad \text{[Integrate by parts]}$$

$$= [\theta . \log \sin\theta]_0^{\pi/2} - \int_0^{\pi/2} 1 . \log \sin\theta \, d\theta$$

$$= -\int_0^{\pi/2} \log \sin\theta \, d\theta = -\left(-\frac{\pi}{2} \log 2\right) = \frac{\pi}{2} \log 2 \qquad \left[\because \lim_{x \to 0} (x \log x) = 0\right]$$

EXERCISE 6.4

Prove that

1. (a) $\int_0^{\pi/2} \log \tan x \, dx = 0$

2. (a) $\int_0^{\infty} \frac{x^2 (1 - x^{20})}{(1 + x^{20})^2} \, dx = 0$

3. (a) $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} \, dx = \frac{\pi}{4}$

4. (a) $\int_0^{\pi/4} \frac{dx}{1 + \sqrt{\tan x}} = \frac{\pi}{8}$

5. (a) $\int_0^{\pi/2} \frac{x \sin x}{\sin x + \cos x} \, dx = \frac{\pi^2}{8}$

6. (a) $\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} \, dx = \frac{1}{2} \left(\frac{\pi}{4}\right)^2$

Evaluate

7. (a) $\int_0^{\pi} \sin^5 x \, dx$

(b) $\int_0^{\pi} \sin^3 x \cos^4 x \, dx$ (U.P.U. 2010)

8. (a) $\int_0^{\pi} x \sin^3 x \, dx$ (U.P.U. 2008)

Prove that

9. (a) $\int_0^{\pi/2} \frac{a \, dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{\pi}{2ab}$

10. (a) $\int_0^{\pi/2} \frac{a \, dx}{a^2 - \cos^2 x} = \frac{\pi}{2\sqrt{a^2 - 1}}$ $(a > 1)$

(b) $\int_0^{\pi} x \log \sin x \, dx = \frac{\pi^2}{2} \log \frac{1}{2}$

(c) $\int_0^{\pi} \log(1) \cos\theta \, d\theta = \frac{\pi}{2} \log 2$ (Kerala 2009)

(d) $\int_0^{\pi/2} \frac{\sin^2 x}{2 - \sin^2 x + \cos^2 x} = \frac{\pi}{4}$

(e) $\int_0^{\pi/2} \frac{\sin^2 x}{\sin^2 x + \cos^2 x} \, dx = \frac{\pi}{4}$

(f) $\int_0^{\pi/2} \frac{\sin x}{1 + \sin x} \, dx = \pi - 2$ (Kerala 2012, 17)

(g) $\int_0^{\pi/2} \frac{x \, dx}{\sin x + \cos x} = \frac{\pi}{2\sqrt{2}} \log(\sqrt{2} + 1)$.

(h) $\int_0^{\pi/2} \sin^2 x \, dx$

(i) $\int_0^{\pi} \sin^6 x \cos^8 x \, dx$

(j) $\int_0^{\pi/2} x \sin^2 x \cos^2 x \, dx$ (Kerala Feb. 2008)

(k) $\int_0^{\pi/2} \frac{a \, dx}{a^2 \cos^2 x + b^2 \sin^2 x} = \frac{\pi}{2ab}$

(l) $\int_0^{\pi/2} \frac{a \sin x \, dx}{1 + \cos^2 x} = \frac{\pi a}{2\sqrt{2}}$

11. $\int_0^{\pi/2} \log(1+\cos\theta)\,d\theta = -\pi\log 2$ (Madras, 2003)

12. (a) $\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx = \frac{\pi}{8}\log 2$ (b) $\int_0^{\pi/2} \frac{\log(\tan\theta+1)}{\tan\theta}\,d\theta = \frac{\pi}{4}\log 2$

13.5 (I) INTEGRAL AS THE LIMIT OF A SUM

We have so far considered integration as inverse of differentiation. We shall now define the definite integral as the limit of a sum :

Def. If $f(x)$ is continuous and single valued in the interval $[a, b]$, then the definite integral of $f(x)$ between the limits a and b is defined by the equation

$$\int_a^b f(x)\,dx = \lim_{h\to 0} h\{f(a) + f(a+h) + f(a+2h) + \ldots + f(a+(n-1)h)\},$$

where $nh = b - a$ ...(1)

(i) Evaluation of limits of series

The summation definition of a definite integral enables us to express the limits of sums of certain types of series as definite integrals which can be easily evaluated. We rewrite (1) as follows :

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} h\sum_{r=0}^{n-1} f(a + rh), \text{ where } nh = b - a.$$

Putting $a = 0$ and $b = 1$, so that $h = 1/n$, we get

$$\lim_{n\to\infty} \frac{1}{n}\sum_{r=0}^{n-1} f\left(\frac{r}{n}\right) = \int_0^1 f(x)\,dx$$

Thus to express a given series as definite integral :

(a) Write the general term T_r or T_{r+1} as the r^{th} or $(r+1)^{th}$ term of $f(r/n)$ i.e., $f(r/n)\cdot 1/n$

(b) Replace r/n by x and $1/n$ by dx,

(c) Integrate the resulting expression, taking

the lower limit = $\lim_{n\to\infty}\left(\frac{0}{n}\right)$ where 0 is in the first term.

and the upper limit = $\lim_{n\to\infty}\left(\frac{k}{n}\right)$ where k is in the last term.

Example 6.18. Find the limit when $n \to \infty$ of the series

$$\frac{n}{n^2} + \frac{n}{n^2+1^2} + \frac{n}{n^2+2^2} + \ldots + \frac{n}{n^2+(n-1)^2}$$

Solution. Here the general term $= T_{r+1} = \frac{n}{n^2+r^2} = \frac{n}{n^2(1+r^2/n^2)} = \frac{1}{n}\cdot\frac{1}{1+(r/n)^2}$

$= \frac{1}{1+x^2}\,dx$ [Putting $r/n = x$ and $1/n = dx$]

Now for the first term $r = 0$ and for the last term $r = n - 1$

$\therefore$ the lower limit of integration $= \lim_{n\to\infty}\left(\frac{0}{n}\right) = 0$

and the upper limit of integration $= \lim_{n\to\infty}\left(\frac{n-1}{n}\right) = \lim_{n\to\infty}\left(1 - \frac{1}{n}\right) = 1$

Hence, the required limit $= \int_0^1 \frac{dx}{1+x^2} = \left|\tan^{-1}x\right|_0^1 = \tan^{-1}(1) - \tan^{-1}(0) = \pi/4$.

To find limit of a product by integration :

Let $P = \lim_{\dots\dots} \dots$ (given product)

Take logs of both sides, we find

$\log P = \lim_{\dots\dots} \dots$ (a series of $\log$'s). Then $P = e^{\log P}$

Example 6.15. Evaluate $\lim_{n\to\infty} \left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\left(1+\frac{3}{n}\right)\cdots\left(1+\frac{n}{n}\right)\right]^{\frac{1}{n}}$. (Nagpur, 2008)

Solution. Let $P = \lim_{n\to\infty} \left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\left(1+\frac{3}{n}\right)\cdots\left(1+\frac{n}{n}\right)\right]^{\frac{1}{n}}$.

Taking logs of both sides,

$$\log P = \lim_{n\to\infty} \frac{1}{n}\left[\log\left(1+\frac{1}{n}\right) + \log\left(1+\frac{2}{n}\right) + \dots + \log\left(1+\frac{n}{n}\right)\right]$$

In general term $= \log\left(1+\frac{r}{n}\right)\frac{1}{n} = \log(1+x)\,dx$ [Putting $\frac{r}{n} = x$ and $\frac{1}{n} = dx$]

Also for first term $r = 1$ and for the last term $r = n$.

$\therefore$ The lower limit of integration $= \lim_{n\to\infty} \left(\frac{r}{n}\right) = 0$ and the upper limit $= \lim_{n\to\infty} \left(\frac{n}{n}\right) = 1$

Hence $\log P = \int_0^1 \log(1+x)\,dx = \int_0^1 \log(1+x)\cdot 1\,dx$ [Integrate by parts]

$$= \left[\log(1+x)\cdot x\right]_0^1 - \int_0^1 \frac{1}{1+x}\cdot x\,dx$$

$$= \log 2 - \int_0^1 \frac{1+x-1}{1+x}\,dx = \log 2 - \int_0^1 dx + \int_0^1 \frac{dx}{1+x}$$

$$= \log 2 - [x]_0^1 + [\log(1+x)]_0^1 = \log 2 - 1 + \log 2$$

$$= \log 2^2 - \log_e e = \log(4/e). \text{ Hence } P = 4/e.$$

PROBLEMS 6.5

Find the limit, as $n \to \infty$, of the series

1. $\frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \frac{1}{(n+3)^2} + \dots + \frac{1}{2n}$ (Nagpur, 2009) 2. $\frac{1}{n^2+1} + \frac{4}{n^2+4} + \frac{9}{n^2+9} + \dots + \frac{n^2}{n^2+n^2} + \dots + \frac{n}{2n}$

3. $\lim_{n\to\infty} \frac{\sqrt{n}}{n^2} + \frac{\sqrt{n}}{\sqrt{(n+1)^3}} + \frac{\sqrt{n}}{\sqrt{(n+2)^3}} + \dots + \frac{\sqrt{n}}{\sqrt{(n+n)^3}}$

Evaluate :

4. $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{(2r-1)^2}$ (Kerala, 2006) 5. $\lim_{n\to\infty} \dots$

6. $\lim_{n\to\infty} \left[\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)\left(1+\frac{3}{n}\right)\cdots\left(1+\frac{n}{n}\right)\right]^{\frac{1}{n}}$ (Nagpur, 2009)

6.10 AREAS OF CARTESIAN CURVES

(1) *Area bounded by the curve $y = f(x)$, the x-axis and the ordinates $x = a$, $x = b$ is* $\int_a^b y\,dx$.

Let AB be the curve $y = f(x)$ between the ordinates $x = a$ and $x = b$ (Fig. 6.1)

Let $P(x, y)$, $Q(x + \delta x, y + \delta y)$ be two neighbouring points on the curve and AM, QN be their respective ordinates.

Let the area $ALMP$ be A, which depends on the position of P or the abscissa is x. Then the area $PMNP' = \delta A$

Complete the rectangles PN and $P'N'$.

Then the area $PNN'P'$ lies between the areas of the rectangles PN' and $P'N$,

i.e., δA lies between $y\delta x$ and $(y + \delta y)\delta x$

$\therefore \quad \frac{\delta A}{\delta x}$ lies between y and $y + \delta y$

Now taking limits as $P' \to P$ i.e., $\delta x \to 0$ and $\therefore \ \delta y \to 0$,

$$\frac{dA}{dx} = y$$

Integrating both sides between the limits $x = a$ and $x = b$, we have

$$\int_a^b dA = \int_a^b y\,dx$$

or Integral of A for $x = b$ − Integral of A for $x = a$ = $\int_a^b y\,dx$

Thus $\text{area } ALMB = \int_a^b y\,dx$.

(2) Interchanging x and y in the above formula, we see that the area bounded by the curve $x = f(y)$, the y-axis and the two abscissas $y = a$, $y = b$ is $\int_a^b x\,dy$. (Fig. 6.2)

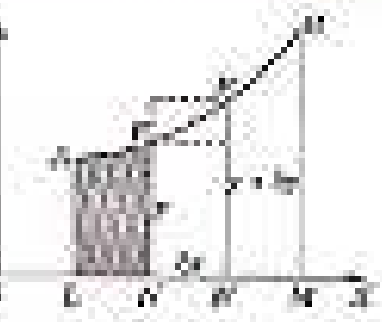

Fig. 6.1

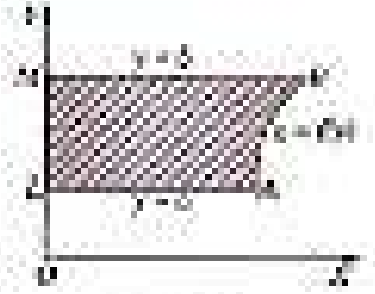

Fig. 6.2

Note 1. The area bounded by a curve, the x-axis and the two ordinates $x = a$ and $x = b$ is the area under the curve. The process of finding the area of plane curves is often called quadrature.

Note 2. Right of the area, the area whose boundaries are described in the right clockwise direction is considered positive and the area whose boundaries are described in the anticlockwise direction is taken as negative.

In Fig. 6.3, the area $ALMB$ $\left[= \int_a^b y\,dx\right]$ which is described in the anti-clockwise direction and lies above the x-axis, will give a positive result.

In Fig. 6.4, the area $ALMB$ $\left[= -\int_a^b y\,dx\right]$ which is described in the clockwise direction and lies below the x-axis, will give a negative result.

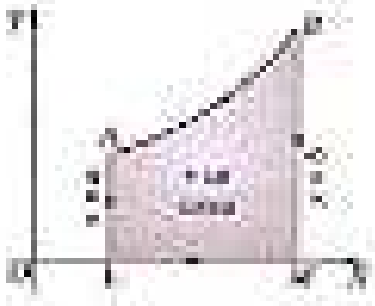

Fig. 6.3

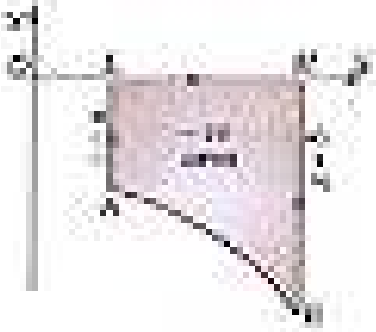

Fig. 6.4

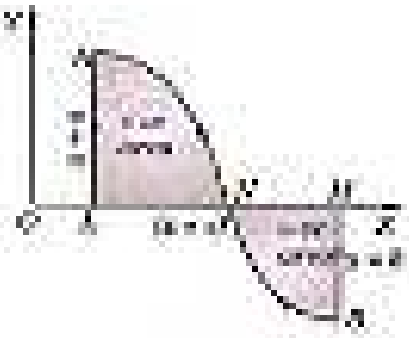

Fig. 6.5

In Fig. 6.5, the area $ALMB$ $\left[= \int_a^b y\,dx\right]$ will be equal to the sum of the area ALN $\left[= \int_a^c y\,dx\right]$ and the area MNB $\left[= \int_c^b y\,dx\right]$, but of the different sign.

Thus to find the total area in such cases the numerical value of the area of each portion must be evaluated separately and then result is added afterwards.

Example 6.20. Find the area of the region bounded by the curve $y^2 = x^2(a - x)$. (V.T.U., 2009; Dec. 2010)

Solution. Let us trace the curve roughly to get the limits of integration.

(1) The curve is symmetrical about x-axis.

(c) It passes through the origin. The tangents at the origin are $ay^2 = cx^2$ or $y = \pm\frac{c}{a}x$. ∴ Origin is a node.

(d) The curve has no asymptotes.

(e) The curve meets the x-axis at (0, 0) and $(a, 0)$. It meets the y-axis at (0, 0) only.

From the equation of the curve, we have $y = \frac{x}{\sqrt{a}}\sqrt{(a - x)}$

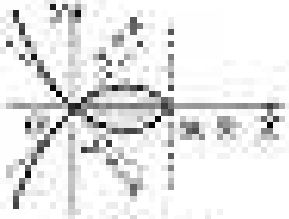

Fig. 4.9

For $x > a$, y is imaginary. Thus no portion of the curve lies to the right of the line $x = a$. Also as $x \to -\infty$, $y \to \infty$.

Thus the curve is as shown in Fig. 4.9.

∴ Area of the loop = 2 (area of upper half of the loop)

$$= 2\int_0^a y\,dx = 2\int_0^a x\sqrt{\left(\frac{a-x}{a}\right)}\,dx = \frac{2}{\sqrt{a}}\int_0^a (x - (a-x))\sqrt{(a-x)}\,dx$$

$$= \frac{2}{\sqrt{a}}\int_0^a \{(a-x)^{3/2} - (a-x)^{1/2}\}\,dx = 2\sqrt{a}\left|\frac{(a-x)^{5/2}}{-5/2}\right|_0^a - \frac{2}{\sqrt{a}}\left|\frac{(a-x)^{3/2}}{-3/2}\right|_0^a$$

$$= -\frac{4}{5}\sqrt{a}(0 - a^{5/2}) + \frac{4}{3\sqrt{a}}(0 - a^{3/2}) = \frac{4}{5}a^3 - \frac{4}{3}a^3 = \frac{8}{15}a^3.$$

Example 4.23. Find the area enclosed between the curve $y^2(2a - x) = x^3$ and its asymptote. (U.P.T.U., 2002)

Solution. The curve is as shown in Fig. 4.10.

Area between the curve and the asymptote

$$= 2\int_0^{2a} y\,dx = 2\int_0^{2a} \sqrt{\left(\frac{x^3}{2a - x}\right)}\,dx$$

Put $x = 2a\sin^2\theta$ so that $dx = 4a\sin\theta\cos\theta\,d\theta$

$$= 2\int_0^{\pi/2} \sqrt{\left|\frac{(2a\sin^2\theta)^3}{2a\cos^2\theta}\right|}\,4a\sin\theta\cos\theta\,d\theta$$

$$= 16a^3\int_0^{\pi/2} \sin^4\theta\,d\theta = 16a^3 \cdot \frac{3\cdot 1}{4\cdot 2}\cdot\frac{\pi}{2} = 3\pi a^2.$$

Example 4.24. Find the area enclosed by the curve $y^2(a^2 + x^2) = a^2(a^2 - x^2)$.

Solution. Let us first find the limits of integration.

(a) The curve is symmetrical about y-axis.

(b) It passes through the origin and the tangents at the origin are $x^2 = 0$ or $x = 0, x = 0$.

∴ There is a cusp at the origin.

(c) The curve has no asymptote.

(d) The curve meets the x-axis at the origin only and meets the y-axis at $(0, \pm a)$. From the equation of the curve, we have

$$y = \pm\frac{a}{x}\sqrt{(a^2 - x^2)}$$

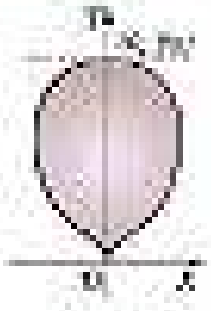

Fig. 4.11

For $x > a$, y is imaginary. Thus the curve entirely lies between $x = 0$ (y-axis) and $x = a$, which is shown in Fig. 4.11.

∴ Area of the curve $= 2\int_0^a x\,dy = \frac{2}{a}\int_0^a y\sqrt{(a^2 + 2a^2 - y^2)}\,dy$

Put $y = 2a\sin^2\theta$ ∴ $dy = 4a\sin\theta\cos\theta\,d\theta$

$$= \frac{2}{a}\int_0^{\pi/2} 2a\sin^2\theta\sqrt{(3a^2 - 2a\sin^2\theta)}\,4a\sin\theta\cos\theta\,d\theta$$

$$= 16a^2\int_0^{\pi/2} \sin^3\theta\cos^2\theta\,d\theta = 16a^2\,\frac{2\cdot 1\cdot 1}{5\cdot 3\cdot 1} = \frac{32}{15}a^2.$$

Example 8.21. *Find the area enclosed between the curve of the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ and its base.* (U.P.T.U., 2005)

Solution. To describe the first arch, θ varies from 0 to 2π i.e., x varies from 0 to $2a\pi$ (Fig. 8.8).

$\therefore$ Required area $= \int_{x=0}^{2a\pi} y\,dx$

where $y = a(1 - \cos\theta)$, $dx = a(1 - \cos\theta)d\theta$.

Fig. 8.8

$$= \int_{\theta=0}^{2\pi} a(1 - \cos\theta) \cdot a(1 - \cos\theta)\,d\theta$$

$$= 2a^2 \int_0^{\pi} (1 - \cos\theta)^2\,d\theta = 4a^2 \int_0^{\pi} \sin^4\frac{\theta}{2}\,d\theta$$

$$= 32a^2 \int_0^{\pi/2} \sin^4\phi\,d\phi, \text{ putting } \theta/2 = \phi \text{ so that } d\theta = 2d\phi$$

$$= 16a^2 \cdot \frac{3 \cdot 1}{4 \cdot 2} \cdot \frac{\pi}{2} = 3\pi a^2.$$

Example 8.22. *Find the area of the loop in the curve of the parabola $x^2 = 4y$ by the line $x - 2y + 4 = 0$.*

Solution. Given parabola is $x^2 = 4y$ ...(i)

and the straight line is $x - 2y + 4 = 0$ ...(ii)

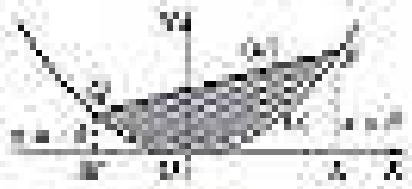

Fig. 8.9

Substituting the value of y from (ii) in (i), we get

$$x^2 = 4x + 8 \text{ or } x^2 - 4x - 32 = 0$$

or $(x - 8)(x + 4) = 0 \quad \therefore \quad x = 8, -4.$

Thus (i) and (ii) intersect at P and Q where $x = 8$ and $x = -4$ (Fig. 8.9)

$\therefore$ Required area PNQMP (i.e., dotted area) = area bounded by straight line (ii) and x-axis from $x = -4$ to $x = 8$ – area bounded by parabola (i) and x-axis from $x = -4$ to $x = 8$.

$$= \int_{-4}^{8} y\,dx, \text{ from (ii)} - \int_{-4}^{8} y\,dx, \text{ from (i)}$$

$$= \int_{-4}^{8} \frac{x+4}{2}\,dx - \int_{-4}^{8} \frac{x^2}{4}\,dx = \frac{1}{2}\left[\frac{x^2}{2} + 4x\right]_{-4}^{8} - \frac{1}{4}\left[\frac{x^3}{3}\right]_{-4}^{8}$$

$$= \frac{1}{2}[32 + 64 - (-8)] - \frac{1}{12}[512 + 64] = 36.$$

Example 8.23. *Find the area common to the parabola $y^2 = 4x$ and the circle $x^2 + y^2 = 12x$.*

Solution. Given parabola is $y^2 = 4x$ ...(i)

and the circle is $x^2 + y^2 = 12x$ ...(ii)

Both these curves are symmetrical about x-axis. Solving (i) and (ii), i.e., we have

$$x^2 + 4x = 12x \text{ or } x(x - 8) = 0$$

or $x = 0, 8.$

Thus the two curves intersect at the points where $x = 0$ and $x = 8$. (Fig. 8.10).

Also (i) meets the x-axis at $A(4a, 0)$.

Area common to (i) and (ii) i.e., the shaded area

$$= 2[\text{Area OAP} + \text{Area APQ}] \qquad \text{(By symmetry)}$$

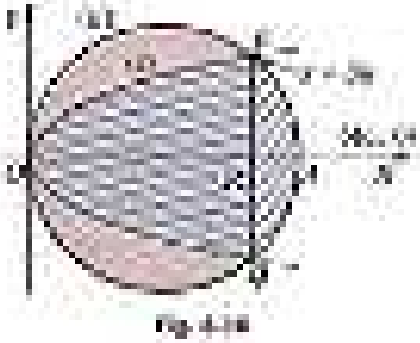

Fig. 8.10

$$= 2\left[\int_0^{4a} y\,dx, \text{ from (i)} + \int_{4a}^{8a} y\,dx, \text{ from (ii)}\right]$$

$$= 2\left[\int_0^{4a} \sqrt{4ax}\,dx + \int_{4a}^{8a} \sqrt{(8ax - x^2)}\,dx\right]$$

$$= 2\sqrt{2}\left[\frac{a^{3/2}}{2/3}\right]_0^a + 2\int_a^{2a}\sqrt{(8ax - 2x - 2ax^2)}\,dx$$

$$= \frac{4\sqrt{2}}{3}2a^{3/2} + 2\left[\frac{1}{2}(x-2a)\sqrt{(8ax-4x-2ax^2)} + \frac{4a^2}{2}\sin^{-1}\frac{x-2a}{2a}\right]_a^{2a}$$

$$= 4\sqrt{2}a^2 + 2\left[0 - \frac{1}{2}(-a)\sqrt{2a^2} + 2a^2\sin^{-1}(0) - \sin^{-1}(-1)\right]$$

$$= 4\sqrt{2}a^2 - \sqrt{2}a^2 + \frac{\pi}{2}a^2 = \left(2\sqrt{2} + \frac{\pi}{2}\right)a^2.$$

Problems 83 to 85

1. (a) Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (Bangalore Univ., 2005)

   (b) Find the area bounded by the parabola $y^2 = 4ax$ and its latus rectum.
2. Find the area bounded by the curve $y = x^2 + 2x + 5$, the x-axis and the ordinates $x = 0$ and $x = 2$.
3. Find the area included between the curve $xy^2 = a^2(a - x)$, the y-axis and the ordinate $x = a$.
4. Find the area of the loop of the curve:

   (a) $3ay^2 = x(x - a)^2$ (Bangalore, 2000) (b) $x(x^2 + y^2) = a(x^2 - y^2)$ (P.T.U., 2010)
5. Find the whole area of the curve

   (a) $a^2y^2 = x^2(2a - x)$ (Nagpur, 2000) (b) $(a^2 - x^2)y^2 = x^2$ (P.T.U., 2004)
6. Find the area included between the curve and its asymptote in each case:

   (a) $y^2 = x^2(x + a)$ (U.P.T.U., 2002) (b) $x^2y^2 = a^2(y^2 - a^2)$ (U.P.T.U., 2003)
7. Show that the area of the loop of the curve $y^2(a + x) = x^2(3a - x)$ is equal to the area between the curve and its asymptote.
8. Find the whole area of the ellipsoid $x^{2/3} + y^{2/3} = a^{2/3}$ (a) by integration with respect to x and (b) with respect to y. (U.P.T.U., 2005)
9. Find the area bounded by the curves $xy^2 = 4(2 - x)$ and the coordinate axes.
10. Find the area enclosed between the parabola $4y = 3x^2$ and the straight line $3x - 2y + 12 = 0$. Find also the ratio in which the line $y = 3$ divides the area between the curve and the line. (Bangalore Univ., 1999)
11. Find the area common to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 4x$.
12. Prove that the area common to the parabolas $y^2 = 4x$ and $x^2 = 4y$ is $\frac{16}{3}$. (U.P.T.U., Computer Science, 2004)
13. Find the area of the ellipse between the circle $x^2 + y^2 = 5a^2$ and the parabola $y^2 = 4x$.
14. Find the area bounded by the parabola $y^2 = 4ax$ and the curve $xy = 2a^2$.
15. Find the area of the segment cut off from the parabola $y = 4x - x^2$ by the straight line $y = x$.

   (U.P.T.U., 2010; P.T.U., 2004)

(3) Areas of polar curves. *Area bounded by the curve $r = f(\theta)$ and the radii vectors $\theta = \alpha$, $\theta = \beta$ is $\frac{1}{2}\int_\alpha^\beta r^2\,d\theta$.*

Let AB be the curve $r = f(\theta)$ between the radii vectors OA ($\theta = \alpha$) and OB ($\theta = \beta$). Let $P(r, \theta)$, $P'(r + \delta r, \theta + \delta\theta)$ be any two neighbouring points on the curve. (Fig. 6.11)

Let the area $OAP = A$ which is a function of θ. Then the area $OAP' = A + \delta A$. Mark circular arcs PQ and $P'Q'$ with centre O and radii OP and OP'.

Evidently area OPP' lies between the sectors OPQ and $OP'Q'$ i.e., δA lies between $\frac{1}{2}r^2\,\delta\theta$ and $\frac{1}{2}(r + \delta r)^2\,\delta\theta$.

$$\therefore \frac{\delta A}{\delta\theta} \text{ lies between } \frac{1}{2}r^2 \text{ and } \frac{1}{2}(r + \delta r)^2$$

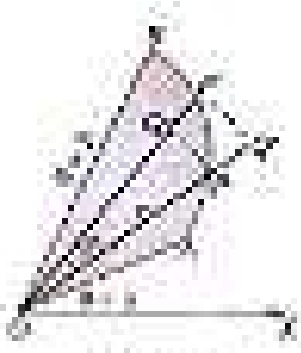

Fig. 6.11

Now taking limits as $\delta\theta \to 0$ i.e. $\delta r \to 0$, $\frac{dA}{d\theta} = \frac{1}{2}r^2$

Integrating both sides from $\theta = \alpha$ to $\theta = \beta$, we get $\int_\alpha^\beta dA = \int_\alpha^\beta \frac{1}{2}r^2\,d\theta$

i.e. Value of A for $\theta = \beta$ − Value of A for $\theta = \alpha$ $= \frac{1}{2}\int_\alpha^\beta r^2\,d\theta$

Hence the required area $OAB = \frac{1}{2}\int_\alpha^\beta r^2\,d\theta$

Example 6.20. *Find the area of the cardioid $r = a(1 + \cos\theta)$.* (V.T.U., 2002)

Solution. The curve is as shown in Fig. 6.12. Its upper half is traced from $\theta = 0$ to $\theta = \pi$.

$\therefore$ Area of the curve $= 2\cdot\frac{1}{2}\int_0^\pi r^2\,d\theta = a^2\int_0^\pi (1+\cos\theta)^2\,d\theta$

$= a^2\int_0^\pi 4\cos^2\theta/2\cdot d\theta = 4a^2\int_0^\pi \cos^4\theta/2\,d\theta$

$= 8a^2\int_0^{\pi/2}\cos^4\phi\,d\phi$, putting $\theta/2 = \phi$ and $d\theta = 2d\phi$

$= 8a^2\cdot\frac{3\cdot1}{4\cdot2}\cdot\frac{\pi}{2} = \frac{3\pi a^2}{2}$.

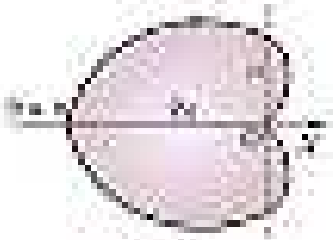

Fig. 6.12

Example 6.21. *Find the area of a loop of the curve $r = a\sin 3\theta$.*

Solution. The curve is as shown in Fig. 6.13. It consists of three loops.

Putting $r = 0$, $a\sin 3\theta = 0$ $\therefore$ $3\theta = 0$ or π i.e., $\theta = 0$ or $\pi/3$ which are the limits for the first loop.

$\therefore$ Area of a loop $= \frac{1}{2}\int_0^{\pi/3} r^2\,d\theta = \frac{1}{2}a^2\int_0^{\pi/3}\sin^2 3\theta\,d\theta = \frac{a^2}{4}\int_0^{\pi/3}(1-\cos 6\theta)\,d\theta$

$= \frac{a^2}{4}\left[\theta - \frac{\sin 6\theta}{6}\right]_0^{\pi/3} = \frac{a^2}{4}\left[\frac{\pi}{3} - 0\right] = \frac{\pi a^2}{12}$.

Obs. The limits of integration for a loop (1) can also be found by finding the two consecutive values of θ for which $r = 0$.

Example 6.22. *Prove that the area of a loop of the curve $x^3 + y^3 = 3axy$ is $\frac{3a^2}{2}$.*

Solution. Changing to polar form by putting $x = r\cos\theta, y = r\sin\theta$, $r = \frac{3a\sin\theta\cos\theta}{\cos^3\theta + \sin^3\theta}$

Putting $r = 0$, $\sin\theta\cos\theta = 0$

$\therefore$ $\theta = 0, \pi/2$, which are the limits of integration for the loop.

$\therefore$ Area of the loop

$= \frac{1}{2}\int_0^{\pi/2} r^2\,d\theta = \frac{1}{2}\int_0^{\pi/2}\frac{9a^2\sin^2\theta\cos^2\theta}{\cos^6\theta + \sin^6\theta}\,d\theta$

$= \frac{9a^2}{2}\int_0^{\pi/2}\frac{\tan^2\theta\sec^2\theta}{(1+\tan^3\theta)^2}\,d\theta$ (Dividing num. and denom. by $\cos^6\theta$)

$= \frac{3a^2}{2}\cdot\int_1^\infty\frac{dz}{z^2}$, putting $1 + \tan^3\theta = t$ and $3\tan^2\theta\sec^2\theta\,d\theta = dt$

$= \frac{3a^2}{2}\left[\frac{t^{-1}}{-1}\right]_1^\infty = \frac{3a^2}{2}(-0+1) = \frac{3a^2}{2}$.

Example 6.23. *Find the area common to the circle $r = a$ and the cardioid $r = a(1 + \cos\theta)$.*

Solution. The equations of the circles are $r = a\sqrt{2}$ ...(1) and $r = 2a\cos\theta$ ...(2)

(1) represents a circle with centre at O, 0 and radius $a\sqrt{2}$. (2) represents a circle symmetrical about OX, with centre at $(a, 0)$ and radius a.

The circles are shown in Fig. 6.13. At their point of intersection P, we have (eliminating r from (1) and (2)),

$$a\sqrt{2} = 2a\cos\theta \text{ i.e., } \cos\theta = 1/\sqrt{2}$$

or $\theta = \pm\pi/4$

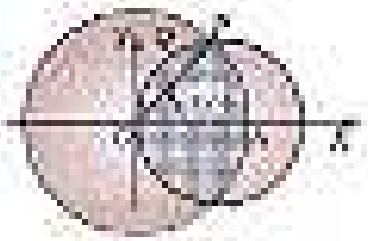

Fig. 6.13

$\therefore$ Required area = 2 × area OAPQ (By symmetry)

$$= 2\left[\frac{1}{2}\int_0^{\pi/4} r^2\,d\theta \text{ for (2)} + \frac{1}{2}\int_{\pi/4}^{\pi/2} r^2\,d\theta \text{ for (1)}\right]$$

$$= \int_0^{\pi/4} (2a\cos\theta)^2\,d\theta + \int_{\pi/4}^{\pi/2} (a\sqrt{2})^2\,d\theta = 2a^2\left[\theta\right]_0^{\pi/4} + 4a^2\int_{\pi/4}^{\pi/2}\frac{1+\cos 2\theta}{2}\,d\theta$$

$$= 2a^2\left[\pi/4 - 0\right] + 2a^2\left[\theta + \frac{\sin 2\theta}{2}\right]_{\pi/4}^{\pi/2} = \frac{\pi a^2}{2} + 2a^2\left[\frac{\pi}{2} - \frac{\pi}{4} - \frac{1}{2}\right] = a^2(\pi - 1)$$

Example 6.20. Find the area included by the cardioids $r = a(1 + \cos\theta)$ and $r = a(1 - \cos\theta)$.

(Kurukshetra, 2009; U.P.T.U., 2006)

Solution. The cardioid $r = a(1 + \cos\theta)$ is AHCOBA and the cardioid $r = a(1 - \cos\theta)$ is OCAA'B'O.

Both the cardioids are symmetrical about the initial line OX and intersect at A and A' (Fig. 6.14).

$\therefore$ Required area (shaded) = 2 area OCAO

= 2 [area OCAO + area AHOA]

$$= 2\left[\left\{\int_0^{\pi/2}\frac{1}{2}r^2\,d\theta\right\}_{r=a(1-\cos\theta)} + \left\{\int_{\pi/2}^{\pi}\frac{1}{2}r^2\,d\theta\right\}_{r=a(1+\cos\theta)}\right]$$

$$= a^2\int_0^{\pi/2}(1-\cos\theta)^2\,d\theta + a^2\int_{\pi/2}^{\pi}(1+\cos\theta)^2\,d\theta$$

$$= a^2\left[\int_0^{\pi/2}(1 - 2\cos\theta + \cos^2\theta)\,d\theta + \int_{\pi/2}^{\pi}(1 + 2\cos\theta + \cos^2\theta)\,d\theta\right]$$

$$= a^2\left[\int_0^{\pi}(1+\cos^2\theta)\,d\theta - 2\int_0^{\pi/2}\cos\theta\,d\theta + 2\int_{\pi/2}^{\pi}\cos\theta\,d\theta\right]$$

$$= a^2\left[\int_0^{\pi}\left(1 + \frac{1+\cos 2\theta}{2}\right)d\theta - 2\left[\sin\theta\right]_0^{\pi/2} + 2\left[\sin\theta\right]_{\pi/2}^{\pi}\right]$$

$$= a^2\left[\left[\frac{3}{2}\theta + \frac{\sin 2\theta}{4}\right]_0^{\pi} - 2(1-0) + 2(0-1)\right] = \left(\frac{3\pi}{2} - 4\right)a^2.$$

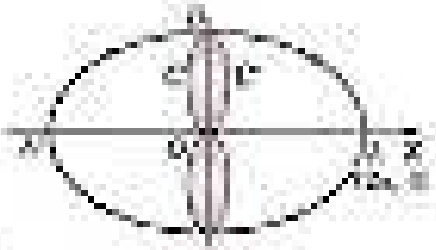

Fig. 6.14

PROBLEMS 6.7

1. Find the whole area of
   (i) the cardioid $r = a(1 + \cos\theta)$ (U.P.T.U., 2004) (ii) the curve $r = a^2\cos 2\theta$. (U.P.T.U., 2006)
2. Find the area above (x-axis) the curve
   (i) $r = a\sin 2\theta$. (ii) $r = a\sin 3\theta$.
3. Show that the area included between the circle $x^2 + y^2 = 4ay$ and the parabola $x^2 = 4ay$ is equal to $\frac{8}{3}$ times of $16a^2$.
4. Prove that the area of the loop of the curve $x^3 + y^3 = 3axy$ is three times the area of the loop of the curve $r^2 = a^2\cos 2\theta$.
5. Find the area inside the cardioid $r = a(1 + \cos\theta)$ and lying outside the circle $r = a(1 - \cos\theta)$. (Ranchi, 2005)
6. Find the area outside the circle $r = 2a\cos\theta$ and inside the cardioid $r = a(1 + \cos\theta)$. (Kurukshetra, 2006)

833 LENGTHS OF CURVES

(1) The length of the arc of the curve $y = f(x)$ between the points where $x = a$ and $x = b$ is

$$\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

Let AB be the curve $y = f(x)$ between the points A and B where $x = a$ and $x = b$ (Fig. 8.35)

Let $P(x, y)$ be any point on the curve and let AP's arc length s be a function of x.

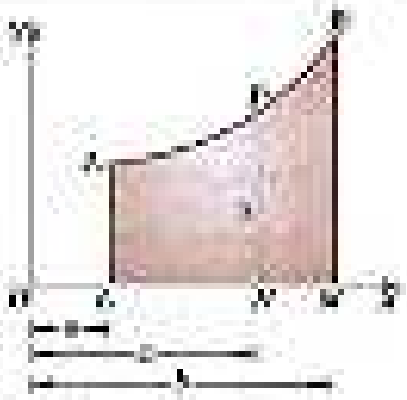

Fig. 8.35

Then $$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$ [(1) of p. 804]

$$\therefore \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx = \int_a^b \frac{ds}{dx}\, dx = [s]_a^b$$

= (value of s for $x = b$) − (value of s for $x = a$) = arc AB − 0

Hence, the arc $AB = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

(2) The length of the arc of the curve $x = f(y)$ between the points where $y = a$ and $y = b$, is

$$\int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$ [Use (2) of p. 805]

(3) The length of the arc of the curve $x = f(t)$, $y = \phi(t)$ between the points where $t = a$ and $t = b$, is

$$\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$ [Use (3) p. 806]

(4) The length of the arc of the curve $r = f(\theta)$ between the points where $\theta = \alpha$ and $\theta = \beta$, is

$$\int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$ [Use (1) of p. 808]

(5) The length of the arc of the curve $\theta = f(r)$ between the points where $r = a$ and $r = b$, is

$$\int_a^b \sqrt{1 + \left(r\frac{d\theta}{dr}\right)^2}\, dr$$ [Use (2) of p. 808]

Example (8.31): *Find the length of the arc of the parabola $x^2 = 4ay$ measured from the vertex to one extremity of the latus rectum.* [Delhi, 2002]

Solution: Let A be the vertex and L one extremity of the latus rectum so that at A, $x = 0$ and at L, $x = 2a$ (Fig. 8.36)

Now $$y = x^2/4a \text{ so that } \frac{dy}{dx} = \frac{1}{4a} \cdot 2x = \frac{x}{2a}$$

$$\therefore \text{ arc } AL = \int_0^{2a} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

$$= \int_0^{2a} \sqrt{1 + \left(\frac{x}{2a}\right)^2}\, dx = \frac{1}{2a}\int_0^{2a} \sqrt{(2a)^2 + x^2}\, dx$$

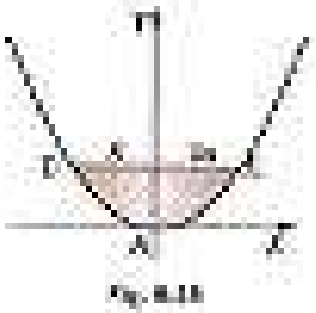

Fig. 8.36

$$= \frac{1}{2a}\left[\frac{x\sqrt{(2ax)^2 + x^2}}{2} + \frac{(2a)^2}{2}\sinh^{-1}\frac{x}{2a}\right]_0^{2a} = \frac{1}{2a}\left[\frac{2a\sqrt{(2a)^2}}{2} + 2a^2\sinh^{-1}1\right]$$

$$= a[\sqrt{2} + \sinh^{-1}1] = a[\sqrt{2} + \log(1 + \sqrt{2})] \quad [\because \sinh^{-1}x = \log(x + \sqrt{(1+x^2)})]$$

Example 6.32. *Find the perimeter of the loop of the curve $3ay^2 = x(a - x)^2$.*

Solution. The curve is symmetrical about the x-axis and the loop lies between the limits $x = 0$ and $x = a$. [Fig. 6.17]

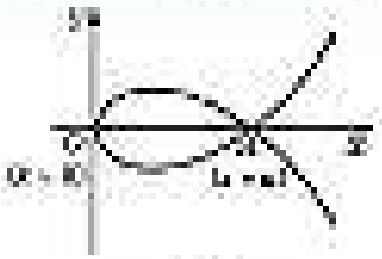

Fig. 6.17

We have $y = \frac{\sqrt{x}(a-x)}{\sqrt{3a}}$

$$\therefore \quad \frac{dy}{dx} = \frac{1}{\sqrt{3a}}\left[\frac{1}{2}x^{-1/2}(a-x) - x^{1/2}\right] = \frac{1}{2\sqrt{3a}}\frac{3x-a}{\sqrt{x}}$$

$\therefore$ Perimeter of the loop $= 2\int_0^a \sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}\,dx$ (By symmetry)

$$= 2\int_0^a \sqrt{\left[1 + \frac{9x^2 - 6ax + a^2}{12ax}\right]}\,dx = 2\int_0^a \frac{\sqrt{(9x^2 + 6ax + a^2)}}{\sqrt{(12ax)}}\,dx$$

$$= \frac{1}{\sqrt{12a}}\int_0^a \frac{3x+a}{\sqrt{x}}\,dx = \frac{1}{\sqrt{12a}}\int_0^a (3x^{1/2} + ax^{-1/2})\,dx$$

$$= \frac{1}{\sqrt{12a}}\left[\frac{3x^{3/2}}{3/2} + a\frac{x^{1/2}}{1/2}\right]_0^a = \frac{1}{\sqrt{12a}}(6a^{3/2}) = \frac{4a}{\sqrt{3}}$$

Example 6.33. *Find the length of one arch of the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$.* (U.P.T.U. 2006; V.T.U. 2006)

Solution. As a point moves from one end O to the other end of an arch, the parameter θ increases from 0 to 2π. [see Fig. 6.18]

Also $\frac{dx}{d\theta} = a(1 - \cos\theta)$, $\frac{dy}{d\theta} = a\sin\theta$.

$\therefore$ Length of an arch $= \int_0^{2\pi} \sqrt{\left[\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2\right]}\,d\theta$

$$= \int_0^{2\pi} \sqrt{[a^2(1 - \cos\theta)^2 + a^2\sin^2\theta]}\,d\theta = a\int_0^{2\pi} \sqrt{[2(1 - \cos\theta)]}\,d\theta$$

$$= 2a\int_0^{2\pi} \sin\theta/2\,d\theta = 2a\left[-\frac{\cos\theta/2}{1/2}\right]_0^{2\pi} = 4a[(-\cos\pi) - (-\cos 0)] = 8a.$$

Example 6.34. *Find the entire length of the cardioid $r = a(1 + \cos\theta)$.*
(U.P.T.U. 2004; Bhopal, 2005; Kurukshetra, 2005)
Also, prove that the upper half is traversed for $0 \le \theta \le \pi$. (Kerala, 2005)

Solution. The cardioid is symmetrical about the initial line and for its upper half, θ increases from 0 to π. [Fig. 6.19]

Also $\frac{dr}{d\theta} = -a\sin\theta$.

$\therefore$ Length of the curve $= 2\int_0^{\pi} \sqrt{\left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]}\,d\theta$

$$= 2\int_0^{\pi} \sqrt{(2a(1+\cos\theta))^2 + (-a\sin\theta)^2}\, d\theta = 2a\int_0^{\pi} \sqrt{2(1+\cos\theta)}\, d\theta$$

$$= 8a\int_0^{\pi} \cos\frac{\theta}{2}\, d\theta = 4a\left|\frac{\sin\theta/2}{1/2}\right|_0^{\pi} = 8a(\sin\pi/2 - \sin 0) = 8a.$$

... Length of upper half of the curve is $4a$. Also length of the arc AP from $\theta = 0$ to $\pi/3$,

$$= a\int_0^{\pi/3} \sqrt{2(1+\cos\theta)}\, d\theta = 2a\int_0^{\pi/3} \cos\theta/2\, \cdot d\theta$$

$$= 4a\left|\sin\theta/2\right|_0^{\pi/3} = 2a = \text{half the length of upper half of the cardioid.}$$

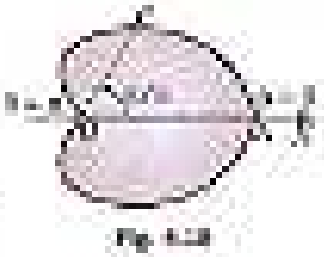
Fig. 5.28

EXERCISE 5.1

1. Find the length of the arc of the semi-cubical parabola $ay^2 = x^3$ from the vertex to the ordinate $x = 4a$.
2. Find the length of the curve (i) $y = \log\sec x$ from $x = 0$ to $x = \pi/3$ (V.T.U. 2005) (ii) $2y = \log(1+x^2) - 2x + \tan^{-1}x$ from $x = 0$ to $x = 2$. (V.T.U. 2005)
3. Find the length of the arc of the parabola $y^2 = 4ax$ from the vertex to one end of the latus rectum.
   (i) Cut off by the line $3y = 4x$. (V.T.U. 2006) (ii) (Mumbai, 2006)
4. Find the perimeter of the loop of the following curves.
   (i) $ay^2 = x^2(a-x)$ (ii) $9y^2 = x(x-3)^2$.
5. Find the length of the curve $y^2 = 4(x+1)^3$ cut off by the line $x = 4$. (V.T.U. 2006)
6. Show that the whole length of the curve $a^2x^2 = y^2(a^2 - y^2)$ is πa.
7. (a) Find the length of one arch of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$.
   (b) In finding the length of the curve show that the curve $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ is divided in the ratio $1:3$ at $\theta = 2\pi/3$. (V.T.U. 2006)
8. Find the whole length of the curve $ax^2 = y(x^2 + y^2)$ (i) (Mumbai, 2006) (ii) (V.T.U., 2010; Kurukshetra, 2006; Rajasthan, 2006)
   Also show that the line $x = a/2$ divides the length of the curve in the first quadrant in the ratio $2:1$. (Mumbai, 2007)
9. Find the length of the loop of the curve $3ay^2 = x(x-a)^2$. (Mumbai, 2007)
10. For the curve $x = a(\cos\theta + \log\tan\theta/2)$, $y = a\sin\theta$ prove that the length intercepted between the origin and any point is equal to the corresponding arc of the circle $x^2 + y^2 = a^2$.
11. Find the length of the curve $x = \frac{1}{2}\left(y + \frac{1}{y}\right)$ from $y = 1$ to $y = 3$. (Kurukshetra, 2006)
12. Find the perimeter of the cardioid $r = a(1 - \cos\theta)$. Also show that the upper half of the curve is bisected by the line $\theta = \pi/3$.
13. Find the whole length of the loop of the curve $3y^2 = x(x-1)^2$.
14. Find the length of the parabola $x = t^2$, $y = 2t$ from the vertex to the point where it is cut by the line $x + y = 4$. (V.T.U. 2006)

5.13 (I) VOLUMES OF REVOLUTION

(a) Revolution about x-axis. *The volume of the solid generated by the revolution about the x-axis, of the area bounded by the curve $y = f(x)$, the x-axis and the ordinates $x = a$, $x = b$ is*

$$\int_a^b \pi y^2\, dx.$$

Let AB be the curve $y = f(x)$ between the ordinates $AA' = a$ and $BB' = b$.

Let $P(x, y)$, $P'(x + \delta x, y + \delta y)$ be two neighbouring points on the curve and M, M' be their respective ordinates (Fig. 5.29).

Let the volume of the solid generated by the revolution about x-axis of the area $AA'PP$ be V, which is clearly a function of x. Then the volume of the solid generated by the revolution of the area $AA'P'P$ is V. Draw PR and $P'Q$.

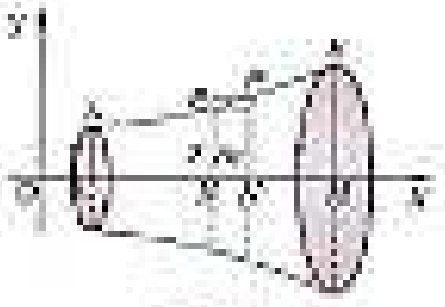
Fig. 5.29

The δV lies between the volumes of the right circular cylinders generated by the revolution of rectangles PN' and PN,

i.e., δV lies between $\pi y^2\,\delta x$ and $\pi(y+\delta y)^2\,\delta x$.

$\therefore$ $\dfrac{\delta V}{\delta x}$ lies between πy^2 and $\pi(y+\delta y)^2$.

Now taking limits as $P' \to P$, i.e., $\delta x \to 0$ and $\therefore$ $\delta y \to 0$, $\dfrac{dV}{dx} = \pi y^2$

$\therefore$ $\displaystyle\int_a^b \frac{dV}{dx}\,dx = \int_a^b \pi y^2\,dx$ or $[V]_a^b = \int_a^b \pi y^2\,dx$

or (value of V for $x = b$) $-$ (value of V for $x = a$)

i.e., volume of the solid obtained by the revolution of the area $ABMA = \displaystyle\int_a^b \pi y^2\,dx$.

Example 8.36. *Find the volume of a sphere of radius a.* (U.P.T.U. 2003)

Solution. Let the sphere be generated by the revolution of the semi-circle ABC, of radius a about its diameter OA (Fig. 8.30).

Taking OA as the x-axis and its mid-point O as the origin, the equation of the circle ABC is $x^2 + y^2 = a^2$.

$\therefore$ Volume of the sphere = 2 (volume of the solid generated by the revolution about x-axis of the quadrant OAB)

$$= 2\int_0^a \pi y^2\,dx = 2\pi\int_0^a (a^2 - x^2)\,dx$$

$$= 2\pi\left[a^2 x - \frac{x^3}{3}\right]_0^a = 2\pi\left[a^3 - \frac{a^3}{3} - (0 - 0)\right] = \frac{4}{3}\pi a^3.$$

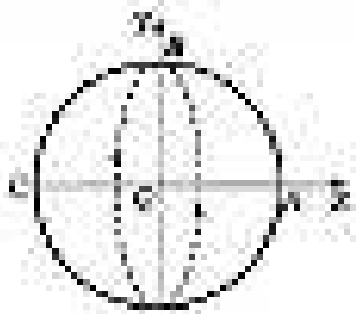

Fig. 8.30

Example 8.37. *Find the volume formed by the revolution of loop of the curve $a^2y^2 = x^2(a^2 - x^2)$ about the x-axis.* (Rohtak U., July 2004)

Solution. The curve is symmetrical about the x-axis, and for the upper half of its loop x varies from 0 to a (Fig. 8.31).

$\therefore$ Volume of the loop $= \displaystyle\int_0^a \pi y^2\,dx = \pi\int_0^a \frac{x^2(a^2 - x^2)}{a^2}\,dx$

$$= \pi\int_0^a \frac{-x^4 + a^2x^2}{a^2}\,dx$$

[Divide the numerator by the denominator]

$$= \pi\int_0^a \left[-x^2 + a^2 - \frac{a^4}{x+a}\right]dx = \pi\left[-\frac{x^3}{3} + a^2x - \frac{a^4}{2} - 4a^3\log(x+a)\right]_0^a$$

$$= \pi\left[\frac{-2a^3}{3} + 2a^3 - 4a^3 - 2a + 4a^3\log 4a - 4a^3\log a\right]$$

$$= \pi a^3\left[-\frac{3}{3} + 4\log 4\right] = \pi a^3(8\log 2 - 2).$$

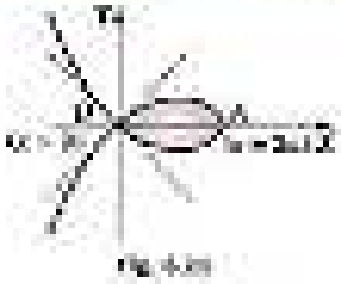

Fig. 8.31

Example 8.38. *Prove that the volume of the solid formed by the revolution of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ about the tangent at the vertex is π^2a^3.* (U.P.T.U. 2003)

Solution. The arch OAB of the cycloid is symmetrical about the y-axis and the tangent at the vertex is the x-axis. For half the cycloid OA, θ varies from 0 to π (Fig. 8.32).

Hence the required volume

$$= 2\int_{x=0}^{x=\pi a} \pi y^2\,dx = 2\pi\int_0^{\pi} a^2(1 - \cos\theta)^2\,a(1 + \cos\theta)\,d\theta$$

$$= 2\pi a^3 \int_0^{2a} (2 - 4a^2 + 6a^2 - 2a^3)\, da$$

$$= 16\pi a^3 \int_0^{2a} (a - a^2 + a^2 - a^3)\, da \qquad [\because b = 2a, \therefore a = 2a]$$

$$= 32\pi a^3 \int_0^{\pi/2} \sin^2\theta \cos^2\theta \, d\theta = 32\pi a^3 \cdot \frac{3.1.1}{8.4.2} \cdot \frac{\pi}{2} = \pi^2 a^3.$$

Example 6.30. *Find the volume of the solid formed by revolving about the x-axis, the area enclosed by the parabola $y^2 = 4ax$ and the ordinate $x = 2a$. Also find the volume of the solid formed by revolving about the same axis the area enclosed by the curve $2ay^2 = x(x - a)^2$ and the ordinate $x = a$.*

Solution. The curve $2ay^2 = x(x - a)^2$ ...(1)

is symmetrical about x-axis and is a semi-cubical parabola with the vertex at $A(a, 0)$. The parabola $y^2 = 4ax$ and (1) intersect at B and C where $2a \cdot 4ax = x(x - a)^2$ or $x^3 - 6ax^2 - 3a^2x - 8a^3 = 0$ which gives $x = -a, -a, 8a$. Since x is not negative, therefore we have $x = 8a$ (Fig. 6.32).

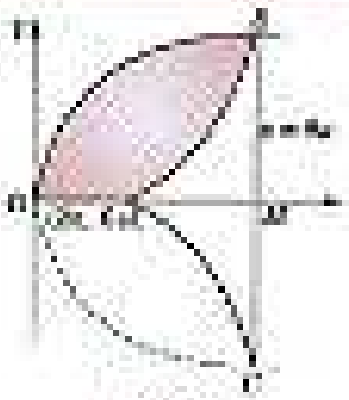

Fig. 6.32

$\therefore$ Required volume = Volume obtained by revolving the shaded area OAB about x-axis = Vol. obtained by revolving area OABO − Vol. obtained by revolving area ADBA

$$= \int_0^{8a} y^2\,(= 4ax)\, dx - \int_a^{8a} y^2\,(\text{for ADB})\, dx$$

$$= 4a \left| \frac{x^2}{2} \right|_0^{8a} - \frac{4a}{2.3a} \left| (x - 3a)^3 \right|_a^{8a}$$

$$= 128a^3 - \frac{4a}{3a} \left| \frac{(x - 3a)^4}{4} \right|_a^{8a}$$

$$= 128a^3 - \frac{8}{3}(5a)^4 = 64a^3.$$

(ii) **Revolution about the y-axis.** In developing x and y in the above formula, we see that *the volume of the solid generated by the revolution about y-axis, of the area bounded by the curve $y = f(x)$, the y-axis and the abscissae $y = a$, $y = b$ is*

$$\int_a^b \pi x^2\, dy.$$

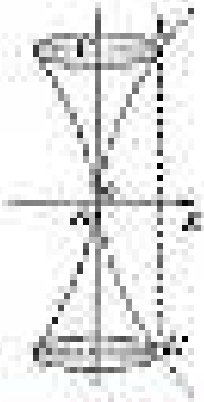

Fig. 6.33

Example 6.31. *Find the volume of the cup shaped solid formed by the revolution about the y-axis of the part of the parabola $y^2 = 4ax$ cut off by the latus rectum.* (Madras, 2003)

Solution. Given parabola is $y^2 = 4ax$.

Let A be the vertex and L one extremity of the latus rectum. For the arc AL, y varies from 0 to $2a$ (Fig. 6.34).

$\therefore$ required volume = Volume generated by the revolution about the y-axis of the area AOL

$$= \pi \int_0^{2a} x^2\, dy = 2\pi \int_0^{2a} \frac{y^4}{16a^2}\, dy = \frac{\pi}{8a^2} \left| \frac{y^5}{5} \right|_0^{2a} = \frac{\pi}{40a^2}(32a^5 - 0) = \frac{4\pi a^3}{5}.$$

(i) **Revolution about any axis.** *The volume of the solid generated by the revolution about any axis OM of the area bounded by the curve AB, the ends AN and BM and the perpendiculars AL, BM on the axis, is*

$$\int_{OL}^{OM} \pi (PN)^2\, d(ON),$$

where O is a fixed point in OM and PN is perpendicular from any point P of the curve AB on OM.

With O as origin, take OLM as the x-axis and OY perpendicular to it as the y-axis (Fig. 6.34).

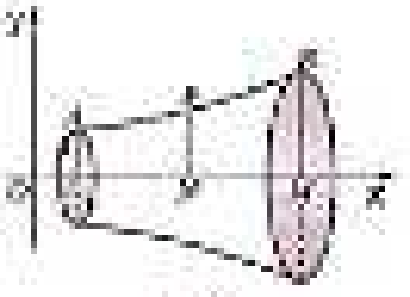

Fig. 6.34

Let the coordinates of P be (x, y) so that $x = OV$, $y = VP$.

If $OL = a$, $OM = b$, then required volume $= \int_b^a \pi y^2\,dx = \int_{OL}^{OM} \pi f(x)^2\,d(x)$.

Example 6.40. Find the volume of the solid formed by rotating the cissoid $y^2(2a - x) = x^3$ about its asymptote. (U.P.T.U., 2009)

Solution. Given curve is $y = \dfrac{x^{3/2}}{\sqrt{2a - x}}$ ...(1)

It is symmetrical about x-axis and the asymptote is $x = 2a$. (See Fig. 6.20). If $P(x, y)$ be any point on it and PN is perpendicular on the asymptote AN then $PN = 2a - x$ and

$$AN = y = \frac{x^{3/2}}{\sqrt{(2a - x)}} \qquad \text{[From (1)}$$

$$\therefore \quad d(AN) = dy = \frac{\sqrt{(2a - x)}\cdot\tfrac{3}{2}x^{1/2} - x^{3/2}\cdot\tfrac{1}{2}(2a - x)^{-1/2}(-1)}{2a - x}\,dx$$

$$= \frac{3\sqrt{(2a)}x - x^{3/2} + x^{3/2}}{2(2a - x)^{3/2}}\,dx = \frac{3ax^{1/2} - x^{3/2}}{(2a - x)^{3/2}}\,dx$$

Required volume $= 2\int_{x=0}^{x=2a} \pi PN^2\,d(AN) = 2\pi\int_0^{2a} (2a - x)^2 \cdot \frac{3ax^{1/2} - x^{3/2}}{(2a - x)^{3/2}}\,dx$

$$= 2\pi\int_0^{2a} \sqrt{(2a - x)}(3ax^{1/2} - x^{3/2})\,dx \qquad \left[\begin{array}{l}\text{Put } x = 2a\sin^2\theta \\ \text{then } dx = 4a\sin\theta\cos\theta\,d\theta\end{array}\right]$$

$$= 2\pi\int_0^{\pi/2} \sqrt{(2a)}\cos\theta(3a \cdot 2a\sin^2\theta\cdot\sqrt{(2a)}\sin\theta - 4a^2\sin^3\theta\cos\theta)\,d\theta$$

$$= 16\pi a^3\left[3\int_0^{\pi/2}\sin^2\theta\cos^2\theta\,d\theta - 2\int_0^{\pi/2}\sin^3\theta\cos^2\theta\,d\theta\right]$$

$$= 16\pi a^3\left[3\cdot\frac{1\cdot1}{4\cdot2}\cdot\frac{\pi}{2} - 2\cdot\frac{2\cdot1\cdot1}{5\cdot4\cdot2}\right] = 2\pi^2a^3.$$

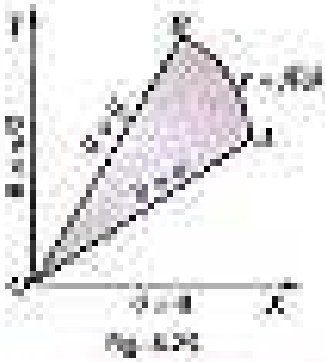

Fig. 6.20

(2) Volumes of revolution (polar curves). The volume of the solid generated by the revolution of the area bounded by the curve $r = f(\theta)$ and the radii vectors $\theta = \alpha$, $\theta = \beta$ (Fig. 6.21)

(a) about the initial line OX: $\theta = \int_\alpha^\beta \frac{2\pi}{3} r^3 \sin\theta\,d\theta$

(b) about the line OY: $\theta = \pi/2$: $= \int_\alpha^\beta \frac{2\pi}{3} r^3 \cos\theta\,d\theta$.

Example 6.41. Find the volume of the solid generated by the revolution of the cardioid $r = a(1 + \cos\theta)$ about the initial line. (U.P.T.U., 2010; Kurukshetra, 2002)

Solution. The cardioid is symmetrical about the initial line and for its upper half θ varies from 0 to π. (Fig. 6.16)

$$\text{required volume} = \int_0^\pi \frac{2\pi}{3} y^3\,dx = \frac{2\pi}{3}\int_0^\pi a^3(1 + \cos\theta)^3 \cdot \sin\theta\,d\theta$$

$$= -\frac{8\pi a^3}{3}\int_0^\pi (1 + \cos\theta)^3(-\sin\theta)\,d\theta = -\frac{2\pi a^3}{3}\left|\frac{(1 + \cos\theta)^4}{4}\right|_0^\pi = -\frac{\pi a^3}{6}(0 - 16) = \frac{8}{3}\pi a^3$$

Example 6.42. Find the volume of the solid generated by revolving the first loop of $r = a\sin 2\theta$ about the line $\theta = \pi/2$. (U.P.T.U., 2005)

Solution. The curve is symmetrical about the pole. For the upper half of the first loop, θ varies from 0 to $\pi/2$. (Fig. 6.17)

— required volume = 2(volume generated by the half loop in the first quadrant)

$$= 2\int_0^{\pi/2} \frac{2}{3}\pi r^3 \sin\theta\, d\theta = \frac{8\pi}{3}\int_0^{\pi/2} a^3(\cos 2\theta)^{3/2}\cos\theta\, d\theta \qquad [\because\ r = a\cos 2\theta]$$

$$= \frac{4\pi a^3}{3}\int_0^{\pi/2} (1-2\sin^2\theta)^{3/2}\cos\theta\, d\theta \qquad \left[\because \sqrt{2}\sin\theta = \sin\frac{\pi}{4}\right.$$
$$\left.\Rightarrow \sqrt{2}\cos\theta\, d\theta = \cos\frac{\pi}{4}\, d\theta\right]$$

$$= \frac{4\pi a^3}{3}\int_0^{\pi/4} (1-\sin^2\theta)^{3/2}\cdot\frac{1}{\sqrt{2}}\cos\frac{\pi}{4}\, d\theta = \frac{4\pi a^3}{3\sqrt{2}}\int_0^{\pi/4}\cos^4\theta\, d\theta = \frac{4\pi}{3\sqrt{2}}\cdot\frac{3.1}{4.2}\cdot\frac{\pi}{2} = \frac{\pi a^3}{4\sqrt{2}}.$$

SOLVED EXAMPLES

1. Find the volume generated by the revolution of the area bounded by $y^2 = x$, the curve $y = x$ and the ordinate $x = 1$, about the axis of x.
2. Find the volume of a spherical segment of height h cut off from a sphere of radius a.
3. Find the volume generated by revolving the portion of the parabola $y^2 = 4ax$ cut off by its latus rectum about the axis of the parabola. (UPTU, 2005)
4. Find the volume generated by revolving the area bounded by the curve $xy = 4$, $x = 1$, $x = 4$ and $y = 0$ about the x-axis.
5. Find the volume of the solid generated by revolving the ellipse $x^2 + 4y^2 = 1$
   (i) about the major axis (Punjab, 2002 C) (ii) about the minor axis (Madras, 2003)
6. Obtain the volume of the frustum of a right circular cone, whose base has radius R, upper base has radius r and altitude is h.
7. Find the volume generated by the revolution of the curve $27ay^2 = x(x-3a)^2$ about the x-axis.
8. Find the volume of the solid formed by the revolution about the x-axis of the loop of the curve
   (i) $y^2 = x^2(a-x)$ (ii) $9y^2 = x(3-x)^2$ (iii) $y^2 = x(2a-x)$.
9. Find the volume obtained by revolving one arch of the cycloid
   (i) $x = a(\theta - \sin\theta)$, $y = a(1-\cos\theta)$, about its base (Rajasthan, 2006; UPTU, 2006)
   (ii) $x = a(\theta + \sin\theta)$, $y = a(1-\cos\theta)$, about the x-axis.
10. Find the volume of the spindle-shaped solid generated by the revolution of the curve $x^{2/3} + y^{2/3} = a^{2/3}$ about the x-axis. (UPTU, 2004; U.P.T.U., 2006)
11. Find the volume of the solid formed by the revolution about the y-axis of the area enclosed by the curves $y^2 = 4x^2$ $(2-x)$ and the y-axis. (U.P.T.U., 2004)
12. Prove that the volume of the solid formed by the revolution of the curve $a^2y^2 = x^2(a^2 - x^2)$ about the x-axis is $\frac{4}{3}\pi a^3$.
13. Find the volume generated by the revolution about the x-axis of the loop of
   (i) $x = 2a\cos\theta - \cos 2\theta$, $y = 2a\sin\theta - \sin 2\theta$ (ii) $x = a(1-\cos\theta)$, $y = a\sin\theta$. (UPTU, 2005)
14. Determine the volume of the solid obtained by revolving the hypocycloid $x = a\cos^3\theta$, $y = a\sin^3\theta$ about the x-axis. (Gorakhpur, 2002)
15. Find the volume of the solid formed by revolving a loop of the lemniscate $r^2 = a^2\cos 2\theta$ about the initial line. (U.P.T.U., 2002; Delhi, 2002)

6.15 SURFACE AREAS OF REVOLUTION

(a) Revolution about x-axis. The surface area of the solid generated by the revolution about x-axis, of the arc of the curve $y = f(x)$, from $x = a$ to $x = b$, is

$$\int_{x=a}^{x=b} 2\pi y\, ds.$$

Let AB be the curve $y = f(x)$ between the ordinates AA' ($x = a$) and BB' ($x = b$). Let $P(x, y)$, $P'(x + \delta x, y + \delta y)$ be two neighbouring points on the curve and PP', PN' be their respective chords (Fig. 6.18).

Let the arc $AP = s$ so that arc $PP' = \delta s$. Let the surface area generated by the revolution about x-axis of the arc AP be S and that generated by the revolution of the arc AP' be $S + \delta S$.

Since δs is small, the surface area δS may be regarded as lying between the curved surfaces of the right cylinders of radii PN and $P'N'$ and of common thickness δs.

Then δs lies between $2\pi y\,\delta s$ and $2\pi(y+\delta y)\,\delta s$

$\therefore \quad \frac{\delta S}{\delta x}$ lies between $2\pi y\frac{\delta s}{\delta x}$ and $2\pi(y+\delta y)\frac{\delta s}{\delta x}$

Taking limits as $P' \to P$, i.e., $\delta x \to 0$ and $\delta y \to 0$, $\frac{dS}{dx} = 2\pi y\frac{ds}{dx}$

$$\therefore \quad \int_{x=a}^{x=b}\frac{dS}{dx}\,dx = \int_{x=a}^{x=b} 2\pi y\frac{ds}{dx}\,dx \quad \text{or} \quad [S]_{x=a}^{x=b} = \int_{x=a}^{x=b} 2\pi y\,ds$$

or $\quad$ (value of S when $x=b$) $-$ (value of S when $x=a$) $= \int_{x=a}^{x=b} 2\pi y\,ds$

or $\quad$ surface area generated by the revolution of the arc $AB = S = \int_{x=a}^{x=b} 2\pi y\,ds$

Hence, the required surface area $= \int_{x=a}^{x=b} 2\pi y\,ds$.

(iii) Practical forms of the formula $S = \int 2\pi y\,ds$

(a) Cartesian form [If the curve is given by $y = f(x)$]

$$S = \int 2\pi y\frac{ds}{dx}\,dx, \text{ where } \frac{ds}{dx} = \sqrt{\left[1+\left(\frac{dy}{dx}\right)^2\right]}$$

(b) Parametric form [If the curve is given by $x = f(t)$, $y = \phi(t)$]

$$S = \int 2\pi y\frac{ds}{dt}\,dt, \text{ where } \frac{ds}{dt} = \sqrt{\left[\left(\frac{dx}{dt}\right)^2+\left(\frac{dy}{dt}\right)^2\right]}$$

(c) [Polar form: If the curve is given by $r = f(\theta)$]

$$S = \int 2\pi y\frac{ds}{d\theta}\,d\theta, \text{ where } y = r\sin\theta \text{ and } \frac{ds}{d\theta} = \sqrt{\left[r^2+\left(\frac{dr}{d\theta}\right)^2\right]}$$

Example 2.43. *Find the surface of the solid formed by revolving the cardioid $r = a(1+\cos\theta)$ about the initial line.* (A.M.I.E., 2000; Rajasthan, 2000; A.M.I.E., 2005)

Solution. The cardioid is symmetrical about the initial line and for its upper half θ varies from 0 to π (Fig. 2.26).

Also $\quad \frac{ds}{d\theta} = \sqrt{\left[r^2+\left(\frac{dr}{d\theta}\right)^2\right]} = \sqrt{[a^2(1+\cos\theta)^2 + a^2\sin^2\theta]}$

$$= a\sqrt{[2(1+\cos\theta)]} = a\sqrt{[2.2\cos^2(\theta/2)]} = 2a\cos(\theta/2)$$

$$\therefore \quad \text{required surface} = \int_0^{\pi} 2\pi y\frac{ds}{d\theta}\,d\theta = 2\pi\int_0^{\pi} r\sin\theta\cdot 2a\cos(\theta/2)\,d\theta$$

$$= 4\pi a\int_0^{\pi}a(1+\cos\theta)\sin\theta\cdot\cos(\theta/2)\,d\theta = 4\pi a^2\int_0^{\pi} 2\cos^2\frac{\theta}{2}\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\cdot\cos\frac{\theta}{2}\,d\theta$$

$$= 16\pi a^2\int_0^{\pi}\cos^4\frac{\theta}{2}\sin\frac{\theta}{2}\,d\theta = 16\pi a^2\left[-\frac{\cos^5(\theta/2)}{5}\right]_0^{\pi} = -\frac{16\pi a^2}{5}(0-1) = \frac{16\pi a^2}{5}.$$

(iv) Revolution about y-axis. In the foregoing study in the above formula, we see that the surface of the solid generated by the revolution about y-axis, of the arc of the curve $x = f(y)$, from $y = a$ to $y = b$ is

$$\int_{y=a}^{y=b} 2\pi x\,ds.$$

Example 2.44. *Find the surface area of the solid generated by the revolution of the asteroid $x^{2/3}+y^{2/3} = a^{2/3}$ about the y-axis.*

Solution. The astroid is symmetrical about the x-axis, and for its portion in the first quadrant t varies from 0 to $\pi/2$. (Fig. 4.20).

Also $$\frac{dx}{dt} = -3a\cos^2 t\sin t, \quad \frac{dy}{dt} = 3a\sin^2 t\cos t$$

$$\therefore \quad \frac{ds}{dt} = \sqrt{\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2\right]} = \sqrt{[9a^2\cos^4 t\sin^2 t + 9a^2\sin^4 t\cos^2 t]}$$

$$= 3a\sin t\cos t\sqrt{(\cos^2 t + \sin^2 t)} = 3a\sin t\cos t$$

$$\therefore \quad \text{required surface} = 2\int_0^{\pi/2} 2\pi y\frac{ds}{dt}\cdot dt = 4\pi\int_0^{\pi/2} a\cos^3 t\cdot 3a\sin t\cos t\, dt$$

$$= 12\pi a^2\int_0^{\pi/2}\cos^4 t\sin t\, dt = 12\pi a^2\frac{\Gamma\frac{5}{2}\Gamma 1}{2\Gamma 3} = \frac{12\pi a^2}{5}.$$

EXERCISE 4.3

1. Find the area of the surface generated by revolving the arc of the catenary $y = c\cosh\frac{x}{c}$ from $x = 0$ to $x = c$ about the x-axis.
2. Find the area of the surface formed by the revolution of $y^2 = 4ax$ about the x-axis, by the arc from the vertex to one end of the latus rectum.
3. Find the surface of the solid generated by the revolution of the ellipse $9x^2 + y^2 = 9$ about the y-axis.
   (Kanpur, 2000; Allahabad, 2002)
4. Find the curved area and surface of the right circular cone formed by the revolution of a right-angled triangle about a side which contains the right angle.
5. Obtain the surface area of a sphere of radius a.
6. Show that the surface area of the solid generated by the revolution of the curve $x = a\cos^3\theta$, $y = a\sin^3\theta$ about the x-axis is $\frac{12\pi a^2}{5}$.
7. The arc of the curve $x^{2/3} + y^{2/3} = a^{2/3}$ in the first quadrant revolves about x-axis. Show that the area of the surface generated is $\frac{6\pi a^2}{5}$.
8. Find the surface area of the solid generated by revolving the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 + \cos\theta)$ about the line.
   (Meerut, 2003; Delhi, 2002; Kurukshetra, 2003)
9. Find the surface area of the solid got by revolving the arch of the cycloid
   $x = a(\theta + \sin\theta)$, $y = a(1 + \cos\theta)$ about the base. (UPTU, 2003)
10. Prove that the surface and volume of the solid generated by the revolution about the x-axis, of the loop of the curve $x = at^2$, $y = at - at^3$ are $\frac{12\pi a^2\sqrt{3}}{5}$ and $\frac{3\pi a^3\sqrt{3}}{10}$ respectively.
11. Prove that the surface of the solid generated by the revolution of the curve $x = a\sin\theta + \frac{a}{2}\log\tan^2\frac{\theta}{2}$, $y = a\cos\theta$ about its axis is $4\pi a^2$.
12. Find the surface area of the solid formed by the revolution of the curve $x = 2a\cos\theta - a\cos 2\theta$, $y = 2a\sin\theta - a\sin 2\theta$ about the initial line. (UPTU, 2004)
13. Find the surface of the solid generated by the revolution of the cardioid $r = a(1 - \cos\theta)$ about the initial line.
14. Find the surface of the solid generated by the revolution of the lemniscate $r^2 = a^2\cos 2\theta$ about the line $\theta = \frac{\pi}{2}$.
    (UPTU, 2004)
15. The part of parabola $y^2 = 4ax$ cut off by the latus rectum revolves about the tangent at the vertex. Find the curved surface of the solid thus formed.

4.14 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS (A)

Choose the correct answer or fill up the blanks in the following problems :

1. If $f(x) = f(2a - x)$ then $\int_0^{2a} f(x)\,dx$ is equal to

(a) $\int_0^{\pi/2} f(2a-x)\,dx$ (b) $\int_0^{a} f(x)\,dx$ (c) $-2\int_0^{a} f(x)\,dx$ (d) 0

8. $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$ is equal to

(a) 0 (b) 1 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

9. The value of definite integral $\int_{-a}^{a} [x]\,dx$ is equal to

(a) a (b) a^2 (c) 0 (d) 2a

10. $\lim_{n\to\infty}\left[\frac{n}{n^2}+\frac{n}{n^2+1^2}+\frac{n}{n^2+2^2}+\cdots+\frac{n}{n^2+(n-1)^2}\right]$ is equal to

(a) $-\frac{\pi}{4}$ (b) 0 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

11. $\int_0^{\pi/2} \frac{\cos 2x}{\sin x+\cos x}\,dx$ is equal to

(a) 1 (b) 0 (c) 2 (d) -1

12. $\lim_{n\to\infty}\left[\frac{1}{n}+\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}\right]$ equals

(a) $\log_e 2$ (b) $2\log_e 2$ (c) $\log_e 2$ (d) $2\log_e 2$

13. $\int_0^{\infty} x e^{-x^2}\left(\frac{x^2}{2}\right)$ is equal to

(a) $\frac{10}{11}$ (b) $\frac{11}{10}$ (c) $\frac{5}{10}e^2$ (d) $\frac{10}{99}$

14. $\int_0^{\pi/2} \sin^{99} x \cos x\,dx$ is equal to

(a) $\frac{1}{100}$ (b) $\frac{1}{100}$ (c) $\frac{99}{100}$ (d) None of these [NDA (I) 2008]

15. The value of $\int_{-\pi/2}^{\pi/2} \sin^3 x\,dx$ is

(a) $\frac{2\pi}{10}$ (b) $\frac{2\pi}{10}$ (c) zero

16. The length of the arc of the cardioid $r = a(1+\cos\theta)$ between the point for which $\theta = 0$ and $\theta = \frac{\pi}{2}$ is

(a) $(\sqrt{2}-1)a$ units (b) $(\sqrt{2}+1)a$ units (c) $(\sqrt{2}-1)a$ units (d) $(\sqrt{2}+1)a$ units

17. The area of the region in the first quadrant bounded by the parabola $y = 4x^2$, $x = 0$, $y = 1$ and $y = 4$ is

(a) $\sqrt{2}$ (b) $\sqrt{2}-1$ (c) $\sqrt{2}-1$ (d) $\sqrt{2}+1$

18. The value of $\int_0^{\pi/2} \sin^{3/2}\theta(1-\sin^{3/2}\theta)\,d\theta$

(a) $\frac{\pi}{32}$ (b) $\frac{\pi}{32}$ (c) $\frac{3\pi}{32}$ (d) $\frac{3\pi}{32}$ (I.F.S. 2008)

19. The entire length of the cardioid $r = a(1+\cos\theta)$ is

(a) $8a$ (b) $4a$ (c) $2a$ (d) a (I.F.S. 2008)

20. The area of the cardioid $r = a(1-\cos\theta)$ is

21. If S_1 and S_2 are respectively the volumes of the solids generated by revolving the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ and $\frac{x^2}{b^2}+\frac{y^2}{a^2} = 1$ about the y-axis. Then

(a) $S_1 = S_2$ (b) $S_1 > S_2$ (c) $S_1 < S_2$ (d) not comparable

22. The area of the loop of the curve $r = a\sin 3\theta$ is

23. If $I_n = \int_0^{\pi/2} \sin^n x\,dx$, then $n I_n = (n-1) I_{n-2}$ is (a) $\int_0^{a} x^2\sqrt{(a^2-x^2)}\,dx$ is

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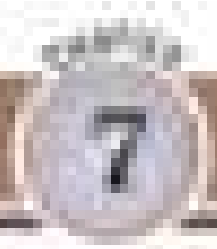

7

Multiple Integrals and Beta, Gamma Functions

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$$[redacted] \tag{1a}$$

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[redacted]

$$[redacted]$$

[redacted]

$$[redacted]$$

Figure 7.1 illustrates this process. Here AB and CD are the two curves whose equations are $y_1 = f_1(x)$ and $y_2 = f_2(x)$. PQ is an elemental strip of width dx.

Then the first rectangle integral means that the integration is along one edge of the strip PQ from P to Q (i.e. vertical integration), while the outer rectangle integral corresponds to the sliding of the edge from AC to BD.

Thus the whole region of integration is the area $ABDC$.

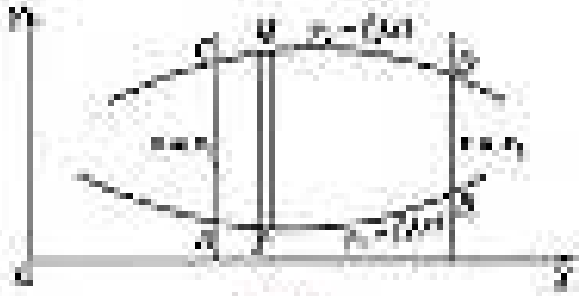

Fig. 7.1

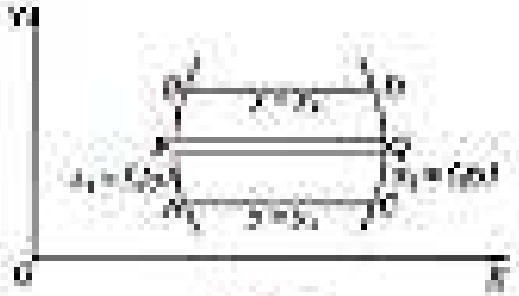

Fig. 7.2

(ii) When x_1, x_2 are functions of y and y_1, y_2 are constants, $f(x, y)$ is first integrated w.r.t. x keeping y fixed, within the limits x_1, x_2 and the resulting expression is integrated w.r.t. y between the limits y_1, y_2, i.e.,

$$I_2 = \boxed{\int_{y_1}^{y_2} \left[\int_{x_1}^{x_2} f(x, y)\, dx \right] dy} \quad \text{which is geometrically illustrated by Fig. 7.2.}$$

Here AB and CD are the curves $x_1 = f_1(y)$ and $x_2 = f_2(y)$. PQ is a horizontal strip of width dy.

Then inner rectangle indicates that the integration is along one edge of the strip from P to Q while the outer rectangle corresponds to the sliding of this edge from AC to BD.

Thus the whole region of integration is the area $ABDC$.

(iii) When both pairs of limits are constants, the region of integration is the rectangle $ABDC$ (Fig. 7.3).

In I_1, we integrate along the vertical strip PQ and then slide it from AC to BD.

In I_2, we integrate along the horizontal strip $P'Q'$ and then slide it from AB to CD.

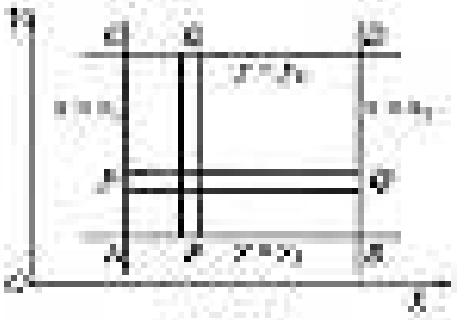

Fig. 7.3

Here obviously $I_1 = I_2$.

Thus for constant limits, it hardly matters whether we first integrate w.r.t. x and then w.r.t. y or vice versa.

Example 7.1. Evaluate $\int_0^a \int_0^a (x^2 + y^2)\, dx\, dy$.

Solution. $$I = \int_0^a dx \int_0^a (x^2 + y^2)\, dy = \int_0^a \left[x^2 y + y \cdot \frac{y^3}{3} \right]_0^a dx = \int_0^a \left[x^2 \cdot x^2 + x \cdot \frac{x^3}{3} \right] dx$$

$$= \int_0^a \left[x^3 + \frac{x^3}{3} \right] dx = \left[\frac{x^4}{4} + \frac{x^4}{12} \right]_0^a = a^4 \left[\frac{1}{4} + \frac{1}{12} \right] = \frac{4a^4}{12} \text{ sq. units.}$$

Example 7.2. Evaluate $\iint_A xy\, dx\, dy$, where A is the domain bounded by x-axis, ordinate $x = 2a$ and the curve $x^2 = 4ay$.

Solution. The line $x = 2a$ and the parabola $x^2 = 4ay$ intersect at $O(0, 0)$ and $P(2a, a)$. Figure 7.4 shows the domain A which is the area OMP.

Integrating first over a vertical strip PQ, i.e., w.r.t. y from $P(y = \frac{x^2}{4a})$ to $Q(y = a)$ lying on the parabola and then w.r.t. x from $x = 0$ to $x = 2a$, we have

$$\iint_A xy\, dx\, dy = \int_0^{2a} dx \int_{\frac{x^2}{4a}}^{a} xy\, dy = \int_0^{2a} x \left[\frac{y^2}{2} \right]_{\frac{x^2}{4a}}^{a} dx$$

Fig. 7.4

$$= \frac{1}{2ka^2}\int_0^{2a} x^3\,dx = \frac{1}{2ka^2}\left|\frac{x^4}{4}\right|_0^{2a} = \frac{a^2}{2}$$

(b) Integration in integrating factor over a horizontal strip RS, i.e., w.r.t. x from R to S, R is $x = 2\sqrt{ay}$ for the parabola $x^2 = 4ay$ and S is $x = 2a$ and then w.r.t. y from $y = 0$ to $y = a$, we get

$$\iint_R xy\,dx\,dy = \int_0^a dy \int_{2\sqrt{ay}}^{2a} xy\,dx = \int_0^a y\left[\frac{x^2}{2}\right]_{2\sqrt{ay}}^{2a} dy$$

$$= 2a\int_0^a (2y - y^2)\,dy = 2a\left[\frac{2y^2}{2} - \frac{y^3}{3}\right]_0^a = \frac{a^3}{3}$$

Example 7.3. Evaluate $\iint_R x^2\,dx\,dy$, where R is the region in the first quadrant bounded by the lines $y = x$, $y = 2x$ and the hyperbola $xy = 32$.

Solution. The line AB ($x = 4$) intersects the hyperbola $xy = 32$ at $A(4, 8)$ while the line $y = x$ intersects this hyperbola at $B(8, 8)$. Figure 7.5 shows the region R of integration which is the area $OLAB$. To evaluate the given integral, we divide this area into two parts OHB and $BLAB$.

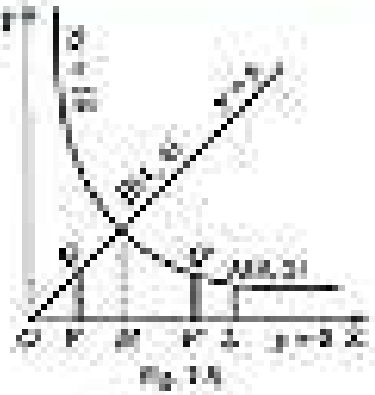

Fig. 7.5

$$\therefore \iint_R x^2\,dx\,dy = \int_{x=0}^{x=4}\int_{y=x}^{y=2x} x^2\,dx\,dy + \int_{x=4}^{x=8}\int_{y=x}^{y=32/x} x^2\,dx\,dy$$

$$= \int_0^4\left[\int_x^{2x} x^2\,dy\right]dx + \int_4^8\left[\int_x^{32/x} x^2\,dy\right]dx$$

$$= \int_0^4 x^2\,dx\,\Big|y\Big|_x^{2x} + \int_4^8 x^2\,dx\,\Big|y\Big|_x^{32/x}$$

$$= \int_0^4 x^2\,dx + \int_4^8 (32x)\,dx = \left|\frac{x^3}{3}\right|_0^4 + 16\left|\frac{x^2}{2}\right|_4^8 = 448$$

7.7 CHANGE OF ORDER OF INTEGRATION

In a double integral with variable limits, the change of order of integration changes the limits of integration. While doing so, it sometimes it is required to split up the region of integration and the given integral is expressed as the sum of a number of double integrals with changed limits. To fix up the new limits, it is always advisable to draw a rough sketch of the region of integration.

The change of order of integration is often facilitated by evaluation of a double integral. The following examples will make the above clear.

Example 7.4. By changing the order of integration of $\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dy\,dx}{\sqrt{a^2-x^2-y^2}}$, prove that

$$\int_0^a \frac{dx}{\sqrt{a^2-x^2}} = \frac{\pi}{2}$$

(U.P.T.U., 2004)

Solution. $\int_0^a \int_0^{\sqrt{a^2-x^2}} e^{-y^2}\sin x\,dy\,dx = \int_0^a \left[\int_0^{\sqrt{a^2-x^2}} e^{-y^2}\sin x\,dx\right]dy$

$$= \int_0^{\infty} \left| -\frac{e^{-(x^2+y^2)}}{x^2+y^2} \right|_0^{\infty} \sin(x^2+y^2)\, dx$$

$$= \int_0^{\infty} \frac{1}{x^2+y^2} \sin(x^2+y^2)\, dx = \left| \tan^{-1}\left(\frac{x}{y}\right) \right|_0^{\infty} = \frac{\pi}{2} \quad \ldots(i)$$

On changing the order of integration, we have

$$\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} \sin(x^2+y^2)\, dx\, dy = \int_0^{\infty} \sin(x^2+y^2) \left[\int_0^{\infty} e^{-(x^2+y^2)}\, dy \right] dx$$

$$= \int_0^{\infty} \sin(x^2+y^2) \left| \frac{e^{-(x^2+y^2)}}{-x} \right|_0^{\infty} dx = \int_0^{\infty} \frac{\sin(x^2+y^2)}{x}\, dx \quad \ldots(ii)$$

Thus from (i) and (ii), we have $\int_0^{\infty} \frac{\sin(x^2+y^2)}{x}\, dx = \frac{\pi}{2}$.

Example 3.8. Change the order of integration in the integral

$$I = \int_0^a \int_{-\sqrt{a^2-y^2}}^{\sqrt{a^2-y^2}} f(x,y)\, dx\, dy.$$

Solution. Here the elementary strip is parallel to x-axis (such as PQ) and extends from $x = 0$ to $x = \sqrt{(a^2-y^2)}$ i.e., to the circle $x^2+y^2 = a^2$ and this strip slides from $y = -a$ to $y = a$. This shaded semi-circular area is, therefore, the region of integration (Fig. 3.11).

On changing the order of integration, we first integrate w.r.t. y along a vertical strip RS which extends from R to $y = -\sqrt{(a^2-x^2)}$ to S to $y = \sqrt{(a^2-x^2)}$. To cover the given region, we then integrate w.r.t. x from $x = 0$ to $x = a$.

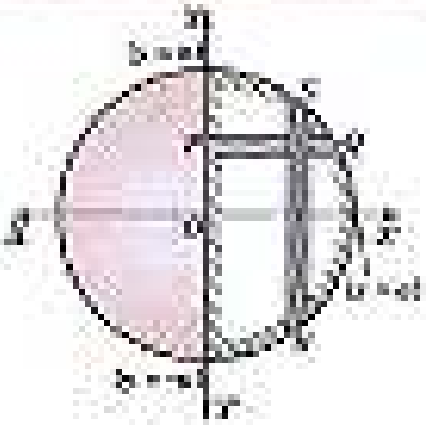

Fig. 3.11

Thus $$I = \int_0^a dx \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} f(x,y)\, dy$$

or $$= \int_0^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} f(x,y)\, dy\, dx.$$

Example 3.9. Evaluate $\int_0^1 \int_x^{x^2} xy(x+y)\, dx\, dy$ by changing the order of integration.

Solution. Here the integration is first w.r.t. y from P to $y = x^2$ to Q on the line $y = x$. Then the integration is w.r.t. x from $x = 0$ to $x = 1$, giving the shaded region OAC (Fig. 3.12).

On changing the order of integration, we first integrate w.r.t. y from R to $y = x$ to S to $y = \log x$ and then w.r.t. x from $y = 1$ to $y = e$.

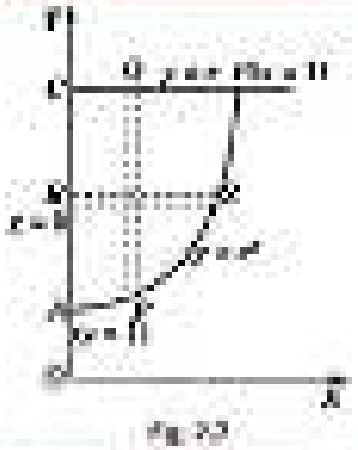

Fig. 3.12

Thus $$\int_0^1 \int_x^{x^2} \frac{xy\, dx\, dy}{\log y} = \int_1^e \int_{\log y}^{y} \frac{xy\, dx\, dy}{\log y}$$

$$= \int_1^e \frac{dy}{\log y} \left| \frac{x^2}{2} \right|_{\log y}^{y} = \int_1^e \frac{y}{2}\, dy = \left| \frac{y^2}{4} \right|_1^e = \frac{e^2-1}{4}.$$

Example 3.10. Change the order of integration in $I = \int_{-a}^{a} \int_{x^2/a}^{\sqrt{(2ax-x^2)}} f(x,y)\, dx\, dy$ and hence evaluate.

(Nagpur, 2000; B.T.U., 2003 S)

Solution. Here integration is first w.r.t. y (i.e. P on the parabola $x^2 = 4ay$ to Q on the parabola $y^2 = 4ax$) and then w.r.t. x from $x = 0$ to $x = 4a$ giving the shaded region of integration (Fig. 2.8).

On changing the order of integration, we first integrate w.r.t. x (i.e. from R to S), then w.r.t. y from $y = 0$ to $y = 4a$, i.e.

$$\therefore \quad I = \int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dx\,dy = \int_0^{4a} dy\,\Big[x\Big]_{x^2/4a}^{2\sqrt{ay}} = \int_0^{4a} (2\sqrt{ay} - y^2/4a)\,dy$$

$$= \left[2\sqrt{a}\,\frac{y^{3/2}}{3/2} - \frac{y^3}{12a}\right]_0^{4a} = \frac{32a^2}{3} - \frac{16a^2}{3} = \frac{16a^2}{3}.$$

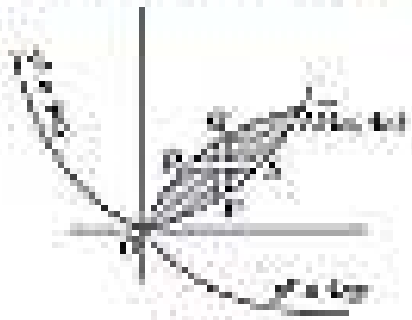

Fig. 2.8

Example 2.6. Change the order of integration and hence evaluate

$$I = \int_0^a \int_{x^2/a}^{x} \frac{x^2\,dx\,dy}{y\sqrt{(a^2 - x^2)(y^2 - a^2)}}$$ (U.P.T.U., 2006 S)

Solution. Here integration is first w.r.t. y from P on the parabola $y^2 = ax$ to Q on the line $y = x$, then w.r.t. x from $x = 0$ to $x = a$, giving the shaded region OAB of integration (Fig. 2.9).

On changing the order of integration, we first integrate w.r.t. x from R to S, then w.r.t. y from $y = 0$ to $y = a$, i.e.

$$\therefore \quad I = \int_0^a \int_0^{y^2/a} \frac{x^2\,dx}{\sqrt{(x^2 - a^2)(y^2 - a^2)}}\,dy = \frac{1}{a}\int_0^a \int_0^{y^2/a} y^2\,dy\,\frac{dx}{\sqrt{(x^2 - a^2)(y^2 - a^2)}}$$

$$= \frac{1}{a}\int_0^a y^2 dy\,\Big[\cosh^{-1}\left(\frac{x}{a}\right)\Big]_0^{y^2/a} = \frac{1}{a}\int_0^a y^2 dy\,\{\cosh^{-1}(1) - \cosh^{-1}(0)\}$$

$$= \frac{\pi}{2a}\int_0^a y^2 dy = \frac{\pi}{2a}\Big[\frac{y^3}{3}\Big]_0^a = \frac{\pi a^2}{6}.$$

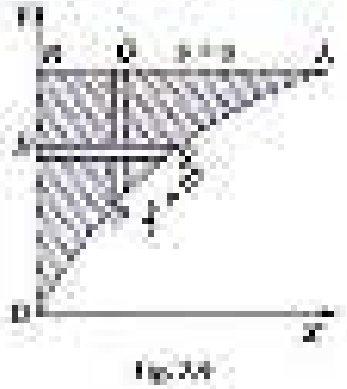

Fig. 2.9

Example 2.8. Change the order of integration in $I = \int_0^1 \int_{x^2}^{2-x} xy\,dx\,dy$ and hence evaluate the same.

(Bhopal, 2000; U.P.T.U., 2005; A.M.I.E., 2003; J.N.T.U., 2002; U.P.T.U., 2006 S)

Solution. Here the integration is first w.r.t. y along a vertical strip PQ which extends from P on the parabola $y = x^2$ to Q on the line $y = 2 - x$. Next, a strip slides from $x = 0$ to $x = 1$, giving the region of integration as the curvilinear triangle OAB (Fig. 2.10).

On changing the order of integration, we first integrate w.r.t. x along a horizontal strip PQ' and then replace the splitting up of the region OAB into two parts by the line AC ($y = 1$), i.e., the curvilinear triangle OAC and the triangle ABC.

For the region OAC, the limits of integration for x are from $x = 0$ to $x = \sqrt{y}$ and those for y are from $y = 0$ to $y = 1$. So the contribution to I from the region OAC is

$$I_1 = \int_0^1 dy \int_0^{\sqrt{y}} xy\,dx$$

For the region ABC, the limits of integration for x are from $x = 0$ to $x = 2 - y$ and those for y are from $y = 1$ to $y = 2$. So the contribution to I from the region ABC is

$$I_2 = \int_1^2 dy \int_0^{2-y} xy\,dx.$$

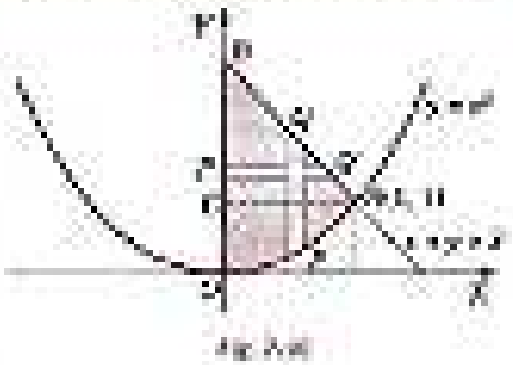

Fig. 2.10

Hence, on reversing the order of integration,

$$I = \int_0^1 dy \int_{y^2}^{2-y} xy\,dx + \int_1^2 dy \int_0^{2-y} xy\,dx$$

$$= \int_0^1 dy \left[\frac{x^2}{2}\cdot y\right]_{y^2}^{2-y} + \int_1^2 dy \left[\frac{x^2}{2}\cdot y\right]_0^{2-y} = \frac{1}{2}\int_0^1 (y^3 - y^5)\,dy + \frac{1}{2}\int_1^2 y(2-y)^2\,dy = \frac{1}{8} + \frac{5}{24} = \frac{2}{3}$$

Example 7.18. Change the order of integration in $I = \int_0^{\sqrt{2}} \int_y^{\sqrt{2-y^2}} \frac{x}{\sqrt{x^2+y^2}}\,dx\,dy$ and hence evaluate it.

(U.P.T.U., 2006; Kerala, 2008)

Solution. Here the integration is first w.r.t. x along PQ which extends from P on the line $y = x$ to Q on the circle $y = \sqrt{2 - x^2}$. Then PQ slides from $y = 0$ to $y = 1$, giving the region of integration OAB as in Fig. 7.31.

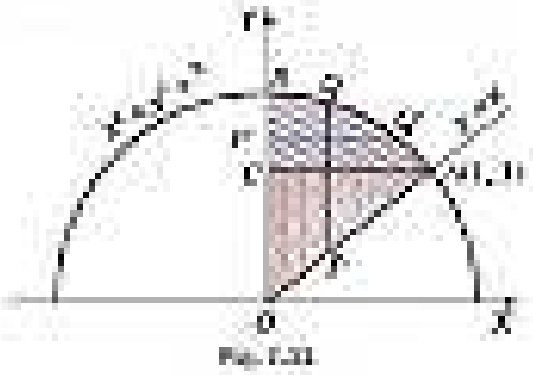

Fig. 7.31

On changing the order of integration, we first integrate w.r.t. x from P' to Q' and that requires splitting the region OAB into two parts OAC and ABC.

For the region OAC, the limits of integration for x are from $x = 0$ to $x = y$ and those for y are from $y = 0$ to $y = 1$. So the contribution to I from the region OAC is

$$I_1 = \int_0^1 dy \int_0^y \frac{x}{\sqrt{x^2+y^2}}\,dx.$$

For the region ABC, the limits of integration for x are 0 to $\sqrt{2 - y^2}$ and those for y are from 1 to $\sqrt{2}$. So the contribution to I from the region ABC is

$$I_2 = \int_1^{\sqrt{2}} dy \int_0^{\sqrt{2-y^2}} \frac{x}{\sqrt{x^2+y^2}}\,dx$$

Hence
$$I = \int_0^1 \left[\sqrt{x^2+y^2}\right]_0^y dy + \int_1^{\sqrt{2}} \left[\sqrt{x^2+y^2}\right]_0^{\sqrt{2-y^2}} dy$$

$$= \int_0^1 (\sqrt{2}-1)\,y\,dy + \int_1^{\sqrt{2}} \sqrt{2-y^2}\,dy = \frac{1}{2}(\sqrt{2}-1)\cdot 1 + \frac{1}{2}\sqrt{2-y^2}\cdot y + \frac{1}{2}\sin^{-1}\frac{y}{\sqrt{2}}\Big]_1^{\sqrt{2}}$$

$$= \frac{1}{2}(\sqrt{2}-1) + \frac{1}{2}\cdot 0 + \frac{1}{2}\cdot\frac{\pi}{2} - \frac{1}{2}\cdot 1\cdot 1 - \frac{1}{2}\cdot\frac{\pi}{4} = \frac{1}{2}(\sqrt{2}-1) + \frac{\pi}{8} - \frac{1}{2} = \frac{\pi}{8} - \frac{1}{\sqrt{2}}.$$

7.13 DOUBLE INTEGRALS IN POLAR COORDINATES

To evaluate $\int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r,\theta)\,dr\,d\theta$, we first integrate w.r.t. r between limits $r = r_1$ and $r = r_2$, keeping θ fixed and the resulting expression is integrated w.r.t. θ from θ_1 to θ_2. In the integral, r_1, r_2 are functions of θ and θ_1, θ_2 are constants.

Figure 7.32 illustrates the process geometrically.

Here AB and CD are the curves $r_1 = f_1(\theta)$ and $r_2 = f_2(\theta)$ bounded by the lines $\theta = \theta_1$ and $\theta = \theta_2$. PQ is a wedge of angular thickness $d\theta$.

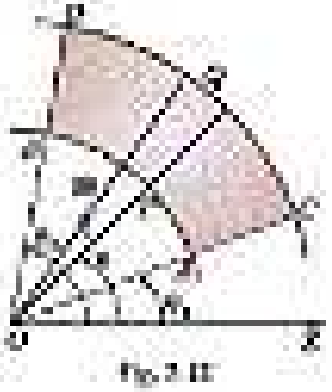

Fig. 7.32

Then $\int_{r_1}^{r_2} f(r,\theta)\,dr$ indicates that the integration is along PQ from P to Q while the integration w.r.t. θ corresponds to the turning of PQ from AC to BD.

Thus the whole region of integration is the area $ACDB$. The order of integration may be changed with appropriate changes in the limits.

Example 7.11. Evaluate $\iint r \sin\theta \, dr \, d\theta$ over the cardioid $r = a(1 - \cos\theta)$ above the initial line.

(Kerala, 2002)

Solution. To integrate first w.r.t. r, the limits are from $r = 0$ to $r = a(1 - \cos\theta)$ and to cover the region of integration, θ varies from 0 to π (Fig. 7.13).

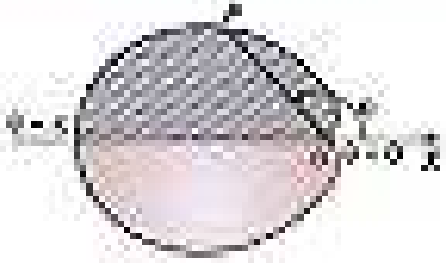

Fig. 7.13

$$\therefore \quad \iint_R r \sin\theta \, dr \, d\theta = \int_0^{\pi} \sin\theta \left[\int_0^{a(1-\cos\theta)} r \, dr \right] d\theta$$

$$= \int_0^{\pi} \sin\theta \left[\frac{r^2}{2} \right]_0^{a(1-\cos\theta)} d\theta = \frac{a^2}{2} \int_0^{\pi} (1 - \cos\theta)^2 \sin\theta \, d\theta$$

$$= \frac{a^2}{2} \left[\frac{(1-\cos\theta)^3}{3} \right]_0^{\pi} = \frac{a^2}{2} \cdot \frac{8}{3} = \frac{4a^2}{3}.$$

Example 7.12. Calculate $\iint r^2 \, dr \, d\theta$ over the area included between the circle $r = 2\sin\theta$ and the cardioid $r = 1 + \cos\theta$.

Solution. Given circle $r = 2\sin\theta$ ...(1)

and $r = 1 + \cos\theta$ ...(2)

are shown in Fig. 7.14. The shaded area between these circles is the region of integration.

If we integrate first w.r.t. r, then its limits are from $PQ = 2\sin\theta$ to $QR = 1 + \cos\theta$ and to cover the whole region, θ varies from 0 to π. Thus the required integral is

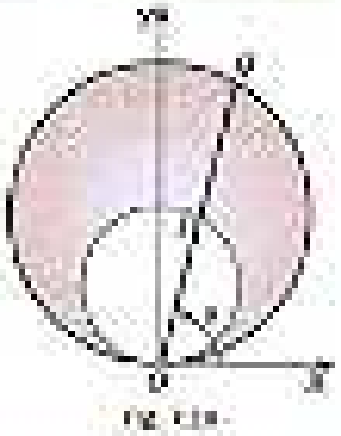

Fig. 7.14

$$I = \int_0^{\pi} d\theta \int_{2\sin\theta}^{1+\cos\theta} r^2 \, dr = \int_0^{\pi} d\theta \left[\frac{r^3}{3} \right]_{2\sin\theta}^{1+\cos\theta}$$

$$= \frac{1}{3} \int_0^{\pi} \left[(1+\cos\theta)^3 - 8\sin^3\theta \right] d\theta = \frac{1}{3} \left[\frac{5}{2} \cdot \frac{\pi}{2} \cdot 2 - 8 \cdot \frac{2}{3} \right] = \frac{5\pi}{12} - \frac{16}{9}.$$

PROBLEMS 7.4

Evaluate the following integrals (1 - 7):

1. $\int_0^1 \int_0^1 xy^2 \, dx \, dy$.

2. $\int_0^a \int_0^x (x^2 + y^2) \, dx \, dy$. (V.T.U., 2006)

3. $\int_0^{\infty} \int_0^{\infty} e^{-(x+y)} \, dx \, dy$. (P.T.U., 2004)

4. $\int_0^{\infty} \int_0^{\infty} \frac{dx \, dy}{(1+x^2)(1+y^2)}$. (Nagarjuna, 2010)

5. $\iint xy \, dx \, dy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$. (Nagarjuna, 2008)

6. $\iint (x+y) \, dx \, dy$ over the area included by the ellipse $9x^2 + 25y^2 = 1$. (Kerala, Marh, 2008 S; C.P.T.U., 2004 S)

7. $\iint xy(x+y) \, dx \, dy$ over the area bounded by $y = x^2$ and $y = x$. (V.T.U., 2010)

Evaluate the following integrals by changing the order of integration (8, 9):

8. $\int_0^a \int_y^a \frac{x \, dx \, dy}{x^2 + y^2}$. (Kerala, Sept. 2005)

9. $\int_0^{\infty} \int_x^{\infty} e^{-y} \, dy \, dx$. (V.T.U., 2008; Anna, 2003 S; Delhi, 2009)

10. $\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{x\,dy\,dx}{\sqrt{x^2+y^2}}$ (P.T.U., 2013) (Kurukshetra, 2009) (U.P.T.U., 2009)

11. $\int_0^{\log 2} \int_0^{\log x} e^{x+y}\,dy\,dx$ (U.P.T.U., 2010)

12. $\int_0^{\pi/2} \int_0^{\sin\theta} r\,dr\,d\theta$ (U.P.T.U., 2010)

13. $\int_0^a \int_{x^2/a}^{\sqrt{2ax-x^2}} xy\,dx\,dy$ (Anna, 2009)

14. $\int_0^a \int_x^a \frac{x^2}{y^2} dy\,dx$ (Kurukshetra, 2009) (U.P.T.U., 2009) (U.P.T.U., 2007)

15. $\int_0^1 \int_0^x e^{x^2+y^2}\,dy\,dx$ (U.P.T.U., 2006) (U.P.T.U., 2009) (U.P.T.U., 2010)

16. Change the order of integration of the following integrals and change the order of integrations:

(i) $\int_0^a \int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} f(x,y)\,dx\,dy$ (Rajasthan, 2006) (ii) $\int_0^{e^{x^2}} \int_{\log y}^{x^2} f(x,y)\,dx\,dy$.

17. Show that $\iint_R e^{-(x^2+y^2)}\,dx\,dy = \pi(1-e^{-a^2})$, where R is the semi-circle $x^2+y^2=a^2$ on the positive side of the x-axis.

18. Evaluate $\iint \frac{x\cos\theta}{\sqrt{1-r^2}}\,dr\,d\theta$ over the loop of the lemniscate $r^2=a^2\cos 2\theta$. (Kurukshetra, 2009) (U.P.T.U., 2010)

19. Evaluate $\iint r^3\,dr\,d\theta$ over the area bounded between the circles $r=2\cos\theta$ and $r=4\cos\theta$.

(Anna, 2009) (Madras, 2010)

7.3 AREA ENCLOSED BY PLANE CURVES

(1) Cartesian coordinates.

Consider the area enclosed by the curves $y=f_1(x)$ and $y=f_2(x)$ and the ordinates $x=x_1$, $x=x_2$ (Fig. 7.10 (a)).

Divide this area into vertical strips of width δx. If $P(x, y)$, $Q(x+\delta x, y+\delta y)$ be two neighbouring points, then the area of the small rectangle $PQ = \delta x\,\delta y$.

$\therefore$ area of strip $KL = \lim_{\delta y \to 0} \sum \delta x\,\delta y$.

Since for all rectangles in this strip δx is the same and y varies from $y=f_1(x)$ to $y=f_2(x)$.

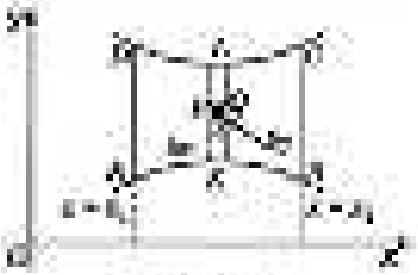

Fig. 7.10 (a)

$$\therefore \text{area of the strip } KL = \lim_{\delta y \to 0} \sum_{f_1(x)}^{f_2(x)} \delta y = \delta x \int_{f_1(x)}^{f_2(x)} dy$$

Now adding up all such strips from $x=x_1$ to $x=x_2$ we get the area $AKCD$

$$= \lim_{\delta x \to 0} \sum_{x_1}^{x_2} \delta x \int_{f_1(x)}^{f_2(x)} dy = \int_{x_1}^{x_2} dx \int_{f_1(x)}^{f_2(x)} dy = \int_{x_1}^{x_2} \int_{f_1(x)}^{f_2(x)} dy\,dx$$

Similarly, dividing the area $AWCD$ (Fig. 7.10 (b)) into horizontal strips of width δy, we get the area $AWCD$.

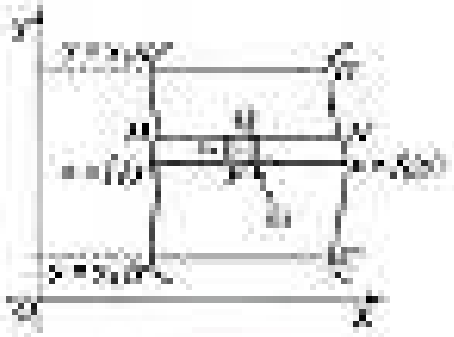

Fig. 7.10 (b)

$$= \int_{y_1}^{y_2} \int_{f_1(y)}^{f_2(y)} dx\,dy$$

(2) Polar coordinates.

Consider an area A enclosed by a curve whose equation is in polar coordinates.

Let $P(r, \theta)$, $Q(r+\delta r, \theta+\delta\theta)$ be two neighbouring points. Mark circular arcs of radii r and $r+\delta r$ meeting OQ in R and OP produced in S (Fig. 7.10).

Since arc $PQ = r\delta\theta$ and $PS = \delta r$,

$\therefore$ area of the curvilinear rectangle $PSQR$ is approximately $= PQ \times PS = r\delta\theta \times \delta r$.

If the whole area A is divided into such curvilinear rectangles, the sum $\sum r\delta\theta\delta r$ taken for all these rectangles, gives in the limit the area A.

Hence $$A = \lim_{\substack{\delta r \to 0 \\ \delta\theta \to 0}} \sum r\delta\theta\delta r = \iint r\,d\theta\,dr$$

where the limits are to be so chosen as to cover the entire area.

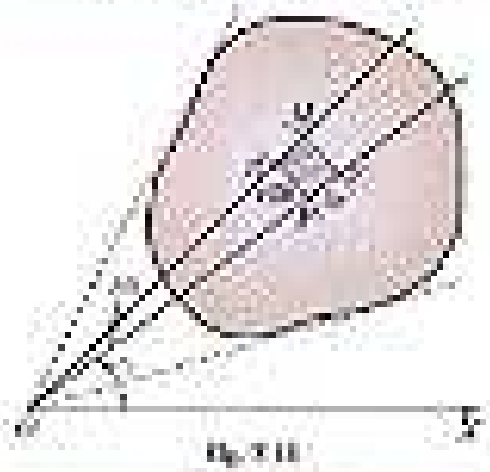

Fig. 3.11

Example 3.13. Find the area of a plate in the form of a quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (V.T.U., 2001; Osmania, 2000)

Solution. Dividing the area into vertical strips of width δx, y varies from $My = 0$ to $Qy = b\sqrt{(1 - x^2/a^2)}$ and then x varies from O to A (Fig. 3.12).

$\therefore$ required area

$$= \int_0^a dx \int_0^{b\sqrt{(1-x^2/a^2)}} dy = \int_0^a dx\,[y]_0^{b\sqrt{(1-x^2/a^2)}}$$

$$= \frac{b}{a}\int_0^a \sqrt{(a^2 - x^2)}\,dx = \text{ans}$$

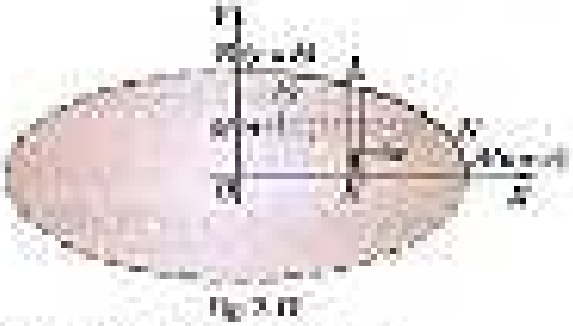

Fig. 3.12

Otherwise, dividing the area into horizontal strips of width δy, x varies from $MQ = 0$ to $NQ = a\sqrt{(1 - y^2/b^2)}$ and then y varies from 0 to b.

$\therefore$ required area

$$= \int_0^b dy \int_0^{a\sqrt{(1-y^2/b^2)}} dx = \int_0^b dy\,[x]_0^{a\sqrt{(1-y^2/b^2)}}$$

$$= \frac{a}{b}\int_0^b \sqrt{(b^2 - y^2)}\,dy = \text{ans}$$

Obs. The change of the order of integration does not in any way affect the value of the area.

Example 3.14. Show that the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$. (Kerala, 2000; Calicut, 2002)

Solution. Solving the equations $y^2 = 4ax$ and $x^2 = 4ay$, it is seen that the parabolas intersect at $O(0, 0)$ and $A(4a, 4a)$. An area for the desired area between these parabolas (Fig. 3.13) is drawn from 0 to $4a$ and y varies from P to Q i.e., from $y = x^2/4a$ to $y = 2\sqrt{(ax)}$. Hence the required area

$$= \int_0^{4a} \int_{x^2/4a}^{2\sqrt{(ax)}} dy\,dx = \int_0^{4a} \left[2\sqrt{(ax)} - \frac{x^2}{4a}\right] dx$$

$$= \left[2\sqrt{a}\cdot\frac{2}{3}x^{3/2} - \frac{1}{4a}\cdot\frac{x^3}{3}\right]_0^{4a} = \frac{32}{3}a^2 - \frac{16}{3}a^2 = \frac{16}{3}a^2$$

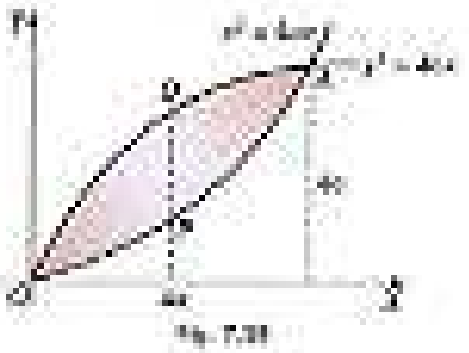

Fig. 3.13

Example 7.31. Calculate the area included between the curve $r = a(1 + \cos\theta)$ and its asymptote.

Solution. The curve is symmetrical about the initial line and has an asymptote $x = a$ or $r\cos\theta = a$ (Fig. 7.18).

Draw any line OP cutting the curve at P and its asymptote at P'. Along this line, θ is constant and r varies from $r = a$ at P' to $r = a(1 + \cos\theta) = 2a\cos^2\frac{\theta}{2}$ at P. Then to get the upper half of the area, θ varies from 0 to $\pi/2$.

$$\therefore \quad \text{required area} = 2\int_0^{\pi/2}\int_{a\sec\theta}^{a(1+\cos\theta)} r\,dr\,d\theta$$

$$= 2\int_0^{\pi/2}\left[\frac{r^2}{2}\right]_{a\sec\theta}^{a(1+\cos\theta)} d\theta$$

$$= a^2\int_0^{\pi/2}(2\cos^2\theta\,d\theta = 6a^2/4.$$

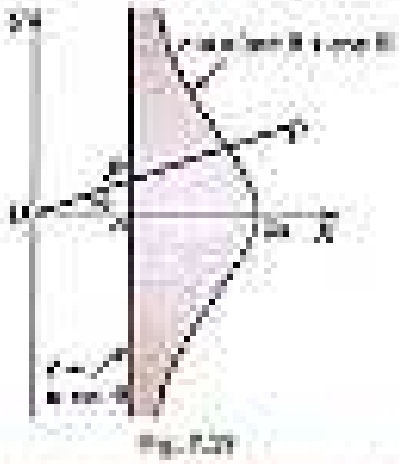

Fig. 7.18

Example 7.32. Find the area lying inside the circle $r = a\sqrt{2}\cos\theta$ and outside the circle $r = a$.

Solution. In Fig. 7.19, $ABCDA$ represents the circle $r = a\sqrt{2}\cos\theta$ and $CEAFC$ is the circle $r = a$.

$$\text{Required area (shaded)} = 2(\text{area } ABCA)$$

$$= 2\int_0^{\pi/4}\int_{a\sec\theta}^{a\sqrt{2}\cos\theta} r\,dr\,d\theta = 2\int_0^{\pi/4}\int_a^{a\sqrt{2}\cos\theta} r\,dr\,d\theta$$

$$= 2\int_0^{\pi/4}\left[\frac{r^2}{2}\right]_a^{a\sqrt{2}\cos\theta} d\theta = a^2\int_0^{\pi/4}(2\cos^2\theta - 1)\,d\theta$$

$$= a^2\int_0^{\pi/4}\cos 2\theta\,d\theta = a^2\left[\frac{1}{2}\sin 2\theta\right]_0^{\pi/4} = \frac{a^2}{2}(1 - 0).$$

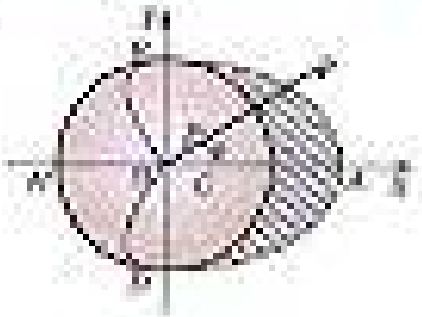

Fig. 7.19

EXERCISE 7.3

1. Find, by double integration, the area lying between the parabola $y = 4x - x^2$ and the line $y = x$.
2. Find the area lying between the parabola $y = x^2$ and the line $x + y = 2$. (Kerala, 2006)
3. By the double integration, find the whole area of the curve $x^2y^2 = a^2(x^2 - y^2)$. (U.P.T.U. 2004)
4. Find, by double integration, the area enclosed by the curves $y = 2x^2$ and $y = 3x + 2$. (U.P.T.U., 2005)
5. Find, by double integration, the area of the lemniscate $r^2 = a^2\cos 2\theta$. (Madras, 2000, S)
6. Find, by double integration, the area lying inside the circle $r = a\cos\theta$ and outside the cardioid $r = a(1 - \cos\theta)$.
(Anna 2006; Madras, 2000)
7. Find the area lying inside the cardioid $r = a(1 + \cos\theta)$ and outside the parabola $r(1 + \cos\theta) = a$.
8. Find the area of loop of the curve $x = at^2, y = at(1 - t^2)$, by double integration. (Madras, 2002)

7.7 TRIPLE INTEGRALS

Consider a function $f(x, y, z)$ defined at every point of the 3-dimensional finite region V. Divide V into n elementary volumes $\delta V_1, \delta V_2, \ldots, \delta V_n$. Let (x_r, y_r, z_r) be any point within the rth subdivision δV_r. Consider the sum

$$\sum_{r=1}^{n} f(x_r, y_r, z_r)\,\delta V_r.$$

The limit of this sum, if it exists, as $n \to \infty$ and $\delta V_r \to 0$ is called the triple integral of $f(x, y, z)$ over the region V and is denoted by

$$\iiint_V f(x, y, z)\,dV.$$

For purposes of evaluation, it can also be expressed as the repeated integral

$$\int_{z_1}^{z_2}\int_{y_1}^{y_2}\int_{x_1}^{x_2} f(x, y, z)\,dx\,dy\,dz.$$

If x_1, x_2 are constants, y_1, y_2 are either constants or functions of x and z_1, z_2 are either constants or functions of x and y, then this integral is evaluated as follows:

First $f(x, y, z)$ is integrated w.r.t. z between the limits z_1 and z_2 keeping x and y fixed. The resulting expression is integrated w.r.t. y between the limits y_1 and y_2 keeping x constant. The result just obtained is finally integrated w.r.t. x from x_1 to x_2.

Thus
$$I = \int_{x_1}^{x_2} \left[\int_{y_1(x)}^{y_2(x)} \left[\int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z)\, dz \right] dy \right] dx$$

where the integration is carried out from the innermost rectangle to the outermost rectangle.

The order of integration may be different for different types of limits.

Example 3.17. Evaluate $\int_1^2 \int_1^3 \int_{-1}^{1} (x + y + z)\, dx\, dy\, dz$. (U.P.T.U., 2002; Gorakhpur, 2005)

Solution. Integrating first w.r.t. y keeping x and z constant, we have

$$I = \int_1^2 \int_{-1}^{1} \left| xy + \frac{y^2}{2} + zy \right|_1^3 dx\, dz = \int_1^2 \int_{-1}^{1} \left[(x + z)(3 - 1) + \frac{3^2}{2} - \frac{1^2}{2} \right] dx\, dz$$

$$= 2 \int_1^2 \left| \frac{x^2}{2} + zx + \frac{x^2}{2} \right|_{-1}^{1} dz = 2 \int_1^2 \left(\frac{1}{2} + z + \frac{1}{2} - \frac{1}{2} + z - \frac{1}{2} \right) dz = 4 \left| \frac{z^2}{2} \right|_1^2 = 6.$$

Example 3.18. Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dz\, dy\, dx$. (U.P.T.U., 2003 A)

Solution. We have

$$I = \int_0^1 x \left[\int_0^{\sqrt{1-x^2}} y \left[\int_0^{\sqrt{1-x^2-y^2}} z\, dz \right] dy \right] dx = \int_0^1 x \left[\int_0^{\sqrt{1-x^2}} y \left| \frac{z^2}{2} \right|_0^{\sqrt{1-x^2-y^2}} dy \right] dx$$

$$= \int_0^1 x \left[\int_0^{\sqrt{1-x^2}} y \cdot \frac{1}{2}(1 - x^2 - y^2)\, dy \right] dx = \frac{1}{2} \int_0^1 x \left| (1 - x^2) \frac{y^2}{2} - \frac{y^4}{4} \right|_0^{\sqrt{1-x^2}} dx$$

$$= \frac{1}{8} \int_0^1 [(1 - x^2)^2 \cdot x - (1 - x^2)^2 \cdot x]\, dx = \frac{1}{8} \int_0^1 (x - 2x^3 + x^5)\, dx$$

$$= \frac{1}{8} \left| \frac{x^2}{2} - \frac{2x^4}{4} + \frac{x^6}{6} \right|_0^1 = \frac{1}{8} \left(\frac{3}{6} - \frac{1}{2} + \frac{1}{6} \right) = \frac{1}{48}$$

EXERCISE 3.8

Evaluate the following integrals:

1. $\int_0^a \int_0^a \int_0^a (x^2 + y^2 + z^2)\, dx\, dy\, dz$. (Ranchi, 2000)

2. $\int_1^2 \int_1^3 \int_{-1}^{1} (x^2 + y^2 + z^2)\, dx\, dy\, dz$. (U.P.T.U., 2000; U.P.T.U., 2006)

3. $\int_0^1 \int_0^1 \int_0^{1-x-y} x\, dz\, dx\, dy$. (Nagpur, 2003)

4. $\int_0^1 \int_0^1 \int_0^1 (x + y + z)\, dx\, dy\, dz$. (U.P.T.U., 2003; Kurukshetra, 2009 S; U.P.T.U., 2006)

5. $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z}\, dz\, dy\, dx$. (Nagpur, 2003)

6. $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz\, dz\, dy\, dx$. (U.P.T.U., 2004; Rohtak, 2006)

7. $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z}\, dz\, dy\, dx$. (U.P.T.U., 2006)

7.13 VOLUMES OF SOLIDS

(1) Volumes as double integrals. Consider a surface $z = f(x, y)$. Let the orthogonal projection on XY-plane of the portion S be the area R (Fig. 7.31).

Divide R into elementary rectangles of area $\delta x\,\delta y$ by drawing lines parallel to X and Y axes. With each of these rectangles as base, erect a prism having its length parallel to OZ.

$\therefore$ volume of this prism between R and the given surface $z = f(x, y)$ is $z\,\delta x\,\delta y$.

Hence the volume of the solid cylinder on R as base, bounded by the given surface with generators parallel to the Z-axis

$$= \lim_{\substack{\delta x \to 0 \\ \delta y \to 0}} \sum z\,\delta x\,\delta y$$

$$= \iint z\,dx\,dy \quad \text{or} \quad \iint f(x, y)\,dx\,dy$$

where the integration is carried over the area R.

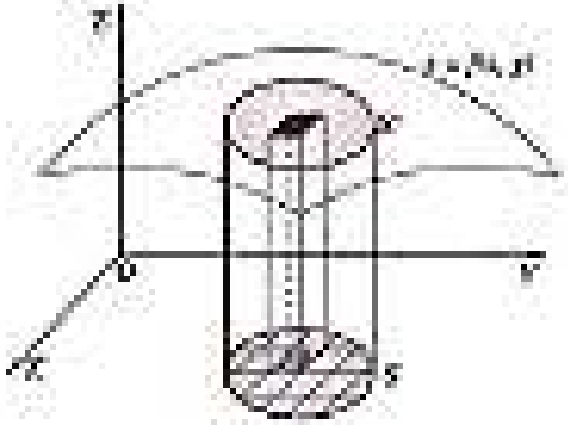

Fig. 7.31

Obs. While using polar coordinates, consider R as the area of $r\,\delta\theta\,\delta r$.

$\therefore$ replacing $dx\,dy$ by $r\,d\theta\,dr$, we get the required volume $= \iint z\,r\,d\theta\,dr$.

Example 7.38. *Find the volume bounded by the cylinder of $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.* (U.P.T.U., 2007; Gorakh., 2008; Madras, 2008 S)

Solution. From Fig. 7.32, it is self evident that $z = 4 - y$ is to be integrated over the circle $x^2 + y^2 = 4$ in the XY-plane. To cover the shaded half of this circle, x varies from 0 to $\sqrt{(4 - y^2)}$ and y varies from -2 to 2.

$\therefore$ Required volume

$$= 2\int_{-2}^{2}\int_{0}^{\sqrt{(4-y^2)}} z\,dx\,dy = 2\int_{-2}^{2}\int_{0}^{\sqrt{(4-y^2)}} (4 - y)\,dx\,dy$$

$$= 2\int_{-2}^{2} (4 - y)\left[x\right]_{0}^{\sqrt{(4-y^2)}}\,dy = 2\int_{-2}^{2} (4 - y)\sqrt{(4 - y^2)}\,dy$$

$$= 2\int_{-2}^{2} 4\sqrt{(4 - y^2)}\,dy - 2\int_{-2}^{2} y\sqrt{(4 - y^2)}\,dy$$

$$= 8\int_{0}^{2} \sqrt{(4 - y^2)}\,dy \qquad \text{[The second term vanishes as the integrand is an odd function.]}$$

$$= 8\left[\frac{y\sqrt{(4 - y^2)}}{2} + \frac{4}{2}\sin^{-1}\frac{y}{2}\right]_{0}^{2} = 16\pi.$$

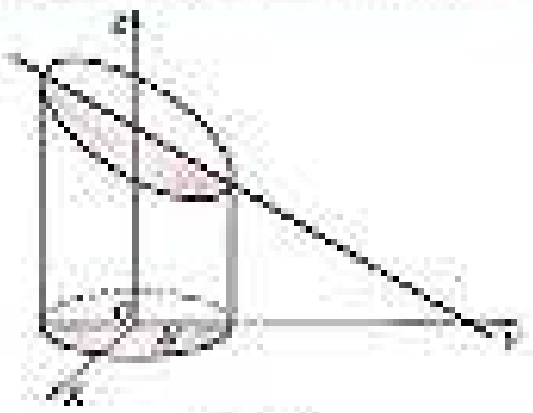

Fig. 7.32

(2) Volume as triple integral.

Divide the given solid by planes parallel to the coordinate planes into rectangular parallelopipeds of volume $\delta x\,\delta y\,\delta z$ (Fig. 7.33).

$\therefore$ the total volume $= \lim_{\substack{\delta x \to 0 \\ \delta y \to 0 \\ \delta z \to 0}} \sum\sum\sum \delta x\,\delta y\,\delta z$

$$= \int\int\int dx\,dy\,dz$$

with appropriate limits of integration.

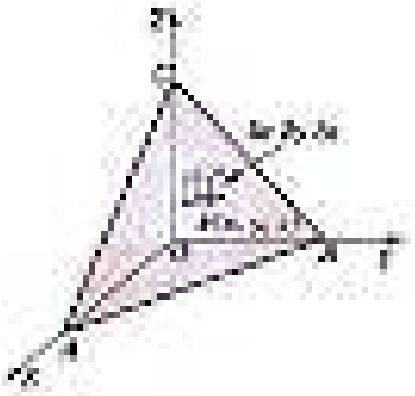

Fig. 7.33

Example 7.20 Calculate the volume of the solid bounded by the planes $x = 0$, $y = 3x$, $x + y + z = 1$ and $z = 0$ (U.E.I., 2005)

Solution. Volume required $= \int_0^{\frac{1}{4}} \int_{3x}^{1-x} \int_0^{1-x-y} dz\,dy\,dx$

$$= \int_0^{\frac{1}{4}} \int_{3x}^{1-x} (1-x-y)\,dy\,dx = \int_0^{\frac{1}{4}} \left| (1-x)y - \frac{y^2}{2} \right|_{3x}^{1-x} dx$$

$$= \int_0^{\frac{1}{4}} \left[(1-x)^2 - \frac{(1-x)^2}{2} \right] dx = \frac{1}{2} \int_0^{\frac{1}{4}} (1-x)^2\,dx = \frac{1}{2} \left| -\frac{(1-x)^3}{3} \right|_0^{\frac{1}{4}} = \frac{37}{128}$$

Example 7.21 Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

(Malaya 2009; U.E.I. 2008; Anticorros 2003)

Solution. Let $OABC$ be the positive octant of the given ellipsoid which is bounded by the planes OAB ($z = 0$), OBC ($x = 0$), OCA ($y = 0$) and the surface ABC, i.e.,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

Divide this region R into rectangular parallelopipeds of volume $\delta x\,\delta y\,\delta z$. Consider such an element at $P(x, y, z)$ (Fig. 7.34)

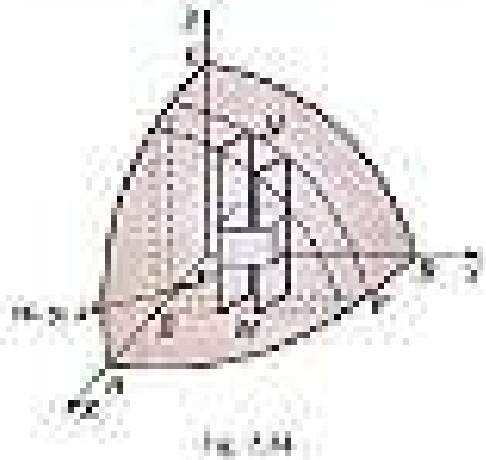

Fig. 7.34

A. The required volume $= \iiint_R dx\,dy\,dz$.

In this region R,

(i) z varies from 0 to MN where

$$MN = z = c\sqrt{(1 - x^2/a^2 - y^2/b^2)},$$

(ii) y varies from 0 to BT, where $BT = b\sqrt{(1 - x^2/a^2)}$ from the equation of the ellipse OAB, i.e.,

$$x^2/a^2 + y^2/b^2 = 1,$$

(iii) x varies from 0 to $OA = a$.

Hence the volume of the whole ellipsoid

$$= 8 \int_0^a \int_0^{b\sqrt{(1-x^2/a^2)}} \int_0^{c\sqrt{(1-x^2/a^2-y^2/b^2)}} dz\,dy\,dx = 8 \int_0^a dx \int_0^{b\sqrt{(1-x^2/a^2)}} dy \left[z \right]_0^{c\sqrt{(1-x^2/a^2-y^2/b^2)}}$$

$$= 8c \int_0^a dx \int_0^{b\sqrt{(1-x^2/a^2)}} \sqrt{(1 - x^2/a^2 - y^2/b^2)}\,dy$$

$$= \frac{8c}{b} \int_0^a dx \int_0^t \sqrt{(t^2 - y^2)}\,dy \quad \text{where } t = b\sqrt{(1 - x^2/a^2)}$$

$$= \frac{8c}{b} \int_0^a dx \left[\frac{y}{2}\sqrt{(t^2 - y^2)} + \frac{t^2}{2} \sin^{-1}\frac{y}{t} \right]_0^t = \frac{8c}{b} \int_0^a \frac{b^2}{2}\left(1 - \frac{x^2}{a^2}\right) \frac{\pi}{2}\,dx$$

$$= 2\pi bc \int_0^a \left(1 - \frac{x^2}{a^2}\right) dx = 2\pi bc \left| x - \frac{x^3}{3a^2} \right|_0^a = \frac{4\pi abc}{3}.$$

Hence, see Problem 37 page 231.

(2) Volumes of solids of revolution

Consider an elementary area lying at the point $P(x, y)$ of a given curve AB (Fig. 7.35)

As this elementary area revolves about x-axis, we get a ring of volume

$$= \pi[(y + \delta y)^2 - y^2]\,\delta x = \pi(2y\,\delta y + \delta y^2)\,\delta x,$$

exactly in the first power of δx

Hence the total volume of the solid formed by the revolution of the area A about x-axis

$$= \iint_R 2\pi y \, dx\,dy.$$

In polar coordinates, the above formula for the volume becomes

$$\iint_R 2\pi r\sin\theta \cdot r\,dr\,d\theta, \text{ i.e. } \int_{\theta_1}^{\theta_2}\int_{r_1}^{r_2} 2\pi r^2 \sin\theta\, dr\,d\theta.$$

Similarly, the volume of the solid formed by the revolution of the area A about y-axis $= \int_{\theta_1}^{\theta_2}\int_{r_1}^{r_2} 2\pi r^2\cos\theta\, dr\,d\theta$.

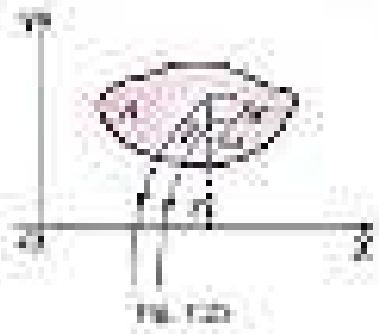

Fig. 7.23

Example 7.30. *Calculate by double integration, the volume generated by the revolution of the cardioid $r = a(1 - \cos\theta)$ about the initial line.*

Solution. Required volume

$$= \int_0^{\pi}\int_0^{a(1-\cos\theta)} 2\pi r^2 \sin\theta\, dr\,d\theta$$

$$= 2\pi\int_0^{\pi}\left[\frac{r^3}{3}\right]_0^{a(1-\cos\theta)}\sin\theta\, d\theta$$

$$= \frac{2\pi a^3}{3}\int_0^{\pi}(1-\cos\theta)^3\sin\theta\, d\theta = \frac{2\pi a^3}{3}\left[\frac{(1-\cos\theta)^4}{4}\right]_0^{\pi} = \frac{8\pi a^3}{3}$$

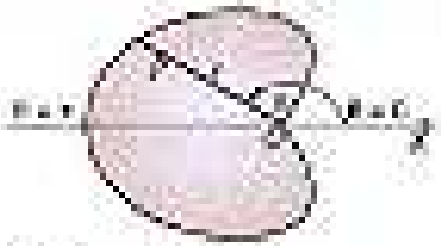
Fig. 7.24

7.13 CHANGE OF VARIABLES

An appropriate choice of co-ordinate system often facilitates the evaluation of a double or a triple integral. By changing the variables, a given integral may be transformed into a simpler integral involving the new variables.

(1) For a double integral, let the variables x, y be changed to the new variables u, v by the transformation

$$x = \phi(u, v),\ y = \psi(u, v)$$

where $\phi(u, v)$ and $\psi(u, v)$ are continuous and have certain first order derivatives in some region R' in the uv-plane which corresponds to the region R in the xy-plane. Then

$$\iint_R f(x, y)\,dx\,dy = \iint_{R'} f[\phi(u, v), \psi(u, v)]\,|J|\,du\,dv \qquad ...(1)$$

where $J = \dfrac{\partial(x, y)}{\partial(u, v)}$ or $\begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix}$

is the Jacobian of transformation* from (x, y) to (u, v) coordinates.

(2) For triple integrals, the formula corresponding to (1) is

$$\iiint_R f(x, y, z)\,dx\,dy\,dz = \iiint_{R'} f[\phi(u, v, w), \psi(u, v, w), \chi(u, v, w)]\,|J|\,du\,dv\,dw$$

where $J = \dfrac{\partial(x, y, z)}{\partial(u, v, w)}$ or $\begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$

is the Jacobian of transformation from (x, y, z) to (u, v, w) coordinates.

Particular cases :

(i) To change rectangular coordinates (x, y) to polar coordinates (r, θ), we have $x = r\cos\theta, y = r\sin\theta$ and

$$J = \frac{\partial(x, y)}{\partial(r, \theta)} = r \qquad \text{[Ex. 6.22, p. 218]}$$

$$\therefore \quad \iint_R f(x, y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta.$$

* For Jacobian, page 218.

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(ii) To change rectangular coordinates (x, y, z) to cylindrical coordinates (r, θ, z)—Fig. 9.27, we have

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$$

and
$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, z)} = r \qquad \text{[Ex. 9.23]}$$

Then $$\iiint_V f(x, y, z)\,dx\,dy\,dz = \iiint_{V'} f(r\cos\theta, r\sin\theta, z)\,r\,dr\,d\theta\,dz$$

(iii) To change rectangular coordinates (x, y, z) to spherical polar coordinates (r, θ, ϕ)—Fig. 9.28, we have

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$$

and
$$J = \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2\sin\theta \qquad \text{[Ex. 9.23]}$$

Then $$\iiint_V f(x, y, z)\,dx\,dy\,dz = \iiint_{V'} f(r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)\,r^2\sin\theta\,dr\,d\theta\,d\phi$$

Example 9.25. Evaluate $\iint_R (x + y)^2\,dx\,dy$, where R is the parallelogram in the xy-plane with vertices $(1, 0)$, $(3, 1)$, $(2, 2)$, $(0, 1)$ using the transformation $u = x + y$ and $v = x - 2y$. *(U.P.T.U., 2005)*

Solution. The region R, i.e., parallelogram $ABCD$ in the xy-plane between the region R', i.e., rectangle $A'B'C'D'$ in the uv-plane as shown in Fig. 9.29, by taking

$$u = x + y \quad \text{and} \quad v = x - 2y \qquad \ldots(1)$$

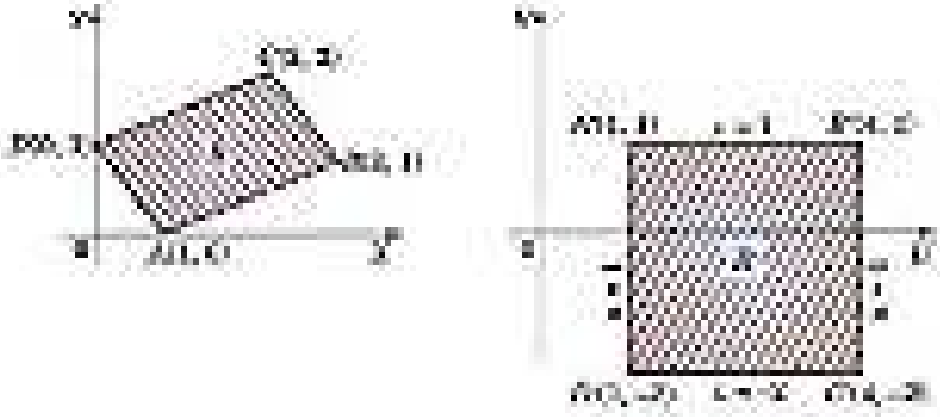

Fig. 9.29

From (1), we have $$x = \frac{1}{3}(2u + v), \quad y = \frac{1}{3}(u - v)$$

$$\therefore \quad \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = -\frac{1}{3}$$

Hence, the given integral

$$= \iint_{R'} u^2 \left|-\frac{1}{3}\right| du\,dv = \int_1^4 \int_{-2}^{1} u^2 \cdot \frac{1}{3}\,dv\,du = \frac{1}{3}\left[\frac{u^3}{3}\right]_1^4 \left[v\right]_{-2}^{1} = 21.$$

Example 9.26. Evaluate $\iint_R xy\sqrt{1 - x - y}\,dx\,dy$, where R is the region bounded by $x = 0$, $y = 0$ and $x + y = 1$ using the transformation $x = u^2 - v^2$, $y = 2uv$. *(Mumbai, 2000)*

Solution. We have $x = r\cos\theta$, $y = r\sin\theta$

$$\therefore \quad J = \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \partial x/\partial r & \partial y/\partial r \\ \partial x/\partial \theta & \partial y/\partial \theta \end{vmatrix} = \begin{vmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{vmatrix} = r$$

Also when $x = 0$, $r\cos\theta = 0$, $\cos\theta = 0$, whence $\theta = \pi/2$; when $x = 2$, $r\cos\theta = 2$, $r = 2\sec\theta$ and when $y = 0$, $r = 1$.

$\therefore$ the limits of r are from 1 to $2\sec\theta$ and limits of θ are from 0 to $\pi/2$.

Then $$\iint_R xy\sqrt{1 - x^2 - y^2}\, dx\, dy = \int_0^{\pi/2}\int_1^{2\sec\theta} r^2\cos\theta\sin\theta\sqrt{1 - r^2}\, dr\, d\theta$$

$$= \int_0^{\pi/2}\int_1^{2\sec\theta} r^3\sin\theta\cos\theta(1 - r^2)^{1/2}\, dr\, d\theta$$

where $x = r\cos\theta$
$dx\, dy = r\, dr\, d\theta$
$x = 0, \theta = \pi/2$
$x = 1, \theta = \pi/3$

$$= \int_1^2 r^3(1 - r^2)^{1/2}\, dr \int_0^{\pi/3}\sin\theta\cos\theta\, d\theta$$

$$= \int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[\frac{r^4}{4}(1 - r^2)^{1/2} - \frac{1}{4}\int r^4\cdot\frac{1}{2}(1 - r^2)^{-1/2}(-2r)\, dr\right]_1^2$$

$$= \frac{1}{2}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[\frac{r^4}{4}(1 - r^2)^{1/2} + \frac{1}{4}\int r^3(1 - r^2)^{1/2}(-2r)\, dr\right]_1^2$$

$$= \frac{1}{4}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[\frac{r^4}{4}(1 - r^2)^{1/2} + \frac{1}{4}\int(1 - r^2)^{3/2}(-2r)\, dr\right]_1^2$$

$$= \frac{1}{4}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[\frac{r^4}{4}(1 - r^2)^{1/2} + \frac{1}{4}\cdot\frac{2}{5}(1 - r^2)^{5/2}\right]_1^2$$

$$= \frac{1}{4}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[\frac{r^4}{4}(1 - r^2)^{1/2} + \frac{1}{10}(1 - r^2)^{5/2}\right]_1^2$$

$$= \frac{1}{4}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[\frac{2^4}{4}(1 - 2^2)^{1/2} + \frac{1}{10}(1 - 2^2)^{5/2}\right]$$

$$= \frac{1}{4}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[-4\sqrt{3} + \frac{1}{10}\cdot 3^2\sqrt{3}\right] = \frac{1}{4}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta\left[-\frac{31}{10}\sqrt{3}\right]$$

$$= -\frac{31}{40}\sqrt{3}\int_0^{\pi/3}\sin\theta\cos\theta\, d\theta = -\frac{31}{40}\sqrt{3}\left[\frac{\sin^2\theta}{2}\right]_0^{\pi/3} = -\frac{31}{40}\sqrt{3}\cdot\frac{1}{2}\left[\frac{3}{4} - 0\right] = -\frac{31}{64}.$$

Example 7.25. Evaluate $\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}\, dx\, dy$ by changing to polar coordinates. (Anna, 2005)

Hence show that $\int_0^\infty e^{-x^2}\, dx = \frac{\sqrt{\pi}}{2}$. (Madras, 2003; U.P.T.U., 2003; J.N.T.U., 2006)

Solution. The region of integration being the first quadrant of the xy plane, r varies from 0 to ∞ and θ varies from 0 to $\pi/2$. Hence,

$$I = \int_0^\infty\int_0^\infty e^{-(x^2+y^2)}\, dx\, dy = \int_{\theta=0}^{\pi/2}\int_{r=0}^{\infty} e^{-r^2}\, r\, dr\, d\theta$$

$$= -\frac{1}{2}\int_0^{\pi/2}\left\{\int_0^\infty e^{-r^2}(-2r)\, dr\right\} d\theta = -\frac{1}{2}\int_0^{\pi/2}\left[e^{-r^2}\right]_0^\infty d\theta = \frac{1}{2}\int_0^{\pi/2} d\theta = \frac{\pi}{4}. \quad \ldots(i)$$

Also $$I = \int_0^\infty e^{-x^2}\, dx\int_0^\infty e^{-y^2}\, dy = \left\{\int_0^\infty e^{-x^2}\, dx\right\}^2 \quad \ldots(ii)$$

Thus, from (i) and (ii), we have $\int_0^\infty e^{-x^2} = \frac{\sqrt{\pi}}{2}$. $\ldots(iii)$

Example 7.26. Find the volume bounded by the paraboloid $x^2 + y^2 = az$, the cylinder $x^2 + y^2 = 2ay$ and the plane $z = 0$.

Solution. The required volume is found by integrating $z = (x^2 + y^2)/a$ over the circle $x^2 + y^2 = 2ay$.

Changing to polar coordinates in the xy-plane, we have $x = r\cos\theta$, $y = r\sin\theta$ so that $r = 2a\sin\theta$ are the polar equation of the circle $x^2 + y^2 = 2ay$ or $r = 2a\sin\theta$.

To cover the circle, r varies from 0 to $2a\sin\theta$ and θ varies from 0 to π (Fig. 7.16).

Hence the required volume

$$= \int_0^\pi\int_0^{2a\sin\theta} z\, r\, dr\, d\theta = \frac{1}{a}\int_0^\pi d\theta\int_0^{2a\sin\theta} r^3\, dr$$

$$= \frac{1}{4a}\int_0^\pi d\theta\left[\frac{r^4}{4}\right]_0^{2a\sin\theta} = 4a^3\int_0^\pi\sin^4\theta\, d\theta = \frac{3\pi a^3}{2}.$$

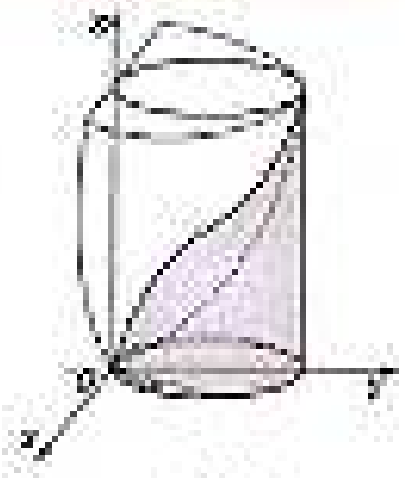

Fig. 7.16

Example 7.37. Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ (Anna, 2004; M.T.U., 2003)

Solution. Changing to polar spherical coordinates by putting

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$$

we have $dx\,dy\,dz = r^2\sin\theta\,dr\,d\theta\,d\phi$.

Also the volume of the sphere is 8 times the volume of its portion in the positive octant for which r varies from 0 to a, θ varies from 0 to $\pi/2$ and ϕ varies from 0 to $\pi/2$.

$\therefore$ volume of the sphere

$$= 8\int_0^{\pi/2}\int_0^{\pi/2}\int_0^a r^2\sin\theta\,dr\,d\theta\,d\phi = 8\left(\int_0^a r^2\,dr\right)\left(\int_0^{\pi/2}\sin\theta\,d\theta\right)\left(\int_0^{\pi/2}d\phi\right)$$

$$= 8\left[\frac{r^3}{3}\right]_0^a\left[-\cos\theta\right]_0^{\pi/2}\left[\phi\right]_0^{\pi/2} = 8\left(\frac{a^3}{3}\right)(1)\left(\frac{\pi}{2}\right) = \frac{4}{3}\pi a^3.$$

Example 7.38. Find the volume of the portion of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 = ay$. (Kerala, 2003)

Solution. The required volume is easily found by changing to cylindrical coordinates (ρ, ϕ, z). We have, here

$$x = \rho\cos\phi,\ y = \rho\sin\phi,\ z = z$$

and

$$J = \frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \rho.$$

Then the equation of the sphere becomes $\rho^2 + z^2 = a^2$ and that of cylinder becomes $\rho = a\sin\phi$.

The volume inside the cylinder bounded by the sphere is twice the volume shown shaded in the Fig. 7.35 for which z varies from 0 to $\sqrt{(a^2 - \rho^2)}$, ρ varies from 0 to $a\sin\phi$ and ϕ varies from 0 to π.

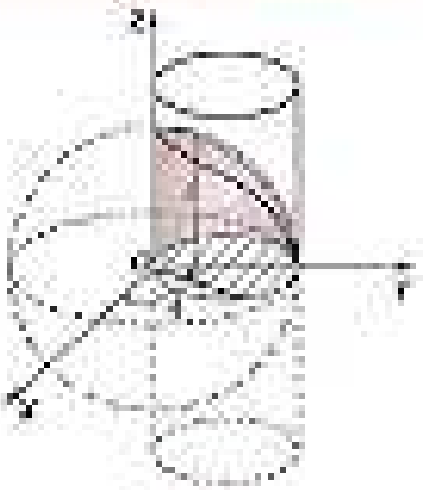

Fig. 7.35

Hence the required volume $= 2\int_0^{\pi}\int_0^{a\sin\phi}\int_0^{\sqrt{(a^2-\rho^2)}}\rho\,dz\,d\rho\,d\phi$

$$= 2\int_0^{\pi}\int_0^{a\sin\phi}\rho\sqrt{(a^2-\rho^2)}\,d\rho\,d\phi = 2\int_0^{\pi}\left[-\frac{1}{3}(a^2-\rho^2)^{3/2}\right]_0^{a\sin\phi}d\phi$$

$$= \frac{2a^3}{3}\int_0^{\pi}(1-\sin^3\phi)\,d\phi = \frac{2a^3}{3}\left(\frac{\pi}{2} - \frac{4}{3}\right).$$

Example 7.39. Evaluate $\int_0^1\int_0^{\sqrt{1-x^2}}\int_{\sqrt{x^2+y^2}}^1 \frac{dx\,dy\,dz}{\sqrt{x^2+y^2+z^2}}$. (U.P.T.U., 2005)

Solution. We change to spherical polar coordinates (r, θ, ϕ), so that

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$$

and

$$J = r^2\sin\theta,\ x^2 + y^2 + z^2 = r^2.$$

The region of integration is bounded by the cone $z^2 = x^2 + y^2$ and the cylinder $x^2 + y^2 = 1$ bounded by the plane $z = 1$ in the positive octant (Fig. 7.36). Hence θ varies from 0 to $\pi/4$, r varies from 1 to $\sec\theta$ and ϕ varies from 0 to $\pi/2$.

$\therefore$ given integral becomes

$$\int_0^{\pi/2}\int_0^{\pi/4}\int_0^{\sec\theta}\frac{1}{r}\,r^2\sin\theta\,dr\,d\theta\,d\phi = \int_0^{\pi/2}d\phi\int_0^{\pi/4}\left[\frac{r^2}{2}\right]_0^{\sec\theta}\sin\theta\,d\theta$$

$$= \frac{\pi}{2}\int_0^{\pi/4}\frac{\sec^2\theta}{2}\sin\theta\,d\theta = \frac{\pi}{4}\int_0^{\pi/4}\sec\theta\tan\theta\,d\theta = \frac{\pi}{4}\left[\sec\theta\right]_0^{\pi/4} = \frac{\pi(\sqrt{2}-1)}{4}.$$

Fig. 7.36

Example 7.80. Find the volume of the solid surrounded by the surface $(x^2/a^2 + y^2/b^2 + z^2/c^2)^2 = x^2/a^2 + y^2/b^2 - z^2/c^2$. (Bhopal, 2005; Kanpur, 2006; Kerala, 2006)

Solution. Changing the variables x, y, z to X, Y, Z where, $x/a = X$, $y/b = Y$, $z/c = Z$ i.e., $x = aX$, $y = bY$, $z = cZ$ so that $J = \partial(x, y, z)/\partial(X, Y, Z) = abc$.

$\therefore$ required volume $= \iiint dx\,dy\,dz = abc \iiint X^2Y^2Z^2\,dX\,dY\,dZ$

taken throughout the sphere $X^2 + Y^2 + Z^2 = 1$. ...(1)

Now change X, Y, Z to spherical polar coordinates r, θ, ϕ so that $X = r\sin\theta\cos\phi$, $Y = r\sin\theta\sin\phi$, $Z = r\cos\theta$, and $dX\,dY\,dZ = r^2\sin\theta\,dr\,d\theta\,d\phi = J\,dr\,d\theta\,d\phi$. To describe the positive octant of the sphere (1), r varies from 0 to 1, θ from 0 to $\pi/2$ and ϕ from 0 to $\pi/2$.

$\therefore$ required volume $= 8abc \times 0 \int_0^{\pi/2}\int_0^{\pi/2}\int_0^1 r^2\sin^2\theta\cos^2\phi \cdot r^2\sin\theta\,dr\,d\theta\,d\phi = r^4\sin^3\theta\cos^2\phi\,dr\,d\theta\,d\phi$, $r^4\sin\theta$ is the Jacobian

$= 8abc \int_0^1 r^4\,dr \int_0^{\pi/2}\sin^3\theta\,d\theta \int_0^{\pi/2}\cos^2\phi\,d\phi = \frac{8abc}{5}\cdot\frac{2}{3}\cdot\frac{\pi}{4} = \frac{4\pi abc}{15}$.

EXERCISE 7.5

Evaluate the following integrals by changing to polar coordinates :

1. $\int_0^a\int_0^{\sqrt{a^2-y^2}} (x^2+y^2)\,dy\,dx$. (Delhi, 2004) 2. $\int_0^a\int_0^{\sqrt{2ax-x^2}} \frac{x\,dx\,dy}{\sqrt{x^2+y^2}}$ (Anna, 2005)

3. $\int_0^{2a}\int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} \frac{x^2-y^2}{x^2+y^2}\,dx\,dy$ (Bharathiar, 2006)

4. $\iint (x^2+y^2)^{1/2}\,dx\,dy$ over the positive quadrant of $x^2+y^2=4$ in the plane $z=0$. (U.P.T.U., 2003)

5. $\int_0^{a/\sqrt{2}}\int_{y}^{\sqrt{a^2-y^2}} y\,dx\,dy$ over the region of the circle $x^2+y^2=a^2$. (Bharathiar, 2007)

6. Transform the following to polar coordinates and then evaluate $\int_0^a\int_y^a \frac{x}{x^2+y^2}\,dx\,dy$. (U.P.T.U., 2006)

7. $\iint x^2\,dx\,dy$ over the area bounded by $x^2+y^2=ax$ and lying above the x-axis. (Bharathiar, 2006)

8. By using the transformation $x+y=u$, $y=uv$, show that $\int_0^1\int_0^{1-x} xy^{n-1}(1-x-y)^{m-1}\,dy\,dx = \frac{1}{n}\beta(m, n+1)$. (U.P.T.U., 2003)

9. Transform $\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} \frac{x^2+y^2}{\sqrt{4a^2-x^2-y^2}}\,dx\,dy$ into polar coordinates and then evaluate it. (U.P.T.U., 2005)

Evaluate the following integrals by changing to spherical coordinates :

10. $\int_0^a\int_0^{\sqrt{a^2-x^2}}\int_0^{\sqrt{a^2-x^2-y^2}} \frac{dx\,dy\,dz}{\sqrt{a^2-x^2-y^2-z^2}}$ (U.P.T.U., 2006; Kerala, 2005)

11. $\iiint_V \frac{dx\,dy\,dz}{x^2+y^2+z^2}$, where V is the volume of the sphere $x^2+y^2+z^2=a^2$. (Anna, 2005)

12. Evaluate $\iiint \frac{dx\,dy\,dz}{(1+x+y+z)^3}$ over the volume of the tetrahedron $x=0, y=0, z=0, x+y+z=1$. (Bharathiar, 2007)

13. Show that $\iiint \frac{dx\,dy\,dz}{xyz\sqrt{1-x^2-y^2-z^2}} = \frac{\pi^3}{8}$, the integral being extended for all the values of the variables for which the expression is real. (U.P.T.U., 2005)

14. $\iiint z^2\,dx\,dy\,dz$, taken over the volume bounded by the surfaces $x^2+y^2=a^2$, $x+y=a$, $x=0$, $y=0$, $z=0$ and $z=h$.

12. Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = y + 4$ and $z = 0$. (8.38 m³, 2003)
13. Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the sphere $x^2 + y^2 + z^2 = 16$. (Approx. 2000)
14. Find the volume bounded by the planes $x + y + z = a$, $x = 0$, $y = 0$ and $z = 0$.
15. Find the volume common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$. (16/3 a³, 2006)
16. Find the volume cut off from the cylinder $x^2 + y^2 = ax$ by the planes $z = 0$ and $z = cx$. (1/3 a²c, 2006)
17. Find the volume enclosed by the cylinders $x^2 + y^2 = 2ay$ and $z^2 = 2ax$. (King's Lynn, 2002)
18. Find the volume of the cylinder $x^2 + y^2 = 2ax$ which is between the paraboloid $x^2 + y^2 = 2az$ and the xy-plane.
19. Find the volume bounded by the cylinder $x^2 + y^2 = a^2$ and the hyperboloid $z^2 = x^2 + y^2 + a^2$.
20. Find the volume of the region bounded by $x = y^2 + z^2$, $y + z = 2$, $x = 0$, $y = 0$ and $z = 0$.
21. Prove, by using a double integral, that the volume generated by the revolution of the circle $x^2 + y^2 = a^2$ about the line $x = b$ ($b > a$) is $2\pi^2 a^2 b$. (12.57 $a^2 b$, 2002)
22. Evaluate $\iiint_V (x + y + z)\,dx\,dy\,dz$ where V is the region of the first octant bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$. (See Fig. 3.34)
23. Find the volume of the tetrahedron bounded by the coordinate planes and the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. (abc/6, 2003)
24. Find, by integration, the volume of the solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = 0$ and $z = 4$. (16π, 2001)
25. Find the volume of the solid bounded by the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ and $z = 0$. (8, 2004)
26. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)
27. Find the volume of the solid bounded by the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ and $z = 0$. (8, 2004)
28. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)
29. Find the volume of the solid bounded by the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ and $z = 0$. (8, 2004)
30. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)
31. Find the volume of the solid bounded by the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ and $z = 0$. (8, 2004)
32. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)
33. Find the volume of the solid bounded by the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ and $z = 0$. (8, 2004)
34. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)
35. Find the volume of the solid bounded by the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ and $z = 0$. (8, 2004)
36. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)
37. Find the volume of the solid bounded by the surface $z = 4 - x^2$ and the planes $x = 0$, $y = 0$, $y = 2$ and $z = 0$. (8, 2004)
38. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)
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100. Find the volume of the solid bounded by the surface $z = 4 - y^2$ and the planes $x = 0$, $x = 1$, $z = 0$ and $y = 0$. (3, 2005)

3.8 AREA OF A CURVED SURFACE

Consider a point P on the surface $S: z = f(x, y)$. Let its projection on the xy-plane be the region A. Divide R into rectangular elements by drawing lines parallel to the axes of X and Y (Fig. 3.31).

On the element $\delta x\,\delta y$ as base, erect a cylinder having generators parallel to OZ and meeting the surface S in an element of area δS.

Area $\delta x\,\delta y$ is the projection of δS on the xy-plane.

$\therefore\ \delta x\,\delta y = \delta S \cdot \cos\gamma$, where γ is the angle between the xy-plane and the tangent plane to S at $P(x, y, z)$, or it is the angle between the Z-axis and the normal to S at P = angle NPN'.

Now since the direction ratios of the normal to the surface $F(x, y, z) = 0$ are proportional to

$$\frac{\partial F}{\partial x},\ \frac{\partial F}{\partial y},\ \frac{\partial F}{\partial z},$$

$\therefore$ the direction cosines of the normal to $S[F = f(x, y) - z]$ are proportional to $-\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}$, 1 and those of the z-axis are 0, 0, 1.

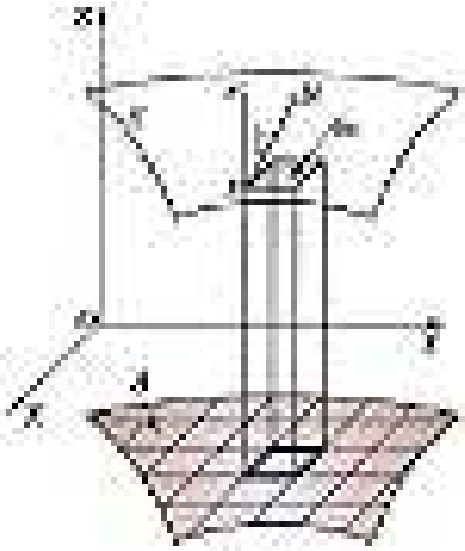

Fig. 3.31

Hence $$\cos\gamma = \frac{1}{\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}} \quad \therefore\ \delta S = \frac{\delta x\,\delta y}{\cos\gamma} = \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}\ \delta x\,\delta y$$

Hence $$S = \lim_{\substack{\delta x \to 0 \\ \delta y \to 0}} \sum \delta S = \iint_R \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}\ dx\,dy$$

Similarly, if R' and R'' be the projections of S on the yz and zx planes respectively, then

$$S = \iint_{R'} \sqrt{\left(\frac{\partial x}{\partial y}\right)^2 + \left(\frac{\partial x}{\partial z}\right)^2 + 1}\ dy\,dz$$

and $$S = \iint_{R''} \sqrt{\left(\frac{\partial y}{\partial z}\right)^2 + \left(\frac{\partial y}{\partial x}\right)^2 + 1}\ dz\,dx.$$

Example 9.21. *Find the area of the portion of the plane $x + y + z = 4a$ lying inside the cylinder $x^2 + y^2 = 4ax$.*

Solution. Figure 9.22 shows one-eighth of the required area. Its projection on the xy-plane is a quadrant of the circle $x^2 + y^2 = 4ax$.

For the cylinder $\quad x^2 + y^2 = 4ax, \qquad \ldots(1)$

we have $\quad \frac{\partial z}{\partial x} = -1, \frac{\partial z}{\partial y} = -1$

so that $\quad \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1 = \frac{x^2 + y^2}{y^2} = \frac{4a}{4a - x}$ [from (1)]

Hence the required surface area = 8 (surface area of the upper portion of (1) lying within the cylinder $x^2 + y^2 = 4ax$ in the positive octant)

$$= 8\int_0^{4a}\int_0^{\sqrt{4ax - x^2}} \frac{2}{\sqrt{4a - x}}\, dx\, dy = 16\int_0^{4a} dx = 32a^2 \text{ sq. units.}$$

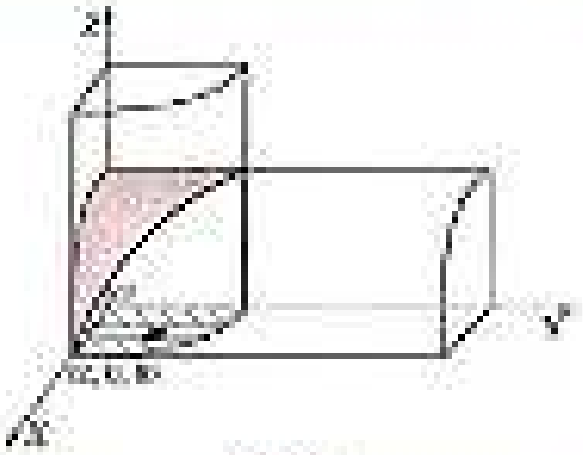

Fig. 9.22

Example 9.22. *Find the area of the portion of the sphere $x^2 + y^2 + z^2 = 9$ lying inside the cylinder $x^2 + y^2 = 3y$.*

Solution. Figure 9.23 shows one-fourth of the required area. Its projection on the xy-plane is the semi-circle $x^2 + y^2 = 3y$ bounded by the Y-axis.

For the sphere

$$x^2 + y^2 + z^2 = 9, \frac{\partial z}{\partial x} = -\frac{x}{z} \text{ and } \frac{\partial z}{\partial y} = -\frac{y}{z}$$

$$\therefore \quad \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1 = \frac{x^2 + y^2 + z^2}{z^2}$$

$$= \frac{9}{9 - x^2 - y^2} = \frac{9}{9 - r^2} \quad \text{where } x = r\cos\theta, y = r\sin\theta.$$

Using polar coordinates, the required area is found by integrating $3\sqrt{\frac{9 - r^2}{9 - r^2}}$ over the semi-circle $r = 3\sin\theta$, in which r varies from 0 to $3\sin\theta$ and θ varies from 0 to $\pi/2$.

Hence the required surface area

$$= 4\int_0^{\pi/2}\int_0^{3\sin\theta} \frac{3}{\sqrt{9 - r^2}}\, r\, dr\, d\theta = -6\int_0^{\pi/2} \left[2\sqrt{9 - r^2}\right]_0^{3\sin\theta} d\theta$$

$$= 36\int_0^{\pi/2} (1 - \cos\theta)\, d\theta = 36\left[\theta - \sin\theta\right]_0^{\pi/2} = 18(\pi - 2) \text{ sq. units.}$$

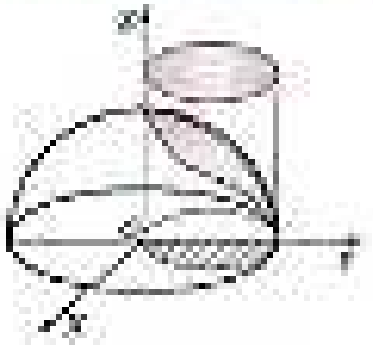

Fig. 9.23

EXERCISE 9.3

1. Show that the surface of the sphere $x^2 + y^2 + z^2 = a^2$ is $4\pi a^2$.
2. Find the area of the portion of the cylinder $x^2 + y^2 = by$ intercepted by the plane $x + y + z = a$.
3. Find the area of the portion of the plane $lx + my + nz = p$ lying in the first octant.
4. Find the area of the surface of the sphere $x^2 + y^2 + z^2 = a^2$ cut out by the cylinder $x^2 + y^2 = ax$.
5. Compute the area of that part of the plane $x + y + z = a$ which lies in the first octant and is bounded by the cylinder $x^2 + y^2 = a^2$. (Bombay, 2003)

9.23 CALCULATION OF MASS

(a) For a plane lamina, if the surface density at the point $P(x, y)$ be $\rho = f(x, y)$ then the elementary mass $\delta F = \rho\,\delta x\,\delta y$.

$\therefore$ total mass of the lamina $= \iint \rho\,dx\,dy$ ...(1)

with integrals embracing the whole area of the lamina.

In polar coordinates, taking $\rho = \delta r, \theta$ at the point $P(r, \theta)$,

total mass of the lamina $= \iint \rho r\,d\theta\,dr$ ...(2)

(b) For a solid, if the density at the point $P(x, y, z)$ be $\rho = f(x, y, z)$, then

total mass of the solid $= \iiint \rho\,dx\,dy\,dz$ with appropriate limits of integration.

Example 9.23. *Find the mass of the tetrahedron bounded by the coordinate planes and the plane* $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, *the variable density* $\rho = kxyz$. *(Anna Univ., 2003; U.P.T.U., 2007)*

Solution. Elementary mass on $F = \rho\,\delta x\,\delta y\,\delta z$.

$\therefore$ the whole mass $= \iiint \rho\,dx\,dy\,dz$,

the integrals embracing the whole volume $OABC$ (Fig. 9.31). The limits here are from 0 to $x = a(1 - y/b - z/c)$.

The limits for y are from 0 to $y = b(1 - z/c)$ and limits for z are from 0 to c.

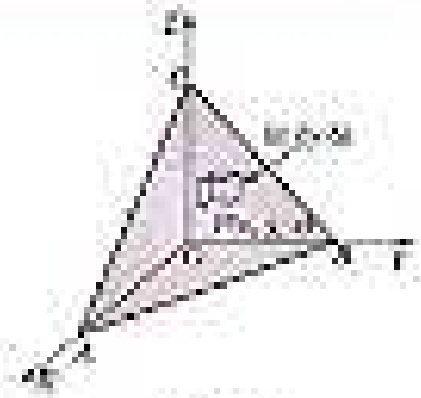

Fig. 9.31

Hence the required mass

$$= \int_0^c \int_0^{b(1-z/c)} \int_0^{a(1-y/b-z/c)} kxyz\,dx\,dy\,dz$$

$$= k\int_0^c \int_0^{b(1-z/c)} yz\left[\frac{x^2}{2}\right]_0^{a(1-y/b-z/c)} dy\,dz$$

$$= k\int_0^c \int_0^{b(1-z/c)} yz\frac{a^2}{2}\left(1-\frac{y}{b}-\frac{z}{c}\right)^2 dy\,dz$$

$$= \frac{ka^2}{2}\int_0^c \int_0^{b(1-z/c)} z\left[\left(1-\frac{z}{c}\right)^2 y - 2\left(1-\frac{z}{c}\right)\frac{y^2}{b} + \frac{y^3}{b^2}\right] dy\,dz$$

$$= \frac{ka^2}{2}\int_0^c z\left[\left(1-\frac{z}{c}\right)^2\frac{y^2}{2} - 2\left(1-\frac{z}{c}\right)\frac{y^3}{3b} + \frac{y^4}{4b^2}\right]_0^{b(1-z/c)} dz$$

$$= \frac{ka^2b^2}{2}\int_0^c z\left[\frac{b^2}{2}\left(1-\frac{z}{c}\right)^4 - \frac{2b^2}{3}\left(1-\frac{z}{c}\right)^4 + \frac{b^2}{4}\left(1-\frac{z}{c}\right)^4\right] dz = \frac{ka^2b^2}{24}\int_0^c z\left(1-\frac{z}{c}\right)^4 dz = \frac{ka^2b^2c^2}{720}.$$

9.24 CENTRE OF GRAVITY

(a) To find the C.G. $(\bar{x}, \bar{y})$ of a plane lamina, take the element of mass $\rho\,\delta x\,\delta y$ at the point $P(x, y)$. Then

$$\bar{x} = \frac{\iint x\rho\,dx\,dy}{\iint \rho\,dx\,dy},\quad \bar{y} = \frac{\iint y\rho\,dx\,dy}{\iint \rho\,dx\,dy}$$ with the integrals embracing the whole lamina.

With the polar coordinates, take the elementary mass as $\rho\,\delta r\,r\,\delta\theta$ at the point $P(r, \theta)$ so that $x = r\cos\theta$, $y = r\sin\theta$.

$$\therefore \quad \bar{x} = \frac{\iint r\cos\theta\,\rho\,r\,dr\,d\theta}{\iint \rho\,r\,dr\,d\theta},\quad \bar{y} = \frac{\iint r\sin\theta\,\rho\,r\,dr\,d\theta}{\iint \rho\,r\,dr\,d\theta}$$

(b) To find the C.G. $(\bar{x}, \bar{y}, \bar{z})$ of a solid, take an element of mass $\rho\,dx\,dy\,dz$ and name the point $P(x, y, z)$. Then

$$\bar{x} = \frac{\iiint_{\text{solid}} x\rho\,dx\,dy\,dz}{\iiint_{\text{solid}} \rho\,dx\,dy\,dz}, \quad \bar{y} = \frac{\iiint_{\text{solid}} y\rho\,dx\,dy\,dz}{\iiint_{\text{solid}} \rho\,dx\,dy\,dz} \text{ and } \bar{z} = \frac{\iiint_{\text{solid}} z\rho\,dx\,dy\,dz}{\iiint_{\text{solid}} \rho\,dx\,dy\,dz}$$

Example 7.34. *Find by double integration, the centre of gravity of the area of the cardioid $r = a(1 + \cos\theta)$.*

Solution. The cardioid being symmetrical about the initial line, its C.G. lies on OX, i.e., $\bar{y} = 0$ (Fig. 7.35).

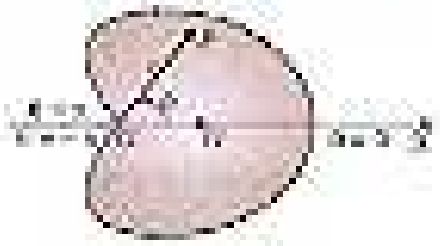

Fig. 7.35

$$\bar{x} = \frac{\iint \rho \times x\,dx\,dy}{\iint \rho\,dx\,dy} = \frac{\int_0^{\pi} \int_0^{a(1+\cos\theta)} r\cos\theta \cdot r\,dr\,d\theta}{\int_0^{\pi} \int_0^{a(1+\cos\theta)} r\,dr\,d\theta}$$

$$= \frac{\int_0^{\pi} \cos\theta \left[\frac{r^3}{3}\right]_0^{a(1+\cos\theta)} d\theta}{\int_0^{\pi} \left[\frac{r^2}{2}\right]_0^{a(1+\cos\theta)} d\theta} = \frac{2a}{3} \frac{\int_0^{\pi} \cos\theta(1+\cos\theta)^3\,d\theta}{\int_0^{\pi} (1+\cos\theta)^2\,d\theta}$$

$$= \frac{2a}{3} \cdot \frac{2\int_0^{\pi} \cos^4\frac{\theta}{2}\cos^2\frac{\theta}{2}\,d\theta}{2\int_0^{\pi} \cos^4\frac{\theta}{2}\,d\theta} \qquad \left[\because \int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx \text{ if } f(2a-x) = f(x)\right]$$

$$= \frac{2a}{3} \cdot \frac{2\int_0^{\pi/2} (2\cos^2\theta - \cos^2\theta)\,d\theta}{2\int_0^{\pi/2} (2\cos^2\theta)^2\,d\theta} \qquad \left[\text{Put } \frac{\theta}{2} = \theta \text{ in the numerator of the first fraction}\right] = \frac{2a}{3} \cdot \frac{\frac{1}{2}\cdot\frac{\pi}{2} + \frac{3}{4}\cdot\frac{\pi}{2}}{\frac{\pi}{2}+\frac{1}{2}\cdot\frac{\pi}{2}} = \frac{5a}{6}$$

Hence the C.G. of the cardioid is at $G\left(\frac{5a}{6}, 0\right)$.

Example 7.35. *Using double integration, find the C.G. of a lamina in the shape of a quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the density being proportional to the product of its coordinates.*

Solution. Let $G(\bar{x}, \bar{y})$ be the C.G. of the lamina OAB (Fig. 7.36), so that

$$\bar{x} = \frac{\iint kxy \cdot x\,dx\,dy}{\iint kxy \cdot dx\,dy} = \frac{\iint x^2 y\,dx\,dy}{\iint xy\,dx\,dy}$$

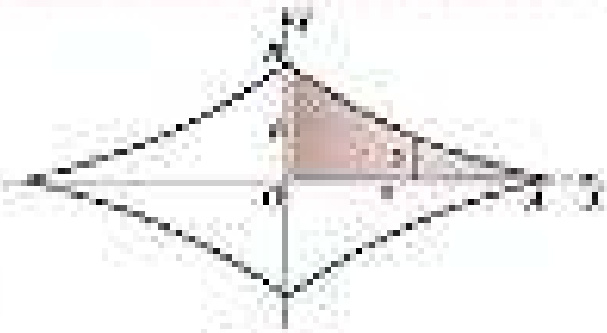

Fig. 7.36

where the integrals are taken over the area OAB so that y varies from 0 to y (to be found from the equation of the curve) and then x varies from 0 to a.

Thus

$$\bar{x} = \frac{\int_0^a \int_0^y x^2 y\,dy\,dx}{\int_0^a \int_0^y xy\,dy\,dx} = \frac{\int_0^a x^2 \left|\frac{y^2}{2}\right|_0^y dx}{\int_0^a x \left|\frac{y^2}{2}\right|_0^y dx} = \frac{\int_0^a x^2 y^2\,dx}{\int_0^a xy^2\,dx}$$

For any point on the curve, we have

$$x = a\cos^3\theta, y = b\sin^3\theta \text{ so that}$$

$$dx = -3a\cos^2\theta\sin\theta\,d\theta$$

Also when $x = 0, \theta = \pi/2$; when $x = a, \theta = 0$.

Hence $$\bar{x} = \frac{\int_{\pi/2}^{\pi} a^3 \cos^3\theta\, b^3 \sin^3\theta\, (-3a\cos^2\theta \sin\theta)\, d\theta}{\int_{\pi/2}^{\pi} a\cos^3\theta\, b^3 \sin^3\theta\, (-3a\cos^2\theta \sin\theta)\, d\theta}$$

$$= a\frac{\int_0^{\pi/2} \sin^7\theta \cos^5\theta\, d\theta}{\int_0^{\pi/2} \sin^7\theta \cos^5\theta\, d\theta} = \frac{128}{429}a$$

Similarly, $$\bar{y} = \frac{\int_0^{\pi} \int_0^{y} dx\, .\, y\, dx dy}{\int_0^{\pi} \int_0^{y} dy\, .\, dx dy} = \frac{128}{429}b$$. Hence the required C.G. is $$G\left(\frac{128}{429}a, \frac{128}{429}b\right)$$.

7.22 CENTRE OF PRESSURE

Consider plane area A immersed vertically in a homogeneous liquid. Take the line of intersection of the given plane with the free surface X of the liquid as the x-axis and any line lying in this plane and perpendicular to it downwards as the y-axis (Fig. 7.37).

If p be the pressure at the point $P(x, y)$ of the area A, then the pressure on an elementary area $\delta x\delta y$ at P is $p\delta x\delta y$ which is normal to the plane.

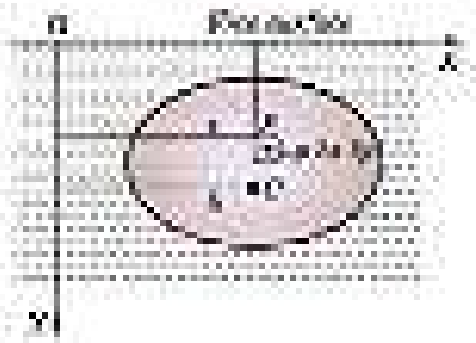

Fig. 7.37

$\therefore$ the resultant pressure on $A = \iint p\delta x\delta y$.

If the resultant pressure acting at $C(\bar{x}, \bar{y})$ is equivalent to pressure at various points such as $p\delta x\delta y$ distributed over the whole area A, then C is called the centre of pressure.

$\therefore$ taking the moment of the resultant pressure at C and the sum of the moments of the individual pressures such as $p\delta x\delta y$ at $P(x, y)$ about the y-axis, we get

$$\bar{x}\iint p\delta x\delta y = \iint x\, .\, p\delta x\delta y, \quad \text{i.e., } \bar{x} = \iint x\, .\, dxdy \Big/ \iint p\delta x\delta y$$

Similarly, taking moments about x-axis, we have

$$\bar{y} = \iint y\, .\, p\delta x\delta y \Big/ \iint p\delta x\delta y$$ with integrals extending the whole of the area A.

While using polar coordinates, replace x by $r\cos\theta$, y by $r\sin\theta$ and $\delta x\delta y$ by $r\, d\theta\, dr$ in the above formulae.

Example 7.26. A horizontal boiler has a flat bottom and its ends are semi-circular and are of diameter $2k$ metres. If it is filled to a depth k metres, show that the depth of the centre of pressure of each end is $\frac{k}{2}\left(1 + \frac{1}{3\pi}\right)$ metres approximately.

Solution. Let the semi-circle $x^2 + y^2 = a^2$ represent an end of the given boiler (Fig. 7.38). By symmetry, its centre of pressure lies on OY.

If w is the weight of water per unit volume, then the pressure p at the point $P(x, y) = w(k - y)$.

$\therefore$ the height $\bar{h}$ of the C.P. above OX, is given by

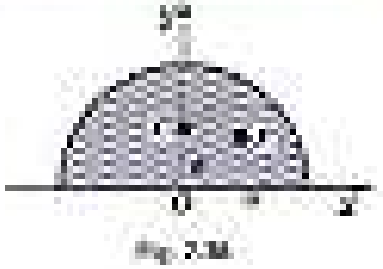

Fig. 7.38

$$\bar{h} = \frac{\iint y\, .\, p\delta x\delta y}{\iint p\delta x\delta y} = \frac{\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} w(k-y)y\, dy\, dx}{\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} w(k-y)\, dy\, dx}$$

$$= \frac{\int_{-a}^{a} wk^2/2 - y^2/2\Big]_0^{\sqrt{a^2-x^2}} dx}{\int_{-a}^{a} wy - y^2/2\Big]_0^{\sqrt{a^2-x^2}} dx} = \frac{\int_{-a}^{a}\left[\frac{1}{2}k^2 - a^2 - \frac{1}{3}y^2 - a^2y^{3/2}\right]dx}{\int_{-a}^{a}\left[wk^2 - a^2y^{3/2} - \frac{1}{2}(a^2 - x^2)\right]dx}$$

Now put $x = a\cos\theta$, so that $dx = -a\sin\theta\,d\theta$.
Also when $x = -a$, $\theta = -\pi/2$, and when $x = a$, $\theta = \pi/2$.

$$\therefore \quad \bar{x} = \frac{\int_{-\pi/2}^{\pi/2} \left|\frac{a^2}{2}\cos^2\theta - \frac{a^2}{3}\cos^3\theta\right| a\cos\theta\, d\theta}{\int_{-\pi/2}^{\pi/2} \left|a^2\cos\theta - \frac{a^2}{2}\cos^2\theta\right| a\cos\theta\, d\theta}$$

$$= \frac{\frac{1}{3}}{\frac{3}{2}} \frac{2\int_0^{\pi/2} (3\cos^2\theta - 2\cos^3\theta)\cos\theta\, d\theta}{2\int_0^{\pi/2} (2\cos^2\theta - \cos^3\theta)\cos\theta\, d\theta} = \frac{2}{9}\left[\frac{16 - 3\pi}{3\pi - 4}\right] = 0.2\text{ nearly.}$$

Hence, the depth of the C.P. = $a - \bar{x} = a - 0.2a = 0.8a$ approximately.

EXERCISE 5.2

1. A function is bounded by the curves $y = x^2 - 4x$ and $y = 2x$. If the density at any point is given by $4xy$, find its mass, considering the area of the function.
2. Find the mass of a function in the form of quadrant $x \geq 0, y \geq 0$ whose density at any point varies as the square of its distance from the origin.
3. Find the mass of a solid in the form of the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$, if the density at any point is xyz.
4. Find the centroid of the area enclosed by the parabola $y^2 = 4ax$, the axis of x and its latus rectum.
5. The density at any point (x, y) of a function is $\rho = x^2y$, where x and y are constants. The function is bounded by the lines $x = 0$, $y = 0$, $x + y = 1$. Find the position of its centre of gravity.
6. Find the centroid of the region of the function $y = x^2$, $x = 0$, $x = 2$.
7. A plane in the form of a quadrant of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is in equilibrium, but varying thickness, the thickness at any point being proportional to the product of the distances of that point from the axes. Show that its centre of gravity is at the centroid of the quadrant of the circle $x^2 + y^2 = a^2$. (Nagpur, 2007)
8. In a semi-circular disc bounded by a diameter DC, the mass per unit area at any point varies as the distance from D. Find the position of the centre of gravity.
9. Find the centroid of the curvilinear triangle bounded by the coordinate axes and the parabola $x = 4 - y^2$, if the density at any point is equal to the distance from the line $x = 4$.
10. Find $\bar{y}$, where $(\bar{x}, \bar{y}, \bar{z})$ is the centroid of the region R bounded by the parabolic cylinder $y = 4 - x^2$ and the planes $z = 0$, $y = 0$, $y + z = 6$. (Assume that the density is constant).
11. If the density at any point of the solid bounded by the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ varies as xyz, find the coordinates of the C.G. of the solid. (V.T.U., 2008)
12. A hydrostatic boiler has a flat bottom and the cross-section of a vertical plane side is bounded by an ellipse whose semi-axis of length b ft. upwards. Determine the depth of the centre of pressure of a door covering the whole of the side if it is just full.
13. A quadrant of a circle is just immersed vertically in a heavy homogeneous liquid with one edge in the surface. Find the centre of pressure.
14. Find the depth of the centre of pressure of a square lamina immersed in the liquid, so that one edge is in the surface and the diagonal vertical.
15. Find the centre of pressure of a triangular lamina immersed in a homogeneous liquid with one side in the free surface. (U.P.T.U., 2002)
16. A vertical cross-section of a lock is a trapezium in which the parallel sides are 10 m and 4 m long and the distance between them is 15 m. If the density at any point of the fluid varies as the depth of the point below the free surface, find the position of the centre of pressure of the lamina.

5.12 (1) MOMENT OF INERTIA

If a particle of mass m of a body be at a distance r from a given line, then mr^2 is called the moment of inertia of the particle about the given line and the sum of similar expressions taken for all the particles of the body, i.e., Σmr^2 is called the moment of inertia of the body about the given line (Fig. 5.26).

If M be the total mass of the body and k is called its moment of inertia $= Mk^2$, then k is called the radius of gyration of the body about the axis.

(2) M.I. of plane lamina. Consider the elementary mass $\rho dxdy$ at the point $P(x, y)$ of a plane area A so that its M.I. about x-axis $= \rho dxdy \cdot y^2$.

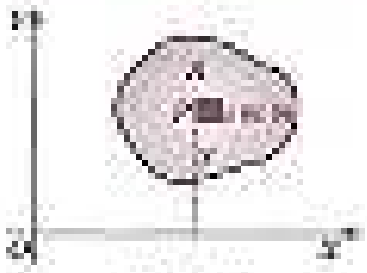

Fig. 6.29

$\therefore$ M.I. of the lamina about x-axis, i.e. $I_x = \iint_A \rho y^2\, dxdy$.

Similarly, M.I. of the lamina about y-axis i.e., $I_y = \iint_A \rho x^2\, dxdy$.

Also M.I. of the lamina about an axis perpendicular to the xy plane, i.e.,

$$I_z = \iint_A \rho(x^2 + y^2)\, dxdy.$$

(3) M.I. of a solid. Consider an elementary mass $\rho dxdydz$ enclosing a point $P(x, y, z)$ of a solid of volume V.

Distance of P from the x-axis is $\sqrt{(y^2 + z^2)}$.

$\therefore$ M.I. of this element about the x-axis $= \rho\, dxdydz\,(y^2 + z^2)$.

Thus M.I. of the solid about x-axis, i.e., $I_x = \iiint_V \rho(y^2 + z^2)\, dxdydz$.

Similarly, the M.I. about y-axis, i.e., $I_y = \iiint_V \rho(z^2 + x^2)\, dxdydz$

and M.I. about z-axis, i.e., $I_z = \iiint_V \rho(x^2 + y^2)\, dxdydz$.

(4) Sometimes we require the moment of inertia of a body about axes other than the principal axes. The following theorems prove useful for this purpose :

I. Theorem of perpendicular axes. *If the moment of inertia of a lamina about two perpendicular axes OX, OY in its plane are I_x and I_y, then the moment of inertia of the lamina about an axis OZ perpendicular to it is given by $I_z = I_x + I_y$.*

The proof follows from the relations giving I_x, I_y and I_z in a plane lamina in (2) above.

II. Steiner's theorem*. *If the moment of inertia of a body of mass M about an axis through its centre of gravity is I then I', moment of inertia about a parallel axis at a distance d from the first axis, is given by $I' = I + Md^2$.*

The proof will be found in any text book on Dynamics of a Rigid Body.

Example 6.25. Find the M.I. of the area bounded by the curve $r^2 = a^2 \cos 2\theta$ about its node.

Solution. Given curve is symmetrical about the pole and for half of the loop in the first quadrant θ varies from 0 to $\pi/4$ (Fig. 6.30).

Elementary area $= r\, dr\, d\theta = \rho\, dr\, d\theta$, if ρ is the surface density, then elementary mass

$$= \rho r\, dr\, d\theta \qquad \ldots(1)$$

$\therefore$ the total mass $M = 4\int_0^{\pi/4} \int_0^{a\sqrt{\cos 2\theta}} \rho\, dr\, d\theta$

$$= 2\rho a^2 \int_0^{\pi/4} \cos 2\theta\, d\theta = \rho a^2 \qquad \ldots(2)$$

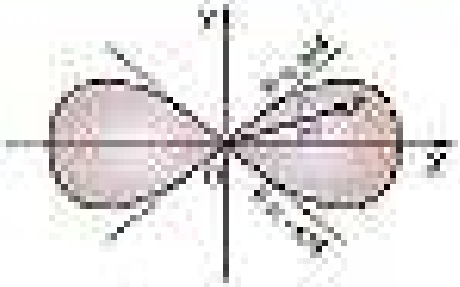

Fig. 6.30

Now M.I. of the elementary mass $\rho r\, dr\, d\theta$ about the node

$$= \rho r\, dr\, d\theta \cdot r^2 = \rho d\theta\, r^4\, dr = \rho d\theta \int_0^{a\sqrt{\cos 2\theta}} r^4\, dr = \rho\, d\theta \left[\frac{r^5}{5}\right]_0^{a\sqrt{\cos 2\theta}}$$

Hence the M.I. of the whole area

$$= 4\int_0^{\pi/4} \int_0^{a\sqrt{\cos 2\theta}} \rho r^4 \sin^2 \theta\, dr\, d\theta = 4\rho \int_0^{\pi/4} \sin^2 \theta \left[\frac{r^5}{5}\right]_0^{a\sqrt{\cos 2\theta}} d\theta$$

$$= \rho a^5 \int_0^{\pi/4} \cos^2 2\theta \cdot \sin^2 \theta\, d\theta = \rho a^5 \int_0^{\pi/4} \cos^2 \theta \cdot \sin^2 \theta\, d\theta = \frac{\rho a^5}{2} \cdot \frac{\pi}{32} \qquad \left[\because \sin 2\theta = 2\sin\theta\cos\theta\right]$$

$$= \frac{\rho a^5}{4} \int_0^{\pi/4} (\cos^2 \theta - \cos^4 \theta)\, d\theta = \frac{\rho a^5}{64}(2\pi - 4) = \frac{Ma^2}{32}(\pi - 2) \qquad [\text{By (2)}]$$

*Named after a Swiss geometer Jacob Steiner (1796–1863) who was a professor at Berlin University.

Example 7.38. *Find the moment of inertia of a sphere of radius a and density ρ about a diameter. Also find its radius of gyration about a diameter.*

Solution. Let the sphere be generated by revolving the circle $x^2 + y^2 = a^2$ about the x-axis. Then the M.I. about the diameter (i.e., x-axis) is

$$I_x = \iiint_V \rho(y^2 + z^2)\,dx\,dy\,dz$$

Changing to polar spherical coordinates, we get

$$I_x = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho(r^2\sin^2\theta\sin^2\phi + r^2\cos^2\theta)\,r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= \rho\left[\int_0^{2\pi}\sin^2\phi\,d\phi\cdot\int_0^{\pi}\sin^3\theta\,d\theta\left|\frac{r^5}{5}\right|_0^a + \int_0^{2\pi}d\phi\int_0^{\pi}\cos^2\theta\sin\theta\,d\theta\left|\frac{r^5}{5}\right|_0^a\right]$$

$$= \frac{8\pi\rho}{15}(a^5 - a^5) = \frac{8}{15}\pi\rho a^5 \text{ sq. units.}$$

Example 7.39. *A solid body of density ρ is in the shape of the solid formed by revolution of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 + \cos\theta)$ about the tangent line. Show that the moment of inertia about a straight line through the point perpendicular to the initial line is $\frac{352}{105}\pi\rho a^5$.* (U.P.T.U. 2002)

Solution. An elementary area δA of the, when revolved about OX generates a circular ring of radius $AP = y = a(1 + \cos\theta)$ (Fig. 7.44).

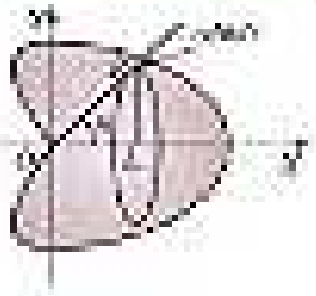

Fig. 7.44

M.I. of this ring about a diameter parallel to OY

$$= \tfrac{1}{2}(\text{density})(\text{area})y^2 = \frac{\rho\cdot 2\pi y\,\delta s\cdot y^2}{2}$$

$\therefore$ M.I. of a ring about a diameter $= \pi\rho y^3\,\delta s$

Now using Pythagoras theorem, we have M.I. of the ring about OY = M.I. of the ring about a diameter AP parallel to OY + Mass of the ring OA $(AP)^2$ [$\because$ $\cos\theta = \frac{AP}{OA}$]

$$= \pi\rho y^3\,\delta s + \pi\rho y\,\delta s\cdot y^2 = \pi\rho y^3\,\delta s(1 + \cos\theta)$$

Hence M.I. of the solid generated by revolution about OY

$$= \pi\rho\int_0^{2\pi}\int_0^{\sin\theta(1+\cos\theta)} y^3\,dx\,d\theta = 2\pi^3\int_0^{2\pi}(1+\cos\theta)^3\sin\theta\,d\theta\,dx$$

$$= \pi\rho\int_0^{2\pi}(\sin^2\theta + 2\sin\theta\cos^2\theta)\,d\theta\int_0^{a(1+\cos\theta)} y^3\,dx$$

$$= \frac{32\pi^3}{5}\int_0^{\pi}\sin^2\frac{\theta}{2}\left(1 - \cos^2\frac{\theta}{2}\right)^2\sin\frac{\theta}{2}\,d\theta \qquad [\because 1 + \cos\theta = 2\cos^2\tfrac{\theta}{2}]$$

$$= \frac{32\pi^3}{5}\int_0^{\pi}\sin^2\frac{\theta}{2}\left(1 + \cos^4\frac{\theta}{2} - 2\cos^2\frac{\theta}{2}\right)\sin\frac{\theta}{2}\,d\theta$$

$$= \frac{32\pi^3}{5}\int_0^{\pi}\left(\sin^2\frac{\theta}{2} + \cos^4\frac{\theta}{2}\sin\frac{\theta}{2} - 2\cos^2\frac{\theta}{2}\sin\frac{\theta}{2}\right)d\theta$$

$$= \frac{32\pi^3\rho a^5}{5}\int_0^{\pi}\left(2\sin^2\frac{\theta}{4}\cos^2\frac{\theta}{4} + 2\cos^4\frac{\theta}{4}\sin\frac{\theta}{4} - 2\cos^2\frac{\theta}{4}\sin\frac{\theta}{4}\right)d\theta$$

$$= \frac{32\pi^3\rho a^5}{5}\int_0^{\pi}\left(\frac{1}{2}\sin^2\frac{\theta}{2} + 2\cos^4\frac{\theta}{4}\sin\frac{\theta}{4} - 2\cos^2\frac{\theta}{4}\sin\frac{\theta}{4}\right)d\theta$$

$$= \frac{32\pi^3\rho a^5}{5}\left|-\frac{\cos^2\frac{\theta}{2}}{2} + \frac{2\cos^5\frac{\theta}{4}}{5} - \frac{2\cos^3\frac{\theta}{4}}{3}\right|_0^{\pi} = \frac{352}{105}\pi\rho a^5$$

Example 7.40. *A circular plate of radius R has a circular hole of radius r cut out of it, the centre of the hole being at a distance a from the centre of the plate. Find the moment of inertia about an axis perpendicular to the plane of the plate and passing through the centre of the plate.*

Solution. M.I. of the given solid about z-axis

$$= \iiint z(x^2 + y^2)\,dx\,dy\,dz$$

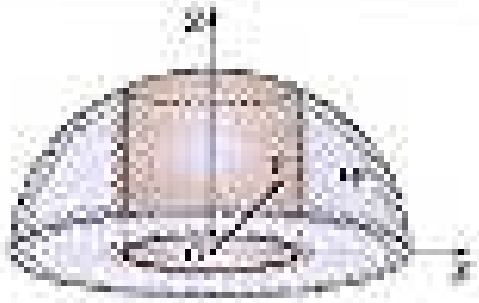

Fig. 3.49

The limits of integration for z are from 0 to $z = \sqrt{(R^2 - x^2 - y^2)}$ found from the equation of the sphere $x^2 + y^2 + z^2 = R^2$. The limits for x and y are taken such as to cover the whole area A in the xy-plane between the concentric circles of radii a and R (Fig. 3.49).

Thus the required M.I. about z-axis

$$= \rho \iint_A \int_0^{\sqrt{(R^2 - x^2 - y^2)}} (x^2 + y^2)\,dx\,dy\,dz$$

$$= \rho \iint_A \left[z(x^2 + y^2) \right]_0^{\sqrt{(R^2 - x^2 - y^2)}} dx\,dy = \rho \iint_A \left[y^2\sqrt{R^2 - x^2 - y^2} + x^2\sqrt{R^2 - x^2 - y^2} \right] dx\,dy$$

Now changing to polar coordinates, we have $x = r\cos\theta$ and $y = r\sin\theta$ and $dx\,dy = r\,dr\,d\theta$.

A is to cover the area A, r varies from a to R and θ varies from 0 to 2π.

Hence the required M.I. about z-axis

$$= \rho \int_0^{2\pi} \int_a^R \left[r^2\sin^2\theta\sqrt{R^2 - r^2} + r^2\cos^2\theta\sqrt{R^2 - r^2} \right] r\,dr\,d\theta$$

$$= \rho \int_0^{2\pi} \int_a^R \left[\frac{1}{2}r^3 (1 - \cos 2\theta)\sqrt{R^2 - r^2} \right] dr\,d\theta \quad [\because \sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)]$$

$$= \rho \int_0^{2\pi} \left[\frac{r^2}{2}\left(0 - \frac{(R^2 - r^2)^{3/2}}{3}\right) + \frac{1}{5}(R^2 - r^2)^{5/2} \right]_a^R d\theta$$

$$= \rho \int_0^{2\pi} 2\pi \left[\frac{a^2}{2} + \frac{R^2 - a^2}{5} \right] \pi(R^2 - a^2)^{5/2}\,d\theta$$

$$= \frac{2\pi}{5} \int_0^{2\pi} (2R^2 + a^2)(R^2 - a^2)^{5/2}\,d\theta \quad [\because \rho, a, R \text{ are constants}]$$

$$= \frac{2\pi}{5} \int_0^{2\pi} Q(R^2 + a^2)(R^2 - a^2)^{5/2}\,d\theta \quad [\because \text{integrands are equal}]$$

$$= \frac{2\pi}{5}\left[(2R^2 + a^2)\frac{(R^2 - a^2)^{7/2}}{7} + \frac{2}{9}(R^2 - a^2)^{9/2} \right] = \frac{2\pi\rho}{3}(R^2 - a^2)^{7/2} + \frac{1}{11}(4R^2 + a^2)$$

$$\left[\because \text{Mass} = \rho \int_0^{2\pi} \int_a^R \int_0^{\sqrt{(R^2 - r^2)}} dz\,r\,dr\,d\theta = \frac{2\pi\rho}{3}(R^2 - a^2)^{3/2} \right]$$

Hence, the radius of gyration $= \sqrt{\{\frac{2}{11}(R^2 + a^2)(R^2 - a^2)^{7/2}\}}$.

3.18 (I) PRODUCT OF INERTIA

If a particle of mass m of a body is at distances x and y from two given perpendicular lines, then Σmxy is called the *product of inertia* of the body about the given lines.

Consider an elementary mass δm of the body enclosing the point $P(x, y, z)$ of volume δV. Then the product of inertia (P.I.) of this element about the axes of x and y = $\rho\delta V\delta x\delta y$.

$\therefore$ P.I. of the solid about x and y axes, i.e., $P_{xy} = \iiint_V xy\rho\,dx\,dy\,dz$

Similarly, $P_{yz} = \iiint_V yz\rho\,dx\,dy\,dz$ and $P_{zx} = \iiint_V zx\rho\,dx\,dy\,dz$.

In particular, for a plane lamina of surface density ρ and covering a region A in the xy-plane,

$$P_{xy} = \iint_A \rho xy\,dx\,dy, \text{ whereas } P_{yz} = P_{zx} = 0.$$

(c) **Principal axes.** The principal axes of a lamina at a given point are that pair of axes in the plane through the given point, about which the product of inertia of the lamina vanishes.

Let $P(x, y)$ be a point of the plane area A referred to rectangular axes OX, OY. Let (x', y') be the coordinates of P referred to another pair of rectangular axes OX', OY' in the same plane and inclined at an angle θ to the first pair (Fig. 7.46).

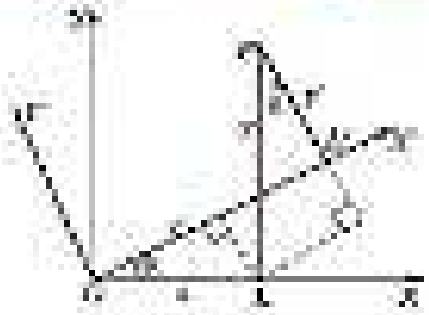
Fig. 7.46

Then $x' = x\cos\theta + y\sin\theta$

$y' = y\cos\theta - x\sin\theta$

If $I_{x'}, I_{y'}$ be the moments of inertia of the area A about OX' and OY' and $P_{x'y'}$ be its product of inertia about these axes, then

$$I_{x'} = \iint_A y'^2\,dA,\ I_{y'} = \iint_A x'^2\,dA,\ P_{x'y'} = \iint x'y'\,dA.$$

$\therefore$ the product of inertia $P_{x'y'}$ about OX' and OY' is given by

$$P_{x'y'} = \int_A x'y'\,dA = \iint_A (x\cos\theta + y\sin\theta)(y\cos\theta - x\sin\theta)\,dx\,dy$$

$$= \sin\theta\cos\theta \iint_A (y^2 - x^2)\,dx\,dy + (\cos^2\theta - \sin^2\theta)\iint_A xy\,dx\,dy$$

$$= \tfrac{1}{2}\sin 2\theta\,(I_y - I_x) + \cos 2\theta\,P_{xy}$$

Now OX', OY' will be the principal axes of the area A if $P_{x'y'}$ vanishes.

i.e., if $\tfrac{1}{2}\sin 2\theta\,(I_y - I_x) + \cos 2\theta\,P_{xy} = 0$

i.e., if $\tan 2\theta = 2P_{xy}/(I_x - I_y)$.

This gives two values of θ differing by $\pi/2$.

Example 7.41. *Show that the principal axes at the vertex of a half loop of the cycloid $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$ are inclined at angles $\tan^{-1}\tfrac{1}{2}$ and $\tfrac{\pi}{2} + \tan^{-1}\tfrac{1}{2}$.*

Solution. Let the element of mass at $P(x, y)$ be ρ (uniform).

Then $I_x = \rho\int_0^{\pi}\int_0^{a(1-\cos\theta)} r^2\sin^2\theta\cdot r\,d\theta\,dr$ (See Fig. 7.47)

$$= \frac{\rho a^4}{4}\int_0^{\pi}\sin^2\theta\cos^4\tfrac{\theta}{2}\,d\theta = \frac{\rho a^4}{20}\left[\frac{\pi}{4} - \frac{2}{3}\right]$$

$$I_y = \rho\int_0^{\pi}\int_0^{a(1-\cos\theta)} r^2\cos^2\theta\cdot r\,d\theta\,dr = \frac{\rho a^4}{16}\left[\frac{\pi}{4} - \frac{2}{3}\right]$$

and $P_{xy} = \rho\int_0^{\pi}\int_0^{a(1-\cos\theta)} r^2\sin\theta\cos\theta\cdot r\,d\theta\,dr = \frac{\rho a^4}{48}.$

Hence the required directions of the principal axes at O are given by

$$\tan 2\theta = \frac{2P_{xy}}{I_x - I_y} = \frac{\rho a^4/24}{(\rho a^4/16)(4/3)} = \frac{1}{2}$$

or by $\theta = \frac{1}{2}\tan^{-1}\frac{1}{2}$ and $\frac{\pi}{2} + \frac{1}{2}\tan^{-1}\frac{1}{2}$.

EXERCISE 7.7

1. Using double integration, find the moment of inertia about the axis of the area enclosed by the lines $x + y = 3$, $x = 0$, $y = 0$. (U.P.T.U. 2006)
2. Find the moment of inertia of a circular disc about a tangent.
3. Find the moment of inertia of the area $y = \sin x$ about $x = 0$ to $x = \pi$ about OX.

1. Find the moment of inertia of a circular lamina of radius a and density ρ about a tangent line.
2. Find the moment of inertia of a uniform solid cone of mass M and radius R about a line through the vertex perpendicular to the axis of the cone.
3. Find the moment of inertia of a uniform solid hemisphere of mass M and radius R about a diameter of the base.
4. Find the moment of inertia of a uniform solid right circular cylinder of mass M and radius R about a diameter of the base.
5. Find the moment of inertia of a solid right circular cylinder about one of its edges.
6. Find the moment of inertia of a solid right circular cone having base radius a and height h about a line through the vertex and perpendicular to the axis of the cone.
7. Find the moment of inertia of a hollow spherical shell of mass M and radius R about a diameter.
8. Find the moment of inertia of a solid sphere of mass M and radius R about a diameter.
9. Find the moment of inertia of a solid right circular cone of mass M and radius R about a diameter of the base.
10. Find the moment of inertia of a uniform solid tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$, $x + y + z = 1$.
11. Find the moment of inertia of an ellipsoid of mass M and density ρ about the z-axis.
12. Find the moment of inertia of a cylinder of mass M and radius R about the z-axis.
13. Show that the principal axes at the origin of the family of curves $ax^2 + 2hxy + by^2 = 1$ are inclined to the x-axis at angles $\frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$ and $\frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right) + \frac{\pi}{2}$.
14. The lengths AA', BB', CC' of the three altitudes of the triangle ABC are a, b, c. Show that the distance of the centre of gravity of the triangle from the centre of the circle ABC is $\frac{1}{3}\sqrt{a^2+b^2+c^2}$.

7.04 BETA FUNCTION

The beta function is defined as

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx \quad \begin{cases} m>0 \\ n>0 \end{cases} \qquad \ldots(1)$$

Putting $x = 1 - y$ in (1), we get $B(m,n) = -\int_1^0 (1-y)^{m-1} y^{n-1}\,dy$

$$= \int_0^1 y^{n-1}(1-y)^{m-1}\,dy = B(n,m)$$

Thus $B(m,n) = B(n,m)$ ...(2)

Putting $x = \sin^2\theta$ in (1), so that $dx = 2\sin\theta\cos\theta\,d\theta$, (1) becomes

$$B(m,n) = \int_0^{\pi/2} \sin^{2m-2}\theta\cos^{2n-2}\theta \cdot 2\sin\theta\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \sin^{2m-1}\theta\cos^{2n-1}\theta\,d\theta \qquad \ldots(3)$$

which is another form of $B(m,n)$.

This function is also Euler's integral of the first kind[1].

7.05 (1) GAMMA FUNCTION

The gamma function is defined as

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\,dx \quad (n>0) \qquad \ldots(1)$$

This integral is known as Euler's integral of the second kind. It defines a function of the positive values of n

[1] After an incredibly creative career Swiss mathematician Leonhard Euler (1707–1783). He studied under Johann Bernoulli and became a professor of mathematics in St. Petersburg, Russia. Even after becoming totally blind in 1771, he remained incredibly productive, producing half of his total lifetime output in the last 17 years of his life. He made important discoveries in fields as diverse as infinite series, differential equations, complex analysis, and number theory. Euler is considered to be one of the two most prolific mathematicians of all time (the other being Leonhard Euler).

In particular, $\Gamma(1) = \int_0^\infty e^{-x}\,dx = \left[-e^{-x}\right]_0^\infty = 1$ ...(vi)

(2) Recursion formula for $\Gamma(n)$.

Since $\Gamma(n+1) = \int_0^\infty e^{-x}x^n\,dx$. By integrating by parts, $= \left[-x^n e^{-x}\right]_0^\infty + n\int_0^\infty e^{-x}x^{n-1}\,dx$

$$\Gamma(n+1) = n\,\Gamma(n) \quad \text{...(vii)}$$

which is the reduction formula for $\Gamma(n)$. From this formula, if the value of $\Gamma(n)$ is known throughout a unit interval say $1 < n \le 2$, then the values of Γ for the points in the next unit interval $2 < n \le 3$ are found, from which the values of $\Gamma(n)$ for $3 < n \le 4$ are determined and so on. In this way, the values of $\Gamma(n)$ for all positive values of $n > 1$ may be found by successive application of (vii).

Also using (vii) in the form

$$\Gamma(n) = \frac{\Gamma(n+1)}{n} \quad \text{...(viii)}$$

We can define $\Gamma(n)$ for values of n for which the definition (i) fails. It gives the values of $\Gamma(n)$ for $0 < n \le 1$ in terms of the values of $\Gamma(n)$ for $1 < n \le 2$. Thus we can define $\Gamma(n)$ for all $n < 0$ provided the value for $1 < n \le 2$ is known. Also if $-1 < n < 0$, (i) gives $\Gamma(n)$ in terms of the values for $0 < n < 1$. This way fails, $\Gamma(n)$ for $-2 < n < -1$ and so on.

The (i) and (ii) together give a complete definition of $\Gamma(n)$ for all values of n except when n is zero or a negative integer. The graph is as shown in Fig. 7.44. The values of $\Gamma(n)$ for $1 < n \le 2$ are given in Table 6, Appendix 2 from which the values of $\Gamma(n)$ for values of n outside the interval $1 < n \le 2$ ($n = 0, -1, -2, -3 \ldots$) may be found.

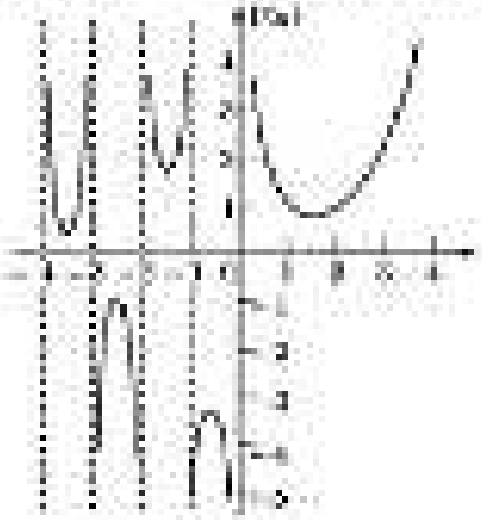

Fig. 7.44

(3) Value of $\Gamma(n)$ in terms of factorials.

Using $\Gamma(n+1) = n\,\Gamma(n)$ successively, we get

$$\Gamma(2) = 1 \times \Gamma(1) = 1!$$
$$\Gamma(3) = 2 \times \Gamma(2) = 2 \times 1 = 2!$$
$$\Gamma(4) = 3 \times \Gamma(3) = 3 \times 2! = 3!$$

In general $\Gamma(n+1) = n!$ provided n is a positive integer. ...(ix)

Taking $n = 0$, (i) defines $0! = \Gamma(1) = 1$.

(4) Value of $\Gamma\left(\frac{1}{2}\right)$. We have

$$\Gamma\left(\tfrac{1}{2}\right) = \int_0^\infty e^{-x}x^{-\frac{1}{2}}\,dx \qquad \left[\text{Put } x = y^2 \text{ so that } dx = 2y\,dy\right]$$
$$= 2\int_0^\infty e^{-y^2}\,dy \text{ which is also } = 2\int_0^\infty e^{-x^2}\,dx$$
$$\left[\Gamma\left(\tfrac{1}{2}\right)\right]^2 = 4\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}\,dx\,dy = 4\int_0^{\pi/2}\int_0^\infty e^{-r^2}\,r\,dr\,d\theta$$
$$= 4\cdot\frac{\pi}{2}\int_0^\infty e^{-r^2}\,r\,dr = 2\pi\left[-\frac{1}{2}e^{-r^2}\right]_0^\infty = \pi$$

whence $\Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi} = 1.772$ ...(x) (U.P.T.U. 2006)

7.13 RELATION BETWEEN BETA AND GAMMA FUNCTIONS

$$\beta(m,n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$$

We have $\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}\,dx$ $\left[\text{Put } x = z^2 \text{ so that } dx = 2z\,dz\right]$

$$= 2\int_0^\infty e^{-x^2} x^{2m-1}\,dx \qquad \ldots(ii)$$

Similarly, $\Gamma(n) = 2\int_0^\infty e^{-y^2} y^{2n-1}\,dy$

$\therefore \quad \Gamma(m)\Gamma(n) = 4\int_0^\infty e^{-x^2} x^{2m-1}\,dx \int_0^\infty e^{-y^2} y^{2n-1}\,dy$

$= 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2m-1} y^{2n-1}\,dx\,dy \qquad \ldots(iii)$ [$\because$ Our limits of integration are constant.]

Now change to polar coordinates by evaluating $x = r\cos\theta$, $y = r\sin\theta$ and $dx\,dy = r\,dr\,d\theta$. To cover the region in (i) which is the entire first quadrant, r varies from 0 to ∞ and θ from 0 to $\pi/2$. Thus (i) becomes

$$\Gamma(m)\Gamma(n) = 4\int_0^\infty \int_0^{\pi/2} e^{-r^2} r^{2(m+n-1)} \cos^{2m-1}\theta \sin^{2n-1}\theta\, r\,dr\,d\theta$$

$$= \left[2\int_0^{\pi/2} \cos^{2m-1}\theta \sin^{2n-1}\theta\,d\theta\right]\left[2\int_0^\infty e^{-r^2} r^{2(m+n)-1}\,dr\right] \qquad \ldots(iv)$$

But by (ii), $2\int_0^\infty e^{-r^2} r^{2(m+n)-1}\,dr = \Gamma(m+n)$

and by (8) of § 7.14, $2\int_0^{\pi/2} \cos^{2m-1}\theta \sin^{2n-1}\theta\,d\theta = \beta(m,n)$.

Thus (4) gives $\Gamma(m)\Gamma(n) = \beta(m,n)\,\Gamma(m+n)$ [U.P.T.U. 2010; Bhopal 2005; P.T.U. 2008 S] which is known as (1) which is extremely useful for evaluating definite integrals in terms of gamma function.

Cor. We can evaluate $\int_0^{\pi/2} \sin^p x \cos^q x\,dx$

$$\int_0^{\pi/2} \sin^p x \cos^q x\,dx = \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right) \qquad \text{[by (8) of § 7.14]}$$

$$= \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)} \qquad \ldots(5)$$

In particular, when $p = 0$ and $q = n$, we have

$$\int_0^{\pi/2} \cos^n x\,dx = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)} \cdot \frac{\sqrt{\pi}}{2}$$

Similarly, $\int_0^{\pi/2} \sin^n x\,dx = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)} \cdot \frac{\sqrt{\pi}}{2} \qquad \ldots(6)$

Example 7.48. Show that

(i) $\Gamma(n) = \int_0^1 \left(\log\frac{1}{y}\right)^{n-1} dy$ $(n > 0)$. [U.P.T.U. 2002; Madras, 2001 S]

(ii) $B(p,q) = \int_0^\infty \frac{x^{p-1}}{(1+x)^{p+q}}\,dx$ [U.P.T.U. 2003; Gauhati, 2000]

$= \int_0^\infty \frac{x^{p-1} + x^{q-1}}{(1+x)^{p+q}}\,dx$ [U.P.T.U. 2006; Osmania, 2001; Rohtak, 2000]

Solution. (i) $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\,dx$ $(n > 0)$

$$= \int_0^1 \left(\log\frac{1}{y}\right)^{n-1} x\left(-\frac{1}{y}dy\right) = \int_0^1 \left(\log\frac{1}{y}\right)^{n-1} dx.$$

Put $y = e^{-x}$
i.e., $x = \log(1/y)$
so that $dx = (-1/y)\,dy$

(3) $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$

$$= \int_\infty^0 \frac{1}{(1+y)^{m-1}}\left(\frac{y}{1+y}\right)^{n-1} \frac{-1}{(1+y)^2}\,dy$$

Put $x = \frac{1}{1+y}$ i.e., $y = \frac{1}{x} - 1$
so that $dx = \frac{-1}{(1+y)^2}\,dy$

$$= \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy = \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}}\,dy + \int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy$$

Now substituting $y = 1/z$ in the second integral, we get

$$\int_1^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy = \int_1^0 \frac{1}{z^{n-1}}\cdot\frac{1}{(1+1/z)^{m+n}}\left(-\frac{1}{z^2}\right)dz = \int_0^1 \frac{z^{m-1}}{(1+z)^{m+n}}\,dz.$$

Hence, $\beta(m,n) = \int_0^1 \frac{y^{n-1}}{(1+y)^{m+n}}\,dy + \int_0^1 \frac{z^{m-1}}{(1+z)^{m+n}}\,dz = \int_0^1 \frac{y^{n-1}+y^{m-1}}{(1+y)^{m+n}}\,dy.$

Example 3.40. Express the following integrals in terms of gamma functions:

(a) $\int_0^a \frac{dx}{\sqrt{a(a-x)}}$ (b) $\int_0^{\pi/2} \sqrt{\tan\theta}\,d\theta$ (Madras, 2003)

(c) $\int_0^\infty \frac{x^4}{4^x}\,dx$ (U.P.T.U., 2006) (d) $\int_0^\infty x^{-1/2}e^{-x}\,dx$

(e) $\int_0^1 x^2(\log(1/x))^3\,dx$ (Madras, 2005)

Solution. (a) $\int_0^a \frac{dx}{\sqrt{a(a-x)}}$

Put $x = a\sin^2\theta$ i.e., $\theta = \sin^{-1}\sqrt{x/a}$
so that $dx = 2a\sin\theta\cos\theta\,d\theta$

$$= \int_0^{\pi/2} \frac{1}{a}\cdot\frac{2a\sin^{1/2}\theta\cos\theta\,d\theta}{\sqrt{a}\cos\theta} = \frac{1}{a}\int_0^{\pi/2} 2a\sin^{1/2}\theta\,d\theta = \frac{1}{a}\cdot\frac{\sqrt{\pi}}{2}\cdot\frac{\Gamma\left(\frac{1}{4}+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{4}+\frac{1}{2}+1\right)} = \frac{\sqrt{\pi}}{a}\cdot\frac{\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)}$$

(b) $\int_0^{\pi/2} \sqrt{\tan\theta}\,d\theta = \int_0^{\pi/2} \sin^{1/2}\theta\cos^{-1/2}\theta\,d\theta$

$$= \frac{\Gamma\left(\frac{1}{4}+\frac{1}{2}\right)\Gamma\left(-\frac{1}{4}+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{4}-\frac{1}{2}+1\right)} = \frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)}{2^{-1}\Gamma\left(\frac{1}{4}\right)} = \frac{1}{2}\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)$$

(c) $\int_0^\infty \frac{x^4}{4^x}\,dx = \int_0^\infty \frac{x^4}{e^{x\log 4}}\,dx$ [$\because a = e^{\log a}$]

$$= \int_0^\infty e^{-x\log 4}\,x^4\,dx$$

Put $x\log 4 = t$ so that $dx = dt/\log 4$

$$= \int_0^\infty e^{-x}\left[\frac{x}{\log e}\right]^{k-1}\frac{dx}{\log e} = \frac{1}{(\log e)^{k-1}}\int_0^\infty e^{-x}x^{k-1}\,dx = \Gamma k\,(\log e)^{k-1}$$

(d) $\int_0^\infty a^{-bx^2}\,dx = \int_0^\infty e^{-bx^2\ln a}\,dx$ $\left[\text{Put } bx^2\ln a = t,\ 2bx\ln a\,dx = dt,\ x = \sqrt{\frac{t}{b\ln a}},\ dx = \frac{dt}{2\sqrt{bt\ln a}}\right]$

$$= \frac{1}{2\sqrt{b\log a}}\int_0^\infty e^{-t}t^{-1/2}\,dt = \frac{\Gamma\left(\frac{1}{2}\right)}{2\sqrt{b\log a}} = \frac{\sqrt{\pi}}{2\sqrt{b\log a}}$$

(e) $\int_0^1 x^2\left(\log\frac{1}{x}\right)^3 dx = \frac{1}{625}\int_0^\infty e^{-t}t^3\,dt$ $\left[\text{Put } x = e^{-t/5} \text{ and so } \log\left(\frac{1}{x}\right) = t/5,\ dx = -\frac{1}{5}e^{-t/5}\,dt\right]$

$$= \frac{\Gamma(4)}{625} = \frac{6}{625}$$

Example 7.44. Evaluate $\int_0^\infty e^{-ax}x^{n-1}\,dx$ by using a suitable result of Gamma functions. (UPTU, 2003)

Solution. We know $\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}\,dx$ [Put $x = ay$, $dx = a\,dy$]

$$= \int_0^\infty e^{-ay}a^{n-1}y^{n-1}a\,dy \quad \text{or} \quad \int_0^\infty e^{-ay}y^{n-1}\,dy = \frac{\Gamma(n)}{a^n} \qquad \ldots(i)$$

Then $I = \int_0^\infty e^{-ax}x^{n-1}\sin bx\,dx = \int_0^\infty e^{-ax}x^{n-1}$ (Imaginary part of e^{ibx})dx

$$= \text{I.P. of } \int_0^\infty e^{-ax}x^{n-1}e^{ibx}\,dx \qquad [\because e^{ibx} = \cos bx + i\sin bx]$$

$$= \text{I.P. of } \int_0^\infty e^{-(a-ib)x}x^{n-1}\,dx \qquad \text{(Using Demoivre's theorem)}$$

$$= \text{I.P. of } \left[\frac{\Gamma(n)}{r^n}(\cos n\theta + i\sin n\theta)\right] \qquad [\because a - ib = r\cos\theta - ir\sin\theta]$$

$$= \frac{\Gamma(n)}{r^n}\sin n\theta, \text{ where } r = \sqrt{(a^2+b^2)},\ \theta = \tan^{-1}\frac{b}{a}.$$

Example 7.45. Prove that $\int_0^a \frac{x^2\,dx}{\sqrt{a^2-x^2}} = \int_0^\infty \frac{dx}{\sqrt{(1+x^4)}} = \frac{\pi}{4\sqrt{2}}$.

Solution. $\int_0^a \frac{x^2\,dx}{\sqrt{a^2-x^2}} = \int_0^{\pi/2}\frac{a\sin\theta}{a\cos\theta}\cdot\frac{a\cos\theta}{2\sqrt{(\sin\theta)}}\,d\theta$ $\left[\text{Putting } x = a\sin\theta, dx = \frac{a\cos\theta\,d\theta}{2\sqrt{(\sin\theta)}}\right]$

$$= \frac{1}{2}\int_0^{\pi/2}\sqrt{\sin\theta}\,d\theta = \frac{1}{4}\beta\left(\frac{3}{4},\frac{1}{2}\right) = \frac{1}{4}\frac{\Gamma(3/4)\Gamma(1/2)}{\Gamma(5/4)} = \frac{\Gamma(3/4)\Gamma(1/2)}{4\Gamma(5/4)}$$

$$\int_0^\infty \frac{dx}{\sqrt{(1+x^4)}} = \int_0^{\pi/2}\frac{\cos^2\theta\,d\theta}{2\sin\theta\cos\theta} \qquad \left[\text{Putting } x^2 = \tan\theta, dx = \frac{\cos^2\theta\,d\theta}{2\sqrt{\tan\theta}}\right]$$

$$= \frac{1}{2}\int_0^{\pi/2}\frac{d\theta}{\sqrt{\tan\theta}} = \frac{1}{2\sqrt{2}}\int_0^{\pi/2}\sin^{-1/2}\theta\,d\theta \qquad \left[\text{Putting } 2\theta = \phi, d\theta = \frac{1}{2}d\phi\right]$$

$$= \frac{1}{4\sqrt{2}}\beta\left(\frac{1}{4},\frac{1}{2}\right) = \frac{1}{4\sqrt{2}}\frac{\Gamma(1/4)\Gamma(1/2)}{\Gamma(3/4)}$$

$$\int_0^1 \frac{x^4\,dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{1}{4\sqrt{2}}\left[\Gamma\left(\frac{1}{4}\right)\right]^2 = \frac{\pi}{4\sqrt{2}}$$

Example 2.46. Prove that $\beta(m,n) = \frac{\sqrt{\pi}}{2^{2n-1}}\,\Gamma(2n)$ **(Duplication Formula)** (P.T.U., 2003)

(P.T.U., 2003; Kerala, M.E., 2003; Madras, 2003 S)

Solution. (i) We know that $\beta(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta\, d\theta$ ...(1)

Putting $n = \frac{1}{2}$, we have $\beta\left(m,\frac{1}{2}\right) = 2\int_0^{\pi/2} \sin^{2m-1}\theta\, d\theta$ ...(2)

Again putting $n = m$ in (1), we get $\beta(m,m) = 2\int_0^{\pi/2} (\sin\theta\cos\theta)^{2m-1}\, d\theta$

$$= \frac{1}{2^{2m-1}}\int_0^{\pi/2} \sin^{2m-1}2\theta\, d\theta$$

$$= \frac{1}{2^{2m-1}}\int_0^{\pi} \sin^{2m-1}\phi\, \frac{d\phi}{2}, \text{ putting } 2\theta = \phi$$

$$= \frac{1}{2^{2m-1}}\cdot 2\int_0^{\pi/2} \sin^{2m-1}\phi\, d\phi$$

or $2^{2m-1}\beta(m,m) = 2\int_0^{\pi/2} \sin^{2m-1}\theta\, d\theta = \beta\left(m,\frac{1}{2}\right)$ [By (2)

(ii) Rewriting the above result in terms of Γ functions, we get

$$2^{2m-1}\frac{\Gamma(m)\,\Gamma(m)}{\Gamma(2m)} = \frac{\Gamma(m)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(m+\frac{1}{2}\right)} \qquad \left[\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}\right]$$

or $\Gamma(m)\Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(2m)}{2^{2m-1}}$

Example 2.47. Prove that

(i) $\iint_R x^{l-1}y^{m-1}\phi(x+y)\,dx\,dy = \frac{\Gamma(l)\,\Gamma(m)}{\Gamma(l+m)}\int_0^a \phi(t)\,t^{l+m-1}\,dt$, where R is the domain $x \geq 0, y \geq 0$ and $x + y \leq a$.

(U.P.T.U., 2004)

(ii) $\iiint_V x^{l-1}y^{m-1}z^{n-1}\,dx\,dy\,dz = \frac{\Gamma(l)\,\Gamma(m)\,\Gamma(n)}{\Gamma(l+m+n+1)}$

where V is the region $x \geq 0, y \geq 0, z \geq 0$ and $x + y + z \leq 1$. This dependent result is known as Dirichlet's Integral.

Solution. (i) Putting $x = X$ and $y = Y$, we have to find the given integral

$$= \iint_R X^{l-1}(aX)^{m-1}\,a^2\,dX\,dY, \text{ where } R' \text{ is the domain } X \geq 0, Y \geq 0 \text{ and } X + Y \leq 1.$$

$$= a^{l+m}\int_0^1\int_0^{1-X} X^{l-1}Y^{m-1}\,dY\,dX = a^{l+m}\int_0^1 X^{l-1}\left[\frac{Y^m}{m}\right]_0^{1-X} dX$$

$$= \frac{a^{l+m}}{m}\int_0^1 X^{l-1}(1-X)^m\,dX = \frac{a^{l+m}}{m}\beta(l,m+1) = \frac{a^{l+m}}{m}\cdot\frac{\Gamma(l)\Gamma(m+1)}{\Gamma(l+m+1)}$$

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*Named after a German mathematician, Peter Gustav Lejeune Dirichlet (1805–1859) who studied under Gauss and succeeded Gauss at Gottingen. He is known for his contributions to Fourier series and number theory.

$$= 2^{l+m+n}\frac{\Gamma(l)\Gamma(m)\Gamma(n)}{\Gamma(l+m+n)} \qquad \text{...(i) by } \Gamma(2m) = 2^{2m-1}\Gamma(m)\Gamma\left(m+\frac{1}{2}\right)$$

(b) Taking $x+y+z=1-w$ in the integral (because integral)

$$= \int_0^1\int_0^{1-w}\int_0^{1-w-y} x^{l-1}y^{m-1}z^{n-1}\,dx\,dy\,dz$$

$$= \int_0^1 x^{l-1}\left[\int_0^{1-x}\int_0^{1-x-y} y^{m-1}z^{n-1}\,dz\,dy\right]dx = \int_0^1 x^{l-1}\frac{\Gamma(n)\Gamma(n)}{\Gamma(m+n+2)}(1-x)^{m+n+1}\,dx \qquad \text{...(Ex 01}$$

$$= \frac{\Gamma(n)\Gamma(n)}{\Gamma(m+n+2)}\int_0^1 x^{l-1}(1-x)^{m+n+1}\,dx = \frac{\Gamma(n)\Gamma(n)}{\Gamma(m+n+1)}\beta(l,m+n+1)$$

$$= \frac{\Gamma(n)\Gamma(n)}{\Gamma(l+m+n+2)}\cdot\frac{\Gamma(l)\Gamma(m+n+1)}{\Gamma(l+m+n+1)} = \frac{\Gamma(l)\Gamma(m)\Gamma(n)\Gamma(n)}{\Gamma(l+m+n+2)}$$

Example 7.45. *Evaluate the integral $\iiint x^{l-1}y^{m-1}z^{n-1}\,dx\,dy\,dz$ taken over the positive octant of the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1$.* *(U.P.T.U., 2005 (S))*

Solution. Put $\frac{x^2}{a^2}=u, \frac{y^2}{b^2}=v, \frac{z^2}{c^2}=w$ so that $x=au^{\frac{1}{2}}, y=bv^{\frac{1}{2}}, z=cw^{\frac{1}{2}}$

$$\therefore dx=\frac{a}{2}u^{-\frac{1}{2}}du \qquad \text{so that } dx=\frac{a}{2}u^{\frac{1}{2}-1}du$$

$$dy=\frac{b}{2}v^{-\frac{1}{2}}dv \qquad \text{so that } dy=\frac{b}{2}v^{\frac{1}{2}-1}dv$$

and $$dz=\frac{c}{2}w^{-\frac{1}{2}}dw \qquad \text{so that } dz=\frac{c}{2}w^{\frac{1}{2}-1}dw$$

Then $$\iiint x^{l-1}y^{m-1}z^{n-1}\,dx\,dy\,dz$$

$$= \iiint (au^{\frac{1}{2}})^{l-1}(bv^{\frac{1}{2}})^{m-1}(cw^{\frac{1}{2}})^{n-1}\frac{a}{2}u^{\frac{1}{2}-1}\frac{b}{2}v^{\frac{1}{2}-1}\frac{c}{2}w^{\frac{1}{2}-1}\,du\,dv\,dw$$

$$= \frac{abc}{8}a^{l-1}b^{m-1}c^{n-1}\iiint u^{\frac{l}{2}-1}v^{\frac{m}{2}-1}w^{\frac{n}{2}-1}\,du\,dv\,dw \text{ taken over } u+v+w\le 1$$

$$= \frac{abc}{8}a^{l-1}b^{m-1}c^{n-1}\frac{\Gamma\left(\frac{l}{2}\right)\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)\Gamma(1)}{\Gamma\left(\frac{l}{2}+\frac{m}{2}+\frac{n}{2}+1\right)} \qquad \text{[By Dirichlet's integral]}$$

Example 7.46. *The plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ cuts the axes in A, B and C. Apply Dirichlet's integral to find the volume of the tetrahedron OABC. Also find the mass of the tetrahedron if the density at any point is $kxyz$.* *(U.P.T.U., 2004)*

Solution. Put $\frac{x}{a}=u, \frac{y}{b}=v, \frac{z}{c}=w$ then the tetrahedron OABC has $u\ge 0, v\ge 0, w\ge 0$ and $u+v+w\le 1$.

$$\therefore \text{ volume of the tetrahedron} = \iiint_V dx\,dy\,dz$$

$$= \iiint_V abc\,du\,dv\,dw \qquad \begin{cases} dx=a\,du, dy=b\,dv, dz=c\,dw \\ \text{for } V: u\ge 0, v\ge 0, w\ge 0 \text{ & } u+v+w\le 1 \end{cases}$$

$$= abc\iiint_V u^{1-1}v^{1-1}w^{1-1}\,du\,dv\,dw$$

$$= abc\frac{\Gamma(1)\Gamma(1)\Gamma(1)\Gamma(1)}{\Gamma(1+1+1+1)} = \frac{abc}{6} \qquad \text{[By Dirichlet's integral]}$$

$$\text{Mass} = \iiint_V kxyz\,dx\,dy\,dz = \iiint_V k(au)(bv)(cw)abc\,du\,dv\,dw$$

$$= ka^2b^2c^2\iiint_V u^{2-1}v^{2-1}w^{2-1}\,du\,dv\,dw$$

$$= ka^2b^2c^2\frac{\Gamma(2)\Gamma(2)\Gamma(2)\Gamma(1)}{\Gamma(2+2+2+1)} = ka^2b^2c^2\cdot\frac{1}{5!} = \frac{k}{120}a^2b^2c^2.$$

PROBLEMS 5.6

1. Derive:
   (a) $\int_0^\infty x^{n-1}e^{-x}\,dx$ (Illinois, 2006) (b) $\Gamma(1/2)$
   (c) $\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{3}{2}\right)$ (d) $\Gamma(3/2)$, 2008 (e) $\Gamma(n+1)$ (f) $\Gamma\left(\frac{5}{2},\frac{7}{2}\right)$ (Madras, 2004)

2. Express the following integrals in terms of gamma functions:
   (a) $\int_0^\infty x^7 e^{-x^2}\,dx$ (b) $\int_0^\infty x^{n-1}e^{-kx}\,dx$, $k>0$ (MAU, 2002; PTU, 2004)
   (c) $\int_0^\infty x^{5/2}e^{-x^2}\,dx$ (d) PTU, 2005 (e) $\int_0^\infty \frac{x^n}{a^{n+1}e^{ax}}\,dx$, $a>0$, $n>-1$

3. Show that:
   (a) $\int_0^\infty x^n e^{-ax}\,dx=\frac{\Gamma(n)}{(\log a)^n}$ (Uttarakhand, 2005)
   (b) $\int_0^{\pi/2}\sqrt{\tan\theta}\,d\theta=\frac{1}{2}\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)$ (Varanasi, 2002(S); PTU, 2003)
   (c) $\int_0^{\pi/2}\sqrt{\sin\theta}\,d\theta\int_0^{\pi/2}\frac{d\theta}{\sqrt{\cos\theta}}=\frac{1}{2}\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right)=\sqrt{2}\pi$ (CAS, 2006)
   (d) $\int_0^{\pi/2}\sqrt{\tan\theta}\,d\theta\int_0^{\pi/2}\frac{d\theta}{\sqrt{\sin\theta}}=\pi$ (UPTU, 2005)

4. Given $\int_0^\infty \frac{x^{n-1}}{1+x}\,dx=\frac{\pi}{\sin n\pi}$, $0<n<1$. Hence find (a) $\Gamma(n)\Gamma(1-n)=\frac{\pi}{\sin n\pi}$ (UPTU, 2006)
   (b) Hence evaluate $\int_0^\infty \frac{dx}{1+x^3}$ (PTU, 2006; CAS, 2006)

5. Prove that:
   (a) $\int_0^\infty \frac{x^{a-1}}{\sqrt{(1+x)}}\,dx=\frac{2}{\sqrt{\pi}}\Gamma\left(a\right)\Gamma\left(a+\frac{1}{2}\right)$ (Kerala, 1998) (b) $\int_0^\infty \frac{x^{a-1}}{(1+x)^{a+b}}\,dx=\frac{1}{\Gamma(a+b)}\Gamma(a)\Gamma(b)$ (UPTU, 2002)
   (c) $\int_0^1 x^{n-1}(1-x)^{n-1}\,dx=\frac{\Gamma(n)\Gamma(n)}{\Gamma(2n)}$ (Uttarakhand, 2004; UPTU, 2006)

6. Show that (a) $\int_0^\infty x^{n-1}e^{-x}\,dx\int_0^\infty x^{n-1}e^{-x}\,dx=\int_0^\infty x^{2n-1}e^{-x}\,dx$, $n>0$ (PTU, 2004; Mumbai, 2006)
   (b) $\int_0^1 \frac{x^{m-1}}{(1+x)^{m+n}}\,dx=\int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ (Kerala, 2004) (c) $\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ (Mumbai, 2004)
   (d) $\int_0^1 \frac{(1-x^{1/n})^{n-1}}{(1+x)^2}\,dx=\frac{1}{2^n}\Gamma\left(\frac{n}{2}\right)\Gamma\left(\frac{n+1}{2}\right)$ (Mumbai, 2003)

7. Prove that $\int_0^\infty x^n(\log x)^n\,dx=\frac{(-1)^n n!}{(m+1)^{n+1}}$, where n is a positive integer and $m>-1$ (UPTU, 2004)
   Hence evaluate $\int_0^1 x(\log x)^3\,dx$ (Nagpur, 2003)

8. Show that $\int_0^\infty x^{n-1}\left(\log\frac{1}{x}\right)^{m-1}dx=\frac{\Gamma(m)}{n^m}$, where $n>0$, $m>0$ (Kerala, 2003(S))

9. Express $\int_0^\infty e^{-x^2}(1-e^{-x^2})^n\,dx$ in terms of gamma functions (Uttarakhand, 2004)
   Hence evaluate (a) $\int_0^1 x(1-x^2)^{3/2}\,dx$ (Nagpur, 2003) (b) $\int_0^1 \frac{dx}{\sqrt{1-x^3}}$ (Anna, 2002)

10. Show that $\int_0^{\pi/2} \frac{dx}{\sqrt{x^2+1}} = \frac{1}{2}\sqrt{\pi}\,\frac{\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)}$ and hence evaluate $\int_0^{\pi/2} \sec^{\frac{1}{2}}\theta\,d\theta$.

11. Prove that $\int_0^{\pi/2} \sin^{n}\theta\,d\theta = \int_0^{\pi/2} \cos^{n}\theta\,d\theta = \frac{\sqrt{\pi}}{2}\,\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n+2}{2}\right)}$ hence evaluate $\int_0^{\pi/2} \sin^{\frac{3}{2}}\theta\,d\theta$, $\int_0^{\pi/2} \cos^{\frac{5}{2}}\theta\,d\theta$ and $\int_0^{\pi/2} \sin^{\frac{7}{2}}\theta\,d\theta$.

(Manchester, 2002)

12. Prove that:

(a) $\frac{\Gamma(m+1)\Gamma(n+1)}{\Gamma(m+n+2)} = \frac{\Gamma(m+1)\Gamma(n)}{\Gamma(m+n+1)} = \frac{\Gamma(m)\Gamma(n)}{(m+n)}$

(b) $\Gamma(n)\Gamma\left(n+\frac{1}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(2n)}{2^{2n-1}\Gamma(n)}$

(c) $\Gamma(n+1) = n\Gamma(n)$ and $\Gamma(n+1) = n!$ when n is a positive integer

(d) $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}$

(Wood, 2003; *J.M.T.C.*, 2006; Stroud, 2003)

13. Show that $\iint x^{m-1}y^{n-1}\,dx\,dy$ over the positive quadrant of the ellipse $\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$ is $\frac{1}{2}a^m b^n \mathrm{B}\left(\frac{m}{2}, \frac{n}{2}\right)$.

14. Show that the area in the first quadrant enclosed by the curve $(x/a)^n + (y/b)^n = 1$, $n > 0$, $a > 0$, $b > 0$, is given by $\frac{ab}{n}\,\mathrm{B}\left(\frac{1}{n}, \frac{1}{n}\right)$ or $\frac{ab}{n}\,\frac{\Gamma(1/n)\Gamma(1/n)}{\Gamma(2/n)}$.

15. Find the mass of the octant of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ if the density at any point is $kxyz$.

(*J.M.T.C.*, 2003)

24.7 (i) ELLIPTIC INTEGRALS

In Applied Mathematics, we often come across integrals of the form $\int_0^x x^{n-1}\,dx$ or $\int_0^{\pi/2} \sin^n\theta\,d\theta$ which cannot be evaluated by any of the standard methods of integration. In such cases, we may find the value to any desired degree of accuracy by expanding their integrands as power series. An important class of such integrals is the *elliptic integrals*.

Def. The integral $F(k,\phi) = \int_0^{\phi} \frac{d\alpha}{\sqrt{(1-k^2\sin^2\alpha)}}$ $(k^2 < 1)$ ... (1)

which is a function of the two variables k and ϕ, is called the *elliptic integral of the first kind*, and the constant k is called the *modulus* of (1).

The integral $E(k,\phi) = \int_0^{\phi} \sqrt{(1-k^2\sin^2\alpha)}\,d\alpha$ $(k^2 < 1)$ ... (2)

is called the *elliptic integral of the second kind* with modulus k and amplitude ϕ.

The name *elliptic integral* arose from its original application in finding the length of an ellipse arc (Fig. 7.45). For instance, consider the ellipse

$$x = a\cos\alpha, \qquad y = b\sin\alpha, \qquad (a > b)$$

The length of its arc

$$AP = \int_0^{\alpha} \sqrt{\left[\left(\frac{dx}{d\alpha}\right)^2 + \left(\frac{dy}{d\alpha}\right)^2\right]}\,d\alpha = \int_0^{\alpha} \sqrt{(a^2\sin^2\alpha + b^2\cos^2\alpha)}\,d\alpha$$

$$= \int_0^{\alpha} \sqrt{a^2(1-\cos^2\alpha) + b^2\cos^2\alpha}\,d\alpha = b\int_0^{\alpha} \sqrt{\left[1 - \left(1 - \frac{a^2}{b^2}\right)\cos^2\alpha\right]}\,d\alpha$$

$$= bE(k,\alpha) \text{ where } k^2 = 1 - a^2/b^2 < 1.$$

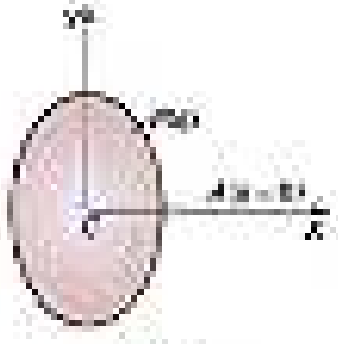

Fig. 7.45

Also the perimeter C of the ellipse

$$= 4b\int_0^{\pi/2}\sqrt{(1-k^2\sin^2\theta)}\,d\theta = 4bE(k,\pi/2).$$

This particular integral with upper limit $\phi = \pi/2$ is called the complete elliptic integral of the second kind and is denoted by $E(k)$.

Thus
$$E(k) = \int_0^{\pi/2}(1-k^2\sin^2\theta)^{1/2}\,d\theta \quad (0 < k < 1) \qquad \ldots(9)$$

Similarly, the complete elliptic integral of first kind is

$$K(k) = \int_0^{\pi/2}\frac{d\theta}{\sqrt{(1-k^2\sin^2\theta)}} \quad (k^2 < 1) \qquad \ldots(10)$$

To evaluate I_2 we expand the integrand in the form

$$(1-k^2\sin^2\theta)^{1/2} = 1 + \frac{k^2}{2}\sin^2\theta + \frac{3k^4}{4}\sin^4\theta + \ldots$$

This series can be shown to be uniformly convergent for $k < 1$, and may, therefore, be integrated term by term (See § 3.10 (3)). Then we have

$$E(k) = \int_0^{\pi/2}\left[1 + \frac{k^2}{2}\sin^2\theta - \frac{3k^4}{8}\sin^2\theta + \frac{5k^6}{16}\sin^6\theta + \ldots\right]d\theta$$

$$= \frac{\pi}{2}\left[1 - \left(\frac{1}{2}\right)^2 k^2 - \left[\frac{1.3}{2.4}\right]^2 k^4 - \left(\frac{1.3.5}{2.4.6}\right)^2 k^6 - \ldots\right] \qquad \ldots(11)$$

This series may be used to compute E for various values of k. In particular, if $k = \sin 15^\circ$, we have

$$K = \frac{\pi}{2}(1 + 0.067124 + 0.004512 + \ldots) = 1.5828 \qquad \ldots(12)$$

In the following tables of the elliptic integrals are constructed. Values of $F(k,\phi)$ and $E(k,\phi)$ are tabulated for $\phi = 0°(1°)90°$, $k = 0.0(0.1)1.0$. (See *Pretoria Abaci* tables).

Example 3.50. Express $\int_0^{\pi/2}\frac{dx}{\sqrt{(2+\cos x)}}$ in terms of elliptic integrals.

Solution. Put $\cos x = \cos^2\phi$ and $dx = \frac{2\cos\phi\,d\phi}{\sqrt{(1+\cos^2\phi)}}$

Then
$$I = \int_0^{\pi/2}\frac{2\cos^2\phi}{\sqrt{(1+\cos^2\phi)}}\,d\phi = \int_0^{\pi/2}\frac{(2+\cos^2\phi)-1}{\sqrt{(2+\cos^2\phi)}}\,d\phi$$

$$= \sqrt{2}\left\{\int_0^{\pi/2}\sqrt{(2+\cos^2\phi)}\,d\phi - \int_0^{\pi/2}\frac{d\phi}{\sqrt{(1-\cos^2\phi)}}\right\} = \sqrt{2}\left\{\int_0^{\pi/2}\sqrt{(2-\sin^2\phi)}\,d\phi - \int_0^{\pi/2}\frac{d\phi}{\sqrt{(2-\sin^2\phi)}}\right\}$$

$$= 2\sqrt{2}\int_0^{\pi/2}\sqrt{(1-\tfrac{1}{2}\sin^2\phi)}\,d\phi - \sqrt{2}\int_0^{\pi/2}\frac{d\phi}{\sqrt{(1-\tfrac{1}{2}\sin^2\phi)}} = 2\sqrt{2}\,E\left(\frac{1}{\sqrt{2}}\right) - \sqrt{2}\,K\left(\frac{1}{\sqrt{2}}\right)$$

(C) Elliptic functions. By putting $\sin\phi = t$ and $\sin\theta = x$, (1) becomes

$$u = \int_0^x\frac{dx}{\sqrt{(1-t^2)(1-k^2t^2)}} \quad (k^2 < 1) \qquad \ldots(13)$$

This is known as Jacobi's form of the elliptic integral of first kind[*] whereas (1) is the Legendre's form[†].

If $k = \sin\alpha$, $0 < k < 1$, is given we write $\sin^{-1}x$ for a fixed, non-zero value of α. This leads to the functions $\operatorname{sn} u = x = \sin\phi$ and $\operatorname{cn} u = \cos\phi$

which are called the Jacobi's elliptic functions.

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[*] See footnote p. 815

[†] A French mathematician Adrien Marie Legendre (1752–1833) who made important contributions to number theory, special functions, calculus of variations and elliptic integrals.

The elliptic functions are analytic and are periodic with a period depending on k and an amplitude equal to unity. These behave somewhat like sin x and cos x. For instance

$$\operatorname{sn}(0,k)=0,\ \operatorname{cn}(0,k)=1 \qquad \text{and} \qquad \operatorname{sn}(-x,k)=-\operatorname{sn}(x,k),\ \operatorname{cn}(-x,k)=\operatorname{cn}(x,k).$$

Example 3.61. Show that $\displaystyle\int_0^{\pi/2}\frac{dx}{\sqrt{(2x-x^2)(x^2-2x)}}=\frac{2}{\sqrt{\pi}}\Gamma\left(\frac{3}{4}\right)$

Solution. Putting $x=\frac{\pi}{2}(1-\sin\theta),\ dx=-\frac{\pi}{2}\cos\theta\,d\theta.$

$$\text{Also } x^2=\frac{\pi^2}{4}(1-\sin\theta)^2 \text{ and } 2x-x^2=\frac{\pi^2}{4}(1+\sin\theta)(1-\sin\theta)$$

Also when $x=0,\ \theta=\pi/2$; **when** $x=\pi/2,\ \theta=0$.

Thus the given integral

$$=\frac{4}{\pi^2}\int_{\pi/2}^{0}\frac{-(\pi/2)\cos\theta\,d\theta}{\sqrt{(1-\sin^2\theta)(1+\sin^2\theta)}}=\frac{2}{2\sqrt{\pi}}\int_0^{\pi/2}\frac{d\theta}{\sqrt{(1+\sin^2\theta)}}=\frac{2}{2\sqrt{\pi}}\Gamma\left(\frac{1}{4}\right)$$

3.18 (3) ERROR FUNCTION OR PROBABILITY INTEGRAL

The error function or the probability integral is defined as

$$\operatorname{erf}(x)=\frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt.$$

This integral arises in the solution of certain partial differential equations of applied mathematics and occurs frequently in the problems in the probability theory.

The complementary error function $\operatorname{erfc}(x)$ **is defined as** $\operatorname{erfc}(x)=1-\operatorname{erf}(x)$.

(i) Properties : (i) $\operatorname{erf}(-x)=-\operatorname{erf}(x)$. (ii) $\operatorname{erf}(0)=0$

(iii) $\operatorname{erf}(\infty)=\frac{2}{\sqrt{\pi}}\int_0^\infty e^{-t^2}\,dt=\frac{2}{\sqrt{\pi}}\frac{\sqrt{\pi}}{2}=1$ [By (3.1), p. 3.8]

This proves that the total area under the Normal or Gaussian error function curve is unity. – § 3.8, 3.9.

PROBLEMS 3.6

1. By means of the substitution $x=\sin\theta$, show that

   (i) $\displaystyle\int_0^1\frac{dx}{\sqrt{(1-x^4)}}=\frac{1}{4\sqrt{2\pi}}\Gamma\left(\frac{1}{4}\right)$,

   (ii) $\displaystyle\int_0^1\sqrt{(1-x^4)}\,dx=\left(\frac{1}{2}\right)\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{5}{4}\right)$

   where $\Gamma(n+1)=n\Gamma(n)=n!$ if n is +ve integer.

   Express the following integrals in terms of elliptic integrals :

2. $\displaystyle\int_0^{\pi/2}\frac{dx}{\sqrt{(1+3\sin^2 x)}}$ (Kerala, III Yr., 2001) 3. $\displaystyle\int_0^{\pi/2}\frac{dx}{\sqrt{(1+\cos^2 x)}}$ 4. $\displaystyle\int_0^{\pi/2}\sqrt{(\sin x)}\,dx$

5. Expand $\operatorname{erf}(x)$ in ascending powers of x. Hence evaluate $\operatorname{erf}(0.1)$. (P.T.U., 2000-01)

6. Compute (i) $\operatorname{erf}(0.2)$, (ii) $\operatorname{erf}(0.5)$, correct to three decimal places.

7. Show that (i) $\operatorname{erf}(x)+\operatorname{erf}(-x)=0$ (ii) $\operatorname{erf}(x)+\operatorname{erfc}(x)=1$.

8. Prove that

   (i) $\frac{d}{dx}\{\operatorname{erf}(ax)\}=\frac{2a}{\sqrt{\pi}}e^{-a^2x^2}$ (Kerala, III Yr., 2000) (ii) $\frac{d}{dx}\{\operatorname{erfc}(ax)\}=-\frac{2a}{\sqrt{\pi}}e^{-a^2x^2}$

9. Prove that $\displaystyle\int_0^x e^{-t^2}\cos 2xt\,dt=\frac{\sqrt{\pi}}{2}e^{-x^2}\{1-\operatorname{erf}(x)\}$

3.19 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 3.19

Fill up the blanks or choose the correct answer from the following problems:

1. $\int_0^1\int_0^1 (x+y)\,dx\,dy =$ ____ 2. $\int_0^1\int_0^{x^2} dy\,dx =$ ____ 3. $\int_0^{\pi/2}\int_0^{\cos\theta} r\,dr\,d\theta =$ ____

4. $\int_0^1\int_0^1 \frac{dx\,dy}{(x+y)} =$ ____ (V.T.U., 2010) 5. $\int_0^{\log 2}\int_0^{\log 2} e^{x+y}\,dx\,dy =$ ____

6. The surface area of the sphere $x^2+y^2+z^2=a^2$ in the first octant is ____

7. $\int_0^1\int_0^2\int_0^3 xyz\,dx\,dy\,dz =$ ____

8. If $u=x+y$ and $v=x-y$, then the area element $dx\,dy$ is replaced by ____ $du\,dv$.

9. In terms of Beta function, $\int_0^{\pi/2} \sin^3\theta\sqrt{\cos\theta}\,d\theta =$ ____

10. The value of $\int_0^{\pi/2}\int_0^{\pi/2} \sin(x+y)\,dx\,dy$ is ____ 11. $\int_0^1\int_0^x e^{x+y}\,dx\,dy =$ ____

12. The area bounded by $x=0$, $y=0$, $x+y=1$ is $2a^2+x^2=a^2$ is ____ (Use a single integral instead)

13. Value of $\int_0^1\int_0^1 xy^2\,dx\,dy$ is equal to

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{12}$ (Bhopal, 2009)

14. $\iint x^2y^2\,dx\,dy$ over the rectangle $0\le x\le 1$ and $0\le y\le 1$ is ____

15. $\int_0^{\pi/2}\int_0^{\cos\theta} r\,dr\,d\theta =$ ____ 16. $\int_0^{\log 2}\int_0^{x+\log 2} e^{x+y}\,dx\,dy =$ ____

17. $\int_0^{\infty}\int_{x^2}^{x^2+1} \frac{dy\,dx}{(x^2+y^2)} =$ ____ 18. $\int_0^1\int_0^{\sqrt{1-y^2}} (x^2+y^2)\,dx\,dy =$ ____

19. To change cartesian coordinates (x, y, z) to spherical polar coordinates (r, θ, ϕ), $dx\,dy\,dz$ is replaced by ____

20. $\int_0^1\int_0^{x^2} e^{x^2}\,dy\,dx =$ ____

21. $\iint xy(x^2+y^2)\,dx\,dy$ over the area bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is ____

22. $\int_0^1\int_0^{\sqrt{1-x^2}} \frac{dy\,dx}{1+x^2+y^2} =$ ____

23. $\iint xy\,dx\,dy$ over the area between $y=x^2$ and $y=x$ is ____

24. Value of $\int_0^{\pi/2}\int_0^{\pi/2} \sin(x+y)\,dx\,dy$ is

(a) zero (b) $\frac{1}{2}(\pi+2)$ (c) $\frac{1}{2}$ (d) 2. (V.T.U., 2010)

25. $\iint dx\,dy$ over the area bounded by $x=0$, $y=0$, $x^2+y^2=a^2$ and $x=a$ is ____

26. $\iint_R x\,dx\,dy$ where R is the region bounded by the parabolas $y^2=4x$ and $x^2=4y$, is ____

27. $\iint (x^2+y^2)\,dx\,dy$ in the positive quadrant for which $x+y\le 1$, is ____

28. Area between the parabolas $y^2=4x$ and $x^2=4y$ is ____

29. Changing the order of integration in $\int_0^a\int_{x^2/a}^{\sqrt{2ax-x^2}} f(x,y)\,dx\,dy =$ ____

30. $\int_0^{\infty} te^{-t}\,dt =$ ____ (V.T.U., 2010) 31. $\int_0^{\pi/2}\sqrt{\tan\theta}\,d\theta =$ ____ 32. $\int_0^1\int_0^{x^2} e^{y/x}\,dy\,dx =$ ____

33. On changing to polar coordinates $\int_0^{2a}\int_0^{\sqrt{2ax-x^2}} dx\,dy$ becomes ____

[illegible]

CHAPTER 8

Vector Calculus and Its Applications

1. Differentiation of vectors. 2. Curves in space. 3. Velocity and acceleration. Tangential and normal acceleration. Relative velocity and acceleration. 4. Scalar and vector point functions—Vector operator del. 5. Del applied to scalar point functions—Gradient. 6. Del applied to vector point functions—Divergence and Curl. 7. Physical interpretations of div F and curl F. 8. Del applied twice to point functions. 9. Del applied to products of point functions. 10. Integration of vectors. 11. Line integral—Circulation—Work. 12. Surface integral—Flux. 13. Green's theorem in the plane. 14. Stoke's theorem. 15. Volume integral. 16. Divergence theorem. 17. Green's theorem. 18. Irrotational and Solenoidal fields. 19. Orthogonal curvilinear coordinates. Del applied to functions in orthogonal curvilinear coordinates. 20. Cylindrical coordinates. 21. Spherical polar coordinates. 22. Objective Type of Questions.

8.1 DIFFERENTIATION OF VECTORS

If a vector $\mathbf{R}$ varies continuously as a scalar variable t changes, then $\mathbf{R}$ is said to be a function of t and is written as $\mathbf{R} = \mathbf{F}(t)$.

Just as in ordinary calculus, we define derivative of a vector function $\mathbf{R} = \mathbf{F}(t)$ as

$$\lim_{\Delta t \to 0} \frac{\mathbf{F}(t + \Delta t) - \mathbf{F}(t)}{\Delta t} \quad \text{and we write it as } \frac{d\mathbf{R}}{dt} \text{ or } \frac{d\mathbf{F}}{dt} \text{ or } \mathbf{F}'(t).$$

2) General rules of differentiation are similar to those of ordinary calculus provided the order of factors in vector products is maintained. Thus, if $\mathbf{F}, \mathbf{G}, \mathbf{H}$ are vector and vector functions of a scalar variable t, we have

(i) $\frac{d}{dt}(\mathbf{F} + \mathbf{G} - \mathbf{H}) = \frac{d\mathbf{F}}{dt} + \frac{d\mathbf{G}}{dt} - \frac{d\mathbf{H}}{dt}$ (ii) $\frac{d}{dt}(\phi\mathbf{F}) = \phi\frac{d\mathbf{F}}{dt} + \frac{d\phi}{dt}\mathbf{F}$

(iii) $\frac{d}{dt}(\mathbf{F} \cdot \mathbf{H}) = \mathbf{F} \cdot \frac{d\mathbf{H}}{dt} + \frac{d\mathbf{F}}{dt} \cdot \mathbf{H}$ (iv) $\frac{d}{dt}(\mathbf{F} \times \mathbf{G}) = \mathbf{F} \times \frac{d\mathbf{G}}{dt} + \frac{d\mathbf{F}}{dt} \times \mathbf{G}$

(v) $\frac{d}{dt}[\mathbf{F}\mathbf{G}\mathbf{H}] = \left[\frac{d\mathbf{F}}{dt}\mathbf{G}\mathbf{H}\right] + \left[\mathbf{F}\frac{d\mathbf{G}}{dt}\mathbf{H}\right] + \left[\mathbf{F}\mathbf{G}\frac{d\mathbf{H}}{dt}\right]$

(vi) $\frac{d}{dt}[\mathbf{F} \times (\mathbf{G} \times \mathbf{H})] = \left[\frac{d\mathbf{F}}{dt} \times \mathbf{G}\right] \times \mathbf{H} + \left[\mathbf{F} \times \frac{d\mathbf{G}}{dt}\right] \times \mathbf{H} + (\mathbf{F} \times \mathbf{G}) \times \frac{d\mathbf{H}}{dt}$

As we illustrate, let us prove (iv), while the others can be proved similarly:

$$\frac{d}{dt}(\mathbf{F} \times \mathbf{G}) = \lim_{\Delta t \to 0} \frac{(\mathbf{F} + \Delta\mathbf{F}) \times (\mathbf{G} + \Delta\mathbf{G}) - \mathbf{F} \times \mathbf{G}}{\Delta t} = \lim_{\Delta t \to 0} \frac{\mathbf{F} \times \Delta\mathbf{G} + \Delta\mathbf{F} \times \mathbf{G} + \Delta\mathbf{F} \times \Delta\mathbf{G}}{\Delta t}$$

$$= \lim_{\Delta t \to 0}\left[\mathbf{F} \times \frac{\Delta\mathbf{G}}{\Delta t} + \frac{\Delta\mathbf{F}}{\Delta t} \times \mathbf{G} + \frac{\Delta\mathbf{F}}{\Delta t} \times \Delta\mathbf{G}\right] = \mathbf{F} \times \frac{d\mathbf{G}}{dt} + \frac{d\mathbf{F}}{dt} \times \mathbf{G} \qquad [\because \Delta\mathbf{G} \to \mathbf{0} \text{ as } \Delta t \to 0]$$

Ex. 1. If $\mathbf{F}(t)$ has a constant magnitude, then $\mathbf{F} \cdot \frac{d\mathbf{F}}{dt} = 0$.

For $P(r)$: $\vec{R}(s) = \vec{r}(s) + \vec{a}$

$$\therefore \quad \vec{R}' = \frac{d\vec{R}}{ds} = \frac{d\vec{r}}{ds} + \vec{0} = \frac{d\vec{r}}{ds}$$

Obviously, $\vec{R}'(s)$ has constant length. Hence, $\vec{R}' = \frac{d\vec{r}}{ds} = \vec{0}$

Let $\vec{R}(s)$ be a unit vector in the direction of $\vec{r}(s)$ so that

$$\vec{r}(s) = r(s)\,\vec{R}(s), \quad \text{where } r(s) = |\vec{r}(s)|$$

$$\therefore \quad \frac{d\vec{r}}{ds} = r\frac{d\vec{R}}{ds} + \frac{dr}{ds}\vec{R} \quad \text{and} \quad \vec{r}\cdot\frac{d\vec{r}}{ds} = r\vec{R}\cdot\left[r\frac{d\vec{R}}{ds} + \frac{dr}{ds}\vec{R}\right]$$

$$\therefore \vec{r}\cdot\vec{R}' = \frac{r^2}{2}\frac{d}{ds}(\vec{R}\cdot\vec{R}) = 0 \qquad [\because \vec{R}\cdot\vec{R} = 1, \text{ a constant}, \ \therefore \frac{d}{ds}(\vec{R}\cdot\vec{R}) = 0]$$

Example 8.1. (i) $\vec{A} = (2t^2 - 3t)\vec{i} - t^3\vec{j} + (t+3)\vec{k}$, $\vec{B} = \sin t\,\vec{i} + \cos t\,\vec{j} - 3t^2\vec{k}$, find (i) $\frac{d}{dt}(\vec{A}\cdot\vec{B})$, (ii) $\frac{d}{dt}(\vec{A}\times\vec{B})$.

Solution. (i) $\frac{d}{dt}(\vec{A}\cdot\vec{B}) = \vec{A}\cdot\frac{d\vec{B}}{dt} + \frac{d\vec{A}}{dt}\cdot\vec{B}$

$$= (2t^2 - 3t)\vec{i} - t^3\vec{j} + (t+3)\vec{k})\cdot(\cos t\,\vec{i} - \sin t\,\vec{j} - 6t\vec{k}) + (4t - 3)\vec{i} - 3t^2\vec{j} + \vec{k})\cdot(\sin t\,\vec{i} + \cos t\,\vec{j})$$

$$= (2t^2\cos t + t\sin t - 6t^2 - \sin t - t\cos t) + (4t\sin t - 4\cos t - 3t^2\sin t + 3\cos t)$$

(ii) $\frac{d}{dt}(\vec{A}\times\vec{B}) = \vec{A}\times\frac{d\vec{B}}{dt} + \frac{d\vec{A}}{dt}\times\vec{B}$

$$= (2t^2 - 3t)\vec{i} - t^3\vec{j} + (t+3)\vec{k})\times(\cos t\,\vec{i} - \sin t\,\vec{j} - 6t\vec{k}) + (4t - 3)\vec{i} - 3t^2\vec{j} + \vec{k})\times(\sin t\,\vec{i} + \cos t\,\vec{j})$$

$$= (2t^2 - 3t)\vec{i} - t^3\vec{j} + (t+3)\vec{k})\times(\cos t\,\vec{i} - \sin t\,\vec{j} - 6t\vec{k}) + (4t - 3)\vec{i} - 3t^2\vec{j} + \vec{k})\times(\sin t\,\vec{i} + \cos t\,\vec{j})$$

$$= (6t^2\sin t + t\cos t - 3t\cos t - t^3\cos t)\vec{i} - (6t^2\cos t - t\sin t + 3t\sin t + t^3\sin t)\vec{j}$$
$$+ (t^3\sin t - 3t^2\cos t + t^2\cos t - 3\sin t)\vec{k} + (4t^2 - 3t)\sin t\,\vec{i} - (4t^2 - 3t)\cos t\,\vec{j} + (4t - 3)\vec{k}$$

8.7 CURVES IN SPACE

(1) Tangent. Let $\vec{R}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ be the position vector of a point P. Then as the scalar parameter t takes different values, the point P traces out a curve in space (Fig. 8.2). If the neighbouring point Q corresponds to $t + \delta t$, then $\vec{PR} = \vec{R}(t + \delta t) - \vec{R}(t)$. $\vec{PR}$ or $\vec{R}(t)$ is directed along the chord PQ. As $\delta t \to 0$, $Q \to P$ and $\vec{PR}$ becomes the tangent vector to the curve at P whenever it exists and is non-zero.

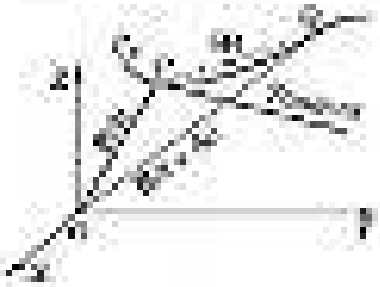

Fig. 8.2.

Then the vector $\vec{R}' = d\vec{R}/dt$ is a tangent to the space curve $\vec{R} = \vec{R}(t)$.

Let P_0 be a fixed point of the curve corresponding to $t = t_0$. If s be the length of the arc P_0P, then

$$\frac{\delta s}{\delta t} = \frac{\Delta s}{|\vec{PR}|}\cdot\frac{|\vec{PR}|}{\delta t} = \frac{\text{arc } PQ}{\text{chord } PQ}\left|\frac{\delta\vec{R}}{\delta t}\right|$$

As $Q \to P$ along the curve QR i.e., $\delta t \to 0$, then arc PQ/chord $PQ \to 1$ and

$$\frac{ds}{dt} = \left|\frac{d\vec{R}}{dt}\right| = |\vec{R}'(t)|.$$

If $\vec{R}(t)$ is continuous, then arc P_0P is given by

$$s = \int_{t_0}^{t} |\vec{R}'|\,dt = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt$$

If we take the parameter in place of t then the magnitude of the tangent vector, i.e., $|d\vec{R}/ds| = 1$. Thus denoting the unit tangent vector by $\vec{T}$, we have

$$\vec{T} = \frac{d\vec{R}}{ds} \qquad \ldots(1)$$

(2) Principal normal. Since $\vec{T}$ is a unit vector, $\vec{T}\cdot\vec{T} = 1$

$$\therefore \quad d(\vec{T}\cdot\vec{T}) = 0$$

i.e., $d\vec{T}$ is perpendicular to $\vec{T}$. Or since $\vec{T}\cdot\vec{T} = 1$, by which even $\vec{T}$ is a constant vector or $\kappa = 0$, the arc length is not to have a fixed direction, i.e., the curve is a straight line.

If we denote a unit normal vector to the curve at P by $\mathbf{N}$ then $d\mathbf{T}/ds$ is in the direction of $\mathbf{N}$ which is known as the principal normal to the space curve at P. The plane of $\mathbf{T}$ and $\mathbf{N}$ is called the osculating plane of the curve at P (Fig. 8.21).

(2) Binormal. A third unit vector $\mathbf{B}$ defined by $\mathbf{B} = \mathbf{T} \times \mathbf{N}$, is called the binormal at P. Since $\mathbf{T}$ and $\mathbf{N}$ are unit vectors, $\mathbf{B}$ is also a unit vector perpendicular to both $\mathbf{T}$ and $\mathbf{N}$ and hence normal to the osculating plane of P.

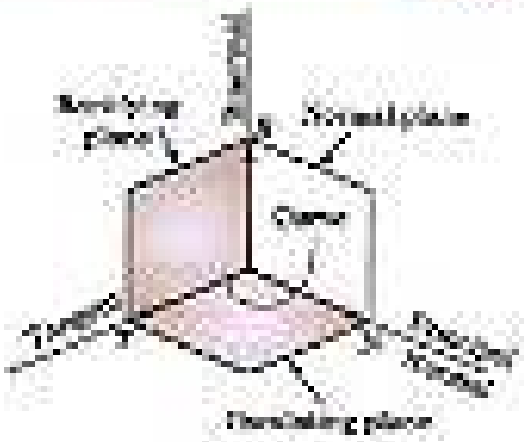

Fig. 8.21

Thus at each point P of a space curve there are three mutually perpendicular unit vectors, $\mathbf{T}$, $\mathbf{N}$, $\mathbf{B}$ which form a moving trihedral such that

$$\mathbf{T} = \mathbf{N} \times \mathbf{B},\ \mathbf{N} = \mathbf{B} \times \mathbf{T},\ \mathbf{B} = \mathbf{T} \times \mathbf{N} \qquad \ldots(5)$$

This moving trihedral determines the following three fundamental planes at each point of the curve :

(1) The osculating plane containing $\mathbf{T}$ and $\mathbf{N}$

(2) The normal plane containing $\mathbf{N}$ and $\mathbf{B}$

(3) The rectifying plane containing $\mathbf{B}$ and $\mathbf{T}$.

(4) Curvature. The arc rate of turning of the tangent (i.e., the magnitude of $d\mathbf{T}/ds$) is called the curvature of the curve and is denoted by κ.

Since $d\mathbf{T}/ds$ is in the direction of the principal normal $\mathbf{N}$, therefore,

$$\frac{d\mathbf{T}}{ds} = \kappa\mathbf{N} \qquad \ldots(6)$$

(5) Torsion. Since $\mathbf{B}$ is a unit vector, we have $\dfrac{d\mathbf{B}}{ds} \cdot \mathbf{B} = 0$

Also $\mathbf{B} \cdot \mathbf{T} = 0$, therefore $\dfrac{d\mathbf{B}}{ds} \cdot \mathbf{T} + \mathbf{B} \cdot \dfrac{d\mathbf{T}}{ds} = 0$.

or $\dfrac{d\mathbf{B}}{ds} \cdot \mathbf{T} + \mathbf{B} \cdot (\kappa\mathbf{N}) = 0$, i.e., $\dfrac{d\mathbf{B}}{ds} \cdot \mathbf{T} = 0$ $\quad [\because \mathbf{B} \cdot \mathbf{N} = 0]$

Hence $d\mathbf{B}/ds$ is perpendicular to both $\mathbf{B}$ and $\mathbf{T}$ and is, therefore, parallel to $\mathbf{N}$.

The arc rate of turning of the binormal (i.e., the magnitude of $d\mathbf{B}/ds$) is called torsion of the curve and is denoted by τ. We may, therefore, write

$$\frac{d\mathbf{B}}{ds} = -\tau\mathbf{N} \qquad \ldots(7)$$

(The negative sign indicates that for $\tau > 0$, $d\mathbf{B}/ds$ has direction of $-\mathbf{N}$).

Now by (6) and (7), we differentiate $\mathbf{N} = \mathbf{B} \times \mathbf{T}$,

$$\therefore \quad \frac{d\mathbf{N}}{ds} = \frac{d\mathbf{B}}{ds} \times \mathbf{T} + \mathbf{B} \times \frac{d\mathbf{T}}{ds} = -\tau\mathbf{N} \times \mathbf{T} + \mathbf{B} \times \kappa\mathbf{N}$$

Using the relations (1), it reduces to $\dfrac{d\mathbf{N}}{ds} = \tau\mathbf{B} - \kappa\mathbf{T}$ $\qquad \ldots(8)$

The equations (5), (6), (7) and (8) are known as the *"Frenet-Serret formulae"* for space curves.

Note 1. $\rho = 1/\kappa$ and $\sigma = 1/\tau$ are respectively called the radius of curvature and torsion.

2. For a plane curve, $\tau = 0$.

Example 8.22. Find the angle between the tangents to the curve $\mathbf{R} = t\mathbf{i} + 2\sin t\,\mathbf{j} + t\mathbf{k}$ at the points $t = \pi/2$, $t = \pi$. *(U.P.T.U., 2010)*

Solution. The tangent at any point t is given by

$$\frac{d\mathbf{R}}{dt} = \mathbf{i} + 2\cos t\,\mathbf{j} + \mathbf{k}$$

$\therefore$ the tangents $\mathbf{T}_1$, $\mathbf{T}_2$ at $t = \frac{\pi}{2}$ and $t = \pi$ are respectively given by

$$\mathbf{T}_1 = \mathbf{i} + 2\mathbf{j} + \mathbf{k},\ \mathbf{T}_2 = \mathbf{i} - 2\mathbf{j} + \mathbf{k},$$

* Named after a French mathematician Jean Frédéric Frenet (1816-1900).

Thus the required angle is given by $\mathbf{T}_1 \cdot \mathbf{T}_2 = |\mathbf{T}_1||\mathbf{T}_2| \cos\theta$, i.e. $2(-2) + 3 \cdot 2 + (-2)(-1) = \sqrt{17}\sqrt{12}\cos\theta$

i.e. $\sqrt{17}\sqrt{12}\cos\theta = 8 \quad \therefore \quad \theta = \cos^{-1}(0.8137)$

Example 6.2. Find the curvature and torsion of the curve $\mathbf{r} = a\cos t\,\mathbf{i} + a\sin t\,\mathbf{j} + ct\,\mathbf{k}$ ($a > 0$).

This curve is known as a circular helix lying on a cylinder making a constant angle and is known as a circular helix.

Solution. The vector equation of the curve is $\mathbf{R} = a\cos t\,\mathbf{i} + a\sin t\,\mathbf{j} + ct\,\mathbf{k}$

$\therefore \quad \dfrac{d\mathbf{R}}{dt} = -a\sin t\,\mathbf{i} + a\cos t\,\mathbf{j} + c\,\mathbf{k}$

Its arc length from P_0 ($t = 0$) to any point P is given by

$$s = \int_0^t |\mathbf{R}'(t)|\,dt = \sqrt{a^2 + c^2}\,t$$

$\therefore \quad \dfrac{ds}{dt} = \sqrt{a^2 + c^2}$

Now $\quad \mathbf{T} = \dfrac{d\mathbf{R}}{ds} = \dfrac{d\mathbf{R}}{dt}\Big/\dfrac{ds}{dt} = \dfrac{-a\sin t\,\mathbf{i} + a\cos t\,\mathbf{j} + c\,\mathbf{k}}{\sqrt{a^2 + c^2}}$

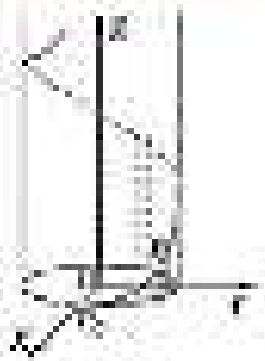

Fig. 6.1

and $\quad \dfrac{d\mathbf{T}}{ds} = \dfrac{d\mathbf{T}}{dt}\Big/\dfrac{ds}{dt} = \dfrac{-a\cos t\,\mathbf{i} - a\sin t\,\mathbf{j}}{a^2 + c^2}$

Thus $\quad \kappa = \left|\dfrac{d\mathbf{T}}{ds}\right| = \dfrac{a}{a^2 + c^2} \quad \ldots(1) \quad$ and $\quad \mathbf{N} = -\cos t\,\mathbf{i} - \sin t\,\mathbf{j}$

Also $\quad \mathbf{B} = \mathbf{T} \times \mathbf{N} = \dfrac{c\sin t\,\mathbf{i} - c\cos t\,\mathbf{j} + a\mathbf{k}}{\sqrt{a^2 + c^2}}$

$\therefore \quad \dfrac{d\mathbf{B}}{ds} = \dfrac{d\mathbf{B}}{dt}\Big/\dfrac{ds}{dt} = \dfrac{c\cos t\,\mathbf{i} + c\sin t\,\mathbf{j}}{a^2 + c^2} = -\dfrac{c}{a^2 + c^2}(-\cos t\,\mathbf{i} - \sin t\,\mathbf{j})$

Hence $\quad \tau = \dfrac{c}{a^2 + c^2} \quad \ldots(2)$

PROBLEMS 6.1

1. Show that, if $\mathbf{R} = A\cos(\omega t + B)\mathbf{i} + C\sin(\omega t + D)\mathbf{j}$, where A, B, C, D and ω are constants, then $\dfrac{d^2\mathbf{R}}{dt^2} = -\omega^2\mathbf{R}$ (Napier 2007)

   and $\mathbf{R} \times \dfrac{d\mathbf{R}}{dt} = \omega AC\sin(B - D)\mathbf{k}$

2. Given $\mathbf{R} = t^2\mathbf{A} + (t-1)\mathbf{B}$, where $\mathbf{A}$, $\mathbf{B}$ are constant vectors, show that $\mathbf{R}$ and $d^2\mathbf{R}/dt^2$ are parallel vectors, then state t when $\mathbf{R} = \mathbf{A}$.

3. If $\mathbf{P} = 3t\mathbf{i} + t^2\mathbf{j} - t^3\mathbf{k}$ and $\mathbf{Q} = \sin t\,\mathbf{i} - \cos t\,\mathbf{j} + 3\mathbf{k}$, find (a) $\dfrac{d}{dt}(\mathbf{P} \cdot \mathbf{Q})$, (b) $\dfrac{d}{dt}(\mathbf{P} \times \mathbf{Q})$.

4. If $\dfrac{d\mathbf{R}}{dt} = \mathbf{W} \times \mathbf{R}$ and $\dfrac{d\mathbf{V}}{dt} = \mathbf{W} \times \mathbf{V}$, prove that $\dfrac{d}{dt}(\mathbf{R} \times \mathbf{V}) = \mathbf{W} \times (\mathbf{R} \times \mathbf{V})$ (Manchester 2008)

5. If $\mathbf{A} = t^2\mathbf{i} - 2t\mathbf{j} + 3t^3\mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} + t\mathbf{j} - 3\mathbf{k}$, find $\dfrac{d^2}{dt^2}(\mathbf{A} \times \mathbf{B})$ at $t = 0$.

6. If $\mathbf{R} = a\cos(\omega t)\mathbf{i} + a\sin(\omega t)\mathbf{j} + ct\mathbf{k}$ is the curve C, find the value of

   (a) $\left|\dfrac{d\mathbf{R}}{dt} \times \dfrac{d^2\mathbf{R}}{dt^2}\right|$ (b) $\left|\dfrac{d\mathbf{R}}{dt}\ \dfrac{d^2\mathbf{R}}{dt^2}\ \dfrac{d^3\mathbf{R}}{dt^3}\right|$ (Salford 2003)

   Also find the unit tangent vector at any point t of the curve.

7. Find the unit tangent vector at any point on the curve $\mathbf{r} = t^2\mathbf{i} + (2t+1)\mathbf{j} - (3t+2)\mathbf{k}$, where t is any variable. Also determine the unit tangent vector at the point $t = 2$.

8. Find the equation of the tangent line to the curve $x = a\sin t$, $y = a\cos t$, $z = at\tan\alpha$ at the point $t = \pi/4$.

9. Find the curvature of the curve $\mathbf{R}(t) = a\sin t\,\mathbf{i} + b\sin t\,\mathbf{j} + c\cos t\,\mathbf{k}$ (a) generally (b) when $a = 2$, $b = -2$, $c = \sqrt{3}$ at the point $t = 1$.

(c) Find the equations of the osculating plane and binormal to the curve
(d) Find the curvature and torsion of the curve
12. A circular helix is given by the equation $\mathbf{R}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$. Find the curvature and torsion of the curve at any point and show that they are constant.
13. Show that for the curve $\mathbf{R} = a(3t - t^3)\mathbf{i} + 3at^2\mathbf{j} + a(3t + t^3)\mathbf{k}$, the curvature equals torsion.

5.5 [3] VELOCITY AND ACCELERATION

Let the position of a particle P at time t on a path (curve) C be $\mathbf{R}(t)$. At time $t + \Delta t$, let the particle be at Q (Fig. 5.13), then $\overrightarrow{PQ} = \mathbf{R}(t + \Delta t) - \mathbf{R}(t)$ or $\overrightarrow{PQ}$ is directed along PQ. As $Q \to P$ along C, the line PQ becomes the tangent at P to C.

$$\lim_{\Delta t \to 0} \frac{\Delta \mathbf{R}}{\Delta t} = \frac{d\mathbf{R}}{dt} = \mathbf{v}$$

is the tangent vector of C at P which is the velocity vector $\mathbf{V}$ of the motion and its magnitude is the speed $v = ds/dt$, where s is the arc length of P from a fixed point P_0 ($s = 0$) on C.

The derivative of the velocity vector $\mathbf{V}(t)$ is called the acceleration vector $\mathbf{A}(t)$, which is given by

$$\mathbf{A}(t) = \frac{d\mathbf{V}}{dt} = \frac{d^2\mathbf{R}}{dt^2}.$$

(2) Tangential and normal accelerations. It is important to note that the magnitude of acceleration is not always the rate of change of $|\mathbf{V}|$ because $\mathbf{A}(t)$ is not always tangential to the path C. In fact

$$\mathbf{V}(t) = \frac{d\mathbf{R}}{dt} = \frac{d\mathbf{R}}{ds}\frac{ds}{dt}, \text{ where } \frac{d\mathbf{R}}{ds} \text{ is a unit tangent vector of } C.$$

$$\mathbf{A}(t) = \frac{d\mathbf{V}}{dt} = \frac{d}{dt}\left[\frac{ds}{dt}\cdot\frac{d\mathbf{R}}{ds}\right] = \frac{d^2s}{dt^2}\cdot\frac{d\mathbf{R}}{ds} + \left(\frac{ds}{dt}\right)^2\frac{d^2\mathbf{R}}{ds^2}$$

Now since $\left|\frac{d\mathbf{R}}{ds}\right| = 1$, $\frac{d^2\mathbf{R}}{ds^2}$ is perpendicular to $\frac{d\mathbf{R}}{ds}$. Hence the acceleration $\mathbf{A}(t)$ is comprised of (i) the tangential component $\frac{d^2s}{dt^2}\cdot\frac{d\mathbf{R}}{ds}$, called the tangential acceleration, and

(ii) the normal component $\left(\frac{ds}{dt}\right)^2\frac{d^2\mathbf{R}}{ds^2}$, called the normal acceleration.

(Note: The acceleration is the time rate change of $|\mathbf{V}|$ only if the normal acceleration is zero, for then $|\mathbf{A}| = \left|\frac{d^2s}{dt^2}\right| = \left|\frac{d|\mathbf{V}|}{dt}\right| = \left|\frac{d^2s}{dt^2}\right|$.)

(3) Relative velocity and acceleration. Let two particles P_1 and P_2 moving along the curves C_1 and C_2 have position vectors $\mathbf{R}_1$ and $\mathbf{R}_2$ at time t. (Fig. 5.14), so that $\mathbf{R} = \overrightarrow{P_1P_2} = \mathbf{R}_2 - \mathbf{R}_1$

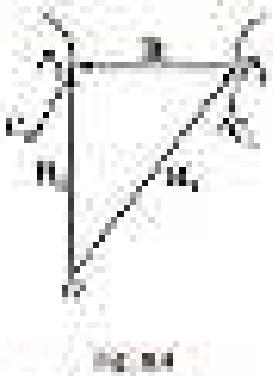

Fig. 5.14

Differentiating w.r.t. t, we get $\frac{d\mathbf{R}}{dt} = \frac{d\mathbf{R}_2}{dt} - \frac{d\mathbf{R}_1}{dt}$ ...(iv)

This defines the relative velocity vector of P_2 w.r.t. P_1 and means that the velocity vector of P_2 relative to P_1 = velocity (vector) of P_2 − velocity (vector) of P_1.

Again differentiating (iv), we have $\frac{d^2\mathbf{R}}{dt^2} = \frac{d^2\mathbf{R}_2}{dt^2} - \frac{d^2\mathbf{R}_1}{dt^2}$ ...(v)

i.e., acceleration (vector) of P_2 relative to P_1 = acceleration (vector) of P_2 − acceleration (vector) of P_1.

Example 5.6. A particle moves along the curve $x^2 = t^2 + 1$, $y = t^2 - 1$, $z = 2t - 3$, where t is the time. Find the components of its velocity and acceleration at $t = 1$ in the direction $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

Solution. Velocity $= \frac{d\mathbf{R}}{dt} = \frac{d}{dt}[(t^2 + 1)\mathbf{i} + (t^2 - 1)\mathbf{j} + (2t - 3)\mathbf{k}]$

$= 2t\mathbf{i} + 2t\mathbf{j} + 2\mathbf{k} = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ at $t = 1$

and acceleration $= \frac{d^2\mathbf{R}}{dt^2} = 2\mathbf{i} + 2\mathbf{j} + 0\mathbf{k} = 2\mathbf{i} + 2\mathbf{j}$ at $t = 1$.

Now unit vector in the direction of $\mathbf{I}+\mathbf{J}+3\mathbf{K}$ is $\dfrac{\mathbf{I}+\mathbf{J}+3\mathbf{K}}{\sqrt{1^2+1^2+3^2}}=\dfrac{1}{\sqrt{11}}(\mathbf{I}+\mathbf{J}+3\mathbf{K})$

$\therefore$ component of velocity at $t=1$ in the direction $\mathbf{I}+\mathbf{J}+3\mathbf{K}$

$$=\frac{(2\mathbf{I}+2\mathbf{J}+6\mathbf{K})\cdot(\mathbf{I}+\mathbf{J}+3\mathbf{K})}{\sqrt{11}}=\frac{2+2+6}{\sqrt{11}}=\sqrt{(11)}$$

and component of acceleration at $t=2$ in the direction

$$\mathbf{I}+\mathbf{J}+3\mathbf{K}=(\mathbf{I}+2\mathbf{J})\cdot(\mathbf{I}+\mathbf{J}+3\mathbf{K})/\sqrt{11}=\frac{6+2}{\sqrt{11}}=\frac{8}{\sqrt{11}}.$$

Example 2.8. *A particle moves along the curve $\mathbf{R}=t^3-4t\mathbf{I}+t^2\mathbf{J}+4t\mathbf{J}+(5t-2t^2)\mathbf{K}$ where t denotes time. Find the magnitudes of acceleration along the tangent and normal at time $t=2$.* (U.P.T.U. 2003 O)

Solution. Velocity $\dfrac{d\mathbf{R}}{dt}=3t^2-4\mathbf{I}+2t+4\mathbf{J}+(5-4t)\mathbf{K}$

and acceleration $\dfrac{d^2\mathbf{R}}{dt^2}=6t\mathbf{I}+2\mathbf{J}-4\mathbf{K}$

$\therefore$ at $t=2$, velocity $\mathbf{V}=8\mathbf{I}+6\mathbf{J}-4\mathbf{K}$ and acceleration $\mathbf{A}=12\mathbf{I}+2\mathbf{J}-4\mathbf{K}$.

Since the velocity is along the tangent to the curve, therefore, the component of $\mathbf{A}$ along the tangent

$$=\mathbf{A}\cdot\frac{\mathbf{V}}{|\mathbf{V}|}=(12\mathbf{I}+2\mathbf{J}-4\mathbf{K})\cdot\frac{8\mathbf{I}+6\mathbf{J}-4\mathbf{K}}{\sqrt{(64+36+16)}}$$

$$=\frac{12\times8+2\times6+(-4)\times(-4)}{12}=16.$$

Now the component of $\mathbf{A}$ along the normal

$$=|\,\mathbf{A}-\text{Resolved part of }\mathbf{A}\text{ along the tangent}\,|$$

$$=\left|12\mathbf{I}+2\mathbf{J}-4\mathbf{K}-16\frac{8\mathbf{I}+6\mathbf{J}-4\mathbf{K}}{12}\right|=\frac{1}{3}|\,4\mathbf{I}-26\mathbf{J}-4\mathbf{K}\,|=2\sqrt{73}.$$

Example 2.9. *The position vector of a particle at time t is $\mathbf{R}=\cos t\,\mathbf{I}+\sin t\,\mathbf{J}+t\mathbf{K}$. Find the condition imposed on a by requiring that at time $t=1$, the acceleration is normal to the position vector.*

Solution. Velocity $\mathbf{v}=\dfrac{d\mathbf{R}}{dt}=-\sin t\,\mathbf{I}+\cos t\,\mathbf{J}+3t^2\mathbf{K}$

Acceleration $\mathbf{a}=\dfrac{d^2\mathbf{R}}{dt^2}=-\cos t\,\mathbf{I}-\sin t\,\mathbf{J}+6t\mathbf{K}=-\mathbf{I}+6a\mathbf{K}$ at $t=1$.

At $t=1$, $\mathbf{R}=\mathbf{I}+a\mathbf{K}$ at $t=1$.

If $\mathbf{R}$ and acceleration at $t=1$ are normal, then their scalar product is zero.

$\therefore$ $(-\mathbf{I}+6a\mathbf{K})\cdot(\mathbf{I}+a\mathbf{K})=0$ or $-1+6a^2=0$

or $a^2=\dfrac{1}{6}$ or $a=1/\sqrt{6}$.

Example 2.10. *Find the radial and transverse accelerations of a particle moving in a plane curve.*
(Kanpur Univ. 2000; Rajasthan, 2000)

Solution. At any time t, let the position vector of the moving particle $P(r,\theta)$ be $\mathbf{R}$ (Fig. 2.3) so that

$$\mathbf{R}=r\mathbf{R}=r\cos\theta\,\mathbf{I}+r\sin\theta\,\mathbf{J}$$

$\therefore$ its velocity $\mathbf{V}=\dfrac{d\mathbf{R}}{dt}=\dfrac{dr}{dt}\mathbf{R}+r\dfrac{d\mathbf{R}}{dt}$

Now $\mathbf{R}=\cos\theta\,\mathbf{I}+\sin\theta\,\mathbf{J}$

and $\dfrac{d\mathbf{R}}{dt}=(-\sin\theta\,\mathbf{I}+\cos\theta\,\mathbf{J})\dfrac{d\theta}{dt}$

Fig 2.3

$\therefore$ $\frac{d\hat{\mathbf{i}}}{dt} \perp \hat{\mathbf{i}}$ and $\left|\frac{d\hat{\mathbf{i}}}{dt}\right| = \frac{d\theta}{dt}$, i.e., if $\hat{\mathbf{U}}$ is a unit vector $\perp \hat{\mathbf{i}}$, then

$$\frac{d\hat{\mathbf{i}}}{dt} = \frac{d\theta}{dt}\hat{\mathbf{U}}$$

$\therefore$ (i) becomes, $\mathbf{V} = \frac{dr}{dt}\hat{\mathbf{i}} + r\frac{d\theta}{dt}\hat{\mathbf{U}}$ ...(ii)

Thus the radial and transverse components of the velocity are $\dot{r}\hat{\mathbf{i}}$ and $r\dot{\theta}\hat{\mathbf{U}}$.

Also $\mathbf{A} = \frac{d\mathbf{V}}{dt} = \frac{d^2r}{dt^2}\hat{\mathbf{i}} + \frac{dr}{d\theta}\frac{d\hat{\mathbf{i}}}{d\theta} + \frac{dr}{d\theta}\frac{d\theta}{dt}\hat{\mathbf{U}} + r\frac{d^2\theta}{dt^2}\hat{\mathbf{U}} + r\frac{d\theta}{dt}\frac{d\hat{\mathbf{U}}}{d\theta}\frac{d\theta}{dt}$

$$= \left[\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2\right]\hat{\mathbf{i}} + \left[2\frac{dr}{dt}\frac{d\theta}{dt} + r\frac{d^2\theta}{dt^2}\right]\hat{\mathbf{U}} \qquad \left[\because \hat{\mathbf{U}} = -\sin\theta\,\hat{\mathbf{i}} + \cos\theta\,\hat{\mathbf{j}} \text{ gives } \frac{d\hat{\mathbf{U}}}{d\theta} = -\frac{d\hat{\mathbf{i}}}{d\theta}\right]$$

Thus the radial and transverse components of the acceleration are

$$\frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2 \text{ and } 2\frac{dr}{dt}\frac{d\theta}{dt} + r\frac{d^2\theta}{dt^2}.$$

Example 8.64 A person going northwards at 4 km per hour finds that the wind appears to blow directly from the west. He then runs at the same speed and the wind seems to come from north-east. Find the actual velocity of the wind.

Solution. Let the actual velocity of the wind be $x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$, where $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ represent velocities of 1 km per hour towards the east and north respectively. As the person is going northwards with a velocity of 4 km per hour, his actual velocity is $4\hat{\mathbf{j}}$.

Then the velocity of the wind relative to the man is $(x\hat{\mathbf{i}} + y\hat{\mathbf{j}}) - 4\hat{\mathbf{j}}$, which is parallel to $-\hat{\mathbf{i}}$, as it appears to blow from the north. Hence $y = 4$ ...(i)

When the velocity of the person becomes $2\hat{\mathbf{j}}$, the velocity of the wind relative to him is $x\hat{\mathbf{i}} + y\hat{\mathbf{j}} - 2\hat{\mathbf{j}}$. But this is parallel to $-(\hat{\mathbf{i}} + \hat{\mathbf{j}})$.

$\therefore$ $(x - 2)\hat{\mathbf{j}} = \lambda(\hat{\mathbf{i}} + \hat{\mathbf{j}})$, which by (i) gives $y = -4$.

Hence the actual velocity of the wind is $4(\hat{\mathbf{i}} - \hat{\mathbf{j}})$, i.e., $4\sqrt{2}$ km. per hour towards the south-east.

EXERCISE 8.3

1. A particle moves along a curve $x = t^3$, $y = 2t^2$ where t is the time variable. Determine the velocity and acceleration vectors and also the magnitudes of velocity and acceleration at $t = 1$.
(I.I.T., 2000; V.T.U., 2004)

2. The position vector of a particle at time t is $\mathbf{R} = \cos\omega t\,\hat{\mathbf{i}} + \sin\omega t\,\hat{\mathbf{j}}$. Find the velocity of particle at any time t proving that at time t, the acceleration is normal to the position vector.

3. A particle moves on the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the components of velocity and acceleration at time $t = 1$ in the direction $\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$. (V.T.U., 2005)

4. A particle moves on the curve whose vectorial equation is given by $\mathbf{R} = 3\cos t\,\hat{\mathbf{i}} + 3\sin t\,\hat{\mathbf{j}} + 4t\,\hat{\mathbf{k}}$. Show that the velocity $\mathbf{V}$ of the particle is perpendicular to $\mathbf{R}$ and $(\mathbf{R} \times \mathbf{V})$ is a constant vector.

5. A particle position vector $\mathbf{R}(t)$ is moving in a circle with constant angular velocity ω. Show by vector methods, that the acceleration is equal to $-\omega^2\mathbf{R}$.

6. (a) Find the tangential and normal accelerations of a point moving in a plane curve. (Gujarat, 2002)

(b) The position vector of a moving particle at time t is $\mathbf{R} = 3\cos t\,\hat{\mathbf{i}} + 4\sin t\,\hat{\mathbf{j}} + 2t\,\hat{\mathbf{k}}$. Find the tangential and normal components of the acceleration at $t = \frac{\pi}{2}$. (Gujarat, 2002)

7. The velocity of a boat relative to water is represented by $3\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ and that of water relative to earth is $\hat{\mathbf{i}} - 3\hat{\mathbf{j}}$. What is the velocity of the boat relative to the earth, if $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ represent one km. due east and north respectively?

8. A vessel is sailing with a velocity of 11 knots per hour in the direction $\hat{\mathbf{i}}$, and a tidal wind $\mathbf{P}$ is blowing with a velocity of 10 knots per hour in a direction $\frac{1}{5}(3\hat{\mathbf{i}} + 4\hat{\mathbf{j}})$. Find the velocity of A relative to B.

9. A person travels up towards the north-east with a velocity of 6 km per hour finds that the wind appears to blow from the north, but when he doubles his speed it seems to come from a direction making an angle $\tan^{-1}\frac{1}{2}$ to the north of east. Show that the actual velocity of the wind is $2\sqrt{5}$ km per hour towards the east.

8.4 SCALAR AND VECTOR POINT FUNCTIONS

(1) If to each point $P(\mathbf{R})$ of a region R in space there corresponds a definite scalar denoted by $f(\mathbf{R})$, then $f(\mathbf{R})$ is called a scalar point function in R. The region R is defined as called a scalar field.

The temperature at any instant, density of a body and potential due to gravitational matter are all examples of scalar point functions.

(2) If to each point $P(\mathbf{R})$ of a region R in space there corresponds a definite vector denoted by $\mathbf{F}(\mathbf{R})$, then it is called the vector point function in R. The region R is defined as called a vector field.

The velocity of a moving fluid at any instant, the gravitational intensity of force are examples of vector point functions.

Differentiation of vector point functions follows the same rules as those of ordinary calculus. Thus if $\mathbf{F}(x, y, z)$ be a vector point function, then

$$\frac{d\mathbf{F}}{dt} = \frac{\partial \mathbf{F}}{\partial x}\frac{dx}{dt} + \frac{\partial \mathbf{F}}{\partial y}\frac{dy}{dt} + \frac{\partial \mathbf{F}}{\partial z}\frac{dz}{dt} \qquad \text{(See (6) p. 284)}$$

and
$$d\mathbf{F} = \frac{\partial \mathbf{F}}{\partial x}dx + \frac{\partial \mathbf{F}}{\partial y}dy + \frac{\partial \mathbf{F}}{\partial z}dz = \left(\frac{\partial}{\partial x}dx + \frac{\partial}{\partial y}dy + \frac{\partial}{\partial z}dz\right)\mathbf{F} \qquad \ldots(5)$$

(3) Vector operator del. The operator on the right side of the equation (1) is in the form of a scalar product of $\mathbf{I}\frac{\partial}{\partial x} + \mathbf{J}\frac{\partial}{\partial y} + \mathbf{K}\frac{\partial}{\partial z}$ and $\mathbf{I}dx + \mathbf{J}dy + \mathbf{K}dz$.

If ∇ (read as del) is defined by the equation $\nabla \equiv \mathbf{I}\frac{\partial}{\partial x} + \mathbf{J}\frac{\partial}{\partial y} + \mathbf{K}\frac{\partial}{\partial z}$ ...(6)

then it may be written as $d\mathbf{F} = (\nabla \cdot d\mathbf{R})\mathbf{F}$ for when $\mathbf{R} = x\mathbf{I} + y\mathbf{J} + z\mathbf{K}$, $d\mathbf{R} = \mathbf{I}dx + \mathbf{J}dy + \mathbf{K}dz$.

8.5 DEL APPLIED TO SCALAR POINT FUNCTIONS—GRADIENT

(1) Def. *The vector function ∇f is defined as the gradient of the scalar point function f and is written as grad f.*

Thus
$$\text{grad } f = \nabla f = \mathbf{I}\frac{\partial f}{\partial x} + \mathbf{J}\frac{\partial f}{\partial y} + \mathbf{K}\frac{\partial f}{\partial z}$$

(2) Geometrical interpretation. Consider the scalar point function $f(\mathbf{R})$, where $\mathbf{R} = x\mathbf{I} + y\mathbf{J} + z\mathbf{K}$.

If the surface $f(x, y, z) = c$ can be drawn through any point $P(\mathbf{R})$ such that at each point on it, the function has the same value as at P, then such a surface is called a *level surface* of the function f through P, *e.g.*, equipotential or isothermal surface [Fig. 8.6].

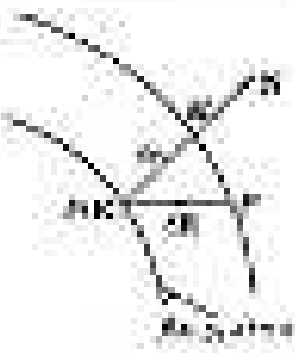

Fig. 8.6

Let $P'(\mathbf{R} + \delta\mathbf{R})$ be a point on a neighbouring level surface $f + \delta f$. Then

$$\nabla f \cdot \delta\mathbf{R} = \left(\mathbf{I}\frac{\partial f}{\partial x} + \mathbf{J}\frac{\partial f}{\partial y} + \mathbf{K}\frac{\partial f}{\partial z}\right) \cdot (\mathbf{I}\delta x + \mathbf{J}\delta y + \mathbf{K}\delta z)$$

$$= \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y + \frac{\partial f}{\partial z}\delta z = \delta f$$

Now if P' lies on the same level surface as P, then $\delta f = 0$, *i.e.*, $\nabla f \cdot \delta\mathbf{R} = 0$. This means that ∇f is perpendicular to any $\delta\mathbf{R}$ lying on the surface. Thus ∇f is normal to the surface $f(x, y, z) = c$.

$$\therefore \qquad \nabla f = |\nabla f|\,\mathbf{N}$$

where $\mathbf{N}$ is a unit vector normal to the surface. If the perpendicular distance δn between the surfaces through P and P' be δn, then the rate of change of f normal to the surface through P

$$= \frac{df}{dn} = \lim_{\delta n \to 0}\frac{\delta f}{\delta n} = \lim_{\delta s \to 0}\nabla f \cdot \frac{\delta\mathbf{R}}{\delta n}$$

$$= |\nabla f| \lim_{\delta s \to 0}\frac{\mathbf{N} \cdot \delta\mathbf{R}}{\delta n} = |\nabla f|, \qquad \left[\because \mathbf{N} \cdot \delta\mathbf{R} = |\delta\mathbf{R}| \cos\theta = \delta n\right]$$

Hence the magnitude of $\nabla f = df/dn$.

Thus grad f is a vector normal to the surface $f =$ constant and has a magnitude equal to the rate of change of f along this normal.

(3) Directional derivative. If δr denotes the length PP' and N is a unit vector in the direction PP', then the limiting value of $\frac{\delta f}{\delta r}$ as $\delta r \to 0$ (i.e., $\frac{df}{dr}$) is known as the directional derivative of f at P along the direction PP'.

Since $\qquad \delta r = \delta r(\cos\alpha\, i + \cos\beta\, j + \cos\gamma\, k) = \delta r N$, N

$\therefore \qquad \frac{df}{dr} = \lim_{\delta r \to 0}\left[N_1 N \frac{\delta f}{\delta r}\right] = N_1 \cdot \frac{\partial f}{\partial n} N = N \cdot \nabla f$

Thus the directional derivative of f in the direction of N is the resolved part of ∇f in the direction N.

Since $\qquad \nabla f \cdot N = |\nabla f| \cos\alpha \le |\nabla f|$

It follows that ∇f gives the maximum rate of change of f.

Example 3.2: *Prove that* $\nabla r^n = n r^{n-2} R$, *where* $R = xi + yj + zk$. *(Bhopal, 2000; Anna, 2002; R, 07 III)*

Solution. We have $f(x, y, z) = r^n = (x^2 + y^2 + z^2)^{n/2}$

$\therefore \qquad \frac{\partial f}{\partial x} = \frac{n}{2}(x^2 + y^2 + z^2)^{\frac{n}{2}-1}(2x) = n x r^{n-2}$. Similarly, $\frac{\partial f}{\partial y} = n y r^{n-2}$ and $\frac{\partial f}{\partial z} = n z r^{n-2}$

Thus $\qquad \nabla r^n = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z} = n r^{n-2}(xi + yj + zk) = n r^{n-2} R$.

Otherwise: The level surfaces for f = constant, i.e., r^n = constant are concentric spheres with centre O and hence unit normal N to the level surface through P is along the radius R

i.e., $\qquad N = \hat{R}$.

$\therefore \qquad \nabla f = \frac{\partial f}{\partial n} \cdot N = \frac{\partial f}{\partial r} R = n r^{n-1} \hat{R}$ $\qquad$ [$\because f = r^n$]

$= n r^{n-1}\left(\frac{R}{r}\right) = n r^{n-2} R$.

Example 3.13: *If* $\nabla(r^n) = n r^{n-2} R$, *find* n. *(Kurukshetra, 2006)*

Solution. We have $\nabla(r^n) = \nabla(x^2 + y^2 + z^2)^{n/2} R$ $\qquad$ [$\because r = \sqrt{(x^2 + y^2 + z^2)}$]

$= 2n(x^2 + y^2 + z^2)^{\frac{n}{2}-1}(xi + yj + zk)$ $\qquad$ ...(1)

But $\qquad \nabla(r^n) = \frac{\partial r^n}{\partial x} i + \frac{\partial r^n}{\partial y} j + \frac{\partial r^n}{\partial z} k$ $\qquad$ ...(2)

Comparing (1) and (2), we get

$$\frac{\partial r^n}{\partial x} = 2n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} x, \frac{\partial r^n}{\partial y} = 2n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} y, \frac{\partial r^n}{\partial z} = 2n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} z$$

Also $\qquad d(r^n) = \frac{\partial r^n}{\partial x} dx + \frac{\partial r^n}{\partial y} dy + \frac{\partial r^n}{\partial z} dz = 2n(x^2 + y^2 + z^2)^{\frac{n}{2}-1}(x dx + y dy + z dz)$

$= 2n r^{n-2}\left(\frac{r}{2}\right)$, taking $x^2 + y^2 + z^2 = r^2$ and $2(x dx + y dy + z dz) = d(r^2)$

Integrating both sides, $r^n = \int r^{n-2} dr + c = \frac{1}{n} r^n + c = \frac{1}{n}(x^2 + y^2 + z^2)^{\frac{n}{2}} + c$

Hence $\qquad n = \frac{1}{2} r^{n-2} + c$.

Example 3.14: *If* $u = x + y + z$, $v = x^2 + y^2 + z^2$, $w = xy + yz + zx$, *prove that grad* u, *grad* v *and grad* w *are coplanar.* *(U.P.T.U., 2000; U.P.T.U., 2001)*

Solution. $\qquad \text{grad}\, u = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right)(x + y + z) = i + j + k$

$\text{grad}\, v = 2xi + 2yj + 2zk$, $\text{grad}\, w = (y + z)i + (z + x)j + (x + y)k$

We know that three vectors are coplanar if their scalar triple product is zero.

Hence $[\text{grad}\,u,\ \text{grad}\,v,\ \text{grad}\,w]$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 2x & 2y & 2z \\ y+z & z+x & x+y \end{vmatrix}$$

$$= 2\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ y+z & z+x & x+y \end{vmatrix} = 2\begin{vmatrix} 1 & 1 & 1 \\ x+y+z & y+z+x & x+y+z \\ y+z & z+x & x+y \end{vmatrix} \qquad [\text{Operate } R_2 + R_3]$$

$$= 2(x+y+z)\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ y+z & z+x & x+y \end{vmatrix} = 0.$$

Hence $\text{grad}\,u$, $\text{grad}\,v$ and $\text{grad}\,w$ are coplanar.

Example 3.12. *Find a unit vector normal to the surface $xy^3z^2 = 4$ at the point $(-1, -1, 2)$.* (Madras, 2000)

Solution. A vector normal to the given surface is $\nabla(xy^3z^2)$

$$= \mathbf{i}\frac{\partial}{\partial x}(xy^3z^2) + \mathbf{j}\frac{\partial}{\partial y}(xy^3z^2) + \mathbf{k}\frac{\partial}{\partial z}(xy^3z^2) = \mathbf{i}(y^3z^2) + \mathbf{j}(3xy^2z^2) + \mathbf{k}(2xy^3z)$$

$$= -4\mathbf{i} - 12\mathbf{j} + 4\mathbf{k} \text{ at the point } (-1, -1, 2).$$

Hence the required unit normal to the surface

$$= \frac{-4\mathbf{i} - 12\mathbf{j} + 4\mathbf{k}}{\sqrt{(-4)^2 + (-12)^2 + 4^2}} = \frac{1}{\sqrt{11}}(\mathbf{i} + 3\mathbf{j} - \mathbf{k}).$$

Example 3.13. *Find the directional derivative of $f(x, y, z) = xy^2 - yz^3$ at the point $(2, -1, 1)$ in the direction of the vector $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.* (Nagpur, 2005; Kerala, Anna, 2006; Madras, 2007)

Solution. Here $\nabla f = \mathbf{i}(y^2) + \mathbf{j}(2xy - z^3) + \mathbf{k}(-3yz^2) = \mathbf{i} - 3\mathbf{j} - 3\mathbf{k}$ at the point $(2, -1, 1)$.

$\therefore$ directional derivative of f in the direction $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$

$$= (\mathbf{i} - 3\mathbf{j} - 3\mathbf{k}) \cdot \frac{\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}}{\sqrt{1^2 + 2^2 + 2^2}} = (1)\cdot\frac{1}{3} + (-3)\cdot\frac{2}{3} + (-3)\cdot\frac{2}{3} = -\frac{9}{3} = -3.$$

Example 3.14. *Find the directional derivative of $f = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ where Q is the point $(5, 0, 4)$. Also find the magnitude of the maximum directional derivative.*

Solution. We have $\nabla f = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right)(x^2 - y^2 + 2z^2) = 2x\mathbf{i} - 2y\mathbf{j} + 4z\mathbf{k}$

$$= 2\mathbf{i} - 4\mathbf{j} + 12\mathbf{k} \text{ at } (1, 2, 3).$$

Also $\quad \vec{PQ} = \vec{OQ} - \vec{OP} = (5\mathbf{i} + 0\mathbf{j} + 4\mathbf{k}) - (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}) = 4\mathbf{i} - 2\mathbf{j} + \mathbf{k} = \vec{A}$ (say)

$\therefore$ unit vector of $\vec{A} = \hat{A} = \frac{\vec{A}}{A} = \frac{4\mathbf{i} - 2\mathbf{j} + \mathbf{k}}{\sqrt{16 + 4 + 1}} = \frac{4\mathbf{i} - 2\mathbf{j} + \mathbf{k}}{\sqrt{21}}$

Thus the directional derivative of f in the direction of $\vec{PQ}$

$$\nabla f \cdot \hat{A} = (2\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}) \cdot (4\mathbf{i} - 2\mathbf{j} + \mathbf{k})/\sqrt{21}$$

$$= (8 + 8 + 12)/\sqrt{21} = 28/\sqrt{21}.$$

The directional derivative of f at a point is in the direction of the normal to the surface f, in the direction of ∇f.

Hence maximum value of the directional derivative

$$= |\nabla f| = |2\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}| = \sqrt{4 + 16 + 144} = \sqrt{164}.$$

Example 5.15. *Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at the point (1, 2, 2) in the direction of the line $\frac{x-1}{2} = \frac{y-3}{-2} = z$.* (Nagpur, 2010; U.P.T.U., 2001)

Solution. We have $\nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$

$= (12xz^2 - 6xy^2z)\vec{i} - 6x^2yz\vec{j} + (12x^3z - 3x^2y^2)\vec{k}$

$= 12\vec{i} - 6\vec{j} - 6\vec{k}$ at $(1, 1, 1)$

Also direction of the given line is $\vec{b} = \frac{2\vec{i} - 2\vec{j} - \vec{k}}{3}$

Hence the required directional derivative

$= \nabla\phi \cdot \vec{b} = (12\vec{i} - 6\vec{j} - 6\vec{k}) \cdot \frac{(2\vec{i} - 2\vec{j} - \vec{k})}{3} = \frac{24 + 12 + 6}{3} = 14\frac{2}{3}$.

Example 5.16. *Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.* (U.P.T.U., 2010; Kottayam, 2006; U.P.T.U., 2003)

Solution. Let $f_1 = x^2 + y^2 + z^2 - 9 = 0$ and $f_2 = x^2 + y^2 - z - 3 = 0$

Then $N_1 = \nabla f_1$ at $(2, -1, 2) = 4\vec{i} - 2\vec{j} + 4\vec{k}$ at $(2, -1, 2) = 4\vec{i} - 2\vec{j} + 4\vec{k}$

and $N_2 = \nabla f_2$ at $(2, -1, 2) = 2x\vec{i} + 2y\vec{j} - \vec{k}$ at $(2, -1, 2) = 4\vec{i} - 2\vec{j} - \vec{k}$

Since the angle θ between the two surfaces at a point is the angle between their normals at that point and N_1, N_2 are the normals at $(2, -1, 2)$ to the given surfaces, therefore

$$\cos\theta = \frac{N_1 \cdot N_2}{|N_1||N_2|} = \frac{(4\vec{i} - 2\vec{j} + 4\vec{k}) \cdot (4\vec{i} - 2\vec{j} - \vec{k})}{\sqrt{(16 + 4 + 16)}\sqrt{(16 + 4 + 1)}}$$

$$= \frac{16 + 4 - 4}{6\sqrt{21}} = \frac{16}{6\sqrt{21}}$$

Hence the required angle $\theta = \cos^{-1}\left[\frac{8}{3\sqrt{21}}\right]$.

Example 5.17. *Find the values of a and b such that the surfaces $ax^2 - byz = (a + 2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1, -1, 2)$.* (Madras, 2004)

Solution. Let $f_1 = ax^2 - byz - (a + 2)x = 0$ ...(1)

and $f_2 = 4x^2y + z^3 - 4 = 0$ ...(2)

Then $\nabla f_1 = (2ax - a - 2)\vec{i} - bz\vec{j} - by\vec{k} = (2a - 3)\vec{i} - 2b\vec{j} + 2b\vec{k}$ at $(1, -1, 2)$,

$\nabla f_2 = 8xy\vec{i} + 4x^2\vec{j} + 3z^2\vec{k} = -8\vec{i} + 4\vec{j} + 12\vec{k}$ at $(1, -1, 2)$.

The surfaces (1) and (2) will cut orthogonally if $\nabla f_1 \cdot \nabla f_2 = 0$, i.e., $-8(2a - 3) + 2b - 24b = 0$

or $-2a - b + 4 = 0$ ...(3)

Also since the point $(1, -1, 2)$ lies on (1) and (2),

$\therefore \quad a + 2b - (a + 2) = 0$ or $b = 1$

From (3), $-2a + 6 = 0$ or $a = 3/2$.

Hence $a = 3/2$ and $b = 1$.

EXERCISE 5.3

1. (a) Find $\nabla\phi$, if $\phi = (x^2 + y^2 + z^2)^{-1/2}$. (Kurukshetra, 2010) (b) If $\phi(x, y, z) = 3x^2y - y^3z^2$, find $\nabla\phi$ at the point $(1, -2, -1)$.
2. Find a unit vector normal to the surface $x^2 + y^2 = 2yz + 4$ at the point $(2, 3, -1)$. (U.P.T.U., 2009)
3. Find the directional derivative of $\phi = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of the vector $2\vec{i} - \vec{j} - 2\vec{k}$. (U.P.T.U., 2002; Kottayam, 2006 (S); U.P.T.U., 2006; U.P.T.U., 2005)
4. Find the directional derivative of $\phi = xy^2z + 4xz^2$ at the point $(1, -2, -1)$ in the direction of $2\vec{i} - \vec{j} - 2\vec{k}$. (U.P.T.U., 2004)

6. Find the values of constants a, b and c so that the directional derivative of $\phi = axy^2 + byz + cz^2x^2$ at $(1, 2, -1)$ has a maximum magnitude of 64 in the direction parallel to the z-axis. (September, 2006)

7. Find the directional derivative of $\phi = x^2y^2 + y^2z^2 + z^2x^2$ at the point $(1, -1, 1)$ in the direction of the normal to the surface $xy + z^2 = 3$ at $(2, -1, 2)$. Also find the maximum directional derivative of ϕ at $(1, -1, 1)$. (March/April, 2006)

8. If the directional derivative of $\phi(x, y, z) = ax^2y + by^2z + cz^2x$ at the point $(1, 1, 1)$ has a maximum magnitude 15 in the direction parallel to the line $\frac{x-1}{2} = \frac{y-3}{-2} = \frac{z-2}{1}$, find the values of a, b and c. (U.P.T.U., 2005)

9. In what direction from $(3, 1, -2)$ is the directional derivative of $\phi = x^2y^2z^4$ a maximum? Find also the magnitude of that maximum. (Madras, 2000)

10. What is the greatest rate of increase of $u = xyz^2$ at the point $(1, 0, 3)$? (Nagpur, 2005)

11. The temperature at points in space is given by $T(x, y, z) = x^2 + y^2 - z$. A mosquito located at $(1, 1, 2)$ desires to fly in such a direction that it will get warmer as fast as possible. In what direction should it move?

12. Calculate the angle between the normals to the surface $xy = z^2$ at the points $(4, 1, 2)$ and $(3, 3, -3)$.

13. Find the angle between the tangent planes to the surface $x \log z = y^2 - 1$, $x^2y = 2 - z$ at the point $(1, 1, 1)$. (U.P.T.U., 2004; Madras, 2000)

14. Find the values of a and b so that the surface $ax^2 - byz = (a + 2)x$ may cut the surface $4x^2 + y^2 = 4$ orthogonally at $(1, -1, 2)$. (Nagpur, 2004)

15. If r and $\mathbf{a}$ are position vectors, prove that the rate of change of $\frac{\mathbf{r}}{r^n}$ in any direction perpendicular to $\mathbf{a}$ is zero. (U.P.T.U., 2004)

3.31 DEL APPLIED TO VECTOR POINT FUNCTIONS

(1) Divergence. *The divergence of a continuously differentiable vector point function* $\mathbf{F}$ *is denoted by* $\operatorname{div}\mathbf{F}$ *and is defined by the equation*

$$\operatorname{div}\mathbf{F} = \nabla \cdot \mathbf{F} = \mathbf{i}\cdot\frac{\partial \mathbf{F}}{\partial x} + \mathbf{j}\cdot\frac{\partial \mathbf{F}}{\partial y} + \mathbf{k}\cdot\frac{\partial \mathbf{F}}{\partial z}$$

If $\qquad \mathbf{F} = f\mathbf{i} + g\mathbf{j} + h\mathbf{k}$

then $\qquad \operatorname{div}\mathbf{F} = \nabla \cdot \mathbf{F} = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right)\cdot(f\mathbf{i} + g\mathbf{j} + h\mathbf{k}) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$

(2) Curl. *The curl of a continuously differentiable vector point function* $\mathbf{F}$ *is defined by the equation*

$$\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F} = \mathbf{i}\times\frac{\partial \mathbf{F}}{\partial x} + \mathbf{j}\times\frac{\partial \mathbf{F}}{\partial y} + \mathbf{k}\times\frac{\partial \mathbf{F}}{\partial z}$$

If $\mathbf{F} = f\mathbf{i} + g\mathbf{j} + h\mathbf{k}$ then curl $\mathbf{F} = \nabla \times \mathbf{F} = \left(\mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}\right) \times (f\mathbf{i} + g\mathbf{j} + h\mathbf{k})$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & g & h \end{vmatrix} = \mathbf{i}\left(\frac{\partial h}{\partial y} - \frac{\partial g}{\partial z}\right) - \mathbf{j}\left(\frac{\partial h}{\partial x} - \frac{\partial f}{\partial z}\right) + \mathbf{k}\left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y}\right).$$

Example 3.19. *If* $\mathbf{R} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, *show that*

(i) $\nabla \cdot \mathbf{R} = 3$ (ii) $\nabla \times \mathbf{R} = 0$. (U.P.T.U., 2008; P.T.U., 2006; U.P.T.U., 2006)

Solution. (i) $\nabla \cdot \mathbf{R} = \frac{\partial}{\partial x}(x\mathbf{i}) + \frac{\partial}{\partial y}(y\mathbf{j}) + \frac{\partial}{\partial z}(z\mathbf{k})$.

(ii) $$\nabla \times \mathbf{R} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \mathbf{i}\left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z}\right) - \mathbf{j}\left(\frac{\partial z}{\partial x} - \frac{\partial x}{\partial z}\right) + \mathbf{k}\left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y}\right)$$

$$= \mathbf{i}(0 - 0) - \mathbf{j}(0 - 0) + \mathbf{k}(0 - 0) = 0.$$

[Remember $\because \mathbf{i} = \mathbf{j} = \mathbf{k} = 0$]

Example 3.16. Find (i) $\text{curl}\,\vec{V}$, (ii) $\text{curl}\,(\text{curl}\,\vec{V})$, where $\vec{V} = \text{grad}\,(x^3 + y^3 + z^3 - 3xyz)$.
(U.P.T.U., 2004; Kurukshetra, 2007; Rajasthan, 2007)

Solution. If $u = x^3 + y^3 + z^3 - 3xyz$, then

$$\vec{V} = \nabla u = i\frac{\partial u}{\partial x} + j\frac{\partial u}{\partial y} + k\frac{\partial u}{\partial z} = i(3x^2 - 3yz) + j(3y^2 - 3xz) + k(3z^2 - 3xy)$$

$$\therefore \quad \text{div}\,\vec{V} = \frac{\partial}{\partial x}(3x^2 - 3yz) + \frac{\partial}{\partial y}(3y^2 - 3xz) + \frac{\partial}{\partial z}(3z^2 - 3xy) = 6x + 6y + 6z$$

$$\text{curl}\,\vec{V} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 - 3yz & 3y^2 - 3xz & 3z^2 - 3xy \end{vmatrix}$$

$$= i(-3x + 3x) - j(-3y + 3y) + k(-3z + 3z) = 0.$$

3.22 (1) PHYSICAL INTERPRETATION OF DIVERGENCE

Consider the motion of the fluid having velocity $\vec{V} = u\,i + v\,j + w\,k$ at a point $P(x, y, z)$. Consider a small parallelopiped with edges $\delta x, \delta y, \delta z$ parallel to the axes in the mass of fluid, with one of its corners at P (Fig. 3.7).

∴ the amount of fluid entering the face PA' in unit time $= \rho u\,\delta y\,\delta z$ and the amount of fluid leaving the face $P'D$ in unit time

$$= \rho_1 u_1\,\delta y\,\delta z = \left(\rho u + \frac{\partial}{\partial x}(\rho u)\,\delta x\right)\delta y\,\delta z \quad \text{nearly}$$

∴ the net decrease of the amount of fluid due to flow across these two faces $= \frac{\partial}{\partial x}(\rho u)\,\delta x\,\delta y\,\delta z$

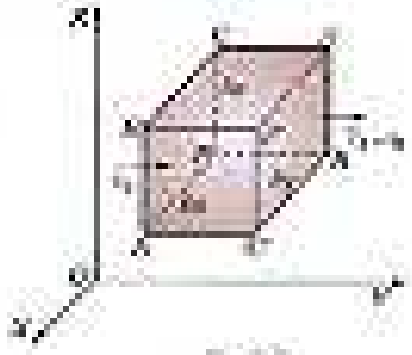

Fig. 3.7

Finding similarly the contributions of other two pairs of faces, we have the total decrease of amount of fluid inside the parallelopiped per unit time $= \left(\frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w)\right)\delta x\,\delta y\,\delta z$.

Thus the rate of loss of fluid per unit volume

$$= \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = \text{div}\,\vec{V}\rho.$$

Hence div $\vec{V}$ gives the rate at which fluid is outgoing at a point per unit volume.

Similarly, if $\vec{V}$ represents an electric flux, div $\vec{V}$ is the amount of flux which diverges per unit volume in unit time. If $\vec{V}$ represents heat flux, div $\vec{V}$ is the rate at which heat is leaving from a point per unit volume. In general, the divergence of a vector point function representing any physical quantity gives at each point, the rate per unit volume at which the physical quantity is coming from that point. These explain the justification for the name divergence of a vector point function.

If the fluid is incompressible, there can be no gain or loss in the volume element. Hence div $\vec{V} = 0$, which is known as Hydrodynamical or the equation of continuity for incompressible fluids.

Def. If the flux entering any element of space is the same as that leaving it, i.e., div $\vec{V} = 0$ everywhere, then such a point function is called a solenoidal vector function.

(2) Physical interpretation of curl. Consider the motion of a rigid body rotating about a fixed axis through O. If $\vec{\omega}$ be its angular velocity, then the velocity $\vec{V}$ of any particle $P(\vec{R})$ of the body is given by $\vec{V} = \vec{\omega} \times \vec{R}$.

(See p. 3.01)

If $\quad \vec{\omega} = \omega_1 i + \omega_2 j + \omega_3 k \quad$ and $\quad \vec{R} = x i + y j + z k$

then $$\vec{V} = \vec{\omega} \times \vec{R} = \begin{vmatrix} i & j & k \\ \omega_1 & \omega_2 & \omega_3 \\ x & y & z \end{vmatrix} = i(\omega_2 z - \omega_3 y) + j(\omega_3 x - \omega_1 z) + k(\omega_1 y - \omega_2 x)$$

$$\therefore \quad \text{curl } \mathbf{V} = \begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \omega_3 y - \omega_2 z & \omega_1 z - \omega_3 x & \omega_2 x - \omega_1 y \end{vmatrix}$$

$$= \mathbf{I}(\omega_2 + \omega_2) + \mathbf{J}(-\omega_3 - \omega_3) + \mathbf{K}(\omega_1 + \omega_1) \qquad [\because \ \omega_1, \omega_2, \omega_3 \text{ are constants}]$$

$$= 2(\omega_1 \mathbf{I} + \omega_2 \mathbf{J} + \omega_3 \mathbf{K}) = 2\Omega. \quad \text{Hence} \quad \Omega = \frac{1}{2} \text{curl } \mathbf{V}$$

Thus the angular velocity of rotation at any point is equal to half the curl of the velocity vector which justifies the name *rotation* used for curl.

In general, the curl of any vector point function gives the measure of the angular velocity at any point of the vector field.

Def. *Any motion in which the curl of the velocity vector is zero is said to be irrotational, otherwise rotational.*

Example 8.30. *Prove that the vector $(x^2 - yz)\mathbf{I} + (y^2 - zx)\mathbf{J} + (z^2 - xy)\mathbf{K}$ is irrotational.*
(U.P.T.U., 2006; U.P.T.U., 2006; 2, C.O., 2006)

Solution. We have $\mathbf{R} = x\mathbf{I} + y\mathbf{J} + z\mathbf{K}$ and $r = \sqrt{(x^2 + y^2 + z^2)}$

$$\therefore \quad \text{curl}(r^n \mathbf{R}) = \nabla \cdot (x^2 + y^2 + z^2)^{n/2} (x\mathbf{I} + y\mathbf{J} + z\mathbf{K})$$

$$= \frac{\partial}{\partial x}[x(x^2 + y^2 + z^2)^{n/2}] - \frac{\partial}{\partial y}[y(x^2 + y^2 + z^2)^{n/2}] + \frac{\partial}{\partial z}[z(x^2 - y^2 + z^2)^{n/2}]$$

$$= 3(x^2 + y^2 + z^2)^{n/2} + nx^2(x^2 + y^2 + z^2)^{n/2 - 1} - ny^2(x^2 + y^2 + z^2)^{n/2 - 1} - nz^2(x^2 + y^2 + z^2)^{n/2 - 1}$$

$$= 3r^n + n(x^2 + y^2 + z^2)^{n/2 - 1} = 3r^n + nr^2 \cdot r^{n-2}$$

Thus $\text{div}(r^n \mathbf{R}) = (n + 3)r^n$

When $n = -3$, $\text{div}(\mathbf{R}/r^3) = 0$ i.e., $\mathbf{R}/r^3$ is solenoidal.

Example 8.31. *Show that $r^n \mathbf{R}$ is irrotational vector for any value of n but is solenoidal if $n = -3$, where $\mathbf{R} = x\mathbf{I} + y\mathbf{J} + z\mathbf{K}$ and r is the modulus of $\mathbf{R}$.* *(U.P.T.U., 2005; Rohtak U., 2005)*

Solution. Let $\mathbf{A} = r^n \mathbf{R} = (x^2 + y^2 + z^2)^{n/2}(x\mathbf{I} + y\mathbf{J} + z\mathbf{K}) = \Sigma\, \mathbf{I}(x^2 + y^2 + z^2)^{n/2}$

$$\therefore \quad \text{curl } \mathbf{A} = \begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x(x^2 + y^2 + z^2)^{n/2} & y(x^2 + y^2 + z^2)^{n/2} & z(x^2 + y^2 + z^2)^{n/2} \end{vmatrix}$$

$$= \Sigma\, \mathbf{I}\left[\frac{\partial}{\partial y}(z^2 + y^2 + z^2)^{n/2} \cdot (2z) - \frac{\partial}{\partial z}(x^2 + y^2 + z^2)^{n/2} \cdot 2y\right] = 0$$

Hence $\mathbf{A}$ is irrotational for any value of n.

But $\text{div } \mathbf{A} = \nabla \cdot r^n \mathbf{R} = (n + 3)r^n$

which is zero for $n + 3 = 0$, i.e., $\mathbf{A}$ is solenoidal if $n = -3$.

8.8 DEL APPLIED TWICE TO POINT FUNCTIONS

∇ and $\nabla \cdot \mathbf{F}$ being vector point functions, we can form their divergence and curl whereas $\nabla \cdot \mathbf{F}$ being a scalar point function, we can have its gradient only. Thus we have the following five formulae :

(1) $\text{div grad } f = \nabla^2 f = \dfrac{\partial^2 f}{\partial x^2} + \dfrac{\partial^2 f}{\partial y^2} + \dfrac{\partial^2 f}{\partial z^2}$

(2) $\text{curl grad } f = \nabla \times \nabla f = 0$

(3) $\text{div curl } \mathbf{F} = \nabla \cdot \nabla \times \mathbf{F} = 0$

(4) curl curl $\vec{F}$ = grad div $\vec{F}$ − $\nabla^2\vec{F}$, i.e., $\nabla\times(\nabla\times\vec{F}) = \nabla(\nabla\cdot\vec{F}) - \nabla^2\vec{F}$

(5) grad div $\vec{F}$ = curl curl $\vec{F}$ + $\nabla^2\vec{F}$, i.e., $\nabla(\nabla\cdot\vec{F}) = \nabla\times(\nabla\times\vec{F}) + \nabla^2\vec{F}$.

Proof. (1) $\nabla\cdot\nabla\phi = \nabla\cdot\nabla\phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)$

$$= \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial z}\right) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\phi$$

$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called the Laplacian operator and $\nabla^2\phi = 0$ is called the Laplace's equation.

(2) $\nabla\times(\nabla\phi) = \nabla\times\left(\vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}\right) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial\phi}{\partial x} & \frac{\partial\phi}{\partial y} & \frac{\partial\phi}{\partial z} \end{vmatrix} = \Sigma\vec{i}\left(\frac{\partial^2\phi}{\partial y\partial z} - \frac{\partial^2\phi}{\partial z\partial y}\right) = \vec{0}$ (V.T.U., 2007)

(3) $\nabla\cdot(\nabla\times\vec{F}) = \left(\Sigma\vec{i}\frac{\partial}{\partial x}\right)\cdot\left(\vec{i}\times\frac{\partial\vec{F}}{\partial x} + \vec{j}\times\frac{\partial\vec{F}}{\partial y} + \vec{k}\times\frac{\partial\vec{F}}{\partial z}\right)$

$$= \Sigma\vec{i}\cdot\left(\vec{i}\times\frac{\partial^2\vec{F}}{\partial x^2} + \vec{j}\times\frac{\partial^2\vec{F}}{\partial x\partial y} + \vec{k}\times\frac{\partial^2\vec{F}}{\partial x\partial z}\right)$$

$$= \Sigma\left(\vec{i}\cdot\vec{i}\times\frac{\partial^2\vec{F}}{\partial x^2} + \vec{i}\cdot\vec{j}\times\frac{\partial^2\vec{F}}{\partial x\partial y} + \vec{i}\cdot\vec{k}\times\frac{\partial^2\vec{F}}{\partial x\partial z}\right) = \Sigma\left(\vec{k}\cdot\frac{\partial^2\vec{F}}{\partial x\partial y} - \vec{j}\cdot\frac{\partial^2\vec{F}}{\partial x\partial z}\right) = 0.$$

(4) $\nabla\times(\nabla\times\vec{F}) = \left(\Sigma\vec{i}\frac{\partial}{\partial x}\right)\times\left(\vec{i}\times\frac{\partial\vec{F}}{\partial x} + \vec{j}\times\frac{\partial\vec{F}}{\partial y} + \vec{k}\times\frac{\partial\vec{F}}{\partial z}\right)$

$$= \Sigma\vec{i}\times\left(\vec{i}\times\frac{\partial^2\vec{F}}{\partial x^2} + \vec{j}\times\frac{\partial^2\vec{F}}{\partial x\partial y} + \vec{k}\times\frac{\partial^2\vec{F}}{\partial x\partial z}\right)$$

$$= \Sigma\left[\left\{\vec{i}\cdot\left(\vec{i}\cdot\frac{\partial^2\vec{F}}{\partial x^2}\right)\vec{i} - \left(\vec{i}\cdot\vec{i}\right)\frac{\partial^2\vec{F}}{\partial x^2}\right\} + \left\{\left(\vec{i}\cdot\frac{\partial^2\vec{F}}{\partial x\partial y}\right)\vec{j} - \left(\vec{i}\cdot\vec{j}\right)\frac{\partial^2\vec{F}}{\partial x\partial y}\right\} + \left\{\left(\vec{i}\cdot\frac{\partial^2\vec{F}}{\partial x\partial z}\right)\vec{k} - \left(\vec{i}\cdot\vec{k}\right)\frac{\partial^2\vec{F}}{\partial x\partial z}\right\}\right]$$

$$= \Sigma\left[\left(\vec{i}\cdot\frac{\partial^2\vec{F}}{\partial x^2}\right)\vec{i} + \left(\vec{i}\cdot\frac{\partial^2\vec{F}}{\partial x\partial y}\right)\vec{j} + \left(\vec{i}\cdot\frac{\partial^2\vec{F}}{\partial x\partial z}\right)\vec{k}\right] - \Sigma\frac{\partial^2\vec{F}}{\partial x^2}$$

$$= \Sigma\vec{i}\frac{\partial}{\partial x}\left(\vec{i}\cdot\frac{\partial\vec{F}}{\partial x} + \vec{j}\cdot\frac{\partial\vec{F}}{\partial y} + \vec{k}\cdot\frac{\partial\vec{F}}{\partial z}\right) - \Sigma\frac{\partial^2\vec{F}}{\partial x^2} = \nabla(\nabla\cdot\vec{F}) - \nabla^2\vec{F}.$$ (Madras, 2006)

(5) is just another way of writing (4) above.

Obs. Any identity of vector calculus may be derived using the above results or some results as given in the exercise appearing at the end of this section.

e.g. i. $\nabla\cdot\nabla\times\vec{F} = 0$ (∵ $\vec{F} = \nabla\phi$)

ii. $\nabla\times\nabla\phi = \vec{0}$ (∵ $\vec{F} = \nabla\phi$)

iii. $\nabla\cdot\vec{r} = 3$ (∵ $\nabla\cdot\vec{r} = 3$)

iv. $\nabla\times(\vec{a}\times\vec{r}) = \nabla(\vec{r}\cdot\vec{b}) - (\vec{a}\cdot\nabla)\vec{r} = \vec{b} - \vec{a}$ is a useful result in vector triple product.

3.8 DEL APPLIED TO PRODUCTS OF POINT FUNCTIONS

In your Own

(1) grad $(\phi\psi)$ = ϕ grad ψ + ψ grad ϕ i.e., $\nabla(\phi\psi) = \phi\nabla\psi + \psi\nabla\phi$

(2) div $(f\vec{G})$ = grad $f\cdot\vec{G} + f$ div $\vec{G}$ i.e., $\nabla\cdot(f\vec{G}) = \nabla f\cdot\vec{G} + f\nabla\cdot\vec{G}$

(3) curl $(f\vec{G})$ = grad $f\times\vec{G} + f$ curl $\vec{G}$ i.e., $\nabla\times(f\vec{G}) = \nabla f\times\vec{G} + f\nabla\times\vec{G}$

(4) grad $(\vec{F}\cdot\vec{G}) = (\vec{F}\cdot\nabla)\vec{G} + (\vec{G}\cdot\nabla)\vec{F} + \vec{F}\times$ curl $\vec{G} + \vec{G}\times$ curl $\vec{F}$

i.e., $\nabla(\vec{F}\cdot\vec{G}) = (\vec{F}\cdot\nabla)\vec{G} + (\vec{G}\cdot\nabla)\vec{F} + \vec{F}\times(\nabla\times\vec{G}) + \vec{G}\times(\nabla\times\vec{F})$

Let us illustrate the application of the above rules in the following examples:

(a) $\nabla \cdot (f\vec{G}) = \nabla f \cdot \vec{G} + f(\nabla \cdot \vec{G})$, $\nabla \cdot (\vec{F} \cdot \vec{G}) = \vec{F} \cdot (\nabla \cdot \vec{G}) - \vec{G} \cdot (\nabla \cdot \vec{F})$

(b) $\nabla(\vec{F} \cdot \vec{G}) = \nabla(\vec{F}_x \cdot \vec{G}) + \nabla(\vec{F}_y \cdot \vec{G}) + \nabla(\vec{F}_z \cdot \vec{G})$

$= \vec{F}_x \times (\nabla \times \vec{G}) + (\vec{F}_x \cdot \nabla)\vec{G} + \vec{F}_y \times (\nabla \times \vec{G}) + (\vec{F}_y \cdot \nabla)\vec{G} + \vec{F}_z \times (\nabla \times \vec{G}) + (\vec{F}_z \cdot \nabla)\vec{G}$

$= \vec{F} \times (\nabla \times \vec{G}) + (\vec{F} \cdot \nabla)\vec{G} + \vec{G} \times (\nabla \times \vec{F}) + (\vec{G} \cdot \nabla)\vec{F}$

(c) $\nabla \times (\vec{F} \times \vec{G}) = \nabla \times (\vec{F}_x \times \vec{G}) + \nabla \times (\vec{F}_y \times \vec{G}) + \nabla \times (\vec{F}_z \times \vec{G}) = (\nabla \cdot \vec{G})\vec{F}_x - (\vec{F}_x \cdot \nabla)\vec{G} + (\vec{G} \cdot \nabla)\vec{F}_x - (\nabla \cdot \vec{F})\vec{G}$

$+ (\nabla \cdot \vec{G})\vec{F}_y - (\vec{F}_y \cdot \nabla)\vec{G} + (\vec{G} \cdot \nabla)\vec{F}_y - (\nabla \cdot \vec{F})\vec{G} + (\nabla \cdot \vec{G})\vec{F}_z - (\vec{F}_z \cdot \nabla)\vec{G} + (\vec{G} \cdot \nabla)\vec{F}_z - (\nabla \cdot \vec{F})\vec{G}$

$= (\nabla \cdot \vec{G})\vec{F} - (\vec{F} \cdot \nabla)\vec{G} + (\vec{G} \cdot \nabla)\vec{F} - (\nabla \cdot \vec{F})\vec{G}$

Example 9.33. Show that $\nabla^2(r^n) = n(n+1)r^{n-2}$. (A.M.I.E., 1998; A.M.I.E., 2000; G.A.T.E., 2002)

Solution. $\nabla^2 r^n = \nabla \cdot (\nabla r^n)$

$= \nabla \cdot \left(nr^{n-1}\frac{\vec{r}}{r}\right) = \nabla \cdot (r^{n-2}\vec{r}) = n\nabla \cdot (r^{n-2}\vec{r}) + \vec{r} \cdot \nabla(r^{n-2})$ [By § 9.9 (2)]

$= n\left[(n-2)r^{n-3}\frac{\vec{r}}{r} \cdot \vec{r} + r^{n-2}(3)\right]$ [Using Ex. 9.18 (2)]

$= n[(n-2)r^{n-4}(r^2) + 3r^{n-2}] = n(n+1)r^{n-2}$ [$\because \vec{r} \cdot \vec{r} = r^2$]

Otherwise, $\nabla^2(r^n) = \frac{\partial^2(r^n)}{\partial x^2} + \frac{\partial^2(r^n)}{\partial y^2} + \frac{\partial^2(r^n)}{\partial z^2}$ [By § 9.8 (1)]

Now $\frac{\partial(r^n)}{\partial x} = nr^{n-1}\frac{\partial r}{\partial x} = nr^{n-1}\frac{x}{r} = nr^{n-2}x$ [$\because r^2 = x^2 + y^2 + z^2$]

$\therefore \frac{\partial^2(r^n)}{\partial x^2} = n\left[r^{n-2} + x(n-2)r^{n-3}\frac{\partial r}{\partial x}\right] = n\left[r^{n-2} + x(n-2)r^{n-3}\frac{x}{r}\right]$

$= n\left[r^{n-2} + (n-2)x^2r^{n-4}\right]$ ...(i)

Similarly, $\frac{\partial^2(r^n)}{\partial y^2} = n\left[r^{n-2} + (n-2)y^2r^{n-4}\right]$ ...(ii)

$\frac{\partial^2(r^n)}{\partial z^2} = n\left[r^{n-2} + (n-2)z^2r^{n-4}\right]$ ...(iii)

Adding (i), (ii) and (iii), we get

$\nabla^2(r^n) = n[3r^{n-2} + (n-2)r^{n-4}(x^2 + y^2 + z^2)]$

$= n[3r^{n-2} + (n-2)r^{n-4}r^2] = n(n+1)r^{n-2}$

In particular $\nabla^2\left(\frac{1}{r}\right) = 0$ (G.A.T.E., 2002; P.T.U., 2003)

Example 9.34. If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ is the position vector of any point and $\vec{F}$ is a vector point function, show that $\vec{F} \cdot \text{curl}\,\vec{F} = 0$.

Solution. Since $\vec{F} = \frac{1}{n}\nabla r^n \Rightarrow \text{curl}\,\vec{F} = \nabla \times \left(\frac{1}{n}\nabla r^n\right)$

or $\text{curl}\,\vec{F} = \nabla\frac{1}{n} \times \nabla r^n + \frac{1}{n}\nabla \times (\nabla r^n)$ [By § 9.9 (10)]

$= \nabla\frac{1}{n} \times \nabla r^n$ [$\because \nabla \times \nabla r^n = 0$]

Hence $\vec{F} \cdot \text{curl}\,\vec{F} = \frac{1}{n}\nabla r^n \cdot \left(\nabla\frac{1}{n} \times \nabla r^n\right) = 0$, for it is a vector triple product in which two factors are equal.

Example 9.35. If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $r = |\vec{r}|$ and $\vec{A}$ is a constant vector, prove that

$\nabla \times \left\{\frac{\vec{A} \times \vec{r}}{r^n}\right\} = \frac{2-n}{r^n}\vec{A} + \frac{n(\vec{A} \cdot \vec{r})}{r^{n+2}}\vec{r}$ (Kerala U. 2000; Kurukshetra, 2000; A.M.I.E.T.E., 2003)

Solution. $\nabla \times \left[r^{-n}(\vec{A} \times \vec{r})\right] = r^{-n}[\nabla \times (\vec{A} \times \vec{r})] + \nabla(r^{-n}) \times (\vec{A} \times \vec{r})$ [By § 9.9 (13)]

$= r^{-n}[\vec{r}(\nabla \cdot \vec{A}) - \vec{A}(\nabla \cdot \vec{r})] + (-nr^{-n-1}\vec{r}) \times (\vec{A} \times \vec{r})$

$$= r^{n-2}[2\mathbf{A} - n\mathbf{A} - nr^{-1}(\mathbf{R}\cdot\mathbf{A})\mathbf{R}] = (2-n)\mathbf{A} - nr^{-2}(\mathbf{R}\cdot\mathbf{A})\mathbf{R}$$

$$= 2\nabla r^{n-2} - \nabla r^{n-2}(\mathbf{R}\cdot\mathbf{R}) - \mathbf{R}\times(\nabla\times\mathbf{A}) - (\mathbf{A}\cdot\nabla)\mathbf{R}$$

$$= \frac{2\mathbf{A}}{r^n} - \frac{n}{r^{n+2}}(\mathbf{R}\cdot\mathbf{A})\mathbf{R} - \mathbf{R}\times(\nabla\times\mathbf{A}) = \frac{2-n}{r^n}\mathbf{A} + \frac{n(\mathbf{A}\cdot\mathbf{R})}{r^{n+2}}\mathbf{R}$$

Example 8.20. If r is the distance of a point (x, y, z) from the origin, prove that $\operatorname{curl}\left(\mathbf{K}\times\operatorname{grad}\frac{1}{r}\right) = \operatorname{grad}\left(\mathbf{K}\cdot\operatorname{grad}\frac{1}{r}\right) = 0$, where $\mathbf{K}$ is the constant vector $a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$. (U.P.T.U., 2003)

Solution. $\operatorname{grad}\frac{1}{r} = \left(\mathbf{I}\frac{\partial}{\partial x} + \mathbf{J}\frac{\partial}{\partial y} + \mathbf{K}\frac{\partial}{\partial z}\right)(x^2+y^2+z^2)^{-1/2}$ $\quad\because\ r = \sqrt{(x^2+y^2+z^2)}$

$$= -\frac{1}{2}(x^2+y^2+z^2)^{-3/2}(2x\mathbf{I} + 2y\mathbf{J} + 2z\mathbf{K})$$

$$= -(x^2+y^2+z^2)^{-3/2}(x\mathbf{I} + y\mathbf{J} + z\mathbf{K})$$

$$\operatorname{curl}\left(\mathbf{K}\times\operatorname{grad}\frac{1}{r}\right) = \nabla\times[-(x^2+y^2+z^2)^{-3/2}(c\mathbf{I} - a\mathbf{K})]$$

$$= \begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ c(x^2+y^2+z^2)^{-3/2} & -a(x^2+y^2+z^2)^{-3/2} & 0 \end{vmatrix}$$

$$= \mathbf{I}\frac{\partial}{\partial y}\left[\frac{a}{(x^2+y^2+z^2)^{3/2}}\right] - \mathbf{J}\frac{\partial}{\partial x}\left[\frac{a}{(x^2+y^2+z^2)^{3/2}}\right]$$

$$- \mathbf{K}\left\{\frac{\partial}{\partial x}\left[(x^2+y^2+z^2)^{-3/2}\right] + \frac{\partial}{\partial y}\left[(x^2+y^2+z^2)^{-3/2}\right]\right\}$$

$$= \frac{-3ayz - 3cxz + (x^2+y^2-2z^2)\mathbf{K}}{(x^2+y^2+z^2)^{5/2}} \quad \ldots(i)$$

$$\operatorname{grad}\left(\mathbf{K}\cdot\operatorname{grad}\frac{1}{r}\right) = \nabla\left[-\mathbf{K}\cdot\frac{(x\mathbf{I} + y\mathbf{J} + z\mathbf{K})}{(x^2+y^2+z^2)^{3/2}}\right]$$

$$= \left(\mathbf{I}\frac{\partial}{\partial x} + \mathbf{J}\frac{\partial}{\partial y} + \mathbf{K}\frac{\partial}{\partial z}\right)\left[\frac{-cz}{(x^2+y^2+z^2)^{3/2}}\right]$$

$$= \frac{3cxz}{(x^2+y^2+z^2)^{5/2}} + \frac{3cyz}{(x^2+y^2+z^2)^{5/2}} + \frac{c(x^2+y^2-2z^2)\mathbf{K}}{(x^2+y^2+z^2)^{5/2}}$$

$$= \frac{3cxz + 3cyz - (x^2+y^2-2z^2)\mathbf{K}}{(x^2+y^2+z^2)^{5/2}} \quad \ldots(ii)$$

Adding (i) and (ii), we get

$$\operatorname{curl}\left(\mathbf{K}\times\operatorname{grad}\frac{1}{r}\right) + \operatorname{grad}\left(\mathbf{K}\cdot\operatorname{grad}\frac{1}{r}\right) = 0.$$

Example 8.21. In electromagnetic theory we have $\nabla\cdot\mathbf{B} = \rho/\varepsilon$, $\nabla\cdot\mathbf{B} = 0$, $\nabla\times\mathbf{D} = \mu\frac{\partial\mathbf{H}}{\partial t}$, $\nabla\times\mathbf{H} = \mathbf{J} + \frac{\partial\mathbf{D}}{\partial t}$. Prove that

(i) $\nabla^2\mathbf{D} = \frac{1}{c^2}\frac{\partial^2\mathbf{D}}{\partial t^2} - \nabla\rho + \frac{1}{c^2}\frac{\partial\mathbf{J}}{\partial t}$ $\quad$ (ii) $\nabla^2\mathbf{H} = \frac{1}{c^2}\frac{\partial^2\mathbf{H}}{\partial t^2} - \nabla\times\mathbf{J}$

Solution. (i) We have $\frac{1}{\mu}\nabla\left[\frac{\partial^2\mathbf{H}}{\partial t^2}+\frac{\partial}{\partial t}(\rho\mathbf{V})\right]=\frac{1}{\mu}\frac{\partial}{\partial t}\left(\frac{\partial\mathbf{H}}{\partial t}+\rho\mathbf{V}\right)$

$$=\frac{1}{\mu}\frac{\partial}{\partial t}(\nabla\times\mathbf{H})=\frac{\partial}{\partial t}\mathbf{J}=\frac{\partial^2\mathbf{D}}{\partial t^2} \qquad \left[\because\ \nabla\times\mathbf{H}=\frac{1}{\mu}\left(\rho\mathbf{V}+\frac{\partial\mathbf{D}}{\partial t}\right)\right]$$

$$=-\nabla\times(\nabla\times\mathbf{D}) \qquad \left[\because\ \nabla\times\mathbf{D}=-\frac{1}{\mu}\frac{\partial\mathbf{H}}{\partial t}\right]$$

$$=-[\nabla(\nabla\cdot\mathbf{D})-\nabla^2\mathbf{D}] \qquad \text{[Using (4.5.(4))}$$

$$=-\nabla\rho+\nabla^2\mathbf{D}$$

Hence $\nabla^2\mathbf{D}-\frac{1}{\mu}\frac{\partial^2\mathbf{D}}{\partial t^2}=\nabla\rho+\frac{1}{\mu}\frac{\partial}{\partial t}(\rho\mathbf{V})$

(ii) L.H.S. $=\nabla^2\mathbf{H}-\frac{1}{\mu}\frac{\partial^2\mathbf{H}}{\partial t^2}=\nabla(\nabla\cdot\mathbf{H})+\frac{1}{\mu}\frac{\partial}{\partial t}\left(-\frac{1}{\mu}\frac{\partial\mathbf{H}}{\partial t}\right)$

$$=\nabla^2\mathbf{H}+\frac{1}{\mu}\frac{\partial}{\partial t}(\nabla\times\mathbf{D}) \qquad \left[\because\ \nabla\times\mathbf{D}=-\frac{1}{\mu}\frac{\partial\mathbf{H}}{\partial t}\right]$$

$$=\nabla^2\mathbf{H}+\frac{1}{\mu}\left(\nabla\times\frac{\partial\mathbf{D}}{\partial t}\right) \qquad \left[\because\ \nabla\times\mathbf{H}=\frac{1}{\mu}\left(\rho\mathbf{V}+\frac{\partial\mathbf{D}}{\partial t}\right)\right]$$

$$=\nabla^2\mathbf{H}+\nabla\times\left(\nabla\times\mathbf{H}-\frac{1}{\mu}\rho\mathbf{V}\right)=\nabla^2\mathbf{H}+\nabla\times\nabla\times\mathbf{H}-\frac{1}{\mu}\nabla\times(\rho\mathbf{V})$$

$$=\nabla^2\mathbf{H}+\nabla(\nabla\cdot\mathbf{H})-\nabla^2\mathbf{H}-\frac{1}{\mu}\nabla\times(\rho\mathbf{V}) \qquad \text{[Using (4.5.(4))}$$

$$=\nabla(\nabla\cdot\mathbf{H})-\frac{1}{\mu}\nabla\times(\rho\mathbf{V}) \qquad [\because\ \nabla\cdot\mathbf{H}=0]$$

$$=-\frac{1}{\mu}\nabla\times\rho\mathbf{V}=\text{R.H.S.}$$

EXERCISE 4.5

1. Evaluate $\phi(\mathbf{F})$ and $\psi(\mathbf{F})$ at the point (1, 2, 2) given $\mathbf{F}=x^2yz\,\mathbf{i}+xy^2z\,\mathbf{j}+xyz^2\,\mathbf{k}$. (U.P.T.U., 2002)
   (ii) $\mathbf{F}=3x^2\mathbf{i}-5xz^2\mathbf{j}+2y\mathbf{k}$ (U.P.T.U., 2003)
   (iii) If $\mathbf{F}=\text{grad}\,(x^3+y^3+z^3-3xyz)$ (U.P.T.U., 2003)
2. If $\mathbf{F}=x(x+y)\mathbf{i}+y(y+z)\mathbf{j}+z(z+x)\mathbf{k}$, show that $\nabla\cdot\mathbf{F}=2(x+y+z)$ and $\nabla\times\mathbf{F}\neq\mathbf{0}$. (Kanpur, 2000)
3. If $\mathbf{F}=(x+y+1)\mathbf{i}+\mathbf{j}-(x+y)\mathbf{k}$, show that $\mathbf{F}$ and $\nabla\times\mathbf{F}$ (U.P.T.U., 2003 (C))
4. Find the value of a if the vector $\mathbf{V}=(x+3y)\mathbf{i}+(y-2z)\mathbf{j}+(x+az)\mathbf{k}$ is solenoidal. Find the potential of the above vector which is also irrotational.
5. Which of the following vectors are solenoidal :
   (i) $(x^2+yz)\mathbf{i}+(y^2+zx)\mathbf{j}+(z^2+xy)\mathbf{k}$ (Kanpur, 2000)
   (ii) $(y^2+z^2)\mathbf{i}+(z^2+x^2)\mathbf{j}+(x^2+y^2)\mathbf{k}$ (Agra, 2000)
6. If $\mathbf{A}$ and $\mathbf{B}$ are irrotational, prove that $\mathbf{A}\times\mathbf{B}$ is solenoidal. (Madras, 2000) (U.P.T.U., 2004)
7. If $r=\sqrt{x^2+y^2+z^2}$ and $\mathbf{V}=r\mathbf{i}+y\mathbf{j}+z\mathbf{k}$, show that $\text{div}\,(r\mathbf{V})=4r$.
8. If $\mathbf{H}=(x^2+y^2+z^2)\mathbf{i}+(x^2+y^2+z^2)\mathbf{j}+(x^2+y^2+z^2)\mathbf{k}$ and $\mathbf{E}=\text{curl}\,\mathbf{H}$, show that $\text{div}\,\mathbf{E}=0$ and $\text{curl}\,\mathbf{E}=\text{curl}\,\text{curl}\,\mathbf{H}$.
   (iii) If $\mathbf{F}=(x+y+1)\mathbf{i}+\mathbf{j}-(x+y)\mathbf{k}$, show that $\mathbf{F}\cdot\text{curl}\,\mathbf{F}=0$. (U.P.T.U., 2000; Kanpur, 2002)
   (iv) $\text{grad}\left[\text{div}\left(\frac{\mathbf{R}}{r^n}\right)\right]=-\frac{n(n+2)}{r^{n+2}}\mathbf{R}$ (U.P.T.U., 2004 (C))
9. If $\mathbf{V}_1$ and $\mathbf{V}_2$ be the resultant forces of the fixed points (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively on a particle (x, y, z), prove that :
   (i) $\text{curl}\,(\mathbf{V}_1+\mathbf{V}_2)=0$, (ii) $\text{curl}\,(\mathbf{V}_1-\mathbf{V}_2)=\mathbf{V}_2-\mathbf{V}_1$
   (iii) $\text{div}\,(\mathbf{V}_1\times\mathbf{V}_2)=2(\mathbf{V}_1-\mathbf{V}_2)$

10. Show that $\nabla\left(\frac{\vec{r}}{r^n}\right) = \frac{3-n}{r^n}$. (Mumbai, 2000)

11. If $\vec{r}(t) = t^2\vec{i} + t\vec{j} - t^3\vec{k}$, prove that $\vec{r}$ and $\frac{d^2\vec{r}}{dt^2}$ are perpendicular for some value of t. (U.P.T.U., 2010; Bhopal, 2006; K.U.T.U., 2006; V.T.U., 2008)

12. If $\vec{A} = xy\vec{i} - z^2\vec{j} + x^2\vec{k}$ and $\vec{B} = 2\vec{i} + y\vec{j} - z\vec{k}$, prove that
(a) $\nabla \cdot (\vec{A} \times \vec{B}) = 2z + x^2$ (b) $\nabla \times (\vec{A} \times \vec{B}) = 2y\vec{i} + (4x + z^2)\vec{j} + y\vec{k}$ (Mumbai, 2000)

13. Prove that $\nabla \times (\vec{A} \times \vec{B}) = \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A}) + (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B}$, where $\vec{A}$ is a constant vector.

14. Calculate the unit tangent vector to the curve $x = t^2 + 1$, $y = 4t - 3$, $z = 2t^2 - 6t$ at $t = 2$. (U.P.T.U., 2008)

15. Find the directional derivative of $\phi = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of the vector $\vec{i} - 2\vec{j} - 2\vec{k}$. (U.P.T.U., 2008; A.M.I.E., 2000)

16. Show that $\vec{A} = (y^2\cos x + z^3)\vec{i} + (2y\sin x - 4)\vec{j} + (3xz^2 + 2)\vec{k}$ is irrotational. Find a scalar function ϕ such that $\vec{A} = \nabla\phi$. (Kerala, 2006; U.P.T.U., 2009)

17. Find the directional derivative of $\nabla \cdot (\nabla\phi)$ at the point $(1, -2, -1)$, where $\phi = 2x^2y - y^2z + xz^2$. (Madras, 2000)

18. Show that $\vec{A} = \left(\frac{x}{r^3}\right)\vec{i} + \left(\frac{y}{r^3}\right)\vec{j} + \left(\frac{z}{r^3}\right)\vec{k}$, where $r = \sqrt{x^2 + y^2 + z^2}$, is solenoidal.

19. Find a solenoidal vector $\vec{B}$, which is perpendicular to $\vec{A} = (x^2 - y)\vec{i} + (y^2 - z)\vec{j} + (z^2 - x)\vec{k}$ at the point $(1, 1, 1)$. (Madras, 2000)

20. Find the constants a, b, c so that $\vec{F} = (x + 2y + az)\vec{i} + (bx - 3y - z)\vec{j} + (4x + cy + 2z)\vec{k}$ is irrotational. Find ϕ such that $\vec{F} = \nabla\phi$. (Kerala, 2006)

3.15 INTEGRATION OF VECTORS

If two vector functions $\vec{F}(t)$ and $\vec{G}(t)$ be such that

$$\frac{d\vec{G}(t)}{dt} = \vec{F}(t),$$

then $\vec{G}(t)$ is called an integral of $\vec{F}(t)$ with respect to the scalar variable t and we write

$$\int \vec{F}(t)\,dt = \vec{G}(t).$$

If $\vec{C}$ be an arbitrary constant vector, we have

$$\vec{F}(t) = \frac{d\vec{G}(t)}{dt} = \frac{d}{dt}\{\vec{G}(t) + \vec{C}\} \quad \text{Hence} \quad \int \vec{F}(t)\,dt = \vec{G}(t) + \vec{C}$$

This is called the indefinite integral of $\vec{F}(t)$ and $\vec{C}$ is called the constant of integration.

$$\int_a^b \vec{F}(t)\,dt = \left[\vec{G}(t) + \vec{C}\right]_a^b = \vec{G}(b) - \vec{G}(a).$$

Example 3.22. If $\vec{R}(t) = t^2\vec{i} + t\vec{j} - t^3\vec{k}$, evaluate $\int_0^1 \left(\vec{R} \times \frac{d^2\vec{R}}{dt^2}\right) dt$.

Solution. $\frac{d}{dt}\left(\vec{R} \times \frac{d\vec{R}}{dt}\right) = \frac{d\vec{R}}{dt} \times \frac{d\vec{R}}{dt} + \vec{R} \times \frac{d^2\vec{R}}{dt^2} = \vec{0} + \vec{R} \times \frac{d^2\vec{R}}{dt^2}$

$$\therefore \quad \int \left(\vec{R} \times \frac{d^2\vec{R}}{dt^2}\right) dt = \vec{R} \times \frac{d\vec{R}}{dt}$$

$$= (t^2\vec{i} + t\vec{j} - t^3\vec{k}) \times (2t\vec{i} + \vec{j} - 3t^2\vec{k})$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ t^2 & t & -t^3 \\ 2t & 1 & -3t^2 \end{vmatrix} = -2t^4\vec{i} + 3t^4\vec{j} - t^2\vec{k}$$

Thus $\int_0^1 \left(\vec{R} \times \frac{d^2\vec{R}}{dt^2}\right) dt = \left[-\frac{2}{5}t^5\vec{i} + \frac{3}{5}t^5\vec{j} - \frac{1}{3}t^3\vec{k}\right]_0^1$

$$= -\frac{2}{5}\vec{i} + \frac{3}{5}\vec{j} - \frac{1}{3}\vec{k}$$

EXERCISE 6.3

1. Show that $\int_{-1}^{1} f(x)\,dx$ [illegible]

2. [illegible] $x(t)$ [illegible] $x(0)$ [illegible] $x(t)$ [illegible] $\frac{dx}{dt}$ [illegible]

3. The acceleration of a particle is [illegible] given by $\frac{d^2x}{dt^2} = -kx$, [illegible] and initially [illegible] find the velocity and displacement at any time.

4. $f(x) = \begin{cases} [illegible], & x \text{ [illegible]} \\ [illegible], & x \text{ [illegible]} \end{cases}$

show that $\int_{-2}^{2} f(x)\,dx = 10$.

6.11 (1) LINE INTEGRAL

Consider a continuous vector function $\mathbf{F}(\mathbf{R})$ which is defined at each point of curve C in space. Divide C into n parts at the points $A = P_0, P_1, \ldots, P_{r-1}, P_r, \ldots, P_n = B$ (Fig. 6.60). Let their position vectors be $\mathbf{R}_0, \mathbf{R}_1, \ldots, \mathbf{R}_{r-1}, \mathbf{R}_r, \ldots, \mathbf{R}_n$. Let $\mathbf{R}_r$ be the position vector of any point P_r on the arc $P_{r-1}P_r$.

Fig. 6.60

Now consider the sum $S = \sum_{r=1}^{n} \mathbf{F}(\mathbf{R}_r) \cdot \delta\mathbf{R}_r$, where $\delta\mathbf{R}_r = \mathbf{R}_r - \mathbf{R}_{r-1}$.

The limit of this sum as $n \to \infty$ in such a way that $|\delta\mathbf{R}_r| \to 0$, provided it exists, is called the **tangential line integral** of $\mathbf{F}(\mathbf{R})$ along C and is symbolically written as

$$\int_C \mathbf{F}(\mathbf{R}) \cdot d\mathbf{R} \quad \text{or} \quad \int_C \mathbf{F} \cdot \frac{d\mathbf{R}}{dt}\,dt.$$

When the path of integration is a closed curve, this fact is indicated by using $\oint$ in place of $\int$.

If $\qquad \mathbf{F}(\mathbf{R}) = X(x, y, z)\,\mathbf{i} + Y(x, y, z)\,\mathbf{j} + Z(x, y, z)\,\mathbf{k}$

and $\qquad d\mathbf{R} = dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}$

then $\qquad \int_C \mathbf{F}(\mathbf{R}) \cdot d\mathbf{R} = \int_C (X\,dx + Y\,dy + Z\,dz)$.

Two physical examples of line integrals are $\int_C \mathbf{F} \cdot d\mathbf{R}$ and $\int_C \mathbf{L} \times d\mathbf{R}$ of which are vectors.

(2) Circulation. If $\mathbf{F}$ represents the velocity of a fluid particle then the line integral $\int_C \mathbf{F} \cdot d\mathbf{R}$ is called the **circulation** of $\mathbf{F}$ around the curve. When the circulation of $\mathbf{F}$ around every closed curve in a region R is zero, $\mathbf{F}$ is said to be irrotational in R.

(3) Work. If $\mathbf{F}$ represents the force acting on a particle moving along an arc AB then the work done in doing the small displacement $d\mathbf{R} = \mathbf{F} \cdot d\mathbf{R}$.

∴ the total work done by $\mathbf{F}$ during the displacement from A to B is given by the line integral $\int_C \mathbf{F} \cdot d\mathbf{R}$.

Example 6.28. If $\mathbf{F} = 3xy\mathbf{i} - y^2\mathbf{j}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{R}$, where C is the curve in the xy-plane $y = 2x^2$ from $(0, 0)$ to $(1, 2)$. (P.T.U., 2003)

Solution. Since the particle moves in the xy-plane, $z = 0$, so $d\mathbf{R} = dx\,\mathbf{i} + dy\,\mathbf{j}$. Thus $\int_C \mathbf{F} \cdot d\mathbf{R}$, where C is the parabola $y = 2x^2$

$$= \int_C (3xy\mathbf{i} - y^2\mathbf{j}) \cdot (dx\,\mathbf{i} + dy\,\mathbf{j}) = \int_C (3xy\,dx - y^2\,dy). \quad \ldots(1)$$

Substituting $y = 2x^2$, where x goes from 0 to 1, (1) becomes

$$= \int_{x=0}^{1} [3x(2x^2) + 4x^2]\,dx + (2x^2)^2\,d(2x^2) = \int_0^1 (6x^3 + 4x^2)\,dx + 8x^4\,dx = \frac{56}{15}.$$

Otherwise, let $x = t$, $y = 2t^2$. Then the parametric equations of C are $x = t, y = 2t^2$. The points $(0, 0)$ and $(1, 2)$ correspond to $t = 0$ and $t = 1$ respectively. Then (1) becomes

$$= \int_{t=0}^{1} [3t(2t^2) + 4t^2]\,dt + (2t^2)^2\,d(2t^2) = \int_0^1 (6t^3 + 4t^2)\,dt + 8t^4\,4t\,dt = \frac{56}{15}.$$

Example 3.20. *A vector field is given by $\vec{F} = (x + y)\hat{i} + 2\hat{j} + (x - y)\hat{k}$. Evaluate $\int \vec{F} \cdot d\vec{R}$ over the circular path given by $x^2 + y^2 = a^2, z = 0$.* (Madras, 2006(S), Anna, 2007)

Solution. As the particle is moving in xy-plane, $\vec{R} = x\hat{i} + y\hat{j}$ so that $d\vec{R} = dx\hat{i} + dy\hat{j}$. Also the circular path is $x = a\cos t, y = a\sin t, z = 0$ where t varies from 0 to 2π.

$$\therefore \quad \int_C \vec{F} \cdot d\vec{R} = \int_C [(x + y)\hat{i} + 2\hat{j} + (x - y)\hat{k}] \cdot (dx\hat{i} + dy\hat{j})$$

$$= \int_C [(x + y)\,dx + 2\,dy] = \int_C [(x\,dx + y\,dx) + 2\,dy] = \int_C (x\,dx + y\,dx + 2\,dy)$$

$$= \int_C [a\cos t(-a\sin t\,dt) + a\sin t\,(a\cos t\,dt) + 2(a\cos t\,dt)] = \int_0^{2\pi} (2a\cos t - a\sin t + a^2\cos^2 t)\,dt$$

$$= \left[2a\sin t + a\cos t + a^2\left(\frac{t}{2} + \frac{\sin 2t}{4}\right)\right]_0^{2\pi} = \frac{a^2}{2}\left[t + \frac{\sin 2t}{2}\right]_0^{2\pi} = \pi a^2.$$

Example 3.21. *Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along*
(a) the straight line from $(0, 0, 0)$ to $(2, 1, 3)$. (Madras, 2003) (Anna, 2002)
(b) the curve defined by $x^2 = 4y, 3x^3 = 8z$ from $x = 0$ to $x = 2$. (Madras, 2002)

Solution. $\int_C \vec{F} \cdot d\vec{R} = \int_C [3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}] \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k})$

$$= \int_C [3x^2\,dx + (2xz - y)\,dy + z\,dz] \qquad \ldots (1)$$

(a) The equations of the straight line from $(0, 0, 0)$ to $(2, 1, 3)$ are $x/2 = y/1 = z/3 = t$ (say).

$\therefore \quad x = 2t, y = t, z = 3t$ are the parametric equations. The points $(0, 0, 0)$ and $(2, 1, 3)$ correspond to $t = 0$ and $t = 1$, respectively.

$$\therefore \quad \text{work done} = \int_C \vec{F} \cdot d\vec{R} = \int_0^1 [3(2t)^2 2\,dt + ((2)(2t)(3t) - t)\,3\,dt + (3t)(3\,dt)]$$

$$= \int_0^1 (24t^2 + 36t)\,dt = 18.$$

(b) Let $x = t$ in $x^2 = 4y, 3x^3 = 8z$. Then the parametric equations of C are $x = t, y = t^2/4, z = 3t^3/8$ and t varies from 0 to 2.

$$\therefore \quad \text{work done} = \int_C \vec{F} \cdot d\vec{R} = \int_0^2 \left[3t^2\,dt + \left[2t\left(\frac{3t^3}{8}\right) - \frac{t^2}{4}\right]d\left(\frac{t^2}{4}\right) + \frac{3t^3}{8}\,d\left(\frac{3t^3}{8}\right)\right]$$

$$= \int_0^2 \left(3t^2 + \frac{t^4}{4} + \frac{9}{64}t^5\right)dt = \left[t^3 + \frac{t^5}{20} + \frac{15}{128}t^6\right]_0^2 = 18.$$

PROBLEMS 3.4

1. Evaluate the line integral $\int_C (x^2 + xy)\,dx + (x^2 + y^2)\,dy$, where C is the square formed by the lines $x = \pm 1, y = \pm 1$.
(Anna, 2005)

2. If $\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along the curve C in the xy plane $y = x^3$ from the point $(1, 1)$ to $(2, 8)$. (U.P.T.U., 2006)

3. Compute the line integral $\int_C (x^2 dx - y^2 dy)$ taken around the triangle whose vertices are $(1, 0)$, $(0, 1)$, $(-1, 0)$.

4. If $\mathbf{A} = (x^2 + xy)\mathbf{i} + (xy + y^2)\mathbf{j}$ and $\mathbf{B} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$, evaluate $\int_C \mathbf{A} \cdot d\mathbf{B}$ from $(0, 0)$ to $(1, 1)$ along the path $y = x^3$. (U.P.T.U., 2004)

5. Evaluate $\int_C (xy + y^2)\,dx$, where C is the arc of the curve $y = 1 - x^2$ in the first quadrant from the point $(1, 0)$ to the point $(0, 1)$. (U.P.T.U., 2005)

6. Find the total work done by the force $\mathbf{F} = (x^2 - y^2 + x)\mathbf{i} - (2xy + y)\mathbf{j}$, moving a particle around the circle $x^2 + y^2 = 4$. (U.P.T.U., 2006)

7. Find the total work done in moving a particle in a force field given by $\mathbf{F} = 3xy\mathbf{i} - 5z\mathbf{j} + 10x\mathbf{k}$ along the curve $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$. (Bhopal, 2006)

8. Using the line integral, compute the work done by the force $\mathbf{F} = 3xy\mathbf{i} - 5z\mathbf{j} + 10x\mathbf{k}$ when it moves a particle from the point $(0, 0, 0)$ to the point $(2, 1, 1)$ along the curve $x = 2t^2$, $y = t$, $z = t^3$. (Bhopal, 2006)

9. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{R}$, where $\mathbf{F} = (x^2 + y - 4)\mathbf{i} + (3x - y^2)\mathbf{j} + (2xz + z^2)\mathbf{k}$, and C is the curve $x = 2\cos t$, $y = 2\sin t$, $z = 3t$ from $t = 0$ to $t = \pi/2$. (U.P.T.U., 2006)

10. If $\mathbf{F} = 3yz\mathbf{i} - xz\mathbf{j} + z^2\mathbf{k}$, evaluate $\int_C \mathbf{F} \cdot d\mathbf{R}$ along the curve $x = t^2$, $y = 2t$, $z = t^3$ from $t = 0$ to $t = 1$.

5.12 (1) SURFACES

As seen in § 5.8, a surface S may be represented by $f(x, y, z) = 0$.

The parametric representation of S is of the form $\mathbf{R}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$ and the continuous functions x, y, z of a real parameter t represent a curve C on this surface S.

For example, the parametric representation of the sphere of radius r is $x^2 + y^2 + z^2 = r^2$, $-1 \le t \le 1$. Similar is the case of right circular cylinder. The equation of the circular helix is $\mathbf{R} = a\cos t\,\mathbf{i} + a\sin t\,\mathbf{j} + bt\mathbf{k}$.

Differentiating $\mathbf{R} = \mathbf{R}(u, v)$, we get $\dfrac{d\mathbf{R}}{dt} = \dfrac{\partial \mathbf{R}}{\partial u}\dfrac{du}{dt} + \dfrac{\partial \mathbf{R}}{\partial v}\dfrac{dv}{dt}$.

The vectors $\partial\mathbf{R}/\partial u$ and $\partial\mathbf{R}/\partial v$ are tangential to S at P and determine the tangent plane of S at P. $\mathbf{N} = \dfrac{\partial \mathbf{R}}{\partial u} \times \dfrac{\partial \mathbf{R}}{\partial v}$ gives a normal vector $\mathbf{N}$ of S at P.

Def. If S has a unique normal at each of its points whose direction depends continuously on the point of S, then the surface S is called a **smooth surface**. If S is not smooth but can be divided into finitely many smooth portions, then it is called a **piecewise smooth surface**.

For instance, the surface of a sphere is smooth while the surface of a cube is piecewise smooth.

Def. A surface S is said to be **orientable** or **two-sided** if the positive normal direction at any point P of S can be continued in a unique and continuous way to the entire surface. If the positive direction of the normal is reversed as we move around a curve on S passing through P, then the surface is non-orientable (i.e., one-sided).

Fig. 5.1

An example of a non-orientable surface is the *Möbius strip*[1]. If we take a long rectangular strip of paper and give a half twist to it and then glue the ends together, the two points A and A' on one point B in Fig. 5.1 coincide, then the surface generated is non-orientable. Such a surface is a model of a Möbius strip.

(2) Surface integral. Consider a continuous function $\mathbf{F}(\mathbf{R})$ and a surface S. Divide S into a finite number of sub-surfaces. Let the surface element surrounding any point $P(\mathbf{R})$ be ΔS which can be regarded as a vector; its magnitude being the area and its direction that of the outward normal to the element.

[1] Named after a German mathematician August Ferdinand Möbius (1790–1868) who was a student of Gauss and professor of astronomy at Leipzig. His important contributions are in projective geometry, theory of surfaces and mechanics.

Consider the curve $ABCDA$, where the intersection points of the curve with the sub-surfaces S_1 and S_2 are A and B. Let S_1 be the surface $ABCDA$ and S_2 be the surface ABA. Then $S = S_1 + S_2$. Now, the surface integral of $\vec{F}$ over S is denoted by

$$\int_S \vec{F} \cdot \vec{dS} \quad \text{or} \quad \int_S \vec{F} \cdot \hat{N}\, dS \text{ where } \hat{N} \text{ is a unit outward normal at } P \text{ to } S.$$

Other types of surface integrals are $\int_S \vec{F} \times \vec{dS}$ and $\int_S f\, dS$ which are both vectors.

Note (i): Only one integration sign is used when there is one differential in dR or dS and two or three signs when there are two or three differentials.

(ii) Flux across a surface. If $\vec{F}$ represents the velocity of a fluid particle then the total outward flux of $\vec{F}$ across a closed surface S is the surface integral $\int_S \vec{F} \cdot \vec{dS}$.

Whereas flux of $\vec{F}$ across any closed surface S is a scalar, $\vec{F}$ is said to be a solenoidal vector point function in S.

It may be noted that $\vec{F}$ could equally well be taken as any other physical quantity e.g., gravitational force, electric force and magnetic force.

Example 8.21: Evaluate $\int_S \vec{F} \cdot \hat{N}\, dS$ where $\vec{F} = 2xy\vec{i} + yz^2\vec{j} + xz\vec{k}$ and S is the closed surface of the region in the first octant bounded by the cylinder $y^2 + z^2 = 9$ and the planes $x = 0, x = 2, y = 0$ and $z = 0$.

Solution. The given closed surface S is generally smooth and is composed of S_1 – the rectangular face $DABS$ in xy-plane; S_2 – the rectangular face $OACC$ in xz-plane; S_3 – the circular quadrant ABC in yz-plane; S_4 – the circular quadrant AED and S_5 – the curved surface $BCDE$ of the cylinder in the first octant (Fig. 8.39).

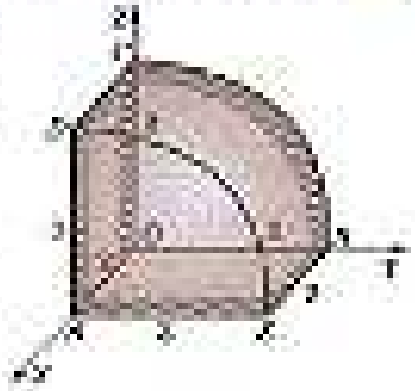

Fig. 8.39

$$\therefore \int_S \vec{F} \cdot \hat{N}\, dS = \int_{S_1} \vec{F} \cdot \hat{N}\, dS + \int_{S_2} \vec{F} \cdot \hat{N}\, dS + \int_{S_3} \vec{F} \cdot \hat{N}\, dS + \int_{S_4} \vec{F} \cdot \hat{N}\, dS + \int_{S_5} \vec{F} \cdot \hat{N}\, dS \quad \ldots(1)$$

Now $\int_{S_1} \vec{F} \cdot \hat{N}\, dS = \int_{S_1} (2xy)\vec{i} - y^2\vec{j} + 4xz^2\vec{k} \cdot (-\vec{k})\, dx\, dy$

$$= -4\int_0^2 \int_0^3 xz^2\, dx\, dy = 0 \quad [\because z = 0 \text{ in the } xy\text{-plane}]$$

Similarly, $\int_{S_2} \vec{F} \cdot \hat{N}\, dS = 0$ and $\int_{S_3} \vec{F} \cdot \hat{N}\, dS = 0$

$$\int_{S_4} \vec{F} \cdot \hat{N}\, dS = \int_{S_4} (2xy^2\vec{i} - y^3\vec{j} + 4xz^2\vec{k}) \cdot \vec{i}\, dS$$

$$= \int_{S_4} 2xy^2\, dS = \int_0^2 \int_0^{\sqrt{9-y^2}} 2xy^2\, dy\, dz\, dx = 4\int_0^2 (9 - z^2)\, dx = 72$$

To find $\hat{N}$ on S_5, we note that $\nabla(y^2 + z^2) = 2y\vec{j} + 2z\vec{k}$

$$\therefore \hat{N} = \frac{2y\vec{j} + 2z\vec{k}}{\sqrt{(4y^2 + 4z^2)}} = \frac{y\vec{j} + z\vec{k}}{3} \quad [\because y^2 + z^2 = 9]$$

and $|\hat{N} \cdot \vec{k}|\, dS = dx\, dy$ or $dz\, dx$ or $dy\, dz$

Thus $\int_{S_5} \vec{F} \cdot \hat{N}\, dS = \int_0^2 \int_0^3 \frac{-y^3 + 4xz^2}{3}\, dy\, dz\, dx = \int_0^2 \int_0^{\pi/2} \left(\frac{-y^3}{3} + 4xz^2\right) y\, dy\, dx$

$$\left[\because y = 3\sin\theta,\ z = 3\cos\theta \atop \therefore dy\, dz = 3\cos\theta\, d\theta\right]$$

$$= \int_0^2 \int_0^{\pi/2} \left|\frac{-27\sin^3\theta}{3}\cos\theta + 4x(9)\cos^2\theta\right| 3\cos\theta\, d\theta\, dx = \int_0^2 \left[-27\times\frac{2}{3} + 108x\times\frac{\pi}{4}\right] dx = 18\pi$$

Hence (i) gives $\int_S \vec{F} \cdot \hat{N}\, dS = 0 + 0 + 0 + 72 + 18\pi = 18\pi$.

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PROBLEMS 4.7

1. If velocity vector is $\vec{V} = x\vec{i} + 3z\vec{j} - y\vec{k}$ prove that the flow of water through the parabolic cylinder $y = x^2$, $0 \le x \le 2$, $0 \le z \le 3$ is $-\frac{56}{5}$.

2. Evaluate $\iint_S \vec{F} \cdot \vec{n}\, dS$ where $\vec{F} = x\vec{i} + y\vec{j} - z\vec{k}$ and S is the triangular surface of Δ with vertices $(2, 0, 0)$, $(0, 2, 0)$ and $(0, 0, 6)$.

3. Evaluate $\iint_S \vec{F} \cdot \vec{N}\, dS$ where $\vec{F} = 18z\vec{i} - 12\vec{j} + 3y\vec{k}$ and S is the portion of the plane $2x + 3y + 6z = 12$ in the first octant.

4. If $\vec{F} = 2xz\vec{i} - 2x^2\vec{j} + z^2\vec{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, show that $\iint_S \vec{F} \cdot \vec{N}\, dS = 192$.

4.13 GREEN'S THEOREM IN THE PLANE*

If $\phi(x, y)$, $\psi(x, y)$, ϕ_x and ψ_y are continuous in a region R of the xy-plane bounded by a closed curve C, then

$$\int_C \phi\, dx + \psi\, dy = \iint_R \left(\frac{\partial \psi}{\partial x} - \frac{\partial \phi}{\partial y}\right) dx\, dy \qquad \ldots(1)$$

Consider the region R bounded by a simple closed curve C which is cut by any line parallel to the axes in the region in two points.

Let R be bounded by $x = a$, $y = \xi_1(x)$, $x = b$ and $y = \eta(x)$, where $\eta \ge \xi_1$ so that C is divided into curves C_1 and C_2 (Fig. 4.11).

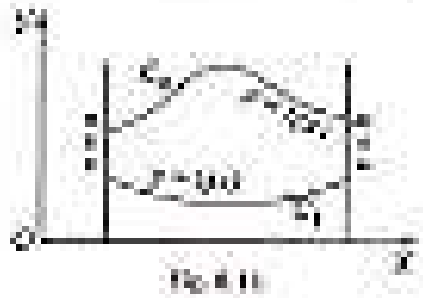

Fig. 4.11

$$\therefore \iint_R \frac{\partial \phi}{\partial y}\, dx\, dy = \int_a^b dx \left[\int_{\xi_1}^{\eta} \frac{\partial \phi}{\partial y}\, dy\right] = \int_a^b dx \,[\phi]_{\xi_1}^{\eta}$$

$$= \int_a^b [\phi(x, \eta) - \phi(x, \xi_1)]\, dx = -\int_a^b \phi(x, \xi_1)\, dx - \int_b^a \phi(x, \eta)\, dx$$

$$= -\int_{C_1} \phi(x, y)\, dx - \int_{C_2} \phi(x, y)\, dx$$

$$= -\int_C \phi(x, y)\, dx \qquad \ldots(2)$$

Similarly, it can be shown that

$$\iint_R \frac{\partial \psi}{\partial x}\, dx\, dy = \int_C \psi(x, y)\, dy \qquad \ldots(3)$$

On subtracting (2) from (3), we get (1).

This result can be extended to regions which may be divided into a finite number of sub-regions such that the boundary of each is cut in the most in two parts by any line parallel to either axis. Applying (1) to each of these sub-regions and adding the results, the surface integrals combine into an integral over the whole region; the line integrals over the common boundaries cancel (for each is covered twice but in opposite directions), whereas the remaining line integrals combine into the line integral over the external curve C.

Obs. This theorem converts a line integral around a closed curve into a double integral and is a special case of Stoke's theorem (Sec Art. 4.14).

Example 4.22. Verify Green's theorem for $\int_C (xy + y^2)\, dx + x^2\, dy$, where C is bounded by $y = x$ and $y = x^2$.

(V.T.U. 2011 ; S.V.T.U. 2009 (Feb-end, 2010))

Solution. Here $\phi = xy + y^2$ and $\psi = x^2$

$$\therefore \int_C (xy + y^2)\, dx + x^2\, dy = \int_{C_1} + \int_{C_2}$$

*Named after the English mathematician George Green (1793–1841) who was, given at Cambridge etc. is known for his work on potential theory in connection with Fourier series, vibrations, elasticity, fluid dynamics and magnetism.

Along C_1, $y = x^2$ and x varies from 0 to 1 (Fig. 8.32)

$$\int_{C_1} = \int_0^1 [(x+x^2)^2 + (x^2)^2]dx + x^2 2x dx]$$

$$= \int_0^1 (2x^2 + x^4 + 4x^3)dx = \frac{19}{30}$$

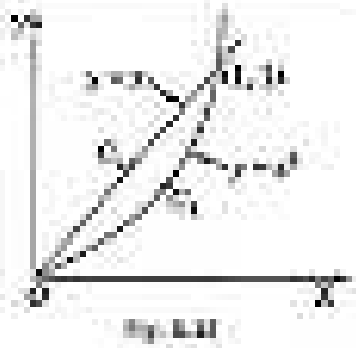

Fig. 8.32

Along C_2, $y = x$ and x varies from 1 to 0.

$$\int_{C_2} = \int_1^0 [(x+x)^2 + (x)^2]dx + x^2 dx] = \int_1^0 6x^2 dx = -2.$$

Then $\int_C (x+y)dx + xy\,dy = \frac{19}{30} - 2 = -\frac{41}{30}$ ...(1)

Also $\iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right]dx\,dy = \iint_R \left[\frac{\partial}{\partial x}(xy) - \frac{\partial}{\partial y}(x+y)\right]dx\,dy$

$$= \int_0^1 \int_{x^2}^x (y - 1)\,dy\,dx = \int_0^1 \left[\frac{1}{2}y^2 - y\right]_{x^2}^x dx = \int_0^1 (x^2 - x^4 - 2x)\,dx = -\frac{41}{30} \quad ...(2)$$

Hence, Green's theorem is verified from the equality of (1) and (2).

Example 8.26. If C is a simple closed curve in the xy-plane not enclosing the origin, then show that

$$\int_C \mathbf{F} \cdot d\mathbf{R} = 0 \quad \text{where} \quad \mathbf{F} = \frac{x\mathbf{i} - y\mathbf{j}}{x^2 + y^2}$$ (U.P.T.U., 2003)

Solution. $\int_C \mathbf{F} \cdot d\mathbf{R} = \int_C \frac{x\mathbf{i} - y\mathbf{j}}{x^2 + y^2}(dx\,\mathbf{i} + dy\,\mathbf{j})$ $\left[\because \mathbf{R} = x\mathbf{i} + y\mathbf{j}\right]$

$$= \int_C \frac{x\,dx - y\,dy}{x^2 + y^2} = \int_C (M\,dx + N\,dy) \text{ where } M = \frac{x}{x^2 + y^2}, N = \frac{-y}{x^2 + y^2}$$

$$= \iint_R \left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right]dx\,dy$$ (By Green's theorem)

$$= \iint_R \left[\frac{-(x^2 + y^2) + x(2x)}{(x^2 + y^2)^2} - \frac{(x^2 + y^2) - y(2y)}{(x^2 + y^2)^2}\right]dx\,dy$$

$$= \iint_R \left[\frac{x^2 - y^2}{(x^2 + y^2)^2} - \frac{x^2 - y^2}{(x^2 + y^2)^2}\right]dx\,dy = 0.$$

Example 8.27. Using Green's theorem, evaluate $\int_C (2y - \sin x)\,dx + \cos x\,dy$, where C is the plane triangle enclosed by the lines $y = 0$, $x = \frac{\pi}{2}$ and $y = \frac{2}{\pi}x$. (U.P.T.U., 2005; Kanpur, 2002)

Solution. Here $\phi = 2y - \sin x$ and $\psi = \cos x$.

By Green's theorem $\int_C (2y - \sin x)\,dx + \cos x\,dy$

$$= \iint_R \left(\frac{\partial \psi}{\partial x} - \frac{\partial \phi}{\partial y}\right)dx\,dy$$

$$= \int_{y=0}^{y=\pi/2} \int_{x=2y/\pi}^{x=\pi/2} (-\sin x - 2)\,dx\,dy = -\int_0^{\pi/2} [\cos x + 2x]_{2y/\pi}^{\pi/2}\,dy$$

$$= -\frac{2}{\pi}\int_0^{\pi/2} (\pi - 2y)\,dy = -\frac{2}{\pi}\left[y - \cos x + x\right]_0^{\pi/2} - \int_0^{\pi/2} (1 - \cos x + x)\,dx$$

$$= -\frac{2}{\pi}\left[\frac{\pi^2}{4} - \left(-\sin x + \frac{x^2}{2}\right)_0^{\pi/2}\right] = -\frac{2}{\pi}\left[\frac{\pi^2}{4} - \left(1 + \frac{\pi^2}{8}\right)\right] = -\left(\frac{\pi}{2} - \frac{2}{\pi}\right)$$

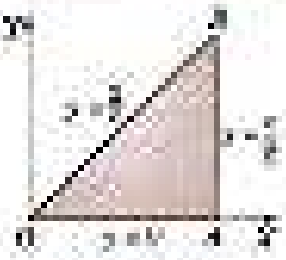

Fig. 8.33

Example 5.33 *Apply Green's theorem to evaluate* $\int_C (2x^2 - y^2)\,dx + (x^2 + y^2)\,dy$, *where* C *is the boundary of the area enclosed by the* x*-axis and the upper half of the circle* $x^2 + y^2 = a^2$. (UPTU, 2002)

Solution. By Green's theorem

$$\int_C (2x^2 - y^2)\,dx + (x^2 + y^2)\,dy$$

$$= \iint_R \left[\frac{\partial}{\partial x}(x^2 + y^2) - \frac{\partial}{\partial y}(2x^2 - y^2)\right] dx\,dy$$

$$= 2\iint_R (x + y)\,dx\,dy, \text{ where } R \text{ is the region of Fig. 5.31}$$

$$= 2\int_{\theta=0}^{\pi}\int_{r=0}^{a} r(\cos\theta + \sin\theta)\,r\,dr\,d\theta$$

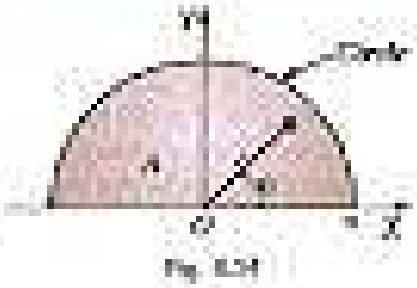

Fig. 5.31

(Changing to polar coordinates i.e., $x = r\cos\theta$, $y = r\sin\theta$, r varies from 0 to a and θ varies from 0 to π)

$$= 2\int_0^a r^2\,dr \int_0^{\pi} (\cos\theta + \sin\theta)\,d\theta = 2\cdot\frac{a^3}{3}\cdot[1 - (-1)] = \frac{4a^3}{3}$$

EXERCISE 5.5

1. Verify Green's theorem for $\int_C (3x - 8y^2)\,dx + (4y - 6xy)\,dy$, where C is the boundary of the region bounded by $y = 0$, $x = 1$ and $x = y = 1$. (Nagpur, 2009; Kerala, 2008; Anna, 2006)
2. Verify Green's theorem for $\int_C (x^2 - \cos hy)\,dx + (y + \sin x)\,dy$, where C is the rectangle with vertices $(0, 0)$, $(\pi, 0)$, $(\pi, 1)$, $(0, 1)$. (Nagpur, 2006; AMIETE, 2006)
3. Verify Green's theorem for $\int_C (y^2\,dx + x^2\,dy)$, where C is the boundary described counter clockwise of the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$. (UPTU, 2010)
4. Apply Green's theorem, to prove that the area enclosed by a plane curve is $\frac{1}{2}\int_C (x\,dy - y\,dx)$.

   Hence find the area of an ellipse whose semi-major and semi-minor axes are of lengths a and b. (Nagpur, 2008; UPTU, 2000 C)
5. Find the area of a circle of radius a using Green's theorem. (Mumbai, 2005)
6. Evaluate $\int_C (3x^2 - 8y^2)\,dx + (4y - 6xy)\,dy$, where C is the region bounded by the lines $x = 2$, $y = x$, $y = 2x$. (UPTU, 2010; Mumbai, 2006)
7. Evaluate $\int_C (x^2 - 2xy)\,dx + (x^2 + y)\,dy$, where C is the boundary of the region defined by $y^2 = 8x$ and $x = 2$.
8. Evaluate by Green's theorem $\int_C \mathbf{F}\cdot d\mathbf{R}$, where $\mathbf{F} = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j}$ and C is the curve $y = x^2$, $x = 0$, $x = b$. (Mumbai, 2006)

5.11 STOKE'S THEOREM[1] (Relation between line and surface integrals)

If S *be an open surface bounded by a closed curve* C *and* $\mathbf{F} = f_1\mathbf{i} + f_2\mathbf{j} + f_3\mathbf{k}$ *be any continuously differentiable vector point function, then*

$$\int_C \mathbf{F}\cdot d\mathbf{R} = \int_S \text{curl}\,\mathbf{F}\cdot\mathbf{N}\,ds$$

where $\mathbf{R} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $d\mathbf{R} = dx\,\mathbf{i} + dy\,\mathbf{j} + dz\,\mathbf{k}$ *is a unit outward normal at any point of* S.

[1] Named after Sir George Gabriel Stokes (1819–1903) who became professor in Cambridge. His important contributions are in fluid dynamics, grading and theory of infinite series.

Writing $d\mathbf{R} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$, it may be reduced to the form

$$\int_C (f\,dx + f_2\,dy + f_3\,dz) = \int_R \left[\left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}\right)\cos\alpha + \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}\right)\cos\beta + \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right)\cos\gamma\right] dS \qquad \ldots(1)$$

Let us first prove that

$$\int_C f_1\,dx = \int_R \left(\frac{\partial f_1}{\partial z}\cos\beta - \frac{\partial f_1}{\partial y}\cos\gamma\right) dS \qquad \ldots(2)$$

Let $z = g(x, y)$ be the equation of the surface S whose projection on the xy-plane is the region R. Then the projection of C on the xy-plane is the curve C' enclosing region R.

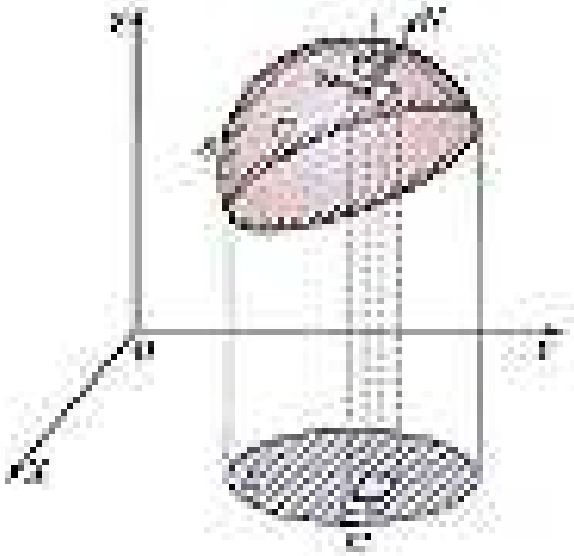

Fig. 3.13

$$\int_C f_1(x, y, z)\,dx = \int_{C'} f_1(x, y, g(x, y))\,dx$$

$$= -\iint_R \frac{\partial}{\partial y} f_1(x, y, g(x, y))\,dx\,dy, \text{ by Green's theorem}$$

$$= -\iint_R \left(\frac{\partial f_1}{\partial y} + \frac{\partial f_1}{\partial z}\frac{\partial g}{\partial y}\right) dx\,dy \qquad \ldots(3)$$

The direction ratios of the normal to the surface $z = g(x, y)$ are given by

$$\frac{\cos\alpha}{-\partial g/\partial x} = \frac{\cos\beta}{-\partial g/\partial y} = \frac{\cos\gamma}{1} \quad \text{(See p. 3.18)} \qquad \ldots(4)$$

Moreover

$dx\,dy$ = projection of dS on the xy-plane

$= dS\cos\gamma$, i.e., $dS = dx\,dy\sec\gamma$

$\therefore$ right side of (2)

$$= \iint_R \left(\frac{\partial f_1}{\partial z}\frac{\cos\beta}{\cos\gamma} - \frac{\partial f_1}{\partial y}\right) dx\,dy = -\iint_R \left(\frac{\partial f_1}{\partial y} + \frac{\partial f_1}{\partial z}\frac{\partial g}{\partial y}\right) dx\,dy \qquad \left[\because \frac{\cos\beta}{\cos\gamma} = -\frac{\partial g}{\partial y} \text{ by (4)}\right]$$

= Left side of (2), by (3).

Thus we have proved (2). Similarly, we can prove the other corresponding relations for f_2 and f_3. Adding these three results, we get (1).

Cor. Green's theorem in a plane as a special case of Stokes theorem. Let $\mathbf{F} = f_1\mathbf{i} + f_2\mathbf{j}$ be a vector function which is continuously differentiable in a region R of the xy-plane bounded by a closed curve C. Then

$$\int_C \mathbf{F}\cdot d\mathbf{R} = \int_C (f_1\,dx + f_2\,dy) \quad (\because d\mathbf{R} = dx\mathbf{i} + dy\mathbf{j}) = \int_C (f_1\,dx + f_2\,dy)$$

and

$$\text{curl}\,\mathbf{F}\cdot\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f_1 & f_2 & 0 \end{vmatrix}\cdot\mathbf{k} = \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}$$

Hence Stoke's theorem takes the form $\int_C (f_1\,dx + f_2\,dy) = \int_R \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y}\right) dx\,dy$ which is Green's theorem in a plane.

Example 3.28. Verify Stoke's theorem for $\mathbf{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$ taken round the rectangle bounded by the lines $x = \pm a$, $y = 0$, $y = b$. (Bhopal, 2005 S; VTU, 2007; JNTU, 2005; GBTU, 2009)

Solution. Let $ABCD$ be the given rectangle as shown in Fig. 3.14.

$$\int_{ABCD} \mathbf{F}\cdot d\mathbf{R} = \int_{AB} \mathbf{F}\cdot d\mathbf{R} + \int_{BC} \mathbf{F}\cdot d\mathbf{R} + \int_{CD} \mathbf{F}\cdot d\mathbf{R} + \int_{DA} \mathbf{F}\cdot d\mathbf{R}$$

and

$$\mathbf{F}\cdot d\mathbf{R} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}\cdot(dx\mathbf{i} + dy\mathbf{j}) = (x^2 + y^2)dx - 2xy\,dy$$

Along AB, $y = 0$, $dy = 0$ and y varies from 0 to b.

$$\therefore \int_{AB} \mathbf{F}\cdot d\mathbf{R} = -2a\int_0^b y\,dy = -2a\cdot\frac{b^2}{2} = -ab^2$$

Similarly, $$\int_{C_2} \mathbf{F}\cdot d\mathbf{R} = \int_{x=a}^{-a} (x^2 + k^2)\,dx = -\frac{2a^3}{3} - 2ak^2.$$

$$\int_{C_3} \mathbf{F}\cdot d\mathbf{R} = 2a\int_{y=0}^{k} y\,dy = -ak^2$$

and $$\int_{C_4} \mathbf{F}\cdot d\mathbf{R} = \int_{-a}^{a} x^2\,dx = \frac{2a^3}{3}.$$

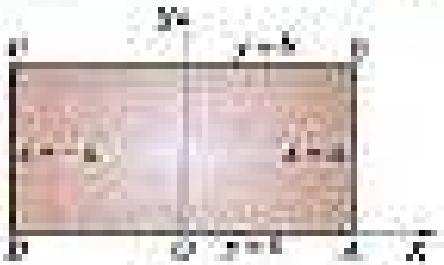

Fig. 3.16

Thus $$\int_{ABCDA} \mathbf{F}\cdot d\mathbf{R} = -4ak^2 \qquad \ldots(i)$$

Also since $\operatorname{curl}\mathbf{F} = -4k\mathbf{j}$

$$\therefore \int_S \operatorname{curl}\mathbf{F}\cdot\mathbf{N}\,ds = \int_{x=-a}^{a}\int_{y=0}^{k} -4k\mathbf{j}\cdot\mathbf{k}\,dx\,dy = -4\int_{-a}^{a}\int_{0}^{k} y\,dx\,dy$$

$$= -4\int_{-a}^{a}\left[\frac{y^2}{2}\right]_0^k dx = -8k\left|\frac{x^2}{2}\right|_{-a}^{a} = -4ak^2 \qquad \ldots(ii)$$

Hence Stoke's theorem is verified from the equality of (i) and (ii).

Example 6.11. Verify Stoke's theorem for the vector field $\mathbf{F} = 2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2z\mathbf{k}$ over the upper half surface of $x^2 + y^2 + z^2 = 1$ bounded by the projection on the xy-plane.

(Nagpur, 2005 ; Madras, 2000 ; B.T.U., 2006)

Solution. The projection of the upper half of the given sphere on the xy-plane is $z = 0$ in the circle $x^2 + y^2 = 1$ (Fig. 3.17).

$$\oint_C \mathbf{F}\cdot d\mathbf{R} = \oint_C (2x - y)\,dx - yz^2\,dy - y^2z\,dz = \oint_C (2x - y)\,dx \qquad [\because z = 0 \text{ in the } xy\text{-plane}]$$

$$= \int_{\theta=0}^{2\pi} (2\cos\theta - \sin\theta)(-\sin\theta)\,d\theta \qquad [\because x = \cos\theta, y = \sin\theta]$$

$$= \int_0^{2\pi} (-\sin 2\theta + \sin^2\theta)\,d\theta = 0 + 4\int_0^{\pi/2} \sin^2\theta\,d\theta = \pi. \qquad \ldots(i)$$

Now $$\operatorname{curl}\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2z \end{vmatrix}$$

$$= (-2yz + 2yz)\mathbf{i} + 0\mathbf{j} + 1\mathbf{k} = \mathbf{k}$$

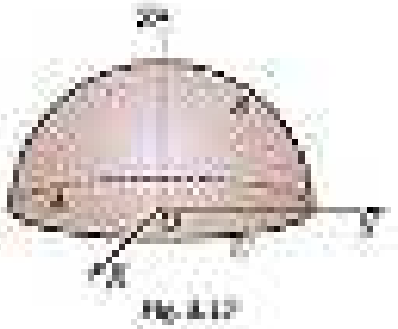

Fig. 3.17

$$\therefore \int_S \operatorname{curl}\mathbf{F}\cdot\mathbf{N}\,ds = \int_R \mathbf{k}\cdot\mathbf{N}\,ds = \int_R \mathbf{k}\cdot\mathbf{N}\frac{dx\,dy}{|\mathbf{N}\cdot\mathbf{k}|}$$

where R is the projection of S on xy-plane and $ds = dx\,dy/|\mathbf{N}\cdot\mathbf{k}|$

$$= \int_R dx\,dy = \text{area of circle } C = \pi \qquad \ldots(ii)$$

Hence, the Stokes theorem is verified from the equality of (i) and (ii).

Example 6.12. Using Stoke's theorem evaluate $\int_C (xy\,dx + yz\,dy + zx\,dz)$, where C is the boundary of the triangle with vertices $(1, 0, 0), (0, 1, 0)$ and $(0, 0, 1)$.

(Nagpur, 2006 ; Kurukshetra, 2008 ; Bangalore, 2005)

Solution. Here $\mathbf{F} = (xy)\mathbf{i} + (yz)\mathbf{j} + (zx)\mathbf{k}$

$$\therefore \operatorname{curl}\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & yz & zx \end{vmatrix} = -y\mathbf{i} - z\mathbf{j} - x\mathbf{k}$$

Also equation of the plane through A, B, C (Fig. 8.10) is

$$\frac{x}{2}+\frac{y}{3}+\frac{z}{6}=1 \text{ or } 3x+2y+z=6$$

Vector N normal to this plane is

$$\nabla(3x+2y+z-6)=3\mathbf{i}+2\mathbf{j}+\mathbf{k}$$

$$\therefore \quad \hat{N}=\frac{3\mathbf{i}+2\mathbf{j}+\mathbf{k}}{\sqrt{3^2+2^2+1^2}}=\frac{1}{\sqrt{14}}(3\mathbf{i}+2\mathbf{j}+\mathbf{k})$$

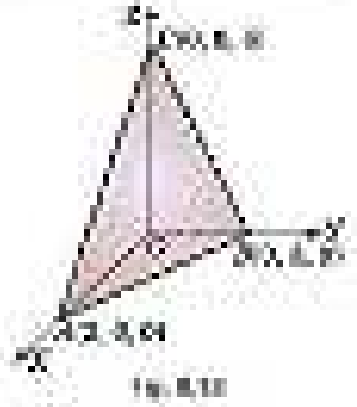

Fig. 8.10

Hence $\int_C (3x-y)\,dx+(2y-z)\,dy+(y-x)\,dz=\int_C \mathbf{F}\cdot d\mathbf{R}$

$$=\int_S \operatorname{curl}\mathbf{F}\cdot\hat{N}\,dS \qquad \text{where } S \text{ is the triangle } ABC$$

$$=\int_S (2\mathbf{i}+\mathbf{k})\cdot\left(\frac{3\mathbf{i}+2\mathbf{j}+\mathbf{k}}{\sqrt{14}}\right)dS=\frac{1}{\sqrt{14}}(6+1)\int_S dS$$

$$=\frac{7}{\sqrt{14}}(\text{Area of } \Delta ABC)=\frac{7}{\sqrt{14}}\cdot\frac{1}{2}\sqrt{56}=7.$$

Example 8.26. *If $\mathbf{F}=3y\mathbf{i}-xz\mathbf{j}+yz^2\mathbf{k}$ and S is the surface of the paraboloid $2z=x^2+y^2$ bounded by $z=2$, evaluate $\iint_S (\nabla\times\mathbf{F})\cdot d\mathbf{R}$ using Stoke's theorem.*

Solution. By Stokes theorem, $I=\iint_S (\nabla\times\mathbf{F})\cdot d\mathbf{R}=\int_C \mathbf{F}\cdot d\mathbf{R}$

where S is the surface $2z=x^2+y^2$ bounded by $z=2$.

$$\therefore \quad I=\int_C \mathbf{F}\cdot d\mathbf{R}=\int_C (3y\mathbf{i}-xz\mathbf{j}+yz^2\mathbf{k})\cdot(dx\mathbf{i}+dy\mathbf{j}+dz\mathbf{k})$$

$$=\int_C (3y\,dx-xz\,dy+yz^2\,dz)$$

$\because$ S: $x^2+y^2=4, z=2$
$\therefore$ Put $x=2\cos\theta, y=2\sin\theta$
$C: x^2+y^2=4, z=0$ to 2π.

$$=\int_0^{2\pi} [6\sin\theta(-2\sin\theta\,d\theta)-4\cos\theta(2\cos\theta)(2\cos\theta\,d\theta)+8\sin\theta(0)]$$

$$=-4\int_0^{2\pi} (3\sin^2\theta+2\cos^2\theta)\,d\theta$$

$$=-4\left[3.\frac{1}{2}\cdot\frac{\pi}{2}+2.\frac{1}{2}.\frac{\pi}{2}\right]=-5\pi.$$

Example 8.27. *Apply Stoke's theorem to evaluate $\int_C (ydx+zdy+xdz)$ where C is the curve of intersection of $x^2+y^2+z^2=a^2$ and $x+y+z=0$.* (Bhopal, 2008)

Solution. The curve C is evidently a circle lying in the plane $x+y+z=0$ and hence AB is a diameter of the circle C (Fig. 8.11).

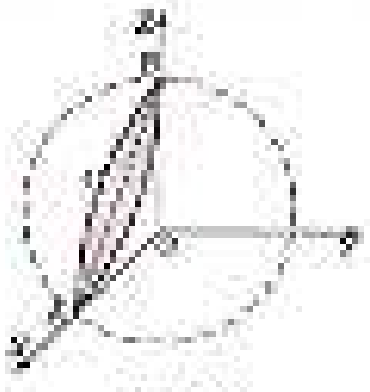

Fig. 8.11

$$\therefore \quad \int_C (x\,dx+y\,dy+z\,dz)=\int_C (y\mathbf{i}+z\mathbf{j}+x\mathbf{k})\cdot d\mathbf{R}$$

$$=\int_S \operatorname{curl}(y\mathbf{i}+z\mathbf{j}+x\mathbf{k})\cdot\hat{N}\,dS$$

where S is the circle on AB as diameter and $\hat{N}=\frac{1}{\sqrt{2}}\mathbf{i}+\frac{1}{\sqrt{2}}\mathbf{k}$

$$=\int_S (-\mathbf{i}-\mathbf{j}-\mathbf{k})\cdot\left(\frac{1}{\sqrt{2}}\mathbf{i}+\frac{1}{\sqrt{2}}\mathbf{k}\right)dS$$

$$=-\frac{2}{\sqrt{2}}\int_S dS=-\sqrt{2}\pi\left(\frac{a}{\sqrt{2}}\right)^2=-\frac{\pi a^2}{\sqrt{2}}$$

Example 8.43. If $\mathbf{F}$ is a constant vector and S is any closed surface, prove that $\int_S \operatorname{curl}(\mathbf{F} \times \mathbf{r})\, d\mathbf{S} = 0$.

Solution. Cut open the surface S by any plane and let S_1, S_2 denote the upper and lower portions. Let C be the common curve bounding both these portions.

$$\int_S \operatorname{curl}(\mathbf{F} \times \mathbf{r})\, d\mathbf{S} = \int_{S_1} \operatorname{curl}(\mathbf{F} \times \mathbf{r})\, d\mathbf{S} + \int_{S_2} \operatorname{curl}(\mathbf{F} \times \mathbf{r})\, d\mathbf{S} = \int_C (\mathbf{F} \times \mathbf{r})\, d\mathbf{R} - \int_C (\mathbf{F} \times \mathbf{r})\, d\mathbf{R} = 0,$$

on applying Stoke's theorem. The second integral is negative because it is traversed in a direction opposite to that of the first.

PROBLEMS 8.9

1. Verify Stoke's theorem for the vector field $\mathbf{F} = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j}$ over the rectangular boundary in the plane $z = 0$ bounded by the lines $x = \pm a$, $y = 0$, $y = b$. (U.P.T.U., 2006) (U.P.T.U., 2007)
   (U.P.T.U., 2008)
2. Verify Stoke's theorem for a vector field defined by $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$, in the region $x^2 + y^2 \le 1$, $z = 0$.
3. Evaluate $\int_S \mathbf{F} \cdot d\mathbf{R}$, where $\mathbf{F} = 2y\mathbf{i} + z\mathbf{j} + y\mathbf{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ in the first octant bounded by $y = 0$, $z = 0$, $y = z$.
   (Madras, 2001)
4. Verify Stoke's theorem for $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$ over the surface of the sphere $x^2 + y^2 + z^2 = 1$ above the xy-plane and bounded by its projection on the xy-plane.
   (Madras, 2000)
5. Evaluate $\int_S \mathbf{F} \cdot d\mathbf{R}$, where $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$, S is the surface of the paraboloid $z = 4 - x^2 - y^2$, $z \ge 0$.
6. Evaluate by Stoke's theorem $\int_C (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz$, where C is the curve $x^2 + y^2 = 1$, $z = y^2$.
   (U.P.T.U., 2005)
7. If S is the surface of the sphere $x^2 + y^2 + z^2 = 1$, prove that $\int_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = 0$.
   (U.P.T.U., 2006)
8. Prove that $\int_C (x + y)\,dx + (2x - z)\,dy + (y + z)\,dz = 2\pi$, where C is the curve $x = \cos\theta + \sin\theta$, $y = \cos\theta$, $z = \sin\theta$ and show that the value of the surface integral $\iint_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n}\, dS$ has the same value over all the surfaces bounded by C.
9. Evaluate by Stoke's theorem $\int_C (x^2 + y)\,dx + (2x + y)\,dy$, where C is the boundary of the rectangle $x = 0, x = 2, y = 0, y = 4$.
   (U.P.T.U., 2002)
10. Use Stoke's theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{R}$, where $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$ and C is the boundary of the region $x^2 + y^2 = a^2$, $z = 0$ above the xy-plane.
    (Kanpur, 2001)
11. Use Stoke's theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{R}$, where $\mathbf{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$ and C is the boundary of the region $x = a$, $y = 0$, $y = b$.
12. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{R}$, where C is the circle $x^2 + y^2 = a^2$, $z = 0$ and $\mathbf{F} = -y\mathbf{i} + x\mathbf{j} + c\mathbf{k}$.

8.15 VOLUME INTEGRAL

Consider a continuous vector function $\mathbf{F}(\mathbf{R})$ and a surface S enclosing the region R. Divide R into finite number of sub-regions $R_1, R_2, \ldots, R_n$. Let δv_i be the volume of the sub-region R_i enclosing any point whose position vector is $\mathbf{R}_i$.

Consider the sum $\quad V = \sum_{i=1}^{n} \mathbf{F}(\mathbf{R}_i)\, \delta v_i$

The limit of this sum as $n \to \infty$ in such a way that $\delta v_i \to 0$, is called the volume integral of $\mathbf{F}(\mathbf{R})$ over R and is symbolically written as $\int_R \mathbf{F}\, dv$.

If $\mathbf{F}(\mathbf{R}) = f(x, y, z)\mathbf{i} + \phi(x, y, z)\mathbf{j} + \psi(x, y, z)\mathbf{k}$ as $\delta v_i \to 0$ then

$$\int_R \mathbf{F}\, dv = \mathbf{i} \iiint_R f\, dx\, dy\, dz + \mathbf{j} \iiint_R \phi\, dx\, dy\, dz + \mathbf{k} \iiint_R \psi\, dx\, dy\, dz.$$

5.10. GAUSS DIVERGENCE THEOREM* [Relation between surface and volume integrals]

Let $\mathbf{F}$ be a continuously differentiable vector function in the region V bounded by the closed surface S. Then

$$\iint_S \mathbf{F} \cdot \mathbf{N}\,dS = \iiint_V \operatorname{div} \mathbf{F}\,dV \qquad (1)$$

where $\mathbf{N}$ is the unit outward normal to S.

If $\mathbf{F}(\mathbf{R}) = f(x, y, z)\mathbf{i} + g(x, y, z)\mathbf{j} + h(x, y, z)\mathbf{k}$ then it is required to prove that

$$\iint_S (f\mathbf{i} + g\mathbf{j} + h\mathbf{k}) \cdot \mathbf{N}\,dS = \iiint_V \left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} \right) dx\,dy\,dz \qquad (2)$$

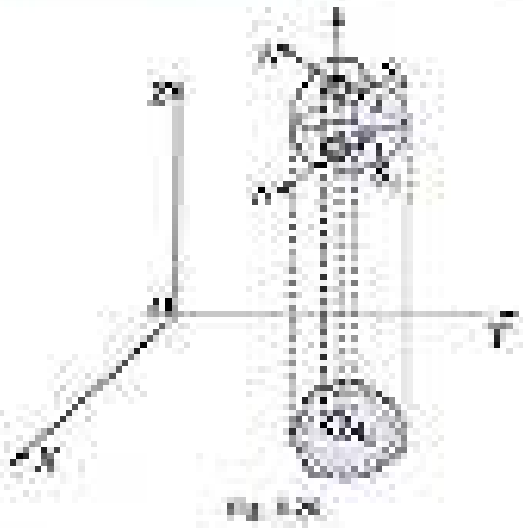

Fig. 5-26

Firstly, consider such a surface S that a line parallel to a coordinate axis cannot meet it in more than two points. Let $S_1(x, y, z_1)$ and $S_2(x, y, z_2)$ (Fig. 5-26).

If S projects into the area A_{xy} on the xy-plane, then

$$\iiint_V \frac{\partial h}{\partial z}\,dx\,dy\,dz = \iint_{A_{xy}} dx\,dy \int_{z_1}^{z_2} \frac{\partial h}{\partial z}\,dz$$

$$= \iint_{A_{xy}} [h(x, y, z_2) - h(x, y, z_1)]\,dx\,dy = \iint_{A_{xy}} h(x, y, z_2)\,dx\,dy - \iint_{A_{xy}} h(x, y, z_1)\,dx\,dy \qquad (3)$$

Let S_1, S_2 be the lower and upper parts of the surface S corresponding to the points P_1 and P_2 respectively and $\mathbf{N}$ be the unit external normal vector at any point of S. As the external normal at any point of S_2 makes an acute angle with the positive direction of z-axis and that at any point of S_1 an obtuse angle, therefore

$$\iint_{A_{xy}} h(x, y, z_2)\,dx\,dy = \iint_{S_2} h\mathbf{k} \cdot \mathbf{N}\,dS \qquad (4)$$

$$\iint_{A_{xy}} h(x, y, z_1)\,dx\,dy = -\iint_{S_1} h\mathbf{k} \cdot \mathbf{N}\,dS \qquad (5)$$

Using (4) and (5), (3) now becomes

$$\iiint_V \frac{\partial h}{\partial z}\,dx\,dy\,dz = \iint_{S_2} h\mathbf{k} \cdot \mathbf{N}\,dS + \iint_{S_1} h\mathbf{k} \cdot \mathbf{N}\,dS = \iint_S h\mathbf{k} \cdot \mathbf{N}\,dS \qquad (6)$$

Similarly, we have

$$\iiint_V \frac{\partial f}{\partial x}\,dx\,dy\,dz = \iint_S f\mathbf{i} \cdot \mathbf{N}\,dS \qquad (7)$$

$$\iiint_V \frac{\partial g}{\partial y}\,dx\,dy\,dz = \iint_S g\mathbf{j} \cdot \mathbf{N}\,dS \qquad (8)$$

Addition of (6), (7) and (8) gives

$$\iiint_V \left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z} \right) dx\,dy\,dz = \iint_S (f\mathbf{i} + g\mathbf{j} + h\mathbf{k}) \cdot \mathbf{N}\,dS \text{ which is same as (1).}$$

Secondly, consider a general region R. Assume that it can be split up into a finite number of sub-regions each of which is such that a line parallel to any axis intersects the surface in not more than two points. Applying (1) to each of these sub-regions and adding the results, the volume integrals will combine to give the volume integral over the whole region R. Also the surface integrals over the common boundaries of two sub-regions cancel because each occurs twice and having corresponding normals in opposite directions whereas the resultant surface integrals reduce to that of the surface integral over the entire surface S.

Finally, consider a region R bounded by two closed surfaces S_1, S_2 (S_2 being outside S_1). Making the outward normal at points of S_1 to be directed towards the interior of S_2 and introducing an additional surface cutting S_1, S_2 so that all parts of R are bounded by a single closed surface, the result of the theorem follows as before. Thus the theorem also holds for regions enclosed by several surfaces.

Here on the theorem is completely established.

*See footnote p. 37.

Example 9.42. Verify Divergence theorem for $\mathbf{F} = (x^2 - yz)\mathbf{i} + (y^2 - zx)\mathbf{j} + (z^2 - xy)\mathbf{k}$ taken over the rectangular parallelopiped $0 \le x \le a$, $0 \le y \le b$, $0 \le z \le c$. (Rohtak 2006 S; Madras 2005 S)

Solution. We have $\operatorname{div} \mathbf{F} = \frac{\partial}{\partial x}(x^2 - yz) + \frac{\partial}{\partial y}(y^2 - zx) + \frac{\partial}{\partial z}(z^2 - xy)$

$= 2(x + y + z)$

$\therefore \quad \int_V \operatorname{div} \mathbf{F}\, dV = 2\int_0^a \int_0^b \int_0^c (x + y + z)\, dx\, dy\, dz$

$= 2\int_0^a dx \int_0^b dy \left[\frac{c^2}{2} + (x + y)c\right]$

$= 2\int_0^a dx \left[\frac{c^2}{2}b + \frac{cb^2}{2} + (x)bc\right]$

$= 2\left[\frac{c^2b}{2}a + \frac{cb^2}{2}a + \frac{b^2}{2}a^2\right]$

$= abc(c + b + a)$ ...(1)

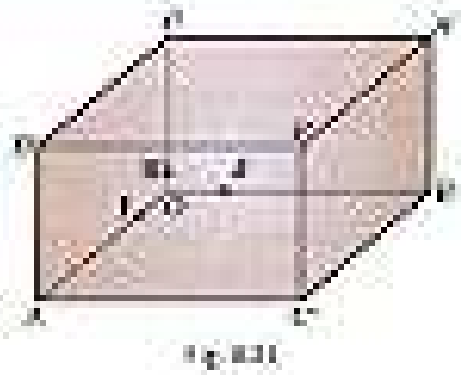

Fig. 9.31

Also $\int_S \mathbf{F} \cdot \mathbf{N}\, dS = \int_{S_1} \mathbf{F} \cdot \mathbf{N}\, dS + \int_{S_2} \mathbf{F} \cdot \mathbf{N}\, dS + \ldots + \int_{S_6} \mathbf{F} \cdot \mathbf{N}\, dS$

where S_1 is the face $OAC'B$, S_2 the face $CFPC'$, S_3 the face $OBFC$, S_4 the face $AC'PA'$, S_5 the face $BFPA$ and S_6 the face $OAPC$ (Fig. 9.31).

Now $\int_{S_1} \mathbf{F} \cdot \mathbf{N}\, dS = \int_{S_1} \mathbf{F} \cdot (-\mathbf{k})\, dx\, dy = -\int_0^a \int_0^b (0 - xy)\, dx\, dy = \frac{a^2b^2}{4}$

$\int_{S_2} \mathbf{F} \cdot \mathbf{N}\, dS = \int_{S_2} \mathbf{F} \cdot \mathbf{k}\, dx\, dy = \int_0^a \int_0^b (c^2 - xy)\, dx\, dy = abc^2 - \frac{a^2b^2}{4}$

Similarly, $\int_{S_3} \mathbf{F} \cdot \mathbf{N}\, dS = \frac{a^2b^2}{4}$, $\int_{S_4} \mathbf{F} \cdot \mathbf{N}\, dS = ab^2c - \frac{a^2b^2}{4}$,

$\int_{S_5} \mathbf{F} \cdot \mathbf{N}\, dS = \frac{a^2b^2}{4}$ and $\int_{S_6} \mathbf{F} \cdot \mathbf{N}\, dS = abc^2 - \frac{a^2b^2}{4}$

Thus $\int_S \mathbf{F} \cdot \mathbf{N}\, dS = abc(c + b + a)$ ...(2)

Hence the theorem is verified from the equality of (1) and (2).

Example 9.43. Evaluate $\int_S \mathbf{F} \cdot \mathbf{n}\, ds$, where $\mathbf{F} = 4x\mathbf{i} - 2y^2\mathbf{j} + z^2\mathbf{k}$ and S is the surface bounding the region $x^2 + y^2 = 4$, $x = 0$, $y = 0$, $z = 0$, $z = 3$. (U.P.T.U. 2007 S; Nasik 2006; A.M.I.E. 2001)

Solution. By divergence theorem,

$\int_S \mathbf{F} \cdot \mathbf{n}\, ds = \int_V \operatorname{div} \mathbf{F}\, dv$

$= \int_V \left[\frac{\partial}{\partial x}(4x) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2)\right] dv$

$= \iiint_V (4 - 4y + 2z)\, dx\, dy\, dz$

$= \int_0^2 \int_0^{\sqrt{(4-x^2)}} \int_0^3 (4 - 4y + 2z)\, dx\, dy\, dz$

$= \int_0^2 \int_0^{\sqrt{(4-x^2)}} \left[4z - 4yz + z^2\right]_0^3 dx\, dy$

$= \int_0^2 \int_0^{\sqrt{(4-x^2)}} (21 - 12y)\, dx\, dy$

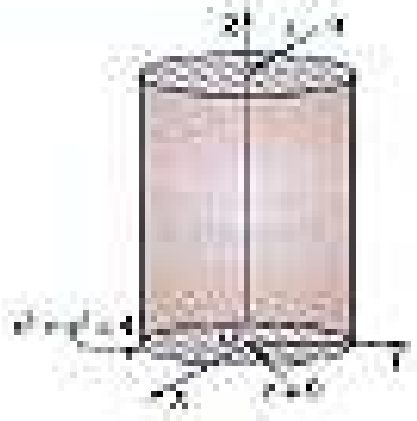

Fig. 9.32

$$= \int_0^a \left[21y - 8y^2\right]_{y=x^2}^{y=x} dx$$

$$= 42\int_0^1 \sqrt{x}(1-x^2)\,dx - 64\int_0^1 x^2(1-x^2)\,dx = 64\left[\frac{x\sqrt{x}(x^2+1)}{7} + \frac{1}{3}\tan^{-1}x\right]_0^1 - 8\pi.$$

Example 3.41. Evaluate $\int_S (yz\,\mathbf{i} + zx\,\mathbf{j} + xy\,\mathbf{k})\cdot d\mathbf{S}$, **where** S **is the surface of the sphere** $x^2 + y^2 + z^2 = a^2$ **in the first octant.** (U.P.T.U., 2003)

Solution. The surface of the region V OABC is given to be bounded by (Fig. 3.23) and is composed of four surfaces (i) S_1 – circular quadrant OBC in the yz-plane,

(ii) S_2 – circular quadrant OCA in the zx-plane,

(iii) S_3 – circular quadrant OAB in the xy-plane,

and (iv) S – surface ABC of the sphere in the first octant.

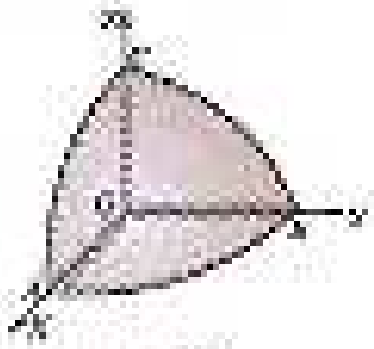

Fig. 3.23

Also $\mathbf{F} = yz\,\mathbf{i} + zx\,\mathbf{j} + xy\,\mathbf{k}$.

By Divergence theorem,

$$\int_V \text{div}\,\mathbf{F}\,dv = \int_{S_1} \mathbf{F}\cdot d\mathbf{S} + \int_{S_2} \mathbf{F}\cdot d\mathbf{S} + \int_{S_3} \mathbf{F}\cdot d\mathbf{S} + \int_S \mathbf{F}\cdot d\mathbf{S} \quad ...(1)$$

Now $\text{div}\,\mathbf{F} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(zx) + \frac{\partial}{\partial z}(xy) = 0$.

For the surface S_1, $x = 0$

$$\therefore \int_{S_1} \mathbf{F}\cdot d\mathbf{S} = \int_0^a \int_0^{\sqrt{a^2-z^2}} (yz\,\mathbf{i} + 0\,\mathbf{j} + 0\,\mathbf{k})\cdot(-dy\,dz\,\mathbf{i}) = -\int_0^a \int_0^{\sqrt{a^2-z^2}} yz\,dy\,dz = -\frac{a^4}{8}$$

Then (1) becomes $0 = -\frac{a^4}{8} + \int_S \mathbf{F}\cdot d\mathbf{S}$ or $\int_S \mathbf{F}\cdot d\mathbf{S} = \frac{a^4}{8}$.

Example 3.42. Apply divergence theorem to evaluate $\int (ax^2\,\mathbf{i} + by^2\,\mathbf{j} + cz^2\,\mathbf{k})\cdot d\mathbf{S}$ **where** S **is the surface of the sphere** $x^2 + y^2 + z^2 = a^2$ **in the first octant.**

Solution. The parametric equations of the sphere are $x = a\sin\theta\cos\phi$, $y = b\sin\theta\sin\phi$, $z = c\cos\theta$ (where θ and ϕ denote the spherical coordinates of the point (x, y, z) on the surface of the sphere).

$$\therefore \int_S (ax^2 + by^2 + cz^2)\,dS = \int_S (ax^2\,\mathbf{i} + by^2\,\mathbf{j} + cz^2\,\mathbf{k})\cdot\mathbf{n}\,dS$$

$$= \int_V \text{div}(ax^2\,\mathbf{i} + by^2\,\mathbf{j} + cz^2\,\mathbf{k})\,dv = 2\int_V (ax + by + cz)\,dv$$

$$= 2\int_0^{\pi/2}\int_0^{\pi/2}\int_0^a (ar\sin\theta\cos\phi + br\sin\theta\sin\phi + cr\cos\theta)\,r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= 2\left(a + b + c\right)\frac{a^4}{4}\left[-\cos\theta\right]_0^{\pi/2}\left[\sin\phi\right]_0^{\pi/2} = \frac{\pi a^4}{4}(a + b + c).$$

Example 3.43. Evaluate $\int_S (x^3 - yz)\,\mathbf{i} - 2x^2y\,\mathbf{j} + 2\,\mathbf{k})\cdot d\mathbf{S}$ **where** S **is the surface of the ellipsoid** $ax^2 + by^2 + cz^2 = 1$.

Solution. Taking $\phi = ax^2 + by^2 + cz^2 - 1 = 0$, $\nabla\phi = 2ax\,\mathbf{i} + 2by\,\mathbf{j} + 2cz\,\mathbf{k}$

$$\therefore \text{Unit vector normal to the ellipsoid} = \mathbf{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{ax\,\mathbf{i} + by\,\mathbf{j} + cz\,\mathbf{k}}{\sqrt{a^2x^2 + b^2y^2 + c^2z^2}}$$

Now $\mathbf{F}\cdot\mathbf{n} = ax^4 + by^3 + cz^3 - yz(by)$ $\therefore \mathbf{F} = ax\,\mathbf{i} + by\,\mathbf{j} + cz\,\mathbf{k}$ in V

Obviously $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ $\quad \because x^2 + y^2 + z^2 = 1$

∴ By Divergence theorem,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V \operatorname{div} \mathbf{F}\, dv = \iiint_V \left[\frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) \right] dv = 3 \iiint_V dv = 3V$$

$$= 3 \cdot \frac{4\pi}{3} \cdot \frac{1}{\sqrt{(a^2+b^2+c^2)}} = \frac{4\pi}{\sqrt{(a^2+b^2+c^2)}}. \qquad \left[\because \text{Vol. of ellipsoid} = \frac{4\pi}{3} \frac{1}{\sqrt{(a^2+b^2+c^2)}} \right]$$

Example 6.45. *If the position vector of any point be $\mathbf{r}$ and r its modulus, find the value of $\iint_S \frac{\mathbf{r}}{r^3} \cdot \mathbf{N}\, dS$ where S is a closed surface not passing through the origin. Discuss the case when S does contain the origin.*

$$\iint_S \frac{\mathbf{r}}{r^3} \cdot \mathbf{N}\, dS = \begin{cases} 0, & \text{if } O \text{ lies outside } S \\ 4\pi, & \text{if } O \text{ lies inside } S \end{cases}$$

Solution. (a) *When O is outside S.* Here $\mathbf{F} = \frac{\mathbf{r}}{r^3}$ is continuously differentiable throughout the volume V enclosed by S. Hence by Divergence theorem, we have

$$\iint_S \frac{\mathbf{r}}{r^3} \cdot \mathbf{N}\, dS = \iiint_V \operatorname{div}\left(\frac{\mathbf{r}}{r^3}\right) dV = 0 \qquad \left[\because \operatorname{div}\left(\frac{\mathbf{r}}{r^3}\right) = 0 \right]$$

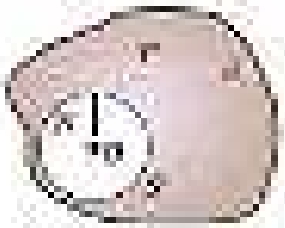

Fig. 6.16

(b) *When O is inside S.* Here $\mathbf{F} = \frac{\mathbf{r}}{r^3}$ has a point of discontinuity at O and as such Divergence theorem cannot be applied to the region V enclosed by S. To remove this point of discontinuity, we enclose O by a small sphere S' of radius ρ.

Now, $\mathbf{F}$ is continuously differentiable throughout the region V' enclosed between S and S'. Therefore applying Divergence theorem to region V', we get

$$\iint_S \frac{\mathbf{R}}{\rho^3} \cdot \mathbf{N}\, dS + \iint_{S'} \frac{\mathbf{R}}{\rho^3} \cdot \mathbf{N}\, dS' = \iiint_{V'} \operatorname{div}\left(\frac{\mathbf{R}}{\rho^3}\right) dV' = 0 \qquad \left[\because \operatorname{div}\left(\frac{\mathbf{R}}{\rho^3}\right) = 0 \right]$$

$$\therefore \qquad \iint_S \frac{\mathbf{R}}{\rho^3} \cdot \mathbf{N}\, dS = -\iint_{S'} \frac{\mathbf{R}}{\rho^3} \cdot \mathbf{N}\, dS' \qquad \ldots(1)$$

Now the outward normal $\mathbf{N}$ on the sphere S' is directed towards the centre O. Therefore $\mathbf{N} = -\frac{\mathbf{R}}{\rho}$ on S' (Fig. 6.16).

$$\therefore \qquad -\iint_{S'} \frac{\mathbf{R}}{\rho^3} \cdot \mathbf{N}\, dS' = -\iint_{S'} \frac{\mathbf{R}}{\rho^3} \cdot \left(-\frac{\mathbf{R}}{\rho}\right) dS' \qquad [\because \text{on } S',\ r = \rho]$$

$$= \iint_{S'} \frac{\rho^2}{\rho^4}\, dS' = \iint_{S'} \frac{1}{\rho^2}\, dS' = \frac{1}{\rho^2} \iint_{S'} dS' = \frac{1}{\rho^2} \cdot 4\pi\rho^2 = 4\pi$$

Hence from (1), $\qquad \iint_S \frac{\mathbf{R}}{\rho^3} \cdot \mathbf{N}\, dS = 4\pi.$

6.17 GREEN'S THEOREM*

If ϕ and ψ are scalar point functions possessing continuous derivatives of first and second orders, then

$$\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi)\, dv = \iint_S \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) dS \qquad \ldots(1)$$

where $\frac{\partial}{\partial n}$ denotes differentiation in the direction of the outward normal to the bounding surface S enclosing the region R.

Applying Divergence theorem, $\iint_S \mathbf{F} \cdot \mathbf{N}\, dS = \iiint_V \nabla \cdot \mathbf{F}\, dv$ to the function $\phi \nabla \psi$, we get

$$\iint_S \phi \nabla \psi \cdot \mathbf{N}\, dS = \iiint_V \nabla \cdot (\phi \nabla \psi)\, dv = \iiint_V (\nabla \phi \cdot \nabla \psi + \phi \nabla^2 \psi)\, dv \qquad [\text{By (1) page 228}]$$

$$= \iiint_V \nabla \phi \cdot \nabla \psi\, dv + \iiint_V \phi \nabla^2 \psi\, dv \qquad \ldots(2)$$

*See footnote on page 203.

Interchanging ϕ and ψ, (1) gives

$$\int_S \psi \nabla\phi \cdot \hat{n}\, dS = \int_V \psi \nabla^2\phi\, dV + \int_V \nabla\psi \cdot \nabla\phi\, dV \quad \ldots(2)$$

Subtracting (2) from (1), we have $\int_S (\phi\nabla\psi - \psi\nabla\phi)\cdot\hat{n}\, dS = \int_V (\phi\nabla^2\psi - \psi\nabla^2\phi)\, dV$

But $\nabla\psi \cdot \hat{N} = \partial\psi/\partial n$ the directional derivative of ψ along the external normal at any point of S. Hence

$$\int_S \left(\phi\frac{\partial\psi}{\partial n} - \psi\frac{\partial\phi}{\partial n}\right) dS = \int_V (\phi\nabla^2\psi - \psi\nabla^2\phi)\, dV$$ which is the required result (1).

Note. Harmonic Function. A scalar point function ϕ satisfying the Laplace's equation $\nabla^2\phi = 0$ at every point of a region R, is called a harmonic function in R.

If ϕ and ψ be both harmonic functions in R, (1) gives

$$\int_S \phi\frac{\partial\psi}{\partial n}\, dS = \int_S \psi\frac{\partial\phi}{\partial n}\, dS$$ which is known as **Green's reciprocal theorem.**

PROBLEMS 3.13

1. Verify divergence theorem for $\vec{F}$ taken over the cube bounded by $x = 0, y = 1, y = 0, y = 1, z = 0, z = 1$ where
   (a) $\vec{F} = 4xi - 2y^2j + z^2k$ (Madras, 2005) (b) $\vec{F} = xi + yj + zk$ (Nagpur, 2006)
2. Verify Gauss divergence theorem for the function $\vec{F} = yi + zj + xk$ over the unit sphere bounded by $x^2 + y^2 + z^2 = 1$.
3. Using divergence theorem, prove that
   (a) $\int_S \vec{R}\cdot\hat{N}\, dS = 3V$ (b) $\int_S \nabla r^2 \cdot \hat{n}\, dS = 4\pi$ (U.P.T.U., 2005)
   where S is any closed surface enclosing a volume V and $r^2 = x^2 + y^2 + z^2$.
4. Using divergence theorem, evaluate $\int_S \vec{R}\cdot\hat{N}\, dS$, where S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.
5. If S is any closed surface enclosing a volume V and $\vec{F} = ai + bj + ck$, prove that
   $$\int_S \vec{F}\cdot\hat{N}\, dS = (a + b + c)V$$ (Madras, 2005)
6. For any closed surface S, prove that $\int_S (ax + by + cz)\, dS = (a + b + c)V$.
7. Use divergence theorem to evaluate $\int_S \vec{F}\cdot\hat{N}\, dS$, where
   (a) $\vec{F} = x^2i + y^2j + z^2k$ and S is the surface of the cylinder $x^2 + y^2 = a^2$, $z = 0$, $z = b$ (U.P.T.U., 2004; U.P.T.U., 2005)
   (b) $\vec{F} = (x^3 - yz)i - 2x^2yj + 2k$ and S is the surface of the cube $|x| = 1, |y| = 1, |z| = 1$. (M.D.U., 2003)
8. Evaluate $\iint_S (ax^2i + by^2j + cz^2k)\cdot\hat{n}\, dS$, over the surface of a sphere of radius r. (Kurukshetra, 2004)
9. Evaluate $\int_S \vec{F}\cdot\hat{N}\, dS$, where $\vec{F} = y^2i + x^2j + z^2k$ and S is the upper part of the surface of $x^2 + y^2 + z^2 = a^2$ above XOY plane.
10. By transforming to triple integral, evaluate $\iint_S (x^3\, dy\, dz + y^3\, dz\, dx + z^3\, dx\, dy)$, where S is the closed surface consisting of the cylinder $x^2 + y^2 = a^2$ and the circular discs $z = 0$ and $z = h$. (Kurukshetra, 2005)
11. Evaluate $\int_S (\nabla \times \vec{F})\cdot\hat{n}\, dS$, where S is the surface of the paraboloid $x^2 + y^2 + z = 1$, $z > 0$ and $\vec{F} = (x^2 + y - 4)i + 3xyj + (2xz + z^2)k$.
12. If $\vec{F} = (2x^2 - 3z)i - 2xyj - 4xk$, then evaluate $\iiint_V \nabla\cdot\vec{F}\, dv$, where V is bounded by $x = 0, x = 1, y = 0, y = 1, z = 0$ and $z = 2$. (Nagpur, 2006)
13. If $\vec{F} = grad\,\phi$ and $\nabla^2\phi = 0$, prove that $\int_S \vec{F}\cdot\hat{N}\, dS = 4\pi\int_V \rho\, dV$, where ρ is a scalar function of coordinates.

9.16 (I) IRROTATIONAL FIELDS

An irrotational field $\mathbf{F}$ is characterized by any one of the following equivalent properties:

(a) $\nabla \times \mathbf{F} = \mathbf{0}$. (b) Circulation $\int \mathbf{F} \cdot d\mathbf{R}$ along every closed curve is zero.

(c) $\mathbf{F} = \nabla\phi$, if the domain is simply connected.*

If $\nabla \times \mathbf{F} = \mathbf{0}$, then by Stoke's theorem,

$$\int_C \mathbf{F} \cdot d\mathbf{R} = \int_S \nabla \times \mathbf{F} \cdot d\mathbf{S} = 0, \text{ i.e., the circulation along every closed curve is zero.}$$

Again since $\nabla \times \nabla\phi = 0$

$\therefore$ In an irrotational field for which $\nabla \times \mathbf{F} = \mathbf{0}$, the vector $\mathbf{F}$ can always be expressed as the gradient of a scalar function ϕ provided the domain is simply connected. That is

$$\mathbf{F} = \nabla\phi.$$

And ϕ is a scalar function ϕ is called the potential. In a rotational field, $\mathbf{F}$ cannot be expressed as the gradient of a scalar potential.

Fig. 9.15

Obs. 1 In an irrotational field, the line integral $\int \mathbf{F} \cdot d\mathbf{R}$ between two points is independent of the path of integration and is equal to the potential difference between those points.

If C and C_1 are any closed curves in an irrotational field $\mathbf{F}$ (Fig. 9.15), then

$$\int_{ABCA} \mathbf{F} \cdot d\mathbf{R} = \int_{ABC_1A} \mathbf{F} \cdot d\mathbf{R} - \int_{AC_1A} \mathbf{F} \cdot d\mathbf{R} = 0$$

or $$\int_{ABC} \mathbf{F} \cdot d\mathbf{R} = \int_{AC_1A} \mathbf{F} \cdot d\mathbf{R}$$

i.e., the value of the line integral is independent of the path joining A and B.

Further, substituting $\mathbf{F} = \nabla\phi$, we have

$$\int_C \mathbf{F} \cdot d\mathbf{R} = \int_C \nabla\phi \cdot d\mathbf{R} = \int_C \left(i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z} \right) \cdot (i\,dx + j\,dy + k\,dz)$$

$$= \int_C \left(\frac{\partial\phi}{\partial x}dx + \frac{\partial\phi}{\partial y}dy + \frac{\partial\phi}{\partial z}dz \right) = \int_C d\phi = \phi_2 - \phi_1$$

Obs. 2 If $\mathbf{F}$ is a force acting on a particle, then $\int_C \mathbf{F} \cdot d\mathbf{R}$ represents the work done in moving the particle around a closed path. [Ans. p. 309]

When $\int_C \mathbf{F} \cdot d\mathbf{R} \neq 0$, the field is said to be non-conservative, i.e., net work is done in displacement of the particle from one point to the field and back to it, and the mechanical energy is converted.

Thus a non-conservative field is rotational.

Obs. 3 The well-known equations of Mr. Poisson and Laplace have good for more complicated fields.

If given $\nabla \cdot \mathbf{F} = \rho(x, y, z)$. Then $\nabla \cdot (\nabla\phi) = \rho(x, y, z)$, i.e., $\nabla^2\phi = \rho(x, y, z)$ ...(1)

which is known as Poisson's equation. Its solution for the electrostatic fields enable us to determine the potential ϕ as a function of the charge distribution $\rho(x, y, z)$.

If $\rho(x, y, z) = 0$ then it reduces to $\nabla^2\phi = 0$ which is the Laplace's equation. The solutions of this equation are of great importance in modern engineering and physics, some of which will be discussed briefly in § 10.31 and 10.32.

(II) Solenoidal Fields. A solenoidal field $\mathbf{F}$ is characterized by any one of the following conditions:

(a) $\nabla \cdot \mathbf{F} = 0$. (b) flux $\int \mathbf{F} \cdot \mathbf{N}\,ds$ across every closed surface is zero. (c) $\mathbf{F} = \nabla \times \mathbf{V}$.

If $\nabla \cdot \mathbf{F} = 0$ then by the Divergence theorem,

$$\int \mathbf{F} \cdot \mathbf{N}\,ds = \int \nabla \cdot \mathbf{F}\,dv = 0, \text{ i.e., the flux across every closed surface is zero.}$$

Again since $\nabla \cdot \nabla \times \mathbf{V} = 0$.

$\therefore$ In a solenoidal field for which $\nabla \cdot \mathbf{F} = 0$, the vector $\mathbf{F}$ can always be expressed as the curl of a vector function $\mathbf{V}$, that is $\mathbf{F} = \nabla \times \mathbf{V}$.

*A domain D is said to be simply connected if every closed curve in D can be shrunk to any point within D.

Example 3.48. *A vector field is given by* $\vec{F} = (x^2 - y^2 + x)\vec{i} - (2xy + y)\vec{j}$. *Show that the field is irrotational and find its scalar potential. Hence, evaluate the line integral from* $(1, 2)$ *to* $(0, 1)$.

Solution. Since $\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y^2 + x & -(2xy + y) & 0 \end{vmatrix} = \vec{0}$

$\therefore$ this field is irrotational and the vector $\vec{F}$ can be expressed as the gradient of a scalar potential.

i.e., $(x^2 - y^2 + x)\vec{i} - (2xy + y)\vec{j} = \vec{\nabla}\phi = \frac{\partial \phi}{\partial x}\vec{i} + \frac{\partial \phi}{\partial y}\vec{j}$

where $\frac{\partial \phi}{\partial x} = x^2 - y^2 + x$ ...(1)

$\frac{\partial \phi}{\partial y} = -(2xy + y)$ ...(2)

Integrating (1) w.r.t. x, keeping y constant, we get $\phi = \frac{x^3}{3} - y^2 x + \frac{x^2}{2} + f(y)$ ...(3)

Similarly integrating (2) w.r.t. y, keeping x constant, we obtain $\phi = -y^2 - \frac{y^2}{2} + g(x)$ ...(4)

Equating (3) and (4), we get $\frac{x^3}{3} - y^2 x + \frac{x^2}{2} + f(y) = -y^2 - \frac{y^2}{2} + g(x)$

$\therefore \quad f(y) = -\frac{y^2}{2}$ and $g(x) = \frac{x^3}{3} + \frac{x^2}{2}$

Hence $\phi = \frac{x^3}{3} - xy^2 + \frac{x^2}{2} - \frac{y^2}{2}$

Since the field is irrotational,

$\therefore \quad \int \vec{F} \cdot d\vec{R}$ from $(1, 2)$ to $(0, 1) = \phi_{(0,1)} - \phi_{(1,2)} = \left(\frac{1}{3} - 1 \times 1 + \frac{1}{2} - \frac{1}{2}\right) - \left(\frac{8}{3} - 2 \times 1 + \frac{4}{2} - \frac{4}{2}\right) = -2\frac{1}{3}$.

Example 3.49. *A fluid motion is given by* $\vec{V} = (y + z)\vec{i} + (z + x)\vec{j} + (x + y)\vec{k}$.
(a) Is this motion irrotational? If so, find the velocity potential. (U.P.T.U., 2006)
(b) Is the motion possible for an incompressible fluid?

Solution. We know $\nabla \times \vec{V} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y + z & z + x & x + y \end{vmatrix} = \vec{i}(1 - 1) - \vec{j}(1 - 1) + \vec{k}(1 - 1) = \vec{0}$

$\therefore$ this motion is irrotational and it is the velocity potential then $\vec{V} = \nabla\phi$. (U.P.T.U.)

i.e., $(y + z)\vec{i} + (z + x)\vec{j} + (x + y)\vec{k} = \frac{\partial \phi}{\partial x}\vec{i} + \frac{\partial \phi}{\partial y}\vec{j} + \frac{\partial \phi}{\partial z}\vec{k}$

$\therefore \quad \frac{\partial \phi}{\partial x} = y + z, \frac{\partial \phi}{\partial y} = z + x, \frac{\partial \phi}{\partial z} = x + y$

Integrating these, we get $\phi = xy + zx + f_1(y, z)$ ...(1)

$\phi = yz + xy + f_2(z, x)$ ...(2)

and $\phi = zx + yz + f_3(x, y)$ ...(3)

Equality of (1), (2) and (3) requires that

$f_1(y, z) = yz$, $f_2(z, x) = zx$, $f_3(x, y) = xy$.

Hence $\phi = yz + zx + xy$.

(b) The fluid motion is possible if $\vec{V}$ satisfies the equation of continuity, which for an incompressible fluid is $\nabla \cdot \vec{V} = 0$. (See I.K.T.U.)

Here $$\nabla \cdot \vec{V} = \frac{\partial}{\partial x}(y+z) + \frac{\partial}{\partial y}(z+x) + \frac{\partial}{\partial z}(x+y) = 0$$

Hence, the fluid motion is possible.

Example 3.22. Find whether $\int_C [(2xy+z^3)dx + (x^2+\cos yz)dy + (3xz^2+y\cos yz)dz]$ is independent of the path joining $(0, 0, 1)$ and $(1, \pi/2, 1)$. Also evaluate this line integral.

Solution. The line integral of $\vec{F}$ is independent of path of integration if $\nabla \times \vec{F} = 0$.

$$= \int_C [(2xy^2+z^3)\vec{i} + (x^2z^2 + z\cos yz)\vec{j} + (2x^2yz + y\cos yz)\vec{k}] \cdot (dx\,\vec{i} + dy\,\vec{j} + dz\,\vec{k}) = \int_C \vec{F} \cdot d\vec{R}$$

and
$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz^2 & x^2z^2 + z\cos yz & 2x^2yz + y\cos yz \end{vmatrix}$$

$$= \vec{i}(2x^2z + \cos yz - yz\sin yz - 2x^2z + \cos yz - yz\sin yz)$$

$$= \vec{i}(0) - \vec{j}(4xyz - 4xyz) + \vec{k}(2xz^2 - 2xz^2) = 0$$

$\therefore$ the given integral is independent of the path C.

Now let $\vec{F} = \nabla\phi$

i.e., $$(2xy^2\vec{i} + (x^2z^2 + z\cos yz)\vec{j} + (2x^2yz + y\cos yz)\vec{k} = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$$

$$\Rightarrow 2xy^2 = \frac{\partial\phi}{\partial x},\ x^2z^2 + z\cos yz = \frac{\partial\phi}{\partial y},\ 2x^2yz + y\cos yz = \frac{\partial\phi}{\partial z}$$

Integrating first w.r.t. x partially, we get

$$\phi = x^2y^2z^2 + \Psi_1(y, z) \quad \ldots(1)$$

Integrating second w.r.t. y partially, we get

$$\phi = x^2y^2z^2 + \sin yz + \Psi_2(x, z) \quad \ldots(2)$$

Integrating third w.r.t. z partially, we get

$$\phi = x^2y^2z^2 + \sin yz + \Psi_3(x, y) \quad \ldots(3)$$

Comparing (1), (2), (3), we have

$\Psi_1(y, z)$ = terms in ϕ independent of $x = \sin yz$

$\Psi_2(x, z)$ = terms in ϕ independent of $y = 0$

$\Psi_3(x, y)$ = terms in ϕ independent of $z = 0$

Thus $$\phi = x^2y^2z^2 + \sin yz$$

Hence the value of the given integral $= [\phi]_{(0,0,1)}^{(1,\pi/2,1)}$

$$= (0 + 0) - (0 + \sin 0) = \sin\frac{\pi}{2} = 1.$$

Example 3.23. Determine constant a if $\vec{F} = (y^2\sin z + x^2)\vec{i} + (2xy\sin z - 4)\vec{j} + (3xy^2\cos z + 2az)\vec{k}$ is a conservative vector field. Find its scalar potential. Also compute the work done in moving the particle from $(0, 1, -2)$ to $(2, -1, 1)$. (Mumbai, 2006)

Solution. $\vec{F}$ is a conservative vector field when $\operatorname{curl}\vec{F} = 0$. Here

$$\operatorname{Curl}\vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2\sin z + x^2 & 2xy\sin z - 4 & 3xy^2\cos z + 2z \end{vmatrix}$$

$$= \vec{i}(0 - 0) - \vec{j}(3y^2\cos z - 3y^2\cos z) + \vec{k}(2y\sin z - 2y\sin z) = 0$$

$\therefore$ $\vec{F}$ is a conservative field.

Now let $\vec{F} = \nabla\phi$

i.e., $(y^2\cos x + z^3)\vec{i} + (2y\sin x - 4)\vec{j} + (3xz^2 + 2)\vec{k} = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$

$\therefore$ $y^2\cos x + z^3 = \frac{\partial\phi}{\partial x}$, $2y\sin x - 4 = \frac{\partial\phi}{\partial y}$, $3xz^2 + 2 = \frac{\partial\phi}{\partial z}$

Integrating first w.r.t. x partially, we get

$$\phi = y^2\sin x + xz^3 + \psi_1(y, z) \quad \ldots(i)$$

Integrating second w.r.t. y partially, we get

$$\phi = y^2\sin x - 4y + \psi_2(z, x) \quad \ldots(ii)$$

Integrating third w.r.t. z partially, we obtain

$$\phi = xz^3 + 2z + \psi_3(x, y) \quad \ldots(iii)$$

Comparing (i), (ii), (iii), we get

$\psi_1(y, z)$ = terms in ϕ independent of $y = -4y + 2z$

$\psi_2(z, x)$ = terms in ϕ independent of $z = xz^3 + 2z$

$\psi_3(x, y)$ = terms in ϕ independent of $x = y^2\sin x - 4y$

Thus $\phi = xz^3 + y^2\sin x - 4y + 2z$

Now work done $= \phi_B - \phi_A$

$$= \phi\left(\frac{\pi}{2}, -1, 2\right) - \phi(0, 1, -1)$$

$$= \left(\frac{\pi}{2}\right)(2)^3 + (-1)^2\sin\frac{\pi}{2} - 4(-1) + 2(2) - [0 + 1 - 4 - 2] = 4\pi + 18.$$

PROBLEMS 6.1

1. If $\vec{r}$ is a solution of the Laplace equation, prove that $\nabla\phi(r)$ is also solution of Laplace equation.
2. Show that the vector field defined by $\vec{F} = (2xy + z^3)\vec{i} + (x^2 + z)\vec{j} + (3xz^2 + y)\vec{k}$ is irrotational and find the scalar potential. Hence evaluate $\int_{(1,-2,1)}^{(3,1,4)} \vec{F}\cdot d\vec{r}$ from $(1, -2, 1)$ to $(3, 1, 4)$.
3. Find the work done by the vector field $\vec{F} = 2y\vec{i} + z\vec{j} + x\vec{k}$ in moving a particle once around the circle C in the plane $z = 3$ whose projection in the xy-plane is the circle $x^2 + y^2 = 4$.
4. Show that the vector field given by $\vec{A} = (y^2\cos x + z^3)\vec{i} + (2y\sin x - 4)\vec{j} + (3xz^2 + 2)\vec{k}$ is irrotational hence conservative. Find the scalar potential such that $\vec{A} = \nabla\phi$.
5. Show that the following vectors are irrotational. Find the scalar potential in each case.
   (i) $\vec{F} = (x^2 + xy^2)\vec{i} + (y^2 + x^2y)\vec{j}$
   (ii) $\vec{F} = y^2\vec{i} + (2xy + e^{3z})\vec{j} + 3ye^{3z}\vec{k}$ (Ans. $\phi = \frac{1}{3}x^3 + xy^2 + \frac{1}{3}e^{3z} + c$)
   (iii) $\vec{F} = (\sin y + z)\vec{i} + (x\cos y - z)\vec{j} + (x - y)\vec{k}$ (Ans. $\phi = x\sin y + xz - yz + c$)
   (iv) $\vec{F} = (2xy + z^3)\vec{i} + x^2\vec{j} + 3xz^2\vec{k}$ (Ans. $\phi = x^2y + xz^3 + c$)
   (v) $\vec{F} = (y^2\cos x + z^3)\vec{i} + (2y\sin x - 4)\vec{j} + (3xz^2 + 2)\vec{k}$ (Ans. $\phi = y^2\sin x - 4y + xz^3 + 2z + c$)
6. Fluid motion is given by $\vec{V} = \omega x\vec{i} + \omega y\vec{j} - 2\omega z\vec{k}$.
   (i) Is the possible fluid motion irrotational? If so, find ϕ.
   (ii) Is the motion possible for an incompressible fluid?
7. Show that the vector field defined by $\vec{F} = (y\sin z - \sin x)\vec{i} + (x\sin z + 2yz)\vec{j} + (xy\cos z + y^2)\vec{k}$ is irrotational and find its scalar potential. (Kerala, 2005)
8. Show that $\vec{F} = (2xy + z^3)\vec{i} + x^2\vec{j} + 3xz^2\vec{k}$ is a conservative vector field and find a function ϕ such that $\vec{F} = \nabla\phi$. Also find the work done in moving an object from $(1, -2, 1)$ to $(3, 1, 4)$. (Nagpur, 2005)
9. If $\vec{F} = (x + y + az)\vec{i} + (bx - 2y - z)\vec{j} + (x + cy + 2z)\vec{k}$, find a, b, c such that curl $\vec{F} = \vec{0}$. Then find ϕ such that $\vec{F} = \nabla\phi$. (U.P.T.U., 2006)
10. Find the constant a so that $\vec{V}$ is a solenoidal vector field, where
    $\vec{V} = (x + 3y)\vec{i} + (y - 2z)\vec{j} + (x + az)\vec{k}$.
    Calculate its scalar potential and work done in moving a particle from $(1, 2, -1)$ to $(4, -4, 2)$ in this field.
    (Mumbai, 2005; Rajasthan, 2006)

6.5C [3] ORTHOGONAL CURVILINEAR COORDINATES

Let the rectangular coordinates (x, y, z) of any point be expressed as functions of (u, v, w) so that

$$x = x(u, v, w),\ y = y(u, v, w),\ z = z(u, v, w) \qquad \ldots(1)$$

Suppose that (1) can be solved for u, v, w in terms of x, y, z to get

$$u = u(x, y, z),\ v = v(x, y, z),\ w = w(x, y, z) \qquad \ldots(2)$$

We assume that the functions in (1) and (2) are single-valued and have continuous partial derivatives so that the correspondence between (x, y, z) and (u, v, w) is unique. Then (u, v, w) are called curvilinear coordinates of (x, y, z).

Each of (u, v, w) has a level surface through an arbitrary point. The surfaces $u = u_1, v = v_1, w = w_1$ are called coordinate surfaces through $P(u_1, v_1, w_1)$. Each pair of these coordinate surfaces intersect in curves called coordinate curves. The curve of intersection of $v = v_1$ and $w = w_1$ will be called the u-curve, for only u changes along it; in same fashion we define v and w-curves.

In vector notation, (1) can be written as $\mathbf{R} = x(u, v, w)\mathbf{i} + y(u, v, w)\mathbf{j} + z(u, v, w)\mathbf{k}$

$$\therefore \qquad d\mathbf{R} = \frac{\partial \mathbf{R}}{\partial u}du + \frac{\partial \mathbf{R}}{\partial v}dv + \frac{\partial \mathbf{R}}{\partial w}dw \qquad \ldots(3)$$

Then $\partial \mathbf{R}/\partial u$ is a tangent vector to the u-curve at P. If $\mathbf{T}_u$ is a unit vector at P in this direction, then $\partial \mathbf{R}/\partial u = h_1\mathbf{T}_u$ where $h_1 = |\partial \mathbf{R}/\partial u|$.

Similarly if $\mathbf{T}_v$ and $\mathbf{T}_w$ be unit tangent vectors to v- and w-curves at P, then

$$\frac{\partial \mathbf{R}}{\partial v} = h_2\mathbf{T}_v \text{ and } \frac{\partial \mathbf{R}}{\partial w} = h_3\mathbf{T}_w$$

where $h_2 = |\partial \mathbf{R}/\partial v|$ and $h_3 = |\partial \mathbf{R}/\partial w|$. h_1, h_2, h_3 are called scalar factors.

Then (3) can be written as

$$d\mathbf{R} = h_1du\,\mathbf{T}_u + h_2dv\,\mathbf{T}_v + h_3dw\,\mathbf{T}_w \qquad \ldots(4)$$

Since ∇u is normal to the surface $u = u_1$ at P, therefore a unit vector in this direction is given by $\mathbf{N}_u = \frac{\nabla u}{|\nabla u|}$.

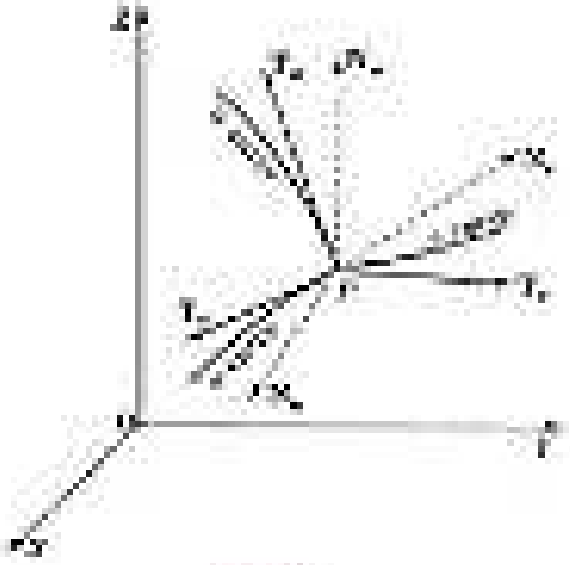

Fig. 6.2(b)

Similarly, the unit vectors $\mathbf{N}_v = \frac{\nabla v}{|\nabla v|}$ and $\mathbf{N}_w = \frac{\nabla w}{|\nabla w|}$ are normal to the surfaces $v = v_1$ and $w = w_1$ at P respectively. Thus at each point P of a curvilinear coordinate system there exist two triads of unit vectors $\mathbf{T}_u, \mathbf{T}_v, \mathbf{T}_w$ tangential to u, v, w-curves and $\mathbf{N}_u, \mathbf{N}_v, \mathbf{N}_w$ normals to the coordinate surfaces (Fig. 6.2(b)).

In particular, when the coordinate surfaces intersect at right angles, the three coordinate curves are then mutually orthogonal and u, v, w are called the orthogonal curvilinear coordinates. In this case $\mathbf{T}_u, \mathbf{T}_v, \mathbf{T}_w$ and $\mathbf{N}_u, \mathbf{N}_v, \mathbf{N}_w$ are mutually perpendicular unit vector triads and hence become identical. Hereafter, we shall refer to orthogonal curvilinear coordinates only.

Multiplying (3) scalarly by ∇u, we get

$$\nabla u \cdot d\mathbf{R} = du = \left(\nabla u \cdot \frac{\partial \mathbf{R}}{\partial u}\right)du + \left(\nabla u \cdot \frac{\partial \mathbf{R}}{\partial v}\right)dv + \left(\nabla u \cdot \frac{\partial \mathbf{R}}{\partial w}\right)dw$$

whence $$\nabla u \cdot \frac{\partial \mathbf{R}}{\partial u} = 1,\ \nabla u \cdot \frac{\partial \mathbf{R}}{\partial v} = 0,\ \nabla u \cdot \frac{\partial \mathbf{R}}{\partial w} = 0$$

Similarly, $$\nabla v \cdot \frac{\partial \mathbf{R}}{\partial u} = 0,\ \nabla v \cdot \frac{\partial \mathbf{R}}{\partial v} = 1,\ \nabla v \cdot \frac{\partial \mathbf{R}}{\partial w} = 0$$

and $$\nabla w \cdot \frac{\partial \mathbf{R}}{\partial u} = 0,\ \nabla w \cdot \frac{\partial \mathbf{R}}{\partial v} = 0,\ \nabla w \cdot \frac{\partial \mathbf{R}}{\partial w} = 1.$$

These relations show that the sets $\partial \mathbf{R}/\partial u, \partial \mathbf{R}/\partial v, \partial \mathbf{R}/\partial w$ and $\nabla u, \nabla v, \nabla w$ constitute reciprocal system of vectors.

$$\therefore \quad \nabla\psi = \frac{\frac{\partial \mathbf{R}}{\partial u}\times\frac{\partial \mathbf{R}}{\partial v}}{\left[\frac{\partial \mathbf{R}}{\partial u}\ \frac{\partial \mathbf{R}}{\partial v}\ \frac{\partial \mathbf{R}}{\partial w}\right]}\frac{\partial \psi}{\partial w} \qquad (h_1\mathbf{T}_1)\times(h_2\mathbf{T}_2)=h_1h_2\mathbf{T}_3$$

$$= \frac{h_1h_2\mathbf{T}_1\times\mathbf{T}_2}{h_1h_2h_3[\mathbf{T}_1\mathbf{T}_2\mathbf{T}_3]}\frac{\partial \psi}{\partial w} = \frac{\nabla_w}{h_3} \qquad \because\ [\mathbf{T}_1\mathbf{T}_2\mathbf{T}_3]=1$$

or
$$\mathbf{T}_3 = h_3\nabla w \qquad (3.35)$$

Similarly $\mathbf{T}_1 = h_1\nabla u$ and $\mathbf{T}_2 = h_2\nabla v$ (3.35)

Also $\quad = \mathbf{T}_1\times\mathbf{T}_2 = h_1h_2\nabla u\times\nabla v$

Similarly $\mathbf{T}_2 = h_3h_1\nabla w\times\nabla u$ and $\mathbf{T}_1 = h_2h_3\nabla v\times\nabla w$ (3.36)

Arc, area and volume elements

(i) *Arc element.* The element of arc length ds is determined from (4)

$$= \qquad ds^2 = d\mathbf{R}\cdot d\mathbf{R} = h_1^2du^2 + h_2^2dv^2 + h_3^2dw^2 \qquad (3.37)$$

The arc length ds_1 along u-curve at P is h_1du for v and w are constants. Therefore the vector arc element along the u-curve is $du = h_1du\mathbf{T}_1$. Similarly vector arc elements along v and w curves at P are $dv = h_2dv\,\mathbf{T}_2$ and $dw = h_3dw\,\mathbf{T}_3$. The arc element ds therefore corresponds to the length of the diagonal of the rectangular parallelopiped of Fig. 3.23.

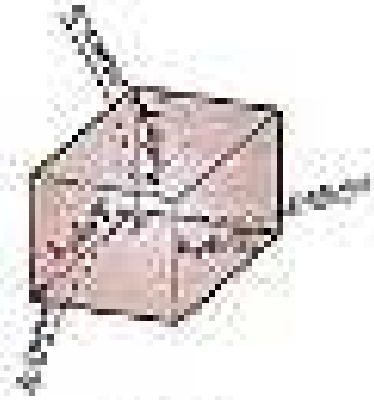

Fig. 3.17

(ii) *Area elements.* The area of the parallelogram formed by du and dv is called the area element on the w surface which is perpendicular to w-curve and we denote it by dS_w. Hence, $dS_w = |du\times dv| = h_1h_2|\mathbf{T}_1\times\mathbf{T}_2|du\,dv$. Similarly, $dS_u = h_2h_3\,|\mathbf{T}_2\times\mathbf{T}_3|dv\,dw$, $dS_v = h_3h_1|\mathbf{T}_3\times\mathbf{T}_1|dw\,du$.

(iii) *Volume element* is the volume of the parallelopiped formed by du, dv, dw.

$$\therefore \qquad dV = |h_1\mathbf{T}_1du,\ h_2\mathbf{T}_2dv,\ h_3\mathbf{T}_3dw| = h_1h_2h_3|\mathbf{T}_1\mathbf{T}_2\mathbf{T}_3|du\,dv\,dw \qquad \because\ |\mathbf{T}_1\mathbf{T}_2\mathbf{T}_3| = 1$$
$$= h_1h_2h_3\,du\,dv\,dw \qquad (3.38)$$

This can also be written as

$$dV = \frac{\partial \mathbf{R}}{\partial u}\cdot\frac{\partial \mathbf{R}}{\partial v}\times\frac{\partial \mathbf{R}}{\partial w}\,du\,dv\,dw = \frac{\partial(x,y,z)}{\partial(u,v,w)}du\,dv\,dw \qquad (3.39)$$

where $\partial(x,y,z)/\partial(u,v,w)$ is called the Jacobian of the transformation from (x, y, z) to (u, v, w) coordinates.

(3) Del applied to Functions in Orthogonal Curvilinear coordinates

Theorem. Show that

(1) $\nabla\psi = \dfrac{\mathbf{T}_1}{h_1}\dfrac{\partial \psi}{\partial u} + \dfrac{\mathbf{T}_2}{h_2}\dfrac{\partial \psi}{\partial v} + \dfrac{\mathbf{T}_3}{h_3}\dfrac{\partial \psi}{\partial w}$

(2) $\nabla\cdot\mathbf{F} = \dfrac{1}{h_1h_2h_3}\left[\dfrac{\partial}{\partial u}(h_2h_3f_1) + \dfrac{\partial}{\partial v}(h_3h_1f_2) + \dfrac{\partial}{\partial w}(h_1h_2f_3)\right]$

(3) $\nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{T}_1 & \mathbf{T}_2 & \mathbf{T}_3 \\ h_1f_1 & h_2f_2 & h_3f_3 \\ \frac{1}{h_1}\frac{\partial}{\partial u} & \frac{1}{h_2}\frac{\partial}{\partial v} & \frac{1}{h_3}\frac{\partial}{\partial w} \\ h_1f_1 & h_2f_2 & h_3f_3 \end{vmatrix}$ where $\mathbf{F} = f_1\mathbf{T}_1 + f_2\mathbf{T}_2 + f_3\mathbf{T}_3$.

(1) Let $\psi(u, v, w)$ be any scalar point function in terms of u, v, w, the orthogonal curvilinear coordinates. Taking u, v, w as functions of x, y, z, we have

$$\frac{\partial \psi}{\partial x} = \frac{\partial \psi}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial \psi}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial \psi}{\partial w}\frac{\partial w}{\partial x} \qquad (3.40)$$

$$\frac{\partial \psi}{\partial y} = \frac{\partial \psi}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial \psi}{\partial v}\frac{\partial v}{\partial y} + \frac{\partial \psi}{\partial w}\frac{\partial w}{\partial y} \qquad (3.41)$$

and
$$\frac{\partial \psi}{\partial z} = \frac{\partial \psi}{\partial u}\frac{\partial u}{\partial z} + \frac{\partial \psi}{\partial v}\frac{\partial v}{\partial z} + \frac{\partial \psi}{\partial w}\frac{\partial w}{\partial z} \qquad (3.42)$$

Multiplying (i) by $f_1(x)$ by $f_2(y)$ by $f_3(z)$ and adding, we have

$$\nabla f = \frac{\partial f}{\partial x}\nabla x + \frac{\partial f}{\partial y}\nabla y + \frac{\partial f}{\partial z}\nabla z \qquad \ldots(ii)$$

$$= \frac{\nabla_1}{h_1}\frac{\partial f}{\partial x} + \frac{\nabla_2}{h_2}\frac{\partial f}{\partial y} + \frac{\nabla_3}{h_3}\frac{\partial f}{\partial z} \qquad \text{[By (6) p. 355]}$$

which is the required result.

(2) Let $\vec{F}(x, y, z)$ be a vector point function such that

$$\vec{F} = f_1\vec{i}_1 + f_2\vec{i}_2 + f_3\vec{i}_3 = \Sigma f_1 h_1 \nabla x \qquad \text{[By (6) p. 355]}$$

$$\therefore \quad \nabla \cdot \vec{F} = \Sigma \nabla \cdot (f_1 h_1 \nabla x) \qquad \ldots(i)$$

$$= \Sigma \{\nabla(f_1 h_1) \cdot (\nabla x \times \nabla a) + (\nabla x \times \nabla a) \nabla(f_1 h_1)\} \qquad \ldots(ii)$$

Now $\nabla \cdot (\nabla x \times \nabla a) = \nabla a \cdot \nabla \times (\nabla x) - \nabla x \cdot \nabla \times (\nabla a) = 0$ [By (6) p. 355]

and $\nabla(f_1 h_1) = \frac{\partial (f_1 h_1)}{\partial x}\nabla x + \frac{\partial (f_1 h_1)}{\partial y}\nabla y + \frac{\partial (f_1 h_1)}{\partial z}\nabla z$ [By (ii) above]

$\therefore$ (i) now becomes

$$\nabla \cdot \vec{F} = \sum (\nabla x \times \nabla a) \cdot \left\{ \frac{\partial (f_1 h_1)}{\partial x}\nabla x + \frac{\partial (f_1 h_1)}{\partial y}\nabla y + \frac{\partial (f_1 h_1)}{\partial z}\nabla z \right\}$$

$$= [\nabla x, \nabla y, \nabla z] \sum \frac{\partial (f_1 h_1)}{\partial x} = \frac{1}{h_1 h_2 h_3} \sum \frac{\partial (f_1 h_1)}{\partial x} \text{ which is the required result.}$$

Cor. Explication. $\nabla f = \nabla \cdot (\nabla f)$

$$= \nabla \cdot \left(\frac{\nabla_1}{h_1}\frac{\partial f}{\partial x} + \frac{\nabla_2}{h_2}\frac{\partial f}{\partial y} + \frac{\nabla_3}{h_3}\frac{\partial f}{\partial z} \right) = \frac{1}{h_1 h_2 h_3} \sum \frac{\partial}{\partial x} \left(\frac{1}{h_1}\frac{\partial f}{\partial x} h_1 \right)$$

(3) Let $\vec{F}(x, y, z)$ be a vector point function such that

$$\vec{F} = f_1\vec{i}_1 + f_2\vec{i}_2 + f_3\vec{i}_3 = f_1 h_1 \nabla x + f_2 h_2 \nabla y + f_3 h_3 \nabla z \qquad \text{[By (1) p. 328]}$$

$$\nabla \times \vec{F} = \Sigma \nabla \times (f_1 h_1 \nabla x) \qquad \text{[Using (3) p. 328]}$$

$$= \Sigma \{\nabla(f_1 h_1) \times \nabla x + f_1 h_1 \nabla \times \nabla x\} = \sum \left[\frac{\partial (f_1 h_1)}{\partial x}\nabla x + \frac{\partial (f_1 h_1)}{\partial y}\nabla y + \frac{\partial (f_1 h_1)}{\partial z}\nabla z \right] \times \nabla x$$

$$= \sum \left[\frac{\partial (f_1 h_1)}{\partial y}\nabla y \times \nabla x + \frac{\partial (f_1 h_1)}{\partial z}\nabla z \times \nabla x \right]$$

$$= \sum \left[\frac{\partial (f_1 h_1)}{\partial y}\left(-\frac{\nabla_3}{h_3 h_2} \right) + \frac{\partial (f_1 h_1)}{\partial z}\left(\frac{\nabla_2}{h_2 h_3} \right) \right]$$

$$= -\frac{\partial (f_1 h_1)}{\partial y}\frac{\nabla_3}{h_3 h_2} + \frac{\partial (f_2 h_2)}{\partial x}\frac{\nabla_3}{h_1 h_3} - \frac{\partial (f_2 h_2)}{\partial z}\frac{\nabla_1}{h_1 h_2} + \frac{\partial (f_3 h_3)}{\partial y}\frac{\nabla_1}{h_2 h_1} - \frac{\partial (f_3 h_3)}{\partial x}\frac{\nabla_2}{h_3 h_1} + \frac{\partial (f_3 h_3)}{\partial z}\frac{\nabla_2}{h_3 h_1}$$

$$= \frac{\nabla_1}{h_2 h_3}\left[\frac{\partial (f_3 h_3)}{\partial y} - \frac{\partial (f_2 h_2)}{\partial z} \right] + \text{two similar terms, whence follows the required result.}$$

TWO SPECIAL CURVILINEAR SYSTEMS

[1] CYLINDRICAL COORDINATES

Any point $P(x, y, z)$ whose projection on the xy-plane is $Q(x, y)$ has the cylindrical coordinates (r, ϕ, z), where $r = OQ$, $\phi = \angle XOQ$ and $z = QP$.

The level surfaces $r = c_1$, $\phi = c_2$, $z = c_3$ are respectively cylinders about the Z-axis, planes through the X-axis and planes perpendicular to the X-axis.

The coordinate curves for r are rays perpendicular to the Z-axis, for ϕ horizontal circles with centres on the Z-axis, for z, lines parallel to the X-axis.

From Fig. 8.58, we have

$$x = r\cos\phi, \; y = r\sin\phi, \; z = z$$

(ii) (As in (i) above),

$$\text{i.e.} \quad \cdots$$

and hence the scale factors are $h_1 = 1, h_2 = \cdots, h_3 = 1$.

(iii) Base vectors are $\mathbf{e}_1 = \cdots$, $\mathbf{e}_2 = \cdots$, $\mathbf{e}_3 = \cdots$ (which are …
$\mathbf{e}_1, \mathbf{e}_2$ in the usual sense) $\mathbf{e}_3$ is a unit vector in the …

(iv) Volume element $dV = \cdots$

(2) Cylindrical coordinates (…)

An arbitrary point P has Cartesian …

Fig. 6.42

$$\text{i.e.} \quad \cdots$$

(i) $\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3$ for the unit vectors at P in the directions of the tangents to the ρ, ϕ, z curves respectively, then

$$\mathbf{T}_1 = \frac{\partial \mathbf{r}/\partial \rho}{|\partial \mathbf{r}/\partial \rho|} = \frac{\cdots}{\sqrt{\cdots}} = \cdots$$

$$\mathbf{T}_2 = \frac{\partial \mathbf{r}/\partial \phi}{|\partial \mathbf{r}/\partial \phi|} = \frac{\cdots}{\sqrt{\cdots}} = \cdots$$

$$\text{and} \quad \mathbf{T}_3 = \frac{\partial \mathbf{r}/\partial z}{|\partial \mathbf{r}/\partial z|} = \mathbf{k}$$

Now $\mathbf{T}_1 \cdot \mathbf{T}_2 = \cdots$, $\mathbf{T}_2 \cdot \mathbf{T}_3 = \cdots$, $\mathbf{T}_3 \cdot \mathbf{T}_1 = \cdots$

$\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3$ form a right-handed system, since $\mathbf{T}_1 \times \mathbf{T}_2 = \cdots$, $\mathbf{T}_2 \times \mathbf{T}_3 = \cdots$, $\mathbf{T}_3 \times \mathbf{T}_1 = \cdots$

(ii) Base vectors coincide with $\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3$ since the cylindrical coordinate system is orthogonal.

Also $\mathbf{e}_1 = \mathbf{T}_1 = \cdots = \mathbf{i}\cos\phi + \mathbf{j}\sin\phi$

$\mathbf{e}_2 = \mathbf{T}_2 = \cdots = -\mathbf{i}\sin\phi + \mathbf{j}\cos\phi$

$\mathbf{e}_3 = \mathbf{T}_3 = \mathbf{k}$

If (ρ_1, ϕ_1, z_1), (ρ_2, ϕ_2, z_2) and (ρ_3, ϕ_3, z_3) where the cylindrical coordinate system is right-handed (i.e. $\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3$) then the …
of the general curvilinear system.

(3) Grad applied to functions in Cylindrical coordinates

With $(x, y, z) = (x_1, x_2, x_3) = (\rho\cos\phi, \rho\sin\phi, z)$, $h_1 = 1$, $h_2 = \rho$, $h_3 = 1$.

Let $\mathbf{T}_1, \mathbf{T}_2, \mathbf{T}_3$ be the unit vectors in the directions of the tangents to the ρ, ϕ, z curves.

(i) Expression for grad f

Hence

$$\nabla f = \cdots$$

i.e.

$$\nabla f = \cdots$$

(ii) Expression for grad $\mathbf{F}$ where $\mathbf{F} = F_1\mathbf{T}_1 + F_2\mathbf{T}_2 + F_3\mathbf{T}_3$

Hence

$$\nabla \cdot \mathbf{F} = \cdots$$

i.e.

$$\nabla \cdot \mathbf{F} = \cdots$$

(iii) Expression for curl $\mathbf{F}$ where $\mathbf{F} = F_1\mathbf{T}_1 + F_2\mathbf{T}_2 + F_3\mathbf{T}_3$.

Hence

$$\nabla \times \mathbf{F} = \begin{vmatrix} \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots \end{vmatrix} = \cdots$$

$$= \cdots$$

(ii) Expression for $\nabla^2 f$

Now $$\nabla^2 f = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2 h_3}{h_1}\frac{\partial f}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_1 h_3}{h_2}\frac{\partial f}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1 h_2}{h_3}\frac{\partial f}{\partial u_3}\right)\right]$$

$$\therefore \quad \nabla^2 f = \frac{1}{\rho}\left[\frac{\partial}{\partial \rho}\left(\rho\frac{\partial f}{\partial \rho}\right) + \frac{\partial}{\partial \phi}\left(\frac{1}{\rho}\frac{\partial f}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(\rho\frac{\partial f}{\partial z}\right)\right] = \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial f}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}.$$

Example 6.32. Express the vector $\vec{F} = 2x\vec{i} + yz\vec{j} - z\vec{k}$ in cylindrical coordinates. (U.P.T.U. 2010-11)

Solution. We have $x = \rho\cos\phi$, $y = \rho\sin\phi$ and $z = z$.

so that $$\vec{R} = x\vec{i} + y\vec{j} + z\vec{k} = \rho\cos\phi\,\vec{i} + \rho\sin\phi\,\vec{j} + z\vec{k}$$

If $\vec{T}_1, \vec{T}_2, \vec{T}_3$ be the unit vectors along the tangents to ρ, ϕ and z curves respectively, then

$$\vec{T}_1 = \frac{\partial\vec{R}/\partial\rho}{|\partial\vec{R}/\partial\rho|} = \frac{\cos\phi\,\vec{i} + \sin\phi\,\vec{j}}{\sqrt{\cos^2\phi + \sin^2\phi}} = \cos\phi\,\vec{i} + \sin\phi\,\vec{j}$$

$$\vec{T}_2 = \frac{\partial\vec{R}/\partial\phi}{|\partial\vec{R}/\partial\phi|} = \frac{-\rho\sin\phi\,\vec{i} + \rho\cos\phi\,\vec{j}}{\sqrt{\rho^2\sin^2\phi + \rho^2\cos^2\phi}} = -\sin\phi\,\vec{i} + \cos\phi\,\vec{j}$$

$$\vec{T}_3 = \frac{\partial\vec{R}/\partial z}{|\partial\vec{R}/\partial z|} = \vec{k}$$

Let the expression for $\vec{F} = x\vec{i} - 2z\vec{j} + y\vec{k}$ in cylindrical coordinates be

$$\vec{F} = f_1\vec{T}_1 + f_2\vec{T}_2 + f_3\vec{T}_3 \qquad \ldots(1)$$

Then $$f_1 = \vec{F}\cdot\vec{T}_1 = x\cos\phi - 2z\sin\phi$$

$$f_2 = \vec{F}\cdot\vec{T}_2 = -x\sin\phi - 2z\cos\phi$$

$$f_3 = \vec{F}\cdot\vec{T}_3 = y$$

Substituting the values of f_1, f_2, f_3 in (1), we get

$$\vec{F} = (x\cos\phi - 2z\sin\phi)\vec{T}_1 - (x\sin\phi + 2z\cos\phi)\vec{T}_2 + y\vec{T}_3$$

$$= (\rho\cos\phi\cos\phi - 2z\sin\phi)\vec{T}_1 - (\rho\sin\phi\cos\phi + 2z\cos^2\phi)\vec{T}_2 + y\sin\phi\,\vec{T}_3$$

Example 6.33. Show that $\nabla^2(\log\rho) = 0$ and $\nabla^2(\rho^n\cos n\phi) = 0$ in cylindrical coordinates.

Solution. (i) $f = \log\rho$ is a function of ρ only. We have to prove that $\nabla^2 f = 0$ (i.e., $\nabla^2(\log\rho) = 0$).

$$\nabla^2 f = \frac{\partial^2 f}{\partial\rho^2} + \frac{1}{\rho}\frac{\partial f}{\partial\rho} + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial\phi^2} + \frac{\partial^2 f}{\partial z^2} = \frac{\partial^2}{\partial\rho^2}(\log\rho) + \frac{1}{\rho}\frac{\partial}{\partial\rho}(\log\rho) + 0 + 0 = -\frac{1}{\rho^2} + \frac{1}{\rho^2} = 0$$

Hence $\nabla(\log\rho)$ is a solenoidal vector.

(ii) $f = \nabla\phi$ is a function of ϕ only. We have to show that $\nabla^2(\nabla\phi) = 0$, i.e., $\nabla^2\phi = 0$.

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial\rho^2} + \frac{1}{\rho}\frac{\partial\phi}{\partial\rho} + \frac{1}{\rho^2}\frac{\partial^2\phi}{\partial\phi^2} + \frac{\partial^2\phi}{\partial z^2} = 0 + 0 + \frac{1}{\rho^2}\frac{\partial^2\phi}{\partial\phi^2} + 0 = 0.$$

Hence the result.

6.20.3 (3) SPHERICAL POLAR COORDINATES

Let $P(x, y, z)$ be any point whose projection on the XY-plane is $Q(x, y)$. Then the spherical polar coordinates of P are (r, θ, ϕ) such that $r = OP$, $\theta = \angle ZOP$ and $\phi = \angle XOQ$.

The level surfaces $r = r_0$, $\theta = \theta_0$, $\phi = \phi_0$ are respectively spheres about O, cones about the Z-axis with vertex at O and planes through the Z-axis.

The coordinate curves for r are rays from the origin; for θ, vertical circles with centre at O (called meridians); for ϕ, horizontal circles with centres on the Z-axis.

From Fig. 6.28, we have

$$x = OQ\cos\phi = OP\cos\angle QOP\cos\phi = r\sin\theta\cos\phi,$$

$$y = OQ\sin\phi = r\sin\theta\sin\phi, \quad z = r\cos\theta.$$

(a) Area element

$$\therefore \quad \text{Area} = \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} \sqrt{\left(\frac{\partial(y,z)}{\partial(\theta,\phi)}\right)^2 + \left(\frac{\partial(z,x)}{\partial(\theta,\phi)}\right)^2 + \left(\frac{\partial(x,y)}{\partial(\theta,\phi)}\right)^2}\, d\theta\, d\phi$$

so that the area element is

$$\delta S = \delta\theta\, \delta\phi = r^2 \sin\theta\, d\theta\, d\phi$$

(b) Area element

$\delta S = \delta^2$ area in $\theta\phi$ plane $= \delta\theta \times \delta\phi$ (Fig. (i))

where $\delta\theta$ is the small change corresponding to the θ direction, etc.

(c) Volume element $\delta V = r^2 \sin\theta\, \delta r\, \delta\theta\, \delta\phi$

(d) **Spherical polar coordinate system is orthogonal**

An area parallel to xy is $\delta x = r\sin\theta\cos\phi$, $\delta y = r\sin\theta\sin\phi$, $\delta z = r\cos\theta$

so that $\delta S = r\sin\theta\cos\phi\, i + r\sin\theta\sin\phi\, j + r\cos\theta\, k$

If ∇_1, ∇_2, ∇_3 be the unit vectors at P in the directions of the tangents to the r, θ, ϕ curves respectively, then

$$\nabla_1 = \frac{\frac{\partial(x,y,z)}{\partial(r,\theta)} \times \frac{\partial(x,y,z)}{\partial(\phi,\theta)}}{\left|\frac{\partial(x,y,z)}{\partial(r,\theta)} \times \frac{\partial(x,y,z)}{\partial(\phi,\theta)}\right|} = \frac{\sin\theta\cos\phi\, i + \sin\theta\sin\phi\, j + \cos\theta\, k}{\sqrt{\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta}}$$

$$= \sin\theta\cos\phi\, i + \sin\theta\sin\phi\, j + \cos\theta\, k$$

$$\nabla_2 = \frac{\frac{\partial(x,y,z)}{\partial(\theta,\phi)} \times \frac{\partial(x,y,z)}{\partial(r,\theta)}}{\left|\frac{\partial(x,y,z)}{\partial(\theta,\phi)} \times \frac{\partial(x,y,z)}{\partial(r,\theta)}\right|} = \frac{r\cos\theta\cos\phi\, i + r\cos\theta\sin\phi\, j - r\sin\theta\, k}{\sqrt{r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta}}$$

$$= \cos\theta\cos\phi\, i + \cos\theta\sin\phi\, j - \sin\theta\, k$$

and $$\nabla_3 = \frac{\frac{\partial(x,y,z)}{\partial(\phi,r)} \times \frac{\partial(x,y,z)}{\partial(\theta,r)}}{\left|\frac{\partial(x,y,z)}{\partial(\phi,r)} \times \frac{\partial(x,y,z)}{\partial(\theta,r)}\right|} = \frac{-r\sin\theta\sin\phi\, i + r\sin\theta\cos\phi\, j}{r\sin\theta} = -\sin\phi\, i + \cos\phi\, j$$

Now $\nabla_1 \cdot \nabla_2 = \sin\theta\cos\theta\cos^2\phi + \sin\theta\cos\theta\sin^2\phi - \cos\theta\sin\theta = 0$

$\nabla_2 \cdot \nabla_3 = -\cos\theta\cos\phi\sin\phi + \cos\theta\sin\phi\cos\phi = 0$

$\nabla_3 \cdot \nabla_1 = -\sin\theta\cos\phi\sin\phi + \sin\theta\sin\phi\cos\phi = 0$

Also $\nabla_1 \times \nabla_2 = \sin\theta\cos\phi\cos\theta\cos\phi\, k + \sin^2\theta\cos\phi\sin\phi\, i - \sin\theta\cos\phi\, j - \sin\theta\sin\phi\cos\theta\sin\phi\, k$

$$- \sin^2\theta\cos\phi\, j + \cos^2\theta\cos\phi\, i - \cos^2\theta\sin\phi\, j$$

$$= -\sin\phi\, i + \cos\phi\, j = \nabla_3$$

$$\nabla_2 \times \nabla_3 = \cos\theta\cos^2\phi\, k + \sin^2\phi\cos\theta\, k + \cos\theta\sin\phi\cos\phi\, i - \sin\theta\cos\phi\, j = \nabla_1$$

and $$\nabla_3 \times \nabla_1 = -\sin\theta\cos^2\phi\, k - \sin\theta\sin^2\phi\, k + \cos\theta\sin\phi\cos\phi\, i - \cos\theta\cos\phi\, j = \nabla_2$$

The above results are satisfied by ∇_1, ∇_2, and ∇_3 show that the spherical polar coordinate system is a right-handed orthogonal curvilinear system. (P.T.U., 2003)

(e) Grad applied to functions in spherical polar coordinates

We have $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$

Let ∇_1, ∇_2, ∇_3 be the unit vectors in the directions of the tangents to the r, θ, ϕ curves.

(i) Expression for grad f

Since $$\nabla f = \frac{\nabla_1}{h_1}\frac{\partial f}{\partial r} + \frac{\nabla_2}{h_2}\frac{\partial f}{\partial \theta} + \frac{\nabla_3}{h_3}\frac{\partial f}{\partial \phi}$$

$$\therefore \quad \nabla f = \frac{\partial f}{\partial r}\nabla_1 + \frac{1}{r}\frac{\partial f}{\partial \theta}\nabla_2 + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\nabla_3$$

(ii) Expression for div $\mathbf{F}$ where $\mathbf{F} = f_1\nabla_1 + f_2\nabla_2 + f_3\nabla_3$

Since $$\nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial}{\partial r}(h_2 h_3 f_1) + \frac{\partial}{\partial \theta}(h_3 h_1 f_2) + \frac{\partial}{\partial \phi}(h_1 h_2 f_3)\right]$$

$$\therefore \quad \nabla \cdot \mathbf{F} = \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}(r^2\sin\theta\, f_1) + \frac{\partial}{\partial \theta}(r\sin\theta\, f_2) + \frac{\partial}{\partial \phi}(r f_3)\right]$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 f_1) + \frac{1}{r\sin\theta}\frac{\partial}{\partial \theta}(f_2\sin\theta) + \frac{1}{r\sin\theta}\frac{\partial f_3}{\partial \phi}$$

(vi) Expression for curl $\vec{F}$ in curvilinear coordinates $\vec{F} = f_1\vec{T}_1 + f_2\vec{T}_2 + f_3\vec{T}_3$.

Since
$$\nabla \times \vec{F} = \begin{vmatrix} \frac{\vec{T}_1}{h_2h_3} & \frac{\vec{T}_2}{h_3h_1} & \frac{\vec{T}_3}{h_1h_2} \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1f_1 & h_2f_2 & h_3f_3 \end{vmatrix}$$

$$\therefore \quad \nabla \times \vec{F} = \begin{vmatrix} \frac{\vec{T}_1}{r^2\sin\theta} & \frac{\vec{T}_2}{r\sin\theta} & \frac{\vec{T}_3}{r} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ f_1 & rf_2 & r\sin\theta f_3 \end{vmatrix}$$

$$= \frac{\vec{T}_1}{r^2\sin\theta}\left\{\frac{\partial}{\partial\theta}(r\sin\theta f_3) - \frac{\partial}{\partial\phi}(rf_2)\right\} - \frac{\vec{T}_2}{r\sin\theta}\left\{\frac{\partial}{\partial r}(r\sin\theta f_3) - \frac{\partial f_1}{\partial\phi}\right\} + \frac{\vec{T}_3}{r}\left\{\frac{\partial}{\partial r}(rf_2) - \frac{\partial f_1}{\partial\theta}\right\}$$

$$= \frac{\vec{T}_1}{r\sin\theta}\left\{\frac{\partial}{\partial\theta}(f_3\sin\theta) - \frac{\partial f_2}{\partial\phi}\right\} + \frac{\vec{T}_2}{r}\left\{\frac{1}{\sin\theta}\frac{\partial f_1}{\partial\phi} - \frac{\partial}{\partial r}(rf_3)\right\} + \frac{\vec{T}_3}{r}\left\{\frac{\partial}{\partial r}(rf_2) - \frac{\partial f_1}{\partial\theta}\right\}$$

(vii) Expression for $\nabla^2 f$

Since
$$\nabla f = \frac{1}{h_1h_2h_3}\left[\frac{\partial}{\partial u_1}\left(\frac{h_2h_3}{h_1}\frac{\partial f}{\partial u_1}\right) + \frac{\partial}{\partial u_2}\left(\frac{h_3h_1}{h_2}\frac{\partial f}{\partial u_2}\right) + \frac{\partial}{\partial u_3}\left(\frac{h_1h_2}{h_3}\frac{\partial f}{\partial u_3}\right)\right]$$

$$\therefore \quad \nabla^2 f = \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial f}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\frac{r\sin\theta}{r}\frac{\partial f}{\partial\theta}\right) + \frac{\partial}{\partial\phi}\left(\frac{r}{r\sin\theta}\frac{\partial f}{\partial\phi}\right)\right]$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2}$$

$$= \frac{\partial^2 f}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2 f}{\partial\theta^2} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\phi^2} + \frac{2}{r}\frac{\partial f}{\partial r} + \frac{\cot\theta}{r^2}\frac{\partial f}{\partial\theta}.$$

Example 3.14. Express the vector $2z\hat{i} - y\hat{j} + 3x\hat{k}$ in spherical polar coordinate system.

Solution. We have $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$

so that $\vec{R} = r\sin\theta\cos\phi\,\hat{i} + r\sin\theta\sin\phi\,\hat{j} + r\cos\theta\,\hat{k}$

If $\vec{T}_1, \vec{T}_2, \vec{T}_3$ be the unit vectors along the tangents to r, θ, ϕ curves respectively, then

$$\vec{T}_1 = \frac{\partial\vec{R}/\partial r}{|\partial\vec{R}/\partial r|} = \frac{\sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} + \cos\theta\,\hat{k}}{\sqrt{\sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi + \cos^2\theta}}$$

$$= \sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} + \cos\theta\,\hat{k}$$

$$\vec{T}_2 = \frac{\partial\vec{R}/\partial\theta}{|\partial\vec{R}/\partial\theta|} = \frac{r\cos\theta\cos\phi\,\hat{i} + r\cos\theta\sin\phi\,\hat{j} - r\sin\theta\,\hat{k}}{\sqrt{r^2\cos^2\theta\cos^2\phi + r^2\cos^2\theta\sin^2\phi + r^2\sin^2\theta}}$$

$$= \cos\theta\cos\phi\,\hat{i} + \cos\theta\sin\phi\,\hat{j} - \sin\theta\,\hat{k}$$

$$\vec{T}_3 = \frac{\partial\vec{R}/\partial\phi}{|\partial\vec{R}/\partial\phi|} = \frac{-r\sin\theta\sin\phi\,\hat{i} + r\sin\theta\cos\phi\,\hat{j}}{\sqrt{r^2\sin^2\theta\sin^2\phi + r^2\sin^2\theta\cos^2\phi}} = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j}.$$

Let the expression for $\vec{F} = 2z\hat{i} - y\hat{j} + 3x\hat{k}$ in spherical polar coordinates be

$$\vec{F} = f_1\vec{T}_1 + f_2\vec{T}_2 + f_3\vec{T}_3 \qquad \ldots(1)$$

Then $f_1 = \vec{F}\cdot\vec{T}_1 = (2r\sin\theta\sin\phi\,\hat{i} - r\cos\theta\,\hat{j} + 3r\sin\theta\cos\phi\,\hat{k})\cdot(\sin\theta\cos\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} + \cos\theta\,\hat{k})$

$$= 2r\sin^2\theta\sin\phi\cos\phi - r\sin\theta\cos\theta\sin\phi + 3r\sin\theta\cos\theta\cos\phi$$

$$f_2 = \mathbf{F}\cdot\mathbf{T}_\theta = (2r\sin\theta\cos\phi)(\cos\theta\cos\phi) - r\cos\theta\sin\phi + 2r\sin\theta\cos\phi(-\sin\phi) = 2r\cos^2\theta\cos^2\phi - r\cos\theta\sin\phi - 2r\sin^2\phi\cos\phi$$

and $$f_3 = \mathbf{F}\cdot\mathbf{T}_\phi = (2r\sin\theta\cos\phi)(-r\sin\theta\sin\phi) - r\cos\theta(r\cos\theta\cos\phi) + 2r\sin\theta\cos\phi(\cos\theta\sin\phi) = -2r\sin\theta\sin^2\phi - r\cos\theta\cos\phi$$

Substituting the values of f_1, f_2, f_3 in (1), we get the desired expression.

Example 8.35. Prove that $\nabla\phi(r) = \frac{\phi'(r)}{r}\mathbf{r}$, $\phi(r) \neq 0$.

Solution. In spherical polar coordinates,

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{T}_r + \frac{1}{r}\frac{\partial f}{\partial\theta}\mathbf{T}_\theta + \frac{1}{r\sin\theta}\frac{\partial f}{\partial\phi}\mathbf{T}_\phi$$

$$\nabla(\phi(r)) = \frac{1}{r}\frac{\partial}{\partial\theta}(\phi(r))\mathbf{T}_\theta = -\frac{1}{r}\phi'(r)\mathbf{T}_\theta \quad \ldots(1)$$

$$\nabla\phi = \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}(\phi(r))\mathbf{T}_\phi = \frac{1}{r\sin\theta}\mathbf{T}_\phi \quad \ldots(2)$$

and $$\nabla\left(\frac{1}{r}\right) = \frac{\partial}{\partial r}(r^{-1})\mathbf{T}_r = -\frac{1}{r^2}\mathbf{T}_r$$

Now from (1) and (2), we get

$$\nabla\phi(r)\times\nabla\frac{1}{r} = -\frac{1}{r^2}\mathbf{T}_\theta\times\mathbf{T}_r = -\frac{1}{r^2}\mathbf{T}_\phi = \nabla\left(\frac{1}{r}\right)$$

Example 8.36. If $\mathbf{F} = r^2\cos\theta\,\mathbf{T}_r - \frac{1}{r}\mathbf{T}_\theta + \frac{1}{r\sin\theta}\mathbf{T}_\phi$, find the value of $\nabla\times\nabla\times\mathbf{F}$.

Solution. In spherical coordinates,

$$\operatorname{curl}\mathbf{F} = \begin{vmatrix} \mathbf{T}_r/r^2\sin\theta & \mathbf{T}_\theta/r\sin\theta & \mathbf{T}_\phi/r \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ f_1 & rf_2 & r\sin\theta f_3 \end{vmatrix}$$

Here $f_1 = r^2\cos\theta, f_2 = -1/r, f_3 = 1/r\sin\theta$.

$$\operatorname{curl}\mathbf{F} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \mathbf{T}_r & r\mathbf{T}_\theta & r\sin\theta\,\mathbf{T}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ r^2\cos\theta & -1 & 1 \end{vmatrix} = r\sin\theta\mathbf{T}_\phi$$

$$\nabla\times\operatorname{curl}\mathbf{F} = \left(r^2\cos\theta\mathbf{T}_r - \frac{1}{r}\mathbf{T}_\theta + \frac{1}{r\sin\theta}\mathbf{T}_\phi\right) = r\sin\theta\mathbf{T}_\phi = -(r^2\sin\theta\cos\theta\mathbf{T}_r + \sin\theta\mathbf{T}_\theta)$$

EXERCISE 8.5

1. Express the following vectors in cylindrical coordinates:
   (i) $2y\mathbf{i} - x\mathbf{j} + 3z\mathbf{k}$ (ii) $3x\mathbf{i} - 2y\mathbf{j} + z\mathbf{k}$ (U.P.T.U., 2006)
2. Express the following vectors in spherical polar coordinates:
   (i) $x\mathbf{i} - 2y\mathbf{j} + z\mathbf{k}$ (ii) $xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$
3. If $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, prove that $\nabla\times\mathbf{r} = 0$.
4. Show that $\nabla\times(r^n\mathbf{r}) = 0$.
5. Prove that $\operatorname{curl}\mathbf{V} = \frac{\cos\theta}{r^2}(\mathbf{T}_r\cot\theta - \mathbf{T}_\theta\cot\theta + 2\mathbf{T}_\phi)$ is solenoidal.
6. Show that (i) $\nabla^2(\log r) = 1/r^2$ (ii) $\nabla\times(\mathbf{a}\times\mathbf{r}) = 2\mathbf{a}$.

7. Prove that $\nabla \times (r \times a) = -2a$ $\left[\because r = xi + yj + zk \text{ and } a = a_1 i + a_2 j + a_3 k \text{ are constant vectors}\right]$.

8. If r_1, r_2, r_3 are non-coplanar vectors and r_1', r_2', r_3' are reciprocal vectors with $\Delta = [r_1\ r_2\ r_3]$ and $\Delta' = [r_1'\ r_2'\ r_3']$, prove that

$$\begin{bmatrix} r_1 \cdot r_1' & r_1 \cdot r_2' & r_1 \cdot r_3' \\ r_2 \cdot r_1' & r_2 \cdot r_2' & r_2 \cdot r_3' \\ r_3 \cdot r_1' & r_3 \cdot r_2' & r_3 \cdot r_3' \end{bmatrix} = \frac{1}{[r_1\ r_2\ r_3]\,[r_1'\ r_2'\ r_3']} = \Delta\Delta'$$

3.22 OBJECTIVE TYPE OF QUESTIONS

PROGRESS TEST

Write the key letter to choose the correct answer from the following problems :

1. A unit tangent vector to the surface $x = t^2 + 1$, $y = 4t - 3$, $z = t^2 - 2t$ at $t = 1$ is _____.
2. The equation of the normal to the surface $2x^2 + y^2 = 2z$ at $(2, 1, 3)$ is _____.
3. If $r = xi + yj + zk$ and $r = |r|$, then the vector whose field is related to the area element only by the relation $\frac{r}{r^3}$ is _____.
4. If $A = 2x^2 i - 3yz j + xz^2 k$, then $\nabla \cdot A =$ _____.
5. $\nabla \times (\nabla \phi) =$ _____.
6. Area bounded by a simple closed curve C is _____.
7. If $A(t)$ is a differentiable vector function of t and $|A| = c$, then $A \cdot \frac{dA}{dt} =$ _____.
8. $\int_C R \cdot N\,ds =$ _____.
9. $\text{div}\, R =$ _____, $\text{curl}\, R =$ _____.
10. If A is such that $\nabla \times A = 0$, then A is called _____.
11. If $\nabla \cdot F = 3$, then $\int_S F \cdot N\,ds$, where S is a surface of a unit sphere, is _____.
12. If $\nabla \cdot F = 3$, then F is called _____.
13. The directional derivative of $\phi(x, y, z) = x^2yz + 4xz^2$ at the point $(1, -2, -1)$ in the direction PQ, where $P = (1, 2, -1)$ and $Q = (-1, 2, 3)$ is _____.
14. If $\phi = x^2y + yz^3$, then $\nabla \phi$ at $(2, -1, 1) =$ _____.
15. $\text{curl}(\nabla \phi) + \text{grad}(\text{div}\, R) =$ _____.
16. If $F = (y + z)i + (z + x)j + (x + y)k$, then $F \cdot (\nabla \times F) =$ _____.
17. If F is a conservative force field, then curl F is _____.
18. If $\phi = 4x^2y - 3yz^2$, then $\frac{\partial \phi}{\partial x}$ at the point $(1, -2, -1)$ is _____.
19. $\text{curl}(\nabla \phi + \psi) =$ _____.
20. A particle is moved by a particle along the square formed by the lines $y = 0$, $y = 1$, $x = 0$, $x = 1$ under the force $F = xi + (xy)j$. Then $\int_C F \cdot dr =$ _____.
21. $\text{Curl}(\text{grad}\, \phi) =$ _____.
22. If A is a constant vector, then $\text{div}(R \times A) =$ _____.
23. If $R = xi + yj + zk$, then $\nabla \log r =$ _____, $\nabla(r^n) =$ _____.
24. A vector function is defined as _____.
25. Unit normal vector to the surface $z = 2xy$ at the point $(2, 1, 4)$ is _____.
26. If the directional derivative of $f = axy + byz + czx$ at $(1, 1, 1)$ has maximum magnitude 4 in direction parallel to x-axis, then the values of a, b, c are _____.
27. Maximum value of the directional derivative of $\phi = x^2 + 3y^2 + xz^2$ at the point $(1, 1, -1)$ is _____.
28. If $r^2 = x^2 + y^2 + z^2$, then $\nabla^2 \cdot (\text{div}\, r) =$ _____.
29. Directional derivative of $f = xyz$ at the point $(1, -1, 3)$ in the direction of the vector $3i - 2j + k$ is _____.
30. If $V = yzi + zxj + xyk$, then $\nabla \cdot (\nabla \times V) =$ _____.
31. If $r = xi + yj + zk$, then $\text{grad}(r^n) =$ _____.

Choose the correct alternative :

32. $\text{grad}\left(\frac{1}{r}\right) =$
(a) $\frac{r}{r^3}$ (b) $-\frac{r}{r^3}$ (c) $\frac{r}{r}$ (d) $-\frac{r}{r}$

33. The value of real constant b, where $\phi = 2x^2 + by^2 + 4z^2$ is
(a) $4x - 4y + 8z$ (b) $4x^2 - 4y^2 + 8z^2$ (c) 1 (d) 2

30. The value of $\int_C (y\,dx + z\,dy + x\,dz)$ from $(0,0,0)$ to $(1,1,1)$ along the curve C is
(a) 1 (b) 3 (c) −1 (d) zero

31. If $\vec{V} = xi + yj + zk$ then $\int \vec{V} \cdot d\vec{s}$ along the curve $x^2 + y^2 + z^2 = 1$ in the first octant is
(a) $\frac{\pi}{2}$ (b) 0 (c) $\frac{\pi}{4}$ (d) none of these

32. A necessary and sufficient condition that $\int \vec{F} \cdot d\vec{r}$ be independent of the path joining any two points is
(a) $\nabla \times \vec{F} = 0$ (b) $\nabla \cdot \vec{F} = 0$ (c) $\nabla \times \vec{F} = 1$ (d) $\nabla \cdot \vec{F} = 1$

33. The value of $\int_S (x\,dy\,dz + z\,dx\,dy)$, where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$ is
(a) 0 (b) 4π (c) $4\pi/3$ (d) 3π

34. If $\phi = x^2 + y^2 + z^2$ and $\vec{V} = xyi + yzj + zxk$, then $\nabla(\phi\vec{V}) =$ _____

35. For any scalar function ϕ, $\nabla \times \nabla\phi =$ _____

36. $\int_C \vec{F} \cdot d\vec{R}$ is interpreted as the path joining any two points P and Q in $F(x, y, z)$ is _____

37. The value of the line integral $\int_C (x^2 dx + z^2 dy)$, where C is the boundary of the square $-1 \le x \le 1, -1 \le y \le 1$ is
(a) 0 (b) $2(x + y)$ (c) 4 (d) 4π (ESE 2000)

38. If $\vec{V}$ is the instantaneous velocity vector of the moving fluid at a point P, then the $\vec{V}$ component is _____

39. The spherical coordinate system is
(a) Orthogonal (b) Cogitative (c) Non-coplanar (d) Non-orthogonal (ESE 2000)

40. The line integration of $\nabla \times \phi$ is zero.

41. The magnitude of the vector drawn perpendicular to the surface $x^2 = 2y^2 + z^2 = 2$ at the point $(2, -1, 2)$ is
(a) 10 (b) 12 (c) 3 (d) 8

42. The value of λ so that the vector $\vec{V} = 3xi + (\lambda x - 2z)j - (x + z)k$ is a solenoidal vector, is
(a) −1 (b) 2 (c) 1 (d) none of these

43. The work done by the force $\vec{F} = yzi + zxj + xyk$, in moving a particle from the point $(1, 1, 1)$ to the point $(2, 3, 4)$ along the curve $x = t$, $y = t^2$, $z = t^3$ is
(a) 21 (b) 19 (c) 0 (d) cannot be found

44. Value of $\int_C (x^2 dx + z^2 dy)$, where C is the boundary of the square $-1 \le x \le 1, -1 \le y \le 1$, is
(a) 4 (b) 0 (c) $2(x + y)$ (d) 4π

45. The directional derivative of $\phi(x, y, z) = 3x^2 - y^2 z$ at $(1, 1, 1)$ in the direction of a vector $2i + 3j + 6k$ is
(a) 45° (b) 60° (c) 135° (d) cannot be found

46. The vector $\vec{V} = r^n (ai + bj + ck)$, $r \neq 0$ is
(a) solenoidal (b) irrotational (c) both (a) and (b)

47. If $\phi = 2x^2 - 3y^2 + z^2$, then $\nabla^2 \phi =$ _____ (Objective Papers)

48. $\vec{F} = (x + 3y)i + (y - 2z)j + (x + pz)k$ is a solenoidal vector, then $p =$ _____ (Objective Papers)

49. $\vec{F} = yzi + zxj - xyk$ is irrotational. (True or False)

CHAPTER 9

Infinite Series

1. Introduction. 2. Sequences. 3. Series : Convergence. 4. General properties. 5. Series of positive terms—6. Comparison tests. 7. Integral test. 8. Comparison of ratios. 9. D'Alembert's ratio test. 10. Raabe's test. Logarithmic test. 11. Cauchy's root test. 12. Alternating series : Leibnitz's rule. 13. Series of positive or negative terms. 14. Power series. 15. Convergence of Exponential, Logarithmic and Binomial series. 16. Procedure for finding interval for convergence. 17. Uniform convergence. 18. Weierstrass's M-test. 19. Properties of uniformly convergent series. 20. Different types of illustrations.

9.1 INTRODUCTION

In finite series convergence is frequently in all types of problems that the necessity of studying that a convergence or divergence is very important. Unless a series employed in an investigation is convergent, it may lead to absurd conclusions. Hence it is essential that the students of engineering, physics and applied sciences should have a thorough knowledge of infinite series.

9.2 SEQUENCES

(1) An ordered set of real numbers. $a_1, a_2, a_3, \ldots, a_n$ is called a sequence and is denoted by $\{a_n\}$. If the number of terms is unlimited, then the sequence is said to be an infinite sequence and a_n is its general term.

For instance (i) $1, 3, 5, 7, \ldots, (2n-1), \ldots$ (ii) $1, 1/2, 1/3, \ldots, 1/n, \ldots$

(iii) $1, -1, 1, \ldots, (-1)^{n-1}, \ldots$ are infinite sequences.

(2) Limit. A sequence is said to tend to a limit l, if for every $\varepsilon > 0$, a number N can be found such that $|a_n - l| < \varepsilon$ for all $n > N$.

We then write $\lim_{n \to \infty} (a_n) = l$ or simply $(a_n) \to l$ as $n \to \infty$.

(3) Convergence. If a sequence $\{a_n\}$ has a finite limit, it is called a convergent sequence. If $\{a_n\}$ is not convergent, it is said to be divergent.

In the above examples, (ii) is convergent, while (i) and (iii) are divergent.

(4) Bounded sequence. A sequence $\{a_n\}$ is said to be bounded, if there exists a number k such that $a_n < k$ for every n.

(5) Monotonic sequence. The sequence $\{a_n\}$ is said to be monotonically or monotone steadily increasing as $a_{n+1} > a_n$ or $a_{n+1} - a_n$ for all values of n. Both increasing and decreasing sequences are called monotonic sequences.

A monotonic sequence always tends to a limit, finite or infinite. Thus, a sequence which is monotonic and bounded is convergent.

(6) Convergence, Divergence and Oscillation. If $\lim_{n \to \infty} (a_n) = l$ is finite and unique then the sequence is said to be convergent.

If $\lim_{n\to\infty} (u_n)$ is infinite (i.e. $\to \infty$), the sequence is said to be divergent.

If $\lim_{n\to\infty} (u_n)$ is not unique, then (u_n) is said to be oscillatory.

Example 2.2. Examine the following sequences for convergence:

(a) $u_n = \dfrac{n^2 + 2n}{3n^2 + n}$ (b) $u_n = 2^n$ (c) $u_n = 3 + (-1)^n$

Solution. (a) $\lim_{n\to\infty}\left(\dfrac{n^2 + 2n}{3n^2 + n}\right) = \lim_{n\to\infty} \dfrac{1 + 2/n}{3 + 1/n} = \frac{1}{3}$ which is finite and unique. Hence the sequence (u_n) is convergent.

(b) $\lim_{n\to\infty} 2^n \to \infty$. Hence the sequence (u_n) is divergent.

(c) $\lim_{n\to\infty} [3 + (-1)^n] = 3 + 1$ or $3 - 1$, i.e. it oscillates between

$u_n = 3 - 1$ or 2, as n is odd

i.e. this sequence doesn't have a unique limit. Hence it oscillates.

PROBLEMS 2.1

Examine the convergence of the following sequences:

1. $u_n = \dfrac{2n - 1}{3 + 2n}$ 2. $u_n = 1 + 2n$ 3. $u_n = 1 + (-1)^n$ 4. $u_n = \sin n\pi$

5. $u_n = \cos n\pi$ 6. $u_n = 1 + (-1)^n/n^2$ 7. $u_n = \left(\dfrac{n}{n-3}\right)$ 8. $u_n = n!$

2.2.3 SERIES

(1) **Defn.** If $u_1, u_2, u_3, \ldots, u_n, \ldots$ is an infinite sequence of real numbers, then

$$u_1 + u_2 + u_3 + \ldots + u_n + \ldots$$

is called an infinite series. An infinite series is denoted by $\sum u_n$ and the sum of its first n terms is denoted by s_n.

(2) **Convergence, divergence and oscillation of a series.**

Consider the infinite series $\sum u_n = u_1 + u_2 + u_3 + \ldots + u_n + \ldots$

and let the sum of the first n terms be $s_n = u_1 + u_2 + u_3 + \ldots + u_n$

Clearly, s_n is a function of n and as n increases (adds on) it may proceed in one of three ways:

(a) If s_n tends to a finite limit as $n \to \infty$, the series $\sum u_n$ is said to be convergent.

(b) If s_n tends to $+\infty$ as $n \to \infty$, the series $\sum u_n$ is said to be divergent.

(c) If s_n does not tend to a unique limit as $n \to \infty$, then the series $\sum u_n$ is said to be oscillatory or non-convergent.

Example 2.3. Discuss the convergence of the series (1) $2 + 2 + 2 + \ldots + n + \ldots \infty$

(2) $2 + 4 + 2 + 4 + 2 + 4 + \ldots + n + \ldots \infty$

Solution. (1) Here $s_n = 1 + 2 + 3 + \ldots + n = \dfrac{n(n+1)}{2}$

$\therefore \lim_{n\to\infty} s_n = \dfrac{1}{2}\lim_{n\to\infty} n(n+1) \to \infty$. Hence this series is divergent.

(2) Here $s_n = 0 + 4 + 1 + 3 + 4 + 1 + 3 + 4 + 1 + \ldots$ n terms

$= 0$, 2 or 1 according as the number of terms is $3n$, $3n + 1$, or $3n + 2$.

Clearly in this case, s_n does not tend to a unique limit. Hence the series is oscillatory.

Example 9.5. Geometric series. Show that the series $1 + r + r^2 + r^3 + \dots$ is (i) convergent if $|r| < 1$, (ii) divergent if $r \geq 1$, and (iii) oscillatory if $r \leq -1$.

Solution. Let $s_n = 1 + r + r^2 + \dots + r^{n-1}$

Case I. When $|r| < 1$, $\lim_{n \to \infty} r^n = 0$

Also $$s_n = \frac{1 - r^n}{1 - r} = \frac{1}{1 - r} - \frac{r^n}{1 - r} \text{ so that } \lim_{n \to \infty} s_n = \frac{1}{1 - r}$$

$\therefore$ The series is convergent.

Case II. (i) When $r > 1$, $\lim_{n \to \infty} r^n \to \infty$

Also $$s_n = \frac{r^n - 1}{r - 1} = \frac{r^n}{r - 1} - \frac{1}{r - 1} \text{ so that } \lim_{n \to \infty} s_n \to \infty$$

$\therefore$ The series is divergent.

(ii) When $r = 1$, then $s_n = 1 + 1 + 1 + \dots + 1 = n$

and $\lim_{n \to \infty} s_n \to \infty$ $\therefore$ The series is divergent.

Case III. (i) When $r = -1$, then the series becomes $1 - 1 + 1 - 1 + 1 - 1 \dots$ which is an oscillatory series.

(ii) When $r < -1$, let $r = -p$ so that $p > 1$. Then $r^n = (-1)^n p^n$

and $$s_n = \frac{1 - r^n}{1 - r} = \frac{1 - (-1)^n p^n}{1 + p} \text{ as } \lim_{n \to \infty} p^n \to \infty$$

$\therefore$ $\lim_{n \to \infty} s_n \to +\infty$ or $-\infty$ according as n is even or odd. Hence the series is oscillatory.

EXERCISE 9.1

Examine the following series for convergence :

1. $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$
2. $1 - \frac{1}{3} + \frac{1}{3^2} - \frac{1}{3^3} + \frac{1}{3^4} - \dots$
3. $8 - 10 + 8 + 8 - 10 + 8 + 8 - 10 + 8 + \dots$
4. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots$ (V.T.U. 2006)
5. A ball is dropped from a height h metres. Each time the ball hits the ground, it rebounds a distance rh times the distance fallen where $0 < r < 1$. It is known that $r = 0.8$. Find the total distance travelled by the ball.

9.4 GENERAL PROPERTIES OF SERIES

The truth of the following properties is self-evident and these may be regarded as axioms :

1. The convergence or divergence of an infinite series remains unaffected by the addition or removal of a finite number of its terms ; for the sum of these terms being the finite quantity does not on addition or removal alter the nature of the sum.

2. If a series in which all the terms are positive is convergent, the series remains convergent no matter in what order the terms are supplied ; for the sum is clearly the greatest when all the terms are positive.

3. The convergence or divergence of an infinite series remains unaffected by multiplying each term by a finite number.

9.5 SERIES OF POSITIVE TERMS

1. An infinite series in which all the terms after some particular term are positive, is a positive term series. e.g., $s_n = 3 - 5 - 2 + 2 + 7 + 12 + 20 + \dots$ is a positive term series as all its terms after the third are positive.

2. A series of positive terms either converges or diverges to $+\infty$; for the sum of its first n terms, counting the negative terms, tends to either a finite limit or $+\infty$.

2. Necessary condition for convergence. If a positive term series Σu_n is convergent, then $\lim_{n\to\infty} u_n = 0$.

(P.T.U., 2009)

Let $s_n = u_1 + u_2 + u_3 + \ldots + u_n$. Since Σu_n is given to be convergent

$\therefore$ $\lim_{n\to\infty} s_n$ is a finite quantity k (say). Also $\lim_{n\to\infty} s_{n-1} = k$

But $u_n = s_n - s_{n-1}$ $\therefore$ $\lim_{n\to\infty} u_n = \lim_{n\to\infty} (s_n - s_{n-1}) = 0$.

Hence the result.

Note. It is important to note that the converse of the result is not true.

Consider, for instance, the series $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \ldots + \frac{1}{\sqrt{n}} + \ldots$

Since the nth term $u_n = \frac{1}{\sqrt{n}}$

$\therefore$ $s_n = 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \ldots + \frac{1}{\sqrt{n}} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{1}} + \ldots + \frac{1}{\sqrt{1}} + \frac{n}{\sqrt{n}}$ i.e., $s_n < \sqrt{n}$

$\therefore$ $\lim_{n\to\infty} s_n < \lim_{n\to\infty} \sqrt{n} = \infty$

Thus the series is divergent even though $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$

Hence $\lim_{n\to\infty} u_n = 0$ is a necessary but not sufficient condition for convergence of Σu_n.

Obs. The above result leads to a simple test for divergence :

If $\lim_{n\to\infty} u_n \neq 0$, the series Σu_n must be divergent.

3.03 COMPARISON TESTS

I. If two positive term series Σu_n and Σv_n be such that

(i) Σv_n converges, (ii) $u_n \leq v_n$ for all values of n, then Σu_n also converges.

Proof: Since Σv_n is convergent,

$\therefore$ $\lim_{n\to\infty} (v_1 + v_2 + v_3 + \ldots + v_n)$ is a finite quantity k (say)

Also since $u_1 \leq v_1$, $u_2 \leq v_2$, $u_3 \leq v_3$, ..., $u_n \leq v_n$

$\therefore$ Adding, $u_1 + u_2 + \ldots + u_n \leq v_1 + v_2 + \ldots + v_n$

$\therefore$ $\lim_{n\to\infty} (u_1 + u_2 + \ldots + u_n) \leq \lim_{n\to\infty} (v_1 + v_2 + \ldots + v_n) = k$.

Hence the series Σu_n also converges.

Obs. If, however, the relation $u_n < v_n$ holds for values of n greater than a fixed number m, then the first m terms of both the series can be ignored without affecting their convergence or divergence.

II. If two positive term series Σu_n and Σv_n be such that :

(i) Σv_n diverges, (ii) $u_n \geq v_n$ for all values of n, then Σu_n also diverges.

Its proof is similar to that of Test I.

III. Limit form

If two positive term series Σu_n and Σv_n be such that

$$\lim_{n\to\infty} \frac{u_n}{v_n} = l$$ (a finite quantity $\neq 0$), then Σu_n and Σv_n converge or diverge together.

Proof: Since $\lim_{n\to\infty} \frac{u_n}{v_n} = l$, a finite number $(\neq 0)$

By definition of a limit, there exists a positive number ε, however small, such that

$$\left| \frac{u_n}{v_n} - l \right| < \varepsilon \qquad \text{for } n \geq m$$

or $-\varepsilon < \frac{S_n}{S_n} - 1 < \varepsilon$ for $n \geq N$

or $1 - \varepsilon < \frac{S_n}{S_n} < 1 + \varepsilon$ for $n \geq N$.

Dividing the first set in terms of both the series, we have

$$1 - \varepsilon < \frac{S_n}{S_n} < 1 + \varepsilon \quad \text{for all } n \qquad \ldots(3)$$

Case I. When Σv_n is convergent, then

$$\lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n) = k, \text{ a finite number} \qquad \ldots(4)$$

Also from (3), $\frac{S_n}{S_n} < 1 + \varepsilon \Rightarrow S_n < (1 + \varepsilon) S_n$ for all n.

$\therefore \lim_{n \to \infty} (u_1 + u_2 + \ldots + u_n) < (1 + \varepsilon) \lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n) = (1 + \varepsilon) k$ [By (2)]

Hence Σu_n is also convergent.

Case II. When Σv_n is divergent, then

$$\lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n) \to \infty \qquad \ldots(5)$$

Also from (3), $1 - \varepsilon < \frac{S_n}{S_n} \Rightarrow (1 - \varepsilon) S_n < S_n$ for all n.

$\therefore \lim_{n \to \infty} (u_1 + u_2 + \ldots + u_n) > (1 - \varepsilon) \lim_{n \to \infty} (v_1 + v_2 + \ldots + v_n) \to \infty$ [By (5)]

Hence Σu_n is also divergent.

5.20 INTEGRAL TEST

A positive term series $f(1) + f(2) + \ldots + f(n) + \ldots$, where $f(n)$ decreases as n increases, converges or diverges according as the integral

$$\int_1^\infty f(x)\,dx \qquad \ldots(\text{finite or infinite})$$

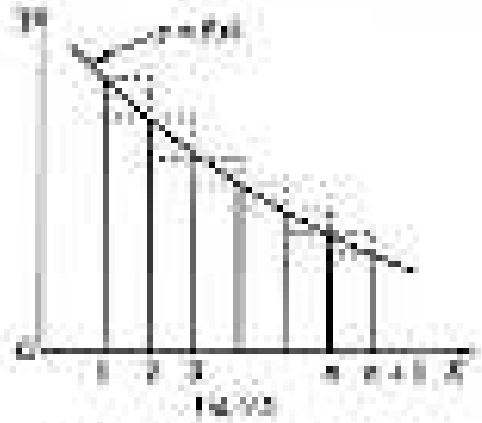

Fig. 5.3

The area under the curve $y = f(x)$ between any two ordinates lies between the set of inscribed and ascribed rectangles formed by ordinates at $x = 1, 2, 3, \ldots$ as in Fig. 5.3. Then

$$f(2) + f(3) + \ldots + f(n) < \int_1^{n+1} f(x)\,dx < f(1) + f(2) + \ldots + f(n)$$

or $s_n > \int_1^{n+1} f(x)\,dx > s_{n+1} - f(1)$

Taking limits as $n \to \infty$, we find from the second inequality that $\lim s_{n+1} < \int_1^\infty f(x)\,dx + f(1)$.

Hence if integral (1) is finite, so is $\lim s_{n+1}$. Similarly, from the first inequality, we see that if the integral (1) is infinite, so is $\lim s_n$. But the given series either converges or diverges to $+\infty$, i.e., $\lim s_n$ is either finite or infinite as $n \to \infty$.

Hence the result follows.

Example 5.4. Test for Convergence. Show that the p-series

$$\sum \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \ldots + \frac{1}{n^p} + \ldots$$

(i) converges for $p > 1$ (ii) diverges for $p \leq 1$. U.P.T.U. 2003; U.P.T.U. 2006; Rohtak 2005

Solution. By the above test, this series will converge or diverge according as $\int_1^\infty \frac{dx}{x^p}$ is finite or infinite.

[illegible]

[illegible]

[illegible]

Example 3.6. Test the convergence of the series :

(i) $\sum \frac{1}{(\sqrt{n+1}+\sqrt{n})\sqrt{n}}$ (V.T.U. 2008) (ii) $\sum \frac{1}{n^n+n^{-n}}$ (iii) $\sum \frac{\sqrt{n^2+1}-n}{\sqrt{n^3+1}}$ (V.T.U. 2008)(O)

Solution. (i) We have $u_n = \frac{\sqrt{n+1}-\sqrt{n}}{(\sqrt{n+1}+\sqrt{n})(\sqrt{n+1}-\sqrt{n})} = \sqrt{n+1}-\sqrt{n}$

$= \sqrt{n}\left(1+\frac{1}{n}\right)^{1/2} - \sqrt{n}$ (Expanding by Binomial Theorem)

$= \sqrt{n}\left[1+\frac{1}{2n}-\frac{1}{8n^2}+\cdots\right]-\sqrt{n} = \frac{1}{2\sqrt{n}}-\frac{1}{8n^{3/2}}+\cdots = \frac{1}{\sqrt{n}}\left(\frac{1}{2}-\frac{1}{8n}+\cdots\right)$

Taking $v_n = 1/\sqrt{n}$, we have

$$\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\left(\frac{1}{2}-\frac{1}{8n}+\cdots\right) = \frac{1}{2}, \text{ which is finite and non-zero.}$$

$\therefore$ both Σu_n and Σv_n converge or diverge together.

But $\Sigma v_n = \Sigma 1/\sqrt{n}$ is known to be divergent. Hence Σu_n is also divergent.

(ii) Where $x = 1$, comparing the given series Σu_n with $\Sigma v_n = \Sigma n^{-n}$,

we get $\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\left(\frac{1}{n^n+n^{-n}}\cdot\frac{1}{n^{-n}}\right) = \lim_{n\to\infty}\frac{1}{n^{2n}+1} = 1$ [$\because n^{2n} = (2\cos n\cdot\cdot\cdot+n)$]

But Σv_n is convergent, so Σu_n is also convergent.

When $x > 1$, comparing Σu_n with $\Sigma v_n = \Sigma x^{-n}$, we get

$$\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\left(\frac{1}{x^n+x^{-n}}\cdot x^n\right) = \lim_{n\to\infty}\frac{1}{1+x^{-2n}} = 1, \qquad [\because x^{2n} = (2\cos n\cdot\cdot\cdot+n)]$$

But Σv_n is convergent, so Σu_n is also convergent.

When $x = 1$, $\Sigma u_n = \frac{1}{2}+\frac{1}{5}+\frac{1}{8}+\cdots+$ which is divergent.

Hence Σu_n converges for $x > 1$ and $x < 1$ but diverges for $x = 1$.

(iii) We have $u_n = \sqrt{\frac{n^2-1}{n^3+2}} = \left(\frac{1}{n}\right)^{3/2}\sqrt{\frac{1-1/n^2}{1+1/n^3}}$

Taking $v_n = \left(\frac{1}{n}\right)^{3/2}$, we get

$$\lim_{n\to\infty}\left(\frac{u_n}{v_n}\right) = \lim_{n\to\infty}\sqrt{\frac{1-1/n^2}{1+1/n^3}} = 1 \neq 0$$

Also since $\Sigma v_n = n^{-p}$ where $p = 3/2 > 1$ is a geometric series having $r = 1$, is divergent.

$\therefore$ Σu_n is also divergent.

Example 3.7. Determine the nature of the series :

(i) $\frac{\sqrt{2}-1}{1^2+1}+\frac{\sqrt{3}-1}{2^2+2}+\frac{\sqrt{4}-1}{3^3+3}+\cdots$ (ii) $\sum \frac{2}{n^2+n}$

(iii) $\sum \frac{(\log n)^p}{n^{n+1}}$ (iv) $\sum \frac{1}{n\log(n+1)}$ to ∞ (V.T.U. 2010)

Solution. (i) We have $u_n = \frac{\sqrt{n+1}-1}{1n+2^2-1} = \frac{\sqrt{n}\left(1+\frac{1}{n}\right)^{1/2}-1}{n^2\left(1+\frac{2}{n}+\frac{1}{n^2}\right)-1}$

Taking $v_n = \frac{1}{n^{3/2}}$, we find that

$$\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\frac{\sqrt{n}\left[\sqrt{n+1}+\sqrt{n-1}-2\sqrt{n}\right]}{\left[(n+1)(n-1)-n^2\right]} = 1 \neq 0$$

Since $\sum v_n$ is convergent, therefore $\sum u_n$ is also convergent.

(d) Since $u_n = \frac{1}{n}\sin\frac{1}{n} = \frac{1}{n}\left[\frac{1}{n} - \frac{1}{3!n^3} + \frac{1}{5!n^5} - \cdots\right] = \frac{1}{n^2}\left[1 - \frac{1}{3!n^2} + \frac{1}{5!n^4} - \cdots\right]$

Taking $v_n = \frac{1}{n^2}$, we have

$$\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\left|1 - \frac{1}{3!n^2} + \frac{1}{5!n^4} - \cdots\right| = 1 \neq 0$$

Since $\sum v_n$ is convergent, therefore $\sum u_n$ is also convergent.

(e) We have $\lim_{n\to\infty}\frac{\log n^2}{n^{1/2}} = 0$, i.e., $\frac{2\log n}{n^{1/2}} < 1$ or $\log n^2 < n^{1/2}$

$$\Rightarrow \quad u_n = \frac{\log n^2}{n^{3/2}} < \frac{n^{1/2}}{n^{3/2}} = \frac{1}{n^{3/2}} = v_n$$

Since $\sum 1/n^{3/2}$ converges by p-series. $\quad \because \ p = 3/2 > 1$

Hence by comparison test, $\sum u_n$ also converges.

(f) Let $f(x) = \frac{1}{x\log x^2}$ so that $f(x) = \frac{(\log x)^{-2}}{2}$

$$\Rightarrow \quad f'(x) = \frac{-2}{2}(\log x)^{-2-1}\cdot\frac{1}{x} = -(\log x)^{-3}\cdot\left(-\frac{1}{x^2}\right) = -\frac{1}{x^2}\left[\frac{0}{(\log x)^{3+1}} + \frac{1}{2x\log x}\right] < 0$$

i.e., $f(x)$ is a decreasing function.

Also $\quad \int_2^\infty f(x)\,dx = \int_2^\infty \frac{dx}{x\log x^2} = \left|\frac{(\log x)^{-1+1}}{-1+1}\right|_2^\infty$

If $p = 1$, then $p - 1 = 0$ & $\log 1 = 0$

$$\therefore \quad \int_2^\infty f(x)\,dx = \left|\frac{\log x}{-0}\right|_2^\infty = \frac{1}{0}[0 + \log 2] \text{ which is finite}$$

Thus by integral test, the given series converges for $p = 1$.

If $p < 1$, then $1 - p > 0$ and $\log x^{1-p} \to \infty$ as $x \to \infty$.

$$\therefore \quad \int_2^\infty f(x)\,dx \to \infty$$

Thus the given series diverges for $p < 1$.

If $p > 1$, then $\int_2^\infty f(x)\,dx = \int_2^\infty \frac{dx}{x\log x} = \left|\log\log x\right|_2^\infty \to \infty$

Thus the given series diverges for $p = 1$.

PROBLEMS

Test the following series for convergence:

1. $1 + \frac{1}{2} + \frac{1}{3^2} + \frac{1}{4^3} + \frac{1}{5^4} + \cdots$ (U.P.T.U. 2005)

2. $\frac{1}{3} + \frac{1}{5} + \frac{2}{7} + \frac{3}{9} + \frac{5}{11} + \cdots$

3. $\frac{2}{1\cdot2} + \frac{3}{2\cdot3} + \frac{4}{3\cdot4} + \cdots$ (Roorkee 2001)

4. $\frac{1}{1\cdot3} + \frac{2}{3\cdot5} + \frac{3}{5\cdot7} + \cdots$ (U.P.T.U. 2005)

5. $1+\frac{x^2}{2!}+\frac{x^4}{4!}+\frac{x^6}{6!}+\cdots$

6. $\frac{2}{1^2}+\frac{3}{2^2}+\frac{4}{3^2}+\frac{5}{4^2}+\cdots$

7. $\frac{1}{1^2}+\frac{1+2}{2^2+2^1}+\frac{1+2+3}{3^2+3^1+3^0}+\cdots$

8. $\frac{1}{1\cdot 2\cdot 3}+\frac{2}{2\cdot 3\cdot 4}+\frac{3}{3\cdot 4\cdot 5}+\cdots$ (UPTU, 2006)

9. $\frac{2}{1}+\frac{4}{2^2}+\frac{6}{3^3}+\frac{8}{4^4}+\cdots$

10. $\sum_{n=1}^{\infty}\frac{n!}{n^n+1}$ (Osmania, 2007)

11. $\sum_{n=1}^{\infty}\frac{2n^2+1}{n^3+1}$

12. $\sum_{n=1}^{\infty}\frac{(n+1)(n+2)}{n^2\sqrt{n}}$ (UPTU, 2006)

13. $\sum_{n=1}^{\infty}\sqrt{n^2+1}-n$ (UPTU, 2006) (Kerala, 2007)

14. $\sum\left(\sqrt{n^2+1}-n\right)$ (UPTU, 2007) (Anna, 2006)

15. $\sum\sqrt{n^2+1}-\sqrt{n^2-1}$

16. $\sum\frac{1}{\sqrt{n}}\sin\frac{1}{n}$

17. $\sum_{n=2}^{\infty}\frac{1}{n(\log n)^2}$

18. $\sum_{n=1}^{\infty}\frac{\sqrt{n+1}-\sqrt{n}}{(n+1)^2-1}$ (UPTU, 2006)

333 COMPARISON OF RATIOS

If Σu_n and Σv_n be two positive term series, then Σu_n converges if (i) Σv_n converges and (ii) from and after some particular term,

$$\frac{u_{n+1}}{u_n} \le \frac{v_{n+1}}{v_n}$$

Let the two series beginning from the particular term be $u_1 + u_2 + u_3 + \ldots$ and $v_1 + v_2 + v_3 + \ldots$

If $\frac{u_2}{u_1} \le \frac{v_2}{v_1}, \frac{u_3}{u_2} \le \frac{v_3}{v_2}, \ldots$

then $u_2 + u_3 + u_4 + \ldots \le u_2\left(1 + \frac{v_3}{v_2} + \frac{v_4}{v_3} + \ldots\right)$

$$\le u_2\left(1 + \frac{v_3}{v_2} + \frac{v_4}{v_3} + \frac{v_5}{v_4} + \ldots\right) \le u_2\left(1 + \frac{v_3}{v_2} + \frac{v_4}{v_3} + \frac{v_5}{v_4} + \ldots\right) \le \frac{u_2}{v_2}(v_2 + v_3 + v_4 + \ldots)$$

Hence, if Σv_n converges, Σu_n also converges.

(ii) A more convenient form of the above test to apply is as follows:

Σu_n converges if (i) Σv_n converges and (ii) from and after a particular term $\frac{u_n}{u_{n+1}} \ge \frac{v_n}{v_{n+1}}$

similarly Σu_n diverges if (i) Σv_n diverges and (ii) from and after a particular term $\frac{u_n}{u_{n+1}} \le \frac{v_n}{v_{n+1}}$

334 GAUSS'S RATIO TEST*

If a positive term series Σu_n if

$\sum_{n=1}^{\infty}\frac{u_{n+1}}{u_n} = 1$, then the series converges for $k < 1$ and diverges for $k > 1$.

Case I. When $\sum_{n=1}^{\infty}\frac{u_{n+1}}{u_n} = k < 1$.

*Called after the French mathematician Karl Friedrich Gauss (1777–1855), who also made important contributions to astronomy.

By definition of a limit, we can find a positive number r (< 1) such that $\frac{a_{n+1}}{a_n} < r$ for all $n \geq m$.

Leaving out the first m terms, let the series be $u_1 + u_2 + u_3 + \ldots$

so that $\frac{u_2}{u_1} < r$, $\frac{u_3}{u_2} < r$, $\frac{u_4}{u_3} < r$, ... and so on. Hence $u_2 < u_1 r$, $u_3 < u_2 r$, $u_4 < u_3 r$, ...

$$< u_1\left(1 + \frac{u_2}{u_1} + \frac{u_3}{u_2}\cdot\frac{u_2}{u_1} + \frac{u_4}{u_3}\cdot\frac{u_3}{u_2}\cdot\frac{u_2}{u_1} + \ldots\right) < u_1(1 + r + r^2 + r^3 + \ldots)$$

$< \frac{u_1}{1-r}$, which is a finite quantity. Hence $\sum u_n$ is convergent. $\quad [\because r < 1]$

Case II. When $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = k > 1$

By definition of limit, we can find m, such that $\frac{a_{n+1}}{a_n} > 1$ for all $n \geq m$.

Leaving out the first m terms, let the series be

$$u_1 + u_2 + u_3 + \ldots \text{ so that } \frac{u_2}{u_1} > 1, \quad \frac{u_3}{u_2} > 1, \quad \frac{u_4}{u_3} > 1 \text{ and so on.}$$

$$\therefore \quad u_1 + u_2 + u_3 + u_4 + \ldots + u_n > u_1\left[1 + \frac{u_2}{u_1} + \frac{u_3}{u_2}\cdot\frac{u_2}{u_1} + \ldots\right]$$

$> u_1(1 + 1 + 1 + \ldots)$ to n terms $> nu_1$

$$\therefore \quad \lim_{n\to\infty}(u_1 + u_2 + \ldots + u_n) > \lim_{n\to\infty} nu_1,$$

which tends to infinity. Hence $\sum u_n$ is divergent.

Note. If $\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = 1$, then the test fails to determine the convergence or divergence of $\sum u_n$.

Here $\lambda = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty}\left[\frac{1}{(1+1/n)^p} \cdot \frac{n^{p-1}}{1}\right] = \lim_{n\to\infty} \frac{1}{(1+1/n)^p} = 1.$

Thus for all values of p, $\lambda = 1$, whereas $\sum 1/n^p$ converges for $p > 1$ and diverges for $p \leq 1$.

Hence $\lambda = 1$ fails for convergence and divergence of $\sum u_n$, which is desired.

Note II. It is necessary to note that the test value is not relative to the comparison of u_{n+1}, u_n but converges only at the limit of the ratio.

For instance in the series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots + \frac{1}{n} + \ldots$, the value $\frac{a_{n+1}}{a_n} = \frac{n}{n+1} < 1$ for all finite values of n, but tends to unity as $n \to \infty$. Hence the Ratio test fails although the series is divergent.

Practical form of Ratio test. Taking reciprocals, the ratio test can be stated as follows:

If the positive term series $\sum u_n$ if $\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = k$, then the series converges for $k > 1$ and diverges for $k < 1$ And fails for $k = 1$.

Example 33.8. Test for convergence the series:

(i) $\frac{1}{2\cdot 1^2} + \frac{x^2}{2\cdot 2^2} + \frac{x^4}{2\cdot 3^2} + \frac{x^6}{2\cdot 4^2} + \ldots$ *(U.P.T.U., 2005; N.T.U., 2003; A.M.I.E., 2003)*

(ii) $1 + \frac{2}{2}x + \frac{6}{3}x^2 + \frac{14}{12}x^3 + \ldots + \frac{2^n - 2}{2^n + 1}x^{n-1} + \ldots \; (x > 0)$ *(U.P.T.U., 2005; N.T.U., 2004)*

Solution. (i) We have $u_n = \frac{x^{2n-2}}{(n+1)!\sqrt{n}}$ and $u_{n+1} = \frac{x^{2n}}{(n+2)!\sqrt{(n+1)}}$

$$\therefore \quad \lim_{n\to\infty}\frac{u_n}{u_{n+1}} = \lim_{n\to\infty}\frac{n^{2n+2}}{(n+1)!n^n}\cdot\frac{(n+2)!\sqrt{(n+1)}}{n^{2n}}$$

$$= \lim_{n\to\infty}\left[\frac{n+2}{n+1}\left(\frac{n+1}{n}\right)^{2n}\right]n^2 = \lim_{n\to\infty}\left[\frac{1+2/n}{1+1/n}\cdot\sqrt{(1+1/n)}\right]e^{2n}\cdot e^{-2n}$$

Hence Σu_n converges if $e^2 > 2$ i.e., for $e^2 < 1$ and diverges for $e^2 > 1$.

If $e^2 = 2$, then $u_n = \frac{1}{(n+1)\sqrt{n}}\cdot\frac{n^{2n}}{e^{2n}} = \frac{1}{n\sqrt{n}}\cdot\frac{1}{1+1/n}$

Taking $v_n = \frac{1}{n\sqrt{n}}$, we get $\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\frac{1}{1+1/n} = 1$, a finite quantity.

$\therefore$ Both Σu_n and Σv_n converge or diverge together. But $\Sigma v_n = \sum\frac{1}{n^{3/2}}$ is a convergent series.

$\therefore$ Σu_n is also convergent. Hence the given series converges if $e^2 \le 2$ and diverges if $e^2 > 2$.

(c) Here $\frac{u_n}{u_{n+1}} = \frac{2^n-1}{2^n+1}\cdot\frac{2^{n+1}+1}{2^{n+1}-1}\cdot\frac{1}{2^n} = \frac{1-\frac{1}{2^n}}{1+\frac{1}{2^n}}\cdot\frac{2+\frac{1}{2^n}}{2-\frac{1}{2^n}}\cdot\frac{1}{2}$

$$\therefore \quad \lim_{n\to\infty}\frac{u_n}{u_{n+1}} = \frac{1-0}{1+0}\cdot\frac{2+0}{2-0}\cdot\frac{1}{2} = 1$$

Thus by Raabe's test, Σu_n converges for $e^{-1} > 1$ i.e., for $e < 1$ diverges for $e > 1$. But it fails for $e = 1$.

When $e = 1$, $\lim_{n\to\infty} u_n = \lim_{n\to\infty}\frac{2^n-1}{2^n+1} = \lim_{n\to\infty}\frac{1-\frac{1}{2^n}}{1+\frac{1}{2^n}} = 1 \neq 0$

$\therefore$ Σu_n diverges for $e = 1$. Hence the given series converges for $e < 1$ and diverges for $e \ge 1$.

Example 8.8. Discuss the convergence of the series

(a) $\sum\frac{n!}{(n+2)^3}$ (P.T.U., 2010) (b) $1+\frac{2}{2^2}+\frac{3}{3^3}+\frac{4}{4^4}+\cdots$ (M.T.U., 2006, 2...)

Solution. (a) We have $u_n = \frac{n!}{(n+2)^3}$ and $u_{n+1} = \frac{(n+1)!}{(n+3)^{n+1}}$

$$\therefore \quad \lim_{n\to\infty}\frac{u_n}{u_{n+1}} = \lim_{n\to\infty}\frac{n!}{(n+2)^n}\times\frac{(n+3)^{n+1}}{(n+1)!}$$

$$= \lim_{n\to\infty}\frac{(n+3)^{n+1}}{(n+2)^n} = \lim_{n\to\infty}\left(\frac{n+3}{n+2}\right)^n\cdot(n+3)$$

$$= \lim_{n\to\infty}\left[\left(1+\frac{1}{n+2}\right)^n\right]\cdot(n+3) = e\cdot\lim_{n\to\infty}(n+3) = \infty$$

Hence the given series is convergent.

(c) Given series is $\Sigma u_n = \sum\frac{n!}{n^n}$. Here $\frac{u_n}{u_{n+1}} = \frac{n!}{n^n}\cdot\frac{(n+1)^{n+1}}{(n+1)!} = \frac{(n+1)^n}{n^n} = \left(1+\frac{1}{n}\right)^n$

$$\therefore \quad \lim_{n\to\infty}\frac{u_n}{u_{n+1}} = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e, \text{ which is } > 1. \text{ Hence the given series is convergent.}$$

Example 8.28. Determine the convergence of the series

(i) $\frac{x}{1+x^2}+\frac{x^2}{1+x^4}+\frac{x^3}{1+x^6}+\cdots$

(ii) $1+\frac{x+2}{6+1}+\frac{(x+2)(x+3)}{(6+1)(6+2)}+\frac{(x+2)(x+3)(x+4)}{(6+1)(6+2)(6+3)}+\cdots$

Solution. (i) Here $u_n=\dfrac{x^n}{1+x^{2n}}$ and $u_{n+1}=\dfrac{x^{n+1}}{1+x^{2n+2}}$

$$\therefore \lim_{n\to\infty}\frac{u_n}{u_{n+1}}=\lim_{n\to\infty}\left[\frac{x^n}{x^{n+1}}\cdot\frac{1+x^{2n+2}}{1+x^{2n}}\right]=\lim_{n\to\infty}\left[\frac{1+x^{2n+2}}{x+x^{2n+1}}\right]$$

$$=\frac{1}{x},\text{ if } x>1 \qquad [\because\ x^{2n+1}\to 0 \text{ as } n\to\infty]$$

Also $$\lim_{n\to\infty}\frac{u_n}{u_{n+1}}=\lim_{n\to\infty}\left[\frac{1+1/x^{2n+2}}{1+x^2/x^{2n+2}}\right]=1 \text{ if } x=1.$$

$\therefore$ by Kummer's test, Σu_n converges for $x>1$ and fails for $x\le 1$.

When $x=1$, $\Sigma u_n=\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\cdots+n$, which is divergent.

Hence the given series converges for $x>1$ and diverges for $x\le 1$.

(ii) Neglecting the first term, we have

$$u_{n+1}=\frac{x+1}{6n+1}$$

$$\therefore \lim_{n\to\infty}\frac{u_n}{u_{n+1}}=\lim_{n\to\infty}\frac{6n+1}{x+1}=\lim_{n\to\infty}\frac{6+1/n}{x/n+1/n}=\frac{6}{0}=\infty$$

By Kummer's test, Σu_n converges for $\frac{1}{x}>1$ or $x<6$, and diverges for $x>6$.

When $x=6$, the series becomes $1+1+1+\cdots+n$, which is divergent.

Hence the given series converges for $0<x<6$ and diverges for $0<6\le x$.

PROBLEMS 8.8

Test for convergence the following series :

1. $x+\frac{x^2}{2}+\frac{x^3}{3}+\cdots$
2. $\frac{1}{1\cdot 2}+\frac{2}{2\cdot 3}x+\frac{3}{3\cdot 4}x^2+\cdots$
3. $1+\frac{x}{2}+\frac{x^2}{3}+\frac{x^3}{4}+\cdots+\frac{x^n}{n+1}+\cdots$
4. $\sum_{n=1}^{\infty}\frac{x^n}{(n+1)(n+2)}$ (V.T.U., 2006)
5. $1+\frac{x^2}{11}+\frac{x^4}{21}+\frac{x^6}{31}+\cdots$ (Kurukshetra, 2006)
6. $\sum_{n=1}^{\infty}\left[\frac{x^n}{n^2}+\frac{1}{n^2}\right]$ (Delhi, 2003)
7. $\sum_{n=1}^{\infty}\frac{n!x^n}{n^n}$ (Kerala, 2008)
8. $\sum_{n=1}^{\infty}\frac{n^2+n}{x^n+n}$
9. $\sum_{n=1}^{\infty}\frac{x^n}{(n^2+1)}$ (U.P.T.U., 2006)
10. $\sum_{n=1}^{\infty}\frac{n^n-n+1}{n!}$ (Madras, 2006)
11. $\frac{2}{2!}+\frac{2\cdot 4}{2\cdot 5\cdot 6}+\frac{2\cdot 4\cdot 6}{2\cdot 5\cdot 7\cdot 8}+\cdots$ (U.P.T.U., 2010)
12. $\left[\frac{1^2}{2}\right]+\left[\frac{1\cdot 2}{3\cdot 4}\right]^2+\left[\frac{1\cdot 2\cdot 3}{3\cdot 4\cdot 5}\right]^3+\cdots$

15. $1+\frac{1^2\cdot 2^2}{1\cdot 2\cdot 3}+\frac{1^2\cdot 2^2\cdot 3^2}{1\cdot 2\cdot 3\cdot 4\cdot 5}+\cdots$ (Delhi 1998)

16. $\frac{4}{9}+\frac{4\cdot 12}{9\cdot 21}+\frac{4\cdot 12\cdot 20}{9\cdot 21\cdot 33}+\cdots$ (Madras 2000)

17. $\frac{1}{2}+\frac{x}{3}+\frac{x^2}{4}+\cdots+\frac{x^{n-1}}{(n+1)^n}+\cdots$ (UP B.Sc. 2000)

18. $\sum_{n=1}^{\infty}\frac{3\cdot 6\cdot 9\cdots 3n}{1\cdot 5\cdot 9\cdots(4n-3)}\frac{x^n}{(n+1)}$ (UP B.Sc. 2000)

19. $1+\frac{\alpha\cdot\beta}{1\cdot\gamma}+\frac{\alpha(\alpha+1)\beta(\beta+1)}{1\cdot 2\cdot\gamma(\gamma+1)}+\frac{\alpha(\alpha+1)(\alpha+2)\beta(\beta+1)(\beta+2)}{1\cdot 2\cdot 3\cdot\gamma(\gamma+1)(\gamma+2)}+\cdots$

8.10 FURTHER TESTS OF CONVERGENCE

When the Ratio test fails, we apply the following tests :

(1) Raabe's test[1]: In the positive term series $\sum u_n$ if $\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right)=k$,

then the series converges for $k>1$ and diverges for $k<1$, but the test fails for $k=1$.

When $k>1$, choose a number p such that $k>p>1$, and compare $\sum u_n$ with the series $\sum\frac{1}{n^p}$ which is convergent since $p>1$.

$\therefore$ $\sum u_n$ will converge, if from and after some term,

$$\frac{u_n}{u_{n+1}}>\frac{(n+1)^p}{n^p}\text{ or }\left(1+\frac{1}{n}\right)^p \quad\text{or}\quad \frac{u_n}{u_{n+1}}>1+\frac{p}{n}+\frac{p(p-1)}{2n^2}+\cdots$$

or if, $n\left(\frac{u_n}{u_{n+1}}-1\right)>p+\frac{p(p-1)}{2n}+\cdots$ or if, $\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}}-1\right)>\lim_{n\to\infty}\left[p+\frac{p(p-1)}{2n}+\cdots\right]$

i.e., if $k>p$, which is true. Hence $\sum u_n$ is convergent.

The other case when $k<1$ can be proved similarly.

(2) Logarithmic test: In the positive term series $\sum u_n$ if $\lim_{n\to\infty}\left(n\log\frac{u_n}{u_{n+1}}\right)=k$,

then the series converges for $k>1$, and diverges for $k<1$, but the test fails for $k=1$.

The proof is similar to that of Raabe's test.

Obs. 1. Logarithmic test is a substitute for Raabe's test and should be applied when either u_n/u_{n+1} or $u_n/u_{n+1}-1$ contains n in the exponent of e or $\lim_{n\to\infty}\frac{u_n}{u_{n+1}}=1$ i.e., when Raabe's test fails on taking logarithm of u_n/u_{n+1}.

Obs. 2. If u_n/u_{n+1} does not involve n as an exponent or a logarithm, the series $\sum u_n$ diverges.

Example 8.11. Test for convergence of the series

(i) $\sum\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots 2n}x^n$ (UPTU, 2002; U.P.T.U., 2006 S) (ii) $\sum\frac{(n+1)^n}{(n+1)!}x^n$

Solution. (i) Here $\frac{u_n}{u_{n+1}}=\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots 2n}x^n\div\frac{1\cdot 3\cdot 5\cdots(2n+1)}{2\cdot 4\cdot 6\cdots(2n+2)}x^{n+1}=\frac{2n+2}{2n+1}\cdot\frac{1}{x}=\left[\frac{1+2/n}{1+1/n}\right]\frac{1}{x}$

$\therefore$ $\lim_{n\to\infty}\frac{u_n}{u_{n+1}}=\frac{1}{x}$.

[1] Called after the Swiss mathematician Joseph Ludwig Raabe (1801–1859).

Thus by Ratio test, the series converges for $\frac{1}{2x} > 1$, i.e., for $x < \frac{1}{2}$ and diverges for $x > \frac{1}{2}$. But it fails for $x = \frac{1}{2}$. ∴ Let us try the Raabe's test.

Now $$\frac{u_n}{u_{n+1}} = \left(1 + \frac{1}{n}\right)\left(1 + \frac{4}{3n}\right)^{-1} \qquad \text{[Expand by Binomial Theorem]}$$

$$= \left(1 + \frac{4}{3n}\right)\left(1 - \frac{4}{3n} + \frac{16}{9n^2} - \ldots\right) = 1 - \frac{1}{3n} + \frac{4}{9n^2} + \ldots$$

$$\therefore \quad n\left(\frac{u_n}{u_{n+1}} - 1\right) = -\frac{1}{3} + \frac{4}{9n} + \ldots \quad \therefore \quad \lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = -\frac{1}{3} < 1 \text{ which is } < 1.$$

Thus by Raabe's test, the series diverges.

Hence the given series converges for $x < \frac{1}{2}$ and diverges for $x > \frac{1}{2}$.

(a) Here $$\frac{u_n}{u_{n+1}} = \left(\frac{n^2}{(n+1)^2}\right)^2 \frac{(2n+1)!}{(2n)!} \cdot \frac{x^{2n}}{x^{2n+2}} = \frac{(2n+1)(2n+2)}{(n+1)^2} \frac{1}{x^2} = \frac{2(2n+1)}{n+1} \frac{1}{x^2}$$

$$\therefore \quad \lim_{n\to\infty} \frac{u_n}{u_{n+1}} = \lim_{n\to\infty} \frac{2(2 + 1/n)}{1 + 1/n} \cdot \frac{1}{x^2} = \frac{4}{x^2}$$

Thus by Ratio Test, the series converges for $x^2 < 4$ and diverges for $x^2 > 4$. But fails for $x^2 = 4$.

When $x^2 = 4$, $$n\left(\frac{u_n}{u_{n+1}} - 1\right) = n\left(\frac{2n+1}{2n+2} - 1\right) = -\frac{n}{2n+2}$$

$$\therefore \quad \lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = -\frac{1}{2} < 1$$

Thus by Raabe's test, the series diverges.

Hence the given series converges for $x^2 < 4$ and diverges for $x^2 \geq 4$.

Example 2.12. Discuss the convergence of the series

$$x + \frac{2^2 x^2}{2!} + \frac{3^3 x^3}{3!} + \frac{4^4 x^4}{4!} + \frac{5^5 x^5}{5!} + \ldots \infty \qquad \text{(P.T.U., 2006; Cochin, 2008; Kerala, 2008)}$$

Solution. Here $$\frac{u_n}{u_{n+1}} = \frac{n^n x^n}{n!} \cdot \frac{(n+1)^{n+1} x^{n+1}}{(n+1)!} = \frac{n^n}{(n+1)^n} = \frac{1}{(1+1/n)^n} < 1$$

$$\therefore \quad \lim_{n\to\infty} \frac{u_n}{u_{n+1}} = \frac{1}{e} < 1$$

Thus by Ratio test, the series converges for $x < 1/e$ and diverges for $x > 1/e$. But it fails for $x = 1/e$. Let us try the log test.

Now $$\frac{u_n}{u_{n+1}} = \frac{1}{(1+1/n)^n}$$

$$\therefore \quad \log \frac{u_n}{u_{n+1}} = \log_e e - n \log\left(1 + \frac{1}{n}\right) = 1 - n\left(\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \ldots\right) = \frac{1}{2n} - \frac{1}{3n^2} + \ldots$$

$$\therefore \quad \lim_{n\to\infty}\left(n \log \frac{u_n}{u_{n+1}}\right) = \frac{1}{2} < 1, \text{ which is } < 1. \text{ Thus by the log test, the series diverges.}$$

Hence the given series converges for $x < 1/e$ and diverges for $x \geq 1/e$.

Example 2.13. Discuss the convergence of the Logarithmic series

$$\frac{x}{1^2} + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \frac{x^4}{4^2} + \ldots \infty \qquad \text{(Mumbai Univ., 2002)}$$

Solution. Neglecting the first term, we have

$$u_{n+1}=u_n\frac{(\alpha+n)(\beta+n)}{(\gamma+n)(\gamma_1+n)}x$$

$$\therefore \quad \lim_{n\to\infty}\frac{u_n}{u_{n+1}}=\lim_{n\to\infty}\frac{(\gamma+n)(\gamma_1+n)}{(\alpha+n)(\beta+n)}\cdot\frac{1}{x}=\lim_{n\to\infty}\frac{(1+\gamma/n)(1+\gamma_1/n)}{(1+\alpha/n)(1+\beta/n)}\cdot\frac{1}{x}=\frac{1}{x}$$

$\therefore$ by Ratio test, the series converges for $1/x>1$, i.e., for $x<1$, and diverges for $x>1$. But it fails for $x=1$.

$\therefore$ let us try the Raabe's one.

$$\lim_{n\to\infty}n\left\{\frac{u_n}{u_{n+1}}-1\right\}=\lim_{n\to\infty}n\left\{\frac{(n+\gamma)(n+\gamma_1)}{(n+\alpha)(n+\beta)}-1\right\}=\lim_{n\to\infty}n\left\{\frac{n(\gamma+\gamma_1-\alpha-\beta)+\gamma\gamma_1-\alpha\beta}{n^2+n(\alpha+\beta)+\alpha\beta}\right\}$$

$$=\lim_{n\to\infty}\left\{\frac{(\gamma+\gamma_1-\alpha-\beta)+(\gamma\gamma_1-\alpha\beta)\frac{1}{n}}{1+(\alpha+\beta)\frac{1}{n}+\alpha\beta\frac{1}{n^2}}\right\}=\gamma+\gamma_1-\alpha-\beta$$

Thus the series converges for $\gamma+\gamma_1-\alpha-\beta>1$, i.e., for $\gamma_1>\alpha+\beta$ and diverges for $\gamma_1<\alpha+\beta$. But it fails for $\gamma_1=\alpha+\beta$. Hence u_n/u_{n+1} does not involve n as an exponent or a logarithm. The series $\sum u_n$ diverges for $\gamma_1=\alpha+\beta$.

Hence the series converges for $x<1$ and diverges for $x>1$. When $x=1$, the series converges for $\gamma_1>\alpha+\beta$ and diverges for $\gamma_1\le\alpha+\beta$.

PROBLEMS

Test the following series for convergence :

1. $\frac{x}{1.2}+\frac{2}{3}\cdot\frac{x^2}{4}+\frac{3}{5}\cdot\frac{x^3}{6}+\frac{4}{7}\cdot\frac{x^4}{8}+\ldots$ $(x>0)$ (Kharagpur, 2008)

2. $\frac{x}{1.2}+\frac{x^2}{2.3}+\frac{x^3}{3.4}+\frac{x^4}{4.5}+\ldots$ (U.P.T.U., 2006; J.N.T.U., 2008)

3. $1+\frac{1}{2}x+\frac{1.3}{2.4}x^2+\frac{1.3.5}{2.4.6}x^3+\ldots$ $(x>0)$ (Nagpur, 2003)

4. $1+\frac{2}{3}x+\frac{4.5}{3.6}x^2+\frac{6.7.8}{3.6.9}x^3+\ldots$ (U.P.T.U., 2005-06)

5. $1+\frac{2}{3}x+\frac{2.4}{3.5}x^2+\frac{2.4.6}{3.5.7}x^3+\ldots$

6. $1+\frac{2}{3.4}x+\frac{2.5}{4.7}x^2+\frac{2.5.8}{4.7.10}x^3+\frac{2.5.8.11}{4.7.10.13}x^4+\ldots$

7. $\frac{x}{1.3}+\frac{x^2}{2.4}+\frac{1.2}{3.5}\frac{x^3}{6}+\frac{1.2.3}{4.6.8}x^4+\ldots$ $(x>0)$ (U.P.T.U., 2006; Nagpur, 2006)

8. $1+\frac{1}{2}\frac{x^2}{4}+\frac{1.3}{2.4.6}x^4+\frac{1.3.5}{2.4.6.8}\frac{x^6}{10}+\ldots$ (Nagpur, 2006; Kharagpur, 2009)

9. $1+\frac{(1!)^2}{2!}x+\frac{(2!)^2}{4!}x^2+\frac{(3!)^2}{6!}x^3+\ldots$ $(x>0)$

10. $\frac{1^2}{1^1}+\frac{(1.2)^2}{2^2}+\frac{3^2}{3^3}+\frac{(1.2.3)^2}{4^4}+\ldots$

11. $\frac{a+x}{1}+\frac{(a+2x)^2}{2!}+\frac{(a+3x)^3}{3!}+\ldots$

12. $e^2+\log 2+e^3+\log 3+e^4+\log 4+\ldots$

13. $\frac{1}{1^2}+\frac{1+2}{1^2+2^2}+\frac{1+2+3}{1^2+2^2+3^2}+\ldots$ (U.P.T.U., 2005)

14. $1+\frac{a}{b}+\frac{a(a+1)}{b(b+1)}x+\frac{a(a+1)(a+2)}{b(b+1)(b+2)}x^2+\ldots$ $(a,b>0,x>0)$

3.11 CAUCHY'S ROOT TEST*

In a positive series Σu_n if $\lim_{n\to\infty} (u_n)^{1/n} = \lambda$,

then the series converges for $\lambda < 1$, and diverges for $\lambda > 1$.

Case (i) When $\lim_{n\to\infty} (u_n)^{1/n} = \lambda < 1$.

By definition of a limit, we can find a positive number r ($\lambda < r < 1$) such that

$$(u_n)^{1/n} < r \text{ for all } n > m, \quad \text{or} \quad u_n < r^n \text{ for all } n > m.$$

Since $r < 1$, the geometric series Σr^n is convergent. Hence, by comparison test, Σu_n is also convergent.

Case (ii) When $\lim_{n\to\infty} (u_n)^{1/n} = \lambda > 1$.

By definition of a limit, we can find a number m, such that

$$(u_n)^{1/n} > 1 \text{ for all } n > m, \quad \text{or} \quad u_n > 1 \text{ for all } n > m.$$

Omitting the first m terms, let the series be $u_1 + u_2 + u_3 + \ldots$ so that $u_1 > 1, u_2 > 1, u_3 > 1$ and so on.

$$\therefore \quad u_1 + u_2 + u_3 + \ldots + u_n + \ldots \text{ and } \lim_{n\to\infty} (u_1 + u_2 + \ldots + u_n) \neq 0$$

Hence the series Σu_n is divergent.

Note. Cauchy's root test fails when $\lambda = 1$.

Example 3.14. Test for convergence of the series

(i) $\Sigma \frac{n^n}{n!}$ (ii) $\Sigma (\log n)^{-n}$ (iii) $\Sigma \left(\frac{n+1}{n}\right)^{n^2}$ (S.V.U. 2000; Kurukshetra 2005)

Solution. (i) We have $u_n = n^n/n!$.

$$\therefore \quad \lim_{n\to\infty} (u_n)^{1/n} = \lim_{n\to\infty} \left[\frac{n^n}{n!}\right]^{1/n} = \lim_{n\to\infty} \frac{(n^n)^{1/n}}{n} = \frac{1}{e} < 1 \qquad \left[\because \lim_{n\to\infty} n^{1/n} = 1\right]$$

Hence the given series converges by Cauchy's root test.

(ii) Here $u_n = (\log n)^{-n}$

$$\therefore \quad \lim_{n\to\infty} (u_n)^{1/n} = \lim_{n\to\infty} (\log n)^{-1} = 0 < 1 \qquad [\because \lim_{n\to\infty} \log n = \infty]$$

Hence, by Cauchy's root test, the given series converges.

(iii) Here $u_n = \left(\frac{n+1}{n}\right)^{n^2}$

$$\therefore \quad (u_n)^{1/n} = \left[\frac{1}{\left(\frac{n+1}{n}\right)^{n^2}}\right]^{1/n} = \frac{1}{\left(\frac{n+1}{n}\right)^n}$$

$$\therefore \quad \lim_{n\to\infty} (u_n)^{1/n} = \lim_{n\to\infty} \frac{1}{\left(\frac{n+1}{n}\right)^n} = \frac{1}{e} < 1, \text{ which is } < 1. \text{ Hence the given series is convergent.}$$

Example 3.15. Discuss the nature of the following series

(i) $\frac{x}{y} + \frac{2}{3}\left(\frac{x}{y}\right)^2 + \left(\frac{3}{5}\right)\left(\frac{x}{y}\right)^3 + \ldots$ (S.V.T.U. 2000)

(ii) $\Sigma \frac{(n-1)^{n-1} x^n}{n^{n+1}}$

(iii) $\left(\frac{x}{y}\right) + \left(\frac{x+1}{y+1}\right)^4 + \left(\frac{x+2}{y+2}\right)^9 + \left(\frac{x+3}{y+3}\right)^{16} + \ldots$ (V.T.U. 2006)

*See footnote p. 164.

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Solution. (i) After leaving the first term, we find that $u_n = \left(\frac{n+1}{n+2}\right)^{n^2} x^n$, so that

$$\lim_{n\to\infty} u_n^{1/n} = \lim_{n\to\infty} \left(\frac{1+1/n}{1+2/n}\right)^n x = x$$

$\therefore$ By Cauchy's root test, the given series converges for $x < 1$ and diverges for $x > 1$.

When $x = 1$, $u_n = \left(\frac{n+1}{n+2}\right)^{n^2} = \frac{1}{\left(1+\frac{1}{n+1}\right)^{n^2}} = \frac{1}{\left(1+\frac{1}{n+1}\right)^{n+1}} \left(1+\frac{1}{n+1}\right)^{-n}$

$\therefore$ $\lim_{n\to\infty} u_n = \frac{1}{e} \neq 0$. Since u_n does not tend to zero, Σu_n is divergent.

Thus the given series converges for $x < 1$ and diverges for $x \geq 1$.

(ii) Here $u_n^{1/n} = \frac{n+2}{n^{1+1/n}} x$

$\therefore$ $\lim_{n\to\infty} (u_n)^{1/n} = \lim_{n\to\infty} \frac{n+2}{n} \cdot \frac{1}{n^{1/n}} x = \lim_{n\to\infty} \left(1+\frac{2}{n}\right)\left(\frac{1}{n^{1/n}}\right) x = x$ $\left[\because \lim_{n\to\infty} n^{1/n} = 1\right]$

$\therefore$ The given series converges for $x < 1$ and diverges for $x > 1$.

When $x = 1$, $u_n = \frac{(n+2)^n}{n^{n+1}} = \frac{1}{n}\left(1+\frac{2}{n}\right)^n$

Taking $v_n = \frac{1}{n}$, $\lim_{n\to\infty} \left(\frac{u_n}{v_n}\right) = \lim_{n\to\infty} \left(1+\frac{2}{n}\right)^n = e^2 \neq 0$ and finite.

$\therefore$ By comparison test both Σu_n and Σv_n behave alike.

But $\Sigma v_n = \Sigma \frac{1}{n}$ is divergent $(\because\ p = 1)$ $\therefore$ Σu_n also diverges. Hence the given series converges for $x < 1$ and diverges for $x \geq 1$.

(iii) Here $u_n = \left[\frac{(n+1)^{n+1}}{n^{n+2}} - \frac{n+1}{n}\right]^n$

$\therefore$ $(u_n)^{1/n} = \left(\frac{n+1}{n}\right)^{-1}\left[\left(\frac{n+1}{n}\right)^{n+1} - 1\right]^{-1} = \left(1+\frac{1}{n}\right)^{-1}\left[\left(1+\frac{1}{n}\right)^{n+1} - 1\right]^{-1}$

$\therefore$ $\lim_{n\to\infty} (u_n)^{1/n} = 1 \cdot (e-1)^{-1} = \frac{1}{e-1} < 1$ $(\because\ e > 2)$

Thus the given series converges.

EXERCISE 5.3(C)

Discuss the convergence of the following series:

1. $\sum \frac{1}{n^p}$ (V.T.U., 2012)

2. $\sum \frac{1}{\log n^n}$ (V.T.U., 2012)

3. $\sum \left[\frac{n}{n+1}\right]^{n^2}$ (V.T.U., 2010)

4. $1 + \frac{2}{1!} + \frac{3^2}{2!} + \frac{4^3}{3!} + \ldots$ to ∞ (V.T.U., 2012)

5. $\sum \left[\frac{n+2}{n+3}\right]^{n^2} x^n$

6. $\sum \frac{(2n+1)x^n}{n^{n+1}}$

7. $\frac{2}{3} + \left(\frac{4}{5}\right)^2 x^2 + \left(\frac{6}{7}\right)^3 x^3 + \ldots$ to ∞ $(x > 0)$ (V.T.U., 2009)

2.15 ALTERNATING SERIES

(1) Definition. An alternating series is one in which the terms are alternately positive or negative is called an alternating series.

(2) Leibnitz's series. An alternating series $u_1 - u_2 + u_3 - u_4 + \dots$ is convergent if its each term is numerically less than its preceding term, and $\lim_{n\to\infty} u_n = 0$.

i.e. if $\lim_{n\to\infty} u_n \neq 0$, the given series is not convergent.

The given series is $\quad u_1 - u_2 + u_3 - u_4 + \dots$

Suppose $\quad u_1 > u_2 > u_3 > u_4 \dots > u_{n-1} > \dots$ ...(1)

and $\quad \lim_{n\to\infty} u_n = 0$ ...(2)

Consider the sum of $2n$ terms. It can be written as

$$s_{2n} = (u_1 - u_2) + (u_3 - u_4) + \dots + (u_{2n-1} - u_{2n}) \quad \dots(3)$$

or as $\quad s_{2n} = u_1 - (u_2 - u_3) - (u_4 - u_5) - \dots - u_{2n} \quad \dots(4)$

By virtue of (1), the expressions within the brackets in (3) and (4) are all positive.

$\therefore$ It follows from (3) that s_{2n} is positive and increases with n.

Also from (4), we note that s_{2n} always remains less than u_1.

Hence s_{2n} must tend to a finite limit s.

Moreover $\quad \lim_{n\to\infty} s_{2n+1} = \lim_{n\to\infty} (s_{2n} + u_{2n+1}) = \lim_{n\to\infty} s_{2n} + 0$ [By (2)]

Thus $\lim_{n\to\infty} s_n$ tends to the same finite limit whether n is even or odd.

Hence the given series is convergent.

When $\quad \lim_{n\to\infty} u_n \neq 0, \lim_{n\to\infty} s_{2n} = \lim_{n\to\infty} s_{2n-1}$ $\therefore$ The given series is oscillatory.

Example 2.18. Discuss the convergence of the series

(i) $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$ (ii) $\frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \dots$

(iii) $\frac{1}{\log 2} - \frac{1}{\log 3} + \frac{1}{\log 4} - \frac{1}{\log 5} + \dots$ (V.T.U., 2010)

Solution. (i) The terms of the given series are alternately positive and negative; each term is numerically less than its preceding term $\left[\because u_n - u_{n-1} = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n-1}} < 0\right]$

Also $\lim_{n\to\infty} u_n = \lim_{n\to\infty} (1/\sqrt{n}) = 0$. Hence by Leibnitz's rule, the given series is convergent.

(ii) The terms of the given series are alternately positive and negative and

$$u_n - u_{n-1} = \frac{2n+2}{2n} - \frac{2n+1}{2n-2} = \frac{-3}{4n(n-1)} < 0 \text{ for } n > 1.$$

i.e., $\quad u_n < u_{n-1}$ for $n > 1$. Also $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{2n+2}{2n} = 1 \neq 0$

Hence by Leibnitz's rule, the given series is oscillatory.

(iii) The terms of the given series are alternately positive and negative.

Also $n + 2 > n + 1$, *i.e.*, $\log(n+2) > \log(n+1)$

i.e., $\quad \frac{1}{\log(n+2)} < \frac{1}{\log(n+1)}$, *i.e.*, $u_{n+1} < u_n$.

and $\quad \lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{\log(n+1)} = 0$

Hence the given series is convergent.

Example 9.17: Examine for convergence of the series

(i) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} n}{2n-1}$ (ii) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n(n+1)}$, $0 < x < 1$

Solution. (i) The terms of the given series are alternately positive and negative, each term is numerically less than its preceding term.

$$\left[\because u_n = u_{n-1} \text{ for } \frac{n}{2n-1} < \frac{n-1}{2n-3} \text{ for } \frac{-1}{(2n-1)(2n-3)} < 0\right]$$

But $\lim_{n \to \infty} u_n = \lim_{n \to \infty} \frac{n}{2n-1} = \lim_{n \to \infty} \frac{1}{2-1/n} = \frac{1}{2}$ which is not zero.

Hence the given series is oscillatory.

(ii) The terms of the given series are alternately positive and negative

$$u_n - u_{n-1} = \frac{x^n}{n(n+1)} - \frac{x^{n-1}}{(n-1)n} = \frac{x^{n-1}[x-(n+1)]}{n(n-1)(n+1)} < 0 \quad \text{for } n \geq 2 \qquad (\because 0 < x < 1)$$

i.e., $u_n < u_{n-1}$ for $n \geq 2$. Also $\lim_{n \to \infty} u_n = \lim_{n \to \infty} \frac{x^n}{n(n+1)} = 0$ $(\because 0 < x < 1)$

Hence the given series is convergent.

EXERCISE 9.3(C)

Examine the convergence of the following series :

1. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots$ (P.T.U. 2003)
2. $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots$ (V.T.U. 2010)
3. $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ (Indore 2005)
4. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{n^2+1}$
5. $\frac{1}{1 \cdot 2} - \frac{1}{3 \cdot 4} + \frac{1}{5 \cdot 6} - \frac{1}{7 \cdot 8} + \ldots$ (Kanpur 2003)
6. $\frac{1}{3} - \frac{2}{10} + \frac{3}{30} - \frac{4}{68} + \frac{5}{208} - \ldots$
7. $1 - 2x + 3x^2 - 4x^3 + \ldots$ $(x < \frac{1}{2})$ (Indore 2002)
8. $\sum_{n=1}^{\infty} \frac{\sin n}{n^2+1}$
9. $\frac{x}{1+x} - \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} - \frac{x^4}{1+x^4} + \ldots$ $(x > 0, x \neq 1)$ (P.T.U. 2004; Delhi 2005)
10. $\left(\frac{1}{2} - \frac{1}{\log 2}\right) - \left(\frac{1}{3} - \frac{1}{\log 3}\right) + \left(\frac{1}{4} - \frac{1}{\log 4}\right) - \left(\frac{1}{5} - \frac{1}{\log 5}\right) + \ldots$

9.11 SERIES OF POSITIVE AND NEGATIVE TERMS

The series of positive terms and the alternating series are special types of these series with arbitrary signs.

Def. (i) If the series of arbitrary terms $u_1 + u_2 + u_3 + \ldots + u_n + \ldots$ be such that the series $|u_1| + |u_2| + |u_3| + \ldots + |u_n| + \ldots$ is convergent, then the series Σu_n is said to be **absolutely convergent.**

(ii) If $\Sigma |u_n|$ is divergent but Σu_n is convergent, then Σu_n is said to be **conditionally convergent.**

For instance, the series $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \ldots$ is absolutely convergent, since the series

$1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \ldots$ is known to be convergent.

Again, since the alternating series $1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\ldots$ is convergent, and the series of absolute values $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\ldots$ is divergent, so the original series is conditionally convergent.

Note 1 An absolutely convergent series is necessarily convergent but not conversely.

Let Σa_n be an absolutely convergent series.

Clearly $a_1, a_2, a_3, a_4, \ldots, a_n, \ldots$

$\Rightarrow |a_1|, |a_2|, |a_3|, |a_4|, \ldots, |a_n|, \ldots$ which is known to be convergent.

Hence the series Σa_n is also convergent.

Note 2 As the series $\Sigma |a_n|$ is of positive terms, the tests already established for positive term series can be applied to examine Σa_n for its absolute convergence. For instance, the nth test can be restated as follows:

The series Σa_n is absolutely convergent if $\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} < 1$,

and is divergent if $\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} > 1$. This test fails when the limit is unity.

Example 3.16. Examine the following series for convergence:

(i) $1+\frac{2}{2^2}-\frac{3}{3^2}+\frac{4}{4^2}+\frac{5}{5^2}+\frac{6}{6^2}-\frac{7}{7^2}+\frac{8}{8^2}+\ldots$ (W.B.U. 2006)

(ii) $\frac{1}{2^2}-\frac{1}{3^2}(1+2)+\frac{1}{4^2}(1+2+3)-\frac{1}{5^2}(1+2+3+4)+\ldots$

Solution. (i) The series of absolute terms is $1+\frac{2}{2^2}+\frac{3}{3^2}+\frac{4}{4^2}+\frac{5}{5^2}+\frac{6}{6^2}+\ldots$ which is evidently convergent.

$\therefore$ The given series is absolutely convergent and hence it is convergent.

Let Here $a_n = (-1)^{n-1}\frac{(1+2+3+\ldots+n)}{(n+1)^2}$

$= (-1)^{n-1}\frac{n(n+1)}{2(n+1)^2} = (-1)^{n-1}\frac{n}{2(n+1)} = (-1)^{n-1}u_n$ (say).

Then $u_n - u_{n+1} = \frac{1}{2}\left[\frac{n}{(n+1)^2} - \frac{n+1}{(n+2)^2}\right] = \frac{1}{2}\frac{n^2+n-1}{(n+1)^2(n+2)^2} > 0$.

i.e., $u_{n+1} < u_n$. Also $\lim_{n\to\infty} u_n = \frac{1}{2}\lim_{n\to\infty}\frac{n}{(n+1)^2} = 0$.

Thus by Leibnitz's rule, Σa_n and therefore Σa_n is convergent.

Also $|a_n| = \frac{1}{2}\frac{n}{n^2+1}$. Taking $v_n = \frac{1}{n}$, we note that

$$\lim_{n\to\infty}\frac{|a_n|}{v_n} = \frac{1}{2}\lim_{n\to\infty}\frac{n^2}{n^2+1} = \frac{1}{2} \neq 0$$

Since Σv_n is divergent, therefore $\Sigma |a_n|$ is also divergent.

i.e., Σa_n is convergent but $\Sigma |a_n|$ is divergent.

Thus the given series Σa_n is conditionally convergent.

Example 3.17. Find whether the following series are absolutely convergent or not:

(i) $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n-1}$ (ii) $\sum_{n=1}^{\infty}\frac{(-1)^n}{n(\log n)^2}$

Solution. (i) Given series is $\Sigma a_n = 1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\ldots$

This is an alternating series of which terms u_n are decreasing and $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n(n+1)} = 0$

$\therefore$ by Leibnitz's rule, $\sum u_n$ converges.

The series of absolute terms is $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots \infty$

Here $u_n = \frac{1}{n(n+1)}$. Taking $v_n = \frac{1}{n}$, we have

$$\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \left[\frac{n}{n(n+1)}\right] = \lim_{n\to\infty} \left(\frac{1}{n+1}\right) = \frac{1}{\infty} = 0 \text{ and finite.}$$

$\therefore$ by Comparison test, $\sum u_n$ diverges $\Rightarrow$ $\sum v_n$ diverges.

Hence the given series converges and the series of absolute terms diverges, therefore the given series converges conditionally.

(ii) The terms of given series are alternately positive and negative. Also each term is numerically less than the preceding term and $\lim_{n\to\infty} |u_n| = \lim_{n\to\infty} \frac{1}{\log n^2} = 0$.

$\therefore$ by Leibnitz's rule, the given series converges.

Also $$\int_2 \frac{dx}{x(\log x)^2} = \left[-\frac{1}{\log x}\right]_2^\infty = \frac{1}{\log 2} = 0 \text{ and finite.}$$

i.e., the series of absolute terms converges.

Hence, the given series converges absolutely.

2.12 POWER SERIES

(i) **Def.** A series of the form $a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n + \ldots$ ...(1)

where $a_0, a_1, a_2, \ldots$ are independent of x, is called a power series in x. Such a series may converge for some or all values of x.

(ii) **Interval of convergence**

In the power series (1), $u_n = a_n x^n$.

$$\therefore \lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{a_{n+1} x^{n+1}}{a_n x^n} = \lim_{n\to\infty} \left[\frac{a_{n+1}}{a_n}\right] \cdot x$$

If $\lim_{n\to\infty} \left[\frac{a_{n+1}}{a_n}\right] = l$, then by Ratio test, the series (1) converges, when it is numerically less than 1, i.e., when $|x| < 1/l$ and diverges for other values.

Thus the power series (1) has an interval $-1/l < x < 1/l$ within which it converges and diverges for values outside this interval. Such an interval is called the **interval of convergence of the power series**.

Example 2.20. Show the values of x for which the following series converge:

(i) $1 - \frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \frac{x^4}{5} - \ldots \infty$ (ii) $\frac{1}{1!} + \frac{x}{2!} + \frac{x^2}{3!} + \frac{x^3}{4!} + \ldots \infty$

Solution. (i) Here $u_n = (-1)^{n-1} \frac{x^n}{n}$ and $u_{n+1} = (-1)^n \frac{x^{n+1}}{n+1}$

$$\therefore \frac{u_{n+1}}{u_n} = -\frac{n}{n+1} x \quad \text{and} \quad \lim_{n\to\infty} \left|\frac{u_{n+1}}{u_n}\right| = \left(\lim_{n\to\infty} \frac{1}{1+1/n}\right) |x| = |x|$$

$\therefore$ by Ratio test the given series converges for $|x| < 1$ and diverges for $|x| > 1$.

Let us examine the series for $x = 2$.

For $x = 2$, the series reduces to $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \ldots$

which is an alternating series and is convergent.

For $x = -2$, the series becomes $-\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \ldots\right)$

which is a divergent series as can be seen by comparison with p-series where $p = 1$.

Hence the given series converges for $-2 < x \leq 2$.

20) Here $a_n = \dfrac{1}{n(1-x)^n}$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{1}{(n+1)(1-x)^{n+1}} \cdot n(1-x)^n\right| = \left|\frac{1}{1-x}\right| \lim_{n\to\infty}\frac{n}{n+1} = \left|\frac{1}{1-x}\right|$$

By Ratio test, $\sum a_n$ converges for $\left|\frac{1}{1-x}\right| < 1$, i.e., $|1-x| > 1$

i.e., $1 - x < -1$ or $1 - x > 1$ or $x < 0$ and $x > 2$.

Let us examine the series for $x = 0$ and $x = 2$.

For $x = 0$, the given series becomes $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots + \frac{1}{n}$ which is a divergent harmonic series.

For $x = 2$, the given series becomes $-1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \ldots + \frac{(-1)^n}{n} + \ldots$

It is an alternating series which is convergent by Leibnitz's rule

$\because \quad a_n < a_{n-1}$ for all n and $\lim_{n\to\infty} a_n = 0$

Hence the given series converges for $x < 0$ and $x \geq 2$.

Example 9.21. Test for convergence $\frac{1}{\sqrt{2}} - \frac{2}{\sqrt{3}} + \frac{3}{\sqrt{4}} - \ldots$ for absolute convergence and conditional convergence. (U.P.T.U., 2010)

Solution. We have $u_n = (-1)^{n-1}\dfrac{n}{\sqrt{(2n+1)}}$ and $u_{n+1} = \dfrac{(-1)^n (n+1)}{\sqrt{(2n+3)}}$

$$\therefore \quad \lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^n (n+1)}{\sqrt{(2n+3)}} \cdot \frac{\sqrt{(2n+1)}}{(-1)^{n-1} n}\right| = \lim_{n\to\infty}\left|(-1)\sqrt{\left(\frac{2n+1}{2n+3}\right)} \cdot \frac{n+1}{n}\right|$$

$$= \lim_{n\to\infty}\left|\sqrt{\left(\frac{2+1/n}{2+3/n}\right)} \cdot \left(1+\frac{1}{n}\right)\right| = 1 \cdot 1 = 1$$

Hence the given series is absolutely convergent for $|x| < 1$ and is divergent for $|x| > 1$ and the test fails for $|x| = 1$.

Now $x = 1$, $\qquad u_n = \dfrac{(-1)^{n-1}}{\sqrt{(2n+1)}}$ Since $2n+1 < 2n+3$ or $\sqrt{(2n+1)} < \sqrt{(2n+3)}$ or $\frac{1}{\sqrt{(2n+1)}} > \frac{1}{\sqrt{(2n+3)}}$

i.e., $\qquad u_n > u_{n+1}$ Also $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{\sqrt{(2n+1)}} = 0$,

$\therefore$ the series is convergent by Leibnitz's test.

But $\qquad \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{5}} + \frac{1}{\sqrt{7}} + \ldots$ here $u_n = \frac{1}{\sqrt{(2n+1)}} > \frac{1}{\sqrt{n}} \cdot \frac{1}{\sqrt{(2+1/n)}}$

On comparing it with $u_n = \frac{1}{\sqrt{n}}$, Σu_n is divergent.

Hence the given series is conditionally convergent for $x = 2$.

For $x = -1$, the series becomes $-\left(\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \ldots\right)$

But we have seen that the series $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \ldots$ is divergent.

Hence, the given series is divergent when $x = -1$.

9.65 (L) CONVERGENCE OF EXPONENTIAL SERIES

The series $1 + x + \frac{x^2}{|2|} + \ldots + \frac{x^n}{|n|} + \ldots$ is convergent for all values of x. (U.P.T.U., 2006)

Here $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \left|\frac{x^n}{|n|} \cdot \frac{|n-1|}{x^{n-1}}\right| = \lim_{n\to\infty} \frac{x}{n} = 0$

Hence the series converges, whatever be the value of x.

(2) Convergence of logarithmic series

The series $x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{n-1}\frac{x^n}{n} + \ldots$ is convergent for $-1 < x \leq 1$.

Here $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{(-1)^{n+1}x^{n+1}}{n+1} \cdot \frac{n}{(-1)^{n-1}x^n} = -x \lim_{n\to\infty} \frac{n}{n+1} = -x \lim_{n\to\infty} \left(\frac{1}{1+1/n}\right) = -x$

Hence the series converges for $|x| < 1$ and diverges for $|x| > 1$.

When $x = 1$, the series being $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots$, is convergent.

When $x = -1$, the series being $-\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots\right)$ is divergent.

Hence the series converges for $-1 < x \leq 1$.

(3) Convergence of binomial series

The series $1 + nx + \frac{n(n-1)}{|2|}x^2 + \ldots + \frac{n(n-1)\ldots(n-r+1)}{|r|}x^r + \ldots \infty$

converges for $|x| < 1$.

Here $u_r = \frac{n(n-1)\ldots(n-r)}{|r-1|}x^{r-1}$ and $u_{r+1} = \frac{n(n-1)\ldots(n-r+1)}{|r|}x^r$

$\therefore \quad \lim_{r\to\infty} \frac{u_{r+1}}{u_r} = \lim_{r\to\infty} \frac{n-r+1}{r}x = \lim_{r\to\infty}\left(\frac{n+1}{r} - 1\right)x = -x$ for $r > n+1$

Hence, the series converges for $|x| < 1$.

EXERCISE 9.D

1. Test the following series for conditional convergence: (i) $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$ (ii) $\sum \frac{(-1)^{n-1}n}{n^2+1}$

2. Prove that the series $\frac{\sin x}{1^2} + \frac{\sin 2x}{2^2} + \frac{\sin 3x}{3^2} + \ldots$ converges absolutely. (Kerala, 2003)

3. Test the following series for conditional convergence:

(i) $1 - \frac{1}{2^p} + \frac{1}{3^p} - \frac{1}{4^p} + \ldots$ (ii) $1 + \frac{x}{2} + \frac{1\cdot 2}{2\cdot 4}x^2 + \frac{1\cdot 2\cdot 3}{2\cdot 4\cdot 6}x^3 + \ldots$

4. Discuss the absolute convergence of $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n+1}$ (Bristol, 2005)

5. $\frac{x}{2} - \frac{x^2}{3} + \frac{x^3}{3^2} + \cdots$ $\frac{1}{(x+1)} - \frac{2}{(2x+1)} + \frac{3}{(3x+2)} - \cdots$

6. Find the values of x for which $\frac{x}{1\cdot 2} - \frac{x^2}{2\cdot 3} + \frac{x^3}{3\cdot 4} - \frac{x^4}{4\cdot 5} + \cdots$ (UWIST, 1994)

7. For what values of x are the following series convergent?

$\frac{x^2}{\sqrt{2}} + \frac{x^3}{\sqrt{3}} + \frac{x^4}{\sqrt{4}} + \cdots$ (UWIST, 2000; Oxford, 2004)

$\frac{x}{2^2} + \frac{x^2}{3^2} + \frac{x^3}{4^2} + \cdots$

8. Find the radius of convergence of the series $\sum \frac{n!}{n^n} x^n$ (Oxford, 2003)

9. Show that $\frac{1}{x} + \frac{1}{x+1} - \frac{1}{x+2} - \frac{1}{x+3} + \frac{1}{x+4} + \frac{1}{x+5} - \cdots$ is a divergent series.

10. Test the series $1 - \frac{1}{3\sqrt{2}} + \frac{1}{5\sqrt{3}} - \frac{1}{7\sqrt{4}} + \cdots$ for

(a) absolute convergence and (b) conditional convergence. (UCE, 2003; Oxford, 2003)

2.12 PROCEDURE FOR TESTING A SERIES FOR CONVERGENCE

First, see whether the given series is

(a) a series with terms alternately positive and negative;

(b) a series of positive terms excluding power series;

or (c) a power series.

For alternating series (a), apply the Leibniz rule (§ 9.13).

For series (b), find u_n and if possible evaluate $\lim_{n\to\infty} u_n$. If $\lim_{n\to\infty} u_n \neq 0$, the series is divergent. If $\lim_{n\to\infty} u_n = 0$, compare $\sum u_n$ with $\sum 1/n^p$ and apply the comparison test (§ 9.8).

If the comparison tests are not applicable, apply the Ratio test (§ 9.10). If $\lim_{n\to\infty} u_{n+1}/u_n = 1$, the ratio test fails, apply Raabe's test (§ 9.11). If Raabe's test fails for a similar reason, apply Logarithmic test (§ 9.30). If this also fails, apply Gauss's test (§ 9.31).

For the power series (c), apply the Cauchy test or its (§ 9.28). If the Ratio test fails, examine the series as in case (b) above.

PROBLEMS 2.9

Test for convergence of the following series:

1. $\sum_{n=1}^{\infty} \frac{n^2 - n}{n^3 + 1} x^n$ $(x > 0)$ (Dundalk, 1995)

2. $1 + \frac{2}{2^2} + \frac{3}{3^2} + \frac{4}{4^2} + \cdots$

3. $\frac{1}{1+\sqrt{2}} - \frac{2}{1+2\sqrt{3}} + \frac{3}{1+3\sqrt{4}} - \cdots$

4. $1 + \frac{x}{2!} + \frac{2!x^2}{3^2} + \frac{3!x^3}{4^3} + \cdots$

5. $\sum_{n=1}^{\infty} \frac{n!}{(n+1)^n}$

6. $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \sqrt{\frac{n}{n+1}} \right)$

7. $\sum_{n=1}^{\infty} \sqrt{\left| \frac{n+1}{n^2+1} \right|}$

8. $\frac{1}{1+1^2} + \frac{2}{1+2^2} + \frac{3}{1+3^2} + \cdots$

9. $\sum_{n=1}^{\infty} \frac{n+1}{n^2+100n+1}$ $(x > 0)$

10. $\sum_{n=1}^{\infty} \frac{n^n}{(2n)!}$

11. $\sum_{n=1}^{\infty} \frac{n^n}{(an+b)^n}$

(i) $\sum_{n=1}^{\infty} \frac{(2n+1)^n}{(n+2)!}$ (ii) $\sum_{n=1}^{\infty} \frac{(n+1)^n}{n^n}$

(iii) $\sum_{n=1}^{\infty} \frac{n+10^n}{n^{10}+n}$ (iv) $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot 1 \cdot \sin n}{n^2}$

(v) $\frac{\sqrt{2}+1}{2^2-1}, \frac{\sqrt{3}+1}{3^2-1}, \frac{\sqrt{4}+1}{4^2-1}, \ldots$ (vi) $\sum_{n=1}^{\infty} \frac{1}{n \log n (\log \log n)^2}$

9.17 UNIFORM CONVERGENCE

Let $s_n(x) = u_1(x) + u_2(x) + \ldots + u_n(x) = \sum_{k=1}^{n} u_k(x)$ ...(1)

be an infinite series of functions each of which is defined in the interval $[a, b]$. Let $s_n(x)$ be the sum of its first n terms, i.e., $s_n(x) = u_1(x) + u_2(x) + \ldots + u_n(x)$.

At some point $x = x_1$, if $\lim_{n \to \infty} s_n(x_1) = s(x_1)$,

then the series (1) is said to converge to sum $s(x_1)$ at that point. This means that for a given positive number ε, we can find a number N such that $|s(x_1) - s_n(x_1)| < \varepsilon$ for $n > N$ ...(2)

Evidently N will depend on ε and generally it will also depend on x_1. Now if we keep the same ε but take some other value x_2 of x for which (1) is convergent, then we may have to change N for the inequality (2) to hold. If we wish to approximate the sum $s(x)$ of the series by its partial sum $s_n(x)$, we shall require different partial sums at different points of the interval and the problem will become quite complicated. If however, we choose an N which is independent of the values of x, the problem becomes simpler. Then the partial sums $s_n(x)$ ($n \geq N$) approximate to $s(x)$ for all values of x in the interval $[a, b]$ and it is uniform throughout this interval. This leads to

Definition. The series $\sum u_n(x)$ is said to be uniformly convergent in the interval $[a, b]$, if for a given $\varepsilon > 0$, a number N can be found independent of x, such that for every x in the interval $[a, b]$,

$$|s(x) - s_n(x)| < \varepsilon \text{ for all } n > N.$$

Example 9.21. Examine the geometric series $1 + x + x^2 + \ldots + x^{n-1} + \ldots$ for uniform convergence in the interval $[-k, k]$, $k < 1$.

Solution. We have $s_n(x) = 1 + x + x^2 + \ldots + x^{n-1} = \frac{1 - x^n}{1 - x}$,

and $s(x) = \lim_{n \to \infty} \frac{1 - x^n}{1 - x} = \frac{1}{1 - x}$ for $|x| < 1$

$\therefore \quad |s(x) - s_n(x)| = \left| \frac{x^n}{1 - x} \right| = \frac{|x^n|}{1 - x} < \frac{|x|^n}{1 - x}$ which will be $< \varepsilon$, if $|x|^n < \varepsilon(1 - x)$.

Since N such that $|x|^N < \varepsilon(1 - x)$

or $N > \log \{\varepsilon(1 - x)\} / \log |x|$ ...(1)

Evidently N increases with the increase of $|x|$ and in the interval $-\frac{1}{2} \leq x \leq \frac{1}{2}$, it assumes a minimum value $N = \log \varepsilon / \log \frac{1}{2}$ at $x = -\frac{1}{2}$ for a given ε.

Then $|s(x) - s_n(x)| < \varepsilon$ for all $n > N$ for every value of x in the interval $[-\frac{1}{2}, \frac{1}{2}]$.

Hence the geometric series converges uniformly in the interval $[-\frac{1}{2}, \frac{1}{2}]$.

Obs. The geometric series though convergent in the interval $(-1, 1)$, is not uniformly convergent in this interval, as we cannot find a fixed number N however x in this interval

$\left[\because N \text{ given by } (1) \to \infty \text{ as } x \to \pm 1 \right]$

8.12 WEIERSTRASS'S M-TEST*

A series $\sum u_n(x)$ is said to be uniformly convergent in an interval $[a, b]$, if there exists a convergent series $\sum M_n$ of positive constants such that $|u_n(x)| \le M_n$ for all values of x in (a, b).

Since $\sum M_n$ is convergent, therefore, for a given $\varepsilon > 0$, we can find a number N, such that $|s - s_n| < \varepsilon$ for every $n > N$,

where $s = M_1 + M_2 + \ldots + M_n + M_{n+1} + \ldots$ and $s_n = M_1 + M_2 + \ldots + M_n$

This implies that $|M_{n+1} + M_{n+2} + \ldots| < \varepsilon$ for every $n > N$.

Since $|u_n(x)| \le M_n$

$\therefore \quad |u_{n+1}(x)| + u_{n+2}(x) + \ldots| \le |u_{n+1}(x)| + |u_{n+2}(x)| + \ldots$

$< M_{n+1} + M_{n+2} + \ldots < \varepsilon$ for every $n > N$

i.e., $|s(x) - s_n(x)| < \varepsilon$ for every $n > N$, where $s(x)$ is the sum of the series $\sum u_n(x)$

Since N does not depend on x, the series $\sum u_n(x)$ is uniformly convergent in $[a, b]$.

Obs. $\sum u_n(x)$ is also absolutely convergent for every x, when $|u_n(x)| \le M_n$

Example 8.22. Show that the following series converge uniformly in every interval:

(i) $\sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$ (Madras, 2006) (ii) $\sum \frac{1}{n^2 + n^2 x^2}$.

Solution. (i) $\left|\frac{\cos nx}{n^2}\right| = \frac{|\cos nx|}{n^2} \le \frac{1}{n^2} = M_n$ for all values of x.

Since $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent for $p > 1$,

$\therefore$ By M-test, the given series converges uniformly for all real values of x and $p > 1$.

(ii) For all values of x, $n^2 + n^2x^2 \ge n^2$

$\therefore \quad \left|\frac{1}{n^2 + n^2x^2}\right| \le \frac{1}{n^2} = M_n$ i.e. $\sum M_n$ being a series with $p = 2$, is convergent

$\therefore$ By M-test, the given series converges uniformly in any interval.

Example 8.23. Examine the following series for uniform convergence:

(i) $\sum_{n=1}^{\infty} \frac{\sin 3nx + n^3}{n^5 + 2x}$ (U.P.T.U., 2002) (ii) $\sum_{n=1}^{\infty} \frac{x}{n^2 + n^4x^2}$ (U.P.T.U., 2003 (C))

Solution. (i) $\left|\frac{\sin(3nx + n^3)}{n^5 + 2x}\right| = \frac{|\sin 3nx + n^3|}{n^5 + 2x} < \frac{1}{n^3} = M_n$ for all real x.

Since $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent, therefore, by M-test, the given series is uniformly convergent for all real values of x.

(ii) For all real values of x, $n^2 + 2n^2x^2 \ge n^2$ i.e., $n^2 + n^2x^2 \ge 0$

i.e., $n^2 + n^2x^2 \ge n^2$ or $\frac{1}{n^2 + n^2x^2} \le \frac{1}{n^2} = M_n$

Since $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent for $p > 1$,

$\therefore$ By M-test, the given series is uniformly convergent for all real values of x and $p > 1$.

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* Named after the great German mathematician Karl Weierstrass (1815–1897) who made basic contributions in Calculus, Approximation theory, Differential geometry and Calculus of variations. He was also one of the founders of Complex analysis.

5.11 PROPERTIES OF UNIFORMLY CONVERGENT SERIES

I. If the series $\sum u_n(x)$ converges uniformly to a sum $s(x)$ in the interval $[a, b]$ and each of the functions $u_n(x)$ is continuous in this interval, then the sum $s(x)$ is also continuous in $[a, b]$.

II. If the series $\sum u_n(x)$ converges uniformly in the interval $[a, b]$ and each of the functions $u_n(x)$ is continuous in this interval, then the series can be integrated term by term

i.e., $$\int_a^b [u_1(x) + u_2(x) + \dots]\,dx = \int_a^b u_1(x)\,dx + \int_a^b u_2(x)\,dx + \dots$$

III. If $\sum u_n(x)$ is a convergent series having continuous derivatives of its terms, and the series $\sum u_n(x)$ converges uniformly, then the series can be differentiated term by term

$$\frac{d}{dx}[u_1(x) + u_2(x) + \dots] = u_1'(x) + u_2'(x) + \dots$$

Example 5.11.1. Prove that $\int_0^1 \left[\sum_{n=1}^{\infty} \frac{x^n}{n^2}\right] dx = \sum_{n=1}^{\infty} \frac{1}{n^2(n+1)}$

Solution. $|x^n| < 1$ for $0 \le x < 1$

$\therefore \quad \left|\frac{x^n}{n^2}\right| < \frac{1}{n^2} = M_n$ for $0 \le x \le 1$. But $\sum M_n$ is a convergent series

$\therefore$ by *M*-test, the series $\sum (x^n/n^2)$ is uniformly convergent in $0 \le x \le 1$. Also x^n/n^2 is continuous in this interval.

$\therefore$ the series $\sum (x^n/n^2)$ can be integrated term by term in the interval $0 \le x \le 1$.

i.e., $$\int_0^1 \left[\sum \frac{x^n}{n^2}\right] dx = \sum \left(\int_0^1 \frac{x^n}{n^2}\,dx\right) = \sum \left[\frac{1}{n^2}\int_0^1 x^n\,dx\right] = \sum \frac{1}{n^2(n+1)}.$$

Exp. 5.11.2. There are convergent series which are not uniformly convergent. In fact, a series may converge absolutely for a certain interval while another series may converge uniformly in the same interval but not absolutely.

For example, the series

$\frac{1}{x^2+1} - \frac{1}{x^2+2} + \frac{1}{x^2+3} - \dots$ can be seen to converge uniformly for $|x| \le$ real any, while the series

$\frac{x^2}{x^2+1} + \frac{x^2}{(x^2+1)^2} + \frac{x^2}{(x^2+2)^2} + \dots$ can be shown to converge absolutely but not uniformly.

EXERCISE 5.11

Test for uniform convergence of the series :

1. $\sum_{n=1}^{\infty} \frac{x^n}{n^{n+1}}$ 2. $\sum \frac{\cos nx}{n^2}$

3. $\frac{\sin x}{1^3} + \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} + \frac{\sin 4x}{4^3} + \dots$ 4. $\frac{\sin x}{1^2} + \frac{\sin 2x}{2^2} + \frac{\sin 3x}{3^2} + \dots$

5. $\frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \frac{\sin 4x}{4^3} + \dots$ 6. $\frac{\sin x}{1^2} + \frac{\sin 2x}{2^2} + \frac{\sin 3x}{3^2} + \dots$

7. $\frac{x}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

8. Show that the series $\sum x^n$ in $[0, 1]$ and $\sum x^n$ on its interval of convergence $[0, 1)$ are not uniformly convergent.

9. Show that $\frac{1}{1+x^2} + \frac{1}{(1+x^2)^2} + \frac{1}{(1+x^2)^3} + \frac{1}{(1+x^2)^4} + \dots$ converges uniformly in the interval $a \le x \le b$ but not absolutely.

10. Prove that $\sum_{n=1}^{\infty} \frac{x}{n(1+nx^2)}$ is uniformly convergent in $0 \le x \le 1$ and value of x.

10. Examine the following series for convergence :

(i) $\sum_{n=1}^{\infty} \frac{1}{n^2+n+1}$ (ii) $\sum_{n=1}^{\infty} \frac{n \sin(1/n) + n^2 \tan(1/n)}{n^3 + n^2 + n}$

11. Show that

(i) $\int_0^{\infty} \left[\sum \frac{\sin x}{n}\right] dx = \frac{1}{2.3.4} + \frac{1}{3.4.5} + \frac{1}{4.5.6} + \ldots$ (ii) $\int_0^{\infty} \left[\sum \frac{\sin x}{n^2}\right] dx = \sum_{n=1}^{\infty} \frac{1}{n(n+1)^2}$

OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 11.1

Choose the correct answer or fill up the blanks in each of the following problems :

1. The series $\frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \ldots$ converges if

(a) $p > 0$ (b) $p > 1$ (c) $p \geq 1$ (d) $0 < p < 1$

2. The series $\sum_{n=1}^{\infty} (-1)^n$ converges if

(a) $-1 < x < 1$ (b) $-\frac{1}{2} < x < \frac{1}{2}$ (c) $-2 < x < 2$ (d) $-\frac{1}{2} \leq x \leq \frac{1}{2}$

3. The series $\frac{1}{2^p} - \frac{1}{3^p} + \frac{1}{4^p} - \frac{1}{5^p} + \ldots$ is

(a) absolutely convergent (b) absolutely convergent
(c) divergent (d) none of the above

4. Which one of the following series is not convergent ?

(a) $\frac{1}{\sqrt{1.2}} + \frac{1}{\sqrt{2.3}} + \frac{1}{\sqrt{3.4}} + \ldots$ (b) $\left(\frac{1}{2} - 1\right) + \left(\frac{2}{3} - 1\right) + \left(\frac{3}{4} - 1\right) + \left(\frac{4}{5} - 1\right) + \ldots$

(c) $\frac{1}{2} - \frac{1}{2} + \frac{2}{3} - \frac{2}{3} + \ldots$ (d) $x + x^2 + (x^3 - x^2) + \ldots$ where $|x| < 1$

5. The sum of the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \ldots$ is

(a) zero (b) unity (c) $\log 2$

6. Let $\{a_n\}$ be a series of positive terms. Given that $\sum a_n$ is convergent and also $\lim_{n \to \infty} \frac{a_{n+1}}{a_n} = \frac{1}{r}$ exists. Then the ratio test is

(a) inconclusive by any of $r > 1$ (b) inconclusive by any of $r < 1$
(c) may be equal to 1 or less than 1 (d) inconclusive by any of $r = 1$

7. $\sum \left[\left(1 + \frac{1}{n}\right)^n - 1\right]$

(a) convergent (b) oscillatory (c) divergent

8. $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \ldots$

(a) oscillatory (b) conditionally convergent
(c) divergent (d) absolutely convergent

9. $1+\frac{1}{2^p}+\frac{1}{3^p}+\frac{1}{4^p}+\frac{1}{5^p}+\frac{1}{6^p}+\frac{1}{7^p}+\frac{1}{8^p}+\cdots$ is

(a) not defined for any integer p (b) convergent

(c) oscillatory (d) divergent

10. $\int_1^\infty \left[\sum_{n=1}^{\infty} \frac{x^n}{n!}\right] dx =$

(a) $\sum_{n=0}^{\infty} \frac{1}{(n+1)!}$ (b) $\sum_{n=1}^{\infty} \frac{1}{n^2(n+1)}$ (c) $\sum_{n=1}^{\infty} \frac{2}{n(n+2)}$ (d) $\sum_{n=1}^{\infty} \frac{1}{n^2(n+1)}$

11. If $\sum a_n$ is a convergent series of positive terms, then $\lim_{n\to\infty} n a_n$ is

(a) 1 (b) $+\infty$ (c) 0 (d) ∞ (CDS (E), 2018)

12. Geometric series $1 + x + x^2 + \ldots + x^{n-1} + \ldots$

(a) converges in the interval ..... (b) converges uniformly in the interval .....

13. The series $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots$ converges in the interval .....

14. If $\lim_{n\to\infty} \frac{a_n}{a_{n+1}} = k$, then $\sum a_n$ converges for k .....

15. A sequence $\{a_n\}$ is said to be bounded, if there exists a number k such that for every n, $a_n \le$ .....

16. The series $1 + (x+1) + (x+2) + (x+3) + \ldots + (x+n) + \ldots$ (Converges/Diverges)

17. The series $1 + \frac{x}{11} + \frac{x^2}{21} + \frac{x^3}{31} + \frac{x^4}{41} + \ldots$ converges for .....

18. If $\lim_{n\to\infty} n\left[\frac{a_n}{a_{n+1}} - 1\right] = k$, then $\sum a_n$ converges for k .....

19. A sequence which is monotonic and bounded is .....

20. The series $\frac{1}{1.2} + \frac{2}{3.4} + \frac{3}{5.6} + \ldots$ is ..... (Converges/Diverges)

21. The series $\frac{x^{2n}}{2!} + \frac{x^{3n}}{3!} + \frac{x^{4n}}{4!} + \ldots$ converges for .....

22. The series $\frac{\sqrt{1}}{\sqrt{4}} + \frac{\sqrt{2}}{\sqrt{10}} + \frac{\sqrt{3}}{\sqrt{18}} + \ldots + n$ ..... (Converges/Diverges)

23. The series $\sqrt{\frac{n^2+1}{n^3-1}}$ is ..... (Converges/Diverges)

24. The series $1 - \frac{1}{3}(x-1) + \frac{1}{5}(x-1)^2 - \ldots + \left(-\frac{1}{2}\right)^n (x-1)^n + \ldots$ converges in the interval .....

25. Is the series $\sum_{n=1}^{\infty} \frac{n \sin^2 n}{1+n^2}$ convergent?

26. The exponential series $1 + \frac{x}{11} + \frac{x^2}{21} + \ldots + \frac{x^n}{n!} + \ldots$ is absolutely convergent. (True/False)

27. The series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \ldots + \frac{1}{n(n+1)} + \ldots$ is ..... (Convergent/Divergent/Oscillating)

28. Is the series $\sum n \sin \frac{1}{n}$ convergent?

29. The series $\sum \frac{1}{n^{1.1}}$ converges for n .....

30. The series $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$ converges uniformly when x lies in the interval .....

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CHAPTER 10

Fourier Series

1. Introduction. 2. Euler's Formulas. 3. Conditions for a Fourier expansion. 4. Functions having points of discontinuity. 5. Change of interval. 6. Odd and even functions. 7. Expansion of even or odd periodic functions. 8. Half-range series. 9. Typical wave forms. 10. Parseval's formula. 11. Complex form of Fourier series. 12. Practical harmonic analysis. 13. Physical meaning of Fourier series.

10.1 INTRODUCTION

In many engineering problems, especially in the study of periodic phenomena* in conduction of heat, electrodynamics and electrical circuits, it is necessary to express a function as a series of sines and cosines. Most of the single-valued functions which occur in applications can be expressed in this form.

$$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + \cdots + b_1 \sin x + b_2 \sin 2x + \cdots$$

within a desired range of values of the variable. Such a series is known as the **Fourier series**†.

10.2 EULER'S FORMULAE

The Fourier series for the function $f(x)$ in the interval c to $c + 2\pi$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where

$$\left.\begin{aligned} a_0 &= \frac{2}{\pi}\int_c^{c+2\pi} f(x)\,dx \\ a_n &= \frac{2}{\pi}\int_c^{c+2\pi} f(x)\cos nx\,dx \\ b_n &= \frac{2}{\pi}\int_c^{c+2\pi} f(x)\sin nx\,dx \end{aligned}\right\} \quad \ldots(1)$$

These values of a_0, a_n, b_n are known as **Euler's formulae**‡.

*Periodic Functions. If a function $f(x)$ is defined for all values of x (except perhaps at certain points) and there exists a positive number T such that $f(x + T) = f(x)$ for all x, then $f(x)$ is called a periodic function having period T; e.g., $\sin x$, $\cos x$ are periodic functions having a period 2π.

†The series $a_0/2$ instead of a_0 is a conventional device to be able to get more symmetrical formulae for the coefficients.

‡Named after the French mathematician and physicist Jean Baptiste Fourier (1768–1830) who was first to use Fourier series in his memorable work "Theorie Analytique de la Chaleur" in which he developed the theory of heat conduction. These series had a deep influence on the further development of mathematics and mathematical physics.

Ans. Refer to p. 396.

To establish these formulae, the following definite integrals will be required :

1. $\int_{\alpha}^{\alpha+2\pi} \cos nx\,dx = \left|\frac{\sin nx}{n}\right|_{\alpha}^{\alpha+2\pi} = 0$ $(n \neq 0)$

2. $\int_{\alpha}^{\alpha+2\pi} \sin nx\,dx = -\left|\frac{\cos nx}{n}\right|_{\alpha}^{\alpha+2\pi} = 0$ $(n \neq 0)$

3. $\int_{\alpha}^{\alpha+2\pi} \cos mx \cos nx\,dx$ $(n \neq 0)$

$$= \frac{1}{2}\int_{\alpha}^{\alpha+2\pi} [\cos(m+n)x + \cos(m-n)x]\,dx$$

$$= \frac{1}{2}\left|\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n}\right|_{\alpha}^{\alpha+2\pi} = 0 \qquad (m \neq n)$$

4. $\int_{\alpha}^{\alpha+2\pi} \cos^2 nx\,dx = \left|\frac{x}{2} + \frac{\sin 2nx}{4n}\right|_{\alpha}^{\alpha+2\pi} = \pi$ $(n \neq 0)$

5. $\int_{\alpha}^{\alpha+2\pi} \sin mx \cos nx\,dx = -\frac{1}{2}\left[\frac{\cos(m-n)x}{m-n} + \frac{\cos(m+n)x}{m+n}\right] = 0$ $(m \neq n)$

6. $\int_{\alpha}^{\alpha+2\pi} \sin nx \cos nx\,dx = \left|\frac{\sin^2 nx}{2n}\right|_{\alpha}^{\alpha+2\pi} = 0$ $(n \neq 0)$

7. $\int_{\alpha}^{\alpha+2\pi} \sin mx \sin nx\,dx = \frac{1}{2}\left|\frac{\sin(m-n)x}{m-n} - \frac{\sin(m+n)x}{m+n}\right|_{\alpha}^{\alpha+2\pi} = 0$ $(m \neq n)$

8. $\int_{\alpha}^{\alpha+2\pi} \sin^2 nx\,dx = \left|\frac{x}{2} - \frac{\sin 2nx}{4n}\right|_{\alpha}^{\alpha+2\pi} = \pi.$ $(n \neq 0)$

[illegible]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \qquad (1)$$

[illegible]

$$\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{2}a_0\int_{-\pi}^{\pi} dx + \int_{-\pi}^{\pi}\left(\sum_{n=1}^{\infty} a_n \cos nx\right)dx + \int_{-\pi}^{\pi}\left(\sum_{n=1}^{\infty} b_n \sin nx\right)dx$$

$$= \left[\text{illegible}\right] \qquad \text{[illegible]}$$

Hence $\quad a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$

[illegible]

$$\int_{-\pi}^{\pi} f(x)\cos mx\,dx = \frac{1}{2}a_0\int_{-\pi}^{\pi}\cos mx\,dx + \int_{-\pi}^{\pi}\left(\sum_{n=1}^{\infty} a_n \cos nx\right)\cos mx\,dx$$

$$= \int_{-\pi}^{\pi}\left(\sum_{n=1}^{\infty} a_n \cos nx\right)\cos mx\,dx \qquad \text{[illegible]}$$

Hence $\quad a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos mx\,dx$

To find b_n, multiply each side of (1) by $\sin nx$ and integrate from $x = c$ to $x = c + 2\pi$. Then,

$$\int_c^{c+2\pi} f(x)\sin nx\,dx = \frac{a_0}{2}\int_c^{c+2\pi}\sin nx\,dx + \int_c^{c+2\pi}\left(\sum_{m=1}^{\infty} a_m \cos mx\right)\sin nx\,dx + \int_c^{c+2\pi}\left(\sum_{m=1}^{\infty} b_m \sin mx\right)\sin nx\,dx$$

$$= 0 + 0 + \pi b_n \qquad \text{(By integrals (2), (3), (5), (7) and (8))}$$

Hence $$b_n = \frac{1}{\pi}\int_c^{c+2\pi} f(x)\sin nx\,dx.$$

Cor. 1. Making $c = 0$, the interval becomes $0 \le x \le 2\pi$, and the formulae (1) reduce to

$$\left.\begin{aligned} a_0 &= \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx \\ a_n &= \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx \\ b_n &= \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx \end{aligned}\right| \quad \ldots(II)$$

Cor. 2. Putting $c = -\pi$, the interval becomes $-\pi \le x \le \pi$, and the formulae (1) reduce to the form :

$$\left.\begin{aligned} a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \\ a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \end{aligned}\right| \quad \ldots(III)$$

Example 10.1. (Obtain the Fourier series for $f(x) = e^{-x}$ in the interval $0 \le x \le 2\pi$.) (U.P.T.U., 2003)

Solution. Let $$e^{-x} = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \ldots(1)$$

Then $$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx = \frac{1}{\pi}\int_0^{2\pi} e^{-x}\,dx = \frac{1}{\pi}\left[-e^{-x}\right]_0^{2\pi} = \frac{1-e^{-2\pi}}{\pi}$$

and $$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\int_0^{2\pi} e^{-x}\cos nx\,dx$$

$$= \frac{1}{\pi(n^2+1)}\left[e^{-x}(-\cos nx + n\sin nx)\right]_0^{2\pi} = \left(\frac{1-e^{-2\pi}}{\pi}\right)\frac{1}{n^2+1}$$

$$\therefore \quad a_1 = \left(\frac{1-e^{-2\pi}}{\pi}\right)\frac{1}{2}, a_2 = \left(\frac{1-e^{-2\pi}}{\pi}\right)\frac{1}{5}, \text{ etc.}$$

Finally, $$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cdot\sin nx\,dx = \frac{1}{\pi}\int_0^{2\pi} e^{-x}\sin nx\,dx$$

$$= \frac{1}{\pi(n^2+1)}\left[e^{-x}(-\sin nx - n\cos nx)\right]_0^{2\pi} = \left(\frac{1-e^{-2\pi}}{\pi}\right)\frac{n}{n^2+1}$$

$$\therefore \quad b_1 = \frac{1-e^{-2\pi}}{\pi}\cdot\frac{1}{2}, b_2 = \left(\frac{1-e^{-2\pi}}{\pi}\right)\frac{2}{5}, \text{ etc.}$$

Substituting the values of a_0, a_n, b_n in (1), we get

$$e^{-x} = \frac{1-e^{-2\pi}}{\pi}\left[\frac{1}{2} + \left(\frac{1}{2}\cos x + \frac{1}{5}\sin 2x + \frac{1}{10}\cos 3x + \ldots\right) + \left(\frac{1}{2}\sin x + \frac{2}{5}\sin 2x + \frac{3}{10}\sin 3x + \ldots\right)\right].$$

Example 5.27. Find a Fourier series to represent $x - x^2$ in $(-\pi, \pi)$. Hence show that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots = \frac{\pi^2}{12}$.
(P.T.U. 2011 ; Roorkee 2006)

Solution. Let $x - x^2 = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$ ...(1)

Then $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} (x - x^2)\,dx = \frac{1}{\pi}\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{-\pi}^{\pi} = -\frac{2\pi^2}{3}$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} (x - x^2)\cos nx\,dx$$

$$= \frac{1}{\pi}\left[(x - x^2)\frac{\sin nx}{n} - (1 - 2x)\left(-\frac{\cos nx}{n^2}\right) + (-2)\left(\frac{-\sin nx}{n^3}\right)\right]_{-\pi}^{\pi}$$

$$= \frac{-4\pi\cdot 1^n}{n^2} \qquad [\because \cos n\pi = (-1)^n]$$

$\Rightarrow$ $a_1 = -4/1^2, a_2 = -4/2^2, a_3 = -4/3^2, a_4 = -4/4^2$ etc.

Finally, $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} (x - x^2)\sin nx\,dx$

$$= \frac{1}{\pi}\left[(x - x^2)\left(-\frac{\cos nx}{n}\right) - (1 - 2x)\left(-\frac{\sin nx}{n^2}\right) + (-2)\left(\frac{\cos nx}{n^3}\right)\right]_{-\pi}^{\pi} = -2(-1)^n/n$$

$\Rightarrow$ $b_1 = 2/1, b_2 = -2/2, b_3 = 2/3, b_4 = -2/4$ etc.

Substituting the values of a_0's and b_n's in (1), we get

$$x - x^2 = -\frac{\pi^2}{3} + 4\left[\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \frac{\cos 4x}{4^2} + \ldots\right] + 2\left[\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \ldots\right]$$

Cor. Putting $x = 0$ (or π) in the above interesting series, we get $-\frac{\pi^2}{3} = 4\left[-\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots\right]$

or $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots = \frac{\pi^2}{12}$ (U.P.T.U. 2010)

Note. In the above example, we have used the result $\sin n\pi = 0$ and $\cos n\pi = (-1)^n$.

Also $\sin\left(n + \frac{1}{2}\right)\pi = (-1)^n$ and $\sin\left(n + \frac{1}{2}\right)\pi = 0$. The reader should remember these results.

Example 5.28. Expand $f(x) = x\sin x$ as a Fourier series in the interval $0 < x < 2\pi$.
(A.M.I.E.T.E., 2008 ; Kanpur, 2008 ; Kolkata, 2006)

Solution. Let $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$ ...(1)

Then $a_0 = \frac{1}{\pi}\int_0^{2\pi} x\sin x\,dx = \frac{2}{\pi}[x(-\cos x) - 1\cdot(-\sin x)]_0^{2\pi} = -2$.

---

* Apply the general rule of integration by parts which states that if u_1, u_2 be two functions of x and $u_1^{(n)}$ denotes the n^{th} derivative and $u_2^{(-1)}$ denotes the integral of u_2 then

$$\int u_1 u_2\,dx = u_1 u_2^{(-1)} - u_1' u_2^{(-2)} + u_1'' u_2^{(-3)} - \ldots$$

In above words : Integral of the product of two functions

= 1st function × integral of 2nd − (1st diff. coeff. of 1st function) × (integral of 2nd) + (2nd diff. coeff. of 1st function) × (integral of integral of 2nd) − ...

$$a_n = \frac{1}{\pi}\int_0^{2\pi} x\sin x\cos nx\,dx = \frac{1}{2\pi}\int_0^{2\pi} x\{\sin(n+1)x + \sin(1-n)x\}\,dx$$

$$= \frac{1}{4\pi}\int_0^{2\pi} x[\sin(n+1)x - \sin(n-1)x]\,dx$$

$$= \frac{1}{2\pi}\left[x\left\{-\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1}\right\} - 1\left\{-\frac{\sin(n+1)x}{(n+1)^2} + \frac{\sin(n-1)x}{(n-1)^2}\right\}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[2\pi\left\{-\frac{\cos 2n\pi}{n+1} + \frac{\cos 2n\pi}{n-1}\right\}\right] = \frac{2}{n^2-1}, (n \neq 1)$$

When $n = 1$, $a_1 = \frac{1}{\pi}\int_0^{2\pi} x\sin x\cos x\,dx = \frac{1}{2\pi}\int_0^{2\pi} x\sin 2x\,dx$

$$= \frac{1}{2\pi}\left[x\left(-\frac{\cos 2x}{2}\right) - 1\left(-\frac{\sin 2x}{4}\right)\right]_0^{2\pi} = -\frac{1}{2}$$

Finally, $b_n = \frac{1}{\pi}\int_0^{2\pi} x\sin x\sin nx\,dx = \frac{1}{2\pi}\int_0^{2\pi} x[\cos(1-n)x - \cos(1+n)x]\,dx$

$$= \frac{1}{2\pi}\left[x\left\{\frac{\sin(1-n)x}{1-n} - \frac{\sin(1+n)x}{1+n}\right\} - 1\left\{-\frac{\cos(1-n)x}{(1-n)^2} + \frac{\cos(1+n)x}{(1+n)^2}\right\}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[\frac{\cos 2n\pi - 1\pi}{(1-n)^2} - \frac{\cos 2n\pi - 1\pi}{(1+n)^2}\right] = 0, (n \neq 1)$$

When $n = 1$, $b_1 = \frac{1}{\pi}\int_0^{2\pi} x\sin x\sin x\,dx = \frac{1}{2\pi}\int_0^{2\pi} x(1-\cos 2x)\,dx$

$$= \frac{1}{2\pi}\left[x\left(x - \frac{\sin 2x}{2}\right) - 1\left(\frac{x^2}{2} + \frac{\cos 2x}{4}\right)\right]_0^{2\pi} = \pi$$

Substituting the values of a_0, a_n and b_n in (1), we get

$$x\sin x = -1 + \pi\sin x - \frac{1}{2}\cos x + \frac{2}{2^2-1}\cos 2x + \frac{2}{3^2-1}\cos 3x + \ldots$$

Example 3.14. Expand $f(x) = \sqrt{1-\cos x}$, $0 \le x \le 2\pi$ in a Fourier series. Hence evaluate

$$\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \ldots$$

(Madras, 2000; JNTU, 2005)

Solution. We have $f(x) = \sqrt{1-\cos x} = \sqrt{2}\sin\frac{x}{2}$.

Let $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos nx + \sum_{n=1}^{\infty} b_n\sin nx \quad \ldots(1)$$

Then $$a_0 = \frac{1}{\pi}\int_0^{2\pi}\sqrt{2}\sin\frac{x}{2}\,dx = \frac{\sqrt{2}}{\pi}\left[-2\cos\frac{x}{2}\right]_0^{2\pi} = \frac{4\sqrt{2}}{\pi}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi}\sqrt{2}\sin\frac{x}{2}\cos nx\,dx = \frac{\sqrt{2}}{2\pi}\int_0^{2\pi} 2\cos nx\sin\frac{x}{2}\,dx$$

$$= \frac{1}{\sqrt{2}\pi}\int_0^{2\pi}\left\{\sin\left(n+\frac{1}{2}\right)x - \sin\left(n-\frac{1}{2}\right)x\right\}dx$$

$$= \frac{1}{\sqrt{2}\pi}\left[-\frac{2}{2n+1}\cos\left(\frac{2n+1}{2}\right)x + \frac{2}{2n-1}\cos\frac{2n-1}{2}x\right]_0^{2\pi}$$

$$= \frac{2}{\sqrt{2}\pi}\left[-\frac{1}{2n+1}\{\cos 2n\pi - 1\pi - 1\} + \frac{1}{2n-1}\{\cos(2n-1)\pi - 1\pi - 1\}\right]$$

$$= \frac{\sqrt{2}}{\pi}\left[\frac{2}{2n+1}\sin\frac{(2n+1)\pi}{2} - \frac{2}{2n-1}\sin\frac{(2n-1)\pi}{2}\right] = -\frac{4\sqrt{2}}{\pi(4n^2-1)} \qquad \because \sin(2n+1)\pi = \sin(2n-1)\pi = 0,\ n = 1$$

$$b_n = \frac{1}{\sqrt{2}\pi}\int_0^{2\pi}\sqrt{2}\sin\frac{x}{2}\sin nx\,dx - \frac{\sqrt{2}}{2\pi}\int_0^{2\pi}2\sin\frac{x}{2}\sin nx\,dx$$

$$= \frac{1}{\sqrt{2}\pi}\int_0^{2\pi}\left[\cos\left(n-\frac{1}{2}\right)x - \cos\left(n+\frac{1}{2}\right)x\right]dx$$

$$= \frac{1}{\sqrt{2}\pi}\left[\frac{2}{2n-1}\sin\left(\frac{2n-1}{2}\right)x - \frac{2}{2n+1}\sin\left(\frac{2n+1}{2}\right)x\right]_0^{2\pi}$$

$$= \frac{\sqrt{2}}{\pi}\left[\frac{1}{2n-1}\{\sin(2n-(n-0)\} - \frac{1}{2n+1}\{\sin(2n+(n-0)\}\right] = 0$$

Substituting the values of a_n's and b_n's in (1), we get

$$\sqrt{1-\cos x} = \frac{2\sqrt{2}}{\pi} - \sum_{n=1}^{\infty}\frac{4\sqrt{2}}{(4n^2-1)\pi}\cos nx$$

When $x = 0$, we have

$$0 = \frac{2\sqrt{2}}{\pi} - \frac{4\sqrt{2}}{\pi}\sum_{n=1}^{\infty}\frac{1}{(4n^2-1)} \qquad \text{i.e.,}\quad \frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \ldots = \frac{1}{2}.$$

PROBLEMS (IV)

1. Obtain a Fourier series to represent e^{-x} from $x = -\pi$ to $x = \pi$. Hence derive series for $\sinh\pi$.

2. Prove that $x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}(-1)^n\frac{\cos nx}{n^2}$, $-\pi < x < \pi$. (VTU, 2009; Anna, 2009; JNTU, 2005)

Hence deduce that (i) $\sum\frac{1}{n^2} = \frac{\pi^2}{6}$ (Anna, 2008; VTU, 2007; Calicut, 2004)

ii) $\sum\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$ iii) $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots = \frac{\pi^2}{12}$ (VTU, 2010)

iv) $\sum\frac{1}{n^4} = \frac{\pi^4}{90}$ (Anna, 2004)

3. If $f(x) = \left(\frac{\pi - x}{2}\right)^2$ in $0 \le x \le 2\pi$, show that $f(x) = \frac{\pi^2}{12} + \sum_{n=1}^{\infty}\frac{\cos nx}{n^2}$. (VTU, 2009; Kerala, 2005)

4. Prove that in the range $-\pi < x < \pi$, $\cos ax = \frac{2a^2}{\pi}\sin a\pi\left[\frac{1}{2a^2} + \sum_{n=1}^{\infty}\frac{(-1)^n}{a^2-n^2}\cos nx\right]$

5. $f(x) = x + x^2$ for $-\pi < x < \pi$ and $f(x) = f(x + 2\pi)$. Expand $f(x)$ in Fourier series.

(Kerala, Marine, 2005; JNTU, 2003)

Hence deduce that $\frac{\pi^2}{12} = \frac{1}{2^2} - \sum_{n=2}^{\infty}(-1)^n\left(\frac{4}{n^2}\cos n\pi - \frac{2}{n}\sin n\pi\right)$

and $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$. (VTU, 2004)

4.68 CONDITIONS FOR A FOURIER EXPANSION

The reader must not be misled by the belief that the Fourier expansion of $f(x)$ as written above shall be valid. The above discussion has merely shown that if $f(x)$ has an expansion, then the coefficients are given by Euler's formulae. The problem concerning the possibility of representing a function by Fourier series and convergence

of this series are unity and zero respectively. Such questions should be left to the curiosity of a pure mathematician. However, almost all engineering applications are covered by the following well known Dirichlet's conditions:

Any function $f(x)$ can be developed as a Fourier series $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$ in the interval $(c, c+2l)$, where a_0, a_n, b_n are constants, provided :

(i) $f(x)$ is periodic, single-valued and finite;

(ii) $f(x)$ has a finite number of discontinuities in any one period;

(iii) $f(x)$ has at the most a finite number of maxima and minima. (Anna, 2008; P.T.U., 2008)

In fact the problem of expressing any function $f(x)$ as a Fourier series depends upon the evaluation of the integrals

$$\frac{1}{\pi}\int f(x)\cos nx\,dx; \quad \frac{1}{\pi}\int f(x)\sin nx\,dx$$

which are the limits $(0, 2\pi)$, $(-\pi, \pi)$ or $(\alpha, \alpha + 2\pi)$ according as $f(x)$ is defined for every value of x in $(0, 2\pi)$, $(-\pi, \pi)$ or $(\alpha, \alpha + 2\pi)$.

SOLVED EXERCISES

Note : Any convenient interval $(c, c+2l)$ may be taken in the following functions which can be expanded as Fourier series in the interval $(c, c+2l)$.

1. $\cos mx$ 2. $\sin kx$ (P.T.U., 2008)

3. $f(x) = (\pi - x)^2$, $0 < x < 2\pi$ 4. $f(x) = x(2\pi - x)$, $0 < x < 2\pi$

13.3 FUNCTIONS HAVING POINTS OF DISCONTINUITY

In deriving the Euler's formulae for a_0, a_n, b_n, it was assumed that $f(x)$ was continuous. Instead a function may have a finite number of points of finite discontinuity i.e., the graph may consist of a finite number of different curves given by different equations. Even then such a function is expressible as a Fourier series.

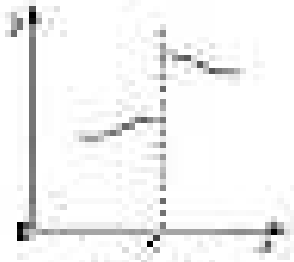

Fig. 13.1

For instance, if in the interval $(c, c + 2\pi)$, $f(x)$ is defined by

$$f(x) = \phi(x), \quad c < x < a$$

$$= \psi(x), \quad a < x < c + 2\pi,$$

$x = a$ is the point of discontinuity, then

$$a_0 = \frac{1}{\pi}\left[\int_c^a \phi(x)\,dx + \int_a^{c+2\pi} \psi(x)\,dx\right]$$

$$a_n = \frac{1}{\pi}\left[\int_c^a \phi(x)\cos nx\,dx + \int_a^{c+2\pi} \psi(x)\cos nx\,dx\right]$$

and $$b_n = \frac{1}{\pi}\left[\int_c^a \phi(x)\sin nx\,dx + \int_a^{c+2\pi} \psi(x)\sin nx\,dx\right]$$

At a point of finite discontinuity $x = a$, there is a finite jump in the graph of function (Fig. 13.1). Both the limits on the left [i.e., $f(a - 0)$] and the right [i.e., $f(a + 0)$] exist and are different. At such a point, Fourier series gives the value of $f(x)$ as the arithmetic mean of these two limits,

i.e., $f(a) = \frac{1}{2}[f(a - 0) + f(a + 0)]$.

Example 13.8. *Find the Fourier series expansion for* $f(x)$, *if*

$$f(x) = -\pi, \quad -\pi < x < 0$$

$$= x, \quad 0 < x < \pi$$

(Bhopal, 2006 S)

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$ (Rohtak, 2002)

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[1] See footnote p. 307

Solution. Let $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$ ...(1)

Then $a_0 = \frac{1}{\pi}\left[\int_{-\pi}^{0} -x\,dx + \int_{0}^{\pi} x\,dx\right] = \frac{1}{\pi}\left[-x\right]_{-\pi}^{0} + \left[x^2/2\right]_{0}^{\pi} = \frac{1}{\pi}\left[-\pi^2 + \frac{\pi^2}{2}\right] = -\frac{\pi}{2}$;

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} -x\cos nx\,dx + \int_{0}^{\pi} x\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[-\left\{\frac{x\sin nx}{n}\right\}_{-\pi}^{0} + \left\{\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right\}_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[0 + \frac{1}{n^2}\cos n\pi - \frac{1}{n^2}\right] = \frac{1}{n^2\pi}(\cos n\pi - 1)$$

$\therefore \quad a_1 = \frac{-2}{\pi \cdot 1^2}, a_2 = 0, a_3 = -\frac{2}{\pi \cdot 3^2}, a_4 = 0, a_5 = -\frac{2}{\pi \cdot 5^2}$ etc.

Finally, $b_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} -x\sin nx\,dx + \int_{0}^{\pi} x\sin nx\,dx\right]$

$$= \frac{1}{\pi}\left[\left\{\frac{x\cos nx}{n}\right\}_{-\pi}^{0} + \left\{-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right\}_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi}{n}(1 - \cos n\pi) - \frac{\pi}{n}\cos n\pi\right] = \frac{1}{n}(1 - 2\cos n\pi)$$

$\therefore \quad b_1 = 3, b_2 = -\frac{3}{2}, b_3 = 1, b_4 = -\frac{1}{4}$, etc.

Hence substituting the values of a_0's and b_n's in (1), we get

$$f(x) = -\frac{\pi}{4} - \frac{2}{\pi}\left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right) + 3\sin x - \frac{\sin 2x}{2} + \frac{3\sin 3x}{3} - \frac{\sin 4x}{4} + \ldots \quad \ldots(2)$$

which is the required result.

Putting $x = 0$ in (2), we obtain $f(0) = -\frac{\pi}{4} - \frac{2}{\pi}\left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\right)$ ...(3)

Now $f(x)$ is discontinuous at $x = 0$. As a matter of fact:

$f(0-0) = -\pi$ and $f(0+0) = 0 \quad \therefore \quad f(0) = \frac{1}{2}\{f(0-0) + f(0+0)\} = -\frac{\pi}{2}$.

Hence (3) takes the form $-\frac{\pi}{2} = -\frac{\pi}{4} - \frac{2}{\pi}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\right)$ whence follows the result.

Example 10.6. Find the Fourier series to represent $f(x) = \begin{cases} 0, & -\pi \le x \le 0 \\ \sin x, & 0 \le x \le \pi \end{cases}$. Hence deduce that $\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} - \ldots = \frac{\pi}{4} - \frac{1}{2}$. (Bombay, 2000; Madras, 2000; S.V.U., 2004)

Solution. Let $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$

Then $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} \sin x\,dx\right] = \frac{2}{\pi}$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0 \cdot dx + \int_{0}^{\pi} \sin x \cos nx\,dx\right]$$

$$= \frac{1}{2\pi}\int_{0}^{\pi} [\sin(1+n)x - \sin(1-n)x]\,dx$$

$$= \frac{1}{2\pi}\left[-\frac{\cos(n+2)x}{n+2} + \frac{\cos(n-1)x}{n-1} + \frac{1}{n+1} - \frac{1}{n-1}\right] \qquad (n \neq 1)$$

$$= \frac{1}{2\pi}\left[\frac{1-(-1)^{n+2}}{n+2} - \frac{1-(-1)^{n-1}}{n-1}\right] = 0, \text{ when } n \text{ is odd}$$

$$= -\frac{2}{\pi(n^2-1)}, \text{ when } n \text{ is even.}$$

When $n = 1$, $a_1 = \frac{1}{\pi}\int_0^{\pi} \sin x \cos x\,dx = \frac{1}{2\pi}\int_0^{\pi} \sin 2x\,dx = \frac{1}{2\pi}\left[-\frac{\cos 2x}{2}\right]_0^{\pi} = 0$

Finally, $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \frac{1}{\pi}\left[\int_{-\pi}^{0} 0\,dx + \int_0^{\pi} \sin x \sin nx\,dx\right]$

$$= \frac{1}{2\pi}\int_0^{\pi} [\cos(n-1)x - \cos(n+1)x]\,dx = \frac{1}{2\pi}\left[\frac{\sin(n-1)x}{n-1} - \frac{\sin(n+1)x}{n+1}\right]_0^{\pi} = 0 \quad (n \neq 1)$$

When $n = 1$, $b_1 = \frac{1}{\pi}\int_0^{\pi} \sin x \sin nx\,dx = \frac{1}{2\pi}\int_0^{\pi} (1-\cos 2x)\,dx = \frac{1}{2\pi}\left[x - \frac{\sin 2x}{2}\right]_0^{\pi} = \frac{1}{2}$

Hence $$f(x) = \frac{1}{\pi} - \frac{2}{\pi}\left[\frac{\cos 2x}{3\cdot 1} + \frac{\cos 4x}{4^2-1} + \frac{\cos 6x}{6^2-1} + \ldots\right] + \frac{1}{2}\sin x \qquad \ldots(1)$$

Putting $x = \frac{\pi}{2}$ in (1), we get $1 = \frac{1}{\pi} - \frac{2}{\pi}\left(-\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} - \frac{1}{5\cdot 7} + \ldots\right) + \frac{1}{2}$

Whence $\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} - \ldots = \frac{1}{4}(\pi - 2)$.

Example 3.3. Find the Fourier series for the function

$$f(x) = \begin{cases} -\pi, & -\pi < x < -\pi/2 \\ 0, & -\pi/2 < x < \pi/2 \\ \pi, & \pi/2 < x < \pi \end{cases}$$

Solution. Let $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \qquad \ldots(1)$$

Then $$a_0 = \frac{1}{\pi}\left[\int_{-\pi}^{-\pi/2} (-\pi)\,dx + \int_{-\pi/2}^{\pi/2} 0\,dx + \int_{\pi/2}^{\pi} \pi\,dx\right]$$

$$= \frac{1}{\pi}\left[-\pi\left[x\right]_{-\pi}^{-\pi/2} + \pi\left[x\right]_{\pi/2}^{\pi}\right] = \frac{1}{\pi}\left[\pi\left(-\frac{\pi}{2} + \pi\right) + \pi\left(\pi - \frac{\pi}{2}\right)\right] = 0$$

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{-\pi/2} (-\pi)\cos nx\,dx + \int_{-\pi/2}^{\pi/2} 0\cdot\cos nx\,dx + \int_{\pi/2}^{\pi} \pi\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi\sin nx}{n}\right]_{-\pi}^{-\pi/2} + \left[\frac{\pi\sin nx}{n}\right]_{\pi/2}^{\pi} = \frac{1}{n}\left[\left(\sin\frac{n\pi}{2} - \sin n\pi\right) + \left(\sin n\pi - \sin\frac{n\pi}{2}\right)\right] = 0$$

and $$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{-\pi/2} (-\pi)\sin nx\,dx + \int_{-\pi/2}^{\pi/2} 0\cdot\sin nx\,dx + \int_{\pi/2}^{\pi} \pi\sin nx\,dx\right]$$

$$= \frac{1}{n}\left[\left[\cos nx\right]_{-\pi}^{-\pi/2} - \left[\cos nx\right]_{\pi/2}^{\pi}\right] = \frac{2}{n}\left(\cos\frac{n\pi}{2} - \cos n\pi\right)$$

$$\therefore \quad b_1 = \frac{2}{\pi}, b_2 = -\frac{2}{\pi}, b_3 = \frac{2}{3\pi}, \text{ etc.}$$

Hence substituting the values of a_0 and b's in (1), we get $f(x) = \frac{2}{\pi}\left[\sin x - \sin 2x + \frac{1}{3}\sin 3x + \ldots\right]$

PROBLEMS 4.10

1. Find the Fourier series to represent the function $f(x)$ given by:
$f(x) = x$ for $0 \le x \le \pi$, and $= 2\pi - x$ for $\pi \le x \le 2\pi$. (C.T.T.C., 2004 / E.P.T.C., 2002 S)

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$. (Madras, 2003 S / E.P.T.C., 2006 S)

2. An alternating current after passing through a rectifier has the form:
$i = I_0 \sin x$ for $0 \le x \le \pi$; $i = 0$ for $\pi \le x \le 2\pi$
where I_0 is the maximum current and the period is 2π. (Fig. 10.2) Express i as a Fourier series and evaluate $\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \ldots$

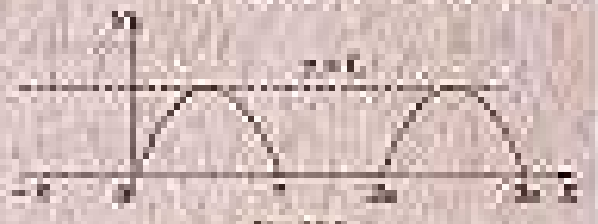


Fig. 10.2

(U.P.T.U., 2003 / Calicut, 2003)

3. Draw the graph of the function $f(x) = -x$, $-\pi \le x \le 0$
$= x^2$, $0 \le x \le \pi$
Find the Fourier series of $f(x)$. Deduce the sum of $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots$ (Mumbai, 2004)

4. Find the Fourier series of the following function:
$f(x) = x^2$, $0 \le x \le \pi$
$= -x^2$, $-\pi \le x \le 0$ (Mumbai, 2004)
(Madras, 2003)

5. Find a Fourier series for the function defined by
$$f(x) = \begin{cases} -k, & \text{for } -\pi < x < 0 \\ 0, & \text{for } x = 0 \\ k, & \text{for } 0 < x < \pi \end{cases}$$
Hence prove that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots = \frac{\pi}{4}$. (U.P.T.U., 2006)

10.5 CHANGE OF INTERVAL

In many engineering problems, the period of the function required to be expanded is not 2π but some other interval, say l. So, in order to apply the foregoing discussion to functions of period 2π, this interval must be converted to the length 2π. This involves only a proportional change in the scale.

Consider the periodic function $f(x)$ defined in $(\alpha, \alpha + 2l)$. To change the problem to period 2π

put $x = \frac{\pi z}{l}$ or $z = \frac{lx}{\pi}$ ...(1)

so that when $x = \alpha$, $z = \alpha\pi/l = \beta$ (say)

when $x = \alpha + 2l$, $z = \beta + 2\pi$ [$\because \alpha + 2l = \beta + 2\pi$]

Thus the function $f(x)$ of period $2l$ in $(\alpha, \alpha + 2l)$ is transformed into the function $f\left(\frac{lz}{\pi}\right) = F(z)$ (say) of period 2π in $(\beta, \beta + 2\pi)$. Hence $f(x)$ or $f(z)$ can be expressed as the Fourier series

$$f\left(\frac{lz}{\pi}\right) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nz + \sum_{n=1}^{\infty} b_n \sin nz \qquad \ldots(2)$$

where
$$\left.\begin{aligned} a_0 &= \frac{1}{\pi}\int_{\beta}^{\beta+2\pi} f\left(\frac{lz}{\pi}\right) dz \\ a_n &= \frac{1}{\pi}\int_{\beta}^{\beta+2\pi} f\left(\frac{lz}{\pi}\right) \cos nz\, dz \\ b_n &= \frac{1}{\pi}\int_{\beta}^{\beta+2\pi} f\left(\frac{lz}{\pi}\right) \sin nz\, dz \end{aligned}\right\} \qquad \ldots(3)$$

Making the inverse substitutions $x = \frac{cz}{\pi}$, $dx = \frac{c}{\pi}dz$ in (3) and (4) the Fourier expansion of $f(x)$ in the interval $(0, c)$, $c = 2\pi$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{c} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{c}$$

where

$$\left.\begin{aligned} a_0 &= \frac{1}{c}\int_0^{0+c} f(x)\,dx \\ a_n &= \frac{1}{c}\int_0^{0+c} f(x)\cos\frac{n\pi x}{c}\,dx \\ b_n &= \frac{1}{c}\int_0^{0+c} f(x)\sin\frac{n\pi x}{c}\,dx \end{aligned}\right| \quad \ldots(6)$$

Cor. By taking $a = 0$ in (6), we get the results for the interval $(0, 2\pi)$. Similarly taking $a = -\pi$ in (6), we get results for the interval $(-\pi, \pi)$.

Example 10.9. Expand $f(x) = e^x$ as a Fourier series in the interval $(-\pi, \pi)$.

(Nov./Dec. 2005, A/M, D., 2006)

Solution. The required series is of the form

$$e^x = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{\pi} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{\pi} \quad \ldots(1)$$

Here $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} e^x\,dx = \frac{1}{\pi}\left[e^x\right]_{-\pi}^{\pi} = \frac{1}{\pi}\left(e^{\pi} - e^{-\pi}\right) = \frac{2\sinh\pi}{\pi}$

and $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} e^x \cos nx\,dx$ $\left[\because \int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx)\right]$

$$= \frac{1}{\pi}\left[\frac{e^x}{1+(n\pi/\pi)^2}\left(-\cos nx + \frac{n\pi}{\pi}\sin nx\right)\right]_{-\pi}^{\pi} = \frac{2(-1)^n\sinh\pi}{\pi^2+(n\pi)^2} \qquad [\because \cos n\pi = (-1)^n]$$

$\therefore$ $a_1 = \frac{-2\sinh\pi}{\pi^2+\pi^2}$, $a_2 = \frac{2\sinh\pi}{\pi^2+2^2\pi^2}$, $a_3 = \frac{-2\sinh\pi}{\pi^2+3^2\pi^2}$ etc.

Finally, $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} e^x \sin nx\,dx$ $\left[\because \int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx)\right]$

$$= \frac{1}{\pi}\left[\frac{e^x}{1+(n\pi/\pi)^2}\left(-\sin nx - \frac{n\pi}{\pi}\cos nx\right)\right]_{-\pi}^{\pi} = \frac{2n(-1)^n\sinh\pi}{\pi^2+n^2\pi^2}$$

$\therefore$ $b_1 = \frac{-2\sinh\pi}{\pi^2+\pi^2}$, $b_2 = \frac{4\sinh\pi}{\pi^2+2^2\pi^2}$, $b_3 = \frac{-6\sinh\pi}{\pi^2+3^2\pi^2}$ etc.

Substituting the values of a_0, a_n and b_n in (1), we get

$$e^x = \sinh\pi\left\{\frac{1}{\pi} - 2\left[\frac{1}{\pi^2+\pi^2}\cos\frac{\pi x}{\pi} - \frac{1}{\pi^2+2^2\pi^2}\cos\frac{2\pi x}{\pi} + \frac{1}{\pi^2+3^2\pi^2}\cos\frac{3\pi x}{\pi} - \ldots\right]\right.$$
$$\left. - 2\left[\frac{1}{\pi^2+\pi^2}\sin\frac{\pi x}{\pi} - \frac{2}{\pi^2+2^2\pi^2}\sin\frac{2\pi x}{\pi} + \frac{3}{\pi^2+3^2\pi^2}\sin\frac{3\pi x}{\pi} - \ldots\right]\right\}$$

Example 10.10. Find the Fourier series expansion of $f(x) = kx - x^2$ in $(0, 2)$ and hence deduce that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots = \frac{\pi^2}{12}$.

(Nov./Dec. 2005)

Solution. The required series is of the form

$$f(x) = x^2 = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l} \quad \text{where } l = 2\pi \qquad \ldots(1)$$

Then $$a_0 = \frac{1}{l}\int_0^{2l} f(x)\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\,dx = \frac{1}{3\pi}\left[x^3\right]_0^{2\pi} = \frac{8\pi^2}{3}$$

$$a_n = \frac{1}{l}\int_0^{2l} f(x)\cos\frac{n\pi x}{l}\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\cos\frac{n\pi x}{\pi}\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\cos\frac{nx}{\pi}\,dx$$

$$= \frac{1}{\pi}\left[x^2\frac{\sin nx}{n} - 2x\left(-\frac{\cos nx}{n^2}\right) + 2\left(-\frac{\sin nx}{n^3}\right)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[\frac{4\pi^2}{n}\sin 2n\pi + \frac{4\pi}{n^2}\cos 2n\pi - 2\left(\frac{\sin 2n\pi}{n^3}\right)\right] = \frac{4}{n^2}$$

$$b_n = \frac{1}{l}\int_0^{2l} f(x)\sin\frac{n\pi x}{l}\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\sin\frac{n\pi x}{\pi}\,dx = \frac{1}{\pi}\int_0^{2\pi} x^2\sin\frac{nx}{\pi}\,dx$$

$$= \frac{1}{\pi}\left[x^2\left(-\frac{\cos nx}{n}\right) - 2x\left(-\frac{\sin nx}{n^2}\right) + 2\left(\frac{\cos nx}{n^3}\right)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[-\frac{4\pi}{n}\cos 2n\pi - \frac{2}{n^3}(\cos 2n\pi - 1)\right] = -\frac{4}{n}$$

Substituting the values of a_0, a_n, b_n in (1), we get

$$2x - x^2 = -\sum_{n=1}^{\infty}\frac{4}{n^2}\cos\frac{2n\pi x}{\pi} + \sum_{n=1}^{\infty}\frac{4}{n}\sin\frac{2n\pi x}{\pi}$$

Putting $x = 2\pi$, we get

$$3 - \frac{8}{4} = -\sum_{n=1}^{\infty}\frac{4}{n^2}\cos 4\pi \quad \text{or} \quad -\sum_{n=1}^{\infty}\frac{\cos 4\pi}{n^2} = \frac{\pi^2}{6} - \frac{3}{4}$$

or $$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{12}.$$

Example 10.18. Obtain Fourier series for the function

$$f(x) = \begin{cases} \pi x, & 0 \le x \le 1 \\ \pi(2-x), & 1 \le x \le 2 \end{cases}$$ (U.P.T.U., 2011; Meerut, 2008; Mumbai, 2009)

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$.

Solution. The required series is of the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\pi x + \sum_{n=1}^{\infty} b_n \sin n\pi x$$

Then $$a_0 = \int_0^2 f(x)\,dx = \int_0^1 \pi x\,dx + \int_1^2 \pi(2-x)\,dx = \pi\left[\frac{x^2}{2}\right]_0^1 + \pi\left[2x - \frac{x^2}{2}\right]_1^2 = \pi\left(\frac{1}{2}\right) + \pi\left[4 - 2 - \left(2 - \frac{1}{2}\right)\right] = \pi$$

$$a_n = \int_0^1 \pi x\cos n\pi x\,dx + \int_1^2 \pi(2-x)\cos n\pi x\,dx$$

$$= \pi\left[x\frac{\sin n\pi x}{n\pi} - 1\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_0^1 + \pi\left[(2-x)\frac{\sin n\pi x}{n\pi} - (-1)\left(-\frac{\cos n\pi x}{n^2\pi^2}\right)\right]_1^2$$

$$= \left(\frac{\cos n\pi}{n^2\pi} - \frac{1}{n^2\pi}\right) - \left(\frac{\cos 2n\pi}{n^2\pi} - \frac{\cos n\pi}{n^2\pi}\right) = \frac{2}{n^2\pi}(-1)^n - 1$$

$$= 0 \text{ when } n \text{ is even}; = -\frac{4}{n^2\pi^2} \text{ when } n \text{ is odd}$$

$$b_n = \int_0^{\pi} x\sin nx\,dx + \int_{\pi}^{2\pi} (2\pi - x)\sin nx\,dx$$

$$= \left|x\left(-\frac{\cos nx}{n}\right) - 1\left(-\frac{\sin nx}{n^2}\right)\right|_0^{\pi} + \left|x(-1)\left(-\frac{\cos nx}{n}\right) + (-1)\left(-\frac{\sin nx}{n^2}\right)\right|_{\pi}^{2\pi}$$

$$= \left(-\frac{\pi\cos n\pi}{n}\right) + \left(\frac{\sin n\pi}{n}\right) = 0$$

Hence $f(x) = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right)$

Putting $x = 0$, $0 = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{\cos 0}{1^2} + \frac{\cos 0}{3^2} + \frac{\cos 0}{5^2} + \ldots\right)$

Whence $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$.

Example 10.13. Find the Fourier series for

$$f(x) = \begin{cases} 1 - x, & -\pi < x < 0 \\ 1 + \sin x, & 0 < x < \pi \end{cases}$$

Solution. Let $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{\pi} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{\pi}$ ...(1)

In $(-\pi, \pi)$, $f(x) = 1 - x$, $f(-x) = 1 + x$ so that $f(x) \neq f(-x)$

Thus $a_0 = \frac{1}{\pi}\left\{\int_{-\pi}^{0}(1-x)\,dx + \int_{0}^{\pi}(1+\sin x)\,dx\right\} = \frac{1}{\pi}\left\|x - \frac{x^2}{2}\right\|_{-\pi}^{0} + \left|x - \frac{\cos x}{1}\right|_{0}^{\pi}$

$$= \frac{1}{\pi}\left\{-\left(-\pi + \frac{\pi^2}{2}\right) + \left(\pi + 1 - \frac{1}{1}\right)\right\} = \frac{1}{2}$$

$a_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0}(1-x)\cos\frac{n\pi x}{\pi}\,dx + \int_{0}^{\pi}(1+\sin x)\cos\frac{n\pi x}{\pi}\,dx\right\}$ Integrate by parts

$$= \frac{1}{\pi}\left\|(1-x)\left(\sin\frac{n\pi x}{\pi}\right)\frac{\pi}{n\pi} - (-1)\left(-\cos\frac{n\pi x}{\pi}\right)\frac{\pi}{n^2\pi^2}\right|_{-\pi}^{0}$$

$$+ \left|(1+\sin x)\left(-\cos\frac{n\pi x}{\pi}\right)\frac{\pi}{n\pi} - \cos x\left(-\cos\frac{n\pi x}{\pi}\right)\frac{\pi}{n^2\pi^2}\right|_{0}^{\pi}$$

$$= \frac{2}{n^2\pi}(1 - \cos n\pi)$$

$$b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{0}(1-x)\sin\frac{n\pi x}{\pi}\,dx + \int_{0}^{\pi}(1+\sin x)\sin\frac{n\pi x}{\pi}\,dx\right\}$$

$$= \frac{1}{\pi}\left\|(1-x)\left(-\cos\frac{n\pi x}{\pi}\right)\frac{\pi}{n\pi} - (-1)\left(-\sin\frac{n\pi x}{\pi}\right)\frac{\pi}{n^2\pi^2}\right|_{-\pi}^{0}$$

$$+ \left|(1+\sin x)\left(-\cos\frac{n\pi x}{\pi}\right)\frac{\pi}{n\pi} - \cos x\left(-\cos\frac{n\pi x}{\pi}\right)\frac{\pi}{n^2\pi^2}\right|_{0}^{\pi}$$

$$= \frac{1}{\pi}\left[\left(\frac{4}{(2n)^2}\sin\frac{2n\pi}{2}\right) - \left(\frac{4}{(2n)^2}\sin\frac{2n\pi}{0}\right)\right] = 0$$

Substituting the values of a_0's and b_n's in (1), we get

$$f(x) = \frac{8}{\pi} + \sum_{n=1}^{\infty}\frac{8}{n^2\pi}\left(1-\cos\frac{n\pi}{2}\right)\cos\frac{n\pi x}{2}.$$

PROBLEMS 3.8

1. Obtain the Fourier series for $f(x) = x$ in $(0, 4, 8)$.
2. (a) Find the Fourier series to represent x^2 in the interval $(0, \pi)$. (Mumbai, 2003)
   (b) Find a Fourier series for $f(x) = x - x^2$ when $-1 < x < 1$. (Mumbai, 2005)
3. If $f(x) = 2x - x^2$ in $0 < x < 2$, show that $f(x) = \frac{2}{3} - \sum_{n=1}^{\infty}\frac{4}{n^2\pi^2}\cos(n\pi x)$. (U.P.T.U., 2006)
4. Find the Fourier series for $f(x) = \begin{cases} x & \text{in } 0 \le x \le 3 \\ 6 - x & \text{in } 3 \le x \le 6 \end{cases}$ (Anna, 2005)
5. A sinusoidal voltage $E\sin\omega t$ is passed through a half-wave rectifier which clips the negative portion of the wave. Develop the resulting periodic function
   $$f(t) = \begin{cases} 0 & \text{when } -\pi/\omega < t < 0 \\ E\sin\omega t & \text{when } 0 < t < \pi/\omega \end{cases}$$
   and find a Fourier series. (Kolhapur, 2004)
6. Find the Fourier series of the function $f(x) = \begin{cases} kx, & 0 < x < \pi \\ 0, & x = \pi \end{cases}$ $f(x + 2\pi) = f(x), -\pi < x < \pi$.
7. Hence show that $\frac{\pi}{4} = \frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$ (Mumbai, 2003)

3.13 (3) EVEN AND ODD FUNCTIONS

A function $f(x)$ is said to be even if $f(-x) = f(x)$,

e.g. $\cos x$, $\cos 2x$, x^2 are all even functions. Graphically an even function is symmetrical about the y-axis.

A function $f(x)$ is said to be odd if $f(-x) = -f(x)$.

Even function Odd function

Fig. 3.31

e.g. $\sin x$, $\tan x$, x^3 are odd functions. Graphically, an odd function is symmetrical about the origin.

We shall be using the following property of definite integrals in the next paragraph :

$$\int_0^a f(x)dx = 2\int_0^a f(x)dx, \text{ when } f(x) \text{ is an even function.}$$

$$= 0, \text{ when } f(x) \text{ is an odd function.}$$

(3) Expansions of even or odd periodic functions. We know that a periodic function $f(x)$ defined in $(-c, c)$ can be represented by the Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{c} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{c},$$

where $a_0 = \frac{1}{c}\int_{-c}^{c} f(x)\,dx$, $a_n = \frac{1}{c}\int_{-c}^{c} f(x)\cos\frac{n\pi x}{c}\,dx$, $b_n = \frac{1}{c}\int_{-c}^{c} f(x)\sin\frac{n\pi x}{c}\,dx$.

Case I. When $f(x)$ is an even function, $b_n = \frac{1}{c}\int_{-c}^{c} f(x)\sin\frac{n\pi x}{c}\,dx = 0$.

Since $f(x)\cos\frac{n\pi x}{c}$ is also an even function,

$$\therefore \quad a_n = \frac{1}{c}\int_{-c}^{c} f(x)\cos\frac{n\pi x}{c}\,dx = \frac{2}{c}\int_{0}^{c} f(x)\cos\frac{n\pi x}{c}\,dx$$

Again since $f(x)\sin\frac{n\pi x}{c}$ is an odd function, $\quad b_n = \frac{1}{c}\int_{-c}^{c} f(x)\sin\frac{n\pi x}{c}\,dx = 0$.

Hence, if a periodic function $f(x)$ is even, its Fourier expansion contains only cosine terms, and

$$\left.\begin{aligned} a_0 &= \frac{2}{c}\int_{0}^{c} f(x)\,dx \\ a_n &= \frac{2}{c}\int_{0}^{c} f(x)\cos\frac{n\pi x}{c}\,dx \end{aligned}\right\} \quad \ldots(1)$$

Case II. When $f(x)$ is an odd function, $a_0 = \frac{1}{c}\int_{-c}^{c} f(x)\,dx = 0$.

Since $\cos\frac{n\pi x}{c}$ is an even function, therefore, $f(x)\cos\frac{n\pi x}{c}$ is an odd function.

$$\therefore \quad a_n = \frac{1}{c}\int_{-c}^{c} f(x)\cos\frac{n\pi x}{c} = 0$$

Again since $\sin\frac{n\pi x}{c}$ is an odd function, therefore, $f(x)\sin\frac{n\pi x}{c}$ is an even function.

$$\therefore \quad b_n = \frac{1}{c}\int_{-c}^{c} f(x)\sin\frac{n\pi x}{c}\,dx = \frac{2}{c}\int_{0}^{c} f(x)\sin\frac{n\pi x}{c}\,dx$$

Thus, if a periodic function $f(x)$ is odd, its Fourier expansion contains only sine terms and

$$b_n = \frac{2}{c}\int_{0}^{c} f(x)\sin\frac{n\pi x}{c}\,dx \quad \ldots(2)$$

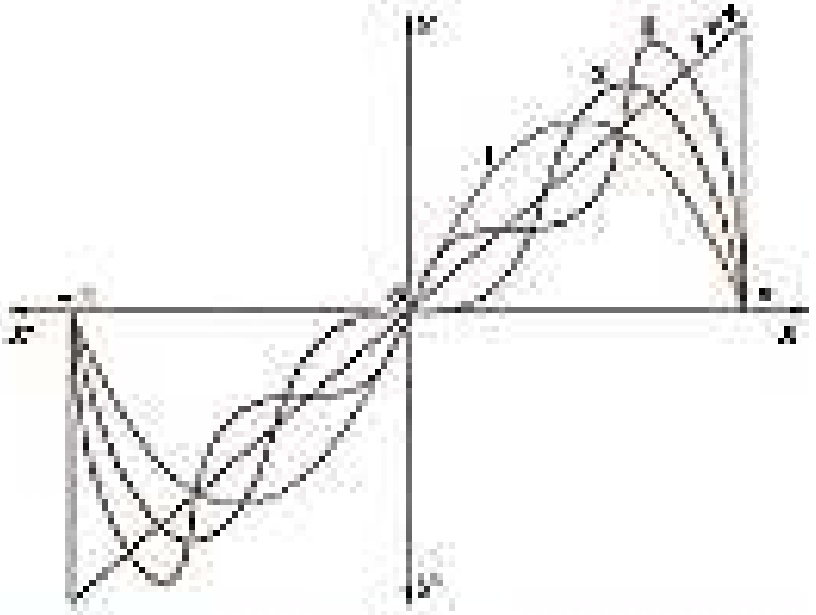


Fig. 10.1

Example 10.1 Express $f(x) = x(2\pi - x)$ as a Fourier series on the interval $0 < x < 2\pi$. (U.P.T.U., 2006)

Solution. Since $\quad f(-x) = -x(2\pi + x) \neq -f(x)$.

$\therefore$ $f(x)$ is an odd function and hence $f(x) = \sum_{n=1}^{\infty} b_n \sin nx$

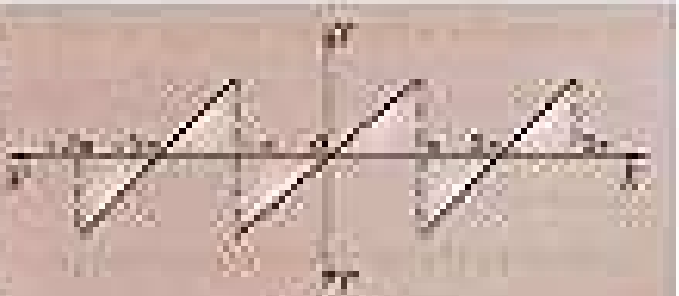
where $b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx = \frac{2}{\pi}\int_0^{\pi} \frac{x}{2}\sin nx\,dx$

$= \frac{1}{\pi}\left[x\left\{-\frac{\cos nx}{n}\right\} - (1)\left\{\frac{-\sin nx}{n^2}\right\}\right]_0^{\pi} = -\frac{\pi\cos n\pi}{n}$

$\therefore$ $b_1 = \pi/1,\ b_2 = -\pi/2,\ b_3 = \pi/3,\ b_4 = -\pi/4$, etc.

Hence the series is $x/2 = \sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \frac{1}{4}\sin 4x + \ldots$ ...(1)

Obs. The graphs of $y = f(x) = x/2$ in $-\pi < x < \pi$ and $y = S(x) = \sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x$ are shown in Fig. 10.6, by the curves 1, 2 and 3 respectively. These illustrate the manner in which the successive approximations to the series (1) approach more and more closely to the function $f(x)$ as the number of terms is increased. Also $S(x) = f(x)$ at $x = \pm\pi$.



As the series has a period 2π, it represents the sawtooth wave function, called sawtooth oscillation, shown in Fig. 10.6. It is important to note that the given function $y = x/2$ is continuous and each term of the series (1) is continuous, but the function represented by the series (1) is not. Further the series (1) is not ...

Example 10.13. Find a Fourier series to represent x^2 in the interval $(-\pi, \pi)$. (U.P.T.U., 2003)

Solution. Since $f(x) = x^2$ is an even function in $(-\pi, \pi)$,

$\therefore$ $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$ ...(1)

Then $a_0 = \frac{2}{\pi}\int_0^{\pi} x^2\,dx = \frac{2}{\pi}\left|\frac{x^3}{3}\right|_0^{\pi} = \frac{2\pi^2}{3}$

$a_n = \int_0^{\pi} x^2 \cos\frac{n\pi x}{\pi}\,dx$ [See footnote, p. 398]

$= \frac{2}{\pi}\left[x^2\left\{\frac{\sin nx}{n}\right\} - 2x\left\{-\frac{\cos nx}{n^2}\right\} + 2\left\{-\frac{\sin nx}{n^3}\right\}\right]_0^{\pi}$

$= 4(-1)^n/n^2$ $\because \cos n\pi = (-1)^n$

$\therefore$ $a_1 = -4/1^2,\ a_2 = 4/2^2,\ a_3 = -4/3^2,\ a_4 = 4/4^2$, etc.

Substituting these values in (1), we get

$$x^2 = \frac{\pi^2}{3} - 4\left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \frac{\cos 4x}{4^2} + \ldots\right)$$

which is the required Fourier series.

Example 10.14. If $f(x) = |\cos x|$, expand $f(x)$ as a Fourier series in the interval $(-\pi, \pi)$.

Solution. As $f(-x) = |\cos(-x)| = |\cos x| = f(x)$, $|\cos x|$ is an even function.

$\therefore$ $f(x) = \frac{a_0}{2} + \sum a_n \cos nx$

where $a_n = \frac{2}{\pi}\int_0^{\pi} |\cos x|\cos nx\,dx = \frac{2}{\pi}\int_0^{\pi/2} \cos x\cos nx\,dx + \int_{\pi/2}^{\pi} -\cos x\cos nx\,dx$

$\because$ $\cos x$ is $-ve$ when $\pi/2 < x < \pi$

$$= \frac{2}{\pi}\left[\frac{\sin nx}{n}\right]_0^{\pi/2} = \left[\frac{\sin nx}{n}\right]_{\pi/2}^{\pi} = \frac{2}{\pi}\left[\frac{1}{n}\sin\frac{n\pi}{2} - 0\right] = \frac{2}{n\pi}\sin\frac{n\pi}{2}$$

and $$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

$$= \frac{2}{\pi}\left[\int_0^{\pi/2} \cos x\cos nx\,dx - \int_{\pi/2}^{\pi} 1\cdot\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[\int_0^{\pi/2} [\cos(n+1)x + \cos(n-1)x]\,dx - \int_{\pi/2}^{\pi} [\cos nx + \cos(n-1)x]\,dx\right]$$

$$= \frac{1}{\pi}\left[\left\{\frac{\sin(n+1)x}{n+1} + \frac{\sin(n-1)x}{n-1}\right\}_0^{\pi/2} - \left\{\frac{\sin nx}{n} + \frac{\sin(n-1)x}{n-1}\right\}_{\pi/2}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[\left\{\frac{\sin(n+1)\frac{\pi}{2}}{n+1} + \frac{\sin(n-1)\frac{\pi}{2}}{n-1}\right\} + \left\{\frac{\sin(n+1)\frac{\pi}{2}}{n+1} + \frac{\sin(n-1)\frac{\pi}{2}}{n-1}\right\}\right]$$

$$= \frac{2}{\pi}\left\{\frac{\cos n\frac{\pi}{2}}{n+1} - \frac{\cos n\frac{\pi}{2}}{n-1}\right\} = \frac{-4\cos n\frac{\pi}{2}}{\pi(n^2-1)} \quad (n \neq 1)$$

In particular $a_1 = \frac{2}{\pi}\left[\int_0^{\pi/2} \cos^2 x\,dx - \int_{\pi/2}^{\pi} \sin^2 x\,dx\right] = 0$

Hence $|\sin x| = \frac{2}{\pi} + \frac{4}{\pi}\left[\frac{1}{1\cdot 3}\cos 2x - \frac{1}{3\cdot 5}\cos 4x + \ldots\right]$

Example 10.18. Obtain Fourier series for the function $f(x)$ given by

$$f(x) = \begin{cases} \pi + 2x/\pi, & -\pi \le x \le 0 \\ \pi - 2x/\pi, & 0 \le x \le \pi \end{cases}$$

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$. (U.P.T.U. 2010; Mumbai 2003)

Solution. Since $f(-x) = 1 - \frac{2x}{\pi} = f(x)$ in $(-\pi, 0) = f(x)$ in $(0, \pi)$

and $f(-x) = 1 + \frac{2x}{\pi} = f(x)$ in $(0, \pi) = f(x)$ in $(-\pi, 0)$

$\therefore$ $f(x)$ is an even function in $(-\pi, \pi)$. This is also clear from the graph.

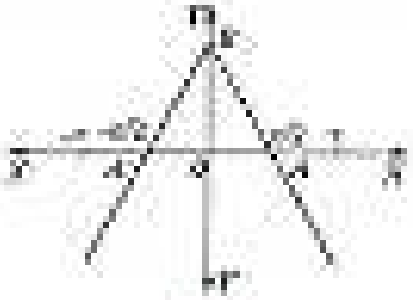


Fig. 10.8

A/B4 (Pg. 1048) which is symmetrical about the y-axis.

$$\therefore \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \qquad \ldots(1)$$

where $$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx = \frac{2}{\pi}\int_0^{\pi}\left(1 - \frac{2x}{\pi}\right)dx = \frac{2}{\pi}\left[x - \frac{x^2}{\pi}\right]_0^{\pi} = 0$$

and $$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_0^{\pi}\left(1 - \frac{2x}{\pi}\right)\cos nx\,dx$$

$$= \frac{2}{\pi}\left[\left(1 - \frac{2x}{\pi}\right)\frac{\sin nx}{n} - \left(-\frac{2}{\pi}\right)\left(-\frac{\cos nx}{n^2}\right)\right]_0^{\pi} = \frac{2}{n\pi}\left(-\frac{2\cos n\pi}{n\pi} + \frac{2}{n\pi}\right) = \frac{4}{n^2\pi^2}[1 - (-1)^n]$$

$\therefore$ $a_1 = 8/\pi^2$, $a^2 = 0$, $a_3 = 8/9\pi^2$, ...

and $a_2 = a_4 = a_6 = \ldots = 0$.

Thus substituting the values of a's in (1), we get

$$f(x) = \frac{8}{\pi^2}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right] \qquad \ldots(2)$$

which is the required Fourier expansion.

Putting $x = 0$ in (1), we get $1 = f(0) = \frac{8}{\pi^2}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\right]$

which gives the desired result.

PROBLEMS 4.5

1. Obtain the Fourier series expansion of $f(x) = x^2$ in $(0, 2\pi)$. Hence show that

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots$$

(Mumbai, 2003; U.P.T.U., 2004)

2. Show that for $-\pi < x < \pi$, $\sin ax = \frac{2\sin a\pi}{\pi}\left[\frac{\sin x}{1^2 - a^2} - \frac{2\sin 2x}{2^2 - a^2} + \frac{3\sin 3x}{3^2 - a^2} - \ldots\right]$

3. Expand the function $f(x) = x \sin x$ as a Fourier series in the interval $-\pi < x < \pi$. (V.T.U., 2008; Anna, 2008)

Deduce that $\frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \frac{1}{7.9} + \ldots = \frac{1}{4}(\pi - 2)$ (U.P.T.U., 2004)

4. Prove that in the interval $-\pi < x < \pi$, $\cos ax = \frac{2a}{\pi}\sin a\pi\left[\frac{1}{2a^2} - \sum_{n=1}^{\infty}\frac{\cos nx}{n^2 - a^2}\right]$ (U.P.T.U., 2004)

5. Find a Fourier series to represent $x - x^2$ in $(-\pi, \pi)$. Hence show that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \ldots$ (Bhopal, 2007; V.T.U., 2004)

Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$ (U.P.T.U., 2003; Kerala, 2003; J.N.T.U., 2003)

6. Find the Fourier series for $x + x^2$ in $(-\pi, \pi)$. Hence show that

(i) $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \ldots = \frac{\pi^2}{12}$ (Mumbai, 2004)

(ii) $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$ (U.P.T.U., 2003)

7. Given $f(x) = \begin{cases} -x, & -\pi \le x \le 0 \\ x, & 0 \le x \le \pi \end{cases}$

Is the function even or odd? Find the Fourier series for $f(x)$ and deduce the value of

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots$$

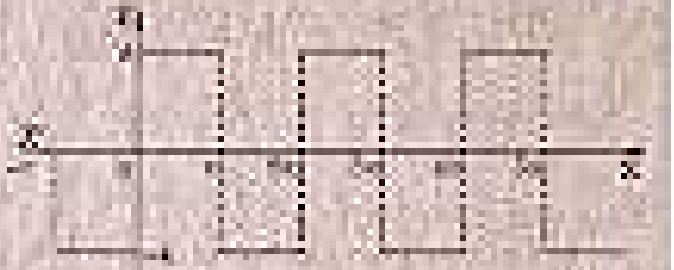


Fig. 4.7

8. Find the Fourier series of the periodic function $f(x)$ of period 2π where $-\pi < x < \pi$ and $f(x) = x + x^2$ in $-\pi < x < \pi$, and $f(x + 2\pi) = f(x)$. Sketch the graph of $f(x)$ and the two periods of it. (See Fig. 4.7)

Deduce that $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n(n-1)} = \frac{\pi}{4}$ (Bhopal, 2007)

9. A function is defined as follows:

$$f(x) = ax + bx^2, \quad -\pi < x < \pi, \quad a \text{ and } b \text{ are constants.}$$

Show that $f(x) = \frac{a\pi}{2} - \frac{4b}{\pi}\left[\frac{1}{1^2}\cos x + \frac{1}{2^2}\cos 2x + \frac{1}{3^2}\cos 3x + \ldots\right] + 2\pi b\left[\frac{1}{1}\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \ldots\right]$

Deduce that $\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$

4.10 HALF RANGE SERIES

Many a time it is required to obtain a Fourier expansion of a function $f(x)$ for the range $0 < x < \pi$, i.e. which is half the period of the Fourier series. As it is immaterial whatever the function may be outside the range $0 < x < \pi$, we extend the function to cover the range $-\pi < x < \pi$ so that the new function may be odd or even. The Fourier expansion of such a function of half the period, therefore, consists of sine or cosine terms only. In such cases the

graphs for the values of x in $(0, c)$ are the same but outside $(0, c)$ have different forms for odd or even functions. That is why we get different forms of series for the same function as is clear from the examples 13.16 and 13.17.

Sine series. If it be required to expand $f(x)$ as a sine series in $0 < x < c$, then we extend the function reflecting it in the origin, so that $f(-x) = -f(x)$.

Then the extended function would be $(-c, c)$ and the expansion will give the desired Fourier sine series :

$$\left.\begin{aligned} f(x) &= \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{c} \\ \text{where} \quad b_n &= \frac{2}{c}\int_0^c f(x)\sin\frac{n\pi x}{c}\,dx \end{aligned}\right\} \quad \ldots(1)$$

Cosine series. If it be required to expand $f(x)$ as a cosine series in $0 < x < c$, we extend the function reflecting it in the y-axis, so that $f(-x) = f(x)$.

Then the extended function is even in $(-c, c)$ and the expansion will give the required Fourier cosine series :

$$\left.\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{c} \\ \text{where} \quad a_0 &= \frac{2}{c}\int_0^c f(x)\,dx \\ \text{and} \quad a_n &= \frac{2}{c}\int_0^c f(x)\cos\frac{n\pi x}{c}\,dx \end{aligned}\right\} \quad \ldots(2)$$

Example 13.16. *Express $f(x) = x$ as a half-range sine series in $0 < x < 2$.* (U.P.T.U., 2010)

Solution. The graph of $f(x) = x$ in $0 < x < 2$ is the line OA. Let us extend the function $f(x)$ in the interval $-2 < x < 0$ shown by the line BO so that the new function is symmetrical about the origin and, therefore, represents an odd function in $(-2, 2)$ (Fig. 13.6)

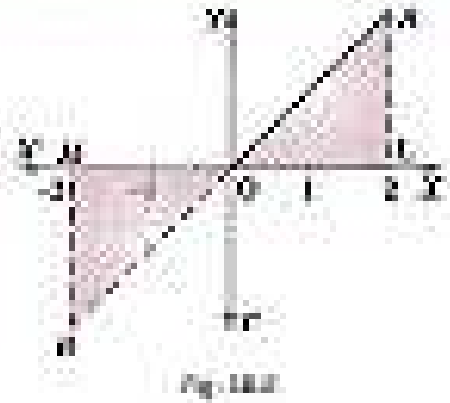


Fig. 13.6

Hence the Fourier series for $f(x)$ over the full period $(-2, 2)$ will contain only sine terms given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{2}$$

where
$$b_n = \frac{2}{2}\int_0^2 f(x)\sin\frac{n\pi x}{2}\,dx = \int_0^2 x\sin\frac{n\pi x}{2}\,dx$$

$$= \left|-\frac{2x}{n\pi}\cos\frac{n\pi x}{2} + \frac{4}{n^2\pi^2}\sin\frac{n\pi x}{2}\right|_0^2 = -\frac{4(-1)^n}{n\pi}$$

Thus $b_1 = 4/\pi$, $b_2 = -4/2\pi$, $b_3 = 4/3\pi$, $b_4 = -4/4\pi$, etc.

Hence the Fourier sine series for $f(x)$ over the half-range $(0, 2)$ is

$$f(x) = \frac{4}{\pi}\left(\sin\frac{\pi x}{2} - \frac{1}{2}\sin\frac{2\pi x}{2} + \frac{1}{3}\sin\frac{3\pi x}{2} - \frac{1}{4}\sin\frac{4\pi x}{2} + \ldots\right).$$

Example 13.17. *Express $f(x) = x$ as a half-range cosine series in $0 < x < 2$.*
(A.M.I.E., 2008 ; Kanpur, 2009 ; Mysore, 2010)

Solution. The graph of $f(x) = x$ in $(0, 2)$ is the line OA. Let us extend the function $f(x)$ in the interval $(-2, 0)$ shown by the line OB so that the new function is symmetrical about the y-axis and, therefore, represents an even function in $(-2, 2)$. (Fig. 13.8)

Hence the Fourier series for $f(x)$ over the full period $(-2, 2)$ will contain only cosine terms given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{2}$$

where $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx = \int_0^{\pi} x\,dx = \pi$

and $a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos\frac{2\pi nx}{\pi}\,dx = \int_0^{\pi} x\cos\frac{2\pi nx}{\pi}\,dx$

$= \left[\frac{x}{n}\sin\frac{2nx}{2} + \frac{1}{n^2}\cos\frac{2nx}{2}\right]_0^{\pi} = \frac{1}{n^2}(1-1^n) = 0$

Thus $a_1 = -4/\pi^2, a_2 = 0, a_3 = -4/9\pi^2, a_4 = 0, a_5 = -4/25\pi^2$ etc.

Hence the desired Fourier series for $f(x)$ over the half-range $0, \pi$ is

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right]$$

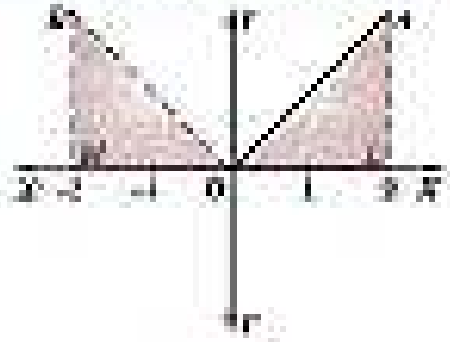


Fig. 20.9

Important Note. It should be clearly understood that we expand a function in a series of sines or cosines only to help us to find the sum of the series at a point in the range. We do not expand a function in order to find the function itself.

Example 20.10. Obtain the Fourier expansion of $f(x) = x$ as a periodic function in $(0, 2\pi)$.

Hence show that $\frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} - \ldots = \frac{\pi^2}{8}$ (*Nov.'02, 2003; Dec.'03, 2004*)

(*Anna, 2001*)

Solution. Let $x = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$

Then $a_0 = \frac{2}{\pi}\int_0^{\pi} x\,dx = \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi} = \pi$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx\,dx = \frac{1}{\pi}\int_0^{\pi} x\left[\cos(n+1)x - \cos(n-1)x\right]dx$$

$$= \frac{1}{\pi}\left[x\left\{\frac{\sin(n+1)x}{n+1} - \frac{\cos(n-1)x}{n-1}\right\} - 1\left\{\frac{-\cos(n+1)x}{(n+1)^2} - \frac{\sin(n-1)x}{(n-1)^2}\right\}\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\left\{\frac{\pi\sin n\pi}{n+1} - \frac{\cos n\pi}{(n+1)^2}\right\} - \left\{\frac{\pi\sin n\pi}{n-1} + \frac{\cos n\pi}{(n-1)^2}\right\}\right] \quad (n \neq 1)$$

When $n = 1, a_1 = \frac{2}{\pi}\int_0^{\pi} x\sin x\cos x\,dx = \frac{1}{\pi}\int_0^{\pi} x\sin 2x\,dx$

$$= \frac{1}{\pi}\left[x\left(\frac{-\cos 2x}{2}\right) - 1\left(\frac{-\sin 2x}{4}\right)\right]_0^{\pi} = \frac{1}{\pi}\left(\frac{-\pi\cos 2\pi}{2}\right) = -\frac{1}{2}$$

Hence $x\sin x = 1 - \frac{1}{2}\cos x - 2\left\{\frac{\cos 2x}{1\cdot 3} + \frac{\cos 3x}{2\cdot 4} + \frac{\cos 4x}{3\cdot 5} + \ldots\right\}$

Putting $x = \pi/2$, we obtain $\frac{\pi}{2} = 1 + 2\left\{\frac{1}{1\cdot 3} - \frac{1}{2\cdot 4} + \frac{1}{3\cdot 5} - \ldots\right\}$

Hence $\frac{1}{1\cdot 3} - \frac{1}{2\cdot 4} + \frac{1}{3\cdot 5} - \ldots = \frac{\pi - 2}{4}$.

Example 20.11. Obtain a half-range cosine series for

$$f(x) = \begin{cases} kx, & 0 \le x \le l/2 \\ k(l-x), & l/2 \le x \le l \end{cases}$$

(*Nagpur, 2004; U.P.T.U., 2004*)

Deduce that the sum of the series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots$ is $\frac{\pi^2}{8}$. (*Kerala, 2006; U.P.T.U., 2003*)

Solution. Let the half-range cosine series be

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l}$$

Here $$a_0 = \frac{2}{\pi}\left[\int_0^{\pi} kx\,dx + \int_{\pi}^{2\pi} k(2\pi - x)\,dx\right] = \frac{2k}{\pi}\left[\left|\frac{x^2}{2}\right|_0^{\pi} + \left|\frac{(2\pi - x)^2}{-2}\right|_{\pi}^{2\pi}\right]$$

$$= \frac{2k}{\pi}\cdot\frac{1}{2}\left[\pi^2 + \left(0 - \frac{\pi^2}{4}\right)\right] = \frac{3k\pi}{4}$$

$$a_n = \frac{2}{\pi}\left[\int_0^{\pi} kx\cos\frac{n\pi x}{\pi}\,dx + \int_{\pi}^{2\pi} k(2\pi - x)\cos\frac{n\pi x}{\pi}\,dx\right]$$

$$= \frac{2k}{\pi}\left[x\left(\frac{\sin\frac{n\pi x}{\pi}}{n\pi/\pi}\right) - 1\left(-\cos\frac{n\pi x}{n\pi/\pi}\right)\right]_0^{\pi} + \left[\right.$$

$$+ \frac{2k}{\pi}\left[\left\{\frac{(2\pi - x)\sin\frac{n\pi x}{\pi}}{n\pi/\pi}\right\} - (-1)\left\{\frac{-\cos\frac{n\pi x}{\pi}}{(n\pi/\pi)^2}\right\}\right]_{\pi}^{2\pi}$$

$$= \frac{2k}{\pi}\left[\left[\frac{\pi^2}{n\pi}\sin\frac{n\pi}{2}\right) + \frac{\pi^2}{n^2\pi^2}\left(\cos\frac{n\pi}{2} - \cos 0\right)\right] - \frac{2k}{\pi}\left[\left[\frac{\pi}{n\pi}\left(-\frac{\pi}{2}\cos\frac{n\pi}{2}\right)\right]\right.$$

$$\left. - \frac{\pi}{n^2\pi^2}\left[\cos 2n\pi - \cos\frac{n\pi}{2}\right]\right]$$

$$= \frac{2k}{\pi}\cdot\frac{\pi^2}{n^2\pi^2}\left[2\cos\frac{n\pi}{2} - 1 - (-1)^n\right] = \frac{2k}{n^2\pi^2}\left\{2\cos\frac{n\pi}{2} - 1 - (-1)^n\right\}$$

Hence the required Fourier series is

$$f(x) = \frac{3k}{4} - \frac{6k}{\pi^2}\left[\frac{1}{1^2}\cos\frac{2\pi x}{3} + \frac{1}{3^2}\cos\frac{6\pi x}{3} + \frac{1}{10^2}\cos\frac{10\pi x}{3} + \ldots\right]$$

Putting $x = 3$, we get

$$0 = \frac{3k}{4} - \frac{6k}{\pi^2}\left(\frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} + \ldots\right)$$

Thus $$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}.$$

Example 10.28. Expand $f(x) = \frac{\pi}{2} - x$, $(0 < x < \pi)$,

$$= x - \frac{3\pi}{2}, \left(\pi < x < \frac{3\pi}{2}\right)$$

as the Fourier series of period $\frac{3\pi}{2}$. (U.P.T.U., 2003) (Academic, 2004)

Solution. Let $f(x)$ be represented as odd function in $(-\pi, \pi)$ so that $f(x) = \sum_{n=1}^{\infty} b_n \sin nx$,

where $$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

$$= 2\left[\int_0^{\frac{\pi}{2}}\left(\frac{\pi}{2} - x\right)\sin nx\,dx + \int_{\frac{\pi}{2}}^{\pi}\left(x - \frac{3\pi}{4}\right)\sin nx\,dx\right]$$

$$= 2\left[\left(\frac{\pi}{2} - x\right)\frac{\cos nx}{-n} - \frac{\sin nx}{n^2(-1)}\right]_0^{\frac{\pi}{2}} + 2\left[\left(x - \frac{3\pi}{4}\right)\frac{\cos nx}{-n} - \frac{\sin nx}{n^2(-1)}\right]_{\frac{\pi}{2}}^{\pi}$$

$$= 2\left[\frac{1}{4n}\cos\frac{n\pi}{2} + \frac{1}{4n} - \frac{\sin n\pi/2}{n^2}\right] + 2\left[-\frac{1}{4n}\cos n\pi - \frac{3}{4n}\cos\frac{n\pi}{2} - \frac{\sin n\pi/2}{n^2}\right]$$

$$= \frac{1}{2n}[1 - (-1)^n] - \frac{4\sin n\pi/2}{n^2\pi^2}$$

Thus $b_1 = \frac{1}{\pi} - \frac{\pi}{4}$; $b_2 = 0$

$b_3 = \frac{1}{3\pi} + \frac{4}{\pi(3^2-1)^2}$; $b_4 = 0$ etc.

Hence $f(x) = \left(\frac{1}{\pi} - \frac{\pi}{8}\right)\sin x + \left(\frac{1}{3\pi} + \frac{4}{\pi 3^2}\right)\sin 3x + \left(\frac{1}{5\pi} - \frac{4}{\pi 5^2}\right)\sin 5x + \cdots$

PROBLEMS ON

1. Show that a function can be expressed in an infinite series $\frac{\pi^2}{3} + 4\left[\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \cdots\right]$ in the range $0 < x < 2\pi$

(Manipal Kunda, 2005; Kerala, 2006)

2. Obtain a cosine and sine series for $f(x) = x$ in the interval $0 < x < \pi$. Hence show that

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$$

(Osmania, 2002 S)

3. Find the half range cosine series for the function $f(x) = x^2$ in the range $0 < x < 2\pi$. (AU, 2000; Kakatiya, 2006)

4. Find the Fourier cosine series of the function $f(x) = (x-1)^2$ in $0 < x < \pi$. Hence show that

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$$

(Pune Bengal, 2004)

5. Find the half range cosine series for the function $f(x) = (x-1)^2$ in the interval $0 < x < 1$.

(VTU, 2010; JNTU, 2006)

Hence show that $\pi^2 = 8\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right)$ (Anna, 2005)

6. Find the half range sine series for the function $f(x) = (x-1)^2$, $0 < x < 1$.

7. Represent $f(x) = x$ in $(0, \pi)$, $0 < x < \pi$ by a half range cosine series. (Mumbai, 2006)

8. Find the half range sine series for $f(x) = x$ over $(0, \pi)$. (Anna, 2005 S)

9. Obtain the half range sine series for $f(x) = \cos x$, $0 < x < \pi$.

10. Find the half range Fourier sine series of $f(x) = x(\pi - x)$, $0 < x < \pi$ and hence deduce that

(a) $\sum_{n=1}^{\infty} \frac{1}{n^3} = \frac{\pi^3}{32}$ (Mumbai, 2003) (b) $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^3} = \frac{\pi^3}{32}$ (Mumbai, 2005)

11. If $f(x) = x$, $0 < x < \pi$

$= 2\pi - x$, $\pi < x < 2\pi$

show that $f(x) = \frac{4}{\pi}\left[\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots\right]$ (Mumbai, 2002; VTU, 2002; JNTU, 2005)

(a) $f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left[\cos 2x + \frac{1}{3^2}\cos 6x + \frac{1}{5^2}\cos 10x + \cdots\right]$ (VTU, 2011)

12. Find the half range cosine series expansion of the function $f(x) = \begin{cases} x, & 0 < x < \frac{1}{2} \\ 1-x, & \frac{1}{2} < x < 1 \end{cases}$ (VTU, 2010)

13. If $f(x) = \cos x$ in $0 < x < \pi$

$= \cos x$ in $-\pi < x < 0$, express $f(x)$ in a series of sines.

14. For the function defined by the graph (OAB in Fig. 10.10), find the half range Fourier sine series.

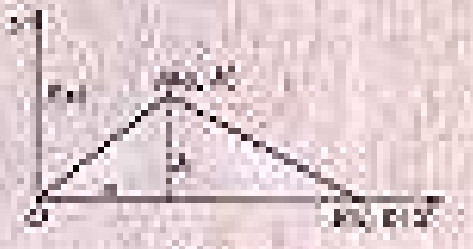


Fig. 10.10

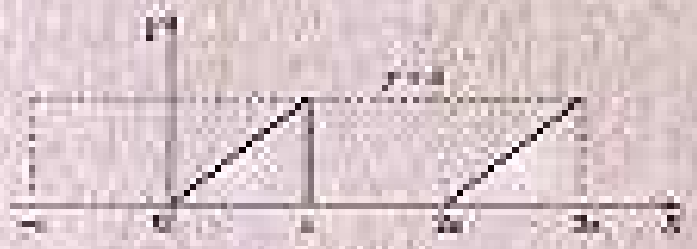


Fig. (11)

10.8 TYPICAL WAVEFORMS

We give below six typical waveforms usually met with in communication engineering :

(1) Square wave form (Fig. 10.9) is an extension of the function of Problem 8, page 413.

(2) Saw tooth or Ramp wave (Fig. 10.5) is an extension of the function in Ex. 10.12, page 408.

(3) Modified saw tooth wave form (Fig. 10.11) is an extension of the function

$$f(x) = 0, \quad -\pi < x \le 0$$
$$= x, \quad 0 < x < \pi$$

Its Fourier equivalent is

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right) + \frac{1}{\pi}\left(\sin 2x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \ldots\right)$$

(4) Triangular wave form (Fig. 10.6) is an extension of the function of Ex. 10.15, page 412.

(5) Half wave rectifier (Fig. 10.2) is an extension of the function of Problem 3, page 412.

(6) Full wave rectifier (Fig. 10.12) is an extension of the function $f(x) = a \sin x$, $0 \le x \le \pi$. Its Fourier equation is

$$f(x) = \frac{2a}{\pi}\left(\frac{1}{2} - \frac{1}{1.3}\cos 2x - \frac{1}{3.5}\cos 4x - \frac{1}{5.7}\cos 6x - \ldots\right)$$

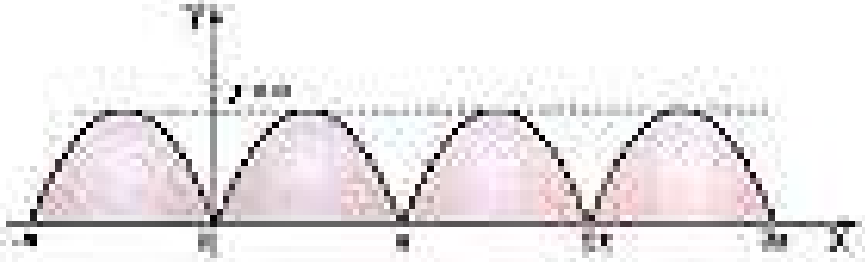


Fig. 10.12

10.9 (1) PARSEVAL'S FORMULA*

To prove that $\int_{-\pi}^{\pi} [f(x)]^2 dx = \pi\left(\frac{1}{2}a_0^2 + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)\right)$,

provided the Fourier series for $f(x)$ converges uniformly in $(-\pi, \pi)$.

The Fourier series for $f(x)$ in $(-\pi, \pi)$ is $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{\pi} + b_n \sin\frac{n\pi x}{\pi}\right)$ ...(1)

Multiplying both sides of (1) by $f(x)$ and integrating term by term from $-\pi$ to π (which is justified as the series (1) is uniformly convergent - p. 368), we get

$$\int_{-\pi}^{\pi} [f(x)]^2 dx = \frac{a_0}{2}\int_{-\pi}^{\pi} f(x)\,dx + \sum_{n=1}^{\infty}\left\{a_n \int_{-\pi}^{\pi} f(x)\cos\frac{n\pi x}{\pi} dx + b_n \int_{-\pi}^{\pi} f(x)\sin\frac{n\pi x}{\pi} dx\right\} \quad ...(2)$$

Now $\int_{-\pi}^{\pi} f(x)\,dx = \pi a_0$

$\int_{-\pi}^{\pi} f(x)\cos\frac{n\pi x}{\pi} dx = \pi a_n$ and $\int_{-\pi}^{\pi} f(x)\sin\frac{n\pi x}{\pi} dx = \pi b_n$, by (4) of p. 405

$\therefore$ (2) takes the form $\int_{-\pi}^{\pi} [f(x)]^2 dx = \pi\left(\frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)\right)$ ...(3)

which is the desired Parseval's formula. [Madras, 2005, S]

*Named after the French mathematician Marc-Antoine Parseval (1755–1836).

Cor. 1. If $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{l} + b_n \sin\frac{n\pi x}{l}\right)$ in $(0, 2l)$, then

$$\int_0^{2l} [f(x)]^2\,dx = l\left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)\right] \quad \ldots(11)$$

Cor. 2. If the half-range cosine series in $(0, l)$ for $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left[\frac{n\pi x}{l}\right], \text{ then}$$

$$\int_0^{l} [f(x)]^2\,dx = \frac{l}{2}\left[\frac{a_0^2}{2} + a_1^2 + a_2^2 + a_3^2 + \ldots\right] \quad \ldots(12)$$

Cor. 3. If the half-range sine series in $(0, l)$ for $f(x) = \sum_{n=1}^{\infty} b_n \sin\left[\frac{n\pi x}{l}\right]$, then

$$\int_0^{l} [f(x)]^2\,dx = \frac{l}{2}(b_1^2 + b_2^2 + b_3^2 + \ldots) \quad \ldots(13)$$

(3) Root mean square (r.m.s) value. The root mean square value of the function $f(x)$ over an interval (a, b) is defined as

$$\text{r.m.s value} = \sqrt{\frac{\int_a^b [f(x)]^2\,dx}{b-a}} \quad \ldots(14)$$

The square of root mean square value of a periodic function is frequently used in the theory of mechanical vibrations and in electrical circuit theory. The r.m.s. value is also known as the effective value of the function.

Example 10.23. Obtain the Fourier series for $y = x^2$ in $-\pi < x < \pi$. Using this find the values of $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \ldots$ and $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \ldots$.

Solution. Let $y = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$

We have $a_0 = 2\frac{\pi^2}{3}$, $a_n = \frac{4}{n^2}(-1)^n$, $b_n = 0$ for all n (See problem 5, p. 406)

If $\bar{y}$ be the r.m.s. value of y in $(-\pi, \pi)$, then

$$\bar{y}^2 = \frac{1}{2\pi}\left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)\right] \qquad \text{(By (11) and (7) of (10.5))}$$

$$= \frac{1}{4}\left(\frac{2\pi^2}{3}\right)^2 + \frac{1}{2\pi}\sum_{n=1}^{\infty}\left[\frac{16}{n^4}(-1)^{2n} + 0\right] = \frac{\pi^4}{9} + 8\sum_{1}^{\infty}\frac{1}{n^4}$$

Also by definition,

$$\bar{y}^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} y^2\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^4\,dx = \frac{\pi^4}{5}$$

Equating the two values of $\bar{y}^2$, we get

$$\frac{\pi^4}{9} + 8\sum_{1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{5} \text{ or } \sum_{1}^{\infty}\frac{1}{n^4} = \frac{\pi^4}{90}.$$

PROBLEMS 10.7

1. By using the first series for $f(x) = x$ in $0 < x < \pi$, show that $\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$

2. From Problem 10.6, $x^2 = \frac{\pi^2}{3} - 4\left(\cos x - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \frac{\cos 4x}{4^2} + \cdots\right)$

   and hence find $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}$

3. If $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l}$ is the half-range cosine series of $f(x)$ in $(0, l)$, then show that the mean square value of $f(x)$ in $(0, l)$ is $\frac{1}{2}\left|\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2\right|$

   Use this result to evaluate $1^2 + 3^2 + 5^2 + \cdots$, from the half-range cosine series of the function $f(x) = x$ period 4 defined in $(0, 2)$.

$$f(x) = \begin{cases} 2x, & 0 < x < 1 \\ 4 - 2x, & 1 < x < 2 \end{cases}$$

10.8 COMPLEX FORM OF FOURIER SERIES

The Fourier series of a periodic function $f(x)$ of period $2l$, is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\frac{n\pi x}{l} + b_n \sin\frac{n\pi x}{l}\right] \quad \ldots(1)$$

Since $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$,

therefore, we can express (1) as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\left(\frac{e^{in\pi x/l} + e^{-in\pi x/l}}{2}\right) + b_n\left(\frac{e^{in\pi x/l} - e^{-in\pi x/l}}{2i}\right)\right]$$

$$= c_0 + \sum_{n=1}^{\infty}\left[c_n e^{in\pi x/l} + c_{-n} e^{-in\pi x/l}\right] \quad \ldots(2)$$

where $c_0 = \frac{1}{2}a_0$, $c_n = \frac{1}{2}(a_n - ib_n)$, $c_{-n} = \frac{1}{2}(a_n + ib_n)$

Now $c_n = \frac{1}{2l}\left[\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}\,dx - i\int_{-l}^{l} f(x)\sin\frac{n\pi x}{l}\,dx\right]$

$$= \frac{1}{2l}\int_{-l}^{l} f(x)\left[\cos\frac{n\pi x}{l} - i\sin\frac{n\pi x}{l}\right]dx = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-in\pi x/l}\,dx$$

and $c_{-n} = \frac{1}{2l}\int_{-l}^{l} f(x)\left[\cos\frac{n\pi x}{l} + i\sin\frac{n\pi x}{l}\right]dx = \frac{1}{2l}\int_{-l}^{l} f(x)e^{in\pi x/l}\,dx$

Combining these, we have $c_n = \frac{1}{2l}\int_{-l}^{l} f(x)e^{-in\pi x/l}\,dx$

where $n = 0, \pm 1, \pm 2, \pm \infty$ ... (3)

Then the series (2) can be compactly written as:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/l}$$

which is the complex form of Fourier series and its coefficients are given by (3).

Obs. The complex form of a Fourier series is especially useful in problems in electrical circuit theory involving alternating periodic voltage.

Example 4.22. Find the complex form of the Fourier series of $f(x) = e^{-x}$ in $(-\pi, \pi)$.
(Madras, 2000 S; Mumbai, 2000 S)

Solution. We have $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}$ ... (1)

where $c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-x} \cdot e^{-inx}\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-(1+in)x}\,dx = \frac{1}{2\pi}\left[\frac{e^{-(1+in)x}}{-(1+in)}\right]_{-\pi}^{\pi} = \frac{e^{-(1+in)\pi} - e^{(1+in)\pi}}{2\pi(1+in)}$

$= \frac{e^{-\pi}\cos n\pi + ie^{-\pi}\sin n\pi - e^{\pi}\cos n\pi - ie^{\pi}\sin n\pi}{2\pi(1+in)} = \frac{e^{-\pi} - e^{\pi}}{2\pi}(-1)^n \cdot \frac{1-in}{1+n^2}$

$= \frac{(-1)^n(1-in)\sinh\pi}{\pi(1+n^2)}$

Hence $e^{-x} = \sum_{n=-\infty}^{\infty} \frac{(-1)^n(1-in)\sinh\pi}{\pi(1+n^2)} e^{inx}$

EXERCISE 4.5(A)

Find the complex form of the Fourier series of the following periodic functions:

1. $f(x) = e^{ax}$, $-\pi < x < \pi$. (Madras, 2001)
2. $f(x) = \cos ax$, $-\pi < x < \pi$, a is not an integer.
3. $f(x) = \cosh ax$ in $(-\pi, \pi)$. (Madras, 2000)
4. $f(x) = \begin{cases} 1 & \text{when } 0 < x < \pi \\ -1 & \text{when } -\pi < x < 0 \end{cases}$ (Anna, 2000; Mumbai, 2000)

4.11 PRACTICAL HARMONIC ANALYSIS

We have discussed at length, the problem of expanding $y = f(x)$ as Fourier series :

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx) \quad \ldots(1)$$

where
$$\left.\begin{aligned} a_0 &= \frac{2}{\pi}\int_0^{2\pi} f(x)\,dx \\ a_n &= \frac{2}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx \\ b_n &= \frac{2}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx \end{aligned}\right\} \quad \ldots(2)$$

So far, the function has always been defined by an explicit function of an independent variable. In practice, however, the function is often given not by a formula but by a graph or by a table of corresponding values. In such cases, the integrals in (2) cannot be evaluated or instead, the following alternative forms of (2) are employed.

Since the mean value of a function $y = f(x)$ over the range (a, b) is $\frac{1}{b-a}\int_a^b f(x)\,dx$.

∴ the equations (2) give,

$$\left.\begin{aligned} a_0 &= 2 \times \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx = 2[\text{mean value of } f(x) \text{ in } (0, 2\pi)] \\ a_n &= 2 \times \frac{1}{2\pi}\int_0^{2\pi} f(x)\cos nx\,dx = 2[\text{mean value of } f(x)\cos nx \text{ in } (0, 2\pi)] \\ b_n &= 2 \times \frac{1}{2\pi}\int_0^{2\pi} f(x)\sin nx\,dx = 2[\text{mean value of } f(x)\sin nx \text{ in } (0, 2\pi)] \end{aligned}\right\} \quad \ldots(3)$$

There are numerous other methods of finding the values of a_0, a_1, b_1 which constitute the field of harmonic analysis.

In (1), the term $a_1 \cos x + b_1 \sin x$ is called the fundamental or first harmonic, the term $a_2 \cos 2x + b_2 \sin 2x$ is called the second harmonic and so on.

Example 10.23. The displacement y of a part of a mechanism is tabulated below which is obtained experimentally. Express y as a Fourier series upto the first harmonic. Hence or otherwise find y when $x = 30°$.

$x°$	0	30	60	90	120	150	180	210	240	270	300	330
y	1.98	1.30	1.05	1.30	0.88	0.25	0.16	0.25	1.30	1.10	1.25	2.01

Solution. Let the Fourier series upto the first harmonic representing y is (1). So be

$$y = \frac{1}{2}a_0 + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x \quad \ldots(1)$$

To evaluate the unknowns, we form the following table.

$x°$	$\cos x$	$\cos 2x$	$\cos 3x$	$\sin x$	$\sin 2x$	$\sin 3x$	y	$y \cos x$	$y \cos 2x$	$y \cos 3x$	$y \sin x$	$y \sin 2x$	$y \sin 3x$
0	0	1	0	0	0	0	1.98	0.00	1.98	0.00	0.00	0.00	0.00
30	0.87	0.50	0.71	0.50	0.87	1	1.30	1.13	0.65	0.00	0.65	1.13	1.30
60	0.50	0.50	0.00	0.87	0.87	0	1.05	0.53	0.53	0.00	0.91	0.91	0.00
90	0.00	0	0	1.00	0	0	0.88	0.00	0.00	0.00	0.88	0.00	0.00
120	−0.50	−0.50	0.00	0.87	0.87	0	0.25	−0.13	−0.13	0.00	0.22	0.22	0.00
150	−0.87	−0.50	0.71	0.50	0.87	−1	0.16	−0.14	−0.08	0.11	0.08	0.14	−0.16
180	−1.00	1	−1	0.00	0	0	0.25	−0.25	0.25	−0.25	0.00	0.00	0.00
210	−0.87	0.50	−0.71	−0.50	0.87	1	1.30	−1.13	0.65	−0.92	−0.65	1.13	−1.30
240	−0.50	−0.50	0.00	−0.87	0.87	0	1.10	−0.55	−0.55	0.00	−0.96	−0.96	0.00
270	0.00	0	0	−1.00	0	0	1.25	0.00	0.00	0.00	−1.25	0.00	0.00
300	0.50	−0.50	−0.71	−0.87	0.87	−1	2.01	1.01	−1.01	−1.42	−1.74	1.74	2.01
330	0.87	0.50	0.71	−0.50	−0.87	0	1.98	1.72	0.99	1.40	−0.99	−1.72	0.00
							15.11	3.30	3.25	1.32	1.23	1.51	−0.32

$$\therefore \quad a_0 = 2 \cdot \frac{15.11}{12} = 2.52; \; a_1 = \frac{1}{6}\Sigma y \cos x = \frac{3.30}{6} = 0.55$$

$$a_2 = \frac{1}{6}\Sigma y \cos 2x = \frac{3.25}{6} = 0.52; \; a_3 = \frac{1}{6}\Sigma y \cos 3x = \frac{-0.62}{6} = -0.1$$

$$b_1 = \frac{1}{6}\Sigma y \sin x = \frac{-1.23}{6} = -0.21$$

$$b_2 = \frac{1}{6}\Sigma y \sin 2x = \frac{-1.51}{6} = -0.25$$

$$b_3 = \frac{1}{6}\Sigma y \sin 3x = \frac{0.32}{6} = 0.053$$

Substituting the values of a's and b's in (1), we get

$$y = 1.26 + 0.55 \cos x + 0.52 \cos 2x - 0.1 \cos 3x - 0.21 \sin x - 0.25 \sin 2x + 0.053 \sin 3x.$$

Example 10.24. The following table gives the variation of periodic current over a period.

t sec	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A amps	1.98	1.30	1.05	1.30	−0.88	−0.25	1.98

Show that there is a direct current part of 0.75 amp in the variable current and obtain the amplitudes of the first harmonic. (U.P.T.U. 2002; U.P.T.U. 2006)

Solution. Here length of the interval is T. i.e. $C = T$. So $2l = 2C$.

Thus $$y = \frac{a_0}{2} + a_1 \cos\frac{2\pi t}{T} + b_1 \sin\frac{2\pi t}{T} + a_2 \cos\frac{4\pi t}{T} + b_2 \sin\frac{4\pi t}{T} + \ldots$$

The derived values are tabulated as follows:

x	$\frac{2\pi x}{6}$	$\cos\frac{2\pi x}{6}$	$\sin\frac{2\pi x}{6}$	y	$y\cos\frac{2\pi x}{6}$	$y\sin\frac{2\pi x}{6}$
0	0	1.0	0.000	1.86	1.860	0.000
1	$\frac{\pi}{3}$	0.5	0.866	1.30	0.650	1.126
2	$\frac{2\pi}{3}$	−0.5	0.866	1.05	−0.525	0.909
3	π	−1.0	0.000	1.36	−1.360	0.000
4	$\frac{4\pi}{3}$	−0.5	−0.866	−0.80	0.400	0.693
5	$\frac{5\pi}{3}$	0.5	−0.866	−0.50	−0.250	0.433
			Σ=	4.5	1.375	3.161

$\therefore$ $a_0 = 2\cdot\frac{1}{6}\Sigma A = \frac{1}{3}(4.5) = 1.5$

$a_1 = 2\cdot\frac{1}{6}\Sigma A\cos\frac{2\pi x}{T} = \frac{1}{3}(1.375) = 0.458$

$b_1 = 2\cdot\frac{1}{6}\Sigma A\sin\frac{2\pi x}{T} = \frac{1}{3}(3.161) = 1.054$

Thus the direct current part is the mean value $= a_0/2 = 0.75$ and amplitude of the first harmonic

$= \sqrt{(a_1^2 + b_1^2)} = \sqrt{[(0.458)^2 + (1.054)^2]} = 1.052$

Example 10.16. Obtain the first three coefficients in the Fourier cosine series for y, where y is given in the following table:

x	0	1	2	3	4	5
y	4	5	10	7	6	2

Solution. Taking the interval as 60°, we have

$\theta =$	0°	60°	120°	180°	240°	300°
$x =$	0	1	2	3	4	5
$y =$	4	5	10	7	6	2

$\therefore$ Fourier cosine series in the interval $(0, 2\pi)$ is

$$y = \frac{a_0}{2} + a_1\cos\theta + a_2\cos 2\theta + a_3\cos 3\theta + \ldots$$

θ	$\cos\theta$	$\cos 2\theta$	$\cos 3\theta$	y	$y\cos\theta$	$y\cos 2\theta$	$y\cos 3\theta$
0°	1	1	1	4	4	4	4
60°	$\frac{1}{2}$	$-\frac{1}{2}$	−1	5	2.5	−2.5	−5
120°	$-\frac{1}{2}$	$-\frac{1}{2}$	1	10	−5	−5	10
180°	−1	1	−1	7	−7	7	−7
240°	$-\frac{1}{2}$	$-\frac{1}{2}$	1	6	−3	−3	6
300°	$\frac{1}{2}$	$-\frac{1}{2}$	−1	2	1	−1	−2
			Σ=	40	−6.5	−5	6

Hence $a_0 = 2\cdot\frac{40}{6} = 13.3$, $a_1 = 2\left(\frac{-6.5}{6}\right) = -2.5$, $a_2 = 2\left(\frac{-5}{6}\right) = -1.6$,

$a_3 = 2\left(\frac{6}{6}\right) = 2.0$.

Example 11.23. The turning moment T Kg-cm for a series of values of the crank angle $\theta°$ are given below:

$\theta°$	0	30	60	90	120	150	180
T	0	5024	8087	5630	3019	1634	0

Obtain the first four terms in a series of sines to represent T and calculate T for $\theta = 75°$.

Solution. Let the Fourier series required T in $(0, 180)$ be

$$T = b_1 \sin\theta + b_2 \sin 2\theta + b_3 \sin 3\theta + b_4 \sin 4\theta + \ldots$$

To evaluate the coefficients, we form the following table :

$\theta°$	T	$\sin\theta$	$\sin 2\theta$	$\sin 3\theta$	$\sin 4\theta$
0	0	0	0	0	0
30	5024	0.500	0.866	1	0.866
60	8087	0.866	0.866	0	−0.866
90	5630	1.000	0	−1	0
120	3019	0.866	−0.866	0	0.866
150	1634	0.500	−0.866	1	−0.866
180	0	0	0	0	0

$$b_1 = \frac{2}{6}\Sigma y \sin\theta = \frac{1}{3}[0 + 5024 + 2(8087)0.5 + 5630 + 3019 \times 0.866 + 1634 \times 0.866 + 0] = 5030$$

$$b_2 = \frac{2}{6}\Sigma y \sin 2\theta = \frac{1}{3}[0 + 5024 \times 0.866 + 8087 \times 0.866 + 0 + 3019(-0.866) + 1634(-0.866) + 0] = 1509$$

$$b_3 = \frac{2}{6}\Sigma y \sin 3\theta = \frac{1}{3}[0 + 5024 - 5630 + 3019 - 1634] = 260$$

$$b_4 = \frac{2}{6}\Sigma y \sin 4\theta = \frac{1}{3}[0 + 5024 \times 0.866 + 8087 \times 0.866 + 0 + 3019(-0.866) + 1634 \times 0.866 + 0] = 1509$$

Hence $T = 5030 \sin\theta + 1509 \sin 2\theta + 260 \sin 3\theta + 1509 \sin 4\theta$

For $\theta = 75°$, $T = 5030 \sin 75° + 1509 \sin 150° + 260 \sin 225° + 1509 \sin 300°$

$= 5030(0.9659) + 1509(0.5) - 260(0.7071) - 1509(0.866) = 4859$.

PROBLEMS 11.3

1. The following values of y give the displacement in a simple harmonic motion of period 2π. Express y as a Fourier series upto the first harmonic.

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y	0	3.2	3.8	1.8	2.3	2.7	0

2. Compute the first two harmonics of the Fourier series of $f(x)$ given in the following table :

x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
$f(x)$	1.0	1.4	1.9	1.7	1.5	1.2	1.0

[Anna, 2002]

3. Obtain the constant term and the coefficients of the first sine and cosine terms in the Fourier expansion of y as given in the following table :

x	0	1	2	3	4	5
y	9	18	24	28	26	20

[T.U.U., 1981] [Anna, 2002]

4. The result on the displacement y of a given point in simple engine is given below :

$\theta°$	0	30	60	90	120	150	180	210	240	270	300	330
y	2.0	1.3	1.2	1.6	2.0	1.7	1.2	0.9	0.7	0.8	1.1	1.6

Find the coefficients of the first five terms in the Fourier series for the above values.

5. Determine the first two harmonics of the Fourier series for the following values :

$x°$	0	30	60	120	150	180	210	240	270	300	330	360
y	1.98	1.30	1.05	1.30	1.88	1.75	1.20	0.88	0.25	0.29	0.83	1.98

[Madras, 2005; Delhi, 2006]

6. The turning moment T in the crank shaft of a steam engine for the crank angle θ degrees is given below :

$\theta°$	0	30	60	90	120	150	180	210	240	270	300	330	360
T	0	5274	7162	7854	6311	3162	0	−3162	−6311	−7854	−7162	−5274	0

Expand T in a series of sines upto the fourth harmonic.

1.10 OBJECTIVE TYPE OF QUESTIONS

Multiple Choice

Fill in the blanks or choose the correct answer in each of the following problems:

1. The period of $\cos 3x + \sin \frac{x}{3}$ is ____.

2. If $a = c$ is the point of discontinuity, then the Fourier series of $f(x)$ in (a, c) is given by $f(x) =$ ____.

3. A function $f(x)$ defined for $0 < x < c$ is to be extended in an odd periodic function in $(-c, c)$. The ____.

4. The mathematical function representing the following graph is ____.

5. Fourier expansion of $y = \sin^2 x$ function has only ____ terms.

6. Point plot for real values of Fourier coefficients for a given set of points (x_i, y_i) $i = 0, 1, 2, \ldots, n$ is the ____.

7. If $f(x) = x^2$ in $(-\pi, \pi)$, then the Fourier coefficient $b_n =$ ____.

8. The period of a constant function is ____.

9. If $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$, then $f(x)$ is an ____.

10. Fourier expansion of a even function $f(x)$ in $(-\pi, \pi)$ has only ____ terms.

11. If $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$, then $f(x)$ is an ____ function in $(-\pi, \pi)$.

12. The smallest period of the function $\sin\left(\frac{2n\pi x}{k}\right)$ is ____.

13. In the Fourier series expansion of $f(x) = |\sin x|$ in $(-\pi, \pi)$, the value of $b_n =$ ____.

14. In the Fourier series for $f(x) = x$ in $-\pi < x < \pi$, $f(x)$ is ____ series involved.

15. If $f(x)$ is an even function in $(-l, l)$, then the value of $b_n =$ ____.

16. If $f(x) = f(x + 2\pi)$ in $-\pi < x < \pi$, $f(x) = x^2$, then a_0 is ____.

17. If $f(x)$ is a periodic function with period $2l$, then the value of the Fourier coefficient $b_n =$ ____.

18. Dirichlet's conditions for the expansion of a function as a Fourier series in the interval $c_1 \le x \le c_2$ are ____.

19. If $f(x) = x \cos x$ in $(-\pi, \pi)$, then the value of $b_n =$ ____.

20. The formulae for finding the half range cosine series for the function $f(x)$ in $(0, l)$ are ____.

21. The half range sine series for $f(x) = \cos x$ in $(0, \pi/2)$ is ____.

22. Period of $\sin x$ is ____.

23. The value of $f(x)$ in the Fourier series of $f(x) = |x|$ in $(-\pi, \pi)$ is ____.

24. If $f(x)$ is defined in $(0, l)$ then the period of $f(x)$ to expand it as a half range sine series is ____.

25. The complex form of Fourier series for $f(x) = \sin^2 x$ in $(-\pi, \pi)$ is ____.

26. $f(x)$ is an odd function in $(-\pi, \pi)$, then the graph of $f(x)$ is symmetric about the ____. (Tamil, I Year)

27. $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$ then $f(x) =$ ____.

28. $f(x) = \begin{cases} -x, & -\pi \le x \le 1 \\ \pi^2 - x, & 1 \le x \le 2\pi \end{cases}$ then a_0 is ____. Function ____ (Anna, 2008)

29. If $f(x)$ is an odd function in $(-c, c)$, then the values of a_0 and a_n are ____.

30. The root mean square value of $f(x) = 2$ in $(0, 4)$ and $f(x) = 0$ in the range $2 < x < 4$ is ____. (Nagpur, 2008)

31. In the Fourier series expansion of the function

$$f(x) = \begin{cases} -kx + k, & -\pi < x < 0 \\ kx - k, & 0 < x < \pi \end{cases}$$ then the value of $a_0/2$ is ____. (CTU, 2008)

32. Let $f(x)$ be defined in $(0, 2\pi)$ by

$$f(x) = \begin{cases} \frac{1 + \cos x}{\pi - x}, & 0 < x < \pi \\ \sin x, & \pi \le x \le 2\pi \end{cases}$$

$f(x) = 2\pi + f(x)$. The value of $f(\pi)$ is ____. (Anna, 2009)

(a) Why cannot we say all f is a convergent Fourier series?

(b) Is f odd, even, or neither? Is f periodic? What is its period? Sketch the graph of f on $[-3\pi, 3\pi]$.

(c) Assume f is the sum of its Fourier series. Explain why $f(0) = 0$.

(d) Find the Fourier coefficients a_n and b_n of f.

(e) Let $S_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} a_n \cos(nx) + b_n \sin(nx)$ be the Nth partial sum of the Fourier series of f. Sketch the graphs of S_1, S_2, S_3, and S_4 on the same axes.

(f) Let $f(x) = \begin{cases} 1 - \frac{2x}{\pi} & \text{if } 0 \le x \le \frac{\pi}{2} \\ -1 & \text{if } \frac{\pi}{2} < x \le \pi \end{cases}$ be defined on $[0, \pi]$. Sketch the graphs of the even and odd extensions of f to $[-\pi, \pi]$.

(g) Let $f(x) = \begin{cases} 1 - \frac{2x}{\pi} & \text{if } 0 \le x \le \frac{\pi}{2} \\ -1 & \text{if } \frac{\pi}{2} < x \le \pi \end{cases}$ be defined on $[0, \pi]$. Sketch the graphs of the even and odd extensions of f to $[-\pi, \pi]$.

CHAPTER 11

Differential Equations of First Order

[redacted]

11.1 [redacted]

11.1 A differential equation is an equation which involves differential coefficients or differentials.

Thus [redacted]

[redacted]

[redacted]

[redacted]

11.2 An ordinary differential equation is one that is free from differential coefficients that have reference to a single independent variable. Thus the equations [redacted] are all examples of ordinary differential equations.

A partial differential equation is one in which there are two or more independent variables and partial differential coefficients with respect to one of them. Thus the equations [redacted] are all partial differential equations.

11.3 The order of a differential equation is the order of the highest derivative occurring in it.

The degree of a differential equation is the degree of the highest differential coefficient occurring in it, after the equation has been expressed in a form free from radicals and fractions so far as the differential coefficients are concerned.

Thus [redacted]

[redacted] is of the first order and first degree.

[redacted] is of the second order and first degree.

[redacted] is of the first order and second degree.

[redacted] is of the second order and second degree.

11.2 PRACTICAL APPROACH TO DIFFERENTIAL EQUATIONS

Differential equations arise from many problems in mechanics of mechanical and electrical systems, bending of beams, conduction of heat, velocity of chemical reactions, etc., and as such play a very important role in all modern scientific and engineering studies.

The approach of an engineering student to the study of differential equations has got its point of difference from that of a student of mathematics, who is only interested in solving the differential equations without knowing as to how the differential equations are formulated and how their solutions are physically interpreted.

Thus for an applied mathematician, the study of a differential equation consists of three phases :

(*i*) *formulation* of differential equations from the given physical situation, called *modelling*,

(*ii*) *solution* of the differential equation, evaluating the arbitrary constants from the given conditions, and

(*iii*) *physical interpretation of the solution*.

11.3 FORMATION OF A DIFFERENTIAL EQUATION

An ordinary differential equation is formed in an attempt to eliminate certain arbitrary constants from a relation in the variables and constants. It will, however, be seen later that the partial differential equations may be formed by the elimination of either arbitrary constants or arbitrary functions. In applied mathematics, every geometrical or physical problem when translated into mathematical symbols gives rise to a differential equation.

Example 11.1. *Form the differential equation of simple harmonic motion given by* $x = A\cos(\omega t + \alpha)$.

Solution. To eliminate the constants A and α differentiating it twice, we have

$$\frac{dx}{dt} = -\omega A\sin(\omega t + \alpha) \quad \text{and} \quad \frac{d^2x}{dt^2} = -\omega^2 A\cos(\omega t + \alpha) = -\omega^2 x$$

Thus $$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

is the desired differential equation which means that the acceleration varies as the distance from the origin.

Example 11.2. *Obtain the differential equation of all circles of radius a and centre* (h, k).

(Anna, 1996)

Solution. Such a circle is $(x-h)^2 + (y-k)^2 = a^2$ ...(1)

where h and k, the coordinates of the centre, and a are the constants.

Differentiating it twice, we have,

$$x - h + (y-k)\frac{dy}{dx} = 0 \quad \text{and} \quad 1 + (y-k)\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$$

Thus $$y - k = -\frac{1 + (dy/dx)^2}{d^2y/dx^2}$$

and $$x - h = -(y-k)\frac{dy}{dx} = \frac{dy}{dx}\frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}{d^2y/dx^2}$$

Substituting these in (1) and simplifying, we get $[1 + (dy/dx)^2]^3 = a^2(d^2y/dx^2)^2$ ...(2)

as the required differential equation.

Writing (2) in the form $\frac{[1 + (dy/dx)^2]^{3/2}}{d^2y/dx^2} = a$,

it states that the radius of curvature of a circle at any point is constant.

Example 11.3. *Obtain the differential equation of the family of circles of the type* $x^2 + y^2 + 2fx + 2gy + c = 0$ *where* f *and* g *are constant parameters but* c *is variable.* (U.P.T.U., 2003)

Solution. We have $x^2 + y^2 + 2ax + c^2 = 0$ ...(1)

Differentiating w.r.t. x, $2x + 2y\frac{dy}{dx} + 2a = 0$

or $$2a = -2\left(x + y\frac{dy}{dx}\right)$$

Substituting in (1), $x^2 + y^2 - 2x\left(x + y\frac{dy}{dx}\right) + c^2 = 0$

or $$2xy\frac{dy}{dx} = y^2 - x^2 + c^2$$

which is the required differential equation.

11.3 (3) SOLUTION OF A DIFFERENTIAL EQUATION

A solution (or integral) of a differential equation is a relation between the variables which satisfies the given differential equation.

For example, $$y = Ax + B \quad \text{is a solution of} \quad \frac{d^2y}{dx^2} + n^2y = 0 \text{ (Example 11.1)} \quad ...(2)$$

The general (or complete) solution of a differential equation is that in which the number of arbitrary constants is equal to the order of the differential equation. Thus (2) is a general solution (2) as the number of arbitrary constants A, B, is the same as the order of (2).

A particular solution is that which can be obtained from the general solution by giving particular values to the arbitrary constants.

For example, $y = A\sin nx + B\cos nx$

is the particular solution of the equation (2) as it can be derived from the general solution (2) by putting $A = B = 0$.

A differential equation may sometimes have an additional solution which cannot be obtained from the general solution by assigning a particular value to the arbitrary constant. Such a solution is called a singular solution and is not of much engineering interest.

Linearly independent solutions. The solutions $y_1(x)$ and $y_2(x)$ of the differential equation

$$\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_2(x)y = 0 \quad ...(3)$$

are said to be linearly independent if $c_1y_1 + c_2y_2 = 0$ only when $c_1 = 0$ and $c_2 = 0$.

If y_1 and y_2 are not both zero, then the two solutions y_1 and y_2 are said to be linearly dependent.

If $y_1(x)$ and $y_2(x)$ are two solutions of (3), then their linear combination $c_1y_1 + c_2y_2$ where c_1 and c_2 are constants, is also a solution of (3).

Example 11.4. *Find the differential equation whose set of independent solutions is (x^2, x^3).*

Solution. Let the general solution of the required differential equation be $y = c_1x^2 + c_2x^3$ ...(4)

Differentiating (4) w.r.t. x, we get

$$y_1 = c_1x^2 + c_2x^3 \quad \text{i.e.} \quad y - y_1 = c_2x^3 \quad ...(5)$$

Again differentiating (4) w.r.t. x, we obtain

$$y_2 = 2c_1x + 3c_2x^2 \quad ...(6)$$

Subtracting (6) from (5), we get

$$y - y_1 - y_2 = 0 \quad \text{or} \quad y - 2y_1 + y_2 = 0$$

which is the desired differential equation.

(2) Geometrical meaning of a differential equation. Consider any differential equation of the first order and first degree

$$\frac{dy}{dx} = f(x, y) \quad ...(1)$$

If $P(x, y)$ be any point, then (1) can be regarded as an equation giving the value of dy/dx at P when the values of x and y are known (Fig. 11.1). Let the value of m at the point $P_0(x_0, y_0)$ derived from (1) be m_0. Take a neighbouring

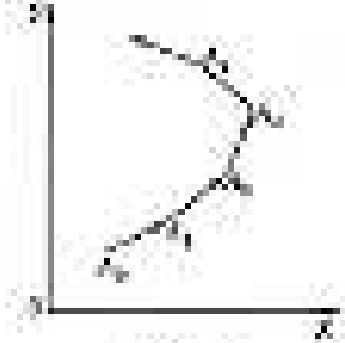


Fig. 11.1

point $A_1(x_1, y_1)$ such that the slope of A_0A_1 is m_0. Let the corresponding value of m at A_1 be m_1. Similarly take a neighbouring point $A_2(x_2, y_2)$ such that the slope of A_1A_2 is m_1 and so on.

If the successive points $A_0, A_1, A_2, A_3, \ldots$ are chosen very near one another, the broken curve $A_0A_1A_2A_3\ldots$ approximates to a smooth curve C: $y = \phi(x)$ which is a solution of (1) associated with the initial point $A_0(x_0, y_0)$. Clearly the slope of the tangent to C at any point and the coordinates of that point satisfy (1).

A different choice of the initial point will, in general, give a different curve with the same property. The equation of such a set of curves is a particular solution of the differential equation (1). The equation of the whole family of such curves is the general solution of (1). The slope of the tangent at any point of each member of this family and the co-ordinates of that point satisfy (1).

Such a simple geometrical interpretation of the solutions of second or higher order differential equations is not available.

PROBLEMS 11.1

From the differential equations from the following equations :

1. $y = ax^2 + bx^2$.
2. $y = C_1 \sin 3x + C_2 \cos 3x$. (Kerala, 2006)
3. $xy = Ax^2 + Bx^{-1} + x^2$. (UPTU, 2002)
4. $y = e^{-x}(A \cos x + B \sin x)$. (UPTU, 2005)
5. $y = ax^2 + bx^{-1} + x^{-2}$.

Find the differential equations of :

6. A family of circles passing through the origin and having centres on the x-axis. (UPTU, 2004)
7. All circles of radius 5, with their centres on the y-axis.
8. All parabolas with vertex at the origin and axis as the x-axis.
9. If $y(x) = ax + by$, $y_1(x) = ax + by$ are two solutions of $y'' + by = 0$, then the $y_1(x)$ and $y_2(x)$ are linearly dependent solutions.
10. Determine the differential equation whose general solution is $(ax + by)^2 = 1$. (UPTU, 2002)
11. Obtain the differential equation of the family of curves $y = Ax^2 + c$ and hence show that the slope of the tangent which passes through $(0, 0)$ at $(1, 1)$ is 2. Justify your answer. (UPTU, 2004)

11.3 EQUATIONS OF THE FIRST ORDER AND FIRST DEGREE

It is not possible to solve such equations in general. We shall, however, discuss some special methods of solution which are applied to the following types of equations :

(i) Equations where variables are separable, (ii) Homogeneous equations,
(iii) Linear equations, (iv) Exact equations.

In other cases, the particular solution may be obtained numerically (Chapter 21).

11.4 VARIABLES SEPARABLE

If in an equation it is possible to collect all functions of x and dx on one side and all the functions of y and dy on the other side, then the variables are said to be separable. Thus the general form of such an equation is $f(y)\,dy = \phi(x)\,dx$.

Integrating, the solution is $\int f(y)\,dy = \int \phi(x)\,dx + c$, where c is the constant of integration.

Example 11.5. Solve $(xy^2 + x)\,dx + (y - x^2y)\,dy = 0$. (UPTU, 2001)

Solution. Given equation is $x(1 + y^2)\,dx + y(1 - x^2)\,dy = 0$.

Integrating, we get $\int \frac{x}{1 - x^2}\,dx + \int \frac{y}{1 + y^2}\,dy = c$

or $-\frac{1}{2}\left[\log(1 - x^2) + \log(1 + y^2)\right] = c$ or $\log(1 - x^2) + \log(1 + y^2) = -2c$.

or $2x^2 \log x - \frac{x^2}{2} + \frac{y^2}{2} = -\cos y + y \sin y + \cos y + c$

Hence the solution is $2x^2 \log x - y \sin y = c$.

Example 11.6. Solve $\frac{dy}{dx} + x^{n-1} y = x^{2n-1} y^n$.

Solution. Given equation is $\frac{dy}{dx} = x^{n-1}(x^n - y^n)$ or $x^n dy = (x^{2n} + y^n) dx$.

Integrating the terms, $\int x^n dy = \int (x^{2n} + y^n) dx + c$

or $\frac{x^{2n}}{2} + \frac{y^{n+1}}{n+1} = \frac{x^{n+1}}{n+1} + c$ or $2x^{2n} + 2y^{n+1} - x^{n+1} = kx$.

Example 11.7. Solve $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$. *(U.P.T.U., 2005)*

Solution. Putting $x + y = z$ so that $1 + \frac{dy}{dx} = \frac{dz}{dx}$ or $\frac{dy}{dx} = \frac{dz}{dx} - 1$.

The given equation becomes $\frac{dz}{dx} - 1 = \sin z + \cos z$

or $\frac{dz}{dx} = 1 + \sin z + \cos z$

Integrating both sides, we get $dx = \int \frac{dz}{1 + \sin z + \cos z} + c$

or $x = \int \frac{2d\theta}{1 + \sin 2\theta + \cos 2\theta}$ + c [Putting $z = 2\theta$]

$= \int \frac{2d\theta}{2\cos^2\theta + 2\sin\theta\cos\theta} + c = \int \frac{\sec^2\theta}{1 + \tan\theta} d\theta + c$

$= \log(1 + \tan\theta) + c$

Hence the solution is $x = \log\left[1 + \tan\frac{1}{2}(x+y)\right] + c$.

Example 11.8. Solve $(4x + y + 1) dx = (x + y + 2) dy$, $y(0) = 2$.

Solution. Putting $4x + y + 1 = z$, we get $4 + \frac{dy}{dx} = \frac{dz}{dx}$.

$\therefore$ the given equation becomes $\frac{dz}{dx} - 4 = \frac{z}{z-3}$ or $\frac{dz}{dx} = 4 + \frac{z}{z-3}$

Integrating both sides, we get $\int \frac{z-3}{4z-3} dz = \int dx + c$

or $\frac{1}{4}\int \frac{4z-3}{4z-3} dz - \frac{3}{4}\int \frac{1}{4z-3} dz = x + c$ or $\frac{1}{4}\log\left[\frac{1}{4}(4z-3)\right] = x + c$.

or $4x + y + 1 = 2\tan(2x + c)$

When $x = 0, y = 2$ $\therefore \frac{1}{2}\tan^{-1}(1) = c$ i.e., $c = \frac{\pi}{8}$.

Hence the solution is $4x + y + 1 = 2\tan(2x + \pi/8)$.

Example 11.9. Solve $\frac{x}{y}\frac{dy}{dx} + \frac{x^2 + y^2 - 1}{2(x^2 + y^2) + 1} = 0$. *(U.P.T.U., 2003)*

Solution. Putting $x^2 + y^2 = z$, we get $2x + 2y\frac{dy}{dx} = \frac{dz}{dx}$ or $\frac{y}{x}\frac{dy}{dx} = \frac{1}{2x}\frac{dz}{dx} - 1$.

Hence here the given equation becomes $\frac{1}{2x}\frac{dv}{dx} - 1 + \frac{v-1}{2v+1} = 0$

or $\frac{1}{2x}\frac{dv}{dx} = 1 - \frac{v-1}{2v+1} = \frac{v+2}{2v+1}$ or $2x\,dx = \frac{2v+1}{v+2}dv$

or $2x\,dx = \left(2 - \frac{3}{v+2}\right)dv$

Integrating, we get $x^2 = 2v - 3\log|v+2| + 2 + c$

or $x^2 + 2y^2 - 3\log|x^2+y^2+2y+c| = 0$ [$\because v = x^2 + y^2$]

which is the required solution.

PROBLEMS 15.5

Solve the following differential equations:

1. $y(2x^2+y^2)\,dx + x(x^2-2y^2)\,dy = 0$.
2. $(x^2+y^2)\,dx - xy\,dy = 0$. (U.P.T.U. 2003)
3. $x\sin\left(\frac{y}{x}\right)\frac{dy}{dx} = y\sin\left(\frac{y}{x}\right) + x$, when $x = 1$, $y = \frac{\pi}{2}$.
4. $x\frac{dy}{dx} = y\log\left(\frac{y}{x}\right)$, when $x = 1$, $y = e$.
5. $(x^2 - xy)\frac{dy}{dx} + y^2 = 0$.
6. $x\frac{dy}{dx} = y + \sqrt{x^2+y^2}$. (U.P.T.U. 2010)
7. $x\frac{dy}{dx} + y\log\left(\frac{y}{x}\right) = 0$, when $x = 1$, $y = e$.
8. $\frac{dy}{dx} = x^{n-1}y^{1-n}$, $y(1) = 0$ when $x = 1$. (U.P.T.U. 2007)
9. $x\frac{dy}{dx} + x\cos\left(\frac{y}{x}\right) = y\sec\left(\frac{y}{x}\right)$, when $x = 1$, $y = \frac{\pi}{2}$.
10. $(x^2+xy)\,dx + (y^2+xy)\,dy = 0$.
11. $\frac{dy}{dx} = x^{n-1}y^{1-n}$, $y(1) = 1$.
12. $y + x\frac{dy}{dx} = x\left(x^2 + \frac{dy}{dx}\right)$.
13. $x + y\frac{dy}{dx} = x^2$. (Bhubaneswar 2006S)
14. $(x^2 - y^2)\,dx + xy\,dy = 0$. (Kanpur Series, 2006)
15. $x^2\log\left(\frac{y}{x}\right)dx + y^2\,dy = xy\,dy$. (U.P.T.U. 2005)
16. $\frac{dy}{dx} = \sin\left(\frac{y}{x}\right) + \frac{y}{x}$. (U.P.T.U. 2004)
17. $x\frac{dy}{dx} = y\cos\left(\log\frac{y}{x}\right) + x$.
18. $\frac{dy}{dx} = \frac{x+y}{x-y}$.
19. $x^2\log\left(\frac{y}{x}\right)dx + y^2\,dy = xy\,dy$. (U.P.T.U. 2005)
20. $\frac{dy}{dx} = \frac{x+y}{x}$, $y(1) = 1$.

15.12 HOMOGENEOUS EQUATIONS

are of the form $\frac{dy}{dx} = \frac{f(x,y)}{\phi(x,y)}$

where $f(x,y)$ and $\phi(x,y)$ are homogeneous functions of the same degree in x and y (see page 206).

To solve a homogeneous equation (i) Put $y = vx$, then $\frac{dy}{dx} = v + x\frac{dv}{dx}$,

(ii) Separate the variables v and x, and integrate.

Example 1. (U.P.T.U. 2004) Solve $(x^2 - y^2)\,dx - xy\,dy = 0$.

Solution. Given equation is $\frac{dy}{dx} = \frac{x^2 - y^2}{xy}$ which is homogeneous in x and y. ...(1)

Put $y = vx$, then $\frac{dy}{dx} = v + x\frac{dv}{dx}$ $\therefore$ (1) becomes $v + x\frac{dv}{dx} = \frac{1 - v^2}{v}$

or $x\frac{dv}{dx} = \frac{1 - v^2}{v} - v = \frac{1 - 2v^2}{v}$,

Separating the variables, $\frac{x}{1-2x^2}\,dx = \frac{dx}{x}$

Integrating both sides, $\int \frac{x\,dx}{1-2x^2} = \int \frac{dx}{x} + c$

or $-\frac{1}{4}\int \frac{-4x}{1-2x^2}\,dx = \int \frac{dx}{x} + c$ or $-\frac{1}{4}\log(1-2x^2) = \log x + c$

or $-4\log x + \log(1-2x^2) = -4c$ or $\log x^{-4}(1-2x^2) = -4c$ [Put $c = \frac{1}{4}\log k$]

or $x^{-4}(1-2x^2) = e^{-4c} = k$

Hence the required solution is $x^2(1-2x^2) = k$.

Example 11.11. Solve $(x\sin^2(y/x) - y)\,dx + x\,dy = 0$. (U.P.T.U., 2006)

Solution. The given equation may be rewritten as

$$\frac{dy}{dx} = \left(\frac{y}{x}\sin^2\frac{y}{x} - \tan\frac{y}{x}\right)\cos^2(y/x) \qquad \ldots(1)$$

which is a homogeneous equation. Putting $y = vx$, (1) becomes $v + x\frac{dv}{dx} = (v\sin^2 v - \tan v)\cos^2 v$

or $x\frac{dv}{dx} = -\tan v\cos^2 v \quad \ldots(2)$

Separating the variables $\frac{\sec^2 v}{\tan v}\,dv = -\frac{dx}{x}$

Integrating both sides $\log\tan v = -\log x + \log c$

or $x\tan v = c$ or $x\tan(y/x) = c$.

Example 11.12. Solve $(1+e^{x/y})dx + e^{x/y}(1-x/y)dy = 0$.

(U.P.T.U., 2006; Rajasthan, 2006; P.T.U., 2007)

Solution. The given equation may be rewritten as

$$\frac{dx}{dy} = -\frac{e^{x/y}(1-x/y)}{1+e^{x/y}} \qquad \ldots(1)$$

which is a homogeneous equation. Putting $x = vy$ so that (1) becomes

$$v + y\frac{dv}{dy} = -\frac{e^v(1-v)}{1+e^v} \quad \text{or} \quad y\frac{dv}{dy} = -\frac{e^v(1-v)}{1+e^v} - v = -\frac{v+e^v}{1+e^v}$$

Separating the variables, we get

$$-\frac{dy}{y} = \frac{1+e^v}{v+e^v}\,dv = \frac{d(v+e^v)}{v+e^v}$$

Integrating both sides, $-\log y = \log(v+e^v) + \log c$

or $y(v+e^v) = c^{-1}$ or $y(x/y + e^{x/y}) = c^{-1}$ [$\because v = x/y$]

which is the required solution.

EXERCISE 4.3

Solve the following differential equations :

1. $(x^2 - y^2)\,dx + xy\,dy = 0$
2. $(x^2+y^2)\,dx - 2xy\,dy = 0$ (U.P.T.U., 2007)
3. $y^2 + x^2\frac{dy}{dx} = xy\frac{dy}{dx}$
4. $(x^2 - xy + y^2)\,dx - xy\,dy = 0$ (Nagpur, 2004)
5. $x\,dy - y\,dx = \sqrt{x^2+y^2}\,dx$ (Nagpur, 2004)
6. $(x^2 - 3y^2)\,dx + 2xy\,dy = 0$ (U.P.T.U., 2007)

Exercise (1.4): Solve the homogeneous equations. Solve each of the following differential equations by reducing it to the homogeneous equation by putting $x = X + h$, $y = Y + k$.

1. $\frac{dy}{dx} = \frac{x+y-1}{x-y-3}$ (V.T.U., 2009) 2. $(x+y+1)dx + (2x+2y-1)dy = 0$ (U.P.T.U., 2006)

3. $(3x+4y+1)dx + (4x-3y-7)dy = 0$ 4. $(x+2y-5)dx + (2x+y-4)dy = 0$

5. $(x+y)\frac{dy}{dx} = (x-y)$ 6. $(x+y+1)\frac{dy}{dx} = (x+y)$

1.13 EQUATIONS REDUCIBLE TO HOMOGENEOUS FORM

The equations of the form $\frac{dy}{dx} = \frac{ax+by+c}{a'x+b'y+c'}$ ...(1)

can be reduced to the homogeneous form as follows:

Case I. When $\frac{a}{a'} \neq \frac{b}{b'}$

Putting $x = X + h$, $y = Y + k$, (h, k being constants)

so that $dx = dX$, $dy = dY$, (1) becomes

$$\frac{dY}{dX} = \frac{aX + bY + (ah + bk + c)}{a'X + b'Y + (a'h + b'k + c')} \quad \text{...(2)}$$

Choose h, k so that (2) may become homogeneous.

Put, $ah + bk + c = 0$, and $a'h + b'k + c' = 0$

so that $$\frac{h}{bc' - b'c} = \frac{k}{ca' - c'a} = \frac{1}{ab' - ba'}$$

or $$h = \frac{bc' - b'c}{ab' - ba'}, k = \frac{ca' - c'a}{ab' - ba'} \quad \text{...(3)}$$

Thus when $ab' - ba' \neq 0$, (2) becomes $\frac{dY}{dX} = \frac{aX + bY}{a'X + b'Y}$ which is homogeneous in X, Y and can be solved by putting $Y = vX$.

Case II. When $\frac{a}{a'} = \frac{b}{b'}$.

In this case $ab' - ba' = 0$, the above method fails as h and k become infinite or indeterminate.

Now $$\frac{a}{a'} = \frac{b}{b'} = \frac{1}{m} \text{ (say)}$$

$\therefore$ $a' = am$, $b' = bm$ and (1) becomes

$$\frac{dy}{dx} = \frac{ax + by + c}{m(ax + by) + c'} \quad \text{...(4)}$$

Put $ax + by = z$, so that $a + b\frac{dy}{dx} = \frac{dz}{dx}$

or $$\frac{dy}{dx} = \frac{1}{b}\left(\frac{dz}{dx} - a\right) \quad \therefore \quad (4) \text{ becomes } \frac{1}{b}\left(\frac{dz}{dx} - a\right) = \frac{z + c}{mz + c'}$$

or $$\frac{dz}{dx} = a + \frac{bz + bc}{mz + c'} = \frac{(a + b)z + ac' + bc}{mz + c'}$$

or Clearly the variables are separable. In this solution, put $ax + by = z$, we get the required solution of (1).

Example (1.13). Solve $\frac{dy}{dx} = \frac{x + y - 2}{x - y + 4}$. (Nov./Dec. 2007)

Solution. Given equation is $\frac{dy}{dx} = \frac{x + y - 2}{x - y + 4}$ $\left[\text{Case } \frac{a}{a'} \neq \frac{b}{b'}\right]$ ...(1)

Putting $x = X + h$, $y = Y + k$, (h, k) being the point of intersection of $x + y - 2 = 0$ and $x - y - 4 = 0$, so that $dx = dX$, $dy = dY$, (1) becomes

$$\frac{dY}{dX} = \frac{Y + X + (k + h - 2)}{Y - X + (k - h - 4)} \quad \ldots(2)$$

But $h + k - 2 = 0$ and $h - k - 4 = 0$ are both true if $h = 3$, $k = -1$

$\therefore$ (2) becomes $\frac{dY}{dX} = \frac{Y + X}{Y - X}$ which is homogeneous in X and Y $\quad \ldots(3)$

$\therefore$ put $Y = vX$, then $\frac{dY}{dX} = v + X\frac{dv}{dX}$

$\therefore$ (3) becomes $v + X\frac{dv}{dX} = \frac{v + 1}{v - 1}$ or $X\frac{dv}{dX} = \frac{v + 1}{v - 1} - v = \frac{1 + 2v - v^2}{v - 1}$

or $\frac{v - 1}{1 + 2v - v^2}dv = \frac{dX}{X}$

Integrating both sides, $-\frac{1}{2}\int \frac{2 - 2v}{1 + 2v - v^2}dv = \int \frac{dX}{X} + c$

or $-\frac{1}{2}\log(1 + 2v - v^2) = \log X + c$

or $\log\left(1 + \frac{2XY}{X^2} - \frac{Y^2}{X^2}\right) + \log X^2 = -2c$

or $\log(X^2 + 2XY - Y^2) = -2c$ or $X^2 + 2XY - Y^2 = e^{-2c} = e^k$ $\quad \ldots(4)$

Putting $X = x - h = x - 3$, $Y = y - k = y + 1$, (4) becomes

$$(x - 3)^2 + 2(x - 3)(y + 1) - (y + 1)^2 = e^k$$

or $x^2 + 2xy - y^2 - 6x + 4y - 18 = e^k$ which is the required solution.

Example 31.14. Solve $(2x + 3y + 1)dx + (4x + 6y + 5)dy = 0$. (Nottingham 2000 S)

Solution. Given equation is $\frac{dy}{dx} = \frac{-2x - 3y - 1}{4x + 6y + 5}$ $\quad \ldots(1)$

Putting $2x + 3y = z$ so that $2 + 3\frac{dy}{dx} = \frac{dz}{dx}$ $\therefore$ (1) becomes $\frac{1}{3}\left(\frac{dz}{dx} - 2\right) = \frac{-z - 1}{2z + 5}$

or $\frac{dz}{dx} = 2 - \frac{2z + 12}{2z + 5} = \frac{3z + 23}{2z + 5}$ or $\frac{2z + 5}{3z + 23}dz = dx$

Integrating both sides, $\int \frac{2z + 5}{3z + 23}dz = \int dx + c$

or $\int\left(\frac{2}{3} - \frac{7}{3}\cdot\frac{1}{3z + 23}\right)dz = x + c$ or $\frac{2}{3}z - \frac{7}{9}\log(3z + 23) = x + c$

Putting $z = 2x + 3y$, we have $3(2x + 3y) - 7\log(6x + 9y + 23) = 9x + 9c$

or $3y - 6y = 7\log(6x + 9y + 23) = c'$ which is the required solution.

EXERCISES 31.8

Solve the following differential equations:

1. $(x + y - 2)dx + (x - y - 4)dy = 0$ (Nottingham 2000)
2. $(2x + y + 2)dx + (x + 2y - 1)dy = 0$ (C.T.I., 2008 S; Madras 1999)
3. $(2x + 3y + 1)dx - (4x + 6y - 1)dy = 0$ (U.A.V.U., 2004)
4. $\frac{dy}{dx} = \frac{x + y + 1}{2x + 2y + 3}$ $\therefore \frac{dy}{dx} = \frac{x + y + 1}{2(x + y) + 3}$
5. $(3x - 6y + 5)dx + (x - 2y - 2)dy = 0$
6. $(x + 2y)dx - (y + 1)dy = 0$ (Nagpur 2000 S; U.P.T.U., 2003)

11.3 LINEAR EQUATIONS

A differential equation is said to be linear if its dependent variable and its differential coefficients occur only in the first degree and not multiplied together.

Thus the standard form of a linear equation of the first order, commonly known as Leibnitz's linear equation*, is

$$\frac{dy}{dx} + Py = Q \quad \text{...(1)}$$

where, P, Q are the functions of x.

To solve the equation, multiply both sides by $e^{\int P dx}$ so that we get

$$\frac{dy}{dx} \cdot e^{\int P dx} + y P e^{\int P dx} = Q e^{\int P dx} \quad \text{i.e.,} \quad \frac{d}{dx}\left(y e^{\int P dx}\right) = Q e^{\int P dx}$$

Integrating both sides, we get $y e^{\int P dx} = \int Q e^{\int P dx} dx + c$ as the required solution.

Note. The factor $e^{\int P dx}$ on multiplying by which the left hand side of (1) becomes the differential coefficient of a single function is called the integrating factor (I.F.) of the linear equation (1).

The above rule is usually stated I.F. $= e^{\int P dx}$

and the solution is y (I.F.) $= \int Q$ (I.F.) $dx + c$.

Example 11.10. Solve $(x+1)\frac{dy}{dx} - ny = e^x (x+1)^{n+1}$.

Solution. Dividing throughout by $(x+1)$, the given equation becomes

$$\frac{dy}{dx} - \frac{n}{x+1} y = e^x (x+1)^n \quad \text{...(1)}$$

which is Leibnitz's equation.

Here $P = -\frac{n}{x+1}$ **and** $\int P dx = -\int \frac{n\,dx}{x+1} = -n \log(x+1) = \log(x+1)^{-n}$

$$\therefore \quad \text{I.F.} = e^{\int P dx} = e^{\log(x+1)^{-n}} = \frac{1}{(x+1)^n}$$

Thus the solution of (1) is y (I.F.) $= \int e^x (x+1)^n \cdot$ (I.F.) $dx + c$

or
$$\frac{y}{(x+1)^n} = \int e^x dx + c = e^x + c \quad \text{or} \quad y = (x+1)^n e^x + c(x+1)^n$$

Example 11.11. Solve $\frac{dy}{dx} - \frac{y}{x} = \frac{x^2}{y^2}$.

Solution. Given equation can be written as $\frac{dy}{dx} + \frac{P}{x} y = \frac{x^2}{y^2}$ ...(1)

$$\therefore \quad \text{I.F.} = e^{\int -\frac{dx}{x}} = e^{-\log x}$$

Thus solution of (1) is y (I.F.) $= \int \frac{x^2}{y^2}$ (I.F.) $dx + c$

or
$$y e^{-\log x} = \int \frac{x^2}{y^2} \cdot e^{-\log x} dx + c$$

or
$$y e^{\log \frac{1}{x}} = \int y^{-2} x\, dx + c \quad \text{or} \quad y \cdot \frac{1}{x} = \frac{x^2}{-2} + c.$$

* See footnote p. 130.

Example 11.17. Solve $(x\log x)\frac{dy}{dx}+y=2\log x$. **(Rajasthan, 2009)**

Solution. Putting $y^2=z$ and $2y\frac{dy}{dx}=\frac{dz}{dx}$, the given equation becomes

$$x(1-x^2)\frac{dz}{dx}+(2x^2-1)z=x^2, \text{ or } \frac{dz}{dx}+\frac{2x^2-1}{x-x^3}z=\frac{x^2}{x-x^3} \quad \ldots(1)$$

which is Leibnitz's equation in z.

$$\therefore \quad \text{I.F.}=\exp\left(\int\frac{2x^2-1}{x-x^3}dx\right)$$

Now $\int\frac{2x^2-1}{x-x^3}dx=\int\left(\frac{1}{x}-\frac{3}{2}\frac{1}{1+x}+\frac{3}{2}\frac{1}{1-x}\right)dx=\log x-\frac{3}{2}\log(1+x)-\frac{3}{2}\log(1-x)$

$$=-\log\left[x\sqrt{(1-x^2)}\right]$$

$$\therefore \quad \text{I.F.}=e^{-\log\left[x\sqrt{(1-x^2)}\right]}=\left[x\sqrt{(1-x^2)}\right]^{-1}$$

Then the solution of (1) is

$$z(\text{I.F.})=\int\frac{x^2}{x-x^3}(\text{I.F.})dx+c$$

or $$\frac{z}{\left[x\sqrt{(1-x^2)}\right]}=\int\frac{x^2}{x(1-x^2)}\frac{1}{x\sqrt{(1-x^2)}}dx+c=\int x(1-x^2)^{-3/2}dx$$

$$=-\frac{1}{2}\int(-2x)(1-x^2)^{-3/2}dx+c=\frac{1}{\sqrt{(1-x^2)}}+c$$

Hence the solution of the given equation is

$$y^2=x^2+cx\sqrt{(1-x^2)}. \qquad [\because z=y^2]$$

Example 11.18. Solve $y\log y\,dx+(x-\log y)dy=0$. **(UPTU, 2006)**

Solution. We have $\frac{dx}{dy}+\frac{x}{y\log y}=\frac{1}{y} \quad \ldots(1)$

which is a Leibnitz's equation in x.

$$\therefore \quad \text{I.F.}=e^{\int\frac{1}{y\log y}dy}=e^{\log(\log y)}=\log y$$

Then the solution of (1) is $x(\text{I.F.})=\int\frac{1}{y}(\text{I.F.})dy+c$

$$x\log y=\int\frac{1}{y}\log y\,dy+c=\frac{1}{2}(\log y)^2+c$$

i.e., $$x=\frac{1}{2}\log y+c(\log y)^{-1}.$$

Example 11.19. Solve $(x+y^2)dy+(y^2-x)dx=0$ **(Kumaun, 2009; UPTU, 2006)**

Solution. This equation contains y^2 and $\tan^{-1}y$ and so, therefore, is not a linear in y, but since only x occurs, it can be written as

$$(1+y^2)\frac{dx}{dy}+\tan^{-1}y=x, \text{ or } \frac{dx}{dy}+\frac{1}{1+y^2}x=\frac{\tan^{-1}y}{1+y^2}$$

which is a Leibnitz's equation in x.

$$\therefore \quad \text{I.F.}=\int\frac{1}{1+y^2}dy=e^{\tan^{-1}y}$$

Then the solution is $x(\text{I.F.})=\int\frac{\tan^{-1}y}{1+y^2}(\text{I.F.})dy+c$

or $$e^{\tan^{-1}y}\cdot y=\int\frac{\ln(1+x^2)}{1+x^2}\cdot e^{\tan^{-1}x}\,dx+c$$ $\left[\text{Put } \tan^{-1}x=t,\ \therefore \frac{dx}{1+x^2}=dt\right]$

$$=\int te^t\,dt+c=te^t-\int 1\cdot e^t\,dt+c \qquad \text{(Integrating by parts)}$$

$$=te^t-e^t+c=e^{\tan^{-1}x}\tan^{-1}x-e^{\tan^{-1}x}+c$$

or $$y=\tan^{-1}x-1+ce^{-\tan^{-1}x}.$$

Example 11.50. Solve $x\cos x\frac{dy}{dx}+y(x\sin x+\cos x)=e^x\cos x$ **or** $(x^2-2x^2)\cos x\,dy+(x\sin x+\cos x)y\,dx=e^x\cos x\,dx$.

Solution. Given equation can be rewritten as

$$x\cos x\frac{dy}{dx}+y(x\sin x+\cos x)=e^x\cos x \qquad \ldots(1)$$

Put $xy=z$ so that $x\frac{dy}{dx}+y=\frac{dz}{dx}$ and $y\sin x=\frac{z}{x}\sin x$.

Then (1) becomes $\cos x\frac{dz}{dx}+\left(\frac{1}{x}\sin x+\cos x\right)z=e^x\cos x$ or $\frac{dz}{dx}+\left(\tan x+\frac{1}{x}\right)z=e^x$

which is a Leibnitz's equation $\therefore$ I.F. $=e^{\int(\tan x+\frac{1}{x})dx}=e^{\log\sec x+\log x}=x\sec x$.

Thus its solution is $z(x\sec x)=\int e^x\cdot x\sec x\,dx+c$

or $$xy\sec x=\frac{1}{2}\int e^x\,2x\,dx+c=\frac{1}{2}e^x+c$$

or $$2xy\cos x=e^x+2c \quad \text{or} \quad x(1+2ce^{-x})=2\cos x.$$

EXERCISE 11.8

Solve the following differential equations :

1. $\cos^2 x\frac{dy}{dx}+y=\tan x$. 2. $x\sin x\frac{dy}{dx}+(x\cos x+\sin x)y=\sin 2x$. (V.T.U., 2012)
3. $(x^2\sin x+1)\,dy+(2xy\sin x+\cos x)\,dx=0$, given $y=1$ when $x=\pi/2$. (V.T.U., 2012)
4. $\sec x\frac{dy}{dx}+2y\tan x=\sin x$, given $y=0$ when $x=\pi/3$. (U.P.T.U., 2004)
5. $(1-x^2)\frac{dy}{dx}-xy=1$. (U.P.T.U., 2004) 6. $(1-x^2)\frac{dy}{dx}+2xy=x\sqrt{(1-x^2)}$. (Nagpur, 2004)
7. $\frac{dy}{dx}=\frac{x+y\sin x}{1+\cos x}$. 8. $(x+2y^3)\frac{dy}{dx}=y$. (U.P.T.U., 2004)
9. $\frac{dy}{dx}+2xy=x^3$. (U.P.T.U., 2005) 10. $(x+y)\frac{dy}{dx}=1$. (Bhubaneswar, 2010)
11. $\frac{dy}{dx}+y\cot x=\operatorname{cosec}^2 x-\cot x$. 12. $y\cos x+x^2=\sin x\frac{dy}{dx}$.
13. $(x+y)\frac{dy}{dx}+(x-y)=e^{\tan^{-1}(y/x)}$. (U.P.T.U., 2005) 14. $e^{\tan^{-1}x}\frac{dy}{dx}+y=\tan^{-1}x$.

11.11 BERNOULLI'S EQUATION

The equation $$\frac{dy}{dx}+Py=Qy^n \qquad \ldots(1)$$

where P, Q are functions of x, is reducible to the Leibnitz's linear equation and is usually called the Bernoulli's equation*.

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*Named after the Swiss mathematician, *Jacob Bernoulli* (1654–1705) who is known for his basic work in probability and also for his theory. He was professor at Basel and had amongst his students his younger brother *Johann Bernoulli* (1667–1748) and his nephew *Nicholas Bernoulli* (1687–1759). Johann is known for his famous controversial series of Calculus with Newton and produced results more on the development of Euler series and probability. His son, *Daniel Bernoulli* (1700–1782) is known for his contributions to Kinetic Theory of gases and Fluid flow.

To solve (1), divide both sides by y^n, so that $y^{-n}\frac{dy}{dx} + Py^{1-n} = Q$ ...(2)

Put $y^{1-n} = z$, so that $(1-n)y^{-n}\frac{dy}{dx} = \frac{dz}{dx}$

i.e. (2) becomes $\frac{1}{1-n}\frac{dz}{dx} + Pz = Q$ or $\frac{dz}{dx} + P(1-n)z = Q(1-n)$,

which is Leibnitz's linear in z and can be solved as in.

Example 11.21. Solve $x\frac{dy}{dx} - y = y^2$.

Solution. Dividing throughout by xy^2, $y^{-2}\frac{dy}{dx} - \frac{1}{x}y^{-1} = 1$ ...(1)

Put $y^{-1} = z$, so that $-1y^{-2}\frac{dy}{dx} = \frac{dz}{dx}$ i.e. (1) becomes $-\frac{dz}{dx} - \frac{z}{x} = 1$

or $\frac{dz}{dx} + \frac{1}{x}z = -1$ which is Leibnitz's linear in z. ...(2)

$$I.F. = e^{\int \frac{1}{x}dx} = e^{\log x} = x$$

$\therefore$ the solution of (2) is $z(I.F.) = \int (-1)(I.F.)dx + c$ or $xz = \int -1 \cdot x\,dx + c$

or $xy^{-1} = -\frac{x^2}{2} + c$ $\left[\because z = y^{-1}\right]$

Dividing throughout by $y^{-1}x^2$, $1 = -\frac{1}{2}x^2y + cxy^2$ which is the required solution.

Example 11.22. Solve $x(1-x^2)\frac{dy}{dx} + 2x^2y = y$. (Degree, Hons)

Solution. Rewriting the given equation as

$$\frac{dy}{dx} - \frac{y}{x(1-x^2)} = \frac{y^2}{x^2}$$

and dividing by y^2, we have

$$y^{-2}\frac{dy}{dx} - \frac{1}{x(1-x^2)}y^{-1} = \frac{1}{x^2} \quad ...(1)$$

Putting $y^{-1} = z$ so that $-y^{-2}\frac{dy}{dx} = \frac{dz}{dx}$ (1) becomes

$$\frac{dz}{dx} + \frac{1}{x(1-x^2)}z = -\frac{1}{x^2}$$ which is Leibnitz's linear in z.

Here $I.F. = e^{\int \frac{1}{x(1-x^2)}dx}$

$\therefore$ the solution is $z(I.F.) = \int (-x^{-2})(I.F.)dx + c$

or $yx^{1/x} = -\int x^{-2}x^{1/x}\,dx + c$ $\left|\begin{array}{l}\text{Put } \frac{1}{x} = t\\ \text{so that } -\frac{1}{x^2}dx = dt\end{array}\right.$

$= -\int te^t dt + c$ [Integrate by parts]

$= -t(e^t) - \int (1)e^t dt] + c = -t[e^t - e^t] + c = e^t(1-t) + c$

or $x = (1-x^2) = ce^{-\frac{1}{x}}$ or $\log x = (1-x^2) + cx^{-\frac{1}{x}}$.

Note. General equation reducible to Leibnitz's linear in z^n by $\frac{dy}{dx} + Py = Qy^n$ ...(1)

where P & Q are functions of x. To reduce (1) to $\frac{dz}{dx} + Pz = Q$

Example 11.23. Solve $\frac{dy}{dx} + y\cos x = y^n \sin 2x$. (U.P.T.U., 2003; Himachal Pradesh, 2004; J.N.T.U., 2005)

Solution. Dividing throughout by $\cos^n y$, we get $\sec^n y \frac{dy}{dx} + 2x \frac{\sin y \cos^n y}{\cos^n y} = x^n$

or $\sec^n y \frac{dy}{dx} + 2x \tan y = x^n$ which is of the form $\frac{dz}{dx} + Pz = Q$ ...(1)

Put $\tan y = z$ so that $\sec^2 y \frac{dy}{dx} = \frac{dz}{dx}$ ∴ (1) becomes $\frac{dz}{dx} + 2xz = x^n$

This is Leibnitz's linear equation in z. ∴ $I.F. = e^{\int 2x\,dx} = e^{x^2}$

∴ the solution is $ze^{x^2} = \int e^{x^2} x^n dx + c = \frac{1}{2}(x^{n-1} - (n-1)e^{x^2}) + c$

Replacing z by $\tan y$, we get $\tan y = \frac{1}{2}(x^{n-1} - (n-1)) + ce^{-x^2}$ which is the required solution.

Example 11.24. Solve $\frac{dy}{dx} + \left(\frac{1}{x}\right)\log y = \frac{1}{x}(\log y)^2$.

Solution. Dividing by y, the given equation becomes

$$\frac{1}{y}\frac{dy}{dx} + \frac{1}{x}\log y = \frac{1}{x}(\log y)^2 \quad ...(1)$$

Put $\log y = z$ so that $\frac{1}{y}\frac{dy}{dx} = \frac{dz}{dx}$ ∴ (1) becomes

$$\frac{dz}{dx} + \frac{1}{x}z = \frac{1}{x}z^2 \text{ or } \frac{1}{z^2}\frac{dz}{dx} + \frac{1}{x}\cdot\frac{1}{z} = \frac{1}{x} \quad ...(2)$$

This is a Bernoulli's equation, put $1/z = v$ so that (2) reduces to

$$-\frac{dv}{dx} + \frac{1}{x}v = \frac{1}{x} \text{ or } \frac{dv}{dx} - \frac{1}{x}v = -\frac{1}{x}$$

This is Leibnitz's linear in v. ∴ $I.F. = e^{-\int(1/x)dx} = 1/x$

∴ the solution is $v\cdot\frac{1}{x} = -\int\frac{1}{x}\cdot\frac{1}{x}dx + c = \frac{1}{x} + c$

Replacing v by $1/z$, we get $1/z^2 = x^{-1} + c$ or $(\log y)^{-1} = 1 + cx$ which is the required solution.

PROBLEMS 4.3

Solve the following equations :

1. $\frac{dy}{dx} + y\tan x = y^2\sec x$ (U.P.T.U., 2002)
2. $\frac{dy}{dx} + y\tan x = y^3\sec x$
3. $\frac{dy}{dx} = \frac{x^2 + y^2 + 1}{2xy}$ (U.P.T.U., 2006)
4. $\frac{dy}{dx} + \frac{\tan y}{1+x} = (1+x)e^x\sec y$ (Kerala, 2003)
5. $\sec^2 y\frac{dy}{dx} + y\tan y = x^2$
6. $\frac{dy}{dx} + y\cos x = \sin x\cos x$ (U.P.T.U., 2011)
7. $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{\cos x}{x}$ (U.P.T.U., 2003)
8. $\frac{dy}{dx} + y\cot x = 2\cos x$ (U.P.T.U., 2002)
9. $(x^2+1)\frac{dy}{dx} + 2xy = 4x^2$
10. $\frac{dy}{dx} + y\tan x = \sec x$ (U.P.T.U., 2003)
11. $\frac{dy}{dx} + \frac{y}{x} = y^2$
12. $x\frac{dy}{dx} + y = y^2\log x$
13. $\frac{dy}{dx} + y\cot x = \cos x$
14. $\frac{dy}{dx} + \frac{y}{x} = x^n$ (U.P.T.U., 2003)
15. $\frac{dy}{dx} + y\cot x = \sin 2x$ (Kerala, 2003)
16. $x\log x\frac{dy}{dx} + y = 2\log x$ (U.P.T.U., 2002)
17. $\frac{dy}{dx} + y\cot x = \cos x$ (U.P.T.U., 2003)
18. $y\log y + x\frac{dy}{dx} - x\log y = 0$ (U.P.T.U., 2003)

[illegible] EXACT DIFFERENTIAL EQUATIONS

1.1 Def. A differential equation of the form $M(x,y)\,dx + N(x,y)\,dy = 0$ is said to be **exact** if the left-hand member is the exact differential of some function $u = f(x,y)$, so that $du = M\,dx + N\,dy = 0$. An equation, therefore, is exact if $M_y = N_x$.

1.2 Theorem. The necessary and sufficient condition for the differential equation $M\,dx + N\,dy = 0$ to be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

[condition is necessary]

The equation $M\,dx + N\,dy = 0$ will be exact if

$$M\,dx + N\,dy = du \tag{1}$$

where u is some function of x and y:

Now $$du = \frac{\partial u}{\partial x}\,dx + \frac{\partial u}{\partial y}\,dy \tag{2}$$

[illegible] equating coefficients of dx and dy in (1) and (2), we get $M = \frac{\partial u}{\partial x}$ and $N = \frac{\partial u}{\partial y}$

[illegible]
$$\frac{\partial M}{\partial y} = \frac{\partial^2 u}{\partial y\,\partial x} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 u}{\partial x\,\partial y}$$

But $$\frac{\partial^2 u}{\partial y\,\partial x} = \frac{\partial^2 u}{\partial x\,\partial y}$$ [illegible]

[illegible] $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, which is the necessary condition for exactness.

[Condition is sufficient] i.e. if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, then $M\,dx + N\,dy = 0$ is exact.

To get $\left[\int M\,dx\right]_{y}$ where y is supposed constant while dx is performing [illegible]

Then $$\frac{\partial}{\partial y}\left[\int M\,dx\right]_{y} = \int \frac{\partial M}{\partial y}\,dx + \phi(y) \tag{3}$$

[illegible] $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ [illegible]

and $\frac{\partial^2 u}{\partial x\,\partial y} = \frac{\partial^2 u}{\partial y\,\partial x}$

[illegible] [illegible] $\frac{\partial u}{\partial y} = \int \frac{\partial M}{\partial y}\,dx + \phi(y)$ [illegible] $\phi(y)$ [illegible]

$$du = \frac{\partial u}{\partial x}\,dx + \phi(y)\,dy \text{ where } \phi(y) \text{ is a function of } y \text{ alone} \tag{4}$$

[illegible] $M\,dx + N\,dy = \frac{\partial u}{\partial x}\,dx + \left[\frac{\partial u}{\partial y} + \phi(y)\right]dy$ [illegible] (3) and (4)

$$= \left[\int \frac{\partial M}{\partial y}\,dx + \int \frac{\partial N}{\partial x}\,dx\right] + \phi(y)\,dy + N\,dy - \int \frac{\partial N}{\partial x}\,dx = \left[\int M\,dx\right]_{y} + \int N\,dy - \int \left[\int \frac{\partial N}{\partial x}\,dx\right]dy \tag{5}$$

[illegible] $M\,dx + N\,dy = du$ [illegible]

1.3 Method of solution. To solve the equation $M\,dx + N\,dy = 0$ [illegible] $\int M\,dx$ [illegible]

Integrating $u = \int M\,dx + \phi(y)$

find $u = \int \frac{\partial}{\partial y}\left[\int M\,dx\right]dy$ and $\phi(y) =$ [illegible] of [illegible] not containing x

[illegible] The solution of $M\,dx + N\,dy = 0$ is

$\left[\int M\,dx\right]_{y} +$ [illegible] of [illegible] not containing x in $\int N\,dy = c$

[illegible]
$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Example 11.25. Solve $(y^2e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 2y)dy = 0$. (U.P.T.U., 2006)

Solution. Here $M = y^2e^{xy^2} + 4x^3$ and $N = 2xye^{xy^2} - 2y$

$\therefore$ $\frac{\partial M}{\partial y} = 2ye^{xy^2} + y^2e^{xy^2} \cdot 2xy = \frac{\partial N}{\partial x}$

This is the equation is exact and its solution is

$$\int_{y\text{ constant}} M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$$

i.e., $$\int_{y\text{ constant}} (y^2e^{xy^2} + 4x^3)\,dx + \int -2y^2\,dy = c \quad \text{or} \quad e^{xy^2} + x^4 - y^2 = c.$$

Example 11.26. Solve $\left[y\left(2 + \frac{1}{x}\right) + \cos y\right]dx + (x + \log x - x\sin y)dy = 0$.

(Rajiv Gandhi, 2009 S, U.P.T.U., 2006)

Solution. Here $M = y\left(2 + \frac{1}{x}\right) + \cos y$ and $N = x + \log x - x\sin y$

$\therefore$ $\frac{\partial M}{\partial y} = 1 + \frac{1}{x} - \sin y = \frac{\partial N}{\partial x}$

Thus the equation is exact and its solution is

$$\int_{y\text{ constant}} M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$$

$$\int_{y\text{ constant}} \left[y\left(2 + \frac{1}{x}\right) + \cos y\right]dx + c \quad \text{or} \quad 2x + \log(xy) + x\cos y = c$$

Example 11.27. Solve $(1 + 2xy\cos x^2 - 2y)dx + (\sin x^2 - 2x)dy = 0$.

Solution. Here $M = 1 + 2xy\cos x^2 - 2y$ and $N = \sin x^2 - 2x$

$\therefore$ $\frac{\partial M}{\partial y} = 2x\cos x^2 - 2 = \frac{\partial N}{\partial x}$

Thus the equation is exact and its solution is

$$\int_{y\text{ constant}} M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$$

i.e., $$\int_{y\text{ constant}} (1 + 2xy\cos x^2 - 2y)\,dx = c \quad \text{or} \quad x + y\left[\int \cos x^2 \cdot 2x\,dx - \int 2x\,dx\right] = c$$

or $$x + y\sin x^2 - yx^2 = c.$$

Example 11.28. Solve $\frac{dy}{dx} + \frac{e^x\cos y + e^y\cos x + xy}{e^x\sin y - e^y\sin x - y} = 0$. (Kanpur Univ., 2005)

Solution. Given equation can be written as

$$(e^x\cos y + e^y\cos x + xy)\,dx + (e^x\sin y - e^y\sin x - y)\,dy = 0.$$

Here $M = e^x\cos y + e^y\cos x + xy$ and $N = e^x\sin y - e^y\sin x - y$

$\therefore$ $\frac{\partial M}{\partial y} = -e^x\sin y + e^y\cos x + x = \frac{\partial N}{\partial x}$

Thus the equation is exact and its solution is

$$\int_{y\text{ constant}} M\,dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$$

i.e., $$\int_{y\text{ constant}} (e^x\cos y + e^y\cos x + xy)\,dx + \int -y\,dy = c \quad \text{or} \quad e^x\sin y + e^y\sin x + \frac{x^2}{2} - \frac{y^2}{2} = c$$

Example 4.126. Solve $(2x^2+2y^2-7)x\,dx-(3x^2+2y^2-8)y\,dy=0$. (U.P.T.U. 2003)

Solution. Given equation can be written as

$$\frac{x\,dx}{x\,dx}=\frac{2x^2+2y^2-7}{3x^2+2y^2-8}$$

or $$\frac{y\,dx+x\,dx}{y\,dx-x\,dx}=\frac{5x^2+5y^2-15}{-x^2+y^2-1}$$ By componendo & dividendo

or $$\frac{x\,dx+y\,dy}{x^2+y^2-8}=\frac{x\,dx-y\,dy}{x^2-y^2-1}$$

Integrating both sides, we get

$$\int\frac{2x\,dx+2y\,dy}{x^2+y^2-8}=5\int\frac{2x\,dx-2y\,dy}{x^2-y^2-1}+c$$

or $$\log(x^2+y^2-8)=5\log(x^2-y^2-1)+\log c$$ Putting $c=\log c'$

or $$x^2+y^2-8=c'(x^2-y^2-1)^5$$

which is the required solution.

EXERCISE 4.7

Solve the following equations:

1. $(x^2-xy)\,dx+(xy-y^2)\,dy=0$
2. $(x^2+y^2-a^2)\,x\,dx+(x^2-y^2-b^2)\,y\,dy=0$ (Kurukshetra, 2010)
3. $(x^2-2xy+y^2+2x-2y+3)\,dx-(x^2-2xy+y^2-2x+2y-1)\,dy=0$
4. $y(x\,dx+2y\,dy)=x\,dy$
5. $(3x^2+6xy^2+3y^4)\,dx+(6x^2y+12xy^3+6y^5)\,dy=0$ (U.P.T.U. 2008)
6. $(3x^2+6xy^2+3y^4)\,dx+(6x^2y+12xy^3+6y^5)\,dy=0$
7. $(x^2+y^2+x)\,dx+(x^2+y^2-y)\,dy=0$
8. $(x+2y-3)\,dx-(2x+y-3)\,dy=0$ (Kurukshetra, 2010)
9. $(2x+3y+4)\,dx+(x+y+1)\,dy=0$ (Nagpur, 2005)

4.12.2 EQUATIONS REDUCIBLE TO EXACT EQUATIONS

Some times a differential equation which is not exact, can be made so on multiplication by a suitable factor called an integrating factor. The rules for finding integrating factors of the equation $Mdx+Ndy=0$ are as follows:

(1) I.F. found by inspection. In a number of cases, the integrating factor can be found after regrouping the terms of the equation and rewriting it as a group of terms equal to zero, each group as being a part of an exact differential. In this connection, the following integrals or combinations prove quite useful:

$$xdy+ydx=d(xy)$$

$$\frac{xdy-ydx}{y^2}=d\left(\frac{x}{y}\right);\ \frac{xdy-ydx}{xy}=d\left[\log\left(\frac{x}{y}\right)\right]$$

$$\frac{xdy-ydx}{y^2}=-d\left(\frac{x}{y}\right);\ \frac{xdy-ydx}{x^2+y^2}=d\left(\tan^{-1}\frac{x}{y}\right)$$

$$\frac{xdy-ydx}{x^2-y^2}=d\left[\frac{1}{2}\log\frac{x+y}{x-y}\right].$$

Example 4.128. Solve $(xy^2+y)\,dx+(x^2y-x)\,dy=0$. (Kurukshetra, 2010)

Solution. It is easy to note that the terms ydx and xdy should be put together.

$$\therefore\qquad (y^2dx-xdy)+(xy^2dx+x^2ydy)=0$$

Now we observe that the term $2xy^2dx$ should not involve y^2. This suggests that $1/y^2$ may be I.F. Multiplying throughout by $1/x^2$, it follows

$$\frac{xy^2dx - x^2dx}{y^2} + 2xdx = 0 \quad \text{or} \quad d\left(\frac{x^2}{y}\right) + 2xdx = 0$$

Integrating, we get $\frac{x^2}{y} + x^2 = c$ which is the required solution.

(2) I.F. of a homogeneous equation. *If $Mdx + Ndy = 0$ be a homogeneous equation in x and y, then $\frac{1}{Mx + Ny}$ is an integrating factor for $Mx + Ny \neq 0$.*

Example 11.21. Solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$. (Rohtak, 2003 S)

Solution. This equation is homogeneous in x and y.

$$\therefore \quad \text{I.F.} = \frac{1}{Mx + Ny} = \frac{1}{(x^3y - 2x^2y^2)x - (x^3 - 3x^2y)y} = \frac{1}{x^2y^2}$$

Multiplying throughout by $1/x^2y^2$, the equation becomes

$$\left(\frac{1}{y} - \frac{2}{x}\right)dx - \left(\frac{x}{y^2} - \frac{3}{y}\right)dy = 0 \text{ which is exact.}$$

$\therefore$ the solution is $\int_{y\,\text{const.}} Mdx + \left[\text{terms of } N \text{ not containing } y\right] = c$ or $\frac{x}{y} - 2\log y + 3\log y = c$

(3) I.F. for an equation of the type $f_1(xy)ydx + f_2(xy)xdy = 0$.

If the equation $Mdx + Ndy = 0$ be of this form, then $\frac{1}{Mx - Ny}$ is an integrating factor ($Mx - Ny \neq 0$).

Example 11.22. Solve $(x + xy)ydx + (x - xy)dy = 0$. (AMIETE, 2009)

Solution. The given equation is of the form $f_1(xy)ydx + f_2(xy)xdy = 0$.

Here $\quad M = (1 + xy)y, N = (1 - xy)x.$

$$\therefore \quad \text{I.F.} = \frac{1}{Mx - Ny} = \frac{1}{(1 + xy)xy - (1 - xy)xy} = \frac{1}{2x^2y^2}$$

Multiplying throughout by $1/2x^2y^2$, it becomes

$$\left(\frac{1}{2x^2y} + \frac{1}{2x}\right)dx + \left(\frac{1}{2xy^2} - \frac{1}{2y}\right)dy = 0, \text{ which is an exact equation.}$$

$\therefore$ the solution is $\int_{y\,\text{const.}} Mdx + \left[\text{terms of } N \text{ not containing } x\right] = c$

or $$\frac{1}{2y}\left(-\frac{1}{x}\right) + \frac{1}{2}\log x - \frac{1}{2}\log y = c \quad \text{or} \quad \log\frac{x}{y} - \frac{2}{xy} = c.$$

(4) In the equation $Mdx + Ndy = 0$,

(a) if $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is a function of x only $= f(x)$ say, then $e^{\int f(x)dx}$ is an integrating factor.

(b) if $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$ is a function of y only $= F(y)$ say, then $e^{\int F(y)dy}$ is an integrating factor.

Example 11.23. Solve $(y^2 - e^{-2x})dx + (xy + e^{-2x})dy = 0$. (AMIETE, 2009; Madras, 2007)

Solution. Here $M = y^2 - e^{-2x}$ and $N = xy + e^{-2x}$.

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{2y - (y - 2e^{-2x})}{xy + e^{-2x}} = \frac{1}{x} \text{ which is a function of } x \text{ only.}$$

$\therefore$ $\quad I.F. = e^{\int \left(\frac{2}{x} - \frac{1}{x}\right)dx} = e^{2\log x - \log x} = e^{\log x} = x$

Multiplying throughout by x, we get $\left(\frac{x^2}{y^2} - \frac{1}{x^2}e^{1/x}\right)dx - \frac{x}{y^2}\,dy = 0$

which is an exact equation.

$\therefore$ the solution is $\int_{(y\text{ const.})} (\text{terms of } N) dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$

or $\quad \int \left(\frac{x^2}{y^2} - \frac{1}{x^2}e^{1/x}\right)dx + 0 = c$

or $\quad -\frac{x^2 e^{-1}}{2} + \frac{1}{2}\int e^{-1}(-2x^{-3})\,dx = c$ or $\frac{1}{2}e^{-1} - \frac{1}{2}\frac{x^2}{y^2} = c$.

Otherwise it can be solved as a Bernoulli's equation (§ 13.10).

Example 11.38. Solve $(xy^2 + y)dx + (2x^2y^2 + x + y^3)dy = 0$.

Solution. Here $M = xy^2 + y$, $N = 2x^2y^2 + x + y^3$.

$\therefore$ $\quad \frac{1}{M}\left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right] = \frac{1}{y(xy^2+1)}(4xy^2 + 1 - 2y^2 - 1) = \frac{1}{y}$, which is a function of y alone.

$\therefore$ $\quad I.F. = e^{\int \frac{1}{y}dy} = e^{\log y} = y$

Multiplying throughout by y, it becomes $(xy^3 + y^2)dx + (2x^2y^3 + xy + y^4)dy = 0$, which is an exact equation.

$\therefore$ the solution is $\int_{(y\text{ const.})} (\text{terms of } M)dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$

or $\quad \int_{(y\text{ const.})} (xy^3 + y^2)dx + \int y^4\,dy = c$ or $\frac{1}{2}x^2y^3 + xy^2 + \frac{1}{5}y^5 = c$.

Example 11.39. Solve $(y\log y)dx + (x - \log y)dy = 0$. *(U.P.T.U., 2000)*

Solution. Here $M = y\log y$ and $N = x - \log y$.

$\therefore$ $\quad \frac{1}{M}\left[\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right] = \frac{1}{y\log y}(1 - \log y - 1) = -\frac{1}{y}$, which is a function of y alone.

$\therefore$ $\quad I.F. = e^{-\int \frac{1}{y}dy} = e^{-\log y} = \frac{1}{y}$

Multiplying the given equation throughout by $\frac{1}{y}$, it becomes

$$\log y\,dx + \frac{1}{y}x - \log y\,dy = 0$$

which is an exact equation $\left[\because \frac{\partial}{\partial y}(\log y) = \frac{\partial}{\partial x}\left(\frac{x - \log y}{y}\right)\right]$

$\therefore$ the solution is $\int_{(y\text{ const.})} (\text{terms of } M)dx + \int (\text{terms of } N \text{ not containing } x)\,dy = c$

or $\quad \log y \int dx + \int \left(\frac{-\log y}{y}\right)dy = c$ or $x\log y - \frac{1}{2}(\log y)^2 = c$.

(II) For the equations of the type

$$x^n y^n (my\,dx + nx\,dy) + x^p y^q (py\,dx + qx\,dy) = 0,$$

on taking making factor as $x^h y^k$

where $\quad \frac{m+h+1}{n+k+1} = \frac{h+1}{k+1}$, $\frac{n+h+1}{m+k+1} = \frac{k+1}{h+1}$,

Example 11.30. Solve $xy(y dx - x dy) + 2(x^4 + y^4)dx = 0$. (Madras, 2003; Kurukshetra, 2005)

Solution. Rewriting the equation as $xy(y dx - x dy) + 2(x^4 + y^4)dx = 0$ and comparing with

$$x^a y^b (mydx + nxdy) + x^{a'} y^{b'} (m'ydx + n'xdy) = 0,$$

we have $a = b = 2$, $c = n = 1$, $a' = 4$, $b' = 0$, $m' = 2$, $n' = -1$.

$$\therefore \quad IF = x^h y^k,$$

where $$\frac{a+b+1}{m} = \frac{h+k+1}{n} = \frac{a'+b'+1}{m'} = \frac{b'+k+1}{n'}$$

i.e. $$\frac{1+1+1}{1} = \frac{1+1+1}{1} = \frac{4+0+1}{2} = \frac{0+k+1}{-1}$$

or $$k - h = 1, 2 + 2h - k = 4$$

Solving these, we get $h = k = -1$. $\therefore IF = (xy)^{-1}$.

Multiplying the equation by $(xy)^{-1}$, it becomes

$$\left(\frac{1}{x^2 y} + \frac{2x}{y}\right)dx + \left(\frac{1}{xy^2} - \frac{1}{y}\right)dy = 0, \text{ which is an exact equation.}$$

$\therefore$ The solution is $\int_{y=\text{const.}} M\,dx + \int$ (terms of N not containing x) $dy = c$

or $$\frac{1}{y}\left(-\frac{1}{x}\right) + 2\log x - \log y = c' \quad \text{or} \quad 2\log x - \log y - \frac{1}{xy} = c.$$

EXERCISE 11.6

Solve the following equations :

1. $xdx + ydy + x(xdy - ydx) = 0$.
2. $y(xy + 2x^2y^2)dx + x(xy - x^2y^2)dy = 0$. (U.P.T.U., 2003)
3. $y(xy + e^x)dx - e^x dy = 0$.
4. $(x^2 + y^2 + x)dx + xydy = 0$.
5. $y(x^2 + y^2 + x)dx + x(x^2 + y^2 - y)dy = 0$.
6. $(x^2 + y^2 + 1)dx - 2xydy = 0$.
7. $(x^2 + y^2 + x)dx + xydy = 0$. (Mumbai, 2004)
8. $x\frac{dy}{dx} - y = x^3(\cos x + \sin x)$.
9. $(x^2 + y^2 + 1)dx - 2xydy = 0$. (Mumbai, 2004)
10. $(x^2 + y^2 + 2x)dx + 2ydy = 0$. (Mumbai, 2004)
11. $(xy \log xy + 1)ydx + (xy \log xy - 1)xdy = 0$. (Madras, 2003)
12. $(x^2 + y^2 + x)dx + xydy = 0$. (Nagpur, 2004)
13. $(xy \log xy + 1)ydx + (xy \log xy - 1)xdy = 0$. (U.P.T.U., 2004)

11.13 EQUATIONS OF THE FIRST ORDER AND HIGHER DEGREE

An equation of first order in higher degree, if it is resolvable into factors each of the form $dy - pdx = 0$, where p is a function of x and y, then the equation is said to be reducible to linear form. Such equations are of the form $f(x, y, p) = 0$. Three cases arise for discussion :

Case I. Equation solvable for p. A differential equation of the first order but of the nth degree is of the form

$$p^n + P_1 p^{n-1} + P_2 p^{n-2} + \ldots + P_n = 0 \qquad \ldots(1)$$

where $P_1, P_2, \ldots P_n$ are functions of x and y.

Splitting up the left hand side of (1) into linear factors, we have

$$[p - f_1(x, y)][p - f_2(x, y)] \ldots [p - f_n(x, y)] = 0.$$

Equating each of the factors to zero,

$$p = f_1(x, y), p = f_2(x, y) \ldots p = f_n(x, y)$$

Solving each of these equations of the first order and first degree, we get the solutions

$$F_1(x, y, c) = 0, F_2(x, y, c) = 0, \ldots F_n(x, y, c) = 0.$$

These n solutions constitute the general solution of (1).
Otherwise, the general solution of (1) may be written as

$$F_1(x, y, c) \cdot F_2(x, y, c) \cdots F_n(x, y, c) = 0$$

Example 13.25. Solve $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$

Solution. Given equation is $p = \frac{1}{p}\left(\frac{x}{y} + \frac{y}{x}\right)$ where $p = \frac{dy}{dx}$ or $p^2 + p\left(\frac{x}{y} - \frac{y}{x}\right) - 1 = 0$.

Factorising $(p + y/x)(p - x/y) = 0$.

Thus we have $p + y/x = 0$ ...(i) and $p - x/y = 0$ ...(ii)

From (i), $\frac{dy}{dx} + \frac{y}{x} = 0$ or $x\,dy + y\,dx = 0$

i.e., $d(xy) = 0$. Integrating, $xy = c_1$.

From (ii), $\frac{dy}{dx} = \frac{x}{y}$ or $y\,dy - x\,dx = 0$

Integrating, $y^2 - x^2 = c_2$. Thus $xy = c_1$ or $x^2 - y^2 = c_2$ constitute the required solutions.

Otherwise, combining these two cases, the required solution can be written as

$$(xy - c_1)(x^2 - y^2 - c_2) = 0$$

Example 13.26. Solve $p^2 + 2py\cot x = y^2$. (Bhopal, 2006; Kanpur, 2007)

Solution. We have $p^2 + 2py\cot x = y^2$ or $p^2 + p^2\cot^2 x = y^2 + y^2\cot^2 x$

or $p^2(1 + \cot^2 x) = y^2(1 + \cot^2 x)$

i.e., $p = y(-\cot x + \operatorname{cosec} x)$ ...(i)

or $p = y(-\cot x - \operatorname{cosec} x)$ ...(ii)

From (i), $\frac{dy}{dx} = y(-\cot x + \operatorname{cosec} x)$ or $\frac{dy}{y} = (\operatorname{cosec} x - \cot x)\,dx$

Integrating, $\log y = \log\tan\frac{x}{2} - \log\sin x + \log c = \log\frac{c\tan x/2}{\sin x}$

or $y = \frac{c}{2\sin^2 x/2}$ or $y(1 - \cos x) = c$ ...(iii)

From (ii), $\frac{dy}{dx} = -y(\cot x + \operatorname{cosec} x)$ or $\frac{dy}{y} = -(\cot x + \operatorname{cosec} x)\,dx$

Integrating, $\log y = -\log\sin x - \log\tan\frac{x}{2} + \log c = \log\frac{c}{\sin x\tan x/2}$

or $y = \frac{c}{2\sin^2 x/2}$ or $y(1 - \cos x) = c$ ...(iv)

Thus combining (iii) and (iv), the required general solution is

$$y(1 \pm \cos x) = c.$$

PROBLEMS 13.9

Solve the following equations:

1. $y\left(\frac{dy}{dx}\right)^2 + (x - y)\frac{dy}{dx} - x = 0$ ...
2. $\left(\frac{dy}{dx}\right)^3 + y^2\left(\frac{dy}{dx}\right)^2 + xy\frac{dy}{dx} + x^2 = 0$. ... (Madras, 2003)

Case II. Equations solvable for x. If the given equation, on solving for y, takes the form

$$y = f(x, p) \quad \ldots(1)$$

then differentiation with respect to x gives an equation of the form

$$p = \frac{dy}{dx} = \phi\left(x, p, \frac{dp}{dx}\right).$$

Now it may be possible to solve this new differential equation for x and p.

Let its solution be $F(x, p, c) = 0$. $\ldots(2)$

The elimination of p from (1) and (2) gives the required solution.

In case elimination of p is not possible, then we may solve (1) and (2) for x and y and obtain

$$x = F_1(p, c), y = F_2(p, c)$$

as the required solution, where p is the parameter.

Obs. This method is successful only for equations which do not contain y explicitly.

Example 11.20. Solve $y = 2px + \tan^{-1}(xp^2)$.

Solution. Given equation is $y = 2px + \tan^{-1}(xp^2)$ $\ldots(1)$

Differentiating both sides with respect to x, $\frac{dy}{dx} = p = 2\left[p + x\frac{dp}{dx}\right] + \frac{p^2 + 2xp\frac{dp}{dx}}{1 + x^2p^2}$

or $p + 2x\frac{dp}{dx} + \left[p + 2x\frac{dp}{dx}\right]\cdot\frac{p^2}{1 + x^2p^2} = 0$ or $\left[p + 2x\frac{dp}{dx}\right]\left[1 + \frac{p^2}{1 + x^2p^2}\right] = 0$

This gives $p + 2x\frac{dp}{dx} = 0$ (Ignoring the other factor)

Separating the variables and integrating, we have $\int\frac{dx}{x} + 2\int\frac{dp}{p} = a$ constant

or $\log x + 2\log p = \log c$ or $\log xp^2 = \log c$

whence $xp^2 = c$ or $p = \sqrt{c/x}$ $\ldots(3)$

Eliminating p from (1) and (3), we get $y = 2\sqrt{cx} + \tan^{-1}c$

or $y = 2\sqrt{cx} + \tan^{-1}c$ which is the general solution of (1).

Obs. The other factor of the form $1 + x^2p^2 = 0$ which we didn't consider, will not be considered here as it is not a relevant solution of (1) whereas we are interested only in finding general solution.

Question. Sometimes one is required to verify that

$$\frac{dy}{dx} = \sqrt{\left(\frac{y}{x}\right)}$$

and is ignoring it we say that the required solution is $y = 2\sqrt{cx} + c$. Such a reasoning is incorrect.

Example 11.21. Solve $y = 2px + p^2$. **(Kanpur 2004)**

Solution. Given equation is $y = 2px + p^2$ $\ldots(1)$

Differentiating it with respect to x, we get

$$\frac{dy}{dx} = p = 2p + 2x\frac{dp}{dx} + 2p\frac{dp}{dx} \quad \text{or} \quad p\frac{dx}{dp} + 2x = -2p^{-1}$$

or $\frac{dx}{dp} + \frac{2x}{p} = -2p^{-2}$ $\ldots(2)$

This is Leibnitz's linear equation in x and p. Here $I.F. = e^{\int\frac{2}{p}dp} = e^{2\log p} = p^2$

∴ the solution of Eq.(i) is

$$x(IF) = \int (1 - x)^{n-1} \cdot (IF) dx + c \quad \text{or} \quad x(1-x)^{n-1} = -n \int y^n dx + c = -\frac{y^{n+1}}{n+1} + c$$

or
$$x = y^n - \frac{(1-x)^{n-1}}{n+1} \qquad \ldots(iv)$$

Substituting the value of x in (i), we get $y = \frac{2n}{p} + \frac{1-n}{1+n} p^2$ $\qquad \ldots(v)$

The equations (iv) and (v) taken together, with parameter p, constitute the general solution of (i).

Obs. In general, the equation of the form $y = px + f(p)$ is known as Lagrange's equation, its solution is obtained in a similar manner as above.

EXERCISE 4.13

Solve the following equations :

1. $y = x - \tan^{-1} p$
2. $y = px + \frac{a}{p}$ (U.T.U. 2003)
3. $y^2 \left(\frac{dy}{dx}\right)^2 - xy \frac{dy}{dx} + y^2 = 0$
4. $x^2 + y = xp$
5. $y = px + p^2$
6. $y = px + \sin^{-1} p$

Case III. Equations solvable for x. If the given equation on solving for x, takes the form

$$x = f(y, p) \qquad \ldots(1)$$

then differentiating with respect to y, gives an equation of the form

$$\frac{1}{p} = \frac{\partial x}{\partial y} + \phi\left(p, x, \frac{dp}{dy}\right)$$

Now it may be possible to solve the new differential equation by usual methods. Let its solution be $F(x, y, c) = 0$.

The elimination of p from (1) and (2) gives the required solution. In case the elimination is not feasible, (1) and (2) may be expressed in terms of p and p may be regarded as a parameter.

Obs. This method is especially useful for equations which are not linear in y.

Example 11.41. Solve $y = 2px - y^2 p^3$ (Bhopal 2008)

Solution. Given equation, on solving for x, takes the form $x = \frac{y + y^2 p^3}{2p}$

Differentiating with respect to y, $\frac{dx}{dy} = \frac{1}{p}$, $\frac{1}{p} = \frac{2p\left[1 - 2y \cdot p^3 - y^2 3p^2 \frac{dp}{dy}\right] - (y + y^2 p^3) 2 \frac{dp}{dy}}{4p^2}$

or
$$2p = p - 2yp^4 - 3y^2 p^2 \frac{dp}{dy} - p - yp^3 = -3yp^2 \frac{dp}{dy}$$

or
$$p + 2yp^3 + 3y^2 p^2 \frac{dp}{dy} + yp^3 = 0 \text{ or } p(1 + 3yp^2) + y \frac{dp}{dy}(1 + 3yp^2) = 0$$

or
$$(p + y \frac{dp}{dy})(1 + 3yp^2) = 0 \text{ This gives } p + y \frac{dp}{dy} = 0, \text{ or } \frac{dp}{dy} + \frac{p}{y} = 0$$

Integrating $\qquad py = c$ $\qquad \ldots(2)$

Eliminating p from the given equation and (2), we get $y = 2x \frac{c}{y} - \frac{c^2}{y^2} y^2$ or $y^3 = 2xc + c^2$

which is the required solution.

EXERCISE 3.11

Solve the following equations :

1. $y^2 - x(x+1)p^2 + 1 = 0$ (Kolhapur, 2006)
2. $y^2 = 2px + p^2$
3. $y = px + ap^2$ Delhi, May 2005
4. $y = \tan\left(x - \frac{dy}{1+dy}\right)$ (U.P.T.U., 2006)

3.13 CLAIRAUT'S EQUATION*

An equation of the form $y = px + f(p)$ is known as Clairaut's equation ...(1)

Differentiating with respect to x, we have $p = p + x\frac{dp}{dx} + f'(p)\frac{dp}{dx}$

or $[x + f'(p)]\frac{dp}{dx} = 0 \quad \therefore \frac{dp}{dx} = 0$, or $x + f'(p) = 0$

$$\frac{dp}{dx} = 0, \text{ gives } p = c \qquad \ldots(2)$$

Thus eliminating p from (1) and (2), we get $y = cx + f(c)$ ...(3)

as the general solution of (1).

Hence the solution of the Clairaut's equation is obtained on replacing p by c.

Obs. If we eliminate p from $x + f'(p) = 0$ and (1), we get an equation involving x and y not containing c. This is the singular solution of (1), which gives the envelope of the family of straight lines (3).

To obtain the singular solution, we proceed as follows :

(i) Find the general solution by replacing p by c, i.e., $y = cx + f(c)$

(ii) Differentiate this w.r.t. c giving $x + f'(c) = 0$ ...(4)

(iii) Eliminate c from (i) and (ii) which will be the singular solution.

Example 3.14. Solve $y = px + \frac{a}{p}$. Also find its singular solution.

Solution. Given equation can be written as

$$ap^2 = y - xp \text{ or } y = px + ap^{-1}p \text{ which is the Clairaut's equation.}$$

$\therefore$ Its solution is $y = cx + ac^{-1}$.

To find the singular solution, differentiate (i) w.r.t. c giving

$$0 = x + \frac{1}{\sqrt{1 - c^2}} \qquad \ldots(ii)$$

To eliminate c from (i) and (ii), we rewrite (ii) as

$$c = \mp\sqrt{x^2 - 1}/x$$

Now substituting this value of c in (i), we get

$$y = \pm x\sqrt{x^2 - 1} \mp x^{-1}\sqrt{x^2 - 1}$$

which is the desired singular solution.

Obs. Equations reducible to Clairaut's form : Many equations of the first order but of higher degree can be easily reduced to the Clairaut's form by making suitable substitutions.

Example 3.15. Solve $(px - y)(py + x) = a(1 + p^2)$. (V.T.U., 2010; A.M.I.E., 2009)

Solution. Put $x^2 = u$ and $y^2 = v$ so that $2xdx = du$ and $2ydy = dv$.

$$\therefore \quad p = \frac{dy}{dx} = \frac{dv}{du}\cdot\frac{du}{dx} = \frac{x}{y}P, \text{ where } P = \frac{dv}{du}$$

*Other than name of a prolific geologist Alexis Claude Clairaut (1713-1765) who first solved this equation. A French mathematician who is also known for his work in astronomy and geodesy.

Then the given equation becomes $\left(\frac{dP}{y}, x - y\right)\left[\left(\frac{dP}{y}, y + x\right)\right] = e^{\frac{dP}{y}}$

or $\frac{dP}{y}(x^2 - y^2) + 2xy\frac{dP}{y} = e^{\frac{dP}{y}}$ or $\frac{dP}{y}(x^2 - y^2 + 2xy) = e^{\frac{dP}{y}}$

or $\frac{dP}{y} = \frac{e^{\frac{dP}{y}}}{x^2 - y^2 + 2xy}$

or $\frac{dP}{y} = e^{\frac{dP}{y}}(x^2 - y^2 + 2xy)$, which is Clairaut's form

∴ its solution is $c = xc - c^2y/c - 2x(x, y^2 = xc^2 - c^2y/c + 2x$

PROBLEMS 12.2

1. Find the general and singular solution of the equation:

(a) $x^2 - yp^2 + x = 0$ (CBSE(D), 2006) (b) $y = 2px + p^2y$

(c) $y = px + \sqrt{a^2p^2 + b^2}$ (CBSE(F), 2004) (d) $\sin px \cos y = \cos px \sin y + p$ (CBSE, 2008)

Solve the following equations:

2. $y = 2x\left(\frac{dy}{dx}\right) + \tan^2\left(\frac{dy}{dx}\right)$ 3. $y = 2x\left(\frac{dy}{dx}\right) - 2\left(\frac{dy}{dx}\right)^2$ 4. $y = -x\left(\frac{dy}{dx}\right) + \sin^{-1}\left(\frac{dy}{dx}\right)$

5. $e^x(p - y) = p^2$ 6. $y = px + p^2$ 7. $px(x + y) + y^2 = 0$

12.3 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 12.3

Fill in the blanks or tick the correct answer in the following problems:

1. $y = px - f(p)$ is the general solution of the differential equation

(a) $(y - px)f'(p) = p$ (b) $y^2 = p$ (c) $y = x$ (d) $(y - px)^2 = y^2(1 + p^2)$

2. The differential equation having $y = 2x + c$ for its solution is also said to satisfy by

(a) $y' = 2x + c$ (b) $y = 2x - 0$ (c) $y' = 2y = 2$ (d) none of these

3. The differential equation $(dy/dx)^2 + 2y^2 = 1$ is

(a) Degree 2 (b) not linear of order 1 and degree 2

(c) not linear of order 1 and degree 2

4. The differential equation $y(dy/dx) + x = c$ represents

(a) a circle (b) a parabola

(c) infinite number of solutions (d) no solution

5. Solution of $(x^2 + y^2)dx + 2xydy = 0$ is ____

6. Solution of $(2x - 2y)dx + (x - y)dy = 0$ is ____

7. Solution of $dy/dx = (x + y + 1)^2$ is

8. The differential equation $y(x^2 + 1)dy + x(y^2 + 1)dx = 0$ is

(a) linear homogeneous (b) not linear

(c) linear but not homogeneous (d) not linear but can be made linear

9. Solution of $dy/dx + x/y = \frac{ay - bx}{y^2 - x^2}$ is ____ 10. Solution of $dy/dx = \frac{x^2 + y^2}{xy}$ is ____

11. The differential equation $(x^2 + xy)dy/dx = y^2 - xy$ is homogeneous of degree

(a) 2 (b) 1 (c) 3 (d) 0

12. Solution of $y(1 - xy)dx - x(1 + xy)dy = 0$ is ____ 13. Solution of $(y^2 - xy + 1)dx = 0$ is ____

14. The differential equation $y = 2px + y^2p^3$ is ____

15. Solution of $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$ is ____

18. The order of the differential equation $(1+y)^{\frac{3}{2}} y_2 + y = 0$ is ____

17. The general solution of $\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{xy} = 0$ is ____

18. Integrating factor of the differential equation $\frac{dy}{dx} + \frac{2x}{x^2+1} = \frac{1}{x^2+1}$ is

(a) x^2+1 (b) x^2 (c) $x-1$ (d) x^2-1 (UPTU, 2009)

19. Solution of the equation $\frac{dy}{dx} = e^{x-y} + e^{x^2-y}$ is

(a) $\log(e^y - \log e^x) = c$ (b) $\log(e^y + \log e^x) = c$
(c) $\log(e^y + \log e^x) = c$ (d) $\log(e^y - \log e^x) = c$ (UPTU, 2009)

20. Solution of $\frac{dy}{\sqrt{1+x^2}} + \frac{dx}{\sqrt{1+y^2}} = 0$ is ____

21. Solution of $(x+y)\frac{dy}{dx} + y = 0$ given $y(1) = 0$ is ____

22. The substitution that transforms the equation $\frac{dy}{dx} = \frac{x+y-3}{2x+2y+2}$ to homogeneous form is ____

23. Integrating factor of $xy' + y = x^3y^6$ is ____

24. Solution of the exact differential equation $(2y+x)dx + (y-6x)dy = 0$ is ____

25. Solution of $(x+2y^2)dx + x^2(x+y)dy = 0$ is ____

26. The general solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = \tan x$ is ____

27. Degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + y = 0$ is

(a) 2 (b) 0 (c) 3 (d) 6 (Kerala, 2009)

28. Integrating factor of the differential equation $\frac{dy}{dx} + y\tan x = \frac{\sin 2x}{2}$ is

(a) $e^{\log(\cos x)}$ (b) $e^{\log(\sin x)}$ (c) $e^{\cos x}$ (d) $\sin x$

29. The differential equation of the family of circles with centre at origin is ____ (Rajasthan, 2008)

30. Solution of $e^{x^2}dx + xydy = 0$ is ____ (Rajasthan, 2008)

31. Solution of $y' + \sin\frac{y}{x} = \frac{y}{x}$ is

(a) $\ln|y| = \frac{y}{x} + \sin^{-1}|x|$ (b) $y = \sin x + \sin^{-1}|x|$ (c) $\ln|y| = \sin x + \sin^{-1}|x|$ (d) $\ln|y| = x + \sin^{-1}|x|$ (UTU, 2012)

32. Differential equation obtained by eliminating A and B from $y = Ae^{2x} + Be^{3x}$ is $d^2y/dx^2 - y = 0$. (True or False)

33. $(x^2 - 2xy^2)dx + (y^2 - 2x^2y)dy = 0$ is an exact differential equation. (True or False)

CHAPTER 12

Applications of Differential Equations of First Order

1. Introduction 2. Geometric applications 3. Orthogonal trajectories 4. Physical applications 5. Simple electric circuits 6. Newton's law of cooling 7. Heat flow 8. Rate of decay of radioactive materials 9. Chemical reactions and solutions 10. Miscellaneous problems

12.1 INTRODUCTION

In this chapter, we shall consider only such practical problems which give rise to differential equations of the first order. The fundamental principles required for the formation of such differential equations are given in each case and are followed by illustrative examples.

12.2 GEOMETRIC APPLICATIONS

(a) Cartesian coordinates. Let $P(x, y)$ be any point on the curve $f(x, y) = 0$ (Fig. 12.1), then (as per 4.6 (1) & 4.13 (1) & (4)), we have

(i) slope of the tangent at P is $\tan\psi = dy/dx$

(ii) equation of the tangent at P is

$$Y - y = \frac{dy}{dx}(X - x)$$

so that its x-intercept is OT

$$= x - y\,dx/dy$$

and y-intercept $= OT' = y - x\,dy/dx$

(iii) equation of the normal at P is $Y - y = -\frac{dx}{dy}(X - x)$

(iv) length of the tangent $= PT = y\sqrt{[1 + (dy/dx)^2]}$

(v) length of the normal $= PN = y\sqrt{[1 + (dx/dy)^2]}$

(vi) length of the sub-tangent $= MT = y\,dx/dy$

(vii) length of the sub-normal $= MN = y\,dy/dx$

(viii) $\frac{dx}{dy} = [1 + (dy/dx)^2]\frac{dy}{dx} = \sqrt{[1 + (dx/dy)^2]}$

(ix) differential of the arc s, $ds = \sqrt{(dx^2 + dy^2)}$

(x) y and x of curvature at $P = \frac{[1 + (dy/dx)^2]^{3/2}}{d^2y/dx^2}$

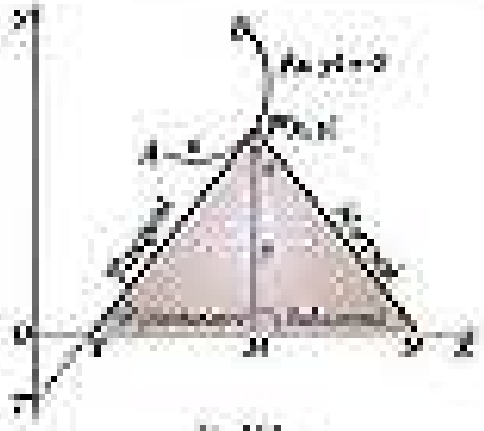

Fig. 12.1

(b) Polar coordinates. Let $P(r, \theta)$ be any point on the curve $r = f(\theta)$ (Fig. 12.3), then (as per § 4.7, 4.8, 4.9 and 4.10) we have

(i) $\psi = \theta + \phi$

(ii) $\tan\phi = r\dfrac{d\theta}{dr}$, $\rho = r\sec\phi$

(iii) $\dfrac{1}{\rho^2} = \dfrac{1}{r^2} + \dfrac{1}{r^4}\left[\dfrac{dr}{d\theta}\right]^2$

(iv) polar subtangent $= r^2\dfrac{d\theta}{dr}$

(v) polar subnormal $= \dfrac{dr}{d\theta}$

(vi) $\dfrac{ds}{d\theta} = \sqrt{1 + \left(r\dfrac{d\theta}{dr}\right)^2}\dfrac{dr}{d\theta} = \sqrt{r^2 + \left(\dfrac{dr}{d\theta}\right)^2}$

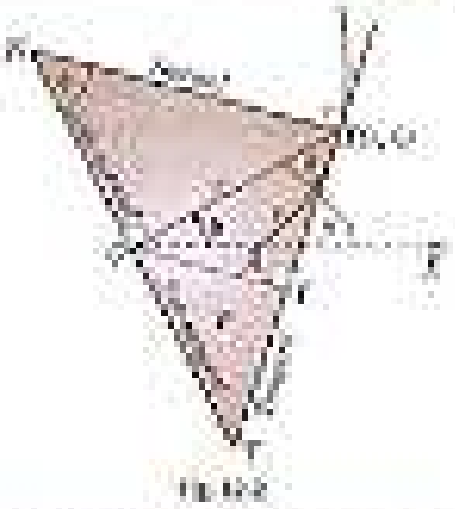

Fig. 12.3

Example 12.1. Show that the curve in which the portion of the tangent intercepted between the coordinate axes is bisected at the point of contact is a rectangular hyperbola.

Solution. Let the tangent at any point $P(x, y)$ of the curve cut the axes at T and T' (Fig. 12.3).

We know that its y-intercept $= OT = y - x\dfrac{dy}{dx}$

and x-intercept $= OT' = x - y\dfrac{dx}{dy}$

$\therefore$ the co-ordinates of T and T' are

$$\left(0, y - x\frac{dy}{dx}\right), \left(x - y\frac{dx}{dy}, 0\right)$$

Since P is the mid-point of TT'

$$\therefore \quad \frac{\left(x - y\dfrac{dx}{dy}\right) + 0}{2} = x$$

or $\quad x = y\dfrac{dx}{dy}$ or $\dfrac{dx}{x} = \dfrac{dy}{y}$ or $\dfrac{dx}{dy} = \dfrac{x}{y}$

or $\quad d(xy) = 0$ On integrating, $xy = c$

which is the equation of a rectangular hyperbola, having x and y axes as its asymptotes.

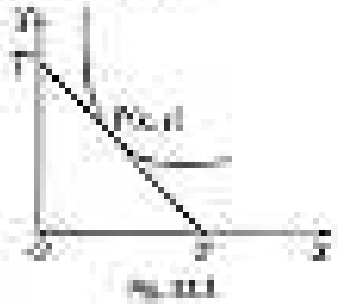

Fig. 12.4

Example 12.2. Find the curve for which the normal at any point is equally inclined to the radius vector and the initial line.

Solution. Let PN and PN' be the tangent and normal at $P(r, \theta)$ of the curve, so that

$$\tan\phi = r\frac{d\theta}{dr}$$

By the condition of the problem,

$$\angle OPN = 90° - \phi = \angle PNT \text{ (Fig. 12.4)}.$$

$$\therefore \quad \theta = \angle PON = 90° - (180° - 2\phi) = 2\phi - 90°$$

or $\quad 90° + \theta \quad \therefore \quad \tan\dfrac{\theta}{2} = \cot\phi = r\dfrac{d\theta}{dr}$.

Here the variables are separable.

$$\therefore \quad \frac{dr}{r} = \frac{\cos\theta/2}{\sin\theta/2}d\theta$$

Integrating both sides $\log r = 2\log\sin\theta/2 + \log c$

or $\quad r = c\sin^2\theta/2 = \dfrac{1}{2}c(1 - \cos\theta)$

This is the curve is the cardioid $r = a(1 - \cos\theta)$.

Example 12.3. *Find the shape of a reflector such that light coming from a fixed source is reflected in parallel rays.*

Solution. Taking the fixed source of light as the origin and the X-axis parallel to the reflected rays, the reflector will be a surface generated by the revolution of the curve $f(x, y) = 0$ about X-axis (Fig. 12.5).

In the XY-plane, let PT be the reflected ray, where P is the point (x, y) on the curve $f(x, y) = 0$.

If QPT' be the tangent at P, then

$\angle$ angle of incidence = angle of reflection,

$\therefore \quad \psi = \angle QPT = \angle PTT' = \angle OTP = \psi$

i.e. $$p = \frac{dy}{dx} = \tan \angle OTP = \tan 2\psi$$

$$= \frac{2 \tan \psi}{1 - \tan^2 \psi} = \frac{2p}{1 - p^2}$$

or $$2xp - yp = y \text{ which is a variable form} \qquad \ldots(1)$$

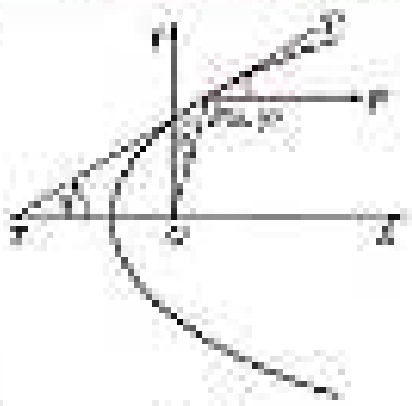

Fig. 13.6

$\therefore$ Differentiating it w.r.t. y, $2\frac{dx}{dy} - p = y\frac{dp}{dy}$

i.e. $$\left(\frac{1}{p} + p\right) + \left(\frac{1}{p^2} + 1\right) p \frac{dp}{dy} = 0 \quad \text{or} \quad \left(\frac{1}{p} + p\right)\left[1 + \frac{p}{y}\frac{dp}{dy}\right] = 0$$

This gives $\qquad dy/y = -dp/p$

Integrating, $\qquad \log y = \log c - \log p, \quad \text{i.e., } py = c \qquad \ldots(2)$

Eliminating p from (1) and (2), we have family of curves $y^2 = 2cx + c^2$.

Hence the reflector is a member of the family of parabolas of revolution $y^2 + z^2 = 2cx + c^2$.

PROBLEMS 13.1

1. Find the equation of the curve which passes through
   (i) the point $(1, -1)$ and has the slope $2x + y$ at the point (x, y) on it,
   (ii) the origin and has the slope $x + y + 1$.
2. At every point on a curve the slope is the sum of the abscissa and the product of the ordinate and the square of the abscissa. The curve passes through $(0, 1)$. Find the equation of the curve.
3. A curve is such that the length of the perpendicular from origin on the tangent at any point P of the curve is equal to the abscissa of P. Prove that the differential equation of the curve is
   $$x^2 - xy\frac{dy}{dx} + y^2 = 0. \text{ Find also the curve.}$$
4. A point moves in the positive half of the x-axis in such a way that the tangent to the curve it describes (except at the origin) always passes through the point $(0, a)$. Find the differential equation of the curve and hence find the curve.
5. A curve is such that the area of the region bounded by the curve, the x-axis and the ordinate of any point P on the curve is equal to the square of the ordinate of P. Find the curve which passes through the point $(1, 2)$. (Bombay, 2000)
6. Determine the curve in which the length of the subnormal is proportional to the square of the ordinate.
7. The tangent at any point of a certain curve passes through the point $(-a, 0)$ and the subtangent is of length a. Find the equation to the curve.
8. Find the curve which is such that the area bounded by the curve, the coordinate axes and the ordinate of any point is twice the cube of that ordinate.
9. Find the curves whose
   (a) polar subnormal is proportional to the sine of the vectorial angle,
   (b) polar subnormal is proportional to the sine of the vectorial angle.
10. Determine the curve for which the angle between the tangent and the radius vector is twice the vectorial angle. (Bangalore, 2000)
11. Find the curve for which the distance of any point P on it from the origin is equal to the length of the intercept cut off by the tangent at P on the y-axis.
12. Find the curve for which the length of the arc is equal to the product of the radius vector and the angle between the radius vector and the tangent. Also find the equation of the curve if it passes through the point $(1, 0)$.

13.5 (I) ORTHOGONAL TRAJECTORIES

Two families of curves such that every member of either family cuts each member of the other family at right angles are called **orthogonal trajectories** of each other (Fig. 13.8).

The concept of the orthogonal trajectories is of wide use in applied mathematics especially in field problems. For instance, in an electric field, the paths along which the current flows are the orthogonal trajectories of the equipotential curves and vice-versa. In fluid flow, the stream lines and the equipotential lines (lines of constant velocity potential) are orthogonal trajectories. Moreover, the lines of heat flow for a body are perpendicular to the isothermal curves. The problem of finding the orthogonal trajectories of a given family of curves depends on the solution of the first order differential equations.

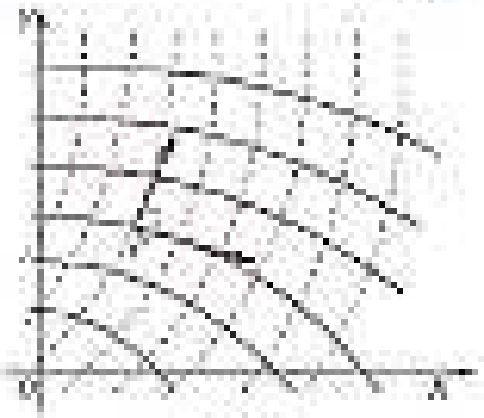

Fig. 12.6

(i) To find the orthogonal trajectories of the family of curves $f(x, y, c) = 0$.

(a) Form an differential equation in the form $f(x, y, dy/dx) = 0$ by eliminating c.

(b) Replace, in the differential equation, dy/dx by $-dx/dy$, so that the product $(dy/dx)(dx/dy)$ is -1 at each point of intersection is -1.

(c) Solve the differential equation of the orthogonal trajectories i.e., $f(x, y, -dx/dy) = 0$.

Example 12.6. Find the orthogonal trajectories of a system of confocal conics. (Bharathiar, 2000)

Solution. Taking the axes as the x-axis, the stream lines of the flow around the corner of the walls is

$$xy = c \qquad \ldots(1)$$

Differentiating, we get, $x\dfrac{dy}{dx} + y = 0$ $\ldots(2)$

as the differential equation of the given family (1).

Replacing $\dfrac{dy}{dx}$ by $-\dfrac{dx}{dy}$ in (2), we obtain $x\left(-\dfrac{dx}{dy}\right) + y = 0$

or $$x\,dx - y\,dy = 0 \qquad \ldots(3)$$

as the differential equation of the orthogonal trajectories.

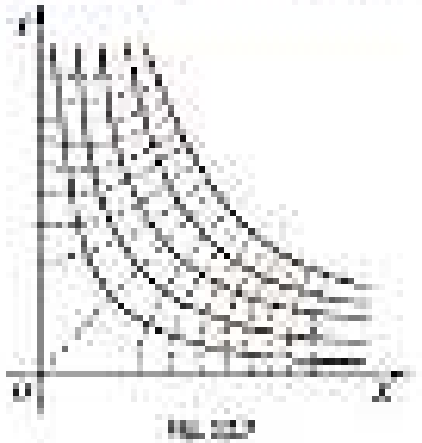

Fig. 12.7

Integrating (3), we get $x^2 - y^2 = c'$ as the required orthogonal trajectories of $xy = c$, the equipotential curves, shown dotted in Fig. 12.7.

Example 12.7. Find the orthogonal trajectories of the family of confocal conics $\dfrac{x^2}{a^2+\lambda} + \dfrac{y^2}{b^2+\lambda} = 1$, where λ is the parameter. (Madras, 2000)

Solution. Differentiating the given equation, we get: $\dfrac{2x}{a^2+\lambda} + \dfrac{2y}{b^2+\lambda}\dfrac{dy}{dx} = 0$

or $$\frac{y}{b^2+\lambda} = -\frac{x}{a^2+\lambda}\frac{dy}{dx} \quad \text{or} \quad \frac{y^2}{b^2+\lambda} = \frac{-xy}{a^2+\lambda}\frac{dy}{dx}$$

Substituting this in the given equation, we get

$$\frac{x^2}{a^2+\lambda} - \frac{xy}{a^2+\lambda}\frac{dy}{dx} = 1 \quad \text{or} \quad x^2 - xy\frac{dy}{dx} = a^2 + \lambda \qquad \ldots(1)$$

which is the differential equation of the given family.

Changing dy/dx to $-dx/dy$ in (1), we get $x^2 + xy\dfrac{dx}{dy} = a^2 + \lambda$ which is a differential equation of the orthogonal trajectories.

Separating the variables and integrating, we obtain

$$\int x\,dy = \int \frac{a^2+\lambda-x^2}{x}\,dx + c \quad \text{or} \quad \frac{1}{2}y^2 = a^2\log x - \frac{1}{2}x^2 + c$$

or $$x^2 + y^2 = 2a^2\log x + c' \qquad \ldots(2)$$

which is the equation of the required orthogonal trajectories.

Example 12.8. Find the orthogonal trajectories of a system of confocal and coaxial parabolas.

Solution. The equation of the family of confocal parabolas having vertex at the origin and axis along the positive direction of x-axis is of the form

$$y^2 = 4a(x + a) \quad \ldots(1)$$

Differentiating, $\quad 2y\dfrac{dy}{dx} = 4a \quad \ldots(2)$

Substituting the value of a from (2) in (1), we get $y^2 = 2y\dfrac{dy}{dx}\left(x + \dfrac{1}{2}y\dfrac{dy}{dx}\right)$

i.e., $\quad y\left(\dfrac{dy}{dx}\right)^2 + 2x\dfrac{dy}{dx} - y = 0$ is the differential equation of the family. $\quad \ldots(3)$

Replacing $\dfrac{dy}{dx}$ by $-\dfrac{dx}{dy}$ in (3), we obtain $y\left(\dfrac{dx}{dy}\right)^2 - 2x\dfrac{dx}{dy} - y = 0$

or $\quad y\left(\dfrac{dx}{dy}\right)^2 + 2x\dfrac{dx}{dy} - y = 0$ which is the same as (3).

Thus we see that a system of confocal and coaxial parabolas is self-orthogonal, i.e., each member of this family cuts every other member of the same family orthogonally.

(2) To find the orthogonal trajectories of the curves $F(r, \theta, c) = 0$.

(i) Form the differential equation in the form $f(r, \theta, dr/d\theta) = 0$ by eliminating c.

(ii) Replace $\dfrac{dr}{d\theta}$ by $-r^2\dfrac{d\theta}{dr}$ in the differential equation.

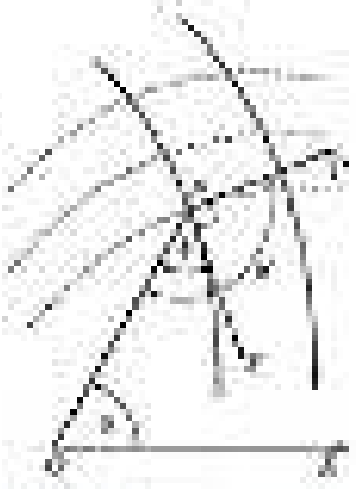

Fig. 12.1

(iii) Solve the differential equation so obtained.

Let the given curve through $P(r, \theta)$ make $\phi = \angle OPT$ with the radius vector OP and let the orthogonal trajectory through P make $\psi = \angle OPT'$ with OP.

Then $\tan\psi = \tan(OPT') = \tan(\theta + \angle PTX) = \tan\left(\theta + \dfrac{\pi}{2}\right) = -\dfrac{1}{\tan\phi}$

$$= -\frac{1}{r\dfrac{d\theta}{dr}} = -\frac{dr}{r\,d\theta}$$

Thus, by getting the differential equation of the orthogonal trajectory

$r\dfrac{d\theta}{dr}$ is to be replaced by $-\dfrac{dr}{r\,d\theta}$

or $\quad \dfrac{dy}{dx}$ is to be replaced by $-\dfrac{dx}{dy}$.

(iv) Solve the differential equation of the orthogonal trajectories.

i.e., $\quad f(r, \theta, -r^2 d\theta/dr) = 0$.

Example 12.9. Find the orthogonal trajectory of the cardioid $r = a(1 - \cos\theta)$. (Dibrugarh 2002)

Solution. Differentiating $\quad r = a(1 - \cos\theta)$ $\quad \ldots(1)$

with respect to θ, we get $\quad \dfrac{dr}{d\theta} = a\sin\theta$ $\quad \ldots(2)$

Eliminating a from (1) and (2), we obtain

$$\frac{dr}{d\theta}\cdot\frac{1}{r} = \frac{a\sin\theta}{1 - \cos\theta} = \cot\frac{\theta}{2}$$ which is the differential equation of the given family.

Replacing $\dfrac{dr}{d\theta}$ by $-r^2\dfrac{d\theta}{dr}$, we obtain

$$\frac{1}{r}\left(-r^2\frac{d\theta}{dr}\right) = \cot\frac{\theta}{2} \quad \text{or} \quad \frac{dr}{r} + \tan\frac{\theta}{2}\,d\theta = 0$$

is the differential equation of orthogonal trajectories. It can be rewritten as

$$\frac{dr}{r} = -\frac{\sin\theta/2\,d\theta}{\cos\theta/2}$$

Integrating, $\log r = -2\log\cos\theta/2 + \log a$

or $r = a\cos^2\theta/2 = \frac{1}{2}a(1+\cos\theta)$ or $r = a'(1+\cos\theta)$

which is the required orthogonal trajectory.

Example 11.8. Find the orthogonal trajectory of the family of curves $r^n = a\cos n\theta$. [U.P.T.U. 2005]

Solution. We have $n\log r = \log a + \log\cos n\theta$.

Differentiating w.r.t. θ, we have

$$\frac{n}{r}\frac{dr}{d\theta} = \frac{-n\sin n\theta}{\cos n\theta} \quad \text{or} \quad \frac{1}{r}\frac{dr}{d\theta} = -\tan n\theta$$

Replacing $d\theta/d\theta$ by $-r^2\,d\theta/dr$, we obtain

$$\frac{1}{r}\left(-r^2\frac{d\theta}{dr}\right) = \tan n\theta \quad \text{or} \quad \cot n\theta\,d\theta = \frac{dr}{r}$$

Integrating, $\int\frac{dr}{r} + \int\frac{\sin n\theta}{\cos n\theta}d\theta = c_1$

i.e., $\log r - \frac{1}{n}\log\cos n\theta = c$ or $\log(r^n/\cos n\theta) = n\log c = \log c^n$ (say)

or $r^n = c^n\cos n\theta$, which is the required orthogonal trajectory.

EXERCISE 11.1

Find the orthogonal trajectories of the family of

1. Parabolas $y^2 = 4ax$. (Meerut 2010) 2. Parabolas $y = cx^2$. (U.P.T.U. 2010)
3. Semi-cubical parabolas $ay^2 = x^3$. (U.P.T.U. 2010)
4. Circles $x^2 + y^2 = 2ax$ touching the y-axis. (U.P.T.U. 2010)
5. Circles $\frac{x^2}{a^2+\lambda} + \frac{y^2}{b^2+\lambda} = 1$, λ being the parameter. (Uttarakhand, 2010)
6. Cardioids $r = a(1+\cos\theta)$. (U.P.T.U. 2010) 7. $r = 2a(\sec\theta + \cos\theta)$. (U.P.T.U. 2010)
8. Confocal and coaxial parabolas $r = 2a(1+\cos\theta)$. (Madras, 2010)
9. Curves $r^n = a^n\sin n\theta$. (U.P.T.U. 2009)
10. $r^n\sin n\theta = a^n$. (U.P.T.U. 2011)
11. Show that the family of circles $x^2 + y^2 = ax$ is self-orthogonal. (Kerala, 2010)
12. Show that the family of curves $r^n = a\sec n\theta$ and $r^n = b\operatorname{cosec} n\theta$ are orthogonal. (Kerala, 2007)
13. The electrical lines of force of two opposite charges of the same strength at the points $(1, 0)$ and $(-1, 0)$ are given by the equation $x^2 + y^2 - 2cy = 1$. Find the equipotential lines (orthogonal trajectories).

   [**Hint.** Differentiating the given equation, we get $x + (y-c)\frac{dy}{dx} = 0$. Eliminating c between this and the given equation, we get $x^2 - y^2 + 1 + 2xy\frac{dy}{dx} = 0$. Replacing $\frac{dy}{dx}$ by $-\frac{dx}{dy}$, we get $x^2 - y^2 + 1 - 2xy\frac{dx}{dy} = 0$, which is the differential equation of the orthogonal trajectories. Solving this equation, we get $x^2 + y^2 + 2cx = 1$, which is the required orthogonal trajectory.]

14. Find the equation of the family of curves $r = a(1+\cos\theta)$, which intersect at $45°$.

11.5 PHYSICAL APPLICATIONS

(1) Let a body of mass m start moving from O along the straight line OZ under the action of a force F. After any time t, let it be moving at P where $OP = x$. Then

(i) the velocity $v = \frac{dx}{dt}$

(ii) the acceleration $f = \frac{dv}{dt} = \frac{dv}{dx}\cdot\frac{dx}{dt} = v\frac{dv}{dx} = \frac{d^2x}{dt^2}$

If, however, the body is moving along a curve, then

(a) its velocity $v = ds/dt$ and

(b) its acceleration $a = \frac{dv}{dt} = v\frac{dv}{ds} = \frac{d^2s}{dt^2}$.

The quantity mv is called the momentum.

(c) Newton's second law states that $F = \frac{d}{dt}(mv)$.

Fig. 11.4

If m is constant, then $F = m\frac{dv}{dt} = ma$, i.e., *force = mass × acceleration*.

(d) Hooke's law[1] states that the tension of a string or spring is proportional to the extension of the string or the spring beyond its natural length.

Thus $T = ke(1)$,

where e is the extension beyond the natural length l and k is the *modulus of elasticity*.

Sometimes for a spring, we write $T = ke$,

where e is the extension beyond the natural length and k is the *stiffness* of the spring.

(e) Systems of units

I. F.P.S. (foot–pound–second) system. If mass m is in pounds and acceleration a is in ft/sec^2, then the force F is measured in poundals.

II. C.G.S. (centimetre–gram–second) system. If mass m is in grams and acceleration a is in cm/sec^2, then the force F is measured in dynes.

III. M.K.S. (metre–kilogram–second) system. If mass m is in kilograms and acceleration a is in m/sec^2, then the force F is measured in newtons (N).

These are called *absolute units*. If g is the acceleration due to gravity and w is the weight of the body, then m/g is the mass of the body (again in kgm/sec^2 units).

$g = 32.2$ ft/sec^2 = 980 cm/sec^2 = 9.9 m/sec^2 approx.

Example 11.9. Motion of a boat across a stream. *A boat moves with a velocity v relative to the water across a stream of width a. If the velocity of the current is always proportional to the product of the distances from the two banks, find the path of the boat and the distance from the point where it lands.*

Solution. Taking the origin as the point from where the boat starts, let the axes be chosen as in Fig. 11.10.

At any time t after the start from O, let the boat be at $P(x, y)$, so that

dy/dt = velocity of the current $= k_1(a - y)y$

dx/dt = velocity with which the boat is being rowed $= v$.

$$\therefore \quad \frac{dy}{dx} = \frac{dy}{dt}\Big/\frac{dx}{dt} = \frac{k}{v}(ay - y^2) \qquad \ldots(1)$$

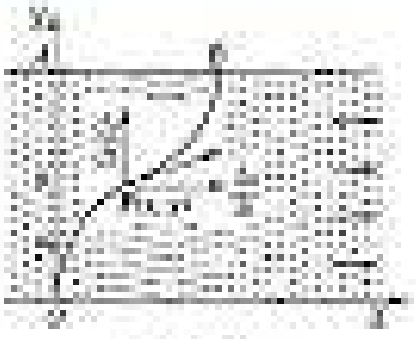
Fig. 11.10

This gives the direction of the resultant velocity of the boat which is also the direction of the tangent to the path of the boat.

Now (1) is of variables separable form and we can write it as

$$\frac{dy}{ay - y^2} = \frac{k}{v}\,dx$$

Integrating, we get $\frac{1}{a}\log\frac{y}{a - y} = \frac{k}{v}x + c$

For $y = 0$, $x = 0$, $\therefore c = 0$.

Hence the equation to the path of the boat is $x = \frac{v}{ak}\log\frac{a}{a - y}$.

Putting $y = a$, we get the distance AB, i.e., the distance where the boat lands $= \infty$.

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[1] Named after an English physicist Robert Hooke (1635–1703) who had discovered the law of gravitation earlier than Newton.

Example 13.10. Resisted motion. *A moving body is opposed by a force per unit mass of value k times the velocity. Find the velocity at any time t if the body starts from rest with a velocity of u cm/sec. Find also the velocity of the particle in terms of x, if it starts from rest.* (Bharathiar, 2003)

Solution. By Newton's second law, the equation of motion of the body is $m\frac{dv}{dt} = -mkv$ [$\because F = -mkv$]

or
$$\frac{dv}{dt} + kv = 0 \qquad \ldots(1)$$

This is Bernoulli's equation. $\therefore$ Put $v^{1-1} = v^0 = 1$ and $\frac{dv}{dt} = 0$ in (1), we get $v = 0$. Hence $v = 0$ is a solution of (1).

Dividing (1) by v, we get
$$\frac{1}{v}\frac{dv}{dt} + k = 0 \qquad \ldots(2)$$

This is Leibnitz's linear equation and $I.F. = e^{\int k\,dt} = e^{kt}$.

$\therefore$ The solution of (2) is $ve^{kt} = -\int 2kx e^{2kx}\,dx + c$ [Integrate by parts]

$$= -2k\left[x\cdot\frac{e^{2kx}}{2k} - \int 1\cdot\frac{e^{2kx}}{2k}\,dx\right] + c = -\frac{kx}{k}e^{2kx} + \frac{1}{2k^2}e^{2kx} + c$$

or
$$v^2 = \frac{1}{2k^2} + ce^{-2kx} - \frac{x}{k} \qquad \ldots(3)$$

Initially $v = u$ when $x = 0$ $\therefore$ $u^2 = \frac{1}{2k^2} + c$.

Thus, substituting $c = u^2 - \frac{1}{2k^2}$ in (3), we get $v^2 = \frac{1}{2k^2}(1 + e^{-2kx}) - \frac{x}{k}$.

Example 13.11. Resisted vertical motion. *A particle falls under gravity in a resisting medium whose resistance varies as the velocity. Find the relation between distance and velocity of acceleration of the particle starting from rest.* (M.U. B.E. 2000)

Solution. After falling a distance x in time t from rest, let v be velocity of the particle. The forces acting on the particle are its weight mg downwards and resistance mkv upwards.

$\therefore$ equation of motion is $m\frac{dv}{dt} = mg - mkv$

or
$$\frac{dv}{dt} = g - kv \quad \text{or} \quad \frac{dv}{g - kv} = dt$$

Integrating, $\int\frac{dv}{g - kv} = \int dt + c$ or $-\frac{1}{k}\log(g - kv) = t + c$

When $t = 0$, $v = u$, $\therefore$ $c = -\frac{1}{k}\log g$

Thus
$$\frac{1}{k}\log\left[\frac{g}{g - kv}\right] = t \quad \text{or} \quad \frac{g - kv}{g} = e^{-kt}$$

or
$$\frac{dx}{dt} = v = \frac{g}{k}(1 - e^{-kt}) \qquad \ldots(1)$$

Integrating, $x = \frac{g}{k}\int(1 - e^{-kt})\,dt + c'$ or $x = \frac{g}{k}\left(t + \frac{1}{k}e^{-kt}\right) + c'$

When $x = 0$, $t = 0$, $\therefore$ $c' = -\frac{g}{k^2}$

Thus
$$x = \frac{g}{k}t + \frac{g}{k^2}(e^{-kt} - 1) \qquad \ldots(2)$$

Eliminating t from (1) and (2), we get

$$x = \frac{g}{k^2}\log\left[\frac{g}{g - kv}\right] - \frac{v}{k}$$

which is the required relation between x and v.

Problem 12.11. A body of mass m is thrown vertically upwards with an initial velocity u. The resistance of the air is proportional to the square of the velocity and is mk^2v^2. If the body reaches a maximum height a, show that the time taken is $\frac{1}{kv}\tan^{-1}\frac{kau}{g}$, where $v^2 = u^2 - 2ga$.

Solution. If the body is moving with the velocity v after having fallen through a distance x, then its equation of motion is

$$mv\frac{dv}{dx} = mg - kv^2 \quad \text{or} \quad mv\frac{dv}{dx} = k(a^2 - v^2). \qquad [\because mg = kv^2] \quad ...(1)$$

or separating the variables and integrating, we get: $\int_u^0 \frac{v\,dv}{a^2 - v^2} = \int_0^x k\,dx + c$

or $$-\frac{1}{2}\log(a^2 - v^2) = kx + c \qquad ...(2)$$

Initially, when $x = 0$, $v = u$, $\therefore -\frac{1}{2}\log a^2 = c \qquad ...(3)$

Subtracting (3) from (2), we have $\frac{1}{2}\log a^2 - \log(a^2 - v^2) = kx$ or $kx = \log\frac{a}{\sqrt{a^2 - v^2}}$

or $$\frac{2kx}{m} = \log\left(\frac{a^2}{a^2 - v^2}\right)$$

Note. When the velocity becomes zero (i.e. the height is a maximum) the resistance becomes zero and gravity remains so that a constant velocity, called the limiting or terminal velocity. From (1), it follows that the terminal velocity is $\sqrt{\frac{g}{k}}$. Thus, the limiting velocity is the greatest velocity which the particle can attain to.

Example 12.12. Velocity of escape from the earth. *Find the initial velocity of a particle which is projected vertically from the earth's surface and is supposed to escape from the earth. Assume that it is acted upon by the gravitational attraction of the earth only.*

Solution. According to Newton's law of gravitation, the acceleration a of the particle is proportional to $\frac{1}{x^2}$ where x is the variable distance of the particle from the earth's centre. Thus

$$a \propto \frac{1}{x^2} \quad \text{or} \quad a = -\frac{\mu}{x^2}$$

where v is the velocity when at a distance x from the earth's centre. The acceleration is negative because v is decreasing. When $x = R$, the earth's radius, then $a = -g$, the acceleration of gravity at the surface.

i.e., $$-g = -\mu R^{-2}, \text{ i.e., } \mu = gR^2 \quad \therefore \quad v\frac{dv}{dx} = -\frac{gR^2}{x^2}$$

Separating the variables and integrating, we obtain $\int v\,dv = -gR^2\int\frac{dx}{x^2} + c$

i.e., $$v^2 = \frac{2gR^2}{x} + 2c \qquad ...(1)$$

On the earth's surface $x = R$ and $v = v_0$ (say), the initial velocity. Then

$$v_0^2 = 2gR + 2c, \text{ i.e., } 2c = v_0^2 - 2gR$$

Inserting this value of $2c$ in (1), we get $v^2 = \frac{2gR^2}{x} + v_0^2 - 2gR$

When x increases, the particle slows down and the velocity v changes from positive to negative and the particle will return to the earth. Thus the velocity will remain positive, if and only if $v_0^2 \geq 2gR$ and then the particle projected from the earth with this velocity will escape from the earth. Hence the minimum velocity of projection $v_0 = \sqrt{2gR}$ is called the velocity of escape from the earth (See Problem 5, page 424).

Example 12.24. Rotating cylinder containing liquid. A cylindrical tank of radius a is filled with water to a depth h. Show the tank is rotated with angular velocity ω about its axis, the liquid just begins to spill over the upper corner (at) from the centre of the tank. Under steady conditions of rotation, show that the surface of the free surface of the water is a paraboloid of revolution, and that the equation of the free surface of the water is given through the origin is the curve

$$y = \frac{\omega^2}{2g}\left(x^2 - \frac{a^2}{2}\right) + h.$$

Solution. Let the figure represent an axial section of the cylindrical tank. Forces acting on a particle of mass m at $P(x, y)$ on the curve, (a) due to the free surface of water, are :

(i) the weight mg acting vertically downwards,

(ii) the reaction R of liquid surrounding particle acting horizontally towards O.

As the particle is steady, P moves just on the surface of the water and, therefore, there is no force along the tangent to the curve. This is the resultant R of mg and mg acts in along the outward normal to the curve.

$\therefore$ $R\cos\psi = mg$ and $R\sin\psi = m\omega^2 x$

$$\text{where} \qquad \frac{dy}{dx} = \tan\psi = \frac{m\omega^2 x}{mg} = \frac{\omega^2 x}{g} \qquad \ldots(1)$$

This is the differential equation of the surface of the rotating liquid.

Integrating (1), we get

$$\int dy = \frac{\omega^2}{g}\int x\,dx + c$$

$$\text{i.e.,} \qquad y = \frac{\omega^2 x^2}{2g} + c \qquad \ldots(2)$$

To find c, we note that the volume of the liquid remains the same in both cases (Fig. 12.11).

When $x = 0$ in (2), $OA = y = c$. When $x = a$ in (2), B is (a, y) $= \frac{\omega^2 a^2}{2g} + c$ $\ldots(3)$

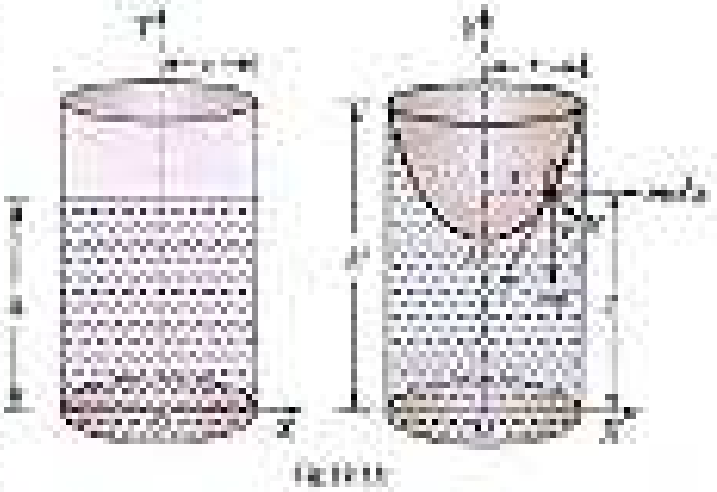


Fig 12.11

Now the volume of the liquid in the initial (static) case $= \pi a^2 h$, and the volume of the liquid in the rotational case

$$= \pi a^2 h - \int_{OA}^{B} \pi x^2\,dy = \pi a^2 h - \frac{\pi\omega^2}{g}\int_0^a (x^3 - a^2 x)\,dx \qquad \text{[From (1)]}$$

$$= \pi a^2 h - \frac{\pi\omega^2}{g}\left[\frac{x^4}{4} - \frac{a^2 x^2}{2}\right]_0^a = \pi a^2\left[\frac{\omega^2 a^2}{4g} + c\right] \qquad \text{[By (2)]}$$

Thus $\pi a^2 h = \pi a^2\left(\frac{\omega^2 a^2}{4g} + c\right)$ whence $c = h - \frac{\omega^2 a^2}{4g}$

$\therefore$ (2) becomes, $y = \frac{\omega^2 x^2}{2g} + h - \frac{\omega^2 a^2}{4g}$ or $y = \frac{\omega^2}{2g}\left(x^2 - \frac{a^2}{2}\right) + h$

which is the desired equation of the curve.

Example 12.25. Discharge of water through a small hole. If the velocity of discharge of water through a small hole at the bottom of a tank is $\sqrt{2gh}$, where h is the height of water level above the hole, find the time required to empty a hemispherical bowl of radius a full of water through a hole at the bottom. Also find the time required to empty the bowl if it is filled to a depth $\frac{1}{2}a$. Take $a = 1$ ft and the area of the hole $= \frac{1}{144}$ sq. ft.

Solution. At any instant, let the height of the water level be y and the radius of the surface be r (Fig. 12.12) so that

$$\frac{r - 0}{y - 0} = \frac{a}{a} \quad \text{or} \quad r = y \qquad \ldots(1)$$

$\therefore$ surface area of the liquid $= \pi r^2 = \pi a^2 (1 - y/h)^2$

Volume of water drained through the hole per unit time

$$= 0.6 \sqrt{2gy}\, A_0 = 0.6 \sqrt{2}\, A_0 \sqrt{y} \qquad [\because\ g = 10]$$

$\therefore$ rate of fall of liquid level $= 0.6 A_0 \sqrt{2y} \div \pi a^2 (1 - y/h)^2$

i.e., $$\frac{dy}{dt} = -\frac{0.6 A_0 \sqrt{2y}}{\pi a^2 (1 - y/h)^2} \qquad [-ve \text{ to indicate that } y \text{ decreases as } t \text{ increases}]$$

Time taken to empty the tank $= T$

$$= -\int_h^0 \frac{\pi a^2 (1 - y/h)^2}{-0.6 A_0 \sqrt{2y}}\, dy = \frac{\pi a^2}{0.6 A_0} \int_0^h (y^{-1/2} - 2y^{1/2}/h + y^{3/2}/h^2)\, dy$$

$$= \frac{\pi a^2}{0.6 A_0} \left[2y^{1/2} - \frac{4}{3h} y^{3/2} + \frac{2}{5h^2} y^{5/2} \right]_0^h = 0.8 \pi a^2 \sqrt{h} / A_0$$

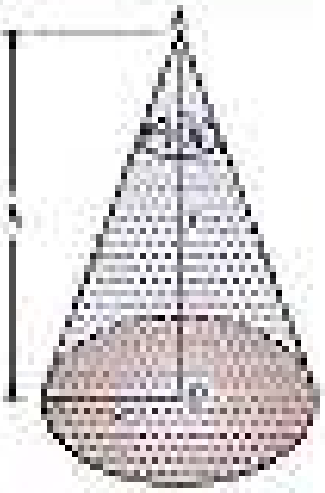


Fig. 12.12

Example 12.16. Atmospheric pressure. Find the atmospheric pressure p at a height z above the sea level, assuming that the temperature remains constant.

Solution. Take a vertical column of air of unit cross-section.

Let p be the pressure at a height z above the sea level and $p + \delta p$ at height $z + \delta z$. Let ρ be the density at a height z. (Fig. 12.13)

Now since the thin column δz of air is being pressed upwards with pressure p and downwards with $p + \delta p$, we get by considering its equilibrium

$$p = p + \delta p + g\rho\delta z \quad .$$

Taking the limit, we get $\qquad dp/dz = -g\rho \qquad \ldots(1)$

which is the differential equation giving the atmospheric pressure at height z.

(i) When the temperature is constant, we have by Boyle's law[1], $p = k\rho \qquad \ldots(2)$

$\therefore$ Substituting the value of ρ from (2) in (1), we get:

$$\frac{dp}{dz} = -g\rho/k \quad \text{or} \quad \int \frac{dp}{p} = -\frac{g}{k} \int dz + c \quad \text{i.e.} \quad \log p = -\frac{g}{k} z + c$$

At the sea level, when $z = 0$, $p = p_0$ (say), then $c = \log p_0$.

$$\therefore \qquad \log p - \log p_0 = -\frac{g}{k} z \quad \text{i.e.,} \quad \log_e p/p_0 = -gz/k$$

Hence p is given by $p = p_0 e^{-gz/k}$.

(ii) When the temperature varies, we have $p = k\rho^\gamma$.

Proceeding as above, we shall find that p is given by $\frac{\gamma}{\gamma - 1} (p_0^{1-1/\gamma} - p^{1-1/\gamma}) = gz^{1-1/\gamma}$.

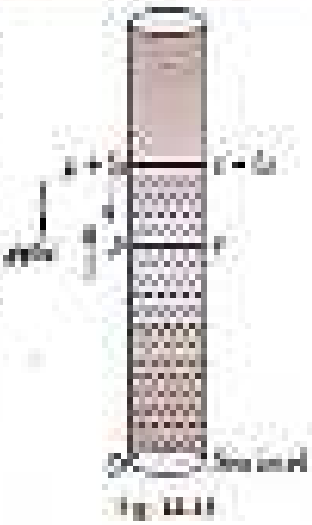


Fig. 12.13

EXERCISE 12.3

1. A particle of mass m is moving in the gravity in a medium which resistance is k times its velocity, where k is a constant. If the particle is projected vertically upwards with a velocity v, show that the time it will fall to the level of the point of projection again is $\frac{v}{g} \log \left| 1 + \frac{kv}{mg} \right|$.

2. A body of mass m falls from rest under gravity in the air, which resists the motion with a force k (velocity). If v is the velocity at time t, show that $v = \frac{mg}{k} (1 - e^{-kt/m})$. (Kurukshetra, 2001)

3. A body of mass m falls from rest under gravity in a fluid whose resistance is mk times the velocity. If v be the velocity of the body and x the distance fallen in time t, prove that $v = g/k (1 - e^{-kt})$ and $x = gt/k + g/k^2 (e^{-kt} - 1)$.

4. A particle is projected in the vertical plane along a smooth horizontal plane in a medium whose resistance per unit mass is k times the velocity. Show that the distance it has described in time t is $\frac{1}{2k} \sqrt{1 + 2kv^2/g} - 1$.

---

[1] Named after the English physicist Robert Boyle (1627–1691) who was one of the founders of the Royal Society.

8. The acceleration of a particle is given by $a = -\omega^2 x$, where ω is a positive constant. Show that the velocity of the particle is given by $v = \omega\sqrt{A^2 - x^2}$, where A is a positive constant. (*Hint:* Use the chain rule in the form $a = v\,dv/dx$.)

9. A projectile is fired from a point on the ground with initial velocity v_0 at an angle θ with the horizontal. Ignoring air resistance, show that the range of the projectile is $R = \frac{v_0^2 \sin 2\theta}{g}$.

10. A ladder of length L rests against a vertical wall. If the bottom of the ladder is pulled away from the wall at a constant rate k, how fast is the top of the ladder moving when the bottom is a distance x from the wall?

11. A trough is 10 ft long and its ends have the shape of isosceles triangles with a base of 3 ft and a height of 1 ft. If the trough is filled with water to within $\frac{1}{2}$ ft of the top, how fast does the water level rise when the water is 1 ft deep? If the trough is filled to within $\frac{1}{2}$ ft of the top and then emptied at a rate of 12 ft³/min, how fast does the water level fall when the water is 1 ft deep?

12. A cylindrical tank with a radius of 5 m is filled with water to a height of 12 m. Water drains from the tank through a hole with a radius of 0.1 m at the bottom. Torricelli's law says that the velocity of the water flowing from the hole is $v = \sqrt{2gh}$, where h is the height of the water above the hole and $g = 9.8$ m/s². Find the rate at which the water level is falling when $h = 8$ m. Then find the time it takes for the tank to drain completely.

13. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

14. A boat is pulled into a dock by a rope attached to the bow of the boat and passing through a pulley on the dock that is 1 m higher than the bow of the boat. If the length of the rope is 8 m, how fast does the boat approach the dock when it is 2 m from the dock?

15. A man starts walking north at 4 ft/s from a point P. Five minutes later a woman starts walking south at 5 ft/s from a point 500 ft due east of P. At what rate are the people moving apart 15 minutes after the woman starts walking?

16. Two sides of a triangle have lengths 12 m and 15 m, and the angle between them is increasing at a rate of 2°/min. How fast is the third side lengthening when the angle between the two sides is 60°?

17. Two cyclists, A and B, start from the point P. Cyclist A rides east at 20 km/h and cyclist B rides southeast at 15 km/h. After how many minutes will the distance between the cyclists be 10 km?

18. A spherical balloon is being inflated and the radius of the balloon is increasing at a rate of 2 cm/s. How fast is the volume increasing when the diameter is 10 cm?

19. Gravel is being dumped from a conveyor belt at a rate of 30 ft³/min and its coarseness is such that it forms a pile in the shape of a cone whose base diameter and height are always equal. How fast is the height of the pile increasing when the pile is 10 ft high?

20. A man 2 m tall walks at a rate of 1 m/s away from a light that is 5 m above the ground. How fast is the length of his shadow increasing when he is 10 m from the light?

21. A particle moves along the curve $y = 2\sin(\pi x/2)$. As the particle passes through the point $(\frac{1}{3}, 1)$, its x-coordinate increases at a rate of $\sqrt{2}$ cm/s. How fast is the distance between the particle and the origin changing at this instant?

22. Two sides of a triangle are 4 m and 5 m in length and the angle between them is increasing at a rate of 0.06 rad/s. Find the rate at which the area of the triangle is increasing when the angle between the sides of length 4 m and 5 m is $\pi/3$.

23. Two carts, A and B, are connected by a rope 39 ft long that passes over a pulley P (see the figure). The point Q is on the floor 12 ft directly beneath P and between the carts. Cart A is being driven away from Q at a speed of 2 ft/s. How fast is cart B moving toward Q at the instant when cart A is 5 ft from Q?

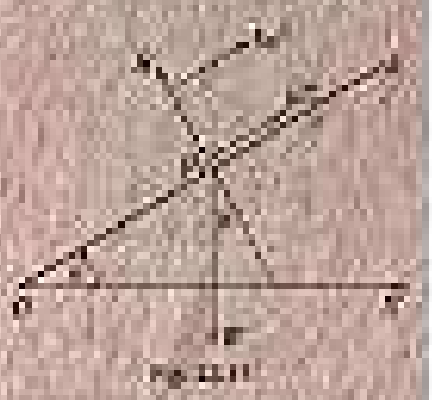


Fig. 23

24. A lamp is located on the ground 10 m from a straight path, and a man 2 m tall walks along the path at a speed of 4.5 km/h. How fast is the length of his shadow changing when he is 5 m from the point on the path closest to the lamp?

25. A plane flies horizontally at an altitude of 1 mi and passes directly over a radar station. When the plane is 2 mi away from the station, it is approaching at a speed of 500 mi/h. How fast is the distance between the plane and the station changing at that instant?

26. A plane flies horizontally at an altitude of 5 km and passes directly over a point P on the ground. When the plane is 13 km away from P, its speed is 780 km/h and is increasing at a rate of 60 km/h. How fast is the distance between the plane and P changing at that instant?

27. A spherical balloon is being inflated at a rate of 10 ft³/min. How fast is the radius of the balloon increasing when the diameter is 20 in?

28. Water is leaking out of an inverted conical tank at a rate of 10,000 cm³/min at the same time that water is being pumped into the tank at a constant rate. The tank has height 6 m and the diameter at the top is 4 m. If the water level is rising at a rate of 20 cm/min when the height of the water is 2 m, find the rate at which water is being pumped into the tank.

29. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

30. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

31. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

32. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

33. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

34. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

35. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

36. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

37. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

38. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

39. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

40. A water tank has the shape of a right circular cone with height 16 ft and radius 4 ft at the top. If water is being pumped into the tank at a rate of 10 ft³/min, find the rate at which the water level is rising when the water is 10 ft deep.

13.5 SIMPLE ELECTRIC CIRCUITS

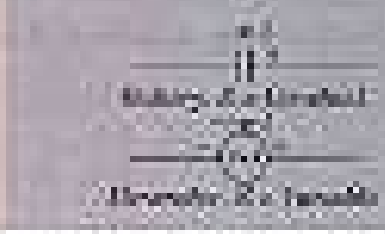
We shall consider circuits made up of

(i) three passive elements—resistance, inductance, capacitance, and

(ii) an active element—voltage source, which may be a battery or a generator.

Symbol	Name	Unit
Q	Quantity of electricity	coulomb (C)
I	Current ($= dQ/dt$)	ampere (A)
R	Resistance	ohm (Ω)
C	Capacitance ($= Q/V$)	farad (F)
L	Inductance	henry (H)

*These are some people only mentioned for the French engineer and physicist Charles Augustin de Coulomb (1736–1806), French physicist André-Marie Ampère (1775–1836), Christian physicist Georg Simon Ohm (1789–1854), Italian physicist Joseph Henry (1797–1878), American physicist Michael Faraday (1791–1867), and the Italian physicist Alessandro Volta (1745–1827).

Element	Symbol	Unit
6. Electromotive force (e.m.f.) or voltage (E)		volt (V)

7. Loop is any closed path formed by passing through two or more elements in series.
8. Nodes are the terminals of any of the two elements.

(2) Basic relations

$Q = \int i\,dt$ or $i = \frac{dQ}{dt}$ where i = current in the unit of flow of electricity

(Basic Law)

(a) Voltage drop across resistance $R = iR$

(b) Voltage drop across inductance $L = L\frac{di}{dt}$

(c) Voltage drop across capacitance $C = \frac{Q}{C}$.

(3) Kirchhoff's laws[*]. The formulation of differential equations for an electrical circuit depends on the following two Kirchhoff's laws which are of vital importance :

I. The algebraic sum of the voltage drops around any closed circuit is equal to the resultant electromotive force in the circuit.

II. The algebraic sum of the currents flowing towards any node is zero.

(4) Differential equations

(a) *R, L series Circuit.* Consider a circuit containing resistance R and inductance L in series with a voltage source of E volts (Fig. 12.33).

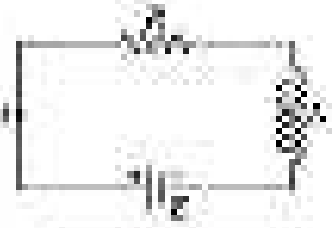


Fig. 12.33

Let i be the current flowing in the circuit at any time t. Then, by Kirchhoff's first law, we have sum of voltage drops across R and L = E

i.e., $$Ri + L\frac{di}{dt} = E \quad \text{or} \quad \frac{di}{dt} + \frac{R}{L}i = \frac{E}{L} \qquad \text{...(1)}$$

This is a Leibnitz's linear equation.

I.F. $= e^{\int \frac{R}{L}dt} = e^{\frac{R}{L}t}$ and therefore, the solution is $i(I.F.) = \int \frac{E}{L}(I.F.)\,dt + c$

or $$i e^{\frac{R}{L}t} = \int \frac{E}{L} e^{\frac{R}{L}t}\,dt + c = \frac{E}{L}\cdot\frac{L}{R}e^{\frac{R}{L}t} + c \quad \text{where } c = \frac{K}{e^{\frac{R}{L}t}} = Ke^{-\frac{R}{L}t} \qquad \text{...(2)}$$

Initially there is no current in the circuit, i.e., $i = 0$, when $t = 0$, we have $c = -\frac{E}{R}$.

Thus (2) becomes $i = \frac{E}{R}\left(1 - e^{-\frac{R}{L}t}\right)$ which shows that i increases with t and attains the maximum value $\frac{E}{R}$.

(b) *R, L, C series circuit.* Now consider a circuit containing resistance R, inductance L and capacitance C all in series with a constant e.m.f. E (Fig. 12.34).

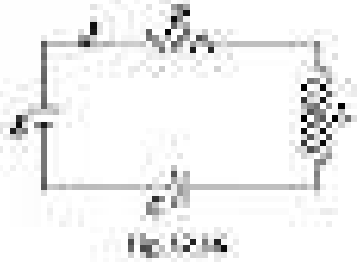


Fig. 12.34

If i be the current in the circuit at time t, then the charge q on the condenser $= \int i\,dt$, $i.e., i = \frac{dq}{dt}$.

Applying Kirchhoff's law, we have, sum of the voltage drops across R, L and C = E.

i.e., $$Ri + L\frac{di}{dt} + \frac{q}{C} = E$$

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[*]Named after Gustav Kirchhoff (1824-1887).

or $$L\frac{d^2q}{dt^2}+R\frac{dq}{dt}+\frac{q}{C}=E$$

This is a second order differential equation of the circuit and will be solved in § 18(h).

Example 18(g). Show that the differential equation for the current i in an electrical circuit containing an inductance L and a resistance R in series and acted on by an electromotive force E is given by the equation $L\frac{di}{dt}+Ri=E\sin\omega t$.

Find the value of the current at any time t, if initially there is no current in the circuit.

(Madras Univ., 2003)

Solution. By Kirchhoff's first law, we have sum of voltage drops across R and L = $E\sin\omega t$

i.e., $$Ri+L\frac{di}{dt}=E\sin\omega t.$$

This is the required differential equation which can be written as $\frac{di}{dt}+\frac{R}{L}i=\frac{E}{L}\sin\omega t$

This is a Leibnitz's equation. Its I.F. $=e^{\int\frac{R}{L}dt}=e^{\frac{Rt}{L}}$

$\therefore$ the solution is $i.e^{\frac{Rt}{L}}=\int\frac{E}{L}\sin\omega t.e^{\frac{Rt}{L}}dt+c$

or $$ie^{\frac{Rt}{L}}=\frac{E}{L}\int e^{\frac{Rt}{L}}\sin\omega t\,dt+c=\frac{E}{L}\frac{e^{\frac{Rt}{L}}}{\sqrt{\left(\frac{R}{L}\right)^2+\omega^2}}\sin\left(\omega t-\tan^{-1}\frac{L\omega}{R}\right)+c$$

i.e., $$i=\frac{E}{\sqrt{(R^2+\omega^2L^2)}}\sin(\omega t-\phi)+ce^{-\frac{Rt}{L}}\quad\text{where }\tan\phi=\frac{L\omega}{R}\qquad\ldots(1)$$

Initially when $t=0$, $i=0$. $\therefore\ 0=\frac{E\sin(-\phi)}{\sqrt{R^2+\omega^2L^2}}+c$, i.e., $c=\frac{E\sin\phi}{\sqrt{(R^2+\omega^2L^2)}}$

Thus (1) reduces to $i=\frac{E\sin(\omega t-\phi)}{\sqrt{(R^2+\omega^2L^2)}}+\frac{E\sin\phi}{\sqrt{(R^2+\omega^2L^2)}}e^{-\frac{Rt}{L}}$

i.e., $$i=\frac{E}{\sqrt{(R^2+\omega^2L^2)}}[\sin(\omega t-\phi)+\sin\phi.e^{-\frac{Rt}{L}}]$$ which gives the current in any time t.

Fig. 18(f)

Note. As t increases in (1), the exponential term $\to 0$ rapidly and so the current i in the circuit soon becomes sinusoidal with the same frequency as that of the impressed voltage (Fig. 18(f)).

EXERCISE 18(g)

1. When a switch is closed in a circuit containing an inductance L, a resistance R and an impressed voltage E, the current i builds up at a rate given by $L\frac{di}{dt}+Ri=E$.
   Find an expression for i in terms of t, if the initial current is zero. Find also the value of i after a long time, if $E=200$ volts and $L=2$ henrys.
2. When a condenser of capacity C is charged through a resistance R by connecting it to a source of E volts, the charge q on one plate of the condenser at any time t is given by the equation $R\frac{dq}{dt}+\frac{q}{C}=E$.
   If $R=10$ ohms and $C=0.1$ farad, find q as a function of t.

3. A thermometer of heat capacity 150 J/K is initially at 20°C and is then placed in a bath of boiling water. Find the temperature indicated by the thermometer 1 s, 2 s and 3 s later. (Manchester, 2000)

4. Explain briefly what is meant by the thermal conductivity of a material. Describe a method for measuring the thermal conductivity of a metal bar in the form of a cylinder.

   A cylindrical vessel of internal radius r_1 and external radius r_2 is filled with a liquid. The inner surface is maintained at a temperature θ_1 and the outer surface at a temperature θ_2. Show that the rate of flow of heat through a length l of the vessel is

   $$\frac{2\pi k l(\theta_1 - \theta_2)}{\ln(r_2/r_1)}$$

   where k is the thermal conductivity of the material of the vessel. (Oxford, 2000)

5. A metal rod of length l and cross-sectional area A is insulated along its length and has its ends maintained at temperatures θ_1 and θ_2 ($\theta_1 > \theta_2$). Derive an expression for the rate of flow of heat through the rod in terms of l, A, θ_1, θ_2 and the thermal conductivity k of the material of the rod.

   The rod is replaced by two rods of the same material, one of length l_1 and area of cross-section A_1, the other of length l_2 and area of cross-section A_2. The two rods are joined end to end and the temperatures at the free ends are maintained at θ_1 and θ_2. Show that the temperature θ at the junction of the rods is given by

   $$\frac{\theta_1 - \theta}{l_1} A_1 = \frac{\theta - \theta_2}{l_2} A_2$$

   Verify that if $A_1 = A_2$ the temperature θ is the mean of θ_1 and θ_2. (London, 2000)

12.6 NEWTON'S LAW OF COOLING*

According to this law, the temperature of a body changes at a rate which is proportional to the difference in temperature between that of the surroundings and that of the body itself.

If θ_0 is the temperature of the surroundings and θ that of the body at any time t, then

$$\frac{d\theta}{dt} = -k(\theta - \theta_0), \text{ where } k \text{ is a constant.}$$

Example 12.6.1. A body originally at 80°C cools down to 60°C in 20 minutes, the temperature of the surroundings being 40°C. Find (a) the temperature of the body after 40 minutes, (b) the time taken for the body to reach 45°C.

Solution. If θ be the temperature of the body at any time t, then

$$\frac{d\theta}{dt} = -k(\theta - 40), \qquad \text{where } k \text{ is a constant.}$$

Integrating, $$\int \frac{d\theta}{\theta - 40} = -k \int dt + \log c, \qquad \text{where } c \text{ is a constant.}$$

or $$\log(\theta - 40) = -kt + \log c, \quad \theta - 40 = ce^{-kt} \tag{1}$$

When $t = 0$, $\theta = 80°$ and when $t = 20$, $\theta = 60°$. $\therefore 40 = c$, and $20 = ce^{-20k}$; $\therefore k = \frac{1}{20}\log 2$.

Thus (1) becomes $\theta = 40 + 40e^{-\left(\frac{1}{20}\log 2\right)t}$.

When $t = 40$ min, $\theta = 40 + 40e^{-2\log 2} = 40 + 40e^{\log(1/4)} = 40 + 40 \times \frac{1}{4} = 50°C$.

12.7 HEAT FLOW

The fundamental principles involved in the problems of heat conduction are

(i) *Heat flows from a higher temperature to the lower temperature.*

(ii) *The quantity of heat in a body is proportional to its mass and temperature.*

(iii) *The rate of heat flow across an area is proportional to the area and to the rate of change of temperature with respect to the distance normal to the area.*

If q (calories) is the quantity of heat that flows across a slab of area a (cm^2) and thickness h in one second, where the difference of temperature at the faces is θ, then by (iii) above

$$q = -ka\frac{d\theta}{dx} \tag{1}$$

where k is a constant depending upon the material of the body and is called the *thermal conductivity*.

Named after the great English mathematical physicist Sir Isaac Newton (1642–1727), whose contributions are of utmost importance. He discovered many physical laws, invented Differential Calculus, designed a Reflecting type Telescope, and carried out brilliant methods of investigating physical problems. He became professor at Cambridge in 1669, but his 'Mathematical Principles of Natural Philosophy' containing a brief statement of classical mechanics was published in 1687.

Example 12.15. A pipe 20 cm in diameter carrying steam at 150°C and is covered with a covering 5 cm thick for which $k = 0.5$ W/m°C. If the temperature of the outer surface of the covering is 40°C, find the temperature at the surface of the steam. The covering under steady state conditions.

Solution. Let q cal/sec be the constant quantity of heat flowing radially through any sheet of the pipe having radius x cm and length l cm (Fig. 12.18). Then the area of the lateral surface $A = 2\pi x l$.

Using the equation $q = -kA\dfrac{dT}{dx}$ we get $dT = -\dfrac{q}{2\pi kl}\cdot\dfrac{dx}{x}$

Integrating, we have

$$T = -\frac{q}{2\pi kl}\log_e x + c$$

Since $T = 150$, when $x = 10$, $\quad 150 = -\dfrac{q}{2\pi kl}\log_e 10 + c$ ...(1)

Again when $T = 90$, when $x = 12.5$, $90 = -\dfrac{q}{2\pi kl}\log_e 12.5 + c$ ...(2)

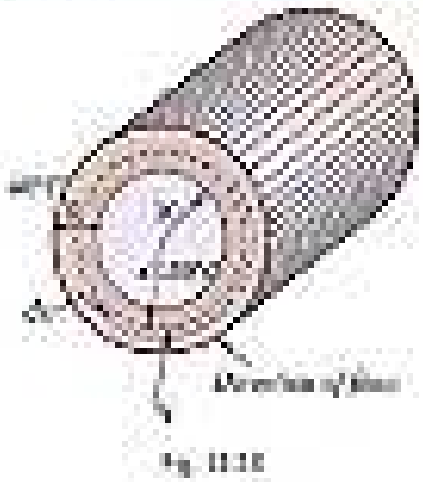

Fig. 12.18

Subtracting (1) from (2), $60 = \dfrac{q}{2\pi kl}\log_e 1.25$ ...(3)

Let $T = t$, when $x = 12.5$ $\quad\therefore\quad t = -\dfrac{q}{2\pi kl}\log_e 12.5 + c$ ...(4)

Subtracting (2) from (4), $t - 150 = -\dfrac{q}{2\pi kl}\log_e 1.25$ ...(5)

Dividing (5) by (3), $\dfrac{t-150}{60} = -\dfrac{\log_e 1.25}{\log_e 1.6}$, where $t = 98.9$°C.

PROBLEMS 12.4

1. If the temperature of the air is 30°C and the radiation varies from 100°C to 70°C in 12 minutes, find when the temperature will be 40°C.
2. If the air is cooled at 30°C and the temperature of the body cools from 100°C to 60°C in 10 minutes, find the temperature of the body after 20 minutes.
3. Two bodies A and B of the same mass have equal temperatures at the same time. A with a mean surface of 100 sq cm radiates to two degrees in 10 minutes while B radiates to one degree in 10 minutes. If the room temperature is 30°C, find the ratio of the specific heats of A and B assuming the Newton's law of cooling. In this ratio which is the heavier body?
4. A pipe 30 cm in diameter contains steam at 200°C. It is covered by a layer of insulation 5 cm thick and thermal conductivity 0.0002 C.G.S. If the temperature of the outer surface is 30°C, find the heat loss per hour from one metre length of the pipe.
5. A steam pipe 20 cm in diameter contains steam at 150°C and is covered with insulation 5 cm thick. The outside temperature is kept at 30°C. (i) how much should the thickness of the covering be increased in order that the rate of heat loss may be reduced by 20%?

12.5 RATE OF DECAY OF RADIO-ACTIVE MATERIALS

The law states that the disintegration of any material is proportional to the amount of material present.

If material is at any time t, then $\dfrac{dx}{dt} = -kx$, where k is a constant.

Example 12.16. Uranium is a radioactive element whose atomic weight is 238. It disintegrates at a rate proportional to the amount present. If M_1 and M_2 grams of uranium are present at the times T_1 and T_2 respectively, find the half life of uranium.

Solution. Let the mass of salt present at any time t be x grams.

Then the equation of disintegration of the substance is $\frac{dx}{dt} = -\mu x$, where μ is a constant. ...(1)

Integrating, we get $\int \frac{dx}{x} = -\mu \int dt + c$, i.e. $\log_e x = -\mu t + c$ ...(2)

Initially, when $t = 0$, $x = M$ (say), so that $0 = \log_e M + c$, or, otherwise, $c = \log_e M - \log_e 1$ ...(3)

Also when $t = T_1$, $x = M_1$, and when $t = T_2$, $x = M_2$

$\therefore$ From (2), we get $\mu T_1 = \log_e M - \log_e M_1$ ...(4)

$\mu T_2 = \log_e M - \log_e M_2$ ...(5)

Subtracting (4) from (5), we get

$$\mu(T_2 - T_1) = \log_e M_1 - \log_e M_2 = \log_e (M_1/M_2), \text{ where } \mu = \frac{\log_e (M_1/M_2)}{T_2 - T_1}$$

Let the mass reduce to half its initial value in time T, i.e. when $t = T$, $x = \frac{1}{2}M$.

$\therefore$ From (2), we get $\mu T = \log_e M - \log_e (M/2) = \log_e 2$

Thus $$T = \frac{1}{\mu} \log_e 2 = \frac{(T_2 - T_1)\log_e 2}{\log_e (M_1/M_2)}.$$

33.7 CHEMICAL REACTIONS AND SOLUTIONS

A type of problems which are especially important to chemical engineers are those concerning relative chemical reactions or chemical solutions. These can be best explained through the following example:

Example 33.11. *A tank initially contains 60 gallons of fresh water. Brine containing 2 pounds per gallon of salt flows into the tank at the rate of 3 gallons per minute and the mixture (well stirred) flows out at the same rate. How long will it take for the quantity of salt in the tank to increase from 60 to 90 pounds?* (Madras, 2001)

Solution. Let the salt content at time t be x lb so that the rate of change is dx/dt

$$= \text{(lb/gal in)} \times \text{(gal/min in)} - \text{(lb/gal out)} \times \text{(gal/min out)}$$

If C be the concentration of the brine at time t, the rate at which the salt content decreases due to the outflow

$$= (\text{gal/min}) \times C \times (\text{lb/gal}) = 3C \text{ lb/min}.$$

$$\therefore \quad \frac{dx}{dt} = 6 - 3C$$

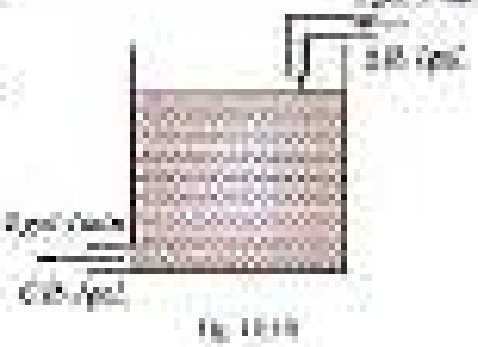

Fig. 33.16

Also since there is no increase in the volume of the liquid, the concentration $C = x/60$.

$$\therefore \text{ Or, otherwise } \frac{dx}{dt} = 6 - 3\frac{x}{60}$$

Separating the variables and integrating, we have

$$\int dt = 20 \int \frac{dx}{120 - x} + A \quad \text{or} \quad t = -20 \log_e (120 - x) + A \qquad \ldots(1)$$

Initially when $t = 0$, $x = 0$ $\therefore$ $0 = -20 \log_e 120 + A$ ...(2)

Eliminating A from (1) and (2), we get $t = 20 \log_e \frac{120}{120 - x}$

Taking $t = t_1$, when $x = 60$ and $t = t_2$, when $x = 90$, we have

$$t_1 = 20 \log_e \frac{120}{60} \text{ and } t_2 = 20 \log_e \frac{120}{30}$$

$\therefore$ The required time $= t_2 - t_1 = 20 \log_e 4 - 20 \log_e 3$

$$= 20 \log_e \frac{4}{3} = 20 \times 0.2877 = 5.75 \text{ min} = 5 \text{ min } 45 \text{ sec}$$

PROBLEMS 11.6

1. The number N of bacteria in a culture grows at a rate proportional to N. The value of N was initially 100 and increased to 332 in one hour. What was N in the value of N after $1\frac{1}{2}$ hours? (Nagarjuna, 2000; JNTU, 2000)
2. The rate at which bacteria multiply is proportional to the instantaneous number present. If the original number doubles in 4 hours, in how many hours will it triple? (JNTU, 2000)
3. Radium decomposes at a rate proportional to the amount present. If the amount of the original mass m is reduced to half in 1000 years, in how many years will it be reduced to $\frac{1}{4}$ th?
4. 12000 of radioactive substance disappeared in 10 days. How long will it take for 90% of it to disappear?
(Madras, 2000)
5. Under certain conditions cane sugar in water is converted into dextrose at a rate which is proportional to the amount unconverted at any time. If 10 gms. of the original 50 gms. are converted during the first 10 minutes, find the amount converted in $1\frac{1}{2}$ hours.
6. In a chemical reaction in which cane sugar is converted, the velocity of conversion at any time t is known to be equal to the product kxy where x and y are the amounts of the two substances that remain unconverted. Find x in terms of t if $x = a$ when $t = 0$ and $x = \frac{a}{2}$ when $t = 60$ minutes.
7. A tank contains 1000 gallons of brine in which 500 lbs. of salt are dissolved. Fresh water runs into the tank at the rate of 10 gallons/minute and the mixture flows out at the same rate. How long will it be before only 100 lbs. of salt is left in the tank?
(Hint: If x is the amount of salt in the tank at any time, then $\frac{dx}{dt} = -\frac{x}{100}$)
8. A tank originally contains 100 gallons of fresh water. Brine containing 2 lbs. of salt per gallon flows into the tank at the rate of 4 gallons per minute and the well-stirred mixture flows out at the same rate. Find the amount of salt in the tank at any time, and the amount present after 10 minutes. How long before the salt in the tank reaches 150 lbs.

EXERCISE 11.6 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 11.6

(Fill up the blanks or choose the correct answer in the following problems)

1. If a cell having a resistance of R ohms and an inductance of L henries is connected to E volts supply, then the current after t seconds is
2. A body at 100°F is placed in a room at 60°F, where the relativity of the body is 0.05. If the time required for the body to cool to 80°F is $\frac{1}{0.05}\log\frac{4}{3}$, then the time required for the body to cool to 70°F is
3. Radium decomposes at a rate proportional to the amount present. If $\frac{1}{4}$ of the original amount disappears in 50 years, then ____ % will remain after 100 years.
4. The curve whose polar subtangent is constant is ____
5. The curve in which the length of the normal is proportional to the square of the ordinate is ____
6. The curve in which the portion of the tangent between the point of contact and the point of intersection with the x-axis is bisected at the origin is ____
7. If the inverse form of a linear vertical circuit element is $v = Ri$, then the appropriate linear element is ____
8. The mathematical model of a system of electrical and mechanical parts can be ____
9. When a cubical tank of volume x cubic feet, the resistance is proportional to $\sqrt{\text{velocity}}$. If it takes 8 seconds to empty with velocity v_1, it will empty in 64 seconds with velocity ____
10. The rate at which fish in a small pond is proportional to the fish present. If in a period of one year the fish increase to two times, then it will triple after ____ years.
11. If an object is moving in a resisting medium whose resistance is proportional to the velocity, then the velocity after time t is ____
12. A man pulls a sled to which one end of a rope is tied and walks along the other end of the rope at a speed of 4 ft/sec. If the rope is 10 feet long, then the speed with which the sled moves along the ground when the man is 6 feet away from the sled is ____
13. The equation $y = ax$ represents a ____ in the polar coordinates.
(a) circle (b) $r = a\sin\theta$ (c) $r = a\cos\theta$ (d) $r = a\tan\theta$

[illegible]

CHAPTER 13

Linear Differential Equations

1. Definitions. 2. Complete solution. 3. Operator D. 4. Rules for finding the Complementary Function. 5. Inverse operator. 6. Rules for finding the particular integral. 7. Working procedure. 8. Two other methods of finding P.I.—Method of variation of parameters ; Method of undetermined coefficients. 9. Cauchy's and Legendre's linear equations. 10. Linear dependence of solutions. 11. Simultaneous linear equations with constant coefficients. 12. Objective Type of Questions.

13.1. DEFINITIONS

Linear differential equations are those in which the dependent variable and its derivatives occur only in the first degree and are not multiplied together. Thus the general linear differential equation of the nth order is of the form

$$\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + \ldots + P_n y = X,$$

where $P_1, P_2, \ldots, P_n$ and X are functions of x only.

Linear differential equations with constant coefficients are of the form

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \ldots + k_n y = X$$

where $k_1, k_2, \ldots, k_n$ are constants. Such equations are most important in the study of electric oscillatory circuits, and other engineering problems.

13.2. (1) THEOREM

If y_1, y_2 are only two solutions of the equation

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + k_2 \frac{d^{n-2} y}{dx^{n-2}} + \ldots + k_n y = 0 \qquad \ldots(1)$$

then $c_1 y_1 + c_2 y_2$ is also its solution.

Since $y = y_1$ and $y = y_2$ are solutions of (1),

$$\therefore \quad \frac{d^n y_1}{dx^n} + k_1 \frac{d^{n-1} y_1}{dx^{n-1}} + k_2 \frac{d^{n-2} y_1}{dx^{n-2}} + \ldots + k_n y_1 = 0 \qquad \ldots(2)$$

and

$$\frac{d^n y_2}{dx^n} + k_1 \frac{d^{n-1} y_2}{dx^{n-1}} + k_2 \frac{d^{n-2} y_2}{dx^{n-2}} + \ldots + k_n y_2 = 0 \qquad \ldots(3)$$

If c_1, c_2 be two arbitrary constants, then

$$\frac{d^n (c_1 y_1 + c_2 y_2)}{dx^n} + k_1 \frac{d^{n-1} (c_1 y_1 + c_2 y_2)}{dx^{n-1}} + \ldots + k_n (c_1 y_1 + c_2 y_2)$$

$$= c_1\left(\frac{d^n y_1}{dx^n} + k_1\frac{d^{n-1}y_1}{dx^{n-1}} + \ldots + k_n y_1\right) + c_2\left(\frac{d^n y_2}{dx^n} + k_1\frac{d^{n-1}y_2}{dx^{n-1}} + \ldots + k_n y_2\right)$$

$$= c_1(0) + c_2(0) = 0 \qquad \text{[By (2) and (3)]}$$

i.e., $$\frac{d^n u}{dx^n} + k_1\frac{d^{n-1}u}{dx^{n-1}} + \ldots + k_n u = 0 \qquad (4)$$

This proves the theorem.

(2) Since the general solution of a differential equation of the nth order contains n arbitrary constants, it follows, from above, that if $y_1, y_2, y_3, \ldots, y_n$ are n independent solutions of (1), then $c_1y_1 + c_2y_2 + \ldots + c_ny_n$ is also the complete solution.

(3) If $y = v$ be any particular solution of

$$\frac{d^n y}{dx^n} + k_1\frac{d^{n-1}y}{dx^{n-1}} + \ldots + k_n y = X \qquad (5)$$

then $$\frac{d^n v}{dx^n} + k_1\frac{d^{n-1}v}{dx^{n-1}} + \ldots + k_n v = X \qquad (6)$$

Adding (4) and (6), we have $\dfrac{d^n(u+v)}{dx^n} + k_1\dfrac{d^{n-1}(u+v)}{dx^{n-1}} + \ldots + k_n(u+v) = X$

This shows that $y = u + v$ is also the complete solution of (5).

The part u is called the complementary function (C.F.) and the part v is called the particular integral (P.I.) of (5).

∴ the complete solution (C.S.) of (5) is y = C.F. + P.I.

Thus in order to solve the equation (5), we have to first find the C.F., i.e., the complete solution of (4), and then the P.I., i.e., a particular solution of (5).

11.3 OPERATOR D

Denoting $\dfrac{d}{dx}, \dfrac{d^2}{dx^2}, \dfrac{d^3}{dx^3}$ etc. by D, D^2, D^3 etc., so that

$\dfrac{dy}{dx} = Dy$, $\dfrac{d^2y}{dx^2} = D^2y$, $\dfrac{d^3y}{dx^3} = D^3y$ etc., the equation (5) above can be written in the symbolic form $D^n y + k_1D^{n-1}y + \ldots + k_n y = X$, i.e., $f(D)y = X$,

where $f(D) \equiv D^n + k_1D^{n-1} + \ldots + k_n$ is a polynomial in D.

Thus the symbol D stands for the operation of differentiation and can be treated much the same as an algebraic quantity, i.e., $f(D)$ can be factorised by ordinary rules of algebra and the factors may be taken in any order. For instance

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y \equiv (D^2 + 2D - 3)y \equiv (D + 3)(D - 1)y \equiv (D - 1)(D + 3)y$$

11.4 RULES FOR FINDING THE COMPLEMENTARY FUNCTION

To solve the equation $$\frac{d^n y}{dx^n} + k_1\frac{d^{n-1}y}{dx^{n-1}} + k_2\frac{d^{n-2}y}{dx^{n-2}} + \ldots + k_n y = 0 \qquad (1)$$

where $k_1, k_2 \ldots k_n$ are constants.

The equation (1) in symbolic form is

$$(D^n + k_1D^{n-1} + k_2D^{n-2} + \ldots + k_n)y = 0 \qquad (2)$$

Its symbolic coefficient equated to zero, i.e.

$$D^n + k_1D^{n-1} + k_2D^{n-2} + \ldots + k_n = 0$$

is called the auxiliary equation (A.E.). Let $m_1, m_2, \ldots, m_n$ be its roots.

Case I. If all the roots be real and unequal, then (2) is equivalent to

$$(D - m_1)(D - m_2)\ldots(D - m_n)y = 0 \qquad (3)$$

Now (1) will be satisfied by the solutions of $(D - m_2)y = 0$, i.e., by $\frac{dy}{dx} - m_2 y = 0$

This is a Leibnitz's linear and I.F. $= e^{-m_2 x}$

$\therefore$ its solution is $y e^{-m_2 x} = c_2$, i.e., $y = c_2 e^{m_2 x}$

Similarly, since the factors in (1) can be taken in any order, it will be satisfied by the solutions of $(D - m_1)y = 0$, $(D - m_3)y = 0$ etc., i.e., by $y = c_1 e^{m_1 x}$, $y = c_3 e^{m_3 x}$ etc.

Thus the complete solution of (1) is $y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \ldots + c_n e^{m_n x}$ ...(4)

Case II. If two roots are equal (i.e., $m_1 = m_2$), then (4) becomes

$$y = (c_1 + c_2) e^{m_1 x} + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$$

$$y = C e^{m_1 x} + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x} \quad [\because c_1 + c_2 = \text{one arbitrary constant } C]$$

It has only $n - 1$ arbitrary constants and is, therefore, not the complete solution of (1). In this case we proceed as follows :

The part of the complete solution corresponding to the repeated root is the complete solution of $(D - m_1)^2 (D - m_3)y = 0$

Putting $(D - m_3)y = v$, it becomes $(D - m_1)^2 v = 0$ or $\frac{d^2 v}{dx^2} - 2m_1 \frac{dv}{dx} + m_1^2 v = 0$

This is a Leibnitz's linear equation and I.F. $= e^{-m_1 x}$. Its solution is $v e^{-m_1 x} = c_1 x + c_2$ or $v = (c_1 x + c_2) e^{m_1 x}$

Thus $(D - m_3)y = v = (c_1 x + c_2) e^{m_1 x}$ or $\frac{dy}{dx} - m_3 y = c_1 x e^{m_1 x} + c_2 e^{m_1 x}$ ...(5)

Its I.F. being $e^{-m_3 x}$, the solution of (5) is

$$y e^{-m_3 x} = \int c_1 x e^{(m_1 - m_3)x} dx + c_2 \int e^{(m_1 - m_3)x} dx + c_3 \text{ or } y = c_1 x e^{m_1 x} + c_2 e^{m_1 x} + c_3 e^{m_3 x}$$

Thus the complete solution of (1) is $y = c_1 x e^{m_1 x} + c_2 e^{m_1 x} + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$

If, however, the A.E. has three equal roots (i.e., $m_1 = m_2 = m_3$), then the complete solution is

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x} + c_3 x^2 e^{m_1 x} + c_4 e^{m_4 x} + \ldots + c_n e^{m_n x}$$

Case III. If one pair of roots be imaginary, i.e., $m_1 = \alpha + i\beta$, $m_2 = \alpha - i\beta$, then the complete solution is

$$y = c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x} + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$$

$$= e^{\alpha x}(c_1 e^{i\beta x} + c_2 e^{-i\beta x}) + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$$

$$= e^{\alpha x}(c_1 \cos \beta x + i c_1 \sin \beta x) + c_2 (\cos \beta x - i \sin \beta x) + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$$

$[\because$ by Euler's Theorem, $e^{i\theta} = \cos \theta + i \sin \theta]$

$$= e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x) + c_3 e^{m_3 x} + \ldots + c_n e^{m_n x}$$

where $C_1 = c_1 + c_2$ and $C_2 = i(c_1 - c_2)$.

Case IV. If two pairs of imaginary roots be equal, i.e., $m_1 = m_2 = \alpha + i\beta$, $m_3 = m_4 = \alpha - i\beta$, then by case II, the complete solution is

$$y = e^{\alpha x}(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x + \ldots + c_n e^{m_n x}.$$

Example 12.1. Solve $\frac{d^2 y}{dx^2} + y\frac{dy}{dx} + x = 0$, given that $y(0) = 1$ and $\left(\frac{dy}{dx}\right)_{x=0} = 0$. [U.P.T.U., 2005]

Solution. Given equation in symbolic form is $(D^2 + D + 1)y = 0$.

Its A.E. is $D^2 + D + 1 = 0$, i.e., $(D + 1)^2 + 1 = 0$ or $(D + 1)^2 = -1$. Solution $D = -1, -1$.

$\therefore$ C.F. is $y = c_1 e^{-x} + c_2 x e^{-x}$ and $\frac{dy}{dx} = -2c_1 e^{-x} - c_2 x e^{-x}$

When $x = 0$, $y = 1$ $\quad \therefore \quad 1 = c_1 + c_2$ ...(1)

When $x = 0$, $dy/dx = 0$ $\quad \therefore \quad 0 = -2c_1 - c_2$ ...(2)

Solving (i) and (ii), $c_1 = 11, c_2 = -36$

Hence the required solution is $x = 33e^{-3t} - 36e^{-4t}$.

Example 12.2. Solve $\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 3x = 0$.

Solution. Given equation in symbolic form is $(D^2 + 4D + 3)x = 0$

$\therefore$ A.E. is $D^2 + 4D + 3 = 0$, i.e., $(D + 3)(D + 1) = 0$ whence $D = -3, -1$.

Hence the C.F. is $x = c_1e^{-3t} + c_2e^{-t}$.

Example 12.3. Solve $(D^4 + 2D^3 + 3D^2 + 4D + 4)y = 0$.

Solution. Here the A.E. is $D^4 + 2D^3 + 3D^2 + 4D + 4 = 0$ i.e., $(D^2 + 2D + 2)(D^2 + 2) = 0$ $\therefore$ $D = -1, \pm 2i$.

Hence the C.F. is $y = e^{-x}(c_1 \cos 2x + c_2 \sin 2x) + (c_3 \cos 2x + c_4 \sin 2x)$

i.e., $y = e^{-x}(c_1 \cos 2x + c_2 \sin 2x) + c_3 \cos 2x + c_4 \sin 2x$.

Example 12.4. Solve (i) $(D^4 - 4D^3 + 4D^2)y = 0$ (Nagpur, 2009)

(ii) $(D^3 + 2D^2 + D)y = 0$ where $D \equiv \frac{d}{dx}$.

Solution. (i) The A.E. equation is $D^4 - 4D^3 + 4D^2 = 0$ or $(D^2 - 2D)^2 = 0$

$\therefore$ $D^2 = 2, 2$ i.e., $D = 0, 0, 2, 2$.

Hence the C.F. is $(c_1 + c_2x)e^{0x} + (c_3 + c_4x)e^{2x}$ [Roots being repeated]

(ii) The A.E. equation is $D^3 + 2D^2 + D = 0$

$\therefore$ $D = 0, (-1, -1)$.

Hence the C.F. is $y = e^{0x}(c_1 + c_2x) + e^{-x}(c_3 + c_4x)$ i.e., $y = c_1 + c_2x + e^{-x}(c_3 + c_4x)$.

Example 12.5. Solve $\frac{d^4x}{dt^4} + 4x = 0$.

Solution. Given equation in symbolic form is $(D^4 + 4)x = 0$

$\therefore$ A.E. is $D^4 + 4 = 0$ or $(D^2 + 2)^2 - 4D^2 = 0$ or $(D^2 + 2D + 2)(D^2 - 2D + 2) = 0$

or $(D^2 + 2D + 2)[(D - 1)^2 + 1] = 0$

$\therefore$ either $D^2 + 2D + 2 = 0$ or $(D - 1)^2 = -1$

whence $D = \frac{-2 \pm \sqrt{(-8)}}{2}$ and $\frac{2 \pm \sqrt{(-4)}}{2}$ i.e., $D = -1 \pm i$ and $1 \pm i$

Hence the required solution is $x = e^{-t}(c_1 \cos t + c_2 \sin t) + e^t(c_3 \cos t + c_4 \sin t)$.

EXERCISE 12.1

Solve :

1. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 13y = 0$ (V.T.U., 2010)
2. $(D^2 - 2D + 10)y = 0$; $y(0) = 1$, $y(\frac{\pi}{2}) = 4$
3. $4\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + y = 0$
4. $\frac{d^2y}{dx^2} + y = 0$ (V.T.U., 2010)
5. $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 0$
6. $\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + 16y = 0$ (V.T.U., 2010)
7. $(D^4 - 8D^3 - 20D^2 + 112D + 96)y = 0$ (V.T.U., 2010)
8. $(D^4 + 1)^2(D - 1)^2y = 0$
9. If $\frac{d^2y}{dx^2} = n^2y$, prove that $y = c_1e^{nx} + c_2e^{-nx}$ or $y = c_3 \sinh nx + c_4 \cosh nx$.

13.5 INVERSE OPERATOR

(1) Definition. $\frac{1}{f(D)}X$ is that function of x, not containing arbitrary constants which when operated upon by $f(D)$ gives X.

i.e., $$f(D)\left\{\frac{1}{f(D)}X\right\} = X$$

Thus $\frac{1}{f(D)}X$ satisfies the equation $f(D)y = X$ and is, therefore, its particular integral.

Obviously, $f(D)$ and $\frac{1}{f(D)}$ are inverse operators.

(2) $$\frac{1}{D}X = \int X dx$$

Let $$\frac{1}{D}X = y \qquad \ldots(i)$$

Operating by D, $$D\frac{1}{D}X = Dy \quad \text{i.e., } X = \frac{dy}{dx}$$

Integrating both sides w.r.t. x, $y = \int X dx$, an integral not having arbitrary constant as (i) does not contain any constant.

Thus $$\frac{1}{D}X = \int X dx$$

(3) $$\frac{1}{D-a}X = e^{ax}\int Xe^{-ax}dx.$$

Let $$\frac{1}{D-a}X = y \qquad \ldots(i)$$

Operating by $D-a$, $(D-a)$, $\frac{1}{D-a}X = (D-a)y$.

or $$X = \frac{dy}{dx} - ay, \text{ i.e., } \frac{dy}{dx} - ay = X \text{ which is a Leibnitz's linear equation.}$$

$\therefore$ I.F. $= e^{\int -a dx} = e^{-ax}$, the solution is

$ye^{-ax} = \int Xe^{-ax}dx$, an integral having added to it (i) doesn't contain any constant.

Thus $$\frac{1}{D-a}X = y = e^{ax}\int Xe^{-ax}dx.$$

13.6 RULES FOR FINDING THE PARTICULAR INTEGRAL

Consider the equation $\frac{d^n y}{dx^n} + k_1\frac{d^{n-1}y}{dx^{n-1}} + k_2\frac{d^{n-2}y}{dx^{n-2}} + \ldots + k_n y = X$

which in symbolic form is $(D^n + k_1D^{n-1} + k_2D^{n-2} + \ldots + k_n)y = X$.

$\therefore$ $$\text{P.I.} = \frac{1}{D^n + k_1D^{n-1} + k_2D^{n-2} + \ldots + k_n}X.$$

Case I. When $X = e^{ax}$

Since $$D(e^{ax}) = ae^{ax}$$
$$D^2(e^{ax}) = a^2e^{ax}$$
$$\ldots\ldots\ldots\ldots$$
$$D^n(e^{ax}) = a^ne^{ax}$$

$\therefore$ $(D^n + k_1D^{n-1} + \ldots + k_n)e^{ax} = (a^n + k_1a^{n-1} + \ldots + k_n)e^{ax}$, i.e., $f(D)e^{ax} = f(a)e^{ax}$

Operating on both sides by $\frac{1}{f(D)}$: $\frac{1}{f(D)}f(D)e^{ax} = \frac{1}{f(D)}f(a)e^{ax}$ or $e^{ax} = f(a)\frac{1}{f(D)}e^{ax}$

∴ dividing by $f(a)$,

$$\frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax} \text{ provided } f(a) \neq 0 \qquad ...(1)$$

If $f(a) = 0$, the above rule fails and we proceed further.

Since a is a root of $A.E.$ $f(D) = D^n + A_1D^{n-1} + \ldots + A_n = 0$.

∴ $D - a$ is a factor of $f(D)$. Suppose $f(D) = (D - a)\phi(D)$, where $\phi(a) \neq 0$. Then

$$\frac{1}{f(D)}e^{ax} = \frac{1}{D-a}\cdot\frac{1}{\phi(D)}e^{ax} = \frac{1}{D-a}\cdot\frac{1}{\phi(a)}e^{ax} \qquad \text{By (1)}$$

$$= \frac{1}{\phi(a)}\cdot\frac{1}{D-a}e^{ax} = \frac{1}{\phi(a)}e^{ax}\int e^{ax}e^{-ax}\,dx \qquad \text{By (1) (3.6 (1))}$$

$$= \frac{1}{\phi(a)}e^{ax}\int dx = x\frac{1}{\phi(a)}e^{ax} \quad \text{i.e.,} \quad \frac{1}{f(D)}e^{ax} = x\frac{1}{f'(a)}e^{ax} \qquad ...(2)$$

$$\left[\because f'(D) = (D-a)\phi'(D) + 1.\phi(D)\right.$$
$$\left.\therefore f'(a) = 0 + \phi(a) = \phi(a)\right]$$

If $f'(a) = 0$, then applying (2) again, we get $\frac{1}{f(D)}e^{ax} = x^2\frac{1}{f''(a)}e^{ax}$, provided $f''(a) \neq 0$ ...(3)

and so on.

Example 3.8. Find the P.I. of $(D^2 + 3D + 4)y = e^x$.

Solution. $\text{P.I.} = \frac{1}{D^2+3D+4}e^x$ [Put $D = 1$] $= \frac{1}{1^2+3.1+4}e^x = \frac{e^x}{12}$.

Example 3.9. Find the P.I. of $(D^2 + 2D - 1)y = e^{2x} + 2\sinh x$.

Solution. $\text{P.I.} = \frac{1}{(D+2)(D-1)}(e^{2x} + 2\sinh x) = \frac{1}{(D+2)(D-1)}(e^{2x} + e^x - e^{-x})$

Let us evaluate each of these terms separately.

$$\frac{1}{(D+2)(D-1)}e^{-2x} = \frac{1}{D+2}\cdot\left[\frac{1}{(D-1)}e^{-2x}\right]$$

$$= \frac{1}{D+2}\cdot\frac{1}{(-2)-1}e^{-2x} = \frac{1}{D+2}\cdot\frac{1}{-3}e^{-2x}$$

$$= \frac{1}{D}\cdot\frac{1}{3}e^{-2x} = \frac{1}{3D}e^{-2x} \qquad \left[\because \frac{d}{dD}(D+2) = 1\right]$$

$$\frac{1}{(D+2)(D-1)^2}e^x = \frac{1}{1+2}\cdot\frac{1}{(D-1)^2}e^x = \frac{1}{3}\cdot x^2\cdot\frac{1}{2}e^x = \frac{x^2}{6}e^x \qquad \left[\because \frac{d^2}{dD^2}(D-1)^2 = 2\right]$$

and

$$\frac{1}{(D+2)(D-1)^2}e^{-x} = \frac{1}{(-1+2)(-1-1)^2}e^{-x} = \frac{e^{-x}}{4}$$

Hence, $\text{P.I.} = \frac{1}{3D}e^{-2x} + \frac{x^2}{6}e^x + \frac{1}{4}e^{-x}$.

Case II. When X is $\sin(ax + b)$ or $\cos(ax + b)$.

Since $D\sin(ax + b) = a\cos(ax + b)$

$D^2\sin(ax + b) = -a^2\sin(ax + b)$

$D^3\sin(ax + b) = -a^2\cos(ax + b)D$

$$D^2\sin(ax+b)=-a^2\sin(ax+b)$$

i.e., $$D^3\sin(ax+b)=(-a^2)D\sin(ax+b)$$

$$(D^2)^2\sin(ax+b)=(-a^2)^2\sin(ax+b)$$

In general $$(D^2)^n\sin(ax+b)=(-a^2)^n\sin(ax+b)$$

$\therefore$ $$f(D^2)\sin(ax+b)=f(-a^2)\sin(ax+b)$$

Operating on both sides by $\frac{1}{f(D^2)}$,

$$\frac{1}{f(D^2)}.f(D^2)\sin(ax+b)=\frac{1}{f(D^2)}f(-a^2)\sin(ax+b)$$

or $$\sin(ax+b)=f(-a^2)\frac{1}{f(D^2)}\sin(ax+b)$$

$\therefore$ Dividing by $f(-a^2)$, $\frac{1}{f(D^2)}\sin(ax+b)=\frac{1}{f(-a^2)}\sin(ax+b)$ provided $f(-a^2)\neq 0$ ...(4)

If $f(-a^2)=0$, the above rule fails. In such case proceed further.

Since $\cos(ax+b)+i\sin(ax+b)=e^{i(ax+b)}$ [Euler's theorem]

$\therefore$ $\frac{1}{f(D^2)}\cos(ax+b)+i\frac{1}{f(D^2)}\sin(ax+b)$ [Since $f(-a^2)=0$ $\therefore$ by (4)]

$=$ I.P. of $\frac{1}{f(D^2)}e^{i(ax+b)}$ [$\because D^2=-a^2$]

$\therefore$ $\frac{1}{f(D^2)}\sin(ax+b)=x\frac{1}{f'(-a^2)}\sin(ax+b)$ provided $f'(-a^2)\neq 0$ ...(5)

If $f'(-a^2)=0$, $\frac{1}{f(D^2)}\sin(ax+b)=x^2\frac{1}{f''(-a^2)}\sin(ax+b)$, provided $f''(-a^2)\neq 0$, and so on.

Similarly, $\frac{1}{f(D^2)}\cos(ax+b)=x\frac{1}{f'(-a^2)}\cos(ax+b)$, provided $f'(-a^2)\neq 0$.

If $f'(-a^2)=0$, $\frac{1}{f(D^2)}\cos(ax+b)=x^2\frac{1}{f''(-a^2)}\cos(ax+b)$, provided $f''(-a^2)\neq 0$ and so on.

Example 1.65. Find the P.I. of $(D^2+4)y=\cos(2x-1)$.

Solution. P.I. $=\frac{1}{D^2+4}\cos(2x-1)$ [Put $D^2=-2^2=-4$]

$=\frac{1}{4(-1)+4}\cos(2x-1)$ [Multiply and divide by $1+4D^2$]

$=\frac{(1+4D^2)}{(1-4D)(1+4D)}\cos(2x-1)=(1+4D^2)\frac{1}{1-16D^2}\cos(2x-1)$ [Put $D^2=-2^2=-4$]

$=(1-4D)\frac{1}{1-16(-4)}\cos(2x-1)=\frac{1}{65}[\cos(2x-1)+4D\cos(2x-1)]$

$=\frac{1}{65}[\cos(2x-1)-8\sin(2x-1)]$.

Example 1.66. Find the P.I. of $\frac{d^2y}{dx^2}+9y=\sin 3x$.

Solution. Given equation in symbolic form is $(D^2+9)y=\sin 3x$

$$\therefore \quad \text{P.I.} = \frac{1}{2(D^2+4)}\sin 2x \qquad [\because D^2+4=0 \text{ if } D^2=-2^2, \text{ i.e., Apply (iv) (ii)}]$$

$$= x\frac{1}{2D}\sin 2x \qquad \left[\because \frac{d}{dD}(D^2+4)=2D\right]$$

$$= x\frac{1}{2(-2^2)}\int \sin 2x = -\frac{x}{8}\cos 2x \qquad [\text{Put } D^2=-2^2=-4]$$

Case III. When $X = x^m$.

Here $$\text{P.I.} = \frac{1}{f(D)}x^m = [f(D)]^{-1}x^m.$$

Expand $[f(D)]^{-1}$ in ascending powers of D as far as the term in D^m and operate on x^m term by term. Since the $(m+1)$th and higher derivatives of x^m are zero, we need not consider terms beyond D^m.

Example 13.10. Find the P.I. of $\frac{d^2y}{dx^2}+\frac{dy}{dx}=x^2+2x+4$.

Solution. Given equation in symbolic form is $(D^2+D)y = x^2+2x+4$.

$$\therefore \quad \text{P.I.} = \frac{1}{D(D+1)}(x^2+2x+4) = \frac{1}{D}(1+D)^{-1}(x^2+2x+4)$$

$$= \frac{1}{D}(1-D+D^2-\ldots)(x^2+2x+4) = \frac{1}{D}(x^2+2x+4-(2x+2)+2)$$

$$= \int (x^2+4)\,dx = \frac{x^3}{3}+4x.$$

Case IV. When $X = e^{ax}V$, V being a function of x.

If u is a function of x, then

$$D(e^{ax}u) = e^{ax}Du + ae^{ax}u = e^{ax}(D+a)u$$

$$D^2(e^{ax}u) = e^{ax}D^2u + 2ae^{ax}Du + a^2e^{ax}u = e^{ax}(D+a)^2u$$

and in general, $$D^n(e^{ax}u) = e^{ax}(D+a)^n u$$

$$\therefore \quad f(D)(e^{ax}u) = e^{ax}f(D+a)u$$

Operating both sides by $\frac{1}{f(D)}$,

$$\frac{1}{f(D)}f(D)(e^{ax}u) = \frac{1}{f(D)}e^{ax}f(D+a)u$$

$$e^{ax}u = \frac{1}{f(D)}e^{ax}f(D+a)u$$

Now put $f(D+a)u = V$, i.e., $u = \frac{1}{f(D+a)}V$, so that $e^{ax}\frac{1}{f(D+a)}V = \frac{1}{f(D)}e^{ax}V$

i.e., $$\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V. \qquad \ldots(1)$$

Example 13.11. Find P.I. of $(D^2-2D+4)y = e^x\cos x$.

Solution. $$\text{P.I.} = \frac{1}{D^2-2D+4}e^x\cos x \qquad [\text{Replace } D \text{ by } D+1]$$

$$= e^x\frac{1}{(D+1)^2-2(D+1)+4}\cos x = e^x\frac{1}{D^2+3}\cos x \qquad [\because D^2=-1^2=-1]$$

$$= e^x\frac{1}{-1+3}\cos x = \frac{1}{2}e^x\cos x.$$

Case V. When X is any other function of x.

Here $\quad P.I. = \frac{1}{f(D)}X$

If $\quad f(D) = (D - m_1)(D - m_2)\ldots(D - m_n)$, resolving into partial fractions,

$$\frac{1}{f(D)} = \frac{A_1}{D - m_1} + \frac{A_2}{D - m_2} + \ldots + \frac{A_n}{D - m_n}$$

$$\therefore \quad P.I. = \left[\frac{A_1}{D - m_1} + \frac{A_2}{D - m_2} + \ldots + \frac{A_n}{D - m_n}\right]X$$

$$= A_1\frac{1}{D - m_1}X + A_2\frac{1}{D - m_2}X + \ldots + A_n\frac{1}{D - m_n}X$$

$$= A_1 e^{m_1 x}\int X e^{-m_1 x}\,dx + A_2 e^{m_2 x}\int X e^{-m_2 x}\,dx + \ldots + A_n e^{m_n x}\int X e^{-m_n x}\,dx \quad \text{[By III.5 ...(iii)]}$$

Note. This method is general in nature and can, therefore, be employed to obtain a particular integral in any given case.

15.9 WORKING PROCEDURE TO SOLVE THE EQUATION

$$\frac{d^n y}{dx^n} + k_1\frac{d^{n-1}y}{dx^{n-1}} + \ldots + k_{n-1}\frac{dy}{dx} + k_n y = X$$

of which the symbolic form is

$$(D^n + k_1 D^{n-1} + \ldots + k_{n-1}D + k_n)y = X.$$

Step I. To find the complementary function

(i) Write the A.E.

i.e., $\quad D^n + k_1 D^{n-1} + \ldots + k_{n-1}D + k_n = 0$ and solve it for D.

(ii) Write the C.F. as follows :

Roots of A.E.	C.F.
1. $m_1, m_2, m_3, \ldots$ (real and different roots)	$c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \ldots$
2. $m_1, m_1, m_3, \ldots$ (two real and equal roots)	$(c_1 + c_2 x)e^{m_1 x} + c_3 e^{m_3 x} + \ldots$
3. $m_1, m_1, m_1, m_4, \ldots$ (three real and equal roots)	$(c_1 + c_2 x + c_3 x^2)e^{m_1 x} + c_4 e^{m_4 x} + \ldots$
4. $\alpha + i\beta, \alpha - i\beta, m_3, \ldots$ (a pair of imaginary roots)	$e^{\alpha x}(c_1 \cos\beta x + c_2 \sin\beta x) + c_3 e^{m_3 x} + \ldots$
5. $\alpha + i\beta, \alpha + i\beta, \alpha - i\beta, \alpha - i\beta, \ldots$ (two pairs of imaginary roots)	$e^{\alpha x}[(c_1 + c_2 x)\cos\beta x + (c_3 + c_4 x)\sin\beta x] + \ldots$

Step II. To find the particular integral

From symbolic form $P.I. = \frac{1}{D^n + k_1 D^{n-1} + \ldots + k_{n-1}D + k_n}X = \frac{1}{f(D)} = \frac{1}{\phi(D^2)}X$

(i) When $X = e^{ax}$

$$P.I. = \frac{1}{f(a)}e^{ax}, \text{ put } D = a. \qquad [f(a) \neq 0]$$

$$= x\frac{1}{f'(a)}e^{ax}, \text{ put } D = a. \qquad [f'(a) = 0, f''(a) \neq 0]$$

$$= x^2\frac{1}{f''(a)}e^{ax}, \text{ put } D = a, \qquad [f''(a) = 0, f'''(a) \neq 0]$$

and so on.

where $\quad f'(D) = \text{diff. coeff. of } f(D) \text{ w.r.t. } D$

$f''(D) = \text{diff. coeff. of } f'(D) \text{ w.r.t. } D$, etc.

(v) When $X = \sin(ax+b)$ or $\cos(ax+b)$

$$\text{P.I.} = \frac{1}{\phi(D^2)}\sin(ax+b)\ (\text{or } \cos(ax+b)), \text{ put } D^2 = -a^2 \qquad [\text{If } \phi(-a^2) \neq 0]$$

$$= x\frac{1}{\phi'(D^2)}\sin(ax+b)\ (\text{or } \cos(ax+b)), \text{ put } D^2 = -a^2 \qquad [\text{If } \phi(-a^2) = 0,\ \phi'(-a^2) \neq 0]$$

$$= x^2\frac{1}{\phi''(D^2)}\sin(ax+b)\ (\text{or } \cos(ax+b)), \text{ put } D^2 = -a^2 \qquad [\text{If } \phi'(-a^2) = 0,\ \phi''(-a^2) \neq 0]$$

and so on.

where $\phi(D^2)$ = Coff. of m in $\phi(D^2) = m$ i.e. $f(D)$,

$\phi'(D^2)$ = diff. coeff. of $\phi(D^2)$ w.r.t. D, etc.

(vi) When $X = e^{ax}$, a being a positive integer,

$$\text{P.I.} = \frac{1}{f(D)}e^{ax} = [f(D)]^{-1}e^{ax}$$

To evaluate it, expand $[f(D)]^{-1}$ in ascending powers of D by Binomial theorem so that D^n and operate on e^{ax} term by term.

(vii) When $X = e^{ax}V$, where V is a function of x,

$$\text{P.I.} = \frac{1}{f(D)}e^{ax}V = e^{ax}\frac{1}{f(D+a)}V$$

and then evaluate $\frac{1}{f(D+a)}V$ as in (i), (ii), and (iii).

(viii) When X is any function of x,

$$\text{P.I.} = \frac{1}{f(D)}X$$

Resolve $\frac{1}{f(D)}$ into partial fractions and operate each partial fraction on X remembering that

$$\frac{1}{D-a}X = e^{ax}\int Xe^{-ax}dx.$$

Step III. To find the complete solution

Then the C.S. is $y = \text{C.F.} + \text{P.I.}$

Example 12.12. Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = x(1+e^x)$.

Solution. Given equation in symbolic form is $(D^2 + D + 1)y = x(1 + e^x)$

(i) To find C.F.

In A.E. is $D^2 + D + 1 = 0$, $\therefore D = \frac{1}{2}(-1 \pm \sqrt{3}i)$

$\therefore$ the C.F. $= e^{-\frac{x}{2}}\left(c_1\cos\frac{\sqrt{3}}{2}x + c_2\sin\frac{\sqrt{3}}{2}x\right)$

(ii) To find P.I.

$$\text{P.I.} = \frac{1}{D^2+D+1}(x - 2x^2 + x^3) + \frac{1}{D^2+D+1}(x^2 - 2x^3 + x^4)$$

$$+ \frac{1}{D^2+D+1}e^{2x} - 2\cdot\frac{1}{1^2+1+1}e^x + \frac{1}{2^2+2+1}e^{2x} = 1 - \frac{2}{3}x^2 + \frac{e^{2x}}{7}$$

(iii) Hence the C.S. is $y = e^{-\frac{x}{2}}\left(c_1\cos\frac{\sqrt{3}}{2}x + c_2\sin\frac{\sqrt{3}}{2}x\right) + 1 - \frac{2}{3}x^2 + \frac{e^{2x}}{7}$.

Example 12.24. Solve $y'' + 4y' + 4y = 8\sin x + 4\cos x$, $y(0) = 2$ and $y'(0) = 0$. *(U.P.T.U. 2003)*

Solution. Given equation in symbolic form is $(D^2 + 4D + 4)y = 8\sin x + 4\cos x$

(i) To find C.F.

Its A.E. is $D^2 + 4D + 4 = 0$ where $D = -2, -2$ $\therefore$ C.F. $= (c_1 + c_2 x)e^{-2x}$.

(ii) To find P.I.

$$P.I. = \frac{1}{D^2 + 4D + 4}(8\sin x + 4\cos x) = \frac{1}{-1 + 4D + 4}(8\sin x + 4\cos x)$$

$$= \frac{4D - 3}{16D^2 - 9}(8\sin x + 4\cos x) = \frac{16D - 8}{-16 - 9}(8\sin x + 4\cos x)$$

$$= \frac{-1}{25}[16(\cos x - 8\sin x) + 4(-8\sin x - 8\cos x)] = \sin x$$

(iii) G.S. is $y = (c_1 + c_2 x)e^{-2x} + \sin x$.

When $x = 0$, $y = 2$, $\therefore$ $2 = c_1$

Also $y' = c_2 e^{-2x} + (c_1 + c_2 x)(-2)e^{-2x} + \cos x$.

When $x = 0$, $y' = 0$, $\therefore$ $0 = c_2 - 2c_1 + 1$ i.e., $c_2 = 3$.

Hence the required solution is $y = (2 + 3x)e^{-2x} + \sin x$.

Example 12.25. Solve $(D - 2)^2(D^2 + 4)y = 8(x^2 + \sin 2x + e^{2x})$.

Solution. (i) To find C.F.

Its A.E. is $(D - 2)^2(D^2 + 4) = 0$, $\therefore$ $D = 2, 2, \pm 2i$

Then C.F. $= (c_1 + c_2 x)e^{2x} + c_3 \cos 2x + c_4 \sin 2x$.

(ii) To find P.I.

$$P.I. = \frac{1}{(D - 2)^2(D^2 + 4)}[8(x^2 + \sin 2x + e^{2x})]$$

Now $\frac{1}{(D - 2)^2(D^2 + 4)}e^{2x} = x^2\frac{1}{2!(D^2 + 4)}e^{2x}$ $\because$ by putting $D = 2$, $(D - 2)^2 = 0, 2D - 2 = 0$

$$= \frac{x^2 e^{2x}}{8}$$

$$\frac{1}{(D - 2)^2(D^2 + 4)}\sin 2x = \frac{1}{D^2 - 4D + 4}\cdot\frac{1}{D^2 + 4}\sin 2x = \frac{1}{1 - 2^2 - 4D + 4}\cdot\frac{1}{D^2 + 4}\sin 2x$$

$$= -\frac{1}{4}\int\sin 2x\,dx = -\frac{1}{4}\left(\frac{-\cos 2x}{2}\right) = \frac{1}{8}\cos 2x$$

and $$\frac{1}{(D - 2)^2}x^2 = \frac{1}{4}\left(1 - \frac{D}{2}\right)^{-2}x^2 = \frac{1}{4}\left[1 + 1 - 2\left(\frac{D}{2}\right) - \frac{1 - 2 - 2i}{2!}\left(-\frac{D}{2}\right)^2 + \ldots\right]x^2$$

$$= \frac{1}{4}\left[1 + D + \frac{3D^2}{4} + \ldots\right]x^2 = \frac{1}{4}\left[x^2 + 2x - \frac{3}{2}\right]$$

Thus $P.I. = \frac{1}{8}x^2e^{2x} + \frac{1}{8}\cos 2x + \frac{1}{4}x^2 + \frac{1}{2}x - \frac{3}{8}$.

(iii) Hence the G.S. is $y = (c_1 + c_2 x)e^{2x} + c_3\cos 2x + c_4\sin 2x + \frac{1}{8}x^2e^{2x} + \frac{1}{8}\cos 2x + \frac{1}{4}x^2 + \frac{1}{2}x - \frac{3}{8}$.

Example 12.26. Find the complete solution of $(D^2 - 2D + 2)y = e^x + e^x\cos x$. *(U.P.T.U. 2002)*

Solution. Given equation in symbolic form is $(D^2 - 2D + 2)y = e^x + e^x\cos x$

(i) To find C.F.

Its A.E. is $D^2 - 2D + 2 = 0$ $\therefore$ $D = \frac{2 \pm \sqrt{4 - 8}}{2} = 1 \pm i$.

Then C.F. $= e^x(c_1\cos x + c_2\sin x)$

(c) To find P.I.

$$\text{P.I.} = \frac{1}{D^2-2D+2}(2x) + \frac{1}{D^2-2D+2}(e^x \cos x)$$

$$= \frac{1}{2}\left[1-\left(D-\frac{D^2}{2}\right)\right]^{-1}(2x) + e^x \frac{1}{(D+1)^2-2(D+1)+2}(\cos x)$$

$$= \frac{1}{2}\left[1+D-\frac{D^2}{2}\right]x + e^x \frac{1}{D^2+1}\cos x \qquad \text{[Case of failure]}$$

$$= \frac{1}{2}(2x+1-0) + e^x \cdot x\frac{1}{2D}\cos x = \frac{1}{2}(2x+1) + \frac{xe^x}{2}\int \cos x\,dx = \frac{1}{2}(2x+1) + \frac{xe^x}{2}\sin x$$

(iii) Hence the G.S. is $y = c_1y_1\cos x + c_2y_2\sin x + \frac{1}{2}(2x+1) + \frac{xe^x}{2}\sin x$.

Example 12.34. Solve $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^{2x} + \sin 2x$.

(U.P.T.U., 2009; Kolaiyan, 2010; C.H.E.C., 2000)

Solution. Given equation in symbolic form is $(D^2-3D+2)y = e^{2x} + \sin 2x$

(i) *To find C.F.*

Its A.E. is $D^2-3D+2=0$ or $(D-1)(D-2)=0$ whence $D=1, 2$.

Then C.F. $= c_1e^x + c_2e^{2x}$

(ii) *To find P.I.*

$$\text{P.I.} = \frac{1}{D^2-3D+2}(e^{2x} + \sin 2x) = \frac{1}{D^2-3D+2}e^{2x}\cdot 1 + \frac{1}{D^2-3D+2}(\sin 2x)$$

$$= e^{2x}\frac{1}{(D+2)^2-3(D+2)+2}(1) + \frac{1}{-4-3D+2}(\sin 2x)$$

$$= e^{2x}\cdot\frac{1}{D^2+3D+2}(1) - \frac{3D-2}{9D^2-4}(\sin 2x) = \frac{e^{2x}}{2}\left[1+\left(\frac{3D+D^2}{2}\right)\right]^{-1} - \frac{(3D-2)}{9(-4)-4}(\sin 2x)$$

$$= \frac{e^{2x}}{2}\left(1-\frac{3D}{2}+\ldots\right)x + \frac{1}{40}(3\cos 2x - 2\sin 2x) = \frac{e^{2x}}{2}\left(x-\frac{3}{2}\right) + \frac{1}{20}(2\sin 2x - 3\cos 2x)$$

(iii) Hence the G.S. is $y = c_1e^x + c_2e^{2x} + e^{2x}\left(\frac{x}{2}-\frac{3}{4}\right) + \frac{1}{20}(2\sin 2x - 3\cos 2x)$.

Example 12.35. Solve $\frac{d^2y}{dx^2} - 4y = x\sinh x$. (Kolaiyan, 2009)

Solution. Given equation in symbolic form is $(D^2-4)y = x\sinh x$.

(i) *To find C.F.*

Its A.E. is $D^2-4=0$, whence $D=\pm 2$.

Then C.F. $= c_1e^{2x} + c_2e^{-2x}$

(ii) *To find P.I.*

$$\text{P.I.} = \frac{1}{D^2-4}x\sinh x = \frac{1}{D^2-4}x\left[\frac{e^x-e^{-x}}{2}\right] = \frac{1}{2}\left[\frac{1}{D^2-4}e^x\cdot x - \frac{1}{D^2-4}e^{-x}\cdot x\right]$$

$$= \frac{1}{2}\left[e^x\frac{1}{(D+1)^2-4}x - e^{-x}\frac{1}{(D-1)^2-4}x\right] = \frac{1}{2}\left[e^x\frac{1}{D^2+2D-3}x - e^{-x}\frac{1}{D^2-2D-3}x\right]$$

$$= \frac{1}{2}\left[\frac{e^x}{-3}\left\{1-\left(\frac{2D}{3}+\frac{D^2}{9}\right)\right\}^{-1}.x - \frac{e^{-x}}{-3}\left\{1+\left(\frac{2D}{3}-\frac{D^2}{9}\right)\right\}^{-1}.x\right]$$

$$= -\frac{1}{6}\left[e^x\left(1+\frac{2D}{3}+\ldots\right)x - e^{-x}\left(1-\frac{2D}{3}+\ldots\right)x\right] = -\frac{1}{6}\left[e^x\left(x+\frac{2}{3}\right)-e^{-x}\left(x-\frac{2}{3}\right)\right]$$

$$= -\frac{x}{3}\left[\frac{e^x-e^{-x}}{2}\right] - \frac{2}{9}\left(\frac{e^x+e^{-x}}{2}\right) = -\frac{x}{3}\sinh x - \frac{2}{9}\cosh x.$$

(iii) Hence the G.S. is $y = c_1e^{3x} + c_2e^{-3x} - \frac{x}{3}\sinh x - \frac{2}{9}\cosh x$.

Example 33.16. Solve $(D^2-1)y = x\sin 2x + \cos x$.

Solution. (i) To find C.F.

Its A.E. is $D^2-1=0$, whence $D=\pm 1$. $\therefore$ C.F. $= c_1e^x + c_2e^{-x}$.

(ii) To find P.I.

$$P.I. = \frac{1}{D^2-1}(x\sin 2x + \cos x) = \frac{1}{D^2-1}\times(\text{I.P. of } e^{2ix}x) + \frac{1}{D^2-1}\cos x$$

$$= \text{I.P. of } \frac{1}{D^2-1}e^{2ix}.x + \frac{1}{(-1)^2-1}\cos x = \text{I.P. of } \left[e^{2ix}\frac{1}{(D+2i)^2-1}x\right] - \frac{\cos x}{2}$$

[Replacing D by $D+2i$]

$$= \text{I.P. of } \left[e^{2ix}\frac{1}{D^2+4iD-5}x\right] - \frac{\cos x}{2}$$

$$= \text{I.P. of } \left[e^{2ix}.\frac{1}{-5}\left\{1-\frac{4iD}{5}-\frac{D^2}{5}\right\}^{-1}x\right] - \frac{\cos x}{2}$$ [Expand by Binomial theorem]

$$= \text{I.P. of } \left[e^{2ix}.\frac{1}{-5}\left\{1+\frac{4iD}{5}+\ldots\ldots\right\}x\right] - \frac{\cos x}{2} = \text{I.P. of } \left[-\frac{e^{2ix}}{5}\left\{x+\frac{4i}{5}\right\}\right] - \frac{\cos x}{2}$$

$$= \text{I.P. of } \left[\frac{-1}{5}(\cos 2x + i\sin 2x)\left(x+\frac{4i}{5}\right)\right] - \frac{\cos x}{2}$$

$$= -\frac{1}{5}\text{I.P. of } \left[\left(x\cos 2x - \frac{4}{5}\sin 2x\right) + i\left(x\sin 2x + \frac{4}{5}\cos 2x\right)\right] - \frac{\cos x}{2}$$

$$= -\frac{1}{5}\left(x\sin 2x + \frac{4}{5}\cos 2x\right) - \frac{\cos x}{2}$$

(iii) Hence the G.S. is $y = c_1e^x + c_2e^{-x} - \frac{1}{10}(2x\sin 2x + 3\cos 2x + 10\cos x)$.

Example 33.17. Solve $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + y = xe^x\sin x$. (U.P.T.U., 2000; U.P.S.C.(I), 2000; U.P.T.U., 2004)

Solution. Given equation in symbolic form is $(D^2-3D+1)y = xe^x\sin x$.

(i) To find C.F.

Its A.E. is $D^2-3D+1=0$, i.e., $(D-1)^2 = D^2$.

$\therefore$ $D=1, 1$. Thus C.F. $= (c_1+c_2x)e^x$.

(ii) To find P.I.

$$P.I. = \frac{1}{(D-1)^2} e^x \cdot x \sin x = e^x \cdot \frac{1}{(D+1-1)^2} x \sin x$$

$$= e^x \frac{1}{D^2} x \sin x = e^x \frac{1}{D} \int x \sin x \, dx \qquad \text{Integrate by parts}$$

$$= e^x \frac{1}{D} \left[x(-\cos x) - \int 1 \cdot (-\cos x) \, dx \right] = e^x \int [-x \cos x + \sin x] \, dx$$

$$= e^x \left[-\left(x \sin x - \int 1 \cdot \sin x \, dx \right) - \cos x \right] = e^x [-x \sin x - \cos x - \cos x]$$

$$= -e^x (x \sin x + 2 \cos x).$$

(iii) Hence the G.S. is $y = c_1 + c_2 x + e^x (-x \sin x + 2 \cos x)$.

Example 4.20. Solve $(D^4 + 2D^2 + 1)y = x^2 \cos x$. (Kolhapur, 2010; Nagpur, December 2010)

Solution. (i) To find C.F.

Its A.E. is $D^4 + 2D^2 + 1 = 0$ whose roots are $D = i, i, -i, -i$

$\therefore$ C.F. $= (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$

(ii) To find P.I.

$$P.I. = \frac{1}{(D^2+1)^2} x^2 \cos x = \frac{1}{(D^2-i^2)^2} x^2 (\text{Re.P. of } e^{ix})$$

$$= \text{Re.P. of} \left(\frac{1}{(D^2+2i)^2} e^{ix} \cdot x^2 \right) = \text{Re.P. of} \left(e^{ix} \frac{1}{(D+i)^2 (D-i)^2} x^2 \right)$$

$$= \text{Re.P. of} \left(e^{ix} \frac{1}{D^2 (D+2i)^2} x^2 \right) = \text{Re.P. of} \left[e^{ix} \left(-\frac{1}{4D^2} \left(1 - \frac{1}{2} \frac{D}{i} \right)^{-2} x^2 \right) \right]$$

$$= \text{Re.P. of} \left[-\frac{1}{4} e^{ix} \cdot \frac{1}{D^2} \left(1 + 2\frac{D}{2i} + 3\left(\frac{D}{2i}\right)^2 + \ldots \right) x^2 \right]$$

$$= \text{Re.P. of} \left[-\frac{1}{4} e^{ix} \cdot \frac{1}{D^2} \left(x^2 + 2x \cdot \frac{1}{i} - \frac{3}{2} \right) \right] = \text{Re.P. of} \left[-\frac{1}{4} e^{ix} \cdot \frac{1}{D^2} \left(x^2 - 2ix - \frac{3}{2} \right) \right]$$

$$= -\frac{1}{4} \text{Re.P. of} \left[e^{ix} \left(\frac{x^4}{12} - \frac{ix^3}{3} - \frac{3}{4} x^2 \right) \right] = \frac{-1}{48} \text{Re.P. of} [\cos x + i \sin x)(x^4 - 4ix^3 - 9x^2)]$$

$$= -\frac{1}{48} (x^4 - 16x^2) \cos x - 4x^3 \sin x].$$

(iii) Hence the G.S. is $y = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x + \frac{1}{48} [16x^2 \cos x - x^2 (x^2 - 24) \cos x]$.

Example 4.21. Solve $(D^2 - 4D + 4)y = 8x^2 e^{2x} \cos 2x$. (U.P.T.U., 2009; G.N.D.U., 2004)

Solution. (i) To find C.F.

Its A.E. is $D^2 - 4D + 4 = 0$ i.e., $(D-2)^2 = 0$ $\therefore$ $D = 2, 2$

$\therefore$ C.F. $= (c_1 + c_2 x) e^{2x}$

(ii) To find P.I.

$$P.I. = \frac{1}{(D-2)^2} 8x^2 e^{2x} \cos 2x = 8e^{2x} \frac{1}{(D+2-2)^2} x^2 \cos 2x$$

$$= 8e^{2x} \frac{1}{D^2} x^2 \cos 2x = 8e^{2x} \cdot \frac{1}{D} \int x^2 \cos 2x \, dx$$

$$= e^{2x}\cdot\frac{1}{D}\left[x^2\left(\frac{-\cos 2x}{2}\right)-\int 2x\left(\frac{-\cos 2x}{2}\right)dx\right]$$

$$= e^{2x}\cdot\frac{1}{D}\left[-\frac{x^2}{2}\cos 2x+\frac{\sin 2x}{2}-\int 1\cdot\frac{\sin 2x}{2}\,dx\right]$$

$$= e^{2x}\int\left[-\frac{x^2}{2}\cos 2x+\frac{x}{2}\sin 2x+\frac{\cos 2x}{4}\right]dx$$

$$= e^{2x}\left[\left(\frac{-x^2}{2}\frac{\sin 2x}{2}-\int -x\frac{\sin 2x}{2}\,dx\right)+\left[\int\frac{x}{2}\sin 2x\,dx\right]+\frac{\sin 2x}{8}\right]$$

$$= e^{2x}\left[\left(\frac{-x^2}{4}+\frac{1}{8}\right)\sin 2x+\int x\sin 2x\,dx\right]$$

$$= e^{2x}\left[\left(\frac{1}{8}-\frac{x^2}{4}\right)\sin 2x+x\left(\frac{-\cos 2x}{2}\right)-\int 1\left(\frac{-\cos 2x}{2}\right)dx\right]$$

$$= e^{2x}\left[\left(\frac{1}{8}-\frac{x^2}{4}\right)\sin 2x-\frac{x\cos 2x}{2}+\frac{\sin 2x}{4}\right]$$

$$= e^{2x}\left[\left(\frac{3}{8}-\frac{x^2}{4}\right)\sin 2x-\frac{x}{2}\cos 2x\right]$$

(ii) To solve the D.E. $(D^2+4)y = e^{2x}[(x^2+1)\sin 2x + 4x\cos 2x]$.

Example Solve $\frac{d^2y}{dx^2}+a^2y = \sec ax$

Solution. Given equation is symbolic form is $(D^2+a^2)y = \sec ax$

(i) To find C.F.

The A.E. is $D^2+a^2=0 \quad \therefore D=\pm ai$,

Then C.F. $= c_1\cos ax + c_2\sin ax$.

(ii) To find P.I.

$$\text{P.I.} = \frac{1}{D^2+a^2}\sec ax = \frac{1}{(D+ai)(D-ai)}\sec ax \qquad \text{[Resolving into partial fractions]}$$

$$= \frac{1}{2ai}\left[\frac{1}{D-ai}-\frac{1}{D+ai}\right]\sec ax = \frac{1}{2ai}\left[\frac{1}{D-ai}\sec ax-\frac{1}{D+ai}\sec ax\right]$$

Now $\quad \frac{1}{D-ai}\sec ax = e^{aix}\int \sec ax\cdot e^{-aix}\,dx \qquad \left[\because \frac{1}{D-a}X = e^{ax}\int Xe^{-ax}\,dx\right]$

$$= e^{aix}\int\frac{\cos ax - i\sin ax}{\cos ax}\,dx = e^{aix}\int(1-i\tan ax)\,dx = e^{aix}\left(x+\frac{i}{a}\log\cos ax\right)$$

Changing i to $-i$, we have

$$\frac{1}{D+ai}\sec ax = e^{-aix}\left(x-\frac{i}{a}\log\cos ax\right)$$

Then $\quad \text{P.I.} = \frac{1}{2ai}\left[e^{aix}\left(x+\frac{i}{a}\log\cos ax\right)-e^{-aix}\left(x-\frac{i}{a}\log\cos ax\right)\right]$

$$= \frac{x}{a}\frac{e^{aix}-e^{-aix}}{2i}+\frac{1}{a^2}\log\cos ax\cdot\frac{e^{aix}+e^{-aix}}{2} = \frac{x}{a}\sin ax+\frac{1}{a^2}\log\cos ax\cdot\cos ax.$$

(iii) The required G.S. is

$$y = c_1\cos ax + c_2\sin ax + \frac{1}{a}x\sin ax + \frac{1}{a^2}\cos ax\log\cos ax.$$

EXERCISE 3.5

Solve :

1. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 4e^{2x} + 3x^2 - 1$ (U.P.T.U. 2006)
2. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = -2\sin 2x$ Also find y when $x = 0$, $\frac{dy}{dx} = 1$ at $x = 0$.
3. $\frac{d^2y}{dx^2} + n^2y = a\cos(nx + \alpha)$
4. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = e^x \tan x$
5. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 4\cos^2 x$ (Nagpur, 2002)
6. $(D^2 - 4D + 3)y = \sin 3x \cos 2x$ (Madras, 2000)
7. $\frac{d^2y}{dx^2} + a^2y = \frac{d^2y}{dx^2} + \frac{dy}{dx} = e^{ax} + \sin 2x$ (U.P.T.U., 2004)
8. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^{2x} + \cos^2 x$ (Delhi, 2000)
9. $(D^3 + 2D^2 + D)y = e^x + \sin x + x^2$ (Nagpur, 2000)
10. $\frac{d^2y}{dx^2} - y = e^x + x^2$ (Nagpur, 2000)
11. $(D^3 - 2D^2 - 5D + 6)y = e^{-x} + \sin x + 2x$ (Madras, 2000)
12. $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x} + \sin 3x + x$ (U.P.T.U., 2006)
13. $(D^2 - 1)y = (4x \sin x + 3)\sin 2x$ (Madras, 2000)
14. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = e^x \sin x + x^2$ (Nagpur, 2000)
15. $(D^2 + D^2 + 1)y = e^{-x} \cos\frac{\sqrt{3}}{2}x$ (Osmania, 2000)
16. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 5y = e^x \sin x$ (U.P.T.U., 2005)
17. $(D^3 + 4D)y = x^2 \sin x + \cos^2 x$ (Nagpur, 2000; Anna, 2001)
18. $\frac{d^2y}{dx^2} + y = x^2 e^{3x} + e^x \sin 2x$
19. $\frac{d^2y}{dx^2} + y = \cos x \cos 2x$
20. $(D^3 + 3D^2 + 3D + 1)y = e^{-x} + x^2 + 1$ (U.P.T.U., 2005)
21. $\frac{d^2y}{dx^2} + y = x\sin 2x$ (U.P.T.U., 2005)
22. $(D^2 - 2D + 1)y = e^x \sin x$ (Rajasthan, 2004)
23. $(D^2 - 1)y = x\sin x + (1 + x^2)e^x$
24. $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^{2x}$ (U.P.T.U., 2005)
25. $(D^2 + a^2)y = \sin ax$ (U.P.T.U., 2005)

3.13 TWO OTHER METHODS OF FINDING P.I.

I. Method of variation of parameters. This method is quite general and applies to equations of the form

$$y'' + Py' + Qy = X \quad \ldots(1)$$

where P, Q, and X are functions of x. If y_1 is a C.F. of (1), then P.I. $= -y_1 \int \frac{y_2 X}{W} dx + y_2 \int \frac{y_1 X}{W} dx$ $\ldots(2)$

where y_1 and y_2 are the solutions of $y'' + Py' + Qy = 0$ $\ldots(3)$

and $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is called the Wronskian[1] of y_1, y_2.

Proof. Let the C.F. of (1) be $y = c_1y_1 + c_2y_2$.

Replacing c_1, c_2 regarded as parameters by unknown functions u of x and v of x, let the P.I. be

$$y = uy_1 + vy_2 \quad \ldots(4)$$

Differentiating (4) w.r.t. x, we get $y' = uy_1' + vy_2' + y_1u' + y_2v'$

[1] Named after the Polish mathematician and philosopher Józef Wroński (1778–1853).

$$= uy_1' + vy_2' \quad \ldots(5)$$

or assuming that $uy_1 + vy_2 = 0$ $\ldots(6)$

Differentiate (6) and substitute in (5). Then noting that y_1 and y_2 satisfy (5), we obtain

$$uy_1' + vy_2' = X \quad \ldots(7)$$

Solving (6) and (7), we get

$$u' = -\frac{y_2 X}{W}, v' = \frac{y_1 X}{W} \qquad \text{where } W = y_1y_2' - y_2y_1'$$

Integrating $u = -\int \frac{y_2 X}{W} dx$, $v = \int \frac{y_1 X}{W} dx$. Substituting these in (4), we get (5).

Example 13.22. Using the method of variation of parameters, solve

$$\frac{d^2y}{dx^2} + 4y = \tan 2x \qquad \text{(V.T.U. 2008 ; Bhopal 2007; R.T.U.C. 2006 S)}$$

Solution. Given equation is symbolically form is $(D^2 + 4)y = \tan 2x$.

(1) To find C.F.

Its A.E. is $D^2 + 4 = 0$, $\therefore$ $D = \pm 2i$

Then C.F. is $y = c_1 \cos 2x + c_2 \sin 2x$.

(2) To find P.I.

Here $y_1 = \cos 2x$, $y_2 = \sin 2x$ and $X = \tan 2x$

$$\therefore \quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix} = 2$$

Then,

$$\text{P.I.} = -y_1 \int \frac{y_2 X}{W} dx + y_2 \int \frac{y_1 X}{W} dx$$

$$= -\cos 2x \int \frac{\sin 2x \tan 2x}{2} dx + \sin 2x \int \frac{\cos 2x \tan 2x}{2} dx$$

$$= -\frac{1}{2}\cos 2x \int \frac{\sin^2 2x}{\cos 2x} dx + \frac{1}{2}\sin 2x \int \sin 2x \, dx$$

$$= -\frac{1}{2}\cos 2x \left[\log|\sec 2x + \tan 2x| - \sin 2x\right] - \frac{1}{4}\sin 2x \cos 2x$$

$$= -\frac{1}{2}\cos 2x \log|\sec 2x + \tan 2x|$$

Hence the C.S. is $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{2}\cos 2x \log|\sec 2x + \tan 2x|$.

Example 13.23. Solve, by the method of variation of parameters $(D^2 - 2D + 1)y = e^x \log x$. $(x > 0)$ (V.T.U. 2009; Bhopal 2007)

Solution. Given equation is $(D^2 - 2D + 1)y = e^x \log x$

A.E. is $D^2 - 2D + 1 = 0$, $D = 1, 1$, $\therefore$ C.F. $= c_1e^x + c_2xe^x$

Here $y_1 = e^x$, $y_2 = xe^x$ and $X = e^x \log x$

$$\therefore \quad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & xe^x \\ e^x & e^x + xe^x \end{vmatrix} = e^{2x} + xe^{2x} - xe^{2x} = e^{2x}$$

Then

$$\text{P.I.} = -y_1 \int \frac{y_2 X}{W} dx + y_2 \int \frac{y_1 X}{W} dx = -e^x \int \frac{xe^x}{e^{2x}} \cdot \frac{e^x \log x}{e^{2x}} dx + xe^x \int \frac{e^x}{e^{2x}} \cdot \frac{e^x \log x}{e^{2x}} dx$$

$$= e^x \int \left(\frac{1}{x} - \frac{1}{1+x^2}\right) dx = e^x \log(1+x^2) = e^x \left[e^x \log x - \int \frac{e^x}{x^2+1} dx\right] = e^{2x} \log(1+x^2)$$

$$= e^x \left[-e^x + \log(x^2+1) - 1\right] = -e^x \log x + e^x \log(x^2+1) - e^x = -e^x \log x + e^x \log(x^2+1) - e^x$$

Hence C.S. is $y = c_1e^x + c_2xe^x - e^x \log x + e^x \log(x^2+1) - e^x \log(x^2+1)$.

Example 13.33. Solve by the method of variation of parameters $y'' + 4y' + 3y = e^{3x}$. (Nagpur, 2006; G.N.D.U., 2003)

Solution. Given equation is $(D^2 + 4D + 3)y = e^{3x}$

A.E. is $D^2 + 4D + 3 = 0$ i.e., $(D + 1)(D + 3) = 0$ $\therefore$ $D = -1, -3$

Here $y_1 = e^{-x}, y_2 = e^{-3x}$ and $X = e^{3x}$

$$\therefore \qquad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & e^{-3x} \\ -e^{-x} & -3e^{-3x} \end{vmatrix} = -2e^{-4x}$$

Thus
$$\text{P.I.} = -y_1 \int \frac{y_2 X}{W}\,dx + y_2 \int \frac{y_1 X}{W}\,dx = -e^{-x} \int \frac{e^{-3x} \cdot e^{3x}}{-2e^{-4x}}\,dx + e^{-3x} \int \frac{e^{-x} \cdot e^{3x}}{-2e^{-4x}}\,dx$$

$$= \frac{1}{2} e^{-x} \int e^{4x}\,dx - \frac{1}{2} e^{-3x} \int e^{2x}\,dx = \frac{1}{8} e^{3x} - \frac{1}{4} e^{3x} = -\frac{1}{8} e^{3x}$$

Hence G.S. is $\qquad y = c_1 e^{-x} + c_2 e^{-3x} - \frac{1}{8} e^{3x}$

Example 13.34. Solve by the method of variation of parameters $y'' - 2y' + y = e^x \log x$. (U.P.T.U., 2005; Kurukshetra, 2005; Madras, 2003)

Solution. Given equation in symbolic form is $(D^2 - 2D + 1)y = e^x \log x$

(i) To find C.F.

Its A.E. is $(D - 1)^2 = 0$, i.e., $D = 1, 1$

Thus C.F. is $y = (c_1 + c_2 x)e^x$

(ii) To find P.I.

Here $y_1 = e^x, y_2 = xe^x$ and $X = e^x \log x$

$$\therefore \qquad W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & xe^x \\ e^x & (1 + x)e^x \end{vmatrix} = e^{2x}$$

Thus
$$\text{P.I.} = -y_1 \int \frac{y_2 X}{W}\,dx + y_2 \int \frac{y_1 X}{W}\,dx$$

$$= -e^x \int \frac{xe^x \cdot e^x \log x}{e^{2x}}\,dx + xe^x \int \frac{e^x \cdot e^x \log x}{e^{2x}}\,dx = -e^x \int x \log x\,dx + xe^x \int \log x\,dx$$

$$= -e^x \left(\frac{x^2}{2} \log x - \int \frac{1}{x} \cdot \frac{x^2}{2}\,dx \right) + xe^x \left(x \log x - \int \frac{1}{x} \cdot x\,dx \right)$$

$$= -e^x \left(\frac{x^2}{2} \log x - \frac{x^2}{4} \right) + xe^x (x \log x - x) = \frac{3}{4} x^2 e^x \left(2 \log x - \frac{3}{2} \right)$$

Hence G.S. is $y = (c_1 + c_2 x)e^x + \frac{3}{4} x^2 e^x \left(2 \log x - \frac{3}{2} \right)$.

II. Method of undetermined coefficients

To find the P.I. of $f(D)y = X$, we assume a trial solution containing unknown constants which are determined by substitution in the given equation. The trial solution to be assumed in each case, depends on the form of X. Thus when (i) $X = ke^{ax}$, trial solution $= ce^{ax}$,

(ii) $X = k \sin bx$, trial solution $= c_1 \sin bx + c_2 \cos bx$

(iii) $X = kx^2$, trial solution $= c_1 x^2 + c_2 x + c_3$

However when $X = \text{sum or product}$, this method fails, since the number of terms obtained by differentiating X is two or more in number.

The above method holds so long as no term in the trial solution appears in the C.F. If any term of the trial solution appears in the C.F., we multiply the trial solution by the lowest positive integral power of x which is large enough so that none of the terms which are thus present, appear in the C.F.

Example 13.21. Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 2x^2 + e^{-x}$ (P.T.U., 2008)

Solution. Here C.F. $= e^{-x}(c_1 \cos x + c_2 \sin x)$

Assume P.I. as $y = ax^2 + bx + c_1 + kx^{-x}$

$\therefore \quad Dy = 2ax + b - ke^{-x}$ and $D^2y = 2a + ke^{-x}$

Substituting these in the given equation, we get

$$2a + 4ax + 2b - 2ke^{-x} + 2ax^2 + 2bx + 2c_1 + ke^{-x} = 2x^2 + e^{-x}$$

Equating corresponding coefficients on both sides, we get

$$2a = 2, 4a + 2b = 0, 2a + 2b + 2c_1 = 0, -k = 1$$

Then $a = 1, b = -2, c_1 = 0, k = -1$. Thus P.I. $= x^2 - 2x - e^{-x}$

$\therefore$ G.S. $y = e^{-x}(c_1 \cos x + c_2 \sin x) + x^2 - 2x - e^{-x}$.

Example 13.22. Solve $(D^2 + 1)y = \sin x \sin 2x$

Solution. Here C.F. $= c_1 \cos x + c_2 \sin x$

We would normally assume a trial solution as $c_1 \cos x + c_2 \sin x$.

However, since these terms appear in the C.F., we multiply by x and assume the trial P.I. as

$$y = x(c_1 \cos x + c_2 \sin x)$$

$\therefore \quad Dy = (c_2 + c_1 x)\cos x + (c_2 - c_1 x)\sin x$ and $D^2y = (2c_2 - c_1 x)\cos x - (2c_1 + c_2 x)\sin x$

Substituting these in the given equation, we get $2c_2 \cos x - 2c_1 \sin x = \sin x \sin 2x$

Equating corresponding coefficients,

$$2c_2 = 0, \quad -2c_1 = 1 \quad \text{so that } c_2 = 0, c_1 = -\frac{1}{2}. \quad \text{Thus P.I.} = -\frac{1}{2} x \sin x$$

$\therefore$ G.S. $y = c_1 \cos x + c_2 \sin x - \frac{1}{2} x \sin x$.

Example 13.23. Solve by the method of undetermined coefficients:

$$\frac{d^2y}{dx^2} + y = x^2 \cos 2x + e^x \sin 2x$$

Solution. Its A.E. is $D^2 + 1 = 0, \quad D = \pm i$

Thus C.F. $= c_1 e^{ix} + c_2 e^{-ix}$

Assume P.I. as $y = e^{2x}(c_1 \cos 2x + c_2 \sin 2x) + e^x(c_3 \cos 2x + c_4 \sin 2x)$

$$\therefore \quad \frac{dy}{dx} = e^{2x}[(2c_1 + 2c_2)\cos 2x + (2c_2 - 2c_1)\sin 2x] + e^x[(2c_3 + 2c_4)\cos 2x + (2c_4 - 2c_3)\sin 2x]$$

and $$\frac{d^2y}{dx^2} = e^{2x}[(2c_1 + 4c_2)\cos 2x + (2c_2 - 4c_1)\sin 2x] + e^x[(3c_3 + 4c_4)\cos 2x + (3c_4 - 4c_3)\sin 2x]$$

Substituting these in the given equation, we get

$$e^{2x}[(3c_1 + 4c_2)\cos 2x + (3c_2 - 4c_1)\sin 2x] + e^x[(3c_3 + 4c_4)\cos 2x + (3c_4 - 4c_3)\sin 2x] = x^2 \cos 2x + e^x \sin 2x$$

Equating corresponding coefficients,

$$3c_1 + 4c_2 = 1, 4c_2 - 3c_1 = 0; 3c_3 + 4c_4 = 0, 3c_4 - 4c_3 = 1$$

whence $c_1 = \frac{3}{25}, c_2 = \frac{4}{25}, c_3 = -\frac{4}{25}, c_4 = \frac{3}{25}$

Thus P.I. $= \frac{1}{25} e^{2x}(3x \cos 2x + 4x \sin 2x) + \frac{1}{25} e^x(3 \cos 2x - 4 \sin 2x)$

Hence G.S. $y = c_1 e^{ix} + c_2 e^{-ix} + \frac{1}{25} e^{2x}(3x \cos 2x + 4x \sin 2x) + \frac{1}{25} e^x(3 \cos 2x - 4 \sin 2x)$.

EXERCISE 153

Solve by the method of variation of parameters:

1. $\frac{d^2y}{dx^2} + y = \tan x$
2. $\frac{d^2y}{dx^2} + y = \sec x$ (U.P.T.U., 2003; Kanpur, 2004)
3. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = e^x \sin x$ (U.P.T.U., 2004)
4. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = \frac{1}{1+x^2}$
5. $y'' + 2y' + y = e^x \tan x$ (U.P.T.U., 2004)
6. $\frac{d^2y}{dx^2} + y = \frac{1}{1+\sin x}$
7. $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$
8. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = e^x \log x$ (U.P.T.U., 2003)
9. $\frac{d^2y}{dx^2} + y = \sec x \tan x$ (U.P.T.U., 2004)
10. $\frac{d^2y}{dx^2} + y = \tan x$ (U.P.T.U., 2004)
11. $\frac{d^2y}{dx^2} + y = \sec^3 x$ (U.P.T.U., 2005)
12. $\frac{d^2y}{dx^2} + 4y = \tan 2x$ (U.P.T.U., 2004)
13. $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x \cot x$ (U.P.T.U., 2005)
14. $(D^2 - 2D + 1)y = e^x \log x$ (U.P.T.U., 2003)
15. $(D^2 - 2D + 2)y = e^x \tan x$ (U.P.T.U., 2004)

15.8 EQUATIONS REDUCIBLE TO LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

Now we shall study two such forms of linear differential equations with variable coefficients which can be reduced to linear differential equations with constant coefficients by suitable substitutions.

1. Cauchy's homogeneous linear equation[6]. An equation of the form

$$x^n\frac{d^ny}{dx^n} + A_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \ldots + A_{n-1}x\frac{dy}{dx} + A_ny = X \qquad \ldots(1)$$

where X is a function of x, is called Cauchy's homogeneous linear equation.

Such equations can be reduced to linear differential equations with constant coefficients by putting

$$x = e^z \quad \text{or} \quad z = \log x. \text{ Then } xD = \frac{d}{dz}$$

$$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{dz}{dx} = \frac{dy}{dz}\cdot\frac{1}{x}, \text{ i.e., } x\frac{dy}{dx} = D_1y$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{1}{x}\frac{dy}{dz}\right] = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d}{dz}\left(\frac{dy}{dz}\right)\frac{dz}{dx} = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d^2y}{dz^2}\frac{dz}{dx} = \frac{1}{x^2}\left(\frac{d^2y}{dz^2} - \frac{dy}{dz}\right)$$

i.e., $x^2\frac{d^2y}{dx^2} = D(D-1)y$. Similarly, $x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y$ and so on.

After making these substitutions in (1), there results a linear equation with constant coefficients, which can be solved as before.

Example 15.30. Solve $x^2\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - y = 2\log x$. (U.P.T.U., 2003)

Solution. This is a Cauchy's homogeneous linear.

[6] See footnote p. 144.

Put $x = e^z$, i.e., $z = \log x$, so that $x\frac{dy}{dx} = D_1 y$, $x^2\frac{d^2y}{dx^2} = D(D-1)y$ where $D \equiv \frac{d}{dz}$

Then the given equation becomes $[D(D-1) - D + 1]y = e^z$ or $(D-1)^2 y = e^z$ ...(i)

which is a linear equation with constant coefficients.

Its A.E. is $(D-1)^2 = 0$ i.e. $D = 1, 1$.

$\therefore$ $C.F. = (c_1 + c_2 z)e^z$ and $P.I. = \frac{1}{(D-1)^2} e^z = (1-D)^{-2} 1 = [1 + 2D + 3D^2 + \ldots]1 = 1 + 2z$.

Hence the solution of (i) is $y = (c_1 + c_2 z)e^z + 1 + 2z$, or, putting $z = \log x$ and $e^z = x$, we get

$y = (c_1 + c_2 \log x)x + \log x + 2$ as the required solution of (i).

Example 18.31. *Solve* $x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = \frac{1}{1-x^2}$. *(U.P.T.U. 2002)*

Solution. Put $x = e^z$ i.e., $z = \log x$ so that $x\frac{dy}{dx} = D_1 y$, $x^2\frac{d^2y}{dx^2} = D(D-1)y$

Then the given equation becomes

$$[D(D-1) + 3D + 1]y = \frac{1}{(1-e^z)^2} \quad \text{or} \quad (D+1)^2 y = \frac{1}{(1-e^z)^2}$$

Its A.E. is $D^2 + 2D + 1 = 0$ or $(D+1)^2 = 0$ i.e., $D = -1, -1$.

$\therefore$ $C.F. = (c_1 + c_2 z)e^{-z} = (c_1 + c_2 \log x)\frac{1}{x}$

$$P.I. = \frac{1}{(D+1)^2}\frac{1}{(1-e^z)^2} = \frac{1}{(D+1)^2} u, \text{ where } u = \frac{1}{(1+e^z)}\cdot\frac{1}{(1-e^z)} \text{ i.e. } \frac{du}{dz} + u = (1-e^z)^{-1}$$

which is Bernoulli's linear equation having $I.F. = e^z$

$\therefore$ its solution is $ue^z = \int \frac{e^z}{(1-e^z)^2} dz = \frac{1}{1-e^z}$ or $u = \frac{e^{-z}}{1-e^{-z}}$

$\therefore$ $P.I. = \frac{1}{(D+1)^2}\left(\frac{e^{-z}}{1-e^{-z}}\right) = e^{-z}\int\frac{1}{1-e^{-z}} dz = \frac{1}{x}\int\frac{dx}{x(1-x)}$

$$= \frac{1}{x}\int\left[\frac{1}{x} + \frac{1}{1-x}\right]dx = \frac{1}{x}[\log x - \log(1-x)] = \frac{1}{x}\log\frac{x}{x-1}$$

Hence the solution is $y = \left[c_1 + c_2 \log x + \log\frac{x}{x-1}\right]\frac{1}{x}$.

Example 18.32. *Solve* $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = \log x \sin(\log x)$.

(Kanpur Univ. 2006; Meerut, 2008 (Old); Rohilkhand, 2005)

Solution. Putting $x = e^z$ i.e., $z = \log x$, the given equation becomes

$$[D(D-1) + D + 1]y = z \sin z \quad \text{i.e.,} \quad (D^2+1)y = z\sin z \quad ...(i)$$

Its A.E. is $D^2 + 1 = 0$ i.e., $D = \pm i$.

$\therefore$ $C.F. = c_1 \cos z + c_2 \sin z$

and $P.I. = \frac{1}{D^2+1} z\sin z = \frac{1}{D^2+2}[1 + D^2/(D^2+1)]z\sin z$

$$= I.P. \text{ of } e^{iz}\frac{1}{(D+i)^2+1} z = I.P. \text{ of } e^{iz}\cdot\frac{1}{D^2+2iD}z$$

$$= \frac{1}{61}(8\sin t + \cos t) + \frac{2}{3721}(27\sin t + 100\cos t)$$

$$\therefore \quad P.I. = e^{t}\left[\frac{1}{6}(1+2t) + \frac{1}{61}(15\sin t + 8\cos t) + \frac{2}{3721}(27\sin t + 100\cos t)\right]$$

Hence $$y = e^{3t}(c_1 e^{2t} + c_2 e^{-4t}) + e^{t}\left[\frac{1}{6}(1+2t) + \frac{1}{61}(15\sin t + 8\cos t) + \frac{2}{3721}(27\sin t + 100\cos t)\right]$$

or $$y = x^{3}(c_1 x^{2} + c_2 x^{-4}) + \frac{1}{x}\left[\frac{1}{6}(1+2\log x) + \frac{1}{61}(15\sin(\log x) + 8\cos(\log x)) + \frac{2}{3721}(27\sin(\log x) + 100\cos(\log x))\right].$$

Example 12.11. Solve $\dfrac{d^2y}{dx^2} + 4x\dfrac{dy}{dx} + 2y = e^{x^2}$ (Kurukshetra, 2008 ; U.P.T.U., 2009)

Solution. Putting $x = e^t$, i.e., $t = \log x$, the given equation becomes

$$[\theta(\theta-1) + 4\theta + 2]y = e^{e^t} \text{ i.e., } (\theta^2 + 3\theta + 2)y = e^{e^t}$$

Its A.E. is $D^2 + 3D + 2 = 0$ whose roots are $D = -1, -2$.

$\therefore$ $C.F. = c_1 e^{-t} + c_2 e^{-2t} = c_1 x^{-1} + c_2 x^{-2}$.

and $$P.I. = \frac{1}{(D^2+3D+2)}e^{e^t} = \frac{1}{(D+1)(D+2)}e^{e^t} = \left(\frac{1}{D+1} - \frac{1}{D+2}\right)e^{e^t}$$

Now $$\frac{1}{D+1}e^{e^t} = \frac{1}{D+1}e^{x}, dx^0 = e^{-t}\frac{1}{(D-1)+1}dx^0$$

$$= e^{-t}\frac{1}{D}dx^0 = e^{-t}\int e^{x}dx^0 = e^{-t}\int e^{x}dx = e^{-t}e^{x}$$

$$\frac{1}{D+2}e^{e^t} = \frac{1}{D+2}e^{x}, e^{0x} = e^{-2t}\frac{1}{(D-2)+2}e^{0x}$$

$$= e^{-2t}\frac{1}{D}e^{0x} = e^{-2t}\int e^{0x}dx$$

$$= e^{-2t}\int e^{x}\,dx \qquad \text{Put } e^{x} = z$$

$$= e^{-2t}(e^{x} - e^{1}) \qquad \text{Integrating by parts}$$

$\therefore$ $P.I. = e^{-t}e^{x} - e^{-2t}(e^{x} - e^{1}) = e^{-t}e^{e^t}$

Hence the required solution is $y = c_1 x^{-1} + c_2 x^{-2} + e^{-t}e^{e^t}$.

II. Legendre's linear equation[1]. An equation of the form

$$(ax+b)^n\frac{d^n y}{dx^n} + k_1(ax+b)^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \ldots + k_n y = X \qquad \ldots(1)$$

where k's are constants and X is a function of x, is called Legendre's linear equation.

Such equations can be reduced to linear equations with constant coefficients by the substitution $ax + b = e^t$, i.e., $t = \log(ax + b)$.

Then, if $$D \equiv \frac{d}{dx}, \frac{dx}{dx} = \frac{dy}{dt}\frac{dt}{dx} = \frac{a}{ax+b}\frac{dy}{dt} \text{ i.e., } (ax+b)\frac{dy}{dx} = aDy$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{a}{ax+b}\frac{dy}{dt}\right) = \frac{-a^2}{(ax+b)^2}\frac{dy}{dt} + \frac{a}{ax+b}\frac{d}{dt}\left(\frac{dy}{dt}\right)\frac{dt}{dx} = \frac{a^2}{(ax+b)^2}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right)$$

---

[1] A French mathematician Adrien Marie Legendre (1752 – 1833) who made important contributions to number theory, special functions, calculus of variations and elliptic integrals.

i.e., $(ax+b)^2\frac{d^2y}{dx^2} = a^2D(D-1)y$. Similarly, $(ax+b)^3\frac{d^3y}{dx^3} = a^3D(D-1)(D-2)y$ and so on.

After making these replacements in (i), there results a linear equation with constant coefficients.

Example 4.33. Solve $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 2\sin[\log(1+x)]$ (i)

(U.P.T.U., 2003; J.N.T.U., 2003; Kerala, 2006)

Solution. This is a Legendre's linear equation.

$\therefore$ put $1+x = e^z$, i.e., $z = \log(1+x)$, so that $(1+x)\frac{dz}{dx} = 1$

and $(1+x)^2\frac{d^2y}{dx^2} = D(D-1)y$, where $D = \frac{d}{dz}$

Then (i) becomes $D(D-1)y + Dy + y = 2\sin z$

or $(D^2+1)y = 2\sin z$ ... (ii)

This is a linear equation with constant coefficients.

Its A.E. is $D^2+1=0$, whence $D = \pm i$ $\therefore$ C.F. $= c_1\cos z + c_2\sin z$

and $P.I. = \frac{1}{D^2+1}2\sin z = 2z\cdot\frac{1}{2D}\sin z$

$= z\int\sin z\,dz = -z\cos z$ [$\because$ on replacing D^2 by -1^2, $D^2+1=0$]

Hence the solution of (ii) is $y = c_1\cos z + c_2\sin z - z\cos z$ and on replacing z by $\log(1+x)$, we get $y = c_1\cos[\log(1+x)] + c_2\sin[\log(1+x)] - \log(1+x)\cos[\log(1+x)]$ as the required solution.

Example 4.34. Solve $(2x-1)^2\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} - 2y = 8x^2 - 2x + 2$ (U.P.T.U., 2004)

Solution. This is a Legendre's linear equation.

$\therefore$ put $2x-1 = e^z$ i.e., $z = \log(2x-1)$ so that $(2x-1)\frac{dz}{dx} = 2$

and $(2x-1)^2\frac{d^2y}{dx^2} = 4D(D-1)y$, where $D = \frac{d}{dz}$

Then the given equation becomes

$$4D(D-1)y + 2Dy - 2y = 8\left[\frac{1+e^z}{2}\right]^2 - 2\left[\frac{1+e^z}{2}\right] + 2$$

or $4D^2y - 2y = e^{2z} + \frac{3}{2}e^z + \frac{1}{2}$ ... (i)

This is a linear equation with constant coefficients.

Its A.E. is $4D^2 - 2 = 0$ whence $D = \pm\frac{1}{\sqrt{2}}$

$\therefore$ C.F. $= c_1e^{z/\sqrt{2}} + c_2e^{-z/\sqrt{2}}$

and $P.I. = \frac{1}{4D^2-2}\left[e^{2z} + \frac{3}{2}e^z + \frac{1}{2}\right] = \frac{1}{4\cdot4-2}e^{2z} + \frac{3}{2}\frac{1}{4\cdot1-2}e^z + \frac{1}{2}\cdot\frac{1}{4\cdot0^2-2}e^{0z}$

[$\because$ on putting $z = 2, 1, 0$ in $4D^2 - 2$]

$= \frac{1}{6}e^{2z} + \frac{3}{2}\cdot\frac{1}{2}e^z - \frac{1}{4} = \frac{1}{6}e^{2z} + \frac{3}{4}e^z - \frac{1}{4}$

Hence the solution of (i) is

$y = c_1e^{z/\sqrt{2}} + c_2e^{-z/\sqrt{2}} + \frac{1}{6}e^{2z} + \frac{3}{4}e^z - \frac{1}{4}$ and on replacing z by $\log(2x-1)$,

$$y = c_1(2x-1) + c_2(2x-1)\log(2x-1) + \frac{1}{2}(2x-1)^2 + \frac{1}{2}(2x-1)\log(2x-1) + x,$$

which is the required solution.

EXERCISE 12.6

Solve:

1. $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = x^2$
2. $x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 4y = x^4$
3. $x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 4y = (1+x)^2$ (B.Tech, 2007)
4. $x^2\frac{d^2y}{dx^2} - \frac{3}{x}\frac{dy}{dx} + \frac{4}{x^2}y = \frac{1}{x^2}$ (B.Tech, 2009)
5. The radial displacement u in a rotating disc is a distance r from the axis is given by $r^2\frac{d^2u}{dr^2} + r\frac{du}{dr} - u = -\rho\omega^2 r^3$, where ρ is density and ω is a constant. Solve this equation for the conditions $u = 0$ when $r = 0$, $u = 0$ when $r = a$.

Solve:

6. $x^2\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = \log x$ (Nagpur, 2008)
7. $x^3\frac{d^3y}{dx^3} + 3x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = x + \log x$ (Nagpur, 2009)
8. $x^2y'' + xy' + y = 2\sin^2(\log x)$ (B.Tech, 2010)
9. $(x+2)^2\frac{d^2y}{dx^2} + 3(x+2)\frac{dy}{dx} + y = 3x + 4$ (B.Tech, 2006; P.T.U., 2008)
10. $x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = \frac{1}{(1-x)^2}$ (U.P.T.U., 2006)
11. $x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} + 4y = x\log x$ (U.P.T.U., 2008)
12. $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} - 9y = x^2\log x$ (Nagpur, 2009)
13. $(3x+2)^2\frac{d^2y}{dx^2} - 3(3x+2)\frac{dy}{dx} + 4y = 6x$ (U.P.T.U., 2006; Kerala, 2006; Anna, 2009)
14. $(x-1)^2\frac{d^2y}{dx^2} + 3(x-1)\frac{dy}{dx} + y = (x-1)^2 + \log(x-1)$ (Nagpur, 2009)
15. $(1+x)^2\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = \sin[2\log(1+x)]$ (U.P.T.U., 2006; U.P.T.U., 2008)
16. $(2x+3)^2\frac{d^2y}{dx^2} + 3(2x+3)\frac{dy}{dx} - 12y = 6x^2 + x + 1$ (Kerala, 2006)

12.8 (1) LINEAR DEPENDENCE OF SOLUTIONS

Consider the initial value problem consisting of the homogeneous linear equation

$$y'' + py' + qy = 0 \quad \ldots(1)$$

with variable coefficients p, q and the initial conditions $y(x_0) = k_0, y'(x_0) = k_1$ $\ldots(2)$

Let the general solution be $y = c_1y_1 + c_2y_2$ $\ldots(3)$

which is made up of two linearly dependent solutions y_1 and y_2.*

If p and q are continuous functions on some open interval I and x_0 is any fixed point on I, then the above initial value problem has a unique solution $y(x)$ on the interval I.

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* As in §12.1, y_1, y_2 are said to be linearly dependent on an interval I, if and only if there exist constants k_1, k_2 not both zero such that $k_1y_1 + k_2y_2 = 0$ for all $x \in I$.

If no such constants exist, i.e., if $k_1y_1 + k_2y_2 = 0$ only when $k_1 = k_2 = 0$, then y_1, y_2 are said to be linearly independent.

(c) Theorem. *If $y_1(x)$ and $y_2(x)$ are continuous functions of x in $[a, b]$ and if $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0$ for some x in $[a, b]$, then the solutions y_1 and y_2 of (1) are linearly independent in $[a, b]$ and conversely if the solutions y_1 and y_2 of (1) are linearly dependent in $[a, b]$ then $W(y_1, y_2) = 0$ for every x in $[a, b]$.*

Proof. If y_1, y_2 are linearly dependent in $[a, b]$ then there exist two constants c_1, c_2 not both zero such that

$$c_1 y_1 + c_2 y_2 = 0 \quad \ldots(1)$$

Differentiating w.r.t. x, $c_1 y_1' + c_2 y_2' = 0$ $\ldots(2)$

Eliminating c_1, c_2 from (1) and (2), we get

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = 0$$

Conversely, suppose $W(y_1, y_2) = 0$ for some $x = x_0$ in $[a, b]$. Then y_1, y_2 are linearly dependent.

Consider the equations

$$\left.\begin{aligned} c_1 y_1(x_0) + c_2 y_2(x_0) &= 0 \\ c_1 y_1'(x_0) + c_2 y_2'(x_0) &= 0 \end{aligned}\right\} \quad \ldots(3)$$

which, on eliminating c_1, c_2 gives $W(y_1, y_2) = \begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = 0$

Hence the system has a solution in which c_1, c_2 are not both zero.

Now consider the function $Y(x) = c_1 y_1(x) + c_2 y_2(x)$

This is a solution of (1) as y_1, y_2 are. By (3), this solution satisfies the initial conditions $y(x_0) = 0$ and $y'(x_0) = 0$. Also since $y = 0$ and $y(x)$ are solutions on $[a, b]$, then by the uniqueness theorem, $y = 0$ is the only possible solution of (1) satisfying the given initial conditions. Hence $Y = c_1 y_1 + c_2 y_2 = 0$ in $[a, b]$. Now since c_1, c_2 are not both zero, it follows that y_1 and y_2 are linearly dependent on $[a, b]$.

Example 13.37. *Show that the functions $e^{ax}\sin bx$ and $e^{ax}\cos bx$ are independent solutions of $y'' + 2ay' + (a^2 + b^2)y = 0$.*

Solution. Substituting $y_1 = e^{ax}\sin bx$ for y and $y_1' = e^{ax}(a\sin bx + b\cos bx)$, $y_1'' = e^{ax}(2ab\cos bx + (a^2 - b^2)\sin bx)$ in the given equation, we find that y_1, y_2 are its solutions.

Also $$W(y_1, y_2) = \begin{vmatrix} e^{ax}\sin bx & e^{ax}\cos bx \\ e^{ax}(a\sin bx + b\cos bx) & e^{ax}(a\cos bx - b\sin bx) \end{vmatrix} = -be^{2ax} \neq 0$$

for any value of a. Hence the solutions y_1, y_2 are linearly independent.

EXERCISE 13.5

Solve :

1. Show that $e^x(\cos x, \sin x)$ are independent solutions of $y'' - 2y' + y = 0$ for any interval.
2. Show that $e^{-x}\cos x, e^{-x}\sin x$ are independent solutions of the equation $y'' + 2y' + 2y = 0$.
3. If y_1, y_2 are two solutions of $y'' + p(x)y' + q(x)y = 0$, show that the Wronskian can be represented as $W(y_1, y_2) = ce^{-\int p\,dx}$. [M.U. 2005]

13.13 SIMULTANEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

Quite often we come across linear differential equations in which there are two or more dependent variables and a single independent variable. Such equations are known as simultaneous linear equations. Here we shall deal with systems of linear equations with constant coefficients only. Such a system of equations is solved by eliminating all but one of the dependent variables and then solving the resulting equations as before. Each of the dependent variables is obtained in a similar manner.

Example 13.38. *Solve the simultaneous equations*[1]

$$\frac{dx}{dt} + 5x - 2y = t, \quad \frac{dy}{dt} + 2x + y = 0$$

being given $x = y = 0$ when $t = 0$. [U.P.T.U. 2004; Kurukshetra, 2002]

[1] See footnote on p. 466.

Solution. Taking $d/dt = D$, the given equations become

$$(D^2 + 3)x - 2y = t \quad \ldots(i)$$

$$2x + (D + 1)y = 0 \quad \ldots(ii)$$

Eliminate x as if D were an ordinary algebraic multiplier. Multiplying (i) by 2 and operating on (ii) by $(D^2 + 3)$ and then subtracting, we get

$$[-4 - (D^2 + 3)(D + 1)]y = 2t \text{ or } (D^3 + D^2 + 5D + 7)y = -2t$$

Its auxiliary equation is $D^3 + D^2 + 5D + 7 = 0$, i.e., $(D + 1)(D^2 + 2^2) = 0$

Hence $D = -1, -2i \quad \therefore \quad C.F. = c_1 e^{-t} + c_2 \cos 2t + c_3 \sin 2t$

and $$P.I. = \frac{1}{(D+1)(D^2+2^2)}(-2t) = -\frac{2}{5}\left[1 + \frac{D^3 + D^2}{5}\right]^{-1} t = -\frac{2}{5}\left[1 - \frac{5D}{5} + \ldots\right]t = -\frac{2t}{5} + \frac{4}{25}$$

Hence $$y = c_1 e^{-t} + c_2 \cos 2t + c_3 \sin 2t - \frac{2t}{5} + \frac{4}{25} \quad \ldots(iii)$$

Note. To find x, either eliminate y from (i) and (ii) and solve the resulting equation or substitute the value of y in (ii). Here, it is more convenient to adopt the latter method.

From (ii), $\quad Dy = c_1 e^{-t} - 2c_2 \sin 2t + 2c_3 \cos 2t - \frac{2}{5}$

$\therefore$ Substituting for y and Dy in (ii), we get

$$x = -\frac{1}{2}(Dy + y) = \left[\left(c_1 - \frac{1}{5}c_2\right) + c_3 t\right]e^{-t} + \frac{1}{10} + \frac{1}{25} \quad \ldots(iv)$$

Equations (iii) and (iv) constitute the solutions of the given equations.

Since $x = y = 0$ when $t = 0$, the equations (iii) and (iv) give

$$0 = c_1 + \frac{4}{25} \text{ and } 0 = c_1 - \frac{1}{5}c_2 + \frac{1}{25} = 0 \text{ whence } c_1 = -\frac{4}{25}, c_2 = -\frac{3}{5}.$$

Hence the desired solutions are

$$x = -\frac{2}{25}(1 + 6t)e^{-t} + \frac{1}{25}(3 + 10t), y = -\frac{2}{25}(2 + 5t)e^{-t} + \frac{2}{25}(2 - 5t).$$

Example 13.30. Solve the simultaneous equations $\frac{dx}{dt} + 2x + y\sin t + z\frac{dy}{dt} = 0$, $2x - y\cos t + z = 0$ given that $x = 0, y = 2$ when $t = \frac{1}{2}\pi$ and $z = 1$.

Solution. Given equations are

$$Dx + 2y = -\sin t \quad \ldots(i), \qquad -2x + Dy + z = \cos t \quad \ldots(ii)$$

Eliminating x by multiplying (i) by 2 and (ii) by D and then adding, we get

$$4y + (D^2 + 1)y + Dz = -2\sin t - \sin t \cos t \text{ or } (D^2 + 1)y = -3\sin t$$

Its A.E. is $D = \pm i \quad \therefore \quad C.F. = c_1 \cos t + c_2 \sin t$

$$P.I. = -3\frac{1}{D^2 + 1}\sin t = -3\frac{1}{-1^2 + 1}\sin t = -\frac{3}{2}t\cos t$$

$$\therefore \quad y = c_1 \cos t + c_2 \sin t - \frac{3}{2}t\cos t \quad \ldots(iii)$$

and $$Dy/dt = -c_1 \sin t + c_2 \cos t - \frac{3}{2}\cos t + \frac{3}{2}t\sin t \quad \ldots(iv)$$

Substituting (iii) and (iv) in (ii), we get

$$2x + Dy - \cos t = -2c_1 \sin t + 2c_2 \cos t - 3\cos t + \frac{3}{2}t\sin t$$

or $$x = -c_2 \sin t + c_1 \cos t + \frac{3}{4}t\sin t \quad \ldots(v)$$

When $t = 0, x = 0, y = 2$, (iii) and (v) give $2 = c_1, 0 = c_2 - 1$.

Hence $x = \cos t - \sin t - \frac{3}{4}t\cos t, y = \cos t + \sin t - \frac{3}{2}t\cos t$

Example 13.31. Solve the simultaneous equations

$$\frac{dx}{dt} + \frac{dy}{dt} + 2y = \sin t, 2\frac{dx}{dt} + \frac{d^2y}{dt^2} + 2x = \sin t \quad \text{(U.P.T.U., 2002)}$$

Solution. Given equations are

$$Dx + (D-2)y = 2\cos t - 7\sin t \quad \ldots(i)$$

$$(D+2)x - 2y = 4\cos t - 5\sin t \quad \ldots(ii)$$

Operating (i) by $(D+2)$ and (ii) by $(D-2)$ and then adding, we get

$$D^2x + 4D - 2(D-2)y = 2(D+2)\cos t - 7(D+2)\sin t + 4(D-2)\cos t - 5(D-2)\sin t$$

or $$2(D^2 - 2)x = -15\cos t - 19\sin t \quad \text{or} \quad (D^2 - 2)x = -9\cos t - 2\sin t$$

Its A.E. is $D^2 - 2 = 0$ or $D = \pm\sqrt{2}$. $\therefore$ C.F. $= c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t}$

$$\text{P.I.} = \frac{1}{D^2-2}(-9\cos t - 2\sin t) = \frac{-9\cos t}{-1-2} + \frac{2\sin t}{-1-2} = 3\cos t + \frac{2}{3}\sin t$$

Hence $x = c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t} + 3\cos t + \frac{2}{3}\sin t$.

Now substituting the value of x in (i), we get

$$2y = (D-2)x - 2\cos t + 4\sin t + 7\cos t$$

$$= c_1\sqrt{2}e^{\sqrt{2}t} + 2c_1 e^{\sqrt{2}t} + c_2(-\sqrt{2}e^{-\sqrt{2}t}) + 2c_2 e^{-\sqrt{2}t} - 3\sin t + \frac{2}{3}\cos t - 4\cos t + 8\sin t + 5\cos t - 4\sin t + \frac{2}{3}\sin t$$

$$= (2+\sqrt{2})c_1 e^{\sqrt{2}t} + (2-\sqrt{2})c_2 e^{-\sqrt{2}t} + 2\cos t + 5\sin t$$

Hence $y = \frac{1}{2}(2+\sqrt{2})c_1 e^{\sqrt{2}t} + \frac{1}{2}(2-\sqrt{2})c_2 e^{-\sqrt{2}t} + \cos t + \frac{5}{2}\sin t$.

Example 13.11. *The oscillations of a certain system with two degrees of freedom are given by the equations*

$$D^2x + 3x - 2y = 0$$

$$D^2y + 2y - 2x + 3y = 0$$

where $D \equiv d/dt$. If $x = y = 0$, $Dx = 2$, $Dy = 3$ when $t = 0$, find x and y when $t = \pi/2$.

Solution. Given equations are $(D^2 + 3)x - 2y = 0$ $\ldots(i)$

$$-2x + (D^2 + 2)y = 0 \quad \ldots(ii)$$

To eliminate x, operate these equations by $D^2 + 2$ and 2 respectively and subtract (i) from (ii). Then

$$(D^2+2)(D^2+3)x + 4x = 2(D^2+2) \quad \text{or} \quad (D^4 + 5D^2 + 2)x = 2$$

Its auxiliary equation is $D^4 + 5D^2 + 2 = 0$ whence $D = \pm i, \pm\sqrt{2}i$

Thus $$x = c_1\cos t + c_2\sin t + c_3\cos\sqrt{2}t + c_4\sin\sqrt{2}t \quad \ldots(iii)$$

To find y, we eliminate y from (i) and (ii).

Operating (i) by $D^2 + 2$ and multiplying (ii) by 2 and adding, we get

$$(D^2 + 1)(D^2 + 4)x = 0. \text{ Thus } x = k_1\cos t + k_2\sin t + k_3\cos 2t + k_4\sin 2t \quad \ldots(iv)$$

To find the relations between the constants in (iii) and (iv), substitute these values of x and y in either of the given equations, say (i). This gives

$$2k_1 - c_1)\cos t + 2(k_2 - c_2)\sin t - 2(3k_3 + c_3)\cos\sqrt{2}t - 2(3k_4 + c_4)\sin\sqrt{2}t = 0$$

which must hold for all values of t.

$\therefore$ Equating to zero the coefficients of $\cos t$, $\sin t$, $\cos\sqrt{2}t$ and $\sin\sqrt{2}t$, we get

$$k_1 = c_1,\ k_2 = c_2,\ 3k_3 = -c_3,\ 3k_4 = -c_4$$

Thus $$x = c_1\cos t + c_2\sin t - \frac{1}{3}c_3\cos\sqrt{2}t + c_4\sin\sqrt{2}t \quad \ldots(v)$$

Now (iii) and (v) must be the solutions of (i) and (ii).

Since $x = y = 0$, when $t = 0$, (iii) and (v) give

$$0 = c_1 + c_3 \text{ and } c_2 - \frac{1}{\sqrt{2}}c_4 = 0 \text{ i.e. } c_1 = -c_3,\ c_2 = \frac{1}{\sqrt{2}}c_4$$

Then (iii) and (v) reduce to

$$\left.\begin{aligned} x &= c_2\cos t + c_4\sin\sqrt{2}t \\ y &= c_2\sin t - \frac{\sqrt{2}}{3}c_4\sin\sqrt{2}t \end{aligned}\right\} \quad \ldots(vi)$$

and

$\therefore$ $Dx = c_1 \cos t - c_2 \sin t$ and $Dy = c_2 \cos t + 2c_1 \sin t$.

Since $Dx = 2$ and $Dy = 2$ when $t = 0$

$\therefore$ $2 = c_1 - c_2$ and $2 = c_2 + 2c_1$ whence $c_2 = 1/5$, $c_1 = \frac{1}{5}$.

Hence equations (i) become $x = \frac{1}{5}(11\cos t + \frac{1}{2}\sin 2t)$, $y = \frac{1}{5}(11\sin t - \sin 2t)$ ...(iii)

$\therefore$ when $t = 32^\circ$, $x = \frac{1}{5}\left[11\sin(0.3) + \frac{1}{2}\sin(1.6)\right] = \frac{1}{5}\left[11(0.4794) + \frac{1}{2}(0.9996)\right] = 1.4622$

and $y = \frac{1}{5}[11\sin(0.3) - \sin(1.6)] = 1.080$.

Example 12.42. Solve the simultaneous equations $\frac{dx}{dt} + 2y = \frac{2t}{dt}$, $2x + \frac{dy}{dt} = 4t$ (U.P.T.U., 2006)

Solution. Differentiating first equation w.r.t. t, $\frac{d^2x}{dt^2} + 2\frac{dy}{dt} = 2(2x)$

Again differentiating equation (i), $\frac{d^2x}{dt^2} + 4\frac{dy}{dt} = 4\left(\frac{dx}{dt}\right)$ ...(ii)

or $(D^2 - 4D)x = 0$

By A.E. is $D^2 - 4D = 0$ or $D(D^2 + 2D + 1) = 0$

or $D = 0, -1 \pm i\sqrt{3}$

$\therefore$ the solution of (i) is $x = c_1 e^{0t} + e^{-t}(c_2 \cos\sqrt{3}t + c_3 \sin\sqrt{3}t)$ ...(iii)

From the first equation, we have $y = \frac{1}{2}\frac{dx}{dt}$

$\therefore$ $y = \frac{1}{2}[2c_1 e^{0t} + (-1)e^{-t}(c_2 \cos\sqrt{3}t + c_3 \sin\sqrt{3}t) + e^{-t}(-\sqrt{3}c_2 \sin\sqrt{3}t + \sqrt{3}c_3 \cos\sqrt{3}t)]$

or $y = c_1 e^{0t} + \frac{1}{2}e^{-t}[(\sqrt{3}c_3 - c_2)\cos\sqrt{3}t - (c_3 + \sqrt{3}c_2)\sin\sqrt{3}t]$ ...(iv)

From the second equation, we have $x = \frac{1}{2}\frac{dy}{dt}$

$\therefore$ $x = \frac{1}{2}2c_1 e^{0t} + \frac{1}{4}[(-1)e^{-t}\{(\sqrt{3}c_3 - c_2)\cos\sqrt{3}t - (c_3 + \sqrt{3}c_2)\sin\sqrt{3}t\}$

$+ e^{-t}\{-\sqrt{3}(c_3 - \sqrt{3}c_2)\sin\sqrt{3}t - \sqrt{3}(c_3 + \sqrt{3}c_2)\cos\sqrt{3}t\}]$

$= c_1 e^{0t} + \frac{1}{4}e^{-t}\{-2c_3 - 2\sqrt{3}c_2\cos\sqrt{3}t + 6c_2 - 2\sqrt{3}c_3\sin\sqrt{3}t\}$

or $x = c_1 e^{0t} - \frac{1}{2}e^{-t}\{(c_2 - \sqrt{3}c_3)\sin\sqrt{3}t + (c_3 + \sqrt{3}c_2)\cos\sqrt{3}t\}$ ...(v)

Hence the equations (iii), (iv) and (v) taken together give the required solution.

PROBLEMS 12.8

Solve the following simultaneous equations :

1. $\frac{dx}{dt} + 2x + y\frac{dy}{dt} + y = 1$, $x + \frac{dy}{dt} + y = 0$, given that $x = 0$ and $y = 0$ when $t = 0$.

2. $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that $x = 2$ and $y = 0$ when $t = 0$.

(Bhopal, 2010; U.P.T.U., 2008) (Answers, page)

[illegible]

OBJECTIVE TYPE OF QUESTIONS

MULTIPLE CHOICE

[illegible]

30. P.I. of $(D^2+4)y=\sin 2x$ is

(a) $\frac{1}{2}x\sin 2x$ (b) $\frac{1}{2}x\cos 2x$ (c) $\frac{1}{4}x\sin 2x$ (d) $\frac{1}{4}x\cos 2x$ (Bharati 2004)

31. By the method of undetermined coefficients, $A_1y'+A_2y+A_3y=12x^2$ is of the form

(a) ax^2+bx+c (b) $x(ax^2+bx+c)$ (c) $x^2(ax^2+bx+c)$ (d) None of these (U.P.T.U. 2006)

32. In the equation $\frac{d^2y}{dx^2}+p_1\frac{dy}{dx}+p_2y=\sin x$, p_1 and p_2 are ..... (Delhi 2004)

33. $a_0y^n+a_1y_{n-1}+a_2y_{n-2}+\ldots+a_ny=0$ is a homogeneous linear differential equation with constant coefficients if ..... .

34. P.I. of $\frac{d^2y}{dx^2}+\frac{dy}{dx}=x^2+e^{-x}+4$ is

(a) $\frac{x^3}{3}+4x$ (b) $\frac{x^3}{3}+4$ (c) $\frac{x^3}{3}+4x^2$ (d) $\frac{x^3}{3}-4x^2$

35. The complete solution of the differential equation $\frac{d^2y}{dx^2}-5\frac{dy}{dx}+6y=e^{2x}$ is given by

(a) $y=C_1e^{-2x}+C_2e^{3x}-\frac{1}{2}e^{2x}$ (b) $y=C_1e^{-3x}+C_2e^{2x}+\frac{1}{2}e^{2x}$

(c) $y=C_1e^{-2x}+C_2e^{3x}+\frac{1}{2}e^{2x}$ (d) $y=C_1e^{-x}+C_2e^{3x}+\frac{1}{2}e^{2x}$

36. The particular integral of the differential equation $(D^2-1)y=e^x+e^{-x}+1$ is $\frac{x}{2}$

(a) $\frac{1}{2}(e^x+e^{-x})$ (b) $\frac{1}{2}(e^x+e^{-x}+1)$ (c) $\frac{1}{2}(e^x-e^{-x}+1)$ (d) $\frac{1}{2}(e^x-e^{-x})$

37. The complementary function of the differential equation $(D^2-2D^2+1)y=\cos x$ is .....

38. The homogeneous linear differential equation of which auxiliary equation has roots $1,-1$ is .....

39. The particular integral of $(D^2-3D+2)y=\sin x$ is ..... (U.P.T.U. 2004)

40. To transform $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}-\frac{1}{x}$ into a linear differential equation with constant coefficients, put $x=e^z$.

41. The particular integral of $(D^2-4)y=\sin 2x$ is

(a) $\frac{1}{8}\sin 2x$ (b) $-\frac{1}{8}\sin 2x$ (c) $\frac{1}{4}\sin 2x$ (d) None of these (U.P.T.U. 2005)

42. The solution of $\frac{d^2y}{dx^2}-3\frac{dy}{dx}+2y=0$ is .....

43. The differential equation whose auxiliary equation has roots $0,-1,-1$ is .....

44. Complementary function of $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+2y=\log x$ is

(a) $C_1x+C_2x\log x$ (b) $(C_1+C_2\log x)x$ (c) $(C_1+C_2\log x)x^2$ (d) $C_1x+C_2\log x^2$ (Bharati 2004)

45. The general solution of $(D^2-3D+2)y=e^x+e^{2x}+e^{3x}$ is (Uttar 2004)

46. $\frac{1}{f(D)}(ax^m)=\frac{1}{f(0)+F(D)}(ax^m)$. (Uttar 2004)

CHAPTER 14

Applications of Linear Differential Equations

1. Introduction. 2. Simple harmonic motion. 3. Free damped motion. Critical and critical damping. 4. Oscillations of a spring. 5. Oscillatory electromotive force. 6. Electromechanical analogies. 7. Deflection of beams. 8. Whirling of shafts. 9. Applications of the methods to nonlinear equations. 10. Objective Type of Questions.

14.1 INTRODUCTION

The linear differential equations with constant coefficients and their many important applications in the study of electrical, mechanical and other linear systems. In fact, such equations play a dominant role by studying the theory of electrical and mechanical oscillatory systems.

We shall begin by explaining the types of oscillations of the mechanical systems and the equivalent electrical circuits. Then we shall study at some length the slightly less striking applications such as deflection of beams and whirling of shafts. At the end, we'll take up some of the applications of simultaneous linear differential equations.

14.2 SIMPLE HARMONIC MOTION

When the acceleration of a particle is proportional to its displacement from a fixed point and is always directed towards it, then the motion is said to be simple harmonic.

If the displacement of the particle at any time t from fixed point O is x (Fig. 14.1), then

Fig. 14.1

$$\frac{d^2x}{dt^2} = -p^2x \quad \text{or} \quad D^2x + p^2x = 0, \qquad \ldots(1)$$

$\therefore$ the solution is $\qquad x = c_1 \cos pt + c_2 \sin pt \qquad \ldots(2)$

$\therefore$ the velocity at t is $\qquad v = \frac{dx}{dt} = -pc_1 \sin pt + pc_2 \cos pt \qquad \ldots(3)$

If the particle starts from rest at A, where $OA = a$,

then from (2), $\quad x = a$ when $t = 0$, $\quad a = c_1$

and from (3), $\quad v = 0$ when $t = 0$, $\quad 0 = pc_2$

Thus $\qquad x = a \cos pt \qquad \ldots(4)$

and $\qquad \frac{dx}{dt} = -a \sin pt = -\sqrt{(a^2 - x^2)} \qquad \ldots(5)$

which give the displacement and the velocity of the particle at any time t.

Nature of motion. The particle starts from A towards O under acceleration directed towards O which gradually decreases until it vanishes at O, where the particle has acquired the maximum velocity. On passing

through O, retardation begins and the particle comes to an instantaneous rest at A', where OA' = OA. It then retraces its path and goes on oscillating between A and A'.

The amplitude or maximum displacement from the centre is a.

The periodic time, i.e., the time of complete oscillation is $2\pi/\mu$, for when t is increased by $2\pi/\mu$, the values of x and $\cos\mu t$ remain unaltered.

The frequency or the number of oscillations per second is

$$\frac{1}{\text{periodic time}} = \frac{\mu}{2\pi}.$$

Example 13.2.1 *In the case of a vertical elastic band or string with one end fixed and a particle of mass m attached to the other, find the equation of motion of the particle given that l is the natural length of the string and e its elongation due to weight mg. Also find the displacement of particle relative to the fixed end.*

Solution. Let OA ($= l$) be the elastic horizontal string with the end O fixed and having a particle of mass m attached to its end A (Fig. 13.2).

Fig. 13.2

At any time t, let the particle be at P where $OP = x$; so that the elongation $AP = x - l$.

Since for the elongation e, tension = mg

$$\therefore \quad \text{for the elongation } x - l, \text{ tension } T = \frac{mg(x-l)}{e}$$

Tension being the only horizontal force, the equation of motion is

$$m\frac{d^2x}{dt^2} = -T \quad \text{or} \quad \frac{d^2x}{dt^2} = -\frac{T}{m} = -\frac{g(x-l)}{e} \qquad \ldots(i)$$

which is the required equation of motion.

Now (i) can be written as $D^2x + g/e(x - l) = 0$, where $D \equiv d/dt$ $\qquad \ldots(ii)$

$\therefore$ the auxiliary equation is $D^2 + g/e = 0$ or $D = \pm i\sqrt{(g/e)}$

$\therefore$ $\qquad C.F. = c_1 \cos\sqrt{(g/e)}t + c_2 \sin\sqrt{(g/e)}t$

and $\qquad P.I. = \frac{1}{D^2 + g/e}.0 = \frac{1}{e}.\frac{1}{D^2 + g/e}e^{0t} = l$

Thus the solution of (i) is

$$x = c_1 \cos\sqrt{(g/e)}t + c_2 \sin\sqrt{(g/e)}t + l \qquad \ldots(iii)$$

When $t = 0$, $x = x_0$ $\quad \therefore \quad x_0 = c_1 + 0 + l \quad$ i.e., $c_1 = x_0 - l$

Again from (iii), $\quad \frac{dx}{dt} = -c_1\sqrt{(g/e)}\sin\sqrt{(g/e)}t + c_2\sqrt{(g/e)}\cos\sqrt{(g/e)}t$

When $t = 0$, $dx/dt = 0$: $\quad \therefore \quad 0 = c_2$

Substituting the values of c_1 and c_2 in (iii), we have

$x = (x_0 - l)\cos\sqrt{(g/e)}t + l$ which is the required result.

Example 13.2.2. *Two particles of masses m_1 and m_2 are tied to the ends of an elastic string of natural length l and modulus λ. They are placed on a smooth table so that the string is just taut and m_2 is projected with any velocity directly away from m_1. Show that the string will become slack after the lapse of time $\pi\sqrt{\{lm_1m_2/\lambda(m_1+m_2)\}}$.*

Solution. Taking O as fixed point of reference, let particle m_1 be at O and m_2 at a distance a from m_1 at time $t = 0$ Fig. 13.3. At any time t, let m_1 be at a distance x from O and m_2 be at a distance y from O. Then the equation of motion of m_1 is

Fig. 13.3

$$m_1\frac{d^2x}{dt^2} = T \qquad \ldots(i)$$

and equation of motion of m_2 is $m_2\frac{d^2y}{dt^2} = -T \qquad \ldots(ii)$

where $T = \lambda(y - a)/a$.

From (i) and (ii), $a(y+x)^2 - a^2(x+l)^2 = -\frac{\lambda}{m_1} - \frac{\lambda}{m_2}$

or $\frac{d^2(y-x)}{dt^2} = -\left(\frac{1}{m_1}+\frac{1}{m_2}\right)\frac{\lambda(y-x)}{a}$ or $\frac{d^2x}{dt^2} = -\frac{\lambda(m_1+m_2)x}{m_1 m_2 a}$ where $x = y - x$

It is a S.H.M. with periodic time $t = 2\pi\sqrt{\left|\frac{am_1m_2}{\lambda(m_1+m_2)}\right|}$

The string will regain its original length i.e. become slack after time $\frac{1}{4}$ of m_1 moving towards m_2 such that

$$t_1 = \frac{1}{4}\cdot\frac{1}{4} = \frac{\pi}{2}\sqrt{\left|\frac{am_1m_2}{\lambda(m_1+m_2)}\right|}.$$

Example 11(2). A particle of mass m is attached to the two points A and B, $AB = 2a$, by two strings each of length a and is constrained to move in a vertical circle through A and B. If the tension in the string AB is λ times the tension in the string BA when the string BA is horizontal, find the velocity of the particle.

Solution. In the equilibrium position, let the particle be at C so that $AC = a + l$ and $BC = a' + l'$, where a, a' are natural lengths of the strings (Fig. 11.4). Then the tensions in the two strings are given by

$$T = \lambda\frac{l}{a} = \lambda'\frac{l'}{a'} \quad \ldots(i)$$

Fig. 11.4

At any time t, let the particle be at P, so that $CP = x$. Then

$$T_1 = \lambda\frac{l+x}{a} \text{ and } T_2 = \lambda\frac{l'-x}{a'}$$

$\therefore$ the equation of motion is $m\frac{d^2x}{dt^2} = T_2 - T_1 = \lambda'\frac{l'-x}{a'} - \lambda\frac{l+x}{a}$

$$= -\left(\frac{\lambda'}{a'}+\frac{\lambda}{a}\right)x = -\left(\frac{T}{a'}+\frac{T}{a}\right)x \quad \text{[By (i)]}$$

or $\frac{d^2x}{dt^2} = -\mu x$, where $\mu = \frac{a+a'}{aa'}\frac{T}{m}$

Hence the periodic time $= \frac{2\pi}{\sqrt{\mu}} = 2\pi\sqrt{\left|\frac{maa'}{T(a+a')}\right|}$.

11(3) (i) SIMPLE PENDULUM

A heavy particle attached by a light string to a fixed point and oscillating under gravity constitutes a simple pendulum.

Let O be the fixed point, l be the length of the string and A be the position of the bob initially (Fig. 11.5). If P be the position of the bob at any time t, such that $AP = s$ and $\angle AOP = \theta$, then $s = l\theta$.

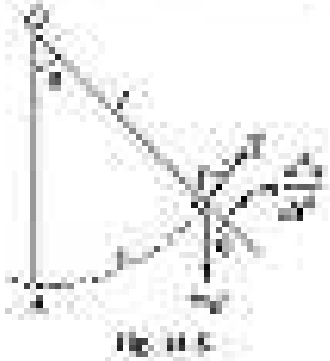

Fig. 11.5

$\therefore$ the equation of motion along PT is $m\frac{d^2s}{dt^2} = -mg\sin\theta$

i.e., $\frac{d^2(l\theta)}{dt^2} = -g\sin\theta$

or $\frac{d^2\theta}{dt^2} = -\frac{g}{l}\sin\theta = -\frac{g}{l}\left(\theta - \frac{\theta^3}{3!} - \ldots\right) = -\frac{g}{l}\theta$ to a first approx.

Now the auxiliary equation being $D^2 + \frac{g}{l} = 0$, we have $D = \pm\sqrt{\frac{g}{l}}i$.

$\therefore$ Its solution is $\theta = c_1\cos\sqrt{\frac{g}{l}}t + c_2\sin\sqrt{\frac{g}{l}}t$.

Thus the motion of the bob is simple harmonic and the time of one oscillation is $2\pi\sqrt{\frac{l}{g}}$.

Thus the increase of the bob from rest to the extreme position of an oscillation and its subsequent return to rest is called a complete oscillation.

The time taken for one beat = $2\sqrt{(l/g)}$.

A second pendulum beats 86400 times a day for there are 86400 seconds in 24 hours.

(2) Gain or loss of oscillations. Let a pendulum of length l which beats in time T, so that

$$T = \text{time of a beat} = \pi\sqrt{(l/g)} \quad \text{or} \quad n = \frac{1}{\pi}\sqrt{(g/l)}$$

Taking logs, $\log n = \log(1/\pi) + \frac{1}{2}(\log g - \log l)$.

Taking differentials of both sides, we get $\frac{dn}{n} = \frac{1}{2}\left[\frac{dg}{g} - \frac{dl}{l}\right]$.

If only g changes, l remaining constant, $\frac{dn}{n} = \frac{dg}{2g}$ ...(1)

If only l changes, g remaining constant, $\frac{dn}{n} = -\frac{dl}{2l}$ ...(2)

Example 3.64. *How fast must a clock which gains 5 seconds per day be made to go if by means of a screw provided on the pendulum its length can be increased by one part in 1000?*

Solution. If the original length of the string be increased to $l + dl$, then

$$\frac{l + dl}{dl} = \frac{501}{500} \quad \therefore \quad \frac{dl}{l} = \frac{501}{500} - 1 = \frac{1}{500}.$$

$\therefore$ using (2) above, we have $\frac{dn}{n} = -\frac{dl}{2l} = -\frac{1}{1000}$.

i.e., $$dn = -\frac{n}{1000} = -\frac{86400}{1000} = -86.$$

Hence dn is negative, the clock will lose 3 seconds per day.

Example 3.65. *A simple pendulum of length l is oscillating in a resisting medium whose resistance per unit mass is k times the velocity. Find the differential equation of the motion. Discuss the motion and find the period of oscillations.*

Solution. Let the position of the bob O (fig. 3.66) at any time t be P and C be the point of suspension such that $OC = l$, $\angle OCP = \theta$ and therefore, arc $OP = s = l\theta$ (Fig. 3.66).

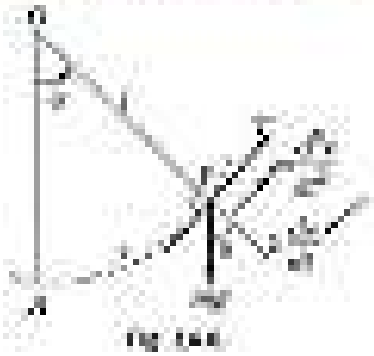

Fig. 3.66

The equation of motion along the tangent PT is

$$m\frac{d^2s}{dt^2} = -mg\sin\theta - k\frac{ds}{dt} \quad \text{where } k \text{ is a constant.}$$

or $$\frac{d^2(l\theta)}{dt^2} + \frac{k}{m}\frac{d(l\theta)}{dt} + g\sin\theta = 0$$

Replacing $\sin\theta$ by θ since θ is small and writing $k/m = 2k$, we get

$$\frac{d^2\theta}{dt^2} + 2k\frac{d\theta}{dt} + \frac{g\theta}{l} = 0 \qquad ...(1)$$

which is the required differential equation.

Its auxiliary equation has roots $D = -k \pm \sqrt{(k^2 - \omega^2)}$ where $\omega = \sqrt{(g/l)}$.

The oscillatory motion of the bob is only possible when $k < \omega$.

Then the roots of the auxiliary equation are $-k \pm i\sqrt{(\omega^2 - k^2)}$.

$\therefore$ the solution of (1) is

$$\theta = e^{-kt}$$

which gives oscillatory motion of period $2\pi/\sqrt{(\omega^2 - k^2)}$.

Example 3.18. A pendulum of length l has one end of its string fastened to a peg on a smooth plane inclined to the horizontal at angle α. With the string taut the weight on the plane, the free end of which is at rest

(i) the pendulum of length l oscillates in one vertical plane through the peg, show that its time of oscillation is $2\pi\sqrt{\frac{l}{g\sin\alpha}}$.

(Calcutta Univ. 2003)

Solution. At any time t, let the bob of mass m be at P and O be the point of suspension so that $OP = l$ and $\angle AOP = \theta$ (Fig. 3.17).

The component of weight along the plane being $mg\sin\alpha$, the equation of motion of the bob along the tangent at P is

$$m\frac{d^2s}{dt^2} = -mg\sin\alpha\sin\theta$$

or $$\frac{d^2(l\theta)}{dt^2} = -g\sin\alpha\sin\theta \qquad \because\ s = l\theta$$

or $$\frac{d^2\theta}{dt^2} = -g\sin\alpha\left(\theta - \frac{\theta^3}{3!} + \ldots\right)$$

or $$\frac{d^2\theta}{dt^2} = -\mu\theta \quad \text{where } \mu = \frac{g\sin\alpha}{l}, \text{ to a first approximation.}$$

$\therefore$ The motion being simple harmonic, the time of oscillation is

$$t = \frac{2\pi}{\mu} = 2\pi\sqrt{\frac{l}{g\sin\alpha}} \qquad \ldots(i)$$

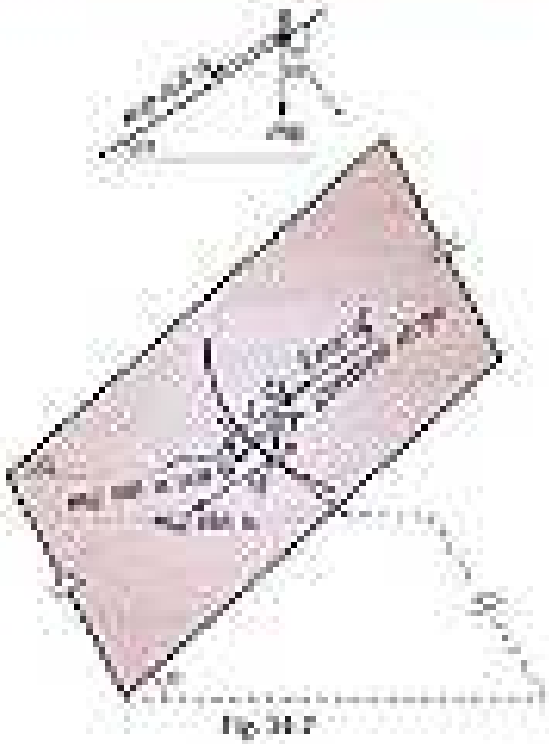

Fig. 3.17

We know that for a pendulum of length l when suspended vertically, the time of oscillation

$$t = 2\pi\sqrt{l/g} \qquad \ldots(ii)$$

$\therefore$ dividing (i) by (ii), we have $\gamma = \sqrt{\left(\frac{1}{g\sin\alpha}\right)}$

or $$t^2 = 4\pi^2/g\sin\alpha \quad \text{or} \quad t = 4\pi^2/(g\sin\alpha).$$

PROBLEMS [4]

1. A particle is moved in a straight line with simple harmonic motion, with amplitude 20 cm and time 4 seconds. Find the time required by the particle in passing between points which are at distances 12 cm and 5 cm from the centre of force and on the same side of it.

2. In the motion of $x = a\cos(nt + \alpha)$, the distances of a point moving with S.H.M. from its mean position are x_1, x_2, x_3. Show that the time of a complete oscillation is

$$\frac{2\pi}{n}\tan^{-1}\left[\frac{x_1 - x_2}{x_2 - x_3}\right]$$

3. An elastic string of natural length l and modulus λ is stretched between two points A and B at the same level and a mass m is attached to the middle point of the string. The mass is released from rest in the position in which each half of the string is inclined at an angle α to the vertical. Show that the mass will perform a simple harmonic motion of period $2\pi\sqrt{\frac{ml\cos\alpha}{2\lambda(1-\cos\alpha)}}$.

4. A particle of mass m is moved in a straight line under the action of force $m\omega^2(x + \frac{a^2}{x^3})$ which is always directed towards fixed point O on the line. If the displacement is zero when $x = a$ and velocity is ωa, where a is the natural unit of length, find the displacement x in terms of the time t given that $x = a\sqrt{2}$ and $t = 0$. Also find the time in which $OP = x$.

5. A point moves in a straight line towards the centre of force which is inversely proportional to the distance from the centre. If the velocity at a distance a from the centre of force is $\sqrt{\frac{1}{2}}$ times the velocity at a distance b, prove that the time of reaching a point c from the centre of force is $\frac{\pi}{2}\sqrt{\frac{a^2+b^2}{2}}$ times the time required to reach the point $\frac{a+b}{2}$ from the point $\sqrt{\frac{a^2+b^2}{2}}$.

(U.P.T.U., 2005)

8. A rod hinged at one end is set vibrating in a vertical plane. Show that the time of one complete oscillation is the same as that of a simple pendulum of length $\frac{2}{3}$ of the length of the rod. (Kanpur, 2000)

9. A block of mass 2 kg is attached to the lower end of a vertical spring whose upper end is fixed. The spring is stretched 20 cm by a force of 3.92 N. If the block is displaced vertically downwards from its equilibrium position and released, find the period of the resulting vibrations. (Kanpur, 2000)

10. A smooth parabolic wire whose plane is vertical and whose latus rectum is $4a$ is fixed with its vertex downwards. A bead of mass m can slide on the wire. If the bead is projected from the vertex with velocity $\sqrt{2ga}$, show that the time of reaching the focus is $\frac{1}{4}\sqrt{\frac{a}{g}}\left\{\sqrt{2}+\log(1+\sqrt{2})\right\}$. (Kanpur, 2000)

11. A particle of mass m is attached to the end of a light elastic string of natural length a and modulus λ, the other end of the string being fixed. The particle is let fall from rest when the length of the string is $2a$. Show that it comes to rest after a time $\sqrt{\frac{a}{g}}\left(\frac{\pi}{2}+2\right)$. (Kanpur, 2000)

12. A body is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

13. A heavy particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

14. A particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

15. A particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

16. A particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

17. A particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

18. A particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

19. A particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

20. A particle is projected vertically upwards from the earth's surface with velocity equal to $\frac{1}{4}$ of the velocity of escape. Neglecting the resistance of the air, show that the greatest height reached by the body is $\frac{1}{7}$ of the radius of the earth. (Kanpur, 2000)

13.13 OSCILLATIONS OF A SPRING

(I) Free oscillations. Suppose a mass m is suspended from the end A of a light spring, the other end of which is fixed at O. (Fig. 13.9)

Let e (= AB) be the elongation produced by the mass m hanging in equilibrium. If λ be the restoring force per unit stretch of the spring due to elasticity, then for the equilibrium at B,

$$mg = \lambda e / a \qquad \ldots(1)$$

At any time t, after the motion commences, let the mass be at P, where BP = x. Then the equation of motion of m is

$$m\frac{d^2x}{dt^2} = mg - \lambda(e+x)/a = mg - \lambda e/a - \lambda x/a \qquad \text{[By (1)]}$$

Or writing $\lambda/ma = \mu^2$, it becomes

$$\frac{d^2x}{dt^2} + \mu^2 x = 0 \qquad \ldots(2)$$

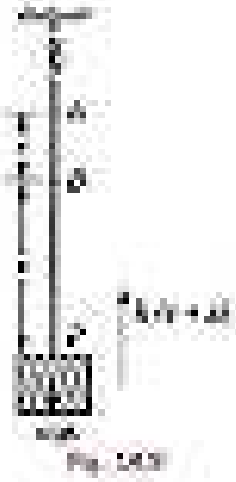

Fig. 13.9

This equation represents the free vibrations of the spring which are of the simple harmonic form having centre of oscillation at B—its equilibrium position and the period of oscillation

$$= \frac{2\pi}{\mu} = 2\pi\sqrt{\left(\frac{a}{\lambda/m}\right)} \qquad \left[\because \frac{\lambda}{ma} = \frac{mg}{\lambda e} = \frac{g}{e}\right], \text{[By (1)]}$$

(II) Damped oscillations. If the mass m be subjected to the damping force proportional to velocity ($= r \times$ velocity) (Fig. 13.10), then the equation of motion becomes

$$m\frac{d^2x}{dt^2} = mg - \lambda(e+x)/a - r\frac{dx}{dt}$$

$$= -\lambda x/a - r\frac{dx}{dt} \qquad \text{[By (1)]}$$

Or writing $r/m = 2k$ and $\lambda/ma = \mu^2$, it becomes

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + \mu^2 x = 0 \qquad \ldots(3)$$

$\therefore$ Its auxiliary equation is

$$D^2 + 2kD + \mu^2 = 0, \text{ whence } D = -k \pm \ldots$$

Case I. When $k >$ ... the roots of the auxiliary equation are real and distinct (say α, β)

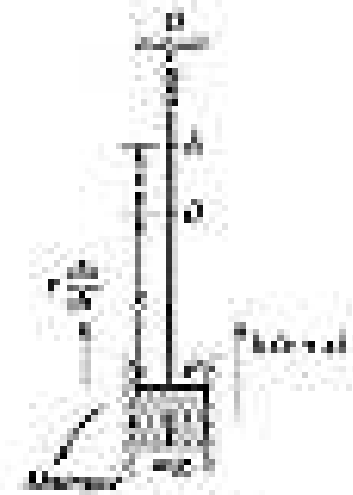

Fig. 13.10

∴ the solution of (i) is $x = c_1 e^{k_1 t} + c_2 e^{k_2 t}$ ...(ii)

To determine c_1, c_2 let the spring be stretched to a length $x = l$ and then released at rest.

$x = l$ and $dx/dt = 0$ at $t = 0$.

∴ From (ii), $l = c_1 + c_2$

Also from $\frac{dx}{dt} = c_1 k_1 e^{k_1 t} + c_2 k_2 e^{k_2 t}$, we get

$0 = c_1 k_1 + c_2 k_2$

whence $c_1 = \frac{-k_2 l}{k_1 - k_2}$ and $c_2 = \frac{k_1 l}{k_1 - k_2}$

Hence the solution of (i) is

$$x = \frac{l}{k_1 - k_2}(k_1 e^{k_2 t} - k_2 e^{k_1 t}) \quad \text{...(iii)}$$

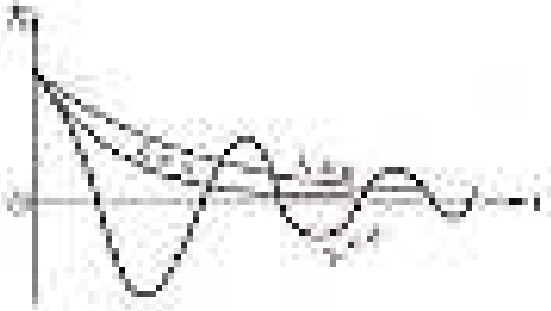

Fig. 14.33

which shows that x is always positive and decreases to zero as $t \to \infty$ (Fig. 14.33).

The restoring force in this case is so great that the motion is non-oscillatory and is, therefore, referred to as over-damped or dead beat motion.

Case II. When $k_1 = k_2$, the roots of the auxiliary equation are real and equal, each being $= -k$.

∴ The general solution of (i) becomes $x = (c_1 + c_2 t)e^{-kt}$.

As in case I, if $x = l$ and $dx/dt = 0$ at $t = 0$, then $c_1 = l$ and $c_2 = kl$.

Hence the solution of (i) is $x = l(1 + kt)e^{-kt}$ which also shows that x is always positive and decreases to zero as $t \to \infty$ (Fig. 14.33).

The nature of motion is similar to that of the previous case and is called the critically damped motion for it separates the non-oscillatory motion of case I from the most interesting oscillatory motion of case III.

Case III. When $k = p$, the roots of the auxiliary equation are imaginary, i.e. $D = -\lambda \pm i\mu$ where $\lambda^2 + \mu^2 = p^2$.

∴ the solution of (i) is $x = e^{-\lambda t}(c_1 \cos \mu t + c_2 \sin \mu t)$

As in case I, $x = l$, $dx/dt = 0$ at $t = 0$, then $c_1 = l$ and $c_2 = \lambda l/\mu$

Thus the solution of (i) becomes $x = le^{-\lambda t}\left(\cos \mu t + \frac{\lambda}{\mu} \sin \mu t\right)$

which can be put in the form $x = l\sqrt{1 + \left(\frac{\lambda}{\mu}\right)^2}\, e^{-\lambda t} \cos\left(\mu t - \tan^{-1}\frac{\lambda}{\mu}\right)$ ...(iv)

Here the presence of the trigonometric factor in (iv) shows that the motion is oscillatory, having

(i) the variable amplitude $= l\sqrt{1 + (\lambda/\mu)^2}\, e^{-\lambda t}$ which decreases with time,

(ii) the periodic time $T = 2\pi/\mu$.

But the periodic time of free oscillations is $T' = 2\pi/p$.

$\therefore \quad \mu = \sqrt{p^2 - \lambda^2} < p$

$\therefore \quad \frac{2\pi}{\mu} > \frac{2\pi}{p}$, i.e. $T > T'$.

This shows that the effect of damping is to increase the period of oscillation and the motion ultimately dies away. Such a motion is termed as damped oscillatory motion.

(iii) **Forced oscillations** (without damping). If the point of the support of the spring is also vibrating with some external periodic force, then the resulting motion is called the *forced oscillatory motion*.

Taking the external periodic force to be $mq \cos nt$, the equation of motion is

$$m\frac{d^2x}{dt^2} = mg - kx - \lambda\frac{dx}{dt} + mq \cos nt$$

$$\text{or} \quad m\frac{d^2x}{dt^2} + \lambda\frac{dx}{dt} + kx = mg + mq \cos nt \quad [\because \ mg = kl] \quad \text{...(v)}$$

Fig. 14.34

Dividing by m, $p^2 = k/m$, (v) takes the form

$$\frac{d^2x}{dt^2} + p^2x = a\sin pt \qquad \ldots(1)$$

Its C.F. $= c_1\cos pt + c_2\sin pt$ and P.I. $= a\dfrac{1}{D^2+p^2}\sin pt$

Now two cases arise :

Case I. When $p \neq n$.

$$\text{P.I.} = \frac{a}{p^2-n^2}\sin nt.$$

$\therefore$ the complete solution of (1) is $x = c_1\cos pt + c_2\sin pt + \dfrac{a}{p^2-n^2}\sin nt.$

On writing $c_1\cos pt + c_2\sin pt$ as $r\sin(pt+\phi)$, we have

$$x = r\sin(pt+\phi) + \frac{a}{p^2-n^2}\sin nt \qquad \ldots(2)$$

This shows that the motion is compounded of two oscillatory motions : the first due to the C.F. gives free oscillations of period $2\pi/p$, and the second due to the P.I. gives forced oscillations of period $2\pi/n$.

Also we observe that if the frequency of free oscillations is very high or n, p is large), then the amplitude of forced oscillations is small.

Case II. When $p = n$.

$$\text{P.I.} = pt\cdot\frac{1}{2D}\sin pt = \frac{pt}{2}\int\sin pt\,dt = \frac{pt}{2}\left(-\frac{\cos pt}{p}\right)$$

$\therefore$ the complete solution of (1) is $x = c_1\cos pt + c_2\sin pt + \dfrac{pt}{2p}\cos pt$

$$= \left[c_2 + \frac{pt}{2p}\right]\sin pt + c_1\cos pt$$

Putting $c_2 + pt/2p = \rho\cos\varphi$ and $c_1 = \rho\sin\varphi$, we get

$$x = \rho\sin(pt+\varphi) \qquad \ldots(3)$$

This shows that the oscillations are of period $2\pi/p$ and amplitude $\rho = \sqrt{(c_2 + pt/2p)^2 + c_1^2}$, which clearly increases with time (Fig. 14.22).

Thus, the amplitude of the oscillations may become abnormally large causing even strong and heavy quantity breakdown of the system. In practice, however, collapse rarely occurs, though the amplitudes may become dangerously large since there is always some resistance present in the system.

This phenomenon of the depressed frequency becoming equal to the natural frequency of the system, is referred to as resonance.

Thus, while designing a machine or a structure, the occurrence of resonance should always be avoided to ensure the stability of the system at any stage. That is why, the soldiers break step while marching over a bridge for the fear that their steps may not be in phase with the natural frequency of the bridge causing its collapse due to 'resonance'.

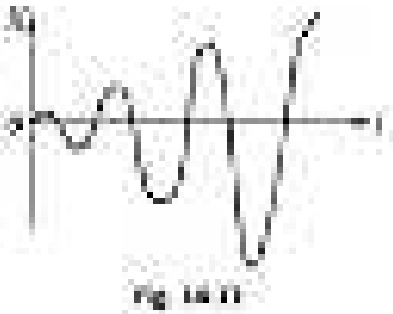

Fig. 14.22

(c) **Forced oscillations with damping.** If, in addition, there is a damping force proportional to velocity (say $-k\dfrac{dx}{dt}$) acting (Fig. 14.23), then the equation of motion becomes

$$m\frac{d^2x}{dt^2} = mg - kx + c(\ldots) + \text{impressed force} - r\frac{dx}{dt}$$

$$= -kx + mg\sin\omega t - r\frac{dx}{dt} \qquad \left[\because\ mg = kx\right]$$

On writing $r/m = 2k$ and $k/m = p^2$, it takes the form

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \mu^2 x = p\cos\omega t \qquad \ldots(12)$$

The auxiliary equation is $D^2 + 2\lambda D + \mu^2 = 0$ whence $D = -\lambda \pm \sqrt{(\lambda^2 - \mu^2)}$.

$$\therefore \quad \text{C.F.} = e^{-\lambda t}\left\{A e^{t\sqrt{(\lambda^2-\mu^2)}} + B e^{-t\sqrt{(\lambda^2-\mu^2)}}\right\}$$

It represents the free oscillations of the system which die out as $t \to \infty$.

After the P.I.

$$= A\frac{1}{D^2 + 2\lambda D + \mu^2}\cos\omega t = B\frac{1}{-\omega^2 + 2\lambda D + \mu^2}\cos\omega t$$

$$= A\frac{\mu^2 - \omega^2 - 2\lambda D}{(\mu^2 - \omega^2)^2 - 4\lambda^2 D^2}\cos\omega t + B\frac{\mu^2 - \omega^2}{(\mu^2 - \omega^2)^2 + 4\lambda^2\omega^2}\cos\omega t + \frac{2\lambda\omega\sin\omega t}{(\mu^2 - \omega^2)^2 + 4\lambda^2\omega^2}$$

Putting $A = \alpha^2 \sin\theta$ and $2\lambda\omega = R\sin\theta$, we get

$$\text{P.I.} = \frac{R}{\sqrt{(\mu^2 - \omega^2)^2 + 4\lambda^2\omega^2}}\cos(\omega t - \theta)$$

which represents the forced oscillations of the system having

(a) a constant amplitude

$$= R/\sqrt{(\mu^2 - \omega^2)^2 + 4\lambda^2\omega^2}$$

and (b) the period $= 2\pi/\omega$ which is the same as that of the impressed force.

Thus with the exception of those, the free oscillations die away while the forced oscillations continue giving the steady state motion.

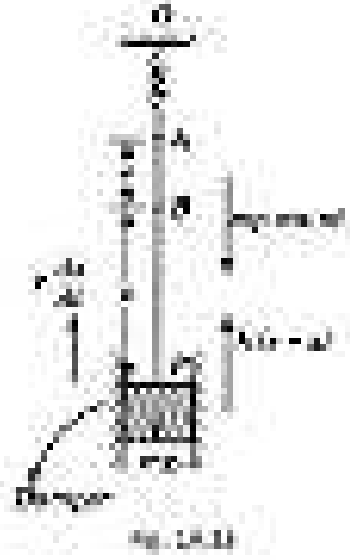

Fig. 14.11

Example 14.2. *A body weighing 20 kg is hung from a spring. A pull of 20 kg wt. will stretch the spring by 10 cm. The body is pulled down 5 cm below the static equilibrium position and then released. Find the displacement of the body from its equilibrium position at time t sec. Also find the maximum velocity and the period of oscillation.*

Solution. Let O be the fixed end and A, the lower end of the spring (Fig. 14.12).

Since a pull of 20 kg wt. at A stretches the spring by 0.1 m.

$$\therefore \quad 20 = T_0 = k \times 0.1, \text{ i.e. } k = 200 \text{ kg/m}.$$

Let B be the equilibrium position when a body weighing $W = 10$ kg is hung from A ; then

$$10 = T_0 = k \times AB$$

$$\text{i.e.,} \quad AB = \frac{10}{200} = 0.05 \text{ m}$$

Now the weight is pulled down to C, where $BC = 0.1$ m. After any time t sec. if the string from C, let the weight be at P where $BP = x$.

Then the tension $T_P = k \times AP = 200(0.1 + x) = 20 + 200x$.

$\therefore$ The equation of motion of the body is

$$\frac{W}{g}\frac{d^2x}{dt^2} = W - T_P, \text{ where } g = 9.8 \text{ m/sec}^2.$$

$$\text{i.e.,} \quad \frac{10}{9.8}\frac{d^2x}{dt^2} = 10 - (20 + 200x) \quad \text{or} \quad \frac{d^2x}{dt^2} = -p^2 x, \quad \text{where } p = 14.$$

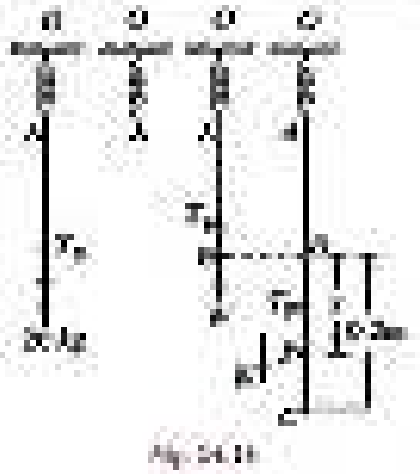

Fig. 14.12

This shows that the motion of the body is simple harmonic about B as centre and the period of oscillation $= 2\pi/p = 0.45$ sec.

Also the amplitude of motion, being $BC = 0.1$ m., the displacement of the body from A at time t is given by $x = 0.1\cos pt = 0.1\cos 14t$ m

and the maximum velocity $= p \times$ amplitude $= 14 \times 0.1 = 2.8$ m/sec.

Example 14.24. *A spring fixed at the upper end supports a weight of 980 gm at its lower end. The spring stretches $\frac{1}{2}$ cm under a load of 10 gm and the resistance (in gm wt) to the motion of the weight is numerically equal to $\frac{1}{10}$ of the speed of the weight in cm/sec. The weight is pulled down $\frac{1}{4}$ cm. below its equilibrium position and then released. Find an expression for the distance of weight from its equilibrium position at the time t during its first upward motion.*

Also find the period before the damping factor is reduced to $\frac{1}{10}$ of its initial value.

Solution. Let O be the fixed end and A the other end of the spring (Fig. 14.12).

Since load of 10 gm attached to A stretches the spring by $\frac{1}{2}$ cm.

$$\therefore \qquad 10 = T_0 = k \cdot \frac{1}{2} \Delta l, \; k = 20 \text{ gm/cm}.$$

Let B be the equilibrium position when 980 gm. weight is attached to A, then

$$980 = T_B = k \times AB, \quad \text{i.e.,} \quad AB = \frac{980}{20} = 49 \text{ cm}.$$

Now the 980 gm weight is pulled down to C, where $BC = \frac{1}{4}$ cm.

After any time t of its release from C, let the weight be at P, where $BP = x$.

Then the tension

$$T = k \times AP = 20(49 + x) = 980 + 20x \text{ and the resistance to motion} = \frac{1}{10}\frac{dx}{dt}.$$

$\therefore$ the equation of motion is

$$\frac{980}{g}\frac{d^2x}{dt^2} = w - T - \frac{1}{10}\frac{dx}{dt} \qquad [\because g = 980 \text{ cm/sec}^2 \text{ (p. 448)}]$$

$$= 980 - 980 - 20x - \frac{1}{10}\frac{dx}{dt} \quad \text{i.e.,} \quad 20\frac{d^2x}{dt^2} + \frac{dx}{dt} + 200x = 0 \qquad \ldots(1)$$

Its auxiliary equation is $20D^2 + D + 200 = 0$,

where $$D = \frac{-1 \pm \sqrt{1 - 4 \times 20 \times 200}}{40} = \frac{-1 \pm \sqrt{-15999}}{40} = -0.025 \pm i(4.99)$$

$\therefore$ the solution of (1) is $x = e^{-0.025t}(c_1 \cos 4.99t + c_2 \sin 4.99t)$ ...(2)

Also $$\frac{dx}{dt} = e^{-0.025t}(-0.025)(c_1 \cos 4.99t + c_2 \sin 4.99t)$$

$$+ e^{-0.025t}(-c_1 \sin 4.99t + c_2 \cos 4.99t)4.99 \qquad \ldots(3)$$

Initially when the mass is at C, $t = 0$, $x = \frac{1}{4}$ cm. and $dx/dt = 0$.

From (2), $c_1 = \frac{1}{4}$, and from (3) $0 = -0.006c_1 + c_2(4.99)$, i.e., $c_2 = 0.003$.

Thus, substituting these values in (2), we get

$$x = e^{-0.025t}(0.25 \cos 4.99t + 0.003 \sin 4.99t)$$

which gives the displacement of the weight from the equilibrium position at any time t.

Here damping factor $= e^{-0.025t}$, where e is a constant of proportionality.

Its initial value $= e^0 = 1$.

It grows after time t, the damping factor $= \frac{1}{10}$. $\quad \therefore \quad e^{-0.025t} = \frac{1}{10}$ or $e^{0.025t} = 10$.

Thus $$t = 20 \log_e 10 = 20 \times 2.3 = 46 \text{ sec}.$$

Example 14.25. *A spring which is stretched by an amount c under a force mg is suspended from a support. A mass m is attached to the lower end. Initially the mass is at rest in the equilibrium position and a speed v is given to it. At a certain instant an impulse is also given to the support O such that at any time $t > 0$ the displacement below the equilibrium position is ve^{-t}. Show that the displacement of the mass below A is given by*

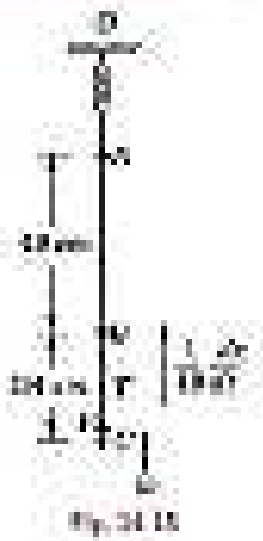

Fig. 14.12

[illegible]

[illegible]

[illegible]

Its amplitude $\left(\frac{kF}{p}\right) = \frac{k}{p}\sqrt{(1+\lambda^2)^2}$, which increases with time. Hence the phenomenon of resonance occurs.

Example 14.33. A spring of negligible weight which stretches 1 inch under tension of 1 lb is fixed at one end and at the other end is attached to a weight of 2 lb as the latter hangs at rest. It is found that resonance occurs when the applied periodic force is $\sin 8t$ lb. Neglecting the weight of the spring and the resistance of the air, the forced vibrations are given by $x = a \sin 8t$, find the values of a and c.

Solution. As a weight of 2 lb is attached to the lower end A of the spring stretched 1 in by $\frac{1}{12}$ ft.

$\therefore$ $\qquad 2 = T = k \cdot \frac{1}{12}$. $\quad$ i.e., $k = 24$ lb/ft.

Let B be the equilibrium position of the weight attached to A (Fig. 14.16), then

$$w = T_B = 2 = AB = 24 \cdot AB$$

$\therefore$ $\qquad AB = \frac{1}{12}$ ft.

At any time t, let the weight be at P, where $BP = x$.

Then the tension T at $P = k \cdot AP = 24\left(\frac{1}{12} + x\right) = 2 + 24x$

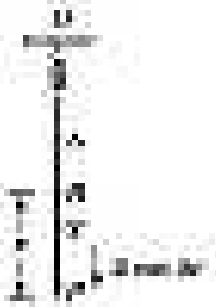

Fig. 14.16

$\therefore$ The equation of motion is

$$\frac{w}{g}\frac{d^2x}{dt^2} = -T + w + 2\sin 8t = -w - 24x + w + 2\sin 8t$$

or $$\frac{1}{16}\frac{d^2x}{dt^2} + 24x = 2\sin 8t \qquad \ldots(1)$$

The phenomenon of resonance occurs when the period of free oscillations is equal to the period of forced oscillations.

Writing (1) as $\frac{d^2x}{dt^2} + p^2x = \frac{2k}{w}\sin 8t$, where $p^2 = 24(16)$, the period of free oscillations is found to be $\frac{2\pi}{p}$ and the period of the force $\frac{2\pi}{8}$ i.e., $\frac{\pi}{4}$.

$\therefore$ $\qquad \frac{2\pi}{p} = \frac{\pi}{4}$ $\quad$ or $\quad \frac{2\pi}{8\sqrt{6}} = \frac{\pi}{4}$ $\quad$ Thus the weight, $w = 6g$.

Taking this value of w, (1) takes the form

$$\frac{d^2x}{dt^2} + 4x = \frac{1}{3}\sin 8t \qquad \ldots(2)$$

We know that the free oscillations are given by the C.F. and the forced oscillations by the P.I.

Thus, when the free oscillations have died out, the forced oscillations are given by the P.I. of (2).

Now P.I. of (2) $= \frac{1}{3}\cdot\frac{1}{D^2+4}\sin 8t = \frac{1}{3}\cdot\frac{1}{-64+4}\sin 8t = -\frac{1}{180}\sin 8t$

Hence $\qquad a = \frac{1}{180}$

EXERCISES

1. An elastic string of natural length a is fixed at one end, and a particle of mass m is hung freely from the other end. The modulus of elasticity is mg. The particle is pulled down a further distance b (also $< a$) and released from rest. Show that the motion of the particle is simple harmonic and find the period.
2. A mass of 4 lb is suspended from a light elastic string and stretches it 6 in. A free particle is then released from rest at a point 3 in above the equilibrium position and falls vertically on the scale pan. Find the amplitude, the period and the maximum velocity of the resulting simple harmonic motion.

2. A light elastic string of unstretched length l has its ends rigidly fixed at points A and B on a horizontal ceiling, the length AB being $2l$. When the mass is in equilibrium it is at a depth $\sqrt{3}l$ below the ceiling. Show that if the mass is displaced vertically downwards a distance $\frac{1}{2}l$ and released from rest, it will return to its maximum upward position after a time $\sqrt{\frac{l}{g}}\ln(2+\sqrt{3})$ seconds.
3. A 4 lb weight on a spring stretches it 6 in. Assuming that a damping force is set up of 2 lb when the instantaneous velocity is 8 ft/sec, and no other weight, show that the motion is over-damped, critically damped or oscillatory according as $k > 4$, $k = 4$ or $k < 4$. Find the period of oscillation when $k = 2.5$.
4. A mass of 200 grammes is hung at the end of a spring which extends by 4 cm under a force of 200 000 dynes. The spring is pulled down and released. Find the displacement t seconds after release if there is a damping force of 800 dynes per cm per second.
5. A body weighing 30 lb is suspended by a spring in a fluid whose resistance is k lb/ft, where k is the speed of the body in ft/sec. A pull of 20 lb on the spring would stretch the spring 10 inches. The body is drawn 3 inches below the equilibrium position in the fluid and then released. Find the period of oscillations and the time taken for the damping factor to be reduced to one-tenth of its initial value. (University of Bombay, 1994)
6. A mass M is suspended from the end of a vertical spring of relaxed length a and of negligible mass. The upper end of the spring is given a displacement $b\cos nt$ and the mass oscillates in a medium whose resistance is k times the velocity. Show that the displacement x of the mass from its equilibrium position satisfies the equation

$$\frac{d^2x}{dt^2} + \frac{k}{M}\frac{dx}{dt} + \frac{g}{a}x = \frac{gb}{a}\cos nt$$

Solve this equation and find the amplitude of the forced oscillations. (University of London, 1977)

7. A vertical spring carrying a 10 lb weight is extended from its natural length of 10 inches to 20 inches. If a force of $\frac{1}{10}v$ lb is applied in the direction of the extension, where v is the velocity in ft/sec, and if the weight is released from rest at its equilibrium position, find the position of the weight at any time. Also find the amplitude, period and frequency of the resulting oscillations.
8. A body weighing 4 lb hangs at rest on a spring providing a restoring force of 1 lb for each inch of extension. The upper end of the spring is now made to execute a vertical simple harmonic motion of amplitude $\frac{1}{2}$ inch and frequency $\frac{1}{2\pi}$ cycles per second. The body is subject to a damping resistance of $\frac{1}{2}v$ lb, where v is the velocity of the body in ft/sec. Show that the differential equation of the motion of the body is $\ddot{x} + 8\dot{x} + 64x = 4\sin 8t$, where x is the displacement of the body, in inches, at time t seconds. Find the displacement of the body at any time t seconds.
9. A body executes simple harmonic motion of period $\frac{2\pi}{n}$ seconds. Show that if a resistive force of magnitude $2k$ times the velocity is introduced, the motion is damped harmonic of period $\frac{2\pi}{\sqrt{n^2-k^2}}$ seconds.
10. A body executes damped forced vibrations given by the equation

$$\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + n^2x = c^2\sin nt$$

Solve the equation for $k < n$ (the case when $k > n$, $k = n$ is left to the reader). (University of London, 2001)

14.5 OSCILLATORY ELECTRICAL CIRCUIT

(i) L-C circuit

Consider an electrical circuit containing an inductance L and capacitance C (Fig. 14.17).

Let i be the current and q the charge on the condenser plate at any time t, so that the voltage drop across

$$L = L\frac{di}{dt} = L\frac{d^2q}{dt^2}$$

and the voltage drop across $C = \frac{q}{C}$.

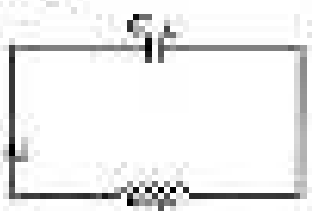

Fig. 14.17

As there is no applied e.m.f. in the circuit, therefore, by Kirchhoff's first law, we have

$$L\frac{d^2q}{dt^2} + \frac{q}{C} = 0$$

Or dividing by L and writing $1/LC = p^2$, we get $\frac{d^2q}{dt^2} + p^2q = 0$ ...(1)

This equation is precisely same as (2) on page 307 and, therefore, represents free electrical oscillations of the current having period $2\pi/p = 2\pi\sqrt{LC}$.

Thus the discharging of a condenser through an inductance L is more or less oscillatory in character at the rate of $\frac{1}{2\pi\sqrt{LC}}$ cycles per second.

(ii) L-C-R circuit

Now consider the discharge of a condenser C through an inductance L and the resistance R (Fig. 14.18). Since the voltage drops across L, C and R are respectively

$$L\frac{d^2q}{dt^2}, \frac{q}{C} \text{ and } R\frac{dq}{dt}$$

$\therefore$ by Kirchhoff's law, we have $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$ ...(1)

On writing $R/L = 2k$ and $1/LC = \mu^2$, we have $\frac{d^2q}{dt^2} + 2k\frac{dq}{dt} + \mu^2 q = 0$

This equation is same as (1) on page 309 and, therefore has the same solution as for the motion of a spring with a damper.

Thus the charging or discharging of a condenser through the resistance R and an inductance L is an electrical analogue of the damped oscillations of a mass on a spring.

(iii) L-C circuit with e.m.f. $E = E_0 \sin pt$.

For equation (1) for an L-C circuit (Fig. 14.19), now because $L\frac{d^2q}{dt^2} + \frac{q}{C} = E_0 \sin pt$.

On writing $1/LC = \mu^2$, we have $\frac{d^2q}{dt^2} + \mu^2 q = \frac{E_0}{L}\sin pt$ ...(2)

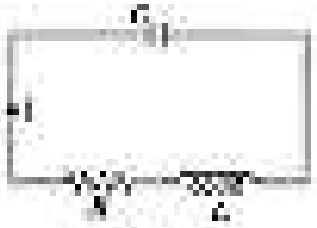

Fig. 14.18

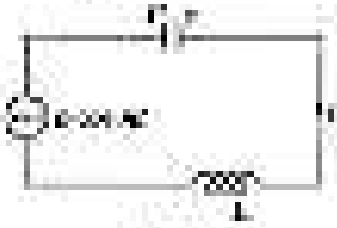

Fig. 14.19

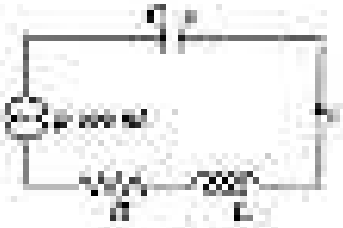

Fig. 14.20

This equation is of the same form as (1) on page 309 and, therefore, has the solution as for the motion of a mass on a spring with external periodic force p acting on it.

Thus the condenser placed in series with source $E = E_0 \sin pt$ is set up and discharging through a coil possessing inductance L is an electrical analogue of the forced oscillations of the mass on a spring.

An electrical instance of resonance phenomena occurs while tuning a radio-station, for the natural frequency of the tuning of L-C circuit is made equal to the frequency of the desired radio-station, giving the maximum output of the receiver at the said receiving station.

(iv) L-C-R circuit with e.m.f. $E = E_0 \sin pt$.

The equation of (1) above, now becomes $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = E_0 \sin pt$. ...(3)

On writing $R/L = 2k$ and $1/LC = \mu^2$ as before, we have

$$\frac{d^2q}{dt^2} + 2k\frac{dq}{dt} + \mu^2 q = \frac{E_0}{L}\sin pt \qquad ...(4)$$

This equation is exactly same as (12) on page 320 and, therefore, in C.F. we have the free oscillations and the term P.I. represents the forced oscillations.

Here also as t increases, the free oscillations die out while the forced oscillations persist giving steady motion.

Thus the L-C-R circuit with a source of alternating e.m.f. is an electrical equivalent of the mechanical phenomena of forced oscillations with resistance.

13.13 ELECTRO-MECHANICAL ANALOGY

We have just seen, how merely by rewording the variables, the differential equations representing the oscillations of a weight on a spring represents an analogous electrical circuit. As electrical circuits are easy to assemble and the current and

voltages are associated with them. This affords a practical method of studying the oscillations of complicated mechanical systems which are very difficult to tackle and, as a result, to tackle by considering an equivalent electrical circuit. While setting up an electrical equivalent of a mechanical system, the following correspondences between the electrical circuit should be kept in mind, making that the circuit may be as simple as possible.

Mechanical System	Electrical circuit	Parallel circuit
Displacement x	Current i	Voltage V
Force or impulse	Voltage E	Current i
Mass or moment of inertia	Inductance L	Capacitance C
Damping force	Resistance R	Conductance $1/R$
Spring stiffness	Elastance $1/C$	Susceptance $1/L$

Example 3.6.12. An inductance L henry and a condenser of capacity C farads are joined in series to a battery E volts. If q coulombs be the charge on the condenser at time t seconds after the circuit is completed, show that $q = \frac{EC}{2}\left\{1 - \cos\frac{t}{\sqrt{LC}}\right\}$ and $i = \frac{E}{2}\sqrt{\frac{C}{L}}\sin\frac{t}{\sqrt{LC}}$. Find also the time when the current is a maximum. (U.P.T.U., 2003)

Solution. If q be the charge on the condenser, the differential equation of the circuit is

$$L\frac{d^2q}{dt^2} + \frac{q}{C} = E\sin\frac{t}{\sqrt{LC}} \qquad ...(1)$$

Its A.E. is $LD^2 + \frac{1}{C} = 0$ i.e. $D = \pm i\sqrt{\frac{1}{LC}}$

$\therefore$ C.F. $= c_1\cos t/\sqrt{LC} + c_2\sin t/\sqrt{LC}$

and P.I. $= \frac{1}{LD^2 + \frac{1}{C}}E\sin\frac{t}{\sqrt{LC}}$ $\left[\text{Putting } D^2 = -\frac{1}{LC}, \text{ denom.} = 0\right]$

$$= E\frac{1}{2LD}\sin\frac{t}{\sqrt{LC}} = \frac{E}{2L}\int\sin\frac{t}{\sqrt{LC}}\,dt = -\frac{E}{2L}\sqrt{LC}\cos\frac{t}{\sqrt{LC}} = -\frac{E}{2}\sqrt{\frac{C}{L}}\cos\frac{t}{\sqrt{LC}}$$

Thus the C.S. of (1) is $q = c_1\cos\frac{t}{\sqrt{LC}} + c_2\sin\frac{t}{\sqrt{LC}} - \frac{E}{2}\sqrt{\frac{C}{L}}\cos\frac{t}{\sqrt{LC}}$

When $t = 0$, $q = 0$, $c_1 = \frac{E}{2}\sqrt{\frac{C}{L}}$

$$\therefore \quad q = c_2\sin\frac{t}{\sqrt{LC}} - \frac{E}{2}\sqrt{\frac{C}{L}}\cos\frac{t}{\sqrt{LC}} \qquad ...(2)$$

Differentiating (2) w.r.t. t, we get

$$\frac{dq}{dt} = \frac{c_2}{\sqrt{LC}}\cos\frac{t}{\sqrt{LC}} - \frac{E}{2}\sqrt{\frac{C}{L}}\left\{\cos\frac{t}{\sqrt{LC}} - \frac{t}{\sqrt{LC}}\sin\frac{t}{\sqrt{LC}}\right\}$$

Also when $t = 0$, $dq/dt = i = 0$,

$$\therefore \quad \frac{c_2}{\sqrt{LC}} - \frac{E}{2}\sqrt{\frac{C}{L}} = 0 \quad \text{or} \quad c_2 = \frac{EC}{2}.$$

Substituting the value of c_2 in (2), q at any time t is given by

$$q = \frac{EC}{2}\left\{1 - \cos\frac{t}{\sqrt{LC}}\right\} - \frac{t}{2\sqrt{LC}}\cos\frac{t}{\sqrt{LC}}\Big\}.$$

Example 3.6.13. In an $LR-C$ circuit the charge q on a plate of a condenser is given by

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = E\sin pt.$$

The circuit is tuned to resonance so that $p^2 = 1/LC$. If initially the current i and the charge q be zero, show that, for small values of R/L, the current in the circuit at time t is given by

$$\frac{Et}{2L}\sin pt. \qquad \text{(U.P.T.U., 2002)}$$

Solution. Given differential equation is $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E\sin pt$ ...(1)

Its auxiliary equation is $LD^2 + RD + \frac{1}{C} = 0$.

which gives $D = \frac{1}{2L}\left[-R \pm \sqrt{\left(R^2 - \frac{4L}{C}\right)}\right] = -\frac{R}{2L} \pm \sqrt{\left(\frac{R^2}{4L^2} - \frac{1}{LC}\right)}$

As R/L is small, therefore, to the first order in R/L,

$$D = -\frac{R}{2L} \pm i\frac{1}{\sqrt{(LC)}} = -\frac{R}{2L} \pm ip \qquad \left[\because p^2 = \frac{1}{LC}\right]$$

$\therefore$ $C.F. = e^{-(R/2L)t}(c_1\cos pt + c_2\sin pt)$

$= (1 - Rt/2L)(c_1\cos pt + c_2\sin pt)$ neglecting terms in R^2/L^2 etc.

and $P.I. = \frac{1}{LD^2 + RD + 1/C}E\sin pt = E\frac{1}{-Lp^2 + RD + 1/C}\sin pt$

$= \frac{E}{R}\int \sin pt\, dt = -\frac{E}{Rp}\cos pt \qquad \left[\because p^2 = \frac{1}{LC}\right]$

Thus the complete solution of (1) is $q = \left(1 - \frac{Rt}{2L}\right)(c_1\cos pt + c_2\sin pt) - \frac{E}{Rp}\cos pt$ ...(2)

$\therefore$ $i = \frac{dq}{dt} = \left(1 - \frac{Rt}{2L}\right)(-c_1\sin pt + c_2\cos pt)p - \frac{R}{2L}(c_1\cos pt + c_2\sin pt) + \frac{E}{R}\sin pt$ ...(3)

Initially, when $t = 0$, $q = 2$, $i = 0$ $\therefore$ from (3), $0 = c_2 - Rc_1p$ $\therefore$ $c_2 = Rc_1p$ and from (2),

$2 = c_1p - Rc_1p/2L = c_1 - c_1Rp/2 = Rc_1p - Rc_1p/2 = Rc_1p/2$

Thus, substituting these values of c_1 and c_2 in (3), we get

$$i = \left(1 - \frac{Rt}{2L}\right)\left(-\frac{E}{Rp}\sin pt + \frac{E}{2Lp^2}\cos pt\right)p - \frac{R}{2L}\left(\frac{E}{Rp}\cos pt + \frac{E}{2Lp^2}\sin pt\right) + \frac{E}{R}\sin pt$$

$$= \frac{Et}{2L}\sin pt. \qquad [\because \text{neglecting } t/L \text{ as } t/L \text{ is small}]$$

EXERCISES 4.3

1. Show that the frequency of free vibrations in a series electrical circuit with inductance L and capacity C is $\frac{1}{2\pi\sqrt{(LC)}}$ per second.
2. The differential equation for a circuit in which are C farads of capacity and impedance measuring each other is $R\frac{dq}{dt} + \frac{1}{C}q = 0$. Find the current i as a function of t given that I is the maximum current and $i = 0$ when $t = 0$.
3. A constant e.m.f. E at $t = 0$ is applied to a circuit consisting of inductance L, resistance R and capacitance C in series. The initial values of the current and the charge being zero, find the current at any time t. If $CR^2 = 4L$, show that the amplitudes of the successive vibrations are in geometrical progression.
4. The damped LCR circuit is governed by the equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = 0$ where L, R, C are positive constants. Find the conditions under which the current in this circuit is not damped but is periodically damped. Find also the natural frequency. [U.P.T.U., 2003]
5. A condenser of capacity C is charged through an inductance L and resistance R. If the current and the charge at time t satisfies the equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$. Show that if $L = 0.25$ henries, $R = 200$ ohms, $C = 8 \times 10^{-6}$ farads, and that at $t = 0$, charge $q = 0.001$ coulombs and the current $i = dq/dt = 0$, then the value of q is $0.0001(5e^{-200t} - 4e^{-800t})$.
6. An e.m.f. E sin pt is applied at $t = 0$ to a circuit containing a capacitance C and inductance L. The current i satisfies the equation $L\frac{di}{dt} + \frac{1}{C}\int i\,dt = E\sin pt$. If $p^2 = 1/LC$ and initially the current i and the charge q are zero, show that the current at time t is $\frac{1}{2}Et\sin pt$, where $t = Rp/E$.

7. For a 2 m × 8 cm × 12 cm steel beam, the bending moment at a point is given by $M = 2\frac{d^2y}{dx^2} + 8\frac{dy}{dx} + \frac{y}{x^2+1}$ N m, where $x = \frac{dy}{dx}$. The moment is found to be zero when $x = 2.5$ m.

If $EI = 8$ N m^2, and initially $y = 0$ and $\frac{dy}{dx} = 0$, show that $y = \frac{8}{3.5}\left[e^{-0.5x}\left(\sin 0.5x + \frac{8}{3.5}\cos 0.5x\right) - \sin x\right]$

where $\alpha^2 = \frac{2}{EI} = \frac{M}{EI}$ (University of Salford)

8. An alternating EMF, E volts, is applied to a circuit at $t = 0$. Given the equation for the current i is $L\frac{di}{dt} + Ri + \frac{1}{C}\int i\,dt = E\sin\omega t$, find the current i when $L = 0.25$ H, $R = 25\ \Omega$, $C = 16$ μF, $E = 200$ V and $\omega = 314$ rad/s.

33.5 DEFLECTION OF BEAMS

Consider a uniform beam made up of fibres running lengthwise. We have to find its deflection under given loadings.

In the bent form, the fibres of the lower half are stretched and those of upper half are compressed. In between these two, there is a layer of unstrained fibres called the neutral surface. The fibre which was initially along the beam (the neutral axis) of the beam now lies in the neutral surface, in the form of a curve called the deflection curve or the elastic curve. We shall now solve differential equations while finding the equation of this curve.

Consider a cross-section of the beam cutting the elastic curve in P and the neutral surface in the line AA'—called the neutral axis of the section (Fig. 33.23).

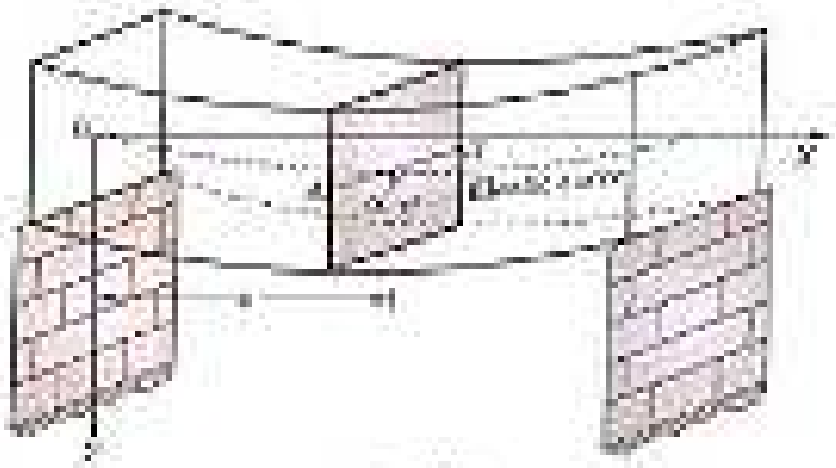

Fig. 33.23

It is well known from consideration of the bending moment M about AA', of all the forces acting on either side of the two portions of the beam separated by this cross-section, is given by the Bernoulli-Euler law

$$M = EI/R$$

where E = modulus of elasticity of the beam,

I = moment of inertia of the cross-section about AA',

and R = radius of curvature of the elastic curve at $P(x, y)$.

If the deflection of the beam is small, the slope of the elastic curve is also small so that we may neglect $(dy/dx)^2$ in the formula,

$$R = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} \Big/ \frac{d^2y}{dx^2}. \quad \text{Thus for small deflections, } R = 1/(d^2y/dx^2).$$

Hence (1) bending moment $M = EI\dfrac{d^2y}{dx^2}$

(2) Shear force $\left(=\frac{dM}{dx}\right) = EI\frac{d^3y}{dx^3}$;

(3) Intensity of loading $\left(=\frac{d^2M}{dx^2}\right) = EI\frac{d^4y}{dx^4}$

(4) *Convention of signs.* The sum of the moments about a section XX' due to external forces on the left of the section, if anti-clockwise is taken as positive and if clockwise (as in Fig. 14.22) is taken as negative.

Fig. 14.21

The deflection y downwards and length x to the right are taken as positive. The slope dy/dx will be positive if the curve rises in the direction of x positive.

(5) *End conditions.* The arbitrary constants appearing in the solution of the differential equation (1) for a given problem are found from the following end conditions :

(i) At a freely supported end (Fig. 14.22), there being no deflection and no bending moment, we have $y = 0$ and $d^2y/dx^2 = 0$.

Fig. 14.22

Fig. 14.23

(ii) At a (horizontal) fixed end (Fig. 14.23), the deflection and the slope of the beam being both zero, we have

$$y = 0 \text{ and } dy/dx = 0.$$

(iii) At a *perfectly free end* (A in Fig. 14.24), there being no bending moment or shear force, we have

$$\frac{d^2y}{dx^2} = 0 \quad \text{and} \quad \frac{d^3y}{dx^3} = 0$$

(6) A member of a structure or a machine which is subjected to end thrusts only is called a *strut* and a vertical strut is called a *column*.

There are four possible ways of the end fixation of a strut.

(i) Both ends fixed, called a *built in* or *encastre* strut.

(ii) One end fixed and the other freely supported, hinged or pin-jointed.

(iii) One end fixed and the other end free, called a *cantilever*.

(iv) Both ends freely supported or pin-jointed.

Example 14.10. *The deflection of a uniform horizontal shaft one end of which is fixed horizontally and the other supported and subjected to end thrust P, satisfies the equation*

$$\frac{d^2y}{dx^2} + a^2y = \frac{a^2R}{P}(x - l).$$

Prove that the deflection curve is $y = \frac{R}{P}\left(\frac{\sin ax}{a} - l\cos ax + (x - l)\right)$, *where* $a^2 = P/EI$.

(U.P.T.U., 2003)

Solution. Given differential equation is $(D^2 + a^2)y = \frac{a^2R}{P}(x - l)$ ...(1)

Its auxiliary equation is $D^2 + a^2 = 0$, whence $D = \pm ai$

$$\therefore \quad \text{C.F.} = \frac{1}{D^2 + a^2}\frac{a^2R}{P}(x - l) = \frac{R}{P}\left(1 + \frac{D^2}{a^2}\right)^{-1}(x - l)$$

$$= \frac{Pl}{p}\left(1 - \frac{k^2}{2!} + \ldots\right)(l - x) = \frac{Pl}{p}(l - x)$$

Thus the complete solution of (1) is $y = c_1 \cosh kx + c_2 \sinh kx + \frac{P}{p}(l - x)$ ...(3)

Also $\frac{dy}{dx} = c_1 k \sinh kx + c_2 k \cosh kx - \frac{P}{p}$ ...(4)

Now as (3) and (4) is true, when $x = 0$, $y = dy/dx = 0$ (see Fig. 14.28), $\therefore y = dy/dx = 0$ at $x = 0$.

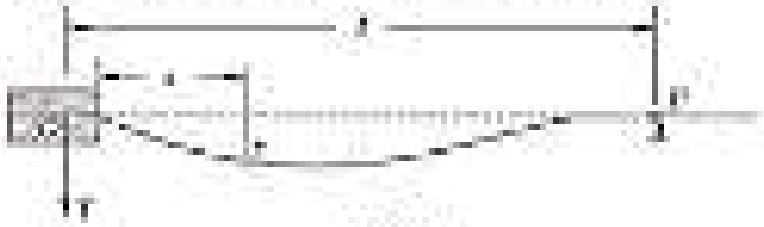

Fig. 14.28

∴ from (3) and (4), we have

$$0 = c_1 - \frac{Pl}{p} \text{ and } 0 = c_2 k - \frac{P}{p}$$

whence $c_1 = \frac{Pl}{p}$ and $c_2 = \frac{P}{kp}$

Thus (3) becomes $y = \frac{Pl}{p}\left(\cosh kx + \frac{\sinh kx}{kl} - \frac{x}{l} + 1\right)$ ...(5)

which is the general equation of the deflection curve.

The end A being freely supported, $y = 0$ when $x = l$ (We don't need the other condition $d^2y/dx^2 = 0$).

$\therefore$ (5) gives $0 = \frac{Pl}{p}\left(\frac{\sinh kl}{kl} - 1 + \cosh kl\right)$ whence $\sinh kl = \cosh kl - 1$.

Example 14.14. *A horizontal beam is freely supported at each end. It carries a uniform load of p per unit length and has a concentrated load P. Find the vertical deflection and the maximum bending moment, taking the origin at one of the ends.*

Solution. Let (14.13) be the given beam of length l (Fig. 14.29).

At each end there is a vertical reaction $R = wl/2$.

The external forces acting to the left of the section XX' are:

(i) the horizontal pull P, (ii) the reaction $R = wl/2$ and (iii) the weight of the portion OX acting midway.

Taking moments about X, we have

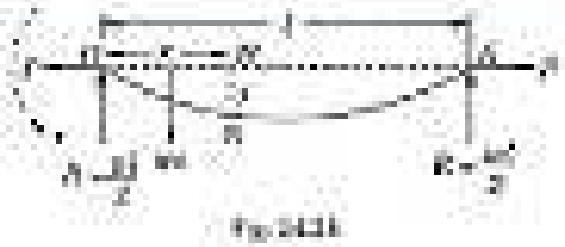

Fig. 14.29

$$EI\frac{d^2y}{dx^2} = Py - \frac{wl}{2}x + w\cdot\frac{x}{2}\cdot\frac{x}{2}$$

or $$EI\frac{d^2y}{dx^2} - Py = \frac{w}{2}(x^2 - lx) \quad \text{or} \quad \frac{d^2y}{dx^2} - a^2y = \frac{w}{2EI}(x^2 - lx), \text{ where } a^2 = \frac{P}{EI} \quad ...(1)$$

This is the differential equation of the elastic curve. Its auxiliary equation is $D^2 - a^2 = 0$, where $D \equiv d/dx$.

$\therefore$ C.F. $= c_1 \cosh ax + c_2 \sinh ax$

and $$\text{P.I.} = \frac{1}{D^2 - a^2}\frac{w}{2EI}(x^2 - lx) = \frac{-w}{2Pa^2}\left(1 - \frac{D^2}{a^2}\right)^{-1}(x^2 - lx)$$

$$= -\frac{w}{2P}\left(1 + \frac{D^2}{a^2} + \ldots\right)(x^2 - lx) = -\frac{w}{2P}\left(x^2 - lx + \frac{2}{a^2}\right)$$

Thus the complete solution of (1) is $y = c_1 \cosh ax + c_2 \sinh ax - \frac{w}{2P}\left(x^2 - lx + \frac{2}{a^2}\right)$ ...(2)

At the end O, $y = 0$ when $x = 0$. [We don't need the other condition $dy/dx = 0$]

$\therefore$ (i) gives $0 = c_1 - \alpha^2 c_2$, or $c_2 = \alpha^2 c_1$ ...(ii)

At the end A, $y = 0$ when $x = l$.

We don't need the trivial solution $c_1 = c_2 = 0$.

$\therefore$ (i) gives $0 = c_1 \cosh \alpha l + c_2 \sinh \alpha l - \dfrac{w}{2P\alpha^2} l^2$ or $c_1 \cosh \alpha l + \dfrac{w}{P\alpha^2}(1 - \cosh \alpha l)$

Hence $c_1 = -\dfrac{w}{P\alpha^2} \tanh \dfrac{\alpha l}{2}$ ...(iii)

Substituting these values of c_1 and c_2 in (i), we get

$$y = \frac{w}{P\alpha^2}\left(\cosh \alpha x - \tanh \frac{\alpha l}{2} \sinh \alpha x\right) - \frac{w}{2P}\left[x^2 - lx + \frac{2}{\alpha^2}\right]$$

which gives the deflection of the beam at N.

Thus the central deflection $= y(l/2) = EI$

$$= \frac{w}{P\alpha^2}\left(\cosh \frac{\alpha l}{2} - \tanh \frac{\alpha l}{2} \sinh \frac{\alpha l}{2} - 1\right) + \frac{wl^2}{8P} = \frac{w}{P\alpha^2}\left(\operatorname{sech} \frac{\alpha l}{2} - 1\right) + \frac{wl^2}{8P}$$

Also the bending moment is maximum at the point of maximum deflection, i.e. $= EI y''$.

$\therefore$ The maximum bending moment

$$= EI \frac{d^2 y}{dx^2}(at\ x = l/2) = EI y'' = \frac{w}{2} lx - \frac{1}{2} wx^2 = \frac{wl^2}{8} - \frac{w}{\alpha^2}\left(\operatorname{sech} \frac{\alpha l}{2} - 1\right)$$

Example 14.33. *A cantilever beam of length l and weighing w per unit length is supported in a horizontal position and compressed by a force P applied at the free end. Taking the origin at the free end and y-axis upwards, establish the differential equation of the beam and hence find the maximum deflection.*

Solution. Let $N(x, y)$ be any point of the beam ordered to move through the free end as shown in Fig. 14.23.

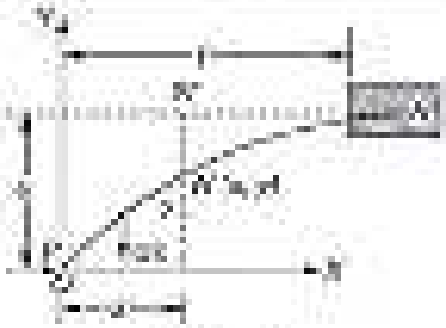

Fig. 14.23

The external forces acting to the left of the section NX', are

(i) the compressive force P,

(ii) the weight of the portion $OX' = wx$ acting midway.

$\therefore$ Taking moments about N, we get $EI \dfrac{d^2y}{dx^2} = -Py - wx \cdot \dfrac{x}{2}$

or $$EI \frac{d^2y}{dx^2} + Py = -\frac{wx^2}{2} \quad ...(i)$$

which is the desired differential equation.

Dividing by EI and taking $P/EI = \alpha^2$, we get

$$\frac{d^2y}{dx^2} + \alpha^2 y = -\frac{wx^2}{2EI} \quad ...(*)$$

Its auxiliary equation is $D^2 + \alpha^2 = 0$, whence $D = \pm \alpha i$.

$\therefore$ C.F. $= c_1 \cos \alpha x + c_2 \sin \alpha x$

$$\therefore \quad \text{P.I.} = \frac{1}{D^2 + \alpha^2}\left[-\frac{wx^2}{2EI}e^{0x}\right] = -\frac{w}{2EI}\left(1 + \frac{D^2}{\alpha^2}\right)^{-1} x^2 = -\frac{w}{2EI}\left(1 - \frac{D^2}{\alpha^2} + \ldots\right)x^2 = \frac{w}{2EI}\left(\frac{2}{\alpha^2} - x^2\right)$$

Thus the complete solution of (i) is $y = c_1 \cos \alpha x + c_2 \sin \alpha x + \dfrac{w}{2EI}\left(\dfrac{2}{\alpha^2} - x^2\right)$ ...(ii)

The boundary conditions at the fixed end are

$x = l$, $y = 0$, the maximum deflection and $dy/dx = 0$.

Using the first condition in (*), $y = 0$, when $x = l$, (ii) gives

$$0 = c_1 \cos \alpha l + c_2 \sin \alpha l + \frac{w}{2EI}\left(\frac{2}{\alpha^2} - l^2\right) \quad ...(iii)$$

Differentiating (iv), we get $\frac{dy}{dx} = n(-c_1 \sin nx + c_2 \cos nx) = \frac{Wx}{P}$

Applying the second condition, it gives $0 = n(-c_1 \sin n\ell + c_2 \cos n\ell) - \frac{W\ell}{P}$ ...(v)

Also imposing the boundary condition for the free end (i.e., $x = 0$, $d^2y/dx^2 = 0$) on

$$\frac{d^2y}{dx^2} = -n^2(c_1 \cos nx + c_2 \sin nx) = \frac{Wx}{P},$$

we get $0 = -n^2c_1 - 0/P$, i.e., $c_1 = -\frac{W}{Pn^2}$.

Substituting this value of c_1 in (v), we get $c_2 = \frac{W\ell}{Pn} \cos n\ell - \frac{1}{Pn^2} \sin n\ell$

Thus, substituting the values of c_1 and c_2 in (iv), we get

the transverse deflection $y = \frac{W}{Pn^3}\left(1 - \frac{P\ell n^2}{2} - \cos nl + n\ell \sin n\ell\right)$.

14.13 WHIRLING OF SHAFTS

(1) Critical or whirling speeds. A shaft seldom rotates about its geometrical axis for there is always some eccentricity of mass in the shaft. In fact, the dead weight of the shaft causes some deflection which tends to become large at certain speeds. Such speeds at which the deflection of the shaft reaches a maximum, whereas the shaft will fracture at the speed is least value are called the critical or whirling speeds of the shaft.

(2) Differential equation of the rotating shaft. Consider a shaft of weight W per unit length which is rotating with a angular velocity ω.

Take its original horizontal position and the vertical downwards through the end O as the axes of x and y (Fig. 14.20). We know that for a uniformly loaded beam, the intensity of loading at $P(x, y)$ is $EI\, d^4y/dx^4$.

Fig. 14.20

∴ the restoring force (i.e. the internal action or upward loading) at $P(x, y) = EI\, d^4y/dx^4$.

Also the centrifugal force per unit length at $P = (\text{mass of } 1 \text{ c.c.}) \times \omega^2 y = \frac{W}{g} \omega^2 y$.

As the restoring force arising out of the rigidity or stiffness of the shaft balances the centrifugal force which causes further deflection,

$$EI \frac{d^4y}{dx^4} = \frac{W}{g} \omega^2 y \quad \text{or} \quad \frac{d^4y}{dx^4} - n^4 y = 0, \text{ where } n^4 = \frac{W\omega^2}{gEI}$$

which is the desired differential equation.

Its auxiliary equation being $D^4 - n^4 = 0$, we have

$$D = \pm n, \pm ni.$$

Hence its solution is $y = c_1 e^{nx} + c_2 e^{-nx} + c_3 \cos nx + c_4 \sin nx$ which may be put in the form

$$y = A \cosh nx + B \sinh nx + C \cos nx + D \sin nx.$$

(3) End conditions. To determine the arbitrary constants A, B, C, D we use the following end conditions:

(i) If an end is a shaft or Rankine Scorner (Fig. 14.21), there being no deflection and also no bending moment, we have

$$y = 0 \text{ and } \frac{d^2y}{dx^2} = 0.$$

Fig. 14.21

Fig. 14.22

(c) At the end of the long arm (load between supports B and C) the deflection and the slope of the shaft (using both conventions) are

$$y = \text{[illegible]} \qquad \theta = \text{[illegible]}$$

(d) At any point in the overhanging arm (load to the left of support A), the deflection and the slope are

$$y = \text{[illegible]} \qquad \theta = \text{[illegible]}$$

Example 14.10. [illegible] Fig. 14.11.

Fig. 14.11

[illegible]

$$\text{[illegible]} \qquad (1)$$

[illegible]

$$\text{[illegible]} \qquad (2)$$

[illegible]

$$\text{[illegible]} \qquad (3)$$

[illegible]

$$\text{[illegible]} \qquad (4)$$

[illegible]

$$\text{[illegible]} \qquad (5)$$

[illegible]

$$\text{[illegible]} \qquad (6)$$

[illegible]

$$\text{[illegible]} \qquad (7)$$

[illegible]

$$\text{[illegible]} \qquad (8)$$

[illegible]

$$\text{[illegible]} \qquad (9)$$

[illegible]

$$\text{[illegible]} \qquad (10)$$

[illegible]

$$\text{[illegible]} \qquad

Again substituting $C=0$ from (iv) and (viii) from (v), we get

$$D\sinh kl + D\sin kl = 0 \text{ and } B\sinh kl - B\sin kl = k, \text{ whence } B = B \text{ and } D = k.$$

Substituting the values of A, B, C and D in (ii), we get

$$y = \frac{k}{2k^3}\left[\frac{\cosh kx}{\cosh kl} + \frac{\cos kx}{\cos kl} - 2\right]$$

Thus the maximum deflection = value of y at the centre ($x = l/2$)

$$= \frac{k}{2k^3}\left[\operatorname{sech}\frac{kl}{2} + \sec\frac{kl}{2} - 2\right].$$

Example 14.15. *The whirling speed of a shaft of length l is given by*

$$\frac{d^4y}{dx^4} - m^4y = 0 \text{ where } m^4 = \frac{\omega^2\rho A}{EI}$$

and y is the displacement at a distance x from one end. If the ends of the shaft are constrained in long bearings, show that the shaft will whirl when one revolution is $\pi\sqrt{\dfrac{EI}{\omega^2\rho A}}$.

Solution. The solution of the given differential equation is

$$y = A\cosh mx + B\sinh mx + C\cos mx + D\sin mx \quad \ldots(i)$$

which on differentiation gives,

$$\frac{dy}{dx} = mA\sinh mx + mB\cosh mx - mC\sin mx + mD\cos mx \quad \ldots(ii)$$

At the end O of the shaft is fixed in long bearings (Fig. 14.32),

$\therefore$ when $x = 0$, $y = 0$, $dy/dx = 0$,

$\therefore$ from (i) and (ii), we have

$$0 = A + C \quad \text{or} \quad C = -A \quad \ldots(iii)$$

and

$$0 = B + D \quad \text{or} \quad D = -B \quad \ldots(iv)$$

Similarly, at the end A, $x = l$, $y = 0$, $dy/dx = 0$.

$\therefore$ From (i) and (ii), we have

$$0 = A\cosh ml + B\sinh ml + C\cos ml + D\sin ml \quad \ldots(v)$$

$$0 = A\sinh ml + B\cosh ml - C\sin ml + D\cos ml \quad \ldots(vi)$$

Substituting the values of C and D in (v) and (vi), we get

$$A(\cosh ml - \cos ml) + B(\sinh ml - \sin ml) = 0$$

and

$$A(\sinh ml + \sin ml) + B(\cosh ml - \cos ml) = 0$$

Eliminating A and B from these equations, we get

$$\frac{\cosh ml - \cos ml}{\sinh ml - \sin ml} = -\frac{B}{A} = \frac{\sinh ml + \sin ml}{\cosh ml - \cos ml}$$

or

$$\cosh^2 ml - 2\cosh ml\cos ml + \cos^2 ml = \sinh^2 ml - \sin^2 ml$$

or

$$-2\cosh ml\cos ml + 2 = 0 \text{ or } \cos ml\cosh ml = 1$$

which is not be satisfied when the shaft whirls.

The solution of this equation gives $ml = 4.73 = 3\pi/2$ radians approximately.

$\therefore$

$$m\sqrt{\frac{k}{\omega^2\rho A}} = m\sqrt{I} = \frac{3\pi}{4}$$

Thus the whirling speed of a shaft with ends as long bearings

$$= \omega = \frac{2\pi}{3\pi}\sqrt{\frac{EI}{\rho A}} \text{ approximately.}$$

Obs. 1. *When the shaft has one long bearing and the other short bearing, the condition to be satisfied is*

$$\tan ml = \tanh ml, \text{ of which the solution is } ml = 3.927.$$

iv $\omega\sqrt{\left(\frac{W}{EI}\right)}\cdot P=\omega V^2=\frac{2\cdot 2617^2}{32\cdot 174}=154.4$ (see (i)).

The required velocity $=v=\frac{154.4}{\omega}\sqrt{\left(\frac{gEI}{W}\right)}$.

(Here ω, where $\omega=\sqrt{\frac{W}{EI}}$, has been chosen because the condition to be satisfied is one of $\omega l=\pi$ ($l=\omega\lambda=2\pi$) but are zero values).

$\therefore\quad \omega\sqrt{\left(\frac{W}{EI}\right)}\cdot P=\omega V^2=k^2$. Thus the required velocity $=v=\frac{k}{\omega}\sqrt{\left(\frac{gEI}{W}\right)}$.

(Here k, where $k=\omega\sqrt{\frac{W}{EI}}$, has been chosen because the condition to be satisfied is one of $\omega l=\pi$).

The actual velocity given $\omega l=1.8924$ (see Example 1 (ii))

$\therefore\quad \omega\sqrt{\left(\frac{W}{EI}\right)}\cdot P=\omega V^2=(1.8924)^2=3.6$ (say). Thus the required velocity $=v=\frac{3.6}{\omega}\sqrt{\left(\frac{gEI}{W}\right)}$.

EXERCISE XXX

1. A horizontal beam of length $2l$ with concentrated load W at the centre and ends freely hinged, satisfies the differential equation $EI\frac{d^2y}{dx^2}-Hy=\frac{W}{2}x$. With usual notation, $x=0$, $y=0$ and $x=l$, $y=0$, prove that the deflection y and the bending moment M at the centre ($x=l$) are given by $y=\frac{W}{2H}\left(\sec\frac{\omega l}{2}-1\right)$ and $M=-\frac{W}{2\omega}\tan\frac{\omega l}{2}$, where $\omega^2=\frac{H}{EI}$.
2. A light horizontal rod AB is freely pivoted at A and B. It is under the action of equal and opposite couples of moment K at its ends and it carries a load W at its centre. Then for $0\le x\le\frac{1}{2}l$, $EI\frac{d^2y}{dx^2}-Hy=\frac{1}{2}Wx-K$. Also $y=0$ at $x=0$ and $\frac{dy}{dx}=0$ at $x=\frac{1}{2}l$.

   Prove that $y=\frac{W}{2\omega^2H}\left(\frac{\sin\omega x}{\cos\omega l/2}+\omega x\right)$ where $\omega^2=\frac{H}{EI}$.
3. A uniform horizontal rod of length l is freely supported at both ends, carries a uniformly distributed load W per unit length, and the thrust at each end is H; prove that the maximum deflection is $\frac{W}{H\omega^2}\left(\sec\frac{\omega l}{2}-1\right)-\frac{Wl^2}{8H}$, where $\frac{H}{EI}=\omega^2$.

   Prove also that the maximum bending moment is of the magnitude $\frac{W}{\omega^2}\left(\sec\frac{\omega l}{2}-1\right)$.
4. The ends of a uniform flexible horizontal rod are fixed at the same level and the rod is just taut when the ends are $2l$ apart. If the rod is given a small deflection y and the tension is T at the centre, then $EI\frac{d^2y}{dx^2}+Ty=-\frac{wx^2}{2}+\frac{wlx}{2}+H$, where H is the tension at a fixed end. Find y in terms of x, given that $y=0$, $\frac{dy}{dx}=0$ at $x=0$ and $y=0$ when $x=l$ or $x=-l$.
5. A light horizontal rod of length l is clamped horizontally at one end and is vertical load W at the free end. If the horizontally fixed end is P, show that the rod satisfies the differential equation

   $EI\frac{d^2y}{dx^2}=W(l-y)(P>0)$, where y is the displacement of a point at a distance x from the fixed end and δ the deflection at the free end.

   Prove that the deflection at the free end is given by $\frac{W}{\omega^2EI}(\sec\omega l-1)$, where $\omega^2=\frac{P}{EI}$.
6. A long elastic rod is vertical with one end fixed and hinged to the other end of a fixed rod of length $2a$ under the action of a thrust P. The free P is applied laterally to the rod in the horizontal plane and the rod is slightly deflected, show that the deflection y satisfies the equation $\frac{d^2y}{dx^2}+\omega^2y=\frac{k\omega^2}{P}(a-x)$, where $k\omega^2=P$. Hence determine the equation of the deflection curve.
7. A uniform beam of length l is fixed at one end and is constrained to rest on the other end. If y is the small deflection at a point distance x from the fixed end, when load P is applied at the free end, find the differential equation satisfied by y and show that the deflection curve is given by $y=\alpha x+\beta(1-\cos\omega x)$ and find the least value of the constant ω in the interval $0<\omega<\pi$ for the free end.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$y = \frac{1}{g}\left(\frac{g}{k} + u\sin\alpha\right)(1 - e^{-kt}) - \frac{gt}{k} \qquad \ldots(iv)$$

Eliminating t from (iii) and (iv), we obtain $y = \left(\frac{g}{k} + u\sin\alpha\right)\frac{x}{u\cos\alpha} + \frac{g}{k^2}\log\left(1 - \frac{kx}{u\cos\alpha}\right)$

which is the required equation of the trajectory.

The projectile will attain the greatest height when $dy/dt = 0$.

i.e., when $e^{-kt} = g/(g + ku\sin\alpha)$, i.e., at time $t = \frac{1}{k}\log\left(1 + \frac{ku\sin\alpha}{g}\right)$. **Ans.(iii)**

Substituting the value of e^{-kt} in (iv), we get the greatest height attained

$$\therefore y = \frac{u\sin\alpha}{k} - \frac{g}{k^2}\log\left(1 + \frac{ku\sin\alpha}{g}\right).$$

Example 14.19. Two particles each of mass m are connected by a light elastic string of modulus λ and natural length l as in Fig. 14.16. After the system comes to rest the lower mass is pulled down a distance a and released. Discuss the motion.

Solution. Let x and y denote the displacements of the upper and lower masses at time t from their respective positions of equilibrium.

Fig. 14.16

Then the stretch of the upper string is y and that of the lower string is $y - x$.

$\therefore$ the restoring force acting on the upper mass

$$= -\lambda x + \lambda(y - x) = \lambda(y - 2x)$$

and that on the lower mass $= -\lambda(y - x)$.

Thus their equations of motion are

$$m\frac{d^2x}{dt^2} = \lambda(y - 2x) \text{ and } m\frac{d^2y}{dt^2} = -\lambda(y - x)$$

i.e. $(mD^2 + 2\lambda)x - \lambda y = 0$ ...(i)

and $-\lambda x + (mD^2 + \lambda)y = 0$ ...(ii)

Operating (i) by $(mD^2 + \lambda)$ and adding to λ times (ii), we get

$$[(mD^2 + 2\lambda)(mD^2 + \lambda) - \lambda^2]x = 0 \text{ or } (D^4 + 3\omega^2D^2 + \omega^4)x = 0, \text{ where } \omega^2 = \lambda/m.$$

Its auxiliary equation is $D^4 + 3\omega^2D^2 + \omega^4 = 0$

which gives $D^2 = \frac{-3\omega^2 \pm \sqrt{(5\omega^4 - 4\omega^4)}}{2} = -\frac{3}{2}\omega^2 \pm \frac{1}{2}\omega^2 = -\omega^2, -2\omega^2$ (say)

or, say $D^2 = -\alpha^2, -\beta^2$.

Thus $x = c_1\cos\alpha t + c_2\sin\alpha t + c_3\cos\beta t + c_4\sin\beta t$ ...(iii)

Also from (i), $y = \left(\frac{D^2}{\lambda} + 2\right)x = (2 - \alpha^2/\omega^2)(c_1\cos\alpha t + c_2\sin\alpha t) + (2 - \beta^2/\omega^2)(c_3\cos\beta t + c_4\sin\beta t)$ ...(iv)

Initially when $t = 0$, $x = y = l$, $dx/dt = dy/dt = 0$.

$\therefore$ from (iii), $l = c_1 + c_3$, $0 = \alpha c_2 + \beta c_4$

and from (iv) $l = (2 - \alpha^2/\omega^2)c_1 + (2 - \beta^2/\omega^2)c_3$ and $0 = (2 - \alpha^2/\omega^2)\alpha c_2 + (2 - \beta^2/\omega^2)\beta c_4$

whence $c_3 = \frac{2\lambda - \beta^2}{\alpha^2 - \beta^2}l$, $c_1 = \frac{2\lambda - \alpha^2}{\beta^2 - \alpha^2}l$, $c_2 = c_4 = 0$.

Substituting these values of constants in (iii) and (iv), we get x and y which show that the motion of the system is a superposition of two simple harmonic motions of periods $2\pi/\alpha$ and $2\pi/\beta$.

Example 14.20. Two tanks of equal cross-section are kept at the same level and connected by a horizontal pipe of length $2l$ and cross-section k times that of the tanks. Show that the period of oscillation of the liquid in the tanks is $\pi\sqrt{(l/g)}$ times the positive root of the equation $k\lambda^2 = \tan\lambda$. Also prove that there is an upper limit to the values of k for which oscillations are possible.

Solution. Let i_1, i_2 amperes be the currents flowing through the primary and secondary coils at time t sec (Fig. 14.35). Then by Kirchhoff's law, we know that sum of the voltage drops across R, L and M = applied voltage.

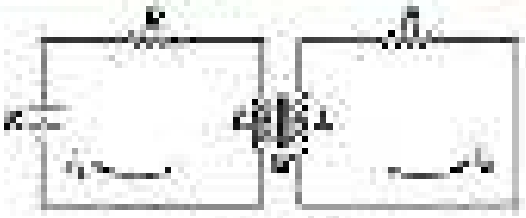
Fig. 14.35

For the primary circuit,

$$Ri_1 + L\frac{di_1}{dt} + M\frac{di_2}{dt} = E \quad \ldots(1)$$

and for the secondary circuit, $Ri_2 + L\frac{di_2}{dt} + M\frac{di_1}{dt} = 0$.

Replacing d/dt by D and rearranging the terms,

$$(LD + R)i_1 + MDi_2 = E \quad \ldots(1)$$

$$MDi_1 + (LD + R)i_2 = 0 \quad \ldots(2)$$

Eliminating i_2, we get $[(LD + R)^2 - M^2D^2]i_1 = (LD + R)E$

i.e., $$(L^2 - M^2)D^2 + 2LRD + R^2)i_1 = RE \quad \ldots(3)$$

Its auxiliary equation is $(L^2 - M^2)D^2 + 2LRD + R^2 = 0$ whose roots are $\frac{-R}{L+M}, \frac{-R}{L-M}$.

As L is usually $> M$, therefore, both values of D are negative and real.

$\therefore$ $C.F. = c_1e^{\frac{-R}{L+M}t} + c_2e^{\frac{-R}{L-M}t}$ and $P.I. = RE \cdot \frac{1}{(L^2 - M^2)D^2 + 2LRD + R^2}e^{0t} = E/R$.

Thus the complete solution of (3) is $i_1 = c_1e^{\frac{-R}{L+M}t} + c_2e^{\frac{-R}{L-M}t} + \frac{E}{R}$ ...(4)

and from (1), we have $i_2 = -\frac{RMi_1}{(LD+R)M}$

$$= -\frac{MD}{LD+R}\left[c_1e^{\frac{-R}{L+M}t} + c_2e^{\frac{-R}{L-M}t}\right] - \frac{RM}{LD+R}\left(\frac{E}{R}\right)$$

$$= -\frac{Mc_1}{L\left(\frac{-R}{L+M}\right)+R}e^{\frac{-R}{L+M}t} - \frac{Mc_2}{L\left(\frac{-R}{L-M}\right)+R}e^{\frac{-R}{L-M}t}$$

$$= c_1e^{\frac{-R}{L+M}t} - c_2e^{\frac{-R}{L-M}t}$$

Initially, when $t = 0$, $i_1 = i_2 = 0$.

$\therefore$ $c_1 + c_2 + E/R = 0$, $c_1 - c_2 = 0$ $\quad \therefore c_1 = c_2 = -E/2R$.

Substituting the values of c_1, c_2 in (4) and (5), we get

$$i_1 = \frac{E}{2R}\left[2 - e^{\frac{-R}{L+M}t} - e^{\frac{-R}{L-M}t}\right] \quad \ldots(6)$$

and $$i_2 = \frac{E}{2R}\left[e^{\frac{-R}{L+M}t} - e^{\frac{-R}{L-M}t}\right] \quad \ldots(7)$$

Eqs. (6) and (7) give the currents at any instant.

EXERCISE 14.3

1. A particle is projected with velocity v_0 at an elevation α. Neglecting air resistance, show that the equation to the path is the parabola $y = x\tan\alpha - \frac{gx^2}{2v_0^2\cos^2\alpha}$. Also find the time of flight and the horizontal range.
2. An inclined plane makes angle α with the horizontal. A projectile is projected from the bottom of the inclined plane with speed V in a direction making angle β with the horizontal. Set up the differential equations and find (i) the range on the inclined plane (ii) the maximum range on the incline.
3. A particle of unit mass is projected with velocity v at an inclination α above the horizon in a medium whose resistance is k times the velocity. Show that the direction will again make an angle α with the horizon after a time $\frac{1}{k}\log\left(1 + \frac{2kv}{g}\sin\alpha\right)$.

4. A particle moving in a plane is attracted to a fixed point O with a force proportional to the distance of the particle from O. Show that the differential equations of motion are of the form $\frac{d^2x}{dt^2} = -\mu^2 x$, $\frac{d^2y}{dt^2} = -\mu^2 y$. Find the curve in which the path of the particle (x, y) is, if $x = a$, $y = 0$, $\frac{dx}{dt} = 0$ and $y = 0$, when $t = 0$.

5. The currents i_1 and i_2 in two coupled circuits are given by the differential equations

$$L_1\frac{di_1}{dt} + Ri_1 + M\frac{di_2}{dt} = E\sin pt, \quad L_2\frac{di_2}{dt} + Ri_2 + M\frac{di_1}{dt} = 0$$

where L_1, L_2, M, R, E, p are constants. Find i_1 and i_2, in terms of t, given that $i_1 = i_2 = 0$ at $t = 0$.

6. The currents i_1 and i_2 in two coupled circuits are given by

$$L\frac{di_1}{dt} + Ri_1 + R(i_1 - i_2) = E(t), \quad L\frac{di_2}{dt} + Ri_2 + R(i_2 - i_1) = 0$$

where L, R, E are constants. Find i_1 and i_2, in terms of t, given that $i_1 = i_2 = 0$ at $t = 0$.

7. The motion of a particle is governed by the equation

$$\frac{d^2x}{dt^2} + a\frac{dx}{dt} + bx = \frac{dy}{dt} + cy, \text{ where } a, b, c \text{ are constants. Find } x \text{ and } y, \text{ in terms of } t.$$

8. Particle motion modelling, the motion of an electron is given by the equations $m\frac{d^2x}{dt^2} + kx\left(\frac{dx}{dt}\right) = -kl$ and $m\frac{d^2y}{dt^2} + ky\left(\frac{dy}{dt}\right) = 0$. Find the path of the electron, if it is released from rest at the origin.

9. Two voltages V and $2V$ are connected to a device in series. The device is modelled by the equation $\frac{dV}{dt} + kV = 3V$, where $k > 0$ is a constant. If $V = V_0$ at $t = 0$ and $V \to 0$ as $t \to \infty$, find the expression for V. Show that $V = V_0$ at $t = \frac{\ln 3}{k}$. Find $\lim_{t \to \infty} V$ and $\lim_{t \to \infty} \frac{dV}{dt}$. Interpret the results.

3.10 OBJECTIVE TYPE OF QUESTIONS

Multiple Choice

Choose the correct or most appropriate answer in the following problems.

1. A particle executing simple harmonic motion of unit mass has a speed of v metres when at a distance of x metres from the centre of the path. The period of the motion of the particle is 2π sec.
   (a) $\sqrt{(x^2 + v^2)}$ sec (b) $x + v$ sec (c) $\frac{x}{v}$ sec (d) $\frac{v}{x}$ sec

2. A ball of mass m is suspended from a fixed point O on a light string of natural length l and modulus of elasticity λ. If the ball is displaced vertically, its motion will be (UPTU 2002 (Part))
   (a) $\frac{d^2x}{dt^2} + \frac{\lambda}{ml}x = 0$ (b) $\frac{d^2x}{dt^2} + \frac{\lambda}{ml}x = g$ (c) $\frac{d^2x}{dt^2} + \frac{\lambda}{ml}x = \frac{\lambda}{ml}$ (d) $\frac{d^2x}{dt^2} + \frac{\lambda}{ml}x = \frac{\lambda}{mg}$

3. The period in time of the motion governed by the differential equation $\frac{d^2x}{dt^2} + 4x = 0$ is
   (a) 2π (b) π (c) $\frac{\pi}{2}$

4. A particle is projected with a velocity u at an angle of $60°$ to the horizontal. The latus rectum of the parabola is equal to
   (a) $\frac{\sqrt{3}u^2}{2g}$ (b) $\frac{\sqrt{3}u^2}{g}$ (c) $\frac{u^2}{4g}$ (d) $\frac{u^2}{g}$

5. A body of 10 kg is suspended by two strings of lengths 5 and 12 metres attached to two points at the same horizontal level whose distance apart is 13 metres. The tensions of the strings are
   (a) 2 kg & 8 kg (b) 2.5 kg & 8 kg (c) 2.56 kg & 8.32 kg (d) 8 kg & 6 kg

6. A particle is projected at an angle of $30°$ to the horizontal with a velocity of 196 cm/sec. Then the time of flight is
   (a) 1 sec (b) 2 sec (c) 3.5 sec (d) 3 sec

7. A point moves in a S.H.M. whose period is 6 seconds. It is 3 cm from one end of a diameter of its motion. How much time will elapse before it is 3 cm from the other end of the diameter?
   (a) $\frac{1}{6}$ second (b) $\frac{2}{3}$ second (c) $\frac{3}{4}$ second (d) $\frac{4}{3}$ second

CHAPTER 15

Differential Equations of Other Types

1. Introduction. 2. Equations of the form $d^2y/dx^2 = f(x)$. 3. Equations of the form $d^2y/dx^2 = f(y)$. 4. Equations which do not contain y. 5. Equations which do not contain x. 6. Equations whose one solution is known. 7. Equations which can be solved by changing the dependent variable. 8. Total differential equation: $Pdx + Qdy + Rdz = 0$. 9. Simultaneous total differential equations. 10. Equations of the form $ax^2 + bxD + cD^2$.

15.1 INTRODUCTION

In this chapter, we propose to study some other important types of ordinary differential equations which require special methods for their solution and have varied applications as illustrated side by side.

15.2 EQUATIONS OF THE FORM $d^2y/dx^2 = f(x)$

Integrating with respect to x, we have: $\frac{dy}{dx} = \int f(x)dx + c_1 = F(x) + c_1$ (say)

Again integrating, we get $y = \int F(x)dx + c_1 x + c_2$ is the required solution.

In general, the solution of the equation of the form $\frac{d^n y}{dx^n} = f(x)$ is obtained by integrating it n times successively.

Example (1). Solve $\frac{d^2y}{dx^2} = e^{2x}$.

Solution. Integrating, we get $\frac{dy}{dx} = \frac{e^{2x}}{2} + c_1 = \frac{1}{2}e^{2x} + c_1$

Again integrating, we get

$$y = \frac{1}{4}e^{2x} + c_1 x + c_2 = \int \frac{1}{2}e^{2x}dx + c_1 x + c_2 = \frac{1}{4}e^{2x} + c_1 x + c_2$$

PROBLEMS 15.1

Solve:

1. $\frac{d^2y}{dx^2} = e^{2x} + \sin x$

2. $\frac{d^2y}{dx^2} = x + \log x$

6. A beam of length $2l$ and weight w per unit length is freely supported at both ends. Prove that the maximum deflection of the beam is $\frac{5wl^4}{384EI}$.

[Hint. Taking the origin at the left end, we have $EI\frac{d^2y}{dx^2} = \frac{wlx}{2} - \frac{wx^2}{2}$ with $y = 0$ at $x = 0$ and $\frac{dy}{dx} = 0$ at $x = l$]

7. For a cantilever beam of length l with a uniform load of w per unit length, show that the maximum deflection at the free end is $\frac{wl^4}{8EI}$, where the symbols have the usual meaning.

3.8 EQUATIONS OF THE FORM $d^2y/dx^2 = f(y)$

Multiplying both sides by $2dy/dx$, we have $2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} = 2f(y)\frac{dy}{dx}$

Integrating with respect to x, $\left(\frac{dy}{dx}\right)^2 = 2\int f(y)\,dy + c = F(y) + c$, say

or $$\frac{dy}{dx} = \sqrt{F(y)+c}$$

Separating the variables and integrating, we get $\int \frac{dy}{\sqrt{F(y)+c}} = x + c$, where c follows the desired solution.

Such equations occur quite frequently in dynamics.

Example 3.8.1. Solve $d^2y/dx^2 = 4y^3 + y$ given that $y = 0$ and $dy/dx = 1$, when $x = 0$. (U.P.T.U., 2003)

Solution. Multiplying by $2\,dy/dx$, the given equation becomes

$$2\frac{dy}{dx}\cdot\frac{d^2y}{dx^2} = 4y^3 + y\cdot 2\frac{dy}{dx}$$

Integrating w.r.t. x, $\left(\frac{dy}{dx}\right)^2 = 4\left[\frac{y^4}{4} - \frac{y^2}{2}\right] + c = y^4 + 2y^2 + c$ ...(1)

At $y = 0$ and $dy/dx = 1$, $\quad \therefore \quad c = 1$

$\therefore$ (1) takes the form $(dy/dx)^2 = y^4 + 2y^2 + 1 = (y^2 + 1)^2$ or $dy/dx = y^2 + 1$

Separating the variables and integrating, we have $\int \frac{dy}{1+y^2} = \int dx + c'$

or $$\tan^{-1} y = x + c' \qquad ...(2)$$

Then (2) becomes $\tan^{-1} y = x$ or $y = \tan x$ which is the required solution.

Example 3.8.2. *A point moves in a straight line towards a centre of force $\mu/(\text{distance})^2$, starting from rest at a distance 'a' from the centre of force. Show that the time taken to reach a point at distance $\frac{1}{2}a$ from the centre is* $\sqrt{\frac{a}{2\mu}}\left(\sqrt{2} + \log(1+\sqrt{2})\right)$ *and that its velocity at that point is* $2\sqrt{\frac{\mu}{a}}$. (U.P.T.U., 2004)

Solution. Let O be the centre of force and A the point of start so that $OA = a$. At any time t, let the point be at P so that $OP = x$ as in the figure.

$$\frac{d^2x}{dt^2} = \frac{-\mu}{x^2} \qquad ...(1)$$

Multiplying both sides by $2\,dx/dt$, we get

$$2\frac{dx}{dt}\cdot\frac{d^2x}{dt^2} = -\frac{\mu}{x^2}\cdot 2\frac{dx}{dt}$$

Fig. 3.8.1

Integrating both sides, we obtain

$$\left(\frac{dx}{dt}\right)^2 = -\mu\int\left(\frac{2}{x^3}\frac{dx}{dt}\right)dt + c = +\frac{\mu}{x^2} + c$$

When $x = a$, velocity $dx/dt = 0$, $\therefore c = -\mu/a^2$.

$$\therefore \quad \left(\frac{dx}{dt}\right)^2 = \mu\left[\frac{1}{x^2} - \frac{1}{a^2}\right] = \frac{\mu(a^2 - x^2)}{a^2x^2} \qquad \ldots(i)$$

At B ($x = b$), velocity $= v$ (say), $\therefore v^2 = \dfrac{\mu(a^2 - b^2)}{a^2b^2}$

Again (i) can be rewritten as $\dfrac{-dx}{\sqrt{(a^2 - x^2)}} = \sqrt{\mu}\,dt$ [$-ve$ is taken as x decreases as P is moving towards O]

Integrating both sides, we get

$$\sqrt{\mu}\int dt = -\int\frac{dx}{\sqrt{(a^2 - x^2)}} + C \quad \text{or} \quad \sqrt{\mu}\,t = \cos^{-1}\frac{x}{a} + C \qquad \ldots(ii)$$

When $t = 0$, $x = a$, $\therefore C = 0$

$$\text{Also } v = \frac{b}{\sqrt{\mu}}\sqrt{(a^2 - b^2)}$$

Hence at B, $(x = b)$, $t = \dfrac{a}{\sqrt{\mu}}\sqrt{(a^2 - b^2)}$.

PROBLEMS 15.5

1. $d^2x/dt^2 = k^2x$; given that $x = a$, $dx/dt = 0$ when $t = 0$.
2. $\dfrac{d^2x}{dt^2} = \dfrac{2k}{x^2}$; given that when $x = a$, $\dfrac{dx}{dt} = 0$, $t = 0$.
3. $(dx/dt)^2 = a/x$; find the value of t in terms of x, if $x = a$ and $dx/dt = \sqrt{a}$ when $t = 0$.
4. The velocity of a particle is k/x where x is the distance from the origin in a straight line. If $x = a$ when $t = 0$, find t in terms of x.
5. A particle is acted upon by a force $\mu(x + a/x^2)$ per unit mass towards the origin, where x is the distance from the origin at time t. If it starts from rest at a distance a, show that it will arrive at the origin in time $\pi/4\sqrt{\mu}$.

15.6 EQUATIONS WHICH DO NOT CONTAIN y

A second order equation of this form is

$$f(d^2y/dx^2, dy/dx, x) = 0$$

On putting $dy/dx = p$ and $d^2y/dx^2 = dp/dx$, it becomes

$$f(dp/dx, p, x) = 0.$$

This is an equation of the first order in x and p and may, therefore, be solved easily.

If the solution is $\phi(p) = dy/dx = \psi(x)$, then $y = \int\psi(x)dx + c$ is the required solution.

Note. This method may be used to reduce any such equation of the second order in which dy/dx is absent. However, the lowest derivative in such an equation is d^2y/dx^2.

e.g. $(x + y)\dfrac{d^2y}{dx^2} + y\left(\dfrac{dy}{dx}\right)^2 = 0$ is not of this form, but $\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^2 = 0$ is of this form.

Example 15.14. Solve $x\dfrac{d^2y}{dx^2} = \sqrt{1 + \left(\dfrac{dy}{dx}\right)^2}$

Solution. Putting $dy/dx = p$ and $d^2y/dx^2 = dp/dx$, the given equation becomes

$$x\frac{dp}{dx} = \sqrt{1 + p^2}.$$

Separating the variables and integrating, we get

$$\int \frac{dp}{\sqrt{1 + p^2}} = \int \frac{dx}{x} + \text{constant}$$

or $$\log\left[p + \sqrt{1 + p^2}\right] = \log x + \log c = \log cx$$

$$\therefore \quad p + \sqrt{1 + p^2} = cx \quad \text{or} \quad \sqrt{1 + p^2} = cx - p$$

or $$(cx - p)^2 = 1 + p^2 \quad \text{or} \quad p = \frac{dy}{dx} = \frac{1}{2}\left(cx - \frac{1}{cx}\right).$$

$\therefore$ Integrating again, we have $y = \frac{1}{2}\left(\frac{cx^2}{2} - \frac{1}{c}\log x\right) + c'$ as the required solution.

Example 5.13. Solve $x^2\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$.

Solution. Putting $d^2y/dx^2 = p\,dp/dy$ and $dy/dx = p$, the given equation becomes $x\frac{dp}{dx} = p^2$.

Integrating w.r.t.x, $\int \frac{dp}{p^2} = \int \frac{dx}{x} + c_1$ or $\log p = \log cy = \log cx + c_1$ or $\log(p - cy) = \log cx + c_1$.

Integrating once more suitably, we get

$$\frac{dy}{dx} = cx\frac{(cx + c_1)^{3/2}}{3/2} + c_2 \quad \text{or} \quad \frac{dy}{dx} = \frac{2\sqrt{2}}{3}\frac{(cx + c_1)^{3/2}}{c^2} + c_2 \quad \text{or} \quad y = \frac{4\sqrt{2}}{15}\frac{(cx + c_1)^{5/2}}{c^3} + c_2\frac{x^2}{2} + c_3 x + c_4$$

Hence $y = \frac{8\sqrt{2}}{15c^3}(cx + c_1)^{5/2} + \frac{1}{2}c_2x^2 + c_3x + c_4$ is the desired solution.

EXERCISE 5.5

Solve the following equations:

1. $x\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + xy = 0$

2. $(1 + x^2)\frac{d^2y}{dx^2} + 1 + \left(\frac{dy}{dx}\right)^2 = 0$

3. $2x\frac{d^2y}{dx^2} - \frac{dy}{dx} = \left(\frac{dy}{dx}\right)^2 - x^2$

4. $x\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = a\frac{d^2y}{dx^2}$

5. A particle of mass m is given a velocity v_0 and moves in a horizontal line on which a particle experiences a resistance proportional to the square of its speed at any instant. Show that the distance it moves in time t is

   (a) $\frac{m}{k}\log\left(\frac{kv_0t}{m} + 1\right)$ (b) the particle moves with exponentially decreasing velocity and never comes to rest in a finite distance.

6. When the value of the resistance against motion of radius r is $\frac{k}{r^2}$ ($r > a$), show that the velocity v acquired in falling from infinity to the point r is $\sqrt{\frac{2k}{a}}$. Hence show that a body projected with velocity $\sqrt{\frac{2k}{a}}$ will escape to infinity.

7. $\frac{d^2y}{dx^2} = \frac{2}{y}\left(\frac{dy}{dx}\right)^2$ 8. Solve the equation $x = c_1y^2 + c_2y + c_3$ and $x = c_1y^2 + c_2y + c_3$ when $c_1 = c_2$.

13.5. EQUATIONS WHICH DO NOT CONTAIN x

A second order equation of the form is

$$f(y'', y', y) = 0, \text{ i.e., } y \text{ is absent.}$$

On putting $\frac{dy}{dx} = p$ and $\frac{d^2y}{dx^2} = \frac{dp}{dx} = \frac{dy}{dx}\cdot\frac{dp}{dy} = p\frac{dp}{dy}$, it becomes

$$f\left(p\frac{dp}{dy}, p, y\right) = 0.$$

This is an equation of the first order in y and p and can, therefore, be solved easily.

Example 13.6. Solve $y\frac{d^2y}{dx^2} + \frac{dy}{dx}\left[\frac{dy}{dx} + a(y)\right] = 0$.

Solution. On putting $dy/dx = p$ and $d^2y/dx^2 = p\,dp/dy$, the given equation becomes

$$py\frac{dp}{dy} + p(p + a(y)) = 0$$

Dividing throughout by $p \neq 0$, of which the solution is $y = c_1$,

or $$\left[y\frac{dp}{dy} + p\right] + a(y) = 0 \quad \text{i.e., } \frac{1}{y}[y(p) + p(y)] + a(y)\,dy = (xy) + a(y)\,dy = 0.$$

Integrating, $xy + \int a(y)\,dy + c_1 = xy + c_1 = 0$

Separating the variables and integrating, we get

$$\int \frac{dy}{y^2 + c_1} = \int dx + c_2 \quad \text{or} \quad \frac{1}{2}\log(y^2 + c_1) = x + c_2 \text{ whence } y^2 + c_1 = e^{2x}$$

Hence the required solution is $y = e^{2x} + c_1 = e^{2x} + c_2 e^{2x}$.

Example 13.7. *Find the curve in which the radius of curvature is twice the normal and is of opposite direction.*

Solution. At any point $P(x, y)$ of a curve, the radius of curvature

$$\rho = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} \Big/ \frac{d^2y}{dx^2}$$

and the length of the normal (PN)

$$= y\sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$$

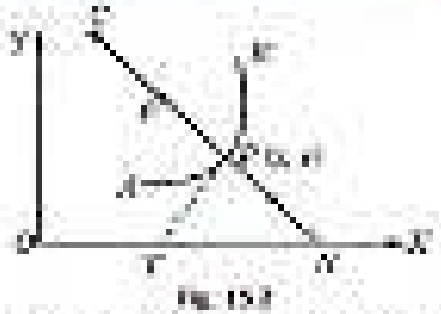

Fig. 13.1

Also we know that ρ is measured towards and the normal is measured outwards, i.e., both of them are positive when measured in opposite directions. So the sign will be positive (or negative) according as y and the curvature are in the opposite (or same) directions.

Thus for the given curve $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} \Big/ \frac{d^2y}{dx^2} = 2y\sqrt{\left[1 + \left(\frac{dy}{dx}\right)^2\right]}$

or $$1 + \left(\frac{dy}{dx}\right)^2 = 2y\frac{d^2y}{dx^2}.$$

On putting $dy/dx = p$ and $d^2y/dx^2 = p\,dp/dy$, the given equation becomes

$$1 + p^2 = 2y \cdot p\frac{dp}{dy}$$

$\therefore$ separating variables and integrating, we have

$$\int \frac{2p\,dp}{1 + p^2} = \int \frac{dy}{y} + \text{constant}$$

or $$\log(1 + p^2) = \log y + \log a = \log ay$$

$\therefore$ $$1 + p^2 = ay \text{ or } p = \pm\sqrt{(ay - 1)}$$

A. Separating the variables and integrating, we get

$$\int dx + A = \int \log y - 1)^{1/2} dy$$

or $$x + A = \frac{2}{3}(\log y - 1)^{3/2} \quad \text{or} \quad x^2(x + 6)^2 = 4(\log y - 1)^3$$

which is required equation of the curve and represents a system of parabolas having axes parallel to y-axis.

EXERCISE 1.11

Solve the following equations :

1. $x\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 + 1 = 0$

2. $x^2\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$

3. $x\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = y\frac{dy}{dx}$

4. $x^2y\frac{d^2y}{dx^2} = (y - x\frac{dy}{dx})\log\left(\frac{dy}{dx}\right)$

5. Find the curve in which the radius of curvature at any point is the normal and is in the same direction.

1.13 EQUATIONS WHOSE ONE SOLUTION IS KNOWN

Consider the equation $d^2y/dx^2 + P(dy/dx) + Qy = R$, where P, Q and R are functions of x only. If $y = u(x)$ is a known solution of this equation, then put $y = ux$ in it. It reduces the differential equation in x and u of first order so as to be solved by method.

One integral belonging to the C.F. can be found by inspection as follows :

(i) If $1 + P + Q = 0$, then $y = e^x$ is a solution.

(ii) If $1 - P + Q = 0$, then $y = e^{-x}$ is a solution.

(iii) If $P + Qx = 0$, then $y = x$ is a solution.

Example 13.1. Solve $x\frac{d^2y}{dx^2} - (2x - 1)\frac{dy}{dx} + (x - 1)y = 0$ (Meerut 2003(P))

Solution. The given equation is $$\frac{d^2y}{dx^2} - \left(2 - \frac{1}{x}\right)\frac{dy}{dx} + \left(1 - \frac{1}{x}\right)y = 0 \quad ...(1)$$

Here $1 + P + Q = 1 - (2 - 1/x) + (1 - 1/x) = 0$

$\therefore$ $y = e^x$ is a part of C.F. ...(i)

Now let $y = ev$ ...(ii)

so that $$\frac{dy}{dx} = e^x v + e^x\frac{dv}{dx} \quad ...(iii) \quad \text{and} \quad \frac{d^2y}{dx^2} = e^x v + 2e^x\frac{dv}{dx} + e^x\frac{d^2v}{dx^2} \quad ...(iv)$$

Substituting (ii), (iii) and (iv) in (i), we get

$$x\left(e^x v + 2e^x\frac{dv}{dx} + e^x\frac{d^2v}{dx^2}\right) - (2x - 1)\left(e^x v + e^x\frac{dv}{dx}\right) + (x - 1)e^x v = 0$$

or removing e^x, it becomes $x\frac{d^2v}{dx^2} + \frac{dv}{dx} = 0$ or $x\frac{dp}{dx} + p = 0$, where $p = \frac{dv}{dx}$

Integrating, we get $\int\frac{dp}{p} = -\int\frac{dx}{x} + c$ or $\log p = -\log x + \log c_1$

i.e., $$p = \frac{c_1}{x} \quad \text{or} \quad \frac{dv}{dx} = \frac{c_1}{x}$$

Again integrating, we obtain $v = c_1\log x + c_2$

Hence the complete solution of (i) is $y = e^x(c_1\log x + c_2)$.

Example 19.8 Solve $(1-x^2)y'' - 2xy' + 2y = 0$, given that $y = x$ is a solution. (U.P.T.U. 2003 S)

Solution. Let $y = vx$ so that $y' = v + x\dfrac{dv}{dx}$

and $$y'' = x\frac{d^2v}{dx^2} + 2\frac{dv}{dx}$$

Substituting these in the given equation, we get

$$(1-x^2)\left(x\frac{d^2v}{dx^2} + 2\frac{dv}{dx}\right) - 2x\left(v + x\frac{dv}{dx}\right) + 2vx = 0$$

or $$(x - x^3)\frac{d^2v}{dx^2} + (2 - 4x^2)\frac{dv}{dx} = 0$$

or $$(1 - x^2)\frac{dp}{dx} + (2 - 4x^2)p = 0 \text{ where } p = \frac{dv}{dx}$$

Integrating, we get $\displaystyle\int\frac{dp}{p} + \int\frac{2 - 4x^2}{x - x^3}\,dx = c$

or $$\log p + \int\frac{2}{x}dx - \int\frac{dx}{1-x} - \int\frac{dx}{1+x} = c$$

or $$\log p + 2\log x - \log|1-x| - \log|1+x| = \log c_1$$

$$px^2(1-x)(1+x) = c_1 \text{ or } \frac{dv}{dx} = \frac{c_1(1+x)}{x^2(1-x)}$$

Again integrating, $v = c_1\int\left(\frac{2}{x} + \frac{1}{x^2} + \frac{2}{1-x}\right)dx + c_2$

or $$v = c_1[2\log|x| - x - \log|x|] + c_2$$

Hence the required complete solution is $y = x[c_1\{\log|x| - x^2 - 2\log|x|\} + c_2]$

Observation $1 + x + x^2 = 0$. That is why $y = x$ is a solution of the given equation.

PROBLEMS SET (19)

1. If $y = e^{ax}\sin bx$ is a solution of $y'' + 2ky' + k^2y = 0$, find a relation between a and b. (U.P.T.U. 2004 S)
2. Solve $y'' - 4y' + 3y = e^x + 2\sin x - 3^{2x}$.
3. Solve $x\dfrac{d^2y}{dx^2} - (2x-1)\dfrac{dy}{dx} + (x-1)y = 0$ given that $y = e^x$ is one solution. (U.P.T.U. 2007 S)
4. Solve $\sin^2 x\dfrac{d^2y}{dx^2} = 2y$, given that $y = \cot x$ is a solution. (Dharwar 2002)
5. Solve $\dfrac{d^2y}{dx^2} + \cot x\dfrac{dy}{dx} - (1 + \cot x)y = x^2e^x + \sin x$.

19.5 SOLUTIONS WHICH CAN BE SOLVED BY CHANGING THE INDEPENDENT VARIABLE

Consider the equation $$\frac{d^2y}{dx^2} + P\frac{dy}{dx} + Qy = R \qquad ...(1)$$

To change the independent variable x to z, let $z = f(x)$

Then $$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{dz}{dx} \qquad ...(2)$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dz}\cdot\frac{dz}{dx}\right) = \frac{dy}{dz}\cdot\frac{d^2z}{dx^2} + \left(\frac{dz}{dx}\right)^2\frac{d^2y}{dz^2} \qquad ...(3)$$

Substituting (2) and (3) in (1), we get $\frac{d^2y}{dz^2} + P_1\frac{dy}{dz} + Q_1y = R_1$ ...(4)

where $P_1 = \left[\frac{d^2x}{dz^2} + P\frac{dx}{dz}\right]\Big/\left(\frac{dx}{dz}\right)^2$, $R_1 = R\Big/\left(\frac{dx}{dz}\right)^2$, $Q_1 = Q\Big/\left(\frac{dx}{dz}\right)^2$

Now equation (4) can be solved by taking Q_1 as constant.

Example 10.10. Solve by changing the independent variable, $x\frac{d^2y}{dx^2} - \frac{dy}{dx} + 4x^3y = x^3$. (U.P.T.U., 2002)

Solution. Given equation is $\frac{d^2y}{dx^2} - \frac{1}{x}\frac{dy}{dx} + 4x^2y = x^2$ ...(1)

Here $P = -1/x$, $Q = 4x^2$ and $R = x^2$.

Choose z so that $Q/(dz/dx)^2$ = constant, or $4x^2/(dz/dx)^2 = 4a^2$ (say).

i.e. $\frac{dz}{dx} = \frac{2x}{a}$ or $z = \frac{x^2}{a}$

Changing the independent variable x to z by $z = x^2/a$, we get

$$\frac{d^2y}{dz^2} + P_1\frac{dy}{dz} + Q_1y = R_1 \quad ...(2)$$

where $P_1 = \left[\frac{d^2y}{dx^2} + P\frac{dy}{dx}\right]\Big/\left(\frac{dz}{dx}\right)^2 = \frac{2}{a} + \left(-\frac{1}{x}\right)\frac{2x}{a}\Big/\frac{4x^2}{a^2} = 0$

$$Q_1 = \frac{Q}{(dz/dx)^2} = \frac{4x^2}{4a^2} = 1, R_1 = \frac{R}{(dz/dx)^2} = \frac{x^2}{4a^2} = \frac{z}{4}$$

$\therefore$ (2) becomes $\frac{d^2y}{dz^2} + y = \frac{z}{4}$ or $(D^2 + 1)y = \frac{z}{4}$

Here A.E. is $D^2 + 1 = 0$, i.e., $D = \pm i$

$\therefore$ C.F. $= c_1\cos z + c_2\sin z$

$$P.I. = \frac{1}{D^2+1}\cdot\frac{z}{4} = \frac{1}{4}(1+D^2)^{-1}z = \frac{1}{4}(1 - D^2 + \ldots)z = \frac{z}{4}$$

Hence the complete solution of (2) is

$$y = c_1\cos z + c_2\sin z + \frac{z}{4} \quad \text{or} \quad y = c_1\cos x^2 + c_2\sin x^2 + \frac{x^2}{4}$$

Example 10.11. Solve $\frac{d^2y}{dx^2} + \cot x\frac{dy}{dx} + y\operatorname{cosec}^2 x = 0$.

Solution. Here $P = \cot x$, $Q = \operatorname{cosec}^2 x$

Choosing z so that $Q\left/\left(\frac{dz}{dx}\right)^2\right.$ = constant or $\operatorname{cosec}^2 x\Big/\left(\frac{dz}{dx}\right)^2$ = constant, say 1

$$\frac{dz}{dx} = \operatorname{cosec} x \text{ or } z = \int \operatorname{cosec} x\,dx = \log\tan\frac{x}{2}$$

Changing the independent variable x to z, we get

$$\frac{d^2y}{dz^2} + P_1\frac{dy}{dz} + Q_1y = R_1 \quad ...(1)$$

where $P_1 = \left[\frac{d^2y}{dx^2} + P\frac{dy}{dx}\right]\Big/\left(\frac{dz}{dx}\right)^2 = \cot x\Big/\operatorname{cosec}^2 x = \sin x\cos x$

$$Q_1 = Q\Big/\left(\frac{dz}{dx}\right)^2 = \frac{\operatorname{cosec}^2 x}{\operatorname{cosec}^2 x} = 1, R_1 = 0$$

(c) It is of the form $\frac{d^2y}{dx^2} + k^2y = 0$

Its solution is $y = c_1 \cos(kx) + c_2 \sin(kx)$

i.e., $y = c_1 \cos(\log x) + c_2 \sin(\log x)$

This is the required complete solution of (i).

EXERCISE 9.5

Solve the following equations by changing the independent variable :

1. $\frac{d^2y}{dx^2} + \tan x \frac{dy}{dx} + y\cos^2 x = 0$ (Anna, 2006) 2. $\frac{d^2y}{dx^2} - \frac{2}{x}\frac{dy}{dx} + \frac{y}{x^2} = x$

3. $\frac{d^2y}{dx^2} - \cot x \frac{dy}{dx} - \sin^2 x\, y = 0$ 4. $x\frac{d^2y}{dx^2} - (4x^2 - 1)\frac{dy}{dx} + 4x^3y = 2x^3$ (A.U. N/D, 2006)

5. $(2x+1)^2\frac{d^2y}{dx^2} + 4(2x+1)\frac{dy}{dx} - 8y = 4x^2 + 8x - 3$ (Anna, 2006)

9.53 TOTAL DIFFERENTIAL EQUATIONS

(1) An ordinary differential equation of the first order and first degree involving three variables is of the form

$$P + Q\frac{dy}{dx} + R\frac{dz}{dx} = 0 \qquad \ldots(1)$$

where P, Q, R are functions of x, y, z and x is the independent variable.

In terms of differentials, (1) can be written as

$$Pdx + Qdy + Rdz = 0 \qquad \ldots(2)$$

which is integrable only if

$$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) = 0 \qquad \ldots(3)$$

(2) Rule to solve $Pdx + Qdy + Rdz = 0$

If the condition of integrability is satisfied, consider any of the variables say z, as constant so that $dz = 0$. Then integrate the equation $Pdx + Qdy = 0$. Replace the arbitrary constant occurring in the integral by $\phi(z)$. Now differentiate the integral just obtained with respect to x, y, z. Finally, compare this result with the given differential equation to determine $\phi(z)$.

Example 9.12. Solve $(y^2 + yz)dx + (z^2 + xz)dy + (y^2 - xy)dz = 0$.

Solution. Here $P = y^2 + yz$, $Q = z^2 + xz$, $R = y^2 - xy$.

$$P\left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right) + Q\left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z}\right) + R\left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right)$$

$$= (y^2 + yz)[2z + y - (-2y - x)] + (z^2 + xz)[-y - z] + (y^2 - xy)[2y + z - z] = 0$$

Hence the condition of integrability is satisfied.

Considering z as constant, the given equation becomes

$$(y^2 + yz)dx + (z^2 + xz)dy = 0, \quad \text{or} \quad \frac{dx}{x(y+z)} + \frac{dy}{y(y+z)} = 0$$

Integrating and noting that z is a constant, we get

$$\frac{1}{z}\int \frac{dx}{x+z} + \frac{1}{z}\int\left(\frac{1}{y} - \frac{1}{y+z}\right)dy = \text{constant}$$

i.e., $\log|x+z| + \log y - \log|y+z| = \text{constant}$.

i.e., $\frac{y(x+z)}{y+z} = \text{constant} = \phi(z)$, say ...(4)

or $yz + xy - (y + x)\,dx/dy = 0$

Multiplying (i) by 1, x, y, z, we obtain

$$yz\,dx + dx/dy + (y + x)\,dx/dy - (y + x)\,dx/dy = 0$$

or $$yz\,dx + (y + x - z)\,dx/dy + (y - (y + x))\,dx/dy = 0 \quad \ldots(ii)$$

Comparing (ii) with the given differential equation, we get

$$\frac{y^2 + yz}{y} = \frac{x^2 + xz}{x + z - \phi(x)} = \frac{x^2 - xy}{y - (y + x)\phi'(xy) - \phi(xy)}$$

The solution $\frac{x^2 + xz}{x} = \frac{x^2 + xy}{x + z - \phi(xy)}$ reduces to $\phi(xy) = z$. It gives no information about $\phi(xy)$.

Taking $\frac{x^2 + xy}{x} = \frac{x^2 - xy}{y - (y + x)\phi'(xy) - \phi(xy)}$, we get

$$x^2 - xy = (y + x)[y - (y + x)\phi'(xy) - \phi(xy)] = y^2 + xy - (y + x)^2\phi'(xy) - (y + x)\phi(xy)$$

$$= y^2 + xy - (y + x)^2\phi'(xy) - xy - x\phi(xy) \qquad \text{[Proved]}$$

$$= y^2 - xy - (y + x)^2\phi'(xy)$$

i.e., $(y - x)^2\phi'(xy) = 0$, $\because \phi'(xy) \neq 0$ so that $\phi(xy) = c$

Hence the required solution is $\phi(xy + y^2) = (x + y)z$. [Proved]

Obs. Sometimes the integral is readily obtained by simply regrouping the terms of the given equation as in the example solved below.

Example 10. Solve $(x^2 + xy)\,dz + xz\,dy + (y + 2z)\,dx = 0$.

Solution. Regrouping the terms, we can write the given equation as

$$x(dz + y\,dy) + z(y\,dy + x\,dy) + 2z\,dx = 0$$

of which the integral is $\frac{z^2}{2} + yz + x^2 = c$.

PROBLEMS 2.13

Solve :

1. $(xz - y^2)\,dx + (xy - z^2)\,dy + (yz - x^2)\,dz = 0$.
2. $yz\,dx + 2xz\,dy - 3xy\,dz = 0$.
3. $(y^2 + z^2 - x^2)\,dx - 2xy\,dy - 2xz\,dz = 0$.
4. $(y^2 + yz)\,dx - (xy + xz)\,dy + (xy - xz)\,dz = 0$.
5. $(xy + 2z^2)\,dx + x(x + 2y)\,dy - z(3x + 4y)\,dz = 0$.
6. $(xy + yz + zx)(dx + dy + dz) = dx + dy + dz$.

2.13 SIMULTANEOUS TOTAL DIFFERENTIAL EQUATIONS

These equations in three variables are given by

$$\left.\begin{aligned} P\,dx + Q\,dy + R\,dz &= 0 \\ P'\,dx + Q'\,dy + R'\,dz &= 0 \end{aligned}\right\} \quad \ldots(1)$$

where P, Q, R and P', Q', R' are any functions of x, y, z.

(i) If each of these equations is integrable and have solutions $f(x, y, z) = c$ and $\psi(x, y, z) = c_1$ respectively, then these taken together constitute the solution of the simultaneous equations (1).

(ii) If neither of both the equations (1) is not integrable, then we write them as follows :

$$\frac{dx}{QR' - Q'R} = \frac{dy}{RP' - R'P} = \frac{dz}{PQ' - P'Q}$$

and solve these by the methods explained below.

2.13.1 EQUATIONS OF THE FORM $dx/P = dy/Q = dz/R$

(1) Method of grouping :

See if it is possible to take two fractions $dx/P = dz/R$ from which y can be cancelled or is absent. Solving equations in x and z only.

Here, integrating it by giving $\phi(x, y) = c$ ...(1)

Again on if one variable say z is absent or can be removed easily be with the help of (1), from the equation $xP + yQ = zR$,

Then, integrate it by giving $\psi(x, z) = c'$ ...(2)

These two independent solutions (1) and (2) taken together constitute the complete solution required.

Example 10.14. Solve $\frac{dx}{x(z^2 - y^2)} = \frac{dy}{y(x^2 + z^2)} = \frac{dz}{z(y^2 - x^2)}$

Solution. Taking the first two fractions and resolving, we get

$$\frac{dx}{x} + \frac{dy}{y} = 0 \quad \text{or} \quad xdx + ydy = 0$$

which on integration gives $x^2 + y^2 = c_1$ ...(3)

Again taking the second and third fractions and resolving, we have

$$\frac{dy}{y^3} = \frac{dz}{z^3}, \text{ i.e., } z^3dy - y^3dz = 0.$$

Its integral is $y^2 - z^2 = c_2$ ...(4)

Thus (3) and (4) taken together constitute the required solution of the given equations.

(2) Method of multipliers

By a proper choice of the multipliers λ, μ, ν which are not necessarily constants, we write

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{\lambda dx + \mu dy + \nu dz}{\lambda P + \mu Q + \nu R} \text{ such that } \lambda P + \mu Q + \nu R = 0$$

Then $\lambda dx + \mu dy + \nu dz = 0$ can be solved giving the integral $\phi(x, y, z) = c$ ...(1)

Again, search for another set of multipliers λ_1, μ_1, ν_1

so that $\lambda_1 P + \mu_1 Q + \nu_1 R = 0$

giving $\lambda_1 dx + \mu_1 dy + \nu_1 dz = 0$,

which on integration gives the relation $\psi(x, y, z) = c'$ ...(2)

Then (1) and (2) taken together constitute the required solution.

Example 10.15. Solve $\frac{dx}{x(y^2 + z^2 - x^2)} = \frac{dy}{y(z^2 + x^2 - y^2)} = \frac{dz}{z(x^2 + y^2 - z^2)}$

Solution. Using the multipliers x, y, z

each fraction $= \frac{xdx + ydy + zdz}{x^2(y^2 + z^2 - x^2) + y^2(z^2 + x^2 - y^2) + z^2(x^2 + y^2 - z^2)} = \frac{xdx + ydy + zdz}{0}$

$\therefore \quad xdx + ydy + zdz = 0$, which on integration gives the solution $x^2 + y^2 + z^2 = c_1$ ...(1)

Again using the multipliers $\frac{1}{x}, -\frac{1}{y}, -\frac{1}{z}$

each fraction $= \frac{\frac{1}{x}dx - \frac{1}{y}dy - \frac{1}{z}dz}{(y^2 - z^2 + x^2) + (z^2 + x^2 - y^2) + (x^2 - y^2 + z^2)} = \frac{\frac{1}{x}dx - \frac{1}{y}dy - \frac{1}{z}dz}{0}$ so that $\frac{dx}{x} - \frac{dy}{y} - \frac{dz}{z} = 0$

which on integration gives $\log x - \log y - \log z = \text{constant}$ or $yz = c_2 x$ ...(2)

Hence the solution of the given equation is $x^2 + y^2 + z^2 = c_1$ or $yz = c_2 x$.

EXERCISE 10.3

Solve :

1. $\frac{dx}{y^2 + z^2} = \frac{dy}{xy} = \frac{dz}{xz}$
2. $\frac{dx}{xz - y} = \frac{dy}{yz - x} = \frac{dz}{1 - z^2}$
3. $\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy}$
4. $\frac{dx}{x(2y^2 + z^2)} = \frac{dy}{y(2x^2 - z^2)} = \frac{dz}{z(z^2 - 2x^2)}$
5. $\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 - x^2)} = \frac{dz}{z(x^2 - y^2)}$
6. $\frac{dx}{x(y^2 - z^2)} = \frac{dy}{y(z^2 + x^2)} = \frac{dz}{z(x^2 + y^2)}$
7. $\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz}$
8. $\frac{dx}{x^2 + y^2} = \frac{dy}{y^2 + z^2} = \frac{dz}{z(x + y)}$
9. $\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy}$

CHAPTER 16

Series Solution of Differential Equations and Special Functions

1. Introduction 2. Validity of series solution 3. Series solution when $x = 0$ is an ordinary point 4. Frobenius method 5. Bessel's equation 6. Recurrence formulas for $J_n(x)$ 7. Equations for J_0 and J_1 8. Value of J_{-n} 9. Generating function for $J_n(x)$ 10. Equations reducible to Bessel's equation 11. Orthogonality of Bessel functions, Fourier-Bessel expansion of $f(x)$ 12. Ber and Bei functions 13. Legendre's equation 14. Rodrigue's formula, Legendre polynomials 15. Generating function for $P_n(x)$ 16. Recurrence formulas for $P_n(x)$ 17. Orthogonality of Legendre polynomials, Fourier-Legendre expansion for $f(x)$ 18. Other special functions 19. Sturm-Liouville problem, Orthogonality of eigen functions 20. Objective Type of Questions.

16.1 INTRODUCTION

Many differential equations arising from physical problems are known but have variable coefficients and do not permit a general solution in terms of known functions. Such equations can be solved by numerical methods (Chapter 20), but in many cases it is easier to find a solution in the form of an infinite convergent series.

The series solution of certain differential equations gives rise to special functions such as Bessel's function, Legendre's polynomial, Laguerre's polynomial, Hermite's polynomial, Chebyshev polynomials. Sturm-Liouville problem based on the orthogonality of functions is also included which shows that Bessel's, Legendre's and other equations can be considered from a common point of view. These special functions have many applications in engineering.

16.2 VALIDITY OF SERIES SOLUTION OF THE EQUATION

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0 \qquad ...(1)$$

can be determined with the help of the following theorems:

Def. I. If $P_0(x) \neq 0$, then $x = a$ is called an ordinary point of (1), otherwise a singular point.

2. A singular point $x = a$ of (1) is called **regular** if when (1) is put in the form

$$\frac{d^2y}{dx^2} + \frac{Q_1(x)}{x-a}\frac{dy}{dx} + \frac{R_1(x)}{(x-a)^2}y = 0,$$

$Q_1(x)$ and $R_1(x)$ possess derivatives of all orders in the neighbourhood of a.

3. A singular point which is not regular is called an **irregular singular point.**

Theorem I. When $x = a$ is an ordinary point of (1), its every solution can be represented in the form

$$y = a_0 + a_1(x-a) + a_2(x-a)^2 + \ldots \qquad ...(2)$$

Theorem II. When $x = a$ is a regular singularity of (1), at least one of the solutions can be expressed in

$$y = (x-a)^m[a_0 + a_1(x-a) + a_2(x-a)^2 + \ldots] \qquad ...(3)$$

Theorem [8]: The series $y(x)$ and $y'(x)$ is convergent on every point within the circle of convergence of its power series and is continuous within the circle (only if $f(x)$ and $g(x)$ is convergent).

8.3.3 SERIES SOLUTIONS WHEN $x = x_0$ IS AN ORDINARY POINT OF THE EQUATION

$$P_0\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2 y = 0 \qquad \ldots(1)$$

where P's are polynomials in x and $P_0 \neq 0$ for $x = x_0$.

(a) Develop the solution in the form $y = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \ldots + a_n(x - x_0)^n + \ldots$ ...(2)

(b) Calculate $y'(x)$, $y''(x)$ from (2) and substitute the values of y, y', y'' in (1).

(c) Equate to zero the coefficients of the various powers of $(x - x_0)$ and successively express $a_2, a_3, a_4, \ldots$ in terms of a_0, a_1.
(The recurrence relation may be used to save all the coefficients of x^n that is called the convergence criterion.)

(d) Substituting the values of $a_2, a_3, a_4, \ldots$ in (2), we get the desired series solution involving a_0, a_1 as the arbitrary constants.

Example 8.3.1: Solve in series about $x = 0$ the equation $\frac{d^2y}{dx^2} + y = 0$. $\qquad$ (U.P.T.U. 2004)

Solution: Since $x = 0$ is an ordinary point, we can assume the solution of $y'' + y = 0$ as

Assume the solution as $y = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots + a_nx^n + \ldots$ ...(1)

Then $\frac{dy}{dx} = a_1 + 2a_2x + 3a_3x^2 + \ldots + na_nx^{n-1} + \ldots$

and $\frac{d^2y}{dx^2} = 2 \cdot 1a_2 + 3 \cdot 2a_3x + 4 \cdot 3a_4x^2 + \ldots + n(n-1)a_nx^{n-2} + \ldots$

Substituting in the given differential equation

$2 \cdot 1a_2 + 3 \cdot 2a_3x + 4 \cdot 3a_4x^2 + \ldots + n(n-1)a_nx^{n-2} + \ldots + a_0 + a_1x + a_2x^2 + \ldots + a_nx^n + \ldots = 0$

or $(2 \cdot 1a_2 + a_0) + (3 \cdot 2a_3 + a_1)x + (4 \cdot 3a_4 + a_2)x^2 + \ldots + [(n+2)(n+1)a_{n+2} + a_n]x^n + \ldots = 0$

Equating to zero the coefficients of the various powers of x:

$a_0 = 0$ $\qquad$ (Coefficient of $x^0 = 0$)

$3 \cdot 2a_3 + a_1 = 0 \Rightarrow a_3 = -\frac{a_1}{3!}$ $\qquad$ (Coefficient of $x^1 = 0$)

$4 \cdot 3a_4 + a_2 = 0 \Rightarrow a_4 = -\frac{a_2}{3!}$ $\qquad$ (Coefficient of $x^2 = 0$)

$5 \cdot 4a_5 + a_3 = 0 \Rightarrow a_5 = -\frac{a_3}{4 \cdot 3} = \frac{a_1}{5!}$ (Coefficient of $x^3 = 0$)

Proceeding in this manner $(n+2)(n+1)a_{n+2} + a_n = 0$ (Coefficient of $x^n = 0$)

or $a_{n+2} = -\frac{a_n}{(n+2)(n+1)}$ ...(2)

which is the recurrence relation.

Putting $n = 0, 1, 2, \ldots$ in (2) we successively get $a_2 = -\frac{a_0}{2 \cdot 1}$, $a_3 = -\frac{a_1}{3 \cdot 2}$, $a_4 = -\frac{a_2}{4 \cdot 3} = \frac{a_0}{4!}$,

$a_5 = -\frac{a_3}{5 \cdot 4} = \frac{a_1}{5!}$, $a_6 = -\frac{a_4}{6 \cdot 5} = -\frac{a_0}{6!}$, and so on.

Substituting these values in (1), we get

$$y = a_0\left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots\right] + a_1\left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\right]$$

which is the required solution.

Example 1.16. Solve in series $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + y = 0$ (B.U. Oct. 2000, U.P.T.U., 2006)

Solution. Here $x = 0$ is an ordinary point since coefficients of y'', y' and y are $1-x^2$, $-x$ and 1 respectively.

Assume the solution of the given equation is

$$y = a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots + a_nx^n + \ldots \quad \ldots(1)$$

Then $$\frac{dy}{dx} = a_1 + 2a_2x + 3a_3x^2 + \ldots + na_nx^{n-1} + \ldots$$

and $$\frac{d^2y}{dx^2} = 2a_2 + 3.2a_3x + 4.3a_4x^2 + \ldots + n(n-1)a_nx^{n-2} + \ldots$$

Substituting in the given equation, we get

$$(1-x^2)[2a_2 + 3.2a_3x + 4.3a_4x^2 + \ldots + n(n-1)a_nx^{n-2} + \ldots] - x[a_1 + 2a_2x + 3a_3x^2 + \ldots + na_nx^{n-1} + \ldots] + [a_0 + a_1x + a_2x^2 + \ldots + a_nx^n + \ldots] = 0$$

Equating to zero the coefficients of the various powers of x,

$$2a_2 + a_0 = 0 \quad \text{i.e.,} \quad a_2 = -\frac{a_0}{2} \qquad [\text{coeff. of } x^0 = 0]$$

$$3.2a_3 - a_1 + a_1 = 0 \quad \text{i.e.,} \quad a_3 = -\frac{1}{3}a_1 \qquad [\text{coeff. of } x^1 = 0]$$

$$4.3a_4 - 2a_2 - 2a_2 + a_2 = 0 \quad \text{i.e.,} \quad a_4 = 0 \qquad [\text{coeff. of } x^2 = 0]$$

$$5.4a_5 - 3.2a_3 - 3a_3 + a_3 = 0 \qquad [\text{coeff. of } x^3 = 0]$$

i.e., $$20a_5 - 8a_3 = 0 \quad \text{i.e.,} \quad a_5 = -\frac{a_3}{3} \text{ and so on.}$$

In general, $(n+2)(n+1)a_{n+2} - n(n-1)a_n - na_n + a_n = 0$

or $$a_{n+2} = \frac{n-1}{n+2}a_n \quad \ldots(2)$$

which is the recurrence relation.

Putting $n = 0, 1, 2, 3, \ldots$ in (2) successively,

$$a_2 = \frac{1}{2}a_0, \quad a_3 = \frac{2}{3}a_1, \quad a_4 = \frac{3}{4}a_2 = \frac{1}{2}a_0, \quad a_5 = \frac{4}{5}a_3 = \frac{8}{15}a_1, \ldots$$

Substituting these values in (1), we get

$$y = a_0\left(1 - \frac{x^2}{2} - \frac{x^4}{2} - \ldots\right) + a_1x\left(1 - \frac{x^2}{3} - \frac{x^4}{5} - \ldots\right)$$

PROBLEMS 1.11

Solve the following equations in series :

1. $\frac{d^2y}{dx^2} + xy = 0$, given $y(0) = 0$, $y'(0) = 1$. (U.P.T.U., 2005)
2. $\frac{d^2y}{dx^2} - x^2y = 0$. (U.P.T.U., 2004)
3. $y'' + xy' + y = 0$, $y(0) = 1$, $y'(0) = 0$. (U.P.T.U., 2004)
4. $(1-x^2)y'' - 2xy' + 2y = 0$, given $y(0) = 1$, $y'(0) = 0$. (U.P.T.U., 2004)
5. $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} + x = 0$. (U.P.T.U., 2005)
6. $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} + 2y = 0$. (U.P.T.U., 2004)

1.24 FROBENIUS* METHOD : Series solution when $x = 0$ is a regular singularity of the equation

$$P_0\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2y = 0 \quad \ldots(1)$$

*A German mathematician F.G. Frobenius (1849-1917) is known for his contributions to the theory of elliptic and group.

(a) Assume the solution to be $y = x^r(a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots)$ ...(1)

(b) Substitute (1) for y, $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ in the given equation.

(c) Equate to zero the coefficient of the lowest degree term in x. It gives a quadratic equation known as the *indicial equation*.

(d) Equating to zero the coefficients of the other powers of x, find the values of $a_1, a_2, a_3, \dots$ in terms of a_0. The complete solution depends on the nature of roots of the indicial equation.

Case I. When roots of the indicial equation are distinct and do not differ by an integer. The complete solution is

$$y = a_0 y_1(x) + a_0 y_2(x)$$

where a_0 is an arbitrary constant.

Procedure (1) to (5) is to be used for solving equations of the form $x^2\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right) + (x^2 - n^2)y = 0$

(Bessel's Equation)

Solution. Here $x = 0$ is a singular point since coefficient of $\frac{d^2y}{dx^2}$ i.e. $x^2 = 0$ at $x = 0$.

Assume the series $y = a_0x^r + a_1x^{r+1} + a_2x^{r+2} + \dots$

$$\frac{dy}{dx} = ra_0x^{r-1} + (r+1)a_1x^r + (r+2)a_2x^{r+1} + \dots + (r+n)a_nx^{r+n-1} + \dots$$

and

$$\frac{d^2y}{dx^2} = r(r-1)a_0x^{r-2} + (r+1)ra_1x^{r-1} + (r+2)(r+1)a_2x^r + \dots + (r+n)(r+n-1)a_nx^{r+n-2} + \dots$$

(a) Substituting these values in the given equation, we obtain

$$[r(r-1)a_0x^r + (r+1)ra_1x^{r+1} + (r+2)(r+1)a_2x^{r+2} + \dots + (r+n)(r+n-1)a_nx^{r+n} + \dots]$$
$$+ [ra_0x^r + (r+1)a_1x^{r+1} + (r+2)a_2x^{r+2} + \dots + (r+n)a_nx^{r+n} + \dots] + [a_0x^{r+2} + a_1x^{r+3} + a_2x^{r+4} + \dots] - n^2[a_0x^r + a_1x^{r+1} + a_2x^{r+2} + \dots] = 0$$

The lowest power of x is x^r. Its coefficient equated to zero gives

$$a_0[r(r-1) + r - n^2] = 0 \quad \text{i.e.} \quad a_0(r^2 - n^2) = 0 \quad \text{or} \quad a_0 \neq 0$$

Thus the roots of the indicial equation are $r = n, -n$. Roots are distinct and do not differ by an integer.

The coefficient of x^{r+1} equated to zero gives $a_1[(r+1)r + (r+1) - n^2] = 0$ i.e. $a_1[(r+1)^2 - n^2] = 0$

$$a_1[2n+1] = 0 \quad \text{i.e.} \quad a_1 = 0 \quad \text{(1)}$$

$$a_2[(r+2)(r+1) + (r+2) - n^2] + a_0 = 0 \quad \text{i.e.} \quad a_2[(r+2)^2 - n^2] + a_0 = 0 \quad \text{(2)}$$

Equating to zero the coefficient of x^{r+n} in general, we get $a_n[(r+n)(r+n-1) + (r+n) - n^2] + a_{n-2} = 0$

$$a_n = -\frac{a_{n-2}}{(r+n)^2 - n^2} \quad \text{i.e.} \quad a_n = -\frac{a_{n-2}}{r^2 + 2nr + n^2 - n^2} \quad \text{i.e.} \quad a_n = -\frac{a_{n-2}}{r^2 + 2nr} \quad \text{...(3)}$$

When $n = 1, 3, 5, \dots$, $a_1 = 0, a_3 = 0, a_5 = 0, \dots$ etc. giving the particular solutions

$$y_1 = a_0x^r\left[1 - \frac{1}{r+2}\left(\frac{x}{2}\right)^2 + \frac{1}{(r+2)(r+4)}\left(\frac{x}{2}\right)^4 - \dots\right]$$

When $n = -n$, the particular solution is

$$y_2 = a_0x^{-n}\left[1 - \frac{1}{2-n}\left(\frac{x}{2}\right)^2 + \frac{1}{(2-n)(4-n)}\left(\frac{x}{2}\right)^4 - \dots\right]$$

Thus the complete solution is $y = c_1y_1 + c_2y_2$

$$y = C_1x^n\left[1 - \frac{1}{n+1}\left(\frac{x}{2}\right)^2 + \frac{1}{(n+1)(n+2)}\left(\frac{x}{2}\right)^4 - \dots\right]$$
$$+ C_2x^{-n}\left[1 + \frac{1}{n-1}\left(\frac{x}{2}\right)^2 + \frac{1}{(n-1)(n-2)}\left(\frac{x}{2}\right)^4 + \dots\right] \quad \text{where } C_1 = c_1a_0, C_2 = c_2a_0$$

Case (ii) When roots of the Auxiliary equations are equal (i.e. $m_1 = m_2 = m$ (say))

$$y = e^{mx}(c_1 + c_2 x) = e^{mx}\left[\frac{D}{D-m}\right]^2_{m=m} 0$$

where c_1, c_2 are the constants

Theorem (iii): [illegible]

Proof: Let $y = f(x) = c_1 x^n + c_2 x^{n-1} + c_3 x^{n-2} + \ldots$ ...(1)

$$\frac{dy}{dx} = nc_1 x^{n-1} + (n-1)c_2 x^{n-2} + (n-2)c_3 x^{n-3} + \ldots$$

and $\frac{d^2y}{dx^2} = n(n-1)c_1 x^{n-2} + (n-1)(n-2)c_2 x^{n-3} + (n-2)(n-3)c_3 x^{n-4} + \ldots$

by the given equation, we obtain

$$\begin{aligned}(\text{constant}) &= a_0[n(n-1)c_1 x^{n-2} + (n-1)(n-2)c_2 x^{n-3} + (n-2)(n-3)c_3 x^{n-4} + \ldots] \\ &+ a_1[nc_1 x^{n-1} + (n-1)c_2 x^{n-2} + (n-2)c_3 x^{n-3} + \ldots] \\ &+ a_2[c_1 x^n + c_2 x^{n-1} + c_3 x^{n-2} + \ldots]\end{aligned}$$

The lowest power of x is x^{n-2}. Its coefficient equated to zero gives $a_0 n(n-1) + a_1 n + a_2 = 0$

$$am^2 + bm + c = 0 \quad \text{or} \quad m = \alpha, \beta$$

The coefficients of $x^{n-1}, x^{n-2}, \ldots$ equated to zero give

$$a_0(n-1) + a_1 = 0, \quad a_0 = 0$$

$$a_0(n-2)c_2 + a_1 c_2 = 0, a_0(n-3)c_3 + a_1 c_3 + a_2 c_2 = 0 \text{ and so on}$$

Clearly $c_1 = c_2 = c_3 = \ldots = 0$.

Also $c_2 = -\frac{a_1}{a_0(n-1)+a_1}c_1 = -\frac{nc_1}{n(n-1)+a_1} = -\frac{nc_1}{n(n-1)+(-2m)}$ etc.

$$y = c_1 x^n\left[1 - \frac{n}{n(n-1)-2mn}x + \frac{n^2}{[n(n-1)-2mn][(n-1)(n-2)-2m(n-1)]}x^2 - \ldots\right] \quad \ldots(2)$$

Putting $n = \alpha$, the first solution is

$$y_1 = c_1 x^\alpha\left[1 - \frac{\alpha}{\alpha - 2m}x + \frac{\alpha^2}{(\alpha-2m)(\alpha-1-2m)}x^2 - \ldots\right] \quad \ldots(3)$$

This gives only one solution instead of two. To get the second solution, differentiate (2) partially w.r.t. n,

$$\frac{\partial y}{\partial n} = y\log x + x^n\left[\frac{x^2}{(\alpha-2m)^2} - \frac{x}{\alpha-2m} + \frac{x^2}{(\alpha-2m)^2(\alpha-1-2m)} + \frac{x^2}{(\alpha-2m)(\alpha-1-2m)}\right] - \ldots$$

The second solution $y_2 = \left[\frac{\partial y}{\partial n}\right]_{n=\alpha}$

$$y_2 = y_1\log x + x^\alpha\left[\frac{1}{(\alpha-2m)^2}x^2 - \frac{1}{\alpha-2m}x\right] + \left[\frac{1}{(\alpha-2m)^2} + \frac{1}{(\alpha-2m)(\alpha-1-2m)}\right]x^2 - \ldots \quad \ldots(4)$$

Hence the complete solution is $y = c_1 y_1 + c_2 y_2$. [illegible]

$$y = AC_1 + C_2 y_1\log x\left[1 - \frac{1}{\alpha-2m}x + \frac{1}{(\alpha-2m)^2}x^2 - \frac{1}{(\alpha-2m)^3}x^3 + \ldots\right]$$

$$+ C_2 x^\alpha\left[\frac{1}{(\alpha-2m)^2}x^2 - \frac{1}{(\alpha-2m)^3}x^3\left[1 + \frac{1}{2}\right] + \frac{1}{(\alpha-2m)^4}x^4\left[1 + \frac{1}{2} + \frac{1}{3}\right] - \ldots\right]$$

where $C_1 = c_1 + c_2$, $C_2 = c_2$

Case (ii). If the roots of the indicial equation are distinct and differ by an integer, multiplying a solution corresponding to the smaller root by $\ln x$ will be necessary in order to obtain a second linearly independent solution. The form of the second solution is given in the following theorem.

Theorem 11.5. *If the roots of the indicial equation are distinct and differ by an integer, multiplying a solution corresponding to the smaller root by $\ln x$ will be necessary in order to obtain a second linearly independent solution. The form of the second solution is given in the following theorem.*

Let r_1 and r_2 be the roots of the indicial equation such that $r_1 - r_2 = N$, where N is a positive integer. Then there exists a solution of the form

$$y = x^{r_1}(1 + a_1 x + a_2 x^2 + \cdots) \qquad x > 0$$

corresponding to the larger root r_1, and a second linearly independent solution of the form

$$y = Cy_1(x)\ln x + x^{r_2}(1 + b_1 x + b_2 x^2 + \cdots) \qquad x > 0$$

where C is a constant that may be zero. Here $y_1(x)$ is the solution corresponding to r_1, and $a_1, a_2, \ldots$ and $b_1, b_2, \ldots$ are coefficients determined by substituting the solution into the differential equation.

Example 11.6. Obtain the general solution of the equation

$$x^2y'' + x\left(\frac{3}{2} + x\right)y' - \frac{1}{2}y = 0$$

Notice that $x = 0$ is a singular point, since coefficient of y' is zero at $x = 0$.

(i) substituting $\quad y = a_0x^r + a_1x^{r+1} + a_2x^{r+2} + \cdots \qquad (1)$

$$\frac{dy}{dx} = ra_0x^{r-1} + (r+1)a_1x^r + (r+2)a_2x^{r+1} + \cdots$$

and $\quad \frac{d^2y}{dx^2} = r(r-1)a_0x^{r-2} + r(r+1)a_1x^{r-1} + (r+1)(r+2)a_2x^r + \cdots$

in the given equation, we obtain

$$x^2[r(r-1)a_0x^{r-2} + r(r+1)a_1x^{r-1} + (r+1)(r+2)a_2x^r + \cdots] + x\left(\frac{3}{2} + x\right)[ra_0x^{r-1} + (r+1)a_1x^r + (r+2)a_2x^{r+1} + \cdots] - \frac{1}{2}[a_0x^r + a_1x^{r+1} + a_2x^{r+2} + \cdots] = 0$$

Equating to zero the coefficients of the lowest power of x, we get $a_0[r(r-1) + \frac{3}{2}r - \frac{1}{2}] = a_0[r^2 + \frac{1}{2}r - \frac{1}{2}] = 0$, $a_0 \neq 0$,

(ii) $\quad r(r-1) + \frac{3}{2}r - \frac{1}{2} = 0$, i.e. $r = 1, -\frac{1}{2}$, i.e. the two roots are distinct and differ by an integer.

Equating to zero the coefficients of successive powers of x, we get

$$(r+1)a_1 + ra_0 + \frac{3}{2}(r+1)a_1 + a_0 = 0, \quad a_1[(r+1)(r+2) + \frac{3}{2}(r+1) - \frac{1}{2}] + a_0(r + \frac{3}{2}) = 0, \text{ and so on.}$$

(iii) $\quad a_1 = -\frac{(r+\frac{3}{2})}{(r+1)}a_0, \quad a_2 = -\frac{(r+\frac{5}{2})}{(r+2)}a_1 = \frac{(r+\frac{3}{2})(r+\frac{5}{2})}{(r+1)(r+2)}a_0, \quad a_3 = -\frac{(r+\frac{7}{2})}{(r+3)}a_2 = -\frac{(r+\frac{3}{2})(r+\frac{5}{2})(r+\frac{7}{2})}{(r+1)(r+2)(r+3)}a_0$, etc.

Thus (1) becomes

$$y = a_0x^r\left[1 - \frac{(r+\frac{3}{2})}{(r+1)}x + \frac{(r+\frac{3}{2})(r+\frac{5}{2})}{(r+1)(r+2)}x^2 - \frac{(r+\frac{3}{2})(r+\frac{5}{2})(r+\frac{7}{2})}{(r+1)(r+2)(r+3)}x^3 + \cdots\right] \qquad (2)$$

Putting $a_0 = 1$ (ignoring all the zero roots) in (2), the first solution is

$$y_1 = x^{1/2}\left[1 - \frac{5}{3}x + \frac{5\cdot7}{3\cdot4}x^2 - \frac{5\cdot7\cdot9}{3\cdot4\cdot5}x^3 + \cdots\right]$$

If we put $a_0 = -\frac{1}{2}$ in (2), the coefficients become infinite.

For the second solution differentiate (2) partially w.r.t. r, we get

$$y = a_0x^r\left[1 - \frac{(r+\frac{3}{2})}{(r+1)}x + \frac{(r+\frac{3}{2})(r+\frac{5}{2})}{(r+1)(r+2)}x^2 - \frac{(r+\frac{3}{2})(r+\frac{5}{2})(r+\frac{7}{2})}{(r+1)(r+2)(r+3)}x^3 + \cdots\right]$$

$$\frac{\partial y}{\partial r} = a_0x^r\log x\left[1 - \frac{(r+\frac{3}{2})}{(r+1)}x + \frac{(r+\frac{3}{2})(r+\frac{5}{2})}{(r+1)(r+2)}x^2 - \frac{(r+\frac{3}{2})(r+\frac{5}{2})(r+\frac{7}{2})}{(r+1)(r+2)(r+3)}x^3 + \cdots\right]$$

$$+ a_0x^r\left[-\frac{1}{(r+1)}x + \frac{(2r+4)}{(r+1)(r+2)}x^2 - \frac{(3r^2+15r+17)}{(r+1)(r+2)(r+3)}x^3 + \cdots\right]$$

(a) $\frac{dy}{dx} + 4y^2 \tan x = \left[\cos x + 4\right] \left[\frac{\sin x}{\cos x} - \frac{\cos x}{\sin x}\right] + 4\cos^2 x \left[\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}\right]^2$

$= 4\cos^2 x \left[\frac{\sin x}{\cos x} + \frac{1}{\sin x \cos x}\right]^2 = \frac{4}{\sin^2 x \cos^2 x} \left[\frac{\sin^2 x + 1}{\sin x}\right]^2$

$= \frac{4}{\sin^2 x \cos^2 x} \left[\frac{\sin^2 x}{\sin x} + \frac{1}{\sin x}\right]^2 = \frac{4}{\sin^2 x \cos^2 x} \left[\sin x + \frac{1}{\sin x}\right]^2$

The general solution is $y = \left[\frac{\sin x}{\cos x} - \frac{\cos x}{\sin x}\right] + \frac{1}{\left[\sin x + \frac{1}{\sin x}\right]}$

$= \frac{\sin^2 x - \cos^2 x}{\sin x \cos x} + \frac{\sin x}{\sin^2 x + 1} = \frac{2\sin^2 x - 1}{\sin x \cos x} + \frac{\sin x}{2 - \cos^2 x}$

Hence the complete solution is $y = \frac{1}{2}\cot 2x + \frac{1}{2}\tan x$

(b) $y = \frac{1}{2}\cot 2x + \frac{1}{2}\left[\frac{\sin x}{2 - \cos^2 x}\right]$

$= \frac{1}{2}\left[\frac{\cos 2x}{\sin 2x} + \frac{\sin x}{2 - \cos^2 x}\right] = \frac{1}{2}\left[\frac{\cos^2 x - \sin^2 x}{2\sin x \cos x} + \frac{\sin x}{1 + \sin^2 x}\right]$

where $C_1 = \frac{1}{2}c_1$, $C_2 = \frac{1}{2}c_2$

Example Each of the following differential equations has a power series solution. Find the first four terms in each case.

(a) $\frac{dy}{dx} = x + y^2$, $y(0) = 1$

(b) $\frac{d^2y}{dx^2} = xy$, $y(0) = 1$, $y'(0) = 0$

Solution (a) Since $x = 0$ is an ordinary point, we may substitute $y = \sum_{n=0}^{\infty} a_n x^n$ in the given equation.

$$y = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots \quad \ldots(1)$$

$$\frac{dy}{dx} = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots \quad \ldots(2)$$

and $\frac{d^2y}{dx^2} = 2a_2 + 3 \cdot 2a_3 x + 4 \cdot 3a_4 x^2 + \cdots \quad \ldots(3)$

so the given equation $x = y'$ gives

$$x = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots - (a_0^2 + 2a_0 a_1 x + (2a_0 a_2 + a_1^2)x^2 + (2a_0 a_3 + 2a_1 a_2)x^3 + \cdots)$$

Equating to zero the coefficients of like powers of x,

$a_1 = 0$, $2a_2 = 1 + a_0^2$, $3a_3 = 2a_0 a_1$, $4a_4 = 2a_0 a_2 + a_1^2$, ...

(a) the results are identical with those obtained in Example 2 above.

Equating to zero the coefficients of x^n, we get

$$(n+2)a_{n+2} = 2a_0 a_n + 2a_1 a_{n-1} + 2a_2 a_{n-2} + \cdots + 2a_{n-1} a_1 + a_n^2 \quad (n \geq 1)$$

or $a_{n+2} = \frac{2}{n+2}(a_0 a_n + a_1 a_{n-1} + a_2 a_{n-2} + \cdots + a_{n-1} a_1) + \frac{1}{n+2} a_n^2 \quad (n \geq 1)$

Putting $n = 1, 2, 3, \ldots$ successively in this recurrence relation

we see that the remaining coefficients $a_3, a_4, a_5, \ldots$ can be expressed in terms of the arbitrary constants a_0 and a_1.

Thus equating to zero the coefficients of successive powers of x, we get

$$\begin{aligned}
a_2 &= \tfrac{1}{2}(a_0^2 + 0) = \tfrac{1}{2}a_0^2 + 0 \\
a_3 &= \tfrac{1}{3}(2a_0 a_1 + 0) = \tfrac{2}{3}a_0 a_1 \\
a_4 &= \tfrac{1}{4}(2a_0 a_2 + a_1^2 + 0) = \tfrac{1}{4}a_0^3 + \tfrac{1}{4}a_1^2 \\
a_5 &= \tfrac{1}{5}(2a_0 a_3 + 2a_1 a_2 + 0) = \tfrac{1}{5}a_0^2 a_1 + \tfrac{1}{5}a_0 a_1^2 \\
a_6 &= \tfrac{1}{6}(2a_0 a_4 + 2a_1 a_3 + a_2^2) = \tfrac{1}{6}a_0^4 + \tfrac{1}{3}a_0^2 a_1^2 + \tfrac{1}{12}a_1^3 + \tfrac{1}{36}a_0^3 a_1 \\
a_7 &= \tfrac{1}{7}(2a_0 a_5 + 2a_1 a_4 + 2a_2 a_3) = \tfrac{1}{7}a_0^3 a_1 + \tfrac{1}{7}a_0 a_1^2 + \tfrac{1}{7}a_0^2 a_1^2 + \tfrac{1}{14}a_1^3 + \tfrac{1}{126}a_0^4 + \tfrac{1}{63}a_0^3 a_1
\end{aligned}$$

(b) $y = a_0 + a_1 x + \frac{1}{2}a_0 x^2 + \frac{1}{6}a_1 x^3 + \frac{1}{12}a_0^2 x^4 + \frac{1}{20}a_0 a_1 x^5 + \left(\frac{1}{30}a_0^3 + \frac{1}{90}a_1^2\right)x^6 + \left(\frac{1}{42}a_0^2 a_1 + \frac{1}{252}a_1^3\right)x^7 + \cdots$

$$a_r = \frac{A}{(r+3)(r+4)(r+4)(r+5)(r+6)} \quad \text{and so on.}$$

Substituting the values in (1), we get

$$y = x^{\frac{1}{2}}\left[a_0 + a_0 x - \frac{a_0}{(1!)(2)(3)}x^2 - \frac{a_0}{(2!)(3)(4)}x^3 + \frac{a_0}{(3!)(4)(5)(6)}x^4 + \frac{a_0}{(4!)(5)(6)(7)(8)}x^5 + \cdots\right]$$

Putting $a_0 = 1$, the complete solution is

$$y = x^{\frac{1}{2}}\left[\left(1 - \frac{x^2}{1\cdot 2} + \frac{x^4}{1\cdot 2\cdot 3\cdot 4} - \cdots\right) + A\left(x - \frac{x^3}{1\cdot 2\cdot 3} + \frac{x^5}{1\cdot 2\cdot 3\cdot 4\cdot 5} - \cdots\right)\right]$$

$$= x^{-\frac{1}{2}}(a_0\cos x + a_1\sin x)$$

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Solve the following equations in power series:

1. $4x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$. (U.P.T.U., 2002)
2. $x\frac{d^2y}{dx^2} - \frac{dy}{dx} + y = 0$.
3. $x\frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + 2y = 0$. (A.M.I.E., 2001)
4. $x^2\frac{d^2y}{dx^2} + (3+x)\frac{dy}{dx} + (1+x)y = 0$. (U.P.T.U., 2004)
5. $x(1-x)\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$.
6. $x\frac{d^2y}{dx^2} + y = 0$. (U.P.T.U., 2002)
7. $x\frac{d^2y}{dx^2} + (1-x)\frac{dy}{dx} + y = 0$. (U.P.T.U., 2003)
8. $x\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - 1)y = 0$. (U.P.T.U., 2003)
9. $x(1-x)\frac{d^2y}{dx^2} + (1-x)\frac{dy}{dx} + y = 0$. (U.P.T.U., 2004)
10. $x(1-x)\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$.
11. $(x+x^2)\frac{d^2y}{dx^2} + (x-2)\frac{dy}{dx} + y = 0$. (Bhopal, 2003)

8.8 BESSEL'S EQUATION*

One of the most important differential equations in applied mathematics is

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)y = 0 \qquad \ldots(1)$$

which is known as Bessel's equation of order n. Its particular solutions are called *Bessel functions of order n*. Many physical problems involving vibrations or heat conduction in cylindrical regions give rise to this equation.

Substituting $y = a_0x^r + a_1x^{r+1} + a_2x^{r+2} + \ldots$

(1) takes the form

$$a_0[r^2 - n^2]x^r + a_1[(r+1)^2 - n^2]x^{r+1} + [a_2[(r+2)^2 - n^2] + a_0]x^{r+2} + \ldots = 0$$

Equating to zero the coefficient of x^r, we obtain the indicial equation $r^2 - n^2 = 0$ (as $a_0 \neq 0$) whence $r = n$ or $-n$.

$$a_1 = a_3 = a_5 = a_7 = \ldots = 0$$

and

$$a_2 = -\frac{a_0}{(r+2)^2 - n^2} + a_4 = -\frac{a_2}{(r+4)^2 - n^2} \text{ etc.}$$

Hence we get

$$y = a_0x^n\left(1 - \frac{1}{(n+2)^2 - n^2}x^2 + \frac{1}{[(n+2)^2 - n^2][(n+4)^2 - n^2]}x^4 - \ldots\right)$$

* Named after the German mathematician and astronomer Friedrich Wilhelm Bessel (1784 – 1846) whose paper on Bessel functions appeared in 1826. He studied Astronomy at home and became director of Königsberg observatory.

For $n = n$, we get

$$y_1 = Ax^n\left\{1 - \frac{1}{4(n+1)}x^2 + \frac{1}{4^2\cdot 2!(n+1)(n+2)}x^4 - \frac{1}{4^3\cdot 3!(n+1)(n+2)(n+3)}x^6 + \ldots\right\} \quad \ldots(9)$$

and for $n = -n$, we have

$$y_2 = Bx^{-n}\left\{1 - \frac{1}{4(-n+1)}x^2 + \frac{1}{4^2\cdot 2!(-n+1)(-n+2)}x^4 - \frac{1}{4^3\cdot 3!(-n+1)(-n+2)(-n+3)}x^6 + \ldots\right\} \quad \ldots(10)$$

Case I. When n is not an integer or zero, the complete solution of (1) is $y = Ay_1 + By_2$.

If we take $A = 2^n\Gamma(n+1)$, then the solution given by (2) is called the Bessel function of the first kind of order n and is denoted by $J_n(x)$. Thus

$$J_n(x) = \left(\frac{x}{2}\right)^n\left\{\frac{1}{\Gamma(n+1)} - \frac{1}{1!\Gamma(n+2)}\left(\frac{x}{2}\right)^2 + \frac{1}{2!\Gamma(n+3)}\left(\frac{x}{2}\right)^4 - \frac{1}{3!\Gamma(n+4)}\left(\frac{x}{2}\right)^6 + \ldots\right\} \quad [\because \Gamma(n+1) = n!]$$

i.e.,
$$J_n(x) = \sum_{r=0}^{\infty}(-1)^r\left(\frac{x}{2}\right)^{n+2r}\frac{1}{r!\Gamma(n+r+1)} \quad \ldots(11)$$

and corresponding to (3), we have $J_{-n}(x) = \sum_{r=0}^{\infty}(-1)^r\left(\frac{x}{2}\right)^{-n+2r}\frac{1}{r!\Gamma(1-n+r+1)}$ $\ldots(12)$

which is called the Bessel function of the first kind of order $-n$.

Hence complete solution of the Bessel's equation (1) may be expressed in the form

$$y = AJ_n(x) + BJ_{-n}(x) \quad \ldots(13)$$

Case II. When n is zero, $y_1 = y_2$ and the complete solution of (1), which reduces to the Bessel's equation of order zero, is obtained as in Example 18.6.

Case III. When n is integer, y_1 fails to give a solution for positive values of n and y_2 fails to give a solution for negative values. Thus another independent integral for Bessel's equation (1) is required to form the general solution. We now proceed to find an independent solution of (1) when n is an integer.

Let $y = uv$ and $J_n(x)$ be a solution of (1). Substituting the values of y, $y' = uv_1' + vJ_n'$ and $y'' = uv_2'' + 2v_1'J_n' + vJ_n''$ in (1), we obtain

$$\begin{aligned} &x^2(uv_2'' + 2v_1'J_n' + vJ_n'') + x(uv_1' + vJ_n') + (x^2 - n^2)uv = 0 \\ \text{or}\quad &x(uv_2'' + 2v_1'J_n') + xJ_n'(v + xv_1') + (x^2 - n^2)uv = 0 \end{aligned} \quad \ldots(14)$$

Now since J_n is a solution of (1), therefore, $xJ_n'' + vJ_n' + (x^2 - n^2)J_n = 0$

$\therefore$ (14) reduces to $x(uv_2'' + 2v_1'J_n') + uv_1' = 0$

Dividing throughout by $x(uJ_n)$, it becomes $\frac{v_2''}{v_1'} + 2\frac{J_n'}{J_n} + \frac{1}{x} = 0$

i.e., $\frac{d}{dx}(\log v_1') + 2\frac{d}{dx}(\log J_n) + \frac{d}{dx}(\log x) = 0$ or $\frac{d}{dx}[\log(v_1'J_n^2x)] = 0$.

Integrating, $\log v_1'J_n^2x = \log A$, where $v_1'J_n^2x = A$.

$$\therefore \quad v_1' = \frac{A}{xJ_n^2} \text{ or } v = A\int\frac{dx}{xJ_n^2} + B$$

Thus $y = AJ_n(x) + BJ_n(x)\int\frac{dx}{xJ_n(x)^2}$.

Hence the complete solution of the Bessel's equation (1) is

$$y = AJ_n(x) + BY_n(x) \quad \ldots(15) \quad (\text{U.T.U., 2006})$$

where $Y_n(x) = J_n(x)\int\frac{dx}{xJ_n(x)^2}$ $\ldots(16)$

$Y_n(x)$ is called the Bessel function of the second kind of order n or Neumann function[*].

[*] Named after the German mathematician and physicist Carl Neumann (1832–1925) whose work on potential theory gave impetus to the development of integral equations by Volterra of Rome, Fredholm of Stockholm, and Hilbert of Gottingen.

[illegible]

[illegible] FELLERIAN FORMULAE FOR $J_\nu(x)$

The following recurrence formulae are readily obtained from the series expansions for $J_\nu(x)$:

(i) $\frac{d}{dx}\{x^\nu J_\nu(x)\} = x^\nu J_{\nu-1}(x)$ (ii) $\frac{d}{dx}\{x^{-\nu} J_\nu(x)\} = -x^{-\nu} J_{\nu+1}(x)$

(iii) $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$ (iv) $J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x)$

(v) $J_\nu(x) = \frac{x}{2\nu}\{J_{\nu-1}(x) + J_{\nu+1}(x)\}$ (vi) $J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x) - J_{\nu-1}(x)$

Many formulae can be readily established by the use of these relations, and it is worth noting the various properties of Bessel functions.

Example. (i) Multiplying by x changes (ii) to (i), so we have

$$\int x^{\nu+1} J_{\nu-1}(x)\,dx = x^{\nu+1} J_\nu(x) + C$$

$$\int x^{-\nu+1} J_{\nu+1}(x)\,dx = -x^{-\nu+1} J_\nu(x) + C$$

Multiplying (iii) by x^ν we have

$$x^\nu J_{\nu-1}(x) = 2\nu x^{\nu-1} J_\nu(x) - x^\nu J_{\nu+1}(x)$$

$$\int x^\nu J_{\nu-1}(x)\,dx = 2\nu \int x^{\nu-1} J_\nu(x)\,dx - x^\nu J_{\nu+1}(x) + C$$

$$=

8.23.2 (1) EXPANSIONS FOR $J_{\frac{1}{2}}$ AND $J_{-\frac{1}{2}}$

We have from (1) of page 551,

$$J_n(x) = 1 - \frac{1}{\lfloor 1 \rfloor \lfloor 2 \rfloor}\left(\frac{x}{2}\right)^2 + \frac{1}{\lfloor 2 \rfloor \lfloor 3 \rfloor}\left(\frac{x}{2}\right)^4 - \frac{1}{\lfloor 3 \rfloor \lfloor 4 \rfloor}\left(\frac{x}{2}\right)^6 + \cdots \quad \ldots(1)$$

and

$$J_n(x) = \frac{x^n}{2^n}\left[\frac{1}{\lfloor n \rfloor} - \frac{1}{\lfloor 1 \rfloor \lfloor n+1 \rfloor}\left(\frac{x}{2}\right)^2 + \frac{1}{\lfloor 2 \rfloor \lfloor n+2 \rfloor}\left(\frac{x}{2}\right)^4 - \frac{1}{\lfloor 3 \rfloor \lfloor n+3 \rfloor}\left(\frac{x}{2}\right)^6 + \cdots\right] \quad \text{(R.P.T.U., 2015)} \ldots(2)$$

Because of their special importance, the values of $J_{\frac{1}{2}}(x)$ and $J_{-\frac{1}{2}}(x)$ are given in Appendix 2 - Table II to four decimal places at intervals of 0.1. With the help of these values, the graphs of $J_{\frac{1}{2}}(x)$ and $J_{-\frac{1}{2}}(x)$ can be drawn as shown in Fig. 8.2, for $x > 0$. Their behaviour would be similar to graphs of cosine and sine functions.

Fig. 8.2

Note : The roots of the equation $J_n(x) = 0$ are useful in some physical problems. The equation has an infinite number but no finite number of real roots. The first four roots are $x = 0.0, 3.142, 6.283, 9.425$ approximately.

8.23.3 VALUE OF $J_{-\frac{1}{2}}$

We may think that $J_{\frac{1}{2}}(x)$ is the simplest of the Bessel functions but $J_{-\frac{1}{2}}(x)$ is simpler, for it can be expressed in a finite form. Taking $n = \frac{1}{2}$ in (1) of page 551, we have

$$J_{\frac{1}{2}}(x) = \left(\frac{x}{2}\right)^{\frac{1}{2}}\left[\frac{1}{\Gamma\left(\frac{3}{2}\right)} - \frac{1}{1!\,\Gamma\left(\frac{5}{2}\right)}\left(\frac{x}{2}\right)^2 + \frac{1}{2!\,\Gamma\left(\frac{7}{2}\right)}\left(\frac{x}{2}\right)^4 - \cdots\right]$$

$$= \left(\frac{x}{2}\right)^{\frac{1}{2}}\left[\frac{1}{\frac{1}{2}\Gamma\left(\frac{1}{2}\right)} - \frac{1}{\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}\left(\frac{x}{2}\right)^2 + \frac{1}{2!\,\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}\left(\frac{x}{2}\right)^4 - \cdots\right]$$

$$= \frac{\sqrt{x}}{\sqrt{\pi\,\Gamma\left(\frac{1}{2}\right)}}\left[\frac{2}{1!} - \frac{2x^2}{3!} + \frac{2x^4}{5!} - \cdots\right]$$

Now multiplying the series by $\sqrt{2}$ and multiplying by $\sqrt{\frac{\pi}{2x}}$, we get

$$J_{\frac{1}{2}}(x) = \frac{\sqrt{2}}{\sqrt{\pi}\sqrt{x}}\left[\frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right] = \sqrt{\left(\frac{2}{\pi x}\right)}\sin x. \quad \ldots(3) \quad \text{(U.P.T.U., 2003 ; J.N.T.U., 2003)}$$

Similarly taking $n = \frac{1}{2}$ in (2) of page 551, it can be shown that

$$J_{-\frac{1}{2}}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\cos x \quad \ldots(4) \quad \text{(Anna, 2005 ; U.P.T.U., 2006 ; V.T.U., 2008)}$$

Example 8.8. *Express $J_{\frac{3}{2}}(x)$ in terms of $J_{\frac{1}{2}}(x)$ and $J_{-\frac{1}{2}}(x)$.* *(Anna, 2006 ; U.P.T.U., 2003)*

Solution. We know that

$$J_{n+1}(x) = \frac{2n}{x}J_n(x) + J_{n-1}(x) \text{ and } J_{n-1}(x) = \frac{2n}{x}J_n(x) - J_{n+1}(x)$$

Putting $n = 1, 2, 3, \ldots$ successively, $J_2(x) = \frac{2}{x}J_1(x) - J_0(x)$ ...(5) $\quad J_3(x) = \frac{4}{x}J_2(x) - J_1(x)$ ...(6)

$$J_1(x) = \frac{2}{x}J_0(x) - J_{-1}(x) \quad \ldots(7) \qquad J_0(x) = \frac{2}{x}J_{-1}(x) - J_{-2}(x) \quad \ldots(8)$$

Substituting the value of $J_2(x)$ in (ii), we have

$$J_3(x) = \frac{4}{x}\left[\frac{2}{x}J_1(x) - J_0(x)\right] - J_1(x) = \left(\frac{8}{x^2} - 1\right)J_1(x) - \frac{4}{x}J_0(x) \quad \text{...(iii)}$$

(M.D.U.T.C., 2005; Madras, 2003)

Now substituting the values of $J_0(x)$ from (i) and $J_1(x)$ from (ii) in (iii), we get

$$J_3(x) = \left[\frac{48}{x^3} - \frac{8}{x}\right]J_0(x) + \left[1 - \frac{24}{x^2}\right]J_1(x) \quad \text{...(iv)} \quad \textit{(P.T.U., 2002)}$$

Finally putting the values of $J_0(x)$ from (i) and $J_1(x)$ from (ii) in (iv), we obtain

$$J_3(x) = \left[\frac{384}{x^4} - \frac{72}{x^2} + 1\right]J_1(x) + \left[\frac{12}{x} - \frac{192}{x^3}\right]J_0(x)$$

Example 16(f). Prove that $J_{-n}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\left\{\frac{n^2 - 1^2}{x^2}\sin\frac{n\pi}{2} - \frac{n}{x}\cos\frac{n\pi}{2}\right\}$ *(U.P.T.U., 2004)*

Solution. We know that $J_{n+1}(x) = \frac{2n}{x}J_n(x) - J_{n-1}(x)$ ...(i)

Putting $n = \frac{1}{2}$, we get $J_{\frac{3}{2}}(x) = \frac{1}{x}J_{\frac{1}{2}}(x) - J_{-\frac{1}{2}}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\left[\frac{\sin x}{x} - \cos x\right]$

(Bhopal, 2000; U.P.T.U., 2006)

Again putting $n = \frac{3}{2}$ in (i), we get $J_{\frac{5}{2}}(x) = \frac{3}{x}J_{\frac{3}{2}}(x) - J_{\frac{1}{2}}(x)$

$$= \frac{3}{x}\left[\sqrt{\left(\frac{2}{\pi x}\right)}\left\{\frac{\sin x}{x} - \cos x\right\}\right] - \sqrt{\left(\frac{2}{\pi x}\right)}\sin x = \sqrt{\left(\frac{2}{\pi x}\right)}\left\{\frac{3 - x^2}{x^2}\sin x - \frac{3}{x}\cos x\right\}$$

which is the required result.

Example 16(g). Prove that

(i) $2J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$ *(U.P.T.U., 2003)*

(ii) $\frac{d}{dx}\{x^n J_n(x)\} = x^n J_{n-1}(x)$ *(U.P.T.U., 2004)*

Solution. (i) We know that $J_n(x) = \frac{1}{2}\{J_{n-1}(x) - J_{n+1}(x)\}$ ...(i)

Differentiating both sides, we get $J_n'(x) = \frac{1}{2}\{J_{n-1}'(x) - J_{n+1}'(x)\}$ ...(ii)

Changing n to $n-1$ in (i), we obtain $J_{n-1}(x) = \frac{1}{2}\{J_{n-2}(x) - J_n(x)\}$ ...(iii)

Changing n to $n+1$ in (i), we have $J_{n+1}(x) = \frac{1}{2}\{J_n(x) - J_{n+2}(x)\}$ ...(iv)

Substituting the values of $J_{n-1}'(x)$ and $J_{n+1}'(x)$ from (iii) and (iv) in (ii), we get

$$J_n' = \frac{1}{4}\{J_{n-2}(x) - 2J_n(x) + J_{n+2}(x)\}$$

(ii) $\frac{d}{dx}\{x^n J_n(x)\} = x^n J_{n-1}(x)$ ...(i)

From (5) of § 16.5, we have $J_n'(x) = \frac{n}{x}J_n(x) - J_{n+1}(x)$ ...(ii)

Changing n to $n+1$ in (i) of page 406, we get $J_{n+1}'(x) = J_n(x) - \frac{n+1}{x}J_{n+1}(x)$ ...(iii)

Now substituting for $u_n(x)$ and $v_n(x)$ to (1), we get

$$\frac{d}{dx}\{x^n J_n(x)\}J_{n-1}(x) = x^n J_n(x)J_{n-1}(x) + x\left[J_n(x)\left\{J_{n-1}(x) - \frac{n-1}{x}J_{n-1}(x)\right\} - \left\{\frac{n}{x}J_n(x) - J_{n-1}(x)\right\}J_{n-1}(x)\right]$$

$$= x\{J_{n-1}^2(x) - J_n^2(x)\}.$$

Example 10.11. Prove that

(a) $\int J_3(x)dx = -J_2(x) - \frac{2}{x}J_1(x)$

(b) $\int xJ_0(x)dx = \frac{1}{2}x^2\{J_0(x) + J_1^2(x)\}$ (U.P.T.U. 2004) (Gaurav Dixit)

Solution. (a) We know that $\frac{d}{dx}\{x^{-n}J_n(x)\} = -x^{-n}J_{n+1}(x)$ [R.M.K.U.] ...(1)

or $\int x^{-n}J_{n+1}(x)dx = -x^{-n}J_n(x)$ ...(2)

$$= \int J_3(x)dx = \int x^3 \cdot x^{-3}J_3(x)dx + c \qquad \text{[Integrate by parts]}$$

$$= x^3 \cdot \int x^{-3}J_3(x)dx - \int 3x^2\left[\int x^{-3}J_3(x)dx\right]dx + c$$

$$= x^3\{-x^{-3}J_2(x)\} - \int 3x^2\{-x^{-3}J_2(x)\}dx + c \qquad \text{[By (a) where } n = 2\text{]}$$

$$= c - J_2(x) + \int \frac{2}{x}J_2(x)dx = c - J_2(x) - \frac{2}{x}J_1(x) \qquad \text{[By (a) where } n = 1\text{]}$$

(b) $\int xJ_0(x)dx = \int J_0(x) \cdot xdx$ [Integrate by parts]

$$= J_0(x)\cdot\frac{1}{2}x^2 - \int J_0'(x)dx\left(\frac{1}{2}x^2\right)dx$$

$$= \frac{1}{2}x^2J_0(x) + \int x^2J_1(x)J_1(x)dx \qquad \text{[By (b) where } n = 0\text{]}$$

$$= \frac{1}{2}x^2J_0(x) + \int xJ_1(x)\cdot\frac{d}{dx}\{xJ_1(x)\}dx \qquad \left[= \frac{d}{dx}\{xJ_1(x)\} = xJ_0(x) \text{ by (10.66-11)}\right]$$

$$= \frac{1}{2}x^2J_0(x) + \frac{1}{2}\{xJ_1(x)\}^2 = \frac{1}{2}x^2\{J_0(x) + J_1^2(x)\}.$$

10.9 GENERATING FUNCTION FOR $J_n(x)$

To prove that $e^{\frac{1}{2}x\left(t-\frac{1}{t}\right)} = \sum_{n=-\infty}^{\infty} t^n J_n(x)$

We have $e^{\frac{1}{2}x\left(t-\frac{1}{t}\right)} = e^{\frac{1}{2}xt} \times e^{-\frac{x}{2t}}$

$$= \left[1 + \left(\frac{xt}{2}\right) + \frac{1}{2!}\left(\frac{xt}{2}\right)^2 + \frac{1}{3!}\left(\frac{xt}{2}\right)^3 + \cdots\right] \times \left[1 - \left(\frac{x}{2t}\right) + \frac{1}{2!}\left(\frac{x}{2t}\right)^2 - \frac{1}{3!}\left(\frac{x}{2t}\right)^3 + \cdots\right]$$

The coefficient of t^n in this product

$$= \frac{1}{n!}\left(\frac{x}{2}\right)^n - \frac{1}{(n+1)!}\left(\frac{x}{2}\right)^{n+2} + \frac{1}{2!(n+2)!}\left(\frac{x}{2}\right)^{n+4} - \cdots = J_n(x).$$

As all the integral powers of t, both positive and negative occur, we have

$$e^{\frac{1}{2}x\left(t-\frac{1}{t}\right)} = J_0(x) + tJ_1(x) + t^2J_2(x) + t^3J_3(x) + \cdots + t^{-1}J_{-1}(x) + t^{-2}J_{-2}(x) + t^{-3}J_{-3}(x) + \cdots$$

$$= \sum_{n=-\infty}^{\infty} t^n J_n(x)$$ (U.P.T.U. 2005)

This shows that Bessel functions of various orders can be derived as coefficients of different powers of t in the expansion of $e^{\frac{1}{2}x\left(t-\frac{1}{t}\right)}$. For this reason, it is known as the *generating function of Bessel functions*.

Example 18.12. Prove that

(i) $J_n(x)=\frac{1}{\pi}\int_0^{\pi}\cos(n\theta-x\sin\theta)\,d\theta$, n being an integer. (P.T.U. 2006)

(ii) $J_n(x)=\frac{1}{\pi}\int_0^{\pi}\cos(x\sin\theta)\cos n\theta\,d\theta$ (Madras, 2006)

(iii) $J_0^2+2(J_1^2+J_2^2+J_3^2+\ldots)=1$ (Gorakhpur M. Tech. 2004; U.P.T.U., 2003; P.T.U., 2004 (S))

Solution. (i) We know that

$$e^{\frac{1}{2}x\left(t-\frac{1}{t}\right)}=J_0(x)+tJ_1(x)+t^2J_2(x)+t^3J_3(x)+\ldots+t^{-1}J_{-1}(x)+t^{-2}J_{-2}(x)+t^{-3}J_{-3}(x)+\ldots$$

Since $J_{-n}(x)=(-1)^nJ_n(x)$

$$\therefore\quad e^{\frac{1}{2}x\left(t-\frac{1}{t}\right)}=J_0+tJ_1(1-1/t)+t^2J_2+(-1)^3J_3+t^4J_4+(-1)^4J_4+\ldots \qquad (1)$$

Now put $t=\cos\theta+i\sin\theta$

so that $t^n=\cos n\theta+i\sin n\theta$ and $1/t^n=\cos n\theta-i\sin n\theta$

giving $t^n+1/t^n=2\cos n\theta$ and $t^n-1/t^n=2i\sin n\theta$.

Substituting these in (1), we get

$$e^{ix\sin\theta}=J_0+2J_2\cos 2\theta+J_4\cos 4\theta+\ldots+2i(J_1\sin\theta+J_3\sin 3\theta+\ldots) \qquad \ldots(2)$$

Since $e^{ix\sin\theta}=\cos(x\sin\theta)+i\sin(x\sin\theta)$.

$\therefore$ equating the real and imaginary parts in (2), we get

$$\cos(x\sin\theta)=J_0+2J_2\cos 2\theta+J_4\cos 4\theta+\ldots \qquad \ldots(3)$$

$$\sin(x\sin\theta)=2(J_1\sin\theta+J_3\sin 3\theta+\ldots) \qquad \ldots(4)$$

which are known as Jacobi series[1]. (U.P.T.U., 2005)

Now multiplying both sides of (3) by $\cos n\theta$ and both sides of (4) by $\sin n\theta$ and subtracting each of the resulting expressions between 0 and π, we obtain

$$\frac{1}{\pi}\int_0^{\pi}\cos(x\sin\theta)\cos n\theta\,d\theta=\begin{cases}J_n(x), & n \text{ even or zero}\\ 0, & n \text{ odd}\end{cases}$$

and

$$\frac{1}{\pi}\int_0^{\pi}\sin(x\sin\theta)\sin n\theta\,d\theta=\begin{cases}0, & n \text{ even}\\ J_n(x), & n \text{ odd}\end{cases}$$

Hence generally, if n is a positive integer,

$$J_n(x)=\frac{1}{\pi}\int_0^{\pi}[\cos(x\sin\theta)\cos n\theta+\sin(x\sin\theta)\sin n\theta]\,d\theta=\frac{1}{\pi}\int_0^{\pi}\cos(n\theta-x\sin\theta)\,d\theta$$

[This is Bessel's original definition of $J_n(x)$ given in 1824 which is known as Poisson's method.]

(ii) Changing θ to $\frac{\pi}{2}+\theta$ in (3), we get

$$\cos(x\cos\theta)=J_0+2J_2\cos 2\theta-2J_4\cos 4\theta+2J_6\cos 6\theta-\ldots$$

Integrating both sides w.r.t. θ from 0 to π, we get

$$\int_0^{\pi}\cos(x\cos\theta)\,d\theta=\int_0^{\pi}[J_0(x)-2J_2(x)\cos 2\theta+2J_4(x)\cos 4\theta-\ldots]\,d\theta$$

$$=\left[J_0(x)\theta-2J_2(x)\cdot\frac{1}{2}\sin 2\theta+2J_4(x)\cdot\frac{1}{4}\sin 4\theta-\ldots\right]_0^{\pi}=\pi J_0(x)$$ (A reference follows the result.

[1] See footnote p. 318.

(c) Squaring (iv) and (v) and integrating w.r.t. θ from 0 to π and noting that $\sin n\pi = 0$, $\cos n\pi = (-1)^n$, we get

$$\int_0^{\pi} \sin^2(x\sin\theta)\,d\theta + \int_0^{\pi} \cos^2(x\sin\theta)\,d\theta = \pi, \qquad (n \neq 0)$$

and

$$\int_0^{\pi} \cos^2(x\sin\theta)\,d\theta - \int_0^{\pi} \sin^2(x\sin\theta)\,d\theta = \pi/2 \cdot \cos n\pi = 0$$

$$4J_1^2(x) + 4J_3^2(x) + 4J_5^2(x) + \ldots = \int_0^{\pi} \cos^2(x\sin\theta)\,d\theta$$

$$4J_0^2(x) + 4J_2^2(x) + 4J_4^2(x) + \ldots = \int_0^{\pi} \sin^2(x\sin\theta)\,d\theta$$

Adding, $\quad 1 + 2J_1^2 + 2J_2^2 + 2J_3^2 + \ldots = \frac{1}{\pi}\int_0^{\pi} d\theta = 1$

Hence $\quad J_1^2 + 2J_2^2 + 2J_3^2 + 2J_4^2 + \ldots = 1$

PROBLEMS 12.3

1. Complete $J_{3/2}(x)$, $J_{-3/2}(x)$ series in closed trigonometric forms.

2. Prove that $(xJ_n)' = \left(\frac{n^2}{x} - \frac{n}{2}\right)J_n(x) + \left(1 - \frac{n^2}{x^2}\right)J_{n+1}(x)$ $\quad (xJ_n)' + xJ_{n+1} = \frac{n}{2}J_n(x)$ $\quad$ (U.P.T.U., 2000)

3. Show that

(a) $J_{n-1}(x) - J_{n+1}(x) = 2J_n'(x)$ $\quad$ (U.P.T.U., 2000)

(b) $J_{n-1}(x) + J_{n+1}(x) = \frac{2n}{x}J_n(x)$ $\quad$ (U.P.T.U., 2000)

(c) $J_{1/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\sin x$ $\quad$ (d) $J_{-1/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\cos x$

(e) $J_{3/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\left(\frac{\sin x}{x} - \cos x\right)$ $\quad$ (f) $J_{-3/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\left(-\frac{\cos x}{x} - \sin x\right)$ $\quad$ (U.P.T.U., 2002)

4. Prove that (a) $\int_0^x J_0(x)\,dx = 1 - J_1(x)$ $\quad$ (b) $\int_0^x J_0(x)\,dx = xJ_0 + \frac{\pi}{2}xH_0(x)$

(c) $\int_0^x x^{n+1}J_n(x)\,dx = x^{n+1}J_{n+1}(x)$ (Meerut, 2000) $\quad$ (d) $\int_0^x J_3(x)\,dx = 2J_2 - J_1$ $\quad$ (U.P.T.U., 2000)

5. Show for the case of transverse vibrations that

(a) $J_{-1/2}(x) = \frac{1}{x}J_{1/2}(x) - J_{3/2}(x)$ $\quad$ (b) $J_{3/2}(x) + J_{-3/2}(x) = \frac{1}{x}J_{1/2}(x)$

(c) $xJ_{-1}(x) + 2J_0'(x) = J_1(x)$ (Varanasi, 2002)

6. Prove that

(a) $\int_0^x xJ_0(x)\,dx = \int_0^x [J_1'(x) - J_{-1}'(x)]\,dx$ $\quad$ (U.P.T.U., 2000; Kanpur, M.E., 2002)

(b) $\int_0^x xJ_0(x)\,dx = J_1(x) + \frac{1}{2}\left[\frac{1}{3}J_3(x) - \frac{1}{5}J_5(x) + \ldots\right]$ $\quad$ (U.P.T.U., 2000; U.P.T.U., 2000 (C.O.))

7. Prove that (a) $\int_0^{\pi/2} \cos(x\sin\theta)\,d\theta = \frac{\pi}{2}J_0(x)$ (U.P.T.U., 2000) $\quad$ (b) $\int_0^{\pi} \sin(x\cos\theta)\,d\theta = \pi J_0(x)$

(c) $\int_0^{\pi} \cos^2(x\cos\theta)\,d\theta = \pi J_0^2(x)$ $\quad$ (U.P.T.U., 2000)

8. Prove that (a) $\int_0^{\pi/2} J_0(x\sin\theta)\cos\theta\,d\theta = \frac{1}{2}J_0(x)$ $\quad$ (b) $\int_0^{\infty} e^{-ax}J_0(bx)\,dx = \frac{1}{\sqrt{a^2+b^2}}$

9. Starting with the series of $J_n(x)$, prove that

$$\frac{2n}{x}J_n(x) = J_{n-1}(x) + J_{n+1}(x) \text{ and } 2J_n'(x) = J_{n-1}(x) - J_{n+1}(x).$$

10. Establish the recurrence relations

$$\cos(b\cos\theta)\cos\theta = J_0 - 2J_2\cos 2\theta + 2J_4\cos 4\theta - \ldots$$

$$\sin(b\cos\theta)\cos\theta = 2J_1\cos\theta - 2J_3\cos 3\theta + 2J_5\cos 5\theta - \ldots$$

(U.P.T.U., 2003 (C.O.))

11. Prove that (a) $J_{n-1}(x) + J_{n+1}(x) = 2J_n'(x)$ $\quad$ (U.P.T.U., 2000 (C.O.))

(b) $\sin x = J_0 - 2J_2 + 2J_4 - \ldots$, $\cos x = 2J_1 - 2J_3 + \ldots$ $\quad$ (Kanpur, M.E., 2002)

(c) $\cos(x\sin\theta) = J_0 + 2J_2\cos 2\theta + 2J_4\cos 4\theta + \ldots$

[illegible] Equations Solvable in Terms of Their Derivatives

In many problems, we obtain equations and differential equations which are easily (or [illegible]) equations and therefore can be solved by means of [illegible] methods.

(1) To reduce the differential equation $F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots, \frac{d^ny}{dx^n}\right) = 0$ to [illegible]

If $y = \int u\,dx$, we have $\frac{dy}{dx} = u$, $\frac{d^2y}{dx^2} = \frac{du}{dx}$, and $\frac{d^ny}{dx^n} = \frac{d^{n-1}u}{dx^{n-1}}$.

Then (1) becomes $F\left(x, \int u\,dx, u, \frac{du}{dx}, \ldots, \frac{d^{n-1}u}{dx^{n-1}}\right) = 0$.

... [illegible]

(a) $y = \int u\,dx$ [illegible]

Hence the solution of (1) is

$y = \int u\,dx + C_1 x^{n-1} + C_2 x^{n-2} + \cdots + C_{n-1}x + C_n$ [illegible]

(a) $y = \int u\,dx + C_1 x^{n-1} + C_2 x^{n-2} + \cdots + C_{n-1}x + C_n$ [illegible]

(2) To reduce the differential equation $F\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots, \frac{d^ny}{dx^n}\right) = 0$ [illegible] (Whatever, [illegible])

(put) $\frac{dy}{dx} = p$.

(so that) $\frac{d^2y}{dx^2} = p\frac{dp}{dy}$, $\frac{d^3y}{dx^3} = p^2\frac{d^2p}{dy^2} + \left(\frac{dp}{dy}\right)^2 p$, and $\frac{d^ny}{dx^n} = p\frac{d}{dy}\left(\frac{d^{n-1}y}{dx^{n-1}}\right)$.

Then (2) reduces the form $\phi\left(y, p, p\frac{dp}{dy}, \ldots\right) = 0$, [illegible] $\phi\left(y, p, \frac{dp}{dy}, \ldots\right) = 0$.

Dividing by the product p^{n-1} and putting for p as $y = \int$.

$$p\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} = 0.$$

By [illegible] $\frac{dy}{dx} = p$, $\frac{d^2y}{dx^2} = p\frac{dp}{dy}$, [illegible]

(a) $p = e^{\int \phi(y)\,dy}$, $\int e^{\int \phi(y)\,dy}\,dy + C_2$ [illegible]

Hence the solution of $\frac{d^2y}{dx^2} + y\left(\frac{dy}{dx}\right)^2 = 0$ [illegible]

(a) $y = \sqrt{2C_1 x + C_2}$, $\sqrt{2C_1 x + C_2} = C_1 x + C_2$ [illegible]

(3) To reduce the differential equation $\phi\left(x, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots, \frac{d^ny}{dx^n}\right) = 0$ [illegible] $\phi(x) = 0$ [illegible], put $x = e^t$, i.e. $t = \log x$.

(so that)
$$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx} = \frac{1}{x}\cdot\frac{dy}{dt} = e^{-t}\frac{dy}{dt},$$

(and)
$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left(\frac{dy}{dx}\right)\cdot\frac{dt}{dx} = e^{-t}\frac{d}{dt}\left(e^{-t}\frac{dy}{dt}\right)\cdot e^{-t} = e^{-2t}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right).$$

Then (3) reduces the form $\phi\left(e^t, e^{-t}\frac{dy}{dt}, e^{-2t}\left(\frac{d^2y}{dt^2} - \frac{dy}{dt}\right), \ldots\right) = 0$, [illegible] $\phi\left(t, \frac{dy}{dt}, \frac{d^2y}{dt^2}, \ldots\right) = 0$.

(a second-[illegible] [illegible]) by $e^{(n-1)t}$, $\phi\left(t, \frac{dy}{dt}, \ldots\right) = 0$, [illegible] $\frac{dy}{dt} = \phi(t, \ldots)$, $\int \phi(t, \ldots)\,dt + C$.

By [illegible] by [illegible] $\frac{dy}{dx} = \frac{1}{x}\frac{dy}{dt}$, [illegible] $\frac{dy}{dt} = x\frac{dy}{dx}$, $\frac{d^2y}{dt^2} = x\frac{dy}{dx} + x^2\frac{d^2y}{dx^2}$.

(and) $y = \int \ldots dx + C$, [illegible] $x\frac{dy}{dx} = \ldots$, $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = \ldots$

Then [illegible] $\frac{d^2y}{dx^2} + a\frac{dy}{dx} + by = 0$ [illegible] as [illegible] to (3).

Hence the solution of (3) is $y = C_1 x^{\alpha_1} + C_2 x^{\alpha_2} + \cdots + C_{n-1}x^{\alpha_{n-1}} + C_n$ [illegible] $x = e^t$ [illegible]

(a) $y = C_1 e^{\alpha_1 t} + C_2 e^{\alpha_2 t} + \cdots + C_{n-1}e^{\alpha_{n-1}t} + C_n$ [illegible]

(when) $x = \frac{1-\alpha}{1+\alpha}$, $\frac{1-\alpha}{1+\alpha} = \frac{\beta}{1+\alpha}$

Example 18.23. Solve the following differential equations :

(i) $x^2y'' + \frac{3}{2}xy' + \left(x - \frac{1}{2x}\right)y = 0$ (ii) $4x^2y'' + 8xy' + y = 0$ (iii) $x(xy'' + y') + x^2y = 0$ (Anna, 2002)

Solution. (i) Rewriting the given equation as $x^2\frac{d^2y}{dx^2} + \frac{3}{2}x\frac{dy}{dx} + \left(x^2 - \frac{1}{2}\right)y = 0$,

and comparing with (1) above, we see that $n = \frac{1}{2}$ and $k = 2/3$.

$\therefore$ The solution of the given equation is $y = c_1J_{1/2}(kx) + c_2J_{-1/2}(kx)$

i.e., $$y = c_1J_{1/2}(2x/3) + c_2J_{-1/2}(2x/3) \quad \ldots(1)$$

(ii) Rewriting the given equation as $\frac{d^2y}{dx^2} + \frac{1}{4x^2}y = 0$ ...(a)

and comparing with (1) above, we find that $n = 0$, $k = 1/2$ and $r = 2$.

$\therefore$ $$n = \frac{1-r}{1+r} = \frac{1}{3}, \quad n_1 = \frac{2}{1+r} = \frac{2}{3} \quad \text{and} \quad \frac{n}{n_1} = \frac{1}{2}.$$

Hence the solution of (a) is $y = x^{1/3}\left[c_1J_{1/2}(kx^{2/3}) + c_2J_{-1/2}(kx^{2/3})\right]$

$$y = \sqrt{x}\left(c_1J_{1/2}(x^{2/3}) + c_2J_{-1/2}(x^{2/3})\right).$$

(iii) Multiplying by x, the given equation becomes

$$x^2y'' + xy' + \frac{1}{4}xy = 0 \quad \ldots(a)$$

Comparing with (2) above, we get $n = 1$, $k = 1/2$ & $r = 1$. $\therefore$ $m = \frac{2}{1+r} = 1$, $n = \frac{1-r}{1+r} = 0$ & $\frac{n}{m} = 0$.

Hence the solution of (a) is

$$y = x^{1/2}\left[c_1J_1\left(\tfrac{1}{2}x^{1/2}\right) + c_2J_{-1}\left(\tfrac{1}{2}x^{1/2}\right)\right] = x^{1/2}\left[c_1J_1\left(\tfrac{1}{2}\sqrt{x}\right) + c_2J_0\left(\tfrac{1}{2}\sqrt{x}\right)\right]$$

i.e., $$y = c_1\sqrt{x}J_1(\sqrt{x}/2) + c_2J_0(\sqrt{x}/2)$$

3.6.5 (L) ORTHOGONALITY OF BESSEL FUNCTIONS

We shall prove that

$$\int_0^1 xJ_n(\alpha x)J_n(\beta x)dx = \begin{cases} 0, & \alpha \neq \beta \\ \frac{1}{2}[J_{n+1}(\alpha)]^2, & \alpha = \beta \end{cases}, \text{ where } \alpha, \beta \text{ are the roots of } J_n(x) = 0.$$

We know that the solution of the equation

$$x^2y'' + xy' + (\alpha^2x^2 - n^2)y = 0 \quad \ldots(1)$$

and $$x^2y'' + xy' + (\beta^2x^2 - n^2)y = 0 \quad \ldots(2)$$

are $y = J_n(\alpha x)$ and $y = J_n(\beta x)$ respectively.

Multiplying (1) by vdx and (2) by udx and subtracting, we get

$$u(x^2v'' + xv') + (\alpha^2x^2 - n^2)uv - v(x^2u'' + xu') - (\beta^2x^2 - n^2)uv = 0$$

or $$\frac{d}{dx}[x(u'v - uv')] + (\alpha^2 - \beta^2)xuv = 0$$

Now integrating both sides from 0 to 1,

$$(\beta^2 - \alpha^2)\int_0^1 xuv\,dx = [x(u'v - uv')]_0^1 = u'(1)v(1) - u(1)v'(1) \quad \ldots(3)$$

Since $$u = J_n(\alpha x),$$

$$\therefore \quad u' = \frac{d}{dx}[J_n(\alpha x)] = \frac{d}{d(\alpha x)}[J_n(\alpha x)]\frac{d(\alpha x)}{dx} = \alpha J_n'(\alpha x)$$

Now let $u = J_n(\beta x)$ and $v' = xJ_n(\alpha x)$. Substituting these values in (3), we get

$$\int_0^1 xJ_n(\alpha x)J_n(\beta x)\,dx = \frac{\alpha J_n(\alpha)J_n'(\beta) - \beta J_n(\beta)J_n'(\alpha)}{\beta^2 - \alpha^2} \quad \ldots(4)$$

If α and β are distinct roots of $J_n(x) = 0$, then $J_n(\alpha) = J_n(\beta) = 0$, and (4) reduces to

$$\int_0^1 xJ_n(\alpha x)J_n(\beta x)\,dx = 0 \quad \ldots(5)$$

This is known as the orthogonality relation of Bessel functions.

When $\beta = \alpha$, the right side of (4) is of 0/0 form. Its value can be found by using Lt'Hospital's rule if α is a root of $J_n(x) = 0$ and $J_n(x)$ is a suitable approximation to it. Then (4) gives

$$\lim_{\beta \to \alpha} \int_0^1 xJ_n(\alpha x)J_n(\beta x)\,dx = \lim_{\beta \to \alpha} \frac{\alpha J_n(\alpha)J_n'(\beta)}{\beta^2 - \alpha^2}$$

or, by L'Hospital's rule, $\int_0^1 xJ_n^2(\alpha x)\,dx = \lim_{\beta \to \alpha} \frac{\alpha J_n(\alpha)J_n'(\beta)}{2\beta} = \frac{1}{2}[J_n'(\alpha)]^2$

$$= \frac{1}{2}[J_{n+1}(\alpha)]^2 \quad \ldots(6) \quad \text{[By (15) of Art. 22.8]}$$

Note: However, the interval in Exs. 1 to 3 is not necessarily unit. Then

$$\int_0^a xJ_n^2\left(\frac{\alpha_m x}{a}\right)dx = \frac{a^2}{2}[J_{n+1}(\alpha_m)]^2 \quad \text{where } \alpha_m \text{ is the } m\text{th root of } J_n(x) = 0. \quad \ldots(7) \text{ [U.P.T.U., 2006]}$$

(2) Fourier-Bessel expansion. If $f(x)$ is a continuous function having finite number of maxima and minima in the interval $(0, a)$, then we can write

$$f(x) = c_1 J_n\left(\frac{\alpha_1 x}{a}\right) + c_2 J_n\left(\frac{\alpha_2 x}{a}\right) + \ldots + c_m J_n\left(\frac{\alpha_m x}{a}\right) + \ldots \quad (8)$$

where $\alpha_1, \alpha_2, \ldots$ are the positive roots of $J_n(x) = 0$.

To determine the coefficients c_m, multiply both sides of (8) by $xJ_n\left(\frac{\alpha_m x}{a}\right)$ and integrate from 0 to a. Then all integrals on the right of (8) vanish by (7), except the term in c_m. This gives

$$\int_0^a xf(x)J_n\left(\frac{\alpha_m x}{a}\right)dx = c_m \int_0^a xJ_n^2\left(\frac{\alpha_m x}{a}\right)dx = c_m \frac{a^2}{2}J_{n+1}^2(\alpha_m) \quad \text{[By (7)]}$$

or $$c_m = \frac{2}{a^2 J_{n+1}^2(\alpha_m)} \int_0^a xf(x)J_n\left(\frac{\alpha_m x}{a}\right)dx$$

Equation (8) is known as the Fourier-Bessel expansion of $f(x)$.

Example 10.14. If $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the positive roots of $J_0(x) = 0$, show that

$$\frac{x^2}{2} = \sum_{m=1}^{n} \frac{2}{\alpha_m J_1(\alpha_m)} J_2(\alpha_m x) + c_0 J_0(\alpha_m x), \text{ for } 0 < x < 1$$

Solution. If $f(x) = c_1 J_0(\alpha_1 x) + c_2 J_0(\alpha_2 x) + \ldots + c_n J_0(\alpha_n x) + \ldots \quad \ldots(1)$

then $$c_m = \frac{2}{a^2 J_{n+1}^2(\alpha_m)} \int_0^a xf(x)J_n\left(\frac{\alpha_m x}{a}\right)dx$$

Taking $f(x) = 1$, $a = 1$ and $n = 0$, we get

$$c_m = \frac{2}{J_1^2(\alpha_m)} \int_0^1 xJ_0(\alpha_m x)\,dx = \frac{2}{\alpha_m J_1^2(\alpha_m)} \left| \frac{xJ_1(\alpha_m x)}{\alpha_m} \right|_0^1 = \frac{2}{\alpha_m J_1(\alpha_m)}$$

From (1), $$1 = \sum_{m=1}^{n} \frac{2}{\alpha_m J_1(\alpha_m)} J_0(\alpha_m x) \quad \text{or} \quad \frac{1}{2} = \sum_{m=1}^{n} \frac{J_0(\alpha_m x)}{\alpha_m J_1(\alpha_m)}$$

Example 10.15. Expand $f(x) = x^3$ in the interval $0 < x < 1$ in terms of $J_3(\alpha_1 x), J_3(\alpha_2 x), \ldots$ where $\alpha_1, \alpha_2, \ldots$ are the roots of $J_3(\alpha) = 0$.

Solution. Let the Fourier-Bessel expansion of $f(x)$ be $x^3 = \sum_{m=1}^{\infty} c_m J_3(\alpha_m x)$

Multiplying both sides by $xJ_n(\alpha_m x)$ and integrating w.r.t. x from 0 to 1, we get

$$\int_0^1 x^{n+1}J_n(\alpha_m x)\,dx = a_m\int_0^1 xJ_n^2(\alpha_m x)\,dx = a_m\frac{[J_{n+1}(\alpha_m)]^2}{2} \quad \text{[By (14)]}$$

or
$$\left[\frac{x^{n+1}J_{n+1}(\alpha_m x)}{\alpha_m}\right]_0^1 = a_m J_{n+1}(\alpha_m)$$

$$a_m = \frac{2}{\alpha_m J_{n+1}(\alpha_m)}$$

Hence
$$x^{n+1} = 2\sum_{m=1}^{\infty}\frac{J_n(\alpha_m x)}{\alpha_m J_{n+1}(\alpha_m)}$$

13.12 ber AND bei FUNCTIONS

Consider the differential equation

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} - iy = 0 \qquad ...(1)$$

which occurs in certain problems of electrical engineering. This is equation (1) of §13.10 with $k = 0$ and $k^2 = -i$, so that its particular solution is

$$y = J_0(kx) = J_0(\sqrt{-i}x) = J_0(i^{1/2}x)$$

Replacing $i^{1/2}x$ in the series for $J_0(x)$ in §13.5, we get

$$y = 1 - \frac{ix^2}{2^2} + \frac{i^2x^4}{(2!)^2 2^4} - \frac{i^3x^6}{(3!)^2 2^6} + \frac{i^4x^8}{(4!)^2 2^8} - \cdots$$

$$= \left[1 - \frac{x^4}{2^2\cdot 4^2} + \frac{x^8}{2^2\cdot 4^2\cdot 6^2\cdot 8^2} - \cdots\right] + i\left[\frac{x^2}{2^2} - \frac{x^6}{2^2\cdot 4^2\cdot 6^2} + \frac{x^{10}}{2^2\cdot 4^2\cdot 6^2\cdot 8^2\cdot 10^2} - \cdots\right] \qquad ...(2)$$

which is complex for x real. The series in the above brackets are taken to define Kelvin's *real* (or *ber*) and Kelvin's *imaginary* (or *bei*) functions, so

Thus
$$\text{ber}\,x = 1 - \sum_{m=1}^{\infty}(-1)^m\frac{x^{4m}}{2^2\cdot 4^2\cdot 6^2\cdots(4m)^2} \qquad ...(3)$$

and
$$\text{bei}\,x = \frac{x^2}{2^2} - \sum_{m=1}^{\infty}(-1)^m\frac{x^{4m+2}}{2^2\cdot 4^2\cdot 6^2\cdots(4m+2)^2} \qquad ...(4)$$

so that
$$y = \text{ber}\,x + i\,\text{bei}\,x \text{ is a solution of (1).}$$

Tables giving numerical values of ber x and bei x are available.

Example 13.12.1 Prove that (i) $\frac{d}{dx}(x\,\text{ber}'x) = -x\,\text{bei}\,x$ (ii) $\frac{d}{dx}(x\,\text{bei}'x) = x\,\text{ber}\,x$.

Solution. We have $x\,\text{ber}'x = x\sum_{m=1}^{\infty}(-1)^m\frac{4mx^{4m-1}}{2^2\cdot 4^2\cdot 6^2\cdots(4m)^2}$

$$= -\sum_{m=1}^{\infty}(-1)^m\frac{x^{4m}}{2^2\cdot 4^2\cdot 6^2\cdots(4m-2)^2\,4m} = -\int_0^x x\,\text{bei}\,x\,dx$$

or
$$\frac{d}{dx}(x\,\text{ber}'x) = -x\,\text{bei}\,x$$

Again
$$\int_0^x x\,\text{ber}\,x\,dx = \frac{x^2}{2} + \sum_{m=1}^{\infty}(-1)^m\frac{x^{4m+2}}{2^2\cdot 4^2\cdot 6^2\cdots(4m)^2(4m+2)}$$

$$= -\sum_{m=1}^{\infty}(-1)^m\frac{x^{4m+2}}{2^2\cdot 4^2\cdot 6^2\cdots(4m-2)^2(4m-2)} = x\,\text{bei}'x \quad \text{or} \quad \frac{d}{dx}(x\,\text{bei}'x) = x\,\text{ber}\,x.$$

EXERCISE 3.4

Obtain the solutions of the following differential equations in terms of Bessel functions:

i. $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+\left(x^2-\frac{1}{9}\right)y=0$ ii. $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+\left(16x^2-\frac{1}{4}\right)y=0$

iii. $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+4(x^2-9)y=0$ iv. $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+4x^2y=0$

v. $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+x^2y=0$

vi. Show that $y=x^{-1/2}\cos x$ is a solution of the equation $x^2\frac{d^2y}{dx^2}+(1+2x)y=0$ (V.T.U., 2007)

vii. Show that $y=x^{1/2}\sin x$ is a solution of the equation $x^2\frac{d^2y}{dx^2}+(1+2x)y=0$.

viii. Show that the transformation $y=x^{1/2}z$ reduces Bessel's equation to the form

$$x^2\frac{d^2z}{dx^2}+x\frac{dz}{dx}+\left(x^2-\frac{1}{4}\right)z=0$$

ix. If the roots of the indicial equation $x=ax^{\alpha}$ are equal, show that the solution of the equation $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+\left(x^2-\frac{1}{4}\right)y=0$ can be written as $y=A\frac{\sin x}{\sqrt{x}}+B\frac{\cos x}{\sqrt{x}}$.

x. Show that $\int_0^{\pi/2}\sin^{m-1}x\cos^{n-1}x\,dx=\frac{1}{2}\beta\left(\frac{m}{2},\frac{n}{2}\right)=\frac{1}{2}\frac{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{m+n}{2}\right)}$.

xi. If $\alpha_1,\alpha_2,\ldots,\alpha_n$ are the positive roots of $J_p(x)=0$, prove that

$$x^p=2\sum_{r=1}^{n}\frac{\alpha_r^{p-1}}{\alpha_r J_{p-1}(\alpha_r)}J_p(\alpha_r x)$$

xii. Express $f(x)=x^3$ in the interval $0<x<1$ in terms of $J_1(\alpha_r x)$ where α_r are the positive roots of $J_3(x)=0$.

3.15 LEGENDRE'S EQUATION*

Another differential equation of importance in Applied Mathematics, particularly in boundary value problems for spheres, is Legendre's equation.

$$(1-x^2)\frac{d^2y}{dx^2}-2x\frac{dy}{dx}+n(n+1)y=0 \qquad ...(1)$$

Here n is a real number. But in most applications only integral values of n are required.

Substituting $\quad y=a_0x^r+a_1x^{r+1}+a_2x^{r+2}+\ldots+a_kx^{r+k}+\ldots$

(1) becomes

$$a_0r(r-1)x^{r-2}+a_1(r+1)rx^{r-1}+\ldots+a_{k+2}(r+k+2)(r+k+1)x^{r+k}+\ldots$$
$$-[2ra_0x^r+2(r+1)a_1x^{r+1}+\ldots+2(r+k)a_kx^{r+k}+\ldots]+n(n+1)[a_0x^r+a_1x^{r+1}+\ldots+a_kx^{r+k}+\ldots]=0$$

Equating to zero the coefficient of the lowest power of x, i.e., of x^{r-2}, we get

$$a_0r(r-1)=0 \Rightarrow a_0=0, \text{ as } a_0\neq 0 \qquad (\because a_0\neq 0)$$

Equating to zero the coefficients of x^{r-1} and x^r, we get $a_1r(r+1)=0$ ...(2)

$$a_{k+2}(r+k+2)(r+k+1)-2a_{k+1}(r+k+1)+a_k(n-r-k)(n+r+k+1)=0 \qquad ...(3)$$

If $a_1\neq 0$, (2) is satisfied and therefore, $a_1\neq 0$. Then (3) gives, taking $k=1, 2, 3, \ldots$ in turn,

$$a_3=-\frac{(n-1)(n+2)}{3!}a_1, \quad a_4=-\frac{(n-2)(n+3)}{3!}a_2$$

$$a_5=\frac{(n-2)(n-3)(n+3)(n+4)}{5!}a_1, \quad a_6=\frac{(n-2)(n-4)(n+3)(n+5)}{6!}a_2$$

$$a_7=-\frac{(n-3)(n-5)(n+4)(n+6)}{7!}a_1, \quad a_8=-\frac{(n-3)(n-5)(n+4)(n+6)}{8!}a_2$$

Hence for $r=0$, there are two independent solutions of (1):

$$y_1=a_0\left[1-\frac{n(n+1)}{2!}x^2+\frac{(n-2)n(n+1)(n+3)}{4!}x^4-\ldots\right] \qquad ...(4)$$

*See footnote p. 400.

$$y_2 = x_1\left[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!}x^5 - \cdots\right] \quad \ldots(9)$$

When $n = 1$, (2) shows that $a_1 = 0$. Therefore, (2) gives

$$a_3 = a_5 = a_7 = \cdots = 0$$

and

$$a_2 = -\frac{(n-1)n}{2!}a_0$$

$$a_4 = \frac{(n-3)(n-1)n(n+2)}{4!}a_0, \text{ etc.}$$

Thus for $n = 1$, we get the solution (8) again. Hence $y = y_1 + y_2$ is the general solution of (1).

If n is a positive even integer, the series (9) terminates at the term in x^n and y_2 becomes a polynomial. Similarly if n is an odd integer, (5) becomes a polynomial of degree n. Thus, whenever n is a positive integer, the general solution of (1) consists of a polynomial solution and an infinite series solution.

Those polynomial solutions, with a_0 or a_1 so chosen that the value of this polynomial is 1 for $x = 1$, are called Legendre polynomials of order n and are denoted by $P_n(x)$. The infinite series solution with a_0 or a_1 properly chosen is called Legendre function of the second kind and is denoted by $Q_n(x)$. (V.T.U., 2006)

15.3 (1) RODRIGUE'S FORMULA*

We shall prove that $P_n(x) = \frac{1}{n!\,2^n}\frac{d^n}{dx^n}(x^2-1)^n$ ...(1)

Let $v = (x^2-1)^n$. Then $v_1 = \frac{dv}{dx} = 2nx(x^2-1)^{n-1}$

i.e., $(1-x^2)v_1 + 2nxv = 0$ ...(2)

Differentiating (2), $(n+1)$ times by Leibnitz's Theorem

$$(1-x^2)v_{n+2} - (n+1)2xv_{n+1} + \frac{1}{2!}n(n+1)2v_n + 2nxv_{n+1} + n(n+1)v_n = 0$$

or $$(1-x^2)\frac{d^2(v_n)}{dx^2} - 2x\frac{d(v_n)}{dx} + n(n+1)v_n = 0$$

which is Legendre's equation and v_n is its solution. Also its finite series solution is $P_n(x)$.

$\therefore$ $$P_n(x) = cv_n = c\frac{d^n}{dx^n}(x^2-1)^n \quad \ldots(3)$$

To determine the constant c, put $x = 1$ in (3). Then

$$1 = c\left[\frac{d^n}{dx^n}(x-1)^n(x+1)^n\right]_{x=1}$$

$= c[n!\,(x+1)^n]_{x=1}$

$+$ terms containing $(x-1)$ and its powers$]_{x=1}$

$= c \cdot n!\,2^n$, i.e., $c = \frac{1}{n!\,2^n}$.

Substituting this value of c in (3), we get (1), which is known as the Rodrigue's formula.

(V.T.U., 2006; Bhopal, 2007; G.B.T.U., 2004)

Ex. 1. Show that (i) roots of $P_n(x) = 0$ are real and lie between -1 and $+1$. (Madras, 2003 S)

(2) Legendre polynomials. Using (1), we get

$P_0(x) = 1$, $P_1(x) = x$,

$P_2(x) = \frac{1}{2}(3x^2-1)$, $P_3(x) = \frac{1}{2}(5x^3-3x)$,

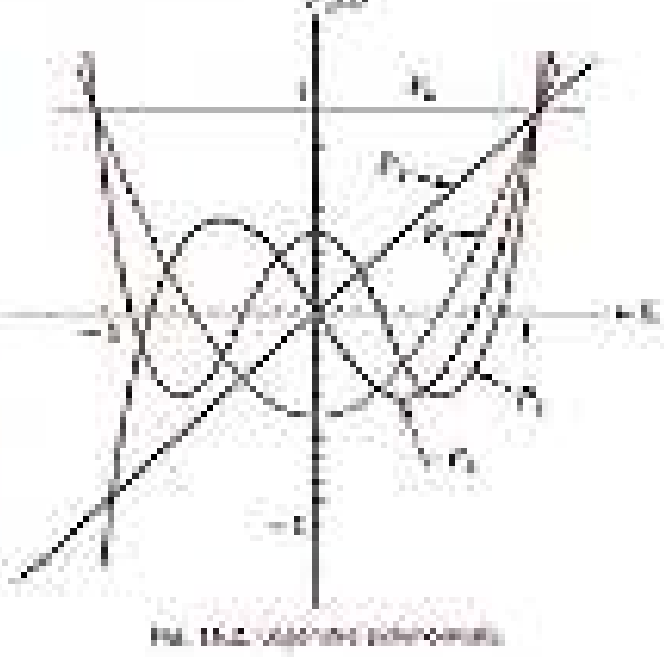

Fig. 15.2. (Legendre polynomials)

* Named after the French mathematician and economist Olinde Rodrigues (1794–1851).

$$P_3(x) = \frac{1}{8}(15x^3 - 30x^2 + 3),\ P_4(x) = \frac{1}{8}(35x^4 - 60x^2 + 15x), \text{ etc.} \qquad \text{(U'97, U'06)}$$

In general, we have $P_n(x) = \sum_{r=0}^{N} \frac{(-1)^r (2n-2r)!}{2^n r!(n-r)!(n-2r)!} x^{n-2r}$ ...(1)

where $N = \frac{1}{2}n$ or $\frac{1}{2}(n-1)$ depending as n is even or odd.

Let us derive (1) from (2).

By Binomial theorem, $(x^2-1)^n = \sum_{r=0}^{n} {}^nC_r (x^2)^{n-r}(-1)^r = \sum_{r=0}^{n} (-1)^r \frac{n!}{r!(n-r)!} x^{2n-2r}$

$\therefore$ By (2), $P_n = \frac{1}{n!2^n} \sum_{r=0}^{n} \frac{(-1)^r n!}{r!(n-r)!} \frac{d^n (x^{2n-2r})}{dx^n} = \sum_{r=0}^{n} \frac{(-1)^r (2n-2r)!}{2^n r!(n-r)!(n-2r)!} x^{n-2r}$

This is same as (1), and the last term $(r = N)$ is such that the power of x is $(n - 2N)$ or this term is either 0 or 1.

Example 10.17. Express $f(x) = x^4 + 3x^3 + x^2 + 5x - 2$ in terms of Legendre polynomials. (U'97, U'01, U'03, U'05, U'07, U'10, U'11)

Solution. Since $P_4(x) = \frac{35}{8}x^4 - \frac{15}{4}x^2 + \frac{3}{8}$ $\therefore x^4 = \frac{8}{35}P_4(x) + \frac{6}{7}x^2 - \frac{3}{35}$

$\therefore$ $f(x) = \left[\frac{8}{35}P_4(x) + \frac{6}{7}x^2 - \frac{3}{35}\right] + 3x^3 + x^2 + 5x - 2$

$= \frac{8}{35}P_4(x) + 3x^3 + \frac{1}{7}x^2 + 5x - \frac{73}{35}$ $\left[\because x^3 = \frac{2}{5}P_3(x) + \frac{3}{5}x,\ x^2 = \frac{2}{3}P_2(x) + \frac{1}{3}\right]$

$= \frac{8}{35}P_4(x) + 3\left[\frac{2}{5}P_3(x) + \frac{3}{5}x\right] + \frac{1}{7}\left[\frac{2}{3}P_2(x) + \frac{1}{3}\right] + 5x - \frac{73}{35}$

$= \frac{8}{35}P_4(x) + \frac{6}{5}P_3(x) + \frac{2}{21}P_2(x) + \frac{24}{5}x - \frac{224}{105}$ $\left[\because x = P_1(x), 1 = P_0(x)\right]$

$= \frac{8}{35}P_4(x) + \frac{6}{5}P_3(x) + \frac{2}{21}P_2(x) + \frac{24}{5}P_1(x) - \frac{224}{105}P_0(x)$

Example 10.18. Show that Rodrigue's formula $P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n$ for Legendre polynomials leads to $\int_{-1}^{1} f(x) P_n(x)\,dx = \frac{1}{2^n n!}\int_{-1}^{1} (1-x^2)^n f^{(n)}(x)\,dx$.

Solution. Using Rodrigue's formula: $P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n$

$$\int_{-1}^{1} f(x) P_n(x)\,dx = \frac{1}{2^n n!}\int_{-1}^{1} f(x)\frac{d^n}{dx^n}(x^2-1)^n\,dx$$ Integrate by parts

$$= \frac{1}{2^n n!}\left[f(x)\cdot\frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n\right]_{-1}^{1} - \int_{-1}^{1} f'(x)\cdot\frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n\,dx$$

$$= \frac{1-1}{2^n n!}\int_{-1}^{1} f'(x)\frac{d^{n-2}}{dx^{n-2}}(x^2-1)^n\,dx$$

$$= \frac{1-1^2}{2^n n!}\int_{-1}^{1} f''(x)\frac{d^{n-3}}{dx^{n-3}}(x^2-1)^n\,dx$$ Again integrating by parts

$$= \frac{1}{2^n n!}\int_{-1}^{1} f^{(n)}(x)\,(x^2-1)^n\,dx \qquad \text{Integrating by parts } n \text{ times}$$

$$= \frac{1}{2^n n!}\int_{-1}^{1} f^{(n)}(x)\,(1-x^2)^n\,dx = \frac{1}{2^n n!}\int_{-1}^{1} f^{(n)}(x)\,(1-x^2)^n\,dx$$

4.3.5 GENERATING FUNCTION FOR $P_n(x)$

To show that $(1-2xt+t^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} t^n P_n(x)$

Now $(1-t)^{-\frac{1}{2}} = 1 + \frac{1}{2}t + \frac{\frac{1}{2}\cdot\frac{3}{2}}{2!}t^2 + \frac{\frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2}}{3!}t^3 + \ldots$

$$= 1 + \frac{1!}{(1!)^2 2^1}t + \frac{3!}{(2!)^2 2^2}t^2 + \frac{5!}{(3!)^2 2^3}t^3 + \ldots$$

$$\therefore \quad (1-t(2x-t))^{-\frac{1}{2}} = 1 + \frac{1!}{(1!)^2 2^1}t(2x-t) + \frac{3!}{(2!)^2 2^2}t^2(2x-t)^2 + \ldots$$

$$+ \frac{(2n-2)!}{[(n-1)!]^2 2^{n-1}}t^{n-1}(2x-t)^{n-1} + \ldots + \frac{(2n)!}{(n!)^2 2^n}t^n(2x-t)^n + \ldots \qquad \ldots(1)$$

The term in t^n from the term containing $t^{n-1}(2x-t)^{n-1}$

$$= \frac{(2n-2)!}{[(n-1)!]^2 2^{n-1}}\,{}^{n-1}C_{n-1}(-1)^{n-1}(2x)^{n-n}$$

$$= \frac{(2n-2)!}{[(n-1)!]^2 2^{n-1}} = \frac{(n-1)!}{n!(n-2)!}(-1)^{n-1}(2x)^{n-n} = \frac{(-1)^{n-1}(2n-2)!}{2^{n-1}n!(n-1)!(n-2)!}x^{n-n}, \text{ say}$$

Collecting all terms in t^n which will occur in the term containing $t^{n-1}(2x-t)^{n-1}$ and the preceding terms, we see that terms in t^n

$$= \sum_{r=0}^{M}\frac{(-1)^r(2n-2r)!}{2^n r!(n-r)!(n-2r)!}x^{n-2r}t^n = P_n(x)t^n$$

where $M = \frac{1}{2}n$ or $\frac{1}{2}(n-1)$ according as n is even or odd.

Hence (1) may be written as $(1-t(2x-t))^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(x)t^n \qquad \ldots(2)$

(Kerala M.E., 2005 ; U.P.T.U., 2009)

This shows that $P_n(x)$ is the coefficient of t^n in the expansion of $(1-2xt+t^2)^{-\frac{1}{2}}$. That is why, it is known as the generating function of Legendre polynomials.

Cor. 1. $P_n(1) = 1$ (U.P.T.U., 2005 S ; Delhi, 2008)

Taking $x = 1$ in (2), we have $(1-2t+t^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} P_n(1)t^n$

i.e., $\sum_{n=0}^{\infty} t^n = (1-t)^{-1} = 1 + t + t^2 + \ldots + t^n + \ldots$

Equating coefficients of t^n, we get $P_n(1) = 1$.

Cor. 2. $P_n(-1) = (-1)^n$. (U.P.T.U., 2005 ; Y.T.U., 2008)

Taking $x = -1$ in (2), we have

$$\sum_{n=0}^{\infty} P_n(-1)t^n = (1+t)^{-1} = 1 - t + t^2 - \ldots + (-1)^n t^n + \ldots$$

Equating coefficients of t^n, we get the desired result.

(ii) Rewriting (i) as

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

$$= (2n+1)P_n(x) + (n+1)xP_n'(x) - xP_{n-1}'(x) \quad \text{By (ii)}$$

$$= (n+1)xP_n'(x) + nP_n(x) - xP_{n-1}'(x)$$

or $$P_{n+1}'(x) = xP_n'(x) + (n+1)P_n(x)$$

Replacing n by $(n-1)$, we get (v).

(iii) Rewriting (ii) and (v) as

$$xP_n'(x) - P_{n-1}'(x) = nP_n(x) \quad \ldots(vi)$$

and $$P_n'(x) - xP_{n-1}'(x) = nP_{n-1}(x) \quad \ldots(vii)$$

Multiplying (vi) by x and subtracting from (vii), we get

$$(1-x^2)P_n'(x) = nP_{n-1}(x) - nxP_n(x)$$

Example 10.15 Show that $(2n+1)(1-x^2)P_n'(x) = n(n+1)[P_{n-1}(x) - P_{n+1}(x)]$.

Solution: We have the recurrence formula

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$$

or $$\frac{(n+1)}{(2n+1)}P_{n+1}(x) = xP_n(x) + \frac{1}{(2n+1)}[nP_{n-1}(x) - (n+1)P_n(x)]$$

or $$(n+1)[xP_n(x) - P_{n+1}(x)] = n[xP_{n-1}(x) - xP_n(x)]$$

$$= (1-x^2)P_n'(x) \quad [\because (1-x^2)P_n'(x) = nP_{n-1}(x) - nxP_n(x)] \quad \ldots(i)$$

or $$xP_n(x) - P_{n+1}(x) = \frac{(1-x^2)P_n'(x)}{n+1} \quad \ldots(ii)$$

Also from (i) $$xP_n(x) - P_{n-1}(x) = \frac{(1-x^2)P_n'(x)}{n} \quad \ldots(iii)$$

From (ii) and (iii) $$P_{n-1}(x) - \frac{(1-x^2)P_n'(x)}{n} = P_{n+1}(x) + \frac{(1-x^2)P_n'(x)}{n+1}$$

or $$n(n+1)P_{n-1}(x) - (n+1)(1-x^2)P_n'(x) = n(n+1)P_{n+1}(x) + n(1-x^2)P_n'(x)$$

or $$(2n+1)(1-x^2)P_n'(x) = n(n+1)[P_{n-1}(x) - P_{n+1}(x)]$$

(1) ORTHOGONALITY OF LEGENDRE POLYNOMIALS

For a full proof see, $$\int_{-1}^{1} P_m(x)P_n(x)\,dx = \begin{cases} 0 & m \neq n \\ \dfrac{2}{2n+1} & m = n \end{cases}$$

We know that the solutions of

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0 \quad \ldots(i)$$

and $$(1-x^2)y'' - 2xy' + m(m+1)y = 0 \quad \ldots(ii)$$

are $P_n(x)$ and $P_m(x)$ respectively.

Multiplying (i) by v and (ii) by u and subtracting, we get

$$(1-x^2)(uv'' - vu'') - 2x(uv' - vu') = -n(n+1)uv + m(m+1)uv$$

or $$\frac{d}{dx}[(1-x^2)(uv' - vu')] = (m-n)(m+n+1)uv$$

Now integrating from -1 to 1, we get

$$(m-n)(m+n+1)\int_{-1}^{1} uv\,dx = \left[(1-x^2)(uv' - vu')\right]_{-1}^{1} = 0$$

Hence $$\int_{-1}^{1} P_m(x)P_n(x)\,dx = 0, \quad m \neq n \quad \ldots(10)$$

This is known as the orthogonality property of Legendre polynomials.

(G.T.U., 2005; Madras, UPTU; P.T.U., 2006)

When $n = n$, we have from Rodrigue's formula,

$$\therefore (2n)!\int_{-1}^{1} P_n^2(x)\,dx = \int_{-1}^{1} D^n(x^2-1)^n \cdot D^n(x^2-1)^n\,dx \qquad \text{[Integrate by parts]}$$

$$= \left[D^n(x^2-1)^n \cdot D^{n-1}(x^2-1)^n\right]_{-1}^{1} - \int_{-1}^{1} D^{n+1}(x^2-1)^n \cdot D^{n-1}(x^2-1)^n\,dx$$

Since $D^{n-1}(x^2-1)^n$ has (x^2-1) as a factor, the first term on the right vanishes for $n = n$, Thus

$$\therefore (2n)!\int_{-1}^{1} P_n^2(x)\,dx = -\int_{-1}^{1} D^{n+1}(x^2-1)^n \cdot D^{n-1}(x^2-1)^n\,dx$$

[Integrate by parts ($n-1$) times]

$$= (-1)^n \int_{-1}^{1} D^{2n}(x^2-1)^n \cdot (x^2-1)^n\,dx = (-1)^n \int_{-1}^{1} (2n)!\,(x^2-1)^n\,dx$$

$$= (2n)!\int_{-1}^{1} (1-x^2)^n\,dx \qquad \text{[Put } x = \sin\theta\text{]}$$

$$= (2n)!\int_{0}^{\pi/2} \cos^{2n+1}\theta\,d\theta = (2n)!\,\frac{2n\cdot(2n-2)\cdots 4\cdot 2}{(2n+1)(2n-1)\cdots 3\cdot 1}$$

$$= (2n)!\,[2^n n!]^2 = (2n)\cdots 4\cdot 2)^2[(2n+1)] = \frac{2}{2n+1}\cdot(2^n n!)^2$$

Hence $$\int_{-1}^{1} P_n^2(x)\,dx = \frac{2}{2n+1} \qquad \ldots(i)$$ [Nagpur, 2004, V.T.U., 2007; J.N.T.U., 2006]

(2) Fourier-Legendre expansion of $f(x)$. If $f(x)$ is a function defined from $x = -1$ to $x = 1$, we can write

$$f(x) = \sum_{n=0}^{\infty} c_n P_n(x) \qquad \ldots(ii)$$

To determine the coefficient c_n, multiply both sides by $P_n(x)$ and integrate from -1 to 1. Then (i) and (ii) give

$$\int_{-1}^{1} f(x)P_n(x)\,dx = c_n \int_{-1}^{1} P_n^2(x)\,dx = \frac{2c_n}{2n+1} \quad \text{or} \quad c_n = \left(n+\frac{1}{2}\right)\int_{-1}^{1} f(x)P_n(x)\,dx$$

Equation (ii) is known as *Fourier-Legendre expansion of* $f(x)$.

Example 10.23. Show that $\displaystyle\int_{-1}^{1} x^2 P_n(x)\,dx = \frac{2n}{4n^2-1}$.

Solution. The recurrence formula (i) can be written as

$$(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}$$

or $$(2n-1)xP_{n-1} = nP_n + (n-1)P_{n-2} \qquad \text{[Changing } n \text{ to } n-1\text{]}$$

Multiplying by P_n, we get $xP_nP_{n-1} = \frac{1}{2n-1}[nP_n^2 + (n-1)P_nP_{n-2}]$

Integrating both sides w.r.t. x from $x = -1$ to $x = 1$, we get

$$\int_{-1}^{1} xP_nP_{n-1}\,dx = \frac{n}{2n-1}\int_{-1}^{1} P_n^2\,dx + \frac{n-1}{2n-1}\int_{-1}^{1} P_nP_{n-2}\,dx$$

$$= \frac{n}{2n-1}\left[\frac{2}{2n+1}\right] + \frac{n-1}{2n-1}(0), \text{ by Rodrigue's property}$$

Hence $$\int_{-1}^{1} xP_n(x)P_{n-1}(x)\,dx = \frac{2n}{4n^2-1}$$

Example 10.24. Show that $\displaystyle\int_{-1}^{1} (1-x^2)\,P_n'(x)P_m'(x)\,dx = \begin{cases} 0 & \text{when } m \neq n \\ \dfrac{2n(n+1)}{2n+1} & \text{when } m = n \end{cases}$

[U.P.T.U., 2002; J.N.T.U., 2006; U.P.T.U., 2006]

Solution. Integrating by parts,

$$\int_{-1}^{1}(1-x^2)P_n'(x)P_m'(x)\,dx = \left|(1-x^2)P_n'(x)P_m(x)\right|_{-1}^{1} - \int_{-1}^{1}\frac{d}{dx}\{(1-x^2)P_n'(x)\}P_m\,dx$$

$$= -\int_{-1}^{1}P_m\{(1-x^2)P_n''(x) - 2xP_n'(x)\}\,dx \qquad \ldots(1)$$

Now $P_n(x)$ being a solution of Legendre's equation

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0, \text{ we have}$$

$$(1-x^2)P_n''(x) - 2xP_n'(x) = -n(n+1)P_n(x)$$

Substituting this in (1), we get

$$\int_{-1}^{1}(1-x^2)P_n'(x)P_m'(x)\,dx = -\int_{-1}^{1}P_m\{-n(n+1)P_n(x)\}\,dx$$

$$= n(n+1)\int_{-1}^{1}P_nP_m\,dx \qquad \ldots(2)$$

When $m \neq n$, $\int_{-1}^{1}P_nP_m\,dx = 0$, by orthogonality property.

$$\therefore \quad \int_{-1}^{1}(1-x^2)P_n'(x)P_m'(x)\,dx = n(n+1),\ 0 = 0 \qquad \text{Hence (a) is}$$

When $m = n$, $\int_{-1}^{1}P_nP_n\,dx = \frac{2}{2n+1}$, by orthogonality property.

$$\therefore \quad \int_{-1}^{1}(1-x^2)P_n'(x)P_n'(x)\,dx = n(n+1)\cdot\frac{2}{2n+1} = \frac{2n(n+1)}{(2n+1)}$$

Example 16.28. Show that $\int_{-1}^{1}x^2P_{n+1}P_{n-1}\,dx = \frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}$.

(U.P.T.U., 2004; Kerala M. Univ., 2001)

Solution. We have from the recurrence relation (3),

$$(2n+3)xP_{n+1} = (n+3)P_{n+2} + nP_{n-1}$$

$$\therefore \quad xP_{n+1} = \frac{1}{2n+3}\{(n+3)P_{n+2} + nP_{n-1}\}$$

and

$$xP_{n-1} = \frac{1}{2n+1}\{(n+2)P_{n+1} + (n+1)P_n\}$$

$$\therefore \quad xP_{n+1}\cdot xP_{n-1} = \frac{1}{(2n+3)(2n+1)}\{(n+3)P_{n+2}P_{n-1} + n(n+1)P_n^2$$

$$+ (n+2)(n+3)P_{n+2}P_{n+1} + n(n-1)P_{n-1}P_n\}$$

Integrating both sides from -1 to 1 and using orthogonality of Legendre polynomials, we get

$$\int_{-1}^{1}x^2P_{n+1}P_{n-1}\,dx = \frac{n(n+1)}{(2n-1)(2n+3)}\int_{-1}^{1}P_n^2\,dx = \frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}$$

Example 16.29. If $f(x) = \begin{cases} -k, & -1 < x < 0 \\ k, & 0 < x < 1 \end{cases}$

show that $f(x) = \frac{3}{2}P_1(x) + \frac{7}{8}P_3(x) + \frac{11}{32}P_5(x) - \frac{25}{128}P_7(x) + \ldots$ (U.P.T.U., 2003)

Solution. Let $f(x) = \sum_{n=0}^{\infty}a_nP_n(x)$

Then a_n is given by $a_n = \left(n+\frac{1}{2}\right)\int_{-1}^{1} f(x)P_n(x)\,dx$

$$= \left(n+\frac{1}{2}\right)\left[\int_{-1}^{0} -P_n(x)\,dx + \int_{0}^{1} xP_n(x)\,dx\right] = \left(n+\frac{1}{2}\right)\int_{0}^{1} xP_n(x)\,dx$$

$\therefore$ $$a_0 = \frac{1}{2}\int_0^1 xP_0(x)\,dx = \frac{1}{2}\int_0^1 x\,dx = \frac{1}{4}$$

$$a_1 = \frac{3}{2}\int_0^1 xP_1(x)\,dx = \frac{3}{2}\int_0^1 x^2\,dx = \frac{1}{2}$$

$$a_2 = \frac{5}{2}\int_0^1 xP_2(x)\,dx = \frac{5}{2}\int_0^1 x\cdot\frac{3x^2-1}{2}\,dx = \frac{5}{4}\left[\frac{3x^4}{4}-\frac{x^2}{2}\right]_0^1 = -\frac{5}{16}$$

$$a_3 = \frac{7}{2}\int_0^1 xP_3(x)\,dx = \frac{7}{2}\int_0^1 x\cdot\frac{5x^3-3x}{2}\,dx = \frac{7}{4}\left[\frac{x^5}{5}-\frac{x^3}{3}\right]_0^1 = 0$$

$$a_4 = \frac{9}{2}\int_0^1 xP_4(x)\,dx = \frac{9}{2}\int_0^1 x\cdot\frac{35x^4-30x^2+3}{8}\,dx$$

$$= \frac{9}{32}\left[\frac{35x^6}{6}-\frac{35x^4}{4}+\frac{3x^2}{2}\right]_0^1 = -\frac{3}{128} \text{ and so on.}$$

Hence $$f(x) = \frac{1}{4}P_0(x) + \frac{1}{2}P_1(x) - \frac{5}{16}P_2(x) - \frac{3}{128}P_4(x) + \ldots$$

PROBLEMS 16.4

1. Show that $P_n(-x) = (-1)^n P_n(x)$ (Edexcel, 2003) (Y/T) (Edexcel, 2004)

2. Prove that (i) $P_{2n}(0) = 0$, (ii) $P_{2n+1}(0) = \frac{(-1)^n(2n)!}{2^{2n}(n!)^2}$, (iii) $P_n(1) = P_n(-1) = (-1)^n\frac{(n+1)(n+2)}{2}$ (Y/T) (Y/T), (Edexcel)

3. Express the following in terms of Legendre polynomials: (i) $3x^2+x$, (ii) x^3+2x^2-x-3. (Edexcel, 2003) (iii) $4x^3+6x^2+7x+2$ (Y/T) (Y/T), (Edexcel)
(iv) $x^4+2x^3-x^2+4x+2$. (Edexcel, 2005) (Edexcel, 2005)

4. Prove that (i) $(2n+1)P_n(x) = P'_{n+1}(x) - P'_{n-1}(x)$,
(ii) $P'_{n+1}(x) = P'_{n-1}(x) + (2n+1)P_n(x)$, (iii) $P_n(x) = P'_{n+1}(x) - P'_{n-1}(x)$, (iv) $\int P_n(x)\,dx = \frac{1}{2n+1}P_{n+1}(x) - \frac{1}{2n-1}P_{n-1}(x)$ (Edexcel, 2002) (Y)

5. Prove that (i) $\int_{-1}^{1} (1-x^2)P'_n(x)P'_m(x)\,dx = 0$, $m \neq n$, $m, n \in \mathbb{Z}^+$ (Y/T) (Y/T), (ii) $\int_{-1}^{1} P_n(x)\,dx = 0$, $n \in \mathbb{Z}^+$.

6. Prove that $\int_{-1}^{1} P_n(x)P_{n+1}(x)\,dx = 0$, $n \in \mathbb{Z}^+$.

7. Show that $\int_{-1}^{1} (1-x^2)P'_n(x)P'_m(x)\,dx = \frac{n(n+1)}{2n+1}$, $m = n$, $m, n \in \mathbb{Z}^+$. (Y/T) (Y/T), (Edexcel)

8. Using Rodrigues's formula, show that $P_n(x)$ satisfies the differential equation
$$\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right] + n(n+1)y = 0.$$

9. Express the following functions in terms of Legendre polynomials in the interval $-1 < x < 1$:
(i) $f(x) = x^3 - 2x^2 + x$, (ii) $f(x) = x^3 + x^2 + 2x^2 + x - 2$. (Y/T) (Y/T), (Edexcel)

10. If $f(x) = 0$, $-1 < x < 0$
$= 1$, $0 < x < 1$, show that $f(x) = \frac{1}{2}P_0(x) + \frac{3}{4}P_1(x) - \frac{7}{32}P_3(x) + \ldots$

16.8 OTHER SPECIAL FUNCTIONS

The following special functions occur in numerous engineering problems. We state below their important properties which can be verified by similar methods:

(1) **Laguerre's polynomials*.** These are the solutions of Laguerre's differential equation

$$xy'' + (1 - x)y' + ny = 0 \qquad ...(1)$$

These polynomials $L_n(x)$ are given by the corresponding Rodrigue's formula

$$L_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x}) \qquad ...(2)$$

In particular, $L_0(x) = 1$; $L_1(x) = 1 - x$, $L_2(x) = 2 - 4x + x^2$; $L_3(x) = 6 - 18x + 9x^2 - x^3$. (Madras, 2000)

Their generating function is given by

$$\frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} \frac{L_n(x)}{n!} t^n \qquad ...(3)$$

The orthogonal property for these polynomials is

$$\int_0^\infty e^{-x} L_m(x) L_n(x)\, dx = \begin{cases} 1, & m = n \\ |n|!, & m = n \end{cases} \qquad ...(4)$$

(2) **Hermite's polynomials†.** These are the solutions of Hermite's differential equation

$$y'' - 2xy' + 2ny = 0 \qquad ...(5)$$

These polynomials $H_n(x)$ are given by the Rodrigue's formula

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}(e^{-x^2}) \qquad ...(6)$$

In particular, $H_0(x) = 1$; $H_1(x) = 2x$; $H_2(x) = 4x^2 - 2$; $H_3(x) = 8x^3 - 12x$. (Madras, 2000)

Their generating function is given by

$$e^{2xt - t^2} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n \qquad ...(7) \quad \text{(Madras, 2002 S)}$$

The orthogonal property of these polynomials is

$$\int_{-\infty}^{\infty} e^{-x^2} H_m(x) H_n(x)\, dx = \begin{cases} 0, & m \neq n \\ 2^n \cdot n! \sqrt{\pi}, & m = n \end{cases} \qquad ...(8)$$

(3) **Chebyshev polynomials‡.** These polynomials denoted by $T_n(x)$ are the solutions of the differential equation

$$(1 - x^2)y'' - xy' + n^2 y = 0 \qquad ...(9)$$

Their generating function is

$$\frac{1 - xt}{1 - 2xt + t^2} = \sum_{n=0}^{\infty} T_n(x)\, t^n \qquad ...(10)$$

and $T_n(x) = \frac{n}{2} \sum_{r=0}^{n} (-2)^r \frac{|n+r-1|}{r!(n-2r)!} (2x)^{n-2r}$ ...(11)

(U.P.T.U., 2006)

Here $N = \frac{1}{2}$. If it is even and $N = \frac{1}{2}(n-1)$, if it is odd.

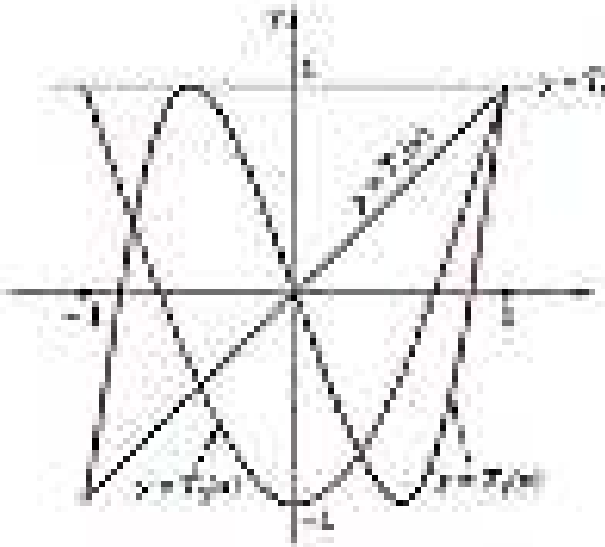

Fig. 16.1: Graphs of $T_0(x)$, $T_1(x)$, $T_2(x)$, $T_3(x)$.

* Named after the French mathematician *Edmond Laguerre* (1834–86) who is known for his work in infinite series and geometry.

† For footnote p. 58

‡ Named after the Russian mathematician *Pafnuti Chebyshev* (1821–1894) who is known for his work in the theory of numbers and approximation theory.

In particular, $T_0(x) = 1, T_1(x) = x, T_2(x) = 2x^2 - 1, T_3(x) = 4x^3 - 3x$. Also, we have the recurrence relation

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x) \quad \text{...(10)} \quad \text{(Bhopal, 2002)}$$

which defines T_{n+1} in terms of T_n and T_{n-1}.

Their orthogonal property is

$$\int_{-1}^{1} (1-x^2)^{-1/2} T_m(x) T_n(x)\,dx = \begin{cases} 0, & m \neq n \\ \frac{\pi}{2}, & m = n, \\ \pi, & m = n = 0 \end{cases} \quad \text{...(11)}$$

Example 16.24. Prove that $\int_{-1}^{1} x^m L_n(x) L_n(x)\,dx = 0$, $m < n$. (Anna, 2006)

Solution. Since $L_n(x)$ and $L_n(x)$ are the solutions of the Laguerre's differential equation (1).

$$\therefore \quad xL_n'' + (1-x)L_n' + nL_n = 0 \quad \text{...(a)}$$

$$xL_n'' + (1-x)L_n' + nL_n = 0 \quad \text{...(b)}$$

Multiplying (a) by L_n and (b) by L_n and subtracting, we get

$$xL_nL_n'' - L_nL_n' + (1-x)(L_nL_n' - L_nL_n') = (n-n)L_nL_n$$

$$\text{or} \quad \frac{d}{dx}(L_nL_n' - L_nL_n') + \frac{1-x}{x}(L_nL_n' - L_nL_n') = \frac{(n-n)L_nL_n}{x}$$

This is Leibnitz's linear equation and its

$$\text{I.F.} = e^{\int \left(\frac{1}{x}-1\right)dx} = e^{\log x - x} = xe^{-x}.$$

$\therefore$ Its solution is $(L_nL_n' - L_nL_n')xe^{-x} = \int_0^x \frac{(n-n)L_nL_n}{x} xe^{-x}\,dx$

$$\text{or} \quad \int_0^x e^{-x} L_nL_n\,dx = \left[\frac{L_nL_n' - L_nL_n'}{n-n}xe^{-x}\right]_0^x = 0 \text{ which proves the result.}$$

Example 16.25. Prove that $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}(e^{-x^2})$.

Solution. The generating function for $H_n(x)$ is $e^{2xt-t^2} = e^{x^2} e^{-(t-x)^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$

Then $$\left[\frac{\partial^n}{\partial t^n} e^{2xt-t^2}\right]_{t=0} = H_n(x) \quad \text{...(a)}$$

Also $$\left[\frac{\partial^n}{\partial t^n} e^{2xt-t^2}\right]_{t=0} = e^{x^2}\left[\frac{\partial^n}{\partial t^n} e^{-(t-x)^2}\right]_{t=0}$$

$$= e^{x^2}\left[\frac{\partial^n}{\partial(-x)^n} e^{-(t-x)^2}\right]_{t=0} = (-1)^n e^{x^2} \frac{d^n}{dx^n}(e^{-x^2}) \quad \text{...(b)}$$

Equating (a) and (b), we get the desired result.

EXERCISE 16.4

1. Using the generating function of Hermite polynomials, show that $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$.

2. Show that: $(1+x)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x)$. (Anna, 2003)

3. Prove that: $(1-x^2)P_n'(x) = (n+1)[xP_n(x) - P_{n+1}(x)]$. (Anna, 2003)

4. Prove that $\int_{-1}^{1} P_n(x)\,dx = \frac{P_{n-1}(-1) - P_{n+1}(-1)}{2n+1}$, $n \geq 1$.

5. Using the generating function of Legendre polynomials, obtain the recurrence formula $nP_n(x) = (2n-1)xP_{n-1}(x) - (n-1)P_{n-2}(x)$.

6. Prove that $\int_{-1}^{1} x^n P_n(x)\,dx = \frac{2^n (n!)^2}{(2n+1)!}$ and $\int_{-1}^{1} x^n P_{n-1}(x)\,dx = 0$. (Madras, 2003)

7. Express $x^4 + 3x^3 + 2x^2 + 1$ in terms of Chebyshev polynomials $T_n(x)$. (U.P.T.U., 2005)

8. Show that $\int_{-1}^{1} T_n(x)\,dx = \begin{cases} 0, & n \text{ odd} \\ \frac{2}{1-n^2}, & n \text{ even} \end{cases}$ (Bhopal, 2003)
(Chennai, 2003)

9. Prove that $\frac{1-t^2}{1-2xt+t^2} = T_0(x) + 2\sum_{n=1}^{\infty} T_n(x)t^n$. (U.P.T.U., 2005)

3.3.0 (1) STURM[*]-LIOUVILLE[†] PROBLEM

Legendre's equation $(1-x^2)y'' - 2xy' + n(n+1)y = 0$ ...(1)

can be written as, $(1-x^2)y'' + \lambda y = 0$ [$\because \lambda = n(n+1)$] ...(2)

Bessel's equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)y = 0$ can be transformed by putting $x = kz$ so that

$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{dz}{dx} = \frac{1}{k}\cdot\frac{dy}{dz}$, $\frac{d^2y}{dx^2} = \frac{1}{k^2}\cdot\frac{d^2y}{dz^2}$ in the form

$$z^2y'' + zy' + (k^2z^2 - n^2)y = 0$$

or $$(xy')' + (kx - n^2/x)y = 0 \quad [\because z = x/k]$$ ...(3)

or $$xy'' + y' + (kx - n^2/x)y = 0$$ ...(3')

Both the equations (1) and (2) are of the form

$$py'' + qy' + (r + \lambda s)y = 0$$ ...(4)

which is known as the Sturm-Liouville's equation. Similarly Laguerre's, Hermite's equations etc. can also be reduced to (4). Thus all the above equations of engineering utility can be considered with a common approach by means of Sturm-Liouville's equation.

Equation (4) considered in some interval $a \leq x \leq b$, satisfying the conditions

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0, \ \alpha_1, \alpha_2 \text{ not both zero and } \beta_1 y(b) + \beta_2 y'(b) = 0$$ ...(5)

with the real constants α_1, α_2 not both zero and β_1, β_2 not both zero. The conditions (5) at the end points are called boundary conditions.

A differential equation together with the boundary conditions is called a **boundary value problem**. Equation (4) together with boundary conditions (5) is called a **Sturm-Liouville's problem**.

Obviously $y = 0$ is a solution of the problem for any value of the parameter λ which is a trivial solution and not useful in practical utility. Any other solution of (4) satisfying (5) is called an **eigen function** of the problem and the corresponding value of λ is called an **eigen value** of the problem.

A special case: Taking $p = r = s = 1$ and $q = 0$ in (4), we get

$$y'' + \lambda y = 0$$ ...(6)

Also if $\alpha_1 = \beta_1 = 1$ and $\alpha_2 = \beta_2 = 0$, then the boundary conditions (5) become

$$y(a) = 0, \quad y(b) = 0$$ ...(6')

Thus (6) and (6') constitute the simplest form of Sturm-Liouville's problem.

(2) Orthogonality. Of the various properties of eigen functions of Sturm-Liouville's problem the orthogonality is of special importance.

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[*] Named after the Swiss mathematician J.C.F. Sturm (1803–1855) who held Ampere's chair at Sorbonne university, Paris.

[†] Named after the French professor Joseph Liouville (1809–1882) who is known for his important contributions to complex analysis, special functions, number theory and differential geometry.

Def. Two functions $y_m(x)$ and $y_n(x)$ defined on some interval $a \le x \le b$, are said to be orthogonal on this interval with respect to the weight function $p(x)$ if $\int_a^b p(x)\, y_m(x) y_n(x)\, dx = 0$ when $m \neq n$

The norm of y_m, denoted by $\| y_m \|$, is defined to be the non-negative square root of $\int_a^b p(x)[y_m(x)]^2\, dx$

Thus $$\| y_m \| = \sqrt{\left\{ \int_a^b p(x)[y_m(x)]^2\, dx \right\}}$$

The functions which are orthogonal and for which $\| y_m \| = 1$ are said to be orthonormal on the interval.

2. Orthogonality of eigenfunctions.

Theorem. *If (1) the functions a, q, r and r' in the Sturm-Liouville equation (1) be continuous in $a \le x \le b$, (2) $y_m(x)$, $y_n(x)$ be two eigenfunctions of the Sturm-Liouville problem corresponding to eigenvalues λ_m and λ_n respectively, then $y_m(x)$ and $y_n(x)$ are orthogonal on that interval with the weight function $p(x)$.*

Proof. Since y_m and y_n satisfy (1) above

$$(ry'_m)' + (\lambda_m p + q) y_m = 0$$
$$(ry'_n)' + (\lambda_n p + q) y_n = 0$$

Multiplying the first equation by y_n and the second by $-y_m$ and adding, we get

$$(\lambda_m - \lambda_n) p y_m y_n = y_m (r y'_n)' - y_n (r y'_m)'$$
$$= \frac{d}{dx} [r y'_n y_m - r y'_m y_n] \text{ after differentiation.}$$

Now integrating both sides w.r.t. x from a to b, we obtain

$$(\lambda_m - \lambda_n) \int_a^b p y_m y_n\, dx = [r y'_n y_m - r y'_m y_n]_a^b$$
$$= r(b)\,[y'_n(b) y_m(b) - y'_m(b) y_n(b)] - r(a)\,[y'_n(a) y_m(a) - y'_m(a) y_n(a)] \quad \ldots (4)$$

The B.V.P. will consist of the boundary conditions are of one of the following forms:

I. $y(a) = y(b) = 0$; II. $y'(a) = y'(b) = 0$; III. $\alpha_1 y(a) + \alpha_2 y'(a) = 0$, $\beta_1 y(b) + \beta_2 y'(b) = 0$ where either α_1 and α_2 is not zero and either β_1 or β_2 is not zero.

Thus in each case (4) reduces to $\int_a^b p y_m y_n\, dx = 0 \quad (m \neq n)$

which shows that the eigenfunctions y_m and y_n are orthogonal on $a \le x \le b$ w.r.t. the weight function $p(x) = 0$.

Obs. The third form of the boundary conditions in fact includes the first two forms as particular cases.

Cor. 1. Orthogonality of Legendre polynomials has already been established directly in § 15.17. But it follows even more from the above theorem.

We have already seen in para (3) that Legendre's equation is Sturm-Liouville equation

$$[(1 - x^2) y']' + \lambda y = 0 \quad \text{[Refer to (1)]}$$

with $r(x) = 1 - x^2$, $p(x) = 1$ and $q(x) = 0$.

Since $y(-1) \neq 0$ and $y(1) \neq 0$ and for $\lambda = 0, 2, 6, \ldots, k = 0, 1, 2, 3, \ldots$, the Legendre polynomials are the solutions of the problem i.e., these are the eigenfunctions. Then it follows by the above theorem, that they are orthogonal on $-1 \le x \le 1$.

Cor. 2. Orthogonality of Bessel functions has also been established directly in § 18.11. But it can easily be seen to follow from the above theorem.

In para (3), we have obtained the Bessel's equation

$$x^2 \frac{d^2 J_n}{dx^2} + x \frac{dJ_n}{dx} + (k^2 x^2 - n^2) J_n(x) = 0$$

or, $[x J'_n(kx)]' + (k^2 x - n^2/x) J_n(kx) = 0$ which is Sturm-Liouville equation with $r(x) = x$, $p(x) = x$, $q(x) = -n^2/x$ and $\lambda = k^2$. Also $r(0) = 0$, it follows from the above theorem that those solutions of $J_n(kx)$ which are zero at $x = 0$ form an orthogonal set on $0 \le x \le R$ with weight function $p(x) = x$.

Example 13.20. For the boundary value problem $y'' + \lambda y = 0$, $y(0) = 0$, $y(l) = 0$, find the eigen functions and show that they are orthogonal.

Solution. For $\lambda = -p^2$, the general solution of the equation is $y(x) = c_1 e^{px} + c_2 e^{-px}$

The above boundary conditions give $c_1 = c_2 = 0$ and $y = 0$ which is not an eigen function.

For $\lambda = p^2$, the general solution is $y(x) = A \cos px + B \sin px$

The first boundary condition gives $y(0) = A = 0$ and the second boundary condition gives $y(l) = B \sin pl = 0$, $pl = 0, \pi, 2\pi, \ldots$ That the eigen values are $\lambda = 0, \pi^2/l^2, 4\pi^2/l^2, \ldots$ and taking $B = 1$, the corresponding eigen functions are

$$y_n(x) = \sin\left(\frac{n\pi x}{l}\right) \qquad n = 0, 1, 2, \ldots$$

From the above theorem, it follows that the solid eigen functions are orthogonal on the interval $0 \le x \le l$.

Obs. This problem concerns an elastic string stretched between fixed points $x = 0$ and $x = l$ and thereby provides the theoretical basis for the vibrations of the deflected string shown in the Lucas Game § 18.41.

PROBLEMS 13.3

Find the eigen functions and the orthogonality relation for the following boundary value problems and verify by direct integration:

1. $y'' + \lambda y = 0$, $y(0) = 0$, $y(\pi) = 0$
2. $y'' + \lambda y = 0$, $y'(0) = 0$, $y(\pi) = 0$
3. $y'' + \lambda y = 0$, $y'(0) = 0$, $y'(l) = 0$
4. $y'' + \lambda y = 0$, $y(0) = 0$, $y(l) = 0$
5. $x^2y'' + xy' + \lambda y = 0$, $y(1) = 0$, $y(e^{\pi}) = 0$

Transform each of the following equations into the Sturm-Liouville equation before using the result in Exercise:

6. Chebyshev's equation $(1 - x^2)y'' - xy' + n^2y = 0$
7. Hermite's equation $y'' - 2xy' + 2ny = 0$

SECTION 13.4 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS 13.4

Either fill in the blanks or choose the correct answer in the following problems:

1. In terms of Legendre polynomials $2 - 3x + 4x^2 =$ ——

2. $P_{-n}(x) =$ —— 3. $\int_{-1}^{1} P_n^2(x)\,dx =$ —— 4. $P_{n+1}(1) =$ ——

5. $\int_{-1}^{1} x^m P_n(x)\,dx =$ —— (m being an integer $< n$)

6. The recurrence relation connecting $P_n(x)$, $P_{n+1}(x)$ and $P_{n-1}(x)$ is ——

7. Orthogonality relation for Bessel functions is ——

8. Bessel's equation of order n is —— 9. $J_{3/2}(x) =$ ——

10. $\frac{d}{dx}[x^n J_n(x)] =$ —— 11. Value of $J_0(x)$ at $x =$ ——

12. $\int_{-1}^{1} P_n(x) P_{n+1}(x)\,dx =$ —— 13. $P_n(-1) = (-1)^n$ (True or False)

14. Rodrigue's formula for $P_n(x)$ is —— 15. $\int_{-1}^{1} [P_n(x)]^2\,dx = 2/(2n+1)$ ——

16. Expansion of $3x^2 + x$ in terms of Legendre polynomials is ——

17. Generating function of $P_n(x)$ is —— 18. $\frac{d}{dx} J_0(x) =$ ——

19. Bessel equation of order 4 is $x^2y'' + xy' + x^2y - 4y = 0$. (True or False)

20. $\int_{-1}^{1} [P_n(x)]^2\,dx = 2/(n+1)$. (True or False)

21. Legendre's polynomial of first degree is —— (True or False)

34. If $x = 0$ is a root of $P_n(x) = 0$, then $P_{n-1}(x)$ and $P_{n+1}(x)$ are of opposite signs. (True or False)
35. $x = 1$ is a singular point of the equation $x(1-x^2)y'' + 2xy' + n(n+1)y = 0$. (True or False)
36. $\sin x = 2J_1 - 3J_3 + 5J_5 - \ldots$ (True or False)
37. If J_1 and J_2 are Bessel functions, then $J_1(x)$ is given by

(a) $-J_2$ (b) $\frac{2}{x}J_2(x) - 3J_3(x)$ (c) $\frac{2}{x}J_2(x) + \frac{3}{x}J_3(x)$

38. If $J_n(x)$ is the Bessel function of first kind, then $\int_0^x x^3 J_2(x)\,dx =$

(a) 2 (b) -2 (c) 0 (d) 1

39. If $a_0 P_0(x) + a_1 P_1(x) + a_2 P_2(x) + a_3 P_3(x) + a_4 P_4(x)$ is identically zero, then

(a) 0 (b) 1 (c) -1 (d) none of these

40. The series $x = \frac{1}{2^2(1!)^2}x^2 + \frac{3^2}{2^4(2!)^2}x^4 - \frac{5^2}{2^6(3!)^2}x^6 + \ldots$ is equals

(a) $J_0(x)$ (b) $J_1(x)$ (c) $xJ_1(x)$ (d) $xJ_0(x)$

41. If $\int_{-1}^{1} P_n(x)\,dx = 0$, then n is

(a) 0 (b) 2 (c) -1 (d) none of these

42. The value of $\int_{-1}^{1} (2x+1)P_2(x)\,dx$, where $P_2(x)$ is the second degree Legendre polynomial, is

(a) 2 (b) -1 (c) 0 (d) 3

43. The value of the integral $\int_{-1}^{1} x^3 P_3(x)\,dx$, where $P_3(x)$ is a Legendre polynomial of degree 3, is

(a) 0 (b) $\frac{2}{35}$ (c) $\frac{4}{35}$ (d) $\frac{16}{35}$

44. The polynomial $2x^3 + x - 3$ in terms of Legendre polynomials is

(a) $\frac{1}{2}(4P_3 - 3P_1 + 12P_0)$ (b) $\frac{3}{2}(4P_3 + 3P_1 - 12P_0)$

(c) $\frac{1}{2}(4P_3 + 3P_1 + 12P_0)$ (d) $\frac{3}{2}(4P_3 - 3P_1 - 12P_0)$

45. If $P_n(x)$ is the Legendre polynomial, then $P_n(-x)$ is equal to

(a) $(-1)^n P_n(x)$ (b) $-P_n(x)$ (c) $(-1)^{n+1} P_n(x)$ (d) $P_n^{-1}(x)$

46. Legendre polynomial $P_3(x) = \frac{1}{2}(5x^3 - 3x^2 + Ax)$ where A is equal to

(a) 10 (b) 12 (c) 15 (d) 30

47. $\int_{-1}^{1} (1-x)\,P_1(x)\,dx$, $(n \neq 1)$, is equal to

(a) $\frac{2}{2n+1}$ (b) $\frac{-2}{2n+1}$ (c) $\frac{-4}{2n+1}$ (d) 0

48. The singular points of the differential equation $x(x-2)^2(x+3)y'' + 3(x+1)y' + 4y = 0$ are ..... (IES, 2006)

CHAPTER 17

Partial Differential Equations

1. Introduction. 2. Formation of partial differential equations. 3. Solutions of a partial differential equation. 4. Equations solvable by direct integration. 5. Linear equations of the first order. 6. Nonlinear equations of the first order. 7. Charpit's method. 8. Homogeneous linear equations with constant coefficients. 9. Rules for finding the complementary function. 10. Rules for finding the particular integral. 11. Working procedure to solve homogeneous linear equations of any order. 12. Non-homogeneous linear equations. 13. Non-linear equations of the second order—Monge's Method. 14. Objective Type of Questions.

17.1 INTRODUCTION

The reader has, already been introduced to the notion of partial differential equations. Here, we shall begin by studying the ways in which partial differential equations are formed. Then we shall investigate the solutions of special types of partial differential equations of the first and higher orders.

In what follows x and y will usually be taken as the independent variables and z, the dependent variable so that $z = f(x, y)$ and we shall employ the following notation:

$$\frac{\partial z}{\partial x} = p, \frac{\partial z}{\partial y} = q, \frac{\partial^2 z}{\partial x^2} = r, \frac{\partial^2 z}{\partial x \partial y} = s, \frac{\partial^2 z}{\partial y^2} = t.$$

17.2 FORMATION OF PARTIAL DIFFERENTIAL EQUATIONS

Unlike the case of ordinary differential equations which arise from the elimination of arbitrary constants, the partial differential equations can be formed either by the elimination of arbitrary constants or by the elimination of arbitrary functions from a relation involving three or more variables. The method is best illustrated by the following examples:

Example 17.1. *Form a partial differential equation by eliminating the constants from the equation*

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2} \qquad \ldots(1)$$

Solution. Differentiating (1) partially with respect to x and y, we get

$$2\frac{\partial z}{\partial x} = \frac{2x}{a^2} \quad \text{or} \quad \frac{1}{a^2} = \frac{1}{x}\frac{\partial z}{\partial x} = \frac{p}{x}$$

and

$$\frac{2\partial z}{\partial y} = \frac{2y}{b^2} \quad \text{or} \quad \frac{1}{b^2} = \frac{1}{y}\frac{\partial z}{\partial y} = \frac{q}{y}$$

Substituting these values of $1/a^2$ and $1/b^2$ in (1), we get

$$2z = xp + yq$$

as the desired partial differential equation of the first order.

Example 17.8. Form the partial differential equation by eliminating the arbitrary functions from
(a) $z = f(x + y) + g(x - y)$ (P.T.U., 2008)
(b) $z = f(x^2 + y^2)$ (P.T.U., 2008) (c) $f(z^2 + y^2, z - xy) = 0$ (R.G.P.V., 2009)

Solution. (a) We have $z = f(x + y) + g(x - y)$ ...(1)

Differentiating z partially with respect to x and y,

$$p = \frac{\partial z}{\partial x} = f'(x + y) + g'(x - y) \quad ...(2)$$

$$q = \frac{\partial z}{\partial y} = f'(x + y) - g'(x - y) \quad ...(3)$$

From (2), $\quad r = \frac{\partial p}{\partial x} = f''(x + y) + g''(x - y)$

From (3), $\quad t = \frac{\partial q}{\partial y} = f''(x + y) + g''(x - y)$

Hence $\quad r = t$ is the required equation.

(b) We have $\quad z = f(x^2 + y^2)$ ...(1)

Differentiating z partially with respect to x and y,

$$\frac{\partial z}{\partial x} = f'(x^2 + y^2) \cdot 2x, \quad \frac{\partial^2 z}{\partial x^2} = f''(x^2 + y^2) \cdot 2x \cdot 2x + f'(x^2 + y^2) \cdot 2 \quad ...(2)$$

$$\frac{\partial z}{\partial y} = f'(x^2 + y^2) \cdot 2y, \quad \frac{\partial^2 z}{\partial y^2} = f''(x^2 + y^2) \cdot 2y \cdot 2y + f'(x^2 + y^2) \cdot 2 \quad ...(3)$$

From (2) and (3) $\quad \frac{\partial^2 z}{\partial x^2} = 4x^2 f''(x^2 + y^2) + 2f'(x^2 + y^2), \quad \frac{\partial^2 z}{\partial y^2} = 4y^2 f''(x^2 + y^2) + 2f'(x^2 + y^2)$

Thus the desired partial differential equation is $\frac{\partial^2 z}{\partial x^2} = \frac{x^2}{y^2} \frac{\partial^2 z}{\partial y^2}$

which is an equation of the second order and (1) is its solution.

(c) Let $x^2 + y^2 = u$ and $z - xy = v$ so that $f(u, v) = 0$.

Differentiating partially w.r.t. x and y, we have

$$\frac{\partial f}{\partial u}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} p\right) + \frac{\partial f}{\partial v}\left[\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z} p\right] = 0$$

or $\quad \frac{\partial f}{\partial u}(2x) + \frac{\partial f}{\partial v}(-y + p) = 0 \quad ...(4)$

and $\quad \frac{\partial f}{\partial u}\left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} q\right) + \frac{\partial f}{\partial v}\left[\frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} q\right] = 0 \quad$ or $\quad \frac{\partial f}{\partial u}(2y) + \frac{\partial f}{\partial v}(-x + q) = 0 \quad ...(5)$

Eliminating $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$ from (4) and (5), we get

$$\begin{vmatrix} 2x & -y + p \\ 2y & -x + q \end{vmatrix} = 0 \quad \text{or} \quad xq - yp = x^2 - y^2.$$

Example 17.9. Find the differential equation of all planes which are at a constant distance k from the origin. (U.P.T.U., Mathematics-I, 2006)

Solution. The equation of the plane in normal form is

$$lx + my + nz = k \quad ...(1)$$

where l, m, n are the d.c.s of the normal from the origin to the plane.

Thus $z^2 + x^2 + y^2 + 2xy\cos\alpha = \sqrt{(1-l^2-m^2)}$

$\therefore$ (i) becomes $lx + my + \sqrt{(1-l^2-m^2)}\,z = a$ ...(iii)

Differentiating partially w.r.t. x, we get

$$l + \sqrt{(1-l^2-m^2)}\,.\,p = 0 \qquad \text{...(iv)}$$

Differentiating partially w.r.t. y, we get

$$m + \sqrt{(1-l^2-m^2)}\,.\,q = 0 \qquad \text{...(v)}$$

Now we have to eliminate l, m from (iii), (iv) and (v).

From (iv), $l = -\sqrt{(1-l^2-m^2)}\,.\,p$ and from (v), $m = -\sqrt{(1-l^2-m^2)}\,.\,q$

Squaring and adding, $l^2 + m^2 = (1-l^2-m^2)(p^2+q^2)$

or $$1^2 + m^2 = (1-l^2-m^2)(1+p^2+q^2) \text{ or } 1-l^2-m^2 = 1 - \frac{p^2+q^2}{1+p^2+q^2} = \frac{1}{1+p^2+q^2}$$

Also $$l = -\frac{p}{\sqrt{(1+p^2+q^2)}} \text{ and } m = -\frac{q}{\sqrt{(1+p^2+q^2)}}$$

Substituting the values of l, m and $1-l^2-m^2$ in (iii), we obtain

$$\frac{-px}{\sqrt{(1+p^2+q^2)}} - \frac{qy}{\sqrt{(1+p^2+q^2)}} + \frac{z}{\sqrt{(1+p^2+q^2)}} = a$$

or $z = px + qy + a\sqrt{(1+p^2+q^2)}$ which is the required partial differential equation.

EXERCISE 5.1

Form the partial differential equation by eliminating the arbitrary constants from

1. $z = ax + by + a^2 + b^2$.
2. $2z = ax^2 + by - b^2 + c^2 + c$. (Kurukshetra, 2005)
3. $(x-a)^2 + (y-b)^2 + z^2 = r^2$ and a, b. (Anna, 2006)
4. $z = x \sin\left[\frac{(y-a)}{(1-ay)}\right]$. (A.M.I.E., 2006)
5. Find the differential equation of all spheres of fixed radius having their centres in the xy-plane. (Madras, 2003)
6. Find the differential equation of all spheres whose centres lie on the z-axis. (Kerala, 2005)

Form the partial differential equations by eliminating the arbitrary functions from

7. $z = y f(x^2 - y^2)$. (S.V.T.U., 2006)
8. $z = f(x^2 + y^2)$. (Anna, 2005)
9. $z = xf(x/y) + g(y/x)$. (U.P.T.U., 2004)
10. $z = x^2 f(y/x) + y^2 g(y/x)$. (Anna, 2005)
11. $z = f(y/x) + e^{x/y}$.
12. $xyz = f(x+y+z)$.
13. $z = f(x/y)\,g(y/x)$.
14. $z = f(x^2 + y^2, z - xy)$. (V.T.U., 2006)
15. $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$. (U.P.T.U., 2005; A.M.I.E., 2003; Madras, 2003)
16. $z = f(y + ax) + g(y - ax)$. (Kurukshetra, 2006)
17. $z = \frac{1}{2}(y-x) - \frac{1}{2}(y-x)^2 + f(y+x)$.
18. $z = f(x+y) + xg(x+y) + x^2 h(x+y)$. (V.T.U., 2006)
19. $z = f(y) + xf(y) + x^2 f(y) + x^3 f(y)$. (U.P.T.U., 2005)
20. If $z = f(x^2 + y^2)$ and $y = \tan x$, prove that $y^2 \frac{\partial z}{\partial x} + (x^2 - y)\frac{\partial z}{\partial y} + x(1 + y^2) = 0$.

5.3 SOLUTIONS OF A PARTIAL DIFFERENTIAL EQUATION

It is clear from the above examples that a partial differential equation can arise in two ways viz. from the elimination of arbitrary constants and from the elimination of arbitrary functions.

The relation $f(x, y, z, a, b) = 0$ ...(1)

of the first order partial differential equation which contains two arbitrary constants is called a complete integral.

A solution obtained from the complete integral by assigning particular values to the arbitrary constants is called a particular integral.

If we put $b = \phi(a)$ in (3) and find the envelope of the family of surfaces $f(x, y, z, a, \phi(a)) = 0$, then we get a solution containing an arbitrary function a, which is called the general integral.

The envelope of the family of surfaces (3), with parameters a and b, if it exists, is called a singular integral. The singular integral differs from the particular integral in that it is not obtained from the complete integral by giving particular values to the constants.

17.2 EQUATIONS SOLVABLE BY DIRECT INTEGRATION

We now consider such partial differential equations which can be solved by direct integration. In place of the usual constants of integration, we get, however, some arbitrary functions of the variable held fixed.

Example 17.2. Solve $\dfrac{\partial^3 z}{\partial x^2 \partial y} = 12x^2y^2 - \cos(2x - y)$. (U.P.T.U., 2004)

Solution. Integrating twice with respect to x (keeping y fixed),

$$\frac{\partial^2 z}{\partial x \partial y} = 4x^3y^2 - \frac{1}{2}\cos(2x - y) = f(y)$$

$$\frac{\partial z}{\partial y} = x^4y^2 - \frac{1}{4}\sin(2x - y) = xf(y) + g(y).$$

Now integrating with respect to y (keeping x fixed),

$$z - x^4y^3 - \frac{1}{4}\cos(2x - y) = \int f(y)\,dy + \int g(y)\,dy + \phi(x)$$

The result may be simplified by writing

$$\int f(y)\,dy = \psi(y) \text{ and } \int g(y)\,dy = \chi(y),$$

Thus $z = \frac{1}{4}\cos(2x - y) + x^4y^3 + x\psi(y) + \chi(y) + \phi(x)$ where ϕ, ψ, χ are arbitrary functions.

Example 17.3. Solve $\dfrac{\partial^2 z}{\partial x^2} + z = 0$, given that when $x = 0$, $z = e^y$ and $\dfrac{\partial z}{\partial x} = e^{-y}$.

Solution. If z were a function of x alone, the solution would have been $z = A\sin x + B\cos x$, where A and B are constants. Since z is a function of x and y, A and B can be arbitrary functions of y. Hence the solution of the given equation is of the form $z = f(y)\sin x + g(y)\cos x$

$\therefore$ $\dfrac{\partial z}{\partial x} = f(y)\cos x - g(y)\sin x$

When $x = 0$, $z = e^y$, $\therefore e^y = g(y)$. When $x = 0$, $\dfrac{\partial z}{\partial x} = e^{-y}$, $\therefore e^{-y} = f(y)$.

Hence the desired solution is $z = \sin x\, e^{-y} + e^y \cos x$.

Example 17.4. Solve $\dfrac{\partial^2 z}{\partial x \partial y} = 4\sin x \sin y$, for which $\dfrac{\partial z}{\partial x} = -2\sin y$, when $x = 0$ and $z = 0$ when y is an odd multiple of $\frac{\pi}{2}$. (U.P.T.U., 2003)

Solution. Given equation is $\dfrac{\partial^2 z}{\partial x \partial y} = 4\sin x \sin y$.

Integrating w.r.t. x, keeping y constant, we get

$$\frac{\partial z}{\partial y} = -4\cos x \sin y + f(y)$$

When $x = 0$, $\frac{\partial z}{\partial x} = -2\sin y$, $\quad \therefore \quad -2\sin y = -\sin y + f(y)$ or $f(y) = -\sin y$

$\therefore$ (1) becomes $\frac{\partial z}{\partial x} = -\cos x \sin y - \sin y$

Now integrating w.r.t. x, keeping y constant, we get

$$z = -\sin x \cos y + x\sin y + g(y) \qquad \ldots(iii)$$

When y is an odd multiple of $\pi/2$, $z = 0$,

$\therefore \quad 0 = -0 + 0 + g(y)$ or $g(y) = 0$ [$\because \sin(2n+1)\frac{\pi}{2} = 1$ & $\cos(2n+1)\frac{\pi}{2} = 0$]

Hence from (iii), the complete solution is $z = -\cos x \sin y + x\sin y$.

EXERCISE 13.2

Solve the following equations:

1. $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + \frac{y}{x}$ 2. $\frac{\partial^2 z}{\partial x^2} = xy$ 3. $\frac{\partial^2 z}{\partial x \partial y} = e^{x+y} + \sin(x+y)$

4. $\frac{\partial^2 z}{\partial x \partial y} = \cos(2x - 3y)$ 5. $\frac{\partial^2 z}{\partial x^2} = 0$, given that when $y = 0$, $z = x^2$ and $\frac{\partial z}{\partial y} = x^3$

6. $\frac{\partial^2 z}{\partial x^2} = a^2 z$, given that when $x = 0$, $\frac{\partial z}{\partial x} = a$ and $z = 0$ and $\frac{\partial z}{\partial y} = 0$.

13.3 LINEAR EQUATIONS OF THE FIRST ORDER

A linear partial differential equation of the first order, commonly known as Lagrange's linear equation*, is of the form

$$Pp + Qq = R \qquad \ldots(1)$$

where P, Q and R are functions of x, y, z. This equation is called a quasi linear equation. When P, Q and R are independent of z, it is known as linear equation.

Such an equation is obtained by eliminating an arbitrary function ϕ from $\phi(u, v) = 0$ $\qquad \ldots(ii)$

where u, v are some functions of x, y, z.

Differentiating (ii) partially with respect to x and y,

$$\frac{\partial \phi}{\partial u}\left[\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z}p\right] + \frac{\partial \phi}{\partial v}\left[\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z}p\right] = 0 \text{ and } \frac{\partial \phi}{\partial u}\left[\frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}q\right] + \frac{\partial \phi}{\partial v}\left[\frac{\partial v}{\partial y} + \frac{\partial v}{\partial z}q\right] = 0.$$

Eliminating $\frac{\partial \phi}{\partial u}$ and $\frac{\partial \phi}{\partial v}$, we get $\begin{vmatrix} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial z}p & \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}q \\ \frac{\partial v}{\partial x} + \frac{\partial v}{\partial z}p & \frac{\partial v}{\partial y} + \frac{\partial v}{\partial z}q \end{vmatrix} = 0$

which on simplification becomes $\left(\frac{\partial u}{\partial y}\frac{\partial v}{\partial z} - \frac{\partial u}{\partial z}\frac{\partial v}{\partial y}\right)p + \left(\frac{\partial u}{\partial z}\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\frac{\partial v}{\partial z}\right)q = \left(\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x}\right)$ $\qquad \ldots(2)$

This is of the same form as (1).

Now suppose $u = a$ and $v = b$, where a, b are constants, so that

$$\frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz = du = 0$$

$$\frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy + \frac{\partial v}{\partial z}dz = dv = 0.$$

*See footnote p. 142.

By cross-multiplication, we have

$$\frac{dx}{\frac{\partial R}{\partial y}\frac{\partial z}{\partial x}-\frac{\partial R}{\partial z}\frac{\partial z}{\partial y}}=\frac{dy}{\frac{\partial P}{\partial z}\frac{\partial z}{\partial x}-\frac{\partial P}{\partial x}\frac{\partial z}{\partial z}}=\frac{dz}{\frac{\partial Q}{\partial x}\frac{\partial z}{\partial y}-\frac{\partial Q}{\partial y}\frac{\partial z}{\partial x}}$$

or $$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R} \qquad \text{(1) [By virtue of (1) and (2)]}$$

The solutions of these equations are $u = a$ and $v = b$.

$\therefore$ $\phi(u, v) = 0$ is the required solution of (1).

Thus to solve the equation $Pp + Qq = R$,

(i) form the subsidiary equations $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$,

(ii) solve these simultaneous equations by the method of §18.10 giving $u = a$ and $v = b$ as the solutions,

(iii) write the complete solution as $\phi(u, v) = 0$ or $u = f(v)$.

Example 17.3. Solve $\frac{x^2}{z}p + yq = x^2$. (Nottingham, 2002)

Solution. Rewriting the given equation as

$$x^2zp + x^2yq = y^2z,$$

the subsidiary equations are $\frac{dx}{x^2z}=\frac{dy}{x^2y}=\frac{dz}{y^2z}$

The first two fractions give $x^2dx = y^2dy$.

Integrating, we get $x^3 - y^3 = a$ ...(1)

Again the first and third fractions give $xdx = zdz$

Integrating, we get $x^2 - z^2 = b$ ...(2)

Hence from (1) and (2), the complete solution is

$$x^3 - y^3 = f(x^2 - z^2)$$

Example 17.4. Solve $(mz - ny)\frac{\partial z}{\partial x} + (nx - lz)\frac{\partial z}{\partial y} = ly - mx$. (UCE 2000, UCE 2003)

Solution. Here the subsidiary equations are $\frac{dx}{mz-ny}=\frac{dy}{nx-lz}=\frac{dz}{ly-mx}$

Using multipliers x, y, and z, we get each fraction $= \frac{xdx + ydy + zdz}{0}$

$\therefore$ $xdx + ydy + zdz = 0$ which on integration gives $x^2 + y^2 + z^2 = a$ ...(1)

Again using multipliers l, m and n, we get each fraction $= \frac{ldx + mdy + ndz}{0}$

$\therefore$ $ldx + mdy + ndz = 0$ which on integration gives $lx + my + nz = b$ ...(2)

Hence from (1) and (2), the required solution is $x^2 + y^2 + z^2 = f(lx + my + nz)$.

Example 17.5. Solve $(y^2 - z^2)p + (z^2 - x^2)q = (x^2 - y^2)$. (UCE 2001, Anna 2000, UCE 2003)

Solution. Here the subsidiary equations are $\frac{dx}{y^2-z^2}=\frac{dy}{z^2-x^2}=\frac{dz}{x^2-y^2}$

From the last two fractions, we have $\frac{dy}{y}=\frac{dz}{z}$

which on integration gives $\log y = \log c + \log z$ or $y/z = c$ ...(1)

Using multipliers x, y and z, we have

$$\text{each fraction} = \frac{xdx + ydy + zdz}{x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2)} \quad \therefore \quad \frac{2xdx + 2ydy + 2zdz}{0} = \frac{dz}{z}$$

which on integration gives $\log(x^2 + y^2 + z^2) = \log a + \log b$

or $$\frac{x^2 + y^2 + z^2}{z} = b \qquad \ldots(ii)$$

Hence from (i) and (ii), the required solution is $x^2 + y^2 + z^2 = z f(yz/x)$.

Example 17.10. Solve $x(y-z)p + y(z-x)q = z(x-y)$. (U.P.T.U., 2000; Bhopal, 2005; U.P.T.U., 2007)

Solution. Here the subsidiary equations are

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

Using the multipliers $1/x$, $1/y$ and $1/z$, we have

$$\text{each fraction} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$\therefore$ $$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$$ which on integration gives

$$\log x + \log y + \log z = \log a \quad \text{or} \quad xyz = a \qquad \ldots(i)$$

Using the multipliers $\frac{1}{x^2}$, $\frac{1}{y^2}$ and $\frac{1}{z^2}$, we get

$$\text{each fraction} = \frac{\frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz}{0}$$

$\therefore$ $$\frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2} = 0$$, which on integrating gives

$$-\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = b \qquad \ldots(ii)$$

Hence from (i) and (ii), the complete solution is

$$xyz = f\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$$

Example 17.11. Solve $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$. (Allahabad, 2004; U.P.T.U., 2005; Madras, 2006)

Solution. Here the subsidiary equations are

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} \qquad \ldots(i)$$

Each of these equations $= \dfrac{x\,dx - y\,dy}{x^2 - y^2 - yz + zx} = \dfrac{x\,dy - y\,dx}{y^2 - x^2 - xz + yz}$

i.e., $$\frac{d(x-y)}{(x+y)(x-y+z)} = \frac{d(x+y)}{(x+y)(y-x+z)} \quad \text{or} \quad \frac{d(x-y)}{x-y+z} = \frac{d(x+y)}{y-x+z}$$

Integrating, $$\log(x-y) = -\log(x+y) + \log c \quad \text{or} \quad \frac{x-y}{x+y} = c \qquad \ldots(ii)$$

Each of the subsidiary equations $(i) = \dfrac{x\,dx + y\,dy + z\,dz}{x^3 + y^3 + z^3 - 3xyz}$

$$= \frac{x\,dx + y\,dy + z\,dz}{(x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)} \qquad \ldots(iii)$$

Also each of the subsidiary equations $= \dfrac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx}$ $\qquad \ldots(iv)$

Equating (i) and (ii) and cancelling the common factor, we get

$$\frac{x\,dx + y\,dy + z\,dz}{x + y + z} = dx + dy + dz$$

or $$\int x\,dx + y\,dy + z\,dz = \int (x + y + z)\,dx + \int (x + y + z)\,dy + \int (x + y + z)\,dz + c'$$

or $$x^2 + y^2 + z^2 = (x + y + z)x + (x + y + z)y + (x + y + z)z + c' \quad \text{or} \quad xy + yz + zx + c' = 0 \qquad \ldots(iv)$$

Combining (iii) and (iv), the general solution is

$$\frac{x - y}{y - z} = f(xy + yz + zx)$$

PROBLEMS 17.5

Solve the following equations:

1. $yz + zy = 2x$
2. $(y - z)p + (z - x)q = x - y$
3. $px + qy = z$
4. $p\tan x + q\tan y = \tan z$
5. $p = z + \log(q + z)$
6. $(y + z)p + (x + z)q = x + y$
7. $(y - z)p + (z - x)q = x - y$
8. $x(y^2 + z^2)p - y(x^2 + z^2)q = z(x^2 - y^2)$
9. $x^2p + y^2q = z^2$ (U.P.T.U., 2000)
10. $(y - z)p + (z - x)q = x - y$ (Anna, 2005)
11. $x(y^2 + z^2)p + y(x^2 + z^2)q = z(x^2 + y^2)$ (U.P.T.U., 2010; Anna, 2006)
12. $y^2zp + x^2zq = y^2x$ (U.P.T.U., 2000)
13. $(x^2 - y^2 - z^2)p + 2xyq = 2xz$ (P.T.U., 2009; U.P.T.U., 2006)
14. $(x^2 - y^2 - z^2)p + 2xyq = 2xz$, (Kerala, 2006)
15. $(mz - ny)p + (nx - lz)q = ly - mx$

17.6 NON-LINEAR EQUATIONS OF THE FIRST ORDER

The equations in which p and q occur either more than in the first degree are called non-linear partial differential equations of the first order. The complete solution of such an equation contains only two arbitrary constants (a or b, equal to the number of independent variables involved) and the particular integral is obtained by giving particular values to the constants.

Here we shall discuss four standard forms of these equations.

Form I. $f(p, q) = 0$, i.e., equations containing p and q only.

Its complete solution is $z = ax + by + c$ $\qquad \ldots(i)$

where a and b are connected by the relation $f(a, b) = 0$ $\qquad \ldots(ii)$

Since from (i), $p = \dfrac{\partial z}{\partial x} = a$ and $q = \dfrac{\partial z}{\partial y} = b$, which when substituted in (ii) gives $f(p, q) = 0$.

Expressing (i) as $z = bx + ay + c$ and substituting this value of b in (ii), we get the required solution as $z = ax + \phi(a)y + c$ in which a and c are arbitrary constants.

Example 17.13. Solve $px + qy = pq$. (Anna, 2008)

Solution. The complete solution is $z = ax + by + c$ where $ab = b$.

Hence $z = ax + \dfrac{x}{a} - by + c$ is the desired solution.

Example 17.14. Solve $p^2x^2 + q^2y^2 = z^2$. (Anna, 2007; Nagpur, 2008; Kerala, 2009; Rohtak Univ, 2010)

Solution. Given equation can be reduced to the above form by writing it as

$$\left(\frac{x}{z}\cdot\frac{\partial z}{\partial x}\right)^2 + \left(\frac{y}{z}\cdot\frac{\partial z}{\partial y}\right)^2 = 1 \qquad \ldots(i)$$

and setting $\dfrac{dx}{x} = du$, $\dfrac{dy}{y} = dv$, $\dfrac{dz}{z} = dw$ which on integration $u = \log x$, $v = \log y$, $w = \log z$.

Then (i) becomes $\left(\dfrac{\partial w}{\partial u}\right)^2 + \left(\dfrac{\partial w}{\partial v}\right)^2 = 1$

i.e., $p^2 + q^2 = 1$ ... (1) where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.

Its complete solution is $z = ax + by + c$ ... (2)

where $a^2 + b^2 = 1$ or $b = \sqrt{1 - a^2}$.

$\therefore$ (2) becomes $z = ax + \sqrt{(1 - a^2)}\, y + c$

i.e., $\log(z - ax - \sqrt{(1 - a^2)}\, y - c) = \log 0$ which is the required solution.

Form II. $f(x, y, z, p, q) = 0$, i.e., equations not containing x and y.

As a trial solution, assume that z is a function of u, i.e., $z = \phi(u)$, where $u = ax + by$, a being an arbitrary constant.

$$\therefore \quad p = \frac{\partial z}{\partial x} = \frac{dz}{du}\cdot\frac{\partial u}{\partial x} = a\frac{dz}{du} \qquad q = \frac{\partial z}{\partial y} = \frac{dz}{du}\cdot\frac{\partial u}{\partial y} = b\frac{dz}{du}$$

Substituting the values of p and q in $f(z, p, q) = 0$, we get

$$f\left(z, a\frac{dz}{du}, b\frac{dz}{du}\right) = 0$$ which is an ordinary differential equation of the first order.

Solving it as $\frac{dz}{du} = \phi(u)$, it can be easily integrated giving

$F(z, u) = a + b$, or $z + cy = b = F(u)$, at which is the desired complete solution.

Note. To solve $f(z, p, q) = 0$,

(i) Assume $u = x + cy$ and substitute $p = dz/du$, $q = c\,dz/du$ in the given equation,

(ii) solve the resulting ordinary differential equation in z and u,

(iii) replace u by $x + cy$.

Example 12.14. Solve $pq = z(p + q)$. (M.D.U. 2005)

Solution. Let $u = x + cy$, so that $p = dz/du$ and $q = c\,dz/du$.

Substituting these values of p and q in the given equation, we have

$$\frac{dz}{du}\left(1 + c\frac{dz}{du}\right) = cz\frac{dz}{du} + cz\frac{dz}{du} = 2cz - 1 \quad \text{or} \quad \int\frac{c\,dz}{cz - 1} = \int du + b$$

or $\log(cz - 1) = u + b$ or $\log(cz - 1) = x + cy + b$

which is the required complete solution.

Example 12.15. Solve $z^2 = a^2(1 + p^2)$. (C.T.U. 2005; Kottaki, 2000)

Solution. Setting $x + cy = u$ and $z = \phi(u)$, we get

$$p = \frac{\partial z}{\partial x} = \frac{dz}{du}\cdot\frac{\partial u}{\partial x} = \frac{dz}{du} \text{ and } q = \frac{dz}{du}\cdot\frac{\partial u}{\partial y} = c\frac{dz}{du}$$

$\therefore$ The given equation becomes $\left(\frac{dz}{du}\right)^2 = a^2\left[1 + \left(\frac{dz}{du}\right)^2\right]$ ... (1)

or $a^2\left(\frac{dz}{du}\right)^2 = a^2 + z^2$ or $\frac{dz}{du} = \frac{\sqrt{(a^2 + z^2)}}{a^2}$

Integrating, $\int\frac{a^2\,dz}{\sqrt{(a^2 + z^2)}} = \int du + c$ or $a\log\{z + \sqrt{(a^2 + z^2)}\} = u + c$

i.e., $a\log(z + \sqrt{a^2 + z^2}) = ax + cy + c$ ... (2) $\because u = x + cy$

The second factor in (1) is $dz/du = 0$. Its solution is $z = c'$.

Example 12.16. Solve $z^2(p^2 + q^2) = x^2$. (Bhopal, 2005 S)

Solution. Given equation can be reduced to the above form by writing it as

$$z^2\left[\left(x\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2\right]=1 \qquad ...(1)$$

Putting $X=\log x$, so that $x\frac{\partial z}{\partial x}=\frac{\partial z}{\partial X}$, (1) takes the standard form

$$z^2\left[\left(\frac{\partial z}{\partial X}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2\right]=1 \qquad ...(2)$$

Let $u=X-ay$ and put $\frac{\partial z}{\partial X}=\frac{dz}{du}$ and $\frac{\partial z}{\partial y}=-a\frac{dz}{du}$ in (2), so that

$$z^2\left[\left(\frac{dz}{du}\right)^2+a^2\left(\frac{dz}{du}\right)^2\right]=1 \quad \text{or} \quad \sqrt{(1+a^2)}\,z\,dz=du$$

Integrating, $\sqrt{(1+a^2)}\,z^2=u+b=X-ay+b$

or $z^2\sqrt{(1+a^2)}=2X\log x-ay+b$

which is the complete solution required.

Form III. $f(x, p)=F(y, q)$ i.e., equations in which x is absent and the terms containing x and p can be separated from those containing y and q.

As usual solution assume that $f(x, p)=F(y, q)=a$, say

Then solving for p, we get $p=\phi(x)$

and solving for q, we get $q=\psi(y)$

Since $dz=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy=p\,dx+q\,dy$

$\therefore$ $dz=\phi(x)\,dx+\psi(y)\,dy$

Integrating, $z=\int\phi(x)\,dx+\int\psi(y)\,dy+b$

which is the desired complete solution containing two constants a and b.

Example 12.17. Solve $p^2+q^2=x+y$. (Nagpur 2006; Kolhapur 2008)

Solution. Given equation is $p^2-x=y-q^2=a$, say

$\therefore$ $p^2-x=a$ gives $p=\sqrt{(x+a)}$

and $y-q^2=a$ gives $q=\sqrt{(y-a)}$

Substituting the values of p and q in $dz=p\,dx+q\,dy$, we get

$$dz=\sqrt{(x+a)}\,dx+\sqrt{(y-a)}\,dy$$

$\therefore$ Integrating gives, $z=\frac{2}{3}(x+a)^{3/2}+\frac{2}{3}(y-a)^{3/2}+b$

which is the required complete solution.

Example 12.18. Solve $z^2(p^2+q^2)=x^2+y^2$. (Nagpur 2008)

Solution. The equation can be reduced to the above form by writing it as

$$\left(z\frac{\partial z}{\partial x}\right)^2+\left(z\frac{\partial z}{\partial y}\right)^2=x^2+y^2 \qquad ...(1)$$

and putting $z\frac{\partial z}{\partial x}=\frac{\partial Z}{\partial x}$, i.e., $Z=\frac{1}{2}z^2$

$\therefore$ $\frac{\partial Z}{\partial x}=\frac{\partial Z}{\partial z}\cdot\frac{\partial z}{\partial x}=z\frac{\partial z}{\partial x}=p'$

and $\frac{\partial Z}{\partial y}=\frac{\partial Z}{\partial z}\cdot\frac{\partial z}{\partial y}=z\frac{\partial z}{\partial y}=q'$

$\therefore$ (i) becomes $P^2 + Q^2 = a^2 + b^2$

or $p^2 - a^2 = b^2 - q^2 = m$, say.

$\therefore$ $p = \sqrt{(x^2 + a)}$ and $q = \sqrt{(y^2 - a)}$.

$\therefore$ $dz = P\,dx + Q\,dy$ gives

$$dz = \sqrt{(x^2 + a)}\,dx + \sqrt{(y^2 - a)}\,dy$$

Integrating, we have $$z = \frac{1}{2}x\sqrt{(x^2 + a)} + \frac{1}{2}a\log\left(x + \sqrt{(x^2 + a)}\right)$$

$$+ \frac{1}{2}y\sqrt{(y^2 - a)} - \frac{1}{2}a\log\left(y + \sqrt{(y^2 - a)}\right) + b$$

or $$z = x\sqrt{(x^2 + a)} + y\sqrt{(y^2 - a)} + a\log\frac{x + \sqrt{(x^2 + a)}}{y + \sqrt{(y^2 - a)}} + 2b$$

which is the required complete solution.

Example 12.10. Solve $(r + s)(p + q) = (p^2 + q^2)(r + t)$. (Bhopal, 2006; Rohtak 2006; VTU, 2008)

Solution. This equation may be reduced to the form $f(x, y) = F(y, z)$ by putting $u = x + y$, $v = x - y$ and taking $z = z(u, v)$.

Then $$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x} = R + Q$$

and $$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y} = T - Q, \text{ where } T = \frac{\partial z}{\partial u}, Q = \frac{\partial z}{\partial v}$$

Substituting these, the given equation reduces to

$$(2R)^2 + (2Q)^2 = 1 \quad \text{or} \quad R^2 + Q^2 = \frac{1}{4} \quad \ldots(i)$$

$$R = \pm\frac{1}{2}\sqrt{\frac{a}{1}}, Q = \pm\frac{1}{2}\sqrt{\frac{1-a}{1}}$$

$\therefore$ $$dz = \frac{\partial z}{\partial u}du + \frac{\partial z}{\partial v}dv = R\,du + Q\,dv$$

$$= \pm\frac{\sqrt{a}}{2}\frac{du}{\sqrt{b}} \pm \frac{\sqrt{1-a}}{2}\frac{dv}{\sqrt{b}}$$

Integrating, we have $$z = \pm\sqrt{a}\,\sqrt{b}\,x \pm \sqrt{1-a}\,\sqrt{b}\,y + b$$

or $$z = \pm\sqrt{a}\sqrt{(x+y)}x \pm \sqrt{(1-a)}\sqrt{(x-y)}y + b$$

which is the required complete solution.

Form IV. $z = px + qy + F(p, q)$ is an equation analogous to the Clairaut's equation $(z = px + qy + f(p, q))$.

Its complete solution is $z = ax + by + f(a, b)$, which is obtained by writing a for p and b for q in the given equation.

Example 12.20. Solve $z = px + qy + \sqrt{(1 + p^2 + q^2)}$. (Delhi, 2002)

Solution. Given equation is of the form $z = px + qy + f(p, q)$ where $f(p, q) = \sqrt{(1 + p^2 + q^2)}$.

$\therefore$ Its complete solution is $z = ax + by + \sqrt{(1 + a^2 + b^2)}$.

EXERCISE 12.1

Obtain the complete solutions of the following equations :

1. $xp + yq = z$		2. $p^2 + q^2 = 1$	(Kurukshetra, 2006)
3. $z = px + qy + pq$	(Anna, 2005; U.P.T.U., 2006)	4. $p(1 - q) = q(1 + z)$	(Anna, 2004)
5. $pq = \log(xy)$		6. $p + q = \sin(x + y)$	

9.8 CHARPIT'S METHOD*

We now explain a general method for finding the complete integral of a non-linear partial differential equation which is due to Charpit.

Consider the equation

$$f(x, y, z, p, q) = 0 \qquad \ldots(1)$$

Since z depends on x and y, we have

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy = p\,dx + q\,dy \qquad \ldots(2)$$

Now if we can find another relation involving x, y, z, p, q such as $\phi(x, y, z, p, q) = 0$ ...(3)

then we can solve (1) and (3) for p and q and substitute in (2). Then will give the solution provided (2) is integrable.

To determine ϕ, we differentiate (1) and (3) with respect to x and y giving

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z}p + \frac{\partial f}{\partial p}\frac{\partial p}{\partial x} + \frac{\partial f}{\partial q}\frac{\partial q}{\partial x} = 0 \qquad \ldots(4)$$

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial z}p + \frac{\partial \phi}{\partial p}\frac{\partial p}{\partial x} + \frac{\partial \phi}{\partial q}\frac{\partial q}{\partial x} = 0 \qquad \ldots(5)$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}q + \frac{\partial f}{\partial p}\frac{\partial p}{\partial y} + \frac{\partial f}{\partial q}\frac{\partial q}{\partial y} = 0 \qquad \ldots(6)$$

$$\frac{\partial \phi}{\partial y} + \frac{\partial \phi}{\partial z}q + \frac{\partial \phi}{\partial p}\frac{\partial p}{\partial y} + \frac{\partial \phi}{\partial q}\frac{\partial q}{\partial y} = 0 \qquad \ldots(7)$$

Eliminating $\frac{\partial \phi}{\partial z}$ between the equations (4) and (5), we get

$$\left[\frac{\partial f}{\partial x}\frac{\partial \phi}{\partial p} - \frac{\partial \phi}{\partial x}\frac{\partial f}{\partial p}\right] + \left[\frac{\partial f}{\partial z}\frac{\partial \phi}{\partial p} - \frac{\partial \phi}{\partial z}\frac{\partial f}{\partial p}\right]p + \left[\frac{\partial f}{\partial q}\frac{\partial \phi}{\partial p} - \frac{\partial \phi}{\partial q}\frac{\partial f}{\partial p}\right]\frac{\partial q}{\partial x} = 0 \qquad \ldots(8)$$

Also eliminating $\frac{\partial \phi}{\partial y}$ between the equations (6) and (7), we obtain

$$\left[\frac{\partial f}{\partial y}\frac{\partial \phi}{\partial q} - \frac{\partial \phi}{\partial y}\frac{\partial f}{\partial q}\right] + \left[\frac{\partial f}{\partial z}\frac{\partial \phi}{\partial q} - \frac{\partial \phi}{\partial z}\frac{\partial f}{\partial q}\right]q + \left[\frac{\partial f}{\partial p}\frac{\partial \phi}{\partial q} - \frac{\partial \phi}{\partial p}\frac{\partial f}{\partial q}\right]\frac{\partial p}{\partial y} = 0 \qquad \ldots(9)$$

Adding (8) and (9) and using $\frac{\partial q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial p}{\partial y}$,

we find that the last terms in both cancel and the other terms, on rearrangement, give

$$\left(\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial p} + \left(\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial q} - \left(p\frac{\partial f}{\partial p} + q\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial z} - \left(\frac{\partial f}{\partial p}\right)\frac{\partial \phi}{\partial x} - \left(\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial y} = 0 \qquad \ldots(10)$$

i.e., $$\left(\frac{\partial f}{\partial p}\right)\frac{\partial \phi}{\partial x} + \left(\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial y} + \left(p\frac{\partial f}{\partial p} + q\frac{\partial f}{\partial q}\right)\frac{\partial \phi}{\partial z} - \left(\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial p} - \left(\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}\right)\frac{\partial \phi}{\partial q} = 0 \qquad \ldots(11)$$

This is Lagrange's linear equation (§ 17.6) with x, y, z, p, q as independent variables and ϕ as the dependent variable. Its solution will depend on the solution of the subsidiary equations

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*Charpit's method was published in the Paris Academy of Sciences in 1784.

$$\frac{dx}{\frac{\partial f}{\partial p}} = \frac{dy}{\frac{\partial f}{\partial q}} = \frac{dz}{p\frac{\partial f}{\partial p} + q\frac{\partial f}{\partial q}} = \frac{dp}{-\left(\frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z}\right)} = \frac{dq}{-\left(\frac{\partial f}{\partial y} + q\frac{\partial f}{\partial z}\right)} = \frac{dt}{0}$$

An integral of these equations involving p or q or both, can be taken as the required relation (2), which alongwith (1) will give the values of p and q to make (1) integrable. Of course, we should take the simplest of the integrals so that it may be easier to solve for p and q.

Example 17.21. Solve $(p^2 + q^2)y = qz$. (U.P.T.U. 2007; Bhopal 2003)

Solution. Let $f(x, y, z, p, q) = (p^2 + q^2)y - qz = 0$ ...(1)

Charpit's subsidiary equations are

$$\frac{dx}{-2py} = \frac{dy}{z - 2qy} = \frac{dz}{-qz} = \frac{dp}{-py} = \frac{dq}{p^2}$$

The last two of these give $pdp + qdq = 0$

Integrating, $p^2 + q^2 = c^2$ ...(2)

Now to solve (1) and (2), put $p^2 + q^2 = c^2$ in (1), so that $qz = c^2y$

Substituting this value of q in (2), we get $p = \frac{1}{z}\sqrt{(c^2z^2 - c^2y^2)}$

Hence $dz = pdx + qdy = \frac{c}{z}\sqrt{(z^2 - y^2)}dx + \frac{c^2}{z}y dy$

or $zdz - c^2y\,dy = c\sqrt{(z^2 - c^2y^2)}dx$ or $\frac{(zdz - c^2y\,dy)}{\sqrt{(z^2 - c^2y^2)}} = c\,dx$

Integrating, we get $\sqrt{(z^2 - c^2y^2)} = cx + a$ or $z^2 = (cx + a)^2 + c^2y^2$ which is the required complete integral.

Example 17.22. Solve $2xz - px^2 - 2xyq + pq = 0$. (Rajasthan 2006)

Solution. Let $f(x, y, z, p, q) = 2xz - px^2 - 2xyq + pq = 0$ ...(1)

Charpit's subsidiary equations are

$$\frac{dx}{x^2 - q} = \frac{dy}{2xy - p} = \frac{dz}{px^2 - 2pxy + 2xyq} = \frac{dp}{2z - 2pz} = \frac{dq}{0}$$

$\therefore$ $dq = 0$ or $q = a$.

Putting $q = a$ in (1), we get $p = \frac{2x(z - ay)}{x^2 - a}$

$\therefore$ $dz = pdx + qdy = \frac{2x(z - ay)}{x^2 - a}dx + ady$ or $\frac{dz - ady}{z - ay} = \frac{2x}{x^2 - a}dx$

Integrating, $\log(z - ay) = \log(x^2 - a) + \log b$

or $z - ay = b(x^2 - a)$ or $z = ay + b(x^2 - a)$

which is the required complete solution.

Example 17.23. Solve $z = px + qy + 2\sqrt{(pq)}$. (U.P.T.U. 2006; Rajasthan 2005)

Solution. Let $f(x, y, z, p, q) = px + qy + 2\sqrt{(pq)} - z$

Charpit's subsidiary equations are

$$\frac{dx}{2p^2} = \frac{dy}{2q^2} = \frac{dz}{-(2p^2 + 2q^2)} = \frac{dp}{2p} = \frac{dq}{2q}$$

From first and fourth ratios,

$$dp = -dx \quad \text{or} \quad p = -x + a$$

Substituting $p = a - x$ in the given equation, we get

$$q = \frac{1}{y}(z - 2x - 2y^2 - 2a - x^2)$$

$$\therefore \qquad dz = p\,dx + q\,dy = kx - a\,dx - \frac{b}{y}(kx - by^2 + cx - a^2)\,dy$$

Multiplying both sides by by^2,

$$by^2\,dz + by(a\,dx + by\,dy) = b^2y^2(x - a)\,dx - by^2(kx - by^2 + cx - a^2)\,dy$$

Integrating $\quad by^2z + \frac{1}{2}b^2y^4 = \frac{1}{3}b^2y^3(x - a)^2 + \frac{1}{2}b^2y^4 + c$

or $\quad y^2(x - a)^2 + 3z = y^3(x - a) = c_1$, which is the desired solution.

PROBLEMS 1.15

Solve the following equations:

1. $z = px + qy + p^2q^2$
2. $z = px + qy + p^2 - q^2$
3. $z(pq + 1) = q(x - y)$
4. $z = px + qy + \sqrt{1 + p^2 + q^2}$
5. $z = px + qy + \sqrt{pq}$
6. $z = px + qy + p^2q$

Answers: 1. $(x + 2y)^2 = 4z$; 2. $z = ax + by + a^2 - b^2$; 3. $z = ax + \frac{a}{a+1}y + c$; 4. $(z - ax - by)^2 = 1 + a^2 + b^2$; 5. $z = ax + by + \sqrt{ab}$; 6. $z = ax + by + a^2b$

1.15 HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

An equation of the form

$$\frac{d^n z}{dx^n} + k_1\frac{d^n z}{dx^{n-1}dy} + \ldots + k_n\frac{d^n z}{dy^n} = f(x, y) \qquad \ldots(1)$$

in which k's are constants, is called a *homogeneous linear partial differential equation of the nth order with constant coefficients*. It is called homogeneous because all terms contain derivatives of the same order.

On writing $\quad \frac{d}{dx} = D$ and $\frac{d}{dy} = D'$, (1) becomes $(D^n + k_1D^{n-1}D' + k_2D^{n-2}D'^2 + \ldots + k_nD'^n)z = f(x, y)$

or briefly $\qquad f(D, D')z = f(x, y) \qquad \ldots(2)$

As in the case of ordinary linear equations with constant coefficients the complete solution of (1) consists of two parts, namely : the complementary function and the particular integral.

The complementary function is the complete solution of the equation $f(D, D')z = 0$, which must contain n arbitrary functions. The particular integral is the particular solution of equation (2).

1.16 RULES FOR FINDING THE COMPLEMENTARY FUNCTION

Consider the equation $\quad \frac{\partial^2 z}{\partial x^2} + k_1\frac{\partial^2 z}{\partial x\partial y} + k_2\frac{\partial^2 z}{\partial y^2} = 0 \qquad \ldots(1)$

which in symbolic form is $\quad (D^2 + k_1DD' + k_2D'^2)z = 0 \qquad \ldots(2)$

The symbolic operator equated to zero, i.e., $D^2 + k_1DD' + k_2D'^2 = 0$ is called the *auxiliary equation* (A.E.)

Let its roots be $\quad DD' = m_1, m_2$

Case 1. If the roots be real and distinct then (2) is equivalent to

$$(D - m_1D')(D - m_2D')z = 0 \qquad \ldots(3)$$

It will be satisfied by the solution of

$$(D - m_1D')z = 0, i.e., p - m_1q = 0$$

This is a Lagrange's linear and the subsidiary equations are

$$\frac{dx}{1} = \frac{dy}{-m_1} = \frac{dz}{0}, \text{ whence } y + m_1x = a \text{ and } z = b.$$

$\therefore$ the solution is $z = \phi(y + m_1x)$.

Similarly (3) will also be satisfied by the solution of

$$(D - m_2D')z = 0, \quad i.e., \quad 2y + z = f(y + m_2x)$$

Hence the complete solution of (3) is $z = \phi(y + m_1x) + \psi(y + m_2x)$.

Case II. If the roots are equal (i.e., $m_1 = m_2$), then (2) is equivalent to

$$(D - m_1)^2 y = 0 \qquad \ldots(4)$$

Putting $(D - m_1)y = u$, (4) becomes $(D - m_1)u = 0$ which gives

$$u = Ce^{m_1 x}$$

$\therefore$ (3) takes the form $(D - m_1)y = Ce^{m_1 x}$ or $py - m_1 y = Ce^{m_1 x}$

This is again Lagrange's linear and the subsidiary equations are

$$\frac{dx}{1} = \frac{dy}{m_1 y} = \frac{dz}{Ce^{m_1 x}}$$

giving $y e^{-m_1 x} = \int Ce^{-m_1 x} dx + A$, i.e., $y = e^{m_1 x}(Cx + A)$

Thus the complete solution of (1) is

$$z = \phi(y + m_1 x) + x e^{m_1 x} \phi_1(y + m_1 x) \quad \text{i.e.,} \quad z = f(y + m_1 x) + x\phi(y + m_1 x)$$

Example 17.24. Solve $\dfrac{\partial^2 z}{\partial x^2} + 2\dfrac{\partial^2 z}{\partial x \partial y} + \dfrac{\partial^2 z}{\partial y^2} = e^{x}$.

Solution. Given equation in symbolic form is $(D^2 + 2DD' + D'^2)z = e^x$.

Its auxiliary equation is $m^2 + 2m + 1 = 0$, where $m = D/D'$.

which gives $m = -1, -1$.

Hence the complete solution is $z = f_1(y - x) + x f_2(y - x) + \frac{1}{4}e^x$.

which may be written as $z = f_1(y - x) + x f_2(y - x) + \frac{1}{4}e^x$.

Example 17.25. Solve $4r + 12s + 9t = e^{x}$. (U.P.T.U. 2010)

Solution. Given equation in symbolic form is $(4D^2 + 12DD' + 9D'^2)z = e^x$

i.e. $r = \dfrac{\partial^2 z}{\partial x^2} = D^2 z$, $s = \dfrac{\partial^2 z}{\partial x \partial y} = DD'z$ and $t = \dfrac{\partial^2 z}{\partial y^2} = D'^2 z$.

$\therefore$ Its auxiliary equation is $4m^2 + 12m + 9 = 0$, whence $m = -3/2, -3/2$.

Hence the complete solution is $z = f_1(y - \tfrac{3}{2}x) + x f_2(y - \tfrac{3}{2}x)$.

17.10 RULES FOR FINDING THE PARTICULAR INTEGRAL

Consider the equation $(D^n + A_1 D^{n-1} + A_2 D^{n-2} + \ldots + A_n)z = f(x, y)$ i.e., $f(D, D')z = F(x, y)$.

$\therefore$ $$P.I. = \frac{1}{f(D, D')} F(x, y)$$

Case I. When $F(x, y) = e^{ax+by}$

Since $D e^{ax+by} = a e^{ax+by}$, $D^2 e^{ax+by} = a^2 e^{ax+by}$

$\therefore$ $D^3 e^{ax+by} = a^3 e^{ax+by}$, $D^n e^{ax+by} = a^n e^{ax+by}$

and $D'^2 e^{ax+by} = b^2 e^{ax+by}$

$\therefore$ $(D^2 + A_1 DD' + A_2 D'^2)e^{ax+by} = a^2 + A_1 ab + A_2 b^2 e^{ax+by}$

i.e., $f(D, D')e^{ax+by} = f(a, b)e^{ax+by}$

Operating (on both sides) by $1/f(D, D')$, we get

$$P.I. = \frac{1}{f(D, D')} e^{ax+by} = \frac{1}{f(a, b)} e^{ax+by}$$

Case II. When $F(x, y)$ is $\sin(ax + by + c)$ or $\cos(ax + by + c)$

Since $D^2 \sin(ax + by + c) = -a^2 \sin(ax + by + c)$

$DD' \sin(ax + by + c) = -ab \sin(ax + by + c)$

and $D'^2 \sin(ax + by + c) = -b^2 \sin(ax + by + c)$

$\therefore$ $f(D^2, DD', D'^2)\sin(ax + by + c) = \phi(-a^2, -ab, -b^2)\sin(ax + by + c)$

Operating both sides by $f(D^2, DD', D'^2)$, we get

$$\text{P.I.} = \frac{1}{f(D^2, DD', D'^2)} \sin(lx + my) = \frac{1}{f(-l^2, -lm, -m^2)} \sin(lx + my)$$

Similarly about the P.I. for $\cos(lx + my)$.

Case III. When $F(x, y) = e^{ax+by}$, we shall be having the formula

$$\text{P.I.} = \frac{1}{f(D, D')} e^{ax+by} \quad [\text{Put } D = a \text{ and } D' = b]$$

To evaluate it, we expand $f(D, D')^{-1}$ in ascending powers of D or D' by Binomial theorem and then operate on e^{ax+by} term by term.

Case IV. When $F(x, y)$ is any function of x and y.

$$\text{P.I.} = \frac{1}{f(D, D')} F(x, y)$$

To evaluate it, we resolve $1/f(D, D')$ into partial fractions treating (D, D') as a function of D alone and operate each partial fraction on $F(x, y)$ remembering that

$$\frac{1}{D - mD'} F(x, y) = \int F(x, c - mx)\, dx$$

where c is replaced by $y + mx$ after integration.

3.11 WORKING PROCEDURE TO SOLVE THE EQUATION

$$\frac{d^n z}{dx^n} + k_1 \frac{d^n z}{dx^{n-1} dy} + \ldots + k_n \frac{d^n z}{dy^n} = F(x, y)$$

Its symbolic form is $D^n + k_1 D^{n-1} D' + \ldots + k_n D'^n z = F(x, y)$

or briefly $f(D, D')z = F(x, y)$

Step I. To find the C.F.

(i) Write the A.E.

i.e., $m^n + k_1 m^{n-1} + \ldots + k_n = 0$ and solve it for m.

(ii) Write the C.F. as follows:

Roots of A.E.	C.F.
1. $m_1, m_2, m_3, \ldots$ (distinct roots)	$f_1(y + m_1x) + f_2(y + m_2x) + f_3(y + m_3x) + \ldots$
2. $m_1, m_1, m_3, \ldots$ (two equal roots)	$f_1(y + m_1x) + xf_2(y + m_1x) + f_3(y + m_3x) + \ldots$
3. $m_1, m_1, m_1, \ldots$ (three equal roots)	$f_1(y + m_1x) + xf_2(y + m_1x) + x^2f_3(y + m_1x) + \ldots$

Step II. To find the P.I.

From the symbolic form, $\text{P.I.} = \frac{1}{f(D, D')} F(x, y)$.

(i) When $F(x, y) = e^{ax+by}$, $\text{P.I.} = \frac{1}{f(D, D')} e^{ax+by}$ [Put $D = a$ and $D' = b$]

(ii) When $F(x, y) = \sin(lx + my)$ or $\cos(lx + my)$

$$\text{P.I.} = \frac{1}{f(D^2, DD', D'^2)} \sin(lx + my) \quad [\text{Put } D^2 = -l^2, DD' = -lm, D'^2 = -m^2]$$

(iii) When $F(x, y) = x^m y^n$, $\text{P.I.} = \frac{1}{f(D, D')} x^m y^n = [f(D, D')]^{-1} x^m y^n$.

Expand $[f(D, D')]^{-1}$ in ascending powers of D or D' and operate on $x^m y^n$ term by term.

(iv) When $F(x, y)$ is any function of x and y, $\text{P.I.} = \frac{1}{f(D, D')} F(x, y)$

Rule for finding P.I. when the partial fraction involving (D, D') is a function of D alone and expansion of each partial fraction in D/C is not terminating after a finite number of terms

$$\frac{1}{D-mD'}F(x,y)=\int F(c,\,y+mc)\,dc \text{ where } c \text{ is replaced by } y+mx \text{ after integration.}$$

Example 17.26. Solve $(D^2+DD'-6D'^2)z=y\cos x+3x$. (Madras, 2000)

Solution. A.E. of the given equation is $m^2+m-6=0$ i.e., $m=2,-3$

$\therefore$ $C.F.=f_1(y+2x)+f_2(y-3x)$

$$P.I.=\frac{1}{D^2+DD'-6D'^2}(y\cos x+3x)\quad [\text{Put } D^2=-1^2, DD'=-2\times 3, D'^2=-3^2$$

$$=\frac{1}{-1-6-6(-9)}\left(y\cos x+\frac{3x}{13}\right)=\frac{1}{13}y\cos x+\frac{3}{13}x.$$

Hence the C.S. is $z=f_1(y+2x)+f_2(y-3x)+\frac{1}{13}y\cos x+\frac{3}{13}x$.

Example 17.27. Solve $\frac{\partial^2 z}{\partial x^2}-\frac{\partial^2 z}{\partial x\partial y}=\sin x+\cos 2y$. (Bhopal, 2005 S)

Solution. Given equation in symbolic form is $(D^2-DD')z=\sin x+\cos 2y$.

Its A.E. is $m^2-m=0$, whose roots are $0, 1$.

$\therefore$ $C.F.=f_1(y)+f_2(y+x)$

$$P.I.=\frac{1}{D^2-DD'}\sin x+\cos 2y=\frac{1}{2}\frac{1}{D^2-DD'}[\cos(x+2y)+\cos(x-2y)]$$

$$=\frac{1}{2}\left[\frac{1}{D^2-DD'}\cos(x+2y)\right. \qquad [\text{Put } D^2=-1, DD'=-2]$$

$$\left.+\frac{1}{D^2-DD'}\cos(x-2y)\right] \qquad [\text{Put } D^2=-1, DD'=2]$$

$$=\frac{1}{2}\left[\frac{1}{-1+2}\cos(x+2y)+\frac{1}{-1-2}\cos(x-2y)\right]=\frac{1}{2}\cos(x+2y)-\frac{1}{6}\cos(x-2y)$$

Hence the C.S. is $z=f_1(y)+f_2(y+x)+\frac{1}{2}\cos(x+2y)-\frac{1}{6}\cos(x-2y)$.

Example 17.28. Solve $\frac{\partial^3 z}{\partial x^3}-2\frac{\partial^3 z}{\partial x^2\partial y}=3x^2+6y$. (U.P.T.U., 2007)

Solution. Given equation in symbolic form is

$$(D^3-2D^2D')z=3x^2+6y$$

Its A.E. is $m^3-2m^2=0$, whose roots are $0, 0, 2$.

$\therefore$ $C.F.=f_1(y)+xf_2(y)+f_3(y+2x)$

$$P.I.=\frac{1}{D^2(D-2D')}(3x^2+6y)=2\frac{1}{D^2(D-2D')}x^2+3\frac{1}{D^2(D-2D')}xy$$

$$=2\frac{1}{D^2-2.D^2D'}x^2+\frac{1}{D^2}\left(1-2D'/D\right)^{-1}xy=\frac{1}{2}x^2+\frac{1}{D^2}\left(1+\frac{2D'}{D}+\frac{4D'^2}{D^2}+\ldots\right)xy$$

$$=\frac{1}{4}x^2+\frac{1}{D^2}\left(xy+\frac{2}{D}x^2.1\right)=\frac{1}{4}x^2+\frac{1}{D^2}\left(xy+\frac{2}{3}x^3\right) \qquad \left[\because \frac{1}{D}f(x)=\int f(x)dx\right]$$

$$=\frac{1}{4}x^2+2y\frac{x^3}{3.4.3}+2.\frac{x^5}{4.5.6} \qquad \left[\because \frac{1}{D^2}f(x)=\int\left[\int f(x)dx\right]dx\right]$$

$$= \frac{x^3}{3} + \frac{x^2y}{2!} + \frac{xy^2}{3!}$$

Hence the C.S. is $z = f_1(y) + xf_2(y) + f_3(y + 2x) + \frac{1}{6}(10x^3 + 3x^2y + xy^2)$

Example 17.20. Solve $(D^3 - 4D^2D' + 4DD'^2)z = e^{2x+y}$.

Solution. Given equation is $\frac{\partial^3 z}{\partial x^3} - 4\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial x \partial y^2} = e^{2x+y}$.

i.e., in symbolic form, $(D^3 - 4D^2D' + 4DD'^2)z = e^{2x+y}$.

Its A.E. is $m^3 - 4m^2 + 4m = 0$, i.e., $m(m-2)^2 = 0$, whence $m = 0, 2, 2$.

$\therefore$ $C.F. = f_1(y) + f_2(y + 2x) + xf_3(y + 2x)$

$$P.I. = \frac{1}{D^3 - 4D^2D' + 4DD'^2} e^{2x+y}$$

The usual rule fails because $D^3 - 4D^2D' + 4DD'^2 = 0$ for $D = 0$ and $D' = 1$.

$\therefore$ to obtain the P.I., we find from $(D^3 - 4D^2D')u = e^{2x+y}$, the solution

$$u = \int e^{2x+(c-mx)} dx = \int e^{(2-m)x+c} dx = \frac{e^{(2-m)x+c}}{2-m} = \frac{e^{2x+y}}{2-m} \qquad [\because y = c - mx \text{ or } c = y + 2x]$$

and from $(D - 2D')v = u = e^{2x+y}/(2-m)$, the solution

$$v = \int \frac{e^{2x+c-mx}}{2-m} dx = \frac{1}{2-m} e^{(2-m)x+c} = \frac{1}{2-m} e^{2x+y} \qquad [\because y = c - mx \text{ or } c = y + 2x]$$

Hence the C.S. is $z = f_1(y) + xf_2(y) + xf_3(y + 2x) + \frac{1}{2} e^{2x+y}$.

Example 17.21. Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = \cos(2x + y)$. (W.S.U. 2010, 2011, 2012, 2013)

Solution. Given equation in symbolic form is $(D^2 + DD' - 6D'^2)z = \cos(2x + y)$

Its A.E. is $m^2 + m - 6 = 0$ or $(m+3)(m-2) = 0$, whence $m = -3, 2$.

$\therefore$ $C.F. = f_1(y - 3x) + f_2(y + 2x)$.

Since $D^2 + DD' - 6D'^2 = -3^2 + (2)(1) - 6(1)^2 = 0$

$\therefore$ P.I. is a case of failure and we have to apply the general method.

$$P.I. = \frac{1}{D^2 + DD' - 6D'^2} \cos(2x + y) = \frac{1}{(D + 3D')(D - 2D')} \cos(2x + y)$$

$$= \frac{1}{D + 3D'} \left[\int \cos(2x + \overline{c - 2x}) dx \right]_{c = y + 2x} = \frac{1}{D + 3D'} \left[\int \cos c \, dx \right]_{c = y + 2x}$$

$$= \frac{1}{D + 3D'} x \cos(y + 2x) = \left[\int x \cos(\overline{c - 3x} + 2x) dx \right]_{c = y + 2x} = \left[\int x \cos(2x + c) dx \right]_{c = y + 2x}$$

$$= \left[\frac{x \sin(2x + c)}{2} + \frac{\cos(2x + c)}{4} \right]_{c = y + 2x} \qquad \text{[Integrating by parts]}$$

$$= \frac{x}{2} \sin(2x + \overline{y - 2x}) + \frac{1}{4} \cos(2x + \overline{y - 2x}) = \frac{x}{2} \sin(2x + y) + \frac{1}{4} \cos(2x + y)$$

Hence the C.S. is

$$z = f_1(y - 3x) + f_2(y + 2x) + \frac{x}{2} \sin(2x + y) + \frac{1}{10} \cos(2x + y)$$

$$\text{or} \quad z = f_1(y - 3x) + f_2(y + 2x) + \frac{x}{2} \sin(2x + y) + \frac{1}{10} \cos(2x + y)$$

Example 17.23. Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = y\cos x$. (Anna. 2005 S; UQP 2012, Anna 2013)

or $r + s - 6t = y\cos x$. (Anna. 2006; UQP 2011, 2008)

Solution. The symbolic form is $(D^2 + DD' - 6D'^2)z = y\cos x$

and the A.E. is $m^2 + m - 6 = 0$, whose roots are $m = -3, 2$.

$\therefore$ C.F. $= f_1(y - 3x) + f_2(y + 2x)$

$$\text{P.I.} = \frac{1}{(D-2D')(D+3D')}y\cos x = \frac{1}{D-2D'}\left[\int (c + 3x)\cos x\,dx\right]_{\text{Put } y = c - mx,\ m = 3}$$

$$= \frac{1}{D-2D'}\left[(c + 3x)\sin x + 3\cos x\right]_{\text{Integrating by parts}}$$

$$= \frac{1}{D-2D'}(y\sin x + 3\cos x) = \int\left[(c - 2x)\sin x + 3\cos x\right]dx\Big]_{c = y+2x}$$

$$= (c - 2x)(-\cos x) + (-2)(-\sin x) + 3\sin x = -y\cos x + 4\sin x$$

Hence the G.S. is $z = f_1(y - 3x) + f_2(y + 2x) + 4\sin x - y\cos x$.

Example 17.24. Solve $4\frac{\partial^2 z}{\partial x^2} - 4\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 16\log(x + 2y)$.

Solution. The symbolic form is $(4D^2 - 4DD' + D'^2)z = 16\log(x + 2y)$

and the A.E. is $4m^2 - 4m + 1 = 0$, $m = 1/2, 1/2$.

$\therefore$ C.F. $= f_1\left(y + \frac{1}{2}x\right) + xf_2\left(y + \frac{1}{2}x\right)$

$$\text{P.I.} = \frac{1}{(2D-D')^2}16\log(x + 2y) = 4\frac{1}{\left(D - \frac{1}{2}D'\right)\left(D - \frac{1}{2}D'\right)}\left[\frac{1}{D - \frac{1}{2}D'}\log(x + 2y)\right]$$

$$= 4\frac{1}{D - \frac{1}{2}D'}\left[\int \log\left[y + x\left(c - \frac{x}{2}\right)\right]dx\right]_{\text{Put } y = c - mx + x^2/2}$$

$$= 4\frac{1}{D - \frac{1}{2}D'}\left[\int \log 2c\,dx\right]_{c = y + x^2/2} = 4\frac{1}{D - \frac{1}{2}D'}(x\log 2c - 2y)$$

$$= 4\left[\int x\log\left[c + x\left(c - \frac{x}{2}\right)\right]dx\right]_{c = y + x^2/2} + 4\left[\log 2c\int x\,dx\right]_{c = y + x^2/2} = 2x^2\log(x + 2y)$$

Hence the G.S. is $z = f_1\left(y - \frac{x^2}{2}\right) + xf_2\left(y - \frac{x^2}{2}\right) + 2x^2\log(x + 2y)$.

PROBLEMS 17.5

Solve the following equations :

1. $\frac{\partial^2 z}{\partial x^2} + 4\frac{\partial^2 z}{\partial x \partial y} + 4\frac{\partial^2 z}{\partial y^2} = 0$.

2. $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = e^{x+y}$. (Madras, 2003)

3. $(D^2 - 3DD' + D'^2)z = e^{x+y}$. (Bharathiar, 2007)

4. $\frac{\partial^2 z}{\partial x^2} - 4\frac{\partial^2 z}{\partial x \partial y} + 4\frac{\partial^2 z}{\partial y^2} = e^{2x+y}$. (Madras, 2003)

5. $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin x \cos 2y$ [$z = f_1(y + x) + x f_2(y + x) - \frac{1}{9}\sin x \cos 2y$]

6. $\frac{\partial^2 z}{\partial x^2} + a^2 \frac{\partial^2 z}{\partial y^2} = 0$ given $y = 0$

7. $\frac{\partial^3 z}{\partial x^3} - \frac{3\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial x \partial y^2} = 2\sin(2x + y)$ [$z = f_1(y) + f_2(y + x) + f_3(y + 4x) + \frac{1}{4}x\cos(2x + y)$]

8. $(D^3 - 3D^2D' - 4DD'^2)z = \cos(x + 2y) + 4$ [$z = f_1(y) + f_2(y - 4x) + f_3(y + x) - \frac{1}{6}x\cos(x + 2y) - \frac{1}{2}x^2$]

9. $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 3\frac{\partial^3 z}{\partial x \partial y^2} - \frac{\partial^3 z}{\partial y^3} = e^{x+y} + \cos(x - 2y)$ [$z = f_1(y) + x f_2(y + x) + x^2 f_3(y + x) + \frac{1}{6}x^3 e^{x+y} + \frac{1}{27}\cos(x - 2y)$]

10. $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = \sin x \cos 2y$ [$z = f_1(y + x) + f_2(y - x) + \frac{1}{6}\sin x \cos 2y$]

11. $(D^3 - 2D^2D' + 4)z = \sin 2y \cos x + \cos x$

12. $(D^3 - D'^3)z = e^{x+y}\sin(x + 2y)$ [$z = f_1(y) + f_2(y + x) + f_3(y + \omega x) + \frac{1}{3}x e^{x+y}\sin(x + 2y)$, ω cube roots of unity]

13. $(D^3 - 3D^2D' + 3DD'^2)z = 24(y + x)$

14. $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = e^{2x+y} + \cos(x + y)$

15. $(D^3 - 2D^2D' - 2DD'^2 + 4D'^3)z = \sin(y - x)$ [$z = f_1(y + x) + f_2(y + 2x) + x f_3(y + 2x) - \frac{1}{2}x\cos(y - x)$]

16. $(D^3 + 2D^2D' - 3DD'^2 - 2D'^3)z = e^{x}\sin 2y$

17. $(D^3 - 3D^2D' + 4D'^3)z = 2e^{x+y} + \cos(x + 2y)$ [$z = f_1(y + x) + f_2(y - 2x) + x f_3(y - 2x) + \frac{1}{3}x^2 e^{x+y} + \frac{1}{25}\cos(x + 2y)$]

12.12 NON-HOMOGENEOUS LINEAR EQUATIONS

If in the equation $f(D, D')z = F(x, y)$ (11)

the polynomial expression $f(D, D')$ is not homogeneous, then (11) is a non-homogeneous linear partial differential equation. As in the case of linear equations in linear partial differential equations, the complete solution $z = C.F. + P.I.$

The methods to find P.I. are the same as those for homogeneous linear equations.

To find the C.F., we factorise $f(D, D')$ into factors of the form $D - mD' - c$. To find the solution of $(D - mD' - c)z = F(x, y)$, we write $R = p - mq$ and $z = u$ (12)

The subsidiary equations are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{cz}$$

An integral is $my + mx = c$ and $z = be^{cx}$.

Taking $b = \phi(my + mx)$, we get $z = e^{cx}\phi(y + mx)$

as the solution of (12). The solution corresponding to various factors added up, give the C.F. of (11).

Example 12.32. Solve $(D^3 + 2D^2D' - D'^3 - 2D)z = \sin(x + 2y)$ [U.P.T.U., 2002]

Solution. Here $f(D, D') = D(D - D')(D + D' - 2)$

Since the solution corresponding to the factor $D - mD' - c$ is known to be

$$z = e^{cx}\phi(y + mx)$$

$$\therefore \quad \text{C.F.} = \phi_1(y - x) + e^{2x}\phi_2(y - x)$$

$$\text{P.I.} = \frac{1}{D^3 + 2D^2D' - D'^3 - 2D - 2D'}\sin(x + 2y)$$

$$= \frac{1}{-1 + 2(-2) - 2(1) + (-8) - 2(-2) - 2(2)}\sin(x + 2y)$$

$$= -\frac{1}{2D - D' + 2}\sin(x + 2y) = -\frac{2D - D' + 2}{4D^2 - D'^2 + 4}\sin(x + 2y)$$

$$= \frac{1}{20}[2\sin(x + 2y) - \cos(x + 2y)]$$

Hence the complete solution is

$$z = \phi_1(y - x) + e^{2x}\phi_2(y - x) + \frac{1}{20}[2\sin(x + 2y) - \cos(x + 2y)]$$

PROBLEMS 9.2

Solve the following equations :

1. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial z}{\partial x} + x + y$
2. $(D - D' - 1)(D - D' - 2)z = (e^x + e^y)$
3. $(D - D' - 1)(D - D' - 2)z = e^{2x+y}$
4. $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y} = e^{x-y} + x^2y$ (Madras, 2005 S)
5. $(D^2 + DD' + D' - 1)z = \cos(x + 2y)$ (U.P.T.U. 2006)
6. $(2D^3 + D^2D' - 8D'^3)z = 8\cos(4x - 2y)$

9.28 NON-LINEAR EQUATIONS OF THE SECOND ORDER

We now give a method due to Monge*, for integrating the equation $Rr + Ss + Tt = V$ ...(1)

in which R, S, T, V are functions of x, y, z, p and q.

Since $dp = \frac{\partial p}{\partial x}dx + \frac{\partial p}{\partial y}dy = r\,dx + s\,dy$, and $dq = s\,dx + t\,dy$,

we have $r = (dp - s\,dy)/dx$ and $t = (dq - s\,dx)/dy$.

Substituting these values of r and t in (1), and rearranging the terms, we get

$$R(dp\,dy + T dq\,dx - V dx\,dy) - s[R dy^2 - S dy\,dx + T dx^2] = 0 \quad ...(2)$$

Let us consider the equations

$$R dy^2 - S dy\,dx + T dx^2 = 0 \quad ...(3)$$

$$R dp\,dy + T dq\,dx - V dx\,dy = 0 \quad ...(4)$$

which are known as Monge's equations.

Since (2) can be factorized, we obtain its integral first. In case the factors are different, we may get two distinct integrals of (2). If their product with (4) is a light an integral of (1). If need be, we may also use the relation $dz = p\,dx + q\,dy$ while solving (3) and (4).

Let $u(x, y, z, p, q) = a$ and $v(x, y, z, p, q) = b$ be the integrals of (3) and (4) respectively. Then $u = a, v = b$ evidently constitute a solution of (2) and therefore, of (1) also. Taking $b = \phi(a)$, we find a general solution of (1) to be $v = \phi(u)$, which should be further integrated by methods of first order equations.

Example 12.24. Solve $(r + s)(xp + yq) = pq(x + y)$ **(U.P.T.U. 2005)**

Solution. Monge's equations are

$$xdy^2 + (x + y)dy\,dx + ydx^2 = 0 \quad ...(1)$$

$$xdpdy + ydqdx - \frac{x+y}{x-y}(p - q)dydx = 0 \quad ...(2)$$

(1) may be factorized as $(xdy + ydx)(dy + dx) = 0$. Hence integrals are $xy = a$ and $x + y = b$.

Taking $xy = a$ and dividing each term of (2) by xdy or its equivalent $-ydx$, we get

$$dp - dq = \frac{dx - dy}{x - y}(p - q) = 0 \quad \text{or} \quad \frac{d(p-q)}{p-q} = \frac{d(x-y)}{x-y}$$

This gives on integration $(p - q)/(x - y) = c$.

Hence a first integral of the given equation is $p - q = (x - y)$ times which is Lagrange's linear equation. Its subsidiary equations are

$$\frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{(x-y)(xy)}$$

From the first two equations, we have $x + y = c$.

Using this, we have

$$dz = -\phi(xy = x^2)\,(x - 2x)\,dx \quad \text{which gives} \quad z = \phi_1(xy - x^2) + b$$

Writing $\phi = \phi_1(x)$ and $b = z = \chi$, we get

$$z = \phi_1(xy) + \chi(x + y).$$

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*Named after Gaspard Monge (1746–1818), Professor at Paris.

Thus, the first two terms of the integral are identical and each term of the first is the sum of the second and third terms of the second integral.

Example 17.22. Solve $y^2z + x(y^2 - 2yz + z)p + y(y^2 - z^2)q = z(y^2 + z^2)$. (Nov/Dec 2002)

Solution. Monge's equations are $y^2dy^2 - 2ydydz + dz^2 = 0$ ...(i)

and $z^2dydy + dzdx - (y + z)dydz = 0$ ...(ii)

(i) gives $(ydy - dz)^2 = 0$ i.e., $y^2 + 2z = c$ ...(iii)

Putting $ydy = -dz$ in (ii), we get

$$ydy - dy + (y + z)dydy = 0 \quad \text{or} \quad (y + z)dy + ydy - dy + (y + z)dy = 0$$

whose integral is $y^2 - y + 2y^2 = a$

Combining this with (iii), we get the integral $xy - y + 2x^2 + 4y^2 = 2x$

The subsidiary equations for this Lagrange's linear equation are

$$\frac{dx}{y} = \frac{dy}{-1} = \frac{dz}{xy^2 + 2x - 2y^2}$$

From the first two equations, we have $y^2 + 2x = c$

Using this, we have $dz + (y^2 + 2x^2)dy = 0$

whose solution is $z + xy(y + 1) - y^2 = b$

Hence the required solution is $z + y^2 - xy(y^2 + 2x) + 4y^2 = 2x$

PROBLEMS 17.6

Solve :

1. $(y + z)p + (z + x)q = (x + y)$
2. $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$
3. $(y^2 + z^2)p - xyq + xz = 0$
4. $x^2p + y^2q = z^2$ (A.M.I.E., 2006)
5. $(x^2 - y^2 - z^2)p + 2xyq = 2xz$
6. $(y - z)p + (z - x)q = x - y$
7. $(x^2 + y^2 + z^2)(xp + yq + z) = 2(x + y + z)$
8. $(x^2 - y^2 - z^2)p + 2xyq = 2xz$
9. $(y^2 + z^2)p - xyq + xz = 0$
10. $(x^2 - y^2 - z^2)p + 2xyq = 2xz$

17.14 OBJECTIVE TYPE OF QUESTIONS

MULTIPLE CHOICE

Fill up the blanks or choose the correct answer in each of the following problems :

1. The equation $\frac{d^2y}{dx^2} + \sin x\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} = 0$ is of order ..... and degree .....
2. The complementary function of $(D^2 - 3DD' + 2D'^2)z = e^{x+y}$ is .....
3. The solution of $\frac{d^2y}{dx^2} = \sin(x) + e^x$ is .....
4. A solution of $(x - 2)p + (y - 1)q = z - y$ is .....
5. The particular integral of $(D^2 - DD')z = \sin(x + y)$ is .....
6. The partial differential equation obtained from $z = ax + by + ab$ by eliminating a and b is .....
7. Solution of $\sqrt{p} + \sqrt{q} = 1$ is .....
8. Solution of $p^2x + q^2y = z^2$ is .....
9. Solution of $p + q = \log(x + y)$ is .....
10. The order of the partial differential equation obtained by eliminating f from $z = f(x^2 - y^2)$ is .....
11. The solution of $z = px + qy + pq$ is .....
12. By eliminating a and b from $z = a(x + y) + b(x - y) + ab$, the partial differential equation formed is .....
13. The solution of $(D^2 - DD' + 2D'^2)z = 0$ is .....
14. By eliminating the arbitrary constants from $z = ax + by + ab$, the partial differential equation formed is .....

(Nov/Dec 2003)

15. A solution of $z_{xy} = 0$ is of the form
16. If $z = x^2 + y^2$ is a solution of $x^2\frac{\partial^2 z}{\partial x^2} + y^2\frac{\partial^2 z}{\partial y^2} = kz$, then $k =$ .....

17. The general solution of $\frac{dy}{dx} = xy$ is
18. The complementary function of $(D^2 - 2D + 1)y = e^x$ is
19. The solution of $\frac{dy}{dx} = xy + x$ is

(a) $y = Ce^{\frac{x^2}{2}} - 1$ (b) $y = Ce^{\frac{x^2}{2}}$ (c) $y = Ce^{\frac{x^2}{2}} + 1$ (d) $y = Ce^{\frac{x^2}{2}} - x$

20. The solution of $(y - xy^2)dx + (x + x^2y)dy = 0$ is
(a) $\frac{1}{xy} + \log\frac{y}{x} = c$ (b) $xy + \log\frac{y}{x} = c$
(c) $\frac{1}{xy} + \log xy = c$ (d) $x^2y^2 + \log\frac{y}{x} = c$

21. The partial differential equation obtained by eliminating a and b from $z = ax + by + ab$ is

(a) $z = p\left[\frac{\partial^2 z}{\partial x^2}\right] + q$ (b) $z = p\left[\frac{\partial^2 z}{\partial y^2}\right] + q$ (c) $z = \frac{1}{p}\left[\frac{\partial^2 z}{\partial x^2}\right] + q$ (d) $z = \frac{1}{q}\left[\frac{\partial^2 z}{\partial y^2}\right] + p$

22. The solution of $(y + z)dx + dy + dz = 0$ is
(a) $\frac{1}{2}(y+z)^2 + \log x = 0$ (b) $\frac{1}{2}(y+z)^2 + x = 0$
(c) $\frac{1}{2}(y+z)^2 + \log x = 0$ (d) None of these

23. Particular integral of $(2D^2 - 3DD' + D'^2)z = e^{x+y}$ is
(a) $\frac{1}{2}e^{x-y}$ (b) $-\frac{1}{2}e^{x+y}$ (c) $2e^{x+y}$ (d) e^{x+y}

24. The solution of $\frac{d^2y}{dx^2} + y = 0$ is
(a) $y = C_1\cos x + C_2\sin x$ (b) $y = C_1\cos x$
(c) $y = C_1\cos x + C_2\sin x + C_3$ (d) $y = C_1 + C_2x + C_3x^2$

25. Particular integral of $(D^2 - D'^2)z = \cos(2x + y)$ is
(a) $\frac{1}{3}\cos(2x + y)$ (b) $-\frac{x}{3}\sin(2x + y)$ (c) $\frac{1}{3}\sin(2x + y)$ (d) $-\frac{x}{3}\cos(2x + y)$

26. The partial differential equation obtained by eliminating the arbitrary constants a and b from $z = (x^2 + a)(y^2 + b)$ is
(a) $pq = xy$ (b) $z = pq + xy$
(c) $z = xy$ (d) $z = (x^2 + p)(y^2 + q)$

27. $xy + yz + zx = c^2$ is a non-linear partial differential equation. (True or False)
28. $xy + yz + zx = 0$ is a non-linear partial differential equation. (True or False)
29. $x + y + z = 0$ is a solution of $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z$. (True or False)
30. $z = x^2 + y^2$ is a solution of $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = z$. (True or False)
31. $z = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} - 2xy$ is a solution of differential equation. (True or False)

CHAPTER 18

Applications of Partial Differential Equations

1. Introduction. 2. Method of separation of variables. 3. Partial differential equations of engineering. 4. Vibrations of a stretched string—Wave equation. 5. One dimensional heat flow. 6. Two dimensional heat flow. 7. Solution of Laplace's equation. 8. Laplace's equation in polar coordinates. 9. Vibrating membranes—Two dimensional wave equation. 10. Transmission lines. 11. Laplace's equation in three dimensions. 12. Solution of three dimensional Laplace's equation. 13. Objective Type of Questions.

18.1 INTRODUCTION

In physical problems, we also generally come across the different types of solutions of the partial differential equations known as the boundary conditions. The differential equation together with these boundary conditions constitute a boundary value problem.

In problems involving ordinary differential equations, we may first find the general solution and then determine the arbitrary constants from the initial values. But the same process is not applicable to problems involving partial differential equations for the general solution of a partial differential equation contains arbitrary functions which are difficult to adjust so as to satisfy the given boundary conditions. Most of the boundary value problems involving linear partial differential equations can be solved by the following method.

18.2 METHOD OF SEPARATION OF VARIABLES

In this we assume a solution which is the product of functions each of which contains only one of the variables. The following example explains the method :

Example 18.1. *Solve by the method of separation of variables*

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial z}{\partial t} + \frac{\partial z}{\partial y} + z \qquad \text{U.P.T.U. 2000 S ; Rewari 2000 ; U.P.T.U. 2000}$$

Solution. Assume the trial solution $z = X(x)Y(y)$ ...(1)

where X is a function of x alone and Y that of y alone.

Substituting this value of z in the given equation, we have

$$X''Y = 2X'Y + XY' + XY \quad \text{where } X' = \frac{dX}{dx}, Y' = \frac{dY}{dy} \text{ etc.}$$

Separating the variables, we get $\frac{X'' - 2X' - X}{X} = \frac{Y'}{Y}$ ...(2)

Since x and y are independent variables, therefore, (2) can only be true if each side is equal to the same constant, c (say).

$$\therefore \quad \frac{X''-2X'}{X} = \alpha^2 \text{ (say), i.e., } X''-2X'-\alpha^2 X = 0 \qquad \ldots(iii)$$

and
$$-T'=\alpha^2 T, \text{ i.e., } T'+\alpha^2 T=0 \qquad \ldots(iv)$$

The solution of the ordinary linear equation (iii), the auxiliary equation is

$$m^2-2m-\alpha^2=0, \quad \text{whence } m=1\pm\sqrt{(1+\alpha^2)}.$$

$\therefore$ the solution of (iii) is $X=c_1 e^{(1+\sqrt{(1+\alpha^2)})x}+c_2 e^{(1-\sqrt{(1+\alpha^2)})x}$

and the solution of (iv) is $T=c_3 e^{-\alpha^2 t}$.

Substituting these values of X and T in (i), we get

$$u=\left(c_1 e^{(1+\sqrt{(1+\alpha^2)})x}+c_2 e^{(1-\sqrt{(1+\alpha^2)})x}\right) c_3 e^{-\alpha^2 t}$$

i.e.,
$$u=\left(A_1 e^{(1+\sqrt{(1+\alpha^2)})x}+A_2 e^{(1-\sqrt{(1+\alpha^2)})x}\right) e^{-\alpha^2 t}$$

which is the required complete solution.

Note In practical problems, the unknown constants α, A_1, A_2 are determined from the given boundary conditions.

Example 16.8. Using the method of separation of variables solve $\frac{\partial u}{\partial x}=2\frac{\partial u}{\partial t}+u$, given $u(x,0)=4e^{-3x}$.
(V.T.U. 2003; Kerala Univ. 2006; Kerala 2008)

Solution. Assume the solution $u(x,t)=X(x)T(t)$

Substituting in the given equation, we have

$$XT'=2XT'+XT \quad \text{or} \quad (T'-T)X=2XT'$$

or
$$\frac{X'-X}{2X}=\frac{T'}{T}=k \text{ (say)}$$

$$\therefore \quad X'-X-2kX=0 \quad \text{or} \quad \frac{X'}{X}=1+2k \quad \ldots(i) \quad \text{and} \quad \frac{T'}{T}=k \qquad \ldots(ii)$$

Solution (i), $\quad \log X=(1+2k)x+\log c \quad \text{or} \quad X=ce^{(1+2k)x}$

Solution (ii), $\quad \log T=kt+\log c' \quad \text{or} \quad T=c'e^{kt}$

Thus $\quad u(x,t)=XT=cc'e^{(1+2k)x}e^{kt} \qquad \ldots(iii)$

Now $\quad 4e^{-3x}=u(x,0)=cc'e^{(1+2k)x}$

$\therefore \quad cc'=4$ and $1+2k=-3 \quad \text{or} \quad k=-2$

Substituting these values in (iii), we get

$u=4e^{-3x}e^{-2t}$ i.e., $u=4e^{-(3x+2t)}$ which is the required solution.

EXERCISE 6.1

Solve the following equations by the method of separation of variables:

1. $y^2u_x+x^2u_y=0$. (V.T.U. 2011; S.V.T.U. 2009) 2. $x\frac{\partial u}{\partial x}=y^2\frac{\partial u}{\partial y}$. (V.T.U. 2006)

3. $\frac{\partial u}{\partial x}+u=\frac{\partial u}{\partial t}$, given that $u(x,0)=6e^{-3x}$. (S.V.T.U. 2006)

4. $4\frac{\partial u}{\partial x}=\frac{\partial u}{\partial t}+u$, given $u=3e^{-5x}-e^{-3t}$ when $x=0$. (S.V.T.U. 2008)

5. $3\frac{\partial u}{\partial x}+2\frac{\partial u}{\partial t}=0$, $u(x,0)=4e^{-x}$. (S.V.T.U. 2008)

6. Find a solution of the equation $\frac{\partial^2 u}{\partial x^2}=\frac{\partial u}{\partial t}+2u$ in the form $u=X(x)Y(t)$. Solve the equation so that in the interval $0<x<\pi$, $u=0$ and $\frac{\partial u}{\partial x}=3+e^{-2t}$, when $x=0$ for all values of t. (Madras 2000)

33.3 PARTIAL DIFFERENTIAL EQUATIONS OF ENGINEERING

A number of problems in engineering give rise to the following well-known partial differential equations:

(i) Wave equation : $\dfrac{\partial^2 y}{\partial t^2} = c^2 \dfrac{\partial^2 y}{\partial x^2}$,

(ii) One dimensional heat flow equation : $\dfrac{\partial u}{\partial t} = c^2 \dfrac{\partial^2 u}{\partial x^2}$,

(iii) The two dimensional heat flow equation which in steady state becomes the two dimensional Laplace's equation : $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$,

(iv) Transmission line equations,

(v) Vibrating membrane, Two dimensional wave equation,

(vi) Laplace's equation in three dimensions.

Besides these, the partial differential equations frequently occur in the theory of Elasticity and Hydrodynamics.

Starting with the method of separation of variables, we find their solutions subject to specific boundary conditions and the combination of such solutions gives the desired solutions. Quite often a certain condition is not applicable. In such cases, the most general solution is written as the sum of the particular solutions already found and the constants are determined using Fourier series so as to satisfy the remaining conditions.

33.4 VIBRATIONS OF A STRETCHED STRING—WAVE EQUATION

Consider a highly stretched elastic string of length l and fixed ends A and B and subjected to constant tension T (Fig. 33.1). The tension T will be considered to be large as compared to the weight of the string so that the effects of gravity are negligible.

Let the string be released from rest and allowed to vibrate. We shall study the subsequent motion of the string, with no external forces acting on it, assuming that each point of the string makes small vibrations at right angles to the equilibrium position AB, all the string motion is in one plane.

Fig. 33.1

Taking the end A as the origin, AB as the x-axis and AY perpendicular to AB as the y-axis, so that the motion takes place entirely in the xy-plane. Figure 33.1 shows the string in the position AQB at time t. Consider the motion of the element PQ of the string between the points P(x, y) and Q$(x + \delta x, y + \delta y)$, where the tangents make angles ψ and $\psi + \delta\psi$ with the x-axis. Clearly the element is moving upwards with the acceleration $\partial^2 y/\partial t^2$. Also the vertical component of the force acting on this element

$$= T\sin(\psi + \delta\psi) - T\sin\psi = T[\sin(\psi + \delta\psi) - \sin\psi]$$

$$= T[\tan(\psi + \delta\psi) - \tan\psi], \text{ since } \psi \text{ is small} = T\left[\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_x\right]$$

If m is the mass per unit length of the string, then by Newton's second law of motion, we have

$$m\delta x \cdot \frac{\partial^2 y}{\partial t^2} = T\left[\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_x\right] \quad \text{i.e.,} \quad \frac{\partial^2 y}{\partial t^2} = \frac{T}{m}\left[\frac{\left(\frac{\partial y}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial y}{\partial x}\right)_x}{\delta x}\right]$$

Taking limits as $Q \to P$ i.e., $\delta x \to 0$, we have $\dfrac{\partial^2 y}{\partial t^2} = c^2 \dfrac{\partial^2 y}{\partial x^2}$, where $c^2 = \dfrac{T}{m}$ ...(1)

This is the partial differential equation giving the transverse vibrations of the string. It is also called the one dimensional wave equation.

(2) **Solution of the wave equation.** Assume that a solution of (1) is of the form

$y = X(x)T(t)$ where X is a function of x and T is a function of t only.

Then $\frac{d^2X}{dx^2} = X''$, T'' and $\frac{d^2T}{dt^2} = X''$, T''

Substituting these in (1), we get $XT'' = c^2X''T$ (i.e., $\frac{X''}{X} = \frac{1}{c^2}\frac{T''}{T}$ ...(2)

Clearly the left side of (2) is a function of x only and the right side is a function of t only. Since x and t are independent variables, (2) can hold good if each side is equal to a constant k (say). Then (2) leads to the ordinary differential equations :

$$\frac{d^2X}{dx^2} - kX = 0 \quad ...(3) \quad \text{and} \quad \frac{d^2T}{dt^2} - kc^2T = 0 \quad ...(4)$$

Solving (3) and (4), we get

(i) When k is positive and $= p^2$, say $X = c_1e^{px} + c_2e^{-px}$; $T = c_3e^{pct} + c_4e^{-pct}$.

(ii) When k is negative and $= -p^2$ say $X = c_1\cos px + c_2\sin px$; $T = c_3\cos pct + c_4\sin pct$

(iii) When k is zero, $X = c_1x + c_2$; $T = c_3t + c_4$.

Thus the various possible solutions of wave-equation (1) are

$$y = (c_1e^{px} + c_2e^{-px})(c_3e^{pct} + c_4e^{-pct}) \quad ...(5)$$

$$y = (c_1\cos px + c_2\sin px)(c_3\cos pct + c_4\sin pct) \quad ...(6)$$

$$y = (c_1x + c_2)(c_3t + c_4) \quad ...(7)$$

Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem. As we will be dealing with problems on vibrations, y must be a periodic function of x and t. Hence that solution must be a trigonometric terms. Accordingly the solution given by (5) & (7) of the form

$$y = (C_1\cos px + C_2\sin px)(C_3\cos pct + C_4\sin pct) \quad ...(8)$$

is the only suitable solution of the wave equation (Nagpur, 2005)

Example 18.5. A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a\sin(\pi x/l)$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by

$y(x, t) = a\sin(\pi x/l)\cos(\pi ct/l)$ (U.P.T.U., 2003; U.P.T.U., 2004; Roorkee, 2000; U.P.T.U., 2002)

Solution. The vibration of the string is given by $\frac{\partial^2y}{\partial t^2} = c^2\frac{\partial^2y}{\partial x^2}$ ...(1)

As the end points of the string are fixed, for all time,

$$y(0, t) = 0 \quad ...(2) \quad \text{and} \quad y(l, t) = 0 \quad ...(3)$$

Since the initial transverse velocity of any point of the string is zero,

therefore, $\left[\frac{\partial y}{\partial t}\right]_{t=0} = 0$ ...(4)

Also $y(x, 0) = a\sin(\pi x/l)$ ...(5)

Now we have to solve (1) subject to the boundary conditions (2) and (3) and initial conditions (4) and (5).

Since the vibration of the string is periodic, therefore, the solution of (1) is of the form

$$y(x, t) = (C_1\cos px + C_2\sin px)(C_3\cos pct + C_4\sin pct) \quad ...(6)$$

By (2), $y(0, t) = C_1(C_3\cos pct + C_4\sin pct) = 0$

For this to be true for all time, $C_1 = 0$.

Hence $y(x, t) = C_2\sin px(C_3\cos pct + C_4\sin pct)$ ...(7)

and $\frac{\partial y}{\partial t} = C_2\sin px(C_3(-pc)\sin pct + C_4pc\cos pct)$

$\therefore$ By (4), $\left[\frac{\partial y}{\partial t}\right]_{t=0} = C_2\sin px \cdot (C_4pc) = 0$, whence $C_2C_4pc = 0$.

If $C_2 = 0$, (7) will lead to the trivial solution $y(x, t) = 0$.

$\therefore$ The only possibility is that $C_4 = 0$.

Then (7) becomes $y(x, t) = C_2C_3\sin px\cos pct$ ...(8)

[illegible]

[illegible]

[illegible]

Hence from (11)

$$c_n \sin n\pi = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

or

$$c_n = \frac{2}{\pi}\left[\frac{\sin nx}{n^2} - \frac{x\cos nx}{n}\right]_0^{\pi} = \frac{2}{\pi}\left(\frac{\sin n\pi}{n^2} - \frac{\pi\cos n\pi}{n}\right) = -\frac{2}{n}\cos n\pi = \frac{2}{n}(-1)^{n+1}$$

Therefore, we have

$$f(x) = 2\left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \cdots + (-1)^{n+1}\frac{\sin nx}{n} + \cdots\right)$$

Hence from (12), the desired solution is

$$y(x,t) = \frac{2c_0}{\pi}\sin\frac{\pi x}{l}\cos\frac{\pi ct}{l} + \frac{2c_1}{2\pi}\sin\frac{2\pi x}{l}\cos\frac{2\pi ct}{l} + \cdots$$

A vertical wire of length l is fixed at its upper end and hangs under its own weight. Find the displacement of the wire when it is released from rest in the vertical position.

Solution. The equation of the motion is

$$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2} \qquad \ldots(1)$$

The boundary conditions are $y(l,t) = 0$, $y(0,t)$ finite $\qquad \ldots(2)$

Since the initial conditions are $y(x,0) = kx$ and $\dot{y}(x,0) = 0$ $\qquad \ldots(3)$

and

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = 0 \qquad \ldots(4)$$

The solution of (1) is of the form

$$y(x,t) = (c_1\cos px + c_2\sin px)(c_3\cos cpt + c_4\sin cpt)$$

By (2), $\qquad y(0,t) = (c_1)(c_3\cos cpt + c_4\sin cpt) = 0$

For this to be true for all values of t, $c_1 = 0$

$$\therefore \qquad y(x,t) = c_2\sin px\,(c_3\cos cpt + c_4\sin cpt)$$

Also by (2) $\qquad y(l,t) = c_2\sin pl\,(c_3\cos cpt + c_4\sin cpt) = 0$ for all t

This is true only if $\sin pl = 0$ or $p = n\pi/l$, n being an integer.

Thus, $\qquad y(x,t) = c_2\sin\frac{n\pi x}{l}\left(c_3\cos\frac{cn\pi t}{l} + c_4\sin\frac{cn\pi t}{l}\right) \qquad \ldots(5)$

$$\frac{\partial y}{\partial t} = c_2\sin\frac{n\pi x}{l}\cdot\frac{cn\pi}{l}\left(-c_3\sin\frac{cn\pi t}{l} + c_4\cos\frac{cn\pi t}{l}\right)$$

By (4) $\qquad \left(\frac{\partial y}{\partial t}\right)_{t=0} = c_2\sin\frac{n\pi x}{l}\cdot\frac{cn\pi}{l}\cdot c_4 = 0$

This is true if $c_4 = 0$

Thus $\qquad y(x,t) = c_2 c_3\sin\frac{n\pi x}{l}\cos\frac{cn\pi t}{l} = b_n\sin\frac{n\pi x}{l}\cos\frac{cn\pi t}{l}$

Letting n take on all values, the general solution of (1) is

$$y(x,t) = \sum_{n=1}^{\infty} b_n\sin\frac{n\pi x}{l}\cos\frac{cn\pi t}{l} \qquad \ldots(6)$$

From (3) $\qquad y(x,0) = kx = \sum_{n=1}^{\infty} b_n\sin\frac{n\pi x}{l}$

where $\qquad b_n = \frac{2}{l}\int_0^l kx\sin\frac{n\pi x}{l}\,dx$, by Fourier half range sine series

$$= \frac{2k}{l}\left[x\left(-\frac{\cos n\pi x/l}{n\pi/l}\right) - \int 1\cdot\left(-\frac{\cos n\pi x/l}{n\pi/l}\right)dx\right]_0^l$$

$$\cdots$$

Hence, from (14) the desired solution is

$$u(x,t) = \frac{4h}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin\frac{n\pi}{2} \cos\frac{n\pi x}{L} \cos\frac{n\pi ct}{L}$$

$$= \frac{8h}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos\frac{(2n-1)\pi x}{L} \cos\frac{(2n-1)\pi ct}{L}.$$

Example 1. This equation of the vibrating string is $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ (13)

The boundary conditions are $u(0, t) = 0$, $u(L, t) = 0$ (14)

And the initial conditions are $u(x, 0) = f(x)$ (15)

and $\left(\frac{\partial u}{\partial t}\right)_{t=0} = v_0(x)$. (16)

Using the separation of the vibrating string problem Theorem 1, the solutions of (13) are of the form

$$u(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos cpt + c_4 \sin cpt)$$

By (14), $u(0, t) = c_1(c_3 \cos cpt + c_4 \sin cpt) = 0$

If $c_1 = 0$ then $u(x, t) = 0$ for all t and c_3, c_4. (17)

Also $u(L, t) = (c_1 \cos pL + c_2 \sin pL)(c_3 \cos cpt + c_4 \sin cpt) = 0$ for all t.

Other than $c_3 = c_4 = 0$, we have $pL = n\pi$, $p = \frac{n\pi}{L}$, n being an integer.

Thus $u(x, t) = c_2 \sin\frac{n\pi x}{L}\left(c_3 \cos\frac{n\pi ct}{L} + c_4 \sin\frac{n\pi ct}{L}\right)$

By (15) $f(x) = c_2 \sin\frac{n\pi x}{L}$ Now call $c_2 c_3 = A$

$u(x, 0) = c_2 c_3 \sin\frac{n\pi x}{L}$ where $c_2 c_4 = B$

Adding all such solutions, the general solution of (13) is

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{L} \cos\frac{n\pi ct}{L} \quad (18)$$

Then $\frac{\partial u}{\partial t} = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{L} \cos\frac{n\pi ct}{L}$

By (16) $v_0(x) = \left(\frac{\partial u}{\partial t}\right)_{t=0} = \sum_{n=1}^{\infty} \frac{n\pi c}{L} B_n \sin\frac{n\pi x}{L}$

$$\int_0^L v_0(x) \sin\frac{n\pi x}{L}\,dx = \sum_{n=1}^{\infty} \frac{n\pi c}{L} B_n \int_0^L \sin\frac{n\pi x}{L} \sin\frac{n\pi x}{L}\,dx \quad (19)$$

$$= \frac{n\pi c}{L} B_n \frac{L}{2} + \frac{2n\pi c}{L} B_n \int_0^{\frac{L}{2}} \sin\frac{n\pi x}{L} \sin\frac{2n\pi x}{L}\,dx + \frac{3n\pi c}{L} B_n \int_0^{\frac{L}{3}} \sin\frac{n\pi x}{L} \sin\frac{3n\pi x}{L}\,dx + \cdots$$

Equating coefficients from both sides, we get

$$\frac{3a_3}{4} = \frac{\pi}{l}b_1, \quad 0 = \frac{2\pi}{l}b_2, \quad -\frac{b_3}{4} = \frac{3\pi}{l}b_3, \ldots$$

$$\therefore \quad a_3 = \frac{8b_1}{3\pi}, \quad b_2 = -\frac{b_1}{12\pi}, \quad b_3 = b_4 = b_5 = \ldots = 0$$

Substituting in (v), the desired solution is

$$y = \frac{lb_1}{12\pi}\left(9\sin\frac{\pi x}{l}\sin\frac{\pi t}{l} - \sin\frac{3\pi x}{l}\sin\frac{3\pi t}{l}\right).$$

Example 18.7. *A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is released from rest from the position given by $y(x, 0) = f(x)$, find the displacement of the string at any distance x from one end at any time t.* (Agra U., 2009; U.P.T.U., 2004)

Solution. The equation of the vibrating string is

$$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2} \qquad \ldots(i)$$

The boundary conditions are $\quad y(0, t) = 0, y(l, t) = 0 \qquad \ldots(ii)$

Also the initial conditions are $\quad y(x, 0) = 0 \qquad \ldots(iii)$

and

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = \lambda x(l - x) \qquad \ldots(iv)$$

As in example 18.6, the general solution of (i) satisfying the conditions (ii) and (iii) is

$$y(x, t) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}\cdot\sin\frac{n\pi ct}{l} \qquad \ldots(v)$$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}\cdot\cos\frac{n\pi ct}{l}\cdot\left(\frac{n\pi c}{l}\right)$$

By (iv), $\quad \lambda x(l - x) = \left(\frac{\partial y}{\partial t}\right)_{t=0} = \frac{n\pi c}{l}\sum_{n=1}^{\infty} nA_n \sin\frac{n\pi x}{l}$

$$\therefore \quad \frac{n\pi c}{l}nA_n = \frac{2}{l}\int_0^l \lambda x(l - x)\sin\frac{n\pi x}{l}\,dx$$

$$= \frac{2\lambda}{l}\left[x(lx - x^2)\left(-\frac{l}{n\pi}\cos\frac{n\pi x}{l}\right) - (l - 2x)\left(-\frac{l^2}{n^2\pi^2}\sin\frac{n\pi x}{l}\right) + (-2)\left(\frac{l^3}{n^3\pi^3}\cos\frac{n\pi x}{l}\right)\right]_0^l$$

$$= \frac{4\lambda l^3}{n^3\pi^3}[1 - \cos n\pi] = \frac{4\lambda l^3}{n^3\pi^3}[1 - (-1)^n]$$

or $\quad A_n = \frac{4\lambda l^3}{n^4\pi^4}[1 - (-1)^n] = \frac{8\lambda l^3}{n^4\pi^4(2m - 1)^4}$ (taking $n = 2m - 1$).

Hence, from (v), the desired solution is

$$y = \frac{8\lambda l^2}{\pi^4}\sum_{m=1}^{\infty}\frac{1}{(2m-1)^4}\sin\frac{(2m-1)\pi x}{l}\cos\frac{(2m-1)\pi ct}{l}.$$

Example 18.8. *The point of trisection of a string are pulled to the height b through the same distance on opposite sides of the position of equilibrium and the string is released from rest. Derive an expression for the displacement of the string at subsequent time and show that the configuration of the string changes its nature at time $t = l/2c$.* (Kerala, 2006)

Solution. Let B and C be the points of the trisection of the string OA ($= l$) (Fig. 18.5). Initially the string is held in the form $OBCA$, where $AB' = CC' = b$ (say).

The displacement $y(x,t)$ of any point of the string is given by

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \qquad \text{(1)}$$

and the boundary conditions are

$$y(0,t) = 0 \qquad \text{(2)}$$

$$y(L,t) = 0 \qquad \text{(3)}$$

$$\left[\frac{\partial y}{\partial t}\right]_{t=0} = 0 \qquad \text{(4)}$$

Fig. 10.1

The remaining condition is that at $t = 0$, the string takes on the form of the broken line OPQ in Fig. 10.1. The equation of OP is $y = \frac{2k}{L}x$ and that of PQ is $y = \frac{2k}{L}(L - x)$; thus the condition at $t = 0$ is

$$y = f(x) = \begin{cases} \frac{2kx}{L} & 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L - x) & \frac{L}{2} < x < L \end{cases} \qquad \text{(5)}$$

and the equation of PQ is $y = \frac{2k}{L}(L - x)$.

Hence the Fourier boundary conditions are

$$\begin{aligned} f(x) &= \frac{2k}{L}x, \quad 0 < x < \frac{L}{2} \\ &= \frac{2k}{L}(L - x), \quad \frac{L}{2} < x < L \end{aligned} \qquad \text{(6)}$$

As in example 10.6, the solution of (1) satisfying the boundary conditions (2), (3) and (4) is

$$y(x,t) = b_1 \sin\frac{\pi x}{L}\cos\frac{\pi ct}{L} \qquad \text{Where } b_1 = \frac{1}{L}\int_0^L f(x)\,dx$$

Adding all such solutions, the most general solution of (1) is

$$y(x,t) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}\cos\frac{n\pi ct}{L} \qquad \text{(7)}$$

Putting $t = 0$, we have $y(x,0) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}$ (8)

So we have to find the coefficients in an expansion of $f(x)$ in a sine series. This equation is the expression of $y(x,0)$ as a Fourier half-range sine series as in the footnote to § 10.3.

So $b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx$

$$= \frac{2}{L}\left[\int_0^{\frac{L}{2}} \frac{2kx}{L}\sin\frac{n\pi x}{L}\,dx + \int_{\frac{L}{2}}^{L} \frac{2k}{L}(L - x)\sin\frac{n\pi x}{L}\,dx\right]$$

$$= \frac{4k}{L^2}\left[\left\{x\left(-\frac{\cos\frac{n\pi x}{L}}{\frac{n\pi}{L}}\right) - 1\left(-\frac{\sin\frac{n\pi x}{L}}{\frac{n^2\pi^2}{L^2}}\right)\right\}_0^{\frac{L}{2}}\right.$$

$$\left. + \left\{(L - x)\left(-\frac{\cos\frac{n\pi x}{L}}{\frac{n\pi}{L}}\right) - (-1)\left(-\frac{\sin\frac{n\pi x}{L}}{\frac{n^2\pi^2}{L^2}}\right)\right\}_{\frac{L}{2}}^{L}\right]$$

$$= \frac{4k}{L^2}\left[-\frac{L}{n\pi}\cos\frac{n\pi}{2} + \frac{L^2}{n^2\pi^2}\sin\frac{n\pi}{2} + \frac{L}{n\pi}\cos n\pi - \frac{L^2}{n^2\pi^2}\sin n\pi + \frac{L}{n\pi}\cos\frac{n\pi}{2} + \frac{L^2}{n^2\pi^2}\sin n\pi\right]$$

$$= \frac{8k}{n^2\pi^2}\sin\frac{n\pi}{2} = \begin{cases} \frac{8k}{n^2\pi^2}(-1)^{\frac{n-1}{2}} & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

$$\cdots$$

$$\cdots$$

$$\cdots$$

Hence the solution of (4) takes the form

$$y(x,t) = \frac{1}{2}[f(x+ct)+k] + \frac{1}{2}[f(x-ct)-k] = f(x+ct) + f(x-ct) \qquad \ldots(5)$$

which is the d'Alembert's solution[1] of the wave equation (1). (V.T.U., 2011)

Obs: The above solution gives a very useful method of solving partial differential equations by the use of its solution.

Example 3.8.3. Find the deflection of a vibrating string of unit length having fixed ends with initial velocities zero and initial deflection $f(x) = k\sin^3 \pi x$. (V.T.U., 2007)

Solution. By d'Alembert's method, the solution is

$$y(x,t) = \frac{1}{2}[f(x+ct) + f(x-ct)]$$

$$= \frac{1}{2}[k\sin^3(x+ct) - \sin 3(x+ct)] + k[\sin^3(x-ct) - \sin 3(x-ct)]$$

$$= k[\sin^3 x\cos 3ct - \sin 3x\cos 3ct]$$

Also $\qquad y(x,0) = k[\sin^3 x - \sin 3x] = f(x)$

and $\qquad \left(\frac{\partial y}{\partial t}\right)_{t=0} = k[3\sin^2 x\cos x\,(3c\sin ct) - 3c\cos 3x\sin 3ct]_{t=0} = 0$

i.e., the given boundary conditions are satisfied.

EXERCISE 3.8

1. Solve completely the equation $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$, representing the displacement of a string of length l, fixed at both ends, given that $y(0,t) = 0 = y(l,t)$, $\forall t$, $\frac{\partial y}{\partial t} = 0$ and $y(x,0) = kx(l-x)$, $0 < x < l$. (Nagpur, 2007; V.T.U., 2007)

2. Solve the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$ under the conditions $y(0,t) = y(l,t) = 0$ and $y(x,0) = \sin\frac{\pi x}{l}$, $\left(\frac{\partial y}{\partial t}\right)_{t=0} = 0$, $0 < x < l$.

3. Find the solution of the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$, corresponding to the initial velocity of the string given by

$$v(x) = \frac{2kx}{l}, \text{ when } 0 < x < \frac{l}{2} \quad \text{and} \quad \frac{2k}{l}(l-x), \text{ when } \frac{l}{2} < x < l$$

and initial displacement zero. (Nagpur, 2006; Kerala, M.E., 2006)

4. A tightly stretched string of length l has its ends fixed at $x = 0$, $x = l$. The midpoint of the string is then taken to height h and then released from rest in this position. Find the lateral displacement of a point of the string at a distance x from one end at time t. (Anna, 2005)

5. A tightly stretched string with fixed end points $x = 0$ and $x = L$ is initially in a position given by

$$y(x) = \begin{cases} x, & 0 \le x \le \frac{L}{2} \\ L - x, & \frac{L}{2} \le x \le L \end{cases}$$

If it is released from this position of rest, determine its displacement at any time $t > 0$.

6. If a string of length l is initially at rest in equilibrium position and each of its points is given a velocity v such that $v = cx(l-x)$, find the deflection.

$v(x,t) = \frac{8cl^2}{\pi^3}\left[\sum_{n=1}^{\infty}\left\{\frac{\sin\frac{n\pi}{2}}{n^3} - \frac{3\sin\frac{n\pi}{2}}{n^3}\right\}\sin\frac{n\pi x}{l}\sin\frac{n\pi ct}{l} - \frac{4l^2}{\pi^3}\sum_{n=1}^{\infty}\frac{(-1)^n}{(2n-1)^3}\sin\frac{(2n-1)\pi x}{l}\sin\frac{(2n-1)\pi ct}{l}\right]$

7. If a string of length l is initially at rest in equilibrium position and each of its points is given a velocity such that $v = cx(l-x)$, find the displacement $y(x,t)$. (Anna, 2005)

8. Using d'Alembert's method, find the deflection of a vibrating string of unit length having its ends fixed and initial velocity zero and initial deflection :

(i) $f(x) = kx(1-x)$ (Gujarat, M. Tech., 2002) (ii) $f(x) = k\sin^3\pi x$.

[1] See footnote on p. 3.28.

4.13.1 [L] [illegible]

[illegible]

Fig. 4.13.1

[illegible]

[illegible]

[illegible]

Example 3.9.10. Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $x > 0$, $t > 0$, boundary conditions $u(x, 0) = 0$ for $x > 0$, $u(0, t) = 0$ and $u(x, t)$ is bounded for $0 < x < \infty$, $t > 0$.

Solution. The solution of the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ ...(1)

is $u(x, t) = c_1 \cos px + c_2 \sin px)\, e^{-p^2 t}$ ...(2)

When $x = 0$, $u(0, t) = c_1 e^{-p^2 t} = 0$ i.e., $c_1 = 0$.

$\therefore$ (2) becomes $u(x, t) = c_2 \sin px\, e^{-p^2 t}$ ...(3)

When $x = \infty$, $u(\infty, t) = c_2 e^{-p^2 t} = 0$ if and only if p is zero or p is an integer, i.e., $p = n\pi$.

$\therefore$ (3) reduces to $u(x, t) = b_n e^{-n^2 \pi^2 t} \sin n\pi x$ where $b_n = c_2$.

Thus the general solution of (1) is $u(x, t) = \sum b_n e^{-n^2 \pi^2 t} \sin n\pi x$ ...(4)

When $t = 0$, $f(x) = u(x, 0) = \sum_{n=1}^{\infty} b_n e^{-n^2 \pi^2 (0)} \sin n\pi x$

Comparing both sides, $b_n = 3$

Hence from (4), the desired solution is

$$u(x, t) = 3 \sum_{n=1}^{\infty} e^{-n^2 \pi^2 t} \sin n\pi x.$$

Example 3.9.11. Solve the one-dimensional equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ for the conduction of heat along a rod of length l with zero temperature, subject to the following conditions:

(i) $u(x, t)$ is bounded for $t > 0$, (ii) $\frac{\partial u}{\partial x} = 0$ for $x = 0$ and $x = l$,

(iii) $u(x, 0) = kx$ for $0 < x < \frac{l}{2}$, $u(x, 0) = k(l - x)$ for $\frac{l}{2} < x < l$. (P.T.U., 2007)

Solution. Substituting $u = X(x)T(t)$ in the given equation, we get

$$XT' = c^2 X''T \quad \text{i.e.,} \quad \frac{X''}{X} = \frac{T'}{c^2 T} = -k^2 \text{ (say)}$$

$$\text{i.e.,} \quad \frac{d^2X}{dx^2} + k^2 X = 0 \text{ and } \frac{dT}{dt} + k^2 c^2 T = 0 \quad \text{...(1)}$$

These solutions are $X = c_1 \cos kx + c_2 \sin kx$, $T = c_3 e^{-k^2 c^2 t}$ ...(2)

If k^2 is changed to $-k^2$, the solutions are

$$X = c_4 e^{kx} + c_5 e^{-kx}, \; T = c_6 e^{k^2 c^2 t} \quad \text{...(3)}$$

If $k^2 = 0$, the solutions are $X = c_7 x + c_8$, $T = c_9$ ...(4)

In (3), $T \to \infty$ as $t \to \infty$ therefore, $u(x, t) \to \infty$ as $t \to \infty$, the given condition (i) is not satisfied. So we reject the solutions (3) while (2) and (4), satisfy this condition.

Applying the conditions (ii) in (4), we get $c_7 = 0$.

$\therefore$ $u = XT = c_7 c_9 + c_8 c_9$ (say) ...(5)

From (2), $\frac{\partial u}{\partial x} = (-c_1 k \sin kx + c_2 k \cos kx) c_3 e^{-k^2 c^2 t}$

Applying the conditions (ii), we get $c_2 = 0$ and $-c_1 \sin kl + c_2 \cos kl = 0$

i.e., $c_2 = 0$ and $kl = n\pi$ (n is an integer)

$\therefore$ $u = c_1 \cos kx \cdot c_3 e^{-k^2 c^2 t} = c_1 \cos\left(\frac{n\pi x}{l}\right) e^{-\frac{n^2 \pi^2 c^2 t}{l^2}}$ ...(6)

Thus the general solution behind the wave of $x = ct$ and $x = 2ct$ is

$$u = a_0 + \sum a_n \cos(n\pi x/l) e^{-n^2\pi^2 c^2 t/l^2} \quad \ldots(17)$$

Now using the condition (iv), we get

$$kx - x^2 = a_0 + \sum a_n \cos(n\pi x/l)$$

This being the expansion of $kx - x^2$ as a half-range cosine series in $(0, l)$, we get

$$a_0 = \frac{1}{l}\int_0^l (kx - x^2)\,dx = \frac{1}{l}\left[\frac{kx^2}{2} - \frac{x^3}{3}\right]_0^l = \frac{kl^2}{6}$$

and

$$a_n = \frac{2}{l}\int_0^l (kx - x^2)\cos\frac{n\pi x}{l}\,dx = \frac{2}{l}\left[(kx - x^2)\left(\frac{l}{n\pi}\sin\frac{n\pi x}{l}\right)\right.$$

$$-(k - 2x)\left(-\frac{l^2}{n^2\pi^2}\cos\frac{n\pi x}{l}\right) + (-2)\left(-\frac{l^3}{n^3\pi^3}\sin\frac{n\pi x}{l}\right)\Bigg]_0^l$$

$$= \frac{2}{l}\left[0 - \frac{l^2}{n^2\pi^2}(k\cos n\pi + k) + 0\right] = -\frac{4l^2}{n^2\pi^2} \text{ when } n \text{ is even, otherwise } 0.$$

Hence taking $n = 2m$, the required solution is

$$u = \frac{kl^2}{6} - \frac{4l^2}{\pi^2}\sum_{m=1}^{\infty}\frac{1}{m^2}\cos\left(\frac{2m\pi x}{l}\right)e^{-4m^2\pi^2c^2t/l^2}$$

Example 11.12. (a) *An insulated rod of length l has its ends A and B maintained at $0°C$ and $100°C$ respectively, until steady-state conditions prevail. If the temperature at B is suddenly reduced to $0°C$ and maintained at $0°C$, find the temperature distribution function $u(x, t)$.* (U.P.T.U., 2003)

(b) *Solve the above problem if the change in temperature at B is a lowering of temperature of B to $50°C$ and subsequent fall of B to $0°C$.* (Madras, 2007)

Solution. (a) Let the equation for the conduction of heat be

$$\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2} \quad \ldots(1)$$

Prior to the temperature change at the end B, when $t = 0$, the heat flow was independent of time (steady state condition). When u depends only on x, (1) reduces to $\partial^2 u/\partial x^2 = 0$.

Its general solution is $u = cx + b$ $\ldots(2)$

Since $u = 0$ for $x = 0$ and $u = 100$ for $x = l$, therefore, (2) gives $b = 0$ and $c = 100/l$.

Therefore the initial condition is expressed by $u(x, 0) = \dfrac{100}{l}x$ $\ldots(3)$

Also the boundary conditions for the subsequent flow are

$$u(0, t) = 0 \text{ for all values of } t \quad \ldots(4)$$

and

$$u(l, t) = 0 \text{ for all values of } t \quad \ldots(5)$$

Thus we have to find a temperature function $u(x, t)$ satisfying the differential equation (1) subject to the initial condition (3) and the boundary conditions (4) and (5).

Now the solution of (1) is of the form

$$u(x, t) = (C_1\cos px + C_2\sin px)e^{-c^2p^2t} \quad \ldots(6)$$

By (4), $\quad u(0, t) = C_1 e^{-c^2p^2t} = 0$, for all values of t.

Hence $C_1 = 0$ and (6) reduces to $u(x, t) = C_2\sin px\, e^{-c^2p^2t}$ $\ldots(7)$

Applying (5), (7) gives $u(l, t) = C_2\sin pl\, e^{-c^2p^2t} = 0$, for all values of t.

This requires $\sin pl = 0$ (i.e., $pl = n\pi$ or $C_2 = 0$, $\quad p = n\pi/l$, where n is any integer.

Hence (7) reduces to $u(x, t) = b_n\sin\dfrac{n\pi x}{l}e^{-c^2n^2\pi^2t/l^2}$, where $b_n = C_2$.

(Hint: use the substitution $x = \cos\theta$ and recognize the resulting differential equation as one of the classical equations corresponding to the eigenvalue $\lambda_n = n^2$.)

Solving all such subproblems, the most general solution of (1) satisfying the boundary conditions (2) is of the form

$$u(x,t) = \sum_{n=0}^{\infty} \left(A_n \cos nx + B_n \sin nx\right) e^{-(n^2+1)t} \qquad \text{(3)}$$

Putting $t = 0$,

$$u(x,0) = \sum_{n=0}^{\infty} \left(A_n \cos nx + B_n \sin nx\right) \qquad \text{(4)}$$

In order that the equations (3) may be satisfied, $f(x)$ and $g(x)$ must be series. This requires that expansions of $f(x)$ and $g(x)$ in a Fourier cosine series in $(0, \pi)$. Then

$$A_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx \quad \text{where } A_0 = \frac{1}{\pi}\int_0^{\pi} f(x)\,dx$$

$$= \frac{2}{\pi}\left[f(x)\left(\frac{\sin nx}{n}\right) - f'(x)\left(\frac{-\cos nx}{n^2}\right) + f''(x)\left(\frac{-\sin nx}{n^3}\right)\right]_0^{\pi} = \frac{2}{\pi}\left[-\frac{1}{n^2}\cos n\pi\right] = \frac{2(-1)^{n+1}}{n^2} \quad (n \neq 0)$$

Similarly we find

$$B_n = \frac{2}{\pi}\int_0^{\pi} g(x)\sin nx\,dx = \frac{2}{n\pi}\left[1 - (-1)^n\right] \qquad \text{(5)}$$

(b) Show that such a condition (indicative) does require $a(x)$ above to satisfy the boundary conditions

$$a(0) = 1 \quad \text{for all values of } t \qquad \text{(6)}$$

$$a'(\pi) = 0 \quad \text{for all values of } t \qquad \text{(7)}$$

In order to do the boundary values (1)–(2). The solution must be the sum of two terms, one of which is that the second solution exists. Since this homogeneous boundary value problem, we have to consider the general case.

We apply to the inhomogeneous boundary value problem (1)–(2) the two-parameter

$$u(x,t) = v_n(x,t) + w_n(x,t) \qquad \text{(8)}$$

where $w_n(x,t)$ is a solution of (1) satisfying a very small boundary value problem (2) and satisfying the boundary conditions (6) and (7) ($w_n(x,t)$ is called a Duhamel's Method for (1)–(2)). Then $v_n(x,t)$ is a solution which satisfies a problem of the form (1)–(2) and $v_n(x,t)$ is zero for a sufficiently large n (a theorem given in the following which determines such a sequence of n).

Since $w_n(x,t) = 0$ and $v_n(x,t) = f(x)$, the boundary conditions (2) are satisfied by

$$v_n(x,0) = f(x) \qquad \text{(9)}$$

Putting $x = 0$ in (1) and (2), we have for v_n

$$v_{nt}(0,t) = a(0)v_{nx}(0,t) + v_n(0,t) + f(0) - f(0) = 0 \qquad \text{(10)}$$

Putting $x = \pi$ in (1) and (2), we have for $v_n(x,t)$

$$v_{nt}(\pi,t) = a(\pi)v_{nx}(\pi,t) + v_n(\pi,t) + f(\pi) - f(\pi) = 0 \qquad \text{(11)}$$

Now $\qquad v_n(x,t) = w(x,t) - w_n(x,t) = \frac{2(-1)^{n+1}}{n^2}\cos nx + \frac{2[1-(-1)^n]}{n\pi}\sin nx \qquad$ (the first term must be)

$$= \frac{2(-1)^{n+1}}{n^2} \quad \text{(12)}$$

Hence (10)–(11) and (12) give the boundary conditions and (12) gives the Laplace transform solution to the transform solutions. Hence the boundary value problem for (10) and (11) are both zero. Accordingly, as in part (a) we have $v_n(x,t) = C_n'$, and so $v = C_n$ also gives $v = e^{-(n^2+1)t}$

We (12),

$$v_n(0,t) = C_n e^{-(n^2+1)t} = 0 \quad \text{for all values of } t.$$

Hence $C_n = 0$ and

$$v_n(x,t) = C_n \sin nx\, e^{-(n^2+1)t} \qquad \text{(13)}$$

Applying (9) to (13) we get

$$v_n(x,0) = C_n \sin nx = f(x) \quad \text{for all values of } x.$$

This requires

$$f(x) = C_n \sin nx \quad (n^2 + 1 \neq 0 \text{ for } n = 0 \text{ or } n = 1 \text{ or } n = 2 \text{ integer})$$

Hence (12) is a solution to $v_n(x,t) = C_n \sin nx\, e^{-(n^2+1)t}$ where $C_n = C_n$

Adding all such solutions, the most general solution of (xxi) satisfying the boundary conditions (xvi) and (xvii) is

$$u_2(x,t) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l} e^{-(n^2\pi^2/l^2)t} \qquad \ldots(xxii)$$

Putting $t = 0$, we have $u_2(x,0) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$ $\ldots(xxiii)$

In order that the condition (xviii) may be satisfied, $u_2(x,t)$ and $u(x,t)$ must be same. This requires the expansion of $f(x) = 20$ as a half range Fourier sine series in $(0, l)$. Thus

$$\frac{80x}{l} - 20 = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l} \quad \text{where} \quad b_n = \frac{2}{l}\int_0^l \left(\frac{80x}{l} - 20\right)\sin\frac{n\pi x}{l}\,dx = -\frac{40}{n\pi}(1 + \cos n\pi)$$

i.e., $b_n = 0$, when n is odd; $= -80/n\pi$, when n is even.

Hence (xxii) becomes $u_2(x,t) = -\sum_{n=2,4,6,\ldots} \left(\frac{-80}{n\pi}\right)\sin\frac{n\pi x}{l}\, e^{-(n^2\pi^2/l^2)t}$ Take $n = 2m$

$$= -\frac{80}{\pi}\sum_{m=1}^{\infty}\frac{1}{m}\sin\frac{2m\pi x}{l}\, e^{-(4m^2\pi^2/l^2)t} \qquad \ldots(xxiv)$$

Finally combining (xxi) and (xxiv), the required solution is

$$u(x,t) = \frac{40x}{l} + 20 - \frac{80}{\pi}\sum_{m=1}^{\infty}\frac{1}{m}\sin\frac{2m\pi x}{l}\, e^{-(4m^2\pi^2/l^2)t}$$

Example 10.11. *The ends A and B of a rod 30 cm long have the temperature at 30°C and 60°C until steady state prevails. The temperature of the ends are changed to 40°C and 80°C respectively. Find the temperature distribution in the rod at time t.*

Solution. Let the heat equation be $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$ $\ldots(i)$

In steady state conditions, u is independent of time and dependent on x only. (i) reduces to

$$\frac{d^2u}{dx^2} = 0 \qquad \ldots(ii)$$

Its solution is $u = ax + b$.

Since $u = 30$ for $x = 0$ and $u = 60$ for $x = 30$, therefore $a = 30$, $b = (60 - 30)/30 = 1/2$.

Thus the initial conditions are expressed by

$$u(x,0) = 30 + \frac{1}{2}x \qquad \ldots(iii)$$

The boundary conditions are $u(0,t) = 30$, $u(30,t) = 60$.

Using (ii), the steady state temperature is

$$u_1(x,t) = 30 + \frac{60 - 30}{30}x = 30 + x \qquad \ldots(iv)$$

To find the temperature u in the intermediate period,

$$u(x,t) = u_1(x) + u_2(x,t)$$

where $u_1(x)$ is the steady state temperature distribution of the form (iv) and $u_2(x,t)$ is the transient temperature distribution which decreases to zero as t increases.

Since $u_2(x,t)$ satisfies one dimensional heat equation

$$\therefore \qquad u(x,t) = 30 + x = \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{l} + b_n\sin\frac{n\pi x}{l}\right)e^{-\lambda^2 t} \qquad \ldots(v)$$

$$u(x,t) = 45 - 15\cos\frac{\pi x}{30} + \sum_{n=1}^{\infty} a_n e^{-\lambda^2 t} \quad \text{where } a_n = 0$$

[illegible]

(a) [illegible]

$$[illegible] \qquad [illegible] \tag{1}$$

Then

$$[illegible] \qquad [illegible] \tag{2}$$

(b)

$$[illegible] \tag{3}$$

Physical [illegible]

$$[illegible] \tag{4}$$

Cosine [illegible]

$$[illegible] \tag{5}$$

(c)

$$[illegible] \tag{6}$$

where $[illegible]$

Hence from (4.5) the desired solution is

$$[illegible] \tag{7}$$

[illegible]

Solution. If the temperature $u(x,t)$ along the bar satisfies the equation

$$[illegible] \tag{8}$$

[illegible] The temperature $u(x,t)$ satisfies the equation (8) and the conditions

$$[illegible] \tag{9}$$

Similarly, $u(x,t)$ satisfies the conditions $[illegible]$

Hence $[illegible]$

Hence the required condition is $[illegible]$ (10)

Now, by substituting (9) in (8) we have $[illegible]$ (11)

Differentiating partially with respect to t, we get

$$[illegible] \tag{12}$$

Putting $[illegible]$

$$[illegible] \tag{13}$$

Putting $[illegible]$

$$[illegible] \tag{14}$$

Hence $[illegible]$

(d)

$$[illegible] \tag{15}$$

Hence $u(x,t) = c_1 \cos\frac{n\pi x}{l} e^{-n^2\pi^2 c^2 t/l^2}$.

∴ the most general solution of (1) satisfying the boundary conditions (2) is

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l} e^{-n^2\pi^2 c^2 t/l^2} + A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l} e^{-n^2\pi^2 c^2 t/l^2} \quad \text{(where } A_0 = a_0\text{)} \quad \ldots(4)$$

Putting $t = 0$, $u = f(x)$, $f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\frac{n\pi x}{l} = a$ By (3)(i)

This expression for $f(x)$ into a half range cosine series in $(0, l)$

Thus $$a = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l} \quad \text{where } a_n = \frac{2}{l}\int_0^l a\cos\frac{n\pi x}{l}\,dx$$

and $$a_n = \frac{2}{l}\left[\int_0^{l/2} x\cos\frac{n\pi x}{l}\,dx + \int_{l/2}^{l}(l-x)\cos\frac{n\pi x}{l}\,dx\right] = \frac{2l}{n^2\pi^2}(2\cos n\pi/2 - 1 - \cos n\pi)$$

$= 0$, when n is even; $= -4l/n^2\pi^2$, when n is odd.

∴ $A_0 = \frac{a_0}{2} = \frac{l}{4}$, and $A_n = a_n = 0$ for n even; $= -4l/n^2\pi^2$ for n odd.

Hence (4) takes the form

$$u(x,t) = \frac{l}{4} - \frac{4l}{\pi^2}\sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{n^2}\cos\frac{n\pi x}{l} e^{-n^2\pi^2 c^2 t/l^2}$$

$$= \frac{l}{4} - \frac{4l}{\pi^2}\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi x}{l} e^{-(2n-1)^2\pi^2 c^2 t/l^2} \quad \ldots(5)$$

This is the required temperature at a point P_1 distant x from one end A at any time t.

FIG. 18.4

Obs. The sum of the temperatures at any two points equidistant from the centre is always 100°C, a constant.

Let P_1, P_2 be two points equidistant from the centre C of the bar so that $CP_1 = CP_2$ (Fig. 18.4).

If $AP_1 = BP_2 = x$ (say), then $AP_2 = l - x$.

Replacing x by $l - x$ in (5), we get the temperature at P_2 as

$$u(l-x,t) = \frac{l}{4} - \frac{4l}{\pi^2}\sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi(l-x)}{l} e^{-(2n-1)^2\pi^2 c^2 t/l^2}$$

$$= \frac{l}{4} + \frac{4l}{\pi^2}\sum_{n=1,3,5,\ldots}^{\infty} \frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi x}{l} e^{-(2n-1)^2\pi^2 c^2 t/l^2} \quad \ldots(6)$$

$$\left[\because \cos\frac{(2n-1)\pi(2l-x)}{l} = \cos\left[2n\pi - \pi - \frac{(2n-1)\pi x}{l}\right] = -\cos\frac{(2n-1)\pi x}{l}\right]$$

Adding (5) and (6), we get $u(x,t) + u(l-x,t) = \frac{l}{2} = 100°C$.

EXERCISE 18.3

1. A homogeneous rod of conducting material of length 100 cm has its ends kept at zero temperature and the temperature initially is

$$u(x,0) = \begin{cases} x, & 0 \le x \le 50 \\ 100 - x, & 50 \le x \le 100 \end{cases}$$

Find the temperature at any time. (U.P.T.U., 2003; U.P.T.U., 2007) (Kurukshetra, 2008)

5. Find the temperature $u(x, t)$ in a bar occupying the interval $0 \le x \le L$ if the bar is insulated laterally, initially the temperature in the bar is $f(x)$, and the ends of the bar are kept at temperature 0°C for all $t > 0$. [P.S.C., 2003]

6. A rod 30 cm long has its ends A and B kept at 20° and 80°C respectively until steady state conditions prevail. The temperatures at ends are then suddenly reduced to 0°C and kept so. Find the resulting temperature function $u(x, t)$ taking $c^2 = 1$ and x. [Anna, 2003]

7. A slab of 10 cm thick, with insulated sides has its ends A and B maintained at temperatures 20°C and 50°C respectively until steady-state conditions prevail. The temperature at A is suddenly raised to 50°C and at the same time that at B is lowered to 0°C. Find the temperature distribution in the bar at time t. [P.S.C., 2003]

Show that the temperature at the middle point of the bar remains unaltered for all time, regardless of the material of the bar.

8. Solve the following boundary value problem:

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L, \quad \frac{\partial u(0,t)}{\partial x} = 0, \quad \frac{\partial u(L,t)}{\partial x} = 0, \quad u(x,0) = x \qquad [\text{U.P.T.U., 2003}]$$

9. The temperature at one end of a very thin rod of length l is kept at 0°C and that at the other end is kept at 100°C until steady state conditions prevail. The two ends are then suddenly insulated so that the temperature gradient is zero at each end thereafter. Find the temperature distribution.

10. Find the solutions of $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$, such that

(a) u is not identically zero for $t \to \infty$, (b) $u = \begin{cases} \frac{\partial u}{\partial x} = 0, & \text{when } x = 0 \\ u = 0, & \text{when } x = l \end{cases}$ for all values of t,

(c) $u = f(x)$, when $t = 0$, for all values of x between 0 and l. [U.P.T.U., 2003]

11. Find the solution of $\frac{\partial^2 u}{\partial x^2} = \frac{1}{k}\frac{\partial u}{\partial t}$, having given that $u = V_0 \cos\frac{\pi x}{2l}$ when $t = 0$ for all values of x between 0 and l, and $u = 0$ when $x = l$ for all values of t.

13.6 TWO-DIMENSIONAL HEAT FLOW

Consider the flow of heat in a metal plate of uniform thickness z (cm), density ρ (gm/cc), specific heat s (cal/gm/deg) and thermal conductivity k (cal/cm sec/deg). Let XOY plane be taken in the surface of the plate (Fig. 13.6). If the temperature u at any point is independent of the z-coordinate and depends only on x, y and time t, then the flow is said to be two-dimensional. In this case, the heat flow is in the XY-plane only and is zero along the normal to the XY-plane.

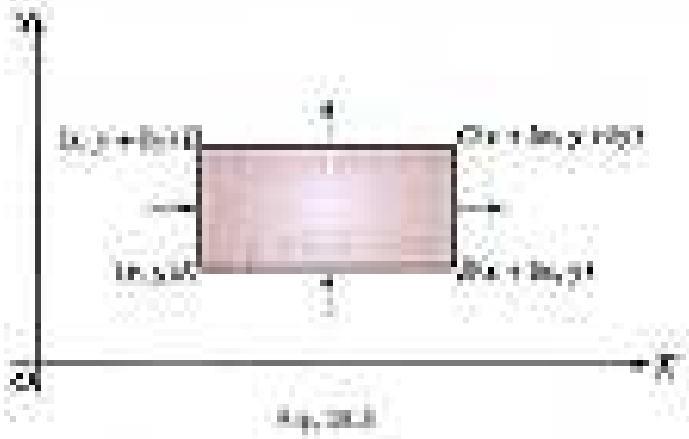


Fig. 13.6

Consider a rectangular element $ABCD$ of the plane with sides δx and δy. By (A) on p. 455, the amount of heat entering the element in 1 sec. from the side AB

$$= -k\delta x\left(\frac{\partial u}{\partial y}\right)_y$$

and the amount of heat entering the element in 1 second from the side $AD = -k\delta y\left(\frac{\partial u}{\partial x}\right)_x$

The quantity of heat flowing out through the side CD per sec. $= -k\delta x\left(\frac{\partial u}{\partial y}\right)_{y+\delta y}$

and the quantity of heat flowing out through the side BC per second $= -k\delta y\left(\frac{\partial u}{\partial x}\right)_{x+\delta x}$

Hence the total gain of heat by the rectangular element $ABCD$ per second

$$= -k\delta x\left(\frac{\partial u}{\partial y}\right)_y - k\delta y\left(\frac{\partial u}{\partial x}\right)_x + k\delta x\left(\frac{\partial u}{\partial y}\right)_{y+\delta y} + k\delta y\left(\frac{\partial u}{\partial x}\right)_{x+\delta x}$$

$$= k\rho\delta y\left[\left(\frac{\partial u}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial u}{\partial x}\right)_x\right] + k\rho\delta x\left[\left(\frac{\partial u}{\partial y}\right)_{y+\delta y} - \left(\frac{\partial u}{\partial y}\right)_y\right]$$

$$= k\rho\delta x\delta y\left[\frac{\left(\frac{\partial u}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial u}{\partial x}\right)_x}{\delta x} + \frac{\left(\frac{\partial u}{\partial y}\right)_{y+\delta y} - \left(\frac{\partial u}{\partial y}\right)_y}{\delta y}\right] \quad ...(1)$$

Also the rate of gain of heat by the element

$$= \rho\delta x\delta y\sigma\frac{\partial u}{\partial t} \quad ...(2)$$

Thus equating (1) and (2),

$$k\rho\delta x\delta y\left[\frac{\left(\frac{\partial u}{\partial x}\right)_{x+\delta x} - \left(\frac{\partial u}{\partial x}\right)_x}{\delta x} + \frac{\left(\frac{\partial u}{\partial y}\right)_{y+\delta y} - \left(\frac{\partial u}{\partial y}\right)_y}{\delta y}\right] = \rho\delta x\delta y\sigma\frac{\partial u}{\partial t}$$

Dividing both sides by $\rho\delta x\delta y$ and taking limits as $\delta x \to 0, \delta y \to 0$, we get

$$k\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = \rho\sigma\frac{\partial u}{\partial t}$$

i.e., $$\frac{\partial u}{\partial t} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \text{ where } c^2 = \frac{k}{\rho\sigma} \text{ is the diffusivity.} \quad ...(3)$$

Hence the equation (3) gives the temperature distribution of the plate in the transient state.
Cor. In the steady state, u is independent of t, so that $\partial u/\partial t = 0$ and the above equation reduces to,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

which is the well known **Laplace's equation** in two dimensions.

Obs. When the element has any motion in space (i.e., the heat flow is three dimensional), we obtain by similar analysis the equation

$$\frac{\partial u}{\partial t} = c^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right)$$

In the steady state, it reduces to $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

which is the three dimensional Laplace's equation.

13.7 SOLUTION OF LAPLACE'S EQUATION

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad ...(1)$$

Let $u = X(x)Y(y)$ be a solution of (1).

Substituting it in (1), we get $\frac{d^2X}{dx^2}Y + X\frac{d^2Y}{dy^2} = 0$

or, separating the variables, $\frac{1}{X}\frac{d^2X}{dx^2} = -\frac{1}{Y}\frac{d^2Y}{dy^2}$ ...(2)

Since x and y are independent variables, (2) can hold good only if each side of (2) is equal to a constant k (say). Then (2) leads to the ordinary differential equations

$$\frac{d^2X}{dx^2} - kX = 0 \text{ and } \frac{d^2Y}{dy^2} + kY = 0$$

Solving these equations, we get

(i) When k is positive and is equal to p^2, say

$$X = c_1 e^{px} + c_2 e^{-px},\ Y = c_3 \cos py + c_4 \sin py$$

(ii) When k is negative, and is equal to $-p^2$, say

$$X = c_1 \cos px + c_2 \sin px,\ Y = c_3 e^{py} + c_4 e^{-py}$$

(iii) When k is zero: $X = c_1 x + c_2$, $Y = c_3 y + c_4$

Thus the various possible solutions of (1) are

$$u = (c_1 e^{px} + c_2 e^{-px})(c_3 \cos py + c_4 \sin py) \quad \ldots(2)$$

$$u = (c_1 \cos px + c_2 \sin px)(c_3 e^{py} + c_4 e^{-py}) \quad \ldots(3)$$

$$u = (c_1 x + c_2)(c_3 y + c_4) \quad \ldots(4)$$

Of these we take that solution which is consistent with the given boundary conditions.

(V.T.U., 2013 S; Ahsanullah, 2005)

Temperature distribution in long plates

Example 10.11.1. An infinitely long plate is enclosed by two parallel edges and one end at right angles to them. The breadth is π. The end is at unit temperature at all points and the edges are at zero temperature. Determine the temperature at any point of the plate in the steady state.

(V.T.U., 2009; A.M.I.E., 2009 S)

Solution. In the steady state (Fig. 10.10), the temperature $u(x, y)$ at any point $P(x, y)$ satisfies the equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \ldots(1)$$

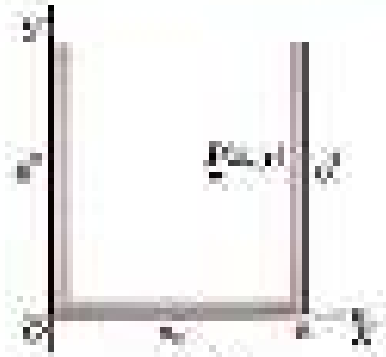


Fig. 10.8

The boundary conditions are $u(0, y) = 0$ for all values of y $\ldots(2)$

$$u(\pi, y) = 0 \text{ for all values of } y \quad \ldots(3)$$

$$u(x, \infty) = 0 \text{ for } 0 < x < \pi \quad \ldots(4)$$

$$u(x, 0) = x \text{ for } 0 < x < \pi \quad \ldots(5)$$

Now the three possible solutions of (1) are

$$u = (c_1 e^{px} + c_2 e^{-px})(c_3 \cos py + c_4 \sin py) \quad \ldots(6)$$

$$u = (c_1 \cos px + c_2 \sin px)(c_3 e^{py} + c_4 e^{-py}) \quad \ldots(7)$$

$$u = (c_1 x + c_2)(c_3 y + c_4) \quad \ldots(8)$$

Of these, we have to choose that solution which is consistent with the physical nature of the problem. The solution (6) cannot satisfy the condition (5) for $u \to 0$ for $y \to \infty$. Hence (6) is not the required solution. The solution (8) cannot satisfy the condition (4). Thus the only possible solution is (7), *i.e.*, of the form

$$u(x, y) = (C_1 \cos px + C_2 \sin px)(C_3 e^{py} + C_4 e^{-py}) \quad \ldots(9)$$

By (2), $\quad u(0, y) = C_1 (C_3 e^{py} + C_4 e^{-py}) = 0$ for all y.

Hence $C_1 = 0$ and (9) reduces to

$$u(x, y) = C_2 \sin px (C_3 e^{py} + C_4 e^{-py}) \quad \ldots(10)$$

By (3), $\quad u(\pi, y) = C_2 \sin p\pi (C_3 e^{py} + C_4 e^{-py}) = 0$, for all y.

This requires $\sin p\pi = 0$, *i.e.*, $p\pi = n\pi$ as $C_2 \neq 0$. $\therefore \quad p = n$, an integer.

Also to satisfy the condition (4), $u \to 0$ as $y \to \infty$, $C_3 = 0$.

Hence (10) takes the form $u(x, y) = b_n \sin nx\, e^{-ny}$, where $b_n = C_2 C_4$.

$\therefore$ the most general solution satisfying (2), (3) and (4) is of the form

$$u(x, y) = \sum_{n=1}^{\infty} b_n \sin nx\, e^{-ny} \quad \ldots(11)$$

Putting $y = 0$, $\quad u(x, 0) = \sum_{n=1}^{\infty} b_n \sin nx \quad \ldots(12)$

In order that the condition (5) may be satisfied, (5) and (12) must be same. This requires the expansion of $u = x$ in a half range Fourier sine series in $(0, \pi)$. Thus

$$u = \sum_{n=1}^{\infty} b_n \sin nx \text{ where } b_n = \frac{2}{\pi}\int_0^{\pi} u_0 \sin nx\,dx = \frac{2u_0}{n\pi}[1-(-1)^n]$$

i.e., $b_n = 0$, if n is even $= \frac{4u_0}{n\pi}$, if n is odd.

Hence the required solution is $u = \frac{4u_0}{\pi}\left[e^{-x}\sin x + \frac{1}{3}e^{-3x}\sin 3x + \frac{1}{5}e^{-5x}\sin 5x + \ldots\right]$.

Temperature distribution in finite plates

Example 10.11. Solve the Laplace's equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to the conditions $u(0, y) = u(\pi, y) = u(x, \infty) = 0$ and $u(x, 0) = u_0$ for $0 < x < \pi$. (AMIETE, June 2007, Dec. 2008, Anna Univ., June 2010, Dec. 2010)

Solution. The three possible solutions are

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \ldots(1)$$

are $u = (c_1 e^{px} + c_2 e^{-px})(c_3 \cos py + c_4 \sin py)$ $\ldots(2.1)$

$u = (c_1 \cos px + c_2 \sin px)(c_3 e^{py} + c_4 e^{-py})$ $\ldots(2.2)$

$u = (c_1 x + c_2)(c_3 y + c_4)$ $\ldots(2.3)$

We have to select the suitable by satisfying the following boundary conditions

$$u(0, y) = 0 \quad ; \quad u(\pi, y) = 0 \qquad \ldots(3.1)$$

$$u(x, 0) = u_0 \quad ; \quad u(x, \infty) = 0 \text{ as } y \to \infty \qquad \ldots(3.2)$$

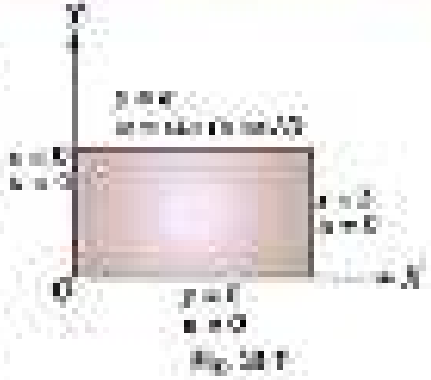


Fig. 10.1

Using (3.1) and (3.2) in (2), we get

$$c_1 = c_2 = 0, \text{ and } c_3 e^{\infty} + c_4 e^{-\infty} = 0$$

Solving these equations, we get $c_3 = c_4 = 0$ which leads to a trivial solution. Similarly, we get a trivial solution by using (3.1) and (3.2) in (2.3). Hence the suitable solution for the present problem is solution (2.1). Using (3.1), we have $c_3(c_1 e^{p\pi} + c_2 e^{-p\pi}) = 0 \quad \because c_3 \neq 0$

$\therefore$ (2.1) becomes $u = c_4 \sin px(c_1 e^{py} + c_2 e^{-py})$ $\ldots(4)$

Using (3.2), we have $c_4 \sin px(c_1 + c_2) = 0$

$\therefore$ either $c_4 = 0$ or $\sin p\pi = 0$

If we take $c_4 = 0$, we get a trivial solution.

The simplest non-trivial solution of $\sin p\pi = 0$ is given by $p = n$, where $n = 0, 1, 2, \ldots$

$\therefore$ (2.1) becomes $u = c_4 \sin nx(c_1 e^{ny} + c_2 e^{-ny})$ $\ldots(5)$

Using (3.2), we have $c_4 \sin nx(c_1 + c_2) = u_0$ or $c_1 + c_2 = \frac{u_0}{c_4}$ or $c_1 = \frac{u_0}{c_4} - c_2$

Thus the suitable solution for this problem is

$$u(x, y) = b_n \sin nx\,e^{-ny} + (u_0 - b_n)\sin nx\,e^{ny} \text{ where } b_n = c_2 c_4$$

Now using the condition (3.2), we have

$$u(x, 0) = u_0 \sin nx = b_n \sin nx + (u_0 - b_n)\sin nx$$

we get $b_n = \frac{1}{(e^{n\pi} - e^{-n\pi})}$

Hence the required solution is

$$u(x, y) = \frac{e^{n\pi} - e^{-n\pi}}{e^{n\pi} - e^{-n\pi}} \sin nx\,e^{-ny} + \frac{\sinh(n\pi - ny)}{\sinh(n\pi)} \sin nx$$

Example 10.12. The function $u(x, y)$ satisfies the Laplace's equation in the region of the xy-plane for which $x \geq 0$, $0 \leq y \leq \pi$ and $u(x, y) = 0$ when $x = 0$ and $0 \leq y \leq \pi$. Also $u(x, 0) = u(x, \pi) = 0$ and $u(x, y) \to 0$ as $x \to \infty$. Obtain a non-trivial solution for $u(x, y)$. (Madras, 2003)

Problem 4.10. The only possible solutions of

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0 \tag{4.1}$$

(a) of the form $\qquad y = e^{mx}$ where $m = m_1$ and $m = m_2$ $\qquad$ (4.2)

The boundary conditions are

$$y(0) + y'(0) = 0 \quad \text{i.e.} \quad m_1 + m_2 = 0 \tag{4.3}$$

$$y(1) = 0 \tag{4.4}$$

$$y(x) = Ae^{m_1 x} + Be^{m_2 x} \quad \text{i.e.} \quad A + B = 0 \tag{4.5}$$

Using (4.3) $\qquad y(x) = A(e^{m_1 x} - e^{m_2 x})$ $\qquad$ (4.6)

Using (4.4) $\qquad y(x) = A(e^{m_1 x} - e^{-m_1 x})$

$\therefore$ $\qquad y(x) = A \sinh m_1 x$

(b) $\qquad y(x) = A \sinh \frac{(2n+1)\pi x}{2}$ where n is an integer $\qquad$ (4.7)

Using (4.5) $\qquad y(x) = A\left(e^{\frac{(2n+1)\pi x}{2}} - e^{-\frac{(2n+1)\pi x}{2}}\right)$

$\therefore$ $\qquad y(x) = A \sinh \frac{(2n+1)\pi x}{2}$

(c) $\qquad y(x) = A \sinh \frac{(2n+1)\pi x}{2} + B \sinh \frac{(2m+1)\pi x}{2}$ where n and m are integers $\qquad$ (4.8)

Using (4.5) $\qquad y(x) = A\left(e^{\frac{(2n+1)\pi x}{2}} - e^{-\frac{(2n+1)\pi x}{2}}\right) + B\left(e^{\frac{(2m+1)\pi x}{2}} - e^{-\frac{(2m+1)\pi x}{2}}\right)$

$\therefore$ $\qquad Ae^{\frac{(2n+1)\pi x}{2}} + Be^{\frac{(2m+1)\pi x}{2}} = 0$ $\quad$ or $\quad Ae^{\frac{(2n+1)\pi}{2}} - Be^{\frac{(2m+1)\pi}{2}} = -\frac{1}{2}A$, where

Using (4.4) $\qquad y(x) = A\left(e^{\frac{(2n+1)\pi x}{2}} - e^{-\frac{(2n+1)\pi x}{2}}\right)$

$$= A_1 \sinh \frac{(2n+1)\pi x}{2} \quad \text{where } A_1 = 2A$$

$\therefore$ The most general solution of (4.1) is

$$y(x) = \sum_{n=0}^{\infty} A_n \sinh \frac{(2n+1)\pi x}{2} \tag{4.9}$$

Using the condition $y = 1$, we have

$$\sinh \frac{\pi}{2} + A_1 \sinh \frac{3\pi}{2} + \cdots = \sum_{n=0}^{\infty} A_n \sinh \frac{(2n+1)\pi}{2} \sin \frac{(2n+1)\pi x}{2}$$

Multiplying $\sin \frac{(2n+1)\pi x}{2}$ and integrating from 0 to 1, we have

$$= \frac{A_n}{2}\left[\cos\left(\frac{(2n+1)\pi}{2} - \frac{(2n+1)\pi x}{2}\right)\right]_0^1 - \frac{A_n}{2}\left[\cos\left(\frac{(2n+1)\pi}{2} + \frac{(2n+1)\pi x}{2}\right)\right]_0^1$$

$$= 0 - 0 + \frac{A_n}{2}(1 - \cos n\pi)$$

$$= \frac{A_n}{2}(1 - \cos n\pi) \quad \text{where } n \text{ is odd, otherwise for even values of } n \text{ it is zero.}$$

Hence from (4.9), the required solution is

$$y(x) = \frac{\sinh \frac{\pi}{2}}{\sinh \frac{3\pi}{2}} \sum_{n=0}^{\infty} \frac{\sinh \frac{(2n+1)\pi}{2} \sin \frac{(2n+1)\pi x}{2}}{\sinh \frac{(2n+1)\pi}{2}}$$

(d) $\qquad y(x) = \frac{\sinh \frac{\pi}{2}}{\sinh \frac{3\pi}{2}} \sum_{n=0}^{\infty} \frac{\sinh \frac{(2n+1)\pi}{2} \sin \frac{(2n+1)\pi x}{2}}{\sinh \frac{(2n+1)\pi}{2}} + \frac{\sinh \frac{\pi}{2}}{\sinh \frac{3\pi}{2}} \sum_{n=0}^{\infty} \frac{\sinh \frac{(2n+1)\pi}{2} \sin \frac{(2n+1)\pi x}{2}}{\sinh \frac{(2n+1)\pi}{2}}$

EXERCISE 18.4

1. A long rectangular plate of width a cm with insulated surfaces is initially at zero temperature throughout. At time $t = 0$, the temperature is suddenly raised to u_0 °C along one short edge and maintained at that temperature while the other edges are kept at 0°C. Show that the steady-state temperature at any point of the plate is

$$u(x, y) = \frac{2u_0}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} e^{-n\pi y/a} \sin \frac{n\pi x}{a}$$ (U.P.T.U., 2002)

2. A rectangular plate with insulated surfaces is 8 cm wide and so long compared to its width that it may be considered infinite in length without introducing any appreciable error. If the temperature along one short edge is given by

$$u(x, 0) = 100 \sin(\pi x/8), \quad 0 < x < 8$$

while the two long edges and the other short edge are kept at 0°C, show that the steady-state temperature at any point of the plate is given by

$$u(x, y) = 100 e^{-\pi y/8} \sin(\pi x/8)$$

3. A rectangular plate with insulated surface is 10 cm wide and so long compared to its width that it may be considered infinite in length without introducing any appreciable error. If the temperature along one short edge is given by

$$u = 20x, \quad \text{for } 0 \le x \le 5$$

$$\text{and} \quad u = 20(10 - x), \quad \text{for } 5 \le x \le 10$$

and the two long edges and the other short edge are kept at 0°C, prove that the temperature at any point is given by

$$u = \frac{40}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{\frac{n-1}{2}}}{n^2} \sin \frac{n\pi x}{10} e^{-n\pi y/10}, \quad n = 1, 3, 5, \ldots$$ (U.P.T.U., 2003)

4. Solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for $0 < x < 2$, $0 < y < \infty$, subject to the conditions given as $u(0, y) = 0$, $u(x, 0) = u(x, \infty) = 0$, $u(2, y) = 3 \sin(\pi x) + 5 \sin(3\pi x)$.

5. A square plate is bounded by the lines $x = 0$, $y = 0$, $x = 20$, and $y = 20$. Its faces are insulated. The temperature along the upper horizontal edge is given by

$$u(x, 20) = x(20 - x), \text{ when } 0 < x < 20$$

while other three edges are kept at 0°C. Find the steady-state temperature in the plate. (Madras, 2002)

6. The temperature at the ends and all along the edges of a square plate of length 100 cm, and the faces insulated, is maintained at 0°C until steady-state conditions prevail. Find an expression for the temperature at any point (x, y). Hence show that for temperature at the centre of the plate

$$= \frac{200}{\pi} \left(\frac{1}{\sinh \pi/4} - \frac{1}{3 \sinh 3\pi/4} + \frac{1}{5 \sinh 5\pi/4} - \cdots \right)$$

7. A square sheet of metal is folded along its diagonal. One half is kept at temperature u_0 and the other half is insulated. The edges are kept at zero temperature. Show that the steady-state temperature at any point (x, y) is given by

$$u(x, y) = \frac{4u_0}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin^2 \frac{(2n-1)\pi}{4} \sin \frac{(2n-1)\pi x}{a} \sinh \frac{(2n-1)\pi y}{a}}{(2n-1) \sinh \frac{(2n-1)\pi b}{a}}$$

8. A rectangular plate has its edges $x = 0$, $x = a$, $y = 0$, $y = b$. Initially the temperature is zero throughout the plate. At time $t = 0$, the edges $x = 0$, $x = a$, $y = 0$ are kept at zero temperature and the edge $y = b$ is kept at temperature u_0 for all t. Find the temperature function $u(x, y, t)$ in the steady-state.

18.8 (1) LAPLACE'S EQUATION IN POLAR COORDINATES

In the study of steady-state temperature distribution in a rectangular plate, it is usually convenient to employ Cartesian coordinates as laid down. However, when Polar coordinates are found to be more useful and the Cartesian form of Laplace's equation is replaced by its polar form :

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} = \frac{\partial^2 u}{\partial \theta^2} = 0$$ (Rgpv, Bu, 8 20, p. 2015-2141)

(2) Solution of Laplace's equation

$$r^2\frac{\partial^2 u}{\partial r^2} + r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0 \qquad ...(1)$$

Assume that a solution of (1) is of the form $u(r, \theta) = R(r) \cdot \Phi(\theta)$ where $R(r)$ is a function of r alone and $\Phi(\theta)$ is a function of θ only.

Substituting it in (1), we get $r^2R''\Phi + rR'\Phi + R\Phi'' = 0$ or $\frac{r^2R'' + rR'}{R} = -\frac{\Phi''}{\Phi}$

Separating the variables $\frac{r^2R'' + rR'}{R} = -\frac{\Phi''}{\Phi}$ ...(2)

Clearly the left side of (2) is a function of r only and the right side is a function of θ alone. Since r and θ are independent variables, (2) can hold good only if each side is equal to a constant λ (say). Then (2) leads to the ordinary differential equations

$$r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} - \lambda R = 0 \qquad ...(3) \qquad \text{and} \qquad \frac{d^2\Phi}{d\theta^2} + \lambda\Phi = 0 \qquad ...(4)$$

Putting $r = e^t$, (3) reduces to $\frac{d^2R}{dt^2} - \lambda R = 0$ ...(5)

Solving (3) and (4), we get

(i) When λ is positive and $= p^2$, say:

$$R = c_1e^{pt} + c_2e^{-pt} = c_1r^p + c_2r^{-p}, \Phi = c_3\cos p\theta + c_4\sin p\theta$$

(ii) When λ is negative and $= -p^2$, say

$$R = c_1\cos p\log r + c_2\sin p\log r = c_1\cos(p\log r) + c_2\sin(p\log r), \Phi = c_3e^{p\theta} + c_4e^{-p\theta}$$

(iii) When λ is zero:

$$R = c_1t + c_2 = c_1\log r + c_2, \Phi = c_3\theta + c_4$$

Thus the three possible solutions of (1) are

$$u = (c_1r^p + c_2r^{-p})(c_3\cos p\theta + c_4\sin p\theta) \qquad ...(6)$$

$$u = (c_1\cos(p\log r) + c_2\sin(p\log r))(c_3e^{p\theta} + c_4e^{-p\theta}) \qquad ...(7)$$

$$u = (c_1\log r + c_2)(c_3\theta + c_4) \qquad ...(8)$$

Of these solutions, we have to take that solution which is consistent with the physical nature of the problem. The general solution will consist of a sum of terms of type (6), (7) or (8). (U.P.T.U., 2005)

Example 18.10. The diameter of a semi-circular plate of radius unity is kept at 0°C and the temperature at the semi-circular boundary is $T°C$. Show that the steady state temperature at the point (r, θ) is

$$u(r, \theta) = \frac{4T}{\pi}\sum_{n=1}^{\infty}\frac{1}{(2n-1)}\left(\frac{r}{a}\right)^{(2n-1)}\sin(2n-1)\theta \qquad \text{(Kerala, M. Tech., 2003)}$$

Solution. Take the centre of the circle as the pole and bounding diameter as the initial line as in Fig. 18.8. Let the steady state temperature at any point $P(r, \theta)$ be $u(r, \theta)$, so that u satisfies the equation

$$r^2\frac{\partial^2 u}{\partial r^2} + r\frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0 \qquad ...(1)$$

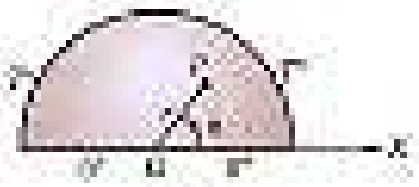


Fig. 18.8

The boundary conditions are:

$$u(r, 0) = 0 \quad \text{for } 0 < r < a \qquad ...(2)$$

$$u(r, \pi) = 0 \quad \text{for } 0 < r < a \qquad ...(3)$$

and

$$u(a, \theta) = T \qquad ...(4)$$

The three possible solutions of (1) are

$$u = (c_1r^p + c_2r^{-p})(c_3\cos p\theta + c_4\sin p\theta) \qquad ...(5)$$

$$u = (c_1\cos(p\log r) + c_2\sin(p\log r))(c_3e^{p\theta} + c_4e^{-p\theta}) \qquad ...(6)$$

$$u = (c_1\log r + c_2)(c_3\theta + c_4) \qquad ...(7)$$

From (3) and (4), $u = 0$ when $r = 0$ i.e., u cannot be finite at the origin. Thus the solutions (6) and (7) are to be rejected. Hence the only suitable solution is (5).

By (iv), $u(x,0) = c_1 e^{0} + c_2 e^{0} = 0$

Hence $c_2 = 0$ and (v) becomes $u(x,t) = c_1 e^{-p^2 t} \sin px$ ...(vii)

By (vi), $u(x,t) = c_1 e^{-p^2 t} \sin px = 0$.

As $c_1 \neq 0$, $\sin pl = 0$, i.e., $p = \frac{n\pi}{l}$, n being a non-zero integer.

Hence (vii) reduces to $u(x,t) = c_1 e^{-n^2\pi^2 t/l^2} \sin \frac{n\pi x}{l}$ ...(viii)

Since $u = 0$, when $x = 0$, $\therefore$ $c_1 = 0$ and (iii) becomes

$u(x,0) = k_n \sin \frac{n\pi x}{l}$, where $k_n = c_1 c_2$

$\therefore$ The most general solution of (i) is of the form

$$u(x,t) = \sum_{n=1}^{\infty} k_n e^{-n^2\pi^2 t/l^2} \sin \frac{n\pi x}{l} \quad \text{...(ix)}$$

Putting $t = 0$, $u(x,0) = \sum_{n=1}^{\infty} k_n \sin \frac{n\pi x}{l}$ ...(x)

In order that (ii) may be satisfied, (x) and (ix) must be same. This requires the expansion of $f(x)$ as a half-range Fourier sine series in $(0, l)$, as $f(x)$.

Thus $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$ where $b_n = \frac{2}{l}\int_0^l f(x) \sin \frac{n\pi x}{l}\,dx = \frac{2T}{n\pi}(1 - \cos n\pi)$ and $k_n = b_n e^{0}$

$\therefore$ $k_n = \frac{b_n}{e^0} = \frac{2T}{n\pi}(1 - \cos n\pi)$

i.e., $k_n = 0$, if n is even

$= \frac{4T}{n\pi}$, if n is odd

Hence (x) gives us, $u(x,t) = \frac{2T}{\pi}\left[\frac{e^{-t/l^2}}{1}\sin\frac{\pi x}{l} + \frac{e^{-9t/l^2}}{3}\sin\frac{3\pi x}{l} + \frac{e^{-25t/l^2}}{5}\sin\frac{5\pi x}{l} + \ldots\right]$

Example 19.18. The heat conduction of a uniform rod of length l is initially at temperature $u = u_0$ throughout. The temperature at the ends is suddenly reduced to zero and kept at zero. Find the temperature function $u(x,t)$.

Solution. We know that $u(x,t)$ satisfies the equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{...(i)}$$

The boundary conditions are, $u(0,t) = 0$, $u(l,t) = 0$ ...(ii)

and $u(x,0) = u_0$ for $0 \le x \le l/2$; $u(x,0) = u_0(l - x)$ for $l/2 \le x \le l$ ...(iii)

As in example 19.16, the most general solution of (i) satisfying the boundary conditions (ii) is of the form

$$u(x,t) = \sum_{n=1}^{\infty} A_n e^{-n^2\pi^2 c^2 t/l^2} \sin \frac{n\pi x}{l} \quad \text{...(iv)}$$

Putting $t = 0$, $u(x,0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}$

In order that the boundary condition (iii) is satisfied, we have $u(x,0) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l}$

where $B_n = A_n = \frac{2}{l}\left[\int_0^{l/2} u_0 \sin\frac{n\pi x}{l}\,dx + \int_{l/2}^{l} u_0(l-x)\sin\frac{n\pi x}{l}\,dx\right]$ ...(v)

$$= \frac{100}{\pi}\left\{0\left(\frac{-\cos n\theta}{n}\right) - 0\left(\frac{-\sin n\theta}{n^2}\right)\Big]_0^{\pi} + \left|r\left(\frac{-\cos n\theta}{n}\right) - 1\cdot 0\left(\frac{-\sin n\theta}{n^2}\right)\right|_{\pi}^{\pi}\right\}$$

$$= \frac{100}{\pi}\left\{-\frac{\pi}{n}\cos\frac{n\pi}{2} + \frac{\sin n\pi/2}{n^2} + \frac{\pi}{n}\cos\frac{n\pi}{2} + \frac{\sin n\pi/2}{n^2}\right\} + \frac{100}{n\pi}\sin n\pi/2$$

When n is even $B_n = 0$, so letting $n = 1, 3, 5$ etc. (i.e. $2r-1$) gives

$$u(r,\theta) = \sum_{n=1,3,5,\ldots}^{\infty}\left(\frac{200}{n\pi}\sin\frac{n\pi}{2}\right)\frac{1}{\pi^n}r^n\sin n\theta$$

$$= \frac{200}{\pi}\sum_{r=1}^{\infty}\frac{1-(-1)^{r-1}}{(2r-1)^2}\left(\frac{r}{\pi}\right)^{2r-1}\sin(2r-1)\theta$$

[Taking $n = 2r-1$, $n = 1, 3, 5, \ldots$ gives $r = 1, 2, 3$, etc. and $\sin n\pi/2 = \sin(2r-1)\pi/2 = (-1)^{r-1}$.] This gives the required temperature function.

PROBLEMS 135

1. A semi-circular plate of radius π has its circumference kept at zero temperature and its boundary diameter is kept at unit temperature. Find the steady-state temperature distribution $u(r,\theta)$ at any point in the plate assuming that the curved surface of the plate is insulated.
2. A semi-circular plate of radius 10 metres has its curved boundary and boundary diameter kept at 0°C and on the circumference the temperature distribution is given by $u(10,\theta) = 100\sin 3\theta - 50\sin 5\theta$. Determine the temperature distribution $u(r,\theta)$ at any point in the plate.
3. A plate in the shape of a quadrant (quarter of a circle) is bounded by $x = 0$, $y = 0$ and $x^2 + y^2 = 9$, $x \geqslant 0$, $y \geqslant 0$. It has its boundary maintained at a fixed temperature of 0°C. The heat in the plane sets up a temperature T°C along the edge $x = 0$, $0 \leqslant y \leqslant 3$ and a temperature $100y$°C along the edge $y = 0$, $0 \leqslant x \leqslant 3$. Determine the temperature distribution.
4. Determine the steady-state temperature at the points in the interior of a long metal plate, if the temperature is maintained at 0°C along the side edges and at a constant temperature 100°C along the curved edges.
5. Find the steady-state temperatures in a circular plate of radius π which has one half of its circumference at 0°C and the other half at 100°C.
6. If the radii of the inner and outer boundaries of a circular annulus are 2 cm and 4 cm and $u(3,\theta) = 15\cos\theta + 6\cos 3\theta - 20\sin\theta$, find the values of $u(r,\theta)$ in the annulus. [Use De Moivre's theorem (see Chapter 1) to obtain expansions of $\cos^3\theta$ and $\sin^3\theta$ in terms of the cosines and sines of multiples of θ.]
7. A plate in the form of a ring is bounded by the circles $r = 1$ and $r = 4$. Its boundaries are insulated and the temperature along the boundaries is

$$u(2,\theta) = 15\sin\theta + 6\sin 3\theta \text{ and } u(4,\theta) = 27\sin\theta + 12\sin 3\theta$$

Find the steady-state temperature $u(r,\theta)$ in the ring.

13.5 (3) VIBRATING MEMBRANE — TWO DIMENSIONAL WAVE EQUATION

We shall now derive the equation for the vibrations of a tightly stretched membrane, such as the membrane of a drum. We shall assume that the membrane is uniform and the tension T in it per unit length is the same in all directions at every point.

Consider the forces acting on an element $ABCD$ of the membrane (Fig. 13.6). Forces $T\delta x$ and $T\delta y$ act on the edges along the tangent to the membrane. Let u be the small displacement perpendicular to the xy plane, so that the forces $T\delta y$ on the opposite edges of length δy make angles α and β to the horizontal. So their vertical components

$= T\delta y\sin\beta - T\delta y\sin\alpha$

$= T\delta y(\tan\beta - \tan\alpha)$ (as α and β are usually small, $\sin\alpha$ and $\tan\alpha$ are almost

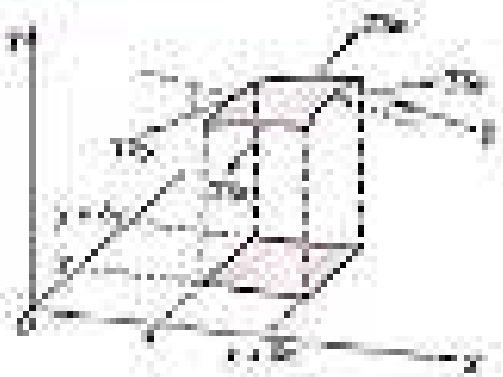


FIG. 13.6

$$\Rightarrow \text{We}\left[\left|\frac{\partial \psi}{\partial x}\right|_{x=a} - \left|\frac{\partial \psi}{\partial x}\right|_{x=b}\right] = \text{We}\int_a^b \frac{\partial^2 \psi}{\partial x^2}\,dx$$ up to a first order integral approximation.

Similarly, the normal component of the force We acting on the edges of length δx is

$$-\text{We}\int_a^b \frac{\partial \psi}{\partial y}\,\delta x$$

(i.e. in the more general case of the coordinates, then the equation of motion of the element is

$$\rho\delta x\delta y\frac{\partial^2 \psi}{\partial t^2} = \sigma\left[\frac{\partial \psi}{\partial x}+\frac{\partial \psi}{\partial x}\right]\delta x\delta y \quad \text{or} \quad \frac{\partial^2 \psi}{\partial t^2} = c^2\left[\frac{\partial^2 \psi}{\partial x^2}+\frac{\partial^2 \psi}{\partial y^2}\right] \quad \text{where} \quad c^2 = \frac{\sigma}{\rho}\,\text{We} \tag{1.1}$$

This is the wave equation in two dimensions.

(ii) Solution of the two-dimensional wave equation — Rectangular membrane. Assume that a solution of (1.1) is of the form $\psi = X(x)\,Y(y)\,T(t)$.

Substituting this in (1.1) and dividing by XYT, we get

$$\frac{1}{X}\frac{d^2X}{dx^2} = \frac{1}{Y}\frac{d^2Y}{dy^2} = \frac{1}{c^2T}\frac{d^2T}{dt^2}$$

Since one independent variable is a constant. Choosing the constants suitably, we have

$$\frac{d^2X}{dx^2} + \alpha^2 X = 0, \quad \frac{d^2Y}{dy^2} + \beta^2 Y = 0 \quad \text{and} \quad \frac{d^2T}{dt^2} + c^2(\alpha^2+\beta^2)T = 0$$

Hence a solution of (1.1) is

$$\psi = (c_1\cos\alpha x + c_2\sin\alpha x)(c_3\cos\beta y + c_4\sin\beta y)\left(c_5\cos c\sqrt{\alpha^2+\beta^2}\,t + c_6\sin c\sqrt{\alpha^2+\beta^2}\,t\right) \tag{1.2}$$

Now suppose the membrane is rectangular and is stretched between the lines $x = 0$, $y = c_1$, $x = b$, $y = 0$. Then the conditions $\psi = 0$ when $x = 0$ gives

$$0 = (c_1\cos\alpha x + c_2\sin\alpha x)c_3 \quad \text{or} \quad \sqrt{\alpha^2+\beta^2}\,x + c_4\sin\sqrt{\alpha^2+\beta^2}\,x = 0 \quad (\because c_3 \neq 0)$$

Then putting $c_1 = 0$ in (1) and applying the condition $\psi = 0$ when $x = b$, we get $\sin\alpha b = 0$ or $\alpha b = n\pi$

$$\Rightarrow \alpha = \frac{n\pi}{b} \tag{1.3}$$

Similarly, applying the conditions $\psi = 0$ when $y = 0$ and $y = c_1$, we obtain

$$\beta = \frac{m\pi}{c_1} \tag{1.4}$$

Thus the solution (2) becomes

$$\psi = (c_1\cos\alpha x + c_2\sin\alpha x)\left(c_3\cos\frac{m\pi}{c_1}y + c_4\sin\frac{m\pi}{c_1}y\right)\left(c_5\cos c\sqrt{\alpha^2+\beta^2}\,t + c_6\sin c\sqrt{\alpha^2+\beta^2}\,t\right)$$

where $\alpha = \frac{n\pi}{b}$, $\beta = \frac{m\pi}{c_1}$ $\tag{1.5}$

Thus we have obtained the wave equation (1) which gives us the displacement of the rectangular membrane. There is a constant $c = \sqrt{\sigma/\rho}$ called the velocity of the wave. ψ_{mn} are the eigen values and the corresponding eigen functions.

Choosing the constants c_5 and c_6 as 0 and 1 (we may write the general solution of the equation (1) as

$$u = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\sin\frac{n\pi x}{a}\sin\frac{m\pi y}{b}\left(A_{mn}\cos\omega_{mn}t + B_{mn}\sin\omega_{mn}t\right) \tag{1.6}$$

If the membrane starts from rest from its initial position $u = f(x, y)$, i.e. $\left(\frac{\partial u}{\partial t}\right)_{t=0} = 0$ then $B_{mn} = 0$.

Also using the condition $u = f(x, y)$ when $t = 0$, we get

$$f(x,y) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_{mn}\sin\frac{n\pi x}{a}\sin\frac{m\pi y}{b}$$

This is a double Fourier series. Multiplying both sides by $\sin\frac{p\pi x}{a}\sin\frac{q\pi y}{b}$ and integrating from $x = 0$ to $x = a$ and $y = 0$ to $y = b$, we get, on the right except one, terms are zero. Hence we obtain

$$\int_0^a \int_0^b f(x,y) \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}\,dy\,dx = \frac{ab}{4} A_{mn} \qquad \ldots(4)$$

which gives the coefficients in the solution and is called the generalised Euler's formula.

Rectangular Membranes

Example 18.20. *Find the displacement $z(x, y, t)$ of the square membrane with sides $x = 0, x = 1$ and $y = 0, y = 1$, if the initial velocity is zero and the initial deflection is $f(x, y) = A \sin \pi x \sin 2\pi y$.*

Solution. Taking $a = b = 1$ and $f(x, y) = A \sin \pi x \sin 2\pi y$, in (3), we get

$$A_{mn} = 4\int_0^1 \int_0^1 A \sin \pi x \sin 2\pi y \sin m\pi x \sin n\pi y\,dx\,dy$$

$$= 4A\int_0^1 \sin \pi x \sin m\pi x\,dx \left[\int_0^1 \sin 2\pi y \sin n\pi y\,dy\right] = 0, \quad \text{for } m \neq 1$$

$$= 4A\left(\frac{1}{2}\right)\int_0^1 \sin 2\pi y \sin n\pi y\,dy, \text{ for } m = 1 \qquad \left[\because \int_0^c \sin mx \sin nx\,dx = \frac{c}{2}\right]$$

Also, $A_{mn} = 2A\int_0^1 \sin 2\pi y \sin n\pi y\,dy = 0$, for $n \neq 2$

$$= 2A\left(\frac{1}{2}\right), \text{ for } n = 2.$$

$\therefore$ $A_{12} = A$. Also from (2), $p_{mn} = \pi\sqrt{(m^2 + n^2)}$ [$\because a = b = 1 = c$]

$\therefore$ $p_{12} = \pi\sqrt{(1^2 + 2^2)} = \sqrt{5}\pi A$

Hence from (1), the required solution is $z(x, y, t) = A \sin \pi x \sin 2\pi y \cos\sqrt{5}\pi At$.

Example 18.21. *Find the solution $z(x, y, t)$ of a rectangular membrane $0 \le x \le a$, $0 \le y \le b$, whose boundaries are kept perfectly fixed and which starts from rest and $z(x, y, 0) = kxy(x - a)(y - b)$.*

Solution. Proceeding as in § 18.9 (2), we have from (1),

$$z = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}(A_{mn} \cos pt + B_{mn} \sin pt), \text{ where } p = \pi\sqrt{\{(m/a)^2 + (n/b)^2\}}$$

Since the membrane starts from rest $\partial z/\partial t = 0$ when $t = 0$.

$\therefore$ $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}(-A_{mn} p \sin pt + pB_{mn} \cos pt) = 0$ when $t = 0$

This gives $B_{mn} = 0$

$\therefore$ $z(x, y, t) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn} \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b} \cos pt \qquad \ldots(5)$

Then $kxy(x - a)(y - b) = z(x, y, 0) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn} \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$

where $A_{mn} = \frac{4}{ab}\int_0^a\int_0^b kxy(x - a)(y - b) \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}\,dy\,dx$

$$= \frac{4k}{ab}\left[\int_0^a (x^2 - ax) \sin\frac{m\pi x}{a}\,dx\right]\left[\int_0^b y(y - b) \sin\frac{n\pi y}{b}\,dy\right]$$

$$= \frac{4k}{ab}\left[\left|(x^2 - ax)\left(\frac{-\cos m\pi x/a}{m\pi/a}\right)\right|_0^a - \int_0^a (2x - a)\left(\frac{-\cos m\pi x/a}{(m\pi/a)^2}\right)dx\right]\left[\left|y(y - b)\left(\frac{-\cos n\pi y/b}{n\pi/b}\right)\right|_0^b - \int_0^b (2y - b)\left(\frac{-\cos n\pi y/b}{(n\pi/b)^2}\right)dy\right]$$

$$= \frac{4k}{ab}\left[\left|(x^2 - ax)\left(\frac{-\cos m\pi x/a}{m\pi/a}\right)\right|_0^a - \left|(2x - a)\left(\frac{-\sin m\pi x/a}{(m\pi/a)^2}\right)\right|_0^a + \int_0^a 2\left(\frac{\sin m\pi x/a}{(m\pi/a)^2}\right)dx\right]$$

$$\times\left[\left|y(y - b)\left(\frac{-\cos n\pi y/b}{n\pi/b}\right)\right|_0^b - \int_0^b (2y - b)\left(\frac{-\sin n\pi y/b}{(n\pi/b)^2}\right)dy\right]$$

$$= \frac{4h}{n^2\pi^2}\frac{2k^2}{c^2n^2\pi^2}\left(1-\cos\frac{n\pi}{2}\right)\left(1-\cos\frac{n\pi}{2}\right)$$

Hence from (5), we get

$$u(x, y, t) = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn}\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}\cos pt$$

where $A_{mn} = \frac{16hk^2}{m^2n^2\pi^4}\left(1-\cos\frac{m\pi}{2}\right)\left(1-\cos\frac{n\pi}{2}\right)$ and $p = \pi c\sqrt{(m/a)^2 + (n/b)^2}$

Circular Membranes*

Example 10.22. A circular membrane of radius c is initially held in the form of the surface $z = f(r)$ and then released from rest. Find the displacement $u(r, t)$ of the membrane at any instant t.

Show that $u(r, t) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r/c)\cos\alpha_n t$ where $A_n = \frac{2}{c^2[J_1(\alpha_n)]^2}\int_0^c r f(r) J_0(\alpha_n r/c)\,dr$

and $\alpha_1, \alpha_2, \alpha_3, \ldots$ are the positive roots of the Bessel function $J_0(\alpha) = 0$.

Solution. The vibrations of a plane circular membrane are governed by 2-dimensional wave equation in polar coordinates r, θ,

$$\frac{\partial^2 u}{\partial t^2} = c^2\left[\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}\right]$$

For a radially symmetric membrane which is one that does not depend on θ, the above equation reduces to

$$\frac{\partial^2 u}{\partial t^2} = c^2\left[\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r}\right] \qquad \ldots(1)$$

For the given membrane fixed along its boundary, the boundary condition is

$$u(1, t) = 0 \quad \text{for all } t > 0 \qquad \ldots(2)$$

For solutions not depending on θ,

initial deflection $u(r, 0) = f(r)$ $\qquad \ldots(3)$

and initial velocity $\left[\frac{\partial u}{\partial t}\right]_{t=0} = 0$ $\qquad \ldots(4)$

which are the initial conditions. We find the solutions $u(r, t) = R(r)T(t)$ $\qquad \ldots(5)$

satisfying the boundary condition (2).

Differentiating and substituting in (1), we get

$$\frac{d^2T/dt^2}{c^2T} = \frac{1}{R}\left(\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr}\right) = -p^2 \text{ (say)}$$

This leads to $\frac{d^2T}{dt^2} + p^2T = 0$ where $p = ck$ $\qquad \ldots(6)$

and $\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr} + p^2R = 0$ $\qquad \ldots(7)$

Now putting $pr = x$ (or $r = x/p$) transforms (7) into $\frac{d^2R}{dx^2} + \frac{1}{x}\frac{dR}{dx} + R = 0$, which is Bessel's equation. Its general solution is $R = c_1J_0(x) + c_2Y_0(x)$ where J_0 and Y_0 are Bessel's functions of the first and second kind of order zero.

Since the deflection of the membrane is always finite, we must have $c_2 = 0$. Thus taking $p = k$, we get

$$R(r) = c_1J_0(kr) = c_1J_0(kr)$$

On the boundary of the circular membrane, we must have $J_0(k) = 0$, which is satisfied for

$$k = \alpha_n, \; n = 1, 2, \ldots$$

*Drums, lakes, bacteria and stars provide examples of circular membranes and are such more quite useful in engineering

Then the solutions of (with are $X(x) = A_n \cos \alpha_n x + B_n \sin \alpha_n x$, $n = 1, 2, \ldots$ and the corresponding solutions of (iv) are $T(t) = A_n \cos p_n t + B_n \sin p_n t$, where $p_n = ck_n = n\pi$.

Hence the general solutions of (i) satisfying (ii) are

$$u(x, t) = (A_n \cos p_n t + B_n \sin p_n t) J_0(k_n x)$$

which are the right functions of the problem and the corresponding right values are k_n.

To find that solution which also satisfies the initial conditions (iii) and (iv), consider the series

$$u(x, t) = \sum_{n=1}^{\infty} (A_n \cos p_n t + B_n \sin p_n t) J_0(k_n x)$$

Putting $t = 0$ and using (iii), we get $u(x, 0) = \sum_{n=1}^{\infty} A_n J_0(k_n x) = f(x)$

Here, the constants A_n must be the coefficients of Fourier-Bessel series [(9.1 page 368)] with $\alpha = 0$, i.e.,

$$A_n = \frac{2}{J_1^2(\alpha_n)} \int_0^1 x f(x) J_0(\alpha_n x)\, dx$$

Using (v), we get $B_n = 0$. Hence the result.

EXERCISE 9.3

1. A right circular cone has a base radius a and height h. The cone is suspended from its vertex and is set into oscillation with a small amplitude. Show that the deflection u satisfies the equation $\frac{\partial^2 u}{\partial t^2} = g\left[\frac{\partial}{\partial x}\left(x \frac{\partial u}{\partial x}\right) + \frac{\partial^2 u}{\partial y^2}\right]$.
2. Find the deflection $y(x, t)$ of the vibrating string of length π when $y(x, 0) = 0$, the initial velocity is zero and the initial deflection is $k(\sin x - \sin^3 x)$.

9.8 TRANSMISSION LINE

Consider a cable l km in length, carrying an electric current with resistance R ohm/km, inductance L henry/km, capacitance C farad/km and leakance G mho/km (Fig. 10.12).

Fig. 10.12

Let the instantaneous voltage and current at any point P distant x km from the sending end O and at time t sec be v volt, i amp and $v + \Delta v$, $i + \Delta i$ respectively. Consider a small length $PQ = \Delta x$ km of the cable.

Now since the voltage drop across the segment is

voltage drop due to resistance + voltage drop due to inductance

$$\therefore \quad -\Delta v = Ri\,\Delta x + L\,\Delta x \cdot \frac{\partial i}{\partial t}$$

and dividing by Δx and taking limits as $\Delta x \to 0$, we get

$$-\frac{\partial v}{\partial x} = Ri + L\frac{\partial i}{\partial t} \qquad \ldots(1)$$

Similarly the current lost between P and Q

= current lost due to capacitance and leakance

$$\therefore \quad -\Delta i = C\frac{\partial v}{\partial t}\Delta x + Gv\Delta x$$ dividing as before, we get

$$-\frac{\partial i}{\partial x} = C\frac{\partial v}{\partial t} + Gv \qquad \ldots(2)$$

Rewriting the above two partial differential equations (1) and (2) as

$$\left(R + L\frac{\partial}{\partial t}\right) i + \frac{\partial v}{\partial x} = 0 \qquad \ldots(3)$$

and $$\frac{\partial V}{\partial t} + \left(C\frac{\partial}{\partial x} + G\right)v = 0 \quad \ldots(30)$$

we shall eliminate v and i in turn.

∴ operating (29) by $\frac{\partial}{\partial x}$ and (30) by $\left(R + L\frac{\partial}{\partial t}\right)$ and subtracting

$$\frac{\partial^2 v}{\partial x^2} - \left(R + L\frac{\partial}{\partial t}\right)\left(C\frac{\partial}{\partial t} + G\right)v = 0$$

or $$\frac{\partial^2 v}{\partial x^2} = LC\frac{\partial^2 v}{\partial t^2} + (LG + RC)\frac{\partial v}{\partial t} + RGv \quad \ldots(31)$$

Again operating (29) by $\left(C\frac{\partial}{\partial t} + G\right)$ and (30) by $\frac{\partial}{\partial x}$ and subtracting

$$\left(C\frac{\partial}{\partial t} + G\right)\left(R + L\frac{\partial}{\partial t}\right)i - \frac{\partial^2 i}{\partial x^2} = 0$$

or $$\frac{\partial^2 i}{\partial x^2} = LC\frac{\partial^2 i}{\partial t^2} + (LG + RC)\frac{\partial i}{\partial t} + RGi \quad \ldots(32)$$

which is (31) with v replaced by i. The equations (31) and (32) are called the *telegraphic equations*.

Cor. 1. If $RG = 0$, the equations (31) and (32) become

$$\frac{\partial^2 v}{\partial x^2} = LC\frac{\partial^2 v}{\partial t^2} \quad \ldots(33) \qquad \frac{\partial^2 i}{\partial x^2} = LC\frac{\partial^2 i}{\partial t^2} \quad \ldots(34)$$

which are known as the *telegraph equations*.

Assuming $V = uw$, $\frac{\partial V}{\partial t} = u\frac{\partial w}{\partial t} + w\frac{\partial u}{\partial t}$, we observe that it is similar to the heat equation (1) p. 611.

Cor. 2. If $RG = 0$ i.e. $R = 0$ or $G = 0$, the equations (31) and (32) become

$$\frac{\partial^2 v}{\partial x^2} = kC\frac{\partial^2 v}{\partial t^2} \quad \ldots(35) \qquad \frac{\partial^2 i}{\partial x^2} = kC\frac{\partial^2 i}{\partial t^2} \quad \ldots(36)$$

which are called the *Kelvin equations*.

Assuming (1) as $\frac{\partial^2 v}{\partial x^2} = k^2\frac{\partial^2 v}{\partial t^2}$ where $k^2 = \frac{1}{LC}$.

The general solution is $v(x, t) = F(x + kt) + G(x - kt)$. [See (4) p. 614]

Similarly from (35), $i(x, t) = F(x + kt) - G(x - kt)$.

Thus the voltage $v(x, t)$ in the conductor is the sum of any pair of arbitrary functions, one of which is obtained by the superposition of a progressive wave and a reversed wave travelling with equal velocity $\frac{1}{\sqrt{LC}}$. This is the case of overhead telegraph wires at high frequencies.

Cor. 3. If $R = G = 0$, *i.e.*, in the case of a submarine cable, then (31) gives

$$\frac{\partial^2 v}{\partial x^2} = LC\frac{\partial^2 v}{\partial t^2} + (LG + RC)\frac{\partial v}{\partial t} + RGv$$

$$\therefore \quad v(x, t) = A\cos(\alpha x)e^{-\beta^2 t} + B\sin(\alpha x)e^{-\beta^2 t} \quad \ldots(37)$$

Now $i(x, t) = -\frac{1}{R}\frac{\partial v}{\partial x} = -\frac{1}{R}[-A\alpha\sin(\alpha x) + B\alpha\cos(\alpha x)]e^{-\beta^2 t}$

$$\therefore \quad i(x, t) = \frac{\alpha}{R}[A\sin(\alpha x) - B\cos(\alpha x)]e^{-\beta^2 t} \quad \ldots(38)$$

If $v(0, t) = v_0$ and $i(0, t) = i_0$ then $v_0 = A$ and $i_0 = -\frac{\alpha}{R}B$.

Hence writing $\frac{\alpha}{R} = \gamma$ and $v(0, t) = v_0$, $i(0, t) = i_0$ in (37) and (38) we get

$$v(x) = v_0\cos(\alpha x) - Ri_0\sin(\alpha x) \quad \ldots(39)$$

and $$i(x) = i_0\cos(\alpha x) - \frac{v_0}{R}\sin(\alpha x) \quad \ldots(40)$$

One-Dimensional Wave Equation We have to find a solution of the one-dimensional wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \qquad \ldots(1)$$

which satisfies the conditions

$$y(0, t) = 0, \quad y(l, t) = 0 \qquad \ldots(2)$$

$$y(x, 0) = f(x) \qquad \ldots(3)$$

and

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = g(x) \qquad \ldots(4)$$

where $f(x)$ and $g(x)$ are given functions.

Solution. Let $y = X(x)\,T(t)$ be a solution of (1), where X is a function of x alone and T is a function of t alone. Then

$$\frac{\partial^2 y}{\partial t^2} = X\frac{d^2 T}{dt^2}, \quad \frac{\partial^2 y}{\partial x^2} = \frac{d^2 X}{dx^2} T$$

Substituting in (1), we get

$$X\frac{d^2 T}{dt^2} = c^2 \frac{d^2 X}{dx^2} T$$

or

$$\frac{1}{c^2}\frac{1}{T}\frac{d^2 T}{dt^2} = \frac{1}{X}\frac{d^2 X}{dx^2}$$

Since x and t are independent variables, this equation can hold good only if each side is equal to a constant, say k. Then

$$\frac{1}{c^2}\frac{1}{T}\frac{d^2 T}{dt^2} = k \quad \text{and} \quad \frac{1}{X}\frac{d^2 X}{dx^2} = k$$

$$\frac{d^2 T}{dt^2} - kc^2 T = 0 \qquad \ldots(5)$$

$$\frac{d^2 X}{dx^2} - kX = 0 \qquad \ldots(6)$$

Case (i) When $k = 0$, the solutions of (5) and (6) are

$$T = c_1 t + c_2 \quad \text{and} \quad X = c_3 x + c_4$$

$$\therefore \quad y = (c_1 t + c_2)(c_3 x + c_4)$$

Since the ends are suddenly grounded, we have the boundary conditions

$$y(0, t) = 0 \Rightarrow (c_1 t + c_2)\,c_4 = 0 \Rightarrow c_4 = 0$$

$$y(l, t) = 0 \Rightarrow (c_1 t + c_2)\,c_3 l = 0 \Rightarrow c_3 = 0$$

$$\therefore \quad y = 0$$

which is a trivial solution.

Case (ii) When $k > 0$, let $k = p^2$. Then the solutions of (5) and (6) are

$$T = c_1 e^{cpt} + c_2 e^{-cpt} \quad \text{and} \quad X = c_3 e^{px} + c_4 e^{-px}$$

$$\therefore \quad y = (c_1 e^{cpt} + c_2 e^{-cpt})(c_3 e^{px} + c_4 e^{-px})$$

Since the ends are suddenly grounded, we have the boundary conditions

$$y(0, t) = 0 \Rightarrow (c_1 e^{cpt} + c_2 e^{-cpt})(c_3 + c_4) = 0$$

$$\Rightarrow c_3 + c_4 = 0 \quad \text{or} \quad c_4 = -c_3$$

$$y(l, t) = 0 \Rightarrow (c_1 e^{cpt} + c_2 e^{-cpt})(c_3 e^{pl} + c_4 e^{-pl}) = 0$$

$$\Rightarrow c_3 e^{pl} + c_4 e^{-pl} = 0 \quad \text{or} \quad c_3 e^{pl} - c_3 e^{-pl} = 0$$

$$\Rightarrow c_3 = 0 \quad \text{or} \quad e^{2pl} = 1$$

$$\Rightarrow c_3 = 0 \quad \text{or} \quad 2pl = 0$$

$$\Rightarrow c_3 = 0 \quad \text{or} \quad p = 0$$

$$\therefore \quad y = 0$$

which is a trivial solution.

Case (iii) When $k < 0$, let $k = -p^2$. Then the solutions of (5) and (6) are

$$T = c_1 \cos cpt + c_2 \sin cpt \quad \text{and} \quad X = c_3 \cos px + c_4 \sin px$$

$$\therefore \quad y = (c_1 \cos cpt + c_2 \sin cpt)(c_3 \cos px + c_4 \sin px)$$

Since the ends are suddenly grounded, we have the boundary conditions

$$y(0, t) = 0 \Rightarrow (c_1 \cos cpt + c_2 \sin cpt)\,c_3 = 0 \Rightarrow c_3 = 0$$

$$y(l, t) = 0 \Rightarrow (c_1 \cos cpt + c_2 \sin cpt)\,c_4 \sin pl = 0$$

$$\Rightarrow c_4 \sin pl = 0 \quad (\because c_3 = 0)$$

$$\Rightarrow \sin pl = 0 \quad (\because c_4 \neq 0)$$

$$\Rightarrow pl = n\pi \Rightarrow p = \frac{n\pi}{l}$$

$$\therefore \quad y = \sin\frac{n\pi x}{l}\left(c_1 \cos\frac{n\pi ct}{l} + c_2 \sin\frac{n\pi ct}{l}\right)$$

Using the initial condition (3), we get

$$c_1 = f(x) \quad (\because c_4 \neq 0)$$

$$\therefore \quad y = \sin\frac{n\pi x}{l}\left(c_1 \cos\frac{n\pi ct}{l} + c_2 \sin\frac{n\pi ct}{l}\right)$$

Using the initial condition (4), we get $c_2 = 0$

$$\therefore \quad y = c_1 \sin\frac{n\pi x}{l}\cos\frac{n\pi ct}{l}$$

Thus the most general solution of (1) is

$$y = \sum_{n=1}^{\infty} c_n \sin\frac{n\pi x}{l}\cos\frac{n\pi ct}{l}$$

Finally by the initial condition (3), we have

$$c_1 \sin\frac{\pi x}{l} + c_2 \sin\frac{2\pi x}{l} + \cdots = \sum c_n \sin\frac{n\pi x}{l}$$

$$\therefore \quad c_1 = c_1, \quad c_2 = c_2, \quad \ldots$$

while all other c's are zero.

Hence

$$y = c_1 \sin\frac{\pi x}{l}\cos\frac{\pi ct}{l} + c_2 \sin\frac{2\pi x}{l}\cos\frac{2\pi ct}{l} + \cdots$$

which is the required solution.

Example 10.2(A). A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If each point is given the velocity $v_0 \sin^3 x$, find the displacement $y(x, t)$.

Solution. Since k and G are negligible, we use the Bernoulli equation

$$\frac{d^2 y}{dx^2} = kG\frac{d^2 y}{dt^2} \qquad \ldots(1)$$

Since the ends are suddenly grounded, we have the boundary conditions

$$y(0, t) = 0, \quad y(l, t) = 0 \qquad \ldots(2)$$

Also the initial conditions are $y(x, 0) = y_0$

and

$$y(x, 0) = y_0 \sin^3 \pi x/l = y_0 \sin\pi x/l \qquad \ldots(3)$$

$$\text{and} \quad \left(\frac{\partial y}{\partial t}\right)_{t=0} = v_0 \sin^3 \pi x/l \qquad \ldots(4)$$

Let $y = X(x)\,T(t)$ be the solution of (1).

Then (1) gives

$$X''T = kGXT''$$

$$\frac{X''}{X} = kG\frac{T''}{T} = -p^2 \text{ (say)}$$

$$\therefore \quad X'' + p^2 X = 0 \quad \text{and} \quad T'' + (p^2/kG)T = 0$$

Solving these equations, we get

$$y = c_1 \cos px + c_2 \sin px\left[c_3 \cos\frac{p}{\sqrt{kG}}t + c_4 \sin\frac{p}{\sqrt{kG}}t\right]$$

Using the boundary conditions (2), we get

$$c_1 = 0 \quad \text{and} \quad p = n\pi/l$$

$$\therefore \quad y = \sin\frac{n\pi x}{l}\left[c_3 \cos\frac{n\pi}{l\sqrt{kG}}t + c_4 \sin\frac{n\pi}{l\sqrt{kG}}t\right]$$

Using the initial condition (3), we get $c_3 = y_0$

$$\therefore \quad y = \sin\frac{n\pi x}{l}\left[y_0 \cos\frac{n\pi}{l\sqrt{kG}}t + c_4 \sin\frac{n\pi}{l\sqrt{kG}}t\right]$$

Thus the most general solution of (1) is

$$y = \sum_{n=1}^{\infty} \sin\frac{n\pi x}{l}\left[y_0 \cos\frac{n\pi}{l\sqrt{kG}}t + c_4 \sin\frac{n\pi}{l\sqrt{kG}}t\right]$$

Finally by the initial condition (4), we have

$$c_4 \sin\frac{\pi x}{l} + c_4 \sin\frac{2\pi x}{l} + \cdots = \sum v_0 \sin^3\frac{\pi x}{l}$$

$$\therefore \quad c_4 = v_0 \quad \text{and} \quad c_4 = 0$$

while all other c's are zero.

Hence

$$y = v_0 \sin\frac{\pi x}{l}\sin\frac{\pi}{l\sqrt{kG}}t + v_0 \sin\frac{3\pi x}{l}\sin\frac{3\pi}{l\sqrt{kG}}t$$

which is the required solution.

Example 10.2(B). A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If each point is given the velocity $v_0 \sin^3 x$, find the displacement $y(x, t)$.

Solution. Since k and G are negligible, we use the Bernoulli equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

Let $u = X(x)T(t)$ be the corresponding solution. Then

$$X''T = a^2 XT'' \quad \text{or} \quad \frac{X''}{X} = a^2 \frac{T''}{T} = -k^2 \quad (\text{say})$$

$$X'' + k^2 X = 0 \quad \text{and} \quad T'' + k^2 a^2 T = 0$$

Solving these equations, we get

$$X = c_1 \sin kx + c_2 \cos kx, \quad T = c_3 e^{ikat} + c_4 e^{-ikat}$$

$$u = (c_1 \sin kx + c_2 \cos kx)(c_3 e^{ikat} + c_4 e^{-ikat}) \quad \ldots(1)$$

When $x = 0$, $y = 0$ for $x = l$, $y = 0$ and $t = 0$, $y = f(x)$.

$$0 = c_2 c_3 + c_2 c_4$$

$$0 = (c_1 \sin kl + c_2 \cos kl)(c_3 e^{ikat} + c_4 e^{-ikat})$$

Putting $c_2 = 0$ and $c_1 \neq 0$ in (1), we have

$$u = c_1 \sin kx \left(A \cos kat + B \sin kat\right) \quad \ldots(2)$$

The most general solution of this problem is

$$u = \sum_{n=1}^{\infty} \sin \frac{n\pi x}{l} \left(A_n \cos \frac{n\pi at}{l} + B_n \sin \frac{n\pi at}{l}\right)$$

Putting $t = 0$, $u = f(x)$, we have

$$f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi x}{l}$$

Multiplying both sides by $\sin \frac{m\pi x}{l}$ and integrating from 0 to l, we get

$$\int_0^l f(x) \sin \frac{m\pi x}{l} dx = \sum_{n=1}^{\infty} A_n \int_0^l \sin \frac{n\pi x}{l} \sin \frac{m\pi x}{l} dx$$

$$= \frac{l}{2} A_m \quad \text{when } m = n$$

$$= 0 \quad \text{when } m \neq n$$

$$\therefore A_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

$$= \frac{2}{l} \left[-\frac{l}{n\pi} f(x) \cos \frac{n\pi x}{l} \right]_0^l + \frac{2}{n\pi} \int_0^l f'(x) \cos \frac{n\pi x}{l} dx$$

$$= \frac{2}{n\pi} \left[f'(x) \frac{l}{n\pi} \sin \frac{n\pi x}{l} \right]_0^l - \frac{2}{n\pi} \frac{l}{n\pi} \int_0^l f''(x) \sin \frac{n\pi x}{l} dx = \frac{2}{n\pi} \frac{l}{n\pi} \left[f'(l) \cos n\pi - f'(0) \right] - \frac{2l^2}{n^2\pi^2} I_2$$

Hence

$$u = \frac{2f(l)}{\pi} + \frac{2l^2}{\pi^2} \sum_{n=1}^{\infty} \frac{\cos n\pi}{n^2} \sin \frac{n\pi x}{l} \cos \frac{n\pi at}{l} \quad \ldots(3)$$

Let $f(x) = \frac{2lx}{\pi^2}$

$$u = \frac{2l}{\pi^2} \left[\frac{\pi x}{2} - \frac{\pi^2}{4} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x}{l} \cos \frac{n\pi at}{l} \right]$$

Let us also suppose that $x = kt$ is a solution of

$$u = -\frac{k}{\pi a} - \frac{2kl}{\pi^2 a} \sum_{n=1}^{\infty} \frac{1}{n^2} \sin \frac{n\pi x}{l} \sin \frac{n\pi at}{l}$$

When $t = 1$ sec., $u = -\frac{k}{\pi a} \left[-\frac{\pi}{2} + \frac{2}{\pi} \left(\frac{1}{1^2} \cos \frac{\pi x}{l} + \frac{1}{3^2} \cos \frac{3\pi x}{l} + \ldots \right) \right]$ $\ldots(4)$

Hence

$$\frac{d^2}{dx^2} \left(\frac{\cos x^2}{\sin x + \sin^{-1} x} \right) = \text{[illegible]}$$

$$= e^{x^2+1} \cdot \cos^{-1} x + \text{[illegible]}$$

Where $\ldots$

Hence from (c), we have

$$i = -\frac{E}{\Omega}\Omega - 0.0538\Omega + 0.0538\Omega^2 - 0.0538\Omega^3 + \ldots)$$

$$= -\frac{E}{\Omega}(\Omega - 1.05\Omega + 0.05\Omega^2 - 0.05\Omega^3 + \ldots)$$

$$= i_0(0.05\Omega) = 5.5\% \text{ of } i_0.$$

Example 11.15. A transmission line 1000 kilometres long is initially under steady state conditions with potential 1200 volts at the sending end ($x = 0$) and 1000 volts at the receiving end ($x = 1000$). The terminal end of the line is suddenly grounded, but the potential at the sending end is kept at 1200 volts. Assuming the leakance and inductance to be negligible, find the potential $v(x, t)$. *(Anna, 2000)*

Solution. The equation of the telegraph line is

$$\frac{\partial^2 v}{\partial x^2} = RC\frac{\partial v}{\partial t} \text{ or } \frac{\partial v}{\partial t} = \frac{1}{RC}\frac{\partial^2 v}{\partial x^2} \quad \ldots(1)$$

v_0 = initial steady voltage satisfying $\frac{\partial^2 v}{\partial x^2} = 0$ or $v = ax + b$ or $v(x, 0) = 1200 - 0.2x$ $\quad \ldots(2)$

v_0' = steady voltage after grounding the terminal end when steady conditions are obtained = $1200 - 1.2x$

$\therefore \quad v(x, t) = v_0' + v_1(x, t)$ where $v_1(x, t)$ is the transient part

$$= 1200 - 1.2x + \sum_{n=1}^{\infty} b_n e^{-n^2\pi^2 t/(RC)^2} \sin\frac{n\pi x}{l} \quad \text{[By (vii), p. 514]} \quad \ldots(3)$$

where $l = 1000$ kilometres.

Putting $t = 0$, we have from (2) and (3)

$$1200 - 0.2x = v(x, 0) = 1200 - 1.2x + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$$

$$\text{i.e.,} \quad 1.2x = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l} \text{ where } b_n = \frac{2}{l}\int_0^l 1.2x \sin\frac{n\pi x}{l}\,dx = \frac{2400}{n\pi}(-1)^{n+1}$$

Hence $$v(x, t) = 1200 - 1.2x + \frac{2400}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n} e^{-n^2\pi^2 t/(1000)^2} \sin\frac{n\pi x}{1000}.$$

EXERCISE 11.5

1. Find the current i and voltage v in a line of length l, if resistance effects are absent and the ends are suitably grounded, given that $v(x, 0) = v_0 \sin(\pi x/l)$, $0 < x < l$, $i(x, 0) = 0$.
   Also find i and v if $L = 0 = G$ (negligible).
2. Show that a transmission line with negligible resistance and leakage propagates waves of current and potential with a velocity equal to $1/\sqrt{LC}$, where L is the coefficient of inductance and C is the capacitance.
3. A steady voltage distribution of 20 volts at the sending end and 12 volts at the receiving end is maintained for a telephone wire of length l. At time $t = 0$, the sending end is grounded. Find the voltage and current if per unit length resistance and leakance are negligible.
4. Obtain the solution of the wave equation

$$\frac{\partial^2 v}{\partial t^2} = c^2\frac{\partial^2 v}{\partial x^2}$$

appropriate to the case where a period T is set up in the line by applying at the end $x = 0$ of the line

9.11 LAPLACE'S EQUATION IN THREE DIMENSIONS

We have seen that the three dimensional heat flow equation in steady state reduces to

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \qquad ...(1)$$

which is the Laplace's equation in three dimensions.

Laplace's equation also arises in the study of gravitational potential at (x, y, z) of a particle of mass m situated at (ξ, η, ζ) given by

$$\frac{Gm}{r}, \text{ where } r = \sqrt{(x-\xi)^2 + (y-\eta)^2 + (z-\zeta)^2}$$

This function is called the potential of the gravitational field and satisfies the Laplace's equation.

If a mass density $\rho(\xi, \eta, \zeta)$ is distributed over a given region R, then the gravitational potential u at an external point (x, y, z) is given by

$$u(x, y, z) = G \iiint_R \frac{\rho}{r}\, d\xi\, d\eta\, d\zeta \qquad ...(2)$$

Since $\nabla^2(1/r) = 0$ and ρ is independent of x, y and z, we get

$$\nabla^2 u = \iiint_R \rho\, \nabla^2 \left(\frac{1}{r}\right) d\xi\, d\eta\, d\zeta = 0.$$

This shows that the gravitational potential defined by (2) also obeys Laplace's equation.

Thus Laplace's equation (1) is one of the most important equations arising in connection with numerous problems of physics and engineering. The theory of its solutions is called the potential theory and its solutions are called the harmonic functions.

In most of the problems leading to Laplace's equation, it is required to solve the equation subject to certain boundary conditions. A proper choice of coordinate system makes the solution of the problem simpler. Now we proceed to take up the solutions of (1) in its other forms.

9.12 SOLUTIONS OF THREE DIMENSIONAL LAPLACE'S EQUATION

(1) **Cartesian form of $\nabla^2 u = 0$ is** $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ ...(1)

Let $u = X(x)Y(y)Z(z)$ ...(2)

be a solution of (1). Substituting (2) in (1) and dividing by XYZ, we obtain

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} = 0 \qquad ...(3)$$

which is of the form $F_1(x) + F_2(y) + F_3(z) = 0$.

As x, y, z are independent, this will hold good only if F_1, F_2, F_3 are constants. Assuming these constants to be $l^2, l^2, -(l^2 + l^2)$ respectively, (3) leads to the equations

$$\frac{d^2X}{dx^2} - l^2X = 0, \quad \frac{d^2Y}{dy^2} - l^2Y = 0, \quad \frac{d^2Z}{dz^2} + (l^2 + l^2)Z = 0$$

Their solutions are $X = c_1 e^{lx} + c_2 e^{-lx}$, $Y = c_3 e^{ly} + c_4 e^{-ly}$

$$Z = c_5 \cos\sqrt{l^2 + l^2}\, z + c_6 \sin\sqrt{l^2 + l^2}\, z$$

Thus a possible solution of (1) is

$$u = (c_1 e^{lx} + c_2 e^{-lx})(c_3 e^{ly} + c_4 e^{-ly})(c_5 \cos\sqrt{l^2 + l^2}\, z + c_6 \sin\sqrt{l^2 + l^2}\, z).$$

Since the three constants could have been taken as $-l^2, -l^2$ and $l^2 + l^2$, an alternative solution of (1) will be

$$u = (c_1 \cos lx + c_2 \sin lx)(c_3 \cos ly + c_4 \sin ly)(c_5 e^{\sqrt{l^2 + l^2}\, z} + c_6 e^{-\sqrt{l^2 + l^2}\, z}).$$

(2) **Cylindrical form of $\nabla^2 u = 0$ is** $\dfrac{\partial^2 u}{\partial r^2} + \dfrac{1}{r}\dfrac{\partial u}{\partial r} + \dfrac{1}{r^2}\dfrac{\partial^2 u}{\partial \theta^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ ...(4)

[illegible]

[illegible]

[illegible]

Example 10(a). *Find the potential in the interior of a sphere of unit radius when the potential on the surface is $f(\theta) = \cos^2\theta$.*

Solution. Take the origin at the centre of the given sphere S. Since the potential is independent of ϕ and θ, so also is the potential at any point. Therefore, the Laplace's equation in spherical co-ordinates reduces to

$$\frac{\partial^2 v}{\partial r^2} + \frac{2}{r}\frac{\partial v}{\partial r} + \frac{1}{r^2}\frac{\partial^2 v}{\partial \theta^2} + \frac{\cot\theta}{r^2}\frac{\partial v}{\partial \theta} = 0 \qquad \ldots(1)$$

Putting $v(r, \theta) = R(r) f(\theta)$ in (1) and proceeding as in § 10.12 (1), we obtain the equations

$$\frac{d^2 f}{d\theta^2} + \cot\theta \frac{df}{d\theta} + n(n+1) f = 0 \qquad \ldots(2)$$

and

$$\frac{1}{R}\left[r^2 \frac{d^2 R}{dr^2} + 2r\frac{dR}{dr}\right] = n(n+1) \qquad \ldots(3)$$

Putting $\cos\theta = x$, (2) takes the form

$$(1 - x^2)\frac{d^2 f}{dx^2} - 2x\frac{df}{dx} + n(n+1) f = 0$$

which is Legendre's equation. Its solutions are

$$f(\theta) = P_n(\cos\theta) \text{ for } n = 0, 1, 2, \ldots$$

The solutions of (3) are $R_n(r) = r^n$, $R'_n(r) = 1/r^{n+1}$.

Hence the equation (1) has the following two sets of solutions :

$$v_n(r, \theta) = r^n P_n(\cos\theta) \text{ for } R_n \text{ and } S_n(r, \theta) = \frac{1}{r^{n+1}} P_n(\cos\theta) \text{ for } R'_{n+1}, \text{ where } n = 0, 1, 2, \ldots$$

For points inside S, we have the general regular solution as $v(r, \theta) = \sum_{n=0}^{\infty} c_n r^n P_n(\cos\theta)$ ...(4)

On the boundary of S, $v(1, \theta) = f(\theta) \quad \therefore \quad f(\theta) = \sum_{n=0}^{\infty} c_n P_n(\cos\theta)$

which is Fourier-Legendre expansion of $f(\theta)$. Hence by (6) p. 500, we have

$$c_n = \left(n + \frac{1}{2}\right)\int_{-1}^{1} f(\theta) P_n(x)\, dx \text{ where } x = \cos\theta$$

$$= \left(n + \frac{1}{2}\right)\int_{-1}^{1} x^2 P_n(x)\, dx \qquad [\because f(\theta) = \cos^2\theta]$$

$$= \left(n + \frac{1}{2}\right)\int_{-1}^{1}\left[\frac{2}{3}P_2(x) + \frac{1}{3}P_0(x)\right]P_n(x)\, dx \qquad \left[\because P_2(x) = \tfrac{1}{2}(3x^2 - 1)\right]$$

Using the orthogonality of Legendre polynomials, we get :

$c_n = 0$, except for $n = 0, 2$. Hence

$$c_0 = \frac{1}{2}\cdot\frac{1}{3}\int_{-1}^{1} P_0^2(x)\, dx = \frac{1}{3}, \; c_2 = \frac{5}{2}\cdot\frac{2}{3}\int_{-1}^{1} P_2^2(x)\, dx = \frac{2}{3}.$$

Substituting in (4), we get $v(r, \theta) = \frac{1}{3} + \frac{2}{3}r^2 P_2(\cos\theta)$ or $v(r, \theta) = \frac{1}{3} + r^2\left(\cos^2\theta - \frac{1}{3}\right)$.

EXERCISE 10(B)

1. Show that a solution of Laplace's equation in spherical coordinates which remains finite at $r = 0$ can be expressed in the form

$$v = \sum_{n=0}^{\infty} A_n r^n P_n(\cos\theta) + B_n \text{ and } v = \sum_{n=0}^{\infty} A_n r^n P_n(\cos\theta) + \frac{B_n}{r^{n+1}}$$

1. The potential on the surface of a unit sphere is $f(\theta) = \cos 2\theta$. Show that the potential at all points of space is given by

$$u(r, \theta) = r^2 (3\cos^2\theta - 1)/2 \quad \text{for } r < 1,$$

and

$$u(r, \theta) = 2r^{-1}(3\cos^2\theta - 1)/2 \quad \text{for } r > 1.$$

2. Show that in spherical polar coordinates (r, θ, ϕ), Laplace's equation possesses solutions of the form

$$(Ar^n + Br^{-n-1})(\mu, \sin m\phi, \cos m\phi),$$

where $\mu = \cos\theta$, and A, B, m, n are constants and $P_n(\mu)$ is Legendre's polynomial of degree n.

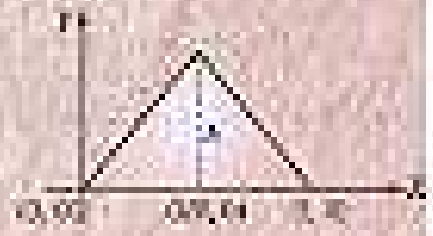
$$(1-\mu^2)\frac{d^2P_n}{d\mu^2} - 2\mu\frac{dP_n}{d\mu} + \left[n(n+1) - \frac{m^2}{1-\mu^2}\right]P_n = 0.$$

OBJECTIVE TYPE OF QUESTIONS

PROBLEMS

Fill up the blanks in each of the following questions :

1. The radio equivalent for the potential and current are _____.
2. The partial differential equation representing the variable heat flow in three dimensions is _____.
3. Temperature gradient is defined as _____.
4. The differential equation $f_{xx} - 2f_{xy} + f_{yy} = 0$ is classified as _____.
5. The partial differential equation of the transverse vibration of a string is _____.
6. The steady state temperature of a rod of length l whose ends are kept at $0°$ and $4°$ is _____.
7. The equation $u_{xx} = u_{tt}$ is classified as _____.
8. The two dimensional steady state heat flow equation in polar coordinates is _____.
9. The mathematical function of the initial form of the string given by the following graph is _____.



10. When a vibrating string is released from rest, the initial velocity u_t for initial conditions are _____.
11. In one dimensional heat flow, the temperature along the normal to the region is _____.
12. If a square plate has its faces and the edges of the sides, the edges $x = 0$ and $x = a$ are kept at zero temperature and the fourth edge is kept at temperature u_0 then the boundary conditions for the problem are _____.
13. If the ends $x = 0$ and $x = l$ are insulated in one dimensional heat flow problem, then the boundary conditions are _____.
14. D'Alembert's solution of the wave equation is _____.
15. The partial differential equation of 2-dimensional heat flow is _____.
16. A rod 10 cm long is insulated at the two faces $x = 0$ and $x = 10$ cm and kept at $50°$ and $100°$ respectively. The steady state temperature distribution in the rod is _____. (Madras, 2006)
17. The three possible solutions of Laplace equation in polar coordinates are _____.
18. Solution of $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$, given $u(0, t) = u(l, t) = 0$, is _____.
19. Solution of $\frac{\partial^2 u}{\partial x^2} + k^2 u = \frac{\partial^2 u}{\partial t^2}$, given $u(0, t) = u(l, t) = 0$, is _____.
20. In the equation $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$, c^2 represents _____.
21. The telegraph equations for potential and current are _____.
22. The general solutions of semi-infinite one dimensional heat flow equation when both ends of the bar are kept at zero temperature is of the form _____.
23. The three possible solutions of Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ are _____.

CHAPTER 19

Complex Numbers and Functions

Definitions. The number $r = +\sqrt{(x^2 + y^2)}$ is called the **modulus** of $z = x + jy$ and is written as $\text{mod}\,z$ or $|z|$ or r.

The angle θ is called the **amplitude** or **argument** of $z = x + jy$ and is written as $\arg z = \theta$ or $\arg(x + jy)$.

Evidently the angle has an infinite number of values. The value of θ which lies between $-\pi$ and $+\pi$ is called the **principal value** of the amplitude. Unless otherwise specified, we shall refer only to the principal value.

Note, $\cos\theta + j\sin\theta$ is usually written as $\angle\theta$ (pronounced as 'cis theta').

If $\bar{z}$ is the conjugate of $z = x + jy$ then $\bar{z} = x - jy$, then

(i) $\overline{(z_1 + z_2)} = \bar{z}_1 + \bar{z}_2$, $\overline{(z_1 - z_2)} = \bar{z}_1 - \bar{z}_2$ (ii) $|z| = \sqrt{(x^2 + y^2)} = |\bar{z}|$

(iii) $z\bar{z} = |z|^2$ (iv) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

(v) $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$ (vi) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$, where $z_2 \neq 0$.

Example 10.1. Express $1 - \cos\alpha + i\sin\alpha$ in modulus-amplitude form.

Solution. Put $1 - \cos\alpha = r\cos\theta$ and $\sin\alpha = r\sin\theta$

$\therefore$ $r = \sqrt{[1 - \cos\alpha)^2 + \sin^2\alpha]} = \sqrt{[2 - 2\cos\alpha]} = \sqrt{[4\sin^2\alpha/2]}$

i.e., $r = 2\sin\alpha/2$

and $\tan\theta = \dfrac{\sin\alpha}{1 - \cos\alpha} = \dfrac{2\sin\alpha/2\cos\alpha/2}{2\sin^2\alpha/2} = \cot\alpha/2$

$= \tan\left[\dfrac{\pi}{2} - \dfrac{\alpha}{2}\right]$ $\therefore \theta = \dfrac{\pi - \alpha}{2}$.

Thus $1 - \cos\alpha + i\sin\alpha = 2\sin\dfrac{\alpha}{2}\left[\cos\dfrac{\pi - \alpha}{2} + i\sin\dfrac{\pi - \alpha}{2}\right]$.

Example 10.2. Find the complex number z if $\arg(z + 1) = \pi/4$ and $\arg(z - 1) = 2\pi/3$.

(Marathwada, 2000)

Solution. Let $z = x + iy$ so that $z + 1 = (x + 1) + iy$ and $z - 1 = (x - 1) + iy$

Now $\arg(z + 1) = \pi/4$, $\therefore \tan^{-1}\left(\dfrac{y}{x + 1}\right) = 45°$

i.e., $\dfrac{y}{x + 1} = \tan 45° = 1$, $\sqrt{3}y = x + 1$ ... (i)

Also since $\arg(z - 1) = 2\pi/3$, $\therefore \tan^{-1}\left(\dfrac{y}{x - 1}\right) = 120°$

i.e., $\dfrac{y}{x - 1} = \tan 120° = -\sqrt{3}$, or $y = -\sqrt{3}x + \sqrt{3}$ or $\sqrt{3}y = -3x + 3$ ... (ii)

Subtracting (ii) from (i), we get $4x - 2 = 0$ i.e., $x = 1/2$

From (i), $\sqrt{3}y = 1/2 + 1$, i.e., $y = \sqrt{3}/2$

Hence $z = \dfrac{1}{2} + \dfrac{\sqrt{3}}{2}i$.

Example 10.3. Find the real values of x, y so that $(x^3 - 3x + 2) + i(y^3 - 12y + 16) = 0$. Hence find the complex number $z = x + iy$ which represents the complex conjugate numbers.

Solution. If $z = -2 + 2\sqrt{3}i$, then $\bar{z} = x^3 + y + 4i$

so that $x = (x^3 + y) - 4i$

$\therefore$ $-2 + 2\sqrt{3}i = x^3 + y - 4i$

Equating real and imaginary parts from both sides, we get

$$-1 = x^2 + y, \ 2xy = -4$$

Eliminating x, $2y + 2y^3 = -4$

or $y^3 + 2y + 2 = 0$ i.e., $(y + 1)(y^2 - y + 2) = 0$

When $y = -1$, $x^2 = -2 - 1 = -3$ or $x = \pm\sqrt{3}i$ which is not feasible

When $y = -4$, $x^2 = 1$ or $x = \pm 1$

Hence $x = 1$, $y = -1$ or $x = -1$, $y = -1$.

10.12 [1] GEOMETRIC REPRESENTATION OF IMAGINARY NUMBERS

Let all the real numbers be represented along $X'OX$, the positive real numbers being along OX and negative ones along OX'. Let OA be equal to one unit of measurement (Fig. 10.1).

Take a point B on OX such that $OB = x$ (OA).

Then B on OX represents the positive real number x and $(-x)$ or $-x = -x$ is represented by a point B' on OX' distant OB from O.

From this we infer that the multiplication of the real number x by -1 is equivalent to the rotation of OB through two right angles in the position OB'.

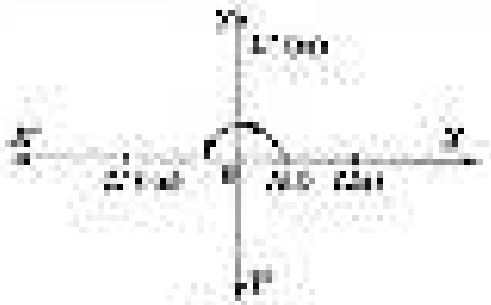


Fig. 10.1

Thus it recently follows that the multiplication of a real number by i is equivalent to the rotation of OB through one right angle in the position OB'.

Hence, if YOY' be a line perpendicular to the real axis $X'OX$, then all imaginary numbers are represented by points on YOY', called the imaginary axis, the positive one along OY and negative one along OY'.

Obs. Geometric interpretation of i^2. From the above, it is clear that i is an operator which when applied to any real number makes it imaginary and causes its rotation through a right angle in the counter clock sense.

(2) Geometric representation of complex numbers†

Consider two lines $X'OX$, YOY' at right angles to each other.

Let all the real numbers be represented by points on the line $X'OX$ (called the *real axis*), positive real numbers being along OX and negative ones along OX'. Let the point B on OX represent the real number x (Fig. 10.2).

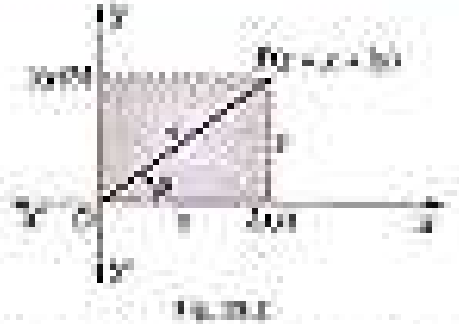


Fig. 10.2

Since the multiplication of a real number by i is equivalent to the rotation of its direction through a right angle. Therefore, let all the imaginary numbers be represented by points on the line YOY' (called the *imaginary axis*), the positive ones along OY and negative ones along OY'. Let the point A on OY represent the imaginary number iy.

Complete the rectangle $OBPA$. Then the point P whose cartesian coordinates are (x, y) uniquely represents the complex number $z = x + iy$ on the complex plane. The diagram in which this representation is carried out is called the **Argand's diagram.**

If (r, θ) be the polar coordinates of P, then r is the modulus of z and θ is its amplitude.

Obs. Since a complex number has a magnitude and direction, therefore, it can be represented like a vector. Hence for our study from now on, we shall refer to the complex numbers as vectors.

Of the points whose coordinates are $(2, 2)$ or $(2, 45°)$ is more convenient than $(2, 90°)$ or $(2, 0°)$ or $(2, 180°)$.

Example 10.1. *The vertices of a regular hexagon lie at the origin and two points represented by $z = 1$ and $z = 2i$ on the Argand diagram. Determine the other vertices.*

---

† The first mathematical idea to propose a geometric representation of imaginary numbers was Book of Beauty (1758–31).

† The generally accepted idea of complex numbers came to be mathematically through the invention of Jean Robert Argand, Paris 1806.

Solution. Let $\vec{OA} = \sqrt{3}i + 1$ so that

$OA = 2$ and $\angle AOM = \tan^{-1}\frac{1}{\sqrt{3}} = 30°$ (Fig. 33.8)

Being a regular hexagon, $OB = OC = 2$

$\angle BOM = 30° + 60° = 90°$

and $\angle COM = 30° + 120° = 150°$

$\therefore \quad \vec{OB} = 2\cos 90° + i\sin 90° = 2i$

$\vec{OC} = 2\cos 150° + i\sin 150° = -\sqrt{3} + i$

Since $\vec{AO}, \vec{BB}, \vec{OF}$ are identical in OA

$\therefore \quad \vec{OD} = -\vec{OA} = -\sqrt{3} - i$

$\vec{OE} = -\vec{OB} = -2i$ and $\vec{OF} = -\vec{OC} = \sqrt{3} - i$.

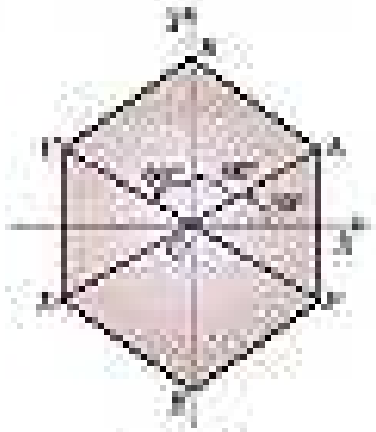


Fig. 33.8

(i) GEOMETRIC REPRESENTATION OF $z_1 + z_2$

Let P_1, P_2 represent the complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ (Fig. 33.9)

Complete the parallelogram OP_1PP_2. Draw P_1L, P_2M and $PN \perp s$ to OX. Also draw $P_1K \perp PN$.

Since $OV = OL + LV = OL + OM = x_1 + x_2$ [$\because LV = P_1K = OM$]

and $NP = NK + KP = LP_1 + MP_2 = y_1 + y_2$

The coordinates of P are $(x_1 + x_2, y_1 + y_2)$ and it represents the complex number

$$z = x_1 + x_2 + i(y_1 + y_2) = (x_1 + iy_1) + (x_2 + iy_2) = z_1 + z_2$$

Thus the point P which is the extremity of the diagonal of the parallelogram having OP_1 and OP_2 as adjacent sides, represents the sum of the complex numbers $P_1(z_1)$ and $P_2(z_2)$ such that

$$|z_1 + z_2| = OP \text{ and } \arg(z_1 + z_2) = \angle XOP.$$

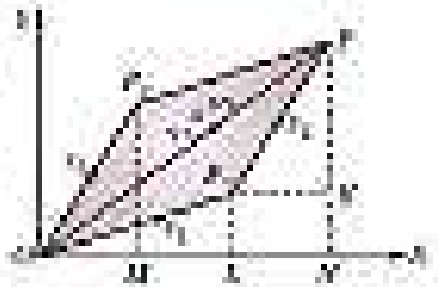


Fig. 33.9

Obs. Vectorially we have $\vec{OP} = \vec{OP_1} + \vec{P_1P} = \vec{OP_1} + \vec{OP_2}$.

(ii) Geometric representation of $z_1 - z_2$

Let P_1, P_2 represent the complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ (Fig. 33.10). Then the subtraction of z_2 from z_1 may be taken as addition of z_1 to $-z_2$.

Produce P_2O backwards to R such that $OR = OP_2$. Then the coordinates of R are evidently $(-x_2, -y_2)$ and so it corresponds to the complex number $-x_2 - iy_2 = -z_2$.

Complete the parallelogram $ORQP_1$, then the sum of z_1 and $(-z_2)$ is represented by OQ i.e. $z_1 - z_2 = \vec{OQ} = \vec{P_2P_1}$.

Hence the complex number $z_1 - z_2$ is represented by the vector $\vec{P_2P_1}$.

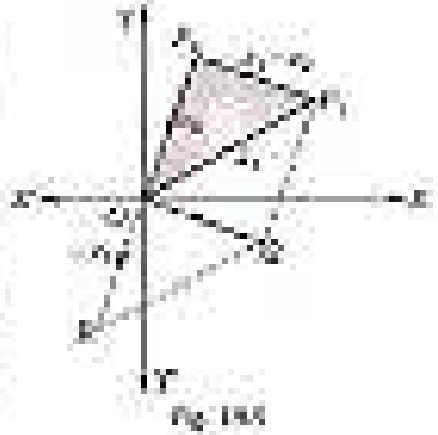


Fig. 33.10

Obs. By means of the relation $\vec{AB} = \vec{OB} - \vec{OA}$, any vector $\vec{AB}$ may be reduced to the origin.

Example 33.3. Find the locus of $P(z)$ when

(i) $|z - c| = k$,

(ii) $\arg(z - a) = \alpha$, where k and α are constants. (Kerala, May 2005)

Solution. Let a, z be represented by A and P in the complex plane, O being the origin (Fig. 33.11).

Then $z - a = \vec{OP} - \vec{OA} = \vec{AP}$

(i) $|z - a| = k$ means that $AP = k$.

Then the locus of $P(z)$ is a circle whose centre is $A(c)$ and radius k.

(ii) $\arg(z-c)=\alpha$, $\arg(\overrightarrow{AP})=\alpha$, means that AP always makes a constant angle with the X-axis.

Thus the locus of $P(z)$ is a straight line through $A(c)$ making an $\angle\alpha$ with OX.

Fig. 15.6

Example 15.16 Interpret the region in the z-plane represented by
(i) $1<|z-2i|\le 2$ (ii) $\mathrm{Re}(z)>3$ (iii) $\pi/6<\arg(z)<\pi/2$.

Solution. (i) $|z-2i|=1$ is a circle with centre $(0,2)$ and radius 1 and $|z-2i|=2$ is a circle with the same centre and radius 2.

Hence $1<|z-2i|\le 2$ represents the region outside the circle $|z-2i|=1$ and inside including circumference of the circle $|z-2i|=2$ [Fig. 15.7]

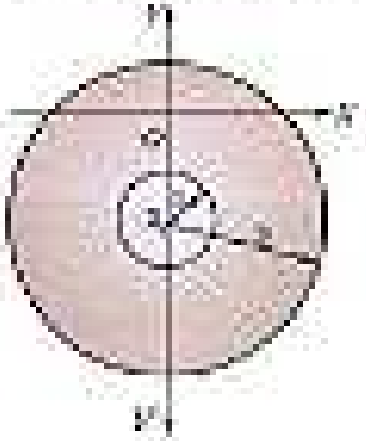


Fig. 15.7

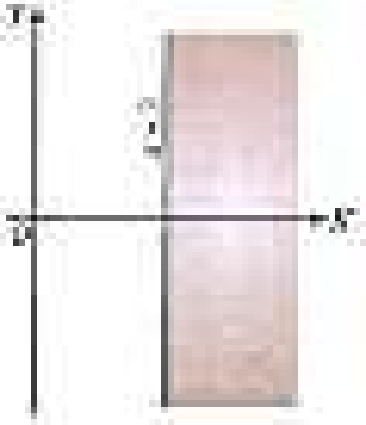


Fig. 15.8

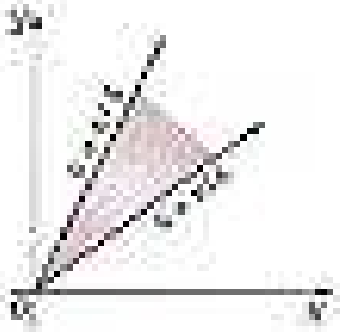


Fig. 15.9

(ii) $\mathrm{Re}(z)>3$ defines all points whose real part is greater than 3. Hence it represents the region of the complex plane to the right of the line $x=3$ [Fig. 15.8].

(iii) If $z=r(\cos\theta+i\sin\theta)$, then $\arg(z)=\theta$.

$\therefore \pi/6<\arg(z)<\pi/2$ defines the region bounded by rays including the lines $\theta=\pi/6$ and $\theta=\pi/2$ [Fig. 15.9].

Example 15.17 If z_1, z_2 be any two complex numbers, prove that
(i) $|z_1+z_2|\le|z_1|+|z_2|$, the modulus of the sum of two complex numbers is less than or equal to the sum of their moduli.
(ii) $|z_1-z_2|\ge|z_1|-|z_2|$, the modulus of the difference of two complex numbers is greater than or equal to the modulus of the difference of their moduli.

Solution. Let P_1, P_2 represent the complex numbers z_1, z_2 [Fig. 15.10]. Complete the parallelogram OP_1PP_2, so that

$$z_1=\overrightarrow{OP_1},\ z_2=\overrightarrow{OP_2}=\overrightarrow{P_1P},$$

and $$z_1+z_2=\overrightarrow{OP}.$$

Now from $\triangle OP_1P$, $OP\le OP_1+P_1P$, the sign of equality corresponding to the case when O, P_1, P are collinear.

Fig. 15.10

Hence $$|z_1+z_2|\le|z_1|+|z_2| \quad \ldots(1)$$

Again $$|z_1|=|(z_1-z_2)+z_2|\le|z_1-z_2|+|z_2|$$

Thus $$|z_1-z_2|\ge|z_1|-|z_2| \quad \ldots(2)$$

Also $|z_1+z_2+z_3|\le|z_1|+|z_2|+|z_3|$

In general $|z_1+z_2+\ldots+z_n|\le|z_1|+|z_2|+\ldots+|z_n|$

Example 15.6. If $|z_1 + z_2| = |z_1 - z_2|$, prove that the difference of amplitudes of z_1 and z_2 is $\pi/2$.
(Bolton 2007)

Solution. Let $z_1 + z_2 = r(\cos\theta + i\sin\theta)$ and $z_1 - z_2 = r(\cos\phi + i\sin\phi)$

Then
$$2z_1 = r(\cos\theta + \cos\phi) + i(\sin\theta + \sin\phi)$$
$$= r\left[2\cos\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2} + i\,2\sin\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2}\right]$$
or
$$z_1 = r\cos\frac{\theta-\phi}{2}\left(\cos\frac{\theta+\phi}{2} + i\sin\frac{\theta+\phi}{2}\right) \quad \text{i.e., } \arg(z_1) = \frac{\theta+\phi}{2} \qquad \ldots(1)$$

Also
$$2z_2 = r(\cos\theta - \cos\phi) + i(\sin\theta - \sin\phi)$$
$$= 2ir\sin\frac{\theta-\phi}{2}\left(-i\sin\frac{\theta+\phi}{2} + i\cos\frac{\theta+\phi}{2}\right)$$
or
$$z_2 = r\sin\frac{\theta-\phi}{2}\left[\cos\left(\frac{\pi}{2} + \frac{\theta+\phi}{2}\right) + i\sin\left(\frac{\pi}{2} + \frac{\theta+\phi}{2}\right)\right]$$
i.e.,
$$\arg(z_2) = \frac{\pi}{2} + \frac{\theta+\phi}{2} \qquad \ldots(2)$$

Hence (2) − (1), gives $\arg(z_2) - \arg(z_1) = \frac{\pi}{2}$.

Example 15.7. Show that the equation of the ellipse having foci at z_1, z_2 and semi-major axis $2a$ is $|z - z_1| + |z - z_2| = 2a$.
Also find its eccentricity.

Solution. Let $P(z)$ be any point on the given ellipse (Fig. 15.11) having foci at $S(z_1)$ and $S'(z_2)$ so that $SP = |z - z_1|$ and $S'P = |z - z_2|$.

We know that $SP + S'P = AA' = 2a$,

i.e.,
$$|z - z_2| + |z - z_1| = 2a$$
which is the desired equation of the ellipse.

Also we know that $SS' = 2ae$, e being the eccentricity.

i.e.
$$\left|\overrightarrow{OS'} - \overrightarrow{OS}\right| = 2ae \quad \text{or} \quad |z_2 - z_1| = 2ae$$
i.e.
$$|z_1 - z_2| = 2ae \text{ where } e = \text{eccentricity}.$$

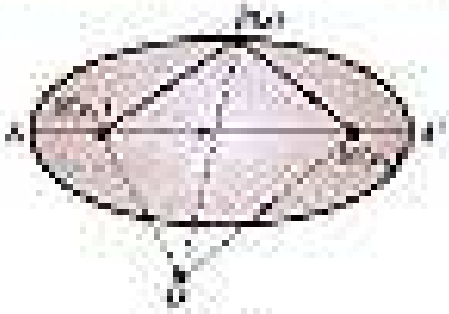


Fig. 15.11

(3) Geometric Representation of $z_1 z_2$. Let P_1, P_2 represent the complex numbers

$$z_1 = x_1 + iy_1 = r_1(\cos\theta_1 + i\sin\theta_1)$$
and
$$z_2 = x_2 + iy_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

Now on OP_1, $OA = 1$ along OX (Fig. 15.12). Construct ΔOP_2P on OP_2 directly similar to ΔOAP_1.

so that
$$\Delta OP_2P \sim \Delta OAP_1 \text{ i.e., } \frac{OP}{OP_2} = \frac{OP_1}{OA}, \quad OP = r_1r_2$$
and
$$\angle AOP = \angle AOP_2 + \angle P_2OP = \angle AOP_2 + \angle AOP_1 = \theta_2 + \theta_1$$

$\therefore$ P represents the number
$$r_1r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)).$$

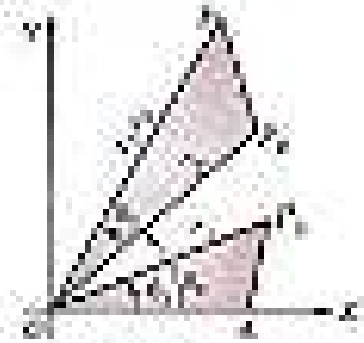


Fig. 15.12

Hence the product of two complex numbers z_1, z_2 is represented by the point P, such that $OP = |z_1z_2| = |z_1| \cdot |z_2|$.

Also $\arg(z_1z_2) = \arg(z_1) + \arg(z_2)$.

Cor. The effect of multiplication of any complex number by $\cos\theta + i\sin\theta$ is to rotate the given line through an angle θ (in the anticlockwise sense) about the origin.

(4) Geometric representation of z_1/z_2.

Let P_1, P_2 represent the complex numbers

$$z_1 = x_1 + iy_1 = r_1(\cos\theta_1 + i\sin\theta_1)$$

and $$z_2 = x_2 + iy_2 = r_2(\cos\theta_2 + i\sin\theta_2)$$

Now since $OA = 1$, construct triangle OAP on OA directly similar to the triangle OP_2P_1 (Fig. 15.13), so that

$$\frac{OP}{OA} = \frac{OP_1}{OP_2} \quad \text{i.e.,} \quad OP = \frac{OP_1}{OP_2} = \frac{r_1}{r_2}$$

and $$\angle POP = \angle P_1OP_2 = \angle AOP_1 - \angle AOP_2 = \theta_1 - \theta_2$$

$\therefore$ P represents the number

$$\frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]$$

Hence the complex number z_1/z_2 is represented by the point P, such that

(i) $|z_1/z_2| = |z_1|/|z_2|$

(ii) $\arg(z_1/z_2) = \arg(z_1) - \arg(z_2)$.

Note. If $P_1(z_1)$, $P_2(z_2)$ and $A(1)$ be any three points, then

$$\arg\left(\frac{z_1 - 1}{z_2 - 1}\right) = \angle P_1AP_2$$

Join O, the origin, to P_1, P_2 and P. Then from the figure 15.14, we have

$$\overrightarrow{OP_1} = z_1 - 0 = z_1 \quad \text{and} \quad \overrightarrow{OP_2} = z_2 - 0 = z_2$$

$$\therefore \quad \arg\left(\frac{z_1 - 0}{z_2 - 0}\right) = \arg\left(\frac{\overrightarrow{OP_1}}{\overrightarrow{OP_2}}\right)$$

$$= \arg(OP_1) - \arg(OP_2) = \theta - \alpha = \angle P_1OP_2$$

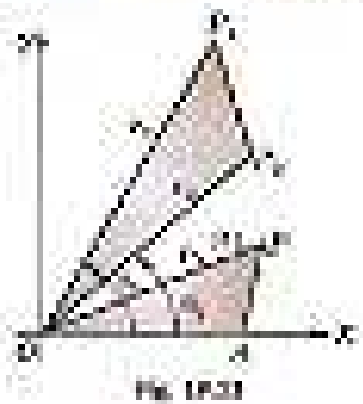


Fig. 15.13

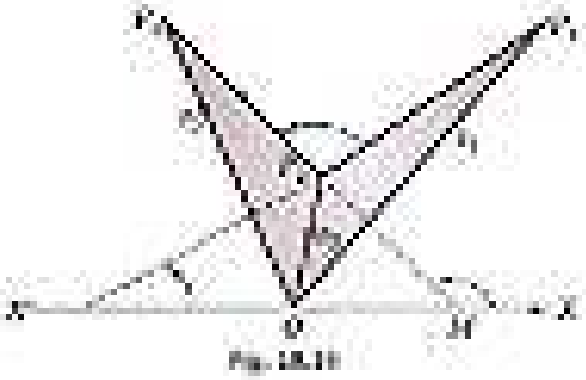


Fig. 15.14

Example 15.18. *Find the locus of the point z if $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$.*

Solution. Let $A(1)$ and $B(-1)$ be any two fixed points on the complex plane and let $P(z)$ be any variable point (Fig. 15.15).

Given $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$

$\therefore$ $\arg(z-1) - \arg(z+1) = \frac{\pi}{4}$

$\therefore$ $\angle PAX - \angle PBX = \frac{\pi}{4}$

$\therefore$ $\angle APB = \frac{\pi}{4}$

Hence the locus of P is the arc APB of the circle passing through A and B such that $\angle APB = \frac{\pi}{4}$.

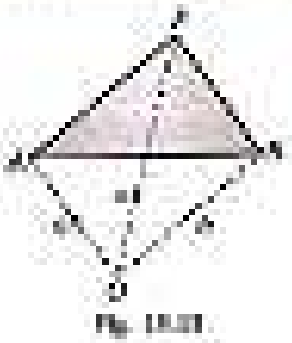


Fig. 15.15

Also, the locus of P is the major arc AQB of the circle passing through A and B such that $\angle AQB = \frac{\pi}{4}$.

Example 15.19. *Find the locus of z if $\arg\left(\frac{z-a}{z-b}\right) = \alpha$, where a and b are two fixed points.*

Solution. Let $A(a)$ and $B(b)$ be any two fixed points on the complex plane and let $P(z)$ be any variable point (Fig. 15.16).

Given $\arg\left(\frac{z-a}{z-b}\right) = \alpha$

$\therefore$ $\arg(z-a) - \arg(z-b) = \alpha$

$\therefore$ $\angle PAX - \angle PBX = \alpha$

$\therefore$ $\angle APB = \alpha$

Hence the locus of P is the arc APB of the circle passing through A and B such that $\angle APB = \alpha$.

Also, the locus of P is the major arc AQB of the circle passing through A and B such that $\angle AQB = \pi - \alpha$, which shows that the locus of P is the arc APB of the same circle.

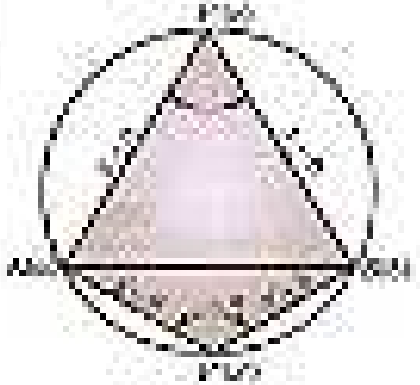


Fig. 15.16

Hint. For all three values of n from $n = 0$ to $n = 2$, the roots represent a family of intersecting circles of radius $\frac{1}{2}$ with centres on the real axis.

Example 19.11. If z_1, z_2 are two complex numbers, show that
$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\{|z_1|^2 + |z_2|^2\}$$

Solution. Let $z_1 = r_1 \cos\theta_1 + ir_1 \sin\theta_1$ and $z_2 = r_2 \cos\theta_2 + ir_2 \sin\theta_2$ so that

$$\begin{aligned} |z_1 + z_2|^2 &= |r_1 \cos\theta_1 + r_2 \cos\theta_2 + i(r_1 \sin\theta_1 + r_2 \sin\theta_2)|^2 \\ &= r_1^2 + r_2^2 + 2r_1 r_2 \cos(\theta_1 - \theta_2) \quad (1) \end{aligned}$$

and
$$\begin{aligned} |z_1 - z_2|^2 &= |r_1 \cos\theta_1 - r_2 \cos\theta_2 + i(r_1 \sin\theta_1 - r_2 \sin\theta_2)|^2 \\ &= r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2) \quad (2) \end{aligned}$$

$$\therefore \quad |z_1 + z_2|^2 + |z_1 - z_2|^2 = 2r_1^2 + 2r_2^2 = 2\{|z_1|^2 + |z_2|^2\}$$

Example 19.12. If z_1, z_2, z_3 be the vertices of an isosceles triangle right angled at z_3, prove that
$$z_1^2 + z_2^2 + 2z_3^2 = 2z_3(z_1 + z_2)$$

Solution. Let A be z_1, B be z_2. C be z_3 be the vertices of $\triangle ABC$ such that $AC = BC$ and $\angle ACB = 90°$. (Fig. 19.37)

Then $|z_1 - z_3| = |z_2 - z_3| = r$ (say)

If $\arg(z_1 - z_3) = \theta$ then $\arg(z_2 - z_3) = 90° + \theta$

$$\therefore \quad z_1 - z_3 = r(\cos\theta + i\sin\theta)$$

and
$$z_2 - z_3 = r\left[\cos\left(\frac{\pi}{2} + \theta\right) + i\sin\left(\frac{\pi}{2} + \theta\right)\right] = r(-\sin\theta + i\cos\theta)$$

$$\text{i.e.,} \quad z_2 - z_3 = ir(\cos\theta + i\sin\theta) = i(z_1 - z_3)$$

or
$$(z_2 - z_3)^2 = -(z_1 - z_3)^2 \text{ or } z_1^2 + z_2^2 - 2z_3^2 = 2z_3(z_1 + z_2)$$

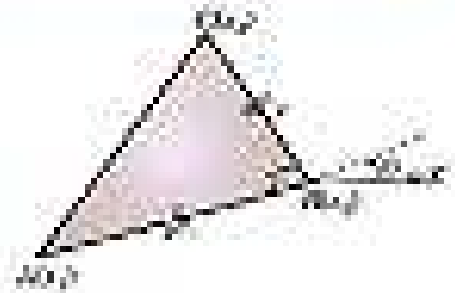


Fig. 19.37

Example 19.13. If z_1, z_2, z_3 be the vertices of an equilateral triangle, prove that
$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$$

Solution. Since $\triangle ABC$ is equilateral, therefore, AC when rotated through $60°$ coincides with BA (Fig. 19.38). But to turn the direction of a complex number through an $\angle\theta$, we multiply it by $\cos\theta + i\sin\theta$.

$$\therefore \quad \overrightarrow{AC}(\cos 60° + i\sin 60°) = \overrightarrow{BA}$$

$$\text{i.e.,} \quad (z_3 - z_1)\left(\frac{1 + i\sqrt{3}}{2}\right) = z_1 - z_2$$

or
$$(1 + i\sqrt{3})(z_3 - z_1) = 2z_1 - z_2 - z_1$$

Squaring
$$-3(z_3 - z_1)^2 = (z_1 - z_2)^2$$

or
$$z_1^2 + z_2^2 + z_3^2 - z_1z_2 - z_2z_3 - z_3z_1 = 0$$

whence follows the required condition.

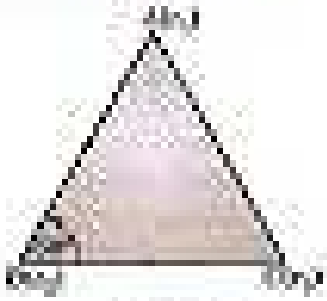


Fig. 19.38

EXERCISE 19.1

1. Express the following in the real $a + ib$ form:

(i) $\frac{1 + \sin\theta + i\cos\theta}{1 + \sin\theta - i\cos\theta}$ (ii) $\frac{1}{(2 + 3i)^2} - \frac{1}{(2 - 3i)^2}$ (VTU, 2011-12)

2. If $\frac{1}{x + iy} = \frac{1}{x + iy} + \frac{1}{x + iy}$ where x_1, y_1, x_2, y_2 are real constants, express x in terms of x_1 and x_2.

3. If a and b are real, solve the equation $\frac{|x|}{x+1} = \frac{2x+ia}{2x+y}$.

4. If $z = \frac{1}{1-i}$, prove that $z^n + z^{n-1}(z+1) + z^{n-2}(z+1)^2 + \dots + (z+1)^n = 2^n$. (Mumbai, 2006)

5. Find the locus of z if $z + z^{-1} = 1$ (Kanpur, 2000)

6. If $\sin\alpha + \sin\beta + \sin\gamma = 0 = \cos\alpha + \cos\beta + \cos\gamma$, show that $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha+\beta+\gamma)$ and $\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha+\beta+\gamma)$.

7. If $|z_1| = |z_2|$ and $\arg z_1 + \arg z_2 = 0$, show that z_1 and z_2 are conjugate complex numbers.

8. A rectangle is constructed in the complex plane and its sides parallel to the axes and its centre is situated at the origin. If one of the vertices of the rectangle is $1 + i\sqrt{3}$, find the complex numbers representing the other three vertices of the rectangle. Find also the area of the rectangle.

9. An equilateral triangle is constructed in the complex plane. Its one vertex is at the point $1 + i\sqrt{3}$. Find the complex numbers representing the other two vertices, if the origin lies in the triangle.

10. The centre of a regular hexagon is at the origin and one vertex is given by $1 + i$. In the Argand diagram, find the remaining vertices.

11. What domain of the z-plane is represented by
   (i) $|z| \le 3$, $|z + 4| > 4$ (ii) $|z| \le 2$
   (iii) $0 < \arg z < \frac{\pi}{4}$ (iv) $|z + 2| > |z - 2|$

12. If $|z^2 - 1| = |z|^2 + 1$, prove that z lies on the imaginary axis. (Mumbai, 2003)

13. What curve is represented by (i) $|z - 1| = |z + 1|$ (ii) $|z - 2| + |z + 2| = 8$ (iii) $|z - 3| + |z + 3| = 4$? Find a cartesian equation in each case.

14. Find the locus of z given by
   (i) $|z| = |z - 2|$ (ii) $|z - 1| = |z - i|$.

15. Find the locus of z if
   (i) $\arg\frac{z+1}{z-1} = \frac{\pi}{2}$ (ii) $\arg\frac{z-1}{z+1} = \frac{\pi}{4}$ (Osmania, 2003)

9.13 DE MOIVRE'S THEOREM*

Statement : *If n is an integer, positive or negative, $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.*
(or) n a fraction, positive or negative, one of the values of $(\cos\theta + i\sin\theta)^n$ is $\cos n\theta + i\sin n\theta$.

Proof. Case I. When n is a positive integer.

By actual multiplication

$$\text{cis}\,\theta_1\,\text{cis}\,\theta_2 = (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) = (\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)$$
$$= \cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2) = \text{cis}(\theta_1 + \theta_2).$$

Similarly $\text{cis}\,\theta_1\,\text{cis}\,\theta_2\,\text{cis}\,\theta_3 = \text{cis}(\theta_1 + \theta_2)\,\text{cis}\,\theta_3 = \text{cis}(\theta_1 + \theta_2 + \theta_3)$

Proceeding in this way,

$$\text{cis}\,\theta_1\,\text{cis}\,\theta_2\,\text{cis}\,\theta_3 \dots \text{cis}\,\theta_n = \text{cis}(\theta_1 + \theta_2 + \theta_3 + \dots + \theta_n)$$

Now putting $\theta_1 = \theta_2 = \theta_3 = \dots = \theta_n = \theta$, we obtain $(\text{cis}\,\theta)^n = \text{cis}\,n\theta$

Case II. When n is a negative integer.

Let $n = -m$, where m is a positive integer.

$$\therefore \quad (\cos\theta + i\sin\theta)^n = (\cos\theta + i\sin\theta)^{-m} = \frac{1}{(\cos\theta + i\sin\theta)^m} = \frac{1}{\cos m\theta + i\sin m\theta} \qquad \text{[By case I]}$$

$$= \frac{\cos m\theta - i\sin m\theta}{(\cos m\theta + i\sin m\theta)(\cos m\theta - i\sin m\theta)}$$

[Multiplying the num. and denom. by $\cos m\theta - i\sin m\theta$]

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*One of the most notable theorems in mathematics, called after the name of the discoverer Abraham De Moivre (1667–1754), a French Mathematician.

$$= \frac{\cos n\theta - i\sin n\theta}{\cos^2 n\theta + \sin^2 n\theta} = \cos n\theta - i\sin n\theta$$

$$= \cos(-n\theta) + i\sin(-n\theta) = \text{cis}(-n\theta) = \text{cis}^n(-\theta) \qquad \because\ -n \text{ is an integer}$$

Case III. When n is a fraction, positive or negative.

Let $n = \frac{p}{q}$, where p is a non-zero integer and q is any integer $\neq 0$ or ± 1.

Now $(\text{cis}\,\theta)^p = \text{cis}\,p\theta$, $(\theta)^q = \text{cis}\,\theta$

$\therefore$ Taking qth root of both sides, $\text{cis}\,\theta$ is one of the q values of $(\text{cis}\,p\theta)^{1/q}$,

i.e., one of the values of $(\text{cis}\,\theta)^{p/q} = \text{cis}\,p\theta$

Raising both sides to power p, $\text{cis}\,n\theta$ is one of the values of $(\text{cis}\,\theta)^p = (\text{cis}\,\theta)^p = \text{cis}\,p\theta$ i.e., one of the values of $\text{cis}\,n\theta = \text{cis}\,n\theta$. (By cases I and II)

Thus the theorem is completely established for all rational values of n.

Cor. 1. $\cos\theta_1 \cos\theta_2 \ldots \cos\theta_n + i\sin\theta_1 \sin\theta_2 \ldots \sin\theta_n$

$= \text{cis}(\theta_1 + \theta_2 + \ldots + \theta_n)$

2. $(\cos\theta - i\sin\theta)^n = \cos n\theta - i\sin n\theta = \cos(-n\theta) + i\sin(-n\theta)$

3. $(\sin\theta + i\cos\theta)^n = \text{cis}\left(\frac{\pi}{2} - n\theta\right) = \cos n\theta + i\sin n\theta$.

Example 10.11. Simplify $\dfrac{(\cos 3\theta + i\sin 3\theta)^4 (\cos\theta - i\sin\theta)^5}{(\cos 2\theta + i\sin 2\theta)^7 (\cos 3\theta + i\sin 3\theta)^{-2}}$.

Solution. We have, $(\cos 3\theta + i\sin 3\theta)^4 = \cos 12\theta + i\sin 12\theta = (\cos\theta + i\sin\theta)^{12}$

$(\cos 4\theta - i\sin 4\theta)^5 = \cos 20\theta - i\sin 20\theta = (\cos\theta + i\sin\theta)^{-20}$

$(\cos 2\theta + i\sin 2\theta)^7 = \cos 14\theta + i\sin 14\theta = (\cos\theta + i\sin\theta)^{14}$

$(\cos 3\theta + i\sin 3\theta)^{-2} = \cos 6\theta - i\sin 6\theta = (\cos\theta + i\sin\theta)^{-6}$

$\therefore$ The given expression $= \dfrac{(\cos\theta + i\sin\theta)^{12}(\cos\theta + i\sin\theta)^{-20}}{(\cos\theta + i\sin\theta)^{14}(\cos\theta + i\sin\theta)^{-6}} = 1.$

Example 10.12. Prove that

$$(1 + \cos\theta + i\sin\theta)^n + (1 + \cos\theta - i\sin\theta)^n = 2^{n+1}\cos^n\left(\frac{\theta}{2}\right)\cos\frac{n\theta}{2}.$$

Solution. Put $1 + \cos\theta = r\cos\alpha$, $\sin\theta = r\sin\alpha$

$\therefore \quad r^2 = (1 + \cos\theta)^2 + \sin^2\theta = 2 + 2\cos\theta = 4\cos^2\frac{\theta}{2}$ i.e., $r = 2\cos\frac{\theta}{2}$

and $\tan\alpha = \dfrac{\sin\theta}{1 + \cos\theta} = \dfrac{2\sin\frac{\theta}{2}\cdot\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \tan\frac{\theta}{2}$ i.e., $\alpha = \frac{\theta}{2}$.

Now L.H.S. $= (r\cos\alpha + ir\sin\alpha)^n + (r\cos\alpha - ir\sin\alpha)^n$

$= r^n(\cos n\alpha + i\sin n\alpha) + r^n(\cos n\alpha - i\sin n\alpha) = r^n(\cos n\alpha + i\sin n\alpha + \cos n\alpha - i\sin n\alpha)$

$= r^n \cdot 2\cos n\alpha$ (Substituting the values of r and α)

$= 2^{n+1}\cos^n\frac{\theta}{2}\cos\frac{n\theta}{2}$.

Example 10.13. If $2\cos\theta = x + \dfrac{1}{x}$, **prove that**

(i) $2\cos n\theta = x^n + \dfrac{1}{x^n}$, (ii) $\dfrac{x^{2n} + 1}{x^{2n} - 1} = \dfrac{\cos n\theta}{\cos n\theta - \cos\theta}$. (Madras, 2000 S)

Solution. Given $x + \frac{1}{x} = 2\cos\theta$ $\therefore \quad x^2 - 2x\cos\theta + 1 = 0$

whence $x = \dfrac{2\cos\theta \pm \sqrt{(4\cos^2\theta - 4)}}{2} = \cos\theta \pm i\sin\theta.$

(i) Taking the + ve sign, $x^n = (\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ (U.P.T.U. 2005)

and $x^{-1} = (\cos\theta + i\sin\theta)^{-1} = \cos n\theta - i\sin n\theta$

Adding $x^n + x^{-n} = 2\cos n\theta$. Similarly with the – ve sign, the same result follows.

(6.27)

$$\begin{aligned}\frac{x^{2n}+1}{x^{n+2}+x^n} &= \frac{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}x^{2n}+1}{x^{n+2}+x^n}\\ &= \frac{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}x^{2n}+1}{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}x^{n+2}+\cos\frac{\pi}{2}x^n+\cos\frac{\pi}{2}}\\ &= \frac{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+1}{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}\\ &= \frac{1+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}\\ &= \frac{1+\cos^2\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}\\ &= \frac{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}\end{aligned}$$

Proof. [illegible]

Solution. Let

$$a = \cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}$$

Then

$$a = b+c+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2} \tag{6.28}$$

(a)

$$\frac{1}{a}+\frac{1}{b}+\frac{1}{c} = \cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}$$

$$= \sum\frac{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}}\cdot\frac{1}{\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}} = \sum\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}$$

$$= \cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2} \tag{Observe}$$

(a)

$$\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+\cos\frac{\pi}{2}+

PROBLEMS 4.8

1. Prove that $\dfrac{\tan(\theta+\alpha)+\tan(\theta-\alpha)}{\tan(\theta+\alpha)-\tan(\theta-\alpha)}=\dfrac{\sin 2\theta+\sin 2\alpha}{\sin 2\theta-\sin 2\alpha}$

   (a) $\left[\dfrac{\cos\theta+i\sin\theta}{\cos\theta-i\sin\theta}\right]^n=\cos n\theta+i\sin n\theta$ (b) $\left[\dfrac{\cos\theta+i\sin\theta}{\sin\theta+i\cos\theta}\right]^n=\cos n\theta+i\sin n\theta$

2. If $p=\cos\theta+i\sin\theta$ and $q=\cos\phi+i\sin\phi$, show that

   (a) $\dfrac{p-q}{p+q}=i\tan\dfrac{\theta-\phi}{2}$ (Dundee, 2000) (b) $\dfrac{(p+q)(pq-1)}{(p-q)(pq+1)}=\dfrac{\sin\theta+\sin\phi}{\sin\theta-\sin\phi}$ (Dundee, 2000)

3. If $a=\cos\theta+i\sin\theta$, $b=\cos\phi+i\sin\phi$ and $c=\cos 2\theta+i\sin 2\theta$, prove that

   (a) $\dfrac{ab}{c}+\dfrac{c}{ab}=2\cos(\theta+\phi-2\theta)$ (b) $\dfrac{a^2b}{c}+\dfrac{bc^2}{a}=2\cos(2\theta+\phi-\theta)$

4. If $z_n=\cos n\theta+i\sin n\theta$, show that $z_1z_2z_3\ldots z_n=1$ (VTU, 2000; Mumbai, 2003)

5. Find the general value of θ which satisfies the equation

   $\sin\theta+\cos\theta+\sin 2\theta+\cos 2\theta+\sin 3\theta+\cos 3\theta=0$

6. Prove that (a) $(x+iy)^{1/n}+(x-iy)^{1/n}=2(x^2+y^2)^{1/2n}\cos\left(\dfrac{1}{n}\tan^{-1}\dfrac{y}{x}\right)$

   (b) $1+2r+2r^2+2r^3+r^4=(1+r)^2(1+r^2)$

7. Simplify (a) $(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta)(\cos\gamma+i\sin\gamma)$ (b) $\dfrac{\cos\alpha-i\sin\alpha}{\cos\beta+i\sin\beta}$

8. Prove that (a) $(1+\cos\theta+i\sin\theta)^n+(1+\cos\theta-i\sin\theta)^n=2^{n+1}\cos^n\dfrac{\theta}{2}\cos\dfrac{n\theta}{2}$ (b) $\left[\dfrac{1+\sin\theta+i\cos\theta}{1+\sin\theta-i\cos\theta}\right]^n=\cos\left(\dfrac{n\pi}{2}-n\theta\right)+i\sin\left(\dfrac{n\pi}{2}-n\theta\right)$ (VTU, 2004)

9. If $\cos\alpha+\cos\beta+\cos\gamma=0$ and $\sin\alpha+\sin\beta+\sin\gamma=0$, show that each of the values of

   (a) $\cos^{2n+1}\alpha+\cos^{2n+1}\beta+\cos^{2n+1}\gamma$ is $3\cos(n\theta+\alpha)$ (VTU, 2003)

   (b) $\dfrac{\cos^n\alpha}{\cos n\alpha}+\dfrac{\cos^n\beta}{\cos n\beta}+\dfrac{\cos^n\gamma}{\cos n\gamma}$ is $3\cos(n\theta+\alpha)$ (Mysore, 2004)

10. If α,β are the roots of $x^2-2x+4=0$, prove that $\alpha^n+\beta^n=2^{n+1}\cos n\dfrac{\pi}{3}$ (VTU, 2002)

11. If α,β are the roots of the equation $x^2\sin^2\theta-x\sin\theta+1=0$, then prove that

    (a) $\alpha^n+\beta^n=2\cos n\theta\csc^n\theta$ (b) $\alpha^n\beta^n=\cos n\theta\csc^n\theta$ (Mumbai, 2003)

12. If $x^2-2x\cos\theta+1=0$, show that $x^{2n}-2x^n\cos n\theta+1=0$.

13. If $x=\cos\alpha+i\sin\alpha$, $y=\cos\beta+i\sin\beta$, $z=\cos\gamma+i\sin\gamma$ and $x+y+z=0$, then prove that

    $x^2+y^2+z^2=0$

14. If $\cos\theta+\cos\phi+\cos\psi=a$ and $\sin\theta+\sin\phi+\sin\psi=b$, prove that

    (a) $\cos 2\theta+\cos 2\phi+\cos 2\psi=0$ (Mumbai, 2000)

    (b) $\cos 3\theta+\cos 3\phi+\cos 3\psi=3\cos(\theta+\phi+\psi)$

    (c) $\cos n\theta+\cos n\phi+\cos n\psi=3\sum\cos n(\alpha+\phi)$

15. If $\cos\alpha+\cos\beta+\cos\gamma=0$ and $\sin\alpha+\sin\beta+\sin\gamma=0$, prove that

    (a) $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=\sin^2\alpha+\sin^2\beta+\sin^2\gamma=\frac{3}{2}$

    (b) $\cos(\alpha+\beta)+2\cos(\gamma+\alpha)\cos(\gamma+\beta)=0$ (Mumbai, 2000; VTU, 2000)

16. If $\sin\alpha+2\sin\beta+3\sin\gamma=0$ and $\cos\alpha+2\cos\beta+3\cos\gamma=0$, prove that (a) $\cos 2\alpha+8\cos 2\beta+27\cos 2\gamma=18\cos(\alpha+\beta)$ and (b) $\cos 3\alpha+8\cos 3\beta+27\cos 3\gamma=36\cos(\alpha+\beta+\gamma)$.

4.25 ROOTS OF A COMPLEX NUMBER

There are q and only q distinct values of $\cos\theta+i\sin\theta$, q being an integer.

Now $\cos\theta=\cos(2n\pi+\theta)$ and $\sin\theta=\sin(2n\pi+\theta)$, where n is any integer.

$\therefore$ $\quad\cos\theta+i\sin\theta=\cos(2n\pi+\theta)$

By De Moivre's theorem one of the values of

$$(\cos\theta + i\sin\theta)^{\frac{1}{q}} = \cos\frac{2n\pi+\theta}{q} + i\sin\frac{2n\pi+\theta}{q} \quad \ldots(1)$$

Giving n the values $0, 1, 2, 3, \ldots, (q-1)$ successively, we get the following q values of $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$:

$$\left.\begin{array}{ll} \cos\frac{\theta}{q} + i\sin\frac{\theta}{q} & \text{(for } n=0\text{)} \\ \cos\frac{2\pi+\theta}{q} + i\sin\frac{2\pi+\theta}{q} & \text{(for } n=1\text{)} \\ \cos\frac{4\pi+\theta}{q} + i\sin\frac{4\pi+\theta}{q} & \text{(for } n=2\text{)} \\ \ldots\ldots\ldots & \ldots\ldots\ldots \\ \cos\frac{2(q-1)\pi+\theta}{q} + i\sin\frac{2(q-1)\pi+\theta}{q} & \text{(for } n=q-1\text{)} \end{array}\right| \quad \ldots(2)$$

Putting $n = q$ in (1), we get a value of $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$ as $\cos\frac{2q\pi+\theta}{q} + i\sin\frac{2q\pi+\theta}{q}$, which is the same as the value at $n = 0$.

Similarly for $n = q+1$, we get a value of $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$ to be $\cos\frac{2\pi+\theta}{q} + i\sin\frac{2\pi+\theta}{q}$, which is the same as the value for $n = 1$ and so on.

Thus, the values of $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$ for $n = q, q+1, q+2$ etc. are the same repetitions of the values obtained in (2).

Moreover, the values given by (2) are clearly distinct from each other, for no two of the angles involved therein are equal or differ by a multiple of 2π.

Hence $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$ has q and only q distinct values given by (2).

Otherwise. $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$ where $\frac{1}{q}$ is a rational fraction in its lowest terms, has also a total only q distinct values, which are obtained by putting $n = 0, 1, 2, \ldots, q-1$ successively in the given form.

Now that $(\cos\theta + i\sin\theta)^{\frac{1}{q}}$ has only q distinct values and not the form $\cos\frac{2n\pi+\theta}{q} + i\sin\frac{2n\pi+\theta}{q}$ for n lesser in value than q.

$\therefore$ *the least positive distinct value of* $\arg(z^{\frac{1}{q}})$ *is* $\frac{\theta}{q}$ *and not* $\frac{2\pi+\theta}{q}$ *or* $\frac{4\pi+\theta}{q}$ *or any other.*

Note. The above discussion can easily be employed for extracting any rational root of a given quantity. We have only to express it in the form $r(\cos\theta + i\sin\theta)$ and proceed as above.

Example 19.18. *Find the cube roots of unity and show that they are represented by vertices of an equilateral triangle.*

Solution. If x be a cube root of unity, then

$$x = (1)^{\frac{1}{3}} = (\cos 0 + i\sin 0)^{\frac{1}{3}} = (\cos 0 + i\sin 0)^{\frac{1}{3}} = (\cos 2n\pi)^{\frac{1}{3}} + i\sin 2n\pi)^{\frac{1}{3}} = \cos\frac{2n\pi}{3} + i\sin\frac{2n\pi}{3}$$

where $n = 0, 1, 2$.

$\therefore$ the three values of x are $\cos 0 + i\sin 0$, $\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$ and $\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}$.

$$\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} = \cos 120^\circ + i\sin 120^\circ = -\frac{1}{2} + i\frac{\sqrt{3}}{2},$$

and

$$\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3} = \cos 240^\circ + i\sin 240^\circ = -\frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

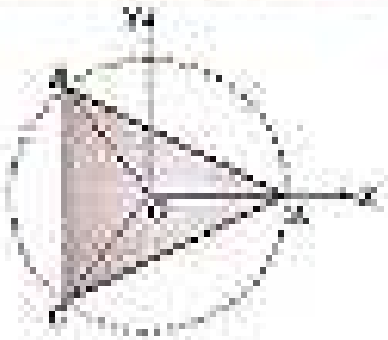

Fig. 19.19

These three cube roots are represented by the points A, B, C on the Argand diagram such that $OA = OB = OC$ and $\angle AOB = 120^\circ$, $\angle BOC = 240^\circ$ (Fig. 19.19).

$\therefore$ these points lie on a circle with centre O and unit radius such that $\angle AOB = \angle BOC = \angle COA = 120^\circ$ i.e., $AB = BC = CA$.

Hence A, B, C form an equilateral triangle.

Example 19.19. *Find all the values of* $\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{\frac{3}{4}}$.

Also show that the continued product of these values is -1. (Rajasthan 2000)

Solution. Put $\frac{1}{2} = r\cos\theta$ and $\frac{\sqrt{3}}{2} = r\sin\theta$ so that $r = 1$ and $\theta = \pi/3$.

$$\therefore \quad \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{\frac{3}{4}} = (\cos\pi/3 + i\sin\pi/3)^{\frac{3}{4}} = (\cos\theta + i\sin\theta)^{\frac{3}{4}}$$

$$= (\cos 2n\pi + \theta)^{\frac{3}{4}} = \cos(2n\pi + \theta)\frac{3}{4}, \text{ where } n = 0, 1, 2, 3.$$

Hence the required values are $\cos\pi/4$, $\cos 3\pi/4$, $\cos 5\pi/4$ and $\cos 7\pi/4$.

$\therefore$ their continued product $= \cos\left(\frac{\pi}{4} + \frac{3\pi}{4} + \frac{5\pi}{4} + \frac{7\pi}{4}\right) = \cos 4\pi = 1$.

Example 19.30. Use De Moivre's theorem to solve the equation $x^5 - x^4 + x^3 - x^2 + x - 1 = 0$. (U.P.T.U., 2004)

Solution. $x^5 - x^4 + x^3 - x^2 + x - 1$ is a G.P. with common ratio $(-x)$, therefore

$$\frac{1 - (-x)^6}{1 - (-x)} = 0, \quad \text{i.e.} \quad 1 + x^6 = 0$$

i.e., $x^6 = -1 = \cos(2n+1)\pi + i\sin(2n+1)\pi$

$\therefore \quad x = [\cos(2n+1)\pi + i\sin(2n+1)\pi]^{1/6} = \cos\frac{(2n+1)\pi}{6} + i\sin\frac{(2n+1)\pi}{6}$, where $n = 0, 1, 2, 3, 4, 5$

Hence the values are $\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}$, $\cos\frac{3\pi}{6} + i\sin\frac{3\pi}{6}$, $\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}$, $\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}$, $\cos\frac{9\pi}{6} + i\sin\frac{9\pi}{6}$, $\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}$

or $\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}, \cos\frac{3\pi}{6} + i\sin\frac{3\pi}{6}, \cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}, -1, \cos\frac{5\pi}{6} - i\sin\frac{5\pi}{6}, \cos\frac{3\pi}{6} - i\sin\frac{3\pi}{6}$

Rejecting the value -1 which corresponds to the factor $x + 1$, the required roots are:

$\cos\frac{\pi}{6} \pm i\sin\frac{\pi}{6}$, $\cos\frac{3\pi}{6} \pm i\sin\frac{3\pi}{6}$, $\cos\frac{5\pi}{6} \pm i\sin\frac{5\pi}{6}$.

Example 19.31. Show that the roots of the equation $(x-1)^n + x^n = 0$, where n is a positive integer are $\frac{1}{2}(1 + i\cot\frac{r\pi}{n})$, where r has the values $1, 2, 3, \ldots, n-1$.

Solution. Given equation is $\left(\frac{x-1}{x}\right)^n = -1$ or $1 - \frac{1}{x} = (-1)^{1/n}$

or $\frac{1}{x} = 1 - (-1)^{1/n} = 1 - \cos\frac{(2r+1)\pi}{n}, r = 0, 1, 2, \ldots (n-1)$ $\quad [\because -1 = \cos(2r+1)\pi]$

or $= \left[1 - \cos\frac{(2r+1)\pi}{n}\right] - i\sin\frac{(2r+1)\pi}{n} = 2\sin^2\frac{r\pi}{n} - 2i\sin\frac{r\pi}{n}\cos\frac{r\pi}{n}$

$\therefore \quad x = \frac{1}{2\sin\frac{r\pi}{n}\left[\sin\frac{r\pi}{n} - i\cos\frac{r\pi}{n}\right]} = \frac{\sin\frac{r\pi}{n} + i\cos\frac{r\pi}{n}}{2\sin\frac{r\pi}{n}}$

$= \frac{1}{2}\left(1 + i\cot\frac{r\pi}{n}\right), r = 1, 2, \ldots (n-1)$ $\quad [\because \cot\theta = \frac{1}{\tan\theta}]$

Hence the roots of the given equation are $\frac{1}{2}(1 + i\cot\frac{r\pi}{n})$ where $r = 1, 2, 3, \ldots (n-1)$.

Example 19.32. Find the n^{th} roots of unity and prove that the sum of the r^{th} powers of these roots is zero or n according as r is or is not a multiple of n, r being any integer, and hence find the sum to n terms of

$1 + 2^r + 3^r + 4^r + \ldots$ to n terms.

(Mumbai, 2003; Kolhapur, 2003)

Solution. We have $(1)^{1/n} = (\cos 2r\pi + i\sin 2r\pi)^{1/n} = \cos\frac{2r\pi}{n} + i\sin\frac{2r\pi}{n}$

Putting $r = 0, 1, 2, 3, 4, 5$, we find that n^{th} roots of unity are $1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5$ where $\alpha = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$.

$\therefore$ sum S of the n^{th} powers of these roots $= 1 + \alpha^r + \alpha^{2r} + \ldots + \alpha^{(n-1)r}$ ...(1)

$= \frac{1 - \alpha^{nr}}{1 - \alpha^r}$, being a G.P. with common ratio α^r

When r is not a multiple of n, $\alpha^{nr} = (\alpha^n)^r = (\cos 2\pi + i\sin 2\pi)^r = 1$.

i.e., $1 - \alpha^{nr} = 0$ and $1 - \alpha^r \neq 0$, as r is not a multiple of n.

Thus $S = 0$.

When r is a multiple of n, $\alpha^r = \alpha^{pn} = (\alpha^n)^p = 1$.

From (1), $\quad S = 1 + (\alpha^r)^1 + (\alpha^r)^2 + \ldots + (\alpha^r)^{n-1} = 1 + 1 + 1 + 1 + \ldots + 1 + 1 = n$.

Example 19.33. Find the equation whose roots are $\cos\theta + i\sin\theta$, $2\cos 2\theta + 2i\sin 2\theta$, $3\cos 3\theta + 3i\sin 3\theta$.

Solution. Let $y = \cos\theta + i\sin\theta$, where $\theta = \pi/7, 2\pi/7, \ldots, 12\pi/7$.

Then $y^5 + (\cos 0 + i \sin 0)^5 + (\cos 5\theta + i \sin 5\theta) = -1$ or $y^5 + 1 = 0$

or $(y+1)(y^4 - y^3 + y^2 - y + 1) = 0$

Leaving the factor $y + 1$ which corresponds to $\theta = \pi$.

We get $y^4 - y^3 + y^2 - y + 1 = 0$ ...(i)

Its roots are $y = \cos\theta$ where $\theta = \pi/5, 3\pi/5, 5\pi/5, 7\pi/5, 9\pi/5, 11\pi/5, 13\pi/5$.

Dividing (i) by y^2, $(y^2 + 1/y^2) - (y + 1/y) + 1 = 0$

or $(y + 1/y)^2 - 2 - (y + 1/y) + 1 = 0$

or $x^2 - x - 2 = 0$ ...(ii)

where $x = y + 1/y = 2\cos\theta$.

Now since $\cos 13\pi/5 = \cos 8\pi/5$, $\cos 11\pi/5 = \cos 3\pi/5$, and $9\pi/5 = \cos 3\pi/5$

Hence the roots of (i) are $2\cos\frac{\pi}{5}, 2\cos\frac{3\pi}{5}, 2\cos\frac{5\pi}{5}$.

EXERCISE 3.1

1. Find all the values of
   (i) $(1)^{1/3}$ (ii) $(-1)^{1/4}$
   (iii) $(-1 + i\sqrt{3})^{1/3}$ (iv) $(1 + i\sqrt{3})^{1/3} + (1 - i\sqrt{3})^{1/3}$.
2. If α is an imaginary root of unity, prove that $(1 + \alpha + \alpha^2) = 0$.
3. Find all the values of $(-1)^{2/5}$.
4. Mark by points on the Argand diagram, all the values of $(1 + i\sqrt{3})^{1/4}$ and verify that they form a parallelogram.
5. Use De Moivre's theorem to solve the following equations :
   (i) $x^5 + 1 = 0$ (ii) $x^5 - x^4 + x^3 + 1 = 0$
   (iii) $x^5 + x^4 - x^3 - 1 = 0$ (S.U. Nov. 2002) (iv) $x^6 - x^4 + x^2 - 1 = 0$.
6. Find the cube roots of (-8) and (-27).
7. Solve the equation $x^7 - 1 = 0$ and find which of its roots satisfy the equation $x^4 + x^2 + 1 = 0$.
8. Prove that the roots of $(x + 1)^n = (x - 1)^n$ are given by $x = i\cot(r\pi/n)$, $r = 1, 2, 3, \ldots$ (Mumbai, 2002)
9. Prove that the nth roots of unity form a geometric progression (Ahmedabad, 2003)
   Also show that the product of these roots is unity and their sum is $(-1)^{n-1}$.
10. Find the equation whose roots are $3 \cos 2\pi/7$, $3 \cos 4\pi/7$, $3 \cos 6\pi/7$.

3.13 (i) TO EXPAND $\sin n\theta$, $\cos n\theta$ AND $\tan n\theta$ IN POWERS OF $\sin\theta$, $\cos\theta$ AND $\tan\theta$ RESPECTIVELY (n BEING A POSITIVE INTEGER)

We know $\cos n\theta + i \sin n\theta = (\cos\theta + i \sin\theta)^n$ (By De Moivre's theorem)

$= \cos^n\theta + {}^nC_1 \cos^{n-1}\theta (i \sin\theta) + {}^nC_2 \cos^{n-2}\theta (i \sin\theta)^2 + {}^nC_3 \cos^{n-3}\theta (i \sin\theta)^3 + \ldots$

(By Binomial theorem)

$= (\cos^n\theta - {}^nC_2 \cos^{n-2}\theta \sin^2\theta + \ldots) + i({}^nC_1 \cos^{n-1}\theta \sin\theta - {}^nC_3 \cos^{n-3}\theta \sin^3\theta + \ldots)$

Equating real and imaginary parts from both sides, we get

$\cos n\theta = \cos^n\theta - {}^nC_2 \cos^{n-2}\theta \sin^2\theta + {}^nC_4 \cos^{n-4}\theta \sin^4\theta - \ldots$ ...(1)

$\sin n\theta = {}^nC_1 \cos^{n-1}\theta \sin\theta - {}^nC_3 \cos^{n-3}\theta \sin^3\theta + {}^nC_5 \cos^{n-5}\theta \sin^5\theta - \ldots$ ...(2)

Replacing every $\sin^2\theta$ by $1 - \cos^2\theta + \ldots$ (1) and every $\cos^2\theta$ by $1 - \sin^2\theta$ in (2), we get the desired expansions of $\cos n\theta$ and $\sin n\theta$.

Dividing (2) by (1),

$$\tan n\theta = \frac{{}^nC_1 \cos^{n-1}\theta \sin\theta - {}^nC_3 \cos^{n-3}\theta \sin^3\theta + {}^nC_5 \cos^{n-5}\theta \sin^5\theta - \ldots}{\cos^n\theta - {}^nC_2 \cos^{n-2}\theta \sin^2\theta - {}^nC_4 \cos^{n-4}\theta \sin^4\theta - \ldots}$$

and dividing numerator and denominator by $\cos^n\theta$, we get

$$\tan n\theta = \frac{{}^nC_1 \tan\theta - {}^nC_3 \tan^3\theta + {}^nC_5 \tan^5\theta - \ldots}{1 - {}^nC_2 \tan^2\theta + {}^nC_4 \tan^4\theta - \ldots}.$$

Example 19.34. Express $\cos 6\theta$ in terms of $\cos\theta$. (Nov/Dec. 2002)

Solution. We know that $\cos n\theta + i\sin n\theta = {}^nC_0\cos^n\theta + {}^nC_1\cos^{n-1}\theta\sin\theta + {}^nC_2\cos^{n-2}\theta\sin^2\theta + \ldots$

Put $n = 6$, then $\cos 6\theta + i\sin 6\theta = \cos^6\theta + {}^6C_1\cos^5\theta\sin\theta + {}^6C_2\cos^4\theta\sin^2\theta + {}^6C_3\cos^3\theta\sin^3\theta + \ldots$

$= \cos^6\theta + 6\cos^5\theta(i\sin\theta) + 15\cos^4\theta(i\sin\theta)^2 + 20\cos^3\theta(i\sin\theta)^3 + 15\cos^2\theta(i\sin\theta)^4 + 6\cos\theta(i\sin\theta)^5 + (i\sin\theta)^6$

$= \cos^6\theta + 6i\cos^5\theta\sin\theta - 15\cos^4\theta\sin^2\theta - 20i\cos^3\theta\sin^3\theta + 15\cos^2\theta\sin^4\theta + 6i\cos\theta\sin^5\theta - \sin^6\theta$

$= (\cos^6\theta - 15\cos^4\theta\sin^2\theta + 15\cos^2\theta\sin^4\theta - \sin^6\theta) + i(6\cos^5\theta\sin\theta - 20\cos^3\theta\sin^3\theta + 6\cos\theta\sin^5\theta)$

$= 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$, $\quad [\because \sin^2\theta = 1 - \cos^2\theta]$

(3) Addition formulae for any number of angles

We have, $\cos(\theta_1 + \theta_2 + \ldots + \theta_n) + i\sin(\theta_1 + \theta_2 + \ldots + \theta_n)$

$= (\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)\ldots(\cos\theta_n + i\sin\theta_n)$

Now $\cos\theta_1 + i\sin\theta_1 = \cos\theta_1(1 + i\tan\theta_1)$, $\cos\theta_2 + i\sin\theta_2 = \cos\theta_2(1 + i\tan\theta_2)$ and so on.

$\therefore \quad \cos(\theta_1 + \theta_2 + \ldots + \theta_n) + i\sin(\theta_1 + \theta_2 + \ldots + \theta_n)$

$= \cos\theta_1\cos\theta_2\ldots\cos\theta_n(1 + i\tan\theta_1)(1 + i\tan\theta_2)\ldots(1 + i\tan\theta_n)$

$= \cos\theta_1\cos\theta_2\ldots\cos\theta_n[1 + i(\tan\theta_1 + \tan\theta_2 + \ldots + \tan\theta_n)$

$+ i^2(\tan\theta_1\tan\theta_2 + \tan\theta_2\tan\theta_3 + \ldots) + i^3(\tan\theta_1\tan\theta_2\tan\theta_3 + \ldots) + \ldots + \ldots]$

$= \cos\theta_1\cos\theta_2\ldots\cos\theta_n(1 + is_1 - s_2 - is_3 + s_4 + \ldots)$

where $s_1 = \tan\theta_1 + \tan\theta_2 + \ldots + \tan\theta_n$, $s_2 = \Sigma\tan\theta_1\tan\theta_2$, $s_3 = \Sigma\tan\theta_1\tan\theta_2\tan\theta_3$, etc.

Equating real and imaginary parts, we have

$\cos(\theta_1 + \theta_2 + \ldots + \theta_n) = \cos\theta_1\cos\theta_2\ldots\cos\theta_n(1 - s_2 + s_4 - \ldots)$

$\sin(\theta_1 + \theta_2 + \ldots + \theta_n) = \cos\theta_1\cos\theta_2\ldots\cos\theta_n(s_1 - s_3 + s_5 - \ldots)$

and by division, we get $\tan(\theta_1 + \theta_2 + \ldots + \theta_n) = \dfrac{s_1 - s_3 + s_5 - \ldots}{1 - s_2 + s_4 - s_6 + \ldots}$.

Example 19.35. If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi/2$, show that $xy + yz + zx = 1$. (NDJM. 2000)

Solution. Let $\tan^{-1}x = \alpha$, $\tan^{-1}y = \beta$, $\tan^{-1}z = \gamma$ so that $x = \tan\alpha$, $y = \tan\beta$, $z = \tan\gamma$.

We know that $\tan(\alpha + \beta + \gamma) = \dfrac{\tan\alpha + \tan\beta + \tan\gamma - \tan\alpha\tan\beta\tan\gamma}{1 - \tan\alpha\tan\beta - \tan\beta\tan\gamma - \tan\gamma\tan\alpha}$

$\therefore \quad \tan\pi/2 = \dfrac{x + y + z - xyz}{1 - xy - yz - zx}$ or $1 - xy - yz - zx = 0$

Hence $xy + yz + zx = 1$.

Example 19.36. If $\theta_1, \theta_2, \theta_3$ be three values of θ which satisfy the equation $\tan 3\theta = \lambda\tan\theta + \alpha$ (where α is a constant and neither of them is a multiple of π), show that $\theta_1 + \theta_2 + \theta_3 = n\pi + \alpha$, where n is an integer.

Solution. Given equation can be written as $\dfrac{2t}{1 - t^2} = \lambda\dfrac{t + \tan\alpha}{1 - t\tan\alpha}$, where $t = \tan\theta$

or $\quad 3t^3 + (\lambda - 3)\tan\alpha\, t^2 + (\lambda - 3)t - \lambda\tan\alpha = 0$

$\therefore \quad \tan\theta_1, \tan\theta_2, \tan\theta_3$ being its roots, we have

$s_1 = \Sigma\tan\theta_1 = -\dfrac{\lambda - 3}{\lambda}\tan\alpha$ $\qquad$ (By § 1.3)

$s_2 = \Sigma\tan\theta_1\tan\theta_2 = \dfrac{\lambda - 3}{\lambda}$ $\quad$ and $\quad s_3 = \tan\alpha$

$\therefore \quad \tan(\theta_1 + \theta_2 + \theta_3) = \dfrac{s_1 - s_3}{1 - s_2} = \dfrac{-(\lambda - 3 + \lambda)\tan\alpha}{\lambda - (\lambda - 3)} = -\tan\alpha = \tan(n\pi - \alpha)$

Thus $\theta_1 + \theta_2 + \theta_3 = n\pi - \alpha$, whence follows the result.

QC To expand $\sin^m\theta$, $\cos^n\theta$ or $\sin^m\theta\cos^n\theta$ in a series of sines or cosines of θ or multiples of θ

If $\quad z = \cos\theta + i\sin\theta$, then $1/z = \cos\theta - i\sin\theta$.

By De Moivre's theorem, $z^n = \cos n\theta + i\sin n\theta$ and $1/z^n = \cos n\theta - i\sin n\theta$

$\therefore \quad z + 1/z = 2\cos\theta$, $z - 1/z = 2i\sin\theta$; $z^n + 1/z^n = 2\cos n\theta$, $z^n - 1/z^n = 2i\sin n\theta$

These results are useful to expand the powers of sin θ or cos θ as their products in a series of sines or cosines of multiples of θ.

Example 10.27. Expand $\cos^6\theta$ in a series of cosines of multiples of θ.

Solution. Let $z = \cos\theta + i\sin\theta$, so that $z^{-1} = \cos\theta - i\sin\theta$ and $z^n + z^{-n} = 2\cos n\theta$.

$$\therefore \quad 2^6\cos^6\theta = \left(z + \frac{1}{z}\right)^6$$

$$= z^6 + {}^6C_1 z^5\cdot\frac{1}{z} + {}^6C_2 z^4\cdot\frac{1}{z^2} + {}^6C_3 z^3\cdot\frac{1}{z^3} + {}^6C_4 z^2\cdot\frac{1}{z^4} + {}^6C_5 z\cdot\frac{1}{z^5} + \frac{1}{z^6}$$

$$= (z^6 + z^{-6}) + 6(z^4 + z^{-4}) + {}^6C_2(z^2 + z^{-2}) + 6(z^2 + z^{-2}) + {}^6C_3(z^0) + 15(z^0) + {}^6C_3$$

$$= 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20.$$

Hence $\cos^6\theta = \frac{1}{32}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)$.

Example 10.28. Expand $\sin^5\theta\cos^3\theta$ in a series of sines of multiples of θ.

Solution. Let $z = \cos\theta + i\sin\theta$

so that $z^n + \frac{1}{z^n} = 2\cos n\theta$, $z^n - \frac{1}{z^n} = 2i\sin n\theta$ and $z^n\cdot\frac{1}{z^n} = 1^n = 1$ (say) (1).

$$\therefore \quad (2i)^5\sin^5\theta\,(2\cos\theta)^3 = \left(z - \frac{1}{z}\right)^5\left(z + \frac{1}{z}\right)^3$$

$$= \left(z^5 - 5z^3\cdot 1 + 10z\cdot 1^2 - \frac{10}{z^3}\cdot 1^3 + \frac{5}{z}\cdot 1^4 - \frac{1}{z^5}\right)\left(z^3 + 3z\cdot 1 + \frac{3}{z}\cdot 1^2 + \frac{1}{z^3}\right)$$

$$= \left(z^8 - \frac{1}{z^8}\right) - 4\left(z^6 - \frac{1}{z^6}\right) + 2\left(z^4 - \frac{1}{z^4}\right) + 8\left(z^2 - \frac{1}{z^2}\right) - 10\left(z - \frac{1}{z}\right)$$

$$= 2i\sin 8\theta - 8i\sin 6\theta + 4i\sin 4\theta + 16i\sin 2\theta - 20i\sin\theta.$$

Hence $i^5 = -i$.

$$\therefore \quad \sin^5\theta\cos^3\theta = -\frac{1}{32}(\sin 8\theta - 4\sin 6\theta + 2\sin 4\theta + 8\sin 2\theta - 10\sin\theta).$$

Obs. The expansion of $\sin^m\theta\cos^n\theta$ is a series of sines or cosines of multiples of θ according as m is odd or even.

EXERCISE 10.5

1. Express $\sin^5\theta\cos^3\theta$ as a sum of multiples of angles. (Purv. Univ. 2010)

2. $\sin^7\theta = \frac{1}{64}(35\sin\theta - 21\sin 3\theta + 7\sin 5\theta - \sin 7\theta)$.

3. $\frac{(1+\cos\theta)^n}{1+\cos n\theta} = \frac{1}{2^n}(\cos\theta + \cos 2\theta + \cos 3\theta + \ldots + \cos n\theta)$, where n is a positive integer. (Madras, 2005)

4. If $2\cos\theta = x + \frac{1}{x}$ and $2\cos\phi = y + \frac{1}{y}$, prove that $x^m y^n + \frac{1}{x^m y^n} = 2\cos(m\theta + n\phi)$.

5. $\cos^6\theta = \frac{1}{32}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)$.

6. If $\cos\theta + \cos\phi + \cos\psi = 0 = \sin\theta + \sin\phi + \sin\psi$, then show that $\cos 3\theta + \cos 3\phi + \cos 3\psi = 3\cos(\theta+\phi+\psi)$.

7. If α, β are the roots of the equation $ax^2 + bx + c = 0$, prove that
$\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma$ is an integral multiple of π unless $a + c = 0$ and $b = 1 - c$. (Purv. Univ. 2011)

8. $\cos^5\theta = \frac{1}{16}(\cos 5\theta + 5\cos 3\theta + 10\cos\theta)$. (Madras, 2005)

9. $\sin^6\theta - \cos^6\theta = -\frac{1}{32}(\cos 6\theta + 15\cos 2\theta)$. (Madras, 2007)

10. $\sin^5\theta\cos^3\theta = \frac{1}{64}(\sin 8\theta - 4\sin 6\theta + 2\sin 4\theta + 8\sin 2\theta - 10\sin\theta)$.

11. $32\cos^5\theta\sin^3\theta = \sin 8\theta - 3\sin 6\theta - 2\sin 4\theta + 2\sin 2\theta$.

12. $\sin^5\theta\cos^2\theta = \frac{1}{64}(\cos 7\theta - 2\cos 5\theta + \cos 3\theta + 2\cos\theta)$. (Madras, 2006)

18. Expand $\sin^6\theta$ in a series of cosines of multiples of θ. (Madras 2003)
19. If $\cos\theta + \cos\phi + \cos\psi = 0 = \sin\theta + \sin\phi + \sin\psi$, find $\cos^3\theta + \cos^3\phi + \cos^3\psi$.
20. If $\sin^6\theta = A_0 + A_1\cos\theta + A_2\cos 2\theta + A_3\cos 3\theta + A_4\cos 4\theta + A_5\cos 5\theta + A_6\cos 6\theta$, prove that
$A_0 + A_6 + 2(A_2 + A_4 + A_6) + 10A_3 = 0$.

3.17 COMPLEX FUNCTION

Definition. *If for each value of the complex variable $z = x + iy$ in a given region R, one or more values of $w = u + iv$, then w is said to be a complex function of z and we write $w = u(x, y) + iv(x, y) = f(z)$ where u, v are real functions of x and y.*

If for each value of z, there corresponds one and only one value of w, then w is said to be a single-valued function of z otherwise a multi-valued function. For example, $w = z^2$ is a single-valued function and $w = \sqrt{z}$ is a multi-valued function of z. The former is defined at all points of the z-plane except at $z = 0$ and the latter assumes two values for each value of z except at $z = 0$.

3.18 EXPONENTIAL FUNCTION OF A COMPLEX VARIABLE

(1) Definition. *When x is real, we are already familiar with the exponential function*

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \ldots + \frac{x^n}{n!} + \ldots \infty.$$

Similarly, we define the exponential function of the complex variable $z = x + iy$, as

$$e^z \text{ or } \exp(z) = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \ldots + \frac{z^n}{n!} + \ldots \infty \qquad \ldots(1)$$

(2) Properties :

I. Exponential form of $z = re^{i\theta}$

Putting $z = r(\cos\theta + i\sin\theta)$ in (1), we get

$$e^{r(\cos\theta + i\sin\theta)} = 1 + \frac{r}{1!}(\cos\theta + i\sin\theta) + \frac{r^2}{2!}(\cos\theta + i\sin\theta)^2 + \frac{r^3}{3!}(\cos\theta + i\sin\theta)^3 + \ldots \infty$$

$$= \left(1 - \frac{r^2}{2!} + \frac{r^4}{4!} - \ldots\right) + i\left(r - \frac{r^3}{3!} + \frac{r^5}{5!} - \ldots\right) = \cos y + i\sin y$$

Thus $e^z = e^x \cdot e^{iy} = e^x(\cos y + i\sin y)$

Also $x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta}$. Thus, $z = re^{i\theta}$

II. e^z is periodic function having imaginary period $2\pi i$. [$\because e^{z + 2\pi i} = e^z \cdot e^{2\pi i} = e^z$]

III. e^z is one-one for any value of z.

Since $e^{z_1} = e^{z_2} \Rightarrow e^{z_1 - z_2} = 1 = e^{0}$ or $e^{z_1} \cdot e^{-z_2} = e^{z_1 - z_2}$

$\therefore$ $z_1 - z_2 = 2\pi i, y = 0, |e^{i\theta}| = 1$.

Thus $|e^z| = |e^x| \cdot |e^{iy}| = e^x \neq 0$

IV. $\overline{e^z} = e^{\overline{z}}$

Since $\overline{e^z} = \overline{e^{x+iy}} = \overline{e^x \cdot e^{iy}} = \overline{e^x(\cos y + i\sin y)}$

$= e^x(\cos y - i\sin y) = e^{\overline{z}}$

3.19 CIRCULAR FUNCTIONS OF A COMPLEX VARIABLE

(1) Definition

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$.

$\therefore$ The circular functions of real angles can be written as

$$\sin y = \frac{e^{iy} - e^{-iy}}{2i}, \cos y = \frac{e^{iy} + e^{-iy}}{2} \text{ and so on.}$$

It is, therefore, natural to define the circular functions of the complex variable z by the equations :

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \cos z = \frac{e^{iz} + e^{-iz}}{2}, \tan z = \frac{\sin z}{\cos z}$$

each name z are a real and z as their respective arguments.

(2) Properties:

I. Circular functions are periodic : $\sin z$, $\cos z$ are periodic functions having real period 2π while $\tan z$, $\cot z$ have period π. i.e. $\sin(z + 2\pi i) = \sin z$, $\tan(z + \pi i) = \tan z$ etc.]

II. Even and odd functions : $\cos z$, $\sec z$ are even functions while $\sin z$, $\tan z$ are odd functions. [$\cos(-z) = \frac{e^{-iz} + e^{iz}}{2} = \cos z$, and $\sin(-z) = \frac{e^{-iz} - e^{iz}}{2i} = -\frac{e^{iz} - e^{-iz}}{2i} = -\sin z$]

III. Zeros of $\sin z$ are given by $z = n\pi$ and zeros of $\cos z$ are given by $z = n\pi + \frac{\pi}{2}$ ($2n + 1)\frac{\pi}{2}$, $n = 0, \pm 1, \pm 2, \ldots$

IV. All the formulae for real circular functions are valid for complex circular functions
e.g., $\sin^2 z + \cos^2 z = 1$, $\sin(z_1 + z_2) = \sin z_1 \cos z_2 + \cos z_1 \sin z_2$

(3) Euler's theorem $e^{iz} = \cos z + i \sin z$.

By definition $\cos z + i \sin z = \frac{e^{iz} + e^{-iz}}{2} + i\frac{e^{iz} - e^{-iz}}{2i} = e^{iz}$ where $z = x + iy$.

Also we have shown that $e^{iy} = \cos y + i \sin y$, where y is real.

Thus $e^{i\theta} = \cos\theta + i \sin\theta$, where θ is real or complex. This is called the Euler's theorem.[1]

Cor. De Moivre's theorem for complex number

Whether θ is real or complex, we have

$$\cos n\theta + i \sin n\theta = (e^{i\theta})^n = e^{in\theta} = \cos n\theta + i \sin n\theta$$

Thus De Moivre's theorem is true for all real or complex n.

Example 10.20. *Prove* (i) $\sin(\theta + i\phi) = \sin\theta \cosh\phi + i \cos\theta \sinh\phi$
(ii) $\cos(\theta + i\phi) = \cos\theta \cosh\phi - i \sin\theta \sinh\phi$.

Solution. (i) R.H.S. $= \sin\theta \cosh\phi + i \cos\theta \sinh\phi = \frac{1}{2}\sin\theta(e^{\phi} + e^{-\phi}) + \frac{i}{2}\cos\theta(e^{\phi} - e^{-\phi})$

$= \frac{1}{2}\sin\theta \cos i\phi + \frac{1}{2}\sin\theta \sin i\phi + \frac{1}{2}\cos\theta \sin i\phi - \frac{1}{2}\cos\theta \cos i\phi$

$= \sin^2\frac{\theta}{2}\cos\frac{i\phi}{2}\cos\frac{i\phi}{2} + 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\sin\frac{i\phi}{2}\cos\frac{i\phi}{2} + \cos^2\frac{\theta}{2}\sin\frac{i\phi}{2}\cos\frac{i\phi}{2} - \sin^2\frac{\theta}{2}\sin\frac{i\phi}{2}\cos\frac{i\phi}{2}$

(ii) R.H.S. $= \sin\alpha \cos i\beta + \cos\alpha \sin i\beta = \sin\alpha \cos\beta - i \cos\alpha \sin\beta$

$= \sin\alpha \cos\alpha \cos^2\beta - i \sin\alpha \cos\alpha \sin^2\beta + \cos\alpha \sin\alpha \cos^2\beta + i \cos\alpha \sin\alpha \sin^2\beta$

$= \sin\alpha \cos\alpha \cos^2\beta - i \sin\alpha \cos\alpha \sin^2\beta$

$= \sin\alpha \cos\alpha \cos^2\beta - i \sin\alpha \cos\alpha \sin^2\beta = \sin\alpha \cos\alpha = \text{L.H.S.}$

Example 10.21. *Show that* $\frac{1}{r} = \frac{\cos\theta}{a\cos\phi} + i\left(\frac{\sin\theta}{a} + \frac{\cos\theta \tan\phi}{a}\right)$, *where a, ϕ, θ are real, express the function Ae^{θ} in the form $A + iB$ where A and B are real.*

Solution. $\frac{1}{r} = \frac{\cos\theta - i\sin\theta}{a\cos\phi} + i\left(\frac{\sin\theta}{a} + \frac{\cos\theta \tan\phi}{a}\right) = \frac{\cos\theta}{a\cos\phi} - \frac{i\sin\theta}{a\cos\phi} + \frac{i\sin\theta}{a} + \frac{i\cos\theta \tan\phi}{a}$

or $r = \frac{a\cos\phi}{\cos\theta - i\sin\theta + i\sin\theta\cos\phi + i\cos\theta\sin\phi} \times \frac{\cos\theta + i\sin\theta - i\sin\theta\cos\phi - i\cos\theta\sin\phi}{\cos\theta + i\sin\theta - i\sin\theta\cos\phi - i\cos\theta\sin\phi}$

$= \frac{a\cos\phi\cos\theta + ia\cos\phi\sin\theta - ia\cos\phi\sin\theta\cos\phi + a\cos\phi\sin\phi\cos\theta}{\cos^2\theta + \sin^2\theta - i\sin\theta\cos\theta\cos\phi - i\cos\theta\sin\theta\cos\phi + i\sin\theta\cos\theta\cos\phi + \sin\theta\cos\theta\sin\phi}$ ... *As $\cos\theta + i\sin\theta = e^{i\theta}$, we have*

[1] See Footnote p. 80.

Equating real and imaginary parts, we have

$$A\cos\theta = \frac{L^2\omega^2 R}{(R - L\omega^2 CR)^2 + (L\omega C)^2} \quad \text{...(i)}$$

$$A\sin\theta = \frac{L\omega(1 - L\omega^2 C)}{(R - L\omega^2 CR)^2 + (L\omega C)^2} \quad \text{...(ii)}$$

Squaring and adding (i) and (ii),

$$A^2 = \frac{L^2\omega^2 R^2 + (L\omega)^2(1 - L\omega^2 C)^2}{[R - L\omega^2 CR]^2 + (L\omega C)^2} \quad \text{or} \quad A = \frac{L\omega}{\sqrt{R^2 - L\omega^2 CR)^2 + (L\omega C)^2}} \quad \text{...(iii)}$$

Dividing (ii) by (i),

$$\tan\theta = \frac{R - L\omega^2 CR}{L\omega} \quad \text{or} \quad \theta = \tan^{-1}\left[\frac{R(1 - L\omega^2 C)}{L\omega}\right] \quad \text{...(iv)}$$

Hence $R + jL\omega = A(\cos\theta + j\sin\theta) = Ae^{j\theta}$

where A and θ are given by (iii) and (iv).

10.10 HYPERBOLIC FUNCTIONS

(i) Definitions. If x is a real variable,

(a) $\frac{e^x - e^{-x}}{2}$ is defined as hyperbolic sine of x and is written as $\sinh x$,

(b) $\frac{e^x + e^{-x}}{2}$ is defined as hyperbolic cosine of x and is written as $\cosh x$,

Thus $\sinh x = \frac{e^x - e^{-x}}{2}$ and $\cosh x = \frac{e^x + e^{-x}}{2}$

Also we define,

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad \coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}, \quad \operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

(ii) Properties

I. Periodic functions : $\sinh x$ and $\cosh x$ are periodic functions having imaginary period $2\pi i$.

[$\because \sinh(x + 2\pi i) = \sinh x \cdot \cos 2\pi i + \cosh x \cdot \sin 2\pi i = \sinh x$; $\cosh(x + 2\pi i) = \cosh x$]

II. Even and odd functions : $\cosh x$ is an even function while $\sinh x$ is an odd function.

III. $\sinh 0 = 0$, $\cosh 0 = 1$, $\tanh 0 = 0$

IV. Relations between hyperbolic and circular functions.

Now for all values of θ, $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$ and $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$

$\therefore$ Putting $\theta = ix$, we have $\sin ix = \frac{e^{i \cdot ix} - e^{-i \cdot ix}}{2i} = \frac{e^{-x} - e^{x}}{2i}$ $[\because i^2 = -1 \Rightarrow i^3 = -i]$

$$= j\frac{e^x - e^{-x}}{2} = j\frac{e^x - e^{-x}}{2} = j\sinh x$$

and, therefore, $\cos ix = \frac{e^{ix} + e^{-ix}}{2} = \cosh x$

Thus $\sin ix = i\sinh x$ ...(v)

$\cos ix = \cosh x$ ...(vi)

and $\therefore$ $\tan ix = i\tanh x$ ...(vii)

Also, $\sinh ix = i\sin x$ ...(viii)

$\cosh ix = \cos x$ ...(ix)

$\tanh ix = i\tan x$ ...(x)

4.5 Properties of Trigonometric Functions

Even-Odd Identities

Pythagorean Identities

Sum and Difference Formulas

Double-Angle Formulas

Half-Angle Formulas

Product-to-Sum Formulas

Sum-to-Product Formulas

Inverse Trigonometric Functions

Trigonometric Equations

35.3 INVERSE HYPERBOLIC FUNCTIONS

(1) Definitions If $\sinh x = y$, then x is called the hyperbolic sine inverse of y and is written as $x = \sinh^{-1} y$. Similarly we define $\cosh^{-1} y$, $\tanh^{-1} y$, etc.

The inverse hyperbolic functions (for other inverse functions are unconvinced, but we shall consider only their principal values).

(2) To show that (i) $\sinh^{-1} x = \log\left(x + \sqrt{x^2 + 1}\right)$ [Mumbai, 2005]

(ii) $\cosh^{-1} x = \log\left(x + \sqrt{x^2 - 1}\right)$, (iii) $\tanh^{-1} x = \frac{1}{2}\log\frac{1+x}{1-x}$

(i) Let $\sinh^{-1} x = u$, then $x = \sinh u = \frac{1}{2}(e^u - e^{-u})$

or $2x = e^u - e^{-u}$ or $e^{2u} - 2xe^u - 1 = 0$

This being a quadratic in e^u, we have

$$e^u = \frac{2x \pm \sqrt{(4x^2 + 4)}}{2} = x \pm \sqrt{(x^2 + 1)}$$

$\therefore$ Taking the positive sign only, we have

$$e^u = x + \sqrt{(x^2 + 1)} \quad \text{or} \quad u = \log\left(x + \sqrt{x^2 + 1}\right)$$

Similarly we can establish (ii)

(iii) Let $\tanh^{-1} x = u$, then $x = \tanh u$

i.e., $$x = \frac{e^u - e^{-u}}{e^u + e^{-u}}.$$

Applying componendo and dividendo, we get $\frac{1+x}{1-x} = \frac{e^{2u}}{1} = e^{2u}$

or $2u = \log\left(\frac{1+x}{1-x}\right)$ which is the result. [U.P.T.U. 2005]

Example 35.11. If $\alpha = \log\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)$, prove that

(i) $\tanh\frac{\alpha}{2} = \tan\frac{\theta}{2}$ [Mumbai, 2003; U.P.T.U. 2004; Madras, 2002]

(ii) $\theta = -i\log\tan\left(\frac{\pi}{4} + \frac{i\alpha}{2}\right)$. [Kurukshetra, 2006]

Solution. We have $\alpha = \log\tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)$ or $e^{\alpha} = \frac{e^{\alpha}}{e^{-\alpha}} = \frac{1 + \tan\frac{\theta}{2}}{1 - \tan\frac{\theta}{2}}$

By componendo and dividendo, we get

$$\frac{e^{\alpha} - e^{-\alpha}}{e^{\alpha} + e^{-\alpha}} = \tan\frac{\theta}{2} \quad \text{i.e.,} \quad \tanh\frac{\alpha}{2} = \tan\frac{\theta}{2} \qquad \ldots(i)$$

or $$\frac{1}{i}\tan\frac{\theta}{2} = \frac{1}{i}\tanh\frac{\alpha}{2} \quad \text{or} \quad \frac{\alpha}{2} = \tanh^{-1}\left(\frac{1}{i}\tan\frac{\theta}{2}\right) = \frac{1}{2}\log\frac{1 + \frac{1}{i}\tan\frac{\theta}{2}}{1 - \frac{1}{i}\tan\frac{\theta}{2}}$$

or $$\alpha = \frac{1}{i}\log\tan\left(\frac{\pi}{4} + \frac{i\alpha}{2}\right) = -i\log\tan\left(\frac{\pi}{4} + \frac{i\alpha}{2}\right). \qquad \ldots(ii)$$

Example 35.12. Show that $\tanh^{-1}(\cos\theta) = \operatorname{cosec}^{-1}(\cosh\phi)$. [Kurukshetra, 2003]

Solution. Let $\tanh^{-1}(\cos\theta) = \phi$ so that $\cos\theta = \tanh\phi$

or $\tanh^2\phi = \cos^2\theta$ or $1 - \operatorname{sech}^2\phi = \cos^2\theta$

or $\operatorname{sech}^2\phi = 1 - \cos^2\theta = \sin^2\theta$ or $\operatorname{sech}\phi = \sin\theta$

or $\cosh\phi = \operatorname{cosec}\theta$ or $\phi = \cosh^{-1}(\operatorname{cosec}\theta)$.

Example 12.33. Find $\tanh x$, if $5\sinh x - 3\cosh x = 1$. (Kurukshetra, 2004)

Solution. We have $5\sinh x - 3\cosh x = 1$

or $(5\sinh x - 3\cosh x)^2 = 1^2$ or $25\sinh^2 x + 9\cosh^2 x - 30\sinh x \cosh x = 1$

or $25\sinh^2 x - 30\sinh x \cosh x + 9(1 + \sinh^2 x) = 1$ or $34\sinh^2 x - 30\sinh x \cosh x + 8 = 0$

or $(5\sinh x - 4\cosh x)(4\sinh x - 2\cosh x) = 0$ $\quad \because \cosh x \neq 0$, $\therefore 5\sinh x = 4\cosh x$ or $2\sinh x = \cosh x$

$\therefore \quad \cosh x = \sqrt{1 + \sinh^2 x} = \frac{5}{3}$ or $-\frac{5}{3}$ $\quad$ If $\cosh x = \frac{5}{3}$ does not satisfy (i)

Hence $\tanh x = \frac{4}{5}$ or $-\frac{3}{5}$.

EXERCISE 12.3

1. Separate into real and imaginary parts:
   (i) $\sin(x + iy)$ where $i^2 = -1$ $\quad$ (ii) $\cos(\theta + i\phi)$ $\quad$ (iii) $\exp(5 + 6i)$.

2. Prove the definitions of $\sinh x$ and $\cosh x$ given above:
   (i) $\cosh 2x = 2\cosh^2 x - 1$ $\quad$ (ii) $\frac{\sinh 2x}{1 + \cosh 2x} = \tanh x$ $\quad$ (iii) $\sinh 3x = 3\sinh x + 4\sinh^3 x$.

3. Prove that $\tan(x + iy) = \frac{\sin 2x}{\cos 2x + \cosh 2y} + i\frac{\sinh 2y}{\cos 2x + \cosh 2y}$.

4. If $x = \cos\theta$, show that $\frac{d^n x}{dx^n} = \cos(n\theta)$.

5. Determine a homogeneous expression in x and y which is equivalent to $\sin(\alpha \log(x + iy))$.

6. If $y = \log(\tan(\frac{\pi}{4} + \frac{\theta}{2}))$, find $\sin\theta$ in terms of y. Hence show that $\sin\theta = \frac{2}{\pi}(\theta^2 - \theta^3)$.

7. If $\tan\alpha = \tan\theta\tan\beta$ and $\tan\alpha = n\tan(\theta - \beta)$, prove that $\tan^2\theta = \frac{n+1}{n-1}\tan^2\beta$. (Kurukshetra, 2004)

8. Prove that
   (i) $\cosh(\alpha + \beta)\cosh(\alpha - \beta) = \cosh^2\alpha - \sinh^2\beta = \cosh^2\beta + \sinh^2\alpha$
   (ii) $\sinh(\alpha + \beta)\sinh(\alpha - \beta) = \frac{1}{2}(\cosh 2\alpha - \cosh 2\beta)$.

9. Prove that (i) $\cosh^2\frac{\theta}{2} = \frac{\cosh\theta + 1}{2}$ and $\sinh^2\frac{\theta}{2} = \frac{\cosh\theta - 1}{2}$ (ii) $\left(\frac{1 + \cosh\theta}{1 + \sinh\theta}\right)^n = \cosh^n\theta + n\sinh\theta\cosh^{n-1}\theta$.

10. Express $\sinh^5\theta$ in terms of hyperbolic functions of multiples of θ.

11. If $\sin\theta = \tanh\phi$, prove that $\sin 2\theta = \tanh 2\phi$.

12. If $\tan\alpha = \tanh\alpha/2$, prove that:
   (i) $\sin\alpha = \sin\theta/(\cosh\alpha + \cos\theta)$ $\quad$ (ii) $x = \log_e\tan(\alpha/4 + \theta/2)$.

13. If $\cosh x = \sec\theta$, prove that:
   (i) $\tanh^2\frac{x}{2} = \tan^2\frac{\theta}{2}$ $\quad$ (ii) $x = \log_e\tan(\pi/4 + \theta/2)$.

14. Show that $\tanh^{-1}x = \frac{1}{2}\log\frac{1+x}{1-x}$.

15. Prove that
   (i) $\sinh^{-1}x + \cosh^{-1}\sqrt{1 + x^2} = \tan^{-1}\frac{x}{\sqrt{1 - x^2}} = \frac{1}{2}\cosh^{-1}\left(\frac{1}{2x^2 - 1}\right)$
   (ii) $\sin\sinh^{-1}x = \tan\tanh^{-1}\frac{x}{\sqrt{1 + x^2}}$.

16. Show that
   (i) $\sinh^{-1}(\tan\theta) = \log\tan(\pi/4 + \theta/2)$ $\quad$ (ii) $\cosh^{-1}(\sec\theta) = \log_e(\sec\theta + \tan\theta)$.

17. Solve the equation $7\cosh x + 8\sinh x = 1$ for real values of x. (Kurukshetra, 2004)

18. Find $\tanh x$ if $\cosh x - \sinh x = 2$.

2.15(2) REAL AND IMAGINARY PARTS OF CIRCULAR AND HYPERBOLIC FUNCTIONS

(1) To express the real and imaginary parts of

(i) $\sin(x+iy)$ (ii) $\cos(x+iy)$, (iii) $\tan(x+iy)$, (iv) $\cot(x+iy)$, (v) $\sec(x+iy)$, (vi) $\operatorname{cosec}(x+iy)$.

Proofs. (i) $\sin(x+iy) = \sin x\cos iy + \cos x\sin iy = \sin x\cosh y + i\cos x\sinh y$.

Similarly, $\cos(x+iy) = \cos x\cosh y - i\sin x\sinh y$

(iii) Let $\alpha + i\beta = \tan(x+iy)$. Then $\alpha - i\beta = \tan(x-iy)$

Adding, $2\alpha = \tan(x+iy) + \tan(x-iy)$

i.e., $$\alpha = \frac{\sin(x+iy+x-iy)}{2\cos(x+iy)\cos(x-iy)} = \frac{\sin 2x}{\cos 2x + \cos 2iy} = \frac{\sin 2x}{\cos 2x + \cosh 2y}$$

Subtracting, $2i\beta = \tan(x+iy) - \tan(x-iy)$

i.e., $$i\beta = \frac{\sin 2iy}{2\cos(x+iy)\cos(x-iy)} = \frac{i\sinh 2y}{\cos 2x + \cosh 2y}$$

or $$\beta = \frac{\sinh 2y}{\cos 2x + \cosh 2y}$$

Similarly, $\cot(x+iy) = \dfrac{\sin 2x - i\sinh 2y}{\cosh 2y - \cos 2x}$.

(v) Let $\alpha + i\beta = \sec(x+iy)$. Then $\alpha - i\beta = \sec(x-iy)$

Adding, $2\alpha = \sec(x+iy) + \sec(x-iy)$

i.e., $$\alpha = \frac{\cos(x-iy) + \cos(x+iy)}{2\cos(x+iy)\cos(x-iy)} = \frac{2\cos x\cosh y}{\cos 2x + \cosh 2y} = \frac{2\cos x\cosh y}{\cos 2x + \cosh 2y}$$

Subtracting, $2i\beta = \sec(x+iy) - \sec(x-iy)$

i.e., $$i\beta = \frac{\cos(x-iy) - \cos(x+iy)}{2\cos(x+iy)\cos(x-iy)} = \frac{2i\sin x\sinh y}{\cos 2x + \cosh 2y} = \frac{2i\sin x\sinh y}{\cos 2x + \cosh 2y}$$

or $$\beta = \frac{2\sin x\sinh y}{\cos 2x + \cosh 2y}$$

Similarly, $\operatorname{cosec}(x+iy) = 2\dfrac{\sin x\cosh y - i\cos x\sinh y}{\cosh 2y - \cos 2x}$.

(2) To express the real and imaginary parts of

(i) $\sin^{-1}(x+iy)$, (ii) $\cos^{-1}(x+iy)$, (iii) $\tan^{-1}(x+iy)$

Proofs. (i) $\sinh(u+iv) = \frac{1}{2}(e^{u+iv} - e^{-u-iv}) = \frac{1}{2}(e^u e^{iv} - e^{-u}e^{-iv})$

$= \frac{1}{2}(e^u\cos v + ie^u\sin v - e^{-u}\cos v + ie^{-u}\sin v) = \frac{1}{2}(\cos v(e^u - e^{-u}) + i\sin v(e^u + e^{-u}))$

$= \cos v\sinh u + i\sin v\cosh u$

Similarly, $\cosh(u+iv) = \cos v\cosh u + i\sin v\sinh u$

(ii) If $\alpha + i\beta = \tan^{-1}(x+iy) = \frac{1}{2}i\log\frac{1-ix-y}{1+ix+y}$

then $\alpha - i\beta = \tan^{-1}(x-iy) = \frac{1}{2}i\log\frac{1+ix+y}{1-ix-y}$

Adding, $2\alpha = \frac{1}{2}i\log\frac{1-x-iy}{1+x+iy} - \frac{1}{2}i\log\frac{1+x-iy}{1-x+iy}$

$$\alpha = \frac{i}{4}\log\frac{1-x+2iy-x^2-y^2}{1+2ix+x^2+y^2}\cdot\frac{1-x-2iy-x^2-y^2}{1+2ix+x^2+y^2} = \frac{i}{4}\log\frac{1-2x-x^2-y^2}{1+2x+x^2+y^2}$$

Subtracting, $2i\beta = \frac{1}{2}i\log\frac{1-x-iy}{1+x+iy} - \frac{1}{2}i\log\frac{1+x-iy}{1-x+iy}$

i.e., $$i\beta = -\frac{i}{4}\log\frac{(1-x-iy)(1-x+iy)}{(1+x+iy)(1+x-iy)}$$

or $$\beta = \frac{\sin 2y}{\cosh 2x + \sin 2y} + \frac{\sin 2y}{\cosh 2x + \cos 2y}$$

Example 10(56). If $\cosh(x+iy) = \sec\alpha + i\tan\alpha$, prove that

$$\frac{x^2}{\sec^2\alpha} + \frac{y^2}{\tan^2\alpha} = 1 \quad \text{and} \quad \frac{x^2}{\sec^2\alpha - \cos^2\alpha} - \frac{y^2}{\sin^2\alpha - \cos^2\alpha} = 1.$$

(Madras, 2000)

Solution. Given $x + iy = \cos(\theta + i\phi)$

$= \cos\theta\cos i\phi - \sin\theta\sin i\phi = \cos\theta\cosh\phi - i\sin\theta\sinh\phi$

$\therefore$ equating real and imaginary parts, we get $x = \cos\theta\cosh\phi$, $y = -\sin\theta\sinh\phi$

i.e., $\frac{x}{\cosh\phi} = \cos\theta$ and $\frac{y}{\sinh\phi} = -\sin\theta$

Squaring and adding, we get the first result.

Again $\frac{x}{\cos\theta} = \cosh\phi$ and $\frac{y}{\sin\theta} = -\sinh\phi$.

$\therefore$ squaring and subtracting, we get the second result.

Example 19.35. If $\tan(\theta + i\phi) = e^{i\alpha}$, show that

$\theta = n\pi + \frac{\pi}{4}$ and $\phi = \frac{1}{2}\log\tan\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$. (S.V.T.U., 2007; R.G.P.V., 2008)

Solution. Given $\tan(\theta + i\phi) = \cos\alpha + i\sin\alpha$ $\therefore$ $\tan(\theta - i\phi) = \cos\alpha - i\sin\alpha$

$\therefore$ $\tan 2\theta = \tan[(\theta + i\phi) + (\theta - i\phi)]$

$= \frac{\tan(\theta + i\phi) + \tan(\theta - i\phi)}{1 - \tan(\theta + i\phi)\tan(\theta - i\phi)} = \frac{2\cos\alpha}{1 - (\cos^2\alpha + \sin^2\alpha)} = \frac{2\cos\alpha}{0} = \infty$

i.e., $2\theta = \frac{\pi}{2}$ or $\theta = n\pi + \frac{\pi}{4}$

Also $\tan 2i\phi = \tan[(\theta + i\phi) - (\theta - i\phi)] = \frac{\tan(\theta + i\phi) - \tan(\theta - i\phi)}{1 + \tan(\theta + i\phi)\tan(\theta - i\phi)}$

or $i\tanh 2\phi = \frac{2i\sin\alpha}{1 + (\cos^2\alpha + \sin^2\alpha)} = i\sin\alpha$ or $\frac{e^{2\phi} - e^{-2\phi}}{e^{2\phi} + e^{-2\phi}} = \sin\alpha$

By componendo and dividendo, we get

$$\frac{e^{2\phi}}{e^{-2\phi}} = \frac{1 + \sin\alpha}{1 - \sin\alpha} = \frac{\sin^2\frac{\alpha}{2} + \cos^2\frac{\alpha}{2} + 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{\sin^2\frac{\alpha}{2} + \cos^2\frac{\alpha}{2} - 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}$$

or $e^{4\phi} = \frac{\left(\cos\frac{\alpha}{2} + \sin\frac{\alpha}{2}\right)^2}{\left(\cos\frac{\alpha}{2} - \sin\frac{\alpha}{2}\right)^2} = \left(\frac{1 - \tan\frac{\alpha}{2}}{1 - \tan\frac{\alpha}{2}}\right)^2$

or $e^{2\phi} = \frac{1 + \tan\frac{\alpha}{2}}{1 - \tan\frac{\alpha}{2}} = \tan\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$. Hence $\phi = \frac{1}{2}\log\tan\left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$.

Example 19.36. If $\sin(\theta + i\phi) = x + iy$, prove that $\frac{x^2}{\sin^2\theta} - \frac{y^2}{\cos^2\theta} = 1$ and $\frac{x^2}{\cosh^2\phi} + \frac{y^2}{\sinh^2\phi} = 1$. (R.G.P.V., 2005)

Solution. Let $x + iy = \tan(\alpha + i\beta)$. Then $x - iy = \tan(\alpha - i\beta)$

Adding, $2x = \tan^{-1}(x + iy) + \tan^{-1}(x - iy) = \tan^{-1}\frac{(x + iy) + (x - iy)}{1 - (x + iy)(x - iy)}$

$\therefore$ $\alpha = \frac{1}{2}\tan^{-1}\frac{2x}{1 - x^2 - y^2}$

Subtracting, $2i\beta = \tan^{-1}(x + iy) - \tan^{-1}(x - iy) = \tan^{-1}\frac{(x + iy) - (x - iy)}{1 + (x + iy)(x - iy)}$

$= \tan^{-1}\frac{2iy}{1 + x^2 + y^2} = i\tanh^{-1}\frac{2y}{1 + x^2 + y^2}$ $\quad$ [$\because$ $\tan^{-1}ix = i\tanh^{-1}x$]

$\therefore$ $\beta = \frac{1}{2}\tanh^{-1}\frac{2y}{1 + x^2 + y^2}$.

Example 19.37. Express $\sin^{-1}(\cos\theta + i\sin\theta)$ in the form $\alpha + i\beta$ and find the principal value of the angle when it is positive.

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* $\tan^{-1}A + \tan^{-1}B = \tan^{-1}\frac{A + B}{1 - AB}$

Solution. Let $\sin^{-1}\frac{x}{a}+\sin^{-1}\frac{y}{b}=\alpha$ ... (i)

Then $\sin^{-1}\frac{x}{a}=\alpha-\sin^{-1}\frac{y}{b}$ or $\sin\left(\sin^{-1}\frac{x}{a}\right)=\sin\left(\alpha-\sin^{-1}\frac{y}{b}\right)$

$\therefore \frac{x}{a}=\sin\alpha\cos\sin^{-1}\frac{y}{b}-\cos\alpha\sin\sin^{-1}\frac{y}{b}$

$\Rightarrow \frac{x}{a}=\sin\alpha\sqrt{1-\frac{y^2}{b^2}}-\frac{y}{b}\cos\alpha$ ... (ii)

Squaring and adding, we have

$$1=\sin^2\alpha\left(1-\frac{y^2}{b^2}\right)+\cos^2\alpha\sin^2\sin^{-1}\frac{y}{b}=\sin^2\alpha\left(1-\frac{y^2}{b^2}\right)+\cos^2\alpha\sin^2\sin^{-1}\frac{y}{b}$$

$$=\sin^2\alpha-\sin^2\alpha\sin^2\sin^{-1}\frac{y}{b}+\cos^2\alpha\sin^2\sin^{-1}\frac{y}{b}$$

or $1-\sin^2\alpha=\sin^2\sin^{-1}\frac{y}{b}$ i.e. $\cos^2\alpha=\sin^2\sin^{-1}\frac{y}{b}$.

Hence from (ii), we have $\sin^2\theta=\cos^2\alpha$, i.e., $\cos^2\alpha=\sin\theta$ because θ being a positive acute angle, $\sin\theta$ is positive.

As α is to be between $-\pi/2$ and $\pi/2$, therefore, we have

$$\cos\alpha=+\sin\sin^{-1}\frac{y}{b} \quad \text{or} \quad \alpha=\cos^{-1}\sin\sin^{-1}\frac{y}{b}$$

The relation (i) then, gives $\sin\sin^{-1}\frac{y}{b}=\sin\theta$ or that $y=b\log\left(\sin\theta+\sqrt{1-\sin^2\theta}\right)$.

EXERCISE 11

1. If $\sin\theta=\sin\alpha\cos\alpha$, prove that

(i) $\frac{x^2}{\cos^2(\theta-\alpha)}+\frac{y^2}{\sin^2(\theta-\alpha)}=1$ (ii) $\frac{x^2}{\sin^2\alpha}-\frac{y^2}{\cos^2\alpha}=1$. [V.T.U., 2005]

2. If $\cos(\theta+\phi)=\cos\theta+\cos\phi$, prove that (i) $\tan\frac{\theta}{2}=\frac{\sin\phi}{\cos\phi+1}$ [Kurukshetra, 2002; Madras, 2003]

(ii) $\phi=\frac{\pi}{2}\log\frac{\sin(\alpha+\theta)}{\sin(\alpha-\theta)}$ [V.T.U., 2004]

3. If $\cos(\theta+\alpha)=\cos\theta+\cos\alpha$, prove that

(i) $\sin^2\theta=\sin\alpha$ [Madras, 2000] (ii) $\cos 2\theta=\cos\alpha\cos 2\alpha$.

4. If $\sin(x+iy)=r(\cos\theta+i\sin\theta)$, prove that

(i) $x^2+y^2=\sin\alpha\cos\alpha=1$ (ii) $x^2+y^2=2y\sin\theta\cos\theta+\cos^2\theta$ (iii) $\cos\theta=\frac{1}{2}\left(r+\frac{1}{r}\right)\cos\alpha$ [Nagpur, 2004; V.T.U., 2004]

5. If $\sin(\theta+i\phi)=\cos\alpha+i\sin\alpha$, prove that $e^{2\phi}=\cot\frac{\alpha}{2}$ and $\theta=\left(n+\frac{1}{2}\right)\pi+\frac{\alpha}{2}$. [Nagpur, 2004; V.T.U., 2004]

6. If $\sin(x+iy)=\tan\alpha+i\sec\alpha$, prove that $\frac{\sin 2x}{\cos 2y}=\frac{\tan\alpha}{\sec\alpha}$. [V.T.U., 2004]

7. If $\sin(\alpha+i\beta)=\cos\theta+i\sin\theta$, prove that (i) $\alpha=\frac{1}{2}(2n\pi+\theta)$ [Mumbai, 2004]

8. If $\cos\alpha+\cos\beta+\cos\gamma=0=\sin\alpha+\sin\beta+\sin\gamma$, prove that $\cos(\alpha-\beta)=\cos(\beta-\gamma)=\cos(\gamma-\alpha)=-\frac{1}{2}$.

9. If $\cos\theta=\cos\alpha\cos\beta$, prove that $e^{i\theta}=\frac{1+i\tan\frac{\alpha}{2}}{1-i\tan\frac{\alpha}{2}}\cdot\frac{1+i\tan\frac{\beta}{2}}{1-i\tan\frac{\beta}{2}}$.

10. Prove that $\tan^{-1}\log(x+iy)=\frac{n\pi}{2}+\frac{\pi}{4}-\frac{i}{2}\log\tan\left[\frac{\pi}{4}+\frac{1}{2}\tan^{-1}\frac{2y}{x^2+y^2-1}\right]$.

11. Separate $\cos^{-1}(\cos\theta+i\sin\theta)$ into real and imaginary parts, where θ is a positive acute angle.

12. If $\cos^{-1}(x+iy)=\alpha+i\beta$, prove that $\sin^2\alpha$ and $\sinh^2\beta$ are the roots of the equation

$$t^2-t(1+x^2+y^2)+x^2=0.$$

13. If $\cos^{-1}(x+iy)=\alpha+i\beta$, prove that

(i) $x^2\sec^2\alpha-y^2\operatorname{cosec}^2\alpha=1$, (ii) $x^2\sinh^2\beta+y^2\cosh^2\beta=1$.

14. Prove that $\tan\left(\frac{\pi}{4}+i\log\sqrt{\frac{a-ib}{a+ib}}\right)=\frac{a}{b}$ [$\because \tan(\theta+i\phi)=\frac{\sin 2\theta+i\sinh 2\phi}{\cos 2\theta+\cosh 2\phi}$].

11.13 LOGARITHMIC FUNCTION OF A COMPLEX VARIABLE

(i) **Definition.** If $a^x=y$ and if a is a real number such that $a^x=y$, then x is said to be the logarithm of y to the base a and it is written as $x=\log_a y$. ...(1)

Also $y=e^{\log_e y}$, $e^{2\pi i}=1$ ...(i)

$\therefore$ $\log y=\log_e y+2\pi i$ ...(2)

i.e., the logarithm of a complex number has an infinite number of values and is, therefore, a multi-valued function.

The general value of the logarithm of z is written as $\operatorname{Log} z$ (beginning with capital L) so as to distinguish it from the principal value which is written as $\log z$. This principal value is obtained by taking $n = 0$ in $\operatorname{Log} z$.

Thus from (vi), $\operatorname{Log}(x + iy) = \log(x + iy) + 2n\pi i$

Obs. If $y = 0$, then $\operatorname{Log} x = \log x + 2n\pi i$.

This shows that the logarithm of a real positive number is also complex. The principal value is real. Both of the values are important.

If we know that the logarithm of a negative quantity has no real value. But we can now explain it as

e.g. $\log(-3) = \log 3 + \log(-1) = \log 3 + \log(\cos\pi + i\sin\pi) = \log 3 + i\pi$ $\left[\because -1 = \cos\pi + i\sin\pi = e^{i\pi}\right]$
$= 0.4771 + (3.1416)i$.

(2) Real and imaginary parts of $\operatorname{Log}(x + iy)$.

$\operatorname{Log}(x + iy) = 2n\pi i + \log(x + iy)$

$= 2n\pi i + \log[r(\cos\theta + i\sin\theta)]$ $\left[\begin{array}{l}\text{Put } x = r\cos\theta,\ y = r\sin\theta \text{ so that}\\ r = \sqrt{x^2 + y^2} \text{ and } \theta = \tan^{-1}(y/x)\end{array}\right.$

$= 2n\pi i + \log(re^{i\theta})$

$= 2n\pi i + \log r + i\theta = \log\sqrt{x^2 + y^2} + i(2n\pi + \tan^{-1}(y/x))$.

(3) Real and imaginary parts of $(\alpha + i\beta)^{x+iy}$.

$(\alpha + i\beta)^{x+iy} = e^{(x+iy)\log(\alpha+i\beta)} = e^{(x+iy)[\log\sqrt{\alpha^2+\beta^2} + i\tan^{-1}(\beta/\alpha)]}$ $\left[\begin{array}{l}\text{Put } \alpha = r\cos\theta, \beta = r\sin\theta \text{ so that}\\ r = \sqrt{\alpha^2 + \beta^2} \text{ and } \theta = \tan^{-1}(\beta/\alpha)\end{array}\right.$

$= e^{[x\log\sqrt{\alpha^2+\beta^2} - y\tan^{-1}(\beta/\alpha)] + i[y\log\sqrt{\alpha^2+\beta^2} + x\tan^{-1}(\beta/\alpha)]}$

$= e^{A+iB} = e^A(\cos B + i\sin B)$,

where $A = x\log r - y(2n\pi + \theta)$ and $B = y\log r + x(2n\pi + \theta)$.

∴ the required real part $= e^A\cos B$ and the imaginary part $= e^A\sin B$.

Example 13(38). Find the principal value of $\log(-1)$.

Solution. $\operatorname{Log}(-1) = 2n\pi i + \log 1 + (-1)$ $\left[\begin{array}{l}\text{Put } 0 = r\cos\theta,\ -1 = r\sin\theta\\ \text{so that } r = 1 \text{ and } \theta = -\pi/2\end{array}\right.$

$= 2n\pi i + \log[r(\cos\theta + i\sin\theta)] = 2n\pi i + \log(re^{i\theta})$

$= 2n\pi i + \log r - i\theta = 2n\pi i + \log 1 + i\left(-\frac{\pi}{2}\right) = i\left(2n - \frac{1}{2}\right)\pi$.

Example 13(39). Prove that (i) $i^i = e^{-(4n+\frac{1}{2})\pi}$ and $\log i^i = -\left(2n + \frac{1}{2}\right)\pi$.

(ii) $(\sqrt{3})^{\sqrt{3}} = e^{\sqrt{3}\log\sqrt{3}}$ and $\log(\sqrt{3})^{\sqrt{3}} = \sqrt{3}\log\sqrt{3}$. (Kerala, 2000)

Solution. (i) By definition, we have

$i^i = e^{i\log i} = e^{i[\log 1 + \log(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})]}$ $\left[\because i = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = \exp(i\pi/2)\right]$

$= e^{i\log(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})} = e^{i\log e^{i\pi/2}}$

Taking logarithms, we get (ii)

(ii) $\log i^i = i\log e^{i\pi/2}$

Now $\sqrt{3}\log\sqrt{3} = \frac{1}{2}\left[\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right]\log\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$= \frac{1}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\log e^{i\pi/3} = \frac{1}{2}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)\frac{i\pi}{3}$

$= \frac{i\pi}{6}\left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\right) = -\frac{\pi}{6\sqrt{2}} + i\frac{\pi}{6\sqrt{2}}$

Hence [illegible]

Example 18.48. If $\cos(\alpha+i\beta)=re^{i\theta}$, prove that each of the following is $\frac{1}{2}\tan^{-1}\left(\frac{2\cos\alpha\sinh\beta}{\cos^2\alpha-\sinh^2\beta}\right)=\log\sqrt{\cos^2\alpha+\sinh^2\beta}$.

Solution. Taking logarithms, $(\alpha+i\beta)\log e=\log r+i\theta$ gives $\log(\alpha+i\beta)=(\alpha+i\beta)\log e$

or $$\frac{1}{2}\log(\alpha^2+\beta^2)+i\tan^{-1}\frac{\beta}{\alpha}=\alpha\log e+i\beta\log e.$$

Equating real and imaginary parts from both sides, we get

$$\frac{1}{2}\log(\alpha^2+\beta^2)=\alpha\log e \quad ...(i), \qquad \beta\tan^{-1}\frac{\beta}{\alpha}=\beta\log e \quad ...(ii)$$

Division of (ii) by (i) gives

$$\beta\lambda=2\tan^{-1}(\beta\lambda)\log(\alpha^2+\beta^2).$$

Example 18.49. If $(\alpha+i\beta)^{\alpha-i\beta}=A+iB$, prove that $\log(A^2+B^2)=2e^{\beta\lambda}\cos(\beta\log(\alpha^2+\beta^2))$ (U.P.T.U., 2005)

Solution. $(\alpha+i\beta)^{\alpha-i\beta}=A+iB$ i.e., $e^{(\alpha-i\beta)\log(\alpha+i\beta)}=A+iB$

or $$A+iB=e^{(\alpha-i\beta)\log(\alpha+i\beta)}=e^{(\alpha-i\beta)\log\left(\alpha+i\beta\right)}$$

$$=\exp\{(\alpha-i\beta)\log(\alpha^2+\beta^2)^{1/2}\}=e^{(\alpha-i\beta)\log}$$

$$=e^{\beta\log}\cdot e^{i\beta\log}=e^{\beta\log}\left(\cos\frac{\beta}{2}+i\sin\frac{\beta}{2}\right)$$

Equating real and imaginary parts, we get

$$A=e^{\beta\log}\cos\frac{\beta}{2} \quad ...(i), \qquad B=e^{\beta\log}\sin\frac{\beta}{2} \quad ...(ii)$$

Division of (ii) by (i) gives $\frac{B}{A}=\tan\frac{\beta}{2}$.

Squaring and adding (i) and (ii), $A^2+B^2=e^{2\beta\log}$.

Example 18.50. Prove that $\log\left(\frac{\alpha+i\beta}{\alpha-i\beta}\right)=2i\tan^{-1}\left(\frac{\beta}{\alpha}\right)$. Hence evaluate $\cos\left\{i\log\left(\frac{\alpha+i\beta}{\alpha-i\beta}\right)\right\}$. (U.P.T.U., 2006)

Solution. Putting $\alpha=r\cos\theta, \beta=r\sin\theta$ so that $\theta=\tan^{-1}\frac{\beta}{\alpha}$, we have

$$\log\left(\frac{\alpha+i\beta}{\alpha-i\beta}\right)=\log\frac{r(\cos\theta+i\sin\theta)}{r(\cos\theta-i\sin\theta)}=\log(e^{i\theta}\cdot e^{i\theta})$$

$$=\log e^{2i\theta}=2i\theta=2i\tan^{-1}\frac{\beta}{\alpha}.$$

Thus $$\cos\left\{i\log\left(\frac{\alpha+i\beta}{\alpha-i\beta}\right)\right\}=\cos\{i(2i\theta)\}=\cos 2\theta=\frac{1-\tan^2\theta}{1+\tan^2\theta}=\frac{1-\beta^2/\alpha^2}{1+\beta^2/\alpha^2}=\frac{\alpha^2-\beta^2}{\alpha^2+\beta^2}.$$

Example 18.51. Express $\log(\cos\theta+i\sin\theta)$ in the form $A+iB$ and hence find the principal value of $\log(\cos\theta+i\sin\theta)$.

Solution. $\log(\cos\theta+i\sin\theta)=\log(\cos\theta+i\sin\theta)\log e=\log(\cos\theta+i\sin\theta)$

$$=\log(\sin\alpha\cosh\beta+i\cos\alpha\sinh\beta)=\log r(\cos\theta+i\sin\theta)$$

where $r\cos\theta=\sin\alpha\cosh\beta$ and $r\sin\theta=\cos\alpha\sinh\beta$.

so that $$r=\sqrt{\sin^2\alpha\cosh^2\beta+\cos^2\alpha\sinh^2\beta}$$

$$=\sqrt{\frac{1-\cos 2\alpha}{2}\cdot\frac{1+\cosh 2\beta}{2}+\frac{1+\cos 2\alpha}{2}\cdot\frac{\cosh 2\beta-1}{2}}=\sqrt{\left|\frac{1}{2}(\cosh 2\beta-\cos 2\alpha)\right|}$$

and $$\theta=\tan^{-1}(\cot\alpha\tanh\beta).$$

Thus $\log(\cos\theta+i\sin\theta)=\log(re^{i\theta})=\log r+i\theta$

$$=\frac{1}{2}\log\left|\frac{1}{2}(\cosh 2\beta-\cos 2\alpha)\right|+i\tan^{-1}(\cot\alpha\tanh\beta).$$

Example 12.44. Find all the roots of the equation
(i) $\sin z = \cosh 4$ (ii) $\cos z = 4$

Solution. (i) $\sin z = \cosh 4 = \cos(4i) = \cos(\pi/2 - 4i)$

$\therefore \quad z = n\pi + (-1)^n (\pi/2 - 4i)$

$\left[\because \text{If } \sin\theta = \sin\alpha \text{ then } \theta = n\pi + (-1)^n \alpha\right]$

(ii) $\cos z = \cosh(4i) = \frac{e^{4i} + e^{-4i}}{2}$

or $e^{4i} - 2e^{2i} + 1 = 0$, i.e., $(e^{2i} - 1)^2 = 0$ i.e., $e^{2i} = 1$

or $z = \frac{1}{2i}\log 1 = \frac{1}{2i}(2n\pi i + \log 1) = n\pi + \log e^{2n\pi i} = n\pi + 2n\pi i = (1+2i)n\pi$.

PROBLEMS 4.17

1. Find the general values of
(i) $\log(3+4i)$ (ii) $\log(\sin\theta + i\cos\theta)$ (iii) $\log(1-i)$ (GGSIP U, 2005)

2. Show that (i) $\log(1+i)\log(1-i) = \log(\text{conjugate of } z)$, where z is a non-zero complex number.

(ii) $\log\frac{x+iy}{x-iy} = 2i\tan^{-1}\frac{y}{x}$

3. If $(x_1 + iy_1)(x_2 + iy_2)\ldots(x_n + iy_n) = A + iB$, prove that
(i) $(x_1^2 + y_1^2)(x_2^2 + y_2^2)\ldots(x_n^2 + y_n^2) = A^2 + B^2$

(ii) $\tan^{-1}\frac{y_1}{x_1} + \tan^{-1}\frac{y_2}{x_2} + \ldots + \tan^{-1}\frac{y_n}{x_n} = \tan^{-1}\frac{B}{A}$

4. Find the modulus and argument of (i) $(1+i)^{2i}$ (GGSIP U, 2005) (ii) $(1+i)^{1-i}$

5. If $(x+iy)^{1/3} = a + ib$, prove that (i) $x/a + y/b = 4(a^2 - b^2)$ (Kurukshetra U, 2005)

6. Prove that $\log\left[\frac{\sin(\theta - i\phi)}{\sin(\theta + i\phi)}\right] = 2i\tan^{-1}(\tan\theta\tanh\phi)$. (Mumbai, 2007)

7. Prove that $\tan\left[i\log\left(\frac{a-ib}{a+ib}\right)^{1/2}\right] = \frac{2ab}{a^2 - b^2}$.

8. If $\log(x+iy) = \log(u+iv)$, show that $\frac{u^2}{x^2} + \frac{v^2}{y^2} = 1$, where $x^2 + y^2 = e^{2\alpha}$.

9. Express $\sin(\alpha + i\beta)$ in real and imaginary parts. Hence find $\sin(\pi/4 + i\log\sqrt{2})$.

10. Find all the roots of the equation (i) $\cos z = 2$, (ii) $\sin z + 3 = 0$.

4.18 SUMMATION OF SERIES – 'C + iS' METHOD

This is the most general method and is applied to find the sum of a series of the form

$$a_0\sin\alpha + a_1\sin(\alpha+\beta) + a_2\sin(\alpha+2\beta) + \ldots$$

or $$a_0\cos\alpha + a_1\cos(\alpha+\beta) + a_2\cos(\alpha+2\beta) + \ldots$$

Procedure. (i) Put the given series = S (say). Converting each term in a series of sines (or cosines).

Then write C (or S) = a similar series of cosines (or sines).

e.g. If $S = a_0\sin\alpha + a_1\sin(\alpha+\beta) + a_2\sin(\alpha+2\beta) + \ldots$

then $C = a_0\cos\alpha + a_1\cos(\alpha+\beta) + a_2\cos(\alpha+2\beta) + \ldots$

(ii) Multiply the series of sines by i and add to the series of cosines, so that

$$C + iS = a_0[\cos\alpha + i\sin\alpha] + a_1[\cos(\alpha+\beta) + i\sin(\alpha+\beta)] + a_2[\cos(\alpha+2\beta) + i\sin(\alpha+2\beta)] + \ldots$$

$$= a_0e^{i\alpha} + a_1e^{i(\alpha+\beta)} + a_2e^{i(\alpha+2\beta)} + \ldots$$

(vi) Sum up the last series using any of the following standard series :

(1) Exponential series i.e., $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots = e^x$

(2) Sine, cosine, sinh or cosh series

i.e., $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots = \sin x$, $\quad 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots = \cos x$

$x + \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots = \sinh x$, $\quad 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \ldots = \cosh x$

(3) Logarithmic series

i.e., $x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots = \log(1+x)$, $\quad -\left[x + \frac{x^2}{2} + \frac{x^3}{3} + \ldots\right] = \log(1-x)$

(4) Gregory's series

i.e., $x - \frac{x^3}{3} + \frac{x^5}{5} - \ldots = \tan^{-1} x$, $\quad x + \frac{x^3}{3} + \frac{x^5}{5} + \ldots = \tanh^{-1} x = \frac{1}{2}\log\frac{1+x}{1-x}$

(5) Binomial series

i.e., $1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{1.2.3}x^3 + \ldots = (1+x)^n$

$1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \ldots = (1+x)^{-n}$

$1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \ldots = (1-x)^{-n}$

(6) Geometric series

i.e., $a + ar + ar^2 + \ldots$ to n terms $= a\frac{1-r^n}{1-r}$, $\quad a + ar + ar^2 + \ldots = a\frac{1}{1-r}$, $|r| < 1$.

(vii) Finally express the sum thus obtained in the form $A + iB$ or this by equating the real and imaginary parts, we get $C = A$ and $S = B$.

Series depending on exponential series

Example 18(d).5. Sum the series $\alpha + \beta\cos\theta + \frac{\beta^2}{2!}\cos(\theta + \theta) + \frac{\beta^3}{3!}\cos(\theta + 2\theta) + \ldots$

Solution. Let $S = \sin\alpha + \beta\sin(\alpha + \theta) + \frac{\beta^2}{2!}\sin(\alpha + 2\theta) + \ldots$

and $C = \cos\alpha + \beta\cos(\alpha + \theta) + \frac{\beta^2}{2!}\cos(\alpha + 2\theta) + \ldots$

$\therefore \quad C + iS = [\cos\alpha + i\sin\alpha] + \beta[\cos(\alpha + \theta) + i\sin(\alpha + \theta)]$

$+ \frac{\beta^2}{2!}[\cos(\alpha + 2\theta) + i\sin(\alpha + 2\theta)] + \ldots$

$= e^{i\alpha} + \beta e^{i(\alpha+\theta)} + \frac{\beta^2}{2!}e^{i(\alpha+2\theta)} + \ldots = e^{i\alpha}\left[1 + \frac{\beta e^{i\theta}}{1!} + \frac{\beta^2 e^{2i\theta}}{2!} + \ldots\right]$

$= e^{i\alpha}e^{\beta e^{i\theta}}$ [expanding $e^{\beta e^{i\theta}}$ as a series in ascending powers of $\beta e^{i\theta}$]

$= e^{i\alpha}e^{\beta(\cos\theta + i\sin\theta)} = e^{\beta\cos\theta + i(\alpha + \beta\sin\theta)}$

Equating imaginary parts from both sides, we have $S = e^{\beta\cos\theta}\sin(\alpha + \beta\sin\theta)$.

Series depending on logarithmic series

Example 18(d).6. Sum the series

$\sin^2\theta - \frac{2}{2}\sin^2(\theta + \theta) + \frac{2^2}{3}\sin^2(\theta + 2\theta) - \frac{2^3}{4}\sin^2(\theta + 3\theta) + \ldots$

(U.P.T.U., 2003) (U.P.T.U., 2004-05)

Solution. Let $S = \sin\theta + \sin 2\theta \sin\theta + \frac{1}{2}\sin 2\theta \sin^2\theta + \frac{1}{2}\sin 3\theta \sin^2\theta + \ldots$ ...(i)

and $C = \cos\theta + \sin 2\theta \sin\theta - \frac{1}{2}\cos 2\theta \sin^2\theta + \frac{1}{2}\cos 3\theta \sin^2\theta - \ldots$ ...(ii)

Adding (i) and (ii), $C + iS = e^{i\theta} + \sin\theta - \frac{e^{i2\theta}\sin^2\theta}{2} + \frac{e^{i3\theta}\sin^2\theta}{2} - \ldots$

$= \log(1 + e^{i\theta}\sin\theta) = \log[1 - i\cos\theta + i\sin\theta\cos\theta]$

$= \log[1 - \cos\theta\sin\theta + i\sin^2\theta]$ [Put $1 + \cos\theta = r\cos\alpha$; $\sin\theta = r\sin\alpha$] ...(iii)

$= \log r\cos\alpha - i\sin\alpha + \log e^{i\alpha} = \log r + i\alpha$

Equating imaginary parts, we have $S = \alpha = \tan^{-1}\left(\frac{\sin^2\theta}{1 + \cos\theta\sin\theta}\right)$. [Refer (iii)]

Newton's Expansion or Binomial Series

Example 53.67. Find the sum to infinity of the series

$1 - \frac{1}{2}\cos\theta + \frac{1.3}{2.4}\cos 2\theta - \frac{1.3.5}{2.4.6}\cos 3\theta + \ldots$ to ∞ $(0 < \theta < \pi)$. (U.P.T.U. 2006)

Solution. Let $C = 1 - \frac{1}{2}\cos\theta + \frac{1.3}{2.4}\cos 2\theta - \frac{1.3.5}{2.4.6}\cos 3\theta + \ldots$

and $S = 0 - \frac{1}{2}\sin\theta + \frac{1.3}{2.4}\sin 2\theta - \frac{1.3.5}{2.4.6}\sin 3\theta + \ldots$

$\therefore C + iS = 1 - \frac{1}{2}e^{i\theta} + \frac{1.3}{2.4}e^{i2\theta} - \frac{1.3.5}{2.4.6}e^{i3\theta} + \ldots$

$= 1 + \left(-\frac{1}{2}\right)e^{i\theta} + \frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)}{1.2}e^{i2\theta} + \frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{1.2.3}e^{i3\theta} + \ldots$

$= (1 + e^{i\theta})^{-\frac{1}{2}} = (1 + \cos\theta + i\sin\theta)^{-\frac{1}{2}} = \left(2\cos^2\frac{\theta}{2} + i.2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)^{-\frac{1}{2}}$

$= \left(2\cos\frac{\theta}{2}\right)^{-\frac{1}{2}}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)^{-\frac{1}{2}} = \left(2\cos\frac{\theta}{2}\right)^{-\frac{1}{2}}\left(\cos\frac{\theta}{4} - i\sin\frac{\theta}{4}\right)$

Equating real parts, we have $C = (2\cos\theta/2)^{-1/2}\cos\theta/4$.

EXERCISE 53.6

Show that the following series are convergent:

1. $\sin\theta + \sin\theta\sin 2\theta + \frac{\sin\theta\sin^2 2\theta}{2!} + \ldots$ to ∞ (P.T.U. 2005)

2. $\frac{\sin(\alpha+\beta)}{1!} + \frac{\sin(\alpha+2\beta)}{2!} + \frac{\sin(\alpha+3\beta)}{3!} + \ldots$ to ∞

3. $\sin\theta + \frac{1}{2}x\sin 2\theta + \frac{1}{3}x^2\sin 3\theta + \ldots$ to ∞ (Kurukshetra, 2005)

4. $\sin\alpha + \frac{1}{2}\sin(\alpha+\beta) + \frac{1}{3}\sin(\alpha+2\beta) + \ldots$ to ∞ (V.T.U. 2006) [Hint: $\sin\alpha + \frac{\sin\beta}{1} + \frac{\sin 2\beta}{2} + \ldots$]

5. $\sin\alpha + \frac{\sin 2\alpha}{2} + \frac{\sin 3\alpha}{3} + \ldots$ to ∞

6. $1 - \frac{1}{2}\cos 2\theta + \frac{1.3}{2.4}\cos 4\theta - \frac{1.3.5}{2.4.6}\cos 6\theta + \ldots$ to ∞ (Kurukshetra, 2005)

7. $x\sin\alpha + \frac{x^2\sin 2\alpha}{2} + \frac{x^3\sin 3\alpha}{3} + \ldots$ to ∞

8. $\cos\alpha\cos\beta + \cos(\alpha+\beta)\cos 2\beta + \cos(\alpha+2\beta)\cos 3\beta + \ldots$ to ∞ (U.P.T.U. 2004)

9. $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots$ to n terms. (Kurukshetra, 2004)

10. $\cos\alpha + \cos\alpha\cos\beta + \cos\alpha\cos^2\beta + \ldots$ to ∞

11. $\sin\alpha + \sin\alpha\cos\beta + \sin\alpha\cos^2\beta + \ldots$ to ∞

12. $1 + x\cos\theta + x^2\cos 2\theta + \ldots + x^n\cos n\theta = \frac{1 - x\cos\theta - x^{n+1}\cos(n+1)\theta + x^{n+2}\cos n\theta}{1 - 2x\cos\theta + x^2}$

33.13 APPROXIMATIONS AND LIMITS

Example 33.10. If $\frac{\sin\theta}{\theta} = \frac{199}{200}$, find an approximate value of θ in radians.

Solution. Since $\frac{\sin\theta}{\theta} = 1 - \frac{1}{200}$ which is nearly equal to 1, $\therefore$ θ must be very small.

We know that $\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \ldots$

$\therefore$ $\frac{\sin\theta}{\theta} = 1 - \frac{\theta^2}{6} - \frac{\theta^4}{5!}$

Omitting θ^4 and higher powers, we have

$\frac{\sin\theta}{\theta} = 1 - \frac{\theta^2}{6} = 1 - \frac{1}{200}$ or $\theta^2 = \frac{1}{300}$. Hence $\theta = 0.1$ radians.

Example 33.11. Solve approximately $\sin\left(\frac{\pi}{3} + \theta\right) = 0.51$.

Solution. Since 0.51 is nearly equal to $\frac{\sqrt{3}}{2}$, which is the value of $\sin\frac{\pi}{3}$, θ must be very small.

$\therefore$ $\sin\left(\frac{\pi}{3} + \theta\right) = \sin\frac{\pi}{3}\cos\theta + \cos\frac{\pi}{3}\sin\theta = \frac{\sqrt{3}}{2}\left(1 - \frac{\theta^2}{2!} + \ldots\right) + \frac{1}{2}\left(\theta - \frac{\theta^3}{3!} + \ldots\right)$

$= \frac{\sqrt{3}}{2} + \frac{1}{2}\theta$, omitting θ^2 and higher powers of θ.

Hence the given equation becomes,

$\frac{\sqrt{3}}{2} + \frac{1}{2}\theta = 0.51$ or $\theta = \frac{1}{50\sqrt{3}}$

or $\theta = \frac{1}{50\sqrt{3}}$ radians $= \frac{\sqrt{3}}{150} = 0.0115$ degrees $= 0.11°$.

EXERCISE 133

1. Given $\frac{\sin\theta}{\theta} = \frac{99\,\pi^2}{100}$, show that $\theta = 0.05\pi$ nearly.
2. If $\frac{\sin\theta}{\theta} = \frac{3\,(10)}{3\,(100)}$, find an approximate value of θ in radians. (Madras, 1963)
3. Given $\theta = \frac{3\,(01)}{3\,(00)}$, find θ approximately.
4. Solve approximately the equation $\cos\left(\frac{\pi}{3} - \theta\right) = 0.49$.

33.14 OBJECTIVE TYPE OF QUESTIONS

EXERCISE 134

Choose the correct answer or fill up the blanks in each of the following questions:

1. If $\theta = \alpha + \sqrt{2}\beta$, then $\sin\theta$ is
   (a) 1 (b) 0 (c) $\frac{1}{2}$ (d) $\sqrt{2}\alpha + \beta$
2. The real part of $\log(\alpha + i\beta)$ is
   (a) $\frac{1}{2}\log(\alpha^2 + \beta^2)$ (b) $\frac{1}{2}\log(\alpha^2 - \beta^2)$ (c) $\tan^{-1}\frac{\beta}{\alpha}$ (d) $\cot^{-1}\frac{\beta}{\alpha}$

2. The number $\sqrt{4}$ is
   (a) a purely imaginary number (b) an irrational number
   (c) a rational number (d) an integer
4. The solution $|3 - x| \leq 2$ is $2 \leq x \leq 5$ represents
   (a) a circle (b) a parabola (c) an ellipse (d) a hyperbola
6. If z is a complex number such that $|z + 3| = 2|z - 3|$, then the value of $|z|$ is
   (a) $\frac{1}{3}\sqrt{65}$ (b) $\frac{1}{3}\sqrt{61}$ (c) $\frac{1}{3}\sqrt{63}$ (d) $\frac{1}{3}\sqrt{67}$
8. If $z = e^{i\theta}$, then the imaginary part of $\left(z + \frac{1}{z}\right)^n$ is
   (a) $\cos n\theta$ (b) $\sin n\theta$ (c) $2\cos n\theta$ (d) $2\sin n\theta$
9. Expression of the form $a\sqrt{-b}$ is called a surd, where a and b are integers is _____.
10. Expression $2\sin\frac{\pi}{6}\cos\frac{\pi}{6}$ in terms of surd is _____.
11. If $f(x) = \frac{x}{x-1}$, then the value of $f(f(x))$ at $x = 2$ is _____.
12. If $z = \cos\theta + i\sin\theta$, then $z^n - \frac{1}{z^n} =$ _____.
13. Imaginary part of $\frac{1}{3} + 2i\sqrt{3}$ is _____.
14. Real part of $\sin(x + iy)$ is _____.
15. If $\frac{x+iy}{x-iy} = \frac{2+3i}{2-3i}$, then $x^2 + y^2 =$ _____ equal to _____.
16. If $\arg(z) = \frac{\pi}{3}$, then the value of $\arg(-z)$ is _____.
17. Imaginary part of $\frac{1}{3+2i}$ is _____.
18. Modulus of $\frac{1}{3+2i}$ is _____.
19. If $\sin\alpha + \sin\beta + \sin\gamma = 3$, then $\cos\alpha + \cos\beta + \cos\gamma =$ _____.
20. $\log(-1) =$ _____.
21. If z is purely real or purely imaginary number, then $z + \bar{z} =$ _____.
22. If $z = 0$, then $\arg z$ is _____.
23. $\cos 3\alpha = C_0\cos^3\alpha - C_1\cos^2\alpha\sin\alpha + C_2\cos\alpha\sin^2\alpha - C_3\sin^3\alpha$, where $C_0 =$ _____, $C_1 =$ _____, $C_2 =$ _____, $C_3 =$ _____.
24. Order of $\sin(x) + \sin(x^2) + \sin(x^3)$ is _____.
25. If $z = 3 + i\sqrt{3}$, then $z^{11} + z^{-11} = 2^{11}$, represents the right triangle, where $a =$ _____ and $b =$ _____.
26. If $\left|\frac{z-1}{z+1}\right| = 1$, then the locus of z is _____.
27. $i > i^{2n}$ is purely real. (True or False)
28. The inequalities $|z| > 1$ and $|z - 2| < |z - 1|$ are equivalent. (Hamilton, 2000) (True or False)
29. Hyperbolic functions are periodic. (True or False)
30. $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$. (True or False)
31. $\sin(ix) = -i\sin hx$. (Marmillan, 2000) (True or False)
32. If the sum and product of two complex numbers are real, then the two numbers must be either real or conjugate. (Marmillan, 2000) (True or False)
33. The sum of a real and the sum of two complex numbers is the sum of their modulus. (True or False)

CHAPTER 20

Calculus of Complex Functions

1. Introduction. 2. Limit and continuity of $f(z)$. 3. Derivative of $f(z)$—Cauchy-Riemann equations. 4. Analytic functions. 5. Harmonic functions— Conjugate systems. 6. Applications to flow problems. 7. Geometric representation of $f(z)$. 8. Some standard transformations. 9. Conformal transformation. 10. Special conformal transformations. 11. Schwarz-Christoffel transformation. 12. Integration of complex functions. 13. Cauchy's theorem. 14. Cauchy's integral formula. 15. Morera's theorem. Cauchy's inequality. Liouville's theorem. Poisson's integral formulae. 16. Series of complex terms—Taylor's series—Laurent's series. 17. Zeros and Singularities of an analytic function. 18. Residues. Residue theorem. 19. Calculation of residues. 20. Evaluation of real definite integrals. 21. Objective Type of Questions.

20.1. INTRODUCTION

In the previous chapter, we have dealt with some elementary complex functions—the exponential, logarithmic, circular and hyperbolic functions, and studied its specific complex values. These functions are useful in the study of fluid mechanics, thermodynamics and electric fields. It, therefore, seems desirable to study the calculus of such functions.

20.2. LIMIT OF A COMPLEX FUNCTION

A function $w = f(z)$ is said to tend to the limit l as z approaches a point z_0 if for every real ε, however small, a corresponding positive real δ can be found such that

$$|f(z) - l| < \varepsilon \quad \text{for} \quad |z - z_0| < \delta$$

i.e., for every z in the δ-disc about z_0 in the z-plane, $f(z)$ lies within the ε-disc about l in the w-plane (Fig. 20.1). In symbols, we write $\lim_{z \to z_0} f(z) = l$.

This definition of limit though similar to that in ordinary calculus, is quite different. For in real calculus x approaches x_0 only along the line whereas here z approaches z_0 from any direction in the z-plane.

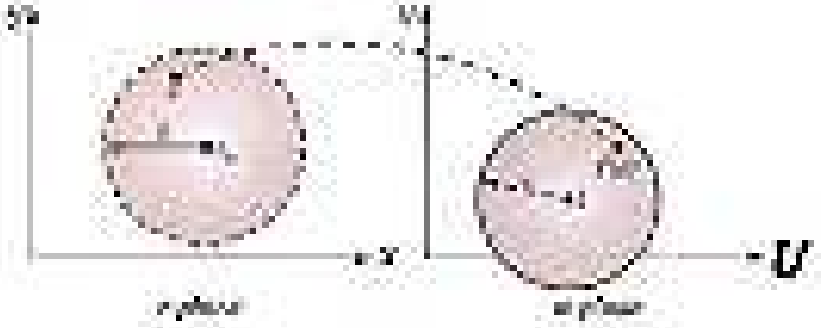

Fig. 20.1

(2) Continuity of $f(z)$. A function $w = f(z)$ is said to be **continuous** at $z = z_0$ if

$$\lim_{z \to z_0} f(z) = f(z_0).$$

Further $f(z)$ is said to be continuous in any region R of the z-plane, if it is continuous at every point of that region.

Also if $w = f(z) = u(x, y) + iv(x, y)$ is continuous at $z = z_0$, then $u(x, y)$ and $v(x, y)$ are also continuous at $z = z_0$, i.e., at $x = x_0$ and $y = y_0$. Conversely if $u(x, y)$ and $v(x, y)$ are continuous at (x_0, y_0), then $f(z) = u + iv$ is also continuous at $z = z_0$. [M.U. 2012, 13 (S)]

20.8 [3] DERIVATIVE OF $f(z)$

Let $w = f(z)$ be a single-valued function of the variable $z = x + iy$. Then we define $df/dz = f'(z)$ as the derivative of $w = f(z)$ and is given by

$$\frac{dw}{dz} = f'(z) = \lim_{\delta z \to 0} \frac{f(z + \delta z) - f(z)}{\delta z},$$

provided the limit exists and has the same value for all the different ways in which δz approaches zero.

Suppose $P(z)$ is fixed and $Q(z + \delta z)$ is a neighbouring point (Fig. 20.2). The point Q may approach P along any straight or curved path in the given region, i.e., it may tend to zero in any manner and dw/dz may not exist. It, therefore, becomes a fundamental problem to determine the necessary and sufficient conditions for dw/dz to exist. The first is provided by the following theorem.

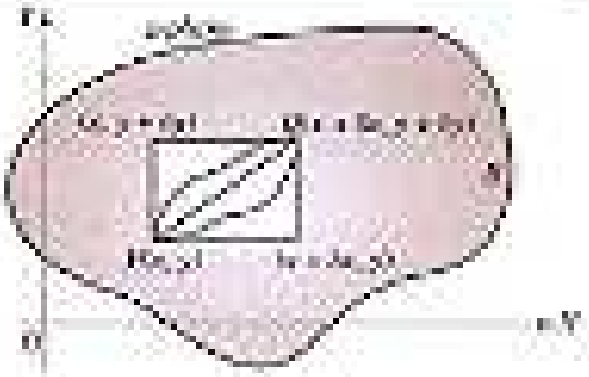

Fig. 20.2

(3) Theorem. *The necessary and sufficient conditions for the derivative of the function $w = u(x, y) + iv(x, y) = f(z)$ to exist for all values of z in a region R, are*

(i) $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ *are continuous functions of x and y in R.*

(ii) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

The relations (ii) are known as **Cauchy-Riemann*** *equations or briefly C-R equations.*

(a) Condition is necessary.

If $f(z)$ possesses a unique derivative at $P(z)$, then

$$f'(z) = \lim_{\delta z \to 0} \frac{f(z + \delta z) - f(z)}{\delta z}$$

$$= \lim_{\substack{\delta x \to 0 \\ \delta y \to 0}} \frac{u(x + \delta x, y + \delta y) + iv(x + \delta x, y + \delta y) - \{u(x, y) + iv(x, y)\}}{\delta x + i\delta y}$$

Since $\delta z \to 0$ in any manner, we can first assume δz to be wholly real and then wholly imaginary. When δz is wholly real, then $\delta y = 0$ and $\delta z = \delta x$.

$$\therefore \quad f'(z) = \lim_{\delta x \to 0} \left\{ \frac{u(x + \delta x, y) - u(x, y)}{\delta x} + i \frac{v(x + \delta x, y) - v(x, y)}{\delta x} \right\} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \quad \ldots(1)$$

When δz is wholly imaginary, then $\delta x = 0$ and $\delta z = i\delta y$.

$$\therefore \quad f'(z) = \lim_{\delta y \to 0} \left\{ \frac{u(x, y + \delta y) - u(x, y)}{i\delta y} + i \frac{v(x, y + \delta y) - v(x, y)}{i\delta y} \right\} = \frac{1}{i} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} = -i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \quad \ldots(2)$$

Now the existence of $f'(z)$ requires the equality of (1) and (2).

---

* Named after Cauchy (p. 344) and the German mathematician Bernhard Riemann (1826–1866) who along with Weierstrass (p. 300) laid the foundations of complex analysis. Riemann introduced the concepts of integration and made basic contributions to number theory and mathematical analysis. He developed the Riemannian geometry which formed the mathematical base for Einstein's relativity theory.

$$\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

Or equating the real and imaginary parts from both sides, we get

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \quad \ldots(2)$$

Thus the necessary conditions for the existence of the derivative of $f(z)$ is that the C-R equations should be satisfied. (V.T.U., 2011)

(b) Condition is sufficient: Suppose $f(z)$ is a single-valued function possessing partial derivatives $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ at each point of the region and the C-R equations (2) are satisfied.

Then by Taylor's theorem for functions of two variables (p. 228)

$$f(z + \delta z) = u(x + \delta x, y + \delta y) + iv(x + \delta x, y + \delta y)$$

$$= u(x, y) + \left[\frac{\partial u}{\partial x}\delta x + \frac{\partial u}{\partial y}\delta y\right] + \ldots + i\left[v(x, y) + \left[\frac{\partial v}{\partial x}\delta x + \frac{\partial v}{\partial y}\delta y\right] + \ldots\right]$$

$$= f(z) + \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)\delta x + \left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right)\delta y$$

(Omitting terms beyond the first powers of δx and δy)

or $$f(z + \delta z) - f(z) = \left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right)\delta x + \left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right)\delta y$$

Now using the C-R equation (2) to replace $\frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y}$ by $-\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$ respectively.

Then $$f(z + \delta z) - f(z) = \left[\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right]\delta x + \left[-\frac{\partial v}{\partial x} + i\frac{\partial u}{\partial x}\right]\delta y = \left[\frac{\partial u}{\partial x} + i\left(\frac{\partial v}{\partial x}\right)\right]\delta x + \left[i\frac{\partial v}{\partial x} - \frac{\partial u}{\partial x}\right]\delta y$$

$$= \left[\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right](\delta x + i\delta y) = \left[\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right]\delta z$$

$$\therefore \quad f'(z) = \lim_{\delta z \to 0}\frac{f(z + \delta z) - f(z)}{\delta z} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \quad \text{or} \quad \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

which by (1) or (1) proves the sufficiency of conditions.

10.4 ANALYTIC FUNCTIONS

A function $f(z)$ which is single-valued and possesses a unique derivative for all values of z in a set of points of a region R is called an **analytic function** *of z in that region. An analytic function is also called a* **regular function** *or an* **holomorphic function**.

A function which is analytic everywhere in the complex plane is known as an **entire function**. *As derivatives of all orders exist at every point, a polynomial of any degree is an entire function.*

A point at which an analytic function ceases to possess a derivative is called a **singular point** *of the function.*

Thm. *If u and v are real single-valued functions of x and y such that $\partial u/\partial x$, $\partial u/\partial y$, $\partial v/\partial x$, $\partial v/\partial y$ are continuous throughout a region R, then the Cauchy-Riemann equations*

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \ldots(1)$$

are both necessary and sufficient conditions for the function $f(z) = u + iv$ to be analytic in R. The derivative of $f(z)$ is then given by (1) of p. 864 or (2) of p. 865.

The real and imaginary parts of an analytic function are called conjugate functions. The relation between two conjugate functions is given by C-R equation (1).

Example 10.11. *Prove that $u = x^2 - y^2$ and $v = -y/(x^2 + y^2)$ cannot be conjugate functions.* (U.P.T.U., 2010; A.M.I.E., 2000)

Solution. We have $u(x, y) = \log(x^2 + y^2)$ and $v(x, y) = 2\tan^{-1}\left(\frac{y}{x}\right)$ [By (1), p. 13.2]

so that $u = \log(x^2 + y^2)$, $v = 2\tan^{-1}\left(\frac{y}{x}\right)$.

$$\therefore \quad \frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}, \frac{\partial v}{\partial y} = \frac{2x}{x^2 + y^2}, \frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2}, \frac{\partial v}{\partial x} = -\frac{2y}{x^2 + y^2}.$$

Since the Cauchy-Riemann equations are satisfied and the partial derivatives are continuous except at $(0, 0)$, hence w is analytic everywhere except at $z = 0$.

$$\therefore \quad \frac{dw}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{2x}{x^2 + y^2} + i\frac{-2y}{x^2 + y^2} = \frac{x - iy}{(x + iy)(x - iy)} = \frac{1}{x + iy} = \frac{1}{z} \text{ or } z \neq 0.$$

Obs. We have used the differentiation of complex functions which are the same as those of real variables. Thus if a complex function is known to be analytic, it can be differentiated just in the ordinary way.

Example 13.11. If $f(z)$ is an analytic function with constant modulus, show that $f(z)$ is constant.
(U.P.T.U., 2008; Mumbai, 2003 S; Madras, 2002; Bhopal, 2005 S)

Solution. If $f(z) = u + iv$ is an analytic function, then

$$|f(z)| = \sqrt{u^2 + v^2} \text{ is constant } \Rightarrow u^2 + v^2 = c^2 \quad \ldots(i)$$

Differentiating (i) partially w.r.t. x and y, we get

$$2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x} = 0; \qquad 2u\frac{\partial u}{\partial y} + 2v\frac{\partial v}{\partial y} = 0$$

or $$u\frac{\partial u}{\partial x} + v\frac{\partial v}{\partial x} = 0 \quad \ldots(ii) \qquad u\frac{\partial u}{\partial y} + v\frac{\partial v}{\partial y} = 0 \quad \ldots(iii)$$

Since $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}; \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ by C-R equations,

$$\therefore \text{ (iii) becomes } -u\frac{\partial v}{\partial x} + v\frac{\partial u}{\partial x} = 0 \quad \ldots(iv)$$

Squaring and adding (ii) and (iv), we obtain

$$u^2\left(\frac{\partial u}{\partial x}\right)^2 + v^2\left(\frac{\partial v}{\partial x}\right)^2 + u^2\left(\frac{\partial u}{\partial y}\right)^2 + v^2\left(\frac{\partial v}{\partial y}\right)^2 = 0$$

or $$(u^2 + v^2)\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right] = 0 \text{ or } \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = 0 \text{ [} \because u^2 + v^2 = c^2 \neq 0\text{]} \quad \ldots(v)$$

Now $$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$$

$$\therefore \quad |f'(z)|^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 = 0 \qquad \text{[By (v)]}$$

or $f'(z) = 0$ or $f(z) =$ constant.

Example 13.12. Show that the function $f(z) = \sqrt{|xy|}$ is not analytic at the origin even though Cauchy-Riemann equations are satisfied there.
(A.M.I.E.T.E., 2005 S; Dibrugarh, 2007)

Solution. If $f(z) = u(x, y) + iv(x, y)$, then $v(x, y) = \sqrt{|xy|}$, $u(x, y) = 0$

At the origin, we have

$$\frac{\partial u}{\partial x} = \lim_{x \to 0} \frac{u(x, 0) - u(0, 0)}{x} = \lim_{x \to 0} \frac{0 - 0}{x} = 0$$

$$\frac{\partial u}{\partial y} = \lim_{y \to 0} \frac{u(0, y) - u(0, 0)}{y} = \lim_{y \to 0} \frac{0 - 0}{y} = 0$$

$$\frac{\partial v}{\partial x} = \lim_{x \to 0} \frac{v(x, 0) - v(0, 0)}{x} = \lim_{x \to 0} \frac{0 - 0}{x} = 0$$

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{x - x}{\Delta x} = 0$$

$$\frac{dy}{dx} = \frac{d}{dx}(x) = \lim_{\Delta x \to 0} \frac{\Delta x}{\Delta x} = 1$$

[illegible] equivalence class modulo $\sim$ the origin

Moreover $$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{\frac{x}{1+x^2} - 0}{x - 0}$$

$$= \lim_{x \to 0} \frac{x}{x(1+x^2)} \quad \text{[illegible] } x \text{ [illegible] along the line } y = x$$

$$= \frac{1}{1+0^2} \quad \text{[illegible] as one converges}$$

[illegible] closed and open. Because $f(x)$ is not equivalent to the origin

Proposition [illegible] Prove that for [illegible] is given by

$$f(x) = \frac{x^2}{1+x^2} \quad \text{for } x \in [illegible]$$

is [illegible] that the [illegible] is not the limit of a sequence $\{x_n\}$ in the [illegible] such that [illegible]

Proof [illegible]

$$\lim_{x \to 0} f'(x) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{\frac{x^2}{1+x^2} - 0}{x} = \lim_{x \to 0} \frac{x}{1+x^2} = 0$$

$$\lim_{x \to 0} f'(x) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{\frac{x^2}{1+x^2} - 0}{x} = \lim_{x \to 0} \frac{x}{1+x^2} = 0$$

$f(x)$ is [illegible]

Thus $\lim_{x \to 0} f'(x) = 0$ [illegible] $x \to 0$ from [illegible] $x \to 0$ from [illegible] [illegible] [illegible]

Hence at $(0, 0)$, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

Thus the C-R equations are satisfied at the origin.

But $f'(0) = \lim_{z \to 0} \frac{f(z) - f(0)}{z} = \lim_{z \to 0} \frac{f(z)}{z} = \lim_{z \to 0} \frac{x^3 - y^3 + i(x^3 + y^3)}{(x^2 + y^2)(x + iy)}$,

If $z \to 0$ along the path $y = mx$, then $f'(0) = \frac{1 - m^3 + i(1 + m^3)}{(1 + m^2)(1 + im)}$

which assumes different values for various m. So $f'(0)$ is not unique at $(0, 0)$ i.e., $f'(0)$ does not exist. Thus $f(z)$ is not analytic at the origin even though it satisfies the C-R equations there.

Example 3.13. Show that the polar form of Cauchy-Riemann equations are

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}. \qquad \text{(U.P.T.U., 2005) (U.P.T.U., 2006)}$$

Deduce that $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$ (Meerut, 2003; Kurukshetra, 2006)

Solution. If (r, θ) be the new system of coordinates of a point whose cartesian coordinates are (x, y), then $x = r\cos\theta$, $y = r\sin\theta$

$$\therefore \quad u + iv = f(z) = f(re^{i\theta})$$

where u and v are now expressed in terms of r and θ.

Differentiating (1) partially w.r.t. r and θ, we have

$$\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r} = f'(re^{i\theta})\, e^{i\theta} \qquad \ldots(1)$$

and $$\frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta} = f'(re^{i\theta})\, ire^{i\theta} = ir\left[\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right]$$

Equating real and imaginary parts, we get

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \qquad \ldots(i) \qquad\qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta} \qquad \ldots(ii)$$

Differentiating (i) partially w.r.t. r, we get

$$\frac{\partial^2 u}{\partial r^2} = -\frac{1}{r^2}\frac{\partial v}{\partial \theta} + \frac{1}{r}\frac{\partial^2 v}{\partial \theta \partial r} \qquad \ldots(iii)$$

Differentiating (ii) partially w.r.t. θ, we have

$$\frac{\partial^2 u}{\partial \theta^2} = -r\frac{\partial^2 v}{\partial r \partial \theta} \qquad \ldots(iv)$$

Dividing (iv) by r and adding to (iii), we get

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = -\frac{1}{r^2}\frac{\partial v}{\partial \theta} + \frac{1}{r}\frac{\partial^2 v}{\partial \theta \partial r} + \frac{1}{r}\left(\frac{1}{r}\frac{\partial v}{\partial \theta}\right) + \frac{1}{r^2}\left(-r\frac{\partial^2 v}{\partial r \partial \theta}\right) = 0 \qquad \left[-\frac{\partial^2 v}{\partial r \partial \theta} = \frac{\partial^2 v}{\partial \theta \partial r}\right]$$

3.25 (I) HARMONIC FUNCTIONS

If $f(z) = u + iv$ be an analytic function in some region of the z-plane, then the Cauchy-Riemann equations are satisfied

i.e., $$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \ldots(i) \qquad \text{and} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \qquad \ldots(ii)$$

Differentiating (i) with respect to x and (ii) with respect to y, we obtain

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \qquad \ldots(iii) \qquad \text{and} \qquad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x} \qquad \ldots(iv)$$

Adding (iii) and (iv) and assuming that $\frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial y \partial x}$, we get

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)=0 \tag{...}$$

Similarly, by differentiating (1) with respect to y and (2) with respect to x and subtracting we obtain

$$\left(\frac{\partial}{\partial x}-\frac{\partial}{\partial y}\right)=0 \tag{...}$$

To solve the system we multiply (...) by i and add to (...) to get

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)=0$$

Thus f is a function of $z=x+iy$ only. We call such functions analytic functions of z. For this reason, they are sometimes also called holomorphic functions. [illegible]

(2) **Orthogonal systems:** Consider the two families of curves

$$\phi(x,y)=c_1 \qquad (1) \qquad \text{and} \qquad \psi(x,y)=c_2 \tag{...}$$

Differentiating (1), we get $\frac{\partial\phi}{\partial x}+\frac{\partial\phi}{\partial y}\frac{dy}{dx}=0$

$$\frac{dy}{dx}=-\frac{\frac{\partial\phi}{\partial x}}{\frac{\partial\phi}{\partial y}}=-\frac{\phi_x}{\phi_y} \tag{...}$$

Similarly (2) gives $\frac{dy}{dx}=-\frac{\psi_x}{\psi_y}$ (...)

$\therefore$ $\phi_x\psi_x+\phi_y\psi_y=0$ (...) which shows the orthogonality.

Hence every analytic function $f(z)=u+iv$ gives rise to two orthogonal families of curves $u(x,y)=c_1$ and $v(x,y)=c_2$ which form an orthogonal system.

[illegible] APPLICATIONS TO FLUID MECHANICS

As the real and imaginary parts of an analytic function are the solutions of the Laplace equation in two variables, the conjugate functions provide solutions to a number of field and flow problems.

As an illustration, consider the two-dimensional motion of an incompressible fluid in the (x,y)-plane. Assuming the flow to be in planes parallel to the xy-plane, the velocity $\vec{V}$ of a fluid particle is represented as

$$\vec{V}=u\vec{i}+v\vec{j} \tag{...}$$

Since the motion is irrotational, therefore, by [illegible] there exists a scalar function $\phi(x,y)$ such that

$$\vec{V}=\nabla\phi=\frac{\partial\phi}{\partial x}\vec{i}+\frac{\partial\phi}{\partial y}\vec{j} \tag{...}$$

The function $\phi(x,y)$ is called the velocity potential and the curves $\phi(x,y)=c$ are known as equipotential lines.

Hence from (...) we get $u=\frac{\partial\phi}{\partial x}$ and $v=\frac{\partial\phi}{\partial y}$ (...)

Also the fluid being incompressible, $\nabla\cdot\vec{V}=0$. By (...) this gives $\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$

Substituting the values of u and v from (...) we get $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}\right)\phi=0$ (...)

which shows that the velocity potential ϕ is harmonic. It follows that there exists a conjugate harmonic function $\psi(x,y)$ such that $\psi=\psi(x,y)$ (...)

is analytic.

Also ψ is constant along any curve of the family $\psi(x,y)=c$ given by

$$\frac{dy}{dx}=-\frac{\partial\psi/\partial x}{\partial\psi/\partial y}=-\frac{v}{u} \qquad \text{[illegible] equation}$$

$$u\,dy-v\,dx=0 \qquad \text{[illegible]}$$

This shows that the velocity at any point (x,y) is along the tangent to the curve $\psi(x,y)=c$. The curves $\psi(x,y)=c$ are called the stream lines. Such a curve can be drawn to represent [illegible] and $\psi(x,y)=c$ is called the stream function. Since the equipotential lines $\phi(x,y)=c$ and the stream lines $\psi(x,y)=c$ are orthogonally

Proof. (a)

$$\begin{aligned} \frac{\partial \phi}{\partial \theta} &= \frac{\partial \phi}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial \phi}{\partial y}\frac{\partial y}{\partial \theta} + \frac{\partial \phi}{\partial z}\frac{\partial z}{\partial \theta} \\ &= v_x - v_y \end{aligned}$$

(b) The energy density of the fluid velocity $\cdots \| \cdots \| \cdots \| \cdots \|$ velocity $\cdots$

The n-by-n (first and n-by-1 fully connected) layer is called the **navigation parameter**. Similarly, the composite polynomial $\alpha(x)$ over the ordinary is expressed, on a different type of $\mathbb{R}$-bundle and k closely flow, by decomposition and generalized fields, the system $\dot{q}_{x,y} + \cdots$ small cycle $\alpha(\cdot) + \cdots$ zero symmetrical form and form of flows. In Euler flow conditions, the momentum $\alpha(\cdot) + \cdots$ and $\alpha(x, y) + \cdots$ zero dynamics as small enough and heat flow theorem integrable by

(a) Let α be $\cdots$ on some field up x, y and $\cdots$ cross.

Theorem (Hilbert). *If α $\cdots$ $\cdots$ the fundamental vector property, a complex case, the physical theorem of $\cdots$ $\cdots$ $\cdots$ is*

$$\cdots \qquad \cdots \qquad \cdots \| \cdots \| \text{ the value, where } \frac{\partial \phi}{\partial x} \text{ gas, some } \cdots$$

Suppose α is a locally connected space, and define the k-algebra of equations

(a) a unit or a non-constant α is locally integrable up positive

$$\frac{\partial \phi}{\partial x} = \frac{\partial \phi}{\partial y} \qquad \cdots \text{on} \qquad \frac{\partial \phi}{\partial x} = \frac{\partial \phi}{\partial y} \tag{1.1}$$

(b) for α,

$$\frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x}\left[x^2 + y^2 + \frac{\lambda}{x^2+y^2} \right] = 2x - \frac{2\lambda x}{(x^2+y^2)^2}$$

Assuming α is $\cdots$ on, we get $\phi = -\frac{\lambda}{x^2+y^2} + \cdots$ $\alpha(x)$ relative up to the $\cdots$ on which α function $\alpha(\cdot)$.

(a) α is given on $\cdots$ on $\alpha = \int_{x_0}^{x} \frac{\partial \phi}{\partial x}\, dx + \alpha(x_0) = \cdots$ for $\alpha = \int_{x_0}^{x} \frac{\partial \phi}{\partial x}\, dx$

Assume $\alpha'(x_0) = 0$, $\cdots$ $\alpha(x) + \cdots$ a $\cdots$ differentiable neighborhood.

Then $\qquad \phi = -\lambda \ln \alpha = \int_{x_0}^{x} \frac{\partial \phi}{\partial x}\, dx + \cdots$

(Hilbert's Hilbert Theorem) is satisfied?

By Taylor $\qquad \frac{\partial \phi}{\partial x} = \frac{\partial \phi}{\partial x} + \left(\frac{\partial \phi}{\partial x} - \frac{\partial \phi}{\partial x} \right) + \frac{1}{2}\left(\frac{\partial \phi}{\partial x} - \frac{\partial \phi}{\partial x} \right) + \left[-2\lambda - \frac{8\lambda x^2}{(x^2+y^2)^3} \right] + \left[2\lambda - \frac{8\lambda y^2}{(x^2+y^2)^3} \right]$

By Hilbert Theorem's method, one sequence decides to become $\alpha(x)$, one neighborhood for x and y, by $\mathbb{R}$.

(a)

$$\frac{\partial \phi}{\partial x} = \left| \frac{\partial \alpha}{\partial x} - \frac{\lambda}{x} \right|$$

Setting satisfies $\alpha(x)$ on α is given $\alpha = \alpha(x)$ $\cdots$ α and α $\cdots$ on α relative α on α on α satisfies α on α and α.

Assume $\alpha = \alpha(x, y)$ is given $\phi = \alpha(x) + \alpha(y)$ on α space

$$\alpha = \int_{x_0}^{x} \alpha(x)\, dx + \alpha_{x_0} \qquad \alpha = \int_{y_0}^{y} \alpha(y)\, dy + \alpha$$

$$\begin{aligned} \phi(x) &= \alpha(x_0) + \alpha(x) - \alpha(x_0) + \phi(x) \\ &= \alpha\left[\frac{x+y}{x} - \frac{x}{x} \right] = \alpha\left[\frac{x+y}{x} - \frac{x}{x} \right] \end{aligned} \tag{2.1}$$

Since α satisfies $\alpha(x)$ on α is $\alpha(x)$ satisfying α on α for α on α on α on α on α and α satisfying α $\cdots$ $\cdots$ α on α

$$\int \alpha(x) = \phi(x) = \alpha(x) + \alpha(x) \tag{2.2}$$

(a) α is α on α on α α α on α α α

Hence $$y = \ln\left[\tan\left(x^2 + \frac{\pi}{2}\right) + \sec\left(x^2 + \frac{\pi}{2}\right)\right] - 2xy + \frac{x}{x^2 + y^2} + c.$$

Example 20.5. Find the stationary points on the curve $\sin 2x + \sinh 2y = 2(x + y)$ and determine their nature. (C.I.T., 2006; Anna, 2003)

Solution. Let $f(x, y) = \sin 2x + \sinh 2y - 2x - 2y$, where $z = \sin 2x + \sinh 2y - 2x - 2y$

$$\therefore \quad f'(x) = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\frac{dy}{dx} = \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\frac{\partial z}{\partial y} \qquad \text{[By C.R. equation 4]}$$

$$= \frac{\cosh 2y - \cos 2x(2\cos 2x - \sin 2x \cdot 0)}{(\cosh 2y - \cos 2x)^2} - \frac{\sin 2x(\sin 2x - 2\cosh 2y)}{(\cosh 2y - \cos 2x)^2}$$

$$= \frac{2\cos 2x \cosh 2y - 2}{(\cosh 2y - \cos 2x)^2} + \frac{2\sin 2x \sinh 2y}{(\cosh 2y - \cos 2x)^2}$$

By 20.4 (c). The curve is stationary, we express $f'(x) = 0$ in terms of x by putting $y = x$ and $y = 0$.

$$\therefore \quad f'(x) = \frac{2\cos 2x - 2}{(1 - \cos 2x)^2} = 0 \Rightarrow \frac{-2}{1 - \cos 2x} = \frac{-2}{2\sin^2 x} = -\operatorname{cosec}^2 x$$

Integrating w.r.t. x, we get $f(x) = \cot x + x$, taking the constant of integration as zero as every curve does not meet the axis of x.

Example 20.6. Determine the stationary points on the curve $f(x) = y$ if $x = \frac{\cos x + \sin x \sinh y - e^{2y}}{2\cos x - \cosh y}$ and $f(x) = 0$. (C.I.T., 2005; Anna, 2004)

Solution. We have $x - y = \dfrac{\cos x + \sin x \sinh y - e^{2y}}{2\cos x - \cosh y}$

$$\therefore \quad \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\frac{dy}{dx} = \frac{\sin x - \cos x \cosh y + 1 - e^{2y}\sin x}{2\cos x - \cosh y)^2} \qquad \ldots(i)$$

and

$$\frac{\partial z}{\partial y} - \frac{\partial z}{\partial x}\frac{dx}{dy} = \frac{(\sin x - \cosh y)e^{2y} + \cos x + \sin x - e^{2y}\sinh y}{2(\cos x - \cosh y)^2}$$

or

$$-\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\frac{dy}{dx} = \frac{\sin x + \cos x \sinh y + e^{2y}(\cos x - \cosh y - \sinh y)}{2(\cos x - \cosh y)^2} \qquad \ldots(ii)$$

Subtracting (ii) from (i), we get

$$2\frac{\partial z}{\partial x} = \frac{\sin x - \cos x \cosh y - \sin x \sinh y + (\cos x)\sinh y + 1}{-e^{2y}(\sin x + \cos x - \cosh y - \sinh y)}$$

Adding (i) and (ii), we have

$$-2\frac{\partial z}{\partial y} = \frac{\sin x - \cos x \cosh y + \sin x \sinh y + \cos x \sinh y + 1}{e^{2y}(1 - \sin x + \cos x - \cosh y - \sinh y)}$$

Thus $$f'(x) = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}\frac{dy}{dx} = \frac{1 - \cos x}{2(1 - \cos x)^2} \qquad \text{[Putting } x = x \text{ and } y = 0\text{]}$$

$$= \frac{1}{2(1 - \cos x)} = \frac{1}{4\sin^2 x/2} = \frac{1}{4}\operatorname{cosec}^2\frac{x}{2} \quad \text{or} \quad f(x) = -\frac{1}{2}\cot\frac{x}{2} + c$$

Since $f(\pi/2) = 0$, $\quad 0 = -\frac{1}{2}\cot\frac{\pi}{4} + c$, where $c = \frac{1}{2}$

Hence $$f(x) = \frac{1}{2}\left(1 - \cot\frac{x}{2}\right)$$

Example 20.28. If $u+iv$ is an analytic function of $z=x+iy$ and $u-v=e^x(\cos y-\sin y)$, find $u+iv$ in terms of z. (Kurukshetra, 2003)

Solution. Let $f(z)=u+iv$. Using C-R equations in polar coordinates (Ex. 20.6),

$$r\frac{\partial v}{\partial r}=\frac{\partial u}{\partial \theta}=-2r^2\sin 2\theta+r\sin\theta \qquad ...(1)$$

$$-\frac{1}{r}\frac{\partial u}{\partial \theta}=\frac{\partial v}{\partial r}=2r\cos 2\theta-\cos\theta \qquad ...(2)$$

$\therefore$ On given, $\quad \dfrac{\partial u}{\partial r}-\dfrac{\partial v}{\partial r}=-2r\sin 2\theta+\sin\theta$

Integrating w.r.t. r,

$$u-v=-r^2\cos 2\theta+r\sin\theta+\phi(\theta) \qquad \text{where } \phi(\theta) \text{ is an arbitrary function.}$$

$\therefore$ $$\frac{\partial u}{\partial \theta}-\frac{\partial v}{\partial \theta}=2r^2\sin 2\theta+r\cos\theta+\phi'(\theta) \qquad ...(3)$$

From (1) and (3), we get

$$-2r^2\cos 2\theta+r\cos\theta+\frac{\partial u}{\partial \theta}=-2r^2\sin 2\theta+r\cos\theta+\phi'(\theta)$$

$\therefore$ $\phi'(\theta)=0 \quad \Rightarrow \quad \phi(\theta)=c$

Thus $u-v=-r^2\sin 2\theta+r\sin\theta+c$ is the conjugate harmonic of v.

Now v will be harmonic if it satisfies the Laplace equation $\dfrac{\partial^2 v}{\partial r^2}+\dfrac{1}{r}\dfrac{\partial v}{\partial r}+\dfrac{1}{r^2}\dfrac{\partial^2 v}{\partial \theta^2}=0$

From (1), $\dfrac{\partial^2 v}{\partial r^2}=-4r\sin 2\theta+\sin\theta$. From (2), $\dfrac{\partial^2 v}{\partial \theta^2}=2r\sin 2\theta$

$$\therefore \quad \frac{\partial^2 v}{\partial r^2}+\frac{1}{r}\frac{\partial v}{\partial r}+\frac{1}{r^2}\frac{\partial^2 v}{\partial \theta^2}=-2r\sin 2\theta+\frac{1}{r}(2r\cos 2\theta-\cos\theta)+\frac{1}{r^2}(-4r^2\sin 2\theta+r\sin\theta)$$

$$=-2r\sin 2\theta+\frac{2}{r}\cos 2\theta-\frac{1}{r}\cos\theta-4r\sin 2\theta+\frac{1}{r}\cos\theta=0$$

Hence v is harmonic.

Example 20.29. (a) Find the orthogonal trajectories of the family of curves
$x^3y-xy^3=c$, where c is a constant.

(b) Show that the curves $x^3-3xy^2=a$ and $3x^2y-y^3=b$ cut orthogonally. (Kurukshetra, 2005; GGSIPU, 2008)

Solution. (a) Let $u=x^3y-xy^3$. This is the family of curves $u(x, y)=$ constant. To find the required trajectories if $f(z)=u+iv$ is analytic.

Now $$\frac{\partial u}{\partial x}=4x^3-12xy^2, \quad \frac{\partial u}{\partial y}=4y^3-12x^2y$$

$\therefore$ $$\frac{\partial v}{\partial y}=\frac{\partial u}{\partial x}=4x^3-12xy^2$$

Integrating, $\quad v=4x^3y-4xy^3+\phi(x)$

Differentiating partially w.r.t. x,

$$12x^2y-4y^3+\frac{d\phi(x)}{dx}=\frac{\partial v}{\partial x}=-\frac{\partial u}{\partial y}=-4y^3+12x^2y$$

$\therefore$ $$\frac{d\phi(x)}{dx}=0 \quad \text{or} \quad \phi=\text{constant}$$

Thus the required orthogonal trajectories are $v=$ constant or $4x^3y-4xy^3=$ constant.

(b) Writing as, $R\cos\theta=x^3-3xy^2=a$ and $R\sin\theta=3x^2y-y^3=b$,

we have $r=R$, $\theta=\tan^{-1}\dfrac{b}{a}$. $\quad a=R\cos\theta=r^3(\cos^3\theta-3\sin^2\theta\cos\theta)=r^3\cos 3\theta$, $b=r^3\sin 3\theta$

That w is an analytic function.

That $f(z) = u + iv$, gives the curves $u = c_1$ and $v = c_2$ which cut orthogonally.

Example 20.13. *Two concentric circular cylinders of radii r_1, r_2 ($r_2 > r_1$) are raised to potentials v_1 and v_2 respectively. Using complex potential show that the capacity per unit length of the capacitor formed by them is $2\pi\varepsilon/\log(r_2/r_1)$. Also find the electric intensity at any point.*

Solution. We have $\phi + i\psi = a \log(x + iy) + c$, where $a = \alpha + i\beta = re^{i\theta}$

$\therefore$ $\qquad \phi = a \log r + c_1 \quad$ and $\quad \psi = c\theta$

so that $\qquad \phi_1 = a \log r_1 + c_1, \quad \phi_2 = a \log r_2 + c_1$

Thus the potential difference $= \phi_2 - \phi_1 = a \log r_2 - \log r_1$

Also the total charge (or flux) $= \int_0^{2\pi} \psi \, d\theta = \int_0^{2\pi} c \, d\theta = 2\pi c$.

The capacitance being the charge required to maintain a unit potential difference : the capacitance without dielectric

$$= \frac{\text{charge}}{\text{potential difference}} = \frac{2\pi c}{a \log r_2 - \log r_1} = \frac{2\pi}{\log(r_2/r_1)}.$$

A medium of dielectric constant ε increases the potential difference to ε times that in vacuum for the same charge. Thus the capacitance with dielectric $= 2\pi\varepsilon/\log(r_2/r_1)$.

Example 20.14. *If $f(z)$ is a regular function of z, prove that*

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2 \qquad \text{(U.P.T.U., 2005; Kanpur, 2001)}$$

or $\qquad \nabla^2 |f(z)|^2 = 4|f'(z)|^2 \qquad \text{(Madras, 2006)}$

Solution. Let $f(z) = u(x, y) + iv(x, y)$ so that $|f(z)|^2 = w^2 = u^2 + v^2 = \phi(x, y)$ (say).

$$\therefore \quad \frac{\partial \phi}{\partial x} = 2u\frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x} \text{ and } \frac{\partial^2 \phi}{\partial x^2} = 2\left[u\frac{\partial^2 u}{\partial x^2} + \left(\frac{\partial u}{\partial x}\right)^2 + v\frac{\partial^2 v}{\partial x^2} + \left(\frac{\partial v}{\partial x}\right)^2\right]$$

Similarly, $\qquad \frac{\partial^2 \phi}{\partial y^2} = 2\left[u\frac{\partial^2 u}{\partial y^2} + \left(\frac{\partial u}{\partial y}\right)^2 + v\frac{\partial^2 v}{\partial y^2} + \left(\frac{\partial v}{\partial y}\right)^2\right]$

Adding, we have

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 2u\left[\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right] + v\left[\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right] + 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2\right] \quad \ldots(1)$$

Since u, v have to satisfy Cauchy-Riemann equations and the Laplace's equation

$$\therefore \quad \left(\frac{\partial u}{\partial x}\right)^2 = \left(\frac{\partial v}{\partial y}\right)^2, \left(\frac{\partial u}{\partial y}\right)^2 = \left(-\frac{\partial v}{\partial x}\right)^2 \text{ and } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}.$$

Thus (1) takes the form $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2\right]$

Hence $\qquad \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2 \quad$ or $\quad \nabla^2 |f(z)|^2 = 4|f'(z)|^2$.

EXERCISE 20.2

1. If $f(z) = \int_0^z \frac{3\zeta^2 + 7\zeta + 2}{\zeta - 3}\, d\zeta$, prove that $f(1 + i) = 10\pi i - 6$ by using a suitable contour from $\zeta = 0$ to $\zeta = 1 + i$ along the curve $\zeta = \zeta^2$.

2. (a) Show that $f(z) = \bar{z}z$ is nowhere differentiable except at the origin. (Anna Univ. 2007)
   (b) $f(z) = x + 2y$ is not an analytic function in the complex plane. (JNTU, 2003)

3. If $f(z)$ is a regular function, show that $|f'(z)|^2 = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ (Nagarjuna, 2003)

4. Prove that the function $f(z) = ax^2 + bxy + cy^2 + idx^2 + exy + fy^2$ is analytic if $a = b = c = d = e = f = 0$. (Nagarjuna, 2008)

5. Show that $f(z)$ is analytic where $f(z) = \frac{1}{z}$. (JNTU, 2007)

6. Determine which of the following functions are analytic:
   (a) $\log(x^2 + y^2)$ (b) $\frac{x - iy}{x^2 + y^2}$ (c) $\sin z$

7. (a) Determine p such that the function $f(z) = \frac{1}{2}\log(x^2 + y^2) + i\tan^{-1}(px/y)$ is an analytic function.
   (Mumbai, 2002; AMIETE, 2007)
   (b) Show that $u(x, y) = \frac{1}{2}\log(x^2 + y^2)$ is harmonic and find its harmonic conjugate function. (UPTU, 2007)

8. Show that each of the following functions is not analytic at any point:
   (a) $\bar{z}$ (UPTU, 2005) (b) $z\bar{z}$

9. Show that $u + iv = (x - iy)/(x^2 + y^2)$ where $u + iv$ is not an analytic function of $x + iy$ even though u and v satisfy Cauchy-Riemann equations.

10. Show that $f(z) = \begin{cases} xy^2(x + iy)/(x^2 + y^4), & z \neq 0 \\ 0, & z = 0 \end{cases}$
    is such that $f(z)$ tends to zero along every radius vector $y = mx$ but $f(z)$ is not analytic at the origin. (UPTU, 2004)

11. Verify $f(z) = \frac{(r^2 + 1)\cos\theta - i(r^2 - 1)\sin\theta}{r}$, $z = re^{i\theta}$ is analytic or not. (UPTU, 2005)

12. Examine the nature of the function $f(z) = \frac{x^3y(y - ix)}{x^6 + y^2}$, $z \neq 0$, $f(0) = 0$. (Nagpur, 2005)

13. Show that $f(z) = \sqrt{|xy|}$ is not analytic at the origin although C-R equations are satisfied at the origin.
    (Delhi and Allahabad Univ. 2005)

14. Determine the analytic function whose real part is
    (a) $x^3 - 3xy^2 + 3x^2 - y^2$ (Nagpur, 2004; Kerala, 2006)
    (b) $e^x(x\cos y - y\sin y)$ (UPTU, 2004; VTU, 2006)
    (c) $e^{-x}(x\sin y - y\cos y)$ (UPTU, 2004)
    (d) $e^x(x\cos 2y - y\sin 2y)$ (VTU, 2004 & Mumbai, 2005; Kerala, 2006)
    (e) $\sin x\cosh y + 2\cos x\sinh y + x$ (UPTU, 2005)
    (f) $\sin 2x/(\cosh 2y - \cos 2x)$ (VTU, 2006; Kerala, 2006)

15. Find the regular function whose imaginary part is
    (a) $e^{-x}(x\sin y - y\cos y)$ (b) $\frac{y}{x^2 + y^2} - 2y$ (c) $\frac{\sin 2x}{\cosh 2y - \cos 2x}$ (Mumbai, 2004)
    (d) $e^x(x\cos y - y\sin y)$ (e) $e^{x^2 - y^2}(x\cos 2xy - y\sin 2xy)$ (UPTU, 2005)

16. Find the analytic function $f(z) = u + iv$, if
    (a) $u - v = e^x(\cos y - \sin y)$ (Mumbai, 2000; VTU, 2004; RGPV, 2005)
    (b) $u + v = \frac{\cos\frac{1}{2}\theta + \sin\frac{1}{2}\theta + r^{-1}}{2\cos\theta - r - r^{-1}}$, where $r = \left|\frac{z}{2}\right| < 1$ (Mumbai, 2003)
    (c) $u + v = \frac{\sin 3x}{e^{2y} + e^{-2y} - 2\cos 3x}$ (VTU, 2004)

17. An electrostatic field in the xy-plane is given by the potential function $u = 3x^2y - y^3$. Find the stream function.

18. If the potential function is $\log(x^2 + y^2)$, find the stream function and the complex potential function.

19. Prove that $u = x^2 - y^2$ and $v = \frac{-y}{x^2 + y^2}$ are harmonic functions, but uv is not harmonic in any region.
    (UPTU, 2004)

28. Show that the function $u = x^2 + y^2$ is harmonic and find a conjugate harmonic function v such that $f(z) = u + iv$ is analytic in terms of z. (Answer: $v = 2xy + c$)

29. Show that $u = x^2 - y^2$ and $v = \frac{y}{x^2 + y^2}$ are harmonic functions in a region not containing the origin, but that uv is not harmonic there.

30. Find the orthogonal trajectories of the family of curves
(a) $x^2 + y^2 = c^2$. (Answer: $y = mx$) (b) $x^2 - y^2 = c^2$. (Answer: $xy = k$) (c) $y = cx^2$. (Answer: $x^2 + 2y^2 = k$)

31. In two dimensional fluid flow, the stream function $\psi(x, y)$ is given. Find the velocity potential ϕ:
(a) $\psi = 2xy$ (b) $\psi = x^2 - y^2$ (c) $\psi = \frac{y}{x^2 + y^2}$

32. Find the analytic function $f(z) = u + iv$, given
(a) $u = x^2 - y^2 + x$ (b) $u = e^{-x}(x \sin y - y \cos y)$ (c) $v = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 3x$

33. If $f(z)$ is an analytic function of z, show that
$$\left(\frac{\partial}{\partial x}|f(z)|\right)^2 + \left(\frac{\partial}{\partial y}|f(z)|\right)^2 = |f'(z)|^2$$
(Hint: Write $f(z) = u + iv$ and use the Cauchy-Riemann equations.)

34. If $f(z)$ is an analytic function of z, prove that
(a) $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\log|f(z)| = 0$ (Harmonic, where $f(z) \neq 0$) (b) $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 2|f'(z)|^2$
(c) $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$ (Harmonic)

35. Prove that $u = 3x^2y - 2x^3 + y^3 - 3y^2$ is harmonic in every region which does not include the point $(1, 2)$. Find a function v such that $u + iv$ is an analytic function of z. Express v in terms of z. (Hint: Let $x = z$ and $y = 0$.)

13.13 GEOMETRICAL REPRESENTATION OF $w = f(z)$

To find the geometrical representation of a function of a complex variable, it requires a departure from the usual practice of cartesian plotting, where we associate a curve to a real function $y = f(x)$.

In the complex domain, the function $w = f(z)$
i.e., $u + iv = f(x + iy)$ (1)
involves four real variables x, y, u, v. Hence a four dimensional region is required to plot (1) in the cartesian fashion. As it is not possible to have 4-dimensional graph papers, we make use of two complex planes, one for the variable $z = x + iy$, and the other for the variable $w = u + iv$. If the points z describe some curve C in the z plane, the point $w = f(z)$ moves along a corresponding curve C' in the w-plane. Given to each point (x, y) there corresponds a point (u, v) (Fig. 13-2). We then, say that a curve C in the z-plane is mapped into the corresponding curve C' in the w-plane by the function $w = f(z)$, which is also a mapping or transformation of the z-plane into the w-plane.

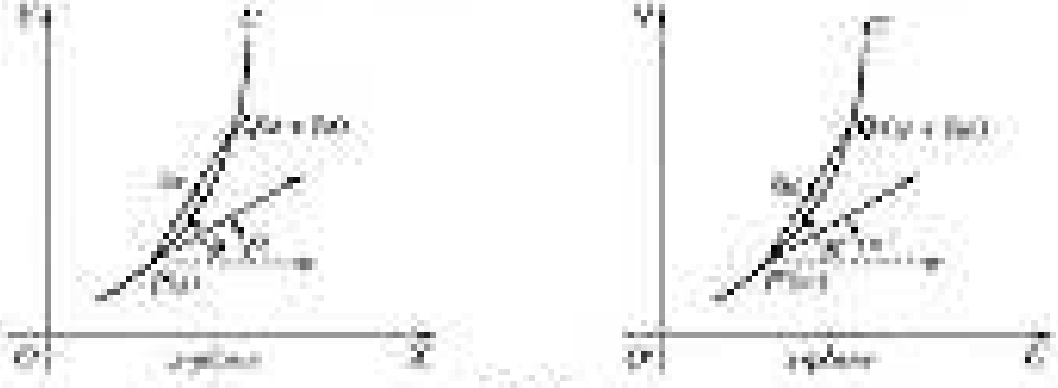

Fig. 13-2

13.14 SOME STANDARD TRANSFORMATIONS

(1) Translation. $w = z + c$, where c is a complex constant.

If $c = c_1 + ic_2$, $z = x + iy$, and $w = u + iv$, then the transformation becomes $u + iv = x + iy + c_1 + ic_2$, where $u = x + c_1$ and $v = y + c_2$, i.e. the point $P(x, y)$ in the z-plane is mapped into the point $P'(x + c_1, y + c_2)$ in the

w-plane. Every point in the z-plane is mapped to a w-plane in the same way. Thus if the z-plane is superposed on the w-plane, figure is shifted through a distance given by the vector c. Accordingly, this transformation maps a figure in the z-plane into a figure in the w-plane of the same shape and size.

In particular, this transformation changes circles into circles.

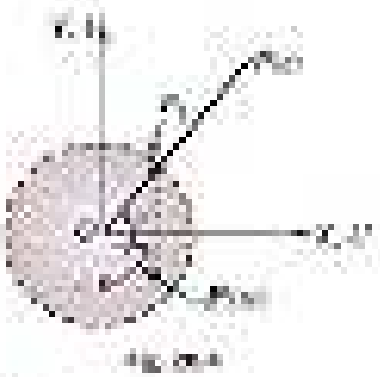

Fig. 18.4

(2) Magnification and rotation. $w = cz$, where c is a complex constant.

If $z = re^{i\theta}$, $c = \rho e^{i\alpha}$ and $w = Re^{i\phi}$, then

$$Re^{i\phi} = \rho r e^{i(\theta+\alpha)}, \quad R = \rho r, \quad \phi = \theta + \alpha$$

where $R = \rho r$ and $\phi = \theta + \alpha$. Or, the point $P(r, \theta)$ in the z-plane is mapped into the point $P'(\rho r, \theta + \alpha)$ in the w-plane. Hence the transformation consists of a magnification or contraction of the radius vector of P by $\rho = |c|$ and its rotation through an angle $\alpha = \arg(c)$. Accordingly any figure in the z-plane is transformed into a geometrically similar figure in the w-plane. In particular, this transformation maps circles into circles.

(3) Inversion and reflection. $w = 1/z$.

For it is convenient to think the w-plane as superposed on z-plane (Fig. 18.4).

If $z = re^{i\theta}$ and $w = Re^{i\phi}$, then $Re^{i\phi} = \frac{1}{r}e^{-i\theta}$

where $R = 1/r$ and $\phi = -\theta$. Thus, if P be (r, θ) and P_1 be (R, ϕ), i.e. P_1 is the inverse* of P w.r.t. the unit circle with centre O, then the reflection P' of P_1 in the real axis represents $w = 1/z$.

Note This transformation is an illustration of a w.r.t. circle and circle $|z - 1| = 2$ followed by reflection of the inverse w.r.t. the real axis.

Ex. 1. Show by the transformation $w = 1/z$ that the interior of the unit circle $|z| = 1$ maps the exterior of the unit circle $|w| = 1$ and the exterior of $|z| = 1$ maps the interior of $|w| = 1$. In particular, the origin $z = 0$ corresponds to the improper point $w = \infty$, while the point at infinity and the image of the improper point $z = \infty$ is the origin $w = 0$.

Sol. The complex transformation $w = 1/z$ is equivalent to the two real transformations $u = \frac{x}{x^2+y^2}$, $v = \frac{-y}{x^2+y^2}$.

To prove this, we write $w = 1/z$ as $u + iv = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2}$

or that $u = \frac{x}{x^2+y^2}$, $v = \frac{-y}{x^2+y^2}$ ...(1)

Now the general equation of a circle in the z-plane is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \text{...(2)}$$

which on substituting from (1) becomes $\frac{u^2}{(u^2+v^2)^2} + \frac{v^2}{(u^2+v^2)^2} + 2g\frac{u}{u^2+v^2} + 2f\frac{-v}{u^2+v^2} + c = 0$

or $cu^2 + cv^2 + 2gu - 2fv + 1 = 0$ ...(3)

This is the equation of a circle in the w-plane. If $c = 0$, the circle (2) passes through the origin and (3) reduces to a straight line. Hence the result.

Ex. 2. Show that the function $w = \frac{1}{z}$ transforms the circle $x^2 + y^2 - 2x = 0$ into a straight line in the w-plane.

Sol. Proceed as in Example 1 above.

(4) Bilinear transformation. The transformation

$$w = \frac{az + b}{cz + d} \quad \text{...(1)}$$

where a, b, c and d are complex constants and $ad - bc \neq 0$ is known as the bilinear transformation.** The restriction $ad - bc \neq 0$ ensures that it is not a constant, i.e., the transformation is conformal. If $ad - bc = 0$ every point of the z-plane is critical point.

The inverse mapping of (1) is

$$z = \frac{-dw + b}{cw - a} \quad \text{...(2)}$$

which is also a bilinear transformation.

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* The inverse of a point A w.r.t. a circle with centre O and radius a is defined as the point B on the line OA such that $OA \cdot OB = a^2$.

** First studied by Möbius in 1831. Hence, sometimes called Möbius transformation.

Obs. 1. From (1), we see that each point in the z-plane (except $z = -d/c$) corresponds to a unique point in the w-plane. Similarly, (2) shows that each point in the w-plane (except $w = a/c$) maps into a unique point in the z-plane, including the images of the two exceptional points (the infinite points) in the two planes. It follows that there is a one-to-one correspondence between all points in the two planes.

Obs. 2. Invariant points of bilinear transformation. The mapping is defined in the z-plane ($z \neq -d/c$) by the relation

$$w = \frac{az+b}{cz+d} \quad \text{or} \quad cwz + dw - az - b = 0$$

The roots of this equation are $z = z_1$ and $z = z_2$ and so the points z_1 and z_2 are invariant points of the bilinear transformation (1). However, if the two roots are equal, the bilinear transformation is said to be parabolic.

Obs. 3. Dividing the numerator and denominator of the right side of (1) by c (or d if $c = 0$) we see that (1) involves only three essential arbitrary constants. Hence three conditions are required to determine a bilinear transformation. For example, three distinct points z_1, z_2, z_3 can be mapped into any three specified points w_1, w_2, w_3.

Two important properties

I. A bilinear transformation maps circles into circles.

By actual division, (1) can be written as $w = \frac{a}{c} + \frac{bc-ad}{c^2} \cdot \frac{1}{z+d/c}$

which is a combination of the transformations

$$z_1 = z + d/c, \quad z_2 = 1/z_1, \quad z_3 = \frac{bc-ad}{c^2} z_2, \quad w = \frac{a}{c} + z_3$$

By these transformations, we successively pass from z-plane to z_1-plane, from z_1-plane to z_2-plane, from z_2-plane to z_3-plane and finally from z_3-plane to w-plane. Now each of these transformations is one or other of the standard transformations, since z_1, z_2 and z_3 are linear and each of these standard transformations maps circles into circles. Hence the bilinear transformation maps circles into circles.

II. A bilinear transformation preserves anharmonic ratio of four points.

Let the points z_1, z_2, z_3, z_4 of the z-plane map onto the points w_1, w_2, w_3, w_4 of the w-plane respectively under the bilinear transformation (1). If these points are finite, then from (1) we have

$$w_2 - w_1 = \frac{az_2+b}{cz_2+d} - \frac{az_1+b}{cz_1+d} = \frac{ad-bc}{(cz_1+d)(cz_2+d)}(z_2 - z_1)$$

Using this relation for $j, k = 1, 2, 3, 4$, we get: $\frac{(w_1 - w_2)(w_3 - w_4)}{(w_1 - w_3)(w_2 - w_4)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)}$

That is the cross-ratio of four points is invariant under bilinear transformation.

This property is very useful in finding a bilinear transformation. If one of the points, say $z_1 = \infty$, the quotient of those two differences which contain z_1 is replaced by 1 i.e.,

$$\frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)} = \frac{z_3 - z_4}{z_2 - z_4}.$$

Example 22.13. Find the bilinear transformation which maps the points $z = 1, i, -1$ onto the points $w = i, 0, -i$ respectively. Hence find (a) the image of $|z| < 1$ (Marathwada, 2005; U.P.T.U., 2002)
(b) the invariant points of this transformation. (U.P.T.U., 2003; P.T.U., 2010)

Solution. Let the points $z_1 = 1, z_2 = i, z_3 = -1$ and $z_4 = z$ map into the points $w_1 = i, w_2 = 0, w_3 = -i$ and $w_4 = w$. Since the cross-ratio remains unchanged under a bilinear transformation,

$$\frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} = \frac{(w_1 - w_3)(w_2 - w_4)}{(w_1 - w_4)(w_2 - w_3)} \quad \text{or} \quad \frac{(1+1)(i-z)}{(1-z)(i+1)} = \frac{(i+i)(0-w)}{(i-w)(0+i)}$$

By componendo dividendo, we get $\frac{2z}{2i} = \frac{(z+1)(i-w) - (i+w)(z-1)}{(z+1)(i-w) - (i-w)(z-1)}$

---

† Ref. If z_1, z_2, z_3, z_4 are four numbers, then $\frac{z_1 - z_3}{z_1 - z_4} \frac{z_2 - z_4}{z_2 - z_3}$ is called the cross-ratio of the four numbers z_1, z_2, z_3, z_4 taken in order.

$$\Rightarrow \qquad w = \frac{1+z}{1-z} \qquad \ldots(2)$$

which is the required Bilinear transformation.

(ii) Rewriting (2) as $\qquad w(1-z) = 1+z$

$$\Rightarrow \qquad \left|\frac{2(1-z)}{1+z}\right| < |z| < 2 \quad \text{or} \quad |z|\,|1-z| < |2|\,|1+z|$$

or $\qquad |1-x-iy| < |2|\,|1+x+iy|$ $\qquad \because\ |z| = 1$

or $\qquad (1-x)^2 + y^2 < (2+x)^2 + y^2$ which reduces to $x > 0$.

Hence the interior of the circle $x^2 + y^2 = 1$ in the z-plane is mapped into the right half of the w-plane to the right of the imaginary axis.

(iii) To find the invariant points of the transformation, we put $w = z$ in (2)

$$\Rightarrow \qquad z = \frac{1+z}{1-z} \quad \text{or} \quad z^2 + z - 1 = 0$$

or $\qquad z = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$

which are the required invariant points.

Example 20.13. Show that $w = \frac{1+z}{1-z}$ maps the real axis of the z-plane onto the circle $|w| = 2$ and the half plane $x > 0$ onto the interior of the circle $|w| = 2$ in the w-plane. (Mumbai, 2007)

Solution. Since $\qquad w = (1+z)/(1-z)$,

$$|w| = 2 \text{ becomes } |1+z| = 2|1-z| \quad \text{or} \quad |x+iy+1| = 2|1-x-iy|$$

or $\qquad |(1+x)+iy| = 2|(1-x)-iy| \quad \text{or} \quad |(1+x)+iy|^2 = 4|(1-x)-iy|^2$

$$\Rightarrow (1+x)^2 + y^2 = 4(1-x)^2 + 4y^2 \quad \text{or} \quad 3x^2 + 3y^2 - 10x + 3 = 0$$

$$\Rightarrow x^2 + y^2 - \frac{10}{3}x + 1 = 0$$

which is the required circle.

Hence the real axis of the z-plane is mapped in the circle $|w| = 2$.

Now for the interior of the circle $|w| = 1$

$$|w| < 1 \quad \text{i.e.,} \quad |1+z| < 2|1-z| \quad \text{or} \quad (1+x)^2 + y^2 < 4(1-x)^2 + 4y^2$$

$$\Rightarrow 3x^2 + 3y^2 - 10x + 3 > 0 \quad \text{i.e.,} \quad x^2 + y^2 - \frac{10}{3}x + 1 > 0$$

Hence the half plane $x > 0$ is mapped into the interior of the circle $|w| = 2$.

PROBLEMS 20.2

1. Find the invariant points of the transformation $w = (z-1)/(z+1)$. (Madras, 2002)
2. Find the transformation which maps the points $z = 1, i, -1$ of the z-plane onto $w = 1, i, -1$ of the w-plane respectively. Also find its invariant points. (UPTU, 2011)
3. Find the bilinear transformation which maps $z = 1, -1, 0$ onto $w = i, -i, 0$ respectively. Find the fixed and critical points of the transformation. (UPTU, 2005; Mumbai, 2005; VTU, 2009)
4. Determine the bilinear transformation that maps the points $z = -2, 2, i$ onto $w = -1, 1, i$ respectively. Find the invariant points. (UPTU, 2003; Delhi, 2004)
5. Find the bilinear transformation that maps
   (i) the points $z = 1, i, -1$ into the points $w = 0, i, \infty$ (UPTU, 2009; Mumbai, 2003)
   (ii) the points $z = 0, 1, \infty$ into the points $w = 1, i, 0$. (UPTU, 2010 (I))
   Also find the invariant points of the transformation. Map the circle $|z| = 1$ in the z-plane into the w-plane.
6. Under the transformation $w = \frac{z-i}{z+i}$, show that the map of the straight lines $x = y$ is a circle and find its centre and radius. (Kerala Mumbai, 2007)

7. Show that the bilinear transformation $w = \frac{iz + 3i}{z + 1}$ maps the circle $x^2 + y^2 - 4x = 0$ onto the line $v = 2u + 3$. (Worked, Part 2, Ex. 37, p. 1000, Higher Eng. Maths)

8. Show that the transformation $w = \frac{z - 1}{z + 1}$ maps the circle $|z| = 1$ onto the imaginary axis in the w plane.

9. Show that the transformation $w = \frac{z - 1}{z + 1}$ maps the circle $|z| = 1$ onto the real axis of the w plane and the interior of the circle $|z| = 1$ onto the upper half of the w plane. (Worked, Part 2, Ex. 37, p. 1000, Higher Eng. Maths)

10. If $R_e(z) \geq 0$ (the right half of the z plane), show that the bilinear transformation $w = \frac{z - 1}{z + 1}$ maps the right half of the z plane onto the interior of the unit circle $|w| = 1$ in the w plane.

20.5 [3] CONFORMAL TRANSFORMATION

Suppose two curves C_1, C_2 in the z plane intersect at the point P and the corresponding curves C' and C_2' in the w plane intersect at P' (Fig. 20.3). If the angle of intersection of the curves at P is the same as the angle of intersection of the curves at P' in magnitude and sense, then the transformation is said to be conformal.

(1) Theorem The transformation effected by any analytic function $w = f(z)$ is conformal at every point of the z plane where $f'(z) \neq 0$.

Let $P(z)$ be a point in the region R of the z plane and $P'(w)$ the corresponding point in the region R' of the w plane (Fig. 20.3). Suppose a curve z on curve C and is mapped on the corresponding curve C'. Let $Q = z + \delta z$ be a neighbouring point on C and $Q'(w + \delta w)$ be the corresponding point on C' so that $\vec{PQ} = \delta z$ and $\vec{P'Q'} = \delta w$. Then δz is a complex number whose modulus r is the length PQ and amplitude θ is the angle which PQ makes with the x axis.

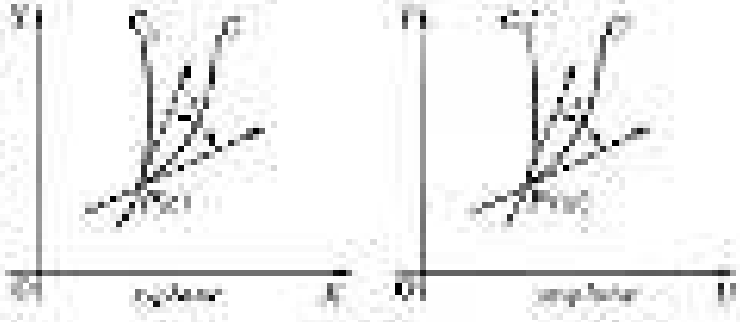

Fig. 20.3

$$\therefore \qquad \delta z = re^{i\theta}$$

Similarly, if the modulus and amplitude of δw be r' and θ', then $\delta w = r'e^{i\theta'}$.

Hence $$\frac{\delta w}{\delta z} = \frac{r'}{r} e^{i(\theta' - \theta)}$$

Now if the tangent at P to the curve C makes an angle ψ with the x axis and the tangent at P' to C' makes an angle ψ' with the u axis, then as $\delta z \to 0$, $\theta \to \psi$ and $\theta' \to \psi'$.

$$\therefore \qquad f'(z) = \frac{dw}{dz} = \lim_{\delta z \to 0} \left(\frac{r'}{r} \right) e^{i(\psi' - \psi)} \qquad \ldots(1)$$

If ρ is the modulus and ϕ the amplitude of the complex number $f'(z)$ which is supposed to be non-zero, then

$$f'(z) = \rho e^{i\phi} \qquad \ldots(2)$$

$\therefore$ from (1) and (2) we have $$\rho = \lim_{\delta z \to 0} \frac{r'}{r} \qquad \ldots(3)$$

and $$\phi = \psi' - \psi. \qquad \ldots(4)$$

Now let C_2 be another curve through P in the z plane and C_2' the corresponding curve through P' in the w plane. If the tangent at P to C_2 makes an angle β with the x axis and tangent at P' to C_2' makes an angle β' with the u axis, then as in (4),

$$\gamma = \beta' - \beta \qquad \ldots(5)$$

Equating (4) and (5), $\psi' - \psi = \beta' - \beta$ or $\beta - \psi = \beta' - \psi' = \gamma$ (Fig. 20.3)

Thus the angle between the curves before and after the mapping is preserved in magnitude and direction. Hence the mapping by the analytic function $w = f(z)$ is conformal at each point where $f'(z) \neq 0$.

Obs. 1. A point at which $f'(z) = 0$ is called a critical point of the transformation.

Obs. 2. The relation (3), $\rho = \lim_{\delta z \to 0} \frac{r'}{r}$, shows that the magnification at P in the curve C is equal to the magnification at P' in the corresponding curve C' under the given transformation.

Obs. 3. The relation (4) shows that in the transformation, curves of any pencil through P in the z plane are rotated through an angle $\phi = \arg f'(z)$, i.e. the argument of the complex number represented by the point $f'(z)$. Consequently the configuration of curves is preserved in direction but not in position.

If $w = f(z)$ is analytic (regular) and $f'(z) \neq 0$ in a region R of the z-plane, then the transformation is conformal at all points of R.

Proof: Let $z = x + iy$ and $w = u + iv$ where $u = u(x, y)$ and $v = v(x, y)$ are real functions of x and y. Then $w = f(z) = u(x, y) + iv(x, y)$.

Let $z_0 = x_0 + iy_0$ be any point in R and $w_0 = u_0 + iv_0$ be the corresponding point in the w-plane. Let C be any curve through z_0 in the z-plane. Let $z_1 = x_1 + iy_1$ be a point on C near z_0. Then $z_1 - z_0 = (x_1 - x_0) + i(y_1 - y_0)$.

Let $w_1 = u_1 + iv_1$ be the point corresponding to z_1 in the w-plane. Then $w_1 - w_0 = (u_1 - u_0) + i(v_1 - v_0)$.

$$\frac{w_1 - w_0}{z_1 - z_0} = \frac{(u_1 - u_0) + i(v_1 - v_0)}{(x_1 - x_0) + i(y_1 - y_0)}$$

Dividing numerator and denominator by $(x_1 - x_0)$ and $(y_1 - y_0)$ and taking limits as $z_1 \to z_0$ (i.e., $x_1 \to x_0$ and $y_1 \to y_0$), we get

$$f'(z_0) = \frac{\left(\frac{\partial u}{\partial x}\right)_0 + i\left(\frac{\partial v}{\partial x}\right)_0}{\left(\frac{\partial u}{\partial y}\right)_0 + i\left(\frac{\partial v}{\partial y}\right)_0} \quad \text{(using C-R equations)}$$

$$= \left(\frac{\partial u}{\partial x}\right)_0 + i\left(\frac{\partial v}{\partial x}\right)_0 \quad \text{(using C-R equations)}$$

$$= \left(\frac{\partial u}{\partial x}\right)_0 - i\left(\frac{\partial u}{\partial y}\right)_0 \quad \text{(using C-R equations)}$$

$$= \left(\frac{\partial v}{\partial y}\right)_0 + i\left(\frac{\partial v}{\partial x}\right)_0 \quad \text{(using C-R equations)}$$

$$= \left(\frac{\partial v}{\partial y}\right)_0 - i\left(\frac{\partial v}{\partial y}\right)_0 \quad \text{(using C-R equations)}$$

$$= \left(\frac{\partial u}{\partial x}\right)_0 + i\left(\frac{\partial v}{\partial x}\right)_0 = \left(\frac{\partial v}{\partial y}\right)_0 - i\left(\frac{\partial u}{\partial y}\right)_0$$

$$= \left(\frac{\partial u}{\partial x}\right)_0 - i\left(\frac{\partial u}{\partial y}\right)_0 = \left(\frac{\partial v}{\partial y}\right)_0 + i\left(\frac{\partial v}{\partial x}\right)_0

Changing to polar coordinates,

$$w = z + \frac{1}{z} = re^{i\theta} + \frac{1}{re^{i\theta}} = r\cos\theta + ir\sin\theta + \frac{1}{r}\cos\theta - \frac{i}{r}\sin\theta$$

$$\phi + i\psi = \left(r + \frac{1}{r}\right)\cos\theta + i\left(r - \frac{1}{r}\right)\sin\theta$$

$$\therefore \quad \phi = \left(r + \frac{1}{r}\right)\cos\theta \quad \text{and} \quad \psi = \left(r - \frac{1}{r}\right)\sin\theta$$

Elimination of θ gives $\dfrac{\phi^2}{\left(r + \frac{1}{r}\right)^2} + \dfrac{\psi^2}{\left(r - \frac{1}{r}\right)^2} = 1$ ...(1)

while the elimination of r gives $\dfrac{\phi^2}{4\cos^2\theta} - \dfrac{\psi^2}{4\sin^2\theta} = 1$ ...(2)

From (1), it follows that the circle $r = \text{constant}$ of the z-plane transforms into a family of ellipses of the w-plane (Fig. 20.70). These ellipses are confocal for $\left(r + \frac{1}{r}\right)^2 - \left(r - \frac{1}{r}\right)^2 = 4$, i.e., a constant.

In particular, the unit circle $r = 1$ in the z-plane flattens out to become the segment $u = -2$ to $u = 2$ of the real axis in w-plane. Thus the exterior of the unit circle in the z-plane maps into the entire w-plane.

(A.M.I.E.T.E., 2005 S)

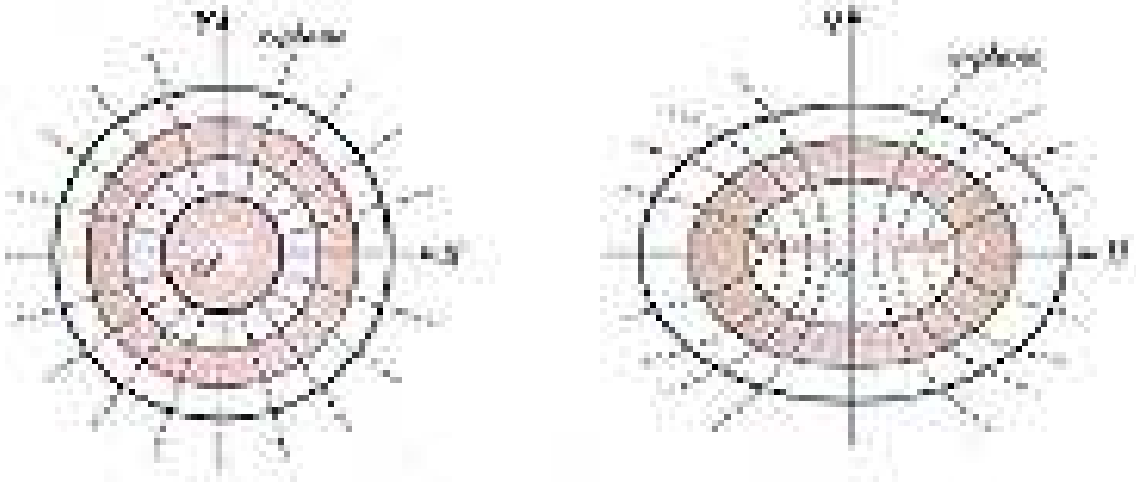

Fig. 20.7

From (2), it follows that the radial line $\theta = \text{constant}$ of the z-plane transforms into a family of hyperbolae which are also confocal (Fig. 20.71).

Note 1. [illegible] the origin, the flow is mostly confined to the x-axis. This can be interpreted as the flow around a circular cylinder of infinite radius having two stagnation points* at $A(-1, 0)$ and $B(+1, 0)$ (Fig. 20.8).

Note 2. This transformation is also used to map the interior of the parallelogram on the interior of a unit circle in the w-plane. These are known as Joukowski aerofoils.

(B) Transformation $w = z^2$.

Writing $w = u + iv$ and $z = re^{i\theta}$, we have $re^{i\theta} = z^2 = r^2e^{2i\theta}$, i.e.,

$$\phi + i\psi = r^2\cos 2\theta + ir^2\sin 2\theta$$

whence $\phi = r^2\cos 2\theta$ ...(1) and $\psi = r^2\sin 2\theta$ ...(2)

From (1), it is clear that the lines parallel to the y-axis (i.e., $x = \text{constant}$) map into circles ($r = \text{constant}$) in the w-plane. Only radii lying less than or equal to $\frac{\pi}{4}$ (according as θ is less than or greater than $\frac{\pi}{4}$) (Fig. 20.9).

(U.P.T.U., 2003)

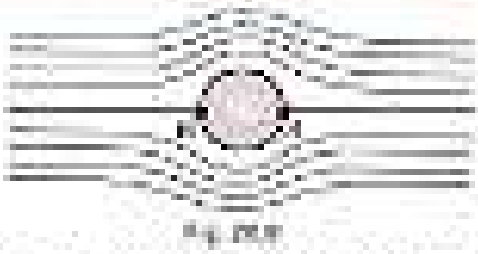

Fig. 20.9

Similarly, it follows from (2) that the lines parallel to the x-axis (i.e., $y = \text{constant}$) map into the radial lines ($\theta = \text{constant}$) of the w-plane. Thus any horizontal strip of height 2π in the z-plane will cover once the entire w-plane.

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*Stagnation points are those at which the fluid velocity is zero.

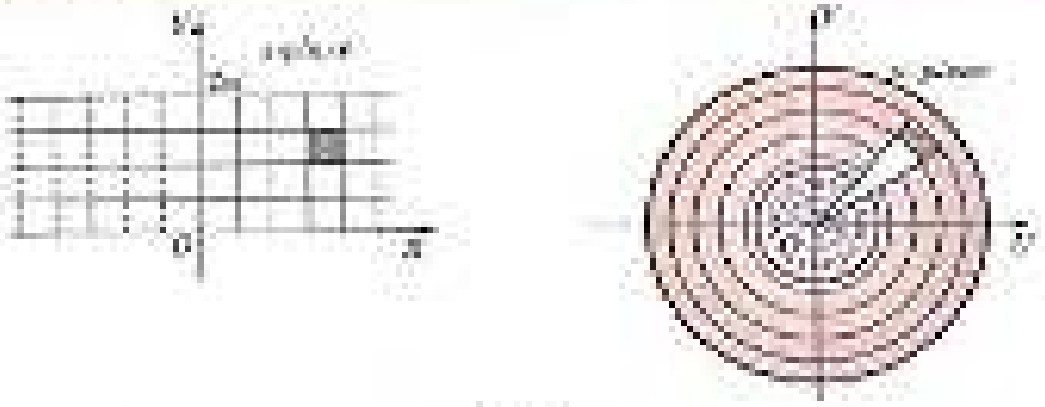

Fig. 103

The rectangular region $a_1 < x < a_2, b_1 < y < b_2$ in the z-plane (shown shaded) transforms into the region $e^{a_1} < u < e^{a_2}, b_1 < v < b_2$ in the w-plane bounded by circles and rays (shown shaded).

(P.C.C., 2005; Kerala, 2008)

Obs. This transformation can be used to obtain the streamlines of liquid around a cylinder obtained by the bent stick fluid due to a charged circular cylinder etc.

(d) Transformation $w = \sin z$.

We have $w = \sin z = \sin(x + iy) = \sin x \cos iy + \cos x \sin iy$ (By (2.10.1), p. 68)

$$u + iv = \sin x \cosh y + i \cos x \sinh y$$

so that $u = \sin x \cosh y$ and $v = \cos x \sinh y$.

Eliminating x from these equations gives

$$\frac{u^2}{\cosh^2 y} + \frac{v^2}{\sinh^2 y} = 1 \qquad \ldots(1)$$

while elimination of y gives $\frac{u^2}{\sin^2 x} - \frac{v^2}{\cos^2 x} = 1 \qquad \ldots(2)$

(i) It shows that the lines parallel to x-axis (i.e., y = constant) in the z-plane map into hyperbolas in the w-plane.

(ii) It shows that the lines parallel to the y-axis (i.e., x = constant) in the z-plane map into ellipses in the w-plane (Fig. 20.10). The rectangular region $a_1 < x < a_2, b_1 < y < b_2$ in the z-plane (shown shaded) transforms into the shaded region in the w-plane bounded by the corresponding hyperbolas and ellipses. (Kerala M. Univ., 2007)

Obs. This transformation can be used

(i) to obtain the streamlines of liquid around an elliptic cylinder,

(ii) to determine the electrostatic field due to a charged conductor,

(iii) to determine the potential between two confocal elliptic (or hyperbolic) cylinders.

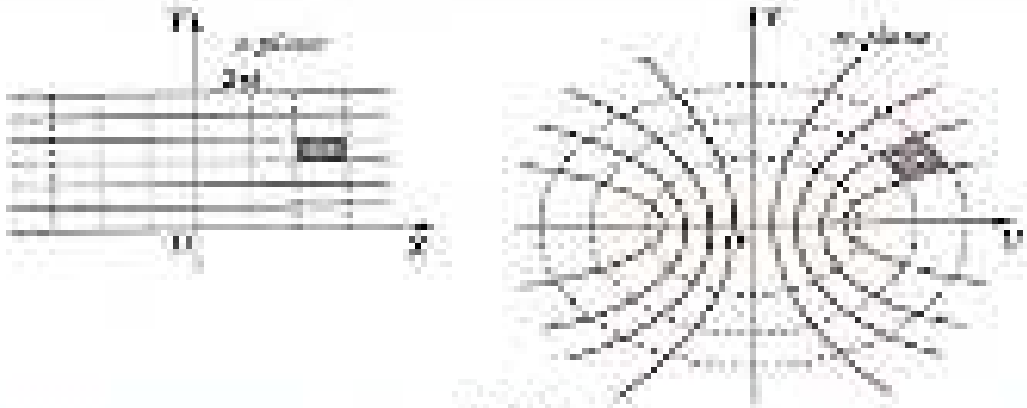

Fig. 20.10

Example 20.10. Show that under the transformation $w = \sin z$, the rectangle (in the z-plane bounded by $x = 0$, $x = \frac{\pi}{2}$, $y = 0$, $y = 1$) which portion of the z-plane corresponds to the interior of the circle $|w| = 1$. (P.T.U., 2003)

Solution. We have $|w| = \left|\frac{z-i}{z+i}\right| = \frac{|z-i|}{|z+i|} = \frac{|x+(y-1)i|}{|x+(y+1)i|}$

$$= \sqrt{x^2+(y-1)^2} \Big/ \sqrt{x^2+(y+1)^2}$$

Now the real axis in z-plane ($i.e.,\ y=0$) transforms into

$$|w| = \sqrt{x^2+1} \Big/ \sqrt{x^2+1} = 1$$

Hence the real axis in the z-plane is mapped into the circle $|w| = 1$.

The interior of the circle, $i.e.,\ |w| < 1$, gives

$$|x^2+(y-1)^2| < |x^2+(y+1)^2| \text{ i.e., } -y < 0 \text{ or } y > 0.$$

Thus the upper half of the z-plane corresponds to the interior of the circle $|w| = 1$.

EXERCISE 6.5

1. Determine the region of the z-plane into which the following regions are mapped by the transformation $w = z^2$:
   (i) first quadrant of the plane (AMIETE, 2006)
   (ii) region bounded by $x = 1,\ y = 1,\ x + y = 1$ (Anna Univ., 2003; AMIETE, 2006 S)
   (iii) the region $1 < x < 2$ and $1 < y < 2$ (Anna Univ., 2007; AMIETE, 2006)
   (iv) circle $|z - 1| = 1$.
2. Find the transformation which maps the triangular region $0 < \arg z < \pi/3$ into the unit circle in w-plane.
3. Discuss the transformation $w = z^2$. Is it conformal at the origin? (Delhi, 2005)
4. Under the transformation $w = 1/z$, find the images of:
   (a) the circle $|z - 2| = 3$ (Bangalore Univ., Anna & Tech., 2000)
   (b) the straight line $y = x + 1 = 0$ (U.P.T.U., 2004)
   (c) the hyperbola $x^2 - y^2 = 1$ (Mumbai, 1999; AMIETE, 2006)
5. Show that under the transformation $w = 1/z$, (a) circle $x^2 + y^2 - 4x = 0$ is transformed into a straight line in the w-plane.
   (b) the circle $x^2 + y^2 = 2$ is transformed into a circle with centre $(0, 0)$ and radius $\sqrt{2}$. (Mumbai, 2000)
6. Show that the transformation $w = 1/z$ transforms all circles and straight lines (i.e. the circles and straight lines in the z-plane. Which circles in the z-plane become straight lines in the w-plane, and which straight lines are transformed into circles? (Anna, 2002)
7. Show that the transformation $w = 1/z$ converts the straight line $x + y = 1$ into a circle passing through the origin. (Rajasthan, 2003 S)
8. Show that the transformation $w = z + (k^2/z)$, $(k > 0)$ maps the circle $|z| = 2k$ in the z-plane into an ellipse $\frac{u^2}{9} + \frac{v^2}{5} = 1$ in the $w = u + iv$ plane.
9. Show that the transformation $w = z + a^2/z$ has a branch point at the origin of the w-plane and circles, centred at the origin in the z-plane.
10. Show that the function $w = Az + (B/z)$ may be used to represent the flow of a perfect fluid past a circular cylinder. Also find the stagnation points.
11. Show that in the solution $u = ce^{-\alpha x}\sin(\alpha y + \beta)$ of the differential equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, α and β are arbitrary constants, u is bounded and periodic in the region between the two branches of the hyperbola $\alpha\beta = k$. (Kerala, 2004)
12. Prove that the transformation $w = \cos z$ maps the boundaries of the rectangle $x = 0$, $x = \pi$, $y = 0$ and $y = a$ onto the two halves of an elliptical conical cross.
13. Discuss the transformation $w = z^2$, and hence find the transformed region between the real axis and the line $y = x$ in the z-plane, into the upper half of the w-plane.
14. Discuss fully the transformation $w = cz$, where c is a real number. What physical problem can we study with the help of this transformation?

6.10 SCHWARZ-CHRISTOFFEL TRANSFORMATION

This transformation maps the interior of a polygon of the z-plane into the upper half of the w-plane and the boundary of the polygon into the real axis. The formula of this transformation is

Fig. 20.11

$$\frac{dw}{dz} = A(z-x_1)^{\frac{\alpha_1}{\pi}-1}(z-x_2)^{\frac{\alpha_2}{\pi}-1}\cdots(z-x_n)^{\frac{\alpha_n}{\pi}-1} \qquad \ldots(1)$$

or
$$w = A\int (z-x_1)^{\frac{\alpha_1}{\pi}-1}(z-x_2)^{\frac{\alpha_2}{\pi}-1}\cdots(z-x_n)^{\frac{\alpha_n}{\pi}-1}\,dz + B \qquad \ldots(2)$$

where $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the interior angles of the polygon having vertices $z_1, z_2, \ldots, z_n$ which correspond to the points $x_1, x_2, \ldots, x_n$ on the real axis of the z-plane (Fig. 20.11). Also A and B are complex constants which determine the size and position of the polygon.

Proof. We have from (1),

$$\arg\left(\frac{dw}{dz}\right) = \arg(A) + \left(\frac{\alpha_1}{\pi}-1\right)\arg(z-x_1) + \left(\frac{\alpha_2}{\pi}-1\right)\arg(z-x_2)$$
$$+ \cdots + \left(\frac{\alpha_n}{\pi}-1\right)\arg(z-x_n) \qquad \ldots(3)$$

As z moves along the real axis from the left towards x_1, suppose that it moves along the side $z_n z_1$ of the polygon towards z_1. As z crosses x_1 from left to right, $\theta_1 = \arg(z-x_1)$ changes from π to 0 while all other terms of (3) remain unaffected. Hence only $\left(\frac{\alpha_1}{\pi}-1\right)\arg(z-x_1)$ decreases by $\left(\frac{\alpha_1}{\pi}-1\right)\pi = \pi - \alpha_1$ i.e. increases by $\pi-\alpha_1$ in the anti-clockwise direction. In other words, $\arg(dw/dz)$ increases by $\pi-\alpha_1$. Thus the direction of z_1 moves through the angle $\pi-\alpha_1$ and is now moving along the side $z_1 z_2$ of the polygon.

Similarly when z passes through x_2, $\theta_2 = \arg(z-x_2)$ and $\theta_1 = \arg(z-x_1)$ change from π to 0 while all other terms remain unchanged. Hence the side $z_1 z_2$ turns through the angle $\pi-\alpha_2$. Proceeding in this way, we see that z moves along sides, as it does the polygon $z_1 z_2 z_3 \ldots z_n$ anti-clockwise.

Example 20.16. Find the transformation which maps the semi-infinite strip in the z-plane (Fig. 20.12) into the upper half of the w-plane. (U.P.T.U., M.C.A. 2004; Dibrugarh, 2003)

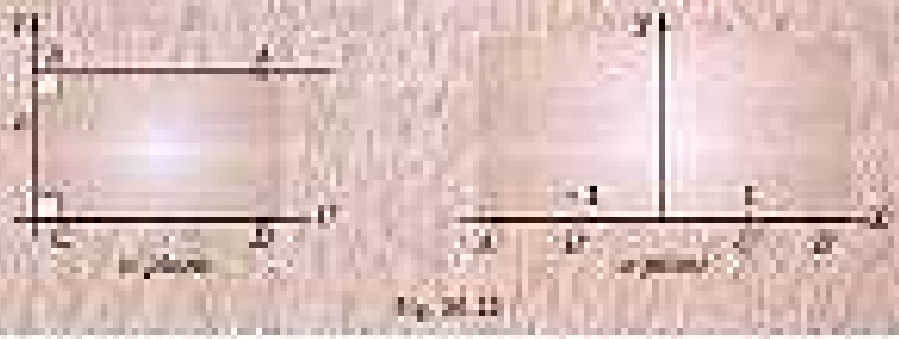

Fig. 20.12

Solution. Consider $ABCD$ as the limiting case of a triangle with two vertices B and C and the third vertex A at B at infinity. Let the vertices B and C map into the points $x=1$ and $x=-1$ of the w-plane. Since the interior angles at B and C are $\pi/4$, we have by the Schwarz-Christoffel transformation,

$$\frac{dz}{dw} = A(w+1)^{\frac{\pi/4}{\pi}-1}(w-1)^{\frac{\pi/4}{\pi}-1} = A/\sqrt{w^2-1}$$

$$\text{i.e.} \qquad w = A\int \frac{dz}{\sqrt{(z^2-1)}} + B = A\cosh^{-1} z + B$$

When $z = 1$, $w = 0$, $\therefore\ 0 = A\cosh^{-1}(1) + B$, i.e., $B = 0$

When $z = -1$, $w = \delta$, $\therefore\ \delta = A\cosh^{-1}(-1) + 0$, i.e., $\cosh(\delta/A) = -1$

or $\cos\dfrac{\delta}{A} = -1 = \cos\pi$, Thus $A = \dfrac{\delta}{\pi}$.

Hence $w = \dfrac{\delta}{\pi}\cosh^{-1} z$ or $z = \cosh\dfrac{\pi w}{\delta}$.

PROBLEMS 20.6

1. Find the transformation which maps the upper half of the z-plane, bounded by the lines $x = 0$, $y = 0$ and $x = 1$ into the upper half of the w-plane.
2. Show how you will find the transformation which maps the semi-infinite strip bounded by the real axis and the lines $x = 0$, $x = 1$ in the z-plane into the upper half of the w-plane.
3. Find the mapping function which maps the semi-infinite strip in the z-plane $-\infty \le x \le \infty$, $y \ge 0$ into the half-plane for which $v \ge 0$, such that the points $z = -2$, $z = 1$ and $z = \infty$ in the z-plane are mapped into the points $w = 1$, $w = 0$ and $w = \infty$ respectively in the w-plane.
4. Find the mapping function which will map the interior of the infinite strip bounded by the lines $x = 0$, $x = \pi$ on to the upper half of the w-plane.

20.13 COMPLEX INTEGRATION

We have already discussed the concept of the line integral as applied to vector fields in §16.11. Now we shall consider the line integral of a complex function.

Consider a single-valued function $f(z)$ of the complex variable $z = x + iy$ defined at all points of a curve C having fixed points A and B. Divide C into n parts at the points

$$A = P_0(x_0, y_0), P_1(x_1, y_1), \ldots, P_r(x_r, y_r), \ldots, P_n(x_n, y_n) = B.$$

Let $\delta z_r = z_r - z_{r-1}$ and ξ_r be any point on the arc $P_{r-1}P_r$. The limit of the sum $\sum_{r=1}^{n} f(\xi_r)\,\delta z_r$ as $n \to \infty$ in such a way that the length of the chord δz_r approaches zero, is called the line integral of $f(z)$ taken along the path C, i.e.,

$$\int_C f(z)\,dz$$

Fig. 20.13

Writing $f(z) = u(x, y) + iv(x, y)$ and noting that $dz = dx + i\,dy$,

$$\int_C f(z)\,dz = \int_C u\,dx - v\,dy + i\int_C v\,dx + u\,dy$$

which shows that the evaluation of the line integral of a complex function may be reduced to the evaluation of two line integrals of real functions.

Note. The value of the integral is independent of the path of integration when the integrand is analytic.

Example 20.17. Prove that

(i) $\displaystyle\int_C \frac{dz}{z-a} = 2\pi i$ (ii) $\displaystyle\int_C (z-a)^n\,dz = 0$ $(n \text{ any integer} \neq -1)$

where C is the circle $|z - a| = r$. (U.P.T.U., 2003)

Solution. The parametric equation of C is $z - a = re^{i\theta}$, where θ varies from 0 to 2π as z describes C once in the positive (anti-clockwise) sense (Fig. 20.14).

(i) $\displaystyle\int_C \frac{dz}{z-a} = \int_0^{2\pi} \frac{1}{re^{i\theta}}\,ire^{i\theta}\,d\theta$ $\left[\because dz = ire^{i\theta}\,d\theta\right]$

$$= i\int_0^{2\pi} d\theta = 2\pi i$$

(c) $\int_C (x - y^2)\,dz = \int_0^{2\pi} (\cos t - \sin^2 t)(-\sin t + i\cos t)\,dt$

$= i\int_0^{2\pi} (\cos t - \sin^2 t)(\cos t + i\sin t)\,dt = i\int_0^{2\pi} (\cos^2 t - \sin^2 t + i\sin t\cos t)\,dt$

$= i\int_0^{2\pi} \cos 2t\,dt + \frac{i}{2}\int_0^{2\pi} \sin 2t\,dt = \frac{i}{2}\left[\sin 2t\right]_0^{2\pi} - \frac{i}{4}\left[\cos 2t\right]_0^{2\pi}$, provided $n \neq -1$

$= \frac{i}{2}(0 - 0) - \frac{i}{4}(1 - 1) = 0$ $\quad [\because \cos 2\pi = 1 = \cos 0]$

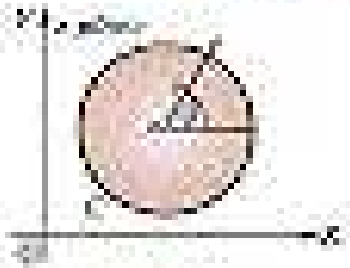

Fig. 20.14

Example 20.10. Evaluate $\int_C (x^2 + iy)\,dz$, along (i) the line $y = x/2$ (ii) the real axis and (iii) the straight line parallel to the imaginary axis and passing through $z = 1$. (Bhopal, 2007; UPTU, 2002; U.P.T.U., 2003; P.T.U., 2004; B.E. Mumbai, 2005)

Solution. (i) Along the line OA, $y = \frac{1}{2}x$, $x = 2 + iy$, $z = 2 - iy$ and $dz = (2 + i)\,dy$ (Fig. 20.15)

$\therefore \quad I = \int_0^{2+i} (x^2 + iy)\,dz = \int_0^1 (2 - y^2 + i\cdot\tfrac{1}{2}y)(2 + i)\,dy$

$= (2 - i)\left[\frac{y^3}{3} + \frac{iy^2}{4}\right]_0^1 = \frac{5}{12}(2 - i)$

(ii) $\quad I = \int_{OB} (x^2 + iy)\,dz + \int_{BA} (x^2 + iy)\,dz$

Now along OB, $y = 0$, $z = x$, $dz = dx$;

and along BA, $x = 2 + iy$, $z = 2 - y$, $dz = -i\,dy$

$\therefore \quad I = \int_0^2 x^2\,dx + \int_0^1 (2 - iy^2)(-i)\,dy = \left[\frac{x^3}{3}\right]_0^2 - \int_0^1 (4y - i4 - y^2)\,dy$

$= \frac{8}{3} + i4 - \frac{4}{3} + i\left[4y - \frac{y^3}{3}\right]_0^1 = \frac{4}{3} + i\left(4 - \frac{1}{3}\right) = \frac{4}{3}(1 + 11i)$

Fig. 20.15

Example 20.11. Evaluate $\int_C (x^2 + iy + 2)\,dz$ where C is the arc of the parabola $x = t^2 + 1$, $y = t^2 + t$ from $t = 0$ to $t = 1$ between the points $(1, 0)$ and $(2, 2)$. (Bhopal, 2006)

Solution. $f(z) = z^2 + 2z + 2$ is analytic in the z-plane being a polynomial. As such, the line integral of $f(z)$ between O and A is independent of the path (Fig. 20.16). We therefore, take the path from O to L and L to A so that

$$\int_C f(z)\,dz = \int_{OL} f(z)\,dz + \int_{LA} f(z)\,dz \qquad ...(1)$$

$\therefore \quad \int_{OL} f(z)\,dz = \int_0^1 (x^2 + 2x + 2)\,dx$

[$\because$ along OL, $y = 0$, $z = x + 0i$ or $dz = dx$, x varies from 0 to 1]

$= \left[\frac{x^3}{3} + \frac{2x^2}{2} + 2x\right]_0^1 = \frac{1}{3}(2 + 6 + 12) = \frac{20}{3}$ $\qquad ...(2)$

and $\quad \int_{LA} f(z)\,dz = \int_0^1 [(2 + y)^2 + i(2 + y) + 2]\,i\,dy$

[$\because$ along LA, $x = 2$, $z = 2 + iy$, $dz = i\,dy$ and y varies from 0 to 1 i.e., L to A]

$= i\left[\frac{(2+y)^3}{3}\right]_0^1 + i\left[\frac{(2+y)^2}{2}\right]_0^1 + 2i\int_0^1 = \frac{i}{3}(3^3 - 2^3) + \frac{i}{2}(3^2 - 2^2) + 2i$ $\qquad ...(3)$

$\therefore$ substituting from (2) and (3) in (1), we get

$\int_C f(z)\,dz = \frac{20}{3} + \frac{i}{3}(27 - 8) + \frac{i}{2}(9 - 4) + 2i = \frac{20}{3} + \frac{19}{3}i + \frac{5}{2}i + 2i = \frac{20}{3} + \frac{29}{6}i + 2i$

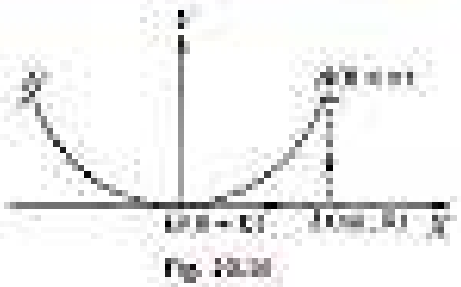

Fig. 20.16

PROBLEMS 2005

1. Evaluate $\int_0^{1+i} (z^2 - z)\,dz$ along the path $y = x^2$ from $z = 0$ to $z = 1 + i$. (UCE O, 2005)

2. Evaluate $\int_C z^2\,dz$ along the line C from $z = 0$ to $z = 2 + i$. (UCE O, 2005)

3. Evaluate $\int_C (z^2 + z)\,dz$ along the line joining $z = 1$ and $z = i$. (UCE O, 2005)

4. Evaluate $\int_C (z^2 + z)\,dz$ around the boundary C of the square whose vertices are $z = 0$, $z = 1$, $z = 1 + i$ and $z = i$. (UCE O, 2005)

5. Evaluate $\int_C (z^2 + z)\,dz$ around the boundary C of the square whose vertices are the points $z = 0$, $z = 1$, $z = 1 + i$ and $z = i$. (UCE O, 2005)

6. Show that for every path between the limits, $\int_0^{1+i} (2z + e^z)\,dz = e - 1$. (UCE O, 2005)

7. Show that $\oint_C \frac{dz}{z - a} = 2\pi i$, where C is the boundary of the square whose vertices are the points $z = 0$, $z = 1$, $z = 1 + i$ and $z = i$. (UCE O, 2005)

8. Evaluate $\int_C (z + 2\bar{z})\,dz$, where C is the circle
   (a) straight line from $z = -1$ to $z = 1$,
   (b) left half of the unit circle $|z| = 1$ from $z = -1$ to $z = 1$,
   (c) sliding vertically $|z + 2| = 3$ clockwise in the left-hand sense. (UCE O, 2005)

9. Find the value of $\int_C (z - y + ix^2)\,dz$
   (a) along the straight line from $z = 0$ to $z = 1 + i$,
   (b) along the path from $z = 0$ to $z = 1$ and then along the path from $z = 1$ to $z = 1 + i$. (UCE O, 2005)

10. Prove that $\int_C f(z)\,dz = 0$ in a clockwise sense around the boundary $|z| = 1$ where n is a positive integer. (Wales, 2005)

11. Evaluate $\int_C (z^2 - z^2)\,dz$, where C is the upper half of the circle $|z| = 1$.
    What is the result if C is the lower half of the circle? (UCE O, 2005)

21.3 CAUCHY'S THEOREM

If $f(z)$ is an analytic function and $f'(z)$ is continuous at each point within and on a closed curve C, then $\oint_C f(z)\,dz = 0$.

Writing $f(z) = u(x, y) + iv(x, y)$ and z along $dz = dx + i\,dy$,

$$\oint_C f(z)\,dz = \oint_C (u\,dx - v\,dy) + i \oint_C (v\,dx + u\,dy) \qquad (1)$$

Since $f'(z)$ is continuous, therefore, $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ are also continuous in the region R enclosed by C. Hence the Green's theorem (p. 338) can be applied to (1), giving:

$$\oint_C f(z)\,dz = -\iint_R \left[\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right] dx\,dy + i \iint_R \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right] dx\,dy \qquad (2)$$

Now $f(z)$ being analytic, u and v necessarily satisfy the Cauchy-Riemann equations and thus the integrands of the two double integrals in (2) vanish identically.

Hence $$\oint_C f(z)\,dz = 0.$$

Note 1. The Cauchy-Riemann equations are precisely the conditions for the existence of $f'(z)$ at all points of the path. Hence the line integral of a function $f(z)$ which is analytic in the region R is independent of the path joining any two points of R.

Note 2. Extension of Cauchy's theorem. If $f(z)$ is analytic in a region R bounded between two simple closed curves C and C_1, then $\oint_C f(z)\,dz = \oint_{C_1} f(z)\,dz$.

To prove this, we need to introduce the cross-cut AB. Then $\oint f(z)dz = 0$ where the path is as indicated by arrows in Fig. 20.17, i.e. along AB—along C_1 in clockwise sense and along BA—along C in anti-clockwise sense.

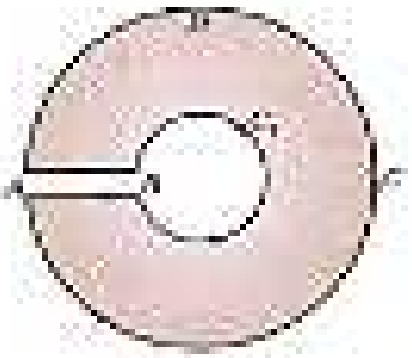

Fig. 20.17(a)

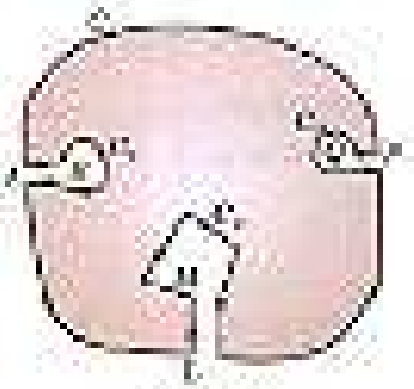

Fig. 20.17(b)

i.e., $$\int_{AB} f(z)dz + \int_{C_1} f(z)dz + \int_{BA} f(z)dz + \int_{C} f(z)dz = 0$$

But since the integrals along AB and along BA cancel, it follows that

$$\int_{C_1} f(z)dz + \int_{C} f(z)dz = 0$$

Reversing the direction of the integral around C_1 and transposing, we get

$$\int_{C} f(z)dz = \int_{C_2} f(z)dz \quad \text{each integral being taken in the anti-clockwise sense.}$$

If $C_1, C_2, C_3, \ldots$ be any number of closed curves as in Fig. 20.17(b), then

$$\oint_{C} f(z)dz = \oint_{C_1} f(z)dz + \oint_{C_2} f(z)dz + \oint_{C_3} f(z)dz + \ldots$$

20.10 CAUCHY'S INTEGRAL FORMULA

If $f(z)$ is analytic within and on a closed curve and $f(a)$ is any point within C, then

$$f(a) = \frac{1}{2\pi i}\oint_{C} \frac{f(z)dz}{z-a}$$

Consider the function $f(z)/(z-a)$ which is analytic at all points within C except at $z = a$. With the point a as centre and radius r, draw a small circle C_1 lying entirely within C.

Now $f(z)/z - a$ being analytic in the region enclosed by C and C_1, we have by Cauchy's theorem,

$$\oint_{C} \frac{f(z)}{z-a}dz = \oint_{C_1} \frac{f(z)}{z-a}dz \qquad \begin{cases} \text{For any point on } C_1, \\ z - a = re^{i\theta} \text{ and } dz = ire^{i\theta}\,d\theta \end{cases}$$

$$= \int_{0}^{2\pi} \frac{f(a + re^{i\theta})}{re^{i\theta}} ire^{i\theta}\,d\theta = i\int_{0}^{2\pi} f(a + re^{i\theta})\,d\theta \qquad \ldots(1)$$

In the limiting form, as the circle C_1 shrinks to the point a, i.e., as $r \to 0$, the integral (1) on the right approaches the value $i\int_{0}^{2\pi} f(a)\,d\theta = f(a)i\int_{0}^{2\pi} d\theta = 2\pi i f(a)$. Thus $\oint_{C} \frac{f(z)}{z-a}dz = 2\pi i f(a)$

i.e., $$f(a) = \frac{1}{2\pi i}\oint_{C} \frac{f(z)}{z-a}dz \qquad \ldots(2)$$

which is the desired Cauchy's Integral formula. (V.T.U., 2011 S)

Cor. Differentiating both sides of (2) w.r.t. a,

$$f'(a) = \frac{1}{2\pi i}\oint_{C} \frac{\partial}{\partial a}\left[\frac{f(z)}{z-a}\right]dz = \frac{1}{2\pi i}\oint_{C} \frac{f(z)}{(z-a)^2}dz \qquad \ldots(3)$$

Similarly, $$f''(a) = \frac{2!}{2\pi i}\oint_{C} \frac{f(z)}{(z-a)^3}dz \qquad \ldots(4)$$

and is given by $$f^n(a) = \frac{n!}{2\pi i}\int_C \frac{f(z)}{(z-a)^{n+1}}\,dz \qquad \ldots(3)$$

That is different from the result (2) is to say that if $f(z)$ is analytic within and on a simple closed curve C then the values of the function and all its derivatives may be found at any point a of C. Therefore, we have established a remarkable fact that an analytic function possesses derivatives of all orders and these are themselves all analytic.

Example 28.30. Evaluate $\int_C \frac{z^2+z+1}{z-1}\,dz$, where C is the circle:

(i) $|z| = 3$, (ii) $|z| = \frac{1}{2}$. (U.P.T.U., 2003)

Solution. (i) Here $f(z) = z^2 + z + 1$ and $a = 1$.

Since $f(z)$ is analytic within and on circle C : $|z| = 3$ and $a = 1$ lies on C.

$\therefore$ by Cauchy's integral formula $\frac{1}{2\pi i}\int_C \frac{f(z)}{z-a}\,dz = f(a) = 1+1+1$, $\int_C \frac{z^2+z+1}{z-1}\,dz = 3\pi i$

(ii) In this case, $a = 1$ lies outside the circle C : $|z| = 1/2$, $\because |z-1| \geq |z| - |1| \Rightarrow |z-1| \geq \frac{1}{2} - 1 \Rightarrow |z-1| \geq \frac{1}{2}$

$\therefore$ by Cauchy's theorem $\int_C \frac{z^2+z+1}{z-1}\,dz = 0$.

Example 28.31. Evaluate, using Cauchy's integral formula

(i) $\int_C \frac{\cos\pi z^2 + \sin\pi z^2}{(z-2)(z-3)}\,dz$, where C is the circle $|z| = 5$ (U.P.T.U., 2005)

(ii) $\int_C \frac{\cos\pi z}{z^2-1}\,dz$, where C is a rectangle with vertices $2+i, 2-i, -2-i, -2+i$

(iii) $\int_C \frac{e^{2z}}{(z-1)^4}\,dz$, where C is the circle $|z| = 3$. (U.P.T.U., 2004)

Solution. (i) $f(z) = \sin\pi z^2 + \cos\pi z^2$ is analytic within the circle $|z| = 5$ and the two singular points $z = 2$ and $z = 3$ lie inside the circle.

$$\therefore \int_C \frac{f(z)\,dz}{(z-2)(z-3)} = \int_C (\sin\pi z^2 + \cos\pi z^2)\left[\frac{1}{z-2} - \frac{1}{z-3}\right]$$

$$= \int_C \frac{(\sin\pi z^2 + \cos\pi z^2)}{z-2}\,dz - \int_C \frac{(\sin\pi z^2 + \cos\pi z^2)}{z-3}\,dz$$

$$= 2\pi i\,[\sin\pi(2)^2 + \cos\pi(2)^2] - 2\pi i\,[\sin\pi(3)^2 + \cos\pi(3)^2]$$

[By Cauchy's integral formula]

$$= 2\pi i\,(0+1) - 2\pi i\,(0-1) = 4\pi i$$

(ii) $f(z) = \cos\pi z$ is analytic in the region bounded by the given rectangle and the two singular points $z = 1$ and $z = -1$ lie inside this rectangle (Fig. 28.10)

$$\therefore \int_C \frac{\cos\pi z}{z^2-1}\,dz = \frac{1}{2}\int_C \left[\frac{1}{z-1} - \frac{1}{z+1}\right]\cos\pi z\,dz$$

$$= \frac{1}{2}\int_C \frac{\cos\pi z}{z-1}\,dz - \int_C \frac{\cos\pi z}{z+1}\,dz$$

$$= \frac{1}{2}\,(2\pi i\cos\pi(1)) - \frac{1}{2}\,(2\pi i\cos\pi(-1)) = 0.$$

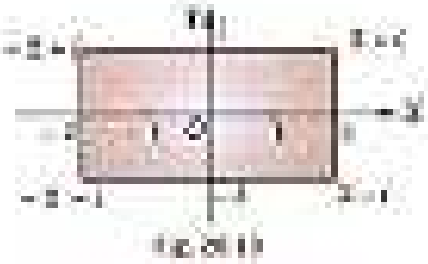

Fig. 28.10

[By Cauchy's integral formula]

Theorem 4.1. Let $f(x)$ be a polynomial with integer coefficients. Then the number of solutions of the congruence $f(x) \equiv 0 \pmod{p}$ is at most the degree of $f(x)$.

Proof. We prove this by induction on the degree of $f(x)$. If $f(x)$ is a constant polynomial, then the theorem is trivially true. Now suppose that the theorem is true for all polynomials of degree $n-1$, and let $f(x)$ be a polynomial of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, then we are done. Otherwise, suppose that a is a solution, so that $f(a) \equiv 0 \pmod{p}$. Then we can write $f(x) = (x-a)g(x) + r$ for some polynomial $g(x)$ of degree $n-1$ and some integer r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$, so $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.2. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.3. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.4. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.5. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.6. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.7. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.8. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.9. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Theorem 4.10. Let p be a prime and let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $p \nmid a_n$. Then the congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

Proof. We proceed by induction on n. If $n = 0$, then $f(x) = a_0$ and the result is clear. Suppose the result holds for all polynomials of degree $n-1$. Let $f(x)$ be of degree n. If $f(x) \equiv 0 \pmod{p}$ has no solutions, we are done. Otherwise, let a be a solution. Then $f(x) = (x-a)g(x) + r$ for some $g(x)$ of degree $n-1$ and some r. Substituting $x = a$ gives $r \equiv 0 \pmod{p}$. Thus $f(x) \equiv (x-a)g(x) \pmod{p}$. By the induction hypothesis, $g(x) \equiv 0 \pmod{p}$ has at most $n-1$ solutions, so $f(x) \equiv 0 \pmod{p}$ has at most n solutions. $\square$

Example 39.33. *Verify Cauchy's theorem by integrating z^n along the boundary of the triangle with the vertices at the points $1 + i$, $-1 + i$ and $-1 - i$.* (U.P.T.U., 2006)

Solution. The boundary of the given triangle consists of three lines AB, BC, CA (Fig. 39.33).

$$\oint_{ABC} z^n\,dz = \int_{AB} z^n\,dz + \int_{BC} z^n\,dz + \int_{CA} z^n\,dz$$

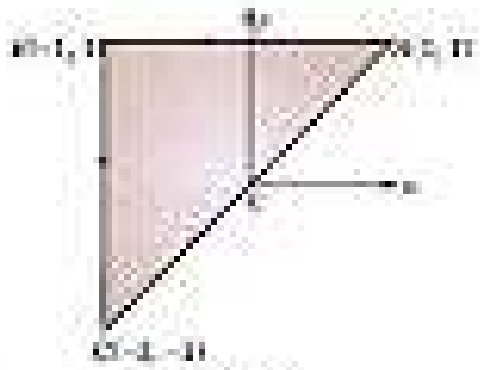

Fig. 39.33

Now $$\int_{AB} z^n\,dz = \int_{1}^{-1} x^{n}(1+i)\,dx \quad \left| \begin{array}{l} \because \text{Along } AB: y = x \\ \therefore z = x + iy = x + ix = x(1+i) \text{ and } dz = (1+i)\,dx \end{array} \right.$$

$$= -\int_{-1}^{1} x^n (1+i)\,dx = -(1+i)\left[\frac{x^{n+1}}{n+1}\right]_{-1}^{1} = -\frac{(1+i)}{n+1}\left[1^{n+1} - (-1)^{n+1}\right]$$

$$\int_{BC} z^n\,dz = \int_{1}^{1} (1+iy)^n\,i\,dy$$

$$\left| \begin{array}{l} \because \text{Along } BC: x = 1 \\ \therefore z = 1 + iy, dz = i\,dy \end{array} \right.$$

$$= i\int_{1}^{-1} (1+iy)^n\,dy = i\left[\frac{(1+iy)^{n+1}}{n+1}\right]_{1}^{-1} = \frac{i}{n+1}\left[(1-i)^{n+1} - 2^{n+1}\right]$$

$$\int_{CA} z^n\,dz = \int_{1}^{-1} (x+ix)^n\,(1+i)\,dx$$

$$\left| \begin{array}{l} \because \text{Along } CA: y = x \\ \therefore z = (1+i)x, dz = (1+i)\,dx \end{array} \right.$$

$$= (1+i)^{n+1}\left[\frac{x^{n+1}}{n+1}\right]_{1}^{-1} = \frac{(1+i)^{n+1}}{n+1}\left[(-1)^{n+1} - 1^{n+1}\right]$$

Thus from (i) $$\oint_{ABC} z^n\,dz = \frac{2^{n+1} - (-1)^{n+1}}{n+1} + \frac{(1-i)^{n+1} - (1+i)^{n+1}}{n+1} + \frac{(-1)^{n+1} - 2^{n+1}}{n+1} = 0 \quad \ldots(ii)$$

Also since $f(z) = z^n$ is analytic everywhere.

$\therefore$ by Cauchy's theorem $\oint_{ABC} f(z)\,dz = 0$ ...(iii)

Hence from (ii) and (iii), Cauchy's theorem is verified.

Example 39.34. *If $F(z) = \oint_C \frac{4z^2 + z + 5}{z - a}\,dz$, where C is the ellipse $9x^2 + 4y^2 = 1$, find the value of (a) $F(3)$ (b) $F'(1)$, (c) $F''(-1)$ and (d) $F'''(-1)$.* (Bhopal, 2009; Kolkata, 2005; Mumbai, 2006)

Solution. (a) $$F(3) = \oint_C \frac{4z^2 + z + 5}{z - 3}\,dz$$

Since $z = 3$ is the only singular point of $(4z^2 + z + 5)/(z - 3)$ and it lies outside the ellipse C, therefore, $(4z^2 + z + 5)/(z - 3)$ is analytic everywhere within C.

Hence by Cauchy's theorem,

$$\oint_C \frac{4z^2 + z + 5}{z - 3}\,dz = 0, \text{ i.e., } F(3) = 0$$

(b) Since $f(z) = 4z^2 + z + 5$ is analytic within C and $\zeta = 1, -1$ and $-i$ all lie within C, therefore, by Cauchy's integral formula

$$f(a) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z - a}\,dz$$

e.g., $$\oint_C \frac{4z^2+z+3}{z-i}\,dz = 2\pi i(4i^2+i+3)$$

i.e., $$F(i) = 2\pi i(4i^2+i+3)$$

or $$F'(z) = 2\pi i(8z+1) \text{ and } F'(i) = 16\pi i$$

Thus $$F(z) = 2\pi i(4z^3+z^2+3z+c) = 2\pi i(z-1)$$

$$F'(-2i) = 2\pi i(8(-2i)+1) = -14\pi i \text{ and } F'(-i) = 10\pi i$$

20.32 (i) CONVERSE OF CAUCHY'S THEOREM: MORERA'S THEOREM*

If $f(z)$ is continuous in a region R and $\oint_C f(z)\,dz = 0$ around every simple closed curve C in R, then $f(z)$ is analytic in R.

Since $\oint_C f(z)\,dz = 0$, then the line integral of $f(z)$ from a fixed point z_0 to a variable point z must be independent of the path and hence must be a function of z only. Thus

$$\int_{z_0}^{z} f(z)\,dz = \phi(z) + c$$

Let $$\phi(z) = U + iV \text{ and } f(z) = u + iv$$

Then $$U + iV = \int_{(x_0,y_0)}^{(x,y)} (u+iv)(dx+i\,dy) = \int_{(x_0,y_0)}^{(x,y)} (u\,dx - v\,dy) + i\int_{(x_0,y_0)}^{(x,y)} (u\,dy + v\,dx)$$

$$\therefore \quad U = \int_{(x_0,y_0)}^{(x,y)} (u\,dx - v\,dy), \; V = \int_{(x_0,y_0)}^{(x,y)} (u\,dy + v\,dx)$$

Differentiating under the integral sign,

$$\frac{\partial U}{\partial x} = u, \; \frac{\partial U}{\partial y} = -v, \; \frac{\partial V}{\partial x} = v, \; \frac{\partial V}{\partial y} = u \quad \therefore \quad \frac{\partial U}{\partial x} = \frac{\partial V}{\partial y}, \; \frac{\partial U}{\partial y} = -\frac{\partial V}{\partial x}$$

Thus U and V satisfy C-R equations.

Also, since $f(z)$ is given to be continuous, u and v and therefore, $\partial U/\partial x$, $\partial U/\partial y$, $\partial V/\partial x$, $\partial V/\partial y$, are also continuous.

$\therefore$ $\phi(z)$ is an analytic function and

$$\phi'(z) = \frac{\partial U}{\partial x} + i\frac{\partial V}{\partial x} = u + iv = f(z)$$

Thus $f(z)$ is the derivative of an analytic function $\phi(z)$. Hence $f(z)$ is analytic by § 20.31 Cor.

(ii) Cauchy's Inequality†. *If $f(z)$ is analytic within and on the circle C: $|z-a| = r$, then*

$$|f^n(a)| \le \frac{M n!}{r^n} \qquad \ldots(3)$$

where M is the maximum value of $|f(z)|$ on C.

From (5) of § 20.31, we get

$$|f^n(a)| = \frac{n!}{2\pi}\left|\oint_C \frac{f(z)\,dz}{(z-a)^{n+1}}\right|$$

$$\le \frac{n!}{2\pi}\cdot\frac{M}{r^{n+1}}\oint_C |dz| \qquad [\because |f(z)| \le M]$$

$$= \frac{n!M}{2\pi r^{n+1}}\oint_C ds = \frac{Mn!}{2\pi r^{n+1}}\,2\pi r = \frac{Mn!}{r^n} \qquad (\because \oint_C ds = 2\pi r)$$

(iii) Liouville's theorem‡. *If $f(z)$ is analytic and bounded for all z in the entire complex plane, then $f(z)$ is a constant.* $(\because |f(z)| \le M, \text{ say})$

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* Named after the Italian mathematician, Giacinto Morera (1856–1909) who worked in Turin.

† See Footnote p. 744.

‡ See Footnote p. 678.

Taking $n = 1$ and replacing ρ by r, this gives

$$|f'(z)| \le M/r$$

As $r \to \infty$, it follows that $f'(z) = 0$ and $f(z)$ is constant for all z.

(b) Poisson's integral formula. (i) Let $f(z)$ be analytic within and on the circle C: $|z| = \rho$ and $z = re^{i\phi}$ is any point within C. Then

$$f(re^{i\phi}) = \frac{1}{2\pi}\int_0^{2\pi} \frac{\rho^2 - r^2}{\rho^2 - 2\rho r\cos(\theta - \phi) + r^2} f(\rho e^{i\theta})\,d\theta$$

By Cauchy's integral formula, $f(z) = \dfrac{1}{2\pi i}\displaystyle\int_C \frac{f(w)}{w - z}\,dw$ ...(1)

As the inverse of the point $z = re^{i\phi}$ in C is the point Z and is given by $\rho^2/\bar{z}$. [See Section 6.10]

$\therefore$ by Cauchy's theorem,

$$0 = \frac{1}{2\pi i}\int_C \frac{f(w)}{w - \rho^2/\bar{z}}\,dw \qquad ...(2)$$

Subtracting (2) from (1), $f(z) = \dfrac{1}{2\pi i}\displaystyle\int_C \left[\frac{1}{w - z} - \frac{1}{w - \rho^2/\bar{z}}\right] f(w)\,dw$

$$= \frac{1}{2\pi i}\int_C \frac{\bar{z}w - \rho^2}{(w\bar{z} - \rho^2) + \rho(w - z)}\,f(w)\,dw \qquad ...(3)$$

Taking $w = \rho e^{i\theta}$ and noting that $z = re^{i\phi}$, (3) gives

$$f(re^{i\phi}) = \frac{1}{2\pi}\int_0^{2\pi} \frac{re^{-i\phi}\cdot\rho e^{i\theta} - \rho^2}{\rho^2 - \rho(re^{-i\phi} - re^{i\phi}) + r^2}\,f(\rho e^{i\theta})\,\rho e^{i\theta}\,d\theta$$

$$= \frac{1}{2\pi}\int_0^{2\pi} \frac{\rho^2 - r^2 + r\rho(e^{i\theta} - e^{-i\phi})}{\rho^2 - r\rho(e^{i\theta} + e^{-i\phi}) + r^2}\,f(\rho e^{i\theta})\,d\theta = \frac{1}{2\pi}\int_0^{2\pi} \frac{(\rho^2 - r^2)f(\rho e^{i\theta})}{\rho^2 - 2\rho r\cos(\theta - \phi) + r^2}\,d\theta \qquad ...(4)$$

This is called Poisson's integral formula for a circle. It expresses the values of a harmonic function u within a circle in terms of its values on the boundary.

We can put $f(re^{i\phi}) = u(r, \phi) + iv(r, \phi)$ and $f(\rho e^{i\theta}) = u(\rho, \theta) + iv(\rho, \theta)$ in (4) and equating real and imaginary parts from both sides, we get the formulae,

$$u(r, \phi) = \frac{1}{2\pi}\int_0^{2\pi} \frac{\rho^2 - r^2}{\rho^2 - 2\rho r\cos(\theta - \phi) + r^2}\,u(\rho, \theta)\,d\theta \qquad ...(5)$$

and

$$v(r, \phi) = \frac{1}{2\pi}\int_0^{2\pi} \frac{\rho^2 - r^2}{\rho^2 - 2\rho r\cos(\theta - \phi) + r^2}\,v(\rho, \theta)\,d\theta \qquad ...(6)$$

PROBLEMS 2.10

1. Evaluate $\displaystyle\int_C \frac{z^2 - 2z}{(z + 1)^2(z^2 + 4)}\,dz$, where C is a simple closed curve and the point $z = 1$ is (i) outside C, (ii) inside C.
2. Evaluate $\displaystyle\int_C \frac{e^{az}}{(z - b)^n}\,dz$, $n = 2, 3, 4, \ldots$, where C is a closed curve containing the point $z = b$.
3. Evaluate (i) $\displaystyle\int_C \frac{z^2 + 1}{z^2 - 1}\,dz$, where C is the circle $|z| = 3/2$ (V.T.U., 2003)

   (ii) $\displaystyle\int_C \frac{e^{2z}}{(z - 1)^2}\,dz$, where C is the circle $|z - 1| = 3$. (U.P.T.U., 2003)

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* Named after the French mathematician Siméon Denis Poisson (1781–1840) who was a professor in Paris and made contributions to partial differential equations, potential theory and probability.

1. Use Cauchy's integral formula to evaluate:

(a) $\int_C \frac{2z+1}{z^2+z-2}dz$, where C is $|z|=1.5$ (P.U. 2000) (b) $\int_C \frac{z^2+1}{z(2z+1)}dz$, where C is $|z|=1$

(c) $\int_C \frac{\cos\pi z+\sin\pi z}{(z-1)(z-2)}dz$, where C is $|z|=4$ (U.P.T.U. 2000)

2. Evaluate (a) $\int_C \frac{z^2-2z+1}{(z-2)^3}dz$, where C is $|z|=3$

(b) $\int_C \frac{e^{2z}}{(z-1)(z-2)^2}dz$, where C is $|z|=3$ (Kerala, 2000)

3. Evaluate, using Cauchy's integral formula:

(a) $\int_C \frac{z}{(z-3)(z+1)}dz$, where C is $|z+1|=\frac{1}{2}$ (U.P.T.U. 2000; Anna, 2000; Madras, 2000)

(b) $\int_C \frac{e^{2z}dz}{(z+1)^4}$, where C is $|z-1|=2$. (Nagpur, 2000)

(c) $\int_C \frac{\cos\pi z}{(z-1)^3}dz$, where C is $|z-1|=\frac{1}{2}$. (U.P.T.U. 2001)

4. Evaluate $f(2)$ and $f'(2)$ where $f(z)=\int_C \frac{3z^2+7z+2}{z-a}dz$ and C is the circle $|z|=3$.

5. If $f(z)=\int_C \frac{3z^2+2z+1}{z-c}dz$, where C is the circle $|z|=1$ find the values of

(a) $f(3)$ (b) $f'(1-i)$ (c) $f(1)+f'(1)$. (Nagpur, 2000)

6. Evaluate $\int_C \frac{z^3+z+1}{z^2-3z+2}dz$, where C is the circle $4x^2+4y^2=1$. (Nagpur, 2000)

7. Verify Cauchy's theorem for the integral of z^2 taken over the boundary of the triangle with vertices $(0,1)$, $(1,i)$ and $(-1,i)$ (triangle is anticlockwise). (U.P.T.U. 2001)

3.13 [3] SERIES OF COMPLEX TERMS

Let $(a_1+ib_1)+(a_2+ib_2)+\ldots+(a_n+ib_n)+\ldots$ ... (1)

be an infinite series of complex terms a_n+ib_n and a_n being real numbers. If the series Σa_n of Σb_n converge to the sums A and B, then series (1) is said to converge to the sum $A+iB$. Also if (1) is a convergent series, then $\lim_{n\to\infty}(a_n+ib_n)=0$.

The series (1) is said to be **absolutely convergent** if the series

$$|a_1+ib_1|+|a_2+ib_2|+\ldots+|a_n+ib_n|+\ldots$$

is convergent. Since $|a_n|$ and $|b_n|$ are both $\le |a_n+ib_n|$, it follows that an absolutely convergent series is convergent.

Next let the series of functions $u_1(z)+u_2(z)+\ldots+u_n(z)+\ldots$ ... (2)

converge to the sum $S(z)$ and $S_n(z)$ be the sum of its first n terms. Then the series (2) is said to be **uniformly convergent** in a region R if corresponding to any positive number ε, however small, there exists a positive number N, depending on ε, but not on z, such that for every z in R

$$|S(z)-S_n(z)|<\varepsilon \text{ for } n>N. \quad \text{[cf. Def. p. 283]}$$

As in the case of real series (p. 284) **Weierstrass M-test** is for the series of complex terms. So the series (2) is uniformly convergent in a region R if there is a convergent series of positive constants ΣM_n such that $|u_n(z)|\le M_n$ for all z in R.

Also a uniformly convergent series of continuous complex functions is itself continuous and may be integrated term by term.

Obs. If a power series $\sum a_n z^n$ converges for $z = z_1$, then it converges absolutely for $|z| < |z_1|$.

Indeed, $\sum a_n z_1^n$ converges by hypothesis, so that $|a_n z_1^n| \to 0$ as $n \to \infty$, and therefore $|a_n z_1^n| < k$ for all n. Thus $|a_n z^n| = |a_n z_1^n| \cdot |z/z_1|^n < k|z/z_1|^n$, where $|z/z_1| < 1$.

But the series $\sum k^n$ converges for $|z| < 1$. Hence the series $\sum a_n z^n$ converges absolutely for $|z| < |z_1|$, i.e. within the circle with center $z = 0$ and radius $|z_1|$ as shown. Also the given series converges absolutely at all points inside the circle.

Such a circle $|z| = R$ within which a power series $\sum a_n z^n$ converges is called the *circle of convergence*, and R is called the *radius of convergence*.

A power series is uniformly convergent in any region which lies entirely within the circle of convergence.

(B) Taylor's series*. *If $f(z)$ is analytic inside a circle C with center at a, then for z inside C*

$$f(z) = f(a) + f'(a)(z-a) + \frac{f''(a)}{2!}(z-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(z-a)^n + \cdots \qquad (10)$$

Proof. Let z be any point inside C. Draw a circle C_1 with center at a enclosing z (Fig. 20.28). Let t be a point on C_1. We have

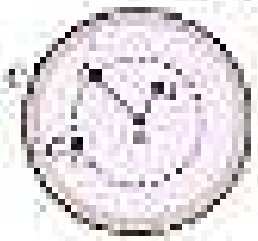
Fig. 20.28

$$\frac{1}{t-z} = \frac{1}{t-a-(z-a)} = \frac{1}{t-a}\left(1 - \frac{z-a}{t-a}\right)^{-1}$$

$$= \frac{1}{t-a}\left[1 + \frac{z-a}{t-a} + \left(\frac{z-a}{t-a}\right)^2 + \cdots + \left(\frac{z-a}{t-a}\right)^n + \cdots\right] \qquad (11)$$

As $|z-a| < |t-a|$, i.e. $|(z-a)/(t-a)| < 1$, this series converges uniformly. By multiplying both sides of (11) by $f(t)$ we can integrate over C_1,

$$\oint_{C_1} \frac{f(t)}{t-z}\,dt = \oint_{C_1} \frac{f(t)}{t-a}\,dt + (z-a)\oint_{C_1} \frac{f(t)}{(t-a)^2}\,dt + \cdots + (z-a)^n \oint_{C_1} \frac{f(t)}{(t-a)^{n+1}}\,dt + \cdots \qquad (12)$$

Since $f(z)$ is analytic on and inside C_1, therefore, applying the formulas (2) in Sec. 20.3 with $n = 0$ and (1), we get (10), which is known as Taylor's series.

Obs. An important remarkable fact is that complex analytic functions can always be represented by power series of the form (10).

(C) Laurent's series†. *If $f(z)$ is analytic in the ring-shaped region R bounded by two concentric circles C and C_2 of radius r and r_2 ($r > r_2$) and center at $z = a$, then for all z in R*

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \cdots + a_n(z-a)^n + \cdots + b_1(z-a)^{-1} + b_2(z-a)^{-2} + \cdots$$

where $a_n = \dfrac{1}{2\pi i}\displaystyle\oint_{C_1} \frac{f(t)}{(t-a)^{n+1}}\,dt$.

C_1 is any circle in R, surrounding C_2, see Fig. 20.29.

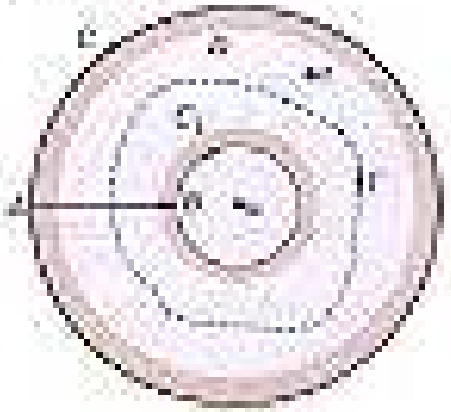
Fig. 20.29

Proof. Introduce cuts on AK, then $f(z)$ is analytic in the region R bounded by AB, C_1 described clockwise, DA and C described anti-clockwise (see Fig. 20.29). Then if z be any point in R, we have

$$f(z) = \frac{1}{2\pi i}\left[\int_{AB} \frac{f(t)}{(t-z)}\,dt + \int_{C_1} \frac{f(t)}{t-z}\,dt + \int_{DA} \frac{f(t)}{(t-z)}\,dt + \int_{C} \frac{f(t)}{t-z}\,dt\right]$$

$$= \frac{1}{2\pi i}\left[\oint_{C_1} \frac{f(t)}{t-z}\,dt - \int_{C_2} \frac{f(t)}{t-z}\,dt\right] \qquad (13)$$

where both C and C_1 are described anti-clockwise in (13) and integrals along AB and DA cancel (Fig. 20.29).

For the first integral in (13), expanding $1/(t-z)$ as in Ex. 20.18 (2), we get

$$\frac{1}{2\pi i}\oint_{C_1} \frac{f(t)}{t-z}\,dt = \sum_{n=0}^{\infty} \frac{(z-a)^n}{2\pi i}\oint_{C_1} \frac{f(t)}{(t-a)^{n+1}}\,dt$$

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* See footnote p. 168.

† Named after the French mathematician Pierre Alphonse Laurent (1813–1854) who published this theorem in 1843.

$$= 2\pi i a_n \text{ where } a_n = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z-a)^{n+1}}\,dz \qquad ...(v)$$

For the second integral in (v), let t lie on C_1. Then we write

$$\frac{1}{t-z} = \frac{1}{(t-a)-(z-a)} = \frac{1}{t-a}\left(1-\frac{z-a}{t-a}\right)^{-1}$$

$$= \frac{1}{t-a}\left[1+\frac{z-a}{t-a}+\left(\frac{z-a}{t-a}\right)^2+\ldots+\left(\frac{z-a}{t-a}\right)^{n-1}+\ldots\right]$$

As $|z-a| < |t-a|$, i.e., $|t-a|/|z-a| > 1$, this series converges uniformly. So multiplying both sides by $f(t)$ and integrating over C_1, we get

$$\frac{1}{2\pi i}\oint_C \frac{f(t)}{t-z}\,dt = \sum_{n=1}^{\infty}\frac{1}{(z-a)^n}\cdot\frac{1}{2\pi i}\oint_C (t-a)^{n-1} f(t)\,dt = \sum_{n=1}^{\infty} a_{-n}(z-a)^{-n} \qquad ...(vi)$$

where $$a_{-n} = \frac{1}{2\pi i}\oint_C \frac{f(t)}{(t-a)^{-n+1}}\,dt$$

Substituting from (v) and (vi) in (iv), we get

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} a_{-n}(z-a)^{-n}. \qquad ...(vii)$$

Now $f(z)/(z-a)^{n+1}$ being analytic in the region between C and Γ, we can take the integral giving a_n over Γ. Similarly we can take the integral giving a_{-n} over Γ. Hence (vii) can be written as

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n \text{ where } a_n = \frac{1}{2\pi i}\oint_\Gamma \frac{f(t)}{(t-a)^{n+1}}\,dt$$

which is known as *Laurent's series*.

(Note 1) $\frac{1}{2\pi i}\oint_C f(t)\,dt$ is not given as the coefficient of $(z-a)^0$ in $f(z)$. However, $f(z)$ is analytic inside C, $a_0 = \frac{1}{2\pi i}\oint_C \frac{f(t)}{(t-a)^{0+1}}\,dt = \frac{f^0(a)}{0!} = f(a)$

(However, $f(z)$ is analytic inside C, then $a_{-n} = 0$, $a_n = \frac{1}{2\pi i}\oint_C \frac{f(t)}{(t-a)^{n+1}}\,dt = \frac{f^n(a)}{n!}$

as in Laurent's series reduces to Taylor's series.

(Note 2) The advantage of Taylor's or Laurent's series, sometimes *(as in binomial theorem instead of finding a_n by complex integration, which is quite complicated)*

(Note 3) Laurent series of a given analytic function $f(z)$ in the annulus of convergence is unique. There may be different Laurent series of $f(z)$ in two annuli with the same centre.

Example 10.25 Show that the series $1 + z + z^2 + \ldots + z^n + \ldots + \frac{1}{z} + \frac{1}{z^2} + \ldots + \frac{1}{z^n} + \ldots$ converges for $|z| = 2$. Determine whether it converges absolutely or not.

Solution. Let the sum of the first n terms of the series be s_n, so that

$$s_n(z) = 1 + z + z^2 + z^3 + \ldots + z^{n-1} + \frac{1}{z} + \frac{1}{z^2} + \ldots + \frac{1}{z^{n-1}} = z^{n-1}$$

For $|z| = 1$, $z^{n-1} \to 0$ as $n \to \infty$

$\therefore$ $\lim_{n\to\infty} s_n(z) = s$, i.e., the given series converges for $|z| = 1$.

$$|s_n(z)| = |s(z) - s| = |z^n(1-z)| + \ldots + |z^n(1-z)|$$

$$= |1-z|\,|1 + |z| + |z|^2 + \ldots + |z|^n|$$

For $|z| < 1$, $\lim_{n\to\infty} |s_n(z)| = |1-z|\frac{|z|}{1-|z|}$ (G.P.)

Hence the given series converges absolutely.

Example 20.20. Expand $\sin x$ in a Taylor's series about $x = \pi/4$. Hence determine the value of $\sin 46^\circ$ correct to four decimal places. (W.T.U. 2008 S)

Solution. Given $f(x) = \sin x, f'(x) = \cos x, f''(x) = -\sin x, f'''(x) = -\cos x, \ldots$

$\therefore f(\pi/4) = 1/\sqrt{2}, f'(\pi/4) = 1/\sqrt{2}, f''(\pi/4) = -1/\sqrt{2}, f'''(\pi/4) = -1/\sqrt{2}$

By Taylor's series about $x = \pi/4$, we have

$$f(x) = f(\pi/4) + \frac{x-\pi/4}{1!}f'(\pi/4) + \frac{(x-\pi/4)^2}{2!}f''(\pi/4) + \frac{(x-\pi/4)^3}{3!}f'''(\pi/4) + \ldots$$

i.e., $$\sin x = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\left(x - \frac{\pi}{4}\right) - \frac{1}{\sqrt{2}}\frac{(x-\pi/4)^2}{2!} - \frac{1}{\sqrt{2}}\frac{(x-\pi/4)^3}{3!} + \ldots$$

Hence $$\sin x = \sum_{n=1}^{\infty} a_n (x - \pi/4)^{n-1}, \text{ where } a_n = \frac{1+(-1)^{n-1}}{\sqrt{2}\,(n-1)!}$$

Since $$\rho = \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{(2n-1)!}{(2n+1)!}\right| = 0$$

The series is convergent if $|x| < \infty$

i.e., the region of convergence of $f(x)$ is all reals.

Example 20.21. Find Taylor's expansion of

(i) $f(x) = \dfrac{1}{(x+1)^2}$ about the point $x = -2$ (W.T.U. 2009 S)

(ii) $f(x) = \dfrac{2x^2+1}{x^3-x}$ about the point $x = 1$. (W.T.U. 2008)

Solution. (i) The expand $f(x)$ about $x = -2$, i.e., in powers of $x + 2$, put $x + 2 = t$. Then

$$f(x) = \frac{1}{(t-1+1)^2} = (1-t)^{-2} = 1 + 2t + 3t^2 + 4t^3 + \ldots = \frac{1}{2}\left[1 - \frac{2t}{1!} + \frac{2^2t^2}{1!} - \frac{2^3t^3}{2!} + \ldots\right]$$

[Expanding by Binomial theorem]

$$= \frac{1}{2}\left[1 + \sum_{n=1}^{\infty}(-1)^n \frac{(n+1)2^n t^n}{n!}\right]$$

(ii) $$f(x) = \frac{2x^2+1}{x(x+1)(x-1)} = 2x - 1 + \frac{2x+1}{x(x+1)} = (2x - 2) + 1 + \frac{1}{x} + \frac{1}{x+1} \quad \ldots(i)$$

[By partial fractions]

The expand $2x - 2$ and $\frac{1}{x} + \frac{1}{x+1}$ about $x = 1$, i.e., put $x - 1 = t$, so that

$$\frac{1}{x} = \frac{1}{(t+1)} = \frac{1}{1}\left(1 + \frac{t}{1}\right)^{-1}$$

[Expanding by Binomial theorem]

$$= \frac{1}{1}\left[1 - \frac{t}{1} + \frac{t^2}{1^2} - \frac{t^3}{1^3} + \frac{t^4}{1^4} - \ldots\right] = 1 - t + t^2 - t^3 + t^4 - \ldots$$

$$= 1 + (x-1) + \sum_{n=2}^{\infty}(-1)^n (x-1)^n \quad \ldots(ii)$$

and $$\frac{1}{x+1} = \frac{1}{1+(t+1)} = \frac{1}{1+t}\left(1 + \frac{t}{2}\right)^{-1}$$

[Expanding by Binomial theorem]

$$= \frac{1}{1+t}\left[1 - \frac{t}{1!} + \frac{t^2}{(1+t)^2} - \frac{t^3}{(1+t)^3} + \frac{t^4}{(1+t)^4} - \ldots\right]$$

$$= \frac{1-t}{2} - \frac{1}{2}\left[\frac{t^2}{(1+t)^2} - \frac{t^3}{(1+t)^3} + \frac{t^4}{(1+t)^4} - \ldots\right] = \frac{1}{2} - \frac{1}{2}\frac{x-1}{2} + \sum_{n=2}^{\infty}(-1)^n \frac{(x-1)^n}{(1+t)^{n+1}} \quad \ldots(iii)$$

Subtracting from (i) and (ii) in (i), we get

$$f(z)=\left(2i-\frac{3}{2}-1+\frac{1}{2}i-\frac{i}{2}\right)+\left(2+1-\frac{1}{2}i\right)(z-1)+\sum_{n=2}^{\infty}(-1)^n\left[\frac{1}{(2i)^{n+1}}+\frac{1}{(1+i)^{n+1}}\right](z-1)^n$$

$$=\left(\frac{1}{2}-\frac{3}{2}i\right)+\left(3+\frac{1}{2}i\right)(z-1)+\sum_{n=2}^{\infty}(-1)^n\left[\frac{1}{(2i)^{n+1}}+\frac{1}{(1+i)^{n+1}}\right](z-1)^n$$

Example 10.28. Expand $f(z)=\frac{z}{(z-1)(z-2)}$ in the regions
(a) $|z|<1$ (b) $1<|z|<2$ (c) $|z|>2$ (d) $0<|z-1|<1$.
(U.P.T.U., 2010; U.P.T.U., 2010 (Mech.), 2009)

Solution. (a) By partial fractions $\frac{z}{(z-1)(z-2)}=\frac{2}{z-2}-\frac{1}{z-1}$ ...(i)

$$=-\frac{1}{2}\left(1-\frac{z}{2}\right)^{-1}+(1-z)^{-1} \quad \text{...(ii)}$$

For $|z|<2$, both $|z/2|$ and $|z|$ are less than 1. Hence (ii) gives on expansion

$$f(z)=-\frac{1}{2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}+\ldots\right)+(1+z+z^2+z^3+\ldots)$$

$$=\frac{1}{2}+\frac{3}{4}z+\frac{7}{8}z^2+\frac{15}{16}z^3+\ldots \text{ which is a Taylor's series.}$$

(b) For $1<|z|<2$, we write (i) as

$$f(z)=-\frac{1}{2}\frac{1}{(1-z/2)}-\frac{1}{z(1-z^{-1})} \quad \text{...(iii)}$$

and we see that both $|z/2|$ and $|z^{-1}|$ are less than 1. Hence (iii) gives on expansion

$$f(z)=-\frac{1}{2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{8}+\ldots\right)-\frac{1}{z}(1+z^{-1}+z^{-2}+z^{-3}+\ldots)$$

$$=\ldots-z^{-4}-z^{-3}-z^{-2}-z^{-1}-\frac{1}{2}-\frac{3}{4}z-\frac{7}{8}z^2-\frac{15}{16}z^3-\ldots$$

which is a Laurent's series.

(c) For $|z|>2$, we write (i) as

$$f(z)=\frac{1}{z(1-2z^{-1})}-\frac{1}{z(1-z^{-1})}$$

$$=z^{-1}(1+2z^{-1}+4z^{-2}+8z^{-3}+\ldots)-z^{-1}(1+z^{-1}+z^{-2}+z^{-3}+\ldots)$$

$$=\ldots+7z^{-4}+3z^{-3}+z^{-2}+\ldots$$

(d) For $0<|z-1|<1$, we write (i) as

$$f(z)=\frac{1}{(z-1)-1}-\frac{1}{z-1}$$

$$=-(z-1)^{-1}-[1-(z-1)]^{-1}$$

$$=-(z-1)^{-1}-[1+(z-1)+(z-1)^2+(z-1)^3+\ldots]$$

Example 10.29. Find the Laurent's expansion of $f(z)=\frac{z^2-2}{(z+1)(z+2)}$ in the region $1<|z+1|<2$.
(U.P.T.U., 2008 (Aero), 2005; U.P.T.U., 2004)

Solution. Writing $z+1=u$, we have

$$f(u)=\frac{(u-1)^2-2}{u(u-1)}=\frac{u^2-2u-1}{u(u-1)}$$

$$=-\frac{1}{u}+\frac{u}{u-1}+\frac{2}{u-1} \quad \text{(splitting into partial fractions)}$$

$$=-\frac{1}{u}+\frac{1}{u(1-1/u)}+\frac{2}{u(1-u^{-1})}=-\frac{1}{u}+\frac{1}{u}\left(1-\frac{1}{u}\right)^{-1}+\frac{2}{u}(1-u^{-1})^{-1}$$

Since $1 < z < 2$ or $|z| > 1$ and $|z^2| > 1$, expanding by Binomial theorem,

$$f(z) = \frac{-2}{z} + \frac{3}{z}\left(2 + \frac{2}{z} + \frac{4}{z^2} + \frac{8}{z^3} + \ldots\right) - \frac{2}{z^2}\left(1 + \frac{2}{z^2} + \frac{4}{z^4} + \frac{8}{z^6} + \ldots\right)$$

$$= -\frac{2}{z} + \frac{6}{z^2} + \frac{6}{z^3} + \ldots - \frac{2}{z^2}\left(1 + \frac{2}{z^2} + \frac{4}{z^4} + \frac{8}{z^6} + \ldots\right)$$

Hence $f(z) = \frac{3}{z+1} + \frac{3}{(z+1)^2} + \frac{3}{(z+1)^3} + \ldots - \frac{2}{z^2}\left[1 - \frac{z+1}{2} + \frac{(z+1)^2}{2^2} + \frac{(z+1)^3}{2^3} + \ldots\right]$

which is valid in the region $1 < |z+1| < 2$.

30.17 (i) ZEROS OF AN ANALYTIC FUNCTION

Def. *A zero of an analytic function $f(z)$ is that value of z for which $f(z) = 0$.*

If $f(z)$ is analytic in the neighbourhood of a point $z = a$, then by Taylor's theorem

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \ldots + a_n(z-a)^n + \ldots \quad \text{where } a_n = \frac{f^n(a)}{n!}$$

If $a_0 = a_1 = a_2 = \ldots = a_{n-1} = 0$ but $a_n \neq 0$, then $f(z)$ is said to have a zero of order n at $z = a$.

When $n = 1$, the zero is said to be simple. In the neighbourhood of zero $z = a$ of order n,

$$f(z) = a_n(z-a)^n + a_{n+1}(z-a)^{n+1} + \ldots$$
$$= (z-a)^n \phi(z) \text{ where } \phi(z) = a_n + a_{n+1}(z-a) + \ldots$$

Then $\phi(z)$ is analytic and non-zero in the neighbourhood of $z = a$.

(ii) Singularities of an analytic function

We have already defined a *singular point* of a function as the point at which the function ceases to be analytic.

(a) **Isolated singularities.** *If $z = a$ is a singularity of $f(z)$ such that $f(z)$ is analytic at each point in the neighbourhood (i.e., there exists a circle with centre a which has no other singularity), then $z = a$ is called an isolated singularity.*

In such a case, $f(z)$ can be expanded in a Laurent's series about $z = a$, giving

$$f(z) = a_0 + a_1(z-a) + a_2(z-a)^2 + \ldots + a_1(z-a)^{-1} + a_2(z-a)^{-2} + \ldots \quad \ldots(1)$$

For example, $f(z) = \cot \pi z$ is not analytic when $\sin \pi z = 0$ i.e. at the points $\pi z = 0, \pm\pi, \pm 2\pi$ or $z = 2k$ ($k = 1, 2, 3, \ldots$).

Thus $z = 1, 1/2, 1/3, \ldots$ are all isolated singularities as there is no other singularity in their neighbourhood.

But when $z = 0$, $\cot \pi z = 0$ is such a singularity that there are infinite number of other singularities in its neighbourhood. Thus $z = 0$ is the non-isolated singularity of $f(z)$.

(a) **Removable singularity.** *If all the negative powers of $(z-a)$ in (1) are zero, then $f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n$.*

Here the singularity can be removed by defining $f(z)$ at $z = a$ in such a way that it becomes analytic at $z = a$. Such a singularity is called a *removable singularity*.

Thus if $\lim_{z \to a} f(z)$ exists finitely, then $z = a$ is a removable singularity.

(b) **Poles.** *If out the negative powers of $(z-a)$ in (1) after the m^{th} are missing, i.e., the singularity of $z = a$ is resolved, a pole of order m.*

A pole of first order is called a simple pole.

(c) **Essential singularity.** *If the number of negative powers of $(z-a)$ in (1) is infinite, then $z = a$ is called an essential singularity.* In this case, $\lim_{z \to a} f(z)$ does not exist.

Example 30.26. *Find the nature and position of singularities of the following functions:*

(i) $\frac{e^z - \sin z - 1}{z^3}$ (ii) $\frac{z}{(z-1)(z-2)}$ (iii) $\frac{z}{\cos z - \sin z}$

Solution: (i) Here $z = 0$ is an isolated singularity.

Also $$\frac{\sin z - \sinh z}{z^3} = \frac{1}{z^3}\left[z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \ldots\right] - \left[z + \frac{z^3}{3!} + \frac{z^5}{5!} + \frac{z^7}{7!} + \ldots\right] = -\frac{2}{3!} - \frac{2z^2}{5!} - \frac{2z^4}{7!} - \ldots$$

Since there are no negative powers of z in the expansion, $z = 0$ is a removable singularity.

(ii) $z \sin\frac{3}{z-2} = (z+2+1)\sin\frac{3}{z-2}$ $\quad$ where $t = z - 2$

$$= (t+3)\left[\frac{3}{t} - \frac{3^3}{3!t^3} + \frac{3^5}{5!t^5} - \ldots\right] = \left[3 - \frac{3^3}{3!t^2} + \frac{3^5}{5!t^4} - \ldots\right] + \left[\frac{9}{t} - \frac{3^4}{3!t^3} + \frac{3^6}{5!t^5} - \ldots\right]$$

$$= 3 + \frac{9}{t} - \frac{3^3}{3!t^2} - \frac{3^4}{3!t^3} + \frac{3^5}{5!t^4} - \ldots = 3 + \frac{9}{z-2} - \frac{27}{6(z-2)^2} - \frac{81}{6(z-2)^3} + \ldots$$

Since there are infinite number of terms in the negative powers of $(z-2)$, $z = 2$ is an essential singularity.

(iii) Poles of $f(z) = \frac{1}{\sin z - \cos z}$ are given by equating the denominator to zero, i.e., by $\sin z - \cos z = 0$ or $\tan z = 1$ or $z = n\pi + \frac{\pi}{4}$. Clearly $z = \frac{\pi}{4}$ is a simple pole of $f(z)$.

Example 30.31. Write the type of singularity for each of the following functions:

(i) $\frac{z}{(z+1)^2}$ $\qquad$ (ii) $\frac{e^z}{(z-1)^3}$ $\qquad$ (iii) $\frac{e^{1/z}}{z^2}$ $\qquad$ (U.P.T.U., 2006)

Solution: (i) Poles of $f(z) = \frac{z}{(z+1)^2}$ are found by equating to zero $1 + e^z = 0$ or $e^z = -1 = e^{(2n+1)\pi i}$

$\therefore \quad z = (2n+1)\pi i \; (n = 0, \pm 1, \pm 2, \ldots)$

Clearly $f(z)$ has a simple pole at $z = (2n+1)\pi i$.

(ii) $\frac{e^z}{(z-1)^3} = \frac{e^{(t+1)}}{t^3} = \frac{e^t}{t^3}e^1$ $\quad$ where $t = z - 1$

$$= \frac{e}{t^3}\left[1 + \frac{2t}{2!} + \frac{(2t)^2}{3!} + \frac{(2t)^3}{4!} + \frac{(2t)^4}{5!} + \ldots\right] = e\left[\frac{1}{t^3} + \frac{2}{t^2} + \frac{2}{t^2} + \frac{4}{3t} + \frac{2}{3} + \frac{4t}{15} + \ldots\right]$$

$$= e\left[\frac{1}{(z-1)^3} + \frac{2}{(z-1)^2} + \frac{2}{(z-1)^2} + \frac{4}{3(z-1)} + \frac{2}{3} + \frac{4}{15}(z-1) + \ldots\right]$$

Since there are finite (4) number of terms containing negative powers of $(z-1)$.

$\therefore$ $z = 1$ is a pole of 4th order.

(iii) $f(z) = \frac{e^{1/z}}{z^2} = \frac{1}{z^2}\left(1 + \frac{1}{1!z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \ldots\right) = z^{-2} + z^{-3} + \frac{z^{-4}}{2} + \ldots$

Since there are infinite number of terms in the negative powers of z, therefore $f(z)$ has an essential singularity at $z = 0$.

EXERCISE 30.3

1. Obtain the expansion of $\frac{1}{z}$ in a Taylor's series in powers of $(z-1)$ and determine the radius of convergence.
2. Find the first three terms of the Taylor's series expansion of $f(z) = 2z^3 + z^2 + 4$ about $z = i$. Also find the region of convergence. (U.P.T.U., 2005)
3. Expand in Taylor's series $f(z) = \sin z \cos z$ about the point $z = \frac{\pi}{4}$. (Kurukshetra, 2005)

4. $\cos z$ about the point $z = \frac{\pi}{2}$. (Kurukshetra, 2005) (U.P.T.U., 2006) (iii) $\frac{1}{z^2(z+4)}$ where z lies $0 < |z| < 4$ (U.P.T.U., 2006)

5. Expand the following functions by Laurent's series:

$f(z) = \frac{1}{z-z^2}$ for (i) $0 < |z| < 1$ (ii) $|z| > 1$ (Bombay, 2005)

(c) $f(z)=\frac{1}{(z-1)(z+2)}$ for $1<|z|<2$ (AMTC, 2000)

(d) $f(z)=\sin\frac{1}{z-2}$ about $z=2$ (UCE, 2000)

6. Find the Laurent's expansion of $f(z)=\frac{e^z}{(z-1)^2}$ about $z=1$. (RAeS, 2000)

Hence find the residue of $f(z)$ at the singularity $z=1$.

7. Expand the following functions in Laurent's series

(a) $\frac{1}{(z-1)(z-2)}$ for $|z-1|>1$ (b) $\frac{1-\cos z}{z^3}$, about $z=0$. (Strath, 2000)

8. Find the Laurent's series expansion of

(a) $\frac{z^2-1}{z^2+2z+3}$ about $z=0$ in the region $2<|z|<3$ (UCE, 2000)

(b) $\frac{z^2-z-1}{(z-1)(z-2)(z+3)}$ in the region $2<|z+2|<5$

(c) $\frac{2z^2-3z-15}{z^2+5z}$ in the region $|z|>5$ (UCE, 2000)

9. Find the Laurent's expansion of $\frac{1}{z^2}\sin^2 z$ valid for (a) $0<|z|<\pi$ (b) $0<|z|<2\pi$ (c) $|z|>2\pi$.

Find the residue and hence state the singularities of the following functions:

10. $\frac{1}{z(z-1)}$ (b) $\frac{z}{(z-1)(z-2)}$ (UCE, 2000) (c) $\frac{1}{z}\sin\left(\frac{1}{z}\right)$ (UCE, 2000)

11. $\frac{e^z-1}{z^2(z-1)}$ (Strathclyde, 2000) (b) $\frac{e^z}{(z-2)^2}$ (c) $\frac{\cos z}{(z-\pi)^2}$ (UCE, 2000)

20.9 (i) RESIDUES

The coefficient of $(z-a)^{-1}$ in the expansion of $f(z)$ around an isolated singularity is called the residue of $f(z)$ at that point. Thus in the Laurent's series expansion of $f(z)$ around $z=a$, i.e. $f(z)=a_0+a_1(z-a)+a_2(z-a)^2+\ldots+a_{-1}(z-a)^{-1}+a_{-2}(z-a)^{-2}+\ldots$, the residue of $f(z)$ at $z=a$ is a_{-1}.

$\therefore$ $$\text{Res}[f(a)]=\frac{1}{2\pi i}\oint_C f(z)\,dz$$

i.e. $$\oint_C f(z)\,dz=2\pi i\,\text{Res}[f(a)] \quad \ldots(1)$$

(ii) Residue Theorem

If $f(z)$ is analytic in a closed curve C except at a finite number of singular points inside C, then $\oint_C f(z)\,dz=2\pi i\times$ (sum of the residues at the singular points inside C).

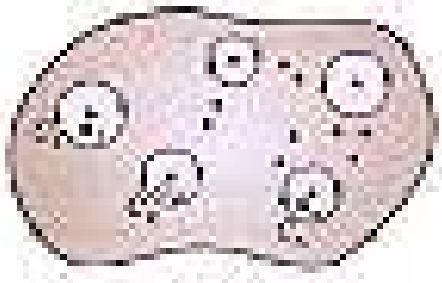


Fig. 20.12

Let us surround each of the singular points $a_1, a_2, \ldots, a_n$ by a small circle such that it encloses no other singular point (Fig. 20.12). Then these circles $C_1, C_2, \ldots, C_n$ together with C, form a multiply connected region in which $f(z)$ is analytic.

$\therefore$ Applying Cauchy's theorem, we have

$$\oint_C f(z)\,dz=\oint_{C_1} f(z)\,dz+\oint_{C_2} f(z)\,dz+\ldots+\oint_{C_n} f(z)\,dz \quad \text{(by (1))}$$

$=2\pi i\{\text{Res}[f(a_1)]+\text{Res}[f(a_2)]+\ldots+\text{Res}[f(a_n)]\}$ which is the desired result.

11.13 CALCULATION OF RESIDUES

(1) If $f(z)$ has a simple pole at $z = a$, then

$$\text{Res}\,f(a) = \lim_{z \to a} (z - a) f(z). \qquad \ldots(1)$$

Laurent's series in this case is

$$f(z) = c_0 + c_1(z - a) + c_2(z - a)^2 + \ldots + c_n(z - a)^n$$

Multiplying throughout by $z - a$, we have

$$(z - a) f(z) = c_0(z - a) + c_1(z - a)^2 + c_2(z - a)^3 + \ldots + c_n$$

Taking limits as $z \to a$, we get

$$\lim_{z \to a} (z - a) f(z) = c_{-1} = \text{Res}\,f(a)$$

(2) Another formula for $\text{Res}\,f(a)$:

Let $f(z) = \phi(z)/\psi(z)$, where $\psi(z) = (z - a) F(z)$, $F(a) \neq 0$.

Then

$$\lim_{z \to a} (z - a) f(z) = \lim_{z \to a} \frac{(z - a)\phi(z)}{\psi(z)}$$

$$= \lim_{z \to a} \frac{(z - a)\phi(z) + \phi(z)}{\psi'(z) + (z - a)\psi''(z) + \ldots}, \quad \text{since } \psi(a) = 0$$

Thus

$$\text{Res}\,f(a) = \frac{\phi(a)}{\psi'(a)}$$

(3) If $f(z)$ has a pole of order n at $z = a$, then

$$\text{Res}\,f(a) = \frac{1}{(n-1)!} \left[\frac{d^{n-1}}{dz^{n-1}} \{(z - a)^n f(z)\} \right]_{z = a}$$

Here

$$f(z) = c_0 + c_1(z - a) + c_2(z - a)^2 + \ldots + c_{n-1}(z - a)^{n-1} + \ldots + c_{-n}(z - a)^{-n}.$$

Multiplying throughout by $(z - a)^n$, we get

$$(z - a)^n f(z) = c_0(z - a)^n + c_1(z - a)^{n+1} + c_2(z - a)^{n+2} + \ldots + c_{-1}(z - a)^{n-1} + c_{-2}(z - a)^{n-2} + \ldots + c_{-n}$$

Differentiating both sides $n - 1$ times and putting $z = a$, we get

$$\left[\frac{d^{n-1}}{dz^{n-1}} \{(z - a)^n f(z)\} \right]_{z = a} = (n - 1)!\, c_{-1}, \text{ whence follows the result.}$$

Note. In many cases, the residue at a pole is evaluated by putting $z = a + h$ in $f(z)$ and expanding it by powers of h where $|h| < 1$ is the required result.

Example 11.13.1. Find the residue of the function $f(z) = \dfrac{\sin z}{z \cos z}$ at its poles inside the circle $|z| = 2$.

(Madurai, 2003)

Solution. $f(z)$ has simple poles at $z = 0, \pm\pi/2, \pm 3\pi/2, \ldots$

Only the poles $z = 0$ and $z = \pm\pi/2$ lie inside $|z| = 2$.

$$\therefore \quad \text{Res}\,f(0) = \lim_{z \to 0} (z - 0) f(z) = \lim_{z \to 0} \left(\frac{\sin z}{\cos z} \right) = 0.$$

$$\text{Res}\,f(\pi/2) = \lim_{z \to \pi/2} \left[\left(z - \frac{\pi}{2} \right) f(z) \right] = \lim_{z \to \pi/2} \left\{ \frac{(z - \pi/2)\sin z}{z \cos z} \right\}$$

$$= \lim_{z \to \pi/2} \frac{(z - \pi/2)\cos z + \sin z}{\cos z - z \sin z} \qquad \left[\text{Taking } \frac{0}{0} \text{ form} \right]$$

$$= \frac{1}{-\pi/2} = -\frac{2}{\pi}$$

and $$\text{Res}(\pi,0) = \lim_{z\to\pi}\left\{\frac{(z-\pi)^2 z\sin z}{z\sin^2 z}\right\} = \lim_{z\to\pi}\frac{(z-\pi)^2\cos z + 2(z-\pi)\sin z}{\sin z + z\cos z} = \frac{-4}{-\pi} = \frac{4}{\pi}$$

Hence sum of residues $= 1 - \frac{3}{\pi} + \frac{4}{\pi} = 0$.

Example 20.33. *Determine the poles of the function* $f(z) = z^3/(z^2 - 3z + 2)$ *and the residue at each pole.* (A.V.C.E.T., 2006) (A.V.C.E.T., 2008)

Hence evaluate $\oint_C f(z)\,dz$, *where* C *is the circle* $|z| = 2.5$.

Solution. Since $\lim_{z\to\infty} f(z) = \lim_{z\to\infty}\frac{z^3}{(z-1)(z-2)} = \infty$,

which is finite and non-zero, the function has a simple pole at $z = \infty$ and $\text{Res}\,f(\infty) = -6$.

Also since $\lim_{z\to\infty} f(z) = \infty$ is finite and non-zero, $f(z)$ has a pole of order two at $z = 1$.

$$\therefore \quad \text{Res}\,f(1) = \frac{1}{1!}\left\{\frac{d}{dz}(z-1)^2 f(z)\right\}_{z=1} = \left[\frac{d}{dz}\left(\frac{z^3}{z-2}\right)\right]_{z=1} = \left[\frac{z^3 - 3z^2}{(z-2)^2}\right]_{z=1} = \frac{-2}{1} = -2.$$

Otherwise writing $z = 1 + t$,

$$f(z) = \frac{(1+t)^3}{t^2 - t - 1} = \frac{1}{-1}(1+t)^3(1-t-t^2)^{-1} = -(1+3t+3t^2+t^3)(1+t+2t^2+\ldots)$$

$$= -(1 + 4t + 9t^2 + \ldots) = -\frac{1}{t^2} - \frac{4}{t} - 9 - \ldots \qquad \ldots(1)$$

$\therefore$ $\text{Res}\,f(1)$ = coefficient of $\frac{1}{t}$ in (1) $= -4$.

Clearly $f(z)$ is analytic on $|z| = 2.5$ and at all points inside except the poles $z = -2$ and $z = 1$. Hence by residue theorem

$$\oint_C f(z)\,dz = 2\pi i\,[\text{Res}\,f(-2) + \text{Res}\,f(1)] = 2\pi i\left[\frac{4}{3} - 4\right] = -\frac{8\pi i}{3}$$

Example 20.34. *Find the residues of* $f(z) = \dfrac{z^3}{(z-1)^2(z-2)(z-3)}$ *at the poles and hence evaluate* $\oint_C f(z)\,dz$, *where* C *is the circle* $|z| = 3.5$. (U.P.T.U., 2004)

Solution. The poles of $f(z)$ are given by $(z-1)^2(z-2)(z-3) = 0$.

$\therefore$ $z = 1$ is a pole of order 2, while $z = 2$ and $z = 3$ are simple poles.

$$\text{Res}\,f(2) = \frac{1}{1!}\frac{d}{dz}\left[(z-2)\cdot\frac{z^3}{(z-1)^2(z-2)(z-3)}\right]_{z=2} = \frac{d}{dz}\left[\frac{z^3}{(z-1)^2(z-3)}\right]_{z=2}$$

$\therefore$ $z = 1$ is a pole of order 2, while $z = 2$ and $z = 3$ are simple poles.

$$\text{Res}\,f(1) = \frac{1}{1!}\frac{d}{dz}\left[(z-1)^2\cdot\frac{z^3}{(z-1)^2(z-2)(z-3)}\right]_{z=1} = \frac{d}{dz}\left[\frac{z^3}{(z-2)(z-3)}\right]_{z=1}$$

$$= \frac{d}{dz}\left[z + 5 - \frac{8}{z-2} + \frac{27}{z-3}\right] = \left[1 - \frac{(z-2)^2 3z^2 - z^3(2z-5)}{(z-2)^2(z-3)^2} + \frac{27(z-3)^2 3z^2 - z^3(2z-6)}{(z-3)^2}\right]_{z=1}$$

$$= -\left[-5 + \frac{27}{25}\right] = \frac{101}{25}$$

$$\text{Res}(f(2)) = \lim_{z\to 2}\left\{(z-2)\cdot\frac{z^2}{(z-1)^2(z-3)(z-2)}\right\} = \lim_{z\to 2}\left\{\frac{z^2}{(z-1)^2(z-3)}\right\} = \frac{4}{(1)^2(-1)} = -4$$

$$\text{Res}(f(3)) = \lim_{z\to 3}\left\{(z-3)\cdot\frac{z^2}{(z-1)^2(z-3)(z-2)}\right\} = \frac{9}{(2)^2\cdot 1} = \frac{9}{4}$$

Now $$\int_C f(z)\,dz = 2\pi i\,[\text{Res}(f(1)) + \text{Res}(f(2))] \qquad [\because\ f(z) \text{ is analytic for } |z|=3 \text{ except at } z=1 \text{ and } z=2]$$

$$= 2\pi i\left[\frac{5}{4} - 4\right] = \frac{-21\pi i}{4}.$$

Example 10.28. Evaluate

$$\int_C \frac{z-3}{z^2+2z+5}\,dz, \text{ where } C \text{ is the circle}$$

(i) $|z+1-i| = 1$ (ii) $|z+1-2i| = 2$ (iii) $|z+1+i| = 2$. (U.P.T.U., 2005)

Solution. The poles of $f(z) = \dfrac{z-3}{z^2+2z+5}$ are given by $z^2+2z+5=0$

i.e., by $$z = \frac{-2\pm\sqrt{4-20}}{2} = -1\pm 2i.$$

(i) Both the poles $z = -1+2i$ and $z = -1-2i$ lie outside the circle $|z|=1$. Therefore, $f(z)$ is analytic everywhere within C.

Hence by Cauchy's theorem, $\int_C \frac{z-3}{z^2+2z+5}\,dz = 0$.

(ii) Here only one pole $z = -1+2i$ lies inside the circle $C: |z+1-i| = 2$. Therefore, $f(z)$ is analytic within C except at this pole.

$$\therefore \qquad \text{Res}(f(-1+2i)) = \lim_{z\to(-1+2i)}[(z-(-1+2i))f(z)] = \lim_{z\to(-1+2i)}\frac{(z+1-2i)(z-3)}{z^2+2z+5}$$

$$= \lim_{z\to(-1+2i)}\frac{z-3}{z+1+2i} = \frac{-4+2i}{4i} = \frac{1}{2}+i.$$

Hence by residue theorem $\int_C f(z)\,dz = 2\pi i\,[\text{Res}(f(-1+2i)) + \text{Res}(f(-1-2i))] = 2\pi i\left[\frac{1}{2}+i-\frac{1}{2}+i\right] = -4\pi$.

(iii) Here only the pole $z = -1-2i$ lies inside the circle $C: |z+1+i| = 2$. Therefore, $f(z)$ is analytic within C except at this pole.

$$\therefore \qquad \text{Res}(f(-1-2i)) = \lim_{z\to(-1-2i)}\frac{(z+1+2i)(z-3)}{z^2+2z+5}$$

$$= \lim_{z\to(-1-2i)}\frac{z-3}{z+1-2i} = \frac{-4-2i}{-4i} = \frac{1}{2}-i$$

Hence by residue theorem, $\int_C f(z)\,dz = 2\pi i\,[\text{Res}(f(-1-2i))] = 2\pi i\left[\frac{1}{2}-i\right] = \pi(1+2i)$.

Example 10.29. Evaluate $\int_C \frac{z^2}{\cos\pi z}\,dz$, **where** C **is the unit circle** $|z|=1$. (Rohtak, 2006)

Solution. $f(z) = \dfrac{z^2}{\cos\pi z}$ has simple poles at $z = \pm\frac{1}{2}, \pm\frac{3}{2}, \pm\frac{5}{2}, \ldots$

Out of these only the poles at $z = \pm\frac{1}{2}$ are inside the given circle $|z|=1$.

$$\therefore \qquad \text{Res}\left(f\left(\frac{1}{2}\right)\right) = \lim_{z\to\frac{1}{2}}\left[\left(z-\frac{1}{2}\right)f(z)\right] = \lim_{z\to\frac{1}{2}}\left|\frac{\left(z-\frac{1}{2}\right)z^2}{\cos\pi z}\right| \qquad \left[\frac{0}{0}\text{ form}\right]$$

$$= \lim_{z \to 2i} \frac{e^z + \left(z - \frac{1}{2}\right)^4}{-8(z+2i)} = \frac{e^{2i}}{-8i}$$

and $$\operatorname{Res}(-2i) = \lim_{z \to -2i} \left[\frac{\left(z - \frac{1}{2}\right)^4}{\cos \pi z}\right] \qquad \left[\frac{0}{0} \text{ form}\right]$$

$$= \lim_{z \to -2i} \frac{e^z + \left(z - \frac{1}{2}\right)^4}{-\pi \sin \pi z} = \frac{e^{-2i}}{8i}$$

Hence $$\int_C \frac{e^z}{\cos \pi z}\,dz = 2\pi i\left\{\operatorname{Res} f\left(\frac{1}{2}\right) + \operatorname{Res} f\left(-\frac{1}{2}\right)\right\}$$

$$= 2\pi i\left(-\frac{e^{2i}}{8i} + \frac{e^{-2i}}{8i}\right) = -\frac{\pi}{4}\left[\frac{e^{2i} - e^{-2i}}{2i}\right] = -\frac{\pi}{4}\sin 2 = -0.2274.$$

Example 21.37 Evaluate $\oint_C \tan z\,dz$ where C is the circle $|z| = 2$. (I.E.E. 2003 (S))

Solution. The poles of $f(z) = \tan z$ are given by $\cos z = 0$ i.e. $z = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \ldots$

Of these poles, $z = \pi/2$ and $-\pi/2$ only are within the given circle.

$$\therefore \quad \operatorname{Res} f\left(\frac{\pi}{2}\right) = \lim_{z \to \frac{\pi}{2}} \frac{\sin z}{\frac{d}{dz}\cos z} = \lim_{z \to \frac{\pi}{2}}\left[\frac{\sin z}{-\sin z}\right] = -1 \qquad \text{(By L'Hôpital's rule)}$$

Similarly $$\operatorname{Res} f\left(-\frac{\pi}{2}\right) = \lim_{z \to -\frac{\pi}{2}} \frac{\sin z}{\frac{d}{dz}\cos z} = -1.$$

Hence by residue theorem,

$$\oint_C f(z)\,dz = 2\pi i\left\{\operatorname{Res} f\left(\frac{\pi}{2}\right) + \operatorname{Res} f\left(-\frac{\pi}{2}\right)\right\} = 2\pi i\{-1 - 1\} = -4\pi i.$$

Example 21.38 Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)}\,dz$ where C is the circle $|z| = 3$.

(V.T.U., 2000; Anna, 2003 (S); U.P.T.U., 2003)

Solution. $$f(z) = \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)}$$

is analytic within the circle $|z| = 3$ excepting the poles $z = 1$ and $z = 2$.

Since $z = 1$ is a pole of order 2,

$$\therefore \quad \operatorname{Res} f(1) = \frac{1}{1!}\left[\frac{d}{dz}\{(z-1)^2 f(z)\}\right]_{z=1} = \left[\frac{d}{dz}\left\{\frac{\sin \pi z^2 + \cos \pi z^2}{(z-2)}\right\}\right]_{z=1}$$

$$= \left[\frac{(z-2)(2\pi z \cos \pi z^2 - 2\pi z \sin \pi z^2) - (\sin \pi z^2 + \cos \pi z^2)}{(z-2)^2}\right]_{z=1}$$

$$= 1 - 1(-2) - 1 - 1 = 2\pi + 1$$

Also $$\operatorname{Res} f(2) = \lim_{z \to 2}\{(z-2) f(z)\} = \lim_{z \to 2} \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2} = 1$$

Hence by residue theorem,

$$\oint_C f(z)\,dz = 2\pi i\{\operatorname{Res} f(1) + \operatorname{Res} f(2)\} = 2\pi i\{2\pi + 1 + 1\} = 4\pi^2 i + 4\pi i.$$

PROBLEM SET 3

1. Expand $f(z) = 1/(z^2 - 1)$ in a Laurent series valid in $1 < |z| < 3$ and hence find the residue there.
2. Find the residue of $f(z) = \cot z/(z - a)^2$ at the pole $z = a$. (U.P.T.U., 2002)
   (a) $\int_C \frac{z^2 dz}{(z^2+a^2)^2}$, $C: |z| = a$. (U.P.T.U., 2003-04)
3. Determine the poles of the following functions and the residue at each pole:
   (a) $\frac{z^2+1}{z^4+13z^2+36}$ (b) $\frac{z^2-2z}{(z+1)^2(z^2+1)}$ (U.P.T.U., 2002-03) (c) $\frac{2z+1}{z^2(z-2)(z^2+2)}$ (U.P.T.U., 2004)
4. Find the residues of the following functions at each pole:
   (a) $\frac{1-e^{2z}}{z^4}$ (b) $\frac{e^{2z}}{(z-1)^2(z+1)}$ (U.P.T.U., 2003) (c) $\frac{\cos z}{z^2}$
5. $\int_C \frac{z^2+4}{(z-2)(z+2)}\,dz$, where C is (a) $|z| = 1$ (b) $|z| = 3$ (c) $|z| = 5$ (Mumbai, 2000)
6. Evaluate the following integrals:
   (a) $\int_C \frac{z^2\,dz}{(z+1)(2z+1)(z-3)}$, $C: |z| = 2$ (U.P.T.U., 2003)
   (b) $\int_C \frac{4z^2-4z+3}{(z-1)(z+2)^2}\,dz$, $C: |z| = 4$
   (c) $\int_C \frac{3z^2+z+1}{(z^2-1)(z+3)}\,dz$, $C: |z| = 2$. (U.P.T.U., 2004)
7. Evaluate
   (a) $\int_C \frac{(3z+1)}{(2z-1)^2}\,dz$, where C is (i) $|z| = 1$ (b) $\int_C \frac{z+3}{z^2+2z+5}\,dz$, where C is (i) $|z+1-i| = 2$
   (c) $\int_C \frac{z^3-2z}{(z+1)^2(z^2+4)}\,dz$, where C is $|z| = 3$. (U.P.T.U., 2005)
8. Evaluate
   (a) $\int_C \frac{\cos\pi z}{(z-1)(z-2)^2}$, $C: |z-2| = \frac{1}{2}$ (Mumbai, 2004)
   (b) $\int_C \frac{z^2+1}{(z-1)(z^2+z+1)}\,dz$, $C: |z+2| = 2$. (Mumbai, 2005)
   (c) $\int_C \frac{dz}{(z-1)^2(z+2)}$, $C: |z-i| = 2$. (Kanpur, 2002; Anna, 2003; Osmania, 2004)
9. Evaluate
   (a) $\int_C \frac{e^z}{z^2}\,dz$, $C: |z| = 1$ (b) $\int_C z^2 \bar{z}\,dz$, $C: |z-1| = 1$
   (c) $\int_C \frac{e^{2z}}{(z+1)^3}\,dz$, $C: |z| = 3$ (d) $\int_C \frac{z^2+z}{z^2+9}\,dz$, $C: |z+3| = 4$. (U.P.T.U., 2000) (e) $\int_C \frac{z^2\,dz}{(z+1)^3}$, $C: |z-1| = 3$
10. Evaluate the following integrals: (a) $\int_C \frac{z^2+1}{(z-1)(z+2)^2}\,dz$, $C: |z| = 3$
    (b) $\int_C \frac{z\sin z}{(z-\pi)^3}\,dz$, $C: |z| = 4$ (c) $\int_C \frac{e^{2z}}{(z-1)^2}\,dz$, $C: |z-1| = 2$. (U.P.T.U., 2003)
11. Evaluate $\int_C \frac{dz}{\sin z\cos 2z}$, where C is the circle $|z| = 2$. (Marathwada, 2000)
12. Obtain Laurent's expansion for $f(z) = 1/(z^2 - 3z)$ and evaluate
    $\int_C \frac{dz}{z^2(z-3)}$, where C is the circle $|z - 1| = 2$. (U.P.T.U., 2000)

2.10 EVALUATION OF REAL DEFINITE INTEGRALS

Many important definite integrals can be evaluated by applying the Residue theorem to properly chosen integrals. The contours chosen will consist of straight lines and circular arcs.

(a) Integration around the unit circle. An integral of the type $\int_0^{2\pi} f(\sin\theta, \cos\theta)\, d\theta$, where the integrand is a rational function of $\sin\theta$ and $\cos\theta$ can be evaluated by writing $e^{i\theta} = z$.

Since $\sin\theta = \frac{1}{2i}\left(z - \frac{1}{z}\right)$ and $\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$, then integral takes the form $\int_C f(z)\,dz$, where $f(z)$ is a rational function of z and C is a unit circle $|z| = 1$.

Hence the integral is equal to $2\pi i$ times the sum of the residues at those poles of $f(z)$ which are within C.

Example 2.10(a). Show that

$$\int_0^{2\pi} \frac{\cos 2\theta\, d\theta}{1 - 2a\cos\theta + a^2} = \frac{2\pi a^2}{1 - a^2}, \quad (a^2 < 1)$$

(Glasgow, 2009; Kolkata, 2008)

Solution. Putting $z = e^{i\theta}$, $d\theta = dz/iz$, $\cos\theta = \frac{1}{2}(z + 1/z)$ and $\cos 2\theta = \frac{1}{2}(e^{2i\theta} + e^{-2i\theta}) = \frac{1}{2}(z^2 + 1/z^2)$

$\therefore$ the given integral

$$I = \int_C \frac{\frac{1}{2}(z^2 + 1/z^2)}{1 - a(z + 1/z) + a^2} \frac{dz}{iz} = \frac{1}{2a}\int_C \frac{(z^4 + 1)\,dz}{z^2(z - az^2 - a + a^2 z)}$$

$$= \frac{1}{2a}\int_C \frac{(z^4 + 1)\,dz}{z^2(z - a)(1 - az)} = \int_C f(z)\,dz \qquad \text{where } C \text{ is the unit circle } |z| = 1$$

Now $f(z)$ has simple poles at $z = a$, $1/a$ and the second order pole at $z = 0$, of which the poles at $z = 0$ and $z = a$ lie within the unit circle.

$\therefore$ $$\text{Res}\,f(a) = \lim_{z\to a}(z - a)f(z) = \frac{1}{a}\lim_{z\to a}\left[\frac{z^4 + 1}{z^2(1 - az)}\right] = \frac{a^4 + 1}{a^2(1 - a^2)}$$

and $$\text{Res}\,f(0) = \lim_{z\to 0}\frac{d}{dz}(z^2 f(z)) = \frac{1}{a}\lim_{z\to 0}\frac{d}{dz}\left[\frac{z^4 + 1}{z - az^2 - a + a^2 z}\right]$$

$$= \frac{1}{a}\lim_{z\to 0}\frac{(z - az^2 - a + a^2 z)(4z^3) - (z^4 + 1)(1 - 2az + a^2)}{(z - az^2 - a + a^2 z)^2} = -\frac{2 + a^2}{a^2}$$

Hence $$I = 2\pi i\,[\text{Res}\,f(a) + \text{Res}\,f(0)] = 2\pi i\left[\frac{a^4 + 1}{a^2(1 - a^2)} - \frac{2 + a^2}{a^2}\right] = \frac{2\pi i a^2}{1 - a^2}.$$

Example 2.10 (b) (University question and solution, modified) $\int_0^{2\pi} \frac{\sin^2\theta}{a + b\cos\theta}\, d\theta$

(U.P.T.U., 2009; U.P.T.U., 2008; Madras, 2001)

Solution. Putting $z = e^{i\theta}$, $d\theta = dz/iz$, $\cos\theta = \frac{1}{2}(z + 1/z)$

and $$\cos(2\theta) = \frac{1}{2}(e^{2i\theta} + e^{-2i\theta}) = \frac{1}{2}(z^2 + 1/z^2).$$

$\therefore$ the given integral $$I = \int_C \frac{\frac{1}{2}(z^2 + 1/z^2)}{a + \frac{b}{2}(z + 1/z)} \frac{dz}{iz}$$

$$= -\frac{1}{ib}\int_C \frac{z^4 + 1}{z^2(z^2 + 2az + 1)}\,dz = -\frac{1}{ib}\int_C \frac{(z^4 + 1)\,dz}{z^2(z + a - \sqrt{a^2 - 1})(z + a + \sqrt{a^2 - 1})}$$

$$= -\frac{1}{2\pi i}\int_C f(z)\,dz, \quad \text{where } C \text{ is the unit circle } |z| = 1.$$

Now $f(z)$ has a pole of order 2 at $z = 0$ and simple poles at $z = \frac{1}{2}$ and $z = 2$. Of these only $z = 0$ and $z = 1/2$ lie within the unit circle.

$$\therefore \quad \text{Res}(f(z))_{z=0} = \lim_{z\to 0}\frac{(z-1)(2z^2+3)}{(2z-1)(z-2)} = \lim_{z\to 0}\left|\frac{z^2+1}{2z^2-5z+2}\right| = -\frac{3}{2}$$

$$\text{Res}(f(z))_{z=1/2} = \frac{1}{(2-1)!}\left|\frac{d^{2-1}}{dz^{2-1}}(z-\tfrac{1}{2})^2 f(z)\right|_{z=1/2}, \quad \text{where } m = 2$$

$$= \frac{1}{2}\left|\frac{d}{dz}\left\{\frac{z^2+1}{z^2(z-2)}\right\}\right|_{z=1/2} = \frac{d}{dz}\left|\frac{(2z^2-5z+2)(3z^2-4z+1)-(z^3+1)(4z-5)}{2z^2(2z^2-5z+2)^2}\right| \text{ at } z=\tfrac{1}{2}$$

$$= \left|\frac{d}{dz}\left\{\frac{8z^4-20z^3+13z^2-4z+1}{2z^2(2z^2-5z+2)^2}\right\}\right|_{z=\frac{1}{2}}$$

$$= \left|\frac{(2z^2-5z+2)^2(32z^3-60z^2+26z-4)-(8z^4-20z^3+13z^2-4z+1)\,2(2z^2-5z+2)(4z-5)}{2z^2(2z^2-5z+2)^4}\right|_{z=\frac{1}{2}}$$

$$= \frac{4(-4)(-3)-20}{2\times 2\times 36} = \frac{28}{72} = \frac{7}{18}$$

Hence $$I = \frac{-1}{2\pi i}\cdot 2\pi i\,[\text{Res}(f(z))_{z=0} + \text{Res}(f(z))_{z=1/2}] = -\pi\left[-\frac{3}{2} + \frac{7}{18}\right] = -\pi\left(-\frac{1}{18}\right) = \frac{\pi}{12}.$$

(c) Integration around a small semi-circle: To evaluate $\int_{-\infty}^{\infty} f(z)\,dz$, we consider $\int_C f(z)\,dz$, where C is the contour consisting of the semi-circle $C_R: |z| = R$, together with the diameter that closes it.

Supposing that $f(z)$ has no singular point on the real axis, we have, by the Residue theorem,

$$\int_{-R}^{R} f(z)\,dz + \int_{C_R} f(z)\,dz = 2\pi i \sum \text{Res}\,f(z).$$

Finally making R tend to ∞, we find the value of $\int_{-\infty}^{\infty} f(x)\,dx$, provided $\int_{C_R} f(z)\,dz \to 0$ as $R \to \infty$.

Example 30.41. Evaluate $\displaystyle\int_{-\infty}^{\infty}\frac{x^2\,dx}{(x^2+2x^2+4)}$. **(U.P.T.U., 2005)**

Solution. Consider $\displaystyle\int_C \frac{z^2\,dz}{(z^2+2)^2+4} = \int_C f(z)\,dz$ ...(1)

where C is the contour consisting of the semi-circle C_R of radius R together with the part of the real axis from $-R$ to R as shown in Fig. 30.33.

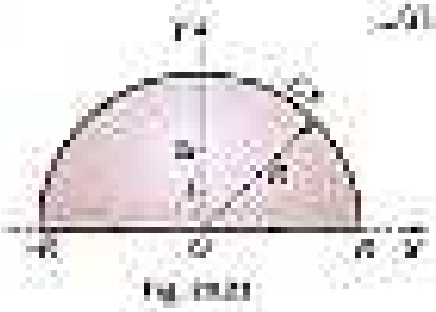


Fig. 30.33

The integrand has simple poles at $z = \pm i, z = \pm 2i$ of which $z = i, 2i$ only lie inside C.

$\therefore$ by the Residue theorem,

$$\int_C f(z)\,dz = 2\pi i\,[\text{Res}(f(i)) + \text{Res}(f(2i))]$$

$$= 2\pi i\left[\lim_{z\to i}(z-i)f(z) + \lim_{z\to 2i}(z-2i)f(z)\right]$$

$$= 2\pi i\left[\frac{z^2}{(z+i)(z^2+4)^2}\right]_{z=i} + \frac{z^2}{(z^2+1)(z+2i)}\Bigg]_{z=2i} = 2\pi i\left[\frac{1}{6} - \frac{1}{3}\right] = -\frac{\pi}{3}$$ ...(2)

Also $$\int_C f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz \qquad ...(iii)$$

Now let $R \to \infty$, we next show that the second integral in (iii) vanishes. For any point on C_2, as $|z| \to \infty$

$$f(z) = \frac{z^2}{(z+e^{i\alpha})(z+4e^{i\alpha})}$$

decreases as $1/z^2$ and tends to zero whereas the length of C_2 increases with z.

Consequently, $\lim_{|z|\to\infty} \int_{C_2} f(z)\,dz = 0$.

Hence from (i), (ii) and (iii), we get $\int_{-\infty}^{\infty} \frac{x^2\,dx}{(x^2+16)(x^2+4)} = \frac{\pi}{3}$

Example 20.40. Evaluate $\int_0^{\infty} \frac{\cos ax}{x^2+1}\,dx$. (U.P.T.U., 2000; Delhi, 2003)

Solution. Consider $\int_C \frac{e^{iaz}}{z^2+1}\,dz = \int_C f(z)\,dz$

where C is the contour consisting of the semi-circle C_R of radius R together with the part of the real axis from $-R$ to R as shown in Fig. 20.23.

The integrand has simple poles at $z = i$ and $z = -i$, of which $z = i$ only lies inside C.

By Residue theorem, $\int_C f(z)\,dz = 2\pi i \,[\text{Res}(f)]_{z=i} = 2\pi i \sum_{z=i} [\text{Residue of } f(z) \text{ at } z = i]$

$$= 2\pi i \lim_{z\to i} \frac{(z-i)e^{iaz}}{z^2+1} = 2\pi i \lim_{z\to i} \frac{e^{iaz}}{z+i} = 2\pi i \frac{e^{-a}}{2i} = \pi e^{-a} \qquad ...(i)$$

Also $$\int_C f(z)\,dz = \int_{-R}^{R} f(x)\,dx + \int_{C_R} f(z)\,dz \qquad ...(ii)$$

Now $|z| = R$ on C_R and $|z^2+1| \geq R^2 - 1$

Also $|e^{iaz}| = |e^{ia(x+iy)}| = |e^{iax}\cdot e^{-ay}| = e^{-ay} < 1 \quad [\because a > 0]$

$$\therefore \left|\frac{e^{iaz}}{z^2+1}\right| < |e^{iaz}|\cdot\frac{1}{|z^2+1|} < 1\cdot\frac{1}{R^2-1}$$

Then $$\int_{C_R} f(z)\,dz = \left|\int_{C_R} \frac{e^{iaz}}{z^2+1}\,dz\right| < \int_{C_R} \frac{1}{R^2-1}|dz| < \frac{\pi R}{R^2-1}, \text{ which } \to 0 \text{ as } R \to \infty \qquad ...(iii)$$

Hence from (i), (ii) and (iii), we get

$$\pi e^{-a} = \int_{-\infty}^{\infty} f(x)\,dx + 0 \quad \text{or} \quad \int_{-\infty}^{\infty} \frac{e^{iax}}{x^2+1}\,dx = \pi e^{-a}$$

Equating real parts from both sides, we obtain

$$\int_{-\infty}^{\infty} \frac{\cos ax}{x^2+1}\,dx = \pi e^{-a}$$

Since $\cos ax/(x^2+1)$ is an even function of x, we have

$$2\int_0^{\infty} \frac{\cos ax}{x^2+1}\,dx = \pi e^{-a} \quad \text{or} \quad \int_0^{\infty} \frac{\cos ax}{x^2+1}\,dx = \frac{\pi}{2}e^{-a}$$

(c) Integration around rectangular contours

Example 20.41. Evaluate $\int_{-\infty}^{\infty} \frac{e^{ax}}{e^x+1}\,dx$.

Solution. Consider $\int_C \frac{e^{az}}{e^z+1}\,dz = \int_C f(z)\,dz$, where C is the rectangle $ABCD$ with vertices at $(R, 0)$, $(R, 2\pi)$, $(-R, 2\pi)$ and $(-R, 0)$, R being positive (Fig. 20.24).

$f(z)$ has simple poles given by

$e^{z}+1=0 \Rightarrow e^{2\pi i}=e^{(2n+1)\pi i}$

or $z=(2n+1)\pi i$, where $n=0,\pm1,\pm2,\pm3,\ldots$

The only pole inside the rectangle is $z=\pi i$.

$\therefore$ by Residue theorem,

$$\int_C f(z)\,dz = 2\pi i \,[\text{Res}\,f(\pi i)]$$

$$= 2\pi i\left[e^{\pi i}/\frac{d}{dz}(e^{z}+1)\right]_{z=\pi i}$$

$$= 2\pi i e^{\pi i}/e^{\pi i} = -2\pi i/e^{\pi i} \qquad [\because e^{\pi i}=-1] \qquad \ldots(1)$$

Also $\int_C f(z)\,dz = \int_{AB} f(z)\,dz + \int_{BC} f(z)\,dz + \int_{CD} f(z)\,dz + \int_{DA} f(z)\,dz$

$$= \int_0^R f(R+iy)\,dy + \int_R^{-R} f(x+i\pi)\,dx + \int_{-R}^{0} f(-R+iy)\,dy + \int_{-R}^{R} f(x)\,dx$$

$[\because z=R+iy$ on AB, $z=x+i\pi$ along BC, $z=-R+iy$ along CD and $z=x$ along $DA.]$

or $$\int_C f(z)\,dz = i\int_0^{2\pi} \frac{e^{R(\cos\theta+i\sin\theta)}}{e^{R(\cos\theta+i\sin\theta)}+1}\,d\theta - \int_{-R}^{R} \frac{e^{x+i\pi}}{e^{x+i\pi}+1}\,dx + i\int_{2\pi}^{0} \frac{e^{-R+iy}}{e^{-R+iy}+1}\,dy + \int_{-R}^{R} \frac{e^{x}}{e^{x}+1}\,dx \qquad \ldots(2)$$

Now for any two complex numbers z_1, z_2

$|z_1| \geq |z_2|$, we have $|z_1+z_2| \geq |z_1|-|z_2|$

so that $|e^{R\cos\theta+iR\sin\theta}+1| \geq e^{R}-1$. Also $|e^{R(\cos\theta+i\sin\theta)}| = e^{R\cos\theta}$

$\therefore$ for the integrand of first integral in (2), we have

$$\left|\frac{e^{R\cos\theta+iR\sin\theta}}{e^{R\cos\theta+iR\sin\theta}+1}\right| \leq \frac{e^{R\cos\theta}}{e^{R}-1} \to 0 \text{ as } R\to\infty \qquad [\because \cos\theta<1]$$

Similarly, for the integrand of the third integral in (2), we get

$$\left|\frac{e^{-R+iy}}{e^{-R+iy}+1}\right| \leq \frac{e^{-R}}{1-e^{-R}} \to 0 \text{ as } R\to\infty \qquad [\because \cos\theta<0]$$

Hence as $R\to\infty$, since the first and third integrals in (2) approach zero, we get

$$\int_C f(z)\,dz = -e^{i\pi x}\int_{-\infty}^{\infty} \frac{e^{x}}{e^{x}+1}\,dx + \int_{-\infty}^{\infty} \frac{e^{x}}{e^{x}+1}\,dx = (1-e^{i\pi x})\int_{-\infty}^{\infty} \frac{e^{x}}{e^{x}+1}\,dx \qquad \ldots(3)$$

Thus from (1) and (3), we obtain $(1-e^{i\pi x})\int_{-\infty}^{\infty} \frac{e^{x}}{e^{x}+1}\,dx = -2\pi i e^{i\pi x}$

$\therefore$ equating real parts, we get $\int_{-\infty}^{\infty} \frac{e^{x}}{e^{x}+1}\,dx = \frac{\pi}{\sin\pi x}$.

Example 20.44. Show that $\int_0^\infty e^{-x^2}\cos 2bx\,dx = \frac{1}{2}\sqrt{\pi}e^{-b^2}$

Solution. Integrate $f(z)=e^{-z^2}$ along the rectangle $ABCDA$ having vertices $A(-R,0)$, $B(R,0)$, $C(R,b)$, $D(-R,b)$ (Fig. 20.28). $f(z)$ has no poles inside this contour. So we get

$$\int_{AB} f(z)\,dz + \int_{BC} f(z)\,dz + \int_{CD} f(z)\,dz + \int_{DA} f(z)\,dz = 0 \qquad \ldots(1)$$

On AB : $z=x$, on BC : $z=R+iy$, on CD : $z=x+ib$ and on DA : $z=-R+iy$.

Therefore, (1) becomes

$$\int_{-R}^{R} e^{-x^2}\,dx + \int_0^b e^{-(R+iy)^2}\,i\,dy + \int_{R}^{-R} e^{-(x+ib)^2}\,dx + \int_b^0 e^{-(-R+iy)^2}\,i\,dy = 0$$

or $$\int_{-R}^{R} e^{-x^2}\,dx - \int_{-R}^{R} e^{-x^2+b^2}(\cos 2bx - i\sin 2bx)\,dx + i\int_0^b e^{-(R+iy)^2}\,dy - i\int_0^b e^{-(-R+iy)^2}\,dy = 0$$

$$\ldots(2)$$

Now let $R \to \infty$. Then the last two integrals

$$= ir^{n+1}\int_0^{\pi} e^{ir}\,[e^{i(n+1)\theta} - e^{-i(n+1)\theta}]\,d\theta - Rr^{n+1}\int_0^{\pi} e^{iR}\sin(n+1)\theta\,d\theta \to 0$$

[$\because$ As $R \to \infty$, $e^{iR} \to$ bounded and $\int_0^{\pi}\sin(n+1)\theta\,d\theta$ is finite]

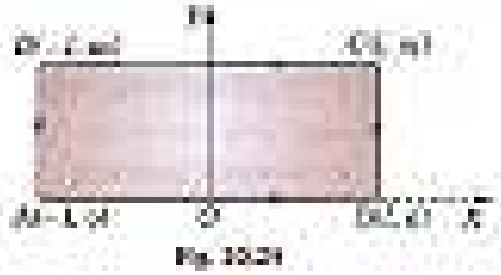


Fig. 10.21

Hence (ii) reduces to

$$\int_{-\infty}^{\infty} e^{ix}\,dx = r^{n+1}\int_0^{\pi} e^{ir\cos\theta}\,e^{ir\sin\theta}\,e^{i(n+1)\theta}\,d\theta \quad \text{... (iii)}$$

Equating real parts, we get

$$r^{n+1}\int_0^{\pi} e^{-r\sin\theta}\cos(r\cos\theta - (n+1)\theta)\,d\theta = \int_{-\infty}^{\infty} e^{ix}\,dx = \sqrt{\pi} \quad \text{[Ever p. 2001]}$$

or $$\int_0^{\pi} e^{-r\sin\theta}\cos(r\cos\theta - n\theta)\,d\theta = \frac{\pi}{2}\sqrt{\pi r^{n+1}}$$

(d) Indenting the contours having poles on the real axis. So far we have considered such cases in which there is no pole on the real axis. When the integrand has a simple pole on the real axis, we differ it from the region by indenting the contour i.e., by drawing a small semi-circle having the pole for the centre. The method will be clear from the following example.

Example 10.11(A): Evaluate $\int_0^{\infty} \frac{\sin mx}{x}\,dx$, m being real > 0. **[U.P.T.U., 2007]**

Solution. Consider the integral $\int_C \frac{e^{imz}}{z}\,dz = \int_C f(z)\,dz$ where C consists of

(i) the real axis from ε to R,
(ii) the upper half of the circle C_R, $|z| = R$,
(iii) the real axis $-R$ to $-\varepsilon$,
(iv) the upper half of the circle C_ε, $|z| = \varepsilon$ (Fig. 10.20).

Since $f(z)$ has no singularity (or is inside C (its only singular point being a simple pole at $z = 0$ which has been cleared by drawing C_ε), we have by Cauchy's theorem :

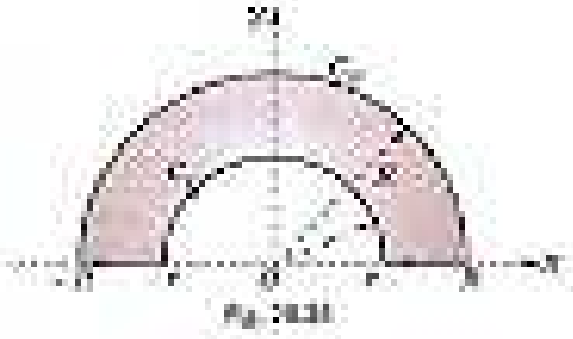


Fig. 10.20

$$\int_{\varepsilon}^{R} f(x)\,dx + \int_{C_R} f(z)\,dz + \int_{-R}^{-\varepsilon} f(x)\,dx + \int_{C_\varepsilon} f(z)\,dz = 0 \quad \text{... (i)}$$

Now $$\int_{C_R} f(z)\,dz = \int_0^{\pi} \frac{e^{imR(\cos\theta + i\sin\theta)}}{Re^{i\theta}}\,iRe^{i\theta}\,d\theta \quad [\because\ z = Re^{i\theta}]$$

$$= i\int_0^{\pi} e^{imR\cos\theta - mR\sin\theta}\,d\theta$$

Now $$|e^{imR\cos\theta - mR\sin\theta}| = |e^{-mR\sin\theta}\cdot e^{imR\cos\theta}| = e^{-mR\sin\theta}$$

$$\therefore \quad \left|\int_{C_R} f(z)\,dz\right| \le \int_0^{\pi} e^{-mR\sin\theta}\,d\theta = 2\int_0^{\pi/2} e^{-mR\sin\theta}\,d\theta$$

$$= 2\int_0^{\pi/2} e^{-mR(2\theta/\pi)}\,d\theta \quad [\because\ \text{For } 0 \le \theta \le \pi/2,\ \sin\theta \ge 2\theta/\pi]$$

$$= \frac{\pi}{mR}(1 - e^{-mR}) \to 0 \text{ as } R \to \infty,$$

Also $$\int_{C_\varepsilon} f(z)\,dz = i\int_{\pi}^{0} e^{im\varepsilon(\cos\theta + i\sin\theta)}\,d\theta \to i\int_{\pi}^{0} d\theta \ (\because \varepsilon \to 0 \Rightarrow m\varepsilon \to 0)$$

Hence as $\varepsilon \to 0$ and $R \to \infty$, we get from (i) $\int_{\varepsilon}^{R} f(x)\,dx + 0 + \int_{-R}^{-\varepsilon} f(x)\,dx - i\pi = 0$

or $$\int_{-R}^{-\varepsilon} f(x)\,dx + \int_{\varepsilon}^{R} f(x)\,dx = i\pi \text{ i.e., } \int_{-R}^{R} \frac{e^{imx}}{x}\,dx = i\pi \quad \text{... (ii)}$$

Equating imaginary parts from both sides,

$$\int_{-R}^{R} \frac{\sin mx}{x}\,dx = \pi. \text{ Hence } \int_0^{\infty} \frac{\sin mx}{x}\,dx = \frac{\pi}{2}.$$

Hint. Equating real parts from both sides of (i), we get

$$\int_0^{\infty} \frac{\sin px}{x}\,dx = \frac{\pi}{2}$$

Example 30.18. Show that $\int_0^{\infty} \frac{x^{p-1}}{1+x}\,dx = \frac{\pi}{\sin p\pi}, 0 < p < 1.$

Solution. Integrate $f(z) = \frac{z^{p-1}}{1+z}$ along the contour consisting of the circles α and γ of radii α and R and the lines AB and PQ along α as in Fig. 30.27). There is a simple pole at $z = -1$ which is within the contour.

$\therefore$ $$2\pi i f(-1) = \lim_{z \to -1} (1+z)\frac{z^{p-1}}{1+z} = e^{(p-1)\pi i} = (-1)^{p-1} = e^{(p-1)\pi i}$$

Then $$\int_{AB} f(z)dz + \int_{\gamma} f(z)dz + \int_{PQ} f(z)dz + \int_{\alpha} f(z)dz = 2\pi i e^{(p-1)\pi i} \quad \ldots(i)$$

On AB, $z = x$ and on PQ, $z = xe^{2\pi i}$

$\therefore$ $$\int_{AB} f(z)dz + \int_{PQ} f(z)dz = \int_{\alpha}^{R} \frac{x^{p-1}}{1+x}dx + \int_{R}^{\alpha} \frac{x^{2\pi i(p-1)}}{1+x}dx e^{2\pi i}$$

$$= \int_{\alpha}^{R} \frac{x^{p-1}}{1+x}\left[1 - e^{2\pi i(p-1)}\right]dx$$

On the circle γ, $z = Re^{i\theta}$. So

$$\int_{\gamma} f(z)dz = \int_{0}^{2\pi} \frac{iR^p e^{ip\theta}}{1+Re^{i\theta}}\,d\theta$$

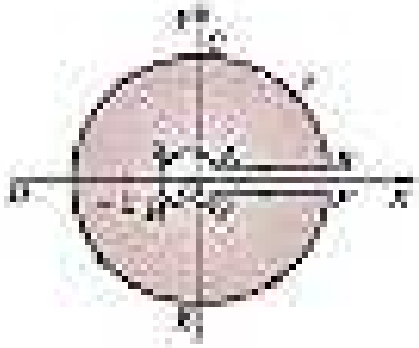


Fig. 30.27

For large R, the integrand is of the order $\frac{R^{p-1}}{1+R}$ i.e. R^{p-2} which tends to zero as $R \to \infty$, $\quad \because \quad p < 1$

Hence $$\int_{\gamma} f(z)dz \to 0 \text{ as } R \to \infty$$

On the circle α, $z = \alpha e^{i\theta}$. So

$$\int_{\alpha} f(z)dz = \int_{2\pi}^{0} \frac{i\alpha^p e^{ip\theta}}{1+\alpha e^{i\theta}} e^{i\theta}\,d\theta$$

For small α, the integrand is of the order α^p which tends to zero as $\alpha \to 0$, $\because \quad p > 0$.

Thus on taking limits as $\alpha \to 0$ and $R \to \infty$, (i) gives

$$\int_0^{\infty} \frac{x^{p-1}}{1+x}\left[1 - e^{2\pi i(p-1)}\right]dx = 2\pi i e^{(p-1)\pi i}$$

or $$\int_0^{\infty} \frac{x^{p-1}}{1+x}dx = \frac{2\pi i e^{(p-1)\pi i}}{1 - e^{2\pi i(p-1)}} = \frac{2\pi i e^{p\pi i}}{e^{p\pi i} - e^{-p\pi i}} = \frac{\pi}{\sin p\pi}$$

Example 30.19. Prove that $\int_0^{\infty} \cos x^2\,dx = \int_0^{\infty} \sin x^2\,dx = \frac{1}{2}\sqrt{\left(\frac{\pi}{2}\right)}$ **(Chennai, 2003)**

Solution. Consider $\int_C e^{-z^2}dz$ where C consists of the real axis from O to A, part of circle AB of radius R and the line $\theta = \frac{\pi}{4}$. (Fig. 30.28).

e^{-z^2} has no singularity within C.

$\therefore$ $$\int_{OA} e^{-z^2}dz + \int_{AB} e^{-z^2}dz + \int_{BO} e^{-z^2}dz = 0 \quad \ldots(1)$$

Fig. 30.28

Ex. Ex. : $\int_{0}^{\infty} e^{-x^2}\,dx = \int_{0}^{\infty} e^{-y^2}\,dy = \frac{\sqrt{\pi}}{2}$ [illegible]

Ex. Ex. : $\int_{0}^{\infty} e^{-x^2}\,dx = \int_{0}^{\pi/4} e^{-\sec^2\theta}\sec\theta\tan\theta\,d\theta$ as $R \to \infty$

[illegible]

Ex. Ex. : $\int_{0}^{\infty} e^{-x^2}\,dx = \int_{0}^{R} e^{-x^2}\cdot e^{-x^2}\,dx = -\int_{0}^{R} e^{-x^2}\frac{1+i}{\sqrt{2}}\,dx$

$\to -\int_{0}^{\infty}(\cos\theta^2 - i\sin\theta^2)\frac{1+i}{\sqrt{2}}\,dx$ as $R \to \infty$

Substituting Ex. in (2), we get

$$\frac{1}{2}\sqrt{\pi} + 0 - \int_{0}^{\infty}(\cos x^2 - i\sin x^2)\left(\frac{1+i}{\sqrt{2}}\right)dx = 0$$

Equating real and imaginary parts, we obtain

$$\int_{0}^{\infty}\cos x^2 + \sin x^2\,dx = \frac{1}{2}\sqrt{\frac{\pi}{2}} \quad \text{and} \quad \int_{0}^{\infty}\cos x^2 - \sin x^2\,dx = 0$$

Hence $\int_{0}^{\infty}\cos x^2\,dx = \int_{0}^{\infty}\sin x^2\,dx = \frac{1}{2}\sqrt{\frac{\pi}{2}}$.

EXERCISE

Ex. Evaluate the following integrals by using Ex. :

1. $\int_{0}^{\pi}\frac{dx}{1-2p\cos x+p^2} = \frac{\pi}{1-p^2}$ $(0 < p < 1)$ (Nagpur, 2007; Mumbai, 2008; Kerala, 2008)
2. $\int_{0}^{\pi}\frac{dx}{1-2p\cos x+p^2} = \frac{\pi}{p(p-1)}$ (P.T.U., 2008; Madras, 2008; Anna, 2008)
3. $\int_{0}^{\infty}\frac{dx}{x^4+1} = \frac{\pi}{\sqrt{2}}$ $(p > 0)$ (P.T.U., 2010) 4. $\int_{0}^{\pi}\frac{\sin^2 x}{5+4\cos x}\,dx = \left(\frac{\pi}{8}\right)$ (U.P.T.U., 2008)
5. $\int_{0}^{\pi}\frac{dx}{a+b\cos x} = \frac{\pi}{\sqrt{a^2-b^2}}$ $(a > b > 0)$ (U.P.T.U., 2007)
6. $\int_{0}^{\infty}\frac{dx}{x^4+4x^2+3} = \frac{\pi}{6}$ 7. $\int_{0}^{\infty}\frac{dx}{(x^2+a^2)^3} = \frac{3\pi}{16a^5}$ $(a > 0)$ (U.P.T.U., 2008)
8. $\int_{-\infty}^{\infty}\frac{x^2\,dx}{(x^2+a^2)(x^2+b^2)} = \frac{\pi}{a+b}$ $(a > 0, b > 0)$ (U.P.T.U., 2007; Mumbai, 2008; Anna, 2008)
9. $\int_{-\infty}^{\infty}\frac{x^2-x+2}{x^4+10x^2+9}\,dx = \frac{5\pi}{12}$ (U.P.T.U., 2008; Delhi, 2008)
10. $\int_{-\infty}^{\infty}\frac{dx}{x^2-x+1} = \frac{2\pi}{\sqrt{3}}$ (U.P.T.U., 2008) 11. $\int_{0}^{\infty}\frac{dx}{(1+x^2)^2} = \frac{\pi}{4}$ (Mumbai, 2008; Kerala, 2008)
12. $\int_{0}^{\infty}\frac{dx}{x^4+1} = \frac{\pi}{4}$ (Kerala, 2008) 13. $\int_{0}^{\infty}\frac{x^2}{x^4+1}\,dx = \frac{\pi}{4\sqrt{2}}$ (Mumbai, 2008)
14. $\int_{-\infty}^{\infty}\frac{\cos 2x}{x^2+x+1}\,dx = \frac{2\pi}{\sqrt{3}}\cos\frac{\pi}{\sqrt{3}}$ 15. $\int_{0}^{\infty}\frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}$ (P.T.U., 2008)
16. $\int_{0}^{\infty}\frac{\sin^2 x}{x^2}\,dx = \frac{\pi}{2}$ (Kerala, 2008) 17. $\int_{-\infty}^{\infty}\frac{\sin ax}{x^2+bx+c}\,dx = -\frac{\pi}{\sqrt{4c-b^2}}\sin\frac{ab}{2}$
18. $\int_{-\infty}^{\infty}\frac{\cos mx}{(x^2+a^2)(x^2+b^2)}\,dx = \frac{\pi}{a^2-b^2}\left[\frac{e^{-mb}}{b}-\frac{e^{-ma}}{a}\right]$ $(a > b > 0, m > 0)$.
19. $\int_{0}^{\infty}\frac{\cos mx}{x^2+a^2}\,dx = \frac{\pi}{2a}e^{-ma}$.

3.12 OBJECTIVE TYPE OF QUESTIONS

Multiple Choice

Select the correct alternative in (1) to (30). Number is equal to one of the following alternatives.

1. The only function which is analytic is
   (a) $f(z) = |z|^2$ (b) $f(z) = \bar{z}$ (c) $f(z) = |z|, z$ (d) $f(z) = z$

2. If $f(z) = u(x, y) + iv(x, y)$ is analytic then $f'(z) =$
   (a) $\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$ (b) $\frac{\partial u}{\partial x} - i\frac{\partial v}{\partial x}$ (c) $\frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$

3. If $2x - y^2 + i(ay^2 + bxy)$ is an analytic function, then a and b are
   (a) 1 (b) 2 (c) 3 (d) 4

4. The analytic function which maps the annular region $0 < r < 1$ onto the upper half plane is
   (a) z^2 (b) $\frac{1}{z}$ (c) z^3 (d) $\frac{z-i}{z+i}$

5. An angular domain in the complex plane is defined by $0 < \arg z < \pi/4$. The mapping which maps this region onto the left half plane is
   (a) $w = z^4$ (b) $w = z^3$ (c) $w = -z^4$ (d) $w = -z^3$

6. The mapping $w = z^2 - 2z - 2$ is
   (a) conformal at all z except $z = 1$ (b) not conformal at $z = 1$
   (c) not conformal at $z = -1$ and $z = 2$ (d) conformal everywhere

7. If $w = f(z)$ is a function of z in some region under the mapping $w(z) = f(z) = u + iv$ is
   (a) $u = 2v$ (b) $u = 2v + u^2$ (c) $u^2 = v - 2v$ (d) $u = v^2$

8. The fixed points of the mapping $w = (5z + 6)/(z + 2)$ are
   (a) 2, 3 (b) 2, -2 (c) -2, -3 (d) -2, 2.5

9. The value of $\int_C (4x^2 + iy)\,dz$ is $0, 1)^2\,dy + i(y^2 - x)\,dx$, where C is a straight line joining $z = -1, 0$ to $1, 2$ is
   (a) 0 (b) 1 (c) 6 (d) -6

10. The value of $\int_C \frac{3z^2 + 2z + 1}{z - 1}\,dz$, where C is $|z| = 1/2$ is
    (a) $2\pi i$ (b) 0 (c) πi (d) $4\pi i$

11. The value of $\int_C \frac{2z + 1}{(z)(z + 1)}\,dz$, where C is the circle $|z| = 2$ is
    (a) $2\pi i$ (b) $3\pi i$ (c) 1 (d) -1

12. The residue of a function can be found (Here poles are isolated singular)
    (a) True (b) False (c) Partially false (d) None of these

13. The value of $\int_C \frac{\sin z}{z - \sin z}\,dz$, where $C : |z| = 4$ is
    (a) $2\pi i$ (b) 0 (c) $3\pi i$ (d) $4\pi i$

14. The value of $\int_C \tan z\,dz$, where $C : |z| = 3$ is
    (a) 0 (b) πi (c) $2\pi i$ (d) $4\pi i$

15. The harmonic conjugate of $u = 2x(1 - y)$ is ... (KTU, 2009)

16. Harmonic conjugate of $e^x \cos y$ is ...

17. The curves $u(x, y) = c$ and $v(x, y) = c'$ are orthogonal if ...

18. The value of $\int_0^{1+i} z^2\,dz$ along the line $x = y$ is ... 19. Residue of $\frac{\cos z}{z}$ at $z = 0$ is ...

20. The critical points of the transformation $w = (z - 1)(z - 3)$ is ...

21. Image of $|z - 1| = 1$ under the mapping $w = 2z$ is ...

22. The poles of $f(z) = (z^2 - 1)(z + 2)^2$ is (are) $z =$ ... 23. $w = \log z$ is analytic except at $z =$ ...

24. If $f(z) = \frac{1}{z - 3}$, $|z| < 1$ then Taylor's series expansion of $f(z)$ about the point $z = 0$ is ...

25. If $|z| < 1$ then Taylor's series expansion of $\log(1 + z)$ about the point $z = 0$ is ...

36. The value of $\int_C \frac{4z^2+z+5}{z-4}dz$, where C is $|z|=3$, is ....

37. The value of $\int z\bar{z}^2 dz$, where C is $|z-1|=2$, is

(a) $4\pi i$ (b) $8\pi i$ (c) $16\pi i$ (d) $32\pi i$

38. If $f(z)$ has a pole of order three at $z=a$ then $\text{Res}[f(z)]_{z=a}$ = ...

39. The value of $\int_C \frac{e^z dz}{(z-1)^3}$, C being $|z-1|=0.5$, is ....

40. The C-R equations for $f(z)=u(x,y)+iv(x,y)$ to be analytic are ....

41. If $f(z)$ is analytic in a simply connected domain D and C is any closed path then $\int_C f(z)dz$ = ...

42. The harmonic conjugate of $e^x \cos y$ is ....

43. The value of $\oint_C \frac{\sin z}{z}dz$, where C is the circle $|z|=1$, is ....

44. The singularity of $f(z)=\sin z - z^3$ is ....

45. The function $f(z)=\bar{z}$ is analytic at ....

46. C-R equations for a function to be analytic in polar form are ....

47. If C is the circle $|z-a|=r$, $\int_C (z-a)^n dz$ is 0, when integer $n \neq -1$, is ....

48. A simply connected region is ....

49. A holomorphic function is ....

50. The poles of the function $f(z)=\frac{z^2}{(z-1)^2(z+3)}$ are at z = ...

51. The residue of the pole z_0 at $z=z_0$ is ....

52. The value of $\int_C \bar{z}\,dz$, where C is the circle represented by the straight line from $z=-i$ to $z=i$, is ....

53. Taylor's series expansion of $\left(\frac{1}{z-2}-\frac{1}{z-1}\right)$ in the region $|z|<2$, is ....

54. The invariant points of the bilinear transformation $w=\frac{2z+3}{z+4}$ are z = ...

55. The residue at $z=0$ of $\frac{\cos z}{z^2+z^3}$ is ....

56. The transformation $w=z^2$ is conformal at ....

57. The residue of $f(z)$ at a pole is ....

58. The value of $\int_C \frac{z^2-z+1}{z-1}dz$ = ...

59. Singular points of $\frac{\sin z}{(z+1)(z-2)}$ are ....

60. Taylor series expansion of $\frac{1}{z-2}$ in $|z|<1$ is ....

61. Evaluate $\int_C \frac{z^2+4z-1}{(z+1)(z+2)}dz$, where C is $|z|=3$. (U.P.T.U., 2005)

62. The poles of $\frac{(z-1)^2}{z(z-2)^2}$ are at z = ...

63. Cauchy's integral theorem states that ....

64. The invariant points of the transformation $w=z+1$ are ....

65. $\int_C \frac{dz}{z^2-1}$, where $|z|=3$, is ....

66. The inverse and standard form of $\frac{z^2+1}{(z+1)^2}$ are ....

67. Residue of $\frac{z}{(z-1)(z-2)}$ at $z=1$ is ....

68. Singularity of $e^{\frac{1}{z}}$ at $z=0$ is of the type ....

69. Cot $(z^2)_{z=0}$ = ...

70. Taylor's series expansion of $\sin z$ about $z=\pi/4$ is ....

71. Laurent's series expansion of $\frac{1}{z-2}$ about $z=1$ is ....

72. The poles of $\frac{z}{z^4+1}$ are ....

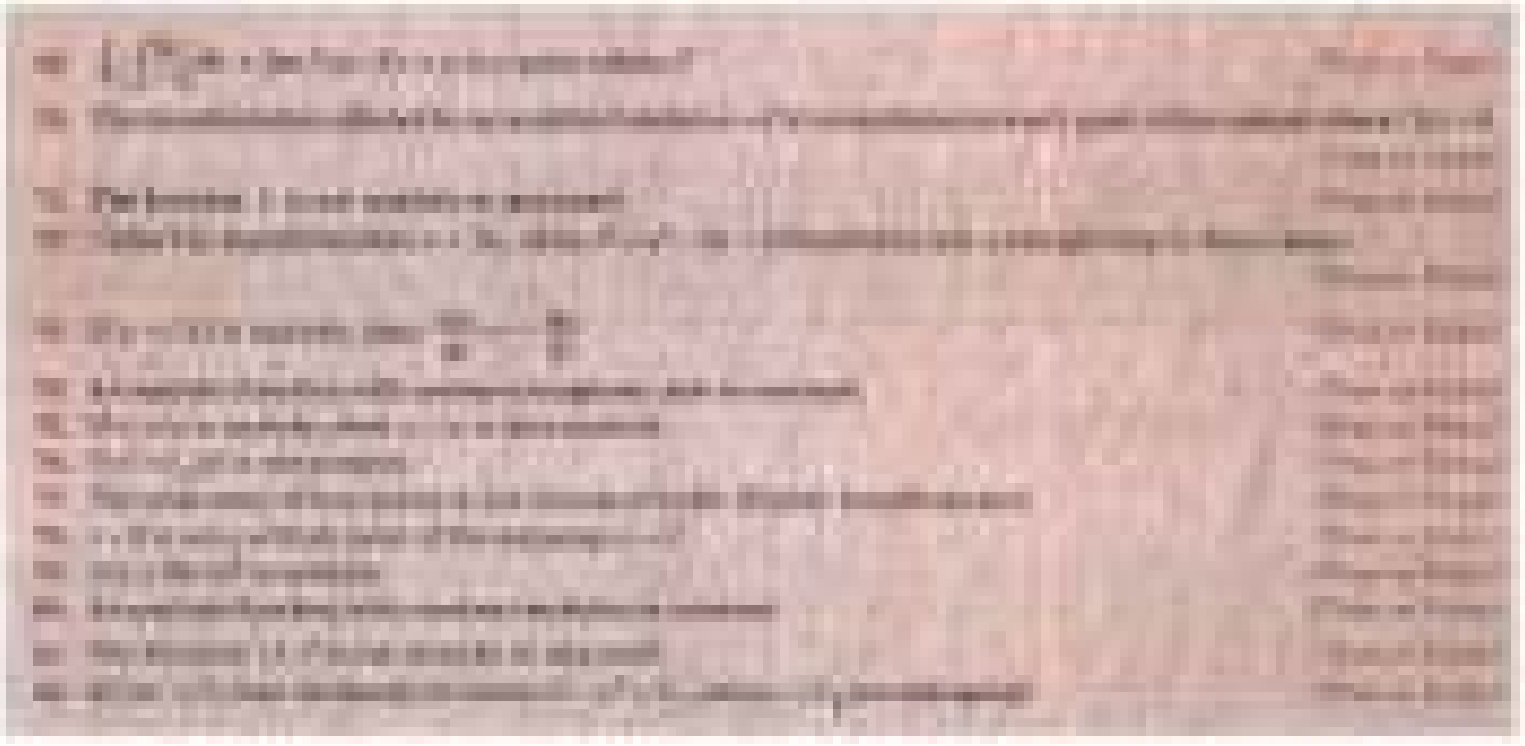
73. If $z=a$ is simple pole, then $\text{Res}[f(z)]_{z=a}$ = ...

74. Bilinear transformation is also known as ....

75. If $f(z)$ and $\overline{f(z)}$ are analytic function, then $f(z)$ is ....

76. The function $u(x,y)=2xy+3x^2-y^2$ is a harmonic function. (True or False) (U.P.T.U., 2003-04)

77. The function $f(z)=\bar{z}$ is harmonic. (True or False)



CHAPTER 21

Laplace Transforms

1. Introduction 2. Definition 3. Conditions for existence 4. Transforms of elementary functions 5. Properties of Laplace transforms 6. Transforms of periodic functions 7. Evaluation of integrals 8. Transforms of derivatives 9. Transforms of integrals 10. Multiplication by t^n 11. Division by t 12. Evaluation of integrals by Laplace transforms 13. Inverse transforms 14. Other methods of finding inverse transforms 15. Convolution theorem 16. Application to differential equations 17. Simultaneous linear equations with constant coefficients 18. Unit step function 19. Unit impulse function 20. Objective Type of Questions

21.1 INTRODUCTION

The knowledge of Laplace transforms has in recent years become an essential part of mathematical background required of engineers and scientists. This is because the transform methods provide an easy and effective means for the analysis of many problems arising in engineering.

This method originated from the operational methods applied by the English engineer Oliver Heaviside (1850–1925) to problems in electrical engineering. Unfortunately, Heaviside's treatment was largely heuristic and lacked rigour, which was placed on sound mathematical footing by Bromwich and Carson (during 1910–17). It was found that Heaviside's operational calculus is best understood by means of a particular type of definite integrals called Laplace transforms.*

The method of Laplace transforms has the advantage of directly giving the solution of differential equations with given boundary values without the necessity of first finding the general solution and then evaluating from it the arbitrary constants. Moreover, the ready tables of Laplace transforms reduce the problem of solving differential equations to mere algebraic manipulations.

21.2 (1) DEFINITION

Let $f(t)$ be a function of t ($t \geq 0$) such that a positive number s exists for which $e^{-st} f(t)$ is integrable over the interval 0 to ∞. Then the Laplace transform of $f(t)$, denoted by $L\{f(t)\}$ is defined by

$$L\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt \qquad \ldots(1)$$

provided that the integral on the right is convergent. Here s is a parameter which may be a real or complex number.

$L\{f(t)\}$ being clearly a function of s, is briefly written as $F(s)$, i.e., $L\{f(t)\} = F(s)$.

Conversely, we also say that $f(t)$ is the inverse Laplace transform of $F(s)$, and we write $f(t) = L^{-1}\{F(s)\}$.

Thus $f(t)$ is called the inverse Laplace transform of $F(s)$. The symbol L^{-1}, which transforms $F(s)$ into $f(t)$, is called the Laplace transformation operator.

*Pierre de Laplace (1749–1827) (See footnote p. 19.) made extensive contributions, mainly during 1770, which developed the theory of probability.

(b) Conditions for the existence

The Laplace transform of $f(t)$ exists if $\int_0^\infty e^{-st} f(t)\,dt$ exists for some $s = a$. If

(i) $f(t)$ is continuous or $\qquad$ (ii) $\int_0^b e^{-st} f(t)\,dt$ is finite

piecewise continuous

for every $b > 0$, then the above conditions are sufficient (not necessary).

For example, $f(t) = \frac{1}{\sqrt{t}}$ exists though $f(t)$ is infinite at $t = 0$.

6.3.2 Properties and Use of Laplace Transforms

The above application of the definition gives the following formulae :

(1) $\mathcal{L}\{k\} = \frac{k}{s}$ $\qquad (s > 0)$

(2) $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$, where $n = 0, 1, 2, 3, \ldots$ $\qquad \left[\because \text{Gamma function } \frac{\Gamma(n+1)}{s^{n+1}}\right]$

(3) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$ $\qquad (s > a)$

(4) $\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$ $\qquad (s > 0)$

(5) $\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2}$ $\qquad (s > 0)$

(6) $\mathcal{L}\{\sinh at\} = \frac{a}{s^2-a^2}$ $\qquad (s > |a|)$

(7) $\mathcal{L}\{\cosh at\} = \frac{s}{s^2-a^2}$ $\qquad (s > |a|)$

Proof: (1) $\mathcal{L}\{k\} = \int_0^\infty e^{-st} \cdot k\,dt = k\left[\frac{e^{-st}}{-s}\right]_0^\infty = -\frac{k}{s}[0 - 1] = \frac{k}{s}$, provided $s > 0$.

(2) $\mathcal{L}\{t^n\} = \int_0^\infty e^{-st} \cdot t^n\,dt = \int_0^\infty e^{-x} \left(\frac{x}{s}\right)^n \frac{dx}{s}$, on putting $st = x$

$= \frac{1}{s^{n+1}} \int_0^\infty e^{-x} \cdot x^n\,dx = \frac{\Gamma(n+1)}{s^{n+1}}$, $\because \Gamma(n+1) = n\,\Gamma(n)$ and $\Gamma(1) = 1$ [Page 6.2]

In particular, $\mathcal{L}\{t^{1/2}\} = \frac{\Gamma(\frac{1}{2}+1)}{s^{\frac{1}{2}+1}} = \frac{\frac{1}{2}\Gamma(\frac{1}{2})}{s^{3/2}}$ $\left(\because \Gamma(\tfrac{1}{2}) = \sqrt{\pi}\right) = \frac{\sqrt{\pi}}{2s^{3/2}}$

(a) is a fractional positive integer, $\Gamma(n+1) = n\,\Gamma(n)$ p. 6.2.

Therefore, $\mathcal{L}\{t^n\} = n!/s^{n+1}$

(3) $\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st} \cdot e^{at}\,dt = \int_0^\infty e^{-s+a)t}\,dt = \frac{[e^{-(s-a)t}]_0^\infty}{-(s-a)} = -\frac{1}{s-a}[0 - 1] = \frac{1}{s-a}$, if $s > a$.

(4) $\mathcal{L}\{\sin at\} = \int_0^\infty e^{-st} \sin at\,dt = \left|\frac{e^{-st}}{s^2+a^2}(-s\sin at - a\cos at)\right|_0^\infty = -\frac{a}{s^2+a^2}$

Similarly, the results (5), (6) & (7).

(5) $\mathcal{L}\{\cos at\} = \int_0^\infty e^{-st} \cos at\,dt = \int_0^\infty e^{-st}\left(\frac{e^{iat} + e^{-iat}}{2}\right)dt$

$= \frac{1}{2}\left[\int_0^\infty e^{-(s-ia)t}\,dt + \int_0^\infty e^{-(s+ia)t}\,dt\right] = \frac{1}{2}\left[\frac{1}{s-ia} + \frac{1}{s+ia}\right] = \frac{s}{s^2+a^2}$ $\quad [\because i^2 = -1]$

Similarly, the results (6) & (7) follow.

29.3 PROPERTIES OF LAPLACE TRANSFORMS

I. Linearity property. If a, b, c be any constants and f, g, h any functions of t, then

$$\mathcal{L}[af(t) + bg(t) - ch(t)] = a\mathcal{L}[f(t)] + b\mathcal{L}[g(t)] - c\mathcal{L}[h(t)]$$

For by definition,

$$\mathcal{L}[af(t) + bg(t) - ch(t)] = \int_0^\infty e^{-st}[af(t) + bg(t) - ch(t)]\,dt$$

$$= a\int_0^\infty e^{-st} f(t)\,dt + b\int_0^\infty e^{-st} g(t)\,dt - c\int_0^\infty e^{-st} h(t)\,dt = a\mathcal{L}[f(t)] + b\mathcal{L}[g(t)] - c\mathcal{L}[h(t)]$$

This result can easily be generalized.

Because of the above property of $\mathcal{L}$, it is called a *linear operator*.

II. First shifting property. If $\mathcal{L}[f(t)] = \bar{f}(s)$, then

$$\mathcal{L}[e^{at} f(t)] = \bar{f}(s-a).$$

By definition, $\mathcal{L}[e^{at} f(t)] = \int_0^\infty e^{-st} e^{at} f(t)\,dt = \int_0^\infty e^{-(s-a)t} f(t)\,dt$

$$= \int_0^\infty e^{-vt} f(t)\,dt, \text{ where } v = s - a = \bar{f}(v) = \bar{f}(s-a).$$

Thus, if we know the transform $\bar{f}(s)$ of $f(t)$, we can write the transform of $e^{at} f(t)$ simply replacing s by $s-a$ to get $\bar{f}(s-a)$.

Application of this property leads us to the following useful results:

(1) $\mathcal{L}[e^{at}] = \dfrac{1}{s-a}$ $\quad\left[\because\ \mathcal{L}[1] = \dfrac{1}{s}\right]$

(2) $\mathcal{L}[e^{at}t^n] = \dfrac{n!}{(s-a)^{n+1}}$ $\quad\left[\because\ \mathcal{L}[t^n] = \dfrac{n!}{s^{n+1}}\right]$

(3) $\mathcal{L}[e^{at}\sin bt] = \dfrac{b}{(s-a)^2 + b^2}$ $\quad\left[\because\ \mathcal{L}[\sin bt] = \dfrac{b}{s^2 + b^2}\right]$

(4) $\mathcal{L}[e^{at}\cos bt] = \dfrac{s-a}{(s-a)^2 + b^2}$ $\quad\left[\because\ \mathcal{L}[\cos bt] = \dfrac{s}{s^2 + b^2}\right]$

(5) $\mathcal{L}[e^{at}\sinh bt] = \dfrac{b}{(s-a)^2 - b^2}$ $\quad\left[\because\ \mathcal{L}[\sinh bt] = \dfrac{b}{s^2 - b^2}\right]$

(6) $\mathcal{L}[e^{at}\cosh bt] = \dfrac{s-a}{(s-a)^2 - b^2}$ $\quad\left[\because\ \mathcal{L}[\cosh bt] = \dfrac{s}{s^2 - b^2}\right]$

where in each case $s > a$.

Example 29.3. *Find the Laplace transforms of*
(1) $\sin t \sin 2t \sin 3t$ (2) $\cos^2 2t$ (3) $\sin^2 3t$

Solution. (1) Since $\sin A \sin 2A = \frac{1}{2}[\cos A - \cos 3A]$,

$$\therefore \quad \mathcal{L}[\sin t \sin 2t \sin 3t] = \frac{1}{2}\mathcal{L}[\sin t \cos t] - \mathcal{L}[\cos t \sin 3t] = \frac{1}{2}\left[\frac{2}{s^2+2^2} - \frac{4}{s^2+4^2}\right] = \frac{24}{(s^2+10)^2 - 24}$$

(2) Since $\cos^2 2t = \frac{1}{2}(1 + \cos 4t)$,

$$\therefore \quad \mathcal{L}[\cos^2 2t] = \frac{1}{2}[\mathcal{L}(1) + \mathcal{L}(\cos 4t)] = \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+16}\right]$$

(a) Since $\sin 3t = 3\sin t - 4\sin^3 t$

or $\sin^3 t = \frac{3}{4}\sin t - \frac{1}{4}\sin 3t$

$\therefore \quad \mathcal{L}(\sin^3 2t) = \frac{3}{4}\mathcal{L}(\sin 2t) - \frac{1}{4}\mathcal{L}(\sin 6t)$

$$= \frac{3}{4}\cdot\frac{2}{s^2+2^2} - \frac{1}{4}\cdot\frac{2}{s^2+6^2} = \frac{48}{(s^2+40)(s^2+36)}$$

Example 32.8. Find the Laplace transform of

(i) $e^{-3t}(2\cos 4t - 3\sin 5t)$ (ii) $t^2\cos 2t$ (iii) $e^{-2t}\sin^2 t$ (U.P.T.U., 2006) (iv) $\sin^3 t$ (U.P.T.U., 2006)

Solution. (i) $\mathcal{L}\{e^{-3t}(2\cos 4t - 3\sin 5t)\} = 2\mathcal{L}(e^{-3t}\cos 4t) - 3\mathcal{L}(e^{-3t}\sin 5t)$

$$= 2\cdot\frac{s+3}{(s+3)^2+4^2} - 3\cdot\frac{5}{(s+3)^2+5^2} = \frac{2s-9}{s^2+6s+34}$$

(ii) Since $\mathcal{L}(\cos 2t) = \frac{1}{2}\{\mathcal{L}(1) + \mathcal{L}(\cos 2t)\} = \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+4}\right]$

$\therefore$ by shifting property, we get

$$\mathcal{L}(e^{2t}\cos^2 t) = \frac{1}{2}\left[\frac{1}{s-2} + \frac{s-2}{(s-2)^2+4}\right].$$

(iii) Since $\mathcal{L}(\sqrt{t}) = \frac{\Gamma(3/2)}{s^{3/2}} = \frac{1/2\cdot\sqrt{\pi}}{s^{3/2}}$

$\therefore$ by shifting property, we obtain $\mathcal{L}(e^{3t}\sqrt{t}) = \frac{\sqrt{\pi}}{2}\frac{1}{(s-3)^{3/2}}$

Example 32.9. If $\mathcal{L}\{f(t)\} = \bar{f}(s)$, then show that

$$\mathcal{L}\{\sinh at\, f(t)\} = \frac{1}{2}[\bar{f}(s-a) - \bar{f}(s+a)]$$

$$\mathcal{L}\{\cosh at\, f(t)\} = \frac{1}{2}[\bar{f}(s-a) + \bar{f}(s+a)]$$

Hence evaluate (i) $\mathcal{L}(t\sinh 2t\sin 2t)$ (ii) $\mathcal{L}(t\cosh 2t\sin 2t)$.

Solution. We have $\mathcal{L}\{\sinh at\, f(t)\} = \mathcal{L}\left[\frac{1}{2}(e^{at} - e^{-at})f(t)\right] = \frac{1}{2}[\mathcal{L}(e^{at}f(t)) - \mathcal{L}(e^{-at}f(t))]$

$$= \frac{1}{2}[\bar{f}(s-a) - \bar{f}(s+a)], \text{ by shifting property.}$$

Similarly, $\mathcal{L}\{\cosh at\, f(t)\} = \frac{1}{2}[\mathcal{L}(e^{at}f(t)) + \mathcal{L}(e^{-at}f(t))]$

$$= \frac{1}{2}[\bar{f}(s-a) + \bar{f}(s+a)], \text{ by shifting property.}$$

(i) Since $\mathcal{L}(\sin 2t) = \frac{2}{s^2+2^2}$, the first result gives

$$\mathcal{L}(\sinh 2t\sin 2t) = \frac{1}{2}\left[\frac{2}{(s-2)^2+2^2} - \frac{2}{(s+2)^2+2^2}\right] = \frac{8s}{s^4+16s^2+64}$$

(ii) Since $\mathcal{L}(\cos 2t) = \frac{s}{s^2+2^2}$, the second result gives

$$\mathcal{L}(\cosh 2t\sin 2t) = \frac{1}{2}\left[\frac{s-2}{(s-2)^2+2^2} + \frac{s+2}{(s+2)^2+2^2}\right] = \frac{2s(s^2-8)}{s^4-16s^2+256}.$$

Example 31.4. Show that

$$\mathcal{L}\{t\sin at\} = \frac{2as}{(s^2+a^2)^2} \quad \text{(Hospital, 2001)} \qquad \text{and} \quad \mathcal{L}\{t\cos at\} = \frac{s^2-a^2}{(s^2+a^2)^2}$$

Solution. Since $\mathcal{L}\{e^{iat}\} = \mathcal{L}\{\cos at + i\sin at\}$

$$\therefore \quad \mathcal{L}\{te^{iat}\} = \frac{1}{(s-ia)^2} = \frac{(s+ia)^2}{[(s-ia)(s+ia)]^2}$$

or
$$\mathcal{L}\{t\cos at + it\sin at\} = \frac{(s^2-a^2)+i(2as)}{(s^2+a^2)^2}$$

Equating the real and imaginary parts from both sides, we get the desired results.

Example 31.5. *Find the Laplace transform of $f(t)$ defined as*

$f(t) = t^2$, when $0 < t < 1$

$= 2$, when $t > 1$ (Kerala, 2002)

$$f(t) = \begin{cases} t^2, & 0 < t < 1 \\ 2, & 1 < t < 2 \\ 0, & t > 2 \end{cases}$$ (U.P.T.U., 2000; U.P.T.U., 2002)

Solution. (i) $\mathcal{L}\{f(t)\} = \int_0^1 e^{-st}\cdot t^2\,dt + \int_1^2 e^{-st}\cdot 2\,dt = \int_0^\infty e^{-st}\cdot t^2\,dt - \int_2^\infty e^{-st}\cdot t^2\,dt + 2\int_1^2 e^{-st}\,dt$

$$= \left|\frac{e^{-st}}{-s}\right|_0^1 + \left|t^2\frac{e^{-st}}{-s}\right|_2^\infty + \int_2^\infty 2t\frac{e^{-st}}{-s}\,dt + 2\left|\frac{e^{-st}}{-s}\right|_1^2$$

$$= \frac{1}{s}\left[\frac{2e^{-2s}-0}{-s} - \left|\frac{e^{-st}}{s^2}\right|_2^\infty\right] + \frac{0-e^{-s}}{-s} + \frac{e^{-2s}}{s} - \frac{e^{-s}}{s} + \frac{1-e^{-2s}}{s^2}$$

$$= \frac{1}{s}\left[\frac{2e^{-2s}}{-s} - \frac{0-e^{-2s}}{s^2}\right] + \frac{e^{-s}}{s} + \frac{e^{-2s}}{s} - \frac{e^{-s}}{s} + \frac{1-e^{-2s}}{s^2}$$

Or
$$\mathcal{L}\{f(t)\} = \int_0^1 e^{-st}\cdot t^2\,dt + \int_1^2 e^{-st}\cdot 2\,dt + \int_2^\infty e^{-st}\cdot 0\,dt$$

$$= \left|\frac{e^{-st}}{-s}\right|_0^1 + \left|t^2\frac{e^{-st}}{-s}\right|_1^2 + \int_1^2 2t\frac{e^{-st}}{-s}\,dt + \left|\frac{e^{-st}}{-s}\right|_1^2$$

$$= \frac{1}{s} - \frac{2e^{-2s}}{s} + \frac{e^{-s}}{s^2} - \frac{e^{-2s}}{s^2}$$

Example 31.6. *Find the Laplace transform of* $\mathrm{D}\left(\frac{1}{\sqrt{t}}\right) = \frac{2}{\sqrt{\pi}}$ (Kerala Univ., 2002)

(i) $\mathrm{D}\frac{\Gamma(n+1)}{s^{n+1}}$ (Madras Univ., 2000)

Solution. (i) Since $(\sqrt{s}-1)^3\sqrt{s} = s^{\frac{3}{2}} - 3s^{\frac{2}{2}} + 3s^{\frac{1}{2}} - s^0$

$$\therefore \quad \mathcal{L}\{\sqrt{t}-1)^3\sqrt{t}\} = \mathcal{L}\{t^{\frac{3}{2}}\} - 3\mathcal{L}\{t^{\frac{2}{2}}\} + 3\mathcal{L}\{t^{\frac{1}{2}}\} - \mathcal{L}\{t^0\}$$

$$= \frac{\Gamma(\frac{3}{2}+1)}{s^{\frac{3}{2}+1}} - 3\frac{\Gamma(\frac{2}{2}+1)}{s^{\frac{2}{2}+1}} + 3\frac{\Gamma(\frac{1}{2}+1)}{s^{\frac{1}{2}+1}} - \frac{\Gamma(0+1)}{s^{0+1}}$$

$$= \frac{\frac{3}{2}\Gamma(\frac{1}{2})}{s^{\frac{3}{2}}} - 3\frac{\Gamma(1)}{s^2} + 3\frac{\frac{1}{2}\Gamma(\frac{1}{2})}{s^{\frac{1}{2}}} - \frac{1}{s}$$

$$= \frac{3}{4}\frac{\sqrt{\pi}}{s^{\frac{3}{2}}} - \frac{3}{s^2} + \frac{3}{2}\frac{\sqrt{\pi}}{s^{\frac{1}{2}}} - \frac{1}{s} \qquad \left[\because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \Gamma\left(\frac{3}{2}\right) = \frac{1}{2}\sqrt{\pi}\right]$$

$$= \frac{\sqrt{\pi}}{2}\left(\frac{3}{s^{\frac{3}{2}}} - \frac{6}{s^{\frac{1}{2}}} + \frac{3}{s^{\frac{1}{2}}} + \frac{2}{s^{\frac{3}{2}}}\right)$$

(iii) We know that $\sin at = \frac{e^{iat} - e^{-iat}}{2i}$

$$\mathcal{L}\{\sin at\} = \frac{1}{2i}\left[\frac{1}{s-ia} - \frac{1}{s+ia}\right] = \frac{a}{s^2+a^2}$$

$$\mathcal{L}\{\sin^2 at\} = \frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+4a^2}\right]$$

$$\mathcal{L}\{\sin^4 at\} = \frac{3}{8}\left[\frac{1}{s} - \frac{s}{s^2+4a^2}\right] + \frac{1}{8}\left[\frac{1}{s} - \frac{s}{s^2+16a^2}\right]$$

Example 2.13 Find the Laplace transform of the periodic function

$$f(t) = \begin{cases} t, & 0 < t < 1 \\ 2-t, & 1 < t < 2 \end{cases}$$

and $f(t+2) = f(t)$.

Solution: $f(t)$ is a periodic function of period 2.

$$\mathcal{L}\{f(t)\} = \int_0^2 e^{-st} f(t)\,dt + \int_2^4 e^{-st} f(t)\,dt + \int_4^6 e^{-st} f(t)\,dt + \cdots$$

$$= \int_0^1 e^{-st} t\,dt + \int_1^2 e^{-st}(2-t)\,dt + \int_2^3 e^{-st} t\,dt + \int_3^4 e^{-st}(2-t)\,dt + \cdots$$

$$= \int_0^1 e^{-st} t\,dt + \int_1^2 e^{-st}(2-t)\,dt + e^{-2s}\int_0^1 e^{-st} t\,dt + e^{-2s}\int_1^2 e^{-st}(2-t)\,dt + \cdots$$

$$= \left(1 + e^{-2s} + e^{-4s} + \cdots\right)\left[\int_0^1 e^{-st} t\,dt + \int_1^2 e^{-st}(2-t)\,dt\right]$$

$$= \frac{1}{1-e^{-2s}}\left[\left(-\frac{te^{-st}}{s} - \frac{e^{-st}}{s^2}\right)_0^1 + \left(-\frac{(2-t)e^{-st}}{s} - \frac{e^{-st}}{s^2}\right)_1^2\right]$$

$$= \frac{1}{1-e^{-2s}}\left[\frac{1}{s} - \frac{e^{-s}}{s} - \frac{e^{-s}}{s^2} + \frac{1}{s^2} + \frac{e^{-s}}{s} - \frac{e^{-2s}}{s^2}\right] = \frac{1}{s^2}\tanh\frac{s}{2}$$

(iv) We know that Heaviside unit step function is defined as

$$H(t-a) = \begin{cases} 0, & t < a \\ 1, & t > a \end{cases}$$

$$\mathcal{L}\{f(t)H(t-a)\} = \int_0^\infty e^{-st} f(t)H(t-a)\,dt = \int_a^\infty e^{-st} f(t)\,dt = e^{-as}\int_0^\infty e^{-st} f(t+a)\,dt$$

$$= e^{-as}\left[\int_0^1 e^{-st} t\,dt + \int_1^2 e^{-st}(2-t)\,dt\right] = e^{-as}\left[\frac{1}{s^2} - \frac{e^{-s}}{s^2}\right]$$

(v) If $\mathcal{L}\{f(t)\} = F(s)$ and a is a constant then $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$

$$\mathcal{L}\{e^{at} f(t)\} = \int_0^\infty e^{-st} e^{at} f(t)\,dt = \int_0^\infty e^{-(s-a)t} f(t)\,dt$$

$$= F(s-a)$$

$$\int_0^\infty |f(t)|\,dt < \infty$$

Example 3.12. Find $L\left[\frac{\sin at}{t}\right]$, given that $L\left[\frac{\cos at}{t}\right] = \log\left(\frac{s}{\sqrt{s^2+a^2}}\right)$.

Solution. By the above property,

$$L\left[\frac{\sin at}{at}\right] = \frac{1}{a}\tan^{-1}\left(\frac{1}{s/a}\right) = \frac{1}{a}\tan^{-1}\left(\frac{a}{s}\right) \text{ i.e., } L\left[\frac{\sin at}{t}\right] = \tan^{-1}\left(\frac{a}{s}\right).$$

PROBLEMS 3.1

Find the Laplace transforms of

1. $e^{at} + at^n + \sin 2t + 3\cos 2t + 5$ (A.M.I.E., 2000)
2. $2\sin 3t + 3\cos 2t$ (Nagarjuna, 2000)
3. $\sin t \cos 2t$
4. $\sin^2 3t$
5. $e^{2t}\sin^2 t - \cos 4t$ (A.M.I.E., 2000)
6. $e^{-t}\cos^2 t$ (Mumbai, 2000)
7. $\cosh at \sin at$ (Delhi, 2000)
8. $t^3 e^{2t}$ (A.M.I.E., 2000)
9. $t - 2t^2 - 3t^3$
10. $\sin(at + b)$
11. $\sin 2t \sin 3t$ (Kottayam, 2000)
12. $\sin^3 2t$ (Bhopal, 2000)
13. $e^{-3t}\sin 2t$
14. $e^{-3t}\cos^2 t$ (A.M.I.E., 2000)
15. $\sinh t \sin t$ (Madras, 2000)
16. $(t+1)^3$
17. $f(t) = \begin{cases} 4, & 0 < t < 2 \\ t^2, & t > 2 \end{cases}$ (U.P.T.U., 2000)
18. $f(t) = \begin{cases} \sin(t - \pi/3), & t > \pi/3 \\ 0, & t < \pi/3 \end{cases}$ (Nagarjuna, 2000)
19. $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & t > \pi \end{cases}$ (Madras, 2000 S)
20. $f(t) = \begin{cases} t, & 0 < t < 2 \\ 4 - t, & 2 < t < 4 \\ 0, & t > 4 \end{cases}$ (Kottayam, 2000)
21. $f(t) = \begin{cases} \sin(t - 2\pi/3), & t > 2\pi/3 \\ 0, & t < 2\pi/3 \end{cases}$
22. $f(t) = \begin{cases} \sin(t - \pi/3), & t > \pi/3 \\ 0, & t < \pi/3 \end{cases}$
23. $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t - 4, & 2 < t < 3 \\ 2, & t > 3 \end{cases}$ (Kottayam, 2000)
24. $L\left[\frac{1}{at^2 + bt + c}\right]$ (Madras, 2000 S)

3.10 TRANSFORMS OF PERIODIC FUNCTIONS

If $f(t)$ is a periodic function with period T, i.e., $f(t + T) = f(t)$, then

$$L[f(t)] = \frac{\int_0^T e^{-st} f(t)\,dt}{1 - e^{-sT}}$$

We have $L[f(t)] = \int_0^T e^{-st} f(t)\,dt + \int_T^{2T} e^{-st} f(t)\,dt + \int_{2T}^{3T} e^{-st} f(t)\,dt + \int_{3T}^{4T} e^{-st} f(t)\,dt + \ldots$

In the second integral put $t = u + T$, in the third integral put $t = u + 2T$, and so on. Then

$$L[f(t)] = \int_0^T e^{-st} f(t)\,dt + \int_0^T e^{-s(u+T)} f(u)\,du + \int_0^T e^{-s(u+2T)} f(u)\,du + \int_0^T e^{-s(u+3T)} f(u)\,du + \ldots$$

$$= \int_0^T e^{-st} f(t)\,dt + e^{-sT}\int_0^T e^{-su} f(u)\,du + e^{-2sT}\int_0^T e^{-su} f(u)\,du + \ldots$$

$$[\because\ f(u) = f(u + T) = f(u + 2T) = \text{etc.}]$$

$$= \int_0^T e^{-st} f(t)\,dt + e^{-sT}\int_0^T e^{-st} f(t)\,dt + e^{-2sT}\int_0^T e^{-st} f(t)\,dt + \ldots$$

$$= (1 + e^{-sT} + e^{-2sT} + \ldots) \int_0^T e^{-st} f(t)\,dt$$

$$= \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t)\,dt$$

(U.P.T.U., 2000; Mumbai, 2000)

Example 21.18. Find the Laplace transform of the function

$$f(t) = \begin{cases} \sin \omega t, & 0 < t < \pi/\omega \\ 0, & \pi/\omega < t < 2\pi/\omega \end{cases}$$

(Kurukshetra, 2005; Madras, 2006)

Solution. Since $f(t)$ is a periodic function with period $2\pi/\omega$

$$\therefore \quad \mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-2\pi s/\omega}} \int_0^{2\pi/\omega} e^{-st} f(t)\, dt$$

$$= \frac{1}{1 - e^{-2\pi s/\omega}} \left[\int_0^{\pi/\omega} e^{-st} \sin \omega t\, dt + \int_{\pi/\omega}^{2\pi/\omega} e^{-st} \cdot 0\, dt \right]$$

$$= \frac{1}{1 - e^{-2\pi s/\omega}} \left| \frac{e^{-st}(-s \sin \omega t - \omega \cos \omega t)}{s^2 + \omega^2} \right|_0^{\pi/\omega} = \frac{\omega e^{-\pi s/\omega} + \omega}{(1 - e^{-2\pi s/\omega})(s^2 + \omega^2)} = \frac{\omega}{(1 - e^{-\pi s/\omega})(s^2 + \omega^2)}$$

Example 21.19. Draw the graph of the periodic function

$$f(t) = \begin{cases} t, & 0 < t < \pi \\ \pi - t, & \pi < t < 2\pi \end{cases}$$

and find its Laplace transform.

(U.P.T.U., 2003)

Solution. Here the period of $f(t) = 2\pi$ and its graph is as in Fig. 21.1.

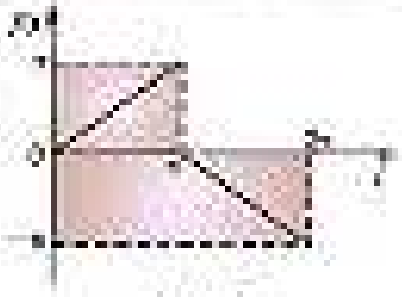


Fig. 21.1

$$\therefore \quad \mathcal{L}\{f(t)\} = \frac{1}{1 - e^{-2\pi s}} \left| \int_0^{\pi} e^{-st} t\, dt + \int_{\pi}^{2\pi} e^{-st} (\pi - t)\, dt \right|$$

$$= \frac{1}{1 - e^{-2\pi s}} \left| t \left(\frac{e^{-st}}{-s} \right) - 1 \left(\frac{e^{-st}}{s^2} \right) \right|_0^{\pi} + \left| (\pi - t) \left(\frac{e^{-st}}{-s} \right) - (-1) \left(\frac{e^{-st}}{s^2} \right) \right|_{\pi}^{2\pi}$$

$$= \frac{1}{1 - e^{-2\pi s}} \left| -\frac{\pi e^{-\pi s}}{s} - \frac{e^{-\pi s}}{s^2} + \frac{1}{s^2} + \frac{\pi e^{-2\pi s}}{s} + \frac{e^{-2\pi s}}{s^2} - \frac{e^{-\pi s}}{s^2} \right|$$

$$= \frac{1}{1 - e^{-2\pi s}} \left| \frac{\pi}{s} (e^{-2\pi s} - e^{-\pi s}) + \frac{1}{s^2} (1 + e^{-2\pi s} - 2e^{-\pi s}) \right|.$$

21.13 TRANSFORMS OF SPECIAL FUNCTIONS

(1) Transform of Bessel functions $J_0(t)$ and $J_1(t)$.

We know that $J_0(t) = 1 - \frac{t^2}{2^2} + \frac{t^4}{2^2 \cdot 4^2} - \frac{t^6}{2^2 \cdot 4^2 \cdot 6^2} + \cdots$ — II 18.7 (1), p. 868]

$$\therefore \quad \mathcal{L}\{J_0(t)\} = \frac{1}{s} - \frac{1}{2^2} \frac{2!}{s^3} + \frac{1}{2^2 \cdot 4^2} \frac{4!}{s^5} - \frac{1}{2^2 \cdot 4^2 \cdot 6^2} \frac{6!}{s^7} + \cdots$$

$$= \frac{1}{s} \left[1 - \frac{1}{2^2} \left(\frac{1}{s^2} \right) + \frac{1 \cdot 3}{2^2 \cdot 4^2} \left(\frac{1}{s^2} \right)^2 - \frac{1 \cdot 3 \cdot 5}{2^2 \cdot 4^2 \cdot 6^2} \left(\frac{1}{s^2} \right)^3 + \cdots \right]$$

$$= \frac{1}{s} \left(1 + \frac{1}{s^2} \right)^{-1/2} = \frac{1}{\sqrt{s^2 + 1}} \quad \ldots (1)$$

Also since $J_1(t) = -J_0'(t)$ — [Problem 6(i), p. 887]

$$\therefore \quad \mathcal{L}\{J_1(t)\} = -\mathcal{L}\{J_0'(t)\} = -[s\mathcal{L}\{J_0(t)\} - 1] = 1 - \frac{s}{\sqrt{s^2 + 1}} \quad \ldots (2)$$

(2) Transforms of Error functions

We know that $\operatorname{erf}(\sqrt{t}) = \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{t}} e^{-u^2}\, du$ — 18.35, p. 880]

$$= \frac{2}{\sqrt{\pi}}\int_0^\infty \left(1 - t^2 + \frac{t^4}{2!} - \frac{t^6}{3!} + \ldots\right)dt = \frac{2}{\sqrt{\pi}}\left(t - \frac{t^3}{3} + \frac{t^5}{5.2!} - \frac{t^7}{7.3!} + \ldots\right)$$

$$\therefore \quad L\{\operatorname{erf}(\sqrt{at})\} = \frac{2}{\sqrt{\pi}}\left[\frac{\Gamma(\frac{1}{2})}{s^{\frac{1}{2}}} - \frac{\Gamma(\frac{3}{2})}{3s^{\frac{3}{2}}} + \frac{\Gamma(\frac{5}{2})}{5.2!s^{\frac{5}{2}}} - \frac{\Gamma(\frac{7}{2})}{7.3!s^{\frac{7}{2}}} + \ldots\right]$$

$$= \frac{1}{s^{\frac{1}{2}}} - \frac{1}{3}\frac{1}{s^{\frac{3}{2}}} + \frac{1.3}{2.4.5}\frac{1}{s^{\frac{5}{2}}} - \frac{1.3.5}{2.4.6.7}\frac{1}{s^{\frac{7}{2}}} + \ldots$$

$$= \frac{1}{s^{\frac{1}{2}}}\left[1 - \frac{1}{3}\cdot\frac{1}{s} + \frac{1.3}{2.4}\cdot\frac{1}{s^2} - \frac{1.3.5}{2.4.6}\cdot\frac{1}{s^3} + \ldots\right]$$

$$= \frac{1}{s^{\frac{1}{2}}}\left[1 + \frac{1}{s}\right]^{-\frac{1}{2}} = \frac{1}{s^{\frac{1}{2}}(s+1)^{\frac{1}{2}}} \qquad \text{(Novelist, 2008)} \quad \ldots(2)$$

(3) Transform of Laguerre's polynomials $L_n(t)$

We know that $L_n(t) = e^t\frac{d^n}{dt^n}(t^n e^{-t})$ (§ 10.18, p. 221)

$$L\{L_n(t)\} = \int_0^\infty e^{-st}e^t\frac{d^n}{dt^n}(t^n e^{-t})\,dt = \int_0^\infty e^{-(s-1)t}\frac{d^n}{dt^n}(t^n e^{-t})\,dt$$

$$= \left[e^{-(s-1)t}\frac{d^{n-1}}{dt^{n-1}}(t^n e^{-t})\right]_0^\infty + \int_0^\infty e^{-(s-1)t}(s-1)\frac{d^{n-1}}{dt^{n-1}}(t^n e^{-t})\,dt$$

$$= (s-1)\int_0^\infty e^{-(s-1)t}\frac{d^{n-1}}{dt^{n-1}}(t^n e^{-t})\,dt \qquad \text{(Integrating by parts)}$$

$$= (s-1)^2\int_0^\infty e^{-(s-1)t}\frac{d^{n-2}}{dt^{n-2}}(t^n e^{-t})\,dt = (s-1)^n\int_0^\infty e^{-st}t^n\,dt$$

$$= (s-1)^n L\{t^n\} = (s-1)^n\cdot\frac{n!}{s^{n+1}}$$

Hence $L\{L_n(t)\} = \frac{n!(s-1)^n}{s^{n+1}}$ $(s > 0)$.

Example 10.11. Find the (i) $L\{e^{-t}J_0(t)\}$ (ii) $L\{\operatorname{erf}(2\sqrt{t})\}$ (Novelist, 2009)

Solution. (i) We know that $L\{J_0(t)\} = \frac{1}{\sqrt{(s^2+a^2)}}$

By shifting property, we get

$$L\{e^{-t}J_0(t)\} = \frac{1}{\sqrt{[(s+1)^2+a^2]}} = \frac{1}{\sqrt{(s^2+2s+2)}}$$

(ii) We know that $L\{\operatorname{erf}(\sqrt{t})\} = \frac{1}{s(s+1)}$

$$\therefore \quad L\{\operatorname{erf}(2\sqrt{t})\} = L\{\operatorname{erf}(\sqrt{4t})\} = \frac{1}{4}\cdot\frac{1}{\frac{s}{4}\left(\frac{s}{4}+1\right)} = \frac{4}{s(s+4)}$$

EXERCISE 10.3

1. Find the Laplace transform of the semi-cardioid wave of period 2π given by $f(t) = a(1+\cos t)$ for $0 < t < 2\pi$. (U.T.U., 2007)
2. Find the Laplace transform of the following periodic function:
   $f(t) = E\sin\omega t$, for $0 < t < \pi/\omega$, having period $2\pi/\omega$.

3. Find the Laplace transform of the square wave (or meander) function of period $2a$ defined as
$$f(t) = k, \quad \text{when } 0 < t < a$$
$$= -k, \quad \text{when } a < t < 2a.$$
(U.P.T.U., 2007)

4. Find the Laplace transform of the rectangular wave of period $2a$ given by
$$f(t) = t, \quad 0 < t < a$$
$$= 2a - t, \quad a < t < 2a.$$
(Bhopal Univ., 2008; U.P.T.U., 2008A; U.P.T.U., 2009)

Find the Laplace transform of the following functions

5. $J_0(at)$ 6. $e^{-at} J_0(bt)$ 7. $e^{at} J_0(\sqrt{t})$

7.13. TRANSFORMS OF DERIVATIVES

(i) If $f'(t)$ be continuous and $L\{f(t)\} = \bar{f}(s)$, then $L\{f'(t)\} = s\bar{f}(s) - f(0)$.

$$L\{f'(t)\} = \int_0^\infty e^{-st} f'(t)\,dt \qquad \text{[Integrate by parts]}$$
$$= \left| e^{-st} f(t) \right|_0^\infty - \int_0^\infty (-se^{-st}) f(t)\,dt.$$

Now assuming $f(t)$ to be such that $\lim_{t\to\infty} e^{-st} f(t) = 0$. When this condition is satisfied, $f(t)$ is said to be *exponential order* s.

Thus, $$L\{f'(t)\} = f(0) + s\int_0^\infty e^{-st} f(t)\,dt$$
$$= -f(0) + s\bar{f}(s).$$

Hence follows the desired result.

(ii) If $f'(t)$ and its $f_1(t), f_2(t), \ldots, f_{n-1}(t)$ derivatives be continuous, then
$$L\{f^n(t)\} = s^n \bar{f}(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \ldots - f^{n-1}(0).$$

Using the general rule of integration by parts (Refer to p. 208),
$$L\{f^n(t)\} = \int_0^\infty e^{-st} f^n(t)\,dt$$
$$= \left| e^{-st} f^{n-1}(t) \right| + s\left| e^{-st} f^{n-2}(t) \right| + s^2\left| e^{-st} f^{n-3}(t) \right| + \ldots$$
$$+ (-1)^{n-1} s^{n-1} \left| e^{-st} f(t) \right|_0^\infty + (-1)^n s^n \int_0^\infty e^{-st} f(t)\,dt$$
$$= -f^{n-1}(0) - sf^{n-2}(0) - s^2 f^{n-3}(0) - \ldots - s^{n-1} f(0) + s^n \int_0^\infty e^{-st} f(t)\,dt$$

Assuming that $\lim_{t\to\infty} e^{-st} f^n(t) = 0$ for $n = 0, 1, 2, \ldots, n-1$.

This proves the required result.

7.14. TRANSFORMS OF INTEGRALS

If $L\{f(t)\} = \bar{f}(s)$, then $L\left\{\int_0^t f(u)\,du\right\} = \frac{1}{s}\bar{f}(s)$.

Let $$\phi(t) = \int_0^t f(u)\,du, \text{ then } \phi(0) = \int_0^0 f(u)\,du = 0 \text{ and } \phi'(t) = f(t)$$

$\therefore$ $$L\{\phi'(t)\} = s\bar{\phi}(s) - \phi(0) \qquad \text{[By § 7.13 (i)]}$$

or $$\bar{f}(s) = \frac{1}{s} L\{f(t)\} \quad \text{i.e.,} \quad L\left\{\int_0^t f(u)\,du\right\} = \frac{1}{s}\bar{f}(s).$$

7.15. MULTIPLICATION BY t^n

If $L\{f(t)\} = \bar{f}(s)$, then

$$L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n}\{\bar{f}(s)\}, \text{ where } n = 1, 2, 3, \ldots$$

We have $\int_0^\infty e^{-st} f(t)\,dt = \bar{f}(s)$

Differentiating both sides with respect to s, $\frac{d}{ds}\left[\int_0^\infty e^{-st} f(t)\,dt\right] = \frac{d}{ds}\{\bar{f}(s)\}$

or By Leibnitz's rule for differentiation under the integral sign (p. XXX)

$$\int_0^\infty \frac{\partial}{\partial s}\left(e^{-st} f(t)\right) dt = \frac{d}{ds}\{\bar{f}(s)\}$$

or $$\int_0^\infty (-t)e^{-st} f(t)\,dt = \frac{d}{ds}\{\bar{f}(s)\} \quad \text{or} \quad \int_0^\infty e^{-st}\{tf(t)\}\,dt = -\frac{d}{ds}\{\bar{f}(s)\}$$

which proves the theorem for $n = 1$.

Now, assume the theorem to be true for $n = m$ (say), so that

$$\int_0^\infty e^{-st} t^m f(t)\,dt = (-1)^m \frac{d^m}{ds^m}\{\bar{f}(s)\}$$

Then, $$\frac{d}{ds}\left[\int_0^\infty e^{-st} t^m f(t)\,dt\right] = (-1)^m \frac{d^{m+1}}{ds^{m+1}}\{\bar{f}(s)\}$$

or By Leibnitz's rule, $\int_0^\infty (-t)e^{-st} t^m f(t)\,dt = (-1)^m \frac{d^{m+1}}{ds^{m+1}}\{\bar{f}(s)\}$

or $$\int_0^\infty e^{-st}(-1)^{m+1} t^{m+1} f(t)\,dt = (-1)^{m+1} \frac{d^{m+1}}{ds^{m+1}}\{\bar{f}(s)\}$$

This shows that, if the theorem is true for $n = m$, it is also true for $n = m + 1$. But it is true for $n = 1$. Hence it is true for $n = 1 + 1 = 2$, and $n = 2 + 1 = 3$ and so on.

Thus, the theorem is true for all positive integral values of n. (U.P.T.U., 2005)

Example 21.18. *Find the Laplace transforms of:*

(i) $t\cos at$ (Nagpur, 2005) (ii) $t^2 \sin at$ (U.P.T.U., 2007)

(iii) $t^2 e^{-3t}$ (Kottayam, 2005) (iv) $t^n \sin 2t$ (Anna Univ., 2005)

Solution. (i) Since $\mathcal{L}\{\cos at\} = s/(s^2 + a^2)$

$$\therefore \quad \mathcal{L}\{t\cos at\} = -\frac{d}{ds}\left(\frac{s}{s^2 + a^2}\right) = -\frac{s^2 + a^2 - s.2s}{(s^2 + a^2)^2}$$

$$= \frac{s^2 - a^2}{(s^2 + a^2)^2}$$ (cf. Example 21.4)

(ii) Since $$\sin at = \frac{a}{s^2 + a^2},$$

$$\therefore \quad \mathcal{L}\{t^2 \sin at\} = (-1)^2 \frac{d^2}{ds^2}\left(\frac{a}{s^2 + a^2}\right) = \frac{d}{ds}\left\{\frac{-2as}{(s^2 + a^2)^2}\right\} = \frac{2a(3s^2 - a^2)}{(s^2 + a^2)^3}$$

(iii) Since $\mathcal{L}\{t^n\} = \Gamma(n+1)/s^{n+1}$,

$$\therefore \quad \mathcal{L}\{t^2 e^{-3t}\} = (-1)^2 \frac{d^2}{ds^2}\left(\frac{1}{s+3}\right) = \frac{(-1)^2\,2!}{(s+3)^{2+1}} = \frac{2}{(s+3)^3}.$$

(iv) Since $\mathcal{L}\{\sin 2t\} = \frac{2}{s^2 + 2^2}$, therefore $\mathcal{L}\{t\sin 2t\} = -\frac{d}{ds}\left(\frac{2}{s^2 + 2^2}\right) = \frac{4s}{(s^2 + 2^2)^2}$

Now using the shifting property (§ 21.4 II), we get

$$\mathcal{L}\{e^{-3t}(t\sin 2t)\} = \frac{2(s+3)}{[(s+3)^2 + 2^2]^2} = \frac{2(s+3)}{(s^2 + 6s + 13)^2}.$$

Example 21.35. Find the Laplace transform of $\frac{\cos at - \cos bt}{t}$. (Madras, 2003)

Solution. (i) Since $\mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$,

$$\therefore \quad \mathcal{L}\{t^n \cos at\} = (-1)^n \frac{d^n}{ds^n}\{\mathcal{L}(\cos at)\} = (-1)^n \frac{d^n}{ds^n}\left(\frac{s}{s^2 + a^2}\right) = \frac{n!}{(s^2 + a^2)^{n+1}}$$

(ii) Since $\mathcal{L}\{1\} = \frac{1}{s}$,

$$\therefore \quad \mathcal{L}\{t^n\} = (-1)^n \frac{d^n}{ds^n}\{\mathcal{L}(1)\} = (-1)^n \frac{d^n}{ds^n}\left(\frac{1}{s}\right) = \frac{n!}{s^{n+1}}$$

(iii) Since $\mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}$,

$$\therefore \quad \mathcal{L}\{t \sin at\} = -\frac{d}{ds}\{\mathcal{L}(\sin at)\} = -\frac{d}{ds}\left(\frac{a}{s^2 + a^2}\right) = \frac{2as}{(s^2 + a^2)^2}$$

(iv) Since $\mathcal{L}\{\sin^2 t\} = \frac{2}{s(s^2 + 4)}$,

$$\therefore \quad \mathcal{L}\{t \sin^2 t\} = -\frac{d}{ds}\{\mathcal{L}(\sin^2 t)\} = -\frac{d}{ds}\left(\frac{2}{s(s^2 + 4)}\right) = \frac{2}{s^2(s^2 + 4)} + \frac{4(3s^2 + 4)}{s^2(s^2 + 4)^2}$$

Thus $\mathcal{L}\{t \sin^2 2t\} = -\frac{d}{ds}\left(\frac{2}{s(s^2 + 4)}\right) = -\frac{d}{ds}\left(\frac{2}{(s^2 + 4s^2)}\right) = \frac{6s^2 + 16}{s^2(s^2 + 4)^2}$.

21.10 Division by t

$\mathcal{L}\{f(t)\} = \bar{f}(s)$, then $\mathcal{L}\left\{\frac{1}{t} f(t)\right\} = \int_s^\infty \bar{f}(s)\,ds$, provided the integral exists.

We have $\bar{f}(s) = \int_0^\infty e^{-st} f(t)\,dt$.

Integrating both sides with respect to s from s to ∞,

$$\int_s^\infty \bar{f}(s)\,ds = \int_s^\infty \left[\int_0^\infty e^{-st} f(t)\,dt\right] ds = \int_0^\infty \int_s^\infty f(t) e^{-st}\,ds\,dt$$

(Changing the order of integration)

$$= \int_0^\infty f(t) \left[\int_s^\infty e^{-st}\,ds\right] dt \qquad [\because \ t \text{ is independent of } s]$$

$$= \int_0^\infty f(t) \left[\frac{e^{-st}}{-t}\right]_s^\infty dt = \int_0^\infty e^{-st} \frac{f(t)}{t}\,dt = \mathcal{L}\left\{\frac{1}{t} f(t)\right\}.$$

Example 21.36. Find the Laplace transform of $\frac{e^{-at} - e^{-bt}}{t}$. (Madras, 2006)

Solution. (i) Since $\mathcal{L}\{1\} = \frac{1}{s}$, $\mathcal{L}\{e^{-t}\} = \frac{1}{s+1}$,

$$\therefore \quad \mathcal{L}\left\{\frac{1 - e^{-t}}{t}\right\} = \int_s^\infty \left(\frac{1}{s} - \frac{1}{s+1}\right) ds = [\log s - \log(s+1)]_s^\infty$$

$$= \left[\log\left(\frac{s}{s+1}\right)\right]_s^\infty = -\log\left[\frac{s}{s+1}\right] = \log\left(\frac{s+1}{s}\right)$$

(ii) Since $\mathcal{L}\{\cos at - \cos bt\} = \frac{s}{s^2 + a^2} - \frac{s}{s^2 + b^2}$ and $\mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}$,

[illegible]

[illegible]

[illegible]

EXERCISE 2.11

1. Find $L\left\{\int_0^t e^{-t}\cos t\,dt\right\}$.

2. Given $L\{f(t)\} = \frac{1}{s^2+1}$, show that $L\{tf(t)\} = \frac{d}{ds}\left(\frac{1}{s^2+1}\right)$. (U.P.T.U. 2010; Madras 2004)

3. Given $L\{f(t)\} = \frac{\sqrt{s}}{s^2+1}$, prove that $L\left\{\frac{f(t)}{t}\right\} = \pi\left(\frac{1}{2}\right) - \tan^{-1}s$. (Madras 2009)

4. Find the Laplace transform of the following functions :

(a) $t^2\sin 3t$ (Meghalaya 2009)

(b) $t^2\cos 3t$ (Madras 2001)

(c) $te^{-t}\sin 2t$ (Madras 2003)

(d) $t^2e^{-3t}\sin 2t$ (Madras 2003)

(e) $t\sin^2 t$ (U.P.T.U. 2003)

(f) $t\cos^2 t$ (Madras 2009)

(g) $t^2\cos^2 t$ (U.P.T.U. 2003)

(h) $t\sin^3 2t$ (U.P.T.U. 2004)

(i) $t\cos 2t\sin 3t$ (U.P.T.U. 2004)

(j) $te^{-t}\cos^2 t$ (U.P.T.U. 2005)

(k) $te^{-t}\sin^2 t$ (Nagarjuna 2009)

(l) $te^{-t}\cos 2t\sin t$ (U.P.T.U. 2004)

(m) $t\sin t\cos 2t$ (Nagarjuna 2009)

(n) $t\sin 2t\cos 3t$ (U.P.T.U. 2004)

(o) $t\cos t\cos 2t$ (U.P.T.U. 2004)

(p) $t\sin t\sin 2t$ (U.P.T.U. 2004)

(q) $te^{-t}\cos t\sin t$ (U.P.T.U. 2004)

(r) $te^{-t}\cos t\cos 2t$ (U.P.T.U. 2004)

(s) $te^{-t}\sin t\sin 2t$ (U.P.T.U. 2004)

(t) $te^{-t}\cos 2t\sin 3t$ (U.P.T.U. 2004)

5. Evaluate : (a) $\int_0^\infty \frac{\cos 3t - \cos 2t}{t}e^{-t}dt$ (Madras 2006; U.P.T.U. 2006)

(b) $\int_0^\infty \frac{e^{-2t}-\cos 4t}{t}dt$ (Madras 2002)

(c) $\int_0^\infty te^{-3t}\sin t\,dt$ (U.P.T.U. 2006)

(d) $\int_0^\infty te^{-3t}\sin^2 t\,dt$

6. Prove that : (a) $\int_0^\infty \frac{e^{-3t}-e^{-2t}}{t}dt = \log\frac{2}{3}$. (U.P.T.U. 2006; Madras 2002; U.P.T.U. 2006)

(b) $\int_0^\infty \frac{e^{-3t}\sin^2 t}{t}dt = \frac{1}{8}\log 5$ (Madras 2002) (c) $\int_0^\infty \frac{e^{-t}\sin^2 t}{t}dt = \frac{1}{4}\log 2$. (U.P.T.U. 2006,S)

(d) $\int_0^\infty \frac{e^{-t}\sin^2 t}{t}dt = \frac{1}{4}\log 2$ (Meghalaya 2009)

7. Evaluate : (a) $L\left\{\int_0^t \frac{\sin t}{t}dt\right\}$ (U.P.T.U. 2005)

(b) $L\left\{\int_0^t e^{-t}\cos t\,dt\right\}$ (c) $L\left\{\int_0^t \frac{e^{-t}\sin t}{t}dt\right\}$ (P.T.U. 2006,S; U.P.T.U. 2005; Nagpur 2009)

8. Show that : (a) $L\{t^n\} = \frac{n}{s^{n+1}}$, $n\Gamma(n) = \Gamma(n+1)$ (b) $\int_0^\infty e^{-st}t^{n-1}dt = \frac{\Gamma(n)}{s^n}$

2.12 INVERSE TRANSFORMS — METHOD OF PARTIAL FRACTIONS

Having found the Laplace transforms of a few functions, let us now determine the inverse transforms of given functions of s. We have seen that $L\{f(t)\}$ in each case, is a rational algebraic function. Hence to find the inverse transformation, we first express the given function as a sum of partial fractions of which each, then, be recognizable as one of the following standard forms :

(i) $L^{-1}\left\{\frac{1}{s}\right\} = 1$; (ii) $L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$.

(10) $L^{-1}\left[\frac{1}{s^n}\right] = \frac{t^{n-1}}{(n-1)!}$, $n = 1, 2, 3, \ldots$ (11) $L^{-1}\left[\frac{1}{(s-a)^n}\right] = \frac{e^{at}t^{n-1}}{(n-1)!}$

(12) $L^{-1}\left[\frac{a}{s^2+a^2}\right] = \frac{1}{a}\sin at$ (13) $L^{-1}\left[\frac{s}{s^2+a^2}\right] = \cos at$

(14) $L^{-1}\left[\frac{a}{s^2-a^2}\right] = \frac{1}{a}\sinh at$ (15) $L^{-1}\left[\frac{s}{s^2-a^2}\right] = \cosh at$

(16) $L^{-1}\left[\frac{a}{(s-b)^2+a^2}\right] = \frac{1}{a}e^{bt}\sin at$ (17) $L^{-1}\left[\frac{s-b}{(s-b)^2+a^2}\right] = e^{bt}\cos at$

(18) $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right] = \frac{t}{2a}\sin at$ (19) $L^{-1}\left[\frac{1}{(s^2+a^2)^2}\right] = \frac{1}{2a^3}(\sin at - at\cos at)$

The reader is strongly advised to remember these results as necessary. The results (11) to (19) follow at once from their corresponding results (1) to (8). As illustrations, we shall prove (11) and (18). Example 21.4 gives

$$A \cdot L^{-1}\{\sin at\} = \frac{2as}{(s^2+a^2)^2} \text{ and } B \cdot L^{-1}\{\cos at\} = \frac{s^2-a^2}{(s^2+a^2)^2}$$

$\therefore$ $t\sin at = 2a\,L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$, whence follows (18)

Also $t\cos at = L^{-1}\left[\frac{s^2-a^2}{(s^2+a^2)^2}\right] = L^{-1}\left[\frac{(s^2+a^2)-2a^2}{(s^2+a^2)^2}\right]$

$$= L^{-1}\left[\frac{1}{s^2+a^2}\right] - 2a^2\,L^{-1}\left[\frac{1}{(s^2+a^2)^2}\right]$$

$$= \frac{1}{a}\sin at - 2a^2\,L^{-1}\left[\frac{1}{(s^2+a^2)^2}\right] \text{, whence follows (19).}$$

Note. To change the order of the factors of a given rational function, see Note (2) of Example 6 in Appendix I.

Example 21.17. *Find the inverse Laplace transforms of*

(i) $\frac{s^2-2s+3}{s^3}$ (ii) $\frac{s+2}{s^2-4s+13}$ (V.T.U., 2006)

Solution. (i) $L^{-1}\left[\frac{s^2-2s+3}{s^3}\right] = L^{-1}\left(\frac{1}{s}\right) - 2L^{-1}\left(\frac{1}{s^2}\right) + 3L^{-1}\left(\frac{2}{s^3}\right) = 1 - 2t + 3 \cdot \frac{t^2}{2!} = 1 - 2t + \frac{3}{2}t^2$.

(ii) $L^{-1}\left[\frac{s+2}{s^2-4s+13}\right] = L^{-1}\left[\frac{s+2}{(s-2)^2+9}\right] = L^{-1}\left[\frac{s-2+4}{(s-2)^2+3^2}\right]$

$$= L^{-1}\left[\frac{s-2}{(s-2)^2+3^2}\right] + 4L^{-1}\left[\frac{1}{(s-2)^2+3^2}\right] = e^{2t}\cos 3t + \frac{4}{3}e^{2t}\sin 3t.$$

Example 21.18. *Find the inverse Laplace transforms of*

(i) $\frac{2s^2-6s+3}{s^3-6s^2+12s-8}$ (V.T.U., 2010; U.P.T.U., 2004)

(ii) $\frac{4s+5}{(s-2)^2(s+3)}$ (Kurukshetra, 2005)

Solution. (a) Here the denominator $= (s-1)(s-2)(s-3)$

Let $\dfrac{3s^2-6s+5}{(s-1)(s-2)(s-3)} \equiv \dfrac{A}{s-1}+\dfrac{B}{s-2}+\dfrac{C}{s-3}$

Then $A = [(s-1)\cdot F(s)]_{s=1} = \left[\dfrac{3s^2-6s+5}{(s-2)(s-3)}\right]_{s=1} = \dfrac{3-6+5}{(-1)(-2)} = \dfrac{1}{2}$

$B = [(s-2)\cdot F(s)]_{s=2} = \left[\dfrac{3s^2-6s+5}{(s-1)(s-3)}\right]_{s=2} = \dfrac{12-12+5}{(1)(-1)} = -5$

and $C = [(s-3)\cdot F(s)]_{s=3} = \left[\dfrac{3s^2-6s+5}{(s-1)(s-2)}\right]_{s=3} = \dfrac{27-18+5}{(2)(1)} = \dfrac{7}{2}$,

$$\therefore \quad \mathcal{L}^{-1}\left\{\frac{3s^2-6s+5}{(s^3-6s^2+11s-6)}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - 5\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \frac{7}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\}$$

$$= \tfrac{1}{2}e^t - 5e^{2t} + \tfrac{7}{2}e^{3t}.$$

(b) Let $\dfrac{4s+5}{(s-1)^2(s+2)} \equiv \dfrac{A}{s-1}+\dfrac{B}{(s-1)^2}+\dfrac{C(s-2s+5)}{(s-2s+5)(s+2)}$

Multiplying both sides by $(s-1)^2(s+2)$, $4s+5 \equiv A(s-1)(s+2)+B(s+2)+C(s-1)^2$

Putting $s = 1$, $9 = 3B$, $\therefore B = 3$.

Equating the coefficients of s^2 from both sides,

$$0 \equiv A+C, \quad \therefore A \equiv -C, \quad \therefore A = -\tfrac{1}{3}.$$

$$\therefore \quad \mathcal{L}^{-1}\left\{\frac{4s+5}{(s-1)^2(s+2)}\right\} = -\frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{(s-1)^2}\right\} - \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$= \tfrac{1}{3}e^t + 3te^t - \tfrac{1}{3}e^{-2t}.$$

Example 11.10. Find the Inverse transform of:

(a) $\dfrac{4s+3}{(s-2)(s^2+2s+5)}$ (Nov/Dec, 2002), (U.P.T.U., 2004)

(b) $\dfrac{s}{s^4+4s^2}$ (Madras, 2003)

Solution. (a) Let $\dfrac{4s+3}{(s-2)(s^2+2s+5)} \equiv \dfrac{A(s+1)+B}{(s-2)(s^2+2s+5)} + \dfrac{Cs+D}{s^2+2s+5}$

Multiplying both sides by $(s-2)(s^2+2s+5)$,

$$4s+3 \equiv A(s+1)(s^2+2s+5)+(Cs+D)(s-2).$$

Equating the coefficients of s^3 from both sides,

$$0 \equiv A+C, \quad \therefore A \equiv -C, \quad \therefore A = -1.$$

Putting $s = 0$, $3 = 5A - 2D$, $\therefore D = -2$.

$$\therefore \quad \mathcal{L}^{-1}\left\{\frac{4s+3}{(s-2)(s^2+2s+5)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \mathcal{L}^{-1}\left\{\frac{-s+2}{s^2+2s+5}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \mathcal{L}^{-1}\left\{\frac{(s+1)-3}{(s+1)^2+2^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+2^2}\right\} + 3\mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2+2^2}\right\}$$

$$= e^{2t} - e^{-t}\cos 2t + \tfrac{3}{2}e^{-t}\sin 2t.$$

(b) Since $s^4+4s^2 = (s^2+2s^2+4)^2 - (2s)^2 = (s^2+2s+4)(s^2-2s+4)$

$$\therefore \quad \text{Let} \quad \frac{s}{s^4+4s^2} \equiv \frac{As+B}{s^2+2s+2^2} + \frac{Cs+D}{s^2-2s+2^2}$$

Multiplying both sides by s^4+4s^2,

$$s \equiv (As+B)(s^2-2s+2^2)+(Cs+D)(s^2+2s+2^2)$$

Equating coefficients of s^3, $0 = A + C$ ...(1)

Equating coefficients of s^2, $0 = -2A + B + 2C + D$ ...(2)

Equating coefficients of s, $1 = 2A - 2B + 2C + 2D$ ...(3)

Putting $s = 0$, $0 = 2A + 2B + 2D$ ...(4)

From (1), $B + C = 0$ ...(5)

$\therefore$ (3) becomes $A + C = 1$, and by (1), we get $A = C = 0$

Then (4) reduces to $D - B = 1/2$ and by (5), $B = -1/4$, $D = 1/4$.

$$= L^{-1}\left(\frac{s}{s^2+4s^2}\right) - \frac{1}{4}L^{-1}\left(\frac{1}{s^2+2s+2^2}\right) + \frac{1}{4}L^{-1}\left(\frac{1}{s^2-2s+2^2}\right)$$

$$= -\frac{1}{4}L^{-1}\left[\frac{1}{(s+s)^2+s^2}\right] + \frac{1}{4}L^{-1}\left[\frac{1}{(s-s)^2+s^2}\right]$$

$$= -\frac{1}{4}\cdot\frac{1}{2}e^{-s}\sin 2t + \frac{1}{4}\cdot\frac{1}{2}e^{s}\sin 2t = \frac{1}{2s^2}\sin 2t\left(\frac{e^{s}-e^{-s}}{2}\right) = \frac{1}{2s^2}\sin 2t\sin 2t.$$

EXERCISE 13.3

Find the inverse Laplace transform of :

1. $\frac{2s-5}{s^2+3s-10}$
2. $\frac{s}{(s-1)(s-2)(s-3)}$ (U.P.T.U., 2000)
3. $\frac{s+2}{s^2-s-6}$ (U.P.T.U., 2001) (U.P.T.U., 2004)
4. $\frac{s-2}{(s+3)(s-1)(s+1)}$ (A.P.T.U., 2001) (A.P.T.U., 2004)
5. $\frac{2s^2-5s+3}{s^3-6s^2+11s-6}$ (U.P.T.U., 2000)
6. $\frac{s-2s}{s+2)(s+1)^2}$
7. $\frac{s}{(s+1)^2(s^2+2)}$
8. $\frac{1}{s^3-s^2}$
9. $\frac{2s+3}{s^2+4s+13}$
10. $\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+1)}$ (Madras, 2000)
11. $\frac{s^2+2s^2}{s^2+4s^2}$
12. $\frac{1}{s^2-5s+6}$
13. $\frac{s^2+s}{(s^2+1)(s^2+2s+2)}$
14. $\frac{s^2}{s^2+s^2}$ (Kerala, 2000)
15. $\frac{s^2+s}{(s^2+1)(s^2+2s+2)}$
16. $\frac{s^2+s}{(s^2+1)(s^2+2s+2)}$
17. $\frac{s}{s^4+s^2+1}$ (Kerala, 2000)
18. (Kerala, 2000)

13.13 OTHER METHODS OF FINDING INVERSE TRANSFORMS

We have seen that the most effective method of finding the inverse transforms is by means of partial fractions. However, various other methods are available which depend on the following important theorems/formulae:

I. Shifting property for inverse Laplace transforms.

If $\mathcal{L}^{-1}[F(s)] = f(t)$, then

$$\mathcal{L}^{-1}[F(s-a)] = e^{at}f(t) = e^{at}\mathcal{L}^{-1}[F(s)]$$

II. If $\mathcal{L}^{-1}[F(s)] = f(t)$ and $f(0) = 0$, then

$$\mathcal{L}^{-1}[sF(s)] = \frac{d}{dt}[f(t)]$$

In general, $\mathcal{L}^{-1}[s^nF(s)] = \frac{d^n}{dt^n}[f(t)]$ provided $f(0) = f'(0) = \ldots = f^{n-1}(0) = 0$.

The above formulae at once follow from the results of § 21.7 (Transforms of derivatives).

III. If $\mathcal{L}^{-1}[F(s)] = f(t)$, then

$$\mathcal{L}^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t f(t)\,dt$$

This result follows from § 21.8 (Transforms of integrals)

Also $$\mathcal{L}^{-1}\left[\frac{F(s)}{s^2}\right] = \int_0^t\left[\int_0^t f(t)dt\right]dt$$

$$\mathcal{L}^{-1}\left[\frac{F(s)}{s^3}\right] = \int_0^t\left[\int_0^t\left(\int_0^t f(t)dt\right)dt\right]dt \text{ and so on.}$$

IV. If $\mathcal{L}^{-1}[F(s)] = f(t)$, then

$$tf(t) = \mathcal{L}^{-1}\left[-\frac{d}{ds}F(s)\right]$$

This result follows from $\mathcal{L}[tf(t)] = -\frac{d}{ds}[F(s)]$ [§ 21.9]

V. The formula of § 21.10, i.e.,

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s)\,ds$$

is useful in finding $f(t)$ when $F(s)$ is given, provided the inverse transform of $\int_s^\infty F(s)\,ds$ can be conveniently calculated.

Example 21.22. Find the inverse Laplace transforms of the following:

(a) $\frac{s^2}{(s-2)^3}$ (b) $\frac{s+3}{s^2-4s+13}$ (c) $\frac{(s-2)^2}{(s^2+4s+8)^2}$ (Hyderabad, 2003)

Solutions. (a) Since $s^2 = (s-2)^2 + 4(s-2) + 4$

$$\therefore \quad \frac{s^2}{(s-2)^3} = \frac{1}{s-2} + \frac{4}{(s-2)^2} + \frac{4}{(s-2)^3}$$

$$\therefore \quad \mathcal{L}^{-1}\left[\frac{s^2}{(s-2)^3}\right] = \mathcal{L}^{-1}\left[\frac{1}{s-2}\right] + 4\mathcal{L}^{-1}\left[\frac{1}{(s-2)^2}\right] + 4\mathcal{L}^{-1}\left[\frac{1}{(s-2)^3}\right]$$

$$= e^{2t} + 4te^{2t} + 2t^2e^{2t}.$$ [using shifting property]

(b) $$\frac{s+3}{s^2-4s+13} = \frac{s-2}{(s-2)^2+3^2} + \frac{5}{(s-2)^2+3^2}$$

$$\therefore \quad \mathcal{L}^{-1}\left[\frac{s+3}{s^2-4s+13}\right] = \mathcal{L}^{-1}\left[\frac{s-2}{(s-2)^2+3^2}\right] + \frac{5}{3}\mathcal{L}^{-1}\left[\frac{3}{(s-2)^2+3^2}\right]$$

$$= e^{3t}\cos 2t + \frac{3}{2}e^{3t}\sin 2t \qquad \text{[Using shifting property (i)]}$$

(iii) $L^{-1}\dfrac{(s+2)^2}{(s^2+4s+8)^2} = L^{-1}\dfrac{(s+2)^2}{(s^2+4s+4+4)^2} = L^{-1}\dfrac{(s+2)^2}{[(s+2)^2+4]^2}$

$$= e^{-2t}L^{-1}\left\{\frac{s^2}{(s^2+4)^2}\right\} = e^{-2t}L^{-1}\left\{\frac{s^2+4-4}{(s^2+4)^2}\right\}$$

$$= e^{-2t}L^{-1}\left\{\frac{1}{s^2+4} - \frac{4}{(s^2+4)^2}\right\} = \frac{e^{-2t}\sin 2t}{2} - 4e^{-2t}L^{-1}\left\{\frac{1}{(s^2+4)^2}\right\}$$

$$= \frac{e^{-2t}\sin 2t}{2} - 4e^{-2t}\left\{\frac{1}{4^2}\left(\frac{\sin 2t}{4} - \frac{t\cos 2t}{2}\right)\right\}$$

$$= e^{-2t}\left\{\frac{\sin 2t}{2} - \frac{\sin 2t}{4} + \frac{t\cos 2t}{2}\right\} = e^{-2t}\left\{\frac{\sin 2t}{4} + \frac{t\cos 2t}{2}\right\}$$

Example 9.53. Find the inverse transform of $\log\left(1+\frac{a^2}{s^2}\right)$ [U.P.T.U. 2003(CO)]
(a) $\frac{2}{t}(1-\cos at)$ (b) $\frac{1}{t}(1-\cos at)$

Solution. (a) Since $L^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \frac{1}{a}\sin at$.

therefore, by formula III above,

$$L^{-1}\left\{\frac{1}{s(s^2+a^2)}\right\} = \int_0^t \frac{1}{a}\sin at\,dt = \frac{1}{a^2}[1-\cos at]_0^t = \frac{1}{a^2}(1-\cos at)$$

(a) $$L^{-1}\left\{\frac{1}{s(s+a)^2}\right\} = L^{-1}\left\{\frac{1}{s(s^2+2as+a^2)}\right\} = e^{-at}L^{-1}\left\{\frac{1}{s(s+a)^2}\right\}$$

Now $$L^{-1}\left\{\frac{1}{s-a}\right\} = e^{at} \quad \therefore \quad L^{-1}\left\{\frac{1}{s(s-a)}\right\} = \int_0^t e^{at}\,dt = \frac{e^{at}}{a} - \frac{1}{a}$$, by III above

$$\therefore \quad L^{-1}\left\{\frac{1}{s(s-a)^2}\right\} = \frac{1}{a}\int_0^t (e^{at}-1)\,dt = \frac{1}{a^2}(e^{at}-at-1)$$

$$L^{-1}\left\{\frac{1}{s(s-a)^3}\right\} = \frac{1}{a^2}\int_0^t (e^{at}-at-1)\,dt = \frac{1}{a^3}\left(e^{at}-\frac{a^2t^2}{2}-at-1\right)$$

Hence $$L^{-1}\left\{\frac{1}{s(s+a)^2}\right\} = e^{-at}\cdot\frac{1}{a^3}\left(e^{at}-\frac{a^2t^2}{2}-at-1\right) = \frac{1}{a^3}\left(1-e^{-at}-at e^{-at}-\frac{a^2}{2}t^2e^{-at}\right)$$

Example 9.54. Find the inverse Laplace transforms of:
(i) $\frac{s}{(s^2+a^2)^2}$ (U.P.T.U. 2002) (ii) $\frac{s^2}{(s^2+a^2)^2}$ (Kanpur Univ. 2000) (iii) $\cot^{-1}\frac{s}{a}$ (U.P.T.U. 2003)

Solution. (i) If $f(t) = L^{-1}\frac{s}{(s^2+a^2)^2}$, then as by formula V above,

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty \frac{s}{(s^2+a^2)^2}ds = \frac{1}{2}\int_s^\infty \frac{2s}{(s^2+a^2)^2}ds = -\frac{1}{2}\left[\frac{1}{s^2+a^2}\right]_s^\infty = \frac{1}{2}\cdot\frac{1}{s^2+a^2}$$

$$\therefore \quad \frac{f(t)}{t} = \frac{1}{2}L^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \frac{\sin at}{2a}$$

Hence, $f(t)=\frac{1}{2a}\sin at$.

Observation: Let $f(t)=L^{-1}\left\{\frac{1}{s^2+a^2}\right\}=\frac{\sin at}{a}$ so that $f'(t)=\frac{1}{s^2+a^2}$

Then by (IV) above, $tf(t)=L^{-1}\left\{-\frac{d}{ds}(F(s))\right\}=L^{-1}\left\{-\frac{d}{ds}\left(\frac{1}{s^2+a^2}\right)\right\}$

or $\frac{t\sin at}{a}=L^{-1}\left\{\frac{2s}{(s^2+a^2)^2}\right\}$. Hence $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}=\frac{1}{2a}t\sin at$.

(vi) In (i), we have proved that

$$L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}=\frac{1}{2a}t\sin at=f(t),\text{ say}$$

Since $f(0)=0$, we get from formula II above, that

$$L^{-1}\left\{\frac{s^2}{(s^2+a^2)^2}\right\}=L^{-1}\left\{s\cdot\frac{s}{(s^2+a^2)^2}\right\}=\frac{d}{dt}(f(t))$$

$$=\frac{d}{dt}\left\{\frac{1}{2a}t\sin at\right\}=\frac{1}{2a}\{t\cos at+\sin at\}$$

(vii) In (i), we have shown that:

$$L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}=\frac{1}{2a}t\sin at=f(t),\text{ say}$$

By formula III above, we have

$$L^{-1}\left\{\frac{1}{s(s^2+a^2)^2}\right\}=L^{-1}\left\{\frac{1}{s}\cdot\frac{s}{(s^2+a^2)^2}\right\}=\int_0^t f(t)\,dt=\int_0^t\frac{t\sin at}{2a}\,dt$$

$$=\frac{1}{2a}\left\{t\cdot\frac{-\cos at}{a}\right\}_0^t-\int_0^t 1\cdot\left(\frac{-\cos at}{a}\right)dt\Bigg\}$$

$$=\frac{1}{2a}\left\{-\frac{t\cos at}{a}+\frac{\sin at}{a^2}\right\}=\frac{1}{2a^3}\{\sin at-at\cos at\}$$

Example 2.20. Find the inverse Laplace transforms of

(i) $\frac{s+2}{s^2(s+1)(s-2)}$ (V.T.U., 2002) (ii) $\frac{s+2}{(s^2+4s+5)^2}$ (V.T.U., 2002) (P.T.U., 2005)

Solution: (i) $L^{-1}\left\{\frac{s+2}{s^2(s+1)(s-2)}\right\}=\frac{4}{3}L^{-1}\left\{\frac{1}{s-2}\right\}-\frac{1}{3}L^{-1}\left\{\frac{1}{s+1}\right\}=\frac{4}{3}e^{2t}-\frac{1}{3}e^{-t}$

By III above, $L^{-1}\left\{\frac{s+2}{s(s+1)(s-2)}\right\}=\int_0^t L^{-1}\left\{\frac{s+2}{(s+1)(s-2)}\right\}dt$

$$=\int_0^t\left\{\frac{4}{3}e^{2t}-\frac{1}{3}e^{-t}\right\}dt=\frac{2}{3}e^{2t}+\frac{1}{3}e^{-t}-1$$

Again by III above, $L^{-1}\left\{\frac{s+2}{s^2(s+1)(s-2)}\right\}=\int_0^t L^{-1}\left\{\frac{s+2}{s(s+1)(s-2)}\right\}dt$

$$=\int_0^t\left\{\frac{2}{3}e^{2t}+\frac{1}{3}e^{-t}-1\right\}dt=\frac{1}{3}e^{2t}-e^{-t}-t$$

(xv) $\int \frac{1}{(x^2+4x+5)} dx = \int \frac{1}{(x+2)^2+1} dx = \tan^{-1}(x+2) + c$

By (i) above, $\int \frac{dx}{(x^2+4x+5)} = \tan^{-1}(x+2) + c$

(xvi) $\int \frac{-(2x+1)}{(x^2+x+1)^2} dx = \frac{1}{x^2+x+1} + c$

(xvii) $\int \frac{x+3}{(x^2+4x+5)^2} dx = \frac{1}{2}\tan^{-1}(x+2) + c$

Example 1: Find the following integrals using inverse trigonometric function.

$\int \frac{1}{x} \log\left(\frac{x+1}{x-1}\right) dx$ (a) $\int \frac{1}{x^2+4x+5} dx$ (b) $\int \frac{x^2+x}{x^2+x+1} dx$ (c) $\int \frac{1}{x^2+2x+3} dx$ (d)

$\int \frac{1}{\sqrt{x}(1+x)} dx$ (e) $\int \frac{1}{\sqrt{x^2+2x+3}} dx$ (f)

Solution (a): $I = \int \frac{1}{x} \log\left(\frac{x+1}{x-1}\right) dx$ (Solved by IV above).

$$= \log\left(\frac{x+1}{x-1}\right) \int \frac{1}{x} dx - \int \left[\frac{d}{dx}\log\left(\frac{x+1}{x-1}\right) \int \frac{1}{x} dx\right] dx = \log\left(\frac{x+1}{x-1}\right) \log x - \int \left[\frac{1}{x+1} - \frac{1}{x-1}\right] \log x \, dx$$

$$= \log x \log\left(\frac{x+1}{x-1}\right) - \int \frac{\log x}{x+1} dx + \int \frac{\log x}{x-1} dx = \log x \log\left(\frac{x+1}{x-1}\right) - I_1 + I_2 + c$$

Where $I_1 = \int \frac{\log x}{x+1} dx$ (solved in (b))

and $I_2 = \int \frac{\log x}{x-1} dx$ (Solved by IV above).

$$I_1 = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_3$$

Where $I_3 = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_3 = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx + \int \frac{1}{x} \log x \, dx$$

$$= -\log x \log(x+1) + \int \frac{1}{x} \log x \, dx - \int \frac{1}{x+1} \log x \, dx = -\log x \log(x+1) + I_4 - I_5$$

Where $I_4 = \int \frac{1}{x} \log x \, dx$ (Solved by I above).

and $I_5 = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_4 = \frac{1}{2}(\log x)^2 + c$$

$$I_5 = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_6$$

Where $I_6 = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_6 = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_7$$

Where $I_7 = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_7 = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_8$$

Where $I_8 = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_8 = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_9$$

Where $I_9 = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_9 = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{10}$$

Where $I_{10} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{10} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{11}$$

Where $I_{11} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{11} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{12}$$

Where $I_{12} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{12} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{13}$$

Where $I_{13} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{13} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{14}$$

Where $I_{14} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{14} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{15}$$

Where $I_{15} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{15} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{16}$$

Where $I_{16} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{16} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{17}$$

Where $I_{17} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{17} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{18}$$

Where $I_{18} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{18} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{19}$$

Where $I_{19} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{19} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{20}$$

Where $I_{20} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{20} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{21}$$

Where $I_{21} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{21} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{22}$$

Where $I_{22} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{22} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{23}$$

Where $I_{23} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{23} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{24}$$

Where $I_{24} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{24} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{25}$$

Where $I_{25} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{25} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{26}$$

Where $I_{26} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{26} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{27}$$

Where $I_{27} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{27} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{28}$$

Where $I_{28} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{28} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{29}$$

Where $I_{29} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{29} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{30}$$

Where $I_{30} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{30} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{31}$$

Where $I_{31} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{31} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{32}$$

Where $I_{32} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{32} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{33}$$

Where $I_{33} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{33} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{34}$$

Where $I_{34} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{34} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{35}$$

Where $I_{35} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{35} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{36}$$

Where $I_{36} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{36} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{37}$$

Where $I_{37} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{37} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{38}$$

Where $I_{38} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{38} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{39}$$

Where $I_{39} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{39} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{40}$$

Where $I_{40} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{40} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{41}$$

Where $I_{41} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{41} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{42}$$

Where $I_{42} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{42} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{43}$$

Where $I_{43} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{43} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{44}$$

Where $I_{44} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{44} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{45}$$

Where $I_{45} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{45} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{46}$$

Where $I_{46} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{46} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{47}$$

Where $I_{47} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{47} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{48}$$

Where $I_{48} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{48} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{49}$$

Where $I_{49} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{49} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{50}$$

Where $I_{50} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{50} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{51}$$

Where $I_{51} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{51} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{52}$$

Where $I_{52} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{52} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{53}$$

Where $I_{53} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{53} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{54}$$

Where $I_{54} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{54} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{55}$$

Where $I_{55} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{55} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{56}$$

Where $I_{56} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{56} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{57}$$

Where $I_{57} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{57} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{58}$$

Where $I_{58} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{58} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{59}$$

Where $I_{59} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{59} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{60}$$

Where $I_{60} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{60} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{61}$$

Where $I_{61} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{61} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{62}$$

Where $I_{62} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{62} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{63}$$

Where $I_{63} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{63} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{64}$$

Where $I_{64} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{64} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{65}$$

Where $I_{65} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{65} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{66}$$

Where $I_{66} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{66} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{67}$$

Where $I_{67} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{67} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{68}$$

Where $I_{68} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{68} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{69}$$

Where $I_{69} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{69} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{70}$$

Where $I_{70} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{70} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{71}$$

Where $I_{71} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{71} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{72}$$

Where $I_{72} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{72} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{73}$$

Where $I_{73} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{73} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{74}$$

Where $I_{74} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{74} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{75}$$

Where $I_{75} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{75} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{76}$$

Where $I_{76} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{76} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{77}$$

Where $I_{77} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{77} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{78}$$

Where $I_{78} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{78} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{79}$$

Where $I_{79} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{79} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{80}$$

Where $I_{80} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{80} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{81}$$

Where $I_{81} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{81} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{82}$$

Where $I_{82} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{82} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{83}$$

Where $I_{83} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{83} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{84}$$

Where $I_{84} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{84} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{85}$$

Where $I_{85} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{85} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{86}$$

Where $I_{86} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{86} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{87}$$

Where $I_{87} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{87} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{88}$$

Where $I_{88} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{88} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{89}$$

Where $I_{89} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{89} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{90}$$

Where $I_{90} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{90} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{91}$$

Where $I_{91} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{91} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{92}$$

Where $I_{92} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{92} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{93}$$

Where $I_{93} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{93} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{94}$$

Where $I_{94} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{94} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{95}$$

Where $I_{95} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{95} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{96}$$

Where $I_{96} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{96} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{97}$$

Where $I_{97} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{97} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{98}$$

Where $I_{98} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{98} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{99}$$

Where $I_{99} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{99} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{100}$$

Where $I_{100} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{100} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{101}$$

Where $I_{101} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{101} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{102}$$

Where $I_{102} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{102} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{103}$$

Where $I_{103} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{103} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{104}$$

Where $I_{104} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{104} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{105}$$

Where $I_{105} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{105} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{106}$$

Where $I_{106} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{106} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{107}$$

Where $I_{107} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{107} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{108}$$

Where $I_{108} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{108} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{109}$$

Where $I_{109} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{109} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{110}$$

Where $I_{110} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{110} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{111}$$

Where $I_{111} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{111} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{112}$$

Where $I_{112} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{112} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{113}$$

Where $I_{113} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{113} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{114}$$

Where $I_{114} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{114} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{115}$$

Where $I_{115} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{115} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{116}$$

Where $I_{116} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{116} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{117}$$

Where $I_{117} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{117} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{118}$$

Where $I_{118} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{118} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{119}$$

Where $I_{119} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{119} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{120}$$

Where $I_{120} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{120} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{121}$$

Where $I_{121} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{121} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{122}$$

Where $I_{122} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{122} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{123}$$

Where $I_{123} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{123} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{124}$$

Where $I_{124} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{124} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{125}$$

Where $I_{125} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{125} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{126}$$

Where $I_{126} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{126} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{127}$$

Where $I_{127} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{127} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{128}$$

Where $I_{128} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{128} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{129}$$

Where $I_{129} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{129} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{130}$$

Where $I_{130} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{130} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{131}$$

Where $I_{131} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{131} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{132}$$

Where $I_{132} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{132} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{133}$$

Where $I_{133} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{133} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{134}$$

Where $I_{134} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{134} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{135}$$

Where $I_{135} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{135} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{136}$$

Where $I_{136} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{136} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{137}$$

Where $I_{137} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{137} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{138}$$

Where $I_{138} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{138} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{139}$$

Where $I_{139} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{139} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{140}$$

Where $I_{140} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{140} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{141}$$

Where $I_{141} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{141} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{142}$$

Where $I_{142} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{142} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{143}$$

Where $I_{143} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{143} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{144}$$

Where $I_{144} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{144} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{145}$$

Where $I_{145} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{145} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{146}$$

Where $I_{146} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{146} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{147}$$

Where $I_{147} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{147} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{148}$$

Where $I_{148} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{148} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{149}$$

Where $I_{149} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{149} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{150}$$

Where $I_{150} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{150} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{151}$$

Where $I_{151} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$I_{151} = \log x \log(x+1) - \int \frac{1}{x} \log(x+1) dx = \log x \log(x+1) - I_{152}$$

Where $I_{152} = \int \frac{1}{x} \log(x+1) dx$ (Solved by IV above).

$$I_{152} = \log(x+1) \log x - \int \frac{1}{x+1} \log x \, dx = \log(x+1) \log x - I_{153}$$

Where $I_{153} = \int \frac{1}{x+1} \log x \, dx$ (Solved by IV above).

$$= L^{-1}\left\{\frac{4s}{(s^2+2s-1)(s^2)}\right\} = L^{-1}\left\{\frac{4s}{(s^2+2s+2)(s^2+2s-2)}\right\}$$

$$= L^{-1}\left\{\frac{1}{s^2-2s+2} - \frac{1}{s^2+2s+2}\right\} = L^{-1}\left\{\frac{1}{(s-1)^2+1} - \frac{1}{(s+1)^2+1}\right\}$$

$$= e^t \sin t - e^{-t}\sin t = 2\sinh t \sin t.$$

21.11 CONVOLUTION THEOREM

If $L^{-1}\{F(s)\} = f(t)$ *and* $L^{-1}\{G(s)\} = g(t)$,

then $L^{-1}\{F(s)\,G(s)\} = \int_0^t f(u)\,g(t-u)\,du = f * g$

If $f * g$ *is called the convolution of* f *and* g.

Let: $\phi(t) = \int_0^t f(u)\,g(t-u)\,du$

$$\mathcal{L}\{\phi(t)\} = \int_0^\infty e^{-st}\left\{\int_0^t f(u)\,g(t-u)\,du\right\}dt = \int_0^\infty \int_0^t e^{-st} f(u)\,g(t-u)\,du\,dt \qquad \ldots(1)$$

The domain of integration for this double integral is the *t*-*u* area lying between the lines $u = 0$ and $u = t$ (Fig. 21.2).

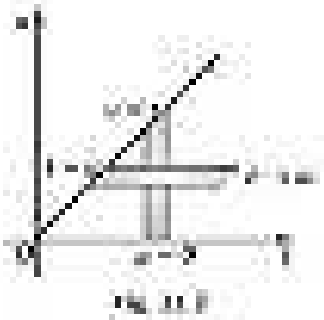


Fig. 21.2

Changing the order of integration, we get

$$\mathcal{L}\{\phi(t)\} = \int_0^\infty \int_u^\infty e^{-st} f(u)\,g(t-u)\,dt\,du$$

$$= \int_0^\infty e^{-su} f(u)\left\{\int_u^\infty e^{-s(t-u)}\,g(t-u)\,dt\right\}du$$

$$= \int_0^\infty e^{-su} f(u)\left\{\int_0^\infty e^{-sv}\,g(v)\,dv\right\}du \text{ on putting } t-u = v$$

$$= \int_0^\infty e^{-su} f(u)\,G(s)\,du = \int_0^\infty e^{-su} f(u)\,du \cdot G(s)$$

$$= f(s)\cdot g(s)$$ which verifies the desired result.

Example 21.10. Apply Convolution theorem to evaluate

(i) $L^{-1}\dfrac{s}{(s^2+a^2)^2}$. (U.P.T.U., 2000)

(ii) $L^{-1}\dfrac{s^2}{(s^2+a^2)(s^2+b^2)}$. (U.P.T.U., 2001; A.M.I.E., 2000; Madras, 2002)

Solution. (i) Since $f(s) = L^{-1}\left\{\dfrac{s}{s^2+a^2}\right\} = \cos at$ and $g(s) = L^{-1}\left\{\dfrac{a}{s^2+a^2}\right\} = \dfrac{1}{a}\sin at$

$\therefore$ by Convolution theorem, we get

$$L^{-1}\left\{\frac{s}{s^2+a^2}\cdot\frac{1}{s^2+a^2}\right\} = \int_0^t \cos au\,\frac{\sin a(t-u)}{a}\,du \qquad \left[\because \begin{array}{l} f(u) = \cos au \\ g(t-u) = \frac{1}{a}\sin a(t-u)\end{array}\right]$$

$$= \frac{1}{2a}\int_0^t [\sin at + \sin(2au - at)]\,du = \frac{1}{2a}\left[u\sin at + \frac{1}{2a}\cos(2au - at)\right]_0^t = \frac{1}{2a}t\sin at$$

Hence $L^{-1}\left\{\dfrac{s}{(s^2+a^2)^2}\right\} = \dfrac{1}{2a}t\sin at$.

(a) Show that $\int_0^{\frac{\pi}{2}} \sin^4 x \cos^2 x \, dx = \frac{\pi}{32}$. [4 marks]

(b) Hence find the exact value of $\int_0^{\frac{\pi}{2}} \sin^4 x \cos^2 x \, dx$. [4 marks]

(c) By Fundamental Theorem, we get

$$\int_0^{\frac{\pi}{2}} \sin^4 x \cos^2 x \, dx = \int_0^{\frac{\pi}{2}} \sin^4 x (1 - \sin^2 x) \, dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^4 x \, dx - \int_0^{\frac{\pi}{2}} \sin^6 x \, dx$$

$$= \left[\frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{5} - \frac{1}{7} = \frac{2}{35}$$

Evaluate $\int \frac{1}{x^2 + 4x + 13} \, dx$. [3 marks]

Hence, $\int \frac{1}{x^2 + 4x + 13} \, dx$ [3 marks]

Evaluate $\int \sin^{-1} x \, dx$. [4 marks]

(a) By Fundamental Theorem, we get

$$\int \sin^{-1} x \, dx = \int \sin^{-1} x \cdot 1 \, dx$$

$$= \sin^{-1} x \cdot x - \int \frac{x}{\sqrt{1 - x^2}} \, dx$$

$$= x \sin^{-1} x + \frac{1}{2} \int (1 - x^2)^{-\frac{1}{2}} (-2x) \, dx$$

$$= x \sin^{-1} x + \sqrt{1 - x^2} + C$$

(b) Hence $\int \sin^{-1} x \, dx = x \sin^{-1} x + \sqrt{1 - x^2} + C$ [1 mark]

Evaluate $\int \frac{x^3}{x^2 + 4x + 13} \, dx$. [5 marks]

(a) By Fundamental Theorem, we get

$$\int \frac{x^3}{x^2 + 4x + 13} \, dx = \int \left(x - 4 + \frac{36 - 3x}{x^2 + 4x + 13} \right) dx$$

$$= \int x \, dx - 4 \int dx + \int \frac{36 - 3x}{x^2 + 4x + 13} \, dx$$

$$= \frac{x^2}{2} - 4x + \int \frac{36 - 3x}{x^2 + 4x + 13} \, dx$$

$$= \frac{x^2}{2} - 4x + \int \frac{36 - 3x}{(x + 2)^2 + 9} \, dx$$

$$= \frac{x^2}{2} - 4x + \int \left[\frac{42}{(x + 2)^2 + 9} - \frac{3x}{(x + 2)^2 + 9} \right] dx$$

$$= \frac{x^2}{2} - 4x + \int \left[\frac{42}{(x + 2)^2 + 9} - \frac{3(x + 2)}{(x + 2)^2 + 9} + \frac{6}{(x + 2)^2 + 9} \right] dx$$

$$= \frac{x^2}{2} - 4x + 14 \int \frac{1}{(x + 2)^2 + 9} \, dx - 3 \int \frac{x + 2}{(x + 2)^2 + 9} \, dx + 6 \int \frac{1}{(x + 2)^2 + 9} \, dx$$

$$= \frac{x^2}{2} - 4x + 20 \int \frac{1}{(x + 2)^2 + 9} \, dx - 3 \int \frac{x + 2}{(x + 2)^2 + 9} \, dx$$

$$= \frac{x^2}{2} - 4x + \frac{20}{3} \tan^{-1} \left(\frac{x + 2}{3} \right) - \frac{3}{2} \ln \left| (x + 2)^2 + 9 \right| + C$$

EXERCISE 2.3

Find the inverse transforms of

1. $\frac{1}{s^2(s+2)}$ (Mumbai, 2003) (S)
2. $\frac{1}{s(s+2)^2}$
3. $\frac{1}{s^2(s^2+1)}$ (Nagpur, 2004) (S)
4. $\frac{1}{(s^2+a^2)^2}$
5. $\frac{1}{s^2(s^2+2)}$
6. $\frac{s+2}{(s^2+4s+13)^2}$ (Mumbai, 2004)
7. $\frac{2s}{(s^2+a^2)^2}$
8. $\frac{s^2}{(s+a)^3}$
9. $\log\left(\frac{s+a}{s-a}\right)$
10. $\log\left(\frac{s+a}{s+b}\right)$ (Mumbai, 2000) (VTU, 2002)
11. $\log\left(\frac{s+2}{(s+1)(s+3)}\right)$
12. $\frac{1}{s}\log\left(\frac{s^2+1}{s^2-1}\right)$ (Mumbai, 2002) (VTU, 2006)
13. $\log\left(1-\frac{a^2}{s^2}\right)$
14. $\log\frac{s^2+1}{s(s-1)}$ (Mumbai, 2000) (S)
15. $\tan^{-1}\left(\frac{a}{s}\right)$ (Mumbai, 2002) (VTU, 2005)
16. $\cot^{-1}(s)$ (VTU, 2006)
17. $\log\frac{s-a}{s+1}$ (Mumbai, 2006)

Using Convolution theorem, evaluate

18. $L^{-1}\left\{\frac{1}{(s+a)(s+b)}\right\}$
19. $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$
20. $L^{-1}\left\{\frac{1}{s^2(s^2+a^2)}\right\}$
21. $L^{-1}\left\{\frac{1}{(s^2+a^2)^2}\right\}$
22. $L^{-1}\left\{\frac{s}{(s+a)(s+b)(s+c)}\right\}$ (Mumbai, 2004)
23. $L^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\}$ (VTU, 2003) (S)
24. $\frac{s}{s(s^2+a^2)}$ (VTU, 2003) (VTU, 2005)
25. $\frac{1}{(s^2+4s+13)^2}$ (Mumbai, 2004)
26. Show that $\cot^{-1}\left(\frac{1}{s}\sin h\frac{1}{s}\right)=1-\frac{1}{3!s^2}+\frac{1}{5!s^4}-\frac{1}{7!s^6}+\ldots$

and $\cot^{-1}\left(\frac{1}{s}\cos h\frac{1}{s}\right)=\frac{1}{s}-\frac{1}{2!s^3}+\frac{1}{4!s^5}-\frac{1}{6!s^7}+\ldots$

2.13 (1) APPLICATION TO DIFFERENTIAL EQUATIONS

The Laplace transform method of solving differential equations yields particular solutions without the necessity of first finding the general solution and then evaluating the arbitrary constants. This method is, in general, shorter than our earlier methods and is specially useful for solving linear differential equations with constant coefficients.

(2) Working procedure to solve a linear differential equation with constant coefficients by transform method :

1. Take the Laplace transform of both sides of the differential equation using the formula of § 2.7, and the given initial conditions.

2. Then upon the terms such as x and y to the right.

3. Divide by the coefficient of y, getting y as a known function of s.

4. Resolve this function of s into partial fractions and take the inverse transform of both sides. This gives us a function of t which is the desired solution satisfying the given conditions.

Example 21.27. Solve by the method of transforms, the equation
$y'' + 2y' - y - 2y = e^t$, given $y(0) = y'(0) = 0$ and $y''(0) = e^t$. (U.P.T.U., 2003 S)

Solution. Taking the Laplace transform of both sides, we get

$$[s^3 Y - s^2 y(0) - s y'(0) - y''(0)] + 2[s^2 Y - s y(0) - y'(0)] - [sY - y(0)] - 2Y = \frac{1}{s-1}$$

Using the given conditions, it reduces to

$$(s^3 + 2s^2 - s - 2)Y = \frac{1}{s-1}$$

or $$Y = \frac{1}{(s-1)(s+1)(s+2)(s-1)} = \frac{1}{(s-1)^2(s+1)(s+2)}$$

$$= \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s+1} + \frac{D}{s+2}$$

On inversion, we get $y = Ae^t + Bte^t + Ce^{-t} + De^{-2t}$ which is the desired result.

Example 21.28. Use transform method to solve

$$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = t^2 e^t, x(0) = 2, \frac{dx}{dt}(0) = -1 \text{ at } t = 0. \quad \text{(Kerala, 2005 S)}$$

Solution. Taking the Laplace transforms of both sides, we get

$$[s^2 X - sx(0) - x'(0)] - 2[sX - x(0)] + X = \frac{2}{(s-1)^3}$$

Using the given conditions, it reduces to

$$(s^2 - 2s + 1)X = \frac{2}{(s-1)^3} + 2s - 5 = \frac{2s^3 - 7s + 6}{(s-1)^3}$$

$$\therefore \quad X = \frac{2s^3 - 7s + 6}{(s-1)^5} = \frac{2}{(s-1)^2} - \frac{3}{(s-1)^4} + \frac{3}{(s-1)^5}$$ on breaking into partial fractions.

On inversion, we obtain $x = 2L^{-1}\left(\frac{1}{(s-1)^2}\right) - 3L^{-1}\frac{1}{(s-1)^4} + 3L^{-1}\frac{1}{(s-1)^5}$

$$= 2te^t - \frac{3t^3}{3!}e^t + \frac{3t^4}{4!}e^t = 2te^t - \frac{1}{2}t^3e^t + \frac{1}{8}t^4e^t.$$

Example 21.29. Solve $(D^2 + n^2)x = a \cos nt$ subject to $x(0) = 0, x'(0) = 0$. (U.P.T.U., 2004 S)

Solution. Taking the Laplace transform of both sides, we get

$$[s^2 X - sx(0) - x'(0)] + n^2 X = a\left(\frac{s}{s^2 + n^2} - \frac{n}{s^2 + n^2}\right), \text{ as } L(\cos nt) = \frac{s}{s^2 + n^2}, L(\sin nt) = \frac{n}{s^2 + n^2}$$

On using the given conditions,

$$(s^2 + n^2)X = a\cos nt \cdot \frac{s}{s^2 + n^2} + a \sin nt \cdot \frac{n}{s^2 + n^2}$$

$$\therefore \quad X = a \cos nt \cdot \frac{1}{(s^2 + n^2)^2} + a \sin nt \cdot \frac{s}{(s^2 + n^2)^2}$$

On inversion, we obtain

$$x = a \cos nt \cdot \frac{1}{2n^2} \int t \sin nt \, dt - a \cos nt \int \frac{t}{2n} \sin nt \, dt + a \sin nt \cdot \frac{t}{2n} \sin nt \, dt \quad \text{[By (1.1) and (1.2), p. 2.11]}$$

$$= a\left[\frac{t}{2n^2} \sin nt \cos nt - \frac{t}{2n} \cos^2 nt + \frac{1}{4n^2} \sin nt \cos nt\right] + a\left[\frac{t}{2n} \sin^2 nt - \frac{1}{4n^2} \sin nt \cos nt\right]$$

Example 21.30. Solve $(D^2 - 2D + 2)y = e^t \sin t$, if $y(0) = 0$, $y'(0) = 1$, $y''(0) = 0$. (U.P.T.U., 2005 S)

Solution. Taking the Laplace transforms of both sides, we get

$$L\{y''\} - s^2 y(0) - s y(0) - y'(0) + 3\{L\{y'\} - s y(0) - y(0)\} + 2\{L\{y\} - y(0)\} - y = \frac{2}{(s-1)^3}$$

Using the given conditions, it reduces to

$$s^2 - 2s + 1 + \frac{2}{(s-1)^3} + \frac{(s-1)^2 - (s-1) - 1}{(s-1)^3} + \frac{2}{(s-1)^3}$$

$$= \frac{1}{s-1} + \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3} + \frac{2}{(s-1)^4}$$

On inversion, we obtain $y = L^{-1}\left\{\frac{1}{s-1}\right\} + L^{-1}\frac{1}{(s-1)^2} - L^{-1}\frac{1}{(s-1)^3} + 2L^{-1}\frac{1}{(s-1)^4}$

$$= e^t\left(1 + t - \frac{1}{2}t^2 + \frac{1}{12}t^3\right).$$

Example 21.31. Solve $\frac{d^2x}{dt^2} + 2x = \cos 2t$, if $x(0) = 2$, $x\left(\frac{\pi}{2}\right) = -1$. (Given $\cos\pi = -1$, $\sin\pi = 0$, $\sin\frac{\pi}{2} = 1$)

Solution. Since $x'(0)$ is not given, we assume $x'(0) = a$.

Taking the Laplace transforms of both sides of the equation, we have

$$L\{x''\} + 2L\{x\} = L\{\cos 2t\}, \quad s^2\bar{x} - sx(0) - x'(0) + 2\bar{x} = \frac{s}{s^2+4}$$

$$(s^2 + 2)\bar{x} = 2s + a + \frac{s}{s^2+4} \quad \text{or} \quad \bar{x} = \frac{2s}{s^2+2} + \frac{a}{s^2+2} + \frac{s}{(s^2+4)(s^2+2)}$$

or
$$\bar{x} = \frac{2s}{s^2+2} + \frac{1}{2}\cdot\frac{a}{s^2+2} + \frac{1}{2}\cdot\frac{s}{s^2+2} - \frac{1}{2}\cdot\frac{s}{s^2+4}$$

On inversion, we get $x = \frac{a}{2}\sin\sqrt{2}t + \frac{5}{2}\cos\sqrt{2}t - \frac{1}{2}\cos 2t$

When $t = \pi/2$, $-1 = -\frac{a}{2} - \frac{1}{2}$ or $\frac{a}{2} = \frac{1}{2}$ $\left[\because \sin\left(\frac{\pi}{2}\right) = 1\right]$

Hence the solution is $x = \frac{1}{2}\cos\sqrt{2}t + \frac{1}{2}\sin\sqrt{2}t - \frac{1}{2}\cos 2t$.

Note. Laplace transform method is not suitable for solving ordinary differential equations with variable coefficients. For example, the differential equation $t\frac{d^2y}{dt^2} + (1-2t)\frac{dy}{dt} - 2y = 0$ becomes $L\left\{t\frac{d^2y}{dt^2}\right\} - 2L\{y\} = 0$.

Example 21.32. Solve $(y^2 - 2x^2 + 3x)\,dx + (2xy - y)\,dy = 0$, given that $y(0) = 2$. (U.P.T.U. 2006)

Solution. Taking Laplace transform of both sides of the equation and noting that

$$L\{y^2(x)\} = -\frac{d}{ds}L\{y^2(x)\}$$ we get

$$-\frac{d}{ds}L\{y^2\} - sy(0) - y'(0) + 2L\{y\} - y(0) - \frac{d}{ds}L\{y\} = \frac{s}{s^2+1}$$

or
$$-\left(s^2\frac{d\bar{y}}{ds} + 2s\bar{y}\right) + sy(0) + 0 + 2\bar{y} - 2y(0) - \frac{d\bar{y}}{ds} = \frac{s}{s^2+1}$$

or
$$(s^2+1)\frac{d\bar{y}}{ds} + 1 = -\frac{s}{s^2+1} \quad \text{or} \quad \frac{d\bar{y}}{ds} = -\frac{1}{s^2+1} - \frac{s}{(s^2+1)^2}$$

On rearranging and taking L^{-1} of $\overline{y}(s) = -1/(1)$, we get

$$-y = -\sin t - \int_0^t y(u)\,du \qquad \text{[See § 21.12 (1)]}$$

or $$y = \frac{1}{2}\left(1 - \frac{d}{dt}\right)\sin t$$ which is the desired solution.

Example 21.20. Solve $t\dfrac{d^2y}{dt^2} + \dfrac{dy}{dt} + ty = 0$, $y(0) = 2$, $y'(0) = 0$.

Solution. Taking Laplace transforms of both sides of the equation, we get

$$L\{ty''\} + L\{y'\} + L\{ty\} = 0$$

or $$-\frac{d}{ds}\{s^2\bar{y} - sy(0) - y'(0)\} + \{s\bar{y} - y(0)\} - \frac{d\bar{y}}{ds} = 0 \quad \text{or} \quad (s^2+1)\frac{d\bar{y}}{ds} + s\bar{y} = 0$$

Separating the variables, $\int \frac{d\bar{y}}{\bar{y}} + \int \frac{s\,ds}{s^2+1} = c$

or $$\log \bar{y} + \frac{1}{2}\log(s^2+1) = \log c' \quad \text{or} \quad \bar{y} = \frac{c'}{\sqrt{(s^2+1)}}$$

Inverting gives $y = c' J_0(t)$

To find c', we have $y(0) = c' J_0(0) = 2$, i.e., $c' = 2$

Hence $y = 2J_0(t)$.

Example 21.21. *An electromotive force* $E(t)$ *is applied to a leaky condenser of capacity* C *and resistance* R *in series. Show by Laplace transform method that the current in the circuit is* $\dfrac{dq}{dt} + \dfrac{q}{CR} = \dfrac{E(t)}{R}$. *Hence find the current in the circuit, when* $E = E_0 \sin \omega t$.

Solution. If i be a current and q the charge at time t in the circuit, then the differential equation is

$$L\frac{di}{dt} + \frac{q}{C} = E(t) \text{ or } R\frac{dq}{dt} + \frac{q}{C} = E(t) \qquad \because\ i = \frac{dq}{dt}$$

Taking Laplace transform of both sides, we get

$$L\{R\bar{q}(s) - Rq(0)\} + \frac{1}{C}L\{q(t)\} = L\{E(t)\} = \frac{E_0\omega}{s^2+\omega^2}$$

But at $t = 0$, $i = 0$ and $q = 0$, i.e., $i(0) = 0$

$$\therefore \qquad L\{R\bar{q}(s)\} + \frac{1}{C}L\{q(t)\} = \frac{E_0\omega}{s^2+\omega^2} \qquad \ldots(1)$$

Also taking Laplace transform of $i = dq/dt$, we get

$$\bar{i}(s) = L\{dq/dt\} = sL\{q\} - q(0)$$

i.e., $$L\{q(t)\} = \frac{\bar{i}(s)}{s} \qquad \because\ q(0) = 0$$

$\therefore$ (1) becomes $L\{R\bar{i}(s)\} + \frac{1}{C}\left\{\frac{\bar{i}(s)}{s}\right\} = \frac{E_0\omega}{s^2+\omega^2}$

or $$\left(R + \frac{1}{Cs}\right)\bar{i}(s) = \frac{E_0\omega}{s^2+\omega^2} \quad \text{or} \quad \bar{i}(s) = \frac{E_0\omega s}{(s^2+\omega^2)(Rs + 1/C)}$$

or $$\bar{i}(s) = \frac{E_0\omega}{R}\cdot\frac{s}{(s^2+\omega^2)\left(s + \frac{1}{RC}\right)} \text{ where } \frac{1}{RC} = \text{constant}$$

$$\bar{i}(s) = \frac{E_0\omega}{R}\left\{\frac{s}{s^2+\omega^2} - \frac{s}{s^2+\frac{1}{R^2C^2}}\right\}$$

Now taking inverse Laplace transforms of both sides, we get

$$x(t) = \frac{Ea}{L(p^2-a^2)}e^{-\frac{R}{2L}t}\left\{\frac{p}{p^2+\omega^2} - \frac{a}{a^2+\omega^2}\right\}$$

or

$$x(t) = \frac{Ea}{L(p^2-a^2)}\left\{\cos\omega t - \frac{a}{\omega}\sin\omega t\right\}$$

PROBLEMS 9.11

Solve the following equations by the Laplace transform method:

1. $y'' + 4y' + 3y = e^{-t}$; $y(0) = y'(0) = 1$ (U.P.T.U., 2005; G. Pondicherry, 2007)
2. $(D^2-1)y = x\sin x$; $y(0) = y'(0) = 0$
3. $y'' + y = 1$; $y(0) = 1, y'(0) = 0$ (Kerala, 2009)
4. $y'' - 3y' + 2y = e^{3t}$, where $y(0) = 1$ and $y'(0) = 0$ (U.P.T.U., 2013)
5. $(D^2 - 2D + 2)y = 4e^{2t}$ and $y(0) = -3$, $y(0) = 5$ (Kerala, 2009)
6. $y'' + 2y' + 10y = 20\sin 3t$ given that $y(0) = 2$, $y'(0) = 3$. (U.P.T.U., 2006)
7. $(D^2 + a^2)y = \sin(at)$, $t > 0$, given that $y = 0$ and $Dy = 1$ at $t = 0$.
8. $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = 4te^{-t}$, $y = \frac{dy}{dt} = 1$ when $t = 0$ (Kerala, 2008) (Kerala, 2010)
9. $\frac{d^2y}{dt^2} - 4y = 8$, where $y(0) = 5$, $y'(0) = y''(0) = y'''(0) = 0$
10. $y^{iv}(t) + 2y''(t) + y(t) = \sin t$, where $y(0) = y'(0) = y''(0) = y'''(0) = 0$
11. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = e^{-t}\sin t$, where $y(0) = 0$ and $y'(0) = 1$. (U.P.T.U., 2015)
12. $y'' - 3y' + 2y = 4e^{2t}$, $y(0) = -3$, $y'(0) = 5$. (U.P.T.U., 2015, 16)
13. $\frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 3\frac{dy}{dt} - y = t^2e^t$, where $y = 1$, $\frac{dy}{dt} = 0$, $\frac{d^2y}{dt^2} = -2$ at $t = 0$.
14. $(D^2 + 1)y = t\sin t$; $y = Dy = 0$ at $t = 0$. (Kerala, 2008) (U.P.T.U., 2016)
15. $y'' + 3y' + 2y = e^{-2t}$, where $y(0) = 1$
16. $y'' - (k_1 + k_2)y' + k_1k_2y = 0$, where $y(0) = 1$, $y'(0) = k_2$ (U.P.T.U., 2005)
17. $y'' + 2y' - y = t$, where $y(0) = 0$, $y'(0) = 1$. (U.P.T.U., 2005)
18. $y'' - y' + 4y = 0$, where $y(0) = 2$, $y'(0) = 0$
19. A body of mass m is projected vertically upwards with velocity v in a medium whose resistance is k times the velocity. Show that the displacement x at time t is given by $x = \frac{mv}{k}(1 - e^{-\frac{k}{m}t})$. (U.P.T.U., 2005)
20. Worked example 11.17(p. 322) by the Laplace transform method.
21. Obtain a complete solution for the forced oscillations of a mass m attached to the lower end of an elastic spring whose upper end is fixed and whose stiffness is k, when the driving force is $F_0\sin\omega t$. Solve this equation, finding the Laplace transform of each term, given that initial velocity and displacement (from equilibrium position) are zero.

[Hint : The equation of motion is $m\frac{d^2x}{dt^2} = -kx + F_0\sin\omega t$ or $\frac{d^2x}{dt^2} + \frac{k}{m}x = \frac{F_0}{m}\sin\omega t$, where x is the displacement at time t.]

9.11.1 SIMULTANEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

The Laplace transform method can also be applied with advantage to the solution of simultaneous linear differential equations.

Example 11.33. Solve the simultaneous equations $\frac{dx}{dt} + 3x - y = 0$ and $\frac{dy}{dt} + x + y = 0$, given that $x = 0$, $y = 1$ when $t = 0$. (Ex. 11.33(i))

Solution. Taking the Laplace transforms of the given equations, we get

$$L\left(\frac{dx}{dt}\right) - x(0) + 5x - 2y = L(e^{2t}) \quad \text{i.e.,} \quad (s+5)x - 2y = \frac{1}{s-2} \qquad \ldots(i) \quad [\because x(0) = 0]$$

and

$$L\left(\frac{dy}{dt}\right) - y(0) + 2x - y = 0 \quad \text{i.e.,} \quad 2x + (s-1)y = 0 \qquad \ldots(ii) \quad [\because y(0) = 0]$$

Solving (i) and (ii) for x, we get

$$x = \begin{vmatrix} \frac{1}{s-2} & -2 \\ 0 & s-1 \end{vmatrix} \div \begin{vmatrix} s+5 & -2 \\ 2 & s-1 \end{vmatrix} = \frac{s-1}{(s-2)(s^2+4s-7)} = \frac{1}{3(s-2)} + \frac{2}{9s} - \frac{2}{3(s-2)^2} - \frac{2}{9(s+7)} - \frac{2}{9(s-2)^2}$$

Substituting the value of x in (ii), we get

$$y = -\frac{2}{(s-2)(s^2+4s-7)} = \frac{4}{3(s-2)} - \frac{2}{9s} - \frac{4}{3(s-2)^2} - \frac{2}{9(s+7)}$$

On inversion, we get

$$x = \frac{1}{3}e^{2t} + \frac{2}{9} - \frac{2}{3}te^{2t} - \frac{2}{9}e^{-7t}, \quad y = \frac{4}{3}e^{2t} - \frac{2}{9} - \frac{4}{3}te^{2t} - \frac{2}{9}e^{-7t}.$$

Example 9.36. *The velocities u and v in a certain theory of plasma satisfy the equations $\frac{du}{dt} + 2u = \frac{dv}{dt}$, $\frac{dv}{dt} + 2v = 4u$. Solve the equations under the conditions $u(0) = 3$, $v(0) = 1$. Hence find x and y when $t = 0$, $x = 3$ and $y = 0$, assuming that $u = \frac{dx}{dt}$, $v = \frac{dy}{dt}$. Also show that the particle describes the curve $4x^2 - 4xy + 3y^2 = 4$.* (U.P.T.U., 2003)

Solution. Taking the Laplace transforms of the given equations and noting that $y(0) = 0$, $x(0) = 3$,

we get

$$L\left(\frac{du}{dt}\right) - u(0) + 2\bar{u} = \frac{s}{s^2+2^2} \quad \text{or} \quad s\bar{u} + 2\bar{v} = \frac{s}{s^2+4} \qquad \ldots(i)$$

and

$$L\left(\frac{dv}{dt}\right) - v(0) + 2\bar{v} = \frac{1}{s^2+2^2} \quad \text{or} \quad s\bar{v} - 4\bar{u} = \frac{1}{s^2+4} + 1 \qquad \ldots(ii)$$

Multiplying (i) by s and (ii) by 2 and subtracting, we get

$$(s^2+8)\bar{v} = -2 \quad \text{or} \quad \bar{v} = -2(s^2+8)^{-1}$$

On inversion,

$$v = -2L^{-1}\left[\frac{1}{s^2+8}\right] = -\sin 2t$$

From the given first equation,

$$2x = \sin 2t - 2y' = \sin 2t - \frac{d}{dt}(1 - \cos 2t)$$

or

$$2x = \sin 2t + 2\cos 2t \quad \text{or} \quad 4x^2 = (\sin 2t + 2\cos 2t)^2 \qquad \ldots(iii)$$

Also

$$4xy = (\sin 2t + 2\cos 2t)(1 - \cos 2t) = 2\sin^2 2t + 3\sin 2t\cos 2t \qquad \ldots(iv)$$

and

$$3y^2 = 3\sin^2 2t. \qquad \ldots(v)$$

Adding (iii), (iv), and (v), we obtain

$$4x^2 + 4xy + 3y^2 = \sin^2 2t + 4\sin 2t\cos 2t + 4\cos^2 2t = 2\sin^2 2t$$
$$= 4\cos 2t\cos 2t + 4\cos^2 2t + 4\cos^2 2t + 4\cos^2 2t = 4$$

Example 9.37. *The small oscillations of a certain system with two degrees of freedom are given by the equations $\frac{d^2x}{dt^2} + 2x - 2y = 0$, $\frac{d^2y}{dt^2} + 2y - 2x = 0$. If $x = 0$, $y = 0$, $\frac{dx}{dt} = 2$, $\frac{dy}{dt} = 0$ when $t = 0$, find x and y when $t = \pi$.* (Example 9.37(1))

Solution. Taking the Laplace transform of both the equations, we get

$$L\left(\frac{d^2x}{dt^2}\right) = s^2\bar{x} - sx(0) - x'(0) + 2\bar{x} - 2\bar{y} = 0 \quad \text{i.e.,} \quad (s^2+2)\bar{x} - 2\bar{y} = 2 \qquad \ldots(i)$$

and

$$L\left(\frac{d^2y}{dt^2}\right) = s^2\bar{y} - sy(0) - y'(0) + 2\bar{y} - 2\bar{x} = 0 \quad \text{i.e.,} \quad -2\bar{x} + (s^2+2)\bar{y} = 0 \qquad \ldots(ii)$$

Solving (i) and (ii) for $\bar{x}$ and $\bar{y}$, we get

$$\bar{x} = \begin{vmatrix} 2 & -2 \\ 0 & s^2+2 \end{vmatrix} \div \begin{vmatrix} s^2+2 & -2 \\ -2 & s^2+2 \end{vmatrix} = \frac{2s^2+8}{s^4+4s^2+0}$$
$$= \frac{11}{4}\cdot\frac{1}{s^2+1} + \frac{1}{4}\cdot\frac{2}{s^2+9}$$

and $$x = \begin{vmatrix} s^2+3 & 3 \\ s^2-3 & 5 \end{vmatrix} \div \begin{vmatrix} s^2+3 & -3 \\ s^2-3 & s^2+3 \end{vmatrix} = \frac{5s^2+15}{s^4+12s^2+36} = \frac{15}{36}\cdot\frac{1}{s^2+1} + \frac{5}{36}\cdot\frac{3}{s^2+9}$$

On inversion, we get $x = \frac{15}{6}\sin t + \frac{1}{12}\sin 3t$; $y = \frac{15}{6}\sin t - \frac{1}{12}\sin 3t$

which are the same as the solution in mid on p. 403.

Obs. The student should compare the earlier solution of the above example with those given here and appreciate the superiority of the transform method over others.

EXERCISE 2.12

Solve the following simultaneous equations by using Laplace transforms:

1. $\frac{dx}{dt} - y = e^t$, $\frac{dy}{dt} + x = \sin t$, given $x(0) = 1$, $y(0) = 0$. (U.P.T.U., 2006; Delhi, 2006)

2. $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that $x = 2$ and $y = 0$ when $t = 0$. (Kerala, 2005; U.P.T.U., 2004)

3. $\frac{d^2x}{dt^2} + n^2 y = \frac{d^2y}{dt^2} + n^2 x = 0$, given that $x = 0$, $\frac{dx}{dt} = n$, $y = 1$, $\frac{dy}{dt} = 0$ when $t = 0$. (P.T.U., 2006-07)

4. $x\frac{dy}{dt} + \frac{dx}{dt} + 2x = t$, $\frac{dy}{dt} + x\frac{dx}{dt} + 2y = 0$, given $x = 0$, $y = 0$ when $t = 0$. (Madras, 2003)

5. (i) $2x - 3y - 0$, (ii) $5x + 3\frac{dx}{dt} - 3\frac{dy}{dt} = 0$, given $x = 2$, $y = 0$ when $t = 0$.

6. The currents i_1 and i_2 in a network are given by the differential equations $L_1\frac{di_1}{dt} + Ri_1 + R(i_1 - i_2) = E\sin\omega t$, $L_2\frac{di_2}{dt} + Ri_2 + R(i_2 - i_1) = 0$. Find the currents i_1 and i_2 by Laplace transforms if $i_1 = i_2 = 0$ at $t = 0$.

2.13 (1) UNIT STEP FUNCTION

At times, we come across such functions of which the inverse transforms cannot be determined by the formulae so far derived. In order to cover such cases, we introduce the unit step function or Heaviside's unit function[*].

Def. The *unit step function* $u(t-a)$ is defined as follows:

$$u(t-a) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t \geq a \end{cases}$$

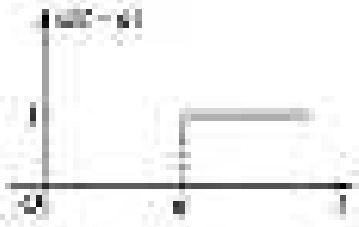


Fig. 2.13

where, a is a unique positive (Fig. 2.13). It is also denoted as $H(t-a)$.

(2) Transform of unit function.

$$\mathcal{L}\{u(t-a)\} = \int_0^\infty e^{-st}\,u(t-a)\,dt = \int_0^a e^{-st}\cdot 0\,dt + \int_a^\infty e^{-st}\cdot 1\,dt = 0 + \left|\frac{e^{-st}}{-s}\right|_a^\infty$$

Thus $$\mathcal{L}\{u(t-a)\} = e^{-as}/s.$$

The product $f(t)\,u(t-a) = \begin{cases} 0 & \text{for } t < a \\ f(t) & \text{for } t \geq a \end{cases}$

The function $f(t-a)\,.\,u(t-a)$ represents the graph of $f(t)$ shifted through a distance a to the right and is of special importance.

Second shifting property: If $\mathcal{L}\{f(t)\} = \bar{f}(s)$, then

$$\mathcal{L}\{f(t-a)\,.\,u(t-a)\} = e^{-as}\bar{f}(s)$$

$$\mathcal{L}\{f(t-a)\,.\,u(t-a)\} = \int_a^\infty e^{-st} f(t-a)\,u(t-a)\,dt$$

---

[*] Named after the British Electrical Engineer Oliver Heaviside (1850–1925).

$$= \int_0^a e^{-st} f(t)\,dt + \int_a^\infty e^{-st} f(t)\,dt \qquad \text{[Put } t = a + u\text{]}$$

$$= \int_0^\infty e^{-s(a+u)} f(a+u)\,du = e^{-as}\int_0^\infty e^{-su} f(a+u)\,du = e^{-as} F(s).$$

Example 22.20. Express the following function (Fig. 22.4) in terms of unit step function and find its Laplace transform. *(U.P.T.U., 2002)*

Solution. We have $f(t) = \begin{cases} 0, & 0 < t < 1 \\ t - 1, & 1 < t < 2 \\ 1, & t > 2 \end{cases}$

or $f(t) = 0 \cdot [u(t-0) - u(t-1)] + (t-1)[u(t-1) - u(t-2)] + 1 \cdot u(t-2)$

$= 0 - 0 + (t-1)u(t-1) - (t-2)u(t-2) + u(t-2)$

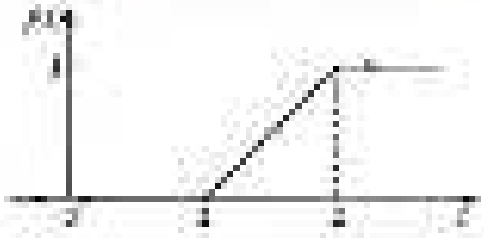


Fig. 22.4

By second shifting property,

$$\mathcal{L}\{f(t)\,u(t-a)\} = e^{-as}\mathcal{L}\{f(t+a)\}.$$

Also $\mathcal{L}\{f(t)\} = \mathcal{L}\{t\} = 1/s^2$.

$\therefore$ $\mathcal{L}\{t-1\} = 1/s^2$ and $\mathcal{L}\{t-2\} = 1/s^2$, $\mathcal{L}\{u(t-2)\} = e^{-2s}/s$

Hence $\mathcal{L}\{f(t)\} = \mathcal{L}\{t\} - \mathcal{L}\{(t-1)u(t-1)\} - \mathcal{L}\{(t-2)u(t-2)\} + \mathcal{L}\{u(t-2)\} = \dfrac{e^{-s} - e^{-2s}}{s^2}$

Example 22.21. Using unit step function, find the Laplace transform of

$$f(t) = \begin{cases} \sin t, & 0 \le t < \pi \\ \cos t, & \pi \le t < 2\pi \\ \sin 2t, & t \ge 2\pi \end{cases}$$ *(U.P.T.U., 2000)*

Solution. $f(t) = \sin t\,[u(t) - u(t-\pi)] + \cos t\,[u(t-\pi) - u(t-2\pi)] + \sin 2t\,u(t-2\pi)$

$= \sin t\,u(t) + (\sin t - \cos t)\,u(t-\pi) + (\sin 2t - \cos t)\,u(t-2\pi)$

Now $\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}\mathcal{L}\{f(t)\}$ and $\mathcal{L}\{\sin(at)\} = \dfrac{a}{s^2 + a^2}$.

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\sin t\} + \mathcal{L}\{(\sin t - \cos t)\,u(t-\pi)\} + \mathcal{L}\{(\sin 2t - \sin 2t)\,u(t-2\pi)\}$$

$$= \frac{1}{s^2+1} + e^{-\pi s}\left(\frac{2}{s^2+4} - \frac{1}{s^2+1}\right) + e^{-2\pi s}\left(\frac{0}{s^2+0} - \frac{2}{s^2+4}\right)$$

Example 22.22. (i) Express the function $f(t) = \sin t - u(t - 2\pi)$ in terms of unit step function and find its Laplace transform. *(U.P.T.U., 2008-09)*

(ii) Obtain the Laplace transform of $e^{-3t} - u(t - 2)$.

Solution. (i) We have $f(t) = \begin{cases} t - 1, & 1 < t < 2 \\ 3 - t, & 2 < t < 3 \end{cases}$

or $f(t) = 0 \cdot [u(t-0) - u(t-1)] + (t-1)[u(t-1) - u(t-2)] + (3-t)[u(t-2) - u(t-3)]$

$= 0 - 0 + (t-1)u(t-1) - (t-2)u(t-2) + u(t-2) + (3-t)u(t-2) - (3-t)u(t-3)$

Since $\mathcal{L}\{f(t-a)\,u(t-a)\} = e^{-as}\mathcal{L}\{f(t)\}$ ...(1)

$\therefore$ $\mathcal{L}\{f(t)\} = e^{-s}\cdot\dfrac{1}{s^2} - 2e^{-2s}\cdot\dfrac{1}{s^2} + e^{-2s}\cdot\dfrac{1}{s^2} + e^{-2s}\cdot\dfrac{1}{s} - e^{-3s}\cdot\dfrac{1}{s^2} = \dfrac{e^{-s} - e^{-2s}}{s^2}$ [$\because f(t) = 1$]

(ii) $\mathcal{L}\{e^{-3t} - u(t-2)\} = \mathcal{L}\{e^{-3t}\} - \mathcal{L}\{u(t-2)\} = \dfrac{1}{s+3} - e^{-2s}\mathcal{L}\{u(t+2)\}$

Taking $f(t) = e^{-t}$, $f(t-a) = \frac{1}{s+2}$ and using Definition,

$$L\{e^{-t}u(t-a)\} = e^{-as}\cdot\frac{1}{s+2}$$

Hence $L^{-1}\left\{\frac{e^{-as}}{s+2}\right\} = u(t-a)e^{-(t-a)}$

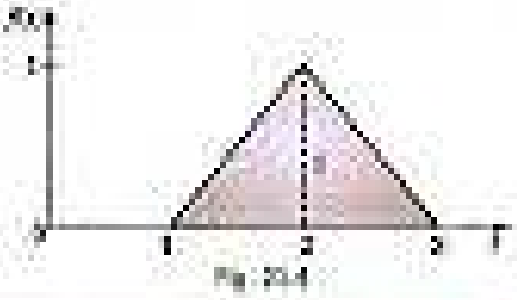


Fig. 21.4

Example 21.11. Using Laplace transforms, evaluate $\int_0^\infty e^{-t}(1+2t-t^2+t^3)H(t-2)\,dt$.

(RGPV, Nov/Dec 2007)

Solution. We have $L\{(1+2t-t^2+t^3)H(t-2)\}$

$$= e^{-2s}L\{1+2(t+2)-(t+2)^2+(t+2)^3\} = e^{-2s}L\{3+2t+2t^2+t^3\}$$

$$= e^{-2s}\left[3\cdot\frac{1}{s}+2\cdot\frac{1}{s^2}+2\cdot\frac{2!}{s^3}+\frac{3!}{s^4}\right] = e^{-2s}\left[\frac{3}{s}+\frac{2}{s^2}+\frac{4}{s^3}+\frac{6}{s^4}\right]$$

By definition, this implies that

$$\int_0^\infty e^{-t}(1+2t-t^2+t^3)H(t-2)\,dt = e^{-2s}\left[\frac{3}{s}+\frac{2}{s^2}+\frac{4}{s^3}+\frac{6}{s^4}\right]$$

Taking $s = 1$, we obtain

$$\int_0^\infty e^{-t}(1+2t-t^2+t^3)H(t-2)\,dt = e^{-2}\left[3+2+4+6\right] = 15e^{-2}.$$

Example 21.12. Evaluate (i) $L^{-1}\left\{\frac{e^{-s}-2e^{-3s}}{s^2}\right\}$ (UTU, Nov/Dec 2009)

(ii) $L^{-1}\left\{\frac{se^{-as}}{s^2-a^2}\right\}$, $s > 0$.

Solution. $$L^{-1}\left\{e^{-s}\cdot\frac{1}{s^2}\right\} = \begin{cases}t-1, & t>1\\ 0, & t<1\end{cases} = (t-1)H(t-1)$$

$$L^{-1}\left\{e^{-3s}\cdot\frac{1}{s^2}\right\} = \begin{cases}t-3, & t>3\\ 0, & t<3\end{cases} = (t-3)H(t-3)$$

$$\therefore \quad L^{-1}\left\{\frac{e^{-s}-2e^{-3s}}{s^2}\right\} = L^{-1}\left\{\frac{e^{-s}}{s^2}\right\} - 2L^{-1}\left\{\frac{e^{-3s}}{s^2}\right\} = (t-1)H(t-1) - 2(t-3)H(t-3)$$

(ii) We know that $L^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \cosh at$

$$\therefore \quad L^{-1}\left\{\frac{se^{-as}}{s^2-a^2}\right\} = \begin{cases}\cosh a(t-a), & t>a\\ 0, & t<a\end{cases}$$

$$= \cosh a(t-a)H(t-a), \text{ by second shifting property.}$$

Example 21.13. Find the inverse Laplace transform of

(i) $\frac{4e^{-s}+2e^{-3s}}{s^2+4}$ (U.P.T.U., 2006) (ii) $\frac{e^{-s}}{s^2(s+1)}$, $s>0$ (Kurukshetra, 2002)

Solution. (i) We have $L^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$, $L^{-1}\left\{\frac{a}{s^2+a^2}\right\} = \sin at$

and $\mathcal{L}^{-1}[e^{-as}\bar{f}(s)] = f(t-a)\,H(t-a)$ ...(2)

$$\therefore \quad \mathcal{L}^{-1}\left[\frac{se^{-\pi s/2}+4e^{-3s}}{s^2+4^2}\right] = \mathcal{L}^{-1}\left[e^{-\pi s/2}\,\frac{s}{s^2+4^2}\right] + \mathcal{L}^{-1}\left[e^{-3s}\,\frac{4}{s^2+4^2}\right]$$

$$= \cos 4\left(t-\frac{\pi}{2}\right)H\left(t-\frac{\pi}{2}\right) + \sin 4(t-3)\,H(t-3)$$

$$= \cos 4\left(t-\frac{\pi}{2}\right)H\left(t-\frac{\pi}{2}\right) + \sin 4(t-3)\,H(t-3)$$

(ii) $\mathcal{L}^{-1}\left[\frac{e^{-s}}{s^2(s+1)}\right] = \mathcal{L}^{-1}\left[e^{-s}\left(-\frac{1}{s^2}+\frac{1}{s}+\frac{1}{s^2}-\frac{1}{s+1}\right)\right]$

Using (2) above, we have

$$\mathcal{L}^{-1}\left[\frac{e^{-s}}{s^2(s+1)}\right] = -\frac{1}{s^2}H(t-1) + \frac{1}{s}H(t-1)\,e^{-(t-1)} + \frac{1}{s^2}H(t-1)\,e^{-(t-1)}(t-1)$$

$$= \frac{1}{s^2}H(t-1) - 1 + e^{-(t-1)} + (t-1)$$

Example 12.46. *A particle of mass m can oscillate about the position of equilibrium under the effect of a restoring force and leaves the equilibrium. If initial jerk and initial force F is felt only for $0 < t < T$ and then ceases. Find the amplitude of the subsequent oscillations.*

Solution. The constant force F acting from $t = 0$ to $t = T$ can be expressed as

$$F[H(t) - H(t-T)],\ 0 < t < T$$

∴ equation of motion of the particle is

$$m\frac{d^2x}{dt^2} = F[H(t) - H(t-T)] - mk^2x \quad \text{or} \quad \frac{d^2x}{dt^2} + k^2x = \frac{F}{m}[H(t) - H(t-T)]$$

Taking Laplace transforms of both sides, we get

$$(s^2 + k^2)\,\bar{x} = \frac{F}{ms}(1 - e^{-sT}) \qquad \because\ x = 0,\ \dot{x} = 0 \text{ at } t = 0$$

or $$\bar{x} = \frac{F}{m}\,\frac{1-e^{-sT}}{s(s^2+k^2)} = \frac{F}{ms}(1-e^{-sT})\cdot\frac{1}{k^2}\left(\frac{1}{s} - \frac{s}{s^2+k^2}\right)$$

$$= \frac{F}{mk^2}\left\{\frac{1}{s} - e^{-sT}\cdot\frac{1}{s} - (1-e^{-sT})\cdot\frac{s}{s^2+k^2}\right\}$$

Taking inverse Laplace transform, we obtain

$$x = \frac{F}{mk^2}[1 - \cos kt - H(t-T) + H(t-T)\cos k(t-T)]$$

i.e., $$x = \frac{F}{mk^2}(1 - \cos kt) \text{ for } 0 < t < T$$

and $$x = \frac{F}{mk^2}(1 - \cos kt) - [1 - \cos k(t-T)] \text{ for } t > T$$

$$= \frac{F}{mk^2}[\cos k(t-T) - \cos kt] \text{ for } t > T$$

or $$x = \frac{2F}{mk^2}\sin\frac{kT}{2}\cdot\sin k\left(t-\frac{T}{2}\right) \text{ for } t > T$$

Hence the amplitude of subsequent oscillations for $t > T$ is $\frac{2F}{mk^2}\sin\frac{kT}{2}$.

Example 12.47. *In an electrical circuit with $e = E_0\sin\omega t$, resistance R and inductance L, the current i builds up at the rate given by*

$$L\frac{di}{dt} + Ri = E_0\sin\omega t \qquad \ldots(1)$$

If the switch is closed at $t = 0$ and disconnected at $t = a$, find the current for any future.

Solution. We have $v = 0$ at $t = 0$ and $E(t) = \begin{cases} E & \text{for } 0 < t < a \\ 0 & \text{for } t > a \end{cases}$

∴ taking the Laplace transform of both sides of the given equation

$$\mathcal{L}(L\dot{i} + Ri) = \int_0^a e^{-st} E(t)\,dt = \int_0^a e^{-st} E\,dt = \frac{E}{s}(1 - e^{-as})$$

or $$I = \frac{E}{s(Ls + R)} - \frac{Ee^{-as}}{s(Ls + R)}$$

On inversion, we get $$i = \mathcal{L}^{-1}\left\{\frac{E}{s(Ls + R)}\right\} - \mathcal{L}^{-1}\left\{\frac{Ee^{-as}}{s(Ls + R)}\right\} \quad \ldots(1)$$

Now $$\mathcal{L}^{-1}\left\{\frac{E}{s(Ls + R)}\right\} = \frac{E}{R}\left\{\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s + R/L}\right\}\right\} = \frac{E}{R}(1 - e^{-Rt/L})$$

and $$\mathcal{L}^{-1}\left\{\frac{Ee^{-as}}{s(Ls + R)}\right\} = \frac{E}{R}(1 - e^{-R(t-a)/L})\,u(t - a)$$ [By the second shifting property]

Thus (1) becomes $$i = \frac{E}{R}(1 - e^{-Rt/L}) - \frac{E}{R}(1 - e^{-R(t-a)/L})\,u(t - a)$$

Hence $$i = \frac{E}{R}(1 - e^{-Rt/L}) \text{ for } 0 < t < a$$

and $$i = \frac{E}{R}(1 - e^{-Rt/L}) - (1 - e^{-Rt/L + Ra/L}) = \frac{E}{R}e^{-Rt/L}(e^{Ra/L} - 1) \text{ for } t > a.$$

Example 21.16. *Calculate the transverse deflection of an isotropic beam (Fig. 21.6), hinged at both ends, carrying a uniformly distributed load of intensity w per unit length.*

Solution. Taking the origin at the end A, we have

$$EI\frac{d^2y}{dx^2} = w\,u(x)$$

where $u(x) = u(x - 0) - u(x - 2l) + u(x - 2l)$

Taking the Laplace transform of both sides (Fig. 21.6), we get

Fig. 21.6

$$EI(p^2\bar{y} - py(0) - y'(0) - y''(0) - y'''(0)) = w\left(\frac{e^{-0p}}{p} - \frac{e^{-2lp}}{p} - \frac{e^{-2lp}}{p}\right)$$

Using the conditions $y(0) = y''(0) = 0$ and taking $y'(0) = c_1$ and $y'''(0) = c_2$, we have

$$EI\bar{y} = w\left(\frac{e^{-0p}}{p^5} - \frac{e^{-2lp}}{p^5}\right) + \frac{c_1}{p^3} + \frac{c_2}{p^4}$$

On inversion, we get $EIy = \frac{w}{24}[x - 0]^4 u(x - 0) - [x - 2l]^4 u(x - 2l)] + \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3$ ...(1)

For $x > 2l$, $$EIy = \frac{w}{24}[x - 0]^4 - [x - 2l]^4] + \frac{1}{2}c_1x^2 + \frac{1}{6}c_2x^3$$

and $$EIy' = \frac{w}{6}[x - 0]^3 - [x - 2l]^3] + c_1x + \frac{1}{2}c_2x^2$$

Using the conditions $y(2l) = 0$ and $y'(2l) = 0$, we get $0 = \frac{w}{24}\left[(2l)^4 - \left(\frac{1}{4}\right)^4\right] + \frac{1}{2}c_1(2l)^2 + \frac{1}{6}c_2(2l)^3$

and $$0 = \frac{w}{6}\left[(2l)^3 - \left(\frac{1}{4}\right)^3\right] + c_1(2l) + \frac{1}{2}c_2(2l)^2$$

whence $c_1 = 21wl^3/192$; $c_2 = -wl/8$.

Thus for $l/4 < x < 3l/4$, M is given by $EI\frac{d^2y}{dx^2} = \frac{W}{2}\left(x - \frac{l}{4}\right) - \frac{Wl}{16}x + \frac{Wl^2}{32} - \frac{Wx^2}{24}$

Hence the maximum deflection is $y(l/2) = \frac{19Wl^3}{4608EI}$

21.18 [3] UNIT IMPULSE FUNCTION

The idea of a very large force acting for a very short time is of frequent occurrence in mechanics. To deal with such and similar ideas, we introduce the so-called *impulse function* (also called *Dirac delta function*[*]).

The unit impulse function is considered as the limiting form of the function $\delta_a(t)$ (Fig. 21.3):

$$\delta_a(t) = \frac{1}{2a} \text{ for } a - a < t < a + a$$
$$= 0, \quad \text{otherwise}$$

as $a \to 0$. It is clear from Fig. 21.3 that as $a \to 0$, the height of the strip increases indefinitely and the width decreases in such a way that its area is always unity.

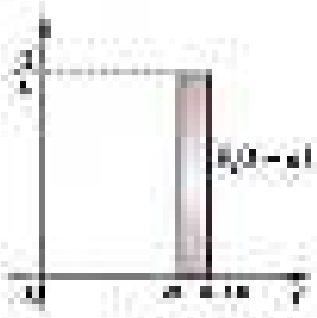


Fig 21.3

Then the unit impulse function (or function $\delta(t-a)$) is defined as follows :

$$\delta(t-a) = \infty \text{ for } t = a \; ; = 0 \text{ for } t \neq a,$$

such that $\int_0^\infty \delta(t-a)\,dt = 1, \quad a \geq 0$

As an illustration, a loading acting at the point $x = a$ of a beam may be considered as the limiting case of uniform loading $w/2a$ per unit length over the portion of the beam between $x = a$ and $x = a + a$. Thus

$$w\delta(x) = w/2a \quad a - a < x < a + a$$
$$= 0, \quad \text{otherwise}$$

i.e., $w\delta(x) = w\delta(x - a)$

(2) Transform of unit impulse function. If $f(t)$ be a function of t continuous at $t = a$, then

$$\int_0^\infty f(t)\delta_a(t-a)\,dt = \int_{a-a}^{a+a} f(t)\cdot\frac{1}{2a}\,dt$$
$$= (a + a - a)f(\eta)\cdot\frac{1}{2a} = f(\eta), \qquad \text{where } a - a < \eta < a + a,$$

by Mean value theorem for integrals.

As $a \to 0$, we get $\int_0^\infty f(t)\delta(t-a)\,dt = f(a)$.

In particular, when $f(t) = e^{-st}$, we have $L\{\delta(t-a)\} = e^{-as}$.

Example 21.27. Evaluate (i) $\int_0^\infty \sin 3t\,\delta(t - \pi/2)\,dt$ **(ii)** $L\left\{\int_0^t \delta(u-a)\,du\right\}$.

Solution. (i) We know that, $\int_0^\infty f(t)\delta(t-a)\,dt = f(a)$

$$\therefore \quad \int_0^\infty \sin 3t\,\delta(t - \pi/2)\,dt = \sin(3\pi/2) = -1$$

(ii) We know that $L\{\delta(t-a)\} = e^{-as}$

$$\therefore \quad L\left\{\int_0^t \delta(u-a)\,du\right\} = \int_0^\infty \left\{\int_0^t \delta(u-a)\,du\right\} e^{-st}\,dt = \int_a^\infty e^{-st}\,dt$$
$$= \left[\frac{e^{-st}}{-s}\right]_a^\infty = \frac{1}{s}e^{-as}$$

[*] After the English physicist P.A.M. Dirac (1902-84) who was awarded the Nobel prize in 1933 for his work in Quantum mechanics.

Example 21.48. An emf $E(t)$ is applied to a series R-L-C circuit consisting of L, R, C in which there is neither initial current nor charge on the capacitor. Find the current $i(t)$ at any subsequent time t. Find the limit of i as $t \to 0$.

Solution. The equation of the circuit governing the current i is

$$L\frac{di}{dt} + Ri + \frac{1}{C}\int_0^t i\,dt = E(t) \qquad \text{where } i = 0, \text{ when } t = 0.$$

Taking Laplace transforms of both sides, we get

$$L\{sI - i(0)\} + RI + \frac{1}{Cs}I = \frac{E}{s} \qquad \text{[Using § 21.7 and 21.8]}$$

or $$\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)I = \frac{E}{L} \quad \text{or} \quad (s^2 + 2as + a^2 + b^2)I = \frac{E}{L},$$

where $R/L = 2a$ and $1/LC = a^2 + b^2$

or $$I = \frac{E}{L}\frac{1}{(s+a)^2 + b^2} = \frac{E}{Lb}\left[\frac{s+a}{(s+a)^2 + b^2} - a\frac{1}{(s+a)^2 + b^2}\right]$$

On inversion, we get

$$i = \frac{E}{Lb}\left[e^{-at}\sin bt - \frac{a}{b}e^{-at}\sin bt\right]$$

Taking limits as $t \to 0$, $i \to 0$.

Although the current $i = 0$ initially, yet a large current will develop instantaneously due to impulsive voltage applied at $t = 0$. In fact, we have determined the limit of this current which is E/L.

Example 21.49. A force is simply supported at its ends $x = 0$ and $x = l$ is displaced at the ends and $x = l/2$ carries a load w at $x = l/2$. Find the resulting deflection at any point.

Solution. The differential equation for deflection is

$$\frac{d^4y}{dx^4} = \frac{w}{EI}\delta\left(x - \frac{l}{2}\right)$$

Taking the Laplace transform, we have $s^4Y - s^3y(0) - s^2y'(0) - sy''(0) - y'''(0) = \frac{w}{EI}e^{-ls/2}$

Using the conditions $y(0) = 0, y''(0) = 0$ and taking $y'(0) = c_1$ and $y'''(0) = c_2$, we get

$$Y = \frac{c_1}{s^3} + \frac{c_2}{s^4} + \frac{w}{EI}\frac{e^{-ls/2}}{s^4}.$$

On inversion, it gives $$y = c_1x + c_2\frac{x^3}{3!} + \frac{w}{EI}\frac{(x - l/2)^3}{3!}\sin(x - l/2)$$

i.e., $$y = c_1x + \frac{1}{6}c_2x^3, \qquad 0 \le x \le l/2$$

and $$y = c_1x + \frac{1}{6}c_2x^3 + \frac{w}{6EI}(x - l/2)^3, \quad l/2 \le x \le l$$ ...(1)

Using the conditions $y(l) = 0$ and $y'(l) = 0$, we get

$$0 = c_1l + \frac{1}{6}c_2l^3 + \frac{w}{6EI}\left(\frac{l}{2}\right)^3 \quad \text{and} \quad 0 = c_1 + \frac{1}{2}c_2l^2 + \frac{w}{2EI}\left(\frac{l}{2}\right)^2$$

whence $$c_1 = \frac{5wl^2}{96EI}, \qquad c_2 = -\frac{11w}{16EI}.$$

Substituting the values of c_1 and c_2 in (1), we get the deflection at any point.

EXERCISE 21.5

1. Represent $f(t) = \sin 3t, 2\pi < t < 4\pi$ and 0 elsewhere, in terms of the unit step function and hence find its Laplace transform. [Madras, 2002]
2. Sketch the graph of the following functions and express them in terms of unit step function. Hence find their Laplace transforms:

(c) $f(t) = 2t^2$ for $0 < t < 4$, $f(t) = 32$ for $t > 4$ (Mumbai, 2000)

(d) $f(t) = t^2$ for $0 < t < 2$, $f(t) = 6$ for $t > 2$

(e) $f(t) = \sin t$ for $0 < t < \pi$, $f(t) = 0$ for $t > \pi$.

5. Express the following functions in terms of unit step function and hence find its Laplace transforms

(a) $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 1, & \pi < t < 2\pi \\ \cos t, & t > 2\pi \end{cases}$ (U.P.T.U., 2003)

(b) $f(t) = \begin{cases} \sin t, & 0 < t < \pi/2 \\ \cos 2t, & \pi/2 < t < \pi \\ \sin 2t, & t > \pi \end{cases}$ (Mumbai, 2000; V.T.U., 2002 S)

(c) $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 0, & t > 4 \end{cases}$ (V.T.U., 2010)

6. Evaluate $\int_0^\infty te^{-3t}\sin t\,dt$ (b) $\int_0^\infty te^{-2t}\cos^2 t\,dt$ (c) $\int_0^\infty e^{-3t}\sin^2 t\,dt$

(d) $\int_0^\infty (t^2 + 2t)e^{-t}\cos t\,dt$ (Mumbai, 2002) (e) $\int_0^\infty e^{-3t}t\sin^2 t\,dt$ (f) $\int_0^\infty e^{-t}\cos^3 t\,dt$.

7. Find the inverse Laplace transform of:

(a) $\frac{e^{-3s}}{s^2+1}$ (b) $\frac{e^{-2s}}{s^2+9s+20}$ (Mumbai, 2000)

(c) $\log\left(\frac{s^2+1}{s(s+1)}\right)$ (V.T.U., 2010) (d) $\cot^{-1}\left(\frac{s+3}{2}\right)$ (V.T.U., 2010 S)

8. Solve using Laplace transform $\frac{dy}{dx} + 3y = f(t)$ if $y(0) = 0$ if $f(t) = \begin{cases} 1, & 0 < t < 1 \\ 0, & t > 1 \end{cases}$ (Mumbai, 2007)

9. Using Laplace transform, solve $y'(t) + y(t) = u(t)$, $y(1) = 0$, $y(0) = 0$

where $u(t) = \begin{cases} 2, & 0 < t < 1 \\ 0, & t > 1. \end{cases}$

10. A beam 1m in length clamped at $x = 0$ and $x = 1$. A concentrated load 'W' acts vertically downwards at the point $x = 1/2$. Find the resulting deflection.

[Hint: The differential equation and the boundary conditions are $\frac{d^2y}{dx^2} = \frac{M}{EI}$, $x = 0$] and

$y(0) = y'(0) = 0$, $y(l) = y'(l) = 0$]

11. A rod of length 1 m is held horizontal at the end $x = 0$ and is free at the end $x = 1$. It carries a load of one kilogram per unit length from $x = 0$ to $x = 1/2$. Calculate the deflection y at any point. (Gondwana, 2000)

[Hint: The differential equation and boundary conditions are

$\frac{d^2y}{dx^2} = \frac{M(x)}{EI}$, $0 < x < 1$ (unit) where $M(x) = \begin{cases} \frac{1}{2}x, & 0 < x < 1/2 \\ \frac{1}{2}, & x > 1/2 \end{cases}$

and $y(0) = y'(0)$, $y(1) = y'(1) = 0$]

12. An airplane flying in still air is oscillating on a spring having a spring constant k. The mass of the airplane is m kg. Assuming a resistance proportional to the velocity of the motion, set up the initial value problem $y(0) = y'(0) = 0$.

[Hint: The equation of motion is $m\frac{d^2y}{dt^2} = f(t) - ky - c\frac{dy}{dt}$.]

3.3.3 OBJECTIVE TYPE OF QUESTIONS

36. Laplace transform of $te^{-t}\sin t$ is

(a) $\frac{2s}{(s+s^2)^2}$ (b) $\frac{4s}{(s+s^2)^2}$ (c) $\frac{4s}{(s^2+4)^2}$ (d) $\frac{4s}{(s-4)^2}$

37. Laplace transform of $\int_0^t te^{-t}\sin t\,dt$ is

(a) $\frac{s+2}{(s+s^2+s^2)^2}$ (b) $\frac{s(s-2)}{(s+s^2+s^2)^2}$ (c) $\frac{2s(s-2)}{(s+s^2+s^2)^2}$ (d) $\frac{(s+s^2)^2}{(s-s^2+s^2)^2}$

38. If $f=\frac{s^2}{s^2+4s^2}$ is

(a) $\frac{1}{2}\sin 2t+\frac{1}{2}\cos 2t$ (b) $\frac{1}{2}\cos 2t+\frac{1}{2}\sin 2t$ (c) $\cos 2t+\frac{1}{2}\sin 2t$ (d) $\frac{1}{2}\sin 2t-\frac{1}{2}\cos 2t$

39. If $f=\frac{1}{4s^2+1}$ is

(a) $1+\sin\frac{t}{2}$ (b) $1-\sin\frac{t}{2}$ (c) $1+\cos\frac{t}{2}$ (d) $1-\cos\frac{t}{2}$

40. If $f(t)=\begin{cases}\sin t & 0<t<\pi\\ 0 & t>\pi\end{cases}$ then $L\{f(t)\}$ is

(a) $\frac{1-e^{-\pi s}}{s^2+1}$ (b) $\frac{1}{s^2+1}$ (c) $\frac{1+e^{-\pi s}}{s^2+1}$ (d) $\frac{e^{-\pi s}}{s^2+1}$ (UPTU, 2010)

41. For a periodic function of period $2a$, $\int_0^{2a} f(t)\,dt=$

(a) $\int_0^a f(t)\,dt$ (b) $\int_{-a}^a f(t)\,dt$ (c) $2\int_0^a f(t)\,dt$ (d) $\int_a^{2a} f(t)\,dt$ (UPTU, 2009)

42. If $L\{f(t)\}=\phi(s)$ then $L\{e^{at}f(t)\}=$ (where a is a real constant)

(a) $\phi(s)$ (b) $e^{as}\phi(s)$ (c) $\phi(s-a)$ (d) $e^{-as}\phi(s)$ (UPTU, 2010)

43. Laplace transform of $e^{3t}\sin 2t$ is

(a) $\frac{2}{s^2+2s}$ (b) $\frac{2}{s^2+2s}$ (c) $\frac{2}{(s-3)^2+4}$ (d) $\frac{2s}{s^2+2s}$ (UPTU, 2009)

44. If $L\{f(t)\}=$

(a) $\frac{1}{(s+1)^2}$ (b) $\frac{1}{(s+1)^2}$ (c) $\frac{1}{(s-1)^2}$ (d) $\frac{1}{(s-2)^2}$ (UPTU, 2011)

45. $\frac{d^n}{ds^n}L\{f(t)\}=L\{t^n f(t)\}$ is (True or False)

46. Laplace transform of $f(t)$ is defined for only real values of t. (True or False)

47. If $L\{f(t)\}=\phi(s)$, then $L\{\int_0^t f(t)\,dt\}=\frac{\phi(s)}{s}$. (True or False)

CHAPTER 22

Fourier Transforms

1. Introduction. 2. Definition. 3. Fourier integrals — Fourier sine and cosine integral — Complex form of Fourier integral. 4. Fourier transforms — Fourier sine and cosine transforms — Fourier transforms and inverse transforms. 5. Properties of Fourier transforms. 6. Convolution theorem for Fourier transforms. 7. Parseval's identity for Fourier transforms. 8. Relation between Fourier and Laplace transforms. 9. Fourier transforms of the derivatives of a function. 10. Inverse Laplace transforms by method of residues. 11. Application of transforms to boundary value problems. 12. Objective Type of Questions.

22.1 INTRODUCTION

In the previous chapter, the reader has already been acquainted with the use of Laplace transforms in the solution of ordinary differential equations. In this chapter, the well known Fourier transforms will be introduced and their properties will be studied which will be used in the solutions of partial differential equations. The choice of a particular transform to be employed for the solution of an equation depends on the boundary conditions of the problem and the ease with which the transform can be inverted. A Fourier transform is often applied to a partial differential equation to reduce the number of its independent variables by one.

The theory of integral transforms afford mathematical devices through which solutions of numerous boundary value problems of engineering may be obtained e.g., conduction of heat, transverse vibrations of a string, transverse and longitudinal vibrations of a beam, free and forced vibrations of a membrane, transmission lines etc. Some applications will be illustrated in the last section.

22.2 DEFINITION

The integral transform of a function $f(x)$ denoted by $I\{f(x)\}$ is defined by

$$I\{f(x)\} = \int_a^b f(x)K(s, x)\,dx$$

where $K(s, x)$ is called the kernel of the transform and is a known function of s and x. The function $f(x)$ is called the inverse transform of $F(s)$.

The three simple examples of integral transforms are as follows :

(i) If we take $K(s, x) = e^{-sx}$, $I\{f(x)\}$ leads to the **Laplace transform** of $f(x)$, i.e.,

$$F(s) = \int_0^\infty f(x)e^{-sx}\,dx. \qquad \text{[Chap. 21]}$$

(ii) If we take $K(s, x) = e^{isx}$, we have the **Fourier transform** of $f(x)$, i.e.,

$$F(s) = \int_{-\infty}^{\infty} f(x)e^{isx}\,dx.$$

(c) When $K(x, s) = x^{s-1}$, (1) gives the Mellin transform of $f(x)$ i.e.,

$$M\{f(x)\} = \int_0^\infty f(x)x^{s-1}\,dx.$$

Other special transforms arise when the kernel is a sine or a cosine function or a Bessel's function. These lead to Fourier sine or cosine transforms and the Hankel transform respectively.

In order to introduce the Fourier transforms, we shall first derive the Fourier integral theorem.

7.2 (A) FOURIER INTEGRAL THEOREM

Consider a function $f(x)$ which satisfies the Dirichlet's conditions (Art. 10.2) in every interval $(-c, c)$ so that, we have

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{c} + b_n \sin\frac{n\pi x}{c}\right) \quad ...(1)$$

where $$a_0 = \frac{1}{c}\int_{-c}^{c} f(x)\,dx, \; a_n = \frac{1}{c}\int_{-c}^{c} f(x)\cos\frac{n\pi x}{c}\,dx, \text{ and } b_n = \frac{1}{c}\int_{-c}^{c} f(x)\sin\frac{n\pi x}{c}\,dx.$$

Substituting the values of a_0, a_n and b_n in (1), it takes the form

$$f(x) = \frac{1}{2c}\int_{-c}^{c} f(x)\,dx + \frac{1}{c}\sum_{n=1}^{\infty}\int_{-c}^{c} f(v)\cos\frac{n\pi(v-x)}{c}\,dv \quad ...(2)$$

Here we assume that $\int_{-\infty}^{\infty} |f(x)|\,dx$ is convergent, the first term on the right hand side of (2) approaches 0 as $c \to \infty$, since

$$\left|\frac{1}{2c}\int_{-c}^{c} f(x)\,dx\right| \le \frac{1}{2c}\int_{-c}^{c} |f(x)|\,dx$$

The second term on the right side of (2) tends to

$$\lim_{c\to\infty}\frac{1}{c}\sum_{n=1}^{\infty}\int_{-c}^{c} f(v)\cos\frac{n\pi(v-x)}{c}\,dv$$

$$= \lim_{\delta\alpha\to 0}\frac{1}{\pi}\sum_{n=1}^{\infty}\delta\alpha\int_{-\infty}^{\infty} f(v)\cos n\delta\alpha(v-x)\,dv, \text{ on writing } \frac{n\pi}{c} = n\delta\alpha$$

This is of the form $\lim_{\delta\alpha\to 0}\sum_{n=1}^{\infty} F(n\delta\alpha)\,\delta\alpha = \int_0^\infty F(\alpha)\,d\alpha$

Therefore $c \to \infty$, (2) becomes $f(x) = \frac{1}{\pi}\int_0^\infty\int_{-\infty}^{\infty} f(v)\cos\alpha(v-x)\,dv\,d\alpha$ ...(3)

which is known as the Fourier Integral of $f(x)$.

Note. We have given a heuristic derivation of the Fourier integral theorem which simply helps in deriving the result (3). It cannot, however, be taken as a rigorous proof for that would involve a proof of the convergence of the Fourier integral which is beyond the scope of this book. When $f(x)$ satisfies the above mentioned conditions, equation (3) holds good at a point of continuity. If however, x is point of discontinuity, we replace $f(x)$ by $\frac{1}{2}[f(x+0) + f(x-0)]$ as in the case of Fourier series.

(i) Fourier sine and cosine integrals. Expanding $\cos\alpha(v-x)$, (3) may be written as

$$f(x) = \frac{1}{\pi}\int_0^\infty \cos\alpha x\int_{-\infty}^{\infty} f(v)\cos\alpha v\,dv\,d\alpha + \frac{1}{\pi}\int_0^\infty \sin\alpha x\int_{-\infty}^{\infty} f(v)\sin\alpha v\,dv\,d\alpha \quad ...(4)$$

If $f(x)$ is an odd function, $f(v)\cos\alpha v$ is also an odd function while $f(v)\sin\alpha v$ is an even. Then the first term on the right side of (4) vanishes and, we get

$$f(x) = \frac{2}{\pi}\int_0^\infty \sin\alpha x\int_0^\infty f(v)\sin\alpha v\,dv\,d\alpha \quad ...(5)$$

which is known as the *Fourier sine integral*.

Similarly, if $f(x)$ is an even function, (4) takes the form

$$f(x) = \frac{2}{\pi}\int_0^\infty \cos\lambda x \int_0^\infty f(v)\cos\lambda v\, dv\, d\lambda \qquad (35)$$

which is known as the *Fourier cosine integral*.

Obs. A function $f(x)$ defined in the interval $(0, \infty)$ can be represented either as a Fourier cosine integral or as a Fourier sine integral, merely by extending the definition of the function to $(-\infty, 0)$ so that the function is even or odd, respectively.

(3) **Complex form of Fourier integrals.** Equation (3) can be written as

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(v)\cos\lambda(v-x)\, dv\, d\lambda \qquad (37)$$

because $\cos\lambda(v-x)$ is an even function of λ. Also since $\sin\lambda(v-x)$ is an odd function of λ, we have

$$0 = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(v)\sin\lambda(v-x)\, dv\, d\lambda \qquad (38)$$

Now multiply (38) by i and subtract it from (37), so that

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(v)e^{i\lambda(v-x)}\, dv\, d\lambda \qquad (39)$$

which is the complex form of the *Fourier integral*.

(4) **Fourier integral representations of a function.**

Using (4), a function $f(x)$ may be represented by a *Fourier integral* as

$$f(x) = \frac{1}{\pi}\int_0^\infty [A(\lambda)\cos\lambda x + B(\lambda)\sin\lambda x]\, d\lambda$$

where $$A(\lambda) = \int_{-\infty}^{\infty} f(t)\cos\lambda t\, dt,\quad B(\lambda) = \int_{-\infty}^{\infty} f(t)\sin\lambda t\, dt \qquad (40)$$

If $f(x)$ is an odd function, then

$$f(x) = \frac{1}{\pi}\int_0^\infty B(\lambda)\sin\lambda x\, d\lambda \text{ where } B(\lambda) = 2\int_0^\infty f(t)\sin\lambda t\, dt \qquad (41)$$

If $f(x)$ is an even function, then

$$f(x) = \frac{1}{\pi}\int_0^\infty A(\lambda)\cos\lambda x\, d\lambda \text{ where } A(\lambda) = 2\int_0^\infty f(t)\cos\lambda t\, dt \qquad (42)$$

Example 32.5. *Express $f(x) = 1$ for $0 \le x \le \pi$*
$= 0$ for $x > \pi$
as a Fourier sine integral and hence evaluate

$$\int_0^\infty \frac{1-\cos\pi\lambda}{\lambda}\sin\frac{\lambda x}{2}\, d\lambda \qquad \text{(Kottayam, 2002; Calicut, 2004)}$$

Solution. The Fourier sine integral is $f(x) = \frac{2}{\pi}\int_0^\infty \sin\lambda x\, d\lambda \int_0^\infty f(v)\sin\lambda v\, dv$

$$= \frac{2}{\pi}\int_0^\infty \sin\lambda x\, d\lambda \int_0^\pi \sin\lambda v\, dv$$

$$= \frac{2}{\pi}\int_0^\infty \sin\lambda x\, d\lambda \left[\frac{-\cos\lambda v}{\lambda}\right]_0^\pi = \frac{2}{\pi}\int_0^\infty \frac{1-\cos\pi\lambda}{\lambda}\sin\lambda x\, d\lambda$$

i.e., $$\int_0^\infty \frac{1-\cos\pi\lambda}{\lambda}\sin\lambda x\, d\lambda = \frac{\pi}{2}f(x) = \begin{cases} \pi/2 & \text{for } 0 < x < \pi \\ 0 & \text{for } x > \pi \end{cases}$$

At $x = \pi$, which is a point of discontinuity of $f(x)$, the value of the above integral

$$= \frac{\pi}{2}\left[\frac{f(\pi-0)+f(\pi+0)}{2}\right] = \frac{\pi}{2}\cdot\frac{1+0}{2} = \frac{\pi}{4}.$$

22.3 (I) FOURIER TRANSFORMS

Rewriting (8) of § 22.2 as

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ix\alpha}\,d\alpha \int_{-\infty}^{\infty} f(\xi)e^{i\xi\alpha}\,d\xi,$$

it follows that if

$$F(\alpha) = \int_{-\infty}^{\infty} f(\xi)e^{i\xi\alpha}\,d\xi \qquad \ldots(1)$$

then

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\alpha)e^{-ix\alpha}\,d\alpha \qquad \ldots(2)$$

The function $F(\alpha)$, defined by (1), is called the Fourier transform of $f(x)$. Also the function $f(x)$, as given by (2), is called the Inverse Fourier transform of $F(\alpha)$. Sometimes we call (1) as an *integral* corresponding to (2).

(i) Fourier sine and cosine transforms. From (1) of § 22.2, it follows that if

$$F_s(\alpha) = \int_0^{\infty} f(x)\sin\alpha x\,dx \qquad \ldots(3)$$

then

$$f(x) = \frac{2}{\pi}\int_0^{\infty} F_s(\alpha)\sin\alpha x\,d\alpha \qquad \ldots(4)$$

The function $F_s(\alpha)$, as defined by (3), is known as the Fourier sine transform of $f(x)$ in $0 < x < \infty$. Also the function $f(x)$, as given by (4), is called the Inverse Fourier sine transform of $F_s(\alpha)$.

Similarly, it follows from (2) of § 22.2 that if

$$F_c(\alpha) = \int_0^{\infty} f(x)\cos\alpha x\,dx \qquad \ldots(5)$$

then

$$f(x) = \frac{2}{\pi}\int_0^{\infty} F_c(\alpha)\cos\alpha x\,d\alpha \qquad \ldots(6)$$

The function $F_c(\alpha)$, as defined by (5), is known as the Fourier cosine transform of $f(x)$ in $0 < x < \infty$. Also the function $f(x)$, as given by (6), is called the Inverse Fourier cosine transform of $F_c(\alpha)$.

(ii) Finite Fourier sine and cosine transforms. These two forms are useful for such a boundary-value problem in which at least two of the boundaries are parallel and separated by a finite distance.

The finite Fourier sine transform of $f(x)$, in $0 < x < c$, is defined as

$$F_s(n) = \int_0^{c} f(x)\sin\frac{n\pi x}{c}\,dx \qquad \ldots(7)$$

where n is an integer.

The function $f(x)$ is then called the inverse finite Fourier sine transform of $F_s(n)$ which is given by

$$f(x) = \frac{2}{c}\sum_{n=1}^{\infty} F_s(n)\sin\frac{n\pi x}{c} \qquad \ldots(8)$$

The finite Fourier cosine transform of $f(x)$, in $0 < x < c$, is defined as

$$F_c(n) = \int_0^{c} f(x)\cos\frac{n\pi x}{c}\,dx \qquad \ldots(9)$$

where n is an integer.

The function $f(x)$ is then called the inverse finite Fourier cosine transform of $F_c(n)$ which is given by

$$f(x) = \frac{1}{c}F_c(0) + \frac{2}{c}\sum_{n=1}^{\infty} F_c(n)\cos\frac{n\pi x}{c} \qquad \ldots(10)$$

Obs. *The finite Fourier sine transform is useful for problems involving boundary conditions at two parallel boundaries, while the finite cosine transform is useful for problems in which the velocity normal to two parallel boundaries are among the boundary conditions.*

22.4 PROPERTIES OF FOURIER TRANSFORMS

(1) Linearity property. *If $F(\alpha)$ and $G(\alpha)$ are Fourier transforms of $f(x)$ and $g(x)$ respectively, then*

$$\mathcal{F}[af(x) + bg(x)] = a\,F(\alpha) + b\,G(\alpha)$$

where a and b are constants.

We have $F(s) = \int_{-\infty}^{\infty} e^{sx} f(x)\,dx$ and $G(s) = \int_{-\infty}^{\infty} e^{sx} g(x)\,dx$

$\therefore F\{af(x) + bg(x)\} = \int_{-\infty}^{\infty} e^{sx}[af(x) + bg(x)]\,dx = a\int_{-\infty}^{\infty} e^{sx} f(x)\,dx + b\int_{-\infty}^{\infty} e^{sx} g(x)\,dx$

$= aF(s) + bG(s)$

(2) Change of scale property. *If $F(s)$ is the complex Fourier transform of $f(x)$, then*

$$F\{f(ax)\} = \frac{1}{a}F\left(\frac{s}{a}\right), a \neq 0$$

We have $F(s) = \int_{-\infty}^{\infty} e^{sx} f(x)\,dx$ ...(1)

$\therefore F\{f(ax)\} = \int_{-\infty}^{\infty} e^{sx} f(ax)\,dx$ $\left|\begin{array}{l}\text{Put } ax = t\\ \text{so that } dx = \frac{dt}{a}\end{array}\right.$

$= \int_{-\infty}^{\infty} e^{st/a} f(t)\frac{dt}{a} = \frac{1}{a}\int_{-\infty}^{\infty} e^{(s/a)t} f(t)\,dt = \frac{1}{a}F\left(\frac{s}{a}\right)$ Hence (2)

Cor. If $F(s)$ and $F(x)$ are the Fourier sine and cosine transforms of $f(x)$ respectively, then

$$F_s\{f(ax)\} = \frac{1}{a}F_s\left(\frac{s}{a}\right) \quad \text{and} \quad F_c\{f(ax)\} = \frac{1}{a}F_c\left(\frac{s}{a}\right)$$

(3) Shifting property. *If $F(s)$ is the complex Fourier transform of $f(x)$, then*

$$F\{f(x-a)\} = e^{isa}F(s)$$

We have $F(s) = \int_{-\infty}^{\infty} e^{isx} f(x)\,dx$ ...(1)

$\therefore F\{f(x-a)\} = \int_{-\infty}^{\infty} e^{isx} f(x-a)\,dx$ $\left|\begin{array}{l}\text{Put } x-a = t\\ \text{so that } dx = dt\end{array}\right.$

$= \int_{-\infty}^{\infty} e^{is(t+a)} f(t)\,dt = e^{isa}\int_{-\infty}^{\infty} e^{ist} f(t)\,dt = e^{isa}F(s)$ Hence (3)

(4) Modulation theorem. *If $F(s)$ is the complex Fourier transform of $f(x)$, then*

$$F\{f(x)\cos ax\} = \frac{1}{2}[F(s+a) + F(s-a)]$$

We have $F(s) = \int_{-\infty}^{\infty} e^{isx} f(x)\,dx$ ...(1)

$\therefore F\{f(x)\cos ax\} = \int_{-\infty}^{\infty} e^{isx} f(x)\cos ax\,dx = \int_{-\infty}^{\infty} e^{isx} f(x)\cdot\frac{e^{iax} + e^{-iax}}{2}\,dx$

$= \frac{1}{2}\left[\int_{-\infty}^{\infty} e^{i(s+a)x} f(x)\,dx + \int_{-\infty}^{\infty} e^{i(s-a)x} f(x)\,dx\right] = \frac{1}{2}[F(s+a) + F(s-a)]$

Cor. If $F_s(s)$ and $F_c(s)$ are Fourier sine and cosine transforms of $f(x)$ respectively, then

(i) $F_s\{f(x)\cos ax\} = \frac{1}{2}[F_s(s+a) + F_s(s-a)]$ Hence, (4)(i)

(ii) $F_s\{f(x)\sin ax\} = \frac{1}{2}[F_c(s+a) - F_c(s-a)]$

(iii) $F_c\{f(x)\sin ax\} = \frac{1}{2}[F_s(s+a) - F_s(s-a)]$

Note. This theorem is of great importance in problems of television where the transmitter receives waves modulated by the signals.

Example 17.23. *Find the Fourier transform of*

$$f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

Hence evaluate $\int_0^{\infty} \frac{\sin x}{x}\,dx$. (U.T.U., 2010; S.V.T.U., 2009; U.P.T.U., 2008)

Solution. The Fourier transform of $f(x)$ is ...

$$F[f(x)] = \int_{-\infty}^{\infty} f(x)e^{isx}\,dx = \int_{-1}^{1} (1)e^{isx}\,dx = \left[\frac{e^{isx}}{is}\right]_{-1}^{1} = \frac{e^{is} - e^{-is}}{is}$$

Thus $F[f(x)] = F(s) = 2\frac{\sin s}{s}$, as $\sin(-\theta) = -\sin\theta$. For $s = 0$, we have $F(s) = 2$.

Now by the inversion formula, we get

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(s)e^{-isx}\,ds = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{2\sin s}{s} e^{-isx}\,ds = \begin{cases} 1 & \text{for } |x| < 1 \\ 0 & \text{for } |x| > 1 \end{cases}$$

Putting $x = 0$, we get

$$\int_{-\infty}^{\infty} \frac{\sin s}{s}\,ds = \pi \quad \therefore \int_{0}^{\infty} \frac{\sin s}{s}\,ds = \frac{\pi}{2}$$ Since the integrand is even.

Example 22.5. *Find the Fourier transform of*

$$f(x) = \begin{cases} 1 - x^2 & |x| \le 1 \\ 0 & |x| > 1 \end{cases}$$

Hence evaluate $\int_0^{\infty} \frac{x\cos x - \sin x}{x^3}\cos\frac{x}{2}\,dx$. (U.P.T.U., 2001 S; Anna, 2000 S; Bhopal, 2000 S)

Solution. $F[f(x)] = \int_{-\infty}^{\infty} f(x)e^{isx}\,dx = F(s)$, say

$$= \int_{-\infty}^{-1} (0)e^{isx}\,dx + \int_{-1}^{1} (1 - x^2)e^{isx}\,dx + \int_{1}^{\infty} (0)e^{isx}\,dx = \left[(1 - x^2)\frac{e^{isx}}{is} - (2x)\frac{e^{isx}}{(is)^2} + (-2)\frac{e^{isx}}{(is)^3}\right]_{-1}^{1}$$

$$= 2\left[\frac{e^{is} + e^{-is}}{-s^2}\right] - 4\left(\frac{e^{is} - e^{-is}}{-is^3}\right) = -\frac{4}{s^2}\cos s - 4i\frac{\sin s}{s^3}$$

Now by inversion formula, we have

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(s)e^{-isx}\,ds$$

or $$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \left(-\frac{4}{s^2}\cos s - 4i\frac{\sin s}{s^3}\right)e^{-isx}\,ds = \begin{cases} 1 - x^2, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$$

Putting $x = 1/2$, we obtain

$$-\frac{1}{2\pi}\int_{-\infty}^{\infty} \left(-\frac{4}{s^2}\cos s - 4i\frac{\sin s}{s^3}\right)e^{-is/2}\,ds = \frac{3}{4}$$

or $$\int_{-\infty}^{\infty} \frac{s\cos s - \sin s}{s^3}\left(\cos\frac{s}{2} - i\sin\frac{s}{2}\right)ds = -\frac{3\pi}{8}$$

or $$\int_{-\infty}^{\infty} \frac{s\cos s - \sin s}{s^3}\cdot\cos\frac{s}{2}\,ds = -\frac{3\pi}{8}$$

or $$\int_{0}^{\infty} \frac{s\cos s - \sin s}{s^3}\cos\frac{s}{2}\,ds = -\frac{3\pi}{16}$$, since the integrand is even.

Example 22.6. *(a) Find the Fourier transform of* $e^{-x^2/2}$, $x \in \mathbb{R}$. *Hence deduce that* $e^{-x^2/2}$ *is self-reciprocal with respect to Fourier transform.* (Madras, 2005; Anna Univ. 2006)

(b) Find Fourier transform of $f(x) = e^{-|x|}$, $x \in \mathbb{R}$ *and hence show that* $\int_0^{\infty} \frac{\cos sx}{1 + x^2}\,dx = \frac{\pi}{2}e^{-s}$.

Solution. Let $F[e^{-x^2/2}] = \int_{-\infty}^{\infty} e^{-x^2/2} \cdot e^{isx}\,dx = \int_{-\infty}^{\infty} e^{-x^2/2 + isx - s^2/2 + s^2/2}\,dx$

$$= \int_{-\infty}^{\infty} e^{-(x - is)^2/2} \cdot e^{s^2/2}\,dx$$

$$= \int_0^\infty e^{-t} \cdot e^{-(\omega^2/4t)} \, dt/\sqrt{t} \qquad \text{[Putting } \omega x = 2\sqrt{t} \Rightarrow x \, dx = dt/\sqrt{t}\text{]}$$

$$= \frac{e^{-\omega^2/4}}{\sqrt{\pi}} \int_0^\infty e^{-t} \, dt = \frac{e^{-\omega^2/4}}{\sqrt{\pi}} \sqrt{\pi} \qquad \left[\because \int_0^\infty e^{-t} \, dt = \sqrt{\pi}\right]$$

Hence $\quad F(e^{-x^2}) = \frac{\sqrt{\pi}}{\pi} e^{-\omega^2/4}$

Taking $x^2 = 2t$, we have

$$F(e^{-x^2/2}) = \frac{\sqrt{\pi}}{\Gamma(1/2)} e^{-\omega^2/2} = \sqrt{2\pi} e^{-\omega^2/2}$$

i.e., Fourier transform of $e^{-x^2/2}$ is a constant times $e^{-\omega^2/2}$. Also the functions $e^{-x^2/2}$ and $e^{-\omega^2/2}$ are the same. Hence it follows that $e^{-x^2/2}$ is self-reciprocal under the Fourier transform.

(ii) Since $e^{-ax^2} = e^{-(\sqrt{a}x)^2/2} = f(\sqrt{a}x)$ where $f(x) = e^{-x^2/2}$

$\therefore$ by change of scale property, $F(f(ax)) = \frac{1}{a} F(\omega/a)$

i.e., $\quad F(e^{-ax^2}) = F(e^{-(\sqrt{a}x)^2/2}) = \sqrt{2\pi} e^{-(\omega/\sqrt{a})^2/2} = \sqrt{\frac{2\pi}{a}} e^{-\omega^2/4a}$

By shifting property $F(f(x-a)) = e^{i\omega a} F(\omega)$

$$\therefore \quad F(e^{-a(x-b)^2}) = e^{i\omega b} \sqrt{\frac{2\pi}{a}} e^{-\omega^2/4a} = \sqrt{\frac{2\pi}{a}} e^{i\omega b - \omega^2/4a} \qquad \ldots(i)$$

Also by modulation theorem,

$$F[f(x)\cos ax] = \frac{1}{2}[F(x+a) + F(x-a)]$$

$$F(e^{-x^2/2}\cos ax) = \frac{1}{2}\sqrt{2\pi}[e^{-(x+a)^2/2} + e^{-(x-a)^2/2}], \qquad \ldots(ii)$$

Example 2.32. Find the Fourier cosine transform of e^{-x^2}. (U.P.T.U. 2015; Rajasthan, 2009)

Solution. We have $F_c(e^{-x^2}) = \int_0^\infty e^{-x^2} \cos \omega x \, dx = I$ (say)

Differentiating under the integral sign w.r.t. ω,

$$\frac{dI}{d\omega} = -\int_0^\infty x e^{-x^2} \sin \omega x \, dx = \frac{1}{2}\int_0^\infty (\sin \omega x)(-2xe^{-x^2}) \, dx$$

$$= \frac{1}{2}\left[\sin \omega x \cdot e^{-x^2}\Big|_0^\infty - \omega \int_0^\infty \cos \omega x \cdot e^{-x^2} \, dx\right]$$

$$= -\frac{\omega}{2}\int_0^\infty e^{-x^2} \cos \omega x \, dx = -\frac{\omega}{2} I \quad \text{or} \quad \frac{dI}{I} = -\int \frac{\omega}{2} d\omega + \log c$$

or $\quad \log I = -\frac{\omega^2}{4} + \log c \Rightarrow \log I = \log(ce^{-\omega^2/4})$

$$\therefore \quad I = ce^{-\omega^2/4} \quad \text{or} \quad \int_0^\infty e^{-x^2} \cos \omega x \, dx = ce^{-\omega^2/4}$$

Putting $\omega = 0$, $\quad c = \int_0^\infty e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}$, i.e., $I = \frac{\sqrt{\pi}}{2} e^{-\omega^2/4}$.

Hence $\quad F_c(e^{-x^2}) = \frac{\sqrt{\pi}}{2} e^{-\omega^2/4}$.

Example 2.33. Find the Fourier cosine transform of e^{-x}.

Hence show that $\int_0^\infty \frac{\cos mx}{1+x^2} dx = \frac{\pi}{2} e^{-m}$, $m > 0$. (U.P.T.U. 2010; U.P.S.C. 2009; Rajasthan, 2008)

Solution. x being positive in the interval $(0, \infty)$, $e^{-|x|} = e^{-x}$.

∴ Fourier sine transform of $f(x) = e^{-x}$ is given by

$$F_s[f(x)] = \int_0^\infty f(x) \sin sx\, dx = \int_0^\infty e^{-x} \sin sx\, dx$$

$$= \left| \frac{e^{-x}}{1+s^2} (-\sin sx - s\cos sx) \right|_0^\infty = \frac{s}{1+s^2}$$

Using Inversion formula for Fourier sine transforms, we get

$$f(x) = \frac{2}{\pi} \int_0^\infty F_s[f(x)] \sin sx\, ds \quad \text{or} \quad e^{-x} = \frac{2}{\pi} \int_0^\infty \frac{s}{1+s^2} \sin sx\, ds$$

or changing x to s, $\quad e^{-s} = \frac{2}{\pi} \int_0^\infty \frac{s \sin sx}{1+s^2} ds \quad \text{or} \quad \frac{2}{\pi} \int_0^\infty \frac{x \sin sx}{1+x^2} dx$

Hence $\int_0^\infty \frac{x \sin sx}{1+x^2} dx = \frac{\pi e^{-s}}{2}$.

Example 3.27. *Find the Fourier sine transform of* $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$ *(A.M.I.E., 2000)*

Solution. Fourier sine transform of $f(x)$ is $F_s[f(x)]$

$$= \int_0^\infty f(x) \sin sx\, dx = \int_0^1 x \sin sx\, dx + \int_1^2 (2-x) \sin sx\, dx + \int_2^\infty 0 \cdot \sin sx\, dx$$

$$= \left| x \frac{\sin sx}{s} - (1) \frac{(-\cos sx)}{s^2} \right|_0^1 + \left| (2-x) \frac{\sin sx}{s} - (-1) \frac{-\cos sx}{s^2} \right|_1^2$$

$$= \left(\frac{\sin s}{s} + \frac{\cos s}{s^2} - \frac{1}{s^2} \right) + \left(-\frac{\cos 2s}{s^2} - \frac{\sin s}{s} + \frac{\cos s}{s^2} \right)$$

$$= \frac{2\cos s}{s^2} - \frac{\cos 2s}{s^2} - \frac{1}{s^2}.$$

Example 3.28. *Find the Fourier sine transform of* $e^{-|x|}$. *(U.P.T.U., 2010 S; U.P.T.U., 2006; Rohtak, 2003)*

Solution. Let $\quad f(x) = e^{-|x|}x$, then its Fourier sine transform

is $\quad F_s[f(x)] = \int_0^\infty f(x) \sin sx\, dx = \int_0^\infty \frac{e^{-|x|}}{x} \sin sx\, dx = \phi(s)$, say

Differentiating both sides w.r.t. s, we get

$$\frac{d}{ds} \phi(s) = \int_0^\infty \frac{e^{-|x|} \cos sx}{x} dx = \int_0^\infty e^{-x} \cos sx\, dx = \frac{1}{s^2+1}$$

Integrating w.r.t. s, we obtain $\phi(s) = \int_0^s \frac{1}{s^2+1} ds = \tan^{-1} \frac{s}{1} + c$

But $\phi(s) = 0$, when $s = 0$ $\quad \therefore \quad c = 0$. Hence $\phi(s) = \tan^{-1} s$, Ans.

Example 3.29. *Find the Fourier sine transform of* $f(x) = x/(1+x^2)$. *(U.P.T.U., 2013 S; Anna, 2005)*
Hence find the Fourier sine transform of $x^n/(1+x^2)$. *(U.P.T.U., 2009 S)*

Solution. $\quad F_s[f(x)] = \int_0^\infty \frac{x \sin sx}{1+x^2} dx = \phi(s)$, say $\quad \text{...(1)}$

$\therefore \quad \frac{d\phi}{ds} = \int_0^\infty \frac{x \cdot x \sin sx}{1+x^2} dx = -\int_0^\infty \frac{x^2 \sin sx}{x(1+x^2)} dx \quad \text{...(2)}$

$$= -\int_0^\infty \frac{\ln(1+x^2)\cdot 1 - \ln x \cdot \frac{2x}{1+x^2}}{\{\ln(1+x^2)\}^2}\,dx = -\int_0^\infty \frac{\ln(1+x^2)}{x}\,dx + \int_0^\infty \frac{\ln x \cdot x}{x\{\ln(1+x^2)\}^2}\,dx$$

or
$$\frac{dI}{dx} = -\frac{\pi}{2} + \int_0^\infty \frac{\ln x \cdot x}{x\{\ln(1+x^2)\}^2}\,dx \qquad \ldots(3)(i)$$

$$\therefore \quad \frac{d^2I}{dx^2} = \int_0^\infty \frac{x\cdot\frac{1}{x}\cdot x}{x\{\ln(1+x^2)\}^2}\,dx = I$$

or
$$\frac{d^2I}{dx^2} - I = 0 \quad \text{or} \quad (D^2 - 1)I = 0, \text{ where } D \equiv \frac{d}{dx}$$

Its solution is
$$I = c_1 e^x + c_2 e^{-x} \qquad \ldots(3)(ii)$$

$$\therefore \quad dI/dx = c_1 e^x - c_2 e^{-x} \qquad \ldots(3)(iii)$$

When $x = 0$, (ii) and (iii) give $c_1 + c_2 = \int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2}$

Also when $x = 0$, (ii) and (i) give $c_1 - c_2 = -\pi/2$.

Solving these, $c_1 = 0$, $c_2 = \pi/2$.

Thus from (ii) and (i), we have $F_c[\ln(x)] = I = (\pi/2)e^{-x}$

Now
$$F_c[\ln(x)] = \int_0^\infty \frac{\ln x \cdot \cos sx}{x}\,dx = -\frac{dI}{ds}, \text{ from (i)}$$

$$= (\pi/2)e^{-s}, \text{ from (v), with } c_1 = 0, c_2 = \pi/2.$$

Example (3.15): Find the Fourier sine and cosine transforms of x^{n-1}, $0 < n < 1$. (Madras, 2005)

Solution. We know that:
$$F_s[x^{n-1}] = \int_0^\infty x^{n-1}\sin sx\,dx \qquad \ldots(i)$$

and
$$F_c[x^{n-1}] = \int_0^\infty x^{n-1}\cos sx\,dx \qquad \ldots(ii)$$

$$\therefore \quad F_s[x^{n-1}] + iF_c[x^{n-1}] = \int_0^\infty (\cos sx + i\sin sx)x^{n-1}\,dx$$

$$= \int_0^\infty e^{isx}\,x^{n-1}\,dx = \int_0^\infty e^{is}\left(-\frac{is}{u}\right)^{n-1}\left(-\frac{is}{u}\right) \qquad [\text{Where } isx = -u]$$

$$= \left(-\frac{1}{is}\right)^n \int_0^\infty e^{-u}u^{n-1}\,du = \frac{(is)^n}{(is)^n}\Gamma(n) = \frac{(i)^n}{s^n}\Gamma(n)$$

$$= \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)^n \Gamma(n)/s^n = \left(\cos\frac{n\pi}{2} + i\sin\frac{n\pi}{2}\right)\Gamma(n)/s^n$$

Equating real and imaginary parts, we get
$$F_s[x^{n-1}] = \frac{\Gamma(n)}{s^n}\cos\frac{n\pi}{2} \quad \text{and} \quad F_c[x^{n-1}] = \frac{\Gamma(n)}{s^n}\sin\frac{n\pi}{2}.$$

Example (3.16): (a) Show that $F_s[xf(x)] = -\frac{d}{ds}F_c[f(x)]$ and $F_c[xf(x)] = \frac{d}{ds}F_s[f(x)]$.

(b) Find the Fourier sine and cosine transforms of xe^{-x}. (Madras, 2005)

Solution. (a)
$$\frac{d}{ds}F_s[f(x)] = \frac{d}{ds}\left[\int_0^\infty f(x)\sin sx\,dx\right] = \int_0^\infty f(x)\frac{\partial}{\partial s}\sin sx\,dx$$

$$= \int_0^\infty [xf(x)]\cos sx\,dx = F_c[xf(x)] \qquad \ldots(i)$$

$$\frac{d}{ds}[F_c(s)] = \frac{d}{ds}\left[\int_0^\infty f(x)\cos sx\,dx\right] = \int_0^\infty f(x)\frac{\partial}{\partial s}\cos sx\,dx$$

$$= -\int_0^\infty [xf(x)]\sin sx\,dx = -F_s[xf(x)] \qquad \ldots(ii)$$

(b) We have $$F(e^{-ax}\cos bx) = \int_0^\infty e^{-ax}\cos bx\, e^{isx}\, dx = \frac{e^{-ax}}{a^2+s^2}\left[-a\cos bx + s\sin bx\right]_0^\infty$$

$$= \frac{a}{a^2+s^2} \quad \text{...(a)}$$

and $$F(e^{-ax}\sin bx) = \int_0^\infty e^{-ax}\sin bx\, e^{isx}\, dx = \frac{e^{-ax}}{a^2+s^2}\left[-a\sin bx + s\cos bx\right]_0^\infty$$

$$= \frac{s}{a^2+s^2} \quad \text{...(b)}$$

Now $$F(xe^{-ax}) = -\frac{d}{ds}\{F(e^{-ax})\} \quad \text{...(c)}$$

$$= -\frac{d}{ds}\left(\frac{a}{a^2+s^2}\right) = \frac{2as}{(a^2+s^2)^2} \quad \text{[By (a)]}$$

$$F(xe^{-ax}\sin bx) = -\frac{d}{ds}F(e^{-ax}\sin bx) \quad \text{[By (c)]}$$

$$= -\frac{d}{ds}\left(\frac{s}{a^2+s^2}\right) = \frac{s^2+a^2-s(2s)}{(a^2+s^2)^2} = \frac{a^2-s^2}{(a^2+s^2)^2}. \quad \text{[By (b)]}$$

Example 9.9.12. *Find the Fourier sine transform of* $f(x) = \begin{cases} 1-x^2, & 0 \le x \le 1 \\ 0, & x > 1 \end{cases}$. *Hence find* $\int_0^\infty \left(\frac{x\cos x - \sin x}{x^3}\right)\cos\frac{x}{2}\,dx$. (UPTU, 2002)

Solution. We know $f(x)$ is an even function. Hence its sine transform is

$$= \frac{2}{\pi}\sum_{n=1}^{\infty} F(n)\sin\frac{n\pi x}{\pi} = \frac{2}{\pi}\sum_{n=1}^{\infty}\left[\frac{1-\cos n\pi}{n^3}\right]\sin nx$$

$$= \frac{2}{\pi}\sum_{n=1}^{\infty}\left[\frac{1-\cos n\pi}{n^3}\right]\sin nx$$

Example 9.9.13. *Solve the integral equation*

$$\int_0^\infty f(x)\cos sx\, dx = \begin{cases} 1-s, & 0 \le s \le 1 \\ 0, & s > 1 \end{cases}$$

Hence evaluate $\int_0^\infty \frac{\sin^2 t}{t^2}\,dt$. (U.P.T.U. 2012 S; Kanpur Univ. 2005)

Solution. We have $$\int_0^\infty f(x)\cos sx\, dx = F_c(s)$$

$$\therefore \quad F_c(s) = \begin{cases} 1-s, & 0 \le s \le 1 \\ 0, & s > 1 \end{cases} \quad \text{...(1)}$$

By the inversion formula, we have

$$f(x) = \frac{2}{\pi}\int_0^\infty F_c(s)\cos sx\, ds = \frac{2}{\pi}\int_0^1 (1-s)\cos sx\, ds \quad \text{[Integrating by parts]}$$

$$= \frac{2}{\pi}\left[(1-s)\frac{\sin sx}{x}\right]_0^1 - \int_0^1 (-1)\frac{\sin sx}{x}\,ds\Bigg] = \frac{2}{\pi x}\left[-\frac{\cos sx}{x}\right]_0^1 = \frac{2(1-\cos x)}{\pi x^2}$$

Now $$F(x) = \int_0^\infty f(x)\cos sx\, dx = \int_0^1 \frac{2(1-\cos x)}{\pi x^2}\cos sx\, dx \quad \text{...(2)}$$

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* Refer to Chapter 10.

∴ From (1) and (2), we have

$$\frac{2}{\pi}\int_0^\infty \frac{1-\cos\theta}{\theta^2}\cos s\theta\, d\theta = \begin{cases} 1-s, & 0 \le s \le 1 \\ 0, & s > 1 \end{cases}$$

Now letting $s = 0$, we get $\frac{2}{\pi}\int_0^\infty \frac{1-\cos\theta}{\theta^2}\, d\theta = 1$ [U.P.T.U., 2006]

or $\int_0^\infty \frac{2\sin^2\theta/2}{\theta^2}\, d\theta = \pi/2$ [Put $\theta/2 = t$, so that $d\theta = 2dt$]

$$\therefore \int_0^\infty \frac{\sin^2 t}{t^2}\, dt = \frac{\pi}{2}.$$

PROBLEMS 9.7

1. Express the function $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$ as a Fourier integral.

   Hence evaluate $\int_0^\infty \frac{\sin\lambda\cos\lambda x}{\lambda}\, d\lambda$. [Kakatiya, 2005]

2. Find the Fourier integral representation for:

   (i) $f(x) = \begin{cases} 1-x^2, & \text{for } |x| < 1 \\ 0, & \text{for } |x| > 1 \end{cases}$ (Mumbai, 2004) (ii) $f(x) = \begin{cases} e^{mx}, & \text{for } x < 0,\ m > 0 \\ e^{-mx}, & \text{for } x > 0,\ m > 0 \end{cases}$

3. Using the Fourier integral representation show that

   (i) $\int_0^\infty \frac{\cos\lambda x}{1+\lambda^2}\, d\lambda = \frac{\pi}{2}e^{-x}$, $(x > 0)$ (ii) $\int_0^\infty \frac{\cos\lambda x}{1+\lambda^2}\, d\lambda = \frac{\pi}{2}e^{-x}$, $(x \ge 0)$ [U.P.T.U., 2004]

   (iii) $\int_0^\infty \frac{\lambda\sin\lambda x}{a^2+\lambda^2}\, d\lambda = \frac{\pi}{2}\cosh ax$, $(0 < x < a)$ (iv) $\int_0^\infty \frac{\sin\lambda x\cos\lambda a}{1-\lambda^2}\, d\lambda = \begin{cases} \frac{\pi}{2}\sin x, & 0 \le x \le a \\ \frac{\pi}{2}\sin a\cos x, & x > a \end{cases}$

4. Find the Fourier transforms of

   (i) $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$ [U.P.T.U., 2004; Bhopal, 2003; P.T.U., 2005]

   Hence evaluate $\int_{-\infty}^{\infty} \frac{\sin ax}{x}\, dx$ [Bhopal, 2004]

   (ii) $f(x) = \begin{cases} x^n, & |x| < a \\ 0, & |x| > a \end{cases}$ [U.P.T.U., 2005]

5. Find the Fourier transform of $f(x) = \begin{cases} a^2 - x^2, & \text{for } |x| < a \\ 0, & \text{for } |x| > a \end{cases}$ [U.P.T.U., 2007]

   Deduce that $\int_0^\infty \frac{x\cos x - \sin x}{x^3}\cos\frac{x}{2}\, dx = \frac{3\pi}{16}$ [Meerut, 2004]

6. Given $f(x) = e^{-x^2/2}$, find the Fourier transform of

   (i) $xe^{-x^2/2}$ (ii) $x^2e^{-x^2/2}$

7. Find the Fourier sine and cosine transforms of $f(x) = \begin{cases} \cos x, & 0 < x < a \\ 0, & x > a \end{cases}$ [U.P.T.U., 2004]

8. Using the Fourier sine transform of e^{-ax}, $a > 0$, show that $\int_0^\infty \frac{x\sin mx}{x^2+a^2}\, dx = \frac{\pi}{2}e^{-am}$, $m > 0$.

   Hence obtain the Fourier sine transform of $x/(x^2+a^2)$. [Kakatiya, 2005; Kashmir, 2003]

9. Find the Fourier cosine transform of e^{-x^2}. [Meerut, 2005]

   Hence evaluate $\int_0^\infty \frac{\cos kx}{x^2+a^2}\, dx$. [U.P.T.U., 2004-05]

10. If the Fourier sine transform of $f(x)$ is e^{-ks}, find $f(x)$. Hence obtain the inverse Fourier sine transform of $\frac{s}{1+s^2}$. [Mumbai, 2004]

12. Find the Fourier sine transform of e^{-ax} and hence evaluate Fourier sine transform of xe^{-ax}.

13. Find the Fourier sine transform of e^{-ax}/x. Hence or otherwise prove that e^{-ax}/x is self-reciprocal under Fourier sine transformation. (Anna, 2005)

14. Find the Fourier sine transform of $\frac{1}{x(x^2+a^2)}$. (MadrasU, 2006)

(a) for $0 < x < a$ (b) for $x > a$ (C.P.T.C., 2008)

15. Obtain Fourier transform of

$f(x) = \begin{cases} \frac{(\sin x)(1+\cos x)}{x}, & 0 < x < \pi \\ 0, & x > \pi \end{cases}$ (Madras, 2005) $\quad (x) f(x) = \begin{cases} a^2 - x^2, & \text{for } 0 < x < a \\ 0, & \text{for } x > a \end{cases}$ (UPTU, 2006)

16. Find the Fourier cosine transform of $1/(1+x^2)$. (UPTU, 2006)

17. Find the Fourier transform $f(x)$ if $f(x) = \begin{cases} \frac{-1}{\pi}, & -\pi < x < 0 \\ \frac{1}{\pi}, & 0 < x < \pi \end{cases}$ where $x > 0$. (UPTU, 2007)

18. Show that the inverse Fourier transform of $F(\alpha) = \frac{1}{\alpha}\left[2\sin\alpha + 3\cos\alpha + 2\sin\frac{\alpha}{2}\right]$ is

$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 3, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$ (U.P.T.U., 2006)

19. Solve the integral equation $\int_0^\infty f(x)\cos\alpha x\,dx = \begin{cases} 1-\alpha, & 0 \le \alpha \le 1 \\ 0, & \alpha > 1 \end{cases}$ (Kottayam, 2005)

20. Solve the integral equation $\int_0^\infty f(x)\cos\alpha x\,dx = e^{-\alpha}$. (U.P.T.U., 2005; Delhi, 2006)

4.8.2 (1) CONVOLUTION

The convolution of two functions $f(x)$ and $g(x)$ over the interval $(-\infty, \infty)$ is defined as

$$f * g = \int_{-\infty}^{\infty} f(x)\, g(x - x)\, dx$$

(2) Convolution theorem for Fourier transforms. *The Fourier transform of the convolution of $f(x)$ and $g(x)$ is the product of their Fourier transforms, i.e.,*

$$F[f(x) * g(x)] = F[f(x)] \cdot F[g(x)]$$

We have $F[f(x) * g(x)] = F\left[\int_{-\infty}^{\infty} f(x)\, g(x - x)\, dx\right]$

$$= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x)\, g(x - x)\, dx\right] e^{ix} dx = \int_{-\infty}^{\infty} f(x) \left[\int_{-\infty}^{\infty} g(x - x)\, e^{ix} dx\right] dx$$

(Changing the order of integration)

$$= \int_{-\infty}^{\infty} f(x) \left[\int_{-\infty}^{\infty} e^{i(x+x)}\, g(x - x)\, dx - x\right] e^{ix} dx$$

$$= \int_{-\infty}^{\infty} e^{ix} f(x) \left[\int_{-\infty}^{\infty} e^{ix}\, g(x)\, dx\right] dx \text{ where } x - x = x$$

$$= \int_{-\infty}^{\infty} e^{ix} f(x)\, dx \cdot F[g(x)] = \int_{-\infty}^{\infty} e^{ix} f(x)\, dx \cdot F[g(x)] = F[f(x)] \cdot F[g(x)]$$

4.8.3 PARSEVAL'S IDENTITY FOR FOURIER TRANSFORMS

If the Fourier transforms of $f(x)$ and $g(x)$ are $F(x)$ and $G(x)$ respectively, then

(1) $\frac{1}{2\pi}\int_{-\infty}^{\infty} F(x)\overline{G(x)}\,dx = \int_{-\infty}^{\infty} f(x)\overline{g(x)}\,dx$ (2) $\frac{1}{2\pi}\int_{-\infty}^{\infty} |F(x)|^2\,dx = \int_{-\infty}^{\infty} |f(x)|^2\,dx$

where $\{a_n\}$ are the complex conjugates.

(c) $\int_{-\infty}^{\infty} f(x)\,g^*(x)\,dx = \int_{-\infty}^{\infty} \left[\sum_n A_n e^{inx}\right]\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} G(\omega)\,e^{-i\omega x}\,d\omega\right] dx$ (Using the inverse transform for Fourier transforms)

$= \frac{1}{2\pi}\sum_n A_n \int_{-\infty}^{\infty} G(\omega)\left[\int_{-\infty}^{\infty} g(x)\,e^{-inx}\,dx\right] d\omega$ (Changing the order of integration)

$= \frac{1}{2\pi}\sum_n A_n$ (Using Parseval's identity for Fourier series)

(c) Putting $g(x) = f(x)$, we get

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega = \sum_n |A_n|^2 \text{ or } \frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega = \sum_n |f(n)|^2$$

(d)(a) The Poisson Processes are one of the most important classes of stochastic processes. Applications of Poisson processes are found in [illegible]

(b) [illegible] (c) [illegible]

(d) [illegible] (e) [illegible]

Consider the [illegible] Poisson [illegible] process $\{X(t), t \geq 0\}$ [illegible]

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(c) [illegible] (d) [illegible]

(e) [illegible] (f) [illegible]

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(w) [illegible] (x) [illegible]

(y) [illegible] (z) [illegible]

(aa) [illegible] (ab) [illegible]

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(ag) [illegible] (ah) [illegible]

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(zk) [illegible] (zl) [illegible]

(zm) [illegible] (zn) [illegible]

(zo) [illegible] (zp) [illegible]

(zq) [illegible] (zr) [illegible]

(zs) [illegible] (zt) [illegible]

(zu) [illegible] (zv) [illegible]

(zw) [illegible] (zx) [illegible]

(zy) [illegible] (zz) [illegible]

Hence using (i) above, we have $\int_0^\infty \frac{x^{n-1}e^{-x}}{\Gamma(n)}\,dx = \int_0^\infty e^{-x}\,dx = 1 - e^{-\infty} = 1$

Thus $\int_0^\infty \frac{x^{n-1}e^{-x}}{\Gamma(n)}\,dx = \frac{1}{\Gamma(n)}\Gamma(n) = 1$

Accordingly, the moment-generating function of the gamma distribution is given by $M_X(t) = \begin{cases} (1-t)^{-\alpha} & \text{for } t < 1 \\ \text{does not exist} & \text{for } t \geq 1 \end{cases}$

We can show that $\int_0^\infty \frac{x^{\alpha-1}e^{-x}}{\Gamma(\alpha)}\,dx = \frac{1}{\Gamma(\alpha)}\int_0^\infty x^{\alpha-1}e^{-x}\,dx = 1$ (continues overleaf)

(Introduction) The moment-generating function of X is $M_X(t) = \int_0^\infty e^{tx}f(x)\,dx = \int_0^\infty e^{tx}\frac{1}{\Gamma(\alpha)}x^{\alpha-1}e^{-x}\,dx$

$= \int_0^\infty \frac{1}{\Gamma(\alpha)}x^{\alpha-1}e^{-x(1-t)}\,dx$ (i)

$= \frac{1}{\Gamma(\alpha)}\int_0^\infty \frac{1}{(1-t)^{\alpha-1}}u^{\alpha-1}e^{-u}\frac{du}{1-t}$ $\left[\begin{array}{l}\text{Put } u = x(1-t) \text{ then } du = dx(1-t) \text{ and } x = \frac{u}{1-t} \\ \text{for } 1-t > 0 \text{ that is } t < 1 \text{ we refer to as the } \textit{domain of convergence}\end{array}\right]$

$= \frac{1}{\Gamma(\alpha)}\left[\frac{1}{(1-t)^{\alpha-1}}\int_0^\infty u^{\alpha-1}e^{-u}\,du\right] = \frac{1}{(1-t)^{\alpha}}\left[\int_0^\infty \frac{1}{\Gamma(\alpha)}u^{\alpha-1}e^{-u}\,du\right]$

(ii) By (i) we can deduce that

$$M_X(t) = \frac{1}{\Gamma(\alpha)}\int_0^\infty e^{tx}x^{\alpha-1}e^{-x}\,dx = \frac{1}{\Gamma(\alpha)}\int_0^\infty \left(\frac{x}{1-t}\right)^{\alpha-1}e^{-\frac{x}{1-t}}\,dx$$

To evaluate $\int_0^\infty \left(\frac{x}{1-t}\right)^{\alpha-1}e^{-\frac{x}{1-t}}\,dx$, put $u = \frac{x}{1-t}$ then $du = \frac{dx}{1-t}$ or $dx = (1-t)\,du$

$$M_X(t) = \frac{1}{\Gamma(\alpha)}\int_0^\infty u^{\alpha-1}e^{-u}(1-t)\,du = \frac{1}{\Gamma(\alpha)}\int_0^\infty \left(\frac{x}{\beta}\right)^{\alpha-1}e^{-\frac{x}{\beta}}\,dx \qquad \left[t = \frac{1}{\beta} \text{ is a necessary assumption}\right]$$

$$\therefore \int_0^\infty \left(\frac{x}{\beta}\right)^{\alpha-1}e^{-\frac{x}{\beta}}\,dx = \Gamma(\alpha)\beta^{\alpha} \qquad \text{(i) } f(x) = \alpha\beta \neq 0$$

(iii) Using Parseval's identity

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} |F(\omega)|^2\,d\omega = \int_{-\infty}^{\infty} |f(x)|^2\,dx$$

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ix\omega}e^{-\frac{x^2}{2}}\,dx\right]^2 d\omega = \int_{-\infty}^{\infty} e^{-x^2}\,dx$$

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\frac{\cos x\omega - i\sin x\omega}{\sqrt{2\pi}}\right]^2 d\omega = 4\int_0^\infty e^{-x^2}\,dx = 4\left[\frac{\sqrt{\pi}}{2}\right]^2 = \pi$$

Putting $\frac{1}{2}\cos\theta$ and $d\theta = \sin\theta\,d\theta$

$$\frac{1}{2\pi}\int_0^{\pi}\left[\frac{\cos x\omega}{\sqrt{2\pi}}\right]^2 d\omega = \frac{1}{2\pi}\int_0^{\pi}\frac{\cos^2 x\omega}{2\pi}\,d\omega \quad \text{or} \quad \frac{1}{2\pi}\int_0^{\pi}\left[\frac{\cos x\omega}{\sqrt{2\pi}}\right]^2 d\omega = \frac{1}{2\pi}\pi$$

Hence

$$\int_0^{\pi}\left[\frac{\cos x\omega}{\sqrt{2\pi}}\right]^2 d\omega = \frac{1}{2}$$

EXERCISES

1. Verify Convolution theorem for $f(x) = g(x) = e^{-x^2}$. (V.T.U., 2008-S)
2. Use Convolution theorem to find the inverse Fourier transform of $\frac{1}{(1+x^2)^2}$, given that $\frac{2}{1+x^2}$ is the Fourier transform of $e^{-|x|}$. (V.T.U., 2010-S)
3. Using Parseval's identity, prove that

(a) $\int_0^\infty \frac{dx}{(x^2+a^2)^2} = \frac{\pi}{4a^3}$ (Madras, 2003) (b) $\int_0^\infty \frac{x^2}{(x^2+a^2)(x^2+b^2)}\,dx = \frac{\pi}{2(a+b)}$ (Madras, 2000)

4. Find the Fourier transform of $f(x) = \begin{cases} 1-x^2, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$

Hence deduce that $\int_0^\infty \left(\frac{\sin x - x\cos x}{x^3}\right)^2 dx = \frac{\pi}{15}$. (Anna, 2009)

5. Evaluate $\int_0^\infty \left(\frac{x\cos x - \sin x}{x^3}\right)^2 dx$.

7.13 RELATION BETWEEN FOURIER AND LAPLACE TRANSFORMS

If $$f(x) = \begin{cases} e^{-px}\phi(x), & x > 0 \\ 0, & x < 0 \end{cases} \quad \ldots(1)$$

then $F[f(x)] = \phi(p+is)$.

We have $$F[f(x)] = \int_{-\infty}^{\infty} e^{isx} f(x)\,dx = \int_{-\infty}^{0} e^{isx}\cdot 0\,dx + \int_0^\infty e^{isx}\cdot e^{-px}\phi(x)\,dx$$

$$= \int_0^\infty e^{-(p-is)x}\phi(x)\,dx = \int_0^\infty e^{-sy}\phi(y)\,dy \quad \text{where } y = x - ia$$

Hence the Fourier transform of $f(x)$ defined by (1) is the Laplace transform of $\phi(x)$.

7.14 FOURIER TRANSFORMS OF THE DERIVATIVES OF A FUNCTION

The Fourier transform of the function $u(x,t)$ is given by

$$F[u(x,t)] = \int_{-\infty}^{\infty} u e^{isx}\,dx$$

Then the Fourier transform of $\partial^2 u/\partial x^2$, i.e.

$$F\left[\frac{\partial^2 u}{\partial x^2}\right] = \int_{-\infty}^{\infty} \frac{\partial^2 u}{\partial x^2} e^{isx}\,dx = \left[e^{isx}\frac{\partial u}{\partial x} - isue^{isx}\right]_{-\infty}^{\infty} + (is)^2\int_{-\infty}^{\infty} ue^{isx}\,dx$$

on applying the general rule of integration by parts (p. 3.6). If u and $\frac{\partial u}{\partial x}$ tend to zero as $x \to \infty$, then

$$F\left[\frac{\partial^2 u}{\partial x^2}\right] = -s^2 F[u] \quad \ldots(1)$$

Similarly in the case of Fourier sine and cosine transforms, we have

$$F_s\left[\frac{\partial^2 u}{\partial x^2}\right] = s[u]_{x=0} - s^2 F_s[u] \quad \ldots(2)$$

and $$F_c\left[\frac{\partial^2 u}{\partial x^2}\right] = -\left[\frac{\partial u}{\partial x}\right]_{x=0} - s^2 F_c[u] \quad \ldots(3)$$

In general, the Fourier transform of the nth derivative of $f(x)$ is given by

$$\mathcal{L}\left[\frac{d^n y}{dx^n}\right] = s^n \mathcal{L}[y(x)] \qquad \ldots(4)$$

provided that $y, y', y'', \ldots, y^{n-1}$ are continuous and y^n is piecewise continuous in $0 \le x \le X$, for any $X > 0$.

For $\mathcal{L}[y'(x)] = \int_0^\infty e^{-sx} y'(x)\,dx$

$$= \left[e^{-sx} y(x)\right]_0^\infty + s\int_0^\infty e^{-sx} y(x)\,dx = -y(0) + s\int_0^\infty e^{-sx} y(x)\,dx$$

by the general rule of integration by parts, where $\mathcal{L}[y(x)]$ exists.

11.10 INVERSE LAPLACE TRANSFORMS BY METHOD OF RESIDUES

Let the Laplace transform of $y(x)$ be $\bar{y}(s)$, so that

$$\bar{y}(s) = \int_0^\infty y(x) e^{-sx}\,dx \qquad \ldots(1)$$

Multiply both sides by e^{sa} and integrate w.r.t. s from $s = -i\infty$ to $s = i\infty$. Then

$$\int_{s=-i\infty}^{s=i\infty} e^{sa}\bar{y}(s)\,ds = \int_{s=-i\infty}^{s=i\infty} e^{sa}\int_0^\infty y(x) e^{-sx}\,dx\,ds \qquad [\text{Put } s = \alpha - i\beta]$$

$$= \int_{\alpha=-\infty}^{\infty} e^{\alpha a} \int_0^\infty y(x) e^{-\alpha x} e^{i\beta(a-x)}\,dx\,(-i\,d\beta) = e^{\alpha a}\int_{-\infty}^{\infty} e^{-i\beta a}\int_0^\infty e^{-\alpha x} y(x) e^{i\beta x}\,dx\,d\beta$$

$$= e^{\alpha a}\int_{-\infty}^{\infty} e^{-i\beta a}\int_0^\infty \phi(x) e^{i\beta x}\,dx\,d\beta$$

where $\phi(x) = \begin{cases} e^{-\alpha x} y(x) & \text{for } x > 0 \\ 0 & \text{for } x < 0 \end{cases}$

Proceeding to limits as $\alpha \to \infty$, we get

$$\int_{s=-i\infty}^{s=i\infty} e^{sa}\bar{y}(s)\,ds = e^{\alpha a}\cdot 2\pi\phi(x), \text{ by (2) of § 22.1 } = 2\pi e^{\alpha a} e^{-\alpha x} y(x) \text{ for } x > 0.$$

Hence $$y(x) = \frac{1}{2\pi i}\int_{\alpha-i\infty}^{\alpha+i\infty} e^{sx}\bar{y}(s)\,ds, \quad x > 0 \qquad \ldots(2)$$

which is called the *complex inversion formula*. It provides a direct means for obtaining the inverse Laplace transform of a given function.

The integration in (2) is performed along a line LM parallel to the imaginary axis in the complex plane $s = \alpha + i\beta$ such that all the singularities of $\bar{y}(s)$ lie to the left* (Fig. 22.1). Let us take a contour C which is composed of the line LM and the semicircle C' (i.e., MNL). Then from (2)

$$\frac{1}{2\pi i}\int_{\alpha-i\infty}^{\alpha+i\infty} e^{sx}\bar{y}(s)\,ds = \frac{1}{2\pi i}\int_C e^{sx}\bar{y}(s)\,ds - \frac{1}{2\pi i}\int_{C'} e^{sx}\bar{y}(s)\,ds$$

The integral over C' tends to zero as $r \to \infty$ (under certain conditions†). Therefore,

$$y(x) = \lim_{r\to\infty}\frac{1}{2\pi i}\int_C e^{sx}\bar{y}(s)\,ds$$

$$= \text{sum of the residues of } e^{sx}\bar{y}(s) \text{ at the poles of } \bar{y}(s) \qquad \ldots(3)$$

[By (20.18)]

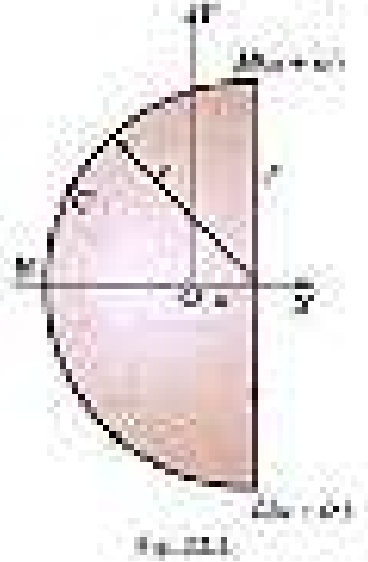


Fig. 22.1

* This can be accomplished in order to ensure the convergence of the integral (2).

† If positive constants A and α can be so found that $|\bar{y}(s)| < A/s^\alpha$, for every point on C', then

$$\lim_{r\to\infty}\frac{1}{2\pi i}\int_{C'} e^{sx}\bar{y}(s)\,ds = 0. \qquad \text{[Jordan's Lemma]}$$

Example 22.15. Evaluate $\mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s^2+1)}\right\}$ by the method of residues.

Solution. Since $\left|\frac{s}{(s-2)(s^2+1)}\right| = \left|\frac{1}{s^2}\right|$ for $|s| \to \infty$, therefore,

$$\mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s^2+1)}\right\} = \text{sum of Res}\left\{\frac{se^{st}}{(s-2)(s^2+1)}\right\} \text{ at the poles } s = 2, \pm i$$

Now
$$(\text{Res})_{s=2} = \lim_{s\to 2}\left\{\frac{(s-2)se^{st}}{(s-2)(s^2+1)}\right\} = \frac{2e^{2t}}{5} \qquad \text{[By (20.10)(a)]}$$

$$(\text{Res})_{s=i} = \lim_{s\to i}\left\{\frac{(s-i)se^{st}}{(s-i)(s^2+1)}\right\} = \frac{ie^{it}}{i+i} = -\frac{1}{2}ie^{it}$$

Changing i to $-i$, we get $(\text{Res})_{s=-i} = \frac{1}{2}ie^{-it}$

$$\therefore \quad \mathcal{L}^{-1}\left\{\frac{s}{(s-2)(s^2+1)}\right\} = \frac{2e^{2t}}{5} - \frac{1}{2}\left\{\frac{e^{it}}{(-i)} + \frac{e^{-it}}{(-i)}\right\} = \frac{1}{5}(2e^{2t} - 4\cos t - 2\sin t)$$

Example 22.16. Prove that $\mathcal{L}^{-1}\left\{\frac{e^{-\sqrt{s}}}{s}\right\} = 1 - \text{erf}\left(\frac{x}{2\sqrt{t}}\right)$.

Solution. By the complex inversion formula,

$$\mathcal{L}^{-1}\left\{\frac{e^{-x\sqrt{s}}}{s}\right\} = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} e^{st}\,\frac{e^{-x\sqrt{s}}}{s}\,ds.$$

Since $s = 0$ is a branch point and the integrand is not single valued, we take a contour $\Lambda\Lambda'\Psi\Psi'\Lambda$ as shown in Fig. 22.2, so that it does not include any such singularity. Therefore, by Cauchy's theorem (5.20.13), we have

$$\int_{\Lambda\Lambda'} + \int_{\Lambda'\Psi} + \int_{\Psi\Psi'} + \int_{\Psi'\Lambda} = \int_{\Lambda\Lambda'} + \int_{\Lambda'\Psi} + \int_{\Psi\Psi'} + \int_{\Psi'\Lambda} = \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st}\,\frac{e^{-x\sqrt{s}}}{s}\,ds = 0 \qquad \ldots(1)$$

If $OX = \rho$ and $OP = s$, then along $\Lambda\Lambda'$, $s = \rho e^{-i\pi}$, therefore,

$$\int_{\Lambda\Lambda'} = \int_{\rho}^{0} e^{-\rho t}\,\frac{e^{-x\sqrt{\rho}\,e^{-i\pi/2}}}{\rho e^{-i\pi}}\,(-d\rho)$$

Similarly along $\Psi\Psi'$, $s = \rho e^{i\pi}$, therefore,

$$\int_{\Psi\Psi'} = \int_{0}^{\rho} e^{-\rho t}\,\frac{e^{-x\sqrt{\rho}\,e^{i\pi/2}}}{\rho e^{i\pi}}\,d\rho$$

Along $\Psi'\Lambda$ on the circle $|s| = \gamma$, $s = \gamma e^{i\theta}$. Also e^{st} and $e^{-x\sqrt{s}}$ are both approximately 1 since s is small. Therefore,

$$\int_{\Psi'\Lambda} = \int_{\pi}^{-\pi} \frac{1}{\gamma e^{i\theta}}\,\gamma e^{i\theta}\,i\gamma e^{i\theta}\,d\theta = -2\pi i \text{ approximately.}$$

For $\rho > 0$, $|e^{-x\sqrt{\rho}\,e^{i\pi/2}}| < |e|^{0}$.

But $\int_{\Lambda\Lambda'}$ and $\int_{\Psi\Psi'}$ both tend to zero as $\gamma \to \infty$.

Thus (1) takes the form

$$\int_{\infty}^{0} \frac{e^{-x\sqrt{\rho}\,i}}{\rho}\,(-d\rho) + \int_{0}^{\infty} e^{-\rho t}\,\frac{e^{-x\sqrt{\rho}\,i} - e^{x\sqrt{\rho}\,i}}{\rho}\,d\rho - 2\pi i = 0$$

Taking limits as $\gamma \to 0$ and $\rho \to \infty$, we get

$$\int_{\infty}^{0} \frac{e^{-x\sqrt{\rho}\,i}}{\rho}\,(-d\rho) + 2\pi i = 2i\int_{0}^{\infty} e^{-\rho t}\,\frac{\sin x\sqrt{\rho}}{\rho}\,d\rho$$

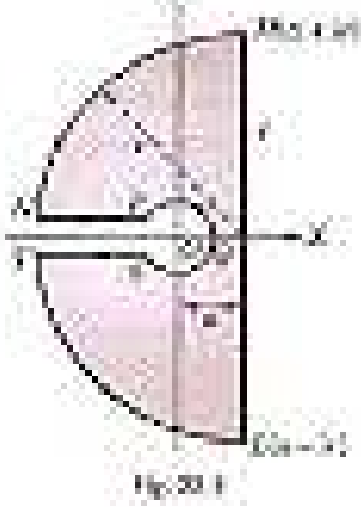


Fig. 22.2

or $$\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{e^{-i\omega t}-e^{-i\omega}}{i\omega}\,d\omega = 1-\frac{2}{\pi}\int_0^{\infty}e^{-x^2}\frac{\sin x t\cos x\sqrt{2}}{x}\,dx,\ \text{where } x=\omega/\sqrt{2}$$

$$= 1-\frac{2}{\pi}\int_0^{\infty}\frac{\sin x t}{x}e^{-x^2}\,dx = \frac{2}{\pi}\tan^{-1}\left(\frac{t}{\sqrt{1-t^2}}\right) \text{ where } 0\le t\le 1$$

PROBLEMS 7.13

Using the method of residues, evaluate the inverse Laplace transform of each of the following:

1. $\dfrac{1}{(s+1)(s-2)^2}$
2. $\dfrac{1}{s(s-1)(s^2+1)}$
3. $\dfrac{1}{s^2(s^2+a^2)}$
4. $\dfrac{1}{(s-1)(s^2+1)}$
5. $\dfrac{1}{(s^2+1)^2}$

(U.P.T.U., 2006 (O))

7.26 APPLICATION OF TRANSFORMS TO BOUNDARY VALUE PROBLEMS

In one dimensional boundary value problems, the partial differential equations can easily be transformed into an ordinary differential equation by applying a suitable transform. The required solution is then obtained by solving the equation and inverting by means of the complex inversion formula or by any other method. In two dimensional problems, it is sometimes required to apply the transform twice and the desired solution is obtained by double inversion.

(i) If in a problem $u(x,t)_{t=0}$ is given then we use infinite sine transforms to remove $\partial u/\partial t$ from the differential equation.

In case $[\partial u/\partial t]_{t=0}$ is given then we employ infinite cosine transforms to remove $\partial^2 u/\partial t^2$.

(ii) If in a problem $u(0,t)$ and $u(l,t)$ are given, then we use finite sine transforms to remove $\partial^2 u/\partial x^2$ from the differential equation.

In case $[\partial u/\partial x]_{x=0}$ and $[\partial u/\partial x]_{x=l}$ are given, then we employ finite cosine transforms to remove $\partial^2 u/\partial x^2$.

The method of solution is best explained through the following examples.

Heat conduction

Example 7.27 (A). *Determine the distribution of temperature in the semi-infinite medium $x \ge 0$, when the end $x = 0$ is maintained at zero temperature and the initial distribution of temperature is $f(x)$.*

(Bhopal U., 2005)

Solution. Let $u(x,t)$ be the temperature at any point x and at any time t. We have to solve the heat flow equation (§ 10.11)

$$\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}\ (x>0,\ t>0) \qquad \ldots(i)$$

subject to the initial condition $u(x,0) = f(x)$ $\qquad \ldots(ii)$

and the boundary condition $u(0,t) = 0$ $\qquad \ldots(iii)$

Taking Fourier sine transform of (i) and denoting $F_s[u(x,t)]$ by $\bar{u}_s$, we have

$$\frac{d\bar{u}_s}{dt} = -c^2\lambda^2\bar{u}_s \quad [\because u(0,t)=0] \qquad \text{(By (12) of § 10.8)}$$

---

* We know that $\int_0^{\infty}e^{-x^2}\cos\lambda x\,dx = \frac{1}{2}\sqrt{\pi}e^{-\lambda^2/4}$ [Example 10.4 (c)]

Replacing λ by α and x by $\beta\sqrt{2}$, we get

$$\int_0^{\infty}e^{-\beta^2}\left\{\frac{\sin 2\alpha\beta}{\beta}\right\}d\beta = \frac{1}{2}\sqrt{\pi}\int_0^{\alpha}e^{-\alpha^2/4}\,d\alpha$$

or $$\int_0^{\infty}e^{-x^2}\frac{\sin x t\sqrt{2}}{x}\,dx = \frac{\pi}{2}\operatorname{erf}\left(\frac{t}{2\sqrt{2}}\right) \qquad \text{[By § 10.10(3)]}$$

or $$\frac{d\bar{u}_1}{dt} + p^2\bar{u}_1 = 0$$ (By 1802, ...(iv))

Also the Fourier sine transform of (ii) is $\bar{u}_1 = \bar{f}(p)$ at $t = 0$ ...(v)

Solving (iv) and using (v), we get $\bar{u}_1 = \bar{f}(p)e^{-p^2t}$

Now taking the inverse Fourier sine transform, we obtain

$$u(x,t) = \frac{2}{\pi}\int_0^\infty \bar{f}(p)e^{-p^2t}\sin px\,dp.$$

Example 2.21. Solve $\partial u/\partial t = \partial^2 u/\partial x^2$, $0 < x < \infty$, $t > 0$, $u(x,0) = e^{-x}$, $x > 0$ and $u(0,t) = 0$, $t > 0$. (Rohtak, 2005)

Solution. Given $\partial u/\partial t = \partial^2 u/\partial x^2$, $x > 0$, $t > 0$ ...(i)

with boundary conditions: $u(0,t) = 0$, $u(x,t)$ is bounded ...(ii)

and initial condition $u(x,0) = e^{-x}$, $x > 0$ ...(iii)

Since $u(0,t)$ is given, we take Fourier sine transform of both sides of (i) so that

$$\int_0^\infty \frac{\partial u}{\partial t}\sin px\,dx = \frac{\partial}{\partial t}\int_0^\infty u\sin px\,dx$$

or $$\frac{d}{dt}\int_0^\infty u(x,t)\sin px\,dx = \left[\frac{\partial u}{\partial x}\sin px\right]_0^\infty - \int_0^\infty \frac{\partial u}{\partial x}\cdot p\cos px\,dx$$ (Integrating by parts)

or $$\frac{d\bar{u}_1}{dt} = -p\int_0^\infty \frac{\partial u}{\partial x}\cos px\,dx$$, if $\frac{\partial u}{\partial x} \to 0$ as $x \to \infty$ where $\bar{u}_1(p,t) = \int_0^\infty u\sin px\,dx$

$$= -p\left[u\sin px\right]_0^\infty - p\int_0^\infty u(x,t)(-\cos px)\,dx$$ (Again integrating by parts)

$$= -p[0 - u(0,t)] - p^2\int_0^\infty u(x,t)\sin px\,dx$$ By (ii) $u(x,t) \to 0$ as $x \to \infty$ by (iii)

$$= p\,u(0,t) - p^2\bar{u}_1$$

or $$\frac{d\bar{u}_1}{dt} = -p^2\bar{u}_1$$ (By (v))

Integrating $\int \frac{d\bar{u}_1}{\bar{u}_1} = -p^2\int dt$ or $\log\bar{u}_1 = \log c = -p^2t$

$\therefore$ $$\bar{u}_1(p,t) = ce^{-p^2t}$$ ...(iv)

Taking Fourier sine transform of both sides of (iii), we get

$$\int_0^\infty u(x,0)\sin px\,dx = \int_0^\infty e^{-x}\sin px\,dx$$

or $$\bar{u}_1(p,0) = \left[\frac{e^{-x}}{1+p^2}(-\sin px - p\cos px)\right]_0^\infty = \frac{p}{1+p^2}$$ ...(v)

Putting $t = 0$ in (iv) and using (v), we obtain $c(1 + p^2) = p$

Then (iv) becomes $\bar{u}_1(p,t) = \frac{p}{1+p^2}e^{-p^2t}$

Now taking inverse Fourier sine transform, we get

$$u(x,t) = \frac{2}{\pi}\int_0^\infty \frac{pe^{-p^2t}}{1+p^2}\sin px\,dp.$$

Example 2.22. Solve the equation $u_t = \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $-\infty < x < \infty$, $t > 0$ subject to the conditions

$u(x,0) = f(x)$ where $-\infty < x < \infty$, $t > 0$ $f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$ ($u(x,t)$ is bounded). (U.P.T.U., 2003-04)

Solution. Since $u(0, t) = 0$, we take Fourier sine transform of both sides of the given equation, we get

$$\int_0^\infty \frac{\partial u}{\partial t} \sin sx\, dx = \int_0^\infty \frac{\partial^2 u}{\partial x^2} \sin sx\, dx$$

$$\frac{\partial}{\partial t}\int_0^\infty u \sin sx\, dx = -s^2 \bar{u}(0) + s\, u(0) \qquad \because u = 0, \text{ when } x = 0$$

or $$\frac{d\bar{u}}{dt} = -s^2\bar{u} \quad \text{or} \quad \frac{d\bar{u}}{dt} + s^2\bar{u} = 0 \quad \text{or} \quad (D^2 + s^2)\bar{u} = 0 \; (\because D = d/dt)$$

$\therefore$ Its solution is $\bar{u}(x, t) = e^{-s^2 t}$ ...(1)

Since $$\bar{u}(x, t) = \int_0^\infty u(x, t) \sin sx\, dx$$

$\therefore$ $$\bar{u}(x, 0) = \int_0^\infty u(x, 0) \sin sx\, dx = \int_0^1 1 \cdot \sin sx\, dx \qquad [\because u(x, 0) = 1]$$

$$= \frac{1 - \cos s}{s} \qquad ...(2)$$

From (1) and (2), $$c = \bar{u}(x, 0) = \frac{1 - \cos s}{s}$$

Thus (1) gives $$\bar{u}(x, t) = \frac{1 - \cos s}{s} e^{-s^2 t}$$

Now taking inverse Fourier sine transform, we get

$$u(x, t) = \int_0^\infty \frac{1 - \cos s}{s} e^{-s^2 t} \sin sx\, ds$$

which is the desired solution.

Example 3.2.21. Using Finite Fourier transform, solve

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

given $u(0, t) = 0$, $u(1, t) = 0$, $u(x, 0) = \sin \pi x + 3 \sin 2\pi x$ and $0 < x < 1$, $t > 0$. (Rajasthan, 2009)

Solution. Since $u(0, t) = 0$, we take finite Fourier sine transform of both sides of the given equation

$$\int_0^1 \frac{\partial u}{\partial t} \sin \frac{n\pi}{1} x\, dx = \int_0^1 \frac{\partial^2 u}{\partial x^2} \sin \frac{n\pi}{1} x\, dx$$

or $$\frac{d}{dt}\bar{u}_n = k\left[\frac{\partial u}{\partial x}\right]_0^1$$

$$= -\frac{n^2\pi^2}{1^2} u_n + \frac{n\pi}{1}[u(1, t) - (-1)^n u(0, t)]$$

$$= -\frac{n^2\pi^2}{1^2} u_n \qquad [\because u(0, t) = 0, u(1, t) = 0]$$

or $$\frac{d\bar{u}_n}{u_n} = -\frac{n^2\pi^2}{1^2} dt$$

Integrating both sides, $$\log \bar{u}_n = -\frac{n^2\pi^2}{1^2} t + c$$

or $$\bar{u}_n(x, t) = Ce^{-\frac{n^2\pi^2}{1^2}t} \qquad ...(1)$$

Putting $t = 0$, $$c = \bar{u}_n(x, 0) = \int_0^1 u(x, 0) \sin \frac{n\pi x}{1} dx \qquad [\because u(x, 0) = 2x]$$

$$= \int_0^1 2x \sin \frac{n\pi x}{1} dx = -\frac{2}{n\pi} \cos n\pi$$

Thus (1) gives, $$u_1(x,t) = -\frac{2u_0}{n\pi}\cos n\pi e^{-n^2\pi^2 t/l^2} = -\frac{2u_0}{n\pi}(-1)^n e^{-n^2\pi^2 t/l^2}$$

Now taking inverse Fourier sine transforms, we get

$$u(x,t) = \frac{2}{l}\sum_{n=1}^{\infty}\frac{2u_0}{n\pi}(-1)^{n+1}e^{-n^2\pi^2 t/l^2}\sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{4u_0}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}e^{-n^2\pi^2 t/l^2}\sin\left(\frac{n\pi x}{l}\right).$$

Example 28.23. *A bar of length l has its ends kept at zero temperature and the temperature initially is $f(x)$ where*

$$f(x) = \begin{cases} kx, & 0 < x < \frac{l}{2} \\ k(l-x), & \frac{l}{2} < x < l \end{cases}$$

determine the temperature at any point x at any time t. (U.P.T.U. 2006) (Kanpur, 2006)

Solution. To determine the temperature $u(x,t)$ at any point at any time, we have to solve the equation

$$\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2} \quad (x > 0) \qquad \ldots(1)$$

subject to the initial condition $u(x,0) = \begin{cases} kx, & 0 < x < \frac{l}{2} \\ k(l-x), & \frac{l}{2} < x < l \end{cases}$ $\ldots(2)$

Taking Fourier sine transform of (1) and denoting $F(x,t)$ by $\bar{u}$, we find

$$\frac{d\bar{u}}{dt} = -c^2\lambda^2\bar{u} \qquad \text{[by (1) of § 28.9]} \quad \ldots(3)$$

Also the Fourier transform of (2) is

$$\bar{u}(x,0) = \int_0^\infty u(x,0)\sin\lambda x\,dx = \int_0^{l/2} kx\sin\lambda x\,dx + \int_{l/2}^{l} k(l-x)\sin\lambda x\,dx \qquad \ldots(4)$$

Solving (3) and using (4), we get $\bar{u} = \frac{2k}{\lambda^2}\sin\frac{\lambda l}{2}e^{-c^2\lambda^2 t}$

Now on taking the inverse Fourier transform, we get

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{2k}{\lambda^2}\sin\frac{\lambda l}{2}e^{-c^2\lambda^2 t}e^{i\lambda x}\,d\lambda = \frac{2k}{\pi}\int_0^\infty\frac{\sin\frac{\lambda l}{2}}{\lambda^2}e^{-c^2\lambda^2 t}(\cos\lambda x + i\sin\lambda x)\,d\lambda$$

$$= \frac{2k}{\pi}\int_0^\infty\frac{\sin\frac{\lambda l}{2}}{\lambda^2}e^{-c^2\lambda^2 t}\cos\lambda x\,d\lambda \qquad \left[\begin{array}{l}\text{The second integral vanishes as}\\ \text{the integrand is an odd function}\end{array}\right]$$

$$= \frac{2k}{\pi}\int_0^\infty e^{-c^2\lambda^2 t}\frac{\sin\frac{\lambda l}{2}\cos\lambda x + \cos\frac{\lambda l}{2}\sin\lambda x - \sin\frac{\lambda l}{2}\cos\lambda x}{\lambda^2}\,d\lambda$$

$$= \frac{2k}{\pi}\int_0^\infty e^{-c^2\lambda^2 t}\left\{\sin\frac{\lambda}{2}\frac{(l-2x)}{c\lambda t}\cdot\sin\frac{\lambda}{2}\frac{(l-2x)}{c\lambda t}\right\}\frac{dx}{\lambda^2}, \text{ where } \lambda^2 = x^2/t^2$$

$$= \frac{2k}{\pi}\left[\pi c^2 t\,\frac{(2x-l)}{2c^2t} + \pi c^2 t\,\frac{(l-2x)}{2c^2t}\right] \qquad \text{[See footnote on p. 268]}$$

Example 28.24. *A bar of length π is at zero temperature. At $t = 0$, the end $x = \pi$ is suddenly raised to temperature u_0 and the ends are kept at zero temperature. Find the temperature at any point x of the bar at any time $t > 0$, assuming that the surface of the bar is insulated.*

Solution. Here we have to solve the differential equation

$$\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2} \quad (0 < x < \pi,\ t > 0) \qquad \ldots(1)$$

subject to the conditions

$$u(x,0) = 0 \quad \ldots(2); \quad u_x(0,t) = 0 \quad \ldots(3) \quad \text{and} \quad u(\pi,t) = u_0 \qquad \text{(Rohtak, 2008)} \quad \ldots(4)$$

The Laplace transform of (1), if $L\{u(x,t)\} = U(x,s)$ is

$$L\left\{\frac{\partial u}{\partial t}\right\} = s\bar{u} - u(x,0) = s\bar{u} - 0 = s\bar{u}, \quad L\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \frac{d^2\bar{u}}{dx^2}$$

Using (3), we get $\dfrac{d^2\bar{u}}{dx^2} - \dfrac{1}{c^2}s\bar{u} = 0$ ...(4)

Similarly the Laplace transforms of (2) and (3) are

$$\bar{u}(0,s) = 0 \quad \text{...(5)} \qquad \bar{u}(a,s) = \frac{b}{s} \quad \text{...(6)}$$

Solving (4), we have $\bar{u} = C_1 e^{\sqrt{s}x/c} + C_2 e^{-\sqrt{s}x/c}$

Using (5), we find $C_1 = -C_2$ so that

$$\bar{u} = C_1\left(e^{\sqrt{s}x/c} - e^{-\sqrt{s}x/c}\right) = 2C_1 \sinh\left(\sqrt{s}x/c\right)$$

Using (6), we have $C = \dfrac{b}{s\sinh\left(\sqrt{s}a/c\right)}$

By the inversion formula (3) § 10.10, we get

$u(x,t)$ = sum of the residues of $\left\{\dfrac{e^{st}\cdot b\sinh\left(x\sqrt{s}/c\right)}{s\sinh\left(a\sqrt{s}/c\right)}\right\}$ at all the poles which occur at $s = 0$

and $\sinh\left(a\sqrt{s}/c\right) = 0$ i.e., at $s = 0, -\dfrac{n^2\pi^2c^2}{a^2} = -\alpha_n^2$ (say), $\alpha_n = \dfrac{n\pi c}{a}$, $n = 1, 2, 3, \ldots$

or at $s = 0, s = -\alpha_n^2 = -\dfrac{n^2\pi^2c^2}{a^2}$, $n = 1, 2, 3, \ldots$

Now $\quad \text{(Res)}_{s=0} = \lim_{s\to 0}\left\{s\cdot\dfrac{be^{st}\sinh\left(x\sqrt{s}/c\right)}{s\sinh\left(a\sqrt{s}/c\right)}\right\} = b$

$$\text{(Res)}_{s=-\alpha_n^2} = b\lim_{s\to -\alpha_n^2}\left\{(s+\alpha_n^2)\cdot\frac{be^{st}\sinh\left(x\sqrt{s}/c\right)}{s\sinh\left(a\sqrt{s}/c\right)}\right\}$$

$$= b\lim_{s\to -\alpha_n^2}\left\{\frac{s+\alpha_n^2}{\sinh\left(a\sqrt{s}/c\right)}\right\}\cdot\lim_{s\to -\alpha_n^2}\left\{\frac{e^{st}\sinh\left(x\sqrt{s}/c\right)}{s}\right\} \quad \left[\frac{0}{0}\text{ form}\right]$$

$$= b\lim_{s\to -\alpha_n^2}\frac{1}{a\cosh\left(a\sqrt{s}/c\right)\cdot\frac{1}{2}s^{-1/2}/c}\cdot\lim_{s\to -\alpha_n^2}\left\{\frac{e^{st}\sinh\left(x\sqrt{s}/c\right)}{s}\right\}$$

$$= \frac{4bc^2}{2n\pi c}e^{-\alpha_n^2 t}\sin\left(i\alpha_n x/c\right) = \frac{2bc}{n\pi}e^{-\alpha_n^2 t}\sin\frac{n\pi x}{a}$$

Thus we get $u(x,t) = u_p + \dfrac{4bc}{\pi}\sum_{n=1}^{\infty}\dfrac{1-(-1)^n}{2n-1}e^{-\alpha_n^2 t}\sin\dfrac{(2n-1)\pi x}{2c}$

Vibrations of a string

Example 10.24. A tightly stretched string with fixed end points $x = 0$ and $x = l$ is initially at rest in its equilibrium position. If it is set vibrating by giving to each of its points a velocity $\lambda x(l-x)$, find the displacement of the string. (U.P.A.M.U. 2005)

Solution. The equation for the vibration of the string is

$$\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2} \quad \text{...(1)}$$

and the initial conditions are

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = \lambda x(x-l), \quad y(x,0) = f(x) \quad \text{...(2)}$$

Multiplying (1) by e^{-st} and integrating w.r.t. t from $t = 0$ to $t = \infty$, we get

$$\frac{d^2Y}{dt^2} + c^2\omega^2 Y = 0 \qquad \text{provided } y \text{ and } \partial y/\partial t \to 0 \text{ as } x \to \infty$$

$\therefore$ a solution of $\frac{d^2Y}{dt^2} + c^2\omega^2 Y = 0$ is $\quad Y = A_1 \cos \omega ct + A_2 \sin \omega ct$ ...(iii)

Also Fourier transforms of $f(x)$ are

$$\partial y/\partial t = 0 \quad \text{and} \quad Y = f(x) \text{ when } t = 0$$

Applying these to (iii), we get

$$A_1 = 0 \quad \text{and} \quad A_2 = f(x)$$

Thus $\qquad Y = f(x) \sin \omega ct$

Now taking inverse Fourier transforms, we get

$$y(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) (\cos \omega ct - i \sin \omega ct) e^{-i\omega x} dx$$

$$= \frac{1}{4\pi i} \int_{-\infty}^{\infty} f(x) \frac{e^{i\omega ct} - e^{-i\omega ct}}{2} e^{-i\omega x} dx$$

$$= \frac{1}{4\pi i} \int_{-\infty}^{\infty} [f(x) e^{-i\omega(x - ct)} - f(x) e^{-i\omega(x + ct)}] dx$$

$$= \frac{1}{2} [f(x - ct) + f(x + ct)] \qquad \text{by } f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

Example 22.23. An infinite string is stretched between two fixed points $x = 0$ and $x = \pi$ and initially at rest along the x-axis. The points of the string are given initial displacement $f(x) = kx(\pi - x)$. Find the displacement of any point of the string at any time.

Solution. Let $y(x, t)$ be the transverse displacement of any point x of the string at any time t. Then we have to solve the wave equation (§ 18.4)

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \qquad (x > 0, t > 0) \qquad ...(i)$$

subject to the conditions: $y(x, 0) = 0$, $y_t(x, 0) = 0$, $y(0, t) = y(\pi, t)$ and the displacement $y(x, t)$ is bounded.

The Laplace transform of (i), writing $Y(p, t) = \mathcal{L}\{y(x, t)\}$ is

$$p^2 Y - py(x, 0) - y_t(x, 0) = c^2 \frac{d^2 Y}{dx^2}$$

Using the first two conditions, we have

$$\frac{d^2 Y}{dx^2} - \left(\frac{p}{c}\right)^2 Y = 0 \qquad ...(ii)$$

Also by the Laplace transforms of the third and fourth conditions are

$$Y(0, t) = Y(\pi, t) \quad \text{or} \quad x = 0 \qquad ...(iii) \quad \text{and} \quad Y(x, t) \text{ is bounded.} \qquad ...(iv)$$

Solving (ii), we get

$$Y(x, p) = C_1 e^{px/c} + C_2 e^{-px/c}$$

To satisfy condition (iv), we must have $C_1 = 0$

Using the condition (iii), we get $C_2 = Y(\pi, t)$

$\therefore \qquad Y(x, p) = Y(\pi, t) e^{-px/c}$

Using the complex inversion formula, we obtain

$$y = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} e^{(t - x/c)p} Y(\pi, t) dp = f(t - x/c)$$

Example 22.24. A tightly stretched flexible string has its ends fixed at $x = 0$ and $x = l$. At time $t = 0$, the string is given a shape defined by $f(x) = \mu x(l - x)$, where μ is a constant and then released. Find the displacement of any point of the string at any time $t > 0$. (U.P.T.U., M.C.A., 2005)

Solution. We have to solve the wave equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$ $\qquad (x > 0, t > 0)$

subject to the conditions $y(0, t) = 0, y(l, t) = 0$

and $y(x, 0) = a\sin\frac{\pi x}{l}, \left(\frac{\partial y}{\partial t}\right)_{t=0} = b$

Now taking Laplace transform, writing $L\{y(x, t)\} = Y(x, s)$, we get

$$s^2Y - sy(x, 0) - \left(\frac{\partial y}{\partial t}\right)_{t=0} = c^2\frac{d^2Y}{dx^2} \quad \ldots(1)$$

where $Y(0, s) = 0, Y(l, s) = 0$ $\ldots(2)$

$\therefore$ (1) reduces to $\frac{d^2Y}{dx^2} - \left(\frac{s}{c}\right)^2 Y = -\frac{a\sin\frac{\pi x}{l} - b}{c^2}$

Its solution is $Y(x, s) = c_1\cosh\left(\frac{sx}{c}\right) + c_2\sinh\left(\frac{sx}{c}\right) + \frac{a\left(\frac{\pi}{l}\right)^2 - b}{s^2} - \frac{2a\left(\frac{\pi}{l}\right)^2}{s^2}$

Applying the conditions (2), we get

$$c_1 = 2a\left(\frac{\pi}{l}\right)^2 \text{ and } c_2 = \frac{2a\left(\frac{\pi}{l}\right)^2}{s^2}\left[\frac{1 - \cosh\left(\frac{sl}{c}\right)}{\sinh\left(\frac{sl}{c}\right)}\right] - \frac{2a\left(\frac{\pi}{l}\right)^2}{s^2}\tanh\left(\frac{sl}{2c}\right)$$

Thus $Y(x, s) = \frac{2a\left(\frac{\pi}{l}\right)^2}{s^2}\left[\frac{\cosh\left(\frac{sl}{2c} - \frac{sx}{c}\right)}{\cosh\left(\frac{sl}{2c}\right)}\right] + \frac{a\left(\frac{\pi}{l}\right)^2 - b}{s^2} - \frac{2a\left(\frac{\pi}{l}\right)^2}{s^2}$

Now using the inversion formula (2) § 22.10, we get

$y(x, t) =$ sum of the residues of

$$2a\left(\frac{\pi}{l}\right)^2\left[e^{st}\frac{\cosh\left(\frac{sl}{2c} - \frac{sx}{c}\right)}{s^2\cosh\left(\frac{sl}{2c}\right)}\right] \text{ at all the poles } + a\sin\frac{\pi x}{l} - bt - a\left(\frac{\pi}{l}\right)^2$$

Proceeding exactly as in Example 22.28, we have,

sum of the residues of $2a\left(\frac{\pi}{l}\right)^2\left[\frac{e^{st}\cosh\left(\frac{sl}{2c} - \frac{isx}{c}\right)}{s^2\cosh\frac{isl}{2c}}\right]$ at all the poles

$$= a\left(\frac{\pi}{l}\right)^2\left[t^2 + \left(\frac{2x - l}{2c}\right)^2 - \left(\frac{l}{2c}\right)^2\right]$$

$$\frac{32a\left(\frac{\pi}{l}\right)^2}{\pi^3}\left(\frac{l}{2c}\right)^2\sum_{n=1}^{\infty}\left[\frac{(-1)^n}{(2n-1)^3}\cos\left\{\frac{(2n-1)\pi(2x-l)}{2l}\right\}\cos\left\{\frac{(2n-1)\pi ct}{l}\right\}\right]$$

$$= c^2\left(\frac{\pi}{l}\right)^2t^2 - a\sin\frac{\pi x}{l} + \frac{8a\left(\frac{\pi}{l}\right)^2}{\pi^3}\sum_{n=1}^{\infty}\left[\frac{1}{(2n-1)^3}\sin\frac{(2n-1)\pi x}{l}\cos\frac{(2n-1)\pi ct}{l}\right]$$

Hence $y(x, t) = \frac{8a\left(\frac{\pi}{l}\right)^2}{\pi^3}\sum_{n=1}^{\infty}\left[\frac{1}{(2n-1)^3}\sin\frac{(2n-1)\pi x}{l}\cos\frac{(2n-1)\pi ct}{l}\right]$

Transmission line theory

Example 22.29. A semi-infinite transmission line of negligible resistance and conductance per unit length is initially at zero potential and current. A constant voltage E_0 is applied at the sending end $x = 0$ for $t > 0$. Find the voltage and current at any point x at time t.

Solution. Let $v(x, t)$ and $i(x, t)$ be the voltage and current at any point x and at any time t. If $G = 0$ and $R = 0$, then the transmission line equations [(11) and (21) of § 18.20] become

$$\frac{\partial v}{\partial x} = -Ri, \frac{\partial i}{\partial x} = -C\frac{\partial v}{\partial t} \text{ i.e., } \frac{\partial^2 v}{\partial x^2} = RC\frac{\partial v}{\partial t} \quad \ldots(1)$$

The boundary conditions are $v(0, t) = E_0$ and $v(x, t)$ is finite for all x and t.

The initial conditions are $v(x, 0) = 0, i(x, 0) = 0$. $\ldots(2)$

Laplace transforms of (i), are

$$\frac{d^2\bar{y}}{dx^2} - s\bar{y}(x) = 0 \quad \text{or} \quad \frac{d^2\bar{y}}{dx^2} - s\bar{y} = 0 \qquad \ldots(iii)$$

Laplace transforms of the conditions in (ii), are

$$\bar{y}(0, s) = \frac{y_0}{s} \text{ at } x = 0 \qquad \ldots(iv)$$

and $\bar{y}(x, s)$ remains finite as $x \to \infty$ $\qquad \ldots(v)$

∴ The solution of (iii) is

$$\bar{y}(x, s) = C_1 e^{\sqrt{s}x} + C_2 e^{-\sqrt{s}x}$$

To satisfy condition (v), we must have $C_1 = 0$.

Using the condition (iv), we get $C_2 = y_0/s$

Thus $$\bar{y}(x, s) = \frac{y_0}{s} e^{-\sqrt{s}x}$$

Using the inversion formula, we obtain

$$y(x, t) = y_0 L^{-1}\left\{\frac{e^{-\sqrt{s}x}}{s}\right\} = y_0 \operatorname{erfc}\left[x\frac{\sqrt{\rho c}}{2\sqrt{kt}}\right] \qquad \text{[By Eq. (2.2.17)]}$$

$$= y_0 \frac{x\sqrt{\rho c}}{2\sqrt{k}} \int_x^\infty e^{-\alpha x^2 \rho c/4kt} \, dx$$

∴ $q(x, t) = -\dfrac{k}{\rho c}\dfrac{\partial y}{\partial x}$, on obtain by differentiation,

$$q(x, t) = \frac{y_0 \rho c}{2\sqrt{\pi}} \sqrt{\frac{k}{t}} e^{-x^2 \rho c/4kt}$$

Example 2.2.20. A two conductor line of length 1 km is initially at temperature and potential. A constant voltage y_0 is applied at the sending end (x = 0) and the line is connected to the receiving end. Assuming the receiving end current and current to be zero, determine the voltage and current.

Solution. For a transmission line with $L = G = 0$, the voltage y and current i are given by the equations

$$\frac{\partial^2 y}{\partial x^2} = RC\frac{\partial y}{\partial t} \text{ and } \frac{\partial i}{\partial x} + Ry = 0 \qquad \ldots(i)$$

The boundary conditions are (for $t > 0$)

$$y = y_0 \text{ at } x = 0 \text{ and } i = \frac{\partial y}{\partial x} = 0 \text{ at } x = l \qquad \ldots(ii)$$

The initial condition is $y = 0$ at $t = 0$ ($x > 0$).

Laplace transforms of (i) and (ii) are

$$\frac{d^2\bar{y}}{dx^2} = RC s\bar{y} \text{ or } \frac{d^2\bar{y}}{dx^2} - RCs\bar{y} = 0 \qquad \ldots(iii)$$

and $$\bar{y} = \frac{y_0}{s} \text{ at } x = 0, \quad \frac{d\bar{y}}{dx} = 0 \text{ at } x = l \qquad \ldots(iv)$$

∴ The solution of (iii) is

$$\bar{y} = c_1 \cosh\sqrt{RCs}\,x + c_2 \sinh\sqrt{RCs}\,x$$

Applying conditions (iv), it gives

$$c_1 = c_1, 0 = c_1 \sinh\sqrt{RCs}\,l + c_2 \cosh\sqrt{RCs}\,l$$

$$\bar{y} = \frac{y_0}{s}\left[\cosh\sqrt{RCs}\,x - \frac{\sinh\sqrt{RCs}\,l}{\cosh\sqrt{RCs}\,l}\cosh\sqrt{RCs}\,x\right]$$

$$= \frac{k_0}{k}\,\frac{\sinh(ap)\sqrt{\frac{s}{k}}}{\sinh p\sqrt{s}} \qquad \text{where } p = \sqrt{\frac{s}{k}}(x-a) \text{ and } p_0 = \sqrt{\frac{s}{k}}(-a)$$

By the inversion formula (5) § 22.10, we get

$$v(x,t) = \text{sum of the residues of } e^{st}\,\bar{v}(x,s) \text{ at all poles of } e^{st}\,\bar{v}(x,s) \qquad \ldots(*)$$

These poles are at $s = 0$ and $p\sqrt{\frac{s}{k}} = \pm i(2n-1)\pi/2 = \pm ip\lambda$ (say)

Now $\quad \text{Res}\left(e^{st}\bar{v}\right)_{s=0} = \lim_{s\to 0} \frac{se^{st}k_0\cosh p\sqrt{\frac{s}{k}}(x-a)}{s\cosh p\sqrt{\frac{s}{k}}a} = k_0$

and $\quad \text{Res}\left(e^{st}\bar{v}\right)_{s=-k\lambda^2} = \lim_{s\to -k\lambda^2} \frac{(s+k\lambda^2)e^{st}k_0\cosh p\sqrt{\frac{s}{k}}(x-a)}{s\cosh p\sqrt{\frac{s}{k}}a} \qquad \left[\frac{0}{0}\text{ form}\right]$

$$= \lim_{s\to -k\lambda^2} \frac{k_0 e^{st}\cosh p\sqrt{\frac{s}{k}}(x-a) + (s+k\lambda^2)k_0(\ldots)}{\cosh p\sqrt{\frac{s}{k}}a + a\sinh p\sqrt{\frac{s}{k}}\cdot\frac{1}{2}ps^{-1/2}}$$

$$= \frac{k_0 e^{-k\lambda^2 t}\cosh(ip\lambda)(x-a)}{\cosh(ip\lambda)a + a\sinh(ip\lambda)\frac{1}{2}ip\lambda^{-1}} = \frac{2k_0 e^{-k\lambda^2 t}\cos p\lambda(x-a)}{ip\lambda\sin p\lambda a}$$

Adding up all the residues, (*) gives

$$v(x,t) = k_0 + \frac{4k_0}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{2n-1}e^{-(2n-1)^2\pi^2 kt/4L^2}\cos\left[(2n-1)\pi/2L\right]$$

$$\left(\because\ p\lambda = (2n-1)\pi/2,\ -\sin ip\lambda = (-1)^n,\ pa\lambda = \tfrac{1}{2}(2n-1)\pi = n\pi,\ a^2 = (2n-1)^2\pi^2/4L^2\right]$$

Also $\quad i = -\frac{1}{R}\frac{\partial v}{\partial x}$. [By (4)]

PROBLEMS 22.4

1. Solve the differential equation by the Laplace transform method: $\frac{\partial v}{\partial t} = k\frac{\partial^2 v}{\partial x^2}$

$x \ge 0, t \ge 0$; $v(0,t) = k_0$; $v(x,0) = 0$; $\lim_{x\to\infty} v(x,t) = 0$ (as $x \to \infty$). [U.P.T.U., 2002]

2. Using double transforms, solve the differential equation $\frac{\partial^2 V}{\partial x^2} = \frac{\partial V}{\partial t}$, $0 < x < \infty$, $t > 0$

if $V(0,t) = 0 = V(\infty,t)$ and $V(x,0) = V_0$ (constant).

3. The initial temperature along the length of an infinite bar is given by $v(x,0) = \begin{cases} T_0, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$. If the temperature $v(x,t)$ satisfies the equation $\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$, $-\infty < x < \infty$, $t > 0$, find the temperature at any point of the bar at any time t.

(Kanpur, 2003)

4. Use the complex form of the Fourier transform to show that

$$v = \frac{1}{2\sqrt{\pi kt}}\int_{-\infty}^{\infty} f(\alpha)e^{-(x-\alpha)^2/4kt}\,d\alpha$$

is the solution of the boundary value problem

$$\frac{\partial v}{\partial t} = k\frac{\partial^2 v}{\partial x^2}, \quad -\infty < x < \infty,\ t > 0,\ v = f(x) \text{ when } t = 0. \qquad [U.P.T.U., 2004(O)]$$

5. A semi-infinite solid $(x \ge 0)$ is initially at temperature zero. At time $t = 0$, a constant temperature v_0 is applied and maintained on the face $x = 0$. Show that the temperature at any point x and at any time t is given by $v(x,t) = v_0\,\text{erfc}\left(\frac{x}{2\sqrt{kt}}\right)$

8. A solid is initially at zero temperature throughout. Its surface $x = 0$ is kept at zero temperature and the temperature at any point of the solid at any time $t > 0$ is required to be found. Determine the temperature at any point of the solid at any time $t > 0$.

9. An infinite string is initially at rest along the x-axis. Its one end which is at $x = 0$ is given a periodic transverse displacement $a \sin \omega t$, $t > 0$. Show that the displacement of any point of the string at any time is given by

$$y(x, t) = \frac{2a\omega}{\pi} \int_0^\infty \frac{\sin \lambda x \sin \lambda ct}{\lambda (\omega^2 - \lambda^2 c^2)} d\lambda$$

where c is the wave velocity.

10. An infinite string has an initial transverse displacement $y(x, 0) = f(x)$, $-\infty < x < \infty$, and is released from rest. Show that

$$y(x, t) = \frac{1}{2}[f(x + ct) + f(x - ct)].$$

11. A semi-infinite transmission line has negligible inductance and leakance per unit length. A voltage v_0 is applied at the sending end $(x = 0)$ which is given by

$$v(0, t) = \begin{cases} v_0, & 0 < t < \tau \\ 0, & t > \tau \end{cases}$$

Show that the voltage at any point $x > 0$ at any time $t > 0$ is given by

$$v(x, t) = v_0 u(t) \left[\frac{x}{2} \operatorname{erfc}\left(\frac{x}{2\sqrt{t}}\right)\right].$$

3.13 OBJECTIVE TYPE OF QUESTIONS

Multiple Choice

Choose the most suitable answer in each of the following problems.

1. Fourier cosine transform of $f(x)$ is _____.
2. Fourier sine transform of $f(x)$ is _____.
3. Convolution theorem for Fourier transform states that _____.
4. If Fourier transform of $f(x)$ is $F(s)$, then the inversion formula is _____.
5. $\mathcal{F}[e^{iax} f(x)] =$ _____.
6. If $\mathcal{F}[f(x)] = F(s)$, then $\mathcal{F}[f(x - a)] =$ _____.
7. Fourier sine integral representation of a function $f(x)$ is given by _____.
8. If $\mathcal{F}_s[f(x)] = F_s(s)$, then $\mathcal{F}_s[f(ax)] =$ _____.
9. Fourier transform of unit impulse at $x = a$, $\delta(x - a)$ is _____.
10. If $f(x) = \begin{cases} 1, & 0 < x < a \\ 0, & x > a \end{cases}$, then Fourier sine integral of $f(x)$ is _____.
11. Fourier sine transform of $f'(x)$ in the interval $(0, \infty)$ is _____.
12. If $F(s)$ is the Fourier transform of $f(x)$, then the Fourier transform of $f(ax)$ is _____.
13. Inverse finite Fourier sine transform of $F_s(n) = \frac{2}{c} \int_0^c f(x) \sin \frac{n\pi x}{c} dx$ for $n = 1, 2, 3, \ldots$ and $0 < x < c$ is _____.
14. If Fourier transform of $f(x) = F(s)$, then Fourier Transform of $f(ax)$ is _____.
15. Fourier cosine transform of e^{-ax} is _____.
16. $f(x)$ is L_2 over a rectangle for expression for a Fourier integral. (True or False)
17. $\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |F(s)|^2 ds$. (True or False)
18. Fourier transform is a linear operator. (True or False)
19. $\mathcal{F}_s[xf(x)] = -\frac{d}{ds} F_s(s)$. (True or False)
20. Kernel of Fourier transform is e^{isx}. (True or False)
21. Finite Fourier cosine transform of $f(x) = 1$ in $(0, \pi)$ is zero. (True or False)

CHAPTER 23

Z-Transforms

1. Introduction. 2. Definition. 3. Some standard Z-transforms. 4. Linearity property. 5. Damping rule. 6. Shifting rule. 7. Z-differentiation. 8. Z-integration. 9. Multiplication by n. 10. Two Basic Theorems. 11. Some more Z-transforms. 12. Initial value theorem. 13. Convolution theorem. 14. Correlation of Z-transforms. 15. Partial fraction method. 16. Inversion of Z-transforms. 17. Application to difference equations. 18. Discrete Fourier Transform.

23.1 INTRODUCTION

The development of the new technique known as discrete analysis, Z-transform plays the same role in discrete analysis as Laplace transform in continuous systems. As we shall see Z-transform has many properties similar to those of the Laplace transform (Ch 22). The main difference is that the Z-transform operates on functions of discrete variables represented by sequences of the discrete instants of time, i.e. $n = 0, 1, 2, \ldots$. The analogy of Laplace transform to Z-transform can be carried further. For every operational rule of Laplace transform, there is a corresponding operational rule of Z-transform and for every application of the Laplace transform, there is a corresponding application of Z-transform. A discrete system is expressible as a difference equation (§ 16.2) and its solutions are found using Z-transforms.

23.2 DEFINITION

If $f(n)$, a function of n, is defined for discrete values of $n = 0, 1, 2, \ldots$ and $f(n) = 0$ for $n < 0$, then its Z-transform is defined to be

$$Z\{f(n)\} = F(z) = \sum_{n=0}^{\infty} f(n) z^{-n} \quad \text{... (1)}$$

The inverse Z-transform is written as $Z^{-1}\{F(z)\} = f(n)$.

If we insert a particular complex number z into the power series (1), the resulting value of $F(z)$ will be a complex number. Thus the Z-transform $F(z)$ is a complex valued function of a complex variable z.

23.3 SOME STANDARD Z-TRANSFORMS

The direct application of the definition gives the following results:

(i) $Z(a^n) = \dfrac{z}{z-a}$ (ii) $Z(n^m) = -z\dfrac{d}{dz}Z(n^{m-1})$, where m is a positive integer.

Proof: (i) By definition, $Z(a^n) = \sum_{n=0}^{\infty} a^n z^{-n}$

$$= 1 + (az) + (az)^2 + (az)^3 + \ldots = \frac{1}{1-(az)} = \frac{z}{z-a} \qquad \text{(S.V.Univ. 2012)}$$

(2) $$Z\{n^p\} = \sum_{n=0}^{\infty} n^p z^{-n} = z\sum_{n=0}^{\infty} n^{p-1}\,(n\,z^{-n-1}) \qquad \ldots(1)$$

Changing p to $p-1$, we get $Z\{n^{p-1}\} = \sum_{n=0}^{\infty} n^{p-1} z^{-n}$

Differentiating w.r.t. z,

$$\frac{d}{dz}[Z\{n^{p-1}\}] = \sum_{n=0}^{\infty} n^{p-1}(-n)z^{-n-1} \qquad \ldots(2)$$

Substituting (2) in (1), we obtain $Z\{n^p\} = -z\frac{d}{dz}[Z\{n^{p-1}\}]$

which is the desired recurrence formula.

In particular, we have the following formulae :

(1) $Z\{1\} = \frac{z}{z-1}$ [Taking $p = 1$ in (1)] (2) $Z\{n\} = \frac{z}{(z-1)^2}$ [Taking $p = 1$ in (2)]

(3) $Z\{n^2\} = \frac{z^2+z}{(z-1)^3}$ (V.T.U., 2006) (4) $Z\{n^3\} = \frac{z^3+4z^2+z}{(z-1)^4}$

(5) $Z\{n^4\} = \frac{z^4+11z^3+11z^2+z}{(z-1)^5}$

2.12 LINEARITY PROPERTY

If a, b, c be any constants and u_n, v_n, w_n be any sequences whose z-transforms exist, then

$$Z\{au_n + bv_n - cw_n\} = aZ\{u_n\} + bZ\{v_n\} - cZ\{w_n\}$$

Proof. By definition, $Z\{au_n + bv_n - cw_n\} = \sum_{n=0}^{\infty} (au_n + bv_n - cw_n)z^{-n}$

$$= a\sum_{n=0}^{\infty} u_n z^{-n} + b\sum_{n=0}^{\infty} v_n z^{-n} - c\sum_{n=0}^{\infty} w_n z^{-n}$$

$$= aZ\{u_n\} + bZ\{v_n\} - cZ\{w_n\}$$

2.13 DAMPING RULE

If $Z\{u_n\} = U(z)$, then $Z\{a^n u_n\} = U(z/a)$

Proof. By definition, $Z\{a^n u_n\} = \sum_{n=0}^{\infty} a^{-n} u_n \cdot z^{-n} = \sum_{n=0}^{\infty} u_n \cdot (az)^{-n} = U(z/a)$. (Madras, 2005)

Cor. $Z\{a^n u_{n-1}\} = U(z/a)$

This theorem holds if $|z/a| > 1$, i.e., if $|z| > |a|$, where the function u_n has the same domain as $a^n u_n$.

2.14 SOME STANDARD RESULTS

The application of the damping rule leads to the following standard results :

(1) $Z\{a^n\} = \frac{az}{(z-a)^2}$ (2) $Z\{a^n n^2\} = \frac{az^2+a^2z}{(z-a)^3}$

(13) $Z(r^n \cos n\theta) = \dfrac{z(z - r\cos\theta)}{z^2 - 2rz\cos\theta + r^2}$ (14) $Z(r^n \sin n\theta) = \dfrac{rz\sin\theta}{z^2 - 2rz\cos\theta + r^2}$

(15) $Z(e^{an}\cos n\theta) = \dfrac{z(z - e^a\cos\theta)}{z^2 - 2ez\cos\theta + e^{2a}}$ (16) $Z(e^{an}\sin n\theta) = \dfrac{ze^a\sin\theta}{z^2 - 2ze^a\cos\theta + e^{2a}}$.

Proofs. (13) We know that $Z(a^n) = \dfrac{z}{(z-a)^2}$. Applying damping rule, we have

$$Z(\cos n\theta + i\sin n\theta) = \frac{z^2 z}{(z^2 z - 1)^2} = \frac{z^3}{(z-e^{i\theta})^2} \qquad \text{(Madras, 2000 S)}$$

(14) We know that $Z(n^2) = \dfrac{z^2 + z}{(z-1)^3}$. Applying damping rule, we have

$$Z(n^2 r^n) = Z(z^{-n}) = \frac{(z^{-1})^2 + z^{-1}}{(z^{-1})^3} = \frac{z(z^2 + z)}{(1-z)^3}.$$

(15) and (16) We know that $Z(1) = \dfrac{z}{z-1}$. Applying damping rule, we have

$$Z(e^{an}) = Z(e^{an})^n \cdot 1 = \frac{z^n}{z^n - 1} = \frac{z}{z - e^{-an}} = \frac{z(z - e^{an})}{(z - e^{-an})(z - e^{an})}$$

$$= \frac{z^2 - z(e^{an} + e^{-an})}{z^2 - z(e^{an} + e^{-an}) + 1} = \frac{z^2 - z\cdot 2\cos n\theta}{z^2 - 2z\cos n\theta + 1}$$

Equating real and imaginary parts, we get (15) and (16). (U.P.T.U., 2015 S; Anna, 2009)

(13) We know that $Z\{\cos n\theta\} = \dfrac{z(z - \cos\theta)}{z^2 - 2z\cos\theta + 1}$. By damping rule, we have

$$Z(r^n \cos n\theta) = \frac{z'(z' - \cos\theta)}{z'^2 - 2z'\cos\theta + 1} = \frac{z(z - r\cos\theta)}{z^2 - 2rz\cos\theta + r^2} \qquad \text{(V.T.U., 2010)}$$

Similarly using (1) above, we get (14).

Example 3.21. *Find $Z(n)$, $Z(n^2)$ and $Z(n^3)$ of the following sequences:*

(1) $\{n\}$ (2) $\{n^2\}$ (3) $\{n^3\}$ *(V.T.U., 2010)*

(4) $\{n(n+1)\}$ *(V.T.U., 2009 S; Kerala, 2005)*

Solution. (1) $Z\{n\} = -z\dfrac{d}{dz}\left\{\dfrac{z}{z-1}\right\} = -z\left\{\dfrac{(z-1)\cdot 1 - z\cdot 1}{(z-1)^2}\right\} = z\dfrac{d}{dz}\left(\dfrac{1}{z-1}\right)$ (By linearity property)

$$= z\cdot\frac{-1}{(z-1)^2} = \frac{z}{(z-1)^2} \qquad \text{(Using formulae for } Z\{1\}, Z\{a^n\}, Z\{n\} \text{ etc)}$$

(2) $Z\{n^2\} = Z\{n\cdot n\} = -z\dfrac{d}{dz}Z\{n\} = -z\dfrac{d}{dz}\left\{\dfrac{z}{(z-1)^2}\right\} = -z\left\{\dfrac{(z-1)^2\cdot 1 - z\cdot 2(z-1)}{(z-1)^4}\right\}$

$$= -z\left\{\frac{(z-1)[(z-1) - 2z]}{(z-1)^4}\right\} = -z\left\{\frac{-(z+1)}{(z-1)^3}\right\} = \frac{z(z+1)}{(z-1)^3}$$

(3) $Z\{n^3\} = Z\{n^2\cdot n\} = -z\dfrac{d}{dz}Z\{n^2\} = -z\dfrac{d}{dz}\left\{\dfrac{z(z+1)}{(z-1)^3}\right\} = -z\left\{\dfrac{(z-1)^3(2z+1) - z(z+1)3(z-1)^2}{(z-1)^6}\right\}$

$$= -z\left\{\frac{(z-1)^2[(z-1)(2z+1) - 3z(z+1)]}{(z-1)^6}\right\} = -z\left\{\frac{-(2z^2+4z+1)}{(z-1)^4}\right\} = \frac{z(2z^2+4z+1)}{(z-1)^4}$$

(4) $Z\{n(n+1)\} = Z\{n^2 + n\} = Z\{n^2\} + Z\{n\} = \dfrac{z(z+1)}{(z-1)^3} + \dfrac{z}{(z-1)^2} = \dfrac{z^2(z+1)}{(z-1)^3}$

(iii) $Z\{\sin(3n+2)\} = Z\{\sin 3n\cos 2 + \cos 3n\sin 2\}$

$= \cos 2\,Z\{\sin 3n\} + \sin 2\,Z\{\cos 3n\}$ (using formulae for $Z\{\sin n\theta\}$, $Z\{\cos n\theta\}$)

$$= \cos 2\cdot\frac{z\sin 3}{z^2 - 2z\cos 3 + 1} + \sin 2\cdot\frac{z(z - \cos 3)}{z^2 - 2z\cos 3 + 1} = z\cdot\frac{z\sin 2 - \sin 3}{z^2 - 2z\cos 3 + 1}.$$

Example 3.22. Find the Z-transform of the following:
(i) a^n (ii) $n(n+1)$ (iii) n^2

Solution. (i) Let $u_n = 1$. $a^n = (a^{-1})^{-n} = k^{-n}$ where $k = a^{-1}$. By damping rule $Z(k^{-n} u_n) = U(kz)$,

$$\therefore \quad Z(a^n) = Z(k^{-n}),\ U(z) = U(kz) = \frac{kz}{kz-1} \qquad \left[\because\ U(z) = Z(1) = \frac{z}{z-1}\right]$$

$$= \frac{a^{-1}z}{a^{-1}z-1} = \frac{z}{z-a}$$

(ii) Let $u_n = n$, $a^n = (a^{-1})^{-n} = k^{-n}$ where $k = a^{-1}$

By damping rule, $Z(a^{-n} \cdot n) = Z(k^{-n} \cdot n) = U(kz)$ where $U(z) = Z(n) = \frac{z}{(z-1)^2}$

$$\frac{kz}{(kz-1)^2} = \frac{z}{(kz-1)k^2} = \frac{a^2 z}{(z-a^2)^2}$$

(iii) Let $u_n = n^2$, $a^n = (a^{-1})^{-n} = k^{-n}$ where $k = a^{-1}$

By damping rule,

$$Z(a^{-n} \cdot n^2) = Z(k^{-n} \cdot n^2) = U(kz) \quad \text{where} \quad U(z) = Z(n^2) = \frac{z^2+z}{(z-1)^3}$$

$$= \frac{(kz)^2 + kz}{(kz-1)^3} = \frac{a^2 z + az}{(z-a)^3} = \frac{az(az+a^2)}{(z-a)^3}.$$

Example 3.23. Find the Z-transform of (i) $\sinh n\theta$ (ii) $\cosh n\theta$ (iii) $a^n \sinh n\theta$ (iv) $a^n \cosh n\theta$

Solution. (i) $Z(\sinh n\theta) = Z\left[\frac{e^{n\theta} - e^{-n\theta}}{2}\right]$

$$= \frac{1}{2}\left[Z\left[(e^{\theta})^n - 1\right] + Z\left[(e^{\theta})^{-n}\right]\right]$$

Apply damping rule to both terms, taking $u_n = 1$.

$$Z(\sinh n\theta) = \frac{1}{2}\left[\frac{z}{z-1} + \frac{z}{e^{\theta}z - 1}\right] \qquad \left[\because\ u(z) = \frac{z}{z-1}\right]$$

$$= \frac{1}{2}\left[\frac{2z^2 - z(e^{\theta} + e^{-\theta})}{z^2 - z(e^{\theta} + e^{-\theta}) + 1}\right] = \frac{z^2 - z\cosh\theta}{z^2 - 2z\cosh\theta + 1}$$

(iii) $Z(a^n \cosh n\theta) = Z(a^{-n} \cdot \cosh n\theta)$ (Apply damping rule using (i))

$$= \frac{(a^{-1}z)^2 - (a^{-1}z)\cosh\theta}{(a^{-1}z)^2 - 2(a^{-1}z)\cosh\theta + 1} = \frac{z(z - a\cosh\theta)}{z^2 - 2az\cosh\theta + a^2}.$$

Example 3.24. Find the Z-transform of
(i) $e^{an}\sin bn$ (Madras, 2003) (ii) $e^{an}\cos bn$, $(a, b > 0)$ (U.P.T.U. 2004 S)

Solution. (i) We know that $Z(\sin bn) = \frac{z\sin b}{z^2 - 2z\cos b + 1}$ ...(A)

$$\therefore \quad Z(e^{an}\sin bn) = Z\left[(e^{-a})^{-n} \cdot \sin bn\right] \qquad \text{[Apply damping rule, using (A)]}$$

$$= \frac{(e^{-a}z)\sin b}{(e^{-a}z)^2 - 2(e^{-a}z)\cos b + 1} = \frac{ze^{a}\sin b}{z^2 - 2ze^{a}\cos b + e^{2a}}$$

(ii) We know $(B.u.)$ $Z(\cos n\theta) = \frac{z^2 - z\cos\theta}{z^2 - 2z\cos\theta + 1}$ ...(B)

$$\therefore \quad Z(a^n \cos bn) = Z\left[(a^{-1})^{-n} \cdot \cos bn\right] \qquad \text{[Apply damping rule, using (B)]}$$

$$= \frac{(a^{-1}z)^2 - (a^{-1}z)\cos b}{(a^{-1}z)^2 - 2(a^{-1}z)\cos b + 1} = \frac{z(z - a\cos b)}{z^2 - 2az\cos b + a^2}$$

Example 8.26. Find the Z-transforms of

(i) $\cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right)$ (ii) $2^n\cos\left(\frac{n\pi}{2}+\theta\right)$. (iii) $\sin\left(\frac{n\pi}{2}+\theta\right)$. (iv) $3^n\cos\left(\frac{n\pi}{2}+\theta\right)$.

Solution. (i) $Z\left[\cos\left(\frac{n\pi}{2}+\frac{\pi}{4}\right)\right]=Z\left[\cos\frac{n\pi}{2}\cos\frac{\pi}{4}-\sin\frac{n\pi}{2}\sin\frac{\pi}{4}\right]$

$=\cos\frac{\pi}{4}.Z\left[\cos\frac{n\pi}{2}\right]-\sin\frac{\pi}{4}.Z\left[\sin\frac{n\pi}{2}\right]$ [Using formulae for $Z(\sin n\theta)$ and $Z(\cos n\theta)$]

$=\frac{1}{\sqrt{2}}\left[\frac{z(z-\cos\pi/2)}{z^2-2z\cos\pi/2+1}-\frac{z\sin\pi/2}{z^2-2z\cos\pi/2+1}\right]=\frac{1}{\sqrt{2}}\left(\frac{z^2}{z^2+1}-\frac{z}{z^2+1}\right)=\frac{z(z-1)}{\sqrt{2}(z^2+1)}$

(ii) $Z\left[\cos\left(\frac{n\pi}{2}+\theta\right)\right]=Z\left[\frac{e^{i(n\pi/2+\theta)}+e^{-i(n\pi/2+\theta)}}{2}\right]=\frac{1}{2}\left[e^{i\theta}Z(e^{in\pi/2})+e^{-i\theta}Z(e^{-in\pi/2})\right]$

Since, $Z(a^n)=\frac{z}{z-a}$, $\therefore Z(e^{in\pi/2})=Z(i^n)=\frac{z}{z-i}$, $Z(e^{-in\pi/2})=\frac{z}{z-e^{-i\pi/2}}=\frac{z}{z+i}$

Thus $Z\left[\cos\left(\frac{n\pi}{2}+\theta\right)\right]=\frac{1}{2}\left[e^{i\theta}\cdot\frac{z}{z-i}+e^{-i\theta}\cdot\frac{z}{z+i}\right]$

$=\frac{z}{2}\left[\frac{ze^{i\theta}+ze^{-i\theta}-ie^{i\theta+i\pi/2}+ie^{-i\theta-i\pi/2}}{z^2-iz^{1+i\pi/2}+iz^{-i\pi/2}+1}\right]=\frac{z^2\cos\theta-z\cos\left(\frac{\pi}{2}-\theta\right)}{z^2+2z\cos\left(\frac{\pi}{2}\right)+1}$.

Example 8.27. Find the Z-transform of

(i) C_p^n (ii) $a^nC_p^n$

Solution. (i) $Z(C_p^n)=\sum_{n=0}^{\infty}\left[{}^nC_p z^{-n}\right]=1+{}^nC_1z^{-1}+{}^nC_2z^{-2}+\ldots+{}^nC_nz^{-n}=(1+z^{-1})^n$

(ii) $Z(a^nC_p^n)=\sum_{n=0}^{\infty}a^n{}^nC_pz^{-n}$

$=1+a^1{}^nC_1z^{-1}+a^2{}^nC_2z^{-2}+a^3{}^nC_3z^{-3}+\ldots\infty$

$=1+(n)(az^{-1})+\frac{(n)(n-1)}{2!}(az^{-1})^2+\frac{(n)(n-1)(n-2)}{3!}(az^{-1})^3+\ldots\infty$

$=1+(n)(az^{-1})+\frac{n(n-1)}{2!}(az^{-1})^2+\frac{n(n-1)(n-2)}{3!}(az^{-1})^3+\ldots\infty$

$=(1-az^{-1})^{-n}$.

Example 8.28. Find the Z-transform of

(i) unit impulse sequence $\delta(n)=\begin{cases}1 & n=0\\ 0 & n\neq 0\end{cases}$ (ii) unit step sequence $u(n)=\begin{cases}1 & n\geq 0\\ 0 & n<0\end{cases}$

Solution. (i) $Z[\delta(n)]=\sum_{n=0}^{\infty}\delta(n)z^{-n}=1+0+0+\ldots=1$

(ii) $Z[u(n)]=\sum_{n=0}^{\infty}u(n)z^{-n}=1+z^{-1}+z^{-2}+z^{-3}+\ldots=\frac{1}{1-z^{-1}}=\frac{z}{z-1}$.

8.8.1 (1) SHIFTING (x_n) TO THE RIGHT

If $Z\{x_n\} = X(z)$, then $Z\{x_{n-k}\} = z^{-k} X(z)$ $(k > 0)$

Proof. By definition,

$$Z\{x_{n-k}\} = \sum_{n=0}^{\infty} x_{n-k} z^{-n} = x_0 z^{-k} + x_1 z^{-(k+1)} + x_2 z^{-(k+2)} + \ldots = z^{-k} \sum_{n=0}^{\infty} x_n z^{-n} = z^{-k} X(z)$$

Obs. This result will be very useful in applications to difference equations.

(2) **Shifting x_n to the left.** If $Z\{x_n\} = X(z)$, then

$$Z\{x_{n+1}\} = z[X(z) - x_0 - x_1 z^{-1} - x_2 z^{-2} - \ldots - x_{n-1} z^{-(n-1)}]$$

Proof. $Z\{x_{n+1}\} = \sum_{n=0}^{\infty} x_{n+1} z^{-n} = z \sum_{n=0}^{\infty} x_{n+1} z^{-(n+1)}$

$$= z \left[\sum_{n=0}^{\infty} x_n z^{-n} - \sum_{n=0}^{1-1} x_n z^{-n} \right]$$

Hence $Z\{x_{n+1}\} = z[X(z) - x_0 - x_1 z^{-1} - x_2 z^{-2} - \ldots - x_{n-1} z^{-(n-1)}]$ (G.N.D.U., 2002)

As particular cases, we have the following standard results:

(1) $Z\{x_{n+1}\} = z[X(z) - x_0]$; (2) $Z\{x_{n+2}\} = z^2[X(z) - x_0 - x_1 z^{-1}]$

(3) $Z\{x_{n+3}\} = z^3[X(z) - x_0 - x_1 z^{-1} - x_2 z^{-2}]$

Example 8.8. Show that $Z\left(\frac{1}{n!}\right) = e^{1/z}$.

Hence evaluate $Z\{1/(n+1)!\}$ *and* $Z\{1/(n+2)!\}$. (Madras, 2005)

Solution. We know $e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots + x^n/n!$

Shifting (1/n!) one unit to the left gives

$$Z\left[\frac{1}{(n+1)!}\right] = z\left[Z\left(\frac{1}{n!}\right) - 1\right] = z(e^{1/z} - 1)$$

Similarly shifting (1/n!) two units to the left gives

$$Z\left[\frac{1}{(n+2)!}\right] = z^2[e^{1/z} - 1 - z^{-1}].$$

8.8.2 MULTIPLICATION BY n

If $Z\{x_n\} = X(z)$, then $Z\{nx_n\} = -z\frac{dX(z)}{dz}$

Proof. $Z\{nx_n\} = \sum_{n=0}^{\infty} n x_n z^{-n} = -z \sum_{n=0}^{\infty} x_n (-n) z^{-n-1} = -z \sum_{n=0}^{\infty} x_n \frac{d}{dz}(z^{-n})$

$$= -z \sum_{n=0}^{\infty} \frac{d}{dz}(x_n z^{-n}) = -z \frac{d}{dz}\left(\sum_{n=0}^{\infty} x_n z^{-n}\right) = -z \frac{d}{dz} X(z).$$

Obs. We have, $Z\{n^2 x_n\} = \left(-z\frac{d}{dz}\right)^2 X(z)$ (Madras, 2005)

In general, $Z\{n^k x_n\} = \left(-z\frac{d}{dz}\right)^k X(z)$

Example 11.12 Show that $\lim_{x \to \infty} \left[\frac{x^2+1}{x+1} - (ax+b) \right] = 0$ if and only if $a = 1$ and $b = -1$.

Solution: (a) We know that $\lim_{x \to \infty} \left[\frac{x^2+1}{x+1} - (ax+b) \right] = 0$ and hence we have $\lim_{x \to \infty} \frac{1-bx-b}{x+1} = a$

$$\lim_{x \to \infty} \frac{1-bx-b}{x+1} = a \Rightarrow \lim_{x \to \infty} \frac{\frac{1}{x}-b-\frac{b}{x}}{1+\frac{1}{x}} = a \Rightarrow -b = a$$

$$\lim_{x \to \infty} \left[\frac{x^2+1}{x+1} - (ax+b) \right] = \lim_{x \to \infty} \frac{x^2+1-ax^2-x-bx-b}{x+1} = \lim_{x \to \infty} \frac{(1-a)x^2-(1+b)x+(1-b)}{x+1}$$

(a) We know that $\lim_{x \to a} f(x) = \frac{1}{\lim_{x \to a} \frac{1}{f(x)}}$

$$\lim_{x \to a} [f(x)]^{g(x)} = \left(\lim_{x \to a} [f(x)]^{g(x)} \right) = \left(\lim_{x \to a} [f(x)] \right)^{\lim_{x \to a} g(x)}$$

$$= \left(\lim_{x \to a} [f(x)] \right)^{\lim_{x \to a} \frac{g(x)[f(x)-1]}{f(x)-1}} = \left(\lim_{x \to a} [f(x)] \right)^{\lim_{x \to a} \frac{g(x)}{f(x)-1}}$$

$$= e^{\lim_{x \to a} \left[\frac{f(x)-1}{g(x)} \cdot \frac{g(x)}{f(x)-1} \right]} = e^{\lim_{x \to a} \frac{f(x)-1}{g(x)}} = e^{\lim_{x \to a} \frac{f'(x)}{g'(x)}}$$

11.13 FIVE BASIC THEOREMS

In applications, we often need the values of $\lim_{x \to a} f(x)$ for $x = a^+$ or $x = a^-$ or $x = \pm\infty$ without actually finding the limit of $f(x)$ as x approaches a or $\pm\infty$. For such limit values we are indebted to the following five basic theorems. The precise relationship is given by the following one-sided and two-sided theorems.

(A Radical value) theorem: If $\lim_{x \to a} f(x) = l$ then $\lim_{x \to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \to a} f(x)}$

Proof: We know that $f(x) = \lim_{x \to a} f(x) = a_0 + a_1 x^{-1} + a_2 x^{-2} + \ldots$
Taking limits as $x \to -\infty$ we get $\lim_{x \to -\infty} f(x) = a_0$ as required.

Similarly additional exhibit can be found concurrently giving:

$$a_0 = \lim_{x \to \infty} [x f(x) - a_1 x] \quad a_1 = \lim_{x \to \infty} [x^2 f(x) - a_0 x - a_2 x^{-1}] \text{ and so on}$$

(B Fixed value) theorem: If $\lim_{x \to a} f(x) = l$ then

$$\lim_{x \to a} k f(x) = k \lim_{x \to a} f(x) = kl$$

Proof: By definition, if $\lim_{x \to a} f(x) = l$ then $\lim_{x \to a} f(x) = \sum_{n=0}^{\infty} a_n x^{-n}$

$$\text{or} \quad \lim_{x \to a} k f(x) = \sum_{n=0}^{\infty} k a_n x^{-n}$$

$$\text{or} \quad \lim_{x \to a} k f(x) = k \lim_{x \to a} \sum_{n=0}^{\infty} a_n x^{-n}$$

$$\text{or} \quad \lim_{x \to a} k f(x) = \sum_{n=0}^{\infty} k a_n x^{-n}$$

Taking limits of both sides as $x \to \pm\infty$, we get

$$\lim_{x \to a} [f(x) \pm g(x)] = \sum_{n=0}^{\infty} a_n x^{-n} \pm \sum_{n=0}^{\infty} b_n x^{-n} = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x)$$

$$= \lim_{z \to \infty} [z_{n+1} - z_n] = z_0 - z_0$$

Hence $z_0 = \lim_{z \to \infty} [z - 1] G(z)$. (June 2008 R)

Example 3.11.8 $G(z) = \frac{2z^2 + 3z + 5}{z - z^2}$, evaluate z_0 and z_1.

Solution: Writing $G(z) = \frac{1}{z^2} \cdot \frac{5 + 3z^{-1} + 2z^{-2}}{(1 - z^{-1})^2}$

By initial value theorem, $z_0 = \lim_{z \to \infty} G(z) = 0$

Similarly, $z_1 = \lim_{z \to \infty} z[G(z) - z_0] = 0$

Now $z_2 = \lim_{z \to \infty} z^2 [G(z) - z_0 - z_1 z^{-1}] = 2 - 3 - 5 = 2$

and $z_3 = \lim_{z \to \infty} z^3 [G(z) - z_0 - z_1 z^{-1} - z_2 z^{-2}] = \lim_{z \to \infty} z^3 [G(z) - 2 - 3z^{-1} - 2z^{-2}]$

$$= \lim_{z \to \infty} z^3 \left[\frac{2z^2 + 3z + 5}{z - z^2} - \frac{2}{z^2} \right] = \lim_{z \to \infty} z^3 \left[\frac{2z^3 + 3z^2 + 5z - 2}{z^2 (z - 1)} \right] = \infty$$

Problems for Practice

1. Find the Z transforms of the following sequences:

   (a) $\frac{a^n}{n!}$ (b) $\sin(n\theta)$ (Nov./Dec. 2008) (c) $\frac{1}{(n+1)!}$ (d) $\sin(n\theta + \alpha)$ and $\cos(n\theta + \alpha)$

2. Using the convolution property, find the Z-transform of the following functions:

   (a) $(2n + 3)\sin(n\pi/2)$ (b) $\frac{1}{2}n(n-1)(n-2)$ (May/June 2009)

   (c) $(n+1)(n+2)$ (d) $\sin(n\pi/2)$ (May/June 2008) (e) $2n^2$ (f) $n(n-1)$ (Nov./Dec. 2011) (R)

3. Show that (a) $Z[\sin n\theta] = \frac{z \sin\theta}{z^2 - 2z\cos\theta + 1}$ (Nov./Dec. 2011) (b) $Z[a^n \sin n\theta] = \frac{az \sin\theta}{z^2 - 2az\cos\theta + a^2}$

4. Show that (a) $Z[n^2 a^n] = \frac{az(az+1)}{(z-a)^3}$, (b) $Z[a^n \sin n\theta] = \frac{az \sin\theta}{z^2 - 2az\cos\theta + a^2}$

   Also evaluate $Z[n^2 \sin n\pi]$ (Nov./Dec. 2007)

5. Using $Z[n^2] = \frac{z^2 + z}{(z-1)^3}$, show that $Z[(n+1)^2] = \frac{z^3 + z^2}{(z-1)^3}$

6. Find the Z-transform of $(n+1)(n+2)$. Given that $Z\left[\frac{1}{n!}\right] = e^{\frac{1}{z}}$ (Model Paper, 2009)

7. Find the Z transform of $\cos(n\pi/2)$ and hence find $Z[\cos(n\pi/2 + \alpha)]$ (June 2009)

8. Find the Z transform of $\cos(n\pi/2)$ and $a^n \cos(n\pi/2)$ (Model, 2009 R)

9. Find the Z transform of the following:

   (a) $a^{|n|}$ (b) $a^{|n-1|}$ (Nov./Dec. 2010) (c) $\sin(n\pi + \alpha)$

10. Show that (a) $Z[1] = \frac{z}{z-1}$, (b) $Z[(n+1)a^n] = \frac{z^2}{(z-a)^2}$

11. Show that $Z[{}^nC_r] = (1 - z^{-1})^{-r-1}$. Using the above rule, evaluate that $Z[{}^nC_2, {}^nC_3]$.

12. If $Z[x_n] = \frac{z}{z-a} + \frac{z}{z^2 - 1}$, find the Z-transform of x_{n+1} (Nov./Dec. 2009)

13. If $G(z) = \frac{2z^2 + 3z + 10}{(z-1)^3}$, find the values of x_0 and x_1

23.3.3 SOME USEFUL Z TRANSFORMS

No.	Sequence x_n [or $x(t)$]	Z transform $X(z)$ or $X(e^{sT})$
1	k	$\frac{kz}{z-1}$, $\lvert z\rvert > 1$
2	k^n	$\frac{z}{z-k}$, $\lvert z\rvert > \lvert k\rvert$
3	n	$\frac{z}{(z-1)^2}$, $\lvert z\rvert > 1$
4	n^2	$\frac{z(z+1)}{(z-1)^3}$, $\lvert z\rvert > 1$
5	n^3	$\frac{z(z^2+4z+1)}{(z-1)^4}$, $\lvert z\rvert > 1$
6	$\delta(n) = \begin{cases} 1 & n=0 \\ 0 & n\neq 0 \end{cases}$	1
7	$u(n) = \begin{cases} 1 & n\geq 0 \\ 0 & n<0 \end{cases}$	$\frac{z}{z-1}$, $\lvert z\rvert > 1$
8	a^n	$\frac{z}{z-a}$, $\lvert z\rvert > \lvert a\rvert$
9	na^n	$\frac{az}{(z-a)^2}$, $\lvert z\rvert > \lvert a\rvert$
10	n^2a^n	$\frac{az(z+a)}{(z-a)^3}$, $\lvert z\rvert > \lvert a\rvert$
11	$\sin n\theta$	$\frac{z\sin\theta}{z^2-2z\cos\theta+1}$, $\lvert z\rvert > 1$
12	$\cos n\theta$	$\frac{z^2-z\cos\theta}{z^2-2z\cos\theta+1}$, $\lvert z\rvert > 1$
13	$a^n\sin n\theta$	$\frac{az\sin\theta}{z^2-2az\cos\theta+a^2}$, $\lvert z\rvert > \lvert a\rvert$
14	$a^n\cos n\theta$	$\frac{z^2-az\cos\theta}{z^2-2az\cos\theta+a^2}$, $\lvert z\rvert > \lvert a\rvert$
15	$\sinh an$	$\frac{z\sinh a}{z^2-2z\cosh a+1}$, $\lvert z\rvert > \lvert e^a\rvert$
16	$\cosh an$	$\frac{z^2-z\cosh a}{z^2-2z\cosh a+1}$, $\lvert z\rvert > \lvert e^a\rvert$
17	$e^{an}\sinh an$	$\frac{ze^a\sinh a}{z^2-2ze^a\cosh a+e^{2a}}$, $\lvert z\rvert > \lvert e^{2a}\rvert$
18	$e^{an}\cosh an$	$\frac{z^2-ze^a\cosh a}{z^2-2ze^a\cosh a+e^{2a}}$, $\lvert z\rvert > \lvert e^{2a}\rvert$
19	$x(n+k)$	$z^k[X(z)-\sum_{r=0}^{k-1}x_r z^{-r}]$
20	x_{n+1}	$z[X(z)-x_0]$
	x_{n+2}	$z^2[X(z)-x_0-x_1z^{-1}]$
	x_{n+3}	$z^3[X(z)-x_0-x_1z^{-1}-x_2z^{-2}]$
21	x_{n-1}	$z^{-1}X(z)$
22	x_{n-2}	$z^{-2}X(z)+x_{-1}z^{-1}$
23	x_{n-3}	$z^{-3}X(z)+x_{-2}z^{-2}+x_{-1}z^{-1}$
24	$\sum_{r=0}^{n}x_r y_{n-r}$	$X(z)Y(z)$

8.28.13 SOME USEFUL INVERSE Z-TRANSFORMS

Sr. No.	$X(z)$	Inverse Z-transform $x_n = Z^{-1}[X(z)]$
1.	$\frac{z}{z-a}$	a^n
2.	$\frac{z}{z+a}$	$(-a)^n$
3.	$\frac{z}{(z-a)^2}$	$n a^{n-1}$
4.	$\frac{z}{(z-a)^3}$	$\frac{1}{2}n(n-1)a^{n-2}$
5.	$\frac{z}{z^2-1}$	$\frac{1}{2}[1-(-1)^n]$
6.	$\frac{z}{z^2+1}$	$\sin\frac{n\pi}{2}$
7.	$\frac{z^2}{(z-1)^2}$	n
8.	$\frac{z^2}{(z-1)^3}$	$\frac{1}{2}n(n+1)$

8.28.2 CONVOLUTION THEOREM

If $Z^{-1}[X(z)] = x_n$ and $Z^{-1}[Y(z)] = y_n$ then

$$Z^{-1}[X(z).Y(z)] = \sum_{r=0}^{n} x_r . y_{n-r} = x_n * y_n$$

where the symbol $*$ denotes the convolution operation.

Proof: We have $X(z) = \sum_{n=0}^{\infty} x_n z^{-n}$, $Y(z) = \sum_{n=0}^{\infty} y_n z^{-n}$

$$\therefore \quad X(z)Y(z) = (x_0 + x_1 z^{-1} + x_2 z^{-2} + \ldots + x_n z^{-n} + \ldots)(y_0 + y_1 z^{-1} + y_2 z^{-2} + \ldots + y_n z^{-n} + \ldots)$$

$$= \sum_{n=0}^{\infty} (x_n y_0 + x_{n-1} y_1 + x_{n-2} y_2 + \ldots + x_0 y_n) z^{-n} = Z(x_n y_0 + x_{n-1} y_1 + \ldots + x_0 y_n)$$

which completes the desired result.

Obs: The convolution theorem plays an important role in the solution of difference equations and in probability problems involving sums of two independent random variables.

Example 8.28.11. Use convolution theorem to evaluate $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right]$

Solution. We know that $Z^{-1}\left[\frac{z}{z-a}\right] = a^n$ and $Z^{-1}\left[\frac{z}{z-b}\right] = b^n$

$$\therefore \quad Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right] = Z^{-1}\left[\frac{z}{z-a}.\frac{z}{z-b}\right] = a^n * b^n$$

$$= \sum_{r=0}^{n} a^r . b^{n-r} = a^n \sum_{r=0}^{n} \left(\frac{b}{a}\right)^r \quad \text{which is a G.P.}$$

$$= a^n . \frac{a\left(\frac{b}{a}\right)^{n+1} - a}{a\left(\frac{b}{a}\right) - 1} = \frac{a^{n+1} - b^{n+1}}{a - b}$$

33.13 CONVERGENCE OF Z-TRANSFORMS

Z-transform operation is performed on a sequence x_n which may exist in the range of integers $-\infty < n < \infty$, and we write

$$X(z) = \sum_{n=-\infty}^{\infty} x_n z^{-n} \qquad \ldots(33)$$

where x_n represents a number in the sequence for $n = \ldots$ integer. The region of the z-plane in which $X(z)$ is convergent absolutely is known as the region of convergence (ROC) of $X(z)$.

We have so far discussed one sided Z transform only for which $n \geq 0$. Here the sequence is always right sided and the ROC is always outside a prescribed circle say $|z| > |a|$ Fig. 33.2 (a). For a left-handed sequence, the ROC is always inside any prescribed circle say $|z| < |b|$ (Fig. 33.2 (c))

33.14 TWO-SIDED Z-TRANSFORM OF x_n IS DEFINED BY

$$X(z) = \sum_{n=-\infty}^{\infty} x_n z^{-n} \qquad \ldots(33)$$

In this case, the sequence is two-sided and the region of convergence for $X(z)$ is the ring-like region $|b| < |z| < |a|$ (Fig. 33.2 (b)). The terms inside the bracket is negative powers of z and the outer circle bounds the terms in positive powers of z. The shaded area is the region of convergence. *It necessarily for the two sided sequence and the Z-transform to exist.*

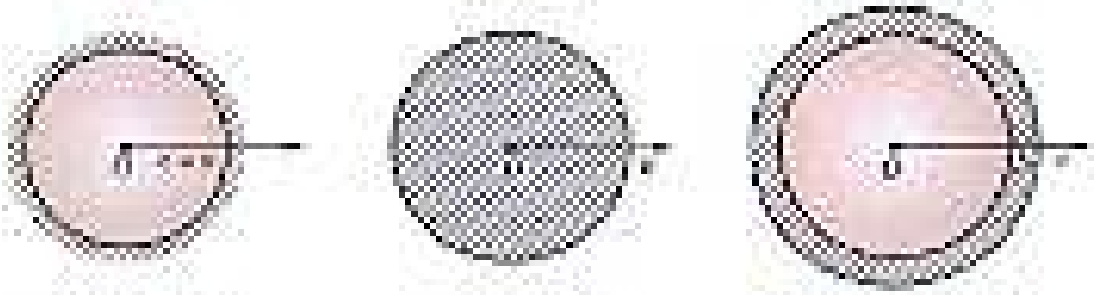


Fig. 33.2

Example 33.15. Find the Z-transform and region of convergence of

(a) $x(n) = \begin{cases} a^n & \text{for } n \geq 0 \\ b^n & \text{for } n < 0 \end{cases}$ (b) $x(n) = a^n$, $n < 0$

Solution. By definition $Z[x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n} = \sum_{n=-\infty}^{-1} b^n z^{-n} + \sum_{n=0}^{\infty} a^n z^{-n}$

Putting $-n = m$ in the first series, we get

$$Z[x(n)] = \sum_{m=1}^{\infty} b^{-m} z^{m} + \sum_{n=0}^{\infty} a^n z^{-n}$$

$$\left[\frac{z}{b} + \frac{z^2}{b^2} + \frac{z^3}{b^3} + \ldots\right] + \left[1 + \frac{a}{z} + \frac{a^2}{z^2} + \ldots\right]$$

$$= \frac{z}{b}\left[1 + \left(\frac{z}{b}\right) + \left(\frac{z}{b}\right)^2 + \ldots\right] + \left[1 + \left(\frac{a}{z}\right) + \left(\frac{a}{z}\right)^2 + \ldots\right] \qquad \ldots(i)$$

$$= \frac{z}{b}\frac{1}{1 - (z/b)} + \frac{1}{1 - (a/z)} = \frac{z}{b}\frac{b}{b - z} + \frac{z}{z - a} = \frac{z(b - a)}{(b - z)(z - a)}$$

Note The Laurent series is (1) being R.H.S will be convergent if $|z/2| < 1$ and $|2/z| < 1$ i.e., $|z| < 4$ and $2 < |z|$ i.e., $2 < z < 4$.

Hence $X(z)$ is convergent if a ring between the annulus as shown shaded in Fig. 3.2. Hence ROC is $2 < z < 4$.

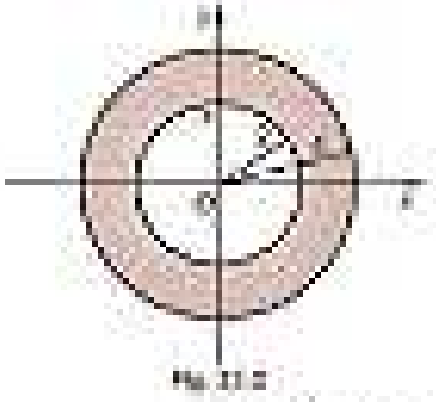


Fig. 3.2

(ii) By definition, $Z[a^n u] = \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n}$

To find the sum of this series, we replace n by $k + r$

$$\text{i.e.,} \quad X(z) = \sum_{k=0}^{\infty} a^{k+r} z^{-(k+r)} = a^r \sum_{k=0}^{\infty} a^k z^{-k}$$

$$= a^r [1 + a^1 z^{-1} + a^2 z^{-2} + \ldots]$$

$$= a^r (1 - az^{-1})^{-1}$$

This series is convergent for $|az^{-1}| < 1$ i.e., for $|z| > |a|$.

Hence ROC is $|z| > |a|$.

Example 3.13 Find the Z-transform and the region of convergence of :

(a) $f(n) = 2^n u(n)$ (b) $f(n) = (n+1)a^n u(n)$ (Madras, 2000)

Solution. (a) Assuming that $f(n) = 0$ for $n \geq 0$ we have

$$X(z) = \sum_{n=-\infty}^{\infty} f(n) z^{-n} = \sum_{n=-\infty}^{0} 2^n z^{-n} = \sum_{m=0}^{\infty} 2^{-m} z^m \quad \text{where } m = -n$$

$$= \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \ldots = \frac{1}{2}\left(1 + \frac{1}{2}z + \frac{1}{4}z^2 + \ldots\right)$$

$$= \frac{1}{2} \cdot \frac{1}{1 - \frac{1}{2}z} = \frac{1}{2 - z}$$

This series being a G.P. is convergent if $|z/2| < 1$ i.e., $|z| < 2$.

Hence ROC is $|z| < 2$.

(b) By definition, $Z[a^n u] = \sum_{n=0}^{\infty} \frac{a^n}{n!} z^{-n} = \sum_{n=0}^{\infty} \frac{(a/z)^n}{n!} = 1 + \left(\frac{a}{z}\right) + \frac{1}{2!}\left(\frac{a}{z}\right)^2 + \frac{1}{3!}\left(\frac{a}{z}\right)^3 + \ldots$

$= e^{a/z}$

The above series is convergent for all values of z.

Hence ROC is the entire z-plane.

PROBLEMS AND SOLUTIONS

Find the Z-transform and the ROC in each of the following sequences :

1. $x(n) = 2^n, n \geq 0$
2. $x(n) = 3^n, n \geq 0$
3. $x(n) = 4^n, 0 \leq n \leq 10$ and $= 0$ for $n \geq 10$
4. $x(n) = (0.5)^n, n \geq 0$
5. $x(n) = \cos(n\omega), n \geq 0$
6. $x(n) = \sin(n\omega), n \geq 0$
7. $x(n) = e^{an}, n \geq 0$

3.3.3 EVALUATION OF INVERSE Z-TRANSFORMS

We can obtain the inverse Z-transform using any of the following three methods :

1. Power series method. This is the simplest of all the methods of finding the inverse Z-transform. If $X(z)$ is expressed as the ratio of two polynomials which cannot be factorized, we simply divide the numerator by the denominator and take the inverse Z-transform of each term in the quotient.

Example 9.14. *Find the inverse Z-transform of* $\log\left(\frac{z+1}{z-1}\right)$ *by power series method.*

Solution. Putting $z = 1/y$, $U(z) = \log\left[\frac{z+1}{z-1}\right] = -\log(1-y) + \log(1+y) = -y + \frac{1}{2}y^2 - \frac{1}{3}y^3 + \ldots$

$$= -z^{-1} + \frac{1}{2}z^{-2} - \frac{1}{3}z^{-3} + \ldots$$

Thus $u_n = \begin{cases} 0 & \text{for } n = 0 \\ \frac{1}{n}(-1)^{n-1} & \text{otherwise} \end{cases}$

Example 9.15. *Find the inverse Z-transform of* $U(z) = \frac{z^2}{z^2 + 2z + 1}$ *by power series method.*

Solution. $U(z) = \frac{z^2}{z^2 + 2z + 1} = z^{-1} - \frac{2z + z^{-1}}{z^2 + 2z + 1}$, by actual division

$$= z^{-1} - 2z^{-2} + \frac{3z^{-2} + 2z^{-3}}{z^2 + 2z + 1} = z^{-1} - 2z^{-2} + 3z^{-3} - \frac{4z^{-3} + 3z^{-4}}{z^2 + 2z + 1}$$

Continuing this process of division, we get an infinite series i.e.,

$$U(z) = \sum_{n=1}^{\infty} (-1)^{n-1} n z^{-n}$$

Thus $u_n = (-1)^{n-1} n$.

II. Partial fractions method. This method is similar to that of finding the inverse Laplace transforms using partial fractions. The method consists of decomposing $U(z)/z$ into partial fractions, multiplying the resulting expression by z and then inverting the terms.

Example 9.16. *Find the inverse Z-transform of*

(i) $\frac{z^2 + 2z}{(z+2)(z-4)}$ (U.T.U., 2004 S, 2007 S, 2009) (ii) $\frac{z^2 + 2z}{(z-2)^2(z-4)}$ (U.T.U., 2010)

Solution. (i) We write $U(z) = \frac{z^2 + 2z}{(z+2)(z-4)}$ so that $\frac{U(z)}{z} = \frac{2z+3}{(z+2)(z-4)} = \frac{A}{z+2} + \frac{B}{z-4}$

where $A = 1/6$ and $B = 11/6$.

$\therefore$ $U(z) = \frac{1}{6}\frac{z}{z-2} + \frac{11}{6}\frac{z}{z-4}$

On inversion, we have

$$u_n = \frac{1}{6}(-2)^n + \frac{11}{6}(4)^n$$ (U.T.U., 2010)

(ii) We write $U(z) = \frac{z^2 - 2z}{(z-2)^2(z-4)}$

so that $\frac{U(z)}{z} = \frac{z^2 - 2z}{(z-2)^2(z-4)} = \frac{A + Bz + Cz^2}{(z-2)^2} + \frac{D}{z-4}$

Readily we get $D = 1/2$. Multiplying throughout by $(z-2)^2(z-4)$, we get

$$z^2 - 2z = (A + 4B + 8C) + (2B + 6C)z + Cz^2 + \frac{1}{2}(z-2)^2$$

Putting $z = 0, 1, -1$ successively and solving the resulting simultaneous equations, we get $A = 0$, $B = 0$, $C = 1/2$.

Thus $U(z) = \frac{1}{2}\cdot\frac{z^2}{(z-2)^2} + \frac{z}{z-4} = \frac{1}{2}\frac{(z-2)^2 + 4z - 4}{(z-2)^2} + \frac{z}{z-4}$

$$= \frac{1}{2}\left[\frac{z}{z-2}\right] + 2\left[\frac{2z^2 + 4z}{(z-2)^2}\right] + \frac{z}{z-4}$$

By inversion, we get $u_n = \frac{1}{2}(4^n + 2\cdot 1^n) = 4^{n-1} + 1$ $\quad$ [Since $|z| > 4$]

$= 2^{2n-1} + 1$.

Example 22.17. Find the inverse Z-transform of

$\frac{2z^2 - 11z + 12}{(z-2)(z-3)^2}$ for $3 < |z| < \infty$

Solution. Splitting into partial fractions, we obtain

$U(z) = \frac{2z^2 - 11z + 12}{(z-2)(z-3)^2} = \frac{A}{z-2} + \frac{B}{z-3} + \frac{C}{(z-3)^2}$ where $A = 2, B = -1, C = 3$

$\therefore \quad U(z) = \frac{2}{z-2} - \frac{1}{z-3} + \frac{3}{(z-3)^2}$

$= \frac{2}{z}\left(1 - \frac{2}{z}\right)^{-1} - \frac{1}{z}\left(1 - \frac{3}{z}\right)^{-1} + \frac{3}{z^2}\left(1 - \frac{3}{z}\right)^{-2}$ $\quad$ as $\left|\frac{3}{z}\right| < 1$ and $\left|\frac{3}{z}\right| < 1$

$= \frac{2}{z}\left(1 + \frac{2}{z} + \frac{4}{z^2} + \frac{8}{z^3} + \cdots\right) - \frac{1}{z}\left(1 + \frac{3}{z} + \frac{9}{z^2} + \frac{27}{z^3} + \cdots\right) + \frac{3}{z^2}\left(1 + \frac{2\cdot 3}{z} + \frac{3\cdot 3^2}{z^2} + \frac{4\cdot 3^3}{z^3} + \cdots\right)$

$\quad$ since $\left|\frac{3}{z}\right| < 1$

$= \left(\frac{2}{z} + \frac{2^2}{z^2} + \frac{2^3}{z^3} + \frac{2^4}{z^4} + \cdots\right) - \left(\frac{1}{z} + \frac{3}{z^2} + \frac{3^2}{z^3} + \frac{3^3}{z^4} + \cdots\right) + \left(\frac{1}{z^2} + \frac{2\cdot 3}{z^3} + \frac{3\cdot 3^2}{z^4} + \frac{4\cdot 3^3}{z^5} + \cdots\right)$

$= \sum_{n=1}^{\infty} 2^{n-1} z^{-n} - \sum_{n=2}^{\infty} 3^{n-1} z^{-n} + \sum_{n=2}^{\infty} (n-1) 3^{n-2} z^{-n}$

By inversion, we get $u_n = 2^{n-1}, n \geq 1$ $\quad$ and $\quad u_n = -3^{n-1} + (n-1)3^{n-2}, n \geq 2$.

III. Inversion Integral methods. The inverse Z transform of $U(z)$ is given by the formula

$u_n = \frac{1}{2\pi i}\int_C U(z) z^{n-1}\, dz$

= sum of residues of $U(z)z^{n-1}$ at the poles of $U(z)$ which are inside the contour C taken anticlockwise in the z plane.

The following examples will illustrate the application of this formula.

Example 22.18. Using the inversion integral method, find the inverse Z transform of

$\frac{z}{(z-1)(z-2)}$ (U.P.T.U., 2010 O)

Solution. Let $U(z) = \frac{z}{(z-1)(z-2)}$. Its poles are at $z = 1$ and $z = 2$.

Using $U(z)$ in the inversion integral, we have

$u_n = \frac{1}{2\pi i}\int_C \frac{z^n}{(z-1)(z-2)}\, dz$,

where C is a circle large enough to enclose both the poles of $U(z)$.

= sum of residues of $U(z)z^{n-1}$ at $z = 1$ and $z = 2$

Now $\quad \text{Res}\left[U(z)z^{n-1}\right]_{z=1} = \lim_{z\to 1}\left[(z-1)\cdot\frac{z^n}{(z-1)(z-2)}\right] = -1$

and $\quad \text{Res}\left[U(z)z^{n-1}\right]_{z=2} = \lim_{z\to 2}\left[(z-2)\cdot\frac{z^n}{(z-1)(z-2)}\right] = 2^n$

Thus the required inverse Z-transform $u_n = 2^n - 1, n = 0, 1, 2, \ldots$

Example 8.31(i). Find the inverse Z-transform of $\frac{z}{(z-1)(z-2)}$, $|z|>2$. (Madras, 2000 S)

Solution. Let $G(z) = \dfrac{z}{(z-1)(z-2)}$. It has simple poles at $z = 1, z = 2$.

Using (2x) in the inversion integral, we have

$$u_n = \frac{1}{2\pi i}\int_C G(z)\, z^{n-1}\, dz$$, where C is a circle large enough to enclose the poles of $G(z)$

= sum of residues of $G(z)\cdot z^{n-1}$ at $z = 1, z = 2$.

Now $\quad \text{Res}\,[G(z)z^{n-1}]_{z=1} = \lim_{z\to 1}\left[(z-1)\dfrac{z^n}{(z-1)(z-2)}\right] = -1$

$$\text{Res}\,[G(z)z^{n-1}]_{z=2} = \lim_{z\to 2}\left[(z-2)\frac{z^n}{(z-1)(z-2)}\right] = \frac{-2^n}{1-2}$$

$$\text{Res}\,[G(z)z^{n-1}]_{z=-1} = \lim_{z\to -1}\left[(z+1)\frac{z^n}{(z-1)(z-2)}\right] = \frac{(-1)^n}{(-1-1)(-1-2)} = \frac{(-1)^n}{6}$$

Hence $\quad u_n = -1 - \dfrac{2^n}{1-2} - \dfrac{(-1)^n}{6}$

PROBLEMS 8.5

Using partial fraction theorem, evaluate the inverse Z-transform of the following:

1. $\dfrac{z^2}{(z-1)(z-2)}$ 2. $\left[\dfrac{z}{(z-1)^2}\right]^2$ (M.D.U., 2000) 3. $\left[\dfrac{z}{(z-2)^2}\right]^2$

4. Show that $\dfrac{1}{z}\cdot\dfrac{1}{z-1}=\dfrac{1}{z^2}\left[\dfrac{z^2}{(z-1)(z-2)}\right]+\dfrac{1}{z-1}\left[\dfrac{z^2}{(z-1)(z-2)}\right]$ and hence find $Z^{-1}\left[\dfrac{z^2}{(z-1)^2(z-2)}\right]$. (Anna, 2004)

Find the inverse Z-transform of the following:

5. $\dfrac{3z}{z^2-4}$, $|z|>2$ (Kottayam, 2006) 6. $\dfrac{5z}{(z-1)(z-3)}$, $|z|>3$ (Madras, 1998)

7. $\dfrac{z}{2z^2-3z+1}$, $|z|>1$ 8. $\dfrac{18z^2}{(3z-1)(z+1)}$, $|z|>\dfrac{1}{3}$ (V.T.U., 2009)

9. $\dfrac{8z-4}{(z-3)^2}$, $|z|>3$ 10. $\dfrac{2z^2-12z+20}{(z-2)(z-3)(z-4)}$, $|z|>4$ (Anna, 2004 S)

11. $\dfrac{4z^2-2z}{z^3-5z^2+8z-4}$, $|z|>2$ (V.T.U., 2010 S) 12. $\dfrac{z^2+2z}{(z-1)^2(z^2-1)}$, $|z|>1$ (Anna, 2009)

13. $\dfrac{(1-e^{-aT})z}{(z-1)(z-e^{-aT})}$

14. Obtain $Z^{-1}\left[\dfrac{z}{(z-2)(z-3)}\right]$ for (i) $|z|>3$ (ii) $|z|<2$ (iii) $2<|z|<3$. (Kurukshetra, 2008)

15. Find $Z^{-1}\left[\dfrac{z^2}{(z-1)^2}\right]$ for $|z|>1$. (Mumbai, 2008)

Using convolution theorem, find the inverse Z-transform of the following functions:

16. $\dfrac{z+3}{(z+2)(z-5)}$ 17. $\dfrac{(2z-1)z}{(2z-1)(z+0.5)}$

18. $\dfrac{1}{(z-1)(z+3)}$, $|z|>3$ (U.P.T.U., 2008) 19. $\dfrac{z^2+z}{(z-1)^3}$ (Madras, 2000)

20. $\dfrac{2z(z^2-3)}{(z^2-1)^2}$

3.24 [3] APPLICATION TO DIFFERENCE EQUATIONS

Just as the Laplace transform is a useful tool in quite effective for solving linear differential equations (§ 3.2.10), the Z-transforms are quite useful for solving linear difference equations.

The performance of discrete systems is expressed by suitable difference equations. Also Z transform plays an important role in the analysis and representation of discrete-time systems. To determine the frequency response of such systems, the solution of difference equations are required for which Z transform method proves useful.

(I) Working procedure to solve a linear difference equation with constant coefficients by Z-transforms :

1. Take the Z-transform of both sides of the difference equation using the formulae of § 3.2.10 and the given conditions.
2. Then express all terms involving $U(z)$ to the right.
3. Divide by the coefficient of $U(z)$, getting $U(z)$ as a function of z.
4. Express this function in terms of the Z-transforms of known functions and take the inverse Z-transform of both sides. This gives u_n as a function of n, which is the desired solution.

Example 3.2.20. Using the Z-transform, solve

$$y_{n+2} - 4y_{n+1} + 3y_n = n^2, \text{ with } y_0 = 0, y_1 = 1. \qquad \text{(U.P.T.U., 2003)}$$

Solution. If $Z[y_n] = U(z)$, then $Z[y_{n+1}] = z[U(z) - y_0]$, $Z[y_{n+2}] = z^2[U(z) - y_0 - y_1 z^{-1}]$

Also $Z[n^2] = \frac{z(z+1)}{(z-1)^3}$

$\therefore$ taking the Z-transforms of both sides, we get

$$z^2[U(z) - y_0 - y_1 z^{-1}] - 4z[U(z) - y_0] + 3U(z) = \frac{z(z+1)}{(z-1)^3}$$

Using the given conditions, it reduces to

$$U(z)(z^2 - 4z + 3) = z + \frac{z(z+1)}{(z-1)^3}$$

$$\therefore \quad \frac{U(z)}{z} = \frac{1}{(z-1)(z-3)} + \frac{1}{(z-3)(z-1)^3} = \frac{3}{8}\frac{1}{z-1} + \frac{3}{2}\frac{1}{(z-1)^3} - \frac{3}{8}\frac{1}{z-3},$$

on breaking into partial fractions.

$$\therefore \quad U(z) = \frac{3}{8}\frac{z}{z-1} + \frac{3}{2}\frac{z}{(z-1)^3} - \frac{3}{8}\frac{z}{z-3}$$

On inversion, we obtain

$$u_n = \frac{3}{8}Z^{-1}\left[\frac{z}{z-1}\right] + \frac{3}{2}Z^{-1}\left[\frac{z}{(z-1)^3}\right] - \frac{3}{8}Z^{-1}\left[\frac{z}{z-3}\right] = \frac{3}{8}(1)^n + \frac{3}{4}n(n-1) - \frac{3}{8}(3)^n.$$

Example 3.2.21. Solve $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using Z-transform. *(U.P.T.U., 2001; Aktu, 2010; U.P.T.U., 2015)*

Solution. If $Z[y_n] = U(z)$, then $Z[y_{n+1}] = z[U(z) - y_0]$, $Z[y_{n+2}] = z^2[U(z) - y_0 - y_1 z^{-1}]$

Also $Z[2^n] = \frac{z}{z-2}$.

Taking Z-transforms of both sides, we get

$$z^2[U(z) - y_0 - y_1 z^{-1}] + 6[zU(z) - y_0] + 9U(z) = \frac{z}{z-2}$$

Since $y_0 = 0$, and $y_1 = 0$, we have $U(z)(z^2 + 6z + 9) = \frac{z}{z-2}$

or $\frac{U(z)}{z} = \frac{1}{(z-2)(z+3)^2} = \frac{1}{25}\left[\frac{1}{z-2} - \frac{1}{z+3} - \frac{5}{(z+3)^2}\right]$, on splitting into partial fractions.

or $U(z) = \frac{z}{25}\left[\frac{z}{z-2} - \frac{z}{z+3} - 5\frac{z}{(z+3)^2}\right]$

On taking inverse Z-transforms of both sides, we obtain

$$y_n = \frac{1}{25}\left[Z^{-1}\left[\frac{z}{z-2}\right] - Z^{-1}\left[\frac{z}{z+3}\right] - 5Z^{-1}\left[-\frac{3z}{(z+3)^2}\right]\right]$$

$$= \frac{1}{25}[2^n - (-3)^n + 5n(-3)^n]$$

$$\left[\because Z^{-1}\left[\frac{az}{(z-a)^2}\right] = na^n\right]$$

Example 4.30.22. Find the sequence of the solution $y_{n+2} - 3y_{n+1} + 2y_n = n$, $n = 0, 1, 2, \ldots$ with $y_0 = 0, y_1 = 1$ using Z-transform method. (U.P.T.U., 2015)

Solution. Taking Z-transform of both sides of the given equation, we get

$$Z[y_{n+2}] - 3Z[y_{n+1}] + 2Z[y_n] = Z[n]$$

$$z^2[Y(z) - y_0 - y_1 z^{-1}] - 3z[Y(z) - y_0] + 2Y(z) = \frac{z}{(z-1)^2}$$

Substituting the values $y_0 = 0, y_1 = 1$, it reduces to

$$(z^2 - 3z + 2)\,Y(z) = \frac{z}{(z-1)^2} + z = \frac{z^3}{(z-1)^2}$$

or

$$\frac{Y(z)}{z} = \frac{z^2}{(z-1)^3(z-2)}$$

$$= \frac{A}{z-1} + \frac{B}{(z-1)^2} + \frac{C}{(z-1)^3} + \frac{D}{z-2} \qquad \text{where } A = -2, B = -1, C = 1, D = 2$$

so that

$$Y(z) = -2\frac{z}{z-1} - \frac{z}{(z-1)^2} + \frac{z}{(z-1)^3} + 2\frac{z}{z-2}$$

On inversion, we obtain

$$y_n = -2 - n + \frac{1}{2}n(n-1) + 2^{n+1}$$

Note. The initial values given in the problem are automatically seen in the answer as $y_0 = 0$, $y_1 = 1$.

Example 4.30.23. Solve the difference equation $y_{n+2} + \frac{1}{4}y_{n+1} = u_n + \frac{1}{2}u_{n-1}$, where u_n is a unit sequence.

Solution. Taking Z-transform of both sides of the given equation, we get

$$Y(z) + \frac{1}{4}z^{-1}Y(z) = U(z) + \frac{1}{2}z^{-1}U(z)$$

or

$$Y(z) = \left(1 + \frac{1}{2}z^{-1}\right)U(z)\left[\frac{1}{1 + \frac{1}{4}z^{-1}}\right] = \left[\frac{z + \frac{1}{2}}{z}\right]\left[\frac{z}{z + \frac{1}{4}}\right]$$

There being only one single pole at $z = -1/4$, consider the contour $|z| = 1/4$.

$$\therefore \quad \text{Res}\left[Y(z)z^{n-1}\right]_{z=-1/4} = \lim_{z \to -1/4}\left[\left(z + \frac{1}{2}\right)\left(z + \frac{1}{4}\right)z^{n-1}\frac{1}{\left(z + \frac{1}{4}\right)}\right]$$

$$= \lim_{z \to -1/4}\left[\left(z + \frac{1}{2}\right)z^{n-1}\right] = \left(-\frac{1}{4} + \frac{1}{2}\right)\left(-\frac{1}{4}\right)^{n-1} = \frac{1}{4}\left(-\frac{1}{4}\right)^{n-1}$$

Hence by inversion integral method, we have

$$y_n = \frac{1}{4}\left(-\frac{1}{4}\right)^{n-1}.$$

Example 4.30.24. Using the Z-transform solve $y_{n+2} - 2y_{n+1} + y_n = 3n + 5$. (U.P.T.U., 2015)

Solution. Given equation is $y_{n+2} - 2y_{n+1} + y_n = 3n + 5$.

Taking the Z-transform of both sides, we get

$$Z[y_{n+2}] - 2Z[y_{n+1}] + Z[y_n] = 3Z[n] + 5Z[1]$$

$$z^2[Y(z) - y_0 - y_1 z^{-1}] - 2z[Y(z) - y_0] + Y(z) = 3\cdot\frac{z}{(z-1)^2} + 5\cdot\frac{z}{z-1}$$

or

$$Y(z)(z^2 - 2z + 1) = \frac{8z^2 - 3z}{(z-1)^2} + y_0 z^2 - 2y_0 z + y_1 z$$

or

$$Y(z) = \frac{8z^2 - 3z}{(z-1)^4} + y_0\frac{z^2 - 2z}{(z-1)^2} + y_1\frac{z}{(z-1)^2}$$

By inversion, we obtain

$$u_n = Z^{-1}\left\{\frac{5z^2 - 2z}{(z-1)^3}\right\} + u_0 Z^{-1}\left\{\frac{z^2 - 2z}{(z-1)^3}\right\} + u_1 Z^{-1}\left\{\frac{z}{(z-1)^3}\right\} \quad \ldots(1)$$

Noting that $Z\{1\} = \frac{z}{z-1}$, $Z\{n\} = \frac{z}{(z-1)^2}$

$$Z\{n^2\} = \frac{z^2 + z}{(z-1)^3}, Z\{n^3\} = \frac{z^3 + 4z^2 + z}{(z-1)^4}$$

We can write $\frac{5z^2 - 2z}{(z-1)^3} = A\frac{z^3 + 4z^2 + z}{(z-1)^4} + B\frac{z^2 + z}{(z-1)^3} + C\frac{z}{(z-1)^2} + D\frac{z}{z-1}$

Equating coefficients of like powers of z, we find

$$A = \frac{1}{2}, B = 1, C = -\frac{3}{2}, D = 0$$

$$\therefore \quad Z^{-1}\left\{\frac{5z^2 - 2z}{(z-1)^3}\right\} = \frac{1}{2}n^3 + n^2 - \frac{3}{2}n = \frac{1}{2}n(n-1)(n+3)$$

Also $Z^{-1}\left\{\frac{z^2 - 2z}{(z-1)^3}\right\} = Z^{-1}\left\{\frac{z}{z-1}\right\} - Z^{-1}\left\{\frac{z}{(z-1)^2}\right\} = 1 - n$

and $Z^{-1}\left\{\frac{z}{(z-1)^3}\right\} = n$.

Substituting these values in (1) above, we get

$$u_n = \frac{1}{2}n(n-1)(n+3) + u_0(1-n) + u_1 n$$

$$= \frac{1}{2}n(n-1)(n+3) + c_0 + c_1 n, \quad \text{where } c_0 = u_0, c_1 = u_1 - u_0$$

Example 8.15. Using z-transforms, solve $y_{n+2} + \frac{1}{2}y_{n+1} - \frac{1}{2}y_n = \frac{1}{2^n}\cos\frac{n\pi}{2}$, $n \geq 0$.

Solution. Taking Z-transform of both sides of the given equation, we get

$$Z\left\{y_{n+2} + \frac{1}{2}y_{n+1} - \frac{1}{2}y_n\right\} = Z\left\{\frac{1}{2^n}\cos\frac{n\pi}{2}\right\}$$

or $Y(z) + \frac{1}{2}z^{-1}Y(z) - \frac{1}{2}y_0 = \frac{z^2}{z^2 + 1/4}$ or $\left(1 + \frac{1}{2}z^{-1}\right)Y(z) = \frac{z^2}{z^2 + \frac{1}{4}}$

or $$Y(z) = \frac{z^2}{\left(1 + \frac{1}{2}z^{-1}\right)\left(z^2 + \frac{1}{4}\right)} = \frac{z^3}{\left(z + \frac{1}{2}\right)\left(z^2 + \frac{1}{4}\right)}$$

There are two poles of second order at $z = i/2$ and $z = -1/2$.

$$\therefore \quad \text{Residue at } z = i/2 = \left[\frac{d}{dz}\left\{\left(z - \frac{i}{2}\right)^2 \frac{z^3}{(z^2 + 1/4)^2}\right\}\right]$$

$$= \left[\frac{d}{dz}\left\{\frac{z^3}{\left(z + \frac{1}{2}\right)\left(z + \frac{i}{2}\right)^2}\right\}\right]_{z=i/2} = \left[\frac{\left(z + \frac{1}{2}\right)^2\left(z + \frac{i}{2}\right)^2 3z^2 - z^3\left\{2\left(z + \frac{1}{2}\right) + 2\left(z + \frac{i}{2}\right)\right\}}{\left(z + \frac{1}{2}\right)^4\left(z + \frac{i}{2}\right)^4}\right]_{z=i/2}$$

$$= \left[\frac{z^3 + z^2 + \frac{3}{4}z + \frac{1}{8} + iz^3 + \frac{3}{2}iz^2 - z^3 - iz^2}{\left(z + \frac{1}{2}\right)^4\left(z + \frac{i}{2}\right)^4}\right]_{z=i/2} = \frac{\left(\frac{i}{2}\right)^3 + \frac{1}{8} + \frac{3}{4}\left(\frac{i}{2}\right) + \frac{1}{8} + i\left(\frac{i}{2}\right)^3 + \frac{3}{2}i\left(\frac{i}{2}\right)^2 - i\left(\frac{i}{2}\right)^2}{\left(\frac{i}{2} + \frac{1}{2}\right)^4\left(\frac{i}{2} + \frac{i}{2}\right)^4}$$

$$= \frac{1}{2}(3k+4)\left(\frac{1}{3}\right)^k + \frac{5}{2}(k+2)\left(\frac{1}{3}\right)^k\left[\cos\frac{k\pi}{2} + \sin\frac{k\pi}{2}\right]^2 = \frac{1}{4}(3k+4)\left(\frac{1}{3}\right)^k\left(\cos\frac{k\pi}{2} + \sin\frac{k\pi}{2}\right) \quad ...(3)$$

Shifting (n – 1) in (3), we have

$$y_{n-1} = -\frac{1}{4}(3k-1)\left(\frac{1}{3}\right)^k\left(\cos\frac{k\pi}{2} - \sin\frac{k\pi}{2}\right) \quad ...(4)$$

Adding (3) and (4), we obtain $y_n = \frac{1}{2}(k+\pi)\left(\frac{1}{3}\right)^k\cos\frac{k\pi}{2}$.

PRACTICE PROBLEMS

Solve the following difference equations using Z-transforms (1 – 8):

1. $8y_{n+2} - 2y_{n+1} - y_n = 0$, given that $y(0) = 3, y(1) = 1$. (Kerala, 2010)
2. $y(k+2) + 3y(k+1) + 2y(k) = 0$, given that $y(0) = 0, y(1) = 1$. (VTU, 2008)
3. $y_{n+2} - 6y_{n+1} + 9y_n = 2^n$, given that $y_0 = 0, y_1 = 1$. (VTU, 2009)
4. $y(k) - 3y(k-1) + 4y(k-2) = 2, k \geq 2$, given that $y(0) = 3, y(1) = 2$. (Madras, 2003)
5. $y_{n+2} - 2y_{n+1} + y_n = 2^n$, given that $y(0) = 2, y(1) = 0$. (Kerala, 2003)
6. $y_{n+2} - 5y_{n+1} + 6y_n = 36$, given that $y(0) = y(1) = 0$. (Anna, 2008)
7. $y_{n+2} - 6y_{n+1} + 9y_n = 0$.
8. $y_{n+2} + 4y_{n+1} + 3y_n = 3^n$.
9. $y_{n+2} + \frac{3}{4}y_{n+1} = \left(\frac{1}{4}\right)^n$, $n \geq 0$, $y_0 = 0$.
10. $u_{n+2} + u_n = 2^n$, given that $u_0 = 1, u_1 = 0$. (Kerala, 2006)
11. $y_{n+2} + 4y_{n+1} + 3y_n = 3^n + n(2)^n$, given that $y_0 = 0, y_1 = 1$. (Madras, 2006)
12. $y_{n+2} - 2y_{n+1} + y_n = 2^n + n(2)^n$, given that $y_0 = 2, y_1 = 1$.
13. $y_{n+2} - 5y_{n+1} + 6y_n = 2^n + 6n$.
14. $y_n + \frac{1}{2(n)}y_{n-1} = \left(\frac{1}{2}\right)^n\cos\frac{n\pi}{2}$, $n \geq 0$.
15. Find the response of the system described by $y_{n+2} + 3y_{n+1} + 2y_n = x_n$, where x_n is a unit step sequence and $y_{-1} = 1$.
16. Find the impulse response of a system described by $y_{n+2} + 3y_{n+1} - 4y_n = y_n = 0$.

11.5 OBJECTIVE TYPE OF QUESTIONS

MORE PROBLEMS

Choose the correct answer or fill in the blanks in each of the following problems:

1. $Z(1) =$ ____
2. $Z\{a^n\} =$ ____ (Kerala, 2010)
3. $Z\{\sin n\pi\} =$ ____
4. $Z\{k^2\} =$ ____
5. $Z\left\{\frac{1}{(n-2)!}\right\} =$ ____
6. If a_n is defined for $n = 0, 1, 2, \ldots$ only, then $Z\{a_{n+k}\} =$ ____
7. $Z\{a^n k\} =$ ____
8. Z-transform of $\{1^n\}$ is ____
9. Linearity property of Z-transform states that ____
10. $Z\left\{\frac{1}{(n+1)!}\right\} =$ ____
11. Initial value theorem on Z-transform states that ____
12. Z-transform is linear. (True or False)
13. If $Z\{a_n\} = A(z)$, then $\lim_{n\to\infty} a_n = \lim_{z\to 1}(z-1)A(z)$. (Gujarat, 2010)
14. Z-transform of the sequence $\{a^n\}$, $k \geq 0$ is $\frac{z}{z-a}$. (True or False)
15. Z-transform of $\{1/n!\}$, $k \geq 0$ is $e^{1/z}$. (True or False)
16. Z-transform of $\{1, 2, 3, 4, 5\}$ is $1 + 2z^{-1} + 3z^{-2} + 4z^{-3} + 5z^{-4}$. (True or False)
17. Z-transform of a unit impulse sequence $\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$ is z. (True or False)
18. Z-transform of a unit step sequence $u(n) = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$ is $\frac{z}{z-1}$. (True or False)

CHAPTER 24

Empirical Laws and Curve-fitting

1. Introduction. 2. Graphical method. 3. Laws reducible to the linear law. 4. Principle of least squares. 5. Method of Least squares. 6. Fitting of other curves. 7. Method of Group averages. 8. Fitting a parabola by Method of Moments. 10. Objective Type of Questions.

24.1 INTRODUCTION

In many branches of applied mathematics, it is required to express a given data, obtained from observations, in the form of a law connecting the two variables involved. Such a law is also referred to as known as **Empirical Law**. For example, it may be desired to obtain the law connecting the length and the temperature of a metal bar. At various temperatures, the length of the bar is measured. Then, by one of the methods explained below, a law is obtained that represents the relationship existing between temperature and length for the observed values. This relation can then be used to predict the length at a definite temperature.

(2) Scatter diagram. To find a relationship between the set of paired observations $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, we plot their corresponding values on the graph taking one of the variables along the x-axis and other along the y-axis. As $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$. The resulting diagram showing a collection of dots is called a **scatter diagram**. A smooth curve that approximates the above set of points is known as the **approximating curve**.

(3) Curve fitting. Several equations of different types can be obtained to express the given data approximately. But the problem is to find the equation of the curve of 'best fit' which may be most suitable for predicting the unknown values. The process of finding such an equation of 'best fit' is known as **curve fitting**.

If there are a pairs of observed values then it is possible to fit the given data to an equation that contains n arbitrary constants. For we can solve n simultaneous equations for n constants. But if we desire to obtain an equation representing the data but having less than n arbitrary constants, then we must have recourse to any of the four methods : *Graphical method, Method of Least squares, Method of Group averages* and *Method of Moments*. The graphical method fails to give the values of the unknowns as accurately as do the other methods do. The method of Least squares is, probably, the best to fit a unique curve to a given data. It is widely used in applications and can be easily implemented on a computer.

24.2 GRAPHICAL METHOD

When the curve representing the given data is a **linear law** $y = mx + c$, it is plotted as follows :

(i) Plot the given points on the graph paper taking a suitable scale.

(ii) Draw the straight line of best fit such that the points are evenly distributed about the line.

(iii) Taking two suitable points (x_1, y_1) and (x_2, y_2) on the line, calculate m, the slope of the line and c, its intercept on y-axis.

When the points representing the observed values do not approximate to a straight line, a smooth curve is drawn through them. From the shape of the graph, we try to infer the form of the curve and then reduce it to the form $y = mx + c$.

31.3 LAWS REDUCIBLE TO THE LINEAR LAW

We give below some of the laws in common use, indicating the way these can be reduced to the linear form by a suitable substitution :

(1) Given the law $y = ax^n + c$.

Taking $y = Y$ and $x^n = X$ the above law becomes $Y = aX + c$.

(2) Given the law $y = ax^n$.

Taking logarithms of both sides, it becomes $\log_{10} y = \log_{10} a + n \log_{10} x$

Putting $\log_{10} x = X$ and $\log_{10} y = Y$, it reduces to the form $Y = aX + c$, where $c = \log_{10} a$.

(3) Given the law $y = ax^n + bx$

Writing it as $\frac{y}{\log x} = a\frac{x^n}{\log x} + b$ and taking $\frac{y}{\log x} = Y$ and $\frac{x^n}{\log x} = X$,

the given law becomes, $Y = aX + b$.

(4) Given the law $y = ae^{bx}$

Taking logarithms, it becomes $\log_{10} y = (b \log_{10} e)x + \log_{10} a$

Putting $x = X$ and $\log_{10} y = Y$, it takes the form $Y = mX + c$ where $m = b \log_{10} e$ and $c = \log_{10} a$.

(5) Given the law $y = ax + bx^2$.

Dividing by x, we have $\frac{y}{x} = b\frac{x}{x} + a$.

Putting $\frac{y}{x} = Y$ and $x = X$, it reduces to the form $Y = bX + a$.

Example 31.1. It is the experience of investors of stock at speed V (in $) that the life N (in hours) of the stock is related to V by the law $N = aV^b$. Using the following data, find N when V = $ 55.

V (in dollars)	20	40	60	80	100
N (in hours)	8	13	21	34	55

We have, Given law is $N = aV^b$ ...(1)

Taking $V = x$ and $N = y$, (1) becomes

$$y = ax^b \quad ...(2)$$

which is a linear law.

Table for the values of x and y is as follows :

x	20	40	60	80	100
y	8	13	21	34	55

Plot these points. Draw the straight line of best fit through these points (Fig. 31.1)

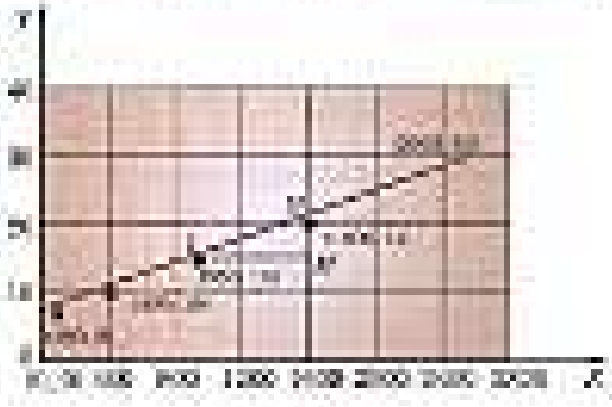


Fig. 31.1

Slope of this line $= AB$

$$= \frac{\Delta Y}{\Delta X} = \frac{21 - 13}{100 - 60} = \frac{8}{40} = 0.0833 \text{ nearly}.$$

Since A $(60, 13)$ lies on (2),

$$13 = a + 0.0833 \times 60,$$

where $$a = 13 - 7.65 = 7.35 \text{ nearly}.$$

Example 31.2. The following values of x and y are supposed to follow the law $y = ax^b + b \log_{10} x$. Find graphically the most probable values of the constants a and b.

x	5.05	5.06	5.08	5.09	5.01	5.12	5.02
y	20.7	20.4	20.1	19.8	20.6	22.5	20.4

Solution. Given law is $y = ax^n + b \log_{10} x$

i.e. $\frac{y}{\log_{10} x} = a\frac{x^n}{\log_{10} x} + b$ ...(1)

Taking $x^n/\log_{10} x = X$ and $y/\log_{10} x = Y$

(1) becomes $Y = aX + b$ ...(2)

This is a linear law. Table for the values of X and Y is as follows:

$X = x^n/\log_{10} x$	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
$Y = y/\log_{10} x$	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Points	P_1	P_2	P_3	P_4	P_5	P_6	P_7

Plot these points and draw the straight line of best fit through these points (Fig. 24.2).

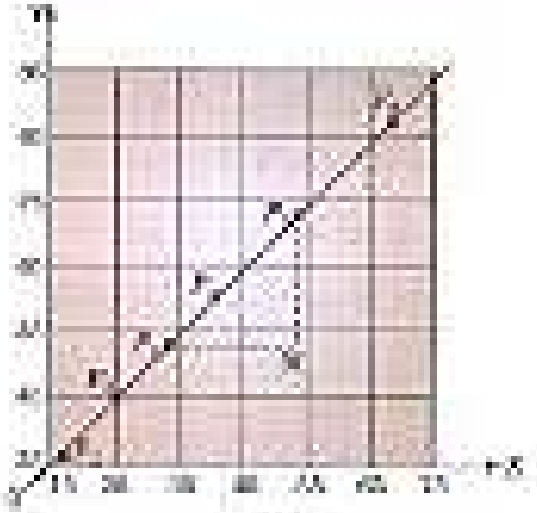


Fig. 24.2

Slope of this line $= \frac{MR}{AR} = \frac{79.65 - 52.50}{3013 - 2048} = \frac{31.15}{3136} = 0.99$

Since P_1 lies on the Y-axis then from (2) $52.50 = 0.99 \times 2048 + b$ whence $b = 32.2$

Hence (2) becomes $y = (0.99)x^n + (32.2)\log_{10} x$

Example 24.6. The [illegible] of a quantity y is measured for various values of x and the following [illegible] $y = ax^n + b\log_{10} x$:

x	2.00	3.00	4.00	4.50	5.00	7.00
y	45.0	58.1	66.5	69.2	82.6	105.0

[illegible] law [illegible] Test graphically the correctness of this law and if it is true [illegible] find the [illegible] values of the constants.

Solution. If the law is $y = ax^n$ ...(1)

Taking logarithms to base 10, we have $\log_{10} y = \log_{10} a + n\log_{10} x$

Putting $x = X$ and $\log_{10} y = Y$, it becomes $y = (n)\log_{10} x + \log_{10} a$ ...(2)

Table for the values of X and Y is as under:

$X = x$	2.00	3.00	4.00	4.50	5.00	7.00
$Y = (\log_{10} y)$	1.65	1.76	1.82	1.84	1.92	2.02
Points	P_1	P_2	P_3	P_4	P_5	P_6

Scale: 1 small division along x-axis = 0.1

1 small division along y-axis = 0.1.

Plot these points and draw the 'best fit' line. As these points are lying almost along a straight line, the given law is nearly accurate (Fig. 24.21).

Slope of this line $= b \log_{10} e$

$$= \frac{BC}{AC} = 0.11$$

whence $$b = \frac{0.11}{\log_{10} e} = 0.11 \times 2.303 = 0.253$$

Since the point (1.11, 1.31) lies on the line, then $1.31 = 0.253 + b \log_{10} c$ whence $c = 17$ nearly.

Hence the curve of best fit is $y = 17e^{0.25x}$.

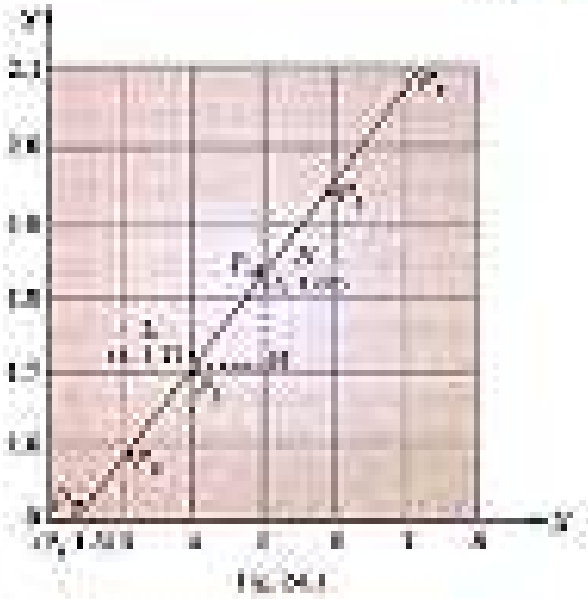


Fig. 24.21

EXERCISE 24.10

1. If y is the pull required to lift the weight by means of a pulley block, find a linear law of the form $y = ax + b$, connecting pull and load, using the following data:

load, x (kg)	20	30	40	50
pull, y (kg)	12	16	20	24

Check graphically that $a = 0.8$.

2. The resistance R of a certain filament lamp was measured at various values of the voltage V and the following observations were made:

Voltage	V	40	50	70	84	91
Resistance	R	75	90.9	109.2	127.8	136.8

Assuming a law of the form $R = \frac{a}{V} + b$, find by graphical methods the best values of a and b.

3. Verify if the values of x and y, obtained as shown in the following table, obey the law $y = ax + b\sqrt{x}$. If so, find graphically the values of a and b.

x	500	1200	2400	4200	5600
y	0.28	0.32	0.36	0.42	0.51

4. The following values of T and l follow the law $T = kl^n$. Test if this is so and find the best values of k and n.

T = 1.0	1.5	2.0	2.5
l = 20	39.9	80	130

5. Find the best values of a and b if $y = ax + b \log_{10} x$ is the form which connects more closely the observed values given below:

x	1	3	4	5	8
y	3.54	5.92	7.47	8.99	13.63

6. To the curve $y = ax^n$ to the following data:

x	2	3	5
y	8.1	12	22.1

(Manchester, 1961)

7. The following are the results of an experiment on friction of leather on iron. The speed being constant, corresponding values of the coefficient of friction and the temperature are shown in the table:

t	100	111	200	300	400	500	600
μ	0.5611	0.5460	0.5011	0.4545	0.4000	0.3614	0.3618

Test if μ are given by the law $\mu = ae^{bt}$. Find the values of a and b by plotting the graph for μ and t.

9.12 PRINCIPLE OF LEAST SQUARES

The graphical method has the obvious drawback of being unable to give a unique curve of fit. The principle of least squares, however, provides an elegant procedure for fitting a unique curve through the data.

Let the curve $y = a + bx + cx^2 + \ldots + kx^{n-1}$ ...(1)

be fitted to the set of n data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$.

Now we have to determine the constants $a, b, c, \ldots, k$ such that it represents the curve of best fit. In case $n = m$, on substituting the values (x_i, y_i) in (1), we get n equations from which n unknown constants can be found. But when $n > m$, we obtain n equations which are more than the m constants and hence cannot be solved for these constants. So we try to determine those values of $a, b, c, \ldots, k$ which satisfy all the equations as nearly as possible and thus may give the best fit. In such cases, we apply the principle of least squares.

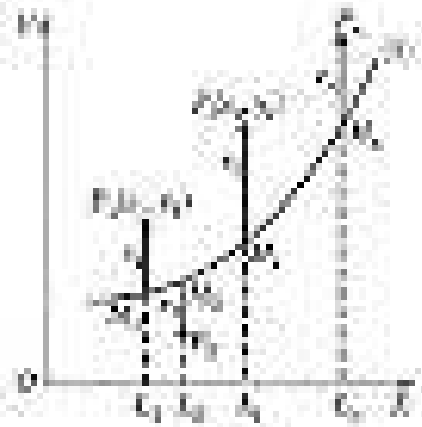


Fig. 9.4

At $x = x_1$, the observed (or experimental) value of the ordinate is $y_1 = P_1Q_1$ and the corresponding value on the fitting curve (1) is $a + bx_1 + cx_1^2 + \ldots + kx_1^{n-1} = R_1Q_1$ (= v_1, say) which is the expected (or calculated) value (Fig. 9.4). The difference of the observed and the expected values ($y_1 - v_1$) or ($v_1 - y_1$) is called the error (or residual) at $x = x_1$. Clearly some of the errors $y_1 - v_1, y_2 - v_2, \ldots, y_n - v_n$ will be positive and others negative. Thus to give equal weightage to each error, we square each of these and form their sum, i.e., $S = e_1^2 + e_2^2 + \ldots + e_n^2$.

The curve of best fit is that for which S is as small as possible, i.e., S is the sum of the squares of the errors is a minimum. This is known as the principle of least squares and was suggested by Legendre* in 1805.

Note. The principle of least squares does not help us to determine the form of the appropriate curve which can fit a given data. It only determines the best possible values of the constants in the equation when the form of the curve is known before hand. The selection of the curve is a matter of experience and graphical considerations.

9.13 (1) METHOD OF LEAST SQUARES

Fitting a straight line by the method of least squares.

Let the straight line to be fitted to the curve be

$$y = a + bx \quad \ldots(1)$$

to a given set of observations $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$. For any x_i, the observed value is y_i and the expected value is $v_i = a + bx_i$ so that the error $e_i = y_i - v_i$.

$\therefore$ the sum of the squares of these errors is

$$E = e_1^2 + e_2^2 + \ldots + e_n^2$$
$$= (y_1 - a - bx_1)^2 + (y_2 - a - bx_2)^2 + \ldots + (y_n - a - bx_n)^2 \quad \text{[See § 2.12 (2)]}$$

For E to be minimum, we have

$$\frac{\partial E}{\partial a} = 0 = 2(y_1 - a - bx_1)(-1) + 2(y_2 - a - bx_2)(-1) + \ldots + 2(y_n - a - bx_n)(-1) \quad \ldots(2)$$

$$\frac{\partial E}{\partial b} = 0 = -2x_1(y_1 - a - bx_1) - 2x_2(y_2 - a - bx_2)$$
$$- \ldots - 2x_n(y_n - a - bx_n) \quad \ldots(3)$$

$$\frac{\partial E}{\partial c} = 0 = -2x_1^2(y_1 - a - bx_1) - 2x_2^2(y_2 - a - bx_2)$$
$$- \ldots - 2x_n^2(y_n - a - bx_n) \quad \ldots(4)$$

Equation (2) simplifies to

$$y_1 + y_2 + \ldots + y_n = na + b(x_1 + x_2 + \ldots + x_n) + c(x_1^2 + x_2^2 + \ldots + x_n^2)$$

i.e., $$\Sigma y_i = na + b\Sigma x_i + c\Sigma x_i^2 \quad \ldots(5)$$

* See footnote on p. 311.

Equation (ii) becomes

$$a(x_1 + x_2 + \dots + x_n) + b(x_1^2 + x_2^2 + \dots + x_n^2) + c(x_1^3 + x_2^3 + \dots + x_n^3) = (x_1y_1 + x_2y_2 + \dots + x_ny_n)$$

i.e., $$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \quad \dots(iv)$$

Similarly (iii) simplifies to $\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$ ...(v)

The equations (iv), (v) and (i) are known as *Normal equations* and can be solved as in the previous exercise to find a, b, c. The values of these constants when substituted in (i) give the desired curve of best fit.

(ii) Working procedure

(a) To fit the straight line $y = a + bx$

(i) Substitute the observed set of values in this equation.

(ii) Form normal equations for each constant

i.e., $$\Sigma y = an + b\Sigma x, \ \Sigma xy = a\Sigma x + b\Sigma x^2$$

The normal equations for the unknowns are obtained by multiplying the equations by the coefficient of a and adding. The normal equation for b is obtained by multiplying the equation by the coefficient of b i.e., x and adding.

(iii) Solve these normal equations as simultaneous equations for a and b.

(iv) Substitute the values of a and b in $y = a + bx$, which is the required line of best fit.

(b) To fit the parabola $y = a + bx + cx^2$

(i) Form the normal equations $\Sigma y = an + b\Sigma x + c\Sigma x^2$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

and $$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

The normal equations for b are obtained by multiplying the equations by the coefficients of b i.e., x and adding.

(ii) Solve these as simultaneous equations for a, b, c.

(iii) Substitute the values of a, b, c in $y = a + bx + cx^2$, which is the required parabola of best fit.

(c) In general, the curve $y = a + bx + cx^2 + \dots + kx^{n-1}$ can be fitted to a given data by writing n normal equations.

Example 3.6.1: Fit a straight line and a parabola to the following data and find the value of y for $x = 1.5$ using the better fit. Which is a better fit? Justify your answer.

x	0	1	2
y	1	1.8	3.4

Solution: The normal equations are

$$\Sigma y = 4a + 3b\Sigma x \Big| \quad \dots(i)$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2 \Big|$$

The values of Σy etc. are calculated for the use of the following table:

x	y	xy	x^2
0	1	0.0	0.0
1	1.8	1.8	1.0
2	3.4	6.8	4.0
3	5.9	17.7	9.0
Total = 6	12	26.3	14.0

The equations (i) become $12 = 4a + 6b$ i.e., $6a + 3b = 12$ ...(iii)

and $$26.3 = 6a + 14b \quad \dots(iv)$$

Multiplying (iii) by 3 and subtracting from (iv), we get

$$b = 0.1571 \quad \text{i.e.,} \quad \text{from (iii)} \quad a = 2.1714$$

Hence the law of the form is

$$P = 0.0769 + 0.357V$$

When $V = 12.5$ kg, $P = 0.0769 + 0.357 \times 12.5 = 4.5394$ kg

Note. The calculations are simplified when the central value of x is zero. It is therefore advisable to make the central value zero if it is not so. This is illustrated by the next example.

Example 2.15. Fit a second degree parabola to the following data:

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

(P.T.U. 2003)

Solution. Let $u = x - 2$ and $v = y$ so that the parabola $y = a + bx + cx^2$ becomes

$$v = A + Bu + Cu^2 \qquad \ldots(1)$$

The normal equations are:

$$\Sigma v = 5A + B\Sigma u + C\Sigma u^2 \quad \text{or} \quad 12.9 = 5A + 10C$$

$$\Sigma uv = A\Sigma u + B\Sigma u^2 + C\Sigma u^3 \quad \text{or} \quad 11.3 = 10B$$

$$\Sigma u^2 v = A\Sigma u^2 + B\Sigma u^3 + C\Sigma u^4 \quad \text{or} \quad 30.8 = 10A + 34C$$

Solving these as simultaneous equations, we get

$$A = 1.44, \qquad B = 1.13, \qquad C = 0.65.$$

Obviously, $v = 1.44 + 1.13u + 0.65u^2$

or $y = 1.44 + 1.13(x - 2) + 0.65(x - 2)^2$

Hence $y = 1.82 - 1.07x + 0.65x^2$.

Example 2.16. Fit a second degree parabola to the following data:

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	1.1	1.3	1.6	2.0	2.7	3.4	4.1

(P.T.U., 2002; Bilaspur, 2006)

Solution. We shift the origin to 2.5, i.e. take 2.5 as the new unit. This amounts to changing the variable x to X, by the relation $X = 2x - 5$.

Let the parabola of fit be $y = a + bX + cX^2$. The values of X etc. are calculated as below:

x	X	y	Xy	X^2	X^2y	X^3	X^4
1.0	−3	1.1	−3.3	9	9.9	−27	81
1.5	−2	1.3	−2.6	4	5.2	−8	16
2.0	−1	1.6	−1.6	1	1.6	−1	1
2.5	0	2.0	0.0	0	0.0	0	0
3.0	1	2.7	2.7	1	2.7	1	1
3.5	2	3.4	6.8	4	13.6	8	16
4.0	3	4.1	12.3	9	36.9	27	81
Total	0	16.2	14.3	28	69.9	0	196

The normal equations are:

$$7a + 28c = 16.2; \qquad 28b = 14.3; \qquad 28a + 196c = 69.9$$

Solving these as simultaneous equations, we get:

$$a = 2.01, b = 0.511, c = 0.161$$

$$\therefore \quad y = 2.07 + 0.511X + 0.161X^2$$

Replacing X by $2x - 5$ in the above equation, we get

$$y = 2.07 + 0.511(2x - 5) + 0.161(2x - 5)^2$$

which simplifies to $y = 1.06 - 0.298x + 0.644x^2$. This is the required parabola of best fit.

Example 4.13. Fit a second degree parabola to the following data :

x	2000	1960	2001	1995	2007	1974	1960	1996	1997
y	352	356	335	348	346	362	343	364	356

(U.P.T.U., 2008)

Solution. Taking $u = x - 1980$ and $v = y - 353$, the equation $y = a + bx + cx^2$ becomes

$$v = A + Bu + Cu^2 \qquad \ldots(1)$$

x	$u = x - 1980$	y	$v = y - 353$	uv	u^2	u^3	u^4	u^2v
2000	20	352	−1	−20	400	8000	160000	−400
1960	−20	356	3	−60	400	−8000	160000	1200
2001	21	335	−18	−378	441	9261	194481	−7938
1995	15	348	−5	−75	225	3375	50625	−1125
2007	27	346	−7	−189	729	19683	531441	−5103
1974	−6	362	9	−54	36	−216	1296	324
1960	−20	343	−10	200	400	−8000	160000	−4000
1996	16	364	11	176	256	4096	65536	2816
1997	17	356	3	51	289	4913	83521	867
Total	$\Sigma u = 11$		$\Sigma v = -15$	$\Sigma uv = -349$	$\Sigma u^2 = 3175$	$\Sigma u^3 = 25013$	$\Sigma u^4 = 1510921$	$\Sigma u^2v = -18359$

The normal equations are

$$\Sigma v = 9A + B\Sigma u + C\Sigma u^2 \quad \text{or} \quad -15 = 9A + 11C$$

$$\Sigma uv = A\Sigma u + B\Sigma u^2 + C\Sigma u^3 \quad \text{or} \quad -349 = 11A + 3175B + 25013C$$

$$\Sigma u^2v = A\Sigma u^2 + B\Sigma u^3 + C\Sigma u^4 \quad \text{or} \quad -18359 = 3175A + 25013B + 1510921C$$

On solving these equations, we get $A = \frac{824}{231}, B = \frac{17}{28}, C = -\frac{247}{824}$

$\therefore$ (1) becomes $v = \frac{824}{231} + \frac{17}{28}u - \frac{247}{824}u^2$

or $y - 353 = \frac{824}{231} + \frac{17}{28}(x - 1980) - \frac{247}{824}(x - 1980)^2$

or $y = 353 - \frac{824}{231} - \frac{17 \times 1980}{28} + \frac{17}{28}x + \frac{247 \times 3960}{824}x - \frac{247}{824}x^2$

or $y = 1 - 134.05 - 2005.23x + 367 + 0.60x + 1038.46x - 0.30x^2$

Hence $y = -1036.28 + 1039.06x - 0.30x^2$.

EXERCISE 4.3

1. By the method of least squares, fit a straight line from the following data :

x	1	2	3	4	5
y	14	27	40	55	68

(U.P.T.U., 2005)

2. Fit a straight line to the following data :

Year x	2002	2003	2004	2005	2006
Production y	5	10	15	20	30

(in thousand tons)

and find the expected production in 2009.

3. A study of apprenticeship system in a certain industrial area of U.P. gives the following figures. Corresponding to various values of X, the number of defectives Y in a batch are noted. The data are given below :

p_i	200	113	100	100	180	208	
T_i	8.00	6.00	8.00	6.70	9.00	8.00	

Calculate the coefficient of correlation between T and p.

4. The results of measurements of electric resistance R of a copper bar at various temperatures T°C are listed below:

t	30	31	30	36	40	41	36
$R(\Omega)$	70	71	73	80	81	80	82

Find a relation $R = a + bt$ between R and t and estimate R for $t = 45$°C.

5. Find the least squares linear regression line $y = a + bx$ for the given data:

x_i	1	2	3	4	5	6	23	24	
y_i	4	2	3	4	4	7	9	8	(U.P.T.U., 2011)

6. Fit a straight line to the following data:

x_i	0	1	2	3	4	5	6	7	8	9
y_i	9	8	10	11	11	13	14	15	15	(Bhopal, 2008)
x_i	0	1	2	3	4	5	6	7	8	9
y_i	2	3	4	4	4	5	6	5	8	(U.P.T.U., 2008)

7. Find the parabola of the form $y = a + bx + cx^2$ which fits the most closely with the observations:

x_i	−3	−2	−1	0	1	2	3
y_i	6.01	5.11	4.03	3.09	4.02	5.10	6.11

(U.P.T.U., 2007; U.P.T.U., 2006)

8. Fit a parabola $y = a + bx + cx^2$ to the following data:

x_i	1	2	3	4	5
y_i	5.07	12.85	31.47	57.38	91.29

(U.P.T.U., 2003)

9. Fit a second degree parabola to the following data:

x_i	1	2	3	4	5	6	7	8	9	10
y_i	10	12	16	20	26	32	38	46	54	62

(U.P.T.U., 2009)

10. The following table gives the results of the measurements of the pressure and volume. If the relation between the pressure and volume is of the form $PV^n = k$, where n and k are constants, find the values of n and k.

V	50	60	70	90	100	120
P	5.4	4.3	3.5	2.7	2.5	1.8

(U.P.T.U., 2005)

11. The velocity V of a liquid is known to vary with the temperature according to the equation $V = a + bT + cT^2$. Find the best values of a, b and c for the following table:

T	1	2	3	4	5	6	7
V	0.53	0.55	0.49	0.38	0.22	0.07	0.01

(U.P.T.U., MCA, 2009)

4.15 FITTING OF OTHER CURVES

(1) $y = ax^b$

Taking logarithms, $\log_{10} y = \log_{10} a + b \log_{10} x$

i.e., $$Y = A + bX \quad \text{where } X = \log_{10} x,\ Y = \log_{10} y \text{ and } A = \log_{10} a \qquad \ldots(1)$$

∴ The normal equations for (1) are $\Sigma Y = nA + b\Sigma X$, $\Sigma XY = A\Sigma X + b\Sigma X^2$

from which A and b can be determined. Then a can be calculated from $A = \log_{10} a$.

(2) $y = ae^{bx}$ (Exponential curve)

Taking logarithms, $\log_{10} y = \log_{10} a + bx \log_{10} e$

i.e., $$Y = A + Bx \text{ where } Y = \log_{10} y,\ A = \log_{10} a \text{ and } B = b \log_{10} e$$

Here the normal equations are $\Sigma Y = nA + B\Sigma x$, $\Sigma xY = A\Sigma x + B\Sigma x^2$

from which A, B can be found and consequently a, b can be calculated.

(3) $xy^n = k$ (Hyperbola)

Taking logarithms, $\log_{10} x + n \log_{10} y = \log_{10} k$ or $\log_{10} y = \frac{1}{n} \log_{10} k - \frac{1}{n} \log_{10} x$.

This is of the form $Y = A + BX$

where $X = \log_{10} x$, $Y = \log_{10} y$, $A = \frac{1}{a}$, $\log_{10} A$, $B = -\frac{b}{a}$.

Here also the method is to find a straight line of best fit through the given data.

Example 24.6. Find the curve of best fit of the form $y = a_0 + a_1 x^2$ to the following data.

x	-2	0	1	2
y	3	1	2	6

(U.P.T.U., 2005)

Solution. Putting $x^2 = X$, we have $y = a_0 + a_1 X$

∴ the normal equations are: $\Sigma y = n a_0 + a_1 \Sigma X$; $\Sigma Xy = a_0 \Sigma X + a_1 \Sigma X^2$.

The values of ΣX, ΣX^2 etc. are calculated below :

x	y	X	X^2	Xy
-2	3	4	16	12
0	1	0	0	0
1	2	1	1	2
2	6	4	16	24
	$\Sigma y = 12$	$\Sigma X = 9$	$\Sigma X^2 = 33$	$\Sigma Xy = 38$

∴ the normal equations become $12 = 4a_0 + 9a_1$; $38 = 9a_0 + 33a_1$

Solving these equations we get, $a_0 = 0.357$, $a_1 = 1.111$.

Hence the curve of best fit is $y = 0.357 + 1.111X$ i.e., $y = 0.357 + 1.111x^2$.

Example 24.7. *An experiment gave the following values :*

v (cm/sec)	350	400	500	700
t (sec)	63	35	7	24

It is known that v and t are connected by the relation $v = at^b$. Find the best possible values of a and b.

Solution. We have $\log_{10} v = \log_{10} a + b \log_{10} t$

or $y = A + bX$, where $X = \log_{10} t$, $y = \log_{10} v$, $A = \log_{10} a$

∴ the normal equations are

$$\Sigma y = 4A + b\Sigma X \quad \ldots(1)$$

$$\Sigma Xy = A\Sigma X + b\Sigma X^2 \quad \ldots(2)$$

Now ΣX etc. are calculated as in the following table :

v	t	$X = \log_{10} t$	$y = \log_{10} v$	Xy	X^2
350	63	1.7993	2.5441	4.5775	3.2377
400	35	1.5441	2.6021	4.0172	2.3841
500	7	0.8451	2.6990	2.2811	0.7141
700	24	1.3802	2.8451	3.9274	1.9054
Total		5.5687	10.6903	14.8032	8.2413

∴ Equations (1) and (2) become

$$4A + 5.569b = 10.690 \quad \ldots(3)$$

$$5.569A + 8.241b = 14.803 \quad \ldots(4)$$

Solving these, $A = 2.845$, $b = -0.1557$

∴ $a = \text{antilog } A = \text{antilog } 2.845 = 698.8$

Example 24.8. *Predict the mean annual rainfall over an altitude of 3000 feet by fitting an exponential curve to the given data :*

Altitude (x)	:	50	100	150	1200	4000	4500	5000
Rainfall (in cm) (y)	:	50	40	30	20	10	05	03

(U.P.T.U., 2007 ; U.P.T.U., 2008)

Solution: Let $y = ab^x$ be the exponential curve.

Then $\log_{10} y = \log_{10} a + x \log_{10} b$

or $Y = A + Bx$ where $Y = \log_{10} y, A = \log_{10} a, B = \log_{10} b$

∴ the normal equations are

$$\Sigma Y = nA + B\Sigma x \quad \ldots(1)$$

$$\Sigma xY = A\Sigma x + B\Sigma x^2 \quad \ldots(2)$$

Now the values are calculated as follows:

x	y	$Y = \log_{10} y$	xY	x^2
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
$\Sigma x = [illegible]$		[illegible]	[illegible]	[illegible]

∴ equations (1) and (2) become

[illegible] $= 7A +$ [illegible]

[illegible] $=$ [illegible] $+$ [illegible]

Solving these equations, we get $A =$ [illegible], $B =$ [illegible]

∴ $\log_{10} y = A + Bx =$ [illegible] $+$ [illegible] x

Hence $y = ab^x =$ [illegible] $\times$ [illegible]^x Ans.

Example 16.11. The pressure and volume of a gas are related by the equation $pv^\gamma = k$, where γ and k are constants. Find the values of γ and k for the following set of observations:

p (kg/cm^2)	0.5	1.0	1.5	2.0	2.5	3.0	[illegible]
v (litres)	1.62	1.00	0.75	0.62	0.52	0.46	[illegible]

Solution. We have $\log_{10} p + \gamma \log_{10} v = \log_{10} k$

or $$\log_{10} v = \frac{1}{\gamma} \log_{10} k - \frac{1}{\gamma} \log_{10} p \quad \text{or} \quad Y = A + BX$$

where $X = \log_{10} p, Y = \log_{10} v, A = \frac{1}{\gamma} \log_{10} k, B = -\frac{1}{\gamma}$

∴ the normal equations are

$$\Sigma Y = nA + B\Sigma X \quad \ldots(1)$$

$$\Sigma XY = A\Sigma X + B\Sigma X^2 \quad \ldots(2)$$

Now the values are calculated as follows:

p	v	$X = \log_{10} p$	$Y = \log_{10} v$	XY	X^2
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Total		[illegible]	[illegible]	[illegible]	[illegible]

∴ equations (1) and (2) become

[illegible] $= 6A +$ [illegible]$B =$ [illegible]

$$\Sigma \log y = 9 \log A + B \Sigma x \quad \text{i.e.,} \quad 9.0414 = 9 \log A + 45 B \quad \ldots(4)$$

Solving these, we get $A = 0.3125$, $B = -0.1998$.

$\therefore \quad \gamma = -B = 1.2256$ and $b = \text{antilog}(\log b) = \text{antilog}\,0.8169 = 1.040$.

Hence the required exponential is $y = 1.040\,(1.2256)^x$.

EXERCISE 2.5

1. If $(1, 1)$, $(2, 2)$, $(3, 3)$ and $(4, 4)$ are connected by a relation of the type $y = ax^b$, find by the method of least squares and hence the help of the following table:

x	10	15	20	25	30	
y	6	20	36	51	80	(Nellore, 2000)

2. Using the method of least squares fit the curve $y = ax^b$ to following observations:

x	1	2	3	4	5
y	15	22	30	37	47

3. Fit the curve $y = ax^b$ to the following data:

x	1	2	3	4	5	6	7	8
y	0.4	0.8	1.3	1.8	2.4	3.2	4.0	4.8

(UPTU, 2005)

4. Fit the curve $y = ax^b$ by fitting a parabola of the form $\log y = A \log x + B$ to the following data:

x	1	2	3	4
y	2	1.6	1	0.8

(UPTU, 2003)

5. Find the least squares curve $y = ax^b$ to fit the following data:

x	1	2	3	4
y	1.5	0.89	0.58	0.44

(Madras, 1997)

6. Fit the curve $y = ax^b$ by fitting a power curve to the given data:

x	1	2	3	4	5	6
y	0.5	4.38	8.43	13.8	19.58	27.0

(UPTU, 2002)

7. Fit the curve of the form $y = ax^b$ to the following data:

x	35	40	45	50	55
y	2.4	3.0	3.8	4.3	4.8

(UPTU, 2011 S; UPTU, 2005)

8. Obtain the least squares fit of the form $y = ax^b + bx^{-2}$ for the data:

x	0.1	0.2	0.3	0.4
y	0.9	0.58	0.44	0.38

(UPTU, 2009)

9. The relation between cost of production y and output x is given by the following table:

x	0	1	2	3	4
y	200	80	50	90	320

Use the method of least squares to fit a curve of the form $y = ax^b$ to the data.

10. Using method of least squares fit a relation of the form $y = ax^b$ to the following data:

x	2	4	6	8	10
y	118	1720	9074	34561	95481

(Delhi Matric, 2001)

2.10 METHOD OF GROUP AVERAGES

Let the straight line $y = a + bx$ ...(1)

fit the set of n observations

$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ equi to closely. (Fig. 2.6)

When $x = x_1$, the observed (or experimental) value of $y = y_1$ and from (1),

$$y = a + bx_1 + \lambda_1 R_1,$$

which is known as the expected (or calculated) value of y at x_1.

Thus y_1 = observed value at x_1 – expected value at x_1

$$= y_1 - (a + bx_1) = R_1 P_1,$$

which is called the error (or residual) at x_1. Similarly the errors for the other observations are

$$e_2 = y_2 - (a + bx_2) = R_2 P_2$$

$$e_n = y_n - (a + bx_n) = R_n P_n$$

Some of these errors may be positive and others negative.

The method of group averages is based on the assumption that the sum of the residuals is zero. To find the constants a and b in (1), we require two equations. As such we divide the data into two groups. The first containing k observations

$$(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k);$$

and the second group having the remaining $n - k$ observations

$$(x_{k+1}, y_{k+1}), (x_{k+2}, y_{k+2}), \ldots, (x_n, y_n).$$

Assuming that the sum of the errors in each group is zero, we get

$$(y_1 - (a + bx_1)) + (y_2 - (a + bx_2)) + \ldots + (y_k - (a + bx_k)) = 0$$

$$(y_{k+1} - (a + bx_{k+1})) + (y_{k+2} - (a + bx_{k+2})) + \ldots + (y_n - (a + bx_n)) = 0$$

On simplification, we obtain

$$\frac{y_1 + y_2 + \ldots + y_k}{k} = a + b\,\frac{x_1 + x_2 + \ldots + x_k}{k} \quad \ldots(2)$$

$$\frac{y_{k+1} + y_{k+2} + \ldots + y_n}{n - k} = a + b\,\frac{x_{k+1} + x_{k+2} + \ldots + x_n}{n - k} \quad \ldots(3)$$

In (2), $\frac{1}{k}(x_1 + x_2 + \ldots + x_k)$ and $\frac{1}{k}(y_1 + y_2 + \ldots + y_k)$ are simply the average values of x and y of the first group. Hence the equations (2) and (3) are obtained from (1) by replacing x and y by their respective averages of the two groups. Solving (2) and (3), we get a and b.

Obs. The main drawback of this method is that a different grouping of the observations will give different values of a and b. To overcome, we divide the data in such a way that each group contains almost an equal number of observations.

Example 10.10. The lateral heat of vaporization (y calories per gram) for various temperatures (x degrees Centigrade) is given in the following table of experimental observations:

x	40	50	60	70	80	90	100	110
y	0.0002	0.0026	0.0042	0.0052	0.0062	0.0072	0.0082	0.0092

For this range of temperature, a relation of the form $y = a + bx$ is known to fit the data. Find the values of a and b by the method of group averages. (Madras, 2002)

Solution. Let us divide the data into two groups each containing four readings. Then we have

x	y	x	y
40	0.0002	80	0.0062
50	0.0026	90	0.0072
60	0.0042	100	0.0082
70	0.0052	110	0.0092
$\Sigma x = 220$	$\Sigma y = 0.0122$	$\Sigma x = 380$	$\Sigma y = 0.0308$

Substituting the averages of x and y of the two groups in the given relation, we get

$$\frac{4363.5}{4} = a + b\frac{720}{8} \quad \text{i.e., } 3090.8 = a + 90b \quad \ldots(i)$$

$$\frac{4308.3}{4} = a + b\frac{800}{8} \quad \text{i.e., } 3039.85 = a + 100b \quad \ldots(ii)$$

Solving (i) and (ii), we obtain

$$a = 1698.26, b = -3.305.$$

15.11 FITTING A PARABOLA

We have applied the method of averages to linear xy involving two constants only. To fit the parabola

$$y = a + bx + cx^2 \quad \ldots(1)$$

which involves three constants, in a set of observations, we proceed as follows :

Let (x_1, y_1) be a point on (1) satisfying the given data so that

$$y_1 = a + bx_1 + cx_1^2$$

Then $$y - y_1 = b(x - x_1) + c(x^2 - x_1^2)$$

or $$\frac{y - y_1}{x - x_1} = b + c(x + x_1)$$

Putting $x_1 = X$ and $(y - y_1)/(x - x_1) = Y$, it takes the linear form

$$Y = b + cX.$$

Now b and c can be found as before.

Example 15.12. The corresponding values of x and y are given in the following table :

x	81.2	83.0	77.6	80.7	84.5	78.9
y	250	255	238	252	262	245

Solution. Taking $x = 84$, $y = 262$ as a particular point on $y = a + bx + cx^2$,

we get $$262 = a + b(84) + c(84)^2 \quad \ldots(i)$$

$$\therefore \quad y - 262 = b(x - 84) + c(x^2 - 84^2)$$

or $$\frac{y - 262}{x - 84} = b + c(x + 84)$$

i.e., $$Y = b + cX \quad \ldots(ii)$$

where $X = x + 84$, $Y = (y - 262)/(x - 84)$.

Now we have the following table of values :

x	y	$X = x + 84$	$Y = (y - 262)/(x - 84)$
81.2	250	165.2	3.333
84.0	262	—	—
77.6	238	161.6	3.387
		$\Sigma X = 326.8$	$\Sigma Y = 6.720$
80.7	252	164.7	3.704
84.5	262	168.5	3.158
78.9	245	162.9	3.158
		$\Sigma X = 496.1$	$\Sigma Y = 9.020$

Substituting the averages of X and Y in (ii), we get

$$\frac{6.060}{3} = b + c\frac{326.3}{2} \quad \text{i.e., } 2.33 = b + 163.15c \quad \ldots(iii)$$

$$\frac{9.020}{3} = b + c\frac{496.1}{3} \quad \text{i.e., } 3.01 = b + 331.1c \quad \ldots(iv)$$

(c)–(b) gives $a = 0.0015$

and (b) gives $b = 2.0997$ i.e., 2.1 nearly

From (2), we get $a = 86.2065$ i.e., 87 nearly.

Hence the parabola of fit is

$$y = 87 + 2.1x + .0015x^2$$

Example 24.14. The point resistance R (ohms) is measured for various velocities V (km/hr) as follows:

V	20	30	40	50	60
R	5	8	14	23	34

If R is related to V by the formula $R = a + bV^2$, find a and b.

Solution. To find a and b we take the following three values of V and R from the data:

$V_1 = 20, \quad V_2 = 40, \quad V_3 = 60$

Then $R_1 = 5, \quad R_2 = 14, \quad R_3 = 34$

$\therefore \quad \Sigma R_1 = n(\Sigma V_1)a + n(\Sigma V_1^2)b = 3a + 120b$

and $\Sigma R_1 V_1 = a(\Sigma V_1) + b(\Sigma V_1^2) = 120a + 5600b$

where $n = 3$

$$a = \frac{R_1 R_2 - R_3^2}{R_1 + R_2 - 2R_3} = 1.63$$

Then $R = 1.63 + bV^2$ or $\log_{10}(R - 1.63) = \log_{10} b + 2\log_{10} V$

i.e., $Y = b + 2X$ ...(1)

where $X = \log_{10} V$, $Y = \log_{10}(R - 1.63)$, $b = \log_{10} b$.

We now have the following table of values:

V	R	$X = \log_{10} V$	$Y = \log_{10}(R - 1.63)$
20	5	1.3010	0.5105
40	14	1.6021	0.9395
60	34	1.7782	1.1847
		$\Sigma X = 4.6813$	$\Sigma Y = 2.6347$
30	8	1.4771	0.7559
50	23	1.6990	1.0334
		$\Sigma X = 3.1761$	$\Sigma Y = 1.7893$

Substituting the averages of ΣX and ΣY in (1), we obtain

$$\frac{1.7893}{2} = b + a\,\frac{3.1761}{2} \quad \text{i.e., } 1.8215 = b + 1.5881a \qquad ...(2)$$

$$\frac{2.6347}{2} = b + a\,\frac{4.6813}{2} \quad \text{i.e., } 1.4130 = b + 1.0518a \qquad ...(3)$$

Solving (2) and (3), we get $a = 2.01$, $b = 1.86$ approx.

$$b = \text{antilog } b = \text{antilog } 1.86 = 72.44$$

Problems 24.4

1. Fit a straight line of the form $y = a + bx$ to the following data by the method of least squares:

x	2	5	10	12	20	25
y	18	15	12	10	6	3

(Ans. $a = 20.01$, $b = -0.81$)

2. The weights of a certain boy recorded at various ages are given below:

Age (years)	1	2	3	5	8	10	11	12	13	14
Weight (kg)	10.5	13.7	16.0	19.2	23.4	31.1	35.2	39.5	45.2	50.8

Find a straight line of best fit.

2. Using the method of averages, fit a parabola $y = ax^2 + bx + c$ to the following data:

x:	78	89	97	98	110	120
y:	24.5	28.6	34.0	37.8	40.8	46.0

3. While finding a correlation curve, the following data is obtained. Use method of group averages to fit the equation $y = ax + bx^2$, where y is the discharge in thousand cubic feet per minute and x is the number of hours. Find the values of the constants a and b by the method of group averages.

x:	2	2.5	3	3.5	4	4.5	5	5.5	6
y:	3.8	17.5	17.5	27	32.5	34.5	35.5	32.5	8

4. By the method of averages, fit a curve of the form $y = ax^b$ to the following data:

x:	5	10	20	30	40	50
y:	10	34	97	185	310	482

(Madras, 2003)

3.15 METHOD OF MOMENTS

Let $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$ be the set of observations such that

$$x_2 - x_1 = x_3 - x_2 = \ldots = x_n - x_{n-1} = h \text{ (say)}$$

We define the moments of the observed values of y as follows:

m_1 the 1st moment $= h\Sigma y$

m_2 the 2nd moment $= h\Sigma xy$

m_3 the 3rd moment $= h\Sigma x^2y$ and so on.

Let the curve fitting the given data be $y = f(x)$. Then the moments of the curve $y = f(x)$ about the y-axis are

$$\mu_1 \text{ the 1st moment} = \int xy\,dx$$

$$\mu_2 \text{ the 2nd moment} = \int x^2y\,dx$$

$$\mu_3 \text{ the 3rd moment} = \int x^3y\,dx \text{ and so on.}$$

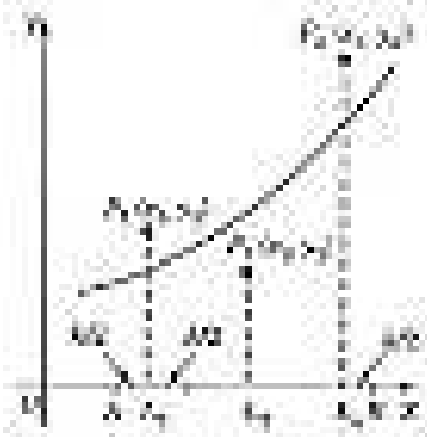


Fig. 3.15

The method is based on the assumption that the moments of the observed values of y are respectively equal to the moments of the unknown values of y i.e., $m_1 = \mu_1$, $m_2 = \mu_2$, $m_3 = \mu_3$ etc. These equations (known as *Moments equations*) are used to determine the constants in $f(x)$.

m's are calculated from the tabulated values of x and y while μ's are computed as follows:

In Fig. 3.15, y_1, the ordinate of P_1 ($x = x_1$), can be taken as the value of y at the mid-point of the interval $(x_1 - h/2, x_1 + h/2)$. Similarly, y_2, the ordinate of P_2 ($x = x_2$), can be taken as the value of y at the mid-point of the interval $(x_2 - h/2, x_2 + h/2)$. If A and B be the points such that

$$OA = x_1 - h/2 \text{ and } OB = x_1 + h/2,$$

then

$$\mu_1 = \int_{x_1-h/2}^{x_1+h/2} xy\,dx = \int_{x_1-h/2}^{x_1+h/2} f(x)\,dx$$

$$\mu_2 = \int_{x_1-h/2}^{x_1+h/2} x^2y\,dx$$

and

$$\mu_3 = \int_{x_1-h/2}^{x_1+h/2} x^3y\,dx.$$

Example 3.16. Fit a straight line $y = a + bx$ for the following data by the method of moments:

x:	1	2	3	4
y:	10	18	27	36

(Madras, 2003)

Solution. Since only two constants a and b are to be found, it is sufficient to calculate the first two moments of the curve. Here $h = 1$.

$$m_1 = h\Sigma y = 1\,(10 + 18 + 27 + 36) = 91$$

$$m_2 = h\Sigma xy = 1\,(1 \times 10 + 2 \times 18 + 3 \times 27 + 4 \times 36) = 227$$

To compute the numerical values of the constants a and b, the limits of integration will be $1 - h/2$ and $4 + h/2$ i.e., 0.5 to 4.5

$$\therefore \quad a_1 = 2\int_{0.5}^{4.5}(ax + bx)\,dx = \left|ax^2 + b\frac{x^3}{3}\right|_{0.5}^{4.5} = 4a + 10b$$

$$a_2 = \int_{0.5}^{4.5}(ax^2 + bx)\,dx = 10a + \frac{91}{3}b$$

Thus, the observation equations $a_r = a_r(r = 1, 2)$ are $4a + 10b = 34$; $10a + \frac{91}{3}b = 120$

Solving these, $a = 10.02$ and $b = 2.28$.

Hence the required equation is $y = 10.02 + 2.28x$.

Example 8.16. Given the following data :

x	0	1	2	3	4
y	1	2	33	82	157

find the parabola of best fit in the sense of least squares.

Solution. Let the parabola of best fit be $y = a + bx + cx^2$ ...(1)

Here three constants are to be found, we calculate the first three moments about zero. Here $h = 1$.

$$m_1 = \Delta(y) = 1 + 1 + 6 + 33 + 82 + 157 = 78$$

$$m_2 = \Delta(xy) = 1 + 2 + 12 + 99 + 328 = 442$$

$$m_3 = \Delta(x^2y) = 1 + 4 + 24 + 297 + 1312 = 1638$$

For computing the moments of the tabulated values of y, the limits of integration are $0 - h/2$ and $4 + h/2$ i.e., -0.5 and 4.5.

$$\therefore \quad a_1 = \int_{-0.5}^{4.5}(ax + bx + cx^2)\,dx = 4a + 20b + 30.4c$$

$$a_2 = \int_{-0.5}^{4.5}x(a + bx + cx^2)\,dx = 10a + 30.4b + 182.2c$$

$$a_3 = \int_{-0.5}^{4.5}x^2(a + bx + cx^2)\,dx = 30.4a + 182.2b + 1093.3c$$

Thus the observation equations $a_r = m_r (r = 1, 2, 3)$ are

$$4a + 20b + 30.4c = 78;\ 10a + 30.4b + 182.2c = 442;\ 30.4a + 182.2b + 1093.3c = 1638$$

Solving these equations, we get $a = 1.4$, $b = 0.15$, $c = 3.4$

Hence the parabola of best fit is $y = 0.4 + 0.15x + 3.4x^2$.

EXERCISE 8.3

1. Use the method of moments to fit the straight line $y = ax + b$ to the data :

x	1	2	3	4
y	0.47	0.71	0.79	0.89

2. Fit a straight line to the following data, using the method of moments :

x	1	2	3	4	5
y	1.5	2.4	4.9	4.7	6.8

(Ans. $y = 1.06x + 0.03$)

3. Fit a parabola of the form $y = a + bx + cx^2$ to the data :

x	1	2	3	4
y	1.7	1.8	2.3	3.2

by the method of moments.

4. By using the method of moments, fit a parabola to the following data :

x	1	2	3	4
y	0.39	0.54	1.28	2.13

(Ans. $y = 0.08 + 0.15x + 0.23x^2$)

OBJECTIVE TYPE QUESTIONS

MULTIPLE CHOICE

Fill in the blanks or choose the correct answer in the following questions:

1. The line $y = ax^2 + bx$ is translated to the origin then it is _____.
2. The pair of equations $px^2 + qy = 0$ and $lx^2 + my = 0$ has only one solution if $p =$ _____ and $l =$ _____.
3. The point (p, q) lies in quadrant number II if _____.
4. $y = ax^2 + bx$ is an even function if _____.
5. To fit the straight line $y = ax + b$ to a given observations, the normal equations are
   (a) $\Sigma y = na + b\Sigma x$, $\Sigma xy = a\Sigma x^2 + b\Sigma x$
   (b) $\Sigma y = na + b\Sigma x$, $\Sigma xy = a\Sigma x^2 + b\Sigma x$
   (c) $\Sigma y = a\Sigma x + nb$, $\Sigma xy = a\Sigma x^2 + b\Sigma x$
6. To fit $y = ab^x$ by least square method, normal equations are _____.
7. The observation which do not fit the straight line by method of moments are _____.
8. The method of group averages is based on the assumption that the sum of the residuals is _____.
9. $y = ax^2 + b\log_{10} x$ reduced to linear is $v = a\log_{10} u + b$ where $u =$ _____.
10. Given $\begin{vmatrix} 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0.1 \end{vmatrix}$, then the straight line of best fit is _____.
11. The method of moments is based on the assumption that _____.
12. If $y = a_0 + a_1x + a_2x^2$, $\Sigma x = 30$, $\Sigma x^2 = 90$, $\Sigma x^3 = 3000$, $\Sigma x^4 = 7500$ and $a_0 = 10$, then $a_1 =$ _____, $a_2 =$ _____.
13. y is called a(n) _____ function of x.
14. If $y = a + bx$ is a straight line, then

x	0	1	2	3	4
y	2	3.4	4.8	5.5	7.3

   then the first normal equation is
   (a) $35 = 5a + 10b$ (b) $34.15 = 5a + 10b$
   (c) $34.15 = 2a + 10b$ (d) $10.15 = 5a + 10b$
15. If $y = 2x + 3$ is the best fit for 4 pairs of observations (x, y) by the method of least squares and $\Sigma x = 100$, then $\Sigma xy =$
   (a) 20 (b) 40 (c) 60 (d) 80

CHAPTER 25

Statistical Methods

1. Introduction. 2. Collection and classification of data. 3. Graphical representation. 4. Construction of frequency distributions. 5. Measures of central tendency. 6. Measures of dispersion. 7. Coefficient of variation. Relations between measures of dispersion. 8. Standard deviation of the combination of two groups. 9. Moments. 10. Skewness. 11. Kurtosis. 12. Correlation. 13. Coefficient of correlation. 14. Lines of regression. 15. Standard error of estimate. 16. Rank correlation. 17. Objective Type of Questions

25.1 INTRODUCTION

Statistics deals with the methods for collection, classification and analysis of numerical data for drawing valid conclusions and making reasonable decisions. It has meaningful applications in practically engineering, in the analysis of experimental data, etc. The importance of statistical methods in engineering is on the increase. As such we shall now introduce the student to this interesting field.

25.2 (1) COLLECTION OF DATA

The collection of data constitutes the starting point of any statistical investigation. Data may be collected for each and every unit of the whole lot (population). for it would ensure greater accuracy. But complete enumeration is prohibitively expensive and time consuming. As such out of a very large number of items, a few of them (a sample) are selected and conclusions drawn on the basis of this sample are taken to hold for the population.

(2) Classification of data. The data collected in the course of an inquiry is not in an easily assimilable form. As such, the proper classification is necessary for making it digest and intelligible. The classification is done by dividing the raw data into a convenient number of groups according to the values of the variable and finding the frequency of the variable in each group.

Let us, for example, consider the raw data relating to marks obtained in Mechanics by a group of 64 students :

78	84	78	80	68	71	88	88
68	86	69	84	96	82	88	98
68	85	88	94	68	85	98	68
98	82	78	81	77	89	94	68
68	82	81	92	68	76	95	88
68	89	87	88	78	82	95	68
68	87	97	79	85	76	85	72
78	89	68	76	96	80	98	98

The data can conveniently be grouped and shown as a tabular form as follows :

Class	Frequency	Cumulative Frequency
50—51	1	1
52—53	2	3
54—55	9	12
56—57	7	19
58—59	8	27
60—61	17	44
62—63	6	50
64—65	8	58
66—67	3	61
68—69	1	62
	Total 62	

It would be seen from the above table that there is one student getting marks between 50—51, two students getting marks between 52—53, nine students getting marks between 54—55 and so on. That the 62 figures have been put in to only 10 groups, called the classes. The width of the class is called the class interval and the number in that interval is called the frequency. The mid point or the mid value of the class is called the class mark. The above table showing the classes and the corresponding frequencies is called a frequency table. Thus a set of raw data is summarised by distributing it into a number of classes along with their frequencies is known as a frequency distribution.

While forming a frequency distribution, the number of classes should not ordinarily exceed 20, and should not, in general, be less than 10. As far as possible, the class intervals should be of equal width.

(3) Cumulative frequency. In many investigations, we require the number of items less than or certain value. We add up the frequencies of the classes upto that value and call this number as the cumulative frequency. In the above table, the third column shows the cumulative frequencies, i.e., the number of students getting less than 54 marks, less than 56 marks and so on.

GRAPHICAL REPRESENTATION

A convenient way of representing a simple frequency distribution is by means of graphs. It gives us at once a general view of the observations and at the same time makes the raw data readily intelligible. We give below the important types of graphs in use :

(1) Histogram. A histogram is drawn by erecting rectangles over the class intervals, such that the areas of the rectangles are proportional to the class frequencies. If the class intervals are of equal size, the height of the rectangles will be proportional to the class frequencies themselves (Fig. 33.1).

(2) Frequency polygon. A frequency polygon for an ungrouped data is obtained by plotting points plotted with the variable values as the abscissae and the frequencies as the ordinates. For a grouped distribution, the abscissae of the points will be the mid-values of the class intervals. In case the intervals are equal, the frequency polygon can be obtained by joining the middle points of the upper sides of the rectangles of the histogram by straight lines as shown by dotted lines in Fig. 33.1. If the class intervals become very very small, the frequency polygon takes the form of a smooth curve called the *frequency curve*.

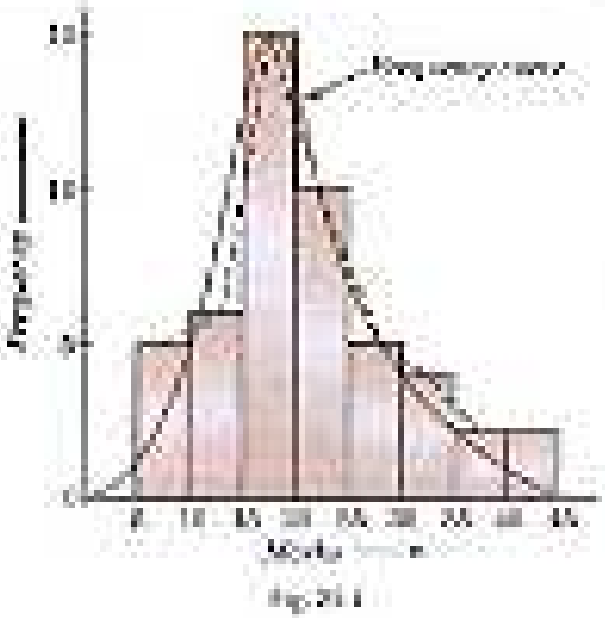


Fig. 33.1

(3) Cumulative frequency curve-Ogive. Very often, it is desired to show in a diagrammatic form, not the relative frequencies in the various intervals, but the cumulative frequencies above or below a given value. For example, we may wish to read off from a diagram the number or proportion of people whose income is not less than any given amount, or proportion of people whose height does not exceed any stated value. Diagrams of

This type are known as cumulative frequency curves or ogives. There are of two kinds 'more than' or 'less than' and typically they look somewhat like a long drawn S (Fig. 26.2)

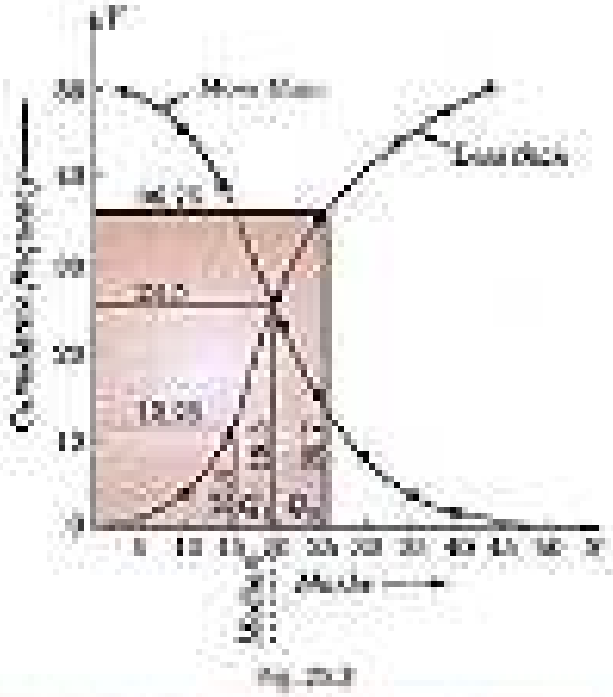


Fig. 26.2

Example 26.1. Draw the histogram, frequency polygon, frequency curve and the ogive (less than) and 'more than' from the following distribution of marks obtained by 80 students :

Class (Marks group)	Frequency (No. of students)	Cumulative frequency (Less than)	(More than)
0—10	3	3	80
10—20	5	8	77
20—30	15	23	72
30—40	18	41	57
40—50	7	48	39
50—60	4	52	32
60—70	2	54	28
70—80	26	80	26

Solution. In Fig. 26.1, the rectangles show the *histogram*, the dotted polygon represents the *frequency polygon* and the smooth curve is the *frequency curve*.

The ogives 'less than' and 'more than' are shown in Fig. 26.2.

26.3 COMPARISON OF FREQUENCY DISTRIBUTIONS

The condensation of data in the form of a frequency distribution is very useful in as far as it brings a large series of observations into a compact form. But in practice, we are generally interested in comparing two or more series. The inherent inability of the human mind to grasp in its entirety even the data in the form of a frequency distribution compels us to seek for certain constants which could reasonably give an insight into the important characteristics of the series. The chief constants which summarise the fundamental characteristics of the frequency distribution are (*i*) *Measures of central tendency*, (*ii*) *Measures of dispersion* and (*iii*) *Measures of skewness*.

26.3.1 MEASURES OF CENTRAL TENDENCY

A frequency distribution in general, shows the scattering of the data around some central value. Finding of this central value or the average is of importance, as it gives a most representative value of the whole group.

Different methods give different averages which are known as the measures of central tendency. The commonly used measures of central value are Mean, Median, Mode, Geometric mean and Harmonic mean.

1) Mean. If $x_1, x_2, x_3, \ldots, x_n$ are a set of n values of a variate, then the arithmetic mean (or simply mean) is given by

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \frac{\Sigma x_i}{n} \quad \ldots (1)$$

In a frequency distribution, if $x_1, x_2, \ldots, x_n$ be the mid-values of the class-intervals having frequencies $f_1, f_2, \ldots, f_n$ respectively, we have

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \ldots + f_n x_n}{f_1 + f_2 + \ldots + f_n} = \frac{\Sigma f x}{\Sigma f} \quad \ldots (2)$$

Calculation of mean. Direct method of computing is quickly when applied to grouped data involves heavy calculations and in order to avoid them, the following formulae are generally used :

A. Short-cut method $\quad \bar{x} = A + \frac{\Sigma f d}{\Sigma f} \quad \ldots (3)$

B. Step-deviation method $\quad \bar{x} = A + h\frac{\Sigma f u}{\Sigma f} \quad \ldots (4)$

where $d = x - A$ and $u = (x - A)/h$, A being an arbitrary origin and h the equal class interval.

Proof. If $x_1, x_2, \ldots, x_n$ are the mid-values of the classes with frequencies $f_1, f_2, \ldots, f_n$, we have

$$\Sigma f x = \Sigma f (A + d) = A\Sigma f + \Sigma f d$$

$$\therefore \quad \bar{x} = \frac{\Sigma f x}{\Sigma f} = A + \frac{\Sigma f d}{\Sigma f}$$

Further $x = A + d$ or $d = hu$. Substituting this value in (3), we get (4).

Note. The appropriate value of the deviation d or u is taken from the same row, for

$$\Sigma f(x - A) = \Sigma f x - A\Sigma f = \Sigma f x - \frac{\Sigma f x}{\Sigma f} \cdot \Sigma f = 0$$

Cor. If $\bar{x}_1, \bar{x}_2$ be the means of two samples of sizes n_1 and n_2 respectively, then the combined mean $\bar{x}$ of the combined sample of size $n_1 + n_2$ is given by

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

For $\quad n_1\bar{x}_1$ = sum of all the observations of the first sample,

and $\quad n_2\bar{x}_2$ = sum of all the observations of the second sample.

$\therefore$ sum of the observations of the combined sample $= n_1\bar{x}_1 + n_2\bar{x}_2$.

Also number of the observations in the combined sample $= n_1 + n_2$.

$\therefore$ mean of the combined sample $= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$.

Example 10(a). The following is the frequency distribution of a random sample of weekly earnings of 200 employees :

Weekly earnings :	10	15	18	19	24	25	30	32	36	38	40	45	48	50	60
No. of employees :	3	4	15	17	34	48	70	80	75	52	34	28	19	11	7

Calculate the average weekly earnings.

Solution. The mid-values are arranged in the following table. The arbitrary origin is generally taken as the value corresponding to the maximum frequency.

x	f	$d = x - 30$	$u = d/5$	fu
10	3	−20	−4	−12
15	4	−15	−3	−12
18	15	−12	−2.4	−36
19	17	−11	−2.2	−37.4
24	34	−6	−1.2	−40.8
25	48	−5	−1	−48
30	70	0	0	0
32	80	2	0.4	32
36	75	6	1.2	90
38	52	8	1.6	83.2
40	34	10	2	68
45	28	15	3	84
48	19	18	3.6	68.4
50	11	20	4	44
60	7	30	6	42
	$\Sigma f = 200$			$\Sigma fu = 255.4$

By direct method, we have

$$\text{Mean } \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{13,210}{200} = 33.16$$

By step-deviation method, we have

$$\bar{x} = A + h\frac{\Sigma fu}{\Sigma f} = 30 + 5 \times \frac{266}{200}$$

$$= 30 + 3.32 = 33.16, \text{ which is same as found above.}$$

Profit earnings	Mid value x	No. of employees f	Calculations		
			fx	$x - (A = 50)/h$	fx
[illegible]	12	3	36	−3	−36
[illegible]	18	5	90	−2	−60
[illegible]	24	16	384	−1	−96
[illegible]	30	18	540	0	0
[illegible]	36	21	756	1	126
[illegible]	42	40	1680	2	160
[illegible]	48	22	1056	3	66
[illegible]	54	10	540	4	40
[illegible]	60	5	300	5	25
[illegible]	66	2	132	6	12
[illegible]	72	1	72	7	7
		$\Sigma f = 145$	$\Sigma fx = 5,611.50$		$\Sigma fx = 284$

(2) Median. If the values of a variable are arranged in the ascending order of magnitude, the value which is the middle item of the number is called median or the mean of the two middle items of the number is even. Thus the median is equal to the mid-value, i.e., the value which divides the total frequency into two equal parts.

For the grouped data,

$$\text{Median} = L + \frac{\left(\frac{1}{2}N - C\right)}{f} \times h$$

where L = lower limit of the median class, N = total frequency,

f = frequency of the median class, h = width of the median class,

and C = cumulative frequency upto the class preceding the median class.

Quartiles. Quartiles are those values which divides the frequency into four equal parts, when the values are arranged in the ascending order of magnitude. The lower quartile (Q_1) is mid-way between the lowest value and the median. The upper quartile (Q_3) is midway between the median and the upper extreme.

For the grouped data, these are calculated by the formulae:

$$Q_1 = L + \frac{\left(\frac{1}{4}N - C\right)}{f} \times h$$

and

$$Q_3 = L + \frac{\left(\frac{3}{4}N - C\right)}{f} \times h$$

where L = lower limit of the class in which Q_1 or Q_3 lies, f = frequency of that class, h = width of the class

and C = cumulative frequency upto the class preceding the class in which Q_1 or Q_3 lies.

The difference between the upper and lower quartiles, i.e., $Q_3 - Q_1$, is called the inter-quartile range.

Note. The quartiles give a ready method of marking out the middle fifty per cent of the items and the quartiles. The inter-quartile range and semi-inter-quartile range are the measures of dispersion (Fig. 28.2).

(3) Mode. The mode is defined as that value of the variable which occurs most frequently, i.e., the value of the maximum frequency.

For a grouped distribution, it is given by the formula

$$\text{Mode} = L - \frac{f_1}{f_1 + f_2} h$$

where L = lower limit of the class containing the mode,

Δ_1 = excess of modal frequency over frequency of preceding class,

Δ_2 = excess of modal frequency over frequency of following class,

and h = size of modal class.

For a frequency curve (Fig. 28.11), the abscissa of the highest ordinate determines the value of the mode. There may be two or more modes in a frequency curve. Curves having a single mode are termed as *unimodal*, those having two modes as *bimodal* and those having more than two modes as *multimodal*.

Thus, for symmetrical distribution, the mean, median and mode coincide. For other distributions, however, they are different and are known to be connected by the empirical relationship :

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median}).$$

Example 28.8. *Calculate median and the lower and upper quartiles from the distribution of marks obtained by 48 students of Example 28.1. Find also the inter-quartile range and the mode.*

Solution. Median (or 40/2 th) falls in the class (15—20) and is given by

$$15 + \frac{\frac{48}{2} - 21}{15} \times 5 = 15 + \frac{3}{5} = 15.6 \text{ marks}.$$

Lower quartile Q_1 (or $48/4 = 12$th) also falls in the class (15—20)

$$\therefore \quad Q_1 = 15 + \frac{\frac{48}{4} - 21}{15} \times 5 = 15 + \frac{3}{5} = 15.6 \text{ marks}$$

Upper quartile (or $\frac{3}{4} \times 48 = 36$th) falls in the class (25—30)

$$\therefore \quad Q_3 = 25 + \frac{36 - 33}{8} \times 5 = 28.75 \text{ marks}.$$

$$\text{Semi-inter-quartile range} = \frac{1}{2}(Q_3 - Q_1) = \frac{28.75 - 15.6}{2} = \frac{13.15}{2} = 6.575.$$

Mode. It is seen that the mode value falls in the class 15—20. Employing the formula for the grouped distribution, we have

$$\text{Mode} = 15 + \frac{15 - 6}{(15 - 6) + (15 - 13)} \times 5 = 15.2 \text{ marks}.$$

Also, in Fig. 28.5, the upper most point whose abscissa is 15.2 which is the median of the distribution. The values for the lower and upper quartiles are similarly seen to be 15.4 (for frequency 12.5) and 28.7 (for frequency 36.5).

Example 28.9. *Given below are the marks obtained by a batch of 30 students in a certain class test in Physics and Chemistry :*

Serial No. of students	Marks in Physics	Marks in Chemistry	Serial No. of students	Marks in Physics	Marks in Chemistry
1	40	39	16	30	18
2	42	36	17	32	17
3	43	38	18	31	15
4	39	36	19	40	19
5	39	34	20	42	20
6	40	37	21	44	24
7	37	34	22	46	26
8	35	36	23	44	28
9	32	32	24	43	27
10	29	33	25	39	25

In which subject is the level of knowledge of the students higher ?

Solution. The subject for which the value of the median is higher will be the subject in which the level of knowledge of the students is higher. To find the median in each case, we arrange the marks in ascending order of magnitude :

Student No.	Marks in Physics	Marks in Chemistry	Student No.	Marks in Physics	Marks in Chemistry
1	38	30	11	40	32
2	58	42	12	42	48
3	50	36	13	48	38
4	44	30	14	54	50
5	36	30	15	62	66
6	62	50	16	56	60
7	54	44	17	54	66
8	68	60	18	60	52
9	46	36	19	66	62
10	42	30	20	52	62

Median marks in Physics = A.M. of marks of 10th and 11th terms

$$= \frac{48 + 48}{2} = 48.0$$

Median marks in Chemistry = A.M. of marks of 10th and 11th terms

$$= \frac{38 + 42}{2} = 40.0$$

Since the median marks in Physics is greater than the median marks in Chemistry, the level of knowledge in Physics is higher.

Example 33.8. An incomplete frequency distribution is given as follows:

Variable x	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency f	12	30	?	65	?	25	18

Given that the total frequency is 229 and median is 46, find the missing frequencies.

Solution. Let f_3, f_5 be the missing frequencies of the classes 30—40 and 50—60 respectively.

Since the median lies in the class 40—50,

$$\therefore \qquad 46 = 40 + \frac{\frac{229}{2} - (12 + 30 + f_3)}{65} \times 10$$

which gives $f_3 = 34.5$ which can be taken as 34.

$$\therefore \qquad f_5 = 229 - (12 + 30 + 34 + 65 + 25 + 18) = 45$$

(d) **Geometric mean.** If $x_1, x_2, \ldots, x_n$ are a set of n observations, then the geometric mean is given by

$$\text{G.M.} = (x_1 x_2 \ldots x_n)^{1/n}$$

or
$$\log \text{G.M.} = \frac{1}{n}(\log x_1 + \log x_2 + \ldots + \log x_n) \qquad \ldots(1)$$

In a frequency distribution, let $x_1, x_2, \ldots, x_n$ be the central values with corresponding frequencies $f_1, f_2, \ldots, f_n$, we have

$$\text{G.M.} = \left[(x_1)^{f_1} (x_2)^{f_2} \ldots (x_n)^{f_n}\right]^{1/N} \quad \text{where } N = \Sigma f$$

or
$$\log \text{G.M.} = \frac{1}{N}(f_1 \log x_1 + f_2 \log x_2 + \ldots + f_n \log x_n) \qquad \ldots(2)$$

Errors (1) and (2) show that logarithm of G.M. = A.M. of logarithms of the values.

(e) **Harmonic means.** If $x_1, x_2, \ldots, x_n$ be a set of n observations, then the harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of the observations. That is

$$\text{H.M.} = \frac{1}{\frac{1}{n}\left[\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_n}\right]}$$

In a frequency distribution, $\text{H.M.} = \dfrac{1}{\frac{1}{N}\left[\frac{f_1}{x_1} + \frac{f_2}{x_2} + \ldots + \frac{f_n}{x_n}\right]}$ where $N = \Sigma f$.

Example 19.8. *Three cities A, B, C are equidistant from each other. A motorist travels from A to B at 30 km/hr, from B to C at 40 km/hr, from C to A at 50 km/hr. Determine the average speed.*

Solution. Let $AB = BC = CA = x$ km

Time taken to travel from A to $B = x/30$

Time taken to travel from B to $C = x/40$

Time taken to travel from C to $A = x/50$

$\therefore$ average time taken $= \frac{1}{3}\left(\frac{x}{30} + \frac{x}{40} + \frac{x}{50}\right)$

Thus the average speed $= \dfrac{x}{\frac{1}{3}\left(\frac{x}{30} + \frac{x}{40} + \frac{x}{50}\right)}$

In other words, the average speed is the harmonic mean of 30, 40, 50 km/hr.

Hence the average speed $= \dfrac{3}{\frac{1}{30} + \frac{1}{40} + \frac{1}{50}} = 39.3$ km/hr.

Q.6. Of the various measures of central tendency, the mean is the most important for a data to be considered good. The median, though more easily calculable, cannot be applied with ease in theoretical analysis. Median is of advantage when there are exceptionally large and small values at the ends of the distribution.

The mode, though most easily calculated, has the least significance. It is particularly misleading in distributions which are small in numbers or highly asymmetrical.

The geometrical mean though difficult to compute, finds applications in cases like population where one unit increases with a certain rate by which changes tend to be directly proportional to the given rate itself.

The harmonic mean is useful in finding out means where numerical values of reciprocals are involved.

PROBLEMS 19.1

1. Given the histogram and frequency polygon for the following distribution. Also calculate the arithmetic mean.

Class interval	0—50	50—100	100—200	200—300	300—400	400—500	500—600	600—700	700—800
Frequency	10	24	144	204	240	80	7	4	

2. The following marks were given by a class teacher:

56	52	45	78	91	81	58	90	51	61
71	63	51	66	58	70	53	59	68	54
58	81	50	59	68	65	73	80	48	52
61	53	62	45	54	43	68	64	75	81
71	62	57	70	62	42	74	66	48	59

Draw a calculative bar chart.

[Hint: Mark the groups as 40—45, 45—50, 50—55, ... Also mark 100 boys and 80 girls for Pass and Third class.]

3. Find the mean, median and mode for the following:

Marks	15	20	25	30	35	40	45	50	55	
Frequency	3	10	18	14	8	6	5	1	1	[Mean = 31.6, Mode = 25.625, Median = 30]

4. Calculate mean, median and mode of the following data relating to weight of 100 students:

Weight (in gm)	50—60	60—70	70—80	80—90	90—100	100—110
No. of students	11	17	26	28	12	6

5. The population of a country was 200 million in 1951, it became 220 million in 1961. Calculate the percentage increase and rate of growth per annum.

[Hint: Use $A = P\left(1 + \frac{r}{100}\right)^n$ to find the growth rate.]

6. The number of divorces per 1000 marriages in the United States increased from 8.8 in 1920 to 19.9 in 1946. Find the annual increase of the process over the period 1920 to 1946.

7. An aeroplane flies along the four sides of a square at speeds of 100, 200, 300 and 400 km/hr respectively. Find the average speed of the plane in its entire journey around the square.

8. A man driving to office 30 km at a uniform speed of 30 km/hr. Find the rate at which he must return in order to average 60 km/hr. What must be his average speed for the second half of the journey if the overall average is to be 40 km/hr?

Solved Problems

11.1 Find the range and coefficient of range for the following data: 63, 64, 75, 84, 86, 89, 92, 95, 96, 97.

Solution:

Range = L − S = 97 − 63 = 34

Coefficient of range = (L − S)/(L + S) = 34/160 = 0.2125

11.2 Find the mean deviation and its coefficient for the following data: 6, 7, 10, 12, 13, 4, 8, 12.

Solution:

x_i	$\lvert x_i - \bar{x} \rvert$	x_i	$\lvert x_i - \bar{x} \rvert$
6	3	12	3
7	2	4	5
10	1	8	1
12	3	12	3
13	4		

$\bar{x} = \frac{\sum x_i}{n} = \frac{6+7+10+12+13+4+8+12}{8} = \frac{72}{8} = 9$

$\text{M.D.} = \frac{\sum \lvert x_i - \bar{x} \rvert}{n} = \frac{24}{8} = 3$

$\text{Coefficient of M.D.} = \frac{\text{M.D.}}{\bar{x}} = \frac{3}{9} = 0.33$

11.3 MEASURES OF DISPERSION

Although measures of central tendency do exhibit one of the important characteristics of a distribution, we have failed to give any idea as to how the individual values differ from the central value. If a number of individuals cluster around the central value or value of maximum frequency, the distribution is said to be concentrated. If the individual values tend to spread about the average (see Fig. 11.1), the magnitude of such a variation is called dispersion. The important measures of dispersion are given below:

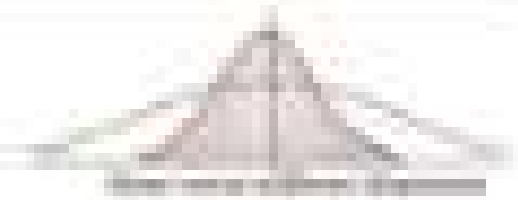


Fig. 11.1

(1) **Range.** The range of the weights of 10 students can be given by the difference between the greatest and the least values in the distribution. If the monthly wages of a group of 5 employees are

₹ 1500, ₹ 1600, ₹ 1800, ₹ 1900, ₹ 2000

then range = ₹ 2000 − ₹ 1500 = ₹ 500.

(2) **Quartile deviation or semi-interquartile range.** One half of the interquartile range is called quartile deviation, or semi-interquartile range. If Q_1 and Q_3 are the first and third quartiles, the semi-interquartile range

$$= \frac{1}{2}(Q_3 - Q_1)$$

(3) **Mean deviation.** The mean deviation is the mean of the absolute differences of the values from the mean, median or mode. The mean deviation = M.D. =

$$= \frac{1}{n}\sum |x_i - A|$$

where A is either the mean or the median or the mode. In the positive and negative differences have equal effects, only the absolute values of differences is taken into account.

(4) **Standard deviation.** The most important and the most powerful measure of dispersion is the standard deviation (S.D.), generally denoted by σ. It is computed as the square root of the mean of the squares of the differences of the various values from their mean.

Thus we obtain the formula (S.D.)

$$\sigma = \sqrt{\left[\frac{\sum (x_i - \bar{x})^2}{n}\right]} \qquad (1)$$

where $\bar{x}$ is the mean (frequency f_i).

Therefore, for the deviations measured from any other value, say A, instead of $\bar{x}$, in which the root-mean-square deviation

The square of the standard deviation is known as the **variance**.

Calculation of S.D. The changes of origin and the changes of scale considerably reduce the labour in the calculation of standard deviation. The formula for the computation of σ are as follows:

I. Short-cut method

$$\sigma = \sqrt{\left[\frac{\Sigma f d'^2}{\Sigma f} - \left[\frac{\Sigma f d'}{\Sigma f}\right]^2\right]} \quad \ldots(i)$$

II. Step deviation method

$$\sigma = h\sqrt{\left[\frac{\Sigma f d'^2}{\Sigma f} - \left[\frac{\Sigma f d'}{\Sigma f}\right]^2\right]} \quad \ldots(ii)$$

where $d_i = x_i - A$ and $d'_i = \frac{x_i - A}{h}$, being the assumed mean and h the equal class interval.

Proof. We know that $d_i = \bar{x} - x_i = A - x_i = -(x_i - A)$

$$\therefore \quad \Sigma f_i(x_i - \bar{x})^2 = \Sigma f_i(x_i - A)^2 = \Sigma f_i d_i^2 + (A - \bar{x})^2 \Sigma f_i - 2(A - \bar{x})\Sigma f_i d_i$$

$$= \Sigma f_i d_i^2 - \frac{(\Sigma f_i d_i)^2}{\Sigma f_i} \qquad \left[\because \bar{x} = A + \frac{\Sigma f_i d_i}{\Sigma f_i}\right]$$

Hence $$\sigma^2 = \frac{\Sigma f_i(x_i - \bar{x})^2}{\Sigma f_i} = \frac{\Sigma f_i d_i^2}{\Sigma f_i} - \left[\frac{\Sigma f_i d_i}{\Sigma f_i}\right]^2$$

Further $d'_i = \frac{x_i - A}{h} = \frac{d_i}{h}$ or $d_i = hd'_i$, then substituting the value in (i) we get (ii).

Obs. The root mean square deviation is the standard deviation measured from the mean.
The root mean square deviation is given by

$$\sigma = \sqrt{\frac{\Sigma f_i d_i^2}{\Sigma f_i}} \text{ and } \sqrt{\frac{\Sigma f_i}{\Sigma f_i}} = \sqrt{A + \frac{\Sigma f_i d_i}{\Sigma f_i}} = \bar{x} = A$$

$$\therefore \text{ from (i), we have } \quad \sigma^2 = \sigma'^2 + (\bar{x} - A)^2 \quad \ldots(1)$$

This shows that σ^2 is always greater than the least value of σ'^2, i.e., when $\bar{x} = A$.

(3) COEFFICIENT OF VARIATION

The ratio of the standard deviation to the mean, is known as the *coefficient of variation*. As this is a ratio having no dimension, it is used for comparing the variation between the two groups with different units. It is also expressed as a percentage.

$$\therefore \quad \text{Coefficient of variation} = \frac{\sigma}{\bar{x}} \times 100$$

(3) Relations between measures of dispersion

(i) Quartile deviation = 2/3 (standard deviation)

(ii) Mean deviation = 4/5 (standard deviation)

STANDARD DEVIATION OF THE COMBINATION OF TWO GROUPS

If $(\bar{x}_1, \sigma_1)$ be the mean and S.D. of a sample of size n_1 and $(\bar{x}_2, \sigma_2)$ be those for a sample of size n_2, then the S.D. σ of the combined sample of size $n_1 + n_2$ is given by

$$(n_1 + n_2)\sigma^2 = n_1\sigma_1^2 + n_2\sigma_2^2 + n_1D_1^2 + n_2D_2^2$$

where $D_1 = \bar{x}_1 - \bar{x}$, $\bar{x}$ being the mean of combined sample.

From (i), we have $n\sigma^2 = n\sigma'^2 + n(\bar{x} - A)^2$ where n is the size of the sample.

i.e., sum of the squares of the deviations from $A = n\sigma'^2 + n(\bar{x} - A)^2$.

Now let us apply this result to the first given sample taking A as $\bar{x}$. Then, sum of the squares of the deviations of x_1 series from $\bar{x} = n_1\sigma_1^2 + n_1(\bar{x}_1 - \bar{x})^2$ ...(ii)

Similarly for the second given sample taking A as $\bar{x}$, sum of the squares of the deviations of x_2 series from $\bar{x} = n_2\sigma_2^2 + n_2(\bar{x}_2 - \bar{x})^2$ ...(iii)

Adding (ii) and (iii), sum of the squares of the deviations of $x_1 + x_2$ series from $\bar{x}$

$$= n_1\sigma_1^2 + n_2\sigma_2^2 + n_1(\bar{x}_1 - \bar{x})^2 + n_2(\bar{x}_2 - \bar{x})^2$$

$$\therefore \quad (n_1 + n_2)\sigma^2 = n_1\sigma_1^2 + n_2\sigma_2^2 + n_1D_1^2 + n_2D_2^2$$

This result can be extended to the combination of any number of samples, giving a result of the form

$$(D_1)\sigma^2 = \Sigma(n_1\sigma_1^2) + \Sigma(n_1D_1^2)$$

Example 22.7. Calculate the mean and standard deviation for the following set of data:

Size of item	6	7	8	9	10	11	12
Frequency	3	6	9	13	8	5	4

Solution. The calculations are arranged as follows:

Size of item x	Frequency f	Deviation $d = x - 9$	fd	fd^2
6	3	−3	−9	27
7	6	−2	−12	24
8	9	−1	−9	9
9	13	0	0	0
10	8	1	8	8
11	5	2	10	20
12	4	3	12	36
	$\Sigma f = 48$		$\Sigma fd = 0$	$\Sigma fd^2 = 124$

$$\therefore \quad \text{mean} = 9 + \frac{\Sigma fd}{\Sigma f} = 9$$

$$\text{Standard deviation} = \sqrt{\left[\frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f}\right)^2\right]} = \sqrt{\left[\frac{124}{48} - \left(\frac{0}{48}\right)^2\right]} = \sqrt{\left(\frac{124}{48}\right)} = 1.607$$

Example 22.8. Calculate the mean and standard deviation of the following frequency distribution:

Weekly wages (£)	No. of men
40–45	4
45–50	24
50–55	27
55–60	21
60–65	9
65–70	6
70–75	5
75–80	3
80–85	1

Solution. The calculations are arranged in the table below:

Wages class £	Mid-value x	No. of men f	Deviation $d = \frac{x - 62.5}{5}$	fd	fd^2
40–45	42.5	4	−4	−16	64
45–50	47.5	24	−3	−72	216
50–55	52.5	27	−2	−54	108
55–60	57.5	21	−1	−21	21
60–65	62.5	9	0	0	0
65–70	67.5	6	1	6	6
70–75	72.5	5	2	10	20
75–80	77.5	3	3	9	27
80–85	82.5	1	4	4	16
		$\Sigma f = 100$		$\Sigma fd = -134$	$\Sigma fd^2 = 478$

$$\therefore \quad \text{mean wage} = 62.5 + 5 \times \frac{\Sigma fd}{\Sigma f} = 62.5 + 5\left(\frac{-134}{100}\right) = £55.80$$

$$\text{Standard deviation} = 5\sqrt{\left[\frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f}\right)^2\right]} = 5\sqrt{\left[\frac{478}{100} - \left(\frac{-134}{100}\right)^2\right]} = £9.6104$$

Example 12.8. The following are marks of two batsmen A and B in a series of innings :

A :	12	115	6	73	7	19	119	36	84	29
B :	47	71	58	42	3	77	77	43	71	0

Who is the better scorer and who is more consistent ? (P.T.U., 2004)

Solution. Let y denote score of A and y that of B.

Taking 81 as the origin, we prepare the following table :

x	$d = (x - 81)$	d^2	y	$(y - 81)$	d^2
12	−69	4761	47	−34	1156
115	34	1156	71	−10	100
6	−75	5625	58	−23	529
73	−8	64	42	−39	1521
7	−74	5476	3	−78	6084
19	−62	3844	77	−4	16
119	38	1444	77	−4	16
36	−45	2025	43	−38	1444
84	3	9	71	−10	100
29	−52	2704	0	−81	6561
Total	−18	−31868		−328	18627

For A,

$$\text{A.M. } \bar{x} = 81 + \frac{\Sigma d}{n} = 81 - \frac{18}{10} = 79.2$$

$$\text{S.D. } \sigma_1 = \sqrt{\left(\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2\right)} = \sqrt{(3186.8 - (-1.8)^2)} = 43.8$$

$$\therefore \text{ coefficient of variation} = \frac{\sigma_1}{\bar{x}} \times 100 = \frac{43.8}{79.2} \times 100 = 55.3\%$$

For B,

$$\text{A.M. } \bar{y} = 81 + \frac{\Sigma d}{n} = 81 - \frac{328}{10} = 49$$

$$\text{S.D. } \sigma_2 = \sqrt{\left(\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2\right)} = \sqrt{(1862.7 - (-32.8)^2)} = 18.8$$

$$\therefore \text{ coefficient of variation} = \frac{\sigma_2}{\bar{y}} \times 100 = \frac{18.8}{49} \times 100 = 38.3\%$$

Since the A.M. of A > A.M. of B. It follows that A is a better score getter i.e., more efficient than B.

Since the coefficient of variation of A < the coefficient of variation of B, it means that B is more consistent than A. That is though A is a better player, he is less consistent.

Example 12.9. The numbers matched, runs scored, wickets and S.D. in each group of matches by three medium bowlers are given below. Find the mean weight and S.D. of the entire data when grouped together :

Match Group	No. Bowled	Mean Wt. Out	S.D. (Kg.)
A	50	150	5
B	60	120	7
C	90	130	6

Solution. We have $n_1 = 50, \bar{x}_1 = 150, \sigma_1 = 5$

$n_2 = 60, \bar{x}_2 = 120, \sigma_2 = 7$

$n_3 = 90, \bar{x}_3 = 130, \sigma_3 = 6$.

If $\bar{x}$ is the mean of the entire data,

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{n_1 + n_2 + n_3} = \frac{50 \times 52 + 80 \times 55 + 90 \times 55}{50 + 80 + 90} = \frac{15200}{220} = 316 \text{ lb.}$$

If σ is the S.D. of the entire data,

$$N\sigma^2 = n_1\sigma_1^2 + n_2\sigma_2^2 + n_3\sigma_3^2 + n_1D_1^2 + n_2D_2^2 + n_3D_3^2$$

where $N = n_1 + n_2 + n_3 = 200$, $D_1 = \bar{x}_1 - \bar{x} = -3$, $D_2 = \bar{x}_2 - \bar{x} = 4$ and $D_3 = \bar{x}_3 - \bar{x} = -1$.

$$\therefore \quad 200\sigma^2 = 50 \times 36 + 80 \times 49 + 90 \times 64 + 50 \times 9 + 80 \times 16 + 90 \times 1$$
$$= 1800 + 3840 + 5760 + 450 + 1280 + 90$$
$$\sigma^2 = \frac{13220}{200} = 80. \quad \text{Hence} \quad \sigma = \sqrt{80} = 7.746 \text{ lb.}$$

EXERCISE 11(B)

1. The breaking strength of 10 pieces of a certain variety of yarn (in pounds) are as follows: 9.2, 9.1, 8.7, 9.5, 9.4, 8.9, 9.7, 9.3 and 9.2. Find the mean breaking strength and the standard deviation.

2. Show that the variance of the first n positive integers is $\frac{1}{12}(n^2 - 1)$. (U.D.C. 2006)

3. The mean of 5 observations is 4 and their variance is 5.2. If three of the observations are 1, 2 and 9, find the other two. (U.D.C. 2005)

4. For the distribution

x	5	6	7	8	9	10	11	12	13	14	15
f	20	15	34	41	101	81	65	25	18	27	12

find the mean, median and lower and upper quartiles, variance and the standard deviation.

5. The following table shows the marks obtained by 100 students in an examination. Calculate the mean, median and standard deviation:

Marks obtained	1—10	11—20	21—30	31—40	41—50	51—60
No. of students	7	10	25	21	15	8

(Annamalai 2000 S; U.D.C. 2007)

6. Calculate the quartile deviation and standard deviation for the following:

x	180—200	170—180	160—170	150—160	140—150	130—140	120—130	110—120
f	15	14	120	130	125	82	50	30

7. Calculate (i) mean deviation about the mean, (ii) mean deviation about the median for the following distribution:

Class	0—4.9	5—9.9	10—14.9	15—19.9	20—24.9	25—29.9	30—34.9	35—39.9
f	5	9	30	42	65	21	8	4

(Madurai 2005)

8. Two doctors bring the following two sets of data which represent blood pressures of the same patients:

I	105.3	103.4	104.9	104.7	104.9	105.0	104.9
II	105.3	103.1	104.8	103.2	104.7	102.6	105.1

Calculate the standard deviation in each case. Which set of data is more reliable? Can the more reliable set be rectified by subtracting the mean from each value?

Hint: The smaller the coefficient of variation, the greater is the reliability or consistency in the data.

9. The height and weight of the 11 boys in a class are given below. (It is observed that height and weight are linearly correlated.)

Height in cm	170	175	166	177	175	171	173	178	170	169
Weight in kg	70	77	56	80	72	70	74	78	74	72

10. The cricket batting of players of two countries A and B in a test match is recorded as given below:

A	114	128	138	150	104	102
B	208	210	228	218	201	196

Which of these batsmen is more reliable?

11. The scores of two golfers A and B in 12 rounds are given below. Who is the better player and who is the more consistent player?

A:	71	78	77	73	77	77	79	83	73	78	72	77
B:	81	84	78	88	80	82	88	82	92	79	86	82

12. The scores obtained by two batsmen A and B in 10 of their innings are given below:

A:	32	41	68	82	60	34	80	18	20	75
B:	34	60	73	80	66	48	60	100	40	80

Find the batting average, S.D. and coefficient of variation for each batsman. Determine who is more efficient and who is more consistent.

13. Find the mean and standard deviation of the following seven numbers (as regards x'):

Number for x_i	f_i	$f_i x_i$	x_i^2
1	61	61	61
2	63	126	126

14. A data is given on the cost of three components with Expectation 500, 500 and 800 having means 50, 60 and 10 and S.D. 3, 4 and 5 respectively. Show that the mean of the combined distribution is 500 and its S.D. is 7.5 approximately.

23.12 (i) MOMENTS

The rth moment about the mean $\bar{x}$ of a distribution is denoted by μ_r and is given by

$$\mu_r = \frac{1}{N}\Sigma f_i (x_i - \bar{x})^r \qquad ...(1)$$

The corresponding moment about any point a is defined as

$$\mu_r' = \frac{1}{N}\Sigma f_i (x_i - a)^r \qquad ...(2)$$

In particular, we have $\mu_0 = \mu_0' = 1$ ...(3)

$$\mu_1 = \frac{1}{N}\Sigma f_i (x_i - \bar{x}) = 0, \mu_1' = \frac{1}{N}\Sigma f_i (x_i - a) = \bar{x} - a = d, \text{say} \qquad ...(4)$$

$$\mu_2 = \frac{1}{N}\Sigma f_i (x_i - \bar{x})^2 = \sigma^2. \qquad ...(5)$$

(2) Moments about the mean in terms of moments about any point.

We have

$$\mu_r = \frac{1}{N}\Sigma f_i (x_i - \bar{x})^r = \frac{1}{N}\Sigma f_i (x_i - a - \bar{x} + a)^r$$

$$= \frac{1}{N}\Sigma f_i (X_i - \bar{X})^r \quad \text{where } X_i = x_i - a, \bar{X} = \bar{x} - a$$

$$= \frac{1}{N}\Sigma f_i \left[X_i^r - {}^rC_1 X_i^{r-1}\bar{X} + {}^rC_2 X_i^{r-2}\bar{X}^2 - \ldots\right]$$

$$= \mu_r' - {}^rC_1 \mu_{r-1}'\mu_1' + {}^rC_2 \mu_{r-2}'\mu_1'^2 - \ldots \qquad ...(6)$$

In particular,

$$\mu_2 = \mu_2' - \mu_1'^2 \qquad ...(7)$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3 \qquad ...(8)$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4 \qquad ...(9)$$

These three results should be committed to memory. It should be noted that in each of these relations, the sum of the coefficients of the various terms on the right side is zero. Also each term on the right side is of the same dimension as the term on the left.

23.13 SKEWNESS

Skewness measures the degree of asymmetry or the departure from symmetry. If the frequency curve has a longer tail to the right, i.e., the mean is to the right of the mode (as in Fig. 23.4 (a)), then the distribution is said to have positive skewness. If the curve is more elongated to the left, then it is said to have negative skewness (Fig. 23.4 (b)).

The following three measures of skewness deserve mention:

(i) Pearson's* coefficient of skewness $= \dfrac{\text{mean} - \text{mode}}{\sigma}$

Fig. 23.4

* After the English statistician and biologist Karl Pearson (1857–1936) who did pioneering work and founded the Englisch school of statistics.

(c) Quartile coefficient of skewness $= \dfrac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$

Its value always lies between -1 and $+1$.

(d) Coefficient of skewness based on third moment $\gamma_1 = \sqrt{\beta_1}$,

where $\beta_1 = \mu_3^2/\mu_2^3$

That $\gamma_1 = \sqrt{\beta_1}$ gives the simplest measure of skewness.

23.23 KURTOSIS

Kurtosis measures the degree of peakedness of a distribution and is given by $\beta_2 = \mu_4/\mu_2^2$.

(a) Leptokurtic (b) Platykurtic (c) Mesokurtic

Fig. 23.3

$\gamma_2 = \beta_2 - 3$ gives the excess of Kurtosis. The curves with $\beta_2 > 3$ are called *Leptokurtic* and those with $\beta_2 < 3$ as *Platykurtic*. The normal curve for which $\beta_2 = 3$, is called *Mesokurtic* (Fig. 23.3).

Example 23.11. The first four moments about the working mean 30.5 of a distribution are 0.294, 7.144, 42.409 and 454.96. Calculate the moments about the mean. Also calculate β_1, β_2 and comment upon the skewness and kurtosis of the distribution. (C.U. B.Com., 2nd Sem., 2010)

Solution. The first four moments about the arbitrary origin 30.5 are $\mu'_1 = 0.294$, $\mu'_2 = 7.144$, $\mu'_3 = 42.409$, $\mu'_4 = 454.96$.

$\therefore \quad \mu'_1 = \dfrac{1}{N}\sum f(x_i - 30.5) = \dfrac{1}{N}\sum f x_i - 30.5 = \bar{x} - 30.5 = 0.294$ or $\bar{x} = 30.794$

$\mu_2 = \mu'_2 - \mu'^2_1 = 7.144 - (0.294)^2 = 7.058$

$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu'^3_1 = 42.409 - 3(7.144)(0.294) + 2(0.294)^3 = 36.351$

$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2\mu'^2_1 - 3\mu'^4_1$

$= 454.96 - 4(42.409)(0.294) + 6(7.144)(0.294)^2 - 3(0.294)^4 = 408.336$

Now $\beta_1 = \mu_3^2/\mu_2^3 = (36.351)^2/(7.058)^3 = 3.757$

$\beta_2 = \mu_4/\mu_2^2 = 408.336/(7.058)^2 = 8.205$.

$\therefore \quad \gamma_1 = \sqrt{\beta_1} = 1.938$, which indicates considerable skewness of the distribution.

$\gamma_2 = \beta_2 - 3 = 5.205$ which shows that the distribution is leptokurtic.

Example 23.12. Calculate the skewness (quartile) and the quartile coefficient of skewness from the following data :

Weights (lbs) :	70–90	90–110	110–130	130–150	150–170	170–190	190–210	210–230
No. of persons :	12	28	32	42	30	42	20	8

Solution. Here total frequency $N = \Sigma f_i = 214$.

The cumulative frequency table is

Weights (lbs) :	70–90	90–110	110–130	130–150	150–170	170–190	190–210	210–230
f :	12	28	32	42	30	42	20	8
cum. f :	12	40	72	114	144	186	206	214

Now $N/2 = 214/2 = 118$th item which lies in 130–150 group.

$\therefore \quad$ median or $Q_2 = L_1 + \dfrac{N/2 - C}{f} \times h = 130 + \dfrac{118 - 114}{30} \times 20 = 133.3$.

Also $3N/4 = 321/4 = 53.5$ i.e. Q_3 is 53.5th item which lies in 90–110 group.

$$\therefore \quad Q_1 = L + \frac{N/4 - C}{f} \times h = 80 + \frac{37.5 - 30}{35} \times 15 = 83.21$$

Similarly, 3/4th of 150 i.e. Q_3 is 122nd item which lies in 120–130 group.

$$\therefore \quad Q_3 = L + \frac{3N/4 - C}{f} \times h = 120 + \frac{112.5 - 107}{48} \times 10 = 121.14$$

$$\text{Bowley's quartile coefficient of skewness} = \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$= \frac{97.85 + 121.14 - 2 \times 111.5}{121.14 - 97.85} = -0.05 \text{ (approx.)}$$

EXERCISE 5.3

1. Calculate the first four moments of the following distribution about the mean:

x:	0	1	2	3	4	5	6	7	8
f:	1	9	26	59	72	52	29	8	1

(Ans. Mean = 4, $\mu_2 = 2$, $\mu_3 = 0$, $\mu_4 = 51$) (B.Com., 2001) (B.Com., 2003)

2. The following table gives the monthly wages of 50 workers in a factory. Compute the standard deviation, quartile deviation, coefficient of variation and skewness. (B.Com., 2003)

Monthly wages (in ₹)	No. of workers	Monthly wages (in ₹)	No. of workers
100–120	12	160–180	4
120–140	10	180–200	3
140–160	9	200–220	1
160–180	11	220–240	1
180–200	3		

3. Find Pearson's coefficient of skewness for the following data:

Class	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90
Frequency	6	9	14	20	15	10	8	4

(P.T.U., 2003 S)

4. Compute the coefficient of skewness for the following distribution:

x:	5–9	10–14	15–19	20–24	25–29	30–34	35–39	40–44
f:	3	30	50	72	90	70	50	5

(Kanpur, 2005) (P.T.U., 2005)

5. Explain the measures of skewness based on the third moment.

6. For a distribution, the mean is 10, the variance is 16, $\beta_2 = 4$ and the coefficient of skewness is 1. Find the values of μ_3 and μ_4. (P.T.U., 2003 S)

7. Compute skewness and kurtosis for the first four moments of a frequency distribution about the values 4 are respectively 2, 4, 10 and 45. (Delhi Univ., 1993)

8. In a certain distribution, the first four moments about a point are 1.5, 17, –30 and 108. Calculate the moments about the mean, β_1 and β_2 and state whether the distribution is leptokurtic or platykurtic?

5.8 CORRELATION

So far we have confined our attention to the analysis of observations on a single variable. There are, however, many phenomena where the changes in one variable are related to the changes in the other variable. For instance, the yield of a crop varies with the amount of rainfall, the price of a commodity increases with the reduction in its supply and so on. Such a simultaneous variation, i.e. when the changes in one variable are associated or followed by changes in the other, is called correlation. Such a data connecting two variables is called bivariate population.

If an increase (or decrease) in the values of one variable corresponds to an increase (or decrease) in the other, the correlation is said to be positive. If the increase (or decrease) in one corresponds to the decrease (or increase) in the other, the correlation is said to be negative. If there is no relationship indicated between the variables, they are said to be independent or uncorrelated.

To obtain a measure of correlation between the two variables, we plot their corresponding values on the graph, taking one of the variables along the x-axis and the other along the y-axis (Fig. 25.6).

Let the origin be shifted to $(\bar{x}, \bar{y})$, where $\bar{x}, \bar{y}$ are the means of x's and y's so that the new co-ordinates are given by

$$X = x - \bar{x}, \qquad Y = y - \bar{y}.$$

Now the points (X, Y) are so distributed over the four quadrants of XY plane that the product XY is positive in the first and third quadrants but negative in the second and fourth quadrants. The algebraic sum of the products can be taken as describing the trend of the data in all the quadrants.

(a) If ΣXY is positive, the trend of the data is through the first and third quadrants.

(b) If ΣXY is negative the trend of the data is in the second and fourth quadrants, and

(c) If ΣXY is zero, the points indicate no trend i.e. the points are evenly distributed over the four quadrants.

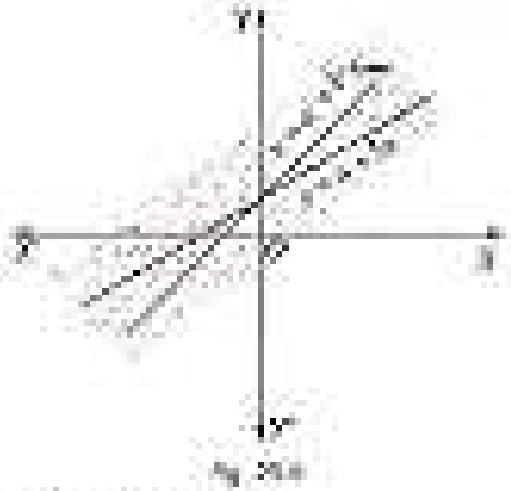


Fig. 25.6

The XY correlation is $\frac{1}{n}\Sigma XY$, i.e., the average of a positive or negative term is taken as a measure of correlation. If we put X and Y in their units, i.e., taking σ_x as the unit for x and σ_y for y, then

$$\frac{1}{n}\Sigma\frac{X}{\sigma_x}\frac{Y}{\sigma_y}, \text{ i.e., } \frac{\Sigma XY}{n\sigma_x\sigma_y}$$

is the measure of correlation.

25.5 COEFFICIENT OF CORRELATION

The numerical measure of correlation is called the coefficient of correlation and is denoted by the symbol r and is given by

$$r = \frac{\Sigma XY}{n\sigma_x\sigma_y}$$

where X = deviation of x from the mean $\bar{x}$, i.e. $X = x - \bar{x}$, Y = deviation of y from the mean $\bar{y}$, i.e. $Y = y - \bar{y}$,

σ_x = S.D. of x-series, σ_y = S.D. of y-series and n = number of values of the two variables.

(Working rules for calculation)

(a) Direct method. By substituting the values of σ_x and σ_y in the above formula, we get

$$r = \frac{\Sigma XY}{\sqrt{(\Sigma X^2)(\Sigma Y^2)}} \qquad (1)$$

Another form of the formula (1) which is quite handy for calculation is

$$r = \frac{n\Sigma xy - \Sigma x\Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}} \qquad (2)$$

(b) Step-deviation method. The direct method becomes very lengthy and tedious if the means of the two series are not integers. In such cases, we take the help of assumed means. If d_x and d_y are step-deviations from the assumed means, then

$$r = \frac{n\Sigma d_x d_y - \Sigma d_x \Sigma d_y}{\sqrt{[n\Sigma d_x^2 - (\Sigma d_x)^2][n\Sigma d_y^2 - (\Sigma d_y)^2]}} \qquad (3)$$

where $d_x = (x - a)/h$ and $d_y = (y - b)/k$.

Note: The change of origin and scale do not affect the value of the correlation coefficient since it is a pure number.

(c) Coefficient of correlation for grouped data. When x and y series are both given as frequency distributions, these can be represented by a two-way table known as the correlation table. It is double-entry table with the x-series along the horizontal and the other along the vertical as shown on page 414. The coefficient of correlation for such a bivariate frequency distribution is calculated by the formula,

$$r = \frac{n\sum fd_x d_y - \sum fd_x \sum fd_y}{\sqrt{\left[n\sum fd_x^2 - \left(\sum fd_x\right)^2\right]\left[n\sum fd_y^2 - \left(\sum fd_y\right)^2\right]}} \quad \ldots(1)$$

where d_x = deviations of the central values from the assumed mean of x-series,

d_y = deviation of the central values from the assumed mean of y-series,

f is the frequency corresponding to the pair (x, y)

and $n = \sum f$ is the total number of frequencies.

Example 33.13. *Ten boys selected from a class of intelligence and engineering ability were applied to 15 students. Here is a record of a grouped data showing intelligence ratio (I.Q.) and engineering ratio (E.R.). Calculate the coefficient of correlation.*

Student	A	B	C	D	E	F	G	H	I	J
I.Q.	105	104	109	102	108	101	105	103	98	103
E.R.	102	108	109	98	102	98	104	102	97	94

(Madras, B.Sc.)

Solution. We construct the following table :

Student	Intelligence ratio (x)		Engineering ratio (y)				
	x	$d_x = x - 100$	y	$d_y = y - 100$	d_x^2	d_y^2	$d_x d_y$
A	105	5	102	2	25	4	10
B	104	4	108	8	16	64	32
C	109	9	109	9	81	81	81
D	102	2	98	−2	4	4	−4
E	108	8	102	2	64	4	16
F	101	1	98	−2	1	4	−2
G	105	5	104	4	25	16	20
H	103	3	102	2	9	4	6
I	98	−2	97	−3	4	9	6
J	103	3	94	−6	9	36	−18
Total	1000	40	1004	24	258	226	149

From this table, mean of x, i.e., $\bar{x} = 100 + 40/10 = 104$ and mean of y, i.e., $\bar{y} = 100 + 24/10 = 102$.

$\sum d_x^2 = 258$, $\sum d_y^2 = 226$ and $\sum d_x d_y = 149$

Substituting these values in the formula (1) p. 746, we have

$$r = \frac{\sum d_x d_y}{\sqrt{(\sum d_x^2)(\sum d_y^2)}} = \frac{92}{\sqrt{(133 \times 140)}} = \frac{92}{136.54} = 0.674$$

Example 33.14. *The correlation table given below shows that the ages of husbands and wife of 50 married couples are highly correlated or not the correlation coefficient between the ages of the husband and that of the wife.* (U.P.T.U., 2003)

Age of Husband	Age of wife						Total
	15–20	20–25	25–30	30–35	35–40	40–45	
20–25	3	2	1	–	–	–	6
25–30	1	12	3	–	–	–	16
30–35	–	4	10	3	–	–	17
35–40	–	–	1	4	1	–	6
40–45	–	–	1	2	1	1	5
45–50	–	–	–	1	2	1	4
Total	4	18	16	10	4	2	54

Solution:

Age of husband	Age of wife (in years)								
	18–20	20–25	25–30	30–35	35–40	40–45	Total	Regression lines
Mean	20	30	40	50	60	70		
Assumed mean	−20	−10	0	10	20	30		
Assumed mean	−2	−1	0	1	2	3		
25–30	30	−20	−2	8 / 1	2 / 1				
30–35	35	−15	−1	8 / 2	15 / 12	3 / 1			
35–40	40	0	0		8 / 2	12 / 10	1 / 1		
40–45	45	15	1			3 / 3	8 / 6	3 / 1	
45–50	50	30	2				4 / 1	16 / 4	8 / 3
50–55	55	40	3					2 / 1	10 / 8
Total				20	37	22	9	9	3
f_x				−40	−37	0	9	18	6
f_x^2				80	37	0	9	36	12
$f(x, y)$				8	14	0	15	34	10

With the help of the above correlation table, we have

$$r = \frac{n\Sigma f_x f_y (x, y) - (\Sigma f_x)(\Sigma f_y)}{\sqrt{[n\Sigma f_x^2 - (\Sigma f_x)^2][n\Sigma f_y^2 - (\Sigma f_y)^2]}}$$

$$= \frac{63 \times 86 - 10 \times 16}{\sqrt{[63 \times 88 - (10)^2][63 \times 93 - (16)^2]}} = \frac{4358}{\sqrt{(5384)(4931)}} = \frac{4358}{4618} = 0.91 \text{ (approx.)}$$

24.6 LINES OF REGRESSION

If it is evidently happens that the dots of the scatter diagram generally tend to cluster along a well defined direction which suggests a linear relationship between the variables under study. Such a line of best fit for the given distribution of data is called the line of regression (Fig. 24.6). In fact there are two such lines, one giving the best possible means values of y for each specified values of x and the other giving the best possible mean values of x for given values of y. The former is known as the line of regression of y on x and the latter as the line of regression of x on y.

Consider first the line of regression of y on x. Let the straight line satisfying the general trend of the data in a scatter diagram be

$$y = a + bx \qquad \ldots(1)$$

We have to determine the constants a and b so that (1) gives for each value of x, the best estimate for the average value of y in accordance with the principle of least squares (see page 316); therefore, the normal equations for a and b are

$$\Sigma y = na + b\Sigma x \qquad \ldots(2)$$

and

$$\Sigma xy = a\Sigma x + b\Sigma x^2 \qquad \ldots(3)$$

(2) gives $\quad \frac{1}{n}\Sigma y = a + b \cdot \frac{1}{n}\Sigma x$ i.e., $\bar{y} = a + b\bar{x}$.

This shows that $(\bar{x}, \bar{y})$ i.e., the means of x and y, lies on (1).

Shifting the origin to $(\bar{x}, \bar{y})$, (3) takes the form:

$$\Sigma(x-\bar{x})(y-\bar{y}) = a\Sigma(x-\bar{x}) + b\Sigma(x-\bar{x})^2, \text{ but } a\Sigma(x-\bar{x}) = 0.$$

$$\therefore \quad b = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2} = \frac{\Sigma XY}{\Sigma X^2} = \frac{\Sigma XY}{n\sigma_x^2} = r\frac{\sigma_y}{\sigma_x} \qquad \left[\because r = \frac{\Sigma XY}{n\sigma_x\sigma_y}\right]$$

Thus the line of best fit becomes $\quad y - \bar{y} = r\frac{\sigma_y}{\sigma_x}(x - \bar{x}) \qquad \ldots(4)$

which is the equation of the line of regression of y on x. Its slope is called the regression coefficient of y on x.

Interchanging x and y, we find that the line of regression of x on y is

$$x - \bar{x} = r\frac{\sigma_x}{\sigma_y}(y - \bar{y}) \qquad \ldots(5)$$

Thus the regression coefficient of y on $x = r\sigma_y/\sigma_x$ $\qquad \ldots(6)$

and the regression coefficient of x on $y = r\sigma_x/\sigma_y$ $\qquad \ldots(7)$

Cor. The correlation coefficient is the geometric mean between the two regression coefficients.

For $\quad r\frac{\sigma_y}{\sigma_x} \times r\frac{\sigma_x}{\sigma_y} = r^2$

Example 32.15. The two regression equations of the variables x and y are $x + 19.13 - 0.87y = 0$ and $y - 11.64 - 0.50x = 0$. Find (i) means of x & y, (ii) value of r, and (iii) the correlation coefficient between x and y.

(C.T.S., 2004; Anna, 2003; Bharathiar, 2002)

Solution. Since the means of x and the means of y lie on the two regression lines, we have

$$\bar{x} = 19.13 - 0.87\bar{y} \qquad \ldots(1)$$

$$\bar{y} = 11.64 - 0.50\bar{x} \qquad \ldots(2)$$

Multiplying (1) by 0.87 and subtracting from (2), we have

$$\bar{x} - 0.87(0.50\bar{x}) = 19.13 - 11.64 \text{ or } 0.57\bar{x} = 7.49 \text{ or } \bar{x} = 13.79$$

$$\therefore \quad \bar{y} = 11.64 - 0.50 \times 13.79 = 4.74$$

$\therefore$ regression coefficient of y on x is -0.50 and that of x on y is -0.87.

Now we use the coefficient of correlation is the geometric mean between the two regression coefficients.

$$\therefore \quad r = \sqrt{(-0.50)(-0.87)} = \sqrt{0.435} = -0.66.$$

[–ve sign is taken since both the regression coefficients are –ve]

Example 32.16. In the following table are given data showing the test scores made by salesmen on an intelligence test and their yearly sales:

Salesmen	1	2	3	4	5	6	7	8	9	10
Test scores	40	50	60	60	50	60	70	70	60	60
Sales (000)	2.5	3.0	4.5	3.0	4.5	3.5	5.5	4.0	4.5	3.5

Calculate the regression lines of sales on test scores and estimate the most probable yearly sales volume of a salesman whose test score is 70.

Solution. With the help of the table below, we have

$\bar{x}$ = mean of x first account = 80 + 0.10 = 80

$\bar{y}$ = mean of y (sales) = 8.5 + 1 = 9.5

Regression line of sales (y) on account (x) is given by

$$y - \bar{y} = b(x, y)(x - \bar{x})$$

where $$b_{xy} = \frac{\sum xy}{\sum y^2} \times \frac{\sigma_y}{\sigma_x} = \frac{\sum xy}{n\sigma_y^2} = \left[\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}\right] \Big/ \left[\sum y_i^2 - \frac{(\sum y_i)^2}{n}\right]$$

$$= \frac{130 - \frac{0 \times (-5)}{10}}{3600 - \frac{0^2}{10}} = \frac{130}{3600} = 0.036$$

∴ the required regression line is

$$y - 4.08 = 0.036(x - 80) \quad \text{or} \quad y = 0.036x + 0.45$$

For $x = 70$, $y = 0.036 \times 70 + 0.45 = 4.85$.

Thus the most probable monthly sales volume for a value of 70 is 4.85.

Test score	Index	Deviation of x from assumed mean $(x-80)$	Deviation of y from assumed average $(y-8.5)$	$d_x \times d_y$	d_x^2	d_y^2
x	y	d_x	d_y			
60	5.0	−20	−3	60	400	9
70	6.0	−10	−2.5	25	100	6.25
80	8.5	0	0	0	0	0
90	8.0	10	−0.5	−5	100	0.25
95	8.5	15	0	0	225	0
60	5.0	−20	−3.5	70	400	12.25
90	9.0	10	0.5	5	100	0.25
80	8.0	0	−0.5	0	0	0.25
65	8.0	−15	−0.5	7.5	225	0.25
		$\sum d_x = 0$	$\sum d_y = -15$	$\sum d_x d_y = 130$	$\sum d_x^2 = 1550$	$\sum d_y^2 = 29.50$

Example 15.17. If b is the angle between the two regression lines, show that

$$\tan\theta = \frac{1-r^2}{r} \cdot \frac{\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2}$$

Explain the significance when $r = 0$ and $r = \pm 1$. (U.P.T.U., 2002; P.T.U., 2003)

Solution. The equations to the lines of regression of y on x and x on y are

$$y - \bar{y} = r\frac{\sigma_y}{\sigma_x}(x - \bar{x}) \quad \text{and} \quad x - \bar{x} = r\frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

∴ their slopes are $m_1 = r\sigma_y/\sigma_x$ and $m_2 = \sigma_y/r\sigma_x$

Thus $$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2} = \frac{\sigma_x(\sigma_x^2 - r^2\sigma_y^2)}{1 + \sigma_y^2/\sigma_x^2} = \frac{1-r^2}{r} \cdot \frac{\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2}$$

When $r = 0$, then $\theta = \pi$ or $\theta = \pi/2$ i.e. when the variables are independent, the two lines of regression are perpendicular to each other.

When $r = \pm 1$, then $\theta = 0$ or π, $\theta = \pi$ or 2π. Thus the lines of regression coincide i.e., there is perfect correlation between the two variables.

Example 15.16. In a partially destroyed laboratory record only the lines of regression of y on x and of x on y are available as $4x - 5y + 33 = 0$ and $20x - 9y = 107$ respectively. Calculate $\bar{x}$, $\bar{y}$ and the coefficient of correlation between x and y. (U.P.T.U., 2004 ; U.P.T.U., 2005 ; U.P.T.U., 2006)

Solution. Since the regression lines pass through $(\bar{x}, \bar{y})$, therefore

$$4\bar{x} - 5\bar{y} + 33 = 0, \quad 20\bar{x} - 9\bar{y} = 107.$$

Solving these equations, we get $\bar{x} = 13$, $\bar{y} = 17$.

Rewriting the line of regression of y on x as $y = \frac{4}{5}x + \frac{33}{5}$, we get

$$b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{4}{5} \quad \ldots(i)$$

Rewriting the line of regression of x on y as $x = \frac{9}{20}y + \frac{107}{20}$, we get

$$b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{9}{20} \quad \ldots(ii)$$

Multiplying (i) and (ii), we get

$$r^2 = \frac{4}{5} \times \frac{9}{20} = 0.36 \quad \therefore \quad r = 0.6$$

Hence $r = 0.6$, the positive sign being taken as b_{yx} and b_{xy} both are positive.

Example 15.17. Establish the formula $r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$.

Hence calculate r from the following data :

x	21	25	26	34	37	38	72	79	87	90	
y	60	77	72	85	110	84	209	90	113	125	(U.P.T.U., 2003)

Solution. (a) Let $u = x - y$ so that $\bar{u} = \bar{x} - \bar{y}$.

Now $u - \bar{u} = (x - \bar{x}) - (y - \bar{y})$

or $(u - \bar{u})^2 = (x - \bar{x})^2 + (y - \bar{y})^2 - 2(x - \bar{x})(y - \bar{y})$

Summing up for n terms, we have

$$\Sigma(x - \bar{x})^2 = \Sigma(x - \bar{x})^2 + \Sigma(y - \bar{y})^2 - 2\Sigma(x - \bar{x})(y - \bar{y})$$

or $$\frac{\Sigma(x - \bar{x})^2}{n} = \frac{\Sigma(x - \bar{x})^2}{n} + \frac{\Sigma(y - \bar{y})^2}{n} - 2\frac{\Sigma(x - \bar{x})(y - \bar{y})}{n}$$

i.e., $$\sigma_x^2 = \sigma_x^2 + \sigma_y^2 - 2r\sigma_x\sigma_y \quad \Big| \quad \therefore r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$$

which is the required result.

(b) To find r, we have to calculate σ_x, σ_y and σ_{x-y}. We make the following table :

x	$(x - \bar{x})$	$(x - \bar{x})^2$	y	$(y - \bar{y})$	$(y - \bar{y})^2$	$(x - y)$	$(x - y)^2$
21	−30	1040	60	−18	1800	39	1521
25	−31	961	77	−13	841	48	2304
26	−34	529	72	−18	196	42	1764
34	−6	1	85	−5	400	51	441
37	−3	9	110	20	100	85	3249
38	−2	16	84	−6	900	54	576
72	32	1024	160	70	0	38	144
79	24	576	90	−8	64	34	196
87	32	1089	113	15	289	26	676
90	36	1296	125	27	1225	45	2025
Total	32	7613		−40	5800	362	13320

$$\therefore \quad \sigma_x^2 = \frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2 = \frac{3604}{10} - \left(\frac{60}{10}\right)^2 = 360.4 - 36 = 324.4$$

$$\sigma_y^2 = \frac{\sum y^2}{N} - \left(\frac{\sum y}{N}\right)^2 = \frac{3604}{10} - \left(\frac{-60}{10}\right)^2 = 360.4 - 36 = 324.4$$

$$\sigma_{x-y}^2 = \frac{\sum (x-y)^2}{N} - \left(\frac{\sum x - \sum y}{N}\right)^2 = 1080.8 - 1296 = 324.8$$

From the above formula,

$$r = \frac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y} = \frac{324.8 + 324.8 - 324.8}{2 \times 324.8 \times 324.8} = 0.5$$

Example 2.5.10 While calculating correlation coefficient between two variables x and y from 10 pairs of observations, the following results were obtained: $n = 20$, $\sum x = 125$, $\sum x^2 = 650$, $\sum y = 100$, $\sum y^2 = 460$, $\sum xy = 508$. Later it was discovered at the time of checking that the pairs of values $\begin{matrix} x & y \\ 8 & 12 \end{matrix}$ were copied twice as $\begin{matrix} x & y \\ 6 & 14 \end{matrix}$. Obtain the correct value of correlation coefficient.

Solution: To get the correct results, we subtract the incorrect values and add the corresponding correct values.

∴ The correct results would be

$$\sum x = 125 - 6 - 6 + 8 + 8 = 129, \quad \sum x^2 = 650 - 6^2 - 6^2 + 8^2 + 8^2 = 714$$

$$\sum y = 100 - 14 - 14 + 12 + 12 = 96, \quad \sum y^2 = 460 - 14^2 - 14^2 + 12^2 + 12^2 = 436$$

$$\sum xy = 508 - 6 \times 14 - 6 \times 14 + 8 \times 12 + 8 \times 12 = 524$$

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}} = \frac{20 \times 524 - 129 \times 96}{\sqrt{[20 \times 714 - 129^2][20 \times 436 - 96^2]}}$$

$$= \frac{58}{\sqrt{(251 \times 344)}} = \frac{2}{3}$$

2.5.13 STANDARD ERROR OF ESTIMATE

The sum of the squares of the deviations of the points from the line of regression of y on x is

$$E(y - a - bx)^2 = \sum (Y - aX)^2, \text{ where } X = x - \bar{x}, \; Y = y - \bar{y}$$

$$= \sum \left(Y - r\frac{\sigma_y}{\sigma_x}X\right)^2 = \sum Y^2 - 2r\frac{\sigma_y}{\sigma_x}\sum XY + r^2\frac{\sigma_y^2}{\sigma_x^2}\sum X^2$$

$$= n\sigma_y^2 - 2r\frac{\sigma_y}{\sigma_x}n.r\sigma_x\sigma_y + r^2\frac{\sigma_y^2}{\sigma_x^2}.n\sigma_x^2 = n\sigma_y^2(1 - r^2).$$

Denoting this sum of squares by $n\sigma_y^2$, we have $\sigma_y = \sigma_y\sqrt{1 - r^2}$ ...(1)

Since σ_y is the root mean square deviation of the points from the regression line of y on x, it is called the *standard error of estimate of y*. Similarly the standard error of estimate of x is given by

$$\sigma_x = \sigma_x\sqrt{1 - r^2} \qquad ...(2)$$

Since the sum of the squares of deviations cannot be negative, it follows that

$$r^2 \le 1 \text{ or } -1 \le r \le 1.$$

i.e., *correlation coefficient lies between –1 and 1.* (*M.K.U.C.*, 2000)

If $r = 1$ or -1, the sum of the squares of deviations from either line of regression is zero. Consequently each deviation is zero and all the points lie on both the lines of regression. These two lines coincide and we say that the correlation between the variables is perfect. The nearer r^2 is to unity the closer are the points to the lines of

represents. Thus the departure of r^2 from unity is a measure of departure from linearity of the relationship between the variables.

9.20(B) RANK CORRELATION

A group of n individuals may be arranged in order to merit with respect to some characteristic. The same group would give different orders for different characteristics. Considering the orders corresponding to two characteristics A and B, the correlation between these n pairs of ranks is called the rank correlation in the characteristics A and B for that group of individuals.

Let x, y be the ranks of the ith individual in A and B respectively. Assuming that no two individuals are bracketed equal in either case, each of the variables x and y taking the values $1, 2, 3, \ldots, n$, we have

$$\bar{x} = \bar{y} = \frac{1+2+3+\ldots+n}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}$$

If X, Y be the deviations of x, y from their means, then

$$\Sigma X_i^2 = \Sigma(x_i - \bar{x})^2 = \Sigma x_i^2 + n\bar{x}^2 - 2\bar{x}\Sigma x_i = \Sigma x^2 + \frac{n(n+1)^2}{4} - n\bar{x}^2 \cdot \frac{n+1}{2} \cdot n$$

$$= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)^2}{4} - \frac{n(n+1)^2}{2} = \frac{1}{12}(n^3 - n)$$

Similarly $\Sigma Y_i^2 = \frac{1}{12}(n^3 - n)$

Now let $d_i = x_i - y_i$ so that $d_i = (x_i - \bar{x}) - (y_i - \bar{y}) = X_i - Y_i$

$\therefore$ $\Sigma d_i^2 = \Sigma X_i^2 + \Sigma Y_i^2 - 2\Sigma X_i Y_i$

or $$\Sigma X_i Y_i = \frac{1}{2}[\Sigma X_i^2 + \Sigma Y_i^2 - \Sigma d_i^2] = \frac{1}{12}(n^3 - n) - \frac{1}{2}\Sigma d_i^2$$

Hence the correlation coefficient between these variables is

$$r = \frac{\Sigma X_i Y_i}{\sqrt{\Sigma X_i^2 \Sigma Y_i^2}} = \frac{\frac{1}{12}(n^3 - n) - \frac{1}{2}\Sigma d_i^2}{\frac{1}{12}(n^3 - n)} = 1 - \frac{6\Sigma d_i^2}{n^3 - n}$$

This is called the rank correlation coefficient and is denoted by ρ.

Example 9.50. Ten participants in a contest are ranked by two judges as follows :

x	1	6	5	10	3	2	4	9	7	8
y	6	4	9	8	1	2	3	10	5	7

Calculate the rank correlation coefficient ρ. (U.P.T.U., 2003)

Solution. If $d_i = x_i - y_i$, then $d_i = -5, 2, -4, 2, 2, 0, 1, -2, 2, 1$

$\therefore$ $\Sigma d_i^2 = 25 + 4 + 16 + 4 + 4 + 0 + 1 + 4 + 4 + 1 = 69$

Hence $$\rho = 1 - \frac{6\Sigma d_i^2}{n^3 - n} = 1 - \frac{6 \times 69}{990} = 0.6 \text{ (nearly)}.$$

Example 9.51. Two judges, A, B, C, give the following marks. Find which pair of judges has common opinion?

A :	1	6	5	10	3	2	4	9	7	8	
B :	2	4	9	8	1	10	3	5	6	7	
C :	10	4	9	8	1	2	3	10	5	7	(U.P.T.U., 2003)

Solution. Here $n = 10$.

A (x)	Rank by d_1 (x)	B (y)	d_1 x − y	d_2 y − x	d_3 x + y	d_1^2	d_2^2	d_3^2
1	1	6	−5	−5	7	25	25	49
2	2	4	−2	−2	6	4	4	36
3	3	7	−4	−4	10	16	16	100
4	4	8	−4	−4	12	16	16	144
5	5	1	0	0	6	0	0	36
6	6	2	4	−4	8	16	16	64
7	7	3	4	−4	10	16	16	100
8	8	5	3	−3	13	9	9	169
9	9	10	−1	1	19	1	1	361
10	10	9	1	−1	19	1	1	361
Total			0	−28		104	104	1320

$$\rho(x,y) = 1 - \frac{6\sum d_1^2}{n(n^2-1)} = 1 - \frac{6\times 104}{10\times 99} = -0.5$$

$$\rho(x,z) = 1 - \frac{6\sum d_2^2}{n(n^2-1)} = 1 - \frac{6\times 104}{10\times 99} = -0.5$$

$$\rho(x,z) = 1 - \frac{6\sum d_3^2}{n(n^2-1)} = 1 - \frac{6\times 80}{10\times 99} = 0.6$$

Since $\rho(x,z)$ is maximum, the pair of judges A and C have the nearest common approach.

EXERCISE 10.4

1. Find the correlation coefficient and the regression lines of y and x from the following data:

x	1	2	3	4	5
y	2	5	3	8	7

(C.T.U., B.Sc.)

2. Find the correlation coefficient between x and y from the given data:

x	78	89	97	69	59	79	68	57	
y	125	137	156	112	107	136	123	108	(C.U.C.E.T., 2005)

3. Find the coefficient of correlation between marks and practical hrs and explain using the following data (Assume all are final year):

Practical (hrs/week, x)	16	10	20	22	19	10	22	
Reports (hrs/week, y)	35	30	35	39	44	25	45	(Madras, 2005)

4. Two people of various heights in an audience were requested to read the letters on a sign 25 yards distance. The number of letters correctly read is given below:

Height (in feet)	5.1	5.5	5.8	5.7	5.3	5.9	5.10	5.11	6.0	6.1
No. of letters	12	25	19	14	9	18	20	9	8	12

Is there any correlation between height and visual power?

5. Using the formula $\rho = \dfrac{\sigma_x^2 + \sigma_y^2 - \sigma_{x-y}^2}{2\sigma_x\sigma_y}$, find ρ from the following data:

x	51	50	47	48	50	57	53	52	52	58
y	60	60	55	57	58	65	62	60	57	57

6. Find the correlation between x (Income in Rs thousands) and y (marks in Engineering Drawing) given in the following data:

x / y	20–40	40–60	60–80	Total
0–30	5	20	—	25
30–60	—	25	5	30
60–90	—	10	35	45
Total	5	55	40	100

7. Find two lines of regression and coefficient of correlation for the data given below.
$$x + 2y - 5 = 0,\ 2x + 3y - 8 = 0,\ 7x^2 - 6y^2 + 31xy - 47 = 0$$ (U.P.T.U., M.C.A., 2006)

8. If the coefficient of correlation between two variables x and y is 0.6 and the angle between their lines of regression is $\tan^{-1}(0.8)$, show that $\sigma_y = \frac{1}{2}\sigma_x$. (U.P.T.U., 2006)

9. For two random variables x and y with the same mean, the two regression lines are $y = ax + b$ and $x = cy + d$. Show that $ac = \frac{1-a}{1-c}$. Find also the common mean. (U.P.T.U., 2010)

10. Two random variables have the regression lines with equations $3x + 2y - 26 = 0$ and $6x + y - 31 = 0$. Find the mean values and the correlation coefficient between x and y. (Madras, 2005)

11. The regression equations of two variables x and y are $x = 0.7y + 5.2$, $y = 0.3x + 2.8$. Find the means of the variables and the coefficient of correlation between them. (Kerala, 2005)

12. In a partially destroyed laboratory data, only the equations giving the two lines of regression of y on x and x on y are available and are respectively, $x - 2y + 3 = 0$, $3x - 4y + 4 = 0$. Calculate the coefficient of correlation, $\bar{x}$ and $\bar{y}$.

13. The following results were obtained from records of age (x) and blood pressure (y) of a group of 10 men:

	$\bar{x}$	$\bar{y}$	σ_x	σ_y	r
Mean	53	142	2.6	2.0	0.9
Variance	110	200			

Find the appropriate regression equation and use it to estimate the blood pressure of a man whose age is 45.

14. Compute the number of terms of a formula N, that the respective heights of the fathers (y) of 10 couples:

Height (y) of the father (inches)	65	63	67	64	68	62	70	66	68	67
Height (x) of the son (inches)	68	66	68	65	69	66	68	65	71	67

15. Calculate the rank correlation coefficient from the following data showing marks of 10 students in two subjects:

Maths	8	6	9	3	7	10	4	5	2	1
Physics	6	7	10	1	5	7	3	4	2	1

16. Find the rank correlation for the following data:

x	68	64	75	50	64	43	56	52	38	46	44	56
y	62	58	68	45	81	48	52	44	35	42	51	49

(U.P.T.U., 2004, 2005, M.C.A., 2005)

9.15 OBJECTIVE TYPE OF QUESTIONS

MULTIPLE CHOICE

Select the correct answer (a), (b), (c), (d) or (e) of the following questions:

1. The median of the numbers 12, 10, 15, 13, 9 is
   (a) 12.5 (b) 12 (c) 13.5 (d) 11
2. The mode of the numbers 5, 7, 5, 9, 10, 11, 11, 11, 13 is
   (a) 11 (b) 10 (c) 7 (d) 5 and 11

4. S.D. is defined as

(a) $\sqrt{\frac{\Sigma f(x-\bar{x})^2}{N}}$ (b) $\frac{\Sigma f(x-\bar{x})}{N}$ (c) $\frac{\Sigma f(x-\bar{x})^2}{N}$

5. Coefficient of Variation is

(a) $\frac{\sigma}{\bar{x}} \times 100$ (b) $\frac{\sigma}{\bar{x}}$ (c) $\sqrt{\frac{\sigma}{\bar{x}}} \times 100$

6. Average scores of three batsmen A, B, C are respectively 40, 45 and 50 and their S.Ds are respectively 9, 21, 19. Which batsman is more consistent ?

(a) A (b) B (c) C

7. The regression lines of regression have one $y = 0.5x + a$ and $x = 0.4y + b$. The correlation coefficient is

(a) $\sqrt{0.2}$ (b) 0.45 (c) $-\sqrt{0.2}$

8. If the correlation coefficient is 0, the two regression lines are

(a) parallel (b) perpendicular (c) coincident (d) an angle of 45° to each other.

9. If r_1 and r_2 are two regression coefficients, then signs of r_1 and r_2 depend on _____

10. Regression coefficient of y on x is 0.3 and that of x on y is 1.2. Is the correlation coefficient r calculated ?

11. The standard deviation of the numbers 21, 22, 23, 24, 25 is _____

12. If $y = ax + b$ and $x = by - 2$ are the two lines of regression then $r =$ _____, $b =$ _____ and $ab =$ _____

13. If the two regression lines are perpendicular to each other, then their coefficient of correlation is _____

14. Spearman's formula is defined as _____

15. The maximum value of correlation coefficient is _____

16. Prediction error of Y is defined as _____

17. If X and Y are independent, then the correlation coefficient between X and Y is _____

18. The point of intersection of the two regression lines is _____

19. The sign of the coefficient of variation, the greater is the _____ in the data

20. The quartile coefficient of skewness is given by _____

21. Karl Pearson measures the _____ of a distribution

22. The equation of the line of regression of y on x is _____

23. Coefficient of correlation = _____

24. The angle between two regression lines is given by _____

25. A frequency curve is said to be Mesokurtic when β_2 is _____

26. Correlation coefficient in the given data is an integer between _____

27. When the variables are independent, the two lines of regression are _____

28. Arithmetic mean of the coefficients of regression is _____ than the coefficient of correlation.

29. If two regression lines are equal, then the coefficient of correlation is _____

30. The rank correlation is given by _____

31. The value of $\Sigma(x_i - \bar{x}) =$ _____

32. The value of coefficient of correlation lies between _____ and _____

33. If the two regression coefficients are -0.4 and -0.9, then the correlation coefficient is _____

34. A distribution with the following constants is positively skew : $Q_1 = 25.5$, median $= 40.8$, $Q_3 = 54.2$

(Tirupur, Feb'20)

35. Quartile coefficient of skewness is $\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$ (Tirupur, Feb'20)

36. Short note on population of Co-operative Institutions. (Tirupur, Feb'20)

CHAPTER 26

Probability and Distributions

[redacted]

Example 36.3. *In how many ways can one select a first, second, third and fourth choice among 12 firms having negotiation agreements.* (J.N.T.U., 2003)

Solution. First choice can be made from any of the 12 firms. Thereafter the second choice can be made from among the remaining 11 firms. Then the third choice can be made from the remaining 10 firms and the fourth choice can be made from the 9 firms.

Therefore the principle of counting, the number of ways in which first, second, third and fourth choices can be effected = $12 \times 11 \times 10 \times 9 = 11880$.

Example 36.4. *Find the number of permutations of all the letters of the word (i) Daughter (ii) Engineering.*

Solution. (i) $n = 8, n_1 = \text{no. of } a = 3, n_2 = \text{no. of } u = 2, n_3 = \text{no. of } h = 2$

$\therefore$ no. of permutations $= \dfrac{n!}{n_1!\,n_2!\,n_3!} = \dfrac{8!}{3!\,2!\,2!} = 5040$.

(ii) $n = 11, n_1 = \text{no. of } g = 3, n_2 = \text{no. of } i = 2, n_3 = \text{no. of } n = 2, n_4 = \text{no. of } e = 2$

$\therefore$ no. of permutations $= \dfrac{11!}{3!\,2!\,2!\,2!} = 277200$

Example 36.5. *From six engineers and five architects a committee is to be formed having three engineers and two architects. How many different committees can be formed if (i) there is no restriction, (ii) two particular engineers must be included, (iii) one particular architect must be excluded.*

Solution. (i) Number of committees $= {}^6C_3 \times {}^5C_2 = \dfrac{6.5.4}{3.2.1} \times \dfrac{5.4}{2.1} = 200$

(ii) Here we have to choose one engineer from the remaining four engineers.

$\therefore$ no. of committees $= {}^4C_1 \times {}^5C_2 = 4 \times \dfrac{5.4}{2.1} = 40$

(iii) Here we have to choose two architects from the remaining four architects.

$\therefore$ no. of committees $= {}^6C_3 \times {}^4C_2 = \dfrac{6.5.4}{3.2.1} \times \dfrac{4.3}{2.1} = 120$

PROBLEMS 36.1

1. If in an exam there are 12 multiple questions, in how many different ways can a student make the non-sequential answer to each question. (J.N.T.U., 2005)
2. How many 4-digit numbers can be formed from the six digits 2, 3, 5, 6, 7 and 8 – when repetition? How many of these numbers will be less than 5000?
3. A student has to answer 7 out of 12 questions. How many choices has he? (i) If he must answer the first two questions (ii) If he must answer at least four of the first five questions.
4. How many car number plates can be made if each plate contains two different letters followed by three different digits? Solve the problem (i) when repetition is allowed (ii) when repetition is not allowed.

36.3 BASIC TERMINOLOGY

(i) Exhaustive events. A set of events is said to be exhaustive, if it includes all the possible events. For example, in tossing a coin there are two exhaustive cases either head or tail and there is no third possibility.

(ii) Mutually exclusive events. If the occurrence of one of the events precludes the occurrence of all the others, then such a set of events is said to be mutually exclusive. Just as tossing a coin, either head comes up or the tail and both can't happen at the same time, i.e., these are two mutually exclusive cases.

(iii) Equally likely events. If one of the events cannot be expected to happen in preference to another then such events are said to be equally likely. For instance, in tossing a coin, the coming of the head or the tail is equally likely.

Then when a die* is thrown, the turning up of the six different faces of the die are exhaustive, mutually exclusive and equally likely.

(ii) **Odds in favour of an event.** If the number of ways favourable to an event A is m and the number of ways not favourable to A is n, then odds in favour of A = $m : n$ and odds against A = $n : m$.

(B) **Definition of probability.** If there are n exhaustive, mutually exclusive and equally likely cases of which m are favourable to an event A, then probability (p) of the happening of A is

$$P(A) = \frac{m}{n}.$$

If there are $n - m$ cases in which A will not happen (denoted by A'), the chance of A' not happening is q or $P(A')$ is (by)

$$q = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - p$$

i.e., $P(A') = 1 - P(A)$ or (i.e.) $P(A) + P(A') = 1$,

i.e., if an event is certain to happen then its probability is unity, while if it is certain not to happen, its probability is zero.

Obs. This definition of probability fails when
(i) number of outcomes is infinite (not exhaustive) and (ii) outcomes are not equally likely.

(C) **Statistical (or Empirical) definition of probability.** If in n trials, an event A happens m times, then the probability (p) of happening of A is given by

$$p = P(A) = \lim_{n \to \infty} \frac{m}{n}.$$

Example 18.1. *Find the chance of throwing (a) five, (b) an even number with an ordinary six-faced die.*

Solution. (a) There are six possible ways in which the die can fall and of these there is only one way of throwing 5. Thus the required chance = $\frac{1}{6}$.

(b) There are six possible ways in which the die can fall. Of these there are only 3 ways of getting 2, 4 or 6. Thus the required chance = $\frac{3}{6} = \frac{1}{2}$.

Example 18.2. *What is the chance that a boy born on the 1st of March will be born on Sunday?* (Madras 2003)

Solution. A leap year consists of 366 days, so that there are 52 full weeks (i.e. 364 days) and two extra days. These two days can be (i) Monday, Tuesday (ii) Tuesday, Wednesday, (iii) Wednesday, Thursday (iv) Thursday, Friday (v) Friday, Saturday (vi) Saturday, Sunday (vii) Sunday, Monday.

Of these 7 cases, the last two are favourable and hence the required probability = $\frac{2}{7}$.

Example 18.3. *A five figure number is formed by the digits 0, 1, 2, 3, 4 without repetition. Find the probability that the number formed is divisible by 4.*

Solution. The first digit can be arranged in 4 ways, out of which 0 will begin with zero.

$\therefore$ total number of 5-figure numbers formed = $4! - 3! = 96$.

These numbers formed will be divisible by 4 which will have two extreme right digits divisible by 4, i.e., numbers ending in 04, 12, 20, 24, 32, 40.

Now numbers ending in $04 = 3! = 6$, numbers ending in $12 = 3! - 2! = 4$,
numbers ending in $20 = 3! = 6$, numbers ending in $24 = 3! - 2! = 4$,
numbers ending in $32 = 3! - 2! = 4$, and numbers ending in $40 = 3! = 6$.

(The numbers having 12, 24, 32 in the extreme right are $3! - 2!$ since the number in question are to the extreme left are to be excluded).

* Die is a small cube. Dots 1, 2, 3, 4, 5, 6 are marked on its six faces. The outcome of throwing a die is the number of dots on its upper face.

$\therefore$ total number of favourable ways $= 6 + 6 + 6 + 4 + 4 + 6 = 32$.

Hence the required probability $= \dfrac{32}{56} = \dfrac{8}{26}$

Example 26.5. A bag contains 15 tickets numbered 1 to 15. Two tickets are drawn at random and arranged in ascending order (x_1, x_2). Find the probability of x_2 being 20.

Solution. Here total number of cases $= {}^{15}C_2$

If $x_2 = 20$, then the tickets x_1 and x_2 must be out of 19 tickets numbered 1 to 19. This can be done in ${}^{19}C_2$ ways.

Then x_2 must come out of the 15 tickets bearing 20 to 101 which can be done in ${}^{15}C_1$ ways.

$\therefore$ favourable number of cases $= {}^{19}C_2 \times {}^{15}C_1$

Hence the probability of x_2 being 20 $= \dfrac{{}^{19}C_2 \times {}^{15}C_1}{{}^{15}C_2} = \dfrac{114}{9120}$.

Example 26.6. A box contains 5 red and 10 black balls. If 6 balls are drawn at random, find the probability that the box contains 2 red and 4 black balls.

Solution. The number of ways in which 6 balls can be drawn out of 15 is ${}^{15}C_6$

The number of ways of drawing 2 red balls is ${}^5C_2$ and corresponding to each of these ${}^5C_2$ ways of drawing 2 red balls, there are ${}^{10}C_4$ ways of drawing 4 black balls.

$\therefore$ the total number of ways in which 2 red and 4 black balls can be drawn is ${}^5C_2 \times {}^{10}C_4$.

$\therefore$ the required probability $= \dfrac{{}^5C_2 \times {}^{10}C_4}{{}^{15}C_6} = \dfrac{100}{429}$

Example 26.7. A committee consisting of 3 students is to be formed from 3 groups, three from 2nd year and four from 3rd year. Three students are to be removed at random. What is the chance that (i) the three students belong to different classes, (ii) two belong to the same class and third to the different class, (iii) three belong to the same class? (V.T.U., 2002 S)

Solution. (i) The total number of ways of choosing 3 students out of 9 is ${}^9C_3$, i.e., 84.

A student can be removed from 2nd year students in 2 ways, from 3rd year in 3 ways and from 3rd year in 4 ways, so that the total number of ways of removing three students, one from each group is $2 \times 3 \times 4$.

Hence the required chance $= \dfrac{2 \times 3 \times 4}{{}^9C_3} = \dfrac{24}{84} = \dfrac{2}{7}$

(ii) The number of ways of removing two from 3rd year students and one from others

$= {}^4C_2 \times {}^5C_1$

The number of ways of removing two from 2nd year students and one from others

$= {}^3C_2 \times {}^6C_1$

The number of ways of removing 3 from 3rd year students and one from others

$= {}^4C_3 \times {}^5C_1$

$\therefore$ the total number of ways in which two students of the same class and third from the others may be removed

$= {}^4C_2 \times {}^5C_1 + {}^3C_2 \times {}^6C_1 + {}^4C_3 \times {}^5C_1 = 3 \times 18 + 20 = 74$.

Hence, the required chance $= \dfrac{74}{84}$.

(iii) Three students can be removed from 2nd year group in ${}^3C_3$, i.e. 1 way and from 3rd year group in ${}^4C_3$, i.e., 4 ways.

$\therefore$ the total number of ways in which three students belong to the same class $= 1 + 4 = 5$.

Hence the required chance $= \dfrac{5}{84}$.

Example 26.10. A box contains 5 tickets and 3 blanks; Ram draws 2 tickets and Shyam 3 tickets. Find the probability that a lottery in which there are 3 prizes and 5 blanks, compare the probability of Ram's success with that of Shyam's success.

Solution. Ram can draw a ticket in ${}^5C_2 = 5$ ways.

The number of cases in which Ram can get a prize is clearly 1.

$\therefore$ the probability of Ram's success $= \frac{1}{5}$.

Again Shyam can draw a ticket in ${}^8C_3 = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$ ways.

The number of ways in which Shyam get all blanks $= {}^3C_3 = \frac{3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = 1$

$\therefore$ the number of ways of getting a prize $= 56 - 1 = 55$.

Thus the probability of Shyam's success $= \frac{55}{56}$.

Hence Ram's probability of success : Shyam's probability of success $= \frac{1}{5} : \frac{55}{56} = 1 : 55$.

26.3 PROBABILITY AND SET NOTATIONS

(1) Random experiment. Experiments which are performed repeatedly under the same conditions and whose results cannot be predicted are known as random experiments, *e.g.*, Tossing a coin or rolling a die are random experiments.

(2) Sample space. The set of all possible outcomes of a random experiment is called sample space for that experiment and is denoted by S.

The elements of the sample space S are called the sample points.

e.g., On tossing a coin, the possible outcomes are the head (H) and the tail (T). Thus $S = \{H, T\}$.

(3) Event. The outcome of a random experiment is called an event. Thus every subset of a sample space S is an event.

The null set ϕ is also an event and is called an impossible event. Probability of an impossible event is zero, *i.e.*, $P(\phi) = 0$.

(4) Axioms

(i) The numerical value of probability lies between 0 and 1,

i.e., for any event A, $0 \le P(A) \le 1$.

(ii) The sum of probabilities of all sample events is unity, *i.e.*, $P(S) = 1$.

(iii) Probability of an event made of two or more sample events is the sum of their probabilities.

(5) Notations

(i) Probability of happening of events A or B is written as $P(A \cup B)$ or $P(A + B)$.

(ii) Probability of happening of both the events A and B is written as $P(AB)$ or $P(A \cap B)$.

(iii) Event A implies (or event A' is expressed as $A - B$.

(iv) Either A and B are mutually exclusive or is expressed as $A \cap B = \phi$.

(v) For any A events A and B,

$$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

Proof. From Fig. 26.1,

$$(A \cap \bar{B}) \cup (A \cap B) = A$$

$$\therefore \quad P[(A \cap \bar{B}) \cup (A \cap B)] = P(A)$$

or $$P(A \cap \bar{B}) + P(A \cap B) = P(A)$$

or $$P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

Similarly, $P(\bar{A} \cap B) = P(B) - P(A \cap B)$

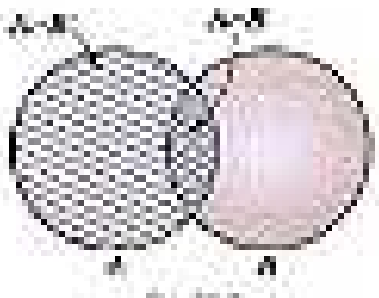


Fig. 26.1

2.8 ADDITION LAW OF PROBABILITY or THEOREM OF TOTAL PROBABILITY

(1) If the probability of an event A happening as a result of a trial is $P(A)$ and the probability of a mutually exclusive event B happening is $P(B)$, then the probability of either of the events happening as a result of the trial is $P(A + B)$ or $P(A \cup B) = P(A) + P(B)$.

Proof. Let n be the total number of equally likely cases and let n_1 be favourable to the event A and n_2 be favourable to the event B. Then the number of cases favourable to A or B is $n_1 + n_2$. Hence the probability of A or B happening as a result of the trial

$$= \frac{n_1 + n_2}{n} = \frac{n_1}{n} + \frac{n_2}{n} = P(A) + P(B).$$

(2) If A, B are any two events (not mutually exclusive), then

$$P(A + B) = P(A) + P(B) - P(AB)$$

or

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If the events A and B are any two events (not mutually exclusive), there are some outcomes which favour both A and B. If n_3 be their number, then these are included in both n_1 and n_2. Hence the total number of outcomes favouring either A or B or both is

$$n_1 + n_2 - n_3$$

Thus the probability of occurrence of A or B or both

$$= \frac{n_1}{n} + \frac{n_2}{n} - \frac{n_3}{n} = \frac{n_1}{n} + \frac{n_2}{n} - \frac{n_3}{n}$$

Hence $P(A + B) = P(A) + P(B) - P(AB)$

or $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Note. When A and B are mutually exclusive, $P(AB) = P(A \cap B) = 0$ and we get

$$P(A + B) = P(A \cup B) = P(A) + P(B)$$

In general, for a number of mutually exclusive events $A_1, A_2, \ldots, A_n$ we have

$$P(A_1 + A_2 + \ldots + A_n) = P(A_1 \cup A_2 \cup \ldots \cup A_n) = P(A_1) + P(A_2) + \ldots + P(A_n)$$

(3) If A, B, C are any three events, then

$$P(A + B + C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(CA) + P(ABC)$$

or

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

Proof. Using the above result for any two events, we have

$$P(A \cup B \cup C) = P[(A \cup B) \cup C]$$

$$= P(A \cup B) + P(C) - P[(A \cup B) \cap C]$$

$$= [P(A) + P(B) - P(A \cap B)] + P(C) - P[(A \cap C) \cup (B \cap C)]$$ (By distributive law)

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) + P(B \cap C) - P(A \cap B \cap C)$$

(\because $(A \cap C) \cap (B \cap C) = A \cap B \cap C$)

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$ (\because $A \cap C = C \cap A$)

Example 2.8(1). *In a race, the odds in favour of the four horses H_1, H_2, H_3, H_4 are 2 : 4, 1 : 5, 1 : 6, 1 : 7 respectively. Assuming that a dead heat is not possible, find the chance that one of them will win the race.*

Solution. Since it is not possible for all the horses to cover the same distance in the same time (a dead heat), the events are mutually exclusive.

If p_1, p_2, p_3, p_4 be the probabilities of winning of the horses H_1, H_2, H_3, H_4 respectively, then

$$p_1 = \frac{2}{2+4} = \frac{1}{3}$$ (\because Odds in favour of H_1 are 2 : 4)

and

$$p_2 = \frac{1}{6} \cdot p_1 = \frac{1}{6} \cdot \frac{1}{3} = \frac{1}{18}$$

Hence the chance that one of them wins $= p_1 + p_2 + p_3 + p_4$

$$= \frac{1}{3} + \frac{1}{18} + \frac{1}{7} + \frac{1}{8} = \frac{502}{1008}$$

Example 20.12. *A bag contains 5 white and 8 red balls. Find the probability of drawing two balls of the same colour.*

Solution. Two balls out of 13 can be drawn in $^{13}C_2$ ways which is the total number of outcomes.

Two white balls out of 5 can be drawn in $^5C_2$ ways. Thus the probability of drawing 2 white balls

$$= \frac{^5C_2}{^{13}C_2} = \frac{10}{78}$$

Similarly 2 red balls out of 8 can be drawn in $^8C_2$ ways. Thus the probability of drawing 2 red balls

$$= \frac{^8C_2}{^{13}C_2} = \frac{28}{78}$$

Hence the probability of drawing 2 balls of the same colour is (either both white or both red)

$$= \frac{10}{78} + \frac{28}{78} = \frac{38}{78}$$

Example 20.13. *Find the probability of drawing an ace or a black card from a deck of cards.*[*]

Solution. The probability of drawing an ace from a deck of 52 cards = 4/52.

Similarly the probability of drawing a card of spades = 13/52 and the probability of drawing an ace of spades = 1/52.

Since the two events (i.e., a card being an ace and a card being of spades) are not mutually exclusive, therefore, the probability of drawing an ace or a spade

$$= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{4}{13}$$

20.5 (L) INDEPENDENT EVENTS

Two events are said to be independent, if happening or failure of one does not affect the happening or failure of the other. Otherwise the events are said to be dependent.

For two dependent events A and B, the symbol $P(A/B)$ denotes the probability of occurrence of A, when B has already occurred. It is known as the conditional probability and is read as 'probability of A given B'.

(1) Multiplication law of probability or Theorem of compound probability. *If the probability of occurrence of A happening as a result of trial is $P(A)$ and after A has happened the probability of occurrence of B happening as a result of another trial i.e., conditional probability of B given A is $P(B/A)$, then the probability of both the events A and B happening as a result of the trials is $P(A \cap B)$ or $P(AB)$ where $P(AB) = P(A) \cdot P(B/A)$.*

Proof. Let n be the total number of outcomes in the first trial out of which n_1 are favourable to the event A so that $P(A) = n_1/n$.

Let n_2 be the total number of outcomes in the second trial of which m_1 are favourable to the event B so that $P(B/A) = m_2/n_2$.

Now each of the n outcomes can be associated with each of the n_2 outcomes. So the total number of outcomes in the combined trial is nn_2. Of these m_1n_2 are favourable to both the events A and B. Hence

$$P(AB) \text{ or } P(A \cap B) = \frac{m_1n_2}{nn_2} = P(A) \cdot P(B/A)$$

Similarly, the conditional probability of A given B is $P(A/B)$.

$\therefore$ $P(AB)$ or $P(A \cap B) = P(B) \cdot P(A/B)$

Thus $P(A \cap B) = P(A) \cdot P(B/A) = P(B) \cdot P(A/B)$

(2) When events A and B are independent, i.e., if the happening of B does not depend in any way on A has happened or not, then $P(B/A) = P(B)$ and $P(A/B) = P(A)$.

$\therefore$ $P(AB)$ or $P(A \cap B) = P(A) \cdot P(B)$.

In general, $P(A_1A_2 \ldots A_n)$ or $P(A_1 \cap A_2 \cap \ldots \cap A_n) = P(A_1) \cdot P(A_2/A_1) \ldots P(A_n/\ldots)$

[*] Each of a pack of cards consists of four suits i.e., Hearts, Diamonds, Spades and Clubs. Each suit has 13 cards : an Ace, a King, a Queen, a Jack and nine cards numbered 2, 3, 4, ..., 10. Hearts and Diamonds are red while Spades and Clubs are black.

Thus, if p_1, p_2 be the probabilities of happening of two independent events, then

(i) the probability that the first event happens and the second fails to happen is $p_1(1-p_2)$

(ii) the probability that both events fail to happen is $(1-p_1)(1-p_2)$

(iii) the probability that at least one of the events happens is

$1-(1-p_1)(1-p_2)$. This is sometimes known as their **cumulative probability**.

In general, if $p_1, p_2, p_3, \ldots, p_n$ are the chances of happening of n independent events, then their cumulative probability, i.e., the chance that at least one of the events will happen is

$$1-(1-p_1)(1-p_2)(1-p_3)\ldots(1-p_n)$$

Example 28.14. Two cards are drawn in succession from a pack of 52 cards. Find the chance that the first is a king and the second a queen if the first card is (i) replaced, (ii) not replaced.

Solution. (i) The probability of drawing a king $= \frac{4}{52} = \frac{1}{13}$

If the card is replaced, the pack will again have 52 cards so that the probability of drawing a queen is $\frac{1}{13}$.

The two events being independent, the probability of drawing both cards in succession $= \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$.

(ii) The probability of drawing a king $= \frac{1}{13}$

If the card is not replaced, the pack will have 51 cards only so that the chance of drawing a queen is $\frac{4}{51}$.

Hence the probability of drawing both cards $= \frac{1}{13} \times \frac{4}{51} = \frac{4}{663}$.

Example 28.15. A pair of dice is tossed twice. Find the probability of scoring 7 points (i) at first and then not (ii) at least once. (Kingston upon Hull Univ., 2000 S; N.E.W.U., 2004)

Solution. In a single toss of two dice, the sum 7 can be obtained in (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) i.e., in 6 ways, so that the probability of getting $7 = \frac{6}{36} = \frac{1}{6}$.

Also the probability of not getting $7 = 1 - \frac{1}{6} = \frac{5}{6}$.

(i) The probability of getting 7 in the first toss and not getting 7 in the second toss $= \frac{1}{6} \times \frac{5}{6} = \frac{5}{36}$.

Similarly, the probability of not getting 7 in the first toss and getting 7 in the second toss $= \frac{5}{6} \times \frac{1}{6} = \frac{5}{36}$.

Since these are mutually exclusive events, addition law of probability applies.

$\therefore$ required probability $= \frac{5}{36} + \frac{5}{36} = \frac{5}{18}$.

(ii) The probability of not getting 7 in either toss $= \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$

$\therefore$ the probability of getting 7 at least once $= 1 - \frac{25}{36} = \frac{11}{36}$.

(i) The probability of getting 7 twice $= \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$.

Example 28.16. There are two groups of subjects, one of which consists of 3 science and 3 engineering subjects, and the other consists of 2 science and 5 engineering subjects. An unbiased die is cast. If the number 3 or a number divisible by 3 appears, a subject is selected at random from the first group, otherwise the subject is selected at random from the second group. Find the probability that an engineering subject is selected ultimately.

Solution. Prob. of throwing up 3 or 6 $= \frac{2}{6} = \frac{1}{3}$

Prob. of selecting an engg. subject from first group $= \frac{3}{6}$

$\therefore$ Prob. of selecting an engg. subject from first group on throwing up 3 or 6

$$= \frac{1}{3} \times \frac{3}{6} = \frac{1}{6}$$

Now prob. of not getting 5 or 6 $= 1 - \frac{1}{3} - \frac{2}{9} = \frac{4}{9}$.

Prob. of selecting an even no. subject from second group $= \frac{6}{8}$

$\therefore$ prob. of selecting an even subject from second group on getting up 5 or 6

$$= \frac{2}{3} \times \frac{6}{8} = \frac{3}{12} \quad \text{...(ii)}$$

Thus the prob. of selecting an even subject

$$= \frac{1}{4} + \frac{3}{12} = \frac{13}{24}$$ [From (i) and (ii)]

Example 28.17. *A box A contains 2 white and 3 black balls. Another box B contains 5 white and 7 black balls. A ball is transferred from the box A to the box B. Then a ball is drawn from the box B. Find the probability that it is white.* (U.P.T.U., 2004)

Solution. The probability of drawing a white ball from box B will depend on whether the transferred ball is black or white.

If a black ball is transferred, its probability is $\frac{3}{5}$. Then, there are now 5 white and 8 black balls in the box B. Thus the probability of drawing white ball from box B is $\frac{5}{13}$.

Thus the probability of drawing a white ball from box B, if the transferred ball is black

$$= \frac{3}{5} \times \frac{5}{13} = \frac{15}{65}$$

Similarly the probability of drawing a white ball from box B, if the transferred ball is white

$$= \frac{2}{5} \times \frac{6}{13} = \frac{12}{65}$$

Hence required probability $= \frac{15}{65} + \frac{12}{65} = \frac{27}{65}$.

Example 28.18. *(a) A coin is tossed until a head appears for the first time. What is the probability that the number of required tosses is odd?* (Mumbai, 2003)

(b) Two persons A and B have an unbiased coin which is tossed by turn. It is understood that the first who gets the head wins. If A starts the game, find their respective chances of winning. (Mumbai, 2000)

Solution. (a) Let p be the probability of getting a head and q the probability of getting a tail in a single toss, so that $p + q = 1$.

Then probability of getting head on an odd toss

= Probability of getting head in the 1st toss

+ Probability of getting head in the 3rd toss

+ Probability of getting head in the 5th toss + ... to ∞

$$= p + qqp + (qq)^2 p + \ldots \infty$$

$$= p\left(1 + q^2 + q^4 + \ldots \infty\right) = p \cdot \frac{1}{1 - q^2} \quad (q < 1)$$

$$= p \cdot \frac{1}{(1 - q)(1 + q)} = p \cdot \frac{1}{p(1 + q)} = \frac{1}{1 + q}$$

(b) Probability of getting a head $= \frac{1}{2}$. Then A can win in 1st, 3rd, 5th, ... throw.

$\therefore$ the chances of A's winning $= \frac{1}{2} + \left(\frac{1}{2}\right)^2 \frac{1}{2} + \left(\frac{1}{2}\right)^4 \frac{1}{2} + \left(\frac{1}{2}\right)^6 \frac{1}{2} + \ldots$

$$= \frac{1/2}{1 - 1/4} = \frac{1}{2} \cdot \frac{4}{3} = \frac{2}{3}.$$

Hence the chances of his winning $= 1 - \frac{2}{3} = \frac{1}{3}$.

Example 30.16 Two cards are selected at random from 10 cards numbered 1 to 10. Find the probability of the sum being odd if

(i) the first card is not replaced

(ii) the first card is replaced after the selection and

(iii) the first card is destroyed after the selection. (UPSEAT, 2001)

Solution. (i) Two cards out of 10 can be selected in ${}^{10}C_2 = 45$ ways. The sum would be even number and the other number is even. The two digit odd numbers (1, 3, 5, ..., 9) and 5 even numbers (2, 4, 6, 8, 10), an odd and an even number be chosen in $5 \times 5 = 25$ ways.

Thus $$p = \frac{25}{45} = \frac{5}{9}.$$

(ii) Two cards out of 10 can be selected one after the other with replacement in $10 \times 9 = 90$ ways.
An odd number is selected in $5 \times 5 = 25$ ways and an even number in $5 \times 5 = 25$ ways

Thus $$p = \frac{25 + 25}{90} = \frac{5}{9}$$

(iii) Two cards can be selected one after the other without replacement in $10 \times 10 = 100$ ways.
An odd number is selected in $5 \times 5 = 25$ ways and an even number in $5 \times 5 = 25$ ways.

Thus $$p = \frac{25 + 25}{100} = \frac{1}{2}$$

Example 30.17 Given $P(A) = 1/4$, $P(B) = 1/3$ and $P(A \cup B) = 1/2$, evaluate $P(A/B)$, $P(B/A)$, $P(A' \cap B)$ and $P(A/B')$.

Solution. (i) Since $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\therefore \quad \frac{1}{2} = \frac{1}{4} + \frac{1}{3} - P(A \cap B) \text{ or } P(A \cap B) = \frac{1}{12}$$

Thus $$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/12}{1/3} = \frac{1}{4}$$

(ii) $$P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1/12}{1/4} = \frac{1}{3}$$

(iii) $$P(A \cap B') = P(A) - P(A \cap B) = \frac{1}{4} - \frac{1}{12} = \frac{1}{6}$$

(iv) $$P(A/B') = \frac{P(A \cap B')}{P(B')} = \frac{1/6}{1 - P(B)} = \frac{1/6}{1 - 1/3} = \frac{1}{4}$$

Example 30.18 The probabilities of A, B and C solving a problem are $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$ respectively. If all of them try to solve the problem, find the probability that

Finally, prob. that all the three are favourable $= \frac{5}{7} \times \frac{3}{7} \times \frac{3}{7} = \frac{45}{343}$

Since they are mutually exclusive events, the required prob.

$$= \frac{80}{343} + \frac{45}{343} + \frac{24}{343} + \frac{48}{343} + \frac{245}{343}.$$

Example 28.22. *Three boys A, B and C are given a target to hit. A is thrice as likely to hit the target as B and B is twice as likely to hit the target as C. If the probability that C can hit the target is $\frac{1}{4}$, find the probability that (i) at least two of them hit the target, (ii) all of them hit the target.* (U.P.T.U. (B.C.A.), 2002; Madras, 2006 S)

Solution. Prob. of A hitting the target $= \frac{3}{7}$, prob. of B hitting the target $= \frac{2}{7}$

Prob. of C hitting the target $= \frac{1}{4}$.

(i) In order that two shots may hit the target, the following cases must be considered :

p_1 = Chance that A and B hit and C fails to hit $= \frac{3}{7} \times \frac{2}{7} \times \left(1 - \frac{1}{4}\right) = \frac{6}{196}$

p_2 = Chance that B and C hit and A fails to hit $= \frac{2}{7} \times \frac{1}{4} \times \left(1 - \frac{3}{7}\right) = \frac{8}{196}$

p_3 = Chance that C and A hit and B fails to hit $= \frac{3}{7} \times \frac{1}{4} \times \left(1 - \frac{2}{7}\right) = \frac{15}{196}$

Since these are mutually exclusive events, the probability that any 2 shots hit

$$= p_1 + p_2 + p_3 = \frac{6}{196} + \frac{8}{196} + \frac{15}{196} = 0.22.$$

(ii) In order that at least two shots may hit the target, we must also consider the case of A, B, C hitting the target in addition to the three cases of (i) for which

p_4 = chance that A, B, C all hit $= \frac{3}{7} \times \frac{2}{7} \times \frac{1}{4} = \frac{6}{196}$

Since all these are mutually exclusive events, the probability of at least two shots hit

$$= p_1 + p_2 + p_3 + p_4 = \frac{6}{196} + \frac{8}{196} + \frac{15}{196} + \frac{6}{196} = 0.22.$$

Example 28.23. *A problem in mathematics is given to three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved.* (U.P.T.U., 2004)

Solution. The probability that A can solve the problem is $\frac{1}{2}$.

The probability that A cannot solve the problem is $1 - \frac{1}{2}$.

Similarly the probabilities that B and C cannot solve the problem are $1 - \frac{1}{3}$ and $1 - \frac{1}{4}$.

$\therefore$ the probability that A, B and C cannot solve the problem is $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)$.

Hence the probability that the problem will be solved, i.e., at least one student will solve it

$$= 1 - \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right) = \frac{3}{4}.$$

Example 28.24. *The students in a class are selected at random, one after the other, for an oral test. Find the probability p that the boys and girls in the class alternate if*

(i) the class consists of 3 boys and 2 girls,

(ii) the class consists of 3 boys and 3 girls. (U.P.T.U., 2001)

Solution. (i) As there are 2 students in the class, the first examined must be a boy.

$\therefore$ prob. that first is a boy $= \frac{4}{7}$

Then the prob. that the second is a girl $= \frac{3}{6}$

$\therefore$ prob. of the next boy $= \frac{3}{5}$

Similarly the prob. that the fourth is a girl $= \frac{2}{4}$,

the prob. that the fifth is a boy $= \frac{2}{3}$,

the prob. that the sixth is a girl $= \frac{1}{2}$

and the last is a boy $= \frac{1}{1}$

Thus $p = \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} \times \frac{1}{2} \times \frac{1}{1} \times \frac{1}{35}$

(iii) If the first student is a boy and the first student is a girl are two mutually exclusive cases. If the first student is a boy, then the probability p_1 that the students alternate is

$$p_1 = \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5} \times \frac{2}{4} \times \frac{2}{3} \times \frac{1}{2} \times \frac{1}{1} = \frac{1}{35}$$

If the first student is a girl, then the probability p_2 that the students alternate is

$$p_2 = \frac{3}{7} \times \frac{3}{6} \times \frac{4}{5} \times \frac{2}{4} \times \frac{2}{3} \times \frac{1}{2} \times \frac{1}{1} = \frac{1}{35}$$

Thus the required prob. $p = p_1 + p_2 = \frac{1}{35} + \frac{1}{35} = \frac{1}{35}$

Example 26.35. *(Engineering problem) A and B throw alternately with a pair of dice. A wins if he throws a total of 6 before B throws a total of 7, and B wins if he throws a total of 7 before A throws a total of 6. If A begins, find his chance of winning.*

Madras, 2000 (A.U.C.E., 2003)

Solution. The various throws are tabulated as follows: (1, 5), (2, 4), (3, 3), (4, 2), (5, 1), (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1).

The probability of A's throwing 6 with 2 dice is $\frac{5}{36}$.

$\therefore$ the probability of A's not throwing 6 is $31/36$.

Similarly the probability of B's throwing 7 with 2 dice is $\frac{1}{6}$.

$\therefore$ the probability of B's not throwing 7 is $5/6$.

Now A can win if he throws 6 in the first, third, fifth, seventh, etc. throws.

$\therefore$ the chance of A's winning

$$= \frac{5}{36} + \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} + \frac{31}{36} \times \frac{5}{6} \times \frac{31}{36} \times \frac{5}{6} \times \frac{5}{36} + \cdots$$

$$= \frac{5}{36}\left[1 + \left(\frac{31}{36} \times \frac{5}{6}\right) + \left(\frac{31}{36} \times \frac{5}{6}\right)^2 + \left(\frac{31}{36} \times \frac{5}{6}\right)^3 + \cdots\right]$$

$$= \frac{5}{36} \cdot \frac{1}{1 - (31/36) \times (5/6)} = \frac{5}{36} \times \frac{36 \times 6}{61} = \frac{30}{61}$$

EXERCISE 26.5

1. A [illegible] P(A) = 2/3, P(A') = 1/3 and P(B) = 3/5, P(B') = 2/5. Find P(A ∩ B). *(Bombay, 2002)*

(ii) Let A and B be two events such that P(A) = 1/2, P(B) = 1/3 and P(A ∩ B') = 1/4. Find P(A/(A ∪ B)) and P(B/(A ∪ B)). *(Madras, 2002) (A.U.C.E., 2002)*

8. In a large shipment of computer chips, suppose that the probability of a chip being defective is 0.05. If a sample of 100 chips is taken, what is the probability that the number of defective chips in the sample is between 3 and 7 (inclusive)?
9. Compare the Poisson distribution with the binomial distribution. Under what conditions can the Poisson distribution be used to approximate the binomial distribution?
10. A certain type of lightbulb has a lifetime that is exponentially distributed with a mean lifetime of 1000 hours. If a lightbulb has already been burning for 500 hours, what is the probability that it will last at least another 1000 hours?
11. In a certain city, the daily consumption of electric power, in millions of kilowatt-hours, can be treated as a random variable having a gamma distribution with $\alpha = 3$ and $\beta = 2$. If the power plant of this city has a daily capacity of 12 million kilowatt-hours, what is the probability that this capacity will be insufficient on a given day?
12. Let Z be a standard normal random variable. Find the value z_0 such that $P(|Z| > z_0) = 0.05$.
13. Let X be a random variable with a uniform distribution on the interval $[0, 10]$. Find the probability that X is within 2 units of its mean. That is, find $P(|X - \mu| \leq 2)$, where $\mu = E(X)$.
14. Let X be a random variable with a gamma distribution with parameters $\alpha = 3$ and $\beta = 2$. Find the probability that X is within 1 standard deviation of its mean. That is, find $P(|X - \mu| \leq \sigma)$, where $\mu = E(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.
15. Let X be a random variable with a Poisson distribution with parameter $\lambda = 4$. Find the probability that X is within 2 standard deviations of its mean. That is, find $P(|X - \mu| \leq 2\sigma)$, where $\mu = E(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.
16. Let X be a random variable with a binomial distribution with parameters $n = 10$ and $p = 0.5$. Find the probability that X is within 1 standard deviation of its mean. That is, find $P(|X - \mu| \leq \sigma)$, where $\mu = E(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.
17. Let X be a random variable with a geometric distribution with parameter $p = 0.2$. Find the probability that X is within 2 standard deviations of its mean. That is, find $P(|X - \mu| \leq 2\sigma)$, where $\mu = E(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.
18. Let X be a random variable with a hypergeometric distribution with parameters $N = 50$, $M = 20$, and $n = 10$. Find the probability that X is within 1 standard deviation of its mean. That is, find $P(|X - \mu| \leq \sigma)$, where $\mu = E(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.
19. Let X be a random variable with a negative binomial distribution with parameters $r = 3$ and $p = 0.4$. Find the probability that X is within 2 standard deviations of its mean. That is, find $P(|X - \mu| \leq 2\sigma)$, where $\mu = E(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.
20. Let X be a random variable with a uniform distribution on the interval $[0, 10]$. Find the probability that X is within 1 standard deviation of its mean. That is, find $P(|X - \mu| \leq \sigma)$, where $\mu = E(X)$ and $\sigma = \sqrt{\mathrm{Var}(X)}$.

1.15 BAYE'S THEOREM

If an event A corresponds to a number of exhaustive events $B_1, B_2, \ldots, B_n$ and $P(B_i)$ and $P(A|B_i)$ are given, then

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum P(B_i)P(A|B_i)}$$

Proof. By the multiplication law of probability,

$$P(AB_i) = P(A)P(B_i|A) = P(B_i)P(A|B_i) \quad \ldots(1)$$

$$\therefore \quad P(B_i|A) = \frac{P(B_i)P(A|B_i)}{P(A)} \quad \ldots(2)$$

Since the event A corresponds to $B_1, B_2, \ldots, B_n$ we have by the addition law of probability,

$$P(A) = P(AB_1) + P(AB_2) + \ldots + P(AB_n) = \sum P(AB_i) = \sum P(B_i)P(A|B_i) \quad \text{[By (1)]}$$

Hence from (2), we have $P(B_i|A) = \dfrac{P(B_i)P(A|B_i)}{\sum P(B_i)P(A|B_i)}$

which is known as the theorem of inverse probability.

Note: The probabilities $P(B_i)$, $i = 1, 2, \ldots, n$ are called prior probabilities because they exist before we get any information from the experiment.

The probabilities $P(B_i|A)$, $i = 1, 2, \ldots, n$ are called posterior probabilities because they are found after the experiment results are known.

Example 1.38. *There are three urns M_1, M_2 and M_3 containing electrical items. Of their respective output 5%, 4% and 2% of items are faulty. On a certain day, M_1 has produced 25% of the total output, M_2 has produced 30% and M_3 the remainder. An item selected at random is found to be faulty. What are the chances that it was produced by the second machine? Which machine output?*

Solution. Let the event of drawing a faulty item from any of the machines be A, and the event that an item drawn at random was produced by M_i be B_i. We have to find $P(B_2|A)$ for which we proceed as follows :

	M_1	M_2	M_3	Details
$P(B_i)$	0.25	0.30	0.45	$\sum P(B_i) = 1$
$P(A\|B_i)$	0.05	0.04	0.02	
$P(B_i)P(A\|B_i)$	0.0125	0.012	0.009	$\sum P(B_i)P(A\|B_i) = 0.0335$
$P(B_i\|A)$	0.3731	0.3582	0.2687	by Baye's theorem

The highest output is from M_1, the required probability $= 0.012/0.0335 = 0.358$

Example 1.39. *There are three bags, first containing 2 white, 3 red, 5 green balls; second 4 white, 1 red, 3 green balls and third 3 white, 2 red, 4 green balls. Two balls are drawn from a bag chosen at random. These are found to be one white and one red. Find the probability that the balls so drawn came from the second bag.*

[A.M.I.E.T.E., 2000]

Solution. Let B_1, B_2, B_3 denote the first, second, third bags chosen and A : the two balls are white and red.

Now $$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}$$

$P(AB_1)$ = P (a white and a red ball are drawn from first bag)

$$= {}^2C_1 \times {}^3C_1 / {}^{10}C_2 = \frac{2}{15}$$

Similarly, $$P(A|B_2) = {}^4C_1 \times {}^3C_1 / {}^8C_2 = \frac{3}{7}, \quad P(A|B_3) = {}^3C_1 \times {}^2C_1 / {}^9C_2 = \frac{1}{6}$$

By Baye's theorem, $$P(B_2|A) = \frac{P(B_2)P(A|B_2)}{P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)}$$

$$= \frac{\frac{1}{3} \cdot \frac{3}{7}}{\frac{1}{3} \cdot \frac{2}{15} + \frac{1}{3} \cdot \frac{3}{7} + \frac{1}{3} \cdot \frac{1}{6}} = \frac{6}{11}$$

PROBLEMS 20.1

1. In a certain college, 4% of the boys and 1% of the girls are taller than 1.8 m. Further, 60% of the students are girls. If a student is selected at random and is found to be taller than 1.8 m, what is the probability that the student is a girl?
2. In a bolt factory, machines A, B and C manufacture, respectively, 25%, 35% and 40% of the total output. Of their output, 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by machines A, B and C? (A: 5/13, B: 28/65, C: 8/13)
3. In a bolt factory, there are two machines A and B. A is manufacturing 65% (by) and 35% of the total output respectively. Of their outputs 1%, 2% and 3% are the ones which are defective bolts. A bolt is chosen at random from the factory's production and it is found defective. What is the probability that this bolt was manufactured by machine A or machine B? (Madras, 2002) (A: 13/27, B: 14/27)
4. There are two bags, one bag contains 3 white and 2 green balls (3 white, 2 red) and the other bag contains 2 white and 4 green balls (2 white, 4 red). One ball is drawn from an unknown bag and is found to be white. What is the probability that the ball was drawn from the first bag? (Kurukshetra, 2002)

20.3 RANDOM VARIABLE

If a real variable X is associated with the events of a random experiment, then since the values which X takes depend on chance, it is called a random variable or a stochastic variable or simply a variate. For instance, if a random experiment E consists of tossing a pair of dice, the sum X of the two numbers which turn up have the values $2, 3, 4, \ldots, 12$ depending on chance. Thus X is the random variable. It is a function whose values are real numbers and depend on chance.

If in a random experiment, the event corresponding to a number a occurs, then the corresponding random variable X is said to assume the value a and the probability of the event is denoted by $P(X = a)$. Similarly the probability of the event X is any integer value in the interval $a \le X \le b$ is denoted by $P(a \le X \le b)$. The probability of the event X is less than or equal to a is $P(X \le a)$.

If a random variable takes a finite set of values, it is called a discrete variate. On the other hand, if it assumes an infinite number of unaccountable values, it is called a continuous variate.

20.4 (1) DISCRETE PROBABILITY DISTRIBUTION

Suppose a discrete variate X is the outcome of some experiment. If the probability that X takes the values x_i is p_i, then

$$P(X = x_i) = p_i = p(x_i) \quad (i = 1, 2, \ldots)$$

where $p(x_i) \ge 0$ for all values of i and $\sum p(x_i) = 1$.

The set of values x_i with their probabilities p_i constitute a **discrete probability distribution** of the discrete variate X.

For example, the discrete probability distribution for X, the sum of the numbers which turn up on tossing a pair of dice is given by the following table:

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
$p(x_i)$	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

(a) There are $6 \times 6 = 36$ equally likely outcomes and therefore, each has the probability $1/36$. We have $X = 2$ for one outcome, *viz.* (1, 1); $X = 3$ for two outcomes (1, 2) and (2, 1); $X = 4$ for three outcomes (1, 3), (2, 2) and (3, 1) and so on.

(2) Distribution function. *The distribution function* $F(x)$ of the discrete variate X is defined by

$$F(x) = P(X \le x) = \sum_{x_i \le x} p(x_i)$$ where x is any integer. The graph of $F(x)$ will be

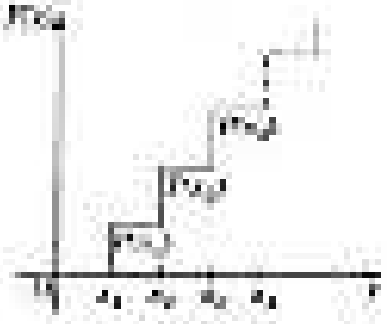


Fig. 20.2

stair-step form (Fig. 20.2). The distribution function is also sometimes called cumulative distribution function.

Example 30.29. A die is rolled three times. A success is getting a 5 or 6 on the first roll. Find the probability of 0, 1, 2 and 3 successes. (U.P.T.U., 2003)

Solution. Probability of a success $= \frac{2}{6} = \frac{1}{3}$, Probability of failure $= 1 - \frac{1}{3} = \frac{2}{3}$

$\therefore$ prob. of no success = Prob. of all 3 failures $= \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{8}{27}$

Probability of one success and 2 failures $= 3p_1 = \frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$

Probability of two successes and one failure $= 3p_2 = \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$

Probability of three successes $= \frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{27}$

Now	$x_i = 0$	1	2	3
	$p_i = 8/27$	4/9	2/9	1/27

$\therefore$ mean $\mu = \Sigma x_i p_i = 0 + \frac{4}{9} + \frac{4}{9} + \frac{1}{9} = 1$

Also $\Sigma x_i p_i^2 = 0 + \frac{4}{9} + \frac{8}{27} + \frac{3}{27} = \frac{5}{3}$

$\therefore$ variance $\sigma^2 = \Sigma x_i p_i^2 - \mu^2 = \frac{5}{3} - 1 = \frac{2}{3}$

Example 30.30. The probability density function of a variate X is

X	0	1	2	3	4	5	6
$p(X)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

(i) Find $P(X < 6)$, $P(X \geq 5)$, $P(3 < X < 6)$. (U.P.T.U., 2003)

(ii) What will be the minimum value of k so that $P(X \leq 2) > 0.3$?

Solution. (i) If X is a random variable, then

$$\sum_{i=0}^{6} p(x_i) = 1 \text{ i.e., } k + 3k + 5k + 7k + 9k + 11k + 13k = 1 \text{ or } k = 1/49.$$

$\therefore$ $P(X < 4) = k + 3k + 5k + 7k = 16k = 16/49.$

$P(X \geq 5) = 11k + 13k = 24k = 24/49.$

$P(3 < X \leq 6) = 9k + 11k + 13k = 33k = 33/49.$

(ii) $P(X \leq 2) = k + 3k + 5k = 9k \geq 0.3$ or $k \geq 1/30$

The minimum value of $k = 1/30$.

Example 30.31. A random variable X has the following probability function :

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

(i) Find the value of k. (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$

(iii) $P(0 < X < 5)$ (U.P.T.U., 2005; A.M.I.E., 2000)

Solution. (i) If X is a random variable, then

$$\sum_{i=0}^{7} p(x_i) = 1, \text{ i.e., } 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$$

i.e., $10k^2 + 9k - 1 = 0$ i.e., $(10k - 1)(k + 1) = 0$ i.e., $k = \frac{1}{10}$

(ii) $P(X < 6) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$

$$= 0 + k + 2k + 2k + 3k + k^2 = 8k + k^2 = \frac{8}{10} + \frac{1}{100} = \frac{81}{100}$$

$$P(X \geq 1) = P(X = 1) + P(X = 2) + P(X = 3) + \ldots + k = \frac{k}{120} = \frac{1}{20} = \frac{19}{380}$$

(iv) $P(1 \leq X \leq 3) = P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$

$$= 0 + \frac{2}{20} + \frac{3}{20} + \frac{4}{20} + \frac{5}{20} = \frac{14}{20} = \frac{7}{10}$$

23.3 (I) CONTINUOUS PROBABILITY DISTRIBUTION

When a variate X takes every value in an interval, it gives rise to *continuous distribution* of X. The distributions defined by the variates like heights of students and weights are continuous distributions.

A major conceptual difference, however, exists between discrete and continuous probabilities. When thinking in discrete terms, the probability associated with an event is meaningful. With continuous events, however, where the number of events is infinitely large, the probability that a specific event will occur is practically zero. For this reason, continuous probability statements must be worded somewhat differently from discrete ones. Instead of finding the probability that x equals some value, we find the probability of x falling in a small interval.

Thus the probability distribution of a continuous variate x is defined by a function $f(x)$ such that the probability of the variate x falling in the small interval $x - \frac{1}{2}dx$ to $x + \frac{1}{2}dx$ is $f(x)dx$. Symbolically it can be expressed as $P\left(x - \frac{1}{2}dx \leq x \leq x + \frac{1}{2}dx\right) = f(x)dx$. Then $f(x)$ is called the probability density function and the continuous curve $y = f(x)$ is called the probability curve.

The range of the variable x can be finite or infinite. But even when the range is finite, it is convenient to consider it as infinite by supposing the density function to be zero outside the given range. Thus if $f(x) = \phi(x)$ be the density function assumed for the variate x in the interval (a, b), then it can be written as

$$f(x) = 0, \quad x < a$$
$$= \phi(x), \quad a < x < b$$
$$= 0, \quad x > b.$$

The density function $f(x)$ is always positive and $\int_{-\infty}^{\infty} f(x)dx = 1$ i.e., the total area under the probability curve and the ordinates at the points corresponding to the extreme values is unity. This corresponds to the fact that the total probability of happening of an event is unity.

(2) Distribution function

$$(i)\ F(x) = P(X < x) = \int_{-\infty}^{x} f(x)\,dx.$$

Here $F(x)$ is defined as the **cumulative distribution function** or simply the **distribution function** of the continuous variate X. It is the probability that the value of the variate X will be $< x$. The graph of $F(x)$ for this case is as shown in Fig. 23.3(b).

The distribution function $F(x)$ has the following properties :

(i) $F(x) = f(x) \geq 0$, so that $F(x)$ is a non-decreasing function.

(ii) $F(-\infty) = 0$; (iii) $F(\infty) = 1$.

(iv) $P(a \leq x \leq b) = \int_a^b f(x)dx = \int_{-\infty}^{b} f(x)dx - \int_{-\infty}^{a} f(x)dx = F(b) - F(a)$.

Example 23.20. (i) If the function $f(x)$ is defined as follows, find the value of k if $f(x)$ is a density function :

$$f(x) = ke^{-kx}, \quad x \geq 0$$
$$= 0, \quad x < 0$$

(ii) For what choice of k does $f(x)$ represent the probability density of a random variable X which can take all positive values ?

(iii) Also find the cumulative probability function $F(x)$.

Solution. (i) $f(x)$ is a density if $f(x) \geq 0$ for every x in $(0, \infty)$ and

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{0} 0\,dx + \int_{0}^{\infty} ke^{-kx}\,dx = 1$$

Hence the function $f(x)$ satisfies the requirements for a density function.

(ii) Required probability $= P(1 \le x \le 2) = \int_1^2 e^{-x}\,dx = e^{-1} - e^{-2} = 0.368 - 0.135 = 0.233$.

This probability is equal to the shaded area in Fig. 3.8.3(a).

(iii) Cumulative probability function $F(2)$

$$\int_{-\infty}^{2} f(x)\,dx = \int_{-\infty}^{0} 0\,dx + \int_{0}^{2} e^{-x}\,dx = 1 - e^{-2} = 1 - 0.135 = 0.865$$

which is shown in Fig. 3.8.3(b).

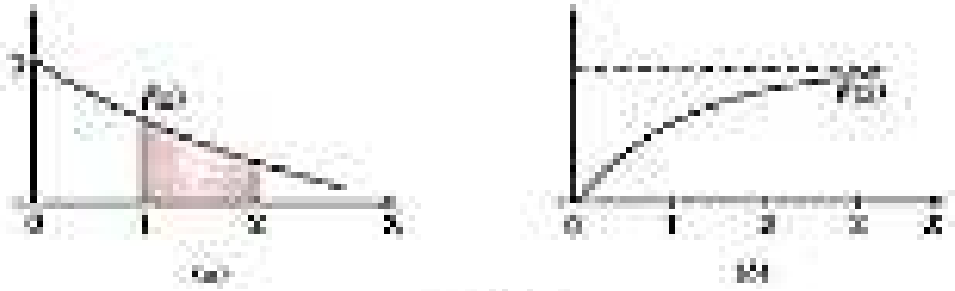


Fig. 3.8.3

Unit 3.8.3 (1) EXPECTATION

The mean value (or) of the probability distribution of a variate X is commonly known as its expectation and is denoted by $E(X)$. If $f(x)$ is the probability density function of the variate X, then

$$\sum_x x\,f(x) \qquad \text{(discrete distribution)}$$

or

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx \qquad \text{(continuous distribution)}$$

In general, expectation of any function $\phi(x)$ is given by

$$E[\phi(x)] = \sum_x \phi(x) f(x) \qquad \text{(discrete distribution)}$$

or

$$E[\phi(x)] = \int_{-\infty}^{\infty} \phi(x) f(x)\,dx \qquad \text{(continuous distribution)}$$

(2) Variance of a distribution is given by

$$\sigma^2 = \sum_x (x - \mu)^2 f(x) \qquad \text{(discrete distribution)}$$

or

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\,dx \qquad \text{(continuous distribution)}$$

where σ is the standard deviation of the distribution.

(3) The rth moment about the mean denoted by μ_r is defined by

$$\mu_r = \sum_x (x - \mu)^r f(x) \qquad \text{(discrete distribution)}$$

or

$$\mu_r = \int_{-\infty}^{\infty} (x - \mu)^r f(x)\,dx \qquad \text{(continuous distribution)}$$

(4) Mean deviation about the mean is given by

$$\sum |x - \mu| f(x) \qquad \text{(discrete distribution)}$$

or by

$$\int_{-\infty}^{\infty} |x - \mu| f(x)\,dx \qquad \text{(continuous distribution)}$$

Example 3.8.1 In a factory, rivets are produced of which 10% are defective. Find (i) the expectation and variance of the number of rivets in a box of 400 rivets.

Solution: Let $x_1, x_2, \ldots, x_n$ be the variables representing the numbers on the first, second, ..., nth ticket. The probability of drawing a ticket out of n tickets being in each case $\frac{1}{n}$, we have

$$E(x_1) = 1\cdot\frac{1}{n} + 2\cdot\frac{1}{n} + 3\cdot\frac{1}{n} + \ldots + n\cdot\frac{1}{n} = \frac{1}{n}(1 + 2 + \ldots + n)$$

$\therefore$ expected value of the sum of the numbers on the tickets drawn

$$= E(x_1 + x_2 + \ldots + x_n) = E(x_1) + E(x_2) + \ldots + E(x_n)$$

$$= n\,E(x_1) = n\cdot\frac{1}{2}n(n+1) = \frac{n}{2}(n+1).$$

Example 28.33. *A continuous random variable has probability density function $f(x)$ given by*

$$f(x) = Ax \quad (0 \le x \le 2)$$
$$= 2A \quad (2 \le x \le 4)$$
$$= -Ax + 6A \quad (4 \le x \le 6)$$

Find A and hence obtain $f(x)$. (U.P.T.U. 2003)

Solution. Since the total probability is unity

$$\int_0^6 f(x)\,dx = 1$$

i.e.,
$$\int_0^2 Ax\,dx + \int_2^4 2A\,dx + \int_4^6 (-Ax + 6A)\,dx = 1$$

or
$$A\left|\frac{x^2}{2}\right|_0^2 + 2A\left|x\right|_2^4 + \left|-\frac{Ax^2}{2} + 6Ax\right|_4^6 = 1$$

or
$$2A + 4A + (-18A + 12A) = 1 \text{ i.e., } A = \frac{1}{6}.$$

$$\text{Mean of } X = \int_0^6 x\,f(x)\,dx$$

$$= \int_0^2 Ax^2\,dx + \int_2^4 2Ax\,dx + \int_4^6 (-Ax^2 + 6Ax)\,dx$$

$$= A\left|\frac{x^3}{3}\right|_0^2 + 2A\left|\frac{x^2}{2}\right|_2^4 + A\left|-\frac{x^3}{3} + 3x^2\right|_4^6$$

$$= A\left(\frac{8}{3} + 24 - \frac{8}{3} - 16 - 72 + 108 + 48 - 108\right) = \frac{1}{6}(64) = 11.$$

Example 28.34. *A random variable X has the probability distribution*

x	-2	6	9
$P(X = x)$	$1/6$	$1/2$	$1/3$

Find $E(X)$ and $E(X^2)$. Hence evaluate $E(2X + 3)^2$.

Solution,
$$E(X) = \sum x\,P(x) = -2\times\frac{1}{6} + 6\times\frac{1}{2} + 9\times\frac{1}{3} = 5.$$

$$E(X^2) = 4\times\frac{1}{6} + 36\times\frac{1}{2} + 81\times\frac{1}{3} = 47$$

$$\therefore E(2X + 3)^2 = E(4X^2 + 12X + 9) = 4E(X^2) + 12E(X) + 9$$
$$= 4(47) + 12(5) + 9 = 257.$$

Example 28.35. *The frequency distribution of a continuous variable is given by $f(x) = x^2$, $0 \le x \le 3$ and $f(x) = 6x - x^2$, $3 \le x \le 6$. Calculate the standard deviation and also the mean deviation about the mean.*

Solution. Total frequency $N = \int_0^1 x^2\,dx + \int_1^2 (2 - x^2)\,dx = \frac{1}{3} + \frac{4}{3} - \frac{7}{3}$

$\therefore$ μ_1' about the origin $= \frac{3}{N}\left[\int_0^1 x \cdot x^2\,dx + \int_1^2 x(2 - x^2)\,dx\right]$

$= 2\left[\left|\frac{x^4}{4}\right|_0^1 + \left|x^2 \cdot \frac{(2 - x^2)}{-2}\right|_1^2 - \left|\frac{(2 - x^2)}{-20}\right|_1^2\right] = 2\left[\frac{1}{4} + \frac{1}{4} + \frac{1}{20}\right] = 1$

μ_2' about the origin $= \frac{1}{N}\left[\int_0^1 x^2 \cdot x^2\,dx + \int_1^2 x^2(2 - x^2)\,dx\right]$

$= 2\left[\left|\frac{x^5}{5}\right|_0^1 + \left|x^2 \cdot \frac{(2 - x^2)}{-2}\right|_1^2 + \frac{1}{2}\int_1^2 x(2 - x^2)\,dx\right]$

$= 2\left\{\frac{1}{5} + \frac{3}{4} + \frac{1}{2}\left[1 + \frac{1}{20}\right]\right\} = \frac{16}{10}$

Hence $\sigma^2 = \mu_2 = \mu_2' - \mu_1'^2 = \frac{1}{12}$

i.e., standard deviation $\sigma = \frac{1}{\sqrt{12}}$

Mean deviation about the mean

$= \frac{1}{N}\left\{\int_0^1 |x - 1|x^2\,dx + \int_1^2 |x - 1|(2 - x^2)\,dx\right\}$

$= 2\left\{\int_0^1 (1 - x)x^2\,dx + \int_1^2 (x - 1)(2 - x^2)\,dx\right\}$

$= 2\left\{\left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{4}{3} - \frac{3}{2} - \frac{1}{20}\right)\right\} = \frac{1}{6}$

2.25 MOMENT GENERATING FUNCTION

(1) The moment generating function $m_x(t)$ of the discrete probability distribution of the random variable X about the origin $x = a$ is defined as the expected value of $e^{t(x-a)}$ and is denoted by $M_x(t)$. Thus

$$M_x(t) = \sum p_x e^{t(x-a)} \qquad \ldots(1)$$

which is a function of the parameter t only.

Expanding the exponential in (1), we get

$$M_x(t) = \sum p_x + t\sum p_x(x - a) + \frac{t^2}{2!}\sum p_x(x - a)^2 + \ldots + \frac{t^r}{r!}\sum p_x(x - a)^r + \ldots$$

$$= 1 + \mu_1' + \frac{t^2}{2!}\mu_2' + \ldots + \frac{t^r}{r!}\mu_r' + \ldots \qquad \ldots(2)$$

where μ_r' is the moment of order r about $x = a$. Thus $M_x(t)$ generates moments and that is why it is called the moment generating function. From (2), we find

μ_r' = the coefficient of $\frac{t^r}{r!}$ in the expansion of $M_x(t)$.

Otherwise differentiating (2) r times with respect to t and then putting $t = 0$, we get

$$\mu_r' = \left|\frac{d^r}{dt^r} M_x(t)\right|_{t=0} \qquad \ldots(3)$$

Thus the moment about any point $x = a$ can be found from (1) or more conveniently from the formula (3).

Denoting $a = 0$ in (1), we get

$$M_0(t) = e^{-at}\sum p_x e^{tx} \quad \text{or} \quad M_0(t) = e^{-at} M_a(t) \qquad \ldots(4)$$

Thus the m.g.f. about the point $x = a$ is e^{-at} (m.g.f. about the origin).

(2) If $f(x)$ is of the discrete function of a continuous variable X, then the moment generating function of the continuous probability distribution about $x = a$ is given by

$$M_a(t) = \int_{-\infty}^{\infty} e^{t(x-a)} f(x)\,dx.$$

Example 8.6. Find the moment generating function of the exponential distribution

$$f(x) = \frac{1}{c} e^{-x/c}, \quad 0 \le x < \infty, c > 0. \text{ Hence find } \mu_1' \text{ and } \mu_2'. \qquad \text{(Kurukshetra, 2008)}$$

Solution. The moment generating function about the origin is

$$M_0(t) = \int_0^{\infty} e^{tx} \cdot \frac{1}{c} e^{-x/c}\,dx = \frac{1}{c}\int_0^{\infty} e^{-x(1/c - t)}\,dx \qquad \left[\because \int_0^{\infty} e^{-ax}\,dx = \frac{1}{a}\right]$$

$$= \frac{1}{c}\left[\frac{e^{-x(1/c - t)}}{-(1/c - t)}\right]_0^{\infty} = \frac{1}{1 - ct} = (1 - ct)^{-1} = 1 + ct + c^2t^2 + c^3t^3 + \ldots$$

$$\mu_1' = \left[\frac{d}{dt} M_0(t)\right]_{t=0} = [c + 2c^2t + 3c^3t^2 + \ldots]_{t=0} = c$$

$$\mu_2' = \left[\frac{d^2}{dt^2} M_0(t)\right]_{t=0} = 2c^2, \text{ and } \mu_2 = \mu_2' - (\mu_1')^2 = 2c^2 - c^2 = c^2.$$

Hence the mean is c and S.D. is also c.

8.10.2 PROBABILITY GENERATING FUNCTION

The probability generating function $\{p_x\}$ or $P(t)$ for a random variable x which can take integral values $0, 1, 2, \ldots$ is defined by

$$P(t) = p_0 + p_1 t + p_2 t^2 + \ldots = \sum_{x=0}^{\infty} p_x t^x = E(t^x)$$

The coefficient of t^x in the expansion of $P(t)$ in powers of t gives $P(x) = p_x$.

$$\frac{dP}{dt} = \sum_{x=0}^{\infty} x p_x t^{x-1} \quad \text{so} \quad \left[\frac{dP}{dt}\right]_{t=1} = \sum x p_x = \mu_1'$$

$$\frac{d^2P}{dt^2} = \sum_{x=0}^{\infty} x(x-1) p_x t^{x-2} \quad \text{so} \quad \left[\frac{d^2P}{dt^2}\right]_{t=1} = \sum x(x-1) p_x = \mu_2' - \mu_1'$$

$$= \mu_2 + \mu_1'^2 - \mu_1' \text{ and so on.}$$

Also $$\left[\frac{d^r P}{dt^r}\right]_{t=1} = \text{coefficient of } t^r \text{ in } P(t).$$

For integral valued variates, we have

$$P_x(t^x) = E(t^x) = \text{coefficient of } t^x \text{ in } P(t).$$

Obs. The moment generating function of the discrete random variable is the product of the probability generating function and e^{at}.

Example 8.7. If x is a random variable which can take only the values $0, 1, 2, \ldots$ such that $p_x = kx$, find the probability generating function of x.

Solution. We have $P(t) = \sum_{x=0}^{\infty} p_x t^x$

(i) Probability generating function of $X = X = \sum_{x=0}^{\infty} p_x t^{x+1} = t \sum_{x=0}^{\infty} p_x t^x = t P_X(t)$

(ii) Probability generating function of $2X = \sum_{x=0}^{\infty} p_x t^{2x} = \sum_{x=0}^{\infty} p_x (t^2)^x = P_X(t^2)$.

EXERCISE 1.4

1. A random variable x has the following probability function :

Values of x	-2	-1	0	1	2	3
$p(x)$	k	k	0.3	$3k$	k	k

Find the value of k and calculate mean and variance. (U.P.T.U. 2007, U.P.T.U. 2008, Madras 2009)

2. Find the standard deviation for the following discrete distribution :

x	0	5	10	20	30
$p(x) = k$	$15k$	$10k$	$30k$	$10k$	$15k$

3. Obtain the distribution of the number of heads occurring in three tosses of an unbiased coin.

4. Show that for any discrete distribution $\sum p_i \geq 1$.

5. Prizes are to be awarded to 2 red and 2 white balls, a coin is to be tossed to decide whether a red ball or a white ball is to be taken first. If head appears Rs. 40 for each red ball will be given and Rs. 30 for each white one. Find the expectation.

6. The probability that a student will pass a certain test is $\frac{2}{3}$. What is the expectation of the number of tests?

7. The diameter of a circle is taken to be a random variable with p.d.f. $f(x) = 6x(1-x), 0 \leq x \leq 1$. Verify that the above is a p.d.f. Also find the mean and variance.

8. A continuous random variable X has the probability function

$$f(x) = 12x^2 - 12x^3 + 3x, \quad 0 \leq x \leq 1$$

(i) Find $P\left(X < \frac{1}{2}\right)$ and $P\left(X > \frac{1}{2}\right)$

(ii) Find a number a such that $P(X < a) = \frac{1}{2}$. (U.P.T.U. 2006)

9. The pressure of steam in an electric boiler is measured by a random variable X and is represented as a continuous variable. The probability density of X is given by $f(x) = \begin{cases} \frac{1}{4} e^{-x/4}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

where x is the average power measured by the meter.

(i) What is the probability that the meter will measure power larger than the power measured on the average ? (ii) What is the probability that the meter will measure power less than the power measured on the average ?

(Kerala, 2006)

10. A function is defined as follows

$$f(x) = 0, \quad x < 0$$
$$= \frac{3}{16}(4x - x^2), \quad 0 \leq x \leq 4$$
$$= 0, \quad x > 4.$$

Show that f is a density function. Find the probability that a variate having this density will fall in the interval $2 \leq x \leq 3$.

11. A continuous distribution (x) variable x has the range $-3 \leq x \leq 3$ is defined as

$$f(x) = \frac{1}{16}(3 + x), \quad -3 \leq x \leq -1$$
$$= \frac{1}{16}(3 - x), \quad -1 \leq x \leq 1$$
$$= \frac{1}{16}(3 - x), \quad 1 \leq x \leq 3$$

Verify that the area under the curve is unity. Find also the probability that ... (Kurukshetra, 2002)

16. The frequency function of a continuous random variable is given by

$$f(x) = x,\ 0 \le x \le 1$$
$$= 2 - x,\ 1 \le x \le 2$$

Find the mean and the variance of x. (Kerala, 2005; AMIE, 2006)

17. The probability density function of a continuous random variable is given by

$$p(x) = ky^2(1 - y),\ -\infty < y < \infty$$

Find the value of k. Find also the mean and variance of the distribution.

(AMIE, 2006; Kurukshetra, 2002; AMIE, 2006)

18. $F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \le x < 2 \\ 1 & x \ge 2 \end{cases}$

is a function of a random variable X. Find $P(X \le 0)$ and $P(0 < X < 2)$.

19. A function is defined as under:

$$f(x) = kx^2,\ 0 \le x \le 3$$
$$= 0,\ \text{elsewhere}$$

Find k so that $f(x)$ is the probability distribution of the variable x. Find also the mean and the variance of the distribution.

4.13 REPEATED TRIALS

We know that the probability of getting a head or a tail in a toss of a coin is $\frac{1}{2}$. If the coin is tossed thrice, the probabilities of getting one head and two tails are represented as $H—T—T, T—H—T, T—T—H$. The probability of each one of these being $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$, i.e., $\left(\frac{1}{2}\right)^3$, their total probability shall be $3\left(\frac{1}{2}\right)^3$.

Similarly if a trial is repeated n times and if p is the probability of a success and q that of a failure, then the probability of r successes and $n - r$ failures is given by $p^r q^{n-r}$.

But these successes and $n - r$ failures can occur in any of the ${}^nC_r$ ways in each of which the probability is same.

Thus the probability of r successes $= {}^nC_r p^r q^{n-r}$.

Cor. The probabilities of at least r successes in n trials

$= $ the sum of the probabilities of $r, r+1, \ldots, n$ successes

$= {}^nC_r p^r q^{n-r} + {}^nC_{r+1} p^{r+1} q^{n-r-1} + \ldots + {}^nC_n p^n$.

4.14 (1) BINOMIAL DISTRIBUTION*

It is concerned with trials of a repetitive nature in which only the occurrence or non-occurrence, success or failure, acceptance or rejection, are of interest to us (and in which the probability of success is known).

If we perform a series of independent trials such that for each trial p is the probability of a success and q that of a failure, then the probability of r successes in a series of n trials is given by ${}^nC_r p^r q^{n-r}$, where r takes any integral value from 0 to n. The probabilities of $0, 1, 2, \ldots, r, \ldots, n$ successes are, therefore, given by

$$q^n, {}^nC_1 pq^{n-1}, {}^nC_2 p^2 q^{n-2}, \ldots, {}^nC_r p^r q^{n-r}, \ldots, p^n.$$

The probability of the number of successes to obtain r is called the **binomial distribution** for the simple reason that the probabilities are the successive terms in the expansion of the binomial $(q + p)^n$.

$\therefore$ the sum of the probabilities

$$= q^n + {}^nC_1 pq^{n-1} + {}^nC_2 p^2 q^{n-2} + \ldots + p^n = (q + p)^n = 1.$$

(2) Constants of the binomial distribution. The moment generating function about the origin is

$$M_0(t) = E(e^{tx}) = \sum {}^nC_x p^x q^{n-x} e^{tx} \qquad (U.P.T.U., 2004; 2011)$$
$$= \sum {}^nC_x (pe^t)^x q^{n-x} = (q + pe^t)^n$$

* It was discovered by a Swiss mathematician Jacob Bernoulli and was published posthumously in 1713.

Differentiating with respect to t and putting $t = 0$ and using (2) of § 35.11, we get the mean

$$\mu_1' = np$$

Since $M_x(t) = e^{-nt} M_x(t)$ the moment of the binomial distribution about the mean $x = np$, is given by

$$M_\mu(t) = e^{-npt}(q + pe^t)^n = (qe^{-pt} + pe^{qt})^n$$

$$= \left(1 + \mu_1 \frac{t}{1!} + \mu_2 \frac{t^2}{2!} + \mu_3 \frac{t^3}{3!} + \mu_4 \frac{t^4}{4!} + \cdots\right)^n$$

or $$1 + n\mu_1 t + n\mu_2 \frac{t^2}{2!} + n\mu_3 \frac{t^3}{3!} + n\mu_4 \frac{t^4}{4!} + \cdots$$

$$= 1 + npqt \frac{t^2}{2!} + npq(q - p)\frac{t^3}{3!} + npq\{1 + 3(n-2)pq\}\frac{t^4}{4!} + \cdots$$

Equating the coefficients of like powers of t on either side, we have

$$\mu_0 = 1, \mu_1 = 0, \mu_2 = npq, \mu_3 = npq(q - p), \mu_4 = npq\{1 + 3(n-2)pq\}.$$

Also $$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(q-p)^2}{npq} = \frac{(1-2p)^2}{npq} \quad \text{and} \quad \beta_2 = \frac{\mu_4}{\mu_2^2} = 3 + \frac{1 - 6pq}{npq}$$

Thus mean $= np$, standard deviation $= \sqrt{npq}$,

skewness $= (1 - 2p)/\sqrt{npq}$, kurtosis $= \beta_2$.

Note. The skewness is positive for $p < \frac{1}{2}$ and negative for $p > \frac{1}{2}$. When $p = \frac{1}{2}$, the skewness is zero, i.e. the probability curve of the binomial distribution is symmetrical about the mean.

At $p = \frac{1}{2}$, the number of points of inflexion is given by $\beta_2 - 3 = 0$, and $\beta_2 = 3$.

(3) Binomial frequency distribution. If n independent trials constitute one experiment and this experiment is repeated N times, then the frequency of r successes is $N \cdot {}^nC_r p^r q^{n-r}$. The possible number of successes together with these expected frequencies constitute the *binomial frequency distribution*.

(4) Applications of Binomial distribution. This distribution is applied to problems concerning :

(a) Number of defectives in a sample from production line.

(b) Estimation of reliability of systems.

(c) Number of rounds fired from a gun hitting a target.

(d) Radar detection.

Example 35.20. The probability that a person will be hired for a company test is $\frac{1}{3}$. If 12 persons take the test, find the probability that

(a) exactly two will be selected (b) at least two will be selected

(c) none will be selected. (U.P.T.U., 2004; Raj. U., 2005)

Solution. The probability of a defective pen is $1/10 = 0.1$

$\therefore$ The probability of a non-defective pen is $1 - 0.1 = 0.9$

(a) The probability that exactly two will be defective

$$= {}^{12}C_2 (0.1)^2 (0.9)^{10} = 0.2301$$

(b) The probability that at least two will be defective

$$= 1 - (\text{prob. that either none or one is non-defective})$$

$$= 1 - [{}^{12}C_0 (0.9)^{12} + {}^{12}C_1 (0.1)(0.9)^{11}] = 0.3410$$

(c) The probability that none will be defective

$$= {}^{12}C_0 (0.9)^{12} = 0.2824.$$

Example 35.21. A coin is tossed 6 times. Find the probability of getting at most four heads. (U.P.T.U., 2003)

Solution. $P(\text{head}) = \frac{1}{2}$ and $P(\text{tail}) = \frac{1}{2}$

By Binomial distribution, probability of 5 heads and 4 tails in 32 trials is

$$P(X=5)={}^{9}C_5\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^4=\frac{126}{256}\cdot\frac{1}{2^9}=\frac{126}{4096}$$

∴ the expected number of such cases in 256 sets

$$=256\times P(X=5)=256\times\frac{126}{4096}=\frac{126}{16}=7.8\approx 8 \text{ (say)}.$$

Example 24.30. In recording a large number of parts manufactured by a machine, the number of defectives in a sample of 20 is 2. Out of 1000 such samples, how many would be expected to contain at least 3 defective parts? (W.B.U.T., 2004)

Solution. Mean number of defectives $= 2 = np = 20p$.

$\therefore$ The probability of a defective part $= p = 2/20 = 0.1$,

and the probability of a non-defective part $= 0.9$

$\therefore$ The probability of at least three defectives in a sample of 20

$= 1 -$ (prob. that either none, or one, or two are non-defective parts)

$= 1 - [{}^{20}C_{20}(0.9)^{20} + {}^{20}C_1(0.1)(0.9)^{19} + {}^{20}C_2(0.1)^2(0.9)^{18}]$

$= 1 - (0.9)^{18}(0.81) = 0.323.$

Thus the number of samples having at least three defective parts out of 1000 samples

$= 1000 \times 0.323 = 323.$

Example 24.31. The following data are the number of seeds germinating out of 10 on damp filter paper for 80 sets of seeds. Fit a binomial distribution to these data :

x	0	1	2	3	4	5	6	7	8	9	10
f	6	20	28	12	8	6	0	0	0	0	0

Solution. Here $n = 10$ and $N = \Sigma f_i = 80$

$\therefore$ $$\text{mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{20 + 56 + 36 + 32 + 30}{80} = \frac{174}{80} = 2.175$$

Now the mean of a binomial distribution $= np$

$\therefore$ $np = 10p = 2.175$ $\Rightarrow$ $p = 0.2175$, $q = 1 - p = 0.7825$.

Hence the binomial distribution to be fitted is

$$N(q+p)^n = 80(0.7825 + 0.2175)^{10}$$

$$= 80\,[{}^{10}C_0(0.7825)^{10} + {}^{10}C_1(0.7825)^9(0.2175) + {}^{10}C_2(0.7825)^8(0.2175)^2 + \ldots + {}^{10}C_{10}(0.2175)^{10}]$$

$$= 6.808 + 19.33 + 25.04 + \ldots + 0.0007 + 0.0000.$$

$\therefore$ the successive terms in the expansion give the expected or theoretical frequencies which are

x :	0	1	2	3	4	5	6	7	8	9	10
f :	6.8	19.3	25.0	17.8	8.8	2.9	0.7	0.1	0	0	0

EXERCISE 24.3

1. Determine the binomial distribution for which mean = 4 and standard deviation = 2. Also find $P(3 \le X \le 5)$. (Kerala, 2004)
2. An ordinary six faced die is thrown four times. What are the probabilities of showing 4, 5, 6 and 1 fives?
3. If the chance that one of the telephone lines is busy at an instant is 0.2,
   (a) What is the chance that 3 of the lines are busy?
   (b) What is the most probable number of busy lines and what is the probability of this number?
   (c) What is the probability that all the lines are busy? (U.P.T.U., 2003)
4. If the probability that a new-born child is a male is 0.5, find the probability that in a family of 5 children there are exactly 2 boys. (Kurukshetra, 2004)

5. If it is an average 3 errors in every 500 observations, find the probability that a sample of 2000 observations will contain exactly 12 errors. (A.M.I.E., 2005)
6. The probability that a bomb dropped from a plane will strike the target is $\frac{1}{5}$. If 6 bombs are dropped, find the probability that (i) exactly 2 will strike the target, (ii) at least two will strike the target.
7. A manufacturer of pencils claims that only 2% of his pencils are defective. If a box of 100 pencils is purchased by a school, find the probability that (i) one pencil will be defective, (ii) more than two pencils will be defective.
8. The probability that an assuming student will graduate is 0.6. Determine the probability that out of 5 students (i) none, (ii) one, (iii) at least one will graduate.
9. (i) A car dealer finds from his records that a new car requires an average of 3 hours of work during the first year after purchase. If he has 50 cars under warranty, what is the probability that exactly 45 will require no warranty work? (ii) If 3% of the cars require a certain repair during the first year after purchase, what is the probability that exactly 2 of 100 cars will require this repair? (U.P.T.U., 2004)
10. If 10 per cent of the items produced by a machine are defective, find the probability that out of 5 items chosen at random (i) none will be defective, (ii) one will be defective, and (iii) at least two will be defective.
11. In a hurdle race which has 10 competitors, it can be assumed that any hurdler will strike the jumps. Two hurdlers are selected at random from the race. How many hurdles one expects to be tripped to give a 90% chance or better of correctly determining the winner. (U.P.T.U., 2003)
12. A product is sold in packets of 20 and is packed by machines of 1% defectives. What percentage of packets are likely to contain more than 3 defectives?
13. If a random variable has the following probability function, find the probability $P(0, 1, 2, 3)$ and hence find the cumulative distribution function of the random variable.

x	1	2	3	4	5	6
$f(x)$	2	14	20	26	32	6

(Bhopal, 2006)

14. In a binomial distribution, the following frequency distribution:

x	0	1	2	3	4	5	6
f	13	28	40	66	30	16	1

(Rajasthan, 2004; U.P.T.U., 2007)

10.15 (1) POISSON DISTRIBUTION*

A Poisson distribution is one in the probabilities of events which are extremely rare, but which have a large number of independent opportunities for occurrence. The number of persons born blind per year is a large city and the number of deaths by horse kick in an army corps are some of the phenomena, in which this law is followed.

This distribution can be derived as a limiting case of the binomial distribution by making n very large and $p = np = m$ (say), keeping np fixed ($= m$, say).

The probability of r successes in a binomial distribution is

$$P(r) = {}^nC_r p^r q^{n-r} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} p^r q^{n-r}$$

$$= \frac{n(n)\left(n-\frac{1}{n}\right)\left(n-\frac{2}{n}\right)\cdots\left(n-\frac{r-1}{n}\right)}{r!} m^r \left(1-\frac{m}{n}\right)^{n-r}$$

As $n \to \infty$, $p \to 0$ (as $np = m$), we have

$$P(r) = \frac{m^r}{r!} \lim_{n\to\infty} \frac{1 \cdot \left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdots\left(1-\frac{r-1}{n}\right)}{\left(1-\frac{m}{n}\right)^r} \cdot \frac{m^r}{r!} e^{-m}$$

so that the probabilities of $0, 1, 2, \ldots, r, \ldots$ successes in a Poisson distribution are given by

$$e^{-m}, me^{-m}, \frac{m^2 e^{-m}}{2!}, \ldots, \frac{m^r e^{-m}}{r!}, \ldots$$

The sum of these probabilities is unity as it should be.

* It was discovered by a French mathematician S.D. Poisson in 1837.

(2) Constants of the Poisson distribution. These constants can easily be derived from the corresponding constants of the binomial distribution simply by making $n \to \infty, p \to 0, q \to 1$ and noting that $np = m$.

Mean = (i) np or m

$\mu_2 = 2nPQ(pq)^2 + nPQ[1 - 3PQ] = m$

$\therefore$ Standard deviation = $\sqrt{m}$

Also $\mu_3 = mQ$, $\mu_4 = mQ + 3m^2$

$\therefore$ Skewness = $\sqrt{\beta_1} = \frac{\mu_3}{\mu_2^{3/2}} = \frac{mQ}{m^{3/2}} = \frac{Q}{\sqrt{m}}$, Kurtosis = $\beta_2 = 3 + \frac{1}{m}$.

Since μ_3 is positive, Poisson distribution is positively skewed and since $\beta_2 > 3$, it is Leptokurtic.

(3) Applications of Poisson distribution. This distribution is applied to problems concerning : (i) Arrival pattern of 'defective vehicles in a workshop', 'patients in a hospital' or 'telephone calls'.

(ii) Demand pattern for certain spare parts.

(iii) Number of fragments from a shell hitting a target.

(iv) Spatial distribution of bacteria etc.

Example 20.43. If the probability of a bad reaction from a certain injection is 0.001, determine the chance that out of 2,000 individuals more than two will get a bad reaction. (U.P.T.U., 2003 ; Kurukshetra, 2007)

Solution. It follows a Poisson distribution as the probability of occurrence is very small.

Mean $= np = 2000 \times 0.001 = 2$

Probability that more than 2 will get a bad reaction

$= 1 - [\text{prob. that no one gets a bad reaction} + \text{prob. that one gets a bad reaction} + \text{prob. that two get bad reaction}]$

$= 1 - \left[e^{-m} + \frac{me^{-m}}{1!} + \frac{m^2 e^{-m}}{2!}\right] = 1 - \left[\frac{1}{e^2} + \frac{2}{e^2} + \frac{2}{e^2}\right]$ $\quad [\because m = 2]$

$= 1 - \frac{5}{e^2} = 0.32$ $\quad [\because e = 2.718]$

Example 20.44. In a certain factory turning out razor blades, there is a small chance of 0.002 for any blade to be defective. The blades are supplied in packets of 10 and use the Poisson distribution to calculate the approximate number of packets containing no defective, one defective and two defective blades respectively in a consignment of 10,000 packets. (Kurukshetra, 2006-S ; Madras, 2006 ; P.T.U., 2005)

Solution. We know that $m = np = 10 \times 0.002 = 0.02$

$e^{-0.02} = 1 - 0.02 + \frac{(0.02)^2}{2!} - \ldots = 0.9802$ approximately

Probability of no defective blade in $e^{-m} = e^{-0.02} = 0.9802$

$\therefore$ no. of packets containing no defective blade is

$10,000 \times 0.9802 = 9802$

Similarly the number of packets containing one defective blade $= 10,000 \times me^{-m}$

$= 10,000 \times 0.02 \times 0.9802 = 196$

Finally the number of packets containing two defective blades

$= 10,000 \times \frac{m^2 e^{-m}}{2!} = 10,000 \times \frac{(0.02)^2}{2!} \times 0.9802 = 2$ approximately.

Example 20.45. Fit a Poisson distribution to the set of observations :

x :	0	1	2	3	4
f :	122	60	15	2	1

(Manipal, 2007-S ; V.T.U., 2006 ; U.P.T.U., 2005)

Solution. Mean $= \frac{\Sigma f x}{\Sigma f} = \frac{60 + 30 + 6 + 4}{200} = 0.5$

∴ mean of Poisson distribution $\lambda = m = 0.6$.

Hence the theoretical frequency for x successes is

$$\frac{Ne^{-m}m^x}{x!} = \frac{200e^{-0.6}(0.6)^x}{x!} \text{ where } x = 0, 1, 2, 3, 4$$

∴ the theoretical frequencies are

x :	0	1	2	3	4
f :	121	81	18	2	0

(V $e^{-0.6} = 0.5488$)

PROBLEMS 9.4

1. If X is a binomial variate with parameters n and p, show that $\lim_{n\to\infty} P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$, where $\lambda = np$. (Mar 2012)
2. If X is a Poisson variable such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the parameter of the distribution. (Mar 2013)
3. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$. Find the probability that X lies between 1 and 3. (Mar 2014)
4. Find the mean and variance of the Poisson distribution $P(X = x) = \frac{e^{-2}2^x}{x!}$, $x = 0, 1, 2, \ldots$. (Mar 2015)
5. Let X be a Poisson variate such that $P(X = 1) = P(X = 2)$. Find $P(X = 0)$ and $P(X = 3)$. (Mar 2016)
6. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X. (Mar 2017)
7. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X. (Mar 2018)
8. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X. (Mar 2019)
9. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X. (Mar 2020)
10. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X. (Mar 2021)
11. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X. (Mar 2022)
12. If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$, find the mean and variance of X. (Mar 2023)

9.5.3 [3] NORMAL DISTRIBUTION*

Now we consider a most important distribution of fundamental importance, namely the normal distribution. Any quantity whose variation depends on random causes is distributed according to the normal law. Its importance lies in the fact that a large number of distributions encountered in the normal distribution.

* In 1801, Karl Friedrich found the distribution which Abraham de Moivre had discovered as early as 1733. See Statistics, p. 343 and 347.

Let us define a random variable X as $X = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ ... (1)

where $\bar{X}$ is the sample mean, μ is the population mean, σ is the standard deviation of the population and n is the sample size. [illegible] X is a standard normal variable with mean 0 and variance 1. [illegible] the distribution of X is known as the standard normal distribution or the unit normal distribution. [illegible] $X \sim N(0, 1)$.

It can be shown that the probability density function of X is given by $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ ... (2)

where $-\infty < x < \infty$. [illegible] the mean and variance of X are 0 and 1 respectively.

(a) Properties of the normal distribution:

1. The normal curve (2) is bell-shaped and is symmetrical about the mean. It is completely determined by the mean μ and the standard deviation σ. The maximum ordinate is $\frac{1}{\sigma\sqrt{2\pi}}$, which is obtained by putting $x = \mu$ in (1).

2. If X is a continuous random variable and $P(x)dx$ represents the probability of X lying in the interval $(x, x+dx)$, then $P(x)dx = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$ ... (3)

[illegible] $P(x)$ is called the probability density function of X and the curve $y = P(x)$ is called the probability curve. [illegible] the total area under the curve is unity.

3. Mean deviation from the mean is

$$= \int_{-\infty}^{\infty} |x - \mu| \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \qquad \left[\because P(x)dx = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx\right]$$

$$= \frac{2}{\sigma\sqrt{2\pi}} \int_{\mu}^{\infty} (x - \mu) e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{2}{\sigma\sqrt{2\pi}} \left[\int_{\mu}^{\infty} -(x-\mu) e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx + \int_{\mu}^{\infty} (x-\mu) e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \right] = \frac{2\sigma}{\sigma\sqrt{2\pi}} \int_{0}^{\infty} z e^{-\frac{z^2}{2}} dz$$

$$= \frac{2\sigma}{\sigma\sqrt{2\pi}} \left| -e^{-\frac{z^2}{2}} \right|_0^{\infty} = \sqrt{\frac{2}{\pi}} \sigma \quad \text{[illegible] } z = \frac{x-\mu}{\sigma} \text{ [illegible] } dz = \frac{1}{\sigma} dx$$

[illegible] Moments about the mean:

$$\mu_{2n+1} = \int_{-\infty}^{\infty} (x - \mu)^{2n+1} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{\sigma^{2n+1}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} z^{2n+1} e^{-\frac{z^2}{2}} dz \quad \text{[illegible] } z = \frac{x-\mu}{\sigma}$$

$= 0$, since the integrand is an odd function.

Hence all odd order moments about the mean vanish.

$$\mu_{2n} = \int_{-\infty}^{\infty} (x - \mu)^{2n} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{\sigma^{2n}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} z^{2n} e^{-\frac{z^2}{2}} dz \qquad \left[\text{Putting } \frac{x-\mu}{\sigma} = z\right]$$

$$= \frac{\sigma^{2n}}{\sqrt{2\pi}} \left[-z^{2n-1} e^{-\frac{z^2}{2}} \right]_{-\infty}^{\infty} + (2n-1) \int_{-\infty}^{\infty} z^{2n-2} e^{-\frac{z^2}{2}} dz \right]$$

$$= \frac{\sigma^{2n}}{\sqrt{2\pi}} (2n-1)(2n-3) \ldots 3 \cdot 1 \sqrt{2\pi} = (2n-1)(2n-3) \ldots 3 \cdot 1 \, \sigma^{2n} = \mu_{2n-2}$$

Repeated application of this relation gives the result μ_{2n} given above.

In particular, $\mu_2 = \sigma^2$, $\mu_4 = 3\sigma^4$.

Hence $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0$ and $\beta_2 = \frac{\mu_4}{\mu_2^2} = 3$

i.e., the coefficient of skewness is zero i.e. the curve is symmetrical and the Kurtosis is 3. This is the basis for the choice of the value 3 in the definitions of platykurtic and leptokurtic (page 414).

IV. The probability of a lying between x_1 and x_2 is given by the area under the normal curve from x_1 to x_2, i.e., $P(x_1 \le x \le x_2)$

$$= \frac{1}{\sigma\sqrt{(2\pi)}} \int_{x_1}^{x_2} e^{-(x-\mu)^2/2\sigma^2}\, dx$$

$$= \frac{1}{\sqrt{(2\pi)}} \int_{z_1}^{z_2} e^{-z^2/2}\, dz \text{ where } z = (x-\mu)/\sigma,\ dz = dx/\sigma \text{ and } z_1 = (x_1-\mu)/\sigma,\ z_2 = (x_2-\mu)/\sigma.$$

$$= \frac{1}{\sqrt{(2\pi)}} \left\{ \int_0^{z_2} e^{-z^2/2}\, dz - \int_0^{z_1} e^{-z^2/2}\, dz \right\} = P_2(z_2) - P_1(z_1)$$

The values of each of the above integrals can be found from the table III Appendix 2, which gives the values of

$$P(z) = \frac{1}{\sqrt{(2\pi)}} \int_0^z e^{-z^2/2}\, dz$$

for various values of z. This integral is called the probability integral or the error function due to its use in the theory of sampling and the theory of errors.

Using this table, we see that the area under the normal curve from $z = 0$ to $z = 1$, i.e., from $x = \mu$ to $x = \mu + \sigma$ is 0.3413.

∴ (i) The area under the normal curve between the ordinates $x = \mu - \sigma$ and $x = \mu + \sigma$ is $0.6826 = 68\%$ nearly. Thus approximately 2/3 of the values lie within the first limits.

(ii) The area under the normal curve between $x = \mu - 2\sigma$ and $x = \mu + 2\sigma$ is $0.9545 = 95.5\%$, which implies that about $4\frac{1}{2}\%$ of the values lie outside these limits.

(iii) 49.93% of the values lie between $x = \mu - 3\sigma$ and $x = \mu + 3\sigma$ i.e., only a quarter % of the values lie outside these limits.

(iv) 50% of the values lie between $x = \mu - 0.67\sigma$ and $x = \mu + 0.67\sigma$ i.e., only 1/2 of the values lie outside these limits.

(v) 80% of the values lie between $x = \mu - 1.28\sigma$ and $x = \mu + 1.28\sigma$ i.e., only 1/5 of the values lie outside these limits.

(vi) 90.0% of the values lie between $x = \mu - 1.65\sigma$ and $x = \mu + 1.65\sigma$.

In other words, a value that deviates more than σ from previous observations in 3 trials. A value that deviates more than 2σ or 3σ from a column about zero in 20 or 400 trials. Almost all values lie within 3σ of the mean.

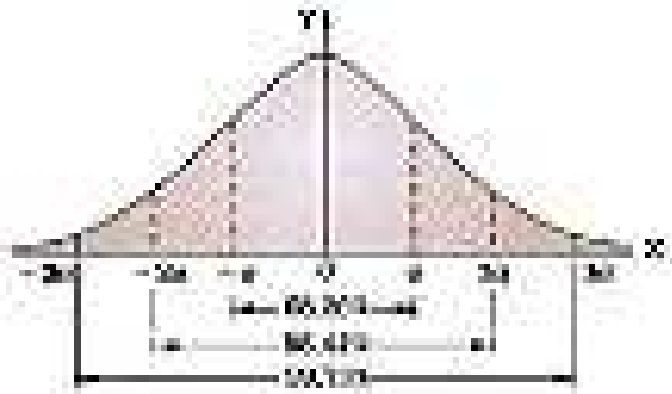


Fig. 28.4

The shape of the standardized normal curve is

$$y = \frac{1}{\sqrt{(2\pi)}} e^{-z^2/2} \text{ where } z = (x-\mu)/\sigma \qquad \ldots(1)$$

and the respective curve are shown in Fig. 28.4. 'y' is called a normal variate.

(3) Normal frequency distribution. We can fit a normal curve to any distribution. If N be the total frequency, μ the mean and σ the standard deviation of the given distribution then the curve

$$y = \frac{N}{\sigma\sqrt{(2\pi)}} e^{-(x-\mu)^2/2\sigma^2} \qquad \ldots(2)$$

will fit the given distribution as best as the data will permit. The frequency of the variate between x_1 and x_2 as given by the fitted curve, will be the area under (2) from x_1 to x_2.

(4) Applications of normal distribution. This distribution is applied to problems concerning:

(i) Calculation of errors made by chance in experimental measurements.

(ii) Computation of the probability of a event.

(iii) Statistical inference in almost every branch of science.

23.12 PROBABLE ERROR

Any bit of statistics we calculate from sample data is subject to sampling variation. In fact, measurement of any physical quantity shows slight error. In general, the two causes of inaccuracy are personal errors and random nature of these errors. If the errors are normally distributed, while quoting a specification of an experimental result, we usually mention the probable error (P.E.). It is such that the probability of an error falling within the limits $\mu - \lambda$ and $\mu + \lambda$ is exactly equal to the chance of an error falling outside these limits, i.e. the chance of an error lying within $\mu - \lambda$ and $\mu + \lambda$ is $\frac{1}{2}$.

$$\therefore \quad \frac{1}{\sigma\sqrt{2\pi}} \int_{\mu-\lambda}^{\mu+\lambda} e^{-(x-\mu)^2/2\sigma^2}\, dx = \frac{1}{2}$$

or
$$\frac{1}{\sqrt{2\pi}} \int_{0}^{\lambda} e^{-z^2/2}\, dz = \frac{1}{4} \qquad \left[z = \frac{x-\mu}{\sigma}\right]$$

The table F, (Appendix 3) gives $\lambda = 0.6745\sigma$.

Hence the probable error $\lambda = 0.6745\sigma = \frac{2}{3}\sigma$.

The quartile deviation $= \frac{1}{2}(Q_3 - Q_1) = \frac{2}{3}\sigma$, Mean deviation $= \frac{4}{5}\sigma$ (p. 624)

$Q_1 : M.D. : S.D. = 10 : 12 : 15$ (Madras 2003)

Example 23.43. If X is a normal variate with mean 30 and S.D. 5, find the probabilities that (i) $26 \le X \le 40$ (ii) $X \ge 45$ and (iii) $|X - 30| > 5$. (U.P.T.U. 2003)

Solution. We have $\mu = 30$ and $\sigma = 5$.

$$\therefore \quad z = \frac{X - \mu}{\sigma} = \frac{X - 30}{5}$$

(i) When $X = 26, z = -0.8$; when $X = 40, z = 2$

$$\therefore \quad P(26 \le X \le 40) = P(-0.8 \le z \le 2)$$
$$= P(-0.8 \le z \le 0) + P(0 \le z \le 2) \qquad \text{(Using Table A3)}$$
$$= P(0 \le z \le 0.8) + 0.4772$$
$$= 0.2881 + 0.4772 = 0.7653$$

(ii) When $X = 45, z = 3$

$$\therefore \quad P(X \ge 45) = P(z \ge 3) = 0.5 - P(0 \le z \le 3)$$
$$= 0.5 - 0.4987 = 0.0013$$

(iii) $\quad P[|X - 30| \le 5] = P[25 \le X \le 35]$
$$= P(-1 \le z \le 1) = 2P(0 \le z \le 1)$$
$$= 2 \times 0.3413 = 0.6826$$

$$\therefore \quad P[|X - 30| > 5] = 1 - P[|X - 30| \le 5]$$
$$= 1 - 0.6826 = 0.3174.$$

Example 23.44. A certain number of candidates were interviewed for some key posts in a firm. The interviewing committee considered the performance characteristics, intelligence, knowledge, etc. Candidates scoring above 60% in aggregate were considered for the posts. The characteristics is known to be normally distributed with mean 50 and standard deviation 10. If the candidates scoring 60% and 70% were found to be over-ambitious.

Solution. Let μ be the mean and σ be the standard deviation of the normal curve. Fig. 23.5.

Now 60% of the candidates were the characteristic below 60, 30% between 50 and 60 and only 8% greater than 60.

Let the area to the left of the ordinate AQ be 0.5 and that between the ordinates PQ and AP be 30% so that the areas to the left of PQ is z_1 and AP is z_2 are 0.5 and 0.30 respectively, i.e., the area of PQP' $= 0.5 - 0.3 = 0.2$ and the area OAP $= 0.3$.

$\therefore$ area corresponding to $z_1 \left(= \frac{90-\mu}{\sigma}\right) = 0.1$

and that corresponding to $z_2 \left(= \frac{60-\mu}{\sigma}\right) = 0.48$

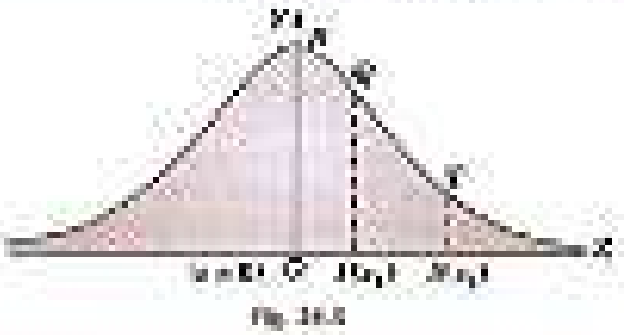


Fig. 26.5

From the table III, we have

$90 - \mu/\sigma = 1.282$ and $60 - \mu/\sigma = 1.645$

whence $\sigma = 3.58$ and $\mu = 84.416$.

Example 26.42. *In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and S.D. of the distribution.* (U.P.T.U., 2000; U.P.T.U., 2002; Kurukshetra, 2007 S)

Solution. Let X be the mean and σ the S.D. 31% of the items are under 45 measured on the left of the ordinate $x = 45$. (Fig. 26.6)

When $x = 45$, let $z = z_1$ so that $z_1 = \frac{45 - \mu}{\sigma}$ ...(1)

$\therefore \quad \int_{-\infty}^{z_1} \phi(z)\,dz = 0.31 \quad$ or $\quad \int_{-\infty}^{0} \phi(z)\,dz - \int_{z_1}^{0} \phi(z)\,dz = 0.31$

Hence $\quad \int_{z_1}^{0} \phi(z)\,dz = \int_{-\infty}^{0} \phi(z)\,dz - 0.31 = 0.5 - 0.31 = 0.19$

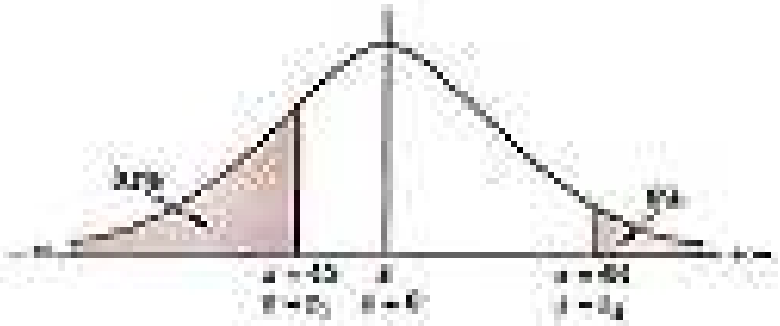


Fig. 26.6

From table III, $z_1 = -0.5$ ...(2)

When $x = 64$, let $z = z_2$ so that $z_2 = (64 - \mu)/\sigma$ ...(3)

$\therefore \quad \int_{z_2}^{0} \phi(z)\,dz = 0.08 \quad$ or $\quad \int_{-\infty}^{0} \phi(z)\,dz - \int_{-\infty}^{z_2} \phi(z)\,dz = 0.08$

Hence $\quad \int_{-\infty}^{z_2} \phi(z)\,dz = \int_{-\infty}^{0} \phi(z)\,dz - 0.08 = 0.5 - 0.08 = 0.42$

From table III, $z_2 = 1.4$ ...(4)

From (1) and (2), $45 - \bar{x} = -0.5\sigma$

From (3) and (4), $64 - \bar{x} = 1.4\sigma$

Solving these equations, we get $\bar{x} = 60$ and $\sigma = 10$.

Example 26.43. *In a test on 2000 electric bulbs, it was found that the life of a particular make was normally distributed with an average life of 2040 hours and S.D. of 60 hours. Estimate the number of bulbs likely to burn for*

(a) more than 2150 hours, (b) less than 1950 hours and

(c) more than 1920 hours but less than 2160 hours. (Bhopal, 2000 S; U.P.T.U., 2006)

Solution. Here $\mu = 2040$ hours and $\sigma = 60$ hours.

(a) For $x = 2150$, $\quad z = \frac{x - \mu}{\sigma} = 1.833$.

$\therefore$ area against $z = 1.65$ in the table $= 0.4505$

We, however, require the area to the right of the ordinate at $z = 1.65$. This area $= 0.5 - 0.4505 = 0.0495$.

Thus the number of bulbs expected to burn for more than 2500 hours

$= 0.0495 \times 2000 = 99$ approximately.

(b) For $x = 1850$, $z = \dfrac{x - \bar{x}}{\sigma} = -1.5$

The area required in this case is to the left of $z = -1.50$

i.e., $= 0.5 - 0.4332$ (table value for $z = 1.50$)

$= 0.0918$.

$\therefore$ the number of bulbs expected to burn for less than 1850 hours

$= 0.0918 \times 2000 = 184$ approximately.

(c) When $x = 1900$, $z = \dfrac{1900 - 2040}{80} = -2$

When $x = 2160$, $z = \dfrac{2160 - 2040}{80} = 2$.

The number of bulbs expected to burn for more than 1900 hours but less than 2160 hours will be represented by the area between $z = -2$ and $z = 2$. This is twice the area from the table for $z = 2$, i.e., $= 2 \times 0.4772 = 0.9544$.

Thus the required number of bulbs $= 0.9544 \times 2000 = 1909$ nearly.

Example 11.10. *If the probability of meeting an error of magnitude x is given by*

$$y = \frac{h}{\sqrt{\pi}} e^{-h^2 x^2},$$

compute the probable error for the following data:

$x_1 = 0.005$; $x_2 = 0.007$; $x_3 = 0.008$; $x_4 = 0.009$;

$x_5 = 0.010$; $x_6 = 0.011$; $x_7 = 0.011$; $x_8 = 0.012$;

$x_9 = 0.013$; $x_{10} = 0.015$. (Kurukshetra, 2003)

Solution. From the given data which is normally distributed, we have

$$\text{mean} = \frac{1}{10}\Sigma x_i = \frac{77.501}{10} = 1.2581$$

and

$$\sigma^2 = \frac{1}{10}\Sigma x_i^2 - (\text{mean})^2$$

$$= \frac{1}{10}[(0.005)^2 + (0.007)^2 + (0.008)^2 + (0.009)^2 + (0.010)^2 + (0.011)^2 + (0.011)^2 + (0.012)^2 + (0.013)^2 + (0.015)^2] - (1.2581)^2$$

$$= 0.000126 - 1.582 = 0.0126.$$

$\therefore$ probable error $= \dfrac{2}{3}\sigma = 0.0842$ approx.

Example 11.11. *Fit a normal curve to the following distribution:*

x:	0	1	2	3	4
f:	7	4	6	4	1

(U.P.T.U., 2003)

Solution.

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{0 + 4 + 12 + 12 + 4}{22} = 2$$

$$\text{S.D.} = \sqrt{\left[\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2\right]} = \sqrt{(3.64 - 2.00)} = 1.28$$

Taking $\mu = 0$ and $\sigma = 2$ and $N = 16$, the equations of the normal curves are

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{and} \quad y = \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x}{2}\right)^2} \qquad \ldots(1)$$

Area under (1) for (x_1, x_2) or (z_1, z_2)

$$= \frac{2}{\sqrt{2\pi}} \int_0^{z_2} e^{-\frac{1}{2}z^2} dz - \frac{1}{\sqrt{2\pi}} \int_0^{z_1} e^{-\frac{1}{2}z^2} dz \qquad \text{where } z = \frac{x-\bar{x}}{\sigma}$$

To evaluate these integrals, we refer to table (II).

(i) Solution :

Mid-x	(x_1, x_2)	(z_1, z_2)	Area under (1) for (z_1, z_2)	Expected frequency
2	(1, 3)	(−0.5, +0.5)	0.6826 − 0.6915	16 × 0.6826 = 6.37
4	(3, 5)	(+0.5, +1.5)	0.8413 − 0.6915	16 × 0.1498 = 2.39
6	(5, 7)	(+1.5, 2.5)	0.9938 − 0.8413	16 × 0.1525 = 2.44
8	(7, 9)	(2.5, 3.5)	0.9998 − 0.9938	16 × 0.0060 = 0.10
10	(9, 11)	(3.5, ∞)	1.0000 − 0.9998	16 × 0.0002 = 0.00

Hence the expected theoretical frequencies corrected to nearest integer are 3, 4, 6, 4, 1 which agree with the observed frequencies. This shows that the normal curve (1) is a proper fit to the given distribution.

PROBLEMS 5(F)

1. Show that the standard deviation for a normal distribution is approximately 20% more than the mean deviation.
2. For a normally distributed variate with mean 1 and S.D. 3, find the probabilities of
   (i) $3.43 \le x \le 6.19$ (ii) $-1.43 \le x \le 6.19$
3. If a normally distributed variate has mean 0 and variance 1, find
   (i) $P(|x| > 1.96)$ (ii) $P(|x| > 3)$ (iii) $P(2 < |x| < 2.5)$.
4. In a certain factory at Howrah, the percentages of casualties due to any cause during a calendar week (6 days) and the corresponding numbers of the average weekly wages paid by the workmen, the coefficient of correlation was found to be 0.8 and the regression equation is $y = 2x + 50$. Find the average weekly wages when the percentage of casualties is 10. Also find the standard deviation of weekly wages if that of casualties is known to be 2. [Kalyani, 2003]
5. A manufacturer of elevators in shopping centres knows from experience that the weights of the passengers are normally distributed with mean 150 pounds and standard deviation 30 pounds. What is the probability that the total weight of 9 passengers will exceed 1500 pounds? Also find the probability that the total weight of 9 passengers will exceed 2000 pounds? [Kalyani, 2003]
6. A manufacturer of elevators in shopping centres knows from experience that the weights of the passengers are normally distributed with mean 150 pounds and standard deviation 30 pounds. What is the probability that the total weight of 9 passengers will exceed 1500 pounds? Also find the probability that the total weight of 9 passengers will exceed 2000 pounds? [Kalyani, 2003]
7. The mean and standard deviation of the marks obtained by 1000 students in a certain subject are respectively 34.5 and 16.5. Assuming the normality of the distribution, find the approximate number of students expected to obtain marks between 30 and 60. [Kalyani, 2003]
8. In an examination taken by 500 candidates, the average and the standard deviation of marks obtained (assumed to be normally distributed) are 40% and 5%. Find approximately
   (i) how many will pass, if 300 is fixed as a minimum ?
   (ii) what should be the minimum if 350 candidates are to pass ?
   (iii) how many have scored marks above 60% ?
9. The mean inside diameter of a sample of 200 washers produced by a machine is 0.502 inch and the standard deviation is 0.005 inch. The purpose for which these washers are intended allows a tolerance in the diameter of 0.496 to 0.508 inch, otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed.
   [Hint: 0.496 in standard units $= (0.496 - 0.502)/0.005 = -1.2$
   0.508 in standard units $= (0.508 - 0.502)/0.005 = 1.2$
   Proportion of normal curve between $z = -1.2$ and $z = 1.2$ is
   $= 0.3849 + 0.3849 = 0.7698$ (approx).
   $\therefore$ percentage of defective washers $= 100 - 77 = 23\%$.]
10. Assuming that the diameters of 1000 brass discs manufactured in a factory form a normal distribution with mean 0.7515 cm. and standard deviation 0.0020 cm., how many of the discs are likely to be rejected if the approved diameter is 0.752 ± 0.004 cm. ? [Bhopal, 2003]

11. It is given that the age of the contents of a particular type of bottle of the cosmetic have a normal law with mean 2 years and S.D. 3 years. 1000 such bottles are sold on every month. How many of these will have to be replaced at the end of the second year.

12. The marks of a group of 10,000 persons are found to be normally distributed with mean Rs. 300 per, and standard deviation of Rs. 50. How many of this group, about 95% had marks between Rs. 400 and 200 and only 5% had marks exceeding Rs. 400. Also find the lowest marks among the students 100. (U.P.T.U., 2002 S)

13. Find the equation of the best fitting normal curve to the following distribution :

x :	0	1	2	3	4	5
y :	22	20	34	22	11	4

14. Obtain the equation of the normal probability curve that must be fitted to the following data :

Variable :	4	6	8	20	22	24	29	33	35	38	40
Frequency :	3	7	22	30	39	43	30	30	15	5	1

15. A factory manufactures articles by mass production and it is found that 10% of the product is rejected. Find the S.D. of the number of rejects and the equation to be assumed to fit to represent the number of rejects.

Hint: $p = 0.1, q = 0.9, n = 100$.

Normal distribution of rejects is given by $y = 10 \cdot 3.14 = \sqrt{npq} = 3$

If the binomial distribution is approximated by a normal distribution, then the equation to the normal curve is

$$y = \frac{100}{3\sqrt{2\pi}} e^{-\frac{x^2}{2(3)^2}} \quad \text{where } x = 0, 1, 2, \ldots, 10.$$

16. Given that the probability of accidents in a year of any driver is p, show that the probability of r accidents is $\frac{\lambda^r}{r!} e^{-\lambda}$, where $\lambda = np$. Also show that the probability of no accidents is $e^{-\lambda}$.

6.16. NORMAL APPROXIMATION TO BINOMIAL DISTRIBUTION

If the number of successes in a Binomial distribution range from x_1 to x_2, then the probability of getting these x successes

$$= \sum_{x=x_1}^{x_2} {}^nC_x \, p^x \, q^{n-x}$$

As the number of trials increases, the Binomial distribution becomes approximated to the Normal distribution. The mean np and the variance npq of the binomial distribution will be quite close to the mean and standard deviation of the approximated normal distribution. Thus for a sufficiently large ($n \geq 30$), the binomial distribution with probability of success p, is approximated by the normal distribution with $\mu = np$, $\sigma = \sqrt{npq}$.

We must however, be careful to get the normal values of x. For any discrete x, we must interval to $(x - 1/2, x + 1/2)$. Hence

$$z_1 = \frac{x_1 - \frac{1}{2} - \mu}{\sigma} = \frac{x_1 - \frac{1}{2} - np}{\sqrt{npq}}, \quad z_2 = \frac{x_2 + \frac{1}{2} - np}{\sqrt{npq}}$$

so that $P(x_1 \leq x \leq x_2) = P(z_1 \leq z \leq z_2) = \int_{z_1}^{z_2} \phi(z)\,dz$ which can be calculated by using table III–Appendix 2.

Example 16.11. In a large population 10% of voters voted in favour. A random sample of 200 voters was selected. What is the probability that in the sample

(a) more than 200 voted in favour ?

(b) between 180 and 200 inclusive voted in favour ?

(c) 200 voted in favour ?

Solution. Here $n = 200, p = 0.8, q = 0.2$

$\therefore \qquad \mu = np = 200 \times 0.8 = 320, \sigma = \sqrt{npq} = \sqrt{40} = 6.928$

(a) $P(x > 180) = P(x > 180.5) = P\left(z > \frac{180.5 - 320}{\sqrt{40}}\right) = P(z > 1.5000) = 0.0668$

(b) $P(100 < x < 120) = P(100.5 < x < 119.5)$

$$= P\left(\frac{100.5 - 108}{\sqrt{48}} < z < \frac{119.5 - 108}{\sqrt{48}}\right) = P(-1.08 < z < 1.66) = 0.8064$$

(c) $P(x = 120) = P(119.5 < x < 120.5)$

$$= P(1.66 < z < 1.73) = 0.0459 - 0.0392 = 0.0067$$

EXERCISE 3.5

1. A pair of unbiased dice are rolled 180 times and their score recorded. Find
   (a) $P(x \le 6)$ (b) $P(7 \le x \le 9)$ (c) $P(10 \le x \le 12)$
2. A multiple-choice test has probability of $\frac{1}{5}$ of getting a correct answer on a single guess. If the student guesses 100 times, what is the probability that it is (a) the expected number of correct answers, (b) between 18 and 30 inclusive, (c) 25 or more?
3. A certain drug is effective in 70% of cases. Given 2000 people are treated with the drug, what is the probability that it will be effective for (a) at least 1400 people, (b) less than 1380 people, (c) 1420 people?

3.23 SOME OTHER DISTRIBUTIONS

Discrete distributions

(1) Geometric distribution. If p is the probability of a success and q is the probability of failure preceding the first success then this distribution is

$$P(k) = q^k p, \quad k = 0, 1, 2, \ldots, q = 1 - p$$

Obviously $\sum_{k=0}^{\infty} P(k) = p \sum_{k=0}^{\infty} q^k = p \cdot \frac{1}{1-q} = 1$.

It may easily be shown that mean $= q/p$, and variance $= q/p^2$.

(2) Negative binomial distribution. This distribution gives the probability that the event occurs for the kth time on the $(r+k)$th trial $(r \ge k)$. If p is the probability of occurrence of an event then

$$P(k) = {}^{r+k-1}C_{r-1}\, p^r q^k.$$

It involves two parameters p and k. If $k = 1$, the Negative binomial distribution reduces to the geometric distribution.

(3) Hypergeometric distribution. Suppose a bag contains m white and n black balls. If r balls are drawn one at a time (without replacement), then the probability that k of them will be white is

$$P(k) = {}^{m}C_k\, {}^{n}C_{r-k} \Big/ {}^{m+n}C_r, \quad k = 0, 1, \ldots, r, \; r \le m, \; r \le n.$$

This distribution is known as *hypergeometric distribution*.

For $\sum_{k=0}^{r} P(k) = 1$, since $\sum_{k=0}^{r} {}^{m}C_k\, {}^{n}C_{r-k} = {}^{m+n}C_r$

This can be proved by equating the coefficient of x^r in

$$(1+x)^m (1+x)^n = (1+x)^{m+n}$$

Continuous distributions

(4) Uniform (or Rectangular) distribution. A random variable X is said to be uniformly distributed over the interval $a \le x \le b$ or (a, b), if its density is given by

$$f(x) = \frac{1}{b-a}, \quad a \le x \le b \qquad \ldots(1)$$

The distribution given by (1) is called a *uniform distribution*. In this distribution, X takes the values with the same probability.

Its mean $\mu = \int_a^b x \cdot f(x)\,dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{a+b}{2}$

and variance $\sigma^2 = \mu_2' - (\mu_1')^2 = \int_a^b x^2 \cdot \frac{1}{b-a}\,dx - \left(\frac{a+b}{2}\right)^2 = \frac{1}{12}(b-a)^2$.

(5) Gamma distribution. This continuous distribution is given by $f(x) = \frac{\lambda}{\Gamma(r)}(\lambda x)^{r-1} e^{-\lambda x}$ for all $x \geq 0$, where r and λ (both > 0) are called the parameters of the gamma distribution. Here mean $= r/\lambda$ and variance $= r/\lambda^2$.

Gamma distribution tends to normal distribution as the parameter r tends to infinity.

(6) Exponential distribution. This distribution is a special case of gamma distribution when $r = 1$ so that $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$, where λ is a parameter.

It can be seen that mean $= 1/\lambda$, standard deviation $= 1/\lambda$.

This distribution plays an important role in the reliability and queuing theory.

(7) Weibull distribution*. This distribution is given by

$$f(x) = \frac{c}{a}\, x^{c-1} e^{-x^c/a}, \quad x \geq 0,\ c > 0$$

where c is a scale parameter and a a shape parameter.

Initially this distribution was used to describe experimentally observed variations in the fatigue resistance of steel and its elastic limits. But it has also been employed to study the variation of length of service of radio service equipment.

Example 15.51. *A dice is rolled until 6 appears. Find (i) the probability that it is necessary to throw the dice 5 times, (ii) the probability that an even number of throws is required.*

Solution. Here probability of getting 6 is $p = \frac{1}{6}$. Thus $q = \frac{5}{6}$.

If X is the number of throws required for the first success, then

$$P(X = x) = q^{x-1} p \text{ for } x = 1, 2, 3, \ldots$$

$\therefore$ required probability $= P(X > 5) = 1 - P(X \leq 5)$

$$= 1 - \sum_{x=1}^{5} \left(\frac{5}{6}\right)^{x-1} \left(\frac{1}{6}\right) = 1 - \frac{1}{6}\left\{1 + \frac{5}{6} + \left(\frac{5}{6}\right)^2 + \left(\frac{5}{6}\right)^3 + \left(\frac{5}{6}\right)^4\right\} = \left(\frac{5}{6}\right)^5$$

Example 15.52. *A random variable X has a uniform distribution over $(-3, 3)$. Find k for which $P(X > k) = \frac{1}{2}$.*

Also evaluate $P(X < 2)$ given $P(|X| > 1) = \frac{1}{3}$.

Solution. (i) Density of $X = f(x) = \frac{1}{b-a} = \frac{1}{3-(-3)} = \frac{1}{6}$

$\therefore$ $P(X > k) = 1 - P(X < k) = 1 - \int_{-3}^{k} f(x)\, dx$

$= 1 - \frac{1}{6}\int_{-3}^{k} dx = 1 - \frac{1}{6}(k + 3) = \frac{1}{2}$ (given)

This gives $k = 1$.

(ii) $P(X < 2) = \int_{-3}^{2} f(x)\, dx = \frac{1}{6}\int_{-3}^{2} dx = \frac{5}{6}$.

(iii) $P(|X - 2| < 2) = P(2 - 2 < X < 2 + 2) = P(0 < x < 4) = \int_{0}^{4} f(x)\, dx = \frac{1}{6}\int_{0}^{4} dx = \frac{4}{6}$.

PROBLEMS 15.8

1. Show that the mode of the geometric distribution $P(x) = \left(\frac{1}{2}\right)^x$, $x = 1, 2, 3, \ldots$ is unity.
2. Show that the rectangular distribution $f(x) = k$, $0 \leq x \leq 1$,

   mean $= \frac{1}{2}$, median $= \frac{1}{2}$ and mode is any value and standard deviation $= \frac{1}{\sqrt{12}}$.

* It was first used by Swedish scientist Weibull in 1951.

c. Find the mean and variance of the uniform distribution given by $f(x) = 1/n$, $x = 1, 2, \ldots, n$.

d. Show that for the symmetrical distribution
$dF = \sin^2 \theta$, $0 < \theta < \pi$,
the mean and SD are both equal to $\pi/2$.

e. Find the mean and variance of the exponential distribution $f(x) = \frac{1}{2} e^{-\frac{1}{2}x}$, $x > 0$. (Physical, SDE)

f. Find the moment generating function for the triangular distribution given by
$f(x) = x$, $0 < x < 1$
$= 2 - x$, $1 < x < 2$.

g. Show that for the Gamma distribution $f(x) = \frac{e^{-x} x^{n-1}}{\Gamma(n)}$, $0 < x < \infty$, the mean and variance are both equal to n.

h. Find the moment generating function of the Gamma distribution $f(x) = \frac{1}{\Gamma\left(\frac{3}{4}\right)} e^{-x} x^{-\frac{1}{4}}$, $x \geq 0$, and hence find its mean.

(I.A.S.T.C., 2008 / Madras, 2005)

i. (Chebyshev's inequality). If x is a continuous random variable with mean μ and variance σ^2, then for any positive real number c,
$$P(|x - \mu| \geq c\sigma) \leq \frac{1}{c^2}, \text{ or } P(|x - \mu| < c\sigma) \geq 1 - \frac{1}{c^2}.$$
This result is known as Chebyshev's inequality. It gives bounds to the probability that the value of the variate (better a standard variable) will differ from its mean by more than $c\sigma$.

ii. For the points on a symmetrical die, prove that Chebyshev's inequality gives
$$P(|x - \mu| > 2.5\sigma) < 0.16,$$
while the actual probability is zero.

iii. For the Geometrical distribution $P(x) = pq^{x-1}$, $x = 1, 2, 3, \ldots$, prove that Chebyshev's inequality gives
$$P(|x - \mu| < k\sigma) \geq 1 - \frac{1}{k^2},$$
while the actual probability is $1 - q^k$.

4.15.2 OBJECTIVE TYPE OF QUESTIONS

Multiple Choice Questions

Select the correct answer out of the alternatives given in each of the following problems :

1. The probability that A happens is $1/3$. The odds against happening of A are
(a) 2 : 1 (b) 1 : 2 (c) 1 : 3 (d) 3 : 1.

2. The odds in favour of an event A are 3 to 5. The probability of occurrence of A is
(a) $\frac{3}{8}$ (b) $\frac{5}{8}$ (c) $\frac{1}{2}$.

3. The probability that A passes a test is $\frac{2}{3}$ and the probability that B passes the same test is $\frac{4}{5}$. The probability that only one of them passes is
(a) $\frac{2}{15}$ (b) $\frac{4}{15}$ (c) $\frac{1}{15}$ (d) $\frac{7}{15}$.

4. A bag contains three tickets on which the chances of winning is 1/10; 5 more tickets on which the chances of winning is 1/20. The chances that one will win one of them is
(a) $\frac{3}{200}$ (b) $\frac{109}{200}$ (c) $\frac{109}{200}$ (d) $\frac{109}{200}$.

5. The probability that a non-leap year should have 53 Tuesdays is ...

6. The probability of getting 2 or 3 or 4 from a throw of single die is ...

7. The mean of the Binomial distribution with n observations and probability of success p is
(a) pq (b) npq (c) $\sqrt{npq}$ (d) $\sqrt{np}$.

8. If the mean of a Poisson distribution is m, then S.D. of this distribution is
(a) m^2 (b) $\sqrt{m}$ (c) m (d) none of these.

* See Solution on page 97.

8. The SD of the binomial distribution is
   (a) $\sqrt{npq}$ (b) $\sqrt{np}$ (c) $\sqrt{npq}$ (d) npq
10. In a Poisson distribution if $2P(1) = 3P(2)$ then the mean is
    (a) 0 (b) 1 (c) 3 (d) 2
11. If the probability of hitting a target by one shot is p and q is the probability that out of two shots, exactly one hits the target is ...
12. For a Poisson variate x, $P(x = 1) = P(x = 2)$. Then the mean of x is ...
13. If $P(A) = 0.65$, $P(A) = 0.50$ and $P(A \cap B) = 0.40$, then $P(A \cup B) =$ ...
14. If A and B are independent, $P(A) = 0.16$ and $P(A \cup B) = 0.36$, then $P(B) =$ ...
15. The probability distribution of the number of heads when two coins are tossed, is
16. The exact statement of law of probability states that ...
17. The area under the standard normal curve which lies between $z = 0.98$ and $z = 1.98$ is ...
    (Given $P(0 < z < 1.98) = 0.4976$, $P(0 < z < 0.98) = 0.3365$)
18. The mean, median and mode of a normal distribution are ...
19. The mean and variance of a Poisson distribution are ...
20. If A and B are two exactly exclusive events, then $P(A \cup B) =$ ...
21. For a normal distribution $\beta_1 =$ ... and $\beta_2 =$ ...
22. The number of ways in which five people can be lined up to get on a bus are ...
23. A bag contains 10 white marbles and 5 black marbles. The number of ways in which one can purchase 4 of these and of which 3 are white is ...
24. The probability of getting a total of 8 when a pair of dice is tossed is ...
25. If $P(A) = 3/4$ and $P(A \cap B) = 3/8$, then $P(A/B) =$ ...
26. If two unbiased dice are thrown simultaneously, the probability that the sum of the numbers on them is 8 or 10 is ...
27. If X is a Poisson variable such that $P(X = 2) = P(X = 3)$, then $P(X = 5) =$ ...
28. An unbiased die is tossed twice. Find the probability of obtaining the score 6 ...
29. The variance of Poisson distribution with parameter $\lambda = 3$ is ...
30. The distribution in which mean, median, mode are equal is ...
31. For the Poisson variate, probability of getting at least one success is ...
32. Total number of events in sampling of attributes is ...
33. If X be normal with mean 10 and variance 4, then $P(X < 10) =$ ...
34. If X is a normal variate with parameters μ and σ, then $\frac{X - \mu}{\sigma}$ is a ...
35. In a normal distribution, mean deviation : standard deviation = ...
36. If A and B are independent and $P(A) = 1/2$, $P(B) = 2/3$ then $P(A \cup B) =$ ...
37. If X is the random variable representing the number of the sixes in a throw of six dice, then $E(X) =$ ...
38. If X is a binomial variate with $p = 1/6$ for the experiment of 180 trials, then the standard deviation is ...
39. The area under the whole statistical curve is ...
40. Given $X \sim B(n, p)$, then the conditions under which X can be a Poisson distribution are ...
41. If A and B are mutually exclusive events then $P(A \cup B) =$
42. The probability of selecting a white balls from a bag containing a white and b red balls is ...
43. The mean of the binomial distribution is ...
44. If A and B are mutually exclusive events, $P(A) = 0.35$, $P(B) = 0.45$, then $P(A \cup B) =$ ... and $P(A \cap B) =$ ...
45. If the mean and variance of a binomial variate are 12 and 4, then the distribution is ...
46. If x is a Poisson variate such that $P(x = 2) = 9P(x = 4) + 90 P(x = 6)$, find the mean = ...
47. p / the theorem stated above the origin is known as the origin is ...
48. The chance of throwing 7 in a single throw with two dice is ...
49. If A and B are any two events with $P(A) = 1/2$, $P(B) = 1/3$ and $P(A \cap B) = 1/4$, then $P(A/B) =$ ...
50. In the roll of six ideal dice, the probability of getting a prime number is ...
51. If A and B are mutually exclusive events, $P(A \cup B) = 0.6$, $P(B) = 0.4$, then $P(A) =$ ...
52. The probability that a leap year should have 53 Sundays is ...
53. The probability density function of a Normal distribution is ...
54. The probable error of ... curve is 0.6745 approximately.
55. For a particular data to be true, the parameters required are ...

81. A card is drawn from a well-shuffled pack of 52 cards, then the probability of drawn card being red and a heart is ...
82. If $P(x) = P(2)$, find the mean of the Poisson distribution is ...
83. Binomial theorem states that ...
84. If x is a Poisson variate such that $P(x = 1) = 0.2$ and $P(x = 2) = 0.3$, then $P(x = 0) = ...$
85. If x and y are events such that $P(X \cup Y) = 3/4$, $P(X \cap Y) = 1/4$, and $P(X) = 2/3$, then $P(Y) =$
86. The inverse of throwing a die is single throw with two outcomes success or failure such as getting a 1 or not a 1 is called a single die. (True or False)
87. If the mean of a Poisson distribution is 9, then its variance is 18. (True or False)
88. If X is a real valued random variable and a and b are real numbers, then $X + b$ is a random variable where X and a are constants. (True or False)
89. If X is a binomial variable with parameters $\lambda = 10$, $p = 0.5$, then its standard deviation is 2.25. (True or False)
90. The mean of a binomial distribution is 5 and 0.25. is ? (True or False)
91. The mean and variance of Poisson distribution are equal. (True or False)
92. The graph of the normal distribution is symmetric with respect to the line $x = \mu$. (True or False)
93. The standard deviation of a binomial distribution is npq. (True or False)
94. $f(x) = 1/n$ for $X = x = 1$ is a valid probability density function. If $k =$
95. If $Var x = 2$. Then $Var 2x =$
96. The p.d.f of an exponential distribution is $\lambda, x, \lambda, x, ...$
97. If X is a binomially distributed $(n = 2, 2)$, then the variance is ...
98. The variance of Poisson distribution with parameter $\lambda = 2$ is ...
99. In Gamma distribution, as the parameter λ, the variable of ...
100. If $f(x) = kx^n$, $0 < x < 1$, $f(x)$ = elsewhere, is a p.d.f, then $k =$ ...
101. The mgf of a random variable X is $(1 - 2t)^{-6}$, then $E(x)$ is ...
102. A random variable Y has F-distribution with (n_1, n_2) degrees of freedom, then $1/Y$ has the same distribution with ... degrees of freedom.
103. If X is a continuous random variable having the p.d.f $f(x)$, then the mgf about the origin is given by ...
104. If $f(x) = 3^x/x$, $x = 0, 1, 2, 3, 4$ is the probability distribution of a discrete random variable, then $k =$
105. The p.d.f of Gamma variable is ...
106. If X is uniformly distributed over $(0, 10)$, then $P(X \leq 2) =$ ...
107. The marks obtained by students were found normally distributed with mean 78 and variance 100. The percentage of students who scored above 80 marks is ...
108. When four unbiased coins are tossed, the probability of getting two tails is ...
109. If $f(x) = \begin{cases} kax^{a-1}, & x > 0 \\ 0, & \text{elsewhere, is the p.d.f of a, then } k = \end{cases}$
110. If f is a uniform distribution defined in the interval $(4, 6)$, then its variance is ...
111. The p.d.f of a continuous random variable is $f(x) = kx^2$, $0 \leq x \leq 3$. If $P(X \leq 3) = 27/64$, find the value of k is
(a) 1 (b) 0 (c) $1/64$ (d) $27/64$
112. The relation between probability density function and cumulative density function of a random variable is ...
113. If X has Poisson distribution with parameter λ, then $P(X)$ is zero =
114. Range of F_x distribution is ...
115. If the p.d.f of x is $f(x) = kx(1 - x)$, $0 < x < 1$, then $k =$
116. The function $f(x) = \begin{cases} kx^2, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere, is a probability function, then } k = ... \end{cases}$
117. If the random variable x is uniformly distributed in $(0, 5)$, then the p.d.f is $f(x) = 0$, $0 < x < 5$, k, elsewhere. (True or False)
118. Exponential distribution $f(x)$ is defined by $f(x) = \alpha e^{-\beta x}$, $x > 0$, $\alpha > 0$, $\beta > 0$, where $\alpha =$ ...
119. The p.d.f of Beta distribution with $\alpha = 2$, $\beta = 4$ is $f(x) =$ ...
120. For a standard normal variate z, $P(-0.32 \leq z \leq 0) =$ ...

CHAPTER 27

Sampling & Inference

1. Introduction 2. Sampling distribution (Standard error) 3. Testing of Hypothesis (Large) 4. Level of significance 5. Tests of significance 6. Confidence limits 7. Simple sampling of attributes 8. Test of significance for large samples 9. Comparison of large samples 10. Sampling of variables 11. Central limit theorem 12. Confidence limits for unknown means 13. Test of significance for means of two large samples 14. Sampling of variables—small samples 15. Student's t-distribution 16. Significance test of a sample mean 17. Significance test of difference between sample means 18. Chi-square test 19. Goodness of fit 20. Fisher's z-distribution 21. Objective Type of Questions

(1) INTRODUCTION

We know that a small section selected from the population is called a sample and the process of drawing a sample is called sampling. It is essential that a sample must be a random selection so that each member of the population has the same chance of being included in the sample. Thus the fundamental assumption underlying the theory of sampling is Random sampling.

A special case of random sampling in which each event has the same probability p of success and the chance of success of different events are independent whether previous trials have been made or not, is known as simple sampling.

The statistical constants of the population such as mean (μ), standard deviation (σ) etc. are called the parameters. Similarly, constants for the sample drawn from the given population i.e., mean ($\bar{x}$), standard deviation (s) etc. are called the statistic. The population parameters are in general, not known and their estimates given by the corresponding sample statistics are used. We use the Greek letters to denote the population parameters and Roman letters for sample statistics.

(2) Objectives of sampling. Sampling aims at gathering the maximum information about the population with the minimum effort, money and time. The object of sampling studies is to obtain the best possible values of the parameters under specific conditions. Sampling determines the reliabilities of these estimates. The logic of the sampling theory is the logic of induction in which we pass from a part (sample) to general population. Such a generalisation from sample to population is called Statistical Inference.

(2) SAMPLING DISTRIBUTION

Consider all possible samples of size n which can be drawn from a given population at random. For each sample, we can compute a statistic. The statistic of the samples will not be identical. If we group these different values according to their frequencies, the frequency distribution so formed is known as sampling distribution of the statistic. Similarly we can have sampling distribution of the standard deviation etc.

While drawing each sample, we put back the previous sample so that the parent population remains the same. This is called sampling with replacement and all the subsequent formulae will pertain to sampling with replacement.

(2) Standard error. The standard deviation of the sampling distribution is called the *standard error* (S.E.). Thus the standard error of the sampling distribution of means is called *standard error of means*. The standard error is used to assess the difference between the expected and observed values. The reciprocal of the standard error is called *precision*.

If $n \geq 30$, a sample is called *large*; otherwise *small*. The sampling distribution of large samples is assumed to be normal.

[3] TESTING A HYPOTHESIS*

To reach decisions about populations on the basis of sample information, we make certain assumptions about the populations involved. Such assumptions, which may or may not be true, are called *statistical hypotheses*. By testing a hypothesis is meant a process for deciding whether to accept or reject the hypothesis. The method consists in assuming the hypothesis as correct and then computing the probability of getting the observed sample. If this probability is less than a certain prescribed value the hypothesis is rejected.

(1) Errors. If a hypothesis is rejected while it should have been accepted, we say that a *Type I error* has been committed. On the other hand, if a hypothesis is accepted when it should have been rejected, we say that a *Type II error* has been made. The statistical testing of hypothesis aims at limiting the Type I error to a prescribed value (say 5% or 1%) and to minimize the Type II error. The only way to reduce both types of errors is to increase the sample size, if possible.

(2) Null Hypothesis. The hypothesis formulated for the sake of rejecting it, under the assumption that it is true, is called the *null hypothesis* and is denoted by H_0. To test whether one procedure is better than another, we assume that there is no difference between the procedures. Similarly to test whether there is a relationship between two variables, we take H_0 that there is no relationship. By accepting a null hypothesis, we mean that on the basis of the statistic calculated from the sample, we do not reject the hypothesis. It however, does not imply that the hypothesis is proved to be true. Nor an rejection implies that it is disproved.

[4] LEVEL OF SIGNIFICANCE

The probability level below which we reject the hypothesis is known as the *level of significance*. The region in which a sample value falling is rejected, is known as the *critical region*. We generally take two critical regions which cover 5% and 1% areas of the normal curve. The shaded portion in the figure corresponds to 5% level of significance. Thus the probability of the value of the variate falling in the critical region is the *level of significance*.

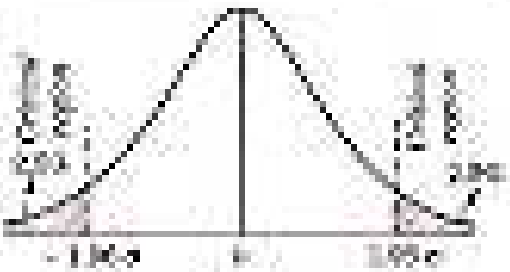


Fig 27.1

Depending on the nature of the problem, we use a single tail test or a double tail test to estimate the significance of a result. In a double tail test, the areas of both the tails of the curve representing the sampling distribution are taken into account whereas in the single tail test, only the area on the right or on the left is taken into consideration. For instance, to test whether a coin is biased or not, double tail test should be used, since a biased coin gives either more number of heads than tails (which corresponds to right tail), or more number of tails than heads (which corresponds to left tail).

(3) Tests of significance. The procedure which enables us to decide whether to accept or reject the hypothesis is called the *test of significance*. Here we test whether the differences between the sample values and the population values for the values given by two samples are so large that they signify evidence against the hypothesis or these differences are accidental to account for fluctuations of sampling.

[5] CONFIDENCE LIMITS**

Suppose that the sampling distribution of a statistic S is normal with mean μ and standard deviation σ. As in Fig 27.1 the sample statistic S can be expected to lie in the interval $(\mu - 1.96\sigma, \mu + 1.96\sigma)$ for 95% cases. Or, we can be confident of finding μ in the interval $(S - 1.96\sigma, S + 1.96\sigma)$ in 95% cases. Because of this, we call

*The Australian statistician J. Neyman (1894–1981) and the English statistician E.S. Pearson (1895–1980) son of Karl Pearson (see footnote p. 813), developed a systematic theory of tests around 1928.

**J. Neyman developed the modern theory and terminology of confidence limits.

$\bar{x} - 1.96\sigma$, $\bar{x} + 1.96\sigma$ the 95% confidence interval for estimation of μ. The ends of this interval i.e. $\bar{x} \pm 1.96\sigma$ are called 95% confidence limits or *fiducial limits* for $\bar{x}$. Similarly $\bar{x} \pm 2.58\sigma$ are 99% confidence limits. The numbers 1.96, 2.58 etc. are called confidence coefficients. The values of confidence coefficients corresponding to various levels of significance can be found from the normal curve area table VI – Appendix 2.

2.10.3 SIMPLE SAMPLING OF ATTRIBUTES

The sampling of attributes may be regarded as the selection of samples from a population whose members possess the attributes R or not R. The presence of R may be called a success and its absence a failure.

Suppose we draw a simple sample of n items. Clearly it is same as a series of n independent trials with the same probability p of success. The probabilities $(q + p)^n$ successes are the terms in the binomial expansion of $(q + p)^n$ where $q = 1 - p$.

We know that the mean of this distribution is np and standard deviation is $\sqrt{(npqp)}$ i.e., the expected value of success in a sample of size n is np and the standard error is $\sqrt{npq}$.

If we consider the proportion of successes, then

(i) the proportion of successes $= np/n = p$.

(ii) standard error of the proportion of successes

$$= \sqrt{\left(n.\frac{p}{n}.\frac{q}{n}\right)} = \sqrt{\left[\frac{pq}{n}\right]}$$

and (iii) prediction of the proportion of successes $= \sqrt{n^2pq}$, which varies as $\sqrt{n}$, since p and q are constants.

2.10.4 TEST OF SIGNIFICANCE FOR LARGE SAMPLES

We know that the binomial distribution tends to normal for large n. Suppose we wish to test the hypothesis that the probability of success in each trial is p. Assuming it to be true, the mean μ and the standard deviation of the sampling distribution of number of successes are np and $\sqrt{npqp}$ respectively.

For a normal distribution, only 5% of the members lie outside $\mu \pm 1.96\sigma$ while only 1% of the members lie outside $\mu \pm 2.58\sigma$.

If x be the observed number of successes in the sample and z is the standard normal variate then $z = (x - np)/\sqrt{npq}$.

Then we have the following test of significance :

(i) If $|z| < 1.96$, difference between the observed and expected number of successes is not significant.

(ii) If $|z| > 1.96$, difference is significant at 5% level of significance.

(iii) If $|z| > 2.58$, difference is significant at 1% level of significance.

Example 17.1 A coin was tossed 400 times and the head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% level of significance. (V.T.U., 2003)

Solution. Suppose the coin is unbiased.

Then the probability of getting the head in a toss $= \frac{1}{2}$

$\therefore$ expected number of successes $= \frac{1}{2} \times 400 = 200$

and the observed value of success is $x = 216$.

Then the excess of observed value over expected value $= 216 - 200 = 16$

Also S.D. of simple sampling $= \sqrt{npq} = \sqrt{\left(400 \times \frac{1}{2} \times \frac{1}{2}\right)} = 10$

Hence $z = \frac{x - np}{\sqrt{npq}} = \frac{16}{10} = 1.6$

As $z < 1.96$, the hypothesis is accepted at 5% level of significance i.e., we conclude that the coin is unbiased at 5% level of significance.

Example 9.12. *A die was thrown 9000 times and a throw of 3 or 4 was observed 3240 times. Show that the hypothesis of the die being unbiased is not consistent with observation.* (U.P.T.U., 2003)

Solution. Suppose the die is unbiased.

Then the probability of throwing 3 or 4 with one die $= \frac{1}{3}$

The expected number of successes $= \frac{1}{3} \times 9000 = 3000$

and the observed value of successes $= 3240$

Thus the excess of observed value over expected value $3240 - 3000 = 240$

Also S.D. of simple sampling $= \sqrt{npq} = \sqrt{\left(9000 \times \frac{2}{3} \times \frac{1}{3}\right)} = 44.72$

Hence $z = \frac{X - np}{\sqrt{npq}} = \frac{240}{44.72} = 5.4$ (nearly).

As $z > 3.29$, the hypothesis has to be rejected at 1% level of significance and we conclude that the die is biased.

Example 9.13. *In locality consisting of 20000 families, a sample of 900 families was selected at random. Of these 900 families, 200 families were found to have a monthly income of ₹ 300 or less. It is desired to estimate how many out of 20000 families have a monthly income of ₹ 300 or less. With what limits would you place your estimate?*

Solution. Here $p = \frac{200}{900} = \frac{100}{450}$ and $q = \frac{317}{450}$

$\therefore$ standard error of the population of families having a monthly income of ₹ 300 or less

$$= \sqrt{\left[\frac{pq}{n}\right]} = \sqrt{\left[\frac{100}{450} \times \frac{317}{450} \times \frac{1}{900}\right]} = .015 \times 1.256$$

Hence taking $\frac{100}{450}$ or 22.5% to be the estimate of families having a monthly income of ₹ 300 or less in the locality, the limits are $(22.5 \pm 3 \times 1.256)\%$ i.e., 26% and 19% approximately.

9.13 COMPARISON OF LARGE SAMPLES

Two large samples of sizes n_1, n_2 are taken from two populations giving proportions of attributes A's as p_1, p_2 respectively.

(i) On the hypothesis that the populations are similar as regards the attribute A, we combine the two samples to find an estimate of the common value of proportion of A's in the population which is given by

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

If σ_1, σ_2 be the standard errors in the two samples then

$$\sigma_1^2 = \frac{pq}{n_1} \text{ and } \sigma_2^2 = \frac{pq}{n_2}$$

If σ be the standard error of the difference between p_1 and p_2, then

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 = pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right) \quad \therefore \quad z = \frac{p_1 - p_2}{\sigma}$$

If $z > 3$, the difference between p_1 and p_2 is real one.

If $z < 2$, the difference may be due to fluctuations of simple sampling.

But if z lies between 2 and 3, then the difference is significant at 5% level of significance.

(ii) If the proportions of A's are not the same in the two populations from which the samples are drawn, but p_1 and p_2 are the true values of proportions of A's and the difference $p_1 - p_2$ is given by

$$\sigma = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}$$

Hence $z = \frac{p_1 - p_2}{\sigma}$ < 2, the difference could have arisen due to fluctuations of simple sampling.

Example 27.5. In a city A, 20% of a random sample of 900 school boys had a certain disease. In another city B, 18.5% of a random sample of 1000 school boys had the same disease. Is the difference between the proportions significant? (V.T.U., 2007 S)

Solution. We have $n_1 = 900, n_2 = 1000$

and $p_1 = \frac{20}{100} = \frac{1}{5}, p_2 = \frac{18.5}{100}$

$$p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{300 + 370}{900 + 1000} = 0.33$$

and $q = 1 - 0.33 = 0.67$

Then $\sigma^2 = pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right) = 0.33 \times 0.67\left(\frac{1}{900} + \frac{1}{1000}\right) = 0.0012$

giving $\sigma = 0.035$ nearly.

Also $p_1 - p_2 = \frac{1.5}{100} = 0.015 \quad \therefore \quad z = \frac{p_1 - p_2}{\sigma} = \frac{0.015}{0.035} = 0.37$

As $z < 1$, the difference between the proportions is not significant.

Example 27.6. In two large populations there are 30% and 25% respectively of fair haired people. Is this difference likely to be hidden in samples of 1200 and 900 respectively from the two populations?

(Calicut Univ., 2001)

Solution. Here $p_1 = 0.3, p_2 = 0.25$ so that $p_1 - p_2 = 0.05$.

$$\therefore \quad \sigma^2 = \frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2} = \frac{0.3 \times 0.7}{1200} + \frac{0.25 \times 0.75}{900}$$

so that $\sigma = 0.0155$

$$\therefore \quad z = \frac{p_1 - p_2}{\sigma} = \frac{0.05}{0.0155} = 2.5 \text{ nearly}$$

Hence it is unlikely that the real difference will be hidden.

EXERCISE 27(C)

1. A die is thrown 1000 times and it falls with 8 upwards 175 times. Is the die biased? (V.T.U., 2000)
2. 12 dice are thrown 4096 times and a throw of 5, 6 is observed 1800 times. Suppose that 1840 throws of 5, 6 were actually observed. Do you think that the observed value deviates from the expected value? If so, may the deviation from the expected value be due to fluctuations of simple sampling?
3. Balls are drawn from a bag containing equal number of black and white balls, each ball being replaced before drawing the next ball. In 2000 drawings 1050 black and 950 white balls have been drawn. Do you suspect some bias on the part of the drawer?
4. A sample of 1000 days is taken from the meteorological records of a certain district and 120 of them are found to be foggy. What are the probable limits to the percentage of foggy days in the district?
5. In a group of 50 fair coins the number of heads is found to be 31. Is the ratio of the observed proportion of heads to that in accordance with the hypothesis that the coins should be in equal proportion?
6. A random sample of 900 apples was taken from a large consignment and 10% were found to be bad. Establish the proportion of bad apples in the consignment as well as the standard error of the estimate. Explain that the percentage of bad apples in the consignment as shown correct to the nearest 0.5 could be 10%.
7. 400 children are chosen at random from a large town and 200 are found to be under weight. Assuming the conditions of simple sampling, estimate the percentage of children who are under weight in the town and assign limits within which the percentage probably lies?

9. A random sample of 50 is picked up from a population and the mean is found to be 60. Find the 95% confidence limits for the mean of the population if the standard deviation of the population is 5.
10. One type of battery is found to have a mean life of 3 years with a standard deviation of 0.5 years, while another type of battery has a mean life of 2.8 years with a standard deviation of 0.4 years. If 100 batteries of each type are tested, find whether the difference in the two types of batteries is significant at the 5% level of significance.
11. A sample of 1000 is taken from a normal population. The mean of the sample is 4.35 and the standard deviation is 2.31. Find the 95% confidence limits for the mean of the population. Is it likely that the population is 4.5? Explain your answer.
12. In a sample of 600 people from a city, 360 are found to be in favour of a proposal to build a new theatre. Can we say that the proposal is supported by the majority of the people at the 5% level of significance?

22.8 (3) SAMPLING OF VARIABLES

We now consider sampling of a variable such as weight, height, etc. Each member of the population gives a value of the variable and the population is a frequency distribution of the variable. Thus a random sample of values from the population is one set of corresponding values of the variables from those of the distribution.

(3) Sampling distribution of the means. *If a population is distributed normally with mean μ and standard deviation σ, then the means of all possible random samples of size n, are also distributed normally with mean equal to μ and standard error of $\sigma/\sqrt{n}$.* This result shows how the precision of a sample mean increases as the sample size increases.

22.10 CENTRAL LIMIT THEOREM

This is a very important theorem regarding the distribution of the mean of a sample *if the parent population is not normal and the sample size is large.*

If the variable X has a non-normal distribution with mean μ and standard deviation σ, then the limiting distribution of

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \quad \text{as } n \to \infty, \text{ is the standard normal distribution (i.e., with mean 0 and unit S.D.)}$$

There is no restriction upon the distribution of X except that it has a finite mean and variance. This theorem holds well for a sample of 30 or more which is regarded as large.

Thus if the population is normal, the sampling distribution of the mean is also normal with mean μ and S.D. $\sigma/\sqrt{n}$, while for large samples the same result holds even if the distribution of the population is not normal. This property is of universal use and throws light on the importance of normal distribution in statistical theory.

22.11 CONFIDENCE LIMITS FOR UNKNOWN MEAN

Let the population from which a random sample of size n is drawn, have mean μ and S.D. σ. If μ is not known, there will be a range of values of μ for which observed mean $\bar{x}$ of the sample is not significant at any assigned level of probability. The relative deviation of $\bar{x}$ from μ is $(\bar{x} - \mu)/\sigma/\sqrt{n}$.

If $\bar{x}$ is not significant at 5% level of probability, then

$$|(\bar{x} - \mu)/\sigma/\sqrt{n}| < 1.96 \quad \text{i.e.,} \quad \bar{x} - 1.96\sigma/\sqrt{n} < \mu < \bar{x} + 1.96\sigma/\sqrt{n}$$

Thus 95% confidence or fiducial limits for the mean of the population corresponding to given sample are $\bar{x} \pm 1.96\sigma/\sqrt{n}$.

Similarly 99% confidence limits for μ are $\bar{x} \pm 2.58\sigma/\sqrt{n}$.

Example 22.6. *A sample of 200 members is found to have a mean of 3.4 cm. Can it be reasonably regarded as a simple random sample from a large population with mean 3.25 cm and S.D. 2.61 cm?*

Solution. Here $\bar{x} = 3.4$ cm, $n = 200$, $\mu = 3.25$ and $\sigma = 1.61$ cm

$$\therefore \quad z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{3.4 - 3.25}{1.61/20} = 2.5$$

As $z > 1.96$, the deviation of this sample mean from the mean of the population is significant at 5% level of significance. Hence it cannot be regarded as a random sample.

Example 27.7. The mean of a certain normal population is equal to the standard error of the mean of the samples of 36 from that population. Find the probability that the mean of the sample of 36 items will exceed the mean by $\frac{1}{2}$ of a standard deviation.

Solution. If μ be the mean and σ the S.D. of the population, then

$$\text{S.E. of the sample means} = \frac{\sigma}{\sqrt{36}} = \frac{\sigma}{6}$$

Also for a sample of size 36, we have $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{36}} = \frac{\bar{x} - \mu}{\sigma/6} = \frac{\frac{1}{2}\sigma}{\sigma/6} = \frac{3}{2}$

Hence z corresponds to $z = \frac{3}{2}$.

$\therefore$ the probability that a normal variate $z < \frac{3}{2}$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{3}{2}} e^{-\frac{1}{2}z^2} dz = \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\frac{1}{2}z^2} dz$$

$= 0.5 + 0.4332 = 0.9332$, from the tables.

Example 27.8. A sample of school boys is taken to study their height. It is found that the arithmetic mean of the sample is 68 inches and the S.D. is 2.5 inches. Find the limits of value of μ that will represent the result with 99% confidence.

Solution. S.E. of the proportion of heads $= \sqrt{\frac{pq}{n}} = \sqrt{\frac{\frac{1}{2} \cdot \frac{1}{2}}{n}} = \frac{1}{2\sqrt{n}}$

50% of results are 68 in. or 67.5 of the total area under the normal curve on each side of the mean.

$\therefore$ the corresponding value of $z = 1.645$, from the tables.

Then $\mu = 1.645\sigma = 2.5 \times 1.645 = 4.11$.

i.e., $67.5 - 1.645 \cdot \frac{1}{2\sqrt{n}} = 0.49$ and $67.5 + 1.645 \cdot \frac{1}{2\sqrt{n}} = 0.51$

where $\frac{1.645}{2\sqrt{n}} = 0.01$ or $\sqrt{n} = \frac{164.5}{2}$ or $n = 6724$ approximately.

27.12 TEST OF SIGNIFICANCE FOR MEANS OF TWO LARGE SAMPLES

(a) Suppose two random samples of sizes n_1 and n_2 have been drawn from the same population with S.D. σ. We wish to test whether the difference between the sample means $\bar{x}_1$ and $\bar{x}_2$ is significant or is merely due to fluctuations of sampling.

If the samples are independent, then the standard error of the difference of their means is given by

$$\sigma^2 = \sigma_1^2 + \sigma_2^2$$

where $\sigma_1 = \sigma/\sqrt{n_1}$, $\sigma_2 = \sigma/\sqrt{n_2}$ are the S.E.s of the means of the two samples.

$\therefore$ $\sigma = \sigma\sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$. Hence $z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

is normally distributed with mean zero and S.D. 1.

Test of significance for n_1, n_2 being large :

If $z < 1.96$, then the difference is significant at 5% level of significance.

If $z > 2.6$ it is highly probable that either the samples have not been drawn from the same population or the sampling is not random.

(b) If the samples are known to be drawn from different populations with means μ_1, μ_2 and standard deviations σ_1 and σ_2. Then the standard error of their means is given by

$$\sigma = \sqrt{\left[\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right]}$$

Assuming that the two populations have the same mean (i.e., $\mu_1 = \mu_2$), the difference of the means of the samples will be normally distributed with mean zero and S.D. σ. Now the same procedure of test of significance is applied.

Example 27.8. *The means of simple samples of sizes 1000 and 2000 are 67.5 and 68.0 cm respectively. Can both samples be regarded as drawn from the same population of S.D. 2.5 cm.* (Madras, 2005)

Solution. We have $\bar{x}_1 = 67.5$, $\bar{x}_2 = 68.0$

$n_1 = 1000$, $n_2 = 2000$

On the hypothesis that the samples are drawn from the same population of S.D. $\sigma = 2.5$, we get

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\left[\frac{1}{n_1} + \frac{1}{n_2}\right]}} = \frac{67.5 - 68.0}{2.5\sqrt{\left[\frac{1}{1000} + \frac{1}{2000}\right]}}$$

$$= \frac{0.5}{2.5 \times 0.0387} = \frac{0.5}{0.0975} = 5.1$$

Here the difference between the sample means i.e., 5.1 is very much greater than 1.96 and is therefore significant. Thus, the samples cannot be regarded as drawn from the same population.

Example 27.10. *A sample of height of 6400 soldiers has a mean of 67.85 inches and a standard deviation of 2.56 inches while a sample of heights of 1600 sailors has a mean of 68.55 inches and a standard deviation of 2.52 inches. Do the data indicate that the sailors are taller than the soldiers?*

Solution. Here $\bar{x}_1 = 67.85$, $\sigma_1 = 2.56$, $n_1 = 6400$

$\bar{x}_2 = 68.55$, $\sigma_2 = 2.52$, $n_2 = 1600$.

∴ S.E. of the difference of the mean heights is

$$\sigma = \sqrt{\left[\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right]} = \sqrt{\left[\frac{2.56^2}{6400} + \frac{2.52^2}{1600}\right]}$$

$$= \sqrt{.001024 + .003969} = 0.068 \text{ nearly}$$

Also difference between the means $= \bar{x}_2 - \bar{x}_1 = 0.7$, which > 1.96. This is highly significant. Hence the data indicate that the sailors are on the average taller than the soldiers.

EXERCISE 27.2

1. A sample of 400 items is taken from a normal population whose mean is 4 and variance 4. If the mean of the sample is 4.05, can the sample be regarded as a simple sample ?
2. To know the opinion regarding rise in T.V. prices in Delhi, a sample of 200 is taken. The mean price is Rs. 4.95 per set with S.D. Rs. 0.25 per set. Can you say that the mean price in Delhi is Rs. 5 per set ? (Hint : Take $\mu = 5$ as the mean price in the population.)
3. A normal population has a mean 0.1 and a S.D. of 2.1. Find the probability that the mean of a simple sample of 900 members is likely to lie between 0.2 and 0.3.
4. If the mean breaking strength of cables supplied by a manufacturer is 1800 lbs. with a standard deviation of 100 lbs., how large a sample should be used in order that there be one chance in 100 that the mean breaking strength of the sample is less than 1750 lbs. ?

   Hint. $|z| = \left|\frac{\bar{x} - \mu}{\sigma}\sqrt{n}\right| = \frac{3}{2.3}$ ...

   Also from table $P(|z| > 2.33) = 0.02$. Hence $n = 43$ nearly.

6. A researcher wishes to estimate mean of a population by using sufficiently large sample. The probability of 95% that sample mean will not differ from the true mean by more than 25% of the S.D. How large a sample should be used ?
7. The density function of a random variable x is $f(x) = kx^{10}e^{-x}$, $0 < x < \infty$. Find the upper 5% point of the distribution of the means of the random samples of size 25 from the above population.
8. The heights of two large samples of 1000 and 2000 members are 165.5 cms. and 170 cms. respectively. Can the samples be regarded as drawn from the same population of standard deviation 2.5 cms.
9. If 100 persons are taken at a given city sample are found to have a mean height of 68.20 inches and 150 persons sample be the of 67.5 inches, is the difference significant that the mean height of the persons in groups does not of the same universe ? Assume the standard deviation of heights to be 2.56 inches.
10. A sample of 100 electric bulbs produced by manufacturer A showed a mean life time of 1200 hours with a standard deviation of 90 hours. A sample of 75 bulbs produced by manufacturer B showed a mean life time of 1250 hours, with a standard deviation of 120 hours. Is there a difference between the mean life time of two brands at a significance level of (i) 0.05 (ii) 0.01.
11. A random sample of 1000 men from North India showed an average height of 67 inches with a S.D. of 2.50. A sample of 1200 men from South India gives a mean height of 68.0 inches with a standard deviation of 2.5. Does the mean height of south differ significantly from that in the north ?

10.23 SAMPLING OF VARIABLES—SMALL SAMPLES

In case of large samples, sampling distributions approaches a normal distribution and values of sample statistics are considered best estimates of the parameters in population. It will no longer be possible to assume that statistics computed from small samples are normally distributed. As such, a new technique has been developed for small samples which involves the concept of 'degrees of freedom' which we explain below.

Number of degrees of freedom is the number of values in a set which may be assigned arbitrarily. For instance, if $x_1 + x_2 + x_3 = 15$ and if we assign any values of two of the variables, say x_1, x_2, then the value of x_3 will be known. The two variables are therefore, free and independent choices for finding the third. Hence there are two degrees of freedom. If there are n observations, the degrees of freedom (d.f.) are $(n - 1)$. In other words, while finding the mean of a small sample, one degree of freedom is used up and $(n - 1)$ d.f. are left to estimate the population variance.

10.24 (i) STUDENT'S t-DISTRIBUTION

Consider a small sample of size n, drawn from a normal population with mean μ and s.d. σ. If $\bar{x}$ and s_1 be the sample mean and s.d., then the statistic 't' is defined as

$$t = \frac{\bar{x} - \mu}{s_1}\sqrt{n} \quad \text{or} \quad t = \frac{\bar{x} - \mu}{s}\sqrt{(n-1)}$$

where $s = s_1\sqrt{\frac{n}{n-1}}$ denotes the S.D. of t. If we calculate t for each sample, we obtain the sampling distribution for t. This distribution known as 'Student's t-distribution', is given by

$$y = \frac{y_0}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}} \qquad \ldots(1)$$

where y_0 is constant such that the area under the curve is unity.

(1) Properties of t-distribution.

1. The curve is symmetrical about the line $t = 0$. For the normal curve, even only even powers of t appear as in (1). But it is more peaked than the normal curve with the same S.D. The t-curve approaches the horizontal axis less rapidly than the normal curve. Also t-curve attains its maximum value at $t = 0$ so that its mode coincides with the curve. [Fig. 10.2]

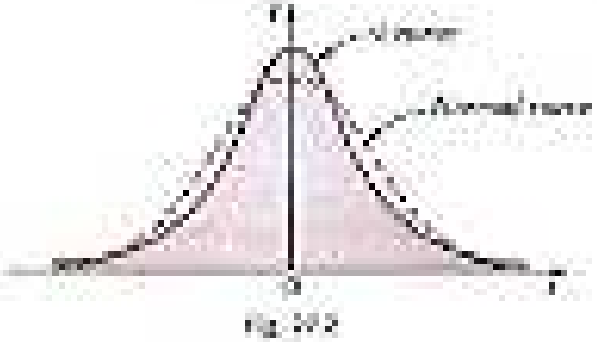


Fig. 10.2

*This distribution was first found by the English statistician W.S. Gosset in 1908 who wrote under the pen-name of 'Student'. R.A. Fisher defined it correctly and found its distribution in 1925.

II. The limiting form of t-distribution when $\nu \to \infty$ is given by $y = y_0 e^{-\frac{1}{2}t^2}$ which is a normal curve. This shows that t is normally distributed for large samples.

III. The probability P that the value of t will exceed t_0 is given by

$$P = \int_{t_0}^{\infty} y\,dt$$

The values of t_0 have been tabulated for various values of P for various values of ν from 1 to 30 (Table IV - Appendix 2).

IV. Moments about the mean

All the moments of odd order about the mean are zero, due to its symmetry about the line $t = 0$.

Moments of even order about the mean are

$$\mu_0 = \frac{\nu}{\nu - 2}, \quad \mu_4 = \frac{3\nu^2}{(\nu-2)(\nu-4)}, \text{ etc.}$$

The t-distribution is often used in tests of hypothesis about the mean when the population standard deviation σ is unknown.

27.15 SIGNIFICANCE TEST OF A SINGLE MEAN

Given a random small sample $x_1, x_2, x_3, \ldots, x_n$ from a normal population, we have to test the hypothesis that mean of the population is μ. For this, we first calculate $t = (\bar{x} - \mu)\sqrt{n}/s$,

where $$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad s^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

Then find the value of P for the given ν from the table.

If the calculated value of $|t| > t_{0.05}$, the difference between $\bar{x}$ and μ is said to be significant at 5% level of significance.

If $t > t_{0.01}$, the difference is said to be significant at 1% level of significance.

If $t < t_{0.05}$, the data is said to be consistent with the hypothesis that μ is the mean of the population.

Example 27.13. A random sample of ten boys had the following I.Q.'s: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do you think that the mean I.Q. of the population is 100? Discuss what can be said about the nature of the population. (U.P.T.U., 2003)

Solution. Let us assume that the population is normal with mean $\mu = 100$ and S.D. σ.

$$\bar{x} = \frac{70 + 120 + 110 + 101 + 88 + 83 + 95 + 98 + 107 + 100}{10} = 100.2$$

$$s^2 = \frac{\sum x^2}{n} - \bar{x}^2 = \frac{1}{10}[70^2 + 120^2 + 110^2 + \ldots + 100^2] - (100.2)^2$$

$$= 97.44, \quad \therefore \quad s = 9.871$$

Now $$t = \frac{\bar{x} - \mu}{s}\sqrt{n-1} = \frac{100.2 - 100}{9.871}\sqrt{10 - 1} = 0.06$$

Here $\nu = n - 1 = 9$.

For $\nu = 11$, $t_{0.05} = 2.2$, from table IV.

Since the $|t| < t_{0.05}$, our assumption is upheld i.e., the students does not increase the I.Q.

Example 27.14. The life time of a sample from the following factory: 42, 37, 39, 42, 45, 37, 49, 43, 51, 46 hours. Can we say that the manufacturer's claim that the average life time of bulb is 45 hours is justified? (U.P.T.U., 2005)

Solution. We find the mean and standard deviation of the sample as follows :

x	$d = x - 45$	d^2
48	3	9
42	−3	9
50	5	25
39	−6	36
44	−1	1
45	0	0
46	1	1
43	−2	4
47	2	4
41	−4	16
Total	0	105

(i) $$\bar{x} = \text{mean} = 45 + \frac{0}{10} = 45 + 0 = 45$$

$$\sigma_x^2 = \frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2 = \frac{105}{10} - \frac{(0)^2}{(10)^2} = \frac{105}{10}$$

(ii) $$\sigma_x = 3.24$$

Hence $$t = \frac{\bar{x} - \mu}{\sigma_x}\sqrt{(n-1)} = \frac{45 - 47.5}{3.24}\sqrt{9} = -2.31$$

Here $$d.f. = 5 - 1 = 4$$

For $\alpha = 5\%$, we get from table IV, $t_{0.05} = 2.132$.

As calculated value of $|t| > t_{0.05}$, the value of t is not significant at 5% level of significance which implies that there is no significant difference between $\bar{x}$ and μ. Thus the test provides no evidence against the population mean being 47.5.

Example 27.33. *A machine is set to produce metal rods with a diameter of 0.7 cm. A random sample of 10 rods shows a mean diameter 0.742 inch with a standard deviation of 0.04 inch. On the basis of this sample, would you say that the work is satisfactory ?* *M.D.U., 2003*

Solution. Here we have $\mu = 0.700$, $\bar{x} = 0.742$, $\sigma_x = 0.040$, $n = 10$.

Taking the hypothesis that the product is satisfactory, there is no significant difference between $\bar{x}$ and μ

$$\therefore \quad t = \frac{\bar{x} - \mu}{\sigma_x}\sqrt{(n-1)} = \frac{0.742 - 0.700}{0.040}\sqrt{(10-1)} = \frac{0.126}{0.040} = 3.15$$

Degrees of freedom $\nu = n - 1 = 9$.

For $\nu = 9$, we get from table IV, $t_{0.05} = 2.262$.

As the calculated value of $t > t_{0.05}$ the value of t is significant at 5% level of significance. This implies that $\bar{x}$ differs significantly from μ and the hypothesis is rejected. Hence the work is inferior. In fact, the work is inferior even at 2% level of significance.

Example 27.34. *Show that 95% confidence limits for the mean μ of the population are $\bar{x} \pm \frac{2\sigma_x}{\sqrt{n}}$.*

Declare that the random samples of 16 values with mean 41.5 inches are from the population with standard deviation 2.5 inches. Also find the 98% confidence limits for the mean of the population.

Solution. (a) The probability P that $|t| < t_{0.05}$ is 0.95. Hence the 95% confidence limits for μ are given by

$$\left|\frac{\bar{x} - \mu}{\sigma_x}\sqrt{n}\right| = t_{0.05}$$

or $$|\bar{x} - \mu| = \frac{\sigma_x}{\sqrt{n}} t_{0.05} \quad \text{or} \quad \mu = \bar{x} \pm \frac{\sigma_x}{\sqrt{n}} t_{0.05}$$ i.e., $\mu = \bar{x} \pm \frac{2\sigma_x}{\sqrt{n}}$

We can therefore say with a confidence coefficient 0.95 that the confidence interval $\bar{x} \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$ contains the population mean μ.

Given, $n = 36$, $\nu = n - 1 = 35$, $s_x = \sqrt{\frac{360}{12}} = 3$.

Also from table 7F, $t_{\alpha/2}$ for $\nu = 30$ is 2.13

$$\therefore \quad \frac{s_x}{\sqrt{n}} t_{\alpha/2} = \frac{3}{6} \times 2.13 = 1.6 \text{ approx.}$$

Hence the required confidence limits are 41.5 ± 1.6, i.e., 39.9 and 43.1 inches.

9.19.10 SIGNIFICANCE TEST OF DIFFERENCE BETWEEN SAMPLE MEANS

Given two independent samples $x_1, x_2, x_3, \ldots, x_{n_1}$ and $y_1, y_2, \ldots, y_{n_2}$ with means $\bar{x}$ and $\bar{y}$ and standard deviations σ_1 and σ_2 from a normal population with the same variance, we have to test the hypothesis that the population means μ_1 and μ_2 are the same.

For this, we calculate $\quad t = \dfrac{\bar{x} - \bar{y}}{\sigma \sqrt{\left[\dfrac{1}{n_1} + \dfrac{1}{n_2}\right]}}$ ...(1)

where $\quad \bar{x} = \dfrac{1}{n_1} \sum_1^{n_1} x_i, \bar{y} = \dfrac{1}{n_2} \sum_1^{n_2} y_i$

and $\quad \sigma^2 = \dfrac{1}{n_1 + n_2 - 2} \left[(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 \right] = \dfrac{1}{n_1 + n_2 - 2} \left[\sum_1^{n_1} (x_i - \bar{x})^2 + \sum_1^{n_2} (y_i - \bar{y})^2 \right]$.

It can be shown that the variate t defined by (1) follows the t-distribution with $n_1 + n_2 - 2$ degrees of freedom.

If the calculated value of $|t| > t_{0.05}$, the difference between the sample means is said to be significant at 5% level of significance.

If $t > t_{0.01}$, the difference is said to be significant at 1% level of significance.

If $t < t_{0.05}$, the data is said to be consistent with the hypothesis, that $\mu_1 = \mu_2$.

Case II : If the two samples are of the same size and the data are paired, then t is defined by

$$t = \frac{\bar{d} - 0}{(\sigma/\sqrt{n})} \quad \text{where} \quad \sigma^2 = \frac{1}{n-1} \sum_1^n (d_i - \bar{d})^2$$

d_i = difference of the ith members of the samples,

$\bar{d}$ = mean of the differences $= \Sigma d_i / n$, and the number of $df = n - 1$.

Example 17.15. Eleven students were given a test in statistics. They were given a month's further tuition and a second test of equal difficulty was held at the end of it. Do the marks given below show that the students have benefited by extra coaching ?

Boys	1	2	3	4	5	6	7	8	9	10	11
Marks in 1st test	23	20	19	21	18	20	18	17	23	16	19
Marks in II test	24	19	22	18	20	22	20	20	23	20	17

(U.P.T.U., 2011)

Solution. We compute the mean and the S.D. of the differences between the marks of the two tests as under :

$$\bar{d} = \text{mean of the } d\text{'s} = \frac{11}{11} = 1, \quad \sigma_d^2 = \frac{\Sigma d^2 - \bar{d}^2}{n-1} = \frac{10}{10} = 1 \quad \therefore \sigma_d = 1.04$$

Assuming that the students have not been benefited by extra coaching, it implies that the means of the differences between the marks of the two tests is zero, i.e., $\mu = 0$.

Then $t = \dfrac{\bar{X}-\mu}{\sigma_x}\sqrt{n} = \dfrac{3-2}{2.24}\sqrt{10} = 1.41$ approximately and $\nu = 11-1 = 10$

Student	X_1	X_2	$d = X_2 - X_1$	$d - \bar{d}$	$(d-\bar{d})^2$
1	23	24	1	5	25
2	20	18	−2	−3	9
3	19	22	3	2	4
4	21	18	−3	−4	16
5	18	20	2	1	1
6	20	22	2	1	1
7	18	20	2	1	1
8	17	16	−1	−2	4
9	23	21	−2	−3	9
10	16	18	2	1	1
11	19	17	−2	−3	9
			$\Sigma d = 1$		$\Sigma(d-\bar{d})^2 = 80$

From table IV, we find that $t_{0.05}$ (for $\nu = 10$) = 2.228. As the calculated value of t is $< t_{0.05}$, the value of t is not significant at 5% level of significance i.e., the test provides no evidence that the students have benefited by extra coaching.

Example 27.14. From a random sample of 10 pigs fed on diet A, the increase in weight in certain period were 10, 6, 16, 17, 13, 12, 8, 14, 15, 9 lbs. For another random sample of 12 pigs fed on diet B, the increase in the same period were 7, 13, 22, 15, 12, 14, 18, 8, 21, 23, 10, 17 lbs. Test whether diets A and B differ significantly as regards their effect on increase in weight?

Solution. We calculate the means and standard deviations of the samples as follows:

	Diet A			Diet B	
x	$x - \bar{x}$	$(x-\bar{x})^2$	y	$y - \bar{y}$	$(y-\bar{y})^2$
10	−2	4	7	−6	36
6	−6	36	13	−0	0
16	4	16	22	7	49
17	5	25	15	0	0
13	1	1	12	−3	9
12	0	0	14	−1	1
8	−4	16	18	3	9
14	2	4	8	−7	49
15	3	9	21	6	36
9	−3	9	23	8	64
			10	−5	25
			17	2	4
120	0	120	180	0	282

$$\bar{x} = \frac{120}{10} = 12 \text{ lbs.}, \quad \bar{y} = \frac{180}{12} = 15 \text{ lbs.}$$

$$s_y^2 = \frac{1}{n_2}\Sigma(y_i - \bar{y})^2 = \frac{1}{n_2}\Sigma(y_i - \bar{y})^2$$

$$= \frac{1}{12}(282 + 2 \times 42 \times 10 + 12 \times 20 = 0 + 4 \times 4 \times 50) = 23.5$$

$\therefore \quad s_y = 4.85$

- Assuming that the samples do not differ in weights so far as the true diets are concerned i.e., $\mu_1 - \mu_2 = 0$.

Hence $$t = \frac{\bar{x} - \bar{y} - 0}{s_y\sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{12 - 15}{4.85\sqrt{\left(\frac{1}{10} + \frac{1}{12}\right)}} = \frac{3}{4.85}\sqrt{\frac{120}{22}} = 1.6 \text{ nearly}$$

Here $d.f. = n_1 + n_2 - 2 = 18 + 12 - 2 = 30$.

For $\nu = 30$, we find $t_{.05} = 2.05$ [From table IV

$\therefore$ the calculated value of $t < t_{.05}$

Hence the difference between the sample means is not significant i.e., the two diets do not differ significantly as regards their effect on increase in weight.

EXERCISE 12.3

1. Find the median for the following set of daily wages (in rupees) for a sample of eight workers : 6, 7, 8, 5, 3, 5, 5, 6 ; taking the wage of the workers to be zero.
2. A random sample of 10 boys had the following I.Q. :
   70, 120, 110, 101, 88, 83, 95, 98, 107, 100.
   Do these data support the assumption of a population mean I.Q. of 120 at 1% level of significance ?
   (U.P.T.U., 2008 ; Calicut, 2005)
3. A sample of 16 values from a normal population has the mean 53 and a standard deviation 3. Is the sample mean 56 significantly different from the population mean ? Use 5% level of significance. (U.P.T.U., 2008)
4. A random sample of size 20 from a normal population has the mean 17.5 and $s.d. = 2.5$. Does this information indicate that the mean of the population is $\mu = 18$? (U.P.T.U., 2008)
5. A process for making certain bearings is under control if the diameter of the bearings have the mean 5 cm. When we test a sample of 10 of these bearings we find a mean diameter of 5.06 cm and a s.d. of 0.04 cm. Is the process under control ?
6. A machine is supposed to produce metal rods of mean diameter 0.12 cm. A sample of 10 rods was found to have a mean diameter of 0.126 cm and standard deviation 0.008. Test whether the machine is working properly at 5% level of significance.
7. Find (a) the median (b) the mode of the sample using the following data : Analysis of results of 15 specimens of 1.8 cm diameter hard-drawn copper wire.

Specimen	1	2	3	4	5	6	7	8	9	10
Breaking Strength (kg)	377	372	379	368	376	378	377	373	376	374

8. Two makes of electric bulbs (A and B) were tested for length of life and following results were obtained :

Bulbs A :	28	30	32	33	33	29	and 34
Bulbs B :	29	30	30	34	37	and 38	

   Test whether you can discriminate between the two brands. (Delhi, 2005 ; Pondicherry, 2004)
9. A group of 10 rats fed on a diet A and another group of 8 rats fed on a different diet B, recorded the following increase in weights :

Diet A :	5	6	8	1	12	4	3	6	8	12	gm
Diet B :	2	3	6	8	10	1	2	8	gm		

   Does it show that superiority of diet A over that of B ? (Mumbai, 2007)
10. A group of boys and girls were given an intelligence test. The mean scores, S.D.'s and numbers in each group are as follows :

	Boys	Girls
Mean	154	148
S.D.	12	15
n	18	14

    Is the mean score of boys significantly different from that of girls ?
11. The means of two random samples of sizes 9 and 7 are 196.42 and 198.82 respectively. The sum of the squares of the deviations from the means are 26.94 and 18.73 respectively. Can the samples be regarded to have been drawn from the same normal population ? (Mumbai, 2004)

9.12.2 (1) CHI-SQUARE [χ^2] TEST

When a fair coin is tossed 80 times, we expect from theoretical considerations that heads will appear 40 times and tail 40 times. But this never happens in practice i.e., the results obtained in an experiment do not agree exactly with the theoretical result. The magnitude of discrepancy between observation and theory is given by the quantity χ^2 (pronounced as chi-square). If $\chi^2 = 0$, the observed and theoretical frequencies completely agree. As the value of χ^2 increases, the discrepancy between the observed and theoretical frequencies increases.

(1) Definition. *If $O_1, O_2, \ldots O_n$ be a set of observed (experimental) frequencies and $E_1, E_2, \ldots E_n$ be the corresponding set of expected (theoretical) frequencies, then χ^2 is defined by the relation*

$$\chi^2 = \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \ldots + \frac{(O_n - E_n)^2}{E_n}$$

$$= \sum \frac{(O - E)^2}{E} \qquad \ldots(1)$$

with $n - 1$ degrees of freedom. [$\because \sum O_i = \sum E_i$ = the total frequency]

(2) Chi-square distribution[1]

If $x_1, x_2, \ldots x_n$ be n independent normal variables with mean zero and unit variance, then it can be shown that $x_1^2 + x_2^2 + \ldots + x_n^2$, is a random variate having χ^2-distribution with n d.f.

The equation of the χ^2 curve is

$$y = y_0 e^{-\chi^2/2} (\chi^2)^{(\nu-2)/2} \qquad \ldots(2)$$

where $\nu = n - 1$ [Fig. 27.1].

(3) Properties of χ^2-distribution

I. If $\nu = 1$, the χ^2-curve (2) reduces to $y = y_0 e^{-\chi^2/2}$, which is the exponential distribution.

II. If $\nu > 1$, this curve is tangential to x-axis at the origin and is positively skewed as the mode at $\chi^2 = \nu - 1$ and exists at $\chi^2 = 2$.

III. The probability P that the value of χ^2 from a random sample will exceed χ_0^2 is given by

$$P = \int_{\chi_0^2}^{\infty} y \, d\chi^2$$

Fig. 27.1

The values of χ_0^2 have been tabulated for various values of P and the values of ν from 1 to 30. (Table-V-Appendix-II)

For $\nu > 30$, the χ^2-curve approximates to the normal curve and we should refer to normal distribution table for significant values of χ^2.

IV. Since the equation of χ^2-curve does not involve any parameters of the population, this distribution does not depend on the form of the population and is therefore, very useful in a large number of problems.

V. Mean = ν and variance = 2ν.

9.12.3 GOODNESS OF FIT

The value of χ^2 is used to test whether the deviations of the observed frequencies from the expected frequencies are significant or not. It is also used to test how well a set of observations fit a given distribution. χ^2 therefore, provides a test of goodness of fit and may be used to examine the validity of some hypothesis about an observed frequency distribution. As a test of goodness of fit, it may be used to study the correspondence between theory and fact.

The χ^2 test is a non-parametric distribution free test since it does not make any assumption about the distribution of the parent population.

---

[1] Karl discovered this distribution in 1900. Karl Pearson rediscovered it independently in 1900 and applied it to test goodness of fit.

Procedure to test significance and goodness of fit.

(1) Set up a 'null hypothesis' and calculate

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

(2) Find the df and read the corresponding values of χ^2 at the prescribed significance level from Table V.

(3) From χ^2-table, we can also find the probability P corresponding to the calculated value of χ^2 for the given df.

(a) If $P < 0.05$, the observed value of χ^2 is significant at 5% level of significance.

If $P < 0.01$, the value is significant at 1% level.

If $P > 0.05$, it is a good fit and the value is not significant.

Example 27.17. *An experiment on pea breeding, the following frequencies of seeds were obtained:*

Round and yellow	Wrinkled and yellow	Round and green	Wrinkled and green	Total
315	101	108	32	556

Theory predicts that the frequencies should be in the proportions 9 : 3 : 3 : 1. Examine the correspondence between theory and experiment.

Solution. The corresponding frequencies are

$$\frac{9}{16} \times 556, \frac{3}{16} \times 556, \frac{3}{16} \times 556, \frac{1}{16} \times 556 \text{ i.e., } 313, 104, 104, 35.$$

Hence
$$\chi^2 = \frac{(315-313)^2}{313} + \frac{(101-104)^2}{104} + \frac{(108-104)^2}{104} + \frac{(32-35)^2}{35}$$

$$= \frac{4}{313} + \frac{9}{104} + \frac{16}{104} + \frac{9}{35} = 0.51 \quad \text{and} \quad df = 4 - 1 = 3.$$

For $\nu = 3$, we have $\chi^2_{0.05} = 7.815$ [From Table V

Since the calculated value of χ^2 is much less than $\chi^2_{0.05}$, there is a very high degree of agreement between theory and experiment.

Example 27.18. *A set of five similar coins is tossed 320 times and the result is*

No. of heads :	0	1	2	3	4	5
Frequency :	6	27	72	112	72	32

Test the hypothesis that the data follow a binomial distribution.

(Kottayam, 2005 ; P.T.U., 2005 ; V.T.U., 2004)

Solution. For $\nu = 5$, we have $\chi^2_{0.05} = 11.07$.

p, probability of getting a head $= \frac{1}{2}$; q, probability of getting a tail $= \frac{1}{2}$.

Hence the theoretical frequencies of getting 0, 1, 2, 3, 4, 5 heads are the successive terms of the binomial expansion $320(p + q)^5$

$$= 320(p^5 + 5p^4q + 10p^3q^2 + 10p^2q^3 + 5pq^4 + q^5)$$

$$= 320\left[\frac{1}{32} + \frac{5}{32} + \frac{10}{32} + \frac{10}{32} + \frac{5}{32} + \frac{1}{32}\right] = 10 + 50 + 100 + 100 + 50 + 10$$

Thus the theoretical frequencies are 10, 50, 100, 100, 50, 10.

Hence
$$\chi^2 = \frac{(6-10)^2}{10} + \frac{(27-50)^2}{50} + \frac{(72-100)^2}{100} + \frac{(112-100)^2}{100} + \frac{(72-50)^2}{50} + \frac{(32-10)^2}{10}$$

$$= \frac{1}{100}[160 + 2300 + 784 + 144 + 968 + 4840] = \frac{7996}{100} = 79.96$$

and $df = 6 - 1 = 5$.

Since the calculated value of χ^2 is much greater than $\chi^2_{0.05}$, the hypothesis that the data follow the binomial law is rejected.

Example 27.19. *Fit a Poisson distribution to the following data and test for its goodness of fit at level of significance 0.05.*

x:	0	1	2	3	4
f:	419	352	154	56	19

$$\therefore \quad \chi^2 = \frac{(5-4.13)^2}{4.13} + \frac{(118-99.66)^2}{99.66} + \frac{(342-365.62)^2}{365.62} + \frac{(527-527.51)^2}{527.51} + \frac{(15-7.43)^2}{7.43}$$

$$= 0.1830 + 0.3379 + 0.2837 + 0.0152 + 0.0837 = 0.8282$$

As regards the number of degrees of freedom (γ), there are three constraints (i) the frequency between total observed and total estimated frequencies (ii) and (iii) mean (m) and standard deviation (σ) have been estimated from the sample data $\quad \therefore \quad \gamma = 5 - 3 = 2$.

For $\gamma = 2$, $\chi^2_{.05} = 5.991$ from table V.

Since $\chi^2 = 0.8282 < 5.991$. Hence the fit is not good.

EXERCISE 8.12

1. Fit the binomial law to the following data and test the goodness of fit at 5% level of significance.

No. of heads appearing in 8 tosses of a coin	0	1	2	3	4	5	6	7	8
Frequency	2	6	29	56	75	50	30	11	1

Find the probability of getting 5 or more heads by chance?

2. Genetic theory states that children having one parent of blood type M and other blood type N will always be one of the three types M, MN, N and that the proportions of these types will be in the ratio 1 : 2 : 1. A sample of 300 such children had 30 in group M, 150 in group MN and 120 in group N. Are these data consistent with the genetic theory?

3. A die was thrown 60 times and the following frequency distribution was observed:

Points	1	2	3	4	5	6
f_i	13	9	8	7	11	12

Test whether the die is unbiased?

4. The following table gives the number of accidents occurred during the various days of the week. Find whether the accidents are uniformly distributed over the week?

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
No. of accidents	14	16	8	12	11	9	14	84

(Madras, 2004)

5. Fit a binomial distribution to the data:

x	0	1	2	3	4	5
f_i	28	108	342	287	128	27

and test the goodness of fit at the level of significance 0.05.

6. In 2000 sets of trials of 10 each for an event of small probability, the frequencies f_i of the number x of successes proved to be

x	0	1	2	3	4	5	6	7
f_i	800	800	350	40	10	0	0	1

Fit a Poisson distribution to the data and test the goodness of fit.

7. The frequencies of accidents prevailing in the number of days it was in existence during eight years in 1000 factories is as follows:

No. of accidents	0	1	2	3	4	5	6	7
No. of days (f)	314	305	208	91	32	8	2	0

Fit a suitable distribution to the data and test the goodness of fit.

8. Obtain the equation of the normal curve that may be fitted to the data and test the goodness of fit.

x_i	4	6	8	10	12	14	16	18	20	22	24	Total
f_i	1	7	15	30	50	60	45	30	12	5	1	256

8.12.2 (1) F-DISTRIBUTION*

Let $x_1, x_2, \ldots, x_{n_1}$ and $y_1, y_2, \ldots, y_{n_2}$ be the values of two independent random samples drawn from the normal population with having equal variances.

* This distribution was introduced by the English statistician Prof. R.A. Fisher (1890–1962) who had greatly influenced the development of modern statistics.

Let $\bar{x}_1$ and $\bar{x}_2$ be the sample means and $s_1^2 = \frac{1}{n_1 - 1}\sum_{1}^{n_1}(x_i - \bar{x})^2$, $s_2^2 = \frac{1}{n_2 - 1}\sum_{1}^{n_2}(y_i - \bar{y})^2$ be the sample variances.

Then we define F by the relation

$$F = \frac{s_1^2}{s_2^2} \qquad (s_1^2 > s_2^2)$$

This gives F-distribution (also known as variance ratio distribution) with $\nu_1 = n_1 - 1$ and $\nu_2 = n - 2$ degrees of freedom. The larger of the variances is placed in the numerator.

(1) Properties. I. The F-distribution curve lies entirely in the first quadrant and is unimodal.

II. The F-distribution is independent of the population variance σ^2 and depends on ν_1 and ν_2 only.

III. $F_\alpha(\nu_1, \nu_2)$ is the value of F for ν_1 and ν_2 such that the area to the right of F_α is α.

IV. It can be shown that the mode of F-distribution is less than unity.

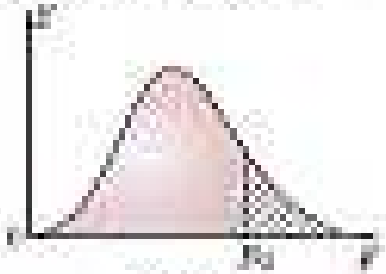


Fig. 17.6

(2) Significance test. Snedecor's F-table gives 5% and 1% points of significance for F. (Table VI – Appendix 2). 5% points of F mean that area under the F-curve to the right of the ordinate at a value of F is 0.05. Clearly value of F at 5% significance is lower than that at 1%. F-distribution is very useful for testing the equality of population means by comparing sample variances. As such it forms the basis of analysis of variance.

Example 17.21. *Two samples of sizes 9 and 8 give the sum of squares of deviations from their respective means equal to 160 inches^2 and 91 inches^2 respectively. Can these be regarded as drawn from the same normal population?* (P.T.U., 2002)

Solution. We have $\Sigma(x - \bar{x})^2 = 160$ and $\Sigma(y - \bar{y})^2 = 91$.

$$\therefore \qquad s_1^2 = \frac{160}{8} = 20$$

and

$$s_2^2 = \frac{91}{7} = 13$$

Hence

$$F = \frac{s_1^2}{s_2^2} = \frac{20}{13} = 1.54 \text{ nearly.}$$

For $\nu_1 = 8$, $\nu_2 = 7$, we have $F_{0.05} = 3.73$ (From Table VI)

Since the calculated value of $F < F_{0.05}$, the population variances are not significantly different. Thus the two samples can be regarded as drawn from two normal populations with the same variance. If the two populations are to be same, their means should also be the same which can be verified by applying t-test provided the sample sizes are known.

Example 17.22. *Measurements on the length of a copper wire were taken by 2 experimenters A and B with results:*

A's measurements (cm)	12.39	12.38	12.40	12.43	12.46	12.39	12.42	12.40	12.43
B's measurements (cm)	12.39	12.40	12.38	12.42	12.43	12.40	12.42	12.41	

Test whether B's measurements are more accurate than A's. (The readings taken in both cases being unbiased).

Solution. Readings in both cases being unbiased, B's measurements will be taken more accurate if its population variance is less than that of A's measurements.

Under the hypothesis that the two populations have the same variance (i.e. $\sigma_1^2 = \sigma_2^2$), we have

$$F = \frac{s_1^2}{s_2^2}$$

with $\nu_1 = n_1 - 1 = 8$ and $\nu_2 = n_2 - 1 = 7$.

We tabulate the results of the two series as follows :

x	$x = 5000 - 15$	x^2	y	$y = 1000 - 25$	y^2
13.00	20	400	12.50	25	625
12.88	08	064	12.54	29	841
13.04	+14	196	12.49	24	576
12.95	05	025	12.52	27	729
12.94	04	016	12.50	25	625
12.96	06	036	12.48	23	529
12.97	07	049	12.50	25	625
12.90	10	100	12.48	23	529
12.97	07	049			
	66	934		201	5049

$$\therefore \quad s_1^2 = \frac{1}{n_1 - 1}\left[\sum x^2 - \frac{(\sum x)^2}{n_1}\right] = \frac{1}{9}\left[934 - \frac{66^2}{9}\right] = \frac{1}{9}[934 - 726] = 23.11$$

$$s_2^2 = \frac{1}{n_2 - 1}\left[\sum y^2 - \frac{(\sum y)^2}{n_2}\right] = \frac{1}{7}\left[5049 - \frac{201^2}{8}\right] = \frac{1}{7}[5049 - 5050.1] = 0.157$$

$$\therefore \quad F = \frac{s_1^2}{s_2^2} = \frac{23.11}{0.157} = 147.2$$

For $\nu_1 = 8$ and $\nu_2 = 7$, from table VI, $F_{0.05} = 3.73$ and $F_{0.01} = 8.84$.

Since the calculated value of F is less than both $F_{0.05}$ and $F_{0.01}$, the result is insignificant at both 5% and 1% level.

Hence there is no reason to say that X's measurements are more accurate than those of Y's.

23.20. (L) FISHER'S z-DISTRIBUTION

Choosing the variable F as z by the substitution $z = \frac{1}{2}\log_e F$ or $F = e^{2z}$ in the F-distribution, we get the Fisher's z-distribution.

It is more widely symmetrical than F-distribution. Table showing the values of z that will be exceeded in sample sampling with probabilities 0.05 and 0.01 have been prepared for various values of ν_1 and ν_2.

(ii) **Significance test.** As variable gives only critical values corresponding to right-hand tail areas, therefore 5% for 1% points simply that the area to the right of the critical value is 0.05 or 0.01. In other words, 5% and 1% points are the ordinates at the 95% and 99% levels of significance respectively.

Example 23.23. Two groups of students are tested for proficiency in reading measurements. One group consists of 21 students with a standard deviation of 2.01 and the other group of 16 students with a standard deviation of 0.88. What is the level of significance of this difference.

(Given that for $\nu_1 = 20$ and $\nu_2 = 15$, $z_{0.05} = 0.366$, $z_{0.01} = 0.702$)

Solution. We have $n_1 = 21$, $\sigma_1 = 2.01$; $n_2 = 16$, $\sigma_2 = 0.88$.

$$\therefore \quad s_1^2 = \frac{n_1}{n_1 - 1}\sigma_1^2 = \frac{21}{20}(2.01)^2 = (2.01)^2 \quad \text{and} \quad s_2^2 = \frac{n_2}{n_2 - 1}\sigma_2^2 = \frac{16}{15}(0.88)^2 = (0.88)^2$$

Hence $F = \left[\frac{2.01}{0.88}\right]^2 = 5.0481$ and $z = \frac{1}{2}\log_e F = 1.0033 \log_{10} 5.0481 = 0.8028$

Since the calculated value of z is just greater than $z_{0.05}$ and less than $z_{0.01}$, the difference between the standard deviations is just significant at 5% level and insignificant at 1% level.

PROBLEMS

1. Two samples of sizes 9 and 8 gave the variances 0.8 and 0.65 respectively. Is the difference significant?
2. Test for the sign test if the given data are consistent with the hypothesis that the median is 0.5 against the alternative that it is greater than 0.5: 0.9, 0.7, 0.5, 0.3, 0.8, 0.6, 0.4, 0.2, 0.1, 0.7, 0.8, 0.9, 0.6, 0.7, 0.8, 0.9, 0.5, 0.6, 0.7, 0.8, 0.9, 0.5, 0.6, 0.7, 0.8, 0.9, 0.5

CHAPTER 28

Numerical Solution of Equations

1. Introduction. 2. Solution of algebraic and transcendental equations—Bisection method, Method of false position, Newton's method. 3. Useful observations to be kept in mind. 4. Application of Newton's method to solution of equations—Numerical value of $\sqrt{N}$. 5. Solution of linear simultaneous equations. 6. Gauss's method of successive elimination method. Gauss-Jordan method. Factorization method. 7. Iterative methods of solution—Jacobi's method. Gauss-Seidal method. Relaxation method. 8. Solution of non-linear simultaneous equations—Newton-Raphson method. 9. Determination of eigen values by iteration. 10. Objective Type of Questions.

28.1 INTRODUCTION

The limitations of analytical methods have led the engineers and scientists to evolve graphical and numerical methods. As seen in § 1.8, the graphical methods, though simple, give results to a low degree of accuracy. Numerical methods are, however, to be relied which are more accurate. With the advent of high-speed digital computers and increasing demand for numerical answers to various problems, numerical techniques have become indispensable tools in the hands of engineers.

Numerical methods are often, of a repetitive nature. These consist in repeated execution of the same process where at each step the result of the preceding step is used. This is known as iteration process and is repeated till the result is obtained to a desired degree of accuracy.

In this chapter, we shall discuss some numerical methods for the solution of algebraic and transcendental equations and also of numerical linear and non-linear equations. We shall close the chapter by describing an iterative method for the evaluation of eigen value problem. For a detailed study of these topics, the reader should refer to author's book *Numerical Methods in Engineering & Science*.

28.2 SOLUTION OF ALGEBRAIC AND TRANSCENDENTAL EQUATIONS

To find the roots of an equation $f(x) = 0$, we start with a known approximate solution and apply any of the following methods :

1) Bisection method. This method consists in locating the root of the equation $f(x) = 0$ between a and b. If $f(x)$ is continuous between a and b, and $f(a)$ and $f(b)$ are of opposite signs then there is a root between a and b. For definiteness, let $f(a)$ be negative and $f(b)$ be positive. Then the first approximation to the root is $x_1 = \frac{1}{2}(a + b)$.

If $f(x_1) = 0$, then x_1 is a root of $f(x) = 0$. Otherwise, the root lies between a and x_1 or x_1 and b according as $f(x_1)$ is positive or negative. Then we bisect the interval as before and continue the process until the root is found to desired accuracy.

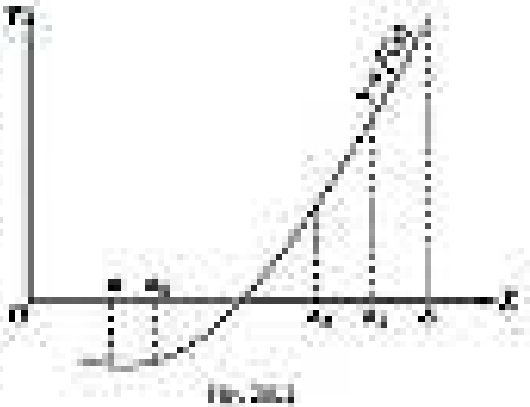


Fig. 28.1

In the Fig. 2.6.1, $f(x_1)$ is +ve, so that the root lies between x_0 and x_1. Then the second approximation to the root is $x_2 = \frac{1}{2}(x_0 + x_1)$. If $f(x_2)$ is –ve, the root lies between x_1 and x_2. Then the third approximation to the root is $x_3 = \frac{1}{2}(x_1 + x_2)$ and so on.

Example 2.6.1. (a) Find a root of the equation $x^3 - 4x - 9 = 0$ using the bisection method correct to three decimal places. (Marathwada, 2003)

(a) Using bisection method, find the negative root of the equation $x^3 - 4x + 9 = 0$. (U.P.T.U., 2004)

Solution. (a) Let $f(x) = x^3 - 4x - 9$

Since $f(2)$ is –ve and $f(3)$ is +ve, a real root lies between 2 and 3.

∴ first approximation to the root is

$$x_1 = \frac{1}{2}(2 + 3) = 2.5$$

Then $f(x_1) = (2.5)^3 - 4(2.5) - 9 = -3.375$ i.e., –ve

∴ the root lies between x_1 and 3. Thus the second approximation to the root is

$$x_2 = \frac{1}{2}(x_1 + 3) = 2.75$$

Then $f(x_2) = (2.75)^3 - 4(2.75) - 9 = 0.7969$ i.e., +ve

∴ the root lies between x_1 and x_2. Thus the third approximation to the root is

$$x_3 = \frac{1}{2}(x_1 + x_2) = 2.625$$

Then $f(x_3) = (2.625)^3 - 4(2.625) - 9 = -1.4121$ i.e., –ve

∴ the root lies between x_3 and x_2. Thus the fourth approximation to the root is

$$x_4 = \frac{1}{2}(x_3 + x_2) = 2.6875$$

Repeating this process, the successive approximations are

$x_5 = 2.71875$, $x_6 = 2.70313$, $x_7 = 2.71094$

$x_8 = 2.70703$, $x_9 = 2.70508$, $x_{10} = 2.70801$

$x_{11} = 2.70654$, $x_{12} = 2.70642$

Hence the root is 2.7064

(b) If α, β, γ are the roots of the given equation, then $-\alpha, -\beta, -\gamma$ are the roots of $(-x)^3 - 4(-x) - 9 = 0$

∴ the negative root of the given equation is the greatest root of $x^3 - 4x - 9 = 0$ which we have found above to be 2.7064.

Hence the negative root of the given equation is –2.7064.

Example 2.6.2. By using the bisection method, find an approximate root of the equation $\sin x = 1 + x$ that lies between $x = 0$ and $x = -0.5$ (measured in radians). Carry out computations upto the 7th stage.

(U.P.T.U., June 03)

Solution. Let $f(x) = \sin x - x - 1$. We know that $1^c = 57.3^\circ$.

Now $f(0) = 0 - \sin 0 - 1 = \sin(57.3^\circ) - 1 = -0.539849$

and $f(-0.5) = -0.5 - \sin(-0.5)^c - 1 = -1.5 + \sin(28.65^\circ) - 1 = 0.479425$;

a root lies between 0 and –0.5.

∴ first approximation to the root is $x_1 = \frac{1}{2}(0 - 0.5) = -0.25$.

Then $f(x_1) = f(-0.25) = \sin(-0.25) - 1 + 0.25 = \sin(-14.325^\circ) - 1 + 0.25 = -0.1584$ and $f(0) < 0$.

∴ a root lies between 0 and $x_1 = -0.25$.

Thus the second approximation to the root is $x_2 = \frac{1}{2}(0 - 0.25) = -0.125$.

Then $f(x_2) = f(-0.125) = \sin(-0.125) - 1 + 0.125 = \sin(-7.163^\circ) - 1 + 0.125 = -0.01809$ and $f(0) < 0$.

∴ a root lies between 0 and $x_2 = -0.125$.

Thus the third approximation to the root is $x_3 = \frac{1}{2}(0 - 0.125) = -0.0625$

Then $f(x_2) = 1.3825 \text{ is } (1.0625) - 1 = 1.0625 \text{ is } (60.86) - 1 = -0.0750 < 0$

and $f(x_0) > 0$, i.e., now the root lies between $x_2 = 1.0625$ and $x_1 = 1.125$.

$\therefore$ fourth approximation to the root is $x_3 = \frac{1}{2}(1.0625 + 1.125) = 1.09375$

Then $f(x_3) = 1.09375 \text{ is } 0.09408 > 0$ and $f(x_2) < 0$,

i.e., the root lies between $x_3 = 1.09375$ and $x_2 = 1.125$.

$\therefore$ fifth approximation to the root is $x_4 = \frac{1}{2}(1.09375 + 1.125) = 1.10937$

Then $f(x_4) = 1.10937 \text{ is } 0.00984 > 0$ and $f(x_3) < 0$

$\therefore$ the root lies between $x_4 = 1.10937$ and $x_3 = 1.125$.

Thus the sixth approximation to the root is

$$x_5 = \frac{1}{2}(1.10937 + 1.125) = 1.11719$$

Then $f(x_5) = 1.11719 \text{ is } 0.00321 > 0$, i.e., $f(x_4) < 0$.

$\therefore$ the root lies between $x_5 = 1.11719$ and $x_4 = 1.10937$.

Thus the seventh approximation to the root is $x_6 = \frac{1}{2}(1.11719 + 1.10937) = 1.11328$

Hence the desired approximation to the root is 1.11328.

(2) Method of false position or Regula-falsi method. This is the oldest method of finding the real root of an equation $f(x) = 0$ and closely resembles the bisection method. Here we choose two points x_0 and x_1 such that $f(x_0)$ and $f(x_1)$ are of opposite signs i.e., the graph of $y = f(x)$ crosses the x-axis between these points (Fig. 25.2). This indicates that a root lies between x_0 and x_1 i.e., we assume $f(x_0)/f(x_1) < 0$.

Equation of the chord joining the points $A(x_0, f(x_0))$ and $B(x_1, f(x_1))$ is

$$y - f(x_0) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) \qquad \ldots(1)$$

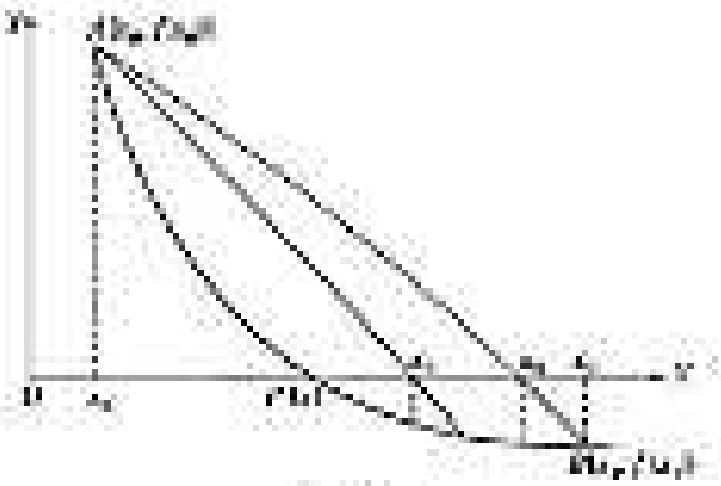


Fig. 25.2

The method consists in replacing the curve AB by means of the chord AB and taking the point of intersection of the chord with the x-axis as an approximation to the root. So the abscissa of the point where the chord cuts the x-axis is $y = 0$ is given by

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0) \qquad \ldots(2)$$

which is an approximation to the root.

If now $f(x_2)$ and $f(x_0)$ are of opposite signs, then the root lies between x_2 and x_0. So replacing x_1 by x_2 in (2), we obtain the next approximation x_3. (The root could as well lie between x_2 and x_1 and we would obtain x_3 accordingly). This procedure is repeated till the root is found to desired accuracy. The iteration process based on (1) is known as the **method of false position.**

Example 25.3. *Find a real root of the equation $x^3 - 2x - 5 = 0$ by the method of false position correct to three decimal places.* (Meerut, 2003)

Solution. Let $f(x) = x^3 - 2x - 5$

so that $f(1) = -6$ and $f(2) = 39$ i.e., A root lies between 2 and 3.

$\therefore$ taking $x_0 = 2, x_1 = 3, f(x_0) = -1, f(x_1) = 16$ in the method of false position, we get

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0) = 2 + \frac{1}{17} = 2.0588 \qquad \ldots(1)$$

Now $f(x_2) = f(2.0588) = -0.3908$ i.e., the root lies between 2.0588 and 3.

$\therefore$ taking $x_0 = 2.0800$, $x_1 = 2.1$, $f(x_0) = -0.2899$, $f(x_1) = 1.6$, in (1), we get

$$x_2 = 2.1000 - \frac{0.0200}{1.8899} \times 1.6 = 2.0841 = 2.0843$$

Repeating the process, the successive approximations are

$$x_3 = 2.0842,\ x_4 = 2.0843,\ x_5 = 2.0843,\ x_6 = 2.0841,\ x_7 = 2.0843 \text{ etc.}$$

Hence the root is 2.084 correct to 3 decimal places.

Example 18.4. Find the root of the equation $\cos x = xe^x$ using the regula-falsi method correct to four decimal places. (Kerala U. 2005)

Solution. Let $f(x) = \cos x - xe^x = 0$

So that $f(0) = 1, f(1) = \cos 1 - e = -2.1789$

i.e., the root lies between 0 and 1.

$\therefore$ taking $x_0 = 0, x_1 = 1, f(x_0) = 1$ and $f(x_1) = -2.1789$ in the regula-falsi method, we get

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0) = 0 - \frac{1}{3.1789} \times 1 = 0.3163 \quad \ldots(1)$$

Now $f(0.3163) = 0.5156$

i.e., the root lies between 0.3163 and 1.

$\therefore$ taking $x_0 = 0.3163, x_1 = 1, f(x_0) = 0.5156, f(x_1) = -2.1789$ in (1), we get

$$x_2 = 0.3163 + \frac{0.6837}{2.6945} \times 0.5156 = 0.4472$$

Now $f(0.4472) = 0.2006$

i.e., the root lies between 0.4472 and 1.

$\therefore$ taking $x_0 = 0.4472, x_1 = 1, f(x_0) = 0.2006, f(x_1) = -2.1789$ in (1), we get

$$x_3 = 0.4472 + \frac{0.5528}{2.3795} \times 0.2006 = 0.4593$$

Repeating the process, the successive approximations are

$$x_4 = 0.4606, \quad x_5 = 0.4623, \quad x_6 = 0.4625$$
$$x_7 = 0.4626, \quad x_8 = 0.4627, \quad x_9 = 0.4627 \text{ etc.}$$

Hence the root is 0.4627 correct to 4 decimal places.

Example 18.5. Find a real root of the equation $x \log_{10} x = 1.2$ using regula-falsi method correct to four decimal places. (U.P.T.U. 2003; U.P.T.U. 2005; Kerala U. 2005)

Solution. Let $f(x) = x \log_{10} x - 1.2$

and $f(1) = -1.2, f(2) = -0.7$ and $f(3) = +0.$

$\therefore$ a root lies between 2 and 3.

Taking $x_0 = 2$ and $x_1 = 3, f(x_0) = -0.6794$ and $f(x_1) = 0.2313$, in the method of false position, we get

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0) = 2.7293 \quad \ldots(1)$$

Now $f(x_2) = f(2.7293) = -0.0109$

i.e., the root lies between 2.7293 and 3.

$\therefore$ taking $x_0 = 2.7293, x_1 = 3, f(x_0) = -0.0109$

and $f(x_2) = 0.2313$ in (1), we get

$$x_3 = 2.7293 + \frac{0.2707}{0.2313 + 0.0109} \times 0.0109 = 2.7402$$

Repeating the process, the successive approximations are

$$x_4 = 2.7401, \quad x_5 = 2.7402 \text{ etc.}$$

Hence the root is 2.7402 correct to 4 decimal places.

Example 18.6. Use the method of false position to find the fourth root of 32 correct to four decimal places.

Solution. Let $x = (32)^{1/4}$ so that $x^4 - 32 = 0$

Take $f(x) = x^3 - 2x$. Then $f(2) = -1$ and $f(3) = 15$, i.e. a root lies between 2 and 3.

∴ taking $x_0 = 2$, $x_1 = 3$, $f(x_0) = -1$, $f(x_1) = 15$ in the method of false position, we get

$$x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0) = 2 + \frac{1}{16} = 2.0625 \qquad (1)$$

Now $f(x_2) = f(2.0625) = -0.5436$, i.e. the root lies between 2.0625 and 3.

∴ taking $x_0 = 2.0625$, $x_1 = 3$, $f(x_0) = -0.5436$, $f(x_1) = 15$

in (1), we get $$x_2 = 2.0625 - \frac{3 - 2.0625}{15 + 0.5436}(-0.5436) = 2.095$$

Now $f(x_2) = f(2.095) = -0.2072$, i.e. the root lies between 2.095 and 3.

∴ taking $x_0 = 2.095$ and $x_1 = 3$, $f(x_0) = -0.2072$ and $f(x_1) = 15$ in (1), we obtain

$$x_2 = 2.095 - \frac{3 - 2.095}{15 + 0.2072}(-0.2072) = 2.1003$$

Repeating this process, the successive approximations are $x_3 = 2.1017$, $x_4 = 2.1022$ etc.

Since $x_3 = x_4$ upto 3 decimal places, we take $(\sqrt{2})^3 = 2.102$.

(iii) Newton–Raphson method. Let x_0 be an approximate root of the equation $f(x) = 0$. If $x_1 = x_0 + h$ is the exact root, then $f(x_1) = 0$.

∴ expanding $f(x_0 + h)$ by Taylor's series

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \ldots = 0$$

Since h is small, neglecting h^2 and higher powers of h, we get

$$f(x_0) + hf'(x_0) = 0 \quad \text{or} \quad h = -\frac{f(x_0)}{f'(x_0)} \qquad (2)$$

∴ a closer approximation to the root is given by

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Similarly, starting with x_1, a still better approximation x_2 is given by

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

In general, $$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad (3)$$

which is known as the *Newton–Raphson formula* or *Newton's iteration formula*.

Note 1. Newton's method is useful in cases of large values of $f'(x)$ (i.e. where the graph of $f(x)$ is changing rapidly near the root) or where the graph of $f(x)$ is nearly vertical.

Note 2. Newton's method has a second order of quadratic convergence. If the error ε_n is first taken to be a small quantity, so that $x_n = x + \varepsilon_n$ and $x_{n+1} = x + \varepsilon_{n+1}$,

then (3) becomes $x + \varepsilon_{n+1} = x + \varepsilon_n - \dfrac{f(x + \varepsilon_n)}{f'(x + \varepsilon_n)}$

i.e. $$\varepsilon_{n+1} = \varepsilon_n - \frac{f(x) + \varepsilon_n f'(x) + \frac{1}{2!}\varepsilon_n^2 f''(x) + \ldots}{f'(x) + \varepsilon_n f''(x) + \ldots} \qquad \text{(by Taylor's expansion)}$$

$$= \varepsilon_n - \frac{\varepsilon_n f'(x) + \frac{1}{2}\varepsilon_n^2 f''(x) - \ldots}{f'(x) + \varepsilon_n f''(x) + \ldots} \qquad [f(x) = 0]$$

$$= \frac{\varepsilon_n^2 f''(x)}{2[f'(x) + \varepsilon_n f''(x)]} \approx \frac{\varepsilon_n^2}{2}\frac{f''(x)}{f'(x)} \qquad \left[\begin{array}{l}\text{neglecting third and}\\ \text{higher powers of } \varepsilon_n\end{array}\right]$$

This shows that the subsequent error at each step is proportional to the square of the previous error and so each iteration is quadratic. (T.D.C. 2006)

Note 3. Geometrical interpretation. Let x_0 be a point near the root x of the equation $f(x) = 0$ (Fig. 28.1). Then the equation of the tangent at A_0 is $y - f(x_0) = f'(x_0)(x - x_0)$

Note 4. Newton's method is named after the English mathematician Joseph Raphson (1648–1715) who suggested a method similar to Newton's method.

Hence, we can write $x_2 = x_1 - \dfrac{f(x_1)}{f'(x_1)}$, which is a first approximation to the root α. If A_2 is the point corresponding to x_2 on the curve, then the tangent at A_2 will cut the x-axis at x_3 which is nearer to α than x_2. Therefore, a second approximation is to the root. Repeating this process, we approach to the root α quite rapidly. Hence the method consists in replacing the part of the curve between the point A_1 and the x-axis by means of the tangent to the curve at A_1.

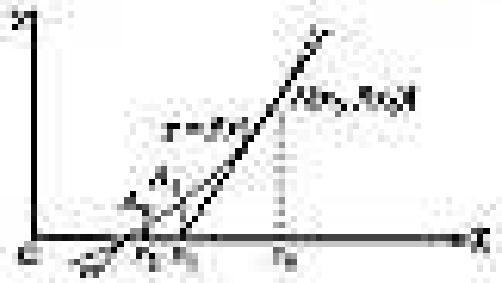


Fig. 28.1

Example 28.7. Find the positive root of $x^3 - x = 10$ correct to three decimal places using Newton-Raphson method. *(U.P.T.U., 2000; Madras, 2006)*

Solution. Let $f(x) = x^3 - x - 10$

So that $f(0) = -10 = -ve, f(2) = 8 - 2 - 10 = -4 = -ve$

$f(3) = 27 - 3 - 10 = 14 = +ve$

$\therefore$ a root of $f(x) = 0$ lies between 2 and 3. Let us take $x_0 = 2$

Also $f'(x) = 3x^2 - 1$

Newton-Raphson's formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad \ldots(1)$$

Putting $n = 0$, the first approximation x_1 is given by

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{f(2)}{f'(2)} = 2 - \frac{-4}{4 \times 2^2 - 1} = 2 + \frac{4}{11} = 1.363$$

Putting $n = 1$ in (1), the second approximation is

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.363 - \frac{f(1.363)}{f'(1.363)}$$

$$= 1.363 - \frac{(1.363)^3 - (1.363) - 10}{4(1.363)^2 - 1} = 1.363 - \frac{0.5636}{20.339} = 1.360$$

Putting $n = 2$ in (1), the third approximation is

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.360 - \frac{(1.360)^3 - (1.360) - 10}{4(1.360)^2 - 1}$$

$$= 1.360 - \frac{0.018}{20.374} = 1.358$$

Since $x_2 = x_3$. Hence the desired is 1.358 correct to three decimal places.

Example 28.8. Find the Newton's method, the real root of the equation $2x = \sin x + 1$. *(U.P.T.U., 2003 / U.P.T.U., 2005)*

Solution. Let $f(x) = 2x - \sin x - 1$

$f(0) = -2 = -ve, f(1) = 2 - 0.8415 - 1 = 1.4597 = +ve$.

So a root of $f(x) = 0$ lies between 0 and 1. It is nearer to 1. Let us take $x_0 = 0.6$.

Also $f'(x) = 2 - \cos x$

$\therefore$ Newton's iteration formula gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{2x_n - \sin x_n - 1}{2 - \cos x_n}$$

$$= \frac{x_n \cos x_n + \sin x_n + 1}{2 + \cos x_n} \qquad \ldots(1)$$

Putting $n = 0$, the first approximation x_1 is given by

$$x_1 = \frac{x_0 \sin x_0 + \cos x_0 + 1}{2 + \cos x_0} = \frac{(0.6)\sin(0.6) + \cos(0.6) + 1}{2 + \cos 0.6}$$

$$= \frac{0.6 \times 0.3729 + 0.03563 + 1}{2 + 0.3729} = 0.6871$$

Putting $n = 2$ in (1), the second approximation is

$$x_2 = \frac{x_1 \sin x_1 + \cos x_1 + 1}{3 + \sin x_1} = \frac{0.6871 \sin(0.6871) + \cos(0.6871) + 1}{3 + \sin(0.6871)}$$

$$= \frac{0.6871 \times 0.6378 + 0.7815 + 1}{3 + 0.6378} = 0.6871 \quad \text{Clearly, } x_1 = x_2$$

Hence the desired root is 0.6871 correct to four decimal places.

Example 4.18. Using Newton's iterative method, find the real root of $x \log_{10} x = 1.2$ correct to four decimal places. (U.P.T.U. 2002; Kerala U. 2000; Bangalore 2003)

Solution. Let $f(x) = x \log_{10} x - 1.2$

$f(1) = -1.2 = -ve$, $f(2) = 2 \log_{10} 2 - 1.2 = 0.6020 - 1.2 = -ve$

and $f(3) = 3 \log_{10} 3 - 1.2 = 1.4314 - 1.2 = 0.2314 = +ve$

So a root of $f(x) = 0$ lies between 2 and 3. Let us take $x_0 = 2$

Also $f'(x) = \log_{10} x + x \cdot \frac{1}{x} \log_{10} e = \log_{10} x + 0.4343$

$\therefore$ Newton's iteration formula gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n \log_{10} x_n - 1.2}{\log_{10} x_n + 0.4343} \quad \ldots(1)$$

Putting $n = 0$, the first approximation is

$$x_1 = \frac{0.4343 \times x_0 + 1.2}{\log_{10} x_0 + 0.4343} = \frac{0.4343 \times 2 + 1.2}{\log_{10} 2 + 0.4343} = \frac{0.8686 + 1.2}{0.3010 + 0.4343} = 2.61$$

Similarly putting $n = 1, 2, 3, 4$ in (1), we get

$$x_2 = \frac{0.4343 \times 2.61 + 1.2}{\log_{10} 2.61 + 0.4343} = 2.741$$

$$x_3 = \frac{0.4343 \times 2.741 + 1.2}{\log_{10} 2.741 + 0.4343} = 2.74384$$

$$x_4 = \frac{0.4343 \times 2.74384 + 1.2}{\log_{10} 2.74384 + 0.4343} = 2.74406$$

$$x_5 = \frac{0.4343 \times 2.74406 + 1.2}{\log_{10} 2.74406 + 0.4343} = 2.74406$$

Clearly $x_4 = x_5$

Hence the required root is 2.7406 correct to five decimal places.

4.18 USEFUL DEDUCTIONS FROM THE NEWTON-RAPHSON FORMULA

(1) Iterative formula to find $1/N$ is $x_{n+1} = x_n (2 - Nx_n)$

(2) Iterative formula to find $\sqrt{N}$ is $x_{n+1} = \frac{1}{2}\left(x_n + \frac{N}{x_n}\right)$

(3) Iterative formula to find $1/\sqrt{N}$ is $x_{n+1} = \frac{1}{2}(x_n - Nx_n^3)$

(4) Iterative formula to find $\sqrt[3]{N}$ is $x_{n+1} = \frac{1}{3}\left(2x_n + \frac{N}{x_n^2}\right)$

Proof. (1) Let $x = 1/N$ or $\frac{1}{x} - N = 0$

Taking $f(x) = \frac{1}{x} - N$, we have $f'(x) = -x^{-2}$

Then Newton's formula gives

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{\frac{1}{x_n} - N}{-x_n^{-2}} = x_n + \left(\frac{1}{x_n} - N\right) x_n^2 = x_n + x_n - Nx_n^2 = x_n(2 - Nx_n)$$

(a) Let $x_0 = 1.5$ and $x_1 = 2.0$. Then $f(x_0) = x_0^3 - 2x_0^2 - 5 = -1.625$ and $f(x_1) = x_1^3 - 2x_1^2 - 5 = -1.0$.

Using (3.3.6) with $n = 1$, we obtain

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)} = 2.0 - \frac{(-1.0)(2.0 - 1.5)}{-1.0 + 1.625} = 2.0 + \frac{0.5}{0.625} = 2.8$$

(b) Let $x_0 = 1.5$ and $x_1 = 2.0$. Then $f'(x) = 3x^2 - 4x$, so that $f'(x_1) = 4.0$.

Using (3.3.7) with $n = 1$, we obtain

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.0 - \frac{-1.0}{4.0} = 2.0 + 0.25 = 2.25$$

(c) Let $x_0 = 1.5$ and $x_1 = 2.0$. Then $f(x_0) = x_0^3 - 2x_0^2 - 5 = -1.625$, so that $f(x_0)f(x_1) < 0$.

Using (3.3.8) with $n = 1$, we obtain

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)} = 2.0 - \frac{(-1.0)(2.0 - 1.5)}{-1.0 + 1.625} = 2.0 + \frac{0.5}{0.625} = 2.8$$

Example 3.3.2 Consider the function $f(x) = x^3 - 2x^2 - 5$ on the interval $[1, 3]$. Since $f(1) = -6$ and $f(3) = 4$, we have $f(1)f(3) < 0$. Thus, $f(x)$ has a root in $[1, 3]$. Let us find this root using the methods of Example 3.3.1.

(a) Using $x_0 = 1.5$ and $x_1 = 2.0$ in the secant formula (3.3.6), we obtain $x_2 = 2.8$ (see Example 3.3.1).

Then

$$x_3 = x_2 - \frac{f(x_2)(x_2 - x_1)}{f(x_2) - f(x_1)} = 2.8 - \frac{(0.392)(0.8)}{0.392 + 1.0} = 2.8 - 0.225 = 2.575$$

$$x_4 = x_3 - \frac{f(x_3)(x_3 - x_2)}{f(x_3) - f(x_2)} = 2.575 - \frac{(-0.0699)(-0.225)}{-0.0699 - 0.392} = 2.575 - 0.032 = 2.543$$

$$x_5 = x_4 - \frac{f(x_4)(x_4 - x_3)}{f(x_4) - f(x_3)} = 2.543 - \frac{(-0.0019)(-0.032)}{-0.0019 + 0.0699} = 2.543 - 0.0009 = 2.5421$$

Since $x_5 \approx x_4$ upto 3 decimal places, we have $f(2.542) \approx 0.0000$.

(b) Using $x_0 = 1.5$ in the Newton-Raphson formula (3.3.7), we obtain $x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right)$.

Since an approximate value of $\sqrt{5}$ is 2.2361, we have

$$x_1 = \frac{1}{2}\left(x_0 + \frac{5}{x_0}\right) = \frac{1}{2}\left(1.5 + \frac{5}{1.5}\right) = 2.417$$

$$x_2 = \frac{1}{2}\left(x_1 + \frac{5}{x_1}\right) = \frac{1}{2}(2.417 + 2.069) = 2.2435$$

$$x_3 = \frac{1}{2}\left(x_2 + \frac{5}{x_2}\right) = \frac{1}{2}(2.2435 + 2.2284) = 2.2360$$

Since $x_3 \approx x_2$ upto 4 decimal places, we take $\sqrt{5} = 2.2360$.

(c) Using $N = 5$ in the Newton-Raphson formula (3.3.7), we obtain $x_{n+1} = \frac{1}{5}\left(4x_n + \frac{5}{x_n^4}\right)$.

Since an approximate value of $\sqrt[5]{5}$ is 1.3797, we have

$$x_1 = \frac{1}{5}\left(4x_0 + \frac{5}{x_0^4}\right) = \frac{1}{5}\left(4 \times 1.5 + \frac{5}{(1.5)^4}\right) = \frac{1}{5}(6 + 1.4815) = 1.4963$$

$$x_2 = \frac{1}{5}\left(4x_1 + \frac{5}{x_1^4}\right) = \frac{1}{5}(5.9852 + 1.3797) = 1.4730$$

$$x_3 = \frac{1}{5}\left(4x_2 + \frac{5}{x_2^4}\right) = \frac{1}{5}(5.8920 + 1.4730) = 1.4730$$

Since $x_3 \approx x_2$ upto 4 decimal places, we take $\sqrt[5]{5} = 1.4730$.

(c) Taking $N = 10$ and $x_0 = 1$, the above formula (3.3.7) becomes $x_{n+1} = \frac{1}{10}\left(9x_n + \frac{5}{x_n^9}\right)$.

Since an approximate value of $(36)^{1/3} = (27)^{1/3} = 3$, we take $x_0 = 3$.

Then $x_1 = \frac{1}{3}(2x_0 + 36/x_0^2) = \frac{1}{3}(6 + 36/9) = 3.33333$

$x_2 = \frac{1}{3}(2x_1 + 36/x_1^2) = \frac{1}{3}(6.6666 + 36/3.33333^2) = 3.30451$

$x_3 = \frac{1}{3}(2x_2 + 36/x_2^2) = \frac{1}{3}(6.6090 + 36/3.30451^2) = 3.3045$

Since $x_2 = x_3$ upto 4 decimal places, we have $(36)^{1/3} = 3.3045$.

(c) Taking $N = 30$ and $k = -3$, the above formula (4) becomes

$$x_{n+1} = \frac{1}{-3}(-8x_n + 30/x_n^3) = \frac{1}{3}(8x_n - 30/x_n^3)$$

Since an approximate value of $(30)^{-1/3} = (32)^{-1/3} = 1/2$, we take $x_0 = 1/2$.

Then $x_1 = \frac{1}{3}(8x_0 - 30x_0^3) = \frac{1}{12}(8 - 30/2^3) = \frac{1}{12}(8 - 30/8) = 0.59375$

$x_2 = \frac{1}{3}(8x_1 - 30x_1^3) = \frac{0.59375}{3}(8 - 30(0.59375)^3) = 0.59375(2.66667 - 3.75) = 0.59375(-1.08333) = -0.64323$

$x_3 = \frac{1}{3}(8x_2 - 30x_2^3) = \frac{-0.64323}{3}(8 - 30(-0.64323)^3) = -0.64323(2.66667 + 3.99999) = -0.64323(6.66666) = -4.28818$

Since $x_2 = x_3$ upto 4 decimal places, we take $(30)^{-1/3} = -0.6432$.

EXERCISE 2.5

1. Find a root of the following equations, using the bisection method correct to three decimal places :
   (a) $x^3 - 4x + 1 = 0$ (U.P.T.U., 2005) (b) $x^3 - x^2 - 1 = 0$ (U.P.T.U., 2004)
   (c) $x^3 + x - 11 = 0$ which lies between 2 and 3 (d) $2x^3 + x^2 - 20x + 12 = 0$.
2. Using the iteration method, find a real root of the following equations correct to three decimal places :
   (a) $\cos x = xe^x$ (Mumbai, 2002) (b) $x \log_{10} x = 1.2$ lying between 2 and 3
   (c) $x^3 + x = 3$ lying between 1 and 1.5 (d) $e^x = 5x + x$
3. Find a real root of the following equations correct to three decimal places by the method of false position :
   (a) $x^3 - x - 1 = 0$ (b) $x^3 - 4x - 9 = 0$ (U.P.T.U., 2003)
   (c) $x^3 - x - 1 = 0$ near $x = 1$ (d) $x^3 - y^3 - z^3 - z = 0$. (Rajasthan, 2001)
4. Using regula-falsi method, compute the real root of the following equations correct to three decimal places :
   (a) $e^x = x + 2$ (U.P.T.U., 2002) (b) $\cos x = 3x - 1$ (c) $x \sin x - 1 = 0$
   (d) $3x + \log x = 0$ (U.P.T.U., 2003) (e) $x e^x = \sin x$ (U.P.T., 2001)
5. Find the fourth root of 12 correct to three decimal places using the method of false position.
6. Find by Newton's method, a root of the following equations correct to 3 decimal places :
   (a) $x^3 - 3x + 1 = 0$ (Anna Univ., 2001) (b) $x^3 - 2x - 5 = 0$ (U.P.T.U., 2002)
   (c) $x^3 - 2x + 3 = 0$ (Madras, 1994)
   (d) $3x^3 - 3x^2 + 1 = 0$ lying between 1 and 2. (Madras, 2003)
7. Find a root of the following equations correct to three places of decimal using Newton's iterative method :
   (a) $x^4 + x^3 - 3x^2 - x + 3 = 0$ lying between 1 and 2 (Rajasthan, 2003)
   (b) $x^3 - 6x^2 + 9 = 0$.
8. Find the negative root of the equation $x^3 - 3x + 3000 = 0$ correct to two decimal places by Newton's method.
9. Using Newton-Raphson method, find a root of the following equations correct to the three decimal places :
   (a) $e^x - 3x = 0$ (U.P.T.U., 2005) (b) $x^3 + 4x - x = 0$ (Rajasthan, 2006)
   (c) $x \tan x + 1 = 0$ which lies near $x = \pi$ (A.M.I.E., 2009, U.P.T.U., 2006)
   (d) $x^3 = x^2 + \sin 20x$ which is near $x = 1.5$. (U.P.T.U., 2005)
10. Find by Newton's method, the root of the equations :
    (a) $\cos x = xe^x$ (A.M.I.E., 2009, U.P.T.U., 2005) (b) $x \log_{10} x = 12.34$ (Madras, 2006)
    (c) $10^x + x - 4 = 0$ (d) $x + \log_{10} x = 0.375$ (Rajasthan, 2002)
11. Develop a recurrence formula for finding $\sqrt[n]{N}$, using Newton-Raphson method and hence compute to three decimal places :
    (a) $\sqrt{23}$ (U.P.T.U., 2004) (b) $\sqrt[3]{20}$ (U.P.T.U., 2005)

13. Find the only root of $4\cos x - e^x + 1 = 0$, using Newton Raphson method. (Madras, 2003)
14. Develop an algorithm using N-R method, to find the fourth root of a positive number N and hence find $(83.45)^{1/4}$. (M.K.U., 2006)
15. Evaluate the following correct to 5 decimal places by using the Newton-Raphson method
(i) $\sqrt{218}$ (ii) $\sqrt[3]{37.2}$ (Anna, 2006) (iii) $\sqrt{15}$ (iv) $\sqrt[3]{68.75}$.

3.16 APPROXIMATE SOLUTION OF EQUATIONS – HORNER'S METHOD

This is the best method of finding approximate values of both rational and irrational roots of a numerical equation. Horner's method consists in determination of the root of an equation by successive digit by digit searching for its roots.

If the root of an equation lies between a and $a + 1$, then the value of this root will be a. $h_1 h_2 h_3 \ldots$, where $h_1, h_2, h_3 \ldots$ are digits is its decimal part. To obtain these, we proceed as follows :

(i) Diminish the roots of the given equation by a so that the root of the new equation is 0. $h_1 h_2 \ldots$

(ii) Then multiply the roots of the transformed equation by 10 so that the root of the new equation is $h_1 . h_2 \ldots$

(iii) Now diminish the root by h_1 and multiply the roots of the resulting equation by 10 so that the root is $h_2 . h_3 \ldots$

(iv) Now diminish the root by h_2 and so on. By continuing this process, the root may be evaluated to any desired degree of accuracy digit by digit. The method will be clear from the following example.

Example 3.11. *Find by Horner's method, the positive root of the equation $x^3 + x^2 + x - 100 = 0$ correct to three decimal places.*

Solution. Step I. Let $f(x) = x^3 + x^2 + x - 100$

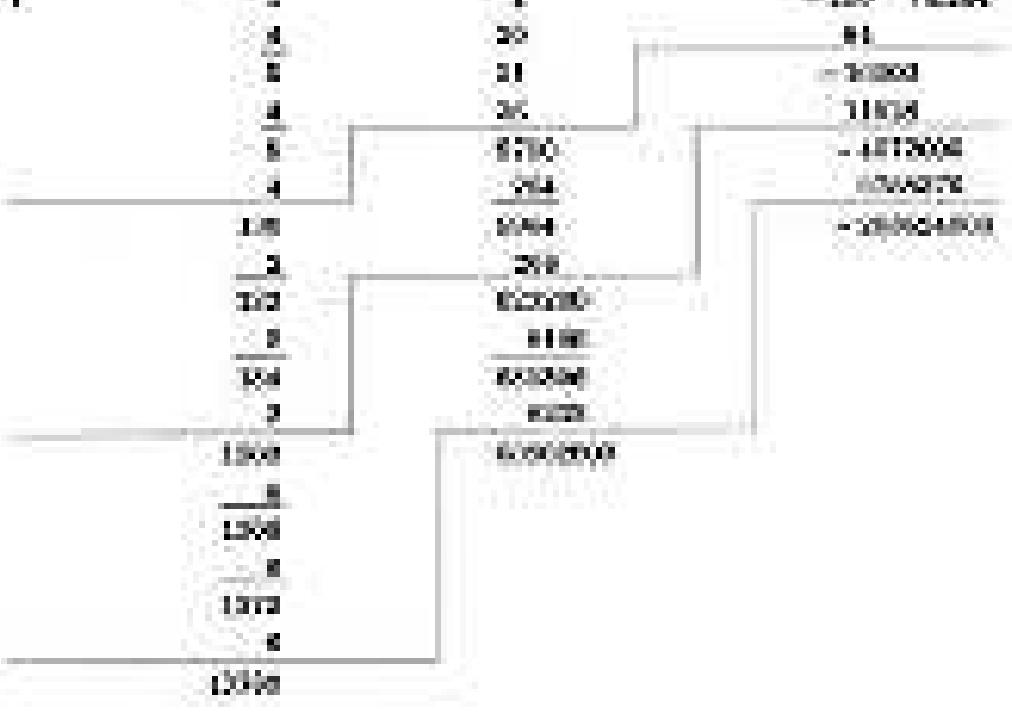
by Descartes' rule of signs, there is only one positive root. Also $f(4) = -$ ve and $f(5) = +$ ve. Therefore, the root lies between 4 and 5.

Step II. Diminish the roots of given equation by 4 so that the transformed equation is

$$x^3 + 13x^2 + 57x - 36 = 0 \quad \ldots(1)$$

Its root lies between 0 and 1. We draw a zig-zag line above the set of figures 1, 13, 57, – 36 which are the coefficients of the terms in (1) as shown below. Now multiply the roots of (1) by 10 for which multiply the second term by 10, the third term by 100 and the fourth term by 1000 (i.e. attach one zero to the second term, two zeros to the third term and three zeros to the fourth term). Then we get the equation

$$f_1(x) = x^3 + 130x^2 + 5700x - 36000 = 0 \quad \ldots(2)$$



Its root lies between 0 and 10.

Clearly $f(2) = -ve$, $f(3) = +ve$

$\therefore$ the root which lies between 2 and 3 i.e., first figure after decimal is 2.

Step III. Diminish the roots of $f(x) = 0$ by 2 so that the required transformed equation is

$$x^4 + 13x^3 + 62x^2 - 602 = 0 \quad \ldots(iii)$$

Its root lies between 0 and 2. (We draw the second sign-change line above the set of figures 13, 62, 0, −602. Multiply the root of (iii) by 10 i.e., annex two zeros to second term, two zeros to third term and three zeros to the fourth term. Then the new equation is

$$f_1(x) = x^4 + 1300x^3 + 620200x - 602000 = 0$$

Its root lies between 0 and 10, which is nearly $= \frac{602000}{620200} = 0$

Its new second figure after decimal place is 9.

Step IV. Diminish the roots of $f_1(x) = 0$ by 9, so that the transformed equation is

$$x^4 + 1336x^3 + 632236x - 203024 = 0$$

Its root lies between 0 and 1. (We draw the third sign-change line above the set of figures 1336, 632236, −203024.) As before multiply the roots by 10 i.e., annex one zero to the second term, two zeros to the third term and three zeros to the fourth term. Then the equation becomes

$$f_2(x) = x^4 + 13360x^3 + 63223600x - 203024000 = 0$$

Its root lies between 0 and 10, which is nearly $= \frac{203024000}{63223600} = 3$. Thus the roots of $f_2(x) = 0$ are to be diminished by 3 i.e., the third figure after decimal place is 3. But there is no need to proceed further as the root is required correct to three decimal places only. Hence the root is 2.391.

Note 1. (*We can adopt a shorter method to apply the principle of trial division in which we divide the last coefficient by the last but one coefficient to get the next integer for which the roots are to be diminished. Then last two coefficients should have opposite signs.*)

Note 2. (*In any stage of the trial division suppose the root we get is zero, then we should annex multiple the roots by 10 and proceed with decimal placing of the root.*)

Example 28.12. *Find the value of $\sqrt[3]{30}$ correct to 3 decimal places using Horner's method.*

Solution. Step I. Let $x = \sqrt[3]{30}$ i.e., $f(x) = x^3 - 30 = 0$

Now $f(3) = -ve$, i.e., $f(4) = +ve$

$\therefore$ the root lies between 3 and 4.

Step II. Diminish the roots of the given equation by 3 so that the transformed equation is

$$x^3 + 9x^2 + 27x - 3 = 0 \quad \ldots(i)$$

Its roots lies between 0 and 1. (We draw a sign-change line above the set of numbers 9, 27, −3 which are the coefficients of the terms in (i). Now multiply the roots of (i) by 10 by annexing a zero to the second term, two zeros to the third term and three zeros to the fourth term. Then we get the equation

$$f_1(x) = x^3 + 90x^2 + 2700x - 3000 = 0 \quad \ldots(ii)$$

Its roots lies between 0 and 10.

Clearly $f_1(1) = -ve$, $f_1(2) = +ve$

$\therefore$ the root of (ii) lies between 1 and 2 i.e., first figure after decimal place is 1.

Step III. Diminish the roots of $f_1(x) = 0$ by 1, so that the required transformed equation is

$$x^3 + 93x^2 + 2883x - 280 = 0 \quad \ldots(iii)$$

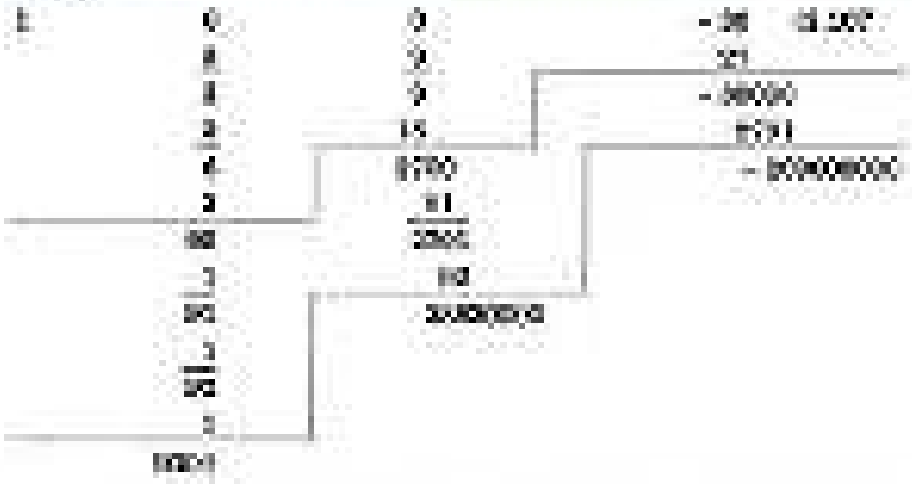
Its root lies between 0 and 1. (We draw a second sign-change line above the set of figures 93, 2883, −280. Multiply the roots of (iii) by 10 i.e., annex one zero to second term, two zeros to third term and three zeros to the fourth term. Then the new equation is

$$f_2(x) = x^3 + 930x^2 + 288300x - 280000 = 0$$

Its root lies between 0 and 10, which is nearly

$$= 280000/288300 = 0.971 = 0 \text{ (nearly)} = 1.$$

Hence second figure after decimal place is 0.



Step IV. Divide (x) the root of $f(x) = 0$ by 10 and then multiply the roots by 10 so that

$f(x) = x^3 - 2000x^2 + 200000x - 2000000 = 0$.

Its root lies between 0 and 10, which is nearly $= 2000/200 = 10 \times 7$ and $= 0$. Therefore roots of $f(x) = 0$ are to be diminished by 7 i.e., the third figure after decimal is 7. Hence the required root is 2.107.

PRACTICE SET

1. Find by Horner's method, the real root correct to three decimal places of the equation
   (i) $x^3 - 2x - 1 = 0$ which lies between 1 and 2
   (ii) $x^3 + x - 1 = 0$ (Coimbatore, 2003) (iii) $x^3 + 4x - 11 = 0$
   (iv) $x^3 + 2x^2 + 3x = 1$ which lies between 1 and 2. (Madras, 2006 S)
2. Using Horner's method, find the largest real root of $x^3 - 4x + 1 = 0$ correct to three decimal places.
3. Show that the root of the equation $x^3 + x^2 - 100 = 0$ lies between 4 and 5. Find its value correct to two decimal places by Horner's method.
4. Find the negative root of the equation $x^3 - 6x^2 + 10 = 0$ correct to two decimal places by Horner's method.
5. Find the value of $\sqrt[3]{20}$ by Horner's method correct to 3 decimal places.

5.5 SOLUTION OF LINEAR SIMULTANEOUS EQUATIONS

Simultaneous linear equations occur in various engineering problems. The student knows that a given system of linear equations can be solved by Cramer's rule or by Matrix method (§ 3.33). But these methods become tedious for large systems. However, there exist other numerical methods of solution which are well-suited for computing machines. We now explain some direct and iterative methods of solution.

5.6 DIRECT METHODS OF SOLUTION

(1) Gauss elimination method*. In this method, the unknowns are eliminated successively and the system is reduced to an upper triangular system from which the unknowns are found by back substitution. The method is quite general and is well adapted for computer operations. Here we shall explain it by considering a system of three equations for the sake of clarity.

Consider the equations

$$\left.\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}\right| \quad \ldots(1)$$

Step I. To eliminate x from second and third equations.

Assuming $a_1 \neq 0$, we eliminate x from the second equation by subtracting $(a_2/a_1) \times$ (the first equation) from the second equation. Similarly we eliminate x from the third equation by subtracting $(a_3/a_1) \times$ the first equation from the third equation. We thus get the new system

*See Section 5.7.

$$\left.\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ b_2y + c_2z &= d_2' \\ b_3y + c_3z &= d_3' \end{aligned}\right] \quad \ldots(3)$$

Here the first equation is called the pivotal equation and a_1 is called the first pivot.

Step II. To eliminate y from third equation in (3).

Assuming $b_2' \neq 0$, we eliminate y from the third equation of (3) by subtracting $\frac{b_3'}{b_2'}$ times the second equation from the third equation. We thus get the new system

$$\left.\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ b_2'y + c_2'z &= d_2' \\ c_3'z &= d_3' \end{aligned}\right] \quad \ldots(4)$$

Here the second equation is the pivotal equation and b_2' is the new pivot.

Step III. To evaluate the unknowns.

The values of x, y, z are found from the reduced system (4) by back substitution.

Note 1. Writing the given equations as $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$ i.e., $AX = D$, this method consists in transforming the coefficient matrix A to upper triangular matrix by elementary row transformations only.

Note 2. Clearly this method will fail if any one of the pivots a_1, b_2', c_3' is zero. In such cases, we rewrite the equations in a different order so that the pivots are non-zero.

Note 3. Partial and complete pivoting: In the first step, the numerically largest coefficient of x is chosen from all the equations and brought as the first pivot by interchanging the first equation with the equation having the largest coefficient of x. In the second step, the numerically largest coefficient of y is chosen from the second and third equations and brought as the second pivot by interchanging the second equation with the equation having the largest coefficient of y. The process is continued till we arrive at the equation with the one unknown. This method is called partial pivoting.

If we are not satisfied about the elimination of x, y, z in a specified order, then we choose as our pivot the numerically largest coefficient of the entire system of coefficients. This requires not only an interchange of equations but also an interchange of the positions of the variables. This method of elimination is called complete pivoting. It is more complicated and does not appreciably improve the accuracy.

Example 20.12. Apply Gauss elimination method to solve the equations $x + 2y + z = 3$, $2x + 3y + 3z = 10$, $3x - y + 2z = 13$. *(Bhopal, 2006)*

Solution. We have

	Check sum	
$x + 2y + z = 3$	7	...(1)
$2x + 3y + 3z = 10$	18	...(2)
$3x - y + 2z = 13$	17	...(3)

Step I. Operate (2) − 2(1) and (3) − 3(1) to eliminate x:

	Check sum	
$-y + 3z = 4$	6	...(4)
$-7y - z = 4$	0	...(5)

Step II. Operate (5) $-\frac{13}{11}$ (4) to eliminate y:

	Check sum	
$\frac{72}{11}z = \frac{96}{11}$	72	...(6)

Step III. By back-substitution, we get

From (6) : $z = \frac{96}{72} = 1.3333$

From (4) : $y = \frac{1}{3}z - \frac{4}{3} = \frac{1}{3}\left(\frac{96}{72}\right) - \frac{4}{3} = -\frac{84}{72} = -1.1666$

From (1): $x = -5 - 4\left(-\frac{91}{11}\right) + \frac{268}{11} = \frac{213}{11} = 1.8678$

Hence $x = 1.8678, y = -1.1488, z = 2.0845$

Note: A useful check is provided by adding the sum of the coefficients and terms on the right, operating on these numbers as on the variables and checking that the derived equation have the correct sum.

Otherwise: We have $\begin{bmatrix}1 & 4 & -3\\ 1 & 3 & -6\\ 3 & -1 & -1\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}-5\\-12\\4\end{bmatrix}$

Operate $R_2 - R_1$ and $R_3 - 3R_1$: $\begin{bmatrix}1 & 4 & -3\\ 0 & -1 & -3\\ 0 & -13 & 8\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}-5\\-7\\19\end{bmatrix}$

Operate $R_3 - \frac{13}{3}R_2$: $\begin{bmatrix}1 & 4 & -3\\ 0 & -1 & -3\\ 0 & 0 & 71/3\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}-5\\-7\\148/3\end{bmatrix}$

Thus, we have $z = 148/71 = 2.0845$,

$-y = 7 - 3z = 7 - 10.4225 = -3.4225$ or, $y = -1.1408$

and $x = -5 - 4y + 3z = -5 + 4(1.1408) + 2.0845 = 1.6478$

Hence $x = 1.6478, y = -1.1408, z = 2.0845$.

Example 20.14. Solve $10x - 2y + 3z + 6w = 9$, $-6x + 6y - z - 4w = 5$, $3x + y + 6z + 11w = 3$, $5x - 5y - 2z + 4w = 7$ **by Gauss elimination method.** (U.P.T.U., 2002)

Solution. We have

	Check sum	
$10x - 2y + 3z + 6w = 9$	17	...(1)
$-6x + 6y - z - 4w = 5$	2	...(2)
$3x + y + 6z + 11w = 3$	23	...(3)
$5x - 5y - 2z + 4w = 7$	9	...(4)

Step I. To eliminate x, operate $\left[(1) - \left(\frac{-6}{10}\right)(2)\right], \left[(3) - \frac{3}{10}(1)\right], \left[(4) - \frac{5}{10}(1)\right]$:

	Check sum	
$3.6y + 1.8z - 1 = 8.8$	12.8	...(5)
$1.2y + 5.1z + 9.8w = 0.2$	16.3	...(6)
$-4.5y - 3.5z + 1.5w = 4$	− 3.5	...(7)

Step II. To eliminate y, operate $\left[(5) - \frac{1.8}{3.6}(6)\right], \left[(7) - \left(\frac{-4.5}{3.6}\right)(6)\right]$:

$$3.2179283z + 10.2107066w = -0.8107960 \quad ...(8)$$

$$-8.3625063z + 0.0625032w = 10.4475063 \quad ...(9)$$

Step III. To eliminate z, operate $\left[(9) - \left(\frac{-8.3625063}{3.2179283}\right)(8)\right]$:

$$9.9218513w = 9.8218377$$

Step IV. By back-substitution, we get

$$w = 1, z = -7, y = 4 \text{ and } x = 5.$$

(2) Gauss-Jordan method*. This is a modification of the Gauss elimination method. In this method, elimination of unknowns is performed not in the equations below but in the equations above also, obviously reducing the system to a diagonal matrix form i.e., each equation involving only one unknown. From these equations the unknowns x, y, z can be obtained readily.

That is in this method, the labour of back-substitution for finding the unknowns is saved at the cost of additional calculations.

*See footnote p. 21

Example 9.8 Apply Gauss-Jordan method to solve the equations
$x + y + z = 9$; $2x - 3y + 4z = 13$; $3x + 4y + 5z = 40$. (UPTU, 2004 / AMIE, 2005)

Solution: We have
$$x + y + z = 9 \quad \ldots(i)$$
$$2x - 3y + 4z = 13 \quad \ldots(ii)$$
$$3x + 4y + 5z = 40 \quad \ldots(iii)$$

Step I. Operate (ii) − 2(i) and (iii) − 3(i) to eliminate x from (ii) and (iii).
$$x + y + z = 9 \quad \ldots(iv)$$
$$-5y + 2z = -5 \quad \ldots(v)$$
$$y + 2z = 13 \quad \ldots(vi)$$

Step II. Operate (vi) + $\frac{1}{5}$ (v) and (iv) + $\frac{1}{5}$ (v) to eliminate y from (vi) and (iv):
$$x + \frac{3}{5}z = 6 \quad \ldots(vii)$$
$$-5y + 2z = -5 \quad \ldots(viii)$$
$$\frac{12}{5}z = 12 \quad \ldots(ix)$$

Step III. Operate (vii) − $\frac{3}{12}$(ix) and (viii) − $\frac{5}{12}$(ix) to eliminate z from (vii) and (viii):
$$x = 1$$
$$-5y = -15$$
$$\frac{12}{5}z = 12$$

Hence the solution is $x = 1, y = 3, z = 5$.

Observation: Rewriting the equations as $\begin{bmatrix} 1 & 1 & 1 \\ 2 & -3 & 4 \\ 3 & 4 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \\ 40 \end{bmatrix}$

Operate $R_2 - 2R_1, R_3 - 3R_1$: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 2 \\ 0 & 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ 13 \end{bmatrix}$

Operate $R_3 + \frac{1}{5}R_2$: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 12/5 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ 12 \end{bmatrix}$

Operate $-R_2, \frac{5}{12}R_3$: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 5 & -2 \\ 0 & 0 & 12 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 60 \end{bmatrix}$

Operate $R_1 + \frac{1}{12}R_3, \frac{2}{12}R_3$: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 15 \\ 5 \end{bmatrix}$

Operate $\frac{1}{5}R_2$: $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 3 \\ 5 \end{bmatrix}$

Operate $R_1 - R_2 - R_3$: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$

Hence $x = 1, y = 3, z = 5$.

Obs: Here the process of elimination of variables amounts to reducing the given coefficient matrix to a diagonal matrix by elementary row transformations only.

Example 10.10: Solve the equations of example 10.9 by Gauss-Jordan method

Solution: We have

$$10x - 7y + 3z + 5u = 8 \quad \ldots(1)$$
$$-6x + 8y - z - 4u = 5 \quad \ldots(2)$$
$$3x + 6y + 4z + 11u = 2 \quad \ldots(3)$$
$$5x + 8y - 3z + 4u = 7 \quad \ldots(4)$$

Step I. To eliminate x, operate $\left[(1)-\left(\frac{-6}{10}\right)(1)\right], \left[(3)-\left(\frac{3}{10}\right)(1)\right], \left[(4)-\left(\frac{5}{10}\right)(1)\right]$:

$$10x - 7y + 3z + 5u = 8 \quad \ldots(5)$$
$$0.8y + 0.8z - u = 6.8 \quad \ldots(6)$$
$$8.1y + 3.1z + 8.5u = 0.2 \quad \ldots(7)$$
$$6.5y - 2.5z + 1.5u = 3 \quad \ldots(8)$$

Step II. To eliminate y, operate $\left[(6)-\left(\frac{-7}{8.1}\right)(7)\right], \left[(8)-\left(\frac{6.5}{8.1}\right)(7)\right], \left[(8)-\left(\frac{-2.5}{8.1}\right)(7)\right]$:

$$10x + 4.4703704z + 3.1358025u = 21.543210 \quad \ldots(9)$$
$$0.8y + 0.8z - u = 6.8 \quad \ldots(10)$$
$$2.4173584z + 10.210370u = -6.4185185 \quad \ldots(11)$$
$$-2.9421582z + 3.9028518u = 10.467285 \quad \ldots(12)$$

Step III. To eliminate z, operate $\left[(11)-\left(\frac{2.417284}{2.4173584}\right)(11)\right]$,

$$\left[(12)-\left(\frac{0.8}{2.4173584}\right)(11)\right], \left[(12)-\left[\frac{-2.9421582}{2.4173584}\right](11)\right]:$$

$$10x - 15.4814815u = 34.2345679$$
$$0.8y - 4.3750426u = 10.8837997$$
$$2.4173584z + 10.2103704u = -6.4185185$$
$$16.0317024u = 32.0634013$$

Step IV. From the last equation $u = 2$ nearly.

Substitution of $u = 2$ in the above three equations gives $x = 5, y = 4, z = -7$.

(3) **Factorization method**[1]. This method is based on the fact that every matrix A can be expressed as the product of a lower triangular matrix and an upper triangular matrix, provided all the principal minors of A are non-singular, i.e., if $A = [a_{ij}]$, then

$$a_{11} \neq 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \neq 0, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \neq 0, \text{ etc.}$$

Also such a factorization if it exists, is unique.

Now consider the equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

which can be written as $AX = B$ $\ldots(13)$

where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

Let $A = LU$ $\ldots(14)$

where $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$

[1] Another name given to this decomposition is Doolittle's method.

Then (1) becomes $LUX = B$ (3)

Writing $UX = Y$, (4)

(3) becomes $LY = B$ which is equivalent to the equations

$$y_1 = b_1 \quad l_{21}y_1 + y_2 = b_2 \quad l_{31}y_1 + l_{32}y_2 + y_3 = b_3$$

Solving these for y_1, y_2, y_3 we have Y. Then (4) becomes

$$u_{11}x_1 + u_{12}x_2 + u_{13}x_3 = y_1 \quad u_{22}x_2 + u_{23}x_3 = y_2 \quad u_{33}x_3 = y_3$$

from which x_1, x_2 and x_3 can be found by back substitution.

To evaluate the matrices L and U, we write (2) as

$$\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Multiplying the matrices on the left and equating corresponding elements from both sides, we obtain

(a) $u_{11} = a_{11}$ $\quad l_{21}u_{11} = a_{21}$ $\quad l_{31}u_{11} = a_{31}$

(b) $l_{21}u_{11} = a_{21}$ or $l_{21} = a_{21}/u_{11}$

$l_{31}u_{11} = a_{31}$ or $l_{31} = a_{31}/u_{11}$

(c) $l_{21}u_{12} + u_{22} = a_{22}$ or $u_{22} = a_{22} - \dfrac{a_{21}}{u_{11}}a_{12}$

$l_{31}u_{13} + u_{23} = a_{23}$ or $u_{23} = a_{23} - \dfrac{a_{21}}{u_{11}}a_{13}$

(d) $l_{31}u_{12} + l_{32}u_{22} = a_{32}$ or $l_{32} = \dfrac{1}{u_{22}}\left[a_{32} - \dfrac{a_{31}}{u_{11}}a_{12}\right]$

(e) $l_{31}u_{13} + l_{32}u_{23} + u_{33} = a_{33}$ which gives u_{33}

We can compute the elements of L and U in the following order:

(1) First row of U, (2) First column of L,

(3) Second row of U, (4) Second column of L, (5) Third row of U.

This procedure can easily be generalized.

Note. This method is important in the numerical solution of linear simultaneous equations when the number of unknowns is large. Computers use this method. Various subroutines are available in the software for computers.

Example 20.11 Apply factorization method to solve the equations

$$3x + 2y + z = 1 \quad 4x + 3y + z = 2 \quad 3x + 4y + 3z = 3 \qquad \text{(Nautical, 2000)}$$

Solution. Let $\begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 3 & 1 \\ 3 & 4 & 3 \end{bmatrix}$ i.e. $LU = A$,

or (by)

(a) $u_{11} = 3, u_{12} = 2, \quad u_{13} = 1$

(b) $l_{21}u_{11} = 4, \quad \therefore l_{21} = 4/3$

$l_{31}u_{11} = 3, \quad \therefore l_{31} = 1$

(c) $l_{21}u_{12} + u_{22} = 3, \quad \therefore u_{22} = 1/3,$

$l_{31}u_{13} + u_{23} = 1, \quad \therefore u_{23} = -2$

(d) $l_{31}u_{12} + l_{32}u_{22} = 4, \quad \therefore l_{32} = 6$

(e) $l_{31}u_{13} + l_{32}u_{23} + u_{33} = 3$

$\therefore \quad u_{33} = 9$

Thus $A = \begin{bmatrix} 1 & 0 & 0 \\ 4/3 & 1 & 0 \\ 1 & 6 & 1 \end{bmatrix}\begin{bmatrix} 3 & 2 & 1 \\ 0 & 1/3 & -2 \\ 0 & 0 & 9 \end{bmatrix}$

Working with $\mathbf{A}^T$, this gives eigenvector components

$$\begin{bmatrix} \\ \\ \end{bmatrix}$$

Solving this system, we have $x_3 = t$,

$$x_2 = \begin{cases} \\ \\ \end{cases} \quad \text{or} \quad x_2 = \begin{cases} \\ \\ \end{cases}$$

Eliminate the original system becomes

$$\begin{bmatrix} \\ \\ \end{bmatrix}$$

(a) $\quad$

By inspection, we have $x = -$ (b) $y =$ and $y =$ (c)

Example 9.9: Finding Scale Factors by Rayleigh's Method for a Two-Degree-of-Freedom System

Solution: Let

$$\begin{bmatrix} & & & & & & & \\ & & & & & & & \\ & & & & & & & \\ & & & & & & & \end{bmatrix} \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \qquad (9.4.\text{--}30)$$

so that

$$\begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

Now

$$\mathbf{A} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

Working with $\mathbf{AC}$ like a power iteration

$$\begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

Solving for the eigenvalue, we get

$$x_1 = , \; x_2 = , \; x_3 = \quad \text{and} \quad x_4 =$$

Where the original system becomes

$$\begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

(a) $\quad$

By inspection, we have $x = -$ (b) $y =$ and $y =$ (c)

EXERCISE 93

Solve the following equations by Gauss elimination method:

1. $2x + y + z = 10$; $3x + 2y + 3z = 18$; $x + 4y + 9z = 16$ (J.M.U., 2000)
2. $2x + 3y + z = 11$; $2x + 5y + 2z = 8$; $5x + 10y - 8z = 15$ (W.M.T.U., 2000)
3. $3x - y - 2z = 8$; $x + y + z = 0$; $x - y + z = 2$ (Nagpur U., 2000)
4. $2x_1 + 4x_2 + x_3 = 3$; $3x_1 + 2x_2 - 3x_3 = 2$; $x_1 + x_2 + x_3 = 0$. (Mysore Univ., 2000)
5. $3x_1 + x_2 + x_3 + x_4 = 4$; $x_1 + 3x_2 + x_3 - x_4 = 13$;
   $x_1 + x_2 + 3x_3 + x_4 = 6$; $x_1 - x_2 + x_3 + 3x_4 = 15$.

Solve the following equations by Gauss-Jordan method:

6. $3x + 6y + 7z = 68$; $2x + y - z = 0$; $x + z = 2y$. (V.T.U., 2000)
7. $2x - 3y + z = -1$; $x + 2y + 2z = 22$; $3x - y + z = 2$. (Nagpur, 2000)
8. $x + 3y + 2z = 26$; $4x + 4y + 3z = 58$; $2x + 3y + 4z = 34$. (Anna, 2000)
9. $3x + y + z = 16$; $x - 2y + z = 0$; $x + 4y - 2z = 25$. (V.T.U., 2000)
10. $2x_1 + x_2 + 2x_3 + x_4 = 6$; $x_1 + x_2 - 2x_3 + 4x_4 = 7$;
    $2x_1 + 3x_2 - 3x_3 + x_4 = 6$; $3x_1 + 4x_2 + 3x_3 - 2x_4 = 8$.

Solve the following equations by factorisation method:

11. $3x + y + z = 12$; $2x + 3y + z = 13$; $2x + 2y + 3z = 14$. (Madras, 2000) (V.T.U., 2007)
12. $x + 2y + 3z = 14$; $2x + 3y + 4z = 20$; $3x + 4y + z = 14$.
13. $5x - 2y + z = 4$; $x + 5y - 2z = 0$; $2x + y + 5z = 8$.
14. $3x_1 + x_2 + x_3 = 7$; $2x_1 - x_2 + x_3 = 3$; $x_1 + 2x_2 - x_3 = 2$; $x_1 + x_2 + 2x_3 = 5$.
15. Find the inverse of the matrix $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 3 \\ 1 & -3 & 3 \end{bmatrix}$ by Gauss's method.

23.7 ITERATIVE METHODS OF SOLUTION

The preceding methods of solving simultaneous linear equations are known as *direct methods* as they yield exact solutions. On the other hand, an iterative method is that in which we start from an approximation to the true solution and obtain better and better approximations from a computational cycle repeated as often as necessary for achieving a desired accuracy.

Simple iterative methods can be devised for systems in which the coefficients of the leading diagonal are large compared to others. We now explain these methods:

(1) Jacobi's iteration method*. Consider the equations

$$\left.\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}\right] \quad \ldots(1)$$

If a_1, b_2, c_3 are large as compared to other coefficients, then solving these for x, y, z respectively, the system can be written in the form

$$\left.\begin{aligned} x &= \lambda_1 - \alpha_1y - \gamma_1z \\ y &= \lambda_2 - \beta_2x - \gamma_2z \\ z &= \lambda_3 - \beta_3x - \gamma_3y \end{aligned}\right] \quad \ldots(2)$$

Let us start with the initial approximations x_0, y_0, z_0 (usually $= 0$) for the values of x, y, z. Substituting these on the right, we get the first approximations $x_1 = \lambda_1, y_1 = \lambda_2, z_1 = \lambda_3$.

Substituting these on the right-hand sides of (2), the second approximations are given by

$$\begin{aligned} x_2 &= \lambda_1 - \alpha_1y_1 - \gamma_1z_1 \\ y_2 &= \lambda_2 - \beta_2x_1 - \gamma_2z_1 \\ z_2 &= \lambda_3 - \beta_3x_1 - \gamma_3y_1 \end{aligned}$$

This process is repeated till the difference between two consecutive approximations is negligible.

*See footnote p. 235.

Example 28.19. Solve by Jacobi's iterative method, the equations $10x + y - z = 11.19$, $x + 10y + z = 28.08$, $x + y + 10z = 35.61$, correct to two decimal places. (Anna, 2003)

Solution. Rewriting the given equations as

$$x = \frac{1}{10}(11.19 - y + z),\ y = \frac{1}{10}(28.08 - x - z),\ z = \frac{1}{10}(35.61 + x + y)$$

We start from an approximation $x_0 = y_0 = z_0 = 0$.

First iteration $x_1 = \frac{11.19}{10} = 1.119,\ y_1 = \frac{28.08}{10} = 2.808,\ z_1 = \frac{35.61}{10} = 3.561$

Second iteration $x_2 = \frac{1}{10}(11.19 - y_1 + z_1) = 1.19$

$y_2 = \frac{1}{10}(28.08 - x_1 - z_1) = 2.24$

$z_2 = \frac{1}{10}(35.61 + x_1 + y_1) = 3.39$

Third iteration $x_3 = \frac{1}{10}(11.19 - y_2 + z_2) = 1.32$

$y_3 = \frac{1}{10}(28.08 - x_2 - z_2) = 2.35$

$z_3 = \frac{1}{10}(35.61 + x_2 + y_2) = 3.43$

Fourth iteration $x_4 = \frac{1}{10}(11.19 - y_3 + z_3) = 1.33$

$y_4 = \frac{1}{10}(28.08 - x_3 - z_3) = 2.38$

$z_4 = \frac{1}{10}(35.61 + x_3 + y_3) = 3.43$

Fifth iteration $x_5 = \frac{1}{10}(11.19 - y_4 + z_4) = 1.33$

$y_5 = \frac{1}{10}(28.08 - x_4 - z_4) = 2.38$

$z_5 = \frac{1}{10}(35.61 + x_4 + y_4) = 3.43$

Hence $x = 1.33,\ y = 2.38,\ z = 3.43$.

Example 28.20. Solve, by Jacobi's iteration method, the equations
$20x + y - 2z = 17,\ 3x + 20y - z = -18,\ 2x - 3y + 20z = 25$ (Anna, 2006)

Solution. We write the given equations in the form

$$\left.\begin{aligned} x &= \frac{1}{20}(17 - y + 2z) \\ y &= \frac{1}{20}(-18 - 3x + z) \\ z &= \frac{1}{20}(25 - 2x + 3y) \end{aligned}\right| \quad \ldots(1)$$

We start from an approximation $x_0 = y_0 = z_0 = 0$.
Substituting these on the right sides of the equations (1), we get

$$x_1 = \frac{17}{20} = 0.85,\ y_1 = -\frac{18}{20} = -0.9,\ z_1 = \frac{25}{20} = 1.25$$

Putting these values on the right of the equations (1), we obtain

$$x_2 = \frac{1}{20}(17 - y_1 + 2z_1) = 1.02$$

$$y_2 = \frac{1}{20}(-18 - 2x_1 + z_1) = -0.990$$

$$z_2 = \frac{1}{20}(25 - 2x_1 + 3y_1) = 1.1510$$

Substituting these values in the right sides of the equations (1), we have

$$x_2 = \frac{1}{20}(17 - y_2 + 2z_2) = 1.0184$$

$$y_2 = \frac{1}{20}(-18 - 3x_2 + z_2) = -0.9864$$

$$z_2 = \frac{1}{20}(25 - 2x_2 + 3y_2) = 1.0002$$

Substituting these values, we get

$$x_3 = \frac{1}{20}(17 - y_2 + 2z_2) = 1.0009$$

$$y_3 = \frac{1}{20}(-18 - 3x_2 + z_2) = -1.0018$$

$$z_3 = \frac{1}{20}(25 - 2x_2 + 3y_2) = 0.9995$$

Putting these values, we have

$$x_3 = \frac{1}{20}(17 - y_3 + 2z_3) = 1.0000$$

$$y_3 = \frac{1}{20}(-18 - 3x_3 + z_3) = -1.0002$$

$$z_3 = \frac{1}{20}(25 - 2x_3 + 3y_3) = 0.9998$$

Again substituting these values, we get

$$x_4 = \frac{1}{20}(17 - y_3 + 2z_3) = 1.0000$$

$$y_4 = \frac{1}{20}(-18 - 3x_3 + z_3) = -1.0002$$

$$z_4 = \frac{1}{20}(25 - 2x_3 + 3y_3) = 1.0000$$

The values in the 3rd and 4th iterations being practically the same, we can stop.

Hence the solution is $x = 1, y = -1, z = 1$.

(3) **Gauss-Seidal iteration method**[*]. This is a modification of the Jacobi's iteration method. As before, we start with initial approximations x_0, y_0, z_0 (each = 0) for x, y, z respectively. Substituting $y = y_0, z = z_0$ in the first of the equations (2) on page 837, we get

$$x_1 = k_1$$

Then putting $x = x_1, z = z_0$ in the second of the equations (2) on page 837, we have

$$y_1 = k_2 - l_2x_1 - m_2z_0$$

Now substituting $x = x_1, y = y_1$ in the third of the equations (2) on page 837, we obtain

$$z_1 = k_3 - l_3x_1 - m_3y_1$$

and so on. As soon as new approximation for an unknown is found, it is immediately used in the next step.

This process of iteration is continued till convergence to the desired degree of accuracy is obtained.

Note 1. Since the most recent approximation of the unknowns are used in proceeding to the next step, the convergence in the Gauss-Seidal method is faster than in Jacobi's method.

Note 2. Gauss-Seidal method converges if in each equation, the absolute value of the largest coefficient is greater than the sum of the absolute values of the remaining coefficients.

[*]See footnote p. 37. *Philipp Ludwig Von Seidel* (1821–1896) of Munich suggested this iterative method.

Example 20.21. Apply Gauss-Seidel iteration method to solve the equations of Ex. 20.20.
(U.P.T.U. 2012 ; Rohtak 2003 ; Madras 2003)

Solution. We write the given equations in the form

$$x = \frac{1}{20}(17 - y + 2z); \; y = \frac{1}{20}(-18 - 3x + z); \; z = \frac{1}{20}(25 - 2x + 3y) \quad ...(1)$$

We start from the approximation $x_0 = y_0 = z_0 = 0$. Substituting $y = y_0, z = z_0$ in the right side of the first of equations (1), we get

$$x_1 = \frac{1}{20}(17 - y_0 + 2z_0) = 0.8500$$

Putting $x = x_1, z = z_0$ in the second of the equations (1), we have

$$y_1 = \frac{1}{20}(-18 - 3x_1 + z_0) = -1.0275$$

Putting $x = x_1, y = y_1$ in the last of the equations (1), we obtain

$$z_1 = \frac{1}{20}(25 - 2x_1 + 3y_1) = 1.0189$$

For the second iteration, we have

$$x_2 = \frac{1}{20}(17 - y_1 + 2z_1) = 1.0048$$

$$y_2 = \frac{1}{20}(-18 - 3x_2 + z_1) = -0.9998$$

$$z_2 = \frac{1}{20}(25 - 2x_2 + 3y_2) = 0.9969$$

For the third iteration, we get

$$x_3 = \frac{1}{20}(17 - y_2 + 2z_2) = 1.0008$$

$$y_3 = \frac{1}{20}(-18 - 3x_3 + z_2) = -1.0000$$

$$z_3 = \frac{1}{20}(25 - 2x_3 + 3y_3) = 1.0000$$

The values in the 2nd and 3rd iterations being practically the same, we can stop.
Hence the solution is $x = 1, y = -1, z = 1$.

Example 20.22. Solve the equations :

$$20x_1 - 2x_2 - x_3 - x_4 = 3$$
$$-2x_1 + 20x_2 - x_3 - x_4 = 18$$
$$-x_1 - x_2 + 10x_3 - 2x_4 = 27$$
$$-x_1 - x_2 - 2x_3 + 10x_4 = -9$$

by Gauss Seidel iteration method. (Bhopal 2009 ; J.N.T.U. 2010)

Solution. Rewriting the given equations as

$$x_1 = 0.15 + 0.1x_2 + 0.05x_3 + 0.05x_4 \quad ...(1)$$
$$x_2 = 1.8 + 0.1x_1 + 0.05x_3 + 0.05x_4 \quad ...(2)$$
$$x_3 = 2.7 + 0.1x_1 + 0.1x_2 + 0.2x_4 \quad ...(3)$$
$$x_4 = -0.9 + 0.1x_1 + 0.1x_2 + 0.2x_3 \quad ...(4)$$

First Iteration
Putting $x_2 = 0, x_3 = 0, x_4 = 0$ in (1), we get $x_1 = 0.15$
Putting $x_1 = 0.15, x_3 = 0, x_4 = 0$ in (2), we obtain $x_2 = 1.98$
Putting $x_1 = 0.15, x_2 = 1.98, x_4 = 0$ in (3), we obtain $x_3 = 2.898$
Putting $x_1 = 0.15, x_2 = 1.98, x_3 = 2.898$ in (4), we get $x_4 = -0.1269$

Second iteration

Putting $x_2 = 1.36, x_3 = 2.068, x_4 = -0.2368$ in (1), we obtain

$x_1 = 0.8687$

Putting $x_1 = 0.8687, x_3 = 2.068, x_4 = -0.2368$ in (2), we obtain

$x_2 = 1.0032$

Putting $x_1 = 0.8687, x_2 = 1.0032, x_4 = -0.2368$ in (3), we have

$x_3 = 2.0081$

Putting $x_1 = 0.8687, x_2 = 1.0032, x_3 = 2.0081$ in (4), we get

$x_4 = -0.0048$

Third iteration

Putting $x_2 = 1.0032, x_3 = 2.0081, x_4 = -0.0048$ in (1), we obtain

$x_1 = 0.9836$

Putting $x_1 = 0.9836, x_3 = 2.0081, x_4 = -0.0048$ in (2), we obtain

$x_2 = 1.0009$

Putting $x_1 = 0.9836, x_2 = 1.0009, x_4 = -0.0048$ in (3), we get

$x_3 = 2.0021$

Putting $x_1 = 0.9836, x_2 = 1.0009, x_3 = 2.0021$ in (4), we get

$x_4 = -0.0012$

Fourth iteration. Proceeding as above

$x_1 = 0.9969, x_2 = 1.0002, x_3 = 2.0005, x_4 = -0.0003$

Fifth iteration is

$x_1 = 0.9994, x_2 = 1.0001, x_3 = 2.0001, x_4 = -0.0001$

Sixth iteration is

$x_1 = 0.9999, x_2 = 1.0000, x_3 = 2.0000, x_4 = -0.0001$

Hence the solution is $x_1 = 1, x_2 = 1, x_3 = 2, x_4 = 0$.

(2) Relaxation method*: Consider the equations

$a_{11}x + a_{12}y + a_{13}z = d_1;\ a_{21}x + a_{22}y + a_{23}z = d_2;\ a_{31}x + a_{32}y + a_{33}z = d_3$

We define the residuals R_1, R_2, R_3 by the relations

$$R_1 = d_1 - a_{11}x - a_{12}y - a_{13}z;\ R_2 = d_2 - a_{21}x - a_{22}y - a_{23}z;\ R_3 = d_3 - a_{31}x - a_{32}y - a_{33}z \quad \ldots(1)$$

To start with we assume $x = y = z = 0$ and calculate the initial residuals. Then the residuals are reduced step by step by giving increments to the variables. For this purpose, we construct the following operation table:

	R_1	R_2	R_3
$\Delta x = 1$	$-a_{11}$	$-a_{21}$	$-a_{31}$
$\Delta y = 1$	$-a_{12}$	$-a_{22}$	$-a_{32}$
$\Delta z = 1$	$-a_{13}$	$-a_{23}$	$-a_{33}$

We note from the equations (1) that if x is increased by 1 keeping y and z constant, R_1, R_2 and R_3 decrease by a_{11}, a_{21}, a_{31} respectively. This is shown in the above table along with the effects on the residuals when y and z are given unit increments. The table is the transpose of the coefficient matrix.

At each step, the numerically largest residual is reduced to zero or almost zero. To reduce a particular residual, the value of the corresponding variable is changed; $e.g.$, to reduce R_1 by p, x should be increased by p/a_{11}.

When all the residuals have been reduced to almost zero, the increments in x, y, z are added separately to give the desired solution.

Obs. As a check, the computed values of x, y, z are substituted in (1) and the residuals are calculated. If these residuals are not all negligible, then there is some mistake and the entire process should be rechecked.

Example 29.23. Solve, by Relaxation method, the equations:

$$20x - 2y + z = 17,\ x + 2y - 20z = 18,\ -2x + 20y + 2z = 19. \qquad \text{(Madras, 2000S)}$$

*This method was originally developed by R.V. Southwell in 1935. It is applicable to structural engineering problems.

Solution. The residuals are given by

$R_1 = 50 - 8x + 2y - z$; $R_2 = 35 - x - 5y + 3z$; $R_3 = 19 + 2x - 2y - 7z$

The operation table is

	$2R_1$	$3R_2$	$5R_3$
$8x = 1$	−8	−1	2
$5y = 1$	2	−5	2
$7z = 1$	−1	−3	−7

The relaxation table is

	R_1	R_2	R_3	
$x = y = z = 0$	50	35	19	−11
$8x = 8$	6	35	35	101
$5y = 4$	1	30	1	−5.6
$5y = 5$	11	3	−13	−1.6
$8x = 1$	3	−1	−7	−5.1
$5y = -1$	3	−4	3	−3.01
$8x = 6.63$	1.4	3	1.6	−0.31
$5y = 3.69$	1.37	0.69	−.89	−0.01
$7z = 2.11$	0	0.69	2.17	.01
$5y = 3.113$	0.006	0	−0.006	−.01

$\Sigma 8x = 8.33$, $\Sigma 5y = 4.31$, $\Sigma 7z = 5.23$.

Thus $x = 8.33$, $y = 4.31$, $z = 5.23$.

[Explanation. In 3.1, 50 is largest residual in 3.0. To reduce R_1 we give an increment to $x = 8$ and the resulting residuals are shown in 3.11. Since $R_2 = 35$ is the largest, and we give an increment to $y = 4$ to get the residuals in 3.12. In 3.13, $R_3 = 4$ is the largest and we give an increment $7z = -14$ to obtain the results in 3.14. Similarly the other steps have been explained.]

Example 15.24. Solve the following system of five equations :

$20x + 2y - 3z + 25t = 228$; $-2x + 20y - 2z + 25t = 153$; $3x - y + 20z + 2t = 72$. (U.P.T.U., 2001 S) (Rohtak, 2003)

Solution. The residuals are given by

$R_1 = 228 - 20x + 2y + 3z + 25t$; $R_2 = 153 + 2x - 20y + 2z + 25t$; $R_3 = 72 - 3x + y - 20z - 2t$.

The operation table is

	$20R_1$	$25R_2$	$25R_3$
$20x = 1$	−20	2	3
$25t = 1$	25	−25	−1
$20z = 1$	3	2	−20

The relaxation table is :

	R_1	R_2	R_3	
$x = y = z = t = 0$	228	153	72	11.4
$20x = 20$	8	155	72	7.75
$25t = 25$	33	6	72	1.32
$25t = 25$	91	48	−2	3.64
$20x = 12$	3	48	18	2.4
$25t = 5$	8	3	25	1.28
$20x = 8$	18	4	3	0.9
$25t = 9$	−8	3	9	0.36
$25t = 1$	−2	12	−2	0.48
$20x = 1$	8	3	−3	0.4

$\Sigma 20x = 8.32$, $\Sigma 25t = 9.00$, $\Sigma 20z = 2.14$.

Hence $x = 0.42$, $y = 0.45$, $z = 0.11$, $t = 0.36$.

EXERCISE 9.8

1. Solve by Jacobi's method, the equations $5x - y + z = 10$; $2x + 4y = 12$; $x + y + 5z = -1$ correct to three decimal places. (Use $x_0 = 2, y_0 = 3, z_0 = 1$.)
2. Solve the equations $27x + 6y - z = 85$; $x + y + 54z = 110$; $6x + 15y + 2z = 72$ by (a) Jacobi's method, (b) Gauss-Seidel method. (Anna, 2006)

Solve the following equations by Gauss-Seidel method.

3. $2x + y + 6z = 9$; $8x + 3y + 2z = 13$; $x + 4y + z = 7$.
4. $20x + 4y - z = 17$; $x + 3y + 20z = 24$; $2x + 13y + 4z = 35$. (Madras, 2005)
5. $10x + y + z = 12$; $2x + 10y + z = 13$; $2x + 2y + 10z = 14$. (U.P.T.U., MCA, 2007)
6. $83x + 11y - 4z = 95$; $7x + 52y + 13z = 104$; $3x + 8y + 29z = 71$. (Bharathiar, 2006)
7. $3x_1 + 0.5x_2 + 0.5x_3 = 2.68$; $0.1x_1 + 2x_2 - 0.2x_3 = -10.8$; $0.3x_1 - 0.4x_2 + 10x_3 = 71.4$. (Madras, 2004)
8. $5.4x + 2.1y + 1.2z = 9.9$; $1.2x + 6.3y + 1.7z = 21.9$; $0.3x + 0.5y + 4.2z = 15.3$.
9. $\begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 5 \end{bmatrix}$

Solve by Relaxation method, the following sets of equations.

10. $20x + y - 2z = 17$; $3x + 20y + z = -18$; $2x - 3y + 20z = 25$. (Madras, 2005)
11. $10x - 2y - 2z = 6$; $-x + 10y - 2z = 7$; $-x - y + 10z = 8$.
12. $-6x + 3y + 12z = 35$; $10x - 2y + 3z = 51$; $x + 9y - 2z = 62$.
13. $54x + y + z = 110$; $2x + 15y + 6z = 72$; $-x + 6y + 20z = 55$. (Madras, 2003)

9.9 SOLUTION OF NON-LINEAR SIMULTANEOUS EQUATIONS—NEWTON-RAPHSON METHOD

Consider the equations

$$f(x, y) = 0, g(x, y) = 0 \qquad \ldots(1)$$

If an initial approximation (x_0, y_0) to a solution has been found by graphical method or otherwise, then a better approximation (x_1, y_1) can be obtained as follows:

Let $x_1 = x_0 + h, y_1 = y_0 + k$ so that $f(x_0 + h, y_0 + k) = 0, g(x_0 + h, y_0 + k) = 0$ $\qquad \ldots(2)$

Expanding each of the functions in (2) by Taylor's series in first degree terms, we get approximately

$$\left.\begin{aligned} f_0 + h\frac{\partial f}{\partial x_0} + k\frac{\partial f}{\partial y_0} = 0 \\ g_0 + h\frac{\partial g}{\partial x_0} + k\frac{\partial g}{\partial y_0} = 0 \end{aligned}\right\} \qquad \ldots(3)$$

where $f_0 = f(x_0, y_0), \dfrac{\partial f}{\partial x_0} = \left(\dfrac{\partial f}{\partial x}\right)_{x_0, y_0}$ etc.

Solving the equations (3) for h and k, we get a new approximation to the root as

$$x_1 = x_0 + h, y_1 = y_0 + k$$

This process is repeated till we get the values to the desired accuracy.

Example 9.9.1. *Solve the system of non-linear equations:*

$$x^2 + y = 11, y^2 + x = 7. \qquad \text{(Anna, 2006)}$$

Solution. An initial approximation to the solution is obtained from a rough graph of the given equations, as $x_0 = 3.5$ and $y_0 = -1.5$.

We have $f = x^2 + y - 11$ and $g = y^2 + x - 7$ so that

$$\frac{\partial f}{\partial x} = 2x, \frac{\partial f}{\partial y} = 1 \quad \text{and} \quad \frac{\partial g}{\partial x} = 1, \frac{\partial g}{\partial y} = 2y.$$

Then Newton Raphson's equations (3) above will be

$$7.0 + h + 0.5k = 0, h - 3.0k + 0.25 = 0$$

Solving these, we get $h = 0.1600, k = 0.2533$

$\therefore$ the better approximation to the root is

$$x_1 = x_0 + h = 2.3636,\ y_1 = y_0 + k = -1.9818$$

Repeating the above process, replacing (x_0, y_0) by (x_1, y_1), we obtain $x_2 = 2.3646$, $y_2 = -1.9832$.

EXERCISE 2.10.2

1. Solve the equations $x^2 + y = 11$, $y^2 + x = 7$.
2. Solve the non-linear equations $x = 2y + \frac{1}{2}(y^2 + xy)$, $y = x + \frac{1}{2}(x^2 + xy)$. Correct to three decimals.
3. Use Newton-Raphson method to solve the equations $x^2 + y^2 + y = 1$, $x^2 - y^2 + x = 1$ correct to four decimals, starting with the approximation $(0.5, 0.5)$.
4. Solve the non-linear equations $x^2 - y^2 = 1$, $x^3 + y^3 = 19$ correct to two decimals, starting with $x_0 = 3$, $y_0 = 2$ using N-R method. Carry out two iterations. (V.T.U., 2010, 2012)
5. Solve the equations $x^2 + 2xy + y^2 + 3 = 0$, $x^2 + 2xy + y^2 + x = 0$. Correct to three decimal places, using Newton-Raphson method, given that $x_0 = -3$ and $y_0 = 3$.

2.12 DETERMINATION OF EIGEN VALUES BY ITERATION

In § 2.11, we came across equations of the type

$$\left.\begin{aligned} (a_{11} - \lambda)x_1 + a_{12}x_2 + a_{13}x_3 &= 0 \\ a_{21}x_1 + (a_{22} - \lambda)x_2 + a_{23}x_3 &= 0 \\ a_{31}x_1 + a_{32}x_2 + (a_{33} - \lambda)x_3 &= 0 \end{aligned}\right\} \quad \ldots(1)$$

which in matrix form, may be written as $[A - \lambda I]X = 0$ or $AX = \lambda X$ $\quad \ldots(2)$

where $A = [a_{ij}]$ and X is the column matrix $[x_i]$.

Equation (1) will have a non-trivial solution if the coefficient matrix vanishes, *i.e.*,

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix} = 0$$

This gives a cubic in λ whose roots are called eigen values of (2) and corresponding to each eigen value, we have a non-zero solution $X = [x_1, x_2, x_3]$ which is called an eigen vector. Such an equation can ordinarily be solved easily.

In some applications, it is required to compute the numerically largest eigen value and the corresponding eigen vector. In such cases, the following iterative method is more convenient which is also well-suited for computing on a computer.

If X_1, X_2, X_3 be the eigen vectors corresponding to the eigen values $\lambda_1, \lambda_2, \lambda_3$, then an arbitrary vector can be written as $X = k_1X_1 + k_2X_2 + k_3X_3$.

Then $\quad AX = k_1AX_1 + k_2AX_2 + k_3AX_3 = k_1\lambda_1X_1 + k_2\lambda_2X_2 + k_3\lambda_3X_3$

Similarly $\quad A^2X = k_1\lambda_1^2X_1 + k_2\lambda_2^2X_2 + k_3\lambda_3^2X_3$

and $\quad A^nX = k_1\lambda_1^nX_1 + k_2\lambda_2^nX_2 + k_3\lambda_3^nX_3$

If $|\lambda_1| > |\lambda_2| > |\lambda_3|$, then the contribution of the term $k_1\lambda_1^nX_1$ to the sum on the right increases with n and therefore, every time we multiply and hence vector by A, it becomes nearer to the eigen vector X_1. Therefore make the largest component of the resulting column vector unity to avoid the factor k_1.

Thus we start with a reasonable vector X which is near the solution as possible and evaluate AX which is written as $X^{(1)}X^{(1)}$ after normalisation. This gives the first approximation $X^{(1)}$ to the eigen vector of $X^{(1)}$ to eigen vector. Similarly we evaluate $AX^{(1)} = X^{(2)}X^{(2)}$ which gives the second approximation. We repeat this process till $X^{(n)} = X^{(n-1)}$ becomes exactly. Then $\lambda^{(n)}$ will be the largest eigen value of A and $X^{(n)}$, the corresponding eigen vector.

This iterative procedure for finding the dominant eigen value of a matrix is known as *Rayleigh's power method*.*

*After the English mathematician and physicist John William Strutt known as Lord Rayleigh (1842-1919) who made important contributions to the theory of sound, fluid dynamics and hydrodynamics. He was professor at Cambridge and London.

[illegible]

Using the power method, determine the largest eigenvalue in magnitude and the corresponding eigenvector of the matrix A given by

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

with $\mathbf{x}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

Then

$$A\mathbf{x}_0 = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 3 \end{bmatrix} = 5 \begin{bmatrix} 0.6 \\ 1 \\ 0.6 \end{bmatrix} = \lambda^{(1)}\mathbf{x}^{(1)}$$

So this is an approximation to the eigenvalue $\lambda^{(1)} = 5$ and the corresponding eigenvector is $\mathbf{x}^{(1)} = \begin{bmatrix} 0.6 \\ 1 \\ 0.6 \end{bmatrix}$.

Now

$$A\mathbf{x}^{(1)} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0.6 \\ 1 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 2.6 \\ 3.8 \\ 2.6 \end{bmatrix} = 3.8 \begin{bmatrix} 0.6842 \\ 1 \\ 0.6842 \end{bmatrix} = \lambda^{(2)}\mathbf{x}^{(2)}$$

Thus this second approximation to the eigenvalue is $\lambda^{(2)} = 3.8$ and the corresponding eigenvector is $\mathbf{x}^{(2)} = \begin{bmatrix} 0.6842 \\ 1 \\ 0.6842 \end{bmatrix}$ (remembering that division by zero is not permitted).

Now

$$A\mathbf{x}^{(2)} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0.6842 \\ 1 \\ 0.6842 \end{bmatrix} = \begin{bmatrix} 2.6842 \\ 3.7368 \\ 2.6842 \end{bmatrix} = 3.7368 \begin{bmatrix} 0.7183 \\ 1 \\ 0.7183 \end{bmatrix} = \lambda^{(3)}\mathbf{x}^{(3)}$$

$$A\mathbf{x}^{(3)} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0.7183 \\ 1 \\ 0.7183 \end{bmatrix} = \begin{bmatrix} 2.7183 \\ 3.8366 \\ 2.7183 \end{bmatrix} = 3.8366 \begin{bmatrix} 0.7086 \\ 1 \\ 0.7086 \end{bmatrix} = \lambda^{(4)}\mathbf{x}^{(4)}$$

$$A\mathbf{x}^{(4)} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0.7086 \\ 1 \\ 0.7086 \end{bmatrix} = \begin{bmatrix} 2.7086 \\ 3.8172 \\ 2.7086 \end{bmatrix} = 3.8172 \begin{bmatrix} 0.7096 \\ 1 \\ 0.7096 \end{bmatrix} = \lambda^{(5)}\mathbf{x}^{(5)}$$

$$A\mathbf{x}^{(5)} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0.7096 \\ 1 \\ 0.7096 \end{bmatrix} = \begin{bmatrix} 2.7096 \\ 3.8192 \\ 2.7096 \end{bmatrix} = \lambda^{(6)}\mathbf{x}^{(6)}$$

Clearly $\lambda^{(5)} \approx \lambda^{(6)}$ and $\mathbf{x}^{(5)} \approx \mathbf{x}^{(6)}$ to three decimal places. Hence the largest eigenvalue in R and the corresponding eigenvector is $\begin{bmatrix} 0.710 \\ 1 \\ 0.710 \end{bmatrix}$.

(c) Let the initial approximation to the required eigenvector be $\mathbf{x}_0 = [1, 1, 1]^T$.

Then

$$A\mathbf{x}_0 = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix} = -2 \begin{bmatrix} 0.5 \\ 1 \\ 0.5 \end{bmatrix} = \lambda^{(1)}\mathbf{x}^{(1)}$$

So this is an approximation to the eigenvalue $\lambda^{(1)} = -2$ and the corresponding eigenvector $\mathbf{x}^{(1)} = [0.5, 1, 0.5]^T$.

Hence

$$A\mathbf{x}^{(1)} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \\ 0.5 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} = \lambda^{(2)}\mathbf{x}^{(2)}$$

Expanding the above process, we get

$$A\mathbf{x}^{(2)} = 0.5 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \lambda^{(3)}\mathbf{x}^{(3)}, \quad A\mathbf{x}^{(3)} = 0.25 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \lambda^{(4)}\mathbf{x}^{(4)}$$

$$A\mathbf{x}^{(4)} = 0.25 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \lambda^{(5)}\mathbf{x}^{(5)}, \quad A\mathbf{x}^{(5)} = 0.25 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \lambda^{(6)}\mathbf{x}^{(6)}, \quad A\mathbf{x}^{(6)} = 0.25 \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \lambda^{(7)}\mathbf{x}^{(7)}$$

Clearly $\lambda^{(5)} \approx \lambda^{(7)}$ and $\mathbf{x}^{(5)} \approx \mathbf{x}^{(7)}$ approximately.

Hence the largest eigenvalue in R is 0.25 and the corresponding eigenvector is $[0.5, -1, 0.5]^T$.

PROBLEMS

1. Find by power method, the largest eigen value of the matrix

(a) $\begin{bmatrix} 1 & 3 \\ 3 & -4 \end{bmatrix}$ (Anna, 2002) (b) $\begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$

2. Obtain the largest eigen value and the corresponding eigen vector for the matrix
$(A - 2I)x_1 - x_2 = 0$, $-x_1 + (2 - 2)x_2 - x_3 = 0$, $-x_2 + (2 - 2)x_3 = 0$
by Rayleigh Quotient method.

3. Find the dominant eigen value and the corresponding eigen vector of the following matrices using the power method:

(a) $\begin{bmatrix} 4 & 2 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ (U.P.T.U., 2003) (b) $\begin{bmatrix} 8 & -2 & 2 \\ -2 & 5 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ (U.P.T.U., 2002)

4. Find the largest eigen value and the corresponding eigen vector of the matrices

(a) $\begin{bmatrix} 1 & -3 & 2 \\ 4 & 2 & 1 \\ 8 & 3 & 1 \end{bmatrix}$ (Madras, 2000) (b) $\begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ (U.P.T.U., 2000)

(c) $\begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & 4 \\ -1 & 4 & 0 \end{bmatrix}$ with initial eigen vector $[1, 1, 0]^T$. (Madras, 2000)

OBJECTIVE TYPE OF QUESTIONS

PROBLEMS

Fill up the blanks or select the correct answer for each of the following problems:

1. Out of Regula-falsi method and Newton-Raphson method, the rate of convergence is faster for ..... .
2. If x_0 is the initial guess, then the Newton-Raphson formula is ..... .
3. In the Regula-falsi method of finding the real root of an equation, the curve AB is replaced by ..... .
4. Newton's iterative formula to find the value of $\sqrt{N}$ is ..... .
5. Newton-Raphson method is also known as ..... .
6. In solving simultaneous equations by Gauss-Jordan method, the coefficient matrix is reduced to ..... matrix.
7. In the regula-falsi method, the convergence is
(a) linear (b) quadratic (c) very slow.
8. The order of convergence in Newton-Raphson method is
(a) 1 (b) 2 (c) 3 (d) zero.
9. The Newton-Raphson algorithm for finding the value of $1/N$ is ..... .
10. The iteration method for finding the root of an equation $f(x) = 0$ is ..... .
11. In Regula-falsi method, the first approximation is given by ..... .
12. The order of convergence in Newton-Raphson method is
(a) 0 (b) 1 (c) 2 (d) none.
13. The iterative formula for finding the reciprocal of N is $x_{n+1} = \ldots$ .....
14. As soon as a new value of a variable is found by iteration, it is used immediately in the following computations, this method is called
(a) Gauss-Jordan method (b) Gauss-Seidel method
(c) Jacobi's method (d) Relaxation method.
15. Out of Regula-falsi method and Newton-Raphson method, the rate of convergence is faster for ..... .
16. The difference between direct and iterative methods of solving a set of simultaneous linear equations is ..... .
17. To which form the coefficient matrix is transformed when $[A|I]$ is solved by Gauss-Jordan method.
18. Jacobi's iteration method can be used to solve a system of non-linear equations. (True or False)
19. The convergence in the Gauss-Seidel method is faster as fast as in Jacobi's method. (True or False)
20. By Gauss elimination method, solve $x + y = 1$ and $2x + 2y = 3$. (Anna, 2003)

CHAPTER 29

Finite Differences and Interpolation

1. Finite differences. 2. Differences of a polynomial. 3. Factorial notation. 4. Relation between the operators. 5. To find the missing terms in a series. 6. Newton's interpolation formulae. 7. Gauss's forward interpolation formula—Gauss's backward interpolation formula. 8. Stirling's formula. 9. Bessel's formula. 10. Everett's formula. 11. Choice of an interpolation formula. 12. Interpolation with unequal intervals. 13. Lagrange's formula. 14. Inverse interpolation. 15. Newton's divided differences formula. 16. Inverse interpolation. 17. Objective Type of Questions.

29.1. FINITE DIFFERENCES

Suppose we are given the following values of $y = f(x)$ for a set of values of x:

x:	x_0	x_1	$x_2 = x_0 + h$
y:	y_0	y_1	$y_2 = y_0 + k$

Then the process of finding the values of y corresponding to any value of $x = x_0$ between x_0 and x_2 is called *interpolation*. That is, interpolation is the technique of estimating the value of a function for any intermediate value of the independent variable while the process of computing the value of the function outside the given range is called *extrapolation*. The study of this interpolation is based on the concept of differences of a function which we proceed to discuss. For a detailed study, the reader should refer to author's book 'Numerical Methods in Engineering and Science'.

Suppose that the function $y = f(x)$ is tabulated for the equally spaced values $x = x_0, x_0 + h, x_0 + 2h, \ldots, x_0 + nh$ giving $y = y_0, y_1, y_2, \ldots, y_n$. To determine the values of $f(x)$ or $f(x)$ for some intermediate value of x, the following three types of differences are found useful :

(1) **Forward differences.** The differences $y_1 - y_0, y_2 - y_1, \ldots, y_n - y_{n-1}$ when denoted by $\Delta y_0, \Delta y_1, \ldots, \Delta y_{n-1}$ respectively are called the first forward differences where Δ is the forward difference operator. Thus the first forward differences are $\Delta y_r = y_{r+1} - y_r$.

Similarly, the second forward differences are defined by

$$\Delta^2 y_r = \Delta y_{r+1} - \Delta y_r$$

In general, $\Delta^n y_r = \Delta^{n-1} y_{r+1} - \Delta^{n-1} y_r$

are the nth forward differences.

These differences are symmetrically set out as follows in what is called a *Forward Difference Table*.

In a difference table, y_0 is called the *leading term* and the first entry in each column is called the *leading term* and $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0$, etc. are called the *leading differences*.

Note. Any higher order forward differences can be expressed in terms of the ordinates.

For, we have $\Delta^2 y_0 = \Delta y_1 - \Delta y_0 = (y_2 - y_1) - (y_1 - y_0) = y_2 - 2y_1 + y_0$

$\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0 = (y_3 - 2y_2 + y_1) - (y_2 - 2y_1 + y_0) = y_3 - 3y_2 + 3y_1 - y_0$

$\Delta^4 y_0 = \Delta^3 y_1 - \Delta^3 y_0 = (y_4 - 3y_3 + 3y_2 - y_1) - (y_3 - 3y_2 + 3y_1 - y_0) = y_4 - 4y_3 + 6y_2 - 4y_1 + y_0$

The coefficients occurring on the right hand side being the binomial coefficients, we have in general,

$$\Delta^n y_0 = y_n - {}^nC_1 y_{n-1} + {}^nC_2 y_{n-2} - \ldots + (-1)^n y_0$$

Forward Difference Table

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5
x_0	y_0					
		Δy_0				
x_0+h	y_1		$\Delta^2 y_0$			
		Δy_1		$\Delta^3 y_0$		
x_0+2h	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$	
		Δy_2		$\Delta^3 y_1$		$\Delta^5 y_0$
x_0+3h	y_3		$\Delta^2 y_2$		$\Delta^4 y_1$	
		Δy_3		$\Delta^3 y_2$		
x_0+4h	y_4		$\Delta^2 y_3$			
		Δy_4				
x_0+5h	y_5					

(2) **Backward differences.** The differences $y_1 - y_0, y_2 - y_1, \ldots, y_n - y_{n-1}$ where denoted by $\nabla y_1, \nabla y_2, \ldots, \nabla y_n$ respectively, are called the first backward differences where ∇ is the backward difference operator. Similarly we define higher order backward differences. Thus we have

$$\nabla y_r = y_r - y_{r-1} \Rightarrow \nabla^2 y_r = \nabla y_r - \nabla y_{r-1}$$
$$\nabla^2 y_r = \nabla^2 y_r - \nabla^2 y_{r-1}, \text{ etc.}$$

The differences are exhibited in the following:

Backward Difference Table

x	y	∇	∇^2	∇^3	∇^4	∇^5
x_0	y_0					
		∇y_1				
x_0+h	y_1		$\nabla^2 y_1$			
		∇y_2		$\nabla^3 y_2$		
x_0+2h	y_2		$\nabla^2 y_2$		$\nabla^4 y_2$	
		∇y_3		$\nabla^3 y_3$		$\nabla^5 y_5$
x_0+3h	y_3		$\nabla^2 y_3$		$\nabla^4 y_3$	
		∇y_4		$\nabla^3 y_4$		
x_0+4h	y_4		$\nabla^2 y_4$			
		∇y_5				
x_0+5h	y_5					

(3) **Central differences.** Sometimes it is convenient to employ another system of differences known as central differences. In this system, the central difference operator δ is defined by the relations:

$$y_1 - y_0 = \delta y_{1/2}, y_2 - y_1 = \delta y_{3/2}, \ldots, y_n - y_{n-1} = \delta y_{n-1/2}$$

Similarly, higher order central differences are defined as

$$\delta y_{1/2} - \delta y_{1/2} = \delta^2 y_1, \delta y_{3/2} - \delta y_{1/2} = \delta^2 y_2, \ldots$$
$$\delta^2 y_1 - \delta^2 y_0 = \delta^3 y_{1/2}, \text{ and so on.}$$

The differences are shown in the following:

Central Difference Table

x	y	δ	δ^2	δ^3	δ^4	δ^5
x_0	y_0					
		$\delta y_{1/2}$				
x_0+h	y_1		$\delta^2 y_1$			
		$\delta y_{3/2}$		$\delta^3 y_{3/2}$		
x_0+2h	y_2		$\delta^2 y_2$		$\delta^4 y_2$	
		$\delta y_{5/2}$		$\delta^3 y_{5/2}$		$\delta^5 y_{5/2}$
x_0+3h	y_3		$\delta^2 y_3$		$\delta^4 y_3$	
		$\delta y_{7/2}$		$\delta^3 y_{7/2}$		
x_0+4h	y_4		$\delta^2 y_4$			
		$\delta y_{9/2}$				
x_0+5h	y_5					

We see from this result that the central differences on the same horizontal line have the same suffix. Also the differences of odd order are known only for half values of the suffix and those of even order for only integral values of the suffix.

It is often required to find the means of adjacent values in the same column of differences. We denote this mean by μ. Thus

$$\mu\delta y_x = \tfrac{1}{2}(\delta y_{x+\frac{1}{2}} + \delta y_{x-\frac{1}{2}}),\quad \mu\delta^2 y_x = \tfrac{1}{2}(\delta^2 y_{x+1} + \delta^2 y_x), \text{ etc.}$$

Note. The reader should now find it a very easy matter to establish such results as the following:

$$y_2 - y_1 = \delta y_{1\frac{1}{2}} = \delta y_{2-\frac{1}{2}} = \mu\delta y_2 - \tfrac{1}{2}\delta^2 y_2$$

Of all the interpolation formulae, those involving central differences are most useful in practice as the coefficients in such formulae decrease in size more rapidly.

Example 6.11. Find an expression for $\Delta\tan^{-1}x$ and hence find $\Delta^2\tan^{-1}x$. (Hence find $\Delta^n\tan^{-1}x$.) (UCE, 2003)

Solution. (i) $\Delta\tan^{-1}x = \tan^{-1}(x+h) - \tan^{-1}x$

$$= \tan^{-1}\left[\frac{x+h-x}{1+x(x+h)}\right] = \tan^{-1}\left(\frac{h}{1+hx+x^2}\right)$$

(ii) $\Delta^2\log 2x = \Delta^{1+1}\log 2x = \Delta^1\log 2(x+h) - \Delta^1\log 2x$

$$= \Delta^{1-1}\log 2(x+h) - \Delta^{1-1}\log 2x + \Delta^{1-1}\log 2(x+h) - \Delta^{1-1}\log 2x$$

$$= \Delta^{1-1}\log\frac{x+h}{x} + \Delta^{1-1}\log\frac{x+2h}{x+h}$$

$$= \Delta^0\left[\log\left(1+\frac{h}{x}\right) + \log\left(1+\frac{h}{x+h}\right)\right]$$

(iii) $\Delta\left(\frac{x^2}{\cos 2x}\right) = \frac{(x+h)^2}{\cos 2(x+h)} - \frac{x^2}{\cos 2x} = \frac{(x+h)^2\cos 2x - x^2\cos 2(x+h)}{\cos 2(x+h)\cos 2x}$

$$= \frac{h^2x + h^3 - x^2(\cos 2x - \cos 2(x+h))}{\cos 2(x+h)\cos 2x}$$

$$= \frac{(2hx+h^2)\cos 2x + 2x^2\sin(h)\sin(2x+h)}{\cos 2(x+h)\cos 2x}$$

(iv) $\Delta^2\cos 2x = \Delta\cos 2(x+h) - \cos 2x$

$$= \cos 2(x+2h) - \cos 2(x+h)$$

$$= [\cos 2x\cos 4h - \sin 2x\sin 4h] - [\cos 2x\cos 2h - \sin 2x\sin 2h]$$

$$= -2\sin(2x+3h)\sin h + 2\sin(2x+h)\sin h$$

$$= -2\sin h[\sin(2x+3h) - \sin(2x+h)]$$

$$= -2\sin h[2\cos(2x+2h)\sin h] = -4\sin^2 h\cos(2x+2h)$$

Example 6.12. Evaluate (i) $\Delta^2\left(\frac{5x+12}{x^2+5x+6}\right)$ (Manchester, 2000) (ii) $\Delta^2\left(\frac{x^2}{(x-1)(x-2)}\right)$ (iii) $\Delta^2\left(\frac{x^2}{(x-1)(x-2)}\right)$ interval of differencing being unity. (Bristol, 2000)

Solution. (i) $\Delta^2\left(\frac{5x+12}{x^2+5x+6}\right) = \Delta^2\left(\frac{5x+12}{(x+2)(x+3)}\right) = \Delta^2\left(\frac{2}{x+2} + \frac{3}{x+3}\right)$

$$= \Delta\left[\Delta\left(\frac{2}{x+2}\right) + \Delta\left(\frac{3}{x+3}\right)\right] = \Delta\left[2\left(\frac{1}{x+3} - \frac{1}{x+2}\right) + 3\left(\frac{1}{x+4} - \frac{1}{x+3}\right)\right]$$

$$= -2\Delta\left(\frac{1}{(x+2)(x+3)}\right) - 3\Delta\left(\frac{1}{(x+3)(x+4)}\right)$$

$$= -2\left(\frac{1}{(x+3)(x+4)} - \frac{1}{(x+2)(x+3)}\right) - 3\left(\frac{1}{(x+4)(x+5)} - \frac{1}{(x+3)(x+4)}\right)$$

$$= \frac{4}{(x+2)(x+3)(x+4)} + \frac{6}{(x+3)(x+4)(x+5)} = \frac{2(3x+26)}{(x+2)(x+3)(x+4)(x+5)}$$

(ii) $\Delta(ab^x) = a\Delta b^x = a(b^{x+1} - b^x) = ab^x(b-1) = a(b-1)b^x$

$\Delta^2(ab^x) = \Delta[a(b-1)b^x] = a(b-1)\Delta b^x$

$= a(b-1)b^{x+1} - a(b-1)b^x = a(b-1)^2 b^x$

(iii) $\Delta x^2 = x^2 + 1 - x^2 = 2x + 1 = 2x^1$

$\Delta^2 x^2 = \Delta(2x^1) = 2\Delta x^1 = 2(x+1-x) = 2 \cdot 1 = 2!$

$\Delta^3 x^2 = \Delta^2(2x^1) = 2\Delta^2 x^1 = 2(x+1-1) = 2(x-1) = 2x^1 - 2 = 2x^2$

Similarly $\Delta x^3 = 3x^2 + 3x + 1 = 3x^2 + 3x^1 + 1$, $\Delta^2 x^3 = 6x + 6 = 6x^1 + 6$, ... and $\Delta^n x^n = n(n-1)\ldots 1 = n!$

3.12 DIFFERENCES OF A POLYNOMIAL

The n^{th} differences of a polynomial of the n^{th} degree are constant and all higher order differences are zero.

Let the polynomial of the n^{th} degree be u_x, so

$$f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + kx + l = u_x$$

$\therefore \quad \Delta f(x) = f(x+h) - f(x)$

$= a(x+h)^n - ax^n + b(x+h)^{n-1} - bx^{n-1} + c(x+h)^{n-2} - cx^{n-2} + \ldots + kh$

$= anhx^{n-1} + b'x^{n-2} + c'x^{n-3} + \ldots + k'x + l'$ ... (1)

where $b', c', \ldots, l'$ are new constant coefficients.

Thus the first difference of a polynomial of the n^{th} degree is a polynomial of degree $(n-1)$.

Similarly $\Delta^2 f(x) = \Delta f(x+h) - \Delta f(x) = \Delta f(x+h) - f(x)$

$= anh[(x+h)^{n-1} - x^{n-1}] + b'[(x+h)^{n-2} - x^{n-2}] + \ldots + k'h$

$= an(n-1)hx^{n-2} + b''x^{n-3} + c''x^{n-4} + \ldots + k''$, By (1)

$\therefore$ the second difference represents a polynomial of degree $(n-2)$.

Continuing this process, for the n^{th} difference we get a polynomial of degree zero i.e.

$$\Delta^n f(x) = an(n-1)(n-2)\ldots 3.2.1.h^n = an!h^n \quad \text{... (2)}$$

which is a constant. Hence the $(n+1)^{th}$ and higher differences of a polynomial of n^{th} degree will be zero.

Obs. The converse of this theorem is also true i.e. if the n^{th} differences of a function is a constant or equal to zero, then the function is a polynomial of degree n. This fact is important in numerical analysis as it enables us to approximate a function by a polynomial of n^{th} degree, if the n^{th} order differences become nearly constant.

Example 3.12.1 If $u_x = ax^3 + bx^2 + cx + d$, show that $\Delta^3 u_x = 6ah^3$.

Solution. $\Delta^3 u_x = a\Delta^3 x^3 + b\Delta^3 x^2 + c\Delta^3 x^1 + d\Delta^3 x^0 = \Delta^3[a(x)^3 + b(x)^2 + c(x)^1 + d(x)^0]$

$= a\Delta^3 x^3 + b\Delta^3 x^2 + c\Delta^3 x^1$ $\quad$ [$\because \Delta^n x^n = 0$ for $n < 0$]

$= a(3!)h^3 + b(0) + c(0)$ $\quad$ [by (2) above]

3.13 (1) FACTORIAL NOTATION

A product of the form $x(x-h)(x-2h)\ldots(x-n+1)h$ is denoted by $[x]^n$ and is called a factorial.

In particular $[x]^1 = x$, $[x]^2 = x(x-h)$

$[x]^3 = x(x-h)(x-2h)$, etc.

In general $[x]^n = x(x-h)(x-2h)\ldots(x-n+1)h$

In case, the interval of differencing is 1, then

$$[x]^n = x(x-1)(x-2)\ldots(x-n+1)$$

which is called a *Factorial polynomial* or *function*.

The factorial notation is of special utility in the theory of finite differences. It helps in finding the successive differences of a polynomial whose x is given by a simple rule of successions.

The result of differencing $[x]^n$ is analogous to that of differentiating x^n.

(2) To express a polynomial in the factorial notation

(a) Arrange the coefficients of the powers of x in descending order, supplying zero for powers by which are missing.

(b) Using detached coefficients divide by x, $x-1$, $x-2$, etc. successively.

(3) Every polynomial of degree n can be expressed as a factorial polynomial of the same degree and conversely.

Example 20.4. Express $y = 2x^3 - 3x^2 + 5x - 10$ in a factorial notation and hence show that $\Delta^3 y = 12$. *(Bhopal, 2007; P.T.U., 2006)*

Solution. First method : Let $y = A(x)^3 + B(x)^2 + C(x) + D$.

Then

	x^3	x^2	x	
1	2	-3	5	$-10 = D$
		2	-1	
2	2	1	$2 = C$	
		4		
3	2	$3 = B$		
	2			
	$2 = A$			

Hence $y = 2(x)^3 + 3(x)^2 + 2(x) - 10$

$\therefore \Delta y = 2 \times 3(x)^2 + 3 \times 2(x) + 2$

$\Delta^2 y = 6 \times 2(x) + 6$

$\Delta^3 y = 12$, which shows that the third differences of y are constant, as they should be.

Note. *The coefficient of the highest power of x remains unchanged while converting a polynomial in factorial notation.*

Second method (Direct working) :

Let $y = 2x^3 - 3x^2 + 5x - 10$

$= 2x(x-1)(x-2) + Bx(x-1) + Cx + D$

Putting $x = 0$, $-10 = D$

Putting $x = 1$, $2 - 3 + 5 - 10 = C + D$

$\therefore C = -5 - D = -5 + 10 = 5$

Putting $x = 2$, $16 - 12 + 10 - 10 = 2B + 2C + D$

$\therefore B = \frac{1}{2}(-2C - D) = \frac{1}{2}(-10 + 10) = 0$.

Hence $y = 2x(x-1)(x-2) + 0x(x-1) + 5x - 10 = 2x^3 - 6x^2 + 7x - 10$

$\therefore \Delta y = 2 \times 3(x)^2 + 0 \times 2(x) + 2$, $\Delta^2 y = 6 \times 2(x) + 6$, $\Delta^3 y = 12$.

Example 20.5. *Find the missing values in the following table :*

x	45	50	55	60	65
y	3.0	—	2.0	—	0.4

(Bhopal, 2007; P.T.U., 2003)

Solution. The difference table is as follows :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
45	$y_0 = 3$			
		$y_1 - 3$		
50	y_1		$A - 5y_1$	
		$2 - y_1$		$3y_1 + y_2 = 9$
55	$y_2 = 2$		$y_1 + y_2 = 4$	
		$y_2 - 2$		$3.0 - y_1 = 3y_2$
60	y_3	$2.0 - y_2$	$0.4 - 2y_2$	
65	$y_4 = 0.4$			

Now only the non-zero and the y_{ij} are given. The function u can be represented by a second degree polynomial.

$$u = \alpha_1^{(e)} + \alpha_2^{(e)} x + \alpha_3^{(e)} y \quad \text{and} \quad v = \alpha_4^{(e)} + \alpha_5^{(e)} x + \alpha_6^{(e)} y$$

$$u_1 = \alpha_1 + \alpha_2 x_1 + \alpha_3 y_1 \qquad u_2 = \alpha_1 + \alpha_2 x_2 + \alpha_3 y_2$$

Solving these we get $\alpha_1 = \frac{1}{2A}(x_2 y_3 - x_3 y_2)u_1 + \frac{1}{2A}(\ldots)u_2$

Differentiate the fully three-node triangle $u = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 x^2 + \alpha_5 xy + \alpha_6 y^2$ and given, the function u can be represented by a second degree polynomial

$$u = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 x^2 + \alpha_5 xy + \alpha_6 y^2$$

$$\frac{\partial u}{\partial x} = \alpha_2 + 2\alpha_4 x + \alpha_5 y = 0 \quad \text{and} \quad \frac{\partial u}{\partial y} = \alpha_3 + \alpha_5 x + 2\alpha_6 y = 0$$

$$u_1 = \alpha_1 + \alpha_2 x_1 + \alpha_3 y_1 + \alpha_4 x_1^2 + \alpha_5 x_1 y_1 + \alpha_6 y_1^2 = 0$$

$$u_2 = \alpha_1 + \alpha_2 x_2 + \alpha_3 y_2 + \alpha_4 x_2^2 + \alpha_5 x_2 y_2 + \alpha_6 y_2^2 = 0$$

$$u_3 = \alpha_1 + \alpha_2 x_3 + \alpha_3 y_3 + \alpha_4 x_3^2 + \alpha_5 x_3 y_3 + \alpha_6 y_3^2 = 0$$

Solving these, we get $\alpha_1 = \frac{1}{2A}(\ldots)u_1 + \frac{1}{2A}(\ldots)u_2 + \frac{1}{2A}(\ldots)u_3$

Example: Determine the shape function for the triangular element shown in figure. Also find the displacement at point P.

Node	x	y	u	v
1	0	0	0	0
2	4	0	0.4	0
3	4	3	0.4	0.2

Solution: We construct the following table from the given data:

i	j	m	x_i	y_i	y_{ij}
1	2	3	0	0	$y_{23} = -3$
2	3	1	4	0	$y_{31} = 3$
3	1	2	4	3	$y_{12} = 0$

Now the values of y_{ij} belong to a polynomial of degree n, the double differences are all constants, that $\Delta^2 y_0 = 0$.

The above double array differences can be given by [1] Then

$$\Delta^2 y_0 = \Delta^2 y_1 - \Delta^2 y_0$$

$$\Delta^2 y_0 = \Delta y_1 + \Delta^2 y_0 = 0 + 16 = 16$$

$$\Delta^2 y_1 = \Delta y_2 + \Delta^2 y_1 = 4 + 24 = 28$$

$$\Delta y_1 = \Delta y_1 + \Delta^2 y_0 = 8 + 20 = 28$$

and $$y_0 = y_1 + \Delta y_0 = 1 + 28 = 29$$

Similarly tabulating with $\Delta^2 y_0 = 16$, we get

$$\Delta^2 y_0 = 64, \Delta^2 y_1 = 256, \Delta y_1 = 64, y_1 = 100$$

Tabulating with $\Delta^2 y_0 = 16$, we obtain

$$\Delta^2 y_0 = 256, \Delta^2 y_1 = 1024, \Delta y_1 = 256, y_1 = 400$$

EXERCISE 2.12

1. Construct the table of differences for the data below:

x	0	1	2	3	4
$f(x)$	1.1	1.2	1.3	1.1	1.5

Evaluate $\Delta^3 f(2)$.

2. If $u_0 = 8$, $u_1 = 12$, $u_2 = 18$, $u_3 = 30.0$, $u_4 = 60$, calculate Δu_0.

3. Show that $\Delta^2 y_n \equiv y_{n+2} - 2y_{n+1} + y_n$.

4. Form the table of the second differences of the function
$f(x) = x^4 - 3x^3 - 5x + 7$ for $x = -1, 0, 1, 2, 3, 4, 5$.

5. Form a table of differences for the function
$f(x) = x^3 + 3x - 7$ for $x = -3, -2, -1, 0, 1, 2, 3, 4, 5$.
Check your table by finding $\Delta^3 f(0)$.

6. Extend the following table to two more terms in all directions by constructing the difference table:

x	−2	−0.8	0.2	1.4	2.6	3.8	5.0
$f(x)$	3.6	5.8	5.4	4.19	7.06	13.2	20.9

7. Show that

(i) $\Delta\left[\frac{1}{f(x)}\right] = \frac{-\Delta^2 f(x)}{f(x)f(x+1)}$ (Bangor 2000) (ii) $\Delta \log f(x) = \log\left[1 + \frac{\Delta f(x)}{f(x)}\right]$

8. Evaluate:

(i) $\Delta(\sin x)$ (ii) $\Delta \tan^{-1}\left[\frac{x-2}{x}\right]$ (iii) $\Delta\left[\frac{1}{(x+a)(x+a+1)}\right]$ (Wrexham 2001)

(iv) $\Delta^2\left[\frac{1}{x^2+3x+2}\right]$ (CAT, 2001)

9. Evaluate:

(i) $\Delta(e^x \log x)$ (ii) $\Delta^2(\cos 2x)$ (iii) $\Delta^2(e^x)$ (Aberdeen 2000) (iv) $\Delta^2\left(\frac{1}{x}\right)$

10. If $f(x) = a^{x+1}$, show that the leading differences form a geometric progression. (Aberdeen, 2002)

11. Prove that

(i) $y_2 = y_1 + \Delta y_1 + \Delta^2 y_0 + \Delta^3 y_0$ (ii) $\nabla^3 y_3 = y_3 - 3y_2 + 3y_1 - y_0$

12. Evaluate

(i) $\Delta^n(ab^x)$ (ii) $\Delta^n(1 - x + 3x^2 - 2x^3)$

(iii) $\Delta^n(1 - x + 3x^2 - 2x^3)$, if the interval of differencing is 2.

13. Express $x^3 - 2x^2 + x + 1$ in factorial polynomials. Hence or otherwise find $\Delta^3 f(x)$. (CAT, 2001)

14. If $f(x) = x^4 - 12x^3 + 21x^2 + 18x + 8$ and the successive differences are calculated, prove that $\Delta^4 f(x) = 24$.

15. Find the first and second differences of $x^3 - 3x^2 + 11x^1 - 2x + 8$ with $h = 1$. Show that the fourth difference is zero.

16. Obtain the function whose first difference is $2x^3 + 3x^2 - 3x + 4$.

17. Find the first term of the series whose second and subsequent terms are $0, 2, 10, 28, \ldots$.

18. If $f(x)$ and $g(x)$ are two functions of x, prove that

Evaluate $\Delta^n(ab^x)$ (Aberdeen, 2000) (ii) $\Delta^n\left[\frac{ab^x}{x!}\right] = \frac{a(b-1)^n b^x}{x(x+1)\ldots(x+n)}$

2.13 OTHER DIFFERENCE OPERATORS

We have already introduced the operators Δ, ∇ and δ. Besides these, there are the operators E and μ, which we define below:

(i) **Shift operator** E is the operator *of increasing the argument* x by h, *i.e.* that

$$Ef(x) = f(x + h), E^2 f(x) = f(x + 2h), E^3 f(x) = f(x + 3h), \text{etc.}$$

The reverse operator E^{-1} is defined by $E^{-1} f(x) = f(x - h)$.

If y_x is the function $f(x)$, then $Ey_x = y_{x+1}$, $E^2 y_x = y_{x+2}$, $E^n y_x = y_{x+n}$,
where n may be any real number.

(vii) Averaging operator μ is defined by the equation $\mu y_x = \frac{1}{2}\left(y_{x+\frac{1}{2}} + y_{x-\frac{1}{2}}\right)$

Obs. In the difference calculus, Δ and E are regarded as the fundamental operators and ∇, δ, μ etc. are expressed in terms of these.

(2) Relations between the operators. We shall now establish the following identities :

(i) $\Delta = E - 1 = \delta E^{\frac{1}{2}}$ (ii) $\nabla = 1 - E^{-1}$

(iii) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$ (iv) $\mu = \frac{1}{2}\left(E^{\frac{1}{2}} + E^{-\frac{1}{2}}\right)$

(v) $E = e^{hD}$.

Examples (i) $\Delta y_x = y_{x+1} - y_x = Ey_x - y_x = (E-1)y_x$.

This shows that the operators Δ and E are connected by the symbolic relation

$$\Delta = E - 1 \quad \text{or} \quad E = 1 + \Delta$$

(ii) $\nabla y_x = y_x - y_{x-1} = y_x - E^{-1}y_x = (1 - E^{-1})y_x$

$\therefore \quad \nabla = 1 - E^{-1} \quad \text{or} \quad E = (1 - \nabla)^{-1}$

(iii) $\delta y_x = y_{x+\frac{1}{2}} - y_{x-\frac{1}{2}} = E^{\frac{1}{2}}y_x - E^{-\frac{1}{2}}y_x = (E^{\frac{1}{2}} - E^{-\frac{1}{2}})y_x$

$\therefore \quad \delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$.

(iv) $\mu y_x = \frac{1}{2}\left(y_{x+\frac{1}{2}} + y_{x-\frac{1}{2}}\right) = \frac{1}{2}(E^{\frac{1}{2}}y_x + E^{-\frac{1}{2}}y_x) = \frac{1}{2}(E^{\frac{1}{2}} + E^{-\frac{1}{2}})y_x$

$\therefore \quad \mu = \frac{1}{2}(E^{\frac{1}{2}} + E^{-\frac{1}{2}})$.

(v) $E\nabla y_x = E(y_x - y_{x-1}) = Ey_x - y_x = \Delta y_x \quad \therefore \quad E\nabla = \Delta$

Also $\nabla E y_x = \nabla y_{x+1} = y_{x+1} - y_x = \Delta y_x \quad \therefore \quad \nabla E = \Delta$

$\delta E^{\frac{1}{2}} y_x = \delta y_{x+\frac{1}{2}} = y_{x+1} - y_x = \Delta y_x \quad \text{or} \quad \delta E^{\frac{1}{2}} = \Delta$

$\therefore \quad \delta E^{\frac{1}{2}} = \Delta$

Hence $\quad \Delta = E\nabla = \nabla E = \delta E^{\frac{1}{2}}$.

(vi) $E^x f(a) = f(x + a)$

$= f(a) + xDf(a) + \frac{x^2}{2!}f''(a) + \ldots$ (By Taylor's series)

$= f(a) + x\Delta f(a) + \frac{x^2}{2!}\Delta^2 f(a) + \ldots = \left(1 + x\Delta + \frac{x^2\Delta^2}{2!} + \frac{x^3\Delta^3}{3!} + \ldots\right) f(a) = e^{x\Delta} f(a)$

$\therefore \quad E^x = e^{x\Delta}$

Cor. 1. $\quad E = 1 + \Delta = e^{hD}$.

i.e. $\quad hD = \log(1 + \Delta) = \Delta - \frac{1}{2}\Delta^2 + \frac{1}{3}\Delta^3 - \ldots$ (Newton, 1669)

Note. A table showing the symbolic relations between the various operators is given below for ready reference. To prove such relations between the operators, always express each operator in terms of the fundamental operator E.

(3) Relations between the various operators

	E	Δ	∇	δ	hD
E	E	$\Delta + 1$	$(1 - \nabla)^{-1}$	$1 + \frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$	e^{hD}
Δ	$E - 1$	Δ	$(1 - \nabla)^{-1} - 1$	$\frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$	$e^{hD} - 1$
∇	$1 - E^{-1}$	$1 - (1 + \Delta)^{-1}$	∇	$-\frac{1}{2}\delta^2 + \delta\sqrt{1 + \frac{1}{4}\delta^2}$	$1 - e^{-hD}$
δ	$E^{\frac{1}{2}} - E^{-\frac{1}{2}}$	$\Delta(1 + \Delta)^{-\frac{1}{2}}$	$\nabla(1 - \nabla)^{-\frac{1}{2}}$	δ	$2\sinh\left(\frac{hD}{2}\right)$
μ	$\frac{1}{2}(E^{\frac{1}{2}} + E^{-\frac{1}{2}})$	$(1 + \frac{1}{2}\Delta)(1 + \Delta)^{-\frac{1}{2}}$	$(1 - \frac{1}{2}\nabla)(1 - \nabla)^{-\frac{1}{2}}$	$\sqrt{1 + \frac{1}{4}\delta^2}$	$\cosh\left(\frac{hD}{2}\right)$
hD	$\log E$	$\log(1 + \Delta)$	$\log(1 - \nabla)^{-1}$	$2\sinh^{-1}\left(\frac{\delta}{2}\right)$	

Example 10.2. Prove that

$$e^x = \left(\frac{\Delta^2}{E}\right)e^x + \frac{\Delta e^x}{E-1}.$$ *(Bhopal, 2006)*

Solution. Since $\left(\frac{\Delta^2}{E}\right)e^x = \Delta^2 E^{-1}e^x = \Delta^2 e^{x-h} = \Delta(\Delta e^{x-h}) = \Delta(e^x - e^{x-h})$

$$\therefore \quad \text{R.H.S.} = e^x - e^{x-h} + \frac{\Delta e^x}{E-1} = e^x - e^{x-h} + e^x - e^{x-h} = \text{L.H.S.}$$

Example 10.3. Prove each of the following identities :

(i) $\Delta[\log f(x)] = \log\left[1 + \frac{\Delta f(x)}{f(x)}\right] = \log f(x+h) - \log f(x)$. *(Kerala, 2005)*

(ii) $(E^{1/2} + E^{-1/2})(1 + \Delta)^{1/2} = 2 + \Delta$. *(Bhopal, 2005 ; U.P.T.U., 2005)*

(iii) $\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{(1 + \delta^2/4)}$ (iv) $\Delta^2 y_{n-1} = \nabla^2 y_n$.

Solution. (i) We know that $e^{hD} = E = 1 + \Delta$ $\therefore$ $hD = \log(1 + \Delta)$.

Also $\Delta[\log f(x)] = \log f(x+h) - \log f(x) = \log\left[\frac{f(x+h)}{f(x)}\right] = \log\left[1 + \frac{\Delta f(x)}{f(x)}\right]$ $[\because E^{-1} = 1 - \nabla]$

We have proved that $\mu = \frac{1}{2}(E^{1/2} + E^{-1/2})$ and $\delta = E^{1/2} - E^{-1/2}$

$$\therefore \quad \mu\delta = \frac{1}{2}(E^{1/2} + E^{-1/2})(E^{1/2} - E^{-1/2}) = \frac{1}{2}(E - E^{-1}) = \frac{1}{2}\Delta(E^{-1}) = \frac{1}{2}\Delta(\Delta + 1)^{-1} = \text{right hand}$$

Or, $hD = \sinh^{-1}(\mu\delta)$.

Hence $hD = \log(1 + \Delta) = -\log(1 - \nabla) = \sinh^{-1}(\mu\delta)$

(ii) $(E^{1/2} + E^{-1/2})(1 + \Delta)^{1/2} = (E^{1/2} + E^{-1/2})E^{1/2} = E + 1 = 2 + \Delta = \text{R.H.S.}$

(iii) $\frac{1}{2}\delta^2 + \delta\sqrt{(1 + \delta^2/4)}$

$$= \tfrac{1}{2}(E^{1/2} - E^{-1/2})^2 + (E^{1/2} - E^{-1/2})\sqrt{[1 + \tfrac{1}{4}(E^{1/2} - E^{-1/2})^2]}$$

$$= \tfrac{1}{2}(E - E^{-1} - 2) + (E^{1/2} - E^{-1/2})\sqrt{[\tfrac{1}{4}(E + E^{-1} + 2)]}$$

$$= \tfrac{1}{2}(E + E^{-1} - 2) + \tfrac{1}{2}(E^{1/2} - E^{-1/2})(E^{1/2} + E^{-1/2})$$

$$= \tfrac{1}{2}(E + E^{-1} - 2) + \tfrac{1}{2}(E - E^{-1}) = \tfrac{1}{2}(2E - 2) = E - 1 = \Delta.$$

(iv) $\Delta^2 y_{n-1} = (E - 1)^2 y_{n-1}$ $[\because \Delta = E - 1]$

$$= (E^2 - 2E + 1)y_{n-1} = y_{n+1} - 2y_n + y_{n-1} \quad \ldots(1)$$

$\nabla^2 y_n = (1 - E^{-1})^2 y_n$ $[\because \nabla = 1 - E^{-1}]$

$$= (1 - 2E^{-1} + E^{-2})y_n = y_n - 2y_{n-1} + y_{n-2} \quad \ldots(2)$$

From (1) and (2), $\Delta^2 y_{n-1} = \nabla^2 y_n$.

10.5. TO FIND ONE OR MORE MISSING TERMS

When one or more values of $y = f(x)$ corresponding to the equidistant values of x are missing, we can find the missing y's by the following two methods :

First method : We assume the missing terms as unknowns, a, b, etc. and form the difference table. Assuming the nth differences as zero, we solve these equations for a, b. These give the missing terms.

Second method : If a tabulated value is given, $f(x)$ can be represented by n – 1 degree polynomial i.e., $\Delta^n y = 0$. Given $x = x_0 + h$, the value $(E - 1)^n y = 0$. Now expanding $(E - 1)^n$ and substituting the given values, we obtain the missing terms.

Example 10.5. Find the missing terms in the following table :

x :	45	50	55	60	65
y :	40.2	49.2	54.6	[illegible]	[illegible]

(U.P.T.U., 2006)

Solution. Let the missing term be a. Then the difference table is as follows :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	$12.91 = y_0$				
		4.2			
1	$17.21 = y_1$		0.7		
		4.9		$a - 19.5$	
2	$22.11 = y_2$		$a - 18.8$		$58.5 - 4a$
		$a - 13.9$		$159.5 - 3a$	
3	$a + y_3$		$140.7 - a$		
		$97.6 - a$			
4	$67.4 + y_4$				

We know that $\Delta^4 y_0 = 0$ i.e., $58.5 - 4a = 0$.

Hence $a = 14.63$.

Otherwise: As only four entries y_0, y_1, y_2, y_3 are given, therefore $y = f(x)$ can be represented by a third degree polynomial.

$\therefore$ $\Delta^4 y$ = constant or $\Delta^4 y_0 = 0$ i.e., $(E - 1)^4 = 0$

i.e., $(E^4 - 4E^3 + 6E^2 - 4E + 1)\, y_0 = 0$ or $y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$

Let the missing entry y_3 be a so that

$$67.4 - 4a + 6(22.11) - 4(17.21) + 12.91 = 0 \text{ or } -4a + 58.5 = 0$$

Hence $a = 14.63$.

Example 10.18. Find the missing values in the following data :

x	45	50	55	60	65
y	3.0	—	2.0	—	1.4

(Meerut, 2005)

Solution. Let the missing values be a and b. Then the difference table is as follows :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
45	$3 = y_0$			
		$a - 3$		
50	$a = y_1$		$b - 2a$	
		$2 - a$		$3a + b - 8$
55	$2 = y_2$		$b + a - 4$	
		$b - 2$		$2.8 - a - b$
60	$b = y_3$		$1.4 - 2b$	
		$1.4 - b$		
65	$1.4 = y_4$			

As only three entries y_0, y_1, y_2 are given, y can be expressed by a second degree polynomial having third differences as zero.

$\therefore$ $\Delta^3 y_0 = 0$ and $\Delta^2 y_0 = 0$

i.e., $3a + b - 8 = 0$, $a + b - 4 = 0$

Solving these, we get $a = 2.2$, $b = 1.8$.

Otherwise: As only three entries $y_0 = 3$, $y_2 = 2$, $y_4 = 1.4$ are given, y can be represented by a second degree polynomial having third differences as zero.

$\therefore$ $\Delta^3 y_0 = 0$ and $\Delta^3 y_1 = 0$

i.e., $(E - 1)^3 y_0 = 0$ and $(E - 1)^3 y_1 = 0$

i.e., $(E^3 - 3E^2 + 3E - 1)\, y_0 = 0$; $(E^3 - 3E^2 + 3E - 1)\, y_1 = 0$

or $y_3 - 3y_2 + 3y_1 - y_0 = 0$; $y_4 - 3y_3 + 3y_2 - y_1 = 0$

or $y_3 + 3y_1 = 9$; $3y_3 + y_1 = 6.6$

Solving these, we get $y_1 = 2.2$, $y_3 = 1.8$.

Example 22.11. If $y_{25} = 24, y_{28} = 32, y_{32} = 35, y_{35} = 40$ and $y_{40} = 57$, find y_{30}. (Madras, 2000)

Solution. Taking y_{25} as y_0, we are required to find y_2, i.e., $x = 2$. Then the difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
$x_0 = 0$	$y_{25} = y_0 = 24$				
		4			
$x_1 = 3$	$y_{28} = y_1 = 32$		2		
		3		1	
$x_2 = 7$	$y_{32} = y_2 = 35$		3		0
		5		1	
$x_3 = 10$	$y_{35} = y_3 = 40$		4		
		17			
$x_4 = 15$	$y_{40} = y_4 = 57$				

Thus
$$y_2 = y_{25+2} = (E^2 - 1)^2 y_0 = (1 - 7\Delta^3 + \ldots) y_0$$
$$= \left(1 - 7\Delta^3 + \frac{12.3}{2}\Delta^6 - \frac{12.3.5}{1.2.3}\Delta^9 + \ldots\right) y_0$$
$$= y_0 - 7\Delta^3 y_0 + 42\Delta^6 y_0 - 210\Delta^9 y_0$$
$$= 57 - 17 \times 9 + 42 \times 2 - 210 \times 0 = 21.$$

Example 22.12. If y_x is a polynomial for which $\Delta^4 y_x = 0$. Advance it one step and $y_1 + y_2 = 2(1 + y_0) + y_3$, $y_2 + y_3 = 3(y_0 + y_1)$, $y_3 + y_4 = 4(y_1 + y_2)$, find y_4. (Madras, 2000)

Solution. Starting with y_1 instead of y_0 we note that $\Delta^4 y_1 = 0$ $\quad [\because \Delta^4 y_x$ is constant.

i.e., $\quad (E-1)^4 y_1 = (E^4 - 4E^3 + 6E^2 - 4E + 1) y_1 = 0$

i.e., $\quad y_5 - 4y_4 + 6y_3 - 4y_2 + y_1 = 0$

or $\quad (y_4 + y_2) - 4(y_3 + y_1) + 6y_2 = y_0 + 1 - 4y_4 + y_1 - 4y_3 = 0$

i.e., $\quad y_4 = \frac{1}{20}[y_0 + y_1 - 4(y_2 + y_3) + 16(y_3 + y_4)]$

$$= \frac{1}{20}[-198 - 4(108) + 16(1008)] = 201.$$

Example 22.13. Prove the following identities :

(i) $y_x + \Delta y_x^2 + \Delta^2 y_x^2 + \ldots = \frac{2}{2-\Delta} y_x + \left(\frac{\Delta}{2-\Delta}\right)^2 \Delta y_x + \left(\frac{\Delta}{2-\Delta}\right)^2 \Delta^2 y_x + \ldots$

(ii) $y_x + \frac{\Delta y_x}{1!} + \frac{\Delta^2 y_x}{2!} + \frac{\Delta^3 y_x}{3!} + \ldots = e^x \left[y_x + x\Delta y_x + \frac{x^2}{2!}\Delta^2 y_x + \frac{x^3}{3!}\Delta^3 y_x + \ldots\right]$

Solution. (i) L.H.S. $= y_x + \Delta^2 y_x + \Delta^4 y_x + \ldots = y_x [1 + \Delta^2 + \Delta^4 + \ldots] y_x$ $\quad [\because \Delta^{2n} y_x = E^{2n} y_x]$

$$= y_x \cdot \frac{1}{1-\Delta^2} y_x \text{ taking sum of infinite G.P.}$$

$$= y_x \left[\frac{1}{1-(E+\Delta)}\right] y_x \qquad [\because E = 1 + \Delta]$$

$$= y_x \left[\frac{1}{1-x-\Delta x}\right] y_x = \frac{x}{1-x}\left[1 - \frac{\Delta x}{1-x}\right]^{-1} y_x = \frac{x}{1-x}\left[1 + \frac{x\Delta}{1-x} + \frac{x^2\Delta^2}{(1-x)^2} + \ldots\right] y_x$$

$$= \frac{x}{1-x} y_x + \frac{x^2}{(1-x)^2}\Delta y_x + \frac{x^3}{(1-x)^3}\Delta^2 y_x + \ldots = \text{R.H.S.}$$

(c) $L.H.S. = e^{x}y_0 + \frac{x}{1!}\Delta e^{x}y_0 + \frac{x^2}{2!}\Delta^2 e^{x}y_0 + \frac{x^3}{3!}\Delta^3 e^{x}y_0 + \ldots$

$$= e^{x}\left(1 + \frac{x\Delta}{1!} + \frac{x^2\Delta^2}{2!} + \frac{x^3\Delta^3}{3!} + \ldots\right)y_0 \quad \because e^{x\Delta}y_0 = y_{x\Delta} = y_x$$

$$= e^{x}e^{x\Delta}y_0 = e^{x}\left(1 + \frac{x\Delta}{1!} + \frac{x^2\Delta^2}{2!} + \frac{x^3\Delta^3}{3!} + \ldots\right)y_0$$

$$= e^{x}\left(y_0 + \frac{x}{1!}\Delta y_0 + \frac{x^2}{2!}\Delta^2 y_0 + \frac{x^3}{3!}\Delta^3 y_0 + \ldots\right) = R.H.S.$$

EXERCISE 5.5 (Q.)

1. Explain the difference between $\left[\frac{x^n}{n}\right]$ and $\frac{x^{(n)}}{n!}$. (Mumbai, 2003)

2. Evaluate the following (in the interval of differencing):

(a) $\sum \frac{x^3}{6}\,\delta x$ (b) $\left[\frac{x^4}{4}\right]$ at $h = 1$ (RTU, 2010)

(c) $\left[\frac{x^4}{6}\right]$ at $x = 5, h = 2$ (d) $(x+1)^{(3)} + (x+1)^{(2)}$

3. With the usual notations, show that:

(a) $\nabla = 1 - e^{-hD}$ (b) $D = \frac{1}{h}\sinh^{-1}\left(\frac{\delta}{2}\right)$

(c) $(1+\Delta)(1-\nabla) = 1$ (d) $\Delta = \frac{\delta^2}{2} + \delta\sqrt{1+\frac{\delta^2}{4}}$ (Mumbai, 1999)

4. Prove that

(a) $(1+\Delta)(1-\nabla) = (1-\nabla)^{-1} = (1+\Delta)^{-1}$ (b) $e^{\delta} = 1 + \frac{\mu\delta}{2}$ (UPTU, 2010)

(c) $\Delta^n(x^{(n)}) = \nabla^n(x^{(n)}) = n!h^n$ (d) $\nabla = \Delta(1+\Delta)^{-1} = e^{-hD} = 1 - E^{-1}$

5. Show that $(1+\Delta)(1-\nabla) = 1$ (b) $\mu = \sqrt{1+\frac{\delta^2}{4}}$ (UPTU, MCA, 2009)

(c) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$ (UPTU, 2010) (d) $\nabla^n = E^{-n}\Delta^n = \Delta^n E^{-n} = \frac{1}{E^n}\Delta^n = \ldots$

6. Prove that

(a) $\nabla^n f_n = \Delta^n f_{n-1}$ (b) $\Delta f_n^2 = 2f_n + f_{n+1} + \Delta f_n$ (UPTU, MCA, 2009)

(c) $\mu = \sqrt{1+\frac{\delta^2}{4}}$ (d) $\Delta^n u_{n-1} = u_n - nu_{n-1} + \frac{n(n-1)}{2!}u_{n-2} - \ldots$

7. Prove that $\Delta^n y_{x-n} = y_x\left(1 - \frac{n}{1}\Delta + \frac{n(n+1)}{2!}\Delta^2 - \ldots + (-1)^n\Delta^n\right)$

8. The following table gives the values of y which is a polynomial of degree 5 in x. It is known that $f(3)$ is in error. Correct the error.

x	0	1	2	3	4	5	6
y	1	2	33	244	1025	3126	7777

(Mumbai, 2000)

9. Estimate the missing terms in the following table:

x	0	1	2	3	4
$f(x)$	1	3	9	—	81

(UPTU, 2007)

10. Find the missing terms of the following data:

x	0	1	2	3	4	5	6
$f(x)$	6	10	18	38	—	132	—

(UPTU, 2009)

11. Find the missing values in the following table:

x	0	1	2	3	4	5	6
y	5	11	32	69		118	

(U.P.T.U. 2004)

12. If $\Delta^3 y_0 = 3$, $\Delta^2 y_0 = 9$, $\Delta y_0 = 17$ and $y_0 = 6$, find y_3. (Mumbai, 2004)

13. Evaluate y_{15} from the following data assuming the common difference to be 1:

$y_0 + y_{10} = 1.8508$, $y_1 + y_9 = 1.8069$, $y_2 + y_8 = 1.7658$, $y_3 + y_7 = 1.7295$ (Mumbai, 2004)

14. Define the method of successive approximation and prove that

$$\lim_{n\to\infty} x_n = x_0 + a_0 + a_0 a_1 + \cdots + a_0 a_1 \cdots a_{n-1} + a_0 a_1 \cdots a_n + \cdots$$

15. Using the method of finite differences, sum the following series:

(i) $1.2.3 + 2.3.4 + 3.4.5 + \cdots$ to n terms.

(ii) $1.2.3 + 2.3.4 + 3.4.5 + \cdots$ to n terms.

16. Prove that $y_x = y_0 + x\Delta y_0 + \frac{x(x-1)}{2!}\Delta^2 y_0 + \frac{x(x-1)(x-2)}{3!}\Delta^3 y_0 + \cdots$

Hence sum the series $1.2 + 2.3 + 3.4 + \cdots$.

22.5 NEWTON'S INTERPOLATION FORMULAE*

We now derive two important interpolation formulae by means of the forward and backward differences of a function. These formulae are often employed in engineering and scientific problems.

(1) Newton's forward interpolation formula. Let the function $y = f(x)$ take the values $y_0, y_1, y_2, \ldots$ corresponding to the values $x_0, x_0 + h, x_0 + 2h, \ldots$ of x. Suppose it is required to evaluate $f(x)$ for $x = x_0 + ph$, where p is any real number.

For any real number p, we have defined E such that

$$E^p f(x) = f(x + ph)$$

i.e., $y_p = f(x_0 + ph) = E^p f(x_0) = (1 + \Delta)^p y_0$ [$\because E \equiv 1 + \Delta$]

$= \left[1 + p\Delta + \frac{p(p-1)}{2!}\Delta^2 + \frac{p(p-1)(p-2)}{3!}\Delta^3 + \ldots\right] y_0$ [Using Binomial theorem]

i.e., $$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \ldots \quad \ldots(1)$$

It is called **Newton's forward interpolation formula** as it contains y_0 and the forward differences of y_0.

Obs: This formula is used for interpolating the values of y near the beginning of a set of tabulated values and extrapolating values of y a little backward (to the left of y_0).

(2) Newton's backward interpolation formula. Let the function $y = f(x)$ take the values $y_0, y_1, y_2, \ldots$ corresponding to the values $x_0, x_0 + h, x_0 + 2h, \ldots$ of x. Suppose it is required to evaluate $f(x)$ for $x = x_n + ph$, where p is any real number. Then we have

$$y_p = f(x_n + ph) = E^p f(x_n) = (1 - \nabla)^{-p} y_n \quad [\because E^{-1} \equiv 1 - \nabla]$$

$= \left[1 + p\nabla + \frac{p(p+1)}{2!}\nabla^2 + \frac{p(p+1)(p+2)}{3!}\nabla^3 + \ldots\right] y_n$ [Using Binomial theorem]

i.e., $$y_p = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \ldots \quad \ldots(2)$$

It is called **Newton's backward interpolation formula** as it contains y_n and backward differences of y_n.

Obs: This formula is used for interpolating the values of y near the end of a set of tabulated values and also for extrapolating values of y a little ahead to the right of y_n.

Example 22.5(1). *The table gives the distances in nautical miles of the visible horizon for the given heights in feet above the earth's surface:*

x = height in feet:	100	150	200	250	300	350	400
y = distance:	10.63	12.53	13.04	14.57	15.63	16.80	17.87

Find the value of y when $x = 218$ ft. (Madras, 2003; U.P.T.U., 2003)

*See footnote on page 22-19.

Solution. The difference table is as under :

x	y	Δ	Δ^2	Δ^3	Δ^4
100	10.63				
		2.13			
150	12.63		−0.29		
		1.84		0.36	
200	13.84		−0.08		−0.01
		1.77		0.35	
250	15.61		−0.43		+0.05
		1.42		0.40	
300	17.03		−0.03		−0.01
		1.39		0.39	
350	18.42		+0.36		
		1.75			
400	21.93				

(i) Here $h = 50$. Also $x_0 = 15.04$, $\Delta y_0 = 1.77$, $\Delta^2 y_0 = -0.15$, $\Delta^3 y_0 = 0.05$ etc.

Now $x = 215$ and $h = 50$, $\therefore \quad p = \dfrac{x - x_0}{h} = \dfrac{15}{50} = 0.30$

$\therefore$ Using Newton's forward interpolation formula, we get

$$y_{215} = y_0 + p\Delta y_0 + \frac{p(p-1)}{1.2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{1.2.3}\Delta^3 y_0 + \dots$$

$$f(215) = 15.04 + 0.30(1.77) - \frac{0.30(-0.70)}{2}(-0.15) + \frac{(0.30)(-0.70)(-1.70)}{6}(0.05) + \dots$$

$$= 15.04 + 0.531 + 0.0158 + 0.0074 + \dots = 15.594 \text{ i.e., } 15.59 \text{ thousand sales}$$

(ii) Since $x = 410$ is near the end of the table, we use Newton's backward interpolation formula.

$\therefore$ Taking $x_n = 400$, $p = \dfrac{x - x_n}{h} = \dfrac{10}{50} = 0.2$

Using the line of backward differences

$$y_n = 21.93,\ \nabla y_n = 1.93,\ \nabla^2 y_n = -0.11,\ \nabla^3 y_n = 0.06 \text{ etc.}$$

$\therefore$ newton's backward formula gives

$$y_{410} = y_{400} + p\nabla y_{400} + \frac{p(p+1)}{2}\nabla^2 y_{400} + \frac{p(p+1)(p+2)}{1.2.3}\nabla^3 y_{400} + \dots$$

$$= 21.93 + 0.2(1.93) + \frac{0.2(1.2)}{2}(-0.11) + \dots = 21.93 \text{ thousand sales.}$$

Example 29.15. *From the following data, estimate the number of students who obtained marks between 40 and 45 :*

Marks	30–40	40–50	50–60	60–70	70–80
No. of Students	31	42	51	35	31

(V.T.U., 2001 S ; A.V.I.T.C., 2007 ; Madras, 2008)

Solution. First we prepare the cumulative frequency table, as follows :

Marks less than (x)	40	50	60	70	80
No. of Students (y)	31	73	124	159	190

Now the difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
40	31				
		42			
50	73		9		
		51		−65	
60	124		−56		37
		35		12	
70	159		−4		
		31			
80	190				

We shall find y_{45} i.e. number of students with marks less than 45.

Taking $x_0 = 40$, $x = 45$, we have $p = \frac{x - x_0}{h} = \frac{5}{10} = 0.5$ [$\because h = 10$]

∴ using Newton's forward interpolation formula, we get

$$y_{45} = y_{40} + p\Delta y_{40} + \frac{p(p-1)}{2}\Delta^2 y_{40} + \frac{p(p-1)(p-2)}{1.2.3}\Delta^3 y_{40} + \ldots$$

$$= 31 + 0.5 \times 82 + \frac{0.5(-0.5)}{2} \times 8 + \frac{0.5(0.5)(-1.5)}{6} \times (-20) + \frac{0.5(-0.5)(-1.5)(-2.5)}{24} \times 37$$

$$= 43.87, \text{ on simplification.}$$

∴ the number of students with marks less than 45 is 43.87 i.e., 44.

But the number of students with marks less than 40 is 31.

Hence the number of students getting marks between 40 and 45 = 44 − 31 = 13.

Example 23.16. *Find the cubic polynomial which takes the following values:*

x	0	1	2	3
$f(x)$	1	2	1	10

Hence or otherwise evaluate $f(4)$. (Nagpur, 2005) (Madras, 2003) (U.P.T.U., 2005)

Solution. The difference table is

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	1			
		1		
1	2		−2	
		−1		10
2	1		8	
		9		
3	10			

We take $x_0 = 0$ and $p = \frac{x - 0}{h} = x$ [$\because h = 1$]

∴ using Newton's forward interpolation formula, we get

$$f(x) = f(0) + \frac{x}{1}\Delta f(0) + \frac{x(x-1)}{1.2}\Delta^2 f(0) + \frac{x(x-1)(x-2)}{1.2.3}\Delta^3 f(0)$$

$$= 1 + x(1) + \frac{x^2 - x}{2}(-2) + \frac{x^3 - 3x^2 + 2x}{6}(10)$$

$$= 2x^3 - 7x^2 + 6x + 1, \text{ which is the required polynomial.}$$

To compute $f(4)$, we take $x_0 = 3$, $x = 4$ so that $p = \frac{x - x_0}{h} = 1$ [$\because h = 1$]

Using Newton's backward interpolation formula, we get

$$f(4) = f(3) + p\nabla f(3) + \frac{p(p+1)}{1.2}\nabla^2 f(3) + \frac{p(p+1)(p+2)}{1.2.3}\nabla^3 f(3)$$

$$= 10 + 9 + 10 + 12 = 41,$$

which is the same value as that obtained by substituting $x = 4$ in the cubic polynomial above.

Obs. The above example shows that if a tabulated function is a polynomial, then interpolation and extrapolation give identical results.

Example 23.17. *In the table below, the values of y are consecutive terms of a series of which 23.6 is the 8th term. Find the first and tenth terms of the series:*

x	3	4	5	6	7	8	9
y	4.8	8.4	14.5	22.6	33.2	45.6	73.3

(Guntur, 2003)

Solution. The difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
3	4.8				
		3.6			
4	8.4		2.6		
		6.2		0.5	
5	14.6		3.0		0
		9.2		0.5	
6	23.8		3.5		0
		12.8		0.5	
7	36.6		4.0		0
		16.8		0.5	
8	53.4		4.5		
		21.3			
9	74.7				

To find the first term, use Newton's forward interpolation formula with $x_0 = 3$, $x = 1$, $h = 1$ and $p = -2$. We have

$$y(1) = 4.8 + \frac{(-2)}{1} \times 3.6 + \frac{(-2)(-3)}{1.2} \times 2.6 + \frac{(-2)(-3)(-4)}{1.2.3} \times 0.5 = 2.1$$

To obtain the tenth term, use Newton's backward interpolation formula with $x_n = 9$, $x = 10$, $h = 1$ and $p = 1$. This gives

$$y(10) = 74.7 + \frac{1}{1} \times 21.3 + \frac{2}{1.2} \times 4.5 + \frac{1.2.3}{1.2.3} \times 0.5 = 100$$

EXERCISE 10.3

1. Using Newton's forward formula, find the value of $f(1.1)$ if

x	1	1.4	1.8	3.2
$f(x)$	3.49	4.82	6.05	12.76

(J.N.T.U. 2006)

2. Apply Newton's interpolation formula to find the root of the equation $f(x) = 0$, given that the following table:

x	1.5	2.0	2.5	3.0	3.5	4.0	4.5
$f(x)$	−3.375	−0.625	1.875	6.000	11.125	17.625	26.875

(Kakatiya, 2005)

3. If $f(1.0) = 1.0000$, $f(1.1) = 1.0488$, $f(1.2) = 1.0954$, $f(1.3) = 1.1402$, $f(1.4) = 1.1833$, find $f(1.05)$.

4. Given $\sin 45° = 0.7071$, $\sin 50° = 0.7660$, $\sin 55° = 0.8192$, $\sin 60° = 0.8660$. Find $\sin 52°$, using Newton's forward formula.

5. From the following table of half-yearly premium for policies maturing at different ages, estimate the premium for policies maturing at age of 46:

Age	45	50	55	60	65
Premium (in rupees)	114.84	96.16	83.32	74.48	68.48

(J.N.T.U., 2002)

6. The area A of a circle of diameter d is given for the following values:

d	80	85	90	95	100
A	5026	5674	6362	7088	7854

Calculate the area of a circle of diameter 105.

7. Estimate the value of $f(22)$ and $f(35)$ from the following table of values:

x	20	25	30	35	40	45
$f(x)$	354	332	314	298	285	274

(J.N.T.U., 2005)

8. From the following table:

x°	20	25	30	35	40	45	50	55
$\sin x$	0.3420	0.4226	0.5000	0.5736	0.6428	0.7071	0.7660	0.8192

Calculate the value of $\sin 38°$ and $\sin 58°$ using Gregory-Newton formula.

(J.N.T.U., 2004)

9. Find the number of men getting wages between Rs. 30 and Rs. 40 from the following data:

Wages in Rs.	0 – 10	10 – 20	20 – 30	30 – 40
No. of men	9	30	35	42

(Anna Univ., 2005)

10. Find the polynomial interpolating the data:

x	0	1	2
$f(x)$	1	2	5

(U.P.T.U., 2005)

11. Construct Newton's forward interpolation polynomial for the following data:

x	4	6	8	10
$y = f(x)$	1	3	8	16

(Madras, 1997)

Hence find $f(5)$ and $f(9)$. Ans. $y = x^2 - 3x + 5$

12. Determine the following table for the following data:

x	0.1	0.2	0.3	0.4	0.5	0.6	0.7
$f(x)$	0.003	0.007	0.149	0.246	0.370	0.530	0.686

Find $\Delta^4 f(x)$. (U.P.T.U., 2003)

13. Estimate from the following table (GRE) to how many years Rama Rao's policy (Degree Kerala backward interpolation formula):

x	0	1	2	3	4
$f(x)$	1	1.5	2.2	3.1	4.4

(U.P.T.U., 2002)

14. The following table gives the population of a town during the last six censuses. Estimate the increase in the population during the period from 1911 to 1915:

Year	1911	1921	1931	1941	1951	1961
Population (in thousands)	12	15	20	27	39	52

(U.P.T.U., 2002)

15. In the following table, the values of y are consecutive terms of a series of which 23.6 is the 6th term. Find the 1st and next terms of the series.

x	0	1	2	3	4	5	6
y	4.8	8.4	14.5	23.6	36.8	52.4	71.2

(U.P.T.U., 2003)

(Anna Univ., 2003)

20. Given $x_0 = 10$, $x_1 = 46$, $x_2 = 82$, find x_3 and x_4.

21. If $x_0 = 10$, $x_1 = 15$, $x_2 = 20$, $x_3 = 25$, find x_4 and x_5.

22. Given $x_0 = 5$, $x_1 = 10$, $x_2 = 15$, $x_3 = 20$, $x_4 = 25$, without forming the difference table, find $\Delta^5 x_0$.

4.5.7 CENTRAL DIFFERENCE INTERPOLATION FORMULAE

In the preceding section, we derived Newton's forward and backward interpolation formulae which are applicable for interpolation near the beginning and end of a tabulated value. Now we shall develop central difference formulae which are best suited for interpolation near the middle of the table.

If we take the values $x_0 - 2h$, $x_0 - h$, x_0, $x_0 + h$, $x_0 + 2h$ and the corresponding values of $y = f(x)$ are $y_{-2}, y_{-1}, y_0, y_1, y_2$, then we can write the difference table in the two notations as follows:

x	y	1st Δy	2nd $\Delta^2 y$	3rd $\Delta^3 y$	4th $\Delta^4 y$
$x_0 - 2h$	y_{-2}				
		$\Delta y_{-2} = y_{-1} - y_{-2}$			
$x_0 - h$	y_{-1}		$\Delta^2 y_{-2} = \Delta y_{-1} - \Delta y_{-2}$		
		$\Delta y_{-1} = y_0 - y_{-1}$		$\Delta^3 y_{-2} = \Delta^2 y_{-1} - \Delta^2 y_{-2}$	
x_0	y_0		$\Delta^2 y_{-1} = \Delta y_0 - \Delta y_{-1}$		$\Delta^4 y_{-2} = \Delta^3 y_{-1} - \Delta^3 y_{-2}$
		$\Delta y_0 = y_1 - y_0$		$\Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1}$	
$x_0 + h$	y_1		$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$		
		$\Delta y_1 = y_2 - y_1$			
$x_0 + 2h$	y_2				

(1) Gauss's forward interpolation formula. The Newton's forward interpolation formula is

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{1.2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{1.2.3}\Delta^3 y_0 + \cdots \quad \ldots (1)$$

We have $\Delta^2 y_0 = \Delta^2 y_{-1} + \Delta^3 y_{-1}$

i.e., $\Delta y_1 = \Delta y_0 + \Delta^2 y_0$ ...(2)

Similarly $\Delta^2 y_1 = \Delta^2 y_0 + \Delta^3 y_0$ ...(3)

$\Delta^3 y_2 = \Delta^3 y_1 + \Delta^4 y_1$ etc. ...(4)

Also $\Delta^2 y_{-1} = \Delta^3 y_{-2} + \Delta^4 y_{-2}$

i.e., $\Delta^3 y_{-1} = \Delta^4 y_{-2} + \Delta^5 y_{-2}$ ...(5)

Similarly $\Delta^4 y_{-1} = \Delta^5 y_{-2} + \Delta^6 y_{-2}$ etc. ...(6)

Substituting for $\Delta^3 y_0, \Delta^4 y_0, \Delta^5 y_0, \ldots$ from (2), (3), (4), ... in (1), we get

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{1.2}(\Delta^2 y_{-1} + \Delta^3 y_{-1}) + \frac{p(p-1)(p-2)}{1.2.3}(\Delta^3 y_{-1} + \Delta^4 y_{-1})$$

$$+ \frac{p(p-1)(p-2)(p-3)}{1.2.3.4}(\Delta^4 y_{-2} + \Delta^5 y_{-2}) + \ldots$$

Hence $y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1}$

$$+ \frac{(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2} + \ldots \text{(Newton (7))}$$

which is called Gauss's forward interpolation formula.

Cor. In the second difference notation, this formula is written as

$$y_p = y_0 + p\delta y_{1/2} + \frac{p(p-1)}{2!}\delta^2 y_0 + \frac{(p+1)p(p-1)}{3!}\delta^3 y_{1/2} + \frac{(p+1)p(p-1)(p-2)}{4!}\delta^4 y_0 + \ldots$$

(Note 1. To employ this formula we select the central line nearest to the value of p and the differences on the central line are chosen as shown below)

(Note 2. This formula is used to interpolate the values of y for $p > 0.25$ or 0.75 nearer to the central line as shown in Note 1)

(3) Gauss's backward interpolation formula. The Newton's forward interpolation formula is

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{1.2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{1.2.3}\Delta^3 y_0 + \ldots \quad ...(1)$$

We have $\Delta y_0 = \Delta y_{-1} + \Delta^2 y_{-1}$

i.e., $\Delta y_0 = \Delta y_{-1} + \Delta^2 y_0$ ...(2)

Similarly $\Delta^2 y_0 = \Delta^2 y_{-1} + \Delta^3 y_{-1}$ ...(3)

$\Delta^3 y_0 = \Delta^3 y_{-1} + \Delta^4 y_{-1}$ etc. ...(4)

Also $\Delta^2 y_{-1} = \Delta^3 y_{-2} + \Delta^4 y_{-2}$

i.e., $\Delta^3 y_{-1} = \Delta^4 y_{-2} + \Delta^5 y_{-2}$ ...(5)

Similarly $\Delta^4 y_{-1} = \Delta^5 y_{-2} + \Delta^6 y_{-2}$ etc. ...(6)

Substituting for $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0, \ldots$ from (2), (3), (4) in (1), we get

$$y_p = y_0 + p(\Delta y_{-1} + \Delta^2 y_{-1}) + \frac{p(p-1)}{1.2}(\Delta^2 y_{-1} + \Delta^3 y_{-1}) + \frac{p(p-1)(p-2)}{1.2.3}(\Delta^3 y_{-1} + \Delta^4 y_{-1})$$

$$+ \frac{p(p-1)(p-2)(p-3)}{1.2.3.4}(\Delta^4 y_{-2} + \Delta^5 y_{-2}) + \ldots$$

$$= y_0 + p\Delta y_{-1} + \frac{(p+1)p}{1.2}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{1.2.3}\Delta^3 y_{-1} + \frac{(p+1)p(p-1)(p-2)}{1.2.3.4}\Delta^4 y_{-2}$$

$$+ \frac{p(p-1)(p-2)(p-3)(p+2)}{1.2.3.4}\Delta^5 y_{-2} + \ldots$$

$$= y_0 + p\Delta y_{-1} + \frac{(p+2)p}{1.2}\Delta^2 y_{-1} + \frac{(p+2)p(p-1)}{1.2.3}(\Delta^3 y_{-1} + \Delta^4 y_{-2}) + \frac{(p+3)p(p-1)(p-2)}{1.2.3.4}(\Delta^5 y_{-2} + \Delta^6 y_{-3}) + \ldots \quad \text{[Using (10) and (11)]}$$

Hence $$y_p = y_0 + p\Delta y_{-1} + \frac{(p+2)p}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-2} + \frac{(p+2)(p+1)p(p-1)}{4!}\Delta^4 y_{-2} + \ldots$$

which is called Gauss's backward interpolation formula.

Case (i) Gauss's central difference notation, this formula is of the

$$y_p = y_0 + p\delta y_{1/2} + \frac{(p+1)p}{2!}\delta^2 y_0 + \frac{(p+1)p(p-1)}{3!}\delta^3 y_{1/2} + \frac{(p+2)(p+1)p(p-1)}{4!}\delta^4 y_0 + \ldots$$

Note 1. This formula uses fourth differences above the central line and even differences on the central line as shown in the figure.

Note 2. It is used to compute the value only for a number of a being between 1 and 0.

Note 3. Gauss's forward and backward formulae are not of much practical use. However, these serve as intermediate steps for obtaining the important formulae of the following methods.

(2) Stirling's Formula. Gauss's forward interpolation formula is

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+2)(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2} + \ldots \quad \ldots(1)$$

Gauss's backward interpolation formula is

$$y_p = y_0 + p\Delta y_{-1} + \frac{(p+1)p}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+2)(p+1)p(p-1)}{4!}\Delta^4 y_{-2} + \ldots \quad \ldots(2)$$

Taking the mean of (1) and (2), we obtain

$$y_p = y_0 + p\left(\frac{\Delta y_0 + \Delta y_{-1}}{2}\right) + \frac{p^2}{2!}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{p^2(p^2-1)}{4!}\Delta^4 y_{-2} + \ldots \quad \ldots(3)$$

which is called Stirling's formula.

Case (ii) Gauss's central difference notation, (3) takes the form

$$y_p = y_0 + p\delta y_0 + \frac{p^2}{2!}\delta^2 y_0 + \frac{p(p^2-1^2)}{3!}\delta^3 y_0 + \frac{p^2(p^2-1^2)}{4!}\delta^4 y_0 + \ldots \quad \ldots(4)$$

i.e.
$$\frac{1}{2}[\Delta y_0 + \Delta y_{-1}] = \frac{1}{2}[\delta y_{1/2} + \delta y_{-1/2}] = \mu\delta y_0$$

$$\frac{1}{2}[\Delta^3 y_{-1} + \Delta^3 y_{-2}] = \frac{1}{2}[\delta^3 y_{-1/2} + \delta^3 y_{-3/2}] = \mu\delta^3 y_0 \text{ etc.}$$

Note 1. (4) is formula involves means of the odd differences just above and below the central line and even differences on the central line as shown in the figure below.

(3) Bessel's formula.[**] Gauss's forward interpolation formula is

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+2)(p+1)p(p-1)}{4!}\Delta^4 y_{-2} + \ldots \quad \ldots(1)$$

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† Journal of the Institute of Actuaries, Vol. 5 (1854), p. 77.

††See footnote p. 320.

We have $\Delta y_0 = \Delta y_{-1} + \Delta y_0$

i.e., $\Delta y_{-1} = \Delta y_0 - \Delta y_0$ ...(1)

Similarly $\Delta y_{-2} = \Delta y_{-1} - \Delta y_{-2}$ etc. ...(2)

Now (1) can be written as

$$y_1 = y_0 + E\Delta y_0 + \frac{p(p-1)}{2}\left(\frac{1}{2}\Delta^2 y_{-1} + \frac{1}{2}\Delta^2 y_{-1}\right) + \frac{p(p^2-1)}{3!}\Delta^3 y_{-1}$$

$$+ \frac{p(p^2-1)(p-2)}{4!}\left(\frac{1}{2}\Delta^4 y_{-2} + \frac{1}{2}\Delta^4 y_{-2}\right) + \dots$$

$$= y_0 + E\Delta y_0 + \frac{1}{2}\frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{1}{2}\frac{p(p-1)}{2!}(\Delta^2 y_0 - \Delta^2 y_{-1}) + \frac{p(p^2-1)}{3!}\Delta^3 y_{-1}$$

$$+ \frac{1}{2}\frac{p(p^2-1)(p-2)}{4!}\Delta^4 y_{-2} + \frac{1}{2}\frac{p(p^2-1)(p-2)}{4!}(\Delta^4 y_{-1} - \Delta^4 y_{-2}) + \dots \quad [\because \Delta^4 y_{-1} = \Delta^4 y_{-2}]$$

$$= y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{p(p-1)}{3!}\left(\frac{p+1}{2} - \frac{1}{2}\right)\Delta^3 y_{-1}$$

$$+ \frac{p(p^2-1)(p-2)}{4!}\cdot\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots$$

Hence $y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{(p-1)p(p+1)}{3!}\Delta^3 y_{-1}$

$$+ \frac{(p+1)p(p-1)(p-2)}{4!}\cdot\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \quad \dots(4)$$

which is known as the Bessel's formula.

Cor. Bessel's central difference solution, in terms of

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p-1)p(p+1)}{3!}\Delta^3 y_{-1} + \frac{(p-1)p(p+1)(p+2)}{4!}\Delta^4 y_{-2} + \dots \quad \dots(5)$$

for $\frac{1}{2}(\Delta y_{-1} + \Delta y_0) = \mu\delta y_{-1/2}$, $\frac{1}{2}(\Delta^2 y_{-1} + \Delta^2 y_0) = \mu\delta^2 y_{-1}$ etc.

Obs. This is very useful formula for interpolation when p lies between $-\frac{1}{4}$ and $\frac{1}{4}$. We should retain differences below those in which the [illegible] of even differences of the [illegible] the first in the series below.

$$y_p = \underset{\text{Mean}}{\left[\begin{matrix}\Delta y_0 \\ \Delta y_{-1}\end{matrix}\right]} \quad \underset{\text{Mean}}{\left[\begin{matrix}\Delta^2 y_{-1} \\ \Delta^2 y_{-1}\end{matrix}\right]} \quad \underset{\text{Mean}}{\left[\begin{matrix}\Delta^3 y_{-1} \\ \Delta^3 y_{-2}\end{matrix}\right]} \quad \text{(central line)}$$

(5) **Everett's formula.** Gauss's forward interpolation formula is

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+2)(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2}$$

$$+ \frac{(p+2)(p+1)p(p-1)(p-2)}{5!}\Delta^5 y_{-2} + \dots \quad \dots(1)$$

We eliminate the odd differences in (1) by using the relations

$$\Delta y_0 = y_1 - y_0, \Delta^2 y_{-1} = \Delta y_0 - \Delta y_{-1}, \Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1} \text{ etc.}$$

Then (1) becomes

$$y_p = y_0 + p(y_1 - y_0) + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}(\Delta^2 y_0 - \Delta^2 y_{-1})$$

$$+ \frac{(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2} + \frac{(p+2)(p+1)p(p-1)(p-2)}{5!}(\Delta^4 y_{-1} - \Delta^4 y_{-2}) + \dots$$

$$= (1-p)y_0 + py_1 - \frac{p(p-1)(p-2)}{3!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_0$$

$$-\frac{(p+1)p^2(p-1)(p-2)}{5!}\Delta^5 y_{-3}+\frac{(p+2)(p+1)p(p-1)(p-2)}{6!}\Delta^6 y_{-3}-\cdots$$

To change the terms with negative signs, put $\log p = 1-q$, we obtain

$$y_p = y_0 + \frac{q(q^2-1^2)}{3!}\Delta^3 y_{-1} + \frac{q(q^2-1^2)(q^2-2^2)}{5!}\Delta^5 y_{-2} + \cdots + \Delta y_0 + \frac{p(p^2-1^2)}{3!}\Delta^3 y_{-1} + \frac{p(p^2-1^2)(p^2-2^2)}{5!}\Delta^5 y_{-2} + \cdots$$

This is known as Everett's formula.

(Note. This formula is symmetrical in form and hence it can be used with equal advantage for interpolating a value near the beginning or the end of the table. It is also known as central difference interpolation formula.)

$$y_p = \frac{y_0 + y_1}{2} - \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\cdot\frac{p}{2} + \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\cdot\frac{p(p^2-1)}{3!} - \frac{\Delta^4 y_{-2} + \Delta^4 y_{-3}}{2}\cdot\frac{p(p^2-1)}{4!} + \cdots$$

12.11 CHOICE OF AN INTERPOLATION FORMULA

The coefficients in the central difference formulae are smaller and converge faster than those in Newton's formulae. After a few terms, the coefficients in the Stirling's formula decrease more rapidly than those of the Bessel's formula and the coefficients of Bessel's formula decrease more rapidly than those of Newton's formula. As much, wherever possible, central difference formulae should be used in preference to Newton's formulae.

The right choice of an interpolation formula however, depends on the position of the interpolated value in the given data.

The following rules will be found useful :

1. To find a tabulated value near the beginning of the table, use Newton's forward formula.
2. To find a value near the end of the table, use Newton's backward formula.
3. To find an interpolated value near the centre of the table, use either Stirling's or Bessel's or Everett's formula.

If interpolation is required for p lying between $-1/4$ and $1/4$, prefer Stirling's formula.

If interpolation is desired for p lying between $1/4$ and $3/4$, use Bessel's or Everett's formula.

Example 12.15. Find $f(22)$ from the Gauss forward formula :

x :	20	25	30	35	40	45	
$f(x)$:	354	332	291	260	231	204	*(M.D.U., 2007)*

Solution. Taking $x_0 = 25$, $h = 5$, we have to find the value of $f(x)$ for $x = 22$.

Here $\quad p = \frac{x - x_0}{h} = \frac{22 - 25}{5} = -0.6$

The difference table is as follows :

x	y	y_p	Δy_p	$\Delta^2 y_p$	$\Delta^3 y_p$	$\Delta^4 y_p$	$\Delta^5 y_p$
20	−1	354 ($= y_{-1}$)					
			−22				
25	0	332 ($= y_0$)		−19			
			−41		38		
30	1	291 ($= y_1$)		19		−57	
			−31		−19		95
35	2	260 ($= y_2$)		0		38	
			−29		19		
40	3	231 ($= y_3$)		19			
			−27				
45	4	204 ($= y_4$)					

Gauss forward formula is

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1}$$
$$+ \frac{(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2} + \frac{(p+2)(p+1)p(p-1)(p-2)}{5!}\Delta^5 y_{-2}$$

i.e. $$f(28) = 302 + (0.6)(-41) + \frac{(-0.6)(-1.6)}{2!}(-20) + \frac{(-0.6)(-1.6)(-2.6)}{3!}(-9)$$
$$+ \frac{(-0.6)(-1.6)(-2.6)(-3.6)}{4!}(-27)$$
$$+ \frac{(-0.6)(-1.6)(-2.6)(-3.6)(-4.6)}{5!}(15)$$
$$= 302 + 24.6 - 8.32 + 3.5392 - 0.5241$$

Hence $f(28) = 247.065$

Example 10.20 Interpolate by means of Gauss's forward formula the population of the town for the year 1974, given that:

Year	1939	1949	1959	1969	1979	1989
Population (in thousands)	20	24	29	36	46	66

(Madras Univ., 2003; Madras, 2003)

Solution. Taking $x_0 = 1969.5 = A$, the population of the town is to be found for $p = \frac{1974 - 1969.5}{10} = 0.5$.

The central difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
1939	−3	27				
			5			
1949	−2	32		3		
			8		2	
1959	−1	40		5		3
			13		4		−13
1969	0	53		9		−2	
			22		−6		
1979	1	75		3			
			25				
1989	2	100					

Gauss's backward formula is

$$y_p = y_0 + p\Delta y_{-1} + \frac{(p+1)p}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-2} + \frac{(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2}$$
$$+ \frac{(p+2)(p+1)p(p-1)(p-2)}{5!}\Delta^5 y_{-3} + \ldots$$

i.e., $$y_{0.5} = 53 + (0.5)(27) + \frac{(0.5)(1.5)}{2}(5) + \frac{(1.5)(0.5)(-0.5)}{6}(3) + \frac{(1.5)(0.5)(-0.5)}{24}(-2)$$
$$+ \frac{(2.5)(1.5)(0.5)(-0.5)(-1.5)}{120}(-13)$$
$$= 53 + 3.5 + 1.875 - 0.3125 + 0.2344 - 0.1172 = 58.3637 \text{ thousands approx.}$$

Example [2020] Given

$\theta°$	0	5	10	15	20	25	30
$\sin\theta$	0	0.0872	0.1736	0.2679	0.3420	0.4226	0.5000

Using Stirling's formula, calculate the value of $\sin 16°$. (Anna, 2005)

Solution. Taking the origin at $\theta = 15°$, $h = 5°$ and $p = \frac{\theta - 15}{5}$, we have the following central difference table :

p	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
-3	0.0000					
		0.0872				
-2	0.0872		−0.0013			
		0.0859		0.0004		
-1	0.1731		−0.0009		−0.0018	
		0.0850		−0.0005		0.0036
0	0.2679		−0.0014		0.0018	
		0.0836		−0.0001		−0.0032
1	0.3515		−0.0015		−0.0014	
		0.0821		−0.0015		0.0004
2	0.4336		−0.0030		−0.0010	
		0.0791		−0.0025		
3	0.5127		−0.0055			
		0.0736				
4	0.5863					

At $\theta = 30°$, $\quad p = \frac{30 - 15}{5} = 3$

Stirling's formula is

$$y_p = y_0 + p\left(\frac{\Delta y_{-1} + \Delta y_0}{2}\right) + \frac{p^2}{2!}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\left(\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2}\right) + \frac{p^2(p^2-1)}{4!}\Delta^4 y_{-2} + \ldots$$

$$\therefore \quad y_{16} = 0.2679 + 0.2\left(\frac{0.0850 + 0.0821}{2}\right) + \frac{0.2^2}{2!}(-0.0015) + \ldots$$

$$= 0.2679 + 0.01657 - 0.00030 + \ldots = 0.2692$$

Hence $\quad \sin 16° = 0.2692$

Example [2020] Employ Stirling's formula to compute $y_{12.2}$ from the following table : (Anna, 2004)

x	10	11	12	13	14
$f(x)$	46.063	48.964	51.774	54.664	57.563

Solution. Taking the origin at $x_0 = 12$, $h = 1$ and $p = x - 12$, we have the following central table :

p	y_p	Δy_p	$\Delta^2 y_p$	$\Delta^3 y_p$	$\Delta^4 y_p$
-2	46.063				
		2.901			
-1	48.964		−0.091		
		2.810		0.006	
0	51.774		−0.085		−0.002
		2.725		0.004	
1	54.499		−0.081		
		2.644			
2	57.143				

At $x = 12.2$, $p = 0.2$. (As p is fractional, Stirling's formula will be quite suitable.)

Stirling's formula is

$$y_p = y_0 + p\frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{p^2}{2!}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \frac{p^2(p^2-1)}{4!}\Delta^4 y_{-2} + \dots$$

When $p = 0.3$, we have

$$\therefore \quad y_{0.3} = 0.31749 + 0.3\left\{\frac{0.00728 + 0.00622}{2}\right\} + \frac{0.09}{2}(-0.00097)$$

$$+ \frac{0.3(0.09-1)}{6}\left\{\frac{0.00068 - 0.00045}{2}\right\} + \frac{(0.09)(0.09-1)}{24}(-0.00013)$$

$$= 0.31750 + 0.00718 - 0.00008 - 0.000002 + 0.0000032 = 0.32459.$$

Example 10.20. *Apply Bessel's formula to obtain y_{25}, given $y_{20} = 2854$, $y_{24} = 3162$, $y_{28} = 3544$, $y_{32} = 3992$.*
(U.P.T.U. 2000; U.P.S.C. 2000 B)

Solution. Taking the origin at $x_0 = 24$, $h = 4$, we have $p = \frac{1}{4}(x - 24)$.

The required difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
20	2854			
		308		
24	3162		74	
		382		−8
28	3544		66	
		448		
32	3992			

Also $x = 25$, $p = \frac{1}{4}(25 - 24) = \frac{1}{4}$. (Step lies between 24 and 28, the use of Bessel's formula will yield accurate result.)

Bessel's formula is

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{(p-1/2)\,p(p-1)}{3!}\Delta^3 y_{-1} + \dots \qquad \dots (1)$$

When $p = 0.25$, we have

$$y_{0.25} = 3162 + 0.25 \times 382 + \frac{0.25(0.25-1)}{2}\left\{\frac{74 + 66}{2}\right\} + \frac{(-0.25)(0.25-1)(0.25-0.5)}{6}(-8)$$

$$= 3162 + 95.5 - 5 - 0.938 - 0.0938 = 3250.575 \text{ approx.}$$

Example 10.21. *Apply Bessel's formula to find the value of $f(25.5)$ from the table:*

x	20	24	28	32	36	40
$f(x)$	0.3420	0.4048	0.4694	0.5371	0.6088	0.6833

(U.P.T.U. 2000)

Solution. Taking the origin at $x_0 = 27$, $h = 1$, we have $p = x - 27$.

The central difference table is

x	p	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
25	−2	0.4226				
			−0.0184			
26	−1	0.4408		0.0012		
			−0.0172		−0.0008	
27	0	0.4580		0.0004		0.0001
			−0.0168		−0.0007	
28	1	0.4748		0.0011		−0.0001
			−0.0157		−0.0002	
29	2	0.4905		0.0009		
			−0.0148			
30	3	0.5053				

At $x = 27.5$, $p = 0.5$ (as p lies between 0.4 and 0.6, the use of Stirling's formula will yield accurate result)
Stirling's formula is

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\right) + \frac{\left(p-\frac{1}{2}\right)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+1)p(p-1)(p-2)}{4!}\left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2}\right) + \dots$$

When $p = 0.5$, we have

$$y_{0.5} = 3.704 - \frac{0.5(0.5-1)}{2}\left(\frac{0.003 + 0.011}{2}\right) + 0 + \frac{(0.5+1)0.5(0.5-1)(0.5-2)}{24}\left(\frac{-0.001 - 0.005}{2}\right)$$

$$= 3.704 - 0.00175 - 0.000006 = 3.703$$

Hence $\sqrt{27.5} = 3.703$.

Example 29.21. Given the table

x	300	310	320	330	340	350
$\log x$	2.47712	2.49136	2.50515	2.51851	2.53148	2.54407

find the value of $\log 337.5$ by Bessel's formula.

Solution. Taking the origin at $y_0 = 320$ and $h = 10$, we have $p = \frac{x - 325}{10}$

∴ The central difference table is

p	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
	2.50515					
		0.01421				
−1	2.51936		−0.00043			
		0.01378		0.00004		
0	2.53314		−0.00039		−0.00005	
		0.01339		0.00001		+0.00004
1	2.54653		−0.00038		0.00002	
		0.01301		0.00003		
2	2.55954		−0.00035			
		0.01266				
3	2.57220					

The required value $\log 337.5$ i.e. for $x = 337.5$, $p = \frac{337.5 - 325}{10} = 0.75$

Here $p > 0.5$ and < 0.75. Bessel's formula will be quite suitable.
Bessel's formula is

$$y_p = \frac{y_0 + y_1}{2} + \left(p - \frac{1}{2}\right)\Delta y_0 + \frac{p(p-1)}{2!}\left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\right) + \frac{p\left(p-\frac{1}{2}\right)(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+1)p(p-1)\left(p-\frac{2}{3}\right)}{4!}\left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2}\right) + \dots$$

$$= 2.53 + 0.01631 + \frac{0.25(0.0625-1)}{2}\times(-0.000035) + \frac{0.25(0.0625-1)(0.0625-4)}{12}\times(-0.00003) + 0.75\times 2.53148 + \frac{0.75(0.5625-1)}{2}\times(-0.00038) + \frac{0.75(0.5625-1)(0.0625-4)}{12}\times(-0.00011)$$

$$= 3.53281 + 0.00002 - 0.00000 + 1.99631 + 0.00002 + 0.00000 = 2.53265 \text{ nearly}$$

EXERCISE 4.3

1. Using Gauss's forward formula, evaluate $f(17.5)$ from the table

x	15	16	17	18	19	20
y	24.143	25.943	27.828	29.803	31.875	34.047

(Kottayam 2002; Madras 2000)

2. Using Gauss's backward difference formula, find $y(32)$ from the following table :

x	25	30	35	40	45
y	7	11	14	18	22

(Kerala 2007)

3. Using Gauss's backward formula, estimate the number of persons earning wages between Rs. 60 and Rs. 70 from the following data :

Wages (in Rs.)	Below 40	40–50	50–60	60–70	70–80
No. of persons – in thousands	250	120	100	70	50

(Madras 2000)

4. From the following table :

x	1.01	1.02	1.03	1.04	1.05	1.06	1.07
$y = e^x$	2.7456	2.7731	2.8011	2.8292	2.8577	2.8864	2.9154

Find $e^{1.03}$, using Gauss forward formula.

(U.P.T.U. 2005)

5. The population of a town corresponding to various years is given in the following data. Find the value of y when $x = 35$.

Year x	10	20	30	40
Population y	15	8	44	78

6. Using Stirling's formula find y_{25} given $y_{20} = 512$, $y_{24} = 608$, $y_{28} = 746$, $y_{32} = 955$.

(Nagarjuna 2003)

where y_x represents the number of persons at age x years in a life table.

7. Employ Bessel's formula to find the value of $f(x)$, $f(25)$, given that

x	17	18	19	20	21	22	23
$f(x)$	0.3376	0.3411	0.3445	0.3479	0.3502	0.3521	0.3531

Which other interpolation formula can be used here ? Which is more appropriate ? Give reasons.

8. Calculate the value of $f(12.5)$ using Bessel's interpolation formula, from the following table :

x	0	5	10	15
$f(x)$	7	11	13	12

(U.P.T.U. 2004)

9. Apply Bessel's formula to calculate y_{25} given $y_{20} = 354$, $y_{24} = 332$, $y_{28} = 291$, $y_{32} = 260$. (U.P.T.U. 2003)

10. Using Bessel's formula, evaluate $f(7.5)$, if $f(7.0) = 3.031$, $f(7.5) = 3.103$, $f(8.0) = 3.186$, $f(8.5) = 3.241$.

(U.P.T.U. 2005)

11. Given the table :

x	310	300	290	280	270	260
$\log x$	2.4914	2.4771	2.4624	2.4472	2.4314	2.4150

Find the value of $\log 285.5$ by Gauss's, Stirling's and Bessel's formula.

4.20 INTERPOLATION WITH UNEQUAL INTERVALS

The various interpolation formulae derived so far possess the disadvantages of being applicable only to equally spaced values of the argument. It is, therefore, desirable to develop interpolation formulae for unequally spaced values of x. Now we shall study two such formulae :

(i) Lagrange's interpolation formula

(ii) Newton's general interpolation formula with divided differences.

4.21 LAGRANGE'S INTERPOLATION FORMULA

If $y = f(x)$ takes the values $y_0, y_1, \ldots, y_n$ corresponding to $x = x_0, x_1, \ldots, x_n$ then

$$f(x) = \frac{(x - x_1)(x - x_2)\ldots(x - x_n)}{(x_0 - x_1)(x_0 - x_2)\ldots(x_0 - x_n)} y_0 + \frac{(x - x_0)(x - x_2)\ldots(x - x_n)}{(x_1 - x_0)(x_1 - x_2)\ldots(x_1 - x_n)} y_1$$

$$+ \ldots + \frac{(x - x_0)(x - x_1)\ldots(x - x_{n-1})}{(x_n - x_0)(x_n - x_1)\ldots(x_n - x_{n-1})} y_n \quad \ldots (1)$$

This is known as *Lagrange's interpolation formula for unequal intervals*.

Proof. Let $y = f(x)$ be a function which takes the values $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$. Since there are $n + 1$ pairs of values of x and y, we can represent $f(x)$ by a polynomial in x of degree n. Let this polynomial be of the form

$$y = f(x) = a_0(x - x_1)(x - x_2)\ldots(x - x_n) + a_1(x - x_0)(x - x_2)\ldots(x - x_n) + a_2(x - x_0)(x - x_1)(x - x_3)\ldots(x - x_n) + \ldots + a_n(x - x_0)(x - x_1)\ldots(x - x_{n-1}) \quad \ldots(1)$$

Putting $x = x_0$, $y = y_0$ in (1), we get

$$y_0 = a_0(x_0 - x_1)(x_0 - x_2)\ldots(x_0 - x_n)$$

$$a_0 = y_0/[(x_0 - x_1)(x_0 - x_2)\ldots(x_0 - x_n)]$$

Similarly putting $x = x_1, y = y_1$ in (1), we have $a_1 = y_1/[(x_1 - x_0)(x_1 - x_2)\ldots(x_1 - x_n)]$

Proceeding the same way, we find $a_2, a_3, \ldots, a_n$.

Substituting the values of $a_0, a_1, \ldots, a_n$ in (1), we get (1).

Note. Lagrange's interpolation formula (1) is a polynomial of degree $(n - 1)$ which is known as Lagrange's polynomial and is very simple to use in numerical computation.

The formula can also be used to express the given function in terms of partial fractions.

For this, dividing both sides of (1) by $y = (x - x_0)(x - x_1)\ldots(x - x_n)$, we get

$$\frac{f(x)}{(x - x_0)(x - x_1)\ldots(x - x_n)} = \frac{y_0}{(x_0 - x_1)(x_0 - x_2)\ldots(x_0 - x_n)}\frac{1}{x - x_0} + \frac{y_1}{(x_1 - x_0)(x_1 - x_2)\ldots(x_1 - x_n)}\frac{1}{x - x_1} + \ldots + \frac{y_n}{(x_n - x_0)(x_n - x_1)\ldots(x_n - x_{n-1})}\frac{1}{x - x_n}$$

Example 10.20. Given the values

x	5	7	11	13	17
$f(x)$	150	392	1452	2366	5202

evaluate $f(9)$ using (i) Lagrange's formula [Anna, 2008]

Solution. (i) Here $x_0 = 5, x_1 = 7, x_2 = 11, x_3 = 13, x_4 = 17$

and $y_0 = 150, y_1 = 392, y_2 = 1452, y_3 = 2366, y_4 = 5202$.

Putting $x = 9$ and substituting the above values in Lagrange's formula, we get

$$f(9) = \frac{(9-7)(9-11)(9-13)(9-17)}{(5-7)(5-11)(5-13)(5-17)} \times 150 + \frac{(9-5)(9-11)(9-13)(9-17)}{(7-5)(7-11)(7-13)(7-17)} \times 392$$

$$+ \frac{(9-5)(9-7)(9-13)(9-17)}{(11-5)(11-7)(11-13)(11-17)} \times 1452 + \frac{(9-5)(9-7)(9-11)(9-17)}{(13-5)(13-7)(13-11)(13-17)} \times 2366$$

$$+ \frac{(9-5)(9-7)(9-11)(9-13)}{(17-5)(17-7)(17-11)(17-13)} \times 5202 = -\frac{50}{3} + \frac{2240}{15} + \frac{3872}{3} - \frac{2366}{3} + \frac{434}{3} = 810.$$

Example 10.21. Fit a polynomial to the following data using Lagrange's formula and hence find $y(3)$. [Anna, 2005]

x	0	1	2	4
$y(x)$	1	3	15	147

Solution. Here $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 4$

and $y_0 = 1, y_1 = 3, y_2 = 15, y_3 = 147$

Lagrange's formula is

$$y = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)}y_0 + \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)}y_1$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)}y_2 + \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)}y_3$$

$$= \frac{(x-1)(x-2)(x-4)}{(0-1)(0-2)(0-4)}(1) + \frac{(x-0)(x-2)(x-4)}{(1-0)(1-2)(1-4)}(3)$$

$$+ \frac{(x-0)(x-1)(x-4)}{(2-0)(2-1)(2-4)}(15) + \frac{(x-0)(x-1)(x-2)}{(4-0)(4-1)(4-2)}(147)$$

Hence $f(x) = x^3 + x^2 - x + 2$

$\therefore f(3) = 27 + 9 - 3 + 2 = 35$

Example 29.27. A curve passes through the points $(0, 18), (1, 10), (3, -18)$ and $(4, 6)$. Find the slope of the curve at $x = 2$. (U.P.T.U. 2006)

Solution. Here $x_0 = 0, x_1 = 1, x_2 = 3, x_3 = 4$ and $y_0 = 18, y_1 = 10, y_2 = -18, y_3 = 30$

Since the values of x are unequally spaced, we use the Lagrange's formula :

$$y = \frac{(x - x_1)(x - x_2)(x - x_3)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} y_0 + \frac{(x - x_0)(x - x_2)(x - x_3)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)} y_1$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_3)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)} y_2 + \frac{(x - x_0)(x - x_1)(x - x_2)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)} y_3$$

$$= \frac{(x-1)(x-3)(x-4)}{(0-1)(0-3)(0-4)}(18) + \frac{(x-0)(x-3)(x-4)}{(1-0)(1-3)(1-4)}(10)$$

$$+ \frac{(x-0)(x-1)(x-4)}{(3-0)(3-1)(3-4)}(-18) + \frac{(x-0)(x-1)(x-3)}{(4-0)(4-1)(4-3)}(30)$$

$$= (-x^3 + 8x^2 - 19x + 12)(18) + (x^3 - 7x^2 + 8x)(10) + (x^3 - 5x^2 + 4x)(-18) + (x^3 - 4x^2 + 3x)(30)$$

i.e., $y = 2x^3 - 15x^2 + 18$

Thus the slope of the curve at $(x = 2) = \left(\frac{dy}{dx}\right)_{x=2}$

$= (6x^2 - 30x)_{x=2} = -36$.

Example 29.28. Using Lagrange's formula, express the function $\frac{3x^2 + x + 1}{(x-1)(x-2)(x-3)}$ as the sum of partial fractions.

Solution. Let us evaluate $y = 3x^2 + x + 1$ for $x = 1, x = 2$ and $x = 3$.

Thus values are

x	$x_0 = 1$	$x_1 = 2$	$x_2 = 3$
y	$y_0 = 5$	$y_1 = 15$	$y_2 = 31$

Lagrange's formula is

$$y = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} y_0 + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} y_1 + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} y_2$$

$$= \frac{(x-2)(x-3)}{(1-2)(1-3)}(5) + \frac{(x-1)(x-3)}{(2-1)(2-3)}(15) + \frac{(x-1)(x-2)}{(3-1)(3-2)}(31)$$

Substituting the above values, we get

$$= \frac{(x-2)(x-3)}{(-1)(-2)}(5) + \frac{(x-1)(x-3)}{(1)(-1)}(15) + \frac{(x-1)(x-2)}{(2)(1)}(31)$$

$$= 2.5x^2 - 25x + 30 - 15(x^2 - 4x + 3) + 15.5(x^2 - 3x + 2)$$

Thus $\frac{3x^2 + x + 1}{(x-1)(x-2)(x-3)} = \frac{2.5(x^2 - 5x + 6) - 15(x^2 - 4x + 3) + 15.5(x^2 - 3x + 2)}{(x-1)(x-2)(x-3)}$

$$= \frac{2.5}{x-1} - \frac{15}{x-2} + \frac{15.5}{x-3}.$$

Example 29.29. Find the distance moved by a particle in 10 seconds from the following data :

Time (sec)	0	2	5	9
Velocity	10	16	17	20

Solution. Since the values of x are not equispaced, we use Lagrange's formula :

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}y_3$$

$$y = \frac{(x-1)(x-2)(x-4)}{(-1)(-3)(-5)}(2) + \frac{(x+1)(x-2)(x-4)}{(1)(-2)(-4)}(1) + \frac{(x+1)(x-1)(x-4)}{(3)(1)(-1)}(1) + \frac{(x+1)(x-1)(x-2)}{(5)(3)(1)}(10)$$

i.e., $y = \frac{1}{15}(-4x^3 + 28x^2 - 40x + 225)$

3. Distance travelled $= \int_0^4 v\,dt = \frac{1}{15}\int_0^4 (-4t^3 + 28t^2 - 40t + 225)\,dt$ $\left[\because v = \frac{ds}{dt}\right]$

$$= \frac{1}{15}\left[-\frac{4t^4}{4} + \frac{28t^3}{3} - \frac{40t^2}{2} + 225t\right]_0^4$$

$$= \frac{1}{15}\left[-256 + \frac{1792}{3} - 320 + 900\right] = 56.8$$

Also acceleration $= \frac{dv}{dt} = \frac{1}{15}(-12t^2 + 56t - 40) = 0$

Hence acceleration at $t = 4$ is $\frac{1}{15}(-192 + 224 - 40) = -3.6$.

EXERCISE PROBLEMS

1. Use Lagrange's interpolation formula to find the value of y when $x = 10$, given the following values are not necessarily in order:

x	5	6	9	11
y	12	13	14	16

(Ans: 15.083)

2. Given $\log_{10} 654 = 2.8156$, $\log_{10} 658 = 2.8182$, $\log_{10} 659 = 2.8189$, $\log_{10} 661 = 2.8202$. Find by using Lagrange's formula the value of $\log_{10} 656$. (Ans: 2.8169)

3. The following are the numbers of deaths in a certain town for each age group. Find approximately the number of deaths between the ages of 45 and 50 years.

Age	20–35	35–45	45–55	55–60
Number	342	406	549	628

(Ans: 155)

4. Using Lagrange's interpolation, calculate the profit in the year 1935 from the following data:

Year	1897	1903	1913	1923
Profit in Rupees	40	43	49	52

(Ans: 46.455)

5. Use Lagrange's formula to find the form of the function

x	0	1	3	4
$f(x)$	−12	−14	24	72

(Madras, 2002)

6. If $f(1) = -3$, $f(3) = 9$, $f(4) = 30$, $f(6) = 132$, find the Lagrange's interpolating polynomial that takes the same values at the given points. (U.P.T.U., 2004)

7. Given $f(0) = 1$, $f(1) = 0$, $f(3) = 0$, $f(4) = 58$, $f(5) = 376$, find $f(2)$. (Nagarjuna, 2000)

8. Find the missing term in the following table using Lagrange's formula:

x	1	2	4	5	6
y	14	15	5	6	9

9. Using Lagrange's formula, express the function $\frac{x^2 + x - 7}{x^3 - 6x^2 + x + 4}$ as sum of partial fractions.

3.11 DIVIDED DIFFERENCES

The Lagrange's formula has the drawback that if another interpolation value were introduced, then the interpolation coefficients are required to be recalculated. The labour of recomputing the interpolation

conflicts is to avoid by using Newton's general interpolation formula which employs what are called 'divided differences'. Before deriving this formula, we shall first define these differences.

If $(x_0, y_0), (x_1, y_1), (x_2, y_2), \ldots$ be given points, then the first divided difference for the arguments x_0, x_1 is defined by the relation $[x_0, x_1] = \frac{y_0 - y_1}{x_0 - x_1}$.

Similarly $[x_1, x_2] = \frac{y_1 - y_2}{x_1 - x_2}$ and $[x_2, x_3] = \frac{y_2 - y_3}{x_2 - x_3}$ etc.

The second divided difference for x_0, x_1, x_2 is defined as $[x_0, x_1, x_2] = \frac{[x_0, x_1] - [x_1, x_2]}{x_0 - x_2}$

The third divided difference for x_0, x_1, x_2, x_3 is defined as

$[x_0, x_1, x_2, x_3] = \frac{[x_0, x_1, x_2] - [x_1, x_2, x_3]}{x_0 - x_3}$ and so on.

Obs. 1. The divided differences are symmetrical in their arguments i.e. independent of the order of the arguments.

For it is easy to write $[x_0, x_1] = \frac{y_0}{x_0 - x_1} + \frac{y_1}{x_1 - x_0} = [x_1, x_0]$ and $[x_0, x_1, x_2]$

$$= \frac{y_0}{(x_0 - x_1)(x_0 - x_2)} + \frac{y_1}{(x_1 - x_0)(x_1 - x_2)} + \frac{y_2}{(x_2 - x_0)(x_2 - x_1)}$$

$= [x_1, x_0, x_2] = [x_2, x_0, x_1]$ and so on.

Obs. 2. The n^{th} divided difference of a polynomial $f(x)$ of the n^{th} degree are constant.

Let the arguments be equally spaced so that $x_1 - x_0 = x_2 - x_1 = \ldots = x_n - x_{n-1} = h$. Then

$$[x_0, x_1] = \frac{y_0 - y_1}{x_0 - x_1} = \frac{\Delta y_0}{h}$$

$$[x_0, x_1, x_2] = \frac{[x_0, x_1] - [x_1, x_2]}{x_0 - x_2} = \frac{1}{2h}\left[\frac{\Delta y_0}{h} - \frac{\Delta y_1}{h}\right]$$

$$= \frac{1}{2!h^2}\Delta^2 y_0 \text{ and in general } [x_0, x_1, x_2, \ldots, x_n] = \frac{1}{n!h^n}\Delta^n y_0$$

If the tabulated function is a n^{th} degree polynomial, then $\Delta^n y_0$ will be a constant. Hence the n^{th} divided differences will also be constant.

4.13.1 NEWTON'S DIVIDED DIFFERENCE FORMULA

Let $y_0, y_1, y_2, \ldots, y_n$ be the values of $y = f(x)$ corresponding to the arguments $x_0, x_1, \ldots, x_n$. Then from the definition of divided differences, we have

$$[x, x_0] = \frac{y - y_0}{x - x_0}$$

or that $y = y_0 + (x - x_0)[x, x_0]$ ...(1)

Again $[x, x_0, x_1] = \frac{[x, x_0] - [x_0, x_1]}{x - x_1}$

or, $[x, x_0] = [x_0, x_1] + (x - x_1)[x, x_0, x_1]$

Substituting this value of $[x, x_0]$ in (1), we get

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x, x_0, x_1] \quad ...(2)$$

Also $[x, x_0, x_1, x_2] = \frac{[x, x_0, x_1] - [x_0, x_1, x_2]}{x - x_2}$

which gives $[x, x_0, x_1] = [x_0, x_1, x_2] + (x - x_2)[x, x_0, x_1, x_2]$

Substituting this value of $[x, x_0, x_1]$ in (2), we obtain

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] + (x - x_0)(x - x_1)(x - x_2)[x, x_0, x_1, x_2]$$

Proceeding in this manner, we get

$$y = f(x) = y_0 + (x - x_0)\,[y_0, x_1] + (x - x_0)(x - x_1)\,[y_0, x_1, x_2] + (x - x_0)(x - x_1)(x - x_2)\,[y_0, x_1, x_2, x_3] + \cdots + (x - x_0)(x - x_1)\cdots(x - x_{n-1})\,[y_0, x_1, x_2, \ldots, x_n] \quad \ldots(5)$$

which is called Newton's general interpolation formula with divided differences.

Example 29.28. Given the values

x :	5	7	11	13	17
$f(x)$:	150	392	1452	2366	5202

evaluate $f(9)$, using Newton's divided difference formula. [U.P.T.U., 2003; U.P.T.U., 2004]

Solution. The divided difference table is

x	y	1st divided differences	2nd divided differences	3rd divided differences
5	150	$\frac{392 - 150}{7 - 5} = 121$		
7	392		$\frac{530 - 121}{13 - 5} = 26$	
		$\frac{1452 - 392}{11 - 7} = 265$		$\frac{32 - 26}{13 - 5} = 1$
11	1452		$\frac{457 - 265}{13 - 7} = 32$	
		$\frac{2366 - 1452}{13 - 11} = 457$		$\frac{32 - 32}{17 - 7} = 0$
13	2366		$\frac{638 - 457}{17 - 11} = 32$	
		$\frac{5202 - 2366}{17 - 13} = 709$		
17	5202			

Taking $x = 9$ in the Newton's divided difference formula, we obtain

$$f(9) = 150 + (9 - 5) \times 121 + (9 - 5)(9 - 7) \times 26 + (9 - 5)(9 - 7)(9 - 11) \times 1$$
$$= 150 + 484 + 104 - 16 = 726$$

Example 29.31. Construct $f(x)$ as a polynomial in x for the following data :

x :	-4	-1	0	2	5
$f(x)$:	1245	33	5	9	1335

[U.P.T.U., 2003]

Solution. The divided differences table is

x	$f(x)$	1st divided differences	2nd divided differences	3rd divided differences	4th divided differences
-4	1245				
		-404			
-1	33		84		
		-28		-14	
0	5		80		1
		4		17	
2	9		98		
		442			
5	1335				

Applying Newton's Divided Difference Formula

$$f(x) = f(x_0) + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2] + \ldots$$
$$= 1245 + (x + 8)(-406) + (x + 4)(x + 8)(241)$$
$$+ (x + 4)(x + 8)(x - 6)(-41) + (x + 8)(x + 4)(x - 6)(x - 1)(1)$$
$$= x^5 - 8x^4 + 6x^3 - 16x + 6.$$

PROBLEMS 3.6

1. Find the third divided difference with arguments 2, 4, 9, 10 of the function $f(x) = x^3 - 5x$. (UPTU, 2002)
2. Use Newton's divided difference method to compute $f(8.5)$ from the following data : (UPTU, 2003)

x	8	3	4	9	5
$f(x)$	1	14	33	9	4

3. Using Newton's divided difference formula, evaluate $f(8)$ and $f(10.5)$ from (UPTU, 2002; AKTU, 2000; VTU, 2001)

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

4. Obtain the forward divided difference interpolation polynomial and hence find $f(8)$ (UPTU, 2002)

x	4	7	9	10
$f(x)$	100	392	722	970

5. Using Newton's divided difference interpolation, find the polynomial of the given data : (Anna, 2005)

x	-2	0	1	2
$f(x)$	-12	1	0	-1

6. For the following table, find $f(x)$ as a polynomial in x using Newton's divided difference formula.

x	2	4	9	11
$f(x)$	12	13	14	18

7. Using the following table, find $f(x)$ as a polynomial in x.

x	-1	0	2	3	5
$f(x)$	2	8	20	222	1841

(UPTU, 2005)

8. Find the missing term in the following table using Newton's divided difference formula.

x	0	1	2	3	4
y	1	3	9	—	81

3.8.3 INVERSE INTERPOLATION

So far, given a set of values of x and y, we have been finding the values of y corresponding to a certain value of x. On the other hand, the process of estimating the value of x for a value of y which is not in the table is called the inverse interpolation.

Lagrange's formula is merely a relation between two variables either of which may be taken as the independent variable. Therefore, on interchanging x and y in the Lagrange's formula, we obtain

$$x = \frac{(y - y_1)(y - y_2)\ldots(y - y_n)}{(y_0 - y_1)(y_0 - y_2)\ldots(y_0 - y_n)}x_0 + \frac{(y - y_0)(y - y_2)\ldots(y - y_n)}{(y_1 - y_0)(y_1 - y_2)\ldots(y_1 - y_n)}x_1$$
$$+ \ldots + \frac{(y - y_0)(y - y_1)\ldots(y - y_{n-1})}{(y_n - y_0)(y_n - y_1)\ldots(y_n - y_{n-1})}x_n \quad \ldots(1)$$

which is used for inverse interpolation.

Example 23.22. The following table gives the values of x and y :

x :	2.2	2.3	2.6	4.1	4.9	5.2
y :	4.5	6.2	8.8	12.4	18.8	19.9

Find the value of x corresponding to $y = 10$, using Lagrange's formula. (PTU, 2009)

Solution. Here $x_0 = 2.2, x_1 = 2.3, x_2 = 2.6, x_3 = 4.1, x_4 = 4.9, x_5 = 5.2$

and $y_0 = 4.5, y_1 = 6.2, y_2 = 8.8, y_3 = 12.4, y_4 = 18.8, y_5 = 19.9$

Taking $y = 12$, the above formula (1) gives

$$x = \frac{(12-6.5)(12-9.5)(12-13.5)(12-15.5)}{(8.2-6.5)(8.2-9.5)(8.2-13.5)(8.2-15.5)}(10) + \frac{(12-8.2)(12-9.5)(12-13.5)(12-15.5)}{(6.5-8.2)(6.5-9.5)(6.5-13.5)(6.5-15.5)}(11)$$

$$+ \frac{(12-8.2)(12-6.5)(12-13.5)(12-15.5)}{(9.5-8.2)(9.5-6.5)(9.5-13.5)(9.5-15.5)}(12) + \frac{(12-8.2)(12-6.5)(12-9.5)(12-15.5)}{(13.5-8.2)(13.5-6.5)(13.5-9.5)(13.5-15.5)}(13)$$

$$+ \frac{(12-8.2)(12-6.5)(12-9.5)(12-13.5)}{(15.5-8.2)(15.5-6.5)(15.5-9.5)(15.5-13.5)}(14)$$

$$= 0.852 + 5.333 + 1.200 + 3.111 + 0.963 + 1.615 = 13.24$$

Example 10.23. *Apply Lagrange's formula inversely to obtain a root of the equation $f(x) = 0$, given that $f(26) = -30$, $f(34) = +23$, $f(30) = 7$ and $f(38) = 18$.* (V.T.U., Dec. 2009)

Solution. Here $x_0 = 26$, $x_1 = 34$, $x_2 = 30$, $x_3 = 42$

and $y_0 = -30$, $y_1 = +23$, $y_2 = 7$, $y_3 = 18$

It is required to find x corresponding to $y = f(x) = 0$.

Taking $y = 0$, the Lagrange's formula gives

$$x = \frac{(y-y_1)(y-y_2)(y-y_3)}{(y_0-y_1)(y_0-y_2)(y_0-y_3)}x_0 + \frac{(y-y_0)(y-y_2)(y-y_3)}{(y_1-y_0)(y_1-y_2)(y_1-y_3)}x_1$$

$$+ \frac{(y-y_0)(y-y_1)(y-y_3)}{(y_2-y_0)(y_2-y_1)(y_2-y_3)}x_2 + \frac{(y-y_0)(y-y_1)(y-y_2)}{(y_3-y_0)(y_3-y_1)(y_3-y_2)}x_3$$

$$= \frac{(0+30)(0-7)(0-18)}{(-30-23)(-30-7)(-30-18)}(26) + \frac{(0+30)(0-7)(0-18)}{(23+30)(23-7)(23-18)}(34) + \frac{(0+30)(0-23)(0-18)}{(7+30)(7-23)(7-18)}(30) + \frac{(0+30)(0-23)(0-7)}{(18+30)(18-23)(18-7)}(42)$$

$$= 1.260 + 6.869 + 29.887 + 9.982 = 27.98$$

Hence the desired root of $f(x) = 0$ is 27.98.

EXERCISE 10.5

1. Apply Lagrange's method to find the value of y when $f(x) = 10$ from the given data.

x	5	6	9	11
$f(x)$	12	13	14	16

(Kerala, 2005)

2. Obtain the value of $f(x)$ when $x = 30$ from the following data, using Lagrange's method :

x	5	7	8	14
y	24.9	49.9	81.9	86.9

10.10 OBJECTIVE TYPE OF QUESTIONS

EXAMPLES

Select the correct answer or fill up the blanks in the following problems :

1. Newton's backward interpolation formula is _____.
2. Newton's forward interpolation formula is valid when x lies between
   (a) x_0 and x_n (b) x_0 and x_1 (c) x_{n-1} and x_n

4. Form the divided difference table for the following data:

x	8	15	18
y	1	10	130

5. Interpolation is the technique of estimating the value of a function for any ____.
6. Bessel's formula for interpolation is ____.
7. The fifth divided difference for $x_0, x_1, x_2, x_3, x_4, x_5$ is ____.
8. Gregory-Newton formula is best suited for ____ interpolation.
9. Newton's general divided difference formula is ____.
10. Given $(x_0, y_0), (x_1, y_1), (x_2, y_2)$, Lagrange's interpolation formula is ____.
11. If $f(0) = 1$, $f(1) = 8$, $f(3) = 35$ and $f(6) = 216$, then $c = $ ____.
12. Gauss forward interpolation formula involves
    (a) even differences above the central line and odd differences on the central line
    (b) even differences below the central line and odd differences on the central line
    (c) odd differences below the central line and even differences on the central line
    (d) odd differences above the central line and even differences on the central line
13. If $y(1) = 4$, $y(3) = 12$, $y(4) = 18$ and $y(x) = 7$ find x using Lagrange's formula.
14. Interpolation is defined as ____.
15. The second divided difference of $f(x) = 1/x$ with arguments a, b, c is ____. (Nov./Dec. 2007)
16. Gauss forward interpolation formula is used to interpolate values of y for
    (a) $0 < p < \frac{1}{4}$ (b) $-1 < p < 0$
    (c) $\frac{1}{4} < p < -\frac{1}{4}$ (d) $-\infty < p < \infty$
17. Given

x	0	1	3	4
y	-32	0	0	32

    Using Lagrange's formula, a polynomial that can be fitted to the data is ____.
18. The nth divided difference of a polynomial of degree n is
    (a) zero (b) a constant
    (c) a variable (d) none of these
19. If h is the interval of differencing, $\Delta x^2 = $ ____.

CHAPTER 30

Numerical Differentiation & Integration

1. Numerical differentiation. 2. Formulae for derivatives. 3. Maxima and minima of a tabulated function. 4. Numerical integration. 5. Newton-Cotes quadrature formula. 6. Trapezoidal rule. 7. Simpson's 1/3rd rule. 8. Simpson's 3/8th rule. 9. Boole's rule. 10. Weddle's rule. 11. Objective Type of Questions.

30.1 NUMERICAL DIFFERENTIATION

It is the process of calculating the value of the derivative of a function at some assigned value of x from the given set of values (x, y). To compute a value, we first replace the function by an easy to integrate interpolating formula and then differentiate the formula. The choice of the interpolation formula to be used, will depend on the assigned value of x at which dy/dx is desired.

If the values of x are equispaced and dy/dx is required near the beginning of the table, we employ Newton's forward formula. If it is required near the end of the table, we use Newton's backward formula. For values near the middle of the table, dy/dx is calculated by means of Stirling's or Bessel's formula. If the values of x are not equispaced, we use Newton's divided difference formula to represent the function.

30.2 NUMERICAL FORMULAE FOR DERIVATIVES

Consider the function $y = f(x)$ which is tabulated for the values $x_i = x_0 + ih$, $i = 0, 1, 2, \ldots, n$.

(1) Derivatives using Forward Difference formula. Newton's forward interpolation formula is

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \ldots$$

Differentiating both sides with respect to p, we have

$$\frac{dy}{dp} = \Delta y_0 + \frac{2p-1}{2!}\Delta^2 y_0 + \frac{3p^2-6p+2}{3!}\Delta^3 y_0 + \ldots$$

Since $p = \dfrac{(x - x_0)}{h}$, therefore $\dfrac{dp}{dx} = \dfrac{1}{h}$

Now
$$\frac{dy}{dx} = \frac{dy}{dp}\cdot\frac{dp}{dx} = \frac{1}{h}\left[\Delta y_0 + \frac{2p-1}{2}\Delta^2 y_0 + \frac{3p^2-6p+2}{6}\Delta^3 y_0 + \frac{4p^3-18p^2+22p-6}{24}\Delta^4 y_0 + \ldots\right] \quad \ldots(1)$$

At $x = x_0$, $p = 0$. Hence putting $p = 0$,

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \frac{1}{5}\Delta^5 y_0 - \frac{1}{6}\Delta^6 y_0 + \ldots\right] \quad \ldots(a)$$

[illegible]

$$\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\left[\frac{\partial}{\partial x}\right]\right]$$

$$-\frac{1}{h^3}\left[\frac{1}{2}u_{i-2}-\frac{3}{2}u_{i-1}+\frac{3}{2}u_{i}-\frac{1}{2}u_{i+1}\right]$$

[illegible]

$$\left[\frac{\partial u}{\partial x}\right]_i=\frac{1}{h}\left[u_{i+1}-u_i-\frac{1}{2}\Delta^2 u_i+\frac{1}{3}\Delta^3 u_i-\frac{1}{4}\Delta^4 u_i+\cdots\right] \tag{24}$$

[illegible]

$$\left[\frac{\partial u}{\partial x}\right]_i=\frac{1}{h}\left[u_i-u_{i-1}+\frac{1}{2}\nabla^2 u_i+\frac{1}{3}\nabla^3 u_i+\frac{1}{4}\nabla^4 u_i+\cdots\right] \tag{25}$$

[illegible]

$$\left[\frac{\partial u}{\partial x}\right]_i=\frac{1}{2h}\left[u_{i+1}-u_{i-1}-\frac{1}{6}\delta^3 u_i+\cdots\right] \tag{26}$$

[illegible]

$$\left[\frac{\partial^2 u}{\partial x^2}\right]_i=\frac{1}{h^2}\left[\Delta^2 u_i-\Delta^3 u_i+\frac{11}{12}\Delta^4 u_i-\frac{5}{6}\Delta^5 u_i+\frac{137}{180}\Delta^6 u_i-\cdots\right] \tag{27}$$

[illegible]

$$\left[\frac{\partial^2 u}{\partial x^2}\right]_i=\frac{1}{h^2}\left[\nabla^2 u_i+\nabla^3 u_i+\frac{11}{12}\nabla^4 u_i+\frac{5}{6}\nabla^5 u_i+\frac{137}{180}\nabla^6 u_i+\cdots\right] \tag{28}$$

[illegible]

$$\left[\frac{\partial^2 u}{\partial x^2}\right]_i=\frac{1}{h^2}\left[\delta^2 u_i-\frac{1}{12}\delta^4 u_i+\cdots\right] \tag{29}$$

[illegible]

$$\left[\frac{\partial^3 u}{\partial x^3}\right]_i=\frac{1}{h^3}\left[\Delta^3 u_i-\frac{3}{2}\Delta^4 u_i+\frac{7}{4}\Delta^5 u_i-\frac{15}{8}\Delta^6 u_i+\cdots\right] \tag{30}$$

[illegible]

$$\left[\frac{\partial^3 u}{\partial x^3}\right]_i=\frac{1}{h^3}\left[\nabla^3 u_i+\frac{3}{2}\nabla^4 u_i+\frac{7}{4}\nabla^5 u_i+\frac{15}{8}\nabla^6 u_i+\cdots\right] \tag{31}$$

[illegible]

$$\left[\frac{\partial^3 u}{\partial x^3}\right]_i=\frac{1}{2h^3}\left[\delta^3 u_i-\frac{3}{2}\delta^5 u_i+\cdots\right] \tag{32}$$

[illegible]

$$\left[\frac{\partial^4 u}{\partial x^4}\right]_i=\frac{1}{h^4}\left[\Delta^4 u_i-2\Delta^5 u_i+\frac{17}{6}\Delta^6 u_i-\cdots\right] \tag{33}$$

[illegible]

$$\left[\frac{\partial^4 u}{\partial x^4}\right]_i=\frac{1}{h^4}\left[\nabla^4 u_i+2\nabla^5 u_i+\frac{17}{6}\nabla^6 u_i+\cdots\right] \tag{34}$$

[illegible]

$$\left[\frac{\partial^4 u}{\partial x^4}\right]_i=\frac{1}{h^4}\left[\delta^4 u_i-\frac{1}{6}\delta^6 u_i+\cdots\right] \tag{35}$$

[illegible]

$$\left[\frac{\partial^5 u}{\partial x^5}\right]_i=\frac{1}{h^5}\left[\Delta^5 u_i-\frac{5}{2}\Delta^6 u_i+\cdots\right] \tag{36}$$

[illegible]

$$\left[\frac{\partial^5 u}{\partial x^5}\right]_i=\frac{1}{h^5}\left[\nabla^5 u_i+\frac{5}{2}\nabla^6 u_i+\cdots\right] \tag{37}$$

[illegible]

$$\left[\frac{\partial^5 u}{\partial x^5}\right]_i=\frac{1}{2h^5}\left[\delta^5 u_i+\cdots\right] \tag{38}$$

[illegible]

$$\left[\frac{\partial^6 u}{\partial x^6}\right]_i=\frac{1}{h^6}\left[\Delta^6 u_i-3\Delta^7 u_i+\cdots\right] \tag{39}$$

[illegible]

$$\left[\frac{\partial^6 u}{\partial x^6}\right]_i=\frac{1}{h^6}\left[\nabla^6 u_i+3\nabla^7 u_i+\cdots\right] \tag{40}$$

[illegible]

$$\left[\frac{\partial^6 u}{\partial x^6}\right]_i=\frac{1}{h^6}\left[\delta^6 u_i+\cdots\right] \tag{41}$$

[illegible]

$$\left[\frac{\partial^7 u}{\partial x^7}\right]_i=\frac{1}{h^7}\left[\Delta^7 u_i-4\Delta^8 u_i+\cdots\right] \tag{42}$$

[illegible]

$$\left[\frac{\partial^7 u}{\partial x^7}\right]_i=\frac{1}{h^7}\left[\nabla^7 u_i+4\nabla^8 u_i+\cdots\right] \tag{43}$$

[illegible]

$$\left[\frac{\partial^7 u}{\partial x^7}\right]_i=\frac{1}{2h^7}\left[\delta^7 u_i+\cdots\right] \tag{44}$$

[illegible]

$$\left[\frac{\partial^8 u}{\partial x^8}\right]_i=\frac{1}{h^8}\left[\Delta^8 u_i-5\Delta^9 u_i+\cdots\right] \tag{45}$$

[illegible]

$$\left[\frac{\partial^8 u}{\partial x^8}\right]_i=\frac{1}{h^8}\left[\nabla^8 u_i+5\nabla^9 u_i+\cdots\right] \tag{46}$$

[illegible]

$$\left[\frac{\partial^8 u}{\partial x^8}\right]_i=\frac{1}{h^8}\left[\delta^8 u_i+\cdots\right] \tag{47}$$

[illegible]

Putting $p = 0$, we obtain

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \frac{137}{180}\nabla^6 y_n + \cdots\right] \quad \ldots(7)$$

Similarly, $$\left(\frac{d^3y}{dx^3}\right)_{x_0} = \frac{1}{h^3}\left[\nabla^3 y_n + \frac{3}{2}\nabla^4 y_n + \cdots\right] \quad \ldots(8)$$

Obs. We know that $1 - \nabla = E^{-1} = e^{-hD}$

$\therefore$ $$-hD = \log(1-\nabla) = -\left[\nabla + \frac{1}{2}\nabla^2 + \frac{1}{3}\nabla^3 + \frac{1}{4}\nabla^4 + \cdots\right]$$

or $$D = \frac{1}{h}\left[\nabla + \frac{1}{2}\nabla^2 + \frac{1}{3}\nabla^3 + \frac{1}{4}\nabla^4 + \cdots\right]$$

$\therefore$ $$D^2 = \frac{1}{h^2}\left[\nabla + \frac{1}{2}\nabla^2 + \frac{1}{3}\nabla^3 + \cdots\right]^2 = \frac{1}{h^2}\left[\nabla^2 + \nabla^3 + \frac{11}{12}\nabla^4 + \cdots\right]$$

Similarly, $$D^3 = \frac{1}{h^3}\left[\nabla^3 + \frac{3}{2}\nabla^4 + \cdots\right]$$

Applying these identities to y_n, we get

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \frac{1}{4}\nabla^4 y_n + \frac{1}{5}\nabla^5 y_n + \frac{1}{6}\nabla^6 y_n + \cdots\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_n} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \frac{137}{180}\nabla^6 y_n + \cdots\right]$$

and $$\left(\frac{d^3y}{dx^3}\right)_{x_n} = \frac{1}{h^3}\left[\nabla^3 y_n + \frac{3}{2}\nabla^4 y_n + \cdots\right]$$

which are the same as (6), (7) and (8).

(II) Derivatives using central difference formulae. Stirling's formula is

$$y_p = y_0 + \frac{p}{2}\left(\frac{\Delta y_0 + \Delta y_{-1}}{2}\right) + \frac{p^2}{2!}\Delta^2 y_{-1} + \frac{p(p^2-1^2)}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{p^2(p^2-1^2)}{4!}\Delta^4 y_{-2} + \cdots$$

Differentiating both sides w.r.t. p, we get

$$\frac{dy}{dp} = \left(\frac{\Delta y_0 + \Delta y_{-1}}{2}\right) + \frac{2p}{2!}\Delta^2 y_{-1} + \frac{3p^2-1}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{4p^3 - 2p}{4!}\Delta^4 y_{-2} + \cdots$$

Since $p = \dfrac{x - x_0}{h}$, $\therefore$ $\dfrac{dp}{dx} = \dfrac{1}{h}$

Now $$\frac{dy}{dx} = \frac{dy}{dp}\cdot\frac{dp}{dx} = \frac{1}{h}\left[\left(\frac{\Delta y_0 + \Delta y_{-1}}{2}\right) + p\Delta^2 y_{-1} + \frac{3p^2-1}{6}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{2p^3 - p}{12}\Delta^4 y_{-2} + \cdots\right]$$

At $x = x_0$, $p = 0$. Hence putting $p = 0$, we get

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h}\left[\frac{\Delta y_0 + \Delta y_{-1}}{2} - \frac{1}{6}\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} + \frac{1}{30}\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} + \cdots\right] \quad \ldots(9)$$

Similarly $$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2}\left[\Delta^2 y_{-1} - \frac{1}{12}\Delta^4 y_{-2} + \frac{1}{90}\Delta^6 y_{-3} - \cdots\right] \quad \ldots(10)$$

[illegible]

[illegible]							
[illegible]							
[illegible]							
[illegible]							
[illegible]							
[illegible]							
[illegible]							
[illegible]							
[illegible]							

[illegible]

$$\left[\frac{\partial \mathbf{R}}{\partial \mathbf{u}_{ij}}\right]_{ij} = \frac{1}{\Delta x}\left[\mathbf{u}_{ij} - \frac{1}{2}\mathbf{u}_{i+1,j} + \frac{1}{2}\mathbf{u}_{i-1,j} + \frac{1}{2}\mathbf{u}_{i,j+1} + \frac{1}{2}\mathbf{u}_{i,j-1} - \frac{1}{2}\mathbf{u}_{i,j+1} + \right] \tag{1}$$

$$\left[\frac{\partial \mathbf{R}}{\partial \mathbf{v}_{ij}}\right]_{ij} = \frac{1}{\Delta x}\left[\mathbf{v}_{ij} + \mathbf{v}_{ij} + \frac{1}{2}\mathbf{v}_{i,j} - \frac{1}{2}\mathbf{v}_{i,j}\right] \tag{2}$$

[illegible]

[illegible]

$$\left[\frac{\partial \mathbf{R}}{\partial \mathbf{u}_{ij}}\right]_{ij} = \frac{1}{\Delta x}\left[\mathbf{u}_{ij} - \frac{1}{2}\mathbf{u}_{i+1,j} + \frac{1}{2}\mathbf{u}_{i-1,j} + \frac{1}{2}\mathbf{u}_{i,j+1} + \frac{1}{2}\mathbf{u}_{i,j-1} - \frac{1}{2}\mathbf{u}_{i,j+1} + \right] \tag{3}$$

$$\left[\frac{\partial \mathbf{R}}{\partial \mathbf{v}_{ij}}\right]_{ij} = \frac{1}{\Delta x}\left[\mathbf{v}_{ij} + \mathbf{v}_{ij} + \frac{1}{2}\mathbf{v}_{i,j} - \frac{1}{2}\mathbf{v}_{i,j}\right] \tag{4}$$

[illegible]

[illegible]

Example 20.8. The following data gives the velocity of a particle for 20 seconds at an interval of 5 seconds. Find the initial acceleration using the entire data.

Time t (sec)	0	5	10	15	20
Velocity v (m/sec)	0	3	14	69	228

(Anna, 2004)

Solution. The difference table is

t	v	Δv	$\Delta^2 v$	$\Delta^3 v$	$\Delta^4 v$
0	0				
		3			
5	3		8		
		11		30	
10	14		41		24
		55		80	
15	69		121		
		159			
20	228				

An initial acceleration $\left(i.e.,\ \frac{dv}{dt}\right)$ at $t = 0$ is required, we use Newton's forward formula:

$$\left(\frac{dv}{dt}\right)_{x=0} = \frac{1}{h}\left(\Delta v_0 - \frac{1}{2}\Delta^2 v_0 + \frac{1}{3}\Delta^3 v_0 - \frac{1}{4}\Delta^4 v_0 + \cdots\right)$$

$$\therefore \quad \left(\frac{dv}{dt}\right)_{x=0} = \frac{1}{5}\left[3 - \frac{1}{2}(11) + \frac{1}{3}(30) - \frac{1}{4}(24)\right] = \frac{1}{5}(3 - 5.5 + 10 - 6) = 1$$

Hence the initial acceleration is $1\ m/sec^2$.

Example 20.9. A motor car is moving along a fixed straight road. Its distance x metres along the road is given below for various values of the time t seconds. Find the velocity of the car and its acceleration when $t = 0.3$ seconds.

t	0	0.1	0.2	0.3	0.4	0.5	0.6
x	30.13	31.62	33.07	34.64	36.30	38.01	39.81

(U.P.T.U., 2003)

Solution. The difference table is :

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
0	30.13						
		1.49					
0.1	31.62		−0.04				
		1.45		−0.04			
0.2	33.07		−0.08		0.00		
		1.37		−0.04		−0.02	
0.3	34.44		−0.12		−0.02		0.00
		1.25		−0.06		0.00	
0.4	35.69		−0.18		0.02		
		1.07		−0.04			
0.5	36.76		−0.22				
		0.85					
0.6	37.61						

As the derivatives are required near the middle of the table, we use Stirling's formula:

$$\left(\frac{dy}{dx}\right)_x = \frac{1}{h}\left[\frac{\Delta y_0 + \Delta y_{-1}}{2}\right] - \frac{1}{6}\left[\frac{\Delta^2 y_{-1} + \Delta^2 y_{-2}}{2}\right] + \frac{1}{30}\left[\frac{\Delta^5 y_{-1} + \Delta^5 y_{-3}}{2}\right] + \cdots \qquad \ldots (1)$$

$$\left(\frac{\partial^2 u}{\partial x^2}\right)_{i,j} = \frac{1}{h^2}\left[u_{i+1,j} - 2u_{i,j} + u_{i-1,j}\right] \qquad (3)$$

Here $h = 0.1$, $k_1 = 0.5$, $k_2 = 0.25$, so $\lambda = k_2/h^2 = 0.25$. Using the boundary conditions we can find the values of u on the grid.

$$\left(\frac{\partial u}{\partial x}\right)_{0,j} = \frac{1}{0.2}\left[\frac{0.44 + 0.17}{2} - \frac{1}{2}(0.81 + 0.68) + \frac{1}{2}\left(\frac{0.68 + 0.27}{2}\right)\right] = \left[-0.\right]$$

$$\left(\frac{\partial u}{\partial x}\right)_{1,j} = \frac{1}{0.2}\left[0.44 - \frac{1}{2}(0.81 + 0.68) + \frac{1}{2}(0.68 + 0.27)\right] = -0.0\ldots$$

Hence the required velocity is 0.38 metres each second since $u = 0.1$ cm/sec.

Problem 10.4 Using Bender–Schmidt's formula find u at the grid points for the following heat equation

$x =$	0.25	0.50	0.75	1.00	1.25	1.50	1.75
$f(x) =$	50.00	75.00	100.00	125.00	150.00	175.00	200.00

Solution: Taking $k_2 = 0.005$, $h = 0.25$, we have $\lambda = k_2/h^2 = \frac{1}{50}$.

The following table is

x	t	$u_{i,j}$	u	u^2	u^3	u^4	u^5	u^6
1.00	−0	0.000						
			0.000					
1.00	−0	0.000		0.000				
			0.000		−0.001			
1.00	−1	0.000		0.000		0.000		
			0.000		−0.001		0.000	
1.00	0	0.000		−0.000		0.000		−0.00
			0.000		0.000		−0.000	
1.00	1	0.000		0.000		0.000		
			0.000		−0.000			
1.00	1	0.000		−0.000				
			0.000					
1.00	1	0.000						

Hence the required values are

$$u_{i,j+1} = u_{i,j} + \lambda\delta^2 u_{i,j} = \frac{u_{i+1,j} + u_{i-1,j}}{2} + \left(\lambda - \frac{1}{2}\right)\delta^2 u_{i,j} + \frac{\left(\lambda - \frac{1}{6}\right)}{2}\delta^4 u_{i,j} + \ldots$$

$$+ \frac{\left(\lambda^2 - \frac{1}{2}\lambda + \frac{1}{12}\right)}{2!}\delta^4 u_{i,j} + \frac{\left(\lambda - \frac{1}{2}\right)\left(\lambda^2 - \frac{7}{6}\lambda + \frac{7}{12}\right)}{3!}\delta^6 u_{i,j} + \ldots$$

$$+ \frac{\left(\lambda^3 - \frac{3}{2}\lambda^2 + \frac{11}{12}\lambda - \frac{1}{4}\right)}{4!}\delta^8 u_{i,j} + \frac{\left(\lambda - \frac{1}{2}\right)\left(\lambda^3 - \frac{5}{3}\lambda^2 + \frac{5}{4}\lambda - \frac{1}{3}\right)}{5!}\delta^{10} u_{i,j} + \ldots \qquad (4)$$

Where $\mu = \frac{1}{2}\delta^2 u_{i,j} = \frac{1}{2}\delta u_{i+\frac{1}{2},j} + \frac{1}{2}\delta u_{i-\frac{1}{2},j}$ and $\frac{\mu\delta}{2} = \frac{1}{2}\left(\frac{\delta}{2}\right)\left(\frac{\delta}{2}\right) + \frac{1}{2}\left(\frac{\delta}{2}\right)\left(\frac{\delta}{2}\right)$

Differentiating (4) w.r.t. t and putting $\mu = 0$, we get

$$\left(\frac{\partial u}{\partial t}\right)_{i,j} = \frac{1}{k}\left(\frac{\partial u}{\partial t}\right)_{i,j+\frac{1}{2}} = \frac{1}{k}\left[\delta u_{i,j} - \frac{1}{24}\delta^3 u_{i,j} + \frac{3}{640}\delta^5 u_{i,j} - \frac{5}{896}\delta^7 u_{i,j} + \ldots\right]$$

$$= \frac{1}{k}\left[\delta u_{i,j} - \frac{1}{24}\delta^3 u_{i,j} + \frac{3}{640}\delta^5 u_{i,j} - \frac{5}{896}\delta^7 u_{i,j} + \ldots\right]$$

$$\left(\frac{dy}{dx}\right)_{x=0} = \frac{1}{0.01}\left[0.0002 - \frac{1}{2}(-0.0001) + \frac{1}{12}(0.0001) + \frac{3}{8}(0.0001)^2 + \frac{1}{24}(-0.0001) + \frac{1}{720}(-0.0001) - \frac{95}{288}(0.0001)^2 + \cdots\right]$$

$$= 0.2 + 0.5 + 0.01666 - 0.00833 + 0.00139 - 0.000.$$

Example 10.6. *Find $f'(3)$ from the following data:*

x	3	5	11	27	34
$f(x)$	−12	20	1084	17315	35606

Solution. As the values of x are not equi-spaced, we shall use Newton's divided difference formula. The divided difference table is

x	$f(x)$	1st div. diff.	2nd div. diff.	3rd div. diff.	4th div. diff.
3	−12	16			
5	20		28		
11	1084	144		8.29	
27	17315	1015	65.86	1.001	0.0002
34	35606	2613	65.84		

Fifth differences being zero, Newton's divided difference formula is

$$f(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x_2) + (x - x_0)(x - x_1)(x - x_2)f(x_0, x_1, x_2, x_3) + (x - x_0)(x - x_1)(x - x_2)(x - x_3)f(x_0, x_1, x_2, x_3, x_4)$$

Differentiating it w.r.t. x, we get

$$f'(x) = f(x_0, x_1) + [2x - x_0 - x_1]f(x_0, x_1, x_2) + [3x^2 - 2x(x_0 + x_1 + x_2) + (x_0x_1 + x_1x_2 + x_2x_0)]f(x_0, x_1, x_2, x_3) + [4x^3 - 3x^2(x_0 + x_1 + x_2 + x_3) + 2x(x_0x_1 + x_1x_2 + x_2x_3 + x_3x_0 + x_0x_2 + x_1x_3) - (x_0x_1x_2 + x_1x_2x_3 + x_2x_3x_0 + x_3x_0x_1)]f(x_0, x_1, x_2, x_3, x_4)$$

Putting $x_0 = 3, x_1 = 5, x_2 = 11, x_3 = 27$ and $x = 30$, we obtain

$$f'(30) = 16 + [2 \times 30 - 3 - 5]65.86 + [3 \times 30^2 - 2 \times 30 \times 46 + 71]1.001 = 855.855.$$

10.8 MAXIMA AND MINIMA OF A TABULATED FUNCTION

Newton's forward interpolation formula is

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots$$

Differentiating it w.r.t. x, we get

$$\frac{dy}{dx} = \Delta y_0 + \frac{2u-1}{2}\Delta^2 y_0 + \frac{3u^2 - 6u + 2}{6}\Delta^3 y_0 + \cdots \quad \ldots(1)$$

For maxima or minima, $dy/dx = 0$. Hence equating the right hand side of (1) to zero and retaining only upto third differences, we obtain

$$\Delta y_0 + \frac{2u-1}{2}\Delta^2 y_0 + \frac{3u^2 - 6u + 2}{6}\Delta^3 y_0 = 0$$

i.e., $$\left(\frac{1}{3}\Delta^3 y_0\right)u^2 + (\Delta^2 y_0 - \Delta^3 y_0)u + \left(\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0\right) = 0$$

Substituting the values of Δy_0, $\Delta^2 y_0$, $\Delta^3 y_0$ from the difference table, we solve this quadratic for p. Then the corresponding values of x will give the abscissa of maximum or minimum.

Example 10.11. *Find the maximum and minimum value of y from the following data:*

x:	0	1	2	3	4	5	6
y:	0	−0.25	0	−0.25	0	18.75	96

(Anna, 2000)

Solution. The difference table is:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
−2	0					
		−0.25				
−1	−0.25		0.5			
		0.25		−3		
0	0		−2.5		9	
		−0.25		6		0
1	−0.25		3.5		9	
		0.00		15		0
2	0		11.5		9	
		18.75		24		
3	18.75		35.5			
		40.25				
4	96					

Taking $x_0 = 0$, we have $y_0 = 0$, $\Delta y_0 = -0.25$, $\Delta^2 y_0 = 0.5$, $\Delta^3 y_0 = 9$, $\Delta^4 y_0 = 6$.

Newton's forward difference formula for the first derivative gives

$$\frac{dy}{dx} = \frac{1}{h}\left[\Delta y_0 + \frac{2p-1}{2!}\Delta^2 y_0 + \frac{3p^2-6p+2}{3!}\Delta^3 y_0 + \frac{4p^3-18p^2+22p-6}{4!}\Delta^4 y_0 + \cdots\right]$$

$$= \frac{1}{1}\left[-0.25 + \frac{2p-1}{2}(0.5) + \frac{3}{6}(3p^2-6p+2)(9) + \frac{1}{24}(4p^3-18p^2+22p-6)(6)\right]$$

$$= \frac{1}{4}\left[-0.25 + 0.5p - 1.25 + 4.5p^2 - 9p + 3 + p^3 - 4.5p^2 + 5.5p - 1.5\right] = p^3 - p$$

For y to be maximum or minimum, $\frac{dy}{dx} = 0$ i.e., $p^3 - p = 0$

i.e., $p = 0, 1, -1$

Now $\frac{d^2y}{dx^2} = 3p^2 - 1 = -ve$ for $p = 0$

$= +ve$ for $p = 1$

$= +ve$ for $p = -1$

Hence $y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \cdots, y(0) = 0$

Thus y is maximum for $x = 0$, and maximum value $= y(0) = 0$.

Also y is minimum for $x = 1$ and minimum value $= y(1) = -0.25$.

EXERCISE 10.3

1. Find $y'(0)$ and $y''(0)$ from the following table:

x:	0	1	2	3	4	5
y:	4	8	15	7	6	2

2. Find the first and second derivatives of $f(x)$ at $x = 1.5$ if

x:	1.5	2.0	2.5	3.0	3.5	4.0
f(x):	3.375	7.000	13.625	24.000	38.875	59.000

(U.P.T.U., 2002)

5. Find the first and second derivatives of the function tabulated below, at the point $x = 1.1$.

x	2.0	1.2	1.4	1.6	1.8	2.0
y	0	0.128	0.544	1.296	2.432	4.000

(U.S.C.T.C., 2019, August, 2020)

6. Given the following table of values of x and y:

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
e^x	2.7183	2.8577	3.0042	3.1582	3.3201	3.4903	3.6693

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.00$ by $(a - 1.05)(a - 1.10)(a - 1.15)$. (U.C.E.D., 2018)

7. For the following values of x and y, find the first derivative at $x = 4$.

x	1	2	3	4	12
y	0	7	5	21	97

(U.S.E.E.D., 2018)

8. From the following table, find the values of $f(x)$ and $f'(x)$ at $x = 3.5$.

x	3.00	3.20	3.40	3.60	3.80
y	0.3033	0.2709	0.2404	0.2113	0.1835

(Mains, 2018)

9. Find the values of x for $f'(x) = 0$ from the following table:

x	1.7	1.71	1.72	1.73	1.74
$\sin x$	0.9916	0.9907	0.9899	0.9890	0.9880

(U.S.T.T., 2018)

10. The distance traveled by an athlete for the 30 months a sport is described as follows:

Time (sec)	0	1	2	3	4	5	6
Distance (meters)	0	32	62	88.5	24.5	158	190

Determine the speed of the athlete at $t = 4$ seconds in kilometers per hour. (U.S.T.T., 2018)

11. The following data gives corresponding values of pressure and specific volume of a superheated steam:

v	2	4	6	8	10
p	600	470	380	310	240

Find the rate of change of p with respect to v when $v = 2$.

(a) Constant with respect to volume when $v = 2$.

(b) Constant with respect to pressure when $p = 300$.

12. The table below records the velocity v of a body during the speed in time t. Find the acceleration at $t = 2.5$.

t	1.0	1.1	1.2	1.3	1.4
v	43.1	47.7	52.1	56.4	60.8

(U.S.T.T., 2018)

13. The following table records the velocity of a body during the speed in time t:

t	0	0.02	0.04	0.06	0.08	0.10	0.12
v	150	153	157	160	251	255	259

Find the gradient of the curve at the middle point.

14. A rod is rotating in a plane. The following table gives the angle θ (in radians) through which the rod has turned for various values of the time t (in seconds).

t	0	0.2	0.4	0.6	0.8	1.0	1.2
θ	0	0.12	0.43	0.89	1.49	2.03	2.67

Calculate the angular velocity and the angular acceleration of the rod when $t = 0.8$ second. (U.T.C., 2018)

15. Find the value of $f'(x)$ at $x = 3.4$ from the following table using Newton's formula:

x	3.00	3.05	3.10	3.15	3.20	3.25
$f(x)$	0.3033	0.3047	0.3061	0.3075	0.3089	0.3104

16. If $x = f(u)$ satisfying the relation $f(x_n + nh)$ is given that, if powers of h above 3^{rd} by neglected

$$\frac{1}{2!}\left[\frac{\Delta^2}{\Delta x^2}\right] - \frac{2}{3!}\left[\frac{\Delta^3}{\Delta x^3}\right](x_n - 1) + \frac{1}{4!}\left[\frac{\Delta^4}{\Delta x^4}\right](x_n - 1)(x_n - 2)$$

(U.C.T.C., 2018)

Hence Differentiate Newton's formula with x_n as given $x = 3$.

17. Find the value of $f'(x)$ for $x = 3.5$ from the table given below:

x	3	3.1	3.2	3.3
$f(x)$	4.804	4.986	5.176	5.373

(U.C.E., 2017)

18. Find the 1^{st} derivative from the following data:

x	0	1	2	3	4	5	6
$f(x)$	4	10	16	22	484	868	1252

(U.S.C.E., 2018) (U.S.C.T., 2018)

(ii) Find the polynomial of degree 3 which takes the following values:

x	0	1	2	3
$f(x)$	1	2	1	10

Hence find the value of x for which $f(x)$ is minimum, using the above.

x	5	7	11	13	17	19
$f(x)$	150	392	1452	2366	5202	6858

Also find the minimum value of $f(x)$.

29.8 NUMERICAL INTEGRATION

The process of evaluating a definite integral from a set of tabulated values of the integrand $f(x)$ is called numerical integration. This process when applied to a function of a single variable, is known as quadrature.

The problem of numerical integration, like that of numerical differentiation, is solved by approximating $f(x)$ by an interpolation formula and then integrating it between the given limits. In this way, we can derive quadrature formula for approximate integration of a function defined by a set of numerical values only.

29.9 NEWTON-COTES QUADRATURE FORMULA

Let $I = \int_{x_0}^{x_n} f(x)\,dx$

where $f(x)$ takes the values $y_0, y_1, y_2, \ldots, y_n$ for $x = x_0, x_1, x_2, \ldots, x_n$. (Fig. 29.1)

Let us divide the interval (x_0, x_n) into n sub-intervals of width h so that $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, $x_3 = x_0 + 3h$, ..., $x_n = x_0 + nh = b$. Then

$$I = \int_{x_0}^{x_0+nh} f(x)\,dx = h\int_0^n f(x_0 + rh)\,dr,$$

putting $x = x_0 + rh$, $dx = h\,dr$

$$= h\int_0^n \left[y_0 + r\Delta y_0 + \frac{r(r-1)}{2!}\Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!}\Delta^3 y_0 + \frac{r(r-1)(r-2)(r-3)}{4!}\Delta^4 y_0 + \frac{r(r-1)(r-2)(r-3)(r-4)}{5!}\Delta^5 y_0 + \cdots \right] dr$$

(By Newton's forward interpolation formula)

Integrating term by term, we obtain

$$\int_{x_0}^{x_0+nh} f(x)\,dx = nh\left[y_0 + \frac{n}{2}\Delta y_0 + \frac{n(2n-3)}{12}\Delta^2 y_0 + \frac{n(n-2)^2}{24}\Delta^3 y_0 \right] + \left[\frac{n^4}{5} - \frac{3n^3}{2} + \frac{11n^2}{3} - 3n \right]\frac{\Delta^4 y_0}{4!} + \left(\frac{n^5}{6} - 2n^4 + \frac{35n^3}{4} - \frac{50n^2}{3} + 10n \right)\frac{\Delta^5 y_0}{5!} + \left[\frac{n^6}{7} - \frac{15n^5}{6} + 17n^4 - \frac{203n^3}{4} + \frac{274n^2}{5} - 60n \right]\frac{\Delta^6 y_0}{6!} + \cdots \quad (A)$$

This is known as Newton-Cotes quadrature formula. From this general formula, we deduce the following important quadrature rules by taking $n = 1, 2, 3, \ldots$

29.10 TRAPEZOIDAL RULE

Putting $n = 1$ in (A) of 29.9, and taking the curve through (x_0, y_0) and (x_1, y_1) as a straight line (i.e., a polynomial of first order so that differences of order higher than first become zero), we get

$$\int_{x_0}^{x_0+h} f(x)\,dx = h\left(y_0 + \frac{1}{2}\Delta y_0\right) = \frac{h}{2}(y_0 + y_1)$$

Similarly $$\int_{x_1}^{x_2} f(x)\,dx = h\left(y_1 + \frac{1}{2}\Delta y_1\right) = \frac{h}{2}(y_1 + y_2)$$

$$\int_{x_{n-1}}^{x_{n-1}+h} f(x)\,dx = \frac{h}{2}(y_{n-1} + y_n)$$

Adding these n integrals, we obtain

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + \cdots + y_{n-1})]$$

This is known as the *trapezium rule*.

Obs. The area of each strip is approximated to trapezoid separately. Thus the arc is replaced by chords and the ordinates y_0 and y_n which appear only once are equal to the areas of the two extreme strips.

4.23 SIMPSON'S ONE-THIRD RULE

Putting $n = 2$ in (A) above and taking the curve through (x_0, y_0), (x_1, y_1) and (x_2, y_2) as a parabola $y = ax^2 + bx + c$, a polynomial of second order so that differences of order higher than second vanish, we get

$$\int_{x_0}^{x_0+2h} f(x)\,dx = 2h\left[y_0 + \Delta y_0 + \frac{1}{8}\Delta^2 y_0\right] = \frac{h}{3}(y_0 + 4y_1 + y_2)$$

Similarly, $$\int_{x_2}^{x_2+2h} f(x)\,dx = \frac{h}{3}(y_2 + 4y_3 + y_4)$$ where

$$\int_{x_{n-2}}^{x_{n-2}+2h} f(x)\,dx = \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n)$$ being even.

Adding all these integrals, we have where n is even

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1}) + 2(y_2 + y_4 + \cdots + y_{n-2})]$$

This is known as the *Simpson's one-third rule* or *simple Simpson's rule* and is most commonly used.

Obs. While applying Simpson's (1/3) rule, the given interval must be divided into even number of equal subintervals also we find the area of two strips at a time.

4.24 SIMPSON'S THREE-EIGHTH RULE

Putting $n = 3$ in (A) above and taking the curve through (x_i, y_i), $i = 0, 1, 2, 3$ as a polynomial of third order so that differences above the third order vanish, we get

$$\int_{x_0}^{x_0+3h} f(x)\,dx = 3h\left(y_0 + \frac{3}{2}\Delta y_0 + \frac{3}{4}\Delta^2 y_0 + \frac{1}{8}\Delta^3 y_0\right)$$

$$= \frac{3h}{8}(y_0 + 3y_1 + 3y_2 + y_3)$$

Similarly,

$$\int_{x_3}^{x_3+3h} f(x)\,dx = \frac{3h}{8}(y_3 + 3y_4 + 3y_5 + y_6)$$ and so on.

Adding all such expressions from x_0 to $x_0 + nh$, where n is a multiple of 3, we obtain

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{3h}{8}[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \cdots + y_{n-2} + y_{n-1}) + 2(y_3 + y_6 + \cdots + y_{n-3})]$$

which is known as *Simpson's three-eighth rule*.

Obs. While applying Simpson's (3/8) rule, the number of subintervals should be taken as multiple of 3.

33.9 BOOLE'S RULE

Putting $n = 4$ in (3) above and neglecting all differences above the fourth, we obtain

$$\int_{x_0}^{x_0+4h} f(x)\,dx = kh\left(y_0 + 2\cdot 2y_0 \frac{2}{2}\Delta^2 y_0 + \frac{2}{3}\Delta^3 y_0 + \frac{7}{90}\Delta^4 y_0\right)$$

$$= \frac{2h}{45}(7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4)$$

Similarly

$$\int_{x_0+4h}^{x_0+8h} f(x)\,dx = \frac{2h}{45}(7y_4 + 32y_5 + 12y_6 + 32y_7 + 7y_8) \text{ and so on.}$$

Adding all these integrals from x_0 to $x_0 + nh$, where n is a multiple of 4, we get

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{2h}{45}(7y_0 + 32y_1 + 12y_2 + 32y_3 + 14y_4 + 32y_5 + 12y_6 + 32y_7 + 14y_8 + \dots$$

This is known as Boole's rule.

Obs: While applying Boole's rule the number of strips is always taken for convenience as a multiple of 4.

33.10 WEDDLE'S RULE

Putting $n = 6$ in (3) above and neglecting all differences above the sixth, we obtain

$$\int_{x_0}^{x_0+6h} f(x)\,dx = \left(y_0 + 3\Delta y_0 + \frac{9}{2}\Delta^2 y_0 + 4\Delta^3 y_0 + \frac{123}{80}\Delta^4 y_0 + \frac{11}{20}\Delta^5 y_0 + \frac{1}{6}\cdot\frac{41}{140}\Delta^6 y_0\right)$$

If we replace $\frac{41}{140}\Delta^6 y_0$ by $\frac{3}{10}\Delta^5 y_0$ the error is of the negligible.

$$\therefore \quad \int_{x_0}^{x_0+6h} f(x)\,dx = \frac{3h}{10}(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6)$$

Similarly

$$\int_{x_0+6h}^{x_0+12h} f(x)\,dx = \frac{3h}{10}(y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12}) \text{ and so on.}$$

Adding all these integrals from x_0 to $x_0 + nh$, where n is a multiple of 6, we get

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{3h}{10}(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6 + 5y_7 + y_8 + \dots)$$

This is known as Weddle's rule.

Obs: While applying Weddle's rule the number of strips is always taken for convenience as a multiple of 6. Weddle's rule is generally more convenient than any of the others. Of the two Simpson's rules, the 3/8 rule is better.

Example 33.11. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using (i) Trapezoidal rule,

(ii) Simpson's 1/3 rule, (Mysore, 2006)

(iii) Simpson's 3/8 rule, (UPTU, 2006)

(iv) Weddle's rule and compare the results with the actual value. (UPTU, 2006)

Solution. Divide the interval $[0, 6]$ into six parts each of width $h = 1$. The values of $f(x) = \frac{1}{1+x^2}$ are given below :

x	0	1	2	3	4	5	6
$f(x)$	1	0.5	0.2	0.1	0.05882	0.03846	0.027
y_r	y_0	y_1	y_2	y_3	y_4	y_5	y_6

(i) By Trapezoidal rule,

$$\int_0^1 \frac{1}{1+x^2} = \frac{h}{2}[(y_0 + y_n) + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$

$$= \frac{1}{2}[(1 + 0.5000) + 2(0.8 + 0.64 + 0.5 + 0.3902 + 0.3077)] = 0.7828.$$

(ii) By Simpson's 1/3 rule,

$$\int_0^1 \frac{1}{1+x^2} = \frac{h}{3}[(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{18}[(1 + 0.5000) + 4(0.9615 + 0.8 + 0.6610) + 2(0.9 + 0.7353)] = 0.7854.$$

(iii) By Simpson's 3/8 rule,

$$\int_0^1 \frac{1}{1+x^2} = \frac{3h}{8}[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)]$$

$$= \frac{3}{8 \times 6}[(1 + 0.5000) + 3(0.9615 + 0.9000 + 0.7353 + 0.6610) + 2(0.8)] = 0.7854.$$

(iv) By Weddle's rule,

$$\int_0^1 \frac{1}{1+x^2} = \frac{3h}{10}[(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6)]$$

$$= 0.3[1 + 4.8075 + 0.9 + 4.8462 + 0.7353 + 3.3050 + 0.5000] = 0.7854.$$

Also, $\int_0^1 \frac{dx}{1+x^2} = [\tan^{-1} x]_0^1 = 0.7854$

Obs. This shows that the value of the integral found by Weddle's rule is the nearest to the actual value followed by the value given by Simpson's 1/3 rd.

Example 20.8. Use the Trapezoidal rule to find an approximate value of the integral $\int_0^1 e^{x^2}$ dx dividing the interval into ten equal parts.

(U.P.T.U., 2004)

Solution. Let $y = e^{x^2}$, $h = 0.2$ and $n = 10$

The values of x and y are as follows:

x	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
y	1	1.0408	1.1735	1.4333	1.8965	2.7183	4.0649	7.0993	12.5846	23.5337	54.5982
	y_0	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}

By Trapezoidal rule, we have

$$\int_0^2 e^{x^2} dx = \frac{h}{2}[(y_0 + y_{10}) + 2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 + y_9)]$$

$$= \frac{0.2}{2}[(1 + 54.5982) + 2(1.0408 + 1.1735 + 1.4333 + 1.8965 + 2.7183$$

$$+ 4.0649 + 7.0993 + 12.5846 + 23.5337)]$$

$$= 17.9681.$$

Hence $\int_0^2 e^{x^2} dx = 17.9681$.

Example 20.9. Use Simpson's 3/8 rd rule to find $\int_0^{0.6} e^{-x^2} dx$ by taking seven ordinates. Check the result by actual integration.

(U.P.T.U., 2012; Kanpur U., 2009)

Solution. Divide the interval [0, 0.6] into six parts each of width $h = 0.1$. The values of $y = f(x) = e^{-x^2}$ are given below:

x	0	0.1	0.2	0.3	0.4	0.5	0.6
x^2	0	0.01	0.04	0.09	0.16	0.25	0.36
y	1	0.9950	0.9801	0.9513	0.9025	0.8253	0.6977
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By Simpson's 1/3rd rule, we have

$$\int_0^{0.6} e^{-x^2}\,dx = \frac{h}{3}[(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{0.1}{3}[1 + 0.6977 + 4(0.99 + 0.9513 + 0.8253) + 2(0.9801 + 0.9025)]$$

$$= \frac{0.1}{3}[1.6977 + 10.7306 + 3.7556] = \frac{0.1}{3}(16.1839) = 0.5351.$$

Example 20.10. Compute the value of $\int_0^1 \log_{10}(x + x^2)\,dx$ using Simpson's $\frac{3}{8}$th rule.

(Marathwada 2005)

Solution. Let $y = \log_{10}(x + x^2)$ and $h = 0.2$, $n = 6$.

The values of y are given below :

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2
y	$-\infty$	-0.9031	-0.5850	-0.3468	-0.1675	0	0.1903
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By Simpson's $\frac{3}{8}$th rule, we have

$$\int_0^1 y\,dx = \frac{3h}{8}[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3]$$

$$= \frac{3}{40}[-0.21 + 7.7336 + 3(3.0431 + 0.1368) + 2(1.3468)] = 1.656$$

Hence $\int_0^1 \log_{10}(x + x^2)\,dx = 1.656$.

Note. Applications of Simpson's rule. If the vertical ordinates in Fig. 20.11 represent equi-spaced cross-sections, then Simpson's rule gives the volume of the solid. As such, Simpson's rule is very useful to civil engineers. For calculating the amount of earth that must be moved to fill a depression or make a cut. Similarly, if the ordinates represent velocities at equal intervals of time, the Simpson's rule gives the distance covered. The following examples illustrate these applications.

Example 20.11. The velocity (in km/hr) of a moving vehicle was noted after every minute, for the first 10 minutes, as follows :

t	0	1	2	3	4	5	6	7	8	9	10
v	30	32	34	36	38	40	42	44	46	48	50

Estimate approximately the distance covered in 10 minutes.

Solution. If s km be the distance covered in t minutes, then $\frac{ds}{dt} = v$

$$\therefore \quad s = \int_0^{10} v\,dt = \frac{h}{3}[(f_0 + f_{10}) + 4(f_1 + f_3 + f_5 + f_7 + f_9) + 2(f_2 + f_4 + f_6 + f_8)], \text{ by Simpson's rule}$$

Here $h = 1$, $f_0 = 30$, $f_1 = 32$, $f_2 = 34$, $f_3 = 36$, $f_4 = 38$, etc.

$$\therefore \quad S = f_0 + f_{10} = 30 + 50 = 80$$

$$O = y_0 + y_6 + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4) = 30 + 86 + 4(32 + 13 + 5) + 2(68)$$
$$E = (y_1 + y_2 + y_3 + y_4) = 4(16 + 25 + 23 + 8) = 92$$

Hence the required distance $= \left| s \int_{1.0}^{1.6} \right| = \frac{0.2}{3}[30 + 4 \times 92 + 2 \times 72]$

$= 80.53$ km.

Example 20.14. The velocity v of a particle at distance s from a point on its linear path is given in the following table:

s (m)	0	2.0	4.0	6.0	8.0	10.0	12.0	14.0	16.0
v (m/sec)	10	15	21	30	40	37.5	33	27	8

Estimate the time taken by the particle to traverse the distance of 16 metres, using Simpson's rule. (U.P.T.U. 2005)

Solution. If t sec be the time taken to traverse a distance s m, then $\frac{ds}{dt} = v$

or $$\frac{dt}{ds} = \frac{1}{v} = y \text{ (say)}$$

$\therefore$ Here $$\left| t \int_{1.0}^{1.6} \right| = \int_{10}^{16} y\,ds$$

Here $h = 2.0$ and $n = 8$

Also $y_0 = \frac{1}{10}, y_1 = \frac{1}{15}, y_2 = \frac{1}{21}, y_3 = \frac{1}{30}, y_4 = \frac{1}{40}, y_5 = \frac{1}{37.5}, y_6 = \frac{1}{33}, y_7 = \frac{1}{27}, y_8 = \frac{1}{8}$

$\therefore$ By Boole's Rule, we have

$$\left| t \int_{1.0}^{1.6} \right| = \int_{10}^{16} y\,ds = \frac{2(2.0)}{45}[7y_0 + 32y_1 + 12y_2 + 32y_3 + 14y_4 + 32y_5 + 12y_6 + 32y_7 + 7y_8]$$

$$= \frac{28.8}{45}\left[7\left(\frac{1}{10}\right) + 32\left[\frac{1}{15}\right] + 12\left(\frac{1}{21}\right) + 32\left(\frac{1}{30}\right) + 14\left[\frac{1}{40}\right] + 32\left(\frac{1}{37.5}\right) + 12\left(\frac{1}{33}\right) + 32\left[\frac{1}{27}\right] + 14\left[\frac{1}{8}\right]\right]$$

$$= \frac{1}{5}[122.137781] = 1.35$$

Hence the required time $= 1.35$ sec.

Example 20.15. A solid of revolution is formed by rotating about the x-axis the curve through the points given by the following table $y = f(x)$, $x = 1$ and x axis through the points with the following co-ordinates:

x	0.00	0.25	0.50	0.75	1.00
y	1.0000	0.9896	0.9589	0.9089	0.8415

Estimate the volume of the solid formed using Simpson's rule. (Kanpur 2006)

Solution. Here $h = 0.25$, $y_0 = 1$, $y_1 = 0.9896$, $y_2 = 0.9589$, etc.

$\therefore$ Required volume of the solid generated

$$= \int_0^1 \pi y^2\,dx = \pi \cdot \frac{h}{3}[(y_0^2 + y_4^2) + 4(y_1^2 + y_3^2) + 2(y_2^2)]$$

$$= (3.14) \cdot \frac{0.25}{3}[(1 + 0.7081) + 4(0.9794^2 + 0.9089^2) + 2(0.9589^2)]$$

$$= \frac{0.25 \times 3.1416}{3}[1.7081 + 7.5218 + 1.8388] = 0.2618(10.0687) = 2.6362$$

PROBLEMS

1. Evaluate $\int_0^6 \frac{dx}{1+x}$ (i) Trapezoidal rule (ii) Simpson's 1/3rd rule (iii) Simpson's 3/8th rule (iv) Weddle's rule

2. Evaluate $\int_0^{\pi/2} \sqrt{\cos\theta}\, d\theta$ using (i) Trapezoidal rule taking $h = \frac{\pi}{8}$

(ii) Simpson's 1/3rd rule taking $h = \frac{\pi}{8}$

(iii) Simpson's 3/8th rule taking $h = \frac{\pi}{8}$

(iv) Weddle's rule taking $h = \frac{\pi}{8}$

Hence compute an approximate value of π in each case.

3. Find an approximate value of $\log_e 2$ by calculating it to 4 decimal places by Simpson's 1/3 rule. $\int_0^1 \frac{dx}{4x+3}$, dividing the range into 10 equal parts

4. Evaluate $\int_0^{\pi/2} \sin x\, dx$ dividing range into 6 equal parts by Trapezoidal rule.

5. Calculate using Trapezoidal rule $\frac{1}{\pi}$ as the value of $\int_0^{\pi} \frac{\sin x}{1+x}\, dx$

(a) $\int_0^{\pi/2} \sin^2 x\, dx$ Take $h = \frac{\pi}{12}$

6. Evaluate using Simpson's 1/3rd rule $\int_0^{\pi/2} \frac{dx}{\sqrt{\cos x}}$, choose step length 0.5°.

7. Evaluate using Simpson's 1/3rd rule, $\int_0^{\pi} \sin x\, dx$ for strips of 6 ordinates.

(a) $\int_0^{\pi/2} \sqrt{\cos\theta}\, d\theta$ (Take 6 ordinates)

8. Evaluate correct to 4 decimal places, by Simpson's $\frac{3}{8}$th rule

(i) $\int_0^{\pi/2} \frac{dx}{1+\sin x}$ (ii) $\int_0^{\pi/2} e^{\sin x}\, dx$

9. Given that

x	4.0	4.2	4.4	4.6	4.8	5.0	5.2
log x	1.3863	1.4351	1.4816	1.5261	1.5686	1.6094	1.6487

Evaluate $\int_4^{5.2} \log x\, dx$

(a) Trapezoidal rule (b) Simpson's 1/3rd rule,

(c) Simpson's 3/8th rule (d) Weddle's rule

10. Use Simpson's rule to compute $\int_0^{\pi/2} \sqrt{\sin x}\, dx$

11. The

t	0	5	7
R	50	75	48

Using Si...

12. A curve i...

x	3.0	3.2
y	2.2	2.7

Evaluate the area enclosed by the c... the x-a...

12. A curve is drawn through the points given by the following table:

x	5	10	15	20	25	30	35	40	45
y	6	9	12	15	17	18	17	14	11

Find approximately the area of the region bounded by the curve, the x-axis and the lines $x = 5$, $x = 45$. (U.P.T.U. 2003)

13. A curve is drawn to pass through the points given by the following table:

x	1	1.5	2	2.5	3	3.5	4
y	2	2.4	2.7	2.6	2	2.5	2.1

Using Weddle's rule, estimate the area bounded by the curve, the x-axis and the lines $x = 1$, $x = 4$. (U.P.T.U. 2011 IV)

14. A tank is in the form of a solid of revolution. The diameter D in cms of the tank at a distance x cms from the vertical axis is given below. Estimate the volume of the tank.

x	0	2.5	5.0	7.5	10.0	12.5	15.0
D	6	5.5	5.0	4.95	4.20	6.5	8.0

15. The velocity v of a particle at distance s from a point on its path is given by the table:

s (ft)	0	10	20	30	40	50	60
v (ft/sec)	47	58	64	65	61	52	38

Estimate the time taken to travel 60 ft. by using Simpson's 1/3 rule. (U.P.T.U. 2005)

Compare the result with Gregory's formula. (Method, 2001)

16. The following table gives the velocity v of a particle at time t:

t (seconds)	0	2	4	6	8	10	12
v (km/s)	4	6	10	14	16	14	10

Find the distance moved by the particle in 12 seconds and also its acceleration at $t = 2$ sec. (U.P.T.U. 2003)

17. A student is knocked down the ground. The acceleration is given every second during the first 10 seconds and is given in the table below. Using Simpson's $\frac{3}{8}$ rule, find the velocity of the student at $t = 10$ seconds.

t sec	0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
f (meters/sec^2)	30	31.63	33.34	35.17	37.91	40.88	43.78	45.09	46.67		

(Mumbai, 2005)

18. A road side Gardening water through stations A, B, C, D, E below for which road width has been measured. Use Simpson's rule to find the area of the garden.

x (ft)	0	15	30	45	60	75
y (ft)	80	86	90	96	104	80

If the area thus found is to be the cost of laying the garden at the rate of Rs. 45 per sq. ft., find the cost.

Further is the area taken for the water level to fall from 80 to 100, given the levels.

27.13 OBJECTIVE TYPE OF QUESTIONS

Multiple Choice

Select the correct answer or fill in the blanks in the following questions:

1. The value of $\int_0^1 \frac{dx}{1+x}$ by Simpson's rule is

(a) 0.6822 (b) 0.69318 (c) 0.69315 (d) 0.70...

2. Using Simpson's $\frac{3}{8}$ rule, the formula for $\int_{x_0}^{x_3} y\,dx$ is

3. The approximate value of $\int_0^{\pi/2} \sin x\,dx$ by the trapezoidal rule with $h = \frac{\pi}{4}$ is

4. $f(x)$ is given by

x	2	2.5	3
$f(x)$	1	2.5	2.5

Then using Trapezoidal rule, the value of $\int_2^3 f(x)\,dx$ is

8. If x | 0 | 0.5 | 1 | 1.5 | 2
$f(x)$ | 0 | 0.25 | 1 | 2.25 | 4

then the value of $\int_0^2 f(x)\,dx$ by Simpson's $\frac{1}{3}$rd rule is ...

9. Simpson's $\frac{1}{3}$rd rule is also known as ...

10. For the data :

x | 1 | 3 | 5 | 7 | 9
$f(x)$ | -1 | 2 | 6 | 8 | A

the value of $\int_1^9 y\,(x)\,dx$ when computed by Simpson's $\frac{1}{3}$rd rule is

(a) 32 (b) 33 (c) 0 (d) 16

11. While evaluating a definite integral by Trapezoidal rule, the accuracy can be increased by taking ...

12. The value of $\int_0^1 \frac{dx}{1+x+x^2}$ by Simpson's $\frac{1}{3}$rd rule (taking $h = 1/4$) is ...

13. For the data

x | 1 | 3 | 5 | 7 | 9
$f(x)$ | 2 | 8 | 9 | 12 |

$\int_1^9 f(x)\,dx$ when found by the Trapezoidal rule is

(a) 36 (b) 72 (c) 30 (d) 60

14. The approximate value of $\left[\int_{x_0}^{x_n} y\,dx\right]$ using Newton-Cotes formula is ...

15. The number of strips required in Weddle's rule is ...

16. The number of strips required in Simpson's $\frac{1}{3}$rd rule is a multiple of

(a) 1 (b) 2 (c) 3 (d) 4

17. If $y_0 = 1, y_1 = \frac{10}{11}, y_2 = \frac{5}{6}, y_3 = \frac{10}{13}, y_4 = \frac{1}{2}$ and $h = \frac{1}{4}$, then using Trapezoidal rule, $\int_0^1 y\,dx =$ ...

18. Using Simpson's $\frac{1}{3}$rd rule, $\int_0^{\pi/2} \sin x\,dx =$ ... (where x is in radians)

19. If $y_0 = 1, y_1 = 0.5, y_2 = 0.2, y_3 = 0.1, y_4 = 0.05, y_5 = 0.03$ and $y_6 = 0.02$, then $\int_0^1 y\,dx$ by Simpson's $\frac{1}{3}$rd rule is ...

20. If $f(0) = 1, f(1) = 2.71, f(2) = 7.39, f(3) = 20.1, f(4) = 54.6$ and $h = 1$, then $\int_0^4 f(x)\,dx$ by Simpson's $\frac{1}{3}$rd rule is ...

21. Simpson's $\frac{1}{3}$rd rule and Trapezoidal rule give the same result if ...

22. To evaluate $\int_{x_0}^{x_n} y\,dx$ by Simpson's $\frac{1}{3}$rd rule as well as Simpson's $\frac{3}{8}$th rule, the number of intervals should be ..... and ..... respectively.

23. Whenever Trapezoidal rule is applicable, Simpson's $\frac{1}{3}$rd rule can also be applied. (True or False)

CHAPTER 31

Difference Equations

1. Introduction 2. Definition 3. Formation of difference equations 4. Linear difference equations 5. Rules for finding complementary function 6. Rules for finding particular integral 7. Simultaneous difference equations with constant coefficients 8. Application to evolution of a bacterial colony 9. Objective Type of Questions

31.1 INTRODUCTION

Difference calculus also forms the basis of Difference equations. These equations arise in all situations in which sequential relation exists between various discrete values of the independent variable. The need is to work with discrete functions arises because there are physical phenomena which are inherently of a discrete nature. In control engineering, it often happens that the input is in the form of discrete pulses of short duration. The radar tracking system receives such discrete pulses from the target which is being tracked. As such difference equations arise in the study of electrical networks, in the theory of probability, in statistical problems and many other fields.

Just as the subject of Differential equations grew out of calculus to become one of the most powerful instruments in the hands of a practical mathematician when dealing with continuous processes in nature, so the subject of Difference equations is coming to the fore for the treatment of discrete processes. Thus the difference equations may be thought of as the discrete counterparts of the differential equations.

31.2 DEFINITION

(1) A difference equation is a relation between the differences of an unknown function of one or more general values of the argument.

Thus $\Delta y_{x+1} + y_x = 2^x$ ...(1) and $\Delta^2 y_{x+2} - 3\Delta y_{x+1} + 2y_x = 1$ ...(2)

are difference equations.

An alternative way of writing a difference equation is as under :

Since $\Delta y_{x+1} = y_{x+2} - y_{x+1}$, therefore (1) may be written as

$$y_{x+2} - y_{x+1} + y_x = 2^x \qquad \ldots(3)$$

Also since, $\Delta^2 y_{x+2} = y_{x+3} - 2y_{x+2} + y_{x+1}$, therefore (2) takes the form :

$$y_{x+3} - 2y_{x+2} + y_{x+1} = 1 \qquad \ldots(4)$$

Quite often, difference equations are met under the name of recurrence relations.

(2) Order of a difference equation is the difference between the largest and the smallest arguments occurring in the difference equation divided by the interval of increment.

Thus (1) above is the second order, for

$$\frac{\text{largest argument} - \text{smallest argument}}{\text{unit of increment}} = \frac{(x+2)-x}{1} = 2,$$

and (2) is of the third order, for $\frac{(x+3)-(x+1)}{1} = 2$.

Note. While finding the order of the difference equation, it is not always necessary to express it in a form free of Δ, the highest power of Δ being equal to the order of the difference equation.

(3) Solution of a difference equation is an expression for y_x, which satisfies the given difference equation.

The **general solution** of a difference equation is that in which the number of arbitrary constants is equal to the order of the difference equation.

A **particular solution** or **particular integral** is a solution which is obtained from the general solution by giving particular values to the constants.

11.3. FORMATION OF DIFFERENCE EQUATIONS

The following examples illustrate the way in which difference equations arise and are formed.

Example 11.1. *Form the difference equation corresponding to the family of curves*

$$y = ax + bx^2 \qquad \ldots(1)$$

Solution. We have $\Delta y = a[(x+1) + b(x+1)^2] - (ax + bx^2) = [a(x+1) + b(x+1)^2] - y$

$$= a + b(2x + 1) \qquad \ldots(2)$$

and $\Delta^2 y = 2b[(x+1) + 1] - 2b = 2b \qquad \ldots(3)$

To eliminate a and b, we have from (3), $b = \frac{1}{2}\Delta^2 y$

and from (2), $a = \Delta y - b(2x + 1) = \Delta y - \frac{1}{2}\Delta^2 y(2x + 1)$

Substituting these values of a and b in (1), we get

$$y = \left[\Delta y - \frac{1}{2}\Delta^2 y(2x + 1)\right]x + \frac{1}{2}\Delta^2 y \cdot x^2$$

or $\quad x^2 \Delta^2 y - 2x\Delta y + 2y = 0$

This is the desired difference equation which may equally well be written in terms of E as

or $\quad x^2 y_{x+2} - (2x^2 + 4x)y_{x+1} + (x^2 + 2x + 2)y_x = 0.$

Example 11.2. *From $y_x = Ax^2 + Bx - 3^x$, derive a difference equation by eliminating the constants.*

Solution. We have $\quad y_x = Ax^2 + Bx - 3^x$, $y_{x+1} = A(x+1)^2 + B(x+1) - 3^{x+1}$

and $\quad y_{x+2} = A(x+2)^2 + B(x+2) - 3^{x+2}$

Eliminating A and B, we get

$$\begin{vmatrix} y_x & x^2 & x \\ y_{x+1} & (x+1)^2 & x+1 \\ y_{x+2} & (x+2)^2 & x+2 \end{vmatrix} + 3 = 0 \quad \text{or} \quad y_{x+2} - 3y_{x+1} + 3y_x = 2$$

which is the desired difference equation.

EXERCISE 11.1

1. Write the difference equation $\Delta^2 y_x + \Delta y_x + y_x = 0$ in the subscript notation.
2. Assuming $\log(1+x) = y_0 + y_1 x + y_2 x^2 + \ldots + y_n x^n + \ldots$, find the difference equation satisfied by y_x.
3. Form a difference equation by eliminating arbitrary constant from $y_x = cx^2$. (Meerut, 2005)
4. Find the difference equation satisfied by
   (i) $y_x = ax + b$ (Kurukshetra, 2003) (ii) $y_x = cx^2 - 2x$
5. Derive the difference equation in each of the following cases :
   (i) $y_x = A2^x + B3^x$ (ii) $y_x = (A + Bx)2^x$ (Bhopal, 2003)
6. Form the difference equations generated by
   (i) $y_x = ax + b$ (ii) $y_x = cx^2 + 3x - 2x^2$ (iii) $y_x = Ax^2 + Bx - 1$

[illegible]

(a) Def: A **linear difference equation** is the recursion $(x_{n+k}, x_{n+k-1}, \ldots, x_n)$ where k is the order of the equation and $a_0 \neq 0$ and $a_k \neq 0$.

(b) Linear difference equation with constant coefficients of the form

$$x_{n+k} + a_{k-1}x_{n+k-1} + \cdots + a_1 x_{n+1} + a_0 x_n = f(n) \quad \text{(1)}$$

where $a_0, a_1, \ldots, a_{k-1}$ are constants.

Note: we shall only work with linear difference equations with constant coefficients. This is analogous to those of linear differential equations with constant coefficients.

(c) Homogeneous properties: If $x_1(n), x_2(n), \ldots, x_k(n)$ for n independent solutions of the equation

$$x_{n+k} + a_{k-1}x_{n+k-1} + \cdots + a_0 x_n = 0 \quad \text{(2)}$$

then the complete solution is $x_n = c_1 x_1(n) + \cdots + c_k x_k(n)$

where $c_1, c_2, \ldots, c_k$ are arbitrary constants.

If x_p is a particular solution of (1), then the complete solution of $x_{n+k} + a_{k-1}x_{n+k-1} + \cdots + a_0 x_n = f(n)$ is called a **complementary function** x_c and the sum x_p is called the **particular integral** (PI) $x = x_c + x_p$.

Thus the complete solution $x = x_c + x_p$ [illegible]

[illegible]

Let x_n denote the nth term in linear difference equation with constant coefficients a_0, a_1, a_2 and $f(n)$ are given.

(1) To begin with, consider any first order linear equation $x_{n+1} + a_0 x_n = f(n)$, where $f(n)$ is a constant.

Rewriting it as $\frac{x_{n+1}}{a^{n+1}} - \frac{x_n}{a^n} = 0$, we have $x_n = \left(\frac{x_0}{a^0}\right) = 0$, which gives $x_n(a)^n = c_1$ is constant.

Thus the solution of $x_{n+1} + a x_n = 0$ is $x_n = c_1 a^n$

(2) Now consider the second order linear equation $x_{n+2} + a_1 x_{n+1} + a_0 x_n = 0$ which is symbolic form as

$$(E^2 + a_1 E + a_0)x_n = 0 \quad \text{(3)}$$

Its symbolic coefficient equated to zero $E^2 + a_1 E + a_0 = 0$

is called the auxiliary equation. Let its roots be λ_1, λ_2.

Case I: If roots are real and distinct, then (3) is equivalent to

$$(E - \lambda_1)(E - \lambda_2)x_n = 0 \quad \text{(4)}$$

$$(E - \lambda_2)(E - \lambda_1)x_n = 0 \quad \text{(5)}$$

If x_n satisfies the substitution equation $(E - \lambda_1)x_n = 0$, then it will also satisfy (4).

Similarly, if x_n satisfies the substitution equation $(E - \lambda_2)x_n = 0$, then it will also satisfy (5).

∴ If both (4) and (5) are equivalent, two independent solutions of (3) by solving the two substitution equations

$$(E - \lambda_1)x_n = 0 \quad \text{and} \quad (E - \lambda_2)x_n = 0$$

These solutions are respectively $x_n = c_1 \lambda_1^n$ and $x_n = c_2 \lambda_2^n$

where c_1 and c_2 are arbitrary constants.

Thus the general solution of (3) is $x_n = c_1 \lambda_1^n + c_2 \lambda_2^n$

Case II: If the roots are equal (repeated) i.e. $\lambda_1 = \lambda_2$ then (3) becomes

$$(E - \lambda_1)^2 x_n = 0 \quad \text{(6)}$$

Let

$$\lambda_1 = \lambda_2 = \lambda_0$$

where λ_0 is a new dependent variable. Then (6) takes the form

$$\begin{aligned} &\lambda_0^{n+2} x_{n+2} - 2\lambda_0(\lambda_0)^{n+1} x_{n+1} + \lambda_0^2 \lambda_0^n x_n = 0 \\ &x_{n+2} - 2\lambda_0 x_{n+1} + \lambda_0^2 x_n = 0 \quad \text{i.e.,} \quad (E - \lambda_0)^2 x_n = 0 \\ &x_n = (c_1 + c_2 n)\lambda_0^n \text{ where } c_1, c_2 \text{ are arbitrary constants} \end{aligned}$$

Thus the solution of (3) becomes $x_n = (c_1 + c_2 n)\lambda_0^n$.

Case III: If the roots are imaginary, i.e., $\lambda_1 = \alpha + i\beta$, $\lambda_2 = \alpha - i\beta$ then the solution of (3) is

$$x_n = c_1(\alpha + i\beta)^n + c_2(\alpha - i\beta)^n \qquad [\because \lambda_1 = \alpha + i\beta \text{ and } \lambda_2 = \alpha - i\beta]$$

$$= r^n \left[c_1(\cos n\theta + i \sin n\theta) + c_2(\cos n\theta - i \sin n\theta)\right]$$

$$= r^n (A_1 \cos n\theta + A_2 \sin n\theta)$$

where A_1, A_2 are arbitrary constants and $r = \sqrt{(\alpha^2 + \beta^2)}$, $\theta = \tan^{-1}(\beta/\alpha)$.

(II) In general, to solve the equation $y_{n+k} + a_1 y_{n+k-1} + \ldots + a_k y_n = 0$ where a's are constants :

(i) Write the equation in the symbolic form $(E^k + a_1 E^{k-1} + \ldots + a_k) y_n = 0$.

(ii) Write down the auxiliary equation $m^k + a_1 m^{k-1} + \ldots + a_k = 0$ and solve it for E.

(iii) Write the solution as follows :

Roots of A.E.	Solution of D.E.
1. $E_1, E_2, E_3, \ldots$ — real and distinct roots	$y_n = c_1 E_1^n + c_2 E_2^n + c_3 E_3^n + \ldots$
2. $E_1, E_1, E_3, \ldots$ — 2 real and equal roots	$y_n = (c_1 + c_2 n) E_1^n + c_3 E_3^n + \ldots$
3. $E_1, E_1, E_1, \ldots$ — 3 real and equal roots	$y_n = (c_1 + c_2 n + c_3 n^2) E_1^n + \ldots$
4. $\alpha + i\beta, \alpha - i\beta, \ldots$ — a pair of imaginary roots	$r^n (c_1 \cos n\theta + c_2 \sin n\theta)$ where $r = \sqrt{(\alpha^2 + \beta^2)}$ and $\theta = \tan^{-1}(\beta/\alpha)$

Example 3.26. Solve the difference equation $y_{n+3} - 3y_{n+2} + 3y_{n+1} - y_n = 0$.

Solution. Given equation in symbolic form is $(E^3 - 3E^2 + 3E - 1) y_n = 0$

$\therefore$ its auxiliary equation is $E^3 - 3E^2 + 3E - 1 = 0$

or $(E-1)^3 = 0$ $\therefore E = 1, 1, 1$

Thus the complete solution is $y_n = c_1 (1)^n + c_2 n (1)^n + c_3 n^2 (1)^n$.

Example 3.27. Solve $y_{n+2} - 2y_{n+1} + y_n = 0$.

Solution. Given difference equation in symbolic form is $(E^2 - 2E + 1) y_n = 0$.

$\therefore$ its auxiliary equation is $E^2 - 2E + 1 = 0$

or $(E-1)^2 = 0$ $\therefore E = 1, 1$

Thus the required solution is $y_n = c_1 + c_2 n$ (1)^n, i.e., $u_n = c_1 + c_2 n$.

Example 3.28. Solve $y_{n+2} - 2y_{n+1} \cos\alpha + y_n = 0$.

Solution. This is a second order difference equation in y_{n+2} which in symbolic form is

$$(E^2 - 2E \cos\alpha + 1) y_n = 0$$

The auxiliary equation is $E^2 - 2E \cos\alpha + 1 = 0$

$$E = \frac{2\cos\alpha \pm \sqrt{(4\cos^2\alpha - 4)}}{2} = \cos\alpha \pm i \sin\alpha$$

Thus the solution is $y_{n+2} = (E^{n+2}) [c_1 \cos(n+2)\alpha + c_2 \sin(n+2)\alpha]$

or $y_n = c_1 \cos n\alpha + c_2 \sin n\alpha$.

Example 3.29. The numbers $0, 1, 1, 2, 3, 5, 8, 13, 21, \ldots$ are said to form a Fibonacci sequence. Form the Fibonacci difference equation and solve it.

Solution. In this sequence, each number beyond the second is the sum of the two previous number. If y_n be the nth number then $y_n = y_{n-1} + y_{n-2}$ for $n > 2$.

or $y_{n+2} - y_{n+1} - y_n = 0$ $(n > 0)$

or $(E^2 - E - 1) y_n = 0$ is the difference equation

Its A.E. is $E^2 - E - 1 = 0$ which gives $E = \frac{1}{2}(1 \pm \sqrt{5})$.

Thus the solution is $y_n = c_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + c_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$, for $n > 0$.

When $n = 1$, $y = 3$

$$c_1\left(\frac{3+\sqrt{5}}{2}\right) + c_2\left(\frac{3-\sqrt{5}}{2}\right) = 0 \quad \ldots(i)$$

When $n = 2$, $y_2 = 8$

$$c_1\left(\frac{3+\sqrt{5}}{2}\right)^2 + c_2\left(\frac{3-\sqrt{5}}{2}\right)^2 = 0 \quad \ldots(ii)$$

Solving (i) and (ii), we get

$$c_1 = \frac{5-\sqrt{5}}{10} \text{ and } c_2 = \frac{5+\sqrt{5}}{10}$$

Hence the required solution is

$$y_n = \frac{5-\sqrt{5}}{10}\left(\frac{3+\sqrt{5}}{2}\right)^n + \frac{5+\sqrt{5}}{10}\left(\frac{3-\sqrt{5}}{2}\right)^n$$

PROGRAMMING

Write the following difference equations

1. $y_{n+1} - 2y_{n-1} + y_n = 3$
2. $y_{n+2} - 3y_{n+1} + y_n = 0$
3. $y_{n+2} - y_{n+1} = 0$ given that $y_0 = 0, y_1 = 1$
4. $(n+4) - 2(n+1) + 3(n) = 0$
5. $y_{n+2} - 2y_{n+1} + 2y_n = 0$, given $y_0 = 1$, $y_1 = 0$ and $y_2 = -2$.
6. $E^3 - 6E^2 + 4E + 13y_n = 0$, given $y(0) = 5$, $y_1 = -2$, $y_2 = 22$.
7. $y_{n+2} - 2y_{n+1} + 2y_n = 0$
8. $y_{n+1} + y_{n-1} + 10y_n = 0$
9. $y_{n+1} + y_{n-1} - 2y_n = 0$
10. $E^3 - 6E + 4 = 0$
11. $y_{n+2} - 10y_{n+1} + 9y_n = 0$

Find $E^3 + - 18 - 10$ for (a) $1, 2, -4$ (b) $3, 3, -2$ (c) $1, 2, 3$ (d) $1, 1, 1$ (e) De Moivre's theorem.

12. Show that the difference equation $f_{n+2} - (a+b)f_{n+1} + f_n + f_{n-1} = 0$ has the solution

$f_n = c_1 \sin(n\alpha) + c_2 \sin((n-1)\alpha)$ where $\cos\alpha = \frac{1}{2}(a+b)$ and $c_1 = f_1$ and $c_2 = f_0 \cos\alpha - f_1$ and $f_n = \sum_{r=0}^{n} u_r v^{n-r}$

13. A series of values of y_n satisfy the relation $y_{n+2} + ay_{n+1} + by_n$.
Given that $y_0 = 0$, $y_1 = 1$, $y_2 = -1$. Show that $y_n = 2^n \sin(n\pi/3)$.
14. A pair of rabbits are placed in a field where one pair did previously inhabit but previously 10 died when two years old. A second is placed in the field and is produced in a planted. Take y_n to be the number of rabbits produced at the end of the nth year. Show that $y_{n+2} - 3y_{n+1} + 3y_n = (n+1)^2 - y_{n+1}$.
Hence show that $y_{n+2} - 3y_{n+1} + 3y_n = 0$ and find y_n.

1.12 RULES FOR FINDING THE PARTICULAR INTEGRAL

Consider the equation $y_{n+r} + a_1 y_{n+r-1} + \ldots + a_r y_n = f(n)$

which in symbolic form is $\phi(E)y_n = f(n)$ ...(1)

where $\phi(E) = E^r + a_1 E^{r-1} + \ldots + a_r$

Then the particular integral is given by $P.I. = \frac{1}{\phi(E)} f(n)$.

Case I. When $f(n) = a^n$

$$P.I. = \frac{1}{\phi(E)} a^n, \text{ put } E = a$$

$$= \frac{1}{\phi(a)} a^n, \text{ provided } \phi(a) \neq 0$$

If $\phi(a) = 0$, then for the equation

$\phi(E) \cdot a^n = a^n$, $\quad P.I. = \frac{n}{\phi'(a)} a^n + \text{soln}^{-1}$

(iv) $(E-a)^2 y_n = a^n$, $\quad P.I. = \frac{1}{(E-a)^2} a^n = \frac{n(n-1)}{2!} a^{n-2}$

(v) $(E-a)^3 y_n = a^n$, $\quad P.I. = \frac{1}{(E-a)^3} a^n = \frac{n(n-1)(n-2)}{3!} a^{n-3}$

and so on.

Example 1.17. Solve $y_{n+2} - 4y_{n+1} + 3y_n = 0$.

Solution. Given equation in symbolic form is $(E^2 - 4E + 3)y_n = 0$.

∴ The auxiliary equation is $E^2 - 4E + 3 = 0$

or $\quad (E-1)(E-3) = 0 \quad \therefore \quad E = 1, 3$

∴ $\quad C.F. = c_1(1)^n + c_2(3)^n = c_1 + c_2 \cdot 3^n$

and $\quad P.I. = \frac{1}{E^2 - 4E + 3} 0^n$ (∵ $a = 0$)

$$= \frac{1}{2^2 - 4.2 + 3} 0^n = \frac{1}{0} \cdot 0^n$$

Thus the complete solution is $\quad y_n = c_1 + c_2 \cdot 3^n + 0 \cdot n$.

Example 1.18. Solve $y_{n+2} - 4y_{n+1} + 4y_n = 3^n$.

Solution. Given equation in symbolic form is $(E^2 - 4E + 4)y_n = 3^n$.

The auxiliary equation is $m^2 - 4E + 4 = 0 \quad \therefore \quad E = 2, 2$.

$$C.F. = (c_1 + c_2 n)2^n$$

$$P.I. = \frac{1}{(E-2)^2} \cdot 3^n = \frac{n(n-1)}{2!} 2^{n-2} = n(n-1)2^{n-3}$$

Hence the complete solution is $y_n = (c_1 + c_2 n)2^n + n(n-1)2^{n-3}$.

Case II. When $f(n) = \sin bn$ or $\cos bn$.

$$P.I. = \frac{1}{\phi(E)} \sin bn = \frac{1}{\phi(E)} \left[\frac{e^{ibn} - e^{-ibn}}{2i} \right] = \frac{1}{2i} \left[\frac{1}{\phi(E)} a^n - \frac{1}{\phi(E)} b^n \right]$$

where $a = e^{ib}$ and $b = e^{-ib}$.

Now proceed as in case I.

(i) When $f(n) = \cos bn$, $\quad P.I. = \frac{1}{\phi(E)} \cos bn = \frac{1}{\phi(E)} \left[\frac{e^{ibn} + e^{-ibn}}{2} \right]$

$$= \frac{1}{2} \left[\frac{1}{\phi(E)} a^n + \frac{1}{\phi(E)} b^n \right] \text{ as before.}$$

Now proceed as in case I.

Example 1.19. Solve $y_{n+2} - 2\cos\alpha \cdot y_{n+1} + y_n = \cos n\alpha$. (Nagpur 2000)

Solution. Given equation in symbolic form is $(E^2 - 2\cos\alpha \cdot E + 1)y_n = \cos n\alpha$.

The auxiliary equation is $m^2 - 2\cos\alpha \cdot m + 1 = 0$.

$$\therefore \quad m = \frac{2\cos\alpha \pm \sqrt{(4\cos^2\alpha - 4)}}{2} = \cos\alpha \pm i\sin\alpha$$

$$\therefore \quad C.F. = (c_1 + c_2)\cos n\alpha + (c_1 - c_2)i\sin n\alpha = c_1 \cos n\alpha + c_2 \sin n\alpha$$

$$P.I. = \frac{1}{E^2 - 2E\cos\alpha + 1} \cos n\alpha$$

$$= \frac{1}{E^2 - 2E\cos\alpha + e^{i\alpha} e^{-i\alpha}} 1 \left(\frac{e^{in\alpha} + e^{-in\alpha}}{2} \right)$$

$$= \frac{1}{2i}\left[\frac{1}{(D-e^{ix})(D-e^{-ix})}e^{ix} - \frac{1}{(D-e^{ix})(D-e^{-ix})}e^{-ix}\right]$$
$$\left[\text{Put } D = e^{ix}\right] \qquad \left[\text{Put } D = e^{-ix}\right]$$

$$= \frac{1}{2i}\left[\frac{1}{(D-e^{ix})}\frac{1}{e^{ix}-e^{-ix}}e^{ix} + \frac{1}{(D-e^{-ix})}\frac{1}{e^{-ix}-e^{ix}}e^{-ix}\right]$$

$$= \frac{1}{4i\sin x}\left[\frac{1}{D-e^{ix}}e^{ix} - \frac{1}{D-e^{-ix}}e^{-ix}\right] = \frac{1}{4i\sin x}\left[e^{ix}\cdot x e^{ix} - e^{-ix}\cdot x e^{-ix}\right]$$

$$= \frac{i}{2\sin x}\left[\frac{e^{2ix}-e^{-2ix}}{2i}\right] = \frac{x\sin 2x}{2\sin x}$$

Hence the complete solution is

$$y = c_1\cos x + c_2\sin x + \frac{x\sin 2x}{2\sin x}$$

Case III. When $f(x) = a^x$. $\quad$ P.I. $= \dfrac{1}{\phi(D)}a^x = \dfrac{1}{\phi(a)}a^x$

(1) Expand $[\phi(D)]^{-1}$ in ascending powers of D by the Binomial theorem as far as the term in D^n.
(2) Expand e^{ax} in the Exponential form (p. 960) and operate on it with each term of the expansion.

Example 9.11 Solve $(D^2 - 4)y = e^{2x} + x - 1$. (Worcester, 1999)

Solution. Given equation is $(D^2 - 4)y = e^{2x} + x - 1$.

The auxiliary equation is $m^2 - 4 = 0 \quad \therefore m = \pm 2$.

$\therefore$ $\quad$ C.F. $= c_1e^{2x} + c_2e^{-2x}$.

$\therefore$ $\quad$ P.I. $= \dfrac{1}{D^2-4}e^{2x} + (x-1) = \dfrac{1}{(D+2)(D-2)}e^{2x} + (x-1)$

$$= \frac{1}{D^2+2D-2}2e^{2x} + (x-1) = -\frac{1}{2}\left[1-\left(\frac{D}{2}+\frac{D^2}{2}\right)\right]^{-1}(2e^{2x}) + (x-1)$$

$$= -\frac{1}{2}\left[1+\left(\frac{D}{2}+\frac{D^2}{2}\right)+\left(\frac{D}{2}+\frac{D^2}{2}\right)^2+\ldots\right](2e^{2x}) + (x-1)$$

$$= -\frac{1}{2}\left[1+\frac{D}{2}+\frac{3}{4}D^2+\ldots\right](2e^{2x}) + (x-1) = -\frac{1}{2}\left[2e^{2x}+e^{2x}+3e^{2x}\right] + (x-1) = -\frac{5}{2}e^{2x} + x - 1$$

Hence the complete solution is $y = c_1e^{2x} + c_2e^{-2x} - \frac{5}{2}e^{2x} + x - 1$.

Case IV. When $f(x) = x^n$, $\phi(x)$, $f(x)$ being a polynomial of finite degree in x.

$$\text{P.I.} = \frac{1}{\phi(D)}x^n = [\phi(D)]^{-1}x^n$$

Note. $\phi(x)$ being a polynomial in x, proceed as in case III.

Example 9.12 Solve $(D^2 - 3D + 1)y = x^2 + 2x$. (Nottingham, 2000)

Solution. Given equation is $(D^2 - 3D + 1)y = x^2 + 2x$.

Its $\quad$ C.F. $= c_1e^{mx} + c_2e^{nx}$

and $\quad$ P.I. $= \dfrac{1}{1D^2 - 3D}x^2 \cdot e^{0x} = e^{0x}\dfrac{1}{1(D^2 - 3D)}x^2 = e^{0x}\dfrac{1}{(1 + 3D)}x^2$

$$= 2^n(1 + 2n)^2 + 2n(n-1) + n + n = 2^n(1 + 4n + 4n^2 - n) + 3n^2 + 1$$
$$= 2^n[4n^2 + 3n] + 4n(n-1) + 3n + 2n + 2$$
$$= 2^n(4n^2 - n) + 8n + 2n = 2^n(4n^2 - 4n + 1) + 6n + 2n$$

Hence the complete solution is $y_n = c_1 + c_2 n + 2^n(4n^2 - 4n + 1)$.

PROBLEMS 5.2

Solve the following difference equations:

1. $y_{n+2} - 3y_{n+1} + 2y_n = 0$; $y_0 = 0, y_1 = 3$. (Madras, 2002)
2. $y_{n+2} + 4y_{n+1} + 3y_n = 2^n$; $y_0 = y_1 = 0$. (U.P.T.U., 2005)
3. $2y_{n+2} - 3y_{n+1} + y_n = 3^n$. (Mumbai, 2002)
4. $y_{n+2} - 2y_{n+1} + 4y_n = 0$, given that $y_0 = 0$ and $y_1 = 4$.
5. $4y^2 - 4y + 3y = 2^n$.
6. $y_{n+2} - 4y_{n+1} + 4y_n = 3.4^n - 5.4^n$.
7. $y_{n+2} - 2y_{n+1} + y_n = n4^n$. (Madras, 2000)
8. $y_{n+2} - \left(2\cos\frac{\pi}{3}\right)y_{n+1} + y_n = 0$.
9. $4y^2 - 4y_{n+1} + y_n = n^2 - 1$.
10. $y_{n+2} + y_n = n^2 + 3y_{n+1} + y_n = y_n = 0$. (U.P.T.U. (C.O.), 2003)
11. $y_{n+2} - 8y_{n+1} + 20y_{n+1} - 16y_n = 2^n + 3n$. (Anna, 2000)
12. $4y^2 - 4y + 3y = 2^n + 3^n$. (Madras, 2000)
13. $y_{n+2} - 6y_{n+1} + 8y_n = 2^n + 3^n$. (Anna, 2000)
14. $y_{n+2} - 4y_{n+1} + 4y_n = 4^n$. (Anna, 2000)
15. $y_{n+2} + 8y_n = 2n + 3^n$.
16. $4y^2 - 4y + 4y_{n+1} = 4^n \left[\frac{1}{2}n - \frac{3}{8}\right] + 2n\sin\left(\frac{n\pi}{2}\right)$.
17. $y_{n+2} - 4y_{n+1} + 4y_n = 4^n(n^2 + 3n + 4)$.

18. A beam of length l supported at points which are a unit distance apart carries a uniform load w per unit length. The bending moments $M_0, M_1, \ldots, M_n$ at the supports satisfy the Clapeyron's equation

$$M_{n-1} + 4M_{n+1} + M_n = -\frac{1}{2}wl^2$$

If a beam weighing 20 kg is supported at its ends and at two other supports dividing it into four equal parts of 1 metre length, show that the bending moment at each of the two middle supports is 3 kg metre.

5.12 SIMULTANEOUS DIFFERENCE EQUATIONS WITH CONSTANT COEFFICIENTS

The method used for solving simultaneous differential equations with constant coefficients also applies to simultaneous difference equations with constant coefficients. The following example illustrates the technique.

Example 5.12. *Solve the simultaneous difference equations*

$$x_{n+1} + y_n - 2x_n = n, \quad 2x_n + x_{n-1} - 2y_n = 2^n$$

subject to the conditions $x_0 = 0, y_0 = 0$.

Solution. Given equations in symbolic form are

$$(E - 2)x_n + y_n = n \quad \ldots(i)$$
$$2x_n + (E - 2)y_n = 2^n \quad \ldots(ii)$$

Operating the first equation with $E - 2$ and subtracting the second from it, we get

$$[(E - 2)(E - 2) - 2]x_n = (E - 2)n - 2^n$$

or
$$(E^2 - 4E + 2)x_n = n - 4n - 4^n$$

Its solution is
$$x_n = c_1 2^n + c_2 4^n - \frac{4}{9}n - \frac{28}{27} + \frac{4^n}{4} \quad \ldots(iii)$$

Substituting the value of x_n from (iii) in (ii), we get

$$y_n = c_1 2^n - 3c_2 4^n - \frac{3n}{2} - \frac{24}{27} - \frac{4^n}{4} \quad \ldots(iv)$$

Taking $n = 0, x_0 = 0, y_0 = 0$ in (iii) and (iv), we obtain

$$c_1 + c_2 = \frac{64}{27}, \quad 2c_1 - 3c_2 = \frac{24}{27}$$

where
$$c_1 = 1.33, c_2 = -0.0741$$

Hence $x_n = 1.3333^n + 0.1667(-0.5)^n - 0.5n(-0.5)^{n-1}$

$x_n = 1.3333^n + 0.1667(-0.5)^n - 0.5n(-0.5)^{n-1}$.

EXERCISE 1.6.5

Solve the following simultaneous linear difference equations:

1. $x_{n+1} = x_n + 2y_n + 3$, $y_{n+1} = x_n - y_n + 1$.
2. $x_{n+1} = x_n + 4y_{n-1} + 5$, $y_{n+1} = x_n + 2y_n + 25$.
3. $x_{n+1} + 5x_n + 3y_n = 0$, $y_{n+1} + x_n + y_n = 0$, given $x_0 = 0$, $y_0 = 1$.
4. $x_{n+1} + x_n = y_n + 2^n$, $y_{n+1} + y_n = x_n + n^2$.

1.6.6 APPLICATION TO DEFLECTION OF A LOADED STRING

Consider a light string of length l stretched tightly between A and B. Let the forces P_i be acting at the equispaced points x_i ($i = 1, 2, \ldots, n-1$) and perpendicular to AB resulting in small transverse displacements y_i at these points (Fig. 31.1). Assuming the angles θ_i made by the portions between x_i and x_{i-1} with the horizontal to be small, we have

$$\sin\theta_i = \tan\theta_i = \theta_i \text{ and } \cos\theta_i = 1$$

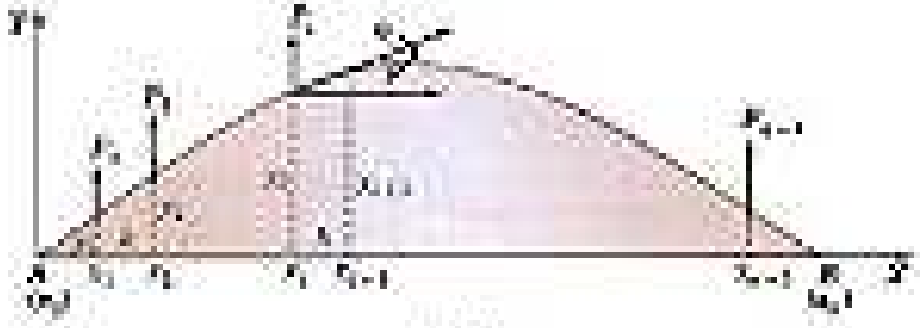

Fig. 31.1

If T be the tension of the string at x_i, then $T\cos\theta_i = T$

i.e., the tension may be taken as uniform.

Taking $x_{i+1} - x_i = h$, we have

$$y_{i+1} - y_i = h\tan\theta_i = h\theta_i \quad \ldots(1)$$

$$y_i - y_{i-1} = h\tan\theta_{i-1} = h\theta_{i-1} \quad \ldots(2)$$

Also resolving the forces in equilibrium at (x_i, y_i) along AY, we get

$$T\sin\theta_i - T\sin\theta_{i-1} + P_i = 0 \text{ i.e., } T(\theta_i - \theta_{i-1}) + P_i = 0 \quad \ldots(3)$$

Eliminating θ_i and θ_{i-1} from (1), (2) and (3), we obtain

$$y_{i+1} - 2y_i + y_{i-1} = -\frac{hP_i}{T} \quad \ldots(4)$$

which is a difference equation and its solution gives the displacements y_i. To obtain the arbitrary constants in the solution, we take $y_0 = y_n = 0$ as the boundary conditions, since the ends A and B of the string are fixed.

Example 31.10. A light string is stretched between two fixed points 200 cm apart and carries 9 equal weights 5 gm each at equal intervals and the tension is 360 gm wt. Obtain the shape of the string in the equilibrium position.

Solution. Taking $h = 10$ cm, $P_i = 5$ gm and $T = 360$ gm wt., the above equation (4) becomes $y_{i+1} - 2y_i + y_{i-1} = -\frac{1}{72}$

i.e., $y_{i+1} - 2y_i + y_{i-1} = -\frac{1}{72}$

Now $x_i = ih = 10i$ cm, $i = 1, 2, \ldots, 9$ $\quad \ldots(5)$ $x_i = x_0 + ih$

and $$P.I. = \frac{1}{(E-1)^2}\left(-\frac{1}{16}\right) = -\frac{1}{16}\frac{1}{(E-1)^2}(1^t) = -\frac{1}{16}\frac{t(t-1)}{2} = \frac{1}{32}t(t-1)$$

Thus the C.S. is $y_t = c_1 + c_2 t + \frac{1}{32}t(t-1)$

Since $y_0 = 0, \Rightarrow c_1 = 0$

and $y_{10} = 3, \Rightarrow c_2 = \frac{11}{32}$.

Hence $y_t = \frac{11}{32}t + \frac{1}{32}t(t-1)$

At the mid-point $t = 5$, we get $y_5 = 1.5$ cm

PROBLEMS

1. A light string of length $l = 10$ is stretched between two fixed points with a tension T. It is loaded with a mass m at its midpoint. If the system starts vibrating with an initial velocity u_0, find the displacement y_t of the t^{th} mass.

11.9 OBJECTIVE TYPE OF QUESTIONS

PROBLEMS

Fill in the blanks or answer the following questions:

1. $y_n = A \cdot 3^n + B \cdot 2^n$ is the solution of the difference equation ____
2. The solution of $\Delta^2 y_n = 0$ is ____.
3. The solution of the difference equation $y_{n+2} - 2y_{n+1} + 5y_n = 0$ is ____
4. The solution of $y_{n+1} - y_n = 2^n$ is ____.
5. The difference equation $y_{n+2} - 2y_{n+1} + (\cos\alpha)y_n = \cdots$ is its solution.
6. The difference equation corresponding to the family of curves $y = ax^2 + bx$ is ____.
7. The particular integral of the equation $(E-1)^2 y_n = 1$.
8. The solution of $u_{n+1} = u_{n-1}$ with $u_0 = 1$, $u_1 = 3$, is ____
9. The equation $\Delta^2 u_n + \frac{1}{2}\Delta^3 u_n = 0$ is of order ____.
10. The difference equation satisfied by $y = a + 2b/x$ is ____.
11. The order of the difference equation $y_{n+2} - 5y_{n+1} + y_n = 0$ is ____.
12. The solution of $y_{n+2} - 4y_{n+1} + 4y_n = 0$ is ____.
13. The particular integral of $y_{n+2} - 6y_{n+1} + 9y_n = 3^n$ is ____.
14. The difference equation generated by $u_n = (a + bn)3^n$ is ____.
15. Solution of $(y_{n+3} - 3y_{n+2} + 3y_{n+1} - y_n) = 3^n + n$ is $y_n = A + Bn + Cn^2 + \frac{1}{6}n^3 + \frac{3^n}{8}$. True or False.

CHAPTER 32

Numerical Solution of Ordinary Differential Equations

1. Introduction. 2. Picard's method. 3. Taylor's series method. 4. Euler's method. 5. Modified Euler's method. 6. Runge's method. 7. Runge-Kutta method. 8. Predictor-corrector methods. 9. Milne's method. 10. Adams-Bashforth method. 11. Simultaneous first order differential equations. 12. Second order differential equations. 13. Boundary value problems. 14. Finite-difference method. 15. Objective Type of Questions.

32.1. INTRODUCTION

The methods of solution so far presented are applicable to a limited class of differential equations. Frequently differential equations appearing in physical problems do not belong to any of these familiar types and are consigned to numerical methods. These methods are of even greater importance since we realize that computing machines are now available which reduce numerical work considerably.

A number of numerical methods are available for the solution of first order differential equations of the form

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \qquad \ldots(1)$$

These methods yield solutions either as a power series or as a form which the values of y can be found by direct substitution, or as a set of values of x and y. The methods of Picard and Taylor series belong to the former class of solutions whereas those of Euler, Runge-Kutta, Milne, Adams-Bashforth etc. belong to the latter class. In these latter methods, the values of y are calculated at short steps for equal intervals of x and are therefore termed as *step-by-step methods*.

Euler and Runge-Kutta methods are used for computing y over a limited range of the values of x while Milne and Adams-Bashforth methods may be applied for finding y over a wider range of x values. These latter methods require starting values which are found by Picard's or Taylor series or Runge-Kutta methods.

The initial condition in (1) is specified at the point x_0. Such problems in which all the initial conditions are given at the initial point only are called **initial value problems**. But there are problems involving second and higher order differential equations in which the conditions may be given at two or more points. These are known as **boundary value problems**. In this chapter, we shall first explain methods for solving initial value problems and then give a method of solving boundary value problems.

32.2. PICARD'S METHOD*

Consider the first order equation $dy/dx = f(x, y)$ ...(1)

*Called after the French mathematician Charles Emile Picard (1856–1941) who became professor in Paris in 1881 and is famous for his researches in the theory of functions.

It is important to find the particular solution of (1) which assumes the value y_0 when $x = x_0$. Geometrically this means that we select, from the infinite family of integral curves represented by the general solution, that one which passes through the point (x_0, y_0). Analytically, we get

$$\int_{y_0}^{y} dy = \int_{x_0}^{x} f(x, y)\, dx \quad \text{or} \quad y - y_0 = \int_{x_0}^{x} f(x, y)\, dx. \tag{2}$$

This is an integral equation equivalent to (1). For its solution the unknown y under the integral sign is replaced by some approximate value y_1 to the solution and we get $y = y_0 + \int_{x_0}^{x} f(x, y_1)\, dx$. Integrating the right hand side, we get

$$y = y_0 + \int_{x_0}^{x} f(x, y_1)\, dx.$$

For a second approximation y_2 we put $y = y_0 + \int_{x_0}^{x} f(x, y_2)\, dx$. And integrating the right hand side, we get

$$y = y_0 + \int_{x_0}^{x} f(x, y_2)\, dx.$$

Similarly, a third approximation to y is $y = y_0 + \int_{x_0}^{x} f(x, y)\, dx$.

Continuing this process, a sequence of functions of x, $y_1, y_2, y_3, y_4, \ldots$ is obtained each giving a better approximation to the desired solution than the preceding one.

Note. Picard's method is not recommended for practical use because the integrations involved in successive approximations may become impossible or very complicated. This method can be considered as a theoretical method for proving the existence of a solution of (1) under certain conditions.

Example 1. Using Picard's process of successive approximations, obtain a solution of $\frac{dy}{dx} = x + y$, $y = 1$ when $x = 0$.

Solution. Let $y = y_0 = 1$ when $x = x_0 = 0$. Then $y = y_0 + \int_{x_0}^{x} (x + y)\, dx$.

First approximation. Put $y = y_1$ in the integrand. Then $y = y_1$ gives

$$y_1 = 1 + \int_0^x (x + 1)\, dx = 1 + \frac{x^2}{2} + x = 1 + x + \frac{x^2}{2}.$$

Second approximation. Put $y = 1 + x + \frac{x^2}{2}$ in the integrand. Then $y = y_2$ gives

$$y_2 = 1 + \int_0^x \left(1 + x + \frac{x^2}{2}\right) dx = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

Third approximation. Put $y = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$ in the integrand. Then $y = y_3$ gives

$$y_3 = 1 + \int_0^x \left(1 + 2x + \frac{x^2}{2} + \frac{x^3}{6}\right) dx = 1 + x + x^2 + \frac{x^3}{6} + \frac{x^4}{24}.$$

Fourth approximation. Put $y = y_3$ in the integrand. Then $y = y_4$ gives

$$y_4 = 1 + \int_0^x \left(1 + 2x + x^2 + \frac{x^3}{6} + \frac{x^4}{24}\right) dx = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{24} + \frac{x^5}{120}.$$

Fifth approximation. Put $y = y_4$ in the integrand. Then $y = y_5$ gives

$$y_5 = 1 + \int_0^x \left(1 + 2x + x^2 + \frac{x^3}{3} + \frac{x^4}{24} + \frac{x^5}{120}\right) dx = 1 + x + x^2 + \frac{x^3}{3} + \frac{x^4}{12} + \frac{x^5}{60} + \frac{x^6}{720}. \tag{3}$$

It is clear that the equation

$$\frac{dy}{dx} = y + x \text{ is a differential equation of degree one.}$$

Now $\frac{dy}{dx} - y = x$, being a linear equation, its integrating factor $= e^{\int -1\, dx} = e^{-x}$. Hence the solution is

$$y e^{-x} = \int x e^{-x}\, dx + c = -x e^{-x} - \int -e^{-x}\, dx + c = -x e^{-x} - e^{-x} + c \qquad \text{(Integrating by parts)}$$

$$y = -x - 1 + c e^{x}.$$

Putting $x = 0$, $y = 1$, we get $1 = -1 + c$, i.e., $c = 2$.

Therefore the desired particular solution is $y = 2e^{x} - x - 1$. $\tag{4}$

Or using the series $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$,

we get $$y = 1 + x + x^2 + \frac{x^3}{2} + \frac{x^4}{12} + \frac{x^5}{60} + \frac{x^6}{360} + \cdots \quad (11)$$

On comparing (10) and (11), it is clear that (10) approximates to the exact particular solution (11) up to the term in x^3.

Now Try Exercise 33.2 page 1014 for a further problem on Euler's method

Example 33.2. Find the value of $y(0.1)$ given $x = 0.1$ by Euler's method, given that

$$\frac{dy}{dx} = \frac{y - x}{y + x}, \quad y(0) = 1 \qquad (C.E.T., 2002)$$

Solution. We have $y = 1 - \int_0^x \frac{y - x}{y + x}\,dx$

First approximation. Put $y = 1$ in the integrand, giving

$$y_1 = 1 + \int_0^x \frac{1 - x}{1 + x}\,dx = 1 + \int_0^x \left(-1 + \frac{2}{1 + x}\right) dx$$

$$= 1 + \left[-x + 2\log(1 + x)\right]_0^x = 1 - x + 2\log(1 + x) \quad (12)$$

Second approximation. Put $y = 1 - x + 2\log(1 + x)$ in the integrand, giving

$$y_2 = 1 + \int_0^x \frac{1 - x + 2\log(1 + x) - x}{1 - x + 2\log(1 + x) + x}\,dx = 1 + \int_0^x \left[1 - \frac{2x}{1 + 2\log(1 + x)}\right] dx$$

which is very difficult to integrate.

Hence we use the first approximation and taking $x = 0.1$ in (12) we obtain

$$y(0.1) = 1 - 0.1 + 2\log 1.1 = 0.9 + 0.1906$$

33.3 TAYLOR'S SERIES METHOD[1]

Consider the first order equation $\frac{dy}{dx} = f(x, y)$ (13)

Differentiating (13), we have

$$\frac{d^2y}{dx^2} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx} \qquad \text{i.e.} \quad y'' = f_x + f_y y' \quad (14)$$

Differentiating this successively, we can get y''', y'''' etc. Putting $x = x_0$ and $y = y_0$, the values of y'_0, y''_0, y'''_0, y''''_0 can be obtained. Hence the Taylor's series

$$y(x_0 + h) = y_0 + hy'_0 + \frac{h^2}{2!}y''_0 + \frac{h^3}{3!}y'''_0 + \frac{h^4}{4!}y''''_0 + \cdots \quad (15)$$

gives the values of y for every value of x for which (15) converges.

On finding the value y_1 for $x = x_1$, then $f(x, y)$, y' can be evaluated at $x = x_1$ by means of (13), (14) etc. They can be expanded about $x = x_1$. In this way, the solution can be extended beyond the range of convergence of series (15).

Example 33.3. Find by Taylor's series method the value of y at $x = 0.1$ and $x = 0.2$, given that y satisfies the differential equation $\frac{dy}{dx} = xy - 1$, $y(0) = 1$. (C.E.T., 2001, 2004, 2006)

Solution. Here $y'_0 = 1$, $y' = xy - 1$, $y''_0 = -1$

Differentiating successively and substituting, we get

$$y'' = 2xy + x^2y', \qquad y''_0 = 0$$
$$y''' = 2y + 4xy' + x^2y'', \qquad y'''_0 = 2$$
$$y'''' = 6y' + 6xy'' + x^2y''', \qquad y''''_0 = -6, \text{ etc.}$$

[1] See footnote p. 1005.

Putting these values in the Taylor's series,

$$y(x) = y_0 + xy'(0) + \frac{x^2}{2!}y''(0) + \frac{x^3}{3!}y'''(0) + \frac{x^4}{4!}y^{iv}(0) + \ldots$$

we have $y(x) = 1 + x - 3\cdot\frac{x^2}{2!}(0) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(-6) + \ldots = 1 + x + \frac{x^2}{2} - \frac{x^4}{4} + \ldots$

Hence $y(0.1) = 0.86033$ and $y(0.2) = 0.80227$.

Example 13.6. Employ Taylor's method to obtain approximate value of y for $x = 0.1$ for the differential equation $\frac{dy}{dx} = 2y + 3x^2$, $y(0) = 1$. Compare the numerical solution obtained with the exact solution. (U.P.T.U., 2003; U.P.T.U., 2005)

Solution. (a) We have $y' = 2y + 3x^2$ $\quad y'(0) = 2y(0) + 3x^2 = 2$.

Differentiating successively and substituting $x = 0$, $y = 1$, we get

$$y'' = 2y' + 6x, \quad y''(0) = 2y'(0) + 0 = 4$$
$$y''' = 2y'' + 6x, \quad y'''(0) = 2y''(0) + 0 = 8$$
$$y^{iv} = 2y''' + 6, \quad y^{iv}(0) = 2y'''(0) + 0 = 16 \text{ etc.}$$

Putting these values in the Taylor's series, we have

$$y(x) = y(0) + xy'(0) + \frac{x^2}{2!}y''(0) + \frac{x^3}{3!}y'''(0) + \frac{x^4}{4!}y^{iv}(0) + \ldots$$

$$= 1 + 2x + \frac{4}{2}x^2 + \frac{8}{6}x^3 + \frac{16}{24}x^4 + \ldots = 1 + 2x + \frac{2}{1}x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + \ldots$$

Hence $y(0.2) = 1.0221 + 1.8667(0.2)^2 + 0.5333(0.2)^3 + 0.6667(0.2)^4 + \ldots = 0.8011$ ...(1)

(b) Now $\frac{dy}{dx} - 2y = 3x^2$ is a Leibnitz's linear in x.

Its I.F. being e^{-2x}, the solution is

$$ye^{-2x} = \int 3x^2 \cdot e^{-2x}\,dx = x^2\cdot\frac{e^{-2x}}{-2} - 2x\cdot\frac{e^{-2x}}{4} + 2\cdot\frac{e^{-2x}}{-8} + c \quad \text{or} \quad y = -\frac{3}{2}x^2 - \frac{3}{2}x - \frac{3}{4} + ce^{2x}$$

Since $y = 1$ when $x = 0$, $\therefore \quad c = \frac{7}{4}$.

Thus the exact solution is $y = \frac{7}{4}e^{2x} - \frac{3}{2}x^2 - \frac{3}{2}x - \frac{3}{4}$.

When $x = 0.2$, $y = \frac{7}{4}e^{0.4} - \frac{3}{2}(0.2)^2 - \frac{3}{2}(0.2) - \frac{3}{4} = 0.8013$ ...(2)

Comparing (1) and (2), the Taylor's method gives approximate to the exact value upto 3 decimal places.

Example 13.7. Solve by Taylor's series method the equation $\frac{dy}{dx} = \log(xy)$ for $y(1.1)$ and $y(1.2)$, given $y(1) = 1$. (Rajasthan, 2009)

Solution. We have $y' = \log x + \log y$ $\quad y'(1) = \log 1 = 0$

Differentiating w.r.t. x and substituting $x = 1$, $y = 1$, we get

$$y'' = \frac{1}{x} + \frac{1}{y}y' \quad y''(1) = 1 + \frac{1}{1}\log 1 = 1$$

$$y''' = -\frac{1}{x^2} + \frac{1}{y}y'' + y''\left(-\frac{1}{y^2}\right)y' \quad y'''(1) = -1 + \frac{1}{1}\left(1 - \frac{1}{1}\log 1\right) - \frac{1}{1}(\log 1)^2$$

Substituting these values in the Taylor's series about $x = 1$, we have

$$y(x) = y(1) + (x-1)y'(1) + \frac{(x-1)^2}{2!}y''(1) + \frac{(x-1)^3}{3!}y'''(1) + \ldots$$

$$= 1 + (x-1)\log 1 + \frac{1}{2}(x-1)^2\left(1 + \frac{1}{1}\log 1\right) + \frac{1}{6}(x-1)^3\left[-\frac{1}{2} + \frac{1}{1}\log 1 - \frac{1}{1}(\log 1)^2\right]$$

$$\therefore \quad y(1.1) = 1 + 0.1\log 1 + \frac{(0.1)^2}{2}\left(1 + \frac{1}{1}\log 1\right) + \frac{(0.1)^3}{6}\left[-\frac{1}{2} + \frac{1}{1}\log 1 - \frac{1}{1}(\log 1)^2\right] = 1.005.$$

$$y(1.2) = 2 + (0.2)\log 2 + \frac{(0.2)^2}{2}\left[3 + \frac{1}{2}\log 2\right] + \frac{(0.2)^3}{6}\left[-\frac{1}{2} + \frac{1}{2}\log 2 - \frac{1}{3}(\log 2)^2\right] = 2.091.$$

EXERCISE 439

1. Using Picard's method, solve $\frac{dy}{dx} = x - y$ given $y(0) = 1$ up to the third approximation. (Hint Ex1, 1997)
2. Using Picard's method, obtain, correct to four places of decimals, solutions of the differential equation $\frac{dy}{dx} = x^2 + y^2$, given $y = 0.1$ when $x = 0$, up to the third approximation. (A.M.I.E., 1997)
3. Obtain Picard's second approximate solution of the initial value problem $y' = x + \sin(y - x)$, $y(0) = 0$. (Nottingham, 1997)
4. Find an approximate value of y when $x = 0.1$, if $\frac{dy}{dx} = x^2 + y$ and $y = 1$ when $x = 0$, using
   (a) Picard's method, (b) Taylor's series. (A.M.I., 1988; Madras, 1997)
5. Solve $y' = x + y$, given $y(1) = 0$. Find $y(1.1)$ and $y(1.2)$ by Taylor's method. Compare the result with the exact solution. (A.M.I.E., 2000; Anna, 2002)
6. Evaluate $y(0.1)$ correct to six places of decimals by Taylor's series method, if $y(x)$ satisfies
   $$y' = y(x + 1), \quad y(0) = 1.$$
7. Solve $y' = 3x + y^2$, $y(1) = 0$. Find $y(1.1)$ and $y(1.2)$ by Taylor's method. Compare the result with the exact solution. (Delhi Ex, 2001)
8. Using Taylor series method, find $y(0.1)$ correct to 4 decimal places given that
   $$\frac{dy}{dx} = x^2 - y^2, \quad y(0) = 1.$$

22.4 EULER'S METHOD*

Consider the equation $\frac{dy}{dx} = f(x, y)$ ... (1)

given that $y(x_0) = y_0$. Its curve of solution through $P(x_0, y_0)$ is shown dotted in Fig. 22.1. Now we have to find the ordinate of any other point Q on this curve.

Let us divide LM into n sub-intervals each of width h at $L_1, L_2, \ldots$ so that h is quite small. In the interval LL_1, we approximate the curve by the tangent at P. If the ordinate through L_1 meets this tangent in $P_1(x_0 + h, y_1)$, then

$$y_1 = L_1P_1 = LP + MP_1$$

$$= y_0 + PR_1 \tan\theta = y_0 + h\left[\frac{dy}{dx}\right]_P$$

$$= y_0 + hf(x_0, y_0)$$

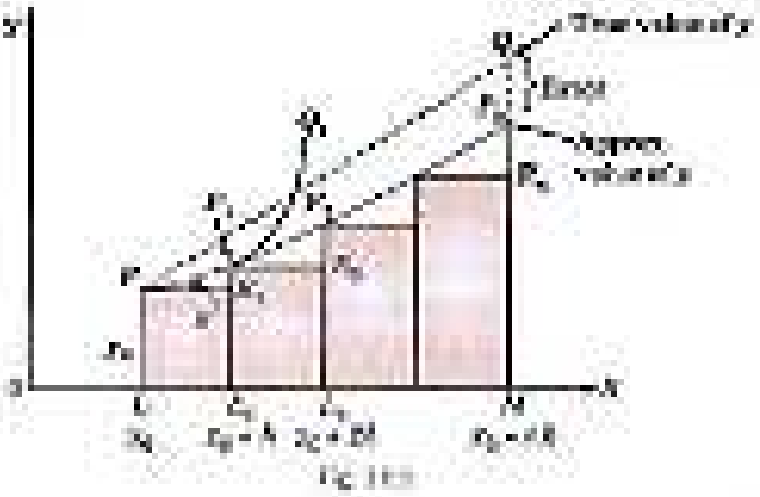

Fig. 22.1

Let P_1Q_1 be the curve of solution through P_1 and let its tangent at P_1 meet the ordinate through L_2 in $P_2(x_0 + 2h, y_2)$. Then

$$y_2 = y_1 + h\,f(x_0 + h, y_1) \quad \ldots (2)$$

Repeating this process n times, we finally reach an approximation MP_n of MQ, i.e. x is $x_0 + nh$ and

$$y_n = y_{n-1} + h\,f(x_0 + (n-1)h, y_{n-1}).$$

This is Euler's method of finding an approximate solution of (1).

Thus, in Euler's method, we approximate the curve of solution by the tangent in each interval, i.e. [illegible] Unless h is small, the error is bound to be quite significant. This sequence of lines may [illegible] from the curve of solution. Hence there is a modification of this method which is given in the next section.

Example 22.5. Using Euler's method, find an approximate value of y corresponding to $x = 1.1$ given that $y = 1$ when $x = 1$ and $y = f(x, y) = x + y$. (Manchester, 2002; Kerala, 2003)

*See Footnote p. 503.

Solution. We take $h = 0.1$ and $h = 0.2$ which is sufficiently small. The various calculations are arranged as follows:

x	y	$h \times$ (slope at x)	y + (increment in y)
0.0	1.00	0.10	1.00 + 0.10 = 1.10
0.1	1.10	0.11	1.10 + 0.11 = 1.21
0.2	1.21	0.12	1.21 + 0.12 = 1.33
0.3	1.33	0.13	1.33 + 0.13 = 1.46
0.4	1.46	0.15	1.46 + 0.15 = 1.61
0.5	1.61	0.17	1.61 + 0.17 = 1.78
0.6	1.78	0.19	1.78 + 0.19 = 1.97
0.7	1.97	0.21	1.97 + 0.21 = 2.18
0.8	2.18	0.24	2.18 + 0.24 = 2.42
0.9	2.42	0.27	2.42 + 0.27 = 2.69
1.0	2.69		

Thus the required approximate value of $y = 2.69$.

Note. In example 11.1, the interval is divided into ten equal parts i.e., $h = 0.1$ whereas by Euler's method $y = 2.71$ and by Picard's method $y = 2.694$. In the above solution, had we chosen $h = 0.05$, the accuracy would have been considerably increased but at the expense of double the labour of computation. Euler's method is useful only when h is taken to be considerably small as we shall see.

Example 11.3. Given $\frac{dy}{dx} = \frac{y - x}{y + x}$ with the initial condition $y = 1$ at $x = 0$, find y for $x = 0.1$ by Euler's method. (U.P.T.U., 2000)

Solution. We divide the interval [0, 0.1] into five steps i.e., we take $n = 5$ and $h = 0.02$. The various calculations are arranged as follows:

x	y	$h \times$ (slope at x) = (increment in y)	y + (increment in y)
0.00	1.0000	0.0200	1.0000 + 0.02 (1.0000) = 1.0200
0.02	1.0200	0.0192	1.0200 + 0.02 (0.9615) = 1.0392
0.04	1.0392	0.0185	1.0392 + 0.02 (0.9259) = 1.0577
0.06	1.0577	0.0179	1.0577 + 0.02 (0.8929) = 1.0756
0.08	1.0756	0.0172	1.0756 + 0.02 (0.8621) = 1.0928
0.10	1.0928		

Hence the required approximate value of $y = 1.0928$.

11.13 MODIFIED EULER'S METHOD

In the Euler's method, the curve of solution in the interval $[x_0, x_1]$ is approximated by the tangent at P_0 [Fig. 11.3] to each finite P_0 we have

$$y_1 = y_0 + hf(x_0, y_0) \quad \ldots(1)$$

Here the slope of the curve of solution through P_0 i.e., $dy/dx]_{P_0} = f(x_0, y_0)$ is computed and the tangent at P_0 to P_0Q_0 is drawn meeting the ordinate through A_1 in $P'(x_0 + h, y_1)$.

Now we find a better approximation $y_1^{(1)} = f(x_1, y_1)$ by taking the slope of the curve as the mean of the slopes of the tangents at P and P_0, i.e.,

$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_0 + h, y_1)] \quad \ldots(2)$$

As the slope of the tangent at P_0 is not known, we take y_1 as found in (1) by Euler's method and insert it on R.H.S. of (2) to obtain the first modified value $y_1^{(1)}$. The equation (1) is therefore, called the *predictor* while (2) serves as the *corrector* for y_1.

Again the corrector is applied and we find a still better value $y_1^{(2)}$ corresponding to x_1, as

$$y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

We repeat this step till two successive values of y agree. This is then taken as the starting point for the next interval x_1x_2.

Once y_1 is obtained to desired degree of accuracy, y corresponding to x_2 is found from the predictor

$$y_2 = y_1 + h[f(x_1, y_1)]$$

and a better approximation $y_2^{(1)}$ is obtained from the corrector

$$y_2^{(1)} = y_1 + \frac{h}{2}[f(x_1, y_1) + f(x_2, y_2)]$$

We repeat this step until y_2 becomes stationary. Then we proceed to calculate y_3 as above and so on.

This is the modified Euler's method which is a predictor-corrector method.

Example 2020. Using modified Euler's method, find an approximate value of y when $x = 0.2$ given that $y(0) = 1$ and $y = f(x, y)$ when $x = 0$. [Anna, (2002); Nagpur, (2003-S); Delhi, (2003)]

Solution. Taking $h = 0.1$, the various calculations are arranged as follows:

n	$x_n = x_0 + nh$	$y_n^{(0)} = y_{n-1} + hf(x_{n-1}, y_{n-1})$	$y_n^{(1)} = y_{n-1} + \frac{h}{2}[f(x_{n-1}, y_{n-1}) + f(x_n, y_n^{(0)})]$	$y_n^{(2)} = y_{n-1} + \frac{h}{2}[f(x_{n-1}, y_{n-1}) + f(x_n, y_n^{(1)})]$
0	0	1		
1	0.1	$1 + 0.1(0 + 1) = 1.1$	$1 + \frac{0.1}{2}[(0 + 1) + (0.1 + 1.1)] = 1.105$	$1 + \frac{0.1}{2}[(0 + 1) + (0.1 + 1.105)] = 1.10525$
2	0.2	$1.105 + 0.1(0.1 + 1.105) = 1.2155$	$1.105 + \frac{0.1}{2}[(0.1 + 1.105) + (0.2 + 1.2155)] = 1.22025$	$1.105 + \frac{0.1}{2}[(0.1 + 1.105) + (0.2 + 1.22025)] = 1.220513$
0	0	1		
1	0.1	$1 + 0.1(0 + 1) = 1.1$	$1 + \frac{0.1}{2}[(0 + 1) + (0.1 + 1.1)] = 1.105$	$1 + \frac{0.1}{2}[(0 + 1) + (0.1 + 1.105)] = 1.10525$
2	0.2	$1.105 + 0.1(0.1 + 1.105) = 1.2155$	$1.105 + \frac{0.1}{2}[(0.1 + 1.105) + (0.2 + 1.2155)] = 1.22025$	$1.105 + \frac{0.1}{2}[(0.1 + 1.105) + (0.2 + 1.22025)] = 1.220513$
0	0	1		
1	0.1	$1 + 0.1(0 + 1) = 1.1$	$1 + \frac{0.1}{2}[(0 + 1) + (0.1 + 1.1)] = 1.105$	$1 + \frac{0.1}{2}[(0 + 1) + (0.1 + 1.105)] = 1.10525$
2	0.2	$1.105 + 0.1(0.1 + 1.105) = 1.2155$	$1.105 + \frac{0.1}{2}[(0.1 + 1.105) + (0.2 + 1.2155)] = 1.22025$	$1.105 + \frac{0.1}{2}[(0.1 + 1.105) + (0.2 + 1.22025)] = 1.220513$

Hence $y(0.2) = 1.2051$ approximately.

Note. In example 2020, the approximate value of y for $x = 0.2$ is obtained as 1.2051 by modified Euler's method and the corresponding value is 1.2051 which is nearer to the true value 1.2052, obtained from the exact solution $y = 2e^x - x - 1$ for $x = 0.2$.

Example 2021. Using modified Euler's method, find $y(0.2)$ and $y(0.4)$ given $y' = xy + y^2$, $y(0) = 1$. [U.P.T.U., 2004]

Solution. We have $f(x, y) = x^2 + y$, $h = 0.2$, $x_0 = 0$, $y_0 = 1$.
The various calculations are arranged as follows :
Table Value of y at $x = 0.2$:

n	x_n	y_n	Mid-values	$f(x_n, y_n)$
0	0	1	—	$0^2 + 1 = 1$
1	0.2	$1 + 0.2 = 1.2$	$\frac{1}{2}(1 + 1.4040) = 1.2020$	$0.2^2 + 1.2020 = 1.2420$
2	0.4	$1.2020 + 0.2484 = 1.4504$	$\frac{1}{2}(1.2020 + 1.4504) = 1.3262$	$0.4^2 + 1.3262 = 1.4862$
3	0.6	$1.3262 + 0.2972 = 1.6234$	$\frac{1}{2}(1.3262 + 1.6234) = 1.4748$	$0.6^2 + 1.4748 = 1.8348$
4	0.8	$1.4748 + 0.3667 = 1.8415$	$\frac{1}{2}(1.4748 + 1.8415) = 1.6581$	$0.8^2 + 1.6581 = 2.2981$

Since the last two values of y are equal, we take $y(0.8) = 1.8415$.
Table Value of y at $x = 1.0$:

n	x_n	y_n	Mid-values	$f(x_n, y_n)$
0	0.8	$1.8415 + 0.4593 = 2.3008$	—	$0.8^2 + 2.3008 = 2.9408$
1	1.0	$2.3008 + 0.5888 = 2.8896$	$\frac{1}{2}(2.3008 + 2.8896) = 2.5952$	$1.0^2 + 2.5952 = 3.5952$
2	1.2	$2.5952 + 0.7190 = 3.3142$	$\frac{1}{2}(2.5952 + 3.3142) = 2.9547$	$1.2^2 + 2.9547 = 4.3947$
3	1.4	$2.9547 + 0.8789 = 3.8336$	$\frac{1}{2}(2.9547 + 3.8336) = 3.3942$	$1.4^2 + 3.3942 = 5.3742$
4	1.6	$3.3942 + 1.0748 = 4.4690$	$\frac{1}{2}(3.3942 + 4.4690) = 3.9316$	$1.6^2 + 3.9316 = 6.5916$
5	1.8	$3.9316 + 1.3183 = 5.2499$	$\frac{1}{2}(3.9316 + 5.2499) = 4.5908$	$1.8^2 + 4.5908 = 7.9108$
6	2.0	$4.5908 + 1.5822 = 6.1730$	$\frac{1}{2}(4.5908 + 6.1730) = 5.3819$	$2.0^2 + 5.3819 = 9.3819$

Since the last two values of y are equal, we take $y(2.0) = 6.1730$.
Hence $y(0.2) = 1.2420$ and $y(1.0) = 6.1730$ approximately.

Example 9.8.10. Solve the following by Runge-Kutta method of order four :

$$\frac{dy}{dx} = \log_{10}(x + y), \quad y(0) = 1$$

Solution. The various calculations are arranged as follows :

n	$\log_{10}(x_n + y_n)$	Mid-values	$h f(x_n, y_n)$
0	$\log_{10} 1 = 0$	—	$0.2 \times 0 = 0.0000$
1	$\log_{10} 1.2 = 0.0792$	$\frac{1}{2}(0.0792 + 0.1041) = 0.0917$	$0.2 \times 0.0917 = 0.0183$
2	$\log_{10} 1.2183 = 0.0858$	$\frac{1}{2}(0.0858 + 0.1106) = 0.0982$	$0.2 \times 0.0982 = 0.0196$
3	0.1106	—	$0.2 \times 0.1106 = 0.0221$
4	$\log_{10} 1.2379 = 0.0928$	$\frac{1}{2}(0.0928 + 0.1176) = 0.1052$	$0.2 \times 0.1052 = 0.0210$
5	$\log_{10} 1.2610 = 0.1006$	$\frac{1}{2}(0.1006 + 0.1254) = 0.1130$	$0.2 \times 0.1130 = 0.0226$

x_n	Exact $y = y(x_n)$	Method	Error $\varepsilon_n = y_n - y(x_n)$
0.4	0.6321	—	[illegible]
0.6	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
0.8	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
0.8	0.8018	—	[illegible]
0.9	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
1.0	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
1.0	0.8647	—	[illegible]
1.1	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
1.2	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
1.2	0.8775	—	[illegible]
1.3	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
1.4	[illegible]	$\frac{1}{2}$[illegible]	[illegible]

Hence $y(1.2) \approx 0.8331$ and $y(1.4) \approx 0.8821$ approximately.

Example 22.13. Using Euler's modified method, obtain a solution of the equation $dy/dx = x + |y|$, $y(0) = 0$ for the range $0 \le x \le 0.6$ in steps of 0.2. (U.P.T.U., 2003)

Solution. The various calculations are arranged as follows :

n	$x_n = x_0 + nh$	Mean slope	$y_{n+1} = y_n +$ (mean slope) $\times h$
0.1	[illegible]	—	[illegible]
0.2	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
0.3	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
0.4	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
0.5	[illegible]	—	[illegible]
0.6	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
0.7	[illegible]	$\frac{1}{2}$[illegible]	[illegible]
0.8	[illegible]	$\frac{1}{2}$[illegible]	[illegible]

x	$y = \int_0^x \sqrt{1+x^3}\,dx$	Mean slope	$y(0) + \frac{1}{2}$(mean slope) $\times$ step
0.5	0.5000	—	0.5000 + 0.5 (1.0353) = 1.0177
0.6	$0.5 + \frac{1}{2}(0.5)\sqrt{1+(0.5)^3} = 1.0177$	$\frac{1}{2}(1.0353 + 1.0911) = 1.0632$	1.0177 + 0.5 (1.0632) = 1.0893
0.7	$0.5 + \frac{1}{2}(0.5)\sqrt{1+(0.6)^3} = 1.0893$	$\frac{1}{2}(1.0911 + 1.1510) = 1.1211$	1.0893 + 0.5 (1.1211) = 1.1499
0.8	$0.5 + \frac{1}{2}(0.5)\sqrt{1+(0.7)^3} = 1.1499$	$\frac{1}{2}(1.1510 + 1.2247) = 1.1879$	1.1499 + 0.5 (1.1879) = 1.1939

Hence $y(0.8) = 1.1939$ approximately.

PROBLEMS 9.3

1. Apply Euler's method to solve $\frac{dy}{dx} = x + y$, $y(0) = 0$, choosing the step length $h = 0.2$. Carry out 5 steps. (Kottayam, 2002)
2. Using an appropriate Euler's method, solve for y at $x = 0.1$ from $\frac{dy}{dx} = x^2 + y$, $y(0) = 1$, taking step size $h = 0.05$.
3. Using Euler's method, find the approximate value of y when $x = 0.1$, if $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$ in five steps. ($h = 0.2$) (Mumbai, 2004)
4. Solve $y' = 1 - x$, $y(0) = 0$ by modified Euler's method and obtain y at $x = 0.1, 0.2, 0.3$. (Anna, 2003)
5. Given $y' = x + y$, $y(0) = 1$. Compute $y(0.1)$ and $y(0.2)$ correct to four decimal places using Euler's modified method. (AMIETE, 2003)
6. Given that $\frac{dy}{dx} = x^2 + y$ and $y(0) = 1$. Find an approximate value of $y(0.1)$ correct to 4 decimal places by modified Euler's method. (AMIETE, 2004)
7. Given $\frac{dy}{dx} = \frac{y - x}{y + x}$ with boundary conditions $y = 1$ when $x = 0$. Find approximately y for $x = 0.1$ by Euler's modified method (4 steps). (AMIETE, 2005)
8. Given that $f(x) = 1 + \sqrt{(xy)}$ and $x = 1$ when $y = 1$. Find approximate value of y at $x = 2$ in steps of 0.2, using Euler's modified method. (Anna, 2005)

9.14 RUNGE'S METHOD*

Consider the differential equation,

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \qquad \ldots(1)$$

Clearly the slope of the curve through $P(x_0, y_0)$ is $f(x_0, y_0)$ (Fig. 9.2).

Integrate both sides of (1) from (x_0, y_0) to $(x_0 + h, y_0 + k)$, we have

$$\int_{y_0}^{y_0+k} dy = \int_{x_0}^{x_0+h} f(x, y)\,dx \qquad \ldots(2)$$

To evaluate the integral on the right, we take N as the mid point of LM and find the values of $f(x, y)$ at the points $x_0, x_0 + h/2, x_0 + h$. For this purpose, we first determine the values of y at these points.

Let the ordinate through N cut the curve PQ in S and the tangent PT in S_1. The value of y_1 is given by the point S_1.

$$\therefore \quad y_1 = RS + LP = RS_1 + LP = y_0 + PU \tan\theta$$

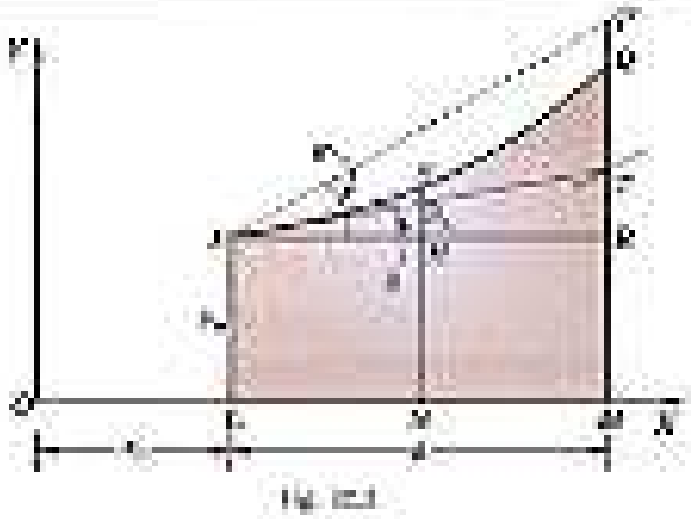

Fig. 9.2

*Called after the German mathematician Carl Runge (1856–1927) who was professor at Gottingen.

$$= y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1)] = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_0 + h, y_0 + hf(x_0, y_0))] \quad \ldots(2)$$

Also $y_0 = MP = AP - AM = y_0 + PN \tan\theta = y_0 + hf(x_0, y_0)$

Now the value of y_0 at $x_0 + h$ is given by the point T where the line through P drawn with slope of $\frac{1}{2}[f(x_0, y_0) + f(x_1, y_1)]$ meets BQ.

$\therefore$ Slope of $PT = \tan\theta = \frac{1}{2}[f(x_0, y_0) + f(x_0 + h, y_0 + hf(x_0, y_0))]$

$\therefore$ $y_0 = RS + ST = y_0 + PT\cos\theta = y_0 + h[f(x_0, y_0) + f(x_0 + h, y_0 + hf(x_0, y_0))]$ $\ldots(3)$

Thus the value of $f(x, y)$ at $P = f(x_0, y_0)$,

the value of $f(x, y)$ at $R = f(x_0 + h/2, y_0)$

and the value of $f(x, y)$ at $Q = f(x_0 + h, y_0)$

where y_0 and y_0 are given by (2) and (3).

Hence from (2), we obtain

$$k = \int_{x_0}^{x_0 + h} f(x, y)\,dx = \frac{h}{6}[y_0 + 4y_0 + y_0] \qquad \text{(By Simpson's rule (p. 1306))}$$

$$= \frac{h}{6}[f(x_0, y_0) + 4f(x_0 + h/2, y_0) + f(x_0 + h, y_0)] \quad \ldots(4)$$

which gives a sufficiently accurate value of k and also of $y = y_0 + k$.

The repeated application of (4) gives the values of y for equispaced points.

Working rule to solve $\frac{dy}{dx} = f(x, y)$ by *Runge's method*:

Calculate successively

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \tfrac{1}{2}h, y_0 + \tfrac{1}{2}k_1\right)$$

$$k_3 = hf(x_0 + h, y_0 + k_2)$$

and $$k_4 = hf(x_0 + h, y_0 + k_3)$$

Finally compute $$k = \frac{1}{6}(k_1 + 4k_2 + k_3).$$

(Note that k is the weighted mean of k_1, k_2 and k_3)

Example 12.12. *Apply Runge's method to find an approximate value of y for $x = 0.2$, given that $\frac{dy}{dx} = x + y$ and $y = 1$ when $x = 0$.*

Solution. Here we have $x_0 = 0, y_0 = 1, h = 0.2, f(x, y) = x + y$

$\therefore$ $k_1 = hf(x_0, y_0) = 0.2(1)$ $= 0.200$

$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = 0.2[0.1, 1.1]$ $= 0.210$

$k_3 = hf(x_0 + h, y_0 + k_2) = 0.2[0.2, 1.2]$ $= 0.220$

and $k_4 = hf(x_0 + h, y_0 + k_3) = 0.2[0.2, 1.220]$ $= 0.244$

$\therefore$ $k = \frac{1}{6}(k_1 + 4k_2 + k_3)$

$= \frac{1}{6}[0.200 + 0.840 + 0.244]$ $= 0.2135$

Hence the required approximate value of y is 1.2135.

12.13. RUNGE-KUTTA METHOD*

The Taylor's series method of solving differential equations numerically is restricted by the labour involved in finding the higher order derivatives. However there is a class of methods known as Runge-Kutta methods which do not require the calculation of higher order derivatives. These methods agree with Taylor's series solution upto the terms in h^r, where r differs from method to method and is called the order of the method. Euler's method, Modified Euler's method and Runge's method are the Runge-Kutta methods of first, second and third order respectively.

*See footnote p. 1317. Named after Wilhelm Kutta (1867 – 1944).

The above order Runge-Kutta method is most commonly used and is often referred to as 'Runge-Kutta method' only.

Working rule for finding the increment k of y corresponding to an increment h of x by Runge-Kutta method (fourth order).

Given the differential equation $\frac{dy}{dx} = f(x, y)$ and the initial conditions $y(x_0) = y_0$, to find the value of y corresponding to $x = x_0 + h$.

Calculate successively

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right)$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right)$$

and $$k_4 = hf(x_0 + h, y_0 + k_3)$$

Finally compute $$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

which gives the required approximate value $y_1 = y_0 + k$.

Note that k is the weighted mean of k_1, k_2, k_3 and k_4.

Note. One of the advantages of this method is that it requires only the knowledge of $f(x, y)$ but not the higher derivatives of y as in Taylor's and Picard's methods.

Example 22.13. Apply Runge-Kutta fourth order method to find an approximate value of y when $x = 0.2$, given that $y(0) = 1$ and $y' = f(x, y) = x + y$. *(U.P.T.U. 2006; P.T.U. 2009; A.M.I.E. 2000)*

Solution. Here $x_0 = 0, y_0 = 1, h = 0.2, f(x_0, y_0) = 1$

$$k_1 = hf(x_0, y_0) = 0.2 \times 1 = 0.2000$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = 0.2 \times f(0.1, 1.1) = 0.2100$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) = 0.2 \times f(0.1, 1.105) = 0.2105$$

and $$k_4 = hf(x_0 + h, y_0 + k_3) = 0.2 \times f(0.2, 1.2105) = 0.2221$$

Also $$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}[0.2000 + 0.4200 + 0.4210 + 0.2221] = \frac{1}{6} \times 1.2631 = 0.2105$$

Hence the required approximate value of y is 1.2105.

Example 22.14. Using Runge-Kutta method of fourth order, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$, $y(0) = 1$ at $x = 0.2$, $x = 0.4$. *(U.P.T.U. 2003; A.M.I.E. 2000; U.P.T.U. 2006)*

Solution. We have $f(x, y) = \frac{y^2 - x^2}{y^2 + x^2}$

To find $y(0.2)$:

Here $x_0 = 0, y_0 = 1, h = 0.2$

$$k_1 = hf(x_0, y_0) = 0.2f(0, 1) = 0.2000$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = 0.2f(0.1, 1.1) = 0.1983$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) = 0.2f(0.1, 1.09835) = 0.1987$$

$$k_4 = hf(x_0 + h, y_0 + k_3) = 0.2f(0.2, 1.1987) = 0.1951$$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \frac{1}{6}[0.2000 + 2(0.1983) + 2(0.1987) + 0.1951] = 0.1970$$

Hence $$y(0.2) = y_0 + k = 1.197.$$

To find $y(0.4)$:

Here $x_0 = 0.2, y_0 = 1.196, h = 0.2$

$k_1 = hf(x_0, y_0) = 0.2f(0.2, 1.196) = 0.2388$

$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{1}{2}k_1\right) = 0.2f(0.3, 1.2959) = 0.2795$

$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{1}{2}k_2\right) = 0.2f(0.3, 1.3377) = 0.2863$

$k_4 = hf(x_0 + h, y_0 + k_3) = 0.2f(0.4, 1.3763) = 0.2968$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.2388 + 2(0.2795) + 2(0.2863) + 0.2968) = 0.2782$$

Hence $y(0.4) = y_0 + k = 1.196 + 0.2782 = 1.4742$.

Example 22.13. Apply Runge-Kutta method to find an approximate value of y for $x = 0.2$ (a) when $y(0) = 1$, $\frac{dy}{dx} = x + y^2$, given that $y = 1$, when $x = 0$. (V.T.U., 2009 ; Rohtak, 2007 ; Kolhan, 2008)

Solution. Here we take $h = 0.1$ and carry out the calculations in two steps.

Step I. $x_0 = 0, y_0 = 1, h = 0.1$

$k_1 = hf(x_0, y_0) = 0.1f(0, 1) = 0.1000$

$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) = 0.1f(0.05, 1.05) = 0.1152$

$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) = 0.1f(0.05, 1.0576) = 0.1166$

$k_4 = hf(x_0 + h, y_0 + k_3) = 0.1f(0.1, 1.1166) = 0.1347$

$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$= \frac{1}{6}(0.1000 + 0.2304 + 0.2332 + 0.1347) = 0.1164$$

Hence $y(0.1) = y_0 + k = 1.1164$.

Step II. $x_1 = x_0 + h = 0.1, y_1 = 1.1164, h = 0.1$

$k_1 = hf(x_1, y_1) = 0.1f(0.1, 1.1164) = 0.1347$

$k_2 = hf\left(x_1 + \frac{1}{2}h, y_1 + \frac{1}{2}k_1\right) = 0.1f(0.15, 1.1838) = 0.1551$

$k_3 = hf\left(x_1 + \frac{1}{2}h, y_1 + \frac{1}{2}k_2\right) = 0.1f(0.15, 1.1940) = 0.1575$

$k_4 = hf(x_1 + h, y_1 + k_3) = 0.1f(0.2, 1.2739) = 0.1803$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.1571$

Hence $y(0.2) = y_1 + k = 1.2735$.

Example 22.14. Using Runge-Kutta method of fourth order, solve for y at $x = 0.2, 0.4$ given that $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$, $y(0) = 1, y(0) = 2$. (Bhopal, 2008)

Solution. We have $f(x, y) = \frac{2xy + x^2}{x^2 + y^2}$

To find $y(0.2)$:

Here $x_0 = 1, y_0 = 2, h = 0.2$

$k_1 = hf(x_0, y_0) = 0.2f\left(\frac{0 + 0}{0 + 0}\right) = 0.1667$

$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.2\left\{\frac{2(1.1 + 0.1667) + (1.1)^2}{(1.1)^2 + (2 + 0.1667)^2}\right\} = 0.1692$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) = 0.2\left[\frac{2(1 + 0.1)(0 + 0.05) - 0^2}{(1 + 0.1)^2 + (0 + 0.05)^2}\right] = 0.1089$$

$$k_3 = hf(x_0 + h/2, y_0 + k_2/2) = 0.2\left[\frac{2(1.1)(0.1089) - 0^2}{(1.1)^2 + (1.1)^2}\right] = 0.1048$$

and
$$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \frac{1}{6}[0.1400 + 0.2178 + 0.2096 + 0.1916]$$
$$= 0.1698$$

Hence $y(1.2) = y_0 + k = 0 + 0.1698 = 0.1698$.

To find $y(1.4)$:

Here $x_1 = 1.2, y_1 = 0.1698, h = 0.2$

$$k_1 = hf(x_1, y_1) = 0.2f(1.2, 0) = 0.1916$$
$$k_2 = hf(x_1 + h/2, y_1 + k_1/2) = 0.2f(1.3, 0.2707) = 0.1969$$
$$k_3 = hf(x_1 + h/2, y_1 + k_2/2) = 0.2f(1.3, 0.2683) = 0.1961$$
$$k_4 = hf(x_1 + h, y_1 + k_3) = 0.2f(1.5, 0.2359) = 0.1888$$

$$\therefore \quad k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \frac{1}{6}[0.1916 + 0.3938 + 0.3922 + 0.1888] = 0.1930$$

Hence $y(1.4) = y_1 + k = 0.1698 + 0.1930 = 0.3628$.

EXERCISE 10.3

1. Use Runge's method to approximate y when $x = 0.1$, given that $x = 0$ when $y = 1$ and $dy/dx = 3x + y^2$.
2. Using Runge-Kutta method of order 4, find $y(0.2)$ given that $dy/dx = 3x + \frac{1}{2}y$, $y(0) = 2$, taking $h = 0.1$.
(U.P.T.U., 2004)
3. Using Runge-Kutta method of order 4, compute $y(0.1)$ and $y(0.2)$ from $10\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$, taking $h = 0.1$.
(Kurukshetra, 2005 ; Gujarat, 2006)
4. Use Runge-Kutta method to find y when $x = 1.2$ in steps of 0.1, given that
$2y\,dx = x^2 + y^2\,dx$ and $y(1) = 1.5$. (Mumbai, 2006)
5. Find $y(0.1)$ and $y(0.2)$ using Runge-Kutta fourth order formula, given that $y' = x^2 - y$ and $y(0) = 1$. (U.P.T.U., 2006)
6. Using 4th order Runge-Kutta method, solve the following equation, taking each step of $h = 0.1$ given $y(0) = 1$, $y'(x) = 0.5y + xy$. Calculate y for $x = 0.1$ and 0.2. (Pune, 2007)
7. Use fourth order Runge-Kutta method to find y at $x = 0.1$, given that $\frac{dy}{dx} = 3x^2 + 2y$, $y(0) = 0$ and $h = 0.1$.
(U.P.T.U., 2006)
8. Find by Runge-Kutta method an approximate value of y for $x = 0.8$, given that $y = 0.41$ when $x = 0.4$ and $dy/dx = \sqrt{x + y}$.
(U.P.T.U., 2005 S)
9. Using Runge-Kutta method of order 4, find $y(0.2)$ for the equation $\frac{dy}{dx} = \frac{y - x}{y + x}$, $y(0) = 1$. Take $h = 0.2$.
(U.P.T.U., 2007 S)
10. Given that $dy/dx = y^2 - 2x/y^2 + x$ and $y = 1$ at $x = 0$, find y at $x = 0.1, 0.2, 0.3$ & 0.4 and 0.5. (Delhi, 2005)

10.13 PREDICTOR-CORRECTOR METHODS

If y_{-3}, y_{-2}, y_{-1} and y_0 be four successive mesh points, we have $x_0 = x_{-1} + h$. In the Euler's method (§ 10.4), we have

$$y_1 = y_{-1} + 2f(x_0) = 2hf(x_0, y_{-1}) : i = 1, 2, 3, \ldots \quad ...(1)$$

The modified Euler's method (1.11.2) gives

$$y_{i+1} = y_{i-1} + \frac{h}{3}[f(x_{i-1}, y_{i-1}) + 4f(x_i, y_i) + f(x_{i+1}, y_{i+1})] \quad \ldots(1)$$

The value of y_{i+1} is first estimated by using (1.) Here this value is incorporated on the right side of (1), giving a better approximation for y_{i+1}. This value of y_{i+1} is again substituted in (1) to find a still better approximation of y_{i+1}. This process is repeated till two successive values of y_{i+1} agree. This technique of finding an initially crude estimate of y_{i+1} by a relatively simple method and then improving it by means of a relatively more accurate formula is known as **predictor-corrector method**. The equation (1.) is called the corrector formula. We now derive a corrector of y_{i+1}.

In the methods so far considered, to arrive at the difference equation involving only one unknown y_{i+1}, we have used the values of y at the beginning of the interval (x_i, x_{i+1}) as initial values. In the present case we use y values at three points x_{i-1}, x_i, x_{i+1}. A parabolic curve is used to approximate the curve $y = f(x)$ in the interval (x_{i-1}, x_{i+1}). A parabolic curve is passed through the points (x_{i-1}, y_{i-1}), (x_i, y_i) and (x_{i+1}, y_{i+1}). Simpson's one-third rule is applied to integrate this curve.

We now derive the three-point corrector formula. Without loss of generality, we take the first point as (x_0, y_0).

[illegible]

Let $y = ax^2 + bx + c$ be a parabola passing through (x_0, y_0), (x_1, y_1) and (x_2, y_2). Then we have $y_0 = ax_0^2 + bx_0 + c$, $y_1 = ax_1^2 + bx_1 + c$ and $y_2 = ax_2^2 + bx_2 + c$. Using Lagrange's interpolation formula, we can find the values of a, b, c in terms of y_0, y_1, y_2 and h. Then we can find the area under the curve between x_0 and x_2 by integrating $\int_{x_0}^{x_2} (ax^2 + bx + c)\,dx$.

The value $y_2 = y(x_2)$ being given, we can interpolate y for x_0, x_1, x_2 by Newton's or Taylor's series method.

Then we can write

$$y_2 = y(x_0 + 2h) = y(x_0) + 2hy'(x_0) + \frac{(2h)^2}{2!}y''(x_0) + \frac{(2h)^3}{3!}y'''(x_0) + \frac{(2h)^4}{4!}y^{iv}(x_0) + \ldots$$

Then we can find $y_1 = y(x_0 + h)$ by substituting $h = \frac{1}{2}$ in the Taylor's series expansion for $y(x_0 + h)$.

$$y(x_0 + h) = y_1 = y(x_0) + \frac{h}{2}y'(x_0) + \frac{h^2}{8}y''(x_0) + \frac{h^3}{48}y'''(x_0) + \ldots$$

We have $y_0 = y(x_0) + \int_{x_0}^{x_0} f(x)\,dx = 0$

$$y_1 = y_0 + \int_{x_0}^{x_0+h} \left[y(x_0) + (x - x_0)y'(x_0) + \frac{(x-x_0)^2}{2}y''(x_0) + \ldots\right]dx \qquad \text{[Put } x - x_0 = z \text{ so that } dx = dz \text{ when } x = x_0,\ z = 0 \text{ when } x = x_0 + h,\ z = h\text{]}$$

$$= y_0 + \int_0^h \left[y(x_0) + zy'(x_0) + \frac{z^2}{2}y''(x_0) + \ldots\right]dz$$

$$= y_0 + \left[zy(x_0) + \frac{z^2}{2}y'(x_0) + \frac{z^3}{6}y''(x_0) + \ldots\right]_0^h$$

$$= y_0 + \left[hy(x_0) + \frac{h^2}{2}y'(x_0) + \frac{h^3}{6}y''(x_0) + \ldots\right]$$

Neglecting fourth and higher order differences and expressing $y(x_0)$, $y'(x_0)$ and $y''(x_0)$ in terms of the function values, we get

$$y_1 = y_0 + \frac{h}{2}[y(x_0) + y(x_0 + h)] \text{ which is called a predictor}$$

Now the first approximation y_1 can be used to find a better approximation $y_2 = f(x_0 + 2h, y_1)$

Then a better value of y_1 is found by Simpson's rule as $\ldots$

$$y_1 = y_0 + \frac{h}{3}[y(x_0) + 4y(x_0 + h) + y(x_0 + 2h)] \text{ which is called a corrector}$$

This corrected value of y_1 is compared with the first approximation and the corrector is applied to find a still better value of y_1. We repeat this step until y_1 remains the same.

Now y_2 and y_1 are obtained to the same degree of accuracy. $y_2 = y(x_0 + 2h)$ is found from the predictor as

$$y_2 = y_0 + \frac{2h}{3}[y(x_0) - y(x_0 + h)]$$

and $y_1 = f(x_0 + h, y_1)$ is calculated. Then a better approximation to the value of y_2 is obtained from the corrector

$$y_1^{(1)} = y_0 + \frac{h}{2}(f_0 + f_1^{(0)}).$$

We repeat this step till y_1 becomes stationary and we, then proceed to calculate y_2 as before.

This is Adam's predictor-corrector method. To ensure greater accuracy, we must first improve the accuracy of the starting values and then sub-divide the intervals.

Example 22.13. *Apply Milne's method to find a solution of the differential equation $\frac{dy}{dx} = x - y^2$ in the range $0 \le x \le 1$ for the boundary condition $y = 0$ at $x = 0$.* (U.P.T.U. 2006; Anna 2005; BAMU 2005)

Solution. Using Picard's method, we have

$$y = y(0) + \int_0^x f(x, y)\,dx, \text{ where } f(x, y) = x - y^2.$$

To get the first approximation, we put $y = 0$ in $f(x, y)$,

giving
$$y_1 = 0 + \int_0^x x\,dx = \frac{x^2}{2}$$

To find the second approximation, we put $y = x^2/2$ in $f(x, y)$.

giving
$$y_2 = \int_0^x \left(x - \frac{x^4}{4}\right)dx = \frac{x^2}{2} - \frac{x^5}{20}$$

Similarly, the third approximation is

$$y_3 = \int_0^x \left[x - \left(\frac{x^2}{2} - \frac{x^5}{20}\right)^2\right]dx = \frac{x^2}{2} - \frac{x^5}{20} + \frac{x^8}{160} - \frac{x^{11}}{4400} \quad \ldots(1)$$

Now let us determine the starting values of the Milne's method from y_3 by choosing $h = 0.2$.

$x_0 = 0.0$, $y_0 = 0.0000$, $f_0 = 0.0000$
$x_1 = 0.2$, $y_1 = 0.1974$, $f_1 = 0.1896$
$x_2 = 0.4$, $y_2 = 0.3795$, $f_2 = 0.3537$
$x_3 = 0.6$, $y_3 = 0.5302$, $f_3 = 0.4690$

Using the predictor, $y_4^{(0)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$

$x = 0.8$, $y_4^{(0)} = 0.6315$, $f_4 = 0.6070$

and the corrector, $y_4^{(1)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4)$, yields

$y_4^{(1)} = 0.6366$, $f_4 = 0.5952$ ...(2)

Again using the corrector, $y_4^{(2)} = 0.6366$, which is same as in (2).

Now using the predictor, $y_5^{(0)} = y_1 + \frac{2h}{3}(2f_2 - f_3 + 2f_4)$,

$x = 1.0$, $y_5^{(0)} = 0.6364$, $f_5 = 0.5950$

and the corrector, $y_5^{(1)} = y_3 + \frac{h}{3}(f_3 + 4f_4 + f_5)$, gives

$y_5^{(1)} = 0.6368$, $f_5 = 0.5946$

Again using the corrector,

$y_5^{(2)} = 0.6368$, a value which is the same as before.

Hence, $y(1) = 0.6368$.

Example 22.14. *Given $y' = xy^2 + y$, $y(0) = 1$, $y(0.1) = 1.1169$, $y(0.2) = 1.2773$ and $y(0.3) = 1.5049$, find $y(0.4)$ by Milne's method and correct to four decimal places.* (Anna, 2007)

Solution. Given $y(0) = 1$ and $h = 0.1$

We have $y'(x) = x(x^2 + y^2)e^{-x^2}$, $y'(0) = 0$

$y''(x) = [3x^2 + xy'](x^2 + y^2)e^{-x^2} + 2x^2 + y^2 + x[2xy + yy'](x^2 + y^2)e^{-x^2}$

$= e^{-x^2}[3x^2 + x^4 + xy^2 + 2x^2 + y^2 + 2x^3y + xyy']$; $y''(0) = 1$

$y'''(x) = -e^{-x^2}[3x^2 + x^4 + xy^2 + 2x^2 + y^2 + 2x^3y + xyy'] + e^{-x^2}[6x + 4x^3 + 2xy^2 + 2x^2y + 2x^3y + yy' + 2xyy' + xy'^2 + xyy'']$ $y'''(0) = 2$

Substituting these values in the Taylor's series,

$$y(x) = y(0) + \frac{x}{1!}y'(0) + \frac{x^2}{2!}y''(0) + \frac{x^3}{3!}y'''(0) + \dots$$

$$y(0.1) = 1 + (0.1)(0) + \frac{1}{2}(0.1)^2(1) + \frac{1}{6}(0.1)^3(2) + \dots$$

$$= 1 + 0.005 - 0.0003 = 1.0047 \text{ i.e., } 1.005$$

Now taking $x = 0.1, y(0.1) = 1.005, h = 0.1$

$y'(0.1) = 0.005, y''(0.1) = 0.040, y'''(0.1) = -1.247$

Substituting these values in the Taylor's series about $x = 0.1$,

$$y(0.2) = y(0.1) + \frac{0.1}{1!}y'(0.1) + \frac{(0.1)^2}{2!}y''(0.1) + \frac{(0.1)^3}{3!}y'''(0.1) + \dots$$

$$= 1.005 + 0.1(0.005) + \frac{(0.1)^2}{2}(0.040) + \frac{(0.1)^3}{6}(-1.247) + \dots$$

$$= 1.010$$

Now taking $x = 0.2, y(0.2) = 1.010, h = 0.1$

$y'(0.2) = 0.170, y''(0.2) = 0.77, y'''(0.2) = 0.810$

Substituting these values in the Taylor's series,

$$y(0.3) = y(0.2) + \frac{0.1}{1!}y'(0.2) + \frac{(0.1)^2}{2!}y''(0.2) + \frac{(0.1)^3}{3!}y'''(0.2) + \dots$$

$$= 1.010 + 0.0170 + 0.0039 + 0.0001 = 1.04$$

Thus the starting values of the Milne's method with $h = 0.1$ are

$x_0 = 0.0$	$y_0 = 1$	$f_0 = y'_0 = 0$
$x_1 = 0.1$	$y_1 = 1.005$	$f_1 = 0.052$
$x_2 = 0.2$	$y_2 = 1.020$	$f_2 = 0.158$
$x_3 = 0.3$	$y_3 = 1.04$	$f_3 = 0.28$

Using the predictor, $y_4^{(1)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$

$$= 1 + \frac{4(0.1)}{3}[2(0.052) - (0.158) + 2(0.28)] = 1.091$$

$\therefore$ $x = 0.4$ $y_4^{(1)} = 1.09$ $f_4 = y'(0.4) = 0.502$

Using the corrector, $y_4^{(2)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4)$

$$y_4^{(2)} = 0.038 + \frac{0.1}{3}(0.158 + 4(0.28) + 0.502) = 1.091$$

Hence $y(0.4) = 1.091$.

Example 22.18. *Using Runge-Kutta method of order 4, find $y(0.2)$ and $y(0.4)$ if $y(0) = 1$, $y'(0) = 0$ and $y'' = x^2y$. Continue the solution at $x = 0.4$ using Milne's method.*

(V.T.U., 2008; S.V.T.U., 2007; Madras, 2005)

Solution. We have $f(x, y) = xy + y^2$.

To find $y(0.1)$:

Here $x_0 = 0, y_0 = 1, h = 0.1$

$k_2 = h\,f(x_0 + \frac{1}{2}h,\ y_0 + \frac{1}{2}k_1) = 0.1\,f(0.05,\ 1.025) = \ldots$

$k_3 = hf\left(x_0 + \frac{1}{2}h,\ y_0 + \frac{1}{2}k_2\right) = 0.1\,f(0.05,\ 1.026) = \ldots$

$k_4 = hf\left(x_0 + \frac{1}{2}h,\ y_0 + \frac{1}{2}k_3\right) = 0.1\,f(0.05,\ 1.0513) = \ldots$

$k_4 = hf(x_0 + h,\ y_0 + k_3) = 0.1\,f(0.1,\ 1.0513) = \ldots$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$= \frac{1}{6}(0.1 + 0.105 + 0.1026 + 0.1053) = \ldots$

Hence $y(0.1) = y_1 = y_0 + k = 1.1033$.

For the next step $x_1 = 0.1$, $y_1 = 1.1033$, $h = 0.1$.

$k_1 = hf(x_1, y_1) = 0.1\,f(0.1,\ 1.1033) = \ldots$

$k_2 = hf\left(x_1 + \frac{1}{2}h,\ y_1 + \frac{1}{2}k_1\right) = 0.1\,f(0.15,\ 1.1543) = \ldots$

$k_3 = hf\left(x_1 + \frac{1}{2}h,\ y_1 + \frac{1}{2}k_2\right) = 0.1\,f(0.15,\ 1.1576) = \ldots$

$k_4 = hf(x_1 + h,\ y_1 + k_3) = 0.1\,f(0.2,\ 1.2109) = \ldots$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

Hence $y(0.2) = y_2 = y_1 + k = 1.2770$.

For the next step $x_2 = 0.2$, $y_2 = 1.2770$, $h = 0.1$.

$k_1 = hf(x_2, y_2) = 0.1\,f(0.2,\ 1.2770) = \ldots$

$k_2 = hf\left(x_2 + \frac{1}{2}h,\ y_2 + \frac{1}{2}k_1\right) = 0.1\,f(0.25,\ 1.3348) = \ldots$

$k_3 = hf\left(x_2 + \frac{1}{2}h,\ y_2 + \frac{1}{2}k_2\right) = 0.1\,f(0.25,\ 1.3384) = \ldots$

$k_4 = hf(x_2 + h,\ y_2 + k_3) = 0.1\,f(0.3,\ 1.4154) = \ldots$

$k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

Hence $y(0.3) = y_3 = y_2 + k = 1.4918$.

Using the starting values $(0.0, 1.0000)$, $(0.1, 1.1033)$, $(0.2, 1.2770)$, $(0.3, 1.4918)$ we have

$x_4 = 0.4$	$y_4 = 1.7379$	$f_4 = 1.7379$
$x_5 = 0.5$	$y_5 = 2.0442$	$f_5 = 2.0442$
$x_6 = 0.6$	$y_6 = 2.4221$	$f_6 = 2.4221$
$x_7 = 0.7$	$y_7 = 2.8798$	$f_7 = 2.8798$

Using the predictor

$$y_{n+1}^{(0)} = y_n + \frac{h}{24}(55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3})$$

$y_4 = 1.4918$ $\qquad y_4^{(0)} = 1.7379$ $\qquad f_4^{(0)} = 1.7379$

and the corrector formula

$$y_{n+1}^{(1)} = y_n + \frac{h}{24}(9f_{n+1}^{(0)} + 19f_n - 5f_{n-1} + f_{n-2})$$

$y_4^{(1)} = 1.4918 + \frac{0.1}{24}(17.3790 + 33.0000 - 6.2132 + 1.0000)$

$= 1.7380$ $\qquad f_4^{(1)} = 1.7380$

Again using the corrector,

$$y_0^{(1)} = 3.3703 + \frac{0.1}{9}[1.0889 + 4(1.2110) + 1.1130]$$
$$= 1.6901 \qquad y_0 = 1.3168 \qquad \text{...10}$$

Again using the corrector

$$y_0^{(2)} = 3.3703 + \frac{0.1}{9}[1.0889 + 4(1.2110) + 1.1130]$$
$$= 1.6901 \text{ which is same as (2).}$$

Hence $y(0.4) = 3.6979$.

EXERCISE 30.4

1. Given $\frac{dy}{dx} = x^2 + y$, $y(0) = 1$. The values $y(0.1) = 1.1172$, $y(0.2) = 1.2649$, and $y(0.3) = 1.4682$ are given by R.K. Method of fourth order. Find $y(0.4)$ by Milne's predictor-corrector method using $h = 0.1$. (Anna, 2003)
2. Given $\frac{dy}{dx} = x(1 + y^2)$ and $y(0) = 1$, $y(0.1) = 1.06$, $y(0.2) = 1.22$, $y(0.3) = 1.51$, evaluate $y(0.4)$ by Milne's predictor-corrector method. (Anna, 2003; Madras, 2003)
3. From the data given below, find $y(x = 1.4)$ using Milne's predictor-corrector formula :

$$\frac{dy}{dx} = x^2 + \frac{y}{2}$$

x	1	1.1	1.2	1.3
y	2	2.2416	2.5044	2.8024

(Anna, 2003)

4. Using Milne's method, find $y(4.4)$ given $5xy' + y^2 - 2 = 0$ given $y(4) = 1$, $y(4.1) = 1.0049$, $y(4.2) = 1.0097$, $y(4.3) = 1.0143$, $y(4.4) = 1.0187$. (Anna, 2003)
5. If $\frac{dy}{dx} = 2e^x - y$, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.040$, $y(0.3) = 2.090$, find $y(0.4)$ using Milne's predictor-corrector method. (A.U., 2003)
6. Using Runge-Kutta method, find the value of $y(0.2)$ and $y(0.4)$ given that $\frac{dy}{dx} = -\frac{2xy}{1+x^2}$, $y(0) = 1$. Hence find the values of $y(0.6)$ and $y(0.8)$ by Milne's method.

30.10 ADAMS-BASHFORTH METHOD

Given $\frac{dy}{dx} = f(x, y)$ and $y_0 = y(x_0)$, we compute

$$y_{-1} = y(x_0 - h), y_{-2} = y(x_0 - 2h), y_{-3} = y(x_0 - 3h)$$

by Taylor's series or Euler's method or Runge-Kutta method.

Next we calculate $f_{-1} = f(x_0 - h, y_{-1})$, $f_{-2} = f(x_0 - 2h, y_{-2})$, $f_{-3} = f(x_0 - 3h, y_{-3})$.

Then to find y_1, we substitute Newton's backward interpolation formula

$$f(x_0 + nh) = f_0 + n\nabla f_0 + \frac{n(n+1)}{2}\nabla^2 f_0 + \frac{n(n+1)(n+2)}{6}\nabla^3 f_0 + \ldots$$

in

$$y_1 = y_0 + \int_{x_0}^{x_0+h} f(x, y)\,dx \qquad \text{...11}$$

i.e.,

$$y_1 = y_0 + \int_0^h \left(f_0 + n\nabla f_0 + \frac{n(n+1)}{2}\nabla^2 f_0 + \cdots\right) dx \qquad [\because x = x_0 + nh,\ dx = h\,dn]$$

$$= y_0 + h\int_0^1 \left(f_0 + n\nabla f_0 + \frac{n(n+1)}{2}\nabla^2 f_0 + \cdots\right) dn$$

$$= y_0 + h\left(f_0 + \frac{1}{2}\nabla f_0 + \frac{5}{12}\nabla^2 f_0 + \frac{3}{8}\nabla^3 f_0 + \cdots\right)$$

Neglecting fourth and higher order differences and expressing ∇f_0, $\nabla^2 f_0$ and $\nabla^3 f_0$ in terms of function values, we get

$$y_1^{(c)} = y_0 + \frac{h}{24}\left[9f_0 - 28f_{-1} + 23f_{-2} - 9f_{-3}\right] \quad \ldots(30)$$

This is called Adams-Bashforth predictor formula.

Having found y_1, we find $f_1 = f(x_1, y_1)$.

Then to find a better value of y_1, we derive a corrector formula by substituting Newton's backward formula for f_1 i.e.,

$$f(x_0 + ph) = f_0 + p\nabla f_0 + \frac{p(p+1)}{2}\nabla^2 f_0 + \frac{p(p+1)(p+2)}{6}\nabla^3 f_0 + \cdots \text{ i.e. (1)}$$

$$\therefore \quad y_1 = y_0 + \int_{x_0}^{x_1}\left(f_0 + p\nabla f_0 + \frac{p(p+1)}{2}\nabla^2 f_0 + \cdots\right)dx \quad [\because x = x_0 + ph, dx = hdp]$$

$$= y_0 + h\int_{-1}^{0}\left(f_0 + p\nabla f_0 + \frac{p^2+p}{2}\nabla^2 f_0 + \cdots\right)dp$$

$$= y_0 + h\left(f_0 - \frac{1}{2}\nabla f_0 - \frac{1}{12}\nabla^2 f_0 - \frac{1}{24}\nabla^3 f_0 - \cdots\right)$$

Neglecting fourth and higher order differences and expressing ∇f_0, $\nabla^2 f_0$ and $\nabla^3 f_0$ in terms of function values, we obtain

$$y_1^{(c)} = y_0 + \frac{h}{24}\left[9f_1 + 19f_0 - 5f_{-1} + f_{-2}\right] \quad \ldots(31)$$

which is called Adams-Moulton corrector formula.

Then an improved value of f_1 is calculated and again the corrector (3) is applied to find a still better value of y_1. This step is repeated till y_1 remains unchanged and then proceed to calculate y_2 as above.

Note. To apply both Milne and Adams-Bashforth methods, we require four starting values of y which are calculated by means of Picard's method or Taylor's series method or Euler's method or Runge-Kutta method. In practice, the Adams corrector (31) is used to obtain upgrades with fourth order Runge-Kutta formulae have been found to be most useful.

Example 12.38. Given $\frac{dy}{dx} = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$, evaluate $y(1.4)$ by Adams-Bashforth method. *(U.T.U., 2003; A.M.I.E., 2000; Anna, 2004)*

Solution. Here $f(x, y) = x^2(1+y)$.

Starting values of the Adams-Bashforth method with $h = 0.1$, are

$x = 1.0$, $y_0 = 1.000$, $f_0 = (1.0)^2(1 + 1.000) = 2.000$

$x = 1.1$, $y_1 = 1.233$, $f_1 = 2.702$

$x = 1.2$, $y_2 = 1.548$, $f_2 = 3.869$

$x = 1.3$, $y_3 = 1.979$, $f_3 = 5.005$

Using (3)

$$y_4^{(p)} = y_3 + \frac{h}{24}\left[55f_3 - 59f_2 + 37f_1 - 9f_0\right]$$

$$= 1.979 + \frac{0.1}{24}\left[55(5.005) - 59(3.869) + 37(2.702) - 9(2.000)\right]$$

$$= 1.979 + \frac{0.1}{24}[275.275 - 228.371 + 99.974 - 18.000]$$

$$= 1.979 + \frac{0.1}{24}(128.878) = 1.979 + 0.537 = 2.516$$

Hence $f_4 = x_4^2(1 + y_4^{(p)}) = (1.4)^2(1 + 2.516) = 1.96(3.516) = 6.892$

Using (4)

$$y_4^{(c)} = y_3 + \frac{h}{24}\left[9f_4 + 19f_3 - 5f_2 + f_1\right]$$

$$= 1.979 + \frac{0.1}{24}\left[9(6.892) + 19(5.005) - 5(3.869) + 2.702\right]$$

$$= 1.979 + \frac{0.1}{24}[62.028 + 95.095 - 19.345 + 2.702]$$

$$= 1.979 + \frac{0.1}{24}(140.480) = 1.979 + 0.585 = 2.564$$

Hence $f_4 = x_4^2(1 + y_4^{(c)}) = (1.4)^2(1 + 2.564) = 1.96(3.564) = 6.986$

Again using (4)

$$y_4^{(c)} = 1.979 + \frac{0.1}{24}\left[9(6.986) + 19(5.005) - 5(3.869) + 2.702\right]$$

$$= 1.979 + \frac{0.1}{24}[62.874 + 95.095 - 19.345 + 2.702]$$

$$= 1.979 + \frac{0.1}{24}(141.326) = 1.979 + 0.589 = 2.568$$

Hence $f_4 = x_4^2(1 + y_4^{(c)}) = (1.4)^2(1 + 2.568) = 1.96(3.568) = 6.993$

Again using (4)

$$y_4^{(c)} = 1.979 + \frac{0.1}{24}\left[9(6.993) + 19(5.005) - 5(3.869) + 2.702\right]$$

$$= 1.979 + \frac{0.1}{24}[62.937 + 95.095 - 19.345 + 2.702]$$

$$= 1.979 + \frac{0.1}{24}(141.389) = 1.979 + 0.589 = 2.568$$

Hence $y(1.4) = 2.568$

Example 33.11 If $\frac{dy}{dx} = 3x^2y$, $y(0) = 0.4$ find, to 4 decimal places, Adams predictor-corrector formula for calculating $y(0.3)$ and $y(0.4)$ using Euler's modified formula. (U.W.S.U. 2003)

Solution. We have $f(x, y) = 3x^2y$.

To find $y(0.1)$:

x	$3x^2y = y'$	Mean slope	$\Delta y = h$(Mean slope) = increment
0.0	0	—	$y + \Delta y = 0.4 + 0.0 = 0.4$
0.1	$3(0.1)^2(0.4) = 0.012$	$\frac{1}{2}(0 + 0.012) = 0.006$	$0.4 + 0.1(0.006) = 0.4006$
0.1	$3(0.1)^2(0.4006) = 0.0120$	$\frac{1}{2}(0 + 0.0120) = 0.0060$	$0.4 + 0.1(0.0060) = 0.4006$
0.1	$3(0.1)^2(0.4006) = 0.0120$	$\frac{1}{2}(0 + 0.0120) = 0.0060$	$0.4 + 0.1(0.0060) = 0.4006$
0.1	$3(0.1)^2(0.4006) = 0.0120$	$\frac{1}{2}(0 + 0.0120) = 0.0060$	$0.4 + 0.1(0.0060) = 0.4006$
0.1	0.012	—	$0.4 + 0.1(0.012) = 0.4012$
0.2	$3(0.2)^2(0.4006) = 0.0481$	$\frac{1}{2}(0.012 + 0.0481) = 0.0301$	$0.4006 + 0.1(0.0301) = 0.4036$
0.2	0.0481	$\frac{1}{2}(0.012 + 0.0481) = 0.0301$	$0.4012 + 0.1(0.0301) = 0.4042$
0.2	0.0481	$\frac{1}{2}(0.012 + 0.0481) = 0.0301$	$0.4012 + 0.1(0.0301) = 0.4042$
0.2	0.0481	$\frac{1}{2}(0.012 + 0.0481) = 0.0301$	$0.4012 + 0.1(0.0301) = 0.4042$
0.2	0.0481	—	$0.4012 + 0.1(0.0481) = 0.4060$
0.3	$3(0.3)^2(0.4042) = 0.1092$	$\frac{1}{2}(0.0481 + 0.1092) = 0.0787$	$0.4042 + 0.1(0.0787) = 0.4121$
0.3	0.1092	$\frac{1}{2}(0.0481 + 0.1092) = 0.0787$	$0.4042 + 0.1(0.0787) = 0.4121$
0.3	0.1092	$\frac{1}{2}(0.0481 + 0.1092) = 0.0787$	$0.4042 + 0.1(0.0787) = 0.4121$
0.3	0.1092	$\frac{1}{2}(0.0481 + 0.1092) = 0.0787$	$0.4042 + 0.1(0.0787) = 0.4121$
0.3	0.1092	—	$0.4042 + 0.1(0.1092) = 0.4151$
0.4	$3(0.4)^2(0.4121) = 0.1978$	$\frac{1}{2}(0.1092 + 0.1978) = 0.1535$	$0.4121 + 0.1(0.1535) = 0.4274$
0.4	0.1978	$\frac{1}{2}(0.1092 + 0.1978) = 0.1535$	$0.4121 + 0.1(0.1535) = 0.4274$
0.4	0.1978	$\frac{1}{2}(0.1092 + 0.1978) = 0.1535$	$0.4121 + 0.1(0.1535) = 0.4274$
0.4	0.1978	$\frac{1}{2}(0.1092 + 0.1978) = 0.1535$	$0.4121 + 0.1(0.1535) = 0.4274$
0.4	0.1978	—	$0.4121 + 0.1(0.1978) = 0.4319$

To find $y(0.4)$ by Adam's method, the starting values with $h = 0.1$ are

$x = 0.0$ $y_0 = 0.4$ $f_0 = 0$

$x = 0.1$ $y_1 = 0.4006$ $f_1 = 0.0481$

$x = 0.2$ $y_2 = 0.4042$ $f_2 = 0.1092$

$x = 0.3$ $y_3 = 0.4121$ $f_3 = 0.1978$

Using the predictor formula

$$y_4^{(p)} = y_3 + \frac{h}{24}(55f_3 - 59f_2 + 37f_1 - 9f_0)$$

$$= 0.4121 + \frac{0.1}{24}(55 \times 0.1978 - 59 \times 0.1092 + 37 \times 0.0481 - 9 \times 0)$$

$$= 0.4283$$

Now $x = 0.4$ $y_4 = 0.4283$ $f_4 = 3x^2y = 0.2055 = 0.0681$

Using the numerical formula,

$$[illegible] \approx \frac{h}{[illegible]} [illegible]$$

$$\approx [illegible] \approx \frac{[illegible]}{[illegible]} [illegible]$$

[illegible]

[illegible]

Solution. [illegible]

The initial [illegible]:

[illegible]

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

[illegible]

The initial [illegible]:

[illegible]

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

[illegible]

The initial [illegible]:

[illegible]

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

$$[illegible] \qquad [illegible]$$

[illegible]

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]

Using the predictor,

$$y_{n+1}^{(0)} = y_n + \frac{h}{24}(55y_n' - 59y_{n-1}' + 37y_{n-2}' - 9y_{n-3}')$$

$n = 0, 1$ $\quad y_4^{(0)} = 0.5081 + \frac{0.2}{24}[55(-0.5081) - 59(-0.3218) + 37(-0.2121) - 9(-0.1)]$

$= 0.7730$ $\quad y_4' = -0.3967$

Using the corrector,

$$y_4^{(1)} = y_3 + \frac{h}{24}[9y_4' + 19y_3' - 5y_2' + y_1']$$

$$y_4^{(1)} = 0.6081 + \frac{0.2}{24}[9(-0.3967) + 19(-0.4458) - 5(-0.3520) - 0.3121] = 0.7730$$

Hence $\quad y(0.8) = 0.7765.$

EXERCISE 10.4

1. Using Adams-Bashforth method, evaluate the solution of $dy/dx = x^2(1 + y)$, given the values

x	0	0.2	0.4	0.6
y	1	1.000	1.026	1.096

(Ans: 1.242)

2. Using Adams-Bashforth formula, determine $y(1.4)$ given the differential equation $dy/dx = \frac{2}{y}$ and the data

x	0	0.2	0.4	0.6
y	1	1.0954	1.1832	1.2649

3. Given $y' = x^2 - y$, $y(0) = 1$ and the starting values $y(0.2) = 0.8213$, $y(0.4) = 0.7027$, $y(0.6) = 0.6122$, obtain $y(0.8)$ using Adams-Bashforth method. (Ans: 0.5379)

4. Using Adams-Bashforth method, find $y(1.4)$ given $5xy' + y^2 - 2 = 0$, $y(1) = 1$, $y(1.1) = 1.0448$, $y(1.2) = 1.0891$, $y(1.3) = 1.1328$. (Ans: 1.1754)

5. Given the differential equation $y'(x) = y^2x - y^2$ and the data

x	1	1.1	1.2	1.3
y	1	1.105	1.2461	1.3979

evaluate $y(1.4)$. (Ans: 1.5735)

6. Using Adams-Bashforth method, find $y(0.4)$ given $dy/dx = x + y^2$, $y(0) = 1$, $y(0.1) = 1.1169$, $y(0.2) = 1.2773$, $y(0.3) = 1.4993$. (Ans: 1.8432)

10.11 SIMULTANEOUS FIRST ORDER DIFFERENTIAL EQUATIONS

The simultaneous differential equations of the type

$$\frac{dy}{dx} = f(x, y, z) \quad \ldots(1)$$

and

$$\frac{dz}{dx} = \phi(x, y, z) \quad \ldots(2)$$

with initial conditions $y(x_0) = y_0$ and $z(x_0) = z_0$ can be solved by the methods discussed in the preceding sections, especially by Picard's or Runge-Kutta methods.

(i) Picard's method gives

$$y_1 = y_0 + \int_{x_0}^{x} f(x, y_0, z_0)\,dx, \quad z_1 = z_0 + \int_{x_0}^{x} \phi(x, y_0, z_0)\,dx$$

$$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1, z_1)\,dx, \quad z_2 = z_0 + \int_{x_0}^{x} \phi(x, y_1, z_1)\,dx$$

$$y_3 = y_0 + \int_{x_0}^{x} f(x, y_2, z_2)\,dx, \quad z_3 = z_0 + \int_{x_0}^{x} \phi(x, y_2, z_2)\,dx$$

and so on.

(ii) Taylor's series method is used as follows :

If h is the step-size, $y_1 = y(x_0 + h)$ and $z_1 = z(x_0 + h)$. Then Taylor's algorithm for (1) and (2) gives

$$y_1 = y_0 + hy_0' + \frac{h^2}{2!}y_0'' + \frac{h^3}{3!}y_0''' + \dots \quad \dots(10)$$

$$z_1 = z_0 + hz_0' + \frac{h^2}{2!}z_0'' + \frac{h^3}{3!}z_0''' + \dots \quad \dots(11)$$

Differentiating (1) and (2) successively, we get y', y'', y''', etc. So the values $y_0', y_0'', y_0''' \dots$ and $z_0', z_0'', z_0''' \dots$ are known. Substituting these in (10) and (11), we obtain y_1, z_1 for the first step.

Similarly, we have the algorithms

$$y_2 = y_1 + hy_1' + \frac{h^2}{2!}y_1'' + \frac{h^3}{3!}y_1''' + \dots \quad \dots(12)$$

$$z_2 = z_1 + hz_1' + \frac{h^2}{2!}z_1'' + \frac{h^3}{3!}z_1''' + \dots \quad \dots(13)$$

Since y_1 and z_1 are known, we can calculate $y_1', y_1'', \dots$ and $z_1', z_1'' \dots$. Substituting these in (12) and (13), we get y_2 and z_2.

Proceeding further, we can calculate the other values of y and z step by step.

(c) Runge-Kutta method is applied as follows :

Starting at (x_0, y_0, z_0) and taking the step sizes for x, y, z to be h, k, l respectively, the Runge-Kutta method gives,

$$k_1 = hf(x_0, y_0, z_0) \qquad l_1 = hg(x_0, y_0, z_0)$$

$$k_2 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1, z_0 + \frac{1}{2}l_1\right) \qquad l_2 = hg\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1, z_0 + \frac{1}{2}l_1\right)$$

$$k_3 = hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2, z_0 + \frac{1}{2}l_2\right) \qquad l_3 = hg\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2, z_0 + \frac{1}{2}l_2\right)$$

$$k_4 = hf(x_0 + h, y_0 + k_3, z_0 + l_3) \qquad l_4 = hg(x_0 + h, y_0 + k_3, z_0 + l_3)$$

Then $y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ and $z_1 = z_0 + \frac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4)$

To compute y_2 and z_2, we simply replace x_0, y_0, z_0 by x_1, y_1, z_1 in the above formulae.

Example 10.33. Using Picard's method find approximate values of y and z corresponding to $x = 0.2$, given that $y = 1$, $z = 1$ when $x = 0$ and $\frac{dy}{dx} = x + z$, $\frac{dz}{dx} = x - y^2$.

Solution. Here $x_0 = 0, y_0 = 1, z_0 = 1$.

$$\frac{dy}{dx} = f(x, y, z) = x + z \qquad \text{and} \qquad \frac{dz}{dx} = \phi(x, y, z) = x - y^2$$

$$\Rightarrow \quad y = y_0 + \int_{x_0}^{x} f(x, y, z)\,dx \qquad \text{and} \qquad z = z_0 + \int_{x_0}^{x} \phi(x, y, z)\,dx.$$

First approximations $y_1 = y_0 + \int_{x_0}^{x} \phi(x, y_0, z_0)\,dx = 1 + \int_0^x (x + 1)\,dx = 1 + x + \frac{1}{2}x^2$

$$z_1 = z_0 + \int_{x_0}^{x} \phi(x, y_0, z_0)\,dx = 1 + \int_0^x (x - 1)\,dx = 1 - x + \frac{1}{2}x^2$$

Second approximations $y_2 = y_0 + \int_{x_0}^{x} f(x, y_1, z_1)\,dx = 1 + \int_0^x \left(x + 1 - x + \frac{1}{2}x^2\right)dx$

$$= 1 + x - \frac{1}{6}x^3 + \frac{x^5}{40}$$

$$z_2 = z_0 + \int_{x_0}^{x} \phi(x, y_1, z_1)\,dx$$

$$= 1 + \int_0^x \left[x - \left(1 + x + \frac{1}{2}x^2\right)^2\right]dx = 1 - x + \frac{3}{2}x^2 - x^3 - \frac{x^4}{4} - \frac{x^5}{20}.$$

First approximation $y_1 = y_0 + \int_{x_0}^{x_1} f(x, y_0)\,dx$

$= 0 + 0.1 - \frac{0.1}{3}x^3 - \frac{0.1}{5}x^5 + \frac{0.1}{7}x^7 - \frac{0.1}{9}x^9 + \frac{0.1}{11}x^{11}$

$y_2 = y_0 + \int_{x_0}^{x_1} f(x, y_1)\,dx$

$= 0 + 0.1 - \frac{0.1}{3}x^3 + \frac{0.1}{5}x^5 - \frac{0.1}{7}x^7 + \frac{0.1}{9}x^9 - \frac{0.1}{11}x^{11} + \frac{0.1}{13}x^{13}$

and so on.

When $x = 0.1$,

$y_1 = 0.1000$, $y_2 = 0.1003$, $y_3 = 0.1003$

$y_1 = 0.1987$, $y_2 = 0.1987$, $y_3 = 0.1987$

Hence $y(0.1) = 0.1987$, $y(0.2) = 0.3951$

to four decimal places.

Theorem If $f(x, y)$ satisfies Lipschitz's condition in the region R defined by $|x - x_0| \le a$, $|y - y_0| \le b$, and if M is the upper bound of $|f(x, y)|$ in R, then the successive approximations $y_1, y_2, \ldots, y_n$ converge to the solution $y(x)$ of the equation $\frac{dy}{dx} = f(x, y)$ for which $y = y_0$ when $x = x_0$, provided that x lies in the interval $x_0 - h \le x \le x_0 + h$, where h is the smaller of the two numbers a and $\frac{b}{M}$.

Proof. Since $f(x, y)$ satisfies Lipschitz's condition in R, we have

$$|f(x, y_1) - f(x, y_0)| \le k|y_1 - y_0| \quad \text{or} \quad |y_1 - y_0| \le k\left|\int_{x_0}^{x} f(x, y_0)\,dx\right|$$

[illegible]

$y_0 = y_0$

$y_1 = y_0 + \int_{x_0}^{x} f(x, y_0)\,dx$

$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1)\,dx$

$y_3 = y_0 + \int_{x_0}^{x} f(x, y_2)\,dx$

$\cdots\cdots\cdots\cdots\cdots\cdots$

$y_n = y_0 + \int_{x_0}^{x} f(x, y_{n-1})\,dx$

$\cdots\cdots\cdots\cdots\cdots\cdots$

Hence $|y_1 - y_0| \le \int_{x_0}^{x} |f(x, y_0)|\,dx \le M|x - x_0|$

$|y_2 - y_1| \le \int_{x_0}^{x} |f(x, y_1) - f(x, y_0)|\,dx \le k\int_{x_0}^{x} |y_1 - y_0|\,dx \le Mk\frac{|x - x_0|^2}{2!}$

$|y_3 - y_2| \le \int_{x_0}^{x} |f(x, y_2) - f(x, y_1)|\,dx \le k\int_{x_0}^{x} |y_2 - y_1|\,dx \le Mk^2\frac{|x - x_0|^3}{3!}$

$\cdots\cdots\cdots\cdots\cdots\cdots$

$|y_n - y_{n-1}| \le \int_{x_0}^{x} |f(x, y_{n-1}) - f(x, y_{n-2})|\,dx \le k\int_{x_0}^{x} |y_{n-1} - y_{n-2}|\,dx \le Mk^{n-1}\frac{|x - x_0|^n}{n!}$

$\cdots\cdots\cdots\cdots\cdots\cdots$

Hence $|y_1| + |y_2 - y_1| + |y_3 - y_2| + \cdots + |y_n - y_{n-1}| \le M|x - x_0| + Mk\frac{|x - x_0|^2}{2!} + Mk^2\frac{|x - x_0|^3}{3!} + \cdots + Mk^{n-1}\frac{|x - x_0|^n}{n!}$

$\cdots\cdots\cdots\cdots\cdots\cdots$

$\cdots\cdots\cdots\cdots\cdots\cdots$

$\cdots\cdots\cdots\cdots\cdots\cdots$

33.32 SECOND ORDER DIFFERENTIAL EQUATIONS

Consider the second order differential equation $\frac{d^2y}{dx^2} = f\left(x, y, \frac{dy}{dx}\right)$

By writing $dy/dx = z$, it can be reduced to two first order simultaneous differential equations

$$\frac{dy}{dx} = z \qquad \frac{dz}{dx} = f(x, y, z)$$

These equations can be solved as explained above.

Example 33.33. Using Runge-Kutta method, solve $y'' - xy' - y^2 = 0$ for $x = 0.2$ given $y = 1$ and $y' = 0.5$ at $x = 0$. Calculate y at $x = 0.2$, $y = 1.2$, $y' = 0$. (AMIE, Winter 2001)

Solution. Let $dy/dx = z = f(x, y, z)$. Then $dz/dx = xz + y^2 = g(x, y, z)$

We have $x_0 = 0, y_0 = 1, z_0 = 0.5, h = 0.2$.

Using $k_1, k_2, \ldots$ for $f(x, y, z)$ and $l_1, l_2, \ldots$ for $g(x, y, z)$, Runge-Kutta formulae become

$$k_1 = hf(x_0, y_0, z_0) = 0.2(0.5) = 0.1$$

$$l_1 = hg(x_0, y_0, z_0) = 0.2(1) = 0.2$$

$$k_2 = hf\left(x_0 + \tfrac{1}{2}h, y_0 + \tfrac{1}{2}k_1, z_0 + \tfrac{1}{2}l_1\right) = 0.2(0.5 + 0.1) = 0.12$$

$$l_2 = hg\left(x_0 + \tfrac{1}{2}h, y_0 + \tfrac{1}{2}k_1, z_0 + \tfrac{1}{2}l_1\right) = 0.2(0.1 + 1) = 0.22$$

$$k_3 = hf\left(x_0 + \tfrac{1}{2}h, y_0 + \tfrac{1}{2}k_2, z_0 + \tfrac{1}{2}l_2\right) = 0.2(0.5 + 0.11) = 0.122$$

$$l_3 = hg\left(x_0 + \tfrac{1}{2}h, y_0 + \tfrac{1}{2}k_2, z_0 + \tfrac{1}{2}l_2\right) = 0.2(0.061 + 1.061) = 0.2244$$

$$k_4 = hf(x_0 + h, y_0 + k_3, z_0 + l_3) = 0.2(0.5 + 0.2244) = 0.1449$$

$$l_4 = hg(x_0 + h, y_0 + k_3, z_0 + l_3) = 0.2(0.1224 + 1.0224) = 0.2290$$

$$\therefore \quad k = \tfrac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = \tfrac{1}{6}(0.1 + 0.24 + 0.244 + 0.1449) = 0.1215$$

$$l = \tfrac{1}{6}(l_1 + 2l_2 + 2l_3 + l_4) = \tfrac{1}{6}(0.2 + 0.44 + 0.4488 + 0.2290) = 0.2296$$

Hence at $x = 0.2$,

$$y = y_0 + k = 1 + 0.1215 = 1.1215$$

and $$y' = z = z_0 + l = 0.5 + 0.2296 = 0.7296$$

Example 33.34. Given $y'' + xy' + y = 0$, $y(0) = 1$, $y'(0) = 0$, obtain y for $x = 0.1$ by any method. Further, compute the solution by Milne's method for $x = 0.4$. Obtain $y(0.4)$. (Anna, 2004; Madras, 2003)

Solution. Putting $y' = z$, the given equation reduces to the simultaneous equations

$$z' + xz + y = 0, \; y' = z \qquad \ldots(1)$$

We employ Taylor's series method to find y.

Differentiating the given equation n times, we get

$$y_{n+2} + xy_{n+1} + (n+1)y_n = 0$$

At $x = 0$, $y_{n+2} + (n+1)y_n = 0$ or $y_{n+2} = -(n+1)y_n$

$\therefore$ $y(0) = 1$, gives $y_2(0) = -1, y_3(0) = 0, y_4(0) = -3, y_5(0) = 0, \ldots$

and $y'(0) = 0$ yields $y_2(0) = y_3(0) = \ldots = 0$.

Expanding $y(x)$ by Taylor's series, we have

$$y(x) = y(0) + xy_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \ldots$$

$\therefore$ $$y(x) = 1 - \frac{x^2}{2!} + \frac{3}{4!}x^4 - \frac{3 \times 5}{6!}x^6 + \ldots \qquad \ldots(2)$$

and $$y_1(x) = y'(x) = -x + \frac{x^3}{3} - \frac{1}{6}x^5 + \ldots = -xy \qquad \ldots(3)$$

From (2), we have

$$y(0.1) = 1 - \frac{(0.1)^2}{2} + \frac{1}{8}(0.1)^4 - \ldots = 0.995$$

$$y(1.2) = 1 - \frac{0.2^2}{2} + \frac{0.2^4}{8} - \ldots = 0.8052$$

$$y(1.3) = 1 - \frac{0.3^2}{2} + \frac{0.3^4}{8} - \frac{0.3^6}{48} + \ldots = 0.7641$$

From (1), we have

$$y'(1.1) = -0.8056,\ y'(1.2) = -0.1236,\ y'(1.3) = -0.5843$$

Also from (1), $y'(x) = -(xy + y)$ $\quad \therefore y'(0.1) = 0.945,\ y'(1.2) = 0.941,\ y'(1.3) = 1.07$

Applying Milne's predictor formula, first two and then to y_4 we obtain

$$y(0.4) = y(0) + \frac{4}{3}[0.1\{2y'(1.1) - y'(1.2) + 2y'(1.3)\}$$

$$= 1 + \left(\frac{0.4}{3}\right)[-1.79 + 0.941 - 1.741] = -0.3992$$

and

$$y(0.4) = y(0) + \frac{4}{3}[0.1\{2y'(1.1) - y'(1.2) + 2y'(1.3)\}$$

$$= 1 + \left(\frac{0.4}{3}\right)[-0.229 + 0.1236 - 0.5734] = 0.9003$$

Also $y'(0.4) = -[y(0.4)x(0.4) + y(0.4)] = -[0.4(-0.3992) + 0.9003] = -0.7704$.

Now applying Milne's corrector formula, we get

$$y(0.4) = y(0.2) + \frac{h}{3}[y'(0.2) + 4y'(0.3) + y'(0.4)]$$

$$= -0.2981 + \left(\frac{0.1}{3}\right)[-0.941 - 0.445 - 0.7704] = -0.3005$$

and

$$y(0.4) = y(0.2) + \frac{h}{3}[y'(0.2) + 4y'(0.3) + y'(0.4)]$$

$$= 0.8002 + \left(\frac{0.1}{3}\right)[-0.104 - 1.1402 - 0.3003] = 0.8002$$

Hence $\quad y(0.4) = 0.8002$ and $y(0.4) = -0.3002$

PROBLEMS

1. Apply Picard's method to find the third approximation to the values of y and z given that $dy/dx = z - x^2y$, $dz/dx = x^2z + y = 0$, given $y = 1, z = \frac{2}{3}$ when $x = 0$.
2. Solve the following differential equations using Taylor series method of the 4th order, for $x = 0.1$ and 0.2.
$\frac{dy}{dx} = xy + 1, \frac{dz}{dx} = x - y$, $y(0) = 0$ and $z(0) = 1$.
3. Find $y(0.1)$, $z(0.1)$, $y(0.2)$ and $z(0.2)$ from the system of equations $y' = x + z$, $z' = x - y^2$ given $y(0) = 0$, $z(0) = 1$ using Runge-Kutta 4th order. (U.P.T.U., 2009)
4. Using Picard's method, obtain the second approximation to the solutions of
$$\frac{d^2y}{dx^2} + x^2\frac{dy}{dx} + xy = 1 \text{ satisfying } y(0) = 1, y'(0) = \frac{1}{2}.$$
5. Use Milne's method to approximate y when $x = 0.4$, given that $\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + y = 0$, $y(0) = 1$, $y'(0) = 0$, $\frac{dy}{dx} = y$ when $x = 0.2$ and $y = 0.96$, $\frac{dy}{dx} = 0.8$ when $x = 0$.
6. Using Runge-Kutta method of order four, solve $y'' + xy' + y = 0$, $y(0) = 1$, $y'(0) = 0$ to find $y(0.2)$ and $y'(0.2)$.
7. Consider the second order ordinary differential equation $y'' - xy^2 + 2y = e^x$ with $y(0) = -1$ and $y'(0) = 0$. Using the fourth order Runge-Kutta method, find $y(0.2)$. (Anna, 2008)

Ex. The equation determining the deflection y of a uniform cantilever beam is given by the equation

$$EI\frac{d^2y}{dx^2} = \frac{1}{2}wlx - \frac{1}{2}wx^2$$

where E is Young's modulus, I is the moment of inertia of the beam about the neutral axis, w is the load per unit length, l is the length of the beam and x is the distance measured from one end of the beam.

32.2 BOUNDARY VALUE PROBLEMS

Such a problem requires the solution of a differential equation in a region R subject to the solution assuming certain values on the boundary of R. Practical applications give rise to many such problems. We shall discuss two-point linear boundary value problems of the following types:

(i) $\frac{d^2y}{dx^2} + f(x)\frac{dy}{dx} + g(x)y = r(x)$ in (a, b) with the conditions $y(a) = \alpha$, $y(b) = \beta$

(ii) $\frac{d}{dx}\left(p(x)\frac{dy}{dx}\right) + q(x)y = r(x)$ in (a, b) with the conditions $y(a) = \beta(a)$ and $y'(a) = \beta'(a)$

Of the three well known numerical methods for solving such boundary value problems, the method of finite differences is most satisfactory to work. We shall explain this method in the next section.

32.3 FINITE-DIFFERENCE METHOD

In this method, the derivatives occurring in the differential equation and the boundary conditions are replaced by their finite difference approximations and the resulting linear system of algebraic equations is solved by any standard procedure. There exists one of the simplest and easiest of the general methods.

The finite difference approximations to the various derivatives are derived as under :

If $y(x)$ and its derivatives are single valued continuous functions of x then by Taylor's theorem we have

$$y(x+h) = y(x) + hy'(x) + \frac{h^2}{2!}y''(x) + \frac{h^3}{3!}y'''(x) + \ldots \quad \ldots(1)$$

and $$y(x-h) = y(x) - hy'(x) + \frac{h^2}{2!}y''(x) - \frac{h^3}{3!}y'''(x) + \ldots \quad \ldots(2)$$

Equation (1) gives $y'(x) = \frac{1}{h}[y(x+h) - y(x)] - \frac{h}{2}y''(x) - \ldots$

or $$y'(x) = \frac{1}{h}[y(x+h) - y(x)] + O(h) \quad \ldots(3)$$

which is the forward difference approximation of $y'(x)$ with an error of the order h.

Equation (2) gives $y'(x) = \frac{1}{h}[y(x) - y(x-h)] + O(h)$

which is the backward difference approximation of $y'(x)$ with an error of the order h.

Subtracting (2) from (1), we obtain

$$y'(x) = \frac{1}{2h}[y(x+h) - y(x-h)] + O(h^2)$$

which is the central difference approximation of $y'(x)$ with an error of the order h^2. Clearly the central difference approximation to $y'(x)$ is better than the formulae of backward difference approximation and is usually adopted for problems.

Adding (1) and (2), we get

$$y''(x) = \frac{1}{h^2}[y(x+h) - 2y(x) + y(x-h)] + O(h^2)$$

which is the central difference approximation of $y''(x)$. Such formulae are also known as central difference approximations to higher derivatives.

Hence the working expressions for the central difference approximations for the first four derivatives of y are as follows:

$$y'_i = \frac{1}{2h}(y_{i+1} - y_{i-1}) \quad \ldots(40)$$

$$y''_i = \frac{1}{h^2}(y_{i+1} - 2y_i + y_{i-1}) \quad \ldots(41)$$

$$y'''_i = \frac{1}{2h^3}(y_{i+2} - 2y_{i+1} + 2y_{i-1} - y_{i-2}) \quad \ldots(42)$$

$$y^{iv}_i = \frac{1}{h^4}(y_{i+2} - 4y_{i+1} + 6y_i - 4y_{i-1} + y_{i-2}) \quad \ldots(43)$$

Note. The accuracy of the method depends on the size of the subinterval h and also on the order of approximation. As in values of h, for accuracy required, but the number of equations to be solved also increases.

Example 22.23. Solve the equation $y'' = x + y$ with the boundary conditions $y(0) = y(1) = 0$ (Anna, 2005)

Solution. We divide the interval $[0, 1]$ into four subintervals so that $h = 1/4$ and the pivotal points are $x_0 = 0$, $x_1 = 1/4$, $x_2 = 1/2$, $x_3 = 3/4$ and $x_4 = 1$.

The differential equation in approximation is

$$\frac{1}{h^2}(y_{i+1} - 2y_i + y_{i-1}) = x_i + y_i$$

or
$$16y_{i-1} - 33y_i + 16y_{i+1} = x_i \quad (i = 1, 2, 3).$$

Given $y_0 = y_4 = 0$, we get the system of equations

$$16y_0 - 33y_1 = \frac{1}{4}$$

$$16y_1 - 33y_2 + 16y_3 = \frac{1}{2}$$

$$-33y_3 + 16y_4 = \frac{3}{4}$$

Their solution gives

$$y_1 = -0.03183, y_2 = -0.05638, y_3 = -0.05606.$$

Note. The exact solution being $y(x) = \frac{\sin 3x}{\sin 4} - x$, the exact values rounded to four decimal places are given in the table:

x	Computed value $y(x)$	Exact value $y(x)$	Error
0.25	−0.03183	−0.03184	0.00001
0.5	−0.05638	−0.05639	0.00001
0.75	−0.05606	−0.05605	0.00001

Example 22.24. Determine the values of $y(x)$ at the pivotal points of the interval $[0, 1]$ if $y(x)$ satisfies the boundary value problem $y'' + 8(y + 0.5x^2) y(0) = y(1) = 0$. Choose $h = 1/3$.

Solution. Here $h = 1/3$ and the pivotal points are $x_0 = 0$, $x_1 = 1/3$, $x_2 = 2/3$, $x_3 = 1$. The corresponding y-values are $y_0 = 0$, y_1, y_2, $y_3 = 0$.

Replacing y'' by the central difference approximation, the differential equation becomes

$$\frac{1}{h^2}(y_{i-1} - 2y_i + y_{i+1}) + 8(y_i + 0.5x_i^2) = 0$$

or
$$y_{i-1} - 2y_i + y_{i+1} + 8h^2 y_i + 4h^2 x_i^2 = 0, \quad i = 1, 2$$

At $i = 1$, $\quad y_0 - 2y_1 + y_2 + 8h^2 y_1 + 4h^2 x_1^2 = 0$

At $i = 2$, $\quad y_1 - 2y_2 + y_3 + 8h^2 y_2 + 4h^2 x_2^2 = 0$

Using $y_0 = y_3 = 0$, we get $\quad -2y_1 + y_2 + 8h^2 y_1 + 4h^2 x_1^2 = 0$ $\ldots(1)$

$$y_1 - 2y_2 + 8h^2 y_2 + 4h^2 x_2^2 = 0 \quad \ldots(2)$$

Expanding the condition $y'_0 + y'_4 = 0$, we have that

$$y'_0 = \frac{1}{2h}(y_{-1} - 0 + y_1) = 0$$

At $i = 0$, $\quad y''_0 = 0(y_1 - 0 + y_{-1}) - 0 + y_{-1}$

or $\quad y_{-1} = y_1$ $\qquad$ (iv) $\quad y_0 + y''_0 = 0$ $\quad$ ...(iii)

At $i = 4$, $\quad y''_4 = 0(y_5 - 2y_4 + y_3) - 2y_4 + y_3$

$y_5 = -y_3$ $\qquad$ (v) $\quad y_4 + y''_4 = 0$ $\quad$ ...(iv)

Using (iii), the equation (i) becomes

$$-4y_1 + 0y_2 = 45 \qquad \text{...(v)}$$

Using (iv), the equation (ii) reduces to

$$0y_1 - 4y_2 = 45 \qquad \text{...(vi)}$$

Solving (v) and (vi), we obtain

$$y_1 = 11.90 \text{ and } y_2 = 9.45.$$

Hence $y(0.5) = 0.1222$ and $y(0.75) = 0.1368$.

Example 22.23. *The deflection of a beam is governed by the equation*

$$\frac{d^2y}{dx^2} + y(x) = \phi(x)$$

where $\phi(x) = x$ and y is given by the table

x	1/3	2/3	1
y	.81	.962	.993

and also $y(0) = y(1) = y''(0) = y''(1) = 0$. Find the deflection at the equally spaced points of the beam using three such intervals.

Solution. Here $h = 1/3$ and the pivotal points are $x_0 = 0, x_1 = 1/3, x_2 = 2/3, x_3 = 1$. The corresponding y-values are $y_0 = 0, y_1, y_2, y_3$.

The given differential equation is approximated to

$$\frac{1}{h^2}(y_{i-1} - 2y_i + y_{i+1}) + y_i = x_i \quad \text{or} \quad y_{i-1} - 2y_i + y_{i+1} + h^2 y_i = h^2 x_i$$

At $i = 1$, $\quad y_0 - 2y_1 + y_2 + h^2 y_1 = h^2 x_1$ $\quad$ ...(i)

At $i = 2$, $\quad y_1 - 2y_2 + y_3 + h^2 y_2 = h^2 x_2$ $\quad$ ...(ii)

At $i = 3$, $\quad y_2 - 2y_3 + y_4 + h^2 y_3 = h^2 x_3$ $\quad$ ...(iii)

We have $\quad y_0 = 0$ $\quad$ ...(iv)

Since $\quad y'' = \frac{1}{2h}(y_{i+1} - y_{i-1})$

at $i = 0$, $\quad 0 = y''_0 = \frac{1}{2h}(y_1 - y_{-1})$ i.e., $y_{-1} = y_1$ $\quad$ ...(v)

Since $\quad y'' = \frac{1}{h^2}(y_{i-1} - 2y_i + y_{i+1})$

at $i = 3$, $\quad 0 = y''_3 = \frac{2}{h^2}(y_4 - 2y_3 + y_2)$ i.e., $y_4 = 2y_3 - y_2$ $\quad$ ...(vi)

Also $\quad y''' = \frac{1}{2h^3}(y_{i+2} - 2y_{i+1} + 2y_{i-1} - y_{i-2})$

at $i = 3$, $\quad 0 = y'''_3 = \frac{1}{2h^3}(y_5 - 2y_4 + 2y_2 - y_1)$

$$y_5 = 2y_4 - 2y_2 + y_1 \qquad \text{...(vii)}$$

Using (v) and (vi), the equation (i) reduces to

$$y_1 - 2y_2 + 2y_3 = 1 \qquad \text{...(viii)}$$

Using (vi) and (vii), the equation (iii) becomes

$$-y_2 + 2y_3 - 2y_4 = 1 \qquad \text{...(ix)}$$

Using (i) and (ii), the equation (iv) reduces to

$$3y_1 - 4y_0 + 2y_1 = 0 \qquad \ldots(v)$$

Solving (iv) and (v), we get

$$y_1 = 0.10,\ y_2 = 2.210,\ y_3 = 3.710.$$

Hence $y(1.1) = 0.1004,\ y(1.2) = 1.4001,\ y(1.3) = 2.1002$

EXERCISE 9.5

1. Solve the boundary value problem for $x = 0.5$:

$$\frac{d^2y}{dx^2} + y + 1 = 0,\ y(0) = y(1) = 0 \qquad \text{Take } h = 0.1$$

2. Find an approximate solution of the boundary value problem

$$y'' + 0.5(y')^2 + y = 0.5,\ y(0) = y(1) = 0 \qquad \text{(Take } h = 0.1\text{)}$$

3. Solve the boundary value problem:

$$2y'' + y = 0,\ y(1) = 2,\ y(2) = 2. \qquad \text{(Take } h = 0.2\text{)}$$

4. Solve the equation

$$y'' - 4y' = 4x + e^{2x},$$ with the conditions $y(0) = 0,\ y'(1) = -2$, taking $h = 0.2$

5. Solve the boundary value problem $y'' - 64y = 10x$, $y(0) = y(2) = 0$ by the Runge-Kutta-Nystrom method. Compare the value of $y(0.5)$ and compare with the true value.

6. Solve the boundary value problem:

$$y'' + xy' + y = 2x^2 + 3,\ y(0) = 3,\ y(1) = 1$$

7. The boundary value problem governing the deflection of a beam of length l carrying a load is given by

$$\frac{d^2y}{dx^2} + \frac{3}{x}\frac{dy}{dx} + \frac{2}{x^2}y = \frac{x}{4},\ y(0) = y'(1) = y(0) = y(1) = 0$$

The beam is built in at the left end $x = 0$ and is hinged at the right end $x = 1$. Determine the deflection at the mid point $x = \frac{1}{2}$ and $x = \frac{3}{4}$.

8. Solve the boundary value problem:

$$\frac{d^2y}{dx^2} + xy = 1,\ y(0) = y(1) = y'(1) = 0 \qquad \text{Take } h = 0.5$$

9. Solve the equation $y'' - y' + y = e^{2x}$ subject to the boundary conditions

$$y(0) = y(1) = 0 \text{ and } y(1) = 2,\ y'(1) = 1 \qquad \text{Take } h = 0.1.$$

SECTION C OBJECTIVE TYPE OF QUESTIONS

EXERCISE 9.6

Select the correct answer or fill in the blanks in the following questions:

1. Which of the following is a single step method?
   (a) Taylor's (b) Adams-Bashforth (c) Picard's (d) None
2. The Euler-Cauchy solution for the equation $2y'' + y = 0$ is ——.
3. If $y' = x + y$, $y(0) = 1$ and $y^{(1)}(x) = 1 + x + x^2/2!$ then by Picard's method, the value of $y^{(2)}(x)$ is ——.
4. The iteration formula of Euler's method for solving $y' = f(x, y)$ with $y(x_0) = y_0$ is ——.
5. Taylor's series is for solving only first order ordinary differential equations in ——.
6. Using Runge-Kutta method of order four, the value of $y(0.1)$ for $y' = x - y$, $y(0) = 1$ taking $h = 0.1$ is
   (a) 0.913 (b) 0.905 (c) 0.9005 (d) none
7. Given y_0, y_1, y_2, y_3. Milne's corrector formula to find y_4 is $y_4 = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4)$ ——.
8. The second order Runge-Kutta formula is ——.
9. Adams-Bashforth predictor formula to solve $y' = f(x, y)$ gives $y_4 = y_3 + \frac{h}{24}(\ldots)$ is ——.
10. The predictor and corrector formulae for solving ordinary differential equations are ——.
11. To predict Adams's corrector formula ——— values of y_1 prior to the desired value are required.
12. Taylor's series solution of $y' = x - y$, $y(0) = 1$ up to x^3 is ——.

CHAPTER 33

Numerical Solution of Partial Differential Equations

1. Introduction. 2. Classification of second order equations. 3. Finite difference approximations to derivatives. 4. Elliptic equations. 5. Solution of Laplace's equation. 6. Solution of Poisson's equation. 7. Parabolic equations. 8. Solution of heat equation. 9. Hyperbolic equation. 10. Solution of wave equation. 11. Objective Type of Questions.

33.1 INTRODUCTION

There are many boundary value problems which involve partial differential equations. Only a few of these equations can be solved by analytical methods. In most cases, we depend on the numerical solution of such partial differential equations. Of the various numerical methods available for solving these equations, the method of finite differences is most commonly used.

In this method, the derivatives appearing in the equation and the boundary conditions are replaced by their finite difference approximations. Thus the given equation is changed to a difference equation which is solved by iterative procedures. This process is slow but gives good results of boundary value problems. An added advantage of this method is that the computations can be done by electronic computer. Here we shall study this method to the solution of important applied partial differential equations. For a detailed study, the reader should refer to standard text books *Numerical Methods in Engineering and Science*.

33.2 CLASSIFICATION OF SECOND ORDER EQUATIONS

The general linear partial differential equation of the second order in two independent variables x and y is

$$A(x,y)\frac{\partial^2 z}{\partial x^2} + B(x,y)\frac{\partial^2 z}{\partial x \partial y} + C(x,y)\frac{\partial^2 z}{\partial y^2} + F\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right) = 0$$

Such a partial differential equation is said to be

(i) **elliptic**, if $B^2 - 4AC < 0$,

(ii) **parabolic**, if $B^2 - 4AC = 0$

and (iii) **hyperbolic**, if $B^2 - 4AC > 0$.

Example 33.1. Classify the following equations:

(i) $(1+x^2)\frac{\partial^2 z}{\partial x^2} + (1+y^2)\frac{\partial^2 z}{\partial y^2} + x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 0$ (ii) $x^2\frac{\partial^2 z}{\partial x^2} + (x+y)^2\frac{\partial^2 z}{\partial y^2} = 0$, $x \neq 0, x \neq -y, x+y \neq 0$ (Madras, 2001)

(iii) $(1+x^2)\frac{\partial^2 z}{\partial x^2} + (1+2x^2)\frac{\partial^2 z}{\partial x \partial y} + (1+x^2)\frac{\partial^2 z}{\partial y^2} = 0$

Solution. 1) Comparing this equation with (1) above, we find that

$$A = 1,\ B = 4,\ C = 4$$

$$\therefore \quad B^2 - 4AC = (4)^2 - 4 \times 1 \times 4 = 0$$

So the equation is parabolic.

(ii) Here $\quad A = x^2,\ B = 0,\ C = 1 - y^2$

$$\therefore \quad B^2 - 4AC = 0 - 4x^2(1 - y^2) = 4x^2(y^2 - 1)$$

For all x between $-\infty$ and $+\infty$, x^2 is positive

For y between -1 and 1, $y^2 < 1 \quad \therefore \quad B^2 - 4AC < 0$

Hence the equation is elliptic.

(iii) Here $A = 1 + x^2,\ B = 2 + 2y^2,\ C = 4 + y^4$

$$\therefore \quad B^2 - 4AC = (2 + 2y^2)^2 - 4(1 + x^2)(4 + y^4) = 2(x^2 - y^2)^2 \geq 0$$

So the equation is hyperbolic.

EXERCISE 30.1

1. What is the classification of the equation $f_{xx} + 2f_{xy} + f_{yy} = 0$?
2. Determine whether the following equations are elliptic or hyperbolic:
   $u_{xx} + 4u_{xy} + 2u_{yy} + 3u_x + 2u_y + u = 0$
3. Classify the equations (i) $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} + 5u_x + 3u = 0$ (Madras, 2000 S)

   (ii) $x^2 \frac{\partial^2 u}{\partial x^2} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ (P.T.U., 2000 S)

   (iii) $x^2 \frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial^2 u}{\partial x \partial y} + (x^2 + y^2) \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + u = 0$ (Anna, 2004)
4. In which parts of the (x, y) plane is the following equation elliptic?
   $y_1 u_{xx} + 2y_2 u_{xy} + u_{yy} + u_x + u_y = 0$ (Anna, 2005)

30.3 FINITE-DIFFERENCE APPROXIMATIONS TO DERIVATIVES

Consider a rectangular region R in the xy plane. Divide this region into a rectangular network of sides $\Delta x = h$ and $\Delta y = k$ as shown in Fig. 30.1. The points of intersection of the dividing lines are called *mesh points, nodal points* or *grid points*.

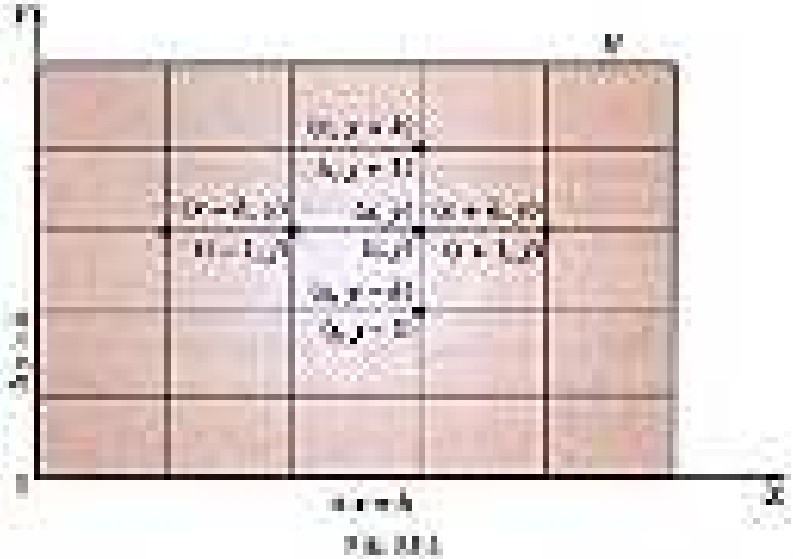

FIG. 30.1

Then we have the finite difference approximations for the partial derivatives in x-direction [§ 22.12]:

$$\frac{\partial u}{\partial x} = \frac{u(x + h, y) - u(x, y)}{h} \quad (30.1)$$

$$= \frac{u(x, y) - u(x - h, y)}{h} + O(h) = \frac{u(x + h, y) - u(x - h, y)}{2h} + O(h^2)$$

and $$\frac{\partial^2 u}{\partial x^2} = \frac{u(x + h, y) - 2u(x, y) + u(x - h, y)}{h^2} + O(h^2)$$

Writing $u(x, y) = u(ih, jh)$ or simply u_{ij}, the above approximations become

$$u_x = \frac{u_{i+1,j} - u_{ij}}{h} + O(h) \quad \ldots(1)$$

$$= \frac{u_{ij} - u_{i-1,j}}{h} + O(h) \quad \ldots(2)$$

$$= \frac{u_{i+1,j} - u_{i-1,j}}{2h} + O(h^2) \quad \ldots(3)$$

and $$u_{xx} = \frac{u_{i-1,j} - 2u_{ij} + u_{i+1,j}}{h^2} + O(h^2) \quad \ldots(4)$$

Similarly we have the approximations for the derivatives w.r.t. y:

$$u_y = \frac{u_{i,j+1} - u_{ij}}{k} + O(k) \quad \ldots(5)$$

$$= \frac{u_{ij} - u_{i,j-1}}{k} + O(k) \quad \ldots(6)$$

$$= \frac{u_{i,j+1} - u_{i,j-1}}{2k} + O(k^2) \quad \ldots(7)$$

and $$u_{yy} = \frac{u_{i,j-1} - 2u_{ij} + u_{i,j+1}}{k^2} + O(k^2) \quad \ldots(8)$$

Replacing the derivatives in any partial differential equation by their corresponding difference approximations (3) to (8), we obtain the finite difference analogues of the given equations.

33.2 ELLIPTIC EQUATIONS

The Laplace's equation

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

and the Poisson's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

are examples of elliptic partial differential equations. Laplace's equation arises in steady state flow and potential problems. Poisson's equation arises in fluid mechanics, electricity and magnetism and torsion problems.

Fig. 33.2

The solution of these equations is a function $u(x, y)$ which is satisfied at every point of a region R subject to certain boundary conditions specified on the closed curve C (Fig. 33.2).

In general, problems governing steady viscous flow, equilibrium stresses in elastic structures etc., lead to elliptic type of equations.

33.3 SOLUTION OF LAPLACE'S EQUATION*

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \ldots(1)$$

* See p. 815

Consider a rectangle region with boundary C. Divide this region into a network of square mesh of side h, as shown in Fig. 23.3 (assume that an exact sub-division of R is possible). Replacing the derivatives in (1) by their difference approximations, we have

$$\frac{1}{h^2}[u_{i-1,j}-2u_{i,j}+u_{i+1,j}]+\frac{1}{k^2}[u_{i,j-1}-2u_{i,j}+u_{i,j+1}]=0$$

or

$$u_{i,j}=\frac{1}{4}[u_{i-1,j}+u_{i+1,j}+u_{i,j-1}+u_{i,j+1}] \quad \ldots(2)$$

This shows that the value of $u_{i,j}$ at any interior mesh point is the average of its values at four neighbouring points to the left, right, above and below. (2) is called the **standard 5-point formula** which is exhibited in Fig. 23.4.

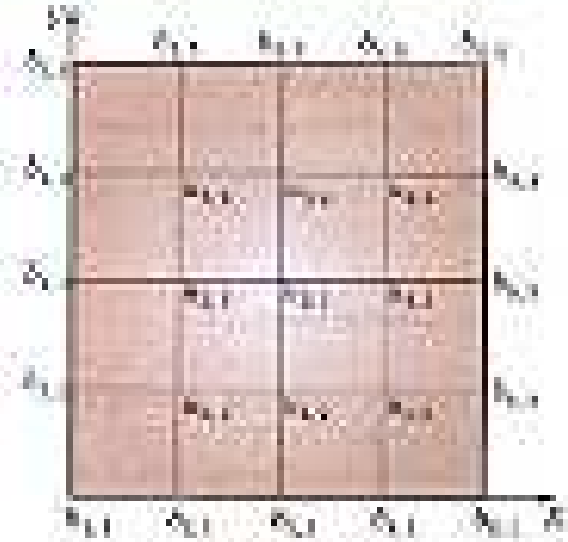

Fig. 23.3

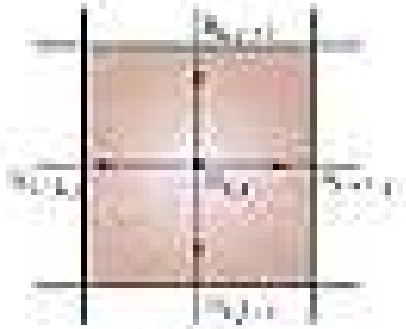

Fig. 23.4

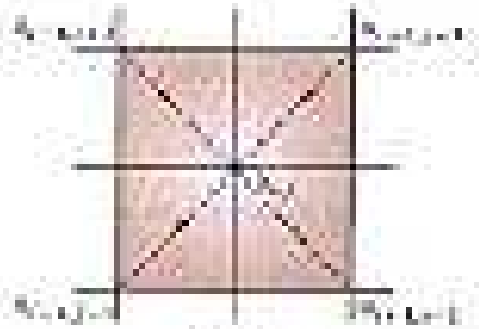

Fig. 23.5

Sometimes a formula similar to (2) is used which is given by

$$u_{i,j}=\frac{1}{8}[u_{i-1,j+1}+u_{i+1,j-1}+u_{i+1,j+1}+u_{i-1,j-1}] \quad \ldots(3)$$

This shows that the value of $u_{i,j}$ is the average of its values at the four neighbouring diagonal mesh points. (3) is called the **diagonal 5-point formula** which is represented in Fig. 23.5. Although (3) is less accurate than (2), yet it serves as a reasonably good approximation for obtaining the starting values at the mesh points.

Now to find the initial values at the interior mesh points, we first use diagonal five point formula (3) and compute $u_{1,1}, u_{2,2}, u_{3,3}, u_{4,4}$ and $u_{3,1}$ in this order. Thus we get,

$$u_{2,2}=\frac{1}{4}[u_{1,3}+u_{3,3}+u_{3,1}+u_{1,1}], \quad u_{4,4}=\frac{1}{4}[u_{3,5}+u_{5,3}+u_{5,5}+u_{3,3}]$$

$$u_{3,3}=\frac{1}{4}[u_{2,4}+u_{4,4}+u_{4,2}+u_{2,2}], \quad u_{4,2}=\frac{1}{4}[u_{3,3}+u_{5,3}+u_{5,1}+u_{3,1}]$$

$$u_{2,4}=\frac{1}{4}[u_{1,5}+u_{3,5}+u_{3,3}+u_{1,3}]$$

The values at the remaining interior points i.e. $u_{1,1}, u_{1,3}, u_{3,1}$ and $u_{3,5}$ are computed by the standard five-point formula (2). Thus, we obtain

$$u_{1,1}=\frac{1}{4}[u_{0,1}+u_{2,1}+u_{1,0}+u_{1,2}], \quad u_{1,3}=\frac{1}{4}[u_{0,3}+u_{2,3}+u_{1,2}+u_{1,4}]$$

$$u_{3,1}=\frac{1}{4}[u_{2,1}+u_{4,1}+u_{3,0}+u_{3,2}], \quad u_{3,5}=\frac{1}{4}[u_{2,5}+u_{4,5}+u_{3,4}+u_{3,6}]$$

Having found all the nine values of $u_{i,j}$ once, their accuracy is improved by repeated application of (2) in the form

$$u^{(n+1)}_{i,j}=\frac{1}{4}[u^{(n+1)}_{i-1,j}+u^{(n)}_{i+1,j}+u^{(n+1)}_{i,j-1}+u^{(n)}_{i,j+1}]$$

This formula utilizes the current iterate as soon as it is available and uses the newly computed values immediately from left to right during each iteration. This process is repeated until the difference of values in successive iterations is within the desired tolerance.

This is known as Gauss-Seidel iteration process.

Example 4.2.3 Solve the simultaneous equations $x_1 + x_2 + x_3 = 3$, $x_1 - x_2 + 2x_3 = 2$, $2x_1 + x_2 - x_3 = 1$ by Gauss-Seidel iteration method. (Perform only 4 iterations) (Madurai, 2008, 11 Marks, April)

Solution: Let x_1, x_2, x_3 be the values of x, y, z after any iteration. Given the iterative values of x, y, z are zero to start with.

$$x_1 = 3 - x_2 - x_3, \quad x_2 = 2 - x_1 + 2x_3, \quad x_3 = 1 - 2x_1 - x_2$$

Put the initial values of x having zero as the first approximation, we get $x_1 = 3$, $x_2 = 2 - 3 = -1$, $x_3 = 1 - 6 + 1 = -4$

Then to get another set of values, we use the recent values of x_1, x_2, x_3 and x_1.

Now we find their suitable values in the following order:

$$x_1 = \frac{1}{3}(3 - x_2 - x_3) = \frac{1}{3}(3 + 1 + 4) = \frac{8}{3} \quad \text{(First Iteration)}$$

$$x_2 = \frac{1}{3}(2 - x_1 + 2x_3) = \frac{1}{3}\left(2 - \frac{8}{3} - 8\right) = -\frac{26}{9} \quad \text{(Second Iteration)}$$

$$x_3 = \frac{1}{3}(1 - 2x_1 - x_2) = \frac{1}{3}\left(1 - \frac{16}{3} + \frac{26}{9}\right) = -\frac{13}{27} \quad \text{(Third Iteration)}$$

$$x_4 = \frac{1}{3}(1 - 2x_1 - x_2) = \frac{1}{3}\left(1 - \frac{16}{3} + \frac{26}{9}\right) = -\frac{13}{27} \quad \text{(Fourth Iteration)}$$

We start to find the iterative process using the formulae:

$$x_1^{(k+1)} = \frac{1}{3}(3 - x_2^{(k)} - x_3^{(k)})$$

$$x_2^{(k+1)} = \frac{1}{3}(2 - x_1^{(k+1)} + 2x_3^{(k)})$$

$$x_3^{(k+1)} = \frac{1}{3}(1 - 2x_1^{(k+1)} - x_2^{(k+1)})$$

$$x_4^{(k+1)} = \frac{1}{3}(1 - 2x_1^{(k+1)} - x_2^{(k+1)})$$

First iteration ($k = 0$):

$$x_1^{(1)} = \frac{1}{3}(3 - x_2^{(0)} - x_3^{(0)}) = \frac{1}{3}(3 - 0 + 0) = 1$$

$$x_2^{(1)} = \frac{1}{3}(2 - x_1^{(1)} + 2x_3^{(0)}) = \frac{1}{3}(2 - 1 + 0) = \frac{1}{3}$$

$$x_3^{(1)} = \frac{1}{3}(1 - 2x_1^{(1)} - x_2^{(1)}) = \frac{1}{3}\left(1 - 2 + \frac{1}{3}\right) = -\frac{2}{9}$$

$$x_4^{(1)} = \frac{1}{3}(1 - 2x_1^{(1)} - x_2^{(1)}) = \frac{1}{3}\left(1 - 2 + \frac{1}{3}\right) = -\frac{2}{9}$$

Second iteration ($k = 1$):

$$x_1^{(2)} = \frac{1}{3}(3 - x_2^{(1)} - x_3^{(1)}) = \frac{1}{3}\left(3 - \frac{1}{3} + \frac{2}{9}\right) = \frac{22}{27}$$

$$x_2^{(2)} = \frac{1}{3}(2 - x_1^{(2)} + 2x_3^{(1)}) = \frac{1}{3}\left(2 - \frac{22}{27} - \frac{4}{9}\right) = \frac{8}{27}$$

$$x_3^{(2)} = \frac{1}{3}(1 - 2x_1^{(2)} - x_2^{(2)}) = \frac{1}{3}\left(1 - \frac{44}{27} - \frac{8}{27}\right) = -\frac{25}{81}$$

$$x_4^{(2)} = \frac{1}{3}(1 - 2x_1^{(2)} - x_2^{(2)}) = \frac{1}{3}\left(1 - \frac{44}{27} - \frac{8}{27}\right) = -\frac{25}{81}$$

Third iteration :

$$u_1^{(3)} = \frac{1}{4}(1000 + 1059 + 500 + 1321) = 970$$

$$u_2^{(3)} = \frac{1}{4}(993 + 970 + 1000 + 1160) = 1031$$

$$u_3^{(3)} = \frac{1}{4}(1000 + 1156 + 970 + 1031) = 1039$$

$$u_4^{(3)} = \frac{1}{4}(1500 + 1269 + 1039 + 1039) = 1212$$

Fourth iteration :

$$u_1^{(4)} = \frac{1}{4}(1000 + 1135 + 970 + 1206) = 1078$$

$$u_2^{(4)} = \frac{1}{4}(1078 + 970 + 1031 + 1181) = 1065$$

$$u_3^{(4)} = \frac{1}{4}(1000 + 1182 + 970 + 1065) = 1054$$

$$u_4^{(4)} = \frac{1}{4}(1500 + 1206 + 1065 + 1054) = 1206$$

Fifth iteration :

$$u_1^{(5)} = \frac{1}{4}(1000 + 1103 + 970 + 1206) = 1070$$

$$u_2^{(5)} = \frac{1}{4}(1070 + 970 + 1065 + 1181) = 1071$$

$$u_3^{(5)} = \frac{1}{4}(1000 + 1182 + 970 + 1071) = 1056$$

$$u_4^{(5)} = \frac{1}{4}(1500 + 1206 + 1071 + 1056) = 1208$$

Similarly, $u_1^{(6)} = 943,\ u_2^{(6)} = 1005,\ u_3^{(6)} = 1000,\ u_4^{(6)} = 1130$
$u_1^{(7)} = 940,\ u_2^{(7)} = 1003,\ u_3^{(7)} = 1000,\ u_4^{(7)} = 1128$
$u_1^{(8)} = 939,\ u_2^{(8)} = 1002,\ u_3^{(8)} = 1000,\ u_4^{(8)} = 1127$
$u_1^{(9)} = 939,\ u_2^{(9)} = 1001,\ u_3^{(9)} = 1000,\ u_4^{(9)} = 1126$

Thus there is negligible difference between the values obtained in the eighth and ninth iterations.
Hence $u_1 = 939,\ u_2 = 1001,\ u_3 = 1000$ and $u_4 = 1126$.

Example 33.3. Given the values of $u(x, y)$ on the boundary of the square in the Fig. 33.7, evaluate the function $u(x, y)$ satisfying the Laplace equation. Use $h = 1$ at the pivotal points of the figure.

(Bhopal, 2009 ; M.D.U., 2008)

Solution. To get the initial values of u_1, u_2, u_3, u_4, we assume $u_4 = 0$. Then

$$u_1 = \frac{1}{4}(1000 + 0 + 2000 + 2000) = 1500 \qquad \text{(Big. formula)}$$

$$u_2 = \frac{1}{4}(1000 + 500 + 1500 + 0) = 625 \qquad \text{(Std. formula)}$$

$$u_3 = \frac{1}{4}(2000 + 0 + 1000 + 500) = 875 \qquad \text{(Std. formula)}$$

$$u_4 = \frac{1}{4}(875 + 0 + 625 + 0) = 375 \qquad \text{(Std. formula)}$$

We now use the successive iterations, using the formulae

$$u_1^{(n+1)} = \frac{1}{4}[2000 + u_2^{(n)} + 2000 + u_3^{(n)}]$$

$$u_2^{(n+1)} = \frac{1}{4}[u_1^{(n+1)} + 500 + 2000 + u_4^{(n)}]$$

$$u_2^{(n+1)} = \frac{1}{4}[1000 + u_1^{(n)} + u_3^{(n+1)} + 500]$$

$$u_3^{(n+1)} = \frac{1}{4}[u_2^{(n+1)} + 0 + u_4^{(n+1)} + 0]$$

First iteration: (put $n = 0$)

$$u_1^{(1)} = \frac{1}{4}(1000 + 625 + 1000 + 625) = 1125$$

$$u_2^{(1)} = \frac{1}{4}(1125 + 500 + 1000 + 500) = 781$$

$$u_3^{(1)} = \frac{1}{4}(1000 + 500 + 1125 + 500) = 1031$$

$$u_4^{(1)} = \frac{1}{4}(1000 + 0 + 781 + 0) = 456$$

Second iteration: (put $n = 1$)

$$u_1^{(2)} = \frac{1}{4}(1000 + 500 + 1000 + 1031) = 1133$$

$$u_2^{(2)} = \frac{1}{4}(1133 + 500 + 1000 + 456) = 772$$

$$u_3^{(2)} = \frac{1}{4}(1000 + 456 + 1133 + 500) = 1022$$

$$u_4^{(2)} = \frac{1}{4}(1000 + 0 + 772 + 0) = 443$$

Third iteration: (put $n = 2$)

$$u_1^{(3)} = \frac{1}{4}(1000 + 500 + 1000 + 1022) = 1131$$

$$u_2^{(3)} = \frac{1}{4}(1131 + 500 + 1000 + 443) = 768$$

$$u_3^{(3)} = \frac{1}{4}(1000 + 443 + 1131 + 500) = 1018$$

$$u_4^{(3)} = \frac{1}{4}(1000 + 0 + 768 + 0) = 442$$

Similarly, $u_1^{(4)} = 1127$, $u_2^{(4)} = 781$, $u_3^{(4)} = 1041$, $u_4^{(4)} = 458$

and $u_1^{(5)} = 1208$, $u_2^{(5)} = 798.5$, $u_3^{(5)} = 1042.5$, $u_4^{(5)} = 458.25$

Thus there is no significant difference between the fourth and fifth iteration values.

Hence $u_1 = 1208$, $u_2 = 798$, $u_3 = 1042$ and $u_4 = 458$.

Example 33.8. Solve the Laplace equation $u_{xx} + u_{yy} = 0$ given in Fig. (33.8).

Solution. We first find the initial values in the following order :

$$u_2 = \frac{1}{4}(0 + 37 + 31 + 32.5) = 33.6 \qquad \text{(Diag. formula)}$$

$$u_3 = \frac{1}{4}(0 + 32.5 + 0 + 17) = 7.4 \qquad \text{(Diag. formula)}$$

$$u_4 = \frac{1}{4}(32.5 + 33.6 + 17 + 31) = 17.53 \qquad \text{(Diag. formula)}$$

$$u_5 = \frac{1}{4}(32.5 + 0 + 0 + 15.31) = 8.33 \qquad \text{(Diag. formula)}$$

$$u_6 = \frac{1}{4}(32.5 + 0 + 31 + 32.5) = 23.63 \qquad \text{(Diag. formula)}$$

$$u_7 = \frac{1}{4}(17 + 32.5 + 7.4 + 17.53) = 23.63 \qquad \text{(Std. formula)}$$

$$u_8 = \frac{1}{4}(17.4 + 8.2 + 0 + 32.5) = 6.02 \qquad \text{(Std. formula)}$$

$$\sigma_{yy}=\begin{vmatrix} 1.7 & 5.2 & 10.7 & 8.4 \\ 3.9 & 2.6 & 6.1 & 11.3 \\ 0.8 & 4.4 & 2.9 & 7.5 \\ 6.0 & 1.3 & 5.5 & 2.2 \end{vmatrix} \quad \text{(kPa, invocation)}$$

$$\sigma_{yy}=\begin{vmatrix} 12.6 & 3.1 & 9.4 & 4.7 \\ 2.5 & 10.8 & 1.6 & 6.3 \\ 7.4 & 0.9 & 11.2 & 3.8 \\ 4.1 & 6.7 & 2.4 & 9.5 \end{vmatrix} \quad \text{(kPa, dependent)}$$

Three successive cast-iron blocks were prepared using the Greenleaf Research Foundry's standard sand-mould procedure. Each block was poured from the same heat of metal, allowed to cool in the mould for 48 hours, and then shaken out and cleaned by the foundry's fettling crew. The castings were numbered GRF-1, GRF-2, and GRF-3 in the order they were poured.

$$\mathbf{K}^{(1,1)}=\begin{bmatrix} 2.0 & 0.5 & 1.3 & 0.2 \\ 0.5 & 1.8 & 0.4 & 1.1 \\ 1.3 & 0.4 & 2.2 & 0.6 \\ 0.2 & 1.1 & 0.6 & 1.5 \end{bmatrix}$$

$$\mathbf{K}^{(1,2)}=\begin{bmatrix} 0.9 & 0.3 & 0.7 & 0.1 \\ 0.3 & 1.0 & 0.2 & 0.6 \\ 0.7 & 0.2 & 1.2 & 0.3 \\ 0.1 & 0.6 & 0.3 & 0.8 \end{bmatrix}$$

$$\mathbf{K}^{(1,3)}=\begin{bmatrix} 0.4 & 0.1 & 0.3 & 0.0 \\ 0.1 & 0.5 & 0.1 & 0.3 \\ 0.3 & 0.1 & 0.6 & 0.2 \\ 0.0 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

$$\mathbf{K}^{(2,1)}=\begin{bmatrix} 0.9 & 0.3 & 0.7 & 0.1 \\ 0.3 & 1.0 & 0.2 & 0.6 \\ 0.7 & 0.2 & 1.2 & 0.3 \\ 0.1 & 0.6 & 0.3 & 0.8 \end{bmatrix}$$

$$\mathbf{K}^{(2,2)}=\begin{bmatrix} 1.8 & 0.5 & 1.2 & 0.3 \\ 0.5 & 1.7 & 0.4 & 1.0 \\ 1.2 & 0.4 & 2.0 & 0.5 \\ 0.3 & 1.0 & 0.5 & 1.4 \end{bmatrix}$$

$$\mathbf{K}^{(2,3)}=\begin{bmatrix} 0.8 & 0.2 & 0.6 & 0.1 \\ 0.2 & 0.9 & 0.1 & 0.5 \\ 0.6 & 0.1 & 1.1 & 0.3 \\ 0.1 & 0.5 & 0.3 & 0.7 \end{bmatrix}$$

$$\mathbf{K}^{(3,1)}=\begin{bmatrix} 0.4 & 0.1 & 0.3 & 0.0 \\ 0.1 & 0.5 & 0.1 & 0.3 \\ 0.3 & 0.1 & 0.6 & 0.2 \\ 0.0 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

$$\mathbf{K}^{(3,2)}=\begin{bmatrix} 0.8 & 0.2 & 0.6 & 0.1 \\ 0.2 & 0.9 & 0.1 & 0.5 \\ 0.6 & 0.1 & 1.1 & 0.3 \\ 0.1 & 0.5 & 0.3 & 0.7 \end{bmatrix}$$

$$\mathbf{K}^{(3,3)}=\begin{bmatrix} 1.6 & 0.4 & 1.1 & 0.2 \\ 0.4 & 1.5 & 0.3 & 0.9 \\ 1.1 & 0.3 & 1.9 & 0.4 \\ 0.2 & 0.9 & 0.4 & 1.3 \end{bmatrix}$$

Where $\mathbf{K}^{(i,j)}$ is the 4×4 cross-block coupling the nodes of element i to those of element j.

$$\mathbf{K}^{(1)}=\begin{bmatrix} 2.0 & 0.5 & 1.3 & 0.2 & 0.9 & 0.3 & 0.7 & 0.1 \\ 0.5 & 1.8 & 0.4 & 1.1 & 0.3 & 1.0 & 0.2 & 0.6 \\ 1.3 & 0.4 & 2.2 & 0.6 & 0.7 & 0.2 & 1.2 & 0.3 \\ 0.2 & 1.1 & 0.6 & 1.5 & 0.1 & 0.6 & 0.3 & 0.8 \\ 0.9 & 0.3 & 0.7 & 0.1 & 0.4 & 0.1 & 0.3 & 0.0 \\ 0.3 & 1.0 & 0.2 & 0.6 & 0.1 & 0.5 & 0.1 & 0.3 \\ 0.7 & 0.2 & 1.2 & 0.3 & 0.3 & 0.1 & 0.6 & 0.2 \\ 0.1 & 0.6 & 0.3 & 0.8 & 0.0 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

$$\mathbf{K}^{(2)}=\begin{bmatrix} 0.9 & 0.3 & 0.7 & 0.1 & 0.8 & 0.2 & 0.6 & 0.1 \\ 0.3 & 1.0 & 0.2 & 0.6 & 0.2 & 0.9 & 0.1 & 0.5 \\ 0.7 & 0.2 & 1.2 & 0.3 & 0.6 & 0.1 & 1.1 & 0.3 \\ 0.1 & 0.6 & 0.3 & 0.8 & 0.1 & 0.5 & 0.3 & 0.7 \\ 0.4 & 0.1 & 0.3 & 0.0 & 1.6 & 0.4 & 1.1 & 0.2 \\ 0.1 & 0.5 & 0.1 & 0.3 & 0.4 & 1.5 & 0.3 & 0.9 \\ 0.3 & 0.1 & 0.6 & 0.2 & 1.1 & 0.3 & 1.9 & 0.4 \\ 0.0 & 0.3 & 0.2 & 0.4 & 0.2 & 0.9 & 0.4 & 1.3 \end{bmatrix}$$

$$\mathbf{K}^{(3)}=\begin{bmatrix} 0.4 & 0.1 & 0.3 & 0.0 & 0.7 & 0.2 & 0.5 & 0.1 \\ 0.1 & 0.5 & 0.1 & 0.3 & 0.2 & 0.8 & 0.1 & 0.4 \\ 0.3 & 0.1 & 0.6 & 0.2 & 0.5 & 0.1 & 0.9 & 0.3 \\ 0.0 & 0.3 & 0.2 & 0.4 & 0.1 & 0.4 & 0.3 & 0.6 \\ 0.8 & 0.2 & 0.6 & 0.1 & 0.4 & 0.1 & 0.3 & 0.0 \\ 0.2 & 0.9 & 0.1 & 0.5 & 0.1 & 0.4 & 0.1 & 0.3 \\ 0.6 & 0.1 & 1.1 & 0.3 & 0.3 & 0.1 & 0.6 & 0.2 \\ 0.1 & 0.5 & 0.3 & 0.7 & 0.0 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

Where $\mathbf{K}^{(e)}$ is the 8×8 contribution of element e to the foundry's assembly.

$$\mathbf{K}^{(1)}=\begin{bmatrix} 2.0 & 0.5 & 1.3 & 0.2 & 0.9 & 0.3 & 0.7 & 0.1 \\ 0.5 & 1.8 & 0.4 & 1.1 & 0.3 & 1.0 & 0.2 & 0.6 \\ 1.3 & 0.4 & 2.2 & 0.6 & 0.7 & 0.2 & 1.2 & 0.3 \\ 0.2 & 1.1 & 0.6 & 1.5 & 0.1 & 0.6 & 0.3 & 0.8 \\ 0.9 & 0.3 & 0.7 & 0.1 & 0.4 & 0.1 & 0.3 & 0.0 \\ 0.3 & 1.0 & 0.2 & 0.6 & 0.1 & 0.5 & 0.1 & 0.3 \\ 0.7 & 0.2 & 1.2 & 0.3 & 0.3 & 0.1 & 0.6 & 0.2 \\ 0.1 & 0.6 & 0.3 & 0.8 & 0.0 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

$$\mathbf{K}^{(2)}=\begin{bmatrix} 0.9 & 0.3 & 0.7 & 0.1 & 0.8 & 0.2 & 0.6 & 0.1 \\ 0.3 & 1.0 & 0.2 & 0.6 & 0.2 & 0.9 & 0.1 & 0.5 \\ 0.7 & 0.2 & 1.2 & 0.3 & 0.6 & 0.1 & 1.1 & 0.3 \\ 0.1 & 0.6 & 0.3 & 0.8 & 0.1 & 0.5 & 0.3 & 0.7 \\ 0.4 & 0.1 & 0.3 & 0.0 & 1.6 & 0.4 & 1.1 & 0.2 \\ 0.1 & 0.5 & 0.1 & 0.3 & 0.4 & 1.5 & 0.3 & 0.9 \\ 0.3 & 0.1 & 0.6 & 0.2 & 1.1 & 0.3 & 1.9 & 0.4 \\ 0.0 & 0.3 & 0.2 & 0.4 & 0.2 & 0.9 & 0.4 & 1.3 \end{bmatrix}$$

$$\mathbf{K}^{(3)}=\begin{bmatrix} 0.4 & 0.1 & 0.3 & 0.0 & 0.7 & 0.2 & 0.5 & 0.1 \\ 0.1 & 0.5 & 0.1 & 0.3 & 0.2 & 0.8 & 0.1 & 0.4 \\ 0.3 & 0.1 & 0.6 & 0.2 & 0.5 & 0.1 & 0.9 & 0.3 \\ 0.0 & 0.3 & 0.2 & 0.4 & 0.1 & 0.4 & 0.3 & 0.6 \\ 0.8 & 0.2 & 0.6 & 0.1 & 0.4 & 0.1 & 0.3 & 0.0 \\ 0.2 & 0.9 & 0.1 & 0.5 & 0.1 & 0.4 & 0.1 & 0.3 \\ 0.6 & 0.1 & 1.1 & 0.3 & 0.3 & 0.1 & 0.6 & 0.2 \\ 0.1 & 0.5 & 0.3 & 0.7 & 0.0 & 0.3 & 0.2 & 0.4 \end{bmatrix}$$

Second invocation (grid $n = 5$):

$$\mathbf{K}^{(1)}=\begin{bmatrix} 4.1 & 1.2 & 2.6 & 0.5 & 1.8 & 0.6 & 1.4 & 0.2 \\ 1.2 & 3.7 & 0.9 & 2.2 & 0.6 & 2.0 & 0.4 & 1.3 \\ 2.6 & 0.9 & 4.4 & 1.3 & 1.4 & 0.4 & 2.4 & 0.7 \\ 0.5 & 2.2 & 1.3 & 3.0 & 0.2 & 1.3 & 0.7 & 1.7 \\ 1.8 & 0.6 & 1.4 & 0.2 & 0.9 & 0.3 & 0.7 & 0.1 \\ 0.6 & 2.0 & 0.4 & 1.3 & 0.3 & 1.0 & 0.2 & 0.6 \\ 1.4 & 0.4 & 2.4 & 0.7 & 0.7 & 0.2 & 1.2 & 0.3 \\ 0.2 & 1.3 & 0.7 & 1.7 & 0.1 & 0.6 & 0.3 & 0.8 \end{bmatrix}$$

$$\mathbf{K}^{(2)}=\begin{bmatrix} 1.8 & 0.6 & 1.4 & 0.2 & 1.7 & 0.5 & 1.3 & 0.2 \\ 0.6 & 2.0 & 0.4 & 1.3 & 0.5 & 1.8 & 0.3 & 1.1 \\ 1.4 & 0.4 & 2.4 & 0.7 & 1.3 & 0.3 & 2.2 & 0.6 \\ 0.2 & 1.3 & 0.7 & 1.7 & 0.2 & 1.1 & 0.6 & 1.5 \\ 0.9 & 0.3 & 0.7 & 0.1 & 0.8 & 0.2 & 0.6 & 0.1 \\ 0.3 & 1.0 & 0.2 & 0.6 & 0.2 & 0.9 & 0.1 & 0.5 \\ 0.7 & 0.2 & 1.2 & 0.3 & 0.6 & 0.1 & 1.1 & 0.3 \\ 0.1 & 0.6 & 0.3 & 0.8 & 0.1 & 0.5 & 0.3 & 0.7 \end{bmatrix}$$

$$\mathbf{K}^{(3)}=\begin{bmatrix} 0.9 & 0.3 & 0.7 & 0.1 & 1.6 & 0.4 & 1.1 & 0.2 \\ 0.3 & 1.0 & 0.2 & 0.6 & 0.4 & 1.5 & 0.3 & 0.9 \\ 0.7 & 0.2 & 1.2 & 0.3 & 1.1 & 0.3 & 1.9 & 0.4 \\ 0.1 & 0.6 & 0.3 & 0.8 & 0.2 & 0.9 & 0.4 & 1.3 \\ 0.4 & 0.1 & 0.3 & 0.0 & 0.7 & 0.2 & 0.5 & 0.1 \\ 0.1 & 0.5 & 0.1 & 0.3 & 0.2 & 0.8 & 0.1 & 0.4 \\ 0.3 & 0.1 & 0.6 & 0.2 & 0.5 & 0.1 & 0.9 & 0.3 \\ 0.0 & 0.3 & 0.2 & 0.4 & 0.1 & 0.4 & 0.3 & 0.6 \end{bmatrix}$$

$$u_1^{(1)} = \frac{1}{4}(0.00 + 12.50 + 31.25 + 16.00) = 14.94$$

$$u_2^{(1)} = \frac{1}{4}(11.56 + 17.50 + 14.06 + 25) = 16.28$$

$$u_3^{(1)} = \frac{1}{4}(0 + 6.50 + 8.7 + 11.06) = 6.56$$

$$u_4^{(1)} = \frac{1}{4}(0.00 + 12.50 + 12.5 + 16.25) = 11.31$$

$$u_5^{(1)} = \frac{1}{4}(11.13 + 16.50 + 12.5 + 17) = 14.28$$

Third iteration (put $n = 2$)

$$u_1^{(2)} = \frac{1}{4}(0 + 11.1 + 0.00 + 12.50) = 7.90$$

$$u_2^{(2)} = \frac{1}{4}(12.50 + 17 + 31.25 + 17.50) = 19.063$$

$$u_3^{(2)} = \frac{1}{4}(12.50 + 16.9 + 16.25 + 21.31) = 17.00$$

$$u_4^{(2)} = \frac{1}{4}(0 + 7.90 + 6.56 + 12.50) = 6.99$$

$$u_5^{(2)} = \frac{1}{4}(6.56 + 19.06 + 11.31 + 16.50) = 11.36$$

$$v_1^{(2)} = \frac{1}{4}(31.90 + 17.06 + 14.30 + 21) = 16.07$$

$$v_2^{(2)} = \frac{1}{4}(0 + 6.66 + 6.7 + 11.18) = 5.31$$

$$v_3^{(2)} = \frac{1}{4}(6.81 + 11.50 + 12.1 + 14.06) = 11.28$$

$$v_4^{(2)} = \frac{1}{4}(11.20 + 16.03 + 12.9 + 17) = 14.28$$

Similarly, $u_1^{(3)} = 7.82, u_2^{(3)} = 12.85, u_3^{(3)} = 17.65, u_4^{(3)} = 6.68, u_5^{(3)} = 11.34, v_1^{(3)} = 16.26,$
$v_2^{(3)} = 6.68, v_3^{(3)} = 11.36, v_4^{(3)} = 14.32$

$u_1^{(4)} = 7.82, u_2^{(4)} = 12.85, u_3^{(4)} = 17.80, u_4^{(4)} = 6.66, u_5^{(4)} = 11.35, v_1^{(4)} = 16.20,$
$v_2^{(4)} = 6.64, v_3^{(4)} = 11.35, v_4^{(4)} = 14.34$

21.5.6 SOLUTION OF POISSON'S EQUATION*

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y) \qquad \ldots(1)$$

This is an elliptic equation which can be solved numerically at the interior mesh points of a square network when the boundary values are known. The standard 5-point formula for (1) takes the form

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = h^2 f(ih, jh) \qquad \ldots(2)$$

By applying (2) at each mesh point, we arrive at linear equations in the pivotal values $u_{i,j}$. These equations can be solved by Gauss-Seidal iteration method (p. 958).

Example 21.5. Solve the Poisson equation $u_{xx} + u_{yy} = x^2 + y^2$ in $0 < x < 3$, $0 < y < 3$ given that $u(0, y) = 0$, $u(3, y) = 9 + y^2$, $u(x, 0) = x^2$, $u(x, 3) = x^2 + 9$ and $h = 1$. (Anna 2002)

Solution. Here $h = 1$, $u_{i-1,j} + u_{i+1,j}$ (Fig. 21.5).

* See p. 862.

The standard 5-point formula for the given equation is

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = h^2 f(x_i, y_j) = h^2(-2x_i y_j) = -\tfrac{1}{8}x_i y_j \quad \ldots(1)$$

For $i = 1, j = 2$, (1) gives $0 + u_2 + u_3 + 100 - 4u_1 = -\frac{1}{8}$

i.e., $$-4u_1 + u_2 + u_3 = -100\tfrac{1}{8} \quad \ldots(2)$$

For $i = 2, j = 2$, (1) gives $u_1 + 200 + u_4 + 100 - 4u_2 = -\frac{1}{4}$

i.e., $$u_1 - 4u_2 + u_4 = -300\tfrac{1}{4} \quad \ldots(3)$$

For $i = 1, j = 1$, (1) gives $0 + u_1 + 0 + u_3 - 4u_3 = -\frac{1}{8}$

i.e., $$u_1 - 4u_3 + u_4 = -\tfrac{1}{8} \quad \ldots(4)$$

For $i = 2, j = 1$, (1) gives $u_3 + 100 + u_2 - 4u_4 = -\frac{1}{4}$

$$u_2 + u_3 - 4u_4 = -100\tfrac{1}{4} \quad \ldots(5)$$

Subtracting (4) from (3), $-3u_2 + 3u_3 = 300$ i.e., $u_3 = u_2 + 100$

Then (2) becomes $3u_2 - 4u_4 = -300$ $\quad \ldots(6)$

and (5) becomes $2u_2 - 4u_4 = -\frac{1}{2}$ $\quad \ldots(7)$

Now (6) $\times$ (1) gives $14u_2 + 4u_4 = -812$ $\quad \ldots(8)$

(8) + (7) gives $-12u_2 = -612$

Thus $$u_2 = 612/12 = 51.0000 = u_4$$

From (6), $$u_1 = \frac{1}{3}(4u_2 - 300) = 101.6667$$

From (4), $$u_3 = \frac{1}{3}\left(4u_4 + \frac{1}{8}\right) = 203.7917.$$

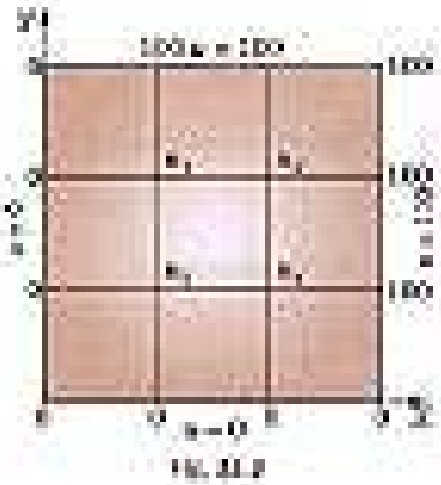

Fig. 21.9

Example 21.6. Solve the partial differential equation $\nabla^2 u = -10(x^2 + y^2 + 10)$ over the square with sides $x = 0$, $y = 0$, $x = 3$, $y = 3$ with $u = 0$ on the boundary and mesh length $= 1$.

(Anna. 2003 ; P.T.U. 2007 ; Delhi 2002)

Solution. Here $h = 1$ (Fig. 21.10)

The standard 5-point formula for the given equation is

$$u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j} = -10(x_i^2 + y_j^2 + 10) \quad \ldots(1)$$

For $i = 1, j = 2$, (1) gives

$$0 + u_2 + 0 + u_3 - 4u_1 = -10(1 + 4 + 10)$$

i.e., $$u_1 = \frac{1}{4}(u_2 + u_3 + 150) \quad \ldots(2)$$

For $i = 2, j = 2$, (1) gives $u_2 = \frac{1}{4}(u_1 + u_4 + 180)$ $\quad \ldots(3)$

For $i = 1, j = 1$, we have $u_3 = \frac{1}{4}(u_1 + u_4 + 130)$ $\quad \ldots(4)$

For $i = 2, j = 1$, we have $u_4 = \frac{1}{4}(u_3 + u_2 + 160)$ $\quad \ldots(5)$

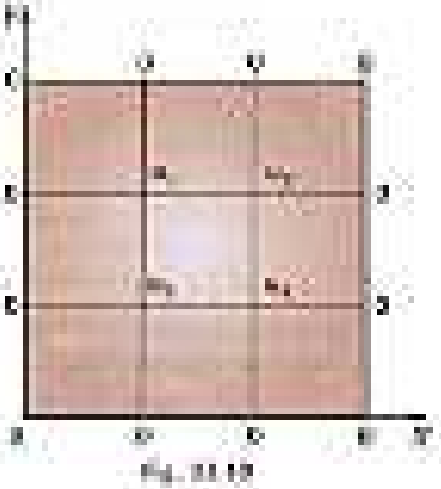

Fig. 21.10

Equations (3) and (5) show that $u_3 = u_4$. Thus the above equations reduce to

$$u_1 = \frac{1}{4}(u_2 + u_3 + 150) \quad \ldots(6)$$

$$u_2 = \frac{1}{2}(u_1 + 90) \quad \ldots(7)$$

$$u_3 = \frac{1}{2}(u_1 + 80) \quad \ldots(8)$$

Now let us solve these equations by Gauss-Seidal iteration method.

First iteration : Starting from the approximations $u_2 = 0$, $u_3 = 0$, we obtain

$$u_1^{(1)} = 37.5. \text{ Then } u_2^{(1)} = \frac{1}{2}(37.5 + 90) = 63\tfrac{1}{2} ; \; u_3^{(1)} = \frac{1}{2}(37.5 + 80) = 58\tfrac{3}{4}$$

Second iteration:

$$u_1^{(2)} = \frac{1}{4}(0 + 48 + 150 + 0) = 50 \quad u_2^{(2)} = \frac{1}{4}(50 + 0 + 150 + 0) = 50 \quad u_3^{(2)} = \frac{1}{4}(0 + 0 + 50 + 50) = 25$$

Third iteration:

$$u_1^{(3)} = \frac{1}{4}(0 + 25 + 150 + 0) = 43.75 \quad u_2^{(3)} = \frac{1}{4}(43.75 + 0 + 150 + 0) = 48.44 \quad u_3^{(3)} = \frac{1}{4}(43.75 + 48.44 + 50 + 50) = 48.05$$

Fourth iteration:

$$u_1^{(4)} = \frac{1}{4}(0 + 24.03 + 150 + 0) = 43.51 \quad u_2^{(4)} = \frac{1}{4}(43.51 + 48.05 + 150 + 0) = 47.89 \quad u_3^{(4)} = \frac{1}{4}(43.51 + 47.89 + 50 + 50) = 47.85$$

Fifth iteration:

$$u_1^{(5)} = \frac{1}{4}(0 + 23.93 + 150 + 0) = 43.48 \quad u_2^{(5)} = \frac{1}{4}(43.48 + 47.85 + 150 + 0) = 47.83 \quad u_3^{(5)} = \frac{1}{4}(43.48 + 47.83 + 50 + 50) = 47.83$$

Since these values are the same as those of the fourth iteration, we have

$u_1 = 75$, $u_2 = 62.5$, $u_3 = 67.5$ and $u_4 = 35$.

EXERCISE 21.3

1. Solve the equation $u_{xx} + u_{yy} = 0$ for the following square mesh with boundary values as shown in Fig. 21.12. Obtain one of the iterations between the successive values at any point to two decimal places. (Delhi, 2005)

2. Solve $\nabla^2 u = 0$ under the conditions $u = 0$ on $x = 0$, $u = 0$ on $y = 0$, $u = 100$ on $x = 4$, $u = 50 + 12.5y$ on $y = 4$ for $0 < x < 4$, $0 < y < 4$. (Nagpur, 2005)

3. Solve the elliptic equation $u_{xx} + u_{yy} = 0$ for the square mesh with boundary values as shown in Fig. 21.13. Iterate until the successive differences between values at any point are less than 0.001. (Delhi, 2006)

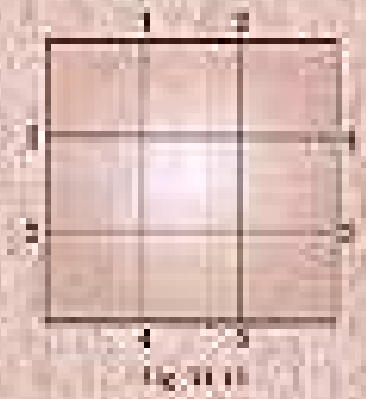

Fig. 21.12

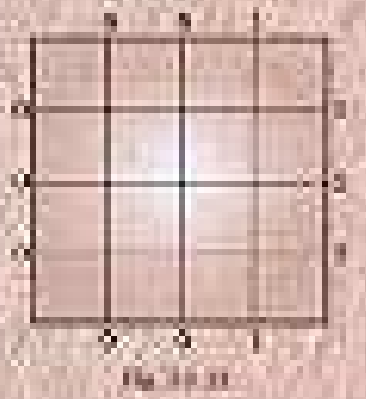

Fig. 21.13

4. Using central difference approximation, solve $\nabla^2 u = 0$ at the nodal points of the square grid of Fig. 21.14 subject to the boundary values indicated. (V.T.U., 2008)

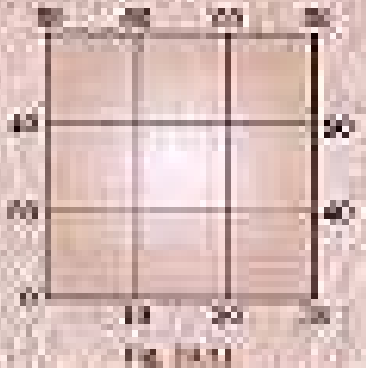

Fig. 21.14

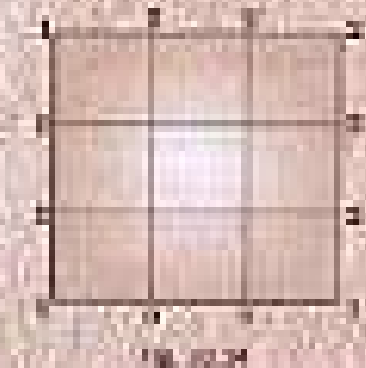

Fig. 21.15

5. Solve $u_{xx} + u_{yy} = 0$ for the square mesh with boundary values as shown in Fig. 21.15. Iterate till the successive values are correct to two decimal places.

6. Solve the Laplace's equation $u_{xx} + u_{yy} = 0$ for the domain of Fig. 22.13.

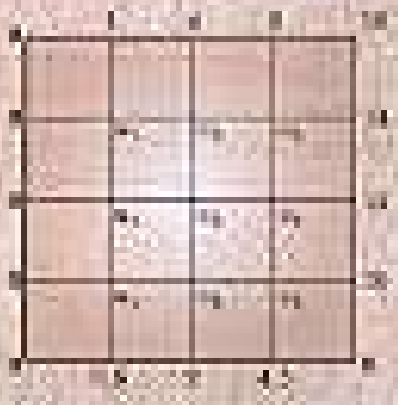

Fig. 22.13

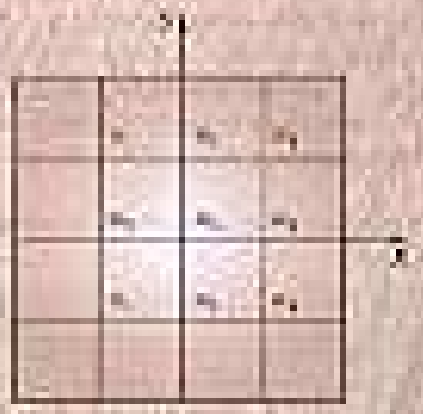

Fig. 22.14

7. Solve the Poisson's equation $\nabla^2 u = 10(x^2 + y^2)$ for the square mesh of Fig. 22.14 with $u(x, y) = 0$ on the boundary and mesh length $h = 1$. ANSWERS (Refer to Author's Book)

Note. Solution of elliptic equations by Relaxation method is given in author's book "Numerical Methods in Engineering and Science".

22.12 PARABOLIC EQUATIONS

The one-dimensional heat conduction equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

is a well known example of parabolic partial differential equations. The solution of this equation is a temperature function $u(x, t)$ which is defined for values of x from 0 to l and for values of time t from 0 to ∞. The solution is not defined over a rectangular domain but advances in one open-ended region from initial values, satisfying the prescribed boundary conditions. (Fig. 22.15).

In general, the study of pressure waves in a fluid, propagation of heat and unsteady state problems lead to parabolic type of equations.

Fig. 22.15

22.13 SOLUTION OF HEAT EQUATION

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \qquad \ldots(1)$$

where $c^2 = k/\rho s$ is the diffusivity of the substance (in cm^2/sec).

Consider a rectangular mesh in the xt plane with spacing h along x direction and k along time t direction. Denoting a mesh point $(x, t) = (ih, jk)$ as simply (i, j), we have

$$\frac{\partial u}{\partial t} = \frac{u_{i,j+1} - u_{i,j}}{k} \qquad \ldots(2)$$

and

$$\frac{\partial^2 u}{\partial x^2} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2} \qquad \ldots(3)$$

Substituting these in (1), we obtain

$$u_{i,j+1} - u_{i,j} = \frac{k^2}{h^2}\left(u_{i-1,j} - 2u_{i,j} + u_{i+1,j}\right)$$

or $$u_{i,j+1} = \alpha u_{i-1,j} + (1 - 2\alpha) u_{i,j} + \alpha u_{i+1,j} \quad \ldots(2)$$

where $\alpha = k/h^2$ is the mesh ratio parameter.

This formula enables us to determine the value of u at the $(i, j+1)$th mesh point in terms of the known function values at the points x_{i-1}, x_i and x_{i+1} at the instant t_j. It is a relation between the function values at the two time levels $j+1$ and j and is therefore, called a 3-level formula. Its schematic form (2) is shown in Fig. 32.14 which is called the Schmidt explicit formula which is valid only for $0 < \alpha \le \frac{1}{2}$.

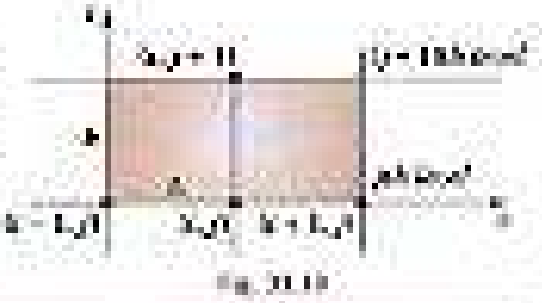

Fig. 32.14

Obs. In particular when $\alpha = \frac{1}{2}$, (2) reduces to

$$u_{i,j+1} = \frac{1}{2}(u_{i-1,j} + u_{i+1,j}) \quad \ldots(3)$$

which shows that the value of u at any mesh point on the $(j+1)$th row is the mean of the values of u at the points on the jth row, one step to the left and one step to the right of the point. This relation known as Bendre-Schmidt recurrence relation, gives the values of u on the interval mesh points on the $(j+1)$th row in terms of boundary conditions.

Note. The other formulae viz. Crank-Nicolson formula and the Bini-Pernicki formula are given in author's book 'Numerical Methods in Engineering and Science'.

Example 32.3. Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ in $0 < x < 5$, $t > 0$ given that $u(x, 0) = 20$, $u(0, t) = 0$, $u(5, t) = 100$, $t > 0$. Compute u for the time step with $h = 1$ by Crank-Nicolson method. (Meerut, 2009)

Solution. Here $c^2 = 1$ and $k = 1$.

Taking h (i.e., $c^2 k/h^2$) $= 1$, we get $k = 1$.

Also we have

x / t	0	1	2	3	4	5
0	0	20	20	20	20	100
1	0	u_1	u_2	u_3	u_4	100

The Crank-Nicolson formula becomes

$$4u_{i,j+1} = u_{i-1,j+1} + u_{i+1,j+1} + u_{i-1,j} + u_{i+1,j}$$

$4u_1 = 0 + 20 + 0 + u_2$ i.e. $4u_1 - u_2 = 20$ ...(1)

$4u_2 = 20 + 20 + u_1 + u_3$ i.e. $4u_1 - 4u_2 + u_3 = -40$ ...(2)

$4u_3 = 20 + 20 + u_2 + u_4$ i.e. $u_2 - 4u_3 + u_4 = -40$ ...(3)

$4u_4 = 20 + 100 + u_3 + 100$ i.e. $u_3 - 4u_4 = -220$ ...(4)

First (1) − 4(2) gives $15u_1 - 4u_2 = 180$ ...(5)

4(2) + (1) gives $-u_2 + 15u_3 = -220$ ...(6)

Then 4(5) − (6) gives $59u_2 = 1220$ i.e. $u_2 = 20.2$

From (6), we get $4u_3 = 20 + 20.2 - 160$ i.e. $u_3 = 20.70$

From (3), $4u_2 = 20 + 20.2$ i.e. $u_1 = 20.05$

From (4), $4u_4 = 20 + 20.70$ i.e. $u_4 = 62.68$

Thus the required values are 10.05, 20.2, 20.70 and 62.68.

Example 32.4. Solve the boundary value problem $u_t = u_{xx}$ under the conditions $u(0, t) = 0$, $u(1, t) = 0$ and $u(x, 0) = \sin \pi x$, $0 \le x \le 1$ using Schmidt method. Take $h = 0.2$ and $k = 0.04$. (Kolhapur, 2003)

Solution. Here $\qquad h = 0.2 \quad$ and $\quad \alpha = \frac{1}{2}$

$\therefore \qquad \alpha = \frac{k}{h^2}$ gives $k = 0.02$

Since $c = 1/2$, we use Bendre-Schmidt relation

$$u_{i,j+1} = \frac{1}{2}(u_{i-1,j} + u_{i+1,j}) \quad \ldots(1)$$

We have $u(0.1, 0) = 0.1, u(0.2, 0) = \sin 36° = 0.5878$

$u(0.4, 0) = \sin 2\pi\theta = 0.9511$, $u(0.5, 0) = \sin 2\pi\theta = 0.9511$

$u(0.6, 0) = \sin 4\pi\theta = 0.9511$, $u(1, 0) = \sin \pi = 0$

The value of u at the mesh points can be obtained by using the recurrence relation (1) as shown in table below :

i / t	$x = 0$	0.2	0.4	0.6	0.8	1.0
j	0	1	2	3	4	5
0	0	0.5878	0.9511	0.9511	0.5878	0
0.02	1	0.4755	0.7695	0.7695	0.4755	0
0.04	2	0.3847	0.6225	0.6225	0.3847	0
0.06	3	0.3113	0.5036	0.5036	0.3113	0
0.08	4	0.2518	0.4074	0.4074	0.2518	0
0.1	5	0.2037	0.3296	0.3296	0.2037	0

Example 33.3. *Find the solution of $u_t = u_{xx}$, $0 < x < 1$, $t > 0$ satisfying the parabolic equation $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$ and the boundary conditions $u(0, t) = 0$ and $u(1, t) = 0$ and $u(x, 0) = \sin \pi x = \frac{1}{2} a^2$ at the points $x = 0, 1, 2, \ldots, 5$ and $t = \frac{1}{2} j$, $j = 0, 1, 2, \ldots, 5$.*

Solution. Here $a^2 = 4$, $h = 1$ and $k = 1/2$. Thus $\alpha = c^2 k/h^2 = 1/2$.

∴ we have Bendre-Schmidt recurrence relation

$$u_{i,j+1} = \frac{1}{2}(u_{i-1,j} + u_{i+1,j}) \quad \ldots(1)$$

Now since $u(0, t) = 0 = u(5, t)$

$\therefore u_{0,j} = 0$ and $u_{5,j} = 0$ for all values of j, i.e., the values in the first and last column are zero.

Similarly, $f(x) = \sin \frac{\pi x}{5} = \frac{1}{2} x^2$

$\therefore$ $u_{i,0} = 4f - \frac{1}{2} i^2 = 0, 2.5, 6, 7.5, 6, 2.5, 0$

for $i = 0, 1, 2, 3, 4, 5, 6, 7$ and $f = 0$

These are the entries of the first row.

i / t	0	1	2	3	4	5	6	7	8
0	0	2.5	6	7.5	6	7.5	6	2.5	0
1	0	3	5.5	7	7.5	7	5.5	3	0
2	0	2.75	5	6.5	7	6.5	5	2.75	0
3	0	2.5	4.625	6	6.5	6	4.625	2.5	0
4	0	2.3125	4.25	5.5625	6	5.5625	4.25	2.3125	0
5	0	2.125	3.9375	5.125	5.5625	5.125	3.9375	2.125	0

Putting $j = 0$ in (*), we have

$$u_{i,1} = \frac{1}{2}(u_{i-1,0} + u_{i+1,0})$$

Taking $i = 1, 2, \ldots, 7$ successively, we get

$$u_{1,1} = \frac{1}{2}(u_{2,0} + u_{0,0}) = \frac{1}{2}(0 + 8) = 4$$

$$u_{2,1} = \frac{1}{2}(u_{3,0} + u_{1,0}) = \frac{1}{2}(8 + 7.0) = 0.0$$

$$u_{3,1} = \frac{1}{2}(u_{4,0} + u_{2,0}) = \frac{1}{2}(6 + 8) = 7$$

$$u_{4,1} = 7.0,\ u_{5,1} = 7,\ u_{6,1} = 5.3,\ u_{7,1} = 3.$$

These are the entries in the second row.

Putting $j = 1$ in (*), the entries of the third row are given by

$$u_{i,2} = \frac{1}{2}(u_{i-1,1} + u_{i+1,1})$$

Similarly putting $j = 2, 3, 4$ successively in (*), the entries of the fourth, fifth and sixth rows are obtained. Hence the values of $u_{i,j}$ are as given in the above table.

Example 33.10. Solve the equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

subject to the conditions $u(x, 0) = \sin \pi x$, $0 \le x \le 1$; $u(0, t) = u(1, t) = 0$. Carry out computations for one time step with $h = 1/3$, $k = 1/36$. (U.T.U. 2006)

Solution. Here $c^2 = 1$, $h = 1/3$, $k = 1/36$ so that

$$\alpha = kc^2/h^2 = 1/4$$

Also $u_{1,0} = \sin \pi/3 = \sqrt{3}/2$, $u_{2,0} = \sin 2\pi/3 = \sqrt{3}/2$

and all other boundary values are zero as shown in Fig. 33.19.

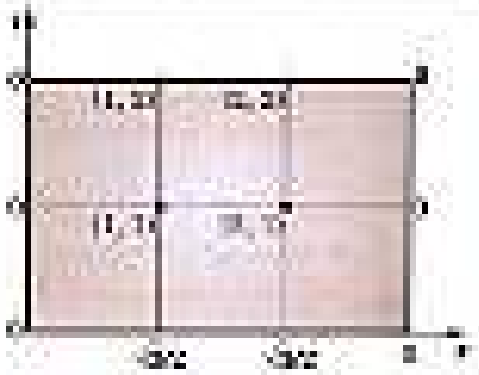

Fig. 33.19

Schmidt's formula (2) of § 33.8

$$u_{i,j+1} = \alpha u_{i-1,j} + (1 - 2\alpha) u_{i,j} + \alpha u_{i+1,j}$$

becomes $$u_{i,j+1} = \frac{1}{4}(u_{i-1,j} + 2u_{i,j} + u_{i+1,j})$$

For $i = 1, 2$; $j = 0$:

$$u_{1,1} = \frac{1}{4}(u_{0,0} + 2u_{1,0} + u_{2,0}) = \frac{1}{4}(0 + \sqrt{3} + \sqrt{3}) = 0.87$$

$$u_{2,1} = \frac{1}{4}(u_{1,0} + 2u_{2,0} + u_{3,0}) = \frac{1}{4}(\sqrt{3} + 2\sqrt{3} + 0) = 0.87$$

For $i = 1, 2$; $j = 1$:

$$u_{1,2} = \frac{1}{4}(u_{0,1} + 2u_{1,1} + u_{2,1}) = 0.65$$

$$u_{2,2} = \frac{1}{4}(u_{1,1} + 2u_{2,1} + u_{3,1}) = 0.65$$

PROBLEMS 33.3

1. Find the solution of the parabolic equation $u_{xx} = 2u_t$, when $u(0, t) = u(4, t) = 0$ and $u(x, 0) = x(4 - x)$, taking $h = 1$. Find the values of $u(x, t)$ for $t = 5$. (Madras 2000)

2. Solve the equation $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ with the conditions $u(x, 0) = 0$, $u(0, t) = 0$, $u(1, t) = t$. Assume $h = 0.2$. Tabulate u for $t = 0.1$, 0.2 and 0.3 (retaining as significant values only). (Anna 2002)

3. Given $\frac{\partial^2 f}{\partial x^2} - \frac{\partial f}{\partial t} = 0$; $f(0, t) = f(5, t) = 0, f(x, 0) = x^2(25 - x^2)$; find the values of f for $x = ih (i = 0,1, ..., 5)$ and $t = jk (j = 0, 1, ..., 6)$ with $h = 1$ and $k = \frac{1}{2}$, using the explicit method.

Replacing the derivatives in (2) by their above approximations, we obtain

$$u_{i,j+1} - 2u_{i,j} + u_{i,j-1} = \frac{c^2k^2}{h^2}(u_{i+1,j} - 2u_{i,j} + u_{i-1,j})$$

or $$u_{i,j+1} = 2(1 - \alpha^2)u_{i,j} + \alpha^2(u_{i+1,j} + u_{i-1,j}) - u_{i,j-1} \quad \ldots(3)$$

where $\alpha = ck/h$.

Now replacing the derivative in (3) by its central difference approximation, we get

$$\frac{u_{i,j+1} - u_{i,j-1}}{2k} = \frac{\partial u}{\partial t} = g(x) \quad \text{[See (2) p. 1068]}$$

or $$u_{i,j+1} = u_{i,j-1} + 2kg(x) \quad \text{at } t = 0 \quad \text{i.e. } u_{i,1} = u_{i,-1} + 2kg(x_i) \text{ for } j = 0 \quad \ldots(4)$$

Also initial condition $u = f(x)$ at $t = 0$ becomes $u_{i,0} = f(x_i)$ $\ldots(5)$

Combining (3) and (4), we have $u_{i,1} = f(x_i) + kg(x_i)$ $\ldots(6)$

Also (2) gives $u_{0,j} = \phi(t)$ and $u_{n,j} = \psi(t)$

Hence (6) gives the values of $u_{i,j+1}$ at the $(j+1)$th level when the initial values $(j-1)$th and jth levels are known from (4) and (5) as shown in Fig. 22.20. Thus (3) gives an explicit scheme for the solution of the wave equation.

A special case : The coefficient of $u_{i,j}$ in (3) will vanish if $\alpha = 1$ or $c = h/k$. Then (3) reduces to the simple form

$$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \quad \ldots(7)$$

This result provides an explicit scheme for the solution of the wave equation.

Obs. 1. *For $\alpha = 1/c$, the solution of (1) is stable and coincides with the solution of (2).*
For $\alpha < 1$, the solution is stable but inaccurate.
For $\alpha > 1$, the solution is unstable.
Obs. 2. *The formula (3) converges for $\alpha \le 1$ i.e. for $k \le h/c$.*

Example 22.21. *Find over the interval of time for the equation $u_{tt} = 16u_{xx}$. Taking $h = 1$ and $k = 0.25$. The boundary conditions are $u(0, t) = 0$, $u(5, t) = 0$, $t > 0$ and $u(x, 0) = x^2(25 - x^2)$, $u_t(x, 0) = 0$, $0 < x < 5$.* (Madras, 2006)

Solution. Here $c^2 = 16$.

$\therefore$ The difference equation for the given equation is

$$u_{i,j+1} = 2(1 - 8\lambda^2)u_{i,j} + 16\lambda^2(u_{i+1,j} + u_{i-1,j}) - u_{i,j-1} \quad \text{where } \alpha = 4\lambda \quad \ldots(1)$$

Taking $\lambda = 1$ and choosing h so that the coefficient of $u_{i,j}$ vanishes, we have $16\lambda^2 = 1$, $i.e.$, $h = 4\lambda = 4$.

$\therefore$ (1) reduces to $$u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1} \quad \ldots(2)$$

which gives a convergent solution for $\lambda h = 1$. Its stability coincides with the solution of the given differential equation.

Now since $u(0, t) = u(5, t) = 0$.

$\therefore$ $u_{0,j} = 0$ and $u_{5,j} = 0$ for all values of j

i.e. the entries in the first and last columns are zero.

Since $u_{i,0} = x_i^2(25 - x_i^2)$

$\Rightarrow$ $u_{i,0} = i^2(25 - i^2) = 0, 15, 16, 15$ for $i = 1, 2, 3, 4$ at $t = 0$.

These are the entries of the first row.

Finally the initial condition $u_t(x, 0) = 0$, becomes

$$\frac{u_{i,j+1} - u_{i,j-1}}{2k} = 0, \text{ when } j = 0, \text{ giving } u_{i,1} = u_{i,-1} \quad \ldots(3)$$

Putting $j = 0$ in (2), $u_{i,1} = u_{i+1,0} + u_{i-1,0} - u_{i,-1}$

$= u_{i+1,0} + u_{i-1,0} - u_{i,1}$ using (3)

or $$u_{i,1} = \frac{1}{2}(u_{i+1,0} + u_{i-1,0}) \quad \ldots(4)$$

Taking $i = 1, 2, 3, 4$ successively, we obtain

$$u_{1,1} = \frac{1}{2}(u_{2,0} + u_{0,0}) = \frac{1}{2}(16 + 0) = 8$$

$$u_{2,1} = \frac{1}{2}(u_{1,0} + u_{3,0}) = \frac{1}{2}(4 + 18) = 11$$

$$u_{3,1} = \frac{1}{2}(u_{2,0} + u_{4,0}) = \frac{1}{2}(32 + 36) = 34$$

$$u_{4,1} = \frac{1}{2}(u_{3,0} + u_{5,0}) = \frac{1}{2}(68 + 0) = 34$$

These are the values of the second row.

Putting $j = 2$ in (11), we get

$$u_{i,2} = u_{i-1,1} + u_{i+1,1} - u_{i,0}$$

Taking $i = 1, 2, 3, 4$ successively, we obtain

$$u_{1,2} = u_{0,1} + u_{2,1} - u_{1,0} = 0 + 11 - 4 = 7$$

$$u_{2,2} = u_{1,1} + u_{3,1} - u_{2,0} = 6 + 34 - 32 = 8$$

$$u_{3,2} = u_{2,1} + u_{4,1} - u_{3,0} = 11 + 34 - 36 = 9$$

$$u_{4,2} = u_{3,1} + u_{5,1} - u_{4,0} = 34 + 0 - 68 = -34$$

These are the values of the third row.

Similarly putting $j = 3, 4$ successively in (11), the values of the fourth, fifth and sixth rows are obtained. Hence the values of $u_{i,j}$ are as shown in the table below :

i / x	0	1	2	3	4	5
0	0	4	32	36	68	0
1	0	11	34	34	0	0
2	0	7	8	9	−34	0
3	0	3	−2	−9	−7	0
4	0	−9	−36	−41	−66	0
5	0	−16	−68	−76	−68	0

Example 11.12. *The two ends displacement of a string of length 2 from each end of an equilibrium position along with the equation $y_{tt} = 4y_{xx}$. If the initial velocity is zero and the initial displacement is $y = 0$ at $x = 0$, $x = 4$ and $y = 0$ and initial conditions $y = x(4 - x)$ and $y_t = 0$ at $t = 0$, $0 \le x \le 4$. Solve this equation numerically for one half period of vibration, taking $h = 1$ and $k = 1/2$.*

Solution. Here, $h/k = 2 = c$.

∴ The difference equation for the given equation is

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1} \qquad ...(1)$$

which gives a convergent solution for $r \le 1$.

Now since $u(0, t) = u(4, t) = 0$,

∴ $u_{0,j} = 0$ and $u_{4,j} = 0$ for all values of j,

i.e., the entries in the first and last columns are zero.

Since $u_{i,0} = x(4 - x)$,

∴ $u_{1,0} = 1(4 - 1) = 3$, $4, 3$ for $i = 1, 2, 3$ or $x = 1, 2, 3$.

These are the entries of the first row.

Also $u_t(x, 0) = 0$ becomes

$$\frac{u_{i,1} - u_{i,-1}}{2k} = 0 \text{ or } u_{i,-1} = u_{i,1}, \text{ giving } u_{i,1} = u_{i,-1} \qquad ...(a)$$

Putting $j = 0$ in (1), $u_{i,1} = u_{i-1,0} + u_{i+1,0} - u_{i,-1}$

$$= u_{i-1,0} + u_{i+1,0} - u_{i,1}, \text{ using (a)}$$

or $$u_{i,1} = \frac{1}{2}(u_{i-1,0} + u_{i+1,0}) \qquad ...(b)$$

Taking $j = 1, 2, 3$ successively, we obtain

$$u_{1,1} = \frac{1}{2}(u_{2,1} + u_{0,1}) = \frac{1}{2}(2 + 0) = 1; \quad u_{2,1} = \frac{1}{2}(u_{3,1} + u_{1,1}) = \frac{1}{2}(0 + 1) = \frac{1}{2}$$

$$u_{3,1} = \frac{1}{2}(u_{4,1} + u_{2,1}) = \frac{1}{2}(0 + \frac{1}{2}) = \frac{1}{4}$$

These are the entries of the 2nd row.

Putting $j = 1$ in (1), $u_{i,2} = u_{i-1,1} + u_{i+1,1} - u_{i,0}$

Taking $i = 1, 2, 3$, successively, we get

$$u_{1,2} = u_{0,1} + u_{2,1} - u_{1,0} = 0 + 2 - 2 = 0$$
$$u_{2,2} = u_{1,1} + u_{3,1} - u_{2,0} = 2 + 0 - 4 = 0$$
$$u_{3,2} = u_{2,1} + u_{4,1} - u_{3,0} = 2 + 0 - 2 = 0$$

These are the entries of the 3rd row and so on.

Now the equation of the vibrating string of length l is $u_{tt} = c^2 u_{xx}$

$\therefore$ Its period of vibration $= \frac{2l}{c} = \frac{2 \times 4}{2} = 4$ sec. [$\because$ $l = 4$ and $c = 2$]

This shows that we have to compute $u_{i,4}$ with $i = 2$.

i.e., similarly we obtain the values of $u_{i,3}$ (3rd row) and $u_{i,4}$ (4th row).

Hence the values of $u_{i,j}$ are as shown in the table below :

j \ i	0	1	2	3	4
0	0	2	4	2	0
1	0	1	2	1	0
2	0	0	0	0	0
3	0	−1	−2	−1	0
4	0	−2	−4	−2	0

Example 10.13. Find the solution of the initial boundary value problem $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1$, $t > 0$, subject to the initial conditions $u(x, 0) = \sin \pi x$, $0 \le x \le 1$ and the boundary conditions $u(0, t) = 0$ and $u(1, t) = 0$, $t > 0$, by using (i) the implicit scheme,

(ii) the explicit scheme. (Anna, 2007)

Solution. (ii) Explicit scheme

Take $h = 0.2$, $k = \frac{h}{c} = 0.2$ [$\because$ $c = 1$]

$\therefore$ We use the formula $u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1}$ ...(1)

Since $u(x, 0) = \sin \pi x$, $u_{0,0} = u_{1,0} = u_{2,0} = u_{3,0} = u_{4,0} = 0$ for all values of j

i.e., the entries in the first and last columns are zero.

Since $u(x, 0) = \sin \pi x$, $u_{i,0} = \sin \pi x$

$\therefore$ $u_{0,0} = 0$, $u_{1,0} = \sin(0.2\pi) = 0.5878$, $u_{2,0} = \sin(0.4\pi) = 0.9511$, $u_{3,0} = \sin(0.6\pi) = 0.9511$, $u_{4,0} = \sin(0.8\pi) = 0.5878$.

These are the entries of the first row.

Since $u_t(x, 0) = 0$ we have $\frac{1}{2h}(u_{i,j+1} - u_{i,j-1}) = 0$, where $j = 0$

i.e., $u_{i,1} = u_{i,-1}$ ...(2)

Putting $j = 0$ in (1), $u_{i,1} = u_{i-1,0} + u_{i+1,0} - u_{i,-1}$

Using (2) $u_{i,1} = \frac{1}{2}(u_{i-1,0} + u_{i+1,0})$

Taking $i = 1, 2, 3, 4$ successively, we obtain the entries of the second row.

Putting $j = 1$ in (1), $u_{i,2} = u_{i-1,1} + u_{i+1,1} - u_{i,0}$

Now taking $i = 1, 2, 3, 4$ successively, we get the entries of the third row.

Similarly taking $j = 2, j = 3, j = 4$ successively, we obtain the entries of the fourth, fifth and sixth rows respectively.

j / i	0	1	2	3	4	5
0	0	0.5878	0.9511	0.9511	0.5878	0
1	0	0.4156	0.6725	0.6725	0.4156	0
2	0	0.1847	0.2988	0.2988	0.1847	0
3	0	0	0.0821	−0.1329	−0.1458	0
4	0	−0.1315	−0.0659	−0.1066	−0.0659	0
5	0	−0.0474	−0.0211	−0.0211	−0.0474	0

(ii) Crank-Nicolson scheme

We have the formula :

$$u_{i,j+1} = 2(1 - \alpha^2 r^2)u_{i,j} + \alpha^2 r^2(u_{i-1,j} + u_{i+1,j}) - u_{i,j-1}, \text{ where } \alpha = h/k \qquad \ldots(1)$$

Here $c^2 = 1$. Take $h = 0.5$ and $k = 0.5$ so that $\alpha = h/k = 2$.

Also, Crank-Nicolson formula

$$u_{i,j+1} = -6u_{i,j} + 4(u_{i-1,j} + u_{i+1,j}) - u_{i,j-1} \qquad \ldots(2)$$

Since $u_{i,0} = \sin ix$

$$\therefore \quad u_{1,0} = 0.5878, u_{2,0} = 0.9511, u_{3,0} = 0.9511$$

These are the entries of the first row.

Since $u_t(x, 0) = 0$, we have $\frac{1}{2k}(u_{i,1} - u_{i,-1}) = 0$, where $j = 0$

$$\therefore \quad u_{i,1} = u_{i,-1} \qquad \ldots(3)$$

Put $j = 0$ and using (3), (2) reduces to

$$u_{i,1} = -2u_{i,0} + 2(u_{i-1,0} + u_{i+1,0})$$

Now taking

$$i = 1, u_{1,1} = -2u_{1,0} + 2(u_{0,0} + u_{2,0}) = 0.1512$$
$$i = 2, u_{2,1} = -2u_{2,0} + 2(u_{1,0} + u_{3,0}) = 0.1716$$
$$i = 3, u_{3,1} = -2u_{3,0} + 2(u_{2,0} + u_{4,0}) = 0.1512$$

These are the entries of the second row.

Put $j = 1$, (2) reduces to

$$u_{i,2} = -6u_{i,1} + 4(u_{i-1,1} + u_{i+1,1}) - u_{i,0}$$

Now taking

$$i = 1, u_{1,2} = -6u_{1,1} + 4(u_{0,1} + u_{2,1}) - u_{1,0} = 0.436$$
$$i = 2, u_{2,2} = -6u_{2,1} + 4(u_{1,1} + u_{3,1}) - u_{2,0} = 0.6302$$
$$i = 3, u_{3,2} = -6u_{3,1} + 4(u_{2,1} + u_{4,1}) - u_{3,0} = 0.436$$

These are the entries of the third row.

Put $j = 2$, (2) reduces to

$$u_{i,3} = -6u_{i,2} + 4(u_{i-1,2} + u_{i+1,2}) - u_{i,1}$$

Now taking

$$i = 1, u_{1,3} = -6u_{1,2} + 4(u_{0,2} + u_{2,2}) - u_{1,1} = 0.1087$$
$$i = 2, u_{2,3} = -6u_{2,2} + 4(u_{1,2} + u_{3,2}) - u_{2,1} = 0.1478$$
$$i = 3, u_{3,3} = -6u_{3,2} + 4(u_{2,2} + u_{4,2}) - u_{3,1} = 0.1087$$

These are the entries of the third row.

Hence the values of $u_{i,j}$ are tabulated below :

i / j	0	1	2	3	4
1	0	0.5878	0.436	0.1087	0
2	0	0.9511	0.6302	0.1478	0
3	0	0.9511	0.436	0.1087	0

PROBLEMS

1. Solve the boundary value problem $u_{xx} = u_t$, with the conditions $u(0,t) = u(1,t) = 0$, $u(x,0) = \frac{1}{2}x(1-x)$ and $u_t(x,0) = 0$, taking $h = \frac{1}{4}$ and $k = \frac{1}{32}$. Determine the solution when $x = \frac{1}{4}$ and $t = 0.5$.
(UCE, 2000)

2. The transverse displacements of points on a flexible wire from its rest position are given by the solution of the equation $\frac{\partial^2 y}{\partial t^2} = 25\frac{\partial^2 y}{\partial x^2}$, with the boundary conditions $y(0,t) = y(2,t) = 0$ and the initial conditions $y(x,0) = \begin{cases} 2x & \text{for } 0 \le x \le 1 \\ 2(2-x) & \text{for } 1 \le x \le 2 \end{cases}$ and $\frac{\partial y}{\partial t}(x,0) = 0$. Solve the equation numerically for one half period of oscillation, taking $h = \frac{1}{2}$, $k = \frac{1}{50}$.

3. Solve $\frac{\partial y}{\partial t} = \frac{\partial^2 y}{\partial x^2}$ upto $t = 0.05$ with a spacing of 0.1 subject to $y(0,t) = 0$, $y(1,t) = 0$, $y(x,0) = \sin \pi x$, $0 \le x \le 1$. Find also $y(x,t)$ at $x = 0.1$ and 0.2.
(Anna, 2004)

4. The function u satisfies the equation

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$$

and the conditions $u(0,t) = u(1,t) = 0$, $t > 0$ and $u(x,0) = \begin{cases} 2x & 0 \le x \le \frac{1}{2} \\ 2(1-x) & \frac{1}{2} \le x \le 1 \end{cases}$.

Use the explicit method to find $u(x,t)$ for $x = 0.1, 0.2, \ldots, 0.9$ and $t = 0.01, 0.02$.

OBJECTIVE TYPE OF QUESTIONS

MULTIPLE CHOICE

Tick (✓) the correct answer (only one) from the following questions:

1. Which of the following equations is parabolic
   (a) $u_{xx} - u_t = 0$ (b) $u_{xx} + u_{yy} + u_{zz} = 0$ (c) $u_{tt} = u_{xx} + u_{yy} + u_{zz}$ (d) none

2. $u_{ij} = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1})$ is Bender-Schmidt's five point formula. (True or False)

3. $u_{xx} + u_{yy} + u_{zz} = 0$ is classified as ____ 4. $\nabla^2 u = f(x,y)$ is known as ____

5. The simplest form of the equation $u_{xx} = c^2 u_{tt}$ is ____ 6. The P.D.E. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is ____

7. Schmidt's explicit difference scheme to solve $u_{xx} = c^2 u_{tt}$ is ____

8. The D'Alembert's solution of the wave equation $u_{tt} = c^2 u_{xx}$ is ____

9. The partial differential equation $(x-1)u_{xx} - 2(x+1)u_{xy} + (x+2)u_{yy} = 0$ is classified as ____

10. $u_{i,j+1} = \frac{1}{2}(u_{i+1,j} + u_{i-1,j})$ is called ____ recurrence relation.

11. Laplace's difference equation $u_{ij} = u_{ij}$ is ____

12. Bender-Schmidt recurrence relation for one dimensional heat equation is ____

13. The diagonal 5 point formula to solve the Laplace equation $u_{xx} + u_{yy} = 0$ is ____

14. In the parabolic equation $u_t = c^2 u_{xx}$, if $h = \Delta x$, $k = \Delta t$, then explicit method is stable if $\lambda = $ ____

15. $k\frac{\partial^2 u}{\partial x^2} + 4\frac{\partial u}{\partial x} + 3\frac{\partial u}{\partial t} = 0$ is classified as ____ (U.P.T.U., 2005)

16. The two unitary conditions of one dimensional wave equation are ____

17. The implicit formula for one dimensional wave equation with $\lambda = \frac{c\Delta t}{\Delta x} = 1$ is ____

18. The general form of Poisson's equation in partial derivatives is ____

19. If a unit for Laplace equation and $u = 100$ on the boundary of a square, the value of u at the interior point is ____

20. The Laplace equation $u_{xx} + u_{yy} = 0$ is called an ____ equation.

21. The equation $u_{xx} + u_{yy} = 0$ is applicable in the region ____

22. To solve $\frac{\partial u}{\partial t} = \frac{1}{2}\frac{\partial^2 u}{\partial x^2}$ by Crank-Nicholson method $\lambda = \frac{k}{h^2}$ is ____

CHAPTER 34

Linear Programming

1. Introduction 2. Formulation of the problem 3. Graphical method 4. Some exceptional cases 5. General linear programming problem 6. Canonical and standard forms of LPP 7. Simplex method 8. Working procedure of the simplex method 9. Artificial variable techniques 10. Method of penalty 11. Two phases method 12. Exceptional cases—Degeneracy 13. Duality concept 14. Duality principle 15. Dual simplex method 16. Transportation problem 17. Working procedure for transportation problems 18. Degeneracy in transportation problems 19. Assignment problem 20. Working procedure to solve assignment problems 21. Objective type of questions

34.1 INTRODUCTION

Linear programming deals with the optimization (maximization or minimization) of linear functions subject to linear constraints. This technique has found its applications in important areas of product mix, blending problems and diet problems. Oil refineries, chemical industries, steel industries and food processing industry are also using linear programming with considerable success.

In this chapter, our purpose is to present the principles of linear programming and the techniques of its application in a manner that will suit the engineering students. Beginning with the graphical method which provides a great deal of insight into the basic concepts, the simplex method of solving linear programming problems is developed. Then the reader is introduced to the Duality concept. Finally a special class of linear programming problems namely : Transportation and Assignment problems is taken up. For a detailed study, the student should refer to author's book *Numerical Methods in Engineering and Science*.

34.2 FORMULATION OF THE PROBLEM

To begin with, a problem is to be presented in a linear programming form which required defining the variables involved, establishing relationships between them and formulating the objective function and the constraints. We illustrate this through a few examples.

Example 34.1. A manufacturer produces two types of models M_1 and M_2. Each M_1 model requires 4 hours of grinding and 2 hours of polishing ; whereas each M_2 model requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works for 40 hours a week and each polisher works for 60 hours a week. Profit on an M_1 model is ₹ 3 and on an M_2 model is ₹ 4. Whatever is produced in a week is sold in the market. How should the manufacturer allocate his production capacity to the two types of models so that he may make the maximum profit in a week.

Solution. Let x_1 be the number of M_1 models and x_2 the number of M_2 models produced per week. Then the weekly profit (in ₹) is

$$Z = 3x_1 + 4x_2 \qquad \ldots(1)$$

To produce these number of models, the total number of grinding hours needed per week

$$= 4x_1 + 2x_2$$

and the total number of polishing hours required per week

$$= 2x_1 + 5x_2$$

Since the number of grinding hours available is not more than 80 and the number of polishing hours is not more than 180, therefore,

$$4x_1 + 2x_2 \leq 80 \quad \ldots(i)$$

$$2x_1 + 5x_2 \leq 180 \quad \ldots(ii)$$

Also since the negative number of models are not produced, obviously we must have

$$x_1 \geq 0 \text{ and } x_2 \geq 0 \quad \ldots(iii)$$

Hence the allocation problem is, to find x_1, x_2 which

maximize $Z = 3x_1 + 4x_2$

subject to $4x_1 + 2x_2 \leq 80$; $2x_1 + 5x_2 \leq 180$; $x_1, x_2 \geq 0$.

Here the constraints that were due to problem are called decision variables.

The function Z showing the total profit is the objective function. The inequalities (i) and (ii) are called the constraints.

The inequalities (iii) $x_1 \geq 0$ and $x_2 \geq 0$ are called the constraints.

The objective function and the constraints being all linear, it is a linear programming problem (LPP). This is an example of a small machine (tool) industry.

Example 3.2. A foundry making castings uses alloys. Alloys in each case must satisfy the following specifications:

	Minimum	Maximum
Carbon	3.25%	3.65%
Silicon	2.25%	2.50%

Typical analysis and costs of various raw materials used for this purpose are given below:

Material	Carbon %	Silicon %	Cost (₹)
Steel scrap	0.5	0.5	200/ton
Cast iron scrap	3.5	2.5	300/ton
Remelt from foundry	3.25	2.25	350/ton

If the total charge of iron must be 4 tons, find the weight in lbs of each raw material that must be used in the optimal mix at minimum cost. (AMIE (S), 1999)

Solution. Let x_1, x_2, x_3 be the amounts (in kg) of three raw materials. The objective is to minimize the cost i.e.,

minimize $\quad Z = \frac{200}{1000}x_1 + \frac{300}{1000}x_2 + \frac{350}{1000}x_3 \quad \ldots(i)$

For iron mix: to have a minimum of 3.25% carbon,

$$0.5x_1 + 3.5x_2 + 3.25x_3 \geq 3.25 \times 4,000 \quad \ldots(ii)$$

For iron mix: to have a maximum of 3.65% carbon,

$$0.5x_1 + 3.5x_2 + 3.25x_3 \leq 3.65 \times 4,000 \quad \ldots(iii)$$

For iron mix: to have a minimum of 2.25% silicon,

$$0.5x_1 + 2.5x_2 + 2.25x_3 \geq 2.25 \times 4,000 \quad \ldots(iv)$$

For iron mix: to have a maximum of 2.50% silicon,

$$0.5x_1 + 2.5x_2 + 2.25x_3 \leq 2.50 \times 4,000 \quad \ldots(v)$$

Also, since the materials added to mix must be equal to the full charge weight of 4 tons,

$$\therefore \quad x_1 + x_2 + x_3 = 4,000 \quad \ldots(vi)$$

Finally since the amounts of raw materials cannot be negative

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0 \quad \ldots(vii)$$

Thus the linear programming problem is to find x_1, x_2, x_3 which

minimize $\quad Z = 0.2x_1 + 0.3x_2 + 0.35x_3$

subject to $\quad 0.5x_1 + 3.5x_2 + 3.25x_3 \geq 13,000$

$$0.5x_1 + 3.5x_2 + 3.25x_3 \leq 13,000$$

$$0.3x_1 + 2.4x_2 + 2.2x_3 \le 9,000$$
$$0.5x_1 + 2.4x_2 + 2.5x_3 \le 8,000$$
$$x_1 + x_2 + x_3 = 4,000$$
$$x_1, x_2, x_3 \ge 0.$$

PROBLEMS 24.1

1. A firm manufactures two types A and B of cycle tyres. It purchases moulds which are then used, heated and pressed. Costings for these A and B are ₹ 5 and ₹ 4 each and are sold at ₹ 8 and ₹ 7 each respectively. Running costs of these machines are ₹ 20, ₹ 16 and ₹ 17.50 per hour respectively. Formulate the problem so that the profit is maximised per hour. Determine the profit? Operation of the machines are

	Type A	Type B
Moulding capacity	20 per hr	40 per hr
Heating capacity	30 per hr	60 per hr
Polishing capacity	60 per hr	60 per hr

2. A firm manufactures 2 products A and B. The profits are ₹ 3, ₹ 5 and ₹ 4 respectively. The firm has two machines M_1 and M_2 and labour in the required processing time is as given below. Each machine can make only one product at a time.

		Product		
		A	B	C
Machine	M_1	4	5	3
	M_2	2	1	4

Machines M_1 and M_2 have 2000 and 2500 machine minutes respectively. The firm must manufacture 100 A's, 200 B's and 50 C's but not more than 300 A's. Set up an L.P.P. to maximise profit. (Kurukshetra, 2003)

3. Three products are processed through three different operations. The time (in minutes) required per unit of each product, the daily capacity of the operations in minutes per day and the profit per unit sold for each product in rupees are as follows:

Operation	Time per unit			Operation capacity
	Product I	Product II	Product III	
1	2	4	3	41
2	3	0	1	4
3	1	4	2	41
Profit (₹)	3	2	1	

The time taken by Product III for operation 2 is not known. The problem is to determine the optimum daily production for these products that maximises the profit. Formulate this problem as a linear programming problem assuming that all units produced are sold.

4. An Airplane can carry a maximum of 200 passengers. A profit of ₹ 400 is made on each First class ticket and a profit of ₹ 300 is made on each economy class ticket. The airline reserves at least 20 seats for First class. However, at least 4 times as many passengers prefer to travel by economy class than by the First class. How many tickets of each class must be sold in order to maximise profit for the airline? Formulate the problem as an L.P.P. model.

(Madurai, 2000)

5. A firm manufactures two kinds of pills in two sizes A and B. Size A contains 3 grains of aspirin, 5 grains of bicarbonate and 2 grains of codeine. Size B contains 5 grains of aspirin, 4 grains of bicarbonate and 6 grains of codeine. It is found by users that it is sufficient if they get from the pills taken at least 70 grains of aspirin, 74 grains of bicarbonate and 24 grains of codeine for providing immediate relief. It is required to determine the least number of pills a patient should take to get immediate relief. Formulate the problem as a standard L.P.P.

6. Consider the following problem faced by a production planner in a soft drink plant. Bottles are filled in two sizes A and B. A is designed for 200 ml bottles and B for 1000 ml bottles. However, each bottle can contain 800 ml of any one kind of beverage. The following data are available:

Machine	Production time per unit	Machine capacity
	(Minutes)	(Hours per day)
A	1	8
B	3	12

The machines can be run 2 times per day, 5 days per week. Profit on a 200 ml bottle is 10 paise and on a 1000 ml bottle is 40 paise. Weekly production of the drink must be at least 2000 litres and the market can absorb 20,000

A metal fabricating company has available 1,500 kg of sheet steel, 2,000 kg of aluminium and 2,500 kg of copper each week. The product requires a mixture of 2 kg of steel, 3 kg of aluminium and 2 kg of copper. Each unit of the product sold yields a profit of £30. Assuming that all the product can be sold, formulate a linear programming model to determine the number of units to be produced each week to maximise profit.

5. A dairy feed company may purchase and mix any one or more of three types of grain containing different amounts of nutritional elements. The data is given in the table below. The problem is to find the least cost for the mixture that will meet at least minimum nutritional requirements and satisfy the feed ratio of 2 : 3 : 1 by mass.

	Per	Per unit weight of Grain 1	Grain 2	Grain 3	Minimum requirement
	A	2	1	7	1,200
Nutritional	B	1	1	0	500
ingredients	C	0	3	0	900
	D	4	10	1	2000
Cost per weight of		£1	£2	£9	

Formulate the problem as a L.P. model.

6. A low precision metal alloy with the following specifications:

(i) specific gravity of 8.35, (ii) thermal expansion coefficient of 19.5×10^{-6} per °C, (iii) melting point of 1054°C

The alloy requires three raw materials A, B and C whose properties are as follows:

Property	Per unit weight of raw materials A	B	C
Specific gravity	8.94	8.00	1.80
Composition	30%	16%	17%
Melting point	1453°C	1083°C	420°C

Find the values of A, B, C (in kg) to be used to make 1 tonne of alloy of desired properties, keeping the raw material costs at the minimum, when they are £100/tonne for A, £400/tonne for B and £200/tonne for C. Formulate the L.P. model for the problem.

34.5 GRAPHICAL METHOD

Linear programming problems involving only two variables can be effectively solved by a graphical technique which provides a pictorial representation of the solution and so gives insight into the basic concepts used in solving large L.P.P.

Working procedure to solve a linear programming problem graphically:

Step 1. Formulate the given problem as a linear programming problem.

Step 2. Plot the given constraints as equalities on $x_1 - x_2$ coordinate plane and determine the convex region* bound by lines.

Step 3. Determine the vertices of the convex region and find the value of the objective function at each vertex. The vertex which gives the optimal (maximum or minimum) value of the objective function gives the desired optimal solution to the problem.

Otherwise. Draw the dotted line through the origin representing the objective function with $Z = 0$. As Z is increased from zero, this line moves to the right remaining parallel to itself. We go on sliding this line (parallel to itself), till it is farthest away from the origin and passes through only one vertex of the convex region. This is the vertex where the maximum value of Z is attained.

*A region or a set of points is said to be convex if the line joining any two of its points lies completely in the region or the set. Figs. 34.1 and 34.2 represent convex regions while Figs. 34.3 and 34.4 do not form convex sets.

Fig. 34.1

Fig. 34.2

Fig. 34.3

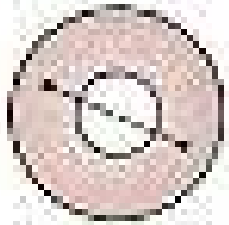

Fig. 34.4

When it is required to minimize Z, value of Z is increased till the dotted line passes through the nearest vertex of the convex region.

Example 3.13: *Solve the L.P.P. Min $Z = 3x_1 + 2x_2$ subject to* (UPTU, 2003)

Solution: The problem is:

Minimize $Z = 3x_1 + 2x_2$ ...(1)

subject to $4x_1 + 2x_2 \geq 80$ ...(2)

$2x_1 + 5x_2 \geq 180$ ...(3)

$x_1, x_2 \geq 0$ ...(4)

Consider $x_1 - x_2$ coordinate system as shown in Fig. 3.15. The non-negativity restrictions (4) imply that the values of x_1, x_2 lie in the first quadrant only.

We plot the lines $4x_1 + 2x_2 = 80$ and $2x_1 + 5x_2 = 180$.

Then any point on or below $4x_1 + 2x_2 = 80$ satisfies (2) and any point on or below $2x_1 + 5x_2 = 180$ satisfies (3). This shows that the desired point (x_1, x_2) must lie somewhere in the shaded convex region $OADC$. This region is called the solution space or region of possible solutions for the given problem. Its vertices are $O(0, 0)$, $A(20, 0)$, $B(25, 30)$ and $C(0, 36)$.

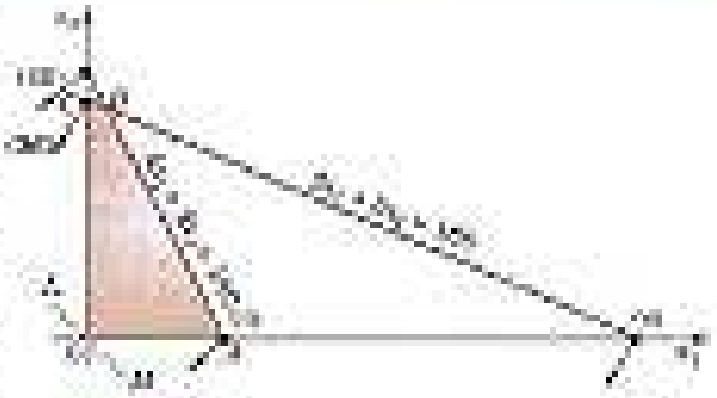

Fig. 3.15

The values of the objective function Z at these points are

$Z(O) = 0$, $Z(A) = 60$, $Z(B) = 147.5$, $Z(C) = 108$.

Thus the maximum value of Z is 147.5 and it occurs at B. Hence the optimal solution to the problem is

$x_1 = 25$, $x_2 = 30$ and $Z_{max} = 147.5$.

Otherwise. One can try to find the point by point in the solution space which maximizes the profit function Z. To do this, we observe that on making $Z = 0$, (1) becomes $3x_1 + 4x_2 = 0$ which is represented by the dotted line OB' through O. As the value of Z is increased, the line AB starts moving parallel to itself towards the right. Larger the value of Z, more will be the company's profit. In this way, we go on shifting OB' till it is farthest away from the origin and passes through one of the corners of the convex region. This is the point where the maximum value of Z is obtained. It is possible, such a line may be one of the edges of the solution space. In that case every point on that edge gives the same maximum value of Z.

Now Z_{min} is attained at $B(25, 30)$. Hence the optimal solution is $x_1 = 25$, $x_2 = 30$ and $Z_{min} = 147.5$.

Example 3.14: *Find the maximum value of $Z = 4x + 3y$ subject to the constraints $x + y \leq 30$, $y \geq 3$, $0 \leq x \leq 15$, $x - y \geq 0$ and $0 \leq y \leq 20$.* (Kashmir, 2008)

Solution: Any point (x, y) satisfying the conditions $x \geq 0$, $y \geq 0$ lies in the first quadrant only. Also since $x + y \leq 30$, $y \geq 3$, $y \leq 15$, $x \geq y$ and $y \leq 20$, the desired point (x, y) lies within the convex region $ABCDE$ below or shaded in Fig. 3.16. Its vertices are $A(3, 3)$, $B(12, 3)$, $C(15, 15)$, $D(15, 15)$, and $E(3, 3)$.

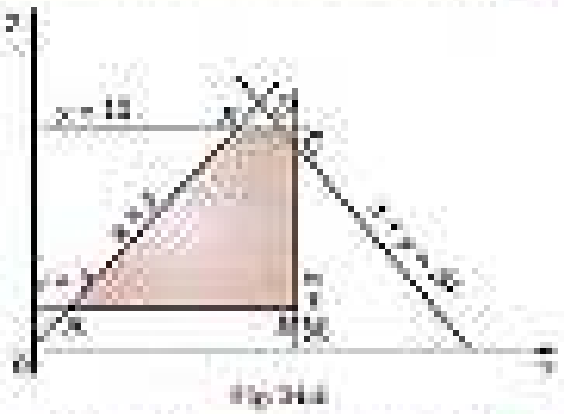

Fig. 3.16

The values of Z at these five vertices are $Z(A) = 15$, $Z(B) = 48$, $Z(C) = 75$, $Z(D) = 75$, and $Z(E) = 21$.

Since the maximum value of Z is 75 which occurs at the vertex C, the solution is the L.P.P. is $x = 15, y = 15$ and maximum $Z = 75$.

Example 3.15: *A company manufactures two types of cloth, using three different colours of wool. One yard length of type A cloth requires 4 oz of red wool, 2 oz of green wool and 2 oz of yellow wool. One yard length of type B cloth requires 3 oz of red wool, 5 oz of green wool and 4 oz of yellow wool. The wool available for manufacture is 1000 oz of red wool, 1000 oz of green wool and 1000 oz of yellow wool. The manufacturer can make a profit of ₹ 8 on one yard of type A cloth and ₹ 5 on one yard of type B cloth. Find the best combination of the production of type A and type B cloth which gives him maximum profit by using the L.P.P. graphically.*

Solution. Let the manufacturer decide to produce x_1 yards of type A cloth and x_2 yards of type B cloth. Then the total income in rupees, from these units of cloth is given by

$$Z = 6x_1 + 3x_2 \quad \ldots(i)$$

To produce these units of two types of cloth, he requires

red wool $= 4x_1 + 2x_2$ oz,

green wool $= 2x_1 + 2x_2$ oz,

and yellow wool $= 0x_1 + 3x_2$ oz.

Since the manufacturer does not have more than 1000 oz of red wool, 1200 oz of green wool and 1200 oz of yellow wool, therefore

$$4x_1 + 2x_2 \le 1000 \quad \ldots(ii)$$

$$2x_1 + 2x_2 \le 1200 \quad \ldots(iii)$$

$$0x_1 + 3x_2 \le 1200 \quad \ldots(iv)$$

Also $x_1 \ge 0, x_2 \ge 0$. $\ldots(v)$

Thus the given problem is to maximize $Z = 6x_1 + 3x_2$ subject to the constraints (ii) to (v). (U.P.T.U., 2006)

Any point satisfying the conditions (v) lies in the first quadrant only. Also the desired point satisfying the constraints (ii) to (iv) lies in the convex region OABCD (Fig. 31.5). Its vertices are O(0, 0), A(500, 0), B(300, 250), C(150, 450), D(0, 600). Now $Z_O = 0$, $Z_A = 3000$, $Z_B = 2550$, $Z_C = 2250$, $Z_D = 1800$.

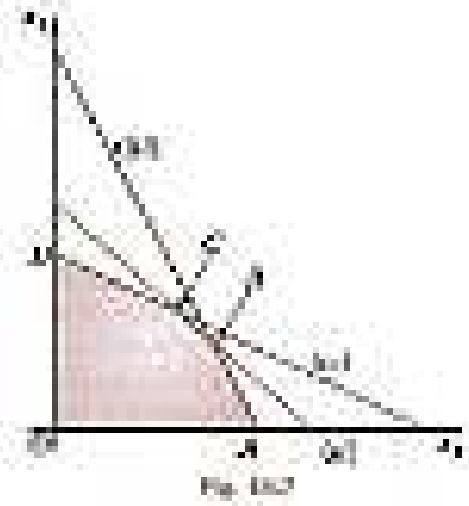

Fig. 31.5

The values of Z at these vertices are given by $Z(O) = 0$, $Z(A) = 6 \times 500 = 3000$, $Z(B) = 1650 + 900 = 2550$, $Z(C) = 900 + 1350 = 2250$ and $Z(D) = 1800$.

Since the maximum value of Z is 3000 which occurs at the vertex A, the solution to the given problem is $x_1 = 500$, $x_2 = 0$ and max. $Z = 3000$.

Hence the manufacturer should produce 500 yards of type A cloth, and 0 yards of type B cloth, to maximize the maximum profit of ₹ 3000.

Example 31.6. *A company making cold drinks has two bottling plants located at towns T_1 and T_2. Each plant produces three drinks A, B and C and their production capacity per day is shown below:*

Out drinks	Plant at	
	T_1	T_2
A	6000	2000
B	1000	2000
C	2000	2000

The monthly requirement of the company for the drinks is 60000 bottles of A, 22,000 bottles of B and 45,000 bottles of C during the month of June. The operating costs per day of plants at T_1 and T_2 are ₹ 5,000 and ₹ 4,000 respectively. Find graphically the number of days for which each plant must be run in June so as to minimize the operating costs while meeting the market demand.

Solution. Let the plants at T_1 and T_2 be run for x_1 and x_2 days. Then the objective is to minimize the operating cost, i.e.,

$$\text{min. } Z = 5000x_1 + 4000x_2 \quad \ldots(i)$$

Constraints on the demand for the three cold drinks are:

for A, $6000x_1 + 2000x_2 \ge 60{,}000$ or $3x_1 + x_2 \ge 40$ $\ldots(ii)$

for B, $1000x_1 + 2000x_2 \ge 22{,}000$ or $x_1 + 2x_2 \ge 22$ $\ldots(iii)$

for C, $2000x_1 + 2000x_2 \ge 45{,}000$ or $x_1 + x_2 \ge 45$ $\ldots(iv)$

Also $x_1, x_2 \ge 0$ $\ldots(v)$

Thus the L.P.P. is to minimize (i) subject to constraints (ii) to (v). (U.P.T.U., 2009-S)

The solution space satisfying the constraints (ii) to (v) is shown shaded in Fig. 31.6. As seen from the direction of the arrows, the solution space is unbounded. The constraint (iv) is over-ruled by the constraints (ii) and (iii) and hence does not affect the solution space. Such a constraint as (iv) is called the redundant constraint.

The vertices of the convex region ABC are $A(32, 0)$, $B(12, 4)$ and $C(8, 12)$.

Values of the objective function Z at these vertices are

$$Z(A) = 192{,}000,\ Z(B) = 88{,}000,\ Z(C) = 104{,}000.$$

Thus the minimum value of Z is ₹ 88,000 and it occurs at B. Hence the solution to the problem is $x_1 = 12$ days, $x_2 = 4$ days, Z_{min} = ₹ 88,000.

Otherwise, Making $Z = 0$, $3x_1 + 20x_2 = 0$ which is represented by the dotted line LM through O. As Z is increased, the line LM moves parallel to itself to the right. Since we are interested in finding the minimum value of Z, value of Z is increased till LM passes through the vertex nearest to the origin of the shaded region, i.e. $B(12, 4)$.

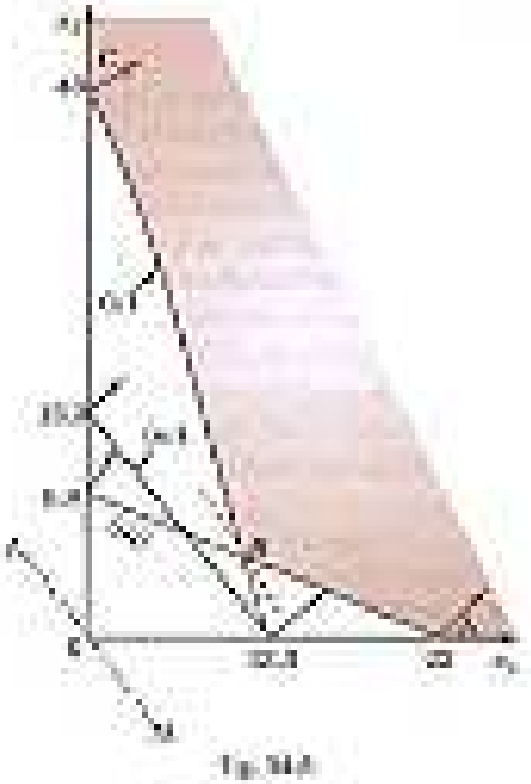

Fig. 13.5

Thus the operating cost will be minimum for $x_1 = 12$ days, $x_2 = 4$ days and

$$Z_{min} = 8000 \times 12 + 10{,}000 \times 4 = ₹\ 88{,}000.$$

Obs. The dotted line parallel to the line LM is called the *isocost line* since it represents all possible combinations of x_1, x_2 which produce the same total cost.

13.8 SOME EXCEPTIONAL CASES

The constraints generally, give region of feasible solution which may be bounded or unbounded. In problems involving two variables and having a finite solution, we observed that the optimal solution existed at a vertex of the feasible region. In fact, this is true for all L.P. problems for which solution exist. Thus it may be stated that *if there exists an optimal solution of an L.P.P., it will be at one of the vertices of the solution space.*

In each of the above examples, the optimal solution was unique. But it is not always so. In fact, *L.P.P. may have*

(i) a unique optimal solution, or (ii) an infinite number of optimal solutions, or (iii) an unbounded solution, or (iv) no solution.

We now give below a few examples to illustrate the exceptional cases (ii) to (iv).

Example 13.7. A firm uses milling machines, grinding machines and lathes to produce two types of parts. The machining hours required for each part, the machining hours available on different machines and the profit on each type of part are given below:

Type of machine	Machining time required for the machine (min.)		Max. time available per week (minutes)
	I	II	
Milling machine	30	4	12,000
Grinding machine	5	5	900
Lathe	6	12	2,400
Profit (in ₹)	180	80	

Determine the number of parts I and II to be manufactured per week to maximize the profit.

Solution. Let x_1, x_2 be the number of parts I and II manufactured per week. Then objective being to maximize the profit, we have, on setting $Z = 180x_1 + 80x_2$ ...(1)

Constraints being on the time available on each machine, we obtain

for milling machines, $30x_1 + 4x_2 \le 12{,}000$ ...(2)

for grinding machines, $5x_1 + 5x_2 \le 900$ ...(3)

for labour, $6x_1 + 12x_2 \leq 3600$ ...(iii)

Also $x_1, x_2 \geq 0$ ...(iv)

Thus the problem is to determine x_1, x_2 which maximize (i) subject to the constraints (ii) to (iv).

The solution space satisfying (ii), (iii), (iv) and meeting the non-negativity restrictions (v) is shown shaded in Fig. 34.9.

Note that (iii) is a redundant constraint as it does not affect the solution space. The vertices of the convex region OABC are

O(0, 0), A(300, 0), B(120, 360), C(0, 250).

Values of the objective function (i) at these vertices are $Z(O) = 0$, $Z(A) = 30{,}000$, $Z(B) = 30{,}000$ and $Z(C) = 10{,}000$.

Thus the maximum value of Z occurs at two vertices A and B.

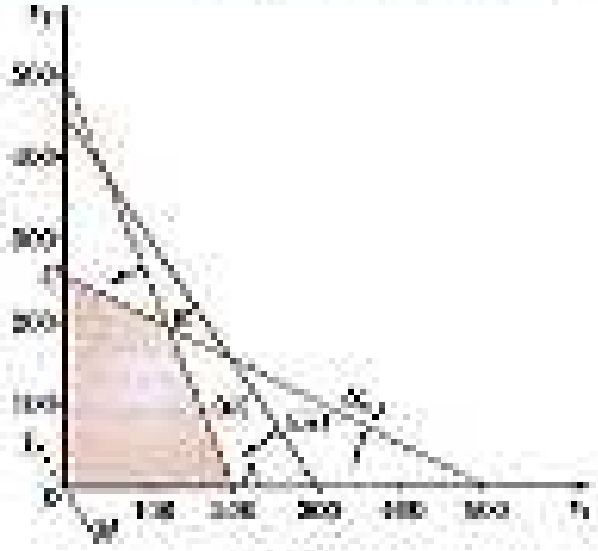

Fig. 34.9

∴ Any point on the line joining A and B will also give the same maximum value of Z i.e. there are infinite number of feasible solutions which yield the same maximum value of Z.

Thus there is no unique optimal solution to the problem and any point on the line AB can be taken to give the profit of ₹ 30,000.

Note. As L.P.P. has more than one optimal solution, it is said to have alternative or multiple optimal solutions. It implies that the resources can be combined in more than one way to maximize the profit.

Example 34.8. Using graphical method, solve the following L.P.P.:

Maximize $Z = 3x_1 + 2x_2$ ...(i)

subject to $x_1 - x_2 \leq 2$ ...(ii)

$x_1 + x_2 \geq 4$ ...(iii)

$x_1, x_2 \geq 0$ (Kurukshetra, 2006; V.T.U., 2009 S) ...(iv)

Solution. Consider $x_1 - x_2$ coordinate system. Any point (x_1, x_2) satisfying the restrictions (ii) to (iv) is in the first quadrant only. The solution space satisfying the constraints (ii) and (iii) is the convex region shown shaded in Fig. 34.10.

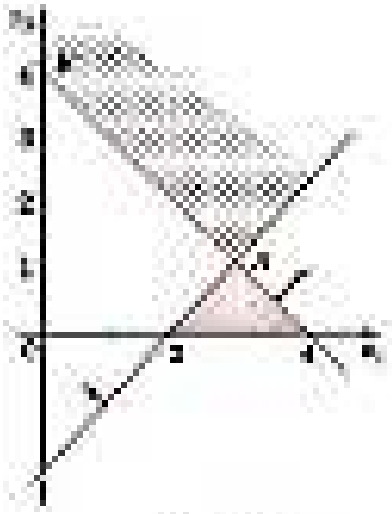

Fig. 34.10

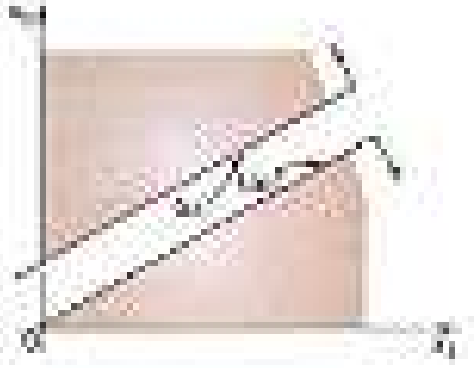

Fig. 34.11

Here the solution space is unbounded. The vertices of the feasible region are A(0, 4) and B(2, 0).

Values of the objective function (i) at these vertices are $Z(A) = 8$ and $Z(B) = 6$.

But there are points in this convex region for which Z will have much higher values. For instance, the point (3, 5) lies in the shaded region and the value of Z thereat is 19. In fact, the maximum value of Z occurs at infinity. Thus the problem has an unbounded solution.

Example 24.8 Solve graphically the following LPP:

Maximize $Z = 6x_1 + 3x_2$ ... (i)

subject to $x_1 + x_2 \geq 2$ ... (ii)

$x_1 + x_2 \leq 0$ ... (iii)

and $x_1, x_2 \geq 0$ ... (iv)

Solution. Consider $x_1 - x_2$ coordinate system. Any point (x_1, x_2) satisfying (iv) lies in the first quadrant only. The two solution spaces, one satisfying (ii) and the other satisfying (iii) are shown shaded in Fig. 24.13.

There being no point (x_1, x_2) common to both the shaded regions, the problem cannot be solved. In fact the solution does not exist since the constraints are inconsistent.

Note. The above problem has not a solution because the constraints were inconsistent. There may be cases in which the constraints are consistent but the problem may still have no feasible solution.

EXERCISE 24.2

Using graphical method, solve the following LP problems:

1. Max. $Z = 3x_1 + 5x_2$
   subject to $x_1 + 2x_2 \leq 2000$, $x_1 + x_2 \leq 1500$, $x_2 \leq 450$, $x_1, x_2 \geq 0$. (Agra Univ., 2007)
2. Max. $Z = 3x_1 + 2x_2$
   subject to $x_1 + x_2 \leq 4$, $3x_1 - 5x_2 \leq 15$, $8x_1 + 5x_2 \leq 40$, and $x_1, x_2 \geq 0$.
3. Min. $Z = 20x_1 + 10x_2$
   subject to $x_1 + 2x_2 \geq 40$, $3x_1 + x_2 \geq 30$, $4x_1 + 3x_2 \geq 60$, $x_1, x_2 \geq 0$.
   (Kurukshetra, 2003; Mumbai, 2001; UPTU, 2004)
4. Max. $z = 3x_1 + 2x_2$, subject to $x_1 + 2x_2 \leq 2000$, $x_1 + x_2 \leq 1500$, $x_2 \leq 600$ and $x_1 \geq 0$, $x_2 \geq 0$. (Kohat, 2000)
5. A firm manufactures two products A and B on which the profit earned per unit are ₹ 3 and ₹ 4 respectively. Each product is processed on two machines M_1 and M_2. Product A requires one minute of processing time on M_1 and 3 minutes on M_2 while B requires one minute on M_1 and one minute on M_2. Machine M_1 is available for not more than 7 hours and 30 minutes while machine M_2 is available for 10 hours during any working day. Find the number of units of products A and B to be manufactured to get maximum profit.
6. Two spare parts X and Y are to be produced by a batch. Each one has to go through two processes A and B. The time required in hours per unit A and B is given below:

	X	Y	Time available in hours
Process A	3	6	36
Process B	9	4	80

   Profit per unit of X and Y are ₹ 4 and ₹ 6 respectively. Find how many number of spare parts of X and Y should be produced at this batch to maximize the profit. Each batch is complete in all respects and one cannot produce fractional units and over the batch.
7. A manufacturer has two products I and II both of which are made to pass by two machines A and B. The process time for each machine for the two products is as given below:

Product	Machine A	Machine B
I	4 hrs	5 hrs
II	6 hrs	4 hrs

   Working hours are restricted to 60 hrs for machine A and 100 hrs for machine B. The contribution for product I is ₹ 30 per unit and for product II is ₹ 5 per unit. The manufacturer is interested in finding the product mix which will yield the best contribution. Formulate the problem and solve by graphical method.
8. A production planning engineer wishes to determine the quantity to be produced per month of products A and B manufactured by him. The data on resources required and availability of resources are given below:

Resources	Requirements		Available per month
	Product A	Product B	
Raw material (kg)	40	120	18,200
Machine hours	8	5	1600
Assembly labour hrs.	5	4	1800
Sales price/unit	₹ 200	₹ 450	

Formulate the problem as a mathematical L.P.P. Find graphically the production schedule that would give maximum profit.

5. A production firm produces two products namely, pineapple and mixed juice. The specific amounts of material, labour and equipment required to produce each product and the availability of each of these resources are shown in the table given below:

	Mixed juice	Pineapple	Available resources
Labour (man hr)	3	20	70 h
Equipment (m/c hrs)	2	10	40
Material (units)	1	14	42

Assuming that unit profit of mixed juice and pineapple are profit margins of ₹ 3 and ₹ 4 respectively. Formulate this as L.P.P. problem and solve it graphically.

Solve the following L.P.P. problems graphically:

(a) Maximize $Z = 3x + 4y$ subject to $2x + y \le 1,\ 3x + 4y \ge 2,\ x \ge 0,\ y \ge 0$. (Mumbai 2008)

(b) Minimize $Z = 6x_1 + 4x_2$ subject to $2x_1 + 0.5x_2 \ge 60,\ 2x_1 + 0.5x_2 \le 160,\ 2x_1 + 1.5x_2 \ge 240$,

and $x_1, x_2 \ge 0$.

(c) Bombardier Inc. have two bottling plants one located at 'G' and other at 'H'. Each plant produces three drinks namely, beer, soda and brandy. The number of bottles produced per day are as follows:

Drink	Plant at 'G'	Plant at 'H'
Whisky	1500	2500
Beer	6000	2000
Brandy	2000	1000

A market survey indicates that during the month of July, there will be a demand of 30,000 bottles of whisky, 40,000 bottles of beer and 14,000 bottles of brandy. The operating cost per day for plants at 'G' and 'H' are ₹ 600 and ₹ 400. For how many days each plant be run in July so as to meet the demand while operating cost is kept to the minimum? Solve graphically.

5.5 GENERAL LINEAR PROGRAMMING PROBLEM

Any L.P.P. problem involving more than two variables may be expressed as follows:

Find the values of the variables $x_1, x_2, \ldots, x_n$ which maximize (or minimize) the objective function

$$Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n \qquad \ldots(1)$$

subject to the constraints

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \le b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \le b_2 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \le b_m \end{aligned}\right\} \qquad \ldots(2)$$

and the non-negativity restrictions

$$x_1, x_2, \ldots, x_n \ge 0 \qquad \ldots(3)$$

Def. 1. A set of values $x_1, x_2, \ldots, x_n$ which satisfies the constraints of the L.P.P. is called its solution.

Def. 2. Any solution of a L.P.P. which satisfies the non-negativity restrictions of the problem is called its feasible solution.

Def. 3. Any feasible solution which maximizes (or minimizes) the objective function of the L.P.P. is called its optimal solution.

Some of the constraints in (2) may be equalities, some others may be inequalities of (≤) type and remaining may inequalities of (≥) type. The inequality constraints are changed to equalities by adding (or subtracting) non-negative variables to (from) the left hand side of such constraints.

Def. 4. If the constraints of a general L.P.P. be

$$\sum_{j=1}^{n} a_{ij}x_j \le b_i \qquad (i = 1, 2, \ldots, k)$$ then the non-negative variables x_j which satisfy

$\sum_{j=1}^{n} a_{ij}x_j + s_i = b_i$, $i = 1, 2, \ldots, m$) are called slack variables.

Def. 5. If the constraints of a general L.P.P. be

$\sum_{j=1}^{n} a_{ij}x_j \geq b_i$, $i = k, k+1, \ldots, l$ then the non-negative variables x_j which satisfy

$\sum_{j=1}^{n} a_{ij}x_j - s_i = b_i$, $i = k, k+1, \ldots, l$, are called surplus variables.

11.3. CANONICAL AND STANDARD FORMS OF L.P.P.

After the formulation of L.P.P., the next step is to obtain its solution. But before any method is used to find its solution, the problem must be presented in a suitable form. As such, we explain its following two forms :

(1) Canonical form. The general L.P.P. can always be expressed in the following form :

Maximize $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$

subject to the constraints $a_{i1}x_1 + a_{i2}x_2 + \ldots + a_{in}x_n \leq b_i$; $i = 1, 2, \ldots, m$

$x_1, x_2, \ldots, x_n \geq 0$,

by making some elementary transformations. This form of the L.P.P. is called its canonical form and has the following characteristics :

(i) Objective function is of maximization type.

(ii) All constraints are of $\leq$ type.

(iii) All variables x_i are non-negative.

The canonical form is a format for a L.P.P. which finds its use in the Duality theory.

(2) Standard form. The general L.P.P. can also be put in the following form :

Maximize $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$

subject to the constraints $a_{i1}x_1 + a_{i2}x_2 + \ldots + a_{in}x_n = b_i$; $i = 1, 2, \ldots, m$

$x_1, x_2, \ldots, x_n \geq 0$.

This form of the L.P.P. is called its standard form and has the following characteristics :

(i) Objective function is of maximization type.

(ii) All constraints are expressed as equations.

(iii) Right hand side of each constraint is non-negative.

(iv) All variables are non-negative.

Rem. Any L.P.P. can be expressed in the standard form.

An inequality $ax_1 + bx_2 + \ldots + kx_n \leq p$ is equivalent to the equality $ax_1 + bx_2 + \ldots + kx_n + s = p$, where $s \geq 0$ is a slack variable.

The objective function can always be expressed in the maximization form.

The inequality constraints can always be converted to equalities by adding or subtracting the slack (or surplus) variables to the left hand sides of such constraints.

So far, the decision variables $x_1, x_2, \ldots, x_n$ have been assumed to be all non-negative. In actual practice, these variables could also be non-negative. If a variable is negative, it can always be expressed as the difference of two non-negative variables e.g. a variable x_j can be written as

$x_j = x_j' - x_j''$ where $x_j', x_j'' \geq 0$.

Example 11.10. Convert the following L.P.P. to its standard form :

Maximize $Z = 3x_1 + 2x_2 + 5x_3$

subject to $2x_1 - 4x_2 \leq 5$, $3x_1 + 2x_2 + 5x_3 \geq 12$, $4x_1 + 3x_2 \leq 8$, $x_1, x_2 \geq 0$.

Solution. As x_3 is unrestricted, let $x_3 = x_3' - x_3''$ where $x_3', x_3'' \geq 0$. Now the given constraints can be expressed as

$2x_1 - 4x_2 \leq 5$,

$3x_1 + 2x_2 + 5x_3' - 5x_3'' \geq 12$

$$\begin{aligned} & \text{[illegible]} \\ & \text{[illegible]} \end{aligned}$$

Introducing the slack/surplus variables, the problem is equivalent to

$$\begin{aligned} \text{Maximize} \quad & Z = [\text{illegible}] \\ \text{subject to} \quad & [\text{illegible}] \\ & [\text{illegible}] \\ & [\text{illegible}] \\ & [\text{illegible}] \end{aligned}$$

Example 2.2.1. [illegible]

$$\begin{aligned} \text{Maximize} \quad & Z = [\text{illegible}] \\ \text{subject to} \quad & [\text{illegible}] \end{aligned}$$

Solution: Here x_3, x_4 are the slack variables and x_1, x_2 are the decision variables. As x_3 is [illegible]

The problem is equivalent to

$$\begin{aligned} \text{Maximize} \quad & Z + 0 = [\text{illegible}] \\ \text{subject to} \quad & [\text{illegible}] \\ & [\text{illegible}] \\ & [\text{illegible}] \\ & [\text{illegible}] \end{aligned}$$

2.3 [illegible]

[illegible] we solve an LPP graphically, the region of feasible solutions was found to be a convex [illegible] the vertices and edges including them. The optimal solution occurred at some vertex. If the optimal solution was not unique, the optimal solutions were on an edge. These observations also hold true for the general LPP. Essentially the problem is that of finding the particular vertex of the convex region which corresponds to the optimal solution. This vertex solution is found out by locating the optimal value at the adjacent vertices. This method consists in moving along the lines that pass through the adjacent vertices. If all the adjacent vertices give values of the objective function less than that of the preceding vertex, as above, this method of comparing the values of the vertices is known as [illegible] vertex as above, the simplex method leads to an optimal vertex in a finite number of steps.

(2) In simplex method, we achieve maximum improvement in the value of Z by [illegible] the entering variable by using the following steps:

(a) Select those row in consideration such as x_r is the variable for being $= 0$. The starting solution is found by setting a sufficient number of variables equal to zero and then solving the remaining equations. Provided the solution obtained is unique. For a set of m equations in n variables ($n > m$), the solution in which $(n - m)$ variables are equated to zero and the values of remaining m variables are found, is called a **basic solution**.

(c) In the [illegible], the variables which [illegible] the non-negative. Some of the basic solutions may contain negative variables. Such solutions are called basic infeasible solutions and others are basic feasible solutions. To achieve this, we choose the [illegible] basic variable which is most negative. The new basic solution would always be non-negative. This is ensured by [illegible] conditions. In [illegible] solution is known as **basic feasible solutions**.

(3) If the variables in the basic feasible solution are positive, then it is called **non-degenerate solution**. If one or more of the variables are zero, it is called **degenerate solution**.

(c) If a non-basic variable which [illegible] the [illegible] then the process can be repeated by [illegible] one of the basic variables to zero and replacing it by a new non-basic variable. The eliminated variable is called the **entering variable** while the new variable is known as the **departing variable**.

The departing variable must maintain the value of the objective function which is ensured by the [illegible] condition. This process is repeated till no further improvement is possible. The resulting solution is called the **optimal basic feasible solution** or simply **optimal solution**.

(2) The simplex method is, therefore, based on the following two conditions:

I. Feasibility condition. It ensures that if the starting solution is basic feasible, the subsequent will also be basic feasible.

II. Optimality condition. It ensures that only improved solutions will be obtained.

(3) Now, we shall elaborate the above terms in relation to the general linear programming problem in standard form, i.e.,

Maximize $$Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n \quad \ldots(1)$$

subject to $$\sum_{j=1}^{n} a_{ij}x_j + x_{n+i} = b_i,\ i = 1, 2, \ldots, m \quad \ldots(2)$$

and $$x_j \geq 0,\ x_{n+i} \geq 0,\ j = 1, 2, \ldots, n \quad \ldots(3)$$

(i) *Solution.* $x_1, x_2, \ldots, x_n$ is a solution of the general L.P.P. if it satisfies the constraints (2).

(ii) *Feasible solution.* $x_1, x_2, \ldots, x_n$ is a feasible solution of the general L.P.P. if it satisfies both the constraints (2) and the non-negativity restrictions (3). The set S of all feasible solutions is called the feasible region. A linear programme is said to be infeasible when the set S is empty.

(iii) *Basic solution* is the solution of the n basic variables where each of these constant variables is equated to zero.

(iv) *Basic feasible solution* is that basic solution which also satisfies the non-negativity restrictions (3).

(v) *Optimal solution* is that basic feasible solution which also optimizes the objective function (1) while satisfying the conditions (2) and (3).

(vi) *Non-degenerate basic feasible solution* is that basic feasible solution which contains exactly m non-zero basic variables. If any of the basic variables becomes zero, it is called a *degenerate basic feasible solution.*

Example 24.12. Find all the basic solutions of the following system of equations, showing in each case the basic and non-basic variables: $2x_1 + x_2 + 4x_3 = 11$; $3x_1 + x_2 + 5x_3 = 14$. (Bharathiar, 2004; V.T.U., 2002 S)

Also, point out which of the basic solutions are degenerate. Are the solutions so obtained basic feasible solutions of the system.

Solution. Since there are $n = 3$ variables and there are $m = 2$ constraints in this problem, a basic solution can be obtained by setting any one variable equal to zero and then solving the resulting equations. Also the total number of basic solutions $= {}^{n}C_{m} = {}^{3}C_{2} = 3$.

The characteristics of the various basic solutions are as given below:

No. of basic sol.	Basic variables	Non-basic variables	Nature of basic solution	Is the sol. feasible? (Are all $x_j \geq 0$)	Is the sol. degenerate?
1.	x_1, x_2	x_3	$2x_1 + x_2 = 11$ $3x_1 + x_2 = 14$ $\therefore x_1 = 3, x_2 = 5$	Yes	No
2.	x_2, x_3	x_1	$x_2 + 4x_3 = 11$ $x_2 + 5x_3 = 14$ $\therefore x_3 = 3, x_2 = -1$	No	Yes
3.	x_1, x_3	x_2	$2x_1 + 4x_3 = 11$ $3x_1 + 5x_3 = 14$ $\therefore x_1 = 34/7, x_3 = 3/7$	Yes	No

The basic feasible solutions are:

(i) $x_1 = 3, x_2 = 5, x_3 = 0$ (ii) $x_1 = 34/7, x_2 = 0, x_3 = 3/7$

which are also non-degenerate basic solutions.

Example 24.13. Find an optimal solution to the following L.P.P. by computing all basic solutions and then finding one that maximizes the objective function:

$$2x_1 + 3x_2 - x_3 = 4,\ x_1 - 2x_2 + 6x_3 = 3;\ x_1, x_2, x_3 \geq 0.$$

$$\text{Max. } Z = 2x_1 + 3x_2 + 4x_3.$$

Solution. Since there are four variables and two constraints, a basic solution can be obtained by setting any two variables equal to zero and then solving the resulting equations. Also the total number of basic solutions = ${}^4C_2 = 6$.

The characterization of the various basic solutions are given below:

No. of basic sol.	Basic variables	Non-basic variables	Values of basic variables	Is the sol. feasible? (Given $x_1, x_2 \geq 0$)	Value of Z	Is the sol. optimal?
1.	x_3, x_4	$x_1 = 0, x_2 = 0$	$3x_1 + 2x_2 + x_3 = 14$ $x_1 - 2x_2 + x_4 = 2$			
			$\Rightarrow x_3 = 14, x_4 = 2$	Yes	0	No
2.	x_2, x_3	$x_1 = 0, x_4 = 0$	$2x_2 + x_3 = 14$ $-x_2 + x_4 = 2$			
			$\Rightarrow x_2 = 4.66, x_3 = 4.68$	No	—	—
3.	x_1, x_3	$x_2 = 0, x_4 = 0$	$3x_1 + x_3 = 14$ $x_1 = 2x_4 + 2$			
			$\Rightarrow x_1 = 2.86, x_3 = 5.56$	Yes	18.3	No
4.	x_1, x_2	$x_3 = 0, x_4 = 0$	$3x_1 + 2x_2 = 14$ $x_1 - 2x_2 = 2$			
			$\Rightarrow x_1 = 4.66, x_2 = 1.33$	Yes	18.3	No
5.	x_2, x_4	$x_1 = 0, x_3 = 0$	$2x_2 + x_4 = 14$ $-x_2 + x_4 = 2$			
			$\Rightarrow x_2 = 4.66, x_4 = 9.33$	Yes	—	—
6.	x_1, x_4	$x_2 = 0, x_3 = 0$	$3x_1 + x_4 = 14$ $x_1 - x_4 = 2$			
			$\Rightarrow x_1 = 4.66, x_4 = 1.33$	Yes	18.3	No

Hence the optimal basic feasible solution is

$x_1 = 0, x_2 = 0, x_3 = 4.67, x_4 = 4.33$ and the maximum value of $Z = 28.5$.

EXERCISE 3.3.5

1. Reduce the following problem to the standard form.
   Minimize $Z = 3x_1 - 2x_2 + 4x_3$
   subject to the constraints $x_1 + 2x_2 + x_3 \leq 4$, $2x_1 + x_2 - x_3 \geq -2$, $x_1, x_2, x_3 \geq 0$
2. Express the following LPP in the standard form:
   Maximize $Z = 3x_1 + 2x_2 + 5x_3$
   subject to $x_1 + 2x_2 + x_3 \leq 430$, $3x_1 + 2x_3 \leq 460$, $x_1 + 4x_2 \leq 420$, $x_1, x_2, x_3 \geq 0$ [Kurukshetra, 2010]
3. Convert the following LPP in standard form:
   Maximize $Z = 3x_1 - 2x_2 + 4x_3$
   subject to $x_1 + 2x_2 + x_3 \leq 10$, $2x_1 - x_2 + x_3 \geq 2$, $2x_1 - 2x_2 - 3x_3 \leq -5$, $x_1, x_2, x_3 \geq 0$ [Kurukshetra, 2010]
4. Obtain all the basic solutions to the following system of linear equations:
   $x_1 + 2x_2 + x_3 = 4$, $2x_1 + x_2 + 3x_3 = 6$.
5. Show that the following system of linear equations has two degenerate basic feasible solutions and also find a non-degenerate basic solution to the problem.
   $2x_1 + 3x_2 + x_3 = 5$, $x_1 + 2x_2 + 3x_3 = 3$. [Kurukshetra, 2010]

Example Find all basic solutions to the following system:

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 4 \\ 2x_1 + x_2 + x_4 &= 3 \end{aligned}$$

Which of these basic solutions are basic feasible solutions? Use the objective function $z = 3x_1 + 4x_2$ to determine which basic feasible solution is optimal.

(Taken from Linear Programming and Network Flows, 2nd edition, by Bazaraa, Jarvis and Sherali)

3.3.3 SIMPLEX PROCEDURE OF THE DANTZIG METHOD

Assuming the existence of a non-degenerate basic feasible solution, an optimal solution to any L.P.P. by simplex method is found as follows:

Step 1: Test whether the objective function is to be maximized or minimized.

If

$$z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

is to be maximized, then convert it into a problem of minimization, by writing

$$\text{Maximize } Z = \text{Minimize } (-Z)$$

(c) Check whether all b_i's are positive.

If any of the b_i's is negative, multiply both sides of that constraint by -1 so as to make the right hand side positive.

Step 2: Express the problem in the standard form.

Express all inequalities of constraints into equations by introducing slack/surplus variables in the constraints and the equations of the form

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \le b_i \Rightarrow a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n + s_i = b_i$$

Step 3: Find an initial basic feasible solution.

If there are m equations and n unknowns, then assign zero values to any $n - m$ of the variables to find the basic solution. Starting with a basic solution, the values x_j ($j = 1, 2, \ldots, n$) are called basic variables. If all $x_j \ge 0$, the basic solution is feasible and non-degenerate. If any of the x_j values are zero, then the solution is degenerate.

The above information is conveniently expressed in the following simplex table:

	c_j		c_1	c_2	c_n	θ	x_B
c_B	Basis		x_1	x_2	x_n	b_i/a_{ij}	$x_B = b$
c_1	x_1		a_{11}	a_{12}	a_{1n}	x_1	$x_1 = b_1$
c_2	x_2		a_{21}	a_{22}	a_{2n}	x_2	$x_2 = b_2$
c_m	x_m		a_{m1}	a_{m2}	a_{mn}	x_m	$x_m = b_m$
			$z_1 - c_1$	$z_2 - c_2$	$z_n - c_n$		$z = z_B$

Basic variables $x_1, x_2, \ldots, x_m$ are called basic variables and variables $x_{m+1}, x_{m+2}, \ldots, x_n$ are called non-basic variables. Basic column is the basic variable x_j for which the coefficient of the variable in the objective function, $z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$, is zero. Basic row is the basic variable and its coefficients in the constraint equations. Entering variable is the variable in the non-basic column having maximum positive coefficient in the objective function. Leaving variable is the basic variable in the row having minimum positive ratio θ. Pivot element is the element in the intersection of the entering variable column and the leaving variable row.

Step 4: Apply optimality test.

Compute $z_j - c_j = \sum_{i=1}^{m} a_{ij}c_i - c_j$, where $z_j = \sum_{i=1}^{m} a_{ij}c_i$.

If $z_j - c_j$ is called net evaluation (or net inflow) for per unit increase in the objective function if the non-basic variable x_j is brought into the solution.

If all $z_j - c_j$ are negative, then the initial basic feasible solution is optimal.

If even one $z_j - c_j$ is positive, then the current feasible solution is not optimal (i.e., can be improved) and proceed to the next step.

Step 5: (a) Identify the incoming and outgoing variables.

If there are more than one positive $z_j - c_j$, then the incoming variable is that one which bears the largest negative coefficient in $z_j - c_j$. The column containing it is known as the key column which is shown marked with an

corner of the tableau. If none of the non-basic variables have the zero coefficient in C_j row of the corresponding variable may be selected arbitrarily as the incoming variable.

Step 4: In this step the algorithm checks for unboundedness. If there is at least one positive element in the key column, then the solution is not unbounded. Otherwise, the solution is unbounded. The ratio between the elements of the solution column and the key column are calculated. The corresponding row is called the key row which is always marked with an arrow on the right end. The element at the intersection of the key row and key column is called the key element which is always marked.

Step 5: If all the ratios are ≤ 0, the pivoting variable can be made as large as we please without violating the feasibility conditions. Hence the problem has an unbounded solution and no further iteration is required.

Step 6: Revise the basic feasible solution.

Drop the outgoing variable and introduce the incoming variable alongside its appropriate value under x_B column. Convert the key element to unity by dividing the key row by the key element. Then make all other elements of the key column zero by subtracting proper multiples of key row from the other rows.

Step 7: Is nothing to do. The procedure is over as the final tableau. The necessary conditions are satisfied.

Step 8: Go to step 4 and repeat the computational procedure until either an optimal or an unbounded solution is obtained.

Example 4.1.1: Solve the following LPP using simplex method.

Maximize $Z = 3x_1 + 5x_2$

subject to $x_1 + 2x_2 + x_3 = 430$ ... (1)

$3x_1 + 2x_4 = 460$ ... (2)

$x_1 + 2x_2 + x_5 = 420$ ... (3)

and $x_1, x_2, x_3, x_4, x_5 \geq 0$

Solution: It consists of the following steps:

Step 1: Check whether the objective function is to be maximized or minimized. If it is to be minimized, then convert it into maximization problem.

The problem being of maximization type, we do not have to change it.

Step 2: Express the problem in the standard form:

By introducing the slack variables x_3, x_4, x_5, the problem is transformed into

Max $Z = 3x_1 + 5x_2 + 0x_3 + 0x_4 + 0x_5$

subject to $x_1 + 2x_2 + x_3 + 0x_4 + 0x_5 = 430$ ... (1)

$3x_1 + 0x_2 + 0x_3 + x_4 + 0x_5 = 460$ ... (2)

$x_1 + 2x_2 + 0x_3 + 0x_4 + x_5 = 420$ ... (3)

and $x_1, x_2, x_3, x_4, x_5 \geq 0$

Step 3: Obtain an initial basic feasible solution.

There are three equations involving five unknowns and for obtaining a solution, any two of the variables are to be taken zero. We shall write a basic solution by setting any two variables $x_1 = 0$ and $x_2 = 0$. These basic solutions correspond to the origin in the graphical method. Substituting $x_1 = x_2 = 0$ in (1), (2) and (3), we get the basic solution

$x_3 = 430,\ x_4 = 460,\ x_5 = 420$

Since all x_3, x_4, x_5 are positive, the basic solution is also feasible and non-degenerate.

- The basic feasible solution is:

  $x_1 = x_2 = 0$ (non-basic) and $x_3 = 430, x_4 = 460, x_5 = 420$ (basic)

- Initial basic feasible solution is given by the following table:

	C_j		3	5	0	0	0	
C_B	Basis	x_B	x_1	x_2	x_3	x_4	x_5	θ
0	x_3	430	1	2	1	0	0	$430/2 = 215$
0	x_4	460	3	0	0	1	0	$460/0 = \infty$
0	x_5	420	1	2	0	0	1	$420/2 = 210$
	$Z_j = \sum C_B x_j$	0	0	0	0	0	0	
	$C_j - Z_j = \Delta_j$		3	5	0	0	0	

[For x_1-column $(j = 1)$, $Z_1 = \sum c_B a_{i1} = (0)1 + (0)2 + (0)1 = 0$

and for x_2-column $(j = 2)$, $Z_2 = \sum c_B a_{i2} = 0 + 1 + 0 = 0$

Similarly $\quad X_3(0) = (0)1 + (0)0 + (0)1 = 0$]

Step 4. Apply optimality test.

As C_j is positive under some columns, the initial basic feasible solution is not optimal (i.e. can be improved) and we proceed to the next step.

Step 5. (i) Identify the incoming and outgoing variables.

The above table shows that x_1 is the incoming variable as its incremental contribution $(C_j = 3)$ is maximum and the column in which it appears is the key column (shown marked by an arrow at the bottom).

Dividing the elements under *b*-column by the corresponding elements of key column, we find minimum positive ratio is 2 in row two. We, therefore, arbitrarily choose the row containing s_2 as the key row (shown marked by an arrow on its right end). The element at the intersection of key row and the key column is a_{21} (= 1), is the key element. s_2 is therefore, the outgoing basic variable which will now become non-basic.

Having decided that x_1 is to enter the solution, we have tried to find as to what maximum value x_1 could have without violating the constraints. On assuming x_1, the new basis will contain x_1, s_1 and s_3 as the basic variables.

(ii) Revise towards the optimal solution

To transform the initial set of equations with a basic feasible solution into an equivalent set of equations with a different basic feasible solution, we make the key element unity. Since the key element is already unity, we retain the key row as it is. Then in order all other elements in key column zero, we subtract proper multiples of key row from the other rows. Here we subtract 2 times the elements of key row from the second row and 3 times the elements of key row from the third row. These become the second and the third rows of the next table. We also change the corresponding values under c_B column from 0 to 3, while replacing s_2 by x_1 under the basis. Thus the second basic feasible solution is given by the following table :

	c_j	3	2	5	0	0	0	
c_B	Basis	x_1	x_2	x_3	s_1	s_2	s_3	θ
0	s_1	1	3	1	0	0	2	
0	s_2	0	−3	−5	1	0	0	
0	s_3	0	2	−3	0	1	0	
	$Z_j = \sum c_B a_{ij}$	0	0	0	0	0	0	
	$C_j = c_j - Z_j$	0	−3	−5	0	0		

As C_j is either zero or negative under all columns, the above table gives the optimal basic feasible solution. This optimal solution is $x_1 = 2$, $s_1 = 1$ and maximum $Z = 12$.

Example 9.11. *A firm produces three products A, B and C. It uses two machines. The relevant data is given below :*

Machine	Time per unit (minutes)			Machine capacity (minutes/day)
	Product A	Product B	Product C	
M_1	2	3	1	430
M_2	3	—	1	420
M_3	1	2	—	460

The profit per unit for products A, B and C is ₹ 3, ₹ 2 and ₹ 5 respectively. Determine the daily number of units to be manufactured for each product. Assume that all the units produced are consumed in the market.

Solution. Let the firm decide to produce x_1, x_2, x_3 units of products A, B, C respectively. Then the L.P. model for this problem is :

Max. $\quad Z = 3x_1 + 2x_2 + 5x_3$

subject to $\quad 2x_1 + 3x_2 + x_3 \le 430,\ 3x_1 + x_3 \le 420,\ x_1 + 2x_2 \le 460,\ x_1, x_2, x_3 \ge 0$ (U.P.T.U., 2004)

Step 1. Check whether the objective function is to be maximized and all b_i are non-negative.

The problem being of maximization type and b_i being ≥ 0, this step is not necessary.

Step 1: Express the problem in the standard form.

By transforming the slack variables x_4, x_5, x_6, the problem is standardized into the following form:

Minimize $Z = -4x_1 - 5x_2 + 0x_3 + 0x_4 + 0x_5 + 0x_6$

subject to $3x_1 + 2x_2 + x_3 + x_4 + 0x_5 + 0x_6 = 10$

$2x_1 + 3x_2 + 0x_3 + 0x_4 + x_5 + 0x_6 = 15$

$x_1 + x_2 + 0x_3 + 0x_4 + 0x_5 + x_6 = 6$

$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$

Step 2: Find out initial basic possible solutions.

The basic non-negative initial feasible solution is:

$x_3 = 10, x_4 = 15, x_5 = 6$ (non-basic)

$x_1 = 0, x_2 = 0$ (basic)

$Z = 0$ (initial basic feasible solution is given by the following table)

	c_j	-4	-5	0	0	0	0		
c_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	b	θ
0	x_3	3	2	1	0	0	0	10	10/2
0	x_4	2	3	0	1	0	0	15	15/3
0	x_5	1	1	0	0	1	0	6	6/1
	z_j	0	0	0	0	0	0		
	$z_j - c_j$	4	5	0	0	0	0		

Step 3: Apply optimality test.

Since $z_j - c_j$ is positive under some columns, the initial basic feasible solution is not optimal and we proceed to the next step.

Step 4: Find the incoming and outgoing variables.

The above table shows that x_2 is the incoming variable while x_5 is the outgoing variable and 1 is the key element.

Find values by the optimal solution.

Since x_2 and variables x_5 with the second pivot value is make x_2 column. Convert the key element to unity and make all other components of key column zero. Then the second pivotal solution is given by the table below:

	c_j	-4	-5	0	0	0	0		
c_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	b	θ
0	x_3	1	0	1	0	-2	0	2	2/1
0	x_4	2	0	0	1	-3	0	3	3/2
-5	x_2	1	1	0	0	1	0	6	6/1
	z_j	-5	-5	0	0	5	0		
	$z_j - c_j$	-1	0	0	0	5	0		

Step 5: As $z_j - c_j$ is positive under the second column, the solution is not optimal and we proceed further. Now x_1 is the incoming variable and x_4 is the outgoing variable and 2 is the key element for the next iteration.

Since x_1 and variables x_4 with the second pivot value 2 make x_1 column. Convert the key element to unity and make all other components of the key column zero. Then the third basic possible solution is given by the following table:

	c_j	-4	-5	0	0	0	0		
c_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	b	θ
0	x_3	0	0	1	-1/2	1/2	0	1/2	
-4	x_1	1	0	0	1/2	-3/2	0	3/2	
-5	x_2	0	1	0	-1/2	5/2	0	9/2	
	z_j	-4	-5	0	-3/2	13/2	0	-51/2	
	$z_j - c_j$	0	0	0	-3/2	13/2	0		

Now since each $x_j \geq 0$, therefore to get max the optimal solution

$x_1 = 0, x_2 = 5000$ ft, $x_3 = 7500$ and $Z_{max} = 15000$ i.e., 15000 ft² region.

Example 3.1.11: Solve the following LPP by simplex method

Maximize $Z = x_1 + 3x_2 + 2x_3$

subject to $3x_1 + x_2 + x_3 \leq 6$, $x_1 - x_2 + 2x_3 \leq 1$, $x_1 + 2x_2 - 4x_3 \geq 8$, $x_1, x_2, x_3 \geq 0$.

(Madurai Univ., 2004, U.P.T.U., 2004)

Solution: The various steps of this interesting simplex method are as follows:

Step 1: Check whether objective function is to be maximized or minimized. If it is minimization problem then convert it into maximization problem by using the result Min. $Z' = -\text{Max. } Z$ where $Z = -Z'$.

As the right hand side of the second constraint is negative, multiply it on both sides by -1 so as to make it positive. The inequality sign of the constraint is changed from $\geq$ to $\leq$. Thus we have

$$-x_1 - x_2 + 2x_3 \leq -8$$

Step 2: Express the problem in the standard form.

For converting the above inequalities into equations, we introduce slack, surplus and artificial variables as follows:

Maximize $Z' = -x_1 - 3x_2 - 2x_3 + 0x_4 + 0x_5 + 0x_6 - Mx_7 - Mx_8$

subject to the constraints
$$3x_1 + x_2 + x_3 + x_4 = 6$$
$$x_1 - x_2 + 2x_3 - x_5 + x_7 = 1$$
$$-x_1 - x_2 + 2x_3 + x_6 - x_8 = -8$$
$$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8 \geq 0$$

Step 3: Find initial basic feasible solution.

The basic non-degenerate feasible solution is

$x_1 = x_2 = x_3 = 0$ (non-basic) $x_4 = 7$, $x_7 = 13$, $x_8 = 18$ (basic)

i.e., the initial basic feasible solution is given by the table below:

C_j		-1	-3	-2	0	0	0	$-M$	$-M$	
C_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	
0	x_4	3	1	1	1	0	0	0	0	$6/1 = 6$
$-M$	x_7	1	-1	2	0	-1	0	1	0	$13/2 = 6.5$
$-M$	x_8	-1	-2	2	0	0	1	0	1	$18/2 = 9$
	Z_j	$-2M$	$3M$	$-4M$	0	M	0	$-M$	$-M$	
$Z_j - C_j = \Delta_j$		$-2M+1$	$3M+3$	$-4M+2$	0	M	0	0	0	
$Z_j - C_j$		$\uparrow$								

Step 4: Apply optimality test.

As Δ_j is positive under several columns, the initial basic feasible solution is not optimal and we proceed further.

Step 5: (i) Identify the incoming and outgoing variables.

The above table shows that x_3 is the incoming variable, x_3 is the outgoing variable and 2 is the key element.

(ii) Revise to obtain the optimal solution.

Divide x_3 row i.e. pivot row by 2 to get the new pivot row. Use this new pivot row to obtain the new elements in the x_3 column. Then use the new pivot row to obtain the remaining new elements. Thus we have the revised basic feasible solution as given by the following table:

C_j		-1	-3	-2	0	0	0	$-M$	$-M$
C_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
0	x_4	$5/2$	2	0	1	$1/2$	0	0	0
-2	x_3	$-1/2$	-1	1	0	$-1/2$	0	$1/2$	0
$-M$	x_8	-3	0	0	0	1	1	-1	1
	Z_j	$1-3M$	$2+3M$	-2	0	$-1+M$	0	$1-M$	$-M$
$Z_j - C_j = \Delta_j$		$-3M$	$3M+5$	0	0	$M-1$	0	$1-M$	0
$Z_j - C_j$			$\uparrow$						

Step-II: As x_2 is negative artificial basic variable, this solution is not optimal and we proceed further. x_2 is the leaving variable and x_1 is the entering variable and $(2,1)$ is the key element.

$\therefore$ Dropping x_2 and introducing x_1 with the transformed value $\frac{1}{2}$ in x_1 column. Therefore the key element is 2 and we find all other elements of the key column zero. Then the third basic feasible solution is given by the table below :

c_j		$-M$	0	$-M$	0	0	0	
c_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	θ
$-M$	x_1	1	0	[illegible]	[illegible]	0	[illegible]	[illegible]
0	x_5	0	0	[illegible]	[illegible]	1	[illegible]	[illegible]
0	x_6	0	0	[illegible]	[illegible]	0	[illegible]	[illegible]
		0	0	[illegible]	[illegible]	0	[illegible]	[illegible]
		0	0	[illegible]	[illegible]	0	[illegible]	

Now since each $z_j - c_j \geq 0$, therefore it gives the optimal solution

$x_1 = [illegible]$, $x_5 = [illegible]$, $x_6 = 4$ (non-basic) and $Z_{max} = [illegible]$

Hence $Z_{max} = [illegible]$

[illegible]

Solution: [illegible] of the following type :

Step-I: [illegible]

[illegible]

Step-II: [illegible]

[illegible] x_3 is a slack variable. By introducing artificial variable x_4 and x_5 this problem is converted into :

Now $\quad [illegible]$

subject to [illegible]

Step-III: [illegible]

[illegible]

Now since feasible solution is

$x_1 = x_2 = x_3 = 0$ (non-basic) $x_4 = [illegible]$, $x_5 = [illegible]$, $x_6 = [illegible]$

$\therefore$ Initial basic feasible solution is given in the table below :

c_j		[illegible]	0	0	0	0	0		
c_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	θ	
0	x_4	[illegible]	[illegible]	[illegible]	1	0	0	[illegible]	[illegible]
0	x_5	[illegible]	[illegible]	[illegible]	0	1	0	[illegible]	[illegible]
0	x_6	[illegible]	[illegible]	[illegible]	0	0	1	[illegible]	[illegible]
$z_j = \sum c_B x_{ij}$		0	0	0	0	0	0		
$z_j - c_j$		[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]		

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Step 4: Apply optimality test.

As C_3 is positive under some columns, the initial basic feasible solution is not optimal and we proceed further.

Step 5: (i) Identify the incoming and outgoing variables.

The above table shows that x_1 is the incoming variable, x_5 is the outgoing variable and (2) is the key element.

(ii) Iterate towards the optimal solution.

Drop x_5 and introduce x_1 with its associated value 337 under x_2 column. Convert key element to unity and make all other elements of the key column zero. Then the second basic feasible solution is given by the following table :

	x_1	337	1	2	3	4	5		
x_3	Basis	x_1	x_2	x_3	x_4	x_5	x_6	$\frac{b}{a}$	θ
3	x_4	0	1509	− 0.6	1	0	1.8	775	− 2514
3	x_5	0	0.8	− 0.5	0	1	− 1.5	0	− 337
337	x_1	1	− 1.5	− 1.0	0	0	4.5	0	0
	Z_j	337	− 505.5	− 1349	0	0	1516		
	$C_j - Z_j$	0	1505	1349	0	0	− 1516		
				↑					

As $C_j - Z_j$ is positive under some columns, the solution is not optimal. Here 1505 being the largest positive value of $C_j - Z_j$, is the incoming variable. But all the values of 0 being ≤ 0, x_4 will not enter the basis. This indicates that the solution to the problem is unbounded.

[Remember that (i) the incoming variable is the non-basic variable corresponding to the largest positive value of $C_j - Z_j$ and

(ii) the outgoing variable is the basic variable corresponding to the least positive ratio θ, obtained by dividing the x_3-column elements by the corresponding key-column elements]

EXERCISE 5(C)

Using simplex method, solve the following L.P.P. (1 to 6) :

1. Maximise $Z = x_1 + 2x_2$
   subject to $x_1 + 2x_2 \leq 30$, $x_1 + x_2 \leq 15$, $x_1, x_2 \geq 0$. (Anna Univ. 2008; M.T.U. 2002)
2. Maximise $Z = 4x_1 + 10x_2$
   subject to $2x_1 + x_2 \leq 20$, $2x_1 + 5x_2 \leq 100$, $2x_1 + 3x_2 \leq 90$, $x_1, x_2 \geq 0$. (Anna Univ. 2006)
3. Maximise $Z = 4x_1 - 5x_2$
   subject to $x_1 - 2x_2 \leq -2$, $-x_1 + x_2 \leq 1$, $2x_1 + 3x_2 \leq 6$, $x_1, x_2 \geq 0$.
4. Maximise $Z = 10x_1 + x_2 + 2x_3$
   subject to $x_1 + x_2 - 2x_3 \leq 10$, $4x_1 + x_2 + x_3 \leq 20$, $x_1, x_2, x_3 \geq 0$.
5. Maximise $Z = 3x_1 - 2x_2 + 3x_3$, subject to $x_1 - 3x_2 + x_3 \leq 10$, $2x_1 - 4x_2 \leq 16$, $x_1 + 4x_3 \leq 40$, $x_1, x_2, x_3 \geq 0$.
   (Chennai 2006)
6. Maximise $Z = 5x_1 + 3x_2 + 4x_3$
   subject to $2x_1 + 3x_2 \leq 8$, $2x_2 + 5x_3 \leq 10$, $3x_1 + 2x_2 + 4x_3 \leq 15$, $x_1, x_2, x_3 \geq 0$. (Madras 2004)
7. Maximise $Z = x_1 + 2x_2 + 3x_3$
   subject to $2x_1 - x_2 + 2x_3 \leq 7$, $-2x_1 + 4x_2 \leq 12$, $-4x_1 + 3x_2 + 8x_3 \leq 10$, $x_1, x_2, x_3 \geq 0$. (Madras 2000)
8. Maximise $Z = 4x_1 + 6x_2 + 4x_3 + 9x_4$
   subject to $x_1 + 2x_2 + x_3 + 4x_4 \leq 60$, $2x_1 + x_2 + x_3 + 3x_4 \leq 40$, $3x_1 + x_2 + x_3 + x_4 \leq 80$, $x_1, x_2, x_3, x_4 \geq 0$.
9. A firm produces products A and B and sells them at a profit of ₹ 2 and ₹ 3 each respectively. Each product is processed on machines Grand M. Product A requires 1 minute on G and 2 minutes on M whereas product B requires 1 minute on G and 1 minute on M. Machine G is not available for more than 6 hrs. 40 minutes whereas machine M is available for 10 hrs. Solve this problem graphically to find the maximising profit.
10. A company makes two types of products. Each product of the first type requires twice as much labour time as the second type. If all products are of second type only, the company can produce a total of 500 units a day. The market limits daily sales of the first and the second type to 150 and 300 units respectively. Assuming that the profit per

unit and 6 lb for type I and 4 oz for type II, determine the number of units of each type to be produced to maximize profit?

11. The owner of a dairy is trying to determine the optimal blend of two types of feed. Both contain various percentages of four essential ingredients. With the following data determine the least cost blend?

Ingredients	Oz per kg of feed		Min requirement per day
	Feed 1	Feed 2	
1	40	20	8
2	10	30	3
3	30	40	7
4	60	20	8
Cost (Rs/kg)	5	3	

12. A manufacturing firm has discontinued production of a certain unprofitable product Z. This created considerable excess production capacity. Management is considering to devote this excess capacity to one or more of three products 1, 2, and 3. The available capacity on various machines and the number of machine-hours required for each unit of the respective products, is given below:

Machine Type	Available Time (Machine-hours/week)	Production Time (Hours/unit)		
		Product 1	Product 2	Product 3
Milling machine	500	9	3	5
Lathe	350	5	4	–
Grinder	300	3	–	1

The unit profit would be ₹ 50, ₹ 20 and ₹ 25 respectively for products 1, 2 and 3. Find how much of each product should be produced in order to maximize profit.

13. The following table gives the various vitamin contents of three types of food and daily requirements of vitamins A, B and C for healthy adult. Find the combination of food for minimum cost.

Vitamin (mg)	Food P	Food Q	Food R	Minimum daily requirement (mg)
A	1	1	10	2
C	100	20	10	20
B	10	100	10	10
Cost/unit (₹)	10	15	5	

14. A farmer has 1,000 pieces of land on which he can grow corn, wheat or soyabeans. Each piece of corn costs ₹ 100 for preparation, requires 7 man-days of work and yields a profit of ₹ 40. Each piece of wheat costs ₹ 120 to prepare, requires 10 man-days of work and yields a profit of ₹ 45. Each piece of soyabeans costs ₹ 70 to prepare, requires 8 man-days of work and yields a profit of ₹ 20. If he never has ₹ 100,000 for preparation and can count on 8,000 man-days of work, how many pieces should be allocated to each crop to maximize profit?

1.13 ARTIFICIAL VARIABLE TECHNIQUES

As we have seen, that the introduction of slack or plus variables provided the initial basic feasible solution. But there are many problems wherein at least one of the constraints is of (≥) or (=) type and slack variables fail to give such a solution. There are two similar methods for solving such problems which we explain below:

(1) Simplex or Method of Penalties. This method is due to A. Charnes and consists of the following steps:

Step 1. Express the problem in standard form.

Step 2. Add non-negative variables to the left hand side of all those constraints which are of (≥) or (=) type. Such new variables are called artificial variables and the purpose of introducing these is just to obtain an initial basic feasible solution. But their addition causes violation of the corresponding constraints. As such, we would

like to get rid of these variables and would not allow them to appear in the final solution. For this purpose we assign a very large penalty $-M$ to these artificial variables in the objective function.

Step 1: Write the modified LPP by adding artificial

(i) any iteration of simplex method, one of the following three cases may arise :

(a) There remains no artificial variable in the basis and the optimality condition is satisfied. Then the solution is an optimal basic feasible solution to the problem.

(b) There is at least one artificial variable in the basis at zero level with some value in its column and the optimality condition is satisfied. Then the problem is degenerate optimal basic feasible solution.

(c) There is at least one artificial variable in the basis at non-zero level with positive value in its column and the optimality condition is satisfied. Then the problem has no feasible solution. The final solution is not optimal since the objective function contains an artificial quantity M. Such a situation usually for some mistake has been committed like objective function and in formulation of the constraints.

(ii) If optimality condition is not satisfied, then we select an optimal basic feasible solution in ordinary manner and the procedure is repeated till we get an optimal basic feasible solution or unbounded solution is indicated.

Note : The artificial variables are used in a simplex technique known as (i) penalty (ii) two phase method. Here we confine ourselves to the penalty method and the two phase method is discussed in the next section.

Example 3.3.1. Use (The two phase method) to

Minimize $z = 4x_1 + x_2$

subject to $3x_1 + x_2 = 3$, $4x_1 + 3x_2 \geq 6$, $x_1 + 2x_2 \leq 4$, $x_1, x_2 \geq 0$.

Solution : The solution is as follows :

Step 1 : Express the problem in standard form.

The inequality constraints are converted into equations by introducing the surplus and slack variables x_3, x_4 respectively.

Now the third and second constraints having $+1$ and (-1) as coefficients have artificial variables A_1, A_2 respectively. Converting the minimization problem to the maximization form, the LPP can be formulated as

Maximize $Z' = -4x_1 - x_2 + 0x_3 + 0x_4 - MA_1 - MA_2$

subject to $3x_1 + x_2 + A_1 = 3$

$4x_1 + 3x_2 - x_3 + A_2 = 6$

$x_1 + 2x_2 + x_4 = 4$

$x_1, x_2, x_3, x_4, A_1, A_2 \geq 0$

Step 2 : Obtain an initial basic feasible solution.

Penalty variable x_3 is not chosen variable since its value is -1. So negative constraint and therefore x_3 must be prevented from appearing in the initial solution. This is done by taking $x_3 = 0$. By setting the other non-basic variables x_1, x_2 and $x_3 = 0$ we obtain the initial basic feasible solution as

$$x_1 = x_2 = 0, x_3 = 0, x_4 = 4, A_1 = 3, A_2 = 6$$

This can be put in the simplex table as

	c_j	-4	-1	0	0	$-M$	$-M$		
c_B	Basis	x_1	x_2	x_3	x_4	A_1	A_2	θ	
$-M$	A_1	3	1	0	0	1	0	3	3/1
$-M$	A_2	4	3	-1	0	0	1	6/3	6/4
0	x_4	1	2	0	1	0	0	4/2	4/1
$Z_j = \sum c_B a_{ij}$		$-7M$	$-4M$	M	0	$-M$	$-M$	$-9M$	
$Z_j - c_j$		$7M - 4$	$4M - 1$	$-M$	0	0	0		

Step 3 : Since x_1 is positive under x_1 and A_2 is leaving, then it will be replaced by x_1.

Step 4 : Repeat the procedure until optimal solution.

Here x_1 and x_4 are x_1 and x_2 in the basis.

∴ The new simplex table is

	c_j	−2	−3	0	0	−M		
c_B	Basis	x_1	x_2	s_1	s_2	A_1	b	θ
−2	x_1	1	1/3	0	2/3	0	1	3
−M	A_1	0	5/3	−1	1/3	1	3	9/5
0	s_1	0	2/3	0	1/3	0	2	6
	Z_j	−2	$-\frac{5}{3}M-\frac{2}{3}$	M	$\frac{1}{3}-\frac{1}{3}M$	−M	$-2-3M$	
	C_j	0	$-\frac{7}{3}+\frac{5}{3}M$	−M	$-\frac{1}{3}+\frac{1}{3}M$	0		

Since C_j is positive under x_2 column, this is not an optimal solution.

∴ Introduce x_2 and drop A_1.

Then the revised simplex table is

	c_j	−2	−3	0	0	
c_B	Basis	x_1	x_2	s_1	s_2	b
−2	x_1	1	0	1/5	2/5	3/5
−3	x_2	0	1	−3/5	1/5	9/5
0	s_2	0	0	1	−1	0
	Z_j	−2	−3	21/5	4/5	−33/5
	C_j	0	0	−21/5	4/5	

Since none of C_j is positive, this an optimal solution. Thus, an optimal basic feasible solution to the problem is

$$x_1 = 3/5, x_2 = 9/5, \text{Max. } Z' = -33/5.$$

Hence the optimal value of the objective function is

$$\text{Min. } Z = -\text{Max. } Z' = -(-33/5) = 33/5.$$

Example 24.15. Minimize $Z = x_1 - 2x_2$

subject to the constraints $2x_1 + x_2 \geq 2$, $x_1 + 3x_2 \leq 5$, $x_1 + x_2 \leq 4$, $x_1, x_2 \geq 0$.

Solution. Consists of the following steps :

Step 1. Express the problem in standard form

The inequalities are converted into equations by introducing the slack and surplus variables s_1, s_2 respectively. Also the second constraint being of $\geq$ type, we introduce the artificial variable A. Thus the L.P.P. can be re-written as

$$\text{Max. } Z' = 2x_1 + 2x_2 + 0s_1 + 0s_2 - MA$$

subject to

$$2x_1 + x_2 - s_1 + 0s_2 + 0A = 2,$$

$$x_1 + 3x_2 + 0s_1 + s_2 + 0A = 15,$$

$$x_1 + x_2 + s_1 + 0s_2 + A = 12,$$

$$x_1, x_2, s_1, s_2, A \geq 0.$$

Step 2. Find an initial basic feasible solution.

Surplus variable s_1 is not a basic variable since its value is -12. Since a negative quantity is not feasible, s_1 would be prevented from appearing in the initial solution. This is done by letting $s_1 = 0$. By taking the other non-basic variables x_1 and x_2 each $= 0$, we obtain the initial basic feasible solution as

$$x_1 = x_2 = s_1 = 0, s_2 = 15, A = 12.$$

∴ The initial simplex table is

	c_j	2	2	0	0	−M		
c_B	Basis	x_1	x_2	s_1	s_2	A	b	θ
0	s_2	2	(1)	−1	0	1	2	2←
−M	A	1	1	0	−1	1	12	12
$Z_j = \Sigma c_B x_{ij}$		−M	−M	0	M	−M	−12M	
$C_j = c_j - Z_j$		2 + M	2 + M	0	−M	0		

Since C_j is positive under some columns, this is not an optimal solution.

Step 3. Iterate for the optimal solution.

Introduce x_2 and drop x_1.

∴ The new simplex table is

	C_j	3	2	1	0	$-M$	
C_B	Basis	x_1	x_2	x_3	x_4	A_1	b
2	x_2	2	1	1	0	1	2
$-M$	A_1	-3	0	-1	-1	1	1
Z_j		$-3+3M$	2	$2-M$	M	$-M$	$4-M$
C_j		$4-3M$	0	$-2+M$	$-M$	0	

Here each C_j is negative and an artificial variable appears in the basis at non-zero level. Thus there exists a pseudo optimal solution to the problem.

(2) Two-phase method. This is another method to deal with the artificial variables wherein the L.P.P. is solved in two phases.

Phase I. *Step 1.* Express the given problem in the standard form by introducing slack, surplus and artificial variables.

Step 2. Form a new artificial objective function

$$Z^* = -A_1 - A_2 \ldots - A_n$$

by assigning -1 cost to each of the artificial variables A_i and zero cost to all other variables.

Step 3. Maximize Z^* subject to the constraints of the original problem using the simplex method. Then there can arise either

(a) Max. $Z^* < 0$ and at least one artificial variable appears in the optimal basis at a positive level.

In this case, the original problem does not possess any feasible solution and the procedure comes to an end.

(b) Max. $Z^* = 0$ and no artificial variable appears in the optimal basis.

In this case, a basic feasible solution is obtained and we proceed to phase II for finding the optimal basic feasible solution to the original problem.

(c) Max. $Z^* = 0$ and at least one artificial variable appears in the optimal basis at zero level.

Here a feasible solution to the auxiliary L.P.P. is also a feasible solution to the original problem with all artificial variables set = 0.

To obtain a basic feasible solution, we prolong phase I for pushing all the artificial variables out of the basis (without proceeding to phase II).

Phase II. The basic feasible solution found at the end of phase I is used as the starting solution for the original problem in this phase i.e., the final simplex table of phase I is taken as the initial simplex table of phase II and the artificial objective function is replaced by the original objective function. Then we find the optimal solution.

Example 3.4 (b). Use two-phase method to

Minimize $Z = 3x_1 - x_2$

subject to the constraints $2x_1 + x_2 - x_3 \geq 2$, $x_1 + x_2 - x_3 \leq 2$,

$x_1, x_2, x_3 \geq 0$.

Phase I. *Step 1.* Express the problem in standard form.

Solution. Introducing surplus variables s_1, s_2 and artificial variables A_1, A_2, the phase I problem in standard form becomes

Max. $Z^* = 0x_1 + 0x_2 - 0x_3 + 0s_1 + 0s_2 - A_1 - A_2$

subject to $2x_1 + x_2 - x_3 - s_1 + 0s_2 + A_1 + 0A_2 = 2$

$x_1 + x_2 - x_3 + 0s_1 - s_2 + 0A_1 + A_2 = 2$

$x_1, x_2, x_3, s_1, s_2, A_1, A_2 \geq 0$.

Step 2. Find an initial basic feasible solution.

Setting $x_1 = x_2 = x_3 = s_1 = s_2 = 0$,

we have $A_1 = 2$, $A_2 = 2$ and $Z^* = -4$.

(c) Initial simplex table is

	c_j	4	6	0	0	0	−1	−1		
c_B	Basis	x_1	x_2	s_1	s_2	s_3	A_1	A_2	θ	
−1	A_1	3	1	−1	0	0	1	0	3	3
−1	A_2	4	3	0	−1	0	0	1	4	4/3
0	s_3	1	2	0	0	1	0	0	2	2
$Z^* = \sum c_B x_B$		−7	−4	1	1	0	−1	−1		
$C_j - Z_j$		7	4	−1	−1	0	0	0		

As C_j in positive under x_1 column, this solution is not optimal.

Step 2: Revise the above to get optimal solution

Dividing key elements (R ratio) and replacing A_2 by x_2 we have the next simplex table :

	c_j	0	0	0	0	0	−1	−1		
c_B	Basis	x_1	x_2	s_1	s_2	s_3	A_1	A_2	θ	
−1	A_1	$\frac{5}{3}$	0	$-\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{5}{3}$	1
0	x_2	$\frac{4}{3}$	1	$-\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{4}{3}$	$\frac{4}{5}$
0	s_3	$\frac{5}{3}$	0	$\frac{2}{3}$	$\frac{2}{3}$	1	$-\frac{2}{3}$	$-\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{5}$
	Z^*	$\frac{5}{3}$	0	$-\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{3}$		
	$C_j - Z_j$	$-\frac{5}{3}$	0	$\frac{1}{3}$	$\frac{1}{3}$	0	$-\frac{4}{3}$	$-\frac{4}{3}$		

Since C_j is positive under x_1 and x_2 columns, the solution is not optimal.

Making key element (5/3) unity and replacing A_1 by x_1 we obtain the revised simplex table :

	c_j	0	0	0	0	0	−1	−1	
c_B	Basis	x_1	x_2	s_1	s_2	s_3	A_1	A_2	θ
0	x_1	1	0	−0.2	0.2	0	0.2	−0.2	0.8
0	x_2	0	1	−0.4	−0.4	0	0.4	0.4	0.8
0	s_3	0	0	0.8	0.8	1	−0.8	−0.8	0.8
	Z^*	0	0	0	0	0	0	0	
	$C_j - Z_j$	0	0	0	0	0	−1	−1	

Since all $C_j \geq 0$, this table gives the optimal solution. Also $A_1^* = A_2^* = 0$ and an artificial variable appears in the basis. So as the optimal basic feasible solution to the original problem and therefore to the original problem, has been obtained.

Phase II : Considering the above final table associated with the original variables, the objective function is

Here, $Z^* = -1.5(0.8) + 0(0.8) + 0(0.8) = -1.2$ or $-Z^* = 1.2$

subject to $0.8x_1 + 0.2x_2 + s_1 = 2$

$0.8x_1 + 0.4x_2 + s_2 = 2$

$0.8x_1 + 0.8x_2 + s_3 = 2$

$x_1, x_2, s_1, s_2, s_3 \geq 0$

The optimal artificial free basic solution thus obtained, will be an optimal basic feasible solution to the original LPP.

Using final table of phase I, the initial simplex table of phase II is as follows :

	c_j	−1.2	0	0	0	0	
c_B	Basis	x_1	x_2	s_1	s_2	s_3	θ
−1.2	x_1	1	0	−0.2	0.2	0	0.8
0	x_2	0	1	−0.4	−0.4	0	0.8
0		0	0	0.8	0.8	1	0.8
		−1.2	0	0.24	0.24	0	−1.2
		0	0	−0.24	−0.24	0	

Since all $C_j \geq 0$, this solution is optimal.

Hence the optimal basic feasible solution to the given problem is

$x_1 = 0.8$, $x_2 = 0.8$, $s_3 = 0.8$ and max $Z = 1.2$

14.10 EXCEPTIONAL CASES

(i) Tie for the incoming variable. When more than one variable has the same largest positive value in $C_j - Z_j$ row in a maximization problem, a tie for the choice of incoming variable occurs. As there is no method to break this tie, we choose any one of the prospective incoming variables arbitrarily. Such an arbitrary choice does not in any way affect the optimal solution.

(ii) Tie for the outgoing variable. When more than one variable has the same least positive ratio under $\frac{b_i}{a_{ij}}$ column, a tie for the choice of outgoing variable occurs. If the equal ratios of old basic variables > 1, choose any one of the prospective leaving variables arbitrarily. Such an arbitrary choice doesn't affect the optimal solution.

If the equal values of ratios are zero, the simplex method fails and we make use of the following degeneracy technique.

(iii) Degeneracy. We know that a basic feasible solution is said to be degenerate if any of the basic variables vanishes. This phenomenon of getting a degenerate basic feasible solution is called degeneracy which may arise

(a) at the initial stage, when at least one basic variable is zero in the initial basic feasible solution

or (b) at any subsequent stage, when the least positive ratios under $\frac{b_i}{a_{ij}}$ are equal for two or more rows.

In this case, an arbitrary choice of one of these basic variables may result in one or more basic variables becoming zero in the next iteration. At times, the successive repetition of simplex iterations is repeated endlessly without improving the solution. There are terms as cycling type of problems. Cycling occurs very rarely. In fact, cycling has seldom occurred in practical problems.

To avoid cycling, we apply the following perturbation procedure :

(a) Divide each element in the tied rows by the positive coefficients of the key column in that row.

(b) Compare the resulting values from left to right (first of unit matrix and then of the body matrix), column by column.

(c) The outgoing variable is in that row which first contains the smallest algebraic ratio.

Example 14.11. Maximize $Z = 6x_1 + 8x_2$
subject to $5x_1 + 10x_2 \le 60$, $10x_1 + 4x_2 \le 40$, $x_1, x_2 \ge 0$.

Solution. Consists of the following steps :

Step 1. Express the problem in the standard form.

Introducing the slack variables s_1, s_2, the problem in the standard form is

$$\text{Max. } Z = 6x_1 + 8x_2 + 0s_1 + 0s_2$$
$$5x_1 + 10x_2 + s_1 = 60$$
$$10x_1 + 4x_2 + s_2 = 40$$
$$x_1, x_2, s_1, s_2 \ge 0$$

Step 2. Find the initial basic feasible solution.

The initial basic feasible solution is

$$x_1 = x_2 = 0 \text{ (non-basic)}$$
$$s_1 = 60, s_2 = 40 \text{ (basic) and } Z = 0.$$

∴ Initial simplex table is

	x_1	6	8	0	0	0		
c_B	Basis	x_1	x_2	s_1	s_2	s_3	b	$\frac{b}{a}$
0	s_1	1	1	1	0	0	3	3
0	s_2	10	2	0	1	0	10	10/2=
0	s_3	5	0	0	0	1	20	20/5
$Z_j = \Sigma c_B a_{ij}$		0	0	0	0	0	0	
$C_j = c_j - Z_j$		6	8	0	0	0		
		↑						

As C_j is positive under x_3 column, this solution is not optimal.

Step 3. Obtain second optimal solution.

x_3 is the incoming variable. But the first two rows have the same ratio under θ-column. Therefore we apply perturbation method.

First column of the unit matrix has 1 and 0 in the first row. Dividing them by the corresponding elements of the key column, we get $\frac{1}{2}$ and $\frac{0}{2}$. a_2 row gives the smaller ratio and therefore x_2 is the first outgoing variable and 2 is the key element.

Thus the new simplex table is

	x_2	8	5	0	3	0		
c_B	Basis	x_1	x_2	x_3	x_4	x_5	b	θ
0	x_5	3	1001	1	−10	0	2	0
5	x_2	1	20	0	15	0	2	2←
0	x_6	0	960	0	−30	1	8	1917
Z_j		5	5	0	1	0	10	
C_j		0	1	0	−1	0		
			↑					

As C_j is positive under x_1 column, this solution is not optimal.

Making key element (30) unity and replacing x_6 by x_3, we obtain the revised simplex table:

	x_3	8	5	0	0	0	
c_B	Basis	x_1	x_2	x_3	x_4	x_5	b
0	x_5	0	0	0.5	−0.5	0	1
5	x_2	1	0	−0.25	0.5	0	3
8	x_1	0	0	−0.25	0.5	1	6
Z_j		8	5	0.5	2.5	0	49
C_j		0	0	−0.5	−2.5	0	

As $C_j \leq 0$ for all the columns, this table gives the optimal solution. Hence an optimal basic feasible solution is $x_1 = 6, x_2 = 3, x_3 = 0$ and $Z_{max} = 81$.

PROBLEMS (M5)

Solve the following LPP problems using Simplex method.

1. Maximize $Z = 3x_1 + 5x_2 + 2x_3$
   subject to $2x_1 + x_2 + x_3 \leq 8, 3x_1 + 4x_2 + 2x_3 \leq 8, x_1, x_2, x_3 \geq 0$
2. Maximize $Z = 4x_1 + x_2 + 3x_3$
   subject to $2x_1 + x_2 + 2x_3 \leq 4, 3x_1 + 4x_2 + 2x_3 \leq 12, x_1, x_2, x_3 \geq 0$
3. Maximize $Z = 6x_1$
   subject to $x_1 + x_2 \leq 5, 2x_1 + 5x_2 \leq 10, x_1, x_2$ are unrestricted.
4. Maximize $Z = 2x_1 - 3x_2 + 3x_3$
   subject to $3x_1 + 5x_2 - x_3 \leq 7, 2x_1 - 4x_2 + 3x_3 \geq 12, x_1, x_2, x_3 \geq 0$. (Madurai, 2000)
5. Maximize $Z = x_1 + 3x_2 - 3x_3 - x_4$
   subject to $x_1 + 3x_2 + 3x_3 \leq 30, 2x_1 + x_2 + 4x_3 \geq 8$,
   $x_1 + 2x_2 + x_3 + x_4 = 15, x_1, x_2, x_3, x_4 \geq 0$. (Madurai, 2000)

Use the graphical method to solve the following LPP problems:

6. Maximize $Z = x_1 + x_2$
   subject to $2x_1 + x_2 \geq 1, x_1 + 3x_2 \geq 2$,
   $x_1, x_2 \geq 0$. (Madurai, 2002)
7. Maximize $Z = 6x_1 + 8x_2$
   subject to $5x_1 + 10x_2 \leq 60, x_1 + x_2 \geq 6$,
   $x_1, x_2 \geq 0$. (Madurai, 2003)

9. Maximize $Z = 3x_1 - 4x_2 + 5x_3$
subject to $2x_1 + 3x_2 - x_3 \geq 8$
$x_1 + 2x_2 + 6x_3 \leq 20$
$x_1, x_2, x_3 \geq 0$. (Madras, 2000)

10. Maximize $Z = 3x_1 + 2x_2 + 5x_3$
subject to $x_1 + x_2 + 2x_3 \leq 30$
$x_1 + 2x_2 + 3x_3 \leq 60$
$x_1, x_2, x_3 \geq 0$

Solve the following Asymmetric L.P.P. problems:

11. Maximize $Z = 3x_1 + 5x_2$
subject to $x_1 + 3x_2 \leq 8$, $2x_1 + x_2 \leq 6$
$x_1, x_2 \geq 0$

12. Maximize $Z = 2x_1 + 3x_2 - 4x_3$
subject to $x_1 + 2x_2 + 3x_3 = 10$
$2x_1 + x_2 + 2x_3 = 8$
$x_1, x_2, x_3 \geq 0$

9.11 (1) DUALITY CONCEPT

One of the most interesting concepts in linear programming is the duality theory. Every linear programming problem has associated with it, another linear programming problem involving the same data and closely related optimal solutions. Both two problems are said to be duals of each other. While one of them is called the primal, the other the dual.

The importance of the duality concept is due to two main reasons. Firstly, if the primal involves a large number of constraints and a smaller number of variables, the labour of computation can be tremendously reduced by converting it into the dual problem and then solving it. Secondly, the interpretation of the dual variables from the cost or economic point of view proves extremely useful in making future decisions in the situation being programmed.

(2) Formulation of dual problem. Consider the following L.P.P.:

Maximize $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$
subject to the constraints
$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \leq b_1$$
$$a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \leq b_2$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \leq b_m$$
$$x_1, x_2, \ldots, x_n \geq 0$$

To construct the dual problem, we adopt the following guidelines:

(i) The maximization problem in the primal becomes the minimization problem in the dual and vice versa.

(ii) (a) type of inequality in the primal becomes (b) type of constraints in the dual and vice versa.

(iii) The coefficients $c_1, c_2, \ldots, c_n$ in the objective function of the primal become $b_1, b_2, \ldots, b_n$ in the objective function of the dual.

(iv) The constants $b_1, b_2, \ldots, b_m$ in the constraints of the primal become $c_1, c_2, \ldots, c_n$ in the constraints of the dual.

(v) If the primal has n variables and m constraints, the dual will have m variables and n constraints. i.e. the transpose of the body matrix of the primal problem gives the body matrix of the dual.

(vi) The variables in both the primal and dual are non-negative.

Then the dual problem will be

Minimize $W = b_1y_1 + b_2y_2 + \ldots + b_my_m$
subject to the constraints
$$a_{11}y_1 + a_{21}y_2 + \ldots + a_{m1}y_m \geq c_1$$
$$a_{12}y_1 + a_{22}y_2 + \ldots + a_{m2}y_m \geq c_2$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$a_{1n}y_1 + a_{2n}y_2 + \ldots + a_{mn}y_m \geq c_n$$
$$y_1, y_2, \ldots, y_m \geq 0.$$

Example 9.11.1. Write the dual of the following L.P.P.:
Maximize $Z = 4x_1 - 2x_2 + 6x_3$
subject to $3x_1 + 5x_2 + 6x_3 \leq 7$, $5x_1 + 3x_2 + 2x_3 \leq 10$,
$x_1 + 8x_2 + 4x_3 \geq 3$, $x_1, x_2, x_3 \geq 0$.

Solution. Since the problem is of minimization, all constraints should be of $\geq$ type. We multiply the third constraint throughout by -1 so that $-x_1 + 2x_2 + x_3 \geq -10$.

Let y_1, y_2, y_3, y_4 and y_5 be the dual variables associated with the above five constraints. Then the dual problem is given by

Maximize $W = 2y_1 + 4y_2 - 10y_3 + 3y_4 + 2y_5$

subject to $2y_1 + 4y_2 - 1y_3 + y_4 + 4y_5 \leq 3$, $3y_1 + y_2 + 2y_3 - 2y_4 + 2y_5 \leq 5$,

$4y_1 + 3y_2 + y_3 + 3y_4 - 2y_5 \leq 8$, $y_1, y_2, y_3, y_4, y_5 \geq 0$.

(ii) Formulation of dual problem when the primal has equality constraints. Consider the problem

Maximize $Z = c_1x_1 + c_2x_2$

subject to $a_{11}x_1 + a_{12}x_2 = b_1$, $a_{21}x_1 + a_{22}x_2 \leq (=, \geq)\, b_2$, $x_1, x_2 \geq 0$.

The equality constraint can be written as

$$a_{11}x_1 + a_{12}x_2 \leq b_1 \text{ and } a_{11}x_1 + a_{12}x_2 \geq b_1$$

or $$a_{11}x_1 + a_{12}x_2 \leq b_1 \text{ and } -a_{11}x_1 - a_{12}x_2 \leq -b_1.$$

Then the above problem can be restated as

Maximize $Z = c_1x_1 + c_2x_2$

subject to $a_{11}x_1 + a_{12}x_2 \leq b_1$, $-a_{11}x_1 - a_{12}x_2 \leq -b_1$

$a_{21}x_1 + a_{22}x_2 \leq b_2$, $x_1, x_2 \geq 0$.

Now we form the dual using y_1', y_1'', y_2 as the dual variables. Then the dual problem is

Minimize $W = b_1(y_1' - y_1'') + b_2y_2$

subject to $a_{11}(y_1' - y_1'') + a_{21}y_2 \geq c_1$, $a_{12}(y_1' - y_1'') + a_{22}y_2 \geq c_2$, $y_1', y_1'', y_2 \geq 0$.

The term $y_1' - y_1''$ appears in both the objective function and all the constraints of the dual. This will always happen whenever there is an equality constraint in the primal. Then the new variable $y_1' - y_1'' = y_1$ becomes unrestricted in sign being the difference of two non-negative variables and the above dual problem takes the form.

Minimize $W = b_1y_1 + b_2y_2$

subject to $a_{11}y_1 + a_{21}y_2 \geq c_1$, $a_{12}y_1 + a_{22}y_2 \leq c_2$, y_1 unrestricted in sign, $y_2 \geq 0$.

In general, if the primal problem is

Maximize $Z = c_1x_1 + c_2x_2 + \ldots + c_nx_n$,

subject to

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ &\cdots\cdots\cdots\cdots\cdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

$x_1, x_2, \ldots, x_n \geq 0$,

then the dual problem is

Minimize $W = b_1y_1 + b_2y_2 + \ldots + b_my_m$

subject to

$$\begin{aligned} a_{11}y_1 + a_{21}y_2 + \ldots + a_{m1}y_m &\geq c_1 \\ a_{12}y_1 + a_{22}y_2 + \ldots + a_{m2}y_m &\geq c_2 \\ &\cdots\cdots\cdots\cdots\cdots \\ a_{1n}y_1 + a_{2n}y_2 + \ldots + a_{mn}y_m &\geq c_n \end{aligned}$$

$y_1, y_2, \ldots, y_m$ all unrestricted in sign.

Thus the dual variables corresponding to equality constraints are unrestricted in sign. Conversely when the primal variables are unrestricted in sign, corresponding dual constraints are equalities.

Example 14.20. Construct the dual of the LPP

Maximize $Z = 3x_1 + 5x_2 - 2x_3$

subject to $2x_1 + 3x_2 + 2x_3 \leq 7$, $3x_1 - 2x_2 + 4x_3 \geq 8$, $x_1, x_2, x_3 \geq 0$.

Solution. Let y_1 and y_2 be the dual variables associated with the first and second constraints. Then the dual problem is

Minimize $W = 7y_1 + 8y_2$

subject to $2y_1 + 3y_2 \geq 3$, $3y_1 - 2y_2 \geq 5$, $2y_1 + 4y_2 \geq -2$, $y_1 \geq 0$, y_2 is unrestricted in sign.

PROBLEMS 34.8

Write the duals of the following problems (1 – 6) :

1. Maximize $Z = 10x_1 + 40x_2 + 18x_3$
   subject to $4x_1 + 2x_2 + 2x_3 \le 200,\ 4x_1 + 6x_2 + 3x_3 \le 300,\ x_1, x_2, x_3 \ge 0$.
2. Minimize $Z = 3x_1 + 4x_2$ (M-I)
   subject to $3x_1 + 4x_2 \ge 12,\ 4x_1 + 2x_2 \ge 8,\ 5x_1 + 3x_2 \ge 15,\ x_1, x_2 \ge 0$.
   $4x_1 + 2x_2 \ge 8,\ x_1, x_2 \ge 0$
3. Maximize $Z = 2x_1 + 5x_2 + 3x_3$
   subject to $x_1 + 2x_2 + x_3 \le 3,\ -x_1 + 3x_2 \le 1,\ 2x_1 + x_2 - 3x_3 \le 4,\ x_1, x_2, x_3 \ge 0$.
4. Minimize $Z = 5x_1 - 8x_2 + x_3$
   subject to $3x_1 - 2x_2 + x_3 \le 10,\ 4x_1 - 5x_2 \ge 0,\ -8x_1 + 5x_2 + 6x_3 \le 4$,
   $x_1, x_2 \ge 0$ and x_3 is unrestricted. (Meerut 2004)
5. Obtain the dual problem of the following LPP :
   Maximize $Z = 5x_1 + 6x_2 + 7x_3$
   subject to $3x_1 + 4x_2 + x_3 \le 10,\ 5x_1 + x_2 + 3x_3 \le 4,\ x_1 + 2x_2 + 5x_3 \le 7$,
   $-2x_1 - 3x_2 + 2x_3 \le 5,\ x_1, x_2, x_3 \ge 0$.
   Also verify that the dual of the dual problem is the primal problem.

34.13 (1) DUALITY PRINCIPLE

If the primal and the dual problems have feasible solutions then both have optimal solutions and the optimal value of the primal objective function is equal to the optimal value of the dual objective function i.e.,

$$\text{Max. } Z = \text{Min. } W$$

This is the fundamental theorem of duality. It suggests that an optimal solution to the primal problem can directly be obtained from that of the dual problem and conversely.

(2) Working rules for obtaining an optimal solution to the primal (dual) problem from that of the dual (primal) :

Suppose we have already found an optimal solution to the dual (primal) problem by simplex method.

Rule I. If the primal variable corresponds to a slack (surplus) variable in the dual problem, then its optimal value is directly given by the coefficients of the slack variable (changed sign in the $C_j - z_j$ row of the optimal dual simplex table and vice-versa.

Rule II. If the primal variable corresponds to an artificial variable in the dual problem, then its optimal value is directly given by the coefficients of the artificial variable, with changed sign, in the $C_j - z_j$ row of the optimal dual simplex table, after deleting the constant M and vice-versa.

On the other hand, if the primal has an unbounded solution, then the dual problem will not have a feasible solution and conversely.

Now we shall verify these two rules to demonstrate the primal-dual relationship.

Example 34.22. Construct the dual of the following problem and solve both the primal and the dual. Also verify Z = Min. W.
Minimize $Z = 20x_1 + 8x_2$
subject to $x_1 + 2x_2 \ge 2,\ 3x_1 + x_2 \ge 3,\ x_1, x_2 \ge 0$. (Kanpur 2000)

Solution. Dual of the primal problem. Since only two variables are involved, it is convenient to solve the problem graphically.

In the x_1, x_2 plane, the first two inequalities show that the point (x_1, x_2) lies within the shaded region OABCD of Fig. 34.12. Values of the objective function $Z = 20x_1 + 8x_2$ at these corners are $Z(0) = 0$, $Z(A) = 6.2.B = 7.2.C = 6$ and $Z(D) = 1$. Hence the optimal solution is $x_1 = 0, x_2 = 1$ and min. $(Z) = 8$.

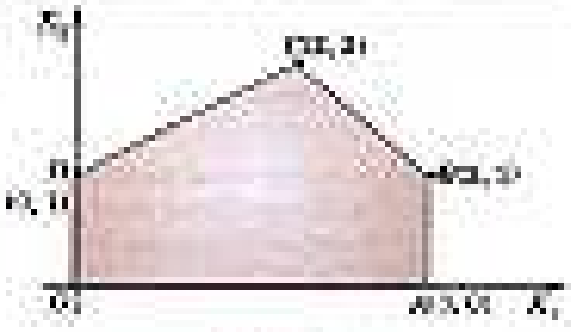


Fig. 34.12

Solution. Obtain the dual problem. The dual problem of the given primal is :

Maximize $W = 4x_1 + 6x_2 + 5x_3$

subject to $-x_1 + 2x_2 + x_3 \le 4$; $3x_1 - x_2 + x_3 \le 5$; $x_1, x_2, x_3 \ge 0$

Step 1. Express the problem in the standard form.

Introducing the slack variables x_4 and x_5 as usual, the linear problem in the standard form is:

Maximize $W - 4x_1 - 6x_2 - 5x_3 + 0x_4 + 0x_5 = 0$

subject to $-x_1 + 2x_2 + x_3 + x_4 = 4$; $3x_1 - x_2 + x_3 + x_5 = 5$;

$x_1, x_2, x_3, x_4, x_5 \ge 0$

Step 2. Find one initial basic feasible solution.

Setting the non-basic variables x_1, x_2, x_3 each equal to zero, we get the initial basic feasible solution as:

$x_1 = x_2 = x_3 = 0$; $x_4 = 4$; $x_5 = 5$; $W = 0$ (Check!)

1st Initial simplex tableau

	c_j	−4	−6	−5	0	0				
c_B	Basis	x_1	x_2	x_3	x_4	x_5			0	θ
0	x_4	−1	2	1	1	0			4	4
0	x_5	3	−1	1	0	1			5	5
	z_j	0	0	0	0	0			0	
	$c_j - z_j$	−4	−6	−5	0	0				

1st x_2 is pivot column, x_4 is pivot row, -1 is pivot element. The second solution is not optimal.

Step 3. Determine the second basic feasible solution.

2nd. Introduce x_2 and drop x_4. Write the new simplex table as:

2nd Simplex tableau

	c_j	−4	−6	−5	0	0				
c_B	Basis	x_1	x_2	x_3	x_4	x_5			0	θ
−6	x_2	−1	2	1	1	0			4	4
0	x_5	2	0	2	1	1			9	9/2
	z_j	6	−6	−6	−6	0			−24	
	$c_j - z_j$	−10	0	1	6	0				

2nd x_1 is pivot column, x_5 is pivot row, 2 is pivot element. The second solution is not optimal.

Step 4. Determine the third basic feasible solution.

3rd. Introduce x_1 and drop x_5. Write the new simplex table as:

3rd Simplex tableau

	c_j	−4	−6	−5	0	0				
c_B	Basis	x_1	x_2	x_3	x_4	x_5			0	θ
−6	x_2	0	2	2	3/2	1/2			13/2	
−4	x_1	1	0	1	1/2	1/2			9/2	
	z_j	−4	−6	−8	−7	−2			−43	
	$c_j - z_j$	0	0	3	7	2				

3rd x_3 is pivot column, x_2 is pivot row, 2 is pivot element. The third solution is not optimal.

Step 5. Determine the fourth basic feasible solution.

4th. Introduce x_3 and drop x_2. Write the new simplex table as:

4th Simplex tableau

	c_j	−4	−6	−5	0	0			
c_B	Basis	x_1	x_2	x_3	x_4	x_5			0
−5	x_3	0	1	1	3/4	1/4			13/4
−4	x_1	1	1	0	1/4	1/4			7/4
	z_j	−4	−9	−5	−11/4	−3/4			−81/4
	$c_j - z_j$	0	3	0	11/4	3/4			

4th x_4 is pivot column, x_3 is pivot row, 3/4 is pivot element. The fourth solution is optimal.

The optimal solution to the linear problem is

$x_1 = 7/4$, $x_2 = 0$, $x_3 = 13/4$, $W = -81/4$ (Ans.)

The reason the optimal basic feasible solution to the primal problem is even from the optimal variables y_1, y_2 was explained in the mathematical meaning of the variables y_1, y_2 contained in the final simplex table of the dual problem. y_1 corresponds to x_4 and y_2 to x_5. Since x_4 and x_5 were basic variables appearing in the W row but with the opposite sign, $y_1 = 11/4$ and $y_2 = 3/4$.

[illegible] an optimal basic feasible solution to the given problem is

$x_1 = 3, x_2 = 1$ and $z = 7$

[illegible] the duality theorem to formulate a dual problem. Also find the optimal solution.

[illegible]

Solution. The dual of the given problem is Max. $W = 4y_1 + 10y_2 + 18y_3 + 10y_4$

subject to $y_1 + 2y_2 + 3y_3 + y_4 \geq 3$, $y_1 + 2y_2 + 2y_3 + 3y_4 \geq 2$, $y_1, y_2, y_3, y_4 \geq 0$.

Step 1. Express the problem in the standard form.

Introducing slack variables, the dual problem in the standard form becomes

Max. $W = 4y_1 + 10y_2 + 18y_3 + 10y_4 + 0s_1 + 0s_2$

subject to $y_1 + 2y_2 + 3y_3 + y_4 - s_1 = 3$;

$y_1 + 2y_2 + 2y_3 + 3y_4 - s_2 = 2$ and $y_1, y_2, y_3, y_4, s_1, s_2 \geq 0$.

Step 2. Find an initial basic feasible solution.

Taking zero as non-basic variables (y_1, y_2, y_3, y_4) each equation gives the basic solution as

$s_1 = -3, s_2 = -2$ (Infeasible) $y_1 = y_2 = y_3 = y_4 = 0$ (Basic)

Since the basic variables $s_1, s_2 < 0$, the initial basic solution is infeasible and non-degenerate.

Hence simplex table is:

	c_j	4	10	18	10	0	0		
c_B	Basis	y_1	y_2	y_3	y_4	s_1	s_2	θ	
0	s_1	1	2	3	1	−1	0	$3/1$	Min
0	s_2	1	2	2	3	0	−1	$2/3$	$\theta = \min(3, 2)$
	z_j	0	0	0	0	0	0	0	
	$z_j - c_j$	−4	−10	−18	−10	0	0		

$\theta = \left(\frac{x_B}{y_{ij}}\right)$ for $y_{ij} > 0$, $\theta = \infty$ for $y_{ij} \leq 0$, Min. ratio is taken for non-optimum.

Step 3. Test for optimum and optimal solution.

$z_j - c_j$ is most negative and largest in y_3. Hence the new simplex table is

	c_j	4	10	18	10	0	0		
c_B	Basis	y_1	y_2	y_3	y_4	s_1	s_2	θ	
0	s_1	1	4/3	0	8/3	−1	1/3	9/8	Min
18	y_3	1/3	2/3	1	1/3	0	−1/3	2/1	$\theta = \min(9/8, 2)$
	z_j	6	12	18	6	0	−6		
	$z_j - c_j$	2	2	0	−4	0	−6		

$\theta = \left(\frac{x_B}{y_{ij}}\right)$ for $y_{ij} > 0$, $\theta = \infty$ for $y_{ij} \leq 0$, Min. ratio is taken for non-optimum.

Step 4. Test for optimum and optimal solution.

$z_j - c_j$ is most negative and largest in y_4. Hence the new simplex table is

	c_j	4	10	18	10	0	0		
c_B	Basis	y_1	y_2	y_3	y_4	s_1	s_2	θ	
10	y_4	9/8	1/2	0	1	−3/8	1/8	9/1	
18	y_3	1/8	1/2	1	0	1/8	−3/8	1/4	Min
	z_j	21/4	14	18	10	−3/4	−3/4		
	$z_j - c_j$	5/4	4	0	0	−3/4	−3/4		

$\theta = \left(\frac{x_B}{y_{ij}}\right)$ for $y_{ij} > 0$, $\theta = \infty$ for $y_{ij} \leq 0$, Min. ratio is taken for non-optimum.

Step 5. Test for optimum and optimal solution.

$z_j - c_j$ is positive or zero for all the columns, this solution is not optimal.

Introducing s_2 and dropping y_3. Then the new simplex table is

	c_j	4	10	18	10	0	0	
c_B	Basis	y_1	y_2	y_3	y_4	s_1	s_2	
10	y_4	3/4	0	−1	1	0	1/2	
0	s_2	1/4	1	2	0	−1/2	1	
	z_j	15/2	10	20	10	−5	0	
	$z_j - c_j$	7/2	0	2	0	−5	0	

As all $C_j \leq 0$, this table gives the optimal solution.

Thus the optimal basic feasible solution is $x_1 = 8, x_2 = 0, s_1 = 20, s_2 = 18$, max. $Z = 24$

Step 4: Write optimal solution to the primal

We now note that the primal variables x_1, x_2 correspond to the slack starting dual variables s_1, s_2 respectively. In the final simplex table of the dual problem, C_j values corresponding to s_1 and s_2 are $-10/3$ and $-2/3$ respectively.

Thus, by rule 1, we conclude that opt. $x_1 = 10/3$ and opt. $x_2 = 2/3$.

Hence an optimal basic feasible solution to the given primal is

$$x_1 = 10/3, x_2 = 2/3, \max. Z = 7.4.$$

Note: To check the validity of the solution obtained, the student is advised to solve the given LPP by simplex method or Graphical method. You must get $Z = 7.4$.

PROBLEMS

Use duality to solve the following problems 1 – 6:

1. Minimize $Z = 2x_1 + 3x_2 + x_3$
   subject to $x_1 + 4x_2 + 2x_3 \geq 5, 3x_1 + 2x_2 + 4x_3 \geq 4$ and $x_1, x_2, x_3 \geq 0$. (A.M.I.E.T.E., 2005)
2. Minimize $Z = 20x_1 + 2x_2$
   subject to $x_1 + 3x_2 \geq 3, 4x_1 + 8x_2 \geq 4, x_1 + x_2 \geq 2$ and $x_1, x_2 \geq 0$. (Delhi Univ. M.Sc., 2005)
3. Maximize $Z = 3x_1 + 5x_2$
   subject to $x_1 + 2x_2 \leq 2, x_1 + 3x_2 \leq 5, 2x_1 + x_2 \leq 3$ and $x_1, x_2 \geq 0$.
4. Maximize $Z = 3x_1 - x_2 - 3x_3$
   subject to $x_1 + 2x_2 - x_3 \geq 4, 3x_1 + 2x_2 + x_3 \geq 6, x_1 + 3x_2 - 2x_3 \geq 5$ and $x_1, x_2, x_3 \geq 0$.

2.15 (I) DUAL SIMPLEX METHOD

In 2.14.3, we have seen that a set of basic variables giving a feasible solution can be found by introducing artificial variables and using M or Two phase method. Using the primal-dual relationships for a problem, we have another method known as Dual simplex method for finding an initial feasible solution. Whereas the regular simplex method starts with a basic feasible (but non-optimal) solution and works towards optimality, the dual simplex method starts with a basic infeasible (but optimal) solution and works towards feasibility. The dual simplex method is quite similar to the regular simplex method. The only difference lies in the criterion used for selecting the incoming and outgoing variables. In the dual simplex method, we first determine the outgoing variable and then the incoming variable which is the exact reverse of what is done in the regular simplex method.

(a) Working procedure for dual simplex method :

Step 1. (i) Convert the problem to maximization form, if it is not so.

(ii) Convert all $\geq$ type constraints (if any) to $\leq$ type by multiplying each such side by -1.

(iii) Express the problem in standard form by introducing slack variables.

Step 2. Find the initial basic solution and express this information in the form of dual simplex table.

Step 3. Test the nature of C_j or $z_j - x_j$:

(i) If all $C_j \leq 0$ and all $b_i \geq 0$, then optimal basic feasible solution has been arrived.

(ii) If all $C_j \leq 0$ and at least one $b_i < 0$, then go to step 4.

(iii) If any $C_j \geq 0$, the method fails.

Step 4. Select the outgoing variable. Select the row that contains the most negative b_i. This will be the key row and the corresponding basic variable is the outgoing variable.

Step 5. Test the nature of key row elements :

(i) If all these elements are ≥ 0, the problem does not have a feasible solution.

(ii) If at least one element < 0, find the ratios of the corresponding elements of C_j row to the elements of the key row, ignoring the ratios when the elements of the key row are positive or zero. The corresponding column is the key column and the associated variable is the incoming variable.

Step-II: Obtain necessary optimal feasible solution. Make the key elements unity. Perform row operations so as to the regular simplex method and repeat the procedure until either an optimal feasible solution or unbounded solution is obtained.

Example 3.13. Use simplex method to solve the following problem :

Maximize $Z = 3x_1 + 5x_2$

subject to $x_1 + 2x_2 \le 2000$, $x_1 + x_2 \le 1500$, $x_2 \le 600$, $x_1 \ge 0$, $x_2 \ge 0$.

Solution : Converting all the inequations into equations :

Step-I : (i) Introduce the slack variables x_3, x_4 and x_5 in the constraints respectively. Then constraints become

$$x_1 + 2x_2 + x_3 = 2000, \quad x_1 + x_2 + x_4 = 1500, \quad x_2 + x_5 = 600$$

(ii) Express the problem in standard form.

Introducing slack variables $x_3, x_4, x_5 \ge 0$, the given problem takes the form :

Maximize $Z = 3x_1 + 5x_2 + 0x_3 + 0x_4 + 0x_5$

subject to $x_1 + 2x_2 + x_3 = 2000$, $x_1 + x_2 + x_4 = 1500$, $x_2 + x_5 = 600$,

$x_1, x_2, x_3, x_4, x_5 \ge 0$.

Step-II : Find the initial basic solution.

Setting the decision variables x_1, x_2 each equal to zero, we get the basic solution

$$x_1 = 0, x_2 = 0, x_3 = 2000, x_4 = 1500, x_5 = 600, Z = 0.$$

Initial solution is given by the table below :

	c_j	3	5	0	0	0	
c_B	Basis	x_1	x_2	x_3	x_4	x_5	b
0	x_3	1	2	1	0	0	2000
0	x_4	1	1	0	1	0	1500
0	x_5	0	1	0	0	1	600
	$z_j = \sum c_B x_B$	0	0	0	0	0	0
	$z_j - c_j$	−3	−5	0	0	0	

Step-I. Find outgoing vector (x_r).

Since $\min\{z_j - c_j : z_j - c_j < 0\} = z_2 - c_2 = -5$, the x_2-column is the key column. Hence x_2 is the incoming vector.

Step-II. Find the outgoing vector.

Since $\min\left\{\frac{b_i}{y_{ir}} : y_{ir} > 0\right\} = \min\left\{\frac{2000}{2}, \frac{1500}{1}, \frac{600}{1}\right\} = \min\{1000, 1500, 600\} = 600$, the x_5-row is the key row and y_{52} is the key element.

Step-III. Make the key element unity and all other elements in the key column zero.

Divide the key row by 1 (the key element) and perform the row operations $R_1 \to R_1 - 2R_3$, $R_2 \to R_2 - R_3$ to make all other elements in the key column zero. We get the following table :

	c_j	3	5	0	0	0	
c_B	Basis	x_1	x_2	x_3	x_4	x_5	b
0	x_3	1	0	1	0	−2	800
0	x_4	1	0	0	1	−1	900
5	x_2	0	1	0	0	1	600
	$z_j = \sum c_B x_B$	0	5	0	0	5	3000
	$z_j - c_j$	−3	0	0	0	5	

Step-I. Find outgoing vector (x_r).

Since $\min\{z_j - c_j : z_j - c_j < 0\} = z_1 - c_1 = -3$, the x_1-column is the key column. Hence x_1 is the incoming vector.

Step-II. Find the outgoing vector.

Since $\min\left\{\frac{b_i}{y_{ir}} : y_{ir} > 0\right\} = \min\left\{\frac{800}{1}, \frac{900}{1}\right\} = \min\{800, 900\} = 800$, the x_3-row is the key row and y_{31} is the key element.

Step-III. Make the key element unity and all other elements in the key column zero.

Since the key element is 1, the key row remains unchanged. Perform the row operations $R_2 \to R_2 - R_1$, $R_4 \to R_4 + 3R_1$ to make all other elements in the key column zero. We get the following table :

	c_j	3	5	0	0	0	
c_B	Basis	x_1	x_2	x_3	x_4	x_5	b
3	x_1	1	0	1	0	−2	800
0	x_4	0	0	−1	1	1	100
5	x_2	0	1	0	0	1	600
	$z_j = \sum c_B x_B$	3	5	3	0	−1	5400
	$z_j - c_j$	0	0	3	0	−1	

Step-I. Find outgoing vector (x_r).

Since $z_j - c_j \ge 0$ for all $j = 1, 2, 3, 4, 5$, the optimal solution is optimal but infeasible. We therefore, proceed further.

Step-II. Make the outgoing vector.

Since b_2 is negative and corresponds to the third row, the third row is the key row and x_4 is the outgoing variable.

Step-III. Find the vector of minimum $\left\{\frac{z_j - c_j}{y_{ij}} : y_{ij} < 0\right\}$ to get the corresponding negative element of the key row.

Since $\min\left\{\frac{z_j - c_j}{y_{ij}} : y_{ij} < 0\right\} = \min\left\{\frac{z_3 - c_3}{y_{23}}, \frac{z_5 - c_5}{y_{25}}\right\} = \min\left\{\frac{3}{-1}, \frac{-1}{-1}\right\} = \min\{-3, 1\} = -3$, the x_3-column is the key column and -1 is the key element.

Step-IV. Divide the key row by the key element.

Divide the key row by -1 (the key element) to get the unity in the key element. Perform the row operations $R_1 \to R_1 + 2R_2$, $R_3 \to R_3 - R_2$ to make all other elements in the key column zero. We get the following table :

	c_j	3	5	0	0	0	
c_B	Basis	x_1	x_2	x_3	x_4	x_5	b
3	x_1	1	0	0	−2	0	1000
0	x_3	0	0	1	−1	−1	100
5	x_2	0	1	0	1	2	500
	$z_j = \sum c_B x_B$	3	5	0	3	3	5300
	$z_j - c_j$	0	0	0	3	3	

Step-I. Find outgoing vector (x_r).

Since $z_j - c_j \ge 0$ for all $j = 1, 2, 3, 4, 5$, the optimal solution is optimal but infeasible. We therefore, proceed further.

Since all C_j values are ≤ 0 and $b_4 = -2$, this solution is optimal but infeasible. We therefore proceed further.

(i) *Mark the outgoing variable.*

Since b_4 is negative, the fourth row is the key row and s_2 is the outgoing variable.

(ii) *Calculate ratio of elements in C_j row to the corresponding negative elements in the key row.*

The ratios are $-2, -\frac{3}{2}$ and 1 (neglecting other ratios corresponding to zero or +ve elements of key row).

∴ x_2 column is the key column and $\left(-\frac{1}{2}\right)$ is the key element.

(iii) *Drop s_2 and introduce x_2 with its associated value -3 under the x_2 column. Convert the key element to unity and make all other elements of the key column zero. Then the third solution is given by the table below:*

	x_1	-2	-3	0	0	0	0	
C_B	Basis	x_1	x_2	s_1	s_2	s_3	s_4	b
0	s_1	0	0	1	0	−1	−1	6
0	s_3	0	0	0	1	1	1	0
−2	x_1	0	1	0	0	0	1	5
−3	x_2	1	0	0	0	−1	−3	4
Z_j		−2	−3	0	0	2	1	−18
C_j		0	0	0	0	−2	−1	

Since all C_j values are ≤ 0 and all b_i's are ≥ 0, therefore this solution is optimal and feasible. Thus the optimal solution is $x_1 = 5, x_2 = 4$ and $Z_{max} = -18$.

Example 4.6.21. *Using dual simplex method, solve the following problem*

Minimize $Z = 2x_1 + 2x_2 + 3x_3 + 4x_4$

subject to $2x_1 + 4x_2 + 3x_3 + 2x_4 \geq 3, 3x_1 + x_2 + 2x_3 + 3x_4 \geq 4, x_1, x_2, x_3, x_4 \geq 0.$

(Kurukshetra, 2009; Kerala, 2010)

Solution. Consists of the following steps:

Step 1. (i) Convert the given problem in maximization form by writing

Maximize $Z' = -2x_1 - 2x_2 - 4x_4$

(ii) Convert the first constraint into '≤' type. This is equivalent to

$$-2x_1 - 3x_2 - 4x_4 \leq -2$$

(iii) Express the problem in standard form.

Introducing slack variables s_1, s_2, s_3, the given problem becomes

$$\text{Max. } Z' = -2x_1 - 2x_2 - 4x_3 + 0s_1 + 0s_2 + 0s_3$$

$$\text{subject to } -2x_1 - 3x_2 - 4x_3 + s_1 + 0s_2 + 0s_3 = -2,$$

$$3x_1 + x_2 + 2x_3 + 0s_1 + s_2 + 0s_3 = 3,$$

$$x_1 + 4x_2 + 3x_3 + 0s_1 + 0s_2 + s_3 = 5,$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \geq 0.$$

Step 2. Find the initial basic solution.

Setting the decision variables x_1, x_2, x_3 each equal to zero, we get the basic solution

$s_1 = s_2 = s_3 = 0, x_1 = -2, x_2 = 3, x_3 = 5$ and $Z' = 0$.

∴ Initial solution is given by the table below:

	C_j	-2	-2	-4	0	0	0	
s_B	Basis	x_1	x_2	x_3	s_1	s_2	s_3	b
0	s_1	−2	(−3)	−4	1	0	0	−2 ←
0	s_2	3	1	2	0	1	0	3
0	s_3	1	4	3	0	0	1	5
	Z_j	0	0	0	0	0	0	0
	C_j	−2	−2	−4	0	0	0	
		↑						

Step 3. Test sign of C_j.

Since all C_j values are ≤ 0 and $b_3 = -2$, the initial solution is optimal but infeasible.

Step 4. Find the outgoing variable.

Since $b_3 = b_2$ the first row is the key row and x_3 is the outgoing variable.

Step 5. Calculate the ratios of the elements of C_j row to the corresponding negative elements of the key row.

These ratios are $-2/-2 = 1$, $-2/-3 = 0.67$, $-1/-1 = 0.8$.

Since 0.67 is the smallest ratio, x_2 column is the key column and -3 is the key element.

Step 6. Form the second optimal feasible solution.

Drop x_3 and introduce x_2 with its associated value -2 under c_j column. Then the revised dual simplex table is

	c_j	-3	-2	-4	0	0	0	
c_B	Basis	x_1	x_2	x_3	s_1	s_2	s_3	b
-3	x_1	$2/3$	1	$1/3$	$-1/3$	0	0	$2/3$
0	s_2	$7/3$	0	$5/3$	$1/3$	1	0	$5/3$
0	s_3	$-1/3$	0	$-1/3$	$2/3$	0	1	$1/3$
	Z_j	-2	-3	-1	1	0	0	-4
	C_j	-1	0	-3	-1	0	0	

Since all $C_j \leq 0$ and all $b_i \geq 0$, this solution is optimal and feasible. Thus the optimal solution is $x_1 = 2/3$, $x_2 = 1/3$, $s_2 = 5/3$ and max. $Z' = -4$ i.e., min. $Z = 4$.

PROBLEMS

Using dual simplex method, solve the following problems:

1. Minimize $Z = 3x_1 + x_2$

   subject to $x_1 + x_2 \geq 1$, $2x_1 - 3x_2 \geq 2$, $x_1, x_2 \geq 0$.

2. Maximize $Z = -2x_1 - x_2$

   subject to $3x_1 + x_2 \geq 3$, $4x_1 + 3x_2 \geq 6$, $x_1 + x_2 \leq 3$, $x_1, x_2 \geq 0$. (Kurukshetra, 2002)

3. Maximize $Z = -x_1 - 2x_2 - 2x_3$

   subject to $2x_1 - x_2 + x_3 \geq 4$, $x_1 + 5x_2 - 3x_3 \geq 7$, $x_1, x_2, x_3 \geq 0$.

4. Maximize $Z = -x_1 - 4x_2 - x_3 - 4x_4$

   subject to $2x_1 + 4x_2 + 2x_3 + x_4 \geq 5$, $2x_1 + x_2 + 3x_3 + 4x_4 \geq 2$,

   $2x_1 + 2x_2 + 3x_3 + 2x_4 \geq 3$, $x_1, x_2, x_3, x_4 \geq 0$.

13.13 (L) TRANSPORTATION PROBLEM

This is a special class of linear programming problems in which the objective is to transport a single commodity from various origins to different destinations at a minimum cost.

(I) Formulation of a transportation problem. There are m plant locations (origins) and n distribution centres (destinations). The production capacity of the ith plant is a_i and the number of units required at the jth destination is b_j. The transportation cost of one unit from the ith plant to the jth destination is c_{ij}. The objective is to determine the number of units to be transported from the ith plant to jth destination so that the total transportation cost is minimum.

Let x_{ij} be the number of units shipped from ith plant to jth destination, then the general transportation problem is:

$$\text{Minimize } Z = \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}\, x_{ij}$$

subject to the constraints

$x_{11} + x_{12} + \ldots + x_{1n} = a_1$ for ith origin $i = 1, 2, \ldots, m$

$x_{1j} + x_{2j} + \ldots + x_{mj} = b_j$ for jth destination $j = 1, 2, \ldots, n$

$x_{ij} \geq 0$.

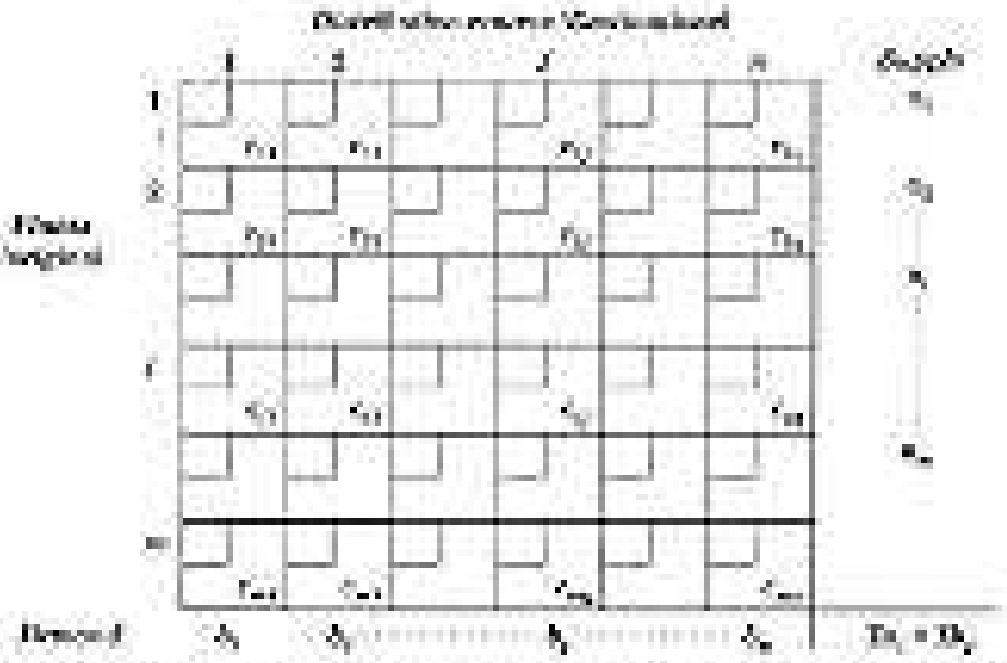
Def. 1. The transportation problem will be consistent if

$$\sum_{i=1}^{m} a_i = \sum_{j=1}^{n} b_j$$

which is the condition for a transportation problem to have a feasible solution. Problems satisfying this condition are called balanced transportation problems.

2. A feasible solution to a transportation problem is said to be a basic feasible solution if it contains at most $(m + n - 1)$ strictly positive allocations, otherwise the solution will degenerate. If the total number of positive (non-zero) allocations is exactly $(m + n - 1)$ then the basic feasible solution is said to be non-degenerate.

3. A feasible solution which minimizes the transportation cost is called an optimal solution. This problem is explicitly represented in the following transportation table :



The row sums are called rim. The per unit cost c_{ij} of transporting from the ith origin to the jth destination is displayed in the lower right side of the (i, j)th cell. Any feasible solution is shown in the table by entering the value of x_{ij} in the small square on the upper left side of the (i, j)th cell. The numbers a_i's and b_j's are called rim requirements. The feasibility of a solution can be verified by summing the values of x_{ij} along the rows and down the columns.

Note 1. The special features of a transportation problem are that

(i) the coefficients of all x_{ij} in the constraints are equal to 1, and

(ii) the total supply is equal to total demand, $\sum a_i = \sum b_j$.

Obs. 2. The objective function and the constraints being all linear, the problem can be solved by simplex method. But the number of variables being larger, there will be too many cycles to obtain. However, the mathematical form of the constraints being unique, we can look for some technique which would be simpler than the simplex method.

34.35 WORKING PROCEDURE FOR TRANSPORTATION PROBLEMS

Step 1. Construct transportation table. Express the supply for the origins a_i, demand at destinations b_j and the unit shipping costs c_{ij} in the form of a matrix, to set up transportation table. If the supply and demand are equal, the problem is balanced.

Step 2. Find the initial basic feasible solution. We find an initial allocation which satisfies the demand of each project site without violating the capacities of the plants (origins) and also meeting the non-negativity

restrictions. There are several methods for finding an initial basic feasible solution such as North-West corner rule, least cost method, Vogel's approximation method. The Vogel's approximation method (VAM) takes into account not only the least cost c_{ij} but also the cost of the next to least cost in the same row and column. Usually VAM gives an initial basic feasible solution closer to the optimal solution than other methods. As such we shall confine ourselves to VAM only which consists of the following steps :

(i) Display the difference between the least and the next to least costs in each row, by enclosing them in brackets to the right of the row. Similarly display the differences for each column at the bottom below that column.

(ii) Identify the row or column with the largest difference among all the rows and columns and allocate as much as possible under the chosen row or column to the lowest cost cell in that row or column. In case of a tie, allocate to the cell associated with the lower cost.

If the greatest difference corresponds to ith row and c_{ij} is the lowest cost in the ith row, allocate as much as possible to (i, j) where $x_{ij} = \min(a_i, b_j)$ in the ith row and cross off the ith row or the jth column.

(iii) Recalculate the row and column differences for the reduced table and go to the previous step.

(iv) Repeat the procedure till all the requirements are satisfied. Note the solution in the upper left corner small squares of the basic cells.

Step 2. Apply optimality check

In the above solution, the number of allocations must be $m + n - 1$, otherwise the basic solution degenerates.

Now to test for optimality, we apply the modified distribution (MODI) method and examine each unoccupied cell to determine whether or not a given initial solution is optimal or not. We find the total transportation cost and then repeat this procedure until lowest possible transportation cost is obtained. This method consists of the following steps :

(i) Write the numbers u_i along the left and v_j along the top of the row or column such that their sum equal to the original costs of occupied cells i.e., solve the equations $u_i + v_j = c_{ij}$ starting initially with row $u_1 = 0$.

(ii) Compute the net evaluations $c_{ij} - u_i + v_j = c_{ij}$ for all the empty cells and enter these in upper right hand corners of the corresponding cells.

(iii) Examine the signs of each w_{ij}. If all $w_{ij} \geq 0$, then the current basic feasible solution is optimal. If even one $w_{ij} < 0$, the solution is not optimal and we proceed further.

Step 3. Move towards optimal solution

(i) Choose the unoccupied cell with the largest w_{ij} and mark it with $\oplus$.

(ii) Draw a closed path consisting of horizontal and vertical lines beginning and ending at the cell and having its other corners at the occupied cells.

(iii) Add and subtract θ alternately to and from the quantities in the loop assigned in the requirements. Assign a maximum value to θ so that one basic variable becomes zero and the other basic variables remain non-negative. Now the basic cell whose allocation has been reduced to zero leaves the basis.

Step 4. Return to step 2 and repeat the process until an optimal (non-degenerate) solution is obtained.

Example 10.20. Solve the following transportation problem :

		Destination				
		A	B	C	D	
Source	X	32	30	22	32	41
	Y	17	20	14	18	26
	Z	24	27	16	20	32
	Requirement	30	30	40	20	

Availability

Solution. Consists of the following steps :

Step 1. Test whether the problem is balanced. Here the total availability and the total requirement being the same i.e., 90, the problem is balanced.

Step 2. Find the initial basic feasible solution. Following Vogel's approach, we find the difference between the smallest and next to the smallest costs in each row and each column are computed and displayed alongside the corresponding row or column as penalties. (Table 1). The largest of these differences is 10 which is associated with the first row. We assign as much as possible to the variable x_{11} subject to the restrictions $x_{11} \le 5$ and $x_{11} \le 7$. Hence we assign $x_{11} = \min(5, 7) = 5$. This satisfies the first row and reduces the availability at the first source to zero. We eliminate the first row and obtain the reduced table (Table 2). The largest difference in Table 2 is 10 which is associated with the fourth column. Hence $x_{24} = 10$ is the largest assignment, we allocate $x_{24} = \min(10, 9) = 9$.

This satisfies the availability at the second source and requirement at the fourth destination. Proceeding in this way, the subsequent reduced transportation tables and differences for the remaining rows and columns are shown in Tables 3, 4 and 5.

Finally the initial basic feasible solution is as shown in Table 6:

Step 3. Apply optimality test

As the number of allocations $= m + n - 1 = 3 + 4 - 1 = 6$, we can apply MODI method.

(a) We know $u_1 + v_1 = 11$, $u_2 + v_1 = 18$, $u_2 + v_2 = 17$

$u_2 + v_3 = 13$, $u_3 + v_3 = 12$, $u_3 + v_4 = 17$

Let $u_1 = 0$. Hence $v_1 = 11$, $v_2 = 10$, $u_2 = 8$, $v_3 = 5$, $v_4 = 9$, $u_3 = 3$

(b) Now evaluate $c_{ij} = u_i + v_j$ for all empty cells we

$c_{12} = 10$, $c_{13} = 5$, $c_{21} = 19$, $c_{23} = 13$, $c_{31} = 14$, $c_{32} = 13$

(c) Since all the net evaluations are negative, the current solution is optimal. Hence the optimal allocation is given by

$x_{11} = 5$, $x_{14} = 2$, $x_{24} = 9$, $x_{31} = 7$, $x_{33} = 5$, $x_{34} = 3$ and $x_{ij} = 0$

∴ The minimum transportation cost

$= 11 \times 5 + 9 \times 2 + 17 \times 9 + 14 \times 7 + 12 \times 5 + 17 \times 3 = 55 + 18 + 153 + 98 + 60 + 51 = 435$.

[illegible]

Solution. Consider the following steps:

Step 1. Construct representation table. Suppose the weights from the factories, measured in tons, and the requirements, in tons, of the following warehouses are given in Table 1. Here the weights from each factory to each warehouse are shown in the corresponding cell.

Table 1

Factories	Warehouse 1	Warehouse 2	Warehouse 3	Warehouse 4	Supply
1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
3	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Demands	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Step 2. Find the initial basic feasible solution.

Using VAM, the initial basic feasible solution is as shown in Table 2. The transportation cost according to this solution is given by

$$[illegible]$$

Step 3. Apply optimality check.

As the number of allocations $= m + n - 1$, the case is [illegible] nondegenerate.

We now compute for each unoccupied cell (i, j) the [illegible] which are exhibited in Table 3. Since the non-occupied cell $(3, 1)$ has a negative [illegible], the solution is not optimal and can be improved.

Table 2

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Table 3

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Step 4. Obtain improved optimal solution.

From inspection,

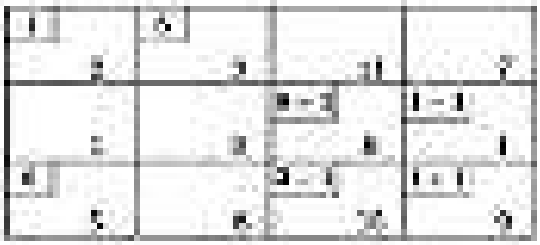
(a) Most basic feasible solution

(b) Obtain the corresponding cell with the maximum c_{ij}. In case of a tie, select the one which [illegible] cost. In Table 3, either (3, 2) or (3, 4) is such a cell. Let us select (3, 2). Construct a [illegible] of these cell (3, 2) that consists of [illegible] cells. Therefore we take [illegible] and the corresponding [illegible] as in [illegible].

(c) Obtain a revised [illegible] by [illegible] as before. Add and subtract [illegible] to the [illegible] so that [illegible] the [illegible] cell (3, 2) [illegible] the [illegible] [illegible]. [illegible] a revised [illegible] which is [illegible] [illegible] [illegible]

Table 4

Table 5



- **Total transportation cost of the revised solution**

$= ₹ (1 \times 2 + 3 \times 0 + 1 \times 6 + 6 \times 0 + 2 \times 10 + 2 \times 9)$ thus $100 \times ₹ 10,100$.

(ii) Optimality check. As the number of allocations in table 5 is $m + n - 1$ (i.e., 6), we can apply MODI method. We compute the net evaluations which are shown in Table 6. Since the cell (1, 3) has a positive value, the current basic feasible solution is not optimal.

Table 6

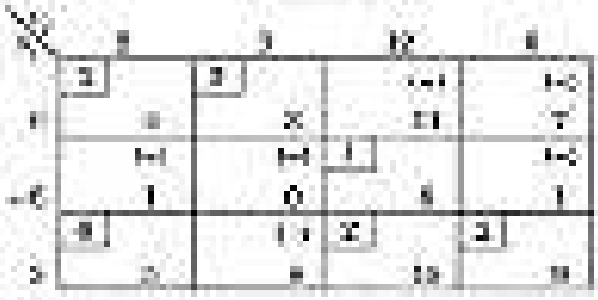
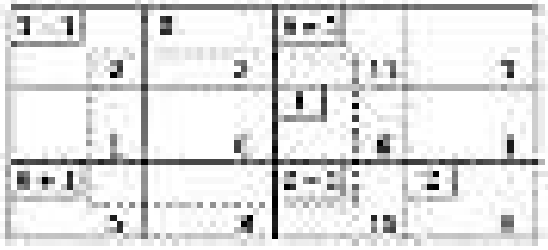


Table 7



Second Iteration :

(i) Find basic feasible solution. In the second basic feasible solution introduce the cell (1, 3) taking $\theta = 1$ and drop the cell (1, 1). Applying Table 5. We can obtain the third basic feasible solution (Table 8).

Table 8

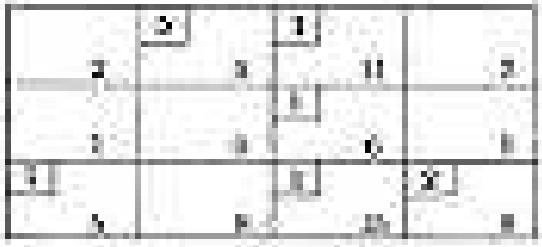
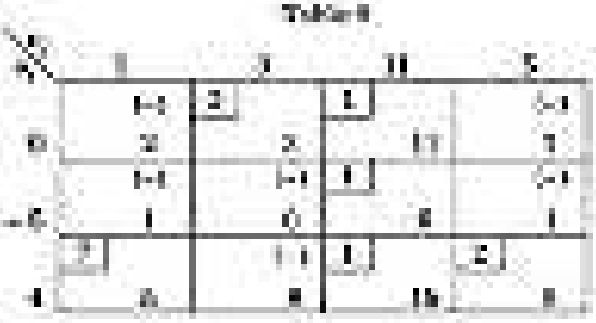


Table 9



(ii) Optimality Check. As the number of allocations in Table 8 is $m + n - 1$ (i.e., 6), we can apply MODI method.

We compute the net evaluations which are shown in Table 9. Since all the net evaluations are ≤ 0, the basic feasible solution is optimal.

Thus the optimal transportation policy is as shown in Table 8 and the optimal transportation cost

$= ₹ (2 \times 2 + 1 \times 11 + 1 \times 8 + 3 \times 6 + 1 \times 15 + 2 \times 9)$ i.e. $100 \times ₹ 10,000$.

4.10.5 DEGENERACY IN TRANSPORTATION PROBLEMS

When the number of basic cells in a transportation table is less than $(m + n - 1)$ the basic solution degenerates. To remove the degeneracy, we assign a small positive value ε to as many unoccupied cells as may be necessary to complete $(m + n - 1)$ basic variables. The cells now assigned ε are then treated like all other occupied cells and the problem is solved in the usual way. This is done till the optimum solution is obtained. Then we let ε tend to 0.

Example 4.1 Solve the following linear fractional programming problem:

Basis							

Solution: Consists of the following steps:

Step 1. Write the given problem in the standard form. Introduce slack variables to convert the inequality constraints into equations.

Step 2. Find the initial basic feasible solution.

Using TORA, the initial basic feasible solution is as shown in Table 1.

Step 3. Apply optimality test. Since the number of basic cells is 6 which is less than $m + n - 1 = 7$, the basic solution is degenerate. In order to complete the basic and thereby ensure degeneracy, we assign an infinitesimally small positive basic variable ε. We select the cell $(3, 3)$ to assign ε as it is the cell with the least cost.

Table 1

We now compute the net evaluations $z_{ij} - c_{ij}$ which are entered in Table 2. Since all $z_{ij} - c_{ij} \leq 0$, the current basic solution is optimal. Hence the optimal solution is:

$x_{12} = 10, x_{14} = 8, x_{21} = 5, x_{23} = 1, x_{24} = 1, x_{33} = \varepsilon, x_{34} = 7$, and $\text{Max } Z = 0$.

The minimum value of the objective function is

Table 2

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1. Obtain an initial basic feasible solution to the following transportation problem:

From	To D	To E	To F	To G	
A	11	13	17	14	250
B	16	18	14	10	300
C	21	24	13	10	400
	200	225	275	250	

2. Solve the following transportation problem:

Suppliers \ Consumers	X	Y	Z	Available
I	6	8	4	16
II	4	8	5	12
III	1	8	6	8
Required	10	14	12	36

3. Consider four factories of a certain firm A_1 and their targets B_1. The loss of factories per month from not having their work done is given in the following table:

	B_1	B_2	B_3
A_1	5	8	10
A_2	15	7	7
A_3	10	9	4
A_4	8	6	4

The cost per month in pounds of the four factories is 150 and is given. The daily requirements of the consumers are 100, 120 and 200. Find the distribution of workers from each factory to each target which minimises the total loss per month for the three targets.

4. A company has factories F_1, F_2, F_3 which supply warehouses at W_1, W_2 and W_3. Weekly factory capacities, weekly warehouse requirements and unit costs of supplying each factory to each warehouse are given in the following table:

From \ To	W_1	W_2	W_3	Supply
F_1	16	20	12	150
F_2	14	8	18	175
F_3	26	24	16	75
Demand	150	100	150	400

Determine the optimal distribution for this company in order to minimise transportation costs.

5. A company has agreed to supply 8, 10 and 12 units of a product at the end of each of the next three months. The supply and demand situation, with unit costs of transportation are given below:

From	Distribution centres				Available
	A_1	B_1	C_1	D_1	
F_1	10	30	20	12	7
F_2	90	20	10	40	9
F_3	40	60	50	20	24
Required	5	8	7	15	

What can be the maximum savings by optimal scheduling?

6. A departmental store wishes to stock the following quantities of a popular product in three types of containers:

Container type	1	2	3
Quantity	150	200	100

Vendors are selected by the store to supply the required quantity. The quantities they can supply are shown in the following table:

Vendor	1	2	3	4
Quantity	120	100	130	200

The store estimates that profit per unit will vary with the vendor as shown below:

Vendor → / Container type ↓	1	2	3	4
1	8	10	6	2
2	6	11	5	10
3	5	9	7	6

Find the maximum profit of the store.

7. Obtain an optimum basic feasible solution to the following transportation problem:

From \ To				
	2	3	4	7
	3	1	2	3
	4	6	5	2
	4	1	6	10

Available

Demand

8. A company has three plants at locations A, B and C which supply to warehouses located at D, E, F, G and H. Monthly plant capacities are 600, 300 and 500 units respectively. Monthly warehouse requirements are 300, 400, 250, 400 and 300 units respectively. Unit transportation costs are given in the following table:

From \ To	D	E	F	G	H
A	5	8	6	6	3
B	8	7	5	4	8
C	9	4	3	6	5

Determine an optimum distribution for the company in order to minimize the total transportation cost.

11.12 (L) ASSIGNMENT PROBLEM

An assignment problem is a special type of transportation problem in which the objective is to assign a number of origins to an equal number of destinations at a minimum cost (or maximum profit).

(L) Formulation of an assignment problem. There are n new machines M_i $(i = 1, 2, \ldots n)$ which are to be installed in a machine shop. There are n vacant spaces S_j $(j = 1, 2, \ldots n)$ available. The cost of installing the machine M_i at space S_j is c_{ij} rupees. Let us formulate the problem of assigning machines to spaces so as to minimize the overall cost.

Let x_{ij} be the assignment of machine M_i to space S_j i.e., let x_{ij} be a variable such that

$$x_{ij} = \begin{cases} 1, \text{ if the machine is installed at } j\text{th space} \\ 0, \text{ otherwise} \end{cases}$$

Since any machine can only be installed at each space, we have

$$x_{i1} + x_{i2} + \ldots + x_{in} = 1, \text{ for machine } M_i\ (i = 1, 2, \ldots n)$$

$$x_{1j} + x_{2j} + \ldots + x_{nj} = 1, \text{ for space } S_j\ (j = 1, 2, \ldots n)$$

Also the total installation cost is $\sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij}$

The above assignment problem can be stated as follows:

Determine $x_{ij} \geq 0$ $(i, j = 1, 2, \ldots n)$ so as to

minimise $\quad Z = \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij}$

subject to the restrictions $\sum_{i=1}^{n} x_{ij} = 1, j = 1, 2, \ldots n$ and $\sum_{j=1}^{n} x_{ij} = 1, i = 1, 2, \ldots n$.

This problem is explicitly represented by the following $n \times n$ cost matrix:

		Jobs				
		J_1	J_2	J_3	...	J_n
Machines	M_1	c_{11}	c_{12}	c_{13}	...	c_{1n}
	M_2	c_{21}	c_{22}	c_{23}	...	c_{2n}
	M_3	c_{31}	c_{32}	c_{33}	...	c_{3n}
	...	...	...	...	...	...
	M_n	c_{n1}	c_{n2}	c_{n3}	...	c_{nn}

Note The assignment problem is sometimes described in terms of determining a minimum cost of assigning n machines to n jobs (or n jobs to n machines) where each machine must be assigned to one and only one job. However, it could be that n machines could do very many jobs and have enough time to do all of the jobs, or that some machines could not do some jobs. Moreover, a more efficient method of solving such problems is available. This is the Hungarian method for solution of assignment problems which we describe below.

WORKING PROCEDURE TO SOLVE AN ASSIGNMENT PROBLEM

Step 1. Reduce the matrix. Subtract the smallest element of each row of the given cost matrix from all elements of that row. See if each row contains a zero now. If not, subtract the smallest element of each column (not containing zero) from all the elements of that column. This gives the reduced matrix.

Step 2. Assign the zeros

(a) Examine rows of the reduced matrix successively until a row with exactly one unmarked zero is found. Make an assignment to this single zero by encircling it. Draw all other zeros in the column of this encircled zero, as these will not be considered for any future assignment. Continue in this way until all the rows have been examined.

(b) Now examine columns successively until a column with exactly one unmarked zero is found. Encircle this zero and make an assignment there. Then cross any other zero in its row. Continue in this way until all the columns have been examined.

In some cases, one may or columns may have more than one unmarked zero, or may have no unmarked zero. In such a case, arbitrarily select any unmarked zero and cross all other zeros in its row or column. Proceed in this way, till no zero is left unmarked.

Step 3. Apply optimality check

Repeat step 2 (a) and (b) until one of the following occurs:

(i) If no row or no column is without an assignment (encircled zero), then the current assignment is optimal.

(ii) If there is some row and/or column without an assignment, then the current assignment is not optimal and we go to next step.

Step 4. Modify the zeros to create a number of lines crossing all zeros.

(a) Tick (✓) the rows which do not have assignments.

(b) Tick (✓) the columns (not already marked) which have zeros in the ticked row.

(c) Tick (✓) the rows (not already marked) which have assignments in ticked columns.

Repeat (b) and (c) until no more ticking is required.

(d) Draw lines through all unticked rows and ticked columns. If the number of lines is equal to the order of the matrix then it is an optimal solution, otherwise not.

Step 5. Revise to get optimal solution.

Select the smallest element and subtract it from all uncovered elements. Add this smallest element to every element lying at the intersection of two lines. The resulting matrix is the revised basic feasible solution.

Step 6. Go to step 3 and repeat the procedure until the optimal solution is attained.

Example 10.21. Four jobs are to be done on four different machines. The cost (in rupees) of assigning i^{th} job on the j^{th} machine is given below.

		Machines			
		M_1	M_2	M_3	M_4
Jobs	J_1	41	27	28	33
	J_2	37	32	33	37
	J_3	34	30	30	32
	J_4	36	35	32	38

Assign the jobs to different machines so as to minimize the total cost.

Solution. Consists of the following steps :

Step 1. Reduce the matrix. Subtract the smallest element 27 of row 1 from all its elements. Similarly subtract 32, 30 and 32 from rows 2, 3 and 4 respectively. The resulting matrix is as shown in Table 1. Columns 1 and 4 do not have any zero element. So subtract the smallest element 4 of Col. 1 from all its elements and element 1 from all elements of Col. 4. The reduced matrix is as given in Table 1.

Table 1

	M_1	M_2	M_3	M_4
J_1	4	0	2	4
J_2	3	0	0	1
J_3	0	0	0	1
J_4	2	3	0	5

Table 2

	M_1	M_2	M_3	M_4
J_1	4	(0)	2	3
J_2	3	×	×	(0)
J_3	(0)	0	×	0
J_4	2	3	(0)	4

Step 2. Assign the zeros. Row 4 has a single uncovered zero in Col. 3. Encircle it and cross all other zeros in Col. 3. Row 3 has a single uncovered zero in Col. 1. Encircle it and cross the other zero in Col. 1. Row 1 has a single uncovered zero in Col. 2. Encircle it and cross the other zero in Col. 2. Finally row 2 has a single uncovered zero in Col. 4. Encircle it in Table 2.

Step 3. Apply optimality criterion. Since we have not encircled zero in each row and in each column, this gives the optimal solution.

∴ The optimal assignment policy is

Job 1 to machine 2, Job 2 to machine 4, Job 3 to machine 1, Job 4 to machine 3.

and the minimum assignment cost = ₹ (31 + 32 + 34 + 32) = ₹ 129.

Example 10.22. A city bicycle manager has 5 cycles and 5 cycle dealers. Considering the capabilities of the salesmen and the nature of dealers, the marketing manager estimates that sales per month (in hundred rupees) for each salesman to each dealer are as follows :

		Cycle dealers				
		A	B	C	D	E
Salesmen	1	32	38	40	28	40
	2	40	24	28	32	41
	3	41	27	33	30	42
	4	22	38	41	34	37
	5	29	33	40	28	32

Find the assignment of salesmen to dealers that will result in maximum sales. (Madras, 2000)

[illegible] Examples of the following steps:

Step 1: Making the matrix. Construct the given administrative problem into a matrix table by writing the existing all the possible negative characteristics ($i = a$ to $i = f$). These characteristics then become the columns of the table. Then subtract the smallest element of each row from the remaining numbers of that column. This gives the reduced matrix (Table 1).

Table 1

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Table 2

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Step 2: Assign the zeroes. Every 0 must be assigned a single unencircled zero in its [illegible] 0 zero) If a row contains a single unencircled zero in row 1, the circle [illegible] zero. Then cross out [illegible] 0 zero) in the same column. Repeat the above process for each [illegible] 0 zero in Table 1 and cross other zeroes.

Step 3: Assign optimal [illegible] to [illegible] assignment. [illegible] solution is not optimal. Therefore we go to next step.

Step 4: Draw minimum number of lines covering all zeroes. Draw the least number of lines to cover all zeroes in the [illegible] Table 2. Now the least number of lines is four and we have [illegible] to order 4. The last number is 12, we get by step 5.

Step 5: Create improved optimal solution. Select the smallest element in the Table 2, not covered by the lines of Step 4 [illegible] it to [illegible] line or two different positions. Interchange the assignments that can be made from [illegible]. In addition, subtract it from all the uncovered elements of the new matrix (Table 3) and add the same to the elements lying at the intersection of two dotted lines. Now there exists a unique set of [illegible] zeroes as in the new matrix. There we draw [illegible] line as in Col. 3 (Table 3).

Table 3

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]		
[illegible]	[illegible]			[illegible]
[illegible]	[illegible]			[illegible]
[illegible]			[illegible]	[illegible]
[illegible]			[illegible]	[illegible]

Here, since the number of [illegible] lines is equal to the order, the [illegible] is optimal solution is obtained.

Finally, to determine [illegible] optimal assignment, we consider only the zero elements (Table 3).

a) Examine successively the rows with exactly one zero. There is no such row.

b) Examine successively the rows with exactly one zero. There is no such row.

c) Continue successively the [illegible] exactly one zero. Col. 3 has one zero, so [illegible] and [illegible] [illegible].

d) Examine successively the rows in question (a, b) and cross off zeroes in row 2 and Col. 1. There are [illegible] unencircled zeroes in row 3. Then encircle arbitrarily the zero in position (3, 2) and cross off zeroes in row 4 and Col. 2. Finally, we see that the remaining unencircled zero is in row 5.

Now, each row and each column has one encircled zero. Hence from the optimal assignment policy is:

Machine-man 1 to Assignment B, 2 to A, 3 to D, 4 to F and 5 to C.

Minimum [illegible] = [illegible] + [illegible] + [illegible] + [illegible] + [illegible] = [illegible] [illegible] hr. [illegible]

PROGRAMMING

1. A firm plans to begin production of three new products and its three plants. The unit cost of production at plants is as given below. Find the assignment that minimizes the total unit cost.

(Rs.)

Products	I	II	III
1	10	8	12
2	15	9	11
3	9	6	7

2. Solve the following assignment problem:

	I	II	III	IV
A	20	12	18	11
B	8	16	7	8
C	10	14	13	11
D	9	10	12	9

3. A machine tool company decides to make four tools assembled through four machines. Each operation in the assembly can be performed by any of the machines. The cost of each tool assembled by each machine is as shown in the table below. The foreman of the company wants to determine an assignment so as to minimize the total cost.

Cost (in Rs.)

Products	A	B	C	D
1	15	13	16	17
2	11	9	12	10
3	13	14	15	13
4	14	12	14	11

4. Four professors are each capable of teaching any one of four different courses. Their preferences in terms of hours for different subjects varies. Your problem is to find out who should teach what course. Each professor is assigned only one course. Find the assignment policy which is to be followed so that total system preference time for all courses is minimum.

Prof.	S.E.	Operating Theory	Discrete Mathematics	Differential equations
A	7	10	9	7
B	15	8	11	8
C	12	11	12	11
D	9	12	13	9

5. Consider the problem of assigning five workers to five jobs. The assignment costs in thousands of rupees are given in the following matrix:

Worker	Job			
	I	II	III	IV
W_1	40	32	50	35
W_2	30	25	20	28
W_3	35	28	30	30
W_4	28	30	40	32

Find the optimal assignment.

6. A company has eight salesmen and is considering six sales areas. The following table gives the estimated annual sales (in Rs. '000) if one salesman is assigned to the sales area. How should the sales be assigned to the salesmen so as to maximize the overall sales?

Markets	Ads					
	I	II	III	IV	V	VI
1	9	22	58	31	55	57
2	68	78	72	89	88	48
3	91	29	95	87	83	84
4	54	45	27	99	36	99
5	56	71	82	36	40	85
6	53	68	63	31	57	58

OBJECTIVE TYPE OF QUESTIONS

FILL IN THE BLANKS

Fill in the blanks for the following questions:

1. Infeasible solution is a feasible solution to a linear programming problem which ____
2. The objective value of the $(C_j - Z_j)$ row in the simplex solution provides the ____
3. The dual of the dual problem is the ____
4. The difference between the transportation problem and the assignment problem is ____
5. The optimal solution of a transportation problem can be ____
6. The method adopted in LPP, if tests that ____
7. The dual problem of the LPP
   Max $Z = 4y_1 - 6y_2 + 3y_3$
   subject to $2y_1 - 3y_2 + 4y_3 \le 3$, $3y_1 - 5y_2 + 6y_3 \le 5$, $y_1, y_2, y_3 \ge 0$ is ____
8. The optimality and feasibility conditions are satisfied in the final simplex table when the problem is ____
9. Possible and feasible solution to a transportation problem must satisfy ____
10. A few questions on problems

The linear programming problem is ____

11. The basic feasible solutions of $2x_1 + x_2 + x_3 = 4$, $3x_1 + x_2 + 2x_3 = 5$ are ____
12. A unit cost table is also known as ____
13. The advantage of dual is applied to ____
14. If the total worth of the project is the same irrespective of the transportation problem is said ____
15. An artificial variable is ____
16. Few conditions on which Cramer's method is based on ____
17. A feasible solution which also satisfies the non-negative restrictions of the problem is ____ solution
18. The dual problem of Maximize $3x_1 + 5x_2$ subject to $x_1 + 2x_2 \le 2$, $x_1 + 4x_2 \le 12$, $3x_1 + 2x_2 \le 18$ and $x_1, x_2 \ge 0$ is ____
19. For a balanced transportation problem with m rows and n columns, the number of basic variables will be ____
20. Using graphical method, Max $Z = 3x_1 + 2x_2$ subject to $2x_1 + x_2 \le 18$, $2x_1 + 3x_2 \le 42$, $3x_1 + x_2 \le 24$, $x_1, x_2 \ge 0$ is ____
21. In a LPP, problem, to be solved, all data is ____
22. Degeneracy in a transportation problem is resolved by ____
23. A feasible solution is said to be non-degenerate if it LPP, when ____
24. The dual of the problem Max $Z = 2x_1 + x_2$ subject to $x_1 + 2x_2 \le 10$, $x_1 + x_2 \le 6$, $x_1 - x_2 \le 2$, $x_1 - 2x_2 \le 1$, $x_1, x_2 \ge 0$ is ____
25. The two methods used to find the initial basic solution of a transportation problem are ____
26. Degeneracy occurring in solving a good by simplex method when any one of ____ for ____ variables

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CHAPTER 35

Calculus of Variations

1. Introduction. 2. Functionals. 3. Euler's equation. 4. Solutions of Euler's equation. 5. Geodesics. 6. Minimal surface of revolution. 7. Several dependent variables. 8. Functionals involving higher order derivatives. 9. Approximate solutions of boundary value problems—Rayleigh-Ritz method. 10. Application to eigenvalue problems—Galerkin method. 11. Hamilton's principle. 12. Lagrange's equations.

35.1 INTRODUCTION

The calculus of variations is a powerful technique for the solution of problems in dynamics of rigid bodies, optimization of orbits and orbital problems. The subject primarily concerns with finding conditions of extreme values of definite integrals involving a certain function. It is concerned to go beyond finding a stationary values of a given function. Only an elementary exposition of the subject is given here with the aim of introducing the student to a topic whose importance is fast growing in science and engineering.

Before proceeding further, the student should review §5.12 concerning maxima and minima of functions of several variables.

35.2 FUNCTIONALS

Consider the problem of finding a curve C through two points $A(x_1, y_1)$ and $B(x_2, y_2)$ whose length is a minimum (Fig. 35.1). It is clear on intuitive grounds that the curve $y = y(x)$ for which $y(x_1) = y_1$, $y(x_2) = y_2$ such that $\int_{x_1}^{x_2} \sqrt{1 + y'^2}\,dx$ is a minimum.

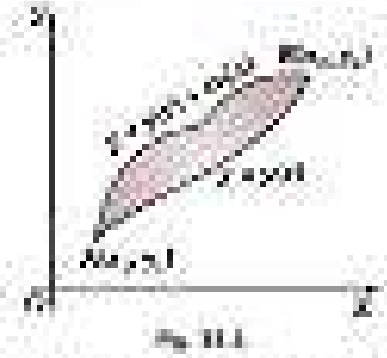

Fig. 35.1

In general terms, we wish to find the curve $y = y(x)$ where $y(x_1) = y_1$ and $y(x_2) = y_2$ such that for a given function $f(x, y, y')$,

$$\int_{x_1}^{x_2} f(x, y, y')\,dx \text{ is a minimum, relative or an extremum.} \qquad (1)$$

An integral of the form (1) which assumes a definite value for functions of the type $y = y(x)$ is called a **functional**.

In differential calculus, we deal with the problem of maxima and minima of functions. The calculus of variations is however, concerned with maximizing or minimizing functionals.

35.3 EULER'S EQUATION

A necessary condition for

$$J = \int_{x_1}^{x_2} f(x, y, y')\,dx \text{ is an extremum is that}$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$$

This is called Euler's equation.

Proof. Let $y = y(x)$ be the curve joining points $A(x_1, y_1)$, $B(x_2, y_2)$ which makes I an extremum. Let

$$y = y(x) + \varepsilon\eta(x) \quad \ldots(1)$$

be a neighbouring curve joining these points so that at A, $\eta(x_1) = 0$ and at B, $\eta(x_2) = 0$. $\ldots(2)$

The value of I along (1) is $I = \int_{x_1}^{x_2} f(x, y(x) + \varepsilon\eta(x), y'(x) + \varepsilon\eta'(x))\,dx$

This being a function of ε, is a maximum or minimum for $\varepsilon = 0$, where

$$\frac{dI}{d\varepsilon} = 0 \text{ i.e., } \varepsilon = 0 \quad \ldots(3)$$

Differentiating I under the integral sign by Leibnitz's rule (p. 1.28), we have

$$\frac{dI}{d\varepsilon} = \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial x}\frac{dx}{d\varepsilon} + \frac{\partial F}{\partial y}\frac{dy}{d\varepsilon} + \frac{\partial F}{\partial y'}\frac{dy'}{d\varepsilon}\right]dx \quad \ldots(4)$$

But x being independent of ε, $\frac{\partial x}{\partial \varepsilon} = 0$. Also from (1), $\frac{\partial y}{\partial \varepsilon} = \eta(x)$ and $\frac{\partial y'}{\partial \varepsilon} = \eta'(x)$.

Substituting these values in (4), we get $\frac{dI}{d\varepsilon} = \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}\eta(x) + \frac{\partial F}{\partial y'}\eta'(x)\right]dx$

Integrating the second term on the right by parts, we have

$$\frac{dI}{d\varepsilon} = \int_{x_1}^{x_2}\frac{\partial F}{\partial y}\eta(x)\,dx + \left[\left(\frac{\partial F}{\partial y'}\right)\eta(x)\right]_{x_1}^{x_2} - \int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\eta(x)\,dx$$

$$= \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]\eta(x)\,dx \quad \text{[By (2)]}$$

Since this has to be zero by (3),

$$\therefore \quad \frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0 \quad \ldots(5)$$

which is the desired Euler's equation.

Obs. 1. Other forms of Euler's equation.

(a) Since F is a function of x, y, y', we have $\frac{dF}{dx} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y}\frac{dy}{dx} + \frac{\partial F}{\partial y'}\frac{dy'}{dx}$

$$= \frac{\partial F}{\partial x} + y'\frac{\partial F}{\partial y} + y''\frac{\partial F}{\partial y'} \quad \ldots(6)$$

and $\frac{d}{dx}\left[y'\frac{\partial F}{\partial y'}\right] = y'\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) + \frac{\partial F}{\partial y'}y'' \quad \ldots(7)$

Subtracting (6) from (7), we get $\frac{\partial F}{\partial x} = \frac{d}{dx}\left[y'\frac{\partial F}{\partial y'}\right] - \frac{dF}{dx} + \frac{\partial F}{\partial y}y' + y'\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)$

or $\frac{d}{dx}\left[y'\frac{\partial F}{\partial y'} - F\right] - \frac{\partial F}{\partial x} + y'\left[\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right] = 0$ [By (5)]

Hence $\frac{d}{dx}\left[y'\frac{\partial F}{\partial y'} - F\right] - \frac{\partial F}{\partial x} = 0 \quad \ldots(8)$

which is another form of (5).

(b) Again since $\frac{\partial F}{\partial y'}$ is also a function of x, y, y', say $= \psi(x, y, y')$,

$$\therefore \quad \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y}\frac{dy}{dx} + \frac{\partial \psi}{\partial y'}\frac{dy'}{dx}$$

$$= \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y'}\right) + \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y'}\right)y' + \frac{\partial}{\partial y'}\left(\frac{\partial f}{\partial y'}\right)y' = \frac{\partial^2 f}{\partial x \partial y'} + y'\frac{\partial^2 f}{\partial y \partial y'} + y''\frac{\partial^2 f}{\partial y'^2}$$

Substituting this in (3), we get $\frac{\partial f}{\partial y} - \frac{\partial^2 f}{\partial x \partial y'} - y'\frac{\partial^2 f}{\partial y \partial y'} - y''\frac{\partial^2 f}{\partial y'^2} = 0$ (TTM; 2003) ...(4)

which is an extended form of (3).

(Note 3) The above problem is mainly to minimize the integral

$$I = \int_{x_0}^{x_n} f(x, y_0, y_1, \ldots, y_n, y_0', y_1', \ldots, y_n')\,dx$$

involving n functions $y_1, y_2, \ldots, y_n$ of x. Then the necessary condition for this integral to be stationary is

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx}\left(\frac{\partial f}{\partial y_i'}\right) = 0, \quad i = 1, 2, \ldots, n \qquad \ldots(5)$$

These are Euler's equations for the n functions.

33.3 SOLUTIONS OF EULER'S EQUATION

Every solution of the Euler's equation which satisfies the boundary conditions, is called an extremal or a stationary function of the problem. The extremals are easily obtained in the following cases:

(1) When f is independent of y:

We have $\partial f/\partial x = 0$ and Euler's equation (3) above becomes $\frac{d}{dx}\left(f - y'\frac{\partial f}{\partial y'}\right) = 0$

Integrating, we get $f - y'\frac{\partial f}{\partial y'} = $ constant. This directly gives a solution of Euler's equation.

(2) When f is independent of y:

We have $\partial f/\partial y = 0$ and Euler's equation (3) reduces to $\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0$

Integrating $\frac{\partial f}{\partial y'}$ is another which gives a solution directly.

(3) When f is independent of y':

We have $\partial f/\partial y' = 0$ and the equation (3) becomes $\frac{\partial f}{\partial y} = 0$ which gives the desired solution.

(4) When f is independent of x and y:

We have $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$ and $\frac{\partial^2 f}{\partial x \partial y'} = 0, \frac{\partial^2 f}{\partial y \partial y'} = 0$

Then the equation (4) above becomes $y''\frac{\partial^2 f}{\partial y'^2} = 0$.

If $\frac{\partial^2 f}{\partial y'^2} \neq 0$, it reduces to $y'' = 0$ which gives a solution of the form $y = ax + b$.

Example 33.2. Find the extremals of the functional $\int_{x_0}^{x_1} y^2(1 - y')^2\,dx$. (TTM; 2002)

Solution. We have $f = y^2(1 - y')^2$ which is independent of y i.e., $\partial f/\partial y = 0$.

Also $$\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = \frac{d}{dx}\left(-2y^2(1 - y')\right) = 2\frac{y^2y' - y \cdot 2yy'}{y^4} = \frac{2}{y^2}(yy' - 2y')$$

$\therefore$ Euler's equation reduces to $\frac{d}{dx}\left(\frac{2}{y^2}(yy' - 2y')\right) = 0$

i.e., $$\frac{2}{y^2}(yy' - 2y') = k \quad \text{or} \quad y'(y - 2) = ky^2$$

Integrating both sides, $\int \frac{y'}{y}\,dx = 2\int \frac{dx}{x} + \log c$

i.e., $\log y' = 2\log x + \log c$ or $y' = cx^2$

Hence $y = cx^3/3 + c'$ or $y = c_1x^3 + c_2$

This is the required solution.

Example 35.2. *Prove that the shortest distance between two points is a straight line.* (U.P.T.U., 2003; Nagpur, 2003)

Solution. Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the given points and s the arc length of a curve connecting them (Fig. 35.2). Then

$$s = \int_{x_1}^{x_2} ds = \int_{x_1}^{x_2} \sqrt{(1 + y'^2)}\,dx$$

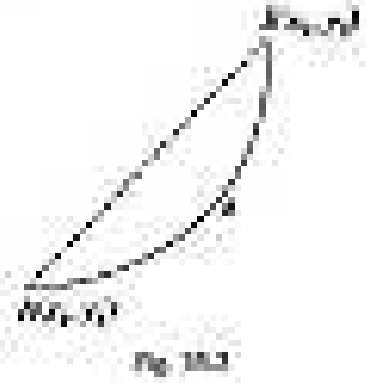

Fig. 35.2

Now s will be minimum if it satisfies Euler's equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0$$

Here $f = \sqrt{(1 + y'^2)}$ which is independent of y i.e., $\partial f/\partial y = 0$

$$\therefore \quad \frac{d}{dx}\left(\frac{\partial}{\partial y'}\sqrt{(1 + y'^2)}\right) = 0 \quad \text{or} \quad \frac{d}{dx}\left(\frac{y'}{\sqrt{(1 + y'^2)}}\right) = 0$$

$\therefore$ On integration, we have $y'/\sqrt{(1 + y'^2)} =$ constant $= c$, or say.

Integrating, we get $y = mx + c_2$ which is a straight line, the constants m and c_2 are determined from the fact that the straight line passes through A and B.

Example 35.3. *Find the curve passing through the points (x_1, y_1) and (x_2, y_2) which when rotated about the x-axis gives a minimum surface area.* (U.P.T.U., 2005)

Solution. In Fig. 35.3, the surface area $s = \int_{x_1}^{x_2} 2\pi y\,ds$

$$= 2\pi \int_{x_1}^{x_2} y\sqrt{(1 + y'^2)}\,dx. \text{ This Lacis to be minimum.}$$

Here $f = y\sqrt{(1 + y'^2)}$ is independent of x, therefore, Euler's equation reduces to

$$f - y'\frac{\partial f}{\partial y'} = \text{constant}, \text{ or say} \qquad \text{[By § 35.6 (1)]}$$

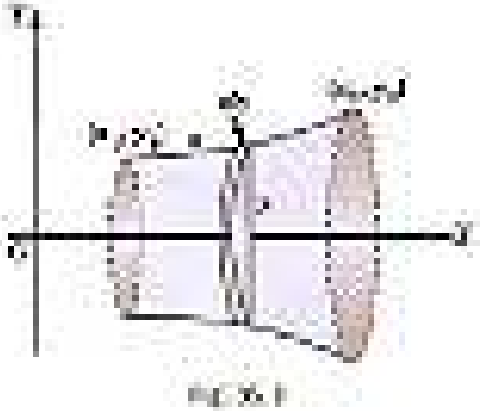

Fig. 35.3

$$\text{or} \quad y\sqrt{(1 + y'^2)} - y'\frac{\partial}{\partial y'}\left\{y\sqrt{(1 + y'^2)}\right\} = c$$

i.e., $$y\sqrt{(1 + y'^2)} - y'\left\{\frac{y}{2}(1 + y'^2)^{-1/2} \cdot 2y'\right\} = c$$

$$\text{or} \quad y(1 + y'^2) - y'^2 y = c\sqrt{(1 + y'^2)} \quad \text{or} \quad y' = \frac{dy}{dx} = \frac{\sqrt{(y^2 - c^2)}}{c}$$

Separating the variables and integrating, we have

$$\int \frac{dy}{\sqrt{(y^2 - c^2)}} = \int \frac{dx}{c} + c' \quad \text{or} \quad \cosh^{-1}\left(\frac{y}{c}\right) = \frac{x + c'}{c}$$

i.e., $$y = c\cosh\left(\frac{x + c'}{c}\right)$$

which is catenary. The constants c and c' are determined from the points (x_1, y_1) and (x_2, y_2).

Note. This problem is also known as minimum revolution problem. Minimal surfaces are known to have shapes which are minimal surface areas.

Example 15.4. *Find the path on which a particle will slide from rest under the action of gravity from one point to another in the shortest time (neglect the action of gravity).* (U.P.T.U., 2004)

Solution. Let the particle start sliding on the curve OP from O with zero velocity (Fig. 15.6). At time t, let the particle be $P(x, y)$ such that arc $OP = s$.

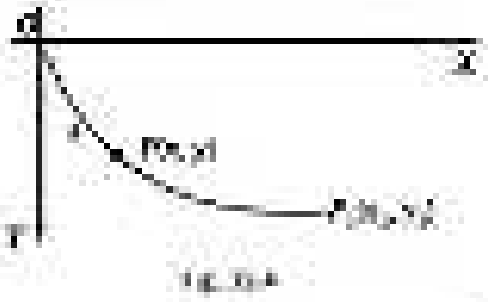

Fig. 15.6

By the principle of work and energy, we have

K.E. at P − K.E. at O = Work done in moving the particle from O to P

$$\frac{1}{2}m\left(\frac{ds}{dt}\right)^2 - 0 = mgy$$

or
$$ds/dt = \sqrt{(2gy)} \qquad \ldots(1)$$

Thus the time taken by the particle to move from O to P_1 is

$$T = \int_0^T dt = \int_0^s \frac{ds}{\sqrt{(2gy)}} = \frac{1}{\sqrt{(2g)}} \int_0^x \frac{\sqrt{(1 + y'^2)}}{\sqrt{y}}\, dx$$

Here $f = \sqrt{(1 + y'^2)/y}$ is independent of x.

$\therefore$ Euler's equation reduces to $f - y' \partial f/\partial y' = \text{constant}, c$: say

i.e.,
$$\frac{\sqrt{(1 + y'^2)}}{\sqrt{y}} - y' \frac{\partial}{\partial y'}\left[\frac{\sqrt{(1 + y'^2)}}{\sqrt{y}}\right] = c \quad \text{or} \quad \frac{\sqrt{(1 + y'^2)}}{\sqrt{y}} - y'\left[\frac{y'}{\sqrt{y}\sqrt{(1 + y'^2)}}\right] = c$$

or
$$\sqrt{y}\sqrt{(1 + y'^2)} - y'^2\sqrt{y} = c\sqrt{(1 + y'^2)}, \text{ say.}$$

Solving for y', we have
$$y' = \frac{dy}{dx} = \sqrt{\left(\frac{y - a}{a}\right)}$$

Separating the variables and integrating, we get

$$\int_0^x dx = \int_0^y \sqrt{\left(\frac{a}{y - a}\right)}\, dy \qquad \text{[Put } y = a\sin^2\theta\text{]} \quad \ldots(2)$$

or
$$x = \int_0^\theta \sqrt{\left(\frac{a\sin^2\theta}{a - a\sin^2\theta}\right)}\, 2a\sin\theta\cos\theta\, d\theta$$

$$x = \int_0^\theta 2a\sin^2\theta\, d\theta = a\int_0^\theta (1 - \cos 2\theta)\, d\theta = \frac{a}{2}(2\theta - \sin 2\theta) \qquad \ldots(3)$$

Writing $a/2 = b$ and $2\theta = \phi$, equations (2) and (3) become $x = b(\phi - \sin\phi)$, $y = b(1 - \cos\phi)$ which is a cycloid. The constant b is found from the fact that the curve passes through (x_1, y_1).

Obs. This is the well-known brachistochrone problem which arises in many forms in Optics work. Newton's account also has such a 'theorem' involving time. It was proposed by John Bernoulli in 1696 and its solution formed the basis for the study of the Calculus of Variations.

Example 15.5. *Find the curves on which the functional* $\int_0^1 (y'^2 + xy)\, dx$ *with* $y(0) = 0$ *and* $y(1) = 1$ *can be extremized.* (U.P.T.U., 2005)

Solution. We have
$$f = y'^2 + 2xy$$

$\therefore$
$$\partial f/\partial y = 2x\,; \quad \frac{\partial f}{\partial y'} = 2y'\,; \quad \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 2y''$$

Hence the Euler's equation $\dfrac{\partial f}{\partial y} - \dfrac{d}{dx}\left(\dfrac{\partial f}{\partial y'}\right) = 0$ becomes

$$2x - 2y'' = 0 \quad \text{i.e.,} \quad y'' = x \qquad \ldots(1)$$

Integrating (i), $y' = 2x^2 + C_1$ ...(ii) and $y = \frac{2}{3}x^3 + C_1x + C_2$ ...(iii)

Using the boundary conditions, when $x = 0, y = 0$ (iii) gives $C_2 = 0$.

When $x = 1, y = 1$, (iii) again gives $C_1 = \frac{1}{3}$.

Hence (ii) reduces to $y = x^3$ which is the only curve on which extremum can be obtained.

Example 12.2. Find the solid curve for which the functional $\int_0^1 (y'^2 - y^2 + 2xy)\,dx$ with $y(0) = 0$ and $y(1) = 0$ can be extremized. (U.P.T.U., 2005)

Solution. Let $f = y'^2 - y^2 + 2xy$ so that $\frac{\partial f}{\partial y} = -2y + 2x$

and $\frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = \frac{d}{dx}(2y') = 2y''$

$\therefore$ Euler's equation $\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0$ becomes

$$-2y + 2x - 2y'' = 0 \quad \text{or} \quad y'' + y = x \quad \text{or} \quad (D^2 + 1)y = x$$

Its A.E. $D^2 + 1 = 0$, gives $D = \pm i$.

$\therefore$ $$\text{C.F.} = c_1 \cos x + c_2 \sin x$$

and $$\text{P.I.} = \frac{1}{D^2 + 1}x = (1 + D^2)^{-1}x = (1 - D^2)x = x$$

Thus $$y = c_1 \cos x + c_2 \sin x + x \qquad ...(i)$$

Using boundary conditions : when $x = 0, y = 0$, (i) gives $c_1 = 0$;

when $x = \pi/2, y = 0$, (i) gives $0 = c_2 + \pi/2$, i.e., $c_2 = -\pi/2$.

Hence (i) reduces to $y = x - \frac{\pi}{2}\sin x$, which is the only curve on which the given functional can be extremized.

Example 12.3. Solve the variational problem

$$\delta \int_0^1 (x^2 y'^2 + 2xy + y^2)\,dx = 0 \text{ given } y(0) = 0, y(1) = 0. \qquad \text{(U.P.T.U., 2006)}$$

Solution. Let $f = x^2y'^2 + 2xy + 2y^2$ so that $\frac{\partial f}{\partial y} = 2x + 4y$, $\frac{\partial f}{\partial y'} = 2x^2y'$

$\therefore$ Euler's equation $\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0$ becomes

$$2x + 4y - \frac{d}{dx}(2x^2y') = 0 \quad \text{or} \quad 2x + 4y - (2x^2y'' + 4xy') = 0$$

or $$x^2y'' + 2xy' - 2y = x$$ This is Cauchy's homogeneous linear eq. (§ 12.8)

Putting $x = e^z$, it reduces to $(D^2 + D - 2)y = e^z$.

Its solution is $$y = c_1e^z + c_2e^{-2z} + \frac{1}{3}e^z \quad \text{or} \quad y = c_1x + c_2\frac{1}{x^2} + \frac{1}{3}x \qquad ...(i)$$

Since $y(1) = 0$, we have $c_1 + c_2 = 0$

and $$y(2) = 0 \text{ gives } 0 = 2c_1 + \frac{1}{4}c_2 + \frac{2}{3}\log 2$$

Solving these equations, we get $c_1 = -c_2 = \frac{8}{21}\log 2$

Putting the values of c_1 and c_2 in (i), we get

$$y = \frac{1}{21}[8\log 2(x^{-2} - x) + 7x\log x]$$

which is the required solution.

23.5 GEODESICS

A geodesic on a surface is a curve along which the distance between any two points of the surface is a minimum. To find the geodesics on a surface is a variational problem involving the conditional constraint. This problem was first studied by *Jacob Bernoulli* in 1698 and the general method of solution was given by *Euler*.

Example 23.8. Show that the geodesics on a plane are straight lines. (U.P.T.U., 2006)

Solution. Let $y = y(x)$ be a curve passing the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the xy-plane. Then the length of the curve joining A and B is given by

$$s = \int_{x_1}^{x_2} \frac{ds}{dx}\,dx = \int_{x_1}^{x_2} \sqrt{[1 + (dy/dx)^2]}\,dx \quad \text{i.e.,} \quad s = \int_{x_1}^{x_2} \sqrt{(1 + y'^2)}\,dx$$

The geodesics on the xy-plane is the curve $y = y(x)$ for which s is minimum.

We have $f(x, y, y') = \sqrt{(1 + y'^2)}$ which depends on y' only. Hence the Euler's equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left[\frac{\partial f}{\partial y'}\right] = 0 \text{ yields}$$

$$\frac{d}{dx}\left[\frac{2y'}{2\sqrt{(1 + y'^2)}}\right] = 0 \quad \text{i.e.,} \quad y'\sqrt{(1 + y'^2)} - \frac{y'(2y')}{2\sqrt{(1 + y'^2)}} = c$$

i.e. $$y'(1 + y'^2) - y'^3 y' = 0 \quad \text{i.e.,} \quad \frac{d^2y}{dx^2} = 0$$

Integrating twice, we get $y = c_1 x + c_2$

which is a straight line.

Hence the geodesics on a plane are straight lines.

Example 23.9. Find the geodesics on a right circular cylinder of radius a.

Solution. In cylindrical coordinates (ρ, ϕ, z) we have

$$x = r\cos\phi,\ y = r\sin\phi,\ z = z. \qquad \text{(p. 357)}$$

$\therefore$ The element of arc on a right circular cylinder of radius a is given by

$$ds^2 = dx^2 + dy^2 + dz^2 = d\rho^2 + (\rho d\phi)^2 + dz^2 = a^2 d\phi^2 + dz^2 \qquad \text{i.e., } \rho = a \text{ and } d\rho = 0$$

or $$ds = \sqrt{[a^2 + (dz/d\phi)^2]}\,d\phi \quad \text{or} \quad s = \int_{\phi_1}^{\phi_2} \sqrt{(a^2 + z'^2)}\,d\phi \qquad \text{...(1)}$$

Now the geodesic for the given cylinder is the curve for which s is minimum. Here $f = \sqrt{(a^2 + z'^2)}$, which is a function of ϕ and z' while z is absent.

$\therefore$ Euler's equation for the function f reduces to

$$\frac{\partial f}{\partial z'} = \text{constant} \quad \text{i.e.} \quad \frac{z'}{\sqrt{(a^2 + z'^2)}} = c_1, \text{ say.}$$

This implies $z'^2 = \text{constant} \times (a^2 + z'^2)$ i.e. $z' = c_1$, say.

Integrating, $z = c_1\phi + c_2$

This is the desired geodesics on a cylinder which is a circular helix. (Example 8.3, p. 353)

Example 23.10. Show that the geodesics on a sphere of radius a are great circles.

Solution. In spherical coordinates (r, θ, ϕ) we have

$$x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta. \qquad \text{(p. 360 ...(1)}$$

$\therefore$ The arc element on a sphere of radius a is given by

$$ds^2 = dx^2 + dy^2 + dz^2 = (a\sin\theta\cos\phi\,d\theta)^2 + (a\sin\theta\sin\phi\,d\phi)^2 + (a\cos\theta\,d\theta)^2 \qquad \text{i.e., } r = a,\ dr = 0$$

or $$ds = a\sqrt{[1 + \sin^2\theta(\sin^2\phi + \cos^2\phi)]}\,d\theta \quad \text{or} \quad s = a\int_{\theta_1}^{\theta_2} \sqrt{(1 + \sin^2\theta\,\theta'^2)}\,d\theta$$

Now the geodesics on the sphere $r = a$ is the curve for which s is minimum. Where $f = a\sqrt{[1 + \sin^2\theta\,\phi'^2]}$ which is a function of θ and ϕ' while ϕ is absent.

$\therefore$ Euler's equation reduces to $\partial f/\partial\phi' = \text{constant}$.

$$\frac{\partial f}{\partial \phi'} = \frac{a\sin^2\theta\cdot\phi'}{\sqrt{[1+\sin^2\theta\cdot\phi'^2]}} = \text{constant}$$

or $$\frac{\sin^2\theta\cdot\phi'}{\sqrt{[1+\sin^2\theta\cdot\phi'^2]}} = c \text{ (say)} \quad \text{or} \quad \sin^2\theta(\sin^2\theta - c^2)\phi'^2 = c^2$$

or $$\frac{d\phi}{d\theta} = \frac{c}{\sin\theta\sqrt{[\sin^2\theta - c^2]}} = \frac{c\operatorname{cosec}^2\theta}{\sqrt{(1-c^2\operatorname{cosec}^2\theta)}}$$

Integrating $$\phi = \int \frac{c\operatorname{cosec}^2\theta\,d\theta}{\sqrt{[(1-c^2)-2c\cot\theta]}} + c' = -\sin^{-1}\left[\frac{c\cot\theta}{\sqrt{(1-c^2)}}\right] + c'$$

or $$\cos(\phi - c') = A\cos\phi + B\sin\phi \quad \text{or} \quad a\cos(\phi - c') = Ax\cos\phi + By\sin\phi \text{ or } ax\cos c' + ay\sin c'$$

or $$z = Ax + By \qquad [\text{By } \gamma) \text{ where } c' = \alpha]$$

This is a plane through the centre $(0, 0, 0)$ of the sphere which cuts the sphere along a great circle. Hence the required geodesics are the arcs of the great circles.

EXERCISE 5(C)

1. Solve the Euler's equation for the following functionals:

(i) $\int_{x_0}^{x_1} (x+y)\,y'\,dx$ (ii) $\int_{x_0}^{x_1} (1+x^2)\,y'^2\,dx$. (U.P.T.U. 2006)

2. Show that the general solution of the Euler's equation for the integral

$$\int_{x_0}^{x_1} \frac{1}{x}\sqrt{\left[1+\left(\frac{dy}{dx}\right)^2\right]}\,dx \text{ is } (x-a)^2 + y^2 = b^2$$

Find the extremals of the following functionals:

3. $\int_{x_0}^{x_1} (y^2 + y'^2 - 2y\sin x)\,dx$. (U.P.T.U. 2004)

4. $\int_{x_0}^{x_1} (y^2 + y'^2 - 2y\cosh x)\,dx$, $y(x_0) = y_0$, $y(x_1) = y_1$. (U.P.T.U. 2005)

5. $\int_{x_0}^{x_1} (y^2 - y'^2 + 2y\cos x)\,dx$, $y(x_0) = 0$, $y(x_1) = 0$. (U.P.T.U. 2005)

6. $\int_{x_0}^{x_1} \frac{1+y'^2}{y^2}\,dx$ 7. $\int_{x_0}^{x_1} y'^2\,dx$, $y(x_0) = 0$, $y(x_1) = 1$

8. $\int_{x_0}^{x_1} \frac{\sqrt{1+y'^2}}{x}\,dx$, $y(x_0) = 0$, $y(x_1) = 1$. (Madras, U.P.T.U. 2005)

9. Solve the variational problem $\delta\int_{x_0}^{x_1} (y'^2 + y^2)\,dx$ under the conditions $y(x_0) = 0$, $y(x_1) = 1$. (U.P.T.U. 2006)

10. A heavy cable hangs freely under gravity between two fixed points. Show that the shape of the cable is a catenary.

11. A particle is moving with a force perpendicular to and proportional to its distance from the line. If it was initially known that the path of projection is a straight line, find the nature of its path.

12. Find the geodesics on a right circular cone of semi-vertical angle α.

11.6 ISOPERIMETRIC PROBLEMS

In certain problems, it is necessary to make a given integral,

$$I = \int_{x_1}^{x_2} f(x, y, y')\, dx \qquad ...(1)$$

a maximum or a minimum while keeping another integral

$$J = \int_{x_1}^{x_2} g(x, y, y')\, dx \qquad ...(2)$$

constant. Such problems involve one or more constraint conditions, just as J = a constant. A typical example of this type is that of finding a closed curve of a given perimeter and maximum area. This being one of the earliest problems to engage attention, we often refer to problems of this type as *isoperimetric problems*.

Such problems are generally solved by the method of Lagrange multipliers. To extremize (1), we multiply (2) by λ and add to (1) where λ is the Lagrange multiplier. Then the necessary condition for the integral $\int_{x_1}^{x_2} H\, dx$ to be an extremum is $\frac{\partial H}{\partial y} - \frac{d}{dx}\left[\frac{\partial H}{\partial y'}\right] = 0$ where $H = f + \lambda g$. The values of the two constants of integration and the parameter λ are determined from the three conditions namely, the two boundary conditions and the integral J having given constant value.

Example 11.11. *Find the plane curve of fixed perimeter and maximum area.* (U.P.T.U. 2006 S)

Solution. Let l be the fixed perimeter of a plane curve between the points with abscissae x_1 and x_2 (Fig. 35.6). Then

$$l = \int_{x_1}^{x_2} \sqrt{(1 + y'^2)}\, dx \qquad ...(1)$$

Also the area between the curve and the x-axis is

$$A = \int_{x_1}^{x_2} y\, dx \qquad ...(2)$$

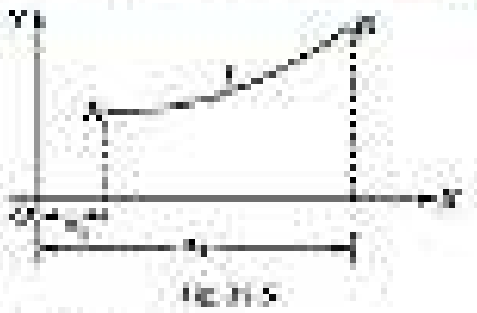

Fig. 35.6

We have to maximize (2) subject to constraint (1).

$\therefore$ Taking $f = y$ and $g = \sqrt{(1 + y'^2)}$, we write $H = f + \lambda g = y + \lambda\sqrt{(1 + y'^2)}$

Now H must satisfy the Euler's equation

$$\frac{\partial H}{\partial y} - \frac{d}{dx}\left[\frac{\partial H}{\partial y'}\right] = 0 \quad \therefore \quad 1 - \frac{d}{dx}\left[\frac{\lambda y'}{\sqrt{(1 + y'^2)}}\right] = 0$$

Integrating w.r.t. x, we have $x - \lambda y'/\sqrt{(1 + y'^2)} = 0$

Solving for y', we get $y' = \frac{x - a}{\sqrt{[\lambda^2 - (x - a)^2]}}$

Integrating again, $y = \sqrt{[\lambda^2 - (x - a)^2]} + b$ i.e., $(x - a)^2 + (y - b)^2 = \lambda^2$ which is a circle.

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Example 11.12. *Prove that the sphere is the solid figure of revolution enclosing given volume and having minimum surface.* (U.P.T.U. 2005; Madras 2000 S)

Solution. Consider the arc PQ of the curve which rotates about the x-axis as shown in Fig. 35.6.

Then surface area $S = \int_{x_1}^{x_2} 2\pi y\, ds = \int_{x_1}^{x_2} 2\pi y \sqrt{(1 + y'^2)}\, dx$

and volume of the solid so formed $V = \int_{x_1}^{x_2} \pi y^2\, dx$.

Now we have to minimise V subject to fixed A. Taking $f = xy^2$ and $g = 2xy\sqrt{(1+y^2)}$, we write $H = f + \lambda g = xy^2 + 2\lambda xy\sqrt{(1+y^2)}$.

Now H has to satisfy Euler's equation. But H does not contain x explicitly.

$$\therefore \quad H - y'\frac{\partial H}{\partial y'} = \text{constant}, \text{ or say}$$

$$\text{or,} \quad xy^2 + 2\lambda xy\sqrt{(1+y^2)} - y' \cdot 2\lambda xy\frac{y'}{\sqrt{(1+y^2)}} = c$$

$$\text{or} \quad xy^2 + \frac{2\lambda xy}{\sqrt{(1+y^2)}} = c \qquad \ldots(1)$$

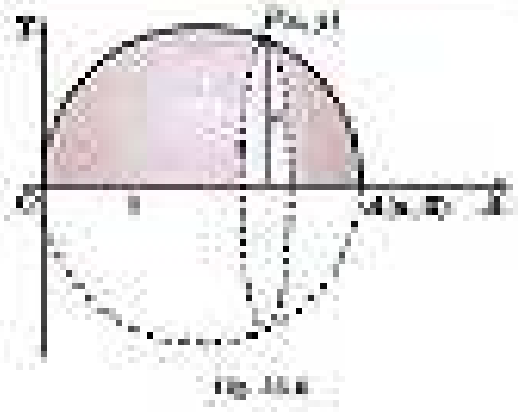

Fig. 33.6

Since the curve passes through O and A for which $y = 0$, x is given $= 0$

$$\therefore \quad y + 2\lambda\sqrt{(1+y^2)} = 0$$

Solving for y', $\quad y'\left[-\frac{dy}{dx}\right] = \frac{\sqrt{(4\lambda^2 - y^2)}}{y}$

Separating the variables and integrating, we get

$$\int dx = \int \frac{y\,dy}{\sqrt{(4\lambda^2 - y^2)}} + k, \quad \text{or} \quad x = k - \sqrt{(4\lambda^2 - y^2)} \qquad \ldots(2)$$

When $x = 0$, $y = 0$ $\quad \therefore \quad k = 2\lambda$

$\therefore$ (2) becomes $(x - 2\lambda)^2 + y^2 = (2\lambda)^2$ which is a circle with centre $(2\lambda, 0)$ and radius 2λ.

Hence the figure formed by the revolution of given arc is a sphere.

EXERCISE 33.2

1. Find a function $y(x)$ for which $\int_0^1 (y'^2 + y^2)\,dx$ is stationary, given that $\int_0^1 y^2\,dx = 2$, $y(0) = 0$, $y(1) = 0$.

(Madras, 2002 A)

2. Find the extremal of the isoperimetric problem $I = \int_0^{\pi/2} y^2\,dx$ given that $\int_0^{\pi/2} y\,dx = 1$, $y(0) = 0$, $y(\pi/2) = 1$.

3. Show that the curve $y = f(x)$, $f(x) \geq 0$, which maximises the curved surface area of the solid generated by the revolution of a curve that passes through two fixed points is a catenary.

(U.P.T.U., 2000 S)

4. Find the extremal of the functional $I = \int_0^{\pi/2} (y'^2 - y^2)\,dx$ under the conditions $y(0) = 0$, $y(\pi/2) = 1$. Find also the value of I for the extremal.

[Hint: $\int_0^{\pi/2} \sin^2 x\,dx = \frac{\pi}{4}$]

5. Prove that the extremal of the isoperimetric problem $I = \int_0^{\pi/2} y'^2\,dx$, $y(0) = 0$, $y(\pi/2) = 0$, subject to the condition $\int_0^{\pi/2} y\,dx = 10$ is a parabola.

(Madras, M.E., 2000)

33.3 SEVERAL DEPENDENT VARIABLES

We now extend the variational problem of § 33.2 to a problem with several variables as functions of a single independent variable x, i.e., J a necessary condition for

$$J = \int_{x_0}^{x_1} f(x, y_1, y_2, \ldots, y_n, y_1', y_2', \ldots, y_n')\,dx \qquad \ldots(3)$$

is the necessary condition for the extremum of I is

$$\frac{\partial F}{\partial y_i} - \frac{d}{dx}\left[\frac{\partial F}{\partial y_i'}\right] = 0, i = 1, 2, \ldots, n \qquad \ldots(1)$$

Let $y_1, y_2, \ldots, y_n$ satisfy the boundary conditions $y_i(x_0) = y_{i0}, y_i(x_1) = y_{i1}$

Consider arbitrary functions $\eta_1(x), \eta_2(x), \ldots, \eta_n(x)$ which are all zero at the boundary i.e.,

$$\eta_i(x_0) = 0 = \eta_i(x_1)$$

Replacing $y_1, y_2, \ldots$ by $y_1 + \varepsilon_1\eta_1, y_2 + \varepsilon_2\eta_2, \ldots$ in (1), we get

$$I(\varepsilon) = \int_{x_0}^{x_1} F(x, y_1 + \varepsilon_1\eta_1, y_2 + \varepsilon_2\eta_2, \ldots, y_n' + \varepsilon_1\eta_1', y_2' + \varepsilon_2\eta_2', \ldots)\, dx \qquad \ldots(2)$$

This is a function of the parameters $\varepsilon_1, \varepsilon_2, \ldots$ and reduces to (1) for $\varepsilon_1 = \varepsilon_2 = \ldots = 0$.

To find the stationary value of (1), we find the stationary value of (2) for $\varepsilon_1 = \varepsilon_2 = \ldots = 0$. (2) will have a stationary value when

$$\frac{\partial I(\varepsilon)}{\partial \varepsilon_1} = 0, \frac{\partial I(\varepsilon)}{\partial \varepsilon_2} = 0, \ldots$$

Writing $\qquad F = F(x, y_1, y_2, \ldots, y_1', y_2', \ldots)$

and $\qquad F^* = F(x, y_1 + \varepsilon_1\eta_1, y_2 + \varepsilon_2\eta_2, \ldots, y_1' + \varepsilon_1\eta_1', y_2' + \varepsilon_2\eta_2', \ldots)$

(2) becomes $I(\varepsilon) = \int_{x_0}^{x_1} F^*\, dx$.

y_1, y_2 being functions independent of $\varepsilon_1, \varepsilon_2$, differentiating under the integral sign, we get

$$\frac{\partial I(\varepsilon)}{\partial \varepsilon_1} = \int_{x_0}^{x_1} \frac{\partial F^*}{\partial \varepsilon_1}\, dx = \int_{x_0}^{x_1} \left(\frac{\partial F^*}{\partial y_1}\eta_1 + \frac{\partial F^*}{\partial y_1'}\eta_1'\right) dx$$

$\therefore \qquad \frac{\partial I(\varepsilon)}{\partial \varepsilon_1} = 0$, when $\varepsilon_1 = \varepsilon_2 = \ldots = 0$ given

$$\int_{x_0}^{x_1} \left(\frac{\partial F}{\partial y_1}\eta_1 + \frac{\partial F}{\partial y_1'}\eta_1'\right) = 0$$

Integrating by parts the second term, we get

$$\int_{x_0}^{x_1} \frac{\partial F}{\partial y_1}\eta_1 + \left|\frac{\partial F}{\partial y_1'}\eta_1\right|_{x_0}^{x_1} - \int_{x_0}^{x_1} \frac{d}{dx}\left(\frac{\partial F}{\partial y_1'}\right)\eta_1(x)\, dx = 0$$

i.e., $$\int_{x_0}^{x_1} \left[\frac{\partial F}{\partial y_1} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_1'}\right)\right]\eta_1(x)\, dx = 0 \qquad \left[\because \eta_1(x_0) = 0 = \eta_1(x_1)\right]$$

Since this equation must hold good for all values of $\eta_1(x)$, we get

$$\frac{\partial F}{\partial y_1} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_1'}\right) = 0$$

Similarly $\frac{\partial I(\varepsilon)}{\partial \varepsilon_2} = 0$ when $\varepsilon_1 = \varepsilon_2 = \ldots = 0$, will give

$$\frac{\partial F}{\partial y_2} - \frac{d}{dx}\left(\frac{\partial F}{\partial y_2'}\right) = 0 \text{ and so on.}$$

All these conditions give a system of Euler's equations (1). A solution of these equations leads to the desired curves.

Example 3.6.13. Show that the functional $\int_0^{\pi/2} (y^2 - y'^2 + x^2)\, dx$ for which $y(0) = 0, y(\pi/2) = 1, y(0) = 0, y(\pi/2) = 1$ is stationary for $y = \sin x$. Is $y = \sin x$ a minimum?

Solution. Euler's equation is $\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$ ...(1)

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0 \qquad \text{...(1)}$$

Since $F = 2xy + x^2 + y^2$, $\therefore \frac{\partial F}{\partial y} = 2x, \frac{\partial F}{\partial y'} = 2y'$; $\frac{\partial F}{\partial x} = 2x, \frac{\partial F}{\partial y'} = 2y'$

(i) Inserting $2x - \frac{d}{dx}(2y') = 0$ i.e., $2x - 2\frac{d^2y}{dx^2} = 0$ or $\frac{d^2y}{dx^2} = x$ ...(2)

(ii) Inserting $2x - \frac{d}{dx}(2y') = 0$ i.e., $2x - 2\frac{d^2y}{dx^2} = 0$ or $\frac{d^2y}{dx^2} = x$ ...(3)

Now to solve these simultaneous differential equations, we differentiate (2) twice.

$$\frac{d^4x}{dt^4} = \frac{d^4y}{dx^4} = x \qquad \text{[By (3)]}$$

or $(D^4 - 1)x = 0$ which is a linear equation with constant coefficients.

Its solution is $x = c_1e^t + c_2e^{-t} + c_3\cos t + c_4\sin t$ ...(4)

Previously, $y = x^2 + c_1e^t + c_2e^{-t} - c_3\cos t - c_4\sin t$ ...(5)

Since $x = 0$ when $t = 0$, $\therefore 0 = c_1 + c_2 + c_3$ ...(6)

$y = 0$ when $t = 0$, $\therefore 0 = c_1 + c_2 - c_3$ ...(7)

Also $x = -1$ when $t = \pi/2$, $\therefore -1 = c_1e^{\pi/2} + c_2e^{-\pi/2} + c_4$ ...(8)

$y = 1$ when $t = \pi/2$, $\therefore 1 = c_1e^{\pi/2} + c_2e^{-\pi/2} - c_4$ ...(9)

Adding (6) and (7), $2(c_1 + c_2) = 0$

Adding (8) and (9), $c_1e^{\pi/2} + c_2e^{-\pi/2} = 0$

Solving these equations, we get $c_1 = c_2 = 0$.

From (6), $c_3 = c_1 + c_2 = 0$. From (8), $c_4 = -1$.

Hence from (4), $x = \cos t - \sin t$ and from (5), $y = \cos t - \sin t$.

33.13 FUNCTIONALS INVOLVING HIGHER ORDER DERIVATIVES

A necessary condition for

$$I = \int_{x_1}^{x_2} f(x, y, y', y'')\,dx \qquad \text{...(1)}$$

to be extremum is $\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) + \frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right) = 0$.

Let $y(x)$ be the function which makes (1) stationary and satisfies the boundary conditions

$$y(x_1) = y_1, y'(x_1) = y_1', y(x_2) = y_2, y'(x_2) = y_2' \text{ and } y''(x_1) = y_1'', y''(x_2) = y_2''$$

Consider the differentiable function $\eta(x)$ such that:

$$\eta(x_1) = 0 = \eta(x_2) \quad \text{and} \quad \eta'(x_1) = 0 = \eta'(x_2) \qquad \text{...(2)}$$

Replacing y by $y + \varepsilon\eta$ in (1), we get:

$$I(\varepsilon) = \int_{x_1}^{x_2} f(x, y + \varepsilon\eta, y' + \varepsilon\eta', y'' + \varepsilon\eta'')\,dx \qquad \text{...(3)}$$

This is a function of the parameter ε and reduces to (1) for $\varepsilon = 0$.

To find the stationary value of (1), we find the stationary value of $I(\varepsilon)$ for $\varepsilon = 0$. But $I(\varepsilon)$ will have a stationary value when $dI/d\varepsilon = 0$.

Writing $f = f(x, y, y', y'')$ and $F = f(x, y + \varepsilon\eta, y' + \varepsilon\eta', y'' + \varepsilon\eta'')$

(3) becomes $I(\varepsilon) = \int_{x_1}^{x_2} F\,dx$

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α_1, α_2 being constants independent of ε, differentiating under the integral sign, we get

$$\frac{dJ(\varepsilon)}{d\varepsilon} = \int_{x_1}^{x_2} \frac{dF}{d\varepsilon}\,dx = \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}\eta + \frac{\partial F}{\partial y'}\eta' + \frac{\partial F}{\partial y''}\eta''\right]dx$$

$$\therefore \qquad \frac{dJ(\varepsilon)}{d\varepsilon} = 0 \text{ when } \varepsilon = 0 \text{ gives } \int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y}\eta + \frac{\partial F}{\partial y'}\eta' + \frac{\partial F}{\partial y''}\eta''\right]dx = 0$$

Integrating by parts over the second term and twice the third term, we get

$$\int_{x_1}^{x_2}\frac{\partial F}{\partial y}\eta\,dx + \left|\frac{\partial F}{\partial y'}\eta\right|_{x_1}^{x_2} - \int_{x_1}^{x_2}\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\eta\,dx + \left|\frac{\partial F}{\partial y''}\eta' - \frac{d}{dx}\left(\frac{\partial F}{\partial y''}\right)\eta\right|_{x_1}^{x_2} + \int_{x_1}^{x_2}\frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right)\eta\,dx = 0$$

or
$$\int_{x_1}^{x_2}\left[\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) + \frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right)\right]\eta\,dx = 0 \qquad \text{...(2)}$$

Since this equation must hold good for all values of $\eta(x)$, we get

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) + \frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right) = 0$$

In general, a necessary condition for the functional $I = \int_{x_1}^{x_2} f(x, y, y', y'', \ldots, y^{(n)})\,dx$ to be extremum is that

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) + \frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right) - \ldots + (-1)^n \frac{d^n}{dx^n}\left(\frac{\partial F}{\partial y^{(n)}}\right) = 0$$

which is called the Euler-Poisson equation and the solutions are called extremals.

Example 11.14. Show that the curve which extremises the functional $I = \int_0^{\pi/2} (y'^2 - y^2 + x^2)\,dx$, under the conditions $y(0) = 0$, $y'(0) = 1$, $y(\pi/2) = y'(\pi/2) = 0$, is $y = \sin x$. [Madras M.E., 2000 S]

Solution. The Euler-Poisson equation is

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) + \frac{d^2}{dx^2}\left(\frac{\partial F}{\partial y''}\right) = 0 \qquad \text{...(1)}$$

Here $F = (y'^2 - y^2 + x^2)$, $\frac{\partial F}{\partial y} = -2y$, $\frac{\partial F}{\partial y'} = 0$, $\frac{\partial F}{\partial y''} = 2y''$

$$\therefore \quad (1) \text{ becomes } -2y + \frac{d^2}{dx^2}(2y'') = 0 \quad \text{or} \quad y'''' - y = 0 \quad \text{or} \quad (D^4 - 1)y = 0$$

Its solution is $\qquad y(x) = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x \qquad \text{...(2)}$

$$\therefore \qquad y'(x) = c_1 e^x - c_2 e^{-x} - c_3 \sin x + c_4 \cos x \qquad \text{...(3)}$$

Applying the given boundary conditions $y(0) = 0$ and $y'(0) = 1$, we get

$$0 = y(0) = c_1 + c_2 + c_3, \quad 1 = y'(0) = c_1 - c_2 + c_4$$

$$\frac{1}{\sqrt{2}} = y(\pi/2) = c_1 e^{\pi/2} + c_2 e^{-\pi/2} + \frac{1}{\sqrt{2}}c_3 + \frac{1}{\sqrt{2}}c_4 \qquad \text{...(4)}$$

$$\frac{1}{\sqrt{2}} = y'(\pi/2) = c_1 e^{\pi/2} - c_2 e^{-\pi/2} - \frac{1}{\sqrt{2}}c_3 + \frac{1}{\sqrt{2}}c_4$$

Solving the equations (4), we get $c_1 = c_2 = c_3 = 0, c_4 = 1$. Hence the required curve is $y = \sin x$.

PROBLEMS 3.5.3

1. Show that the functional $\int_0^1 [(y')^2 + \left(\frac{dy}{dx}\right)^2 + y^2 + \frac{dy}{dx}]\,dx$ is such that $y(0) = 1$, $y(1) = 2$, $y(2) = 3$, $y(3) = 4$ is a minimum if $y = x + 1$.

2. Find the extremals of the functional $\int_0^{\pi/2} (2xy - 2y^2 + y^2 - 2y^2)\,dx$. (U.P.T.U., M.E., 2006)

3. Find a function $y(x)$ such that $\int_0^1 y^2\,dx$ is a relative extremum and $\int_0^1 y^2\,dx$ is a minimum if $y(x) = 0$, $y(1) = 0$, $y(2) = 0$, $y(3) = 0$.

4. Find the extremals of the functional $\int_0^{\pi/2} (y'^2 - y^2 + x^2)\,dx$ so that it satisfies the conditions $y(0) = 1$, $y'(0) = 0$, $y(\pi/2) = -1$, $y'(\pi/2) = -1$. (Madras, M.E., 2005)

5. Find the extremals of the functional $\int_{x_1}^{x_2} [(y + x)^2 + y'^2]\,dx$ so that satisfies the boundary conditions $y(x_1) = y_1$, $y(x_2) = y_2$, $y(x_1) = 0$, $y(x_2) = 0$.

6. Find the extremals of the following functional $\int_0^{\pi/2} (4y^2 - y'^2 + x^2)\,dx$. (Nagpur, 2000) (U.P.T.U., 2004) $\int_0^{\pi/2} (4y - y'^2)\,dx$

3.5.4 APPROXIMATE SOLUTION OF BOUNDARY VALUE PROBLEMS — Rayleigh-Ritz Method

In § 3.5.3, we have seen that the solution of Euler's differential equation subject to boundary conditions amounts to extremizing a certain definite integral. This fact provides a technique of solving a boundary value problem approximately by assuming a trial solution satisfying the given boundary conditions and then extremizing the integral whose integrand is found from the given differential equation.

To solve a boundary value problem of Rayleigh-Ritz method, we try to write the given differential equation as the Euler's equation of some variational problem. Then we reduce the variational problem to a minimizing problem assuming an approximate solution in the form

$$y(x) = y_0(x) + \sum c_i \phi_i(x) \qquad \ldots(1)$$

where the trial functions $\phi_i(x)$ satisfy the boundary conditions and $\phi_i(x) = 0$ on the boundary C of the region R.

Let the integral to be extremized be $I = \int_a^b f(x, y, y')\,dx$ $\qquad \ldots(2)$

such that $y(x) = A$ and $y(b) = B$.

Substituting (1) in (2) by replacing y in f, giving I as a function of the unknowns c_i. There are infinite parameters which are so determined as to extremize I. This requires

$$\frac{\partial I}{\partial c_i} = 0, \qquad i = 1, 2, \ldots$$

Solving these equations, we get the values of c_i which when substituted in (1) give the desired solution.

Example 3.5.13. Solve the boundary value problem $y'' - y + x = 0$, $0 < x < 1$, $y(0) = 0 = y(1)$ by Rayleigh-Ritz method.

Solution. Given differential equation is $y'' - y + x = 0$ $\qquad \ldots(1)$

Its solution is equivalent to determining the integral $I = \int_0^1 F(x, y, y')\,dx$

where $F(x, y, y') = 2xy - y^2 - y'^2$, $\qquad \ldots(2)$

since the Euler's equation $\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$ gives (1).

Assume that the trial function is $\bar{y}(x) = c_0 + c_1 x + c_2 x^2$ ...(4)

To satisfy $y(0) = 0, y(1) = 0$, we take $c_0 = 0, c_2 = -c_1$.

$\therefore$ (4) becomes $\bar{y}(x) = c_1 x(1 - x)$ ...(5)

Substituting $\bar{y}$ and $\bar{y}'$ in I, we have

$$I = \int_0^1 [2c_1 - x^2 - c_1^2 x^2] dx = \int_0^1 [2c_1(x^2 - x^3) - c_1^2 x^2 + x^4] = c_1^2 [\tfrac{1}{3} - \tfrac{1}{4}] + [\tfrac{1}{3} - \tfrac{1}{5}]$$

$$= \frac{1}{6} c_1^2 - \frac{11}{30} c_1^2$$

Its stationary values are given by $dI/dc_1 = 0 \quad \therefore \quad \frac{1}{6} - \frac{11}{15} c_1 = 0 \quad \text{i.e.,} \quad c_1 = \frac{5}{22}$.

Thus the approximate solution is $\bar{y}(x) = \frac{5}{22} x(1 - x)$ ...(6)

3.38 WEIGHTED RESIDUAL METHOD – Galerkin's Method

The starting point of this method is to guess a solution to the differential equation which satisfies the boundary conditions. This trial solution will contain certain parameters which are to be adjusted later into the errors so that the trial solution is as close to the exact solution as possible.

Consider the boundary value problem

$$y'' = f(x, y, y'), a \le x \le b \text{ with } y(a) = A \text{ and } y(b) = B \quad ...(1)$$

We write the differential equation as $R = y'' - f(x, y, y') = 0$ ...(2)

where R is the residual of the equation. If $R = 0$ for the exact solution $y(x)$ only which will satisfy the boundary conditions).

Consider the trial solution as $\bar{y}(x) = c_1 \phi_1(x) + c_2 \phi_2(x) + \ldots$

where $\bar{y}(a) = A$ and $\bar{y}(b) = B$. The trial solution is differentiated twice and is substituted in (2).

To find $c_1, c_2, \ldots$, we weight the residual by trial functions $\phi_1(x), \phi_2(x), \ldots$ and set the integrals to zero. Thus we have $\int_a^b R\, \phi_1(x)\, dx = 0, \int_a^b R\, \phi_2(x)\, dx = 0, \ldots$

These lead to simultaneous equations in the unknowns.

Having found $c_1, c_2, \ldots$ the approximate solution $\bar{y}(x)$ is obtained.

Example 3.16. Use Galerkin's method to solve the boundary value problem of Example 3.15. Compare your approximate solution with the exact solution.

Solution. The residual is $R = y'' - y + x$ ...(1)

To find the trial solution which satisfies the boundary conditions, we derive a Lagrange's polynomial $\bar{y}(x)$ which passes through the points

x	:	0	1/2	1
y	:	0	c	0

The resulting polynomial is $\bar{y}(x) = 4cx(1 - x)$, so that $\bar{y}(x) = c(1 - x)$.

Substituting $\bar{y}(x)$, $\bar{y}''(x) = -8c$ in (1), we get $R = -4cx^2 + 4(1 - 2c)x - 8c$

Then $\int R\, \phi(x)\, dx = 0$ gives $\int_0^1 [-4cx^2 + 4(1 - 2c)x - 8c](1 - x)\, dx = 0$ whence $c = 5/66$.

Hence the approximate solution is $\bar{y}(x) = \frac{5}{22} x(1 - x)$ which is same as obtained in Example 3.15.

Exact solution. Rewriting the given equation as $(D^2 - 1)y = -x$,

we find the solution as $y = c_1 e^x + c_2 e^{-x} + x$

Since $y(0) = 0$ and $y(1) = 0$, therefore $c_2 = -c_1 = 1/(e - e^{-1})$

Hence the exact solution is $y = \frac{e^x - e^{-x}}{e - e^{-1}}$

The approximate and the exact solutions for some values of x are given below for comparison :

x	Approx. Sol.	Exact Sol.
0.25	0.340	0.332
0.50	0.697	0.697
0.75	1.163	1.05

Obs. To obtain a trial solution containing two unknown parameters, we derive a Legendre polynomial which passes through the points :

x	0	0.5	1
y	0	c_1	c_2	0

More the undetermined parameters, the more accurate is the solution, but it involves more labour to find their values.

Example 3.8.11 Find the approximate deflection of a simply supported beam under a uniformly distributed load w per unit length. Neglect the weight of the beam.

Solution. The differential equation governing the deflection $y(x)$ of the beam is

$$EI\frac{d^2y}{dx^2} = -w, \quad 0 < x < l \qquad \text{...(1)}$$

Fig. 3.7

The boundary conditions are

$$y(0) = 0 = y(l) = 0 \quad \text{(Deflection is zero at ends)} \qquad \text{...(2)}$$

$$\frac{d^2y}{dx^2}(x = 0) = \frac{d^2y}{dx^2}(x = l) = 0 \quad \text{(Bending moment zero at ends)} \qquad \text{...(3)}$$

We assume the trial solution $y(x) = c_1 \sin(\pi x/l) + c_2 \sin(2\pi x/l)$, which satisfies the boundary conditions (2) and (3).

Substituting the trial solution in (1), we obtain the residual

$$R = EIc_1 (\pi/l)^2 \sin(\pi x/l) + 4EIc_2 (\pi/l)^2 \sin(2\pi x/l) - w$$

Then $\int_0^l R \cdot \sin(\pi x/l)\,dx = 0$ and $\int_0^l R \cdot \sin(2\pi x/l)\,dx = 0$

Computing these integrals, we get,

$$EIc_1 (\pi/l)^2 l/2 - wl = 0, \quad 4EIc_2 (\pi/l)^2 l/2 - wl = 0$$

Solving these, we obtain $c_1 = \frac{4wl^2}{\pi^3 EI}$ and $c_2 = \frac{4wl^2}{8\pi^3 EI}$.

Hence the deflection of the beam is given by

$$y(x) = \frac{4wl^2}{\pi^3 EI}\left[\sin\left(\frac{\pi x}{l}\right) + \frac{1}{8}\sin\left(\frac{2\pi x}{l}\right)\right]$$

EXERCISE 3.8

1. Solve the boundary value problem
   $y'' + y = x$, $0 < x < 1$, $y(0) = y(1) = 0$ by
   (i) Rayleigh-Ritz method (ii) Galerkin method. Compare your results with the exact solution.
2. Using Galerkin's method, solve the boundary value problem $y'' + 2y + 4y = x$, $y(0) = 0$, $y(1) = 1$.
3. Apply Galerkin's method to the boundary value problem $y'' + y = x$, $0 < x < 1$, $y(0) = 0$, $y(1) = 0$. Also find the coefficients of the approximate solution $y(x) = c_1 x(x-1) + c_2 x^2(x-1)$.

Hint: Substituting $y = y_1 + y_2$ in the given equation and solving y_2'' for x, we get a modified BVP $y_2'' = f_1(x, y_2)$, $y_2(0) = 0 = y_2(1)$.

Then $$\int_0^1 \left[\tfrac{1}{2}(y_2')^2 + \tfrac{1}{2}y_2^2 - y_2\right]dx = I[y_2] = \int_0^1 F(x, y_2, y_2')\,dx$$

Computing these integrals, we get

$$\frac{2}{3!} + \frac{2}{3^3} + \frac{1}{5!} - \frac{2}{5^3} + \frac{12}{7!} - \frac{1}{7^3}$$

Solving these, we obtain $c_1 = 0.4980$, $c_2 = 0.0417$.

4. Complete the first, find an approximate solution of the problem $y'' + y = x$, $y(0) = y(1) = 0$, $y(1) = x(1) = x$.

5. Solve the boundary value problem $(x^2 + 1)y'' + xy' - y = 0$, $y(0) = 0$, $y(1) = 1$ by using the approximate solution $Y(x) = c_1(x - x^3) + c_2(x^2 - x^3)$ and obtaining $I[Y]$ by the Ritz method. Find the exact solution.

6. Given the boundary value problem $y'' + xy = 0$, $y(0) = 0$, $y(1) = 1$.

Use Galerkin's method to estimate $y(0.5)$, taking the trial solution

$$y = 1 - 1.5x + c_1 x(1 - x) + c_2 x^2(1 - x)$$

7. Using Galerkin's method, obtain an approximate solution of the boundary value problem

$$\frac{d}{dx}\left[(1 + x^2)\frac{dy}{dx}\right] + y = x, \quad y(0) = 0, \quad y(1) = 1,$$

in the form $y(x) = x + c_1(1 - x)x + c_2(1 - x)x^2$.

8. Of all the paraboloids of the type $z = ax^2 + by^2$ through $(0, 0, 0)$ and $(1, 1, 2)$, determine the one which when rotated about the z-axis, generates a solid of revolution with least possible moment of inertia about the z-axis. $I = \int\int (x^2 + y^2)\,dz\,dx\,dy$.

Hint: Take the paraboloid as $z = x^2 + ky^2$ ($k \neq 1$).

9. Using Rayleigh-Ritz method, find the potential energy stored due to the applied values of x and y.

Hint: Potential at a distance r from the centre of the sphere $x^2 + y^2 + z^2 = a^2$ is given by $\frac{k}{r}$ where $k > 0$ is such that $r \to a$ as $r \to \infty$.

Electrostatic field due to charged sphere being conservative, electrostatic intensity $E = \frac{k}{r^2}$.

Also potential energy for unit volume $= \frac{1}{2}E^2$

Total potential energy of the field in the region R outside the sphere is given by

$$V = \frac{1}{8\pi}\iiint_R E^2\,dx\,dy\,dz = \frac{1}{8\pi}\iiint_R \left[\left(\frac{\partial \phi}{\partial x}\right)^2 + \left(\frac{\partial \phi}{\partial y}\right)^2 + \left(\frac{\partial \phi}{\partial z}\right)^2\right] dx\,dy\,dz$$

$$= \frac{1}{8\pi}\int_{\alpha}^{\beta}\int_{\gamma}^{\delta}\int_{a}^{\infty}\left(\frac{\partial \phi}{\partial r}\right)^2 r^2 \sin\theta\,dr\,d\theta\,d\phi$$

Potential is $\phi = \frac{k}{r}$. Hence $\frac{\partial \phi}{\partial r} = -\frac{k}{r^2}$, $V = \frac{1}{8\pi}\int_{\alpha}^{\beta}\int_{\gamma}^{\delta}\int_{a}^{\infty}\frac{k^2}{r^4} r^2 \sin\theta\,dr\,d\theta\,d\phi$

$= \frac{k^2}{8\pi}\int_{\alpha}^{\beta}\int_{\gamma}^{\delta}\int_{a}^{\infty}\frac{1}{r^2}\sin\theta\,dr\,d\theta\,d\phi$

Evaluate the following integrals using Rayleigh-Ritz method.

10. $I = \int_0^1 (y^2 + x^2)\,dx$, $y(0) = 1$, $y(1) = 2$

11. $I = \int_0^{\pi/2} (y^2 - x^2)\,dx$, $y(0) = 0$, $y\left(\frac{\pi}{2}\right) = 1$

13.15 HAMILTON'S PRINCIPLE*

An important concept of mathematical physics is Hamilton's principle which states that the time integral of the difference between the kinetic and potential energy of a dynamical system is minimum.

Consider a particle of mass m moving from a fixed origin O under the effect of force F (Fig. 13.8). At any time, let its position vector be R. Then by Newton's second law,

$$\frac{m d^2\mathbf{R}}{dt^2} = \mathbf{F} \qquad \text{...(1)}$$

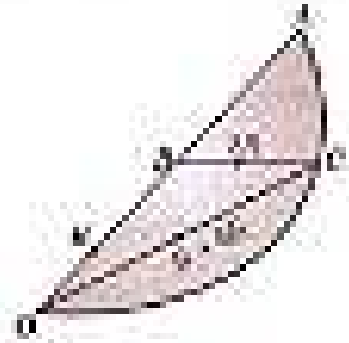

Fig. 13.8

* Named after the Irish mathematician William Rowan Hamilton (1805–1865) who is known for his contributions to dynamics.

Let the natural path OBA of the particle be changed to another path DCA, and points D and C are arbitrarily close. Let this variation in path, often called a *virtual displacement*, be $\delta\mathbf{R}$. Then the work done during this displacement is

$$\delta T = \mathbf{F} \cdot \delta\mathbf{R} = \frac{m\mathrm{d}^2\mathbf{R}}{\mathrm{d}t^2} \cdot \delta\mathbf{R} \qquad \text{[Eq (1)]}$$

Also the kinetic energy of the particle is

$$T = \frac{1}{2}m\left(\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t}\right)^2$$

$$\therefore \quad \delta T = m\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t} \cdot \delta\left(\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t}\right) = m\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t} \cdot \frac{\mathrm{d}}{\mathrm{d}t}(\delta\mathbf{R})$$

Thus $$\delta T + \delta V = m\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t} \cdot \frac{\mathrm{d}}{\mathrm{d}t}(\delta\mathbf{R}) + m\frac{\mathrm{d}^2\mathbf{R}}{\mathrm{d}t^2} \cdot \delta\mathbf{R} - m\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t} \cdot \delta\mathbf{R}\right)$$

Integrating both sides w.r.t. t from t_1 to t_2, we get

$$\int_{t_1}^{t_2}(\delta T + \delta V)\,\mathrm{d}t = m\left[\frac{\mathrm{d}\mathbf{R}}{\mathrm{d}t} \cdot \delta\mathbf{R}\right]_{t_1}^{t_2} = 0 \qquad \text{... (1)} \quad \text{[since } \delta\mathbf{R} = 0 \text{ at } t_1 \text{ and } t_2\text{]}$$

If the force field is conservative, there exists a potential V such that $\mathbf{F} = -\nabla V$. Then (1) takes the form

$$\int_{t_1}^{t_2}(\delta T - \delta V)\,\mathrm{d}t = 0 \quad \text{or} \quad \int_{t_1}^{t_2}\delta(T - V)\,\mathrm{d}t = 0 \quad \text{i.e.,} \quad \int_{t_1}^{t_2}(T - V)\,\mathrm{d}t \qquad \text{... (2)}$$

is stationary. This proves the Hamilton's principle for a particle. Its derivation can be extended to a system of particles by summation and to a rigid body by integration. Hence the principle is true for any dynamical system.

Obs. It can be easily shown that for integral (2) to be a minimum along the natural path as compared to that along any other path joining O to A.

Def. The energy difference $T - V$ is called the **kinetic potential** or the **Lagrangian function**.

35.3 LAGRANGE'S EQUATION

In a dynamical system, the position of a body can be specified by the quantities $q_1, q_2, \ldots, q_n$ which are called the *generalized coordinates*.

The potential energy V, being a function of position only depends on the generalized coordinates q_i. The kinetic energy T, however, depends upon q_i and the velocities $\dot{q}_i$ (or $\mathrm{d}q_i/\mathrm{d}t$) i.e., q_i's $(i = 1, 2, \ldots, n)$. Therefore, Lagrangian function $L = T - V$ is also a function of q_i and $\dot{q}_i$, $i = 1, 2, \ldots, n$.

Thus by Hamilton's principle, the system moves so that $\int_{t_1}^{t_2} L\,\mathrm{d}t$ is stationary.

$\therefore$ Euler's equation must hold good, i.e., $\dfrac{\partial L}{\partial q_i} - \dfrac{\mathrm{d}}{\mathrm{d}t}\left(\dfrac{\partial L}{\partial \dot{q}_i}\right) = 0, i = 1, 2, \ldots, n.$

These are called *Lagrange's equations* which determine the motion of the system.

Example 35.1(b). A mass, suspended at the end of a light spring (mass of spring neglected), is set into vertical motion. Use Lagrange's equation to find the equation of motion of the mass.

Solution. At any instant, let the displacement of m from the equilibrium position O be x (Fig. 35.3). Then the kinetic energy of P is

$$T = \frac{1}{2}m\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2$$

Also the work done during its fall from O to P is

$$W = \int_0^x (mg - kx)\,\mathrm{d}x = mgx - \frac{1}{2}kx^2$$

If V is the potential energy of the mass at P, then

$$V = -W = \frac{1}{2}kx^2 - mgx$$

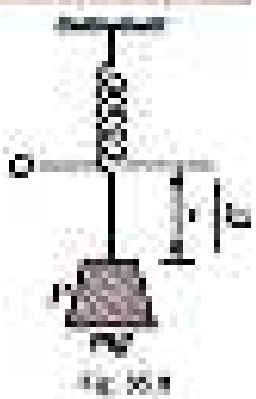

Fig. 35.3

∴ Lagrangian $L = T - V = \frac{1}{2}m_1\dot{x}^2 + m_2\dot{x}\dot{\theta}l\cos\theta - \frac{1}{2}kx^2$

Now the Lagrange's equation

$$\frac{\partial L}{\partial x} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = 0$$

becomes $(m_1+m_2)\ddot{x} + m_2 l\ddot{\theta}\cos\theta - m_2 l\dot{\theta}^2\sin\theta + kx = 0$ or $\ddot{x} + \frac{k}{(m_1+m_2)}x = m_2 l\dot{\theta}^2\sin\theta - m_2 l\ddot{\theta}\cos\theta$

which is the required equation of motion.

Similarly, for the second degree of freedom we take the generalized coordinate θ as shown in the figure. The velocity of the bob is given by

$$v_2 = v_1 + v_{21} = \dot{x}\hat{i} + l\dot{\theta}\sin\theta\,\hat{i} - l\dot{\theta}\cos\theta\,\hat{j}$$

and

$$v_2^2 = \dot{x}^2 + l^2\dot{\theta}^2 - 2l\dot{x}\dot{\theta}\cos\theta$$

The kinetic energy is

$$T = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2(\dot{x}^2 + l^2\dot{\theta}^2 - 2l\dot{x}\dot{\theta}\cos\theta)$$

The potential energy is

$$V = \frac{1}{2}kx^2 - m_2 gl\cos\theta$$

∴ Lagrangian $L = T - V = \frac{1}{2}(m_1+m_2)\dot{x}^2 + \frac{1}{2}m_2 l^2\dot{\theta}^2 - m_2 l\dot{x}\dot{\theta}\cos\theta - \frac{1}{2}kx^2 + m_2 gl\cos\theta$

Now the Lagrange's equation corresponding to x is

$$\frac{\partial L}{\partial x} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = 0 \text{ becomes}$$

$$-m_2 l\dot{\theta}^2\sin\theta - \ddot{x}(m_1+m_2) + m_2 l\ddot{\theta}\cos\theta - \frac{k}{(m_1+m_2)}(m_1+m_2)x + m_2 l\dot{\theta}^2\sin\theta = 0$$

or $$\ddot{x} + \frac{k}{(m_1+m_2)}x - m_2 l\ddot{\theta}\cos\theta + m_2 l\dot{\theta}^2\sin\theta = 0 \quad \ldots(iii)$$

Similarly, now w.r.t. Lagrange's equation corresponding to θ, i.e.,

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = 0$$

becomes $$m_2 l\dot{x}\dot{\theta}\sin\theta - m_2 gl\sin\theta - \frac{d}{dt}(m_2 l^2\dot{\theta} - m_2 l\dot{x}\cos\theta) = 0$$

or $$\ddot{x}\cos\theta + l\ddot{\theta} - g\sin\theta = 0 \quad \ldots(iv)$$

Now $\ddot{x}$ and $\ddot{\theta}$ being small, neglecting their order terms we get from (iii) and (iv) as follows as

$$m_2 l\ddot{\theta}\cos\theta = m_2 l\dot{\theta}^2\sin\theta - \ddot{x}(m_1+m_2) - \frac{k}{(m_1+m_2)}(m_1+m_2)x$$

and $$l\ddot{\theta} = g\sin\theta - \ddot{x}\cos\theta$$

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CHAPTER 36

Integral Equations

1. Introduction, 2. Definitions, 3. Conversion of a linear differential equation to an integral equation and vice versa, 4. Conversion of boundary value problems to integral equations using Green's function, 5. Solution of an integral equation, 6. Integral equations of the convolution type, 7. Abel's integral equation, 8. Integro-differential equations, 9. Integral equations with separable kernels, 10. Solution of Fredholm equations with separable kernels, 11. Solution of Fredholm and Volterra equations by the method of successive approximations.

36.1 INTRODUCTION

Integral equations play an effective role in the study of boundary value problems. Such equations also occur in many fields of mechanics and mathematical physics. Integral equations may be obtained directly from physical problems (e.g., radiation transfer problems and neutron diffusion problems). They also serve as representation formulae for the solutions of differential equations. A differential equation can be reduced to an integral equation with the help of initial and boundary conditions.

Integral equations were first encountered in the theory of Fourier integrals. In 1903, another integral equation was obtained by Abel. Actual development of the theory of integral equations began with the works of the Italian mathematician V. Volterra (1860) and the Swedish mathematician I. Fredholm (1900).

36.2 DEFINITION

An integral equation is an equation in which an unknown function appears under the integral sign. We shall take up integral equations in which only linear functions of the unknown functions are involved. The general type of linear integral equations is of the form

$$g(x)\,u(x) = f(x) + \lambda \int_a^b K(x,t)\,u(t)\,dt$$

where $f(x)$ and $g(x)$ are known functions of x which is to be determined. The function $K(x,t)$ is called the *kernel* of the integral equation.

If a and b are constants, the equation is known as *Fredholm integral equation.*

If the parameter λ is a variable, it is called a *Volterra integral equation.*

36.3 CONVERSION OF A LINEAR DIFFERENTIAL EQUATION TO AN INTEGRAL EQUATION AND VICE VERSA

To make this transformation, the use of the following formula is necessary:

$$\int_a^x \int_a^x f(x)\,dx\,dx = \int_a^x (x-t) f(t)\,dt \qquad \text{...(1)}$$

In general, $$\int_a^x \int_a^x \cdots \int_a^x f(x)\,dx^n = \frac{1}{(n-1)!}\int_a^x (x-t)^{n-1} f(t)\,dt \qquad \text{...(2)}$$

Proof. Let $I_n = \int_0^x (x-t)^{n-1} f(t)\,dt$ ...(1)

where n is a positive integer and x is a constant.

Differentiating both sides of (1) w.r.t. x, using Leibnitz's rule (p. 1.96)

$$\frac{dI_n}{dx} = \int_0^x \frac{\partial}{\partial x}(x-t)^{n-1} f(t)\,dt + [(x-t)^{n-1} f(t)]_{t=x} \cdot 1 - [(x-t)^{n-1} f(t)]_{t=0} \cdot 0$$

$$= \int_0^x (n-1)(x-t)^{n-2} f(t)\,dt + 0 - 0 = (n-1) I_{n-1} \quad \text{...(2)}$$

Again, differentiating (2) w.r.t. x

$$\frac{d^2 I_n}{dx^2} = (n-1)\frac{d}{dx} I_{n-1} = (n-1)(n-2) I_{n-2}, \text{ using (2)}$$

Proceeding in this way, we get

$$\frac{d^{n-1} I_n}{dx^{n-1}} = (n-1)(n-2)\cdots 1 \cdot I_1 = (n-1)! I_1 \quad \text{...(3)}$$

Now taking $n = 1$ in (1), we get

$$I_1 = \int_0^x f(t)\,dt = \int_0^x f(x_1)\,dx_1 \quad \text{...(4)}$$

Putting $n = 1$ in (3), we obtain

$$\frac{d I_1}{dx} = f(x) \quad \text{i.e.,} \quad I_1' = f(x)$$

Taking $n = 2$ in (1), we get $I_2 = \int_0^x f(x_1)\,dx_1$ (i) $I_2(0) = 0$

$$= \int_0^x \int_0^{x_1} f(x_2)\,dx_2\,dx_1 \quad \text{(Using (4))} \quad \text{...(5)}$$

Putting $n = 2$ in (2), we have $\frac{dI_2}{dx} = I_1(x)$

i.e., $I_2 = \int_0^x I_1(x)\,dx$ (ii) $I_2(x) = 0$

$$= \int_0^x \int_0^{x_1} \int_0^{x_2} f(x_3)\,dx_3\,dx_2\,dx_1 \quad \text{(Using (5))}$$

Proceeding in this way, we get

$$I_n = (n-1)! \int_0^x \int_0^{x_1} \cdots \int_0^{x_{n-1}} f(x_n)\,dx_n\,dx_{n-1}\cdots dx_1$$

i.e., $$\int_0^x \int_0^{x_1} \cdots \int_0^{x_{n-1}} f(x_n)\,dx_n\,dx_{n-1}\cdots dx_1 = \frac{1}{(n-1)!}\int_0^x (x-t)^{n-1} f(t)\,dt$$

If $x_1, x_2, \ldots, x_n$ be the same as x, we get the formula (1) above.

Example 36.2. Convert the differential equation $y''(x) - 2y'(x) + y(x) = xe^x$, $y(0) = 1$, $y'(0) = 2$ into an integral equation.

Solution. Integrating both sides of the given differential equation, we get

$$[y'(x) - y'(0)] - 2[y(x) - y(0)] + \int_0^x y(t)\,dt = \frac{x^3}{3!} - \cos x$$

Since $y'(0) = 2$ and $y(0) = 1$, it becomes

$$y'(x) + 3 - 2y(x) + 2 + \int_0^x y(t)\,dt = \frac{x^3}{3!} - \cos x$$

i.e., $$y'(x) - 2y(x) + 2\int_0^x y(t)\,dt = -6\sin x$$

Integrating again as before, we have

$$y(x) - y(0) - 2\int_0^x y(t)\,dt + 2\int_0^x\int_0^x y(t)\,dt = -6\sin x$$

or $$y(x) - 1 - 2\int_0^x y(t)\,dt + 2\int_0^x (x-t)\,y(t)\,dt = -6\sin x \qquad \text{[Using (ii) above]}$$

or $$y(x) + \int_0^x [2x - 1 - 2]\,y(t)\,dt = 1 - 6\sin x$$

which is the desired integral equation.

Example 10.2. *Find the IVP equivalent to the integral equation*

$$y(x) = \int_0^x (x+t)\,y(t)\,dt + 1 \qquad \ldots(i)$$

is equivalent to the differential equation

$$y''(x) - 2y'(x) - 2y(x) = 0,\ x > 0;\ y(0) = 1,\ y'(0) = 0 \qquad \text{(Ayres, Jr. and Mendelson, 2008)}$$

Solution. Differentiating (i) by Leibnitz's rule (p. 1160), we have

$$\frac{dy}{dx} = \int_0^x \frac{\partial}{\partial x}(x+t)\,y(t)\,dt + (x+x)\,y(x)\frac{d}{dx}(x) - (x+0)\,y(0)\frac{d}{dx}(0)$$

$$= \int_0^x y(t)\,dt + 2xy(x) = \int_0^x y(t)\,dt + 2xy(x) \qquad \ldots(ii)$$

Differentiating (ii) again w.r.t. x, we get

$$\frac{d^2y}{dx^2} = y(x) + 2[y'(x)x + 1 \cdot y(x)] = 2xy'(x) + 3y(x)$$

or $$\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} - 3y(x) = 0 \qquad \ldots(iii)$$

Putting $x = 0$ in (i), we obtain

$$y(0) = \int_0^0 (x+t)\,y(t)\,dt + 1 \quad \text{i.e.,} \quad y(0) = 1$$

and putting $x = 0$ in (ii), we get $y'(0) = 0$.

Hence (i) is equivalent to (iii) with initial conditions $y(0) = 1, y'(0) = 0$.

Example 10.3. *Show that the integral equation*

$$y(x) = \int_0^x (x-t)\,y(t)\,dt + \frac{x^3}{3} \qquad \ldots(i)$$

is equivalent to the differential equation

$$\frac{d^2y}{dx^2} + y = x \text{ with suitable conditions } y(0) = y'(0) = 0.$$

Solution. Differentiating (i) w.r.t. x by Leibnitz's rule (p. 1160), we have

$$\frac{dy}{dx} = \int_0^x \frac{\partial}{\partial x}(x-t)\,y(t)\,dt + (x-x)\,y(x)\frac{d}{dx}(x) - (x-0)\,y(0)\frac{d}{dx}(0)$$

$$= \int_0^x (x-t)\,y(t)\,dt + x\,y(x) - 0 = x\int_0^x y(t)\,dt - \int_0^x t\,y(t)\,dt \qquad \ldots(ii)$$

Differentiating (ii) w.r.t. x, we get

$$\frac{d^2y}{dx^2} = 1 \cdot \int_0^x y(t)\,dt + x\left[\int_0^x \frac{\partial}{\partial x}y(t)\,dt + y(x)\frac{d}{dx}(x) - 0\right] - [x\,y(x)\cdot 1] = 1 - xy(x)$$

or $\frac{d^2y}{dx^2} + \lambda y = 1$, which is the differential equation corresponding to (i)

Also $y(0) = 0$ and $y'(0) = 0$

Example 20.4. Find the integral equation corresponding to the boundary value problem $y''(x) + \lambda y(x) = \lambda \qquad y(0) = y(1) = 0$

Solution. Integrating both sides of the given differential equation w.r.t. x over $(0, x)$, we get

$$y'(x) - y'(0) + \lambda \int_0^x y(x)\,dx = 0$$

or $$y'(x) = c - \lambda \int_0^x y(x)\,dx, \text{ taking } y'(0) = c \qquad \ldots(i)$$

Again integrating (i) w.r.t. x in $(0, x)$, we obtain

$$y(x) - y(0) = cx - \lambda \int_0^x \int_0^x y(x)\,(dx)^2$$

$$y(x) = cx - \lambda \int_0^x (x - t)\,y(t)\,dt \qquad \ldots(ii)$$

[$\because$ using (i) in [20.3 and writing t in x] and noting that $y(0) = 0$]

Putting $x = 1$ in (ii), we get

$$y(1) = c - \lambda \int_0^1 (1 - t)\,y(t)\,dt \qquad \because \quad y(1) = 0$$

$$\therefore \quad c = \lambda \int_0^1 (1 - t)\,y(t)\,dt \qquad \ldots(iii)$$

Substituting the value of c from (iii) in (ii), we get

$$y(x) = \lambda x \int_0^1 (1 - t)\,y(t)\,dt - \lambda \int_0^x (x - t)\,y(t)\,dt$$

$$= \lambda x \left\{ \int_0^x (1 - t)\,y(t)\,dt + \int_x^1 (1 - x)\,y(t)\,dt \right\} - \lambda \int_0^x (x - t)\,y(t)\,dt$$

$$= \lambda \int_0^x x(1 - t)\,y(t)\,dt + \lambda \int_x^1 x(1 - t)\,y(t)\,dt$$

$$= \lambda \left\{ \int_0^x K(x, t)\,y(t)\,dt + \int_x^1 K(x, t)\,y(t)\,dt \right\}$$

where $$K(x, t) = \begin{cases} t(1 - x) & \text{when } t < x \\ x(1 - t) & \text{when } t > x \end{cases}$$

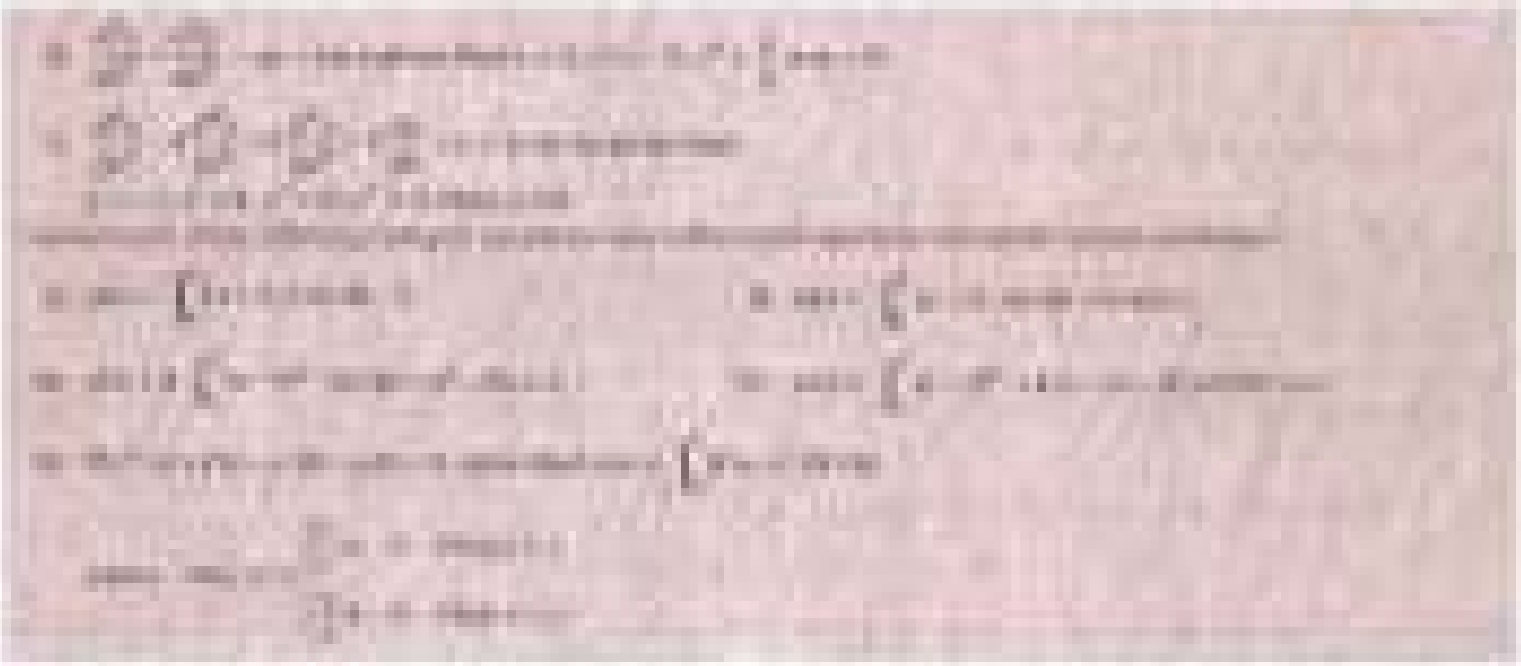
Hence $$y(x) = \lambda \int_0^1 K(x, t)\,y(t)\,dt$$

which is a Fredholm integral equation with a symmetric kernel $K(x, t)$.

EXERCISE 20.1

Transform each of the following boundary value problems into an integral equation:

1. $y'' + xy' + y = 0$, given that $y(0) = 1$, $y'(0) = 0$ (Madras, M A, 1999)
2. $y'' + xy' + y = 0$, given that $y = 1$, $\frac{dy}{dx} = -1$ when $x = 0$
3. $y'' + xy' + 1 = 0$, given that $y(0) = y(1) = 0$ (Madras, 1998)
4. $y'' + (1 - x)y' + 2y = x^2$, given that $y = 1$, $\frac{dy}{dx} = 1$ when $x = 0$
5. $\frac{d^2y}{dx^2} + xy = 0$, given that $y(0) = 1$, $y'(0) = 0$ (Kerala, M Sc, 2000)



[illegible] Conversion of boundary value problems to integral equations using Green's functions

Consider the linear homogeneous differential equation

$$L[y] = y'' + p(x)y' + q(x)y = 0 \tag{1}$$

where $L[y] = \left[\frac{d^2}{dx^2} + p(x)\frac{d}{dx} + q(x)\right]y = 0$ and $p(x)$ and $q(x)$ are continuous on $[a, b]$.

Suppose now that the homogeneous boundary conditions of the form

$$\alpha y + \beta \frac{dy}{dx} = 0 \tag{2}$$

There [illegible] $y_1(x)$ and $y_2(x)$ which the [illegible] given by $\Gamma(x)$ where $x = t$ and by $\Gamma_2(x)$ where $x = t$ and which [illegible] the following properties:

I. Γ_1 and Γ_2 satisfy the equation $L[\Gamma] = 0$ for $x \neq t$ [illegible] $L[\Gamma_1] = 0$ [illegible] $L[\Gamma_2] = 0$ where $x > t$.

II. Γ_1 and Γ_2 satisfy the boundary conditions of the form [illegible] $x = a$ and $x = b$ respectively.

III. $\Gamma_1(x) = \Gamma_2(x)$ [illegible] $x = t$, $\Gamma_2'(t) - \Gamma_1'(t) = \frac{1}{p(t)}$

IV. The derivative of Γ is continuous at every point except the point $x = t$ [illegible] $\Gamma_2'(t) - \Gamma_1'(t) = \frac{1}{p(t)}$

Proof: [illegible]

If this is [illegible] then the solutions of the given problem can be transformed to the integral equations

$$y(x) = \int_a^b \Gamma(x, t)\, q(t)\, y(t)\, dt \tag{3}$$

$y_1 = \Gamma_1(x)$ and $y = \Gamma_2(x)$ for the [illegible] extension of the equation $L[y] = 0$ [illegible] the homogeneous conditions $\alpha y + \beta y' = 0$ and $\gamma y + \delta y' = 0$ respectively.

The above conditions I and II are satisfied if we write

$$G = \begin{cases} \Gamma_2(x,t) & \text{when } a \le x \le t \\ \Gamma_1(x,t) & \text{when } t \le x \le b \end{cases} \tag{4}$$

Applying the condition III we [illegible] we get

$$\Gamma_2(t,t) - \Gamma_1(t,t) = k \tag{5}$$

Applying the condition IV we [illegible] we have

$$\Gamma_2'(t,t) - \Gamma_1'(t,t) = \frac{1}{p(t)} \tag{6}$$

Equations (6) and (7) give a unique solution, if

$$\begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t)y_2'(t) - y_2(t)y_1'(t) \neq 0$$

By Abel's formula*, we find that

$$y_1(t)y_2'(t) - y_2(t)y_1'(t) = C\phi(t) \quad \ldots(8)$$

Now $G(t) = y_1(t) - (t) \times y_1(t)$ gives on using (8),

$$C_1 = -\frac{1}{C}y_2(t) \text{ and } C_2 = -\frac{1}{C}y_1(t)$$

Then (5) reduces to $\quad G(x,t) = \begin{cases} -\frac{1}{C}y_1(t)y_2(x), & x < t \\ -\frac{1}{C}y_2(t)y_1(x), & x > t \end{cases} \quad \ldots(9)$

Obviously it can be shown that the integral equation

$$y(x) = \int_a^b G(x,t)\phi(t)\,dt$$

where $G(x,t)$ is as defined by (9), satisfies the differential equation $L(y) + \phi(x) = 0$ together with the associated boundary conditions.

Example 1.61. Determine the Green's function for the differential equation $y'' + y = 0$, $y(0) = 0$, $y'(\pi) = 0$ and hence solve the integral equation $\phi(x) = f(x) + \int_0^{\pi} K(x,t)\phi(t)\,dt$, where $K(x,t)$ is the Green's function. (Madras, B.E., 2000 S)

Solution. Given equation is $\quad L(y) + \phi(x) = 0 \quad \ldots(1)$

with the conditions $y(0) = 0$, $y'(\pi) = 0$, where $L(y) \equiv y''$ and $\phi(x) \equiv y$ ... (2)

The associated equation $\quad L(y) = 0$ i.e., $y'' = 0 \quad \ldots(3)$

Its solution is $\quad y = C_1 x + C_2 \quad \ldots(4)$

Now $y_1(x)$ is a particular solution of (3) satisfying the condition $y_1(0) = 0$.

$\therefore$ Taking $C_2 = 0$ and $C_1 = 1$, we get $y_1(x) = x \quad \ldots(5)$

$y_2(x)$ is another solution of (3) satisfying the condition $y_2'(\pi) = 0$.

$\therefore$ From (4), $y_2'(\pi) = C_1 = 0$. Then $y_2(x) = C_2$

Taking $C_2 = 1$, a particular solution is $y_2(x) = 1$.

The constant C is found from $y_1(x)y_2'(x) - y_2(x)y_1'(x) = \frac{C}{p(x)}$

Since $L(y) \equiv y'' + p(x)y' + q(x)y = y''$ ($p(x) = 0$), $\therefore \quad p = 1$.

Then $C = y_1(x)y_2'(x) - y_2(x)y_1'(x) = x \cdot 0 - 1 \cdot 1 = -1$.

$\therefore$ Green's function is given by

$$G(x,t) = \begin{cases} -\frac{y_1(t)y_2(x)}{C}, & x < t \\ -\frac{y_1(x)y_2(t)}{C}, & x > t \end{cases} = \begin{cases} t, & x < t \\ x, & x > t \end{cases} \quad \ldots(6)$$

Hence the equation (1) is equivalent to the integral equation

$$y(x) = \int_0^{\pi} G(x,t)\phi(t)\,dt = \int_0^{\pi} G(x,t)(f(t) - y(t))\,dt$$

$$= \int_0^{\pi} G(x,t)y(t)\,dt - \left\{\int_0^x x \cdot f(t)\,dt + \int_x^{\pi} t \cdot f(t)\,dt\right\}$$

---

* The conditions that $y_1(x)$ and $y_2(x)$ satisfy the equations

$L(y) = 0$ i.e., $y'' + py' + qy = 0 \quad \ldots(5)$ $\quad$ and $\quad L(y_1) = y_1'' + py_1' + qy_1 = 0 \quad \ldots(6)$

$L(y_2) = y_2'' + py_2' + qy_2 = 0 \quad \ldots(7)$, gives on $y_2 \times (5) - y \times (7)$ gives on using (6),

$y_2(y'' + py' + qy) - y(y_2'' + py_2' + qy_2) = 0$

or $\quad [(y_2y' - yy_2')]' + p(y_2y' - yy_2') = 0 \quad$ or $\quad y_2y' - yy_2' = Ce^{-\int p\,dx}$

$$= \int_0^1 G(x,t)\,y(t)\,dt - \frac{1}{2}\left|\frac{t^2}{2}\right|_0^x + \left|\frac{t^2}{2}\right|_x^1$$

$$= \int_0^1 G(x,t)\,y(t)\,dt = \frac{1}{2}x^2 + 2t$$

where $G(x,t)$ is given by (*).

Example 10.6: Find the Green's function for the boundary value problem
$y''(x) + y(x) = 0, \quad y(0) = 0 = y(1)$

Solution. We observe that the solution $y_1 = \sin x$ of $y'' + y = 0$ satisfies the boundary condition $y(0) = 0$ and the solution $y_2 = \sin(x - 1)$ satisfies the second condition $y(1) = 0$. Also both these solutions are linearly independent.

The constant C is found from $y_1 y_2' - y_2 y_1' = C(x)$ (*)

Since $f(x) = y'' + y' + xy = y''$ (given), $\therefore \quad p = 1$

$\therefore \quad C = y_1 y_2' - y_2 y_1' = \sin x \cos(x-1) - \sin(x-1)\cos x = \sin[x - (x-1)] = \sin 1$ (constant)

Hence the Green's function is given by

$$G(x,t) = \begin{cases} -\dfrac{y_1(x)\,y_2(t)}{C}, & x < t \\ -\dfrac{y_2(x)\,y_1(t)}{C}, & x > t \end{cases} = \begin{cases} \dfrac{\sin x \cos(t-1) - \sin(x-1)\cos t}{\sin 1}, & x < t \\ \dfrac{\sin t \cos(x-1) - \sin(t-1)\cos x}{\sin 1}, & x > t \end{cases}$$

PROBLEMS 10.5

1. Transform the problem $y'' + y = x, \; y(0) = 0 = y(1)$ into an integral equation, finding the corresponding Green's function.
2. Transform the problem $y'' + y = x, \; y(0) = 1, y'(1) = 0$ into an integral equation using Green's function.
3. Obtain the integral equation corresponding to the boundary value problem
$$\frac{d^2y}{dx^2} + x^2 y = x, \; y(0) = 0, \; y(1) = 1$$
4. Find the Green's function for the boundary value problem $y''(x) - y = 0$ with $y(0) = y(1) = 0$.
5. Transform the boundary value problem $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} + (x^2 - 1)y = 0, \; y(0) = 0, \; y(1) = 1$, into an integral equation.

Hint. $y(x) = x, \; y_2(x) = \dfrac{1}{x} - x \; \text{and} \; C = -2$

10.5 SOLUTION OF AN INTEGRAL EQUATION

The solution of the integral equation $y(x) = F(x) + \lambda \int_a^b K(x,t)\,y(t)\,dt$ is a function $y(x)$ which when substituted in the equation reduces it to an identity in x.

Example 10.7: Show that $y(x) = 2 - x$ is a solution of the integral equation
$$\int_0^1 e^{x-t}\,y(t)\,dt = e^x + x - 1 \qquad (*)$$

Solution. Since $y(t) = 2 - t$

$$\therefore \quad \int_0^1 e^{x-t}\,y(t)\,dt = \int_0^1 e^{x-t}(2 - t)\,dt$$

$$= 2e^x \int_0^1 e^{-t}\,dt - e^x \int_0^1 t\,e^{-t}\,dt$$

$$= 2e^{x}\left[-e^{-x}\left\{x(-e^{-x})-\int_0^x 1\cdot(-e^{-x})\,dx\right\}\right] - e^{x}\left\{x^2(-e^{-x})-\int_0^x 2x(-e^{-x})\,dx\right\}$$

$$= 2e^{x}(-xe^{-x}+1-e^{-x}) - e^{x}(x^2e^{-x}-2xe^{-x}-2e^{-x}+2)$$

$$= -2x+2e^{x}-2+2x+x^2-2x+2 = x^2-2x+2.$$

Thus (1) is identically satisfied by $y(x) = 2-x$. Hence $y(x) = 2-x$ is a solution of (1).

Example 3.6. Show that the function $y(x) = (1+x^2)^{-3/2}$ is a solution of the Fredholm integral equation

$$y(x) = \frac{1}{1+x^2} - \int_0^1 \frac{x^2}{1+x^2}\, y(t)\,dt. \qquad (1)$$

Solution. Substituting $y(x) = (1+x^2)^{-3/2}$ in the R.H.S. of (1), we have

$$\frac{1}{1+x^2} - \int_0^1 \frac{x^2}{1+x^2}\,\frac{1}{(1+t^2)^{3/2}}\,dt$$

$$= \frac{1}{1+x^2} - \frac{1}{1+x^2}\left[\frac{t}{(1+t^2)^{1/2}}\right]_0^1$$

$$= \frac{1}{1+x^2} - \frac{1}{1+x^2}\cdot\frac{1}{(1+1^2)^{1/2}} = \frac{1}{1+x^2} - \frac{1}{1+x^2}\cdot\frac{1}{\sqrt{2}} = \frac{1}{(1+x^2)^{3/2}} = y(x)$$

Thus $y(x) = (1+x^2)^{-3/2}$ is a solution of the integral equation (1).

Example 3.7. Show that the function $y(x) = x^2$ is a solution of the integral equation

$$y(x) = \sin x + 2\int_0^x \cos(x-t)\,y(t)\,dt. \qquad (1)$$

Solution. Substituting $y(t) = t^2$ in the R.H.S. of (1), we have

$$\sin x + 2\int_0^x \cos(x-t)\,t^2\,dt$$

$$= \sin x + 2\left[\cos x\int_0^x t^2\cos t\,dt + \sin x\int_0^x t^2\sin t\,dt\right] \qquad \text{Integrating by parts}$$

$$= \sin x + 2\cos x\left[\frac{t^2}{2}(\sin t+\cos t) - \frac{1}{2}\int 2t(\sin t+\cos t)\,dt\right]$$

$$+ 2\sin x\left[\frac{t^2}{2}(\sin t-\cos t) - \frac{1}{2}\int 2t(\sin t-\cos t)\,dt\right]$$

$$= \sin x + x^2(\cos^2 x + \cos x\sin x - \sin^2 x + \sin x\cos x)$$

$$- \cos x\int t^2(\cos t+\sin t)\,dt + \sin x\int t^2(\sin t-\cos t)\,dt$$

$$= \sin x + x^2 - \cos x\left[t^2\sin t - \int 2t\sin t\,dt\right] + \sin x\left[t^2\cos t - \int 2t\cos t\,dt\right]$$

$$= \sin x + x^2 - t^2\cos x\sin t + 2\cos x\int t\sin t\,dt + t^2\sin x\cos t - 2\sin x\int t\cos t\,dt$$

$$= \sin x + x^2 - t^2\cos(x-t) + 2\int t\sin(x-t)\,dt$$

$$= \sin x + x^2 - t^2\cos(x-t) + 2\left[t\int\sin(x-t)\,dt - \int\left\{\frac{d}{dt}(t)\int\sin(x-t)\,dt\right\}dt\right]$$

$$= \sin x + x^2 - t^2\cos(x-t) + 2\left[t\cos(x-t) - \int 1\cdot\cos(x-t)\,dt\right]$$

$$= \sin x + x^2 - t^2\cos(x-t) + 2t\cos(x-t) - 2\sin(x-t)$$

$$= \sin x + x^2 - x^2\cos 0 + 2x\cos 0 - 2\sin x = x^2 = y(x).$$

Thus $y(x) = x^2$ is a solution of the integral equation (1).

3.3. INTEGRAL EQUATIONS OF THE CONVOLUTION TYPE

$$y(x) = f(x) + \int_0^x K(x-t)\,y(t)\,dt$$

is an integral equation of convolution type and can be written as

$$y(x) = f(x) + K(x) * y(x) \qquad \text{[Roop. 2481]}$$

It is a special integral equation of importance in applications.

Taking Laplace transform of both sides, assuming that $\mathcal{L}\{y(t)\} = \bar{y}(s)$ and $\mathcal{L}\{K(t)\} = \bar{K}(s)$ both exist, and using convolution theorem

$$\bar{y}(s) = \bar{f}(s) + \bar{K}(s)\cdot\bar{y}(s) \quad \text{or} \quad \bar{y}(s) = \frac{\bar{f}(s)}{1-\bar{K}(s)}$$

Now taking the inverse transform of both sides, we get the required solution.

Example 28.10. Solve the following equation

$$y(x) = 2x^2 + \int_0^x y(t)\sin(x-t)\,dt$$

Solution. Given integral equation can be written as

$$y(x) = 2x^2 + y(x) * \sin(x-1) \qquad \ldots(1)$$

Taking Laplace transform of both sides and using the convolution theorem (p. 740), we get

$$\bar{y} = \frac{4}{s^3} + \bar{y}\cdot\frac{1}{s^2+1} \quad \text{or} \quad \bar{y} = \frac{4(s^2+1)}{s^5} = 4\left(\frac{1}{s^3} + \frac{1}{s^5}\right)$$

On inversion, we get $\qquad y = 4\left(\frac{x^2}{2!} + \frac{x^4}{4!}\right) = 2x^2 + \frac{x^4}{6}$

which is the required solution of (1).

Obs. The above solution can also be verified by direct substitution in the given integral equation.

Example 28.11. Solve $y(x) = x + 2\int_0^x \cos(x-t)\,y(t)\,dt$

Solution. The given equation can be written as

$$y(x) = x + 2\cos(x) * y(x)$$

Taking Laplace transform of both sides and using convolution theorem, we get

$$\bar{y}(s) = \frac{1}{s^2} + 2\frac{s}{s^2+1}\,\bar{y}(s) \quad \text{or} \quad \bar{y}\left(1 - \frac{2s}{s^2+1}\right) = \frac{1}{s^2}$$

or $$\bar{y} = \frac{s^2+1}{s^2(s-1)^2} = \frac{2}{s} + \frac{1}{s^2} - \frac{2}{s-1} + \frac{2}{(s-1)^2}$$

On inversion, we obtain $\qquad y = 2 + x - 2e^x + 2xe^x$

Hence $y = x + 2 + 2(x-1)e^x$ is the desired solution.

Example 28.12. Solve the integral equation $\int_0^x y(t)\,y(x-t)\,dt = x^3$, $x > 0$.

Solution. The given integral equation can be written as

$$y(x) * y(x) = x^3 \qquad \ldots(1)$$

Taking Laplace transform of both sides and using the convolution theorem, we get

$$[\bar{y}(s)]^2 = \frac{3!}{s^4} \quad \text{or} \quad \bar{y} = \frac{6}{\sqrt{s^4}} = \frac{6}{s^2}$$

On inversion and noting that $\mathcal{L}^{-1}\left\{\frac{\Gamma(n+1)}{s^{n+1}}\right\} = x^n$, $n > -1$, we get

$$y = 6\,\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = 6\,\mathcal{L}^{-1}\{\bar{f}(s)\}$$

Thus $y = 6f(x)$. Obviously $y = -6f(x)$ is also a solution of (1).

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33.10 ABEL'S INTEGRAL EQUATION

This is integral equation $\int_0^t \frac{y(t)}{\sqrt{t-\tau}}\,d\tau = g(t)$

where $g(t)$ is given and α is a constant $(0 < \alpha < 1)$, is known as Abel's integral equation. This is an important integral equation of the convolution type. An application of this equation is in finding the shape of a smooth wire lying in a vertical plane such that a bead placed anywhere on the wire slides to the lowest point in the same time. This is the well known *tautochrone problem* and the shape of the wire is a cycloid.

Example 33.13. Solve the Abel's integral equation $\int_0^t \frac{y(\tau)}{\sqrt{t-\tau}}\,d\tau = t + 2t^2 - t^3$

Solution. The given equation can be written as

$$y(t) * t^{-1/2} = t + 2t^2 - t^3$$

Taking the Laplace transform of both sides and using convolution theorem, we get

$$\mathcal{L}\{y(t)\}\,\mathcal{L}\{t^{-1/2}\} = \mathcal{L}\{t + 2t^2 - t^3\}$$

or $$Y(s)\frac{\Gamma(1/2)}{s^{1/2}} = \frac{1}{s^2} + \frac{2}{s^3} - \frac{3!}{s^4} \quad \text{or} \quad Y = \frac{1}{\Gamma(1/2)}\left[\frac{1}{s^{5/2}} + 2\cdot\frac{1}{s^{7/2}} - 3!\frac{1}{s^{9/2}}\right]$$

On inversion, and noting that $\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{\Gamma(n+1)}$, we have

$$y = \frac{1}{\Gamma(1/2)}\frac{t^{-1/2}}{\Gamma(1/2)} + 2\cdot\frac{t^{3/2}}{\Gamma(3/2)} - 3!\frac{t^{5/2}}{\Gamma(5/2)}$$

$$= \frac{1}{\pi}\left[t^{-1/2} + 4t^{3/2} - \frac{8}{3}t^{5/2}\right]$$

Hence $$y = \frac{1}{3\pi\sqrt{t}}(3 + 12t - 8t^2)$$ is the desired solution.

33.11 INTEGRO-DIFFERENTIAL EQUATIONS

An integral equation in which various derivatives of the unknown function $y(t)$ are also present, is called an *integro-differential equation*. An example of such an equation is:

$$y'(t) + y(t) = \cos t + \int_0^t \sin(t-\tau)\,y(\tau)\,d\tau$$

The solution of integro-differential equation, subject to given initial conditions can also be found by Laplace transforms as illustrated below :

Example 33.14. Solve $\frac{dy}{dt} + 3y + 2\int_0^t y\,dt = t$, given $y(0) = 1$.

Solution. Given equation can be written as $y'(t) + 3y(t) + 2\int_0^t y\,dt = t$

Taking Laplace transforms of both sides, we get

$$\mathcal{L}\{y'(t)\} + 3\mathcal{L}\{y(t)\} + 2\mathcal{L}\left\{\int_0^t y(t)\,dt\right\} = \mathcal{L}\{t\}$$

or $$[sy(s) - y(0)] + 3y(s) + 2\frac{1}{s}y(s) = \frac{1}{s^2}$$ (Using (33.8))

or $$s^2Y + 3sY + 2Y = 1 + \frac{1}{s^2}$$ (∵ $y(0) = 1$)

or $$Y = \frac{1+s^2}{s(s^2+2s+2)} = \frac{1+s^2}{s(s+1)(s+1)} = \frac{1}{2s} - \frac{2}{s+1} + \frac{5}{2(s+2)}$$

On inversion, we obtain $y = \frac{1}{2}L^{-1}\left(\frac{1}{s}\right) - 2L^{-1}\left(\frac{1}{s+1}\right) + \frac{5}{2}L^{-1}\left(\frac{1}{s+2}\right)$

Hence $y = \frac{1}{2} - 2e^{-x} + \frac{5}{2}e^{-2x}$ is the required solution.

Example 18.13. Solve $\frac{dy}{dx} = 2\int_0^x \cos 2(x-t)\, y(t)\, dt + 2$, given $y(0) = 1$.

Solution. Given equation can be written as

$$y'(x) = 2\cos 2x * y(x) + 2$$

Taking Laplace transforms of both sides, we get

$$L[y'(x)] = 2L[\cos 2x] \cdot y(x) + \frac{2}{s}$$

$$sY(s) - y(0) = \frac{2sY(s)}{s^2+4} + \frac{2}{s} \quad \text{or} \quad Y = \frac{s^2+2(s^2+4)}{s^2(s^2+2)} \qquad [\because y(0) = 1]$$

$$= \frac{A}{s} + \frac{B}{s^2} + \frac{Cs}{s^2+2} + \frac{D}{s^2+2}$$

On inversion, we obtain $y = 1 + 2x - 2\cos\sqrt{2}x - \sqrt{2}\sin\sqrt{2}x$ which is the required solution.

Note. The above integral differential equation can be converted into the form of initial value problem by differentiating it twice and using $y(0) = 1$.

$$y(x) = 2x + 1 + 2\int_0^x (x-t)\cos 2(x-t)\, y(t)\, dt$$

Exercise 18.3

1. Show that $y(x) = \frac{1}{2}x^2$ is a solution of the integral equation $\int_0^x y(t)\,dt = x$.

2. Show that $y(x) = 1$ is a solution of the Fredholm integral equation

$$y(x) = \frac{3}{2} + \int_0^{\pi/2} \sin(x-t)\, y(t)\, dt$$

3. Show that $y(x) = \frac{1}{\sqrt{x}}$ is a solution of the integral equation $\int_0^x \frac{y(t)}{\sqrt{x-t}}\,dt = 1$.

4. Show that $y(x) = e^x + x^2 + 2x$ is a solution of the Fredholm integral equation

$$y(x) = 1 + \int_0^x e^{x-t}\, y(t)\, dt + \frac{x^3}{3}$$

Solve each of the following integral equations :

5. $y(x) = x + \frac{1}{6}\int_0^x (x-t)^3 y(t)\,dt$.

6. $y(x) = e^x + \int_0^x y(t)\sin(x-t)\,dt$.

7. $\int_0^x y(t)\,dt = ay(x) + b$.

8. $\int_0^x y(t)\,dt = x + \int_0^x (x-t)\,y(t)\,dt$.

9. Find a solution of the integral equation $y(x) = \frac{1}{2}\sin x + \int_0^x y(t)\,y(x-t)\,dt$.

Solve the following integral equations :

10. $\frac{dy}{dx} + 4y + 3\int_0^x y(t)\,dt = e^{-x}$, $y(0) = 0$.

11. $\frac{dy}{dx} + 2y = \int_0^x y(t)\,dt$, $y(0) = 1$. (Mumbai, 2005)

[illegible]

[illegible] INTEGRAL EQUATIONS AND ITS GENERALIZED NUMERICAL

A function $\phi(x)$ of the integral equation is used to be represented on the basis of a finite number of terms, each of which is the product of a function of x alone and a function of [illegible]

$$\phi(x) = \sum_{i=1}^{n} c_i \psi_i(x)$$

[illegible]

[illegible] SOLUTION OF FREDHOLM EQUATIONS WITH SYMMETRIC KERNELS

Consider the integral equation

$$\phi(x) = \lambda \int_a^b K(x,t)\,\phi(t)\,dt + f(x) \tag{1}$$

where $K(x,t) = \sum_{i=1}^{n} a_i(x)\,b_i(t)$ [illegible] $K(x,t) = K(t,x)$.

Substituting (2) in (1), we get

$$\phi(x) = \lambda \int_a^b \left[\sum_{i=1}^{n} a_i(x)\,b_i(t)\right]\phi(t)\,dt + f(x)$$

$$= \lambda \sum_{i=1}^{n} a_i(x) \int_a^b b_i(t)\,\phi(t)\,dt + f(x) \tag{3}$$

If we write $\int_a^b b_i(t)\,\phi(t)\,dt = k_i$, a constant, and [illegible] for different values of i, we get from (3)

$$\phi(x) = \lambda \sum_{i=1}^{n} k_i\,a_i(x) + f(x) \tag{4}$$

If we put $x = x_1, x_2, \ldots, x_n$ in (4), we get n equations in $k_1, k_2, \ldots, k_n$ as follows.

$$\int_a^b \phi(x_1)\,b_1(x_1)\,dx = \lambda \sum_{i=1}^{n} k_i \int_a^b a_i(x_1)\,b_1(x_1)\,dx + \int_a^b f(x_1)\,b_1(x_1)\,dx$$

Writing $\int_a^b \phi(x_1)\,b_1(x_1)\,dx = k_1$ and $\int_a^b a_i(x_1)\,b_1(x_1)\,dx = \beta_{i1}$, the above equation becomes

$$k_1 = \lambda \sum_{i=1}^{n} k_i \beta_{i1} + \gamma_1$$

$$= \lambda k_1 \beta_{11} + \lambda k_2 \beta_{21} + \cdots + \lambda k_n \beta_{n1} + \gamma_1$$

Taking $x = 1, 2, \ldots, n$, we get the following equations in $k_1, k_2, \ldots, k_n$:

$$\begin{aligned}
(1 - \lambda\beta_{11})k_1 - \lambda\beta_{21}k_2 - \cdots - \lambda\beta_{n1}k_n &= \gamma_1 \\
-\lambda\beta_{12}k_1 + (1 - \lambda\beta_{22})k_2 - \cdots - \lambda\beta_{n2}k_n &= \gamma_2 \\
\cdots\cdots\cdots

Now the following cases arise :

I. When $F(x) \neq 0$, (1) is said to be a homogeneous integral equation and all four cases arise.

(i) If $\Delta \neq 0$, the only solution of (1) is the trivial solution $C_1 = C_2 = \ldots = C_n = 0$. Then $y(x) = 0$ is the trivial solution of (1).

(ii) If $\lambda = 0$, at least one of the C's can be assigned any value and the remaining C's can be found accordingly. Then (1) gives infinitely many solutions. The values of λ for which $\Delta = 0$ are known as the eigen values. Any non trivial solution of the homogeneous integral equation for a certain value of λ is called the corresponding eigen function. The solutions corresponding to eigen values of λ can be expressed as arbitrary multiples of eigen functions.

II. When $F(x) \neq 0$. Let us assume that $\int_a^b A_{rs}(x) F(x) dx = 0$ so that $\beta_r = 0$

(i) If $\Delta \neq 0$, the only solution of (1) is the trivial solution $C_1 = C_2 = \ldots = C_n = 0$.
Then $y = F(x)$ is the desired solution of (1).

(ii) If $\Delta = 0$, at least one of the C's can be given any value and the remaining C's can be found.
Then (1) gives infinitely many solutions.

III. When at least one of the $\beta_r \neq 0$

(i) If $\Delta \neq 0$, then equations (2) give a unique solution of the constants C.
Hence there is a unique solution of (1).

(ii) If $\Delta = 0$, then equations (2) will be inconsistent.

$\therefore$ Either there is no solution or infinitely many solutions of (1) exist.

Example 11.15. Find (i) eigen values and eigen functions of the following homogeneous integral equation with degenerate kernel :

$$y(x) = \lambda \int_0^{2\pi} (2\sin x + 4x^2 \cos t) y(t)\, dt \quad (ii)\ y(x) = \frac{x}{e^x - 1} \int_0^1 e^{xy} y(y)\, dy$$

Solution. (i) Given equation may be written as

$$y(x) = \lambda \left[2\sin x \int_0^{2\pi} y(t)\, dt - 4x^2 \int_0^{2\pi} y(t) \cos t\, dt \right]$$

or
$$y(x) = \lambda [2\sin x\, C_1 - 4x^2 C_2] \qquad \ldots (1)$$

where $C_1 = \int_0^{2\pi} y(t)\, dt, C_2 = \int_0^{2\pi} y(t) \cos t\, dt$

Substituting $y(x)$ in C_1, C_2 we get

$$C_1 = \int_0^{2\pi} \lambda (2\sin x C_1 - 4x^2 \cos x C_2)\, dx$$

$$C_2 = \int_0^{2\pi} \lambda (2\sin x C_1 - 4x^2 \cos^2 t C_2)\, dx$$

or
$$C_1 \left[1 - 2\lambda \int_0^{2\pi} \sin x\, dx \right] - C_2 \left[4\lambda \int_0^{2\pi} x^2 \cos x \right] = 0$$

$$C_1 \left[-2\lambda \int_0^{2\pi} \cos x\, dx \right] + C_2 \left[1 + 4\lambda \int_0^{2\pi} x^2 \cos^2 x \right] = 0$$

or
$$C_1 \left[1 - \frac{8\pi^2}{3} \right] \lambda C_2 = 0 \text{ i.e. } -2C_1 + C_2 (1 + 4\pi^2) = 0 \qquad \ldots (2)$$

$\therefore$ The determinant of eigen values will be

$$\begin{vmatrix} 1 - 8\pi^2 \lambda & \lambda \\ -\lambda & 1 + 4\pi^2 \lambda \end{vmatrix} = 0 \text{ or } 4\pi^2 \lambda^2 + 3\lambda^2 = 0$$

$\therefore$ The eigen values are $\lambda_1 = -3, \lambda_2 = -1$

For $\lambda_1 = \lambda_2 = -3$, the equations (2) reduce to

$$2C_1 - 2C_2 = 0 \text{ i.e. } C_1 = C_2$$

$\therefore$ From (1) $y(x) = -6C_2(x - 2\pi^2) = -x + 2\pi^2$ if $C_2 = -2\pi$

Eliminate the constant from the corresponding to $x_1 = x_2 = \cdots = 0$, to

$$y(x) = 1 - \frac{x^2}{2!}$$

(ii) Volterra integral equation may be written as

$$y(x) = \int_0^x \int_0^t y(t')\,dt'\,dt = \int_0^x (x-t)\,y(t)\,dt \qquad \text{...(ii)}$$

where $t' = \int_0^t y(t')\,dt'$

Differentiating the value of $y(x)$ from (i) to (ii), we get

$$y(x) = \frac{d}{dx}\left[\int_0^x \int_0^t y(t')\,dt'\,dt - \int_0^x (x-t)\,y(t)\,dt\right]$$

or $$y(x) = \left[\int_0^x y(t')\,dt' + \int_0^x y(t)\,dt - \int_0^x y(t)\,dt\right] + 0 \quad \text{i.e.} \quad y(x) = y(x)$$

Hence the $y(x) = 1$

Hence it shows that the given integral equation has exist and real solutions.

Hence it does not evaluate any eigen values or eigen functions.

Example: Solve the given equation by the following form:

$$y(x) = \sin x + \lambda \int_0^{\pi} \sin(x+t)\,y(t)\,dt$$

Solution: We bring the given equation in the following form:

$$y(x) = \sin x + \lambda \left[\sin x \int_0^{\pi} \cos t\,y(t)\,dt + \cos x \int_0^{\pi} \sin t\,y(t)\,dt\right]$$

or $$y(x) = \sin x + \lambda A_1 \sin x + \lambda A_2 \cos x \qquad \text{...(i)}$$

where $A_1 = \int_0^{\pi} \cos t\,y(t)\,dt, \quad A_2 = \int_0^{\pi} \sin t\,y(t)\,dt$

Putting the value of $y(x)$ in A_1 and A_2, we get

$$A_1 = \int_0^{\pi} \cos t\,[\sin t + \lambda A_1 \sin t + \lambda A_2 \cos t]\,dt$$

$$A_2 = \int_0^{\pi} \sin t\,[\sin t + \lambda A_1 \sin t + \lambda A_2 \cos t]\,dt$$

or $$\begin{aligned} A_1 &= \int_0^{\pi} \sin t \cos t\,dt + \lambda A_1 \int_0^{\pi} \sin^2 t\,dt + \lambda A_2 \int_0^{\pi} \cos^2 t\,dt \\ A_2 &= \int_0^{\pi} \sin^2 t\,dt + \lambda A_1 \int_0^{\pi} \sin t \cos t\,dt + \lambda A_2 \int_0^{\pi} \sin t \cos t\,dt \end{aligned} \qquad \text{...(ii)}$$

On evaluating each of the integrals in (ii), we get

$$A_1 = \frac{\pi}{2}\lambda A_1 + 0 + \frac{\pi}{2}\lambda A_2 \quad \text{or} \quad A_2 = 0 + \frac{\pi}{2}\lambda A_1 + 0 \qquad \text{...(iii)}$$

The determinant of the equations (iii) is given by

$$\begin{vmatrix} 1 & -\frac{\pi}{2}\lambda \\ -\frac{\pi}{2}\lambda & 1 \end{vmatrix} = 1 + \frac{\pi^2}{4}\lambda^2 \neq 0$$

Thus the equations (iii) have a unique solution

$$A_1 = \frac{2\pi\lambda}{4 + \pi^2\lambda^2}, \quad A_2 = \frac{4\pi^2\lambda^2}{4 + \pi^2\lambda^2}$$

Substituting these values of C_1 and C_2 in (1), we obtain the required solution

$$y(x) = \cos x + \frac{\lambda}{4 + \lambda^2 \pi^2}(2\pi \cos x + \lambda \pi^2 \sin x)$$

or

$$y(x) = \frac{\lambda}{4 + \lambda^2 \pi^2}(2\pi \cos x + \lambda \pi^2 \sin x)$$

PROBLEMS 11.4

Determine the characteristic values and characteristic functions for the following homogeneous integral equations (integral equations with zero right-hand side):

1. $y(x) = \lambda \int_0^{2\pi} \sin(x+t)\, y(t)\, dt$
2. $y(x) = \lambda \int_0^{\pi} (\sin^2 x \cos^2 t + \sin 2x \sin 2t)\, y(t)\, dt$
3. $y(x) = \lambda \int_0^{2\pi} \cos^2 x \, y(t)\, dt$
4. $y(x) = \lambda \int_0^{\pi} \sin x \cos t\, y(t)\, dt$ (Madras B.Sc., 2000)
5. $y(x) = \lambda \int_0^{\pi} \sin x \cos t\, y(t)\, dt$
6. $y(x) = \lambda \int_0^{2\pi} \cos(x+t)\, y(t)\, dt$ (Madras B.Sc., 2000)

Solve the following integral equations:

7. $y(x) = x + \lambda \int_0^{1} (1 - 3xt)\, y(t)\, dt$
8. $y(x) = x + \lambda \int_0^{1} (1 + x + t)\, y(t)\, dt$
9. $y(x) = x + \lambda \int_0^{1} (1 + xt + x^2 t^2)\, y(t)\, dt$
10. $y(x) = 1 + \lambda \int_0^{1} (x + t)\, y(t)\, dt$
11. $y(x) = \sin x + \lambda \int_0^{\pi/2} \sin x \cos t\, y(t)\, dt$
12. $y(x) = x + \lambda \int_0^{1} (x + t + 1)\, y(t)\, dt$
13. Obtain the solution $y(x) = 1 + \lambda \int_0^{x} e^{x-t}\, y(t)\, dt$ in the form $y(x) = 1 + \frac{\lambda x}{1!} + \frac{\lambda^2 x^2}{2!} + \frac{\lambda^3 x^3}{3!} + \cdots$

What happens when $\lambda = 1$?

14. For the integral equation

$$y(x) = F(x) + \lambda \int_0^{2\pi} \sin(x+t)\, y(t)\, dt$$

find the eigen values of λ and the corresponding eigen functions.

15. Obtain the characteristic numbers of the equation

$$y(x) = F(x) + \lambda \int_0^{\pi} \sin(x+t)\, y(t)\, dt$$

where $F(x) = x$ and $F(x) = \sin x$ under the assumption that $\lambda \neq \pm 2/\pi$.

11.11 SOLUTION OF FREDHOLM INTEGRAL EQUATION BY THE METHOD OF SUCCESSIVE APPROXIMATIONS

Consider the Fredholm equation $y(x) = F(x) + \lambda \int_a^b K(x,t)\, y(t)\, dt$ ...(1)

where $F(x)$ is continuous in $a \le x \le b$ and $K(x,t)$ is finite and continuous in the rectangle $a \le x \le b$ and $a \le t \le b$.

Replacing y under the integral sign by an initial approximation $y^{(0)}$, we get the first approximation

$$y^{(1)}(x) = F(x) + \lambda \int_a^b K(x,t)\, y^{(0)}(t)\, dt$$ ...(2)

Replacing y under the integral sign in (1) by $y^{(1)}$, we get the next approximation

$$y^{(2)}(x) = F(x) + \lambda \int_a^b K(x,t)\, y^{(1)}(t)\, dt$$ ...(3)

Proceeding in this manner, we get the general formula for successive approximations as

$$y^{(n)}(x) = f(x) + \lambda \int_a^b K(x,t)\, y^{(n-1)}(t)\, dt \qquad \ldots(6)$$

We now, obtain the condition for the convergence of the sequence $y^{(n)}(x)$.

Replacing x in (2) by a dummy variable t, (2) becomes

$$y^{(n)}(t) = f(t) + \lambda \int_a^b K(t, t_1)\, y^{(n-1)}(t_1)\, dt_1$$

Then (2) takes the form

$$y^{(n)}(x) = f(x) + \lambda \int_a^b K(x,t) \left\{ f(t) + \lambda \int_a^b K(t, t_1)\, y^{(n-2)}(t_1)\, dt_1 \right\} dt$$

$$= f(x) + \lambda \int_a^b K(x,t)\, f(t)\, dt + \lambda^2 \int_a^b K(x,t) \left\{ \int_a^b K(t, t_1)\, y^{(n-2)}(t_1)\, dt_1 \right\} dt \qquad \ldots(8)$$

Writing $K^{(2)}(x,t) = \int_a^b K(x,z)\, K(z,t)\, dz$, the equations (1), (2) and (8) become

$$y(x) = f(x) + \lambda K^{(1)} y(x)$$

$$y^{(1)}(x) = f(x) + \lambda K^{(1)} y^{(0)}(x)$$

$$y^{(2)}(x) = f(x) + \lambda K^{(1)} f(x) + \lambda^2 K^{(2)} y^{(0)}(x)$$

Similarly $y^{(3)}(x) = f(x) + \lambda K^{(1)} f(x) + \lambda^2 K^{(2)} f(x) + \lambda^3 K^{(3)} y^{(0)}(x)$

In general $y^{(n)}(x) = f(x) + \lambda K^{(1)} f(x) + \lambda^2 K^{(2)} f(x) + \ldots + \lambda^n K^{(n)} y^{(0)}(x)$

As $n \to \infty$, we get

$$y(x) = \lim_{n \to \infty} y^{(n)}(x) = f(x) + \lim_{n \to \infty} \left(\lambda K^{(1)} f(x) + \lambda^2 K^{(2)} f(x) + \ldots \right) \qquad \ldots(9)$$

Now $f(x)$ and $K(x,t)$ being continuous for all values of x and t in (a,b), $f(x) \le M$ and $|K(x,t)| \le n$ where M, n are their respective maximum values in (a,b).

$$\therefore \quad |K^{(1)} f(x)| = \left| \int_a^b K(x,t)\, f(t)\, dt \right|$$

$$\le nM \left| \int_a^b dt \right| \le nM(b-a)$$

Similarly $K^{(2)} f(x) \le n^2 M (b-a)^2$

Then $|\lambda^2 K^{(2)} f(x)| < |\lambda^2|\, n^2 M (b-a)^2$

$$\le M\, |\lambda n (b-a)|^2$$

i.e. $\sum_1^\infty \lambda^n K^{(n)} f(x) \le M \sum_1^\infty |\lambda n (b-a)|^n$

Now the series on R.H.S. being a geometric series, converges for $|\lambda| \, n(b-a) < 1$.

Thus by comparison test, $\sum_1^\infty \lambda^n K^{(n)} f(x)$ is also convergent for $|\lambda| \, n(b-a) < 1$.

i.e., for $|\lambda| < \dfrac{1}{n(b-a)}$ $\qquad \ldots(7)$

Hence the given integral equation (1) will have a continuous solution when the condition (7) is satisfied.

Note 1. To calculate the successive terms in the series (9) we evidently have to evaluate $K^{(n)}(x,t) = f(x)$.

Obs. 2. Volterra integral equations can also be solved by following exactly similar procedure as above (See Example 36.10).

Example 36.10. Solve, by using the method of successive approximations, the integral equation

$$y(x) = f(x) + \lambda \int_0^x e^{x-t}\, y(t)\, dt \qquad \ldots(1)$$

Solution. Taking the initial approximation $y^{(0)}(x) = 1$ and substituting it in the R.H.S. of (i), we get

$$y^{(1)}(x) = 1 + \lambda \int_0^x (x - 2t)\,dt = 1 + \lambda x \left| \frac{t^2}{2} \right|_0^x = 1 + \frac{\lambda x^2}{2}$$

Substituting $y^{(1)}(x)$ in the R.H.S. of (i), we have

$$y^{(2)}(x) = 1 + \lambda \int_0^x x\left(1 + \lambda \frac{t^2}{2}\right) dt = 1 + \lambda x \int_0^x \left(1 + \frac{\lambda t^2}{2}\right) dt$$

$$= 1 + \lambda x \left| t + \frac{\lambda}{2} \frac{t^3}{3} \right|_0^x = 1 + \lambda x^2 + \frac{\lambda^2 x^4}{6}$$

Substituting $y^{(2)}(x)$ in the R.H.S. of (i), we get

$$y^{(3)}(x) = 1 + \lambda \int_0^x x\left(1 + \frac{\lambda x^2}{2} + \frac{\lambda^2 t^4}{6}\right) dt = 1 + \lambda x \left| t + \frac{\lambda}{2} \frac{t^3}{3} + \frac{\lambda^2}{6} \frac{t^5}{5} \right|_0^x$$

$$= 1 + \frac{\lambda x^2}{2} + \frac{\lambda^2 x^4}{8} + \frac{\lambda^3 x^6}{48} = 1 + \frac{\lambda x^2}{2}\left[1 + \frac{\lambda}{4} + \frac{\lambda^2}{24} + \cdots\right]$$

Hence the solution of (i) is

$$y(x) = 1 + \frac{\lambda x^2}{2}\left[1 + \frac{\lambda}{4} + \left(\frac{\lambda}{4}\right)^2 + \left(\frac{\lambda}{4}\right)^3 + \cdots\right]$$

As the number of terms tends to infinity, the exact solution is

$$y(x) = 1 + \frac{\lambda x^2}{2}\left[1 + \frac{\lambda}{4} + \left(\frac{\lambda}{4}\right)^2 + \left(\frac{\lambda}{4}\right)^3 + \cdots\right]$$

$$= 1 + \frac{\lambda x^2}{2} \frac{1}{1 - \lambda/4} \qquad \text{[Summing up the G.P. which converges for } |\lambda/4| < 1]$$

or

$$y(x) = 1 + \frac{2\lambda x^2}{2(4 - \lambda)} \quad \text{only if } |\lambda| < 4.$$

Example 24.10. Using the method of successive approximations, solve the following integral equation

$$y(x) = 1 + x + \int_0^x (x - t)\,y(t)\,dt$$

Solution. Taking the initial approximation $y^{(0)}(x) = 1 + x$ and substituting it in the R.H.S. of (i), we get

$$y^{(1)}(x) = 1 + x + \int_0^x (x - t)(1 + t)\,dt$$

$$= 1 + x + x\left|t + \frac{t^2}{2}\right|_0^x - \left|\frac{t^2}{2} + \frac{t^3}{3}\right|_0^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}$$

Substituting $y^{(1)}(x)$ in the R.H.S. of (i), we obtain

$$y^{(2)}(x) = 1 + x + x \int_0^x (x - t)\left(1 + t + \frac{t^2}{2} + \frac{t^3}{6}\right) dt$$

$$= 1 + x + x\left|t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{24}\right|_0^x - \left|\frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{12} + \frac{t^5}{120}\right|_0^x$$

$$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120}$$

Proceeding in this manner, we get

$$y^{(n)}(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots + \frac{x^n}{n!}$$

And so, the exact solution of (2) is

$$y(x) = \frac{1}{e^{-x}}\left[1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}\right] = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + x^n$$

PROBLEMS 55 to 58

Apply the method of successive approximations to solve the following Fredholm integral equations (55 to 58):

55. $y(x) = 1 - \lambda\int_0^1 xt\,y(t)\,dt$

56. $y(x) = \cos x + \lambda\int_0^{\pi} \sin x \cos t\,y(t)\,dt$

57. $y(x) = 1 + \lambda\int_0^1 (1 - 2xt)\,y(t)\,dt$

Using Calculus of Variations, solve the following Volterra integral equations (58 to 60):

58. $y(x) = 1 + \int_0^x y(t)\,dt$ (Madras U., 2005)

59. $y(x) = 1 + x - \int_0^x y(t)\,dt$

60. $y(x) = x - \int_0^x (x - t)\,y(t)\,dt$

61. $y(x) = 2x^2 + x^3 - \int_0^x 3x\,y(t)\,dt$

62. Choosing the initial approximation $y^{(0)}(x) = 0$, find the solution of the integral equation

$$y(x) = \int_0^{\pi/2} \sin(x + t)\,y(t)\,dt + \frac{x^2}{2}$$ (Show that $y^{(1)}(x) = \frac{x^2}{2} - \frac{\pi}{2}x$)

63. Starting with the initial approximation $y^{(0)}(x) = 1$, find the solution of the integral equation

$$y(x) = 1 + \int_0^x (x + t)\,y(t)\,dt$$

(Show that) $y^{(3)}(x) = 1 + \frac{3x^2}{2} + \frac{7x^3}{6} + \frac{5x^4}{24}$ (Madras U., 2002)

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CHAPTER 37

Discrete Mathematics

I. Set Theory: 1. Sets 2. Set operations 3. Laws of set theory 4. Principle of inclusion, duality.

II. Algebra of Logic: 5. Introduction, Propositions 6. Arithmetic Compound statements 7. Logical operations 8. Statements generated by n-set 9. Tautology, Contradiction 10. Equivalence 11. Duality law, Tautology and Contra. 12. Arguments 13. Predicates 14. Quantifiers 15. Normal forms 16. Inference theory.

III. Boolean Algebra: 17. Introduction, Boolean function 18. Duality 19. Boolean identities 20. Minimal Boolean functions 21. Disjunctive normal form 22. Conjunctive normal form 23. Switching circuits, Simplification of circuits.

IV. Fuzzy Sets: 24. Fuzzy logic, Fuzzy set 25. Fuzzy set operations 26. Truth values 27. Algebraic operations, Properties of fuzzy sets 28. Generation of rules for fuzzy problems 29. Classification of fuzzy propositions 30. Applications of fuzzy sets.

I. SET THEORY

37.1 SETS

(I) The concept and the language of sets play a very important role in expressing mathematical ideas accurately and precisely. It was Cantor* who first introduced and developed the notion of sets in mathematical investigations. It is, therefore, essential for a student of engineering to grasp the basic ideas of Set Theory.

Def. A collection of objects defined by some property is called a **set**. The objects belonging to a set are called its **elements or members**.

Example: (a) A set may be the set of positive integers less than 20. (b) The set of pages in a book. (c) The set of all the students in a college.

A set is denoted by a single capital letter e.g. A, B, ..., S, X, Y and the elements of a set are generally denoted by small letters e.g. $a, b, c, \ldots, x, y, z$.

If a is an element of a set S, we write $a \in S$ and read as "a belongs to S". When a is not an element of S, we write $a \notin S$.

Example: If X be a set of odd integers, $3 \in X$ but $4 \notin X$, $11 \in X$.

(2) Representation of a set

(a) **Tabular form of a set.** In this, the elements are enclosed in curly brackets after separating them by commas, e.g., the set of positive integers less than 10 is written as $A = \{1, 2, 3, \ldots, 9\}$ and the set of prime numbers between 4 and 24 is $T = \{5, 7, 11, 23\}$.

(b) **Symbolic form of a set.** In this, the set is written as $S = \{x : P(x)\}$ where x is a typical element of the set and $P(x)$ is the property satisfied by this element. In symbolic form, the above two sets are

$$S = \{x : x \text{ is a positive even integer} < 10\}$$

$$T = \{y : y \text{ is a prime number between } 4 \text{ and } 24\}$$

*The great German mathematician Georg Cantor (1845–1918), the creator of Set theory.

(3) Empty set or null set. *A set which does not contain any element is called an empty set or the null set and is denoted by the symbol* ϕ.

(4) Finite and infinite sets. *A set is said to be finite if it has a finite number of elements. Otherwise it is said to be infinite.*

The number of distinct elements in a finite set A is called the cardinality and is denoted by $|A|$.

For instance, the set of days in a week is finite, the set of points in a line is an infinite set.

(5) Subset. *If every element of a set A is also an element of set B, then A is called a subset of B and this relationship is denoted by* $A \subset B$ *or* $B \supset A$ *which is read as A is contained in B.*

Another definition : If A and B are two sets such that

$$x \in A \Rightarrow x \in B,$$

then A is called a subset of B.

The notation $\Rightarrow$ stands for the word 'implies'.

For instance, the set V of vowels is a subset of the set A of the English alphabets and we write $V \subset A$.

(6) Power set. *For a set A, collection of all subsets of A is called the power set of A and is denoted by P(A).*

If $A = \{1, 2, 3\}$ *then P(A) consists of* 2^3 *i.e. 8 elements* ϕ, $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{2, 3\}$, $\{3, 1\}$ *and* $\{1, 2, 3\}$.

In general, if A has n elements, then P(A) has 2^n *elements.*

(7) Equality of sets. *Two sets A and B are said to be equal if the elements of both are the same i.e., if each element of A is also an element of B and vice versa, and we write* $A = B$.

In other words, if A and B are subsets such that

$$A \subset B \quad \text{and} \quad B \subset A \Leftrightarrow A = B.$$

Here $\Leftrightarrow$ *stands for 'implies and is implied by' or 'if and only if'.*

For instance, $\{2, 3, 5\} = \{3, 5, 3, 2\} = \{5, 3, 2, 2\}$, *since the change in the order of elements or the repetitions does not disturb the essential set of those sets with the same elements 2, 3, 5.*

(8) Proper and improper subsets. *When the set B contains all the elements of A and some others, A is said to be a proper subset of B and is denoted by* $A \subset B$.

i.e., $A \subset B$ *and* $A \neq B$ *then* $A \subset B$.

If $A \subset B$ *and every element of B is also an element of A i.e.,* $B \subset A$, *then A is said to be an* **improper subset** *of B i.e.,* $A = B$.

For instance, the set of positive even integers and the set of positive even integers are both proper subsets of the set of natural numbers.

(9) Universal set is that which is a superset of all the sets under consideration in a discussion or in a universe. It is generally denoted by 'U'.

For instance the set of all letters of English alphabet is a universal set of the set of the letters $\{a, b, c, d, \ldots, x, y, z\}$ *etc.*

Example 1.7.1. *If* A, B, C *are sets such that* $A \subset B$ *and* $B \subset C$, *then show that* $A \subset C$.

Solution. Let x be any element of A.

Now $A \subset B$ i.e., all the elements of A belong to B.

$$\therefore \quad x \in A \Rightarrow x \in B \qquad \ldots(i)$$

Again as $B \subset C$ i.e., all elements of B belong to C.

$$x \in B \Rightarrow x \in C \qquad \ldots(ii)$$

$\therefore$ It follows from (i) and (ii) that $x \in A \Rightarrow x \in C$

i.e., $A \subset C$.

Example 1.7.2. *Which of the following sets are equal?*

$A_1 = \{x : x \text{ is a letter in the word 'FOLLOW'}\}$

$A_2 = \{x : x \text{ is a letter in the word 'WOLF'}\}$

$A_3 = \{x : x \text{ is a letter in the word 'FLOW'}\}$

Solution. Here $A_1 = \{L, O, W, F\} = \{F, L, O, W\}$

$A_2 = \{x : x - 1)^2 = 0\} = \{1\}$, $A_3 = \{F, L, O, W\}$

$A_4 = \{x : (x - 1)(x - 2)(x - 3) = 0\} = \{1, 2, 3\}$

From (i) we find that A_1, A_3 are equal.

1.3.1 SET OPERATIONS

(1) Union of two sets A and B is the set of all those elements which belong to A or to B or to both. It is denoted by $A \cup B$ and is read as 'A union B' and is represented by the shaded portion in Fig. 37.1.

Symbolically $A \cup B = \{x : x \in A \text{ or } x \in B\}$.

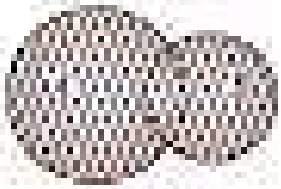


Fig. 37.1

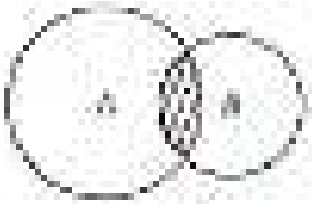


Fig. 37.2

(2) Intersection of two sets A and B is the set of elements which are common to both A and B. It is denoted by $A \cap B$ and is read as 'A intersection B' and is represented by the shaded portion in Fig. 37.2.

Symbolically $A \cap B = \{x : x \in A \text{ and } x \in B\}$

Such diagrams as Figs. 37.1 and 37.2 which exhibit the various relations between the sets are known as **Venn diagrams.**

(3) Disjoint sets. *If the sets A and B have no common elements, they are called* **disjoint sets.** Their intersection is an empty set.

For instance, if A be the set of boys in a college and B be the set of girls in the same college, then A and B are disjoint sets i.e. $A \cap B = \phi$.

(4) Complement of a set. *If $B \subset A$, the set of elements of A which are not in B is called* **the complement** of B in A and is denoted by B' or $A - B$. It is also known as the difference $A - B$ of sets A and B. Thus

$$B' \text{ in } A = \{x : x \in A \text{ and } x \notin B\}$$

which is shown shaded in Fig. 37.3 (i).

If U be a universal set, then the set $U - A$ is called the complement of A and is denoted by A', which is shown shaded in Fig. 37.3 (ii).

For instance, if $U = \{1, 2, 3, 4, 5, \ldots\}$ and $A = \{1, 3, 5, 7, \ldots\}$, then $A' = \{2, 4, 6, 8 \ldots\}$.

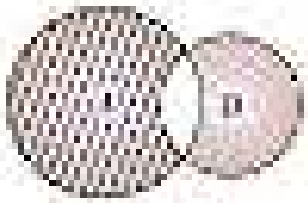


Fig. 37.3 (i)

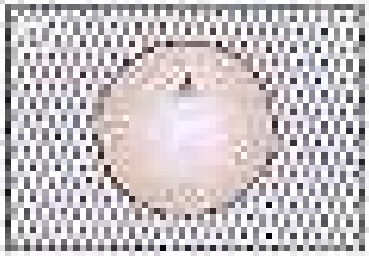


Fig. 37.3 (ii)

(5) Cartesian product *of two sets A and B denoted by $A \times B$ is defined to be the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$ i.e.,*

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

For instance if $A = \{1, 2, 3\}$, $B = \{1, 2, 3\}$, then $A \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$, $B \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$ $\therefore$ $A \times B \neq B \times A$.

Example 37.1. *If $A = \{2, 3, 4, 5\}$, $B = \{3, 4, 5, 6\}$, $C = \{1, 2, 3, 4\}$, then find $A \cap (B \cup C)$ and $(A \cap B) \cup (A \cap C)$.*

Solution. Here $B \cup C = \{1, 2, 3, 4, 5, 6\}$

$\therefore$ $A \cap (B \cup C) = \{2, 3, 4, 5\}$ ...(i)

Again $A \cap B = \{3, 4, 5\}$,

$A \cap C = \{2, 3, 4\}$

$\therefore$ $(A \cap B) \cup (A \cap C) = \{2, 3, 4, 5\}$ ...(ii)

Hence from (2) and (3), we get

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Example 37.4. With the help of Venn diagram, show that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

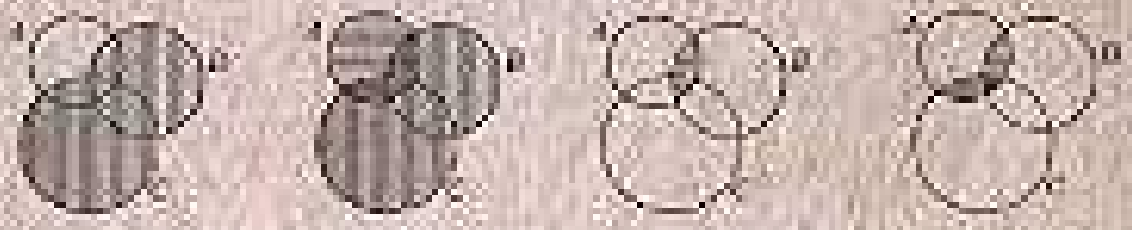


Fig. 37.4

Solution. First we draw vertical lines in the whole area of B and C so as to represent $B \cup C$ [Fig. 37.4 (i)].

Now draw horizontal lines in the whole area A. Then the double hatched area in [Fig. 37.4 (ii)] represents area common to A and $B \cup C$ i.e., $A \cap (B \cup C)$.

Again we draw horizontal lines in the area common to A and B so as to represent $A \cap B$ [Fig. 37.4 (iii)].

Now draw vertical lines in the area common to A and C so as to represent $A \cap C$. Then the whole hatched area in [Fig. 37.4 (iv)] represents $(A \cap B) \cup (A \cap C)$.

Hence we observe that the double hatched area in Fig. 37.4 (ii) is equal to the total hatched area in Fig. 37.4 (iv).

Example 37.5. Prove that (i) $A - (B \cap C) = (A - B) \cup (A - C)$

(ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$. (Gujarat, 2003)

Solution. (i) Let x be an arbitrary element of the set $A - (B \cap C)$, then

$$\begin{aligned} x \in A - (B \cap C) &\Rightarrow x \in A \text{ and } x \notin (B \cap C) \qquad [\because x \notin (A - B) \Rightarrow x \in A \text{ and } x \notin B] \\ &\Rightarrow x \in A \text{ and } (x \notin B \text{ or } x \notin C) \\ &\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in A \text{ and } x \notin C) \\ &\Rightarrow x \in (A - B) \text{ or } x \in (A - C) \\ &\Rightarrow x \in (A - B) \cup (A - C) \end{aligned}$$

$$\therefore \quad A - (B \cap C) \subset (A - B) \cup (A - C) \qquad \ldots(1)$$

Again if x be an arbitrary element of the set $(A - B) \cup (A - C)$, then

$$\begin{aligned} x \in (A - B) \cup (A - C) &\Rightarrow x \in (A - B) \text{ or } x \in (A - C) \\ &\Rightarrow (x \in A \text{ and } x \notin B) \text{ or } (x \in A \text{ and } x \notin C) \\ &\Rightarrow x \in A \text{ and } (x \notin B \text{ or } x \notin C) \\ &\Rightarrow x \in A \text{ and } x \notin B \cap C \\ &\Rightarrow x \in A - (B \cap C) \end{aligned}$$

$$\therefore \quad (A - B) \cup (A - C) \subset A - (B \cap C) \qquad \ldots(2)$$

From (1) and (2), we get $A - (B \cap C) = (A - B) \cup (A - C)$.

(ii) $(x, y) \in A \times (B \cap C)$

$$\begin{aligned} &\Rightarrow x \in A \text{ and } y \in (B \cap C) \\ &\Rightarrow (x \in A \text{ and } y \in B) \text{ and } (x \in A \text{ and } y \in C) \\ &\Rightarrow (x, y) \in (A \times B) \text{ and } (x, y) \in (A \times C) \\ &\Rightarrow (x, y) \in (A \times B) \cap (A \times C) \end{aligned}$$

Hence $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

3.3.3 LAWS OF SET THEORY

1. **Commutative Law**

$$A \cup B = B \cup A \; ; \; A \cap B = B \cap A$$

2. **Associative Law**

$$A \cup (B \cup C) = (A \cup B) \cup C$$
$$A \cap (B \cap C) = (A \cap B) \cap C$$

3. **Distributive Law**

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

4. **Complement Law**

$$A \cup A' = U \; ; \; A \cap A' = \phi$$

5. **Identity Law**

$$A \cup \phi = A = \phi \cup A$$
$$A \cap U = A = U \cap A$$

6. **Absorption Law**

$$A \cup (A \cap B) = A \; ; \; A \cap (A \cup B) = A$$

7. **De Morgan's Law**

$$(A \cup B)' = A' \cap B' \; ; \; (A \cap B)' = A' \cup B'$$

8. **Involution Law**

$$(A')' = A$$

3.3.4 PRINCIPLE OF INCLUSION

(I) *If A and B be sets, then* $|A \cup B| = |A| + |B| - |A \cap B|$.

Proof: The number of common elements in A and B is $|A \cap B|$. Each of these elements is counted twice in $|A| + |B|$, once in $|A|$ and once in $|B|$. This should be adjusted by subtracting the term $|A \cap B|$ from $|A| + |B|$.

Hence $|A \cup B| = |A| + |B| - |A \cap B|$.

Obs: Using the above theorem, we can extend the above result for three sets A, B, C as follows:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

(UPTU, 2002)

For $|A \cup B \cup C| = |(A \cup B) \cup C|$

$$= |A \cup B| + |C| - |(A \cup B) \cap C|$$
$$= |A| + |B| + |A \cap B| + |C| - |(A \cap C) \cup (B \cap C)|$$
$$= |A| + |B| + |C| - |A \cap B| - |(A \cap C) + |B \cap C| - |A \cap B \cap C|$$

which follows the result.

(II) *Duality:* If A be any identity involving sets and operations $\cup, \cap$, complement, or inequalities $\subset$ and $\supset$, and if a set A^* is obtained by replacing $\cup$ by $\cap$, $\cap$ by $\cup$, ϕ by U and U by ϕ in S, then the statement A^* is true and is called the dual of the statement A.

For instance, the dual of $A \cup (B \cap U) = A$ is $A \cap (B \cup \phi) = A$.

Example 3.7.1: *In a survey conducted on 200 persons, it was found that 150 drink tea and 75 drink coffee and 50 take both tea and coffee. How many drink atleast one beverage and how many drink neither?*

Solution: Let A be the set of tea drinkers and B be set of coffee drinkers. Then

$$|A \cup B| = |A| + |B| - |A \cap B| = 150 + 75 - 50 = 200$$

Hence 200 persons drink at least one beverage and $250 - 200 = 50$ persons drink neither tea nor coffee.

Example 27.5. *How many integers between 1 and 450 are divisible by 3 but not by 5 ?*

Solution. Number of integers between 1 and 450 which are divisible by $3 = \left[\frac{450}{3}\right] = 150$

Number of integers between 1 and 450 which are divisible by 3 and $5 = \left[\frac{450}{3 \times 5}\right] = 30$

Hence the number of integers between 1 and 450 divisible by 3 but not by $5 = 150 - 30 = 120$.

Example 27.6. *How many integers are between 1 and 200 which are divisible by any one of the integers 2, 5 and 8 ?*

Solution. Let A_1, A_2, A_3 denote the set of integers between 1 and 200 which are divisible by 2, 5, 8 respectively.

$$|A_1| = \left[\frac{200}{2}\right] = 100, |A_2| = \left[\frac{200}{5}\right] = 40, |A_3| = \left[\frac{200}{8}\right] = 25$$

$$|A_1 \cap A_2| = \left[\frac{200}{2 \times 5}\right] = 20, |A_1 \cap A_3| = \left[\frac{200}{2 \times 8}\right] = 12$$

$$|A_2 \cap A_3| = \left[\frac{200}{5 \times 8}\right] = 5, |A_1 \cap A_2 \cap A_3| = \left[\frac{200}{2 \times 5 \times 8}\right] = 2$$

Hence $|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_2 \cap A_3| - |A_1 \cap A_3| + |A_1 \cap A_2 \cap A_3|$

$$= 100 + 40 + 25 - 20 - 5 - 12 + 2 = 130$$

EXERCISE 27.1

1. Show that the following sets are equal :
   $A = \{2, 3\}, B = \{1, 2, 3, 4\} - \{1, 4\}, C = \{x : x^2 - 5x + 6 = 0\}$
2. Which of the following statements are true ? Give reasons to support your answer :
   (i) $0 \in \{a, b, c\}$ (ii) $a \in \{a, b, c\}$ (iii) $a \subseteq \{a, b, c\}$
   (iv) $\{a, b\} \subseteq \{a, b, c\}$ (v) $\{a, b\} \in \{a, b, c\}$ (vi) $a \in \{a, \{b, c\}\}$
3. Prove that :
   (i) $A - B$ is a subset of A (ii) $B - A = B \cap A'$ (Madras 2004)
   (iii) If $A \subseteq B$, $B \subseteq C$, $C \subseteq A$ then $A = B = C$.
4. If $A = \{a, b, c, d, e\}$, $B = \{a, c, e, g\}$ and $C = \{a, e, f, g\}$, prove that
   (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (ii) $(A \cap B) \cup (B \cap C) \cup (A \cap C) = (A \cup B) \cap (B \cup C) \cap (A \cup C)$
5. If $A = A \cap B$, then prove that $B = A \cup B$.
6. Prove that $A \cup B = B$ iff $A \subseteq B$.
7. With the help of the Venn diagram, prove that
   (i) $(A \cup B)' = A' \cap B'$ and (ii) $(A \cap B)' = A' \cup B'$ (Madras 2002)
   (iii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. (V.T.U., 2001)
8. If $B \subseteq A$, prove that
   (i) $B \cap C \subseteq A \cap C$ (ii) $B \cup C \subseteq A \cup C$.
9. (i) If $A \cup A = A \cap A$, show that $A = \phi$.
   (ii) If $A \cup B = A \cup C$ and $A \cap B = A \cap C$, show that $B = C$.
10. (i) Prove that (a) $A - B = A \cap A'$ (b) $A \cap B = B \cap A$ (V.T.U. 2001; Madras 2002)
    (ii) $A - (B \cup C) = (A - B) \cap (A - C)$.
11. Show that for any two sets A and B
    (i) $A = B$ iff $A \cap B' = \phi$ (ii) $A \subseteq B$ iff $B' \subseteq A'$
    (iii) $A \cup B = (A \cap B') \cup (B \cap A') \cup (A \cap B)$.
12. If A, B, C be such that $A \cup B = A \cup C$, $A \cap B = A \cap C$, show that $B = C$.
13. Show that $A \cup (B \cap C)' = (A \cup B') \cap (A \cup C')$

14. Prove that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$. (Madurai, 2007)
15. If S is any set and $P(S)$ is its power set and A and B belong to $P(S)$, prove that $A \cap (A \cup B) = A \cup (A \cap B)$.
16. If A and B are finite sets then prove that $A \cup B$ and $A \cap B$ are finite sets and $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. (Madurai, 2008)
17. In a survey conducted on 200 people, it was found that 140 are smokers while 60 are alcoholics and 40 are both smokers and alcoholics. Find how many are neither smokers nor alcoholics.
18. How many integers between 1 and 450 are divisible by 3 but not by 5?
19. How many integers are between 1 and 250 which are divisible by any of the integers 2, 3, and 5?
20. Out of a class of 120 students 36 have taken History, 48 have taken Geography, 52 have taken Economics and 12 have taken Geography and History, 16 have taken History and Economics, 20 have taken Geography and Economics and 8 have taken all the three subjects. How many of the students have not taken any of these three subjects? Use a Venn diagram.

II. ALGEBRA OF LOGIC

2.1. INTRODUCTION

(1) Logic is concerned with all types of reasoning such as valid statements, mathematical proofs, valid conclusions etc. Logical reasoning is used to prove theorems, to verify the correctness of computer programs and to draw conclusions from experiments. Later on, we shall observe that the algebra of sets and logic is analogous to the algebra of switching circuits which is similar to 'Boolean Algebra'.

(2) Propositions and Statements. A proposition is a declarative sentence which is either true (T) or false (F). Sometimes we use T and F respectively for 1 and 0. The truth or falsity of a proposition is defined to its truth value.

All the declarative sentences to which it is possible to assign one and only one of the two possible truth values are called statements.

Example 2.1.1. Which of the following are statements: (a) Agra is in India. (b) $2 + 3 = 5$. (c) Where do you live? (d) Do you speak Hindi?

Solution. (a) and (b) are statements that happen to be true and (c) is false.

(c) and (d) are questions as they are not declarative.

(3) Compound statements. The statements which is composed of sub-statements and logical connectives is called a compound statement.

e.g. "It is raining and it is cold" is a compound statement as it is composed of two sub-statements "It is raining" and "It is cold".

(4) Truth table. The truth value of a compound statement is completely determined by the truth value of its substatements. A convenient way to represent a compound statement is by means of the truth table showing the values of a compound statement are specified for all possible choices of the values of the sub statements.

We shall use the numbers 0 and 1 to denote the false and true statements. Also we use letters $p, q, \ldots, s$ to represent a proposition or logical variable.

2.2. LOGICAL OPERATORS

(1) Conjunction. If p and q are two statements then their conjunction p and q is written as $p \wedge q$, is defined by the truth table 1.

Table 1

p	q	$p \wedge q$
1	1	1
2	0	0
3	1	0
4	0	0

Table 2

p	q	$p \vee q$
1	1	1
2	0	1
3	1	1
4	0	0

For example, the conjunction of p: It is raining and q: I am sick is $p \wedge q$: It is raining and I am sick.

(3) Disjunction. If p and q are any two statements, then their *disjunction* $p \vee q$ written as $p + q$ is defined by the truth table 1.2.

For example, the disjunction of p: today is Friday and q: It is raining today is $p \vee q$: Today is Friday or it is raining today, or 2 is an odd integer or 2 is an even integer.

(4) Negation. If p is any given statement and the negation "not p" written as $\sim p$ or $\overline{p}$ or Tp is defined by the following truth table:

p	$\sim p$
0	1
1	0

For example, the negation of the following statement

(a) p: $2 + 3 = 5$ is $\sim p$: $2 + 3 \neq 5$

(b) q: It is hot is $\sim q$: It is not hot.

Example 1.2.10. If p: Ice is hot and q: Sun is rising, then write each of the following statements in symbolic form:

(a) $\sim p$ (b) $\sim p \wedge q$ (c) $\sim (p \wedge q)$

Solution. (a) It is not the case that ice is hot.

(b) Ice is not hot and sun is rising.

(c) It is not the case that ice is hot and sun is rising.

(4) Conditional operator. The conditional statement "if p then q" written as $p \to q$ is defined by the truth table 3:

Table 4

p	q	$p \to q$
1	1	1
1	0	0
0	1	1
0	0	1

Table 5

p	q	$p \leftrightarrow q$
1	1	1
1	0	0
0	1	0
0	0	1

Obs: The interpretation of conditional statement $p \to q$ is the statement $\sim p \vee q$.

(5) Biconditional operator. If p and q be two statements, then the statement p if and only if q denoted by $p \leftrightarrow q$ and abbreviated as p iff q is called a *biconditional statement*. The truth table for biconditional statement is table 5.

Example 1.2.11. Construct the truth table for

(a) $(p \to q) \wedge q$ (b) $(p \vee q) \to p$

(c) $(p \to q) \wedge (q \to p)$ (d) $p \to (q \vee r) \wedge (p \to q)$

Solution. (a) The truth table is

p	q	$p \to q$	$(p \to q) \wedge q$
1	1	1	1
1	0	0	0
0	1	1	1
0	0	1	0

(d) The truth table is

p	q	$q \vee r$	$p \to q$	$p \to (q \vee r) \wedge (p \to q)$
1	1	1	1	1
1	0	1	0	0
0	1	1	1	1
0	0	0	1	0

(c) The truth table is

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \wedge (p \wedge q)$
1	1	1	1	1
1	0	1	0	0
0	1	1	0	0
0	0	0	0	0

(d) The truth table in this case is

p	q	$p \vee q$	$\sim q$	$p \wedge \sim q$	$(p \vee q) \wedge \sim (p \wedge \sim q)$	$(p \wedge \sim q) \vee [(p \vee q) \wedge \sim (p \wedge \sim q)]$
1	1	1	0	0	1	1
1	0	1	1	1	0	1
0	1	1	0	0	1	1
0	0	0	1	0	0	0

STATEMENTS GENERATED BY A SET

(1) *If S be a set of statements, then any valid combination of statements in S with conjunction, disjunction or negation is a statement generated by S.*

A statement generated by a set S need not include each element of S in its expression.

For example, if p, q, r are statements and then

(i) $(p \vee q) \wedge r$ (ii) $\sim q \wedge r$ (iii) $p \vee (q \wedge \sim r)$

are all statements generated by S. Their truth tables are :

p	q	r	$p \vee q$	$(p \vee q) \wedge r$ (i)	$\sim q$	$p \vee (q \wedge \sim r)$ (iii)	$\sim q \wedge r$ (ii)
1	1	1	1	1	0	1	0
1	1	0	1	0	0	1	0
1	0	1	1	1	1	1	1
0	1	1	1	1	0	0	0
1	0	0	1	0	1	1	0
0	1	0	0	0	0	0	0
0	0	1	0	0	1	1	1
0	0	0	0	0	1	0	0

(2) Tautology is an expression involving logical variables which is true for all cases in its truth table. It is also called a logical truth.

(3) Contradiction is an expression involving logical variables which is false for all cases in its truth table. Obviously, the negation of a contradiction is a tautology.

In other words, a statement formula which is a tautology is identically true, while a formula which is a contradiction is identically false.

Obs. The conjunction of two tautologies is also a tautology.

Example 37.13. Show that $(p \wedge q) \rightarrow p$ is a tautology. Or $p \rightarrow (p \vee q)$ is a tautology.

Solution. (i) The truth table is

p	$\sim p$	$p \vee \sim p$
1	0	1
0	1	1

Hence $p \vee \sim p$ is a tautology.

(b) The truth table is

p	q	$p \to q$	$\neg q$	$\neg p \vee q$	$(p \to q) \leftrightarrow (\neg p \vee q)$
T	T	T	F	T	T
T	F	F	T	F	T
F	T	T	F	T	T
F	F	T	T	T	T

Hence $(p \to q) \leftrightarrow (\neg p \vee q)$ is true.

SECTION 1.13 EQUIVALENCE

(1) If p and q are statements governed by the set of connectives S, then p and q are equivalent if $p \leftrightarrow q$ is a tautology which is denoted by $p \Leftrightarrow q$.

If $p \to q$ is a tautology, then we say that p implies q and write it as $p \Rightarrow q$.

Note: All tautologies are equivalent to each other and all contradictions are equivalent to each other.

(2) Equivalent formulas. Some basic equivalent formulas are given below which can be proved by using truth tables.

1. $p \vee p \Leftrightarrow p$	$p \wedge p \Leftrightarrow p$	*Idempotent laws*
2. $p \vee q \Leftrightarrow q \vee p$	$p \wedge q \Leftrightarrow q \wedge p$	*Commutative laws*
3. $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$	$(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$	*Associative laws*
4. $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$	*Distributive laws*
5. $p \vee \neg p \Leftrightarrow T$	$p \wedge \neg p \Leftrightarrow F$	*Negation laws*
6. $p \vee F \Leftrightarrow p$	$p \wedge T \Leftrightarrow p$	*Identity laws*
7. $p \vee T \Leftrightarrow T$	$p \wedge F \Leftrightarrow F$	*Null laws*
8. $p \vee (p \wedge q) \Leftrightarrow p$	$p \wedge (p \vee q) \Leftrightarrow p$	*Absorption laws*
9. $\neg (p \wedge q) \Leftrightarrow \neg p \vee \neg q$	$\neg (p \vee q) \Leftrightarrow \neg p \wedge \neg q$	*De Morgan's laws*
10. $p \Leftrightarrow p \vee q$	$q \Leftrightarrow p \vee q$	*Disjunctive addition*
11. $p \wedge q \Rightarrow q$	$p \wedge q \Rightarrow p$	
12. $(p \vee q) \wedge \neg p \Rightarrow q$	$(p \vee q) \wedge \neg q \Rightarrow p$	
13. $(p \to q) \wedge (q \to r) \Rightarrow (p \to r)$		*Chain rule*
14. $p \to q \Leftrightarrow \neg p \vee q$		*Conditional equivalence*
15. $p \leftrightarrow q \Leftrightarrow (p \to q) \wedge (q \to p)$		*Biconditional equivalence*

Example 1.13.1 Show that
(a) $(p \to q) \wedge (p \to r) \Leftrightarrow p \to (q \wedge r)$
(b) $\vdash (p \to (q \vee r)) \Leftrightarrow ((p \wedge \neg q) \to r)$

Solution. (a) By conditional equivalence $p \to r \Leftrightarrow \neg p \vee r$

Replacing $q \to r$ by $\neg q \vee r$, we get $p \to (\neg q \vee r)$ which is equivalent to $\neg p \vee (\neg q \vee r)$ by the same rule.

Thus	$\neg p \vee (\neg q \vee r)$	$\Leftrightarrow (\neg p \vee \neg q) \vee r$	By (3)
		$\Leftrightarrow \neg (p \wedge q) \vee r$	By (9)
		$\Leftrightarrow (p \wedge q) \to r$	By (14)
(b)	$\vdash p \to (q \vee r) \Leftrightarrow (p \wedge \neg q) \to r$		
		$\Leftrightarrow \vdash p \to (\neg q \vee r) \Leftrightarrow (p \wedge \neg q) \to r$	By (14)
		$\Leftrightarrow \vdash p \to \neg q \vee r \Leftrightarrow \neg (p \wedge \neg q) \vee r$	By (3)
		$\Leftrightarrow \vdash \neg p \vee (\neg q \vee r) \Leftrightarrow \neg (p \wedge \neg q) \vee r$	By (9)
		$\Leftrightarrow \vdash \neg (p \wedge \neg q) \vee (p \wedge \neg q) \vee r$	By (10)
		$\Leftrightarrow \vdash T \vee r$	By (5)
		$\Leftrightarrow \vdash T$	By (6)

1.23 DUALITY LAW

(1) Two formulae A and A* are said to be duals of each other if either can be obtained from the other by replacing ∧ by ∨ and ∨ by ∧.

If the formula A contains special variables T or F, then its dual A* is obtained by replacing T by F and F by T.

e.g. (i) Dual of $(p \vee q) \wedge r$ is $(p \wedge q) \vee r$.

(ii) Dual of $p \vee q \vee (r \wedge s)$ is $p \wedge q \wedge (r \vee s)$.

(2) Tautology implications. A statement A is said to tautologically imply a statement B if and only if $A \rightarrow B$ is a tautology which is read as "A implies B".

The implications listed below have important applications which can be proved by truth tables :

1. $p \wedge q \Rightarrow p$ $\quad$ $p \Rightarrow p \vee q$
2. $\neg p \wedge (p \vee q) \Rightarrow q$ $\quad$ $q \Rightarrow p \vee q$
3. $\neg(p \rightarrow q) \Rightarrow p$ $\quad$ $\neg(p \rightarrow q) \Rightarrow \neg q$
4. $p \wedge (p \rightarrow q) \Rightarrow q$ $\quad$ $\neg p \wedge (p \vee q) \Rightarrow q$
5. $(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r$

1.24 ARGUMENTS

(1) An argument is an assertion that a given set of propositions $p_1, p_2, \ldots, p_n$ (called premises) yields another proposition q (called conclusion). The argument is symbolically written as "$p_1, p_2, \ldots, p_n \vdash q$".

An argument $p_1, p_2, \ldots, p_n \vdash q$ is true provided q is true whenever all the premises $p_1, p_2, \ldots, p_n$ are true. An argument which is true is said to be "valid argument". Otherwise it is called a fallacy.

Example 1.24.1. Show that

(i) the argument $p \leftrightarrow q, q \Rightarrow p$ is valid.

(ii) the argument $p \rightarrow q, q \Rightarrow p$ is a fallacy.

Solution. (i) Let us first prepare the truth table as follows :

p	q	$p \leftrightarrow q$
1	1	1
1	0	0
0	1	0
0	0	1

Since $p \leftrightarrow q$ is true in cases (rows) 1 and 4, and q is true in cases 1 and 3, therefore $p \leftrightarrow q$ and q both are true in row 1 only when p is also true. This shows that the given argument is valid.

(ii) Let us first prepare the truth table below :

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1

This table shows that $p \rightarrow q$ and q both are true in case 3 only while the conclusion p is false. Hence the given argument is a fallacy.

(3) Theorem. *The argument $p_1, p_2, \ldots, p_n \vdash q$ is valid if and only if the proposition $(p_1 \wedge p_2 \wedge p_3 \wedge \ldots \wedge p_n) \rightarrow q$ is a tautology.*

The propositions $p_1, p_2, \ldots, p_n$ are also true exactly true if and only if the proposition $p_1 \wedge p_2 \wedge \ldots \wedge p_n$ is true i.e., if the proposition $(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \rightarrow q$ is a tautology.

Obs: The validity of an argument depends upon the particular form of the argument and not on the truth values of the statements appearing in the argument.

Example 3.7.18. Examine validity of the following argument :

S_1 : If 5 is less than 3, then 5 is not a prime number.

S_2 : 5 is not less than 3.

S_3 : 5 is a prime number.

Solution. Let '5 is less than 3' be p and '5 is a prime number' be q. Then the given argument is of the form $p \to \sim q, \sim p \vdash q$.

But in the last line of the truth table, the premises $p \to \sim q$ and $\sim p$ are true but the conclusion q is false. Therefore the argument is invalid.

p	q	$\sim q$	$p \to \sim q$	$\sim p$
T	T	F	F	F
T	F	T	T	F
F	T	F	T	T
F	F	T	T	T

EXERCISE 3.2

1. If p, q have the truth values T, F respectively, find the truth values of the following compound statements :
   (a) $\sim p \to q$ (b) $p \to \sim q$ (c) $\sim p \to \sim q$ (d) $\sim p \leftrightarrow q$
2. Construct the following compound statements :
   (a) If it does not rain or if the streets are not icy, then I will go to the movies.
   (b) You can go to the beach only if you work hard all week. (U.P.T.U. 2004)
3. Prepare truth tables for the following compound statements : (a) $(p \to q) \to \sim q$ (b) $(p \to q) \to (q \to p)$
4. Write down the truth table of
   (a) $p \to q$ (Madras, 1997) (b) $(p \to q) \to q$ (Madras, 1997)
5. Verify that the following are tautologies :
   (a) $p \to (q \to (p \wedge q))$ (b) $(p \to q) \to (\sim q \to \sim p)$ (c) $(p \to q) \to (\sim p \to q)$ (d) $(p \to q) \to (\sim p \to \sim q)$ (e) $(p \to q) \to (\sim q \to p)$ (f) $(p \to q) \to (\sim q \to \sim p)$ (g) $(p \to q) \to (\sim p \to q)$ (h) $(p \to q) \to (\sim p \to \sim q)$
6. Show that $(p \to q) \to r \Leftrightarrow p \to (q \to r)$ is a tautology. (Madras, 1994 S; Madras, 1995)
7. Show the validity of the following arguments :
   (a) If x is prime and $x > 2$, then x is odd.
   x is prime and $x > 2$.
   Therefore x is odd.
   (b) What are the truth values of these propositions ?
   (c) Which of the three propositions is the negation of the other ?
8. Given the propositions over the natural numbers :
   p : If $x < 4$, y : If $x > 10$ and z : If x is a divisor of 18, what are the truth values of
   (a) y (b) $y \to z$
   (c) x (d) $(x \wedge y) \to z$ (Madras, 1999)
9. Show that $(p \to q) \wedge p \to q \Leftrightarrow T$ (Madras, 2000)
10. $(\sim p \to q) \wedge (q \to p) \Leftrightarrow p \wedge q$ (Madras, 2001)
11. Construct the truth table for $(p \to q) \to q) \to p$. (Osmania, May 2001)
    (b) $(p \to q) \wedge (p \to r) \Leftrightarrow p \to (q \wedge r)$ (Madras, 1999)
12. Prove that the following statement is a contradiction :
    $\sim (p \to q) \wedge (p \vee q) \wedge (q \vee p) \wedge (\sim p \vee \sim q)$
13. If p, q, r are three statements, then prove that
    (a) $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$ (b) $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
    (c) $\sim (p \wedge q) \Leftrightarrow \sim p \vee \sim q$
14. Define tautology, contradiction, conditional and biconditional with examples.

15. Show that $(p \wedge q) \vee (\neg p \wedge \neg q)$ is logically equivalent to $(p \vee \neg q) \wedge (\neg p \vee q)$. (IGNOU, 2007)
16. Write the dual of $(p \vee q) \wedge (r \vee s)$.
17. Show that $p \rightarrow q$ is logically equivalent to $\neg q \rightarrow \neg p$. (Burdwan, 2006; Kharagpur, 2007)
18. Let $P(n)$ be "$n^2 - n$ is an odd prime". Prove that $P(n)$ is a tautology.
19. Prove that $P \rightarrow (Q \vee R) \Rightarrow (P \rightarrow Q) \vee (P \rightarrow R)$ is a tautology. (Kharagpur, 2005)
20. Prove that $p \rightarrow (q \rightarrow r) \Rightarrow (p \rightarrow q) \rightarrow (p \rightarrow r)$. (Kharagpur, 2005)
21. Show that $R \rightarrow S$ follows logically from the premises $P \vee Q$, $Q \rightarrow R$, $P \rightarrow M$ and $\neg M \rightarrow S$.

3.3 PREDICATES

Statements involving variables such as $x > 7$ and $x = y + 7$ are neither true nor false as long as the values of the variables x, y are not specified. We now, discuss the ways that propositions can be created from such statements. The statement $x > 7$ has two parts. First part—"the variable x" is the subject of the statement; Second part—"is greater than 7" is the predicate which refers to the property that the subject of statement can have. If $P(x)$ denotes the statement "$x > 7$" then P is the predicate and x is the variable. The statement $P(x)$ is also known as the value of the propositional function P at x. The predicate are denoted by capital letters and the objects by the variable (denoted by small letters in brackets).

Thus predicates are simple statements which turn out to be propositions involving variables whose values are not explicitly specified.

In other words, predicate is a variable statement which becomes specific when particular values are assigned to the variables.

There are statements which involves more than one variable consider the statement "$x = y + 7$" which is denoted by $Q(x, y)$ where Q is the predicate and x, y are the variables. When values are assigned to the variables x, y, the statement $Q(x, y)$ has the truth values.

Similarly $R(x, y, z)$ denotes the statement of the type "$x + y = z$". When values are assigned to x, y, z this statement has a truth value.

For example, consider the statements (i) Rose is fair, (ii) Moon is fair.

Here is fair is the predicate while Rose and Moon are the objects. If we denote the predicate by F and the objects by r and s, then the above statements can be symbolically represented as (i) $F(r)$, (ii) $F(s)$.

Now consider the statement Rose is fairer than the Moon is pink.

Writing "the heavenly bodies" as $P(R)$, the given statement can be expressed as $F(r) \wedge P(R)$.

3.3.1 QUANTIFIERS

(1) If in a propositional function, when all the variables are assigned values, the resulting statement has a truth value. However, there is another method to create a proposition from a propositional function which is called quantification. It is of two types: Universal quantification and Existential quantification.

(2) Universal quantification. Many statements assert that a property is true for all values of a variable in a particular domain. This domain sometimes called the universe of discourse, and such statements can be expressed using universal quantification.

The universal quantification of $P(x)$ is the proposition "$P(x)$ is true for all values of x in the universe of discourse".

The universal quantification of $P(x)$ is denoted by $\forall x P(x)$. The symbol $\forall$ is called the universal quantifier.

Obs. When it is possible to list all the elements in the universe of discourse $x_1, x_2, \ldots, x_n$, then the universal quantification $\forall x P(x)$ is same as the conjunction $P(x_1) \wedge P(x_2) \wedge \ldots \wedge P(x_n)$ for this conjunction is true if and only if $P(x_1)$, $P(x_2)$, ..., $P(x_n)$ are all true.

Example 3.3.1. *What is the truth value of the quantification $\forall x P(x)$ where*
(a) $P(x)$ is the statement "$x + 2 > 3$" and the universe of discourse is the set of real numbers.
(b) $P(x)$ is the statement "$x^2 < 10$" and the universe of discourse consists of positive integers not exceeding 3?

Solution. *(a) For instance, $2(1) > 3$ is false. Therefore $P(x)$ is not true for all real numbers x.*
Thus $\forall x P(x)$ is false.

(b) The statement $\forall x P(x)$ is taken as the conjunction $P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5)$.

Here the universe of discourse is 1, 2, 3, 4, 5 and $P(5)$ is the statement '$5^2 < 25$' which is false. Hence $\forall x P(x)$ is also false.

(3) Existential quantification. Many statements assert that there is an element with a certain property. To express such statements we use existential quantification. In other words, we form a proposition which is true if and only if $P(x)$ is true for at least one value of x in the universe of discourse.

Then the *existential quantification of $P(x)$* is the proposition: 'There exists an element x in the universe of discourse such that $P(x)$ is true'.

The notation $\exists x P(x)$ is used for the existential quantification whereas $\exists$ is called the existential quantifier.

Note When it is possible to list all the elements in the universe, say $x_1, x_2, \ldots, x_n$, then the existential quantification $\exists x P(x)$ is same as the disjunction $P(x_1) \vee P(x_2) \vee \ldots \vee P(x_n)$.

Example 17.37. What is the truth value of the quantification $\exists x P(x)$, where

(a) $P(x)$ is the statement '$x + 2 = 5$' and universe of discourse is the set of real numbers.

(b) $P(x)$ is the statement '$x^2 > 10$' and the universe of discourse consists of positive integers not exceeding 3?

Solution. (a) Since '$x + 2 = 5$' is true, say : for $x = 6$, $5 + 6 = 11$.

$\therefore \quad \exists x P(x)$ is true.

(b) The statement $\exists x P(x)$ is same as the disjunction $P(1) \vee P(2) \vee P(3)$.

The universe of discourse is 1, 2, 3, 4, 5 and $P(x)$ is the statement '$x^2 > 10$', which is true.

Hence $\exists x P(x)$ is also true.

Example 17.38. *Find the truth value of each of the following statements :*

(a) $\exists x, x^2 = 1$ (b) $\forall x, \lfloor x \rfloor = x$ (c) $\exists x, x + 4 = x$.

Solution. (a) If $x_0 = 1$, then $x_0^2 = 1$, therefore the given statement is true.

(b) If $x_0 = 5$, then $\lfloor x_0 \rfloor \neq x_0$, therefore the given statement is false.

(c) As there is no solution to $x + 4 = x$, the given statement is false.

Example 17.39. *Given $P = \{2, 3, 5, 7\}$, state the truth values of each of the following statements :*

(a) $\forall x \in P, (x + 3 < 10)$ (b) $\forall x \in P, (x + 2 \leq 7)$

(c) $\exists x \in P, (x + 2 = 10)$ (d) $\exists x \in P, (x + 3 = 7)$

Solutions. (a) True, for each number in P satisfies $x + 3 < 10$.

(b) False, for if $x_0 = 5$, then $x_0 + 2$ is not ≤ 7.

(c) False, for no number in P is a solution to $x + 2 = 10$.

(d) True, for if $x_0 = 2$ then $x_0 + 5 = 7$ is a solution.

Example 17.40. *Negate each of the following statements :*

(a) $\forall x, x^2 = x$ (b) $\forall x, x + 4 = x$ (c) $\exists x, \lfloor x \rfloor = x$.

Solution. (a) $\sim \forall x, x^2 = x \equiv \exists x \sim (x^2 = x) \equiv \exists x, x^2 \neq x$

(b) $\sim \forall x, x + 4 = x \equiv \exists x \sim (x + 4 = x) \equiv \exists x, x + 4 \neq x$.

(c) $\sim \forall x, \lfloor x \rfloor = x \equiv \exists x \sim (\lfloor x \rfloor = x) \equiv \exists x, \lfloor x \rfloor \neq x$.

Example 17.41. *Symbolize using quantifiers :*

(a) *Every even number is divisible by 2.*

(b) *There is no prime number between 23 and 29.* (UPTU, MCA, 2004-05)

Solution. (a) $E(x)$: x is even number ; $D(x, y)$: x is divisible by y.

$\forall x) (E(x) \rightarrow D(x, 2)]$

(b) If p denotes the set of prime numbers, then $\exists x_0 \in p \; (23 < x < 29)$.

Example 37.32. Express in symbolic form: *All the world loves a lover.* (Hinton, 2001)

Solution. Let $p(x)$: x is a person;
$L(x)$: x is a lover;
and $Q(x, y)$: x loves y.

Then the required expression is

$$\forall x\,[p(x) \to (\forall y)(p(y) \wedge L(y) \to Q(x, y))].$$

Summary. (a) $\forall x\, Q(x)$ means that the predicate $Q(x)$ is true for all values of the universe of x.

(b) $\exists x Q(x)$ means that the predicate $Q(x)$ is satisfied if there is at least one value in the universe of x.

37.33 NORMAL FORMS

(1) For the given variables $p_1, p_2, \ldots, p_n$ we may form a statement $S(p_1, p_2, \ldots, p_n)$. The truth table for S will contain 2^n rows for all possible truth values of the n variables. The expression S may have the truth value T in all cases or may have the truth value F in all cases or have the truth value T for at least one combination of truth values assigned to the n variables. (Here S is said to be satisfiable). The problem of finding in a finite number of steps whether a given expression is a tautology or a contradiction or at least satisfiable is known as a decision problem. As the formation of a truth table requires enormous work, we prefer an alternate approach called *normal form*.

In this approach, we use the word 'sum' in place of disjunction and 'product' in place of conjunction.

A sum of the variables and their negations is called an *elementary sum*. Similarly a product of the variables and their negation is called an *elementary product*.

(2) Disjunctive normal form of a given formula is the formula which is equivalent to the given formula and which consists the sum of the elementary products. The disjunctive normal form of a given form is not unique. In fact, several disjunctive normal forms can be obtained for a given formula by applying the distributive laws in different ways.

A given formula is however, identically false if every elementary product appearing in its disjunctive normal form is identically false.

Example 37.33. Obtain the disjunctive normal forms of
(i) $p \leftrightarrow (q \wedge r)$ (ii) $(p \wedge q) \vee (\neg p \wedge r)$

Solution. (i) $p \leftrightarrow (q \wedge r) \Leftrightarrow (p \to (q \wedge r)) \wedge ((q \wedge r) \to p) \Leftrightarrow (\neg p \vee (q \wedge r)) \wedge (\neg (q \wedge r) \vee p)$
which is the desired disjunctive normal form.

$$\begin{aligned}
&\text{And} \quad (\neg p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee \neg r) \wedge (p \vee q \vee r) \Leftrightarrow (\neg p \vee (q \wedge r)) \wedge (\neg (q \wedge r) \vee p) \qquad [\because A \leftrightarrow B \Leftrightarrow (A \to B) \wedge (B \to A)] \\
&\Leftrightarrow (\neg p \vee (q \wedge r)) \wedge ((\neg q \vee \neg r) \vee p) \\
&\Leftrightarrow (\neg p \vee (q \wedge r)) \wedge (\neg (q \wedge r) \vee p) \\
&\Leftrightarrow (\neg p \wedge q \wedge r) \vee (p \wedge \neg (q \wedge r)) \\
&\Leftrightarrow (\neg p \wedge q \wedge r) \vee (p \wedge (\neg q \vee \neg r)) \\
&\Leftrightarrow (\neg p \wedge q \wedge r) \vee (p \wedge \neg q) \vee (p \wedge \neg r)
\end{aligned}$$

which is the desired disjunctive normal form.

(3) Conjunctive normal form of a given formula is that formula which is equivalent to the given formula and consists the product of elementary sums.

Example 37.34. Find a conjunctive normal form of $(\neg p \vee q) \leftrightarrow (p \wedge q)$.

Solution. $\neg (p \vee q) \leftrightarrow (p \wedge q) \Leftrightarrow (\neg (p \vee q) \to (p \wedge q)) \wedge ((p \wedge q) \to \neg (p \vee q))$

$$\begin{aligned}
&\Leftrightarrow (p \vee q) \vee (p \wedge q)) \wedge (\neg (p \wedge q) \vee \neg (p \vee q)) \qquad \text{[By material equivalence]} \\
&\Leftrightarrow (p \vee q \vee p) \wedge (p \vee q \vee q) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee \neg q) \\
&\Leftrightarrow (p \vee q) \wedge (p \vee q) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee \neg q)
\end{aligned}$$

which is the required conjunctive normal form.

(4) Principal disjunctive normal form. Consider a formula for the propositions p and q using conjunction $p \wedge q$, $p \wedge \neg q$, $\neg p \wedge q$, $\neg p \wedge \neg q$. These terms are called minterms or Boolean conjunction of p and q.

An equivalent formula for a given formula, consisting of disjunctions of minterms only is called the principal disjunctive normal form (pdf) or sum of products canonical form.

Procedure to obtain the principle disjunctive normal form : (i) Replace the conditional and biconditional by their equivalent formulae containing $\wedge, \vee, \neg$ only.

(ii) Using DeMorgan's laws, apply negation to the variables.

(iii) Apply the distribution laws.

(iv) Introduce the missing factors to obtain minterms in the disjunctions.

(v) Delete identical minterms appearing in the disjunctions.

Example 27.25: Obtain the pdnf for

(i) $p \vee q$ (ii) $\neg(p \wedge q)$ (iii) $p \rightarrow q$, $\neg p \vee q$

Solution. (i) $p \vee q \Leftrightarrow (p \wedge \neg p) \vee (\neg q \wedge q) \vee (p \wedge \neg p)$

$\Leftrightarrow (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q) \vee (p \wedge \neg p)$

$\Leftrightarrow (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$

(ii) $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q \Leftrightarrow (\neg p \wedge (q \vee \neg q)) \vee ((p \vee \neg p) \wedge \neg q)$

$\Leftrightarrow (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$

$\Leftrightarrow (\neg p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$

(iii) $\neg p \vee q \Leftrightarrow (\neg p \wedge (q \vee \neg q)) \vee ((p \vee \neg p) \wedge q)$

$\Leftrightarrow (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (p \wedge q) \vee (\neg p \wedge q)$

$\Leftrightarrow (\neg p \wedge q) \vee (p \wedge q) \vee (\neg p \wedge \neg q)$

(3) Principal conjunctive normal form. Consider a formula for the propositions p and q using disjunctions as $p \vee q$, $\neg p \vee q$, $p \vee \neg q$, $\neg p \vee \neg q$. These formulae are called maxterms.

An equivalent formula for a given formula, consisting of conjunctions of the maxterms only is called the principal conjunctive normal form (pcnf) or product of sum canonical form.

Procedure for obtaining pcnf for a given formula is similar to the one for pdnf as all assertions made for pdnf can be made for pcnf using duality principle.

Example 27.26: Obtain the principal disjunctive and conjunctive normal forms of

$p \rightarrow (p \rightarrow q) \wedge \neg(\neg q \vee \neg p)$ (November, 2014)

Solution. (i) $p \rightarrow [p \rightarrow (q \wedge \neg q)] \wedge \neg(\neg q \vee \neg p)$

$\Leftrightarrow \neg p \vee [\neg p \vee (q \wedge \neg q)] \wedge (q \wedge p)$

(Using DeMorgan's law and equivalence $p \rightarrow q \Leftrightarrow \neg p \vee q$)

$\Leftrightarrow \neg p \vee [\neg p \vee q] \wedge [\neg p \vee \neg q] \wedge (q \wedge p)$

$\Leftrightarrow \neg p \vee (q \wedge p)$

$\Leftrightarrow [\neg p \vee q] \wedge [\neg p \vee p]$

$\Leftrightarrow (\neg p \vee q) \wedge (p \vee \neg p) \wedge (q \vee p)$

This is the desired pdnf.

(ii) $p \rightarrow [p \rightarrow (q \wedge \neg q)] \wedge \neg(\neg q \vee \neg p) \Leftrightarrow \neg p \vee q$

$\Leftrightarrow \neg p \vee [\neg p \vee (q \wedge \neg q)] \wedge (q \wedge p)$

$\Leftrightarrow [\neg p \vee \neg p \vee q] \wedge [\neg p \vee \neg q] \wedge (q \wedge p)$

$\Leftrightarrow (\neg p \vee q) \wedge (\neg p \vee \neg q) \wedge (\neg p \vee p) \wedge (\neg p \vee q)$

$\Leftrightarrow \neg p \vee q$

This is the desired pcnf.

Example 27.27: Obtain the principal conjunctive normal form for $(Q \rightarrow P) \wedge (\neg P \wedge Q)$. (Madras, 2006)

Solution. $(Q \rightarrow P) \wedge (\neg P \wedge Q) \Leftrightarrow (\neg Q \vee P) \wedge (\neg P \wedge Q)$

$\Leftrightarrow (\neg Q \vee P \vee (R \wedge \neg R)) \wedge (Q \wedge \neg P \wedge (R \vee \neg R))$

$\Leftrightarrow (\neg Q \vee P \vee R) \wedge (\neg Q \vee P \vee \neg R) \wedge (Q \wedge \neg P \wedge R) \wedge (Q \wedge \neg P \wedge \neg R)$

$\Leftrightarrow (\neg Q \vee P \vee R) \wedge (\neg Q \vee P \vee \neg R) \wedge (Q \vee \neg P \vee R) \wedge (Q \vee \neg P \vee \neg R)$

3.7.15 INFERENCE THEORY

Inferring the conclusion from certain premises is known as the inference theory. When conclusion is reached from a set of premises by using the accepted rules of reasoning, then this process is called a deduction or a formal proof. A proof of a theorem is a valid argument.

The criteria for finding whether an argument is valid' are called rules which are expressed in terms of premises and conclusions or in terms of statement formulae.

The proofs are of two types : Direct or Indirect.

(i) To a proof the truth of the premises directly shows the truth of the conclusion, then it is called a direct proof.

(ii) An indirect proof proceeds by assuming that p is true and also C is false, and then deduce a contradiction using $p \wedge \sim C$, along with other premises.

The rule of Elimination between assumptions in a direct proof or an indirect proof is the negated conclusion.

Rules for deriving direct and indirect proofs :

(i) the proof should have finite steps only ;

(ii) each step is an hypothesis, a premise or a proposition which is implied from previous steps using valid equivalence or implication ;

(iii) the last step for a direct proof is not known while for an indirect proof it must be a contradiction.

Example 37.28. Show whether the conclusion C follows logically from the premises R and S.

(a) $R : p \rightarrow q$, $R : p$, $C : q$

(b) $R : p$, $S : p \rightarrow q$, $C : \sim (p \rightarrow q)$

(c) $R : p \rightarrow q$, $S : p$, $C : q$

Solution. Let us first form the following truth table :

p	q	$\sim p$	$\sim q$	$R : p \rightarrow q$	$\sim (p \rightarrow q)$	$p \wedge \sim q$
1	1	0	0	1	0	0
1	0	0	1	0	1	0
0	1	1	0	1	0	0
0	0	1	1	1	0	0

(a) Here only the first row of premises R and S contains 1 but not the conclusion C.

Hence C is not valid.

(b) Only the first row of both R and S contains 1 but not the conclusion C.

Hence C is not valid.

(c) Here both R and S contain 1 only in the first row and the conclusion C also has 1 in that row. Hence our conclusion is valid.

Example 37.29. Find the direct and indirect proofs of $p \rightarrow (q \rightarrow r), \sim r \vee s, \sim s \wedge t, p \Rightarrow \sim q$.

Solution. *Direct proof*

(1) $\sim r \vee s$	(premise)	(2) r	(another premise)
(3) $\sim \sim r$	[By (1) and (2)]	(4) $p \rightarrow (q \rightarrow r)$	(premise)
(5) $q \rightarrow r$	[By (3) and (4)]	(6) q	(premise)
(7) r	[By (5) and (6), which is a conclusion.		

Indirect proof

(1) $\sim (q \rightarrow r)$	(negation of conclusion)
(2) $\sim r \vee r$	(By conditional equivalence)
(3) $\sim r \rightarrow p$	(premise)
(4) $\sim \sim r$ and (2)	[By contrapositive simplification]
(5) p	[By disjunctive simplification]
(6) $p \rightarrow (q \rightarrow r)$	(premise)

(xiii) $q \to r$ [By (xi) and (ix)]
(xiv) q [premise]
(xv) r [By (xiii) and (xiv)]
(xvi) $\sim p \wedge r$ [By (x) and (xv)]
This is a contradiction.

EXERCISES 1.3

1. If $X = \{1, 2, 3, 4, 5\}$ be the universal set, determine the truth values of each of the following statements:
   (i) $(\forall x \in X)(x + 3 < 10)$ (ii) $(\exists x \in X)(x + 1 = 10)$
2. Negate each of the following statements:
   (i) $p \wedge (q \vee r)$ (ii) $(\forall x)(x \in A \to x \in B)$
   (iii) Some students are fifteen years old. (iv) All students like to do boards.
3. What is the truth value of $\forall x, P(x)$ where $P(x)$ is a statement '$x + 10 > 10$' and the universe of discourse consists of positive integers not exceeding 4?
4. Use universal quantifier to state the sum of any two rational numbers is rational.
5. Over the universe of real numbers, use quantifier to say that the equation $ax + b = 0$ has a solution for all values of a and b.
6. Translate the following statements in ordinary (non-symbolic) form of speech:
   (i) All materials are rods. (ii) No materials are rods.
   (iii) Some materials are rods. (iv) Some materials are not rods.
7. Show that $Q \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$ is a tautology.
8. Convert $\sim(A \cup (B \cap \sim C)) \cap \sim A$ into DNF. (V.T.U., BCA, 2005)
9. Without constructing truth table, obtain the product of sums canonical form of the formula $(P \to Q) \wedge (Q \leftrightarrow P)$. Hence find the sum of products canonical form. (Kerala, BCA, 2007)
10. Find the dual of $(p \wedge r) \wedge (q \vee s) \vee (p \wedge q) \vee (r \wedge s)$.
11. Prove that $P \to Q, Q \to R, R \to S, P \to S$ by using indirect method. (U.P.T.U., 2007)
12. Define quantifier negation laws and verify them.
13. State whether the conclusion C follows logically from the premises P and Q:
   (i) $P: (p \vee q) \to r, \; Q: \sim p, \; C: q \to r$ (ii) $P: p \to q, \; Q: q \to r, \; C: \sim p \to r$

III. BOOLEAN ALGEBRA

1.13 INTRODUCTION

(I) The concept of Boolean algebra was first introduced by George Boole* in 1854 through his paper 'An investigation of the laws of thought'. It is basically two values i.e., 0, 1 etc. Earlier it had applications to statements and sets which are either true or false. In 1938 Claude Shannon showed that basic rules given by Boole could be used to design circuits. These days however, Boolean algebra has wide applications in switching circuits, electrical networks and electronic computers.

Basically there are three operations in the Boolean algebra (i) AND, (ii) OR, and (iii) NOT, which are symbolically represented by $\wedge$, $\vee$ and $'$ respectively. Some authors, use the symbols $\cdot$ (.), $+$ and $\bar{}$ for the same operations.

Here $'$ denotes the complement of an element and is defined for $0' = 1$ and $1' = 0$.

The operator $\wedge$ (i.e., 'AND') has the following values $1 \wedge 1 = 1, 1 \wedge 0 = 0, 0 \wedge 1 = 0, 0 \wedge 0 = 0$; while the operator $\vee$ (i.e., OR) has the values $1 \vee 1 = 1, 1 \vee 0 = 1, 0 \vee 1 = 1, 0 \vee 0 = 0$.

Def. *Any non-empty set B with the binary operations '$\wedge$' and '$\vee$' and the unary operation '$'$' is called the Boolean algebra $(B, \wedge, \vee, ')$ if the following axioms hold where a, b, c are elements in B:*

1. Commutative laws: $a \wedge b = b \wedge a$; $a \vee b = b \vee a$
2. Associative laws: $a \wedge (b \wedge c) = (a \wedge b) \wedge c$
   $a \vee (b \vee c) = (a \vee b) \vee c$

*A British mathematician George Boole (1815–1881) who created Boolean algebra.

3. Distributive laws: $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

$a + (b \cdot c) = (a + b) \cdot (a + c)$

4. Complement laws: $a \cdot a' = 0$, $a + a' = 1$

The operations $\cdot$, $+$ and $'$ are called sum, product, and complement respectively. We shall follow the usual practice that $'$ has precedence over $\cdot$, and $\cdot$ has precedence over $+$ unless guided by brackets. e.g., $a \cdot b + c$ means $(a \cdot b) + c$ not $a \cdot (b + c)$ while $a + b'$ implies $a + b'$ not $(a + b)'$.

(2) Boolean function. The variable x is called a Boolean variable if it assumes values only from $\{0, 1\}$.

Def. *A function from the set $\{x_1, x_2, \ldots, x_n\}$ where $x_i \in \{0, 1\}$, $1 \le i \le n$ to $\{0, 1\}$ is called a Boolean function of degree n.* Boolean functions can be represented by expressions composed of variables and Boolean operations.

x_i, 0, 1, $x_1 \cdot x_2$, $x_1 + x_2$, ... x_n are Boolean expressions in the variables x_i $(1 \le i \le n)$. If p and q are Boolean expressions then $p \cdot q$, $p + q$ and p' are also Boolean expressions and each represents a Boolean function.

By substituting 0 and 1 for the variables in the expression, the values of the function can be found.

(3) *If f and g be Boolean functions of degree n, then*

(i) Complement of f is the function f' where

$$f'(x_1, x_2, \ldots, x_n) = [f(x_1, x_2, \ldots, x_n)]'$$

(ii) *f and g are equal if $f(x_1, x_2, \ldots, x_n) = g(x_1, x_2, \ldots, x_n)$*

(iii) Boolean sum $f + g$ is

$$(f + g)(x_1, x_2, \ldots, x_n) = f(x_1, x_2, \ldots, x_n) + g(x_1, x_2, \ldots, x_n)$$

(iv) Boolean product $f \cdot g$ is

$$(f \cdot g)(x_1, x_2, \ldots, x_n) = f(x_1, x_2, \ldots, x_n) \cdot g(x_1, x_2, \ldots, x_n)$$

(4) *Powers in a Boolean function:* $x^0 = x' + x = 1$

$$x^1 = x^0 \cdot x = 1 \cdot x = x, \ldots, x^n = x$$

Similarly, $x \cdot x_1 \cdot x_2 = x$, etc.

Example 11.20. *Find the values of the Boolean function $f = (x \cdot y) + z'$*

Solution. *f* being third degree Boolean function has 2^3 i.e. 8 values which are shown in the following table :

x	y	z	z'	$x \cdot y$	$(x \cdot y) + z'$
1	1	1	0	1	1
1	1	0	1	1	1
1	0	1	0	0	0
1	0	0	1	0	1
0	1	1	0	0	0
0	1	0	1	0	1
0	0	1	0	0	0
0	0	0	1	0	1

11.6.1 DUALITY

(1) The dual of any Boolean function is obtained by interchanging Boolean sums and Boolean products along with the interchange of zeros and ones.

For example the dual of $x \cdot (y + 0)$ is $x + (y \cdot 1)$.

The dual of any theorem of a Boolean algebra is also its theorem. This implies that the dual of any theorem in Boolean algebra is always true.

(2) Principle of duality. *The dual of any theorem (or property) in Boolean algebra is also a theorem (or property).*

37.17 BOOLEAN IDENTITIES

There are many identities in Boolean algebra which are quite useful in simplifying electrical circuits. Some of the important ones are given below :

1. Identity law :	$x \vee 0 = x$; $x \wedge 1 = x$	
2. Dominance law :	$x \vee 1 = 1$; $x \wedge 0 = 0$	
3. Complement law :	$x \vee x' = 1$; $x \wedge x' = 0$	
4. Idempotent law :	$x \vee x = x$; $x \wedge x = x$	
5. Double complement law :	$(x')' = x$	
6. Commutative law :	$x \vee y = y \vee x$; $x \wedge y = y \wedge x$	
7. Associative law :	$x \vee (y \vee z) = (x \vee y) \vee z = x \vee y \vee z$	
	$x \wedge (y \wedge z) = (x \wedge y) \wedge z = x \wedge y \wedge z$	
8. Distributive law :	$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$	
	$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$	
9. De Morgan's law :	$(x \wedge y)' = x' \vee y'$; $(x \vee y)' = x' \wedge y'$	(Rajat, 2006)
10. Absorption law :	$x \vee (x \wedge y) = x$	

Example 37.21. Let B be a Boolean algebra. Show that B is a unique complemented. (Andhra, 2004)

Solution. Let b and c be two complements of a.

Then $b = b \wedge 1$ [$\because$ 1 is an additive identity]

$= b \wedge (a \vee c)$ [$\because$ c is complement of a]

$= (b \wedge a) \vee (b \wedge c) = (a \wedge b) \vee (b \wedge c)$

$= 0 \vee (b \wedge c)$ [$\because$ $a \wedge b = a \wedge a' = 0$]

$= b \wedge c$ ...(1)

Similarly $c = c \wedge 1 = c \wedge (a \vee b)$ [$\because$ $a \vee b = a \vee a' = 1$]

$= (c \wedge a) \vee (c \wedge b)$

$= (a \wedge c) \vee (b \wedge c)$ [$\because$ $a \wedge c = a \wedge a' = 0$]

$= 0 \vee (b \wedge c)$

$= b \wedge c$ ...(2)

From (1) and (2), we find that $b = c$.

Thus the complement of a is unique.

Example 37.22. In a Boolean algebra, show that
(i) $x \vee (x' \wedge y) = x \vee y$ (ii) $x \wedge (x' \vee y) = x \wedge y$ (Rajat, 2006)

Solution. (i) $x \vee (x' \wedge y) = (x \vee x') \wedge (x \vee y) = 1 \wedge (x \vee y)$ [$\because$ $x \vee x' = 1$]

$= x \vee y$ [By identity law]

$= x \vee (y \wedge 1)$ [By commutative law]

$= x \vee y$ [$\because$ $y \wedge 1 = y$]

$= x$

(ii) $x \wedge (x' \vee y) = (x \wedge x') \vee (x \wedge y)$

$= 0 \vee (x \wedge y)$ [$\because$ $x \wedge x' = 0$]

$= x \wedge y$ [By distributive law]

$= x \wedge (y \vee 1)$ [By commutative law]

$= x \wedge 1$ [$\because$ $y \vee 1 = 1$]

$= x$.

Example 37.23. Simplify the following :

(i) $(x \vee y) \cdot x' \cdot y'$ (ii) $x \cdot y \cdot z + x \cdot y \cdot z' + x \cdot y' \cdot z$ (iii) $x \vee y \vee (x \wedge y) \vee x'$

Solution. (i) $(x \wedge y) \vee (x' \wedge y') = (x \wedge y) \vee (x' \wedge y')$

$= (y \wedge x) \vee (x \wedge y')$ [By the Absorption law]

$= x$ [$\because p \vee p' = 1$]

(ii) $x \vee y \vee (x \wedge y \vee z) \wedge (x \vee y \vee z') = (y \vee x) \vee [(x \wedge y) \vee (x \wedge y) \vee z)]$ [By commutative law]

$= (y \vee x) \vee [(x \wedge y) \vee (x \wedge y) \vee z)]$ [By distributive law]

$= y \vee [x \wedge (x \wedge y) \vee z]$

$= y \vee [x \wedge (x \wedge y) \vee z]$ [By associative law]

$= y \vee (x \wedge 1)$ [$\because x \wedge x' = 1$]

$= y \vee x$ [$\because x \wedge 1 = x$]

(iii) $x \vee y \vee [(x \wedge y') \vee y]' = x \vee y \vee [y \vee (x \wedge y')]'$ [By commutative law]

$= x \vee y \vee [(y \vee x) \wedge (y \vee y')]'$ [By distributive law]

$= (x \vee y) \wedge [(y \vee x) \wedge 1]'$ [$\because y \vee y' = 1$]

$= (x \vee y) \wedge (x \vee y)'$.

Example 3.3.36 Show that

(i) $(x \wedge y \wedge z) \vee (x' \wedge y \wedge z) \vee (x \wedge y' \wedge z) \vee (x \wedge y \wedge z') = x \vee y \vee z$

(ii) $(x \wedge y) \vee (x' \wedge z) \vee (y \wedge z) = (x \vee z) \wedge (x' \vee y)$. (Anna, 2004)

Solution. (i) R.H.S. $= (x \wedge y \wedge z) \vee (x \wedge y \wedge z') \vee (x' \wedge y \wedge z) \vee (x \wedge y' \wedge z)$

$= (x \wedge y) \wedge (z \vee z') \vee (x \wedge y') \wedge (z \vee z')$ [By commutative law]

$= (x \wedge y) \vee (x \wedge y')$ [By distributive law]

$= [x \wedge (y \vee y')] \wedge [(x \wedge y) \vee (x \wedge y')] \wedge [(x \vee y) \vee z]$

$= (x \wedge 1) \wedge [(x \wedge y) \vee (x \wedge y')] \wedge [(x \vee y) \vee z]$ [$\because x \vee x' = 1$ and $x \wedge 1 = x$]

$= (x \wedge 1) \wedge [(x \wedge y) \vee (x \wedge y')] \wedge [(x \vee y) \vee z]$

$= (x \wedge 1) \wedge (x \wedge 1) \wedge [(x \vee y) \vee z]$ [$\because p \wedge p = p$]

$= (x \wedge y) \wedge (x \wedge z) \wedge [(x \vee y) \vee z]$ [By commutative law]

$=$ L.H.S.

(ii) L.H.S. $= (x \wedge y) \vee (x' \wedge z)$

$= [(x \vee x') \wedge (x \vee z)] \wedge [(y \vee x') \wedge (y \vee z)]$ [By distributive law]

$= [1 \wedge (x \vee z)] \wedge [(y \vee x') \wedge (y \vee z)]$

$= (x \vee z) \wedge [(y \vee x') \wedge (y \vee z)]$

$= (x \vee z) \wedge [(x' \vee y) \wedge (z \vee y)]$

$= (x \vee z) \wedge [(x' \vee y) \wedge (y \vee z)]$

$= (x \vee z) \wedge (x' \vee y) \wedge (y \vee z) \wedge (y \vee z)$

$= (x \vee z) \wedge (x' \vee y) \wedge (y \vee z)$

$= (x \vee z) \wedge (x' \vee y) \wedge [(x \wedge x') \vee (y \vee z)]$

$= (x \vee z) \wedge (x' \vee y) \wedge [(x \vee y \vee z) \wedge (x' \vee y \vee z)]$

$= (x \vee z) \wedge (x' \vee y) \wedge (x \vee y \vee z)$

$= (x \vee z) \wedge [(x' \vee y) \wedge 1]$

$= (x \vee z) \wedge (x' \vee y) =$ R.H.S.

Example 3.3.37 Show that

(i) $(x \vee y) \wedge (x' \vee y') = (x \wedge y') \vee (x' \wedge y)$

(ii) $(x \wedge y') \vee (y \wedge z') \vee (x' \wedge z) = (x \vee y \vee z) \wedge (x' \vee y' \vee z')$.

Solution. (i) $(x \vee y) \wedge (x' \vee y') = [(x \vee y) \wedge x'] \vee [(x \vee y) \wedge y']$

$= [(x \wedge x') \vee (y \wedge x')] \vee [(x \wedge y') \vee (y \wedge y')]$ [By distributive law]

$= [0 \vee (y \wedge x')] \vee [(x \wedge y') \vee 0]$ [$\because x \wedge x' = 0$]

$= (x' \wedge y) \vee (x \wedge y')$

(ii) $[x \wedge (x' \vee y)] \vee [x' \wedge (x \vee y)]$

$= [(x \wedge x') \vee (x \wedge y)] \vee [(x' \wedge x) \vee (x' \wedge y)]$

$= [0 \vee (x \wedge y)] \vee [0 \vee (x' \wedge y)]$ [$\because x \wedge x' = 0$]

$= (x \vee 0) \wedge (y \vee 0) \wedge (x \vee y) \wedge (x' \vee y)$.

Example 37.26. If $x \wedge y = x \wedge z$ and $x \vee y = x \vee z$, then prove that $y = z$.

Solution: We have $x \wedge y = x \wedge z$ and $x \vee y = x \vee z$

$\therefore$ $(x \wedge y) \vee (x \wedge y') = (x \wedge z) \vee (x \wedge z')$

i.e., $x \wedge (y \vee y') = x \wedge (z \vee z')$ [By distributive law]

or $x \wedge 1 = x \wedge 1$ $\because x \vee x' = 1$

or $x = x$.

Example 37.27. In Boolean algebra $(B, +, \cdot, ')$ show that

$(x \cdot y' + z \cdot y)' \cdot x \cdot z = 0$ (Meerut, 2010; M.D.U., 2007)

Solution: $(x \cdot y' + z \cdot y)' \cdot x \cdot z = (x \cdot y' + z \cdot y)' \cdot (x \cdot z)$

$= (x' + (y')' \cdot z') \cdot (y' + y \cdot z') \cdot (x \cdot z)$

$= (x' + y \cdot z') \cdot (y' + y \cdot z') \cdot (x \cdot z)$

$= (x' \cdot y' + y \cdot z' \cdot y' + x' \cdot z' + y \cdot z' \cdot z') \cdot (x \cdot z)$

$= (x' \cdot y' + 0 + x' \cdot z' + y \cdot z' \cdot 0) \cdot (x \cdot z)$

$= (x' \cdot y' + x' \cdot z' + 0) \cdot (x \cdot z)$

$= x' \cdot y' \cdot x \cdot z + x' \cdot z' \cdot x \cdot z$

$= y' \cdot x' \cdot x \cdot z + x' \cdot x \cdot z \cdot z'$

$= y' \cdot 0 \cdot z + 0 \cdot z \cdot z'$

$= 0 + 0 = 0$.

EXERCISE 37.4

1. Find the truth table for the Boolean function $f(x, y, z) = (x \cdot y \cdot z) + (x \cdot y' \cdot z) + (x \cdot y \cdot z')$.
2. Write the dual of the Boolean expression $x \cdot (y + z)$. (Madras, 2004)
3. Simplify the following:
   (i) $(x \cdot y \cdot z)'$ (ii) $(x \cdot y \cdot z) + (x' \cdot y \cdot z) + (x \cdot y' \cdot z) + (x \cdot y \cdot z')$
4. In a Boolean algebra $(B, \wedge, \vee, ')$, prove that
   (i) $(a \wedge b \wedge c)' = a' \vee b' \vee c'$ (Meerut, 2009) (ii) $a' \wedge (a \vee b \vee c)' = a' \wedge b' \wedge c'$
5. Show that $x \cdot y + x \cdot y' + x' \cdot y = x + y$ for any Boolean algebra $(B, +, \cdot, ')$.
6. In a Boolean algebra $(B, \wedge, \vee, ')$, show that
   (i) $(a \wedge b)' = a' \vee b'$ (ii) $(a \vee b) \wedge (a' \wedge b') = 0$
7. In Boolean algebra, prove that
   (i) $a \vee (a' \wedge b) = a \vee b$ (ii) $a \wedge (a' \vee b) = a \wedge b$
8. Show that $(x \cdot y) + (x \cdot y') + (x' \cdot y) + (x' \cdot y') = 1$.
9. If B is a Boolean algebra and $x, y, z \in B$, prove that:
   (i) $x \cdot y + x \cdot y' + x' \cdot y + x' \cdot y' = 1$
   (ii) $x \cdot y \cdot z + x \cdot y \cdot z' + x \cdot y' \cdot z + x' \cdot y \cdot z + x' \cdot y \cdot z' + x' \cdot y' \cdot z + x' \cdot y' \cdot z' = 1$
10. Prove that $x \cdot y + x \cdot y' + x' \cdot y + x' \cdot y' = 1$ in a Boolean algebra.
11. In a Boolean algebra B, prove that $a \wedge b' = 0$ if and only if $a \wedge b = a$. (Madras, 2005)
12. In any Boolean algebra, show that $a = b$ if and only if $(a \wedge b') \vee (a' \wedge b) = 0$. (Madras, 2004)
13. In Boolean algebra, show that
    (i) $(a \vee b) \wedge (a \vee b') \wedge (a' \vee b) = a \wedge b$
    (ii) $(a \wedge b) \vee (a \wedge b') \vee (a' \wedge b) = a \vee b$ (Madras, 2005)
14. Consider a truth table for the Boolean function $f(x, y, z)$ determined by the polynomial
    $f(x, y, z) = x_0 x_1 x_2 + x_0 x_1 x_2' + x_0 x_1' x_2 + x_0 x_1' x_2'$. (U.P.T.U., 2003)

37.15 MINIMAL BOOLEAN FUNCTION

Def. A minimal Boolean function is a variable or the product of $x_1, x_2, \ldots, x_n$. It is also called literal.

Here, we have considered x', y' are their complementary variables respectively. Hence $x \vee y, x' \vee y, x \vee y', x' \vee y'$ are called minimal Boolean functions.

Similarly there are 2^2 i.e. 4 minimal Boolean functions in the three variables x, y, z i.e., $x \wedge y \wedge z$, $x' \wedge y \wedge z$, $x \wedge y' \wedge z$, $x \wedge y \wedge z'$, $x \wedge y' \wedge z'$, $x' \wedge y \wedge z'$, $x' \wedge y' \wedge z$, $x' \wedge y' \wedge z'$.

In general, there are 2^n minimal Boolean functions for minimum n variables.

Similarly the join of the variables $x_1, x_2, \ldots, x_n$ is called minterm and there will be 2^n minterms.

3.12.2 DISJUNCTIVE NORMAL FORM

(i) Def. *A Boolean function which can be represented as sum of minimal Boolean functions is called a* **Disjunctive normal form or minterm normal form or Canonical form.**

(ii) *If the number of distinct terms in a disjunctive normal form of Boolean function in n variables are 2^n, then it is called a* **complete disjunctive normal form.**

(iii) Complement function of a disjunctive normal form *function f is the sum of all those terms of a complete disjunctive normal form which are not present in the disjunctive normal form of f. The complement of f is denoted by f'.*

For example: If $f = (x \wedge y) \vee (x \wedge (y \vee z))$

then its complete disjunctive normal form in two variables only is

$$(x \wedge y) \vee (x' \wedge y) \vee (x \wedge y') \vee (x' \wedge y')$$

$\therefore$ The complement function of this disjunctive normal form is $f(x, y) = (x' \wedge y') \vee (x' \wedge y)$.

Example 3.12.2: *Find the value of the complete disjunctive normal form in three variables x, y, z.*

Solution. The complete disjunctive normal form in three variables x, y, z is

$$f(x, y, z) = (x \vee y \vee z) \wedge (x \vee y \vee z') \wedge (x \vee y' \vee z) \wedge (x' \vee y \vee z)$$

$$= (x \vee y' \vee z') \wedge (x' \vee y \vee z) = (x' \vee y' \vee z) \wedge (x \vee y' \vee z')$$

$$= [(x \vee y) \vee z] \wedge [(x \wedge x') \vee z'] = [(x \vee y) \wedge z] \vee [(x \wedge x') \wedge z']$$

$$= [(x \vee y) \wedge z] \vee [(x \vee y) \wedge z'] = [(x \wedge y) \vee (x \wedge z)] \vee [(x' \vee y) \wedge (x' \vee z)]$$

$$= [(x \wedge y) \vee (x \wedge z)] \vee [(x' \wedge x) \vee (x' \vee y) \wedge (z \vee x) \vee (z \vee y)]$$

$$= [(x \wedge y) \vee (y \wedge y')] \vee [x' \vee (y \wedge z)]$$

$$= (x \wedge y) \vee x' \vee (y \wedge z)$$

3.12.3 CONJUNCTIVE NORMAL FORM

Def. *If a Boolean function $f(x_1, x_2, \ldots, x_n)$ is expressed in the form of factors and each factor is the sum of all the n variables, then such a function is called a* **conjunctive normal form or maxterm normal form or dual canonical form.**

(ii) *If a conjunctive normal of a function of n variables consists all the 2^n distinct factors, then such a function is called a* **complete conjunctive normal form.**

(iii) Complement function of a conjunctive normal form *function f is a Boolean function which is the product of all those terms of complete conjunctive normal form which are not present in conjunctive normal form of f.*

The complement of conjunctive normal form f is denoted by f'.

For example, if $f(x, y) = (x \wedge y) \vee (x \wedge y')$

then its complete conjunctive normal form in two variables only is

$$(x \wedge y) \vee (x \wedge y') \vee (x' \wedge y) \vee (x' \wedge y')$$

$\therefore$ The complement function of this conjunctive normal form is

$$f'(x, y) = (x' \wedge y') \vee (x' \wedge y)$$

Example 3.12.3: *Given Boolean expression f where $f(x_1, x_2, x_3) = (x_1' \wedge x_2) \vee (x_1 \wedge x_2) \vee (x_2 \wedge x_3)$ formally find the complete sixteen the function and obtain the minterm normal form.* (Bharathiar, 2002)

Solution. Given $f(x_1, x_2, x_3) = (x_1' \wedge x_2) \vee (x_1 \wedge x_2) \vee (x_2 \wedge x_3)$

$= (x_2 \wedge x_1') \vee (x_2 \wedge x_1) \vee (x_2 \wedge x_3)$ [By commutative law]

$= (x_2 \wedge x_3) \vee (x_2 \wedge x_1) \vee (x_1' \wedge x_2)$

$= (x_2 \wedge (x_3 \vee x_1')) \vee (x_1' \wedge x_2)$ [By distributive law]

$= (x_2 \wedge 1) \vee (x_2' \wedge x_3)$ (by $p \vee p' = 1$)
$= x_2 \vee (x_2' \wedge x_3)$ (By Identity law)

Minterm normal form of $f(x_1, x_2, x_3)$

$= (x_2' \wedge x_3 \wedge x_1) \vee (x_2' \wedge x_3 \wedge x_1') \vee (x_2' \wedge x_3) \vee (x_2 \wedge x_3 \wedge x_1) \vee (x_2 \wedge x_3 \wedge x_1')$
$= (x_2' \wedge x_3 \wedge x_1) \vee (x_2' \wedge x_3 \wedge x_1') \vee (x_2' \wedge x_3 \wedge x_2) \vee (x_2' \wedge x_3 \wedge x_2') \vee (x_2 \wedge x_3 \wedge x_1) \vee (x_2 \wedge x_3 \wedge x_1')$
$= (x_2' \wedge x_3 \wedge x_1) \vee (x_2' \wedge x_3 \wedge x_1') \vee (x_2' \wedge x_3 \wedge x_2) \vee (x_2' \wedge x_3 \wedge x_2') \vee (x_2 \wedge x_3 \wedge x_1)$

Example 17.40: Express the following functions in the complete conjunctive normal form:
(i) $f(x, y)$ (ii) $f(x, y, z) = (x' \vee y)$

Solution: (i) $f(x, y) = x \vee y = (x \wedge 1) \vee (y \wedge 1) = (x \wedge (y \vee y')) \vee (y \wedge (x \vee x'))$

(ii) $f(x, y, z) = (x \vee y) \vee (x' \wedge y' \wedge z) = (x \vee y \vee 0) \vee (x' \wedge y' \wedge z)$
$= (x \vee y \vee (z \wedge z')) \vee (x' \wedge y' \wedge z)$
$= (x \vee y \vee z) \wedge (x \vee y \vee z') \vee (x' \wedge y' \wedge z)$
$= (x \vee y \vee z) \wedge (x \vee y \vee z') \wedge (x' \vee y' \vee z) \wedge (x' \vee y' \vee z')$

Example 17.41: The function $f(x, y, z) = (x \wedge y \wedge z) \vee (x \wedge y' \wedge z) \vee (x \wedge y' \wedge z') \vee (x' \wedge y \wedge z)$ is in conjunctive normal form. Write its complement?

Solution: The complete conjunctive normal form in three variables x, y, z is $(x \wedge y \wedge z) \vee (x \wedge y \wedge z') \vee (x \wedge y' \wedge z) \vee (x \wedge y' \wedge z') \vee (x' \wedge y \wedge z) \vee (x' \wedge y \wedge z') \vee (x' \wedge y' \wedge z) \vee (x' \wedge y' \wedge z')$.

The complement of the given function F is

$F' = (x' \wedge y \wedge z') \vee (x' \wedge y' \wedge z') \vee (x' \wedge y' \wedge z)$.

EXERCISE 17.3

1. Find the value of the complete disjunctive normal form in
   (i) two variables x, y (ii) three variables x, y, z
2. Express the following functions into disjunctive normal form:
   (i) $x \vee y$ (ii) $x \wedge (x' \vee y)$
3. Express the Boolean function $f = x \wedge (x' \vee y)$ in sum of minterms.
4. Obtain the function $x \wedge y'$ in disjunctive normal form in three variables x, y, z.
5. Express the function $f = (x \vee y \vee z) \wedge (x \vee y \vee z') \wedge (x \vee y' \vee z)$ into disjunctive normal form in which maximum number of variables are used.
6. Write the complement of the conjunctive normal form Boolean function $(x \vee y \vee z) \wedge (x \vee y \vee z') \wedge (x \vee y' \vee z) \wedge (x' \vee y \vee z)$.

17.11 SWITCHING CIRCUITS

(1) A switching network is an arrangement of wires and switches for getting current to some circuits. A switch can be either closed or open. A closed switch permits and an open switch stops flow of current.

(2) If p denotes a switch, then p' denotes that switch which is open when p is closed and p' is closed when p is open.

If x denotes the state of the switch p, then x' represents the state of the switch p'. x is called the Boolean variable which is a binary variable.

If $x = 1$ denotes the switch is closed or current flows, then $x = 0$ denotes that the switch is open or current stops.

(3) Two switches p_1 and p_2 are in the series connection is series (represented by $\wedge$) or connected in parallel (or over-written by $\vee$).

They are shown as follows:

(i) $p_1 \wedge p_2$ is series $(p_1 \wedge p_2)$

(ii) $p_1 \vee p_2$ is parallel $(p_1 \vee p_2)$

If $(B, \cdot)$ is a non-empty set and $+, \cdot, '$ are the operations on B, then the system $(B, \cdot, +, ', 0, 1)$ is usually called Boolean switching algebra.

(4) Simplification of circuits. The simplification of a circuit means the least complicated circuit with minimum cost and best results. This depends on the cost of the equipment, number of switches and the type of the material used. Thus the simplification of circuits implies the use of fewer number of switches which can be achieved by using different properties of Boolean algebra. In other words, the simplification of switching circuits is equivalent to simplification of the corresponding Boolean function.

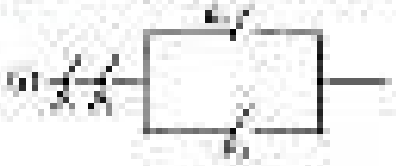
Example 13.20. Draw the circuits which represent the Boolean functions :

(i) $(x_1 \cdot x_2) + (x_1 \cdot x_3) + x_2'$ (ii) $(x_1 \cdot x_2) + x_3' + (x_2 \cdot x_3')$

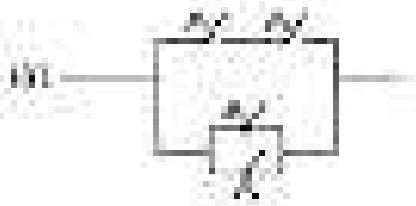
Solution. Here $x_1 \cdot x_2$ is a series circuit while $x_3' + x_2$ is a parallel circuit.

The required circuits are as follows :

(i)



(ii)



Example 13.21. Write the Boolean function representing the following circuit.

Also draw the circuit diagram which would be the replacement of the circuit in (i).

Solution. (i) The given circuit is represented by the Boolean function :

$$f = x_1 \cdot [x_2 \cdot x_3' + x_3] + [x_2 \cdot x_3 + x_2'] \cdot x_1$$

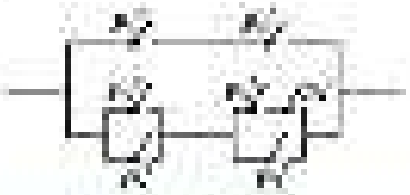
(ii) The Boolean function for the given circuit is

$$f = [x_1 \cdot x_2' + x_3] \cdot x_1 + [x_2 \cdot x_3 + x_2'] \cdot x_1$$

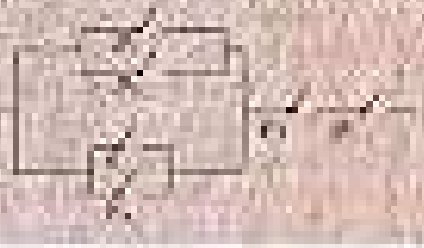
The complement of f is,

$$\begin{aligned} f' &= [x_1 \cdot x_2' + x_3]' + [x_2 \cdot x_3 + x_2']' + x_1' \\ &= [(x_1 \cdot x_2')' \cdot x_3'] + [(x_2 \cdot x_3)' \cdot x_2''] + x_1' \\ &= (x_1' + x_2'') \cdot x_3' + (x_2' + x_3') \cdot x_2 + x_1' \end{aligned}$$

Its circuit diagram is as follows :



Example 13.22. Simplify the following circuit and draw the diagram of the simplified circuit.



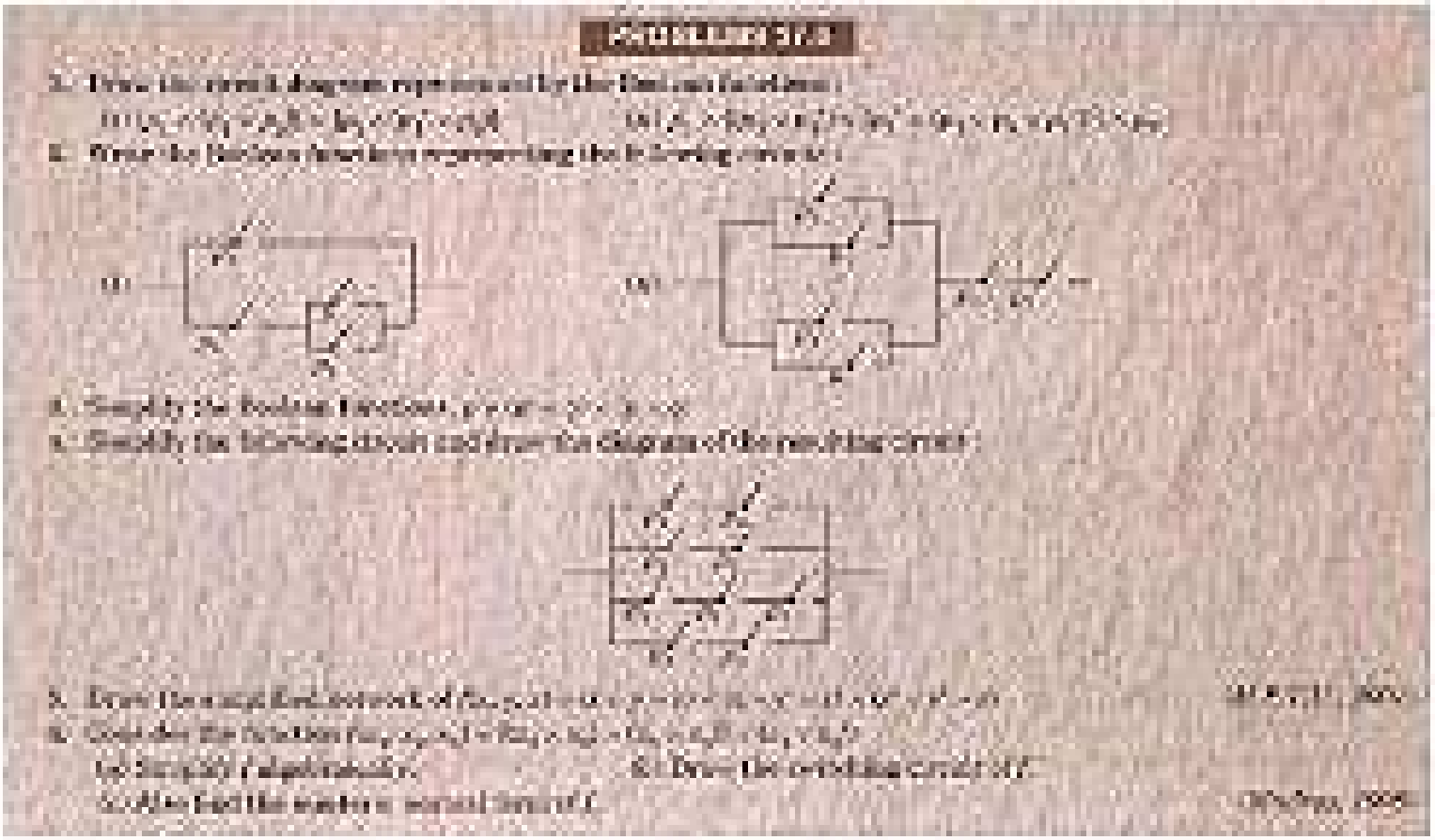
Solution. The given circuit is represented by the Boolean function f

$$= (p_1 \wedge p_2 \wedge \sim p_3) \vee (p_1 \wedge p_2 \wedge p_3) \vee (p_1 \wedge \sim p_2 \wedge p_3) \vee (\sim p_1 \wedge p_2 \wedge p_3)$$

$$= (p_1 \wedge p_2 \wedge q) \vee (p_1 \wedge q \wedge p_3) \vee (q \wedge p_2 \wedge p_3) \quad \text{(By distributive law)}$$

$$= (p_1 \wedge p_2) \vee (p_1 \wedge p_3) \vee (p_2 \wedge p_3) \quad \text{(By absorption law)}$$

Resultant diagram is



IV. FUZZY SETS

3.125 FUZZY LOGIC

We have so far dealt with the fundamentals of classical logic. Besides this, we have many logics which deals with propositions that are neither true or false. There is however another type of logic which includes not only the crisp values but all the values between true (1) and false (0). But there is some degree of vagueness about the exact value between 0, 1. The logic to find a definite outcome from such vague premises is called fuzzy logic.

Fuzzy set. To provide a mathematical model for fuzzy logic, L.A. Zadeh introduced the theory of 'Fuzzy sets' in 1965 on the basis of a membership function. The theory of 'fuzzy sets' is now fully developed.

Def. *A fuzzy set F of a non-empty set X is defined as $F = \{x, \mu_F(x) : x \in X\}$*

Here $\mu_F : X \to [0, 1]$ is a function called the membership function of F and $\mu_F(x)$ is the degree of membership of $x \in X$ in F.

In particular $\mu_F(x) = 1$ implies full membership

$\mu_F(x) = 0$ implies non-membership

and $0 < \mu_F(x) < 1$ means intermediate membership.

A fuzzy set F is, however, a set of pairs consisting of a particular element of the universe X and its degree of membership in it, such a is assigned a value in the range $[0, 1]$ indicating the extent to which x has the attribute F. It can also be represented as $F = \{x_1, \mu_F(x_1)\}, \{x_2, \mu_F(x_2)\}, \ldots, \{x_n, \mu_F(x_n)\}$

For example, if x is the number of cars in a lane, 'small' may be taken as a particular value of the fuzzy variable x and to each x is assigned a number in the range $[0, 1]$ that is the membership function.

Example 33.13. In a car race, all the cars complete the race in four groups. Some cars are new, some are old, some have long lives and some have short lives. If we note the four values for each car in a group, it will give us a distribution of cars. Now let us find the variation of the race based on engine power, car speed and road conditions. Each of these variables may further be divided into:

(i) low, medium and high for the variable engine power,

(ii) slow, moderate and fast for the variable car speed,

(iii) rough, bumpy and smooth for the variable road conditions.

Now we try to predict on some basis, in which of the four groups the car will finish, if it has low engine power, moderate speed and rough road.

Thus the distribution for the engine power would correspond to the membership function for low, medium and high. Similarly the distribution for the speed would depend on the membership function for slow, moderate and fast, while the distribution for road conditions would depend on membership function for rough, bumpy and smooth.

33.13 FUZZY SET OPERATIONS

(1) A fuzzy set is said to be normalized when the largest element of the set in the unit interval is unity.

For instance, the set of numbers $\{0, 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$ is normalized to $\{0.2, 0.1, 0.8, 0.9, 1\}$ by dividing each number by 2, the largest one in the set.

The normalization of a fuzzy set F is expressed as $\sup_x F(x) = 1$.

(2) Complement. The complement of a fuzzy set F to the set F' is the degree of non-membership of an element in F' equal to one minus the degree of membership of this element in F. (Fig. 33.5)

For example, if $F = \{0.4/\text{Ram}, 0.6/\text{John}, 0.8/\text{Jack}, 0.9/\text{Hari}\}$
is a set of intelligent students, then $F' = \{0.6/\text{Ram}, 0.4/\text{John}, 0.2/\text{Jack}, 0.1/\text{Hari}\}$
is a set of non-intelligent students.

Fig. 33.5

Fig. 33.6

(3) Intersection. The intersection of two fuzzy sets F and G is the set $F \cap G$, where the degree of membership of an element in $F \cap G$ is the minimum of the degrees of membership of this element in F and G. (Fig. 33.6)

(4) Union. The union of two fuzzy sets F and G is the set $F \cup G$, where the degree of membership of an element in $F \cup G$ is the maximum of this element in F and G. (Fig. 33.7)

For example, if
$$F = \{0.5/\text{Ram}, 0.2/\text{Sachin}, 0.4/\text{Ankit}, 0.8/\text{Sushil}\}$$
is a set of intelligent, and
$$G = \{0.1/\text{Ram}, 0.8/\text{Sachin}, 0.3/\text{Ankit}, 0.5/\text{Sushil}\}$$
is a set of diligent then
$$F \cap G = \{0.1/\text{Ram}, 0.2/\text{Sachin}, 0.3/\text{Ankit}, 0.5/\text{Sushil}\}$$
and
$$F \cup G = \{0.5/\text{Ram}, 0.8/\text{Sachin}, 0.4/\text{Ankit}, 0.8/\text{Sushil}\}$$

(5) Equality. Two fuzzy sets F and G are said to be equal if $F(x) = G(x)$ for all x in X.

(6) Subset. The fuzzy set F is said to be a subset of the fuzzy set G (i.e., $F \subseteq G$) if and only if $F(x) \le G(x)$ for all $x \in X$.

(7) Double negation. If F is a fuzzy set, then $(F')' = F$.

(8) De Morgan's Laws. If F and G are two fuzzy sets, then
$$(F \cup G)' = F' \cap G' \quad ; \quad (F \cap G)' = F' \cup G'.$$

Example 3.7.10. *Let the membership functions for the fuzzy sets F and G are as in the following table:*

x	1	2	∞	4	5	6	7	8	9	10
F	0	0	0.1	0.5	0.9	1	0.9	0.5	0	0
G	0	0	0	0	0.1	0.5	0.9	1	1	1
F'	1	1	0.9	0.5	0.1	0	0.1	0.5	1	1

Then the corresponding

$$F \cap G = \{0, 0, 0, 0.1, 0.5, 0.9, 1, 0.9, 0.5, 0, 0\}$$

$$F \cup G = \{0, 0, 0.1, 0.5, 0.9, 1, 0.9, 0.5, 1, 1, 1\}$$

Clearly F is not a subset of G and G is not a subset of F.

3.12.4 TRUTH VALUE

1) Truth value of the negation of a proposition in fuzzy logic is one minus the truth value of the proposition.

For example, if the truth value of the statement 'Ram is happy' is 0.8, then the truth value of 'Ram is not happy' is 0.2.

2) Truth value of the conjunction of two propositions in the fuzzy logic is the minimum of the truth values of the two propositions.

3) Truth value of the disjunction of two propositions in the fuzzy logic is the maximum of the truth values of two propositions.

For example, if the truth value of 'Ram is tall' is 0.8, and the truth value of 'John is tall' is 0.6, then find the truth value of the statement

(a) 'Ram and John are tall' is 0.6.

(b) 'Ram or John is tall' is 0.8.

(c) 'It is false that Ram or John is tall' is negation of maximum of truth value of 'Ram is tall' i.e., 0.8 and truth value of 'John is tall' i.e., 0.6 i.e., 0.8 is 0.2.

(d) 'Ram is not tall or John is not tall' is maximum of 0.4 and 0.2 i.e., 0.4.

3.12.5 ALGEBRAIC OPERATIONS ON FUZZY SETS

(1) Algebraic sum of two fuzzy sets F and G is defined by the membership function

$$\mu_{F+G}(x) = \mu_F(x) + \mu_G(x) - \mu_F(x) \cdot \mu_G(x)$$

and is written as $F + G$.

Algebraic product of two fuzzy sets F and G is defined by the membership function

$$\mu_{FG}(x) = \mu_F(x) \cdot \mu_G(x)$$

and is written as FG.

(2) Properties of fuzzy set operations which are same as in crisp set operations, are as under

1. Idempotent : $F \cup F = F$, $F \cap F = F$
2. Identity : $F \cup \phi = F$, $F \cap U = F$
3. Commutative : $F \cup G = G \cup F$, $F \cap G = G \cap F$
4. Distributive : $F \cap (G \cap H) = (F \cap G) \cup (F \cap H)$, $F \cup (G \cap H) = (F \cup G) \cap (F \cup H)$
5. Associative : $(F \cup G) \cup H = F \cup (G \cup H)$, $(F \cap G) \cap H = F \cap (G \cap H)$
6. Absorption : $F \cup (F \cap G) = F$, $F \cap (F \cup G) = F$.

3.12.6 GENERATION OF RULES FOR FUZZY PROBLEMS

We should know before hand all possible input-output relations while dealing with problems concerning fuzzy engineering/fuzzy methods. These input output relations are then expressed with 'if ... then' statements.

For instance, if F_i and G_i are inputs of fuzzy problems and R_i are the action taken for each rule, then the set of 'if ... then' rules with two input variables F_i and G_i and the various values are shown in table 3.

i.e., If F_1 and/or G_1, then R_{11}, else
If F_2 and/or G_1, then R_{21}, else
If F_1 and/or G_2, then R_{12}, else
If F_2 and/or G_2, then R_{22}.

Table 1

F_2	R_{21}	R_{22}
F_1	R_{11}	R_{12}
	G_1	G_2

In case, the fuzzy statements have more variables, then 'If ... then' rules becomes more complicated to tabulate.

However such a tabulation can be simplified by following a decomposition process as follows :

The decomposition process of three fuzzy variables F, G and H with relation R is taken as shown below :

Table 2

F_1	R_{21}	R_{11}
F_2	R_{22}	R_{12}
	H_2	H_1

Table 3

N_1	R_{221}	R_{211}	R_{121}	R_{111}
N_2	R_{222}	R_{212}	R_{122}	R_{112}
	G_2	G_1	G_2	G_1

Here the state events F_1 and G_1 and H_1 then R_{111} is decomposed as 'If F_1 and G_1 then R_{11}' and 'If N_1 and N_1 then R_{111}'.

Similarly R_{211} and R_{221} then R_{2211}.

This decomposition process can easily be extended to any number of input variables.

3.8.2 FUZZY PROPOSITIONS

(1) *A fuzzy number is a fuzzy set $R \to [0, 1]$.* We can easily extend classical two-valued logic to three-valued logic. Fuzzy logic, however is an extension of multi-valued logic. It provides foundations for approximate reasoning with imprecise fuzzy propositions using fuzzy set theory.

The classical propositions are statements which are either true or false. In fuzzy logic, the truth or falsity of fuzzy propositions is assigned different degrees, i.e., the truth and falsity are expressed by numbers in [0, 1].

A variable whose values are 'words' or 'sentences' in a natural language is called a linguistic variable. For example 'height' is a linguistic variable and its values are tall, very tall, quite tall, not tall, short, not very short, not quite tall etc.

(2) **Classification of fuzzy propositions.** The classical or propositions are statements which are either true or false. In fuzzy logic, the truth or falsity of fuzzy propositions is assigned different degrees i.e., the truth and falsity are expressed by numbers in [0, 1].

The fuzzy propositions of simple nature can be classified into the following four types. In each case, we introduce the relevant canonical form and then discuss its interpretation.

Type 1. *Unconditional and unqualified propositions.*

The standard canonical form of this type of proposition is expressed as

$$x \text{ is } F \qquad \ldots(1)$$

Here x is the variable that takes values in a given crisp universal set U and F is a fuzzy set on U which represents a fuzzy predicate such as young, tall, low, high etc. Given a particular value u for x, i.e., $x = u$, this value belongs to F with membership grade $F(u)$. This membership grade is then interpreted as the degree of truth for the proposition

$$x \text{ is } F \text{ is } F(u) \qquad \ldots(2)$$

Here F is a fuzzy set on U, which assigns the membership grade $F(u)$ to each value u of x.

In some fuzzy propositions, values of variable x in (2) are assigned to individuals in a given set U (i.e., variable x becomes a function $x : U \to u$, where $x(u)$ is the value of x for individuals in U. Accordingly the canonical form (2) is modified to the form

$$p : x(u) \text{ is } F \text{ is } F(x(u)) \qquad \ldots(3)$$

Example 3.8.1. *Consider a set U of persons each person in U is characterized by its age. Let a fuzzy set representing the predicate young is given. Interpret the variable by 'age' and fuzzy set by 'young', the canonical form is*

$$p : \text{age } (u) \text{ is young}$$

Solution. The degree of truth of this proposition $T(p)$ is now determined for each person x in X by means of the equation $T(p) = \mu_{young}(x)$.

Example 5.5. At a particular place on the earth, over the course of a year, the temperature is $T(x)$ for x days. Let $F(x)$ represent the membership function for predicate 'high'. Assuming that all values of x and $T(x)$ are known, the corresponding fuzzy proposition is expressed as:

p : Today $(x = x_0)$ is a high $T(x)$

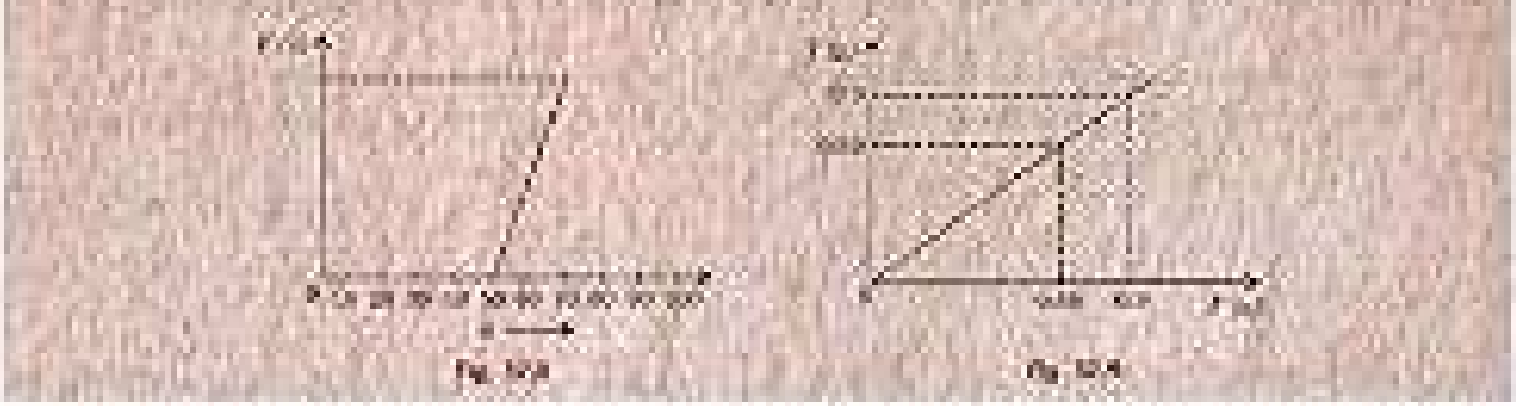


Fig. 5.9 Fig. 5.10

Solution. The degree of truth $T(p)$ depends upon the actual value of the temperature and on the nature of predicate 'high' which is defined by the membership function F in Fig. 5.9.

e.g., If $x = 75$ then $F(75) = 0.65$ and $T(p) = 0.65$.

Type II: Conditional and unqualified propositions

A proposition p of this type is expressed by the canonical form

$$p : \text{If } x \text{ is } F \text{ then } y \text{ is } G \qquad \ldots(4)$$

where x, y are variables whose values are in the sets X, Y and F, G are fuzzy sets on X and Y respectively. These propositions may also be viewed as propositions of the form

$$(x, y) \text{ is } R \qquad \ldots(5)$$

where R is a fuzzy set on $X \times Y$ which is determined for each $x \in X$ and each $y \in Y$ by the formula

$$\mu_R(x, y) = \mu(F(x), G(y)), \qquad \ldots(6)$$

where μ is a binary operation on $[0, 1]$ representing a suitable fuzzy implication.

Type III: Categorical and qualified propositions

A proposition of this type is expressed by either of the following canonical forms:

$$p : a \text{ is } F \text{ is } S' \qquad (7)$$

or

$$p : \text{Prob}(U \text{ is } F) \text{ is } P \qquad \ldots(8)$$

where a is a variable that takes values u from some universal set U and F is a fuzzy set on U which represents a fuzzy predicate such as small, young, daughter etc.

Prob. $(U$ is $F)$ is the probability of fuzzy set event 'u is F'; S is the fuzzy truth qualifier and P is the fuzzy probability qualifier. S and P are both represented by fuzzy sets on $[0, 1]$.

Example 5.6. (a) Mary is 'young' is true (or false) when the predicate 'young' and the truth qualifier 'very true' are represented by the membership functions shown in Fig. 5.10.

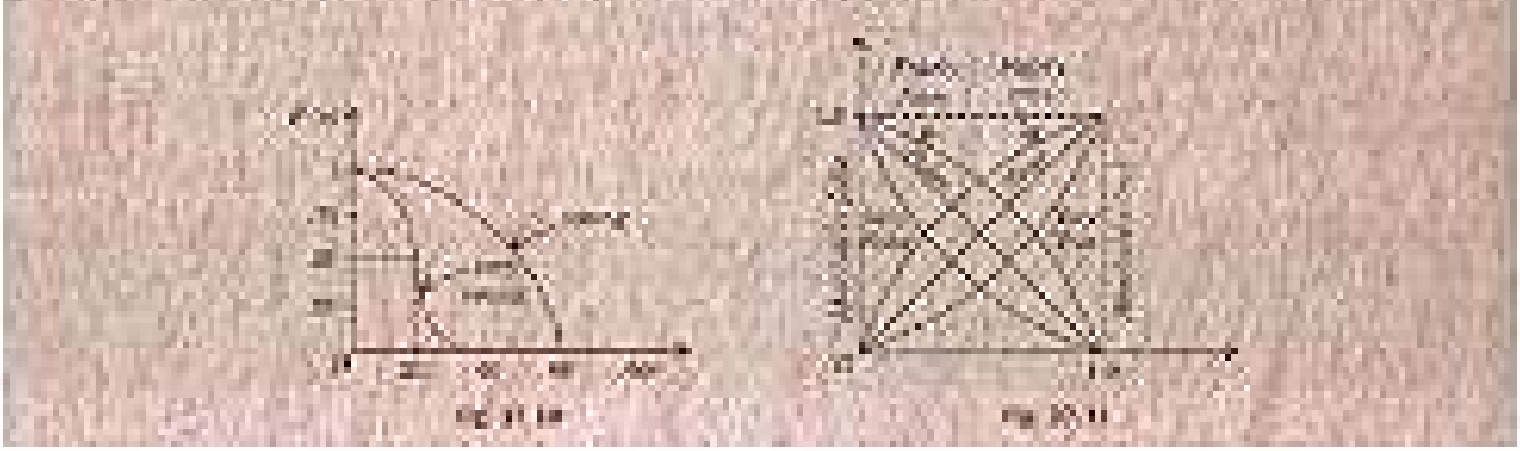


Fig. 5.10 Fig. 5.11

Solution. The degree of truth $T(p)$ of any truth-qualified proposition p is given for each $x \in U$ by $T(p) = \mu_F(x)$.

Assuming that Mary's age is 22, she belongs to the set representing the predicate 'very young' with membership grade 0.80, so proposition belongs to the set of propositions which are 'very true' with membership grade 0.80 as shown in Fig. 17.11. This implies that the degree of truth of our truth-qualified proposition is 0.8.

If the proposition be modified by changing the predicate to 'young' or the truth qualifier to 'fairly true', we would obtain the corresponding degree of truth of such propositions by the same method.

Type IV. *Conditional and qualified propositions*

This type of propositions are either be expressed in the canonical form

p: If x is F, then y is G is S

or p: Prob [x is F] is G is F,

where Prob [x is F] is G] is conditional probability.

SOME APPLICATIONS OF FUZZY SETS

The concept of fuzzy sets has already influenced all engineering disciplines in various degrees.

Electrical engineering is the first such discipline where the utility of fuzzy logic and fuzzy sets has been recognized by developing controllers. Electronic circuits for fuzzy image processing have also been developed.

Some ideas regarding the application of fuzzy sets in civil engineering emerged around 1970. In the construction of bridges, dams, buildings etc. a designer has to take into account the safety factor for which the fuzzy theory has a definite role to play. Fuzzy set theory has also proved quite useful for executing the job of existing constructions.

In *mechanical engineering* design problems, the utility of fuzzy set theory was realized during mid 1960's. The membership functions is expressed in terms of thermal expansion or corrosion or cost of different materials etc.

With the utility of fuzzy controllers was increasingly felt around mid 1980's, the need for computer hardware to implement the various operations involving fuzzy logic, had been recognized. In digital mode, fuzzy sets have been expressed as vectors of 0, 1 numbers.

Fuzzy control and *fuzzy decision making* are two well-developed areas of fuzzy set theory. These are directly relevant to industrial engineering problems. The utility of fuzzy sets has also been recognized for estimating the service life of given equipment under various conditions.

Modern Reliability theory has also been developed on the assumption of fuzzy sets. At any given time, an engineering product may be in functioning state to some degree or in failed state to another degree. The behaviour of an engineering product with respect to its functioning state and failed state has been characterized as based on fuzzy set theory.

The use of fuzzy set theory in *Robotics* includes approximate reasoning, fuzzy controllers, fuzzy pattern recognition and fuzzy data bases.

EXERCISE 17.1

1. Given fuzzy sets P_1 = {0.1/Delhi, 0.8/Kanpur, 0.3/Patna, 0.5/Mumbai}
   P_2 = {0.2/Mumbai, 0.3/Delhi, 0.8/Kanpur, 0.9/Patna}
   and P_3 = {0.6/Patna, 0.2/Mumbai, 0.3/Delhi, 0.4/Kanpur}
   Which of the above two sets are equal?
2. Write down explicitly each of the fuzzy sets P: 'Tall' in {Ram, Hari, Jadu, Shyam}
3. If P = {0.2/x_1, 0.7/x_2, 0.9/x_3, 0.6/x_4} and Q = {0.4/x_1, 0.6/x_2, 0.3/x_3, 0.8/x_4}
   Determine fuzzy sets corresponding to $P \cup Q$ and $P \cap Q$.
4. State whether each of the negation of the following propositions
   (i) The value of $\sqrt{2}$ is rational.
   (ii) The value of $\sqrt{3}$ is rational.
   (iii) The value of $\sqrt{5}$ is irrational.
5. Let the membership functions of fuzzy sets P and Q be as follows:
   X = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

CHAPTER 38

Tensor Analysis

1. Introduction. 2. Summation Convention. 3. Transformation of coordinates. Tensor of order zero. 4. Kronecker Delta. 5. Contravariant vectors. Covariant vectors. 6. Tensors of higher order. 7. Symmetric and skew-symmetric tensors. 8. Addition of tensors. 9. Outer product of two tensors. 10. Contraction of tensors. 11. Inner product of two tensors. 12. Quotient Law. 13. Reciprocal tensor (pure). Metric tensor. 14. Conjugate tensor. 15. Associated vectors. 16. Length of a vector. Angle between two vectors. 17. Christoffel symbols. 18. Transformation of Christoffel symbols. 19. Covariant differentiation of covariant vector. Covariant differentiation of a contravariant vector. 20. Gradient, Divergence, Curl.

38.1 INTRODUCTION

Many physical quantities are specified by their magnitude only or by those by their magnitude and direction. But certain quantities are associated with two or more directions. Such a quantity is called a tensor. The stress at a point of an elastic solid is an example of a tensor which depends on two directions—one normal to the area and other that of the force on it.

The properties of tensors are independent of the frames of reference used to describe them. That is why Einstein found tensors as a convenient tool for formulation of his theory of relativity. Since then, the subject of tensor analysis has come into prominence and is of great use in the study of Riemannian geometry, mechanics, elasticity, electromagnetic theory and numerous other fields of science and engineering. The main purpose of tensor calculus as a systematic subject is due to Ricci and his student Levi-Civita.

38.2 SUMMATION CONVENTION

Consider a sum of the type

$$a_1 x_1 + a_2 x_2 + \ldots + a_n x_n \quad \text{i.e.,} \quad \sum_{r=1}^{n} a_r x_r \qquad \ldots(1)$$

In tensor analysis, the subscripts of the symbols $x_1, x_2, \ldots x_n$ are replaced by superscripts and we write these as $x^1, x^2, \ldots x^n$. The superscripts do not stand for the various powers of x but serve as labels to distinguish different symbols. The power of a symbol say x will be indicated as $(x)^2$, $(x)^3$ etc. Hence (1) is written as

$$\sum_{r=1}^{n} a_r x^r. \qquad \ldots(2)$$

A still simpler notation is to drop the Σ sign and write (2) as $a_r x^r$ $\qquad \ldots(3)$

In (3) it is repeated twice as a subscript and a superscript. The repeated index r takes the values $1, 2, \ldots n$ and the expression (3) represents the sum of all such terms. The repeated index r over which the summation is to be done, is called a dummy index since it does not appear in the final result. This notation, known as **Einstein's summation convention**, is due to Einstein. We shall adopt this convention throughout this chapter and take the same where a letter appears twice in a term once as a subscript and once as a superscript.

Example [3.2] Write the covariant tensor of rank 3 whose components are $a_{ijk} = i + j + k$.

Solution. Since the indices are covariant both as subscript and as a superscript, we find that each of i, j, k is 1 or 2.

$\therefore$ $$A = a_{ijk}\, x^i x^j x^k = a_{111}x^1x^1x^1 + a_{112}x^1x^1x^2 + a_{121}x^1x^2x^1 + a_{122}x^1x^2x^2 + a_{211}x^2x^1x^1 + a_{212}x^2x^1x^2 + a_{221}x^2x^2x^1 + a_{222}x^2x^2x^2$$

Now each term in A has to be summed up over all repeated indices i, j, k from 1 to 2.

$\therefore$ $$A = a_{111}x^1x^1x^1 + a_{112}x^1x^1x^2 + a_{121}x^1x^2x^1 + a_{122}x^1x^2x^2 + a_{211}x^2x^1x^1 + a_{212}x^2x^1x^2 + a_{221}x^2x^2x^1 + a_{222}x^2x^2x^2$$

$$= a_{111}(x^1)^3 + a_{112}(x^1)^2x^2 + a_{121}(x^1)^2x^2 + a_{122}x^1(x^2)^2 + a_{211}x^2(x^1)^2 + a_{212}x^1x^2x^1 + a_{221}(x^2)^2x^1 + a_{222}(x^2)^3.$$

Example [3.3] If $f(x^1, x^2, \ldots, x^n)$ is a function of n variables $x^1, x^2, \ldots, x^n$ then find df.

Solution. Since $f = f(x^1, x^2, \ldots, x^n)$

$\therefore$ From Calculus, we have

$$df = \frac{\partial f}{\partial x^1}dx^1 + \frac{\partial f}{\partial x^2}dx^2 + \ldots + \frac{\partial f}{\partial x^n}dx^n = \frac{\partial f}{\partial x^i}dx^i.$$

[3.3] TRANSFORMATION OF COORDINATES

In a 3-dimensional space, the coordinates of a point are (x^1, x^2, x^3) referred to a particular frame of reference. Similarly in an n-dimensional space, the coordinates of a point are n independent variables $(x^1, x^2, \ldots, x^n)$ with respect to a certain frame of reference. Let $(\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n)$ be the coordinates of the same point referred to another frame of reference. Suppose, $\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n$ are independent single-valued functions of $x^1, x^2, \ldots, x^n$ so that

$$\bar{x}^1 = \phi^1(x^1, x^2, \ldots, x^n)$$
$$\bar{x}^2 = \phi^2(x^1, x^2, \ldots, x^n)$$
$$\cdots\cdots\cdots\cdots\cdots$$
$$\bar{x}^n = \phi^n(x^1, x^2, \ldots, x^n)$$

or more briefly $$\bar{x}^i = \phi^i(x^1, x^2, \ldots, x^n) \quad \ldots(1)$$

We can solve these equations uniquely to express x^i as functions of $\bar{x}^i$ so that

$$x^i = \psi^i(\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n) \quad \ldots(2)$$

The equations (1) and (2) are said to define a transformation of the coordinates from one frame of reference to another.

(1) Scalars or invariants. A function $\phi(x^1, x^2, x^3)$ is called a scalar or an invariant if its numerical value does not change upon transformation of coordinates from x^1, x^2, x^3 to $\bar{x}^1, \bar{x}^2, \bar{x}^3$.

i.e., $$\phi(x^1, x^2, x^3) = \phi(\bar{x}^1, \bar{x}^2, \bar{x}^3)$$

A scalar or invariant is also called a *tensor of order zero* or *rank zero*.

[3.4] KRONECKER DELTA*

The quantity δ^i_j defined by the relations

$$\delta^i_j = 0, \quad \text{when } i \neq j$$

and $$\delta^i_j = 1, \quad \text{when } i = j \text{ (no sum)}$$

Evidently $$\delta^1_1 = \delta^2_2 = \delta^3_3 = \ldots = \delta^n_n = 1$$

while $$\delta^1_2 = \delta^2_1 = \delta^3_1 = \ldots = 0$$

*Called after the German mathematician Leopold Kronecker (1823-91) who made important contributions in number theory, algebra and group theory.

We note that by summing up w.r.t. the repeated index j,

$$a_{ij}b^j = a_{i1}b^1 + a_{i2}b^2 + a_{i3}b^3 + a_{i4}b^4 + \dots$$
$$= 0 + a_{i2} + 0 + 0 + \dots = a_{i2}$$

In general, $$a_{ij}b^j = a_{i1}b^1 + a_{i2}b^2 + \dots + a_{ij}b^j + \dots + a_{in}b^n$$
$$= 0 + 0 + \dots + a_{ij} \cdot 1 + \dots + 0 = a_{ij}$$

Example 10.2. Show that $a_{ij}A^iA^j = a_{ij}A^jA^i$ where a_{ij} is a skew-symmetric tensor of order two and A^i are contravariant vectors. (UPTU, 2003)

Solution. By expansion of determinants, we have

$$a_{11}A^1 + a_{12}A^2 + a_{13}A^3 = A$$
$$a_{21}A^1 + a_{22}A^2 + a_{23}A^3 = B$$
$$a_{31}A^1 + a_{32}A^2 + a_{33}A^3 = C$$

which may be compactly written as

$$a_{1j}A^j = A, a_{2j}A^j = B, a_{3j}A^j = C$$

Using Kronecker delta notation, these can be combined into a single equation

$$a_{ij}A^j = B_i^i$$

Similarly $$a_{ij}A^i = B_j^i, a_{ij}A^i = B_j^i$$

All these nine equations are included in $a_{ij}A^iA^j = B_j^i$.

Example 10.3. $\phi(x^1, x^2)$ is a function of two independent coordinates x^1, x^2. Show that

$$\frac{\partial \phi}{\partial x^1}\frac{\partial x^1}{\partial x^2} = \frac{\partial \phi}{\partial x^2}$$

Solution. The partial derivatives of ϕ in the two coordinate systems are different and are connected by the following formula of Differential Calculus :

$$\frac{\partial \phi}{\partial x^1} = \frac{\partial \phi}{\partial x^1}\frac{\partial x^1}{\partial x^1} + \frac{\partial \phi}{\partial x^2}\frac{\partial x^2}{\partial x^1} + \frac{\partial \phi}{\partial x^3}\frac{\partial x^3}{\partial x^1} = \frac{\partial \phi}{\partial x^1}\frac{\partial x^1}{\partial x^1}$$

In particular, when $\phi = x^j$, we have $$\frac{\partial x^j}{\partial x^i} = \frac{\partial x^j}{\partial x^k}\frac{\partial x^k}{\partial x^i} \quad \dots(1)$$

Since x^j is independent of x^i, $\partial x^j/\partial x^i = 0$, when $j \neq i$
$= 1$, when $j = i$ $\quad \dots(2)$

Hence the result follows from (1) and (2).

10.3. (1) CONTRAVARIANT VECTORS

Let $A^1, A^2, \dots A^n$ be n in a function of the coordinate system $x^1, x^2, \dots, x^n$ (i.e., x^i). If these transform to another system of coordinates $\bar{x}^1, \bar{x}^2, \dots \bar{x}^n$ (i.e., $\bar{x}^i$) according to the law

$$\bar{A}^i = \frac{\partial \bar{x}^i}{\partial x^j}A^j \quad \dots(3)$$

then A^i are called components of a contravariant vector or contravariant tensor of order one.

An example of a contravariant vector. Let us take n for the coordinates of a point x^i in x^i in an n-dimensional space.

Since x is a function of x^i (i.e., $x_1, x_2, x_3, \dots, x_n$), then we

$$dx^1 = \frac{\partial x^1}{\partial x^1}dx^1 + \frac{\partial x^1}{\partial x^2}dx^2 + \dots + \frac{\partial x^1}{\partial x^n}dx^n$$

$$= \frac{\partial x^1}{\partial x^j}dx^j$$ using the summation convention.

Comparing this with (1), it follows that the set of differentials $dx^1, dx^2, \ldots, dx^n$ is an example of a contravariant vector. That is why the coordinates of a point are written with superscripts and not by subscripts.

(2) Covariant vectors. Let A_i be a set of n functions of the coordinate system x^i. If they transform to another system of coordinates X^i according to the law

$$A_i = \frac{\partial x^j}{\partial X^i} A_j \qquad \ldots(2)$$

then A_i are called the components of a covariant vector or covariant tensor of order one.

An example of a covariant vector. Let ϕ be a function which has a fixed value at each point of space independent of the coordinate system employed. Therefore, ϕ is a function of the coordinates x^i in the first system and a function of the coordinates X^i in the second system. By the chain rule

$$\frac{\partial \phi}{\partial x^1} = \frac{\partial \phi}{\partial x^1}\frac{\partial x^1}{\partial x^1} + \frac{\partial \phi}{\partial x^2}\frac{\partial x^2}{\partial x^1} + \ldots + \frac{\partial \phi}{\partial x^n}\frac{\partial x^n}{\partial x^1} = \frac{\partial \phi}{\partial x^1}\frac{\partial x^1}{\partial x^1} + \frac{\partial \phi}{\partial x^2}\frac{\partial x^2}{\partial x^1} + \ldots + \frac{\partial \phi}{\partial x^n}\frac{\partial x^n}{\partial x^1}$$

Comparing this equation with (2), it follows that the set of derivatives

$$\partial\phi/\partial x^1, \partial\phi/\partial x^2, \ldots, \partial\phi/\partial x^n$$

form a covariant vector.

Example 20.6. *A contravariant vector has components $A_1 = x^2, A_2 = 2x^1$ in rectangular coordinates. Find its covariant components in spherical coordinates.*

Solution. Here
$$x^1 = x, x^2 = y, x^3 = z \qquad \ldots(1)$$
and
$$X^1 = r, X^2 = \theta, X^3 = \phi$$

Let
$$A_1 = xy, A_2 = 2y - z^2, A_3 = zx \qquad \ldots(2)$$

According to the law of transformation, we have $\bar{A}_i = \dfrac{\partial x^j}{\partial X^i} A_j \quad (i = 1, 2, 3)$

and we wish to evaluate $\bar{A}_1, \bar{A}_2, \bar{A}_3$ where A_1, A_2, A_3 are known.

We know that $x = r\sin\theta\cos\phi, y = r\sin\theta\sin\phi, z = r\cos\theta$ $\ldots(3)$

Now
$$\bar{A}_1 = \frac{\partial x^1}{\partial X^1}A_1 + \frac{\partial x^2}{\partial X^1}A_2 + \frac{\partial x^3}{\partial X^1}A_3 = \frac{\partial x}{\partial r}xy + \frac{\partial y}{\partial r}(2y - z^2) + \frac{\partial z}{\partial r}zx \qquad \text{[From (1) and (2)]}$$

$$= \sin\theta\cos\phi \cdot r\sin\theta\cos\phi \cdot r\sin\theta\sin\phi + \sin\theta\sin\phi\,(2r\sin\theta\sin\phi - r^2\cos^2\theta) + \cos\theta \cdot r\sin\theta\cos\phi \cdot r\cos\theta \qquad \text{[From (3)]}$$

Similarly
$$\bar{A}_2 = \frac{\partial x^1}{\partial X^2}A_1 + \frac{\partial x^2}{\partial X^2}A_2 + \frac{\partial x^3}{\partial X^2}A_3 = r\cos\theta\cos\phi \cdot r\sin\theta\cos\phi \cdot r\sin\theta\sin\phi$$

$$+ r\sin\theta\sin\phi\,(2r\sin\theta\sin\phi - r^2\cos^2\theta) + (-r\sin\theta) \cdot r\sin\theta\cos\phi \cdot r\cos\theta$$

and
$$\bar{A}_3 = \frac{\partial x^1}{\partial X^3}A_1 + \frac{\partial x^2}{\partial X^3}A_2 + \frac{\partial x^3}{\partial X^3}A_3$$

$$= -r\sin\theta\sin\phi \cdot r\sin\theta\cos\phi \cdot r\sin\theta\sin\phi$$

$$+ r\sin\theta\cos\phi\,(2r\sin\theta\sin\phi - r^2\cos^2\theta) + 0$$

20.5. TENSORS OF HIGHER ORDER

Let i and j be each given values 1 or 2, then the symbol A^{ij} will give rise to n^2 functions.

If A^{ij} be a set of n^2 functions of the coordinates $x^1, x^2, \ldots, x^n$ which correspond to $\bar{A}^{ij}$ in another system of coordinates $X^1, X^2, \ldots, X^n$, according to the law

$$\bar{A}^{ij} = \frac{\partial X^i}{\partial x^r}\frac{\partial X^j}{\partial x^s}A^{rs} \qquad \ldots(1)$$

then the functions A^{ij} are said to be components of a contravariant tensor of the second order.

(2) If A_{ij} be a set of n^2 functions of $x^1, x^2, \ldots, x^n$ which transform to $\bar{A}_{ij}$ in another system of coordinates, $\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n$ according to the law

$$\bar{A}_{ij} = \frac{\partial x^r}{\partial \bar{x}^i} \frac{\partial x^s}{\partial \bar{x}^j} A_{rs} \qquad \ldots(2)$$

then the functions A_{ij} are said to be the components of a covariant tensor of the second order.

(3) If A_j^i be a set of n^2 functions of $x^1, x^2, \ldots, x^n$ which transform to $\bar{A}_j^i$ in another system of coordinates, $\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n$, according to the law

$$\bar{A}_j^i = \frac{\partial \bar{x}^i}{\partial x^r} \frac{\partial x^s}{\partial \bar{x}^j} A_s^r \qquad \ldots(3)$$

then A_j^i are said to be the components of a mixed tensor of the second order. It transforms like a contravariant vector with respect to the index i and like a covariant vector with respect to the index j. That is why i is placed as a superscript and j as subscript.

We can similarly define tensors of the orders higher than two.

(4a) To show that the sum (or difference) of two tensors of the same type and order is a tensor of the same type and order.

Example 9.6. Show that δ_j^i the Kronecker delta is a mixed tensor of order two.

Solution. If δ_j^i transforms to $\bar{\delta}_j^i$ in the coordinate system $\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n$ by the law for mixed tensors of order two, then

$$\bar{\delta}_j^i = \frac{\partial \bar{x}^i}{\partial x^r} \frac{\partial x^s}{\partial \bar{x}^j} \delta_s^r = \frac{\partial \bar{x}^i}{\partial x^r} \cdot \frac{\partial x^r}{\partial \bar{x}^j} = \frac{\partial \bar{x}^i}{\partial \bar{x}^j} = \delta_j^i \qquad [\because \delta_s^r \text{ will be } 1 \text{ if } r = s]$$

Hence δ_j^i is a mixed tensor of order two, having the same components in every coordinate system.

Example 9.7. Show that the velocity of a fluid at any point is a contravariant tensor of order one.

Solution. Let $dx^1/dt, dx^2/dt, dx^3/dt$ be the components of fluid velocity at the point (x^1, x^2, x^3) i.e., dx^i/dt is the component of velocity in the coordinate system x^i. Suppose the corresponding components of velocity in the coordinate system $\bar{x}^i$ are $d\bar{x}^i/dt$. Then $\bar{x}^1, \bar{x}^2, \bar{x}^3$ being the functions of x^1, x^2, x^3 which in turn are functions of t, we can write

$$\frac{d\bar{x}^1}{dt} = \frac{\partial \bar{x}^1}{\partial x^1} \frac{dx^1}{dt} + \frac{\partial \bar{x}^1}{\partial x^2} \frac{dx^2}{dt} + \frac{\partial \bar{x}^1}{\partial x^3} \frac{dx^3}{dt}$$

or

$$\frac{d\bar{x}^i}{dt} = \frac{\partial \bar{x}^i}{\partial x^r} \frac{dx^r}{dt} \qquad \ldots(1)$$

Now according to the law of tensor transformation, (1) shows that the velocity of a fluid is a contravariant tensor of rank one.

Example 9.8. Show that there is no distinction between contravariant and covariant vectors if the transformation law is of the form $\bar{x}^i = a_{ij} x^j + b^i$, where a's and b's are constants such that $|a_{ij}| \neq 0$.
(Rajasthan, 2000)

Solution. Given transformation $\qquad \bar{x}^i = a_{ij} x^j + b^i \qquad \ldots(1)$

yields $\qquad \frac{\partial \bar{x}^i}{\partial x^j} = a_{ij} \qquad \ldots(2)$

Also from (1), $\qquad a_{ij} x^j = a_{ij} a_{jk} x^k + a_{ij} b^j$

$$= \delta_{ik} x^k + a_{ij} b^j = x^i + a_{ij} b^j$$

$\therefore \qquad \frac{\partial x^i}{\partial \bar{x}^j} = a_{ij}^{-1}$ i.e., $\frac{\partial x^i}{\partial \bar{x}^j} = a_{ij}^{-1} \qquad \ldots(3)$

From (3) and (4), it is clear that any vector with components a, b, c will not transform in the same manner as the components whether transformation is over to a contravariant vector or over to a covariant vector. Thus in this case, there is no distinction between the two.

3.15 SYMMETRIC AND SKEW-SYMMETRIC TENSORS

(1) A tensor is said to be symmetric with respect to two contravariant (or two covariant) indices if its components remain unchanged on an interchange of the two indices.

Thus the tensor A^{ij} is symmetric if $A^{ij} = A^{ji}$ for every i and j.

(2) A tensor is said to be **skew-symmetric** with respect to two contravariant (or covariant) indices if its components change sign on interchange of the two indices.

Thus the tensor A^{ij} is skew-symmetric if $A^{ij} = -A^{ji}$ for every i and j.

In general, the tensor A^{ij}_{kl} is said to be symmetric or skew-symmetric in i and j according as

$$A^{ij}_{kl} = A^{ji}_{kl} \quad \text{or} \quad -A^{ji}_{kl}$$

Example 3.15. *Show that (a) a symmetric tensor of the second order has only $\frac{1}{2}n(n+1)$ different components.*

(b) A skew-symmetric tensor of the second order has only $\frac{1}{2}n(n-1)$ different non-zero components.

Solution. (a) Let A^{ij} be a symmetric tensor of order two so that $A^{ij} = A^{ji}$.

If each of the indices i and j take the values 1 to n, then A^{ij} will have n^2 components. Out of these n^2 components, n components $A_{11}, A_{22}, \ldots A_{nn}$ are independent.

Thus the remaining components are $(n^2 - n)$ which can be taken in pairs i.e., $A_{12} = A_{21}, A_{13} = A_{31}$ etc.

Hence the total number of independent components

$$= n + \frac{1}{2}(n^2 - n) = \frac{1}{2}n(n+1).$$

(b) Let A^{ij} be a skew-symmetric tensor of order two so that $A^{ij} = -A^{ji}$. As above, A^{ij} will have n^2 components. Out of these, n components $A^{11}, A^{22}, \ldots A^{nn}$ are all zero i.e., $A^{11} = -A^{11}$.

Omitting these, there are $(n^2 - n)$ components. Since $A^{12} = -A^{21}, A^{13} = -A^{31}$ etc., therefore ignoring the signs, $(n^2 - n)$ components can be taken in pairs.

Hence the total number of independent non-zero components

$$= \frac{1}{2}(n^2 - n) = \frac{1}{2}n(n-1).$$

EXERCISE 3.1

1. Write down the following sums using the summation convention :

   (a) $\frac{dy}{dx} = \frac{\partial y}{\partial x^1}\frac{dx^1}{dx} + \frac{\partial y}{\partial x^2}\frac{dx^2}{dx} + \cdots + \frac{\partial y}{\partial x^n}\frac{dx^n}{dx}$ (b) $s^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

2. Write out in full the following :

   (a) $a_{ij}x^i x^j$ $(i, j = 1, 2, 3)$ (b) $a_{ij}x^i x^j$ $(i, j = 1, 2, 3)$

   (c) $\delta_{ij}a^{ij}$.

3. (a) Show that δ^i_j is an invariant. (Meerut 2003)

   (b) If a_{ij} is a tensor, show that $a_{ij}x^i x^j$ is an invariant.

4. Show that (a) $\frac{\partial x^i}{\partial x^j} = \delta^i_j$ (b) $\frac{\partial x^i}{\partial x^j}\frac{\partial x^j}{\partial x^k} = \delta^i_k$.

5. If the n variables independent functions of $x^1, x^2, \ldots x^n$ and $x^1, x^2, \ldots x^n$ are functions of $x^1, x^2, \ldots x^n$, show that

$$\frac{\partial x^i}{\partial x^j}\frac{\partial x^j}{\partial x^k} = \delta^i_k$$

6. Write down the law of transformation for the scalars
   (a) A_i^j (b) Γ_{ijk}^i
7. A quantity $A(i, j, k, l, m)$ which is a function of the coordinates of tensors in another coordinate system $\bar{x}^i$ according to the law
$$A(r, s, t, u, v) = \frac{\partial \bar{x}^r}{\partial x^i}\frac{\partial \bar{x}^s}{\partial x^j}\frac{\partial x^k}{\partial \bar{x}^t}\frac{\partial x^l}{\partial \bar{x}^u}\frac{\partial x^m}{\partial \bar{x}^v} A(i, j, k, l, m)$$
   is this quantity a tensor? If so, express it suitably and state its nature and rank?
8. If the components of two tensors are equal in one coordinate system, show that they are equal in all coordinate systems.
9. A covariant tensor has components $2xz, -yz, y^2$ in cylindrical coordinate system. Find its components in
   (a) rectangular coordinate system (Bhopal, M.E., 2008) (b) spherical coordinates.
10. If g_{ij} denotes the metric tensor of a two-dimensional manifold of rank two, show that the product g_{ij} dx dx is an invariant under
    (Delhi, 2008)
11. A third-order and second-order tensors have components $xz, 4y, z$ in rectangular coordinates. Find the components in spherical coordinates.
12. Prove that $A_{ij}B^{ij}$ is an invariant, if B^{ij} and C^{ij} are contravariant vectors and A_{ij} is a covariant tensor.
    (Madras, M.E., 2005)
13. Show that $\delta_{ij}\delta^{ij}$ is not a tensor even though δ_{ij} is a covariant tensor of rank one. (Bhopal, 2008)
14. If a tensor A^{ijk}_{lm} is skew-symmetric with respect to the indices i and j in one coordinate system, show that it is skew-symmetric with respect to i and j in any coordinate system.

3.8 ADDITION OF TENSORS

The sum (or difference) of two tensors of the same order and type is another tensor of the same order and type.

Let A_{ij} and B_{ij} be two tensors of the same order and same type. Their components in the coordinate system $\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n$ are $\bar{A}_{ij}$ and $\bar{B}_{ij}$, such that
$$\bar{A}_{ij} = \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j} A_{kl} \quad \text{and} \quad \bar{B}_{ij} = \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j} B_{kl}$$
$$\therefore \quad \bar{A}_{ij} + \bar{B}_{ij} = \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j}(A_{kl} + B_{kl}) \quad \text{i.e.,} \quad \bar{C}_{ij} = \frac{\partial x^k}{\partial \bar{x}^i}\frac{\partial x^l}{\partial \bar{x}^j} C_{kl}$$

Thus C_{ij} transforms in exactly the same manner as A_{ij} and B_{ij} and is, therefore, a tensor of the same order and same type.

3.9 OUTER PRODUCT OF TWO TENSORS

If A^{ij} is a contravariant tensor of order two and B_{kl} is a covariant tensor of order two then their product is a mixed tensor C^{ij}_{kl} of order four such that
$$\bar{C}^{ij}_{kl} = \bar{A}^{ij}\bar{B}_{kl} = \left(\frac{\partial \bar{x}^i}{\partial x^r}\frac{\partial \bar{x}^j}{\partial x^s} A^{rs}\right)\left(\frac{\partial x^t}{\partial \bar{x}^k}\frac{\partial x^u}{\partial \bar{x}^l} B_{tu}\right) = \frac{\partial \bar{x}^i}{\partial x^r}\frac{\partial \bar{x}^j}{\partial x^s}\frac{\partial x^t}{\partial \bar{x}^k}\frac{\partial x^u}{\partial \bar{x}^l} A^{rs}B_{tu} = \frac{\partial \bar{x}^i}{\partial x^r}\frac{\partial \bar{x}^j}{\partial x^s}\frac{\partial x^t}{\partial \bar{x}^k}\frac{\partial x^u}{\partial \bar{x}^l} C^{rs}_{tu}$$

But this is the law of transformation of a mixed tensor of order four. Therefore, D^{ij}_{kl} is a mixed tensor of order four. Such products are called outer products of two tensors.

3.10 CONTRACTION OF A TENSOR

Consider a mixed tensor A^{ij}_{kl} of order four. By the law of transformation, we have
$$\bar{A}^{ij}_{kl} = \frac{\partial \bar{x}^i}{\partial x^r}\frac{\partial \bar{x}^j}{\partial x^s}\frac{\partial x^t}{\partial \bar{x}^k}\frac{\partial x^u}{\partial \bar{x}^l} A^{rs}_{tu}$$

In this, put the covariant index $l = n$ contravariant index k so that

$$A^{ij}_{\ \ k} = \frac{\partial \xi^i}{\partial x^r}\frac{\partial x^s}{\partial \xi^j}\frac{\partial x^t}{\partial \xi^k}\frac{\partial \xi^l}{\partial x^n}A^{rs}_{\ \ tn} = \frac{\partial \xi^i}{\partial x^r}\frac{\partial x^s}{\partial \xi^j}\frac{\partial x^t}{\partial \xi^k}A^{rs}_{\ \ tn}$$

$$-\frac{\partial \xi^i}{\partial x^r}\frac{\partial x^s}{\partial \xi^j}\frac{\partial^2 x^t}{\partial \xi^k \partial \xi^l}A^{rs}_{\ \ tn} - \frac{\partial \xi^i}{\partial x^r}\frac{\partial x^s}{\partial \xi^j}\frac{\partial^2 x^t}{\partial \xi^k \partial \xi^l}A^{rs}_{\ \ tn}$$

This shows that $A^{ij}_{\ \ k}$ is a contravariant tensor of order two.

The process of getting a tensor of lower order by 2 by putting a contravariant index equal to a covariant index and performing the summation indicated is known as contraction.

The tensors $A^{ij}_{\ \ k}$ and $A^{ij}_{\ \ l}$ obtained from contraction of the same tensor $A^{ij}_{\ \ kl}$ are generally different from each other unless the tensor $A^{ij}_{\ \ kl}$ is symmetric with respect to the pair of indices, $A^{ij}_{\ \ k} = A^{ji}_{\ \ k}$.

3.19 INNER PRODUCT OF TWO TENSORS

Given the tensors $A^{ij}_{\ \ k}$ and $B^{lm}_{\ \ n}$, *we first form their outer product $A^{ij}_{\ \ k}B^{lm}_{\ \ n}$ and contract this by putting $p = k$, then the result is $A^{ij}_{\ \ k}B^{lm}_{\ \ k}$ which is also a tensor, called the inner product of the given tensors.*

Hence the inner product of two tensors is obtained by first taking their outer product and then by contracting it. We can get several inner products for the same two tensors by contracting in different ways.

Example 3.11. *Show that any inner product of the tensors $A^{ij}_{\ \ k}$ and $B^{lm}_{\ \ n}$ is a tensor of rank three.*

Solution. The transformation laws for $A^{ij}_{\ \ k}$ and $B^{lm}_{\ \ n}$ are

$$\bar{A}^{ij}_{\ \ k} = \frac{\partial \xi^i}{\partial x^r}\frac{\partial \xi^j}{\partial x^s}\frac{\partial x^t}{\partial \xi^k}A^{rs}_{\ \ t} \quad \ldots(1) \qquad \text{and} \qquad \bar{B}^{lm}_{\ \ n} = \frac{\partial \xi^l}{\partial x^r}\frac{\partial \xi^m}{\partial x^s}\frac{\partial x^t}{\partial \xi^n}B^{rs}_{\ \ t} \quad \ldots(2)$$

∴ Inner product of $\bar{A}^{ij}_{\ \ k}$ and $\bar{B}^{lm}_{\ \ n}$ is

$$\bar{A}^{ij}_{\ \ k}\bar{B}^{lm}_{\ \ n} = \left(\frac{\partial \xi^i}{\partial x^r}\frac{\partial \xi^j}{\partial x^s}\frac{\partial x^t}{\partial \xi^k}\right)\left(\frac{\partial \xi^l}{\partial x^u}\frac{\partial \xi^m}{\partial x^v}\frac{\partial x^w}{\partial \xi^n}\right)A^{rs}_{\ \ t}B^{uv}_{\ \ w}$$

$$= \frac{\partial \xi^i}{\partial x^r}\frac{\partial \xi^j}{\partial x^s}\frac{\partial \xi^l}{\partial x^u}\frac{\partial \xi^m}{\partial x^v}\frac{\partial x^w}{\partial \xi^n}A^{rs}_{\ \ t}B^{uv}_{\ \ w} \qquad \left| \because \frac{\partial x^t}{\partial \xi^k}\frac{\partial \xi^l}{\partial x^u} = \frac{\partial x^t}{\partial x^u} = \delta^t_u \right.$$

$$= \frac{\partial \xi^i}{\partial x^r}\frac{\partial \xi^j}{\partial x^s}\frac{\partial x^w}{\partial \xi^n}A^{rs}_{\ \ w}B^{lm}_{\ \ n}$$

Hence the inner product of $A^{ij}_{\ \ k}$ and $B^{lm}_{\ \ n}$ is a tensor of rank 3.

Similarly putting $p = 1$ in the product of (1) and (2) and noting that

$$\frac{\partial x^t}{\partial \xi^1}\frac{\partial \xi^1}{\partial x^u} = \frac{\partial x^t}{\partial x^u} = \delta^t_u,$$

$A^{ij}_{\ \ 1}B^{lm}_{\ \ n}$ is found to be a tensor of rank 3.

Similarly, $A^{ij}_{\ \ p}B^{lm}_{\ \ p}$ can also be shown to be a tensor of rank 3.

3.20 QUOTIENT LAW

To ascertain that a set of given functions forms the components of a tensor, we have to verify if the functions obey the tensor transformation laws. But this is a very tedious job. A simple test is provided by the quotient law which states that if the inner product of a set of functions with an arbitrary tensor is a tensor, then the set of functions are the components of a tensor.

The proof of this law is given below for a particular case.

Example 3.9.1. Show that the quotient $A(i,j,k)$ is a tensor if it is the inner product with an arbitrary tensor B_{jk} is a tensor.

Solution. Let $$A(i,j,k)B_{jk} = C_i \quad \text{...(1)}$$

where C_i is a tensor. In the coordinate system x^i, let (1) transform to

$$\bar{A}(p,q,r)\bar{B}^{qr} = \bar{C}_p \quad \text{...(2)}$$

where $\bar{B}^{qr}$ and $\bar{C}_p$ are the components of the tensors B_{jk} and C_i. Expressing $\bar{B}^{qr}$ in terms of B_{jk} and $\bar{C}_p$ in terms of C_i, (2) takes the form

$$\bar{A}(p,q,r)\frac{\partial x^j}{\partial \bar{x}^q}\frac{\partial x^k}{\partial \bar{x}^r}B_{jk} = \frac{\partial x^i}{\partial \bar{x}^p}C_i \quad \text{...(3)}$$

Multiplying (1) by $\frac{\partial \bar{x}^p}{\partial x^s}\frac{\partial \bar{x}^q}{\partial x^t}$ and subtracting from (3), we get

$$\left[\bar{A}(p,q,r)\frac{\partial \bar{x}^p}{\partial x^s}\frac{\partial \bar{x}^q}{\partial x^t}\frac{\partial \bar{x}^r}{\partial x^k} - A(s,t,k)\frac{\partial \bar{x}^p}{\partial x^s}\frac{\partial \bar{x}^q}{\partial x^t}\right]B_{jk} = 0$$

Now B_{jk} being an arbitrary tensor, the quantity within the brackets must be identically zero, i.e.,

$$\bar{A}(p,q,r)\frac{\partial \bar{x}^p}{\partial x^s}\frac{\partial \bar{x}^q}{\partial x^t}\frac{\partial x^k}{\partial \bar{x}^r} - A(s,t,k)\frac{\partial \bar{x}^p}{\partial x^s}\frac{\partial \bar{x}^q}{\partial x^t}$$

or $$\bar{A}(p,q,r) = \frac{\partial x^s}{\partial \bar{x}^p}\frac{\partial x^t}{\partial \bar{x}^q}\frac{\partial x^k}{\partial \bar{x}^r}A(s,t,k)$$

But this is the law of tensor transformation. Hence $A(i,j,k)$ is a tensor of order 3 covariant with i, j as covariant indices and k as contravariant index.

EXERCISE 3.9

1. Prove that if a tensor equation is true for one coordinate system, it is true for all coordinate systems.
2. Show that every tensor can be expressed as the sum of two tensors, one of which is symmetric and the other skew-symmetric.
3. If A^{ij} and B^{ij} are tensors, prove that their sum and difference are also tensors.
4. Prove that A^i_j is a tensor. If its inner product with an arbitrary mixed tensor B^j_k is a tensor.
5. Prove that (i) the contraction of the tensor A^i_j is an invariant
   (ii) the contraction of the inner product of the tensors A^i and B_j is also an invariant.
6. Let A^{ij}_{kl} be a tensor; choose p and q and show that A^{ij}_{pq} is also a tensor. What is its rank?

3.10. (1) RIEMANNIAN SPACE

The distance ds between two adjacent points whose rectangular Cartesian coordinates are (x, y, z) and $(x + dx, y + dy, z + dz)$ is given by $ds^2 = dx^2 + dy^2 + dz^2$.

Riemann extended the concept of distance to a space of n dimensions and defined the distance ds between two adjacent points x^i and $x^i + dx^i$ $(i = 1, 2, \ldots, n)$ by the relation

$$ds^2 = a_{11}(dx^1)^2 + a_{22}(dx^2)^2 + \ldots + a_{nn}(dx^n)^2 + a_{12}dx^1dx^2 + \ldots + a_{ij}dx^idx^j + \ldots + a_{n-1,n}dx^{n-1}dx^n$$
$$= a_{ij}dx^idx^j, \text{ using summation convention.} \quad \text{...(1)}$$

The coefficients a_{ij} are the functions of the coordinates x^i. The quadratic form (1) is called a Riemannian metric and any space in which the distance is given by a such a metric is called a Riemannian space.*

If in a particular coordinate system X^i, the quadratic form (1) reduces to the form

$$ds^2 = dX^1{}^2 + dX^2{}^2 + \ldots + dX^n{}^2,$$

then (1) is called a Euclidean metric and the corresponding space is called the Euclidean space.

*See footnote on p. 103.

Note The geometry based on the Riemannian metric is called the Riemannian geometry and that based on the Euclidean metric is called the Euclidean geometry.

(2) Metric tensor. As in the physical space, the distance ds in the n-dimensional space is assumed to be independent of the coordinate system, i.e. a scalar invariant or a tensor of order zero. In the relation (1), dx^i and dx^j being displacements are components of a contravariant vector or a tensor of order one. Therefore, their outer product $dx^i dx^j$ is a contravariant tensor of order two. By the quotient law, the functions g_{ij} must be components of a covariant tensor of order two.

Let us write $\quad g_{ij} = s_{ij} + k_{ij} \quad$ where $s_{ij} = \frac{1}{2}(g_{ij} + g_{ji})$ and $k_{ij} = \frac{1}{2}(g_{ij} - g_{ji})$.

Interchanging i and j, we have $s_{ji} = \frac{1}{2}(g_{ji} + g_{ij}) = s_{ij}$ and $k_{ji} = \frac{1}{2}(g_{ji} - g_{ij}) = -k_{ij}$

$\therefore$ s_{ij} is symmetric and k_{ij} is skew-symmetric. Thus (1) takes the form

$$ds^2 = s_{ij}dx^i dx^j + k_{ij}dx^i dx^j$$

Now $k_{ij}dx^i dx^j$ is zero, since k_{ij} is skew-symmetric. Hence $ds^2 = g_{ij}dx^i dx^j$ where g_{ij} is a covariant symmetric tensor of order two. It is called the **metric tensor** or the **first fundamental tensor**.

3.3.3 CONJUGATE TENSOR

Let g be the determinant $|g_{ij}|$ and G_{ij} denote the cofactors of g_{ij} in g. Define the function of g^{ij} by the relation $g^{ij} = G_{ij}/g$ ...(2)

Since the functions g_{ij} and G_{ij} are symmetric in the subscripts, the functions g^{ij} will be symmetric in the superscripts. Now

$$g_{ij}g^{ij} = g_{ij}\frac{G_{ij}}{g} = \frac{g}{g} = 1 \quad \text{and} \quad g_{ij}g^{ij} = g_{ij}\frac{G_{ij}}{g}$$
$$= 1, \text{ if } i = j = 0, \text{ if } i \neq j$$

Thus $\quad g_{ij}g^{jk} = \delta_i^k$ ...(3)

If a^j is an arbitrary contravariant tensor, then its inner product with the tensor g_{ij} will be an arbitrary covariant tensor due to contraction, i.e.,

$$g_{ij}a^j = a_i \qquad ...(4)$$
$$\therefore \quad g^{ki}a_i = g^{ki}g_{ij}a^j = a^k$$

which is a contravariant tensor of order one. Therefore by quotient law, g^{ij} are the components of a contravariant tensor of order two. Here g^{ij} is a symmetric contravariant tensor which is called the **conjugate tensor** or the **second fundamental tensor**.

Note In view of (3), the relation between g_{ij} and g^{ij} is reciprocal. As such they are also called **reciprocal tensors** or **associated tensors**.

Example 3.3.1 Find the components g^{ij} and g_{ij} of the fundamental tensors in rectangular coordinates.

Solution. Let (x^1, x^2, x^3) be the rectangular cartesian coordinates and $(\bar{x}^1, \bar{x}^2, \bar{x}^3)$ be the spherical coordinates of a point so that

$$x^1 = x, x^2 = y, x^3 = z, \text{ and } \bar{x}^1 = r, \bar{x}^2 = \theta, \bar{x}^3 = \phi \qquad ...(5)$$

and

We know that $\quad x = r\sin\theta\cos\phi, y = r\sin\theta\sin\phi, z = r\cos\theta \qquad ...(6)$

Let g_{ij} and $\bar{g}_{ij}$ be the metric tensors in cartesian and spherical coordinates respectively.

Then $\quad ds^2 = dx^2 + dy^2 + dz^2 = g_{ij}dx^i dx^j$

$\therefore \quad g_{11} = 1 = g_{22} = g_{33}$ and $g_{12} = 0 = g_{23} = g_{31}$ etc. ...(7)

On transformation

$$\bar{g}_{ij} = \frac{\partial x^1}{\partial \bar{x}^i}\frac{\partial x^1}{\partial \bar{x}^j}g_{11} + \frac{\partial x^2}{\partial \bar{x}^i}\frac{\partial x^2}{\partial \bar{x}^j}g_{22} + \frac{\partial x^3}{\partial \bar{x}^i}\frac{\partial x^3}{\partial \bar{x}^j}g_{33} \qquad ...(8)$$

Putting $i = j = 1$ in (3), we have

$$g_{11} = \left(\frac{\partial x^1}{\partial \xi^1}\right)^2 g_{11} + \left(\frac{\partial x^2}{\partial \xi^1}\right)^2 g_{22} + \left(\frac{\partial x^3}{\partial \xi^1}\right)^2 g_{33}$$

$$= \left(\frac{\partial x}{\partial r}\right)^2 + \left(\frac{\partial y}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial r}\right)^2 \qquad \text{[By (1) and (2)]}$$

$$= \sin^2\theta\cos^2\phi + \sin\theta\cos\theta\sin\phi^2 + \cos^2\theta = \sin^2\theta + \cos^2\theta = 1 \qquad \text{[By (4)]}$$

Putting $i = j = 2$ in (3), we have

$$g_{22} = \left(\frac{\partial x^1}{\partial \xi^2}\right)^2 g_{11} + \left(\frac{\partial x^2}{\partial \xi^2}\right)^2 g_{22} + \left(\frac{\partial x^3}{\partial \xi^2}\right)^2 g_{33}$$

$$= \left(\frac{\partial x}{\partial \theta}\right)^2 + \left(\frac{\partial y}{\partial \theta}\right)^2 + \left(\frac{\partial z}{\partial \theta}\right)^2 \qquad \text{[By (1) and (2)]}$$

$$= r^2\cos\theta\cos\phi^2 + r^2\cos\theta\sin\phi^2 + r^2\sin^2\theta = r^2\cos^2\theta + r^2\sin^2\theta = r^2$$

Similarly $g_{33} = \left(\frac{\partial x^1}{\partial \xi^3}\right)^2 g_{11} + \left(\frac{\partial x^2}{\partial \xi^3}\right)^2 g_{22} + \left(\frac{\partial x^3}{\partial \xi^3}\right)^2 g_{33} = r^2\sin^2\theta$

and $g_{12} = 0 = g_{13} = g_{21} = g_{23} = g_{31} = g_{32}$

Hence the first fundamental tensor, written in matrix form, is

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{vmatrix}$$

or $g = \begin{vmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{vmatrix} = r^4\sin^2\theta$

and the cofactors in g are given by

$$G_{11} = r^4\sin^2\theta, G_{22} = r^2\sin^2\theta, G_{33} = r^2; G_{12} = 0 = G_{13} = G_{21} = G_{23} = G_{31} = G_{32}$$

The components of the second fundamental tensor are given by $g^{ij} = G_{ij}/g$. Hence the second fundamental tensor in matrix form, is $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1/r^2 & 0 \\ 0 & 0 & 1/r^2\sin^2\theta \end{vmatrix}$.

Example 19(D). *Find the components of the metric tensor and the conjugate tensor in cylindrical coordinates.*

Solution. Let (x^1, x^2, x^3) be the cartesian coordinates and (ξ^1, ξ^2, ξ^3) be the cylindrical coordinates of a point so that

$$x^1 = x, x^2 = y, x^3 = z \quad \text{and} \quad \xi^1 = \rho, \xi^2 = \phi, \xi^3 = z \qquad \ldots(1)$$

We know that $x = \rho\cos\phi, y = \rho\sin\phi, z = z \qquad \ldots(2)$

Let a_{ij} and $\bar{a}_{ij}$ be the metric tensors in cartesian and cylindrical coordinates respectively.

Then $d\bar{s}^2 = (dx^1)^2 + (dx^2)^2 + (dx^3)^2 = a_{ij}dx^i dx^j$

$\therefore \quad a_{11} = 1 = a_{22} = a_{33}$ and $a_{12} = 0 = a_{13} = a_{23}$ etc. $\qquad \ldots(3)$

On transformation,

$$\bar{a}_{ij} = \frac{\partial x^1}{\partial \xi^i}\frac{\partial x^1}{\partial \xi^j}a_{11} + \frac{\partial x^2}{\partial \xi^i}\frac{\partial x^2}{\partial \xi^j}a_{22} + \frac{\partial x^3}{\partial \xi^i}\frac{\partial x^3}{\partial \xi^j}a_{33} \qquad \ldots(4)$$

Putting $i = j = 1$ in (i), we have

$$g_{11} = \left(\frac{\partial x^1}{\partial y^1}\right)^2 + \left(\frac{\partial x^2}{\partial y^1}\right)^2 + \left(\frac{\partial x^3}{\partial y^1}\right)^2 = \left(\frac{\partial x}{\partial r}\right)^2 + \left(\frac{\partial y}{\partial r}\right)^2 + \left(\frac{\partial z}{\partial r}\right)^2$$

$$= \cos^2\theta + \sin^2\theta + 0 = 1 \qquad \text{[By (i) and (ii)]}$$

Putting $i = j = 2$ in (i), we have

$$g_{22} = \left(\frac{\partial x^1}{\partial y^2}\right)^2 + \left(\frac{\partial x^2}{\partial y^2}\right)^2 + \left(\frac{\partial x^3}{\partial y^2}\right)^2 = \left(\frac{\partial x}{\partial \theta}\right)^2 + \left(\frac{\partial y}{\partial \theta}\right)^2 + \left(\frac{\partial z}{\partial \theta}\right)^2$$

$$= (-r\sin\theta)^2 + (r\cos\theta)^2 + 0 = r^2 \qquad \text{[By (i) and (ii)]}$$

Similarly $g_{33} = \left(\frac{\partial x^1}{\partial z^1}\right)^2 + \left(\frac{\partial x^2}{\partial z^2}\right)^2 + \left(\frac{\partial x^3}{\partial z^3}\right)^2 = 0 + 0 + 1 = 1$

and $g_{12} = 0 = g_{21} = g_{13} = g_{31} = g_{23} = g_{32}$

Hence the metric tensor, written in matrix form, is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{i.e.,} \qquad g = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = r^2$$

Also cofactors in g are given by

$$G_{11} = r^2, G_{22} = 1, G_{33} = r^2, G_{12} = 0 = G_{21} = G_{13} = G_{31} = G_{23} = G_{32}$$

The components of the conjugate tensor are given by $g^{ij} = G_{ij}/g$.

Hence the conjugate tensor in matrix form is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/r^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1.12 ASSOCIATED TENSORS

From (3) of § 1.6.3A we have $g^{ij} A_j = A^i$ ... (1)

i.e., the inner product of the tensor a^j with the fundamental tensor g_{ij} is another tensor a_i which is called the associated tensor of a^j.

Similarly, we have $a_i g^{ij} = a^j$ ... (2)

Hence a^j is the associated tensor of a_i.

This fact of lowering of one tensor can be lowered or raised by forming its inner product with one of the fundamental tensors g_{ij} or g^{ij} as in (1) or (2) above.

1.13 (1) LENGTH OF A VECTOR

The vector A is given by

$$A = A_i a^i \quad \text{or} \quad A = A_j a^j \qquad \ldots (1)$$

Also we have the associated vector $A_i = g_{ij} A^j$ ... (2)

or $A^j = g^{ij} A_i$ ... (3)

$\therefore$ Length of vector $|A| = (A \cdot A)^{1/2} = (A_i a^i \cdot A_j a^j)^{1/2}$ [By (1)]

$= (g_{ij} A^i A^j)^{1/2}$ [$\because A_i = g_{ij} A^j$] [By (2)]

$= (A_j A^j)^{1/2}$ [By (3)]

Also length of vector $A = (A_i a^i \cdot A^j a_j)^{1/2} = (A_i g^{ij} \cdot A_j a^j)^{1/2} = (g^{ij} A_i A_j)^{1/2} = (A_i A^i)^{1/2}$ [By (3)]

Hence the magnitude or length of the vector $A = \sqrt{(g_{ij} A^i A^j)} = \sqrt{(g^{ij} A_i A_j)} = \sqrt{(A_i A^i)}$ ... (4)

which is an invariant.

Obs. The length of a vector A^i in N-dimensional space is $\sqrt{g_{ij}A^iA^j}$. Similarly the length of the vector $0, A^2, 0$ is $\sqrt{g_{22}A^2}$ and the length of the vector $0, 0, A^3$ is $\sqrt{g_{33}A^3}$. Hence the physical components of a vector A^i are $\sqrt{g_{11}}A^1, \sqrt{g_{22}}A^2, \sqrt{g_{33}}A^3$.

(2) Angle between two vectors. Let $\mathbf{A}$ and $\mathbf{B}$ be the given vectors such that

$$\mathbf{A} = A^i\mathbf{g}_i \text{ and } \mathbf{B} = B^j\mathbf{g}_j \qquad ...(1)$$

$$\therefore \quad \mathbf{A}\cdot\mathbf{B} = |\mathbf{A}|\,|\mathbf{B}|\cos\theta$$

or
$$\cos\theta = \frac{\mathbf{A}\cdot\mathbf{B}}{|\mathbf{A}||\mathbf{B}|} = \frac{A^iB^jg_{ij}}{\sqrt{(g_{ij}A^iA^j)}\sqrt{(g_{ij}B^iB^j)}} \qquad \text{[Using (1) and (2)]}$$

In terms of associated vectors, we have

$$\cos\theta = \frac{A_iB^i}{\sqrt{A^iA_i}\sqrt{B^iB_i}}.$$

EXERCISE 11.3

1. If $ds^2 = 6dx^2 + 4dy^2 + 3dz^2 - 2dxdy + 4dxdz$, find the value of g_{ij} and g^{ij}.
2. Find ds^2 corresponding to the metric $a^{ij} = \frac{a^{ij}}{1 - a^{ij}x^ix^j} + dx^1dx^1 + dx^2dx^2 + dx^3dx^3$.
3. The non-zero contravariant components of a vector A in plane polar coordinate system are $A^1 = r^2$ and $A^2 = \theta^2$. Using the transformation, obtain the non-zero components in polar coordinates $r = x^1$ and $\theta = x^2$.
4. Prove that the angles θ_{12}, θ_{23} and θ_{31} between the coordinate curves in a 3-dimensional coordinate system are given by $\cos\theta_{12} = \frac{g_{12}}{\sqrt{g_{11}g_{22}}}, \cos\theta_{23} = \frac{g_{23}}{\sqrt{g_{22}g_{33}}}, \cos\theta_{31} = \frac{g_{31}}{\sqrt{g_{33}g_{11}}}$.
5. Prove that for an orthogonal coordinate system
$$(i) g_{12} = g_{23} = g_{31} = 0 \qquad (ii) g^{11} = 1/g_{11}, g_{22} = 1/g^{22}, g_{33} = 1/g^{33}$$

11.15 CHRISTOFFEL SYMBOLS

Christoffel symbol of the first kind is denoted by $[ij, k]$ and is defined by

$$[ij, k] = \frac{1}{2}\left[\frac{\partial g_{jk}}{\partial x^i} + \frac{\partial g_{ki}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k}\right] \qquad ...(1)$$

where g_{ij} are the components of the metric tensor.

Christoffel symbol of the second kind is denoted by $\left\{\begin{matrix} l \\ ij \end{matrix}\right\}$ and is defined by

$$\left\{\begin{matrix} l \\ ij \end{matrix}\right\} = g^{lk}[ij, k] \qquad ...(2)$$

Some authors write Christoffel symbol of the second kind as $\{ij, l\}$ or Γ^l_{ij}.

Obs. 1. Here summation is understood in (2) for the dummy suffix k. The Christoffel symbol of the first kind is symmetric in the first two indices i and j but not necessarily symmetric in all the three indices. Similarly Christoffel symbol of the second kind is symmetric in the lower two indices i and j.

Obs. 2. It is evident from (1) and (2) that the Christoffel symbols of both kinds are symmetric in the indices i and j i.e.,
$$[ij, k] = [ji, k] \text{ and } \left\{\begin{matrix} l \\ ij \end{matrix}\right\} = \left\{\begin{matrix} l \\ ji \end{matrix}\right\}.$$

Example 11.16 If $ds^2 = 16dx^2 + y^2(dy^2 + z^2dz^2 + 2\cos\alpha\, dydz)$, find the values of

(a) $[22, 2]$ and $[23, 3]$ (b) $\left\{\begin{matrix} 1 \\ 22 \end{matrix}\right\}$ and $\left\{\begin{matrix} 3 \\ 12 \end{matrix}\right\}$.

Solution. In the 3-dimensional space in spherical coordinates such that

$$x^1 = r,\ x^2 = \theta \text{ and } x^3 = \phi$$

Clearly $g_{11} = 1, g_{22} = r^2, g_{33} = r^2 \sin^2\theta$ and $g_{ij} = 0$ for $i \neq j$. ...(i)

Also $g^{11} = 1, g^{22} = 1/r^2 = 1/g^{22}$ ...(ii) [See Ex. 3.8(12)]

(a) Christoffel symbols of the first kind are given by

$$[ij, k] = \frac{1}{2}\left[\frac{\partial g_{jk}}{\partial x^i} + \frac{\partial g_{ki}}{\partial x^j} - \frac{\partial g_{ij}}{\partial x^k}\right]_{i,j,k = 1,2,3} \quad \text{...(iii)}$$

Taking $i = 2, j = 2$, and $k = 1$ in (iii), we get

$$[22, 1] = \frac{1}{2}\left[\frac{\partial g_{21}}{\partial x^2} + \frac{\partial g_{21}}{\partial x^2} - \frac{\partial g_{22}}{\partial x^1}\right] = \frac{1}{2}\left[\frac{\partial(0)}{\partial\theta} + \frac{\partial(0)}{\partial\theta} - \frac{\partial(r^2)}{\partial r}\right] = -r \quad \text{...(iv)}$$

Putting $i = 1, j = 2$ and $k = 3$ in (iii), we obtain

$$[12, 3] = \frac{1}{2}\left[\frac{\partial g_{31}}{\partial x^2} + \frac{\partial g_{32}}{\partial x^1} - \frac{\partial g_{12}}{\partial x^3}\right] = \frac{1}{2}\left[\frac{\partial(r^2\sin^2\theta)}{\partial\theta} + \frac{\partial(0)}{\partial r} - \frac{\partial(0)}{\partial\phi}\right] = r\sin^2\theta \quad \text{...(v)}$$

(b) Christoffel symbols of the second kind are defined by

$$\left[\begin{matrix} k \\ ij \end{matrix}\right] = g^{k1}[ij, 1] + g^{k2}[ij, 2] + g^{k3}[ij, 3] \quad \text{...(vi)}$$

$$\left[\begin{matrix} 1 \\ 22 \end{matrix}\right] = g^{11}[22, 1] + g^{12}[22, 2] + g^{13}[22, 3] = [22, 1] + 0[22, 2] + 0[22, 3]$$

$$= -r \qquad \text{[By (iv)]}$$

$$\left[\begin{matrix} 3 \\ 12 \end{matrix}\right] = g^{31}[12, 1] + g^{32}[12, 2] + g^{33}[12, 3]$$

$$= 0[12, 1] + 0[12, 2] + \frac{1}{r^2\sin^2\theta}[12, 3] \qquad \text{[By (v)]}$$

$$= \frac{1}{r^2\sin^2\theta} \cdot r\sin^2\theta = \frac{1}{r}. \qquad \text{[By (v)]}$$

Example 3.8 (13) Prove that

(a) $\dfrac{\partial g^{ij}}{\partial x^k} = -[ik, j] - [jk, i]$ (b) $\dfrac{\partial^2 g^{ij}}{\partial x^k \partial x^l} = -g^{ir}\dfrac{\partial}{\partial x^l}\left[\begin{matrix} j \\ rk \end{matrix}\right] - g^{jr}\left[\begin{matrix} i \\ rk \end{matrix}\right]$

Solution. (a) By definition of Christoffel symbol of the first kind, we have

$$[ik, j] = \frac{1}{2}\left[\frac{\partial g_{ij}}{\partial x^k} + \frac{\partial g_{jk}}{\partial x^i} - \frac{\partial g_{ki}}{\partial x^j}\right] \quad \text{...(i)}$$

and

$$[jk, i] = \frac{1}{2}\left[\frac{\partial g_{ji}}{\partial x^k} + \frac{\partial g_{ik}}{\partial x^j} - \frac{\partial g_{kj}}{\partial x^i}\right] \quad \text{...(ii)}$$

Since g_{ij} is a symmetric tensor, $\therefore g_{ij} = g_{ji}, g_{jk} = g_{kj}, g_{ki} = g_{ik}$

Adding (i) and (ii), we get the required result.

(b) We know that $g^{ij}g_{jk} = \delta^i_k$ [Refer to § 3.8 (12)]

Differentiating w.r.t. x^l, we get

$$g^{ij}\frac{\partial g_{jk}}{\partial x^l} + g_{jk}\frac{\partial g^{ij}}{\partial x^l} = 0 \qquad \left[\because \frac{\partial \delta^i_k}{\partial x^l} = 0\right]$$

Multiplying by g^{kr} and transposing, we have

$$g^{ij}g^{kr}g_{jk}\frac{\partial g^{ij}}{\partial x^l} = -g^{ij}g^{kr}\frac{\partial g_{jk}}{\partial x^l} \text{ or } \delta^i_r\frac{\partial g^{ij}}{\partial x^l} = -g^{ij}g^{kr}[lk, j] + [lk, i] \qquad \text{[From (i)]}$$

or $$\frac{\partial^2 T}{\partial t^2} = c^2\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) + \frac{1}{\rho}\frac{\partial p}{\partial t} + \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right)$$

(neglecting u and $\frac{\partial}{\partial t}$) to obtain the desired result.

1.1.2 TRANSFORMATION OF DERIVATIVES; OPERATORS

We first determine the values of $\frac{\partial}{\partial x}$, $\frac{\partial}{\partial y}$ and $\frac{\partial}{\partial z}$ as functions of the coordinates x, y, z and these quantities in the $\frac{\partial}{\partial \xi}$, $\frac{\partial}{\partial \eta}$ and $\frac{\partial}{\partial \zeta}$ in another coordinate system ξ, η, ζ.

(a) Value of the differentiation of the complex quantity of the form

Let $f(z)$ which is a function of z, transform in $\phi(\zeta)$ to another coordinate system ζ. Then

$$f(z) = \frac{1}{2}\left[\frac{\partial \phi}{\partial \xi} + \frac{\partial \phi}{\partial \eta} - i\left(\frac{\partial \phi}{\partial \xi} - \frac{\partial \phi}{\partial \eta}\right)\right] \tag{1.1}$$

Since $f(z)$ is a complex variable of order two, we have

$$f(z) = \frac{\partial \phi}{\partial \xi}\frac{\partial \xi}{\partial z} + \frac{\partial \phi}{\partial \eta}\frac{\partial \eta}{\partial z} \tag{1.2}$$

Differentiating both sides w.r.t. z, we get

$$\frac{\partial f}{\partial z} = \left(\frac{\partial^2 \phi}{\partial \xi^2}\frac{\partial \xi}{\partial z} + \frac{\partial^2 \phi}{\partial \xi \partial \eta}\frac{\partial \eta}{\partial z}\right)\frac{\partial \xi}{\partial z} + \frac{\partial \phi}{\partial \xi}\frac{\partial^2 \xi}{\partial z^2} + \left(\frac{\partial^2 \phi}{\partial \eta \partial \xi}\frac{\partial \xi}{\partial z} + \frac{\partial^2 \phi}{\partial \eta^2}\frac{\partial \eta}{\partial z}\right)\frac{\partial \eta}{\partial z} + \frac{\partial \phi}{\partial \eta}\frac{\partial^2 \eta}{\partial z^2} \tag{1.3}$$

Since $f(z)$ is a function of z only, $\frac{\partial f}{\partial z}$ is independent of ζ. But ξ is a function of z and ζ only, and η is a function of z and ζ only (i.e. $\xi = \xi(z, \zeta)$, $\eta = \eta(z, \zeta)$).

Interchanging ζ and z in (1.3) and also in the last terms of (1.3), we have

$$\frac{\partial f}{\partial \zeta} = \left(\frac{\partial^2 \phi}{\partial \xi^2}\frac{\partial \xi}{\partial \zeta} + \frac{\partial^2 \phi}{\partial \xi \partial \eta}\frac{\partial \eta}{\partial \zeta}\right)\frac{\partial \xi}{\partial \zeta} + \frac{\partial \phi}{\partial \xi}\frac{\partial^2 \xi}{\partial \zeta^2} + \left(\frac{\partial^2 \phi}{\partial \eta \partial \xi}\frac{\partial \xi}{\partial \zeta} + \frac{\partial^2 \phi}{\partial \eta^2}\frac{\partial \eta}{\partial \zeta}\right)\frac{\partial \eta}{\partial \zeta} + \frac{\partial \phi}{\partial \eta}\frac{\partial^2 \eta}{\partial \zeta^2} \tag{1.4}$$

Similarly interchanging z and ζ in (1.3), we get

$$\frac{\partial^2 f}{\partial z^2} = \left(\frac{\partial^2 \phi}{\partial \xi^2}\frac{\partial \xi}{\partial z} + \frac{\partial^2 \phi}{\partial \xi \partial \eta}\frac{\partial \eta}{\partial z}\right)\frac{\partial \xi}{\partial z} + \frac{\partial \phi}{\partial \xi}\frac{\partial^2 \xi}{\partial z^2} + \left(\frac{\partial^2 \phi}{\partial \eta \partial \xi}\frac{\partial \xi}{\partial z} + \frac{\partial^2 \phi}{\partial \eta^2}\frac{\partial \eta}{\partial z}\right)\frac{\partial \eta}{\partial z} + \frac{\partial \phi}{\partial \eta}\frac{\partial^2 \eta}{\partial z^2} \tag{1.5}$$

Subtracting the values from (1.3), (1.4) and (1.5) in $\frac{\partial f}{\partial z}$, we obtain

$$\frac{\partial f}{\partial z} = \frac{\partial^2 \phi}{\partial \xi^2}\frac{\partial \xi}{\partial z}\frac{\partial \xi}{\partial \zeta} + \frac{\partial^2 \phi}{\partial \eta^2}\frac{\partial \eta}{\partial z}\frac{\partial \eta}{\partial \zeta} + \frac{\partial \phi}{\partial \xi}\frac{\partial^2 \xi}{\partial z \partial \zeta} + \frac{\partial \phi}{\partial \eta}\frac{\partial^2 \eta}{\partial z \partial \zeta} \tag{1.6}$$

This is the required form of transformation of Cauchy-Riemann equations.

(b) Value of transformation of the complex quantity of the second kind

Let $\phi^2(\xi, \eta)$ transform to $\phi^2(\zeta)$.

Since ϕ^2 is a complex variable of order two,

$$\phi^2 = \frac{\partial \phi^2}{\partial \xi^2}\frac{\partial \xi}{\partial z} + \frac{\partial \phi^2}{\partial \eta^2}\frac{\partial \eta}{\partial z} \tag{1.7}$$

Hence ϕ_z becomes $\frac{\partial \phi^2}{\partial \xi^2}\frac{\partial \xi}{\partial z}\frac{\partial \xi}{\partial \zeta} + \frac{\partial \phi^2}{\partial \eta^2}\frac{\partial \eta}{\partial z}\frac{\partial \eta}{\partial \zeta}$ (1.8)

(c) Applying the respective value of ϕ^2 and ϕ_z, we get

$$\frac{\partial \phi}{\partial \xi}\frac{\partial^2 \xi}{\partial z \partial \zeta} + \frac{\partial \phi}{\partial \eta}\frac{\partial^2 \eta}{\partial z \partial \zeta} = \frac{\partial \phi^2}{\partial \xi^2}\frac{\partial \xi}{\partial z}\frac{\partial \xi}{\partial \zeta} + \frac{\partial \phi^2}{\partial \eta^2}\frac{\partial \eta}{\partial z}\frac{\partial \eta}{\partial \zeta} \tag{1.9}$$

Hence $\frac{\partial \phi}{\partial \xi}\frac{\partial^2 \xi}{\partial z \partial \zeta} + \frac{\partial \phi}{\partial \eta}\frac{\partial^2 \eta}{\partial z \partial \zeta} = 0$

and $$A_j g^{ij}[pq, r] + g^{ij}[pq, r] = \begin{bmatrix} i \\ pq \end{bmatrix}$$

$$\therefore \quad \begin{bmatrix} l \\ qj \end{bmatrix} \frac{\partial x^q}{\partial \bar{x}^k \partial \bar{x}^j} + \frac{\partial \bar{x}^l}{\partial x^p} \frac{\partial x^p}{\partial \bar{x}^k} + \frac{\partial x^p}{\partial \bar{x}^k} \frac{\partial x^q}{\partial \bar{x}^j} \frac{\partial \bar{x}^l}{\partial x^r} \begin{bmatrix} i \\ pq \end{bmatrix}$$

or $$\begin{bmatrix} l \\ kj \end{bmatrix} = \frac{\partial^2 x^r}{\partial \bar{x}^k \partial \bar{x}^j} \frac{\partial \bar{x}^l}{\partial x^r} + \frac{\partial x^p}{\partial \bar{x}^k} \frac{\partial x^q}{\partial \bar{x}^j} \frac{\partial \bar{x}^l}{\partial x^r} \begin{bmatrix} r \\ pq \end{bmatrix} \quad \ldots(10)$$

This is the law of transformation of Christoffel symbol of the second kind.

Note 1. From (10), we obtain the following important relation :

$$\frac{\partial x^r}{\partial \bar{x}^k \partial \bar{x}^j} = \frac{\partial \bar{x}^l}{\partial x^r} \begin{bmatrix} l \\ kj \end{bmatrix} - \frac{\partial x^p}{\partial \bar{x}^k} \frac{\partial x^q}{\partial \bar{x}^j} \begin{bmatrix} r \\ pq \end{bmatrix} \quad \ldots(11)$$

Note 2. It is evident from (9) and (10) that the Christoffel symbols are not tensors. These symbols therefore possess only for linear transformation of coordinates.

Example 3.8.16. *Prove that* $\begin{bmatrix} j \\ ij \end{bmatrix} = \frac{\partial}{\partial x^i} \log \sqrt{g}$.

Solution. Let G_j be the cofactor of g_{ij} in g so that $g = g_{ij} G_{ij}$ (summation over i and j)

$$\therefore \quad \frac{\partial g}{\partial g_{ij}} = G_{ij} \qquad [\because \ G_{ij} \text{ does not contain } g_{ij} \text{ [why?]}]$$

Also $$\frac{\partial g}{\partial x^i} = \frac{\partial g}{\partial g_{kl}} \frac{\partial g_{kl}}{\partial x^i} = G_{kl} \frac{\partial g_{kl}}{\partial x^i} \quad \text{(summation over } k \text{ and } l\text{)} \quad \ldots(1)$$

We know that $$g^{jk} = \frac{G_{jk}}{g} \quad \ldots(2)$$

Substituting the value of G_{jk} from (2) in (1), we get

$$\frac{\partial g}{\partial x^i} = g g^{jk} \frac{\partial g_{jk}}{\partial x^i} \text{ or } \frac{1}{g} \frac{\partial g}{\partial x^i} = g^{jk} \frac{\partial g_{jk}}{\partial x^i}$$

or $$\frac{\partial}{\partial x^i} (\log g) = g^{jk} [ik, j] + [ij, k] \qquad \text{[By Eq. 3.8.15(a)]}$$

$$= g^{jk} [ik, j] + g^{jk} [ji, k] = \begin{bmatrix} k \\ ij \end{bmatrix} + \begin{bmatrix} j \\ ij \end{bmatrix} = 2 \begin{bmatrix} j \\ ij \end{bmatrix}$$

Hence $$\begin{bmatrix} j \\ ij \end{bmatrix} = \frac{1}{2} \frac{\partial}{\partial x^i} (\log g) = \frac{\partial}{\partial x^i} \log \sqrt{g} \quad \ldots(3)$$

3.8.17 (i) COVARIANT DIFFERENTIATION OF A COVARIANT VECTOR

Let A_j and $\bar{A}_k$ be the components of a covariant vector i.e., a tensor of first order in the coordinate systems x^i and $\bar{x}^i$ respectively. Let us investigate the tensor character of the partial derivatives of A_j w.r.t. the variables x^i. From the law of transformation, $\bar{A}_k = \frac{\partial x^r}{\partial \bar{x}^k} A_r$

Differentiating w.r.t. $\bar{x}^l$, we have $\frac{\partial \bar{A}_k}{\partial \bar{x}^l} = \frac{\partial^2 x^r}{\partial \bar{x}^l \partial \bar{x}^k} A_r + \frac{\partial x^r}{\partial \bar{x}^k} \frac{\partial A_r}{\partial x^s} \frac{\partial x^s}{\partial \bar{x}^l} \quad \ldots(1)$

(Note that A_r is not directly a function of $\bar{x}^l$)

Due to presence of the first term on the R.H.S. of (1), it is evident that $\frac{\partial A_r}{\partial x^s}$ is not a tensor.

On replacing k in terms by $\frac{\partial^2 x^k}{\partial x^i \partial x^j} A_k$ and multiplying by the covariant derivative form (10) of § 35.10, we get

$$\frac{\partial A_i}{\partial x^j} = \frac{\partial x^r}{\partial \bar{x}^i} \frac{\partial x^s}{\partial \bar{x}^j} \bar{A}_r \begin{Bmatrix} l \\ j\ s \end{Bmatrix} - \frac{\partial x^r}{\partial \bar{x}^i} \frac{\partial x^s}{\partial \bar{x}^j} A_k \begin{Bmatrix} k \\ rs \end{Bmatrix} + \frac{\partial^2 x^k}{\partial \bar{x}^i \partial \bar{x}^j} \frac{\partial \bar{x}^r}{\partial x^k} \bar{A}_r$$

or $$\frac{\partial \bar{A}_i}{\partial \bar{x}^j} - \bar{A}_r \begin{Bmatrix} l \\ i\ j \end{Bmatrix} = \frac{\partial x^r}{\partial \bar{x}^i} \frac{\partial x^s}{\partial \bar{x}^j} \left[\frac{\partial A_r}{\partial x^s} - A_k \begin{Bmatrix} k \\ rs \end{Bmatrix} \right]$$

This shows that the expression $\frac{\partial \bar{A}_i}{\partial \bar{x}^j} - \bar{A}_r \begin{Bmatrix} l \\ i\ j \end{Bmatrix}$

is a covariant tensor of the second order. This is called the covariant derivative of $\bar{A}_i$ w.r.t. $\bar{x}^j$ and is denoted by $\bar{A}_{i,j}$.

(2) Covariant differentiation of a contravariant vector. Let A^i and $\bar{A}^i$ be the components of a contravariant vector in the coordinate systems x^i and $\bar{x}^i$. From the law of transformation $A^i = \frac{\partial x^i}{\partial \bar{x}^r} \bar{A}^r$.

Differentiating w.r.t. $\bar{x}^j$, we have $\frac{\partial A^i}{\partial \bar{x}^j} = \frac{\partial^2 x^i}{\partial \bar{x}^r \partial \bar{x}^j} \bar{A}^r + \frac{\partial x^i}{\partial \bar{x}^r} \frac{\partial \bar{A}^r}{\partial \bar{x}^j}$

Substituting for the second derivative from (10) of § 35.10, we get

$$\frac{\partial A^i}{\partial x^k} \frac{\partial x^k}{\partial \bar{x}^j} = \frac{\partial x^i}{\partial \bar{x}^r} \begin{Bmatrix} l \\ r\ j \end{Bmatrix} \bar{A}^r - \frac{\partial x^i}{\partial \bar{x}^r} \frac{\partial x^s}{\partial \bar{x}^j} \begin{Bmatrix} k \\ s\ t \end{Bmatrix} \bar{A}^r + \frac{\partial x^i}{\partial \bar{x}^r} \frac{\partial \bar{A}^r}{\partial \bar{x}^j}$$

Interchanging the dummy indices k in the first term on the R.H.S. and putting $\frac{\partial x^i}{\partial \bar{x}^r} \bar{A}^r = A^i$ in the second term, we obtain

$$\frac{\partial x^i}{\partial \bar{x}^r} \frac{\partial \bar{A}^r}{\partial \bar{x}^j} = \frac{\partial x^i}{\partial \bar{x}^r} \begin{Bmatrix} r \\ j\ k \end{Bmatrix} \bar{A}^k - \frac{\partial x^i}{\partial x^k} A^k \begin{Bmatrix} k \\ j\ s \end{Bmatrix} + \frac{\partial x^i}{\partial \bar{x}^r} \frac{\partial \bar{A}^r}{\partial \bar{x}^j}$$

Transposing the second term on the R.H.S. to the L.H.S., we get

$$\frac{\partial x^i}{\partial \bar{x}^r} \left[\frac{\partial \bar{A}^r}{\partial \bar{x}^j} + A^k \begin{Bmatrix} r \\ j\ k \end{Bmatrix} \right] = \frac{\partial x^i}{\partial x^k} \left[\frac{\partial A^k}{\partial x^j} + A^s \begin{Bmatrix} k \\ j\ s \end{Bmatrix} \right]$$

or $$\frac{\partial \bar{A}^r}{\partial \bar{x}^j} + A^k \begin{Bmatrix} r \\ j\ k \end{Bmatrix} = \frac{\partial \bar{x}^r}{\partial x^i} \frac{\partial x^s}{\partial \bar{x}^j} \left[\frac{\partial A^i}{\partial x^s} + A^k \begin{Bmatrix} i \\ s\ k \end{Bmatrix} \right]$$

This shows that $\frac{\partial A^i}{\partial x^j} + A^k \begin{Bmatrix} i \\ j\ k \end{Bmatrix}$ is a mixed tensor of the second order. This is called the covariant derivative of A^i w.r.t. x^j and is denoted by $A^i_{,j}$.

Obs. The following laws hold good for covariant differentiation:

(i) Covariant derivative of the sum (or difference) of two tensors = sum (or difference) of their covariant derivatives.

(ii) Covariant derivative of the product of two tensors = covariant derivative of first tensor × second tensor + first tensor × covariant derivative of second tensor.

Example 35.17. Prove that the covariant derivative of g^{ij} is zero.

Solution. Let A_k denote a covariant vector which is arbitrary and is not zero. Then

$$A_k \text{ or } \frac{\partial A_k}{\partial x^j} - A_r \begin{Bmatrix} r \\ j\ k \end{Bmatrix} = 0 \qquad \ldots (1)$$

Let $\phi = g^{ij} A_i A_j$ so that ϕ is a scalar invariant. Differentiating ϕ w.r.t. x^k, we have

$$\frac{\partial \phi}{\partial x^k} = \frac{\partial g^{ij}}{\partial x^k} A_i A_j + g^{ij} \frac{\partial A_i}{\partial x^k} A_j + g^{ij} A_i \frac{\partial A_j}{\partial x^k}$$

$$= \frac{\partial g^{ij}}{\partial x^k} A_i A_j + g^{ij} \left\{ \begin{matrix} l \\ i\,k \end{matrix} \right\} A_l A_j + g^{ij} \left\{ \begin{matrix} l \\ j\,k \end{matrix} \right\} A_i A_l \qquad \text{[By (3)]}$$

Interchanging i and l in the second term and j and l in the last term on the right, we get

$$\frac{\partial \phi}{\partial x^k} = \left[\frac{\partial g^{ij}}{\partial x^k} + g^{lj} \left\{ \begin{matrix} i \\ l\,k \end{matrix} \right\} + g^{il} \left\{ \begin{matrix} j \\ l\,k \end{matrix} \right\} \right] A_i A_j$$

Since $\partial \phi / \partial x^k$ is a covariant vector, the expression

$$\frac{\partial g^{ij}}{\partial x^k} + g^{lj} \left\{ \begin{matrix} i \\ l\,k \end{matrix} \right\} + g^{il} \left\{ \begin{matrix} j \\ l\,k \end{matrix} \right\}$$

is a tensor of the third order by quotient law. Thus it is the covariant derivative $g^{ij}{}_{,k}$.

$$\therefore \quad g^{ij}{}_{,k} = \frac{\partial g^{ij}}{\partial x^k} + g^{lj} \left\{ \begin{matrix} i \\ l\,k \end{matrix} \right\} + g^{il} \left\{ \begin{matrix} j \\ l\,k \end{matrix} \right\}$$

$$= -g^{lj} \left\{ \begin{matrix} i \\ l\,k \end{matrix} \right\} - g^{il} \left\{ \begin{matrix} j \\ l\,k \end{matrix} \right\} + g^{lj} \left\{ \begin{matrix} i \\ l\,k \end{matrix} \right\} + g^{il} \left\{ \begin{matrix} j \\ l\,k \end{matrix} \right\} \qquad \text{[By Ex. 38.15 (ii)]}$$

$$= -g^{lj} \left\{ \begin{matrix} i \\ l\,k \end{matrix} \right\} + g^{lj} \left\{ \begin{matrix} i \\ l\,k \end{matrix} \right\} - g^{il} \left\{ \begin{matrix} j \\ l\,k \end{matrix} \right\} + g^{il} \left\{ \begin{matrix} j \\ l\,k \end{matrix} \right\} = 0$$

[Changing the dummy indices in (2)]

38.20 (1) GRADIENT

If ϕ be a scalar function of the coordinates, then the gradient of ϕ is denoted by $\text{grad}\,\phi = \frac{\partial \phi}{\partial x^i}$ which is a covariant vector.

(2) Divergence. The divergence of the contravariant vector A^i is defined by

$$\text{div}\, A^i = \frac{\partial A^i}{\partial x^i} + A^l \left\{ \begin{matrix} i \\ l\,i \end{matrix} \right\} \text{ which is sometimes written as } A^i{}_{,i}$$

The divergence of the covariant vector A_i is defined by $\text{div}\, A_i = g^{ij} A_{ij}$.

(3) Curl. Let A_i be a covariant vector, then

$$A_{ij} = \frac{\partial A_i}{\partial x^j} - A_m \left\{ \begin{matrix} m \\ j\,i \end{matrix} \right\} \text{ and } A_{ji} = \frac{\partial A_j}{\partial x^i} - A_m \left\{ \begin{matrix} m \\ i\,j \end{matrix} \right\} \text{ are covariant tensors.}$$

$$\therefore \quad A_{ij} - A_{ji} = \frac{\partial A_i}{\partial x^j} - \frac{\partial A_j}{\partial x^i}$$

is a covariant tensor of second order, which is called curl of A_i.

Thus $\text{curl}\, A_i = A_{ij} - A_{ji}$.

Obs 1. Curl A_i is a skew-symmetric tensor.

Since $A_{ij} - A_{ji} = -(A_{ji} - A_{ij})$.

Obs 2. Curl is a tensor and not a vector. In 3-dimensional space, however, curl has only three independent components and it can, therefore, be taken as a vector.

Example 38.15. Prove that

(a) $\text{div}\, A_i = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i} (\sqrt{g}\, A^i)$ (b) $\nabla^2 \phi = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i} \left(\sqrt{g}\, g^{ij} \frac{\partial \phi}{\partial x^j} \right)$

Solution. (a) Using $\left[\begin{matrix} l \\ ij \end{matrix}\right] = \frac{\partial x^l}{\partial u^k} \frac{\partial^2 x^k}{\partial u^i \partial u^j}$ By Eq. (4.33)

$$\frac{\partial A^l}{\partial u^j} = \frac{\partial A^l}{\partial u^j} + A^k \left[\begin{matrix} l \\ kj \end{matrix}\right] = \frac{\partial A^l}{\partial u^j} + \frac{A^k}{\sqrt{g}} \frac{\partial \sqrt{g}}{\partial u^k} = \frac{\partial A^l}{\partial u^j} + \frac{A^k}{\partial u^k} \frac{\partial \sqrt{g}}{\partial u^j}$$

$$= \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^j} (\sqrt{g} A^l) \quad \text{...(i)}$$

(b) We know $\nabla \phi = \text{div grad } \phi$ ...(ii)

and $\text{grad } \phi = \frac{\partial \phi}{\partial x^i}$, which is a covariant vector.

The contravariant vector associated with $\partial\phi/\partial x^i$ is

$$A^k = g^{ik} \frac{\partial \phi}{\partial x^i}.$$

Then from (i) and (ii), $\nabla \phi = \text{div} \left(g^{ik} \frac{\partial \phi}{\partial x^i} \right) = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^k} \left(\sqrt{g} g^{ik} \frac{\partial \phi}{\partial x^i} \right)$

EXERCISE 4.4

1. Determine Christoffel symbols of the first and second kind for an orthogonal curvilinear coordinate system.
2. Determine the Christoffel symbols of the first kind in (i) rectangular, (ii) cylindrical and (iii) spherical coordinates.
3. Evaluate the Christoffel symbols of the second kind in (i) rectangular, (ii) cylindrical and (iii) spherical coordinates.
4. Prove that (i) $[ij, k] = [ji, k]$ (ii) $\frac{\partial g_{ij}}{\partial x^k} = [ik, j] + [jk, i]$
5. If $ds^2 = a(dx^1)^2 + b(dx^2)^2 + c(dx^3)^2$, find the values of
   (i) $[12, 1]$ and $[13, 3]$ (ii) $\left[\begin{matrix} 1 \\ 22 \end{matrix}\right]$ and $\left[\begin{matrix} 2 \\ 12 \end{matrix}\right]$
6. If $ds^2 = (dx^1)^2 + r^2(dx^2)^2 + r^2 \sin^2 x^2 (dx^3)^2$, find the values of
   (i) $[22, 1]$ and $[33, 3]$ (ii) $\left[\begin{matrix} 1 \\ 33 \end{matrix}\right]$ and $\left[\begin{matrix} 3 \\ 33 \end{matrix}\right]$
7. Show that the tensors g_{ij}, g^{ij} and δ^i_j are covariant with respect to coordinate transformation.
8. Write the covariant derivative of all the tensors A^{ij} and A^i_{jk}.
9. Show that the covariant derivative of g_{ij} is zero. (Madras M.Sc. 2000)
10. Find the covariant derivative of A_{ij}, A^{ij}_k with respect to x^k.
11. If the metric ds^2 is $(dx^1)^2 + (dx^2)^2 + (dx^3)^2$, find the values of $[ij, k]$.
12. Obtain the Laplacian equation in (i) cylindrical, (ii) spherical coordinates.
13. If A_{ij} is the tensor of second rank skew-symmetric, prove that
    $$A_{ij,k} + A_{jk,i} + A_{ki,j} = 0.$$ (Madras M.Sc. 2000 (S))
14. Using tensor notation, show that
    (i) $\text{div curl } A = 0$ (ii) $\text{curl grad } \phi = 0$

APPENDIX

1

Useful Information

BASIC DATA

1. Basic Constants

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]

2. Conversion Factors

[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

3. Systems of Units

Quantity	[illegible] system	[illegible] system	[illegible] system
Length	[illegible]	[illegible]	[illegible]
Mass	[illegible]	[illegible]	[illegible]
Time	[illegible]	[illegible]	[illegible]
Force	[illegible]	[illegible]	[illegible]

Note: The [illegible] system is also known as the [illegible] system of units (SI system).

4. Greek Letters Used

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
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[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

5. Some Notations

[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]		

Factorials: $n! = n(n-1)(n-2)\ldots 3.2.1$

Double factorials: $(2n)!! = 2n.2n-2.2n-4\ldots 6.4.2$

$(2n-1)!! = (2n-1)(2n-3)(2n-5)\ldots 5.3.1$

Stirling's approximation: When n is large, $n! = \sqrt{(2\pi)}\, n^{n+\frac{1}{2}} e^{-n}$.

II. ALGEBRA

1. Quadratic equation: $ax^2 + bx + c = 0$. Roots are

$$\alpha = \frac{-b + \sqrt{(b^2 - 4ac)}}{2a}, \beta = \frac{-b - \sqrt{(b^2 - 4ac)}}{2a}$$

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}.$$

Roots are equal if	$b^2 - 4ac = 0$
Roots are real and distinct if	$b^2 - 4ac > 0$
Roots are imaginary if	$b^2 - 4ac < 0$

2. Cubic equation: $x^3 + bx^2 + cx + d = 0$

Cardan's method:

(i) Remove x^2 term by putting $x = y - (b/3)$.

(ii) Equate coeffs. in the new equation and $y^3 - 3\alpha\gamma y - (\alpha^3 + \gamma^3) = 0$ [i.e. $y = \alpha + \gamma$]

(iii) Find α^3 and γ^3. Then find α and γ.

(iv) Get $y = \alpha + \gamma$ and $x = y - (b/3)$.

3. Biquadratic equation: $ax^4 + bx^3 + cx^2 + dx + e = 0$

I. Ferrari's method:

(i) Combine x^4 and x^2 terms to form a perfect square by adding a term in λ.

(ii) Make R.H.S. a perfect square to find λ.

(iii) Solve resulting quadratic equations.

II. Descartes's method:

(i) Remove x^3 term by putting $x = y - (b/4a)$.

(ii) Equate transformed expression to $(y^2 + py + q)(y^2 - py + q)$.

(iii) Equate coeffs. of like powers from both sides.

(iv) Find p, q and q' and solve resulting quadratics.

4. Cross-multiplication: $a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$

Then $\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$

5. Method of least squares:

(i) Straight line of best fit: $y = a + bx$.

Normal equations: $\Sigma y = na + b\Sigma x$, $\Sigma xy = a\Sigma x + b\Sigma x^2$.

To find a, b, solve these equations.

(ii) Parabola of best fit: $y = a + bx + cx^2$.

Normal equations: $\Sigma y = na + b\Sigma x + c\Sigma x^2$,

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$, $\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$.

To find a, b, c, solve these equations.

6. Progressions:

(i) Numbers $a, a + d, a + 2d, \ldots$ are said to be in Arithmetic progression (A.P.)

The nth term $T_n = a + (n-1)d$ and sum $S_n = \frac{n}{2}[2a + (n-1)d]$.

(ii) Numbers $a, ar, ar^2, \ldots$ are said to be in Geometric progression (G.P.)

The nth term $T_n = ar^{n-1}$ and sum $S_n = \frac{a(1-r^n)}{1-r}$, $S_\infty = \frac{a}{1-r}$ $(r < 1)$.

(iii) Numbers $U_1, U_2 = a, U_3 = a + d, U_4 = a + 2d, \ldots$ are said to be in Harmonic progression (H.P.) (i.e., a sequence is said to be in H.P. if its reciprocals are in A.P.)

nth term $T_n = 1/[a + (n-1)d]$

(iv) If a and b are two numbers then

Arithmetic mean $= \frac{1}{2}(a+b)$, Geometric mean $= \sqrt{ab}$, Harmonic mean $= 2ab/(a+b)$

(v) Natural numbers are $1, 2, 3, \ldots, n$

$$\Sigma n = \frac{n(n+1)}{2}, \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}, \Sigma n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

7. Permutations and Combinations

$$^nP_r = \frac{n!}{(n-r)!}; {}^nC_r = \frac{n!}{r!(n-r)!} = \frac{{}^nP_r}{r!}; {}^nC_{n-r} = {}^nC_r; \quad {}^nC_0 = 1 = {}^nC_n$$

8. Binomial theorem

(i) When n is a positive integer

$$(1+x)^n = 1 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \ldots + {}^nC_n x^n$$

(ii) When n is a negative integer or a fraction

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1.2}x^2 + \frac{n(n-1)(n-2)}{1.2.3}x^3 + \ldots \infty$$

(iii) Also in (ii), ${}^nC_r = \frac{n!}{r!(n-r)!}$

9. Logarithms

(i) Naperian logarithm: $\log_e x$ has base e and is denoted by $\ln x$.

Common logarithm $\log_{10} x = M \log_e x$ where $M = \log_{10} e = 0.4343$

(ii) $\log_a 1 = 0$, $\log_a a = 1$, $\log_a a^n = n$ $(a > 0)$, $\log_a a = 1$.

(iii) $\log(mn) = \log m + \log n$; $\log(m/n) = \log m - \log n$; $\log(m^n) = n \log m$.

10. Partial Fractions

A fraction of the form $\frac{P_0x^m + P_1x^{m-1} + \ldots + P_m}{Q_0x^n + Q_1x^{n-1} + \ldots + Q_n}$

in which m and n are positive integers, is called a rational algebraic fraction. If in this m is less than n, i.e. the numerator is of lower degree than the denominator, it is called a *proper fraction*.

To resolve a proper fraction into partial fractions, we first factorise the denominator into its real factors. These will be of the form $ax + b$ or $ax^2 + bx + c$, and some factors repeated. Then the proper fraction is resolved into a sum of partial fractions such that:

(i) to each non-repeated linear factor $(ax + b)$ in the denominator corresponds a partial fraction of the form $\frac{A}{ax+b}$;

(ii) to a repeated linear factor $(ax + b)^r$ in the denominator corresponds the sum of r partial fractions of the form $\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \frac{A_3}{(ax+b)^3} + \ldots + \frac{A_r}{(ax+b)^r}$;

(iii) to a non-repeated quadratic factor $(ax^2 + bx + c)$ in the denominator, corresponds a partial fraction of the form $\frac{Ax + B}{ax^2 + bx + c}$;

(iv) to a repeated quadratic factor $(ax^2 + bx + c)^r$ in the denominator, corresponds the sum of r partial fractions of the form $\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \ldots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$

There are now to be determined the unknown constants A, A_1, B_1 etc.

To obtain the partial fraction corresponding to the non-repeated linear factor $x - a$ in the denominator, put $x = a$ everywhere in the given fraction except in the factor $x - a$ itself.

In all the above cases, equating the given fraction to a sum of suitable partial fractions in accordance with (i) to (iv) above, having found the partial fractions corresponding to the non-repeated linear factors by the above rule. Then multiply both sides by the denominator of the given fraction and equate the coefficients of the powers of one variable or by substituting numerical values of the variable. Finally solve the simplest of the resulting equations to find the unknown constants.

11. Matrices

(i) A matrix of mn numbers arranged in a rectangular array of m rows and n columns is said to be a matrix of order $m \times n$.

In particular if $m = n$, it is called a square matrix of order n.

(ii) Two matrices of the same order can be added or subtracted by adding or subtracting the corresponding elements.

(iii) Product of a matrix A by a scalar k is a matrix whose each element is k times the corresponding element of A.

(iv) Two matrices can be multiplied only when the number of columns in the first is equal to the number of rows in the second. If A and B are two matrices of order $m \times p$, then the product AB is a matrix of order $m \times p$, obtained by multiplying and adding the row elements of A with the corresponding column elements of B.

(v) The transpose of a matrix A is the matrix obtained by interchanging its rows and columns and is denoted by A'.

A square matrix A is said to be symmetric if $A = A'$ and skew-symmetric if $A = -A'$.

(vi) If A and B are two square matrices such that $AB = I$ (i.e., a unit matrix), then B is called the inverse of A and is denoted by A^{-1}. Thus $AA^{-1} = A^{-1}A = I$.

(vii) Rank of a matrix is the largest order of any non-vanishing minor of the matrix.

(viii) Consistency of a system of equations in n unknowns.

With each of the coefficient matrix A and that of the augmented matrix B (say), then

(a) the equations are inconsistent i.e. there is no solution if $\rho < \rho'$,

(b) the equations are consistent when $\rho = \rho'$,

(c) the equations are consistent and there are infinite number of solutions when $\rho = \rho' < n$.

(ix) Eigen values. If A is any square matrix of order n, then the determinant of the matrix $A - \lambda I$, equated to zero is called the Characteristic equation of A and its roots are called the eigen values of A.

(x) Cayley-Hamilton theorem: Every square matrix satisfies its own characteristic equation.

12. Determinants

(i) A determinant is defined for a square matrix A and is denoted by $|A|$. It is a number it has a single value e.g.,

$$|A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

In this way, determinants can be expanded in terms of any row or column.

(ii) **Properties:**

I. A determinant remains unaltered if the rows and columns are interchanged.

II. A determinant vanishes if two of its rows (or columns) are identical in every detail.

III. If each element of a row (or column) consists of n terms, the determinant can be expressed as the sum of n determinants.

IV. If to each element of a row (or column) be added equal multiples of the corresponding elements of one or more rows (or columns), the determinant remains unaltered.

V. If the elements of a determinant A are functions of x and two parallel lines are identical when $x = a$, then $x - a$ is a factor of A.

III. GEOMETRY

1. Coordinates of a point : Given (x, y) and polar (r, θ)

$$x = r\cos\theta, \qquad y = r\sin\theta$$

or $$r = \sqrt{(x^2 + y^2)}, \qquad \theta = \tan^{-1}(y/x) + k\pi, (k = 0, 1)$$

Distance between two points (x_1, y_1) and $(x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Point of division of the line joining (x_1, y_1) and (x_2, y_2) in the ratio $m_1 : m_2$ is

$$\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)$$

In a triangle having vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3)

(i) Area $= \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

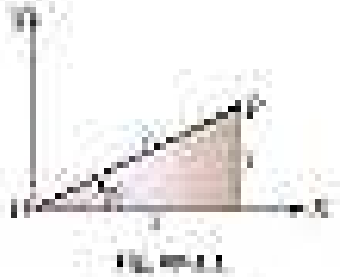


Fig. III-1.1

(ii) Centroid (point of intersection of medians) is

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

(iii) In-centre (point of intersection of the internal bisectors of the angles) is

$$\left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}\right)$$

where a, b, c are the lengths of the sides of the triangle.

(iv) Circum-centre is the point of intersection of the right bisectors of the sides of the triangle.

(v) Ortho-centre is the point of intersection of the perpendiculars drawn from the vertices to the opposite sides of the triangle.

2. Straight Line

(i) Slope of the line joining the points (x_1, y_1) and $(x_2, y_2) = \frac{y_2 - y_1}{x_2 - x_1}$

Slope of the line $ax + by + c = 0$ is $-\frac{a}{b}$ i.e., $-\frac{\text{coeff. of } x}{\text{coeff. of } y}$

(ii) Equations of a line

(a) having slope m and cutting an intercept c on y-axis is $y = mx + c$.

(b) cutting intercepts a and b from the axes is $\frac{x}{a} + \frac{y}{b} = 1$.

(c) passing through (x_1, y_1) and having slope m is $y - y_1 = m(x - x_1)$

(d) passing through (x_1, y_1) and making angles α and β with the x-axis is $\frac{x - x_1}{\cos\alpha} = \frac{y - y_1}{\sin\alpha}$ or

(e) through the point of intersection of the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is $a_1x + b_1y + c_1 + k(a_2x + b_2y + c_2) = 0$

(iii) Angle between two lines having slopes m_1 and m_2 is $\tan^{-1}\frac{m_1 - m_2}{1 + m_1m_2}$

Two lines are parallel if	$m_1 = m_2$
Two lines are perpendicular if	$m_1m_2 = -1$
Any line parallel to the line	$ax + by + c = 0$ is $ax + by + k = 0$
Any line perpendicular to	$ax + by + c = 0$ is $bx - ay + k = 0$

(iv) Length of the perpendicular from (x_1, y_1) to the line $ax + by + c = 0$, is $\frac{|ax_1 + by_1 + c|}{\sqrt{(a^2 + b^2)}}$.

3. Circle

(i) Equation of the circle having centre (h, k) and radius r is $(x - h)^2 + (y - k)^2 = r^2$

(ii) Equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle having centre $(-g, -f)$ and radius $= \sqrt{g^2 + f^2 - c}$.

(vi) Equation of the tangent at the point (x_1, y_1) to the circle $x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$.

(vii) Condition for the line $y = mx + c$ to touch the circle $x^2 + y^2 = a^2$ is $c = \pm a\sqrt{(1 + m^2)}$.

(viii) Length of the tangent from the point (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $\sqrt{(x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c)}$.

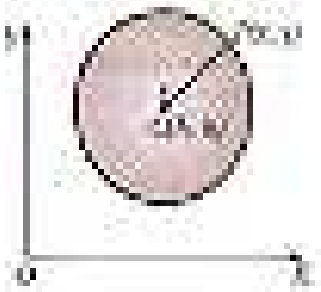


Fig. 10.13

4. Parabola

(i) Standard equation of the parabola $y^2 = 4ax$.

Its parametric equations are $x = at^2, y = 2at$.

Latus rectum $LL' = 4a$. Focus is $S(a, 0)$.

Directrix ZM is $x + a = 0$.

Focal distance of any point $P(x_1, y_1)$ on the parabola

$$y^2 = 4ax \text{ is } SP = x_1 + a$$

Equation of the tangent at (x_1, y_1) to the parabola $y^2 = 4ax$ is

$$yy_1 = 2a(x + x_1)$$

Condition for the line $y = mx + c$ to touch the parabola

$$y^2 = 4ax \text{ is } c = a/m.$$

Equation of the normal to the parabola $y^2 = 4ax$ in terms of its slope m is

$$y = mx - 2am - am^3.$$

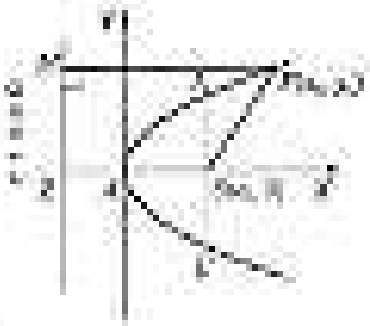
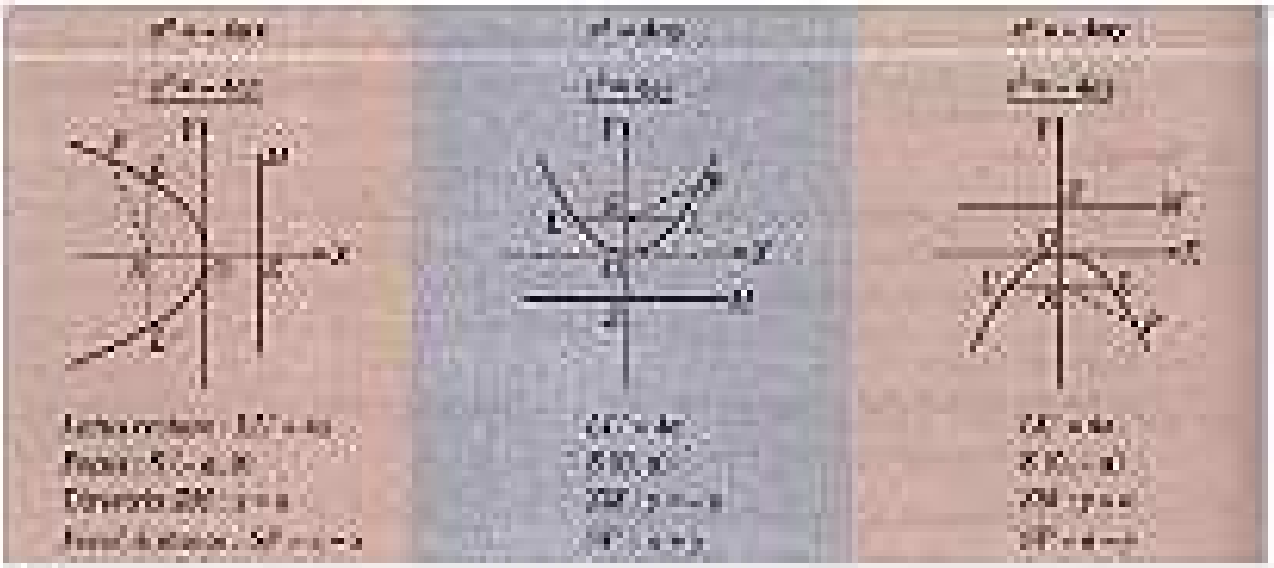


Fig. 10.14

(ii) Other standard forms of parabola



5. Ellipse

(i) Standard equation of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be $a > b > 0$.

Its parametric equations are $x = a\cos\theta, y = b\sin\theta$.

Eccentricity $e = \sqrt{(1 - b^2/a^2)}$.

Latus rectum $LL' = 2b^2/a$.

Foci $S(ae, 0)$ and $S'(-ae, 0)$.

Directrices ZM is $x = a/e$ and $Z'M'$ is $x = -a/e$.

Sum of the focal distances of any point on the ellipse is equal to the major axis i.e.,

$$SP + S'P = 2a.$$

Equation of the tangent at the point (x_1, y_1) to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1.$$

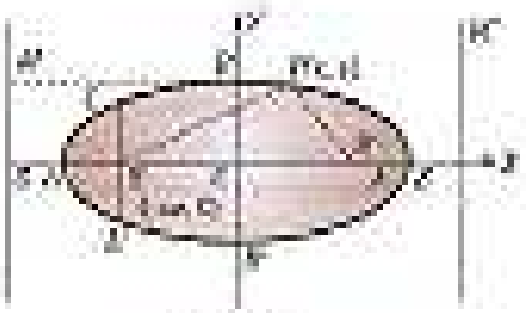


Fig. 10.15

Condition for the line $y = mx + c$ to touch the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c = \pm\sqrt{a^2m^2 + b^2}$.

(ii) Another standard form of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a > b > 0)$

Vertices $A(a, 0), A'(-a, 0)$

Foci : $S(ae, 0), S'(-ae, 0)$

Directrices : $MM' : x = a/e$, $ZZ' : x = -a/e$

Latus rectum : $LSL' = 2b^2/a$

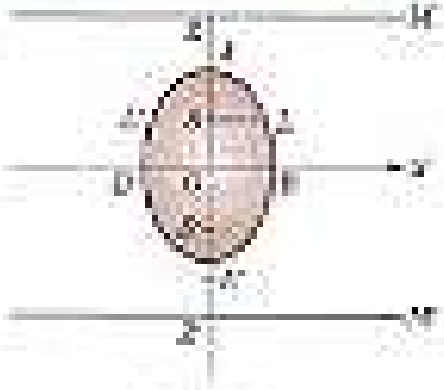


Fig. AP-I.A10

8. Hyperbola

(i) Standard equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Its parametric equations are

$x = a \sec\theta$, $y = b \tan\theta$

Eccentricity $e = \sqrt{(1 + b^2/a^2)}$,

Latus rectum $= 2b^2/a = 2a(e^2 - 1)$.

Directrices

$MM' (x = a/e)$ and $ZZ' (x = -a/e)$

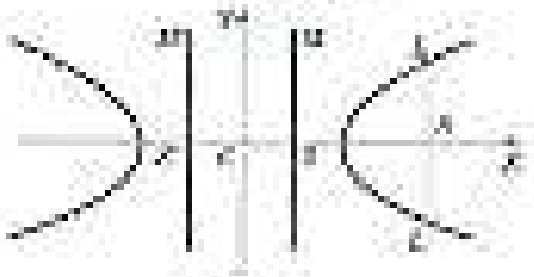


Fig. AP-I.A11

(ii) Equation of the tangent at the point (x_1, y_1) on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is } \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

(iii) Condition for the line $y = mx + c$ to touch the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is } c = \pm\sqrt{a^2m^2 - b^2}$$

(iv) Asymptotes of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $\frac{x}{a} + \frac{y}{b} = 0$ and $\frac{x}{a} - \frac{y}{b} = 0$.

(v) Equation of the rectangular hyperbola with asymptotes as axes is $xy = c^2$.

Its parametric equations are $x = ct$, $y = c/t$.

9. Nature of a conic

The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents

(i) a pair of lines, if $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \Delta = 0$

(ii) a circle, if $h = 0$, $a = b$, $a \neq 0$

(iii) a parabola, if $h^2 - ab = 0$, $\Delta \neq 0$.

(iv) an ellipse, if $h^2 - ab < 0$, $\Delta \neq 0$.

(v) a hyperbola, if $h^2 - ab > 0$, $\Delta \neq 0$.

and a rectangular hyperbola if in addition $a + b = 0$.

IV. SOLID GEOMETRY

1. (i) If l, m, n be the direction cosines of a line then $l^2 + m^2 + n^2 = 1$.

If a, b, c be the direction ratios of a line then $l = \frac{a}{\sqrt{(a^2+b^2+c^2)}}$, $m = \frac{b}{\sqrt{(a^2+b^2+c^2)}}$, $n = \frac{c}{\sqrt{(a^2+b^2+c^2)}}$

(ii) If θ be the angle between the lines having direction cosines l, m, n and l', m', n', then

$$\cos\theta = ll' + mm' + nn'$$

Lines are perpendicular if $ll' + mm' + nn' = 0$

Lines are parallel if $\frac{l}{l'} = \frac{m}{m'} = \frac{n}{n'}$

(vi) Projection of the line passing the point (x_1, y_1, z_1) and (x_2, y_2, z_2) on a line having d.r.'s a_3, b_3, c_3 is $\left|\begin{matrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_3 & b_3 & c_3 \end{matrix}\right|$

b. Plane:

(i) Different forms of equations of plane:

—General form: $ax+by+cz+d=0$

—where a, b, c are the d.r.'s of a normal to the plane.

—Normal form: $lx+my+nz=p$

—Intercept form: $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$

—Any plane passing through the point (x_1, y_1, z_1) is $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$

(ii) Angle between the planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ is given by

$$\cos\theta = \frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

Planes are perpendicular if $a_1a_2+b_1b_2+c_1c_2=0$

Planes are parallel if $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

(iii) Any plane passing through the intersection of $ax+by+cz+d=0$ and $a'x+b'y+c'z+d'=0$ is $(ax+by+cz+d)+k(a'x+b'y+c'z+d')=0$

c. Straight line:

(i) Equation of the line through the point (x_1, y_1, z_1) and having d.r.'s a, b, c is

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$ (Symmetrical form)

(ii) Equation of the line through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$$ (Two point form)

(iii) Angle between the lines $\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$ is

$$\cos\theta = \frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$$

Lines are parallel if $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

Lines are perpendicular to each other if $a_1a_2+b_1b_2+c_1c_2=0$

(iv) Coplanar lines:

Two lines $\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$

are coplanar if $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$

and equation of the plane containing these lines is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}=0$$

(v) Shortest distance between two skew lines:

$$\frac{x-x_1}{a_1}=\frac{y-y_1}{b_1}=\frac{z-z_1}{c_1} \text{ and } \frac{x-x_2}{a_2}=\frac{y-y_2}{b_2}=\frac{z-z_2}{c_2}$$

is $\left|\frac{(x_2-x_1)(b_1c_2-b_2c_1)+(y_2-y_1)(c_1a_2-c_2a_1)+(z_2-z_1)(a_1b_2-a_2b_1)}{\sqrt{(b_1c_2-b_2c_1)^2+(c_1a_2-c_2a_1)^2+(a_1b_2-a_2b_1)^2}}\right|$

where (x_1, y_1, z_1) and (x_2, y_2, z_2) are points on the lines and a_1, b_1, c_1 and a_2, b_2, c_2 are d.r.'s of the lines.

Equation of the line of S.D. is given by

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l & m & n \end{vmatrix} = 0 \text{ and } \begin{vmatrix} x-x_2 & y-y_2 & z-z_2 \\ l_2 & m_2 & n_2 \\ l & m & n \end{vmatrix} = 0$$

4. Spheres

(i) Equation of the sphere having centre (a, b, c) and radius r is

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$$

(ii) Equation $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ represents a sphere having centre $(-u, -v, -w)$ and radius $= \sqrt{u^2+v^2+w^2-d}$

(iii) Equation of the sphere having the points (x_1, y_1, z_1) and (x_2, y_2, z_2) as the ends of a diameter is $(x-x_1)(x-x_2) + (y-y_1)(y-y_2) + (z-z_1)(z-z_2) = 0$

(iv) Equation of a circle, section of a sphere $S = 0$ by the plane $U = 0$ is given by $S = 0$ and $U = 0$ is called a great circle.

(v) Equation of any sphere through the circle, intersection of the sphere $S = 0$ and the plane $U = 0$ is $S + kU = 0$.

(vi) Tangent plane at any point (x_1, y_1, z_1) of the sphere

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \text{ is}$$

$$xx_1 + yy_1 + zz_1 + u(x+x_1) + v(y+y_1) + w(z+z_1) + d = 0$$

(vii) Two spheres $x^2 + y^2 + z^2 + 2u_1x + 2v_1y + 2w_1z + d_1 = 0$ and $x^2 + y^2 + z^2 + 2u_2x + 2v_2y + 2w_2z + d_2 = 0$ cut orthogonally if $2u_1u_2 + 2v_1v_2 + 2w_1w_2 = d_1 + d_2$.

5. Cone

(i) Equation of a cone with vertex at the origin is a homogeneous equation of the second degree in x, y, z.

(ii) Enveloping cone of the sphere $S = 0$ with vertex (x_1, y_1, z_1) is $SS_1 = T^2$ where $S = x^2 + y^2 + z^2 - a^2$, $S_1 = x_1^2 + y_1^2 + z_1^2 - a^2$, $T = xx_1 + yy_1 + zz_1 - a^2$

6. Quadric surfaces

(i) Ellipsoid: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$

(ii) Hyperboloid of one sheet: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$

Hyperboloid of two sheets: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = -1$

(iii) Cone: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 0$

(iv) Elliptic paraboloid: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{2z}{c}$

Hyperbolic paraboloid: $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = \dfrac{2z}{c}$

7. Volumes and surface areas

Solid	Volume	Curved surface area	Total surface area
Cube (side a)	a^3	$4a^2$	$6a^2$
Cuboid (length l, breadth b, height h)	lbh	$2(l+b)h$	$2(lb+bh+hl)$
Sphere (radius r)	$\frac{4}{3}\pi r^3$	—	$4\pi r^2$
Cylinder (base radius r, height h)	$\pi r^2 h$	$2\pi rh$	$2\pi r(r+h)$
Cone (base radius r, height h)	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(r+l)$
Slant height l is given by $l = \sqrt{r^2+h^2}$.			

5. TRIGONOMETRY

1.

θ =	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	π
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	0
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	-1
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	0

2. Signs and variations of ratios

Quadrant	$\sin\theta$	$\cos\theta$	$\tan\theta$
I	$+$ $(0\to 1)$	$+$ $(1\to 0)$	$+$ $(0\to\infty)$
II	$+$ $(1\to 0)$	$-$ $(0\to -1)$	$-$ $(-\infty\to 0)$
III	$-$ $(0\to -1)$	$-$ $(-1\to 0)$	$+$ $(0\to\infty)$
IV	$-$ $(-1\to 0)$	$+$ $(0\to 1)$	$-$ $(-\infty\to 0)$

3. Any trigonometric ratio of an angle θ is equal to the same ratio of θ, where n is an integer.
   or the co-ratio of θ, where n is odd.
   The sign of the result is determined from the quadrant in which the angle θ lies.

   e.g., $\sin 570^\circ = \sin(6\times 90^\circ - 30^\circ) = -\sin 30^\circ = -\frac{1}{2}$,

   $\tan 315^\circ = \tan(4\times 90^\circ - 45^\circ) = -\tan 45^\circ = -1$

4. $\sin(A\pm B) = \sin A\cos B \pm \cos A\sin B$

   $\cos(A\pm B) = \cos A\cos B \mp \sin A\sin B$

   $$\sin 2A = 2\sin A\cos A = 2\tan A/(1+\tan^2 A)$$

   $$\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = 2\cos^2 A - 1 = \frac{1-\tan^2 A}{1+\tan^2 A}$$

   $$\tan(A\pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}, \quad \tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

5. $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

   $\cos A\sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$

   $\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$

   $\sin A\sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$

6. $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$

   $\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$

   $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$

   $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$

7. $\sin 2A = 2\sin A\cos A = 4\sin A\sin(60^\circ - A)\sin(60^\circ + A)$, $\cos 2A = 4\cos^2 A - 2\cos A$, $\tan 2A = \dfrac{2\tan A - 3\tan^3 A}{1 - 3\tan^2 A}$

8. $a\sin\alpha + b\cos\beta = c$, $a\cos\alpha + b\sin\beta = c$ and $\alpha + \beta = 90^\circ$

$a\cos\alpha + b\sin\beta = c$, $a\sin\alpha - b\cos\beta = c$ and $\alpha - \beta = 90^\circ$

where $a = r\cos\theta$, $b = r\sin\theta$ so that $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}(b/a)$

9. In $\triangle ABC$:

(i) $a/\sin A = b/\sin B = c/\sin C$ (Sine formula)

(ii) $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ (Cosine formula)

(iii) $a = b\cos C + c\cos B$ (Projection formula)

(iv) Area of $\triangle ABC = \dfrac{1}{2}bc\sin A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \dfrac{1}{2}(a + b + c)$

10. Series

(i) Exponential series: $e^x = 1 + \dfrac{x}{1!} + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots \infty$

(ii) Sine, cosine, sinh, cosh series

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \infty \qquad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \infty$$

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \infty \qquad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \infty$$

(iii) Log series

$$\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \infty \qquad \log(1 - x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots \infty\right)$$

(iv) Gregory series

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots \infty \qquad \tanh^{-1} x = \frac{1}{2}\log\frac{1+x}{1-x} = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots \infty$$

11. (i) Complex number: $z = x + iy = r\cos\theta + i\sin\theta = re^{i\theta}$ See Fig. AP-L11

(ii) Euler's theorem: $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$, $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$

(iii) De moivre's theorem: $\cos n\theta + i\sin n\theta = (\cos\theta + i\sin\theta)^n$

12. (i) Hyperbolic functions: $\sinh x = \dfrac{e^x - e^{-x}}{2}$; $\cosh x = \dfrac{e^x + e^{-x}}{2}$;

$$\tanh x = \frac{\sinh x}{\cosh x};\ \operatorname{cosech} x = \frac{\cosh x}{\sinh x};\ \operatorname{sech} x = \frac{1}{\cosh x};\ \coth x = \frac{1}{\tanh x}$$

(ii) Relations between hyperbolic and circular functions:

$\sin ix = i\sinh x$; $\cos ix = \cosh x$; $\tan ix = i\tanh x$

(iii) Inverse hyperbolic functions:

$$\sinh^{-1} x = \log\left[x + \sqrt{(x^2 + 1)}\right];\ \cosh^{-1} x = \log\left[x + \sqrt{(x^2 - 1)}\right];\ \tanh^{-1} x = \frac{1}{2}\log\frac{1+x}{1-x}$$

VI. DIFFERENTIAL CALCULUS

1. Standard limits:

(i) $\lim_{x\to a}\dfrac{x^n - a^n}{x - a} = nx^{n-1}$, n any rational number

(ii) $\lim_{x\to 0}\dfrac{\sin x}{x} = 1$

(iii) $\lim_{x\to 0}(1 + x)^{1/x} = e$

(iv) $\lim_{x\to\infty} e^{-x} = 0$

(v) $\lim_{x\to a}\dfrac{a^x - 1}{x} = \log_e a$

2. Differentiation

(i) $\frac{d}{dx}[cf(x)] = c\frac{df}{dx} = cf'(x)$

$\frac{dy}{dx} = \frac{dy}{dt}\cdot\frac{dt}{dx}$ (Chain Rule)

(ii) $\frac{d}{dx}(e^x) = e^x$

$\frac{d}{dx}(\log_e x) = \frac{1}{x}$

(iii) $\frac{d}{dx}(\sin x) = \cos x$

$\frac{d}{dx}(\tan x) = \sec^2 x$

$\frac{d}{dx}(\sec x) = \sec x \tan x$

(iv) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{(1-x^2)}}$

$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{(x^2-1)}}$

(v) $\frac{d}{dx}(\sinh x) = \cosh x$

$\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

(vi) $D^n(\sin x) = \sin\left(x + \frac{n\pi}{2}\right)$, $D^n(\log(ax+b)) = (-1)^{n-1}\frac{(n-1)!\,a^n}{(ax+b)^n}$

$D^n(e^{ax}) = a^n e^{ax}$ $D^n(a^{mx+b}) = a^{mx+b}(\log a)^n\, m^n$

$D^n\begin{bmatrix}\sin(ax+b)\\ \cos(ax+b)\end{bmatrix} = a^n\begin{bmatrix}\sin\left(ax+b+\frac{n\pi}{2}\right)\\ \cos\left(ax+b+\frac{n\pi}{2}\right)\end{bmatrix}$

$D^n\begin{bmatrix}\sin(ax+b)\\ \cos(ax+b)\end{bmatrix} = a^n(a^2+b^2)^{n/2}\begin{bmatrix}\sin(2x+c+n\tan^{-1}b/a)\\ \cos(2x+c+n\tan^{-1}b/a)\end{bmatrix}$

(vii) Leibnitz's theorem: $(uv)_n = u_n v + {}^nC_1 u_{n-1}v_1 + {}^nC_2 u_{n-2}v_2 + \cdots + {}^nC_r u_{n-r}v_r + \cdots + uv_n$

3. (i) Maclaurin's series: $f(x) = f(0) + \frac{x}{1!}f'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ldots$

(ii) Taylor's series: $f(a+x) = f(a) + \frac{x}{1!}f'(a) + \frac{x^2}{2!}f''(a) + \frac{x^3}{3!}f'''(a) + \ldots$

4. Curvature

(i) Radius of curvature $\rho = \frac{[1+y_1^2]^{3/2}}{y_2}$, $\rho = \frac{(x'^2+y'^2)^{3/2}}{x'y'' - y'x''}$, $\rho = \frac{ds}{d\psi}$

(ii) Centre of curvature: $\alpha = x - \frac{y_1(1+y_1^2)}{y_2}$, $\beta = y + \frac{1}{y_2}(1+y_1^2)$

(iii) **Evolute** is the locus of the centre of curvature of a curve. The curve is called the involute of the evolute.

(iv) **Envelope** of a curve $f(x, y, a) = 0$ is obtained from

$$f(x, y, a) = 0 \text{ and } \frac{\partial f}{\partial a}(x, y, a) = 0.$$

The envelope of the normals to a curve is its evolute.

6. Asymptotes

(i) Investigate the possibility of a vertical asymptote by equating to zero the coefficient of the highest power of x in the equation, provided that it is not merely a constant.

Investigate a possible $y = mx + c$ asymptote by substituting $y = mx + c$ in the equation, provided that it is not merely a constant.

(ii) Oblique asymptotes are determined as follows:

Take $y = mx + c$ as the [illegible] equation, [illegible]

Find $y_1 - (mx + c)$ and find the limit of it.

Find c from $c = \lim_{x \to \infty} [y_1 - mx]$

If m or values of m are not found, there [illegible]

$$\lim_{x \to \infty} [y_1 - (mx + c)] = \lim_{x \to \infty} [y_1 - y_{m,c}] = 0$$

The asymptotes to $y = mx + c$

(iii) Asymptotes of polar curves $r(\theta) = f(\theta)$ are $r\sin(\theta - \alpha) = f(\theta)$ where α is a root of $f'(\theta) = 0$

8. Singular Points

(i) A point is a [illegible] if there is only [illegible] passing through it.

(ii) A point is a [illegible] if there is only [illegible] passing through it.

(iii) A curve is [illegible] about the line $y = x$ if [illegible] and y, the equation remains unchanged.

(iv) A curve passes through the origin, if there is no constant term in the equation.

(v) Tangents to curves at the origin are found by equating to zero the lowest degree terms.

7. Partial Differentiation

(i) Euler's Theorem: If u_n is a homogeneous function in x and y of degree n, then

$$x\frac{\partial u_n}{\partial x} + y\frac{\partial u_n}{\partial y} = nu_n \quad x^2\frac{\partial^2 u_n}{\partial x^2} + 2xy\frac{\partial^2 u_n}{\partial x \partial y} + y^2\frac{\partial^2 u_n}{\partial y^2} = n(n-1)u_n$$

(ii) Chain rule: $\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$ where $x = f(u, v)$, $y = g(u, v)$

(iii) $\frac{\partial(x,y)}{\partial(u,v)} = \frac{\partial(u,v)}{\partial(x,y)}^{-1}$ where $u = f(x, y)$, $v = g(x, y)$

(iv) Jacobians: $J = \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$

If $u = f(x, y)$, $v = g(x, y)$ and $J' = \frac{\partial(u,v)}{\partial(x,y)}$ and $J = \frac{\partial(x,y)}{\partial(u,v)}$ then $JJ' = 1$

$$\frac{\partial(u,v)}{\partial(x,y)} = \frac{\partial(u,v)}{\partial(r,\theta)} \cdot \frac{\partial(r,\theta)}{\partial(x,y)}$$

(v) Taylor's series: $f(x + h, y + k) = f(x, y) + \left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)f(x, y) + \frac{1}{2!}\left(h\frac{\partial}{\partial x} + k\frac{\partial}{\partial y}\right)^2 f(x, y) + \ldots$

(vi) Maxima: Where $f_x = 0$, $f_y = 0$

(vii) $\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 > 0$ for maximum $\frac{\partial^2 f}{\partial x^2} < 0$ for minimum

(viii) Gradient is Max: $\frac{\partial}{\partial x}\left[\int_{\phi_1(x)}^{\phi_2(x)} f(x, y)\,dy\right] = \int_{\phi_1(x)}^{\phi_2(x)} \frac{\partial f(x,y)}{\partial x}\,dy$

where $f(x, y)$ and $\frac{\partial f(x,y)}{\partial x}$ are continuous functions of x and y in $a \le x \le b$, $\phi_1(x) \le y \le \phi_2(x)$.

VII. INTEGRAL CALCULUS

1. Integration

(*i*) $\int x^n\,dx = \dfrac{x^{n+1}}{n+1}\ (n \neq -1)$ $\qquad$ $\int \dfrac{1}{x}\,dx = \log_e x$

$\int e^x\,dx = e^x$ $\qquad$ $\int a^x\,dx = a^x/\log_e a$

(*ii*) $\int \sin x\,dx = -\cos x$ $\qquad$ $\int \cos x\,dx = \sin x$

$\int \tan x\,dx = -\log \cos x$ $\qquad$ $\int \cot x\,dx = \log \sin x$

$\int \sec x\,dx = \log(\sec x + \tan x)$ $\qquad$ $\int \operatorname{cosec} x\,dx = \log(\operatorname{cosec} x - \cot x)$

$\int \sec^2 x\,dx = \tan x$ $\qquad$ $\int \operatorname{cosec}^2 x\,dx = -\cot x$

(*iii*) $\int \dfrac{dx}{a^2+x^2} = \dfrac{1}{a}\tan^{-1}\dfrac{x}{a}$ $\qquad$ $\int \dfrac{dx}{\sqrt{(a^2-x^2)}} = \sin^{-1}\dfrac{x}{a}$

$\int \dfrac{dx}{a^2-x^2} = \dfrac{1}{2a}\log\dfrac{a+x}{a-x}$ $\qquad$ $\int \dfrac{dx}{\sqrt{(a^2+x^2)}} = \sinh^{-1}\dfrac{x}{a}$

$\int \dfrac{dx}{x^2-a^2} = \dfrac{1}{2a}\log\dfrac{x-a}{x+a}$ $\qquad$ $\int \dfrac{dx}{\sqrt{(x^2-a^2)}} = \cosh^{-1}\dfrac{x}{a}$

(*iv*) $\int \sqrt{(a^2-x^2)}\,dx = \dfrac{x\sqrt{(a^2-x^2)}}{2} + \dfrac{a^2}{2}\sin^{-1}\dfrac{x}{a}$

$\int \sqrt{(a^2+x^2)}\,dx = \dfrac{x\sqrt{(a^2+x^2)}}{2} + \dfrac{a^2}{2}\sinh^{-1}\dfrac{x}{a}$

2. Lengths of curves

(a) Length of curve $y = f(x)$ between $x = a$, $x = b$: $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$

(b) Length of curve $x = f(y)$ between $y = c$, $y = d$: $\int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\,dy$

(c) Length of curve $x = f(t)$, $y = g(t)$ between $t = t_1$, $t = t_2$: $\int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$

(d) Length of curve $r = f(\theta)$ between $\theta = \alpha$, $\theta = \beta$: $\int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta$

3. Areas of curves

(a) Area bounded by $y = f(x)$, x-axis and $x = a$, $x = b$: $\int_a^b y\,dx$

(b) Area bounded by $x = f(y)$, y-axis and $y = c$, $y = d$: $\int_c^d x\,dy$

(c) Area bounded by $r = f(\theta)$ and lines $\theta = \alpha$, $\theta = \beta$: $\frac{1}{2}\int_\alpha^\beta r^2\,d\theta$

4. Volumes of revolution

(a) Volume of revolution about x-axis of area bounded by $y = f(x)$, x-axis and $x = a$, $x = b$:

$$\int_a^b \pi y^2\,dx$$

(b) Volume of revolution about y-axis of area bounded by $x = f(y)$, y-axis and $y = c$, $y = d$:

$$\int_c^d \pi x^2\,dy$$

(c) Volume of revolution bounded by $r = f(\theta)$ and $\theta = \alpha$, $\theta = \beta$:

(a) about OC $= \int_\alpha^\beta \frac{2\pi}{3} r^3 \cos\theta\,d\theta$ (b) about OT $= \int_\alpha^\beta \frac{2\pi}{3} r^3 \sin\theta\,d\theta$

5. Surface areas of revolution

(a) Surface area of revolution about x-axis of curve $y = f(x)$ from $x = a$ to $x = b$:

$$S = \int_{x=a}^{x=b} 2\pi y\,ds$$

Cartesian form: $S = \int_a^b 2\pi y \frac{ds}{dx}\,dx$ where $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

Parametric form: $S = \int 2\pi y \frac{ds}{dt}\,dt$ where $\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$

Polar form: $S = \int 2\pi y \frac{ds}{d\theta}\,d\theta$ where $\frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$

(b) Surface area of revolution about y-axis is $\int 2\pi x\,ds$

6. Multiple integrals

(a) $\int_{x_1}^{x_2}\int_{y_1}^{y_2} dx\,dy$; Volume $= \int_{x_1}^{x_2}\int_{y_1}^{y_2} z\,dx\,dy = \int_{x_1}^{x_2}\int_{y_1}^{y_2}\int_{z_1}^{z_2} dx\,dy\,dz$

(b) C.G. of a plate bounded by $x = x_1$, $x = x_2$, $y = y_1$, $y = y_2$: $\bar{x} = \frac{\int\int x\rho\,dx\,dy}{\int\int \rho\,dx\,dy}$, $\bar{y} = \frac{\int\int y\rho\,dx\,dy}{\int\int \rho\,dx\,dy}$

(c) C.G. of a solid $\bar{x} = \frac{\iiint \rho x\,dx\,dy\,dz}{\iiint \rho\,dx\,dy\,dz}$, $\bar{y} = \frac{\iiint \rho y\,dx\,dy\,dz}{\iiint \rho\,dx\,dy\,dz}$, $\bar{z} = \frac{\iiint \rho z\,dx\,dy\,dz}{\iiint \rho\,dx\,dy\,dz}$

(d) Centre of pressure $\bar{x} = \frac{\int_a^b \int_0^{y} \rho x\,dx\,dy}{\int_a^b \int_0^{y} \rho x\,dx\,dy}$, $\bar{y} = \frac{\int_a^b \int_0^{y} \rho y\,dx\,dy}{\int_a^b \int_0^{y} \rho\,dx\,dy}$

(e) M.I. about x-axis $= I_x = \iiint_V \rho(y^2 + z^2)\,dx\,dy\,dz$

M.I. about y-axis $= I_y = \iiint_V \rho(z^2 + x^2)\,dx\,dy\,dz$

M.I. about z-axis $= I_z = \iiint_V \rho(x^2 + y^2)\,dx\,dy\,dz$

3. Gamma function $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1}\,dx = (n-1)!$, $\Gamma(n+1) = n\Gamma(n) = n!$, $\Gamma(0) = \infty$

Beta function $\beta(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$ $(m > 0, n > 0)$

VIII. VECTORS

1. (a) If $\mathbf{R} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ then $r = |\mathbf{R}| = \sqrt{x^2 + y^2 + z^2}$

(b) $\overrightarrow{PQ}$ = Position vector of Q – position vector of P.

2. If $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$, $\mathbf{B} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$, then

(a) Scalar product: $\mathbf{A} \cdot \mathbf{B} = ab\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$

(b) Vector product: $\mathbf{A} \times \mathbf{B} = ab\sin\theta\,\hat{\mathbf{N}}$ = Area of the parallelogram having $\mathbf{A}$ and $\mathbf{B}$ as sides

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

(c) If $\mathbf{A} \perp \mathbf{B}$ i.e. $\mathbf{A} \cdot \mathbf{B} = 0$ and $\mathbf{A}$ is parallel to $\mathbf{B}$ if $\mathbf{A} \times \mathbf{B} = \mathbf{0}$

3. (a) Scalar triple product $[\mathbf{A}\,\mathbf{B}\,\mathbf{C}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ = Volume of parallelopiped

(b) $[\mathbf{A}\,\mathbf{B}\,\mathbf{C}] = 0$, then $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are coplanar.

(c) Vector triple product $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\,\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\,\mathbf{C}$

$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\mathbf{C} \cdot \mathbf{A})\,\mathbf{B} - (\mathbf{C} \cdot \mathbf{B})\,\mathbf{A}$

4. (a) $\text{grad}\,f = \nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$

(b) $\text{div}\,\mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$

$$\text{curl}\,\mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} \quad \text{where } \mathbf{F} = f_1\mathbf{i} + f_2\mathbf{j} + f_3\mathbf{k}$$

(c) If $\text{div}\,\mathbf{F} = 0$, then $\mathbf{F}$ is called a solenoidal vector.

(d) If $\text{curl}\,\mathbf{F} = \mathbf{0}$ then $\mathbf{F}$ is called an irrotational vector

5. Velocity $= d\mathbf{R}/dt$; Acceleration $= d^2\mathbf{R}/dt^2$; Tangential vector $= d\mathbf{R}/ds$; Normal vector $= \nabla\phi$

6. Green's theorem: $\int_C \eta\,dx + \varphi\,dy = \iint_R \left(\frac{\partial \varphi}{\partial x} - \frac{\partial \eta}{\partial y}\right) dx\,dy$

Stokes's theorem: $\int_C \mathbf{F}\cdot d\mathbf{R} = \int_S \operatorname{curl}\mathbf{F}\cdot\mathbf{N}\,dS$

Gauss divergence theorem: $\int_S \mathbf{F}\cdot\mathbf{N}\,dS = \int_V \operatorname{div}\mathbf{F}\,dV$

3. Coordinate systems

	Polar coordinates (r,θ)	Cylindrical coordinates (ρ,ϕ,z)	Spherical polar coordinates (r,θ,ϕ)
Cartesian transformations	$x = r\cos\theta$ $y = r\sin\theta$	$x = \rho\cos\phi$ $y = \rho\sin\phi$ $z = z$	$x = r\sin\theta\cos\phi$ $y = r\sin\theta\sin\phi$ $z = r\cos\theta$
Jacobian Line length ds^2	$\dfrac{\partial(x,y)}{\partial(r,\theta)} = r$ $ds^2 = dr^2 + r^2 d\theta^2$ (plane) Area $= r\,dr\,d\theta$	$\dfrac{\partial(x,y,z)}{\partial(\rho,\phi,z)} = \rho$ $ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2$ Volume $= \rho\,d\rho\,d\phi\,dz$	$\dfrac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = r^2\sin\theta$ $ds^2 = dr^2 + r^2 d\theta^2 + r^2\sin^2\theta\,d\phi^2$ Volume $= r^2\sin\theta\,dr\,d\theta\,d\phi$
Volume element		$dV = \rho\,d\rho\,d\phi\,dz$	$dV = r^2\sin\theta\,dr\,d\theta\,d\phi$

IX. DIFFERENTIAL EQUATIONS

1. Equations of first order

(i) Variables separable: $f(y)\,dy/dx = \phi(x)$, or $\int f(y)\,dy = \int \phi(x)\,dx + c$.

(ii) Homogeneous equation $dy/dx = f(x,y)/g(x,y)$ where $f(x,y)$ and $g(x,y)$ are of the same degree. Put $y = vx$ so that $dy/dx = v + x\,dv/dx$.

(iii) Equations reducible to homogeneous form: $\dfrac{dy}{dx} = \dfrac{ax + by + c}{a'x + b'y + c'}$

When $ab' \neq ba'$, put $x = X + h$, $y = Y + k$.

When $ab' = ba'$, put $ax + by = v$.

(iv) Leibnitz's linear equation: $\dfrac{dy}{dx} + Py = Q$ where P, Q are functions of x.

$IF = e^{\int P\,dx}$, the solution is $y\,(IF) = \int Q\,(IF)\,dx + c$.

(v) Bernoulli's equation: $dy/dx + Py = Qy^n$, reducible to Leibnitz's equation by writing it as

$$y^{-n}\frac{dy}{dx} + Py^{1-n} = Q \text{ and putting } y^{1-n} = z.$$

(vi) Exact equation: $M(x,y)\,dx + N(x,y)\,dy = 0$

Solution is $\int_{(y=\text{const.})} M\,dx + \int$ (terms of N not containing x)$\,dy = c$, provided $\partial M/\partial y = \partial N/\partial x$.

(vii) Clairaut's equation: $y = px + f(p)$, where $p = dy/dx$.

Solution is obtained on replacing p by c.

2. Linear equations with constant coefficients

$$\frac{d^n y}{dx^n} + k_1\frac{d^{n-1}y}{dx^{n-1}} + \cdots + k_{n-1}\frac{dy}{dx} + k_n y = X$$

Symbolic form: $D^n y + k_1 D^{n-1} y + \cdots + k_{n-1} D y + k_n y = X$.

I. To find C.F.

Roots of A.E.	C.F.
(a) $m_1, m_2, m_3, \ldots$ —	$c_1e^{m_1x} + c_2e^{m_2x} + c_3e^{m_3x} + \ldots$
(b) $m_1, m_1, m_3, \ldots$	$(c_1 + c_2x)e^{m_1x} + c_3e^{m_3x} + \ldots$
(c) $\alpha \pm i\beta, m_3, m_4, \ldots$	$e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x) + c_3e^{m_3x} + \ldots$
(d) $\alpha \pm i\beta, \alpha \pm i\beta, \ldots$	$e^{\alpha x}[(c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x] + \ldots$

II. To find P.I.

(a) $f(D)y = e^{ax}$, P.I. $= \frac{1}{f(D)}e^{ax}$, put $D = a$, provided $f(a) \neq 0$

$= x\frac{1}{f'(D)}e^{ax}$, put $D = a$, provided $f'(a) = 0, f'(a) \neq 0$

$= x^2\frac{1}{f''(D)}e^{ax}$, put $D = a$, provided $f''(a) = 0, f''(a) \neq 0$

(b) $f(D)y = \sin(ax + b)$ or $\cos(ax + b)$

P.I. $= \frac{1}{\phi(D^2)}\sin(ax + b)$ or $\cos(ax + b)$, put $D^2 = -a^2$, [$\phi(-a^2) \neq 0$]

$= x\frac{1}{\phi'(D^2)}\sin(ax + b)$ or $\cos(ax + b)$, put $D^2 = -a^2$, [$\phi'(-a^2) \neq 0$]

$= x^2\frac{1}{\phi''(D^2)}\sin(ax + b)$ or $\cos(ax + b)$, put $D^2 = -a^2$, [$\phi''(-a^2) = 0, \phi''(-a^2) \neq 0$]

(c) $f(D)y = x^n$, P.I. $= \frac{1}{f(D)}x^n = [f(D)]^{-1}x^n$. Expand $[f(D)]^{-1}$ in ascending powers of D so far as D^n and operate on x^n term by term.

(d) $f(D)y = e^{ax}V$, P.I. $= \frac{1}{f(D)}e^{ax}V = e^{ax}\frac{1}{f(D+a)}V$.

III. Complete Solution = C.F. + P.I.

1. Homogeneous linear equation $x^n\frac{d^ny}{dx^n} + k_1x^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \ldots + k_{n-1}x\frac{dy}{dx} + k_ny = X$.

Reduce to linear equation with constant coefficients by putting

$x = e^z$, $x\frac{dy}{dx} = Dy$, $x^2\frac{d^2y}{dx^2} = D(D-1)y$, $x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y$

4. Lagrange's linear partial differential equation

$Pp + Qq = R$, P, Q, R being functions of x, y, z.

To solve (1) (1) form the subsidiary equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

(a) solve these equations giving $u = a, v = b$

(b) Complete solution is $\phi(u, v) = 0$ or $u = f(v)$.

5. Homogeneous linear partial differential equations with constant coefficients:

$$\frac{\partial^n z}{\partial x^n} + k_1\frac{\partial^n z}{\partial x^{n-1}\partial y} + \ldots + k_n\frac{\partial^n z}{\partial y^n} = F(x, y)$$

Symbolic form: $(D^n + k_1D^{n-1}D' + \ldots + k_nD'^n)z = F(x, y)$

To find C.F.

Roots of A.E.	C.F.
(i) $m_1, m_2, m_3, \ldots$	$f_1(y+m_1x)+f_2(y+m_2x)+f_3(y+m_3x)+\ldots$
(ii) $m_1, m_2, m_3, \ldots$	$f_1(y+m_1x)+xf_2(y+m_2x)+x^2f_3(y+m_3x)+\ldots$
(iii) $\alpha+i\beta, \alpha-i\beta, m_3, \ldots$	$e^{\alpha x}[f_1(y+\beta x)+xf_2(y+\beta x)]+f_3(y+m_3x)+\ldots$

To find P.I.

(i) $F(x,y)=e^{ax+by}$, P.I. $=\dfrac{1}{f(D,D')}e^{ax+by}$, put $D=a, D'=b$.

(ii) $F(x,y)=\sin(ax+by)$ or $\cos(ax+by)$

$$P.I.=\frac{1}{f(D^2,DD',D'^2)}\sin(ax+by) \text{ put } D^2=-a^2, DD'=-ab, D'^2=-b^2$$

(iii) $F(x,y)=x^my^n$, P.I. $=[f(D,D')]^{-1}x^my^n$. Expand $[f(D,D')]^{-1}$ and operate on x^my^n.

(iv) $F(x,y)$ is any function of x and y, P.I. $=\dfrac{1}{f(D,D')}F(x,y)$.

Resolve $[f(D,D')]^{-1}$ into partial fractions considering $f(D,D')$ as a function of D alone and operate each partial fraction on $F(x,y)$ remembering that $\dfrac{1}{D-mD'}F(x,y)=\int F(x,c-mx)\,dx$.

Complete solution: C.S. is $y=$ C.F. + P.I.

X. INFINITE SERIES

1. *Basic test:* If $\lim_{n\to\infty} u_n \neq 0$ then the series Σu_n diverges.

2. *G.P. Series:* $1+r+r^2+r^3+\ldots$ converges if $|r|<1$, diverges if $r\geq 1$ and oscillates if $r\leq -1$.

3. *p series:* $\dfrac{1}{1^p}+\dfrac{1}{2^p}+\dfrac{1}{3^p}+\ldots$ converges for $p>1$, diverges for $p\leq 1$.

4. *Comparison test:* Given positive term series Σu_n and Σv_n be such that $\lim_{n\to\infty}\dfrac{u_n}{v_n}$ is finite quantity $\neq 0$, then Σu_n and Σv_n converge or diverge together.

5. *Ratio test:* In the positive term series Σu_n if $\lim_{n\to\infty}\dfrac{u_n}{u_{n+1}}=k$, then the series converges for $k>1$, diverges for $k<1$ and fails for $k=1$.

6. *Raabe's test:* In the positive term series Σu_n, if $\lim_{n\to\infty} n\left(\dfrac{u_n}{u_{n+1}}-1\right)=k$, then the series converges for $k>1$, diverges for $k<1$ and fails for $k=1$.

7. *Logarithmic test:* In the positive term series Σu_n, if $\lim_{n\to\infty}\left(n\log\dfrac{u_n}{u_{n+1}}\right)=k$, then the series converges for $k>1$, diverges for $k<1$ and fails for $k=1$.

8. If u_n/u_{n+1} does not tend to one as the exponent of a logarithm, then the series Σu_n diverges.

9. *Cauchy's root test:* In a positive term series Σu_n if $\lim_{n\to\infty} u_n^{1/n}=k$, then the series converges for $k<1$, diverges for $k>1$ and fails for $k=1$.

10. *Integral test:* A positive term series $f(n)$ converges or diverges according as $\int_1^\infty f(x)\,dx$ is finite or infinite where $f(x)$ is continuous in $1<x<\infty$ and decreases as x increases.

11. *Leibnitz's test for alternating series:* An alternating series $u_1-u_2+u_3-u_4+\ldots$ converges if each term is numerically less than the previous term and $\lim_{n\to\infty} u_n=0$.

If $\lim_{n\to\infty} u_n\neq 0$, then the given series is not convergent.

12. *General Rule for* $\sum a_n$: If $\sum a_n$ is an arbitrary series and $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = |R|$, then $\sum a_n$ is absolutely convergent if $|R| < 1$ and divergent if $|R| > 1$ and the test fails if $|R| = 1$.

XI. FOURIER SERIES

1. $f(x) = \frac{1}{2}a_0 + a_1\cos x + a_2\cos 2x + \ldots + b_1\sin x + b_2\sin 2x + \ldots$ to $[0, 2\pi]$

   where $a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx$, $a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx$, $b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$

2. $f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n\cos\frac{n\pi x}{c} + \sum_{n=1}^{\infty} b_n\sin\frac{n\pi x}{c}$ over the interval $(0, 2c)$

   where $a_0 = \frac{1}{c}\int_0^{2c} f(x)\,dx$, $a_n = \frac{1}{c}\int_0^{2c} f(x)\cos\frac{n\pi x}{c}\,dx$, $b_n = \frac{1}{c}\int_0^{2c} f(x)\sin\frac{n\pi x}{c}\,dx$

3. For even function $f(x)$, Fourier expansion reduces to only cosine terms

   i.e., $a_0 = \frac{2}{c}\int_0^{c} f(x)\,dx$, $a_n = \frac{2}{c}\int_0^{c} f(x)\cos\frac{n\pi x}{c}\,dx$, $b_n = 0$.

   For odd function $f(x)$, Fourier expansion reduces to only sine terms

   i.e., $a_0 = 0, a_n = 0, b_n = \frac{2}{c}\int_0^{c} f(x)\sin\frac{n\pi x}{c}\,dx$.

XII. TRANSFORMS

1. Laplace Transform: $L\{f(t)\} = \int_0^{\infty} e^{-st} f(t)\,dt$

   (i) $L\{1\} = \frac{1}{s}$

   (ii) $L\{t^n\} = \frac{n!}{s^{n+1}}$

   (iii) $L\{e^{at}\} = \frac{1}{s-a}$

   (iv) $L\{\cos at\} = \frac{s}{s^2+a^2}$

   (v) $L\{\sin at\} = \frac{a}{s^2+a^2}$

   (vi) $L\{\cosh at\} = \frac{s}{s^2-a^2}$

   (vii) $L\{\sinh at\} = \frac{a}{s^2-a^2}$

   (viii) $L\{e^{at}f(t)\} = F(s-a)$

   (ix) $L\{f'(t)\} = sL\{f(t)\} - f(0)$

   (x) $L\{t^n f(t)\} = (-1)^n\frac{d^n}{ds^n}[F(s)]$

   (xi) $L\left\{\int_0^t f(u)\,du\right\} = \frac{1}{s}F(s)$

   (xii) $u(t-a) = \begin{cases} 0 \text{ when } t < a \\ 1 \text{ when } t > a \end{cases}$

   (xiii) $L\{u(t-a)\} = \frac{e^{-as}}{s}$

   (xiv) $L\{u(t-a)\} = e^{-as}$

   (xv) $L\{f(t)\} = \frac{\int_0^T e^{-st} f(t)\,dt}{1-e^{-sT}}$ where $f(t)$ is a periodic function of period T.

2. Inverse Laplace Transforms:

   (i) $L^{-1}\left\{\frac{1}{s}\right\} = 1$

   (ii) $L^{-1}\left\{\frac{1}{s^n}\right\} = \frac{t^{n-1}}{(n-1)!}$

   (iii) $L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}$

   (iv) $L^{-1}\left\{\frac{1}{s^2+a^2}\right\} = \frac{1}{a}\sin at$

   (v) $L^{-1}\left\{\frac{s}{s^2+a^2}\right\} = \cos at$

   (vi) $L^{-1}\left\{\frac{s}{s^2-a^2}\right\} = \frac{1}{a}\sinh at$

(viii) $L^{-1}\left[\frac{s}{(s^2-a^2)^2}\right] = \frac{t}{2a}\sinh at$ (ix) $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right] = \frac{t}{2a}\sin at$

(x) $L^{-1}\left[\frac{s^2-a^2}{(s^2+a^2)^2}\right] = t\cos at$

6. Fourier Transform : $F(s) = \int_{-\infty}^{\infty} f(x)e^{isx}\,dx$

Fourier sine transform : $F_s(s) = \int_0^{\infty} f(x)\sin sx\,dx$

Fourier cosine transform : $F_c(s) = \int_0^{\infty} f(x)\cos sx\,dx$

$$F\left(\frac{d^2y}{dx^2}\right) = -s^2F(s)$$

7. Z Transform : $Z(a_n) = \sum_{n=0}^{\infty} a_n z^{-n}$

(i) $Z(1) = \frac{z}{z-1}$ (ii) $Z(n) = \frac{z}{(z-1)^2}$

(iii) $Z(n^2) = \frac{z^2+z}{(z-1)^3}$ (iv) $Z(a^n) = \frac{z}{z-a}$

(v) $Z(na^n) = \frac{az}{(z-a)^2}$ (vi) $Z(\sin n\theta) = \frac{z\sin\theta}{z^2-2z\cos\theta+1}$

(vii) $Z(\cos n\theta) = \frac{z^2-z\cos\theta}{z^2-2z\cos\theta+1}$ (viii) $Z(\sinh n\theta) = \frac{z\sinh\theta}{z^2-2z\cosh\theta+1}$

(ix) $Z(\cosh n\theta) = \frac{z^2-z\cosh\theta}{z^2-2z\cosh\theta+1}$

XIII. STATISTICS AND PROBABILITY

1. A.M. $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$ 2. S.D. $\sigma = \sqrt{\frac{\sum f_i(x_i-\bar{x})^2}{\sum f_i}}$

3. Moment about the mean : $\mu_0 = 1, \mu_1 = 0, \mu_2 = \sigma^2, \mu_r = \frac{\sum f_i(x_i-\bar{x})^r}{\sum f_i}$

4. Coeff. of skewness = (mean − median) / standard deviation between −1 and 1.

5. Kurtosis : $\beta_2 = \mu_4/\mu_2^2$.

6. Coeff. of correlation $r = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{[n\sum x_i^2 - (\sum x_i)^2][n\sum y_i^2 - (\sum y_i)^2]}}$, $-1 \le r \le 1$

7. Line of regression of y on x : $y - \bar{y} = r\frac{\sigma_y}{\sigma_x}(x-\bar{x})$

Line of regression of x on y : $x - \bar{x} = r\frac{\sigma_x}{\sigma_y}(y-\bar{y})$

8. **Probability** $p(A) = \frac{\text{No. of ways favourable to } A}{\text{Total no. of equally likely ways}}$, $0 \le p \le 1$.

(i) $p(A \text{ or } B) = p(A) + p(B)$ (ii) $p(A \text{ and } B) = p(A)\cdot p(B)$

9. **Binomial distribution** : $p(r) = {}^nC_r\, p^r q^{n-r}$

Mean $= np$, Variance $(\sigma^2) = npq$

18. Poisson distribution $p(r) = \frac{m^r}{r!} e^{-m}$

Mean $= m$, Variance $(\sigma^2) = m$.

19. Normal distribution $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, Standard variate $= \frac{x-\mu}{\sigma}$

(i) Probable error $\lambda = 0.6745\sigma$.

(ii) 50% of values lie between $x = \mu - \lambda$ and $x = \mu + \lambda$

50% of values lie between $x = \mu - 0.6745\sigma$ and $x = \mu + 0.6745\sigma$

95% of values lie between $x = \mu - 1.96\sigma$ and $x = \mu + 1.96\sigma$

XIV. NUMERICAL TECHNIQUES

1. Solution of equations

(i) Bisection method: $x_2 = \frac{1}{2}(x_0 + x_1)$

(ii) Method of false position: $x_2 = x_0 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0)$

(iii) Newton-Raphson method: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

(iv) Derived formula to find $\sqrt{N}$ is $x_{n+1} = x_n(3 - Nx_n^2)$

(v) Derived formula to find $\sqrt{N}$ is $x_{n+1} = \frac{1}{2}(x_n + Nx_n)$

2. Solution of Linear Simultaneous equations

(i) Matrix inversion method. For the equations

$$a_1x + b_1y + c_1z = d_1,\ a_2x + b_2y + c_2z = d_2,\ a_3x + b_3y + c_3z = d_3$$

if $$A = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

then $$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{|A|} \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} \times \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

where A_1, B_1, etc., are the co-factors of a_1, b_1, etc., in the determinant $|A|$.

(ii) Gauss-Jordan method. In this method the coefficient matrix is transformed to a unit matrix.

(iii) Gauss-Seidal method. In this method the coefficient matrix is transformed to upper triangular matrix.

(iv) Gauss-Jordan method of finding the inverse of a matrix A. The matrices A and I are written side by side and the same row transformations are performed on both till A is reduced to I. Then the other matrix represents A^{-1}.

3. Finite differences and Interpolation

(i) Forward difference $\quad \Delta y_x = y_{x+1} - y_x$

Backward difference $\quad \nabla y_x = y_x - y_{x-1}$

Central difference $\quad \delta y_{x+\frac{1}{2}} = y_{x+1} - y_x$

(ii) Relations between operators:

$$\Delta = E - 1; \nabla = 1 - E^{-1}; \delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$$

$$\mu = \frac{1}{2}(E^{\frac{1}{2}} + E^{-\frac{1}{2}}); \Delta = E\nabla = \nabla E = \delta E^{\frac{1}{2}}; E = e^{hD}$$

(iii) Newton's forward interpolation formula $\quad y_x = y_0 + x\Delta y_0 + \frac{x(x-1)}{2!}\Delta^2 y_0 + \frac{x(x-1)(x-2)}{3!}\Delta^3 y_0 + \ldots$

Newton's backward interpolation formula $\quad y_x = y_0 + x\nabla y_0 + \frac{x(x+1)}{2!}\nabla^2 y_0 + \frac{x(x+1)(x+2)}{3!}\nabla^3 y_0 + \ldots$

(ii) Newton's forward interpolation formula

$$y_p = y_0 + p\,\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \ldots \text{ where } p = (x - x_0)/h.$$

(iii) Newton's backward interpolation formula:

$$y_p = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \ldots \text{ where } p = (x - x_n)/h.$$

(iv) Stirling's formula:

$$y_p = y_0 + p\left(\frac{\Delta y_0 + \Delta y_{-1}}{2}\right) + \frac{p^2}{2!}\Delta^2 y_{-1} + \frac{p(p^2-1)}{3!}\left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right) + \frac{p^2(p^2-1)}{4!}\Delta^4 y_{-2} + \ldots$$

(v) Bessel's formula:

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\left(\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2}\right) + \frac{\left(p - \frac{1}{2}\right)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+1)p(p-1)(p-2)}{4!}\left(\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2}\right) + \ldots$$

(vi) Lagrange's interpolation formula:

$$y = \frac{(x-x_1)(x-x_2)\ldots(x-x_n)}{(x_0-x_1)(x_0-x_2)\ldots(x_0-x_n)}y_0 + \frac{(x-x_0)(x-x_2)\ldots(x-x_n)}{(x_1-x_0)(x_1-x_2)\ldots(x_1-x_n)}y_1 + \ldots + \frac{(x-x_0)(x-x_1)\ldots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\ldots(x_n-x_{n-1})}y_n$$

(vii) Newton's divided difference formula:

$$y = f(x) = y_0 + (x-x_0)[x_0,x_1] + (x-x_0)(x-x_1)[x_0,x_1,x_2] + (x-x_0)(x-x_1)(x-x_2)[x_0,x_1,x_2,x_3] + \ldots$$

where $[x_0,x_1] = \frac{y_1-y_0}{x_1-x_0}$, $[x_0,x_1,x_2] = \frac{[x_1,x_2]-[x_0,x_1]}{x_2-x_0}$ and so on.

4. Numerical differentiation

(i) Forward difference formulae:

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h}\left[\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \ldots\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 + \ldots\right] \text{ and so on.}$$

(ii) Backward difference formulae:

$$\left(\frac{dy}{dx}\right)_{x_n} = \frac{1}{h}\left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \frac{1}{4}\nabla^4 y_n + \ldots\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_n} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \ldots\right] \text{ and so on.}$$

(iii) Central difference formulae:

$$\left(\frac{dy}{dx}\right)_{x_0} = \frac{1}{h}\left[\frac{\delta y_{\frac{1}{2}} + \delta y_{-\frac{1}{2}}}{2} - \frac{1}{6}\frac{\delta^3 y_{\frac{1}{2}} + \delta^3 y_{-\frac{1}{2}}}{2} + \frac{1}{30}\frac{\delta^5 y_{\frac{1}{2}} + \delta^5 y_{-\frac{1}{2}}}{2} + \ldots\right]$$

$$\left(\frac{d^2y}{dx^2}\right)_{x_0} = \frac{1}{h^2}\left[\delta^2 y_0 - \frac{1}{12}\delta^4 y_0 + \frac{1}{90}\delta^6 y_0 + \ldots\right]$$

5. Numerical integration

(i) Trapezoidal rule:

$$\int_{x_0}^{x_0+nh} f(x)\,dx = \frac{h}{2}\left[(y_0 + y_n) + 2(y_1 + y_2 + \ldots + y_{n-1})\right]$$

(a) Simpson's 1/3 Rule:

$$\int_{x_0}^{x_0+nh} f(x)dx = \frac{h}{3}[(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2})]$$

(Number of sub-intervals should be taken as even)

(b) Simpson's 3/8 Rule:

$$\int_{x_0}^{x_0+nh} f(x)dx = \frac{3h}{8}[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3})]$$

(Number of sub-intervals should be taken as a multiple of 3)

(c) Weddle's rule:

$$\int_{x_0}^{x_0+6h} f(x)dx = \frac{3h}{10}(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 + 5y_7 + \dots)$$

(Number of sub-intervals should be taken as multiple of 6)

5. Numerical solution of ordinary differential equations

(i) Picard's method: $y_1 = y_0 + \int_{x_0}^{x} f(x, y_0)dx$

$$y_2 = y_0 + \int_{x_0}^{x} f(x, y_1)dx \text{ etc.}$$

(ii) Taylor's method:

$$y = y_0 + (x - x_0)y'_0 + \frac{(x - x_0)^2}{2!}y''_0 + \frac{(x - x_0)^3}{3!}y'''_0 + \dots$$

(iii) Euler's method: $y_1 = y_0 + hf(x_0, y_0)$

Repeat this process till y_1 is stationary. Then calculate y_2 and so on.

(iv) Modified Euler's method: $y_1 = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_0 + h, y_0 + hf(x_0, y_0))]$

Repeat this step till y_1 is stationary. Then calculate y_2 and so on.

(v) Runge-Kutta method: $y_1 = y_0 + k$ where $k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

such that $k_1 = hf(x_0, y_0)$, $k_2 = hf(x_0 + h/2, y_0 + k_1/2)$

$k_3 = hf(x_0 + h/2, y_0 + k_2/2)$, $k_4 = hf(x_0 + h, y_0 + k_3)$

(vi) Milne's method:

Predictor formula: $y_1 = y_0 + \frac{4h}{3}(2Y_1 - Y_2 + 2Y_3)$

Corrector formula: $y_1 = y_2 + \frac{h}{3}(Y_2 + 4Y_3 + Y_4)$

(Four prior values are required to find the next value)

(vii) Adams-Bashforth method:

Predictor formula: $y_1 = y_0 + \frac{h}{24}(55Y_0 - 59Y_{-1} + 37Y_{-2} - 9Y_{-3})$

Corrector formula: $y_1 = y_0 + \frac{h}{24}(9Y_1 + 19Y_0 - 5Y_{-1} + Y_{-2})$

(Four prior values are required to find the next value)

7. Numerical solution of partial differential equations

(i) Classification of a second order equation:

$$A(x, y)\frac{\partial^2 u}{\partial x^2} + B(x, y)\frac{\partial^2 u}{\partial x \partial y} + C(x, y)\frac{\partial^2 u}{\partial y^2} + \left[f(x, y, u)\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right] = 0$$

is said to be: elliptic if $B^2 - 4AC < 0$

parabolic if $B^2 - 4AC = 0$

hyperbolic if $B^2 - 4AC > 0$

(a) Laplace equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Standard five-point formula: $u_{i,j} = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1})$

Diagonal five-point formula: $u_{i,j} = \frac{1}{4}(u_{i+1,j+1} + u_{i-1,j+1} + u_{i+1,j-1} + u_{i-1,j-1})$

(b) Poisson's equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = g(x, y)$

Standard five-point formula:

$$u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j} = h^2 g(ih, jh)$$

(c) One-dimensional heat equation: $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

Richardson explicit method: $u_{i,j+1} = (1 - 2\alpha^2)u_{i,j} + \alpha^2(u_{i+1,j} + u_{i-1,j})$ where $\alpha = k/h^2$ where $\alpha = 1/2$ is known as

Richardson's Crank–Nicolson method: $u_{i,j+1} = \frac{1}{2}(u_{i+1,j} + u_{i-1,j})$

(d) Wave equation: $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

We get the explicit difference equation

$$u_{i,j+1} = 2(1 - \alpha^2)u_{i,j} + \alpha^2(u_{i+1,j} + u_{i-1,j}) - u_{i,j-1} \quad \text{where } \alpha = k/h$$

It is an elementary matter to find the coefficient of $u_{i,j}$ in terms of α. It is clear that the above equation can be rearranged to give

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j-1}$$

APPENDIX

2

Tables

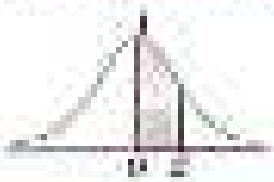
Table I : Gamma Function, $\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt$

x	Γ(x)	x	Γ(x)	x	Γ(x)	x	Γ(x)
1.00	1.000000	1.26	0.904397	1.51	0.886623	1.76	0.920637
1.01	0.994326	1.27	0.902906	1.52	0.886227	1.77	0.921857
1.02	0.988846	1.28	0.901627	1.53	0.886046	1.78	0.923190
1.03	0.983553	1.29	0.900560	1.54	0.886080	1.79	0.924636
1.04	0.978438	1.30	0.899704	1.55	0.886320	1.80	0.926196
1.05	0.973504	1.31	0.899061	1.56	0.886766	1.81	0.927869
1.06	0.968744	1.32	0.898631	1.57	0.887408	1.82	0.929657
1.07	0.964152	1.33	0.898413	1.58	0.888244	1.83	0.931560
1.08	0.959725	1.34	0.898405	1.59	0.889270	1.84	0.933577
1.09	0.955418	1.35	0.898627	1.60	0.890480	1.85	0.935708
1.10	0.951351	1.36	0.899041	1.61	0.891877	1.86	0.937953
1.11	0.947385	1.37	0.899666	1.62	0.893456	1.87	0.940312
1.12	0.943590	1.38	0.900480	1.63	0.895218	1.88	0.942785
1.13	0.939962	1.39	0.901495	1.64	0.897157	1.89	0.945371
1.14	0.936490	1.40	0.902709	1.65	0.899277	1.90	0.948069
1.15	0.933174	1.41	0.904121	1.66	0.901572	1.91	0.950880
1.16	0.930005	1.42	0.905731	1.67	0.904044	1.92	0.953803
1.17	0.926984	1.43	0.907538	1.68	0.906690	1.93	0.956837
1.18	0.924102	1.44	0.909541	1.69	0.909511	1.94	0.959983
1.19	0.921361	1.45	0.911742	1.70	0.912507	1.95	0.963240
1.20	0.918759	1.46	0.914139	1.71	0.915675	1.96	0.966607
1.21	0.916277	1.47	0.916732	1.72	0.919015	1.97	0.970085
1.22	0.913936	1.48	0.919521	1.73	0.922526	1.98	0.973672
1.23	0.911715	1.49	0.922507	1.74	0.926206	1.99	0.977369
1.24	0.909626	1.50	0.925687	1.75	0.930054	2.00	1.000000
1.25	0.907660	1.51	0.929062	1.76	0.934070		

Table 3 | Bessel Functions

x	$J_0(x)$	$Y_0(x)$	x	$J_1(x)$	$Y_1(x)$
0.0	1.0000	0.0000	0.0	0.0000	−∞
0.1	0.9975	0.0893	0.1	0.0499	−1.5342
0.2	0.9900	0.1784	0.2	0.0995	−0.7652
0.3	0.9776	0.2667	0.3	0.1483	−0.5033
0.4	0.9604	0.3541	0.4	0.1960	−0.3715
0.5	0.9385	0.4404	0.5	0.2423	−0.2919
0.6	0.9120	0.5252	0.6	0.2867	−0.2394
0.7	0.8812	0.6084	0.7	0.3290	−0.2027
0.8	0.8463	0.6901	0.8	0.3688	−0.1754
0.9	0.8075	0.7699	0.9	0.4059	−0.1541
1.0	0.7652	0.8461	1.0	0.4401	−0.1366
1.1	0.7196	0.9208	1.1	0.4709	−0.1218
1.2	0.6711	0.9918	1.2	0.4983	−0.1089
1.3	0.6201	1.0588	1.3	0.5220	−0.0973
1.4	0.5669	1.1215	1.4	0.5419	−0.0867
1.5	0.5118	1.1797	1.5	0.5579	−0.0768
1.6	0.4554	1.2329	1.6	0.5699	−0.0674
1.7	0.3980	1.2810	1.7	0.5778	−0.0583
1.8	0.3400	1.3241	1.8	0.5815	−0.0496
1.9	0.2818	1.3623	1.9	0.5812	−0.0411
2.0	0.2239	1.3957	2.0	0.5767	−0.0329
2.1	0.1666	1.4245	2.1	0.5683	−0.0247
2.2	0.1104	1.4490	2.2	0.5560	−0.0166
2.3	0.0555	1.4695	2.3	0.5399	−0.0086
2.4	0.0025	1.4860	2.4	0.5202	−0.0007
2.5	−0.0484	1.4987	2.5	0.4971	+0.0071
2.6	−0.0968	1.5079	2.6	0.4708	+0.0148
2.7	−0.1424	1.5137	2.7	0.4416	+0.0224
2.8	−0.1850	1.5162	2.8	0.4097	+0.0297
2.9	−0.2243	1.5155	2.9	0.3754	+0.0367

Table 11 | Area under the Normal curve



z	0	1	2	3	4	5	6	7	8	9
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2258	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993

Table IV: Values of t for the probability P and degrees of freedom ν

ν \ P	0.001	0.01	0.05	0.10	0.20
1	636.619	63.657	12.706	6.314	3.078
2	31.599	9.925	4.303	2.920	1.886
3	12.924	5.841	3.182	2.353	1.638
4	8.610	4.604	2.776	2.132	1.533
5	6.869	4.032	2.571	2.015	1.476
6	5.959	3.707	2.447	1.943	1.440
7	5.408	3.499	2.365	1.895	1.415
8	5.041	3.355	2.306	1.860	1.397
9	4.781	3.250	2.262	1.833	1.383
10	4.587	3.169	2.228	1.812	1.372
11	4.437	3.106	2.201	1.796	1.363
12	4.318	3.055	2.179	1.782	1.356
13	4.221	3.012	2.160	1.771	1.350
14	4.140	2.977	2.145	1.761	1.345
15	4.073	2.947	2.131	1.753	1.341
16	4.015	2.921	2.120	1.746	1.337
17	3.965	2.898	2.110	1.740	1.333
18	3.922	2.878	2.101	1.734	1.330
19	3.883	2.861	2.093	1.729	1.328
20	3.850	2.845	2.086	1.725	1.325
21	3.819	2.831	2.080	1.721	1.323
22	3.792	2.819	2.074	1.717	1.321
23	3.768	2.807	2.069	1.714	1.319
24	3.745	2.797	2.064	1.711	1.318
25	3.725	2.787	2.060	1.708	1.316
26	3.707	2.779	2.056	1.706	1.315
27	3.690	2.771	2.052	1.703	1.314
28	3.674	2.763	2.048	1.701	1.313
29	3.659	2.756	2.045	1.699	1.311
30	3.646	2.750	2.042	1.697	1.310
40	3.551	2.704	2.021	1.684	1.303
60	3.460	2.660	2.000	1.671	1.296
120	3.373	2.617	1.980	1.658	1.289
∞	3.291	2.576	1.960	1.645	1.282

Table V : Values of χ^2 for a probability P and n'

n' \ P	0.99	0.98	0.95	0.90	0.80	0.70	0.50	0.30
1	0.000157	0.000628	0.00393	0.0158	0.064	0.148	0.455	1.074
2	0.0201	0.0404	0.103	0.211	0.446	0.713	1.386	2.408
3	0.115	0.185	0.352	0.584	1.005	1.424	2.366	3.665
4	0.297	0.429	0.711	1.064	1.649	2.195	3.357	4.878
5	0.554	0.752	1.145	1.610	2.343	3.000	4.351	6.064
6	0.872	1.134	1.635	2.204	3.070	3.828	5.348	7.231
7	1.239	1.564	2.167	2.833	3.822	4.671	6.346	8.383
8	1.646	2.032	2.733	3.490	4.594	5.527	7.344	9.524
9	2.088	2.532	3.325	4.168	5.380	6.393	8.343	10.656
10	2.558	3.059	3.940	4.865	6.179	7.267	9.342	11.781
11	3.053	3.609	4.575	5.578	6.989	8.148	10.341	12.899
12	3.571	4.178	5.226	6.304	7.807	9.034	11.340	14.011
13	4.107	4.765	5.892	7.042	8.634	9.926	12.340	15.119
14	4.660	5.368	6.571	7.790	9.467	10.821	13.339	16.222
15	5.229	5.985	7.261	8.547	10.307	11.721	14.339	17.322
16	5.812	6.614	7.962	9.312	11.152	12.624	15.338	18.418
17	6.408	7.255	8.672	10.085	12.002	13.531	16.338	19.511
18	7.015	7.906	9.390	10.865	12.857	14.440	17.338	20.601
19	7.633	8.567	10.117	11.651	13.716	15.352	18.338	21.689
20	8.260	9.237	10.851	12.443	14.578	16.266	19.337	22.775
21	8.897	9.915	11.591	13.240	15.445	17.182	20.337	23.858
22	9.542	10.600	12.338	14.041	16.314	18.101	21.337	24.939
23	10.196	11.293	13.091	14.848	17.187	19.021	22.337	26.018
24	10.856	11.992	13.848	15.659	18.062	19.943	23.337	27.096
25	11.524	12.697	14.611	16.473	18.940	20.867	24.337	28.172
26	12.198	13.409	15.379	17.292	19.820	21.792	25.336	29.246
27	12.879	14.125	16.151	18.114	20.703	22.719	26.336	30.319
28	13.565	14.847	16.928	18.939	21.588	23.647	27.336	31.391
29	14.256	15.574	17.708	19.768	22.475	24.577	28.336	32.461
30	14.953	16.306	18.493	20.599	23.364	25.508	29.336	33.530
31	15.655	17.042	19.281	21.434	24.255	26.440	30.336	34.598
32	16.362	17.783	20.072	22.271	25.148	27.373	31.336	35.665
33	17.074	18.527	20.867	23.110	26.042	28.307	32.336	36.731
34	17.789	19.275	21.664	23.952	26.938	29.242	33.336	37.795
35	18.509	20.027	22.465	24.796	27.836	30.178	34.336	38.859
36	19.233	20.783	23.269	25.643	28.735	31.115	35.336	39.922
37	19.960	21.542	24.075	26.492	29.635	32.053	36.336	40.984
38	20.691	22.304	24.884	27.343	30.537	32.992	37.335	42.045
39	21.426	23.069	25.695	28.196	31.441	33.932	38.335	43.105
40	22.164	23.838	26.509	29.051	32.345	34.872	39.335	44.165
41	22.906	24.610	27.326	29.907	33.251	35.813	40.335	45.224
42	23.650	25.384	28.144	30.765	34.157	36.755	41.335	46.282
43	24.398	26.162	28.965	31.625	35.065	37.698	42.335	47.339
44	25.148	26.942	29.787	32.487	35.974	38.642	43.335	48.396
45	25.901	27.725	30.612	33.350	36.884	39.586	44.335	49.452
46	26.657	28.511	31.439	34.215	37.795	40.531	45.335	50.507
47	27.416	29.300	32.268	35.081	38.706	41.477	46.335	51.562
48	28.177	30.091	33.098	35.949	39.619	42.424	47.335	52.616
49	28.941	30.885	33.930	36.818	40.532	43.371	48.335	53.670
50	29.707	31.681	34.764	37.689	41.446	44.319	49.335	54.723
51	30.475	32.480	35.600	38.560	42.361	45.268	50.335	55.775
52	31.246	33.282	36.437	39.433	43.277	46.218	51.335	56.827
53	32.018	34.085	37.276	40.308	44.193	47.168	52.335	57.878
54	32.793	34.891	38.116	41.183	45.111	48.118	53.335	58.929
55	33.570	35.699	38.958	42.060	46.029	49.069	54.335	59.979
56	34.350	36.509	39.801	42.937	46.948	50.021	55.335	61.029
57	35.131	37.321	40.646	43.816	47.868	50.973	56.335	62.078
58	35.913	38.135	41.492	44.696	48.788	51.926	57.335	63.127
59	36.698	38.951	42.339	45.577	49.710	52.878	58.335	64.176
60	37.485	39.769	43.188	46.459	50.632	53.832	59.335	65.224

Table VI: 5% and 1% points of F

APPENDIX

3

Answers to Problems

Problems 1.1, page 6

1. $x^4 - 6x^3 + 3x^2 + 12x - 20 = 0$
2. $(x) = 2, (-1) = 26, (-2) = 2$
3. $(x) = 0, (-1) = 0$
4. $(x) = 0, (-1) = 0$
5. Two roots between $(1, 2)$ and $(-2, -3)$
6. $2, 3, -\frac{1}{2}$
7. -2
8. $x = 2, x = 1$
9. $-1, -1, 2$
10. $-1, 2, 6$
11. $-1, -1, 2, 2$
12. $\frac{1}{2}(3 + \sqrt{5}), \frac{1}{2}(3 - \sqrt{5})$
13. $1, -5, 7$
14. $-1, \frac{1}{2}, 3$
15. $(x) = 0, (-2) = 0, (3) = 0, (-4) = 0$
16. $2, -\frac{1}{2}, -3$
17. $(x) = 0, (-2) = 0, (4) = 0$
18. $(x) = 0, (-2) = 0, (3) = 0, (5) = 0$
19. $x^3 - 6x^2 + 11x - 6 = 0$

Problems 1.2, page 9

1. $x^3 + 6x^2 - 3x + 27 = 0$
2. $6x^5 - 5x^4 - 13x^3 + 8x^2 - 2 = 0$
3. $20x^5 + 8x^4 - 5x^3 - 2x + 1 = 0$
4. $-\frac{1}{2}, \frac{1}{3}, 2$
5. $(x) = 0, (-2) = 0, (3) = 0, (-4) = 0$;
   $(x) = 0, x^2 - 7 = 0$
6. $x^4 + 13x^3 + 53x - 36 = 0$
7. $x^5 - 4x^4 + 3x^3y + 3x^2y - 4xy^4 + y^5 = 0$
8. $2x^3 - 13x^2 + 17x - 5 = 0$
9. $x^3 - 4x^2 + 3xy - y^3 = 0$
10. $3x^2y^2 + 4xy - 4x - 3 = 0$; $4x^4 + 4x^3y^2 - 24x^2y + y^4 = 0$; $3x^2y^2 + 2y^3 = x$
11. $y^3 - 3x^2 + 12xy - 20 = 0$
12. (a) $\frac{-3 \pm \sqrt{21}}{2}, \frac{3 \pm \sqrt{29}}{2}$
    (b) $2, 2, 10, 10$
    (c) $3, -3, 4, -10, 10$;
    (d) $-1, -1, 2, 3, -10, 10$;
    (e) $\pm 1, -3, -10, \frac{3 \pm \sqrt{5}}{2}$
13. $\frac{1}{2}(1 \pm \sqrt{29}), \frac{1}{2}(1 \pm \sqrt{5})$
14. $-1, -3, -5, -7$

Problems 1.3, page 11

1. $-5, 3, 3$
2. $4, \frac{1}{2}(-5 \pm i\sqrt{3})$
3. $4, -3 \pm 2\sqrt{-3}$
4. $-1, -2, \frac{1}{2}$
5. $\frac{1}{2}, -\frac{3}{2}(1 \pm i\sqrt{3})$
6. $\frac{1}{2}(1 \pm \sqrt{5})$
7. $2\cos\frac{2\pi}{9}, 2\cos\frac{8\pi}{9}, 2\cos\frac{14\pi}{9}$
8. $\frac{1}{2}, \frac{-7 \pm 5i\sqrt{3}}{4}$

Problems 1.1, page 12

1. $1, 2, 3, 4$
2. $-3, 1, +3$
3. $4, -2, -1 \pm i$
4. $-1, 3, 3 \pm \sqrt{55}$
5. $3 \pm \sqrt{5}, 2 \pm \sqrt{3}$
6. $1 \pm 2i, -1 \pm \sqrt{2}$
7. $2 \pm \sqrt{3}, -2 \pm i\sqrt{2}$
8. $2, 4, 2 \pm 2i\sqrt{2}$.

Problems 1.2, page 15

1. 1.82
2. 2.29
3. 0.45
4. $10.0.77$ rad
5. 3.81 rad
6. 0.28.

Problems 1.3, page 18

1. (d)
2. (c)
3. (c)
4. (c)
5. (e)
6. (e)
7. (d)
8. (b)
9. (c)
10. (a)
11. (e)
12. minus
13. $\$ 35.1(\%)$
14. pq
15. 31
16. -3 and -2
17. Conjugate pairs
18. $f(x)$ is not bounded in $[a, b]$
19. $x^4 - 9x^3 + 20x - 21 = 0$
20. $3, 5, -2$
21. $x^2 + 20x - 1500 = 0$
22. pv
23. $x^4 + 2x^3 - x^2 - 4x - 4 = 0$
24. 8
25. infinite
26. $pq = r$
27. $1, \frac{1}{2}(-1 \pm \sqrt{3})$
28. (iv)
29. $1, 1, -3$
30. $x^3 - 7x^2 + 12x - 10 = 0$
31. Zero and 2
32. 21
33. True
34. True.

Problems 2.1 (Skip 20)

5. (d) $1 \pm 3i$, 0
12. $(x - 1)(x - 3)(x - (2 - i))$
14. $(x - 3)(x - (2 + i))(x - (2 - i))(x + 1)(x - 1)$
15. $(x - 3)(x - (2 + i))(x - (2 - i))(x + 1)(x - 1)(x + 1)$
16. $(x - 3)(x - (2 + i))(x - (2 - i))(x + 1)(x - 1)(x + 1)(x - 1)$
17. $y = 0, 2 \pm \sqrt{(x^2 + 2^2 + x^2 - xz - 4x - cz)}$
18. $2, -\frac{1}{2}$.

Problems 2.2, page 31

1. $x = -3, 2$
2. $x = -\frac{1}{2}, y = -2, z = -4, w = 3$
3. $x = \frac{1}{2}, y = 0$
4. $-2\begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$
8. $AB = \begin{bmatrix} 8 & 4 & 7 & 10 \\ 10 & 7 & 11 & 21 \\ 6 & 3 & 3 & 9 \end{bmatrix}$
7. (c) $(x^3 + 4y^3 + z^3 + 3yz + 3xz + 3xy)$
(d) $\begin{bmatrix} 6 & 7 \\ 121 & 104 \\ -385 & -331 \end{bmatrix}$ (e) $\begin{bmatrix} 3 & 6 \\ -15 & -12 \\ 33 & 18 \end{bmatrix}$
10. 31
11. $\begin{bmatrix} -6 & 1 & 3 \\ 0 & 4 & 4 \\ 2 & 6 & -6 \end{bmatrix}$
18. $\begin{bmatrix} 1 & 0 & 0 \\ 200 & 1 & 0 \\ 300 & 43/15 & 1 \end{bmatrix}\begin{bmatrix} 3 & -2 & 1 \\ 0 & 30/3 & -330 \\ 0 & 0 & 287/15 \end{bmatrix}$

Problems 1.31 (page 31)

3. (i) $\begin{bmatrix} 3 & 0 & 3.5 \\ 0 & 7 & 1.5 \\ 0.6 & 3.8 & 3 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & -2 & 5 \\ 3 & 6 & -3.5 \\ -0.8 & 2.8 & 0 \end{bmatrix}$ (iii) $\begin{vmatrix} 4 & \frac{3}{2}(x-y) & \frac{1}{2}(x+y) \\ \frac{3}{2}(x-y) & 5 & \frac{3}{2}(x+y) \\ \frac{1}{2}(x+y) & \frac{1}{2}(x+y) & x \end{vmatrix} + \begin{vmatrix} 0 & \frac{1}{2}(x-y) & \frac{1}{2}(x-y) \\ \frac{1}{2}(x-y) & 0 & \frac{1}{2}(x-y) \\ \frac{1}{2}(x-y) & \frac{1}{2}(x-y) & 0 \end{vmatrix}$

4. (i) $\begin{bmatrix} 2 & 4.6 & 0.8 \\ 3 & -4.8 & -3.6 \\ -1 & 3.6 & 0.4 \end{bmatrix}$ (ii) $\begin{bmatrix} 1.03 & -4.33 & 2.12 \\ -4.66 & 14.66 & 13.33 \\ 2.11 & 13.33 & -2.33 \end{bmatrix}$

6. $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

8. (i) $\begin{bmatrix} 2 & -1 & 0 \\ 3 & 5 & -4 \\ -2 & 3 & -3 \end{bmatrix}$.

Problems 2.4 (page 40)

1. 3 2. 2 3. 3 4. 3 6. 2

8. No value of k is possible.

7. (i) $\begin{bmatrix} 3 & -1 & -2 \\ -5 & 2 & 3 \\ 10 & -3 & -4 \end{bmatrix}$

(ii) $\begin{bmatrix} -19 & 39 & 29 \\ 29 & -19 & 29 \\ 29 & 29 & -19 \end{bmatrix}$

(iv) $\frac{1}{21}\begin{bmatrix} 1 & 20 & -7 \\ 1 & -21 & 14 \\ -3 & 22 & 0 \end{bmatrix}$

(vi) $\begin{bmatrix} 0.2 & -1.6 & 1.2 \\ -1.2 & 3 & -1 \\ 0.2 & -1.2 & 1.2 \end{bmatrix}$

8. $P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & 2.5 & -1.5 & 1.5 \\ 0 & -1.5 & 0 & 2.5 \\ 0 & 0 & -1.5 & 1.5 \\ 0 & 0 & -1.5 & 0 \end{bmatrix}$, Rank $[A] = 3$

9. $\begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 5 & -3 \end{bmatrix}$

10. (i) $P = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ \frac{1}{2} & 1 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$,

(ii) $P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

11. (i) 3 (ii) 3 (iii) 3 (iv) 3.

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Problems 2.5 (Page 43)

1. $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

2. $\frac{1}{5}\begin{bmatrix} 2 & 2 & -2 \\ -3 & 11 & 3 \\ 1 & -7 & -1 \end{bmatrix}$

3. $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -1 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{bmatrix}$

4. $\begin{bmatrix} 7 & -3 & 0 & -5 \\ 3 & 1 & -2 & -11 \\ -5 & 0 & 1 & 6 \\ 10 & 6 & -6 & 26 \end{bmatrix}$

Problems 2.6 (page 49)

1. $x = 1, y = 2, z = 3$

2. $x = 2, y = -3, z = 12$

3. $x = 1.2, y = 2.2, z = 3.3$

4. $x = y = z = w^2$

5. $x = 1, y = 37, z = 10$
6. $x_1 = 1, x_2 = 0, x_3 = 3$
7. $x = 2, y = 1, z = 0$
8. $x = 167, y = 109, z = 57$
9. $x = y = z = 3$
10. $x_1 = 5, x_2 = 1, x_3 = -2, x_4 = 3$
11. $x = 1, y = -1, z = 1$
12. $I_1 = 1.4, I_2 = 2.3$
13. $I_1 = 289/125, I_2 = 39/20, I_3 = 73/50$
14. $x_1 = 8, x_2 = 19, x_3 = 3, x_4 = 48$

Problems 3.7 (page 50)

1. Consistent : $x = 1, y = 2k - 2, z = k$ for all k
2. $k = 1, x = -3c, y = 2c + 1$; $k = 3, x = 1 - 2c, y = 2c$
3. (i) $k = 2, \mu \neq 10$; (ii) $k \neq 3$; (iii) $k = 5, \mu = 10$
4. (i) Equations are inconsistent ; (ii) Consistent ; $x = 1, y = 1, z = 2$;
   (iii) Equations are inconsistent ; (iv) Consistent : $x = 2, y = 1, z = -4$
5. If $a = -1, b \neq 0$, equations will be consistent and have infinite number of solutions.
   If $a = -1, b \neq 0$, equations will be inconsistent ;
   If $a \neq -1, b$ having any value, equations will be consistent and will have a unique solution.
6. $k = -3, x = 47, y = -89, z = 0$; $\lambda = -3, x = \frac{1}{7}(8 - 6k), y = \frac{1}{7}(13k - 8), z = k$ for all k
7. $k = -1, 1, 12$; $x = -1/11, y = 10/11$; $x = 0, y = 1$; $x = \frac{3}{2}, y = 1$
8. $k = 3$ is the only real value for which $x = y = z$
9. $k = 1, x_1 = 3t - 4, x_2 = t, x_3 = t$; $k = -2, x_1 = -t, x_2 = 3t, x_3 = t$
10. $k = 1, -9$. For $\lambda = 2$, and $k = 1$, $x = -3, y = 2$
    For $k = -9$, and $\mu = \lambda$, $x = 3t, y = 3t, z = 3t$
11. (i) Have infinite number of non-trivial solutions : $x = 1 - 6\mu, y = 1 - 4\mu, z = 2, w = \mu$ for all values of λ and μ ; (ii) $x = 13k_2 + 8k_1, y = -8k_2 - 5k_1, z = k_2, w = k_1$ where k_1, k_2 are arbitrary constants.

Problems 3.8 (page 64)

1. $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 11 & -3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$
2. $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}, A^{-1} = A' = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$
3. $Z = 1650$, where $M = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 3 & -1 \\ -3 & 14 & -1 \end{bmatrix}$
4. $x_1 = 18y_1 - 9y_2 + 2y_3 ; x_2 = -9y_1 + 5y_2 - y_3 ; x_3 = -2y_1 + y_3$
5. $a = 1, \frac{1}{12}, b = 1, \frac{1}{10}, c = 1, \frac{1}{13}$
6. (i) $3x$ (ii) $3x$ (iii) The $2x_1 - 12x_2 + 5x_3 - 5x_4 = 0$

Problems 3.9 (page 80)

1. 10 ; 20
2. (a) (i) π ; (ii) $(1, 0, 0, -3)$; (iii) $-3, 0$; (iv) $\pi, \pi, 12, -6$
3. (a) (i) $2, 3, 0$; (ii) $2, 20, 12, 1, -20, 12, -2, 11$ (b) (i) $1, 2, 3$; (ii) $(1, 0, -1), (2, 1, 0), (3, 0, 1)$
   (c) (i) $8, -3, -3$; (ii) $(1, 3), -1), (2, -1, 1), (3, 1, 0)$
   (d) (i) $2, 2, 2$; (ii) $2, -1, 1$; (iii) $(1, 0, -1), (0, 1, 1), (1, 1, 0)$ (e) (i) $2, 3, -1$; (ii) $(1, 1, 1), (1, -1, 0), (1, 0, 1)$
4. (a) (i) $8, 72, 8$ (ii) $(1, 0, 0), (0, 1, 0), (0, 0, 1)$
5. $1, 1, 20$
6. (a) $\begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}$; (b) $\frac{1}{8}\begin{bmatrix} 8 & 0 & 0 \\ 3 & 3 & -7 \\ 2 & -1 & -1 \end{bmatrix}$; (c) $\frac{1}{8}\begin{bmatrix} 12 & 2 & 8 \\ -5 & -1 & -3 \\ -1 & -1 & -1 \end{bmatrix}$; (d) $\begin{bmatrix} 1 & 1/2 & -3/2 \\ 0 & -1/2 & 0 \\ 0 & 1 & 3/2 \end{bmatrix}$
10. $\lambda^3 - 4\lambda^2 - 20\lambda - 35 = 0$; $\frac{1}{35}\begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 55 \\ 7 & 1 & -20 \end{bmatrix}$

11. (i) $\frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ (ii) $\frac{1}{3}\begin{bmatrix} -3 & -3 & 3 \\ 6 & 3 & -3 \\ -6 & -3 & 5 \end{bmatrix}$ (iii) $\frac{1}{29}\begin{bmatrix} 1 & 10 & -6 \\ -9 & 1 & 10 \\ 10 & -6 & 1 \end{bmatrix}$

12. 6207

13. $\lambda^3 - 15\lambda^2 + 30\lambda - 8 = 0$, $\begin{bmatrix} 1024 & -1022 & 1023 \\ -510 & 508 & -509 \\ 765 & -762 & 764 \end{bmatrix}$

14. $\begin{bmatrix} 125 & 0 & 0 \\ 0 & 125 & 0 \\ 0 & 0 & 125 \end{bmatrix}$

$A^{-1} = \frac{1}{64}\begin{bmatrix} 1 & 79 & 79 \\ -21 & 85 & 85 \\ 21 & -149 & -149 \end{bmatrix}$

15. $A^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 9 & 8 \\ -1 & 6 & 5 \\ 1 & -18 & -15 \end{bmatrix}$, $A^{-1} = \frac{1}{264}\begin{bmatrix} 3 & -18 & -18 \\ -5 & 13 & -6 \\ 5 & 1 & 23 \end{bmatrix}$,

16. $\begin{bmatrix} 1 & 0 & 0 \\ 20 & 1 & 0 \\ 20 & 0 & 1 \end{bmatrix}$.

Problems 3.10, page 87

1. $\begin{bmatrix} -5 & 0 \\ 0 & 2 \end{bmatrix}$

2. $A^n = \begin{bmatrix} 2^n + 3^n & -3 \cdot 2^n + 3^n \\ -2^n + 3^n & 3 \cdot 2^n + 3^n \end{bmatrix}$; $A^{-1} = \begin{bmatrix} 870 & 930 \\ 930 & 990 \end{bmatrix}$

3. $\begin{bmatrix} 361 & -406 & 225 \\ -406 & 821 & -415 \\ 225 & -406 & 221 \end{bmatrix}$

9. $(1, 1, -1), (1, 2, -1), (2, -1, 1)$; $4x^2 + y^2 + z^2$

8. $x^2 + y^2 - 2z^2$

10. (a) 2, 3, 4; $(2, 0, 0), (0, 1, 1), (0, 1, -1)$; $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

(b) $x_1^2 + 2x_2^2 + 4x_3^2$

$\begin{bmatrix} -1/\sqrt{2} & 0 & 2/\sqrt{6} \\ 1/\sqrt{2} & 1/\sqrt{2} & 2/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{2} & 2/\sqrt{6} \end{bmatrix}$, positive definite;

(c) $2y_1^2 + 10y_2^2$, $\begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$, positive semidefinite

11. 2, 1

12. Indefinite.

Problems 3.11, page 91

8. $\begin{bmatrix} 3 & 2 & 2 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} + i\begin{bmatrix} 2 & 2 & -2 \\ 2 & -2 & 2 \\ -2 & 2 & 0 \end{bmatrix}$

Problems 3.12, page 95

1. (c)	2. (a)	3. (c)	4. (c)	5. (c)	6. (a)
7. (a)	8. (b)	9. (d)	10. (a)	11. (d)	12. (c)
13. (c)	14. (d)	15. (c)	16. V	17. sum	18. 0, 3

19. 2 20. $\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}$ 21. 0

22. All the eigenvalues are > 0 and at least one eigenvalue is zero.

23. (a) $x = p$, (b) $x = y$, $z = q$ 24. 0 25. (a)

Answers to Problems 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |

Problems 3.8, page 101

1. $2x - y + 3z = 9$
2. $3i - 12i + 2k + 3i = 5$
3. $\frac{1}{2}(2i + j + 3k)$
4. 3
5. $4x - 3y + 2z = 3$
6. $x - 6y - 2z + 7 = 0$
7. $2x + 2y - 3z = 0$
8. $\lambda_1 - 4\lambda_2)x - \frac{1}{2}(\lambda_1 + \lambda_2)z = 0$
9. $y = 2$
10. $2x + 4y - 6z = 0$
11. $d = 10\sqrt{2}$; $5x - 15y - 21z = 34$
12. $19x - 2y + 6z + 5 = 0$
13. $\cos^{-1}(\sqrt{2}/3)$
14. (i) $\frac{5}{\sqrt{83}}, \frac{-7}{\sqrt{83}}, \frac{-8}{\sqrt{83}}$ (ii) $63.7°$ (iii) $6x - 7y - 8z + 7 = 0$
15. $6x + 3y - 2z = 16$; $2x - 3y - 6z = 8$
16. $x^2 + y^2 + z^2 = 9r^2$
17. $xyz = 48^3$
18. (i) $25x + 17y + 62z - 78 = 0$; (ii) $x - 35y - 18z - 150 = 0$; (iii) Do not lie on the same side.

Problems 3.9, page 106

1. $\frac{x-5}{3} = \frac{y-3}{2} = \frac{z-4}{-2}$
2. $45°$
3. $90°$
4. $x = z = \frac{x-3}{2} = \frac{z-4}{2}$
5. 2
6. $\frac{x-1}{1} = \frac{y-2}{-2} = \frac{z+2}{2}$
7. $\frac{x-1}{1} = \frac{y+2}{-15} = \frac{z-3}{35}$; $\frac{x-1}{11} = \frac{y+1}{15} = \frac{z-3}{-21.5}$
8. (3, 4, 6)
9. 4.1
10. 8.67
11. (3, 4, 6); $(\pm 1/\sqrt{82}, -15/\sqrt{82}, 1/\sqrt{82})$
12. $41°10'$
13. $23x - 17y - 28z = 65$
14. $2 - x = y + 1 = 0 + 1/2$

Problems 3.9, page 107

1. $3x - 2y - 3z = 0$
2. $2x + 3y + 6z = 28$
3. $12x + 12y - 8z = 0$
4. $3y - z = 3$
5. $\frac{x-4}{7} = \frac{y-6}{-10} = \frac{z+20}{6}$
6. $x + y + 3z = 1$, $x + y = 2$; $(2, 0, -1)$, $x = 1$
7. $\frac{x+4/3}{-11} = \frac{y-2/3}{9} = \frac{z}{11}$
8. $\frac{x-2}{2} = \frac{y-1}{-1} = \frac{z-1}{1}$

Problems 3.10, page 110

1. $x - 2y + z = 0$
2. $(2, -7, 0)$
3. $\frac{ax + by + cz + d}{\sqrt{a^2 + b^2 + c^2}} = \frac{a'x + b'y + c'z + d'}{\sqrt{a'^2 + b'^2 + c'^2}}$
4. $(1, 1, 2)$; $4x + y - 2z + 2 = 0$
5. $-\frac{1}{6}(x - 5) = y - 2 = \frac{1}{2}(z - 13)$
6. $\frac{x-2}{7} = \frac{y-3}{4} = \frac{z-1}{-3}$
7. $(2, 3, -3)$; $(0, 1, 2)$; $\sqrt{66}$.

Problems 3.10, page 113

1. $5\sqrt{3}$; $12x + 3y - 2z + 13 = 0$; $7x + y - 3z + 7 = 0$
2. $(11/7)$; $\frac{2}{3}(x - 2) = \frac{2}{3}(y - 1) = \frac{2}{4}(z - 7)$; $(2, 3, 7)$; $(-1, -2, -1)$
3. $\frac{1}{\sqrt{3}}(2x - 2y + 4z - 1 = 0 = x + 2y$

Problems 3.11, page 115

4. Pivot and screwed planes at slant $y = 2x$, $z = -\frac{1}{2}(y + 12)$, $x = 0$

Problems 3.12, page 118

1. $x^2 + y^2 + z^2 - 2x + 4y - 2z + 5 = 0$; $(2, -2, 1)$; 3
2. $x^2 + y^2 + z^2 - 2x + 2y - 4z = 0$; $11, -1, 20$; $\sqrt{6}$
3. (a) $3x^2 + y^2 + z^2 - 4x - 4y - 4z + 5 = 0$ (b) $9x^2 + y^2 + z^2 - 2x + y + z - 1 = 0$
5. (a) $x^2 + y^2 + z^2 - xz - 2y - z = 0$

   (b) $x^2 + y^2 + z^2 - xz - 3y - zz = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$; $\left[\frac{ab^{-2} + c^{-2}}{2a^{-2}}, \frac{bc^{-2} + a^{-2}}{2b^{-2}}, \frac{ca^{-2} + b^{-2}}{2c^{-2}}\right]$

8. $(1, 3, 4)$; $\sqrt{3}$ 10. $x^2 + y^2 + z^2 + 2x + y + z - 11 = 0$
11. $10x^2 + y^2 + z^2 - 20x - 20y + 10z + 116 = 0$
12. $3x^2 + y^2 + z^2 - 6x - 8y + z + 16 = 0$ 13. $x^2 + y^2 + z^2 + 2y - 8z + 24 = 0$.

Problems 3.13, page 120

1. (1) $x + 3 = 2$, $x - 2 = 0$ (2) $x + 2y + 3z = 9$, $3x + y - 2z = 5$
2. $x^2 + y^2 + z^2 + 2x + 4y + 6z - 11 = 0$ and $8x^2 + y^2 + z^2 - 4x - 8y - 32z - 33 = 0$
3. (1) $x^2 + y^2 + z^2 - 10y - 10z - 31 = 0$ 4. $16x^2 + y^2 + z^2 - 14x + y + z + 14 = 0$
6. $\frac{x - 3.5}{-2} = \frac{y - 2.5}{1} = \frac{z}{0.5}$ 8. $2\sqrt{3}, \sqrt{6}$
9. $2x + y + z + 3 = 0$ 10. (1) (A, 4, 90)

Problems 3.14, page 124

1. $(2x - y)^2 = 4x(x - z) + 2(z - y)$
2. $108x^2 + 300y^2 + 75z^2 - 108xy - 360yz + 300zx + 780x + 1300y - 6436 = 0$
3. $5x^2 + 3y^2 + z^2 - 2xy - 6yz - 4zx + 6x + 8y + 10z - 26 = 0$
4. $x^2 + y^2 + 3z^2 + 2x + 2y + 6z + 1 = 0$ 5. $x^2 + y^2 + z^2 \cos^2 \alpha$
6. $x^2 + 7y^2 + z^2 + 6xy + 6yz - 10zx = 0$ 7. $4x^2 + 9y^2 + 25z^2 - 12xy - 12yz + 20zx = 0$
8. $x^2 - y^2 + z^2 + 4y - 4z = 0$ 9. $yz + zx + xy = 0$, $\cos^{-1}(1/\sqrt{3})$; $x = y^2 + t + z/\sqrt{2}$
10. $\cos^{-1} 4/\sqrt{161}$; $25x^2 + 36y^2 - 36z^2 = 1$ 11. $4x^2 + 4y^2 - z^2 + 8x - 8z = 0$
12. $x = y\sqrt{2} + z/\sqrt{2} + 1$; $y\sqrt{2} + 2xy = z$ 14. $-2x^2 + y^2 - 2z^2 + 4yz - 4zx + 8x + 2z - 10y + 6z - 3 = 0$.

Problems 3.15, page 126

1. $5x^2 + 5y^2 + 5z^2 + 4yz + 8zx - 4xy - 36 = 0$
2. $3x^2 + 6y^2 + 3z^2 + 8yz - 2zx + 4x - 24y - 16z + 20 = 0$
3. $42x^2 + 45y^2 + 12z^2 + 12xy + 28yz - 24zx - 42x - 112y - 126z - 252 = 0$
4. $x^2 + y^2 + z^2 - yz - zx - xy = a^2$
5. $3x^2 + 5y^2 + 3z^2 + 12yz + 6yx - 30x - 30y - 32z + 38 = 0$; a circle
6. $x^2 + y^2 - 2x - 4y - 11 = 0$
7. $a(yz - zx)^2 + 2b(zx - xy)(xy - yz) + c(xy + yz - 2zx)^2 + 2g(yz - zx) + 2h(xy - yz) + 2k(xy + yz) + abc^2 = 0$.

Problems 3.16, page 133

1. $2\frac{1}{2}$ quad, 20π sr
2. Hyperboloid of revolution of one sheet; Hyperbola $5x^2 - y^2 = 4$; 5π sr/sr

3. Right circular cylinder with axis along z-axis
4. Hyperbolic paraboloid
5. Hyperboloid of two sheets
6. Parabolic cylinder
7. Right circular cylinder
8. Cone with vertex at the origin
9. Hyperbolic paraboloid
10. Hyperboloid of two sheets.

Problems 3.17, page 130

1. (b)	2. (a)	3. (c)	4. (b)	5. (b)	6. (a)
7. (c)	8. (d)	9. (c)	10. (c)	11. (a)	12. (c)
13. (b)	14. (a)	15. (c)	16. (c)	17. (c)	18. (b)
19. (c)	20. (c)	21. (b)	22. (c)	23. (c)	24. (c)

25. (b) 26. (a) 27. $1\sqrt{3}, 1\sqrt{3}, 1\sqrt{3}$ 28. 0, 0, 1 29. $x = 3, y = 0$

30. $1 - 3, 3, -1$ 31. $8x + 3y = 30$ 32. $\begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix} = 0$

33. $y^2 + z^2 = a(x+a)^2$ 34. $12x + 33y - 10z = 90$ 35. 1001.6

36. $\cos^{-1}(1/\sqrt{22})$ 37. $\frac{x^2}{4} + \frac{y^2}{9} + z^2 + 4z + y + z^2$

38. 13, 5, 7, 15, 6, 11 39. $-4x - 11 = y - 5 = \frac{1}{3}(z - 3)$ 40. $x^2 + y^2 + z^2 + z - 6y - 3z + 5 = 0$

41. $ax + by + cz + d = 0, cx_1 + dy_1 + ez_1 + d = 0$ 42. $2\sqrt{26}$

43. $15\hat{i} - 3.3\hat{j} + 3\hat{k}$ 44. $\sqrt{14}$ 45. $\frac{x+2}{3} = \frac{y+2}{-2} = \frac{z}{1}$

46. Parabolic cylinder 47. Hyperboloid of two sheets

48. $\cos^{-1}\left(\frac{7}{\sqrt{64}}\right)$ 49. 8, −4, 12 50. 8

51. True 52. True 53. True

54. Elliptic cylinder 55. $4x^2 + y^2 + z^2 + 6xy + yz + zx = 0$

56. $3 - 5i - 6k\sqrt{38}$

Problems 4.1, page 135

1. $5e^{2t} - t^2$ 2. $-\frac{3}{2}$ 3. $\sin\frac{\pi}{6}\cos^{-1}1$.

Problems 4.2, page 139

1. $(-1)^{n-1}(2n-1)!2^n[(2n+1)^n + (2n-1)^n]$ 2. $(-1)^n\left[\frac{n!}{(2n+1)^{n+1}} - \frac{(n-1)!}{(2n+1)^n} + \frac{(n-1)!}{(2n-1)^n}\right]$

3. $\frac{1}{16}[8\cos(2x - \sin 2x) + 3^n\sin(3x + n\pi/2) - 5^n\sin(5x + n\pi/2)]$

4. $\frac{1}{250}[5^n\cos(2\theta + n\pi/2) + 10.7^n\cos(3\theta + n\pi/2) + 20.5^n\cos(4\theta + n\pi/2) + 61.3^n\cos(5\theta + n\pi/2) + 120\cos n\theta]$

5. $\frac{(\sqrt{2})^{n+1}}{2}k^{2n}\sin(2x + n\tan^{-1}2) - e^{-4x}\sin(4x - n\tan^{-1}2)$

6. $\frac{1}{2}e^{2x}[(13)^{n/2}\cos(3x + n\tan^{-1}0.67) + (29)^{n/2}\cos(5x + n\tan^{-1}0.4)]$

7. $\frac{(-1)^n n!}{3}\left\{\frac{1}{(x-1)^{n+1}}-\frac{1}{(x+2)^{n+1}}\right\}$ 8. $(-1)^n n!\left\{\frac{2(2)^{n-1}}{(2x+3)^{n+1}}-\frac{1}{(x+3)^{n+1}}\right\}$

9. $\frac{(-1)^n n!}{3}\left\{\frac{1}{(x+1)^{n+1}}+\frac{(-2)}{1!}\frac{1}{(x+1)^{n+1}}+\frac{(-2)(-3)}{2!}\frac{1}{(x-1)^{n+1}}\right\}$

10. $\frac{(-1)^n n!\sin(n\theta+(n+1)\beta)}{(x^2+a^2)^{(n+1)/2}}$ where $\theta=\tan^{-1}(a/x)$

11. $(-1)^{n-1}(n-1)!\sin^n\alpha\sin n\alpha$ where $\alpha=\cot^{-1}x$.

Problems 4.1, page 143

1. (i) $\frac{(-1)^{n-1}(n-2)!}{x^n}\{(n-1)(n-2)+n(3-n)x^2\}$;

(ii) $\frac{1}{2(x)_n}\{(\log 2)^n 2^x\cos(3x+n\pi/2)+9\cos(3x+n\pi/2)+3\cos(3x+n\pi/2)+64\cos(3x+n\pi/2)+128\cos(3x+n\pi/2)\}+{}^nC_1\{(\log 2)^{n-1}2^x\cos(3x+2\pi/2)+9\cos(3x+2\pi/2)+3\cos(3x+2\pi/2)+32\cos(3x+2\pi/2)+64\cos(3x+2\pi/2)+128\cos(3x+2\pi/2)\}+\ldots\ldots+2^n\{\cos(3x+n\pi/2)+9\cos(3x+n\pi/2)+3\cos(3x+n\pi/2)+32\cos(3x+n\pi/2)+64\cos(3x+n\pi/2)+128\cos(3x+n\pi/2)\}$

4. $y_n(0)=0, y_{n+1}(0)=(-1)^n\cdot(2n)!$

7. $D_{2n}(x)=0$, if n is even

$=n!(1^2-n^2)(3^2-n^2)\ldots\ldots(2n-1)^2-n^2]$, if n is odd

8. $D_{2n}(x)=e^{x\sin\alpha}\cos n\alpha(1^2+\sin^2\alpha)(4^2+\sin^2\alpha)\ldots\ldots(2n-2)^2+\sin^2\alpha]$

$D_{2n+1}(x)=e^{x\sin\alpha}\sin(n+1)\alpha(1^2+\sin^2\alpha)(3^2+\sin^2\alpha)\ldots\ldots(2n-1)^2+\sin^2\alpha]$

11. $\{n^2-(n-2)^2\}\{n^2-(n-4)^2\}\ldots\ldots n^2-2^2\}n^2$, n even

$\{n^2-(n-2)^2\}\{n^2-(n-4)^2\}\ldots\ldots(n^2-1^2)n$, n odd.

Problems 4.2, page 150

2. $x=\pm(2n-3)\pi/2(2n+2n-1)$

3. (i) $c=1.154, 0.548$; (ii) $c=\pi/2$; (iii) $c=e-1$. (iv) $c=0.5613$

4. 0.58 12. $\theta=0.58$.

Problems 4.3, page 160

1. $x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\ldots\ldots\infty$ 2. $x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\ldots\ldots\infty$

3. $1+x-\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{24}+\ldots\ldots$ 4. $x+\frac{1}{2}\cdot\frac{x^3}{3}+\frac{1.3}{2.4}\cdot\frac{x^5}{5}+\frac{1.3.5}{2.4.6}\cdot\frac{x^7}{7}+\ldots\ldots\infty$

5. $1-\frac{x^2}{2}+\frac{x^4}{12}-\frac{x^6}{7}+\ldots\ldots\infty$ 6. $\frac{x^2}{2}+\frac{x^4}{12}+\frac{x^6}{45}+\ldots\ldots$

24. $\frac{\cos(\alpha+\theta)}{1!}-\frac{\sin(\alpha+2\theta)}{2!}\sin^2\theta+\frac{\sin(\alpha+3\theta)}{3!}\sin^3\theta-\ldots\ldots$

25. $4+22(x-1)+31(x-1)^2+13(x-1)^3+2(x-1)^4$ 26. $\log 2\left\{1+\frac{(x-1)}{1!}+\frac{(x-1)^2}{2!}+\frac{(x-1)^3}{3!}+\ldots\ldots\right\}$

(vi) $\frac{5}{4}+\frac{1}{2}(x-1)-\frac{1}{8}(x-1)^2+\frac{1}{12}(x-1)^3$ 29. $1-\frac{(x-\pi/2)^2}{2!}+\frac{(x-\pi/2)^4}{4!}-\ldots\ldots=0.968$

38. $\log(0.01) = \sqrt{3}(x - a)(1 - 2x - a)(3 - a)^2 - \frac{5\sqrt{2}}{3}(x - a)^2 + \ldots$

38. 0.8462 38. (a) 0.8121, (b) 1.12

Problems 4.8, page 156

1. $\log_a(ac^2)$	2. $-1/2$	3. $1/3$	4. $a \log a$	5. 1	6. $1/10$
7. $1/2$	8. $1/12$	9. $3/2$	10. 0	11. $1/60$	12. 1
13. $1/3$	14. 2	15. 1	16. 1	17. 2	18. $13a/24$

19. $x = 1$; $\frac{1}{2}$ 20. $x = 6, \frac{1}{2}$ $x = -6$ 21. $x = 3, y = 9, x = \frac{1}{3}$

Problems 4.9, pages 159–160

1. $-1/3$	2. $3/2$	3. -2	4. $-1/2$	5. $2/3$	6. $1/2$
7. ∞	8. 1	9. e	10. $1/\sqrt{e}$	11. $1/e$	12. 0
13. 0	14. 1	15. $e^{-1/2}$	16. e	17. $e^{1/2}$	18. $e^{1/2}$

19. $-\frac{1}{2}$ 20. $y(x) = \frac{1}{2}$

Problems 4.8, page 163

1. $x - 2y = 7$; $2x + y = 14$ 2. (a, b) 3. $\pi/4$

4. $T = 2a \sin \theta/2$; $N = 2a \tan \theta/2 \sin \theta/2$; 5. $T = a \sin t$; 6. $N = 2a \sin^2 \theta/2 \tan \theta/2$

7. $a \sin^2 \theta \tan \theta$.

Problems 4.9, page 165

7. (a) $\pi/2$; (b) $\pi/2$

Problems 4.10, pages 169–170

4. $r^2 = 2ap^2$	5. $r^2 = a/p$	6. $pr^{n-1} = r^{n-1} + 1$	7. $r^{n-1} = a\sqrt{2}/p$
8. $(2 - a^2)p^2 = r^2$	9. (a) $\sqrt{2} + 2\sqrt{4\pi}$; (b) $\cosh a\pi$		10. $a\theta$
11. (a) $2a \sin \theta/2$;	(b) $a\sqrt{\cos 2\theta}$;	(c) $r\sqrt{\theta^2 - 2\theta}$.	

Problems 4.11, page 172

1. (a) $2a(2 - 1)^{3/2}$;	(b) $y\sqrt{x}$	(c) $(2 - a^{3/2})\pi a^2$
6. (a) $(a^2 \sin^2 \theta + b^2 \cos^2 \theta)^{3/2} d\theta$;	(b) $4\pi \sin \theta/2$;	(c) πr
11. (a) $2\pi a$;	(b) 1 ;	(c) $\sqrt{2\pi}$
12. (a) $\frac{4\pi}{3} \sin \frac{\theta}{2}$;	(b) $\pi^2 3a + 1)\pi^{n-1}$	13. $2\sqrt{(r^2 + 2a)}$

Problems 4.12, page 174

1. $a\sqrt{2} + 3\sqrt{2}$, $-4\sqrt{2}\pi a^{3/2}$ 4. (a) $x = at - \sin t$, $y = 2a - a(1 + \cos t)$, (b) $x = a \cos \theta$, $y = a \sin \theta$

8. $(x + y)^{2/3} + (x - y)^{2/3} = 2a^{2/3}$.

7. (a) $(x - 2a)(x^2 + (y + 2a)^2 = a^2)$ (b) $x^2 + y^2 - \frac{2}{3}(x - y) + \frac{4\sqrt{2}}{128} = 0$

11. $y^2 = 4ax$ 12. $1x(y^2 + (y\sqrt{2})^2 = 1$ 13. $2(3y)^2 = 4(x - 2a)^3$.

14. $(x/a)^2 + (y/b)^2 = 1$ 18. $y = \frac{x^2}{2c} - \frac{ac^2}{2b^2}$

16. (i) $\sqrt{x} + \sqrt{y} = \sqrt{c}$; (ii) $4xy = c^2$; 17. $x^{2/3} + y^{2/3} = c^{2/3}$

Problems 4.11, page 1131

2. $a = 1, b = 3/4$; Point of minimum 4. $x = 0.532$ 5. $x = (\sin^2 2\alpha)^{1/3}$

6. $\theta = \frac{\pi}{4} - \frac{\alpha}{2}$; $(1 - \sin\alpha)(1 + \sin\alpha)$ 10. Sq. with side $\sqrt{2}a$

13. $\theta = 3\sqrt{7}, \varphi = 2\sqrt{7}, r = \sqrt{7}$ 14. Depth is half the width 15. $(a^{2/3} + b^{2/3})^{3/2}$

16. $3\sqrt{3}a/4$ 25. 2.5 km/hr.

Problems 4.12, page 1163

1. $x + y + z = 0$ 2. $x = a, y = b, z = c$ 3. $x = a, y = b$

4. $y = 1$; $x + 1 = 0$; $x + y = 0$ 5. $y = x, y + 2x = 0, y + 2x + 1 = 0$

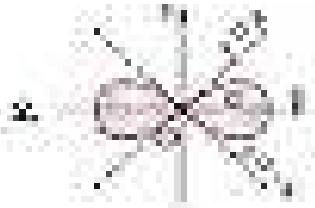
8. $x + a = 0$; $x - a = 0$; $x - y + \sqrt{2}a = 0$; $x + y - \sqrt{2}a = 0$

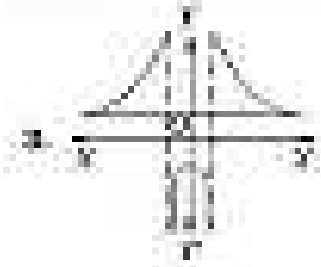
9. $x + 3y + 2 = 0$, $x + y = a\sqrt{2}$ 10. $r \cos\theta = a$; $r \cos\theta = -a$ 11. $r \cos\theta = 0$; $r \sin\theta = 2a$

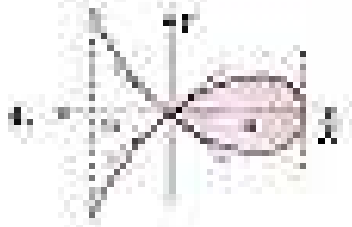
12. $r \sin\theta = 2$ 13. $r \sin\theta - r\cos\theta = a\theta$ cos $r\theta$.

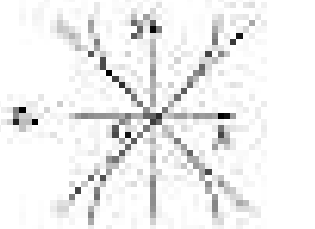
Problems 4.13, page 1181

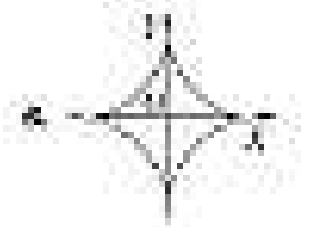
1.

2. 

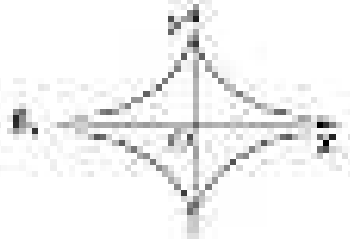
3. 

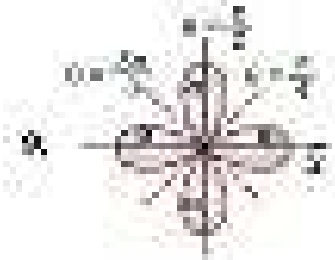
4. 

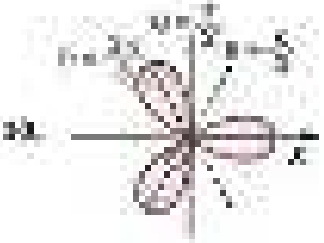
5. 

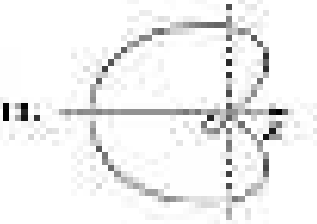
6. 

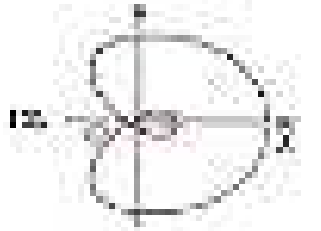
7.

8. 

9. 

10. 

11. 

12. 

Problems 4.10, page 194

1. c
2. $x^2 - 4y = 0$
3. (d)
4. (c)
5. (a)
6. (b)
7. (c)
8. (b)
9. (d)
10. (c)
11. (b)
12. (c)
13. (c)
14. (b)
15. $x^2 = 4y$
16. effective height

17. $x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$
18. $\frac{1}{8}\left[e^x \cos\left(2x + n\frac{\pi}{2}\right) + e^x \cos\left(4x + n\frac{\pi}{2}\right) + e^x \cos\left(6x + n\frac{\pi}{2}\right)\right]$

19. $-2000a$
20. True
21. $-c$
22. $Bx(1 + x^2)^{n/2}$
23. $(x - a)^2 + (y - b)^2 = k^2$
24. envelope
25. $xy = c^2$
26. a
27. $2a$
28. $x^3 - 12x^2 - 36x - 40$
29. (c)
30. $(x/a)^2 + (y/b)^2 = 1$
31. (b)
32. $c = 2.5$
33. $x = y$
34. circle
35. Four loops of $r = a \sin 2\theta$ and Three loops of $r = a \cos 3\theta$
36. $y = x$
37. $x = 4$
38. 45
39. (a)
40. $r = a$
41. (b)
42. (d)
43. (c)

Problems 5.1, page 226

1. 201
2. Does not exist
3. Zero
4. Does not exist
5. Discontinuous

Problems 5.2, page 232

1. (a) $xy(2 - \cos xy - \sin xy)\,(x^2 + 1 + \cos xy)$;
   (b) $2x(3x^2 + y^2)$; $2y(3x^2 + y^2)$;
   (c) $(x^2 + 2xy - y^2)(x^2 + y^2 + 2x + y^2)$; $(x^2 + 2xy - y^2)(3x^2 + y^2 + 2x + y^2)$;
   (d) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} = \frac{z}{1 - z}$
2. $x = 2, -3$
3. $x^2(6y^2 + 2y) + 2xy + 1$

Problems 5.3, page 238

1. $2x$

Problems 5.5, page 251

1. $4x^2y^2 + 2z$
2. $-2x(x^2 + y^2)^{-2}$
3. zero
4. Both are $8x^3$.

Problems 5.6, page 258

1. $0 \le 0$.

Problems 5.7, page 268

2. zero
3. $z(yz + 1 - xz) + x - 2xz$
4. 0; $x = \sin\theta$
5. $x^2 - y^2 = 8x$

Problems 5.8, page 209

1. $4x^2 + y^2 + z^2 = 9$, $\frac{x-2}{4} = y-1 = z+3$ 2. $3x^2 + 3z^2 - x - 3 = 0$, $1 - x = \frac{y-2}{3} = \frac{z+1}{x}$

3. $\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 3$; $z_1(x - x_1) + y_1(y - y_1) = z_1(z - z_1)$

4. $7x - 3y + 6z = 30$; $\frac{x-3}{7} = \frac{y+1}{-3} = \frac{z-2}{6}$

5. $(-1, 2, 3)$ 7. $\frac{x-x_1}{x_1} = \frac{y-y_1}{y_1} = \frac{z-z_1}{z_1}$

Problems 5.9, page 214

1. (i) $z = \frac{1}{2}x^2 + 3xy^2$

(ii) $\frac{1}{2!}\left[1 + (3x + 1)(x - 1) + (y - 1)^2\right] + \frac{1}{3!}\left[(3x + 1)^3 - 3(3x + 1)^2(y - 1)^2 + (y - 1)^4\right]$

$+ \frac{1}{5!}\left[(3x + 1)^5 + 5(3x + 1)^4(y - 1)^2 - 10(3x + 1)^3(y - 1)^4 + 5(3x + 1)(y - 1)^6 + (y - 1)^8\right] + \ldots\ldots\Big]$

(iii) $1 + x - \frac{1}{2!}(x^2 - y^2) + \frac{1}{3!}(x^3 - 3xy^2) + \ldots\ldots$

2. $1 + (x - 1) + (x - 1)(y - 1) + \frac{1}{2}(x - 1)^2(y - 1) + \ldots$

3. -0.8532 4. -4500 cubic 6. 2% 7. 2%

11. $\frac{\pi}{2}$ cubic 12. Rs. 41.20 13. $(x - 3y - 1)$ 14. $-\frac{1}{3}$ 15. π

Problems 5.10, page 220

1. (i) 2π, origin is saddle point if $a < 0$ and minimum if $a > 0$

(ii) Min. at (a, a), a (iii) Max. at $(4, 0)$; Min. at $(0, 0)$

(iv) Max. at $(-1, 0)$; Min. at $(0, -1)$ (v) Max. at $(0.5, 0.5)$; Min. at $(0.5, 0.5)$

2. 4, 2, 3 3. (i) $2a^3$; (ii) $p^2(p^2 + q^2 + r^2)$; (iii) $2a^3$ 4. $12 \times 12 \times 6$ cm

6. $(0, 0, \pm 2)$ 8. 4, 1 9. 60

10. 4, 8, 12 11. Two stationary values of x are given by $\frac{a^2}{ax + 1} + \frac{b^2}{by + 1} + \frac{c^2}{cz + 1} = 1$.

Problems 5.11, page 226

1. $\frac{1}{2b^2}\tan^{-1}\frac{x}{a} + \frac{y}{2a^2(a^2 + x^2)}$ 2. $\frac{1 - 2^n x^n}{1 - x} + 1$

3. $x\log\left[\frac{1}{2} + \frac{1}{2}\sqrt{1 - x^2}\right]$ 4. $-\pi(2a^2 - b^2)$

Problems 5.12, page 228

1. none	2. (a)	3. 1	4. (b)	5. (c)	6. (a)
7. (c)	8. (c)	9. (d)	10. (d)	11. (c)	12. (c)
13. (a)	14. (a)	15. (c)	16. (b)	17. zero	18. $2\pi z + y$

18. $\pi - e^2 < 1$ 20. $\frac{1}{2}$ 21. 4π 22. $\partial f/\partial x, \partial f/\partial y, \partial f/\partial z$

23. $f_x(x,0) = 0, f_y(x,y) = 1$ if $x = 0$ 24. $\frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$

25. 24 27. $-\frac{1}{2}$ 28. $\frac{1}{2}$ 29. equal

30. False.

Problems 9.7, page 396

1. (a) $\frac{1}{2}$; (b) $\frac{1}{2}$ 2. (a) $\frac{1}{2}$; (b) $\frac{1}{2}\ln 2$

3. (a) $\frac{12x - 20}{12x^2 - 20x - 42} = \frac{6x - 10}{6x^2 - 10x - 21} = \frac{3}{3x + 7} + \frac{1}{2x - 3}$ (b) $\frac{1}{6}\left(\frac{\pi}{2} - \frac{3}{5}\right)$ 4. $\frac{35}{18}\pi$

5. (a) $\frac{3}{2}\pi$; (b) $\frac{1}{2}\pi$ 6. (a) $\frac{5}{2}\pi$; (b) $\frac{5}{2}\pi$ 7. (a) $\frac{3}{2}\pi\sqrt{3}$; (b) $\frac{5}{2}\pi\sqrt{3}$

8. (a) $\frac{1}{2}\pi$; (b) $\frac{1}{2}\pi$

Problems 9.8, page 397

1. (a) $\frac{1}{2}\tan^2 x - \frac{1}{2}\ln|\sec x| + C$ (b) $-\frac{1}{4}\cot^2 x + \frac{1}{2}\ln|\csc x| + C$

2. $\frac{1}{2}\log 2 - \frac{1}{4}$ 4. $\frac{\pi}{4} - \frac{2}{3}$ 5. $I_n = \frac{(2)^{n-2}}{n-1} + \frac{n-2}{n-1}I_{n-1}$

6. (a) $\frac{1}{6}\sec^3 x \tan x + \frac{4}{15}(\sec^3 x + 2)\tan x$ (b) $\frac{3\sqrt{2}}{4} + \frac{3}{8}\log(2 + \sqrt{2})$

(c) $-\frac{1}{4}\csc x \operatorname{cosec} x - \frac{3}{8}\cot x \operatorname{cosec} x + \frac{3}{8}\log|\operatorname{cosec} x - \cot x|$

7. $\left(\frac{27\sqrt{2}}{16} + \frac{3}{16}\log(1 + \sqrt{2})\right)a^4$ 8. $\frac{x^4}{8} - \frac{x^2}{2} + C = \frac{1}{8}\sin^4 t$

Problems 9.9, page 403

1. $x^4(3 - x + x^2 - y^2 + y^4)$

3. $\int x^n (\log x)^2\,dx = \frac{x^{n+1}}{n+1}(\log x)^2 - \frac{2}{n+1}\int x^n (\log x)^{n-1}\,dx$; $\int_0^1 x^3 (\log x)^2\,dx = \frac{1}{32}$

5. $\frac{1}{2}\pi^2$ 6. $2e^2 - 14$ 7. $\frac{1}{2}x^2 - 15x^2 + 15$

11. $I_n = \frac{e^{ax}\cos^{n-1} bx(a\cos bx + nb\sin bx)}{a^2 + n^2b^2} + \frac{n(n-1)}{a^2 + n^2b^2}I_{n-2}$; $\int_0^{\pi} e^{2x}\cos^3 x\,dx = \frac{3}{65}(2e^{\pi} - 1)$

12. $\frac{1}{2}\pi\sin\theta$.

Problems 9.10, page 408

1. (a) $\frac{1}{2}\pi$; (b) $\frac{1}{2}\pi$; (c) $\frac{1}{2}\pi\sqrt{3}$; (d) $\frac{1}{2}\pi\sqrt{3}$ 3. (a) $\frac{1}{2}\pi$; (b) $\frac{1}{2}\pi$

Problems 9.11, page 410

1. $\log 2$ 2. $\frac{1}{2}\log 2$ 3. $\frac{1}{2}\pi$ 4. $\pi/2$ 5. $\frac{1}{4}\log 2$ 6. $2^{n-1}n!$

Problems 9.12, page 410

1. (a) $\pi/6$; (b) $\frac{1}{6}\pi\sqrt{3}$ 2. $21\frac{1}{2}\pi$ 3. $3\pi^2$

4. (i) $6a\sqrt{10}$, (ii) $\frac{1}{2}$ 5. (i) $8a^2$, (ii) $\frac{\sqrt{2}}{3}a^2$ 6. (i) $2a^2$, (ii) $4a^2$
8. $3a^2$ 9. $a\sqrt{6}$ 10. $3a^2$; a^2
11. $\frac{21a}{3} = 7a$ 13. $3a = 3.1a$ 14. $4a/3$
16. $3\frac{1}{2}$.

Problems 6.7 (page 222)

1. (i) $2a\sqrt{2}$, (ii) a^2 2. (i) $a\sqrt{10}$, (ii) $a\sqrt{13}$ 3. $11 = 9.9a^2$ 4. $a\sqrt{2}$.

Problems 6.8 (page 228)

1. $32\frac{13}{24}a$ 3. (i) $\log(3 + \sqrt{3})$, (ii) $\log_e(e + 3e)$
5. (i) $a\left[\sqrt{2} + \log(1 + \sqrt{2})\right]$, (ii) $(15/8 + \log 2)a$ 4. (i) $4a\sqrt{3}$, (ii) $4\sqrt{3}$
6. 22.36 7. (i) $8a$ 8. $8a$
9. $4a\sqrt{3}$ 11. $2 + \frac{1}{2}\log 3$ 12. $8a$
13. $\sqrt{3}a\left[1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1\cdot 3}{2\cdot 4}\right)^2\left(\frac{1}{2}\right)^2 + \left(\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\right)^2\left(\frac{1}{2}\right)^2 + \ldots\right]$ 14. $2a\left[\sqrt{2} + \log(\sqrt{2} + 1)\right]$.

Problems 6.9 (page 233)

1. $a^2(1 + \sinh 1 \cosh 1)$ 2. $\pi k^2 a - k^2 a^2$ 3. $2\pi a^2$
4. $\pi a^2/12$ 6. (i) $\frac{4}{3}\pi a^2$; (ii) $\frac{4}{3}\pi a^2$ 8. $\frac{8\pi}{15}(a^2 + ab + b^2)$
7. $48\pi a^2$ 8. (i) $2\pi a^2(2\log 2 - 3/5)$; (ii) $\pi a^2/24$; (iii) $\pi a^2/48$
9. (i) $6\pi^2 a^2$; (ii) $2\pi^2 a^2$ 10. $2\pi a^2/105$ 11. $4\pi^2 a^2$
13. (i) $\frac{4}{3}\pi a^2$; (ii) $\frac{8}{3}\pi a^2$ 14. $\frac{\pi}{3}a(a^2 + b^2)$ 15. $\frac{\pi a^2}{4}\left[\frac{3}{\sqrt{2}}\log(\sqrt{2} + 1) - \frac{1}{3}\right]$.

Problems 6.10 (page 237)

1. $\frac{\pi a^2}{2}(1 + \sinh 2)$ 2. $\frac{2\pi a^2}{3}(2\sqrt{2} - 1)$ 3. $2\pi a\left[\frac{b}{a} + \frac{a}{\sqrt{(a^2 - b^2)}}\sin^{-1}\left\{\sqrt{(a^2 - b^2)}/a\right\}\right]$
4. $\frac{1}{2}\pi r l$, $(= \pi r\sqrt{(r^2 + h^2)})$, where r is the base radius and h the height of the cone
5. $8\pi a^2$ 6. $\frac{64}{3}\pi a^2$ 11. $\frac{64}{3}\pi a^2$ 12. $8\pi a^2$
13. $\frac{20}{9}\pi a^2$ 14. $4\pi a^2(11 - 4\sqrt{2})$ 15. $\pi a^2\left[2\sqrt{2} - \log(1 + \sqrt{2}) + 2\right]$.

Problems 6.11 (page 237)

1. (c) 2. (c) 3. (b) 4. (c) 5. (b) 6. (c)
7. (c) 8. (d) 9. (b) 10. (d) 11. (c) 12. (c)
13. (a) 14. $\frac{3\pi a^2}{2}$ 15. (d) 16. $\pi a^2/12$ 17. 1 18. $7\pi/3$

29. (a) 30. (a) 31. $abc\sqrt{3}+c\sqrt{2}+c\sqrt{3}$ 32. (c) 33. (a)
34. (b) or (c) 35. (c)

Problema 7.1, página 288

1. 12 2. 3888 3. $\frac{1}{2}(e-1)$ 4. $\frac{1}{2}+\log(1+\sqrt{2})$
5. e^{100} 6. $\frac{3}{4}\pi(a^2+b^2)$ 7. 3008 8. $\sin 1$
9. 2410.80 10. $1-\sqrt{3}/2$ 11. $\frac{28\pi}{4}(\log e-\frac{1}{2})$ 12. 1024
13. $\frac{2}{3}\pi^3$ 14. πe^{25} 15. 1

16. (a) $\int_0^{2a}\int_{\frac{1}{2}x}^{3a} f(x,y)\,dx\,dy-\int_0^{a}\int_{\frac{1}{2}x}^{\sqrt{2a^2-y^2}} f(x,y)\,dx\,dy$. (c) $\int_0^{\pi/2}\int_0^{\sin\theta} f(r,\theta)\,r\,dr\,d\theta$
17. $4\sqrt{3}$ 18. $45\pi/2$.

Problema 7.2, página 293

1. 4.5 2. 36 3. $2a^2$ 4. $\frac{1}{2}\log 2-\frac{1}{8}$
5. e^2 6. $e^2(1-e^{24})$ 7. 45 8. e^{24}.

Problema 7.3, página 304

1. $\frac{abc}{3}(a^2+b^2+c^2)$ 2. $\frac{\pi}{2}abc(a^2+b^2+c^2)$ 3. 4725
4. $\frac{1}{5}e^{10}-\frac{1}{2}e^{20}+e^{40}-\frac{1}{8}$ 5. $\frac{\pi}{2}\log 2-\frac{13}{16}$ 6. $\frac{1}{2}(13-4e+e^2)$
7. $2\pi\sqrt{3}\pi$

Problema 7.4, página 297

1. $a\pi$ 2. $a\pi$ 3. $\pi\left(\frac{5}{2}-\frac{2}{e}\right)a^2$ 4. $\frac{a^{n+2}}{n+2}$
5. $\frac{\pi}{4}-\frac{1}{2}$ 6. 0 7. $\frac{128\pi a^3}{35}$ 8. π
9. $\pi^2 a^2$ 10. $4a\pi$ 11. $\frac{3}{2}\left(\log 2-\frac{3}{8}\right)$ 12. $a\sqrt{2}\pi$
13. 2π 14. 4π 15. $a\sqrt{2}-\sqrt{2}\,a$ 16. $32a/35$
17. $a\sqrt{2}\pi$ 18. $128\sqrt{2}\pi$ 19. $8\pi a^2$ 20. $4\sqrt{2}\,\pi$
21. $8a^2\pi$ 22. $\frac{1}{2}\pi k a$

Problema 7.5, página 300

2. 64 3. $32\pi-8\sqrt{3}$ 4. $8\pi^2$ 5. $\frac{2\pi a^2}{3}$

Problems 7.0, page 207

1. $183\frac{2}{24}$ h
2. 21 mph 532
3. 23 328
4. $\left[\frac{30}{29}, \frac{30}{19}\right]$
5. $\left[\frac{x(4x+35)}{49x+61}, \frac{5(2x+3)}{49x+61}\right]$
6. $\left(\frac{\ln 2}{3}, 0\right)$
7. $\dot{x} = 6x^2$, $\dot{y} = 5x^4$; t where $x = 0$
8. (19, 19, 20)
9. $x = 24$
10. $\left(\frac{35}{12}, \frac{105}{32}, \frac{105}{32}\right)$
11. $\frac{23}{24}$ metres
12. $\left(\frac{2a}{5}, \frac{2a^2}{15}\right)$
13. Problem the x agreed in the ratio 7 : 3
14. $\left(\frac{3}{5}a, \frac{3}{5}a\right)$ where a is the base, h the depth
15. C.F. has as its radius L is the bounding diameter at a depth $3(a)/16$ from the centre.

Problems 7.1, page 301

1. $\pi h^2/12$
2. $\pi h(a^2+b^2)$
3. $2\pi^2 b$
4. $\frac{1}{3}\pi h(a^2+b^2)$
5. $(31/32)\ \text{kg m}^2$
6. $\frac{2}{3}\pi R^3$
7. $\frac{1}{12}\pi h^3$; $\frac{1}{12}\pi h^2(3h^2+4R^2)$
8. (a) $\frac{3\pi h^2}{10}$; (b) $\frac{2\pi}{105}(r^2+4h^2)$; (c) $\frac{\pi}{105}(2r^2+3h^2)$
9. 1348.8 cm^3
10. $\frac{1}{28}$
11. $\frac{8\rho a b c(a^2+b^2)}{15}$
12. $\frac{\rho\pi R^5}{4}$

Problems 7.2, page 309

1. (a) 0.2236; (b) 31.4928; (c) $(1/\sqrt{3})$; (d) 0.3804; (e) 10.1327
2. (a) $\sqrt{\pi}/2$; (b) 0.8862; (c) $\sqrt{\pi}/2$; (d) $2^{p+q-1}B(p,q)$
3. $\frac{\Gamma\left(\frac{2p+1}{n}\right)\Gamma(q+1)}{n\Gamma\left(\frac{2}{n}+p+1+\frac{1}{n}\right)}$; (b) $\frac{1}{2520}$; (c) $\frac{\sqrt{\pi}}{n}\cdot\frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{1}{n}+\frac{1}{2}\right)}$
4. $\pi\sqrt{2}$
5. -2π
6. $\frac{\pi\sqrt{2}}{4}$
7. $\frac{\pi^2}{48}$.

Problems 7.3, page 318

2. $\frac{1}{2}K(1,\sqrt{3}/2)$
3. $\frac{2}{\sqrt{3}}\left[F\left(\sqrt{\frac{3}{10}}, \frac{1}{2}\pi\right) - F\left(\sqrt{\frac{3}{7}}, \frac{1}{4}\pi\right)\right]$
4. $\frac{1}{2}\sqrt{2}\left[K(1/\sqrt{2}) - K(\sqrt{3}/2)\right]$
5. $\text{sn}(x) = \frac{x}{2}\left(1 - \frac{x^2}{8} + \frac{x^4}{32} - \frac{x^6}{128} + \ldots\right)$; $\text{cn}(x) = 1$; (b) 0.2048; (c) 0.2024.

Problems 7.4, page 325

1. 4
2. Area of the triangle having vertices $(0,0)$, $(2,1)$, $(1,4)$
3. $\sqrt{\pi}/2$
4. 2.1416
5. $16\sqrt{\pi}/9$
6. 24
7. -120
8. $192\beta(4,3/4)$
9. 1

11. 500 | 12. $\frac{\pi r^2}{8}$ | 13. 6 | 14. 2/3

15. $\pi\sqrt{3}/2$ | 16. -10 | 17. — | 18. 64/105

19. $e^2 \sin b - e \sin a$ | 20. $e^2 - 1$ | 21. $\frac{1}{4}\pi ab(a^2 + b^2)$ | 22. $\frac{1}{4}\pi \ln(1 + \sqrt{2})$

23. 32/63 | 24. 1/4 | 25. $\frac{\pi}{30} + \frac{\sqrt{3}}{2}$ ln $\frac{3}{2}$ | 26. 4/45

27. 1/6 | 28. 3/20 | 29. $\int_0^1 \int_{x^2}^{\sqrt{2-x^2}} f(x, y)\,dy\,dx$

30. $\sqrt{2}\pi$ | 31. $3\pi/256$ | 32. 1/2 | 33. $\int_0^{\pi/2} \int_0^{2\cos\theta} r\,dr\,d\theta$

34. $\frac{73}{12}$ | 35. $\left(\frac{2}{20}, \frac{2}{18}\right)$ | 36. 18 | 37. $16\pi/3\sqrt{3}$

38. 1 | 39. $\ln 2$ | 40. $\frac{1}{2}\sqrt{\pi}$ | 41. (c) | 42. (c)

Problems 6.1, page 319

2. (a) $\rho^2 \sin\phi + 3\rho^2 \cos\phi + 2\rho \sin\phi - 10\rho$; (b) $12\rho^2 + \rho \sin\phi - \cos\phi = 0$

$-2\rho^2 \cos\phi + 2\rho \sin\phi + 7\rho^2 = 0$; (c) $\rho \sin\phi + 2\rho^2 \cos\phi + 10 \cos\phi = 0$

3. $-4\mathbf{i} + 3\mathbf{k}$ | 4. (a) $\tan^2 \sec \alpha$; (b) $\tan^2 \alpha \tan \alpha$; (c) $\sec \beta - \sin\beta \tan\alpha \cos\alpha + \sin\alpha$

7. $14\mathbf{i} + 9\mathbf{j} + 6\mathbf{k} = 3\sqrt{61}\left\langle \frac{2}{\sqrt{61}}, \frac{-12}{\sqrt{61}}, \frac{11}{\sqrt{61}} \right\rangle$; $\frac{1}{3}(2\mathbf{i} + 2\mathbf{j} + \mathbf{k})$

8. $(x - a)\sqrt{2} + y - a\sqrt{2} = \left(z - \frac{\pi a}{4} \tan\alpha\right)\sqrt{2} \tan\alpha$

9. (a) $c^2 b^2 + a^2 b^2 + a^2 c^2 - b^2 \cos^2(\theta/2)$; (b) $3\sqrt{2}/2$

10. (a) $\mathbf{R} = c_1 \mathbf{i} + c_2 \mathbf{j} + c_3 \mathbf{k} + 2c_4 \mathbf{K}$; $\frac{2\mathbf{M} - \mathbf{K}}{\sqrt{3}}$

(b) $\mathbf{R} = p\mathbf{i} + p\mathbf{j} + 2p\mathbf{k}$; $(p + q)\mathbf{i} + (p + q)\mathbf{j} + (p - q)\mathbf{k}$; $(2\mathbf{K} - \mathbf{I} - \mathbf{J})/\sqrt{3}$

Problems 6.2, page 330

1. $\mathbf{v}(t) = \mathbf{i} + \sqrt{3}\mathbf{j}$, $\mathbf{v}(\pi/3) = \mathbf{i} + \sqrt{3}\mathbf{j}$

2. $\mathbf{v} = \mathbf{i} + 2t\sqrt{3}\mathbf{j}$

3. $6\sqrt{5}\mathbf{N}$; $-\sqrt{3}\mathbf{G}\mathbf{N}$

4. (a) $t^2 + 6t + 9$; (b) $\sqrt{2}t$; (c) $t = 0$; 3

5. $\sqrt{2}\sqrt{5}$ m p.h. in the direction $\tan^{-1}(1/3)$ North of East

6. 21.28 km/hr in the direction 78°41′ South of East

Problems 6.3, page 336

1. (a) $\mathbf{v} = 2\mathbf{i} + 4\mathbf{j} + 2\mathbf{K}$; $\mathbf{a} = 2\mathbf{K}$; (b) $\mathbf{v} = \mathbf{i} + 2\mathbf{j} + \mathbf{K}$; $\mathbf{a} = \mathbf{0}$

2. (a) $\mathbf{v} = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{K}$; $\mathbf{a} = \mathbf{0}$; (b) $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{K}$; $\mathbf{a} = \mathbf{0}$

3. (a) $\mathbf{v} = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{K}$; $\mathbf{a} = \mathbf{0}$; (b) $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{K}$; $\mathbf{a} = \mathbf{0}$

4. (a) $\mathbf{v} = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{K}$; $\mathbf{a} = \mathbf{0}$; (b) $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{K}$; $\mathbf{a} = \mathbf{0}$

5. (a) $\mathbf{v} = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{K}$; $\mathbf{a} = \mathbf{0}$; (b) $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{K}$; $\mathbf{a} = \mathbf{0}$

6. (a) $\mathbf{v} = 2\mathbf{i} + 2\mathbf{j} + 2\mathbf{K}$; $\mathbf{a} = \mathbf{0}$; (b) $\mathbf{v} = \mathbf{i} + \mathbf{j} + \mathbf{K}$; $\mathbf{a} = \mathbf{0}$

Problems 6.5, page 355

1. (a) $\mathbf{i} + 2\mathbf{j} + 3\mathbf{K}$; $\sqrt{14}$; (b) $\frac{1}{\sqrt{14}}(\mathbf{i} + 2\mathbf{j} + 3\mathbf{K})$; (c) $\frac{1}{\sqrt{14}}(\mathbf{i} + 2\mathbf{j} + 3\mathbf{K})$

2. $-3 + 3\mathbf{i} + 2\mathbf{K}$; $\sqrt{14}$

3. $15\frac{1}{3}$ | 4. $18\sqrt{17}$ | 5. $a = 5, b = 24, c = -5$

6. $-2\mathbf{i}/61$; $\sqrt{11/60}$ | 7. $a = \pm\frac{2\mathbf{i}}{11}$, $b = \pm\frac{6\mathbf{j}}{11}$, $c = \pm\frac{9\mathbf{k}}{11}$ | 8. $9\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$; $3\sqrt{19}$

9. 9 | 10. $\frac{1}{3}(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ | 11. $\cos^{-1}(2/\sqrt{22})$

12. $\cos^{-1}(-2/\sqrt{30})$ | 13. $a = -4, b = -16$

Problems 8.4, page 293

1. (a) 32 ; (b) $-18i + 18k$; (c) $3i + 3j + 3k$; (d) $3i + 3j + 3k$; (e) $-3i - 3j - 3k$; (f) $-3i - 3j - 3k$
4. $a = -2$; $4xz\,i - xy\,j + (y - 2yz + 4xy^2)\,k$; $(2x^2 + y^2 - z^2)\,i + 2xy\,j - 2xz\,k$
13. (a) $2\,i$; (b) $2i + 2j + 2k$; (c) $i + j + 2k$ 14. (a) $2xe^{2x}\,i - 12x^2y^2\,j + x^2z^2\,k$; (b) $x = \pm 1$
16. (a) $5y^4 + 3x^2y - 6y^2z\,i + (5x^2y^3 + x^3 - 6xy^2z)\,j + (5xy^3 + x^3 - 3xy^2)\,k$; (b) $2xyz$
15. $2\sqrt{21}$.

Problems 8.5, page 295

1. $35\,i + 35j - 42\,k$ 2. $e^x - x + 2z\,i + (1 - z^2)\,j + (x - x\cos z - 2y)\,k$
3. $V = 4\sin 2\theta + 9(\cos\theta\sin\theta - 1)\theta + 9\cos^2\theta$; $R = 3(1 - \cos 2\theta) - 2\sin 2\theta - \frac{9\pi^2}{2}\,k$.

Problems 8.6, page 296

1. 0 2. 33 3. -300 4. 0 5. $\frac{R^2\sqrt{2}\,\pi}{2}$ 6. zero

7. 300 8. $1\frac{1}{30}$ 9. 99 10. $\left(2 - \frac{\pi}{4}\right)i - 4x - \frac{1}{4}\,k$.

Problems 8.7, page 300

2. $1\frac{1}{3}$ 3. 0.

Problems 8.8, page 301

2. πab 4. πa^2 5. $2\pi ca$ 6. $12\pi b$ 7. $36\pi a^2\sqrt{16}$.

Problems 8.9, page 305

2. $-2\pi b^3$ 8. $\frac{10}{3}\pi$ 9. $2\pi ma$ 10. 2 11. 0 12. π

Problems 8.10, page 310

4. 108π 7. (a) $\frac{27}{2}\pi a^3$ (b) $12\pi a - a^3$ 8. $4\pi a^2$

9. $\frac{4\pi a^2}{32}$ 10. $\frac{5}{4}\pi\sqrt{6}$ 11. -4π 12. 4π

Problems 8.11, page 314

2. $14\frac{2}{3}$ 4. $x^2y - y^2z^2 + z^3$

5. (a) $\frac{1}{3}x^3 + y^3 + z^3 - 3xyz$; (b) $x^2y + y^2z + z$;
(c) $xz^2 - yz + 3x^2y$. (d) $x^2y^2 + y^2z^2 + xyz = 0$

6. (a) Yes, $\frac{x^2}{2}z^2 + y^2 - 2x^2z$; (b) Yes

7. $xy\sin z + \cos xz + y^2z + c$ 8. $x^2y + xz^2 = 282$

9. $a = 1$, $b = 2$, $c = -1$ 10. $x = 0$; $2x^2y - xz^2$; 47.

Problems 8.12 (page 202)

1. (a) $(2\cos 2\theta - r\sin\theta)\mathbf{T}_r - (2r\sin 2\theta + r\cos\theta)\mathbf{T}_\theta + 2r\cos\phi\,\mathbf{T}_\phi$
   (b) $(2r\cos^2\phi - 2r^2\sin^2\phi)\mathbf{T}_r - r^2\sin 2\phi\,\mathbf{T}_\theta + 2r\cos\phi\,\mathbf{T}_\phi$
2. (a) $r\sin\theta\,\big|\,(\sin\theta + \sin^2\phi)\mathbf{T}_r + \sin\theta\cos\theta\sin\phi\,\mathbf{T}_\theta$
   $+ (\cos\theta(1+\sin^2\phi) - r\sin\theta(\cos\theta\sin\phi)\mathbf{T}_\phi + \sin\theta\cos\phi\,\mathbf{T}_\phi\,\big|$
   (b) $r^2\sin\theta\,\big|(\sin^2\theta\cos^2\phi\sin\phi + r\sin\theta\cos\theta\sin^2\phi + r\cos^2\theta\cos\phi)\mathbf{T}_r$
   $+ (\sin\theta\cos\theta\cos^2\phi\sin\phi - r\sin^2\theta\sin^2\phi + r\cos^2\theta\sin\phi)\mathbf{T}_\theta$
   $+ (\cos\theta\cos\phi\sin\phi - r\sin\theta\cos^2\phi + r\sin\phi)\mathbf{T}_\phi\,\big|$
3. $pr\sin 2\theta\,\mathbf{T}_r + pr\cos 2\theta\,\mathbf{T}_\theta + \frac{1}{2}p^2\sin 2\theta\,\mathbf{T}_\phi$

Problems 8.13 (page 203)

1. $11\sqrt{13}\,;\ 22\sqrt{13}\,;\ 2014$
2. $\frac{1}{4}(x-2)+y-1=x+3$
3. $\mathbf{v}\cdot\mathbf{n}=\frac{1}{\sqrt{3}}\,\mathbf{v}\cdot\mathbf{n}_0$
4. $4x-3y+3z$
5. $\nabla\cdot\mathbf{v}$
6. $\int_C \mathbf{v}\cdot\mathbf{n}\,ds$
7. ∇V
8. $3\,;3$
9. Irrotational
10. 4π
11. solenoidal
12. $-2\nabla\sqrt{5}$
13. zero.
14. $-(\mathbf{i}+2\mathbf{j}+3\mathbf{k})$
15. $\frac{\partial V_1}{\partial x}+\frac{\partial V_2}{\partial y}+\frac{\partial V_3}{\partial z}$.
16. zero
17. $-(12\mathbf{i}-6\mathbf{j}+4\mathbf{k})$
18. zero
19. zero
20. zero
21. zero
22. $8\mathbf{i}-2\mathbf{j}+4\mathbf{k}$
23. $y(x\mathbf{i}+z\mathbf{k})$
24. $\frac{1}{\sqrt{21}}(2\mathbf{i}+4\mathbf{j}-\mathbf{k})$
25. $\mathbf{i}-2\mathbf{j}$
26. $\mathbf{i}\frac{\sqrt{3}}{3}$
27. $2\mathbf{v}$
28. $\nabla\phi$
29. zero
30. $\mathbf{k}\mathbf{i}$
31. $2\mathbf{i}$
32. $\mathbf{i}\mathbf{v}$
33. $\mathbf{i}\mathbf{i}$
34. $\mathbf{k}\mathbf{i}$
35. $(\mathbf{c})$
36. $\mathbf{i}\mathbf{a}$
37. zero
38. Irrotational field
39. $(\mathbf{c})$
40. the case in which field is originating at P per unit volume.
41. $\mathbf{k}\mathbf{i}$
42. A gives the maximum rate of change of ϕ
43. $\mathbf{i}\mathbf{v}$
44. $\mathbf{k}\mathbf{i}$
45. $(\mathbf{c})$
46. $\mathbf{i}\mathbf{v}$
47. $\mathbf{i}\mathbf{i}$
48. $\mathbf{v}\mathbf{v}$
49. zero
50. True
51. True

Problems 8.5 (page 209)

1. Convergent
2. Convergent
3. Convergent
4. Divergent
5. Divergent
6. Convergent
7. Convergent
8. Divergent

Problems 8.6 (page 217)

1. Convergent
2. Convergent
3. Oscillatory
4. Convergent
5. 10 m

Problems 8.7 (page 222)

1. Convergent
2. Convergent
3. Divergent
4. Divergent
5. Conv. for $p>2$; Div. for $p\le 2$.
6. Conv. for $p>2$; Div. for $p\le 2$.
7. Divergent
8. Convergent
9. Convergent
10. Convergent
11. Convergent

12. Divergent 13. Divergent 14. Convergent 15. Convergent
16. Convergent 17. Divergent 18. Convergent

Problems 9.4, page 378

1. Conv. for $x < 1$; Div. for $x \geq 1$ 2. Conv. for $x < 3$; Div. for $x \geq 3$
3. Conv. for $x \leq 3$; Div. for $x > 3$ 4. Conv. for $x > 3$; Div. for $x \leq 3$
5. Convergent for all values of p 6. Convergent 7. Divergent
8. Convergent 9. Conv. for $x < 3$; Div. for $x \geq 3$ 10. Convergent
11. Convergent 12. Convergent 13. Divergent
14. Convergent 15. Conv. for $x < 3$; Div. for $x \geq 3$; Conv. for $p > 1$ and Div. for $p \leq 1$
16. Divergent 17. Conv. if $3 > a > 3$; Div. if $a \leq 3 > 3$.

Problems 9.5, page 388

1. Conv. for $x \leq 1$; Div. for $x > 1$ 2. Conv. for $x \leq 1$; Div. for $x > 1$
3. Conv. for $x < 1$; Div. for $x > 1$ 4. Conv. for $x < 3$; Div. for $x > 3$
5. Conv. for $x < e$; Div. for $x \geq e$ 6. Conv. for $x \leq 1$; Div. for $x > 1$
7. Conv. for $x < 3$; Div. for $x > 1$ 8. Conv. for $x^2 < 1$; Div. for $x^2 > 1$
9. Conv. for $x^2 < 4$; Div. for $x^2 \geq 4$ 10. Convergent
11. Conv. for $x < 10$; Div. for $x \geq 10$ 12. Conv. for $x < 1$; Div. for $x \geq 1$
13. Divergent
14. Conv. for $x < 1$; Div. for $x \geq 1$; When $p = 1$, Conv. for $k - a > 1$, Div. for $k - a \leq 1$.

Problems 9.6, page 393

1. Convergent 2. Convergent 3. Convergent
4. Convergent 5. Conv. for $x < 1$; Div. for $x > 1$
6. Conv. for $x < \frac{1}{10}$; Div. for $x \geq \frac{1}{2}$ 7. Convergent.

Problems 9.7, page 399

1. Oscillatory 2. Convergent 3. Convergent 4. Convergent
5. Convergent 6. Oscillatory 7. Convergent 8. Convergent
9. Convergent 10. Oscillatory.

Problems 9.8, page 402

1. (a) and (c) conditionally convergent
2. (a) Conditionally convgt. for $3 < p \leq 1$; (b) Conditionally conv.
4. Absolutely convergent for (a) $0 < x < 3$; (b) $-1 < x < 1$; (c) $|x| < 2$.
5. Divergent for $x \leq 1$ and not convergent for $x > 1$
6. (a) $-3 < x < 3$; (b) $-1 < x < 1$;
7. $-e < x \leq e$ 8. (a) Absolutely convergent (b) convergent
8. Absolutely convergent

Problems 9.9, page 408

1. Conv. for $x > 3$; Div. for $x \leq 1$ 2. Convergent 3. Divergent
4. Convergent 5. Divergent 6. Conv. for $x < 2$; Div. for $x \geq 1$
7. Conv. for $x > 3$; Div. for $x > 1$ 8. Conv. for $x < 1$; Div. for $x > 1$
8. Conv. for $x < 24$; Div. for $x \geq 24$ 10. Conv. for $x < 2$; Div. for $x \geq 2$

11. Convergent for all x. 12. Conv. for $x < 2$; Div. for $x \geq 1$ 13. Convergent.
14. Absolutely convergent 15. Convergent.
16. Convergent for $p > 1$; divergent for $p \leq 1$.

Problems 9.10, page 1001

1. Uniformly convergent for $0 < x < 1$. 2. 1a
3. Uniformly convergent for all real values of x 4. Uniformly convergent for $0 < x < 1/a$
5. (1) and (2) Both converge uniformly for all real values of x.

Problems 9.11, page 1005

1. (c) 2. (d) 3. (c) 4. (b) 5. (c) 6. (d)
7. (c) 8. (c) 9. (c) 10. (d) 11. (c)
12. $(-1, 1)$ 13. $(-1/2, 1/2)$ 14. $-2 < x \leq 2$ 15. $a_n < 2$ 16. Oscillatory
17. All values of x 18. $x < 1$ 19. Convergent 20. Divergent 21. $p = x + 1$
22. Divergent 23. Convergent. 24. $3 < x < 4$ 25. yes 26. True 27. Convergent
28. Divergent 29. $x > 1$ 30. $3 < x < 1$ 31. (b) 32. (c) 33. (c)
34. (c) 35. True.

Problems 10.1, page 1010

1. $\dfrac{2\sinh \pi x}{\pi}\left\{\left[\dfrac{1}{2x} - \dfrac{x\cos x}{1^2 + x^2} + \dfrac{x\cos 2x}{2^2 + x^2} - \cdots\right] - \left[\dfrac{\sin x}{1^2 + x^2} - \dfrac{2\sin 2x}{2^2 + x^2} + \dfrac{3\sin 3x}{3^2 + x^2} - \cdots\right]\right\}$

$\dfrac{\pi}{\sinh \pi} = 2\left[\dfrac{1}{2^2 - 1} - \dfrac{3}{3^2 + 1} + \dfrac{5}{5^2 + 1} - \cdots\right]$

Problems 10.2, page 1021

1. 2π 2. 2π 3. Yes.

Problems 10.3, page 1034

1. $\dfrac{1}{2}\pi - \dfrac{4}{\pi}\left(\cos x + \dfrac{\cos 3x}{3^2} + \dfrac{\cos 5x}{5^2} + \cdots\right)$ 2. $\dfrac{A_0}{2} + \dfrac{1}{2}A_1 \sin x - \dfrac{2A_2}{\pi}\sum_{n=1}^{\infty}\dfrac{n\sin 2nx}{4n^2 - 1}$, $n \neq 1$

3. $\dfrac{\pi^2}{3} - 4\left[\cos x - \dfrac{\cos 2x}{2^2} + \dfrac{\cos 3x}{3^2} - \cdots\right] - \dfrac{2}{\pi}\left[\left(\dfrac{2}{1^3} - \dfrac{\pi^2}{1}\right)\sin x - \left(\dfrac{2}{2^3} - \dfrac{\pi^2}{2}\right)\sin 2x + \cdots\right]$

4. $2\left(\pi - \dfrac{\pi}{2}\right)\sin x - \pi \sin 2x + \dfrac{2}{3}\left(\pi - \dfrac{\pi}{3}\right)\sin 3x - \dfrac{\pi}{2}\sin 4x + \cdots$

5. $\dfrac{\pi}{2}\left(\sin x + \dfrac{\sin 2x}{2} + \dfrac{\sin 3x}{3} + \cdots\right)$

Problems 10.4, page 1041

1. $-\dfrac{2}{\pi}\sum_{n=1}^{\infty}\dfrac{n\sin nx}{m}$

2. (a) $\frac{a^2}{3}-\frac{a^2}{12}\left(\frac{1}{1^2}\cos\frac{2\pi x}{a}+\frac{1}{2^2}\cos 4\pi\frac{x}{a}+\cdots\right)$ (b) $\frac{a^2}{8}\left[\frac{1}{1^2}\cos\frac{2\pi x}{a}+\frac{1}{3^2}\cos\frac{6\pi x}{a}+\cdots\right]$

(c) $f(x)=\frac{2}{3}+\frac{4}{\pi^2}\left(\cos\pi x-\frac{\cos 2\pi x}{2^2}+\frac{\cos 3\pi x}{3^2}-\cdots\right)$

4. $\frac{2}{3}-\frac{12}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\left[\frac{\cos(2n-1)\pi x}{2}\right]$

5. $\frac{\pi}{4}+\frac{\pi}{2}\cos x=\frac{8\pi}{\pi}\left[\frac{1}{1\cdot3}\cos 2\pi x+\frac{1}{3\cdot5}\cos 4\pi x+\frac{1}{5\cdot7}\cos 6\pi x+\cdots\right]$

6. $f(x)=\frac{\pi}{2}+2\sum_{n=1}^{\infty}(-1)^{n+1}\sin nx$, para $x=\pm\pi$.

Problemas 10.6, página 632

1. $\frac{\pi^2}{3}+\sum_{n=1}^{\infty}\frac{4n^2}{n^2\pi^2}(-1)^n\cos\frac{n\pi x}{\pi}$

2. $1-\frac{1}{2}\cos x-\frac{2}{3\cdot2}\cos 2x+\frac{2}{4\cdot3}\cos 3x-\frac{2}{5\cdot4}\cos 4x-\cdots$

3. $\frac{\pi}{2}=\frac{4}{\pi}\left(\cos x+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+\cdots\right)$

4. (a) $\frac{2}{\pi}-\frac{4}{\pi}\left(\frac{\cos 2x}{3}+\frac{\cos 4x}{15}+\cdots+\frac{\cos 2nx}{4n^2-1}+\cdots\right)$ (b) $\frac{2}{\pi}+\sum_{n=1}^{\infty}\frac{2(-1)^{n+1}}{(4n^2-1)\pi}\sin\frac{2n\pi x}{\pi}$

5. $\frac{\pi}{4}+1-\frac{4}{\pi}\left(\cos x+\frac{1}{3^2}\cos 3x+\frac{1}{5^2}\cos 5x+\cdots\right)=\frac{\pi^2}{6}$

6. $\frac{4\pi}{\pi}\left(\sin x+\frac{\sin 3x}{3}+\frac{\sin 5x}{5}+\cdots\right)$.

Problemas 10.8, página 670

2. $\frac{\pi}{2}-\frac{4}{\pi}\left[\cos x+\frac{1}{3^2}\cos 3x+\frac{1}{5^2}\cos 5x+\cdots\right]$; $2\left(\sin x-\frac{1}{2}\sin 2x+\frac{1}{3}\sin 3x-\cdots\right)$

3. $\frac{\pi^2}{3}-4\left[\cos x-\frac{1}{2^2}\cos 2x+\frac{1}{3^2}\cos 3x-\frac{1}{4^2}\cos 4x+\cdots\right]$

4. $\sum_{n=1}^{\infty}\frac{1}{n}\sin 2n\pi$

5. $\frac{1}{2}+\frac{1}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin n\pi x$

6. $\frac{8}{\pi^2}\left(\frac{\sin\pi x}{1^2}+\frac{\sin 3\pi x}{3^2}+\frac{\sin 5\pi x}{5^2}+\cdots\right)$

7. $\frac{1}{\pi x}(3-|-1|)$

8. $-\frac{1}{2}\sin\pi x+\sum_{n=2}^{\infty}\frac{1-(-1)^n}{n^2-1}\sin n\pi x$

9. $\sum_{n=1}^{\infty}\frac{2n\pi}{1+n^2\pi^2}(1-\pi\cos n\pi)\sin n\pi x$

10. $\sum_{n=1}^{\infty}\frac{1}{n^2\pi}[1-(-1)^n]\sin n\pi x$

11. $\frac{1}{2}+\sum_{n=1}^{\infty}\frac{2}{(n\pi)^2}\left[2\cos\frac{n\pi}{2}-1-(-1)^n\right]\sin\frac{n\pi x}{1}$

12. $\frac{8}{\pi}\cos\frac{\pi}{4}\left[\frac{\sin 2x}{1\cdot3}-\frac{\sin 6x}{5\cdot7}+\frac{\sin 10x}{9\cdot11}+\cdots\right]$

14. $\frac{2(3)}{\pi(3^2-2\pi^2)}\left[\sin\frac{\pi x}{3}\sin\frac{\pi x}{3}+\frac{1}{2^2}\sin\frac{2\pi x}{3}\sin\frac{2\pi x}{3}+\frac{1}{3^2}\sin\frac{3\pi x}{3}\sin\frac{3\pi x}{3}+\cdots\right]$

Problem 1001 (page 418)

2. $\pi/4$

Problem 1002 (page 420)

1. $\sum_{n=1}^{\infty} \frac{(4n)\sin n\pi - 1\cos n\pi}{(4n^2+\pi^2)^2}\sin nx$

2. $\frac{2}{\pi}\left[1 - \frac{e^{2\pi}+e^{-2\pi}}{1\cdot 3} - \frac{e^{3\pi}+e^{-3\pi}}{3\cdot 5} - \frac{e^{4\pi}+e^{-4\pi}}{5\cdot 7} - \cdots\right]$

3. $\frac{\pi}{2}\sin\alpha x - \sum_{n=1}^{\infty}\frac{(-1)^n n^2\sin n\pi}{\alpha^2 - n^2}$ 4. $\sin kx = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}2n\sin n\pi}{4k^2-4n^2}\sin nx$

5. $\frac{\pi}{2} - \frac{4}{\pi}\left[\sin x + \frac{1}{3^2}\sin 3x + \frac{1}{5^2}\sin 5x + \cdots\right]$ where $x = \sin\theta$.

Problem 1003 (page 423)

1. $11.733 - 7.703\cos 2x - 2.303\cos 4x + \cdots - 1.568\sin 2x - 0.156\sin 4x + \cdots$
2. $1.45 + (-0.57\cos x + 0.17\sin x) - (0.1\cos 2x + 0.05\sin 2x)$
3. $a_0 = 41.66, a_1 = -8.33.5, = -1.25$ 4. -3.37%
5. $y = 2.103 + 0.034(\cos x + 1.403\sin x + 0.034\cos 2x + 0.143\sin 2x$
6. $7.0\sin\theta + 1.5\sin 3\theta - 0.2\sin 5\theta + 11.5\sin 4\theta - \cdots$.

Problem 1004 (page 424)

1. $2\pi^2$ 2. $\frac{1}{2}[f(x-\pi) + f(x+\pi)]$ 3. -1, it is odd that $f(x) = -f(-x)$
4. $f(x) = A$ when $0 < x < \pi$ and $f(x) = -A$ when $\pi < x < 2\pi$ 5. Even
6. $\{1, 2, 3\}$ 7. Zero 8. not defined 9. odd
10. Cosine 11. even 12. $y = Ax$ 13. Zero
14. Cosine 15. Zero 16. $\int_0^2 x^3\cos\frac{2n\pi x}{2}dx$ 17. $\frac{1}{T}\int_0^{-2\pi} f(x)\sin\frac{2n\pi}{T}dx$
18. $\{1, 3\}$ 19. Zero 20. $a_0 = \frac{2}{T}\int_0^T f(x)dx$, $a_n = \frac{2}{T}\int_0^T f(x)\cos\frac{2n\pi}{T}dx$
21. $\frac{2}{\pi}\left[\sin x + \frac{\sin 2x}{2} + \frac{\sin 3x}{3} + \cdots\right]$ 22. π 23. Zero
24. 21 25. $\sum_{n=-\infty}^{\infty}(-1)^n\frac{(1-in\pi)}{1+n^2\pi^2}\sinh(\pi)e^{inx}$ 26. Polar
27. $-\pi/2$ 28. odd 29. Sine 30. 2.303Δ
31. zero 32. $-\pi/2$ 33. $\frac{1}{2}a_0$ 34. $\frac{\pi^2}{6}$
35. $\frac{\pi}{2} - \frac{4}{\pi}\left[\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots\right]$ 36. $f(x) = \frac{\pi}{2\pi} + \frac{1}{2}\cos 2x + \frac{1}{2}\cos 4x$
37. $x^2 - x$ 38. $x(1+x)$ 39. Even 40. Polar
41. Polar

Problems 11.1, page 429

1. $x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = 0$
2. $\frac{d^2y}{dx^2} + 4y = 0$
3. $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy = 0$
4. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$
5. $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 6y = 0$
6. $2xy\frac{dy}{dx} + x^2 - y^2 = 0$
7. $(x^2 - 2)\left(\frac{dy}{dx}\right)^2 + y^2 = 0$
8. $y\frac{dy}{dx} = 2x$
10. $y''' - 3y'' + 3y' - y = 0$.

Problems 11.2, page 433

1. $(1 - y^2) + (1 - y^2) = c$
2. $\log\frac{x}{y} = \frac{1}{x} - \frac{1}{y} + c$
3. $\tan x \tan y = 1$
4. $(1 + x^2)(3 - 3x)(1 + y^2) = c$
5. $\sin x \sin y = c(1 - e^x)$
6. $2e^{-y} = e^{x^2} + 1$
7. $x = 2\sin y$
8. $(x^2 + 1)(y^2 + 1) = c$
9. $3e^{2x} - 2e^{3y} + 6e^x + c$
10. $(2 - cy)(3x + y) = xy$
11. $(x + 1)(2 - e^y) = c$
12. $a\ln\left(\frac{x - y + a}{x - y - a}\right) = 2y + c$.
13. $y = \tan^{-1}(x + y + 2) + c$
14. $\ln\sin(x + y) = \sec(x + y) + x + c$
15. $x = \operatorname{cosec}(x + y + 2) - \cot(x + y + 2) + c$
16. $\log\sin(y - x) = \frac{1}{2}x^2 + cx$
17. $\cos xy + \frac{2}{2a^2} = c$.

Problems 11.3, page 435

1. $x(x^3 - 3y^3) = c$
2. $cy^2 = x(x^{-4y})$
3. $(x/y)^2 = 2\log cy$
4. $y + \sqrt{(x^2 + y^2)} = c$
5. $y^2 - 2xy + x\log(x/a) = 0$
6. $x\frac{dy}{dx} + y = xy^2$
7. $y = 2x\tan^{-1}(x/a)$
8. $x^{2y} = y + c$
9. $3\log y - \frac{x^2}{4y^2}\left(2\log\frac{x}{y} + 1\right) = c$
10. $\log x = \frac{1}{2a}\left[\frac{x}{y} - \frac{1}{2}\sin\left(\frac{2y}{x}\right)\right] + c$
11. $xy\cos(x/y) = c$.

Problems 11.4, page 436

1. $(X^2 + 2Y^2)^3 = c\left(\frac{-2X - X^2}{2X^2 + X}\right)$ where $X = x + 1$, $Y = y - 1$
2. $(x - y)^2 = x(y + x - 2)$
3. $(x + y)^2 = x(x - y - 2)^2$
4. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
5. $2(2y - x) + \log(2x + 2y + 4) = c$
6. $x - y + \frac{7}{2}\log(3x - 12y - 6) = c$
7. $\log(x + y + \frac{1}{2}) + \frac{1}{2}(2y - x) = c$.

Problems 11.5, page 438

1. $y = cx^{\tan x} - \tan x - 1$
2. $y = \log x + c\log x$
3. $x\sec^2 x = \tan x - 2$
4. $y\cos 2x = x + \frac{\pi}{2}\cos^2 x$
5. $y\sqrt{(1 - x^2)} = \sin^{-1}x + c$
6. $x = c(3 - x^2) + (1 - x^2)$
7. $y(1 + \sin x) = c - x^2/2$
8. $2x\sec^2\theta + \tan^2\theta = c$
9. $yx^2 = 2x + c$

10. $x = y^2 + cy$ 13. $x = \sin^{-1} y - 1 + ce^{-\sin^{-1} y}$ 15. $xy^{-1} = c - e^{xy}$

12. $e^{\sin^{-1} x} = \tan^{-1} y + c$ 14. $e^{xy} = c + \ln(xy)$

Problems 11.8, page 439

1. $y^{-1} \cos x = \ln \sec x + c$ 3. $2y = \sin x \cos x + c \cos x$ 5. $x^2 + (2x^2 + c)y^2 = 0$

4. $3y = x^3 - 2 + ce^{-x^3/3}$ 8. $y^2 = x^2 + cx - 1$ 9. $yx = \ln xy + c$

7. $\sec y = (1 + c)(e^x + c)$ 11. $x^{n+1} = \frac{1}{2} e^{2x} + c$ 12. $\ln xy = e^x - 2e^{-x} + 6x - 8 + ce^{-x}$

10. $\cos y = \cos x \ln \ln x + c$ 13. $\sqrt{x} = \sqrt{y} 2 \sin \sqrt{y} + c$ 15. $y^{-1} = \frac{1}{2} \ln x - \frac{1}{4} + cx^{-2}$

Problems 11.9, page 443

1. $x^2 + y^2 - 2xy = c$ 2. $x^3 + 2x^2y^2 - y^3 - 3x^2a^2 - 2y^2a^2 = c$

3. $x^3 - 6x^2y - 6xy^2 + y^3 = c$ 4. $\frac{x^4}{4} - x^2y^2 + xy^3 + \cos xy = c$ 5. $x^2 + y^2 = c$

6. $x^3 + x^2y^2 - x^2y^3 - y^3 = c$ 7. $x^3 + 2x^2y^2 + y^3 = c$ 8. $x^3 - y^3 = cy^2$

9. $2y \cos 2x + 6y + 2y^3 = c$ 10. $e^x + \sec x \ln xy = c$ 11. $x^2y + xy - x \ln xy + \ln xy = c$

Problems 11.10, page 445

1. $cx = \tan^{-1} yx = c$ 2. $x^3 + y^3 - 3a^2 \tan^{-1}(y/x) = c$ 3. $y - cx + \log xy + 1 = 0$

4. $2 \log xy - (y/x)^2 = c$ 5. $\log(y/x) + \frac{1}{2}(x/y)^2 = c$ 6. $xy + \log(xy) - (1/xy) = c$

7. $(y + 2)e^x \ln x + y^2 = c$ 8. $4x^2y + 4x^3y^2 - y^4 = c$ 9. $2 \cos(xy) + y^2 = c$

10. $\log(xy) = c + xy$ 11. $(x/y) + e^{x^2} = c$ 12. $4xy^{3/2} - \frac{3}{2}(x/y)^{4/3} = c$

13. $4y \log x + y^2 = c$.

Problems 11.11, page 446

1. $(x - y + 1)(x^2 + y^2 + c) = 0$ 2. $(2y - x^2 + c)(y + x + cx^{-1} - 1) = 0$

3. $x^2 + y^2 = cx$ 4. $(x - cx)(y^2 - x^2 - cx) = 0$

5. $(y - cx)(y + x^2 - c)(xy + cy + 1) = 0$.

Problems 11.12, page 448

1. $x + c = \frac{p}{2}\left[\log \frac{p^2 + 1}{4(1 + p^2)} - \tan^{-1} p\right]$, with the given solution

2. $xp = e^{p^2} + c$ 3. $y = 2\sqrt{(ax)} + c^2$ 4. $2xy = e^{2x^2} + 1$

5. $x = a \log p - p + c(p - 1)^2$, with the given solution

6. $x = a \sin p + c$, with the given solution.

Problems 11.13, page 449

1. $y = cx - c^2$ 2. $y^2 = 2cx + c^2$

3. $(x - a)p(x)(p^2 - 1) + x \cosh^{-1} p = c$, with the given solution

4. $y + (1 + p^2)^{1/2} = c$, with the given solution.

Problems 11.12, page 420

1. (i) Gen. sol. : $y = cx + c/(c^2)$; Singular sol. : $27x^2 + 32y = 0$
   (ii) Gen. sol. : $x = \log(cy - y)$; Singular sol. : $y = 0$ $\log x = 0$
   (iii) Gen. sol. $y = cx + \sqrt{(a^2c^2 + b^2)}$; Singular sol. $y = \sqrt{(1 - x^2)}$ or 0
   (iv) Gen. sol. $y = cx - a(c^2 + 1)$; Singular sol. $y = \sqrt{(x^2 + 1)} - a\log\left(\frac{\sqrt{(x^2+1)}}{x}\right)$
2. $y = cx - 2c^2$
3. $(y - cx)(x - 2c) = 0$
4. $(y - cx)(x + 3) = cx^2 + 0$
5. $y^2 = cx^2 + c^2$ [Hint : Put $x^2 = u, y^2 = v$]
6. $xy = cx - c^2$ [Hint : Put $x = u, y = v$]
7. $y^2 = cx^2 - \frac{2c}{1 + c}$

Problems 11.13, page 420

1. 42
2. 150
3. 0150
4. 20
5. $\log y - x = c^2(2y)^2$
6. $y^2 = x^2 + c$
7. $x^2 + xy + cy = 0$.
8. 6120
9. $x^2 + y^2 + 2\tan^{-1}y/x = c$
10. $\log x + c = y^2/2x^2$
11. 10
12. $y^2 + 2x + ce^{-x^2/2} = 2$
13. $y = cx + ca^2$
14. $x = \log(cx - y)$
15. $xy = c$ or $x^2 - y^2 = c$
16. 2
17. $xy = c$
18. 60
19. 30
20. $(1 + x^2)^{3/2} + (1 + y^2)^{3/2} = c$
21. $y = 3x^{2/3}$
22. $x + y = c$
23. x^{-3}
24. $\frac{1}{2}(1 + 1)(3)$
25. $4x^2y^2 + 2(x^2 + y^2) = c$
26. $\sin^{-1}(y/x) = cx$
27. 101
28. 60
29. $x + y + \log(x/y) = 0$
30. $e^{-x^2} + 2\cos y = c$
31. 161
32. False
33. False.

Problems 12.1, page 434

1. (i) $3y + 4x^2 = 3$; (ii) $2(x + 3y + 3) = 2(1 - e^{x})$
2. $y + 1 = 2e^{x/2}$
3. $x^2 + y^2 = cx$
4. $y = \sqrt{(x^2 - a^2)} + a\log\left(\frac{x - \sqrt{(x^2 - a^2)}}{a}\right) + c$
5. $y^2 = cx$
6. $y = ax^2$
7. $y = ax + c$
8. $x = 2y^2$
9. $(1 + x)(y - x) = c$; (ii) $x = a(\cos\theta + \theta\sin\theta)$
10. $x^2 = x^2\sin 2\theta$
11. $x^2y^2 = 2y + 2$
12. $r = a(e^{\theta\cot\alpha})$.

Problems 12.2, page 437

1. $2x^2 + y^2 = c$
2. $x^2 + 2y^2 = c^2$
3. $2y^2 + 2x^2 = c^2$
4. $x^2 + y^2 + 2y - c = 0$
5. This system is self-orthogonal
6. $r = a(1 - \cos\theta)$
7. $r = b(\cos\theta + \sin\theta)$
8. $r = 2a\cos(\theta - \alpha)$
9. $r^2 = a^2\sin 2\theta$
10. $r^2 = a\cos 2\theta = b$
11. $x^2 + y^2 + cx - 1 = 0$
12. $y = cx$.

Problems 12.3, page 450

1. $V = \sqrt{\left[\left(\frac{mg}{k}\right)\tan h\left(\frac{gt}{v/m} + c\right)\right]}$
2. $\frac{1}{k}\log_e 2$
3. $2x/\sqrt{3}$
4. $s^2 = 2gr - \frac{2}{3}v^2$
5. $y = 14(68 - 0.0012286)^2$; $t_1 = 40$ min. 1 sec., $t_2 = 1$ hr. 16 min. 31 sec., $t_3 = 1$ hr. 24 min. 13 sec
6. 17 min. 4 sec.

Problems 12.2, page 485

1. 0.0000001 sec
2. $\frac{10}{s^2+9}(3\sin t - s\cos t + 2e^{-3t})$
3. $\frac{1}{s^2}$
4. $\frac{1}{s^2}$
5. $\frac{1}{s^2}$
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Problems 12.3, page 489

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Problems 12.4, page 492

1. $\frac{1}{s^2}$
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Problems 12.5, page 494

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Problems 12.6, page 496

1. $\frac{1}{s^2}$
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Problems 12.7, page 499

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Problems 13.1, page 504

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Problems 13.2, page 506

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Problems 13.3, page 508

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Problems 13.4, page 510

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Problems 13.5, page 512

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Problems 13.6, page 514

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Problems 13.7, page 516

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Problems 13.8, page 518

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Problems 13.9, page 520

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Problems 13.10, page 522

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Problems 13.11, page 524

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Problems 13.12, page 526

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Problems 13.13, page 528

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Problems 13.14, page 530

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Problems 13.15, page 532

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Problems 13.16, page 534

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Problems 13.17, page 536

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Problems 13.18, page 538

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Problems 13.19, page 540

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8. $y = c_1 e^x + c_2 e^{2x} + \frac{1}{650}(130\cos 3x - 13\sin 3x) + \frac{1}{10}(\sin x + 2\cos x)$

9. $y = c_1 + c_2 x + c_3 e^{-x} - \frac{x^2}{2}e^{-x} + \frac{3}{50}\cos 2x - \frac{4}{25}\sin 2x$

10. $y = c_1 + c_2 e^x + c_3 e^{-x} + \frac{1}{2} - \frac{1}{5}(2\sin 2x + \cos 2x)$

11. $y = c_1 + c_2 e^x + c_3 e^{2x} + \frac{3}{8}\ln e^{2x} - x(x+1)$ 12. $y = c_1 e^x + c_2 e^{-x} + \frac{e^x}{12}(3x^3 - 3x^2 + 3x)$

13. $y = c_1 + c_2 e^x + c_3 e^{-3x} + xe^x - (x^2 + x) - 2\cos x$

14. $y = e^{2x}(c_1 \cos 4x + c_2 \sin 4x) + \frac{1}{17}e^{4x} + \frac{2}{1000}(53\sin x + 11\cos x) + \frac{3}{25} + \frac{3}{125}x$

15. $y = c_1 + c_2 e^x \cos x + c_3 e^x \sin x + e^x - 2\ln e^x + 2x + \frac{1}{200}\sin 4x - \frac{1}{8}\sin 2x$

16. $y = c_1 e^{-3x} + c_2 e^{-2x} - \frac{x^2 e^{-2x}}{20}(\cos 2x + 2\sin 2x)$

17. $y = e^{-x/2}\left[c_1 + x\left(c_2 \cos\frac{\sqrt{3}}{2}x + c_3 \sin\frac{\sqrt{3}}{2}x\right)\right] + e^{x/2}\left[c_4 \cos\sqrt{3}x + c_5 \sin\sqrt{3}x\right]$

18. $y = e^{-3x}(c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x) + \frac{e^x}{12}(4\sin x + 3\cos x)$

19. $y = c_1 e^{x/2} + c_2 e^{-3x} - \frac{e^x}{2}(\sin x + 2\cos x) + \frac{e^{2x}}{52}\left(x - \frac{9}{13}\right)$

20. $y = c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x + \frac{e^{2x}}{13}\left(x^3 - \frac{12}{13}x + \frac{58}{169}\right) + \frac{e^x}{17}(4\sin 2x - \cos 2x)$

21. $y = c_1 e^{2x} + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - (3x)\cos x + \sin x$

22. $y = c_1 + c_2 + c_3 e^{-x} + \frac{2x^3}{125}x^3 - \frac{3x}{25} + \frac{13}{625}) + \frac{1}{100}(3\sin 2x + 4\cos 2x)$

23. $y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{6}\left(x\sin 3x - \frac{2}{3}\cos 3x\right)$

24. $y = c_1 + c_2 e^x + \frac{3}{2}\cos x + \frac{1}{2}(x-1)\sin x$

25. $y = c_1 e^x + c_2 e^{-x} - \frac{1}{2}(x\sin x + \cos x) + \ln(e^x + 1)(\cos x - 2x + 3)$

26. $y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x} + e^{-3x} - e^{-x}$

27. $y = c_1 \cos\ln x + c_2 \sin\ln x - \frac{1}{e^x}\cos\ln x \ln(\sec\ln x + \tan\ln x)$

Additional Exercises for Chapter 4

1. $y = c_1 - \ln|\cos x| + c_2 + (\ln|\sec x + \tan x| - \sin x)\cos x + \sin x$
2. $y = c_1 \cos x + c_2 \sin x + \cos x \ln|\sec x + \tan x| + x\sin x$
3. $y = c_1 \cos x + c_2 \sin x + \cos x \ln|\sec x + \tan x|$

4. $y = c_1 \cos x + c_2 \sin x + \frac{3}{2}x\sin x - \frac{x^2}{2}\cos x$ 5. $y = c_1 + c_2 e^x + xe^x \ln x$

6. $y = 2e^x + e^{2x}(\ln|1 + e^x| + c_1) + c_2 - 1 - e^x(1 + e^x) + (c_3 - e^x)e^{2x}$

7. $y = e^x(c_1 \cos x + c_2 \sin x) - e^x \cos x \ln(\sec x + \tan x)$

8. $y = c_1 + c_2 e^{2x} - \frac{1}{2}e^x \sin x$ 9. $y = c_1 \cos x + c_2 \sin x + \sin x \ln|1 + \sin x| - x\cos x - 1$

10. $y = c_1 e^x + c_2 e^{-x} + \frac{1}{2}(2e^x + 2x + 3.5 - 2e^{x-1})$ 11. $y = c_1 \cos x + c_2 \sin x - x\sin x$

12. $y = c_1 e^{2x} + c_2 e^{3x} + \sin^2 x + \frac{1}{10}\sin x + \cos x$ 13. $y = c_1 e^{x} + c_2 e^{-2x} - \frac{3}{4}(2x+1) - \frac{1}{20}(\sin x + 2\cos x)$

14. $y = e^{-x}(c_1 \cos\sqrt{2}x + c_2 \sin\sqrt{2}x) + \frac{1}{20}(8x^3 + 16x^2 + 6x - 6) + \frac{1}{5}(\cos x - 4\sin x)$

15. $y = c_1 + c_2 e^{2x} - \frac{1}{2}e^x \sin x$.

Problema 134, pag. 456

1. $y = c_1 e^{2x} + c_2 e^{-x} - e^x \log x$ 2. $y = c_1 x^2 + c_2 x^{-1} + \frac{x^2}{6}\log x$

3. $y = (c_1 + c_2 \log x)x^2 + \frac{1}{4} + 2x + \frac{1}{2}x^2(\log x)^2$

4. $y = c_1 e^{2x} + c_2 e^{-x} + \frac{1}{2}(2x^2 - 2x)\log x$ 5. $y = \frac{1}{8}(4x^2 - e^{2x})$

6. $c_1 e^{-x} + c_2 e^{-2x} + \frac{1}{2}(\log x - \frac{3}{4})$

7. $y = c_1 e^{-x} + \sqrt{2}e^{-x}(c_2 \cos(\sqrt{2}\log x) + c_3 \sin(\sqrt{2}\log x)) + c_4 \sin(\sqrt{2}\log x) - \frac{1}{2}x + \log x$

8. $y = c_1 e^{-x} + e^{\frac{x}{2}}(c_2 \cos(\frac{\sqrt{3}}{2}\log x) + c_3 \sin(\frac{\sqrt{3}}{2}\log x)) + 8\cos(\log x) - \sin(\log x)$

9. $y = c_1 e^{-x} + (c_2 \cos(\log x) + c_3 \sin(\log x))x + 2x + 10\log x$

10. $y = \frac{1}{x}(c_1 + c_2 \log x) + \frac{1}{x}\log\frac{\sqrt{x}}{1-x}$ 11. $y = x^{-1}(c_1 + c_2 \log x) + \frac{x}{2}(\log x - \frac{3}{2})$

12. $y = c_1 x^2 + c_2 x^{-4} + \frac{x^2}{294}\log x(7\log x - 2)$

13. $y = c_1(2x+3)^2 + c_2(2x+3)^{\frac{1}{2}} - \frac{1}{25}(2x+3)\log(2x+3)$, con $a, b = \frac{1 \pm \sqrt{37}}{4}$

14. $y = c_1(x-1) + c_2(x-1)^2 + c_3(x-1)^{-1} + \log(x-1) + 1$

15. $x = c_1 \cos(\log(1+t)) + c_2 \sin(\log(1+t)) - \frac{1}{5}\sin(2\log(1+t))$

16. $y = c_1(3x+2)^{\frac{1}{3}} + c_2(3x+2)^{-1} + \frac{1}{21}\left[\frac{1}{10}(3x+2)^{\frac{7}{3}} + (3x+2)^{\frac{4}{3}} + \frac{1}{4}(3x+2) - 1\right]$

Problema 135, pag. 459

1. $x = (c_1 + c_2 t)e^{2t}$, $y = (c_1 - c_2)e^{2t}$

2. $x = e^t + e^{-t}$, $y = e^t - e^{-t} + \sin t$ 3. $x = c_1 e^t + c_2 e^{-2t} + \frac{1}{3}e^{2t}$, $y = c_1 e^t - c_2 e^{-2t} + \frac{1}{3}e^{2t}$

4. $x = e^{2t}(c_2 \cos t + c_3 \sin t)$, $y = e^{2t}((c_2 - c_3)\cos t + (c_2 + c_3)\sin t)$

5. $x = \frac{1}{2}e^t + \frac{3}{2}e^{-t} - c_1 \sin 2t + c_2 \cos 2t$, $y = \frac{1}{2}e^t + \frac{1}{2}e^{-t} + c_1 \cos 2t + c_2 \sin 2t$

6. $x = c_1 e^t + c_2 e^{-2t} + \frac{2}{3}e^{3t} - \frac{2}{25}t - \frac{13}{225}$, $y = c_1 e^t - c_2 e^{-2t} - \frac{4}{3}e^{3t} - \frac{2}{5}t - \frac{12}{25}$

7. $x = -t - \frac{3}{2}$, $y = \frac{1}{2}t^2 + \frac{3}{2}t + 1$ 8. $y = c_1 e^t + c_2 e^{2t} + 2e^{3t}$, $z = 2c_1 e^t + 2c_2 e^{2t} + 3e^{3t}$

9. $x = c_1 e^{-t} + c_2 e^{3t} - \frac{1}{5}(\cos t - 2\sin t)$, $y = 2c_1 e^{-t} - 2c_2 e^{3t} - \frac{1}{5}(\sin t + 2\cos t)$

10. $x = \frac{3}{2}(t + \frac{1}{2})$, $y = \frac{1}{2}(-t + \frac{1}{4})$

11. $x = c_1 e^{-t} + c_2 e^{2t} - \frac{1}{6}(\cos t - 2\sin t)$, $y = 2c_1 e^{-t} - 2c_2 e^{2t} + \frac{2}{5}\cos t - \frac{1}{5}\sin t$

12. $x = c_1 + c_2(t+1)e^{-t} + c_3 + c_4(t+1)e^t$, $y = -\frac{1}{2}c_1 + c_2(2-t)e^{-t} + c_3(t-1)e^t + \frac{1}{2}c_4(1-t) - c_2 e^{-t}$

12. $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{4} x \cos x + \frac{1}{4} x \sin x$

$y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + \frac{1}{4}(2 + x)\sin x - x \cos x$

14. $x = \frac{1}{3}\left(1 - \cos\frac{3}{2}t\right), y = \frac{1}{2}t - \frac{1}{3}\sin\frac{3}{2}t$

Problems 10.3, page 600

1. $y = c_1 e^{x^2} + c_2 e^{-x^2} + c_3 \cos x^2 + c_4 \sin x^2$

2. $\frac{1}{24}(3\sin 2x + 4\cos 2x)$ 3. 1/8 4. $e^x(x-1)$

5. 20 6. $y = c_1 + c_2 x + c_3 x^2 + c_4 x^3 + e^{4x}$ 7. (a)

8. $y = e^x(c_1 + c_2 \cos 2x + c_3 x + c_4 x)\sin 2x$ 9. $y = x\cos x + 2\sin x$

10. 2(a) 11. $y = c_1 + c_2 x + (c_3 \cos x + c_4 x)\sin x$

12. $\frac{1}{10}\cos 2x$ 13. $y = x\log x + 4$

14. $y = c_1 + c_2 x + c_3 x^{-2} + c_4 x^{-2}\ln x$ 15. $\frac{1}{6}x^3 e^x$

16. $\frac{1}{2}e^{2x}$ 17. $x\sin 2x$ 18. $\frac{1}{2}x^2 e^x$

19. $y = c_1 + c_2 x + c_3 x^{-2} + c_4$ 20. (c) 21. (a)

22. $x = e^{-t}$ 23. $\frac{d^4y}{dx^4} + 2y = 2x^4$ 24. (c)

25. (a) 26. (b)

27. $y = c_1 x + c_2 \log x + c_3$ 28. $x^3 y''' + xy' - y = 0$ 29. $\frac{1}{2}\log 2$

30. e^x 31. (c)

32. $y = c_1 e^{x^2} + c_2 e^{x(1+\sqrt{2})} + c_3 e^{x(1-\sqrt{2})}$ 34. $\frac{d^2Q}{dt^2} + 2$

34. $\frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$ 35. False 36. False.

Problems 10.4, page 608

1. 28 sec 4. $x = \frac{mv_0}{c\sqrt{1-\lambda^2}}\sin\left(\sqrt{1-\lambda^2}\,ct\right)$

6. It must be shortened by 1/6000 of its length

7. It must be increased by 0.0004 ft/sec^2 8. 4321.0274

9. $d^2 + 4p_1 0 = c_1 e^{\frac{(-1+\sqrt{5})}{2}x} + c_2 e^{\frac{-(1+\sqrt{5})}{2}x}$ where $\lambda = \sqrt{(k^2 - 4)/2}$

$d^2 + 4p_1 0 = (c_1 + c_2 x)e^{-x/2}$

$d^2 < 4p_1 0 = c_1 e^{-x/2}\cos\left(\frac{\sqrt{4p - d^2}}{2}x + c_2\right)$

10. $x = \frac{F_0}{2m^2}(\sin\omega t - \omega\cos\omega t)$

Problems 1-5 (page 513)

2. $1.5\ \text{kg}$; $\sqrt{80}\ \text{rad/s}$; $4\sqrt{5}\ \text{N/m}$
3. $x = e^{-3t}\left(\cos\sqrt{3200}t + 2\sqrt{200}\sin\sqrt{200}t\right)$
4. $\omega = \sqrt{7}$
5. $0.45\ \text{sec}$; $1.39\ \text{sec}$
6. $x = \frac{10}{21}e^{t} - \frac{37}{37}e^{-6t} - \frac{\sqrt{2}}{2}\sin\left(2t + \frac{\pi}{4}\right)$; $\frac{\sqrt{2}}{2}$; $\frac{2\pi}{2}$; $\frac{5\pi}{6}$ sec; $\frac{6}{\pi}$ cycles/sec
7. $0.5\left(2\sin 4t - \cos 4t\right)$
8. (a) $x = Ae^{-3t}\cos\left(\sqrt{(\omega_0^2 - A^2)}\,t + \phi\right)$; $\sqrt{\omega_0^2 - A^2}$; $e^{-3t}\sqrt{A^2 + B^2}$; $\omega = \sqrt{11}$ rad/s

(b) $x = Ae^{-3t}\cos\left(\sqrt{(\omega_0^2 - A^2)}\,t + \phi\right)$; $\sqrt{\omega_0^2 - A^2}$; $-3e^{-3t}\sqrt{A^2 + B^2}$; $\omega = \sqrt{11}$ rad/s

Problems 6-10 (page 517)

2. $I = I_0\sin\left(\frac{\pi}{RC}t\right)$
3. $I = 20\sqrt{2}e^{-0.5t}\sin(0.5\sqrt{2}t)$; when $\lambda = \sqrt{\left(\frac{4L - CR^2}{C}\right)}$
4. $R^2 > 4L/C$ for over damping; $R^2 = 4L/C$ for critical damping; $R^2 < 4L/C$ for under damping; critical resistance $= 2\sqrt{(L/C)}$
5. $q = e^{-0.05t}(0.002\cos 1000t + 0.0005\sin 1000t)$
6. (a) $I = Ae^{-\alpha t}\cos(\beta t + \phi)$; (b) $I = Ae^{-\alpha t}\sin(2\pi\nu t + \phi + \frac{\pi}{2})$; $\cos\phi\sin(\beta t + \phi)$

where $\alpha = -\frac{R}{2L}$, $\beta = \sqrt{\left[\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}\right]}$ and $\phi = \tan^{-1}(1) = 45°$ or $\frac{\pi}{4}$ rad.

Problems 1-5 (page 520)

4. $y = \frac{wL^2}{24EI}\cosh\frac{2x}{L}\cos\left(\omega t - \frac{\pi L^2}{2}\right) - \frac{wL}{2\omega^2}\cos\frac{\pi t}{2} + \frac{w}{2P}(x^2 - Lx)$
5. $y = \frac{F}{\rho\omega}\left(\omega t\sin\omega t - \frac{1}{2}\sin\omega t\cos\omega t + t\right)$
6. $\omega^2 = 6EI/ml^3$
7. $\frac{W}{2g}\left(\omega\sin\omega\frac{\pi t}{2} - \omega\cos\omega t\right)$

Problems 1-5 (page 523)

1. $\frac{2x\sin\alpha}{g}$; $\frac{x^2\sin 2\alpha}{g}$
2. (a) $\frac{2v^2\sin\alpha\cos\alpha - g\cos^2\alpha}{g\sin^2\beta}$; (b) $\frac{v^2}{g(1+\sin\beta)}$
3. $\omega = \sqrt{6EI/ml^3}$
4. $4x^2 + 9y^2 = 4$
5. $x_1 = \frac{m}{p^2 + m^2}\sin pt$; $x_2 = \frac{m}{p^2 + m^2}\cos pt$.
6. $x_1 = \frac{F}{K}\left(\frac{2}{3} - \frac{1}{2}e^{-3t} - \frac{1}{6}e^{-6t}\right)$; $x_2 = \frac{F}{K}\left(\frac{1}{3} - \frac{1}{3}e^{-3t} + \frac{1}{3}e^{-6t}\right)$
7. $x = a(\omega t - \sin\omega t)$; $y = a(1 - \cos\omega t)$
8. $x = \frac{K}{10\omega}(1 - \cos\omega t)$; $y = \frac{K}{10\omega}(\omega t - \sin\omega t)$, where $\omega = \omega_0/\omega_n$

Problems 14.6, page 500

1. (b)
2. (c)
3. (c)
4. (c)
5. (b)
6. (c)
7. (b)
8. 60 cm
9. $30\sqrt{6}$ cm
10. 0.0014 cm
11. maximum
12. $8V\dfrac{d^2y}{dx^2} - 8y = \dfrac{x}{2}(x^2 - 10)$
13. $y = 0$ and $\dfrac{dy}{dx} = 0$

Problems 15.1, page 521

1. $y = -x^2\sin x - 4x\cos x + c_1x + c_2$
2. $y = \dfrac{x^4}{24} + \dfrac{x^3}{6}\log x - \dfrac{11}{36}x^3 + c_1x^2 + c_2x + c_3$

Problems 15.2, page 535

1. $D(x^{n+1} - 1) = n$
2. $y(y^2 - 6y + 4)\sin^{-1}\frac{1}{2}y - (y - 3)\sqrt{} = 3x$
3. $r = \dfrac{(b^2 - a^2\cos^2\theta)}{ab}\sin\theta\left[ab + \sin\theta^{-1}\dfrac{ab}{\sqrt{(b^2 - a^2\cos^2\theta)}}\right]$
4. $r = \dfrac{ab\sin\theta}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}\left[\cos^{-1}\sqrt{\dfrac{b}{a}} - \sqrt{\left[\dfrac{ab - a^2}{b}\right]}\right]$

Problems 15.3, page 554

1. $y = c_1 - e^{2x} - c_2e^{3x}$
2. $y = c_1e^x + (c_2 - 3)\log(x - c_3) + c_4$
3. $18y^2p = 12p^3 + 9x^2p^2 + c_1p + c_2$
4. $x^2 + y^2 = c^2$
5. $k = \dfrac{2\pi}{pn}\log\left[1 + \dfrac{\text{natural}}{cn}\right]$
6. $k = \dfrac{3}{c_1 - c_2}\left[\log y - \log x - \dfrac{1}{2}\log(x + c_1)(x + c_2)\right]$

Problems 15.4, page 558

1. $y = 2x - 2\log(1 - e^{x/2}) + c_1$
2. $y^2 = x^2 + c_1x + c_2$
3. $\log y = c_1e^x + c_2e^{-x}$
4. $(\log c)^2 - 2\log c + c_1^2 = 1$
5. $(x - a)^2 + y^2 = c^2$, circles whose centres are on the x-axis

Problems 15.5, page 567

1. $y = e^x(c_1x + c_2)$
2. $y = e^x(c_1 - x + c_2x^2 + c_3x^3)$
3. $y = e^x(c_1\log x + c_2x)$
4. $c_1y + 2x + 15 = c_2\sin x$
5. $y = \left[c_1 - \frac{1}{2}\cos x - \frac{1}{2}c_2x^{-1/2}\cos x - 2c_2\sin x\right]e^x$.

Problems 15.6, Page 570

1. $y = c_1\cos(\sin x) + c_2\sin(\sin x)$
2. $y = c_1\cos(15x) + c_2\sin(15x)$
3. $y = c_1e^x + c_2e^{-1} - 1$ where $k = \cos\alpha$
4. $y = c_1\cos(12\tan^{-1}x) + c_2\sin(12\tan^{-1}x)$
5. $y = c_1e^{x\cos\alpha} + c_2e^{-x\cos\alpha} - \frac{1}{2}\sin^2\alpha$

Problema 1317, página 660

1. $xy = c$
2. $x^2 + y^2 + z^2 = cx$
3. $xy^2 = cz^3$
4. $x^2 + y^2 - 2xz = cz$
5. $yz + zx = cx(x + y + z)$
6. $x - y - z + \log(xyz) = c$

Problema 1318, página 661

1. $x^2 - y^2 = c_1,\ x^2 - z^2 = c_2$
2. $3x + xy + xz = c_1,\ x^2 + y^2 + z^2 = c_2$
3. $\frac{x-y}{y-z} = c_1,\ \frac{x-z}{y-z} = c_2$
4. $x^2 - y - 2xy = c_1,\ x^2 - y^2 - z^2 = c_2$
5. $xyz = c_1,\ x^2 + y^2 + z^2 = c_2$
6. $y = c_1 x,\ x^2 + y^2 + z^2 = c_2 x$

Problema 1319, página 664

1. $y = c_0\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)$

2. $y = c_0\left(1 - \frac{x^4}{4 \cdot 3} + \frac{x^8}{8 \cdot 7 \cdot 4 \cdot 3} - \frac{x^{12}}{12 \cdot 11 \cdot 8 \cdot 7 \cdot 4 \cdot 3} + \cdots\right)$
$+ c_1\left(x - \frac{x^5}{5 \cdot 4} + \frac{x^9}{9 \cdot 8 \cdot 5 \cdot 4} - \frac{x^{13}}{13 \cdot 12 \cdot 9 \cdot 8 \cdot 5 \cdot 4} + \cdots\right)$

3. $y = c_0\left(1 - \frac{x^3}{3} + \frac{x^6}{3 \cdot 6} - \frac{x^9}{3 \cdot 6 \cdot 9} + \cdots\right) + c_1\left(x - \frac{x^4}{4} + \frac{x^7}{4 \cdot 7} - \frac{x^{10}}{4 \cdot 7 \cdot 10} + \cdots\right)$

4. $y = c + 2x - 4x^2 - \frac{8}{3}x^3 - \frac{x^4}{2} - \frac{x^5}{5} - \cdots$

5. $y = c_0\left(1 + \frac{x^2}{2} - \frac{x^4}{8} + \frac{x^6}{24} - \frac{5x^8}{320} + \cdots\right) + c_1 x$

6. $y = c_0(1 - x^2 + \frac{1}{3}x^3 - \frac{1}{6}x^4 + \cdots) + c_1 x$

Problema 1320, página 666

1. $y = c_1 \operatorname{sen} \sqrt{3}x + c_2 \cos \sqrt{3}x$

2. $y = c_0\left(1 - x^2 + \frac{x^4}{4} - \cdots\right) + c_1\left(x - \frac{x^3}{3} + \frac{2x^5}{15} - \cdots\right)$

3. $y = (c_1 + c_2 \log x)\left(1 + x + \frac{3}{(3!)^2}x^3 + \frac{5}{(5!)^2}x^5 + \cdots\right) - 2c_2\left(x + \frac{1}{(3!)^2}(1+4)x^3 + \frac{3}{(5!)^2}(1+\frac{1}{2}+\frac{1}{3})x^5 + \cdots\right)$

4. $y = c_1(1 + x + x^2/2! + x^3/3! + \cdots) + c_2 x^{1/2}(1 + \frac{1}{2}x + x^2/2! \cdot \frac{1}{2^2} + \cdots)$

5. $y = c_0\left(1 - 2x + \frac{2}{3}x^2 - \frac{4}{15}x^3 + \cdots\right) + c_1\left(y_1 \log x + x\left(3x - \frac{13}{4}x^2 + \cdots\right)\right)$

6. $y = c_0 x^{1/2}\left(1 + \frac{x}{5} + \frac{x^2}{70} + \cdots\right) + c_1 x^{-1/2}\left(1 - x - \frac{x^2}{2} + \cdots\right)$

7. $y = c_1 x^{-\frac{1}{2}}\left(1 + \frac{x}{2} + \frac{x^2}{48} + \cdots\right) + c_2 x^{1/2}\left(1 + \frac{x}{10} + \frac{x^2}{600} + \cdots\right)$
$+ c_3 x^{1/2}\log x\left(1 + x^2/10 \cdot 2 + x^4/10 \cdot 8 \cdot 4 \cdot 7 + x^6/10 \cdot 8 \cdot 6 \cdot 4 \cdot 7 \cdot 11 + \cdots\right)$

8. $y = c_0 x^{1/2}(1 - x) + c_1 x^{1/2}\left(1 - 2x + \frac{2x^2}{1 \cdot 3} - \frac{2x^3}{3 \cdot 5} + \frac{2x^4}{5 \cdot 7} + \cdots\right)$

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8. $y = c_0[1 - \frac{1}{2}x + \frac{1}{3}x^2 + \frac{1}{6}x^3 - \frac{1}{12}x^4 + \cdots] + c_1[x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots]$

10. $y = c_0\left[1 - 2x^2 + \frac{2}{3}x^4 - \cdots\right] + c_1 x^{1/2}\left[1 - \frac{x^2}{6} + \frac{1 \cdot 3}{6 \cdot 10}x^4 - \frac{1 \cdot 3 \cdot 5}{6 \cdot 10 \cdot 14}x^6 + \cdots\right]$

Problems 9.3, page 337

1. 0.254, 0.44

Problems 9.4, page 343

1. $y = c_1 J_{1/2}(x) + c_2 J_{-1/2}(x)$

2. $y = c_1 J_{3/2}(x) + c_2 J_{-3/2}(x)$

3. $y = x^{1/2}[c_1 J_{1/3}(x) + c_2 J_{-1/3}(x)]$ where $x = \frac{1}{2}(1-x)$

4. $y = x[c_1 J_1(x) + c_2 Y_1(x)]$

5. $y = c_1 x J_1(x) + c_2 x Y_1(x)$

7. $y = c_1\sqrt{x} J_{1/4}(x) + c_2\sqrt{x} J_{-1/4}(x)$

11. $y^2 = \sum_{n=0}^{\infty} \frac{2}{x_n^2} \cdot \frac{1}{J_1^2(x_n)}[J_0(x_n x_1)J_0(x_n x_2) + J_1(x_n x_1)J_1(x_n x_2)]$

Problems 10.1, page 370

2. (a) $2P_2 = 4P_1$

(c) $\frac{5}{8}P_3 - 4P_2 + \frac{17}{2}P_1 + 4$

(e) $\frac{2}{5}P_2 + \frac{2}{3}P_1 - \frac{2}{5}P_1 - \frac{2}{3}P_0$

(f) $\frac{8}{63}P_4 - \frac{4}{5}P_3 - \frac{2}{21}P_2 + \frac{34}{5}P_1 - \frac{224}{105}P_0$

3. (a) $f(x) = -\frac{7}{3}P_3(x) - \frac{2}{3}P_2(x) + \frac{5}{3}P_1(x) + \frac{2}{3}P_0(x)$

(c) $f(x) = -\frac{32}{15}P_3(x) - \frac{4}{5}P_2(x) - \frac{48}{21}P_1(x) + \frac{2}{3}P_0(x) + \frac{8}{35}P_2(x)$

Problems 10.2, page 382

1. $x^2 = \frac{1}{4}(2T_2 + T_0)$

Problems 10.3, page 395

1. $y_n(x) = \sin nx, n = 1, 2, \ldots$

2. $y_n(x) = \cos nx, n = 0, 1, 2, \ldots$

3. $y_n(x) = \sin \frac{n\pi x}{2}, n = 0, 1, 2, \ldots$

4. $1, \sin x, \cos x, \sin 2x, \cos 2x, \ldots$

5. $y_n(x) = \sin\left[(2n+1)\frac{\pi}{2}\ln|x|\right], n = 0, 1, 2, \ldots$

8. $[x(x^2y')' + xe^{-x}y = 0, y_0(x) = e^{-x}$

9. $[x^{-1}y']' + 2xe^{-x^2}y = 0, y_0(x) = e^{-x^2}$

Problems 10.4, page 410

1. $\frac{1}{2}(3 - 3P_1 + 5P_2)$

2. $\sqrt{2}(T_0 + 2T_1)\cos x$

3. $\frac{2}{(2n+1)}$

4. $\sin\pi x$

5. $\cos\pi x$

6. $J_0(x) = \frac{2}{\pi}[J_{-1/2}(x) + J_{1/2}(x)]$

7. $\int_0^1 x J_1(\alpha_1 x) J_1(\alpha_2 x)\,dx = 0$

8. $x\frac{d^2y}{dx^2} + \frac{dy}{dx} + \alpha y = 0$

9. $\sqrt{2}(1 + \cos x)\sin x$

18. $x^2 y_{n+1}(x)$ 19. $\frac{1}{2}(3x^2-1)$ 20. zero

21. True 22. $P_n(x)=\frac{1}{n!2^n}\frac{d^n}{dx^n}(x^2-1)^n$ 23. $\alpha+\beta$

24. $2P_1=4P_1$ 25. $(1-2xt+t^2)^{-1/2}=\sum_{n=0}^{\infty}t^n P_n(x)$

26. $-J_1(x)$ 27. Bessel 28. True 29. True 30. True 31. True
32. False 33. (b) 34. (c) 35. (a) 36. (d) 37. (b)
38. (c) 39. (b) 40. (c) 41. (c) 42. (c) 43. (c)
44. 0, 1

Problems 17.1, page 529

1. $x=pz+qy+p^2+q^2$ 2. $z(p^2+q^2+1)=c^2$ 3. $pz+qz=\tan(x+y)$
4. $p+q=px+qy$ 5. $z^2(p^2+q^2+1)=1$ 6. $py-qx=0$
7. $py+qx=0$ 8. $qz-py=x+y$ 9. $z(p+q)=px+qy-z$
10. $xyz=\frac{1}{3}(px+qy-2z)$ 11. $\frac{\partial^2 z}{\partial x^2}=\frac{\partial z}{\partial y}$ 12. $xy-xp-yq-z(q+1)=xz-yz$
13. $z\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial z}{\partial x}\cdot\frac{\partial z}{\partial y}$ 14. $p+q=pq$ 15. $px^2+qy=2z^2$
16. $\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial x\partial y}-6\frac{\partial^2 z}{\partial y^2}=0$ 17. $\frac{\partial^2 z}{\partial x^2}=\frac{a^2}{c^2}\frac{\partial}{\partial x}\left(c^2\frac{\partial z}{\partial x}\right)$ 18. $\frac{\partial^2 z}{\partial x^2}-2\frac{\partial^2 z}{\partial x\partial y}+\frac{\partial^2 z}{\partial y^2}=0$
19. $p(x-2z)+q(2z-y)=y-x$ 20. $(y-z)p+(z-x)q=x-y$

Problems 17.2, page 531

1. $z=\frac{x^2}{2}\log(x+ay)+\phi(x+ay)+\psi(x-ay)$ 2. $z=\frac{1}{2}xy+x\phi(y+4y)+\psi(y)$
3. $z=-e^x\sin x+\phi(x+y)+\psi(x)$ 4. $z=f(x)+x\phi(x)+\psi(x)-\frac{1}{2}x\cos(2x+2y)$
5. $z=e^x\cos y+e^x\sin x$ 6. $z=x\sin x+x^2\cos x$

Problems 17.3, page 536

1. $z=x^2f(x+y)$ 2. $\sqrt{x}-\sqrt{y}=f(\sqrt{x}-\sqrt{z})$ 3. $x^2+y^2+z^2=f(x+y+z)$
4. $\log(x-y)+\sin(x+y)=\phi(x^2+\tan\left(\frac{x+y}{2}+\frac{z}{4}\right)\Big]$ 5. $x^2-y^2+z^2=f(y^2-z^2)$
6. $\phi\left(\frac{\sin x}{\sin y},\frac{\sin y}{\sin z}\right)=0$ 7. $x\log(x+y)-z=f(x+y)$ 8. $x^2+y^2+2z=2xy\tan(x+y)$
9. $x^2+y^2-z^2=f(x+y+z)$ 10. $x-y-z=f(yz)$ 11. $\phi(x^2+y^2+z^2,xyz)=0$
12. $x^2+y^2=f(x^2-yz)$ 13. $\phi(xz,x^2+y^2+z^2)=0$ 14. $x^2+y^2+z^2=f(y^2-2yz-z^2)$
15. $f\left(\frac{x}{z},\frac{z}{x}-\frac{x}{z}+x^2\right)=0$

Exercise 2.7.3 | Level 3 (Challenging)

1. $\int \cos x \cdot \cos(\sin x) \cdot \cos(\sin(\sin x)) \, dx$
2. $\int \sec^2 x \cdot \tan^2 x \, dx$
3. $\int \sin(\ln x) \cdot \cos(\ln x) \cdot \sin(e^{\ln x}) \, dx$
4. $\int \frac{1}{x \ln x \ln(\ln x)} \, dx$
5. $\int \frac{\ln(\ln x)}{x \ln x} \, dx$
6. $\int \frac{x^2 + 1}{x^4 + 1} \, dx$
7. $\int \frac{1}{x^{2n} + 1} \, dx$
8. $\int \frac{x^{n-1}}{x^{2n} + 1} \, dx$
9. $\int \frac{x^{n-1}}{x^{2n} + a^2} \, dx = \frac{1}{a^2} \cdot \frac{a}{n} \tan^{-1}\left(\frac{x^n}{a}\right) + \frac{1}{a^2} \cdot \frac{a}{n} \cdot \frac{x^n}{a} - \frac{1}{a^2} \cdot \frac{a}{n} \cdot \frac{x^n}{a} + \frac{1}{a^2} \cdot \frac{a}{n} \tan^{-1}\left(\frac{x^n}{a}\right) + C$
10. $\int \frac{1}{\sin x \cos x} \, dx$
11. $\int \frac{1}{\sin x} \, dx$
12. $\int \frac{1}{\sin x \cos x \cdot \ln(\tan x)} \, dx$
13. $\int \frac{1}{\sin x \cos x \cdot \ln(\tan x) \cdot \ln(\ln(\tan x))} \, dx$
14. $\int \frac{1}{\sin x \cos x \cdot \ln(\tan x) \cdot \ln(\ln(\tan x)) \cdot \ln(\ln(\ln(\tan x)))} \, dx$

Exercise 2.7.4 | Level 4 (Olympiad)

1. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x}} \, dx$
2. $\int \frac{x^{n-1}}{(x^n + a)(x^n + b)} \, dx$
3. $\int \left| \frac{1}{x} \right| \left| \frac{1}{x^2} - \frac{1}{x^3} \right| \left| \frac{1}{x^4} - \frac{1}{x^5} \right| \, dx$
4. $\int \frac{\ln(\ln x)}{x} \, dx$
5. $\int \frac{\ln x \cdot \ln(\ln x)}{x} \, dx$
6. $\int \frac{\ln(\ln x)}{x \ln x} \, dx$
7. $\int \frac{1}{x^{\ln x}} \, dx$

Exercise 2.7.5 | Level 5 (Research)

1. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x}} \, dx$
2. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x}} \, dx$
3. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x}} \, dx$
4. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x}} \, dx$
5. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x}} \, dx$
6. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x}} \, dx$
7. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x}} \, dx$
8. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x}} \, dx$
9. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x}} \, dx$
10. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x}} \, dx$
11. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x} + \sqrt[14]{x}} \, dx$
12. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x} + \sqrt[14]{x} + \sqrt[15]{x}} \, dx$
13. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x} + \sqrt[14]{x} + \sqrt[15]{x} + \sqrt[16]{x}} \, dx$
14. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x} + \sqrt[14]{x} + \sqrt[15]{x} + \sqrt[16]{x} + \sqrt[17]{x}} \, dx$
15. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x} + \sqrt[14]{x} + \sqrt[15]{x} + \sqrt[16]{x} + \sqrt[17]{x} + \sqrt[18]{x}} \, dx$
16. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x} + \sqrt[14]{x} + \sqrt[15]{x} + \sqrt[16]{x} + \sqrt[17]{x} + \sqrt[18]{x} + \sqrt[19]{x}} \, dx$
17. $\int \frac{1}{x + \sqrt{x} + \sqrt[3]{x} + \sqrt[4]{x} + \sqrt[5]{x} + \sqrt[6]{x} + \sqrt[7]{x} + \sqrt[8]{x} + \sqrt[9]{x} + \sqrt[10]{x} + \sqrt[11]{x} + \sqrt[12]{x} + \sqrt[13]{x} + \sqrt[14]{x} + \sqrt[15]{x} + \sqrt[16]{x} + \sqrt[17]{x} + \sqrt[18]{x} + \sqrt[19]{x} + \sqrt[20]{x}} \, dx$

Problems 12.7 (page 582)

1. $z = e^{x}\phi_1(y) + e^{2x}\phi_2(y) + x^{3} - \frac{3x^{4}}{2}$
2. $z = x^{2}\phi_1(y) + x\phi_2(y) + e^{x}\phi_3(y) - x^{2} + \frac{1}{2}x^{3}y$
3. $z = x\phi_1(y) - x^{2} + e^{2x}\phi_2(y - 2x) + x^{2}y + 3y + 1$
4. $z = x\phi_1(y) + e^{-2x}\phi_2(y + x) + \frac{1}{2}x^{3} - x^{2} + xy^{2} + 8x$
5. $z = e^{x}\phi_1(y) + e^{-x}\phi_2(y + x) + \frac{1}{2}\cos(2x) + 2x$
6. $z = f_1(y) + e^{2x}f_2(2x - y) + \frac{x}{80}\left[16\cos(2x - 2y) + 5\sin(2x - 2y)\right]$

Problems 12.8 (page 585)

1. $z = \phi_1(x) + \phi_2(x + y) + x^{2}$
2. $z = \phi_1(y) + x\phi_2(x) + \phi_3(y) - x\sin x$
3. $z = \phi_1(xy^{2}) + \phi_2(x^{2}y)$
4. $z = \phi_1(y\sin x) + \phi_2(x^{2} + y^{2}) + xy$
5. $y = \phi_1(x) + e^{x}\phi_2(x)$
6. $y = \phi_1(x + y) + x\phi_2(x + y) + x^{2}$

Problems 12.9 (page 588)

1. order two & degree one
2. $z = f_1(y) + 2x\phi + xf_2(y) + 2x^{2}$
3. $z = -x^{2}\sin(y) + \phi_1(x) + \phi_2(y)$
4. $x^{2} + y^{2} + z^{2} = f(x + y + z)$
5. $-\frac{1}{2}\sin(x + y)$
6. $xp + yq = z$
7. $z = ax + by - \sqrt{a^{2} + b^{2}}$
8. $\sqrt{x} - \sqrt{y} = f(\sqrt{x} - \sqrt{z})$
9. $x\log(z) = y\log z + f(x + y)$
10. First
11. $z = 2x + y\log x + f(xy)$
12. $\partial z/\partial x = 6x^{2}y$
13. $z = f_1(x) - f_2(y + x) + f_3(y + 2x)$
14. $6^{2}y = y^{3}$
15. $u = \int f(y)\sin x + \phi(x)$
16. $z = 1$
17. $u = \frac{1}{6}x^{2}y + x^{2}\phi(x) + \psi(x)$
18. $f(y - x) + f(y + 2x)$
19. (c)
20. (a), (c)
21. (c), (d)
22. (c), (d)
23. (a), (c)
24. (c)
25. (c)
26. (a), (d)
27. False
28. False
29. True
30. True
31. False.

Problems 13.1 (page 601)

1. $z = e^{4x^{2}} + e^{-4x^{2}}$
2. $u = e^{x}\phi(y) + \psi(y)$
3. $u = \phi e^{-(x + y)}$
4. $u = 2e^{x} + \ldots + \phi e^{-2x}$
5. $u = 2e^{-2x} + \phi_1(y)e^{-2x} + \phi_2(y)e^{-3x}$
6. $u = \frac{1}{2\sqrt{2}}\sin(\sqrt{2}x) + e^{-2x}\sin x$

Problems 13.2 (page 610)

1. $y(x, t) = \sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{l}\cos\frac{n\pi at}{l}$, where $A_n = \frac{2}{l}\int_0^{l} f(x)\sin\frac{n\pi x}{l}\,dx$
2. $y(x, t) = \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi at}{l} + b_n\sin\frac{n\pi at}{l}\right)\sin\frac{n\pi x}{l}$, where

$$a_n = \frac{2}{l}\int_0^{l} f(x)\sin\frac{n\pi x}{l}\,dx, \quad b_n = \frac{2}{n\pi a}\int_0^{l} g(x)\sin\frac{n\pi x}{l}\,dx$$

3. $y = \frac{8h}{\pi^{2}}\left(\sin\frac{\pi x}{l}\cos\frac{\pi at}{l} - \frac{1}{3^{2}}\sin\frac{3\pi x}{l}\cos\frac{3\pi at}{l} + \ldots\right)$
4. $y = \frac{8h}{\pi^{2}}\left[\sin\frac{\pi x}{l}\cos\frac{\pi at}{l} - \frac{1}{3^{2}}\sin\frac{3\pi x}{l}\cos\frac{3\pi at}{l} + \ldots\right]$

8. $y(x,t) = \frac{4l^2 v}{\pi a^2}\left[\sin\frac{\pi x}{l}\sin\frac{\pi a t}{l} - \frac{1}{2^3}\sin\frac{2\pi x}{l}\sin\frac{2\pi a t}{l} + \cdots\right]$

9. $u(x,t) = u_0 - u_0 e^{-\alpha^2 \lambda_1^2 t}\cos\lambda_1 x$, $\lambda_1 = \frac{\pi}{2l}$, $\lambda_1 = \cos\lambda_1 l \sin\lambda_1 l$.

Problems 13.3, page 517

1. $u(x,t) = \frac{100}{\pi^2}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n(n+2)^2}e^{-\alpha^2(n\pi/100)^2 t}\sin\frac{n\pi x}{100}$

2. $\sum_{n=1,3,5,\ldots}^{\infty}\frac{8u_0}{n\pi}\sin\frac{n\pi x}{l}e^{-\alpha^2(n\pi/l)^2 t}$ 3. $u(x,t) = \frac{40}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}\sin\left(\frac{n\pi x}{20}\right)e^{-\alpha^2(n\pi/20)^2 t}$

4. $u(x,t) = -2x + 101 = \frac{400}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi x}{100}e^{-\alpha^2(n\pi/100)^2 t}$

5. $u(x,t) = \frac{5}{2} - \frac{41}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi x}{l}e^{-\alpha^2((2n-1)\pi/l)^2 t}$

6. $u(x,t) = 50 - \frac{400}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{(2n-1)^2}\cos\frac{(2n-1)\pi x}{20}e^{-\alpha^2((2n-1)\pi/20)^2 t}$

7. $k = \frac{48u_0}{\pi}\left[e^{-\alpha^2(\pi/32)^2 t}\cos\frac{\pi x}{32} - \frac{1}{3}e^{-\alpha^2(3\pi/32)^2 t}\cos\frac{3\pi x}{32} + \frac{1}{5}e^{-\alpha^2(5\pi/32)^2 t}\cos\frac{5\pi x}{32} - \cdots\right]$

8. $V = V_0 e^{-\alpha^2(\pi/l)^2 t}\sin\frac{\pi x}{l}$, $\alpha^2 = \frac{k}{\sigma\rho}$, $\sigma = 0.11$, $\rho = 10.5$.

Problems 13.4, page 526

1. $u = -\frac{8}{\pi}\sum_{n=1,3,5,\ldots}^{\infty}\frac{\sin\frac{n\pi}{2}\cos\frac{n\pi x}{2}\cosh\frac{n\pi y}{2}}{n(n^2-4)\cosh\frac{n\pi}{2}}$ 2. $u(x,y) = \frac{3600}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin\frac{(2n-1)\pi x}{20}\sinh\frac{(4-3)\pi y}{20}}{(2n-1)^2\sinh(2n-1)\pi}$

3. $u(x,y) = u_0\cosh\frac{\pi x}{a}\sinh\frac{\pi}{a}(b-y)\operatorname{csch}\frac{\pi b}{a}$.

Problems 13.5, page 535

1. $u(r,\theta) = \frac{8u_0}{\pi}\sum_{n=1,3,5,\ldots}^{\infty}\left(\frac{r}{a}\right)^{n-1}\frac{\sin(2n-1)\theta}{(2n-1)^2}$ 2. $u(r,\theta) = \frac{2500}{\pi^2}\sum_{n=1}^{\infty}\left(\frac{r}{100}\right)^{2n-1}\frac{\sin(2n-1)\theta}{(2n-1)^2}$

3. $u(r,t) = \frac{2}{\pi}\sum_{n=1,2,3,\ldots}^{\infty}\left(\frac{r}{c}\right)^{2n}\frac{e^{2n}-e^{-2n}}{e^{2n}-e^{-2n}}\frac{\sin 2n\theta}{n^2}$ 4. $u(r,\theta) = \sum\frac{2k}{n\pi(n^2-k^2)}\left[(1-\cos n\pi)\sin n\pi\theta\right]$

5. $u(r,\theta) = 50 - \frac{200}{\pi}\sum_{n=1}^{\infty}\frac{1}{2n-1}\left(\frac{r}{a}\right)^{2n-1}\cos(2n-1)\theta$

6. $u(r,\theta) = \cos\theta\left(\frac{200}{r} - \frac{r}{25}\right) - \sin\theta\left(\frac{r}{r} - \frac{200}{r}\right)$ 7. $u(r,\theta) = 4\cos\theta(r - 1/r) + 4\sin\theta(r - 3/r)$.

Problems 13.6, page 539

1. $u = \sum_{n=1}^{\infty}\frac{2\sqrt{2}(-1)^{n-1}}{n\pi\sqrt{n^2\pi^2 - 2}}\cos\alpha_n t\,\sin\alpha_n x$, $\alpha_n = \frac{n\pi}{l}$.

Problems 10.2, page 634

1. $u = u_0 \sin\frac{\pi x}{l}\cos\frac{at}{l}$, $u = u_0 \sin\left(\frac{\pi}{l}\right)\cos\frac{\pi at}{l}$, $u = u_0 \sin\frac{\pi x}{l}\cos\frac{\pi at}{l}$

2. $u = \frac{100}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n}\sin\frac{n\pi x}{l}\exp\left(-n^2\pi^2 t/400l^2\right)$

$u = \frac{200}{\pi}+\frac{200}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n}\sin\frac{n\pi x}{l}\exp\left(-n^2\pi^2 t/400l^2\right)$

4. $u = V_0 \cos(\omega t - \pi x\sqrt{CLG})$

Problems 10.3, page 638

1. $\frac{\partial^2 u}{\partial x^2} = 4\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial t^2}+4\frac{\partial u}{\partial t}$ 2. $\frac{\partial u}{\partial t} = c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\right)$

3. If $u(x,t)$ is the temperature, then temperature gradient at a point is $\partial u/\partial x$ for all t.

4. Elliptic

5. $y^2 dx^2 - x^2 dy^2 = x^2 y^2 dz^2$

6. $u = \frac{13}{7}x + 50$

7. parabolic partial differential equation

8. $x^2\frac{\partial^2 u}{\partial x^2}+x\frac{\partial u}{\partial x}+\frac{\partial^2 u}{\partial y^2} = 0$

9. $y = \frac{20}{7}x, 0 \le x \le \frac{7}{2}$; $y = \frac{7}{20}(20-x), \frac{7}{2} \le x \le 7$

10. $y(x,1) = 0, y(x,1) = 0, \left(\frac{\partial y}{\partial t}\right)_{t=0} = 0$

11. zero

12. $u(0,y) = 0, u(a,y) = 0, 0 < y < \infty$; $u(x,0) = 0$ for all x and $u(x,a) = u$ for $0 < x < a$

13. $\frac{\partial u(0,t)}{\partial x} = 0, \frac{\partial u(l,t)}{\partial x} = 0$ for all t

14. $y(x,t) = f(x+ct)+f(x-ct)$

15. $\frac{\partial u}{\partial t} = c^2\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$ where c^2 is thermal diffusivity

16. $u(x) = e^x + 60$

17. $f(x) = 60, (0, 120)$

18. $u = 6e^{10x-5}$

19. $u = 4e^{2x-1}$

20. $\alpha^2 = k/\rho\sigma$ is called the diffusivity of the substance (in ft²/sec)

21. $\frac{\partial^2 y}{\partial x^2} = 2c^2\frac{\partial y}{\partial x}, \frac{\partial^2 y}{\partial x^2} = 2c^2\frac{\partial y}{\partial t}$

22. $u(x,t) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}e^{-n^2\pi^2 kt/l^2}$

23. $y(x) = 10x - 120, y(x) = 10$

24. Pulse.

Problems 10.4, page 650

1. $(1)\ \sqrt{2}\left(\cos\frac{\pi}{4}+\sin\frac{\pi}{4}\right)\cos\left(\frac{\pi}{4}-\frac{\pi}{3}\right), (1) = 0.965$

2. $\frac{z^2}{x^2+y^2-2x+1}$

3. $x = a = 1.0, y = a = 2$

4. A circle centered at $(-1, 3)$ radius $\sqrt{2}$

5. $-1+i\sqrt{3}, -1-i\sqrt{3}$; $1-i\sqrt{5}, 4\sqrt{5}$

6. $-2+3i, 1-i\sqrt{3}$

7. $-1-i, \sqrt{2}(\cos 15^\circ + i\sin 15^\circ), \sqrt{2}(\cos 165^\circ + i\sin 15^\circ)$

11. (i) Annular region between the circles of radii 3 and 4 with centre $(-3, 2)$ including boundary of lower circle; (ii) Region of complex plane above the line $y = 2$; (iii) Infinite region bounded by the rays $\theta = \pi/3$ and $\theta = \pi/2$; (iv) Real axis and region above it; Between $y = x$ & $y = x + 2$

13. (i) Ellipse with foci $z = \pm 4$ and major axis = 10
(ii) (a) Right bisector of the line joining $z = 2$ and $z = -1$.
(b) Circle through the points $z = 2$ and $z = -1$.

14. (i) Right bisector of the points 3 and 2; (ii) Circle through the points $\frac{1}{2}$ and 1

16. (i) A straight line; (ii) Circle with centre $(1, 1/2)$ and radius $\sqrt{5}/2$.

Problems 11.2 (page 1356)

5. $4\cos\theta(\cos\theta + i\sin\theta)$ 7. $2^{n+1}\sin^n\frac{\theta-\pi}{2}\left(\cos n\left[\frac{\theta+\pi}{2}\right] + i\sin n\left[\frac{\theta+\pi}{2}\right]\right)$

Problems 11.3 (page 1363)

1. (i) $\pm 2(\cos 45° + i\sin 45°)$; $2(\cos 45° - i\sin 45°)$
(ii) $\pm 2\left(\cos\frac{4n+3}{10}\pi + i\sin\frac{4n+3}{10}\pi\right)$, where $n = 0, 1, 2, 3, 4$; $\pm(\sqrt{3} + i)$
(iii) $2^n\sqrt{2}\left(\cos n\frac{\pi}{4} + i\sin n\frac{\pi}{4}\right)$, where $n = 1, 2$ or 3

3. $\pm 2i$; $\pm\frac{1}{2}(\sqrt{3} + i)$, $\pm\frac{1}{2}(-\sqrt{3} + i)$

5. (i) $1, -1, \cos\frac{1}{3}\pi \pm i\sin\frac{1}{3}\pi$, $\cos\frac{2}{3}\pi \pm i\sin\frac{2}{3}\pi$
(ii) -1; $\frac{1}{2}(1 \pm i\sqrt{3})$; $\pm\frac{1}{2}(1 + i)\sqrt{2}$; $\pm\frac{1}{2}(-1 + i)\sqrt{2}$
(iii) $\frac{1 \pm i}{\sqrt{2}}$, $\frac{-1 \pm i}{\sqrt{2}}$; $\pm\left(\cos\frac{2\pi}{9} \pm i\sin\frac{2\pi}{9}\right)$, $\pm\left(\cos\frac{4\pi}{9} \pm i\sin\frac{4\pi}{9}\right)$.
(iv) $\cos 18n + i\sin 18n$, where $n = 0, 1, 2, 3, 4$; $\pm(\cos 3n + i\sin 3n)$, where $n = 0, 1, 2$

6. $1 - i$; $\frac{1}{2}(\sqrt{3} \pm i)$; $\pm\frac{1}{2}(1 - i)\sqrt{3}$

7. $n = 3$; $\pm\left(\cos\frac{\pi}{8} \pm i\sin\frac{\pi}{8}\right)$, $\pm\left(\cos\frac{3\pi}{8} \pm i\sin\frac{3\pi}{8}\right)$; last four values

8. $x^4 + x^3 - 3x - 1 = 0$.

Problems 11.4 (page 1370)

1. $32\cos^5\theta - 24\cos^3\theta + 6\cos\theta$
15. $-32\sin^5 120 - 2\sin 108 - 8\sin 30 + 10\sin 60 + 5\sin 48 - 20\sin 30$
14. $\sin^5\theta = A\sin\theta - B\sin 3\theta + C\sin 5\theta$.

Problems 11.5 (page 1381)

1. (i) $e^{x^2-y^2}\cos 2xy$, $e^{x^2-y^2}\sin 2xy$; (ii) $\frac{1}{2}e^{x^2}$; $\frac{1}{2}e^{x^2}\cos 2y$, $e^{x^2}\sin 2y$

6. $32xy^5 - 32x^3y^3 + 8x^5y + 32xy^5 - 8x^3y^3 + 4x^5 - 8x^3y^2 + 8x^5y - 8x^3y^3$

10. $\frac{1}{64}(\cos 7\theta + 7\cos 5\theta + 21\cos 3\theta + 35\cos\theta)$

17. $-4\pi^2$ 18. $-32\sqrt{2}$.

Problems 13.8, page 664

10. $\frac{\pi}{4} + \frac{1}{2}\log\frac{1+\sin\theta}{1-\sin\theta}$ ascending powers of $\sin\theta$ or $\sec\theta + \tan\theta$

11. $\sinh^{-1}\left[\sqrt{\sin\theta}\right] + \frac{1}{2}\log\left[\sin\theta + \sqrt{\sin\theta}\right]$.

Problems 13.9, page 668

1. (a) $\log_e 10 + \frac{1}{2}(\tan^{-1}(\ln 2) + \frac{1}{2}\ln 5)$; (b) $\log_e 5 + \frac{1}{2}(\ln 5 + 2\pi i)$

6. (a) $\sqrt{2}e^{-(2n-1)\pi}$, $\frac{1}{4}\pi + \log\sqrt{2}$; (b) $e^{-\pi/8}$, $\frac{1}{4}\log_e 5$

9. $\sqrt{\frac{1}{2}}(\cos 2x + \cosh 2y)$; $-\frac{1}{2}\sin^{-1}(\tanh x \tanh y)$

10. (a) $2\sinh x \log(2\cosh x + \sqrt{2}\cosh 2x)$; (b) $-\frac{1}{2}\log 2\cos x + \frac{1}{2}\pi x$.

Problems 13.10, page 669

1. $e^{\ln x \ln\sin\theta}\cos(\ln x\cos\theta) + i\sin(\ln x\cos\theta)$

2. $\sin\alpha\cos\alpha\cosh(\beta)\cosh(\ln\alpha\beta) - \cos\alpha\sin\alpha\sinh(\beta)\sinh(\ln\alpha\beta)$

3. $\tan^{-1}\frac{x\sin\alpha}{1+x\cos\alpha}$, except when $x=1$ and $\alpha=(2n+1)\pi$ 4. $\log(2\cos\theta)$

5. $-\frac{1}{2}\tan^{-1}(\cos\beta\operatorname{cosec}\alpha)$ 8. $\frac{1}{2}\tan^{-1}\frac{2x\sin\alpha}{1-x^2}$

7. $(2\sin\theta)^{-\sin\theta}\cos\theta\pi$ 8. $\cos\frac{n(n-1)}{2}\pi(2\sin\alpha)^{n}\cos n\alpha$

9. $\cos\left(\alpha+\frac{n-1}{2}\beta\right)\sin\frac{n\beta}{2}\operatorname{cosec}\frac{\beta}{2}$ 10. $\frac{\cos\left(\alpha+\frac{2}{3}(n-1)\beta\right)\sin n\frac{\beta}{2}}{\sin\frac{1}{2}\beta}$

11. $\frac{\sin\alpha(\sin\alpha-\sin\beta\cos\alpha)}{1-\sin 2\alpha\sin\beta+\sin^2\alpha}$ 12. $\frac{1-x\cos\theta-x^n\cos n\theta+x^{n+1}\cos(n-1)\theta}{1-2x\cos\theta+x^2}$.

Problems 13.11, page 670

3. 0.065 radians 8. 178° 4. 284°.

Problems 13.12, page 670

1. (b)
2. (c)
3. (b)
4. (c)
5. (b)
6. (d)
7. odd
8. $32\cos^5\theta - 40\cos^3\theta + 10\cos\theta - 1$
9. $8(1-3i)$
10. $2i\sin n\theta$
11. $\frac{1}{25}(-8+15i)$
12. $\cosh x\cosh y$
13. $\frac{1}{10}\cosh 2x\cos x$
14. 1
15. $-\cos x\sinh y$
16. $e^{-\pi/4}$
17. $\alpha+\beta+\gamma$
18. $2i\pi$
19. real
20. $\sinh\frac{1}{2}$
21. $\cosh\frac{1}{2}\log 2 + \sinh\frac{1}{2}\log 2$
22. $16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$
23. an equilateral
24. $\pi/3$
25. $x=a, y=-a$.
26. a circle
27. True
28. True
29. False
30. True
31. False
32. True

Problems 9.2.3 (page 9.2.3)

8. $a = 2, b = -6, c = 1, d = 2, e = 4$
9. (a) analytic (b) Not analytic (c) Analytic
7. $p = -2$

11 & 12. $f(z)$ is not analytic at origin although C-R equations are satisfied there

14. (a) $x^3 + 3x^2 + 1 + k$ (b) $\cos x$ (c) $\log x$ (d) $\sin x$ (e) $x^2 + 2x - 3$ (f) $x^{n+1} + c$ (g) $\frac{1}{2}\log x$ (h) $\frac{1}{3}x^3 + x$

15. (a) $1 + \frac{1}{2}x + c$ (b) $\sin x + c$ (c) $\log x + c$ (d) x^3 (e) $2x^{\frac{3}{2}} + c$ (f) $2 + \tan^{-1} x$ (g) $\frac{1}{2}\sin x \cos x + \frac{1}{2}x$

16. (a) $\frac{1}{2}x - \frac{1}{4}\sin 2x$ (b) $\frac{\sin x}{x} + c$ (c) $\left(\frac{1+2x}{2}\right)e^x + c$

17. $u = 2xy^2 - x^2 + y$

18. $2\tan^{-1}(y/x)$ (b) $2\log x + c$ 20. $v = e^x - e^{2x}\cos(x^2 - y^2)$; $f(z) = e^z - e^{z^2}$

22. (a) $x^2 - 4xy^2 + y^2 + c$ (b) $x^2 - y^2 + 2e^x \cos y + c$ (c) $e^x \sin 2y + c$

23. (a) $x^2 + y^2$ (b) $\frac{1}{2}\log(x^2 + y^2)$

24. (a) $c(1 + \cos\theta + i\sin\theta)\log r$ (b) $r(\cos\theta + i\sin\theta) - \frac{1}{2}r(\cos\theta - i\sin\theta) + c$

25. $-2\tan^{-1}(y/x) - 2\log r + c$

Problems 9.2.4 (page 9.2.7)

1. $x = \cos\theta$ 2. $u = -2y$

3. $w = (2z + 3i)(z - 2i)$; fixed points $w = 1, 2i$; invariant points

4. $w = \frac{22 + 18z - 6z^2 + 3iz^3}{12 + 2z - 11z^2 - 3iz^3}$

5. (a) $w = 4(1 - \sqrt{3}i + z)$ (b) $w = \frac{(2z - 3i)(z - 2i)}{2z + 1 + 3i}$ (c) $w = 1 - \sqrt{3}i + z$

Problems 9.2.5 (page 9.2.11)

1. (a) $Re(z) > 0$ (b) Region bounded by the parabolas $v^2 = 4(1 + u)$ and $v^2 = 1 - 4u$ (c) Region bounded by parabolas $v^2 = \frac{1}{4} + u$, $v^2 = 9(1 + u)$ (d) Region boundary $x = 2\sqrt{5}\cos\frac{\theta}{2} + 6$.

2. $w = z^2$

3. Lines parallel to x and y axes correspond to two families of rectangular hyperbolas in the w plane which cut each other at right angles. Lines passing through origin in the z plane map into two families of parabolas in the w plane which cut each other orthogonally. It is conformal at $z = 0$

4. (a) Line $4x + 1 = 0$

5. (b) Every circle through the origin in z plane transforms into a straight line not passing through the origin in w plane. (c) A line passing through $z = 0$ maps into a line through $w = 0$. (d) Circle with centre $(3/2, 1/2)$ and radius $(1/\sqrt{2})$ (e) Parabola $y^2 = 4a(x + 2y)$

10. $z = 1 + w$ 11. $Re(\frac{1}{w}) \geq 1/2$ 14. $Re(\frac{1}{w}) \geq 1/4$

Problems 9.2.6 (page 9.2.15)

1. $\frac{1}{2}(1 + \cos 2\theta)$ 3. $w = \sin z$ 4. $w = \log z$

Problems 9.2.7 (page 9.2.19)

1. (a) $\frac{1}{2}(1 - \cos 2\theta)$ (b) $\frac{1}{2}\sin 2\theta$ (c) $\frac{1}{2}(1 + \cos 2\theta)$

2. (a) $\frac{1}{4}(3\sin\theta - \sin 3\theta)$ (b) $\frac{1}{4}(\cos 3\theta + 3\cos\theta)$

3. $-\frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta$

6. (a) $\frac{1}{2}$ (b) $\frac{1}{2}$ (c) 0

7. (a) $\frac{1}{32}(\cos 6\theta + 6\cos 4\theta + 15\cos 2\theta + 10)$ (b) $\frac{1}{32}(\sin 6\theta + 6\sin 4\theta + 15\sin 2\theta)$

8. (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{4}$.

Problems 2008, page 703

3. (a) 0; (b) $2\pi i$
4. (a) $8\pi i$; (b) $\pi i/(a^3b^3)$
5. (a) $6\pi i$; (b) 1
6. $3\pi i - 10\pi i \sin(0.5)$ Exact
7. (a) 0; (b) $2\pi i + 3\pi i \cdot \sin(2\pi)$
8. zero
9. (a) zero; (b) zero
10. $(1+4\pi i)\exp(2\pi i)\exp(-1) = -e$

Problems 2009, page 704

1. $\sum_{n=1}^{\infty} (-1)^{n+1} n(z-1)^n$; Circle of radius unity and centre $(1, 0)$
2. $f(z) = \frac{1}{2} - \frac{1}{4}(z+1) - \frac{1}{8}(z+1)^2 + \frac{1}{16}(z+1)^3 + \ldots$ Region of convergence is $|z+1| < 2$
3. (a) $\frac{1}{2}(z-3) - \frac{1}{2^2}(z-3)^2 + \frac{1}{2^3}(z-3)^3 - \ldots$; $f(1) = \frac{1}{2}$, $f(2) = \frac{1}{4}$, $f(3) = \frac{1}{8}$, $f(4) = \frac{1}{16}$, $f(5) = \frac{1}{32}$, ...
   (b) $\ln(z+1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n}(z-1)^n - \frac{1}{2!}\sum_{n=0}^{\infty}\left(\frac{z-1}{4}\right)^n$ in the region $|z-1| < 4$. Also $\ln(z+1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n}(z-1)^n$ in the region $|z-1| < 2$
   $\ln(z) = -\frac{1}{8}\sum_{n=0}^{\infty}\left(\frac{z-3}{2}\right)^n + \frac{1}{16}\sum_{n=0}^{\infty}\left(\frac{z-3}{2}\right)^n$ in the region $|z-3| < 2$. Also $\ln(z-1) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n}(z-2)^n$ in the region $|z-2| < 1$
4. (a) $\frac{1}{z+1} + \frac{1}{(z+1)^2} + \frac{1}{(z+1)^3} + \ldots + \frac{1}{2}\left[1 + \frac{z+3}{2} + \frac{(z+1)^2}{2^2} + \frac{(z+3)^3}{2^3}\right]$
   (b) $\frac{1}{4}\left[1 + \frac{3}{z} + \frac{3}{z^2} + \ldots\right] - \frac{3}{12}\left[1 - \frac{4}{z} + \frac{4^2}{z^2} - \ldots\right]$; $\ln(z) = \ln(z-2) + \ln 2 = \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(z-2)^n}{2^n n}$
5. (a) $\frac{1}{2}\left[(z-1)^2 + (z-1)^3 + \frac{1}{2!} + \frac{1}{3!}(z-1) + \frac{1}{4!}(z-1)^2 + \ldots\right]$
   (b) $(3z-1)^2 + 3(3z-1)^3 + (3z-1)^4 + (3z-1)^5 + \ldots$; $\frac{1}{z} = \frac{4}{3} + \frac{2}{3}(z-1) + \ldots$
6. (a) $\sum_{n=1}^{\infty} (-1)^{n+1} n \cdot 2^n (z-1)^n$ for $|z-1| < \frac{1}{2}$; (b) $\frac{1}{z} = \sum_{n=0}^{\infty} \frac{(-1)^n (z-1)^n}{2^{n+1}}$
7. (a) $z = \frac{2}{3}\left(1 - \frac{2}{z} - \frac{2^2}{z^2} - \frac{2^3}{z^3} + \ldots\right) - \frac{2}{3}\left(2 - \frac{4}{3} + \frac{4^2}{3^2} - \frac{4^3}{3^3} + \ldots\right)$
   (b) $\frac{2}{z+2} + \frac{3}{(z+2)^2} + \frac{5^3}{(z+2)^3} + \ldots + \frac{1}{2}\left[1 + \frac{z+2}{2} + \frac{(z+2)^2}{2^2} + \frac{(z+2)^3}{2^3} + \ldots\right]$
   (c) (a) $\frac{7}{z} - \frac{8}{z^2} - \frac{49}{z^3} - \frac{81}{z^4} + \ldots$ (b) $\frac{5}{2(z-3)} + \frac{1}{12} - \frac{z-3}{24} - \frac{9(z-3)^2}{432} + \frac{9(z-3)^3}{1944} + \ldots$
8. (a) $\frac{1}{4} - \frac{5z^2}{16} + \frac{21}{64}z^4 - \ldots$; (b) $\frac{1}{2}\left(\frac{1}{z^2} - \frac{1}{z^4} + \frac{1}{4z^6} - \frac{1}{8z^8} + \ldots\right)$; (c) $\frac{2}{z^2} - \frac{2}{z^4} + \ldots$
9. $z = 0, z = 2$ are the isolated singularities
10. $z = 0$ is an isolated essential singularity
11. $z = 0$ is a non-isolated essential singularity
12. $z = 1$ is a pole of order 2
13. $z = 1$ is a pole of order 4
14. $z = n$ is a removable pole and $z = 0, \pm 1, \pm 2, \ldots$ are simple poles

Problems 2010, page 715

1. $-\frac{1}{2} - 2(z+2) + 2(z+2)^2 + 4(z+2)^3 + \ldots$, where $z = (z+2) = -1$

3. (a) $\{x \mid x \neq 0\}$ (b) $\{x \mid x \neq 0, x \neq -1\}$ (c) $\{x \mid x \neq 0, x \neq -1, x \neq 1\}$

4. (a) $\{x \mid x \neq 0\}$; $\{y \mid y \neq 0\}$; $y = \frac{1}{x}$ (b) $\{x \mid x \neq 0\}$; $\{y \mid y \neq 0\}$; $y = \frac{1}{x}$

(c) $\{x \mid x \neq 1\}$; $\{y \mid y \neq 1\}$; $y = \frac{x+1}{x-1}$

5. (a) $\{x \mid x \neq 0\}$; $\{y \mid y \neq 0\}$; $y = \frac{1}{x}$; $\{x \mid x \neq 0\} \to \{y \mid y \neq 0\}$; $x \to \frac{1}{x}$; one-to-one and onto

(b) $\{x \mid x \neq 0\}$; $\{y \mid y \neq 0\}$; $y = \frac{1}{x}$; $\{x \mid x \neq 0\} \to \{y \mid y \neq 0\}$; $x \to \frac{1}{x}$; one-to-one and onto

6. (a) $\frac{x^2+1}{x}$ (b) $\{x \mid x \neq 0\}$ (c) $\{x \mid x \neq 0, x \neq \pm 1\}$ (d) $\{x \mid x \neq 0\}$; $\{y \mid y \neq 2\}$; $y = \frac{x^2+1}{x}$; $\{x \mid x \neq 0\} \to \{y \mid y \neq 2\}$; $x \to \frac{x^2+1}{x}$; not one-to-one

7. (a) $-\frac{1}{x(x+h)}$ (b) $\{x \mid x \neq 0\}$ (c) $\{x \mid x \neq 0, x \neq -h\}$

8. (a) $-\frac{2x+h}{x^2(x+h)^2}$ (b) $\{x \mid x \neq 0\}$ (c) $\{x \mid x \neq 0, x \neq -h\}$

9. (a) $-\frac{2x+h}{x^2(x+h)^2}$ (b) $\{x \mid x \neq 0\}$; $\{y \mid y \neq 0\}$; $y = \frac{1}{x^2}$; $\{x \mid x \neq 0\} \to \{y \mid y \neq 0\}$; $x \to \frac{1}{x^2}$; not one-to-one

10. (a) $\frac{x^2+2xh+h^2-1}{x^2+2xh+h^2}$ (b) $\{x \mid x \neq 0, x \neq -h\}$ (c) $\{x \mid x \neq 0, x \neq \pm 1, x \neq -h, x \neq 1-h\}$

11. $\frac{1}{x} - \frac{1}{x+1} + \frac{1}{x^2} - \frac{1}{x^2+1} + \frac{1}{x^3} - \cdots - \frac{1}{x+1}$

Chapter 1 Review Exercises

1. (a) -1 **2.** (a) $\frac{1}{2}$ **3.** (a) $\frac{1}{2}$ **4.** (a) $\frac{1}{2}$ **5.** (a) $\frac{1}{2}$ **6.** (a) $\frac{1}{2}$

7. (a) $\frac{1}{2}$ **8.** (a) $\frac{1}{2}$ **9.** (a) $\frac{1}{2}$ **10.** (a) $\frac{1}{2}$ **11.** (a) $\frac{1}{2}$ **12.** (a) $\frac{1}{2}$

13. (a) $\frac{1}{2}$ **14.** (a) $\frac{1}{2}$ **15.** $\{(x, y) \mid x \neq 0, y \neq 0, x \neq -y\}$

16. $\{x \mid x \neq 0, x \neq \pm 1\}$ **17.** $x = -1$ is an undefined fraction **18.** $\{x \mid x \neq 0\}$

19. $\{x \mid x \neq 0\}$ **20.** $y = \frac{1}{2}(x + 6)$ **21.** $\{x \mid x \neq 1, x \neq 0\}$

22. $\{x \mid x \neq 1, \frac{1}{2}(x + \sqrt{3})\}$ **23.** $\{x \mid x \neq 0\}$ **24.** $\{x \mid x \neq 0\}$

25. $x = \frac{x^3}{2!} + \frac{x^5}{3!} + \frac{x^7}{4!} + \cdots$ **26.** $\{x \mid x \neq 0\}$ **27.** $\{x \mid x \neq 0\}$

28. $\frac{1}{2}\left[\frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} - \cdots\right] = \frac{1}{2}(1 - \cos x)$ **29.** $\{x \mid x \neq 0\}$ **30.** $\frac{x^2}{2} - \frac{x^4}{4} + \frac{x^6}{6} - \cdots$

31. $\{x \mid x \neq 0\}$ **32.** $x^2 \sin x$ **33.** $\{x \mid x \neq 0\}$

34. $\{x \mid x \neq 0\}$ **35.** $\{x \mid x \text{ is not a multiple of } \pi\}$ **36.** $\frac{x^2}{2!} - \frac{x^4}{4!} + \frac{x^6}{6!} - \cdots$

37. $\{x \mid x \neq 0\}$

38. A complex environment engineer is now on refresh upon closest terms. Quaternion works within π, and the removal on a point is the set giving rise of the region.

39. Which is unclear or irregular **40.** $x = 1, 2$

41. $a_0 + a_1 h + a_2 h^2 + a_3 h^3 + a_4 h^4 + \cdots$ **42.** $\frac{1}{2}$

43. $\frac{1}{2} - \frac{x}{2^2} + \frac{x^2}{2^3} + \frac{x^3}{2^4} + \cdots$ **44.** $-\frac{1}{2}$

45. $\frac{1}{2}$ **46.** Regular functions and notations

47. The coefficients of $(x_0 - x)^{-1}$ in the expansion of $\frac{1}{x}$ in powers of $(x - x_0)$ is called the residue of $\frac{1}{x}$ at $x = x_0$. $x = x_0$ is called the residue of $\frac{1}{x}$ at $x = x_0$ if x_0 is a finite point.

48. $\{x \mid x \neq 0\}$ **49.** $\{x \mid x \neq 0\}$ **50.** $\{x \mid x \neq 1, 2\}$

51. $\frac{1}{2}\left[\frac{x}{2!} + \frac{x^3}{3!} + \frac{x^5}{4!} + \frac{x^7}{5!} + \cdots\right]$ **52.** $\{x \mid x \neq 0\}$ **53.** $\{x \mid x \neq 0, 1\}$

54. $\{x \mid x \neq 0\}$ **55.** $\{x \mid x \neq 0\}$ **56.** $\{x \mid x \neq 0\}$

57. Remove removable $\{x \mid x \neq 1, x \text{ irregularities for } x(x) = 1\}$ **58.** $\frac{1}{2}$

58. exponential singularity 59. zero

61. $\sin z = \frac{1}{\sqrt{2}}\left[1 + \left(z - \frac{\pi}{4}\right) - \frac{\left(z - \frac{\pi}{4}\right)^2}{2!} - \frac{\left(z - \frac{\pi}{4}\right)^3}{3!} + \frac{\left(z - \frac{\pi}{4}\right)^4}{4!} + \cdots\right]$

62. a circle with center $(1, 2)$ and radius 2 in w-plane 63. $n\pi, n$ an integer

64. $\cos(\frac{\pi}{2} + iy) = -i\sinh y$	65. circles	66. True	67. False	68. True	69. True
70. False	71. True	72. True	73. True	74. True	75. True
76. True	77. False	78. False	79. False	80. True	81. True
82. True					

Problem 6.2 | page 703

1. $\frac{1}{z-2} - \frac{2i}{z^2} + \frac{2z-2i}{z^2+2}$

2. $\frac{2}{3} + \frac{\sqrt{2}i}{z^2+2} + \left(\frac{2}{z}\right)$

3. $\frac{2z-20}{z^2-25}$

4. $\frac{x\cosh b - a\sinh b}{x^2+a^2}$

5. $\frac{z^2-2z+5}{z(z^2+5)}$

6. $\frac{20z^3-5i}{(z^2+1)(z^2+25)}$

7. $\frac{4(z-e^{(2+i)z})}{2z^2+2}$

8. $\frac{2}{3}\left[\frac{1}{z^2+1} - \frac{2(z)}{z^2+2} - \frac{1(z)}{z^2+25}\right]$

9. $\frac{az^2+2ai}{(z^2+a^2)^2}$

10. $\frac{5}{(z+5i)^2-5^2}$

11. $\frac{60}{z-2} - \frac{z-2}{z^2-4z+20}$

12. $\frac{2(6z+i)}{(z^2+6)(z+2)^2+6z-20}$

13. $\frac{2}{(z+1)(z^2+2z+2)}$

14. $\frac{2}{3}\left[\frac{2}{z-2} - \frac{4(z-2)}{z^2-4z+6} - \frac{z-4}{z^2-6z+12}\right]$

15. $\frac{z(z^2+2z^2)}{z^2+2z^2}$

16. $\frac{2}{3}\left[\frac{1}{z^2-2} + \frac{z^2-2z}{z^2-10z^2+100}\right]$

17. $\frac{2}{2z+2z^2}$

18. $\frac{2}{z^2} + \frac{2}{(z-1)^2} + \frac{6}{2z+2z^2} + \frac{6}{(z-2)^2}$

19. $\frac{4(14z^2+4z-2)}{(4z^2+1)^2}$

20. $\frac{2}{z} - \frac{2z^2}{z^2}$

21. $\frac{1+e^{-2z}}{z^2+1}$

22. $\frac{e^{-2z+2}}{z^2+2}$

23. $e^{-2\tanh 2}\frac{z}{z^2+1}$

24. $\frac{2}{z^2} - \frac{e^{-2z}}{z^2}(2+2z+2z^2) + \frac{e^{-2z}}{z^2}(2z-2)$

25. $\frac{6}{(z-1)(z^2-2z+2)}$

Problem 6.3 | page 708

1. $(2z^2)^2 - e^{-z}(2z)(1-e^{-z})$

2. $(2z\cos 2z^2 + z^2)\left(\frac{1}{2}\sinh(2z\cos 2z)\right)$

3. $(\sin z)(\sin z)\sin(2z)$

4. $(2z^2)\tanh\frac{1}{2}z^2$

5. $\frac{1}{z(z^2+z^2)}$

6. $(z^2-2az+a^2+b^2)^{-1}$

7. $\frac{2}{(z-2)(2z+1)}$

Problem 6.4 | page 710

1. $\frac{z-1}{z(z^2-2z+2)}$

4. $\frac{2(2z^2+4)}{z^2(z^2+2)^2}$

5. $\frac{25}{(z^2+z)^2}$

4. $\frac{2x^3 - 6x^2 + 1}{(x^2 + x)^2}$

7. $\frac{2x}{(x^2 - 1)^2}$

8. $\frac{6(x-2)}{(x^2 - 4x + 13)^2}$

9. $\frac{48x + 32}{x^2 - 4x + 20}$

10. $\frac{21x^3 + 63x^2 + 59x + 20}{(x^2 + 4x + 5)^2}$

11. $\log|x| + 6\log|x| + c$

12. $\cot^{-1}|x|$

13. $\frac{1}{3}\log|3x^2 + 36x + 101|$

14. $\frac{1}{2}\log\left|\frac{x^2 + 1}{x - x^2}\right|$

15. $\cot^{-1}(x + 1)$

16. $\frac{1}{2}\log\left|\frac{x^2 + 2}{x^2}\right|$

17. $\cot^{-1}x - \frac{1}{2}x\log(1 + x^{-2})$

18. $\frac{1}{x - \log 2} + \frac{2x}{(x^2 + 3)^2} + \frac{1}{2}\log\left|\frac{x^2 + 3}{x^2 + 1}\right|$

22. (i) $\log 2$; (ii) $\log 3$; (iii) $\frac{1}{2}\log 3$; (iv) $\frac{1}{2}\log 2$; (v) $\frac{1}{2}\log 3$; (vi) $\frac{8(x+1)}{x(x^2 + 2x + 17)}$

23. (i) $\frac{1}{x}\cot^{-1}(x)$; (ii) $\frac{1}{x} - \frac{x+2}{x^2 + 2x + 2}$; (iii) $\frac{\cot^{-1}(x+1)}{x}$.

Problems 11.4, page 740

1. $\frac{2}{3}\left(\sin\frac{3x}{2} - \sin\frac{3x}{2}\right) + 4\sin 3x + 8\sin 2x$

2. $e^{2x} - e^{2x}$

3. $2e^{4x} - \frac{1}{2}e^{2x}$

4. $e^{4x} + 2e^{-4x}$

5. $\frac{1}{3}(8e^{3x} - e^{x})$

6. $\cosh x$

7. $e^{x} - e^{-2x} - 2e^{3x}$

8. $\frac{1}{2}e^{2x} - \frac{1}{2}e^{-2x} - \frac{1}{2}e^{-4x}$

9. $\frac{3}{8}e^{x} - e^{-2x} + \frac{3}{8}e^{-6x}$

10. $\frac{1}{2}x\sinh 2x$

11. $\frac{1}{2}(3e^{x} - e^{-3x})$

12. $\frac{1}{13}(3e^{2x} - 2\cos 3x + 3\sin 3x)$

13. $\frac{1}{2}(\sin x - 2x^{2})$

14. $\frac{1}{2}(\sin x\sinh x + \cosh x\sinh x)$

15. $\left(\frac{1}{2}x^{2}\right)e^{x} + e^{-x}\left[\cos\left(\frac{\sqrt{3}}{2}x\right) + \frac{\sqrt{3}}{3}\sin\left(\frac{\sqrt{3}}{2}x\right)\right]$

16. $\frac{1}{2}(3\sin x - \sin 2x)$

17. $\frac{1}{5}e^{-2x}(6\sin 2x - 3\cos 2x)$

18. $\frac{1}{5}(1 + e^{-2x})\sin x + \frac{2}{5}(1 - e^{-2x})\cos x$

19. $\frac{1}{2}e^{-x}(\cos x + \sin 2x)$

20. $\frac{1}{2}(3\sin x + \sin 2x)$

21. $\cos x\sin 2x\sin 3x$.

Problems 11.5, page 743

1. $\frac{3}{10}(e^{2x} + 2e - 3)$

3. $\frac{1}{6} - \frac{1}{4}\left(x^{2} + x + \frac{1}{2}\right)e^{-2x}$

4. $\frac{2}{x^{3}}\cos\left(\frac{\pi x}{4}\right)$

5. $\frac{1}{a^{2}}\left(1 - \frac{1}{2}ax\sin ax\right)$

6. $\frac{1}{2}e^{x}(\sin x + \cos x + 1)$

7. $\frac{1}{2}e^{-2x}\sin 2x$

8. $x\sin x\cos x$

9. $\frac{1}{2}(3e^{2x} - 4xe^{x} + 2)e^{-2x}$

10. $\frac{1 - x^{2}}{x}$

11. $\frac{1}{4}(e^{-2x} - e^{-4x})$

12. $e^{x} - e^{2x} - e^{3x}$

13. $\frac{1}{2}(\sin ax\sin bx - \cos ax\cos bx)$

14. $\frac{2}{3}(e^{x} - \cos x)$

15. $\frac{\sin 2x}{x}$

16. $\frac{\sin 2x}{x}$

17. $\frac{\sin ax\sinh bx - x\sinh bx}{x^{2}}$

18. $\frac{e^{-2x} - e^{-4x}}{x - 3}$

17. $\frac{1}{2a^3}(\sin at - at\cos at)$ 18. $\frac{1}{a^2}(at - \sin at)$ 19. $\frac{1}{2}(e^t + 1) + \frac{1}{2}(e^t - 1)$

20. $\frac{1}{18}(t^3 - e^{-3t} - 4te^{-3t})$ 21. $\frac{1}{13}(3\sin 2t + 2\cos 2t - 2e^{-3t})$ 22. $\frac{t^n}{n!} + \cos t - 1$

23. $\frac{e^{-3t}}{64}(\sin 2t - 2t\cos 2t)$

Problems 8.6, page 254

1. $y = \frac{7}{4}e^{-t} - \frac{5}{4}e^{-3t} - \frac{1}{2}te^{-t}$ 2. $x = \frac{50}{9}\sinh 3t$ 3. $y = 1 - 3\sin t + \cos t$

4. $y = 2t + 2 + \frac{1}{2}(e^{2t} - e^{t} - 2te^{2t})$ 5. $y = 4e^{2t}(1 + t) - 5t^2$ 6. $y = 2\cos 3t + 3\sin 3t$

7. $y = \frac{1}{200}\sin 4t$ 8. $y = \frac{1}{8}e^t - \frac{3}{40}e^{-4t} - \frac{1}{10}(2\cos t + \sin t)$

9. $y = \frac{1}{2}(\cosh 3t + \cos 3t)$ 10. $y = \frac{1}{9}(2 - t^2)\sin t - 2t\cos t$ 11. $y = \frac{11}{2}e^{-t}\sin t + \sin 2t$

12. $y = -\frac{17}{6} + \frac{13}{5}e^{-t}\cos 2t + \frac{7}{12}e^{-t}\sin 2t$ 13. $y = e^{2t}(t^2 - 3t + 13) - e^{t}(3t^2 + 5t + 11)$

14. $x = \frac{5}{16}\sin 2t - \frac{3}{8}\sin t - \frac{1}{2}t\cos 2t$ 15. $y = \frac{1}{2}\left(\frac{3\sin t}{t} - \cos t\right)$

16. $y = e^{2t}$ 17. $y = t$ 18. $y = 3J_0(3t)$

21. $\ln\left|\frac{t + a}{t - a}\right| + \ln\left|\frac{t + b}{t - b}\right|$; $\ln\left|\frac{t^2 - a^2}{t^2 - b^2}\right|$, $a^2 \neq b^2$

Problems 8.7, page 258

1. $x = \frac{1}{2}(e^t + \cos t + 2\sin t - t\cos t)$, $y = \frac{1}{2}(t\sin t - e^t + \cos t - \sin t)$

2. $x = e^t + e^{-t}$, $y = e^{-t} - e^t + \sin t$ 3. $x = 2 + e^{2t}(4t)$, $y = -1 - te^{2t}$

4. $x = \frac{1}{20}(1 - 2e^{-t} - 3e^{-4t})$, $y = \frac{1}{8}(e^{-t} - e^{-4t})$ 5. $x = e^{2t}(1 + 2t) + 2e^{3t}$, $y = e^{2t}\sin t + \cos t - e^{-2t} - e^{3t}$

6. $I_1 = \frac{e^{-t}}{p + 1}(\sin t + \sin 2t)$; $I_2 = \frac{e^{-t}}{p - 1}(\sin t - \cos t)$

Problems 8.8, page 262

1. $\frac{2}{p^2 + 4}(e^{-2\pi p} - e^{-4\pi p})$

2. (i) $1 - 2\cos t - \sin t + 2\sin t\cos t$, $\frac{2}{p^2} + \left(\frac{1 - 2p}{p} - \frac{2}{p^2}\right)e^{-pt}$

(ii) $e^t\sin 2t - \sin t - 2t$, $\frac{2(1 - e^{-2\pi p})}{p^2} - \frac{4e^{-2\pi p}(1 + \pi p)}{p^2}$

(iii) $\frac{1}{2}\sin 2t - \sin t - 2t\cos 2t + t$; $\frac{2}{p^2} - \frac{2}{p^2 + 1} - \frac{2p}{(p^2 + 4)^2} + \frac{1}{p^2}$

3. (i) $\frac{p}{p^2 + 1} + \left(\frac{2}{p} + \frac{p}{p^2 + 1}\right)e^{-\pi p} - \left(\frac{1}{p} - \frac{p}{p^2 + 1}\right)e^{-2\pi p}$

(ii) $\frac{1}{p^2 + 1} + e^{-\pi p}\left(\frac{p}{p^2 + 1} - \frac{p}{p^2 + 4}\right) + e^{-2\pi p}\left(\frac{p}{p^2 + 1} - \frac{p}{p^2 + 4}\right)$

(iii) $\frac{2}{p^2}(1 + e^{-\pi p}(2\pi^2 - 1 - 2\pi^2 p)(1 + e^{-\pi p}))$

1. (a) $\frac{1}{2}(x+1)^2(x-3)$; (b) $(x+3)(x-3)(x+1)$; (c) $\frac{1}{2}(x+3)(x-3)(x+1)$; (d) $\frac{1}{2}(x+3)(x-3)(x+1)$; (e) $(x+1)^2(x-3)$; (f) $(x+3)(x-3)(x+1)$
2. $\frac{1}{2}(x+3)^2$
3. (a) $\frac{1}{2}(x+3)(x+1)(x-3)$; (b) $\frac{1}{2}(x+3)(x+1)(x-3)$; (c) $\frac{1}{2}(x+3)(x+1)(x-3)$; (d) $\frac{1}{2}(x+3)(x+1)(x-3)$; (e) $\frac{1}{2}(x+3)(x+1)(x-3)$; (f) $\frac{1}{2}(x+3)(x+1)(x-3)$
4. $y=\begin{cases}\frac{1}{2}\sin 2x+\frac{1}{2}(1-\cos 2x)-\frac{\pi}{2}x+\cos x-2x+\pi-\frac{1}{2}, & x\geq\frac{\pi}{2}\\ \frac{1}{2}\sin 2x+\frac{1}{2}(1-\cos 2x)+\frac{\pi}{2}x+\cos x-2x+\pi-\frac{1}{2}, & x<\frac{\pi}{2}\end{cases}$
5. $a=0$; $b=\frac{1}{2}\sin 1+\frac{1}{2}(1-\cos 1)-\frac{\pi}{2}+\cos 1-2+\pi-\frac{1}{2}$
6. $f(x)=\begin{cases}\frac{1}{2}\sin 2x+\frac{1}{2}(1-\cos 2x)-\frac{\pi}{2}x+\cos x-2x+\pi-\frac{1}{2}, & x\geq\frac{\pi}{2}\\ \frac{1}{2}\sin 2x+\frac{1}{2}(1-\cos 2x)+\frac{\pi}{2}x+\cos x-2x+\pi-\frac{1}{2}, & x<\frac{\pi}{2}\end{cases}$
7. $f(x)=\frac{\sqrt{2}}{2}x^2-\frac{\sqrt{2}}{2}x^2+

38. $\int_0^{\pi} f(x)\,dx$ 39. $\ln 2$ 41. $\ln 4$ 42. $\ln 3$

43. Time 44. Value 45. Pulse

Problems 20.1, page 770

1. $\frac{2}{\pi}\int_0^{\pi}\frac{\sin 4x\cos 3x}{x}\,dx$; $\frac{\pi}{2}$ for $|x|<1$, $\frac{\pi}{4}$ for $|x|=1$, 0 for $|x|>1$

2. (a) $\frac{4}{\pi}\int_0^{\infty}\frac{\sin\omega\pi-\omega\cos\omega\pi}{\omega^2}\cos\omega x\,d\omega$ (c) $\frac{2}{\pi}\int_0^{\infty}\cos\omega x\,\frac{\omega}{\omega+\omega^2}\,d\omega$

3. (a) $\frac{2\sin\omega\pi}{\pi}$; π; (c) $\frac{2}{\pi}\int_0^{\infty}\frac{2\omega\cos\omega\pi+2\sin\omega\pi-\omega\pi\sin\omega\pi}{\omega^2}\cos\omega x\,d\omega$

5. $\frac{4}{\pi}\int_0^{\infty}\frac{\omega\sin\omega\pi-\omega\pi\cos\omega\pi}{\omega^2}\sin\omega x\,d\omega$ 6. (a) $\frac{1}{\pi}\int_0^{\infty}\frac{\cos\omega x}{1+\omega^2}\,d\omega$ (c) $\frac{\sqrt{2}}{\pi}\int_0^{\infty}\frac{\omega\sin\omega x+\cos\omega x}{1+\omega^2}\,d\omega$

7. $\frac{1-\cos 2\omega}{\omega}+\frac{\sin 2\omega}{\omega}$ 8. $\frac{1}{\pi}\int_0^{\infty}\frac{2\omega}{1+\omega^2}\cos\omega x\,d\omega$ 9. $\frac{\omega}{\omega^2+\beta^2}+\frac{\beta}{\omega^2+\beta^2}e^{-\omega x}$

10. 1 11. (a) $\frac{\pi}{2\omega}\left(1-e^{-\omega\pi}\right)\cos\omega x\,d\omega$; $\frac{1}{\omega\sqrt{2\pi}}e^{-\omega^2/4}\cos\omega x\,d\omega$; $\frac{2}{\omega^2\sqrt{2\pi}}e^{-\omega^2/4}\cos\omega x\,d\omega$

14. (a) $\frac{1}{2}\left[\frac{\sin\pi(1-\omega)}{1-\omega}+\frac{\sin\pi(1+\omega)}{1+\omega}\right]\cos\omega x\,d\omega$; $\frac{2\omega\sin\omega\pi-\pi\omega^2\cos\omega\pi-2\sin\omega\pi}{\omega^3}\cos\omega x\,d\omega$

15. $\frac{2}{\pi}\sin\omega\pi$ 16. $F(\omega)=-\frac{\omega}{\pi}\left(1-2\cos\omega\pi\right)$; $F(\omega)=\frac{2}{\pi}\frac{1-\cos\omega\pi}{\omega^2}$

17. $\frac{1}{\pi}\sin\omega\pi\cos\omega x$ 19. $f(x)=\frac{1}{2}+\frac{2}{\pi}\sin\pi x-\frac{4}{\pi}\cos 2\pi x+\frac{2}{3\pi}\sin 3\pi x$ 20. $\frac{2}{\pi}\sin(1+\omega^2)$

Problems 20.2, page 778

2. $\frac{1}{4}\int_{-\infty}^{\infty}e^{-|\omega|}\cos\omega(x-1)\,d\omega$ 4. $\frac{2}{\pi}\left(\frac{1-\cos\omega\pi}{\omega^2}\right)\sin\omega x\,d\omega$

Problems 20.3, page 788

1. $\frac{2}{\pi}\left[\omega^2+\omega^4\right]^{-1}+\omega^4\left(\frac{2}{\pi}-\frac{1}{\omega^2}\right)$ 3. $\frac{1}{\pi}\left(e^{-\omega\pi}-2\sin\omega\pi-\cos\omega\pi\right)$ 5. $\frac{2}{\pi}\left(\frac{\omega\sin\omega\pi-\omega^2\cos\omega\pi}{\omega^2}\right)$

4. $\frac{2}{\pi}\left[\omega^2(1-2\omega)+\cos\omega\pi\right]$ 5. $\frac{1}{\pi}\left(\frac{\omega\sin\omega\pi-\pi\cos\omega\pi}{\omega^2}\right)$

Problems 20.4, page 793

1. $y=30e^{-2x}\sin 5x$ 3. $\sum_{n=1}^{\infty}\frac{8}{n\pi}\left(1-\cos n\pi\right)e^{-n^2\pi^2 t/100}\sin\frac{n\pi x}{10}$

3. $u(x,t)=\frac{2}{\pi}\int_0^{\infty}e^{-\omega^2 t}\left(\frac{\omega\sin\omega-\omega^2\cos\omega+\omega\cos\omega-\sin\omega}{\omega^2}\right)\sin\omega x\,d\omega$

5. $u(x,t)=\frac{A_0}{2}\operatorname{erf}\left(\frac{x}{2\sqrt{kt}}\right)+A_0\sum_{n=1}^{\infty}(-1)^n\left[\operatorname{erf}\frac{nL-x}{2\sqrt{kt}}-\operatorname{erf}\frac{nL+x}{2\sqrt{kt}}\right]$.

Problems 1 to 10 | page 600

1. $F(\omega) = \int_{-\infty}^{\infty} f(t) \sin \omega t \, dt$ 2. $\sqrt{2\pi}$

3. The Fourier transform of the convolution of $f(x)$ and $g(x)$ is the product of their Fourier transforms.

4. $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{-i\omega x} \, d\omega$ 5. $\int_{-\infty}^{\infty} e^{-x} f(t) e^{-st} \, dt$ 6. $e^{-as} F(s)$

7. $\frac{2}{\pi} \int_0^{\infty} \sin \lambda x \, dx \int_0^{\infty} f(t) \sin \lambda t \, dt$ 8. $\frac{1}{s}$ 9. $-s^2 \mathcal{F}(s)$

10. $\frac{2}{\pi} \int_0^{\infty} \cos \lambda x \, dx \int_0^{\infty} \cos \lambda t \, f(t) \, dt$ 11. $-\frac{\pi k}{2} \int_0^{\infty} f(x) \sin \frac{kx}{T} \, dx$ 12. $\frac{1}{a} F\left(\frac{\lambda}{a}\right)$

13. $f(x) = \frac{2}{\pi^2} \sum_{n=1}^{\infty} \left(\frac{1 - \cos n\pi}{n^2} \right) \sin nx$ 14. $\frac{1}{2} f(x/2)$

15. $\mathcal{L}\{t^2 + 2\}$ 16. True 17. False 18. True

19. True 20. False 21. True

Problems 11 to 20 | page 608

1. (i) e^{4s}; (ii) $\frac{1}{s(s^2 - 3)}$; (iii) $\frac{s}{s - e^{as}}$

2. (i) $\frac{2s}{(s - 2)^2} + \frac{s\sqrt{2}}{s^2 - \sqrt{2}s + 1} + \frac{s}{s - 1}$ (ii) $\frac{s^3 - 2s^2 + 4s}{(s - 1)^4}$; (iii) $\frac{s(s + 1)}{(s - 1)^3} + \frac{2s}{(s - 1)^2} + \frac{2s}{s - 1}$

3. (i) $\frac{s^2 \sin \theta}{s^2 - 2s \cos \theta + 1}$ (ii) $\cos \alpha \frac{s(s - \cos \alpha \beta)}{s^2 - 2s \cos \alpha \beta + 1} + \sin \alpha \frac{s \sin \alpha \beta}{s^2 - 2s \cos \alpha \beta + 1}$

4. $\frac{s(s - \cos \theta)}{s^2 - 2s \cos \theta + 1}, |s| > 1$; $\frac{s(s^2 \cos \theta - 2s + \cos \theta)}{(s^2 - 2s \cos \theta + 1)^2}$ 5. $\frac{s^2}{s^2 + 1}, |s| > 1$; $\frac{s^3}{s^4 + s^2}$

6. (i) $\frac{s}{s - s^{-1}}$; (ii) $\frac{s^{-1}}{(s - s^{-1})^2}$; (iii) $\frac{(s + s^{-1})s^{-1}}{(s - s^{-1})^2}$

7. $\frac{s^2(1 + 3s^2)}{(1 - s^2)(1 + s^2)}$ 8. $a_0 = 2, a_1 = 1$

Problems 21 to 30 | page 614

1. $\frac{s}{s - 4}; |s| > 4$ 2. $\frac{s}{2 - s}; |s| < 2$ 3. $\frac{2s}{(4 - s)(s - 2)}; 2 < |s| < 4$

4. $\frac{6s}{(4 - s^2)^2}; |s| > 2$ 5. $-\log(1 - 2s); |s| < \frac{1}{2}$ 6. s^{2n}, ROC is s plane

7. $(1 - s^2)/s^2; |s| > |s^2|$.

Problems 31 to 40 | page 622

1. $\frac{1}{2}(s^{n+1} - 1)$ 2. $(s + 1)s^n$ 3. $\frac{1}{2}s(s - 1)$

4. $4s^3$ 5. $(1/3)^n - 3^n$ 6. s

7. $\frac{3}{4}\left[\frac{1}{|2s^2 - 1|} + \frac{1}{|s - 4|^2}\right]$ 8. $(s^2 + 5s + 4)/(s^2 - 1)$

12. $a_n = 12(3^{n-1}) + 12(3^{n-2}) + (3^{n-2})(n \ge 0)$ 11. $5 + 12(3^n) + 20n = 31(3^n), n \ge 11$

13. $6 - 11^{n+1} - 2n + 6n + 6n^2$ 15. $1 - e^{-2n}$

14. (a) $\left(-\frac{1}{3} - \frac{x}{3^2} - \frac{x^2}{3^3} - \frac{x^3}{3^4} - \cdots\right) + \left[\frac{1}{2} + \frac{x}{2^2} + \frac{x^2}{2^3} + \frac{x^3}{2^4} + \cdots\right]$ (c) $(-1)^{n-1}(n^2), n \ge 0$

(b) $(3^{n-1} - 2^{n-1})(n^{n-1}), n \ge 1, 0, n < 0$

12. $\frac{1}{2}(n+1)(n+2)(2^{n-2}), n \ge 2$ and $= 0, n = 2$ 16. $2(n-1)^{n-1} + (-2)^{n-1}$

13. $\frac{1}{3} - \frac{1}{3}\left(-\frac{2}{2}\right)^n$ 18. $\frac{1}{3} - \frac{2}{3}\left(-\frac{1}{2}\right)^{n-1}$

19. $a_n = 1 + \frac{1}{2}[2^{n-1} + (-2)^{n-1}]$ 20. $2n$ and $(n+1)(n+2), n = 0, 1, 2, \ldots$

Problems 10.4, page 611

1. $y_n = \frac{2}{5}\left(\frac{1}{2}\right)^n - \frac{2}{5}\left(-\frac{1}{2}\right)^n$ 2. $y(n) = (n-1)(n-1)^{n-1}y(n-2) = 2^n$

3. $y_n = 2^{n-1} + (-2)^{n-1}$ 4. $f(n) = 2 + (-1)^n$

5. $y_n = \frac{2}{3}[2(-1)^n + (2)^n]$ 6. $2n\left[\frac{1}{2}(2)^n - \left(\frac{1}{2}\right)^n\right]$

7. $y_n = (c_1 + c_2 n)(2)^n + \frac{3}{2}n(n-1)2^{n-2}$ 8. $y_n = c_1 + c_2 \cdot 2^n + 2n^2$

9. $y_n = 3\left[\left(\frac{1}{4}\right)^n - \left(-\frac{1}{4}\right)^n\right]$ 10. $2^n 2^n$

11. $y_n = \frac{1}{8}(-1)^n - \frac{3}{8}(-3)^n + \frac{1}{10}(2)^n$ 12. $y_n = 3 - 2n + 2^n$

13. $y_n = c_1 3^n + \left(c_2 - \frac{1}{2}\right)2^n + 2n - \frac{3}{2}$ 14. $y_n = \frac{3}{8}(n+2)\frac{1}{2^n}\cos\frac{\pi n}{2}$

15. $y_n = \left[\frac{1}{4} - \frac{5}{4}(-3)^n\right]\mu(n)$ 16. $y_n = 1 - 2^{n-1}, (n \ge 1)$

Problems 10.5, page 621

1. $x(z) = 1$ 2. $\sum_{n=0}^{\infty} a_n z^{-n}$ 3. $x(z) = z^n$ 4. $x(z) = z^n$

5. $\frac{z \sin \theta}{z^2 - 2z\cos\theta + 1}$ 6. $z^{1/2}$ 7. $(z^2 + z)(z-1)^3$

8. $3(a_{n+3} + b_{n+2} + a_{n+1}) + a^2(a_{n+2} + b_{n+1} + a_n) + a^3(a_{n+1} + b_n + a_{n-1})$ 9. z^{n-1}

10. $1 - (z^{-1})^n$ 11. $a_n = \sum_{k=0}^{n} (2)(a_k)$ 12. False 13. False

14. True 15. True 16. False 17. False

18. False

Problems 10.6, page 631

1. $a = 2.28, b = 0.1670, y = 68.46$ 2. $a = 1128, b = 55.1$ 3. $a = 0.5, b = 0.0864$

4. $a = 0.8818, a = 0.8$ 5. $a = 0.318, b = 31.8$ 6. $a = 1, b = 0.43$

7. $a = 0.6666, b = -1.32$

Problems 8.12, page 163

1. $y = 3.6x$
2. 18.2 thousand tons
3. $T = 0.031P + 0.115$
4. $\hat{H} = 30.052 + 0.202t$
5. $a = 0.545, b = 0.658$
6. $\log y = 4.3662 + 1.113x$
7. $y = 1.243 - 0.504x + 0.822x^2$
8. $y = 0.24 - 2.73t + 0.59t^2$
9. $y = 15.866 + 66.355t - 4.333$
10. $\hat{R} = 3.46 - 0.0627P + 0.00264P^2$
11. $\hat{Y} = 2.263 - 0.2697 + 0.0621T^2$.

Problems 8.13, page 165

1. $0.32, \hat{A} = 3.2083$
2. $a = 1.02, b = 0.49$
3. $a = 3, b = 2$
4. $y = 7.163 + 3.301\frac{1}{x}$; 4.988
5. $a = 0.1998, b = 0.978$
6. $y = 3.978x^{-2.1649}$; 0.8705
7. $a = 0.2028, b = 0.0021$
8. $f(t) = 0.638e^{-.08} + 0.362e^{-.3t}$
9. $a = 149.3, b = -0.173$
10. $a = 93.95, b = 1.2$

Problems 8.14, page 168

1. $a = 31.1, b = 0.71$
2. $y = 48.06 + 0.1x$
3. $a = 0.0026, b = 3.31, c = 5.16$
4. $a = 15.9, b = 2.1, r = -0.8$
5. $a = 1.429, b = 0.042$.

Problems 8.15, page 169

1. $y = 0.12 + 0.47x$
2. $y = 2.164 + .32x$
3. $y = 1.55 + 0.062x + 0.074x^2$
4. $y = 0.465 + 0.395x + 0.126x^2$.

Problems 8.16, page 170

1. $Y = aX + b$ where $X = x, Y = y/x$
2. $Y = A + BX$, where $X = \log_{10} x$, $Y = \log_{10} y$, $A = \frac{1}{3}\log A, B = -1/3$
3. 154.4
4. $y = aX + c$, where $X = x^2$
5. (d)
6. $\Sigma y = nA + B\Sigma x, \Sigma xy = A\Sigma x + B\Sigma x^2$ where $y = \log_{10} y, A = \log_{10} a, B = \log_{10} b$
7. $ 84.19
8. Zero
9. $y = aX + b$ where $X = x^2/\log_{10} x$, $Y = y/\log_{10} x$
10. $a = 3.3167, b = 1.06$.
11. The numerals of the observed values only are respectively equal to the numerals of the calculated values of y.
12. $a = 1.7, b = 1.95$
13. $y = ax + bx$ where $x = 1/x, y = 1/y$.
14. (e)
15. (c)

Problems 15.1, page 337

1. 335.78
2. 64% get more than 50 marks; median at 5, $Q_1 = 4\frac{1}{2}$, $Q_3 = 8\frac{1}{2}$
3. Mean = 17.9 ; Median = 20.68 ; Mode = 23.85
4. Mean = 12.56 ; Median = 12.8 ; Mode = 13.3
5. 2.1%
6. 1.3%
7. 152 km/hr
8. 60 km/hr
9. 28.4 ; 36.2
10. Median = 12.3 days; Mode = 11.4 days

Problems 15.2, page 339

1. 0.43, 0.39
2. 4, 7
3. 12.34, 10.12, 11.63, 8.54, 2.35

5. 35, 33.5, 12.5 6. $Q_1 = 10.5$, $Q_3 = 18.25$ 7. 1.045 ; 1.0379
8. 0.63, 1.34 ; first, you 9. Height 10. A
11. B is a better player and more consistent.
12. A is more efficient, B is more consistent. 13. 161.5, 5.55.

Problems 2.3.2 (page 068)

1. $\bar{x}_1 = \bar{x}_2 = 0$, $\mu_3 = 2$, $\mu_4 = 11$: $\beta_1 = 0$, $\beta_2 = 2.72$ 2. 6.65 : 5.25 : 3.32 : 1.39
3. −0.5934 4. 0.35 ; 2.857 5. 0 ; 2.5
7. $\beta_1 = 0.495$; $\beta_2 = 0.655$; platykurtic.

Problems 2.3.3 (page 074)

1. $r = 0.81$; $x = 0.5y + 0.5$, $y = 1.2x + 1.1$ 2. $r = 0.96$
3. $r = 0.92$ 4. $r = -0.095$ 5. 0.7273
6. $r = 0.4627$ 7. $r = 0.632$; $y = 0.497 + 0.8x$, $x = 0.187 + 0.5y$
8. $w = 0 - 0.95z - 01$ 10. $\bar{X} = 4$, $\bar{y} = 7$, $r = -0.8$ 11. $\bar{X} = 9.08$, $\bar{y} = 6.92$, $r = 0.46$
12. $r = 0.7586$; $\bar{x} = -0.1034$; $\bar{y} = 0.6173$ 13. 354.6
14. 1.26 inch 15. 0.6646 16. 0.502.

Problems 2.3.4 (page 080)

1. (d) 2. (d) 3. (a) 4. (a)
5. (b) 6. (a) 7. (b) 8. Coefficient of correlation
9. 36 10. 13.83 11. $\bar{X} = 2$, $\bar{y} = 3$, $r = \sqrt{0.5}$
12. Zero 13. $\frac{1}{2}(Q_3 - Q_1)$ 14. −1
15. 1.25 3 16. $\frac{\Sigma XY}{n\sigma_x\sigma_y}$ 17. (x, y)
18. Reliability or consistency 19. $\sqrt{\frac{\Sigma}{n}}$ 20. degree of peakedness
21. $y - \bar{y} = r\frac{\sigma_y}{\sigma_x}(x - \bar{x})$ 23. 100 or 2 24. $\tan^{-1}\left[\frac{1 - r^2}{r}\cdot\frac{2\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2}\right]$
24. 0 25. Two regression coefficients 26. perpendicular
27. greater 28. ± 1 29. $r = -\frac{8\Sigma d_1^2}{n^3 - n}$
30. Coefficient of standard deviation 31. zero
32. −1 and 1 33. −2.5 34. False
35. True 36. False

Problems 2.3.5 (page 083)

1. 4896 2. 360 ; 120 3. 120 ; 116 4. (a) uncorrelated (b) statistically

Problems 2.3.7 (page 088)

1. (i) 0.15 (ii) $P(A/B) = 0.6$, $P(A \cup B) = 0.72$, $P(A' \cap B') = 0.28$
2. (a) 1/66 (b) (i) 2/3. Yes 3. 35 : 33 : 28
4. 2/5 5. (i) 2/3, (ii) 2/3, (iii) 1/18 6. 20/1024

3. 3/28 8. 3/61 9. (i) 3/10 (ii) 3/10 ; (iii) 2/10 (iv) 2/5
10. 0.11 12. (i) 0.729 ; (ii) 0.036 13. (i) 0.134 ; (ii) 0.0021143
13. 3/80 14. 3/91 15. $1 - (1 - p_1)(1 - p_2)(1 - p_3)$; 0.216
16. 10/17 17. 1/5 18. 5/12 19. (i) 63/130 (ii) 25/65
20. $1 - 2(n - 1)$ 21. 7/20 22. (i) 1/6 ; (ii) 5/324 23. 81/85
24. 0.72 28. 0.2353 26. 1/36

Problems 3.3 (page 107)

1. 3/11 2. 35/52, 35/51, 15/52 3. 0.1175, 0.0254 4. 15/34.

Problems 3.4 (page 111)

1. $k = 1$; $\mu = 3$, $\sigma^2 = 2.222$ 2. 3/5
3. $P(x) = 0$, $x = a + b$
$= 1/6$, $0 \le x \le 1$
$= 1/3$, $1 < x < 2$
$= 7/6$, $2 \le x \le 3$
$= 1$, $3 \le x < \infty$ 4. 7/30

6. 2 7. $f(x)$ is a p.d.f. $\mu = \frac{1}{2}$, $\sigma^2 = \frac{1}{80}$ 8. (i) 3/16, 5/16 ; (ii) $k = 0.45$

9. (i) 0.37 ; (ii) 0.63 10. 4/9
12. $p_x = 2^x$; Mean = 1 ; Variance = 1/3. 13. 0.2.
14. 2/3 ; 2/3 15. $P(x) = 0$, $x \le x_1$; $(n - x_1)/n$, $x_1 \le x < x_2$; $(n - x_2)/n$, $x_2 \le x < x_3$

Problems 3.5 (page 115)

1. $x = 1$, $p = q = \frac{1}{2}$; $\frac{15}{16}$ 2. ${}^{5}C_x (1/3)^{5-x} (2/3)^x$; $x = 0, 1, 2, 3, 4$
3. (i) 0.000378 ; (ii) 0.81351 ; (iii) 3.938×10^{-7} 4. 0.3439
5. 0.0229880000 6. (i) 0.266 ; (ii) 0.348

7. (i) ${}^{20}C_3 (1/2)^{20} (1/2)^{17}$; (ii) $\sum_{r=0}^{3} {}^{20}C_r (1/2)^r (1/2)^{20-r}$ (iii) 20

8. (i) 0.36 ; (ii) 0.50 ; (iii) 0.32 9. (i) 3/20 ; (ii) 3/5 ; (iii) 3/20
10. (i) 0.36419 ; (ii) 0.32768 ; (iii) 0.01143 11. 11
12. 99/50 13. 0.0355, 0.3731, 0.3907 ; 0.0006
14. 0.37 15. $100 (0.25 + 0.75^2)$
16. $500 (0.654 + 0.445^2)$.

Problems 3.6 (page 120)

1. (i) 2 ; (ii) $\frac{2}{3} e^{-2}$ 2. $P(0) = 0.2636$, $P(1) = 0.3543$, $P(> 2) = 0.1506$
4. $(2e)^2 e^{-2}/16.1 = 0.035$ 4. 0.08
6. (i) 0.2231 (ii) 0.1912 7. 0.0000
8. $m = 0.91 = \sigma^2$; Poisson frequencies of 0, 1, 2, 3, 4 accidents are 183.1, 81.31, 35.16, 4, 0.9
9. 0.6 11. Theoretical frequencies are 46, 43, 21, 7, 1
12. Theoretical frequencies are 108, 140, 91, 39, 10, 1, 1, 0, 0, 0, 0, 0.

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98. [illegible]
99. [illegible]
100. [illegible]

79. $-\frac{1}{2}$ 80. $\frac{e^{-x}x^{n-1}}{\Gamma(n)}, 0 < x < \infty$ 81. $\frac{1}{3}(b-a)$

82. 0.99 83. 3/8 84. 1 85. 3/4

86. (a) 87. $A(x) = \int_{-\infty}^{x} f(x)\,dx$ 88. $\frac{(2\gamma)^{n-1}}{(n-1)!}$ 89. $-\infty < x < \infty$

90. 6 91. 1/6 92. False 93. 2

94. $4(1-e^{-1})$ 95. 0.266

Problems 27.1 (page 931)

1. Die is biased 2. No 3. Yes

4. 8.81% and 35.07% 5. Coefficient 6. $p = 0.2640$, S.E. $= 0.015$

7. 37.6%; 34.8 and 44.7 respectively 8. No

9. Difference is not significant

10. $Z = 0.56$ so that the difference is insignificant 11. No

Problems 27.2 (page 934)

1. No 2. Mean weight lies between 63.8 and 68.4 lbs

3. 0.0734 4. 62 5. 0.636 6. No

7. No 8. (i) Yes, (ii) No 9. Yes

Problems 27.3 (page 938)

1. 0.30 2. $t = 0.82$, Yes 3. 11.057 and 12.153 cm

4. Pool the data 5. Process is not under control 6. No

7. Sample mean = 170.3 kg., S.E. = 2.78 kg 8. Accept null hypothesis

9. Yes with 75% confidence. 10. No 11. No

12. No

Problems 27.4 (page 944)

1. 0.43 2. Hypothesis is correct

3. Significant at 5% level 4. Yes

5. f_i: 33.82 162.79 315.89 333.68 158.56 55.4

$\chi^2 = 7.97$. Binomial distribution gives a good fit at 5% level

6. f_i: 26.8 106 210 88 18 12

f_i: 20.2 103.4 216.6 88.7 36 7.9; $\chi^2 = 3.897$

Poisson distribution gives a good fit at 5% level

7. f_i: 211 305 204 53 20 12

f_i: 206 304.2 215.3 88.8 26.1 6.5

Poisson distribution can be fitted to the data

8. $\chi^2 = 1.2$. The fit is quite good at 5% level.

Problems 27.5 (page 947)

1. First variance cannot be regarded as significantly greater than the second

2. Not significant as $F = 2.1$ and $F_{0.05} = 4.18$ 4. Not significant as $F = 2.4$ and $F_{0.05} = 3.2$

5. Product of firm B cannot be said to be of better quality than that of firm A.

6. Not significant at 1% level as just significant at 5% level as $F = 3$, $F_{0.05} = 3.00$ and $F_{0.01} = 3.01$
7. $F = 1.19$; Not significant 8. $F = 1.905$; Yes

Problems 27.3, page 980

1. $27.50 2. $27.15 3. $27.0151
4. We are testing the hypothesis that one process is better than another
5. $27.11 6. 3 7. 80 8. $(3, \infty)$
9. Ω 10. $\emptyset$; Ω 11. $x = x - 3$ 12. False
13. Zero from 50 14. $27.17 15. $-\infty < x < \infty$ 16. Ω
17. True

Problems 28.1, page 988

1. (1) 2.087, (2) 1.45, (3) 2.831, (4) 2.015 2. (1) 0.519, (2) 2.015, (3) 1.146, (4) 0.260
3. (1) −0.686, (2) 2.705, (3) 0.686, (4) 2.4088
4. (1) 0.903, (2) 0.400, (3) 2.306, (4) 2.776, (5) −0.903
5. 1.891 6. (1) 3.250, (2) 2.093, (3) 1.328, (4) 1.328
7. (1) 1.895 (2) 2.155 (3) 1.852 8. −16.06
9. (1) 0.903, (2) −1.8331, (3) 2.7638, (4) 4.043
10. (1) 0.258, (2) 0.053, (3) 0.395, (4) 3.311 11. $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$; (2) 3.805 (3) 3.302
12. 3.4852 13. 2.3084
14. (1) 0.056 (2) 0.258 (3) 0.4343

Problems 28.2, page 998

1. (1) 1.324, (2) 0.684, (3) 2.16, (4) 1.184 2. 1.6914
3. 2.831 4. −1.355 5. 2.584

Problems 28.3, page 1008

1. $x = 7, y = -5, z = 6$ 2. $x = -3.14, y = 1.195, z = 3.96$ 3. $x = 1, y = 2, z = 3$
4. $x_1 = 2, x_2 = -1, x_3 = 3$ 5. $x_1 = 1, x_2 = 2, x_3 = -1, x_4 = 3$ 6. $x = 1, y = 2, z = 3$
7. $x = 5.7, y = 8.3, z = -1.3$ 8. $x = 1, y = 2, z = 3$ 9. $x = 2, y = -3, z = 6$
10. $x_1 = 2, x_2 = 0.5, x_3 = 0.5, x_4 = 4.5$ 11. $x = y = z = 1$ 12. $x = 1, y = 2, z = 3$
13. $x = 3.916, y = 2.215, z = 5.28$ 14. $x_1 = -1, x_2 = 0, x_3 = 1, x_4 = 2$ 15. $\begin{bmatrix} 1.2 & -0.4 & 0.2 \\ -0.2 & -0.1 & 0.3 \\ -0.1 & 0.3 & 0.1 \end{bmatrix}$

Problems 28.4, page 1018

1. $x = 2.068, y = 1.722, z = -2.068$
2. (1) $x = 2.426, y = 3.573, z = 1.536$ (2) $x = 2.668, y = 3.033, z = 1.968$
3. $x = 1, y = 2, z = 3$ 4. $x = 0.966, y = 3.725, z = 2.034$
5. $x = 1.062, y = 1.559, z = 1.965$ 6. $x = -13.225, y = 16.765, z = -2.306$
7. $x = 1, y = 2, z = 3, w = 4$ 8. $x = 1.98, y = 3.103, z = 2.812$
9. $x = y = z = 1$ 10. $x = 28.5, y = 44.5, z = 58.7$
11. $x = 1.98, y = 3.09, z = 2.12$

Problems 8.1.5, page 343

1. $x = 2, y = 1$
2. $x = -1.853, y = -1.067$
3. $x = 0.7874, y = 0.8268$
4. $x = -0.131, y = 2.862$
5. $x = 0.185, y = 0.43$

Problems 8.2.5, page 349

1. (a) 0.638, $\begin{bmatrix} 0.45 \\ 1 \\ 0.8 \end{bmatrix}$; (b) 4.416, $\begin{bmatrix} 1 \\ 0.618 \end{bmatrix}$
2. 3.47; $[0.74, -1, 0.57]^T$
3. (a) 6, $[1, 1, -1]^T$ (b) 8, $[1, -0.5, 1.5]^T$
4. (a) 7; $[0.0667, 0.4667, 1]^T$ (b) 12.8, $[0.2, 1, 0.4375, 0.0625]^T$ (c) 11.84, $[0.855, 0.422, 1.000]$.

Problems 8.3, page 360

1. Newton-Raphson method
2. $x_n = x_{n-1} - \dfrac{f(x_{n-1})}{f'(x_{n-1})}$
3. Closed AB
4. $x_{n+1} = \dfrac{1}{2}\left(x_n + N/x_n\right)$
5. initial approximation x_0 is chosen sufficiently close to the root
6. diagonal
7. (c)
8. (a)
9. $x_{n+1} = \dfrac{1}{2}\left(2x_n + N/x_n^2\right)$
10. $x_{n+1} = \dfrac{1}{2}\left(x_n + x_{n-1}\right)$
11. $x_3 = x_2 - \dfrac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$
12. (a)
13. $x_{n+1} = x_n(2 - Nx_n)$
14. (b)
15. Newton-Raphson method
16. § 8.6.6
17. Upper triangular matrix
18. False
19. True
20. $x = 1, y = 1$.

Problems 8.4.11, page 368

1. 2.4
2. -0.828
3. 2.29
4. 4.48, 2.24, 0.8, 0.448
5. (a) $1 - \frac{1}{2}\sin 2x + \frac{1}{20}\sin 3x$; (b) $\tan^{-1}(0.2\sin^2 x)$;
   (c) $\frac{1}{360}(x+4)(x+3)(x+2)(x+1)x + \frac{1}{18}(x+3)(x+2)(x+1)x + \frac{1}{4}(x+2)(x+1)x + (x+1)x$
6. (a) $e^{2x}(x^2\log(1+2x) + x^2 - 1)\log 2$; (b) $(x+1)^2(3-x)(1+x)$
   (c) $4x - 1^2x^2$; (d) $(x+1)^2(x-1)^2/(2!2!) + 2(x+2)_{-1}(x+1)_3$.
7. (a) -30; (b) $36 + 2^{10} + 101$
8. $x = x^3 - 6x^2 + 13x^2 + x + 9$
9. $4x^3 - 12x^2 + 8x + 1$; $13x(x-1)$
10. $\frac{1}{12}x^3 + 0x^2 + 4x + c$
11. $y(4) = 74, y(5) = 201$
12. 15

Problems 8.5.5, page 383

1. $\left[\dfrac{\Delta^2}{h^2}\right]u_x = u_{x+2} - 2u_{x+1} + u_x$; $\dfrac{d^2y_x}{dx^2} = \dfrac{y_{x+2} - 2y_{x+1} + y_x}{h^2}$
2. (a) $2(\cos h - 1)\cos x$; (b) $\frac{1}{2}x$; (c) $120\cos h - 1)[\cos(x+h)+1]$; (d) 8.
3. Error = 10
4. 21
5. $f(1.5) = 3.375, f(5) = 25.062$
6. $y(4) = 74, y(5) = 201$
7. -10
8. $y_2 = 1$ approx
9. (a) $\frac{1}{2}(3x^2 + 6x + 1)$; (b) $\dfrac{3h^2 + 2(1x + 2)(x + 2)}{4}$
10. $2(1 - x^2)$.

Problems 20.1 (page 556)

1. 5.64
2. 0.35
3. 1.1313
4. 0.756
5. ₹ 110.62
6. 9664
7. 332, 329
8. 0.9823, 0.0306
9. 24 approx
10. $f(x) = 8x - 4x^2$
11. 1.635
12. 0.1958
13. 4.229
14. 2558
15. $x_1 = 0.1, y_{10} = 300$
16. $x_5 = 45, x_6 = 49$
17. 18, 22
18. 756.

Problems 20.2 (page 563)

1. 19.4
2. 19.606
3. 84000
4. 3.8219
5. 0.0078
6. 306
7. 3.267
8. 8
9. 3268.876
10. 2.8284 by all formulae.

Problems 20.3 (page 574)

1. 16.95
2. 2.8168
3. 0.89
4. 300
5. $600 + 20x - x^2$
6. $x^3 - 3x^2 + 3x - 6$
7. $x^3 - 6x^2 + 16x^2 - x^2 + 20x - 12$
8. 3
9. $\frac{0.5}{x-1} - \frac{3.5}{x+1} + \frac{1}{x-2}$

Problems 20.4 (page 577)

1. 3
2. 3189
3. 618.3184
4. 332.18
5. $f(x) = \frac{1}{24}x^4 - 2x^3 - 24 - \frac{1}{6}x^3 + \frac{26^2}{90}x - 32.6$
6. $f(x) = \frac{1}{60}x^5$
7. $f(x) = x^4 - 5x^3 + 5x^2 - 1$
8. 31.

Problems 20.5 (page 579)

1. 11.6
2. 0.6908
3. 37.59

Problems 20.6 (page 583)

1. 4.7.3
2. (i)
3.

x	$f(x)$	I D.D.	II D.D.
3	7		
		29	
10	86		0.91
22	160	12.7	

4. Intermediate value of the variables.
5. 37.8
6. $\frac{(x_1 - x_2)(x_1 - x_3)y_0 + (x_0 - x_1)(x_0 - x_2)y_3}{x_3 - x_2}$
7. $-\frac{1}{4}$ and $\frac{3}{4}$
8. 37.14
9. $f(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$
10. $\frac{13}{5}$
11. (i)
12. 1.667
13. Abscissa is the process of estimating the value of a function outside the given range of values.
14. $(x+1)(x+2)$
15. (a)
16. $x^3 - 3x^2 - 16x - 12$
17. (b)
18. $(x^2)(x+3)$

Problems 23.1, page 807

1. $-37.0, 113.0^\circ$
2. 4.53, 0
3. 0.63, 0.0
4. (a) 0.426, -1.165 (b) 0.4473, -0.1865; (c) 0.4062, -0.0643
5. 0.0326
6. $-0.08, 0.5$
7. 0.135
8. 13.33 meters
9. (a) $-59.4, 0.0 - 0.0191$
10. 44.93
11. 0.095
12. 3.92 radians, 6.92 radians^2
13. 0.3361
14. 0.0586
15. 138
16. $x_{max}(1) = 0.38$, $x_{min}(0) = 0$
17. $M_{max}(10.04) = 1040.00$.

Problems 23.2, page 825

1. (a) 0.926 (b) 0.926 (c) 0.653
2. (a) 0.7854 (b) 0.7854, (c) 3.7854, (d) 0.7854
3. 1.91
4. -8.48
5. (a) 16.39 (b) 0.024
6. 0.6285
7. (a) 0.0909; (b) 1.1473
8. (a) 1.1240 (b) 0.891
9. (a) 1.8271831, .0001824; (b) 1.8278478, .0000063; (c) 1.8278470, .0000035; (d) 1.8278474, .0000001
10. 1.9999
11. 400.67
12. 3.75
13. 710 sq ft
14. 3.022
15. 400.0 cub. cm.
16. 1.063 sec; 1.063 sec
17. 500 m; 3 m/s^2
18. 30.87 m/sec
19. 20 cu.s nearly.

Problems 23.3, page 836

1. (a)
2. $\frac{1}{6}\left[y'(a) - \frac{1}{2}h^2 f(a) + \frac{1}{3}h^3 f(a) - \cdots\right]$
3. A rule should be used
4. 0.795
5. $\frac{2}{3}\frac{\pi}{3}$
6. 20.8
7. (a)
8. Larger number of sub-intervals
9. 0.7854
10. 60
11. $y'_x \frac{1}{h} = \left[\nabla y_x + \frac{1}{2}\nabla^2 y_x + \frac{1}{3}\nabla^3 y_x + \cdots\right]$
12. a multiple of 6
13. 60
14. 0.761
15. 0.69
16. 1.08138
17. 1.36
18. If the entire curve is itself a parabola
19. even and multiples of 3
20. False

Problems 23.4, page 839

1. $y_{x+2} - 10y_{x+1} + 9y_{x-1} = 0$
2. $\Delta y_x = (-1)^x \Delta y_0$
3. $u_{x+1} - 2u_x = 3$
4. (a) $\Delta^2 y - 2\Delta y_{x-1} - 3\Delta^2 y + 4y_{x+1} + 3y_x = 0$; (b) $\Delta^3 y + 3\Delta y_{x-1} - 6\Delta^2 y_{x-1} + 6\Delta y_{x-1} + \Delta^3 y_{x-1} + 2y_x = 0$
5. (a) $y_{x+2} - 8y_{x+1} + 16y_x = 0$; (b) $y_{x+2} - 8y_{x+1} + 4y_x = 0$
6. (a) $\Delta^2 y - 2\Delta y_{x-1} - 6\Delta^2 y - 2\Delta^2 y_{x-1} + 2y_x = 0$; (b) $y_{x+2} - 4y_x = 0$;
   (c) $y_{x+3} - 9y_{x+2} + 24y_{x+1} - 16y_x = 0$.

Problems 23.5, page 1000

1. $u_p = (c_1 + c_2 p)3^p$
2. $y_p = c_1 \cos\frac{2\pi p}{3} + c_2 \sin\frac{2\pi p}{3}$
3. $u_p = c_1 \cos\pi p/2 + c_2 \sin\pi p/2$
4. $y_p = c_1 \cdot 2^p + c_2 \cdot 3^p$.
5. $y_p = (2^{p-1} + 1 - 2^{p-1})$
6. $u_p = c_1 1 - 2^p + c_2 + c_3 p 2^p$.
7. $f(x) = y_p + c_1(1 - 2^p + c_2 \cdot 2^p$
8. $u_p = 2p + 1 - 2^p$.
9. $y_p = 6^p + (p - 3)2^p$
10. $u_p = 2^{p/2}(c_1 \cos p\pi/4 + c_2 \sin p\pi/4)$
11. $y_p = 2^p \left\{c_1 \cos\frac{p\pi}{4} + c_2 \sin\frac{p\pi}{4} + c_3 \cos\frac{3p\pi}{4} + c_4 \sin\frac{3p\pi}{4}\right\}$
12. $y_p = c_1(-1)^p + c_2(3)^p$

Problems 11.2, page 1004

1. $y_n = c_1(-1)^n + c_2 3^n - 2n/3$
2. $y_n = \left(\frac{n}{20} - \frac{1}{20}\right)(-3)^n + \frac{3^n}{25}$
3. $y_n = c_1 + c_2 p + c_3 p^2 + \frac{2}{6} p(p-1)(p-2)$
4. $y_n = 2^n\left(\frac{2}{\sqrt{3}} \sin\frac{\pi n}{3} - 2\cos\frac{\pi n}{3}\right) + 2$
5. $y_n = c_1 + c_2 \cdot 2^n + \frac{1}{2} n \cdot 2^{n-1}$
6. $y_n = c_1 + c_2(-1)^n + 3n/4 - 3/16 - 2 \cdot 4^{n-1}$
7. $y_n = c_1 + c_2(-2)^n + \frac{2}{3}\frac{\cos\left[\frac{\pi}{2}(n-1)\right] - \cos\frac{\pi}{2}}{1-\cos\frac{\pi}{2}}$
8. $y_n = c_1 \cos\frac{\pi n}{2} + c_2 \sin\frac{\pi n}{2} + \frac{\cos\left(n-\frac{3}{2}\right)}{2\sin\frac{1}{2}}$
9. $y_n = c_1 + 3^n + c_2(-2)^n - \frac{1}{25}(5n^2 + 2n + 11)$
10. $y_n = c_1(-1)^n + c_2 \cos\frac{n\pi}{2} + c_3 \sin\frac{n\pi}{2} + \frac{n}{3} - \frac{1}{2}\cos\frac{n\pi}{2}$
11. $y_n = c_1 + c_2(-1)(2)^n + c_3 2^n - 3^n + \frac{3}{8}(2^n - 2^{n-3})$
12. $y_n = c_1 + c_2(-1)^n + c_3 \cdot 2^n + \frac{2^n}{5} + n(n-1)\left(\frac{1}{2}\right)^{n-1}$
13. $y_n = c_1(-2)^n + c_2 \cdot 3^n + \frac{8}{15} - \frac{7}{144}$
14. $y_n = c_1 + c_2(-1)^n + \frac{2^n}{20}(3n-2) + \frac{2}{4}n^{n-1} + \frac{7}{16}$
15. $y_n = c_1(-2)^n + 2^n[c_2 \cos n\pi/2 + c_3 \sin n\pi/2] + \frac{3}{10}(2^n + 2^{n-1}(2n+2))$
16. $y_n = \left[c_1 + c_2 n + \frac{1}{48} n(n-1)^2(n-2)\right]2^n$
17. $y_n = c_1 \cdot 2^n + c_2 \cdot 3^n + \frac{n^2}{2}(3^n - 12n + 51)$
18. $y_n = 2^n\left[c_1 + A\cos\frac{n\pi}{2} + c_2 \sin\frac{n\pi}{2}\right]$.

Problems 11.3, page 1008

1. $y_n = a + b(-2)^n + x \cdot z_n$, $z_n = a + b(-2)^{n-1} - 4z_n + 2$
2. $y_n = [a + bn](-1)^n - \frac{3}{8} \cdot 3^{n-1}$, $z_n = \frac{3^n}{8} - (-3)^n(3n + 6)x - \frac{1}{2}$
3. $w_n = 2 \cdot 4^n - 2\frac{1}{2}\sin n - 11$, $z_n = 4^n + 2 \cdot \frac{1}{n} + \sin n + 11$
4. $y_n = -2n + 84 - 2^n - x + \frac{1}{2x}(2 - x)x$, $z_n = a + c + b(-2)^n + \frac{1}{2}\cos n - 1$.

Problems 11.5, page 1020

1. $y_{n+1} - 2y_n + y_{n-1} = -\frac{h}{p}y_n$. Solve it for y_n.

Problems 11.6, page 1030

1. $y_{n+2} - 2y_{n+1} + 5y_n = 0$
2. $y_n = c_1 + c_2 n + c_3 n^2$
3. $y_n = c_2 + c_1(-2)^n + c_3 3^n$
4. $y_n = c + 2^n$
5. $y_n = c(2)^n - 4n + 1$
6. $(a^2 + a)y_{n+2} - (2a^2 + 4a)y_{n+1} + (a^2 + 3a + 2)y_n = 0$
7. $y_n = c(2)^n + 1$
8. $y_n = 12^{n+1} + c - 2^{n-1}$
9. Blank

10. $(n+2)y_{n+2} - 2(n+1)y_{n+1} + ny_n = 0$ 11. Second

12. $y_n = (C_1 + C_2 n)2^n$ 13. $\frac{1}{2}y_{n+2} - 3(1.5)^{n+2}$ 14. $y_{n+2} - 6y_{n+1} + 9y_n = 0$

15. True.

Problems 33.1, page 1013

1. $y = 1 - \frac{x^2}{2} + \frac{x^4}{8} - \frac{x^6}{48}$ 2. 0.0214 3. $y = \frac{1}{2}x^2 - \frac{1}{24}x^4 + \cdots$

4. $y(0.5) = 0.5128, y(1.0) = 0.1536$

5. $y(0.1) = 0.1105, y(0.2) = 0.04824$. Result $y(0.3) = 0.1183$ and $y(0.4) = 0.2428$

6. 1.006383 7. 1.3272 8. 1.106.

Problems 33.2, page 1017

1. 0.3681048 2. 1.1448 3. 4.5569

4. $y(0.1) = 1.0080, y(0.2) = 1.0321, y(0.3) = 1.0728$ 5. $y(0.2) = 1.2048, y(0.4) = 1.4864$

6. 2.2362 7. 1.0928 8. 1.351.

Problems 33.3, page 1021

1. 1.1218 2. 1.1748 3. 1.1207, 1.0428

4. 2.1009 5. $y(0.1) = 1.0962, y(0.2) = 1.1918$

6. $y(0.1) = 2.0810, y(0.2) = 2.0827$ 7. 1.1467 8. 1.1169

9. 1.1574 10. 1.0812, 1.1871, 1.2532, 1.2868, 1.334.

Problems 33.4, page 1026

1. 1.208 2. 1.2707 3. $y(1.4) = 1.0784$

4. $y(1.5) = 1.1621$ 5. $y(0.4) = 1.368$ 6. $y(1.1) = 0.861$.

Problems 33.5, page 1030

1. 0.9128 2. 1.0468 3. 0.8367

4. $y(1.6) = 1.009$ 5. 1.5703 6. $y(1.4) = 0.945$.

Problems 33.6, page 1034

1. $y_2 = 1 + \frac{1}{2}x + \frac{3}{40}x^5 - \frac{1}{48}x^6 - \frac{1}{196}x^8$, $y_3 = \frac{1}{2}x + \frac{3}{8}x^3 + \frac{1}{10}x^5 - \frac{3}{64}x^7 + \frac{7}{360}x^9 + \frac{1}{256}x^{11}$

2. $y(0.1) = 0.105, z(0.1) = 0.005$; $y(0.2) = 0.22, z(0.2) = 0.02$

3. $y(0.1) = 0.0948, z(0.1) = 0.0997$ 4. $y_2 = 1 + \frac{1}{2}x + \frac{3}{40}x^5$

5. $y(0.1) = 0.0975$ 6. $y(0.2) = 0.0963, y(0.3) = -0.090$

7. -0.5189 8. $y(0.2) = 0.0967, (\text{error})_2 = 2.8\times10^{-5}$

Problems 33.7, page 1038

1. 0.14631 2. $y(0.2) = y(0.3) = 2.6, y(0.4) = 3.2$

3. $y(1.25) = 1.3513, y(1.5) = 1.625, y(1.75) = 1.9506$

4. $y(1.25) = -0.03472, y(1.5) = -0.08594, y(1.75) = -0.1587$

5. $n = 2, y(0.5) = 0.1368$. true value $= 0.1465$; $n = 4, y(0.5) = 0.147$

6. $y(0.2) = 0.082, y(0.4) = 0.33, y(0.6) = 0.868$
7. $y(1) = 1.0171, y(2) = 1.091$ 8. $y(0.2) = 1.1003, y(0.4) = 1.1023, y(1) = 1.4818$

Problems 32.2 (page 1000)

1. (c) 2. $y_{i+2} - 2y_i + y_{i-2} = h^2 y_i''$ 3. $1 + x + x^2 + x^3/3$
4. 3.014 5. 2.013 6. (a)
7. $y_4 = y_2 + \frac{h}{3}(y_2 + 4y_3 + y_4)$ 8. Modified Euler's method
9. $y_4 = y_3 + \frac{h}{24}(55y_3 - 59y_2 + 37y_1 - 9y_0)$
10. Milne's method and Adam-Bashforth method
11. four 12. $y = 1 + \frac{x^2}{2} + \frac{x^3}{6}$ 13. (b)
14. $y_4 = y_2 + \frac{4h}{3}(2y_3 - y_2 + 2y_1)$ 15. $y_4 = y_3 + \frac{h}{24}(9y_4 + 19y_3 - 5y_2 + y_1)$
16. 1.1035 17. $\frac{dy}{dx} = y\frac{dy}{dx} + y(1 + y) = 0$ 18. 1.1017
19. starting values
20. Picard's and Runge-Kutta methods
21. It agrees with Taylor's series solution upto the term in h^3
22. (a) 23. $y_{i+1} + 2y_i^2 - 3y_i + y_{i-1} = 0$ 24. (a)
25. True 26. False 27. False

Problems 33.1 (page 1041)

1. Parabolic 2. Hyperbolic
3. (a) Parabolic (b) Elliptic (c) Elliptic 4. Outside the ellipse $(x/3)^2 + (y/2)^2 = 1$

Problems 33.2 (page 1050)

1. $u_1 = 1.000, u_2 = 2.000, u_3 = 3.000$
2. 0.00, 0.48, 0.80, 0.96, 1.00, 0.96, 0.80, 0.48, 0.00
3. $u_1 = 10.156, u_2 = 0.6, u_3 = 1.356, u_4 = 0.55, u_5 = 0.035, u_6 = 1.25$
4. $u_1 = 20.00, u_2 = 35.00, u_3 = 45.00, u_4 = 40.00$
5. $u_1 = 0.50, u_2 = 1.00, u_3 = 0.50$
6. $u_1 = 1.57, u_2 = 2.72, u_3 = 3.57, u_4 = 3.86, u_5 = 4.66, u_6 = 6.50, u_7 = 5, u_8 = 4.00, u_9 = 3$
7. $u_1 = -3, u_2 = -3, u_3 = -2$

Problems 33.3 (page 1064)

1.

j \ i	0	1	2	3	4
4	0	2	4	6	8
3	0	1	3	5	7
2	0	1.0	2	4.0	6
1	0	1	1.0	3	5
0	0	0.0	1	2.0	4
	0	0	0.0	1	2

2.

t / x	0	1	2	3	4	5	6	7	8	9	10
0	0	0.09	0.18	0.27	0.36	0.45	0.54	0.63	0.72	0.81	0
1	0	0.068	0.135	0.203	0.271	0.338	0.406	0.473	0.541	0.608	0
2	0	0.0473	0.095	0.142	0.189	0.236	0.284	0.331	0.378	0.426	0
3	0	0.027	0.0533	0.080	0.107	0.133	0.160	0.187	0.213	0.240	0

3.

t / x	0	1	2	3	4	5
0	0	64	96	160	192	0
1	0	48	96	128	112	0
2	0	32	80	96	72	0
3	0	16	64	80	48	0
4	0	8	32.00	48.00	28.00	0
5	0	20.0	36.00	40.00	24.00	0
6	0	18.00	32.00	37.00	22.00	0

4.

t / x	0	1	2	3	4
0	0	0.5	1	0.5	0
1	0	0.5	0.5	0.5	0
2	0	0.25	0.5	0.25	0
3	0	0.25	0.25	0.25	0

Problems 1201 (page 1302)

1.

t / x	0.1	0.2	0.3	0.4	0.5
Explicit	0.00	0.01	0.06	0.0503	0.20
Implicit	0.00	0.01	0.06	0.0503	0.20

3.

t / x	0	1	2	3	4	5
0	0	20	10	10	5	0
1	0	15	10	10	5	0
2	0	−5	−2.5	10	5	0
3	0	−5	−10	−2.5	5	0
4	0	−5	−10	−15	−7.5	0
5	0	−5	−10	−15	−20	0

3.

t / x	0	1	2	3	4	5	6	7	8	9	10
0	0	0.09	0.18	0.26	0.34	0.42	0.50	0.58	0.66	0.74	0
1	0	0.09	0.18	0.26	0.34	0.42	0.50	0.58	0.66	0.74	0
2	0	0.09	0.18	0.26	0.34	0.42	0.50	0.58	0.66	0.74	0
3	0	0.09	0.18	0.26	0.34	0.42	0.50	0.58	0.66	0.74	0
4	0	0.09	0.18	0.26	0.34	0.42	0.50	0.58	0.66	0.74	0
5	0	0	0	0	0	0	0	0	0	0	0

4.

t / x	0	0.2	0.4	0.6	0.8	1.0
0.0	0	0.307	0.47	0.865	1.135	0.135
0.2	0	0.201	0.369	0.865	1.001	0.001
0.4	0	0.102	0.042	0.867	0.897	0.098
0.6	0	0.002	0.003	0.001	0.997	0.004
0.8	0	0	0	0	0	0

Problems 14.1, page 1045

1. (b)
2. False
3. a hyperbolic equation
4. Poisson's equation
5. $u_{i,j+1} = u_{i+1,j} + u_{i-1,j} - u_{i,j-1}$
6. $\frac{1}{k}(u_{i,j+1} - 2u_{i,j} + u_{i,j-1}) = u_{i+1,j} - 2u_{i,j} + u_{i-1,j}$
7. $\frac{1}{4}(u_{i+1,j+1} + u_{i+1,j-1} + u_{i-1,j+1} + u_{i-1,j-1})$
8. hyperbolic
9. Bender-Schmidt
10. $u_{i,j+1} = 2(1 - 6r^2)u_{i,j} + 6r^2(u_{i+1,j} + u_{i-1,j}) + u_{i,j-1}$ where $r = k/h$.
11. $u_{i,j-1} = \frac{1}{2}(u_{i-1,j} + u_{i+1,j})$
12. 303.303
13. $\lambda = \frac{1}{2}$
14. Elliptic
15. $u(0, t) = 0 = u(1, t), t > 0$; $u(x, 0) = 8x(1 - x), 0 < x < 1$; $\partial u/\partial t(x, 0) = 0, 0 < x < 1$
16. $u_{i,j+1} = u_{i+1,j} - u_{i-1,j} - u_{i,j-1}$
17. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$
18. 0.80
19. $\frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{k^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2} = 0$
20. $y = 0$
21. $k = 1/4$

Problems 14.2, page 1063

1. Max $Z = 1.2x_1 + 1.4x_2$; subject to $40x_1 + 25x_2 \le 1800$, $30x_1 + 50x_2 \le 1800$, $20x_1 + 30x_2 \le 900$ and $x_1, x_2 \ge 0$
2. Max $Z = 3x_1 + 2x_2 + 4x_3$; subject to $4x_1 - 3x_2 + 5x_3 \le 3000$, $2x_1 + 3x_2 + 4x_3 \le 5000$, $300 \le x_1 \le 550$, $200 \le x_2$ and $50 \le x_3$
3. Max $Z = 3x_1 + 5x_2 + x_3$; subject to $3x_1 + 4x_2 - 2x_3 \le 42$, $6x_1 + 3x_2 \le 45$, $2x_2 + 4x_3 - 3x_2 \le 61$ and $x_1, x_2, x_3 \ge 0$.
4. Max $Z = 600x + 300y$; subject to $x + y \le 500$, $x \ge 50$, $y \ge 40x$, $x \ge 0$, $y \ge 0$
5. Min $Z = x_1 + x_2$; subject to $2x_1 + x_2 \ge 12$, $3x_1 + 4x_2 \ge 24$, $x_1 + 4x_2 \ge 24$ and $x_1, x_2, x_3 \ge 0$.

6. Max. $Z = 0.15x_1 + 0.25x_2$; subject to
$3x_1 + 5x_2 \le 600{,}000$, $x_1 + 4x_2 \le 320{,}000$,
$8x_1 + 15x_2 \le 960{,}000$, $0 \le x_1 \le 85{,}000$ and $0 \le x_2 \le 75{,}000$
7. Min. $Z = 4x_1 + 3x_2 + 9x_3$; subject to
$2x_1 + 3x_2 + 5x_3 \ge 1250$, $x_1 + x_3 \le 350$, $4x_1 + 2x_2 \le 900$,
$4x_1 + 2x_2 + x_3 \ge 1225$ and $x_1, x_2, x_3 \ge 0$
8. Min. $Z = 100x_1 + 200x_2 + 300x_3$; subject to
$0.3x_1 + x_2 + 1.0x_3 \le 0.9x$, $10x_1 + 15x_2 + 27x_3 \ge 3x$,
$400x_1 + 500x_2 + 650x_3 \ge 600$, $x_1 + x_2 + x_3 = 1$ and $x_1, x_2, x_3 \ge 0$

Problems 3.4.2, page 1060

1. $x_1 = 300, x_2 = 60$: max. $Z = 660$
2. $x_1 = 6.15, x_2 = 12.5$, max. $Z = 24.5$
3. $x_1 = 35, x_2 = 0$; max. $Z = 280$
4. $x_1 = 1050, x_2 = 600$; max. $Z = 5850$
5. 450 units of product B only : max. profit = ₹ 1800
6. $X = 5, Y = 4.5$; max. profit = ₹ 85
7. $A = 1.33$ units, $B = 0.33$ units : max. profit = ₹ 14.00 approx.
8. 2000/11 units of product A and 1000/11 units of B ; max. profit = ₹ 11,000
9. $x_1 = 4, x_2 = 5$: max. $z = 8$
10. Unbounded solution
11. $x_1 = 2, x_2 = 4$; min. $Z = 44$
12. Production cost will be min. if 6 and 3 runs for 12 and 4 days respectively.

Problems 3.5.1, page 1074

1. Max. $Z = 3x_1 + 5x_2 + 8x_3$; subject to $2x_1 - 3x_2 + s_1 = 3$,
$2x_1 + 3x_2 + s_2 - s_3 = 8$, $3x_1 + 4x_2 + s_3 = 2$;
$x_1, x_2, x_3, s_1, s_2, s_3 \ge 0$
2. Max. $Z = 2x_1 + 3x_2 + 5x_3$ subject to $-3x_1 + 2x_2 + s_1 = 3$,
$2x_1 + 2x_2 + 4x_3 - s_2 = 7$, $2x_1 + 5x_2 + s_3 \ge 0$
$x_1, x_2, x_3, s_1, s_2, s_3 \ge 0$.
3. Max. $Z = 3x_1 - 2x_2 + 5x_3 - 4x_4$; subject to
$x_1 + 2x_2 + x_3 - x_4 + s_1 = 8$, $2x_1 - x_2 + x_3 - s_2 - s_3 = 2$,
$-4x_1 + 2x_2 + 3x_3 - 5x_4 \ge 0$; $x_1, x_2, x_3, x_4, s_1, s_2, s_3 \ge 0$
4. (i) $x_1 = 2, x_2 = 1$ (Basic); $x_3 = 0$ (Non-basic) (ii) $x_1 = 3$,
$x_2 = -1$ (Basic); $x_3 = 1$ (Non-basic) (iii) $x_1 = 8/3, x_2 = 5/3$ (Basic) ; $x_3 = 0$ (Non-basic). All the three have solutions are non-negative.
5. Basic solutions are (i) $x_1 = 2, x_2 = 1$ (Basic) and $x_3 = 0$;
(ii) $x_1 = x_2 = 1$ (Basic) and $x_3 = 2$; (iii) $x_1 = -1, x_2 = 2$ (Basic) and $x_3 = 0$
(a) First two solutions are non-degenerate basic feasible solutions
(b) First solution is optimal and $Z_{max} = 5$.

Problems 3.5.2, page 1081

1. $x_1 = 2, x_2 = 3$, max. $Z = 11$
2. $x_1 = 0, x_2 = 20$; max. $Z = 200$
3. $x_1 = 700, x_2 = 400$; max. $Z = 30$
4. $x_1 = 5, x_2 = x_3 = 0$; max. $Z = 50$
5. $x_1 = 0, x_2 = 100, x_3 = 200$; max. $Z = 1200$
6. $x_1 = 8941, x_2 = 5941, x_3 = 6941$; max. $Z =$ ₹ 18,641
7. $x_1 = 1, x_2 = 6, x_3 = 0$; min. $Z = -31$
8. $x_1 = 20/13, x_2 = 0, x_3 = 20/13, x_4 = 100/13$; max. $Z = 220/13$
9. $x_1 = 0, x_2 = 400$ units ; max. profit = ₹ 1200
10. $x_1 = 125, x_2 = 250$ units ; max. profit = ₹ 2250
11. $x_1 = 10$ (post), $x_2 = 0$; min. cost = ₹ 3
12. $x_1 = 0, x_2 = x_3 = 30$; max. profit = ₹ 300

12. $x_1 = 2.5, x_2 = x_3 = 5.00$ units; min. cost = ₹ 5.00
14. Acreages for corn, wheat, soybeans are 250, 500, zero respectively to achieve a max. profit of ₹ 22,500

Problems 33.11, page 1033

1. $x_1 = 3, x_2 = 2, x_3 = 0$; max. $Z = 4$
2. $x_1 = 5, x_2 = 2, x_3 = 0$; max. $Z = 5$
3. $x_1 = x_2 = -5/3$; max. $Z = -10/3$
4. $x_1 = 25/3, x_2 = 5, x_3 = 0$; max. $Z = 65/3$
5. $x_1 = x_2 = x_3 = 3/4, x_4 = 0$; max. $Z = 15$
6. $x_1 = 21/13, x_2 = 10/13$; max. $Z = 31/13$
7. Infeasible
8. $x_1 = 10/3, x_2 = 5, x_3 = 0$; max. $Z = 40/3$
9. $x_1 = 55/7, x_2 = 20/7, x_3 = 0$; max. $Z = 135/7$
10. $x_1 = 2, x_2 = 5$; max. $Z = 16$
11. Degenerate solution ($x_1 = 0$ is not basic) ; $x_2 = 1, x_3 = 0$ (basic) ; max. $Z = 3$.

Problems 34.6, page 1091

1. Max. $W = 20y_1 + 5y_2$; subject to $6y_1 + 4y_2 \geq 30$
$5y_1 + 2y_2 \geq 15, 3y_1 + 6y_2 \geq 15$; $y_1, y_2, y_3 \geq 0$
2. Max. $W = 11y_1 + 9y_2 + y_3 + 5y_4$; subject to $2y_1 + 5y_2 - y_3 + 3y_4 \leq 5$,
$4y_1 + 3y_2 + 2y_3 + 3y_4 \leq 4, y_1 - 3y_2 + 3y_3 + 2y_4 \leq 3$;
$y_1, y_2, y_3, y_4 \geq 0$
3. Max. $W = -2y_1 + y_2 + 4y_3$; subject to $y_1 + 3y_2 - 2y_3 \leq -3$,
$y_1 + y_2 \geq 10, y_1 - 2y_2 + y_3 \leq -2$;
$y_1, y_2 \geq 0, y_3$ unrestricted in sign
4. Max. $W = -5y_1 + 3y_2 + 6y_3$; subject to $-2y_1 + 4y_2 - 6y_3 \leq 3$,
$3y_1 - 2y_2 + 6y_3 \leq -2, -y_1 + 3y_2 = 1$;
$y_1, y_2 \geq 0, y_3$ unrestricted
5. Min. $y = 3y_1 + 4y_2 + y_3 + 6y_4$; subject to $5y_1 - 3y_2 + y_3 - 3y_4 \geq 2$,
$6y_1 + y_2 - 5y_3 - 3y_4 \geq 6, -y_1 + 4y_2 - 3y_3 + 7y_4 \geq 6, y_1, y_2, y_3, y_4 \geq 0$.

Problems 34.7, page 1094

1. $x_1 = x_2 = 15, x_3 = 30$; max. $Z = 225$
2. $x_1 = 4, x_2 = 2$; max. $Z = 34$
3. $x_1 = 7, x_2 = 8$, max. $Z = 15$
4. $x_1 = 0, x_2 = 100, x_3 = 200$; max. $Z = 1200$

Problems 34.8, page 1097

1. $x_1 = 0, x_2 = 2$; max. $Z = -1$
2. $x_1 = 300, x_2 = 600$; max. $Z = 150$
3. $x_1 = 0, x_2 = 2, x_3 = 8$; min. $Z = 10$
4. $x_1 = 83/28, x_2 = 0, x_3 = 20/7, x_4 = 0$; min. $Z = 215/28$.

Problems 34.9, page 1101

1. $x_{11} = 200, x_{12} = 50, x_{21} = 175, x_{22} = 125, x_{31} = 325, x_{32} = 125$; min. cost = ₹ 15075
2. $x_{11} = 14, x_{12} = 6, x_{21} = 5, x_{22} = 3, x_{31} = 5$; min. cost = 143
3. $x_{11} = 50, x_{12} = 300, x_{21} = 150, x_{22} = 150, x_{31} = 300, x_{32} = 50$; min. tonnage = 1300
4. $x_{11} = 140, x_{12} = 60, x_{21} = 40, x_{23} = 150, x_{32} = 50$; min. cost = ₹ 9260
5. $x_{11} = 6, x_{12} = 2, x_{23} = 8, x_{24} = 7, x_{34} = 6, x_{35} = 13$;
min. cost = ₹ 705 and maximum saving = ₹ 281
6. $x_{11} = 150, x_{12} = 30, x_{23} = 180, x_{24} = 40, x_{32} = 50, x_{33} = 90$; max. profit = 4530
7. $x_{11} = 2, x_{12} = 1, x_{23} = 3, x_{31} = 4, x_{32} = 1$; min. cost = 35
8. $x_{11} = 0, x_{12} = 800, x_{21} = 600, x_{23} = 300, x_{31} = 400, x_{33} = 200$,
$x_{34} = 300, x_{35} = 300$; min. cost = 8500

Problems 33.10 (page 1108)

1. $x_{23} = x_{11} + x_{22} = 5$; max. cost = £ 18
2. $A \to 2, B \to 3, C \to 4, D \to 1$; min. $Z = 33$
3. $I \to B, N \to A, M \to D, V \to C$; min. cost = £ 50
4. $A \to$ Bus; Paving, $N \to$ Queuing Th., $C \to$ Reg. Analysis, $D \to$ L.P.; min. time = 38 hrs
5. (i) $A \to 1, B \to 3, C \to 2, D \to IV$; (ii) $A \to 1, B \to 3, C \to 4, D \to IV$
6. $1 \to IV, 2 \to II, 3 \to VI, 4 \to I, 5 \to III, 6 \to V$; max. profit = £ 320

Problems 33.11 (page 1110)

1. § 34.8 Def. 2
2. It provides an optimality test
3. § 34.13
4. § 34.15 (1)
5. § 34.18
6. § 34.6 (1)
7. Max. $W = 3y_1 + 4y_2$, subject to $2y_1 + 5y_2 \le 4$,
   $3y_1 - 2y_2 \le 6, 3y_1 + 4y_2 \le 3, y_1 \ge 0, y_2$ is unrestricted in sign
8. § 34.12 (2)
9. § 34.14
10. Maximise $Z = (2x_{11} + 3x_{12} + 11x_{13} + 4x_{14}) + (5x_{21} + 6x_{22} + 8x_{23} + 7x_{24})$,
    subject to $x_{11} + x_{12} + x_{13} + x_{14} = a_1$ (= 18), $x_{21} + x_{22} + x_{23} + x_{24} = a_2$ (= 20),
    $x_{11} + x_{21} = b_1$ (= 10), $x_{12} + x_{22} = b_2$ (= 8), $x_{13} + x_{23} = b_3$ (= 12), $x_{14} + x_{24} = b_4$ (= 8) and $x_{ij} \ge 0$.
    IV: $2b_1 - 2b_2 = 60$
11. (i) $x_1 = 3, x_2 = 5, x_3 = 0$; (ii) $x_1 = 0.6, x_2 = 0, x_3 = 2.5$
12. § 34.8 Def. 4)
13. § 34.13
14. balanced
15. § 34.8
16. § 34.5 (2)
17. optimal
18. Maximise $y = 5y_1 - 3y_2$, subject to $y_1 + y_2 = 5, 2y_1 - 5y_2 \le 6, y_2 \ge 0$ and y_1 unrestricted
19. 5
20. Max. $Z = 870$
21. § 34.7
22. § 34.16
23. § 34.10
24. Max. $W = 3y_1 + 4y_2 + 2y_3$, subject to $-y_1 + y_2 + y_3 \le 3, 2y_1 + y_2 + 3y_3 \le 4, y_1, y_2, y_3 \ge 0$
25. North west corner rule and Vogel's approximation method
26. Slack or surplus variables

Problems 33.1 (page 1113)

1. (i) $y = \frac{1}{4}x^2 + c_1x + c_2$; (ii) $y = c_1x^{-1} + c_2$
2. $y = x \sin x + 2$
3. $y = (x + c_1)\sin x$
4. $y = c_1x^2 + c_2x^{-1} + \frac{2x^2}{7}$
5. $y = x^2 - 1$
6. $(x + c_1)^2 + y^2 = 4^2$
7. $y = c_1 \sinh(x + c_2)$
8. $y = 2\sin x$
9. The family of the family $y = c_1 \sin(x + c_2) \sin x + c_3$

Problems 33.2 (page 1120)

1. $y = x + 2n\pi$, where n is an integer
2. $y = Ax^2 + Bx + 6$, where A, B, C are determined from the isoperimetric and boundary conditions.
3. $y(x) = \frac{1}{2}(1 - \cos x) + \frac{1}{4}(2 - e)\sin x$

Problems 33.3 (page 1124)

2. $y = \sin x_1 + c_2(1 + \cos x) + c_3 + c_4 \cos x$,
   $z = 4c_1 + c_2(1 + \cos x) + c_3 + c_4 \cos x - 2c_1 \sin x + 2c_2 \cos x$
3. $y = c_2 \sin(n\pi x)$, $n = 1, 2, 3 \ldots$
4. $y = \cos x$
5. $y = -\frac{\lambda}{2\mu_0}(x^2 - a^2)$

6. (a) $y = c_1 y^{(1)} + c_2 y^{(2)} + c_3 \cos 2x + c_4 \sin 2x$;
(b) $y = c_1 + c_2 y + c_3 y^2 + c_4 y^3 + c_5 y^4 + c_6 y^5 + y^{(1)}$.

Problems 10.4, page 1026

1. (a) and (b) $y = \frac{5}{16} x(1 - x)$ 3. $y = \frac{3}{4}(3x - 1)$ 5. $y = 0.05 + 0.27x$
6. (a) and (b) $r_1 = 0.618, r_2 = -3.23$ 8. (0.0) 7. $r_1 = 0.27, r_2 = -3.69$
9. $y = \frac{1}{2}(6x^2 - 2x)$.

Problems 10.5, page 1034

1. $y(x) = 6x - 5 + \int_0^x (x - t)(6t + 8)\, y(t)\, dt$.
2. $y(x) = \cos x + x^2(x - 1)x - \int_0^x [\sin(x - t) - t^2(x - t)]\, y(t)\, dt$
3. $y(x) = \int_0^x (x - t) y(t)\, dt + \frac{1}{2} x^2$
4. $y(x) + \int_0^x [1 + x - 2x + (x - t)e^{x-t}] y(t)\, dt = \frac{x^5}{20} - \frac{5x^3}{6} + x - 3$
5. $y(x) = \cos x = \frac{1}{2} x^2 - \frac{1}{8} x^4 - \frac{3}{2} \int_0^x (x - t)^2 y(t)\, dt$
6. $y(x) = \sin x = -\frac{3}{2} x + \int_0^x \left[\frac{1}{12}(x - t)^3 - 1\right] y(t)\, dt$
7. $y(x) = \int_0^x [8 - 6(x - t) + 2(x - t)^2 - \frac{1}{3}(x - t)^3] y(t)\, dt = \frac{1}{4}(\cos 2x - \frac{28}{32} + \frac{23}{8} x - \frac{50}{8} x^2 + \frac{20}{3} x^3$
8. $y''(x) - 6y'(x) - 3y(x) = 0$; $y(0) = 1, y'(0) = 0$
9. $y''(x) - y(x) + 2\sin x = 0$; $y(0) = 2, y'(0) = 0$
10. $y'''(x) + 6y(x) = 0$; $y(0) = 4, y'(0) = -3, y''(0) = 2$
11. $y^{(4)}(x) - 3y'''(x) + 6y''(x) + 3y'(x) + e^{-x} = 0$; $y(0) = 1, y'(0) = 2, y''(0) = 3$.

Problems 10.6, page 1050

1. $y(x) = \int_0^x G(x, t) f(t)\, dt + \frac{2}{3} x(x - 1)$, where $G(x, t) = t(1 - x)$ and $= x(1 - t)$, $x > t$
2. $y(x) = \int_0^x G(x, t) \cos t\, dt + \frac{1}{8} x^2 - 2x + 1$, where $G(x, t) = t - x$ and $= x$, $x > t$
3. $y(x) = \int_0^x G(x, t)\, e^t\, dt + \frac{3}{8} e^x + x$, where $G(x, t) = \frac{1}{4}(3 - 2t)(x + 1)$ and $= \frac{1}{4}(3 - 2x)(t + 1)$, $x > t$
4. $G(x, t) = \frac{\sinh x \sinh(t - 1)}{\sinh 1}$, $x < t$ and $= \frac{\sinh t \sinh(x - 1)}{\sinh 1}$, $x > t$
5. $y(x) = \lambda \int_0^x G(x, t)\, t e^t\, dt$, where $G(x, t) = \frac{1}{\pi} \sin\left(\frac{1}{\pi} x\right)$, $x < t$ and $= \frac{1}{\pi} \sin\left(\frac{1}{\pi} t\right)$, $x > t$

Problems 10.7, page 1061

2. $y(x) = \frac{1}{2} \ln(\sec x + \tan x)$ 4. $y(x) = x^2 + \frac{2}{12} x^4$ 7. $y(x) = 1$
8. $y(x) = x \tanh(x)$ 9. $y(x) = x^2, (2x^2 \ \ (x > 0)$

12. $y(x) = \frac{3}{2}e^{2x} + \cos x + \frac{1}{2}\sin x - \frac{3}{2}e^{-x}$ 13. $y(x) = 1 - x\ln x^2 + 1.5x\sin x$

14. $y(x) = 1 + x^2 + x^2\ln x$ 15. $y(x) = 1 - x^2/2$ 16. $y(x) = 1/2$

17. $y(x) = \frac{3\sqrt{3}}{4\pi}x^{1/2}(2x+3)$

Problems 30.4, page 1143

1. Eigenvalues and eigenfunctions
2. Eigenvalue $\lambda = 1/4$; eigenfunction is $y(x) = x^{1/2} + 2x^{3/2}$
3. Eigenvalue $\lambda = 4(2n-1)^2$; eigenfunction is $y(x) = \sin^2(x)$
4. Eigenvalue $\lambda = 1/e$, $y = x\ln x$ 5. Eigenvalues or eigenfunctions
6. Eigenvalues are $\lambda = n\pi$; eigenfunctions are $y(x) = \cos nx - \sin nx$, $y(x) = \cos nx - \sin nx$

7. $y(x) = x + \frac{3.18 - 3.1x - 4I}{18 + 3^2}$ 8. $y(x) = x + \frac{1}{12(11 - 30) - 1^2}\cdot 11(8 + 13 + 6)x$

9. $y(x) = x + \frac{\frac{3}{4}}{6 - 2\pi\left(1 - \frac{1}{2}\cdot 3\pi\right) + 4\lambda^2}\left[2\pi x + \frac{1}{2}\pi^2\left(1 - \frac{1}{2}\cdot 6\right) + \pi(1 - 3x)(\cos x\right]$

10. $y(x) = 2x - x + \frac{\pi^2\sin^2 x}{\pi - 3}$ 11. $y(x) = \frac{3}{3 - x}\cdot 4\ln x, \lambda = 3$

12. $y(x) = x + \frac{20\pi}{1 + 20^2\pi^2}\cdot 0.4x - 4x\sin x + \cos x$

13. There is no solution to the integral equation when $\lambda = 3$

14. $\lambda_1 = 3$, $\lambda_2 = -2$; $y_1(x) = 1 - x$, $y_2(x) = 1 - 2x$

15. (a) When $f(x) = x$, solution is $y(x) = x + \lambda\left(\frac{2\pi^2}{3\pi^2 - 1}\sin\pi x + \frac{2\pi}{3\pi^2 - 1}\cos\pi x\right)$

(b) When $f(x) = 1$, solution is $y(x) = 1$.

Problems 30.5, page 1149

1. $y(x) = 1 - \frac{2\pi x}{2\pi + 2}$, $|\lambda| < 3$ 8. $y(x) = \frac{\pi\ln x}{1 + \lambda\pi}$, $\sin(\ln x) \neq 1$, $\lambda \neq \frac{1}{\pi}$

3. $y(x) = \frac{1 + 2\lambda(2 - 3\pi)}{4 - \lambda^2}$, $|\lambda| < 2$ 4. $y = x^2$

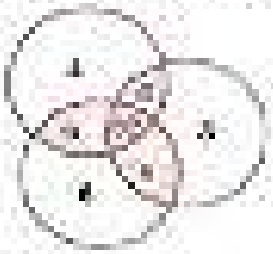
5. $y(x) = 1$ 6. $y = \sin x$ 7. $y(x) = 2$

Problems 30.6, page 1154

2. (a) True, since $\{a\}$ is a subset of the set $\{a, b, c\}$. (b) not true. False, since the element a cannot be a subset of the set $\{a, b, c\}$. (c) True, since the set $\{a, b\}$ is a subset of the set $\{a, b, c\}$. (d) False, since the set $\{a, b\}$ is not an element of the set $\{a, b, c\}$. (e) True, since the null set φ is a subset of every set.

17. 23 18. 300 19. 126

20. Number of students not taking any of these courses is 31.



Problems 37.2, page 1160

1. (a) It is not true that Sam is a teacher and John is an honest boy; (b) Sam is a teacher and John is not an honest boy; (c) Sam is not a teacher if John is an honest boy; (d) If Sam is a teacher then John is not an honest boy.

2. (a) $(p \vee q) \wedge r$ where p = I have no car, q = I do not want good drive, r = I am sad, (b) $r \wedge (p \vee q)$

3.

p	q	$\neg q$	$p \vee q$	$p \vee (\neg q)$
T	T	F	T	T
T	F	T	T	T
F	T	F	T	F
F	F	T	F	T

(b)

p	q	r	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	F	T	F
F	F	F	F	F	F

7. (a) $T_p = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31\}$
$T_q = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29\}$
(b) $T_r \subseteq T_q$

10. (a)

p	q	$p \wedge q$	$(p \wedge q) \vee p$
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F

13. (a) Double negative law (b) Double negative law (c) Double negative law

Problems 37.3, page 1168

1. (a) $\forall x \in A (x + 2 < 13) \wedge (\exists x \in A)(x + 2 = 20)$

2. (a) $\forall x, (x^2 \neq x)$ (b) $\neg x, (x + 5 < x)$
(c) None of the students are 28 or older (d) Some circles do not lie in the hemisphere

3. $\forall x$ $P(x)$ is false 4. $\forall (x_1, x_2)\, Q(x_1 + x_2)$

5. $\forall x_1, \exists Q (x_1 + x = 0)$

6. (a) $\forall x \exists Q (x) \rightarrow P(x)$ (b) $\neg \forall x (Q(x) \rightarrow P(x))$ (c) $\exists x (Q(x) \wedge P(x))$ (d) $\exists x (Q(x) \wedge \neg P(x))$

8. $(\neg A \vee \neg B) \wedge (A \vee B) \vee (A \wedge \neg A) \wedge (\neg A \vee C) \wedge (B \vee C)$

10. 1. $p \vee q$ (Premise),
2. $\neg p \rightarrow q$ (conditional equivalence)
3. $q \rightarrow r$ (Premise)
4. $\neg p \rightarrow r$ (2, 3 chain rule)
5. $p \vee r$ (Premise)
6. $\neg r \rightarrow p$ (4, conditional equivalence)
7. $\neg r \rightarrow p$ (6, double negation)
8. $r \vee r$ (5, conditional equivalence)

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12. (a) $\exists x (R(x) \wedge \neg Q(x))$ 13. (a) Conclusion is not valid (b) Conclusion is not valid

Problems 3.2.4, page 3.130

1.

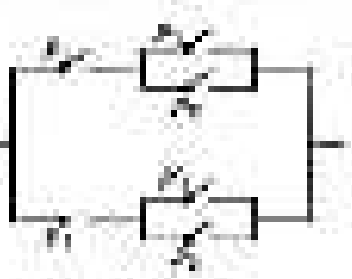
3.

Problems 3.2.5, page 3.132

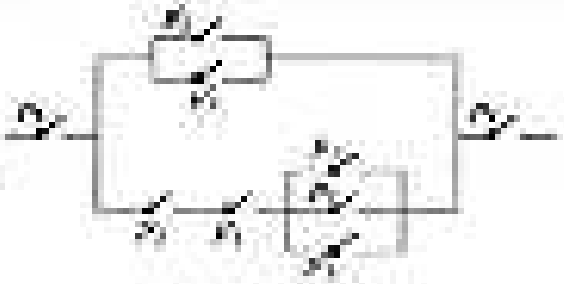
1. (a) (b)
3. (a) (b)
4. (a) (b)
6. (a) (b)

Problems 3.2.6, page 3.134

1. (a)



(b)



3.

8. (a) $x_1 \wedge x_2'$ (b)

(c) $(x_1 \vee x_2' \vee x_3) \wedge (x_1 \vee x_2' \vee x_3') \wedge (x_1 \vee x_2' \vee x_3) \wedge (x_1 \vee x_2' \vee x_3')$

Problems 3.3.3 (page 115)

1. $F_0 = F_2$ 2. F = {0.2 Ram, 0.3 Shan, 0.4 John, 1 Hari}
3. $F = \{0.3/x_1, 0.6/x_2, 0.5/x_3, 0.9/x_4\}$
   $G = \{0.3/x_1, 0.6/x_2, 0.1/x_3, 0.8/x_4\}$
4. (a) Truth value of 'Ram is not rich' is 0.2
   (b) Truth value of 'Hari is not fat' is 0.4
   (c) Truth value of 'Many is not beautiful' is 0.3
5. (a) $F \neq G$ (c) F is not a subset of G, G is not a subset of F.
   (d) $F = \{1, 1, 1, 1, 0.9, 0.7, 0.5, 0.3, 0.0\}$
   $F' = G = \{0, 0.1, 0.3, 0.5\}$
   $F \cup G = \{1, 1, 0.9, 0.5, 1, 0.9, 0.5, 1, 1\}$
6. (a) Truth value of the statement of 'Latif and John are good players' is 0.6.
   (b) Truth value of the statement of 'Latif and John are good players' is 0.7.
8. Membership and its degree of membership.

Problems 3.3.4 (page 120)

1. (i) $\frac{\partial \tilde{A}}{\partial x'} = \frac{\partial \tilde{A}}{\partial x^1} \frac{\partial x^1}{\partial x'}$; (ii) $x^i x^j$

2. (i) $a_{11}(x^1)^2 + a_{12}x^1x^2 + a_{21}x^2x^1 + a_{22} + a_{13}x^1x^3 + a_{31}x^3x^1 + a_{23}x^2x^3 + a_{32}x^3x^2 + a_{33}x^3x^3$
   (ii) $g_{11}\dot{x}^1\dot{x}^1 + g_{12}\dot{x}^1\dot{x}^2 + g_{21}\dot{x}^2\dot{x}^1 + g_{22}\dot{x}^2\dot{x}^2 + g_{13}\dot{x}^1\dot{x}^3 + g_{31}\dot{x}^3\dot{x}^1 + g_{23}\dot{x}^2\dot{x}^3 + g_{32}\dot{x}^3\dot{x}^2 + g_{33}\dot{x}^3\dot{x}^3$.
   (iii) $g_{ij} = g_{ij}x^ix^j + \cdots + g_{ij}x^ix^j$

3. (i) ξ_1 ; (ii) ξ_1^2 6. (a) $A_{jk}^{i'} = \frac{\partial x^{i'}}{\partial x^i} \frac{\partial x^i}{\partial x^j} \frac{\partial x^j}{\partial x^{j'}} A_{jk}^i$; (b) $C_{jk} = \frac{\partial x^{i'}}{\partial x^i} \frac{\partial x^i}{\partial x^j} C_{ik}$

7. Yes, $A_{jk}^{i'}$ contravariant order 1, covariant order 2, Rank 3.
8. (a) $2p\cos^2\phi - r\cos\theta + p^2\sin^2\phi\cos^2\theta - p^2\sin 2\theta + p\sin\phi + p^2\sin\theta\cos^2\theta, p\sin\phi$;
   (b) $3r^2\sin^2\theta\cos^2\phi - r\sin\theta\cos\theta\cos\phi + r^2\sin^2\theta\sin^2\phi\cos^2\phi - r^2\sin\theta\cos^2\theta\sin\phi$;
   $3r^2\sin\theta\cos\theta\cos^2\phi - r^2\cos^2\theta\cos\phi + r^2\sin^2\theta\cos\theta\sin^2\phi\cos^2\phi - r^2\sin^2\theta\cos\theta\sin\phi$;
   $-3r^2\sin^2\theta\sin\phi\cos\phi + r^2\sin\theta\cos\theta\sin\phi + r^2\sin^2\theta\sin\phi\cos^2\phi$
11. $\lambda = \cos\alpha + \sin\alpha\sin\phi\sin\theta + \cos\alpha\cos\theta$; $[\lambda\cos\phi + \lambda\sin\phi)\cos\theta - r\sin\theta]^2$; $\lambda\cos\phi - \alpha\sin\phi)^2\sin\theta$.

Problems 3.3.5 (page 123)

8. Rank = 1

Problems 3.3.6 (page 126)

1. $a = 4, g^{11} = 0, g^{12} = 0, g^{13} = 1.5, g^{21} = 3, g^{22} = -2.5, g^{23} = -1.5$
2. $g_{11} = 1, g_{12} = r^2, g_{13} = 1, g_{21} = 0$; $g^{11} = 1, g^{12} = -r^{-2}, g^{13} = 1, g^{21} = 0$; $\lambda = 2$
3. $g = r^4\sin^2\theta(1 - r^2\cos^2\theta)$; $g^{11} = 1 - r^2\cos^2\theta, g^{12} = 0, g^{13} = r^2\sin\theta\cos\theta, g^{21} = 0$; $\lambda = 2$

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Higher Engineering Mathematics

Dr. B.S. Grewal

The book provides a clear exposition of essential tools of applied mathematics from a modern point of view and meets complete requirements of engineering and computer science students. Every effort has been made to keep the presentation at once simple and lucid. It is written with the firm conviction that a good book is one that can be read with interesting guidance from the instructor. To achieve this, more than the usual number of solved examples, followed by properly graded problems have been given. Many of the examples and problems have been selected from recent papers of various university and other engineering examinations. Basic Concepts and Useful Information has been given in an Appendix. However, the subject matter has been set in eight main units :

- *Algebra & Geometry* : Solution of equations, Linear algebra, Determinants, Matrices, Vector algebra and Solid geometry.
- *Calculus* : Differential calculus, Partial differentiation, Integral calculus, Multiple integrals, Vector calculus.
- *Series* : Infinite series and Fourier series.
- *Differential Equations* : Differential equations of first order and their applications, Linear differential equations and their applications, Differential equations of different types, Series solution of differential equations and special functions, Partial differential equations and their applications.
- *Complex Analysis* : Complex numbers and functions, Calculus of complex functions.
- *Transforms* : Laplace transforms, Fourier transforms and Z-transforms.
- *Numerical Techniques* : Empirical Laws and Curve fitting, Statistics, measures, Probability and Distributions, Sampling and Inference, Numerical methods, Finite differences and Interpolation, Difference equations, Numerical solution of Ordinary and Partial differential equations, Linear programming.
- *Special Topics* : Calculus of variations, Integral equations, Boundary value problems, Tensors.

An exhaustive list of Objective Type of Questions has been given at the end of each chapter. Standard Tables, Answers to Problems, and a fairly comprehensive Index is given at the end.

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